Materials Science and Engineering

Structure, Thermodynamics, Kinetics, Properties, Processing and Design

The whole discipline: why materials have the structures they have, how thermodynamics and kinetics set them, how structure produces properties, and how materials are processed, characterized, modelled and selected.

The Studio of Sadiq · Materials Science and Engineering

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Part I — Structure of Materials

Every engineering property starts with the arrangement of atoms. A steel plate that stays tough in an Arctic winter, a single-crystal turbine blade that survives gas hotter than its own melting point, a zirconia hip head that resists cracking, a polyethylene pipe that holds pressure for fifty years and a piezoelectric ceramic that turns a voltage into ultrasound all owe their behaviour to structure: which atoms are present, how they are bonded, how they are arranged in crystals, glasses or chains, which imperfections that arrangement contains, and how grains and phases fill the component. Part I builds the language for describing that structure and the tools for reasoning about it. It introduces the central idea of the discipline, that **processing sets structure, structure sets properties, and properties set performance**, and then works down from the whole field to the atom and back up to the microstructure.

By the end of this part you should understand:

- what distinguishes materials science (explaining properties from structure) from materials engineering (designing structure through processing), and how the structure–processing–property–performance tetrahedron organizes both;
- how structure is distributed over length scales from bonds to components and over time scales from atomic vibrations to design lives, and why thermal activation links them;
- the property packages of metals, ceramics, glasses, polymers, composites, semiconductors and biomaterials, and how bonding explains them;
- what isotropy, anisotropy and homogeneity mean, and how property charts and performance indices turn a design requirement into a ranking of materials, with cost, criticality and embodied carbon as additional objectives;
- the main crystal structures of metals, ceramics, semiconductors and silicates, Pauling's rules, polymorphism, and how to name directions and planes with Miller and Miller–Bravais indices;
- how polycrystals, texture and grain size, glasses, semicrystalline polymers, liquid crystals and quasicrystals depart from the perfect single crystal;
- the symmetry operations, point groups, Bravais lattices and space groups of crystals, how to read the International Tables, and how stereographic projections and pole figures display orientation;
- the reciprocal lattice and diffraction, structure factors and peak broadening, and how symmetry constrains tensor properties such as conductivity, piezoelectricity and elasticity;
- the families of defects, from vacancies and dislocations to grain boundaries and precipitates, how they are quantified, and why they control the structure-sensitive properties.

Chapter	Central question	What it delivers
1. The Materials Discipline	What is the materials discipline, and how is one material chosen rationally over another?	The tetrahedron; length and time scales; the material classes with typical property values; bonding as the origin of the classes; anisotropy and effective properties; performance indices and property charts; economics, criticality and sustainability
2. Crystal Structures and Non-Crystalline Solids	How are atoms arranged in metals, ceramics, silicates, glasses and polymers, and how are directions and planes named?	FCC, BCC, HCP and stacking; ionic and covalent structure types and Pauling's rules; silicates and polymorphism; Miller and Miller–Bravais indices with linear and planar densities; polycrystals, texture and grain size; glasses and radial distribution functions; previews of polymer crystallinity, liquid crystals and quasicrystals
3. Symmetry and Space Groups	Which symmetries can a crystal have, and what do they imply for properties and transformations?	Symmetry operations; the 32 point groups and 7 crystal systems; the 14 Bravais lattices; the 230 space groups and Wyckoff positions; stereographic projection and pole figures; symmetry breaking and variants; ordering and antiphase boundaries; quasicrystals and incommensurate structures
4. Reciprocal Space, Diffraction and Tensor Properties	How does diffraction reveal structure, and how does symmetry shape directional properties?	The reciprocal lattice and Brillouin zones; Bragg, Laue and Ewald; structure factors, absences and intensities; size and strain broadening; Neumann's principle and tensors of rank 2, 3 and 4; why cubic crystals are isotropic in conduction but not in elasticity
5. Defects and Microstructure	What imperfections do real materials contain, and how do they control properties?	Point defects and their equilibrium concentration; defect chemistry in ionic crystals; dislocations; grain, twin and phase boundaries; pores, inclusions and precipitates; quantitative stereology; an overview of defect-controlled properties

Part I is the foundation for everything that follows. Part II (thermodynamics and kinetics) explains *why* a given structure forms and how fast it changes. Part III builds on slip planes, grain size, dislocations and ordering to explain strength, creep and fracture. Parts IV and V apply the same language to metals, ceramics, cement, polymers and composites; Parts VI and VII use symmetry and tensors for thermal, electrical, magnetic and optical properties; Parts VIII and IX extend it to advanced materials and degradation. Part X shows how processing sets these structures, and Part XI returns to them through characterization, computation and selection.

Chapter 1 — The Materials Discipline

Part I · Structure of Materials

On 16 January 1943 the tanker SS *Schenectady*, newly completed and lying at her fitting-out pier in Portland, Oregon, broke in two with a noise heard a mile away. The sea was calm, the ship was unloaded and the air was a little below freezing. She was one of several thousand Liberty ships and T2 tankers built in the United States during the Second World War, the first large ships assembled by welding rather than riveting, and over the following years hundreds of them developed serious fractures; a dozen or so broke completely. The steel met its specification for tensile strength. What nobody had specified was how that steel behaved at a few degrees below zero with a notch in it. Plain carbon steels of that era, with high sulphur and phosphorus and a low manganese-to-carbon ratio, change from tough to brittle over a narrow temperature range that happened to sit near the temperature of the North Atlantic. Welding made each hull one continuous piece of metal, so a crack that started at a sharp hatch corner or a weld defect had no rivet seam to stop it. The cure came from three directions at once: chemistry and grain size that pushed the ductile–brittle transition below service temperatures, a design that removed sharp corners and added crack-arresting strakes of tougher steel, and a test (the notched-bar impact test, now routine) that measured the property that had been missing.

The *Schenectady* is the materials discipline in one story. A material is not a name or a single number on a datasheet; it is a structure, set by its chemistry and by how it was made, and that structure fixes a whole family of properties at once. Performance in service depends on all of them, including the ones nobody checked.

> **What is the materials discipline, and how does it connect the arrangement of atoms to the performance of a bridge, an aircraft or an implant?**
> **How do we organize the tens of thousands of engineering materials so that one can be chosen rationally over another, and what limits that choice beyond the properties themselves?**

Concept	Engineering destination
Science versus engineering of materials; the history of materials	Reading specifications critically; recognizing that each era's limiting technology was a material
Structure–processing–property–performance tetrahedron	Failure analysis (Chapter 44); process specifications (Chapters 46–49)
Length and time scales	Choosing the right model and test; multiscale simulation (Chapter 56)
Material classes and property ranges	First screening of candidates for any component
Bonding as the origin of property classes	Predicting stiffness, melting point and expansion from chemistry
Isotropy, anisotropy, homogeneity	Rolled sheet, wood, fibre composites, single-crystal blades
Property charts and selection logic	Systematic material selection (Chapter 58)
Economics, criticality, sustainability	Supply-chain risk, cost, embodied carbon, circular design (Chapters 40, 58–59)

Roadmap. What the discipline is and how it grew (1.1) → its organizing idea, the tetrahedron (1.2) → length and time scales (1.3) → material classes and their property ranges (1.4) → bonding as the reason the classes differ (1.5) → direction dependence and spatial uniformity (1.6) → property charts and selection (1.7) → materials in the economy and the environment (1.8).

1.1 Materials science and materials engineering

Two questions about the same objects

Materials science asks *why*. Why does steel become brittle in the cold while aluminium does not? Why is a single-crystal turbine blade more creep resistant than a cast blade of the same alloy? Why does a ceramic hip head survive millions of cycles in the body when a glass of similar stiffness would shatter? The answers run from the arrangement of atoms and electrons to a

measurable property. The science borrows from chemistry the nature of the bond and the thermodynamics and kinetics of change, and from physics the behaviour of electrons and lattice vibrations in a solid. What it adds is attention to the *real* arrangement of atoms, including its imperfections, and to how that arrangement changes during processing and in service.

Materials engineering asks *how*. How do we make a steel plate for a ship hull that is tough at −40 °C and still weldable? How do we cast a nickel alloy blade as one crystal, with internal cooling passages, at an acceptable scrap rate? How do we make a concrete that reaches 100 MPa yet still flows into congested formwork? The engineer uses the explanations of science to design a structure, then designs a manufacturing route that produces that structure reliably and cheaply.

The jet-engine turbine blade shows the two halves working together. The *science* was the discovery that nickel alloys gain their high-temperature strength from a dense dispersion of ordered $Ni_3(Al,Ti)$ precipitates (the γ′ phase), and that creep at high temperature proceeds largely along grain boundaries, especially those lying across the stress. The *engineering* was a sequence of processing routes that removed those boundaries: conventionally cast equiaxed blades, then directionally solidified blades with columnar grains along the blade axis, and from the early 1980s single-crystal blades with no grain boundaries at all. Each step added a few tens of kelvin of allowable metal temperature; together with ceramic thermal-barrier coatings and internal cooling, they let turbine-entry temperatures rise from about 1100 K in early jet engines to over 1900 K today, well above the melting point of the blade alloy itself (Chapters 10, 16 and 41).

A datasheet value is an engineering fact about one grade made by one route; a scientific explanation tells you which way it moves when the grade or route changes.

> **Boundary.**
> The types of chemical bond, the laws of thermodynamics and reaction chemistry belong to the Advanced Chemistry book; the quantum mechanics of electrons in solids belongs to Engineering Physics. This book uses those results, recapping them where needed, and concentrates on how the arrangement of atoms in a solid, and its imperfections, produce engineering properties. Parts VI and VII treat electrical, magnetic, semiconductor and photonic materials in depth.

A short history of materials ages

Historians name the early ages of humanity after their dominant material, and the names are well chosen: in each age, what people could build was limited by what they could make materials do.

- **Stone and natural materials.** Flint, obsidian, wood, bone, hide and fibre. Obsidian is a natural volcanic glass; with no crystal planes it fractures conchoidally and gives the sharpest edge of any natural material. Fired clay pottery, the first synthetic ceramic, appears around 18 000 years ago in East Asia.
- **Copper and bronze** (from about 5000 and 3300 BCE in the Near East). Copper was smelted from its ores and alloyed, first with arsenic and then with tin. Bronze (Cu with about 10 % Sn) is harder than copper and casts better because tin lowers the melting point: the first deliberate alloy design.
- **Iron** (from about 1200 BCE). Iron ore is far more abundant than copper and tin, but iron melts at 1538 °C, beyond ancient furnaces, so it was reduced in the solid state and forged. Smiths learned empirically that carburizing, quenching and tempering made iron hard: heat treatment, the control of structure by processing, three thousand years before anyone could see a microstructure.
- **Steel, cement and the industrial age** (from the nineteenth century). Portland cement (1824) and later reinforced concrete made the modern city; the Bessemer converter (1856) made cheap tonnage steel, and with it railways, ships and skyscrapers. The Hall–Héroult process (1886) turned aluminium from a precious metal into a structural one, and the discovery of age hardening in Al–Cu alloys (Alfred Wilm, 1906) made metal aircraft frames possible.
- **Polymers** (from 1907). Bakelite, the first fully synthetic plastic, was followed by polyethylene and nylon in the 1930s and the explosion of commodity plastics after 1950, from about 2 million tonnes a year in 1950 to more than 400 million tonnes today.
- **Silicon and electronic materials** (from 1947). The transistor and the integrated circuit depend on single crystals of a purity never needed before (Chapter 32).
- **Composites and designed materials** (from the 1960s). Carbon and aramid fibres, superalloys and single-crystal blades, high-performance concrete, biomaterials designed to be accepted or resorbed by the body, and materials designed atom by atom by computation (Chapters 55–57).

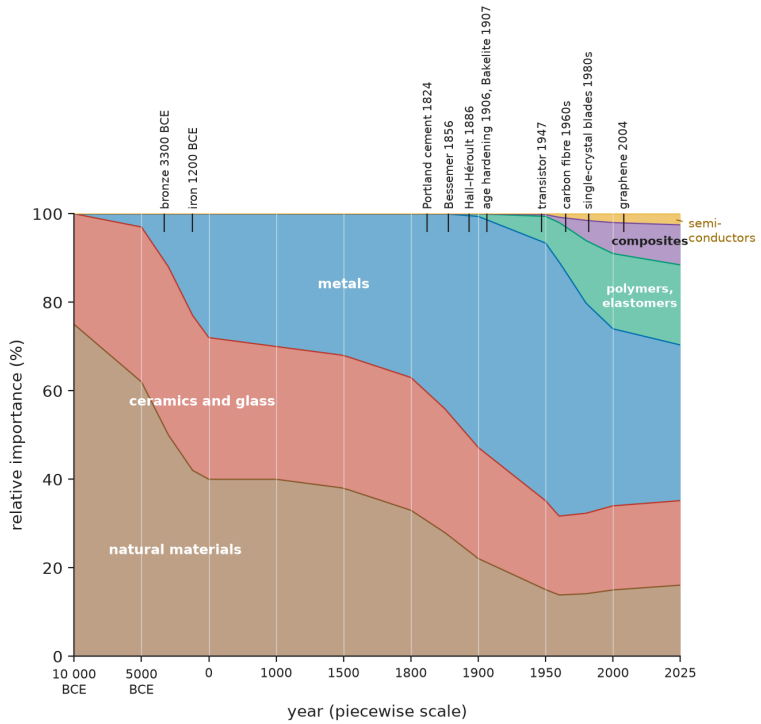

Figure 1.1 — The relative importance of material classes over ten thousand years of engineering, with milestones marking the processes that changed it.

Figure 1.1 sketches the changing mix. Metals peaked in the mid-twentieth century; since then polymers, composites and technical ceramics have taken share, while steel and concrete remain by far the largest materials by tonnage. The problem has shifted from finding *a* material that works to choosing the *best* one under many constraints (Sections 1.7–1.8).

1.2 The structure–processing–property–performance tetrahedron

The four elements

- **Structure** is the arrangement of the material's constituents at every length scale: which atoms are present and how they are bonded; how they are arranged in a crystal or a glass; which defects that arrangement contains; how grains and phases are arranged in space; and the shape of the component.
- **Properties** are the material's responses to stimuli. A stress produces a strain (elastic modulus) or permanent deformation (yield strength); a temperature gradient produces a heat flow (thermal conductivity); a corrosive environment produces a reaction rate. A property is always a *relation between a stimulus and a response*, which is why it can depend on direction (Section 1.6).
- **Processing** is everything done to give the material its shape and structure: solidification, rolling and forging, heat treatment, sintering, polymerization and moulding, curing of concrete, deposition of thin films, joining, additive manufacturing. Processing is the engineer's control knob.
- **Performance** is the behaviour of the material *in a specific application*: the deflection of a floor beam, the number of flights a turbine disc survives, the wear of a hip-joint liner after twenty years, the life of a bridge deck exposed to de-icing salt. Performance combines properties with geometry, loading and environment.

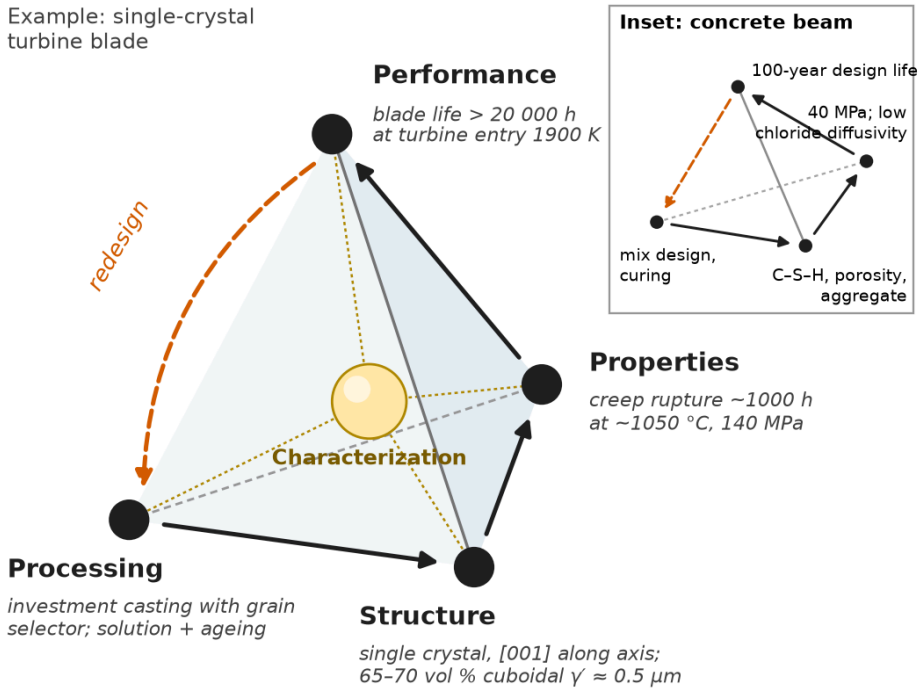

Figure 1.2 — The materials tetrahedron: processing, structure, properties and performance, with characterization at the centre, illustrated by a single-crystal turbine blade.

The chain and its direction

The four ideas are usually drawn as a tetrahedron (Figure 1.2) or as a chain:

$$\text{processing} \longrightarrow \text{structure} \longrightarrow \text{properties} \longrightarrow \text{performance}$$

The chain has a definite direction of cause. Processing does not act on properties directly; it acts by changing structure. Quenching a steel does not "make it harder" by magic; it traps carbon in a distorted crystal structure (martensite) that resists dislocation motion. That is why a single structural change usually moves several properties at once, often in opposite directions from the designer's point of view. Cold-rolling an aluminium sheet multiplies its dislocations, doubling its yield strength but reducing its formability. Adding glass fibres to nylon triples its stiffness but makes it anisotropic and abrasive to moulding tools. Lowering the water–cement ratio of concrete raises its strength and its resistance to chloride penetration, but makes it harder to place and more prone to early-age shrinkage cracking. An engineer who asks for "more strength" is, without saying so, also asking for different ductility, toughness and manufacturability.

Design runs the chain backwards: from the performance required to the properties needed, to a structure that gives those properties, to a processing route that produces that structure. The step from properties to structure is the hard one, because many structures give the required value of one property but differ in the others.

Characterization: closing the loop

The chain is useless unless each link can be measured. **Characterization** measures structure (optical and electron microscopy, X-ray and neutron diffraction, spectroscopy, tomography) and properties (mechanical, thermal, electrical and corrosion tests). It sits at the centre of the tetrahedron because it connects every link, and Part XI (Chapters 52–54) is devoted to it. The Liberty-ship failures were understood only after impact tests on plates from fractured ships showed that plates in which fractures started had transition temperatures well above the water temperature.

Structure-insensitive and structure-sensitive properties

A useful distinction separates

- **structure-insensitive** properties, fixed mainly by chemistry and crystal structure, which vary by only a few per cent between grades: density, elastic modulus of a fully dense polycrystal, melting point, heat capacity, thermal expansion,

band gap; and
- **structure-sensitive** properties, controlled by defects and microstructure, which can vary by orders of magnitude between grades of the same material: yield strength, fracture toughness, fatigue limit, creep rate, corrosion rate, electrical resistivity at low temperature, magnetic coercivity, the permeability of concrete.

The Young's modulus of every carbon and low-alloy steel lies within a few per cent of 210 GPa; their yield strengths range from about 200 MPa to above 2000 MPa. Most properties that limit the life of a component belong to the second group, which is why this book spends Chapters 2–5 on structure and defects and Chapters 6–10 on the thermodynamics and kinetics that set them before it reaches mechanical behaviour.

> **Status of the principle: the structure–processing–property–performance paradigm**
> *Empirical.* It is an organizing framework distilled from practice rather than a law; it holds because properties are functions of structure and structure is a function of processing history, but it does not by itself predict any number.

1.3 Length and time scales from electrons to structures

The ladder of length scales

"Structure" exists on many scales at once, and different properties are controlled at different scales (Figure 1.3).

Electronic and atomic scale (about 0.01–1 nm). Electrons forming bonds, the crystal structure and its unit cell, and point defects such as vacancies and solute atoms. The nearest-neighbour distance in iron is 0.248 nm; the Si–O bond in quartz and glass is 0.162 nm. Structure-insensitive properties are decided here.

Nanoscale (about 1–100 nm). Dislocation cores, strengthening precipitates (γ′ in superalloys, Guinier–Preston zones in aluminium alloys), the calcium silicate hydrate gel that binds concrete, the crystalline lamellae of polyethylene (10–20 nm thick), the mineral platelets in bone, thin films and coatings. When a structural dimension becomes comparable to a physical length of the electrons, phonons or dislocations that carry a property, the property itself changes (Chapter 38).

Microscale (about 0.1–100 μm). Grains, the phases of an alloy such as the lamellae of pearlite in rail steel, pores in a sintered ceramic, the 7 μm fibres of a carbon composite, spherulites in a polymer, wood cells, osteons in bone. This is what metallurgists usually mean by "microstructure"; strength, toughness and fatigue resistance are mostly set here.

Meso- and macroscale (above about 0.1 mm). Plies of a laminate, aggregate in concrete, weld beads and heat-affected zones, texture across a rolled sheet, cracks, and the shape of the component. Performance is finally determined here, by geometry combined with properties.

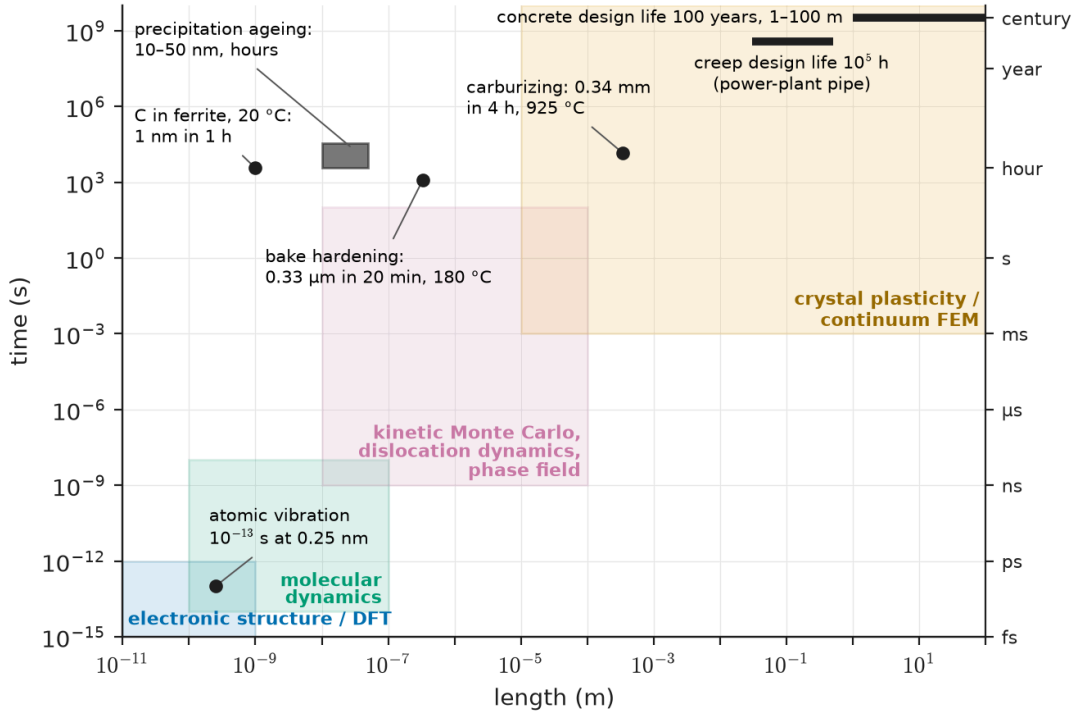

Figure 1.3 — Length and time scales of materials structure and processes, from electrons to engineering structures, with the modelling methods that cover each range.

Time scales

Structure also lives on many *time* scales. Atoms vibrate about their sites with periods of about 10^{-13} s. A vacancy in copper near its melting point jumps roughly a billion times a second; at room temperature, far less often than once a second. Phase transformations take from microseconds (martensite) to hours (precipitation in aluminium alloys) to years (the slow ageing of polymers). Engineering design lives at the long end: a steam pipe is designed for 10^5 hours of creep life, a bridge for 100 years. Between an atomic vibration and a century lie 22 orders of magnitude.

The bridge between atomic and engineering times is usually **thermal activation**. A process that requires atoms to overcome an energy barrier Q runs at a rate proportional to $\exp(-Q/RT)$, where $R = 8.314\ \mathrm{J\,mol^{-1}\,K^{-1}}$. For diffusion,

$$D = D_0 \exp\left(-\frac{Q}{RT}\right)$$

and the typical distance an atom moves in time t is of order $\sqrt{Dt}$. Chapter 8 derives both results from a random walk of atomic jumps.

> **Status of the principle: Arrhenius temperature dependence**
> *Approximate.* It follows from the Boltzmann probability of crossing a single energy barrier; it describes diffusion, creep and many reactions well over limited temperature ranges, but fails when the mechanism, and hence Q, changes with temperature.

Worked example — Why carbon moves in a furnace but steel still ages at room temperature. *Carburizing.* Gears are hardened by heating in a carbon-rich atmosphere so that carbon diffuses into austenite (face-centred cubic iron, Chapter 2), with $D_0 = 2.3 \times 10^{-5}\ \mathrm{m^2\,s^{-1}}$ and $Q = 148\ \mathrm{kJ\,mol^{-1}}$. At 925 °C ($T = 1198$ K),

$$D = 2.3 \times 10^{-5} \exp\left(-\frac{148\,000}{8.314 \times 1198}\right) = 8.1 \times 10^{-12}\ \mathrm{m^2\,s^{-1}}$$

and in 4 hours ($t = 14\,400$ s) the characteristic depth is $\sqrt{Dt} = \sqrt{8.1 \times 10^{-12} \times 14\,400} = 3.4 \times 10^{-4}$ m $= 0.34$ mm, the right order for a case-hardened gear tooth (the error-function solution of Chapter 8 gives case depths about twice this).

Strain ageing. Room temperature looks too cold for anything to happen, but carbon dissolved in ferrite (body-centred cubic iron) has a lower barrier: $D_0 = 1.1 \times 10^{-6}\ \mathrm{m^2\,s^{-1}}$, $Q = 87.4\ \mathrm{kJ\,mol^{-1}}$. At 20 °C,

$$D = 1.1 \times 10^{-6} \exp\left(-\frac{87\,400}{8.314 \times 293.15}\right) = 2.9 \times 10^{-22}\ \mathrm{m^2\,s^{-1}}$$

In one hour $\sqrt{Dt} = \sqrt{2.9 \times 10^{-22} \times 3600} = 1.0 \times 10^{-9}$ m; in one day about 5 nm. A few nanometres is all a carbon atom needs to reach a nearby dislocation and pin it (Chapter 12). That is why mild-steel sheet stored for weeks after rolling develops a sharp yield point and shows stretcher-strain markings when pressed into a car panel. *Bake-hardening* steels turn the same physics to advantage: in the 20-minute paint bake at about 180 °C, D rises to $9.3 \times 10^{-17}\ \mathrm{m^2\,s^{-1}}$, carbon moves about $\sqrt{9.3 \times 10^{-17} \times 1200} = 0.33$ µm, the dislocations introduced by pressing are pinned, and the yield strength of the finished panel rises by 30–60 MPa, improving dent resistance. One atomic mechanism links a furnace cycle, weeks of storage and a paint oven.

Why scales interact

Three consequences recur throughout the book. **Properties are set at the scale of their carriers**: a crack a millimetre long sets the strength of a ceramic; precipitates tens of nanometres apart set the strength of an alloy. **Models are scale-specific**: electronic-structure calculations handle hundreds of atoms for picoseconds, continuum mechanics whole structures for decades; linking them is multiscale modelling (Chapter 56). **Tests are scale-specific**: when the feature that controls failure is rarer than a small specimen captures, small-specimen data overestimate strength and statistics enter (Chapters 11 and 14).

1.4 Material classes and their property ranges

Why classify

Each class carries a recognizable *package* of properties. Knowing that a candidate is a polymer tells you, before you look at a datasheet, that it is light, insulating, poor at conducting heat, compliant, expands a lot on heating, is limited to a few hundred degrees Celsius and may creep at room temperature.

Metals and alloys are held together by metallic bonding, with valence electrons shared by the whole crystal (Section 1.5). The delocalized electrons give high electrical and thermal conductivity and opacity. The bonds are non-directional, so atoms pack densely (Chapter 2) and planes slide over one another by dislocation motion (Chapters 12–13), making metals ductile, tough and formable. Their strengths can be raised enormously by alloying and heat treatment. Steels are the backbone of civil and mechanical engineering; aluminium, titanium and magnesium alloys dominate where mass matters; nickel superalloys serve at the highest temperatures (Chapters 15–16).

Ceramics are compounds of metals with non-metals (oxides, nitrides, carbides), bonded ionically, covalently or by a mixture. They are stiff, hard, refractory, chemically stable and mostly insulating, but brittle, because dislocations hardly move in them; their tensile strength is controlled by the largest flaw and is therefore statistical (Chapter 14). Alumina and zirconia make cutting tools, hip heads and dental crowns; silicon carbide and nitride make bearings, seals and armour; refractories line furnaces; cement-based materials are ceramics too (Chapters 17–18).

Glasses are non-crystalline solids, usually silicate-based, though metallic and polymeric glasses exist. They share the stiffness, durability and brittleness of ceramics but, lacking crystals, are isotropic, can be transparent and are shaped as viscous liquids. Tempering and chemical strengthening, which put the surface in compression, multiply their usable strength (Chapter 17).

Polymers and elastomers are long-chain molecules, mostly carbon-based, bonded covalently along the chain and by weak van der Waals and hydrogen bonds between chains, or by occasional crosslinks in thermosets and elastomers. The weak interchain bonds explain their package: low density, low stiffness (typically 0.2–4 GPa; 1–10 MPa for elastomers), low softening temperatures, high expansion, low thermal conductivity and strongly time-dependent behaviour. Their advantages are easy processing into complex shapes, corrosion resistance and low cost (Chapters 19–21).

Composites combine materials so that the result has properties neither has alone. CFRP outperforms aluminium on stiffness and strength per unit mass; GFRP makes wind-turbine blades and boat hulls; reinforced concrete puts steel where concrete would crack in tension; wood and bone are natural composites. Their properties depend on the *arrangement* of the constituents as much as on their amounts, and they are usually strongly anisotropic (Section 1.6, Chapters 22–23).

Semiconductors are covalent solids whose band gap ($\sim$0.1–4 eV) lets their conductivity be controlled over many orders of magnitude by doping, temperature, light or field. Mechanically they behave like ceramics. Silicon dominates electronics and photovoltaics; SiC and GaN serve power electronics. Uniquely, they are used at near-perfect crystalline quality (Chapters 31–33).

Biomaterials are defined by function, not bonding: materials intended to interact with the body, drawn from every other class. Titanium alloys and cobalt–chromium make joint replacements; alumina, zirconia, hydroxyapatite and bioactive glass make bearing surfaces and bone substitutes; ultra-high-molecular-weight polyethylene makes joint liners; biodegradable polyesters make sutures and scaffolds. What distinguishes them is the extra requirement of biocompatibility in a corrosive, cyclically loaded environment (Chapter 39).

Representative property values

Table 1.1 collects typical room-temperature values; the property chapters give fuller data.

Class	Material	Density (g cm^{-3})	Young's modulus (GPa)	Strength (MPa)	Fracture toughness (MPa m$^{1/2}$)	Thermal conductivity (W m^{-1} K^{-1})	CTE (10^{-6} K^{-1})	Max. service T (°C, typical)
Metal	Structural steel S355	7.85	210	355 (yield)	50–150	45–50	12	350–400
Metal	Al alloy 6061-T6	2.70	69	275 (yield)	25–30	167	23.6	150
Metal	Ti-6Al-4V	4.43	114	880 (yield)	55–75	6.7	8.6	400
Metal	Ni superalloy (IN718)	8.19	200	1030 (yield)	90–110	11	13	650
Ceramic	Alumina (99.5 %)	3.9	370	300–380 (flexure)	3.5–4.5	30	8	1700
Ceramic	Silicon carbide (sintered)	3.15	410	400–550 (flexure)	3–4	80–120	4.0	1600
Ceramic	Concrete (C30/37)	2.4	33	30–40 (compression), 2–3 (tension)	0.2–1.4	1.5–2.5	10–12	300
Glass	Soda-lime glass (annealed)	2.5	70	30–70 (flexure)	0.75	1.0	9	500
Polymer	HDPE	0.95–0.96	0.9–1.2	20–30 (yield)	1–3	0.45–0.5	120–200	80
Polymer	Epoxy (cast)	1.2	2.5–3.5	60–80 (tension)	0.5–0.8	0.2	55–65	130
Elastomer	Natural rubber	0.92	0.001–0.005	20–30 (tension)	—	0.13	220	80
Composite	CFRP, unidirectional (along fibres)	1.55	135–140	1500–2000 (tension)	30–45 (laminate)	5–10	−0.5 to 0.5	120–180
Composite	Douglas fir (along grain)	0.50	13	50 (compression)	0.4–0.6 (across grain)	0.12 (across)	3–5	100
Semiconductor	Silicon (single crystal)	2.33	130–188 (by direction)	flaw-controlled	0.8–0.9	148	2.6	—
Biomaterial	Cortical bone	1.9–2.0	15–20	100–150 (tension)	2–12	0.3–0.6	—	—

Two features stand out. First, the ranges between classes are enormous: Young's modulus runs from about a megapascal for rubber to 410 GPa for silicon carbide, five orders of magnitude, and the expansion coefficients of materials that may be bonded together in one assembly (silicon 2.6, steel 12, epoxy about 60 ppm/K) differ by a factor of more than twenty, the root of the thermal-stress failures of Chapter 24. Second, within each class properties cluster: metals stiff and tough, ceramics stiff, hard and brittle, polymers compliant and light. Section 1.7 turns that clustering into a tool for selection.

Exceptions show the logic

Classification is a starting point, not a law, and each exception has a structural reason. Diamond is an electrical insulator that conducts heat five times better than copper, because its light, stiff lattice scatters lattice vibrations little. Transformation-

toughened zirconia reaches 8–12 MPa $m^{1/2}$ because a stress-induced phase change absorbs energy at a crack tip (Chapter 14). Drawn polyethylene fibres are stiffer than aluminium because their chains are loaded through their covalent backbones. Metallic glasses reach yield strengths near 2 GPa because they have no dislocations.

1.5 Bonding as the origin of property classes

The bond types in brief

The Advanced Chemistry book develops chemical bonding; here we need only its consequences for solids.

- **Metallic.** Valence electrons are delocalized over the crystal. The bond is non-directional, so atoms pack densely and can change neighbours without the solid falling apart: hence ductility, conductivity and opacity.
- **Ionic.** Electrons are transferred and oppositely charged ions attract. The bond is strong and non-directional, but any shear that brings like ions into contact is resisted: hence stiffness, hardness, brittleness, high melting points and electrical insulation.
- **Covalent.** Atoms share electron pairs in bonds with definite directions. In a three-dimensional network (diamond, silicon, SiC, silica) the result is the hardest and stiffest of solids; in a polymer the strong bonds run only along the chain.
- **Secondary.** Van der Waals and hydrogen bonds between molecules are ten to a hundred times weaker. They hold polymer chains together, bind the layers of graphite and clays and govern the softening of thermoplastics.

Real materials usually mix bond types; the Si–O bond in silica is roughly half ionic, half covalent.

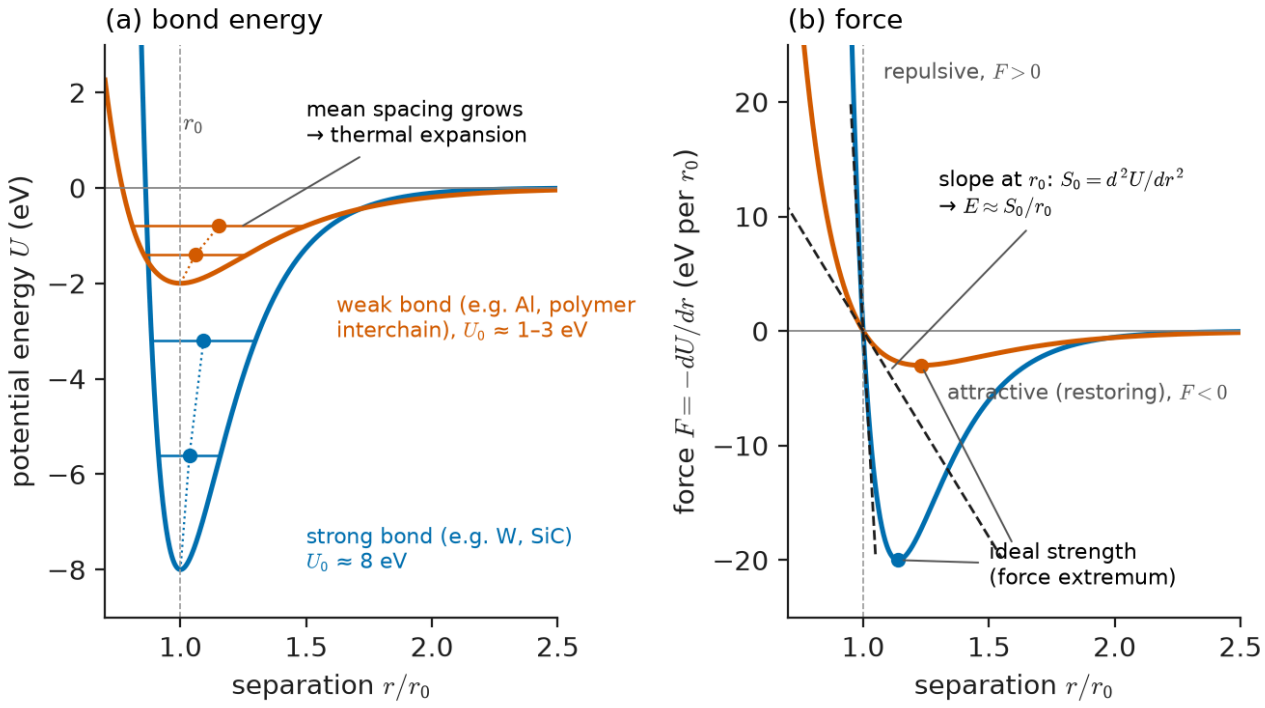

Figure 1.4 — Interatomic potential energy and force against separation for a strong, stiff bond and a weak, compliant bond, showing why stiffness, melting point and thermal expansion are linked.

Stiffness from the bond curve

Figure 1.4 shows the potential energy $U(r)$ of two atoms against their separation r. Attraction dominates at large r and repulsion at small r; the minimum, at the equilibrium spacing r_0, has depth U_0, the bond energy. Near the minimum the curve is close to a parabola, and the force $F = -dU/dr$ is proportional to the displacement, with stiffness

$$S_0 = \left(\frac{d^2U}{dr^2}\right)_{r=r_0}$$

If each bond carries $F = S_0(r - r_0)$ and there is one bond per area r_0^2, the stress is F/r_0^2 and the strain $(r - r_0)/r_0$, so

$$E \approx \frac{S_0}{r_0}$$

Typical bond stiffnesses are 20–200 N/m for covalent bonds, 30–100 N/m for metallic bonds, 8–25 N/m for ionic bonds and only 0.5–2 N/m for van der Waals bonds; with $r_0 \approx 0.2$–0.3 nm these give moduli of hundreds of gigapascals for primary bonds and a few gigapascals for secondary bonds, just the spread between ceramics and metals on one hand and polymers on the other.

> **Status of the principle: modulus from bond stiffness, $E \approx S_0/r_0$**
> *Approximate.* It treats the solid as an array of independent springs; it gives the right order of magnitude and ranking for fully dense solids, but ignores bond angles, crystal anisotropy and, for elastomers, the entropic origin of stiffness (Chapters 11 and 20).

Melting point and thermal expansion

A solid melts when thermal energy becomes comparable to a fraction of the bond energy, so deep wells mean high melting points. For metals the cohesive energy per atom divided by $k_B T_m$ ($k_B = 8.617 \times 10^{-5}$ eV/K) is roughly 30: 29.8 for copper (3.49 eV, 1358 K), 27.4 for iron (4.28 eV, 1811 K), 28.0 for tungsten (8.90 eV, 3695 K), with outliers such as aluminium (42).

The curve is asymmetric: it rises steeply on compression and gently on extension. As an atom vibrates with more energy, the midpoint of its oscillation moves to larger r (Figure 1.4a), so the solid expands on heating. Deep, symmetric wells expand little; shallow, asymmetric ones a lot. For many metals the product αT_m is nearly constant at about 0.02: 0.022 for copper, 0.021 for iron, 0.023 for nickel and 0.017 for tungsten. That is why refractory metals and ceramics have low expansion coefficients and polymers high ones (Chapter 24).

> **Status of the principle: correlations of bond energy with melting point and expansion ($E_{\mathrm{coh}}/k_B T_m \approx 30$, $\alpha T_m \approx 0.02$)**
> *Empirical.* Both are trends fitted across many metals and rationalized by the shape of the bond-energy curve; they hold to within about ±30 % for close-packed metals and fail for directional bonding or for materials whose liquid is bonded differently from the solid.

From bond to class

Bond type	Typical cohesive energy (eV per atom)	Example solids	Consequences for properties
Metallic	1–9	Na 1.1, Al 3.4, Cu 3.5, Fe 4.3, W 8.9	Conductive, ductile, dense packing, moderate to high stiffness
Ionic	3–8 (averaged over the formula)	NaCl 3.3, MgO 5.2, Al_2O_3 about 6	Stiff, hard, brittle, insulating, high melting point
Covalent network	4–8	Si 4.6, SiC 6.3, diamond 7.4	Hardest and stiffest solids, brittle, low density
Covalent chain + secondary	Chain bonds 3.5–4; interchain 0.05–0.5	Polyethylene, nylon, epoxy	Compliant, light, low softening point, high expansion
Van der Waals	0.02–0.1	Solid Ar 0.08; graphite interlayers	Very soft, low melting point
Hydrogen	0.1–0.5	Ice; between nylon or cellulose chains	Raises stiffness and T_g of polymers; water uptake

Table 1.2 explains why the classes are classes, and why an engineer who needs a combination that no single bond type offers designs a composite.

1.6 Isotropy, anisotropy and homogeneity

Properties as tensors

Many properties connect one vector or tensor to another: a temperature gradient to a heat flux, a stress to a strain. If the response is parallel to the stimulus and independent of direction,

$$\mathbf{q} = -k\,\nabla T$$

the material is **isotropic**. In general each component of the response depends on every component of the stimulus,

$$q_i = -\sum_{j} k_{ij} \frac{\partial T}{\partial x_j}, \qquad i, j \in \{1, 2, 3\}$$

and the nine numbers k_{ij} form a second-rank tensor. If they are not equivalent in all directions the material is **anisotropic**: heat flow is generally not parallel to the gradient. Thermal expansion and electrical conductivity are also second-rank tensors; elastic stiffness, relating stress to strain, is fourth-rank. Chapter 4 develops the tensor description and Neumann's principle; one consequence is worth stating now. Cubic crystals are isotropic in second-rank properties such as conductivity and expansion but anisotropic in elasticity: single-crystal iron has a Young's modulus of about 130 GPa along a cube edge and about 280 GPa along a body diagonal.

Sources of anisotropy

- **Crystal structure.** Zinc expands four to five times more along its hexagonal c axis than in the basal plane; graphite conducts heat several hundred times better within its layers than across them.
- **Crystallographic texture.** Processing can align the grains of a polycrystal. Textured aluminium sheet forms "ears" when deep-drawn into a beverage can; zirconium-alloy fuel cladding is given a texture so that brittle hydrides precipitate in a harmless orientation (Chapters 3 and 46).
- **Microstructural alignment.** Inclusions strung out by rolling make steel plate weaker through its thickness (a cause of lamellar tearing under welds); columnar grains form in castings and additively manufactured parts (Chapter 48).
- **Architecture.** Fibres in a composite, cells in wood, plies in a laminate, osteons in bone. Wood is about 10–20 times stiffer along the grain than across it.

Anisotropy is a resource when the loads are known: a CFRP wing skin puts its fibres along the load paths, and a single-crystal blade grown along its compliant $\langle 001 \rangle$ direction has lower thermal stresses.

Homogeneity depends on the scale of observation

A material is **homogeneous** if its properties are the same at every point. The definition makes sense only once a scale is chosen. A steel beam is heterogeneous at the scale of micrometres, where grains and the ferrite and pearlite phases are visible, but homogeneous at the scale of millimetres. Concrete is heterogeneous at the scale of its aggregate and is treated as homogeneous only for volumes several times larger. The smallest volume that contains a representative sample of the microstructure is the **representative volume element** (RVE); over larger volumes the material has *effective properties*.

The idea fails when the component approaches the RVE: a stent strut 80 µm thick may hold only a few grains across its section, and such parts scatter from one to the next.

Homogeneity and isotropy are independent:

	Isotropic	**Anisotropic**
Homogeneous	Glass; annealed polycrystalline steel (at mm scale)	Single-crystal turbine blade or sapphire window; textured sheet
Heterogeneous	Concrete; polymer with randomly dispersed particles	Unidirectional CFRP; wood; bone; reinforced concrete

Effective properties: the two simplest arrangements

For two phases with stiffnesses (or conductivities) P_1 and P_2 and volume fractions f_1 and $f_2 = 1 - f_1$: when layers or fibres lie **parallel** to the load, both phases see the same strain and the responses add, like springs in parallel,

$$P_{\parallel} = f_1 P_1 + f_2 P_2$$

and when layers lie **perpendicular** to the load, the same stress passes through each and the strains add, like springs in series,

$$\frac{1}{P_{\perp}} = \frac{f_1}{P_1} + \frac{f_2}{P_2}$$

Any other arrangement of the same phases at the same volume fractions lies between these two.

> **Status of the principle: Voigt–Reuss (Wiener) bounds**
> *Derived.* They follow from equilibrium and compatibility (equal strain or equal stress), or from flux conservation for conductivity; they are exact for ideal laminates and bound any two-phase arrangement with given volume fractions, provided the interfaces are perfect.

Worked example — Anisotropy of a carbon-fibre ply. A unidirectional CFRP ply contains $V_f = 0.60$ carbon fibre with $E_f = 230$ GPa in epoxy with $E_m = 3.5$ GPa. Along the fibres,

$$E_1 = 0.60 \times 230 + 0.40 \times 3.5 = 138 + 1.4 = 139.4 \text{ GPa}$$

Across the fibres,

$$\frac{1}{E_2} = \frac{0.60}{230} + \frac{0.40}{3.5} = 0.00261 + 0.1143 = 0.1169 \text{ GPa}^{-1}, \qquad E_2 = 8.6 \text{ GPa}$$

The ply is 16 times stiffer along its fibres than across them, though its composition is the same in every direction; only the arrangement differs. Measured values are about 135–140 GPa and 8–10 GPa. Two refinements explain the residual difference: carbon fibres are themselves anisotropic, with a transverse modulus of only about 15 GPa (lowering the series estimate to 6.5 GPa), and real fibres are not stacked in layers, so the transverse modulus lies above the series bound (the Halpin–Tsai equations of Chapter 22 interpolate). Laminates therefore stack plies at 0°, ±45° and 90° to give stiffness in every direction needed (Chapter 23), at the cost of a lower stiffness along any one direction, about 50–60 GPa for a quasi-isotropic layup.

1.7 Property charts and the logic of selection

Function, constraints, objectives, free variables

Choosing a material is a design decision, made by stating clearly what is required:

- **Function**: what the component does. Support a load; contain a pressure; transmit heat; replace a joint.
- **Constraints**: requirements that must be met but bring no credit when exceeded. The beam must not deflect more than span/360; the implant must be biocompatible; the part must operate at 300 °C.
- **Objectives**: quantities to minimize or maximize. Mass, cost, embodied carbon, volume.
- **Free variables**: parameters the designer may adjust, such as a cross-section.

Constraints *screen*; objectives *rank*, through a performance index.

Performance indices

Take a beam of square section $b \times b$ and length L that must have bending stiffness at least S, with b free and mass to be minimized:

$$S = \frac{C\,E\,I}{L^3}, \qquad I = \frac{b^4}{12}, \qquad m = \rho\, b^2 L$$

where C depends on loading and supports ($C = 48$ for a simply supported beam loaded at mid-span). Eliminating b,

$$m = \left(\frac{12\,S}{C}\right)^{1/2} L^{5/2}\,\frac{\rho}{E^{1/2}}$$

The mass separates into a factor set by the design and a factor set by the material alone, so the best material maximizes the **performance index**

$$M_{\text{beam}} = \frac{E^{1/2}}{\rho}$$

The same procedure gives E/ρ for a stiff tie, $E^{1/3}/\rho$ for a stiff panel of free thickness and $\sigma_f^{2/3}/\rho$ for a strong beam. A tie uses its material uniformly; a beam or panel gains stiffness faster than mass by getting deeper, so a light material that can be made thicker is favoured.

> **Status of the principle: performance index**
> *Derived.* It follows algebraically from the model of the component (here elastic bending of a given shape); it is only as valid as that model and ranks materials for one objective under the stated constraints, ignoring any requirement not written into the equations.

Property charts

A **property chart** (Ashby chart) plots one property against another for all materials on logarithmic axes (Figure 1.5). Because classes cluster, each occupies a recognizable bubble. On log axes an index becomes a straight line: $E^{1/2}/\rho =$ constant is $\log E = 2\log\rho +$ constant, slope 2; E/ρ gives slope 1 and $E^{1/3}/\rho$ slope 3. Materials on a line perform equally well; those above it, better. Selection becomes geometric: slide the line upward until only a handful of candidates remain above it.

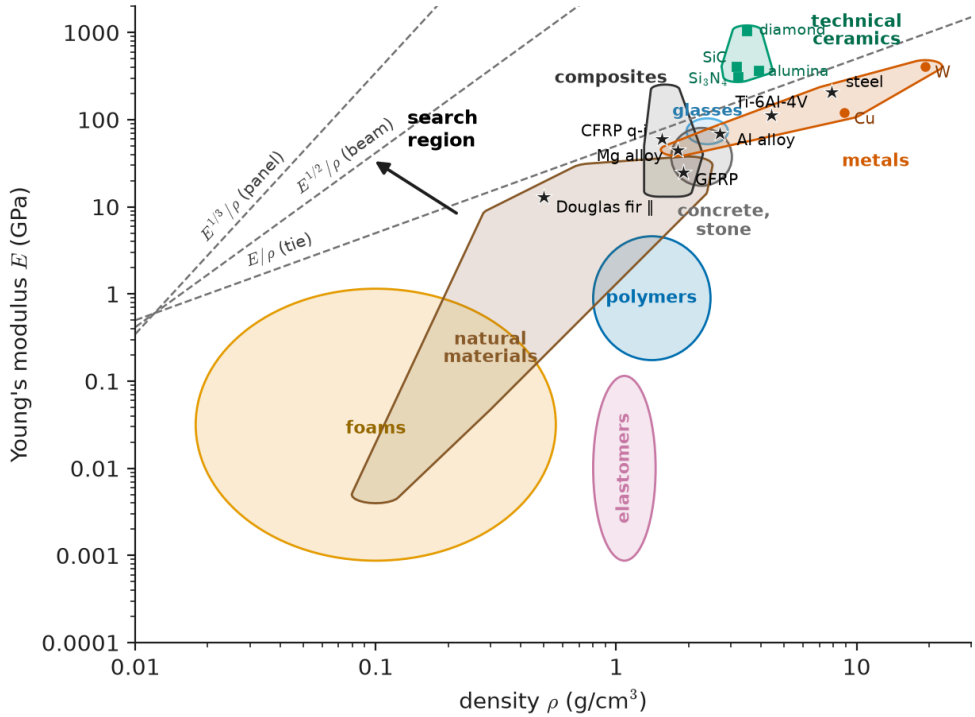

Figure 1.5 — Property chart of Young's modulus against density for engineering materials, with guide lines for minimum-mass ties, beams and panels.

Worked example — Choosing a light, stiff beam, and asking what it costs. A simply supported beam of span $L = 3$ m with square section must have mid-span stiffness $S = 1.0 \times 10^6$ N/m ($C = 48$). For steel ($E = 210$ GPa, $\rho = 7850$ kg m^{-3}),

$$I = \frac{SL^3}{CE} = \frac{1.0\times10^6 \times 27}{48 \times 210 \times 10^9} = 2.68 \times 10^{-6}\ \text{m}^4, \qquad b = (12I)^{1/4} = 75.3\ \text{mm}$$

$$m = 7850 \times (0.0753)^2 \times 3 = 133.5\ \text{kg}$$

Repeating for the other candidates, and multiplying mass by indicative prices and cradle-to-gate embodied carbon (both vary widely with grade, region and recycled content):

Material	E (GPa)	ρ (kg m^{-3})	$E^{1/2}/\rho$ (GPa$^{1/2}$ per Mg m^{-3})	b (mm)	Mass (kg)	Price (USD/kg)	Cost (USD)	CO_2 (kg/kg)	CO_2 (kg)
Steel	210	7850	1.85	75.3	134	0.8	107	1.9	254
Al alloy	70	2700	3.10	99.1	80	2.5	199	12	954
Ti-6Al-4V	114	4430	2.41	87.7	102	20	2045	40	4091
Mg alloy	45	1800	3.73	110.7	66	4	265	25	1653
GFRP	25	1900	2.63	128.2	94	4	375	6	562
CFRP (quasi-isotropic)	60	1550	5.00	103.0	49	40	1973	25	1233
Douglas fir (along grain)	13	500	7.21	151.0	34	1.5	51	0.3	10

The masses follow the index exactly: any two masses are in the inverse ratio of their indices (steel to CFRP, $5.00/1.85 = 2.7$). Three lessons follow. First, timber is the lightest of all for a stiffness-limited beam, which is why wood, bamboo and bone are so efficient in bending; titanium, a "light" metal, loses to aluminium and magnesium here because its modulus is not high enough for its density. Second, the ranking changes completely with the objective: on cost, timber and steel win and titanium and CFRP lose by a factor of about 20; on embodied carbon, timber wins by a factor of 25 over steel, and primary aluminium is nearly four times worse than steel. Third, the index ignores everything not written into it: timber needs a 151 mm section, creeps under sustained load and is vulnerable to fire and moisture; CFRP needs different joining. An aircraft floor beam, a bicycle frame and a building joist face the same equations but weigh mass, cost and carbon differently, and so choose differently.

The general method

The procedure, developed fully in Chapter 58, is: translate the requirement into function, constraints, objectives and free variables; screen out every material that violates a constraint; rank the survivors by the index on a property chart; examine the leaders for secondary properties, manufacturability, joining, reliability, cost and supply; decide, and document why. When objectives conflict, as mass and cost do above, no single material is best. The useful set is the **Pareto front**: materials for which no other candidate is better in one objective without being worse in another (Chapter 58).

1.8 Materials and society: economics, criticality, sustainability

The scale of materials use

Humanity uses materials in staggering quantities, dominated by a few cheap structural materials (Table 1.3).

Material	Annual global production (order of magnitude)	Typical price (USD/kg, indicative)	Embodied CO_2 (kg CO_2e per kg, typical cradle-to-gate)
Concrete	25–30 billion tonnes	0.05–0.1	0.1–0.2
Cement	about 4 billion tonnes	0.1	0.6–0.9
Steel	about 1.9 billion tonnes	0.6–1	1.4–2.3 (primary); 0.4–0.7 (scrap-based electric arc)
Plastics	about 0.4 billion tonnes	1–3 (commodity)	2–4
Glass	about 0.15 billion tonnes	0.5–1	0.7–1.2
Aluminium (primary)	about 70 million tonnes	2–3	8–17 (depends on electricity source)
Copper (refined)	about 26 million tonnes	8–10	3–5
Carbon fibre	about 0.1–0.2 million tonnes	20–50	20–40

Cement and steel production each account for roughly 7–8 % of global anthropogenic CO_2 emissions; calcining limestone ($CaCO_3 \rightarrow CaO + CO_2$) and reducing iron oxide with carbon release CO_2 whatever the fuel (Chapters 18 and 59).

Economics

Three economic facts shape material choice. **Material cost is often a small part of component cost**: for a turbine blade or an implant, processing, qualification and certification dominate; for a building, the material is a large share. **Prices are volatile**: copper, nickel, cobalt and lithium have each doubled or halved within a few years, which pushes designers toward substitutable materials. **Mass has a lifetime cost in transport**: removing a kilogram from a car saves fuel worth a few dollars over its life; from a commercial aircraft, hundreds to thousands of dollars. This value of mass saving decides whether expensive light materials pay off, and explains why CFRP is common in aircraft and rare in buildings.

Criticality

A material is **critical** when it is economically important and its supply is at risk, because mining or refining is concentrated in a few countries, substitutes are poor, recycling is low or it is produced only as a by-product of another metal. Most rare-earth separation (about 90 %), needed for the neodymium–iron–boron magnets of wind turbines and electric motors, takes place in China; about 70 % of cobalt for battery cathodes comes from the Democratic Republic of the Congo; gallium is almost entirely a by-product of alumina refining; about 90 % of niobium for high-strength low-alloy steels comes from Brazil. Criticality assessments plot supply risk against economic importance (Figure 1.6).

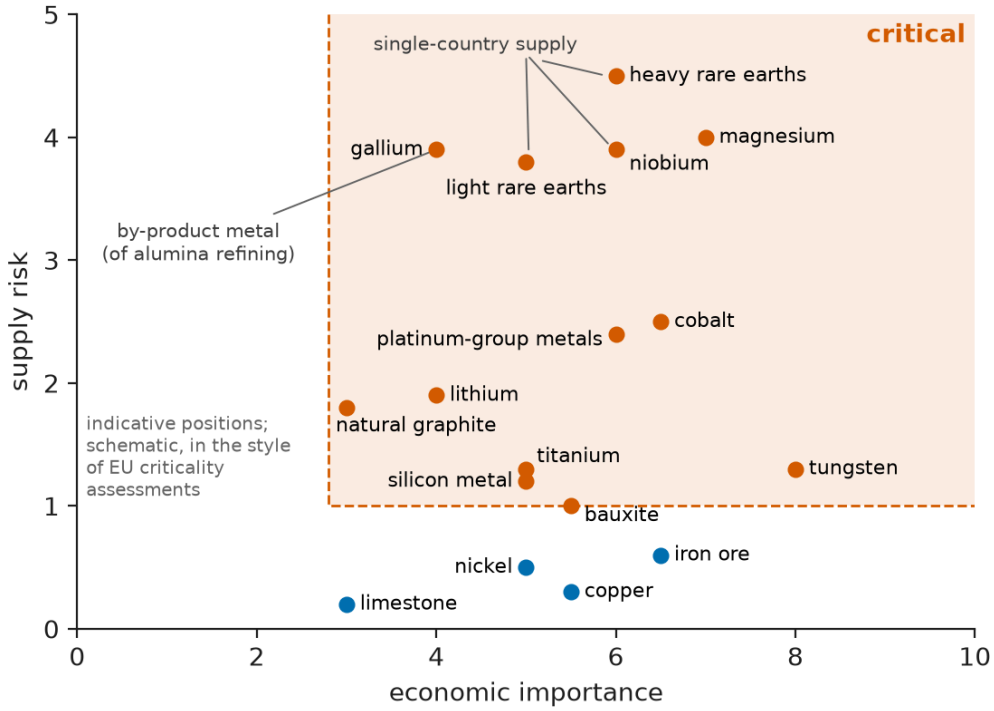

Figure 1.6 — A criticality assessment places each raw material by economic importance and supply risk; materials in the upper right are critical.

Materials engineering responds by **substitution** (magnets that need less dysprosium because it is diffused only into grain boundaries; iron-phosphate battery cathodes instead of cobalt-rich ones), **thrifting** (thinner platinum catalyst layers) and **recycling** (Chapter 59). Each depends on knowing which structural feature delivers the property, so that the scarce element is used only where it matters.

Sustainability

Sustainability adds the whole lifecycle to selection: extraction, processing, manufacture, use and end of life. Its first quantitative tools are **embodied energy** and **embodied carbon** per kilogram (Table 1.3) and the **lifecycle assessment** that combines them with the use phase (Chapter 59). The beam example of Section 1.7 shows how much the ranking changes when carbon is the objective. The use phase may dominate: a lighter car or aircraft that burns less fuel over its life can justify aluminium or CFRP, while a component sitting in a building wall for 60 years gains nothing from lightness and should minimize embodied carbon.

Three principles connect sustainability back to structure. **Durability** divides embodied impacts over more years: a bridge deck whose concrete resists chloride ingress for 100 years rather than 40 is more sustainable even if it uses more cement. **Recyclability** depends on composition: aluminium can be remelted with about 5 % of the energy of primary production, but mixing alloys accumulates iron and copper that are hard to remove, and thermoset composites are hard to separate at all. **Design for disassembly** and alloy simplification keep materials in circulation. The circular economy is ultimately a materials-engineering problem: designing structures that can be recovered without losing the properties that made them valuable.

> **Status of the principle: embodied-carbon and lifecycle figures**
> *Empirical.* They are inventory data from industrial surveys; they vary by a factor of two or more with region, electricity mix, recycled content and system boundary, so they rank options and identify hot spots rather than serving as precise constants.

What remains open

Open question	Where it is answered
How are atoms arranged in metals, ceramics, silicates, glasses and polymers, and how are directions and planes named?	Chapter 2
What symmetry do crystals have, and why do only certain lattices exist?	Chapter 3
How are tensor properties formulated, and why are cubic crystals elastically anisotropic?	Chapter 4
What defects do real materials contain, and why do structure-sensitive properties depend on them?	Chapter 5
How do thermodynamics and kinetics decide which structure forms, and how are carburizing depths calculated?	Chapters 6–10 (diffusion in Chapter 8)
Why does steel become brittle in the cold, and how is toughness measured?	Chapters 11 and 14
What controls strength, and how do pinned dislocations cause strain ageing?	Chapters 12–13
How are composite stiffnesses predicted beyond the simple bounds, and laminates designed?	Chapters 22–23
How are thermal stresses from expansion mismatch calculated?	Chapter 24
How are single-crystal blades and other high-temperature materials designed?	Chapters 16 and 41
How are multiple objectives, shape and processing combined in selection?	Chapter 58
How are embodied carbon, lifecycle assessment, criticality and recycling handled quantitatively?	Chapter 59

Next: Chapter 2 describes the atomic arrangements on which everything else is built: lattices and unit cells, the crystal structures of metals, ceramics and silicates, the notation for directions and planes, polycrystals and texture, and the non-crystalline and partly ordered solids — glasses, polymers, liquid crystals and quasicrystals.

Chapter 2 — Crystal Structures and Non-Crystalline Solids

Part I · Structure of Materials

Zirconia (ZrO_2) looks like an ideal engineering ceramic: it melts at about 2700 °C, resists chemical attack and is tougher than almost any other oxide. Yet a pure zirconia component sintered at 1500 °C cracks to pieces as it cools. The cause is a change of crystal structure. Near 950 °C on cooling (1170 °C on heating, with hysteresis), the tetragonal form of zirconia transforms to a monoclinic form whose volume is about 4–5 % larger, and the strain shatters the part. Engineers solved the problem in two quite different ways, both by controlling crystal structure. Adding about 8 mol % yttria stabilizes the cubic fluorite structure at all temperatures; this "fully stabilized" zirconia conducts oxygen ions and is the electrolyte of the oxygen sensor in every car exhaust and of solid-oxide fuel cells. Adding only about 3 mol % yttria keeps fine tetragonal grains *metastable* at room temperature; when a crack approaches, the stress triggers the transformation locally, the expansion squeezes the crack shut, and the toughness rises several-fold. That material makes dental crowns, hip-joint heads and cutting blades. One chemical compound, three crystal structures, three industries.

Every property traced in this book starts from the arrangement of atoms. This chapter builds the vocabulary for describing it: lattices and unit cells, the structures of metals, ceramics and silicates, the notation for directions and planes, the aggregates of crystals that make up real parts, and the solids, from window glass to polyethylene to liquid crystals, that are not perfectly periodic at all.

> **How are atoms arranged in metals, ionic and covalent compounds and silicates, and what decides which structure a compound adopts?**
> **How do we name directions and planes in a crystal, and how do real materials depart from the single perfect crystal: as polycrystals, glasses, semicrystalline polymers, liquid crystals and quasicrystals?**

Concept	Engineering destination
Lattice, basis, unit cell; coordination and packing	Density, porosity checks, interstitial solubility (Chapters 5, 8)
FCC, BCC, HCP and stacking	Ductility of metals, slip systems (Chapters 12–13), steel heat treatment (Chapter 15)
Ionic and covalent structures; Pauling's rules	Ceramics, refractories, oxide fuels, battery cathodes, semiconductors, piezoelectrics
Silicates and polymorphism	Glass, cement, clays, quartz inversion in refractories, zirconia toughening
Directions and planes; Miller and Miller–Bravais indices	Slip planes, cleavage, X-ray diffraction (Chapter 4), wafer and blade orientation
Polycrystals, texture, grain size	Strength, formability, creep, magnetic and corrosion behaviour
Glasses and radial distribution functions	Window and fibre glass, metallic glasses, amorphous coatings
Polymer crystallinity; liquid crystals; quasicrystals	Polymer stiffness and barrier properties, aramid fibres, displays, coatings

Roadmap. Lattice, basis and cell (2.1) → metallic structures (2.2) → ionic and covalent structures and Pauling's rules (2.3) → silicates and polymorphism (2.4) → naming directions and planes (2.5) → polycrystals, texture and grain size (2.6) → glasses (2.7) → polymer crystallinity (2.8) → liquid crystals and quasicrystals (2.9).

2.1 Lattices, bases and unit cells

Crystal = lattice + basis

A **crystal** is a solid in which atoms are arranged in a pattern that repeats periodically in three dimensions. Its description separates two ideas.

- The **lattice** is an infinite array of mathematical points, each with identical surroundings, generated by three non-coplanar **lattice vectors** **a**, **b**, **c**:

$$\mathbf{R} = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c}, \qquad n_1, n_2, n_3 \text{ integers}$$

- The **basis** (or motif) is the group of atoms attached to each lattice point. In copper it is one Cu atom; in silicon two Si atoms; in rock salt one Na and one Cl; in polyethylene, two chain segments containing four CH_2 groups.

Crystal structure is lattice plus basis. Different materials can share a lattice while having very different bases (copper, silicon and NaCl all have face-centred cubic lattices), and properties such as piezoelectricity depend on the symmetry of the whole structure, not of the lattice alone. Why atoms form periodic arrays at all is a matter of thermodynamics: at low temperature the periodic arrangement lets every atom adopt the same, optimal set of neighbours. Chapter 3 classifies the possible lattices (seven crystal systems, fourteen Bravais lattices) by symmetry; here we need only their geometry.

Unit cells and lattice parameters

The **unit cell** is a parallelepiped built on the lattice vectors that fills space when repeated by translation. It is specified by six **lattice parameters**: edge lengths a, b, c and interaxial angles α (between **b** and **c**), β and γ. A *primitive* cell contains one lattice point; a *conventional* cell is often larger so that it shows the full symmetry. Atoms on corners, edges and faces are shared with neighbouring cells and count as $1/8$, $1/4$ and $1/2$ respectively; this bookkeeping gives the number n of atoms (or formula units) per cell.

Coordination and packing

Two numbers summarize how tightly a structure is built. The **coordination number** (CN) is the number of nearest neighbours of an atom. The **atomic packing factor** (APF) is the fraction of the cell volume occupied by atoms modelled as hard spheres of radius R:

$$\mathrm{APF} = \frac{n \cdot \tfrac{4}{3}\pi R^3}{V_c}$$

And because the cell repeats, its mass divided by its volume is the density of a perfect crystal:

$$\rho = \frac{n\,A}{V_c\,N_A}$$

where A is the atomic (or formula) mass and $N_A = 6.022 \times 10^{23}\ \mathrm{mol^{-1}}$. Comparing this theoretical density with a measured one is the standard way to find the porosity of a sintered ceramic or a powder-metallurgy part.

> **Status of the principle: hard-sphere model of crystal structures**
> *Approximate.* Atoms are treated as rigid touching spheres; it predicts lattice parameters and densities to within a few per cent for close-packed metals and many ionic solids, but fails for covalent crystals, where bond directions rather than packing set the structure.

2.2 Metallic structures: FCC, BCC, HCP and stacking sequences

Metallic bonds are non-directional, so most metals adopt one of three simple, dense structures (Figure 2.1).

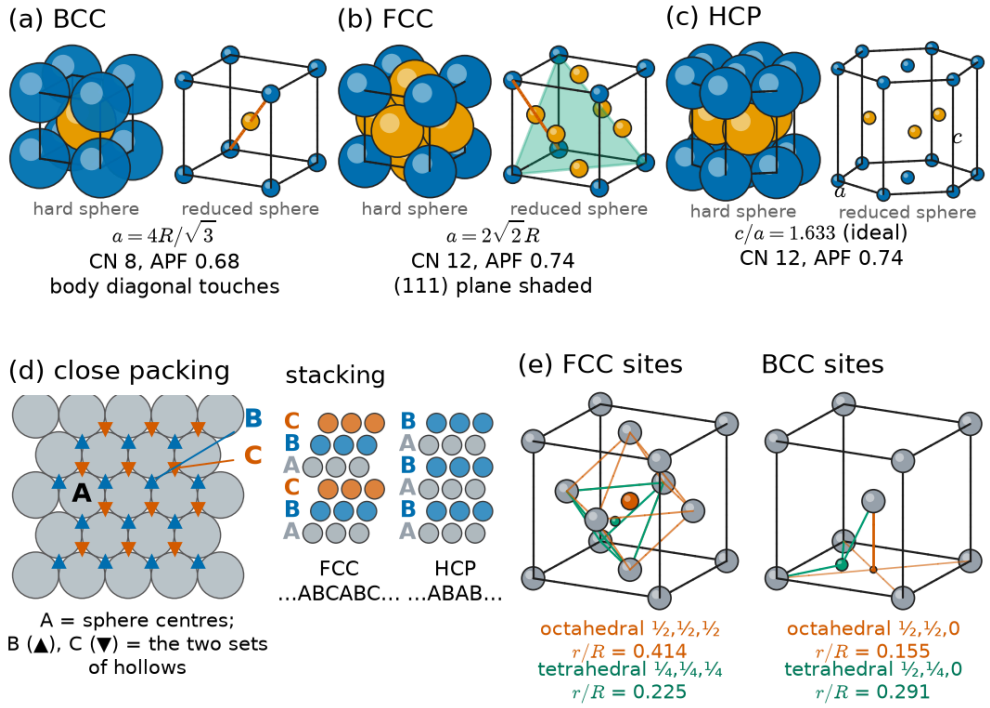

Figure 2.1 — The metallic structures: body-centred cubic, face-centred cubic and hexagonal close-packed, with the stacking of close-packed layers and the interstitial sites.

Body-centred cubic (BCC)

Atoms at the cube corners and one at the centre: $n = 8 \times \tfrac{1}{8} + 1 = 2$. Atoms touch along the body diagonal, so $\sqrt{3}\,a = 4R$ and

$$a = \frac{4R}{\sqrt{3}}, \qquad \text{CN} = 8, \qquad \text{APF} = \frac{\sqrt{3}\,\pi}{8} = 0.680$$

Iron at room temperature (α-Fe, ferrite), chromium, molybdenum, tungsten, vanadium, niobium and β-titanium are BCC. BCC metals are strong and often become brittle at low temperature (Chapter 12), the root of the Liberty-ship fractures in Chapter 1.

Face-centred cubic (FCC)

Atoms at the corners and the centre of each face: $n = 8 \times \tfrac{1}{8} + 6 \times \tfrac{1}{2} = 4$. Atoms touch along the face diagonal, $\sqrt{2}\,a = 4R$:

$$a = 2\sqrt{2}\,R, \qquad \text{CN} = 12, \qquad \text{APF} = \frac{\pi}{3\sqrt{2}} = 0.740$$

Aluminium, copper, nickel, silver, gold, platinum, lead and austenitic stainless steel are FCC, as is iron between 912 °C and 1394 °C (γ-Fe, austenite). FCC metals stay ductile down to cryogenic temperatures, which is why austenitic stainless steels and aluminium alloys make tanks for liquefied natural gas and liquid hydrogen.

Hexagonal close-packed (HCP) and stacking sequences

Equal spheres in a plane pack most densely in a triangular array. Stacking such layers, each new layer sits in hollows of the one below, and there are two sets of hollows, so layer positions can be labelled A, B and C. The sequence $\ldots ABABAB \ldots$ gives **hexagonal close-packed**; $\ldots ABCABC \ldots$ gives FCC, whose close-packed layers are the $\{111\}$ planes (Section 2.5). Both have CN 12 and APF 0.740, the densest possible packing of equal spheres.

The HCP conventional cell is a hexagonal prism containing $n = 12 \times \tfrac{1}{6} + 2 \times \tfrac{1}{2} + 3 = 6$ atoms, with an ideal axial ratio

$$\frac{c}{a} = \sqrt{\frac{8}{3}} = 1.633$$

Real HCP metals deviate: magnesium (1.624) is nearly ideal, titanium (1.587) compressed and zinc (1.856) elongated. HCP metals have fewer easy slip systems than FCC metals and are less formable at room temperature, which is why magnesium sheet

is usually formed warm (Chapter 13).

Stacking matters beyond metals. A single error in the FCC sequence, $\dots ABCAB \mid ABC \dots$, is a **stacking fault**, a planar defect whose energy controls how copper and stainless steel work-harden (Chapters 5 and 12); a mirror reversal, $\dots ABCACBA \dots$, is a **twin boundary**. The polytypes of silicon carbide (3C, 4H, 6H) differ only in the stacking of their Si–C layers.

Interstitial sites

Between the spheres of a close-packed structure lie holes. In FCC each atom has one **octahedral** site (surrounded by six atoms, at the cell centre and edge midpoints) and two **tetrahedral** sites (surrounded by four). The largest sphere that fits without strain has radius $r = 0.414R$ in an octahedral site and $r = 0.225R$ in a tetrahedral site. BCC, though less densely packed, has smaller holes: its octahedral sites are distorted and admit only $r = 0.155R$, its tetrahedral sites $r = 0.291R$. These numbers matter for steel. In austenite at 912 °C ($R = 0.129$ nm) the octahedral hole has radius 0.053 nm; in ferrite ($R = 0.126$ nm) the largest hole is 0.037 nm. A carbon atom (radius about 0.077 nm) is a tight fit in either, but far tighter in ferrite, which is why austenite dissolves up to 2.1 wt % carbon and ferrite at most 0.022 wt %, the fact on which all steel heat treatment rests (Chapters 7 and 15).

Worked example — Density of iron and the volume change at 912 °C. α-Fe is BCC with $n = 2$, $A = 55.85\ \text{g mol}^{-1}$, $a = 0.2866$ nm at room temperature:

$$\rho = \frac{2 \times 55.85}{(2.866 \times 10^{-8}\ \text{cm})^3 \times 6.022 \times 10^{23}} = \frac{111.7}{2.354 \times 10^{-23} \times 6.022 \times 10^{23}} = 7.88\ \text{g cm}^{-3}$$

against a measured 7.87 g cm^{-3}. At 912 °C the BCC and FCC forms coexist, with lattice parameters of about 0.2904 nm and 0.3647 nm. The volume per atom is

$$\frac{a_\alpha^3}{2} = \frac{0.2904^3}{2} = 0.012245\ \text{nm}^3, \qquad \frac{a_\gamma^3}{4} = \frac{0.3647^3}{4} = 0.012127\ \text{nm}^3$$

so iron *contracts* by $(0.012127 - 0.012245)/0.012245 = -0.97\,\%$ in volume on transforming from BCC to FCC on heating, and expands by the same amount on cooling. The hard-sphere radii, $R_\alpha = \sqrt{3}a_\alpha/4 = 0.1257$ nm and $R_\gamma = a_\gamma/(2\sqrt{2}) = 0.1289$ nm, show that the atoms themselves are effectively about 2.5 % larger in the 12-coordinated structure; the denser packing (0.74 against 0.68) wins. This dilatation is measured in a dilatometer to find transformation temperatures, and its larger cousin, the expansion when austenite becomes martensite, causes quench distortion and cracking in heat-treated parts (Chapter 15).

Metal	Structure	Lattice parameters at 20 °C (nm)	Atoms per cell	Density (g cm^{-3})
Aluminium	FCC	$a = 0.4050$	4	2.70
Copper	FCC	$a = 0.3615$	4	8.96
Nickel	FCC	$a = 0.3524$	4	8.91
α-Iron	BCC	$a = 0.2866$	2	7.87
Tungsten	BCC	$a = 0.3165$	2	19.3
Magnesium	HCP	$a = 0.3209$, $c = 0.5211$	6	1.74
α-Titanium	HCP	$a = 0.2951$, $c = 0.4683$	6	4.51
Zinc	HCP	$a = 0.2665$, $c = 0.4947$	6	7.14

2.3 Ionic and covalent structures

Building ionic structures: radius ratio

In an ionic compound the larger ions, usually the anions, form a close-packed or nearly close-packed array and the smaller cations sit in its holes. Each cation wants as many anion neighbours as possible, but only as long as it still touches them; if the hole is larger than the cation, the cation "rattles" and the structure is unstable. Geometry then fixes the minimum cation-to-anion radius ratio r_c/r_a for each coordination:

Coordination number	Polyhedron	Range of r_c/r_a	Examples
3	Triangle	0.155–0.225	B in B_2O_3 glass
4	Tetrahedron	0.225–0.414	Si in SiO_2, Zn in ZnO
6	Octahedron	0.414–0.732	Mg in MgO, Al in Al_2O_3
8	Cube	0.732–1.0	Cs in CsCl, Ca in CaF_2
12	Cuboctahedron	≥ 1.0	Ba in $BaTiO_3$

The boundaries are pure geometry: $2/\sqrt{3} - 1 = 0.155$, $\sqrt{3/2} - 1 = 0.225$, $\sqrt{2} - 1 = 0.414$ and $\sqrt{3} - 1 = 0.732$.

> **Status of the principle: radius-ratio rules**
> *Approximate*. They follow from treating ions as hard spheres of fixed radius; they predict the coordination of strongly ionic compounds about two times in three, and fail where covalency (ZnO, tetrahedral despite a ratio of about 0.53 with the sixfold Zn radius) or ion polarizability matters.

The common structure types

Figure 2.2 shows the important types.

- **Rock salt** (NaCl type): FCC anions with cations in every octahedral hole; CN 6:6. NaCl, MgO, CaO, NiO, FeO, and the hard interstitial compounds TiC, TiN and ZrC used in cutting-tool coatings.
- **Caesium chloride** (CsCl type): anions at the cube corners and a cation at the centre; CN 8:8. Although it looks like BCC, the two sites hold different atoms, so the lattice is *simple cubic* with a two-atom basis. Ordered intermetallics adopt it: NiAl (a bond coat for turbine blades), β-brass CuZn and FeAl (Chapter 3).
- **Fluorite** (CaF_2 type): FCC cations with anions in all tetrahedral holes; CN 8:4. CaF_2, UO_2 (the nuclear fuel in most power reactors), ThO_2, CeO_2 (a catalyst support), and cubic yttria-stabilized ZrO_2. The empty octahedral holes make the structure open enough to accommodate fission products and oxygen vacancies.
- **Zincblende** (sphalerite, ZnS type): FCC anions with cations in half the tetrahedral holes; CN 4:4. ZnS, GaAs, InP, cubic SiC and cubic boron nitride. If both sites hold the same atom, the result is the **diamond cubic** structure of diamond, Si and Ge.
- **Wurtzite**: the hexagonal (AB-stacked) analogue of zincblende; ZnO, GaN, AlN, BeO. Its unique *c* axis is polar, making these materials piezoelectric.
- **Perovskite** (ABO_3): a large A cation at the cube corners, a small B cation at the centre inside an octahedron of six oxygens at the face centres. $BaTiO_3$ and PZT (capacitors, sensors, ultrasound transducers), $SrTiO_3$, the high-pressure mineral bridgmanite $MgSiO_3$ (the most abundant mineral in the Earth's mantle) and the halide perovskites of emerging solar cells.
- **Spinel** (AB_2O_4): an FCC array of oxygen with A cations in one-eighth of the tetrahedral holes and B cations in half the octahedral holes, eight formula units in a cubic cell of edge about 0.81–0.84 nm. $MgAl_2O_4$ (transparent armour, refractories), magnetite Fe_3O_4, ferrites, and $LiMn_2O_4$ battery cathodes, whose lithium moves through the empty sites.
- **Corundum** (α-Al_2O_3): a nearly HCP array of oxygen with Al in two-thirds of the octahedral holes; trigonal. Alumina substrates and wear parts, sapphire windows, ruby (with Cr^{3+}), and the protective Cr_2O_3 and Al_2O_3 scales on stainless steels and superalloys (Chapter 42).

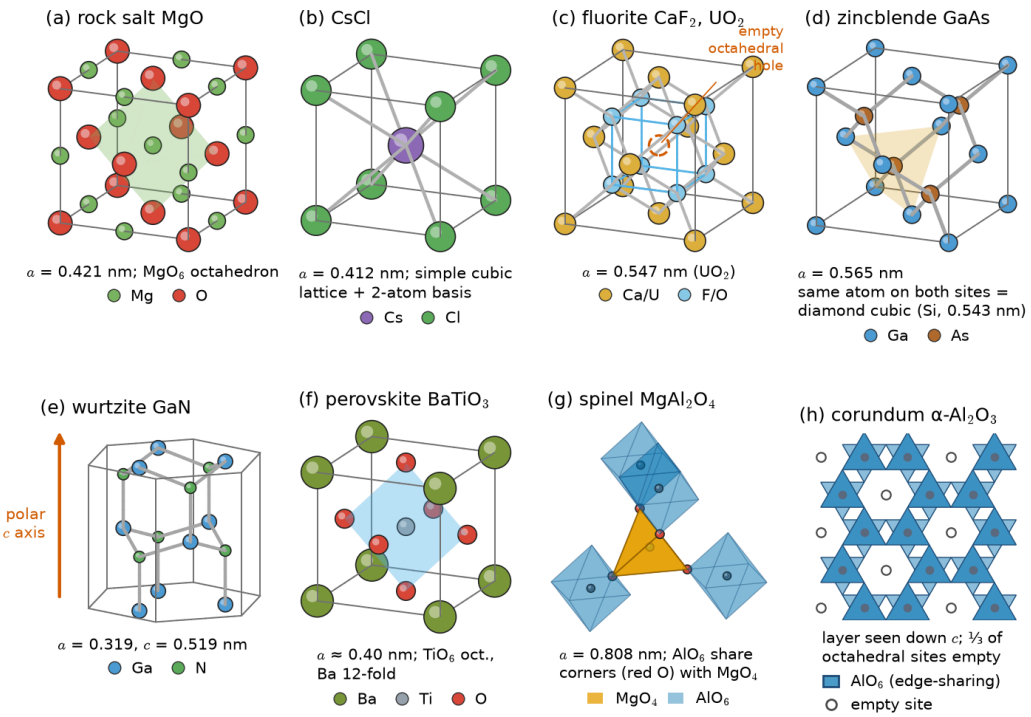

Figure 2.2 — Common ionic and covalent structure types: rock salt, CsCl, fluorite, zincblende/diamond, wurtzite and perovskite, with spinel and corundum shown as polyhedral sketches.

Compound	Structure type	Lattice parameters (nm)	Formula units per cell	Density (g cm^{-3})
NaCl	Rock salt	$a = 0.5640$	4	2.16
MgO	Rock salt	$a = 0.4212$	4	3.58
CsCl	CsCl	$a = 0.4123$	1	3.99
UO_2	Fluorite	$a = 0.5470$	4	10.96
ZrO_2 (cubic, Y-stabilized)	Fluorite	$a \approx 0.514$	4	about 6.0
Si	Diamond cubic	$a = 0.5431$	8 atoms	2.33
GaAs	Zincblende	$a = 0.5653$	4	5.32
ZnO	Wurtzite	$a = 0.3250$, $c = 0.5207$	2	5.68
$BaTiO_3$ (tetragonal)	Perovskite	$a = 0.3994$, $c = 0.4038$	1	6.01
$MgAl_2O_4$	Spinel	$a = 0.8083$	8	3.58
α-Al_2O_3	Corundum	$a = 0.4759$, $c = 1.2991$ (hexagonal setting)	6	3.99

Pauling's rules

Linus Pauling summarized the principles of ionic structures in five rules (1929), still the best qualitative guide to oxide and silicate structures.

1. **Coordination polyhedra.** A polyhedron of anions forms around each cation; the cation–anion distance is the sum of the radii and the coordination number is set by the radius ratio.
2. **Electrostatic valence.** The strength of each bond from a cation of charge z and coordination CN is $s = z/\mathrm{CN}$; in a stable structure the bond strengths reaching each anion sum to the magnitude of its charge.

3. **Sharing of polyhedron elements.** Sharing edges, and especially faces, between polyhedra lowers stability, because it brings the cations closer together.
4. **Highly charged cations avoid one another.** In crystals with different cations, those with high charge and low coordination tend not to share polyhedron elements.
5. **Parsimony.** The number of essentially different kinds of site tends to be small.

Rule 2 is quantitative and easy to check. In perovskite $BaTiO_3$, each oxygen is bonded to 2 Ti (each bond $s = 4/6 = 2/3$) and 4 Ba ($s = 2/12 = 1/6$): $2 \times \frac{2}{3} + 4 \times \frac{1}{6} = 2$, exactly the oxygen charge. In spinel $MgAl_2O_4$ each oxygen bonds to one tetrahedral Mg ($2/4 = \frac{1}{2}$) and three octahedral Al ($3/6 = \frac{1}{2}$ each): total 2. In fluorite each F^- bonds to four Ca ($2/8 = \frac{1}{4}$): total 1. And in silicates, each Si–O bond has $s = 4/4 = 1$, so an oxygen can bond to at most *two* silicon atoms: the SiO_4 tetrahedra can share corners but never edges or faces (rule 3), which is the founding principle of silicate structures (Section 2.4).

> **Status of the principle: Pauling's rules**
> *Empirical.* They were distilled from known structures and rationalized by electrostatics; they hold well for predominantly ionic oxides, silicates and halides and become unreliable for covalent, metallic or highly polarizable compounds.

Perovskite stability: the tolerance factor

For the perovskite structure, the ideal cubic geometry requires the A–O distance to equal $\sqrt{2}$ times the B–O distance. Goldschmidt's **tolerance factor**

$$t = \frac{r_A + r_O}{\sqrt{2}\,(r_B + r_O)}$$

measures how well the ions fit. Cubic perovskites form for t close to 1; for $t > 1$ the B cation is too small for its octahedron and tends to displace off-centre, giving ferroelectricity; for $t < 1$ the octahedra tilt and the symmetry drops to orthorhombic or rhombohedral; below about 0.8, other structures form.

> **Status of the principle: Goldschmidt tolerance factor**
> *Empirical.* A hard-sphere geometric criterion fitted to known perovskites; it is a useful screen (perovskites form for roughly $0.8 < t < 1.05$) but cannot predict the detailed distortions, which need the full energetics (Chapter 55).

Worked example — From ionic radii to structure and density. *MgO.* With Shannon radii $r_{Mg^{2+}} = 0.072$ nm (sixfold) and $r_{O^{2-}} = 0.140$ nm, $r_c/r_a = 0.072/0.140 = 0.514$, inside the octahedral range 0.414–0.732, so rock salt is predicted, as observed. Cations and anions touch along the cube edge, so $a = 2(r_c + r_a) = 2(0.212) = 0.424$ nm, within 0.7 % of the measured 0.4212 nm. With four MgO units (40.30 g mol^{-1}) per cell,

$$\rho = \frac{4 \times 40.30}{(4.212 \times 10^{-8})^3 \times 6.022 \times 10^{23}} = 3.58 \text{ g cm}^{-3}$$

in agreement with dense magnesia refractories; a fired MgO brick of bulk density 3.0 g cm^{-3} therefore contains $1 - 3.0/3.58 = 16\,\%$ porosity.

Titanates. With $r_{Ti^{4+}} = 0.0605$ nm (sixfold) and twelvefold A-site radii of 0.161 nm (Ba), 0.144 nm (Sr) and 0.134 nm (Ca):

$$t_{BaTiO_3} = \frac{0.161 + 0.140}{\sqrt{2} \times (0.0605 + 0.140)} = \frac{0.301}{0.2836} = 1.062$$

and similarly $t = 1.002$ for $SrTiO_3$ and 0.966 for $CaTiO_3$. The ranking matches the behaviour: $SrTiO_3$ is cubic at room temperature; $BaTiO_3$, with its too-small Ti, is ferroelectric with Ti displaced along c (tetragonal, $c/a = 1.011$); $CaTiO_3$, with its too-small Ca, has tilted octahedra and orthorhombic symmetry. Three numbers from a table of radii sort a capacitor dielectric, a substrate crystal and a mineral.

2.4 Silicate structures and polymorphism

The SiO_4 tetrahedron and its connectivity

Silicon and oxygen are the two most abundant elements in the Earth's crust, and silicates make up glass, cement, clay, brick, porcelain, refractories, rocks and sand. All are built from the same unit, the SiO_4 tetrahedron (Si–O 0.162 nm), with a radius ratio (about 0.19 for Si^{4+}, borderline between threefold and fourfold) and partly covalent bonding that both favour fourfold coordination. Following Pauling's second and third rules, tetrahedra link only through shared corners, and each shared ("bridging") oxygen joins exactly two tetrahedra. Silicates are classified by how many corners of each tetrahedron are shared, written Q^n with $n = 0$–4 bridging oxygens (Figure 2.3).

Class	Shared corners (Q^n)	Silicate unit	Example materials
Orthosilicates (nesosilicates)	0	Isolated SiO_4^{4-}	Forsterite Mg_2SiO_4 (refractories), zircon $ZrSiO_4$, Ca_2SiO_4 and Ca_3SiO_5 in cement clinker
Pyrosilicates (sorosilicates)	1	$Si_2O_7^{6-}$	Melilite in slags
Rings and single chains	2	$(SiO_3)_n^{2n-}$	Beryl (rings); pyroxenes such as enstatite $MgSiO_3$ in steatite ceramics (chains)
Double chains	2 and 3	$Si_4O_{11}^{6-}$	Amphiboles, including the asbestos minerals
Sheets (phyllosilicates)	3	$Si_2O_5^{2-}$	Kaolinite $Al_2Si_2O_5(OH)_4$ (clay, porcelain), talc, mica
Frameworks (tektosilicates)	4	SiO_2	Quartz, cristobalite, feldspars (with Al substituting for Si), zeolites

The connectivity explains engineering behaviour. Sheet silicates cleave into flakes and, as clays, slide past each other when wet, which makes clay mouldable before firing. Chain silicates such as asbestos form fibres. Frameworks are hard and refractory. Cement clinker phases are orthosilicates whose isolated tetrahedra react readily with water to form the calcium silicate hydrate that binds concrete (Chapter 18). In glass, adding Na_2O breaks bridging oxygens into non-bridging ones, lowering connectivity, viscosity and melting temperature (Section 2.7 and Chapter 17).

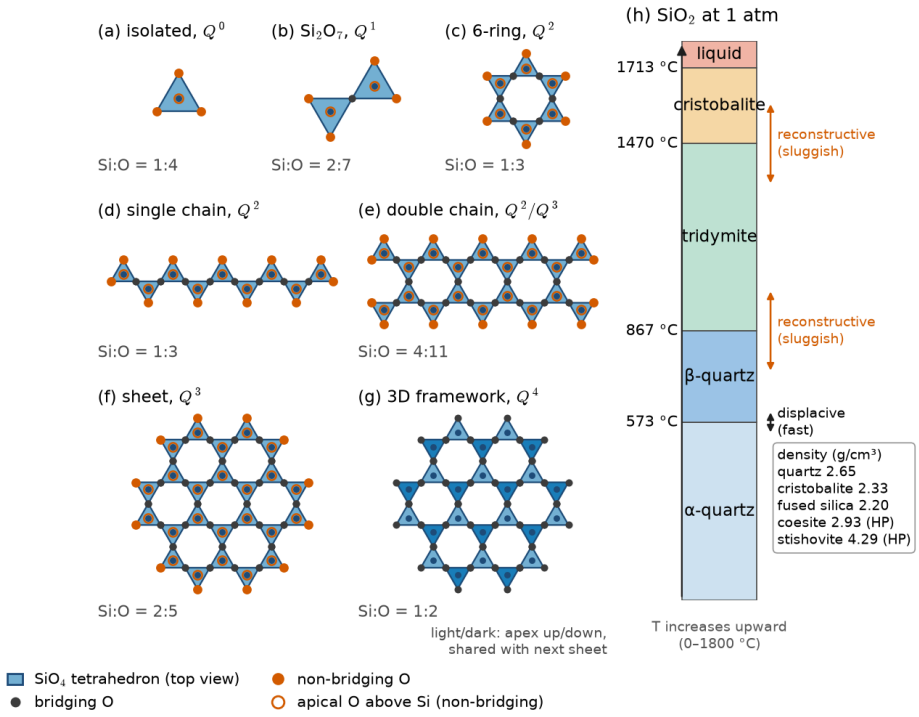

Figure 2.3 — Silicate structures built from corner-sharing SiO4 tetrahedra, from isolated tetrahedra to sheets and frameworks, and the polymorphs of silica.

Polymorphism

Many materials change crystal structure with temperature or pressure; this is **polymorphism** (called *allotropy* for elements). Two kinds of transformation differ in speed. **Displacive** transformations need only small atomic shifts and bond-angle changes; they happen almost instantly and cannot be suppressed. **Reconstructive** transformations require bonds to break and re-form; they are sluggish and the high-temperature form can often be retained metastably.

Material	Polymorphs (at 1 atm unless stated)	Engineering consequence
Iron	α (BCC) → γ (FCC) at 912 °C → δ (BCC) at 1394 °C → liquid at 1538 °C	Heat treatment of steels (Chapters 7, 10, 15)
Titanium	α (HCP) → β (BCC) at 882 °C	α, α+β and β alloys (Chapter 16)
Silica	α-quartz → β-quartz at 573 °C (displacive) → tridymite 867 °C → cristobalite 1470 °C (reconstructive) → liquid 1713 °C; coesite and stishovite at high pressure	Silica refractories and porcelain must be heated slowly through 573 °C, where quartz changes volume by roughly 1–2 %
Zirconia	Monoclinic → tetragonal at about 1170 °C → cubic at about 2370 °C	4–5 % volume change; stabilization and transformation toughening (opening of this chapter, Chapter 14)
Carbon	Graphite (stable) and diamond (metastable at ambient conditions)	Synthetic diamond at high pressure and temperature; graphite as lubricant and electrode
Tin	β (body-centred tetragonal, 7.29 g cm^{-3}) → α (diamond cubic, 5.77 g cm^{-3}) below 13.2 °C	"Tin pest" disintegration of lead-free solders and tin objects in the cold
TiO_2	Anatase, brookite (metastable) → rutile on heating	Photocatalytic coatings (anatase) versus white pigment (rutile)

Polymorphism is one of the most powerful tools of processing. A structure stable at high temperature can be retained, transformed partly, or transformed into a fine mixture by controlling cooling; Chapters 7, 9 and 10 develop the thermodynamics and kinetics, and Chapter 3 shows how the symmetry relationship between two polymorphs determines how many orientation variants form.

2.5 Directions, planes, Miller and Miller–Bravais indices

Directions

A crystallographic **direction** is written $[uvw]$: the components of a vector parallel to it, expressed in multiples of a, b, c and reduced to the smallest integers, with bars for negative values ($[1\bar{1}0]$ means $u = 1$, $v = -1$, $w = 0$). Directions related by symmetry form a **family** in angle brackets: in a cubic crystal $\langle 100 \rangle$ stands for the six cube-edge directions and $\langle 111 \rangle$ for the eight body diagonals. Points inside the cell are given as fractional coordinates, such as $\tfrac{1}{2}\tfrac{1}{2}\tfrac{1}{2}$ for the body centre.

In cubic crystals only, directions behave as Cartesian vectors, and the angle between two directions is

$$\cos\theta = \frac{u_1u_2 + v_1v_2 + w_1w_2}{\sqrt{u_1^2 + v_1^2 + w_1^2}\,\sqrt{u_2^2 + v_2^2 + w_2^2}}$$

For $[100]$ and $[111]$, $\cos\theta = 1/\sqrt{3}$ and $\theta = 54.74^\circ$; for $[110]$ and $[111]$, 35.26°; between two body diagonals, 70.53°.

Planes and Miller indices

A crystallographic **plane** always means a whole family of parallel, equally spaced planes passing through lattice points. Its **Miller indices** (hkl) (Figure 2.4) are found by taking the intercepts of one plane on the three axes in units of a, b, c (shifting the origin if the plane passes through it), taking reciprocals (a plane parallel to an axis has intercept ∞ and index 0), and clearing fractions to the smallest integers. A plane with intercepts $1, 2, \infty$ has reciprocals $1, \tfrac{1}{2}, 0$ and indices (210). Families of symmetry-equivalent planes are written in braces: $\{100\}$ for the six cube faces, $\{111\}$ for the eight octahedral planes. Reciprocals are used because (hkl) are the components of the plane normal in the *reciprocal lattice*, the natural language of diffraction (Chapter 4).

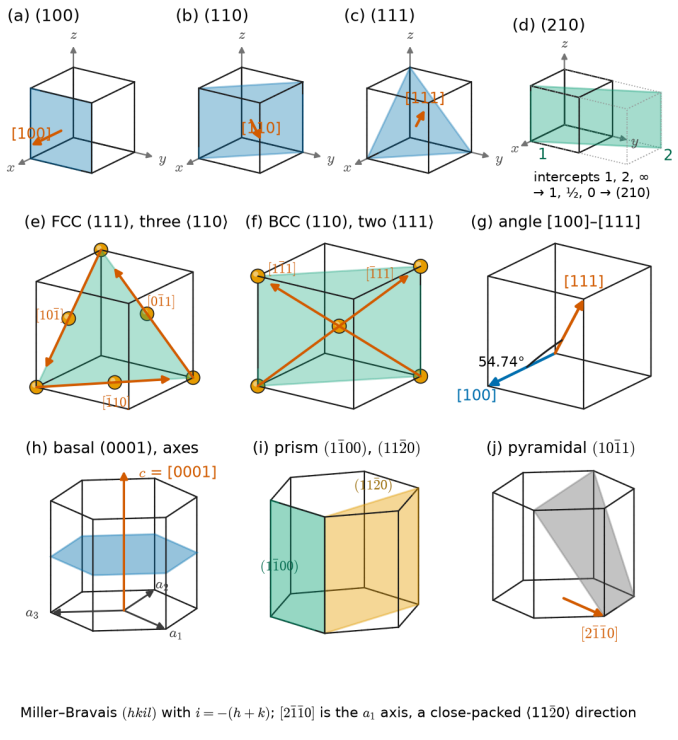

Figure 2.4 — Miller indices of planes and directions in a cubic cell, the close-packed directions and planes of FCC and BCC, and the four-index system for hexagonal crystals.

Three results for cubic crystals

The direction $[hkl]$ is perpendicular to the plane (hkl), so the normal to a (111) plane is the body diagonal.

The interplanar spacing is

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

For copper, $d_{111} = 0.3615/\sqrt{3} = 0.2087$ nm, $d_{200} = 0.1808$ nm and $d_{220} = 0.1278$ nm. Diffraction measures exactly these spacings (Chapter 4).

The zone rule. A direction $[uvw]$ lies in a plane (hkl) when

$$hu + kv + lw = 0$$

So $[1\bar{1}0]$ lies in (111); this is how one finds the slip directions lying in a slip plane or the line along which a plane meets a surface.

> **Status of the principle: interplanar spacing and zone rule**
> *Derived.* Both are pure lattice geometry: the spacing formula holds for cubic lattices only (Chapter 4 gives the general form), and the zone rule holds in every crystal system when (hkl) and $[uvw]$ refer to the same axes.

Linear and planar densities

The **linear density** of a direction is the number of atoms centred on it per unit length; the **planar density** of a plane is the number of atoms centred on it per unit area. They identify the most densely packed directions and planes, which are the slip directions and slip planes of metals (Chapter 13), the planes along which crystals cleave and the surfaces that are slowest to etch or oxidize.

For FCC, the (100) face contains a face-centre atom and four quarter-corner atoms in area a^2, so $\mathrm{PD}_{100} = 2/a^2$; the (110) rectangle $a \times \sqrt{2}a$ contains 2 atoms, $\mathrm{PD}_{110} = \sqrt{2}/a^2$; the (111) triangle of side $\sqrt{2}a$ and area $(\sqrt{3}/2)a^2$ contains $3 \times \frac{1}{6} + 3 \times \frac{1}{2} = 2$ atoms, $\mathrm{PD}_{111} = 4/(\sqrt{3}a^2)$. Atoms touch along $\langle 110 \rangle$.

Worked example — Close-packed planes and directions in copper and iron. For copper ($a = 0.3615$ nm, $a^2 = 0.1307$ nm^2):

$$\mathrm{PD}_{100} = \frac{2}{0.1307} = 15.3, \qquad \mathrm{PD}_{110} = \frac{1.414}{0.1307} = 10.8, \qquad \mathrm{PD}_{111} = \frac{4}{1.732 \times 0.1307} = 17.7 \ \text{atoms nm}^{-2}$$

The $\{111\}$ planes are both the densest and the most widely spaced ($d_{111} = 0.209$ nm against $d_{200} = 0.181$ nm), and the linear density along $\langle 110 \rangle$ is $2/(\sqrt{2}a) = 3.91$ nm^{-1}, just $1/2R$. Copper therefore slips on $\{111\}\langle 110 \rangle$: 4 planes × 3 directions = 12 slip systems.

For BCC iron ($a = 0.2866$ nm), atoms touch along the body diagonal: $\mathrm{LD}_{111} = 2/(\sqrt{3}a) = 4.03$ nm^{-1}. The densest plane is $\{110\}$, a rectangle $a \times \sqrt{2}a$ containing 2 atoms:

$$\mathrm{PD}_{110} = \frac{\sqrt{2}}{a^2} = \frac{1.414}{0.08214} = 17.2, \qquad \mathrm{PD}_{100} = \frac{1}{a^2} = 12.2, \qquad \mathrm{PD}_{111} = \frac{1}{\sqrt{3}a^2} = 7.0 \ \text{atoms nm}^{-2}$$

Iron slips along $\langle 111 \rangle$, mainly on $\{110\}$. But no BCC plane is truly close-packed (17.2 atoms nm^{-2} against 18.7 for an ideal close-packed layer of iron-sized atoms), so dislocations in BCC metals face a strong lattice resistance that rises steeply as temperature falls; that is one root of the ductile–brittle transition of ferritic steels (Chapters 12 and 14).

Miller–Bravais indices for hexagonal crystals

In hexagonal crystals three indices hide the sixfold symmetry: equivalent planes get indices that look unrelated. The **Miller–Bravais** system uses three basal axes $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$ at 120° and $\mathbf{c}$ perpendicular to them. A plane is written $(hkil)$ with

$$i = -(h + k)$$

so that the six prism faces become the family $\{1\bar{1}00\}$. The important planes for HCP metals (Mg, Ti, Zn, Zr) and wurtzite crystals are the **basal** plane (0001), the **prism** planes $\{1\bar{1}00\}$ and $\{11\bar{2}0\}$, and the **pyramidal** planes such as $\{10\bar{1}1\}$. Directions are written $[uvtw]$ and converted from the three-index form $[UVW]$ by

$$u = \tfrac{1}{3}(2U - V), \qquad v = \tfrac{1}{3}(2V - U), \qquad t = -(u + v), \qquad w = W$$

The a_1 axis, $[100]$ in three indices, becomes $\frac{1}{3}[2\bar{1}\bar{1}0]$, i.e. the $\langle 11\bar{2}0 \rangle$ family, the close-packed slip directions of HCP metals. Titanium alloys deform mainly by prism slip, magnesium by basal slip; their scarcity of slip systems, and the textures that rolling gives them, are why both are anisotropic and harder to form than aluminium (Chapters 13 and 16).

2.6 Polycrystals, texture and grain size

Single crystals and polycrystals

In a **single crystal** the periodic arrangement extends uninterrupted through the whole piece. Large single crystals must be grown deliberately from a seed: silicon wafers, sapphire windows, quartz resonators, laser crystals and the single-crystal turbine blades of Chapter 1. Most crystalline materials are instead **polycrystalline**: aggregates of many small crystals (**grains**) with different orientations, joined at **grain boundaries** about 0.5–1 nm thick in which the atomic arrangement is disturbed. Polycrystals form whenever crystals nucleate at many places at once: in a casting, in a sintered powder, in a recrystallizing rolled sheet, in a weld or in an additively manufactured track. Grain boundaries are described in Chapter 5; here we need their number and the orientations of the grains they separate.

Texture

If grain orientations are random, a polycrystal is isotropic on scales much larger than the grain size, even when each grain is anisotropic. If the orientations are not random, the polycrystal has a **texture**. Textures are written as $\{hkl\}\langle uvw \rangle$, meaning that the $\{hkl\}$ planes of most grains lie parallel to the sheet surface and their $\langle uvw \rangle$ directions point along the rolling direction; a **fibre texture** has one crystal direction aligned with an axis and random rotation about it. Representative cases from across engineering:

- **Steel for deep drawing** is processed to have $\{111\}$ planes parallel to the sheet (the γ-fibre). This resists thinning, so car-body panels and food cans can be drawn deep without splitting.

- **Aluminium can stock** combines rolling and recrystallization textures in balance; if one dominates, the drawn cup forms four or six "ears" that must be trimmed off.
- **Grain-oriented electrical steel** has the Goss texture $\{110\}\langle 001\rangle$, putting the easy magnetization direction along the strip (Chapter 29).
- **Zirconium-alloy fuel cladding** is rolled so that basal-plane normals point radially, which makes hydrides precipitate circumferentially, where they do little harm.
- **Drawn wire and electroplated or sputtered films** have fibre textures; in bone, the *c* axes of hydroxyapatite crystals align with the collagen fibres.
- **Additively manufactured metals** grow columnar grains with $\langle 100\rangle$ along the build direction, making their properties anisotropic (Chapter 48).

Chapter 3 introduces the pole figure, the standard way to display and measure texture.

Measuring grain size

Grain size is measured on a polished and etched section. The **mean linear intercept** $\bar{\ell}$ is found by laying test lines of total length L_T on a micrograph and counting the P grain-boundary intersections: $\bar{\ell} = L_T / P$. A result of stereology links it exactly to the boundary area per unit volume:

$$S_V = 2 P_L = \frac{2}{\bar{\ell}}$$

where $P_L = P / L_T$; the factor 2 comes from averaging over the angles at which random lines cross a surface.

> **Status of the principle:** $S_V = 2P_L$
> *Derived.* A theorem of geometric probability, exact for isotropic random test lines whatever the grain shape; for elongated or textured grains the test lines must be oriented in several directions and the results averaged.

The **ASTM grain-size number** G (standard E112) is defined so that at a magnification of 100× there are

$$N = 2^{G-1}$$

grains per square inch of image. Larger G means finer grains, and each step of 1 halves the grain area.

Worked example — From ASTM number to boundary area. A structural steel plate is rated $G = 8$. At 100×, $N = 2^{7} = 128$ grains per square inch of image. One square inch at 100× corresponds to $(25.4/100)^2 = 0.0645$ mm^2 of specimen, so there are $128 / 0.0645 = 1984$ grains per mm^2, an average grain section area of 504 μm^2 and an equivalent square side of $\sqrt{504} = 22.5$ μm. The E112 relation between G and mean intercept gives $\bar{\ell} \approx 20$ μm. The boundary area per unit volume is then

$$S_V = \frac{2}{20 \times 10^{-6}\ \text{m}} = 1.0 \times 10^{5}\ \text{m}^{-1}$$

so every cubic centimetre of this steel contains about 0.1 m^2 of grain boundary. Refining to $G = 12$ (about 5 μm) multiplies the boundary area by about four. Because boundaries obstruct dislocations, such refinement is the main strengthening mechanism of controlled-rolled high-strength low-alloy steels, with yield strength rising roughly as $d^{-1/2}$ (the Hall–Petch relation, Chapter 13); it also lowers the ductile–brittle transition temperature, the change that cured the Liberty ships. For creep at high temperature the opposite holds: boundaries slide and carry fast diffusion, so turbine blades remove them altogether.

2.7 Amorphous solids and glasses

Order without periodicity

An **amorphous** solid has no long-range periodic order, but it is not random. Each atom keeps nearly the same number of neighbours at nearly the same distance as in the crystal, because these are fixed by bonding; what is lost is the regular repetition of that environment over large distances. In silica glass, every Si is surrounded by four O and every O bridges two Si, exactly as in quartz, but the Si–O–Si angle between tetrahedra varies from about 120° to 180° (mean about 144°), so the network never repeats. This is Zachariasen's **continuous random network** (1932), which also gave rules for which oxides form glasses: the cation should have low coordination (3 or 4), oxygen should bridge no more than two cations, polyhedra should share corners

rather than edges or faces, and at least three corners of each should be shared. SiO_2, B_2O_3, GeO_2 and P_2O_5 satisfy them and are **network formers**; Na_2O and CaO break the network and are **modifiers** (Chapter 17).

> **Status of the principle: Zachariasen's rules for glass formation**
> *Empirical.* They are structural criteria inferred from oxide glasses; they explain why SiO_2 and B_2O_3 form glasses readily, but they say nothing about kinetics and do not apply to metallic or polymer glasses.

The radial distribution function

The structure of a non-crystalline solid is described statistically by the **pair distribution function** $g(r)$: the average density of atoms at distance r from any atom, divided by the mean density ρ_0. The number of atoms in a spherical shell between r and $r + dr$ is $4\pi r^2 \rho_0 g(r)\,dr$, a quantity called the **radial distribution function**, and the area under its first peak is the mean coordination number:

$$N_1 = \int_{\text{first peak}} 4\pi r^2 \rho_0\, g(r)\, dr$$

In a crystal $g(r)$ is a series of sharp peaks continuing to arbitrary distance. In a glass the first peak is almost as sharp as in the crystal, the second is broader and beyond three or four neighbour distances $g(r)$ approaches 1: short-range order without long-range order (Figure 2.5). For silica glass the peaks lie at 0.162 nm (Si–O), 0.265 nm (O–O, the tetrahedron edge) and about 0.31 nm (Si–Si, consistent with $2 \times 0.162 \times \sin 72^\circ = 0.308$ nm for a 144° bridge), and the first-peak area gives four O around each Si. Diffraction from a glass shows broad humps instead of sharp Bragg peaks; Fourier transforming them yields $g(r)$, the basis of the pair-distribution-function analysis of Chapter 53.

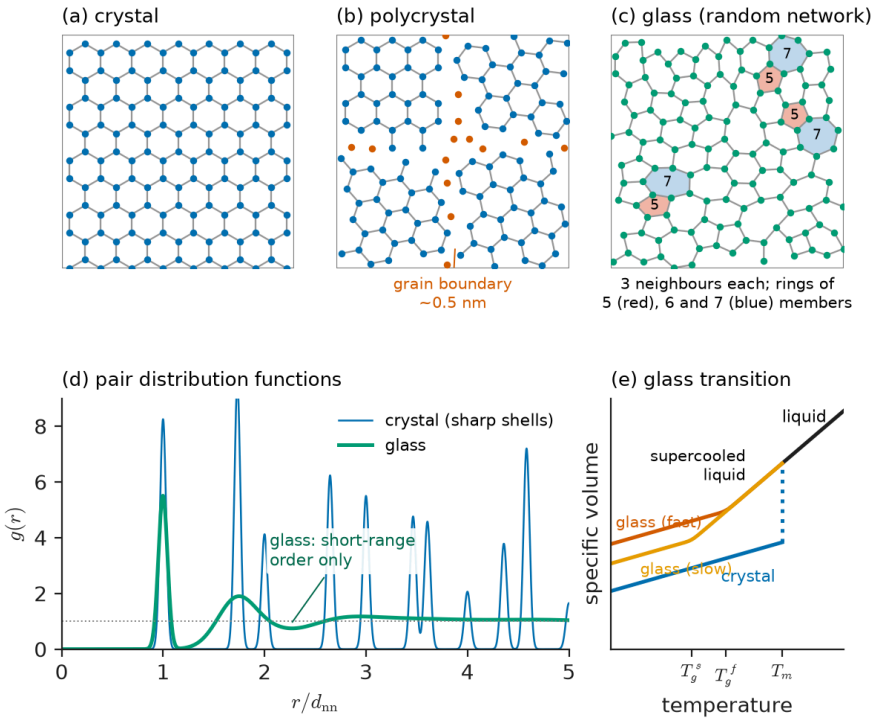

Figure 2.5 — Crystal, polycrystal and glass in two dimensions, with their pair distribution functions and the volume–temperature behaviour that defines the glass transition.

How glasses form and the glass transition

A liquid cooled below its melting point T_m must rearrange its atoms to crystallize, which takes time. If it is cooled faster than it can rearrange, or its viscosity rises so steeply that rearrangement stalls, the disordered structure is frozen in. The specific volume then shows no jump at T_m; it follows the supercooled-liquid line until, over a narrow range, its slope falls to that of a solid. The intersection defines the **glass transition temperature** T_g (Figure 2.5), which shifts slightly with cooling rate, so the glass transition is kinetic, not a true phase transition. Typical values: fused silica about 1200 °C, soda-lime glass about 550 °C, polystyrene about 100 °C, PMMA about 105 °C, epoxy 120–180 °C, polyethylene about −120 °C.

Glass-forming ability varies enormously:

- **Silicate glasses** form at ordinary cooling rates because their networks of strong directional bonds make the liquid extremely viscous.
- **Polymers** with irregular chains (atactic polystyrene, PMMA, polycarbonate, epoxies) cannot crystallize at all and are always glassy or rubbery.
- **Metals** crystallize so readily that metallic glasses needed cooling rates of about 10^5–10^6 K s^{-1}, achieved by melt spinning ribbon 20–30 μm thick; multicomponent alloys such as Zr–Ti–Cu–Ni–Be now form "bulk" glasses centimetres thick at about 1 K s^{-1}. Their atoms are arranged in a dense random packing (packing fraction about 0.64, compared with 0.74 for FCC); with no dislocations, they combine high strength with large elastic strain, used in golf-club faces, precision springs and transformer cores.
- **Thin films** deposited from vapour at low temperature are often amorphous, because arriving atoms are buried before they find crystal sites: sputtered oxide coatings, diamond-like carbon, and amorphous silicon.

Amorphous materials are isotropic, have no grain boundaries to act as leakage or diffusion paths, and can be formed as viscous liquids. Their disadvantage is that they are not in equilibrium: given time at a high enough temperature, they crystallize ("devitrify").

2.8 Polymer chain packing and crystallinity (preview)

A polymer chain is a long, flexible covalent backbone, and whether it can crystallize depends on its regularity. Linear polyethylene, polytetrafluoroethylene, nylons and isotactic polypropylene have regular chains that can pack side by side in a lattice; atactic polystyrene and PMMA, with randomly placed side groups, cannot. Even regular polymers never crystallize completely, because their chains are entangled in the melt. The result is a **semicrystalline** solid: thin crystalline **lamellae**, typically 10–20 nm thick, in which chains fold back and forth, separated by amorphous layers and linked by **tie molecules** that pass from one lamella to the next. On cooling from the melt, lamellae grow radially from nuclei into **spherulites** 1–100 μm across (Figure 2.6). The amorphous regions give toughness and, above T_g, flexibility; the crystals give stiffness, solvent resistance and a barrier to gas permeation.

Polymer crystals are ordinary crystals with an unusual basis. Polyethylene is orthorhombic with $a = 0.741$ nm, $b = 0.494$ nm and $c = 0.2547$ nm along the chain, two chains passing through each cell and four CH_2 groups per cell. The chain axis is a screw axis (Chapter 3): in the planar zigzag of polyethylene, each CH_2 unit is rotated 180° about the axis and advanced $c/2$; isotactic polypropylene forms a helix with three repeat units per turn.

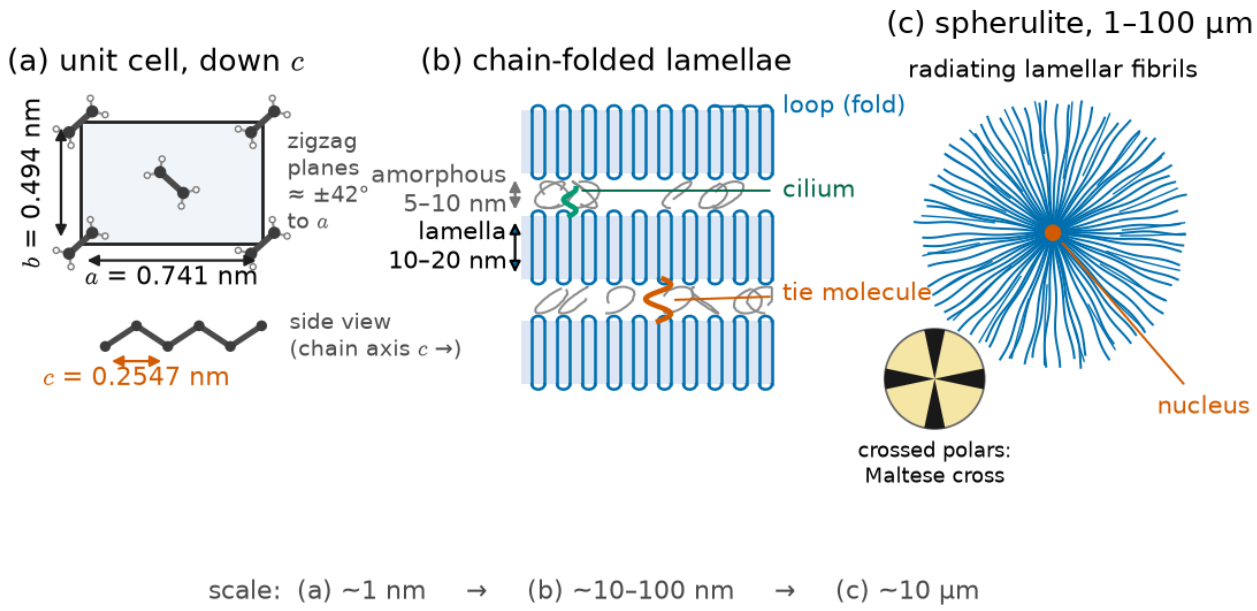

Figure 2.6 — Chain packing in crystalline polyethylene, chain-folded lamellae and the spherulite, from the unit cell to the micrometre scale.

Degree of crystallinity

Because crystals pack chains more densely than the amorphous phase, density measures crystallinity. If ρ_c and ρ_a are the densities of fully crystalline and fully amorphous polymer, a two-phase model gives the mass fraction crystallinity of a sample of density ρ:

$$x_c = \frac{\rho_c\,(\rho - \rho_a)}{\rho\,(\rho_c - \rho_a)}$$

> **Status of the principle: two-phase crystallinity from density**
> *Approximate*. It treats the polymer as a mixture of perfect crystal and bulk amorphous phase; it ignores the interphase between them and voids or fillers, so it agrees with calorimetric and X-ray measurements to within a few per cent for unfilled polymers.

Worked example — Crystallinity of polyethylenes. The crystal density follows from the unit cell: four CH_2 units of 14.03 g mol^{-1} in a volume abc,

$$\rho_c = \frac{4 \times 14.03}{(0.741 \times 0.494 \times 0.2547) \times 10^{-21}\ \text{cm}^3 \times 6.022 \times 10^{23}} = \frac{56.11}{0.09323 \times 602.2} = 1.00\ \text{g cm}^{-3}$$

With $\rho_a = 0.855$ g cm^{-3}, a high-density polyethylene of density 0.960 g cm^{-3} has

$$x_c = \frac{1.00 \times (0.960 - 0.855)}{0.960 \times (1.00 - 0.855)} = \frac{0.105}{0.1392} = 0.754$$

about 75 % crystalline by mass, while a low-density polyethylene of 0.920 g cm^{-3}, whose short-chain branches disrupt packing, is only 49 % crystalline. The difference of 4 % in density marks a factor of three to four in stiffness, much better barrier properties in the HDPE milk bottle and gas pipe, and much better flexibility and clarity in the LDPE film. Processing changes crystallinity too: a PET bottle preform quenched from the melt is clear and amorphous, and stretching it during blow moulding induces crystallinity that stiffens the bottle and slows carbon-dioxide loss. Chapters 19–21 develop polymer structure in full.

2.9 Liquid crystals and quasicrystals (preview)

Liquid crystals

Between the perfect order of a crystal and the disorder of a liquid lie **liquid crystals**: fluids whose rod-like or disc-like molecules keep some orientational, and sometimes positional, order. In the **nematic** phase the molecules point on average along a common direction, the *director* **n**, but their centres are random. The degree of alignment is measured by the **order parameter**

$$S = \left\langle \frac{3\cos^2\theta - 1}{2} \right\rangle$$

where θ is the angle between a molecule and the director and the brackets denote an average; $S = 1$ for perfect alignment and $S = 0$ for an isotropic liquid, with typical nematics between about 0.3 and 0.8. **Smectic** phases add layering, and **cholesteric** phases twist the director into a helix whose pitch can match the wavelength of light. *Thermotropic* liquid crystals form within a temperature window: the display compound 5CB is nematic between about 24 °C and 35 °C. *Lyotropic* ones form in solution above a critical concentration.

The engineering uses are significant. Liquid-crystal displays exploit the optical anisotropy of nematics and the ease of turning the director with a few volts (Chapter 37). Aramid fibres (Kevlar) are spun from a lyotropic liquid-crystalline solution in sulphuric acid; the chains are already aligned in the fluid, and the fibre reaches a modulus above 100 GPa. Liquid-crystalline polymers mould into thin-walled connectors with very low shrinkage (Chapter 20). Cell membranes and many biological tissues are liquid-crystalline too.

Quasicrystals

In 1982 Dan Shechtman, examining rapidly solidified Al–Mn alloys in an electron microscope, saw diffraction patterns with sharp spots, proof of long-range order, arranged with tenfold symmetry. Such symmetry is forbidden to any periodic crystal (Chapter 3 proves that only two-, three-, four- and sixfold rotations are compatible with a lattice). The material was a **quasicrystal**: ordered but not periodic. Its atomic arrangement is the three-dimensional analogue of a Penrose tiling, in which two kinds of tile cover the plane with fivefold local symmetry but never repeat, and in which the ratio of the numbers of the two tiles tends to the golden ratio $\tau = (1+\sqrt{5})/2 = 1.618$. Quasicrystals can be described as slices through periodic structures in higher-dimensional space. Their discovery forced the International Union of Crystallography to redefine a crystal in 1992 as any solid with an essentially discrete diffraction pattern, and earned Shechtman the 2011 Nobel Prize in Chemistry.

Quasicrystals are hard, have low surface energy and friction, and conduct heat and electricity poorly for metallic alloys. They have been used as non-stick and wear-resistant coatings and as fine strengthening particles in some precipitation-hardened maraging steels and magnesium alloys. Chapter 3 returns to them, together with incommensurately modulated structures, the other important family of ordered but non-periodic solids.

What remains open

Open question	Where it is answered
What symmetry operations relate the atoms of a crystal, and why are only fourteen lattices and 230 space groups possible?	Chapter 3
How is texture measured and displayed with stereographic projections and pole figures?	Chapter 3
How do diffraction patterns reveal lattice spacings, structures and grain size?	Chapter 4
What do grain boundaries, stacking faults, vacancies and dislocations look like, and how do they control properties?	Chapter 5
Why does a given structure, polymorph or glass form under given conditions?	Chapters 6, 7 and 9
How does carbon dissolve and diffuse in the interstitial sites of iron?	Chapters 8 and 15
How do slip systems and grain size set the strength of metals?	Chapters 12–13
How does transformation toughening make zirconia tough?	Chapters 14 and 17
How do cement silicates hydrate to bind concrete?	Chapter 18
How do polymer chains crystallize, and how do lamellae and T_g set polymer properties?	Chapters 19–20
How do liquid crystals work in displays and actuators?	Chapter 37
How are glasses and amorphous structures measured by pair-distribution-function analysis?	Chapter 53

Next: Chapter 3 makes the idea of a periodic crystal precise through symmetry: the operations that leave a crystal unchanged, the point groups, Bravais lattices and space groups they generate, the stereographic projection, and the ordered superlattices, quasicrystals and modulated structures that extend the classical picture.

Chapter 3 — Symmetry and Space Groups

Part I · Structure of Materials

Every medical ultrasound probe, every car parking sensor and every sonar array contains a piezoelectric ceramic, almost always lead zirconate titanate (PZT). Above its Curie temperature of about 350 °C, PZT is a cubic perovskite and is not piezoelectric at all. A cubic crystal has a centre of symmetry, and a centrosymmetric crystal cannot produce an electric polarization when it is squeezed: any polarization would have to point in some direction, and inversion would turn it into its opposite without changing the crystal. Only when PZT cools through its Curie point and the titanium and zirconium ions shift off-centre, lowering the symmetry to tetragonal or rhombohedral, does piezoelectricity become possible. Even then a freshly sintered ceramic shows almost none: each grain splits into domains whose polarizations point along the six (tetragonal) or eight (rhombohedral) symmetry-equivalent directions, and they cancel. The manufacturer must *pole* the part by applying a strong field at elevated temperature to align the domains. Whether a material can be piezoelectric, how many domain orientations it forms and how it must be processed all follow from symmetry alone, before any property has been measured.

Symmetry is the most economical description of a crystal: instead of listing every atom, one lists a few atoms and the operations that generate the rest. It is also a powerful constraint on properties, a guide to phase transformations and the key to reading the crystallographic databases that every materials engineer eventually uses. This chapter develops the theory at the level an engineer needs.

> **Which symmetry operations can a crystal have, and why are only certain lattices, point groups and space groups possible?**
> **How do symmetry and its breaking explain piezoelectricity, domains, ordering, texture and the existence of quasicrystals?**

Concept	Engineering destination
Symmetry operations, including screw axes and glide planes	Reading structure files; predicting diffraction absences (Chapter 4)
32 point groups and 7 crystal systems	Which tensor properties exist: piezo-, pyro- and ferroelectricity (Chapters 4, 27)
14 Bravais lattices	Choosing cells; recognizing equivalent descriptions; lattice-based calculations (Chapters 4, 55)
230 space groups; International Tables	Atomic positions, Wyckoff sites, databases and simulation inputs (Chapters 55–57)
Stereographic projection and pole figures	Texture control in sheet, wire, cladding and additively manufactured parts (Chapters 46, 48, 52)
Symmetry of real structures; symmetry breaking	Domains in ferroelectrics, variants in martensite and shape-memory alloys (Chapters 10, 37)
Superlattices, ordering, antiphase boundaries	Superalloys, intermetallics, magnetic alloys (Chapters 12, 16)
Quasicrystals and incommensurate structures	Coatings, modulated martensites, the modern definition of a crystal

Roadmap. Symmetry operations (3.1) → the point groups and crystal systems they allow (3.2) → the fourteen Bravais lattices (3.3) → space groups and the International Tables (3.4) → stereographic projection and pole figures (3.5) → symmetry of real structures and its breaking (3.6) → ordering and antiphase boundaries (3.7) → quasicrystals and incommensurate structures (3.8).

3.1 Symmetry operations

What a symmetry operation is

A **symmetry operation** is a rigid motion (one that preserves all distances) that carries an object into a position indistinguishable from the original. The set of points, lines or planes left fixed by the operation is its **symmetry element**. For a finite object such as a molecule, or for the arrangement of atoms around one point of a crystal, the operations leave at least one point fixed; for an infinite crystal they may also include translations.

Every rigid motion can be written as $\mathbf{x}' = \mathbf{W}\mathbf{x} + \mathbf{w}$, where $\mathbf{W}$ is a 3×3 orthogonal matrix (a rotation or a rotation combined with inversion) and $\mathbf{w}$ a translation. Crystallographers write the operations in the compact notation of the International Tables: the operation $(x, y, z) \to (-y, x, z)$ is a fourfold rotation about z; $(x, y, z) \to (-x, -y, -z)$ is inversion.

Point operations

Operations that leave at least one point fixed are:

- **Rotation** by $360^\circ/n$ about an axis, an n-fold axis written n (Hermann–Mauguin notation) or C_n (Schoenflies). Applying it n times returns the object to its starting position.
- **Reflection** in a mirror plane, written m (Schoenflies σ). It turns a right hand into a left hand.
- **Inversion** through a centre, written $\bar{1}$ (Schoenflies i): $\mathbf{x} \to -\mathbf{x}$. A crystal with this element is **centrosymmetric**.
- **Rotoinversion**, rotation by $360^\circ/n$ followed by inversion, written $\bar{n}$. $\bar{1}$ is inversion itself; $\bar{2}$ is identical to a mirror perpendicular to the axis; $\bar{3}$ equals a threefold axis plus a centre; $\bar{4}$ is a genuinely new operation, present in the tetrahedral sites of zincblende; $\bar{6}$ equals a threefold axis with a perpendicular mirror. (Schoenflies notation uses the equivalent *rotoreflections* S_n instead.)

Rotations are **proper** operations (determinant of $\mathbf{W}$ equal to $+1$): they can be performed physically on a real object. Reflections, inversion and rotoinversions are **improper** (determinant -1): they convert an object into its mirror image. A crystal whose symmetry contains only proper operations can exist in left- and right-handed forms; quartz is the classic example, and the two hands rotate polarized light in opposite senses.

Operations with translation: screw axes and glide planes

Figure 3.1 summarizes these operations. An infinite crystal can also be invariant under operations combining a point operation with a translation that is a *fraction* of a lattice vector.

- A **screw axis** n_p combines a rotation of $360^\circ/n$ with a translation of p/n of the lattice repeat along the axis. A 2_1 axis rotates by 180° and translates by half a repeat; 3_1 and 3_2 rotate by 120° and translate by one-third and two-thirds, giving right- and left-handed helices; 6_3 rotates by 60° and translates by half the c repeat, which is exactly how the A and B layers of HCP are related. The polymer chains of Chapter 2 are built on screw axes: the planar zigzag of polyethylene has a 2_1 axis, and isotactic polypropylene a threefold helix.
- A **glide plane** combines reflection with a translation of half a lattice vector parallel to the plane: an a, b or c glide translates along that axis, an n glide along a face diagonal and a d ("diamond") glide by a quarter of a face diagonal.

Screw axes and glide planes are invisible in the external shape of a crystal and in its macroscopic properties, because over macroscopic distances the fractional translation is negligible; a 2_1 axis and a 2 axis produce the same symmetry of a property tensor. But they are clearly visible in diffraction, where they cause certain reflections to vanish (*systematic absences*, Chapter 4), which is how they are detected.

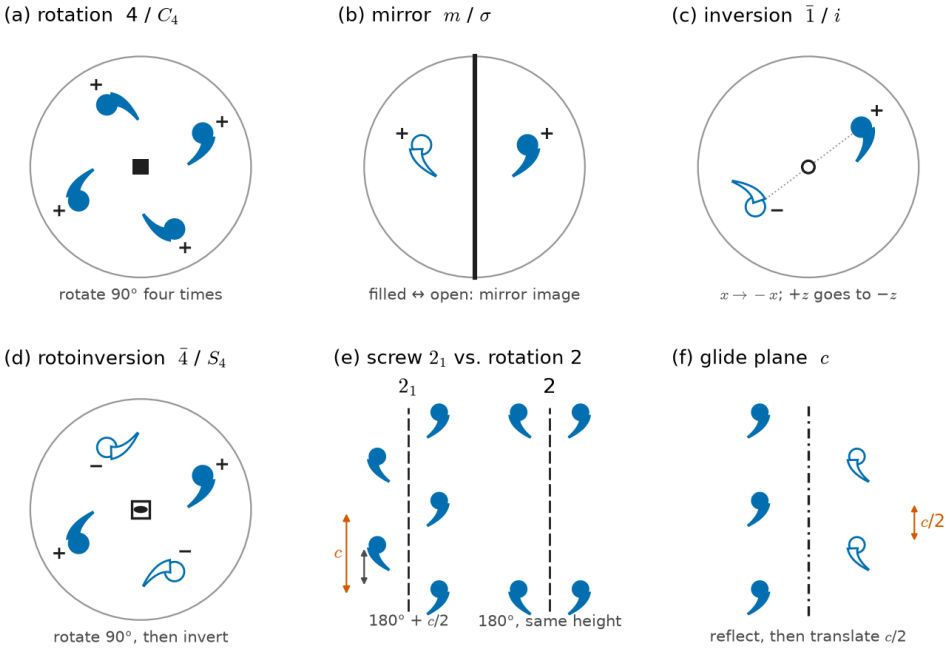

Figure 3.1 — The crystallographic symmetry operations acting on a motif: rotation, reflection, inversion, rotoinversion, screw axis and glide plane.

Groups

The symmetry operations of an object always form a **group** in the mathematical sense: the combination of any two is again a symmetry operation, there is an identity (do nothing), each operation has an inverse, and combination is associative. Group theory is what makes symmetry so economical: a few **generators** produce all the operations, and the number of operations, the **order** of the group, is a key quantity. The point group of a cube, for example, has order 48 and can be generated from a fourfold axis, a threefold axis and inversion.

3.2 The 32 point groups and the 7 crystal systems

The crystallographic restriction

A lattice can have only certain rotational symmetries. Suppose a lattice has an n-fold axis. In a basis of lattice vectors, the matrix of the rotation must have integer entries, because it maps lattice points onto lattice points. The trace of a matrix does not depend on the basis, and in a Cartesian basis the trace of a rotation by ϕ is $1 + 2\cos\phi$. Hence

$$1 + 2\cos\frac{2\pi}{n} \in \mathbb{Z} \quad \Longrightarrow \quad 2\cos\frac{2\pi}{n} \in \{-2, -1, 0, 1, 2\}$$

which allows only $n = 1, 2, 3, 4, 6$ (traces $3, -1, 0, 1, 2$). For $n = 5$ the trace would be $1 + 2\cos 72^\circ = 1.618$, the golden ratio; for $n = 7$, 2.247; for $n = 8$, 2.414. None is an integer.

> **Status of the principle: crystallographic restriction theorem**
> *Derived.* It follows rigorously from the requirement that a rotation map a three-dimensional lattice onto itself; it forbids five-, seven- and higher-fold axes in any periodic crystal, but not in quasicrystals, which are ordered without being periodic (Section 3.8).

A geometric picture makes the same point (Figure 3.2): squares, equilateral triangles and hexagons tile the plane, but regular pentagons leave gaps.

The 32 crystallographic point groups

The **point group** of a crystal is the set of point operations (rotations, mirrors, inversion, rotoinversions) that leave the crystal's *pattern of directions* unchanged; it is obtained from the full symmetry by discarding all translations, so screw axes count as plain rotations and glides as plain mirrors. Combining the allowed axes 1, 2, 3, 4, 6 and their rotoinversions in every self-consistent way gives exactly **32 crystallographic point groups** (also called crystal classes). They are the possible symmetries of crystal shapes and of crystal properties.

The point groups are sorted into seven **crystal systems** according to their characteristic symmetry, the minimum symmetry that a crystal of that system must have:

Crystal system	Characteristic symmetry	Lattice parameters (conventional cell)	Point groups (Hermann–Mauguin)	Number	Engineering examples
Triclinic	None, or only $\bar{1}$	$a \neq b \neq c$; $\alpha \neq \beta \neq \gamma$	1, $\bar{1}$	2	Plagioclase feldspar, CuSO$_4\cdot5H_2$O
Monoclinic	One 2-fold axis or mirror	$a \neq b \neq c$; $\alpha = \gamma = 90^\circ \neq \beta$	2, m, $2/m$	3	Monoclinic ZrO_2, gypsum, many molecular crystals
Orthorhombic	Three perpendicular 2-fold axes or mirrors	$a \neq b \neq c$; all 90°	222, $mm2$, mmm	3	Polyethylene, cementite Fe_3C, olivine, $YBa_2Cu_3O_7$
Tetragonal	One 4-fold axis	$a = b \neq c$; all 90°	4, $\bar{4}$, $4/m$, 422, $4mm$, $\bar{4}2m$, $4/mmm$	7	β-Sn, rutile TiO_2, tetragonal $BaTiO_3$ and ZrO_2, martensite
Trigonal	One 3-fold axis	$a = b = c$, $\alpha = \beta = \gamma \neq 90^\circ$ (rhombohedral) or hexagonal axes	3, $\bar{3}$, 32, $3m$, $\bar{3}m$	5	α-quartz, sapphire, calcite, $LiNbO_3$, Bi_2Te_3
Hexagonal	One 6-fold axis	$a = b \neq c$; $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$	6, $\bar{6}$, $6/m$, 622, $6mm$, $\bar{6}m2$, $6/mmm$	7	Mg, Ti, Zn, ice, graphite, ZnO, GaN, hydroxyapatite
Cubic	Four 3-fold axes (body diagonals)	$a = b = c$; all 90°	23, $m\bar{3}$, 432, $\bar{4}3m$, $m\bar{3}m$	5	Cu, Fe, Al, Si, NaCl, GaAs, spinels, garnets

The equalities of lattice parameters are consequences of the symmetry, not its definition. A crystal with a fourfold axis must have $a = b$; but a crystal whose a and b happen to be equal to within measurement error is not tetragonal unless it has the fourfold axis.

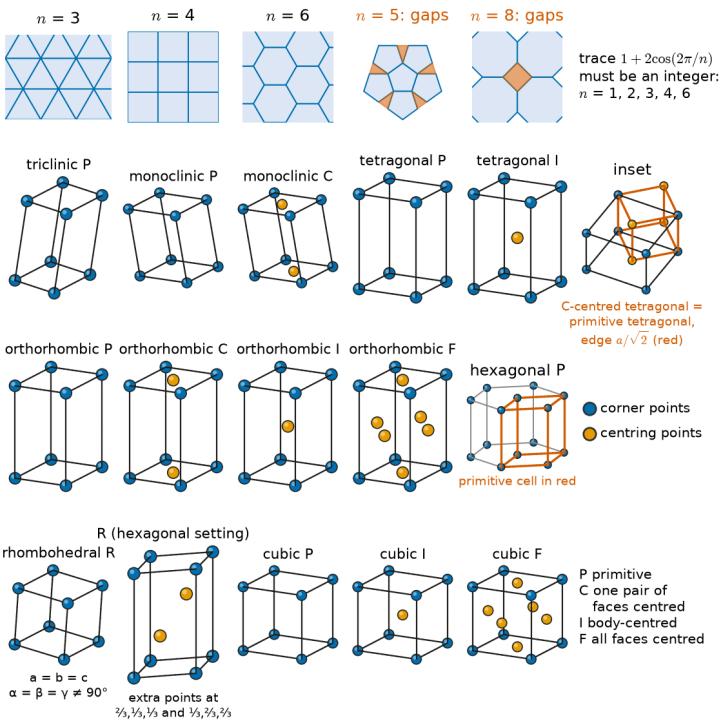

Figure 3.2 — Why only 1-, 2-, 3-, 4- and 6-fold axes are compatible with a lattice, and the 14 Bravais lattices.

Laue classes and the properties symmetry allows

Point groups decide which physical properties a crystal can have, through **Neumann's principle**: the symmetry of any physical property must include the symmetry of the crystal's point group (developed with tensors in Chapter 4).

> **Status of the principle: Neumann's principle**
> *Fundamental.* A property cannot change when the crystal is carried into an indistinguishable configuration, so every symmetry operation of the point group must leave every property tensor unchanged.

Three consequences sort the 32 point groups.

- **Eleven point groups are centrosymmetric** ($\bar{1}$, $2/m$, mmm, $4/m$, $4/mmm$, $\bar{3}$, $\bar{3}m$, $6/m$, $6/mmm$, $m\bar{3}$, $m\bar{3}m$). These are the **Laue classes**: because diffraction intensities normally add a centre of symmetry, a diffraction pattern reveals only the Laue class of a crystal, not its full point group.
- **The other 21 lack a centre of symmetry**, and 20 of them (all except 432) allow **piezoelectricity**, the polarization produced by stress. Quartz (32), GaN and ZnO ($6mm$), GaAs ($\bar{4}3m$) and poled PZT ($4mm$ or $3m$) are piezoelectric; silicon, iron and alumina ($m\bar{3}m$, $m\bar{3}m$, $\bar{3}m$) are not.
- **Ten of the 20 are polar** (1, 2, m, $mm2$, 4, $4mm$, 3, $3m$, 6, $6mm$): they have a direction that no symmetry operation reverses, so they can carry a spontaneous polarization. These crystals are **pyroelectric**: their polarization changes with temperature, the basis of infrared detectors. If the spontaneous polarization can be switched by a field, the crystal is **ferroelectric** ($BaTiO_3$, PZT, $LiNbO_3$; Chapter 27).

Property	Symmetry requirement	Number of point groups	Examples
Enantiomorphism (left- and right-handed forms)	No improper operations	11	Quartz, sugar crystals
Piezoelectricity	No centre of symmetry (and not 432)	20	Quartz, AlN, ZnO, PZT, PVDF (β-phase)
Pyroelectricity, ferroelectricity	Polar point group	10	Tourmaline, $LiTaO_3$, $BaTiO_3$, PZT
Isotropic second-rank tensors (conductivity, expansion)	Cubic	5	Cu, Fe, Si, MgO
Uniaxial second-rank tensors	Tetragonal, trigonal, hexagonal	19	Sapphire, Zn, Ti, GaN, graphite

The opening problem now has its formal answer. Cubic PZT has point group $m\bar{3}m$, centrosymmetric, so its piezoelectric tensor must vanish. Tetragonal PZT has point group $4mm$, polar, so it can be piezoelectric and ferroelectric.

3.3 The 14 Bravais lattices and why only these exist

Lattice systems and centring

A lattice is itself an object with symmetry, and its point symmetry must be one of the *holohedral* (maximal) point groups of the seven systems: $\bar{1}$, $2/m$, mmm, $4/mmm$, $\bar{3}m$, $6/mmm$ and $m\bar{3}m$. Taking the simplest (primitive, P) lattice of each system gives seven lattices. Others arise by **centring**: adding lattice points at the centre of the cell (body-centred, I), at the centres of all faces (F) or at the centres of one pair of faces (A, B or C). The trigonal system has its own rhombohedral lattice (R), equivalent to a hexagonal cell with two extra points inside.

A centred cell is a new Bravais lattice only if two conditions hold: the centred arrangement must keep the full symmetry of its system, and it must not be describable by a smaller cell of the same system. Applying these tests to every possible centring of every system leaves exactly **14 Bravais lattices** (Figure 3.2):

System	Bravais lattices	Lattice points per conventional cell
Triclinic	P	1
Monoclinic	P, C	1, 2
Orthorhombic	P, C, I, F	1, 2, 2, 4
Tetragonal	P, I	1, 2
Trigonal (rhombohedral)	R	1 (3 in hexagonal setting)
Hexagonal	P	1
Cubic	P, I, F	1, 2, 4

> **Status of the principle: the 14 Bravais lattices**
> *Derived.* They follow from enumerating the centrings compatible with the seven lattice point symmetries and removing duplicates (Bravais, 1848); the list is complete for three-dimensional periodic lattices.

Why the others are missing

Each missing combination fails one of the tests.

- **C-centred cubic** would put extra points on only one pair of faces, destroying the threefold axes along the body diagonals; the result would no longer be cubic, but tetragonal.

- **C-centred tetragonal** is identical to a smaller primitive tetragonal lattice. In a C-centred tetragonal cell of side a, the face-centre point and the corners together form a square net of side $a/\sqrt{2}$ rotated by 45°; the new cell has the same c, edge $a/\sqrt{2}$, half the volume and only corner points.
- **F-centred tetragonal** is similarly identical to an I-centred tetragonal cell with edge $a/\sqrt{2}$.
- **Body-centred or face-centred hexagonal** would break the sixfold symmetry, and **base-centred hexagonal** either reduces to a smaller primitive hexagonal cell (centring at $\tfrac{2}{3},\tfrac{1}{3},0$) or, with ordinary C-centring at $\tfrac{1}{2},\tfrac{1}{2},0$, destroys the sixfold axis and gives an orthorhombic lattice.

Worked example — Two descriptions of one lattice. A tetragonal crystal is reported with a C-centred cell $a = b = 0.600$ nm, $c = 0.400$ nm. Its conventional cell contains $8 \times \tfrac{1}{8} + 2 \times \tfrac{1}{2} = 2$ lattice points in volume $a^2 c = 0.1440$ nm^3, i.e. 0.0720 nm^3 per point. The primitive tetragonal cell spanned by $\tfrac{1}{2}(\mathbf{a}+\mathbf{b})$, $\tfrac{1}{2}(\mathbf{a}-\mathbf{b})$ and $\mathbf{c}$ has edges $0.600/\sqrt{2} = 0.4243$ nm, perpendicular to each other, and $c = 0.400$ nm, so its volume is $0.4243^2 \times 0.400 = 0.0720$ nm^3 with one lattice point: the same lattice, described by a cell of half the size, and still tetragonal. The reverse effect occurs in FCC metals. The primitive cell of copper is a rhombohedron with edge $a/\sqrt{2} = 0.2556$ nm and interaxial angles of 60°, volume $a^3/4 = 0.01181$ nm^3; it is perfectly valid but hides the cubic symmetry, which is why the four-point conventional cube is used. Neither description is "the" lattice; one is chosen for convenience and the conventions of the International Tables fix the choice so that databases agree.

Lattice systems, crystal systems and crystal families

Two subtleties often confuse. First, the classification of *lattices* and of *point groups* differ in the trigonal/hexagonal case: a crystal of the trigonal system may have either a rhombohedral (R) lattice (sapphire, calcite) or a hexagonal (P) lattice (quartz). Strictly, there are seven crystal systems, seven lattice systems and six crystal families. Second, the lattice can have higher symmetry than the crystal: tetragonal $BaTiO_3$ has a lattice that is very nearly cubic ($c/a = 1.011$), but its point group, which includes the displaced atoms, is unmistakably tetragonal.

3.4 The 230 space groups; reading the International Tables

From point groups to space groups

Combining the 32 point groups with the 14 Bravais lattices, and with all the ways of replacing rotations by screw axes and mirrors by glides, gives exactly **230 space groups** (enumerated independently by Fedorov and Schoenflies in 1891). Every periodic crystal belongs to one of them.

Crystal system	Number of space groups	Space-group numbers
Triclinic	2	1–2
Monoclinic	13	3–15
Orthorhombic	59	16–74
Tetragonal	68	75–142
Trigonal	25	143–167
Hexagonal	27	168–194
Cubic	36	195–230

Seventy-three of these are **symmorphic**, containing no screw axes or glides (for example $Pm\bar{3}m$, $Fm\bar{3}m$); the other 157 contain at least one. Eleven pairs differ only in handedness (such as $P3_121$ and $P3_221$, the two hands of α-quartz).

> **Status of the principle: the 230 space groups**
> *Derived.* They are the complete set of symmetry groups of three-dimensional periodic patterns, obtained by group-theoretical enumeration; the count assumes perfect periodicity and is exceeded by magnetic (1651 Shubnikov groups) and modulated (superspace) structures.

The space-group symbol

The **Hermann–Mauguin symbol** begins with the lattice letter (P, A, B, C, I, F, R) and then lists the symmetry along characteristic directions, which depend on the system: for cubic, the cube axis, the body diagonal and the face diagonal; for tetragonal, c, a and $[110]$; for hexagonal, c, a and $[1\bar{1}0]$; for orthorhombic, a, b and c. Thus:

- $Fm\bar{3}m$ (No. 225): F-centred cubic, mirrors perpendicular to the cube axes, $\bar{3}$ along the body diagonals, mirrors perpendicular to the face diagonals. Copper, aluminium, nickel, NaCl, MgO, CaF_2, UO_2.
- $Im\bar{3}m$ (No. 229): the same point group on a body-centred lattice. α-Fe, W, Mo, Cr.
- $P6_3/mmc$ (No. 194): hexagonal, a 6_3 screw axis along c with a perpendicular mirror, mirrors and c-glides parallel to c. Mg, Ti, Zn, Co; graphite.
- $Fd\bar{3}m$ (No. 227): cubic with diamond glides. Diamond, Si, Ge, and spinels such as $MgAl_2O_4$.
- $F\bar{4}3m$ (No. 216): zincblende GaAs and cubic SiC, which lack the centre of diamond.
- $P6_3mc$ (No. 186): wurtzite GaN, ZnO, AlN, polar.
- $Pm\bar{3}m$ (No. 221): cubic perovskite ($BaTiO_3$ above its Curie point), CsCl, and ordered alloys such as Cu_3Au, Ni_3Al and NiAl.
- $P4mm$ (No. 99): tetragonal $BaTiO_3$. $R\bar{3}c$ (No. 167): corundum. $Pnma$ (No. 62): cementite, many orthorhombic perovskites, polyethylene. $P2_1/c$ (No. 14): monoclinic ZrO_2, and the most common space group of all for molecular crystals.

Reading a page of the International Tables

The *International Tables for Crystallography, Volume A* devotes a page or two to each space group. The entries an engineer needs are:

- **Symbols and number**: short and full Hermann–Mauguin symbols, Schoenflies symbol, number, point group and crystal system.
- **Diagrams**: the positions of the symmetry elements in the cell, and a stereogram-like diagram of the equivalent positions generated from one general point.
- **Origin choice**: some groups have two standard origins (for example, origin choice 1 of $Fd\bar{3}m$ at a site of $\bar{4}3m$ symmetry, origin choice 2 at a centre of inversion). Structure files must say which is used, or atoms land in the wrong places.
- **Symmetry operations**, as coordinate triplets such as $(-y, x-y, z+\tfrac{1}{2})$.
- **Wyckoff positions**: the list of positions, each given a **multiplicity** (how many equivalent points it generates in the conventional cell), a **Wyckoff letter**, a **site symmetry** (the point symmetry of that position) and its coordinates. The *general position* has site symmetry 1 and the largest multiplicity, equal to the order of the point group times the number of lattice points per cell; *special positions* lie on symmetry elements, have higher site symmetry and lower multiplicity, and some have fixed coordinates while others have free parameters such as (x, x, x).
- **Reflection conditions**: the systematic absences produced by centring, screw axes and glides (Chapter 4).
- **Maximal subgroups and minimal supergroups**: the symmetry relations used to analyse phase transitions (Section 3.6).

A crystal structure is then completely specified (Figure 3.3) by its space group, its lattice parameters and the Wyckoff positions (with any free parameters) of the atoms in the **asymmetric unit**. This is exactly the content of a crystallographic information file (CIF) in databases such as the ICSD, and of the input to electronic-structure and molecular-dynamics codes (Chapters 55–57).

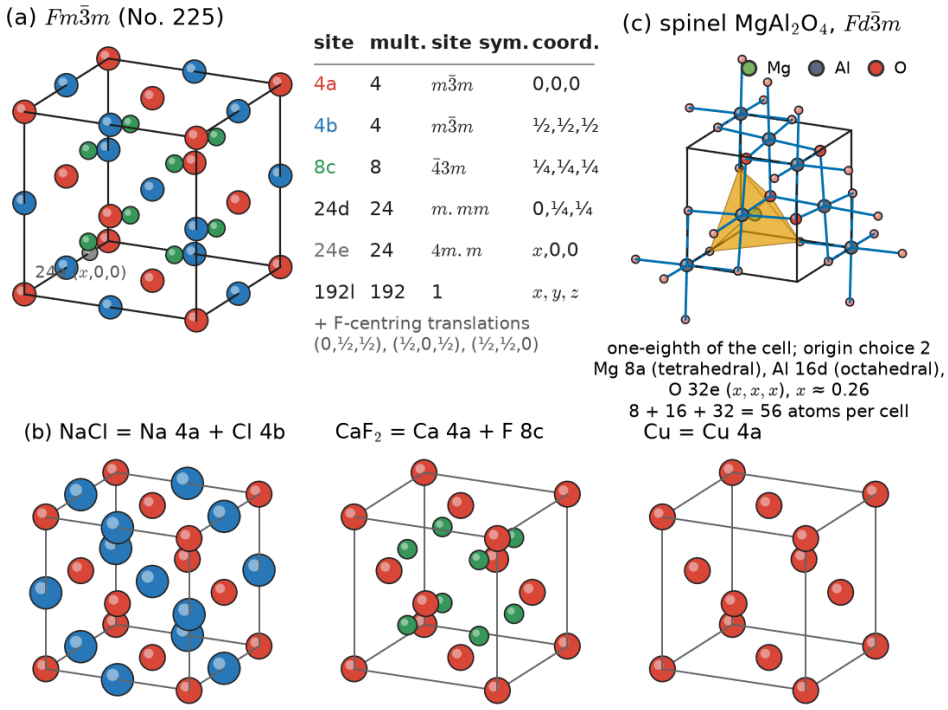

Figure 3.3 — Building structures from Wyckoff positions: rock salt, fluorite and spinel in space groups Fm-3m and Fd-3m.

Worked example — Building three structures from the tables. *Rock salt.* $Fm\bar{3}m$ has point group order 48 and 4 lattice points per cell, so its general position 192*l* has multiplicity $48 \times 4 = 192$. The special positions 4*a* at $(0,0,0)$ and 4*b* at $(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2})$ each have site symmetry $m\bar{3}m$ and multiplicity 4: the four points are the origin plus the F-centring translations $(0,\tfrac{1}{2},\tfrac{1}{2})$, $(\tfrac{1}{2},0,\tfrac{1}{2})$, $(\tfrac{1}{2},\tfrac{1}{2},0)$. Placing Na at 4*a* and Cl at 4*b* gives the whole structure: 4 Na and 4 Cl per cell, each ion octahedrally surrounded by six of the other. With $a = 0.5640$ nm,

$$\rho = \frac{4 \times 58.44}{(5.640 \times 10^{-8})^3 \times 6.022 \times 10^{23}} = 2.16\ \mathrm{g\,cm^{-3}}$$

Changing only the anion site to 8*c* at $(\tfrac{1}{4},\tfrac{1}{4},\tfrac{1}{4})$ (site symmetry $\bar{4}3m$) gives fluorite: 4 Ca + 8 F, the CaF_2 stoichiometry, every F tetrahedrally coordinated. One line of a table distinguishes table salt from nuclear fuel (UO_2).

Silicon. In $Fd\bar{3}m$ (origin choice 1) silicon occupies 8*a*, whose two representative points $(0,0,0)$ and $(\tfrac{1}{4},\tfrac{1}{4},\tfrac{1}{4})$ plus F-centring give the 8 atoms of the diamond cell, each on a site of symmetry $\bar{4}3m$, the tetrahedral symmetry of the sp^3 bond.

Corundum. In $R\bar{3}c$ (hexagonal setting, 3 lattice points per cell) Al occupies 12*c* at $(0,0,z)$ with $z \approx 0.352$ and O occupies 18*e* at $(x,0,\tfrac{1}{4})$ with $x \approx 0.306$. The cell therefore holds $12 + 18 = 30$ atoms, six Al_2O_3 formula units, and two free parameters describe the slight buckling of the oxygen layers and displacement of the Al ions from ideal octahedral sites. These two numbers, refined by diffraction, are all that distinguishes real sapphire from an idealized close-packed model.

3.5 Stereographic projection and pole figures

Projecting directions onto a plane

Crystallography constantly needs angles between directions and planes in three dimensions: between slip planes and a loading axis, between a wafer surface and an etch facet, between grain orientations and a sheet's rolling direction. The **stereographic projection** maps directions onto a flat diagram while preserving angles.

Imagine the crystal at the centre of a **reference sphere** of radius R. Each plane is represented by its normal, extended until it meets the sphere at a **pole**. Each pole in the northern hemisphere is joined by a straight line to the south pole of the sphere, and the point where that line crosses the equatorial plane is the projection. A pole at angle ϕ from the north pole (the projection axis) lands at distance

$$r = R \tan\frac{\phi}{2}$$

from the centre. Poles at the north pole plot at the centre, poles on the equator ($\phi = 90°$) on the **primitive circle** of radius R. Planes containing the projection axis plot as straight lines through the centre; all other planes plot as arcs of circles (great circles). Small circles of constant angle about any pole also project as circles: the projection is *conformal*, which is why it is used instead of simpler projections. Angles are measured with a **Wulff net**, a stereographic grid of great and small circles.

> **Status of the principle: stereographic projection**
> *Derived.* Pure spherical geometry: the mapping $r = R\tan(\phi/2)$ preserves angles locally and maps circles on the sphere to circles in the plane, but it does not preserve areas, which is why texture intensities are normalized on the sphere, not on the diagram.

The standard projection of a cubic crystal

A **standard projection** shows the poles of the low-index planes of a crystal with one chosen pole at the centre (Figure 3.4). For a cubic crystal viewed down $[001]$:

- the $\{100\}$ poles lie at the centre and at four points on the primitive circle;
- the $\{110\}$ poles lie at $45°$ from the centre ($r = R\tan 22.5° = 0.414R$) and on the primitive circle;
- the $\{111\}$ poles lie at $54.74°$ ($r = R\tan 27.37° = 0.518R$), at azimuths of $45°$, $135°$, $225°$ and $315°$;
- $\{112\}$ poles at $35.26°$ ($r = 0.318R$) and $\{012\}$ at $26.57°$ ($r = 0.236R$).

The cubic symmetry divides the projection into 48 identical spherical triangles. One of them, bounded by 001, 011 and 111, the **standard stereographic triangle**, contains every distinct orientation of a direction in a cubic crystal, and is used to plot loading axes in single-crystal experiments (Chapter 13) and in inverse pole figures.

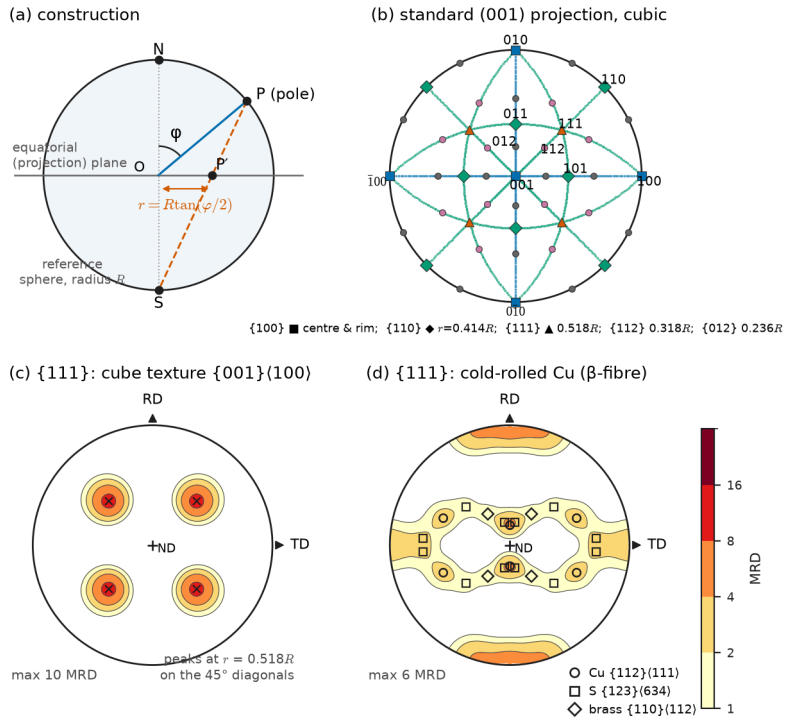

Figure 3.4 — The stereographic projection of a cubic crystal and the {111} pole figures of rolled and recrystallized FCC sheet.

Worked example — Locating texture components on a pole figure. An aluminium sheet recrystallized after rolling has the **cube texture** $\{001\}\langle 100\rangle$: most grains have a cube face in the sheet plane and a cube edge along the rolling direction (RD). A $\{111\}$ pole figure is plotted with the sheet normal (ND) at the centre, RD at the top and the transverse direction (TD) at the right. Where do the peaks appear?

In each cube-oriented grain, [001] is parallel to ND and [100] to RD. The four ⟨111⟩ poles in the upper hemisphere, [111], [$\bar{1}$11], [1$\bar{1}$1] and [$\bar{1}\bar{1}$1], all make 54.74° with [001], so they lie at

$$r = R\tan\frac{54.74^\circ}{2} = R\tan 27.37^\circ = 0.518R$$

from the centre. Their projections on the sheet plane point along [110], [$\bar{1}$10], [1$\bar{1}$0] and [$\bar{1}\bar{1}$0], that is, at 45° between RD and TD. The pole figure therefore shows four sharp peaks at about half the radius, on the diagonals. If instead the texture were the Goss component {110}⟨001⟩ of electrical steel, with [110] along ND and [001] along RD, the ⟨111⟩ poles would lie at 35.26° from ND ($r = 0.318R$) in the RD–ND plane and at 90° (on the rim) at ±35.26° from TD: a quite different, immediately recognizable pattern. Reading which ideal orientations a measured pole figure contains is the first step in controlling texture.

Pole figures and texture

A **pole figure** is a stereographic (or equal-area) map, in the *sample* reference frame (RD, TD, ND for sheet), of the density of poles of one family of crystal planes, summed over all grains. It is measured by X-ray or neutron diffraction, tilting and rotating the sample with the detector fixed on one Bragg reflection, or calculated from orientation maps obtained by electron backscatter diffraction (Chapter 52). Intensities are expressed in **multiples of a random distribution** (MRD): 1 everywhere for a random polycrystal, peaks of 5–20 for strongly textured sheet.

Because one pole figure shows only one family of planes, the full description of texture is the **orientation distribution function** (ODF), the probability density of grain orientations expressed in three Euler angles, computed from several pole figures. Its complement, the **inverse pole figure**, plots the density of crystal directions parallel to one sample axis inside the standard triangle; it is the natural display for wires, fibres, thin films and additively manufactured parts, whose textures are fibres about one axis.

Texture is not an academic curiosity. In deep-drawing steel the γ-fibre ({111} ∥ ND) raises the plastic strain ratio r (the ratio of width to thickness strain in a tensile test) above 1.5, so the sheet resists thinning. In aluminium can stock the balance of rolling and cube components sets the height of ears on the drawn cup. In titanium and zirconium alloys, basal textures control anisotropy of strength and hydride orientation. In additively manufactured nickel alloys, the ⟨100⟩ fibre along the build direction lowers the modulus along that direction by about a third compared with a random polycrystal. Chapters 13, 46 and 48 develop how processing produces these textures.

3.6 Symmetry of real structures

Symmetry and properties

Section 3.2 showed that symmetry determines which properties can exist. It also determines how many independent numbers are needed to describe them: a cubic crystal needs one thermal conductivity and three elastic constants, a hexagonal crystal two conductivities and five elastic constants, a triclinic crystal six and twenty-one (Chapter 4). The effective symmetry of a polycrystal is set by its texture: a random polycrystal is isotropic whatever the crystal symmetry; a rolled sheet usually has orthorhombic sample symmetry (*mmm*, with mirrors perpendicular to RD, TD and ND); a fibre-textured wire has cylindrical symmetry about its axis. **Curie's principle** generalizes the idea to effects: the symmetry of an effect (such as the strain of a crystal under a field) contains the common symmetry elements of the cause and of the medium.

> **Status of the principle: Curie's principle**
> *Fundamental.* An effect cannot be less symmetric than the combination of its causes and the medium; asymmetry in an effect must therefore come from the cause, the medium or a spontaneous symmetry breaking (as in a phase transition).

Symmetry breaking at phase transitions

Many important transformations lower the symmetry of a crystal: the cubic-to-tetragonal ferroelectric transition of $BaTiO_3$ and PZT, the tetragonal-to-monoclinic transformation of zirconia, the cubic-to-tetragonal Bain distortion in steel martensite, the cubic (B2) to monoclinic martensite of NiTi shape-memory alloys, the disorder–order transitions of Section 3.7. In each case the low-symmetry phase has a space group that is a **subgroup** of the parent's, and the International Tables list the maximal subgroups that guide such analyses.

When symmetry is lost, the lost operations relate several equally good orientations of the new phase. A transformation from a parent point group of order $|G|$ to a product of order $|H|$ creates

$$N = \frac{|G|}{|H|}$$

orientation variants (or domain states), which form side by side in different regions of the parent crystal. They are separated by **domain walls** or **twin boundaries**. A second, translational kind of variant appears when the lattice repeat increases, as in ordering (Section 3.7), producing antiphase domains.

> **Status of the principle: number of orientation variants $|G|/|H|$**
> *Derived.* A consequence of group theory (Lagrange's theorem): the parent operations not retained in the product generate distinct but energetically equivalent orientations; it counts the possible variants, not which ones form, which depends on stress, field and nucleation.

Worked example — Domains in ferroelectrics and variants in martensite. Cubic $BaTiO_3$ has point group $m\bar{3}m$ of order 48. Tetragonal $BaTiO_3$ has point group $4mm$ of order 8. Hence

$$N = \frac{48}{8} = 6$$

ferroelectric domain states, with polarization along $[100]$, $[\bar{1}00]$, $[010]$, $[0\bar{1}0]$, $[001]$ or $[00\bar{1}]$. The rhombohedral phase ($3m$, order 6) has $48/6 = 8$ states, polarized along the eight body diagonals; the orthorhombic phase ($mm2$, order 4) has 12. PZT near its morphotropic phase boundary, where tetragonal and rhombohedral phases coexist, offers $6 + 8 = 14$ possible polarization directions, which helps explain why compositions near that boundary pole most completely and give the highest piezoelectric coefficients (Chapter 27).

The same arithmetic counts martensite variants. If the cubic-to-tetragonal transformation of steel kept only the tetragonal holohedry $4/mmm$ (order 16), there would be $48/16 = 3$ Bain variants, one for each cube axis that becomes the c axis. The cubic-to-monoclinic transformation of NiTi gives $48/4 = 12$ lattice-correspondence variants (24 habit-plane variants when the two twin solutions per correspondence are counted). The rich variety of variants is what lets shape-memory alloys accommodate the transformation strain by arranging variants into self-accommodating groups, and then reorient them under stress (Chapters 10 and 37).

3.7 Superlattices, ordering and antiphase boundaries

Disorder and order in alloys

In a random substitutional solid solution, each lattice site is occupied by A or B atoms at random, with probabilities equal to the composition. Such an alloy has the symmetry of the simple metallic structure. Below a critical temperature, many alloys with an attraction between unlike atoms **order**: A and B atoms segregate onto distinct sublattices. Because the sites are no longer all equivalent, the lattice repeat or the centring changes, and the ordered structure is called a **superlattice** (or superstructure).

Ordered structure	Parent	Example alloys	Ordering temperature (typical)
L1$_2$ (Cu_3Au type), $Pm\bar{3}m$	FCC	Cu_3Au, Ni_3Al (γ′ in superalloys), Ni_3Fe, Al_3Sc, Al_3Li	390 °C (Cu_3Au); Ni_3Al ordered to its melting point
L1$_0$ (CuAu type), $P4/mmm$	FCC	CuAu, FePt, CoPt, TiAl (γ-TiAl turbine blades)	FePt about 1300 °C
B2 (CsCl type), $Pm\bar{3}m$	BCC	CuZn (β-brass), NiAl, FeAl, FeCo, NiTi austenite	460 °C (CuZn); NiAl ordered to melting
D0$_3$ (Fe_3Al type), $Fm\bar{3}m$	BCC	Fe_3Al, Fe_3Si	about 550 °C

In L1$_2$ Cu_3Au, Au occupies the cube corners and Cu the face centres. The F-centring of the disordered FCC lattice is lost, and the lattice becomes simple cubic with a four-atom basis. In B2 CuZn, Cu sits at the corners and Zn at the body centre; the I-

centring of BCC is lost. The degree of ordering is measured by the **long-range order parameter**

$$S = \frac{p_A - x_A}{1 - x_A}$$

where p_A is the probability that a site of the A sublattice is occupied by an A atom and x_A the overall atom fraction of A; $S = 1$ for perfect order and $S = 0$ for a random alloy. The thermodynamics of ordering, and why S falls continuously to zero in some alloys (B2) but discontinuously in others (L1$_2$), are developed in Chapter 6.

Ordering matters for properties. Ordered Ni_3Al precipitates are what make nickel superalloys strong at high temperature (Chapter 16); L1$_0$ FePt has enormous magnetocrystalline anisotropy, used in high-density magnetic recording; ordering in FeCo and Ni_3Fe (permalloy) changes magnetic softness; long-range ordering lowers electrical resistivity (Cu_3Au drops several-fold on ordering) and generally raises strength and hardness.

Detecting order: superlattice reflections

When the centring is lost, reflections forbidden for the disordered lattice reappear weakly: the **superlattice reflections**. Their amplitude is proportional to S times the difference in scattering power of the two atoms, while the fundamental reflections depend on the average (Chapter 4 derives this from the structure factor).

Worked example — Why X-rays see order in Cu_3Au but hardly in β-brass. For L1$_2$ Cu_3Au, approximate the X-ray atomic scattering factors at low angle by the atomic numbers: $f_{\mathrm{Au}} \approx 79$, $f_{\mathrm{Cu}} \approx 29$. The fundamental reflections (all indices odd or all even, as in FCC) have structure-factor amplitude $f_{\mathrm{Au}} + 3f_{\mathrm{Cu}} = 166$; the superlattice reflections (mixed indices, such as 100 and 110) have $S(f_{\mathrm{Au}} - f_{\mathrm{Cu}}) = 50S$. The intensity ratio is

$$\frac{I_{\mathrm{super}}}{I_{\mathrm{fund}}} = \left(\frac{50S}{166}\right)^2 = 0.091\,S^2$$

so a fully ordered alloy shows superlattice peaks about 9 % as strong as the fundamentals (before angle-dependent corrections), easily measured; at $S = 0.5$ they fall to 2.3 %. For B2 CuZn, $f_{\mathrm{Cu}} \approx 29$ and $f_{\mathrm{Zn}} \approx 30$ differ by one electron, and the ratio is $(1/59)^2 = 2.9 \times 10^{-4}$: the order is practically invisible to X-rays. Neutrons, which scatter from nuclei with lengths unrelated to atomic number ($b_{\mathrm{Cu}} = 7.72$ fm, $b_{\mathrm{Zn}} = 5.68$ fm), give a ratio of $\left(\frac{2.04}{13.40}\right)^2 = 0.023$, eighty times better, which is why the ordering of β-brass was first measured by neutron diffraction (Chapter 53). For Ni_3Al with X-rays, $(28 - 13)/(13 + 3 \times 28) = 0.155$, a ratio of 0.024: the γ′ superlattice reflections are weak but routine.

Antiphase boundaries

Ordering nucleates independently in different regions of a crystal. Because the A atoms can choose any one of the equivalent sublattices, neighbouring ordered regions may be perfectly aligned crystallographically yet out of step in their A–B sequence. Where they meet, an **antiphase boundary** (APB) forms: a surface across which A atoms face A atoms and B face B, with a displacement between the domains equal to a vector of the disordered lattice that is not a vector of the ordered one ($\frac{1}{2}\langle 110\rangle$ in L1$_2$, $\frac{1}{2}\langle 111\rangle$ in B2). An ordered alloy that nucleated from many points becomes a mosaic of **antiphase domains**; their number of distinct translational variants equals the ratio of the volume of the ordered primitive cell to that of the disordered one: 4 for L1$_2$, 2 for B2.

APBs cost energy, because they create wrong A–A and B–B bonds; typical APB energies are 0.1–0.3 J m^{-2} for Ni_3Al on {111}. They are also created whenever a dislocation of the disordered lattice passes through an ordered crystal: its Burgers vector shifts the sublattices. A single such dislocation would leave a sheet of APB behind it, so dislocations in ordered alloys (Figure 3.5) travel in pairs, **superdislocations**, the second restoring order behind the first. This is the central mechanism of precipitation strengthening by γ′ and of the anomalous rise in strength of Ni_3Al with temperature (Chapters 12, 13 and 16).

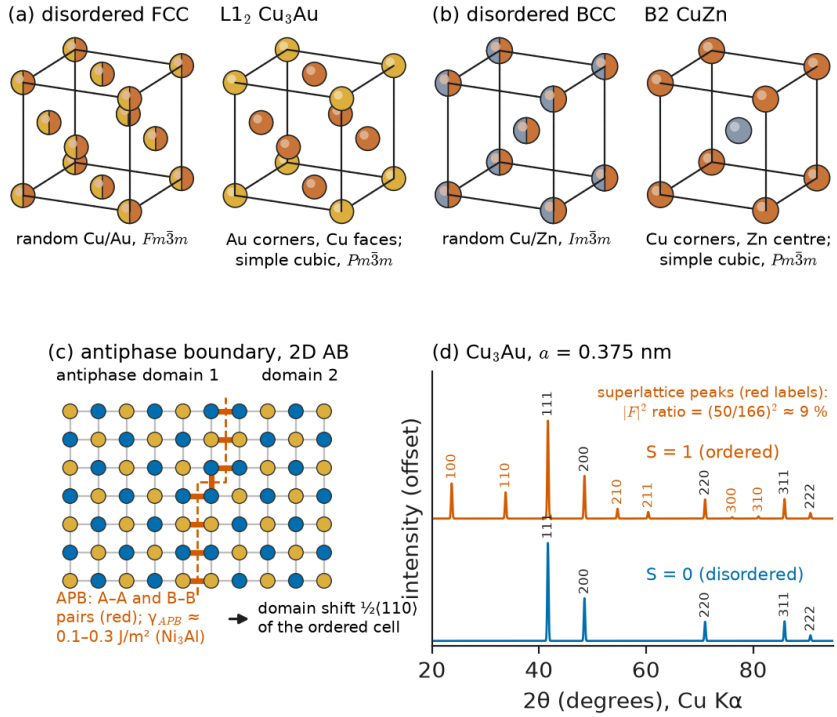

Figure 3.5 — Order–disorder in alloys: the L1₂ and B2 superlattices, an antiphase boundary, and superlattice reflections in diffraction.

Artificial superlattices

The word superlattice is also used for **artificial multilayers**: periodic stacks of two materials deposited layer by layer, such as GaAs/AlAs in quantum-cascade lasers, Co/Pt and Fe/Cr in magnetic devices (the Fe/Cr multilayers in which giant magnetoresistance was discovered), and TiN/NbN or TiN/AlN hard coatings, whose hardness peaks at bilayer periods of a few nanometres because dislocations cannot cross the many coherent interfaces. They impose a new, longer period on the parent lattice exactly as chemical ordering does, and they too produce satellite reflections around the fundamental Bragg peaks.

3.8 Quasicrystals and incommensurate structures

Quasicrystals

Section 3.2 proved that a periodic crystal cannot have fivefold symmetry, yet Shechtman's Al–Mn alloys gave sharp diffraction spots with icosahedral symmetry (Chapter 2). The resolution is that sharp diffraction requires *long-range order*, not periodicity. A quasicrystal is **quasiperiodic**: its structure can be generated as a three-dimensional slice through a periodic lattice in six-dimensional space (for icosahedral quasicrystals) or five dimensions (for decagonal ones, which are periodic in one direction and quasiperiodic in the plane perpendicular to it). Because the slice is at an irrational angle to the higher-dimensional lattice, the pattern never repeats, but every local configuration recurs infinitely often.

The two-dimensional model is the **Penrose tiling** (Figure 3.6a), made from a "fat" rhombus (angles 72° and 108°) and a "thin" rhombus (36° and 144°) assembled under matching rules. Its local fivefold symmetry and its scaling properties involve the golden ratio:

$$\tau = \frac{1+\sqrt{5}}{2} = 1.6180\ldots, \qquad \tau^2 = \tau + 1$$

The ratio of fat to thin tiles tends to τ, and the pattern is self-similar under scaling by τ. The one-dimensional analogue, the Fibonacci chain of long (L) and short (S) segments generated by repeatedly replacing L → LS and S → L, makes the point most simply: the numbers of L and S segments are successive Fibonacci numbers, their ratio converges to τ (after 16 generations it is 1.618034, against $\tau = 1.618034$), and the chain is ordered but never periodic. Diffraction from such structures gives sharp peaks indexed by six integers (icosahedral) instead of three, with intensities forming self-similar patterns related by powers of τ.

Hundreds of stable quasicrystals are now known, in Al–Cu–Fe, Al–Pd–Mn, Zn–Mg–RE, Ti–Zr–Ni and other systems; a natural one was found in a meteorite. Their properties reflect their unusual structure: high hardness, low friction and surface energy (hence non-stick and wear-resistant coatings), low thermal and electrical conductivity for metallic alloys, and brittleness at room temperature. Closely related **approximants** are periodic crystals with very large cells whose local structure mimics the quasicrystal; they bridge the two descriptions.

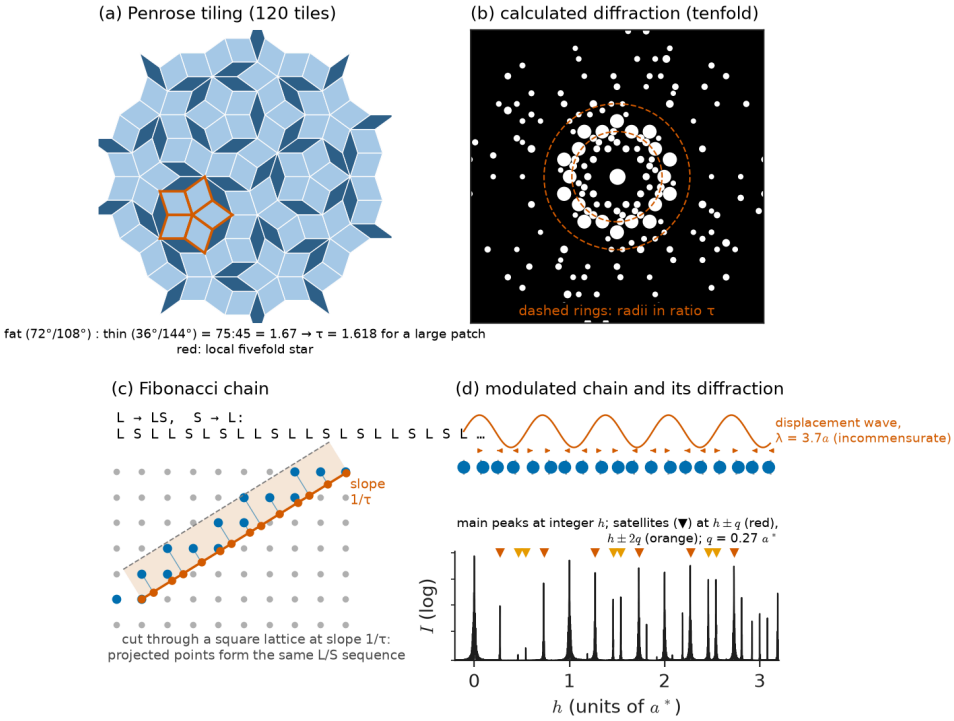

Figure 3.6 — Order without periodicity: a Penrose tiling and its tenfold diffraction pattern, the Fibonacci chain, and an incommensurately modulated structure with its satellite reflections.

Incommensurate and modulated structures

A second family of ordered but non-periodic solids are **incommensurately modulated structures**. Start from an ordinary crystal (the *basic structure*) and impose a periodic distortion, a displacement of atoms, a variation in site occupancy or a rotation of magnetic moments, whose wavelength is not a rational multiple of the lattice repeat. The modulation wave vector is written

$$\mathbf{q} = \alpha\,\mathbf{a}^* + \beta\,\mathbf{b}^* + \gamma\,\mathbf{c}^*$$

with at least one of α, β, γ irrational, where $\mathbf{a}^*$, $\mathbf{b}^*$, $\mathbf{c}^*$ are reciprocal-lattice vectors (Chapter 4). The diffraction pattern then shows the main Bragg reflections of the basic structure surrounded by **satellite reflections** (Figure 3.6d) at $\pm\mathbf{q}$, $\pm 2\mathbf{q}$, …, all sharp. Such structures are described by **superspace groups** in $(3+d)$ dimensions, where d is the number of independent modulation vectors.

Engineering and scientific examples include:

- **Modulated martensites** in Ni–Mn–Ga magnetic shape-memory alloys (the "5M" and "7M" structures), whose modulation makes twin boundaries so mobile that a magnetic field produces strains of several per cent (Chapter 37);
- the narrow **incommensurate phase of quartz**, about 1.4 K wide, between the α and β forms near 573 °C;
- **charge-density waves** in layered compounds such as $NbSe_2$ and TaS_2, and **spin-density waves** and helical magnetic structures in chromium and rare-earth metals;
- **composite (misfit) crystals** in which two interpenetrating substructures have incommensurate repeats along a common direction, as in the misfit-layer sulphides and in the host–guest high-pressure phases of barium and bismuth.

When the modulation locks into a rational value, the structure becomes an ordinary but larger-celled superstructure, a **commensurate** phase; many systems pass through incommensurate phases on the way between two commensurate ones as temperature changes.

> **Status of the principle: the crystal as a solid with essentially discrete diffraction**
> *Empirical.* The 1992 IUCr definition was adopted because experiment found sharp diffraction from quasicrystals and modulated phases; it replaces periodicity, which remains the special case covered by the 230 space groups, with long-range order detectable by diffraction.

Quasicrystals and modulated phases show that the classical crystallography of Sections 3.1–3.4 is the most important special case of a more general idea of order. For the engineer, the classical case covers almost every structural material; the generalization matters for coatings, shape-memory alloys, some magnetic and electronic materials, and the interpretation of unexpected diffraction spots.

What remains open

Open question	Where it is answered
How are reciprocal lattices, structure factors and systematic absences derived from symmetry, and how are superlattice intensities calculated exactly?	Chapter 4
How do point-group symmetries reduce the number of independent components of conductivity, piezoelectric and elastic tensors?	Chapter 4
How are domain walls, twin boundaries and antiphase boundaries classified among planar defects?	Chapter 5
What thermodynamics controls order–disorder transitions, and why are some continuous and some discontinuous?	Chapter 6
How do martensite variants form and accommodate transformation strain?	Chapter 10
How do superdislocations and APB energy strengthen γ′ and other ordered intermetallics?	Chapters 12, 13 and 16
How do texture and the plastic strain ratio govern formability?	Chapters 13 and 46
How do ferroelectric domains, poling and the morphotropic phase boundary set piezoelectric performance?	Chapter 27
How are shape-memory and magnetic shape-memory alloys designed?	Chapter 37
How are pole figures, ODFs and orientation maps measured in practice?	Chapter 52
How do neutrons and synchrotron X-rays reveal ordering, modulations and light atoms?	Chapter 53
How do electronic-structure and database tools use space groups and Wyckoff positions?	Chapters 55 and 57

Next: Chapter 4 turns from real space to reciprocal space: the reciprocal lattice, Bragg's law and the Ewald sphere, structure factors and intensities, peak broadening, and the tensor description of properties that symmetry constrains.

Chapter 4 — Reciprocal Space, Diffraction and Tensor Properties

Part I · Structure of Materials

A batch of carburized and quenched transmission gears fails a bench test: several gear teeth pit and spall after a fraction of their design life. Hardness readings are on specification and the metallographic sections look normal. The laboratory then puts a tooth in an X-ray diffractometer. Alongside the broad peaks of martensite sit small, sharp peaks at angles where body-centred iron has no reflections at all; they belong to face-centred austenite. Comparing integrated intensities gives about 30 % retained austenite in the case, three times the specified limit. Under contact stress the austenite slowly transforms, the case swells and the residual compressive stress that protects the surface is lost. The diffractometer found in twenty minutes what a microscope could not see, and the same pattern also showed, through the width of the martensite peaks, how heavily strained that martensite was.

Elsewhere, an engineer specifying single-crystal nickel-superalloy turbine blades finds that they are always grown with ⟨001⟩ along the blade axis. The reason is stiffness: Young's modulus of a nickel crystal along ⟨001⟩ is less than half its value along ⟨111⟩, and a lower modulus means lower thermal stress. Yet the thermal and electrical conductivities of the same cubic crystal are identical in every direction. Both facts follow from one principle of symmetry.

> **How does a crystal's periodic structure turn into the diffraction pattern we measure, and what can the positions, intensities and widths of the peaks tell us?**
> **Why do some properties of a crystal depend on direction and others do not, and how much direction dependence does the crystal's symmetry allow?**

Concept	Engineering destination
Reciprocal lattice, Brillouin zones	Electron bands (Ch 31), phonons and heat capacity (Ch 24), diffraction indexing
Bragg's law, Laue condition, Ewald sphere	X-ray, electron and neutron diffraction (Ch 52–53), phase identification, texture (Ch 46)
Structure factor, systematic absences	Identifying FCC, BCC and ordered phases; γ′ in superalloys; retained austenite in steels
Intensity factors	Quantitative phase analysis, Rietveld refinement, temperature measurement from the Debye–Waller factor
Peak broadening (Scherrer, Williamson–Hall)	Crystallite size of catalysts and nanopowders; dislocation density in cold-worked and milled metals
Neumann's principle	Deciding which properties a crystal or a textured part can possibly show
Rank-2 tensors	Thermal expansion of HCP metals and ceramics, anisotropic conduction in graphite, permittivity of sapphire substrates
Rank-3 tensors	Piezoelectric sensors and actuators: quartz, PZT, AlN (Ch 27, 37)
Rank-4 tensors	Anisotropic stiffness of single-crystal blades, silicon MEMS, textured sheet, composites and wood (Ch 11, 22)

Roadmap. The reciprocal lattice and Brillouin zones (4.1) → the diffraction condition in three equivalent forms (4.2) → which reflections appear: the structure factor (4.3) → how strong they are (4.4) → how wide they are (4.5) → symmetry and properties: Neumann's principle (4.6) → tensor properties of rank 2, 3 and 4 (4.7) → why cubic crystals conduct isotropically but deform anisotropically (4.8).

4.1 The reciprocal lattice and Brillouin zones

Why a second lattice is useful

Chapter 2 described a crystal by its direct lattice: basis vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ and a motif. Many questions, however, are about *which waves fit the crystal*. X-rays, neutrons, lattice vibrations and conduction electrons all respond to the periodicity of families of lattice planes, each characterised by a normal and a spacing d_{hkl}. Both fit into one vector, normal to the planes and of length inversely proportional to their spacing. The set of all such vectors forms a lattice of its own, the **reciprocal lattice**.

Definition

The reciprocal basis vectors are

$$\mathbf{b}_1 = 2\pi \,\frac{\mathbf{a}_2 \times \mathbf{a}_3}{V_c}, \qquad \mathbf{b}_2 = 2\pi \,\frac{\mathbf{a}_3 \times \mathbf{a}_1}{V_c}, \qquad \mathbf{b}_3 = 2\pi \,\frac{\mathbf{a}_1 \times \mathbf{a}_2}{V_c}$$

where $V_c = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)$ is the volume of the direct unit cell. They satisfy $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\,\delta_{ij}$. A general reciprocal-lattice vector is

$$\mathbf{G}_{hkl} = h\,\mathbf{b}_1 + k\,\mathbf{b}_2 + l\,\mathbf{b}_3$$

with integer h, k, l, and it has two properties that make it the natural language of diffraction:

1. $\mathbf{G}_{hkl}$ is perpendicular to the lattice planes with Miller indices (hkl);
2. its length is $|\mathbf{G}_{hkl}| = 2\pi / d_{hkl}$.

Crystallographers often drop the 2π, so that $|\mathbf{G}| = 1/d$; this chapter keeps it.

> **Status of the principle: reciprocal-lattice properties**
> *Derived.* Both properties follow from the definition by vector algebra; they hold exactly for any lattice, of any symmetry.

For a cubic lattice of edge a, $\mathbf{b}_i = (2\pi/a)\,\hat{\mathbf{x}}_i$, so $|\mathbf{G}_{hkl}| = (2\pi/a)\sqrt{h^2+k^2+l^2}$ and

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

the plane-spacing formula of Chapter 2, now obtained in one line. For copper, $a = 0.3615$ nm and $d_{111} = 0.2087$ nm, so $|\mathbf{G}_{111}| = 30.1\ \text{nm}^{-1}$. Every point of the reciprocal lattice is one family of planes; every family of planes is one point.

The reciprocal of FCC is BCC

Take the primitive vectors of the FCC lattice, $\mathbf{a}_1 = \tfrac{a}{2}(0,1,1)$, $\mathbf{a}_2 = \tfrac{a}{2}(1,0,1)$, $\mathbf{a}_3 = \tfrac{a}{2}(1,1,0)$. Working through the definition gives $\mathbf{b}_1 = \tfrac{2\pi}{a}(-1,1,1)$ and its permutations, which are the primitive vectors of a **body-centred cubic** lattice with conventional cube edge $4\pi/a$. Conversely, the reciprocal of BCC is FCC. The hexagonal lattice is its own reciprocal type, rotated by 30° about the c axis. These pairs matter because they tell us, without calculation, the pattern of allowed reflections from FCC and BCC metals (Section 4.3).

Brillouin zones

The Wigner–Seitz cell of the reciprocal lattice — the region of reciprocal space closer to one lattice point than to any other — is the **first Brillouin zone**. Its faces are the perpendicular bisector planes of the shortest reciprocal-lattice vectors, so every point $\mathbf{k}$ on a zone face satisfies

$$\mathbf{k} \cdot \hat{\mathbf{G}} = \tfrac{1}{2}|\mathbf{G}|$$

That is precisely the condition, derived in Section 4.2, for a wave of wavevector $\mathbf{k}$ to be Bragg-reflected by the planes $\mathbf{G}$. A wave travelling in the crystal with $\mathbf{k}$ on a zone face is reflected back on itself and forms a standing wave. For electrons this opens the energy gaps that separate metals, semiconductors and insulators (Chapter 31); for lattice vibrations it caps the phonon wavevector and fixes the Debye cut-off of the heat capacity (Chapter 24).

For FCC metals, whose reciprocal lattice is BCC, the first Brillouin zone is a **truncated octahedron**: eight hexagonal faces normal to $\langle 111 \rangle$ and six square faces normal to $\langle 100 \rangle$ (Figure 4.1). For BCC metals it is a rhombic dodecahedron with twelve

faces normal to $\langle 110 \rangle$. High-symmetry points carry standard labels (Γ at the centre, X and L at the square and hexagonal face centres) used in band and phonon diagrams.

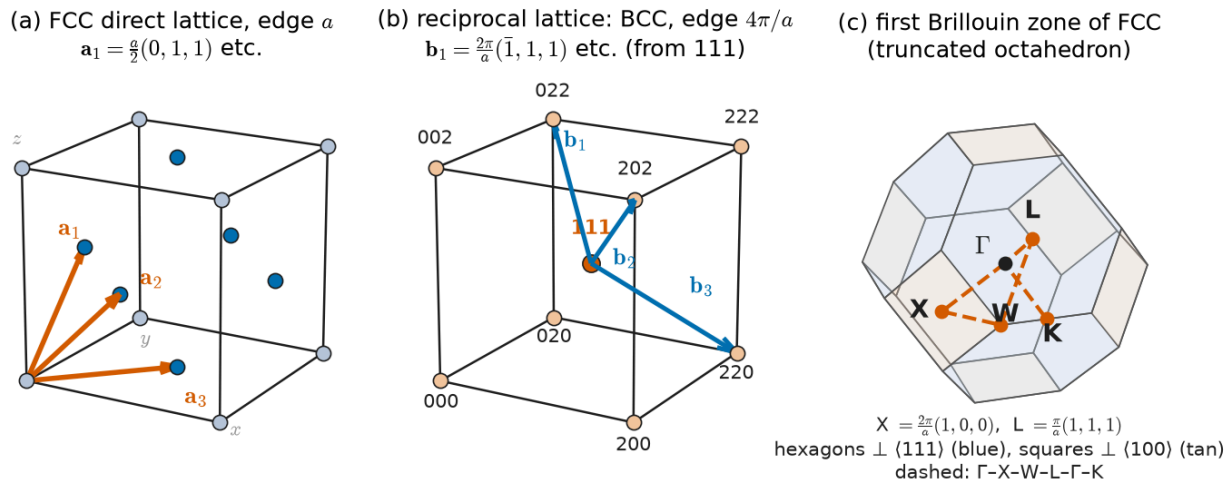

Figure 4.1 — The FCC direct lattice, its BCC reciprocal lattice, and the first Brillouin zone (truncated octahedron) with high-symmetry points.

Status of the principle: Brillouin-zone construction
Fundamental. It expresses the translational symmetry of the crystal in wavevector space (Bloch's theorem, treated in the Engineering Physics book); it holds for any wave in any perfectly periodic crystal and blurs, but does not vanish, in alloys and defected crystals.

Boundary.
The quantum mechanics of electrons in a periodic potential, and the derivation of Bloch's theorem, belong to the Engineering Physics book. This chapter uses the reciprocal lattice as a geometric tool; Chapters 24 and 31 use it for phonons and electrons.

4.2 Bragg's law, the Laue condition and the Ewald sphere

Bragg's law

In 1913 W. L. Bragg pictured diffraction as reflection of X-rays from successive lattice planes. Rays reflected from adjacent planes of spacing d travel paths that differ by $2d\sin\theta$, where θ is the glancing angle between the beam and the planes. They reinforce when that difference is a whole number of wavelengths:

$$n\lambda = 2d\sin\theta$$

Because a reflection of order n from planes (hkl) is geometrically identical to first-order reflection from planes $(nh\,nk\,nl)$ with spacing d/n, the order is usually absorbed into the indices and Bragg's law written $\lambda = 2d_{hkl}\sin\theta$. The detector, which sits at angle 2θ from the incident beam, records peaks at the "two-theta" positions of the pattern.

Status of the principle: Bragg's law
Derived. It follows from constructive interference of waves scattered by a periodic array of planes; it fixes peak positions exactly for an ideal infinite crystal, and is modified slightly only by refraction (significant at grazing incidence) and by the finite size of real crystals (Section 4.5).

Bragg's law has two immediate consequences. First, since $\sin\theta \le 1$, only planes with $d \ge \lambda/2$ can diffract. With Cu Kα_1 radiation, $\lambda = 0.15406$ nm, the smallest accessible spacing is 0.0770 nm; for copper this admits reflections up to $h^2+k^2+l^2 = 22$. Second, the wavelength must be comparable to interatomic spacings. Laboratory X-ray tubes (Cu, Co, Mo targets, $\lambda = 0.07$–

0.18 nm), thermal neutrons from reactors and spallation sources (0.1–0.4 nm), and electrons in a transmission microscope (2.51 pm at 200 kV) all qualify.

Probe	Typical wavelength	Scatters from	Penetration in metals	Typical engineering use
Laboratory X-rays (Cu Kα)	0.154 nm	Electron density, $\propto Z$	1–50 µm	Phase identification, lattice parameters, residual stress, texture
Synchrotron X-rays (high energy)	0.01–0.1 nm	Electron density	mm to cm	Strain mapping inside components, in-situ processing (Ch 53)
Thermal neutrons	0.1–0.4 nm	Nuclei (and magnetic moments)	cm	Residual stress in welds and rails, hydrogen and lithium positions, magnetic order
Electrons (200 kV TEM)	2.5 pm	Electrostatic potential	10–200 nm	Phases, orientation and defects in individual grains and precipitates (Ch 52)

The Laue condition

Max von Laue's earlier description treats each lattice point as a scatterer. An incident plane wave of wavevector $\mathbf{k}$ (magnitude $2\pi/\lambda$) is scattered elastically into $\mathbf{k}'$ with $|\mathbf{k}'| = |\mathbf{k}|$. Waves scattered from lattice points separated by a lattice vector $\mathbf{R}$ differ in phase by $(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}$; they all add in phase only if this is a multiple of 2π for every $\mathbf{R}$. Since $\mathbf{G} \cdot \mathbf{R}$ is a multiple of 2π exactly when $\mathbf{G}$ is a reciprocal-lattice vector, the condition for diffraction is

$$\Delta\mathbf{k} = \mathbf{k}' - \mathbf{k} = \mathbf{G}_{hkl}$$

The scattering vector must equal a reciprocal-lattice vector. Taking magnitudes, $|\Delta\mathbf{k}| = 2k\sin\theta = (4\pi/\lambda)\sin\theta$ and $|\mathbf{G}| = 2\pi/d$, which reproduces Bragg's law. Squaring $\mathbf{k}' = \mathbf{k} + \mathbf{G}$ with $k' = k$ gives $2\mathbf{k}\cdot\mathbf{G} + G^2 = 0$, the zone-face condition used in Section 4.1: a wave is diffracted exactly when its wavevector ends on a Brillouin-zone face.

> **Status of the principle: Laue condition**
> *Derived.* It is equivalent to Bragg's law but more general, because it treats all three dimensions at once and leads directly to the structure factor; it assumes elastic, single (kinematic) scattering.

The Ewald sphere

Paul Ewald turned the Laue condition into a drawing (Figure 4.2). Place the end of the incident wavevector $\mathbf{k}$ at the origin of reciprocal space and draw a sphere of radius $k = 2\pi/\lambda$ centred at its tail. Every scattered wavevector $\mathbf{k}'$ of the same length ends on this sphere. Diffraction occurs for each reciprocal-lattice point that lies on the sphere, and the diffracted beam points from the sphere's centre to that point.

The construction explains how every diffraction experiment works.

- **Single crystal, monochromatic beam.** In general no point lies exactly on the sphere; rotating the crystal sweeps points through it one at a time (four-circle diffractometers).
- **Single crystal, white beam (Laue method).** A range of wavelengths gives a shell of spheres, and every point inside the shell diffracts. Laue patterns check the orientation of single-crystal turbine blades and silicon boules to a fraction of a degree.
- **Powder or polycrystal.** Randomly oriented grains spread each $\mathbf{G}$ over a sphere of radius $|\mathbf{G}|$; its intersection with the Ewald sphere gives a cone of diffracted rays of half-angle 2θ. Every family with $d \geq \lambda/2$ diffracts at once, which makes powder diffraction the workhorse of phase identification.
- **Electron diffraction.** At 200 kV, $k = 2\pi/\lambda \approx 2500$ nm^{-1}, eighty times $|\mathbf{G}_{111}|$ of copper. The sphere is almost flat, and the reciprocal-lattice points of a thin foil are stretched into rods along the foil normal, so a whole plane of points appears at once as a spot pattern.

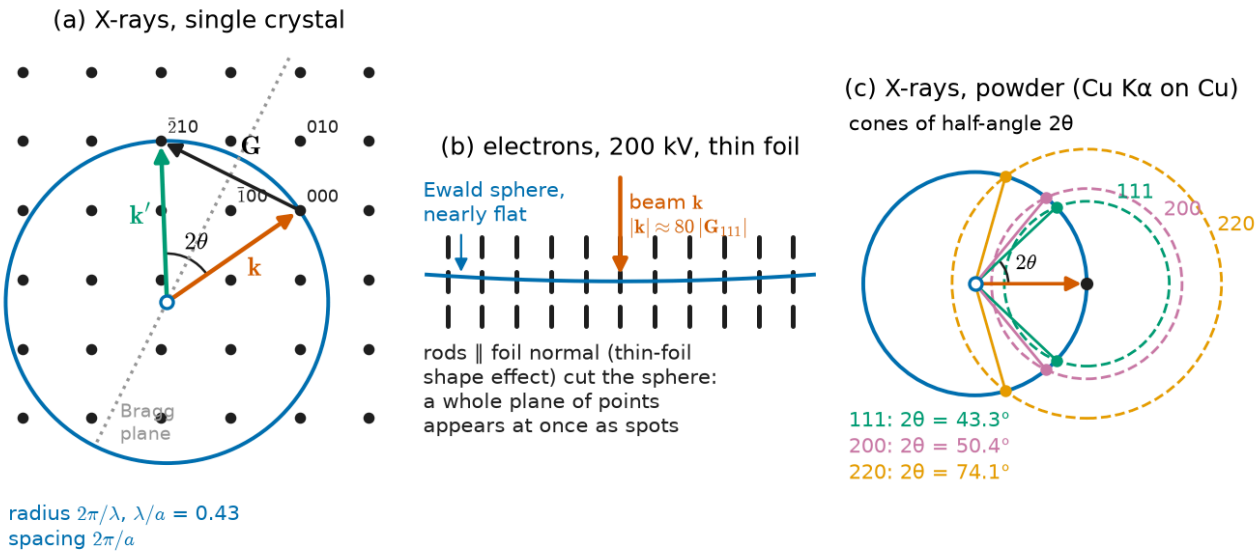

Figure 4.2 — The Ewald construction: diffraction occurs when a reciprocal-lattice point lies on the sphere of radius $2\pi/\lambda$; X-ray and electron cases compared.

> **Status of the principle: kinematic (single-scattering) approximation**
> *Approximate.* It assumes each photon, neutron or electron scatters once; it is excellent for X-rays and neutrons in powders and small crystals and fails for electrons in all but very thin foils and for large, perfect crystals, where multiple (dynamical) scattering redistributes intensity between beams.

Indexing a powder pattern

For a cubic crystal, Bragg's law and the plane-spacing formula combine to

$$\sin^2\theta = \frac{\lambda^2}{4a^2}\,(h^2 + k^2 + l^2)$$

so the values of $\sin^2\theta$ for successive peaks stand in the ratio of integers $N = h^2 + k^2 + l^2$. Simple cubic gives 1, 2, 3, 4, 5, 6, 8, ...; BCC gives 2, 4, 6, 8, 10, ... (equivalently 1, 2, 3, 4, 5); FCC gives 3, 4, 8, 11, 12, 16, ... The pattern of ratios identifies the Bravais lattice and the first peak gives a. Why particular values of N are missing is the subject of the next section.

4.3 The structure factor and systematic absences

Adding up the motif

The Laue condition says where diffraction *can* occur. Whether it does, and how strongly, depends on what sits at each lattice point. If the unit cell contains atoms j at fractional coordinates (x_j, y_j, z_j), each with atomic scattering factor f_j (Section 4.4), the amplitude scattered into reflection hkl is proportional to the **structure factor**

$$F_{hkl} = \sum_j f_j \,\exp\!\left[2\pi i\,(hx_j + ky_j + lz_j)\right]$$

and the intensity to $|F_{hkl}|^2$. The phases of the waves from different atoms in the cell either reinforce or cancel.

> **Status of the principle: structure factor**
> *Derived.* It is the Fourier transform of the electron (or nuclear) density of one unit cell, sampled at the reciprocal-lattice points; it holds within the kinematic approximation and is the basis of all crystal-structure determination.

Centred lattices

Body-centred cubic. Treat BCC as a simple cubic cell with atoms at $(0,0,0)$ and $(\tfrac12,\tfrac12,\tfrac12)$:

$$F = f\left[1 + e^{\pi i(h+k+l)}\right] = \begin{cases} 2f & h+k+l \text{ even} \\ 0 & h+k+l \text{ odd} \end{cases}$$

The (100) reflection of α-iron is absent because the body-centre atoms lie on planes exactly halfway between the cube faces and scatter in antiphase.

Face-centred cubic. With atoms at $(0,0,0)$, $(\tfrac12,\tfrac12,0)$, $(\tfrac12,0,\tfrac12)$, $(0,\tfrac12,\tfrac12)$:

$$F = f\left[1 + e^{\pi i(h+k)} + e^{\pi i(h+l)} + e^{\pi i(k+l)}\right] = \begin{cases} 4f & h,k,l \text{ all even or all odd} \\ 0 & \text{mixed} \end{cases}$$

So FCC copper, aluminium, nickel and austenite show 111, 200, 220, 311, 222, …, and never 100, 110 or 210. These **systematic absences** are what the reciprocal lattice already predicted: the allowed points of FCC form a BCC array, and vice versa.

Diamond cubic. Silicon and germanium have an FCC lattice with a two-atom motif, at $(0,0,0)$ and $(\tfrac14,\tfrac14,\tfrac14)$. The structure factor is the FCC result multiplied by $1 + e^{\pi i(h+k+l)/2}$, which vanishes when $h+k+l = 4n+2$. Silicon therefore lacks 200 and 222 as well as all mixed reflections; its powder pattern with Cu Kα shows 111 (28.44°), 220 (47.30°), 311 (56.12°), 400 (69.13°) and 331 (76.37°). **Hexagonal close-packed.** With atoms at $(0,0,0)$ and $(\tfrac13,\tfrac23,\tfrac12)$, $F = 0$ when $h + 2k = 3n$ and l is odd. The (0001) basal reflection of titanium, magnesium and zinc is therefore absent; (0002) is present.

Compounds and ordered alloys

When the atoms in the motif differ, cancellation becomes partial. In rock-salt NaCl, with Na on the FCC sites and Cl displaced by $(\tfrac12,0,0)$,

$$F = \begin{cases} 4\,(f_{\mathrm{Na}} + f_{\mathrm{Cl}}) & h,k,l \text{ all even} \\ 4\,(f_{\mathrm{Na}} - f_{\mathrm{Cl}}) & h,k,l \text{ all odd} \\ 0 & \text{mixed} \end{cases}$$

The all-odd reflections are weak because $\mathrm{Na^+}$ (10 electrons) and $\mathrm{Cl^-}$ (18) partly cancel. In KCl the ions are isoelectronic, the odd reflections almost vanish, and the pattern mimics a simple cubic crystal of half the true lattice parameter.

Order–disorder transformations (Chapter 3) are detected the same way. In disordered β-brass (CuZn) the two sites of the CsCl-type cell are occupied at random and the pattern is BCC. Below about 460 °C copper orders onto the corners and zinc onto the body centres; reflections with $h+k+l$ odd appear with $F \propto \eta\,(f_{\mathrm{Cu}} - f_{\mathrm{Zn}})$, where η is the long-range order parameter. These **superlattice reflections** have intensity proportional to η^2. In nickel-base superalloys the γ′ phase $\mathrm{Ni_3(Al,Ti)}$ is an ordered L1$_2$ version of the FCC matrix; its 100 and 110 superlattice spots are how γ′ precipitates are imaged and sized (Chapters 10 and 16).

Lattice or structure	Allowed reflections	First reflections ($N = h^2+k^2+l^2$)	Examples
Simple cubic	all	100, 110, 111, 200 (1, 2, 3, 4)	α-Po; ordered B2 superlattice lines
BCC	$h+k+l$ even	110, 200, 211, 220 (2, 4, 6, 8)	α-Fe, W, Mo, Cr, β-Ti
FCC	h,k,l unmixed	111, 200, 220, 311 (3, 4, 8, 11)	Cu, Al, Ni, γ-Fe, Au
Diamond cubic	unmixed, and not $h+k+l = 4n+2$	111, 220, 311, 400 (3, 8, 11, 16)	Si, Ge, diamond
Rock salt	unmixed; odd set weak	111, 200, 220, 311	NaCl, MgO, NiO, TiC
HCP	not ($h+2k = 3n$ with l odd)	$10\bar10$, 0002, $10\bar11$	Ti, Mg, Zn, Zr

Retained austenite, again

The opening failure can now be read. Martensite in a carburized case is body-centred tetragonal, close to BCC: its strong peaks with Cu Kα are near 44.7° (110) and 65.0° (200), and α-iron with $a = 0.2867$ nm gives 44.67°, 65.02°, 82.33° and 98.94° for 110, 200, 211 and 220. Austenite is FCC with $a \approx 0.359$ nm, and its 111, 200 and 220 reflections fall at about 43.6°, 50.8° and 74.7°. Because the two lattices have different absences, their peaks interleave, and the ratio of integrated intensities, corrected by the factors of Section 4.4, gives the volume fraction of each phase. The same principle measures monoclinic zirconia in a toughened ceramic or anatase versus rutile in a titania photocatalyst.

4.4 Diffracted intensities

The integrated intensity of a powder reflection — the area under its peak — can be written

$$I_{hkl} = S\,|F_{hkl}|^2\,p_{hkl}\,L_p(\theta)\,e^{-2M}\,A(\theta)$$

where S is a scale factor common to all peaks of one phase. Each remaining factor has a physical origin.

Atomic scattering factor

X-rays are scattered by electrons. In the forward direction an atom scatters as Z electrons; at higher angles, waves from different parts of the electron cloud get out of phase and the **atomic scattering factor** f falls. It depends on $\sin\theta/\lambda$ and is tabulated (International Tables) or computed from a four-Gaussian fit. For copper, $f = 29$ at zero angle, about 22 at the 111 reflection ($\sin\theta/\lambda = 0.240$ Å^{-1}) and about 15 at 311 (0.459 Å^{-1}). Close to an absorption edge of the scattering atom, small "anomalous" corrections f' and f'' appear; Cu Kα on copper or iron is close enough to their edges that $f' \approx -2$ to -4, and iron-containing samples also fluoresce strongly under Cu radiation, which is why steels are usually measured with Co or Cr tubes.

Neutrons scatter from nuclei, which are points: the scattering length does not fall with angle and varies irregularly with Z, so neutron diffraction locates hydrogen in hydrides and lithium in battery cathodes, and distinguishes neighbours such as Fe, Co and Ni.

Multiplicity

In a powder, all planes of a family $\{hkl\}$ diffract at the same angle. The **multiplicity** p counts them: in cubic crystals $p = 6$ for $\{100\}$, 12 for $\{110\}$, 8 for $\{111\}$, 24 for $\{hk0\}$ and $\{hhl\}$, and 48 for general $\{hkl\}$. Lower symmetry splits families, which is why the 200 peak of tetragonal martensite or $BaTiO_3$ splits into a 2:1 doublet.

Lorentz–polarization factor

The **Lorentz factor** accounts for the time each reciprocal-lattice point spends in reflecting position and the fraction of each cone intercepted by the detector; the **polarization factor** $(1+\cos^2 2\theta)/2$ for the partial polarization of an unpolarized beam on scattering. For a powder diffractometer they combine to

$$L_p = \frac{1+\cos^2 2\theta}{\sin^2\theta\,\cos\theta}$$

which is large at low angles and passes through a minimum near $2\theta \approx 100^\circ$.

Debye–Waller factor

Atoms vibrate about their sites. The vibrations smear the electron density and reduce the coherently diffracted intensity by the **Debye–Waller factor**

$$e^{-2M} = \exp\!\left(-2B\,\frac{\sin^2\theta}{\lambda^2}\right), \qquad B = 8\pi^2\langle u^2\rangle$$

where $\langle u^2\rangle$ is the mean-square displacement of an atom along the scattering vector. For copper at room temperature $B \approx 0.55$ Å^2, a root-mean-square displacement of $\sqrt{0.55/8\pi^2} = 0.083$ Å. The lost intensity reappears as diffuse background. Because $\langle u^2\rangle$ rises roughly linearly with temperature, high-angle peaks fade on heating, which in-situ diffraction must correct for; static displacements from solutes and defects add to B in the same way.

> **Status of the principle: Debye–Waller factor**
> *Approximate*. The Gaussian form is exact for harmonic vibrations; it needs corrections for strongly anharmonic solids near melting and for static disorder that is not Gaussian.

Absorption

In the symmetric (Bragg–Brentano) geometry with a fully absorbing sample, the irradiated volume is independent of angle and A is a constant; thin films and grazing-incidence geometry need explicit corrections. Penetration depth also sets what is measured: in steel, most of the signal comes from the top few micrometres, exactly the region of the gear tooth that failed.

Worked example — The powder pattern of copper. Cu Kα_1 ($\lambda = 0.15406$ nm) on FCC copper ($a = 0.3615$ nm). Take f from the four-Gaussian fit for neutral Cu, $B = 0.55$ Å^2, and $F = 4f$ for the allowed reflections.

For 111: $d = 0.3615/\sqrt{3} = 0.2087$ nm, $\sin\theta = 0.15406/(2 \times 0.2087) = 0.3691$, $\theta = 21.66°$, $2\theta = 43.32°$. Then $\sin\theta/\lambda = 0.240$ Å^{-1}, $f = 22.1$, $F = 88.3$, $|F|^2 = 7.80 \times 10^3$, $p = 8$, $L_p = (1 + \cos^2 43.32°)/(\sin^2 21.66° \cos 21.66°) = 12.08$, $e^{-2M} = \exp(-2 \times 0.55 \times 0.240^2) = 0.939$. Product: $7.80 \times 10^3 \times 8 \times 12.08 \times 0.939 = 7.08 \times 10^5$.

Repeating for the next reflections and normalising to 111:

hkl	d (nm)	2θ (°)	f	p	L_p	e^{-2M}	Calculated I	Reference pattern I
111	0.2087	43.32	22.1	8	12.08	0.939	100	100
200	0.1807	50.45	20.7	6	8.55	0.919	46	46
220	0.1278	74.12	16.8	12	3.71	0.845	24	20
311	0.1090	89.93	14.8	24	2.83	0.793	27	17
222	0.1044	95.14	14.2	8	2.74	0.777	8	5

The positions are exact; the intensities agree at low angles and are 20–50 % high at higher angles. The anomalous correction $f' \approx -2$ lowers the calculated 220 and 311 values to 22 and 24; surface roughness and residual texture account for the rest. No single factor dominates, so quantitative work (phase fractions, Rietveld refinement, Chapter 52) must model them all.

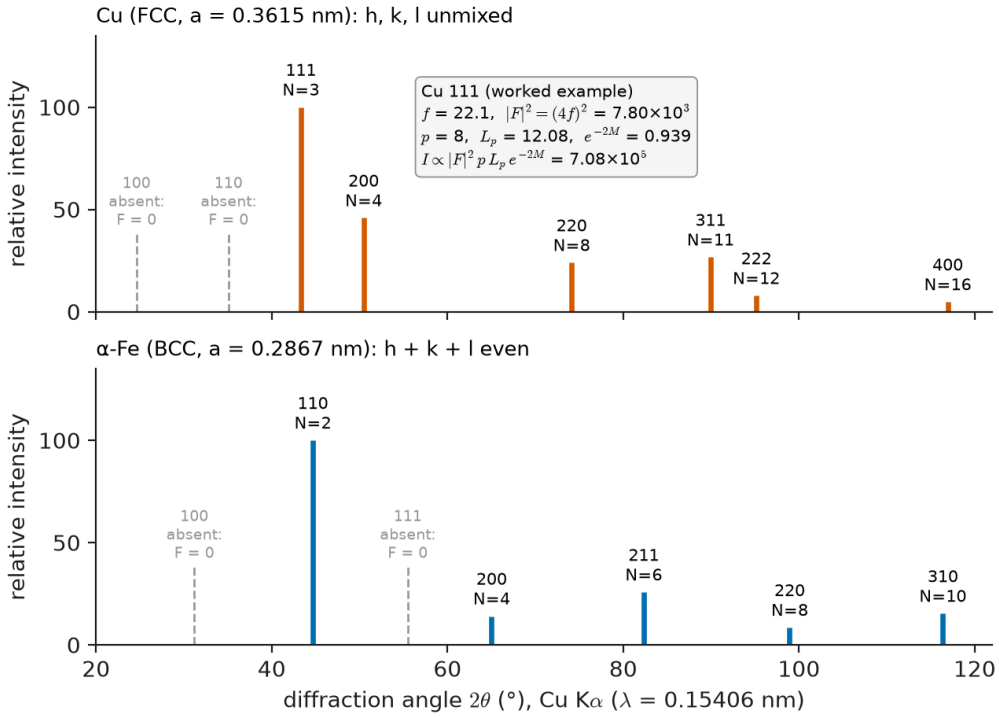

Figure 4.3 — Calculated powder patterns of FCC copper and BCC α-iron with Cu Kα radiation: the different systematic absences make the two lattices instantly distinguishable.

4.5 Peak broadening: size and strain

An infinite, perfect crystal would give infinitely sharp peaks (apart from the natural width of the X-ray line). Real peaks have widths, and the widths carry information about the microstructure. Three contributions add.

1. **Instrumental broadening** from the finite source size, slits, wavelength spread and detector resolution, measured with a strain-free coarse-grained standard (LaB_6, Si or corundum powders).
2. **Size broadening** from crystallites (coherently diffracting domains) that are too small for destructive interference to be complete just off the Bragg angle.
3. **Strain broadening** from non-uniform elastic strain — different crystallites, or different parts of one crystallite, having slightly different d-spacings.

A *uniform* strain does not broaden a peak; it shifts it. That is the basis of residual-stress measurement by the $\sin^2\psi$ method, in which the shift of a high-angle peak is measured as the sample is tilted (Chapter 52).

The Scherrer equation

A crystallite with N planes of spacing d in the diffraction direction behaves like a diffraction grating with N lines: the angular width of its peak is inversely proportional to N. The result, due to Scherrer (1918), is

$$D = \frac{K\lambda}{\beta \cos\theta}$$

where β is the integral breadth or full width at half maximum of the size-broadened peak in radians of 2θ, D is a volume-weighted thickness of the crystallites normal to the diffracting planes, and K is a shape factor of about 0.9 (0.89–0.94 for common shapes and width definitions).

> **Status of the principle: Scherrer equation**
> *Approximate.* It follows from the interference of a finite stack of planes; it measures coherently diffracting domain size rather than grain size, is meaningful only for domains below about 100–200 nm (above that the broadening is lost in the instrumental width), and overestimates size whenever strain broadening is ignored.

Worked example — Crystallite size of a ceria catalyst support. Cerium dioxide (fluorite structure, $a = 0.5411$ nm) is the oxygen-storage component of automotive three-way catalysts; its crystallite size controls surface area and sinters during engine ageing. The 111 reflection sits at $2\theta = 28.55^\circ$ with Cu Kα. A fresh support shows a FWHM of 0.80°; the instrument, measured on a LaB_6 standard, contributes 0.10°.

Assuming Gaussian profiles, widths add in quadrature: $\beta = \sqrt{0.80^2 - 0.10^2} = 0.794^\circ = 0.01385$ rad. With $\theta = 14.27^\circ$, $\cos\theta = 0.969$:

$$D = \frac{0.9 \times 0.15406\ \text{nm}}{0.01385 \times 0.969} = 10.3\ \text{nm}$$

If the profiles are instead treated as Lorentzian, widths subtract linearly, $\beta = 0.70^\circ$, and $D = 11.7$ nm. The 13 % difference is typical of the systematic uncertainty of a single-peak Scherrer estimate; it is still ample to track the crystallite growth that degrades oxygen storage when the catalyst is aged at high temperature.

Strain broadening and the Williamson–Hall plot

If the lattice spacing varies from place to place with a characteristic microstrain ε (a width of the distribution of $\Delta d/d$), differentiating Bragg's law gives a broadening

$$\beta_\varepsilon = 4\,\varepsilon \tan\theta$$

Size broadening varies as $1/\cos\theta$ and strain broadening as $\tan\theta$, so measuring several reflections separates them. Adding the two contributions (the simplest assumption, valid for Lorentzian profiles) and multiplying by $\cos\theta$ gives the **Williamson–Hall** relation

$$\beta\cos\theta = \frac{K\lambda}{D} + 4\varepsilon\sin\theta$$

A plot of $\beta\cos\theta$ against $4\sin\theta$ is a straight line with intercept $K\lambda/D$ and slope ε.

> **Status of the principle: Williamson–Hall analysis**
> *Approximate.* It assumes additive size and strain widths and isotropic strain; in elastically anisotropic metals the strain broadening depends on hkl (the plot scatters), which the modified Williamson–Hall and Warren–Averbach methods correct using dislocation contrast factors.

Worked example — Ball-milled iron powder. Mechanical alloying in a ball mill produces nanocrystalline metal powders for oxide-dispersion-strengthened steels used in fusion and fission reactor cladding. A milled α-iron powder measured with Cu Kα gives instrument-corrected widths (FWHM, degrees of 2θ):

hkl	2θ (°)	β (°)	$\beta\cos\theta$ (rad)	$4\sin\theta$
110	44.67	0.63	0.01017	1.520
200	65.02	0.81	0.01192	2.150
211	82.33	1.02	0.01340	2.633
220	98.94	1.29	0.01463	3.040
310	116.37	1.71	0.01573	3.399

A least-squares line through the last two columns has slope 0.00297 and intercept 0.00560 rad. Thus $\varepsilon = 0.30\,\%$ and

$$D = \frac{0.9 \times 0.15406}{0.00560} = 24.7\ \text{nm}$$

Had the 110 peak alone been put into the Scherrer equation, the answer would have been 13.6 nm; from the 310 peak alone, 8.8 nm. The disagreement is the signature of strain broadening, and ignoring it would understate the crystallite size by a factor of two to three. A microstrain of 0.3 % in iron corresponds, through the models of Chapter 12, to a dislocation density of order 10^{16} m^{-2}, typical of severely deformed metal.

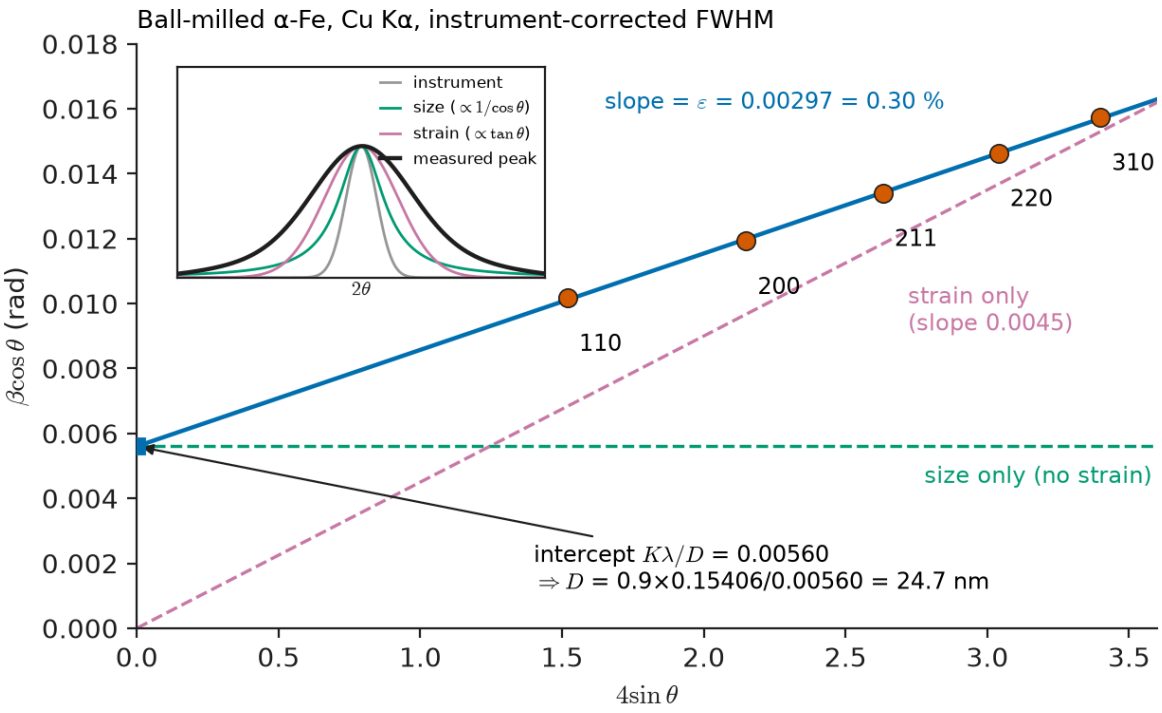

Figure 4.4 — Williamson–Hall plot for ball-milled α-iron: the intercept gives crystallite size, the slope microstrain.

In every case the number is a *coherent-domain* size: a grain divided into subgrains, or containing twins and stacking faults, diffracts as several smaller domains, so diffraction sizes are usually smaller than grain sizes measured by microscopy (Chapter 5).

4.6 Neumann's principle

From symmetry of structure to symmetry of properties

The second half of this chapter turns from diffraction to properties. A crystal of quartz conducts heat faster along its c axis than across it; a copper crystal conducts heat equally in all directions but is three times stiffer along $\langle 111 \rangle$ than along $\langle 100 \rangle$. To decide which properties *may* depend on direction, and how, we need a rule connecting the symmetry of the structure to the symmetry of a property. It was stated by Franz Neumann in the nineteenth century:

> *The symmetry elements of any physical property of a crystal must include the symmetry elements of the point group of the crystal.*

If a crystal is unchanged by a rotation, then any measurement on it must also be unchanged by that rotation, because nothing physical distinguishes the rotated crystal from the original. The property may have *more* symmetry than the crystal, but never less.

> **Status of the principle: Neumann's principle**
> *Fundamental.* It follows from the invariance of physical laws under coordinate transformations; it holds for any equilibrium property of a perfect crystal, and constrains (but does not determine) the property's values.

Consequences

Neumann's principle has three kinds of consequence, all used routinely.

- **Properties that must vanish.** A centre of symmetry inverts every vector. Any property that relates a vector to a symmetric second-rank quantity, such as piezoelectricity (polarization from stress), changes sign under inversion and so must be zero in the 11 centrosymmetric point groups. Pyroelectricity, a spontaneous polarization vector, is allowed only in the 10 polar point groups.
- **Properties that must be equal.** A threefold axis along [111] in a cubic crystal permutes $x \to y \to z$, so any property must have equal components along the three cube axes.
- **The number of independent constants.** For each crystal system, the principle fixes how many numbers must be measured to characterise a property completely (Sections 4.7–4.8).

The principle applies to *any* object with a symmetry, not only single crystals. A random polycrystal is statistically isotropic, so it has no piezoelectric effect even if each grain is piezoelectric; a sintered PZT ceramic must be **poled** in a strong field, lowering its symmetry to ∞m, before it becomes a transducer. A rolled sheet has orthorhombic sample symmetry, which governs its stiffness and the earing of deep-drawn cans (Chapter 46); a unidirectional fibre composite is transversely isotropic (Chapter 22); wood is orthotropic (Chapter 18).

> **Boundary.**
> The mathematics of tensors and their transformation belongs to The Mathematics Universe; this chapter states what materials engineers need and applies it.

4.7 Tensor properties

Properties as tensors

Many properties relate one directional quantity to another. Thermal conductivity relates the heat-flux vector $\mathbf{q}$ to the temperature-gradient vector ∇T; in general the two need not be parallel:

$$q_i = -k_{ij}\,\frac{\partial T}{\partial x_j}$$

(summation over repeated indices). The nine coefficients k_{ij} form a **second-rank tensor**: on rotation of axes by a matrix a_{ij} they transform as $k'_{ij} = a_{ik}a_{jl}k_{kl}$. The **rank** of a property tensor equals the number of indices, which is the sum of the ranks of the two quantities it connects: vector–vector gives rank 2; vector–rank-2 (polarization from stress) gives rank 3; rank-2–rank-2 (stress from strain) gives rank 4.

Rank	Property	Relation	Examples
0 (scalar)	Density, heat capacity	—	Any crystal
1 (vector)	Pyroelectric coefficient	$\Delta P_i = p_i\,\Delta T$	$LiTaO_3$, PVDF, tourmaline
2	Electrical and thermal conductivity, permittivity, permeability	$J_i = \sigma_{ij}E_j$, $q_i = -k_{ij}\,\partial_j T$, $D_i = \varepsilon_{ij}E_j$	All crystals
2	Thermal expansion	$\varepsilon_{ij} = \alpha_{ij}\,\Delta T$	All crystals
3	Piezoelectric coefficients	$P_i = d_{ijk}\,\sigma_{jk}$	Quartz, PZT, AlN, ZnO, PVDF
4	Elastic stiffness and compliance	$\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl}$	All crystals
4	Electrostriction, photoelasticity	$\varepsilon_{ij} = M_{ijkl}E_kE_l$	PMN relaxors, glass stress birefringence

Rank 2: conductivity, permittivity and thermal expansion

The conductivity, permittivity and thermal-expansion tensors are **symmetric** ($k_{ij} = k_{ji}$, which for transport coefficients follows from Onsager's reciprocal relations), so they have at most six independent components. Any symmetric tensor can be diagonalised by choosing **principal axes**; along those axes, flux and gradient are parallel and the property takes its three principal values. The value along an arbitrary unit direction $\mathbf{n}$ is

$$k(\mathbf{n}) = n_i \, k_{ij} \, n_j = k_1 n_1^2 + k_2 n_2^2 + k_3 n_3^2$$

and the surface $k_{ij} x_i x_j = 1$ is an ellipsoid, the **representation quadric**. Neumann's principle then fixes its shape:

Crystal system	Independent components	Shape of quadric	Examples
Cubic	1	Sphere (isotropic)	Cu, Fe, Si, NaCl, MgO, spinel
Tetragonal, hexagonal, trigonal	2 ($k_\parallel$, $k_\perp$ to the unique axis)	Ellipsoid of revolution	Zn, Ti, Mg, quartz, sapphire, rutile, graphite, β-Sn
Orthorhombic	3	General ellipsoid, axes fixed by symmetry	Olivine, aragonite, U (α)
Monoclinic	4	One axis fixed	Gypsum, monoclinic ZrO_2
Triclinic	6	Fully general	Plagioclase feldspars

Typical anisotropies are large. Representative room-temperature values:

Material	Property	Parallel to c	Perpendicular to c	Consequence
α-quartz	Thermal conductivity (W m^{-1} K^{-1})	about 10.4	about 6.2	Oriented cuts for oscillator and sensor plates
α-quartz	Thermal expansion (10^{-6} K^{-1})	about 7.1	about 13.2	Cracking of quartz grains in fired ceramics and in concrete aggregate during fires
Zinc	Thermal expansion (10^{-6} K^{-1})	about 60	about 15	Thermal ratcheting and roughening of zinc coatings and castings
β-tin	Thermal expansion (10^{-6} K^{-1})	about 30	about 15	Grain-scale stresses in Sn-based solder joints under thermal cycling
Sapphire (α-Al_2O_3)	Relative permittivity	about 11.6	about 9.4	Microwave substrates and LED wafers must be cut and specified by orientation
Graphite	Electrical conductivity	about 10^3 times lower than in-plane	about 2×10^6 S m^{-1}	Anisotropic electrodes, heat spreaders, pyrolytic-graphite components

Worked example — Heat flow in a quartz plate. A quartz plate is cut with its normal at 45° to the c axis, and a temperature gradient of 1000 K m^{-1} is imposed across its thickness. With $k_\parallel = 10.4$ and $k_\perp = 6.2$ W m^{-1} K^{-1}, the conductivity along the plate normal is

$$k(45^\circ) = k_\perp \sin^2 45^\circ + k_\parallel \cos^2 45^\circ = \tfrac{1}{2}(6.2 + 10.4) = 8.3 \ \text{W m}^{-1}\text{K}^{-1}$$

The heat flux, however, is not parallel to the gradient. Resolving the gradient into components along and across c, the flux components are $10.4 \times 707 = 7350$ and $6.2 \times 707 = 4380$ W m^{-2}, so the flux is tilted at $\arctan(4380/7350) = 30.8^\circ$ from the c axis, 14.2° away from the plate normal towards c. In a plate heated on one face, the hottest region on the opposite face is displaced sideways. The same off-axis flow appears in graphite heat spreaders and in anisotropic composites (Chapter 22), and it is why thermal-property data for non-cubic materials must state the direction.

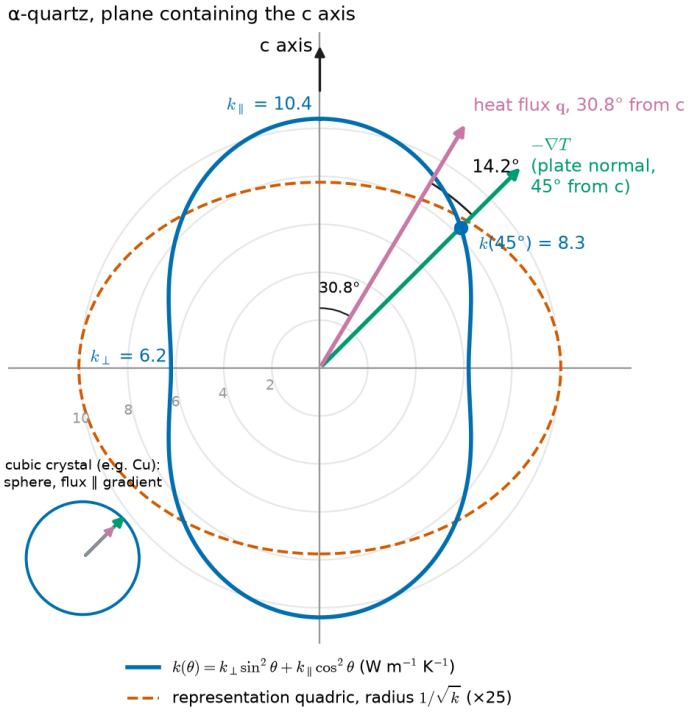

Figure 4.5 — Direction dependence of a rank-2 property: thermal conductivity of α-quartz in a plane containing the c axis, and the tilt of heat flux away from the temperature gradient.

Thermal expansion can even be negative along some directions: cordierite ($Mg_2Al_4Si_5O_{18}$) contracts along c and expands across it. In extruded cordierite honeycombs for catalytic converters, aligned grains and microcracks between them bring the bulk expansion down to about 1×10^{-6} K^{-1}, low enough to survive cold-start thermal shock. The volume coefficient is the trace, $\alpha_V = \alpha_1 + \alpha_2 + \alpha_3$; for quartz, $7.1 + 2 \times 13.2 = 33.5 \times 10^{-6}$ K^{-1}. Anisotropic expansion also stresses non-cubic polycrystals: neighbouring grains mismatch by $\Delta\alpha\,\Delta T$, and coarse-grained alumina and TiB_2 microcrack on cooling while fine-grained versions do not, because stored elastic energy scales with grain volume but crack energy with boundary area (Chapter 17).

> **Status of the principle: representation of rank-2 properties by an ellipsoid**
> *Derived.* It follows from the symmetry of the tensor and holds for any linear relation between two vectors (or a scalar and a symmetric tensor); it fails when the response becomes non-linear, as in conduction at very high fields.

Rank 3: piezoelectricity

In a piezoelectric crystal a stress produces an electric polarization (the direct effect) and an electric field produces a strain (the converse effect), with the same coefficients:

$$P_i = d_{ijk}\,\sigma_{jk}, \qquad \varepsilon_{jk} = d_{ijk}\,E_i$$

Because σ_{jk} is symmetric, the 27 components of d_{ijk} reduce to 18, which are written in **Voigt notation** as a 3×6 matrix $d_{i\lambda}$ with the index pairs $11 \to 1$, $22 \to 2$, $33 \to 3$, $23 \to 4$, $13 \to 5$, $12 \to 6$. Neumann's principle removes most of them. All 11 centrosymmetric point groups have $d = 0$, and so does the non-centrosymmetric cubic class 432, leaving **20 piezoelectric point groups**. α-Quartz (trigonal, 32) has only two independent coefficients, $d_{11} = 2.3$ pC N^{-1} and $d_{14} \approx 0.7$ pC N^{-1}. A poled ceramic (∞m) has three, d_{33}, d_{31} and d_{15}, the same set as wurtzite AlN and ZnO (6mm).

Material	Symmetry	d_{33} or main coefficient (pC N^{-1})	Uses
α-quartz	32	$d_{11} = 2.3$	Frequency standards, force and pressure sensors
AlN (thin film)	6mm	about 5	RF filters in phones, MEMS microphones
ZnO	6mm	about 12	Surface-acoustic-wave devices, energy harvesters
PVDF (poled polymer)	mm2	about −25 to −33	Flexible sensors, hydrophones, wearable devices
PZT ceramic (soft)	∞m	300–600	Ultrasonic transducers, fuel injectors, actuators, sonar
PMN–PT single crystal	4mm (poled)	1500–2500	Medical ultrasound arrays

Worked example — A quartz force sensor and a PZT actuator. A quartz load washer in a machine-tool dynamometer is cut so that a compressive force along its x_1 axis produces charge on the faces normal to x_1 through d_{11}. Because the charge per area is $d_{11}\sigma$ and $\sigma = F/A$, the total charge is independent of area: $Q = d_{11}F$. For a cutting force of 100 N, $Q = 2.3 \times 10^{-12} \times 100 = 2.3 \times 10^{-10}$ C. Across the sensor's own capacitance of 10 pF this would give 23 V, but the charge leaks away through any finite resistance, which is why piezoelectric force sensors use charge amplifiers and measure dynamic rather than static loads. Run the converse effect in a soft PZT actuator with $d_{33} = 400$ pC N^{-1} under a field of 2 kV mm^{-1}: the strain is $400 \times 10^{-12} \times 2 \times 10^{6} = 8 \times 10^{-4}$, so a 10 mm multilayer stack extends by 8 µm. That stroke, delivered in microseconds against kilonewton forces, drives piezoelectric diesel injectors and the nanopositioners of scanning-probe microscopes.

> **Status of the principle: linear piezoelectric equations**
> *Approximate.* They are the leading term of the coupled electromechanical response; valid for quartz to very high stresses, but in ferroelectric ceramics such as PZT only at low fields, beyond which domain switching makes the response non-linear and hysteretic (Chapter 27).

> **Boundary.**
> Ferroelectric domains, poling and the electrical side of piezoelectric devices are treated in Chapter 27; actuators and sensors return in Chapter 37.

Rank 4: elasticity

Hooke's law for a crystal relates two symmetric second-rank tensors, stress and strain:

$$\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl}, \qquad \varepsilon_{ij} = S_{ijkl}\,\sigma_{kl}$$

The stiffness tensor C_{ijkl} has 81 components. Symmetry of stress and strain reduces them to 36, written as a 6×6 Voigt matrix $C_{\lambda\mu}$. The existence of a strain-energy function, $W = \tfrac{1}{2}C_{\lambda\mu}\varepsilon_\lambda\varepsilon_\mu$, makes that matrix symmetric, leaving at most **21 independent elastic constants** (triclinic). Neumann's principle reduces the count further:

Symmetry	Independent elastic constants	Engineering examples
Triclinic	21	Some minerals; general anisotropic composites
Monoclinic	13	Monoclinic zirconia, some polymers
Orthorhombic (orthotropic)	9	Rolled sheet, wood, laminated composites in their principal axes
Tetragonal	6	β-Sn, tetragonal zirconia, TiO_2
Hexagonal (transversely isotropic)	5	Ti, Mg, Zn, ZnO, unidirectional fibre composites, columnar coatings
Cubic	3 (C_{11}, C_{12}, C_{44})	Cu, Al, Ni and superalloys, Fe, W, Si, GaAs, MgO
Isotropic	2 (E and ν, or K and G)	Glasses, random polycrystals, amorphous polymers

> **Status of the principle: generalized Hooke's law with 21 independent constants**
> *Fundamental* in structure (it follows from the symmetry of stress and strain and the existence of a strain-energy function), *approximate* in its linear form, which holds for small strains, typically below 0.1–0.5 % in metals and ceramics (Chapter 11).

4.8 Why cubic crystals are isotropic in conduction but anisotropic in elasticity

Second rank: the sphere

Apply Neumann's principle to a rank-2 tensor in a cubic crystal. A fourfold (or twofold) axis along z sends $x \to y$, $y \to -x$; requiring k_{ij} to be unchanged forces $k_{11} = k_{22}$ and $k_{12} = k_{13} = k_{23} = 0$. The threefold axis along $[111]$ then forces $k_{33} = k_{11}$. All cubic point groups contain four threefold axes, so for every cubic crystal

$$k_{ij} = k\,\delta_{ij}$$

The representation quadric is a sphere. Conductivity, permittivity, susceptibility, diffusivity and thermal expansion of copper, iron, silicon or spinel are the same in every direction — exactly, by symmetry alone. Textured copper conductors and silicon wafers of any orientation have the same resistivity, and cubic ceramics such as MgO and stabilised zirconia do not microcrack from expansion anisotropy.

> **Status of the principle: isotropy of rank-2 properties in cubic crystals**
> *Derived.* A direct consequence of Neumann's principle and cubic point symmetry; it holds for every linear rank-2 property and fails only for higher-rank (or non-linear) properties.

Fourth rank: the cube

The same symmetry operations applied to C_{ijkl} leave three independent constants, and the stiffness matrix of any cubic crystal is

$$C = \begin{pmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{44} \end{pmatrix}$$

An isotropic solid has the same form but with only two independent constants, because isotropy additionally requires $C_{44} = (C_{11} - C_{12})/2$. Nothing in cubic symmetry enforces that relation. The two shear moduli of a cubic crystal — C_{44} for shear on $\{100\}$ planes and $(C_{11} - C_{12})/2$ for shear on $\{110\}$ planes along $\langle 1\bar{1}0 \rangle$ — are set by different bond geometries and are generally different. Their ratio is the **Zener anisotropy ratio**

$$A = \frac{2C_{44}}{C_{11} - C_{12}}$$

with $A = 1$ for an isotropic solid. Physically, a fourth-rank tensor contains direction products of the form $n_1^2 n_2^2$, which a threefold axis cannot average away. The direction-dependent Young's modulus of a cubic crystal follows from the compliances S_{11}, S_{12} and S_{44}:

$$\frac{1}{E(\mathbf{n})} = S_{11} - 2\left(S_{11} - S_{12} - \tfrac{1}{2}S_{44}\right)\left(n_1^2 n_2^2 + n_2^2 n_3^2 + n_3^2 n_1^2\right)$$

The bracketed orientation function is 0 along $\langle 100 \rangle$, $1/4$ along $\langle 110 \rangle$ and $1/3$ along $\langle 111 \rangle$, so E has its extreme values along the cube axes and the body diagonals.

> **Status of the principle: directional Young's modulus of a cubic crystal**
> *Derived.* It follows from transforming the cubic compliance tensor to an arbitrary direction; it is exact within linear elasticity for a single crystal under uniaxial stress.

Crystal	C_{11} (GPa)	C_{12} (GPa)	C_{44} (GPa)	A	$E_{\langle 100\rangle}$ (GPa)	$E_{\langle 110\rangle}$ (GPa)	$E_{\langle 111\rangle}$ (GPa)
Tungsten	522	204	161	1.01	407	410	411
Aluminium	108	61	28.5	1.22	64	73	76
Silicon	166	64	79.6	1.56	130	169	188
α-Iron	231	135	116	2.41	132	220	283
Nickel	247	147	125	2.51	136	233	304
Gold	193	164	41.5	2.85	43	81	115
Copper	168	121	75.4	3.21	67	130	191
NaCl	49.1	12.8	12.8	0.71	44	35	33

Values are typical room-temperature handbook constants. Tungsten is almost isotropic by coincidence; NaCl has $A < 1$, so its stiffest direction is $\langle 100\rangle$.

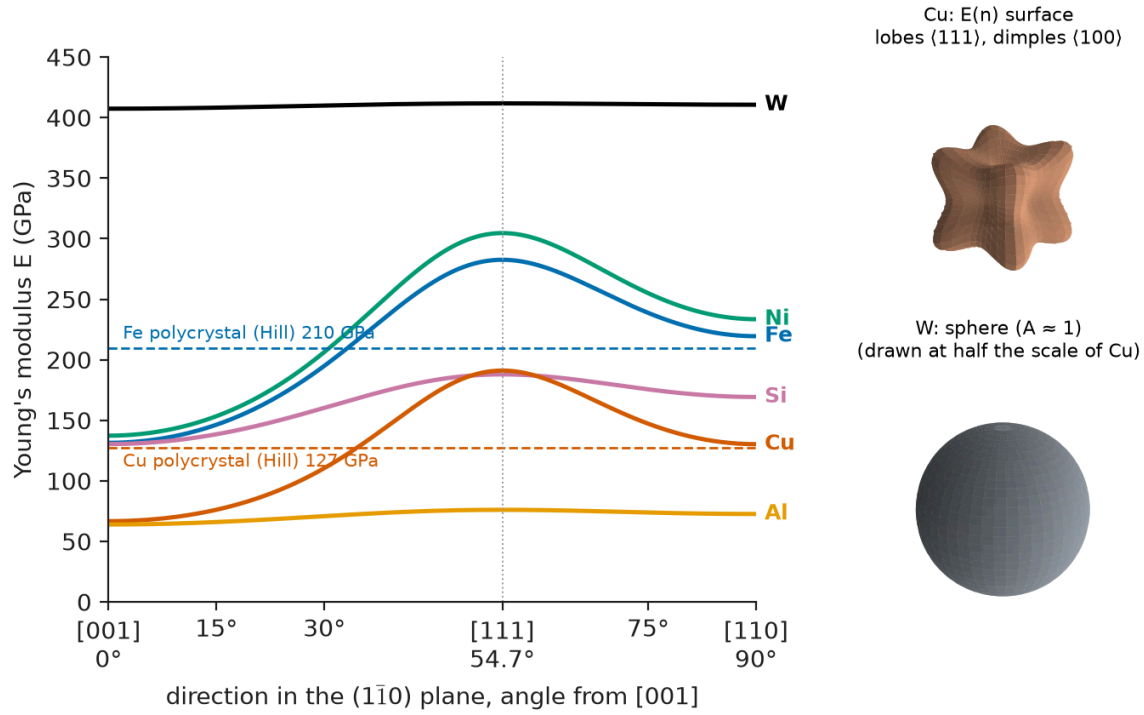

Figure 4.6 — Young's modulus versus direction in cubic crystals, in the (1-10) plane from [001] through [111] to [110]: cubic symmetry permits strong elastic anisotropy.

Worked example — Stiffness of a copper crystal and of copper polycrystal. From the constants in the table, inverting the cubic stiffness matrix gives $S_{11} = 0.01500$, $S_{12} = -0.00628$ and $S_{44} = 1/C_{44} = 0.01326$ GPa^{-1}. Then $S_{11} - S_{12} - \tfrac{1}{2}S_{44} = 0.01500 + 0.00628 - 0.00663 = 0.01465$ GPa^{-1} and

$$E_{\langle 100\rangle} = \frac{1}{0.01500} = 66.7 \text{ GPa}, \qquad E_{\langle 111\rangle} = \frac{1}{0.01500 - \tfrac{2}{3}\times 0.01465} = 191 \text{ GPa}$$

A factor of 2.9 between the softest and stiffest directions. What does a randomly oriented polycrystal show? Assuming every grain has the same strain (Voigt bound) gives shear modulus $G_V = (C_{11} - C_{12} + 3C_{44})/5 = 54.6$ GPa; assuming the same stress (Reuss bound) gives $G_R = 5/[4(S_{11} - S_{12}) + 3S_{44}] = 40.0$ GPa. The bulk modulus is the same in both, $K = (C_{11} + 2C_{12})/3 = 137$ GPa, because a cubic crystal compresses isotropically under hydrostatic pressure. With $E = 9KG/(3K + G)$, the bounds are $E_V = 145$ GPa and $E_R = 109$ GPa, and their mean (the Hill estimate) is 127 GPa, close to the 125–130 GPa measured on untextured copper. A polycrystal is isotropic only on average: individual grains still carry stresses that differ by tens of percent from the average, which concentrates stress at grain boundaries in fatigue (Chapter 14) and scatters ultrasound in coarse-grained austenitic welds, making them notoriously difficult to inspect.

Where elastic anisotropy is engineered

- **Single-crystal turbine blades.** For nickel, $E_{\langle 001\rangle} = 136$ GPa against $E_{\langle 111\rangle} = 304$ GPa. A blade grown with $\langle 001\rangle$ along its axis (the natural growth direction of dendrites in directional solidification, Chapter 9) has the lowest available stiffness, so a given thermal strain $\alpha\Delta T$ produces the lowest thermal stress. Thermal-fatigue life improves several-fold over randomly oriented cast blades (Chapters 16 and 41).
- **Silicon MEMS.** In a standard (100) wafer, beams along $\langle 110\rangle$ have $E = 169$ GPa and beams along $\langle 100\rangle$ have 130 GPa. Resonant frequencies, which scale as $\sqrt{E}$, differ by 14 % between the two orientations; accelerometer and gyroscope designers must use the orientation-dependent modulus, not a handbook "isotropic" value.
- **Textured sheet and films.** Steel sheet with a strong $\{111\}$ texture is stiffer and draws more deeply than random sheet; copper films with $\langle 111\rangle$ fibre texture develop higher thermal stress on silicon than random films (Chapter 49).

The same logic applies to every rank-4 property: photoelasticity, electrostriction and magnetostriction are also anisotropic in cubic crystals (Chapter 28). Plastic yielding, governed by slip systems and Schmid factors (Chapter 13), is direction-dependent too.

What remains open

Open question	Where it is answered
What are the vacancies, dislocations and grain boundaries that broaden peaks and limit coherent domain size?	Chapter 5 (defects); Chapter 12 (dislocations and their strain fields)
How are elastic constants related to bonding, and how is anisotropic Hooke's law used in design?	Chapter 11
How do electron waves form bands at Brillouin-zone boundaries?	Chapter 31; foundations in the Engineering Physics book
How do phonons on the Brillouin zone set heat capacity, expansion and thermal conductivity?	Chapter 24
How are orthotropic and transversely isotropic stiffness matrices handled for composite plies and wood?	Chapters 22 and 18
How do piezoelectric and ferroelectric materials work as devices?	Chapters 27 and 37
How is diffraction done in practice: Rietveld refinement, residual stress, texture, EBSD?	Chapter 52
What do synchrotron, neutron, PDF and 4D-STEM methods add?	Chapter 53
How do textures form during rolling, recrystallization and additive manufacturing?	Chapters 10, 46 and 48

Next: Chapter 5 turns from the ideal crystal to the real one — the point, line, planar and volume defects that diffraction detects and that control almost every property in this book.

Chapter 5 — Defects and Microstructure

Part I · Structure of Materials

On 19 July 1989 the fan disc of the tail engine of a DC-10 burst in cruise over Iowa. Fragments cut all three hydraulic systems, and the aircraft crash-landed at Sioux City. The investigators traced the fracture to a fatigue crack that had grown for years from a single small region in the titanium-alloy disc: a **hard-alpha inclusion**, a pocket of titanium enriched in nitrogen during melting of the ingot, which had stabilised the brittle α phase over a volume a few millimetres across. The rest of the forging met every specification. One volume defect, a few cubic millimetres in a component weighing more than 100 kg, decided its life. Titanium melting practice changed as a result: cold-hearth melting and ultrasonic inspection of billets were introduced to remove and detect exactly this class of defect.

The same disc also illustrates the opposite truth. Its strength came from defects too: from aluminium and vanadium atoms dissolved on titanium sites, from the boundaries between millions of fine grains, and from the interfaces between two phases deliberately produced by forging and heat treatment. A perfect crystal of titanium would yield at a small fraction of the alloy's strength. Defects are not the exception in engineering materials; they are the means by which structure is controlled, and the most common reason structure fails.

> **What kinds of imperfections do real crystalline materials contain, and how many of each? How are defects described, counted and measured, and which properties does each kind control?**

Concept	Engineering destination
Vacancies and interstitials, equilibrium concentration	Diffusion and heat treatment (Ch 8), creep (Ch 14), radiation damage (Ch 41)
Defects in ionic crystals, Kröger–Vink notation	Solid electrolytes and fuel cells (Ch 25, 40), oxide nuclear fuel, catalysts, gas sensors, corrosion films (Ch 42)
Non-stoichiometry	UO_{2+x} fuel, CeO_{2-x} catalyst supports, FeO scales, n-type oxides
Dislocations, Burgers vector	Plasticity and strengthening (Ch 12–13), threading dislocations in semiconductors (Ch 34)
Grain boundaries, CSL, twins	Hall–Petch strengthening, intergranular corrosion and grain-boundary engineering (Ch 43), creep (Ch 14)
Stacking faults	Work hardening of austenitic steels, TWIP and TRIP steels (Ch 15)
Phase boundaries	Precipitation hardening (Ch 10), superalloys (Ch 16), thin-film epitaxy (Ch 49)
Pores, inclusions, precipitates	Fatigue of castings and AM parts (Ch 14, 48), clean steel, ceramic strength (Ch 17), concrete durability (Ch 18)
Stereology	Grain-size specifications, phase fractions, porosity limits, quality control (Ch 52)

Roadmap. Point defects in metals and their equilibrium number (5.1) → point defects in ionic crystals and defect chemistry (5.2) → line defects (5.3) → planar defects: grain boundaries, twins, stacking faults, phase boundaries (5.4) → volume defects (5.5) → measuring microstructure: stereology (5.6) → how defects control properties (5.7).

5.1 Point defects

Classification by dimension

Defects are classified by their dimensionality, and the classification is the skeleton of this chapter.

Dimension	Defect	Typical size	Equilibrium?	Example
0	Vacancy, interstitial, substitutional solute, antisite	One atomic site	Yes (vacancies, solutes)	Vacancies in hot copper; carbon in steel; Y in ZrO_2
1	Dislocation	Core ~1 nm wide, lines μm to m long	No	Dislocations in cold-worked aluminium
2	Grain boundary, twin, stacking fault, phase boundary, surface	~0.5–1 nm thick, μm to mm across	No (except some interfaces in multiphase equilibrium)	Grain boundaries in a steel; γ/γ′ interfaces in a superalloy
3	Pore, crack, inclusion, precipitate	nm to mm	Precipitates can be; pores and inclusions are not	Gas pores in a casting; MnS in steel; θ′ in Al–Cu

The column "equilibrium?" is the key physical distinction. A defect is an equilibrium defect if introducing a few of them *lowers* the Gibbs energy of the crystal. Only point defects satisfy this, because only they gain enough configurational entropy to pay for their energy (Section 5.1, below). Dislocations, grain boundaries and pores raise the Gibbs energy and survive only because the atomic motion needed to remove them is too slow. That is why their numbers depend on processing history while vacancy numbers depend only on temperature.

Vacancies and self-interstitials

A **vacancy** is an empty lattice site. A **self-interstitial** is a host atom squeezed into a site between normal positions. In close-packed metals the interstitial costs three to four times as much energy as a vacancy (about 3 eV against 1.3 eV in copper) because it must push its neighbours far apart; in thermal equilibrium vacancies outnumber self-interstitials by many orders of magnitude. Interstitials matter mainly when they are produced athermally, by irradiation. A fast neutron striking an atom in a reactor pressure vessel or a fusion first wall knocks it out of its site with tens of keV of energy and starts a **displacement cascade**, leaving behind equal numbers of vacancies and interstitials (Frenkel pairs). Their accumulation and clustering cause the swelling, hardening and embrittlement of nuclear materials (Chapter 41).

Solutes: substitutional and interstitial

A foreign atom can replace a host atom (**substitutional solute**) or occupy a gap between host atoms (**interstitial solute**). Substitution is favoured when the atoms are similar in size, crystal structure, electronegativity and valence — the Hume-Rothery rules (Chapter 7). Copper and nickel mix in all proportions; zinc dissolves in copper up to about 38 wt % to make α-brass; chromium, nickel and molybdenum substitute for iron in stainless steels.

Interstitial solutes are small atoms — H, B, C, N, O — that fit, with distortion, into the octahedral and tetrahedral holes of the host lattice. In FCC metals the largest hole (octahedral) admits a sphere of radius $0.414R$ without distortion, where R is the host atom radius; in BCC the largest hole (tetrahedral) admits only $0.291R$, and the octahedral one only $0.155R$. Carbon in iron, with a radius ratio near 0.57, fits better in FCC austenite than in BCC ferrite, so austenite dissolves up to 2.1 wt % carbon and ferrite only 0.022 wt %. The resulting supersaturation when austenite is quenched is the origin of martensite and of the whole heat-treatment practice of steels (Chapters 10 and 15). Hydrogen, the smallest interstitial, diffuses through steel even at room temperature and gathers at defects, which makes it the agent of hydrogen embrittlement in high-strength fasteners, pipelines and pressure vessels (Chapters 40 and 43).

Every solute distorts the lattice around it: a large substitutional atom pushes neighbours out, a small one pulls them in, and an interstitial in a BCC octahedral site produces a tetragonal (non-spherical) distortion. These strain fields interact with dislocations, which is the basis of solid-solution strengthening (Chapter 13), and scatter electrons and phonons, which lowers electrical and thermal conductivity (Section 5.7).

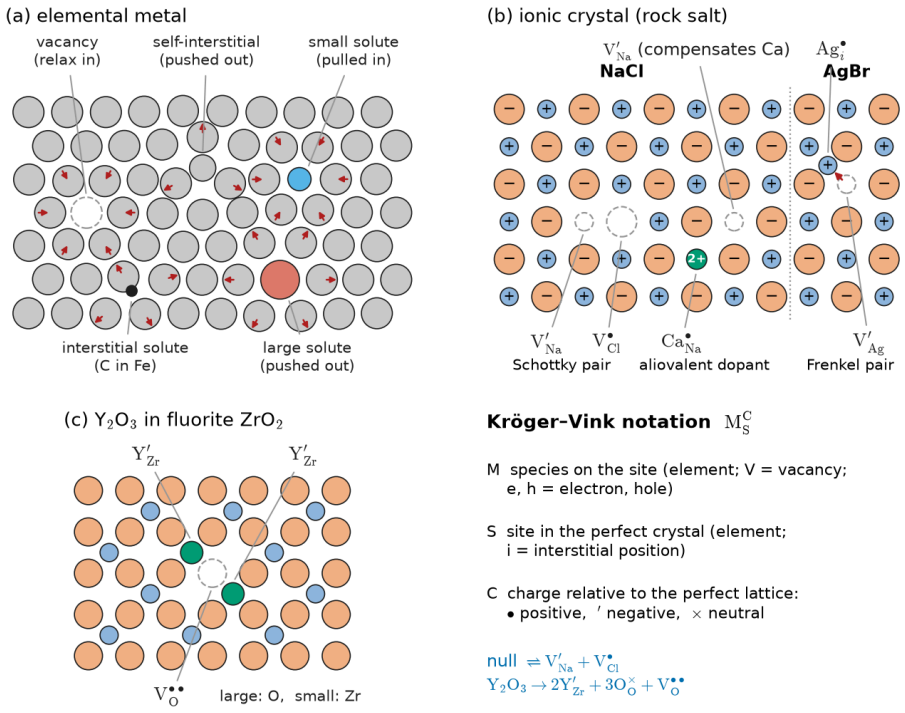

Figure 5.1 — Point defects in an elemental metal and in an ionic crystal, with Kröger–Vink labels.

The equilibrium concentration of vacancies

Why should a crystal at equilibrium contain any vacancies at all? Creating n vacancies among N sites costs an enthalpy $n\,\Delta H_v$, but the vacancies can be arranged in $N!/[n!(N-n)!]$ ways, and that configurational entropy, together with a small vibrational entropy ΔS_v per vacancy, lowers the Gibbs energy. Minimising G with respect to n gives the equilibrium vacancy fraction

$$c_v = \frac{n}{N} = \exp\!\left(\frac{\Delta S_v}{k_B}\right)\exp\!\left(-\frac{\Delta H_v}{k_B T}\right)$$

with $k_B = 8.617 \times 10^{-5}$ eV K^{-1}. Chapter 6 derives this result from the chemical potential of the vacancy; here we use it. ΔS_v is typically 1–2 k_B, so the pre-exponential factor is 3–7.

> **Status of the principle: equilibrium vacancy concentration**
> *Derived.* It follows from minimising the Gibbs energy of a crystal containing dilute, non-interacting vacancies; it is reached in practice only where atoms are mobile, roughly above 0.4–$0.5\,T_m$, and fails at very high concentrations where vacancies cluster into divacancies and voids.

Metal	Structure	ΔH_v (eV)	c_v at T_m (with $\Delta S_v = 0$)	Measured c_v near T_m
Aluminium	FCC	0.67	2.4×10^{-4}	about 9×10^{-4}
Copper	FCC	1.28	1.7×10^{-5}	about 2×10^{-4}
Nickel	FCC	1.79	6.0×10^{-6}	about 10^{-4} (estimated)
Tungsten	BCC	3.6	1.2×10^{-5}	about 10^{-4} (estimated)

Values of ΔH_v are typical experimental figures (±10 %). The striking regularity is that every metal at its melting point contains roughly one vacancy per 10^3–10^4 sites, because ΔH_v scales with the cohesive energy and hence with T_m: ΔH_v is about 8–12 $k_B T_m$ for these metals. The difference between the "$\Delta S_v = 0$" and measured columns is the vibrational-entropy factor, plus the contribution of divacancies near melting.

Worked example — Vacancies in copper during annealing. Copper has $a = 0.3615$ nm, so its atomic density is $4/a^3 = 8.47 \times 10^{28}$ m^{-3}. Take $\Delta H_v = 1.28$ eV and $\Delta S_v = 1.5\,k_B$, so $e^{\Delta S_v/k_B} = 4.48$.

At an annealing temperature of 1000 °C ($T = 1273$ K), $k_B T = 0.1097$ eV and $\Delta H_v / k_B T = 11.67$:

$$c_v = 4.48 \times e^{-11.67} = 4.48 \times 8.56 \times 10^{-6} = 3.8 \times 10^{-5}$$

or $3.8 \times 10^{-5} \times 8.47 \times 10^{28} = 3.2 \times 10^{24}$ vacancies per cubic metre. At room temperature ($T = 300$ K), $\Delta H_v / k_B T = 49.5$ and $c_v = 1.4 \times 10^{-21}$: a cubic micrometre of copper, 8.5×10^{10} sites, would contain on average 10^{-10} vacancies — in effect none. Between 300 K and 1273 K the equilibrium fraction rises by a factor of about 3×10^{16}.

Two practical consequences follow. First, because diffusion in copper proceeds by atoms jumping into neighbouring vacancies, the diffusion rate inherits this enormous temperature dependence (Chapter 8): annealing, sintering, creep and homogenisation all happen at high temperature and essentially stop at low temperature. Second, a copper or aluminium alloy quenched from a high temperature keeps a large non-equilibrium vacancy population for minutes to hours, which accelerates room-temperature diffusion and makes natural ageing of aluminium alloys possible (Chapter 10).

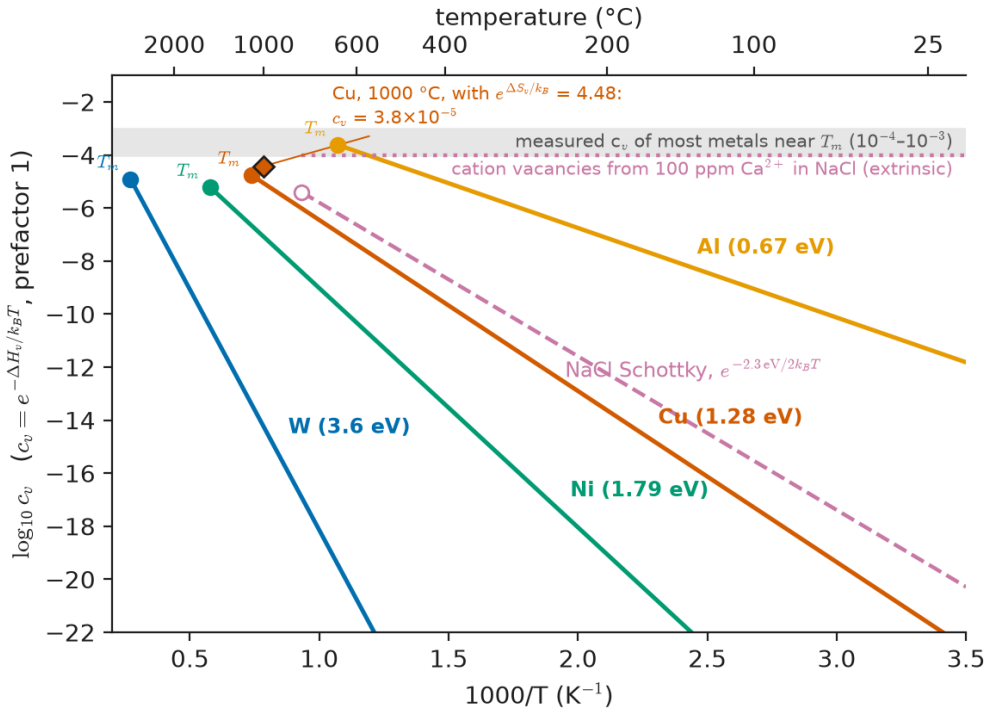

Figure 5.2 — Equilibrium vacancy fraction versus inverse temperature for several metals, and Schottky-pair fraction in NaCl, compared with an extrinsic (doping-controlled) level.

5.2 Defects in ionic crystals

Charge neutrality and the two classic disorders

In an ionic crystal every defect carries an effective charge, and the crystal as a whole must remain neutral. Point defects therefore appear in charge-compensating combinations.

- A **Schottky defect** is a set of cation and anion vacancies in the stoichiometric ratio: one Na^+ and one Cl^- vacancy in NaCl, one Mg^{2+} and one O^{2-} vacancy in MgO, one Zr^{4+} and two O^{2-} vacancies in ZrO_2. Schottky disorder dominates when the ions are of similar size and the structure is close-packed (alkali halides with the rock-salt structure, MgO, NiO).
- A **Frenkel defect** is an ion moved from its site to an interstitial position, leaving a vacancy. It dominates when one ion is small and the structure has open interstitial space: cation Frenkel defects in AgBr and AgCl, anion Frenkel defects in fluorite-structure oxides (CaF_2, UO_2, CeO_2), where large empty cubes in the anion array accommodate interstitial anions.

For Schottky pairs in a 1:1 compound, the same entropy argument as for vacancies gives a pair fraction $\exp(-\Delta H_S / 2 k_B T)$, where ΔH_S is the enthalpy to form one *pair*; the factor 2 appears because each pair produces two independently distributed defects.

Kröger–Vink notation

Defect chemistry needs a bookkeeping system that keeps track of sites, species and charges. **Kröger–Vink notation** writes each defect as $\mathrm{M}_{\mathrm{S}}^{\mathrm{C}}$:

- M is the species on the site: an element symbol, V for a vacancy, e or h for an electron or hole;
- S is the site normally occupied in the perfect crystal: an element symbol, or i for an interstitial position;
- C is the charge *relative to the perfect crystal*: $\bullet$ for each unit of positive charge, $'$ for each negative unit, $\times$ for zero.

So $\mathrm{V}'_{\mathrm{Na}}$ is a sodium vacancy (a missing Na^{+} leaves an effective -1), $\mathrm{V}_{\mathrm{O}}^{\bullet\bullet}$ an oxygen vacancy in an oxide, $\mathrm{Ca}_{\mathrm{Na}}^{\bullet}$ a calcium ion on a sodium site, $\mathrm{Y}'_{\mathrm{Zr}}$ an yttrium ion on a zirconium site, $\mathrm{O}_{\mathrm{i}}''$ an interstitial oxide ion, and $\mathrm{O}_{\mathrm{O}}^{\times}$ a normal oxygen ion on its own site.

Defect reactions are written like chemical equations and must balance in three ways: **mass** (atoms on each side), **site ratio** (the ratio of cation to anion sites of the host must be preserved; sites can be created or destroyed only in that ratio), and **charge** (effective charges sum to the same value). The Schottky and Frenkel reactions in NaCl and AgBr are

$$\mathrm{null} \;\rightleftharpoons\; \mathrm{V}'_{\mathrm{Na}} + \mathrm{V}_{\mathrm{Cl}}^{\bullet}, \qquad \mathrm{Ag}_{\mathrm{Ag}}^{\times} \;\rightleftharpoons\; \mathrm{Ag}_{\mathrm{i}}^{\bullet} + \mathrm{V}'_{\mathrm{Ag}}$$

Aliovalent doping

Adding ions of a different valence forces compensating defects. Dissolving $\mathrm{CaCl_2}$ in NaCl places each Ca^{2+} on a sodium site and requires one sodium vacancy for neutrality:

$$\mathrm{CaCl_2} \xrightarrow{\mathrm{NaCl}} \mathrm{Ca}_{\mathrm{Na}}^{\bullet} + \mathrm{V}'_{\mathrm{Na}} + 2\,\mathrm{Cl}_{\mathrm{Cl}}^{\times}$$

The same logic, applied to yttria in zirconia, produces oxygen vacancies:

$$\mathrm{Y_2O_3} \xrightarrow{\mathrm{ZrO_2}} 2\,\mathrm{Y}'_{\mathrm{Zr}} + 3\,\mathrm{O}_{\mathrm{O}}^{\times} + \mathrm{V}_{\mathrm{O}}^{\bullet\bullet}$$

One oxygen vacancy per two yttrium ions. In 3 mol % yttria-stabilised zirconia (3Y-TZP), a formula unit with 0.97 $\mathrm{ZrO_2}$ and 0.03 $\mathrm{Y_2O_3}$ contains 1.03 cations and 2.06 oxygen sites, of which 0.03 are vacant: 1.5 % of the oxygen sites are empty. In 8YSZ the figure is $0.08/2.16 = 3.7$ %. The vacancies stabilise the tetragonal and cubic forms of zirconia at room temperature, which is what makes 3Y-TZP a tough ceramic for hip-joint heads, dental crowns and cutting blades (Chapters 17 and 39); and they let oxide ions hop from vacancy to vacancy, which makes 8YSZ the electrolyte of solid-oxide fuel cells and of the oxygen sensors in every petrol-engine exhaust (Chapters 25 and 40).

> **Status of the principle: electroneutrality and site balance in defect reactions**
> *Fundamental*. They express conservation of charge and of the crystal structure's site ratio; they hold for every defect reaction, and they — not the details of energetics — fix which compensating defects must form.

Worked example — Intrinsic or extrinsic? Vacancies in salt. For Schottky pairs in NaCl take $\Delta H_S = 2.3$ eV (measured values 2.2–2.5 eV). The intrinsic vacancy fraction on each sublattice is $\exp(-\Delta H_S / 2k_B T)$. At 800 K, $2k_B T = 0.1379$ eV and

$$c_{\mathrm{intrinsic}} = e^{-2.3/0.1379} = e^{-16.68} = 5.7 \times 10^{-8}$$

Even at the melting point (1074 K) it is only 4.0×10^{-6}. Now suppose the salt contains 100 ppm of Ca^{2+} on Na sites, a typical impurity level. Each Ca creates one sodium vacancy, so $c_{\mathrm{extrinsic}} = 10^{-4}$, independent of temperature. The intrinsic value would equal this only at

$$T = \frac{\Delta H_S}{2k_B \ln(10^4)} = \frac{2.3}{2 \times 8.617 \times 10^{-5} \times 9.21} = 1449\ \mathrm{K}$$

which is above the melting point. In practice, therefore, the cation-vacancy concentration — and with it the ionic conductivity — of nominally pure NaCl is set by impurities over its whole solid range. The same is true for most oxides: in alumina, MgO and zirconia the intrinsic Schottky energies are 5–7 eV per defect set, and impurity or dopant levels of tens of ppm dominate. Defect chemistry in ceramics is essentially the chemistry of aliovalent impurities, deliberate or not.

Non-stoichiometry and the atmosphere

Ionic crystals can also exchange atoms with their surroundings, which changes their composition. An oxide in a reducing atmosphere can lose oxygen:

$$\mathrm{O_O^{\times}} \rightleftharpoons \mathrm{V_O^{\bullet\bullet}} + 2e' + \tfrac{1}{2}\mathrm{O_2(g)}$$

The oxygen vacancy is compensated by two electrons, released into the conduction band or onto reducible cations ($\mathrm{Ti^{4+}} \rightarrow \mathrm{Ti^{3+}}$, $\mathrm{Ce^{4+}} \rightarrow \mathrm{Ce^{3+}}$). The oxide becomes oxygen-deficient, $\mathrm{MO_{2-x}}$, and an n-type electronic conductor. Applying the law of mass action to this reaction (Chapter 6) gives $[\mathrm{V_O^{\bullet\bullet}}] \propto p_{\mathrm{O_2}}^{-1/6}$ when the vacancies and electrons compensate each other, so the deviation from stoichiometry and the conductivity both rise as the oxygen pressure falls.

Other oxides take up extra oxygen. Wüstite is never stoichiometric FeO; it exists as $\mathrm{Fe_{1-x}O}$ with x from about 0.05 to 0.15, its cation vacancies compensated by $\mathrm{Fe^{3+}}$ ions (holes, in effect):

$$2\,\mathrm{Fe_{Fe}^{\times}} + \tfrac{1}{2}\mathrm{O_2(g)} \rightleftharpoons 2\,\mathrm{Fe_{Fe}^{\bullet}} + \mathrm{V_{Fe}''} + \mathrm{O_O^{\times}}$$

In $\mathrm{Fe_{0.95}O}$, 5 % of the iron sites are vacant and 10 % of the remaining iron is $\mathrm{Fe^{3+}}$. Those vacancies are why iron ions diffuse quickly outward through the FeO layer of a steel oxide scale, and why FeO-rich scales grow fast above 570 °C (Chapter 42).

Non-stoichiometry is engineered across the discipline:

Oxide	Defect	Engineering role
$\mathrm{UO_{2+x}}$	Oxygen interstitials $\mathrm{O_i''}$	Nuclear fuel: x changes thermal conductivity, creep and fission-gas release (Ch 41)
$\mathrm{CeO_{2-x}}$	$\mathrm{V_O^{\bullet\bullet}}$ + $\mathrm{Ce^{3+}}$	Three-way catalysts store and release oxygen as engine exhaust swings rich and lean
$\mathrm{SnO_{2-x}}$, $\mathrm{ZnO_{1-x}}$, $\mathrm{In_2O_3}$:Sn	$\mathrm{V_O^{\bullet\bullet}}$, donors	Gas sensors, transparent conducting electrodes for solar cells and displays
$\mathrm{Fe_{1-x}O}$, $\mathrm{Ni_{1-x}O}$	Cation vacancies + holes	Growth of oxide scales on steels and nickel alloys (Ch 42)
$\mathrm{La_{1-x}Sr_xCoO_{3-\delta}}$	$\mathrm{V_O^{\bullet\bullet}}$	Mixed-conducting cathodes and oxygen-separation membranes (Ch 40)
$\mathrm{BaTiO_{3-\delta}}$	$\mathrm{V_O^{\bullet\bullet}}$	Degradation of capacitor dielectrics fired in reducing atmosphere

> **Boundary.**
> The thermodynamics of defect equilibria — the law of mass action, defect formation energies and Brouwer diagrams — is developed in Chapter 6, and the electrical consequences in Chapter 25. The underlying chemical equilibrium theory belongs to the Advanced Chemistry book.

5.3 Line defects: dislocations

Why they must exist

A perfect crystal sheared by sliding one plane rigidly over the next would need a shear stress of roughly $G/2\pi$ to $G/30$ — several gigapascals for a metal. Annealed metal single crystals yield at a thousandth of that or less. The explanation, given independently by Orowan, Polanyi and Taylor in 1934, is that slip proceeds by the motion of a line defect, the **dislocation**, which breaks and remakes bonds one row at a time. Chapter 12 develops dislocation theory in depth; this section introduces the geometry needed for the rest of the book.

Edge, screw and mixed dislocations

- An **edge dislocation** is the boundary of an extra half-plane of atoms inserted into the crystal. The lattice above the line is compressed and below it stretched. Under shear, the half-plane shifts by one atomic spacing at a time, and when the dislocation leaves the crystal it leaves a step on the surface.
- A **screw dislocation** turns the stack of atomic planes around the line into a single helical ramp, like a spiral car park: walking once around the line takes you up one atomic plane.
- A **mixed dislocation** has both characters. A general dislocation line is curved; a loop in a slip plane is pure edge at two points, pure screw at two others, and mixed in between.

The Burgers vector

A dislocation is characterised by its **Burgers vector b**. Draw a closed circuit of atom-to-atom steps around the line in the real crystal; repeat the same sequence of steps in a perfect reference crystal; the circuit fails to close, and the closure vector is **b** (Figure 5.3). For an edge dislocation **b** is perpendicular to the line; for a screw dislocation, parallel. The Burgers vector is constant along the whole dislocation however its line wanders, so a dislocation cannot end inside a perfect crystal: it ends at a surface or grain boundary, meets other dislocations at a node, or closes on itself as a loop.

Dislocations choose the shortest available Burgers vector, because their line energy is approximately $\tfrac{1}{2}Gb^2$ per unit length:

Structure	Burgers vector	Magnitude	Examples (b in nm)
FCC	$\frac{a}{2}\langle 110\rangle$	$a/\sqrt{2}$	Cu 0.256, Al 0.286, Ni 0.249
BCC	$\frac{a}{2}\langle 111\rangle$	$a\sqrt{3}/2$	α-Fe 0.248, W 0.274
HCP	$\frac{a}{3}\langle 11\bar{2}0\rangle$	a	Ti 0.295, Mg 0.321
Diamond cubic	$\frac{a}{2}\langle 110\rangle$	$a/\sqrt{2}$	Si 0.384
Rock salt	$\frac{a}{2}\langle 110\rangle$	$a/\sqrt{2}$	MgO 0.298

A dislocation moves by **glide** in the plane containing its line and **b**, which needs only shear stress, and an edge dislocation can also **climb** out of that plane by absorbing or emitting vacancies, which needs diffusion and becomes important above about $0.4\,T_m$ (creep, Chapter 14). Glide on the densest planes along the densest directions defines the **slip systems** — 12 $\{111\}\langle 1\bar{1}0\rangle$ systems in FCC metals, for example — whose number and ease of operation largely decide whether a material is ductile (Chapter 13).

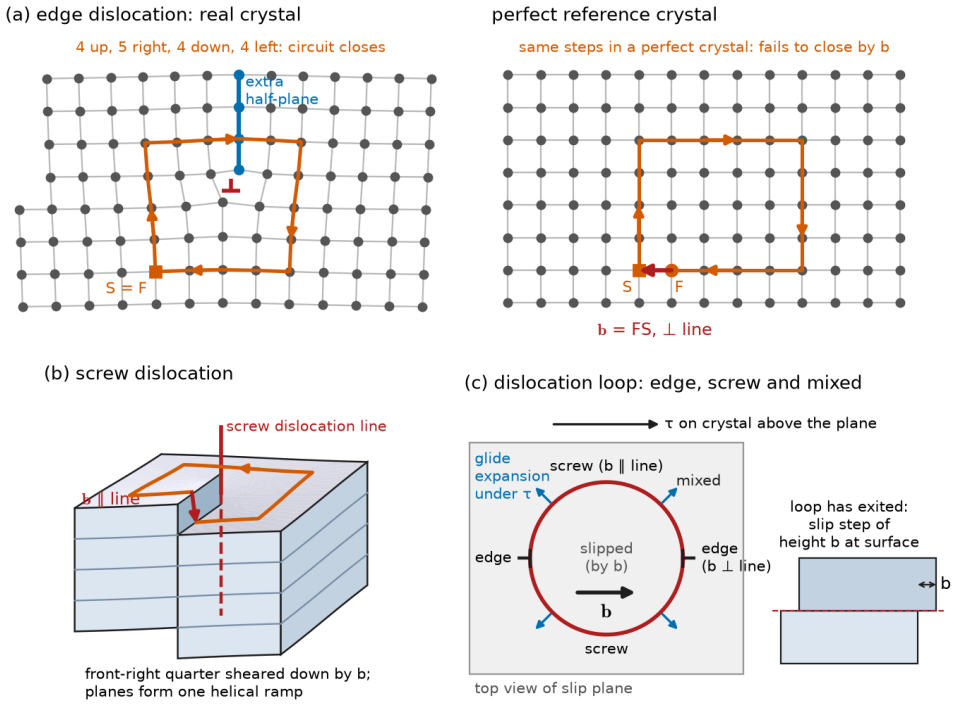

Figure 5.3 — Edge, screw and mixed dislocations with Burgers circuits.

How many dislocations?

The **dislocation density** ρ is the total line length per unit volume (m m^{-3} = m^{-2}). Typical values span ten orders of magnitude:

Material and condition	Dislocation density
Dislocation-free silicon wafer	~0
Well-annealed metal	10^{10}–10^{12} m^{-2}
As-cast or hot-rolled metal	10^{12}–10^{13} m^{-2}
Heavily cold-worked metal, martensite	10^{15}–10^{16} m^{-2}
Additively manufactured stainless steel (as built)	10^{14}–10^{15} m^{-2}
GaN epitaxial layer on sapphire	10^{12}–10^{13} m^{-2} (10^{8}–10^{9} cm^{-2})

Dislocation densities enter the flow stress through the Taylor relation $\tau \propto Gb\sqrt{\rho}$ (Chapter 13); they are measured by TEM, by etch pits and, as Chapter 4 showed, by X-ray peak broadening.

> **Status of the principle: dislocation line energy** $\approx \frac{1}{2}Gb^2$
> *Approximate.* It follows from the elastic field around the line with the logarithmic factor absorbed into the constant; good to within about a factor of two, and enough to show that dislocations are never equilibrium defects (several eV per atomic length of line, with negligible entropy gain).

5.4 Planar defects

Grain boundaries

Most engineering solids are **polycrystals**: aggregates of grains, each a single crystal, joined at **grain boundaries** where the lattice orientation changes abruptly. A boundary is specified by five macroscopic degrees of freedom — three for the misorientation (an axis and an angle) and two for the orientation of the boundary plane — and its structure and energy depend on all five.

Low-angle boundaries. If two grains differ by a small rotation θ, the boundary can be built from an array of dislocations. A symmetric tilt boundary is a wall of edge dislocations of spacing

$$D = \frac{b}{\theta}$$

For aluminium ($b = 0.286$ nm) and $\theta = 2^\circ = 0.0349$ rad, $D = 8.2$ nm; at 0.5° it is 33 nm. Low-angle boundaries form when dislocations rearrange into walls during recovery (subgrains, Chapter 10) and in hot-worked metals and directionally solidified castings. Their energy, from the elastic energy of the dislocation array, follows the **Read–Shockley** relation

$$\gamma = \gamma_0\,\theta\,(A - \ln\theta)$$

which rises steeply from zero, flattens out and, beyond about 15°, merges into the high-angle value as the dislocation cores overlap.

> **Status of the principle: Read–Shockley energy of low-angle boundaries**
> *Derived.* It follows from linear elasticity of a dislocation wall; valid below about 10–15°, where the dislocation cores are separate, and not applicable to high-angle boundaries.

High-angle boundaries. Above about 15° the boundary is a disordered slab about 0.5–1 nm thick, with extra free volume and distorted bonds. Its energy is roughly constant, about a third of the surface energy: typically 0.3 J m^{-2} in aluminium, 0.6 in copper, 0.8 in iron and about 1 J m^{-2} in nickel. High-angle boundaries are fast diffusion paths (Chapter 8), sites for segregation, precipitation and corrosion (intergranular attack in sensitized stainless steel, Chapter 43), sources and sinks of vacancies, and obstacles to dislocations (Hall–Petch strengthening, Chapter 13).

Special boundaries and the coincidence site lattice. At certain misorientations a fraction of the lattice sites of the two grains coincide. If the two lattices are imagined to interpenetrate, the coinciding points form a **coincidence site lattice** (CSL),

and the boundary is labelled Σn, where $1/n$ is the fraction of sites in coincidence. Low-Σ boundaries pack well and often have low energy, low diffusivity and high resistance to segregation and corrosion. Real boundaries deviate slightly from exact coincidence; a boundary is usually counted as Σn if its deviation is below the **Brandon criterion** $\Delta\theta = 15^\circ/\sqrt{n}$, which accommodates the deviation by a network of secondary dislocations.

Boundary	Misorientation (cubic)	Brandon tolerance
$\Sigma 1$ (low-angle)	$< 15^\circ$ about any axis	—
$\Sigma 3$ (twin)	60° about $\langle 111 \rangle$	8.7°
$\Sigma 5$	36.87° about $\langle 100 \rangle$	6.7°
$\Sigma 7$	38.21° about $\langle 111 \rangle$	5.7°
$\Sigma 9$	38.94° about $\langle 110 \rangle$	5.0°
$\Sigma 11$	50.48° about $\langle 110 \rangle$	4.5°

> **Status of the principle: Brandon criterion**
> *Empirical.* A convention based on the spacing of secondary dislocations; it is widely used to classify boundaries measured by EBSD, but low Σ alone does not guarantee low energy, because the boundary-plane orientation also matters.

Grain-boundary engineering exploits these differences. Repeated cycles of light deformation and annealing in low-stacking-fault-energy alloys (nickel alloys, austenitic stainless steels, lead alloys) raise the fraction of $\Sigma 3^n$ boundaries ($\Sigma 3$, $\Sigma 9$, $\Sigma 27$) above 50–70 % and break up the connectivity of the remaining random boundaries. Intergranular corrosion, stress-corrosion cracking and creep cavitation, which need a continuous path of susceptible boundaries, are then strongly reduced. Lead–acid battery grids, nuclear steam-generator tubing and superalloy components have been produced this way.

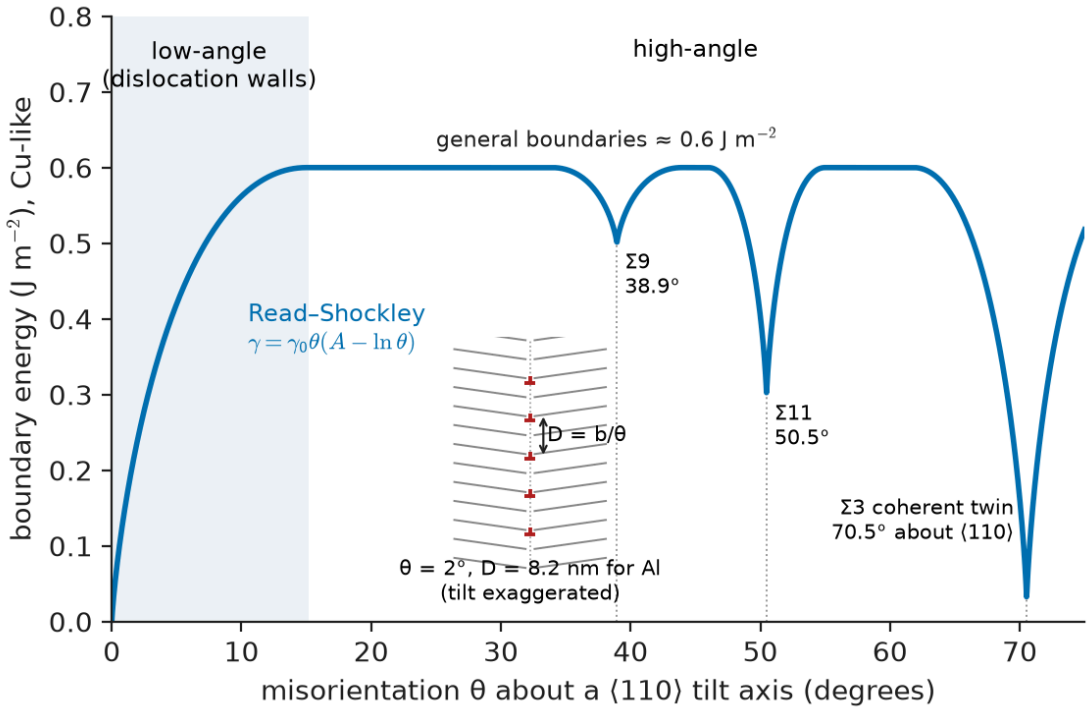

Figure 5.4 — Grain-boundary energy versus misorientation: Read–Shockley rise for low-angle boundaries, a plateau for general high-angle boundaries, and cusps at special CSL boundaries.

Twin boundaries

A **twin boundary** is a mirror plane across which the lattice is reflected. In FCC metals the coherent $\Sigma 3$ twin lies on $\{111\}$ and simply reverses the stacking sequence, …ABCABCBACBA…, with no broken nearest-neighbour bonds; its energy is tiny (about

0.02–0.04 J m^{-2} in copper, around 5 % of a general boundary). **Annealing twins** form during recrystallization and grain growth of low-stacking-fault-energy FCC metals (copper, brass, austenitic stainless steel, nickel alloys); they appear in micrographs as straight parallel bands crossing grains. **Deformation twins** form under stress where slip is difficult: in HCP metals (magnesium, titanium, zinc), in BCC iron at low temperature or high strain rate, and in TWIP steels, where a fine mesh of deformation twins subdivides grains during straining and gives exceptional work hardening (Chapter 15). Nanotwinned copper, electrodeposited with twin spacings of tens of nanometres, reaches strengths near 1 GPa while keeping nearly the conductivity of pure copper, because twins block dislocations but barely scatter electrons.

Stacking faults

In close-packed structures a local error in the stacking sequence of close-packed planes is a **stacking fault**: an intrinsic fault in FCC changes ...ABCABC... to ...ABCBCA..., inserting a thin slab of HCP-like stacking. It costs an energy per unit area, the **stacking-fault energy** γ_{SF}. Stacking faults are bounded by **partial dislocations**: a perfect FCC dislocation $\frac{a}{2}[\bar{1}10]$ can dissociate into two Shockley partials $\frac{a}{6}[\bar{2}11] + \frac{a}{6}[\bar{1}2\bar{1}]$, lowering its energy from $a^2/2$ to $2 \times a^2/6 = a^2/3$ (in units of $G/2$), at the price of a ribbon of fault between them. The equilibrium ribbon width balances the elastic repulsion of the partials against the fault energy; in isotropic elasticity it is roughly

$$d \approx \frac{G\, b_p^2}{2\pi\, \gamma_{\mathrm{SF}}}$$

where $b_p = a/\sqrt{6}$ is the partial Burgers vector (with a correction factor of order one that depends on the dislocation's character).

> **Status of the principle: partial-dislocation separation** $d \approx G b_p^2 / 2\pi\gamma_{\mathrm{SF}}$
> *Approximate.* It balances isotropic elastic repulsion against fault energy; good for comparing metals, but character-dependent factors and elastic anisotropy change d by tens of percent.

Worked example — How wide are the stacking faults? Using typical values:

Metal	G (GPa)	a (nm)	b_p (nm)	γ_{SF} (mJ m^{-2})	d (nm)
Aluminium	26	0.4050	0.165	166	0.7
Nickel	76	0.3524	0.144	125	2.0
Copper	48	0.3615	0.148	45	3.7
Silver	30	0.4086	0.167	16	8.3
Austenitic stainless steel (316)	77	0.359	0.147	about 20	13

For copper, $d = 48 \times 10^9 \times (0.1476 \times 10^{-9})^2 / (2\pi \times 0.045) = 3.7 \times 10^{-9}$ m, about 14 nearest-neighbour distances. In aluminium the partials are about 2.5 nearest-neighbour distances apart: effectively undissociated. The consequence is large. A widely dissociated dislocation must be constricted back to a perfect one before its screw segments can cross-slip onto another plane, which is hard when d is large. Low-SFE metals therefore cannot easily bypass obstacles by cross-slip; they store dislocations in planar arrays, work-harden strongly, twin readily and keep high strength after cold work. This is why austenitic stainless steel and TWIP steel work-harden so much more than aluminium, why aluminium recovers rapidly during hot working while copper and stainless steel recrystallise, and why stacking-fault energy is a design variable in steels (Chapters 12, 13 and 15).

Phase boundaries

When the two sides of an interface are different phases, the interface is a **phase boundary** (interphase boundary). Its structure is classified by how well the lattices match. If the two phases have lattice parameters a_α and a_β along the interface, the **misfit** is

$$\delta = \frac{a_\beta - a_\alpha}{a_\alpha}$$

- **Coherent**: lattice planes continue across the interface, and any small misfit is taken up by elastic strain. Interface energy is low, typically below 0.2 J m^{-2}, but coherency strain energy grows with particle volume. Examples: GP zones and θ″ plates in Al–Cu alloys, γ′ precipitates in nickel superalloys (misfit typically -0.1 to -0.5 %), Al_3Sc in aluminium.

- **Semicoherent**: misfit too large to be accommodated elastically is relieved by **misfit dislocations** at a spacing $S \approx b/|\delta|$. Interface energy is 0.2–0.5 J m^{-2}. For a γ/γ′ interface with $b = 0.254$ nm and $|\delta| = 0.3$ %, $S \approx 85$ nm; when γ′ particles grow beyond a few hundred nanometres, or after long creep exposure, such networks appear around them. The θ′ phase in Al–Cu, semicoherent on its broad faces, is another example.
- **Incoherent**: no lattice continuity; the interface resembles a high-angle grain boundary, with energy 0.5–1 J m^{-2} or more. Examples: equilibrium θ (Al_2Cu) particles, carbides on steel grain boundaries, oxide inclusions, fibre–matrix interfaces in metal-matrix composites.

The sequence coherent → semicoherent → incoherent is the natural life history of a precipitate as it grows: small particles minimise interface energy by staying coherent, large ones minimise strain energy by losing coherency. It explains the precipitation sequence of age-hardening alloys and why their strength peaks and then falls (Chapter 10). The same classification applies to epitaxial thin films on substrates (Chapter 49), where the misfit-dislocation spacing and the threading dislocations that accompany them set the quality of LED, power-device and solar-cell layers (Chapter 34).

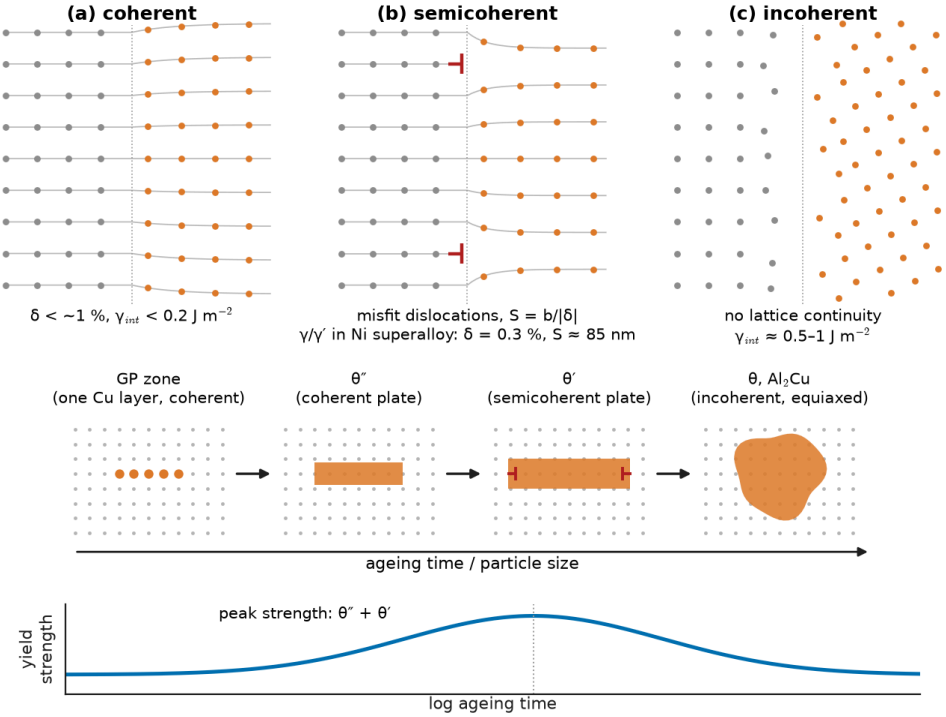

Figure 5.5 — Coherent, semicoherent and incoherent phase boundaries, and the Al–Cu precipitate sequence that passes through all three.

5.5 Volume defects: pores, inclusions and precipitates

Pores and voids

Pores are the most common volume defect in cast, sintered, welded and additively manufactured parts, and in concrete.

- **Castings** contain gas porosity (hydrogen rejected from solidifying aluminium, round pores 10–500 μm) and shrinkage porosity (irregular interdendritic cavities where liquid feeding failed). In cast aluminium suspension and engine components, fatigue cracks start at the largest pore near the surface; fatigue strength falls roughly with the square root of the pore's projected area (Chapters 14 and 46).
- **Sintered ceramics and powder metals** retain residual porosity between particles. Strength and stiffness fall steeply with porosity P; useful empirical fits are $\sigma \approx \sigma_0\, e^{-bP}$ with $b \approx 4$–7 for strength, and $E \approx E_0(1 - 1.9P + 0.9P^2)$ for modulus. For $b = 7$, 5 % porosity costs 30 % of the strength but only about 9 % of the stiffness, because strength is controlled by the largest flaw and stiffness by the average section.
- **Additive manufacturing** produces lack-of-fusion pores (irregular, between unmelted powder layers) and keyhole pores (spherical, trapped vapour), which dominate the fatigue scatter of printed titanium and nickel parts and are closed by hot isostatic pressing (Chapter 48).

- **Concrete** contains capillary pores from excess mixing water, which control strength and permeability, and deliberately **entrained air** — 4–7 % by volume of 10–100 μm bubbles — which gives freezing water room to expand and is essential for freeze–thaw resistance of bridge decks and pavements (Chapter 18).
- **Irradiation and creep voids.** Vacancies produced by neutron irradiation condense into voids that swell austenitic steels by several percent (Chapter 41); vacancies flowing to grain boundaries under creep nucleate cavities that link into creep cracks (Chapter 14).

> **Status of the principle: exponential porosity–strength relation**
> *Empirical.* A fit to data for ceramics and powder metals over porosities of roughly 0–30 %; the constant depends on pore shape and connectivity, and the relation has no meaning for isolated large flaws, which must be treated by fracture mechanics (Chapter 14).

Inclusions

Inclusions are particles of a foreign phase, mostly non-metallic, introduced during processing and usually unwanted. In steel they are oxides (alumina from deoxidation, silicates), sulfides (MnS) and nitrides (TiN). Their effects depend on size, shape and hardness:

- Hard, brittle oxides tens of micrometres across initiate fatigue cracks in rolling bearings, springs and gears; the revolution in bearing life since the 1970s came largely from vacuum degassing and ladle metallurgy that reduced oxygen content below 10 ppm.
- Soft MnS inclusions are elongated into stringers by rolling. They improve machinability (free-cutting steels add sulfur deliberately), but they make toughness and ductility through the plate thickness poor; welded structural plates loaded through the thickness can fail by **lamellar tearing** along the stringers, which is why offshore and bridge steels specify low sulfur and calcium treatment to keep sulfides globular.
- Hard-alpha inclusions in titanium (the opening example) and ceramic particles from crucibles in nickel superalloy powder are the classic rogue flaws of rotating aerospace parts; probabilistic damage-tolerance design now explicitly accounts for their rare presence.

Precipitates and dispersoids

Precipitates form within the material from a supersaturated solid solution, during heat treatment or service. They range from clusters of a few atoms to micrometre particles and can be coherent, semicoherent or incoherent (Section 5.4). Deliberately produced fine precipitates — θ″/θ′ in Al–Cu aerospace alloys, γ′ in turbine discs and blades, carbides in tempered martensite, NbC and VC in microalloyed steels — are the most powerful strengthening tool in metallurgy because they block dislocations (Chapter 13) while consuming little alloying. **Dispersoids** are fine particles that do not dissolve at all: Y–Ti–O nanoclusters in oxide-dispersion-strengthened steels for reactors, Al_2O_3 in dispersion-strengthened copper for welding electrodes, and Al–Mn–Fe dispersoids that control grain structure in aluminium alloys. Unwanted precipitates are equally important: chromium carbides on grain boundaries of sensitized stainless steel cause intergranular corrosion (Chapter 43), and σ phase in duplex steels and superalloys embrittles them.

5.6 Quantitative microstructure: stereology

The problem

Microstructures are three-dimensional, but they are almost always observed on two-dimensional sections (polished metallographic or ceramographic sections, SEM images, EBSD maps) or one-dimensional traverses. **Stereology** is the set of exact geometric-probability relations that recover three-dimensional quantities — volume fractions, interface areas, line lengths — from measurements on sections, without assuming particle shapes. Specifications for grain size, inclusion content, porosity and phase fraction in steels, aluminium alloys, ceramics, welds and AM parts are all written in stereological terms.

Notation and the fundamental relations

Stereology uses a compact notation: V_V is a volume per unit volume (volume fraction), S_V a surface area per unit volume, L_V a line length per unit volume; A_A an area fraction on a section, L_A a line length per unit area, P_A a number of points per unit area;

L_L a line fraction on a test line, P_L a number of intersections per unit test-line length; and P_P the fraction of test points falling in a phase. The basic relations, valid for isotropic, uniform random sections, are:

$$V_V = A_A = L_L = P_P$$

$$S_V = 2P_L, \qquad L_V = 2P_A$$

The first set (the Delesse–Rosiwal–Glagolev principle) says that the volume fraction of a phase equals its area fraction on a random section, its length fraction along a random line, and the fraction of random points that fall in it. The second set says that the area of interface per unit volume is twice the number of intersections per unit length of test line, and that the length of lines (dislocations, triple junctions) per unit volume is twice the number of their intersection points per unit area of section.

> **Status of the principle: fundamental stereological relations**
> *Derived.* They follow from geometric probability and are exact, without any assumption about particle shape, provided the sections are random (or the structure isotropic and uniform); for textured or elongated structures, measurements on several oriented sections are required.

The quantities that are *not* obtainable from sections without shape assumptions are numbers and sizes of particles: a random section cuts large particles more often than small ones and cuts each at less than its full diameter. Converting section-size distributions into particle-size distributions (the Saltykov or Schwartz–Saltykov methods) requires assuming a shape, usually spheres. For monosized spheres of diameter D_s, the mean intercept length is $\bar{L} = 2D_s/3$ and the mean section diameter is $\pi D_s/4$.

Grain size

For a single-phase polycrystal, the most robust grain-size measure is the **mean linear intercept** $\bar{L}$: lay test lines of total length L_T (corrected for magnification) on the section, count the number N of grain-boundary intersections, and

$$\bar{L} = \frac{L_T}{N} = \frac{1}{P_L}, \qquad S_V = \frac{2}{\bar{L}}$$

The ASTM grain-size number G, still the most common specification in industry, is related to the mean intercept in millimetres by

$$G = -6.6439 \log_{10} \bar{L} - 3.288$$

and to the number of grains per square millimetre of section, N_A, by $G = 3.3219 \log_{10} N_A - 2.954$. Larger G means finer grains: $G = 5$ corresponds to $\bar{L} \approx 56$ µm, $G = 8$ to 20 µm, $G = 10$ to 10 µm. Electron backscatter diffraction (EBSD) measures grain size directly from orientation maps, and also counts twins, low-angle boundaries and CSL boundaries separately — distinctions that matter, because whether twins are counted as grain boundaries changes the measured size by a factor of up to two in annealed copper and austenitic steel.

Material and condition	Typical mean intercept	ASTM G (approx.)
Cast aluminium, unrefined	0.5–2 mm	below 0
Hot-rolled structural steel (ferrite)	10–30 µm	7–10
Annealed austenitic stainless steel	30–100 µm	3.5–7
Microalloyed (HSLA) steel, controlled rolling	3–8 µm	10.5–13.5
Superalloy turbine disc (fine grain)	10–20 µm	8–10
Sintered alumina, dense	1–5 µm	12–16.5
Nanocrystalline electrodeposit	10–50 nm	—

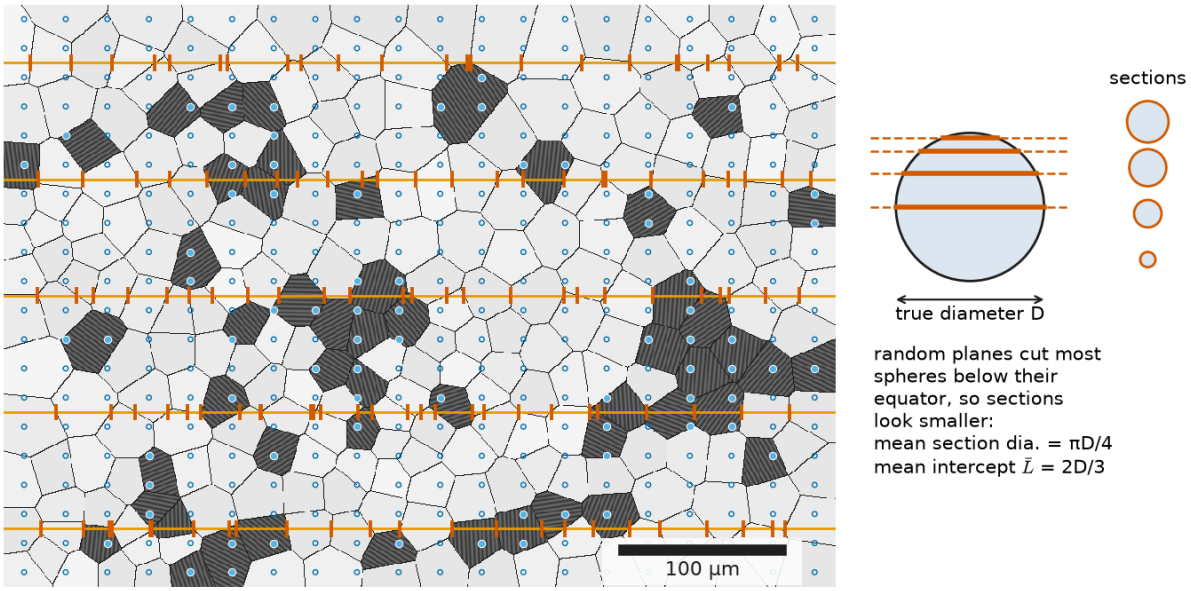

Figure 5.6 — Stereological measurement on a section: line intercepts for grain size and interface area, point counting for volume fraction.

Worked example — Grain size, interface area and phase fraction of a structural steel. A polished and etched section of a ferrite–pearlite steel is photographed at 200× magnification. Five test lines, each 100 mm long on the print, are laid across it: $L_T = 500\ \text{mm}/200 = 2.5\ \text{mm}$ on the specimen. They cross 125 grain boundaries.

$$P_L = \frac{125}{2.5\ \text{mm}} = 50\ \text{mm}^{-1}, \qquad \bar{L} = \frac{1}{50}\ \text{mm} = 20\ \mu\text{m}$$

The grain-boundary area per unit volume is $S_V = 2P_L = 100\ \text{mm}^{-1} = 1.0 \times 10^5\ \text{m}^2\,\text{m}^{-3}$, and the ASTM number is

$$G = -6.6439 \log_{10}(0.020) - 3.288 = -6.6439 \times (-1.699) - 3.288 = 8.0$$

A grid of 400 points is then laid on the same section; 72 fall on pearlite colonies. The pearlite volume fraction is $V_V = P_P = 72/400 = 0.18$. If the points are independent, the standard error is $\sqrt{P_P(1-P_P)/P_T} = \sqrt{0.18 \times 0.82/400} = 0.019$, so $V_V = 0.18 \pm 0.02$; systematic grids spaced wider than the pearlite colonies do somewhat better, and several fields must be counted to cover variation across the plate. A pearlite fraction of 0.18 corresponds, by the lever rule (Chapter 7), to roughly 0.15 wt % carbon, consistent with a structural grade such as S275 or A36.

The interface area has a direct physical meaning. With a high-angle boundary energy of about 0.5 J m^{-2}, the stored boundary energy is $1.0 \times 10^5 \times 0.5 = 5 \times 10^4$ J m^{-3}, or about 0.35 J mol^{-1} of iron — tiny compared with transformation energies, but enough to drive grain growth if the steel is overheated during welding or forging (Chapter 10).

> **Status of the principle: ASTM grain-size relations**
> *Empirical* as a convention (the number G is defined by a historical rule of grains per square inch at 100×), but its relation to $\bar{L}$ and N_A is exact geometry for equiaxed grains; it is meaningless for strongly elongated or bimodal grain structures, which must be characterised by size distributions.

5.7 How defects control properties: an overview

Almost every chapter that follows is, at bottom, about one or more of the defects introduced here. The table collects the main connections; each is developed in the chapter cited.

Property	Controlling defects	Mechanism	Engineering example	Chapter
Yield strength of metals	Dislocations, solutes, grain boundaries, precipitates	Obstacles to dislocation glide (Taylor, solid-solution, Hall–Petch, Orowan)	Cold-drawn wire, HSLA steel, 7075-T6 aluminium, superalloy discs	12, 13
Ductility and toughness	Inclusions, pores, grain-boundary films	Void nucleation, crack initiation, intergranular fracture	Lamellar tearing of plate; temper embrittlement of steel	14, 15
Fatigue life	Pores, inclusions, surface defects	Crack initiation at the largest flaw	Cast Al wheels, AM brackets, bearings	14, 40
Creep resistance	Vacancies, dislocations, grain boundaries, precipitates	Diffusion, climb, boundary sliding, particle pinning	Single-crystal blades (no boundaries), ODS steels	14, 16
Fracture strength of ceramics	Pores, surface flaws, large grains	Griffith flaws	Alumina substrates, zirconia hip heads, glass	14, 17
Diffusion and sintering	Vacancies, grain boundaries, dislocations	Lattice and short-circuit transport	Carburizing, powder metallurgy, solder-joint ageing	8, 38
Electrical conductivity of metals	Solutes, vacancies, dislocations, boundaries	Electron scattering (Matthiessen's rule)	Conductor alloys, interconnects	25
Ionic conductivity	Aliovalent dopants, oxygen vacancies	Vacancy hopping	YSZ fuel-cell electrolytes, lambda sensors	25, 32
Semiconductor behaviour	Dopants, deep-level impurities, dislocations	Carrier generation and recombination	Solar-cell lifetime, LED efficiency	26
Thermal conductivity	Point defects, boundaries, pores	Phonon scattering	Thermal-barrier coatings, thermoelectrics	24
Magnetic softness	Inclusions, precipitates, boundaries, stress	Domain-wall pinning	Electrical steel, amorphous cores	27
Corrosion resistance	Grain-boundary carbides, inclusions, segregation	Local anodes; breakdown of passive film	Sensitized stainless steel; MnS pits	34, 35
Radiation tolerance	Sinks: boundaries, interfaces, dispersoids	Recombination of Frenkel pairs	ODS and nanostructured reactor steels	33
Optical transparency	Pores, second phases, birefringent grains	Light scattering	Transparent alumina and spinel armour	28

Three general lessons emerge from the table.

First, **the same defect can help or harm**. Grain boundaries strengthen at low temperature and weaken at high temperature, where they slide and cavitate; dopant oxygen vacancies make YSZ a useful electrolyte and make $BaTiO_3$ capacitors degrade; precipitates harden aluminium alloys and, on grain boundaries, ruin stainless steel's corrosion resistance. The designer's question is never whether to have defects but which ones, how many, and where.

Second, **properties differ in their sensitivity to defect populations**. Elastic modulus, density and heat capacity are *structure-insensitive*: they change by at most a few percent with defect content. Yield strength, toughness, fatigue life, conductivity of semiconductors, creep rate, coercivity and corrosion rate are *structure-sensitive*: they can change by factors of ten to a million. Structure-sensitive properties are the ones processing can control, and the ones that show scatter.

Third, **the weakest-link defects are rare**. Strength-limiting flaws in ceramics, rogue inclusions in turbine discs and the largest pore in a casting are, by definition, the extremes of a distribution. Average microstructural measures from stereology cannot see them; extreme-value statistics, Weibull analysis and non-destructive inspection are needed (Chapters 11, 14 and 52).

What remains open

Open question	Where it is answered
How are the equilibrium concentrations of vacancies and ionic defects derived, and how do they depend on oxygen pressure?	Chapter 6
How do point defects carry atoms through a crystal, and how fast?	Chapter 8
How do precipitates nucleate, grow, change coherency and coarsen?	Chapters 9 and 10
How do recovery, recrystallization and grain growth change defect populations?	Chapter 10
What are the stress fields, forces, interactions and sources of dislocations?	Chapter 12
How do dislocations, solutes, boundaries and precipitates combine to set strength?	Chapter 13
How do pores and inclusions control fatigue and fracture?	Chapter 14
How do defects set electrical, thermal, magnetic and optical behaviour?	Chapters 24–36
How are defects created by irradiation, and how are they managed in reactors?	Chapter 41
How are defects imaged and measured: TEM, EBSD, atom probe, positron annihilation, tomography?	Chapters 52–54

Next: Part II begins with Chapter 6, which supplies the thermodynamics behind the phases, solutions, interfaces and point defects met so far.

Part II — Thermodynamics and Kinetics

Part I described what materials are made of: crystal structures, their symmetry, the diffraction that reveals them and the defects that every real crystal contains. It did not explain why a particular structure forms, why a steel quenched from 900 °C is hard while the same steel cooled slowly is soft, or why an aluminium alloy grows stronger for a week after heat treatment and then stops. Those questions have two halves. **Thermodynamics** says which state a material is trying to reach: which phases, of which compositions, in which proportions, are stable at a given temperature, pressure and chemical environment. **Kinetics** says how fast it gets there, by which path, and what it leaves behind if it is stopped on the way. Almost every microstructure an engineer uses — tempered martensite, age-hardened aluminium, a directionally solidified turbine blade, a sintered ceramic, a glass — is a kinetic compromise frozen somewhere between the starting state and equilibrium. Part II supplies the tools to predict and control that compromise.

By the end of this part you should understand:

- how Gibbs energy and chemical potential govern equilibrium in solids and solutions, and how free-energy–composition curves decide which phases coexist;
- how miscibility gaps, ordering, interfaces and point defects enter the same thermodynamic framework, and how Ellingham diagrams predict oxidation and reduction;
- how to read and use unary, binary, ternary and computed (CALPHAD) phase diagrams, including the Fe–C and key ceramic systems;
- how atoms move through solids by diffusion, how fast, along which paths, and how diffusion is exploited in carburizing, doping and homogenization;
- how new phases nucleate and grow, how solidification sets casting microstructures and segregation, and how rapid cooling produces metallic glasses;
- how precipitation, spinodal decomposition, pearlite, bainite and martensite form in the solid state, how recrystallization and grain growth reset microstructure, and how TTT and CCT diagrams turn all of this into heat-treatment practice.

Chapter	Central question	What it delivers
6 Thermodynamics of Materials	Which state of a solid or solution has the lowest Gibbs energy, and why?	Solution models, activity, common-tangent construction, miscibility gaps and spinodals, sublattice models, interface thermodynamics, defect equilibria, Ellingham diagrams
7 Phase Diagrams	Which phases are present, of what composition and in what amounts, at each temperature and composition?	Phase rule, unary to ternary diagrams, lever rule, Fe–C and ceramic diagrams, CALPHAD, use of diagrams in alloy and process design
8 Diffusion in Solids	How fast do atoms move in a solid, and by what mechanisms?	Fick's laws and solutions, atomic mechanisms, Arrhenius law, Kirkendall effect, short-circuit and ionic diffusion, carburizing and doping calculations
9 Nucleation, Growth and Solidification	How does a new phase start, how fast does it grow, and what structure does a casting inherit?	Nucleation theory, JMAK kinetics and TTT construction, constitutional supercooling, dendrites, Scheil segregation, eutectic and rapid solidification
10 Phase Transformations in Solids	How do solid-state transformations produce the microstructures used in engineering alloys?	Precipitation and age hardening, coarsening, spinodal decomposition, pearlite, bainite, martensite and shape memory, recrystallization and grain growth, TTT and CCT diagrams

The chapters build on one another in a deliberate order. Chapter 6 provides the free-energy curves from which Chapter 7 constructs phase diagrams. Phase diagrams say where a material is going; Chapter 8 says how quickly atoms can carry it there. Chapters 9 and 10 combine both — a thermodynamic driving force and a kinetic rate — into the theory of transformations, first from the liquid and then within the solid.

This part is the hinge of the book. It turns the structural vocabulary of Part I into microstructures that can be designed, and it underlies everything that follows. The strength of metals in Part III depends on the precipitates, grain sizes and martensite produced by the transformations of Chapters 9 and 10. The steels, aluminium, titanium and nickel alloys, ceramics and cements of Part IV are defined by their phase diagrams and heat treatments. Polymer blends and crystallization in Part V reuse the same free-energy and nucleation arguments. Corrosion and oxidation in Part IX start from Ellingham and Pourbaix diagrams and from diffusion through oxide scales. Casting, welding, additive manufacturing and thin-film growth in Part X are applied solidification and diffusion. And the CALPHAD, phase-field and kinetic Monte Carlo methods of Part XI are computational versions of the thermodynamics and kinetics introduced here. The laws of thermodynamics themselves are taken as known from the Advanced Chemistry book; Part II applies them to solids, solutions, interfaces and defects.

Chapter 6 — Thermodynamics of Materials

Part II · Thermodynamics and Kinetics

A duplex stainless-steel valve body in a refinery has run for twelve years at about 320 °C without trouble. During a shutdown it is struck by a dropped tool and cracks like cast iron. Its Charpy impact energy, more than 100 J when new, has fallen below 20 J. No corrosion, no fatigue and no overload explain it. The cause is inside the ferrite: at these temperatures iron–chromium solid solutions are not stable as a single phase. Over years the ferrite has separated, on a scale of a few nanometres, into iron-rich and chromium-rich regions, and the resulting internal stresses pin dislocations. The phenomenon, "475 °C embrittlement", is predicted by a free-energy curve with a hump in it — a miscibility gap — and it is why design codes cap the service temperature of duplex and ferritic stainless steels near 250–300 °C.

In the same plant, stainless-steel tube leaving a bright-annealing furnace at 1100 °C emerges dull green instead of mirror-bright: the hydrogen atmosphere was wet enough to oxidise chromium. The maximum permissible water content of the hydrogen can be read off an Ellingham diagram in a minute. Both problems, one taking twelve years and one taking twelve minutes, are questions about which state of a material has the lowest Gibbs energy.

> **How do Gibbs energy and chemical potential decide which phases, compositions and defect populations a material adopts?**
> **How do solution models, interfaces and gas atmospheres change that answer, and how is it read off a diagram?**

Concept	Engineering destination
Gibbs energy, chemical potential, equilibrium	Every phase diagram (Ch 7) and transformation (Ch 9–10)
Ideal, regular and real solutions; activity	Alloy design, slag–metal refining, braze and solder chemistry, carbon potential of furnace atmospheres
Gibbs–Duhem equation	Obtaining one component's activity from another's; consistency of thermodynamic databases (CALPHAD, Ch 7, 56)
Free-energy curves, common tangent	Phase boundaries, lever rule, driving forces for precipitation
Miscibility gaps and spinodals	475 °C embrittlement, Alnico and Fe–Cr–Co magnets, phase-separated glasses, polymer blends (Ch 20)
Sublattice models, ordering	Interstitial steels, intermetallics (Ni_3Al, TiAl), ordered superalloy phases
Surface and interface energy, Gibbs–Thomson, wetting	Coarsening (Ch 10), nanoparticle melting (Ch 38), brazing and soldering (Ch 47), liquid-phase sintering (Ch 17)
Thermodynamics of point defects	Vacancy equilibrium, oxide conductivity, fuel-cell and sensor ceramics (Ch 25, 40)
Ellingham diagrams	Extractive metallurgy, heat-treatment atmospheres, oxidation resistance (Ch 42), brazing atmospheres

Roadmap. Gibbs energy and equilibrium (6.1) → solutions and activity (6.2) → free-energy curves and the common tangent (6.3) → miscibility gaps and spinodals (6.4) → sublattices and ordering (6.5) → surfaces and interfaces (6.6) → point defects (6.7) → oxidation and reduction: Ellingham diagrams (6.8).

6.1 Gibbs energy, chemical potential and equilibrium

The criterion

Materials processing takes place at fixed temperature and pressure — in a furnace, a mould, a reactor, the atmosphere. Under those conditions the second law of thermodynamics takes a simple form: a closed system evolves spontaneously towards lower

Gibbs energy

$$G = H - TS$$

and is in equilibrium when G is a minimum. H is the enthalpy (for condensed phases at ordinary pressure, essentially the internal energy of bonding) and S the entropy (configurational and vibrational). At low temperature the enthalpy term dominates and the most strongly bonded, most ordered state wins; at high temperature the $-TS$ term rewards disorder: mixing, vacancies, liquids, gases. Every phase transformation in this part of the book is a competition between these two terms.

The equilibrium may be **stable** (the global minimum), **metastable** (a local minimum separated from the global one by a barrier: martensite, a glass, diamond at room temperature, a supersaturated aluminium alloy) or **unstable** (no barrier at all: the inside of a spinodal, Section 6.4). Engineering materials are very often metastable; thermodynamics tells us the direction in which they would change, and kinetics (Chapters 8–10) whether they do so in a millisecond or a millennium.

Chemical potential

For a system containing several components, the Gibbs energy depends on the amounts n_i of each:

$$dG = V\,dP - S\,dT + \sum_i \mu_i\,dn_i, \qquad \mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_{j\neq i}}$$

The **chemical potential** μ_i is the increase in G when one mole of component i is added at constant temperature, pressure and amounts of everything else. It plays the role for matter that temperature plays for heat: atoms of i flow from regions or phases of high μ_i to regions of low μ_i. Equilibrium between phases $\alpha, \beta, \ldots$ therefore requires

$$\mu_i^{\alpha} = \mu_i^{\beta} = \cdots \qquad \text{for every component } i$$

together with equal temperature and pressure. This condition, applied to free-energy curves, generates every phase diagram in Chapter 7. Chemical-potential gradients, not concentration gradients, are what drive diffusion (Chapter 8); the two usually point the same way, but not always, as "uphill" diffusion of carbon in welded dissimilar steels shows.

> **Status of the principle: minimum Gibbs energy at constant T and P**
> *Fundamental.* A consequence of the second law of thermodynamics for a closed system exchanging heat and work with surroundings at fixed temperature and pressure; it identifies the equilibrium state but says nothing about the rate of approach.

> **Boundary.**
> The laws of thermodynamics, the definitions of H, S and G, standard states and reaction equilibrium belong to the Advanced Chemistry book. This chapter recaps them only as far as needed and applies them to solids, solutions, interfaces and defects.

Pure substances: the G–T diagram

For a pure substance, $(\partial G/\partial T)_P = -S$, so G falls with temperature, and it falls faster for the phase with the higher entropy. The liquid, with more entropy than the solid, has a steeper $G(T)$ curve; the two curves cross at the melting point, where $\Delta G = \Delta H_f - T_m \Delta S_f = 0$, so $\Delta S_f = \Delta H_f / T_m$. For most metals $\Delta S_f \approx R$, about 8–10 J mol^{-1} K^{-1} (Richard's rule). Just below T_m, the driving force for solidification is approximately

$$\Delta G_v \approx \frac{\Delta H_f\,\Delta T}{T_m}$$

per mole, where ΔT is the undercooling. This simple linear result is the starting point of nucleation theory (Chapter 9). The same construction for two solid phases explains allotropy: iron is BCC below 912 °C, FCC between 912 and 1394 °C, and BCC again up to melting at 1538 °C, because the magnetic and vibrational entropy contributions of the BCC phase change with temperature differently from those of FCC.

6.2 Solutions: ideal, regular and real

Mixing

When two components A and B are mixed to form a single solid or liquid solution with mole fractions $x_A = 1 - x$ and $x_B = x$, the Gibbs energy per mole of solution is

$$G = x_A G_A^\circ + x_B G_B^\circ + \Delta G_{\text{mix}}, \qquad \Delta G_{\text{mix}} = \Delta H_{\text{mix}} - T\,\Delta S_{\text{mix}}$$

The first two terms are the weighted Gibbs energies of the pure components (a straight line in a plot of G against x); everything interesting is in ΔG_{mix}.

The ideal solution

If A and B atoms are distributed at random on the sites of one lattice and the bond energies do not depend on which atoms are neighbours, $\Delta H_{\text{mix}} = 0$ and the only change on mixing is the configurational entropy. Counting the arrangements of N_A and N_B atoms on N sites with Boltzmann's formula $S = k_B \ln W$ and Stirling's approximation gives

$$\Delta S_{\text{mix}}^{\text{id}} = -R\,(x_A \ln x_A + x_B \ln x_B), \qquad \Delta G_{\text{mix}}^{\text{id}} = RT\,(x_A \ln x_A + x_B \ln x_B)$$

$\Delta G_{\text{mix}}^{\text{id}}$ is always negative, with a maximum depth of $-RT\ln 2$ at $x = 0.5$ (-5.76 kJ mol^{-1} at 1000 K), and its slope is infinite at both ends. That infinite slope has a profound consequence: the first small addition of any solute always lowers the Gibbs energy, however unfavourable its bonding. **No substance is perfectly pure in equilibrium.** Ultra-pure silicon, zone-refined to parts per billion (Chapter 9), fights against this entropy term at every step.

> **Status of the principle: ideal entropy of mixing**
> *Derived.* It follows from statistical counting of random arrangements; it is exact for random mixing on equivalent sites and is the configurational part of every solution model, though real solutions also show short-range order, which lowers the entropy below the ideal value.

The regular solution

Real atoms do care about their neighbours. In the simplest model, the **regular solution**, the atoms still mix randomly (so the entropy is ideal), but each A–B bond has an energy different from the average of A–A and B–B bonds. Counting bonds in a random mixture with coordination number z gives

$$\Delta H_{\text{mix}} = \Omega\, x_A x_B, \qquad \Omega = zN_A\left[\varepsilon_{AB} - \tfrac{1}{2}(\varepsilon_{AA} + \varepsilon_{BB})\right]$$

so that

$$\Delta G_{\text{mix}} = \Omega\, x_A x_B + RT\,(x_A \ln x_A + x_B \ln x_B)$$

The **interaction parameter** Ω summarises the chemistry. If unlike bonds are stronger (more negative) than the average of like bonds, $\Omega < 0$: mixing is exothermic, the atoms prefer unlike neighbours and, at low temperature, may order (Section 6.5) or form compounds. If unlike bonds are weaker, $\Omega > 0$: mixing is endothermic and, at low enough temperature, the solution splits into two (Section 6.4). The dimensionless ratio Ω/RT decides the shape of the curve (Figure 6.1).

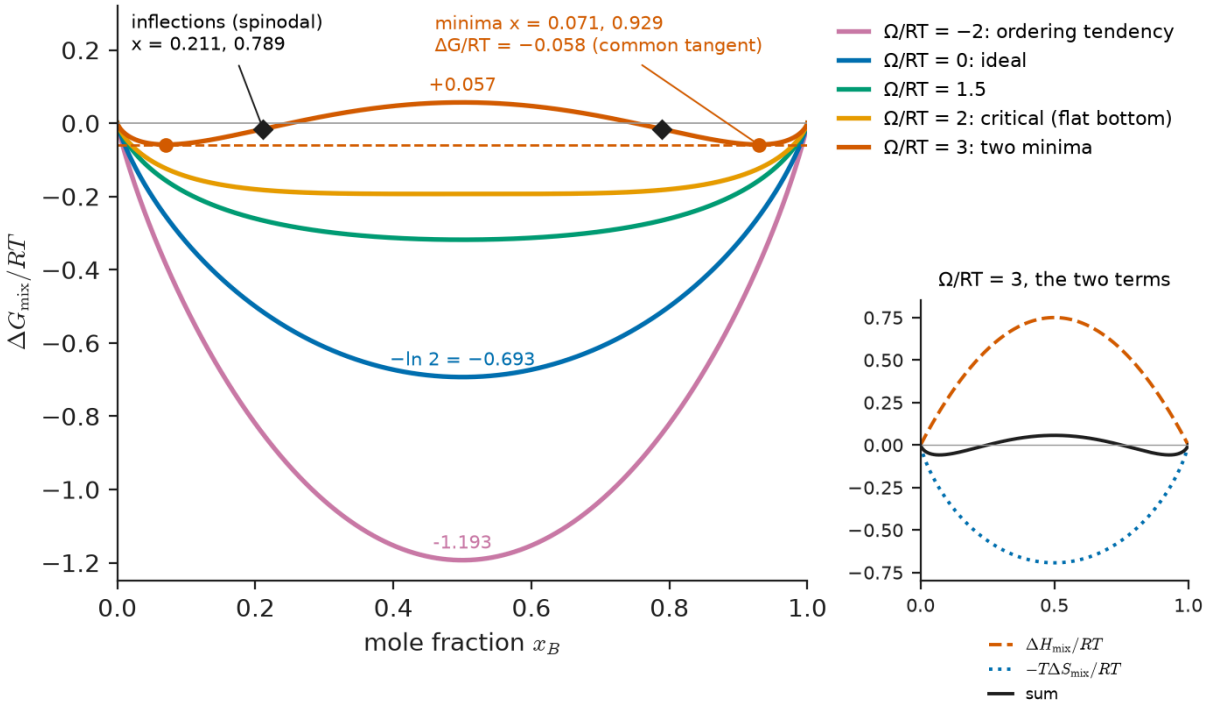

Figure 6.1 — Gibbs energy of mixing of a regular solution for several values of Ω/RT: a single minimum for Ω/RT < 2, two minima and a central hump for Ω/RT > 2.

> **Status of the principle: regular-solution model**
> *Approximate.* It assumes random mixing, pairwise nearest-neighbour bonds independent of composition and temperature, and no size mismatch; it captures miscibility gaps and ordering tendencies qualitatively and fits some liquid and solid alloys well, but real interaction parameters vary with composition and temperature.

Real solutions and excess functions

For real systems the deviation from ideality is collected into an **excess Gibbs energy**, $G^{\mathrm{ex}} = \Delta G_{\mathrm{mix}} - \Delta G^{\mathrm{id}}_{\mathrm{mix}}$. The regular solution has $G^{\mathrm{ex}} = \Omega x_A x_B$. Computational thermodynamics (CALPHAD, Chapter 7) generalises this with a **Redlich–Kister** polynomial,

$$G^{\mathrm{ex}} = x_A x_B \sum_{v=0}^{n} L_v \,(x_A - x_B)^v, \qquad L_v = a_v + b_v T$$

whose coefficients are fitted to measured phase boundaries, enthalpies of mixing and activities. Two to four terms usually suffice for a binary. The quasi-chemical model goes one step further and allows the distribution of atoms to depart from random (short-range order), which matters in strongly interacting liquids such as slags and molten salts.

> **Status of the principle: Redlich–Kister excess Gibbs energy**
> *Empirical.* A flexible polynomial fit with no physical derivation, used because it is thermodynamically consistent and extrapolates well into ternary and higher systems; its reliability is limited to the range of data fitted.

Activity

The chemical potential of a component in a solution is written relative to a standard state (usually the pure component in the same phase) as

$$\mu_i = \mu_i^{\circ} + RT \ln a_i, \qquad a_i = \gamma_i \, x_i$$

where a_i is the **activity** and γ_i the **activity coefficient**. For an ideal solution $\gamma_i = 1$ (**Raoult's law**). For a regular solution,

$$RT\ln\gamma_A = \Omega\,x_B^2, \qquad RT\ln\gamma_B = \Omega\,x_A^2$$

Positive Ω gives $\gamma > 1$ (the component is "less at home" in the solution than pure, and escapes more readily: higher vapour pressure, more reactive). Negative Ω gives $\gamma < 1$. In very dilute solutions γ_B approaches a constant γ_B^{∞} (**Henry's law**), which is why dilute solutes — carbon, nitrogen and hydrogen in steel, dopants in silicon, impurities in copper — are often described with a 1 wt % Henrian standard state in metallurgical practice.

Activity is the quantity that controls chemical exchange with the surroundings. The carbon activity of a furnace atmosphere sets the carbon content reached in a carburized gear (Chapters 8, 15); the activity of zinc in brass determines how much zinc boils off during casting and welding; the activity of oxygen in liquid steel determines how much aluminium is needed to deoxidise it; the activity of chromium in stainless steel enters the bright-annealing calculation of Section 6.8.

The Gibbs–Duhem equation

The chemical potentials of the components of a solution are not independent. Because $G = \sum n_i\mu_i$ and also $dG = \sum \mu_i\,dn_i$ at constant T and P, it follows that

$$\sum_i x_i\,d\mu_i = 0 \qquad \text{or, for a binary,} \qquad x_A\,d\ln a_A + x_B\,d\ln a_B = 0$$

If one component's activity is measured over a composition range — for example the vapour pressure of zinc over brass, or the EMF of a solid-electrolyte cell sensitive to one species — the activity of the other follows by integration. The equation also tests the consistency of tabulated data: any model in which the activities violate it is wrong.

> **Status of the principle: Gibbs–Duhem equation**
> *Derived.* A mathematical consequence of G being a homogeneous first-order function of the amounts of the components; it holds exactly for any phase in equilibrium at fixed T and P.

Worked example — Activities in a gold–nickel alloy. Gold and nickel are both FCC and mix completely at high temperature, but the large size difference (atomic radii 0.144 and 0.125 nm) makes the mixing enthalpy positive and opens a miscibility gap at low temperature, with a maximum near 810 °C (1083 K). Treat the system as a regular solution with that critical temperature; as Section 6.4 shows, $T_c = \Omega/2R$, so

$$\Omega = 2RT_c = 2 \times 8.314 \times 1083 = 18.0\ \text{kJ mol}^{-1}$$

At 1200 K, for an alloy with $x_{\text{Ni}} = 0.20$:

$$\ln\gamma_{\text{Ni}} = \frac{\Omega\,x_{\text{Au}}^2}{RT} = \frac{18\,008 \times 0.64}{8.314 \times 1200} = 1.155, \qquad \gamma_{\text{Ni}} = 3.17, \qquad a_{\text{Ni}} = 0.635$$

$$\ln\gamma_{\text{Au}} = \frac{18\,008 \times 0.04}{8.314 \times 1200} = 0.0722, \qquad \gamma_{\text{Au}} = 1.075, \qquad a_{\text{Au}} = 0.860$$

An alloy containing only 20 % nickel has a nickel activity of 0.64: the nickel behaves almost as if it were two-thirds pure, so it oxidises, evaporates and dissolves into a contacting phase far more readily than its mole fraction suggests. Differentiating the two expressions confirms that $x_{\text{Au}}\,d\ln a_{\text{Au}} + x_{\text{Ni}}\,d\ln a_{\text{Ni}} = 0$ at every composition, as Gibbs–Duhem requires. Gold–nickel alloys near Au–18Ni (wt %) are high-reliability brazes for aerospace and vacuum hardware; the positive deviation explains both why they melt at a minimum temperature lower than either metal and why their solidified microstructure can phase-separate during long service at intermediate temperatures. (The real miscibility gap is asymmetric, with its maximum near 70 at.% Ni; the symmetric regular model captures its size but not its shape, which is why CALPHAD uses Redlich–Kister terms.)

6.3 Free-energy–composition curves and the common tangent

Molar Gibbs energy and chemical potentials

A plot of molar Gibbs energy against composition at fixed temperature carries all the thermodynamic information of a binary system. Its most useful geometric property is the **tangent construction**: the tangent to the $G(x)$ curve at composition x intersects the vertical axes at $x_B = 0$ and $x_B = 1$ at the chemical potentials μ_A and μ_B of the solution of that composition.

$$\mu_A = G - x_B \frac{dG}{dx_B}, \qquad \mu_B = G + (1 - x_B)\frac{dG}{dx_B}$$

> **Status of the principle: tangent-intercept construction**
> *Derived.* It follows directly from the definition of partial molar quantities for a binary; it holds for any $G(x)$ curve, measured or modelled.

Two-phase equilibrium

Suppose two phases, α and β, each have their own $G(x)$ curve. An alloy of overall composition x_0 can exist as a single phase, or it can split into α of composition x_α and β of composition x_β, with mole fractions given by mass balance (the **lever rule**):

$$f_\beta = \frac{x_0 - x_\alpha}{x_\beta - x_\alpha}, \qquad f_\alpha = 1 - f_\beta$$

The Gibbs energy of the two-phase mixture lies on the straight line joining the points (x_α, G^α) and (x_β, G^β). The lowest such line is the **common tangent** to the two curves. At the tangent points, both phases have the same μ_A and the same μ_B, which is exactly the equilibrium condition of Section 6.1. Therefore:

- for $x_0 < x_\alpha$, single-phase α has the lowest G;
- for $x_\alpha < x_0 < x_\beta$, the equilibrium state is a mixture of α at x_α and β at x_β, in proportions given by the lever rule;
- for $x_0 > x_\beta$, single-phase β is stable.

The tangent compositions, traced as temperature changes, become the phase boundaries (solvus, solidus, liquidus) of a phase diagram (Chapter 7, Figure 6.2).

> **Status of the principle: common-tangent construction**
> *Derived.* It is the geometric statement of equal chemical potentials in coexisting phases; exact within whatever $G(x)$ curves are used, and the basis of every computed phase diagram.

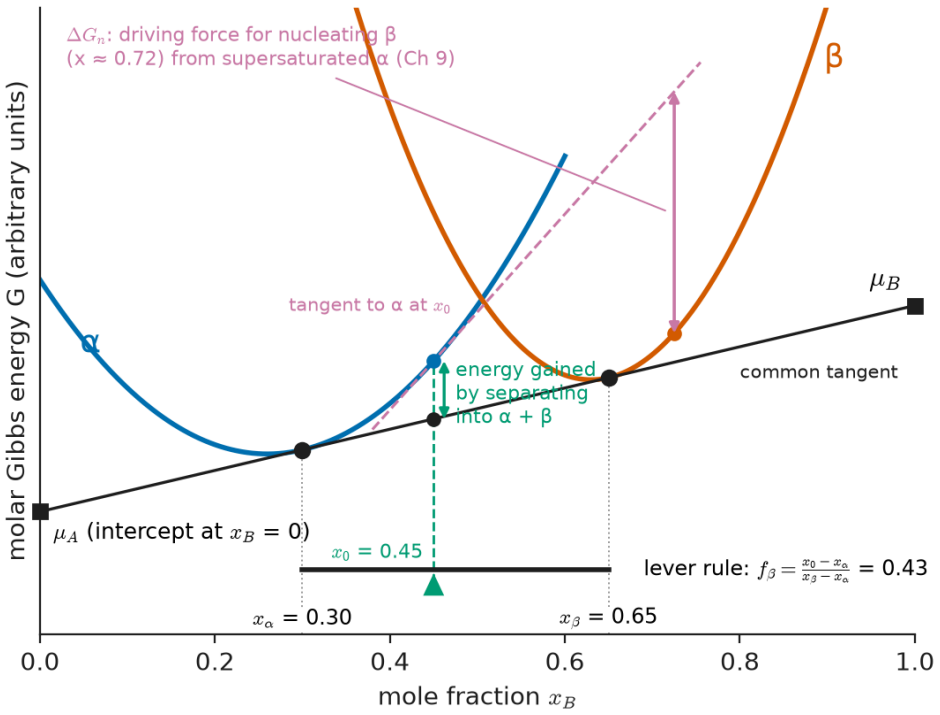

Figure 6.2 — Common-tangent construction: two phases coexist at the compositions where a single straight line is tangent to both free-energy curves; the tangent's intercepts give the chemical potentials.

Driving forces

The same diagram measures the **driving force** for a transformation. A supersaturated α solution of composition x_0 (an aluminium–copper alloy just quenched from the solution treatment, for example) has a higher G than the equilibrium α + β mixture. The vertical distance between its G and the common tangent is the total energy that precipitation can release; a parallel-tangent construction gives the driving force per mole of the first β nucleus. These quantities, typically a few hundred to a few thousand J mol^{-1}, set nucleation rates (Chapter 9) and precipitation sequences (Chapter 10). The same diagram also shows why metastable phases form first: a phase whose curve lies above the equilibrium β curve can still lie below the supersaturated α, so its precipitation is downhill, and if it nucleates more easily (coherent, low interface energy) it wins the race.

6.4 Miscibility gaps and spinodals

The gap

When $\Omega > 0$ the enthalpy term $\Omega x(1-x)$ has a maximum at $x = 0.5$, while the entropy term has a minimum there. At high temperature entropy wins and $G(x)$ has a single minimum. Below a **critical temperature** T_c the curve develops a central hump with two minima, and alloys between them lower their Gibbs energy by separating into two solutions of the same crystal structure but different compositions. The region of compositions and temperatures where this happens is a **miscibility gap**, bounded by the **binodal** (the locus of common-tangent points).

For a symmetric regular solution, the common tangent is horizontal, the binodal compositions are the minima of G, and setting $dG/dx = 0$ gives

$$\ln\frac{x}{1-x} = -\frac{\Omega\,(1-2x)}{RT}$$

The spinodal

Inside the gap there are two kinds of region. Where the curve is concave upward ($\partial^2 G/\partial x^2 > 0$) a homogeneous solution is metastable: a small fluctuation in composition *raises* G, and separation can only start by nucleation of a distinctly different composition. Where the curve is concave downward ($\partial^2 G/\partial x^2 < 0$) the solution is unstable: any small fluctuation lowers G and grows spontaneously, with no nucleation barrier. The boundary between them, $\partial^2 G/\partial x^2 = 0$, is the **spinodal**. For the regular solution,

$$\frac{\partial^2 G}{\partial x^2} = -2\Omega + \frac{RT}{x(1-x)} = 0 \quad\Rightarrow\quad x(1-x) = \frac{RT}{2\Omega}$$

The binodal and spinodal meet at the top of the gap, where $x = \tfrac{1}{2}$ and

$$T_c = \frac{\Omega}{2R}$$

> **Status of the principle: critical temperature $T_c = \Omega/2R$ of a regular solution**
> *Derived.* Exact within the regular-solution model; real miscibility gaps are often asymmetric and their critical temperatures are lowered by short-range order, and in crystalline solids the elastic energy of coherent composition waves depresses the observable ("coherent") spinodal below the chemical one.

Worked example — The miscibility gap of a regular solution with $\Omega = 18.0$ kJ mol^{-1}. With the Au–Ni value from Section 6.2, $T_c = 18\,008/(2\times 8.314) = 1083$ K. Solving the binodal equation numerically and the spinodal equation directly:

T (K)	Binodal x	Spinodal x
500	0.015 and 0.985	0.133 and 0.867
700	0.063 and 0.937	0.203 and 0.797
900	0.169 and 0.831	0.295 and 0.705
1000	0.268 and 0.732	0.362 and 0.638
1083	0.5	0.5

At 700 K, the spinodal composition follows from $x(1-x) = RT/2\Omega = 8.314 \times 700/36\,016 = 0.1616$, giving $x = 0.203$. Now consider two alloys aged at 700 K. An alloy with $x = 0.15$ lies between binodal and spinodal: it is metastable and decomposes only by nucleation and growth of discrete particles of composition near 0.94. An alloy with $x = 0.30$ lies inside the spinodal: it decomposes spontaneously by the growth of a fine, interconnected composition wave whose wavelength is set by the balance of chemical driving force and the gradient-energy penalty for sharp composition changes (the Cahn–Hilliard picture, Chapter 10). In both cases the final state is a mixture of $x = 0.063$ and $x = 0.937$, and by the lever rule the alloy with $x = 0.30$ ends with a fraction $(0.30 - 0.063)/(0.937 - 0.063) = 0.27$ of the nickel-rich phase.

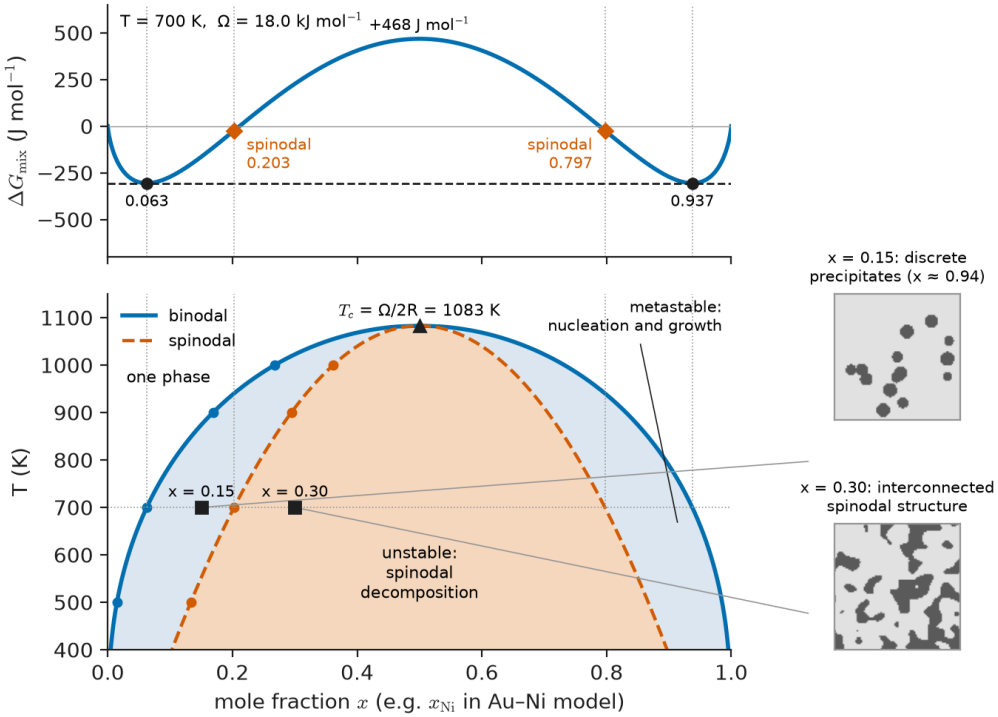

Figure 6.3 — Miscibility gap of a regular solution (Ω = 18.0 kJ mol⁻¹): free-energy curve at 700 K above, binodal and spinodal in temperature–composition space below.

Where miscibility gaps matter

System	Gap	Engineering consequence
Fe–Cr (ferrite)	Below about 500–550 °C, α (Fe-rich) + α′ (Cr-rich)	475 °C embrittlement of ferritic and duplex stainless steels and of cast duplex pipework in nuclear plants; service limits near 250–300 °C
Fe–Cr–Co, Alnico (Fe–Ni–Al–Co)	Spinodal decomposition into elongated magnetic and non-magnetic phases	Permanent magnets whose coercivity comes from the spinodal shape anisotropy (Ch 29)
Al–Zn	Gap below about 350 °C	Rapid natural ageing of Al–Zn alloys; Zn-rich zones in 7xxx aerospace alloys
Cu–Ni–Sn	Spinodal hardening	High-strength, high-conductivity spring and bearing alloys used in place of Cu–Be
Na_2O–B_2O_3–SiO_2 glass	Liquid–liquid immiscibility	Vycor process: leach the borate phase, sinter the silica network into near-pure silica glass
Polymer blends	Upper or lower critical solution temperatures	Phase-separated toughened plastics, blend compatibility (Ch 20)
Au–Ni, Au–Pt	Solid-solution gap	Ageing of braze joints and thermocouple alloys

6.5 Sublattice and compound-energy models; ordering

Why a single-lattice model is not enough

The regular solution places all atoms on one set of equivalent sites. Many important phases do not work that way. In austenite, iron occupies the FCC sites while carbon occupies some of the octahedral interstices, a second set of sites. In an ionic oxide, cations and anions live on separate sublattices, and vacancies may sit on either. In an ordered intermetallic such as Ni_3Al, nickel prefers the face-centre sites and aluminium the corners, but with some interchange. The **sublattice model** treats each set of sites separately.

A phase is described by a formula such as $(\mathrm{Fe})_1(\mathrm{C},\mathrm{Va})_1$ for austenite (iron on the metal sublattice; carbon or vacancy on the interstitial sublattice, one per metal atom) or $(\mathrm{Ni},\mathrm{Al})_3(\mathrm{Al},\mathrm{Ni})_1$ for γ′. Compositions are expressed as **site fractions** $y_i^{(s)}$, the fraction of sites on sublattice s occupied by species i. Mixing is assumed random within each sublattice, so the configurational entropy is a sum over sublattices:

$$S_{\mathrm{conf}} = -R \sum_s a_s \sum_i y_i^{(s)} \ln y_i^{(s)}$$

where a_s is the number of sites of sublattice s per formula unit.

The compound-energy formalism

For a two-sublattice phase $(A,B)_a(C,D)_c$, the reference Gibbs energy is a weighted average over the four "end-member compounds" AC, AD, BC, BD that result from filling each sublattice with one species:

$$G = \sum_{i,j} y_i^{(1)} y_j^{(2)} G^{\circ}_{i:j} - T S_{\mathrm{conf}} + G^{\mathrm{ex}}$$

This **compound-energy formalism** is the workhorse of CALPHAD databases. It handles interstitial solutions (the end member Fe:Va is pure iron and Fe:C a hypothetical "FeC" with every interstice filled), ionic phases with charged species and vacancies (where only charge-neutral combinations are physically accessible), and ordered intermetallics within one consistent framework. A practical consequence of the interstitial description is that the configurational entropy of carbon in austenite is counted per interstitial site, so the carbon activity rises steeply as the site fraction approaches saturation; this is why furnace carbon potentials must be controlled carefully near the Acm line in carburizing (Chapter 15).

> **Status of the principle: compound-energy formalism**
> *Approximate.* It assumes random mixing within each sublattice, with short-range order absorbed into fitted excess terms; it is highly successful for interstitial, ionic and ordered phases and is the standard model in thermodynamic databases.

Ordering

When $\Omega < 0$, unlike neighbours are favoured, and at low temperature the atoms arrange themselves into an **ordered** structure in which A and B occupy distinct sublattices: CuZn (β-brass, B2), Cu_3Au and Ni_3Al (L1$_2$), FeCo (B2), TiAl (L1$_0$). Order is measured by the **long-range order parameter** η, which is 1 for perfect order and 0 for random occupation of the two sublattices. In the simplest mean-field (Bragg–Williams) treatment of an equiatomic B2 alloy, the sublattice occupations are $y_A^{(1)} = \tfrac{1}{2}(1+\eta)$ and $y_A^{(2)} = \tfrac{1}{2}(1-\eta)$, and minimising G with respect to η gives the self-consistent condition

$$\eta = \tanh\!\left(\frac{T_c}{T}\,\eta\right)$$

where T_c is proportional to the ordering energy. Below T_c there is a non-zero solution; η falls continuously to zero at T_c — a second-order (continuous) transition, with no latent heat but a peak in heat capacity. Solving numerically: $\eta = 0.96$ at $T/T_c = 0.5$, 0.71 at 0.8, 0.53 at 0.9 and 0.17 at 0.99.

> **Status of the principle: Bragg–Williams model of ordering**
> *Approximate.* A mean-field theory that ignores correlations between neighbouring sites; it predicts the existence and continuous character of B2 ordering correctly but overestimates T_c and gives the wrong shape of $\eta(T)$ close to T_c, and it wrongly predicts continuous ordering for L1$_2$ structures, which in fact order discontinuously.

Ordered phases are central to high-temperature engineering. In β-brass, ordering at about 460 °C is detected by the superlattice reflections of Chapter 4. Ni_3Al remains ordered almost to its melting point, so a dislocation crossing it must create an **antiphase boundary** (Chapter 3); this is why γ′ precipitates strengthen superalloys up to about 1000 °C (Chapters 13 and 16). FeCo alloys order near 730 °C and become brittle, which limits their use in high-saturation electrical machines unless vanadium is added. Long-range order also changes physical properties: the electrical resistivity of Cu_3Au drops sharply when it orders, because the periodic arrangement no longer scatters electrons.

6.6 Surfaces and interfaces

Surface energy

Atoms at a surface have lost some of their neighbours. The excess free energy per unit area, the **surface energy** γ (J m^{-2}, numerically equal to the surface tension in N m^{-1} for liquids), is the work needed to create new surface reversibly. A broken-bond estimate is instructive. In copper each atom has 12 nearest neighbours, and the enthalpy of sublimation, 337 kJ mol^{-1}, breaks all of them (each bond shared by two atoms). An atom in a $\{111\}$ surface has lost 3 of its 12 bonds, so it carries an excess energy of $\tfrac{3}{12} \times 337\ \text{kJ mol}^{-1}/N_A = 9.3 \times 10^{-20}$ J. With 1.77×10^{19} atoms m^{-2} on that plane,

$$\gamma_{\{111\}} \approx 0.25 \times 5.60 \times 10^{-19}\ \text{J} \times 1.77 \times 10^{19}\ \text{m}^{-2} = 2.5\ \text{J m}^{-2}$$

The measured average for solid copper is about 1.8 J m^{-2}. The estimate is 35 % high, because real bonds are not pairwise and surface atoms relax, but it explains the trends: surface energy scales with cohesive energy, close-packed planes have the lowest γ (and dominate equilibrium crystal shapes), and surfaces of refractory metals are high-energy while those of polymers are low.

Material	Surface energy (J m^{-2}, typical)
Tungsten (solid)	about 2.8–3.3
Iron (solid)	about 2.4
Copper (solid)	about 1.8
Aluminium (solid)	about 1.1
Liquid iron at melting point	about 1.85
Liquid copper	about 1.3
Liquid aluminium	about 0.9
Alumina (solid)	about 0.9–1.5
Silica glass	about 0.3
Water at 20 °C	0.073
Polyethylene	about 0.03
PTFE	about 0.019
High-angle grain boundary in Cu (for comparison)	about 0.6

> **Status of the principle: broken-bond estimate of surface energy**
> *Approximate*. It counts nearest-neighbour bonds only; it gives the right order of magnitude and correct trends with cohesive energy and plane packing, but overestimates real values by 20–50 % because bonding is not pairwise and surfaces relax and reconstruct.

The Gibbs–Thomson effect

A curved interface raises the Gibbs energy of the material inside it. For a spherical particle of radius r, interface energy γ and molar volume V_m, the extra pressure is $2\gamma/r$ (Laplace) and the chemical potential is raised by

$$\Delta\mu = \frac{2\gamma V_m}{r}$$

Two engineering consequences follow.

Solubility of small particles. A small precipitate is in equilibrium with a higher solute concentration in the surrounding matrix than a large one:

$$c(r) = c_\infty \exp\!\left(\frac{2\gamma V_m}{rRT}\right)$$

Solute therefore diffuses from small particles to large ones, and the large grow at the expense of the small. This **Ostwald ripening** coarsens precipitates in aged aluminium alloys, γ′ in superalloys during creep, carbides in tempered steels and catalyst particles in exhaust converters (Chapter 10).

Melting-point depression. For a solid particle of radius r in its own melt, the melting point falls by

$$\Delta T_m \approx \frac{2\gamma_{sl}\, T_m\, V_m}{\Delta H_f\, r}$$

where γ_{sl} is the solid–liquid interface energy. This is the basis of low-temperature sintering of nanoparticle inks and pastes (silver and copper for printed electronics and die attach) and of the size-dependent properties of nanomaterials (Chapter 38). It also sets the critical nucleus size in solidification (Chapter 9), which is the same equation solved for r.

> **Status of the principle: Gibbs–Thomson relation**
> *Derived.* It follows from the Laplace pressure of a curved interface and equality of chemical potential; valid when the radius is large compared with atomic dimensions (above roughly 2 nm) and γ is independent of size and orientation, which fails for the smallest clusters and for strongly faceted particles.

Worked example — Small precipitates and small solder particles. (a) A precipitate in an aluminium alloy aged at 300 °C (573 K) has an interface energy of 0.2 J m^{-2} and $V_m = 1.0 \times 10^{-5}$ m^3 mol^{-1}. The exponent is $2\gamma V_m/(rRT) = 4.0 \times 10^{-6}/(r \times 4764)$. For $r = 5$ nm it is 0.168, so $c(r)/c_\infty = e^{0.168} = 1.18$: the matrix next to a 5 nm particle holds 18 % more solute in equilibrium than next to a very large one. For $r = 2$ nm the factor is 1.52; for $r = 20$ nm only 1.04. The large concentration differences around the smallest particles are why coarsening is fast at first and then slows (Chapter 10).

(b) Tin nanoparticles have been proposed for low-temperature solder pastes. With $\gamma_{sl} \approx 0.06$ J m^{-2}, $T_m = 505$ K, $V_m = 16.3 \times 10^{-6}$ m^3 mol^{-1} and $\Delta H_f = 7.03$ kJ mol^{-1}:

$$\Delta T_m = \frac{2 \times 0.06 \times 505 \times 16.3 \times 10^{-6}}{7030 \times r} = \frac{1.41 \times 10^{-7}\ \mathrm{K\,m}}{r}$$

For $r = 10$ nm, $\Delta T_m = 14$ K; for $r = 5$ nm, 28 K; for $r = 50$ nm, under 3 K. Useful depression needs particles below about 10 nm in radius, which are difficult to keep from oxidising and agglomerating — the reason such pastes remain niche.

Wetting and Young's equation

When a liquid drop rests on a solid, three interfaces meet at the contact line: solid–vapour (γ_{sv}), solid–liquid (γ_{sl}) and liquid–vapour (γ_{lv}). Balancing their tensions parallel to the solid surface gives **Young's equation** for the equilibrium contact angle θ:

$$\gamma_{sv} = \gamma_{sl} + \gamma_{lv}\cos\theta$$

A contact angle below 90° means the liquid **wets** the solid; $\theta \to 0$ means it spreads, which requires the spreading coefficient $S = \gamma_{sv} - \gamma_{sl} - \gamma_{lv}$ to be non-negative. Wetting decides whether solder or braze flows into a joint (Sn–Ag–Cu on clean copper, $\theta \approx 20$–40°; on oxidised copper, no wetting at all — hence fluxes and reducing atmospheres), whether a liquid metal infiltrates a ceramic preform to make a metal-matrix composite (aluminium on alumina is non-wetting below about 900 °C), whether paints, adhesives and coatings adhere (Chapter 45), and whether a surface is self-cleaning (water on PTFE or textured surfaces, $\theta > 110^\circ$).

> **Status of the principle: Young's equation**
> *Derived.* It follows from minimising total interfacial energy at fixed liquid volume; exact for an ideal, smooth, chemically homogeneous, non-reactive surface, and modified by roughness (Wenzel, Cassie–Baxter), contact-angle hysteresis and reaction at the interface, which in brazing and soldering usually dominates.

Dihedral angles: where a second phase sits

The same force balance at a grain boundary meeting a second phase gives the **dihedral angle** ϕ. Where a grain boundary of energy γ_{gb} meets a liquid (or a second solid phase) with interface energy γ_{sl},

$$\gamma_{gb} = 2\gamma_{sl}\cos\frac{\phi}{2}$$

If $\gamma_{gb}/\gamma_{sl} = 1$, $\phi = 120^\circ$ and the liquid forms isolated pockets at triple junctions. If the ratio is 1.8, $\phi = 52^\circ$ and the liquid forms channels along triple lines. If $\gamma_{gb} \geq 2\gamma_{sl}$, $\phi = 0$ and the liquid penetrates every boundary as a continuous film. This single ratio controls the microstructure of liquid-phase-sintered ceramics and hard metals (WC–Co, silicon nitride), the high-temperature strength of materials with grain-boundary glass, and the catastrophic **liquid-metal embrittlement** of aluminium by gallium, of steels by zinc during galvanizing and welding, and of nickel alloys by bismuth or lead.

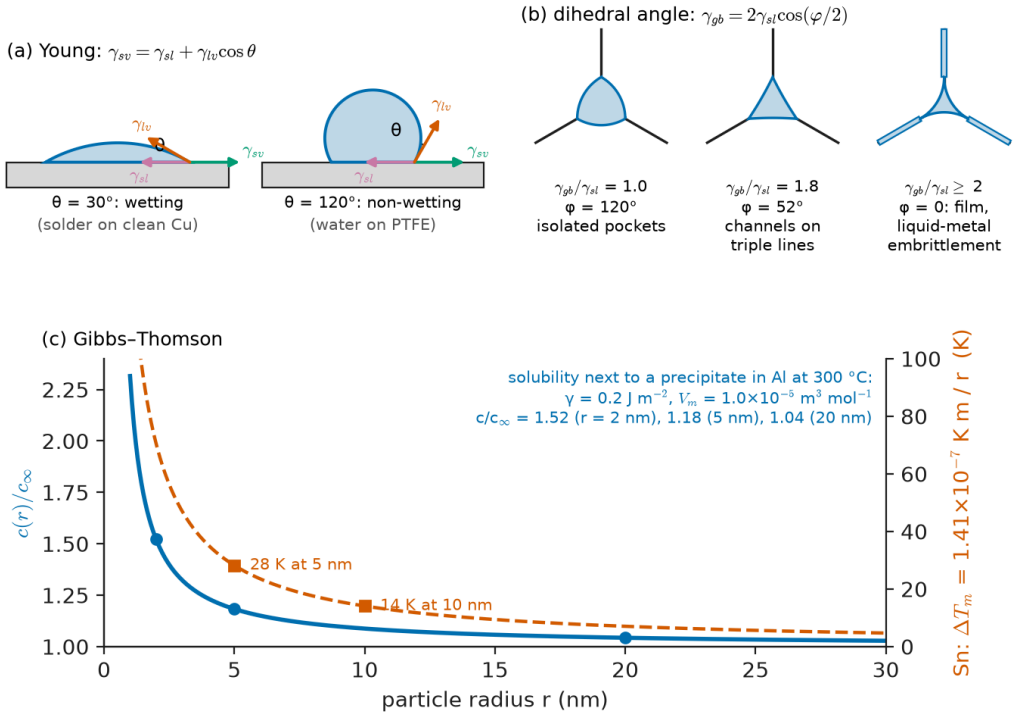

Figure 6.4 — Interface thermodynamics: Young's contact angle, the dihedral angle at a grain boundary, and the Gibbs–Thomson rise of solubility around small particles.

6.7 Thermodynamics of point defects

Vacancies as a component with a chemical potential

Chapter 5 used the equilibrium vacancy fraction; here is where it comes from. Introduce n vacancies into a crystal of N atoms. Each costs an enthalpy ΔH_v and adds a vibrational entropy ΔS_v (the atoms around a vacancy vibrate at lower frequencies). The configurational entropy is $k_B \ln\{(N+n)!/(N!\,n!)\}$. The Gibbs energy change is

$$\Delta G = n\,(\Delta H_v - T\Delta S_v) - k_B T \ln \frac{(N+n)!}{N!\,n!}$$

Using Stirling's approximation and setting $\partial \Delta G/\partial n = 0$ gives

$$\frac{n}{N+n} = c_v = \exp\left(\frac{\Delta S_v}{k_B}\right)\exp\left(-\frac{\Delta H_v}{k_B T}\right)$$

An equivalent and more general view treats vacancies as a component with a chemical potential,

$$\mu_v = \Delta G_v + k_B T \ln \frac{c_v}{1 - c_v}$$

Vacancies can be created and destroyed at sources and sinks (surfaces, grain boundaries, dislocations by climb), so at equilibrium no free energy is gained by creating more: $\mu_v = 0$, which reproduces the result. The formulation pays off away from equilibrium. A crystal quenched from high temperature holds a vacancy supersaturation with $\mu_v > 0$, which drives vacancies to sinks, makes dislocations climb and condenses vacancies into loops and voids. Under irradiation, both vacancies and interstitials are supersaturated, and their chemical potentials drive swelling (Chapter 41). Under creep stress, the vacancy chemical potential differs between boundaries in tension and compression, which drives diffusional creep (Chapter 14).

> **Status of the principle: equilibrium vacancy concentration from minimum G**
> *Derived.* Exact for dilute, non-interacting vacancies with a well-defined formation free energy; corrections are needed for divacancies near the melting point, for interactions with solutes, and in compounds where the vacancy chemical potential depends on the chemical potentials of the components.

Defect equilibria in compounds: the law of mass action

In an ionic compound, defects are created by reactions that also exchange atoms with the environment, and their concentrations are linked to the activities (partial pressures) of those atoms. Consider an oxide MO with oxygen vacancies, compensating electrons and holes. Two equilibria and one neutrality condition fix the defect populations:

$$\mathrm{O_O^\times} \rightleftharpoons \mathrm{V_O^{\bullet\bullet}} + 2e' + \tfrac{1}{2}\mathrm{O_2} \qquad K_R = [\mathrm{V_O^{\bullet\bullet}}]\, n^2\, p_{\mathrm{O_2}}^{1/2}$$

$$\mathrm{null} \rightleftharpoons e' + h^\bullet \qquad K_i = np$$

and, if the oxide is doped with an acceptor A' ($\mathrm{Y'_{Zr}}$ in zirconia, $\mathrm{Gd'_{Ce}}$ in ceria, $\mathrm{Mg'_{Al}}$ in alumina), electroneutrality

$$2[\mathrm{V_O^{\bullet\bullet}}] + p = n + [A']$$

Each mass-action constant is an exponential of the standard Gibbs energy of its reaction, $K = \exp(-\Delta G^\circ / RT)$, which is how defect formation energies enter. Solving the set in limiting regions, where one pair of defects dominates the neutrality condition, gives power laws in oxygen partial pressure:

Region	Neutrality approximation	Dependences
Strongly reducing	$n \approx 2[\mathrm{V_O^{\bullet\bullet}}]$	n and $[\mathrm{V_O^{\bullet\bullet}}] \propto p_{\mathrm{O_2}}^{-1/6}$
Intermediate, acceptor-doped	$2[\mathrm{V_O^{\bullet\bullet}}] \approx [A']$	$[\mathrm{V_O^{\bullet\bullet}}]$ constant; $n \propto p_{\mathrm{O_2}}^{-1/4}$; $p \propto p_{\mathrm{O_2}}^{+1/4}$
Strongly oxidizing	$p \approx [A']$ (holes compensate the acceptor)	p constant; $[\mathrm{V_O^{\bullet\bullet}}] \propto p_{\mathrm{O_2}}^{-1/2}$

A log–log plot of defect concentrations against $p_{\mathrm{O_2}}$, built from these straight-line segments, is a **Brouwer diagram** (Figure 6.5). It is the map engineers use to choose operating windows for oxide materials. A YSZ electrolyte must work in the middle region, where the vacancy concentration (and so the ionic conductivity) is fixed by doping and electronic conduction is negligible, across the oxygen pressures between air and fuel (10^{-21} to 0.21 atm at 800 °C). A ceria-based anode or catalyst is used precisely because it enters the reducing region easily and becomes a mixed conductor. A capacitor dielectric fired in a reducing atmosphere sits on the $-1/4$ or $-1/6$ line and becomes semiconducting unless acceptors are added. A UO_2 fuel pellet's oxygen-to-metal ratio, set by the oxygen potential during sintering and changed by fission, controls its thermal conductivity and creep.

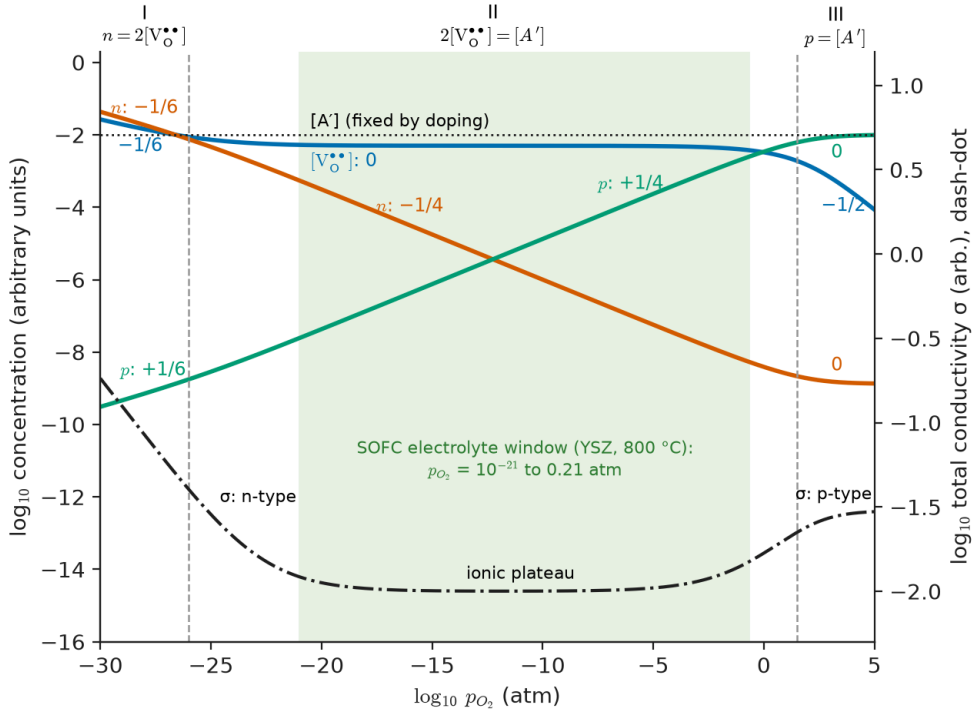

Figure 6.5 — Brouwer diagram for an acceptor-doped oxide: defect concentrations versus oxygen partial pressure, built from mass-action laws and limiting neutrality conditions.

Status of the principle: mass-action treatment of defect equilibria
Approximate. Exact thermodynamics for dilute, non-interacting defects with ideal (concentration-based) activities; it fails at high defect concentrations (above roughly 1 %), where defects associate into clusters and the simple power laws bend, as in heavily doped YSZ and strongly reduced ceria.

6.8 Ellingham diagrams for oxidation and reduction

The diagram

Whether a metal oxidises or its oxide can be reduced depends on the standard Gibbs energy of the oxidation reaction. To compare metals fairly, each reaction is written per mole of O_2:

$$\tfrac{2x}{y}\mathrm{M} + \mathrm{O_2} \;\rightarrow\; \tfrac{2}{y}\mathrm{M}_x\mathrm{O}_y, \qquad \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

Over temperature ranges without phase changes, ΔH° and ΔS° are nearly constant, so ΔG° is close to a straight line in T. Harold Ellingham plotted these lines for many oxides on one chart in 1944 (Figure 6.6). Its features follow from simple thermodynamics.

- **Slopes.** The slope is $-\Delta S^\circ$. Forming a solid oxide from a solid metal consumes a mole of gas, so $\Delta S^\circ \approx -$ (entropy of O_2) and almost all metal lines rise with roughly the same slope, about $+0.12$–0.22 kJ mol^{-1} K^{-1}. Kinks appear where the metal or oxide melts or boils (boiling of magnesium at 1090 °C steepens its line markedly).
- **Carbon.** $2\mathrm{C} + \mathrm{O_2} \rightarrow 2\mathrm{CO}$ *creates* a mole of gas, so its line *falls* with temperature. $\mathrm{C} + \mathrm{O_2} \rightarrow \mathrm{CO_2}$ conserves gas moles and is nearly flat. This is why carbon becomes a stronger reducing agent the hotter it gets — the foundation of iron- and steelmaking.
- **Position.** The lower a line, the more stable the oxide. Any metal can reduce the oxide of any metal whose line lies above its own at that temperature: aluminium reduces chromium oxide (the aluminothermic process), magnesium reduces titanium tetrachloride (the related chloride chart underlies the Kroll process for titanium sponge).
- **Equilibrium oxygen pressure.** With pure metal and oxide (activities 1), the reaction is at equilibrium when $\Delta G^\circ = RT\ln p_{\mathrm{O_2}}$. Each line therefore also gives the oxygen partial pressure below which the oxide is reduced; nomographic scales of $p_{\mathrm{O_2}}$, $\mathrm{H_2/H_2O}$ and $\mathrm{CO/CO_2}$ ratios are printed around the edge of the chart.

Status of the principle: Ellingham diagram
Derived from reaction thermodynamics ($\Delta G^\circ = -RT\ln K$) with *approximate* linear fits for $\Delta G^\circ(T)$. It assumes pure condensed phases at unit activity and says nothing about rates: aluminium, chromium and silicon are thermodynamically far less noble than iron, yet resist oxidation better because their thin oxide films are protective (Chapter 42).

Representative linear fits (typical handbook values, per mole of O_2, valid over roughly 300–1500 K between phase changes):

Reaction (per mol O_2)	ΔG° (kJ mol^{-1})	ΔG° at 1000 K	p_{O_2} at 1000 K (atm)
$4\mathrm{Cu} + \mathrm{O_2} \rightarrow 2\mathrm{Cu_2O}$	$-337 + 0.140\,T$	-197	5×10^{-11}
$2\mathrm{Ni} + \mathrm{O_2} \rightarrow 2\mathrm{NiO}$	$-471 + 0.172\,T$	-299	2×10^{-16}
$2\mathrm{Fe} + \mathrm{O_2} \rightarrow 2\mathrm{FeO}$	$-519 + 0.125\,T$	-394	3×10^{-21}
$2\mathrm{H_2} + \mathrm{O_2} \rightarrow 2\mathrm{H_2O(g)}$	$-493 + 0.109\,T$	-384	—
$2\mathrm{C} + \mathrm{O_2} \rightarrow 2\mathrm{CO}$	$-223 - 0.175\,T$	-399	—
$\mathrm{C} + \mathrm{O_2} \rightarrow \mathrm{CO_2}$	$-394 - 0.001\,T$	-395	—
$\tfrac{4}{3}\mathrm{Cr} + \mathrm{O_2} \rightarrow \tfrac{2}{3}\mathrm{Cr_2O_3}$	$-747 + 0.170\,T$	-577	8×10^{-31}
$\mathrm{Si} + \mathrm{O_2} \rightarrow \mathrm{SiO_2}$	$-902 + 0.174\,T$	-728	9×10^{-39}
$\mathrm{Ti} + \mathrm{O_2} \rightarrow \mathrm{TiO_2}$	$-941 + 0.178\,T$	-763	1×10^{-40}
$\tfrac{4}{3}\mathrm{Al} + \mathrm{O_2} \rightarrow \tfrac{2}{3}\mathrm{Al_2O_3}$	$-1117 + 0.208\,T$	-909	3×10^{-48}
$2\mathrm{Mg} + \mathrm{O_2} \rightarrow 2\mathrm{MgO}$	$-1202 + 0.216\,T$ (below 1363 K)	-986	3×10^{-52}

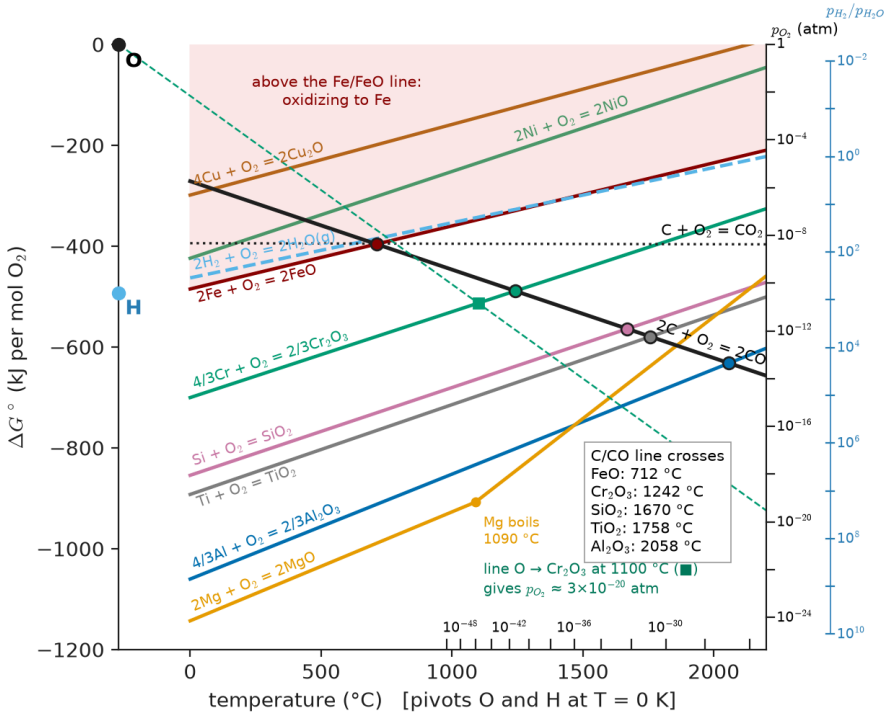

Figure 6.6 — Ellingham diagram for selected oxides: standard Gibbs energy of oxidation per mole of O2 versus temperature, with the C/CO line falling across the metal lines.

Reading the diagram

Three readings cover most uses.

Smelting. The C/CO line crosses the Fe/FeO line at

$$-519.2 + 0.1251\,T = -223.4 - 0.1753\,T \quad \Rightarrow \quad T = \frac{295.8}{0.3004} = 985\ \mathrm{K}\ (712\ ^\circ\mathrm{C})$$

Above this temperature, carbon at unit activity reduces FeO with CO as the product — the chemistry of the blast furnace. The C/CO line crosses Cr_2O_3 near 1240 °C, SiO_2 near 1670 °C, TiO_2 near 1760 °C and Al_2O_3 only above 2000 °C. Ferrochromium and silicon are therefore made by carbothermic reduction in electric-arc furnaces at very high temperature, while aluminium is too stable to reduce with carbon at practical temperatures (and forms carbides instead) and is produced by electrolysis of alumina

dissolved in molten cryolite — one of the most energy-intensive processes in materials production (Chapter 59). Titanium, likewise, would form carbides and dissolve oxygen, so it is made via the chloride route.

Protective atmospheres. For heat treatment, brazing and sintering, the atmosphere must hold the oxygen potential below the line of the most reactive alloying element that must stay metallic. Gas mixtures fix the oxygen potential: in H_2/H_2O, the ratio p_{H_2O}/p_{H_2} does; in CO/CO_2, the ratio p_{CO_2}/p_{CO}. Because water vapour content is measured as a **dew point**, furnace specifications are written as dew points.

Oxidation resistance. The metals whose lines lie lowest — Al, Si, Cr — form the most stable oxides, and if their oxides are dense and slow-growing they protect the metal underneath. Stainless steels (chromia formers), nickel superalloys and aluminide coatings (alumina formers), and silicon carbide and silicon nitride (silica formers) are designed around this principle. Selective oxidation also has a downside: in a steel or alloy heat-treated in an atmosphere that reduces iron oxide but not chromium, manganese or silicon oxides, those elements oxidise internally just below the surface, which affects galvanizing of advanced high-strength steels (Chapter 15).

Worked example — Bright annealing of stainless steel. Stainless tube is annealed at 1100 °C (1373 K) in hydrogen and must emerge without a chromium-oxide tint. What is the maximum allowable water vapour content?

The relevant reaction combines the Cr_2O_3 and H_2O lines (subtracting the first from the second):

$$\tfrac{2}{3}\,Cr_2O_3 + 2H_2 \;\rightleftharpoons\; \tfrac{4}{3}\,Cr + 2H_2O$$

At 1373 K, $\Delta G^\circ_{Cr_2O_3} = -746.8 + 0.1702 \times 1373 = -513.1$ kJ and $\Delta G^\circ_{H_2O} = -492.9 + 0.1094 \times 1373 = -342.7$ kJ per mole of O_2. The reduction reaction therefore has $\Delta G^\circ = -342.7 - (-513.1) = +170.4$ kJ, and

$$K = \frac{a_{Cr}^{4/3}\,(p_{H_2O}/p_{H_2})^2}{1} = \exp\!\left(-\frac{170\,400}{8.314 \times 1373}\right) = 3.3 \times 10^{-7}$$

For pure chromium ($a_{Cr} = 1$, pure Cr_2O_3), the threshold is $p_{H_2O}/p_{H_2} = \sqrt{3.3 \times 10^{-7}} = 5.7 \times 10^{-4}$. In 1 atm of hydrogen the water partial pressure must then stay below $5.7 \times 10^{-4} \times 101\,325 = 58$ Pa, a dew point of about -26 °C.

In a stainless steel with about 18 % Cr, the chromium activity is roughly 0.2. Since $(p_{H_2O}/p_{H_2})^2 = K/a_{Cr}^{4/3}$ at equilibrium, diluting the chromium *raises* the tolerable ratio by $0.2^{-2/3} = 2.9$, to 1.7×10^{-3} (170 Pa, dew point about -15 °C). Industrial lines nevertheless specify dew points of -40 °C or lower, for three reasons that the same diagram explains: silicon and manganese in the steel form oxides more stable than Cr_2O_3; the threshold ratio falls steeply with temperature — at 800 °C it is only 2.6×10^{-5} for pure chromium and tubes discolour if they cool slowly through the cooler zones of the furnace; and a margin is needed for leaks and moisture released from the charge.

The same chart answers related questions in other fields: the CO/CO_2 ratio needed to sinter iron powder compacts without oxidation, the oxygen pressure at which nickel anodes of solid-oxide fuel cells oxidise (about 5×10^{-11} atm at 1000 °C, from the Ni/NiO line), why copper can be brazed in mildly reducing "exothermic" gas while titanium must be brazed in high vacuum, and why silicon wafers can be oxidised in steam while their metal contacts are protected.

What remains open

Open question	Where it is answered
How are free-energy curves turned into complete binary and ternary phase diagrams, and how does CALPHAD compute them?	Chapter 7
How fast do atoms move down chemical-potential gradients, including uphill diffusion?	Chapter 8
How does the driving force from a common tangent translate into nucleation rates and growth?	Chapter 9
How does spinodal decomposition develop in time (Cahn–Hilliard), and how does Ostwald ripening proceed (LSW)?	Chapter 10
How do surface energy, adhesion and wetting control friction, coatings and joining?	Chapters 45 and 47
How do defect equilibria set ionic and electronic conductivity in ceramics?	Chapters 25 and 40
How do oxide scales grow, and why do some protect while others do not?	Chapter 42
How are Gibbs energies of solid phases calculated from first principles and stored in databases?	Chapters 55, 56 and 57
What is the environmental cost of the high-temperature reductions read from the Ellingham diagram?	Chapter 59

Next: Chapter 7 uses free-energy curves and the phase rule to build and read the phase diagrams on which alloy and process design depend.

Chapter 7 — Phase Diagrams

Part II · Thermodynamics and Kinetics

A foundry pours a batch of aluminium–copper aerospace castings and sends them for solution treatment. The furnace controller is set 15 °C higher than the specification, because the binary Al–Cu diagram in an old handbook shows that the alloy stays fully solid until well above 570 °C, and a hotter soak dissolves the copper faster. The castings come out with blistered surfaces and grain boundaries that look, under the microscope, like dried river beds: small pools of liquid had formed at the boundaries and then frozen. The alloy contained magnesium as well as copper, and the ternary Al–Cu–Mg system has a eutectic about 40 °C below the binary one, and some 65 °C below the alloy's binary solidus. The engineer read a correct diagram for the wrong system. A few kilometres away, a refractory supplier is solving a mirror-image problem: which alumina–silica brick will survive the roof of a glass-melting furnace at 1600 °C without sagging. And at a dental laboratory a technician is choosing how much yttria to put into a zirconia crown so that it stays tetragonal at body temperature, and tough.

All three questions are answered by the same kind of map. A phase diagram records, for each temperature, pressure and overall composition, which phases are present at equilibrium, what their compositions are and how much of each there is.

> **How many phases can coexist, and how do we read their compositions and amounts from a diagram? How does the diagram predict the microstructure that forms on cooling, and where does that prediction stop being reliable?**

Concept	Engineering destination
Gibbs phase rule	Counting what can be fixed in a furnace, a reactor or a crystal grower
Unary diagrams (water, iron, carbon, silica)	Pressure–temperature routes: diamond synthesis, refractory firing, steel heat treatment
Binary isomorphous systems	Cupronickel condenser tubes, Ni–Cu Monel, Si–Ge, solid-solution ceramics
Lever rule	Phase fractions in castings, steels, solders and refractories
Eutectic, peritectic, monotectic, eutectoid reactions	Casting alloys, solders, bearing bronzes, pearlite in steel
Intermediate compounds	Intermetallic strengthening phases, solder-joint interfaces, spinel refractories
Fe–Fe_3C diagram	Every steel and cast-iron heat treatment
Ceramic diagrams	Refractories, zirconia implants and thermal-barrier coatings, fuel-cell electrolytes
Ternary diagrams	Stainless steels, lead-free solders, slags and cement clinker
CALPHAD	Multicomponent alloy design, process windows, integrated computational materials engineering

Roadmap. The phase rule (7.1) → one-component diagrams (7.2) → two components with complete solubility (7.3) and the lever rule (7.4) → invariant reactions and compounds (7.5) → microstructure development on cooling (7.6) → the Fe–C diagram (7.7) → ceramic systems (7.8) → three components (7.9) → computed diagrams (7.10) → reading real diagrams for design (7.11).

7.1 The Gibbs phase rule

Phases, components and degrees of freedom

A **phase** is a region of material that is homogeneous in structure and composition and is separated from other regions by an interface: liquid aluminium, a grain of FCC austenite, a crystal of cementite, a glass. Many grains of the same crystal structure and composition are one phase. A **component** is one of the independent chemical species needed to describe the compositions of all phases: Cu and Ni in cupronickel, Fe and C in steel, Al_2O_3 and SiO_2 in a refractory (oxides are chosen as components when

the oxygen content follows from them). The **degrees of freedom** F are the intensive variables (temperature, pressure, phase compositions) that can be changed independently without changing the number of phases present.

Equilibrium between phases requires that temperature, pressure and the chemical potential of every component are equal in all phases (Chapter 6). Counting variables and constraints gives the **Gibbs phase rule**:

$$F = C - P + 2$$

where C is the number of components and P the number of phases. Each phase contributes $C - 1$ independent composition variables, so there are $P(C-1)$ composition variables plus T and p; equality of chemical potentials imposes $C(P-1)$ constraints. Subtracting gives the rule.

Most materials processing takes place at atmospheric pressure, and condensed phases are insensitive to modest pressure changes. With pressure fixed, the **condensed phase rule** is

$$F = C - P + 1$$

> **Status of the principle: Gibbs phase rule**
> *Fundamental.* It is a counting consequence of equilibrium thermodynamics; it fails only when the system is not at equilibrium, or when extra variables (surface energy of nanoparticles, electric or magnetic fields, stress) are significant and must be added to the count.

What the rule tells an engineer

In a binary alloy at fixed pressure ($C = 2$), a single-phase region has $F = 2$: temperature and composition can both be varied. A two-phase region has $F = 1$: once the temperature is chosen, the compositions of both phases are fixed (they are the ends of the tie line, Section 7.4). Three phases can coexist only at $F = 0$, an **invariant point** at a single temperature with three fixed compositions. That is why a eutectic alloy freezes at one temperature, with a flat arrest on a cooling curve, while an off-eutectic alloy freezes over a range. The rule also forbids four phases from coexisting in a binary at equilibrium; if a micrograph shows four, the specimen is not at equilibrium, which is common and informative (Section 7.6).

The rule says nothing about *which* phases form or at what temperatures; that is the job of the Gibbs-energy curves of Chapter 6. And it says nothing about the *amounts* of each phase; that is the job of mass balance (Section 7.4).

7.2 Unary diagrams

With one component, $F = 3 - P$: single-phase areas, two-phase lines, triple points (Figure 7.1). The axes are pressure and temperature, and the slope of each two-phase line is set by the **Clausius–Clapeyron equation**,

$$\frac{dp}{dT} = \frac{\Delta H}{T\,\Delta V}$$

where ΔH and ΔV are the molar enthalpy and volume changes of the transformation.

> **Status of the principle: Clausius–Clapeyron equation**
> *Derived.* It follows exactly from equality of the Gibbs energies of the two phases along the coexistence line; integrated forms that treat ΔH as constant or the vapour as an ideal gas are approximations.

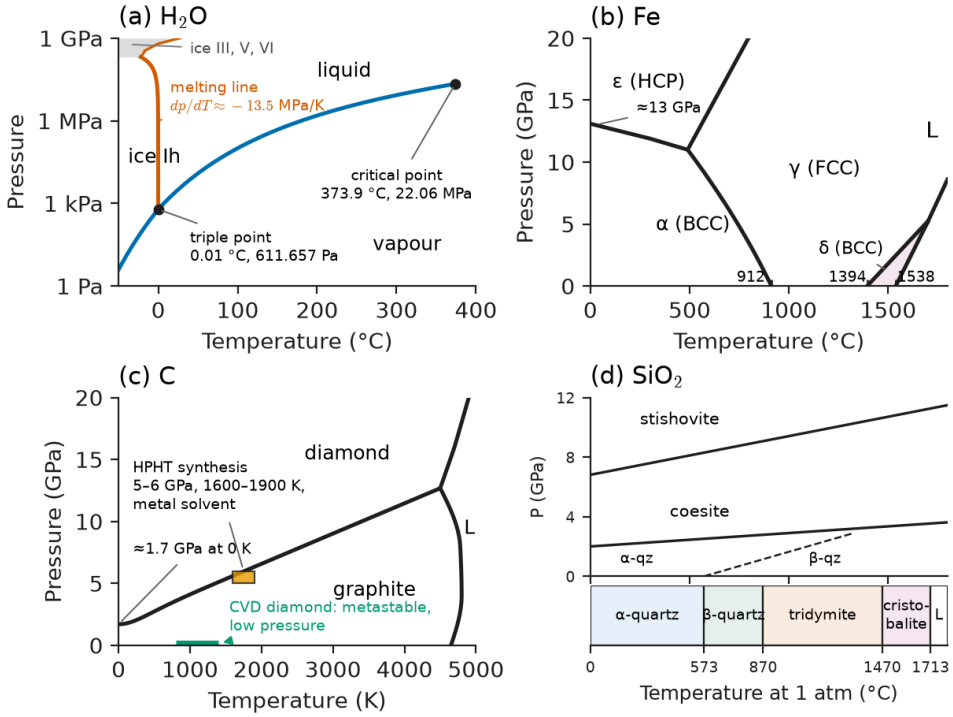

Figure 7.1 — Unary phase diagrams of water, iron, carbon and silica, showing the stable phases as functions of temperature and pressure.

Water

Water is the familiar example, and its oddity is instructive. Ice Ih is less dense than liquid water, so ΔV for melting is negative and the melting line slopes backwards. With $\Delta H_m = 6.01$ kJ/mol, molar volumes of 19.65 cm³/mol for ice and 18.02 cm³/mol for water at 0 °C, Clausius–Clapeyron gives

$$\frac{dp}{dT} = \frac{6010}{273.15 \times (-1.63 \times 10^{-6})} \approx -13.5 \text{ MPa/K}$$

Lowering the melting point by 1 K needs 13.5 MPa (about 135 atm). The same expansion on freezing is what bursts water pipes and spalls concrete in freeze–thaw cycles (Chapter 18). The triple point (0.01 °C, 611.657 Pa) is a defining point of the temperature scale and the operating window of freeze-drying of pharmaceuticals and foods: below 611 Pa, ice sublimes directly to vapour.

Iron

Pure iron at atmospheric pressure is BCC (α, ferrite) up to 912 °C, FCC (γ, austenite) from 912 °C to 1394 °C, BCC again (δ) up to the melting point at 1538 °C. The α→γ change on heating is accompanied by a volume contraction of about 1 %, because FCC packs more densely; dilatometers detect this to measure transformation temperatures in steels. At about 13 GPa at room temperature, iron transforms to hexagonal ε-iron; this matters in shock loading of armour steels and in the physics of the Earth's core. Every heat treatment of steel exploits the α/γ change, and the alloying additions of Section 7.7 and Chapter 15 move its temperatures.

Carbon

Graphite is the stable form of carbon at ambient pressure; diamond is metastable, persisting indefinitely because converting it requires breaking and reforming strong covalent bonds. The graphite–diamond equilibrium line starts near 1.7 GPa at 0 K and rises roughly linearly, reaching about 6 GPa near 1700 K. Industrial **high-pressure, high-temperature (HPHT)** synthesis of the abrasive diamond used in saw blades, drill bits and grinding wheels operates at 5–6 GPa and 1300–1600 °C, with molten iron, nickel or cobalt as a solvent–catalyst: the diagram says where diamond is stable, and the molten metal supplies a fast kinetic path. **Chemical-vapour-deposited (CVD)** diamond, used for heat spreaders and optical windows, grows at low pressure where it is not stable at all; atomic hydrogen etches graphitic carbon faster than diamond, so kinetics, not thermodynamics, select the product.

Silica

Silica (SiO_2) is the most important polymorphic ceramic. At atmospheric pressure the sequence is α-quartz → β-quartz at 573 °C → tridymite at 870 °C → cristobalite at 1470 °C → liquid at 1713 °C. The α–β quartz change is **displacive** (small rotations of SiO_4 tetrahedra) and therefore fast and unavoidable; it carries a volume change of about 0.8 %. Porcelain and whiteware bodies containing free quartz must be cooled slowly through 573 °C or they crack ("dunting"), and quartz-bearing concrete aggregate expands abruptly near the same temperature in a fire. The quartz → tridymite → cristobalite changes are **reconstructive** (bonds break) and so sluggish that impurities and kinetics decide what actually forms; silica refractory bricks are fired to convert as much quartz as possible to tridymite and cristobalite before service. High-pressure coesite and stishovite complete the diagram. Amorphous silica, fused silica glass, is outside the equilibrium diagram altogether (Chapter 17).

7.3 Binary isomorphous systems

Liquidus, solidus and the two-phase field

When two components have the same crystal structure, similar atomic sizes and electronegativities (the Hume-Rothery conditions of Chapter 5), they can be completely miscible in the solid as well as the liquid. The binary diagram, drawn at fixed pressure with temperature against composition, then has just three fields: liquid L, a single solid solution α, and a lens-shaped two-phase field L + α between them. The upper boundary of the lens is the **liquidus** (above it, all liquid); the lower boundary is the **solidus** (below it, all solid).

Cu–Ni is the textbook case: both FCC, radii 0.128 and 0.125 nm. The lens runs from the melting point of copper (1085 °C) to that of nickel (1455 °C). Other isomorphous systems include Si–Ge (used in heterojunction transistors and thermoelectrics), Au–Ag, Nb–Ti (the superconducting wire in MRI magnets), Al_2O_3–Cr_2O_3 (ruby is the dilute end) and NiO–MgO.

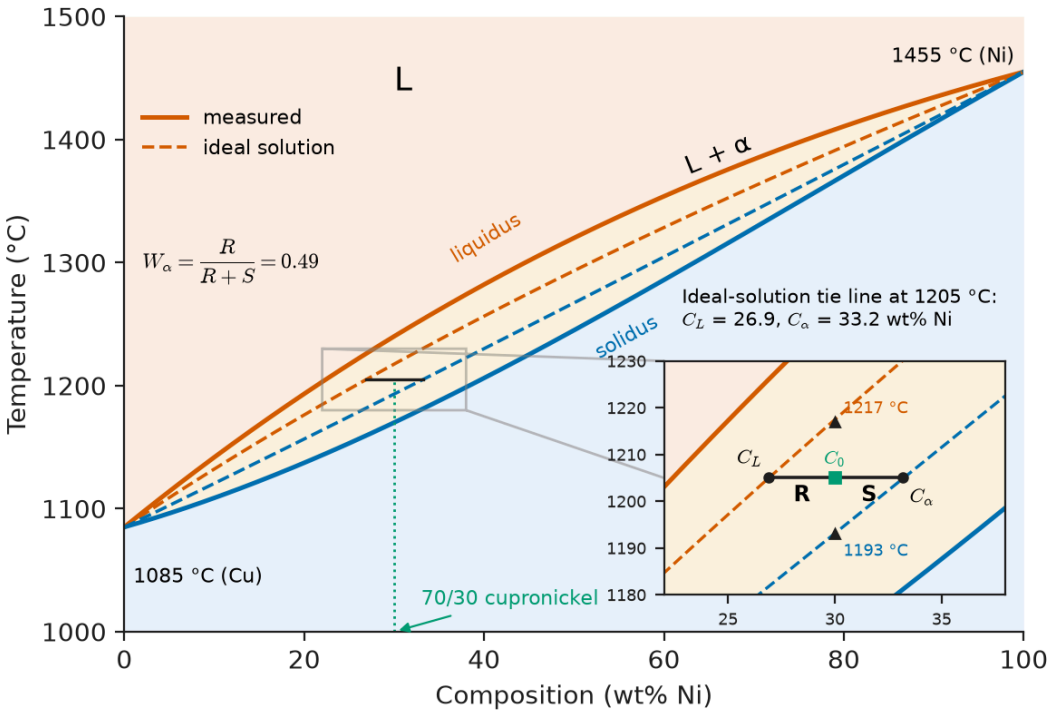

Figure 7.2 — The Cu–Ni isomorphous diagram with a tie line through Cu–30 wt% Ni, comparing the measured diagram with the ideal-solution prediction.

Where the lens comes from

Chapter 6 showed that the phase diagram is the projection of common tangents between Gibbs-energy curves. For two components that form ideal solutions in both liquid and solid, the condition that each component has the same chemical potential in both phases gives, for component i,

$$\ln \frac{x_i^S}{x_i^L} = \frac{\Delta H_{m,i}}{R}\left(\frac{1}{T} - \frac{1}{T_{m,i}}\right)$$

where x_i^S and x_i^L are mole fractions in solid and liquid and $\Delta H_{m,i}$, $T_{m,i}$ the enthalpy and temperature of melting of pure i. Writing this once for each component and using $x_A + x_B = 1$ in each phase gives two equations in two unknowns at each temperature: the solidus and liquidus compositions.

> **Status of the principle: Ideal-solution liquidus and solidus**
> *Approximate.* Exact for ideal liquid and solid solutions with temperature-independent enthalpies of melting; real systems deviate as their mixing enthalpies depart from zero, and when ΔH_{mix} is strongly positive the solid develops a miscibility gap.

The ideal model reproduces the shape of the Cu–Ni lens but makes it too narrow (Figure 7.2). Real Cu–Ni solid solutions have a slight positive mixing enthalpy, enough to produce a miscibility gap below about 350 °C, which is too sluggish to matter in practice. The gap between the dashed and solid curves is a measure of the non-ideality that CALPHAD assessments (Section 7.10) capture with excess Gibbs-energy terms.

Partition coefficient

At any temperature in the two-phase field, solid and liquid have different compositions. Their ratio, the **equilibrium partition coefficient**

$$k = \frac{C_S}{C_L}$$

measures how strongly a solute is rejected by the growing solid. In Cu–Ni, Ni is enriched in the solid ($k_{Ni} > 1$) because it raises the melting point; Cu is rejected ($k_{Cu} < 1$). For most impurities in metals and semiconductors $k < 1$, which is the basis of segregation, zone refining and dopant non-uniformity in crystal growth (Chapter 9).

7.4 The lever rule

Tie lines

In a two-phase field at temperature T, draw a horizontal line (an isotherm) through the overall alloy composition C_0 until it meets the phase boundaries on either side. This **tie line** connects the compositions of the two coexisting phases. Its ends give C_L and C_α; moving C_0 along the tie line changes only the amounts of the phases, not their compositions.

Mass balance

Let W_L and W_α be the mass fractions of liquid and solid. Conservation of total mass and of solute gives

$$W_L + W_\alpha = 1, \qquad W_L C_L + W_\alpha C_\alpha = C_0$$

so that

$$W_\alpha = \frac{C_0 - C_L}{C_\alpha - C_L}, \qquad W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$$

This is the **lever rule**: the fraction of a phase is the length of the tie-line segment on the *opposite* side of C_0, divided by the whole tie line. If compositions are in weight percent, the fractions are mass fractions; with atomic percent, they are mole fractions. Converting to volume fractions, which is what an image of a microstructure shows, requires the phase densities.

> **Status of the principle: Lever rule**
> *Derived.* It is pure mass conservation applied to the tie-line compositions; it holds whenever the phases have the compositions the tie line gives, which requires equilibrium (or at least local equilibrium) within each phase.

Worked example — Freezing a cupronickel condenser tube. Cu–30 wt% Ni (70/30 cupronickel) is used for seawater condenser and heat-exchanger tubes in ships and power stations because of its resistance to biofouling and erosion–corrosion. Use the ideal-solution model with $T_{m,Cu} = 1358$ K, $\Delta H_{m,Cu} = 13.26$ kJ/mol, $T_{m,Ni} = 1728$ K, $\Delta H_{m,Ni} = 17.48$ kJ/mol to find the phases at 1205 °C.

Partition ratios. At $T = 1478$ K,

$$\frac{x^S_{Cu}}{x^L_{Cu}} = \exp\left[\frac{13260}{8.314}\left(\frac{1}{1478} - \frac{1}{1358}\right)\right] = 0.909, \qquad \frac{x^S_{Ni}}{x^L_{Ni}} = \exp\left[\frac{17480}{8.314}\left(\frac{1}{1478} - \frac{1}{1728}\right)\right] = 1.228$$

Compositions. With $x^S_{Ni} = 1.228\,x^L_{Ni}$ and $1 - x^S_{Ni} = 0.909\,(1 - x^L_{Ni})$, solving gives $x^L_{Ni} = 0.285$ and $x^S_{Ni} = 0.350$, which with molar masses 63.55 (Cu) and 58.69 g/mol (Ni) are $C_L = 26.9$ wt% Ni and $C_\alpha = 33.2$ wt% Ni.

Amounts.

$$W_\alpha = \frac{30 - 26.9}{33.2 - 26.9} = 0.49, \qquad W_L = 0.51$$

At 1205 °C the alloy is about half solid, and the solid is 3 wt% richer in Ni than the alloy. The ideal model puts the liquidus of this alloy at 1217 °C and the solidus at 1193 °C, a freezing range of 24 °C; the measured range is closer to 1240–1170 °C, about 70 °C. The model explains the diagram; it cannot set a furnace. Section 7.6 shows what that composition difference does to the microstructure of a real casting.

7.5 Invariant reactions and intermediate phases

Limited solubility and the eutectic

When the components differ in crystal structure or atomic size, solid solubility is limited. The simplest result is the **eutectic** diagram: two terminal solid solutions α and β, each with a solvus beyond which the solute no longer dissolves, and two liquidus lines falling from the melting points of the pure components to meet at the eutectic point. At the eutectic temperature T_E three phases coexist:

$$L \;\longrightarrow\; \alpha + \beta \qquad \text{(on cooling, at } T_E\text{)}$$

Al–Si, the basis of most aluminium casting alloys, is a nearly ideal example. Aluminium melts at 660 °C; silicon at 1414 °C. The eutectic lies at 577 °C and 12.6 wt% Si. Silicon dissolves in solid aluminium up to 1.65 wt% at 577 °C (falling to almost zero at room temperature); aluminium is practically insoluble in silicon, so β is essentially pure Si. The eutectic composition melts at a temperature 83 °C below pure Al and has a narrow freezing range, the two features that make Al–Si alloys flow well and fill thin moulds: engine blocks, cylinder heads, wheels, pump housings.

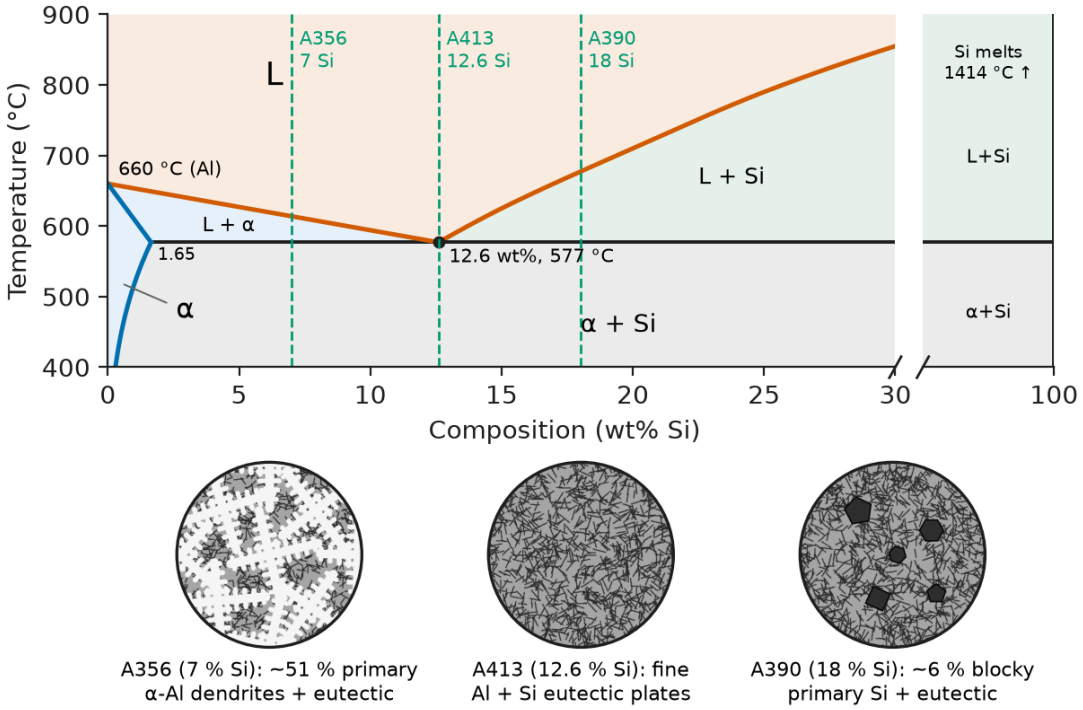

Figure 7.3 — The Al–Si eutectic diagram with microstructures of hypoeutectic, eutectic and hypereutectic alloys.

Other eutectics of engineering importance: Pb–Sn at 183 °C and 61.9 wt% Sn (the classic solder), Sn–Ag–Cu near 217 °C (lead-free solder), Ag–Cu at 779 °C and 28 wt% Cu (silver brazing alloy BAg-8), Au–Si at 363 °C (die attach in microelectronics), Fe–C at 1147 °C (cast irons), Al_2O_3–ZrO_2 near 1860 °C (eutectic ceramics for abrasives), and the CaO–SiO_2–Al_2O_3 eutectics that govern slag fluidity in steelmaking.

The other invariant reactions

Every invariant reaction in a binary involves three phases at a horizontal line. They are classified by which phase lies above or below the junction:

Reaction	On cooling	Engineering example	Typical temperature
Eutectic	$L \rightarrow \alpha + \beta$	Al–12.6Si casting alloys; Sn–Pb, Sn–Ag–Cu solders	577 °C; 183 °C; ~217 °C
Peritectic	$L + \alpha \rightarrow \beta$	Fe–C at 0.17 wt% C; Cu–Zn brasses; Pt–Ag	1495 °C; 902 °C (α-brass)
Monotectic	$L_1 \rightarrow \alpha + L_2$	Cu–Pb leaded bronze bearings; Al–Bi, Al–Pb	955 °C (Cu–Pb)
Eutectoid	$\gamma \rightarrow \alpha + \beta$	Fe–C pearlite; Cu–Al aluminium bronze	727 °C; ~565 °C
Peritectoid	$\alpha + \beta \rightarrow \gamma$	Formation of ordered intermetallics in Ni–Mo, Ti–Al	various

The **eutectoid** and **peritectoid** are the all-solid analogues of the eutectic and peritectic. Because they take place in the solid, where diffusion is slow, their products are fine and their kinetics matter as much as their thermodynamics (Chapter 10).

The **peritectic** reaction consumes a solid and the liquid to make a new solid. The product forms as a shell on the primary solid and then has to grow by diffusion through itself, so peritectic reactions rarely go to completion during normal cooling. In continuous casting of steel, grades with 0.09–0.17 wt% C pass through the Fe–C peritectic at 1495 °C; the δ→γ change contracts the newly solidified shell, which pulls away from the mould and causes surface depressions and cracks. Casters call these "peritectic grades" and run them with special mould powders and slower casting speeds.

In a **monotectic**, a liquid splits into a solid and a second liquid. Cu–Pb has a liquid miscibility gap: at 955 °C a liquid containing about 36 wt% Pb forms solid copper and a lead-rich liquid with about 87 wt% Pb, which finally freezes at 326 °C. The result is a copper matrix with lead pools, a self-lubricating bearing alloy used in engines and pumps. The two liquids differ in density and tend to separate under gravity, which makes such alloys hard to cast uniformly.

Intermediate phases and compounds

Many binaries contain phases with crystal structures different from either component. Those with a narrow composition range are **line compounds** or stoichiometric **intermetallics**; those with a wide range are **intermediate solid solutions**. They divide a complicated diagram into simpler sub-diagrams.

- **Congruently melting** compounds melt to a liquid of the same composition, at a temperature maximum on the liquidus. Mg_2Si (melting near 1085 °C) is the strengthening phase of 6xxx aluminium alloys; NiAl (near 1640 °C) is a coating phase on turbine blades; $MgAl_2O_4$ spinel is the ceramic analogue (Section 7.8). The diagram on either side of such a compound can be read as an independent binary.
- **Incongruently melting** compounds decompose on heating into a liquid and a different solid, through a peritectic. Cu_6Sn_5, the intermetallic that forms between solder and copper pads, melts incongruently near 415 °C; mullite in the Al_2O_3–SiO_2 system is debated but is usually drawn as melting incongruently near 1890 °C.
- **Ordered phases** such as Ni_3Al (γ′), the strengthening precipitate of nickel superalloys, and Fe_3Al or CuZn (β′ brass) are intermediate phases whose atoms occupy distinct sublattices (Chapter 6).

Intermetallics are usually hard, brittle and of high melting point. That is useful when they are fine precipitates (Chapters 10 and 13) and harmful when they are thick continuous layers: a solder joint whose Cu_6Sn_5 layer has grown to 10 µm fractures at the interface under drop impact.

> **Status of the principle: Binary diagram topology rules**
> *Derived*. The rules that two single-phase fields are separated by a two-phase field, that three-phase equilibria are horizontal lines, and that boundaries meet at invariant points with extrapolations falling into two-phase fields (the "180° rule") follow from the phase rule and the shape of Gibbs-energy curves; they are used to check experimentally drawn diagrams.

7.6 Microstructure development on cooling

The phase diagram tells what phases are present at each temperature. Following an alloy down a vertical line through the diagram predicts the sequence of reactions and therefore the microstructure, which in turn sets strength, ductility and corrosion behaviour.

Isomorphous alloys: equilibrium and coring

Cool the Cu–30Ni alloy of Section 7.4 slowly. At the liquidus the first solid forms, richer in Ni than the liquid. As the temperature falls, the tie-line ends move down the liquidus and solidus: the liquid becomes richer in Cu, and so must the solid, which requires Ni to diffuse out of the solid already formed. At the solidus, at equilibrium, the last liquid disappears and the solid has the overall composition everywhere.

Real castings cool too fast for diffusion in the solid to keep up (Chapter 8 gives the distances). Each layer of solid keeps roughly the composition it had when it formed. Dendrite centres are Ni-rich; the regions between dendrite arms, which froze last, are Cu-rich. This **coring** or **microsegregation** shows up after etching as concentration contours, lowers the effective solidus (the last liquid can be far richer in the low-melting component than the overall alloy) and causes local differences in corrosion potential. It is removed by a **homogenization** anneal, whose time is calculated in Chapter 8. Chapter 9 quantifies the non-equilibrium limit with the Scheil equation.

Eutectic alloys

Follow three Al–Si alloys down Figure 7.3.

Eutectic composition (12.6 wt% Si). The liquid remains liquid down to 577 °C and then freezes at constant temperature into a two-phase eutectic mixture of α-Al and Si. Because both solids form together from the liquid, with solute diffusing sideways only a short distance, the eutectic is fine: lamellae, rods or plates a few micrometres apart.

Hypoeutectic (less than 12.6 wt% Si). Cooling into the L + α field produces **primary** α-Al dendrites. The remaining liquid becomes richer in Si along the liquidus, reaching the eutectic composition at 577 °C, where it freezes as eutectic. The final microstructure is primary α dendrites in a eutectic matrix.

Hypereutectic (more than 12.6 wt% Si). The primary phase is now silicon, which forms as hard faceted blocks, followed by eutectic.

The lever rule, applied just below and just above the eutectic temperature, gives the amounts of each constituent.

Worked example — Constituents in two aluminium casting alloys. A356 (Al–7 wt% Si, plus Mg for age hardening) is used for automotive wheels and suspension parts; A390 (Al–18 wt% Si) for unlined engine blocks. Use the Al–Si eutectic at 12.6 wt% Si, the α solvus at 1.65 wt% Si and treat Si as pure.

A356, primary α. Just above 577 °C the alloy is α (1.65 wt% Si) plus liquid of eutectic composition (12.6 wt% Si):

$$W_{\alpha,\text{primary}} = \frac{12.6 - 7}{12.6 - 1.65} = 0.51$$

The remaining 0.49 is liquid, which becomes eutectic on further cooling.

A356, total silicon phase. Just below 577 °C, across the whole α + Si field,

$$W_{Si} = \frac{7 - 1.65}{100 - 1.65} = 0.054$$

All of this 5.4 % silicon lies inside the eutectic (it is $0.054/0.49 = 11\,\%$ of the eutectic).

A390, primary Si. For the hypereutectic alloy, just above 577 °C the alloy is pure Si plus eutectic liquid:

$$W_{Si,\text{primary}} = \frac{18 - 12.6}{100 - 12.6} = 0.062$$

and the total silicon phase is $(18 - 1.65)/(100 - 1.65) = 0.17$. The A390 block therefore contains about 6 % blocky primary silicon (by mass; by volume slightly more, since Si at 2.33 g/cm³ is less dense than Al at 2.70 g/cm³) dispersed in a eutectic. These primary crystals, not the total silicon, determine wear resistance and machinability, and their size is controlled by adding phosphorus to nucleate them (Chapter 9).

Beyond equilibrium

Two warnings attach to these predictions. First, the eutectic in unmodified Al–Si forms coarse brittle silicon plates; adding 0.01–0.02 wt% strontium or sodium changes the growth mode to a fine fibrous structure and raises ductility several-fold, without changing the phase diagram at all. Second, an alloy with less solute than the maximum solubility (say Al–1 wt% Si) should freeze entirely as α, but under real casting conditions coring enriches the last liquid enough that some non-equilibrium eutectic appears between the dendrites. Phase diagrams predict which phases are possible and their equilibrium amounts; morphology and departures from equilibrium belong to Chapters 9 and 10.

7.7 The Fe–C (Fe–Fe_3C) diagram

Stable and metastable diagrams

Steels and cast irons are the most used engineering materials, and the iron–carbon diagram is the most used phase diagram. Strictly, the stable carbon-rich phase in contact with iron is graphite. But in steels the formation of graphite is so slow that the carbide **cementite**, Fe₃C (6.70 wt% C), forms instead and persists for practical purposes indefinitely. The diagram used for steels is therefore the **metastable Fe–Fe₃C diagram**; the stable Fe–graphite diagram, with slightly shifted lines, governs grey and ductile cast irons, where silicon promotes graphite.

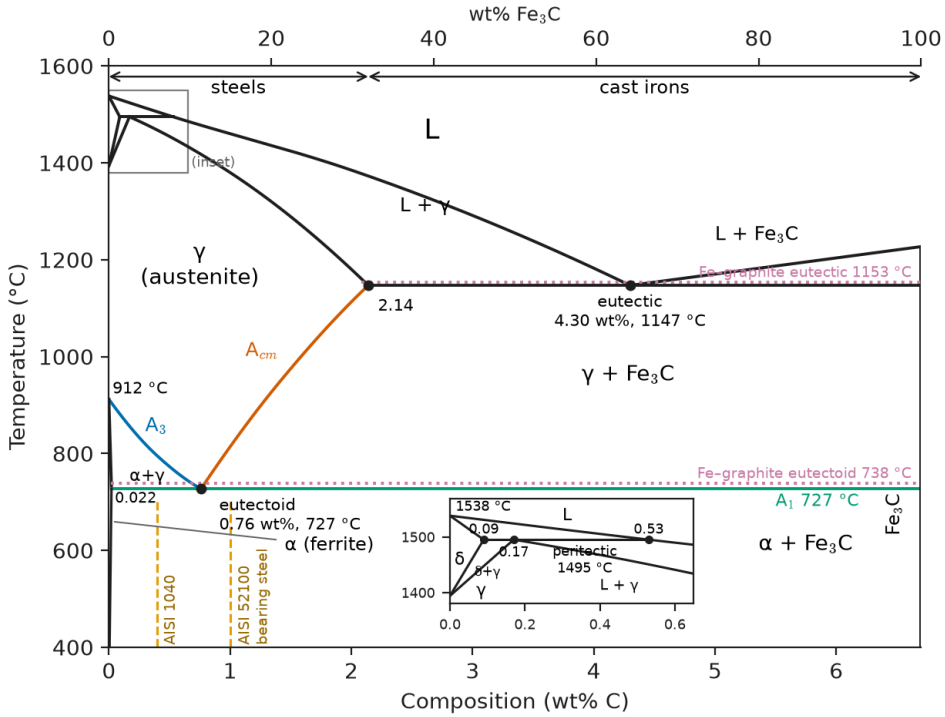

Figure 7.4 — The metastable Fe–Fe3C phase diagram with the steel and cast-iron regions and the key invariant points.

The landmarks

The key features of Figure 7.4, which every metallurgist knows by heart:

Feature	Temperature	Compositions (wt% C)
Melting of pure iron	1538 °C	0
Peritectic: $\delta + L \rightarrow \gamma$	1495 °C	δ 0.09, γ 0.17, L 0.53
Eutectic: $L \rightarrow \gamma + Fe_3C$ (ledeburite)	1147 °C	L 4.30, γ 2.14, Fe3C 6.70
Eutectoid: $\gamma \rightarrow \alpha + Fe_3C$ (pearlite)	727 °C	γ 0.76, α 0.022, Fe3C 6.70
α→γ in pure iron (A3 at 0 % C)	912 °C	0

Austenite dissolves far more carbon (up to 2.14 wt%) than ferrite (at most 0.022 wt%), because carbon sits in the larger octahedral interstices of FCC iron (Chapter 2). That difference is the whole basis of steel heat treatment: heat into γ to dissolve carbon, then cool so that the carbon must come out again, either slowly, as cementite in pearlite, or not at all, trapping it in martensite (Chapter 10). The conventional boundary between steels and cast irons is 2.14 wt% C: steels can be made fully austenitic and hot-worked; cast irons always contain eutectic and are shaped by casting.

The critical temperature lines carry their own names: **A1** is the eutectoid isotherm at 727 °C, **A3** the boundary between γ and α + γ in hypoeutectoid steels, and **Acm** the boundary between γ and γ + Fe3C in hypereutectoid steels. Heat-treatment specifications are written in these terms: full annealing and hardening of hypoeutectoid steels at about A3 + 30–50 °C, of hypereutectoid steels between A1 and Acm.

Microstructures of slowly cooled steels

A **eutectoid steel** (0.76 wt% C) transforms entirely at 727 °C to **pearlite**, alternating plates of ferrite and cementite. The lever rule across the α + Fe3C field gives the cementite fraction within pearlite: $(0.76 - 0.022)/(6.70 - 0.022) = 0.11$. Pearlite is about 89 % ferrite and 11 % cementite by mass, which, because the two phases have similar densities, is about 7:1 in plate thickness.

A **hypoeutectoid** steel forms **proeutectoid ferrite** at the γ grain boundaries between A3 and A1; the remaining austenite reaches 0.76 wt% C at 727 °C and becomes pearlite. A **hypereutectoid** steel forms proeutectoid cementite on the austenite

grain boundaries between Acm and A1, then pearlite.

Worked example — Ferrite, pearlite and cementite in two steels. Consider AISI 1040 (0.40 wt% C), a medium-carbon steel for shafts, axles and connecting rods, and a 1.20 wt% C tool steel, both slowly cooled.

1040, constituents. Just above 727 °C the steel is α (0.022 wt% C) plus γ (0.76 wt% C):

$$W_{\alpha,\text{pro}} = \frac{0.76 - 0.40}{0.76 - 0.022} = 0.49, \qquad W_{\text{pearlite}} = 0.51$$

1040, phases. Just below 727 °C, across α + Fe3C,

$$W_{\alpha} = \frac{6.70 - 0.40}{6.70 - 0.022} = 0.94, \qquad W_{\text{Fe}_3\text{C}} = 0.06$$

1.20 wt% C steel. Proeutectoid cementite, just above 727 °C between γ (0.76) and Fe3C (6.70):

$$W_{\text{Fe}_3\text{C,pro}} = \frac{1.20 - 0.76}{6.70 - 0.76} = 0.074$$

and total cementite $(1.20 - 0.022)/(6.70 - 0.022) = 0.18$.

The 1040 steel is half ferrite grains, half pearlite colonies, which gives a tensile strength of roughly 520–620 MPa in the normalized condition with good ductility. The tool steel contains about 7 % cementite as a network around the former austenite grains; if left continuous, that brittle network embrittles the steel, so such steels are given a spheroidizing anneal before machining and are hardened from between A1 and Acm, where some cementite remains undissolved to resist wear (Chapter 15). Note that the constituent fractions (ferrite vs pearlite) and the phase fractions (ferrite vs cementite) answer different questions; mixing them up is a common reading error.

Alloying elements and the limits of the diagram

Real steels contain manganese, silicon, chromium, nickel, molybdenum and other elements, and the binary diagram is only a guide. Austenite stabilizers (Ni, Mn, C, N) lower A3 and A1 and enlarge the γ field; at 8 % Ni plus 18 % Cr, austenite is retained to room temperature in 304 stainless steel. Ferrite stabilizers (Cr, Si, Mo, V, Ti) raise the critical temperatures and close the γ field ("γ loop"). More importantly, the diagram says nothing about what happens when cooling is fast: the bainite and martensite that give high-strength steels their properties are not on it at all. They live on the TTT and CCT diagrams of Chapters 9 and 10, which add time as an axis.

> **Status of the principle: Fe–Fe3C diagram as a guide to steel microstructure**
> *Approximate.* It is a metastable equilibrium diagram, accurate for plain-carbon steels cooled slowly enough for carbon diffusion to keep pace; alloying shifts its lines, and fast cooling produces phases it does not contain.

7.8 Ceramic phase diagrams

Ceramic diagrams are read in exactly the same way, with oxides as components. Three systems cover refractories, structural and biomedical ceramics, and energy devices.

Al_2O_3–SiO_2: refractories and porcelains

The alumina–silica system contains one intermediate compound, **mullite**, nominally 3Al2O3·2SiO2 (71.8 wt% Al2O3), with a solid-solution range toward alumina. The two eutectics are between silica and mullite, near 1587 °C and 5–6 wt% Al2O3, and between mullite and corundum, near 1840–1890 °C.

The diagram tells a refractory engineer exactly what to buy. Fireclay bricks with 25–45 wt% Al2O3 sit in the silica–mullite field: above 1587 °C they contain liquid, and the lever rule gives its amount. A brick with 40 wt% Al2O3 at 1600 °C contains roughly half its mass as viscous liquid and will sag under load. Silica bricks (about 95 % SiO2 or more) avoid this only if the alumina impurity is kept very low, because a small amount of alumina moves the composition into the liquid-forming region near the steep silica-rich liquidus; even 1 % Al2O3 produces a noticeable liquid fraction at 1600 °C. High-alumina bricks (above 72 wt%

Al_2O_3) contain no liquid until about 1840 °C. Porcelain relies on the same diagram, with K_2O from feldspar as a third component supplying the bonding liquid during firing.

ZrO_2–Y_2O_3: transformation-toughened and ion-conducting zirconia

Pure zirconia is monoclinic up to about 1170 °C, tetragonal up to about 2370 °C, cubic (fluorite) above that, and melts near 2710 °C. The tetragonal→monoclinic change on cooling is martensitic and expands the volume by about 4–5 %, enough to shatter a sintered pure zirconia part. Adding yttria stabilizes the higher-symmetry phases to lower temperature; the diagram (drawn in mol% $YO_{1.5}$ or Y_2O_3) shows the tetragonal and cubic fields widening as yttria increases, with a broad two-phase t + c field.

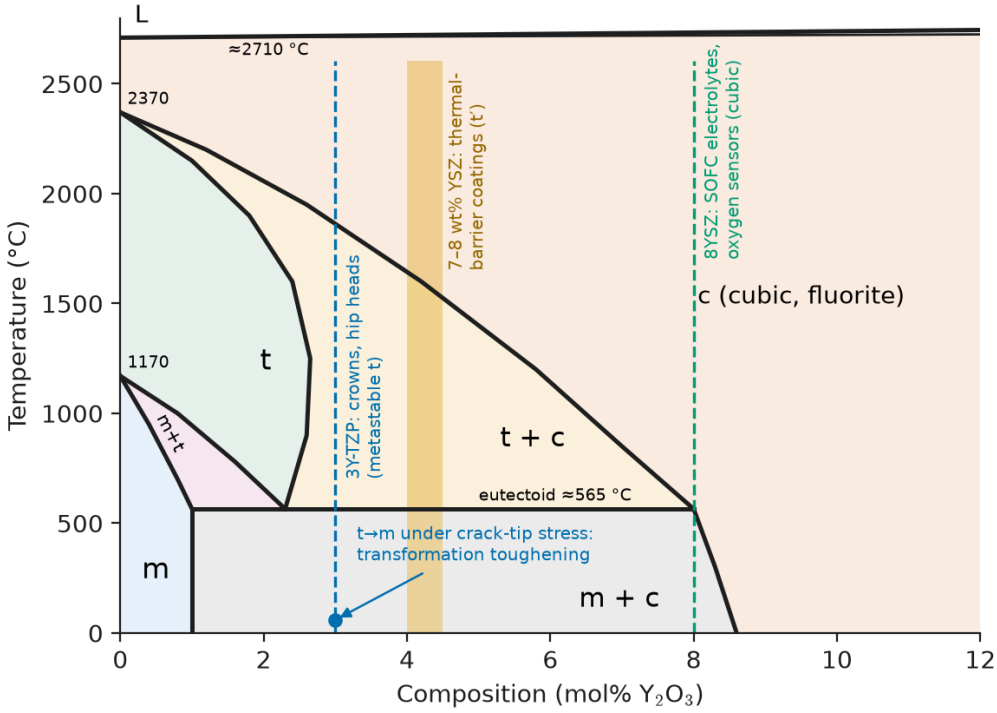

Figure 7.5 — The zirconia-rich end of the ZrO2–Y2O3 diagram, with the compositions used for dental and implant ceramics, thermal-barrier coatings and fuel-cell electrolytes.

Three compositions illustrate the engineering use of one diagram (Figure 7.5):

- **3 mol% Y_2O_3 (about 5.4 wt%), 3Y-TZP.** Fine-grained tetragonal zirconia polycrystal, metastable at room temperature because the grains are too small to transform spontaneously. At a crack tip the stress triggers the t→m change; the expansion squeezes the crack shut. This **transformation toughening** gives fracture toughness of 5–10 MPa√m, two to three times that of alumina, and makes zirconia suitable for dental crowns, bridges and femoral heads of hip replacements (Chapters 14 and 39). The same metastability is a weakness: in warm moisture the surface slowly transforms ("low-temperature degradation").
- **About 4–4.5 mol% (7–8 wt%) Y_2O_3.** Plasma-sprayed or electron-beam-deposited thermal-barrier coatings on turbine blades are quenched into a non-transformable tetragonal phase (t′) that resists both destabilization and the t→m change during thermal cycling (Chapter 41).
- **8 mol% Y_2O_3 (about 13.7 wt%), 8YSZ.** Fully cubic. Each pair of Y^{3+} ions replacing Zr^{4+} creates one oxygen vacancy, giving high oxide-ion conductivity at 700–1000 °C: the electrolyte of solid-oxide fuel cells and of lambda sensors in car exhausts (Chapters 8 and 40).

MgO–Al_2O_3: spinel

MgO (melting near 2830 °C) and Al_2O_3 (2054 °C) form the compound **spinel**, $MgAl_2O_4$ (71.7 wt% Al_2O_3), which melts congruently near 2105 °C and dissolves excess alumina at high temperature. Eutectics lie on both sides, near 1995 °C toward MgO and near 1925 °C toward Al_2O_3. Because spinel melts congruently, the diagram splits into two simple eutectic sub-systems. Steel ladles are lined with magnesia–spinel or alumina–spinel castables, chosen because spinel resists penetration by basic slags

and because its thermal expansion is moderate. Transparent polycrystalline spinel serves as infrared windows and armour (Chapter 36).

7.9 Ternary diagrams

The Gibbs triangle

With three components and fixed pressure, two composition variables and temperature are needed, so a full ternary diagram is a triangular prism. Composition is plotted in an equilateral **Gibbs triangle**, whose corners are the pure components. The perpendicular distance from a point to the side opposite corner A, as a fraction of the triangle's height, is the fraction of A; the three fractions sum to one. In practice, gridlines parallel to each side are read directly.

Two geometric rules make the triangle useful. Any point on a line from a corner to the opposite side has a constant ratio of the other two components; adding or removing one component moves the composition along that line. And the **centre-of-gravity rule**: if an alloy of composition P splits into phases of compositions α, β, γ, then P lies inside the triangle $\alpha\beta\gamma$, and the phase fractions are the weights that place the centre of gravity at P. Two-phase mixtures obey the ordinary lever rule along tie lines.

> **Status of the principle: Ternary phase rule and centre-of-gravity rule**
> *Derived.* With $C = 3$ at fixed pressure, $F = 4 - P$: single-phase volumes, two-phase regions with tie lines, three-phase tie triangles at each temperature, and four-phase invariant planes; the centre-of-gravity rule is mass conservation in three components.

Sections and projections

A three-dimensional diagram is communicated by two-dimensional slices (Figure 7.6):

- **Isothermal sections** cut the prism at one temperature. They show single-phase areas, two-phase areas crossed by tie lines (which are not generally parallel and must be measured or computed), and three-phase tie triangles. Phase fractions can be read with the lever and centre-of-gravity rules. The 900 °C or 1100 °C sections of Fe–Cr–Ni are used to design stainless steels and to predict σ-phase formation in welds.
- **Vertical sections** (isopleths) cut the prism along a line of composition, for example constant 18 wt% Cr in Fe–Cr–Ni, or the line from Fe–C to a fixed Mn content. They look like binary diagrams but are not: tie lines generally do not lie in the plane, so the lever rule *cannot* be applied on them. They show which phases appear on heating or cooling a given alloy, which is what a heat-treatment engineer needs.
- **Liquidus projections** show the liquidus surface from above, with isotherms and the **valleys** (monovariant lines) where two solids crystallize together, meeting at ternary invariant points such as a **ternary eutectic** $L \rightarrow \alpha + \beta + \gamma$. Arrows on the valleys show the direction of falling temperature, which is the path followed by the last liquid during solidification.

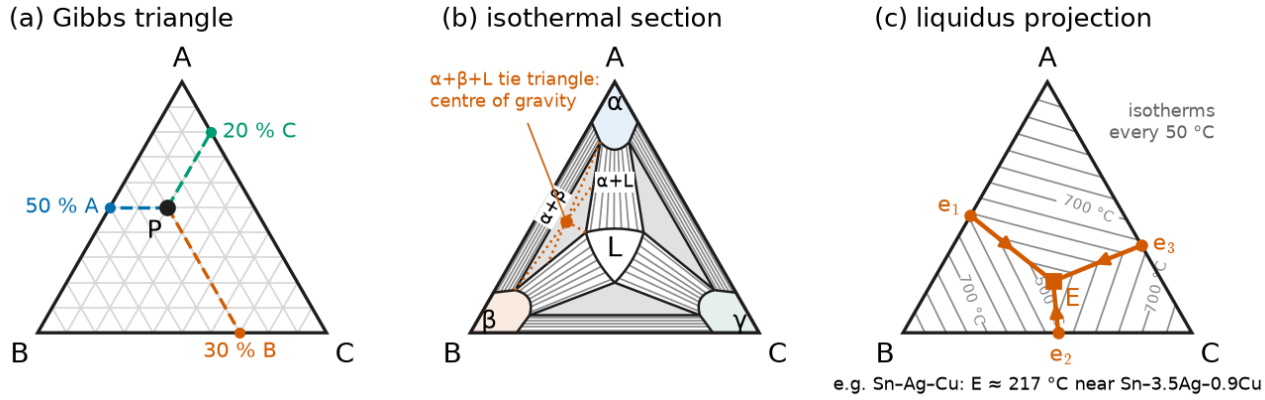

Figure 7.6 — Reading a ternary system: the Gibbs triangle with a composition point, an isothermal section with tie lines and a tie triangle, and a liquidus projection with a ternary eutectic.

Engineering ternaries

Lead-free solder. Sn–Ag–Cu has a ternary eutectic near 217 °C, close to Sn–3.5Ag–0.9Cu. Commercial SAC305 (Sn–3.0Ag–0.5Cu) lies just off it and melts over a range of a few degrees. At room temperature the solder consists of tin with small particles of the intermetallics Ag_3Sn and Cu_6Sn_5.

Worked example — Phase fractions in a lead-free solder joint. Estimate the intermetallic content of SAC305 at room temperature, assuming that Ag and Cu are entirely in Ag_3Sn and Cu_6Sn_5 and that tin dissolves neither.

Composition of the compounds. With molar masses Ag 107.87, Cu 63.55, Sn 118.71 g/mol:

$$w_{Ag}(\mathrm{Ag_3Sn}) = \frac{3(107.87)}{3(107.87)+118.71} = 0.732, \qquad w_{Cu}(\mathrm{Cu_6Sn_5}) = \frac{6(63.55)}{6(63.55)+5(118.71)} = 0.391$$

Centre of gravity. All silver is in Ag_3Sn, so $W_{\mathrm{Ag_3Sn}} = 0.030/0.732 = 0.041$. All copper is in Cu_6Sn_5, so $W_{\mathrm{Cu_6Sn_5}} = 0.005/0.391 = 0.013$. The tin fraction is $1 - 0.041 - 0.013 = 0.946$.

About 4 % Ag_3Sn and 1.3 % Cu_6Sn_5 by mass, fine particles that strengthen the tin matrix much as precipitates strengthen an aluminium alloy. The composition lies inside the Sn–Ag_3Sn–Cu_6Sn_5 tie triangle, so these three are the only phases expected; the joint additionally forms a Cu_6Sn_5 interfacial layer by reaction with the copper pad, which the bulk calculation does not include.

Stainless steels. The Fe–Cr–Ni system decides whether a stainless steel is ferritic (Fe–17Cr, 430), austenitic (Fe–18Cr–8Ni, 304), duplex (Fe–22Cr–5Ni, 2205) or martensitic. For fast-solidifying weld metal, welders use the empirical Schaeffler or WRC-1992 diagrams in chromium and nickel *equivalents* (Chapter 47).

Oxide systems. The CaO–Al_2O_3–SiO_2 liquidus projection is the map of slags in blast furnaces and steelmaking, of Portland cement clinker (which lies in the field of tricalcium silicate, Chapter 18), of glass compositions and of the mineral wool used in building insulation.

7.10 CALPHAD: computed phase diagrams

Why compute?

A commercial nickel superalloy contains ten or more elements; a high-entropy alloy five in nearly equal amounts. Measuring their phase equilibria experimentally, at every composition and temperature of interest, is impossible. The **CALPHAD** (CALculation of PHAse Diagrams) method instead models the Gibbs energy of every phase as a function of temperature, pressure and composition, fits the model parameters to the experimental and first-principles data for the binary and ternary subsystems, and then predicts multicomponent equilibria by minimizing the total Gibbs energy.

The models

For a substitutional solution phase φ of components i, the molar Gibbs energy is written

$$G_m^{\varphi} = \sum_i x_i \, {}^{\circ}G_i^{\varphi} + RT \sum_i x_i \ln x_i + {}^{ex}G_m^{\varphi}$$

The first term is the weighted sum of the pure-element Gibbs energies in the structure of φ (taken from standard "lattice-stability" databases, which give, for example, the Gibbs energy of hypothetical FCC chromium); the second is ideal mixing entropy; the third is the excess term, usually a **Redlich–Kister polynomial** for each pair,

$${}^{ex}G_m^{\varphi} = \sum_{i<j} x_i x_j \sum_{\nu=0}^{n} L_{ij}^{(\nu)} (x_i - x_j)^{\nu}$$

where the interaction parameters $L^{(\nu)}$ are linear functions of temperature. With only $L^{(0)}$ constant, this is the regular-solution model of Chapter 6, which already predicts a symmetric miscibility gap with critical temperature $T_c = L^{(0)}/2R$ (for $L^{(0)} = 20$ kJ/mol, $T_c \approx 1200$ K). Ternary terms are added where binaries alone do not reproduce ternary data. Ordered phases, carbides, oxides and intermetallics are described with the **sublattice (compound-energy) models** introduced in Chapter 6, in which different species mix on distinct sublattices, for example $(\mathrm{Fe},\mathrm{Cr})_3(\mathrm{C})_1$ for cementite with chromium substitution, or $(\mathrm{Ni},\mathrm{Al})_3(\mathrm{Al},\mathrm{Ni})_1$ for γ′.

The equilibrium state at a given T, p and overall composition is the set of phases, fractions and phase compositions that minimizes the total Gibbs energy; graphically, this is the multidimensional common-tangent construction of Chapter 6. Software packages (Thermo-Calc, FactSage, Pandat, and open-source tools such as OpenCalphad and pycalphad) perform this minimization and draw sections, projections, phase-fraction curves and property diagrams.

> **Status of the principle: CALPHAD phase-equilibrium predictions**
> *Approximate.* Equilibrium is exact thermodynamics, but the Gibbs-energy functions are fitted models; predictions are as good as the database coverage of the relevant subsystems, and extrapolations into untested composition ranges or to metastable phases carry uncertainties of tens of degrees.

What CALPHAD delivers

- **Phase-fraction diagrams** for a single alloy as a function of temperature: the solvus of γ′ in a superalloy, the temperature where σ phase appears in a duplex stainless steel, the amount of liquid at each temperature in a brazing filler.
- **Solidification paths** using the Scheil–Gulliver approximation (Chapter 9), which predict the non-equilibrium freezing range and the last phases to form; essential for castability, weldability and the design of alloys for additive manufacturing (Chapter 48).
- **Thermodynamic inputs to kinetics**: driving forces for nucleation, chemical potentials and thermodynamic factors for diffusion simulations (Chapter 8) and precipitation models (Chapter 10).
- **Alloy design**: searching composition space for alloys with a target phase constitution. Computationally designed ultra-high-strength gear and landing-gear steels, developed this way (integrated computational materials engineering, ICME), reached aerospace qualification far faster than trial-and-error alloys (Chapters 55 and 57).

The essential skill for the engineer is to know which subsystems are well assessed. A prediction for an alloy whose key ternary has never been measured is a hypothesis to be checked, not a design value.

7.11 Reading real diagrams for alloy and process design

Real diagrams in handbooks and databases are dense with lines, two composition scales (weight and atomic percent), dashed uncertain boundaries, and notes on phases that are metastable or ordered. A short checklist turns them into engineering decisions.

1. **Check the system.** Is it the right number of components? The opening example of this chapter failed at this step: 2024 aluminium is Al–4.4Cu–1.5Mg–0.6Mn, and the binary Al–Cu diagram is the wrong map.
2. **Check the composition scale and the axis units.** Weight percent for alloying, atomic or mole percent for crystal chemistry and for oxides; the same eutectic can look very different on the two scales.
3. **Identify the invariant reactions** that the alloy's vertical line crosses; they are where liquid appears or disappears and where microstructure changes abruptly.
4. **Apply the lever rule** at the temperatures that matter: service, process soak, just above and below each invariant.
5. **Ask whether equilibrium is reached.** Is the process time long compared with the diffusion time (Chapter 8)? If not, expect coring, non-equilibrium eutectic, retained or metastable phases.

Worked example — The solution-treatment window of an Al–Cu alloy. A designer wants to dissolve all the copper of an Al–4.4 wt% Cu alloy into the α solid solution before quenching and ageing (Chapter 10). The Al–Cu binary has a eutectic at 548 °C and 33.2 wt% Cu, a maximum solubility of Cu in Al of 5.65 wt% at 548 °C, and the α solvus passes through about 4.4 wt% Cu near 500 °C.

Lower limit. The alloy must be above its solvus, about 500 °C, for all θ (Al_2Cu) to dissolve.

Upper limit from the binary. Approximating the solidus as a straight line from 660 °C (pure Al) to 548 °C (5.65 wt% Cu), the solidus at 4.4 wt% Cu is

$$T_{sol} \approx 660 - \frac{660 - 548}{5.65} \times 4.4 = 573\ ^\circ\mathrm{C}$$

The binary window appears to be about 500–573 °C, a comfortable 70 °C wide.

Correction for the real alloy. The alloy has 1.5 % Mg. The Al–Cu–Mg ternary contains a eutectic involving α, θ and the S phase (Al_2CuMg) near 507 °C, and the coring present in a casting or a hot-worked product leaves small regions rich enough in Cu and Mg to reach it. Commercial 2024 is therefore solution-treated at about 495 °C, with a tolerance of a few degrees, and incipient melting is checked metallographically. The window that the binary showed as 70 °C wide is, in the real alloy, about 10 °C. A furnace that overshoots produces exactly the grain-boundary melting described at the start of this chapter.

The same reading habit serves elsewhere:

- **Brazing.** The Ag–Cu eutectic (779 °C, 28 wt% Cu) melts at a single temperature, which makes BAg-8 ideal for vacuum brazing in a furnace where the filler must flow all at once. Fillers with a wide melting range liquate: the low-melting liquid runs out first (Chapter 47).
- **Hot shortness.** The Fe–FeS eutectic at 988 °C leaves liquid grain-boundary films during hot rolling at 1100–1200 °C; manganese ties sulphur up as high-melting MnS.
- **Heat treatment of steels.** A3 and Acm locate the austenitizing temperature; the eutectoid composition tells whether proeutectoid ferrite or cementite will form; alloy additions move both (Chapter 15).
- **Refractory selection.** Read the liquid fraction at the service temperature with the lever rule (Section 7.8); a few percent of liquid is tolerable in a castable, tens of percent is not.
- **Crystal growth and purification.** The partition coefficient, read from the slopes of liquidus and solidus near the pure component, predicts how dopants distribute along a Czochralski silicon ingot and how efficiently zone refining removes impurities (Chapter 9).
- **Liquid-phase sintering.** Cemented carbides (WC–Co) and tungsten heavy alloys are sintered just above a binder eutectic so that a small, controlled liquid fraction speeds densification (Chapter 46).

> **Boundary.**
> Phase diagrams describe equilibrium. How long it takes to approach equilibrium, and what happens when it is not approached, is the subject of the next three chapters: diffusion (Chapter 8), nucleation and growth (Chapter 9) and solid-state transformations (Chapter 10). The thermodynamic laws used to build the diagrams are recapped in Chapter 6 and belong in full to the Advanced Chemistry book.

What remains open

Open question	Where it is answered
Where do the Gibbs-energy curves behind each diagram, the common tangent and the miscibility gap come from?	Chapter 6
How fast does a cored casting homogenize, and how far does carbon move during carburizing?	Chapter 8
Why does a liquid undercool before freezing, and what sets dendrite spacing, segregation and the non-equilibrium (Scheil) freezing path?	Chapter 9
What microstructures form when steels and aluminium alloys are cooled too fast for the equilibrium diagram, and what are TTT and CCT diagrams?	Chapter 10
How do the phases on a diagram translate into strength through precipitation and grain size?	Chapter 13
How are steels and cast irons designed with the Fe–C diagram and alloying additions?	Chapter 15
Which non-ferrous alloy systems (Al, Ti, Ni, Cu, Mg) rely on which invariant reactions?	Chapter 16
How are zirconia, alumina and silicate ceramics processed and toughened?	Chapter 17
How do clinker phases in the $CaO–SiO_2–Al_2O_3$ system become cement?	Chapter 18
How do welding and brazing use eutectics, and how is weld-metal ferrite controlled?	Chapter 47
How are phase diagrams measured by thermal analysis and diffraction?	Chapter 52
How are first-principles energies and machine learning combined with CALPHAD?	Chapters 55 and 57

Next: Chapter 8 supplies the clock that phase diagrams lack, deriving how atoms move through solids and how far they travel in the time a process allows.

Chapter 8 — Diffusion in Solids

Part II · Thermodynamics and Kinetics

A gearbox manufacturer for wind turbines changes its heat-treatment supplier. The new supplier carburizes the gears at the same temperature, with the same furnace atmosphere, for the specified time. Six months later several gearboxes in an offshore park show pitting on the gear teeth. Sectioning shows that the hardened case is about 0.85 mm deep instead of the 1.0 mm on the drawing. The supplier's thermocouple read 25 °C high, so the parts were actually treated 25 °C colder than they should have been. That small temperature error cut the carbon diffusion coefficient by about a quarter, and the case depth, which grows with the square root of the diffusion coefficient times time, fell short. Diffusion turned a thermocouple error into a field failure.

The same physics operates in a silicon wafer where boron must penetrate exactly 1 µm, in a solid-oxide fuel cell whose power depends on how fast oxide ions hop through zirconia, in a smartphone cover glass strengthened by exchanging sodium for potassium, in a carbon-fibre aircraft panel that slowly absorbs water, and in a solder joint where unequal diffusion rates leave a line of voids. Phase diagrams (Chapter 7) tell where a system wants to go; diffusion tells how long it takes to get there.

> **How fast do atoms move through a solid, and how far do they get in the time a process or a service life allows?**
> **Why is diffusion so sensitive to temperature, to the crystal and its defects, and to the chemistry of the diffusing species?**

Concept	Engineering destination
Fick's first law, flux and permeability	Hydrogen permeation through pipeline steel, gas barriers in packaging, membranes
Fick's second law and its solutions	Case depth in carburizing and nitriding, dopant profiles, homogenization time
Atomic mechanisms	Why carbon moves 10^5–10^9 times faster than iron in steel
Random walk and Arrhenius law	Temperature sensitivity of every heat-treatment specification
Kirkendall effect, Darken equations	Voids in solder joints and wire bonds, diffusion bonding, coatings
Short-circuit paths	Low-temperature creep, interconnect electromigration, grain-boundary embrittlement
Ionic diffusion, Nernst–Einstein	Fuel-cell electrolytes, oxygen sensors, battery electrodes, oxidation scales
Diffusion in polymers and glasses	Moisture uptake of composites, barrier packaging, chemically strengthened glass
Uphill diffusion	Carbon redistribution across dissimilar welds, spinodal decomposition

Roadmap. Flux and Fick's first law (8.1) → Fick's second law and its classic solutions (8.2) → how atoms jump (8.3) → random walks and the Arrhenius law (8.4) → self-diffusion, interdiffusion and the Kirkendall effect (8.5) → fast paths (8.6) → ionic solids (8.7) → polymers and glasses (8.8) → diffusion driven by chemical potential (8.9) → engineering applications (8.10).

8.1 Fick's first law

Flux

Diffusion is the net transport of atoms by random thermal motion. Its rate is measured by the **flux** J, the number of atoms (or moles, or kilograms) crossing unit area per unit time, with units such as atoms $\mathrm{m^{-2}\ s^{-1}}$ or $\mathrm{kg\ m^{-2}\ s^{-1}}$. In 1855 Adolf Fick proposed, by analogy with heat conduction, that the flux is proportional to the concentration gradient:

$$J = -D\,\frac{\partial C}{\partial x}$$

The **diffusion coefficient** (diffusivity) D has units of $\mathrm{m^2/s}$. The minus sign says that atoms flow from high to low concentration. In three dimensions $\mathbf{J} = -D\nabla C$, and in an anisotropic crystal D is a second-rank tensor (Chapter 4); in cubic

crystals it reduces to a scalar.

> **Status of the principle: Fick's first law**
> *Approximate.* It is a linear-response law, valid when the concentration gradient is the only driving force and is small on the scale of the jump distance; the deeper driving force is the gradient of chemical potential (Section 8.9), and D itself can depend on concentration.

Steady state and permeation

When the concentration at every point stops changing, the flux is uniform and the concentration profile across a slab is linear. This is the situation in a membrane separating two gas reservoirs. Hydrogen dissolving in the wall of a steel pipeline or a pressure vessel, nitrogen through a polymer liner of a hydrogen tank, and oxygen through a food-packaging film are all steady-state permeation problems. For a slab of thickness h with surface concentrations C_1 and C_2,

$$J = D\,\frac{C_1 - C_2}{h}$$

For a gas that dissolves according to a solubility law (Henry's law $C = Sp$ in polymers; Sieverts' law $C = S\sqrt{p}$ for diatomic gases that dissociate in metals, such as H_2 in steel), the flux is written in terms of pressure, and the product of diffusivity and solubility is the **permeability** $P = DS$. Barrier materials need low D, low S or both. Section 8.8 returns to polymers, where this is the main design variable.

A classic industrial use of steady-state diffusion is hydrogen purification: Pd–23Ag membranes a few micrometres thick pass hydrogen (which dissolves and diffuses rapidly as atoms) and block every other gas, giving hydrogen of 99.9999 % purity for semiconductor fabs and fuel cells.

8.2 Fick's second law and its solutions

The diffusion equation

When the concentration changes with time, mass conservation links the time derivative of concentration to the divergence of flux: what flows into a thin slice and does not flow out accumulates there. Combining this with Fick's first law gives **Fick's second law**:

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x}\left(D\,\frac{\partial C}{\partial x}\right) \xrightarrow{D\ \text{const}} \frac{\partial C}{\partial t} = D\,\frac{\partial^2 C}{\partial x^2}$$

> **Status of the principle: Fick's second law**
> *Derived.* It is Fick's first law plus conservation of matter; the constant-D form is an approximation that fails when D depends strongly on composition, as it does for carbon in austenite and for water in polymers.

This is the same equation as transient heat conduction (Figure 8.1 shows its two most used solutions), so the whole mathematical toolkit of heat transfer applies. One result follows by dimensional analysis alone: the only length scale that can be built from D and t is $\sqrt{Dt}$. Every diffusion distance is some multiple of it, so **doubling a diffusion depth takes four times as long**. A useful rule of thumb is that atoms penetrate a distance of order

$$x \approx \sqrt{Dt} \quad \text{to} \quad 2\sqrt{Dt}$$

Three classes of solution cover most engineering problems.

Thin-film (instantaneous source) solution

A fixed amount Q of solute per unit area is deposited on the surface of a semi-infinite solid and allowed to diffuse inward with no further supply. The solution is a half-Gaussian:

$$C(x,t) = \frac{Q}{\sqrt{\pi D t}}\exp\!\left(-\frac{x^2}{4Dt}\right)$$

The total amount stays constant, the surface concentration falls as $t^{-1/2}$ and the profile broadens. (For a film sandwiched between two semi-infinite blocks, the solution has $2\sqrt{\pi Dt}$ in the denominator, and the Gaussian extends to both sides.) This is the profile used to measure diffusion coefficients with radioactive tracers: deposit a thin layer, anneal, section and fit $\ln C$ against x^2; the slope is $-1/(4Dt)$. It is also the profile of a semiconductor **drive-in** step, where an implanted dose redistributes during a high-temperature anneal.

Error-function solution (constant surface concentration)

If instead the surface is held at a constant concentration C_s by a gas, salt bath or liquid, and the solid initially contains C_0 uniformly, the solution is

$$\frac{C(x,t)-C_0}{C_s-C_0} = 1-\operatorname{erf}\!\left(\frac{x}{2\sqrt{Dt}}\right) = \operatorname{erfc}\!\left(\frac{x}{2\sqrt{Dt}}\right)$$

where $\operatorname{erf}(z)=\frac{2}{\sqrt{\pi}}\int_0^z e^{-u^2}du$. Carburizing, nitriding, decarburization, boriding, dopant predeposition and the joining of two semi-infinite blocks of different composition (a diffusion couple, where the same function appears with C_s replaced by the mean of the two compositions) all follow this form.

z	$\operatorname{erf}(z)$	z	$\operatorname{erf}(z)$
0.10	0.1125	0.80	0.7421
0.20	0.2227	0.90	0.7969
0.30	0.3286	1.00	0.8427
0.40	0.4284	1.20	0.9103
0.50	0.5205	1.50	0.9661
0.60	0.6039	2.00	0.9953
0.70	0.6778	2.50	0.9996

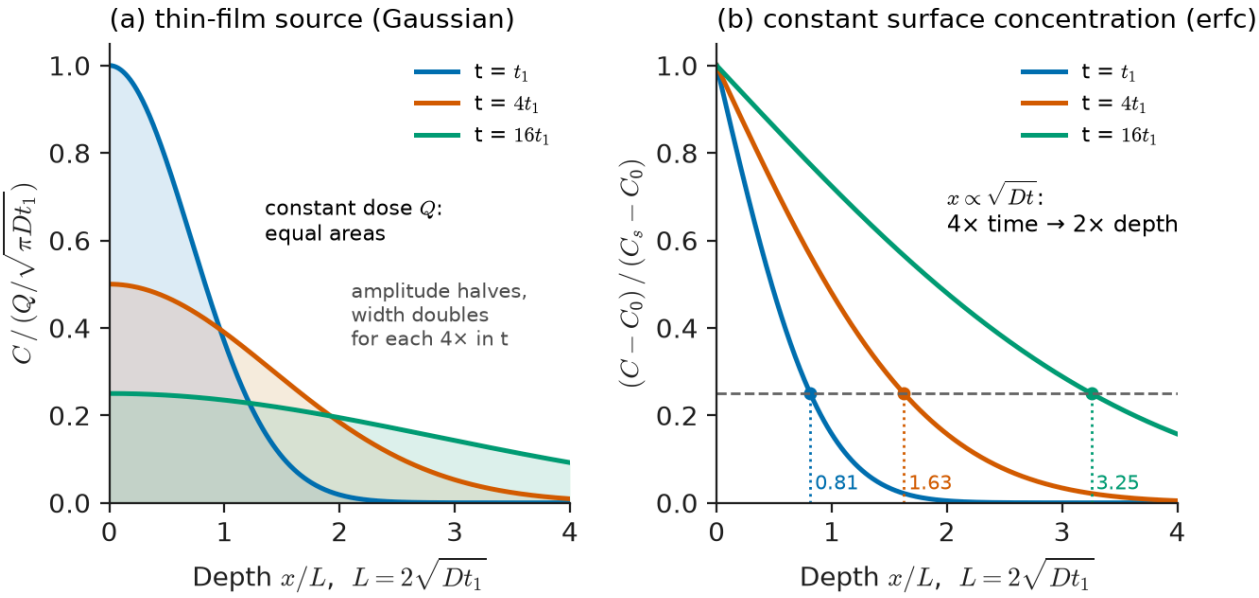

Figure 8.1 — The two classic solutions of Fick's second law: a thin-film source spreading as a Gaussian and a constant-surface-concentration (error-function) profile, each at three times.

Separation of variables: finite bodies and homogenization

For finite geometries the solution is a sum of decaying spatial modes. Consider a cored casting in which the solute concentration varies sinusoidally between dendrite centres and the interdendritic regions, with half-wavelength l (half the dendrite arm spacing):

$$C(x,0) = \bar{C} + \beta_0 \sin\frac{\pi x}{l}$$

Substituting $C = \bar{C} + \beta(t)\sin(\pi x/l)$ into Fick's second law gives

$$\beta(t) = \beta_0 \exp\left(-\frac{t}{\tau}\right), \qquad \tau = \frac{l^2}{\pi^2 D}$$

Any initial profile can be expanded as a Fourier series, and each harmonic decays with its own $\tau_n = l^2/(n^2\pi^2 D)$; the higher harmonics vanish first, so after a short time only the fundamental remains. The same method gives the classic slab, cylinder and sphere solutions for degassing, drying, outgassing of vacuum parts and moisture uptake of a composite panel (Section 8.8).

> **Status of the principle: Classic solutions of the diffusion equation**
> *Derived.* Exact solutions of Fick's second law for constant D and idealized boundary conditions (semi-infinite body, fixed surface concentration or fixed dose); they are approximations wherever D varies with composition, the surface concentration drifts, or the body is not effectively semi-infinite ($x \gtrsim$ thickness).

8.3 Atomic mechanisms

Atoms in a crystal vibrate about their sites at about 10^{13} Hz. Occasionally, a fluctuation gives one enough energy to squeeze past its neighbours into an adjacent site. How it does that depends on where it sits.

Vacancy mechanism. A substitutional atom (an iron atom in iron, a zinc atom in copper) can move only if a neighbouring lattice site is empty. It exchanges places with the vacancy, and the vacancy moves the opposite way. The atom's jump rate is therefore the product of the probability that a neighbour site is vacant (the equilibrium vacancy fraction of Chapter 6, $x_v = \exp(S_f/k)\exp(-H_f/kT)$, typically 10^{-4} near the melting point of a metal) and the rate at which a vacancy jumps once present. The activation energy for diffusion is the sum of vacancy formation and migration enthalpies, $Q = H_f + H_m$; in copper, for example, $H_f \approx 1.3$ eV and $H_m \approx 0.7$ eV give $Q \approx 2.0$ eV (about 190 kJ/mol), close to the measured 2.1–2.2 eV for self-diffusion.

Interstitial mechanism. Small atoms that sit in the interstices between host atoms (H, C, N, O, B in metals) jump from one interstice to the next. Neighbouring interstices are almost always empty, so no defect needs to be formed and $Q = H_m$ only. This is why carbon diffuses in iron about 10^5 to 10^9 times faster than iron itself at 500–1000 °C (the ratio depends on temperature and phase), and why hydrogen diffuses in ferritic steel at room temperature about as fast as solutes diffuse in a liquid.

Interstitialcy mechanism. A self-interstitial (an extra host atom squeezed into an interstice, usually as a dumbbell sharing a site) pushes a neighbouring lattice atom into an interstitial position and takes its place. In metals self-interstitials are too costly to form in equilibrium, but they are created in large numbers by neutron irradiation in nuclear reactors and dominate radiation-enhanced diffusion (Chapter 41). In silicon, dopants such as boron and phosphorus diffuse largely through interstitialcy-type mechanisms, which is why oxidizing the surface (injecting self-interstitials) speeds up boron diffusion, a well-known process effect in chip manufacturing.

Figure 8.2 compares the three mechanisms. Other mechanisms occur in special cases: direct exchange or ring rotation (energetically costly, not observed in practice), and the **dissociative** or kick-out mechanisms of fast-diffusing metals such as Cu, Ni and Au in silicon, which switch between substitutional and interstitial sites and contaminate whole wafers in minutes.

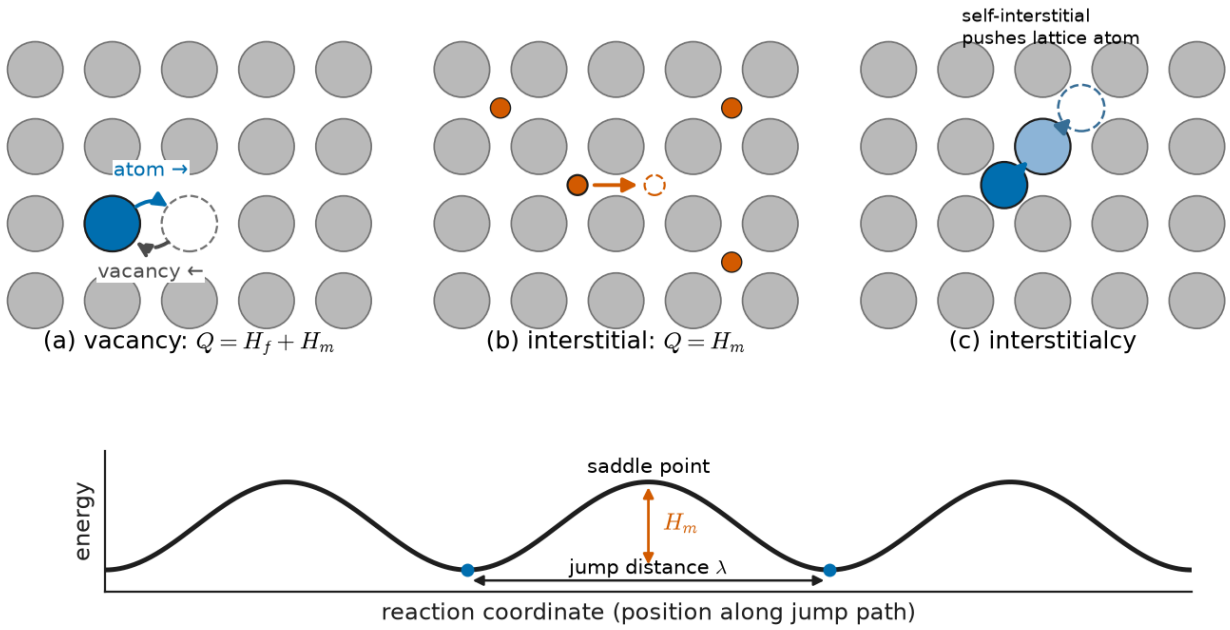

Figure 8.2 — Atomic mechanisms of diffusion: vacancy exchange, interstitial jumping and the interstitialcy mechanism, each with its energy barrier.

8.4 Random walk and the Arrhenius law

From jumps to D

Each atomic jump is random: an atom does not know where the concentration is high. Yet random jumps produce a net flux down a concentration gradient, simply because more atoms are available to jump from the crowded side. Consider two adjacent atomic planes, 1 and 2, separated by the jump distance λ, containing n_1 and n_2 solute atoms per unit area. If each atom jumps Γ times per second in random directions, and in a cubic lattice a fraction $1/6$ of jumps go from plane 1 to plane 2 (one of six directions), the net flux is

$$J = \tfrac{1}{6}\Gamma\,(n_1 - n_2)$$

With concentrations $C = n/\lambda$ and $n_1 - n_2 = -\lambda^2\,\partial C/\partial x$,

$$J = -\tfrac{1}{6}\Gamma\lambda^2\,\frac{\partial C}{\partial x} \qquad \Longrightarrow \qquad D = \tfrac{1}{6}\Gamma\lambda^2$$

Fick's law has emerged from random motion, and the diffusivity is a jump frequency times a jump distance squared. The same picture gives the mean-square displacement of a single atom after $N = \Gamma t$ random jumps in three dimensions,

$$\langle r^2 \rangle = N\lambda^2 = 6Dt$$

which is the microscopic meaning of the $\sqrt{Dt}$ length. For vacancy diffusion, successive jumps of a tracer atom are not independent (an atom that has just exchanged with a vacancy is likely to jump straight back), and a **correlation factor** f (0.78 for FCC, 0.73 for BCC) multiplies D.

Thermal activation

To jump, an atom must pass over a saddle point of energy ΔG_m above its site. The fraction of time it has enough energy is a Boltzmann factor, so

$$\Gamma = z\,\nu_0 \exp\!\left(-\frac{\Delta G_m}{kT}\right)$$

where z is the number of neighbouring sites and $\nu_0 \approx 10^{13}$ s^{-1} the attempt frequency. For vacancy diffusion, the vacancy probability adds another Boltzmann factor. Writing $\Delta G = \Delta H - T\Delta S$ and collecting the temperature-independent terms gives

the **Arrhenius law**

$$D = D_0 \exp\left(-\frac{Q}{RT}\right)$$

with pre-exponential factor D_0 (typically 10^{-6} to 10^{-4} m²/s) and activation energy Q per mole (or E_a per atom, in eV, with k in place of R).

> **Status of the principle: Arrhenius law for diffusion**
> *Derived.* It follows from transition-state theory for thermally activated jumps with a single dominant mechanism; it is approximate when two mechanisms or paths compete (curved Arrhenius plots), near magnetic or structural transitions, and at low temperatures where defect concentrations are frozen in rather than at equilibrium.

Representative data

Diffusing species in host	D_0 (m²/s)	Q (kJ/mol)	T	$D(T)$ (m²/s)
H in α-Fe (interstitial)	7×10^{-8}	6	25 °C	6×10^{-9}
C in α-Fe (interstitial)	1.1×10^{-6}	87	500 °C	1.4×10^{-12}
C in γ-Fe (interstitial)	2.3×10^{-5}	148	950 °C	1.1×10^{-11}
Fe in α-Fe (vacancy)	2.8×10^{-4}	251	500 °C	3×10^{-21}
Fe in γ-Fe (vacancy)	5.0×10^{-5}	284	1000 °C	1.1×10^{-16}
Cu in Cu	7.8×10^{-5}	211	500 °C	4×10^{-19}
Zn in Cu	2.4×10^{-5}	189	500 °C	4×10^{-18}
Al in Al	2.3×10^{-4}	144	500 °C	4×10^{-14}
Cu in Al	6.5×10^{-5}	136	500 °C	4×10^{-14}
B in Si	7.6×10^{-5}	334 (3.46 eV)	1100 °C	1.5×10^{-17}
P in Si	3.9×10^{-4}	353 (3.66 eV)	1100 °C	1.4×10^{-17}

Values are typical literature values; measured parameters scatter by tens of percent between sources. Two patterns stand out. Interstitial diffusion is fast and has low activation energy. Substitutional diffusion in metals has Q that scales with the melting point (an empirical rule puts $Q \approx 18RT_m$ for many FCC and BCC metals), so that at the same **homologous temperature** T/T_m most metals have similar diffusivities, about 10^{-12} m²/s near the melting point. Figure 8.3 collects these lines. That is why aluminium alloys are heat-treated near 500 °C and steels near 1000 °C, and why creep (which is diffusion-controlled) becomes a design concern above roughly $0.4T_m$ (Chapter 14).

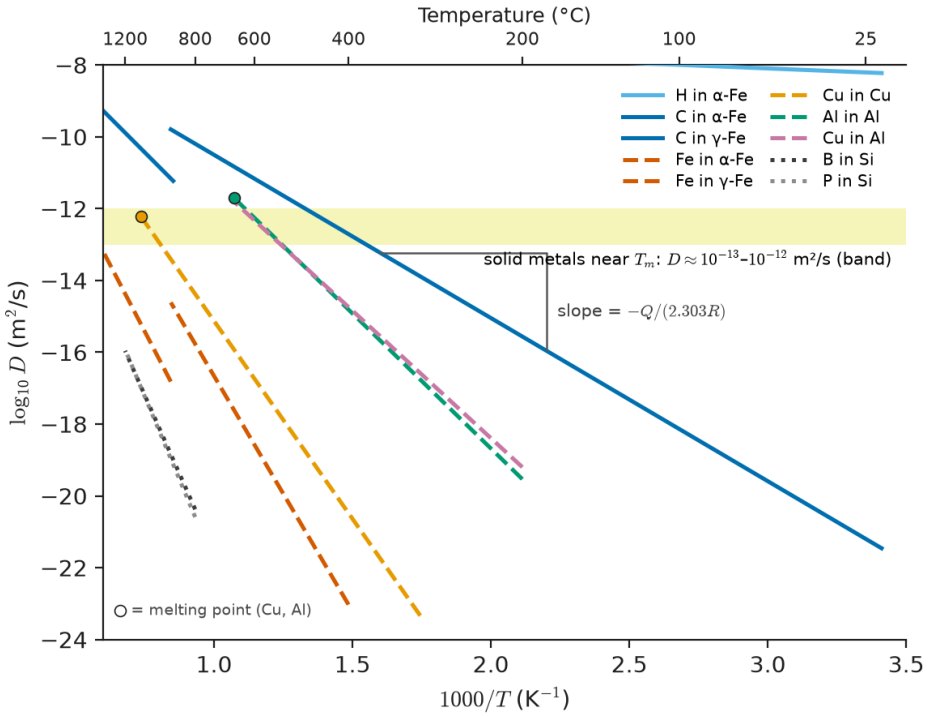

Figure 8.3 — Arrhenius plot of diffusion coefficients for interstitial and substitutional diffusion in metals and dopant diffusion in silicon.

Worked example — Jumps and distances for carbon in austenite. At 1000 °C, D for carbon in γ-iron is $2.3 \times 10^{-5}\exp[-148000/(8.314\times1273)] = 1.95\times10^{-11}$ m²/s. Carbon jumps between octahedral sites of FCC iron separated by $\lambda = a/\sqrt{2}$, with $a \approx 0.365$ nm at this temperature, so $\lambda = 0.258$ nm.

Jump frequency.

$$\Gamma = \frac{6D}{\lambda^2} = \frac{6\times1.95\times10^{-11}}{(2.58\times10^{-10})^2} = 1.8\times10^{9}\ \text{s}^{-1}$$

Each carbon atom jumps almost two billion times per second.

Distance in one second. The root-mean-square displacement is $\sqrt{6Dt} = \sqrt{6\times1.95\times10^{-11}\times1} = 1.1\times10^{-5}$ m, about 11 µm. The total path length travelled is $\Gamma t\lambda = 1.8\times10^{9}\times2.58\times10^{-10} = 0.45$ m.

Temperature sensitivity. For carbon in ferrite, D at 900 °C (1.4×10^{-10} m²/s) is about 4×10^{11} times larger than at 20 °C (3.5×10^{-22} m²/s). At room temperature a carbon atom in ferrite moves about $\sqrt{6Dt}\approx 0.26$ µm per year.

The atom travels half a metre to get 11 µm, which is what a random walk means. And the Arrhenius factor makes processes that take minutes at heat-treatment temperatures effectively frozen at room temperature; that is why a quenched or tempered microstructure is stable in service, while the same steel used above 400–500 °C in a power plant slowly changes over decades.

8.5 Self-diffusion, interdiffusion and the Kirkendall effect

Self-diffusion and tracer diffusion

In a pure metal, atoms continually exchange places with vacancies even though there is no concentration gradient. This **self-diffusion** is measured by depositing a radioactive isotope (^{64}Cu on copper, ^{59}Fe on iron) and using the thin-film solution; the result is the tracer diffusivity D^{*}. Self-diffusion controls high-temperature creep, sintering, grain growth and the climb of dislocations.

Intrinsic diffusion and interdiffusion

In a binary alloy with a composition gradient, each component has its own **intrinsic diffusivity**, D_A and D_B, measured relative to the crystal lattice. They are generally different. In α-brass, zinc diffuses faster than copper. If, across a diffusion couple of copper and brass, zinc atoms flow into the copper faster than copper atoms flow back, more atoms cross one way than the other. With the vacancy mechanism, the net flux of atoms is balanced by an opposite flux of vacancies, so lattice planes are created on one side and destroyed on the other. The lattice itself moves.

The Kirkendall experiment

In 1947 Smigelskas and Kirkendall wound molybdenum wires around a block of α-brass (Cu–30 wt% Zn), electroplated copper over it and annealed the couple at 785 °C. The inert Mo wires, marking the original interface, moved toward the brass: the brass side lost volume as zinc left faster than copper arrived. This **Kirkendall effect** (Figure 8.4) proved that diffusion in metals proceeds by a vacancy mechanism (a direct exchange mechanism would give equal fluxes), and overturned the then-common view.

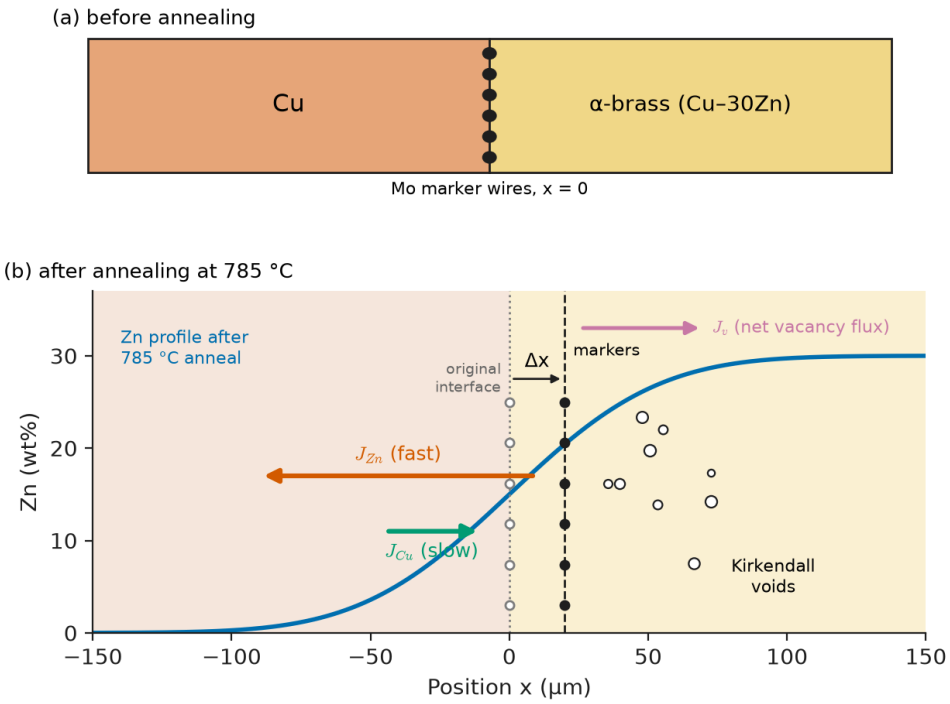

Figure 8.4 — The Kirkendall effect in a Cu/Cu–Zn diffusion couple: unequal intrinsic fluxes shift inert markers and can leave voids on the fast-diffuser side.

Darken's equations

Darken analysed the experiment by separating the motion into diffusion relative to the lattice and the motion of the lattice relative to the ends of the specimen. The **marker velocity** is

$$v = (D_A - D_B)\,\frac{\partial X_A}{\partial x}$$

and the composition evolves with a single **interdiffusion coefficient**

$$\tilde{D} = X_B D_A + X_A D_B$$

where X_A and X_B are mole fractions. $\tilde{D}$ is what is measured from the concentration profile of a diffusion couple (by the Boltzmann–Matano analysis, which extracts $\tilde{D}$ as a function of composition from one experiment). The intrinsic diffusivities are related to the tracer diffusivities by the **thermodynamic factor** $\phi = 1 + d\ln\gamma_A/d\ln X_A$ (Chapter 6): $D_A = D_A^{*}\phi$, so that

$$\tilde{D} = (X_B D_A^{*} + X_A D_B^{*})\,\phi$$

> **Status of the principle: Darken equations**
> *Approximate*. They assume vacancies are kept at equilibrium everywhere by sources and sinks, and they neglect correlation and vacancy-wind effects (Manning's corrections of a few tens of percent); they capture the marker shift and the composition dependence of $\tilde{D}$ well.

Engineering consequences of unequal diffusion

When vacancies flowing toward the faster diffuser's side cannot all be absorbed by dislocations and boundaries, they condense into **Kirkendall voids**. Examples are common and costly:

- **Wire bonds.** Gold wire bonded to aluminium pads forms Au–Al intermetallics; at 150–200 °C unequal diffusion leaves a line of voids that weakens and eventually opens the bond ("purple plague").
- **Solder joints.** Between copper pads and tin-based solder, Cu_3Sn grows at the copper side, and Kirkendall voids in it, especially with impurities in electroplated copper, cause brittle interfacial fracture under drop impact (Chapter 51).
- **Coatings.** Aluminide coatings on nickel superalloys degrade by interdiffusion with the substrate; voids at the interface promote spallation (Chapter 41).
- **Deliberate use.** Oxidizing or sulphidizing metal nanoparticles, where the metal diffuses outward faster than oxygen or sulphur diffuses inward, produces hollow nanoshells for catalysis and drug delivery (Chapter 38).

8.6 Short-circuit paths: grain boundaries, dislocations and surfaces

Why fast paths matter

A grain boundary is a slab of disordered material about $\delta \approx 0.5$ nm thick. Atoms there are less tightly bound, and vacancy-like free volume is abundant, so the activation energy for boundary diffusion Q_{gb} is typically 0.4–0.6 times the lattice value and D_{gb} is enormously larger. Dislocation cores act as pipes of radius about 0.5 nm (**pipe diffusion**), and free surfaces are faster still, with Q_s about a quarter to a half of the lattice value. The ordering for the same material and temperature is

$$D_{lattice} \ll D_{dislocation} < D_{gb} < D_{surface}$$

Because the activation energies differ, the ratio between them grows rapidly as temperature falls. At high temperature the volume of the lattice dominates total transport, since fast paths are a tiny fraction of the cross-section; below roughly 0.5–$0.6\,T_m$, fast paths carry most of the flux. That is why grain-boundary diffusion controls low-temperature sintering of powders, Coble creep of fine-grained materials, grain-boundary oxidation and embrittlement, the interdiffusion of thin-film stacks in microelectronics (Chapter 49), and electromigration in copper interconnects, where atoms move along interfaces with the cap layer (Chapter 50).

Harrison's A, B and C regimes

When a tracer diffuses into a polycrystal from the surface, the shape of the profile depends on how far atoms leak out of the boundaries into the grains, $\sqrt{D_l t}$, compared with the grain size d and the boundary width δ. Harrison (1961) classified three kinetic regimes (Figure 8.5):

- **Type A** ($\sqrt{D_l t} \gtrsim d$): lattice diffusion fields of neighbouring boundaries overlap. The polycrystal behaves as a homogeneous medium with an effective diffusivity $D_{eff} = g D_{gb} + (1-g) D_l$, where $g \approx 3\delta/d$ is the volume fraction of boundary material. High temperature, long times, fine grains.
- **Type B** ($\delta \ll \sqrt{D_l t} \ll d$): each boundary is an isolated fast channel that leaks into the grain on either side. The profile has a near-surface lattice part and a long tail whose slope, analysed with the Whipple–Suzuoka solution, gives the product $s\delta D_{gb}$ (s is a segregation factor). Intermediate temperatures; the most commonly measured regime.
- **Type C** ($\sqrt{D_l t} < \delta$): essentially no leakage; transport is confined to the boundaries, and D_{gb} is measured directly. Low temperatures, requiring very sensitive detection.

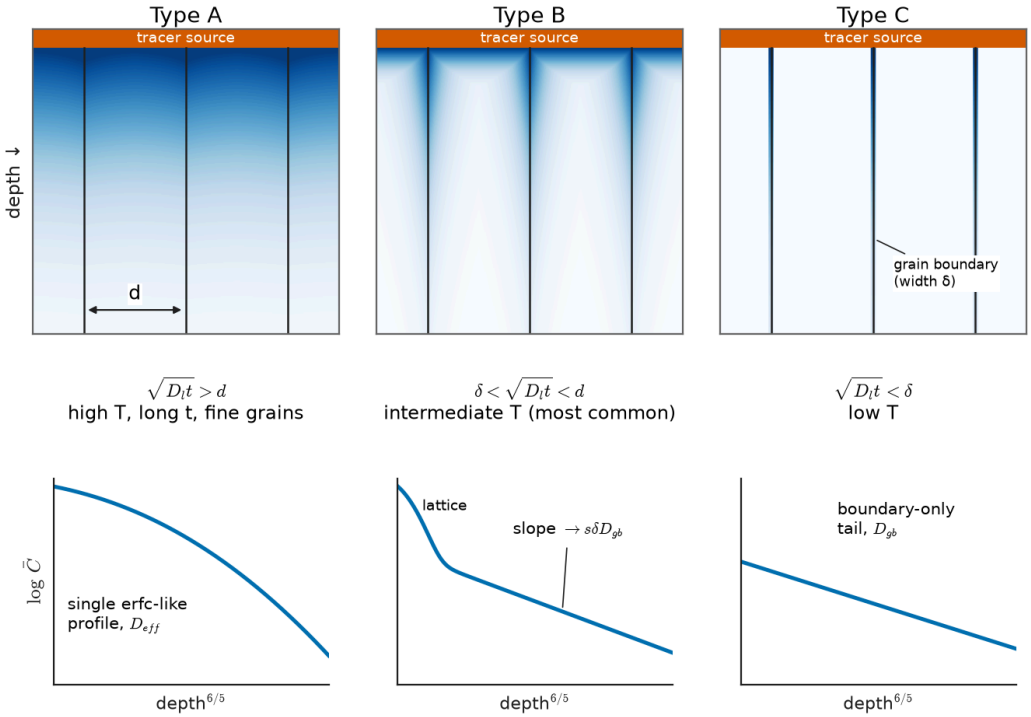

Figure 8.5 — Harrison's A, B and C regimes of grain-boundary diffusion in a polycrystal, with the corresponding penetration profiles.

Worked example — Grain boundaries versus lattice in copper at 400 °C. Take lattice self-diffusion in copper $D_l = 7.8 \times 10^{-5}\exp(-211000/RT)$ and an illustrative boundary diffusivity $D_{gb} = 1 \times 10^{-5}\exp(-105000/RT)$ m²/s (activation energy half the lattice value). At $T = 673$ K,

$$D_l = 3.3 \times 10^{-21}\ \mathrm{m^2/s}, \qquad D_{gb} = 7.1 \times 10^{-14}\ \mathrm{m^2/s}$$

a ratio of about 2×10^7. After 1 h ($t = 3600$ s) the characteristic distances are $\sqrt{D_l t} = 3.4$ nm in the lattice and $\sqrt{D_{gb} t} = 16$ µm along boundaries. For a grain size of 50 µm, $\delta\,(0.5\ \mathrm{nm}) < \sqrt{D_l t}\,(3.4\ \mathrm{nm}) \ll d$: this is the type-B regime. Even after 1000 h, lattice diffusion moves atoms only about 0.1 µm, while boundary transport covers about 0.5 mm.

The result explains a common observation: a copper part annealed at 400 °C can show oxidation, solder-element penetration or tracer uptake along grain boundaries hundreds of micrometres deep while the grain interiors are untouched. The specific numbers depend on the boundary parameters chosen, but the order-of-magnitude conclusion holds for most metals below about half their melting temperature.

8.7 Diffusion in ionic solids

Charged defects and conductivity

In an ionic crystal, diffusion of cations and anions is mediated by point defects that carry charge: Schottky pairs (vacancies on both sublattices), Frenkel pairs (a vacancy and an interstitial on one sublattice), and defects created by aliovalent dopants (Chapter 5). Because the moving species are charged, diffusion and electrical conduction are two views of the same jumps. A field E biases the random walk, producing a drift velocity $v = uE$, where u is the mobility. The **Nernst–Einstein relation** links the mobility to the diffusivity:

$$\frac{u}{D} = \frac{zq}{kT} \qquad \Longrightarrow \qquad \sigma = \frac{n\,(zq)^2 D}{kT}$$

where n is the concentration of mobile species of charge zq and σ the ionic conductivity.

> **Status of the principle: Nernst–Einstein relation**
> *Derived.* It follows from requiring that drift and diffusion fluxes balance in equilibrium (Einstein relation); tracer and conductivity diffusivities differ by the correlation factor and a "Haven ratio" (typically 0.3–1), and the relation breaks down when ions move in correlated groups.

Defect-controlled diffusion

Ionic diffusion is controlled by the concentration of defects, which can be set by temperature (**intrinsic** regime) or fixed by dopants (**extrinsic** regime). In NaCl, adding $CdCl_2$ creates one cation vacancy per Cd^{2+} and raises sodium diffusion at moderate temperatures by orders of magnitude; on an Arrhenius plot a "knee" separates the extrinsic region (slope H_m only) from the intrinsic region (slope $H_m + H_f/2$).

The same principle, defect engineering by doping, is the basis of solid electrolytes. In yttria-stabilized zirconia (Chapter 7), each Y_2O_3 unit replacing two ZrO_2 creates one oxygen vacancy, and oxide ions hop through these vacancies. In 8YSZ, $Zr_{0.852}Y_{0.148}O_{1.926}$, about 3.7 % of oxygen sites are empty.

Worked example — Vacancy diffusivity in a fuel-cell electrolyte. 8YSZ has a fluorite cell with lattice parameter $a = 0.514$ nm and four cations per cell; its ionic conductivity is about 3 S/m at 800 °C. Find the diffusivity of oxygen vacancies.

Vacancy concentration. There are 0.074 vacancies per cation, so per cell $4 \times 0.074 = 0.296$, and

$$n_v = \frac{0.296}{(0.514 \times 10^{-9})^3} = 2.18 \times 10^{27}\ \text{m}^{-3}$$

Nernst–Einstein, with vacancy charge $2q = 3.20 \times 10^{-19}$ C and $kT = 1.381 \times 10^{-23} \times 1073 = 1.48 \times 10^{-20}$ J:

$$D_v = \frac{\sigma kT}{n_v (2q)^2} = \frac{3 \times 1.48 \times 10^{-20}}{2.18 \times 10^{27} \times (3.20 \times 10^{-19})^2} = 2.0 \times 10^{-10}\ \text{m}^2/\text{s}$$

The vacancies are almost as mobile as carbon in austenite at 1000 °C. Spread over all oxide ions ($n_O = 5.7 \times 10^{28}$ m⁻³), the equivalent oxygen diffusivity is about 8×10^{-12} m²/s.

This is why an SOFC electrolyte must be thin (5–20 μm in anode-supported cells): its area-specific resistance is h/σ, so a 10 μm 8YSZ layer at 800 °C contributes $10^{-5}/3 \approx 3 \times 10^{-6}\ \Omega\,\text{m}^2$ (0.03 Ω cm²), small next to the electrodes. Below 600 °C, the Arrhenius fall in D makes YSZ too resistive, which drives the search for gadolinia-doped ceria and other electrolytes (Chapter 40). The same physics governs the oxide scales that protect high-temperature alloys: alumina and chromia grow slowly because ion diffusion through them is slow (Chapter 42), and lithium-ion battery electrodes are limited by Li diffusion over particle radii of micrometres (Chapter 40).

8.8 Diffusion in polymers and glasses

Polymers

Small molecules (water, oxygen, CO_2, solvents, plasticizers) diffuse through polymers by jumping between transient holes created by the thermal motion of chain segments. Diffusivity therefore depends on the **free volume**, which increases sharply above the glass transition temperature T_g (Chapter 20). Typical values at room temperature range from about 10^{-9} m²/s for small gases in rubbers to 10^{-13} m²/s for water in glassy epoxies.

For packaging and barrier applications, the property that matters is permeability $P = DS$:

Polymer	O_2 permeability (relative, LDPE = 1)	Typical use
Silicone rubber (PDMS)	~200	Contact lenses, gas-exchange membranes
LDPE	1	Films, bags
HDPE	~0.3	Bottles, pipes
PET	~0.02	Drink bottles
PVDC	~0.002	Food barrier films
EVOH (dry)	~0.0002	Barrier layer in multilayer packaging

Semicrystalline polymers are better barriers than amorphous ones because crystallites are nearly impermeable and lengthen the diffusion path; a PET bottle loses CO_2 slowly enough for a shelf life of months, while an LDPE bottle would not keep carbonation for days. Multilayer films with a thin EVOH core (protected from moisture, which plasticizes it) combine low cost and excellent barrier properties.

Moisture in composites and electronics. Epoxy matrices in carbon- and glass-fibre composites absorb 1–3 wt% water at saturation, which lowers T_g and matrix-dominated strength. Uptake follows Fickian kinetics in a plate of thickness h exposed on both faces:

$$\frac{M_t}{M_\infty} \approx \frac{4}{h}\sqrt{\frac{Dt}{\pi}} \qquad (\text{for } M_t/M_\infty \lesssim 0.6)$$

With $D = 1 \times 10^{-12}$ m²/s and $h = 2$ mm, half saturation takes $t = \pi(0.5h/4)^2/D = 2.0 \times 10^5$ s, about 2.3 days, and near-saturation takes several weeks ($h^2/D \approx 46$ days is the characteristic time). Aerospace qualification of composite parts uses accelerated conditioning (70 °C, 85 % relative humidity) to reach saturation before testing hot–wet properties. The same equation governs moisture in IC mould compounds, which turns to steam during reflow soldering and cracks packages ("popcorning").

Case II diffusion. When a solvent diffuses into a glassy polymer and plasticizes it, a sharp front moves at constant velocity rather than as $\sqrt{t}$; this non-Fickian behaviour causes solvent crazing of polycarbonate and PMMA and is used in controlled drug release from hydrogels (Chapter 39).

Inorganic glasses

In silicate glasses, network-forming silicon and oxygen move very slowly, but network-modifying alkali ions (Na^+, K^+, Li^+) hop relatively easily through the open network. This gives soda-lime glass its small ionic conductivity and makes **chemical strengthening** possible: a sheet of aluminosilicate glass is immersed in molten KNO_3 at about 400–450 °C, and K^+ ions from the salt exchange with Na^+ ions in the glass. The larger potassium ions are stuffed into sites sized for sodium, creating a compressive surface layer of several hundred MPa. With an interdiffusion coefficient of order 10^{-14} m²/s, the exchanged layer reaches $2\sqrt{Dt} \approx 34$ μm in 8 h, the order of the tens of micrometres used in phone cover glass. Because cracks must first overcome the compression, strength and scratch-induced fracture resistance rise several-fold (Chapter 17).

Gas permeation through glasses matters in vacuum and cryogenic systems: helium permeates fused silica fast enough to limit the life of some sealed devices, while hydrogen and water diffusion into silica optical fibres raises their optical loss.

8.9 Uphill diffusion and chemical-potential gradients

The true driving force

Fick's law says atoms flow down concentration gradients. But at equilibrium what must be uniform is not concentration but chemical potential (Chapter 6). The general flux equation is

$$J_i = -M_i C_i \frac{\partial \mu_i}{\partial x}$$

where M_i is a mobility. For an ideal dilute solution, $\mu_i = \mu_i^\circ + RT\ln C_i$ and this reduces to Fick's law with $D_i = M_i RT$. For non-ideal solutions it gives Fick's law with the thermodynamic factor of Section 8.5. When the thermodynamic factor becomes negative, or when a second element changes the activity of the first, atoms can diffuse **up** their own concentration gradient.

> **Status of the principle: Chemical-potential-driven diffusion**
> *Fundamental.* Linear irreversible thermodynamics identifies the chemical-potential gradient as the driving force for mass transport; Fick's law is its special case for ideal or dilute solutions with no cross effects.

Darken's uphill-diffusion experiment

In 1949 Darken welded together two steels with nearly the same carbon content (about 0.45 and 0.48 wt% C), one of which contained 3.8 wt% Si. Silicon raises the activity of carbon in austenite. After annealing at 1050 °C for 13 days, carbon had flowed out of the silicon-bearing steel into the other, building a large step in carbon concentration at the weld: carbon diffused up its concentration gradient but down its chemical-potential gradient. Silicon itself, a slow substitutional diffuser, barely moved.

Engineering consequences

- **Dissimilar welds.** In welds between low-alloy ferritic steel and austenitic stainless steel or nickel alloys in power plants, chromium in the stainless side lowers carbon activity. Carbon migrates across the fusion line during service at 500–600 °C, leaving a soft decarburized band in the ferritic steel and a hard carbide-rich band on the other side; creep cracks start there (Chapter 47).
- **Spinodal decomposition.** Inside a miscibility gap, where $\partial^2 G/\partial X^2 < 0$, the effective interdiffusion coefficient is negative: small composition fluctuations grow instead of decaying. This is how Fe–Cr stainless steels embrittle at 475 °C and how Alnico magnets and some glass-ceramics form their fine two-phase structures (Chapter 10).
- **Stress-driven diffusion.** A gradient of hydrostatic stress changes chemical potential: hydrogen accumulates in the tensile region ahead of a crack tip in high-strength steel and causes hydrogen embrittlement (Chapter 44). Under an electric current, the "electron wind" adds a force that drives electromigration in interconnects.
- **Thermal gradients (Soret effect)** drive species separation in nuclear fuel pellets and in solders under steep temperature gradients.

8.10 Engineering applications

Carburizing

Gears, camshafts, bearings and transmission shafts need a hard, wear- and fatigue-resistant surface on a tough core. Low-carbon steels (0.15–0.25 wt% C, such as 8620 or 20MnCr5) are heated into the austenite field, typically 900–950 °C, in an atmosphere whose carbon potential holds the surface at 0.8–1.0 wt% C. Carbon diffuses inward following the error-function solution; after quenching the high-carbon case becomes hard martensite (about 60 HRC) and the core stays tough. Case depth is commonly defined as the depth to 0.40 wt% C or to a hardness of 50 HRC.

Worked example — Carburizing time and the cost of a cold furnace. A 0.20 wt% C gear steel is carburized at 950 °C with the surface held at 1.0 wt% C. How long does it take to reach 0.40 wt% C at a depth of 1.0 mm, and what case depth results if the furnace is actually at 925 °C?

Diffusivity. $D = 2.3\times10^{-5}\exp[-148000/(8.314\times1223)] = 1.10\times10^{-11}$ m²/s.

Error function.

$$\frac{C_x - C_0}{C_s - C_0} = \frac{0.40-0.20}{1.0-0.20} = 0.25 = 1-\operatorname{erf}(z) \;\Rightarrow\; \operatorname{erf}(z) = 0.75,\; z = 0.813$$

Time. With $z = x/(2\sqrt{Dt})$,

$$t = \frac{x^2}{4Dz^2} = \frac{(1.0\times10^{-3})^2}{4\times1.10\times10^{-11}\times0.813^2} = 3.44\times10^{4}\ \text{s} \approx 9.5\ \text{h}$$

Cold furnace. At 925 °C (1198 K), $D = 8.1 \times 10^{-12}$ m²/s, 26 % lower. After the same 9.5 h, the depth to 0.40 wt% C is $x = 2z\sqrt{Dt} = 2 \times 0.813 \times \sqrt{8.1 \times 10^{-12} \times 3.44 \times 10^{4}} = 0.86$ mm.

Hotter furnace. At 1000 °C, $D = 1.95 \times 10^{-11}$ m²/s and the same 1.0 mm case takes 5.4 h, which is why vacuum (low-pressure) carburizing at 980–1050 °C is attractive, limited by austenite grain growth.

A 25 °C error removed 14 % of the case depth (Figure 8.6), the failure described at the start of the chapter. In practice the case-depth reduction is somewhat worse than the error-function estimate, because D of carbon in austenite increases with carbon content, and the case is defined by hardness rather than carbon alone; carburizing simulation software solves Fick's second law with concentration-dependent D and a surface mass-transfer coefficient. Related thermochemical treatments are nitriding (500–570 °C, in ferrite, forming hard nitrides, no quench needed, for crankshafts and extrusion dies), carbonitriding and boriding.

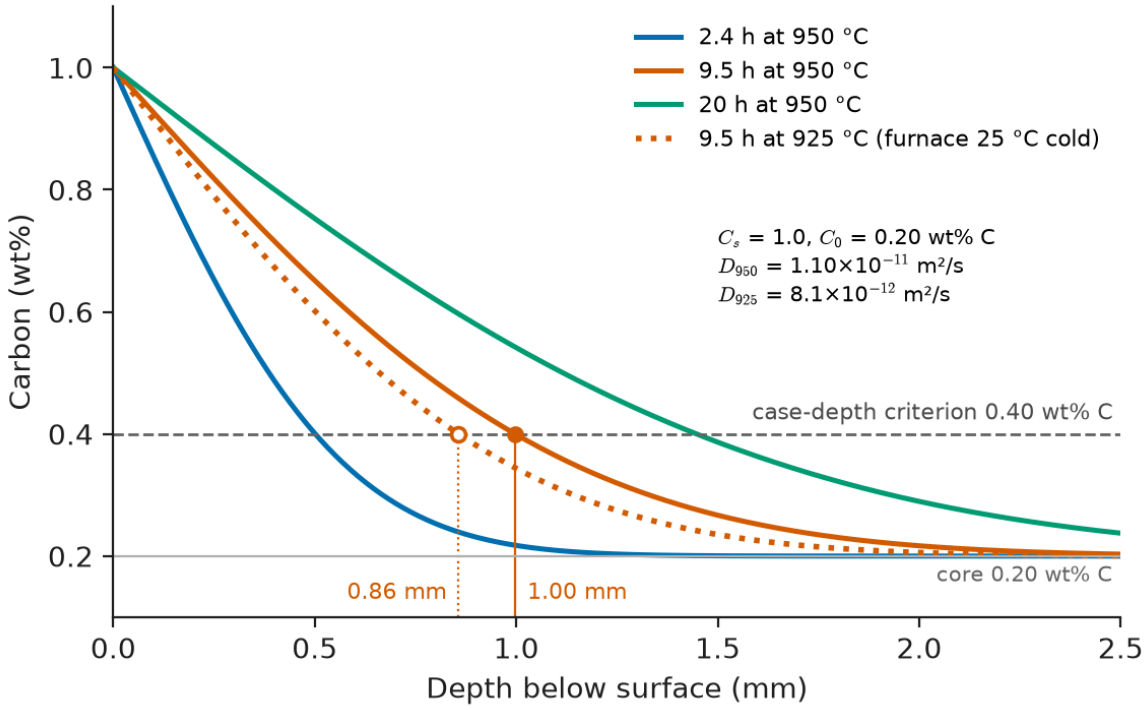

Figure 8.6 — Carbon profiles in a carburized 0.20 wt% C steel at 950 °C for several times, and the effect of a 25 °C temperature error on case depth.

Doping semiconductors

Before ion implantation became dominant, silicon devices were doped by diffusion in two steps: a **predeposition** at constant surface concentration (erfc profile), then a **drive-in** with the source removed (Gaussian profile). Today ions are implanted and then annealed, and the anneal still redistributes the dopant by diffusion.

At 1100 °C boron has $D \approx 1.5 \times 10^{-17}$ m²/s (table above). A 1 h drive-in gives $2\sqrt{Dt} = 0.47$ µm. If a dose $Q = 10^{15}$ cm⁻² (10^{19} m⁻²) is driven in, the Gaussian solution gives a surface concentration $Q/\sqrt{\pi Dt} = 2.4 \times 10^{19}$ cm⁻³ and the depth at which boron falls to a background of 10^{16} cm⁻³ (the p–n junction) is

$$x_j = \sqrt{4Dt \ln \frac{C_{surface}}{C_B}} = 1.3\ \mu\text{m}$$

Modern transistors need junctions only tens of nanometres deep, so thermal budgets ($\sum Dt$ over all hot steps) are minimized with millisecond flash and laser anneals that activate dopants while moving them only a few nanometres (Chapters 32 and 50).

Homogenization

Cast ingots of aluminium, copper, nickel and steel alloys are cored (Chapter 7). Before hot working, they are homogenized: held at high temperature until the segregation decays. The separation-of-variables solution gives the time directly.

Worked example — Homogenizing an aluminium–copper ingot. A direct-chill-cast Al–Cu ingot has a secondary dendrite arm spacing of 50 µm, so the segregation half-wavelength is $l = 25$ µm. It is homogenized at 500 °C, where $D_{Cu\ in\ Al} = 6.5 \times 10^{-5} \exp[-136000/(8.314 \times 773)] = 4.2 \times 10^{-14}$ m²/s. How long does it take to reduce the segregation amplitude to 1 % of its initial value?

Relaxation time.

$$\tau = \frac{l^2}{\pi^2 D} = \frac{(25 \times 10^{-6})^2}{9.87 \times 4.2 \times 10^{-14}} = 1.5 \times 10^3 \text{ s}$$

Time to 1 %. $\beta/\beta_0 = \exp(-t/\tau) = 0.01$, so $t = \tau \ln 100 = 4.6\tau = 6.9 \times 10^3$ s ≈ 1.9 h. At 530 °C the time falls to about 0.9 h.

In practice industrial homogenization of aluminium alloys takes 4–24 h. The difference is instructive: real ingots have coarser spacing near the centre (time scales as l^2: doubling the spacing quadruples the time), the slowest-diffusing elements such as Mn and Cr are the ones that must be homogenized, and the soak also dissolves non-equilibrium eutectic particles and precipitates dispersoids. The simple solution gives the scaling; the specification adds the metallurgy.

Other applications at a glance

Process or failure	Diffusing species	Typical conditions
Nitriding of tool and crankshaft steels	N in α-Fe	500–570 °C, 10–80 h, case 0.1–0.6 mm
Decarburization during forging	C out of steel	1100–1250 °C, minutes; soft surface, fatigue loss
Diffusion bonding of Ti fan blades	Ti self-diffusion, creep	900–950 °C, a few MPa, hours
Sintering of ceramics and powder metals	Boundary and lattice diffusion	0.6–0.9 T_m
Hydrogen embrittlement of high-strength bolts	H in α-Fe	Room temperature, hours to days
Interdiffusion in thermal-barrier coatings	Al, Ni, Cr	1000–1150 °C, thousands of hours
Dopant diffusion and gettering in Si wafers	B, P, As, Fe, Cu	800–1100 °C
Ion exchange in cover glass	K^+/Na^+	400–450 °C, hours

Boundary.
The statistical-mechanical basis of the Boltzmann factor and of transition-state theory belongs to the Engineering Physics and Advanced Chemistry books; this chapter uses their results. Diffusion of charge carriers (electrons and holes) in semiconductors follows the same mathematics and is treated in Chapter 32.

What remains open

Open question	Where it is answered
What sets the equilibrium vacancy concentration and the chemical potentials that drive diffusion?	Chapters 5 and 6
What phases and compositions does diffusion move a system toward?	Chapter 7
How does diffusion control the growth of nuclei, dendrites and solidification segregation?	Chapter 9
How do precipitation, coarsening, spinodal decomposition and pearlite growth depend on diffusion?	Chapter 10
How does diffusion produce creep, and how are creep mechanisms mapped?	Chapter 14
How are carburizing, nitriding and heat treatments specified for steels?	Chapter 15
How do ionic conductors serve in fuel cells and batteries?	Chapter 40
How do oxide scales grow by diffusion, and how is hydrogen embrittlement controlled?	Chapters 42 and 44
How are diffusion profiles measured (SIMS, EPMA, tracer methods)?	Chapters 52 and 53
How are jump barriers computed from first principles and diffusion simulated atom by atom?	Chapters 55 and 56

Next: Chapter 9 combines the driving forces of Chapter 6 with the atomic mobility of this chapter to explain how new phases are born and grow, from the first nucleus in an undercooled melt to the grains of a casting.

Chapter 9 — Nucleation, Growth and Solidification

Part II · Thermodynamics and Kinetics

An aluminium rolling mill receives a batch of direct-chill cast slabs that crack along their edges during the first hot-rolling pass. The fracture surfaces show grains several centimetres long, columnar crystals that grew inward from the mould wall like the bristles of a brush, with brittle films of the last liquid between them. The previous batch, from the same alloy and the same caster, rolled without trouble; its grains were equiaxed and about 150 μm across. The difference was a reel of thin rod that an operator had forgotten to feed into the launder: aluminium–titanium–boron master alloy, added at about 1 kg per tonne of melt. Its job is to scatter roughly a million tiny TiB_2 particles through every cubic centimetre of liquid, each one a site where a new grain can start with almost no undercooling. Without them, the few grains that nucleated at the mould wall grew unchecked across the whole slab.

The phase diagram (Chapter 7) said the alloy would solidify as α-aluminium. It did not say how many grains, of what shape, with what composition differences between their centres and edges. Those depend on **nucleation**, the birth of a new phase, and **growth**, its advance into the old one. This chapter builds both from the thermodynamics of Chapter 6 and the diffusion of Chapter 8, then applies them to solidification, the most widespread processing step in materials engineering: casting of metals, welding, crystal growth of silicon and turbine blades, additive manufacturing, glass formation and even the freezing of water in clouds.

> **Why does a liquid cool below its melting point before it freezes, and what makes some particles such good starting points for crystals?**
> **What sets the shape, size and composition variations of the crystals that grow, and how fast must a melt be cooled to avoid crystallizing at all?**

Concept	Engineering destination
Driving force and undercooling	Supercooled water in clouds and aircraft icing; precipitation in alloys
Homogeneous nucleation	Limits of undercooling; why pure clean melts and droplets resist freezing
Heterogeneous nucleation and inoculation	Grain refiners in Al and Mg, inoculation of cast iron, cloud seeding, weld-metal toughness
Interface- and diffusion-controlled growth	Crystal-growth rates, precipitate growth, polymer spherulites
JMAK kinetics and TTT diagrams	Annealing and recrystallization schedules, polymer moulding cycles, glass formation
Constitutional supercooling	Planar growth of semiconductor crystals; dendritic growth in castings and welds
Scheil equation and zone refining	Microsegregation, freezing range, purification of silicon and germanium
Eutectic and peritectic solidification	Al–Si and cast-iron microstructures, permanent magnets, continuous casting
Rapid solidification and metallic glasses	Atomized powders, additive manufacturing, amorphous transformer cores, sporting goods

Roadmap. Driving force (9.1) → homogeneous nucleation (9.2) → heterogeneous nucleation and inoculation (9.3) → growth (9.4) → overall kinetics and TTT diagrams (9.5) → solidification fronts and constitutional supercooling (9.6) → segregation (9.7) → eutectic, peritectic and casting structures (9.8) → rapid solidification and metallic glasses (9.9).

9.1 Driving force for transformation

Undercooling a liquid

At the melting temperature T_m, liquid and solid have equal Gibbs energies; nothing drives either to become the other. Below T_m the solid has the lower Gibbs energy. Per unit volume of solid formed, the change is

$$\Delta G_v = \Delta H_v - T\Delta S_v$$

where ΔH_v is the latent heat released per unit volume (taken as positive here, so ΔG_v is the magnitude of the drop). At T_m, $\Delta G_v = 0$, so $\Delta S_v = \Delta H_v / T_m$. If ΔH and ΔS change little with temperature over the range of interest,

$$\Delta G_v \approx \frac{\Delta H_v \, \Delta T}{T_m}$$

where $\Delta T = T_m - T$ is the **undercooling** (or supercooling).

> **Status of the principle: Linear driving force with undercooling**
> *Approximate.* It assumes the enthalpy and entropy of fusion are independent of temperature, which holds within about 20 % for undercoolings up to roughly $0.2\,T_m$ in metals; for deeply undercooled glass-forming liquids, the heat-capacity difference between liquid and crystal must be included.

The driving force is zero at the melting point and rises linearly with undercooling. For copper ($\Delta H_v = 1.88 \times 10^9$ J/m³, $T_m = 1358$ K), 100 K of undercooling gives $\Delta G_v = 1.4 \times 10^8$ J/m³, or about 1 kJ/mol, small compared with $RT \approx 10$ kJ/mol. The transformation is driven by a modest free-energy difference, which is why it can be resisted, for a while, by the cost of creating an interface.

Solid-state driving forces

The same idea applies to any transformation. For precipitation from a supersaturated solid solution of solute concentration C_0 when the equilibrium solubility is C_e, the driving force per unit volume of precipitate is, for a dilute ideal solution, roughly

$$\Delta G_v \approx \frac{RT}{V_m} \ln \frac{C_0}{C_e}$$

where V_m is the molar volume of the precipitate. The driving force grows with supersaturation, and supersaturation grows as the alloy is cooled further below its solvus. For eutectoid and allotropic transformations, ΔG_v scales with undercooling below the transformation temperature, as for solidification. In each case a CALPHAD database (Chapter 7) supplies the exact value.

9.2 Homogeneous nucleation

The energy of an embryo

Small clusters of solid-like atoms form and dissolve continually in a liquid by thermal fluctuation. Consider a spherical cluster of radius r forming inside the liquid, far from any wall or particle (**homogeneous** nucleation). Its volume lowers the Gibbs energy by ΔG_v per unit volume, but its surface raises it by the solid–liquid interfacial energy γ_{SL} per unit area:

$$\Delta G(r) = -\frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma_{SL}$$

For small r the surface term ($\propto r^2$) dominates and the cluster is unstable; for large r the volume term ($\propto r^3$) wins. $\Delta G(r)$ passes through a maximum at the **critical radius**

$$r^* = \frac{2\gamma_{SL}}{\Delta G_v}$$

where the barrier is

$$\Delta G^* = \frac{16\pi \gamma_{SL}^3}{3\,\Delta G_v^2} = \frac{16\pi \gamma_{SL}^3 T_m^2}{3\,\Delta H_v^2\,\Delta T^2}$$

Figure 9.1 plots the two terms and their sum for copper. Clusters smaller than r^* (**embryos**) are more likely to shrink than grow; clusters larger than r^* (**nuclei**) are more likely to grow. The critical radius is the same Gibbs–Thomson condition met in Chapter 6: a particle of radius r^* is in (unstable) equilibrium with the undercooled liquid because its curvature lowers its melting point by exactly ΔT.

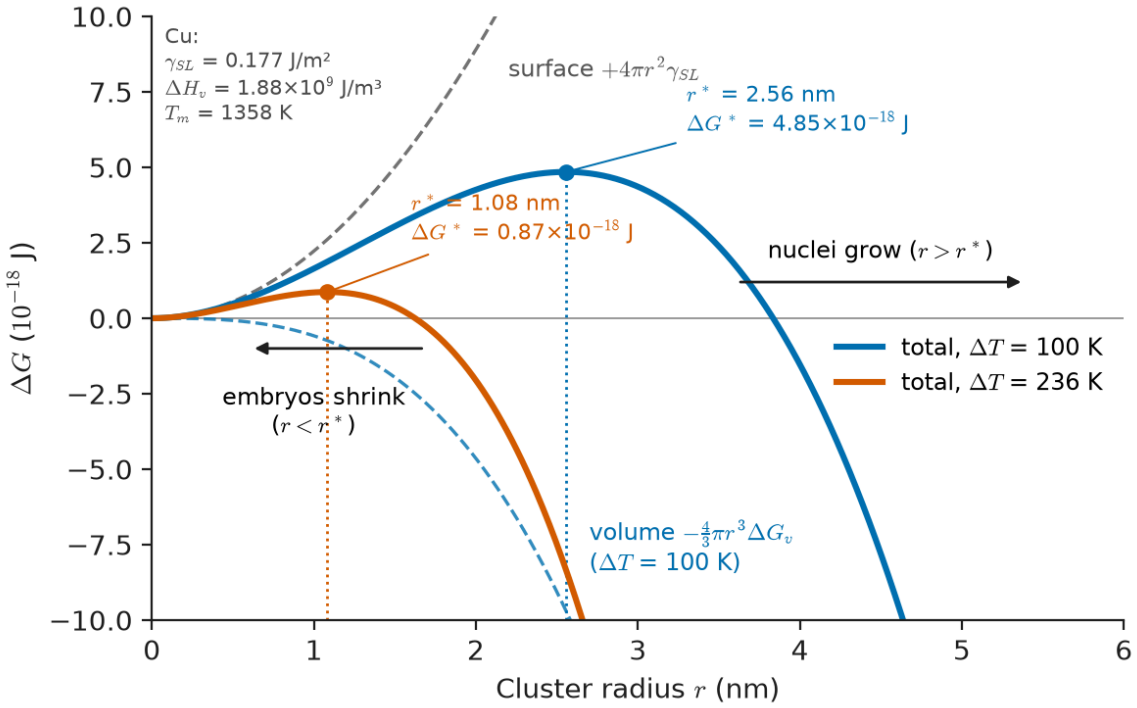

Figure 9.1 — Gibbs energy of a spherical solid cluster in undercooled liquid copper versus radius, showing the surface and volume terms and the critical nucleus at two undercoolings.

The nucleation rate

The equilibrium number of critical clusters per unit volume follows a Boltzmann distribution, $N^* = N_0 \exp(-\Delta G^*/kT)$, where N_0 is the number of atoms per unit volume. Each critical cluster becomes a growing nucleus when one more atom joins it, at a rate set by atomic jumps across the interface (a frequency ν that contains its own activation energy for atomic transport, as in Chapter 8). The steady-state **nucleation rate** is therefore

$$I = I_0 \exp\left(-\frac{\Delta G^*}{kT}\right), \qquad I_0 \approx N_0\,\nu \sim 10^{41}\text{–}10^{42}\ \mathrm{m^{-3}\,s^{-1}}\ \text{(metal melts)}$$

with a Zeldovich factor (about 0.01–0.1) and the temperature dependence of ν absorbed into I_0 for a liquid metal, whose atomic mobility is high.

> **Status of the principle: Classical nucleation theory**
> *Approximate.* It treats a cluster of a few hundred atoms as a sphere with bulk properties and a sharp interface of macroscopic energy; it predicts trends and the extreme sensitivity to undercooling correctly but can miss absolute rates by many orders of magnitude, mainly because γ_{SL} is uncertain and enters cubed.

Worked example — How far can liquid copper undercool? Take for copper $\gamma_{SL} = 0.177\ \mathrm{J/m^2}$, $\Delta H_v = 1.88 \times 10^9\ \mathrm{J/m^3}$, $T_m = 1358\ \mathrm{K}$, atomic volume $1.18 \times 10^{-29}\ \mathrm{m^3}$, and $I_0 = 10^{42}\ \mathrm{m^{-3}\,s^{-1}}$.

At $\Delta T = 100\ K$. $\Delta G_v = 1.88 \times 10^9 \times 100/1358 = 1.38 \times 10^8\ \mathrm{J/m^3}$, so

$$r^* = \frac{2 \times 0.177}{1.38 \times 10^8} = 2.56\ \mathrm{nm}, \qquad \Delta G^* = \frac{16\pi (0.177)^3}{3(1.38 \times 10^8)^2} = 4.85 \times 10^{-18}\ \mathrm{J}$$

The nucleus contains $\frac{4}{3}\pi r^{*3}/\Omega \approx 5900$ atoms, and $\Delta G^*/kT = 4.85 \times 10^{-18}/(1.381 \times 10^{-23} \times 1258) = 279$. The rate is $10^{42} e^{-279} \approx 10^{-79}\ \mathrm{m^{-3}\,s^{-1}}$: effectively zero.

At $\Delta T = 236\ K$. $r^* = 1.08$ nm (about 450 atoms), $\Delta G^* = 8.7 \times 10^{-19}$ J, $\Delta G^*/kT = 56$, and $I \approx 10^{42} e^{-56} \approx 5 \times 10^{17}\ \mathrm{m^{-3}\,s^{-1}}$: a droplet of 1 mm³ would nucleate within a few nanoseconds.

Onset. Solving $I = 10^{6}$ m^{-3} s^{-1} (one nucleus per cubic centimetre per second) gives $\Delta T \approx 190$ K, or $0.14\,T_m$. Changing ΔT by ±20 K around this value changes I from about 10^{-2} to 10^{12} m^{-3} s^{-1}.

The theory predicts that copper must undercool by roughly 200 K before homogeneous nucleation is noticeable, and that the onset is abrupt: below the threshold nothing happens, and a few tens of kelvin further the liquid freezes instantly. Turnbull's classic experiments on small droplets, each dispersed so that most contained no impurity particle, found maximum undercoolings of about $0.18\,T_m$ for many metals (236 K for copper), consistent with this estimate. In ordinary casting, metals freeze with undercoolings of only a few kelvin. Something other than homogeneous nucleation is at work.

Water, clouds and aircraft

Water shows the same behaviour. Pure micrometre-sized cloud droplets remain liquid down to about −38 °C, the limit of homogeneous nucleation of ice. Clouds of supercooled droplets between 0 and about −20 °C are a serious aviation hazard: when an aircraft flies through them, the droplets freeze on impact on the wing leading edges, which become heterogeneous nucleation sites. Ice-protection systems (heated leading edges, pneumatic boots, icephobic coatings) are designed around this metastability.

9.3 Heterogeneous nucleation

The spherical cap

Nucleation almost always begins on a pre-existing surface: a mould wall, an oxide film, an inclusion, a deliberately added particle. Consider a solid nucleus forming as a spherical cap on a flat substrate, with contact (wetting) angle θ set by the balance of interfacial tensions (Young's equation, Chapter 6):

$$\gamma_{LS'} = \gamma_{SS'} + \gamma_{SL}\cos\theta$$

where S' denotes the substrate. Repeating the calculation of Section 9.2 for the cap shows that the critical radius of curvature is unchanged, $r^{*} = 2\gamma_{SL}/\Delta G_v$, but the cap contains fewer atoms than a full sphere, and the barrier is reduced by a geometric factor:

$$\Delta G^{*}_{het} = \Delta G^{*}_{hom}\, f(\theta), \qquad f(\theta) = \frac{(2+\cos\theta)(1-\cos\theta)^{2}}{4}$$

> **Status of the principle: Spherical-cap model of heterogeneous nucleation**
> *Approximate*. It inherits the assumptions of classical nucleation theory and assumes a flat, chemically uniform substrate with a macroscopic contact angle; real potent nucleants act through atomic-scale lattice matching, which the contact angle only parametrizes.

Contact angle θ	$f(\theta)$	Undercooling relative to homogeneous, $\sqrt{f}$
10°	0.00017	0.013
20°	0.0027	0.052
30°	0.013	0.11
60°	0.16	0.40
90°	0.50	0.71
120°	0.84	0.92
180°	1	1 (no help)

Figure 9.2 shows how steeply $f(\theta)$ falls with decreasing contact angle. Because $\Delta G^{*} \propto f(\theta)/\Delta T^{2}$, achieving the same barrier on a substrate requires an undercooling smaller by about $\sqrt{f(\theta)}$ (the third column, which neglects the smaller number of sites available on surfaces compared with atoms in the volume). With $\theta = 30^{\circ}$ the 190 K estimated for copper becomes about 22 K; with $\theta = 20^{\circ}$, about 10 K. A substrate that the solid wets well reduces the undercooling to the few kelvin observed in practice.

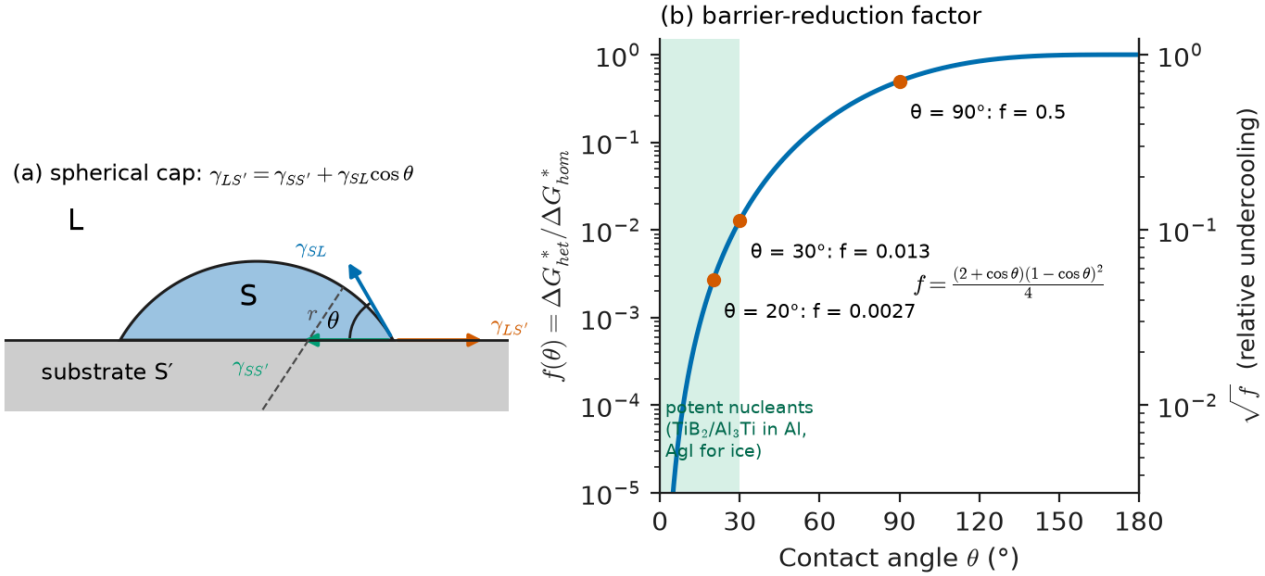

Figure 9.2 — Heterogeneous nucleation: a spherical-cap nucleus on a substrate, and the barrier-reduction factor f(θ) versus contact angle.

What makes a good nucleant

A low contact angle corresponds to a low interfacial energy between the nucleating solid and the substrate, which in turn requires good **lattice matching** and chemical affinity. The **planar disregistry** (lattice mismatch) between low-index planes of substrate and nucleus is a useful empirical guide: below about 6 % a substrate is usually a potent nucleant, above about 12 % rarely. Examples across engineering:

- **Aluminium grain refinement.** Al–5Ti–1B master alloy introduces TiB_2 particles of about 1–2 µm, on which a monolayer of Al_3Ti forms; Al_3Ti matches α-Al closely, and grains nucleate at undercoolings below 1 K. Grain size falls from millimetres to 100–200 µm, reducing hot tearing and porosity and making ingots workable, as in the opening example. Zirconium plays the same role in magnesium alloys.
- **Cast iron inoculation.** Ferrosilicon containing small amounts of Ca, Ba, Sr or Al, added to the ladle, creates oxide and sulphide particles that nucleate graphite and prevent the formation of hard, brittle iron carbide ("chill") in thin sections.
- **Cloud seeding.** Silver iodide has a hexagonal structure with a lattice mismatch of about 1–2 % with ice Ih; dispersed in clouds it nucleates ice at around −5 to −10 °C, far above the −38 °C homogeneous limit. Bacterial proteins (from *Pseudomonas syringae*) are even more potent and are used in snow-making.
- **Weld-metal toughness.** Fine titanium-oxide inclusions in steel weld metal nucleate acicular ferrite inside austenite grains, giving a fine interlocking microstructure with high toughness (Chapter 10).
- **Polymers.** Nucleating agents (sodium benzoate, sorbitol derivatives) in polypropylene raise the crystallization temperature during moulding, shortening cycle times and, with very fine spherulites, making the parts clear.
- **Pharmaceuticals and foods.** Seeding controls which crystal polymorph of a drug forms, which matters because polymorphs differ in solubility and bioavailability; tempering of chocolate seeds the desired cocoa-butter polymorph.

Conversely, **avoiding** heterogeneous nucleation is how deep undercooling is studied: levitated droplets, droplets in emulsions and fluxed melts (molten glass encasing the metal) remove surface sites.

9.4 Growth: interface-controlled and diffusion-controlled

Once a nucleus exceeds r^*, it grows. Two steps are in series: atoms must cross the interface and attach to the growing crystal, and (if the new phase has a different composition) solute must be carried to or from the interface by diffusion, while latent heat must be removed. The slower step controls the rate.

Interface-controlled growth

When the composition does not change (pure metals, allotropic and massive transformations, crystallization of glasses and polymers) only attachment matters. The net rate at which atoms attach is the difference between forward and backward jumps across the interface, each thermally activated. For a driving force per atom Δg,

$$v = \lambda \nu \exp\!\left(-\frac{Q_i}{kT}\right)\left[1 - \exp\!\left(-\frac{\Delta g}{kT}\right)\right] \approx M\,\Delta G$$

where the second form, with interface mobility M, holds for small driving forces. Growth velocity is constant in time and, at small undercooling, proportional to ΔT. Rough metallic interfaces attach atoms at almost every site, so their kinetic undercooling is a fraction of a kelvin even at growth speeds of m/s. **Faceted** crystals (silicon, many intermetallics and oxides, organic molecules) have smooth, atomically flat interfaces; atoms can attach only at steps, which must be created by two-dimensional nucleation or supplied by screw dislocations. Their growth is slower and strongly anisotropic, which gives faceted shapes such as the plates of silicon in Al–Si eutectics. Polymer spherulites grow at a constant radial rate (micrometres per second), also an interface-controlled process.

Diffusion-controlled growth

When the growing phase has a different composition from its matrix (a precipitate rich in solute, a ferrite grain that must reject carbon), solute must diffuse over growing distances, and growth slows with time. For a planar precipitate whose interface keeps local equilibrium, the thickness grows as

$$x = \alpha \sqrt{D t}$$

and for a spherical precipitate in a dilute matrix

$$r = \lambda \sqrt{D t}, \qquad \lambda \approx \sqrt{2\,\frac{C_0 - C_m}{C_p - C_m}}$$

where C_0 is the far-field matrix concentration, C_m the matrix concentration in equilibrium with the precipitate and C_p the precipitate composition. With $C_0 = 0.02$, $C_m = 0.01$ and $C_p = 0.30$ (atom fractions), $\lambda = 0.26$: in the time that solute diffuses a distance $\sqrt{Dt}$, the precipitate radius grows by only about a quarter of that distance.

> **Status of the principle: Parabolic diffusion-controlled growth**
> *Derived.* It follows from Fick's laws with local equilibrium at a moving interface and a constant far-field composition; it fails when the diffusion fields of neighbouring particles overlap (soft impingement), when the interface is not at equilibrium (mixed control), and in the later stage where coarsening takes over (Chapter 10).

Heat is the analogue of solute in solidification: the latent heat must flow away, either into the cold mould (columnar growth into a positive temperature gradient) or into an undercooled liquid (free equiaxed growth). Both diffusions, of heat and of solute, decide the shapes of crystals growing from the melt (Section 9.6).

Many transformations have **mixed control**, with interface reaction dominating at first and diffusion later. The growth of ferrite from austenite in low-alloy steels, and of oxide scales (interface-controlled linear kinetics early, parabolic later, Chapter 42), are examples.

9.5 Overall transformation kinetics: JMAK and TTT

Extended volume and the Avrami equation

A transformation in a real part involves many nuclei appearing at different times and growing until they meet. Johnson and Mehl, Avrami, and Kolmogorov solved this counting problem. Imagine first that nuclei form and grow in the *whole* volume, ignoring the fact that some of them would form in already transformed material and that grains overlap. For constant nucleation rate I and constant growth velocity v of spheres, this **extended** transformed fraction is

$$f_{ext} = \int_0^t I\,\frac{4}{3}\pi v^3 (t-\tau)^3\, d\tau = \frac{\pi}{3} I v^3 t^4$$

Only the untransformed fraction $(1 - f)$ of any increment of extended volume is real, so $df = (1 - f)\, df_{ext}$, which integrates to

$$f = 1 - \exp(-f_{ext}) = 1 - \exp(-kt^n)$$

This is the **Johnson–Mehl–Avrami–Kolmogorov (JMAK)** equation. The time exponent n (the **Avrami exponent**) reflects the nucleation and growth modes:

Conditions	Avrami exponent n
Constant nucleation rate, 3D interface-controlled growth	4
Site saturation (all nuclei at $t = 0$), 3D interface-controlled growth	3
Constant nucleation rate, 3D diffusion-controlled growth	2.5
Site saturation, 3D diffusion-controlled growth	1.5
Site saturation on grain boundaries, after boundary saturation	1
Growth of plates or needles of fixed length, diffusion-controlled thickening	1–1.5

> **Status of the principle: JMAK equation**
> *Derived.* Exact for randomly located nuclei in an infinite volume with constant (or simply time-dependent) nucleation and growth rates; it fits many recrystallization, crystallization and precipitation data well, but nucleation on non-random sites, changing driving forces and soft impingement make fitted exponents lose their mechanistic meaning.

A plot of $\ln[-\ln(1 - f)]$ against $\ln t$ is a straight line of slope n. Measured values are often non-integer, and the exponent should be interpreted cautiously.

Worked example — Recrystallization of cold-rolled aluminium. Hardness measurements on cold-rolled aluminium sheet annealed at 300 °C indicate 10 % recrystallization after 10 min and 90 % after 60 min. Find the Avrami parameters and the time for 50 % and 99 % recrystallization.

Exponent. From $-\ln(1 - f) = kt^n$ at the two points,

$$n = \frac{\ln[-\ln 0.1] - \ln[-\ln 0.9]}{\ln 60 - \ln 10} = \frac{\ln 2.303 - \ln 0.1054}{\ln 6} = \frac{3.084}{1.792} = 1.72$$

Rate constant. $k = -\ln 0.9 / 10^{1.72} = 2.0 \times 10^{-3}\ \text{min}^{-1.72}$.

Times. $t_{50} = (\ln 2/k)^{1/n} = 30$ min and $t_{99} = (\ln 100/k)^{1/n} = 90$ min.

The exponent of 1.7, well below the ideal value of 3 or 4, is typical of recrystallization, where nuclei form preferentially at prior grain boundaries and shear bands and the driving force (stored dislocation energy, Chapter 10) is consumed unevenly. The fitted equation is nevertheless a reliable interpolation tool for setting continuous-annealing line speeds.

From rate to TTT diagram

The JMAK rate constant k depends steeply on temperature through both I and v. Just below the equilibrium temperature, the driving force is small, ΔG^* is huge and nucleation is negligible. Far below it, the driving force is large but atomic mobility (proportional to $\exp(-Q/RT)$, Chapter 8) is small. The overall rate is the product of a factor rising with undercooling and a factor falling with it, and therefore has a maximum at an intermediate temperature.

Plotting, for each temperature, the time to reach a given fraction transformed (commonly 1 % and 99 %) gives a **time–temperature–transformation (TTT)** diagram with characteristic C-shaped curves. The nose of the C is the temperature of fastest transformation; to avoid the transformation entirely, a part must be cooled past the nose faster than the nose time. This construction applies equally to the formation of pearlite in steel (Chapter 10), precipitation in aluminium alloys during quenching, crystallization of polymers during injection moulding and the crystallization of glass-forming melts, where it gives the critical cooling rate for glass formation (Section 9.9).

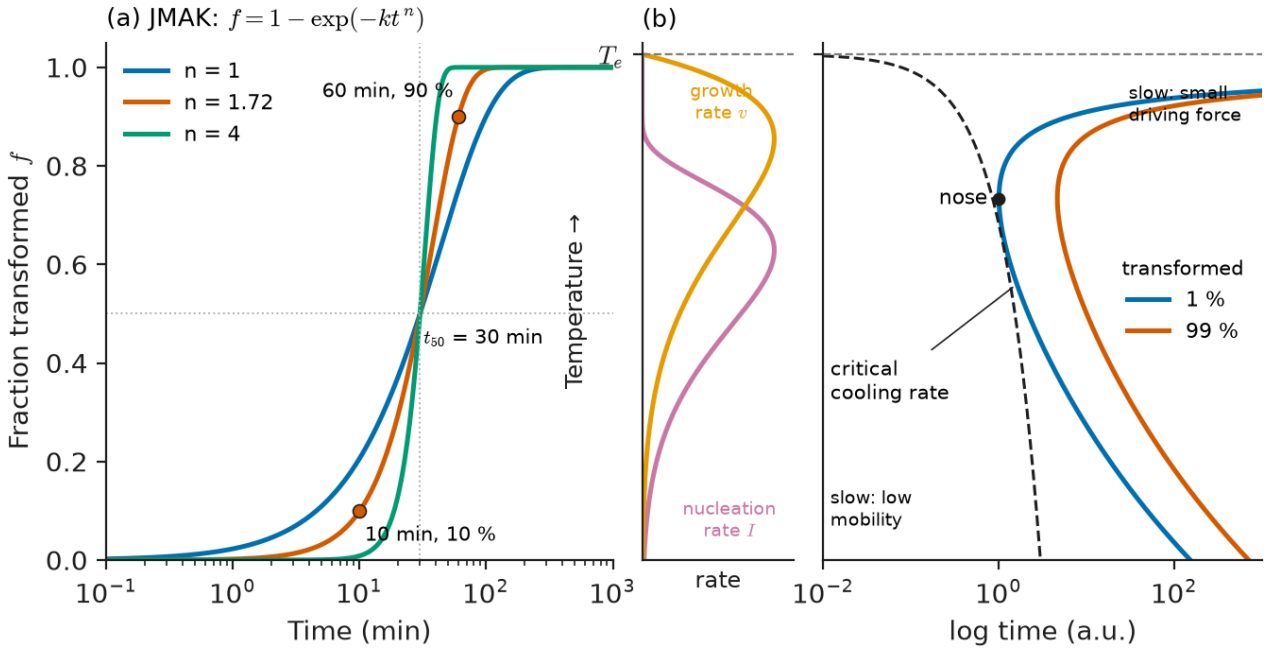

Figure 9.3 — Overall transformation kinetics: JMAK curves for several Avrami exponents, and the construction of a C-shaped TTT curve from nucleation and growth rates.

9.6 Solidification: fronts, dendrites and constitutional supercooling

Pure metals: thermal supercooling

When a pure metal solidifies inward from a cold mould wall, heat flows from the liquid through the solid to the mould: the temperature gradient in the liquid ahead of the interface is **positive** (temperature rises into the liquid). A bump on the interface pokes into hotter liquid, melts back and the front stays planar. In an undercooled liquid, by contrast, such as an equiaxed crystal growing freely in the centre of a casting, the gradient in the liquid is negative; a bump finds itself in colder liquid, grows faster and becomes a branched **thermal dendrite**, with primary arms along preferred crystallographic directions (⟨100⟩ in cubic metals) and secondary and tertiary branches.

Alloys: constitutional supercooling

Alloys behave differently even in a positive gradient. With $k < 1$, the growing solid rejects solute, which piles up in a boundary layer of the liquid ahead of the interface. At steady state for a planar front moving at velocity V, the solid has composition C_0, the liquid at the interface has C_0/k, and the excess decays into the liquid over a distance of about D_L/V:

$$C_L(x') = C_0\left[1 + \frac{1-k}{k}\exp\!\left(-\frac{V x'}{D_L}\right)\right]$$

where x' is distance ahead of the interface. Each liquid composition has its own liquidus temperature; the liquid near the interface, rich in solute, freezes at a lower temperature than the liquid farther away. If the actual temperature gradient G is shallower than the gradient of this local liquidus temperature at the interface, a zone of liquid ahead of the front is below its own liquidus: it is **constitutionally supercooled**, and a planar front becomes unstable. The criterion for stability is

$$\frac{G}{V} \;\geq\; \frac{\Delta T_0}{D_L}, \qquad \Delta T_0 = \frac{m\,C_0(1-k)}{k}$$

where m is the magnitude of the liquidus slope and ΔT_0 is the equilibrium freezing range of the alloy (the gap between liquidus and solidus at C_0).

> **Status of the principle: Constitutional-supercooling criterion**
> *Approximate.* It compares two temperature gradients and ignores the stabilizing effect of interfacial energy and the latent-heat distribution; the more complete Mullins–Sekerka stability analysis confirms it as an excellent approximation at normal growth rates and adds absolute stability at very high velocities.

As G/V falls below the critical value, the front evolves from **planar** to **cellular** (an array of parallel rounded protrusions) to **columnar dendritic** (cells develop side branches), and finally to **equiaxed dendritic** grains (Figure 9.4) nucleating in the supercooled liquid ahead of the front. The product GV (the cooling rate, K/s) sets the scale of the structure: higher cooling rates give finer cells and dendrites.

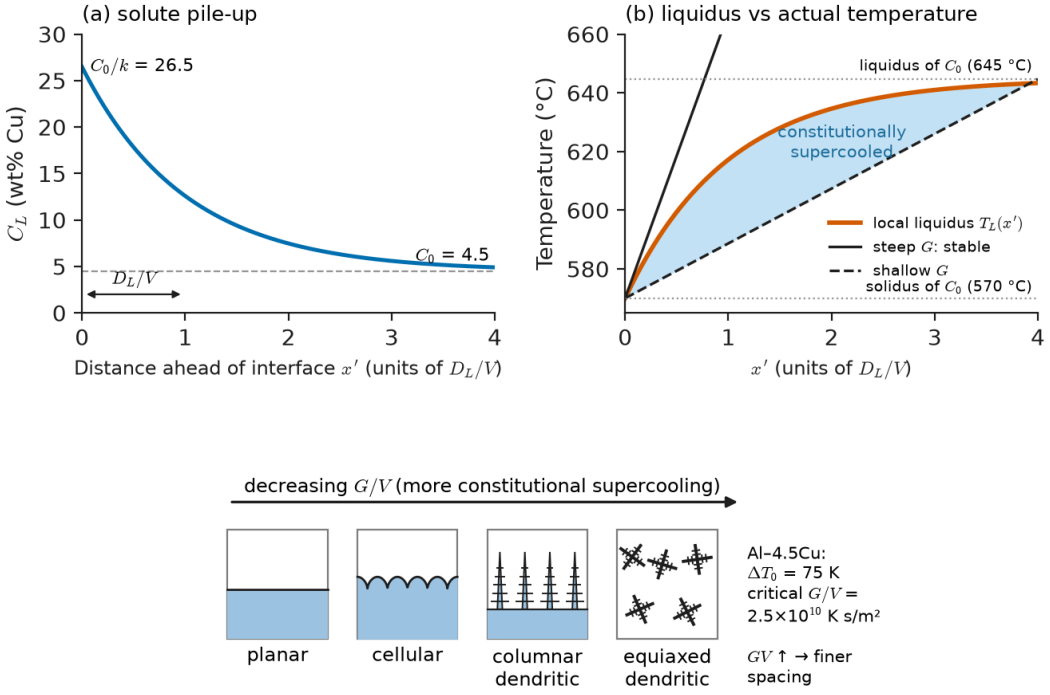

Figure 9.4 — Constitutional supercooling ahead of a solidifying alloy front, and the sequence of interface morphologies as G/V decreases.

Worked example — Can an aluminium–copper alloy freeze with a planar front? For Al–4.5 wt% Cu, take the liquidus slope $m = 3.4$ K per wt% Cu, $k = 0.17$ and $D_L = 3 \times 10^{-9}$ m²/s.

Freezing range.

$$\Delta T_0 = \frac{3.4 \times 4.5 \times (1 - 0.17)}{0.17} = 75 \text{ K}$$

Critical ratio. $G/V \geq \Delta T_0 / D_L = 75/(3 \times 10^{-9}) = 2.5 \times 10^{10}$ K s/m².

Speed limit. In a steep laboratory gradient of $G = 10$ K/mm (10^4 K/m), a planar front requires $V \leq 10^4/2.5 \times 10^{10} = 4 \times 10^{-7}$ m/s, or 0.4 µm/s (1.4 mm per hour). Even at 100 K/mm, the limit is only 4 µm/s.

Planar solidification of a typical engineering alloy is possible only in slow laboratory crystal growth. Every casting, weld and additively manufactured part of such an alloy solidifies dendritically. Directionally solidified and single-crystal nickel-superalloy turbine blades, grown at 3–10 K/mm and a few mm/min, are still dendritic; "single crystal" means that all the dendrites belong to one crystal, selected by a spiral "grain selector" at the base of the mould, so that no grain boundaries are available for creep cavitation (Chapter 41). Semiconductor crystals are the exception: silicon is nearly pure, dopant concentrations are tiny, so ΔT_0 is small and Czochralski growth at about 1 mm/min proceeds with a planar (or smoothly curved) interface.

Dendrite arm spacing

The spacing of secondary dendrite arms, λ_2, coarsens during solidification because small arms dissolve in favour of large ones (a ripening process, Chapter 10). It is found empirically to depend mainly on the local solidification time t_f:

$$\lambda_2 \approx A\, t_f^{1/3}$$

with $A \approx 10$ μm s$^{-1/3}$ for many aluminium alloys. A thin-walled pressure die casting freezing in about 1 s has $\lambda_2 \approx 10$ μm; a sand casting freezing in 1000 s has about 100 μm. Strength, ductility and fatigue resistance of cast aluminium improve as λ_2 decreases, because porosity and brittle interdendritic phases are finer, and homogenization time scales as λ_2^2 (Chapter 8).

> **Status of the principle: Secondary dendrite arm spacing law**
> *Empirical.* The one-third exponent is supported by coarsening theory, but the constant A is fitted for each alloy family; it is used to estimate local cooling rates from micrographs of failed castings.

9.7 Segregation: the Scheil equation and zone refining

Two limits

During solidification of an alloy with $k < 1$, the first solid is purer than the alloy and the last liquid is richer. How the solute ends up distributed depends on how fast it can move in each phase. Two limits bracket reality:

- **Equilibrium (lever rule).** Complete diffusion in both liquid and solid: at each temperature, solid and liquid have the tie-line compositions of Chapter 7. The final solid is uniform.
- **Scheil–Gulliver.** Complete mixing in the liquid (by convection and fast liquid diffusion), *no* diffusion in the solid, and local equilibrium at the interface. Each layer of solid keeps the composition it froze with.

Solute balance for the Scheil case gives, as fraction solid f_s grows by df_s,

$$(C_L - C_S)\, df_s = (1 - f_s)\, dC_L$$

and with $C_S = kC_L$ and constant k,

$$C_S = kC_0(1 - f_s)^{k-1}, \qquad C_L = C_0(1 - f_s)^{k-1}$$

This is the **Scheil equation**. For $k < 1$, C_L rises without limit as $f_s \to 1$: the last liquid always reaches the eutectic composition (or the lowest-melting composition available), however dilute the alloy.

> **Status of the principle: Scheil equation**
> *Approximate.* It assumes perfect liquid mixing, no solid diffusion, local interface equilibrium and constant k; it is close to reality for substitutional solutes in castings, but overestimates segregation for fast-diffusing interstitials such as carbon in steel, for which back-diffusion corrections are added.

Worked example — Non-equilibrium eutectic in an Al–Cu casting. For Al–4.5 wt% Cu with $k = 0.17$ and eutectic composition 33.2 wt% Cu, compare equilibrium and Scheil solidification.

Equilibrium. The maximum solubility of Cu in Al is 5.65 wt% (Chapter 7). An alloy with 4.5 wt% Cu should freeze entirely as α and contain no eutectic.

Scheil. The liquid reaches the eutectic composition when

$$1 - f_s = \left(\frac{C_E}{C_0}\right)^{1/(k-1)} = \left(\frac{33.2}{4.5}\right)^{-1/0.83} = 0.090$$

so 9.0 % of the casting freezes as eutectic at 548 °C.

Coring. The solid composition rises from $kC_0 = 0.77$ wt% Cu at the dendrite centres to 1.36 wt% at $f_s = 0.5$ and 5.2 wt% at $f_s = 0.9$.

The two limits differ qualitatively (Figure 9.5): equilibrium predicts a single-phase alloy with a solidus near 573 °C, Scheil predicts about 9 % eutectic and a final freezing temperature of 548 °C. Real castings lie close to the Scheil limit; metallographers routinely find interdendritic Al2Cu in such alloys, and it must be dissolved by homogenization before the alloy can be solution-

treated (Chapters 7, 8 and 10). The Scheil prediction of a wider freezing range also explains **hot tearing**: in the last stages, a network of dendrites holding thin liquid films cannot accommodate thermal contraction and tears.

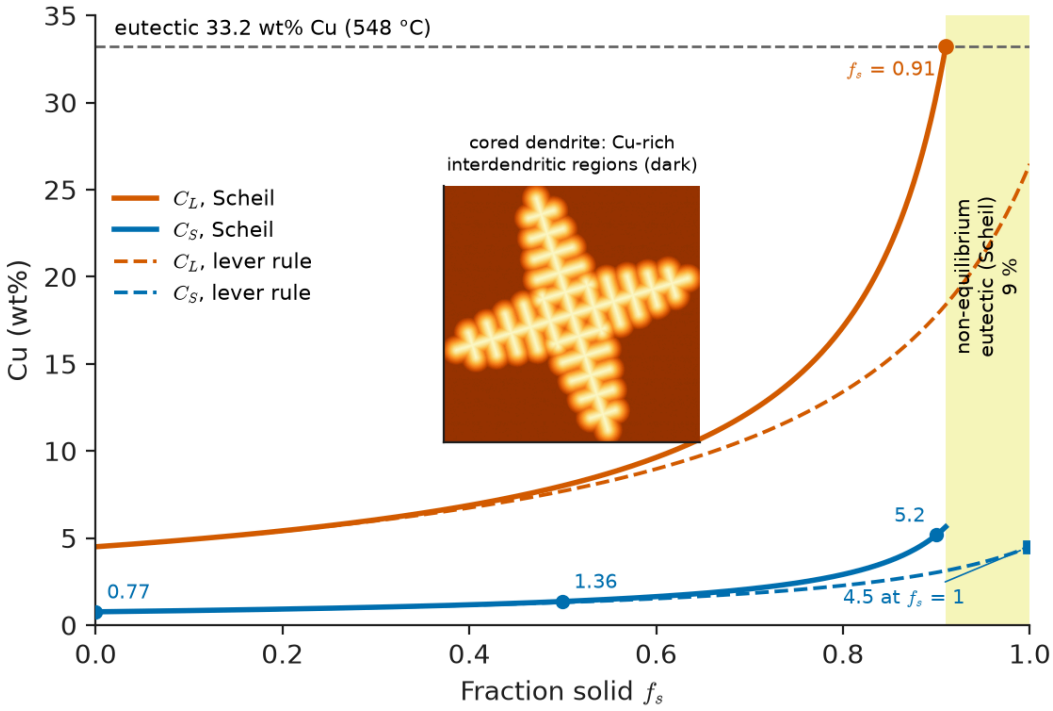

Figure 9.5 — Solid composition and liquid composition versus fraction solid for Al–4.5 wt% Cu under equilibrium (lever rule) and Scheil solidification.

Macrosegregation and segregation in crystal growth

Microsegregation occurs over dendrite arm spacings and can be removed by homogenization. **Macrosegregation**, composition variation over the scale of a whole ingot, arises when solute-rich liquid is moved by convection, shrinkage flow or buoyancy; it produces "A-segregates" and centre-line segregation in large steel ingots for forgings and in continuously cast slabs, which cannot be removed by heat treatment and must be controlled by casting practice.

In crystal growth the Scheil equation, called the **normal-freezing** equation, gives the axial dopant distribution. For a Czochralski silicon ingot pulled from a melt doped with boron ($k = 0.8$), the dopant concentration rises from $0.80 C_0$ at the seed end to $0.92 C_0$ at half length and $1.27 C_0$ at 90 % solidified; with phosphorus ($k = 0.35$) it rises from $0.35 C_0$ through $0.55 C_0$ to $1.56 C_0$. Because resistivity tracks dopant concentration, wafers from different parts of a phosphorus-doped ingot differ in resistivity by a factor of more than four, and only part of each ingot meets a tight specification (Chapter 32).

Zone refining

Scheil's equation also describes **purification**. Pfann's **zone refining** (1952) passes a short molten zone of length L along a solid bar. Impurities with $k < 1$ are swept along with the zone toward the far end. After one pass, the solid at distance x from the start has

$$\frac{C_S}{C_0} = 1 - (1 - k)\exp\!\left(-\frac{k x}{L}\right)$$

For $k = 0.1$ the first zone length is purified tenfold ($C_S/C_0 = 0.10$ at $x = 0$, 0.19 at $x = L$, 0.26 at $x = 2L$). Multiple passes multiply the effect, and the method produced the germanium and silicon of the first transistors with impurity levels below one part per billion, and is still used for detector-grade germanium and ultra-pure metals. Its industrial descendant, **float-zone** growth of silicon, uses an RF-heated molten zone with no crucible, giving the oxygen-free high-resistivity wafers used in power devices and radiation detectors.

9.8 Eutectic and peritectic solidification; casting microstructures

Coupled eutectic growth

At a eutectic, two solids grow side by side from one liquid. Solute rejected by one phase is the solvent needed by the other, so atoms diffuse sideways along the interface between neighbouring lamellae or rods. Close spacing shortens the diffusion distance; but each lamella adds interface, whose energy has to be paid. The balance (Jackson–Hunt theory) gives a spacing λ that varies with growth velocity as

$$\lambda^2 V = \text{constant}$$

so doubling the growth rate reduces the spacing by $\sqrt{2}$. Lamellar structures form when both phases have comparable volume fractions; when one phase is a minor fraction (roughly below 0.3) rods are favoured, as they have less interface. Directionally solidified eutectics of Ni–TaC, Al_2O_3–ZrO_2 or Al_2O_3–YAG form aligned fibre composites in situ, which have been studied for turbine and cutting-tool applications.

> **Status of the principle: Jackson–Hunt eutectic spacing**
> *Derived.* It follows from lateral solute diffusion and interface curvature for regular (non-faceted/non-faceted) eutectics with the extremum assumption for the operating spacing; it applies poorly to irregular eutectics with a faceted phase.

Irregular eutectics occur when one phase is faceted. In Al–Si alloys, silicon grows as coarse, brittle plates whose sharp edges initiate cracks. Adding about 0.01–0.02 wt% strontium or sodium poisons the growth steps of silicon (by inducing twinning) and turns the plates into a fine coral-like fibrous network, raising elongation from a few percent to over 10 %. Grey cast iron is an Fe–graphite eutectic in which graphite grows as interconnected flakes, giving good damping and machinability but low tensile ductility. Treating the melt with about 0.04 wt% magnesium (or cerium) removes sulphur and oxygen, changes the growth of graphite and produces **spheroidal graphite** (ductile) iron, with elongations of 10–20 %: the material of modern crankshafts, wind-turbine hubs and water pipes (Chapter 15).

Peritectic solidification

In a peritectic $L + \alpha \rightarrow \beta$, the product β forms first at the α–liquid interface and then separates the reactants; the reaction can continue only by diffusion through β, so it is rarely completed. Three engineering examples:

- **Steel continuous casting.** "Peritectic" steels (about 0.09–0.17 wt% C) undergo δ→γ near 1495 °C in the thin solid shell, with a contraction that pulls the shell from the mould and leads to uneven shell thickness, depressions and surface cracks (Chapter 7). Casters use special mould fluxes and reduced casting speeds.
- **Nd–Fe–B magnets.** The magnetic phase $Nd_2Fe_{14}B$ forms peritectically from primary iron and liquid. Ingot casting leaves soft-magnetic primary α-iron dendrites that degrade the magnet; rapid **strip casting** (cooling rates of order 10^3 K/s) suppresses primary iron and gives the fine, uniform alloy used for sintered magnets in electric vehicle motors and wind turbines (Chapter 29).
- **YBCO superconductors.** $YBa_2Cu_3O_{7-\delta}$ forms peritectically near 1000 °C from Y_2BaCuO_5 and liquid. Slow "melt-texturing" through the peritectic grows large aligned grains for levitation magnets and current leads.

Casting macrostructure

A metal poured into a mould typically develops three zones (Figure 9.6):

1. **Chill zone.** A thin layer of fine equiaxed grains forms against the cold mould wall, where undercooling is high and the wall provides abundant nucleation sites.
2. **Columnar zone.** Grains whose fast-growth direction (⟨100⟩ for cubic metals) is nearly parallel to the heat flow outgrow their neighbours and form long columnar grains, a natural texture.
3. **Equiaxed zone.** In the centre, the liquid becomes constitutionally supercooled; crystals nucleated on inoculant particles or formed from dendrite fragments detached by convection grow freely as equiaxed grains.

The balance between columnar and equiaxed zones is a central control in casting. Large columnar grains give anisotropic properties, hot tearing and poor workability (the opening example); inoculation, low pouring temperature, stirring (mechanical or electromagnetic) and alloying with growth-restricting solutes (such as Ti in Al, quantified by a "growth-restriction factor" proportional to $mC_0(1-k)$) promote equiaxed grains. Conversely, directional solidification deliberately eliminates equiaxed

grains to produce columnar or single-crystal turbine blades whose grain boundaries are aligned with, or absent from, the stress axis.

Solidification time itself, which governs dendrite arm spacing and grain size, is estimated for sand castings by **Chvorinov's rule**, $t_s = B\,(V/A)^2$, where V/A is the casting modulus (volume over cooling surface area) and B is a mould constant. Feeders (risers) are designed to have a larger modulus than the section they feed, so that they freeze last and supply liquid to compensate for solidification shrinkage of 3–7 % in most metals (Chapter 46).

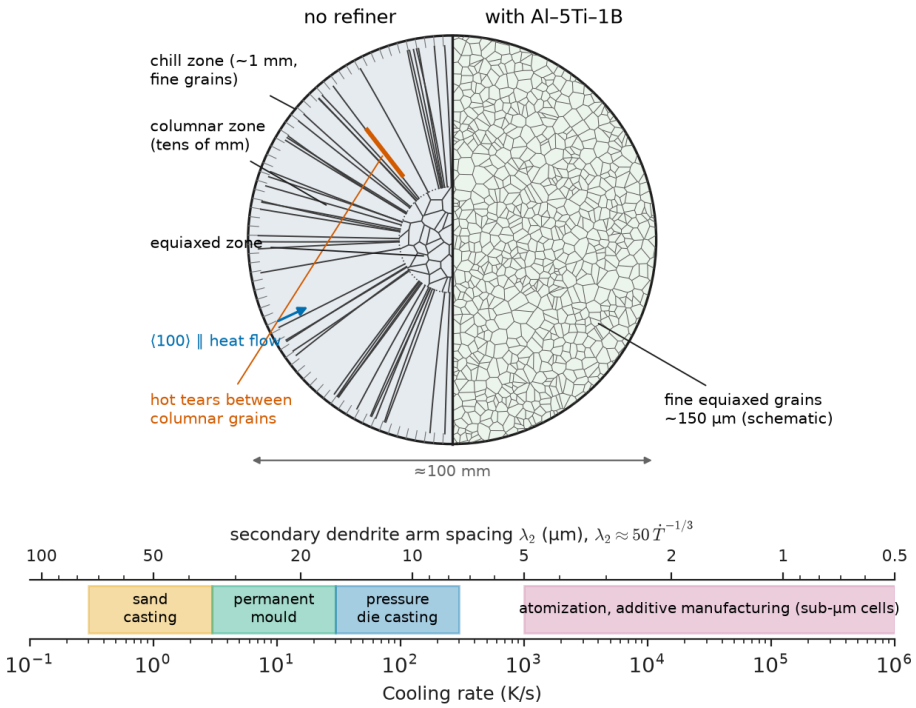

Figure 9.6 — Macrostructure of a cast ingot: chill zone, columnar zone and equiaxed zone, with the effect of grain-refiner addition.

9.9 Rapid solidification and metallic glasses

Cooling faster

Conventional casting cools at 0.1–100 K/s. Rapid solidification processes reach 10^3–10^8 K/s by making at least one dimension small and extracting heat into a good conductor:

Process	Typical cooling rate (K/s)	Product
Sand casting	0.1–10	Large castings
Pressure die casting	10–10^3	Thin-walled Al and Mg parts
Strip casting	10^2–10^4	Nd–Fe–B flake, thin steel strip
Gas atomization	10^2–10^5 (finer powders faster)	Metal powders for AM, PM and thermal spray
Laser powder-bed fusion	10^4–10^7	Additively manufactured parts
Melt spinning	10^5–10^6	Amorphous and nanocrystalline ribbons 20–50 µm thick
Splat quenching, laser surface glazing	10^6–10^8	Foils, surface layers

The effects of rapid solidification follow from this chapter's results. Higher cooling rates mean finer dendrites and eutectics (Sections 9.6 and 9.8), less segregation, **extended solid solubility** (solute trapped in the solid at concentrations far above the equilibrium solvus, as interface velocities approach the speed of solute diffusion and $k \to 1$), and **metastable phases** that

nucleate faster than the stable ones. Additively manufactured aluminium and nickel alloys owe their high as-built strength to sub-micrometre cells, and also their susceptibility to solidification cracking, to these effects (Chapter 48).

Bypassing crystallization: metallic glasses

If a liquid is cooled fast enough to pass the nose of its crystallization TTT curve (Figure 9.3), it never crystallizes. Its viscosity rises continuously until, at the **glass transition temperature** T_g, the atomic structure freezes into an amorphous solid. Silicate liquids do this at ordinary cooling rates because their viscosity near the melting point is already enormous (Chapter 17). Simple metallic liquids have viscosities like water and crystallize within microseconds, so a glass forms only above a **critical cooling rate** R_c given approximately by the nose of the TTT curve:

$$R_c \approx \frac{T_l - T_n}{t_n}$$

where T_l is the liquidus, and T_n and t_n are the temperature and time of the nose.

The first metallic glass, $Au_{75}Si_{25}$, was made by splat quenching in 1960 by Duwez, with R_c of order 10^6 K/s. Since the thickness that can be cooled at a given rate is limited by heat conduction, the critical thickness scales roughly as $x_c \approx \sqrt{\alpha\,\Delta T / R_c}$, with thermal diffusivity $\alpha \approx 2 \times 10^{-6}$ m²/s and a temperature drop $\Delta T \approx 600$ K from liquidus to glass transition. That gives about 35 μm for $R_c = 10^6$ K/s (the thickness of melt-spun ribbon), about 1 mm for 10^3 K/s and about 35 mm for 1 K/s: an order-of-magnitude estimate that matches the observed hierarchy.

Glass-forming ability is improved by anything that slows crystallization:

- **Deep eutectics.** Alloys near deep eutectics have a low liquidus, so the liquid reaches high viscosity sooner. Turnbull's **reduced glass-transition temperature** $T_{rg} = T_g / T_l$ is the usual measure; $T_{rg} \gtrsim 2/3$ indicates easy glass formation.
- **Confusion principle (Inoue's rules).** At least three elements, atomic size differences above about 12 %, and negative heats of mixing; complex liquids find it hard to rearrange into the crystal structures available.
- **Clean melts.** Heterogeneous nucleation on oxides must be avoided, often by fluxing or high-purity processing.

> **Status of the principle: Glass-forming-ability criteria (T_{rg}, Inoue's rules)**
> *Empirical.* They summarize the nucleation and growth physics of this chapter and correlate well with measured critical thicknesses, but many exceptions exist, and no single parameter predicts glass formation reliably.

Bulk metallic glasses with R_c of 1–100 K/s can be cast into parts millimetres to centimetres thick. Vitreloy 1 ($Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10}Be_{22.5}$) has $T_g \approx 625$ K and $T_l \approx 993$ K, so $T_{rg} \approx 0.63$, and an R_c of about 1 K/s; Pd–Cu–Ni–P glasses form rods several centimetres in diameter.

Properties and uses

Having no crystal lattice and therefore no dislocations, metallic glasses yield at stresses near their theoretical limit: Zr-based glasses reach about 1.8–2 GPa with an elastic strain limit of about 2 %, compared with 0.5 % or less for crystalline alloys, and store large elastic energy per unit volume (Chapter 11). Their weakness is that deformation localizes in narrow **shear bands** with no work hardening, so in tension they fail with almost no plastic strain; composites containing ductile crystalline dendrites improve toughness. Because they soften above T_g like a polymer, they can be thermoplastically moulded with nanometre precision.

Application	Material	Property exploited
Distribution-transformer cores	Fe–Si–B ribbon (melt-spun)	Core loss about 60–75 % lower than grain-oriented Si steel; no grain boundaries or magnetocrystalline anisotropy (Chapter 29)
Anti-theft tags, magnetic sensors	Co- and Fe-based ribbons	Soft magnetism, magnetostriction
Golf club faces, springs, watch parts	Zr- and Ti-based bulk glasses	High elastic strain and resilience
Precision gears for micromotors, surgical blades	Zr-, Ni-based glasses	Net-shape moulding, sharp edges, wear
Nanocrystalline alloys (Finemet)	Fe–Si–B–Nb–Cu annealed glass	Controlled crystallization to 10 nm grains: very soft magnets

The last row closes a circle: annealing a metallic glass just above T_g triggers copious homogeneous nucleation, with Cu clusters acting as heterogeneous sites and Nb restricting growth. The result is a nanocrystalline magnet with 10–15 nm grains, an example of using the nucleation and growth theory of this chapter as a design tool rather than a hazard.

> **Boundary.**
> The thermodynamics of interfaces (interfacial energy, Gibbs–Thomson, Young's equation) are developed in Chapter 6 and the statistical mechanics of fluctuations belongs to the Engineering Physics book. Heat-transfer analysis of castings and welds is part of the thermal-engineering curriculum; this chapter takes cooling rates and gradients as inputs.

What remains open

Open question	Where it is answered
Where do the driving forces, interfacial energies and the Gibbs–Thomson effect come from?	Chapter 6
Which phases and compositions does solidification follow, and what are the eutectic and peritectic points?	Chapter 7
How fast does solute diffuse, and how long does homogenization take?	Chapter 8
How do nucleation and growth operate in solid-state transformations: precipitation, pearlite, bainite, martensite, recrystallization?	Chapter 10
How do grain size, dendrite spacing and porosity control strength and fatigue?	Chapters 13 and 14
How are cast irons and steels inoculated, cast and heat-treated?	Chapter 15
How are glasses formed and glass-ceramics crystallized?	Chapter 17
How do polymers crystallize, and how is crystallinity controlled in moulding?	Chapter 20
How are silicon crystals grown and doped for devices?	Chapter 50
How are casting, welding and additive manufacturing processes designed around solidification?	Chapters 46, 47 and 48
How are nucleation and dendritic growth simulated by phase-field and atomistic methods?	Chapter 56

Next: Chapter 10 carries nucleation and growth into the solid state, where strain energy, coherency and diffusionless shear produce the precipitates, pearlite, bainite and martensite that give engineering alloys their strength.

Chapter 10 — Phase Transformations in Solids

Part II · Thermodynamics and Kinetics

A batch of aluminium-alloy wing ribs arrives at an aircraft assembly line with the right chemistry, the right dimensions and a yield strength 30 % below specification. The fault is traced to a furnace controller that ran the ageing treatment 25 °C too hot. Nothing melted and the parts look identical to good ones; what changed was the size and nature of particles a few nanometres across, formed inside the solid aluminium. Elsewhere, a railway replaces rail that wore out in five years with a head-hardened grade of almost the same chemistry that lasts fifteen, the difference lying in how fast the rail head was cooled through 700–500 °C; and a nickel–titanium stent crimped onto a catheter at room temperature springs open at body temperature. All three are phase transformations in the solid state — the transformations heat treatment exploits, and the source of most of the metallurgist's control over properties. This chapter applies the nucleation and growth ideas of Chapter 9 to them.

> **How can a solid change its crystal structure or composition without melting, and what decides whether it does so by diffusion or by a coordinated shear of the lattice?**
> **Why do fine, metastable precipitates and non-equilibrium phases such as martensite give the highest strengths, and how are time and temperature chosen to produce them?**

Concept	Engineering destination
Classification of transformations	Choosing heat treatments; predicting whether a quench "freezes in" a phase
Precipitation and age hardening	Aircraft aluminium alloys, maraging steels, nickel superalloys, Cu–Be springs
Ostwald ripening (LSW)	Creep life of turbine blades, over-ageing, stability of dispersions
Spinodal decomposition	475 °C embrittlement of duplex stainless steel, Cu–Ni–Sn springs, porous Vycor glass
Pearlite and bainite	Rails, bridge-cable and tyre-cord wire, austempered components
Martensite and shape memory	Hardened tools and gears, NiTi stents and actuators, zirconia toughening
Recovery, recrystallization, grain growth	Annealing of sheet and wire, grain-oriented electrical steel, weld HAZ grain size
Massive and discontinuous transformations	Brass and TiAl processing, unwanted cellular precipitates in Mg and Ni alloys
TTT and CCT diagrams	Quench design, hardenability, austempering, glass and titanium processing

Roadmap. Classification (10.1) → diffusional precipitation and age hardening (10.2) → coarsening (10.3) → spinodal decomposition as the nucleation-free alternative (10.4) → the eutectoid products of steel, pearlite and bainite (10.5) → diffusionless martensite and shape memory (10.6) → transformations driven by stored defects: recovery, recrystallization, grain growth (10.7) → massive and discontinuous reactions (10.8) → TTT and CCT diagrams as engineering tools (10.9).

10.1 Classifying solid-state transformations

A phase transformation in a solid needs a driving force (Chapter 6) and a kinetic path (Chapters 8 and 9), as solidification does. What differs is the cost of the path: atoms move only by vacancy or interstitial jumps whose rate falls exponentially on cooling, and every new particle must fit inside a crystal that resists its change of shape and volume. Slow diffusion and elastic constraint explain most of what is distinctive here.

Diffusional and diffusionless

The most useful single division is by how atoms move:

- **Diffusional (reconstructive) transformations** need long-range atomic transport, so composition can change and the product can have any structure. Precipitation, pearlite, recrystallization and grain growth belong here; their rates follow the Arrhenius temperature dependence of diffusion.

- **Diffusionless transformations** move every atom by less than one interatomic spacing, in a coordinated way, so the product keeps the parent composition. Martensite is the prime example; it can proceed where diffusion is frozen, at speeds approaching that of sound.

Displacive and order–disorder

A second division is by what happens to the lattice. In a **displacive** transformation the lattice is homogeneously deformed by a small strain plus small atomic shuffles. The coordinated motion produces a macroscopic shape change, and the parent–product interface is glissile, gliding like a dislocation array. Martensite is displacive; so, in most researchers' view, is the ferrite in bainite. Christian called the two extremes *civilian* (atoms move individually, like a crowd dispersing) and *military* (atoms move in step, like soldiers changing formation).

In an **order–disorder** transformation the lattice sites stay put but the atomic species rearrange among them. β-brass orders from random BCC to the CsCl-type B2 structure below about 460 °C; Cu_3Au orders to $L1_2$ below 390 °C; γ′ in nickel superalloys is $L1_2$-ordered Ni_3Al. Ordering may be continuous (second-order, as in β-brass) or first-order (with latent heat, as in Cu_3Au). Chapters 3 and 6 gave the structures and thermodynamics.

Nucleated versus continuous

Most transformations need nucleation: a small region of new phase must overcome a barrier before it grows (Chapter 9). Some proceed **continuously**, by the growth of small fluctuations present everywhere from the start; spinodal decomposition (Section 10.4) and continuous ordering are of this type.

Table 10.1 — A practical classification of solid-state transformations.

Transformation	Atom movement	Nucleation	Composition change	Rate controlled by	Example
Precipitation	Long-range diffusion	Yes	Yes	Bulk diffusion	Al–Cu ageing, γ′ in superalloys
Spinodal decomposition	Long-range (uphill) diffusion	No	Yes	Bulk diffusion	Fe–Cr at 475 °C, Cu–Ni–Sn
Eutectoid (pearlite)	Diffusion at the interface	Yes	Yes (cooperative)	Volume or boundary diffusion	Fe–0.8C at 550–700 °C
Bainite	Displacive plus carbon diffusion	Yes	Carbon only	Carbon partitioning	Austempered steels
Martensite	Coordinated shear, under one spacing	Yes (heterogeneous, athermal)	No	Undercooling below M_s	Quenched steel, NiTi
Massive	Short-range jumps across the interface	Yes	No	Interface mobility	Cu–Zn β→α, γ-TiAl
Order–disorder	Short-range exchange	Sometimes	No	Short-range diffusion	β-brass, Cu_3Au, Ni_3Al
Recrystallization, grain growth	Short-range jumps across boundaries	Recrystallization yes; growth no	No	Boundary mobility	Annealing of cold-worked sheet

Real heat treatments combine rows: a quenched-and-tempered steel forms martensite without diffusion, then precipitates carbides from it by diffusion.

10.2 Precipitation and age hardening

Supersaturation and the barrier in a solid

Aluminium with 4 wt% copper dissolves all its copper in FCC α above about 500 °C, but at room temperature the equilibrium solubility is a small fraction of a percent and the rest should be θ (Al_2Cu). **Solution-treated** at 540 °C and **quenched**, the alloy becomes a **supersaturated solid solution** (SSSS) with many quenched-in vacancies. On holding at room or modestly elevated temperature it precipitates and hardens: **age hardening**, discovered by Wilm in 1906 in the alloy that became Duralumin.

The nucleation theory of Chapter 9 carries over with one new term. Forming a precipitate of volume V releases chemical free energy $V\Delta G_v$ but costs interfacial energy $A\gamma$ and, because the precipitate rarely fits its hole exactly, an elastic **misfit strain energy** $V\Delta G_s$:

$$\Delta G = -V(\Delta G_v - \Delta G_s) + A\gamma$$

For a spherical nucleus the critical barrier becomes

$$\Delta G^* = \frac{16\pi\gamma^3}{3(\Delta G_v - \Delta G_s)^2}$$

The strain energy subtracts directly from the driving force. A precipitate that matches the matrix lattice closely has small ΔG_s and, more importantly, can keep a **coherent** interface — one in which lattice planes run continuously from matrix into precipitate — with an interfacial energy of only 5–200 mJ/m², compared with 500–1000 mJ/m² for an incoherent interface. Because ΔG^* scales as γ^3, this matters enormously.

> **Status of the principle: Classical nucleation barrier with misfit strain**
> *Approximate.* Extends the capillarity model of Chapter 9 by adding an elastic term; it treats the nucleus as a uniform particle with a sharp interface and bulk properties, which is questionable for nuclei of a few tens of atoms, but it orders the competing phases correctly.

Coherent, semicoherent and incoherent precipitates

The three interface types of Chapter 5 decide which phase forms first. A **coherent** precipitate shares the matrix lattice with a small misfit $\delta = (a_p - a_m)/a_m$; its **coherency strain** field extends beyond the particle and makes it an effective obstacle to dislocations (Chapter 13). Because strain energy grows with volume and the cost of misfit dislocations only with area, a growing particle eventually becomes **semicoherent**. An unrelated structure gives an **incoherent** interface, high in energy but free of long-range strain.

The precipitation sequence in Al–Cu

The equilibrium θ phase (tetragonal, $I4/mcm$) cannot fit coherently into FCC aluminium, so its nucleation barrier is enormous at low ageing temperatures. The alloy takes a staircase of metastable steps, each with a lower barrier than the next:

$$\text{SSSS} \;\rightarrow\; \text{GP zones} \;\rightarrow\; \theta'' \;\rightarrow\; \theta' \;\rightarrow\; \theta\ (Al_2Cu)$$

Guinier–Preston (GP) zones, detected by X-rays in 1938, are fully coherent copper-rich monolayers on $\{100\}_\alpha$, forming within hours at room temperature with the help of quenched-in vacancies. **θ″** is a coherent ordered phase with large coherency strains; **θ′** is semicoherent, often nucleated on dislocations; **θ** is incoherent and coarse (Table 10.2). This is the Ostwald step rule: the phase that forms first is the easiest to nucleate, not the most stable. Each metastable phase has its own solvus; the GP-zone solvus for Al–4Cu lies at about 150–200 °C, so ageing above it skips the zones.

Table 10.2 — The precipitation sequence in Al–4 wt% Cu (typical).

Stage	Structure	Interface	Typical size	Where it nucleates	Effect on hardness
GP zones	Cu monolayers on {100}	Coherent	2–10 nm diameter, 1–2 atoms thick	Homogeneously (vacancy-assisted)	Rises from ~40 to ~90 HV
θ″	Ordered tetragonal, coherent	Coherent, large strain	10–100 nm plates	On or from GP zones	Peak, ~130 HV
θ′	Tetragonal Al_2Cu	Semicoherent	0.1–1 μm plates	Dislocations, θ″	Declining from peak
θ	Tetragonal Al_2Cu ($I4/mcm$)	Incoherent	Micrometres	Grain boundaries, θ′	Over-aged, ~60–70 HV

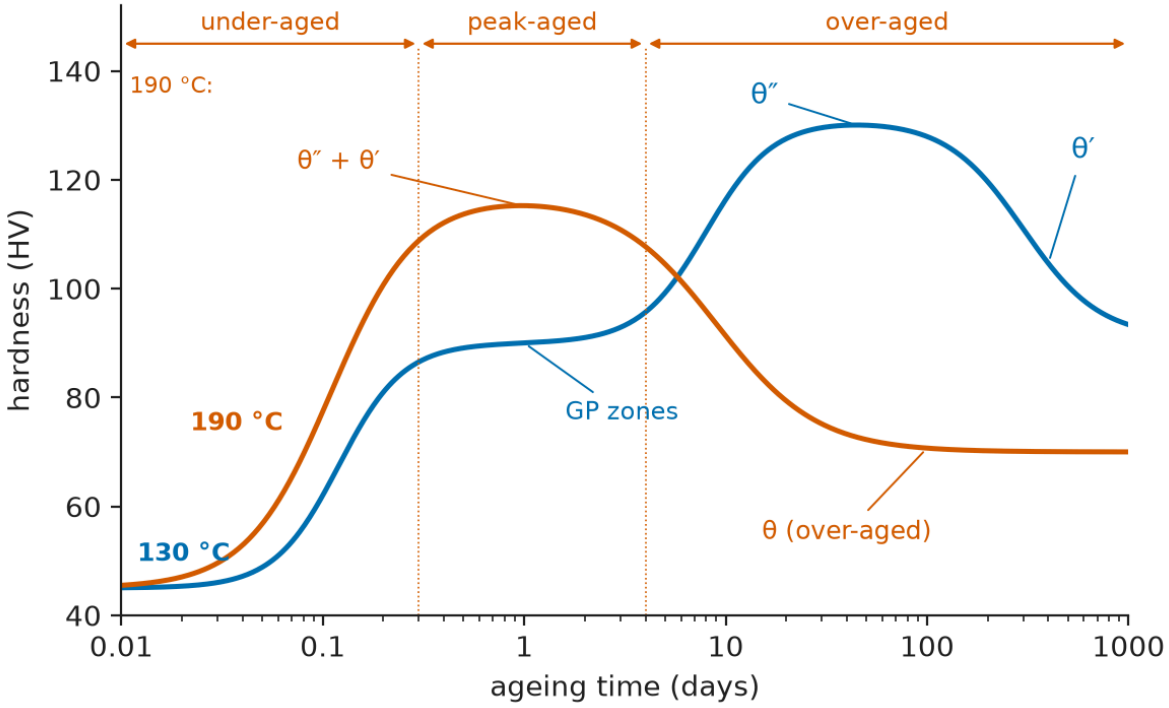

Figure 10.1 — Hardness versus ageing time for Al–4 wt% Cu at 130 °C and 190 °C, with the dominant precipitate labelled along each curve.

Figure 10.1 shows the consequence. At a low ageing temperature hardness rises in two steps to a high peak that takes weeks; at a higher temperature the zones never form and a lower peak arrives within about a day, because fewer, coarser nuclei form. Beyond the peak the alloy **over-ages**: precipitates coarsen, lose coherency and become so widely spaced that dislocations loop around them (Orowan bypass, Chapter 13).

Other precipitation-hardening systems

The same logic drives every age-hardenable alloy: Al–Mg–Si (6xxx, β″ needles, hardened during automotive paint baking at about 180 °C); Al–Zn–Mg–Cu (7xxx, η′, 7075-T6 at about 500 MPa yield for wing spars); nickel superalloys (coherent L12 γ′ with misfit 0.1–0.5 % at up to 70 vol% in turbine blades); maraging steels ($Ni_3(Ti,Mo)$ in carbon-free Fe–Ni martensite, 1.7–2.4 GPa, for rocket-motor cases and landing gear); Cu–2Be springs above 1 GPa; and 17-4PH stainless steel for pump shafts.

Grain boundaries complicate the picture. Solute and vacancies drain to them, leaving a soft **precipitate-free zone** 0.1–1 μm wide, and coarse boundary precipitates can provide paths for intergranular fracture and corrosion. Some alloys show **quench sensitivity**: in a slow quench, solute precipitates coarsely on the way down and is lost to ageing, which is why thick 7xxx plate uses 7050 rather than 7075. Aluminium tempers encode these steps: T4 (naturally aged), T6 (artificially aged to peak), T7 (over-aged for stress-corrosion resistance, Chapter 43), T8 (cold-worked before ageing so that dislocations nucleate a finer precipitate).

Worked example — Shifting an ageing schedule. Alloy 6061 reaches peak strength (T6) in about 8 h at 175 °C. A heat-treater wants to know the equivalent time at 160 °C, 190 °C and 205 °C. If the rate-limiting step is diffusion of Mg and Si with an activation energy $Q \approx 130$ kJ/mol, the time to reach the same microstructural state scales as $t \propto \exp(Q/RT)$:

$$t_2 = t_1 \exp\left[\frac{Q}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

With $T_1 = 448.15$ K and $t_1 = 8$ h:

- At 160 °C ($T_2 = 433.15$ K): $\frac{Q}{R}\left(\frac{1}{433.15} - \frac{1}{448.15}\right) = 15\,636 \times 7.73 \times 10^{-5} = 1.21$, so $t_2 = 8e^{1.21} = 27$ h.
- At 190 °C ($T_2 = 463.15$ K): exponent $= 15\,636 \times (-7.23 \times 10^{-5}) = -1.13$, so $t_2 = 8e^{-1.13} = 2.6$ h.
- At 205 °C ($T_2 = 478.15$ K): exponent $= -2.19$, so $t_2 = 0.90$ h.

The estimate brackets practice (published T6 schedules include about 18 h at 160 °C). The scaling assumes the *same* precipitate sequence at both temperatures, so it fails across a solvus, and a one-hour treatment at 205 °C gives a lower peak, not the same one. Further, a 25 °C overshoot (200 °C instead of 175 °C) cuts the time to peak by a factor of about six ($e^{-1.84} = 0.16$), so a part held for the nominal time spends most of it over-ageing — exactly the 30 % strength loss observed.

10.3 Coarsening: Ostwald ripening and LSW theory

The Gibbs–Thomson driving force

Even when the precipitate has reached its lever-rule volume fraction, the microstructure still carries excess interfacial energy, which falls if small particles dissolve and large ones grow. By the **Gibbs–Thomson effect** (Chapter 6), the solubility next to a particle of radius r exceeds the flat-interface value c_∞:

$$c(r) = c_\infty \exp\!\left(\frac{2\gamma V_m}{rRT}\right) \approx c_\infty\left(1 + \frac{2\gamma V_m}{rRT}\right)$$

where V_m is the molar volume of the precipitate. Solute diffuses from small particles to large ones: **Ostwald ripening**, which operates in every dispersion, from alloy precipitates to liquid-phase-sintered ceramics (Chapter 17) and emulsions.

LSW theory

Lifshitz and Slyozov (1961) and Wagner (1961) solved the problem for a dilute dispersion of spherical particles whose coarsening is controlled by diffusion through the matrix. Their **LSW theory** predicts that, after a transient, the system reaches a self-similar state in which the size distribution keeps a fixed shape while the mean radius $\bar{r}$ grows as

$$\bar{r}^{\,3} - \bar{r}_0^{\,3} = k\,t, \qquad k = \frac{8\gamma D c_\infty V_m}{9RT}$$

for a precipitate that is almost pure solute, with c_∞ here the equilibrium mole fraction of solute in the matrix and D the solute diffusivity. The number of particles falls as t^{-1}, and the distribution is cut off sharply at $1.5\,\bar{r}$. When the controlling step is transfer across the interface rather than diffusion through the matrix, $\bar{r}^2 \propto t$ instead.

> **Status of the principle: LSW coarsening law**
> *Approximate.* Derived exactly for a vanishing volume fraction of spherical, stress-free particles with diffusion-controlled growth; the $t^{1/3}$ law is widely obeyed, but at finite volume fractions the rate constant rises (by a factor of 2–3 at 30 % volume fraction) and the size distribution broadens, and elastic interactions in coherent systems alter particle shapes and rates.

Designing against coarsening

The rate constant identifies the levers available to an alloy designer. Each factor in k can be made small:

- **Low interfacial energy.** Coherent, low-misfit precipitates coarsen slowly; this is a principal reason γ′ in nickel superalloys survives thousands of hours at 1000 °C.

- **Low solubility** c_∞. Oxide dispersions in oxide-dispersion-strengthened (ODS) alloys have practically zero solubility in the metal matrix and do not coarsen even near the melting point; Y_2O_3 in ODS steels for fission and fusion reactors (Chapter 41) is the example.
- **Slow diffusion.** Adding slowly diffusing elements — Re, W and Ta in superalloys, Sc and Zr in aluminium (Al_3Sc and Al_3Zr precipitates stay a few nanometres in size up to about 300 °C) — slows the step that controls k.

Worked example — How long does γ′ take to coarsen? Estimate the coarsening of γ′ in a binary Ni–Al alloy at 1000 °C (1273 K). Take the diffusivity of Al in Ni as $D = D_0 \exp(-Q/RT)$ with $D_0 = 1.87 \times 10^{-4}$ m²/s and $Q = 268$ kJ/mol; the γ/γ′ interfacial energy $\gamma = 0.02$ J/m²; the equilibrium Al mole fraction in γ, $c_\infty = 0.15$; and $V_m = 7.0 \times 10^{-6}$ m³/mol.

Diffusivity: $Q/RT = 268\,000/(8.314 \times 1273) = 25.32$, so $D = 1.87 \times 10^{-4} \times e^{-25.32} = 1.88 \times 10^{-15}$ m²/s.

Rate constant:

$$k = \frac{8 \times 0.02 \times 1.88 \times 10^{-15} \times 0.15 \times 7.0 \times 10^{-6}}{9 \times 8.314 \times 1273} = 3.3 \times 10^{-27}\ \mathrm{m^3/s}$$

Time for the mean radius to grow from 0.20 μm to 0.40 μm:

$$t = \frac{(0.40 \times 10^{-6})^3 - (0.20 \times 10^{-6})^3}{3.3 \times 10^{-27}} = \frac{5.6 \times 10^{-20}}{3.3 \times 10^{-27}} = 1.7 \times 10^{7}\ \mathrm{s} \approx 4700\ \mathrm{h}$$

At 900 °C (1173 K), $D = 2.2 \times 10^{-16}$ m²/s and $k = 4.2 \times 10^{-28}$ m³/s, so the same coarsening takes about eight times longer: an engine run 100 K hotter consumes microstructural life roughly an order of magnitude faster. The cube law is self-limiting — at fixed k, growing from r to $2r$ takes seven times as long as growing from zero to r — and real superalloys coarsen more slowly still because slow diffusers (Ta, W, Re) control the rate. The absolute number is an order-of-magnitude estimate; the scaling is the robust part. Under stress, γ′ coarsens directionally into plates (**rafting**), which changes creep behaviour (Chapter 14).

10.4 Spinodal decomposition

Decomposition without nucleation

A solution with a miscibility gap (Chapter 6) has a central region where $\partial^2 G/\partial c^2 < 0$, bounded by the inflection points, the **chemical spinodal**. Between spinodal and binodal the solution is metastable and a new phase must nucleate; inside the spinodal any fluctuation lowers G and grows, with no barrier. Growth is by **uphill diffusion** (Chapter 8): the effective interdiffusion coefficient $\tilde{D} \propto \partial^2 G/\partial c^2$ is negative, so solute flows towards regions already rich in it.

The Cahn–Hilliard picture

If the effective diffusivity is negative, why do fluctuations not grow infinitely sharp, at the atomic scale? Cahn and Hilliard (1958) answered that a sharp composition gradient carries an energy of its own. They wrote the free energy of an inhomogeneous solution as

$$F = \int_V \left[f(c) + \kappa \left(\nabla c \right)^2 \right] dV$$

where $f(c)$ is the free energy per unit volume of homogeneous material and $\kappa > 0$ is the **gradient-energy coefficient**, representing the extra bonds broken across a composition gradient. Taking the variational derivative gives a generalized chemical potential $\mu = f'(c) - 2\kappa \nabla^2 c$, and requiring that flux follow $-\nabla \mu$ gives the **Cahn–Hilliard equation**

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[M \, \nabla \left(f'(c) - 2\kappa \nabla^2 c \right) \right]$$

with M an atomic mobility. Linearizing about the mean composition c_0 and inserting a sinusoidal fluctuation of wavenumber $\beta = 2\pi/\lambda$, $c - c_0 \propto e^{R(\beta) t} \cos \beta x$, gives the amplification factor

$$R(\beta) = -M\beta^2 \left(f'' + 2\kappa \beta^2 \right)$$

Inside the spinodal ($f'' < 0$), long wavelengths grow, short ones decay because of the gradient term, and the fastest growth is at

$$\lambda_{\max} = \sqrt{2}\,\lambda_c, \qquad \lambda_c = \sqrt{\frac{8\pi^2\kappa}{-f''}}$$

where λ_c is the shortest wavelength that can grow at all. Deep inside the spinodal $-f''$ is large and the wavelength is short, typically 2–10 nm in metals. In a crystal the lattice parameter usually changes with composition, which adds an elastic energy and shifts the instability to a lower **coherent spinodal**; the elastic anisotropy also aligns the modulations along elastically soft directions, $\langle 100 \rangle$ in most cubic metals.

> **Status of the principle: Cahn–Hilliard theory**
> *Derived.* Follows from a gradient expansion of the free energy and linear irreversible thermodynamics; the linearized solution describes only the early stage, while later coarsening needs the full nonlinear equation, which is now the basis of phase-field modelling (Chapter 56).

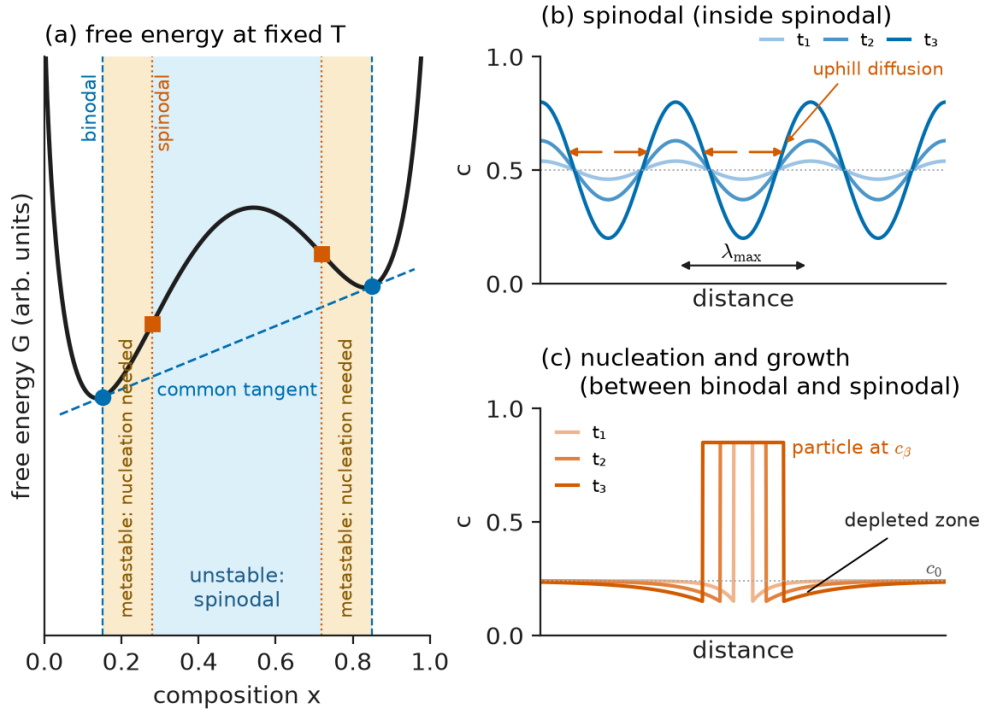

Figure 10.2 — Spinodal decomposition compared with nucleation and growth: free-energy curve with binodal and spinodal points, and the evolution of composition profiles in each regime.

Figure 10.2 contrasts the paths: nucleation makes a sharp-edged particle of final composition that grows in size; spinodal decomposition makes a diffuse periodic modulation that grows in amplitude.

Where spinodal decomposition matters

- **Fe–Cr and duplex stainless steels.** Ferrite with more than about 15 % Cr decomposes into Fe-rich α and Cr-rich α′ at 300–500 °C. The nanometre modulation hardens and embrittles it — **475 °C embrittlement** — limiting duplex steels in chemical and nuclear plant to about 250–300 °C.
- **Cu–Ni–Sn spring alloys** (Cu–15Ni–8Sn) reach yield strengths over 1 GPa as a beryllium-free replacement for Cu–Be in aerospace bearings and connectors.
- **Alnico magnets** owe their coercivity to Fe–Co-rich rods formed spinodally in a magnetic field (Chapter 29).
- **Glasses and polymer blends.** Sodium borosilicate glass separates into interconnected silica-rich and borate-rich networks; leaching the borate leaves the porous silica of the Vycor process. Quenched polymer blends give co-continuous morphologies for membranes (Chapter 20).

Near the spinodal line the nucleation barrier becomes very small, so in practice the two regimes merge.

10.5 Pearlite and bainite

The eutectoid reaction and pearlite

At the eutectoid point of the Fe–C diagram (727 °C, 0.76 wt% C; Chapter 7), austenite (FCC γ) decomposes into almost carbon-free ferrite (BCC α) and cementite (Fe_3C, 6.67 wt% C). The two products grow **cooperatively**: alternating lamellae advance together into the austenite, with carbon diffusing sideways over about half the lamellar spacing from the ferrite front to the cementite front. The product, **pearlite**, nucleates at austenite grain boundaries and grows as nodules until they impinge.

The **interlamellar spacing** λ is set by a balance. A finer spacing shortens the diffusion distance and speeds growth, but it creates more ferrite–cementite interface, whose energy must be paid out of the driving force. Zener's analysis gives a minimum possible spacing at which all the driving force is consumed by interface,

$$\lambda_{\min} = \frac{2\gamma_{\alpha\theta} V_m T_E}{\Delta H \, \Delta T}$$

where T_E is the eutectoid temperature, ΔT the undercooling below it and ΔH the enthalpy of transformation; the observed spacing is typically about twice $\lambda_{\min}$, near the spacing that maximizes the growth rate. The essential result is

$$\lambda \propto \frac{1}{\Delta T}$$

> **Status of the principle: Zener–Hillert spacing and growth of pearlite**
> *Approximate.* From a steady-state diffusion analysis with a maximum-growth-rate (or maximum entropy-production) criterion; the inverse dependence of spacing on undercooling is well confirmed, but absolute rates need the right mix of volume and interface diffusion and the effect of alloying elements partitioning at the front.

Pearlite formed just below 727 °C is **coarse** (spacing about 0.5–1 μm, hardness about 180–220 HB); formed at 550–600 °C it is **fine** (0.1–0.2 μm, 300–400 HB). Yield strength rises roughly as $\lambda^{-1/2}$, a Hall–Petch relation with spacing in place of grain size (Chapter 13). Head-hardened **rails** are cooled with forced air or water mist to spacings near 0.1 μm and 350–400 HB (standard grades 260–300 HB), multiplying wear life in heavy-haul curves. **Patented wire** is transformed at about 550 °C to fine pearlite and cold-drawn to true strains of 3–4, reaching 1770–1960 MPa for bridge cables and over 3500 MPa for tyre cord.

Bainite

Below about 550 °C in a eutectoid steel, austenite transforms instead to **bainite** (described by Davenport and Bain in 1930), a non-lamellar aggregate of ferrite plates or laths and carbides. In **upper bainite** (about 550–400 °C), carbon rejected by parallel ferrite laths precipitates as cementite films between them, which limits toughness. In **lower bainite** (about 400–250 °C), finer plates contain fine carbides at 55–60° to the plate axis, giving strength with good toughness, often better than tempered martensite of equal hardness.

The mechanism is still debated: surface relief and a martensite-like orientation relationship suggest displacive growth with carbon partitioning afterwards (Bhadeshia's view); others argue for diffusional ledge growth controlled by carbon (Aaronson's school). The engineering behaviour is well established either way.

The bainite-start temperature is estimated by empirical formulae such as that of Steven and Haynes,

$$B_s(^\circ\mathrm{C}) = 830 - 270\,\mathrm{C} - 90\,\mathrm{Mn} - 37\,\mathrm{Ni} - 70\,\mathrm{Cr} - 83\,\mathrm{Mo}$$

with compositions in weight percent. For a 4340 steel (0.40 C, 0.70 Mn, 1.80 Ni, 0.80 Cr, 0.25 Mo) it gives $B_s \approx 516$ °C.

> **Status of the principle: Empirical B_s and M_s formulae**
> *Empirical.* Linear regressions on measured start temperatures for low-alloy steels; accurate to about ±25 °C inside the composition range of the data, unreliable outside it.

Silicon (1.5–2 wt%) suppresses cementite, so rejected carbon stabilizes the austenite between the laths. This **carbide-free bainite** is used in rail crossings, automotive steels (Chapter 15) and, transformed at 200–250 °C for days, "nanostructured" bainite with 20–40 nm plates and 2.3–2.5 GPa strength for armour. **Austempering** (quench to a salt bath above M_s, hold until bainite forms) is used for springs and for austempered ductile iron (ADI) gears and crankshafts.

10.6 Martensitic transformations

A transformation without diffusion

Quench a 0.4 % carbon steel from 850 °C into water and the austenite has no time to form pearlite or bainite. Below the **martensite start** temperature M_s, regions of the crystal shear into a new structure in microseconds, without change of composition. The product, **martensite**, is a supersaturated solution of carbon in body-centred tetragonal (BCT) iron, the hardest constituent of steel and the basis of every quenched-and-tempered gear, axle, bolt and tool.

The features that define a martensitic transformation are:

1. **Diffusionless.** The product inherits the parent composition and the order of the atoms; neighbours remain neighbours.
2. **Displacive, with a shape change.** A polished surface tilts where a martensite plate forms. The lattice change is a homogeneous deformation that the surrounding parent must accommodate.
3. **Crystallographically specific.** Each plate has a definite **habit plane** (the parent plane on which it lies) and a definite **orientation relationship** with the parent.
4. **Athermal in most steels.** The amount of martensite depends on how far below M_s the material is cooled, not on how long it is held there. Transformation stops if cooling stops, and finishes at the **martensite finish** M_f.

The Bain correspondence

How can FCC become BCC by a small homogeneous strain? Bain (1924) noticed that an FCC lattice already contains a body-centred tetragonal cell: two adjacent FCC cells share a BCT cell with axes $a_\gamma/\sqrt{2}$, $a_\gamma/\sqrt{2}$, a_γ and axial ratio $\sqrt{2}$. Compressing this cell by about 17–20 % along its c axis and expanding it by about 12–13 % along the other two converts it into the BCC (or, with carbon trapped in octahedral sites along c, the slightly tetragonal BCT) lattice of martensite. This is the **Bain strain** (Figure 10.3), and it requires the smallest atomic movements of any FCC → BCC correspondence.

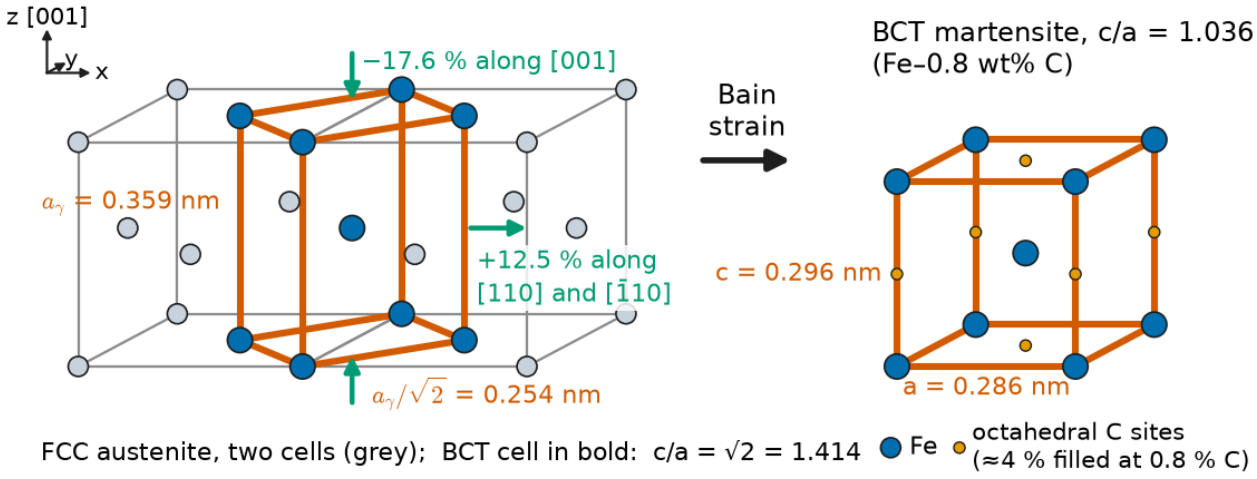

Figure 10.3 — The Bain correspondence: a body-centred tetragonal cell outlined within two FCC austenite cells is compressed along c and expanded along a to give the BCT martensite cell, with interstitial carbon on octahedral sites along c.

Worked example — Bain strains and volume change in a 0.8 % C steel. The lattice parameters vary approximately linearly with carbon content w (wt%): $a_\gamma = 0.3555 + 0.0044w$ nm for austenite, and for martensite $c = 0.2866 + 0.0116w$ nm and $a = 0.2866 - 0.0013w$ nm. At $w = 0.8$:

$a_\gamma = 0.3590$ nm, $c = 0.2959$ nm, $a = 0.2856$ nm; tetragonality $c/a = 1.036$.

The Bain cell has edges $a_\gamma/\sqrt{2} = 0.2539$ nm (twice) and $a_\gamma = 0.3590$ nm. The principal Bain stretches are therefore

$$\eta_3 = \frac{c}{a_\gamma} = \frac{0.2959}{0.3590} = 0.824, \qquad \eta_1 = \eta_2 = \frac{\sqrt{2}\,a}{a_\gamma} = \frac{1.4142 \times 0.2856}{0.3590} = 1.125$$

a compression of 17.6 % and two expansions of 12.5 %. The volume per iron atom changes from $a_\gamma^3/4 = 0.01157$ nm³ in austenite to $ca^2/2 = 0.01207$ nm³ in martensite, a ratio of 1.043: an expansion of about 4.3 %.

Strains of 12–18 % cannot be accommodated elastically, so the transformation must keep the macroscopic shape change small (below). The 4 % volume expansion produces large stresses when the surface of a quenched part transforms before its core; this is why water-quenched high-carbon parts crack and why alloy steels are quenched in oil or polymer.

Invariant planes, habit planes and orientation relationships

The Bain strain leaves no plane of the parent undistorted and unrotated, yet a martensite plate needs such an **invariant plane** as its interface. The **phenomenological theory of martensite crystallography** (Wechsler, Lieberman and Read; Bowles and Mackenzie, 1953–54) resolved this: the shape deformation is the Bain strain plus a **lattice-invariant shear** (slip or twinning inside the plate) plus a rigid rotation, which together give an **invariant-plane strain** — a shear of about 0.2 parallel to the habit plane with a small normal dilatation. From lattice parameters alone, with no adjustable constants, the theory predicts habit plane, orientation relationship and shape change in good agreement with measurement.

Orientation relationships in steels are close to **Kurdjumov–Sachs**, $\{111\}_\gamma \parallel \{110\}_{\alpha'}$ with $\langle 110\rangle_\gamma \parallel \langle 111\rangle_{\alpha'}$, or **Nishiyama–Wassermann** (rotated by about 5°). Habit planes depend on carbon content:

Table 10.3 — Martensite morphologies in Fe–C and Fe–Ni steels (typical).

Carbon content (wt%)	Morphology	Habit plane	Lattice-invariant shear	M_s (°C)	Character
< 0.6	Lath (packets of parallel laths, ~0.2 μm thick)	$\{557\}_\gamma$ (near $\{111\}_\gamma$)	Slip, high dislocation density	350–520	Tough after tempering
0.6–1.0	Mixed lath and plate	—	—	200–350	Intermediate
> 1.0 (or Fe–30Ni)	Lenticular plate with midrib	$\{225\}_\gamma$, $\{259\}_\gamma$	Fine internal twins	< 200	Hard, brittle, microcracks

The start temperature and the amount of martensite

Alloying elements that stabilize austenite lower M_s. Andrews' widely used linear formula is

$$M_s(^\circ\mathrm{C}) = 539 - 423\,\mathrm{C} - 30.4\,\mathrm{Mn} - 17.7\,\mathrm{Ni} - 12.1\,\mathrm{Cr} - 7.5\,\mathrm{Mo}$$

Carbon dominates. Since the transformation is athermal, the fraction transformed depends on the undercooling below M_s; for steels the **Koistinen–Marburger** relation fits well:

$$f = 1 - \exp\left[-0.011\,(M_s - T)\right]$$

> **Status of the principle: Koistinen–Marburger relation**
> *Empirical.* Fitted to measured martensite fractions in plain-carbon and low-alloy steels; the rate constant 0.011 K^{-1} varies somewhat with composition, and the relation does not describe isothermal martensite or the burst transformations of some Fe–Ni alloys.

Worked example — Retained austenite after quenching. For 4340 steel (0.40 C, 0.70 Mn, 1.80 Ni, 0.80 Cr, 0.25 Mo):

$M_s = 539 - 169.2 - 21.3 - 31.9 - 9.7 - 1.9 = 305$ °C.

Quenched to 25 °C: $f = 1 - \exp(-0.011 \times 280) = 1 - e^{-3.08} = 0.954$. About 95 % martensite and 4.6 % **retained austenite**, which is acceptable for a structural shaft.

Now a bearing steel, 52100 (1.0 C, 0.35 Mn, 1.45 Cr), hardened from a temperature at which about 0.6 % C is in solution (the rest remains as undissolved carbides that resist wear). Taking 0.6 C and 1.0 Cr in the austenite: $M_s = 539 - 253.8 - 10.6 - 12.1 = 262$ °C. At 25 °C, $f = 1 - \exp(-0.011 \times 237) = 0.927$: about 7 % retained austenite. In a precision bearing this austenite can transform slowly in service, expanding by 4 % and changing dimensions. A **sub-zero treatment** to −80 °C raises the martensite fraction to 0.977, and to −196 °C to 0.994, which is why gauge blocks and aerospace bearings are cryogenically

treated before tempering. Had all 1.0 % C dissolved, M_s would fall to about 88 °C and about half the austenite would survive the quench — the reason high-carbon steels are hardened with some carbide left undissolved.

Why martensite is hard

Fresh martensite is strong because trapped carbon produces an extreme tetragonal distortion that interacts with every dislocation, the laths or plates are very fine, and the lattice-invariant shear leaves 10^{15}–10^{16} m^{-2} of dislocations or fine twins. As-quenched hardness rises steeply with carbon — about 50 HRC at 0.2 % C, 57–58 HRC at 0.4 %, 64–65 HRC at 0.6 % and above — and hardly depends on alloy content: alloying makes martensite easier to form (Section 10.9), not harder. It is **tempered** at 150–650 °C to restore toughness (Chapter 15).

Martensite is widespread beyond steel: hexagonal α′ in quenched titanium alloys, FCC → HCP cobalt near 420 °C, and the tetragonal → monoclinic transformation of zirconia, whose 4–5 % volume increase, triggered at a crack tip, gives **transformation toughening** (Chapter 14).

Shape memory and superelasticity

In some alloys the transformation is **thermoelastic**: the interface is highly mobile, the transformation strain is accommodated elastically, and the transformation reverses crystallographically on heating. The prime example is equiatomic **NiTi** (Nitinol), B2 austenite ↔ monoclinic B19′ martensite; others are Cu–Zn–Al, Cu–Al–Ni and Fe–Mn–Si.

- **Shape-memory effect.** Below M_f, martensite forms as self-accommodating twin variants with almost no shape change. Deformation reorients the variants by moving twin boundaries, giving up to about 8 % recoverable strain in NiTi. Heating above the austenite finish A_f restores the single-orientation parent and the original shape.
- **Superelasticity.** Just above A_f, stress induces martensite. The material deforms at a nearly constant plateau stress (400–600 MPa for NiTi at body temperature) to 6–8 % strain, and recovers on unloading.

The stress needed to induce martensite rises with temperature according to a **Clausius–Clapeyron relation**,

$$\frac{d\sigma}{dT} = -\frac{\Delta H}{T\,\varepsilon_{tr}}$$

with ΔH the transformation enthalpy per unit volume (negative for austenite → martensite) and ε_{tr} the transformation strain. For NiTi, $\Delta H \approx 20$ J/g (multiplied by the density 6450 kg/m³, 1.3×10^8 J/m³), $T \approx 300$ K and $\varepsilon_{tr} \approx 0.06$ give about 7 MPa/K, in line with measured slopes of 5–8 MPa/K.

> **Status of the principle: Clausius–Clapeyron relation for stress-induced martensite**
> *Fundamental.* Follows from equating the Gibbs energies of parent and martensite under stress (thermodynamics of a first-order transformation with stress and strain as conjugate variables); real materials deviate because of hysteresis and texture.

Superelastic NiTi is used for self-expanding stents and orthodontic arch wires; the shape-memory effect for pipe couplings, thermal actuators and deployable space structures (Chapters 37 and 39).

10.7 Recovery, recrystallization and grain growth

Stored energy as a driving force

Not every solid-state transformation changes phase. When a metal is cold-worked, most of the mechanical work becomes heat, but 1–10 % is stored as dislocations and point defects. The energy per unit volume stored by a dislocation density ρ is about $\frac{1}{2}\rho G b^2$ (the line energy of Chapter 12 times the length per unit volume). For copper ($G = 48$ GPa, $b = 0.256$ nm) cold-worked to $\rho = 10^{15}$ m^{-2},

$$\tfrac{1}{2}\rho G b^2 = \tfrac{1}{2} \times 10^{15} \times 48 \times 10^{9} \times (0.256 \times 10^{-9})^2 \approx 1.6 \times 10^{6}\ \mathrm{J/m^3}$$

about 1.6 MJ/m³, or 11 J/mol — tiny compared with chemical driving forces, but enough to drive a complete reconstruction of the grain structure when atoms become mobile. On heating, the stored energy is released in three overlapping stages (Figure

10.4).

Recovery

Recovery annihilates and rearranges dislocations without moving high-angle boundaries: excess vacancies anneal out, opposite dislocations annihilate, and the rest form low-energy walls by glide and climb — **polygonization** into subgrains. Hardness falls only slightly, but resistivity and residual stress fall substantially, which is why stress-relief anneals of welded structures and cold-formed springs are recovery treatments. High-stacking-fault-energy metals (aluminium, ferritic iron) recover easily because their dislocations climb and cross-slip readily (Chapter 12); in copper, brass and austenitic steel recovery is limited.

Recrystallization

Recrystallization replaces deformed grains with new, nearly dislocation-free grains bounded by high-angle boundaries that sweep through the deformed matrix. Its "nuclei" are pre-existing subgrains at sites of high misorientation (grain boundaries, deformation bands, large particles). The fraction recrystallized follows the **JMAK equation** of Chapter 9, $X = 1 - \exp(-kt^n)$, with n typically 1–3. For a half-time of 10 min and $n = 2$, $k = \ln 2 / t_{0.5}^2$, so 5 % takes $(-\ln 0.95/k)^{1/2} = 2.7$ min and 95 % takes 20.8 min; curves at other temperatures shift by an Arrhenius factor.

Empirically, recrystallization needs a minimum strain of a few percent; more strain lowers the temperature needed and refines the new grains; purity lowers the temperature, and solutes and fine particles raise it. The **recrystallization temperature** (complete in one hour) is roughly 0.3–0.5 of the absolute melting temperature for heavily deformed commercial metals.

Table 10.4 — Approximate recrystallization temperatures after heavy cold work (1 h anneal, commercial purity).

Metal	T_m (°C)	Recrystallization temperature (°C)	Consequence
Lead, tin	327, 232	Below room temperature	Cannot be work-hardened at 20 °C
Aluminium	660	150–300	Annealed (O temper) at 345–415 °C
Copper	1085	200–300	Wire annealed in-line at 400–600 °C
Brass (Cu–30Zn)	~930	350–450	Cold-rolled and annealed in cycles
Low-carbon steel	1538 (Fe)	450–600	Batch or continuous annealing of sheet
Tungsten	3422	~1200–1400	Lamp filament grain structure stabilized by K dopant

Hot and **cold working** are defined by recrystallization, not absolute temperature: rolling lead at room temperature is hot working, rolling tungsten at 1000 °C is not (Chapter 46).

Grain growth

After recrystallization the grain boundaries themselves still store energy (0.3–1 J/m² for metals). Boundaries migrate toward their centres of curvature, large grains with many sides grow at the expense of small ones, and the mean grain size increases. The driving pressure on a boundary of radius of curvature R is about $2\gamma_{gb}/R$: for $\gamma_{gb} = 0.6$ J/m² and $R = 10$ µm it is 1.2×10^5 J/m³, an order of magnitude below the recrystallization driving force computed above, which is why grain growth is slower. If the boundary velocity is proportional to the driving pressure and R is proportional to the mean grain size d, then

$$d^2 - d_0^2 = k_g\, t$$

> **Status of the principle: Parabolic grain-growth law**
> *Approximate.* Derived for curvature-driven migration with a constant mobility (Burke and Turnbull); real metals usually give $d^n - d_0^n = kt$ with n between 2 and 4, because solutes and particles slow boundaries as they move.

Grain growth matters wherever metal is held hot: weld heat-affected zones (Chapter 47), austenitizing before quenching, and sintering of ceramics, where it competes with pore removal (Chapter 17).

Zener pinning

Fine second-phase particles resist boundary migration. A boundary that meets a particle of radius r loses the area it would otherwise have, and must pull free of it to move on; the maximum restraining force per particle is $\pi r \gamma_{gb}$. Summed over the particles that intersect a unit area of boundary for a volume fraction f of randomly distributed particles, the **Zener pinning pressure** is

$$P_Z = \frac{3 f \gamma_{gb}}{2 r}$$

Grain growth stops when the curvature driving force falls to this value. With a driving pressure of $\sim 2\gamma_{gb}/R$ and R related to grain size, Zener's classic estimate of the limiting grain size is

$$d_Z = \frac{4 r}{3 f}$$

> **Status of the principle: Zener pinning and limiting grain size**
> *Approximate.* From a simple force balance with randomly placed spherical particles and a planar boundary; the numerical prefactor varies between models and simulations (from about 0.2 to 1.3 times Zener's), but the proportionality to r/f is robust.

Worked example — Grain-size control in a microalloyed steel. A plate steel contains a volume fraction $f = 0.001$ of NbC precipitates of radius $r = 10$ nm. The limiting austenite grain size is

$$d_Z = \frac{4 \times 10 \times 10^{-9}}{3 \times 0.001} = 1.33 \times 10^{-5}\ \text{m} \approx 13\ \mu\text{m}$$

The pinning pressure is $P_Z = 3 \times 0.001 \times 0.6/(2 \times 10 \times 10^{-9}) = 9.0 \times 10^{4}$ J/m³, comparable to the grain-growth driving pressure but more than an order of magnitude below the stored energy of heavy deformation ($\sim 10^{6}$ J/m³). The particles that stop grain growth therefore cannot stop recrystallization after heavy rolling (though NbC precipitating on dislocations during rolling retards it, the basis of controlled rolling). If the particles coarsen to $r = 50$ nm at the same f (Section 10.3), d_Z rises to 67 μm; if they dissolve, as NbC does near the fusion line of a weld, pinning vanishes, which is why the more stable TiN is used in steels for high-heat-input welding. Only 0.05–0.1 wt% Nb, Ti or V is needed — the "micro" in microalloyed.

Abnormal grain growth

When most boundaries are pinned, a few grains with an advantage — larger size, high-mobility boundaries, a local absence of particles — can escape and consume the fine matrix. This **abnormal grain growth** usually harms toughness and fatigue strength (in carburized gears, for example). It is exploited in **grain-oriented electrical steel** (Fe–3 % Si) for transformer cores: MnS and AlN inhibitors pin normal growth, and Goss-oriented $\{110\}\langle 001\rangle$ grains grow abnormally to centimetre size, aligning the easy magnetization direction with the rolling direction and roughly halving core losses (Chapter 29).

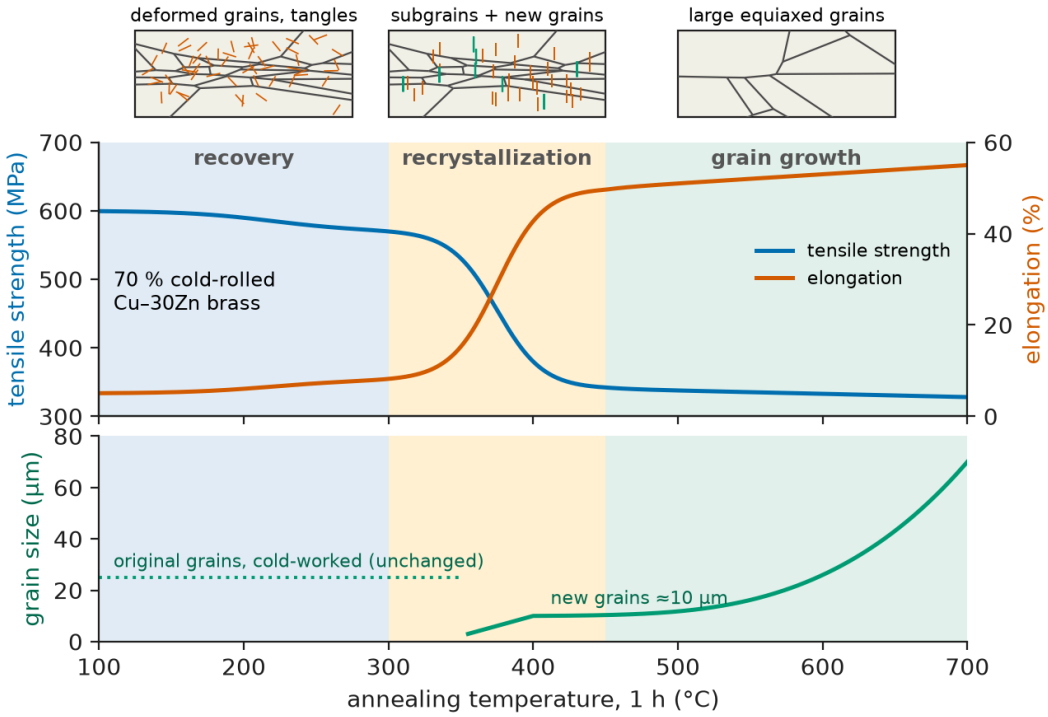

Figure 10.4 — Changes in properties and microstructure during annealing of a cold-worked metal: recovery, recrystallization and grain growth.

Figure 10.4 brings the three stages together for cold-rolled brass, showing why a recrystallization anneal is the tool that resets strength and ductility between cold-rolling passes.

10.8 Massive and discontinuous transformations

Massive transformations

Some alloys, cooled quickly enough to suppress long-range diffusion but not fast enough (or not with the right crystallography) to form martensite, transform by a third route. The product has the **same composition** as the parent but a different crystal structure, and it grows by the rapid thermally activated migration of an incoherent interface, with atoms jumping individually across it. This is a **massive transformation**: diffusional in its atomic mechanism (civilian), but without long-range diffusion and therefore without composition change. Growth rates can reach centimetres per second.

A massive transformation can occur only below the temperature T_0 at which parent and product of the same composition have equal Gibbs energy (Chapter 6). Examples: β brass quenched from about 850 °C transforms to blocky "massive α" (FCC) of the same composition; rapidly cooled very-low-carbon steels form massive ferrite; and in γ-TiAl alloys for low-pressure turbine blades a massive α → γ transformation on fast cooling refines the microstructure and improves ductility.

Discontinuous (cellular) precipitation

In ordinary, **continuous** precipitation, particles form throughout each grain and the matrix composition falls gradually everywhere. In **discontinuous** or **cellular precipitation**, a grain boundary migrates into the supersaturated grain and leaves behind a two-phase lamellar colony of depleted matrix and precipitate, much like pearlite. The matrix composition changes abruptly ("discontinuously") at the moving boundary, and solute diffuses along that boundary rather than through the lattice, which is why the reaction can run at temperatures where volume diffusion is slow:

$$\alpha' \;\rightarrow\; \alpha + \beta \quad \text{(at a moving grain boundary)}$$

where α' is the supersaturated matrix and α the depleted one. It is common in Mg–Al ($Mg_{17}Al_{12}$ lamellae in AZ91 die castings), Cu–Ag, Ni–In and Pb–Sn, and is usually undesirable: the coarse colonies are weaker than a fine continuous dispersion and can lower creep and corrosion resistance.

10.9 Transformation diagrams in practice

Isothermal transformation (TTT) diagrams

A **time–temperature–transformation (TTT)** diagram shows, for a given composition and starting condition, how long transformation takes to start and finish at each constant temperature. Small specimens are quenched from the austenitizing temperature into baths at different temperatures, held, quenched again and examined by metallography or dilatometry. The start (1 %) and finish (99 %) lines form **C-curves**: near the equilibrium temperature the driving force is small, at low temperature diffusion is slow, and in between lies a **nose** of fastest transformation (Chapter 9).

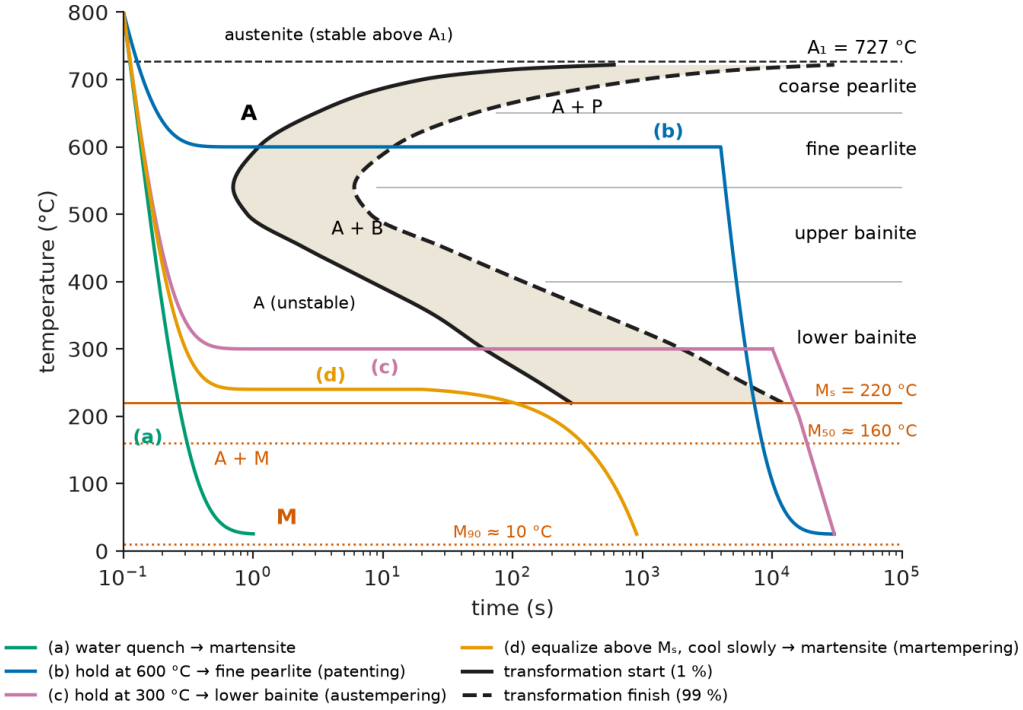

Figure 10.5 — TTT diagram for a eutectoid (0.77 wt% C) plain-carbon steel with pearlite and bainite regions, martensite lines and four cooling paths.

Figure 10.5 is the TTT diagram for a plain eutectoid steel. Its nose lies at about 540 °C and less than one second, which is why only thin sections of plain-carbon steel can be quenched fast enough to miss it and form martensite throughout. The paths drawn on it are standard treatments: **patenting** (hold above the nose for fine pearlite), **austempering** (hold in the bainite range; low distortion, high toughness) and **martempering** (equalize just above M_s, then cool slowly so the whole section transforms together, reducing distortion and quench cracking).

Alloying elements that must partition between ferrite and carbide (Cr, Mo, Mn, Ni) shift the pearlite C-curve to longer times by factors of 10–1000, and Mo and Cr separate pearlite and bainite into distinct C's. Coarser austenite grains delay transformation (fewer boundary sites), and 10–30 ppm boron, segregated to austenite boundaries, delays ferrite nucleation strongly.

Continuous-cooling transformation (CCT) diagrams

Real parts cool continuously, at different rates at different depths. A **continuous-cooling transformation (CCT)** diagram, usually measured by dilatometry, gives start and finish lines for a range of cooling curves. Its curves lie at lower temperatures and longer times than the TTT curves, because the steel spends only part of the time near the nose.

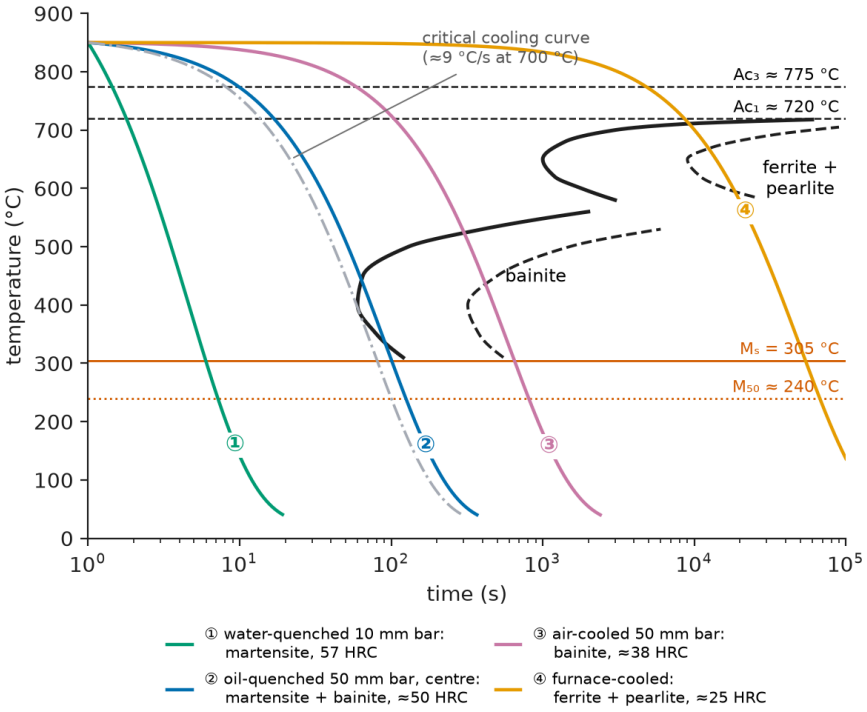

Figure 10.6 — Continuous-cooling transformation diagram for 4340 steel with cooling curves for different section sizes and the resulting microstructures and hardnesses.

The most important reading of a CCT diagram is the **critical cooling rate**: the slowest cooling curve that misses all the diffusional transformation curves and therefore gives fully martensitic structure. For a plain eutectoid steel it is of the order of 100–200 °C/s at 700 °C, attainable only at the surface of small parts with a brine or water quench; for 4340 (Figure 10.6) it is of the order of 1–10 °C/s, attainable in the centre of an oil-quenched bar several centimetres across. The ability of a steel to form martensite at depth is its **hardenability**, measured by the Jominy end-quench test and developed in Chapter 15. Hardenability is what alloying elements are mainly for: they are added not to make martensite harder but to make it possible in thick sections with gentle quenches that avoid cracking.

Transformation diagrams beyond steel

The same tools appear wherever a transformation competes with cooling: quench-sensitivity C-curves limit the furnace-to-tank transfer of 7075 sheet to about 10–15 s; CCT diagrams of Ti–6Al–4V show when cooling from the β field gives α′ martensite, basket-weave α or colony α, which matters for forged and additively manufactured aerospace parts (Chapter 48); crystallization TTT curves set critical cooling rates for glasses and bulk metallic glasses (Chapter 9); and weld-cycle CCT diagrams predict heat-affected-zone hardness and hydrogen-cracking risk (Chapter 47).

Table 10.5 — Reading a transformation diagram: questions and answers.

Engineering question	What to read on the diagram
Will this section harden through?	Compare the centre cooling curve with the nose of the diffusional curves (CCT)
What hardness will the core have?	Hardness values printed at the end of each cooling curve
How long can a part wait before quenching?	Time to the start curve at the transfer temperature (C-curve for quench sensitivity)
At what temperature should I austemper, and for how long?	Bainite region of the TTT diagram; hold beyond its finish curve
How much retained austenite to expect?	Position of M_s and M_f relative to the final temperature (Koistinen–Marburger)
Is the diagram valid for my material?	Composition, austenitizing temperature and grain size stated with the diagram

> **Status of the principle: TTT and CCT diagrams**
> *Empirical.* Measured for a specific composition, austenitizing temperature and grain size; they are reliable for interpolation within a steel grade but must not be transferred between grades or used outside their stated starting conditions. Computed diagrams from CALPHAD-based kinetic models are increasingly used for screening but are calibrated against the same kind of data.

What remains open

Open question	Where it is answered
How exactly do coherent precipitates, GP zones and over-aged particles impede dislocations, and what are cutting and Orowan bypass?	Chapter 13
How do dislocations climb, cross-slip and arrange into walls during recovery, and why does stacking-fault energy matter?	Chapter 12
How are steels heat-treated in practice: tempering stages, hardenability, Jominy, alloy and advanced high-strength steels?	Chapter 15
Which aluminium, titanium, nickel and copper alloys use these transformations, and in which tempers?	Chapter 16
How does γ′ coarsening and rafting set the creep life of turbine blades?	Chapters 14 and 41
How is transformation toughening used in zirconia and how does it give R-curves?	Chapter 14
How are shape-memory and superelastic alloys designed as actuators and medical devices?	Chapters 37 and 39
How do Goss texture and grain size control losses in electrical steels?	Chapter 29
How are recrystallization and grain growth used in rolling, forging and thermomechanical processing, and in welds?	Chapters 46 and 47
How are precipitates and spinodal modulations a few nanometres across actually observed?	Chapters 52 and 53 (TEM, atom-probe tomography)
How are Cahn–Hilliard and microstructure evolution simulated quantitatively?	Chapter 56 (phase-field modelling)
How were the driving forces, nucleation barriers and JMAK kinetics used here derived?	Chapters 6 and 9 (back references)

Next: Chapter 11 opens Part III by asking how a solid responds to force — from the elastic stretching of bonds to the tests that measure yield, strength, hardness and toughness — the properties these transformations exist to control.

Part III — Mechanical Behaviour

Every engineered object carries load: a bridge girder carries traffic, a turbine disc the pull of its blades at high temperature, a hip implant several times body weight a million times a year. Part II explained how thermodynamics and kinetics set a material's structure. Part III asks what that structure does when a force is applied: how much the material stretches, when it stops springing back, how far it can deform before it breaks, and how its behaviour changes with time, temperature and repeated loading.

The answers run through a single chain of ideas. Elastic stiffness comes from the curvature of the interatomic bond and is almost insensitive to microstructure. Plastic flow in crystals comes from the motion of dislocations, and is therefore extremely sensitive to microstructure: every grain boundary, solute atom, precipitate and other dislocation in the way raises the strength. Fracture comes from cracks, and is governed by the competition between the energy released when a crack grows and the energy the material can absorb at the crack tip. Creep, viscoelasticity and fatigue come from thermally activated processes and damage accumulated cycle by cycle. With that chain in hand, a steel bolt, a polymer housing, a zirconia hip head and a jet-engine disc become applications of the same physics.

By the end of this part you should understand:

- How stress and strain are described as tensors, how elastic constants are linked to bonding and crystal symmetry, and what the standard mechanical tests actually measure.
- Why real crystals yield at a thousandth of their theoretical strength, and how the stress fields, forces, interactions and multiplication of dislocations explain it.
- How slip on specific crystal planes produces macroscopic yielding, and how work hardening, solutes, grain boundaries and precipitates raise strength — and how these mechanisms combine in commercial alloys.
- How materials deform slowly under sustained load (creep) and how polymers combine elastic and viscous response.
- How cracks control fracture through K, G and J, why some materials change from ductile to brittle on cooling, how fatigue cracks start and grow, and why brittle strength must be treated statistically.

Chapter	Central question	What it delivers
11. Elasticity and Mechanical Testing	How does a solid respond to small loads, and how are stiffness, strength, hardness and toughness measured?	Stress and strain tensors, Hooke's law and anisotropic elastic constants, the bonding origin of moduli, entropic elasticity (preview), tensile, hardness, impact, bend, compression and shear tests, the statistical nature of strength
12. Dislocation Theory	Why are crystals so much weaker than their bonds, and how do dislocations behave?	Theoretical versus real strength, stress fields and line energy, Peach–Koehler force, glide and climb, jogs, kinks and junctions, Frank–Read sources, partial dislocations and stacking-fault energy, dislocations in FCC, BCC, HCP and ceramics, how they are observed
13. Plasticity and Strengthening Mechanisms	How does slip on crystal planes become macroscopic yielding, and how is strength raised?	Schmid's law and CRSS, single- and polycrystal plasticity, Tresca and von Mises criteria, Taylor work hardening, solid-solution, Hall–Petch and precipitate strengthening, twinning, superplasticity, combining mechanisms
14. Creep, Viscoelasticity and Fracture	How do materials fail over time, under cycles and from cracks?	Creep mechanisms and deformation-mechanism maps, Larson–Miller, viscoelastic models and time–temperature superposition, LEFM and elastic–plastic fracture, ductile–brittle transition, fatigue (S–N, strain–life, Paris law), ceramic toughening, Weibull statistics

Chapter 11 sets up the continuum language and the test methods that every later chapter uses. Chapter 12 descends to the defect that controls crystal plasticity, building on the introduction to dislocations in Chapter 5. Chapter 13 climbs back to the macroscopic scale, turning dislocation behaviour into yield criteria and strengthening rules that alloy designers use. Chapter 14 adds what the earlier chapters hold fixed — time and cracks — and so reaches the failure modes that end real components' lives.

How this part connects to the rest of the book. Part III depends on Parts I and II: crystal structures and symmetry (Chapters 2–4) fix slip systems and elastic anisotropy; defects (Chapter 5) introduce dislocations and grain boundaries; diffusion (Chapter 8) controls climb and creep; and the transformations of Chapter 10 create the precipitates, martensite and grain structures whose strengthening effects are explained here. In turn, Part III is the mechanical foundation for the materials classes of Parts IV and V — steels, non-ferrous alloys, ceramics, concrete, polymers and composites are each judged by the properties defined here — and for extreme environments (Chapter 41), degradation and failure analysis (Chapter 44), tribology (Chapter 45), forming and joining (Chapters 46–48), mechanical characterization (Chapters 52 and 54), dislocation dynamics and crystal-plasticity simulation (Chapter 56) and materials selection (Chapter 58).

Chapter 11 — Elasticity and Mechanical Testing

Part III · Mechanical Behaviour

In 2007 a steel-truss highway bridge over the Mississippi in Minneapolis collapsed at rush hour. The investigation found that gusset plates at several joints were half the thickness they should have been, a design error that had stood unnoticed for forty years while extra concrete deck and construction equipment added load. Nothing about the steel was wrong: its modulus, yield strength and toughness were what the specification said. What failed was the chain that links those properties to a structure — the calculation of the stress each plate carried and its comparison with what the material could bear. A few years earlier, batches of zirconia femoral heads for hip replacements had been recalled after several fractured in patients; there the stresses had been calculated correctly, but a change in the sintering process had altered the population of flaws, and with it the strength distribution that the design relied upon.

Both cases turn on the vocabulary of this chapter: how stress and strain are described, how a material resists them elastically, and how the numbers that engineers put into calculations — modulus, yield strength, tensile strength, hardness, toughness — are actually measured, and with what scatter.

> **How does a solid respond to small loads, and why do stiffnesses span seven orders of magnitude from rubber to diamond?**
> **What do the standard mechanical tests measure, how should their numbers be interpreted, and why must brittle strength be treated as a statistical quantity?**

Concept	Engineering destination
Stress and strain tensors, principal stresses	Finite-element analysis, pressure vessels, shafts, bone implants
Hooke's law, elastic constants	Deflection of beams, bridges and aircraft wings; bolt preload; stiffness design
Anisotropic elasticity	Single-crystal turbine blades, silicon MEMS, wood, composite laminates
Bonding origin of moduli	Choosing stiff, light materials; porous ceramics and foams
Entropic elasticity	Seals, tyres, vibration mounts (developed in Chapter 20)
Tensile test, true stress–strain	Specifications, forming limits, crash simulation
Hardness and nanoindentation	Quality control of heat treatment, coatings and thin films
Impact and toughness tests	Ship plate, pipelines, pressure vessels, cold-climate structures
Bend, compression and shear tests	Ceramics, concrete, adhesives, composite laminates
Statistics of strength	Ceramic implants, glass panels, design allowables, concrete grades

Roadmap. Describing stress and strain (11.1) → linear elasticity, isotropic and anisotropic (11.2) → why moduli have the values they do (11.3) → entropic elasticity of rubbers (11.4) → the tensile test (11.5) → hardness (11.6) → impact and toughness (11.7) → bend, compression and shear tests (11.8) → strength as a statistical property (11.9).

11.1 Stress and strain and their tensors

Traction and the stress tensor

A force applied to a body is transmitted through it from atom to atom. To describe how intensely the material is loaded at a point, imagine cutting the body by a plane through that point with unit normal **n**. The force per unit area acting across the cut is the **traction t**. It depends on the orientation of the cut: a rod in tension carries traction F/A on a cross-section, but none on a plane parallel to its axis. Cauchy showed that the traction on every plane through a point is determined by nine numbers, the components of the **stress tensor** σ_{ij}:

$$t_i = \sigma_{ij}\, n_j$$

(summation over the repeated index). σ_{ij} is the i component of force per unit area on a plane whose normal is along j. The diagonal terms are **normal stresses** (positive in tension); the off-diagonal terms are **shear stresses**, often written τ_{xy}, τ_{yz}, τ_{zx}. Balance of moments on a small cube shows that the tensor is symmetric, $\sigma_{ij} = \sigma_{ji}$, so six components are independent. Stresses are measured in pascals; engineering values are usually given in MPa (equal to N/mm²).

> **Status of the principle: Cauchy stress tensor**
> *Fundamental.* Follows from force and moment equilibrium of an infinitesimal element of a continuum; it applies to any material, elastic or not, at length scales large compared with the atomic spacing.

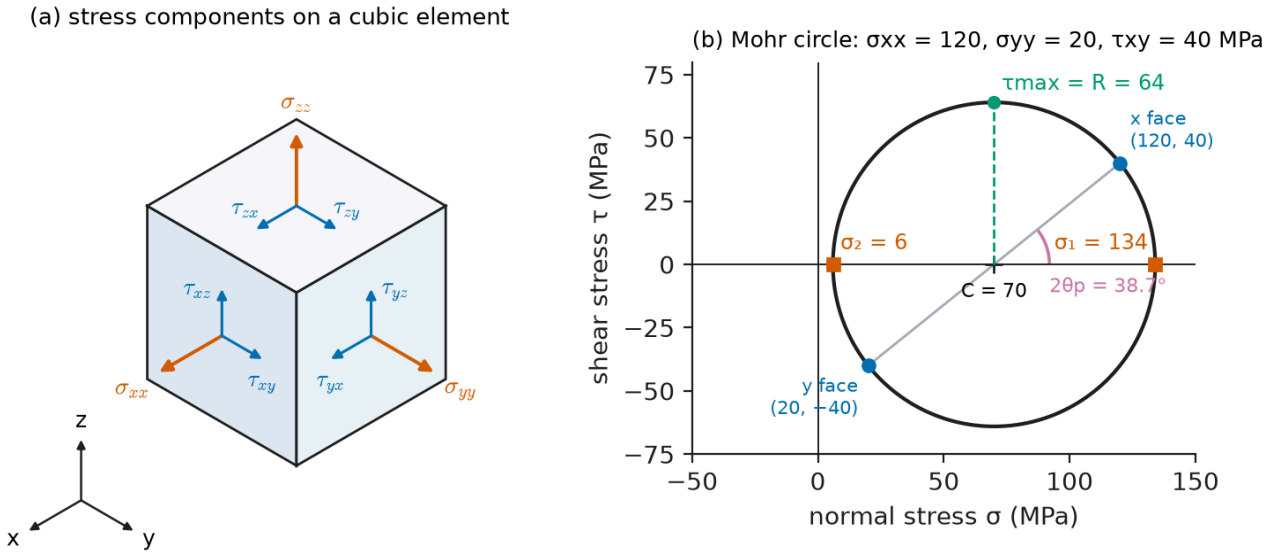

Figure 11.1 — The stress tensor: components acting on the faces of a cubic element, and the Mohr circle for a two-dimensional stress state showing principal stresses and maximum shear.

Principal stresses, hydrostatic and deviatoric parts

Because σ_{ij} is symmetric, there is always a set of three mutually perpendicular directions — the **principal axes** — on whose planes the shear stresses vanish. The normal stresses on them, the **principal stresses** $\sigma_1 \geq \sigma_2 \geq \sigma_3$, are the eigenvalues of the tensor. In two dimensions they are

$$\sigma_{1,2} = \frac{\sigma_{xx} + \sigma_{yy}}{2} \pm \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}$$

and the construction of Mohr's circle (Figure 11.1) displays the normal and shear stress on every plane through the point. For the example in the figure ($\sigma_{xx} = 120$, $\sigma_{yy} = 20$, $\tau_{xy} = 40$ MPa) the centre is at 70 MPa and the radius $\sqrt{50^2 + 40^2} = 64$ MPa, so $\sigma_1 = 134$ MPa, $\sigma_2 = 6$ MPa, and the maximum in-plane shear stress is 64 MPa. The largest principal stress drives brittle fracture; the maximum shear stress drives plastic yielding (Chapter 13).

Any stress state can be split into a **hydrostatic** part, the mean stress $\sigma_m = \tfrac{1}{3}(\sigma_{11} + \sigma_{22} + \sigma_{33})$ acting equally in all directions, and a **deviatoric** part $s_{ij} = \sigma_{ij} - \sigma_m \delta_{ij}$. The hydrostatic part changes volume; the deviatoric part changes shape. The split matters because crystals yield by shear: plastic flow of metals is driven almost entirely by the deviatoric stress, while the hydrostatic stress governs the growth of voids in ductile fracture and the closing of cracks in compression.

Common special states include **uniaxial** tension (a tensile bar, a cable), **pure shear** (a shaft in torsion), **biaxial** tension (a thin-walled pressure vessel, with hoop stress pr/t twice the axial stress $pr/2t$; a thin film on a substrate), and **plane stress** ($\sigma_{3j} = 0$, valid in thin sheets) versus **plane strain** ($\varepsilon_{3j} = 0$, valid deep inside thick sections, where the surrounding material prevents through-thickness contraction).

The strain tensor

Deformation is described by the displacement $\mathbf{u}(\mathbf{x})$ of each material point. For small displacement gradients, the **infinitesimal strain tensor** is

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

Its diagonal components are the fractional changes of length along the axes; the off-diagonal components are half the change of angle between two lines originally along the axes. Engineers commonly use the **engineering shear strain** $\gamma_{ij} = 2\varepsilon_{ij}$ $(i \neq j)$, the full change of angle, so care is needed with factors of two. The trace $\varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33} = \Delta V / V$ is the volumetric strain. The antisymmetric part of the displacement gradient, excluded from ε_{ij}, is a rigid rotation and produces no stress.

> **Status of the principle: Infinitesimal strain**
> *Approximate.* Exact as the displacement gradients tend to zero; adequate for elastic strains of metals and ceramics (below about 1 %), but large-deformation problems — metal forming, rubbers, soft tissue — need finite-strain measures such as the logarithmic (true) strain or the Green–Lagrange tensor.

11.2 Linear elasticity

The generalized Hooke's law

For small strains, the stress in an elastic solid is a linear function of the strain:

$$\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl}, \qquad \varepsilon_{ij} = S_{ijkl}\,\sigma_{kl}$$

C_{ijkl} is the **stiffness tensor** and S_{ijkl} the **compliance tensor**, both of rank 4 (Chapter 4). Symmetry of stress and strain reduces their 81 components to 36, and the existence of a strain-energy function $W = \tfrac{1}{2} C_{ijkl}\varepsilon_{ij}\varepsilon_{kl}$ reduces them to 21 for the least symmetric (triclinic) crystal. It is convenient to use **Voigt notation**, replacing index pairs by single indices ($11 \to 1$, $22 \to 2$, $33 \to 3$, $23 \to 4$, $13 \to 5$, $12 \to 6$), so that stiffness becomes a symmetric 6×6 matrix C_{mn} acting on the vector $(\varepsilon_1, \ldots, \varepsilon_6)$ with $\varepsilon_4 = \gamma_{23}$ and so on.

> **Status of the principle: Generalized Hooke's law**
> *Approximate.* The linear term of an expansion of stress about the unstrained state, with the symmetries guaranteed by energy conservation (which are *fundamental*); accurate for metals and ceramics to strains of about 0.1–0.5 %, but not for rubbers at large strain or for polymers whose response depends on time (Chapter 14).

Crystal symmetry, through Neumann's principle (Chapter 4), reduces the number of independent constants further:

Table 11.1 — Independent elastic constants by symmetry.

Symmetry	Independent constants	Examples
Triclinic	21	Some minerals, feldspars
Orthotropic (orthorhombic)	9	Wood, rolled sheet with texture, unidirectional composite lamina (in general)
Tetragonal	6	Sn, TiO_2 (rutile), martensite
Hexagonal / transversely isotropic	5	Mg, Ti, Zn, ZnO; unidirectional fibre composite; cortical bone (approximately)
Cubic	3 (C_{11}, C_{12}, C_{44})	Fe, Cu, Al, Ni, W, Si, NaCl, MgO
Isotropic	2 (E, ν or K, G or λ, μ)	Glasses, untextured polycrystals, amorphous polymers

Isotropic solids

For an isotropic material, two constants suffice. In uniaxial tension along x,

$$\varepsilon_{xx} = \frac{\sigma_{xx}}{E}, \qquad \varepsilon_{yy} = \varepsilon_{zz} = -\nu\,\frac{\sigma_{xx}}{E}$$

which define **Young's modulus** E and **Poisson's ratio** ν. In general,

$$\varepsilon_{ij} = \frac{1+\nu}{E}\,\sigma_{ij} - \frac{\nu}{E}\,\sigma_{kk}\,\delta_{ij}$$

The **shear modulus** G ($\tau = G\gamma$) and **bulk modulus** K ($\sigma_m = K\,\Delta V/V$) are

$$G = \frac{E}{2(1+\nu)}, \qquad K = \frac{E}{3(1-2\nu)}$$

> **Status of the principle: Isotropic elastic relations**
> *Derived.* Exact consequences of linear elasticity with no preferred direction; they fail for single crystals, textured polycrystals, wood and fibre composites.

Positive-definiteness of the strain energy requires $K > 0$ and $G > 0$, so $-1 < \nu < 0.5$. Most metals have $\nu \approx 0.26$–0.35, ceramics and glasses 0.15–0.27, concrete 0.15–0.2, glassy polymers 0.33–0.40 and rubbers almost exactly 0.5, the incompressible limit at which $K \gg G$. Materials with $\nu < 0$, which expand laterally when stretched, exist as engineered **auxetic** foams and lattices, used in protective padding and some stents.

For steel with $E = 207$ GPa and $\nu = 0.30$: $G = 79.6$ GPa and $K = 172.5$ GPa. Two further combinations recur. A thin film or plate constrained in its plane deforms with the **biaxial modulus** $E/(1-\nu)$; a plate in bending, a wide beam or a material under plane strain with the **plate modulus** $E/(1-\nu^2)$.

Table 11.2 — Elastic constants of engineering materials (room temperature, typical).

Material	E (GPa)	ν	Density (Mg/m³)	E/ρ (GPa per Mg/m³)
Diamond	1000–1100	0.07–0.10	3.52	~300
Tungsten carbide (WC–Co)	550–650	0.22	14–15	~42
Silicon carbide	400–440	0.15–0.19	3.15–3.2	~130
Tungsten	411	0.28	19.3	21
Alumina (99.5 %)	370–390	0.22	3.9	~97
Silicon nitride	300–320	0.27	3.2	~97
Steels (carbon, low-alloy)	200–210	0.29–0.30	7.85	26
Nickel and superalloys (polycrystal)	200–220	0.30–0.31	8.2–8.9	~25
Copper	120–130	0.34	8.96	14
Titanium alloys (Ti–6Al–4V)	110–120	0.34	4.43	26
Aluminium alloys	68–72	0.33	2.7–2.8	26
Soda-lime glass	70–74	0.22	2.5	29
Magnesium alloys	44–45	0.29	1.8	25
Concrete	25–40	0.15–0.20	2.3–2.4	~13
Cortical bone (longitudinal)	15–20	~0.3	1.9–2.0	~9
Spruce, along grain (across grain)	10–13 (0.5–0.8)	—	0.40–0.45	~27
CFRP, unidirectional, along fibres	130–180	0.3	1.55–1.6	~100
Epoxy	2.5–3.5	0.35	1.2	~2.5
PMMA	2.4–3.3	0.37	1.18	~2.5
HDPE	0.8–1.2	0.42	0.95	~1
Natural rubber (vulcanized)	0.001–0.005	0.4999	0.92–1.0	~0.003

A striking feature of the table is that steel, aluminium, titanium and magnesium alloys all have almost the same **specific modulus** $E/\rho \approx 25$ GPa per Mg/m³. Substituting one metal for another does not make a stiff structure lighter unless the geometry changes; the large gains come from ceramics and from carbon-fibre composites, which is why aircraft structures moved from aluminium to CFRP (Chapters 22 and 58).

Anisotropic crystals: the cubic case

A cubic crystal has three independent constants, C_{11}, C_{12} and C_{44}. It would be isotropic only if $C_{44} = (C_{11} - C_{12})/2$; the **Zener anisotropy ratio**

$$A = \frac{2C_{44}}{C_{11} - C_{12}}$$

measures the departure. Chapter 4 showed why cubic symmetry forces a second-rank property such as conductivity to be isotropic but leaves a fourth-rank property free to vary with direction. The Young's modulus along a direction with direction cosines l_1, l_2, l_3 relative to the cube axes is

$$\frac{1}{E_{\langle hkl \rangle}} = S_{11} - 2\left(S_{11} - S_{12} - \tfrac{1}{2}S_{44}\right)\left(l_1^2 l_2^2 + l_2^2 l_3^2 + l_3^2 l_1^2\right)$$

The orientation factor $l_1^2 l_2^2 + \ldots$ is 0 along $\langle 100 \rangle$, 1/4 along $\langle 110 \rangle$ and 1/3 along $\langle 111 \rangle$, so these three directions are the extremes and the intermediate.

> **Status of the principle: Directional modulus of a cubic crystal**
> *Derived.* Exact transformation of the fourth-rank compliance tensor under rotation of axes, given the cubic symmetry.

Worked example — How anisotropic is copper? The single-crystal stiffnesses of copper are $C_{11} = 168.4$, $C_{12} = 121.4$, $C_{44} = 75.4$ GPa. Inverting the cubic stiffness matrix gives the compliances $S_{11} = 14.99$, $S_{12} = -6.28$ and $S_{44} = 13.26$ TPa^{-1}.

Anisotropy ratio: $A = 2 \times 75.4/(168.4 - 121.4) = 150.8/47.0 = 3.21$.

Moduli: $E_{\langle 100\rangle} = 1/S_{11} = 1/14.99$ TPa$^{-1} = 66.7$ GPa. The bracket $S_{11} - S_{12} - \tfrac{1}{2}S_{44} = 14.99 + 6.28 - 6.63 = 14.64$ TPa^{-1}, so

$$\frac{1}{E_{\langle 111\rangle}} = 14.99 - 2 \times 14.64 \times \tfrac{1}{3} = 5.23 \text{ TPa}^{-1}, \qquad E_{\langle 111\rangle} = 191 \text{ GPa}$$

and similarly $E_{\langle 110\rangle} = 130$ GPa. A copper crystal is almost three times stiffer along the cube diagonal than along the cube edge.

An untextured polycrystal averages over all orientations. The Voigt bound (uniform strain) gives $G_V = (C_{11} - C_{12} + 3C_{44})/5 = 54.6$ GPa, the Reuss bound (uniform stress) $G_R = 5/[4(S_{11} - S_{12}) + 3S_{44}] = 40.0$ GPa; with $K = (C_{11} + 2C_{12})/3 = 137.1$ GPa, the Hill average $G_H = 47.3$ GPa gives $E = 9KG/(3K + G) = 127$ GPa and $\nu = 0.345$ — in agreement with the 120–130 GPa measured on annealed copper. The same calculation gives $A = 2.4$ and $E_{\langle 100\rangle} = 132$ GPa, $E_{\langle 111\rangle} = 283$ GPa for iron; for aluminium, with $A = 1.2$, the range is only 64–76 GPa, and tungsten ($A = 1.01$) is isotropic to within 1 %.

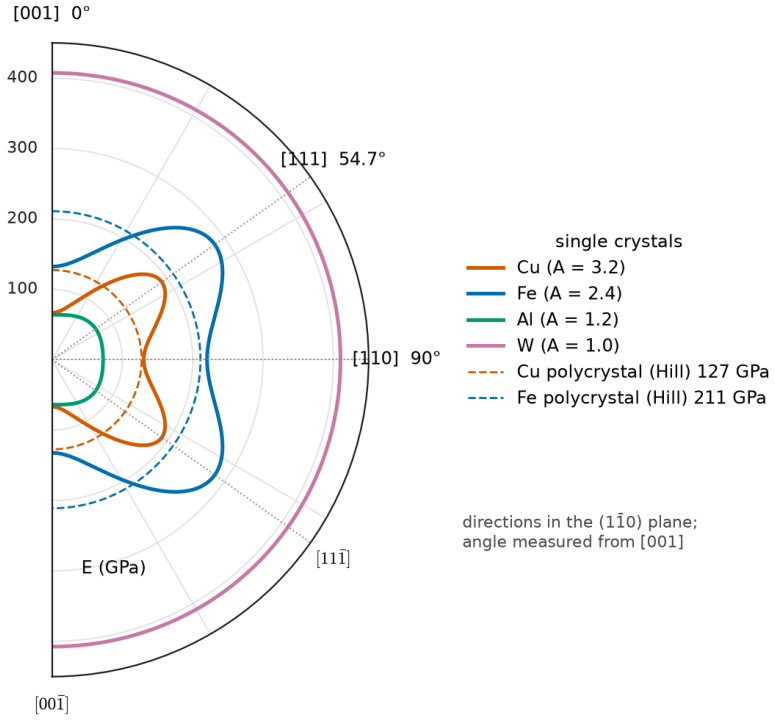

Figure 11.2 — Directional Young's modulus of cubic single crystals of copper, iron, aluminium and tungsten in the (1-10) plane, compared with the isotropic polycrystal values.

Elastic anisotropy has direct engineering consequences, shown in Figure 11.2:

- **Single-crystal turbine blades** of nickel superalloys are grown with $\langle 001\rangle$ along the blade axis. Their modulus in that direction, about 125–130 GPa at room temperature, is roughly 60 % of a polycrystal's (about 210–220 GPa), so thermal strains during engine start-up produce proportionately lower thermal stresses and the blades survive more thermal-fatigue cycles (Chapter 41).
- **Silicon MEMS** resonators and cantilevers have moduli from 130 GPa ($\langle 100\rangle$) to 188 GPa ($\langle 111\rangle$) depending on the direction they are etched along.
- **Textured sheet** has a modulus varying by several percent in its plane, which matters for spring-back in forming.

Orthotropic and transversely isotropic materials

Many materials are anisotropic because of microstructure rather than crystal symmetry. **Wood** is orthotropic, spruce being 15–25 times stiffer along the grain than across it; a **unidirectional fibre composite** is transversely isotropic with five constants (Chapters 22 and 23); **cortical bone** is approximately transversely isotropic. For all of them the full compliance matrix, not a single E, must enter the analysis.

11.3 Origins of elastic moduli in bonding

The interatomic potential

Elastic deformation stretches interatomic bonds without breaking them. Two atoms at separation r have a potential energy $U(r)$ with a minimum at the equilibrium spacing r_0, and the force between them is $F = -dU/dr$. Near r_0 the force is linear in displacement, with a bond stiffness $S_0 = (d^2U/dr^2)_{r_0}$. Treating a solid as an array of such springs, one per area r_0^2 in parallel and one per length r_0 in series, gives

$$E \approx \frac{S_0}{r_0}$$

> **Status of the principle: Bond-stiffness estimate of the modulus**
> *Approximate*. Captures the magnitude and ranking of moduli across bond types; it ignores crystal geometry, angular (bond-bending) forces and many-body effects, and it does not apply to rubbers, whose stiffness is entropic (Section 11.4).

The shape of $U(r)$ is a question of chemical bonding (Advanced Chemistry); the consequences for stiffness are summarized in Table 11.3. Covalent bonds are strong and directional and resist both stretching and bending, giving the stiffest solids. Ionic bonds are strong but non-directional; metallic bonds intermediate. Polymers are covalently bonded along each chain but held to their neighbours by van der Waals or hydrogen bonds 10–100 times less stiff, and these weak links in series dominate the modulus of an unoriented polymer. Drawing the chains parallel, as in ultra-high-molecular-weight polyethylene fibre (about 100 GPa) or aramid fibre (70–140 GPa), loads the covalent backbone directly and raises the modulus by two orders of magnitude.

Table 11.3 — Bond type and stiffness.

Bond type	Typical bond stiffness S_0 (N/m)	Typical E (GPa)	Examples
Covalent (C–C, Si–C)	50–180	200–1000	Diamond, SiC, carbon fibre (axial)
Ionic / ionic–covalent	8–30	30–400	Alumina, MgO, NaCl (~40)
Metallic	15–75	40–410	Mg, Al, Cu, Fe, W
Hydrogen bond	2–5	2–10	Ice (~9), cellulose cross-links
Van der Waals	0.5–2	1–4	Unoriented polymers, graphite between layers

Correlation with melting temperature

The depth of the potential well sets the melting temperature and its curvature sets the modulus; since both scale with bond strength, E correlates with T_m. A useful estimate is $E \approx 100\,kT_m/\Omega$, where Ω is the atomic volume. For copper ($\Omega = 1.18 \times 10^{-29}$ m³, $T_m = 1358$ K) it gives 159 GPa against a measured 124; for iron 212 against 211; for aluminium 78 against 70; for tungsten 322 against 411. The estimate is correct to a factor better than 1.3 across a factor of six in modulus. It also explains why modulus falls with temperature: as atoms vibrate with larger amplitude in an asymmetric well, the mean spacing increases and the curvature at the mean position decreases. Metals typically lose 10–20 % of their room-temperature modulus by half their melting temperature, and steels about 15 % by 500 °C — an effect included in the design of boilers, turbine casings and fire-exposed structures.

Microstructure, porosity and cellular solids

Modulus is called **structure-insensitive** because cold work, grain size and modest alloying change it by only a few percent. Two microstructural features do change it strongly:

- **Second phases** in large volume fractions, as in composites, where the modulus follows rule-of-mixtures bounds (Chapter 22). Tungsten carbide particles raise the modulus of a cobalt matrix to 550–650 GPa in cemented carbides.
- **Porosity.** A widely used empirical fit for isolated pores is $E = E_0(1 - 1.9p + 0.9p^2)$, so 5 % porosity lowers the modulus of a sintered ceramic or powder-metallurgy part by 9 % and 10 % porosity by 18 %. For highly porous **cellular solids** — polymer and metal foams, balsa, trabecular bone, additively manufactured lattices — bending of cell walls dominates and $E/E_s \approx (\rho/\rho_s)^2$, so a foam of 10 % relative density has only about 1 % of the solid's stiffness. Bone-replacement scaffolds and porous titanium implant surfaces are designed with this scaling to match the stiffness of bone and avoid **stress shielding**, the bone loss that occurs when an implant much stiffer than bone carries most of the load (Chapter 39).

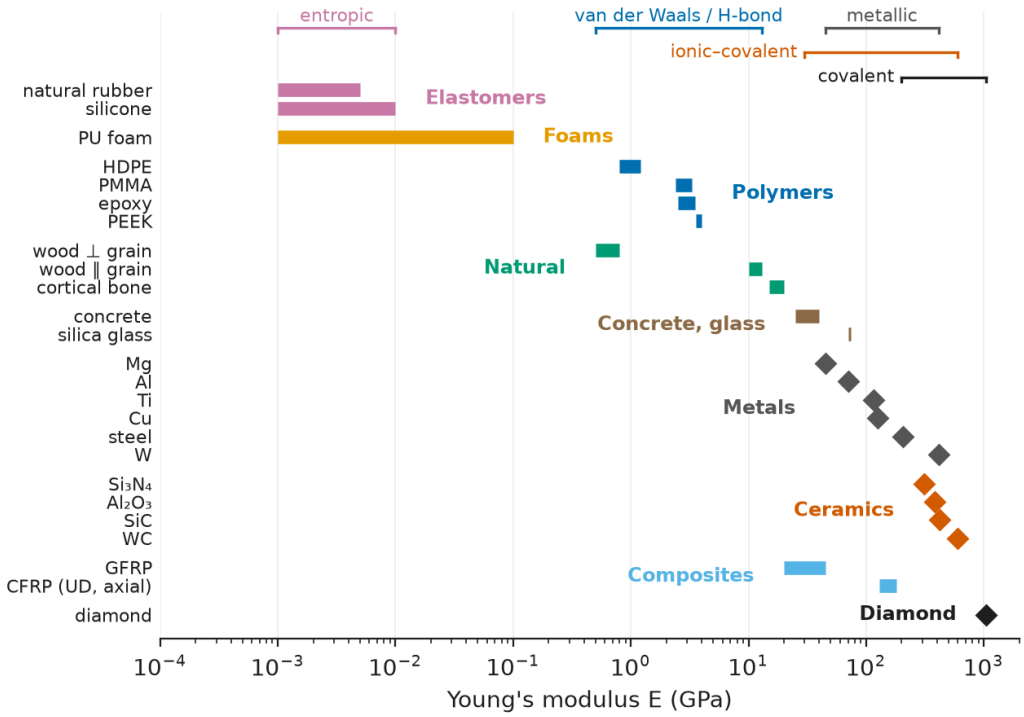

Figure 11.3 — Young's modulus of engineering materials grouped by class and bond type, on a logarithmic scale from 10^{-4} to 10^{3} GPa.

Figure 11.3 summarizes the result: moduli span about seven orders of magnitude, and the main divisions follow bond type, with entropy taking over at the soft end.

11.4 Rubber and entropic elasticity (preview)

A rubber band behaves in ways that bond stretching cannot explain. Its modulus is a few MPa, a thousand times lower than a glassy polymer; it stretches reversibly to 5–10 times its length; it becomes *stiffer*, not softer, when heated; and it warms when stretched quickly. Rubber elasticity is **entropic**. Above its glass-transition temperature, a lightly cross-linked polymer consists of long, flexible chain segments between cross-links, each writhing among an enormous number of conformations. Stretching the network aligns the chains, reduces the number of available conformations and therefore the entropy. The restoring force is the thermodynamic tendency to return to the more probable, randomly coiled state — the same origin as the pressure of an ideal gas.

The simplest network theory (Chapter 20) gives a shear modulus $G = NkT$ and, since rubber is incompressible ($\nu \approx 0.5$), a Young's modulus

$$E = 3NkT$$

where N is the number of network chains per unit volume. For $N = 10^{26}$ m^{-3} (a typical vulcanized natural rubber), $E = 3 \times 10^{26} \times 1.38 \times 10^{-23} \times 300 = 1.24$ MPa at 300 K, rising to 1.45 MPa at 350 K. The modulus is proportional to the cross-link density, which is how vulcanization with sulfur turns a sticky gum into a tyre, seal or vibration mount (Chapter 21).

> **Status of the principle: Entropic (affine network) rubber elasticity**
> *Approximate.* Derived from the statistics of ideal Gaussian chains deforming affinely with the network; it predicts the magnitude and temperature dependence of rubber moduli well at moderate strains but underestimates stiffening at large extensions (finite chain length) and ignores entanglements and fillers.

Rubber stress–strain curves are nonlinear from the start, so a single E is only a small-strain description; hyperelastic models (neo-Hookean, Mooney–Rivlin, Ogden) are used in the finite-element design of seals, bearings for bridges and buildings, and soft robotics. Biological tissues such as arteries, skin and the elastin in lungs combine entropic elasticity with stiff collagen fibres that straighten at larger strains, giving their characteristic J-shaped curves.

11.5 The tensile test

Specimens and measurement

The **tensile test** is the most widely used mechanical test and the source of most datasheet properties. A specimen with a reduced, uniform **gauge section** (round or flat, with gauge length L_0 often $5.65\sqrt{A_0}$ in ISO practice or $4d_0$ in ASTM) is pulled at a controlled rate while the force F and the extension of the gauge length ΔL are recorded (ISO 6892-1 and ASTM E8 for metals; ISO 527 and ASTM D638 for plastics; ASTM D3039 for composites). Engineering stress and strain are

$$\sigma = \frac{F}{A_0}, \qquad e = \frac{\Delta L}{L_0}$$

An **extensometer** clipped to the gauge length is essential for measuring the modulus and yield point: the crosshead displacement includes the elastic deformation of the grips and load frame, which can exceed that of the specimen in the elastic range. Strain rate matters for polymers and for metals near their melting point, and must be reported.

Reading the engineering stress–strain curve

Figure 11.4 shows the features of a curve for a ductile metal:

- The **linear elastic region**, with slope E. (Moduli from tensile tests are usually less accurate than those from ultrasonic or resonance methods, which measure the sound velocity $v = \sqrt{E/\rho}$ — about 5100 m/s in steel and aluminium, 9800 m/s in alumina.)
- **Yield.** Most metals depart gradually from linearity, so the **0.2 % offset yield strength** $R_{p0.2}$ is defined as the stress at which a line parallel to the elastic slope, offset by 0.2 % strain, meets the curve. Low-carbon steels instead show a sharp **upper yield point** followed by a lower yield plateau, during which deformation propagates along the specimen as **Lüders bands**; the cause is carbon and nitrogen atoms pinning dislocations (Chapter 13), and the visible stretcher-strain markings it leaves on pressed sheet are why automotive panels use interstitial-free or temper-rolled steel.
- **Uniform plastic deformation with work hardening.** The stress rises as dislocations multiply and interact.
- The **ultimate tensile strength** (UTS, R_m), the maximum engineering stress, at the **uniform elongation**. Beyond it a **neck** forms, deformation localizes and the engineering stress falls until fracture.
- **Ductility**, reported as the percentage **elongation after fracture** (over a stated gauge length, because the neck contributes a fixed extension) and the **reduction of area** $RA = (A_0 - A_f)/A_0$, which reflects the local ductility in the neck.

Yield strength and UTS are **structure-sensitive**: they depend strongly on grain size, dislocation density, solutes and precipitates, which is the subject of Chapter 13. The area under the curve to fracture is the **tensile toughness** (Section 11.7); the area under the elastic part to yield, $\sigma_y^2/2E$, is the **modulus of resilience**, the elastic energy a spring can store per unit volume.

True stress and true strain

Once plastic deformation begins, the cross-section shrinks. The **true stress** uses the current area, $\sigma_T = F/A$, and the **true (logarithmic) strain** integrates increments of length relative to the current length:

$$\varepsilon_T = \int_{L_0}^{L} \frac{dL}{L} = \ln \frac{L}{L_0} = \ln(1+e)$$

Plastic deformation conserves volume ($AL = A_0 L_0$), so up to the onset of necking

$$\sigma_T = \sigma(1+e)$$

True strains are additive for successive deformation steps — a property essential in metal-forming calculations — whereas engineering strains are not. Beyond necking the area must be measured directly, and the true strain at fracture is $\ln(A_0/A_f)$. The stress in the neck is also no longer uniaxial; the **Bridgman correction** accounts for the triaxial stress state the neck geometry creates.

The plastic part of the true curve for many metals follows the **Hollomon** relation

$$\sigma_T = K\,\varepsilon_T^{\,n}$$

with strength coefficient K and strain-hardening exponent n: about 0.4–0.5 for annealed copper and austenitic stainless steel, 0.2–0.25 for low-carbon steel, 0.05–0.15 for high-strength and age-hardened alloys.

> **Status of the principle: Hollomon power law**
> *Empirical.* A two-parameter fit to true stress–strain data over roughly a decade of strain; refined forms (Ludwik, Swift, Voce) are used where the fit near yield or at large strains matters.

Necking: the Considère criterion

Necking starts when the increase in strength from work hardening can no longer compensate for the decrease in area. At the load maximum $dF = d(\sigma_T A) = 0$; with $dA/A = -d\varepsilon_T$ from volume constancy,

$$\frac{d\sigma_T}{d\varepsilon_T} = \sigma_T$$

For the Hollomon law this gives necking at $\varepsilon_T = n$: the strain-hardening exponent equals the uniform true strain. High-n materials can be stretched further before they localize, which is why austenitic stainless steels and TWIP steels (Chapter 15) are valued for deep drawing and crash-absorbing structures.

> **Status of the principle: Considère criterion**
> *Derived.* Follows exactly from the load maximum and volume constancy; its accuracy is that of the hardening law inserted, and it ignores strain-rate sensitivity, which delays necking in superplastic alloys and polymers.

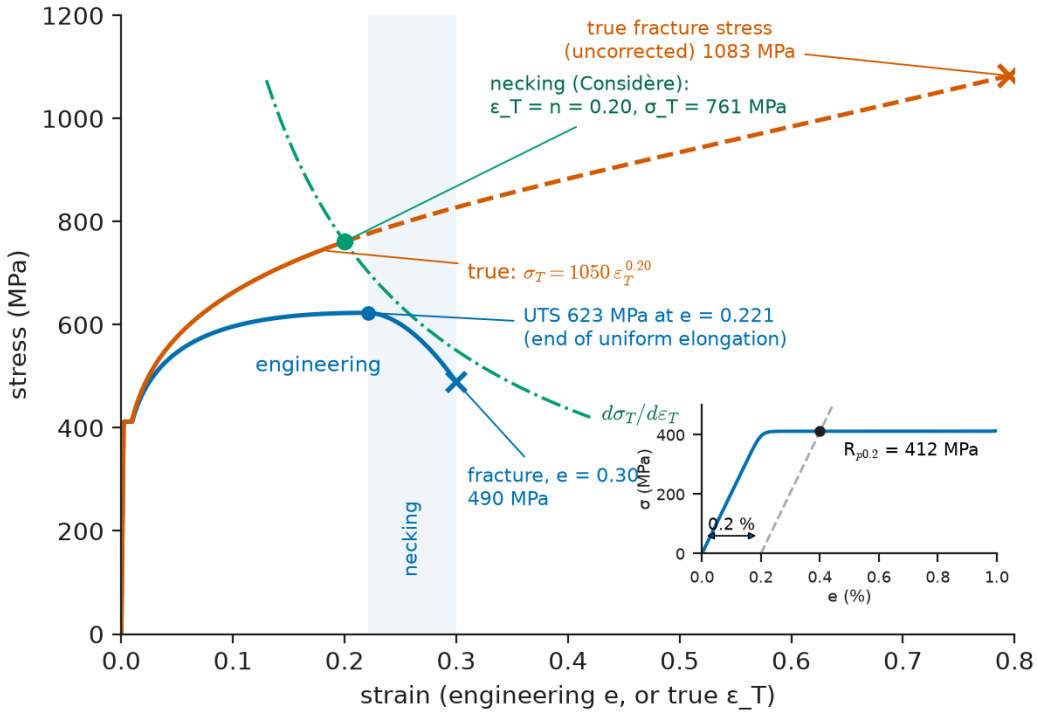

Figure 11.4 — Engineering and true stress–strain curves for a normalized medium-carbon steel (worked example), showing yield, uniform elongation, necking, fracture and the Considère construction.

Worked example — Reducing tensile-test data. A round bar of normalized medium-carbon steel (similar to 1045) with $d_0 = 12.5$ mm and $L_0 = 50$ mm is tested. The record gives: 0.2 % offset yield load 50.5 kN; load 67.1 kN at engineering strain 0.05; 75.6 kN at 0.15; maximum load 76.4 kN; load at fracture 60.0 kN; final gauge length 65.0 mm; diameter at the fracture 8.4 mm.

Area: $A_0 = \pi(12.5)^2/4 = 122.7\ \text{mm}^2$.

Engineering properties: $R_{p0.2} = 50\,500/122.7 = 412$ MPa; UTS $= 76\,400/122.7 = 623$ MPa; elongation $= (65.0 - 50.0)/50.0 = 30$ %; reduction of area $= 1 - (8.4/12.5)^2 = 54.8$ %.

True stress and strain at the two intermediate points: at $e = 0.05$, $\sigma = 547$ MPa, $\sigma_T = 547 \times 1.05 = 574$ MPa, $\varepsilon_T = \ln 1.05 = 0.0488$; at $e = 0.15$, $\sigma = 616$ MPa, $\sigma_T = 708$ MPa, $\varepsilon_T = 0.1398$.

Hollomon fit through the two points:

$$n = \frac{\ln(708.5/574.1)}{\ln(0.1398/0.0488)} = \frac{0.2103}{1.0524} = 0.200, \qquad K = \frac{574.1}{0.0488^{0.200}} = 1050\ \text{MPa}$$

Consistency check with Considère: necking at $\varepsilon_T = n = 0.200$, i.e. engineering strain $e^{0.200} - 1 = 0.221$, where $\sigma_T = 1050 \times 0.2^{0.2} = 760$ MPa and the engineering stress is $760/1.221 = 623$ MPa — the measured UTS.

True fracture strain $= \ln(A_0/A_f) = 2\ln(12.5/8.4) = 0.795$; true fracture stress $= 60\,000/(\pi \times 4.2^2) = 1083$ MPa (before Bridgman correction).

Two practical lessons. The true strain at fracture, 0.795, is nearly four times the uniform strain: most of the ductility of this steel is consumed locally in the neck, which is why reduction of area is a better guide to fracture resistance in a forming operation than elongation. And the elastic extension at a load of 20 kN is only $\sigma/E \times L_0 = (163\ \text{MPa}/205\ \text{GPa}) \times 50\ \text{mm} = 40\ \mu\text{m}$ — comparable to the slack and elastic deflection of grips and load train, which is why modulus cannot be taken from crosshead displacement.

Table 11.4 — Tensile properties of engineering materials (room temperature, typical).

Material and condition	$R_{p0.2}$ (MPa)	UTS (MPa)	Elongation (%)
Structural steel S355 (EN 10025)	≥ 355	470–630	≥ 20–22
AISI 1045, normalized	410–430	620–680	20–25
AISI 4340, quenched and tempered (425 °C)	1350–1500	1500–1650	10–12
Austenitic stainless 304, annealed	205–290	515–620	40–60
Aluminium 6061-T6	275	310	10–12
Aluminium 7075-T6	500	570	10–11
Ti–6Al–4V, mill-annealed	880–950	950–1000	10–15
Copper, annealed	50–70	220–240	40–50
Polycarbonate	60–65	65–70	80–120
PMMA	— (brittle)	50–75	2–5
Alumina	—	250–300 (tensile)	~0
CFRP UD, along fibres	—	1500–2500	1–1.8

11.6 Hardness tests

Indentation hardness

Hardness is resistance to localized plastic deformation, measured by pressing a hard indenter into the surface and measuring the size or depth of the impression. It is quick, nearly non-destructive, works on small regions and finished parts, and correlates with strength. It is the everyday quality-control test for heat treatment, case hardening and welds.

- **Brinell (HB, HBW)**: a 10 mm hard-metal ball pressed with 3000 kgf (for steels) or 500–1500 kgf (softer metals) for 10–15 s; $HB = 2F/[\pi D(D - \sqrt{D^2 - d^2})]$ with F in kgf and D, d (indent diameter) in mm. The large indent averages over coarse microstructures, such as cast iron and castings.
- **Vickers (HV)**: a square-based diamond pyramid with 136° between opposite faces; $HV = 1.8544\,F/d^2$, with d the mean diagonal in mm. Loads from 10 gf (microhardness) to 100 kgf cover everything from single phases and thin coatings to bulk steel, on one continuous scale.
- **Knoop (HK)**: an elongated pyramid giving a shallow indent, for thin coatings and brittle materials; $HK = 14.229\,F/L^2$.
- **Rockwell (HRC, HRB, …)**: the depth of penetration under a major load after a minor preload, read directly from a dial. HRC uses a diamond cone and 150 kgf for hardened steels; HRB a 1/16 in ball and 100 kgf for softer metals. Fast and automated, it is the standard on heat-treatment lines.
- **Shore durometers (A, D)**: spring-loaded indenters for elastomers and soft plastics.

Hardness and strength: Tabor's relation

Under the indenter the plastically deforming material is constrained by the surrounding elastic material, so it must be stressed to about three times its uniaxial flow stress before the indenter sinks in. Slip-line field analysis of a rigid-plastic solid gives

$$H \approx 3\,\sigma_y$$

with H in pressure units (1 HV = 9.807 MPa). Because the material under a Vickers or Brinell indenter is strained by roughly 8 %, σ_y here is the flow stress at that **representative strain**, which for work-hardening metals lies between the yield strength and the UTS. Empirical conversions such as UTS $\approx 3.45\,HB$ (MPa) or UTS ≈ 3.2–$3.3\,HV$ for steels, and standard conversion tables between scales (ASTM E140), are used routinely, but they are material-specific.

> **Status of the principle: Tabor's relation**
> *Approximate.* From plasticity theory for a rigid-perfectly-plastic material with a constraint factor near 3; it fails for brittle materials (cracking), for materials with low E/σ_y such as polymers (elastic accommodation lowers the constraint) and for strongly work-hardening metals unless a representative strain is used.

Worked example — From an indent to a strength. A quenched-and-tempered 4140 shaft is checked with a Vickers indenter at 30 kgf. The mean diagonal is 0.400 mm.

$HV = 1.8544 \times 30/0.400^2 = 1.8544 \times 30/0.160 = 348$ HV, or $H = 348 \times 9.807 = 3.41$ GPa. (This corresponds to about 35 HRC.)

Tabor gives a flow stress $H/3 \approx 1.14$ GPa at about 8 % strain. The steel conversion UTS $\approx 3.3\,HV$ gives about 1150 MPa, consistent with the 1100–1200 MPa expected for 4140 tempered to this hardness; its 0.2 % yield would be roughly 950–1050 MPa. The hardness test therefore checks the heat treatment of every shaft in seconds, while a tensile test on a sample from each batch confirms the conversion.

Table 11.5 — Representative hardness values.

Material	Hardness
Lead	~5 HV
Pure aluminium, annealed	20–25 HV
Copper, annealed	40–50 HV
Aluminium 6061-T6	~105 HV (95 HB)
Low-carbon steel, hot-rolled	120–150 HV
Ti–6Al–4V	330–360 HV
Bearing and tool steels, hardened	60–65 HRC (700–850 HV)
Cemented carbide (WC–Co)	1300–1800 HV
Alumina	1500–1900 HV
Silicon carbide	2500–2800 HV
Cubic boron nitride	~4500 HV
Diamond	8000–10 000 HV
Natural rubber, vulcanized	40–70 Shore A
HDPE / PMMA	60–70 / 85–90 Shore D

Nanoindentation

Nanoindentation (instrumented indentation) records load P against depth h continuously, at loads from micronewtons to hundreds of millinewtons and depths of tens of nanometres to a few micrometres, usually with a three-sided Berkovich pyramid. Instead of imaging the indent, the **Oliver–Pharr method** extracts properties from the unloading curve (Figure 11.5). The initial unloading slope $S = dP/dh$ is elastic and gives the **reduced modulus**

$$E_r = \frac{\sqrt{\pi}}{2\beta}\frac{S}{\sqrt{A}}, \qquad \frac{1}{E_r} = \frac{1-\nu^2}{E} + \frac{1-\nu_i^2}{E_i}$$

where A is the projected contact area ($A \approx 24.5\,h_c^2$ for an ideal Berkovich tip at contact depth h_c), $\beta \approx 1.03$–1.05 a geometry correction, and the subscript i refers to the diamond indenter ($E_i = 1141$ GPa, $\nu_i = 0.07$). Hardness is $H = P_{\max}/A$.

> **Status of the principle: Oliver–Pharr analysis**
> *Approximate.* Based on elastic contact solutions (Sneddon) applied to the unloading curve; accurate for materials that sink in around the indenter, but it overestimates H and E by up to 30 % for soft, non-hardening materials that pile up, and requires careful calibration of the tip area function.

Nanoindentation is the standard tool for thin films and coatings (keeping the depth below about 10 % of the film thickness so that the substrate does not dominate), individual phases, ion-irradiated surface layers in nuclear materials (Chapter 41), bone lamellae and biomaterials. At small depths, hardness rises as the indent shrinks — the **indentation size effect**, caused by geometrically necessary dislocations (Chapter 12) — so nanoindentation hardnesses of metals are not directly comparable with macroscopic values.

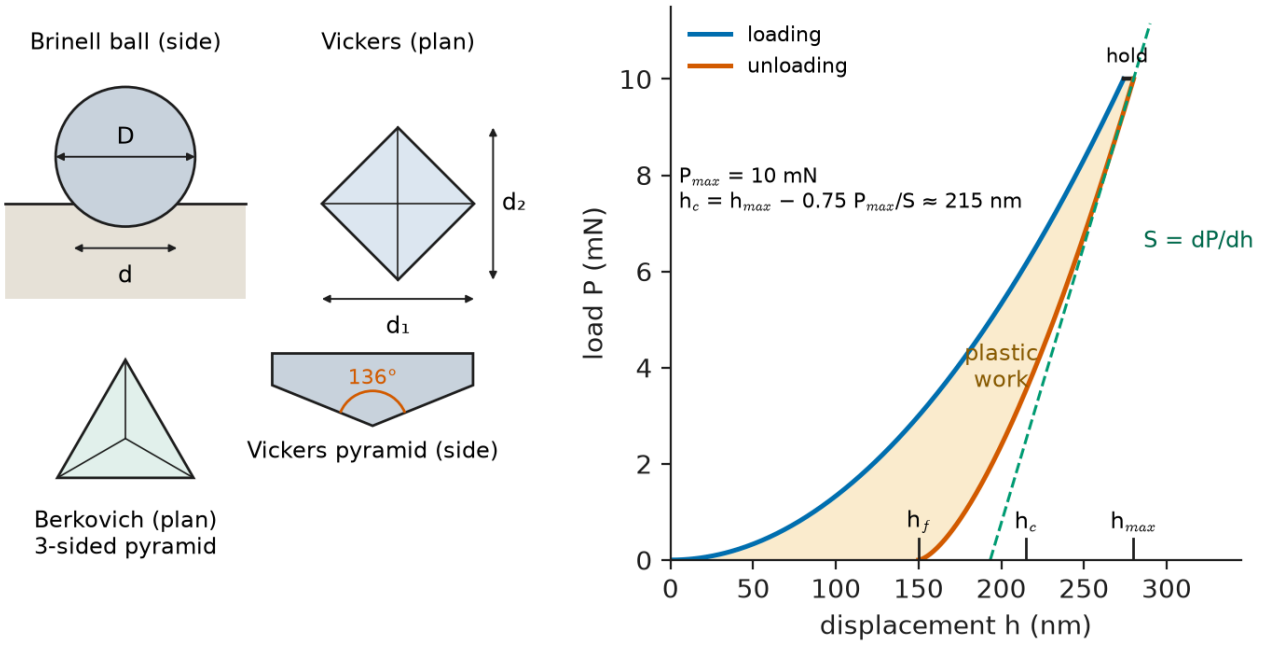

Figure 11.5 — Hardness indenter geometries and a nanoindentation load–displacement curve with the Oliver–Pharr quantities.

11.7 Impact and toughness testing

What "toughness" means

Toughness is the ability to absorb energy before fracture. Three distinct measures share the name:

1. **Tensile toughness**, the area under the stress–strain curve, $U_T = \int_0^{\varepsilon_f} \sigma \, d\varepsilon$, in J/m³. For the steel of the worked example (average flow stress about 560 MPa over an engineering strain of 0.30) it is roughly 1.7×10^8 J/m³; for alumina, $\tfrac{1}{2} \times 300$ MPa $\times\ 0.0008 \approx 1.2 \times 10^5$ J/m³, over a thousand times less.
2. **Impact energy**, absorbed in breaking a notched bar at high rate (this section). It is comparative and specimen-dependent, but cheap and sensitive to embrittlement.
3. **Fracture toughness** K_{Ic} or J_{Ic}, the resistance to propagation of an existing sharp crack, a genuine material property that enters design calculations through fracture mechanics (Chapter 14).

The Charpy and Izod tests

The **Charpy V-notch test** (ISO 148, ASTM E23) breaks a bar 10 × 10 × 55 mm with a 2 mm deep, 45° V-notch in three-point bending by a swinging pendulum striking at about 5 m/s. The energy absorbed is the difference between the pendulum's potential energy at release and at the top of its follow-through swing: a 20 kg pendulum released from 1.5 m (294 J available) that rises to 0.40 m after breaking a specimen has absorbed $20 \times 9.81 \times (1.5 - 0.40) = 216$ J. **Izod** tests use a cantilevered notched specimen and are common for plastics (ISO 180, ASTM D256), where results are often reported per unit width (J/m) or

area (kJ/m²). **Instrumented** strikers record force against time and separate the energy for crack initiation from that for propagation.

The notch, the high strain rate and the bending all promote brittle behaviour, which is the point: the test is a sensitive detector of conditions under which a material that is ductile in a slow tensile test will fracture with little energy.

The ductile–brittle transition

The most important use of impact testing is to find the **ductile-to-brittle transition** of ferritic (BCC) steels (Figure 11.6). At high temperature the fracture surface is fibrous and the energy is high (the **upper shelf**, often 100–300 J); at low temperature it is crystalline and shiny, made of cleavage facets, and the energy falls to a **lower shelf** of a few joules. Between them a transition range spans 30–80 °C. Specifications quote a transition temperature at a fixed energy, such as $T_{27\mathrm{J}}$, or a minimum energy at the service temperature: ship-plate grades, for example, require 27–34 J at 0, −20 or −40 °C depending on grade.

The engineering record makes the importance clear. Of the roughly 2700 welded Liberty ships built during the Second World War, several hundred suffered serious brittle fractures, and about a dozen broke completely in two, many in cold North Atlantic water; the plate steels had transition temperatures near or above the sea temperature. The cause in BCC metals is the steep temperature dependence of their lattice friction for dislocations (Chapter 12), which lets cleavage win over plastic blunting at low temperature. FCC metals — austenitic stainless steel, aluminium, copper, nickel — have no such transition, which is why they are used for cryogenic tanks for liquefied natural gas and liquid hydrogen (Chapter 41). Polymers show a related transition near their glass-transition temperature, and impact tests at −20 or −40 °C qualify plastic housings and pipes for cold climates.

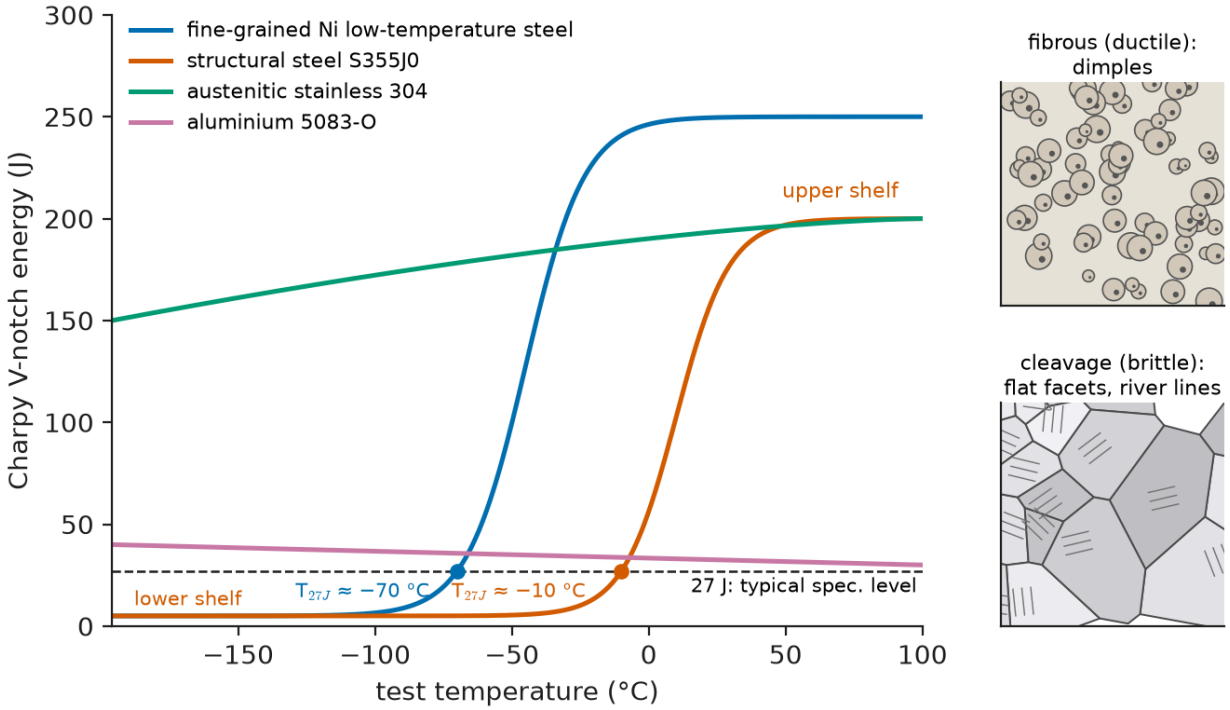

Figure 11.6 — Charpy V-notch impact energy versus test temperature for a ferritic structural steel, a fine-grained low-temperature steel, austenitic stainless steel and an aluminium alloy.

Impact energies cannot be converted directly into fracture toughness, but empirical correlations between Charpy energy and K_{Ic} are used for screening when fracture-toughness data are unavailable, and the **master-curve** method (ASTM E1921) links small-specimen toughness tests to the transition behaviour of reactor pressure-vessel steels, whose transition temperature shifts upward with neutron irradiation (Chapter 41).

11.8 Bend, compression and shear testing

Bend (flexural) tests

Brittle materials are hard to test in tension: gripping a ceramic bar crushes or misaligns it, and a slight bending moment from misalignment raises the stress at one surface. They are therefore tested in **bending** (ASTM C1161 / ISO 14704 for ceramics; ASTM D790 / ISO 178 for plastics; ASTM D7264 for composites). For a rectangular beam of width b and depth d on supports a span L apart:

$$\sigma_{3\text{pt}} = \frac{3FL}{2bd^2}, \qquad \sigma_{4\text{pt}} = \frac{3F(L - L_i)}{2bd^2}$$

for three-point loading (central load F) and four-point loading (total load F on two inner rollers a distance L_i apart). The result is the **flexural strength** or **modulus of rupture**. The flexural modulus follows from the load–deflection slope m as $E = L^3 m / (4bd^3)$ in three-point bending. Because only a small volume near the loading point is highly stressed in three-point bending, three-point strengths of brittle materials are systematically higher than four-point strengths, which in turn are higher than tensile strengths — a size effect quantified in Section 11.9.

Compression tests

Compression testing is used for materials that are mainly loaded in compression or that are brittle in tension: concrete, masonry, ceramics, rock, foams, bone and polymer-matrix composites. Concrete strength classes are defined by compression of 150 mm cubes or 150 × 300 mm cylinders at 28 days; a C30/37 concrete has a characteristic cylinder strength of 30 MPa, so a cylinder fails at about $30 \times 10^6 \times \pi \times 0.075^2 = 530$ kN (Chapter 18). Ceramics are 5–15 times stronger in compression than in tension, because cracks close rather than open.

Platen friction restrains lateral expansion, causing **barrelling** and raising apparent strength, and slender specimens **buckle**; lubrication and a height-to-diameter ratio of 1.5–2 control both. Because compression avoids necking, hot compression tests supply the large-strain flow curves used in forging design (Chapter 46).

Shear and torsion tests

Torsion of a solid round bar of diameter d by torque T produces pure shear, with maximum shear stress at the surface

$$\tau_{\max} = \frac{Tr}{J} = \frac{16T}{\pi d^3}$$

($J = \pi d^4/32$ is the polar second moment of area). A 20 mm steel drive shaft carrying 200 N m therefore sees 127 MPa at its surface. Torsion tests give the shear modulus directly from the twist angle and, because no necking occurs, flow stresses to very large strains for hot-working studies.

Other shear tests suit specific materials: **lap-shear** for adhesive joints (an average over a joint whose real stress peaks at the ends; Chapter 47), **short-beam shear** for the interlaminar strength of composites, and **V-notched** specimens for their in-plane shear.

Table 11.6 — Which test for which material and question.

Material or question	Preferred test	Reason
Ductile metal: specification properties	Tensile	Yield, UTS, elongation, RA in one test
Heat-treatment check on finished parts	Hardness (Rockwell, Vickers)	Fast, local, almost non-destructive
Coatings, thin films, phases	Nanoindentation	Microscale volume, gives H and E
Ceramics, glass	Four-point bend	Avoids gripping; defined stressed volume
Concrete, masonry, rock, foams	Compression	Service loading; tension strength low
Steels for cold service	Charpy V-notch at temperature	Detects ductile–brittle transition
Adhesive joints	Lap shear, peel	Mimics joint loading
Composite laminates	Tensile, compression, short-beam and in-plane shear	Anisotropy requires several tests

11.9 The statistical nature of strength

Why strength scatters

Repeat a tensile test on twenty bars of a ductile steel from the same heat and the yield strengths will typically fall within ±3–5 %. Repeat a bend test on twenty alumina bars and the strengths may range over ±25 % or more. The difference has a physical cause. Ductile failure is a collective process involving the whole cross-section, so it averages over many microstructural features. Brittle fracture starts at the single worst flaw — a pore, an inclusion, a machining scratch — in the stressed region, and flaw sizes vary from specimen to specimen. Strength is then a property of the flaw population, not of the material alone, and must be described statistically.

The Weibull distribution

If a body fails when its weakest element fails (the **weakest-link** hypothesis), and the probability that a small volume contains a flaw critical at stress σ rises as a power of σ, the probability of failure of a body of volume V under uniform stress is

$$P_f(\sigma) = 1 - \exp\left[-\frac{V}{V_0}\left(\frac{\sigma}{\sigma_0}\right)^{m}\right]$$

where σ_0 is the **characteristic strength** of the reference volume V_0 (the stress at which 63.2 % of such specimens fail) and m is the **Weibull modulus**. A large m means little scatter: typical values are 5–10 for glass and traditional ceramics, 10–20 for technical ceramics, about 20–30 for well-controlled silicon nitride, and effectively above 50 for metals. Taking logarithms twice gives $\ln\ln[1/(1-P_f)] = m\ln\sigma + \text{const}$, so test results ranked and plotted this way fall on a straight line of slope m; fitting by maximum likelihood is preferred (ASTM C1239, ISO 20501).

> **Status of the principle: Weibull weakest-link statistics**
> *Empirical.* The weakest-link form follows from independent flaws and a power-law flaw-size tail, but m and σ_0 are fitted to tests, and extrapolation to failure probabilities far below those tested (one in a million for a medical implant) relies on the assumed form; multiple flaw populations give kinked Weibull plots.

Two consequences follow. **Scatter**: with $m = 10$, a stress of $0.8\sigma_0$ gives $P_f = 1 - \exp(-0.8^{10}) = 10$ %, so a design must sit far below the characteristic strength. **Size effect**: at equal P_f, $\sigma \propto V^{-1/m}$, so a larger part is weaker. For non-uniform stress, V is replaced by an **effective volume** V_E, the volume that under the peak stress would give the same failure probability. For a rectangular bar of volume V between the outer supports, $V_E = V/[2(m+1)^2]$ in three-point bending and $V_E = V(m+2)/[4(m+1)^2]$ in quarter-point four-point bending.

Worked example — From bend bars to a ceramic component. Alumina bars 3 mm deep and 4 mm wide are broken in three-point bending over a 40 mm span. The characteristic load is 230 N, so the characteristic flexural strength is

$$\sigma_0 = \frac{3 \times 230 \times 0.040}{2 \times 0.004 \times 0.003^2} = 383 \text{ MPa}$$

and the Weibull modulus is $m = 10$. The volume between supports is $V = 4 \times 3 \times 40 = 480$ mm³.

Effective volumes: three-point, $V_E = 480/(2 \times 11^2) = 1.98$ mm³; four-point (quarter-point), $V_E = 480 \times 12/(4 \times 11^2) = 11.9$ mm³. The four-point characteristic strength is predicted to be $383 \times (1.98/11.9)^{1/10} = 383 \times 0.836 = 320$ MPa — a 16 % difference from the same material, due only to the test geometry.

A component stresses 10 cm³ (10 000 mm³) in uniform tension. Its characteristic strength is $383 \times (1.98/10\,000)^{1/10} = 383 \times 0.426 = 163$ MPa. If the acceptable failure probability is 10^{-4} rather than 63.2 %, the allowable stress falls by a further factor $[-\ln(1 - 10^{-4})]^{1/10} = 0.398$, to 65 MPa.

A material whose bend bars average nearly 400 MPa can therefore be used at only about 65 MPa in a large, uniformly stressed part. Raising m — through cleaner powders, hot isostatic pressing and better surface finishing — narrows the gap far more than raising the mean strength. It is also why ceramic hip heads are individually **proof-tested** to a load above the service load: the test breaks the parts that contain the worst flaws and truncates the low-strength tail of the distribution (Chapters 14 and 39).

Design values from scattered data

Other fields handle scatter with their own conventions. Aerospace design uses **A-basis** allowables (a value exceeded by 99 % of the population with 95 % confidence) for single load paths and **B-basis** (90 % with 95 % confidence) for redundant structure, derived from many tests on several batches. Concrete design uses the **characteristic strength** f_{ck}, the 5 % fractile; for a normal distribution with standard deviation s, $f_{ck} = f_{cm} - 1.645\,s$, so a mix with mean cylinder strength 38.5 MPa and $s = 5$ MPa ($38.5 - 8.2 = 30.3$ MPa) just achieves a C30 class. Steel specifications quote guaranteed minimum yield strengths, with actual mill values usually 10–20 % higher. Fatigue lives scatter by factors of ten at a given stress, and their statistics return in Chapter 14; reliability statistics in service are the subject of Chapter 44.

What remains open

Open question	Where it is answered
Why do crystals yield at a tiny fraction of their theoretical strength?	Chapter 12
How do slip systems, Schmid's law and polycrystal constraints set the yield stress, and what are the Tresca and von Mises criteria?	Chapter 13
What mechanisms raise yield strength and work hardening, and why do low-carbon steels show a yield point?	Chapter 13
How is fracture toughness defined and measured, and how are K, G and J used in design?	Chapter 14
What causes the ductile–brittle transition, and how do fatigue and creep shorten life below the static strength?	Chapter 14
How do time and temperature change polymer stiffness (viscoelasticity)?	Chapters 14 and 21
How does rubber elasticity follow from chain statistics?	Chapter 20
How are the stiffnesses of fibre composites and laminates built from their constituents?	Chapters 22 and 23
How are concrete strength classes, structural-steel grades and timber stiffness specified?	Chapter 18
How are mechanical tests performed at small scales and with what uncertainty?	Chapters 52 and 54
How are tensor properties and Neumann's principle defined?	Chapter 4 (back reference)
How are modulus, strength and density combined to select materials?	Chapter 58

Next: Chapter 12 asks why real crystals yield at stresses thousands of times below the strength of their bonds, and finds the answer in the dislocation — its stress field, the forces on it, and how it moves, multiplies and interacts.

Chapter 12 — Dislocation Theory

Part III · Mechanical Behaviour

Silicon wafers for microchips are grown as single crystals 300 mm in diameter and nearly free of dislocations. During one high-temperature furnace step, a production line found that wafers loaded too quickly into the hot tube developed lines of "slip" visible under a bright light: rows of tiny steps on the polished surface, radiating from the edges where the wafer had touched its quartz support. The thermal stress had never approached the theoretical strength of silicon — it was a few megapascals — yet the wafers had deformed plastically, and the slip lines ruined the devices that crossed them. On the other side of the materials world, a copper wire that yields at 60 MPa when annealed yields at 350 MPa after cold drawing, with no change in composition at all. A steel bridge cable, a jet-engine disc, an aluminium can and a silicon wafer all owe their plastic behaviour to the same defect: the dislocation.

Chapter 5 introduced dislocations as line defects with a Burgers vector. This chapter develops their theory: why they are needed to explain real strengths, what stress fields they carry, what forces act on them, how they move, multiply and interact, how they split into partials, how their behaviour differs between crystal structures, and how they are seen.

> **Why do crystals yield at stresses thousands of times smaller than the strength of their bonds? What controls how easily dislocations move and multiply, and why do FCC, BCC and HCP metals and ceramics behave so differently?**

Concept	Engineering destination
Theoretical versus real strength	Whiskers, nanopillars, defect-free silicon; why strengthening is possible at all
Stress fields, line energy, line tension	Solute atmospheres, dislocation walls, bowing between precipitates
Peach–Koehler force; glide and climb	Yield under complex loads, creep at high temperature
Jogs, kinks, junctions, forest hardening	Work hardening of cold-worked wire, sheet and forgings
Frank–Read and other sources	Dislocation multiplication in forming; slip in wafers
Partials and stacking-fault energy	Cross-slip, recovery, TWIP/TRIP steels, stress-corrosion behaviour
Peierls stress and crystal structure	Ductile–brittle transition of ferritic steels, limited ductility of Mg, Ti and ceramics
Observing dislocations	TEM, etch pits, X-ray topography, EBSD in failure analysis and process control

Roadmap. Theoretical versus real strength (12.1) → geometry recap (12.2) → stress fields and energy (12.3) → forces, glide and climb (12.4) → interactions and forest hardening (12.5) → multiplication (12.6) → partial dislocations and stacking-fault energy (12.7) → dislocations in different crystal structures (12.8) → how they are observed (12.9).

12.1 Theoretical shear strength versus real strength

Frenkel's estimate

Imagine shearing a perfect crystal by sliding one plane of atoms rigidly over the next. The shear stress needed must be periodic in the displacement x, with period equal to the interatomic spacing b in the slip direction, and it must vanish both at $x = 0$ (the perfect crystal) and at $x = b/2$ (an unstable position halfway between sites). The simplest such function is a sine:

$$\tau = \tau_{\max} \sin \frac{2\pi x}{b}$$

For small displacements this must reduce to Hooke's law, $\tau = G\gamma = Gx/d$, where d is the spacing between the planes. Matching slopes at $x = 0$ gives Frenkel's theoretical shear strength (1926):

$$\tau_{\max} = \frac{G}{2\pi}\,\frac{b}{d} \approx \frac{G}{2\pi}$$

Status of the principle: Frenkel theoretical shear strength
Approximate. Assumes rigid planes and a sinusoidal force law; more realistic interatomic potentials and the existence of metastable stacking positions (stacking faults) lower the estimate to about $G/30$, but the order of magnitude, a few percent of G, is confirmed by atomistic calculations and by experiments on defect-free crystals.

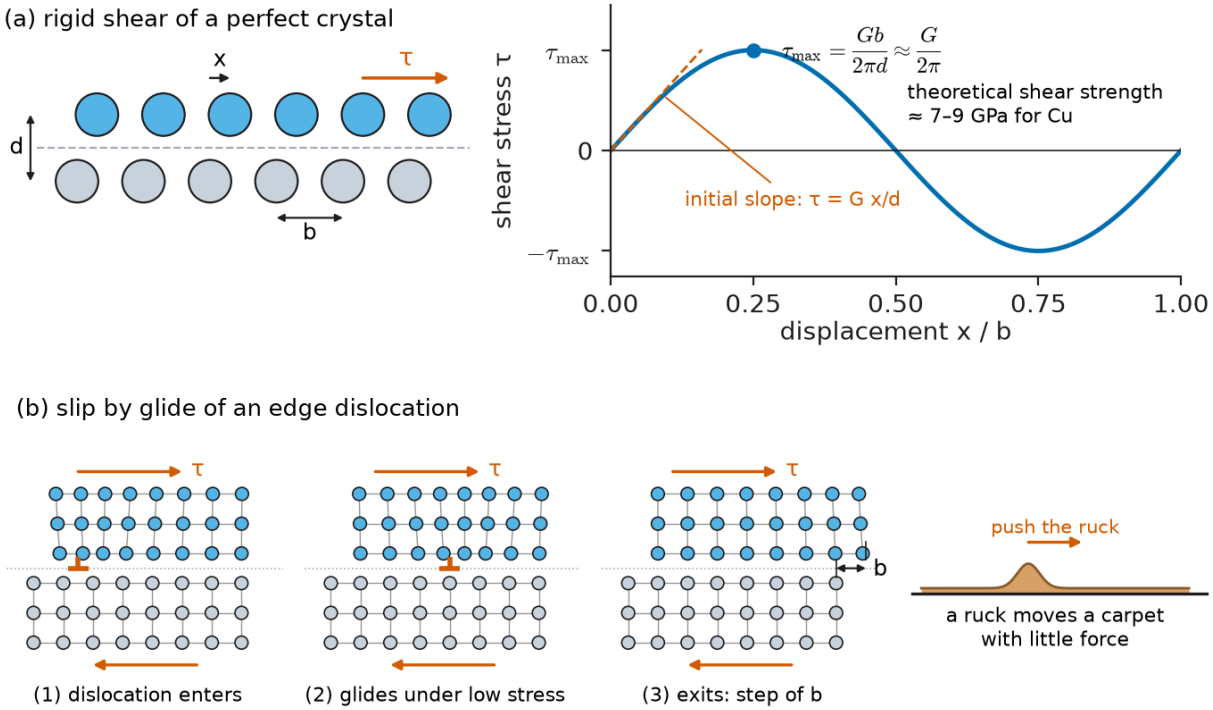

Figure 12.1 — Rigid shear of a perfect crystal compared with slip by dislocation motion.

Worked example — How weak is a real copper crystal? For copper, the polycrystalline shear modulus is $G = 48$ GPa. Slip occurs on $\{111\}$ planes, spaced $d = a/\sqrt{3} = 0.3615/1.732 = 0.2087$ nm apart, in $\langle 110 \rangle$ directions, where the repeat distance is $b = a/\sqrt{2} = 0.2556$ nm. Then

$$\tau_{\max} = \frac{48 \times 10^9}{2\pi} \times \frac{0.2556}{0.2087} = 7.64 \times 10^9 \times 1.225 = 9.4 \text{ GPa}$$

Measured critical resolved shear stresses of well-annealed high-purity copper single crystals are 0.5–1 MPa. The discrepancy is a factor of roughly 10^4. With $G/30$ instead, the theoretical value is still 1.6 GPa, three orders of magnitude above experiment. No refinement of the force law can close a gap of this size; a different mechanism must operate.

The resolution: dislocations

In 1934 Taylor, Orowan and Polanyi independently proposed that slip does not occur by rigid sliding but by the motion of a line defect — the **dislocation** — that separates slipped from unslipped regions of the plane. As the dislocation moves, only the bonds along its line are rearranged at any instant, each by a fraction of an atomic spacing, so the stress needed is a tiny fraction of the theoretical strength. When the dislocation reaches the surface, the crystal above the slip plane has been displaced by one Burgers vector, exactly as if the planes had slid rigidly (Figure 12.1). A ruck in a carpet makes the same point: the whole carpet can be moved a short distance by pushing the ruck across it with a small force, whereas dragging it all at once takes a large one.

Two lines of evidence confirm the picture. Dislocations were seen directly by transmission electron microscopy in 1956 (Section 12.9). And crystals that contain no mobile dislocations do approach theoretical strength: micrometre-sized metal **whiskers** grown from the vapour reach elastic strains of 2–5 % and shear strengths near $G/30$; copper and gold nanopillars below about 100 nm, and defect-free silicon nanowires, reach gigapascal strengths. Bulk dislocation-free silicon wafers are strong only until a dislocation is nucleated at an edge chip or scratch, after which slip spreads easily at high temperature — the opening problem of this chapter.

The engineering consequence is profound. Because real crystals are weak *because of* the mobility of dislocations, they can be made strong by **impeding** that mobility. Every strengthening mechanism in Chapter 13 — work hardening, solutes, grain boundaries, precipitates — works by placing obstacles in the path of dislocations.

12.2 Dislocation geometry: a recap

Chapter 5 defined the essentials, summarized here for use below.

- A dislocation is characterized by its **line direction** $\boldsymbol{\xi}$ (a unit vector tangent to the line) and its **Burgers vector** $\mathbf{b}$, the closure failure of a circuit drawn around it in the crystal compared with the same circuit in a perfect crystal.
- In an **edge** dislocation $\mathbf{b} \perp \boldsymbol{\xi}$; in a **screw** dislocation $\mathbf{b} \parallel \boldsymbol{\xi}$; in general a dislocation is **mixed**, and its character changes along a curved line while $\mathbf{b}$ stays constant.
- A dislocation cannot end inside a crystal: it ends at a surface or grain boundary, forms a closed loop, or meets other dislocations at a **node**, where the Burgers vectors are conserved ($\sum \mathbf{b}_i = 0$ for lines all pointing into the node).
- The plane containing both $\mathbf{b}$ and $\boldsymbol{\xi}$ is the **slip plane**. An edge dislocation has a unique slip plane; a screw dislocation, whose $\mathbf{b}$ and $\boldsymbol{\xi}$ are parallel, can glide on any plane containing its line.
- A **perfect** dislocation has a Burgers vector equal to a lattice translation and leaves a perfect crystal behind it. A **partial** dislocation has a smaller Burgers vector and leaves a stacking fault (Section 12.7).

Table 12.1 — Burgers vectors and slip systems of common structures.

Structure	Burgers vector	$\lvert\mathbf{b}\rvert$	Usual slip plane	Examples ($\lvert\mathbf{b}\rvert$ in nm)
FCC	$\tfrac{a}{2}\langle 110\rangle$	$a/\sqrt{2}$	$\{111\}$	Cu 0.256, Al 0.286, Ni 0.249, austenite ~0.25
BCC	$\tfrac{a}{2}\langle 111\rangle$	$a\sqrt{3}/2$	$\{110\}$, $\{112\}$, $\{123\}$	Fe 0.248, W 0.274, Nb 0.286
HCP	$\tfrac{a}{3}\langle 11\bar{2}0\rangle$ ($\langle a\rangle$)	a	(0001), $\{10\bar{1}0\}$, $\{10\bar{1}1\}$	Mg 0.321, Ti 0.295, Zn 0.266
HCP	$\tfrac{1}{3}\langle 11\bar{2}3\rangle$ ($\langle c+a\rangle$)	$\sqrt{a^2+c^2}$	Pyramidal $\{11\bar{2}2\}$, $\{10\bar{1}1\}$	Mg 0.61, Ti 0.55
Rock salt	$\tfrac{a}{2}\langle 110\rangle$	$a/\sqrt{2}$	$\{110\}$	NaCl 0.399, MgO 0.298
Diamond cubic	$\tfrac{a}{2}\langle 110\rangle$	$a/\sqrt{2}$	$\{111\}$	Si 0.384, Ge 0.400
Corundum	$\tfrac{1}{3}\langle 11\bar{2}0\rangle$	a	(0001) (high temperature)	Al_2O_3 0.476

12.3 Stress fields, line energy and line tension

The screw dislocation

The elastic field of a straight dislocation in an infinite isotropic solid can be found exactly outside a small **core** region of radius $r_0 \approx b$ to $4b$, where strains are too large for linear elasticity. The screw dislocation is simplest. Cut a cylinder along a radial plane, displace one face of the cut parallel to the axis by b and reglue it. The displacement $u_z = b\theta/2\pi$ produces pure shear, with the only non-zero stress component

$$\tau_{\theta z} = \frac{Gb}{2\pi r}$$

There is no hydrostatic stress: a screw dislocation changes shape but not volume, so it interacts only weakly with solute atoms that simply dilate the lattice.

The edge dislocation

For an edge dislocation along z with $\mathbf{b}$ along x and slip plane $y = 0$, the stresses in Cartesian coordinates are

$$\sigma_{xx} = -D\,\frac{y(3x^2+y^2)}{(x^2+y^2)^2}, \qquad \sigma_{yy} = D\,\frac{y(x^2-y^2)}{(x^2+y^2)^2}, \qquad \sigma_{xy} = D\,\frac{x(x^2-y^2)}{(x^2+y^2)^2}$$

with $\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy})$ and

$$D = \frac{Gb}{2\pi(1-\nu)}$$

The edge dislocation has both shear and normal stresses (Figure 12.2). The mean (hydrostatic) stress is

$$\sigma_m = \tfrac{1}{3}\sigma_{kk} = -\frac{2}{3}(1+\nu)\,D\,\frac{y}{x^2+y^2}$$

compressive above the slip plane ($y > 0$), where the extra half-plane is squeezed in, and tensile below it. Large solute atoms are therefore attracted to the tensile side and small ones to the compressive side, forming **Cottrell atmospheres** that pin dislocations — the origin of the yield point and strain ageing of low-carbon steels (Chapter 13).

> **Status of the principle: Elastic fields of straight dislocations**
> *Derived.* Exact solutions of isotropic linear elasticity for an infinite straight line, valid outside the core ($r \gtrsim b$); anisotropic elasticity changes the angular form, and the $1/r$ decay is modified near free surfaces and by neighbouring dislocations.

All components fall as $1/r$, so the field is long-ranged. In copper ($G = 48$ GPa, $b = 0.256$ nm, $\nu = 0.34$), the shear stress on the slip plane 10 nm from an edge dislocation is about $D/r = Gb/[2\pi(1-\nu)r] = 296$ MPa — far above the yield strength of annealed copper. Neighbouring dislocations therefore interact strongly even when tens of nanometres apart.

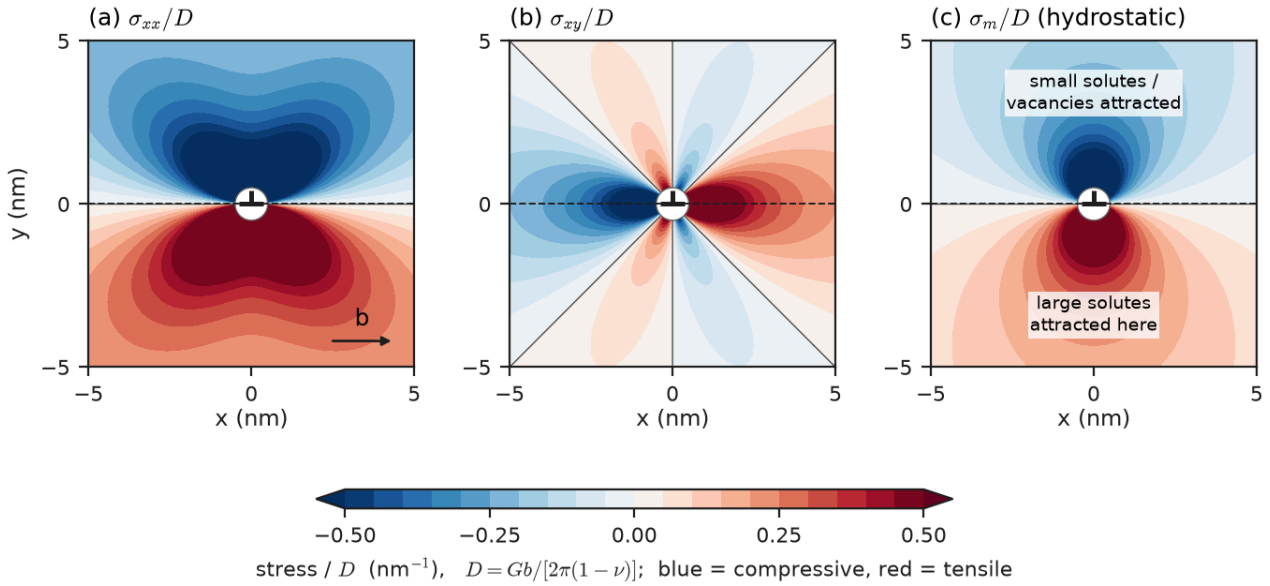

Figure 12.2 — Stress field of an edge dislocation: contour maps of σxx, σxy and the hydrostatic stress around a dislocation at the origin.

Line energy

The elastic energy stored per unit length of dislocation is found by integrating the strain-energy density from the core radius r_0 to an outer cut-off R:

$$E_{\text{screw}} = \frac{Gb^2}{4\pi}\ln\frac{R}{r_0}, \qquad E_{\text{edge}} = \frac{Gb^2}{4\pi(1-\nu)}\ln\frac{R}{r_0}$$

The logarithm diverges for an isolated dislocation in an infinite body; in a real crystal R is set by the distance to neighbouring dislocations of opposite sign, typically $\rho^{-1/2}$, or by the grain size. The core adds a further energy of about $Gb^2/10$. Because the logarithm varies slowly, a convenient approximation for the total line energy is

$$E_L \approx \alpha\, Gb^2, \qquad \alpha \approx 0.5\text{–}1$$

> **Status of the principle: Line energy $E_L \approx \alpha G b^2$**
> *Approximate.* Follows from integrating the elastic field with physically chosen cut-offs; the numerical factor depends on character, cut-off radii and core structure, but the proportionality to Gb^2 is robust.

Worked example — The energy of a dislocation in copper. Take $G = 48$ GPa, $b = 0.2556$ nm, $\nu = 0.34$, $r_0 = b$ and $R = 1$ μm (the spacing at a density near 10^{12} m^{-2}). Then $\ln(R/r_0) = \ln(3912) = 8.27$ and

$$E_{\text{screw}} = \frac{48 \times 10^9 \times (0.2556 \times 10^{-9})^2}{4\pi} \times 8.27 = 2.1 \times 10^{-9}\ \text{J/m}, \qquad E_{\text{edge}} = \frac{2.1 \times 10^{-9}}{0.66} = 3.1 \times 10^{-9}\ \text{J/m}$$

The approximation $\tfrac{1}{2}Gb^2 = 1.57 \times 10^{-9}$ J/m is within a factor of two. Expressed per atomic length b of line, $\tfrac{1}{2}Gb^2 \times b = 4.0 \times 10^{-19}$ J $= 2.5$ eV.

Three consequences follow from these numbers. First, dislocations are **not equilibrium defects**: a few electron-volts per atom length of line is far more than the configurational entropy gained ($k_B T \ln$(few), a fraction of 0.1 eV at room temperature per site), so unlike vacancies, dislocations raise the free energy and are present only because of the crystal's growth and deformation history. Second, since $E \propto b^2$, dislocations with the **shortest Burgers vectors** are favoured: this is why slip in metals occurs along close-packed directions, and why a dislocation with Burgers vector $\mathbf{b}_1$ tends to dissociate into $\mathbf{b}_2 + \mathbf{b}_3$ if $b_2^2 + b_3^2 < b_1^2$ (**Frank's rule**, used in Section 12.7). Third, the stored energy of a cold-worked metal, $\sim \tfrac{1}{2}\rho G b^2$, is about 1.6 MJ/m³ at $\rho = 10^{15}$ m^{-2} in copper — the driving force for recovery and recrystallization calculated in Chapter 10.

Line tension

Because energy is proportional to length, a curved dislocation behaves like a stretched string with **line tension** $T \approx E_L \approx \tfrac{1}{2}Gb^2$. To hold a dislocation segment curved to radius R against this tension, a shear stress τ on its slip plane must supply a force τb per unit length, so

$$\tau = \frac{T}{bR} \approx \frac{Gb}{2R}$$

> **Status of the principle: Line-tension approximation**
> *Approximate.* Treats the dislocation as a string of constant tension, ignoring the long-range interaction between different parts of a curved line and the dependence of energy on character; it captures the scaling of bowing, source and bypass stresses to within a factor of about two.

This simple relation underlies the stress needed to bow a dislocation between pinning points: a dislocation pinned at points L apart can bow out no further than a semicircle of radius $L/2$, which requires $\tau \approx Gb/L$. It is the basis of the Frank–Read source (Section 12.6) and of the Orowan stress for bypassing precipitates (Chapter 13).

12.4 Forces on dislocations: glide and climb

The Peach–Koehler equation

A dislocation moves because the applied stress does work when it moves. If a segment of length l glides a distance s, the crystal above the swept area ls is displaced by $\mathbf{b}$ relative to that below; the work done by the shear stress τ on the slip plane in the direction of $\mathbf{b}$ is $\tau b l s$. The **force per unit length** on the dislocation is therefore τb, acting perpendicular to the line in the slip plane. For an arbitrary stress tensor, Peach and Koehler (1950) showed that the force per unit length is

$$\mathbf{F} = (\boldsymbol{\sigma} \cdot \mathbf{b}) \times \boldsymbol{\xi}$$

> **Status of the principle: Peach–Koehler force**
> *Derived.* Follows from the work done by a stress field on a moving dislocation in linear elasticity; it gives both the glide and climb components and applies to applied, internal and image stresses alike.

The component of **F** in the slip plane is the **glide force** $\tau_{\text{rss}}\, b$, where τ_{rss} is the shear stress resolved on the slip plane in the direction of **b** (Schmid's law, Chapter 13). For a resolved shear stress of 10 MPa in copper the glide force is $10^{7} \times 2.556 \times 10^{-10} = 2.6 \times 10^{-3}$ N/m. The component perpendicular to the slip plane, which exists only for edge character, is the **climb force**, produced by normal stress along **b**: for an edge dislocation, a compressive σ_{xx} pushes the extra half-plane to shrink (climb up) and a tensile one to grow (climb down).

Glide

Glide (slip) is motion in the slip plane. It is **conservative**: atoms only rearrange locally and no mass transport is required, so glide can occur at any temperature at which the applied stress overcomes the lattice resistance and obstacles. The relation between dislocation motion and macroscopic strain is the **Orowan equation**

$$\dot{\gamma} = \rho_m\, b\, \bar{v}$$

where ρ_m is the density of mobile dislocations and $\bar{v}$ their average velocity. A strain rate of 10^{-3} s^{-1} in a copper crystal with $\rho_m = 10^{12}$ m^{-2} requires an average velocity of only $10^{-3}/(10^{12} \times 2.556 \times 10^{-10}) = 3.9$ μm/s, though in practice dislocations move in jerks between obstacles at much higher local speeds. Velocities rise steeply with stress, often as $v \propto \tau^{m}$ with m from 2–5 in FCC metals to 20–40 in some BCC metals and ionic crystals, and are limited at very high stress by phonon drag and ultimately by the shear-wave speed.

> **Status of the principle: Orowan equation**
> *Derived.* A kinematic identity relating plastic strain rate to the flux of Burgers vector; it holds whatever controls dislocation velocity, provided the mobile density and average velocity are defined consistently.

Cross-slip

A screw dislocation has no unique slip plane and can switch from one glide plane to another that contains its Burgers vector. This **cross-slip** lets screw dislocations bypass obstacles, annihilate with screws of opposite sign on nearby planes, and spread slip out of a single plane. In FCC metals the planes are two {111} planes sharing a ⟨110⟩ direction; how easily cross-slip occurs depends strongly on the stacking-fault energy (Section 12.7). **Double cross-slip** — out of the primary plane and back onto a parallel one — is also a source of new dislocations (Section 12.6).

Climb

An edge dislocation can move out of its slip plane only by adding or removing atoms at the edge of its extra half-plane: absorbing vacancies makes it climb up, emitting vacancies (or absorbing interstitials) makes it climb down. **Climb** is therefore **non-conservative** and requires diffusion. It is negligible at low temperature but becomes important above about 0.3–$0.4\,T_m$, where it lets dislocations climb over obstacles they cannot cut or bypass by glide, and lets them rearrange into low-energy walls. Climb is the rate-controlling step in **recovery** (Chapter 10) and in **dislocation creep** of metals at high temperature (Chapter 14), which is why creep rates have activation energies close to that of self-diffusion.

The climb force can come from an applied normal stress or from a non-equilibrium vacancy concentration: a supersaturation c/c_0 produces an **osmotic** (chemical) force of $(kT/\Omega)\, b \ln(c/c_0)$ per unit length, where Ω is the atomic volume. Quenched-in vacancies, or those produced by irradiation, therefore drive climb and form prismatic loops — an important source of swelling and hardening in reactor materials (Chapter 41).

Image forces

Near a free surface, a dislocation is attracted towards it, because its strain energy can be reduced by relaxing the stresses at the surface. The **image force** is that of an imaginary dislocation of opposite sign mirrored across the surface, about $Gb^2/[4\pi(1-\nu)\lambda]$ for an edge dislocation at distance λ. Near a stiffer layer, such as an oxide film or a hard coating, the force is repulsive. Image forces pull dislocations out of thin films and nanopillars, contributing to their "dislocation starvation" and high strength.

12.5 Interactions: junctions, jogs, kinks and forest hardening

Interactions between parallel dislocations

Two parallel dislocations interact through their stress fields; the force on one is given by the Peach–Koehler equation using the stress of the other. Two parallel screws with the same Burgers vector repel with force $Gb^2/(2\pi r)$ per unit length; screws of opposite sign attract. For edge dislocations on parallel slip planes a distance y apart, the glide force on one from the other is

$$F_x = \frac{Gb^2}{2\pi(1-\nu)}\,\frac{x(x^2-y^2)}{(x^2+y^2)^2}$$

For like-signed edges, F_x vanishes and is stable at $x = 0$: they line up one above another, forming a **tilt wall** — a low-angle grain boundary with misorientation $\theta \approx b/h$ for spacing h. This is **polygonization**, the basic process of recovery (Chapter 10). For opposite-signed edges the stable position is at $x = \pm y$, forming a **dipole**. Walls and dipoles screen each other's long-range stresses, which is why dislocations in deformed metals arrange themselves into **cells** with tangled walls and relatively clean interiors at strains above a few percent.

Junctions and locks

When two dislocations gliding on intersecting planes meet, they may react to form a third segment along the line of intersection, provided this lowers the energy by Frank's rule. In FCC metals the most important are the **Lomer** and **Lomer–Cottrell locks**, formed when dislocations on two $\{111\}$ planes combine to give a segment with Burgers vector $\frac{a}{2}\langle 110\rangle$ lying on neither glide plane — or, after dissociation, a sessile configuration with a stair-rod partial at the apex. Such junctions cannot glide and become strong obstacles that anchor the dislocations that formed them. **Glissile junctions** and **Hirth locks** are other products. Dislocation-dynamics simulations (Chapter 56) show that the strength of these junctions, more than the elastic interaction of distant dislocations, sets the forest-hardening stress.

Jogs and kinks

When two dislocations cut through each other, each acquires a step equal to the other's Burgers vector. A step that lies **in** the slip plane of the dislocation is a **kink**; it can glide with the line and is readily removed. A step that takes the dislocation **out** of its slip plane is a **jog**. A jog on an edge dislocation lies in a plane that contains $\mathbf{b}$, so it can usually glide along with the line. A jog on a **screw** dislocation has edge character with a slip plane perpendicular to the screw's motion, so it can follow only by climb — producing vacancies or interstitials as it is dragged. Such **jog drag** helps explain why screw dislocations move more slowly than edges in many metals, and the point defects produced during deformation accelerate diffusion in cold-worked material.

Each jog costs energy, about $Gb^3/10$ (roughly 0.5 eV in copper), so cutting through a "forest" of dislocations threading the slip plane requires work. Together with the junction reactions, this is the origin of the major part of work hardening.

Forest hardening: the Taylor equation

A dislocation gliding on its slip plane is obstructed by the other dislocations that thread that plane, the **forest**, with average spacing $l \approx \rho^{-1/2}$. To pass between them, it must bow out and form junctions or jogs; using the line-tension result $\tau \approx Gb/l$ and lumping the details into a constant gives the **Taylor equation**

$$\tau = \alpha\, G\, b\, \sqrt{\rho}$$

with α typically 0.2–0.5. In a polycrystal the tensile flow stress is $\sigma = M\tau$, with the Taylor factor $M \approx 3.06$ for FCC metals (Chapter 13).

> **Status of the principle: Taylor forest-hardening equation**
> *Approximate.* Derived from the line tension and the mean spacing of forest obstacles, with α fitted to experiment or obtained from dislocation-dynamics simulations; the $\sqrt{\rho}$ dependence is confirmed over five orders of magnitude of density in FCC metals.

Worked example — Work hardening of copper from dislocation density. Take $\alpha = 0.3$, $G = 48$ GPa, $b = 0.2556$ nm and $M = 3.06$. Then $\alpha G b = 3.68$ N/m, and:

Table 12.2 — Forest hardening in copper from the Taylor equation.

Condition	ρ (m^{-2})	Spacing $\rho^{-1/2}$	$\tau = \alpha G b \sqrt{\rho}$ (MPa)	$\sigma = M\tau$ (MPa)
Well-annealed single crystal	10^{10}	10 µm	0.37	1.1
Annealed polycrystal	10^{12}	1 µm	3.7	11
Lightly cold-worked (~10 %)	10^{14}	100 nm	37	113
Heavily cold-drawn wire	10^{15}	32 nm	116	356
Severe plastic deformation	10^{16}	10 nm	368	1130

The fourth row reproduces the 300–400 MPa yield strength of hard-drawn copper wire. The first two rows show that the forest contribution is negligible in annealed material, whose yield strength of 40–70 MPa comes mainly from grain boundaries and impurities; the last shows why severely deformed metals reach gigapascal strengths, until grain-boundary effects and dynamic recovery limit further increases. Figure 12.3 compares the relation with measurements.

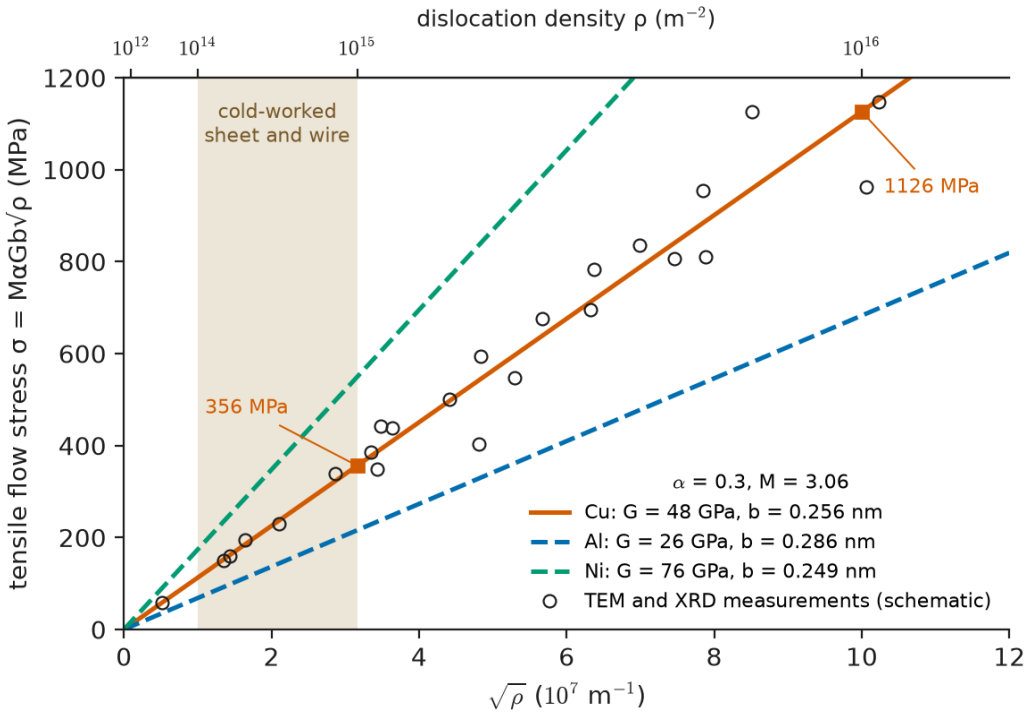

Figure 12.3 — Flow stress versus square root of dislocation density for FCC metals, showing the Taylor relation.

12.6 Multiplication: Frank–Read and other sources

Why multiplication is needed

An annealed metal contains perhaps 10^{10}–10^{12} m of dislocation line per cubic metre. If each dislocation simply glided out of the crystal, producing a step of b, the total available plastic strain would be tiny — of order $\rho b L$ for dislocations travelling a distance L, about $10^{12} \times 2.6 \times 10^{-10} \times 10^{-4} \approx 0.03$ for $L = 100$ µm, and the initial dislocations would soon be exhausted. Yet metals can be deformed to strains of 1 or more, and their dislocation density *rises* to 10^{15}–10^{16} m^{-2}. Dislocations must therefore multiply during deformation.

The Frank–Read source

Frank and Read (1950) proposed the classic mechanism (Figure 12.4). A dislocation segment of length L is pinned at both ends — by nodes of the network, by precipitates or by points where the line leaves the slip plane. Under a resolved shear stress, it

bows out. Its radius of curvature decreases as the stress rises, until it becomes a semicircle of radius $L/2$; this configuration requires the maximum stress,

$$\tau_{\mathrm{FR}} \approx \frac{Gb}{L}$$

Beyond it the loop becomes unstable and expands at decreasing stress, wrapping around the pinning points. The two parts of the loop that meet behind the pinning points have opposite character and annihilate, leaving a complete loop that expands outward and a regenerated segment between the pins, which starts the process again. A single source can emit hundreds of loops, each contributing one Burgers vector of slip, until the back stress of the piled-up loops (at a grain boundary, for example) stops it.

> **Status of the principle: Frank–Read source stress**
> *Approximate.* From the line-tension approximation; more exact calculations with the full self-interaction of the bowing line give $\tau \approx (Gb/2\pi L)\ln(L/r_0)$ for the critical stress, within a factor of about two of Gb/L for typical L.

For copper with $L = 1$ μm, $\tau_{\mathrm{FR}} \approx 48 \times 10^9 \times 2.556 \times 10^{-10}/10^{-6} = 12$ MPa; for $L = 10$ μm, 1.2 MPa; for $L = 0.1$ μm, 120 MPa. Long segments operate first; as deformation proceeds and the forest densifies, the free segment lengths shorten, and higher stresses are needed — another way of seeing work hardening. Frank–Read sources were first observed directly as spiral etch figures and as concentric loops decorated with copper in silicon (Dash, 1956).

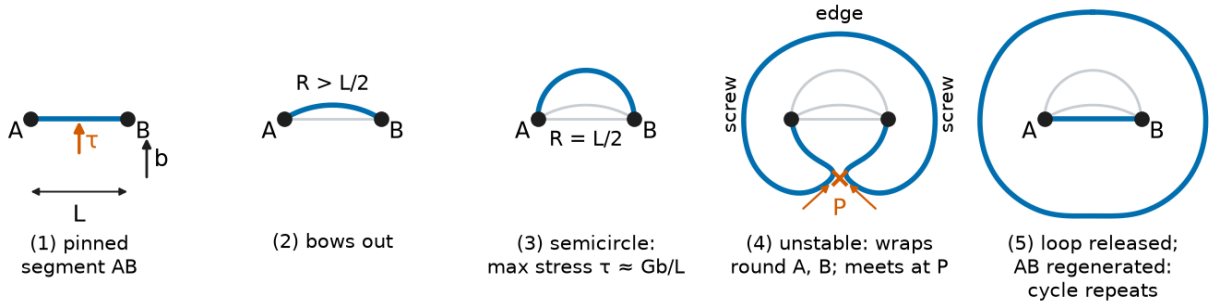

Figure 12.4 — Operation of a Frank–Read source: successive configurations of a pinned dislocation segment under increasing shear stress, ending in an emitted loop and a regenerated segment.

Other sources

- **Double cross-slip.** A screw segment cross-slips onto a parallel primary plane; the two short segments on the cross-slip plane act as pinning points for a new Frank–Read-type source. This mechanism, observed in LiF by Johnston and Gilman, accounts for the rapid multiplication and broad slip bands in materials with easy cross-slip.
- **Bardeen–Herring climb sources.** An edge segment pinned at both ends and exposed to a vacancy supersaturation climbs into a spiral or concentric prismatic loops, the climb analogue of a Frank–Read source. It operates during quenching and in irradiated metals.
- **Grain boundaries and interfaces.** Boundary ledges, triple junctions and particle–matrix interfaces with thermal-expansion mismatch emit dislocations, the dominant sources in fine-grained and nanocrystalline metals and around reinforcing particles in metal-matrix composites (Chapter 23).
- **Surfaces and stress concentrations.** Scratches, edge chips and crack tips nucleate dislocations where local stresses approach the theoretical strength. In dislocation-free silicon this is the only route to slip; wafer handling therefore minimizes edge damage and contact stresses, and furnace ramps are limited to keep thermal stresses below the threshold for slip from the edge.
- **Homogeneous nucleation**, at stresses near $G/30$, occurs only in defect-free volumes — under a nanoindenter, in whiskers and in nanopillars — where it produces "pop-in" events in the load–displacement curve.

12.7 Partial dislocations, stacking faults and stacking-fault energy

Dissociation in FCC metals

In an FCC crystal the $\{111\}$ planes are stacked in the sequence ABCABC. A perfect dislocation with $\mathbf{b} = \frac{a}{2}[\bar{1}01]$ on (111) moves an atom in a B position over to the next B position — a path that climbs over an A atom. The easier path goes via the "valley" through a C position, in two steps. This corresponds to splitting the perfect dislocation into two **Shockley partial dislocations**:

$$\tfrac{a}{2}[\bar{1}01] \;\rightarrow\; \tfrac{a}{6}[\bar{2}11] + \tfrac{a}{6}[\bar{1}\bar{1}2]$$

Each partial has $|\mathbf{b}_p| = a/\sqrt{6}$. By Frank's rule the dissociation is favourable: $b^2 = a^2/2$ before and $2 \times a^2/6 = a^2/3$ after. Between the two partials the stacking sequence is changed: the region swept by the leading partial alone becomes ABCA**C**ABC — an **intrinsic stacking fault**, a thin layer of HCP-like stacking — which has an energy per unit area γ_{SF}, the **stacking-fault energy** (SFE) (Figure 12.5). The partials repel each other elastically, and the fault pulls them together; they settle at an equilibrium separation that, in the isotropic approximation and averaging over character,

$$d \approx \frac{G b_p^2}{2\pi \gamma_{\mathrm{SF}}}$$

> **Status of the principle: Equilibrium width of a dissociated dislocation**
> *Approximate.* A balance between the isotropic elastic repulsion of two parallel partials and the fault tension; the exact width also depends on character (screws are about two to three times narrower than edges), on elastic anisotropy and on the applied stress, which can widen or constrict the ribbon.

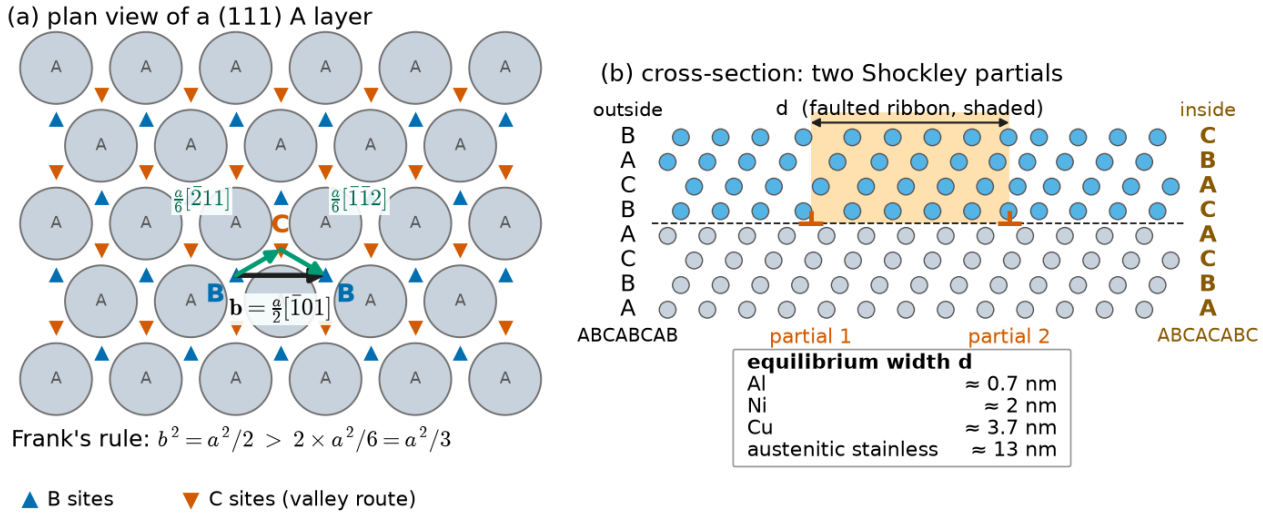

Figure 12.5 — Dissociation of a perfect FCC dislocation into two Shockley partials bounding an intrinsic stacking fault.

Worked example — Partial separations in FCC metals. For copper, $a = 0.3615$ nm, so $b_p = a/\sqrt{6} = 0.1476$ nm; $G = 48$ GPa and $\gamma_{\mathrm{SF}} = 45$ mJ/m². Then

$$d \approx \frac{48 \times 10^{9} \times (0.1476 \times 10^{-9})^2}{2\pi \times 0.045} = \frac{1.046}{0.283} \times 10^{-9}\ \mathrm{m} = 3.7\ \mathrm{nm}$$

about 14 perfect Burgers vectors. Repeating the calculation:

Table 12.3 — Stacking-fault energies and dissociation widths of FCC metals (typical).

Metal	γ_{SF} (mJ/m²)	G (GPa)	d (nm)	d/b	Typical behaviour
Aluminium	160–200 (166 used)	26	0.7	2.4	Easy cross-slip, wavy slip, extensive recovery, cells
Nickel	120–130	76	2.0	8	Cross-slip, cells
Copper	40–50	48	3.7	14	Intermediate; annealing twins
Silver	16–22	30	8.3	29	Planar slip, many annealing twins
Austenitic stainless steel (316)	~20	77	13	52	Planar slip, deformation twins or ε-martensite
Brass Cu–30Zn	7–14 (7 used)	~37	~19	~70	Very planar slip, high work hardening

Why stacking-fault energy matters

A dissociated screw dislocation cannot cross-slip as two separate partials, because the partials' Burgers vectors are not in the cross-slip plane. The ribbon must first **constrict** to a perfect screw over a short length, which costs energy that rises with the dissociation width. Stacking-fault energy therefore controls a whole family of behaviours:

- **Cross-slip and recovery.** High-SFE metals (Al, Ni, ferritic iron) cross-slip easily, form well-defined cell structures and recover readily on annealing; low-SFE metals (Ag, brass, austenitic steels) show planar slip, tangled arrays, and recrystallize rather than recover (Chapter 10).
- **Work hardening.** Planar slip keeps dislocations confined to their planes, where they pile up and interact; low-SFE alloys therefore work-harden faster and to higher stresses, with higher strain-hardening exponents n. This is why brass and austenitic stainless steels are favoured for deep drawing.
- **Twinning and phase transformation.** At SFE below about 20 mJ/m², partials separate so widely that deformation twinning or the formation of HCP ε-martensite competes with slip. **TWIP** (twinning-induced plasticity) steels, Fe–(15–25)Mn–C alloys with SFE of 20–40 mJ/m², and **TRIP** steels with lower SFE exploit this to combine strengths of 800–1000 MPa with elongations of 50–60 % for automotive crash structures (Chapter 15).
- **Annealing twins** are abundant in low-SFE metals (copper, brass, austenitic steel) and rare in aluminium.
- **Environmental cracking.** Planar slip concentrates deformation in narrow bands that rupture protective films, one reason austenitic stainless steels are susceptible to chloride stress-corrosion cracking (Chapter 43).

Solute additions change the SFE: zinc and aluminium lower it in copper, nickel raises it in iron–manganese steels, which is how alloy designers select the deformation mode.

Other partials and faults

A **Frank partial**, $\mathbf{b} = \frac{a}{3}\langle 111\rangle$, bounds a stacking fault formed by inserting or removing part of a close-packed layer — for example, when a disc of condensed vacancies collapses after quenching or irradiation. Its Burgers vector is normal to the fault plane, so it is **sessile** and can move only by climb. Frank loops are a principal irradiation defect in austenitic steels used in reactor cores (Chapter 41). At the intersection of two faults on different $\{111\}$ planes, short **stair-rod** partials such as $\frac{a}{6}\langle 110\rangle$ form, as in the Lomer–Cottrell lock. The **Thompson tetrahedron** is the standard bookkeeping device for all these Burgers vectors in FCC crystals.

12.8 Dislocations in FCC, BCC, HCP and ceramic crystals

The Peierls stress

A dislocation moving through a perfect lattice still experiences a periodic resistance, because its core energy varies as it passes from one lattice valley to the next. The stress needed to overcome this lattice friction at 0 K is the **Peierls (Peierls–Nabarro) stress**. The classical estimate for a dislocation of core width w is

$$\tau_P \approx \frac{2G}{1-\nu}\exp\!\left(-\frac{2\pi w}{b}\right)$$

with $w \approx d/(1-\nu)$ for an edge dislocation on planes spaced d apart.

> **Status of the principle: Peierls–Nabarro stress**
> *Approximate.* From a continuum model of the core with a sinusoidal interplanar force; it correctly predicts that wide cores on widely spaced, close-packed planes have negligible lattice friction and narrow, compact cores have large friction, but absolute values can be wrong by orders of magnitude, and non-planar cores (as in BCC screws) need atomistic calculation.

The exponential is the essential feature. With $\nu = 0.3$, a core of width $w = 0.5b$ gives $\tau_P \approx 0.12G$; $w = b$ gives $5 \times 10^{-3}G$; $w = 1.5b$ gives $2.3 \times 10^{-4}G$. A change of core width by a factor of three changes the lattice friction by a factor of about 500. Wide cores and widely spaced slip planes — both features of close-packed metals — mean low friction; narrow, directional, covalent cores mean high friction. This explains the differences among structures.

FCC metals

In FCC metals, slip on the widely spaced $\{111\}$ planes by dissociated dislocations with planar cores meets very little lattice resistance: Peierls stresses are of order $10^{-5}G$ or less, well below 1 MPa. The critical resolved shear stress of a pure FCC crystal is therefore set by other dislocations and impurities, and depends only weakly on temperature and strain rate. FCC metals stay ductile down to liquid-helium temperatures and show no ductile–brittle transition — the basis of their use in cryogenic tanks and LNG carriers (Chapter 41).

BCC metals

BCC metals have no truly close-packed plane. Their $\frac{a}{2}\langle 111\rangle$ screw dislocations have a compact, **non-planar core** spread over three $\{110\}$ planes of the $\langle 111\rangle$ zone, and must transform the core to glide. The result is a high Peierls stress (of order $10^{-2}G$ at 0 K) and a strongly temperature-dependent flow stress. At low temperature, screws move by thermally activated nucleation of **kink pairs** — short segments that jump into the next valley and then spread sideways — so the flow stress rises steeply on cooling (Figure 12.6) below about 0.15–$0.2\,T_m$, which for iron is around room temperature. Edge dislocations, by contrast, are mobile, so plasticity in BCC metals at low temperature is controlled by long, straight, slow screws. Slip in BCC metals occurs on $\{110\}$, $\{112\}$ and $\{123\}$ planes sharing the $\langle 111\rangle$ direction, producing wavy "pencil glide" traces, and the screw-core structure causes the yield stress to depend on the sense of shear (twinning–antitwinning asymmetry) and to violate Schmid's law.

The steep rise of flow stress on cooling is the physical root of the **ductile–brittle transition** of ferritic steels and refractory metals (Chapters 11 and 14): at low temperature the stress needed to move dislocations and blunt a crack tip exceeds the stress needed for cleavage. Tungsten, molybdenum and chromium have transition temperatures above room temperature unless very pure and fine-grained, which is why tungsten plasma-facing components in fusion reactors must be operated hot.

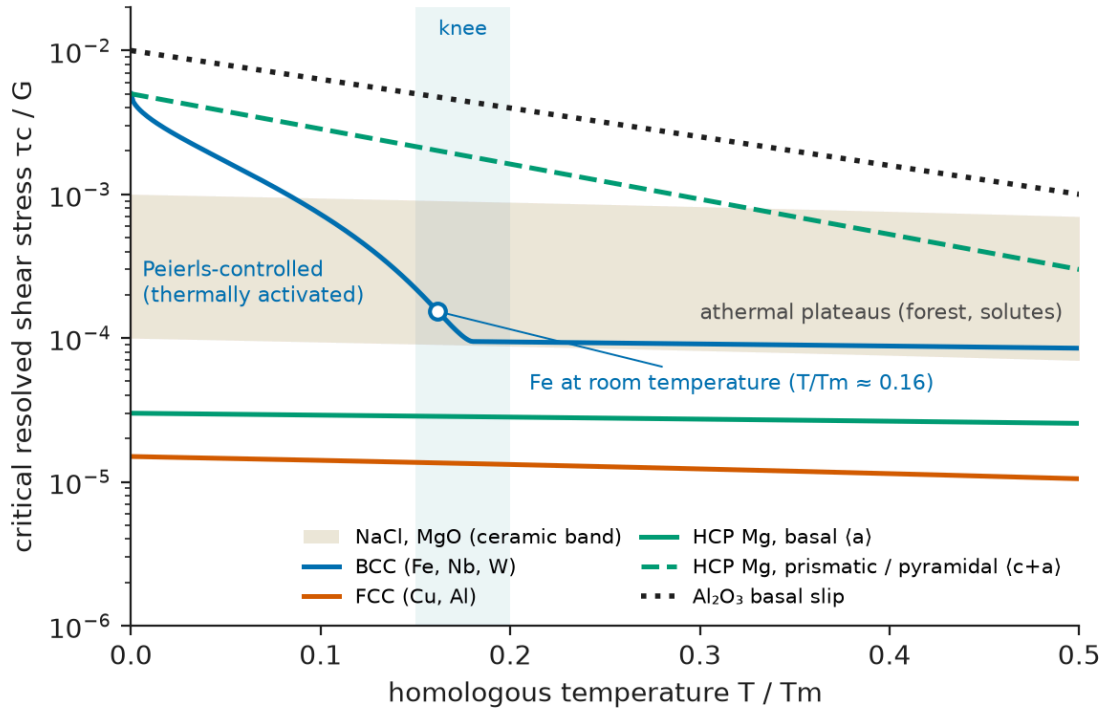

Figure 12.6 — Temperature dependence of the critical resolved shear stress of FCC, BCC and HCP single crystals, normalized by shear modulus.

HCP metals

In HCP metals the $\langle a \rangle$ Burgers vector $\frac{a}{3}\langle 11\bar{2}0 \rangle$ is the shortest, and the easiest slip plane depends on the axial ratio c/a (ideal value 1.633) and on the electronic structure. Magnesium ($c/a = 1.624$), zinc (1.856) and cadmium (1.886) slip mainly on the basal plane; titanium (1.587) and zirconium (1.593) slip mainly on prismatic $\{10\bar{1}0\}$ planes. In either case, $\langle a \rangle$ slip cannot produce strain along the c axis. That requires either pyramidal slip of $\langle c + a \rangle$ dislocations, whose Burgers vector is about twice as long and whose critical stress at room temperature is typically 10–100 times that for basal slip in magnesium, or deformation **twinning**.

The consequences are large. HCP polycrystals lack five independent easy slip systems (the von Mises requirement developed in Chapter 13), so they show limited ductility, strong anisotropy, tension–compression asymmetry and strong textures after rolling. Magnesium alloy sheet for automotive and aerospace use is difficult to form at room temperature and is usually warm-formed at 200–300 °C, where non-basal slip becomes easier. Titanium alloys are more ductile because prismatic, basal and pyramidal $\langle a \rangle$ slip and twinning together supply enough systems, and their texture is exploited in aerospace forgings and Zircaloy fuel cladding, whose properties along the tube axis and through the wall are tailored by texture.

Ordered intermetallics

In ordered alloys such as Ni3Al (γ′), a perfect FCC lattice translation $\frac{a}{2}\langle 110 \rangle$ is not a translation of the ordered structure: a single such dislocation leaves an **antiphase boundary** (APB) behind it. Dislocations therefore travel in pairs, as **superdislocations** joined by a ribbon of APB. The need to cut γ′ particles with pairs of dislocations is a major source of strength in nickel superalloys (Chapter 13), and the cross-slip of superdislocation segments onto cube planes, where they become locked (Kear–Wilsdorf locks), gives Ni3Al the anomalous property of a yield strength that *increases* with temperature up to about 700 °C.

Ceramics and covalent crystals

In ionic and covalent crystals dislocations exist, but they move with difficulty, for three reasons. **Long Burgers vectors**: the lattice translation must connect ions of the same sign, so b is large (0.399 nm in NaCl; 0.476 nm in Al2O3 for the shortest $\frac{1}{3}\langle 11\bar{2}0 \rangle$ vector), and energy and lattice friction rise with b. **Electrostatic constraints**: in rock-salt crystals, slip on $\{110\}\langle 1\bar{1}0 \rangle$ avoids bringing like ions face to face, but this gives only two independent systems; slip on $\{100\}$, needed for general deformation, is much harder. **Directional bonds**: in covalent solids such as silicon, SiC and diamond, moving a dislocation requires breaking and re-forming directional bonds, giving very high Peierls stresses.

As a result, ceramic single crystals such as MgO and NaCl can deform plastically at room temperature in favourable orientations, but polycrystalline ceramics fracture before enough slip systems activate. Alumina deforms by basal slip only above about 900–1000 °C. Silicon is brittle below about 500–600 °C but deforms plastically at higher temperatures — the slip that ruined the wafers in this chapter's opening, generated by thermal stresses during furnace processing, which is why wafer boats, ramp rates and furnace profiles are designed to keep the resolved shear stress below the threshold at every temperature. In semiconductors, dislocations are also electrically active, introducing levels in the band gap that act as recombination centres; GaN light-emitting diodes are remarkably tolerant of threading dislocation densities of 10^{8}–10^{9} cm⁻² (10^{12}–10^{13} m⁻²) that would destroy GaAs devices (Chapter 33). Ceramics are toughened therefore not by encouraging dislocations but by crack bridging and transformation toughening (Chapter 14).

Table 12.4 — Dislocation behaviour by crystal class.

Class	Typical Peierls stress	Temperature dependence of flow stress	Ductility at room temperature	Engineering consequence
FCC metals (Cu, Al, Ni, austenite)	$\lesssim 10^{-5}G$	Weak	High, to cryogenic temperature	Cryogenic and formable alloys
BCC metals (Fe, W, Mo, Nb)	~$10^{-2}G$ (screws, 0 K)	Strong below ~0.2 T_m	Good above transition, brittle below	Ductile–brittle transition of ferritic steels
HCP metals (Mg, Ti, Zn, Zr)	Low for $\langle a \rangle$ on easy plane; high for $\langle c+a \rangle$	Strong for hard modes	Limited, anisotropic	Warm forming of Mg; texture control in Ti, Zr
Ordered intermetallics (Ni_3Al, TiAl)	Moderate; superdislocations	Anomalous in Ni_3Al	Low in polycrystals	High-temperature strength of γ′; brittleness of TiAl
Ionic ceramics (NaCl, MgO)	~$10^{-3}G$	Moderate	Single crystals only	Too few independent systems in polycrystals
Covalent and ionic–covalent (Si, SiC, Al_2O_3)	~10^{-2}–$10^{-1}G$	Very strong	Brittle below ~0.5 T_m	Slip only at high temperature; brittle fracture

12.9 Observing dislocations

Dislocations were postulated in 1934 and became directly visible only two decades later. Today a range of techniques images them across scales from the atomic core to the whole wafer.

Etch pits and decoration

Where a dislocation meets a surface, the strained core region dissolves faster in a suitable etchant, leaving a small **etch pit**. Counting pits gives the dislocation density in units of lines per area (cm⁻²), and repeated etching after small amounts of loading tracks individual dislocations — the method Johnston and Gilman used in LiF (1959) to measure dislocation velocity as a function of stress. Etch-pit density (EPD) is still the routine specification of crystal quality for semiconductor substrates: silicon wafers are essentially dislocation-free, GaAs substrates have EPD of 10^{2}–10^{4} cm⁻², and SiC substrates for power devices are graded by their densities of threading and basal-plane dislocations. **Decoration** precipitates atoms along dislocation lines so that they can be seen by optical or infrared microscopy inside transparent crystals (silver halides, copper-decorated silicon).

Transmission electron microscopy

The **transmission electron microscope** (TEM) made dislocations directly visible in 1956, when Hirsch, Horne and Whelan in Cambridge and Bollmann in Geneva independently observed moving dislocations in thin metal foils. Dislocations are imaged by **diffraction contrast**: the strain field near the line bends the diffracting planes locally, changing the intensity of the diffracted beam, so the line appears as a dark (or bright) band in an image formed with one strong reflection **g**. If the diffracting

planes are not distorted by the dislocation, there is no contrast; for a screw dislocation, and approximately for an edge, the **invisibility criterion** is

$$\mathbf{g}\cdot\mathbf{b}=0$$

By finding two reflections for which a dislocation is invisible, its Burgers vector is determined as their cross product. **Weak-beam dark-field** imaging narrows the image of a dislocation to about 1.5 nm, enough to resolve the two partials of a dissociated dislocation and measure stacking-fault energies from their separation. Densities are measured by counting intersections of lines with a test grid: $\rho = 2N/(L_t\,t)$ for N intersections with test lines of total length L_t in a foil of thickness t. **In-situ** straining and heating stages show sources operating, dislocations cutting precipitates and boundaries migrating (Chapter 54).

High-resolution TEM and aberration-corrected STEM image the core itself as an atomic column arrangement, so that a Burgers circuit can be drawn directly around it, and atom-probe tomography (Chapter 53) reveals the solute atmospheres segregated to dislocations.

X-ray, electron-backscatter and other methods

- **X-ray topography** (Lang, 1958) images individual dislocations over whole wafers and large crystals with micrometre resolution, using the strain contrast in a diffracted X-ray beam; synchrotron topography is used to map slip in silicon wafers and defects in SiC and diamond.
- **X-ray and neutron line broadening** (Chapter 4) gives average dislocation densities in bulk polycrystals from the strain broadening of diffraction peaks — the source of the densities in Table 12.2.
- **Electron backscatter diffraction** (EBSD) maps local lattice misorientation in the SEM. Gradients of misorientation give the density of **geometrically necessary dislocations** required to accommodate lattice curvature, around indents, cracks and grain boundaries; **electron channelling contrast imaging** (ECCI) images near-surface dislocations in bulk samples without thin-foil preparation (Chapter 52).
- **Simulation** complements observation: discrete dislocation dynamics follows millions of dislocation segments with the fields and interactions of this chapter, and atomistic simulation resolves cores, junctions and kink-pair mechanisms (Chapters 55 and 56).

Table 12.5 — Typical dislocation densities.

Material and condition	ρ (m^{-2})	Method usually used
Float-zone or Czochralski silicon	$< 10^6$ (essentially dislocation-free)	Etch pits, X-ray topography
GaAs substrate	10^6–10^8 (10^2–10^4 cm^{-2})	Etch pits
GaN on sapphire (LED epilayers)	10^{12}–10^{13}	TEM, cathodoluminescence
Well-annealed metal	10^{10}–10^{12}	TEM, etch pits
Cold-worked metal (moderate)	10^{14}–10^{15}	TEM, XRD
Heavily cold-drawn wire, martensite	10^{15}–10^{16}	XRD line broadening
Severely deformed (HPT, ECAP) metals	up to ~10^{16}	XRD, TEM

What remains open

Open question	Where it is answered
How does the resolved shear stress on slip systems determine yield in single crystals and polycrystals (Schmid's law, Taylor factor, von Mises criterion)?	Chapter 13
How do solutes, grain boundaries and precipitates obstruct dislocations quantitatively (Hall–Petch from pile-ups, cutting versus Orowan bypass)?	Chapter 13
How do twinning and superplasticity provide deformation where slip is insufficient?	Chapter 13
How does climb-controlled dislocation creep differ from diffusional creep, and how is it mapped?	Chapter 14
How does the competition between crack-tip dislocation emission and cleavage set the ductile–brittle transition and fracture toughness?	Chapter 14
How do dislocations rearrange during recovery and recrystallization?	Chapter 10 (back reference)
How do TWIP and TRIP steels and nickel superalloys exploit stacking-fault energy and superdislocations?	Chapters 15 and 16
How do irradiation loops and dislocations cause swelling and hardening in reactor materials?	Chapter 41
Why do nanocrystalline metals soften at the smallest grain sizes?	Chapter 38
How are dislocations imaged in practice and measured with uncertainty?	Chapters 52, 53 and 54
How are dislocation ensembles simulated (discrete dislocation dynamics, crystal plasticity)?	Chapter 56

Next: Chapter 13 scales up from single dislocations to the plasticity of real crystals and polycrystals, deriving Schmid's law, yield criteria and the strengthening mechanisms that let engineers set yield strength over two orders of magnitude.

Chapter 13 — Plasticity and Strengthening Mechanisms

Part III · Mechanical Behaviour

A car maker wants to take 40 kg out of a body-in-white without losing crash performance. The request to the materials group is one line: "same part, thinner sheet, 50 % more yield strength, and it must still press-form." The materials group has four levers, and each costs something. Cold work raises strength but eats the ductility the press needs. Solute raises strength a little and weldability problems a lot. Grain refinement raises strength and, unusually, toughness too, but a rolling mill can only make grains so fine. Fine precipitates raise strength strongly, but only at the right size, and they coarsen if the part is heated. The same levers set the strength of a 7075 aluminium wing spar, a nickel turbine disc, a titanium bone plate, a gas pipeline and a copper connector spring.

Chapter 12 built the theory of a single dislocation. This chapter scales it up to the yield strength and work hardening of real crystals and polycrystals, and to the yield criteria engineers use when stresses are multiaxial.

> **Why does a metal crystal yield at a stress a thousand times below its theoretical strength, and why does that stress depend on the direction of loading?**
> **How do solutes, grain boundaries, precipitates and other dislocations raise the yield strength, by how much, and how do their contributions combine in a real alloy?**

Concept	Engineering destination
Slip systems, Schmid's law, CRSS	Single-crystal turbine blades, texture effects in rolled sheet, magnesium formability
Stages I–III hardening	Formability, cold rolling schedules, springback
Taylor factor, von Mises criterion for slip	Why HCP metals are hard to form; texture design in sheet
Tresca and von Mises yield criteria	Design of shafts, pressure vessels, pipelines, bolted joints
Taylor work-hardening equation	Cold-drawn wire, cold-rolled strip, temper designations
Solid-solution strengthening	Brasses, Al–Mg marine alloys, interstitial strengthening of steel, strain ageing
Hall–Petch grain-boundary strengthening	Controlled-rolled pipeline and ship steels, fine-grained Ti bone plates
Precipitate cutting and Orowan bypass	Age-hardened Al aircraft alloys, maraging steels, Ni superalloy discs
Twinning	Magnesium sheet, TWIP steels, low-temperature deformation of BCC and HCP metals
Superplasticity	Superplastic forming of Ti fan blades and Al car panels
Combining mechanisms	Alloy design for automotive, pipeline, aerospace and electrical conductors

Roadmap. Slip systems and Schmid's law (13.1) → single-crystal hardening stages (13.2) → from one crystal to many: Taylor factor and the five-slip-system condition (13.3) → yield criteria for multiaxial stress (13.4) → work hardening (13.5) → solid-solution (13.6), grain-boundary (13.7) and precipitate strengthening (13.8) → twinning and superplasticity (13.9) → adding the mechanisms together in real alloys (13.10).

13.1 Slip systems, Schmid's law and the critical resolved shear stress

Slip planes and slip directions

Plastic deformation of a crystalline metal at ordinary temperatures proceeds by the glide of dislocations (Chapter 12). Glide is shear on a particular crystallographic plane in a particular direction. The combination of a **slip plane** and a **slip direction** lying in it is a **slip system**. Two rules select them, both consequences of the Peierls stress discussed in Section 12.8:

1. **The slip direction is the shortest lattice translation**, because the Burgers vector is a lattice vector, the line energy scales with b^2, and the shortest **b** is the cheapest dislocation.

2. **The slip plane is usually the most densely packed plane** containing that direction, because widely spaced, atomically smooth planes give the lowest Peierls barrier.

In FCC metals these rules give unambiguous answers: slip on $\{111\}$ planes along $\langle 110\rangle$ directions, with $\mathbf{b} = \tfrac{a}{2}\langle 110\rangle$. There are four $\{111\}$ planes, each containing three $\langle 110\rangle$ directions, for twelve slip systems. In BCC metals the slip direction is always the close-packed $\langle 111\rangle$, but no plane is as densely packed as $\{111\}$ in FCC, and slip occurs on $\{110\}$, $\{112\}$ and $\{123\}$; the screw dislocations have non-planar cores (Section 12.8), so slip lines are often wavy ("pencil glide"). In HCP metals the close-packed direction is $\langle 11\bar{2}0\rangle$ in the basal plane, and whether the easiest plane is the basal plane (Mg, Zn, Cd, Co) or the prism plane (Ti, Zr) depends on the axial ratio c/a and on the electronic structure.

Table 13.1 — Principal slip systems and typical critical resolved shear stresses (high-purity single crystals, room temperature).

Structure	Example metals	Slip system	Number of systems	CRSS (MPa)
FCC	Cu, Al, Ni, Ag, Au, γ-Fe	$\{111\}\langle 1\bar{1}0\rangle$	12	Cu 0.5–1.0; Al 0.5–1.0; Ni 3–7; Ag 0.4–0.6
BCC	α-Fe, Mo, W, Nb, Ta	$\{110\}\langle\bar{1}11\rangle$, $\{112\}\langle 11\bar{1}\rangle$, $\{123\}\langle 11\bar{1}\rangle$	12 + 12 + 24	α-Fe 15–30; Mo 50–70; strongly temperature-dependent
HCP basal	Mg, Zn, Cd, Co	$(0001)\langle 11\bar{2}0\rangle$	3 (2 independent)	Mg 0.5–0.8; Zn 0.2–0.9; Cd 0.6
HCP prismatic	Ti, Zr	$\{10\bar{1}0\}\langle 11\bar{2}0\rangle$	3 (2 independent)	Ti 15–90 (O-dependent); Mg 40–50
HCP pyramidal ⟨c+a⟩	Mg, Ti, Zn	$\{11\bar{2}2\}\langle 11\bar{2}\bar{3}\rangle$	6 (5 independent)	Mg 40–100+; Ti several hundred
Rock salt	NaCl, LiF, MgO	$\{110\}\langle 1\bar{1}0\rangle$	6 (2 independent)	LiF 1–6; MgO ~100 (high purity)

These CRSS values measure a nearly perfect lattice; engineering alloys are one or two orders of magnitude stronger, and the rest of this chapter is about where that extra strength comes from.

Resolving the applied stress

A rod-shaped single crystal is pulled with a force F along its axis. Let the slip-plane normal make an angle ϕ with the tensile axis and the slip direction an angle λ with it. The force component in the slip direction is $F\cos\lambda$; the area of the slip plane is $A_0/\cos\phi$. The **resolved shear stress** on the slip system is therefore

$$\tau_R = \frac{F\cos\lambda}{A_0/\cos\phi} = \sigma\cos\phi\cos\lambda = m\,\sigma$$

where $m = \cos\phi\cos\lambda$ is the **Schmid factor** (Figure 13.1). Its maximum value is 0.5, reached when $\phi = \lambda = 45^\circ$; it vanishes when the slip plane is parallel or perpendicular to the load.

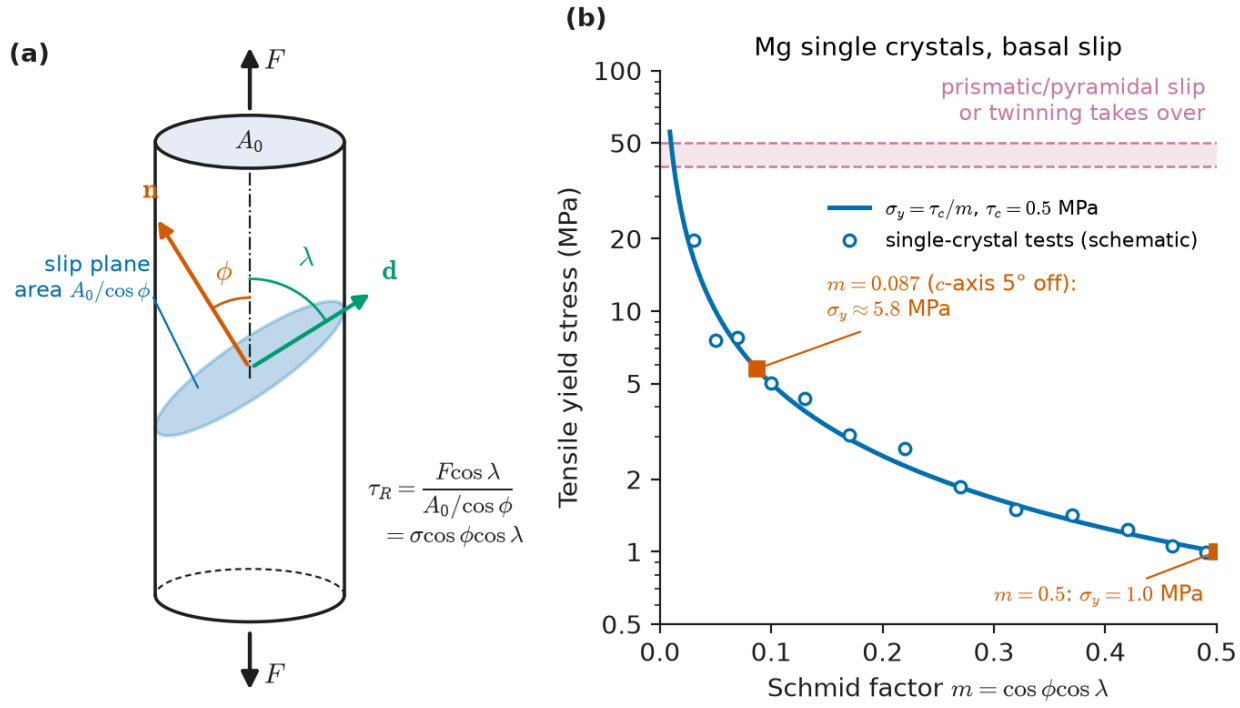

Figure 13.1 — Geometry of Schmid's law: a cylindrical single crystal under tension, with the slip-plane normal at angle phi and the slip direction at angle lambda to the tensile axis; inset of yield stress versus Schmid factor for magnesium basal slip.

Schmid's law

Experiments on single crystals of zinc, cadmium, magnesium and copper in the 1920s and 1930s showed that the tensile yield stress varies by orders of magnitude with orientation, but the resolved shear stress on the active system at yield is almost constant. That constant is the **critical resolved shear stress** τ_c (CRSS). Yielding begins on the system with the largest Schmid factor when

$$\sigma_y = \frac{\tau_c}{m_{\max}}$$

> **Status of the principle: Schmid's law**
> *Approximate.* It follows from the fact that only the shear stress resolved on the slip plane in the slip direction does work on a planar-core dislocation; it holds well for FCC and basal-slip HCP crystals, but fails for BCC metals at low temperature, where the non-planar screw cores make the yield stress depend on the sense of shear and on stress components other than τ_R (non-Schmid effects).

Behind it is the Peach–Koehler glide force $\tau_R b$ per unit length (Section 12.4); normal stress does no work on a gliding dislocation, so hydrostatic pressure does not cause yield (Section 13.4).

Worked example — Orientation and yield of a copper and a magnesium crystal. A copper single crystal ($\tau_c = 0.6$ MPa) is loaded in tension along $[123]$. The Schmid factors of the twelve $\{111\}\langle 110\rangle$ systems are found from $m = (\hat{\mathbf{t}}\cdot\hat{\mathbf{n}})(\hat{\mathbf{t}}\cdot\hat{\mathbf{d}})$ with $\hat{\mathbf{t}} = [123]/\sqrt{14}$. For the system $(\bar{1}11)[101]$:

$$\cos\phi = \frac{-1+2+3}{\sqrt{14}\sqrt{3}} = \frac{4}{6.481} = 0.617, \qquad \cos\lambda = \frac{1+0+3}{\sqrt{14}\sqrt{2}} = \frac{4}{5.292} = 0.756$$

so $m = 0.617 \times 0.756 = 0.467$ ($\phi = 51.9^\circ$, $\lambda = 40.9^\circ$). Checking all twelve systems shows this is the largest; the next largest is 0.350. Yield begins on $(\bar{1}11)[101]$ alone at

$$\sigma_y = \frac{0.6}{0.467} = 1.29 \text{ MPa}$$

— about 130 g hung from each square millimetre of cross-section, two hundred times less than the yield strength of annealed polycrystalline copper. Along $[001]$, eight systems share the Schmid factor $1/\sqrt{6} = 0.408$, so yield occurs at 1.47 MPa on many

systems at once.

For a magnesium crystal with basal CRSS 0.5 MPa, loading at $\phi = \lambda = 45^\circ$ gives $\sigma_y = 0.5/0.5 = 1.0$ MPa. Loading with the c-axis tilted only 5° from the tensile axis ($\phi = 5^\circ$, $\lambda = 85^\circ$, $m = \cos 5^\circ \cos 85^\circ = 0.087$) would need $\sigma_y = 5.8$ MPa for basal slip; before that is reached in a polycrystal, other mechanisms — twinning or prismatic slip, with CRSS some 50–100 times higher — must operate instead. This is the origin of the strong texture dependence of magnesium sheet (Section 13.9) and the reason single-crystal turbine blades are cast with a controlled $\langle 001 \rangle$ orientation (Chapter 16).

13.2 Single-crystal stress–strain curves: stages I–III

The resolved shear stress–shear strain curve

If a single crystal is deformed and the results are plotted as resolved shear stress τ_R against resolved shear strain γ, curves from crystals of very different orientation collapse onto a characteristic shape. For an FCC crystal oriented for single slip (such as [123]), the curve has three stages (Figure 13.2).

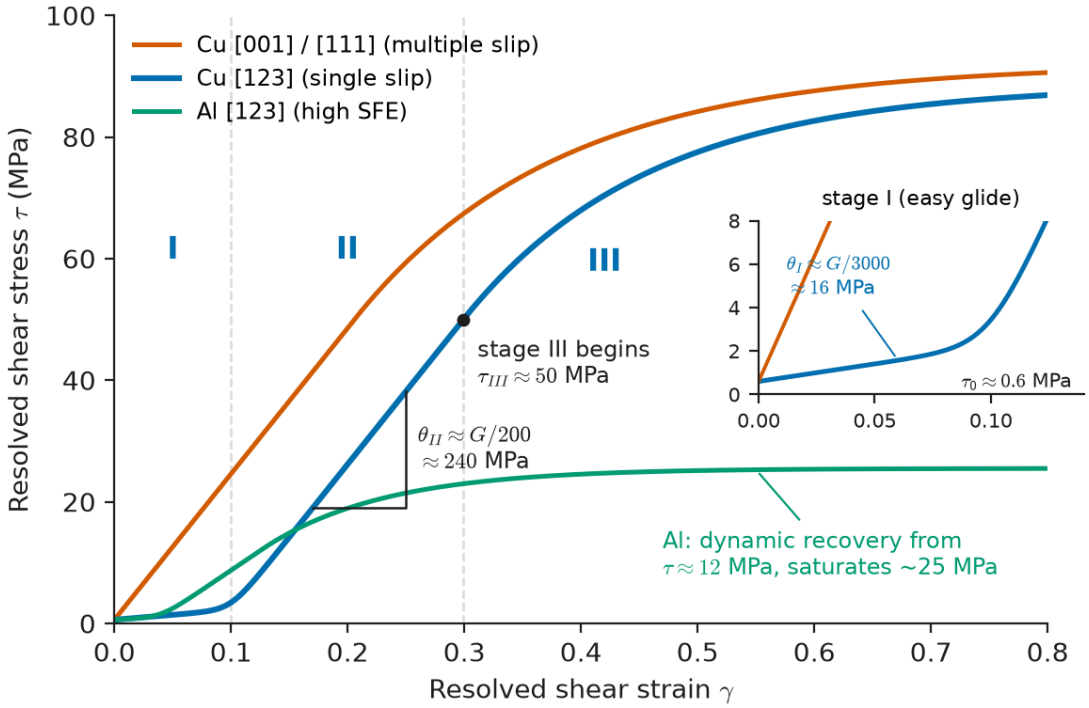

Figure 13.2 — Resolved shear stress versus resolved shear strain for FCC single crystals: stages I (easy glide), II (linear hardening) and III (dynamic recovery), for copper in single-slip and multiple-slip orientations and for aluminium.

Stage I — easy glide. Only the primary system is active; dislocations glide long distances on parallel planes, interact little, and many leave at the surface. The hardening rate $\theta_I = d\tau/d\gamma$ is of order 10^{-4}–$3 \times 10^{-4}\, G$ (about 5–16 MPa for copper). Stage I is longest for orientations such as [123] deep inside the stereographic triangle and absent for [001] and [111], where several systems are equally stressed. As a crystal in single slip stretches, it rotates toward the slip direction until a second ("conjugate") system catches up.

Stage II — linear hardening. Once two or more slip systems operate, dislocations on intersecting planes react to form sessile junctions (Lomer–Cottrell locks in FCC, Section 12.5), forests thread every slip plane, and the mean free path of a gliding dislocation shrinks as the dislocation density rises. The hardening rate is high and remarkably constant, $\theta_{II} \approx G/200$ to $G/300$, nearly independent of temperature, metal and orientation. For copper ($G \approx 48$ GPa) this is about 160–240 MPa per unit shear strain.

Stage III — dynamic recovery. At higher stress, screw dislocations can cross-slip around obstacles and annihilate with screws of opposite sign. The hardening rate falls progressively, and the stress at which stage III begins, τ_{III}, falls with rising temperature and with rising stacking-fault energy. Cross-slip requires that the partial dislocations of a dissociated screw be

recombined into a perfect dislocation over a short length (Section 12.7); the wider the stacking fault, the harder this is. Aluminium (stacking-fault energy about 160 mJ/m²) enters stage III almost at once and work-hardens weakly; copper (about 45 mJ/m²) has a long stage II; silver and austenitic stainless steels (about 15–20 mJ/m²) harden strongly and for a long time. At still larger strains a stage IV of low, constant hardening and a stage V of saturation are sometimes distinguished, but these matter mainly for severe deformation processes such as wire drawing to high reduction.

> **Status of the principle: Three-stage hardening of FCC single crystals**
> *Empirical.* The stages are robust observations explained qualitatively by dislocation theory; the hardening rates, and especially the onset of stage III, depend on stacking-fault energy, temperature, strain rate and purity and are not predicted quantitatively by simple models.

BCC and HCP crystals show less clear stages. BCC crystals at room temperature have a high, temperature-dependent yield stress set by the Peierls barrier for screw dislocations, followed by relatively low hardening. HCP crystals in basal slip can show stage I extending to shear strains of several hundred percent: a zinc crystal can be drawn into a ribbon because a single slip system operates with almost no hardening.

Stacking-fault energy is therefore one of the most useful single numbers in physical metallurgy. Low-stacking-fault-energy metals (austenitic stainless steel, 70/30 brass) work-harden strongly, form planar slip bands and twin easily; high-stacking-fault-energy metals (aluminium, nickel, ferrite) recover easily and form dislocation cells. The deep-drawn stainless kitchen sink and the drawn-and-ironed aluminium can are both stacking-fault-energy decisions.

13.3 Polycrystal plasticity: the Taylor factor and the von Mises criterion

Compatibility between grains

If each grain of a polycrystal slipped only on its own favoured system, grains would open gaps or overlap at their boundaries. Instead each grain must undergo, approximately, the shape change of the whole specimen. An arbitrary shape change has six strain components; volume conservation leaves five independent ones, and each slip system supplies one shear. Von Mises showed in 1928 that a crystal can accommodate an arbitrary plastic shape change only if it possesses at least **five independent slip systems**.

> **Status of the principle: Von Mises (Taylor) criterion for polycrystal ductility**
> *Derived.* A geometric consequence of volume-conserving plastic strain having five independent components; a polycrystal whose grains have fewer independent slip systems cannot deform homogeneously and is usually brittle or strongly anisotropic, unless twinning, grain-boundary sliding or diffusion supplies the missing modes.

FCC and BCC metals, with twelve or more systems each yielding five independent ones, satisfy the criterion easily and are ductile as polycrystals. HCP metals with only basal slip have two independent systems; adding prismatic slip gives four; only the pyramidal $\langle c+a \rangle$ systems (or twinning) supply strain along the c-axis. This is why polycrystalline magnesium is difficult to form at room temperature but forms well above about 200 °C, where $\langle c+a \rangle$ slip becomes easy; why zinc sheet cracks if bent across the wrong direction; and why titanium alloys, in which $\langle c+a \rangle$ slip and twinning are available at room temperature, are much more formable than magnesium. Ionic crystals with rock-salt structure slip on $\{110\}\langle 1\bar{1}0 \rangle$, which gives only two independent systems: a single crystal of MgO or LiF can be bent, but a polycrystal is brittle.

The Taylor factor

In a polycrystal, each grain must activate at least five systems, and not all of them have a large Schmid factor. G. I. Taylor (1938) assumed every grain undergoes the macroscopic strain and chose, among all combinations of five systems that achieve it, the one minimizing the total slip $\sum |\gamma_i|$. The average over random orientations gives

$$\sigma_y = M\,\tau_c, \qquad d\sigma = M\,d\tau, \qquad d\varepsilon = \frac{d\gamma}{M}$$

with the **Taylor factor** $M = 3.06$ for FCC polycrystals with random texture and about 2.75–3.0 for BCC (depending on the assumed slip planes). An alternative lower-bound estimate (Sachs), which lets each grain slip on its single most favoured system,

gives $M \approx 2.2$; real behaviour lies close to the Taylor value. The hardening rate transforms as $d\sigma/d\varepsilon = M^2\, d\tau/d\gamma$, so the stage II rate of $G/200$ becomes about $9.4 \times G/200 \approx G/21$ in tensile terms — about 2.3 GPa for copper.

> **Status of the principle: Taylor factor**
> *Approximate.* An upper-bound estimate from the iso-strain assumption; it neglects grain-to-grain stress variations and boundary effects, but reproduces the ratio of polycrystal to single-crystal flow stress for cubic metals within about 10 %.

Texture and plastic anisotropy

The Taylor factor depends on texture, so a rolled sheet whose grains share a few orientations yields and thins differently in different directions. The **Lankford coefficient** $r = \varepsilon_w/\varepsilon_t$ (width to thickness strain in a sheet tensile test) measures this: interstitial-free deep-drawing steels with a strong $\{111\}$ texture reach $r \approx 1.8$–2.5 and draw deeply, while aluminium can stock has $r < 1$ and forms "ears" around a drawn cup. Texture control returns in Chapters 46 and 56.

13.4 Yield criteria: Tresca and von Mises

The problem

A tensile test gives σ_y under uniaxial stress, but a drive shaft is in torsion and bending, a pressure vessel is biaxial, and a weld root is triaxial. A **yield criterion** predicts yielding under principal stresses $\sigma_1 \geq \sigma_2 \geq \sigma_3$ from σ_y alone. For isotropic crystalline metals it must be shear-driven, insensitive to hydrostatic stress, and symmetric in tension and compression.

The Tresca criterion

The simplest criterion follows directly from Schmid's law: yield occurs when the largest shear stress anywhere in the material reaches a critical value. The maximum shear stress is $(\sigma_1 - \sigma_3)/2$; in uniaxial tension it equals $\sigma_y/2$. Hence

$$\sigma_1 - \sigma_3 = \sigma_y$$

The **Tresca** (maximum shear stress) criterion predicts a shear yield strength $k = \tau_y = \sigma_y/2$.

The von Mises criterion

A smoother criterion uses all three principal shear stresses. Yield occurs when the **von Mises equivalent stress** reaches σ_y:

$$\sigma_{\mathrm{eq}} = \sqrt{\tfrac{1}{2}\left[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2\right]} = \sigma_y$$

In terms of the general (non-principal) components,

$$\sigma_{\mathrm{eq}} = \sqrt{\tfrac{1}{2}\left[(\sigma_{xx}-\sigma_{yy})^2 + (\sigma_{yy}-\sigma_{zz})^2 + (\sigma_{zz}-\sigma_{xx})^2\right] + 3\left(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2\right)}$$

The criterion has three equivalent interpretations: yield when the second invariant of the deviatoric stress $J_2 = \sigma_{\mathrm{eq}}^2/3$ reaches a critical value; when the elastic distortion (shape-change) energy reaches a critical value (Hencky); and when the shear stress on the octahedral planes reaches a critical value. It predicts $k = \sigma_y/\sqrt{3} = 0.577\sigma_y$ in pure shear.

> **Status of the principle: Tresca and von Mises yield criteria**
> *Approximate.* Both encode pressure-independent, isotropic, tension–compression-symmetric yielding; von Mises fits experiments on most ductile metals better (measured k/σ_y is typically 0.55–0.60), Tresca is slightly conservative and easier to use by hand. Neither applies to textured sheet (anisotropic criteria such as Hill's are needed), to polymers and soils (pressure-dependent criteria such as Drucker–Prager or Mohr–Coulomb) or to brittle materials, which fail by the largest principal tensile stress.

Yield surfaces

In principal stress space, the von Mises criterion is a circular cylinder whose axis is the hydrostatic line $\sigma_1 = \sigma_2 = \sigma_3$; the Tresca criterion is a regular hexagonal prism inscribed in it. Cutting at $\sigma_3 = 0$ (plane stress, as in thin sheet and thin-walled tubes) gives an ellipse and an inscribed hexagon (Figure 13.3). The two agree in uniaxial tension and equibiaxial tension, and differ most in pure shear, by $2/\sqrt{3} = 1.155$.

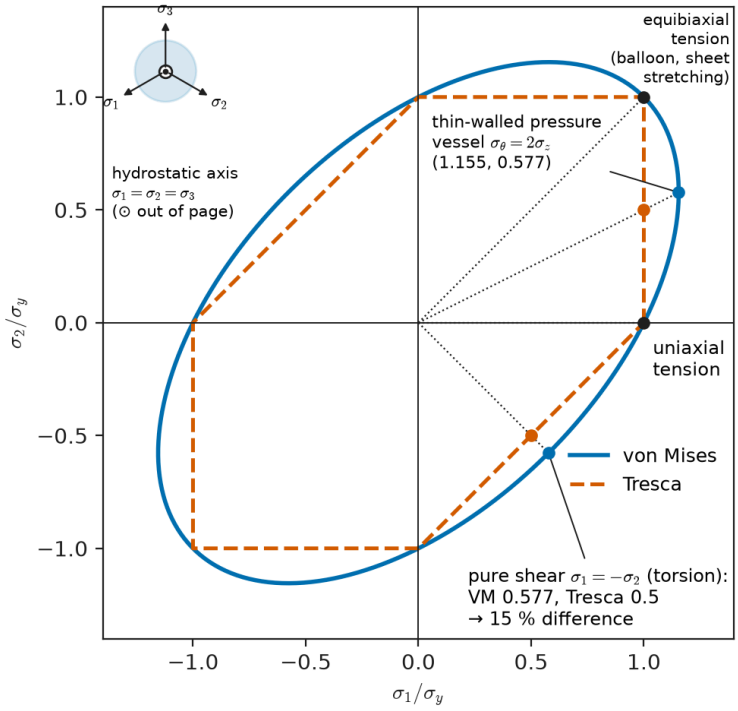

Figure 13.3 — Tresca hexagon and von Mises ellipse in the plane-stress section (sigma3 = 0), normalized by the uniaxial yield strength, with characteristic loading paths.

Worked example — A drive shaft in combined bending and torsion. A solid steel shaft for a wind-turbine yaw drive carries, at its most stressed surface point, a bending stress $\sigma = 150$ MPa and a torsional shear stress $\tau = 100$ MPa. The steel is a quenched-and-tempered 42CrMo4 with $\sigma_y = 350$ MPa at the elevated-temperature, large-section condition the designer uses. Is the shaft safe against yield, and by what factor?

For a state with one normal and one shear component, the principal stresses are $\sigma/2 \pm \sqrt{(\sigma/2)^2 + \tau^2}$:

$$\sigma_{1,3} = 75 \pm \sqrt{75^2 + 100^2} = 75 \pm 125 \text{ MPa} \quad \Rightarrow \quad \sigma_1 = 200,\ \sigma_2 = 0,\ \sigma_3 = -50 \text{ MPa}$$

Tresca: $\sigma_1 - \sigma_3 = 250$ MPa; safety factor $350/250 = 1.40$.

Von Mises: $\sigma_{\text{eq}} = \sqrt{\sigma^2 + 3\tau^2} = \sqrt{150^2 + 3 \times 100^2} = \sqrt{52\,500} = 229.1$ MPa; safety factor $350/229.1 = 1.53$.

The 9 % difference is Tresca's conservatism in shear-dominated loading. For a fatigue-loaded shaft this static check is only a first step (Section 14.8).

Plastic flow after yield

A yield criterion also fixes the *direction* of plastic flow. With plastic strain increments normal to the yield surface (the **associated flow rule**), von Mises gives the Lévy–Mises equations $d\varepsilon^p_{ij} = d\lambda\, s_{ij}$, with s_{ij} the deviatoric stress; in uniaxial tension they give a plastic Poisson's ratio of exactly 0.5. Crash, forming and pipeline finite-element codes implement exactly this: a von Mises (or anisotropic Hill) surface, an associated flow rule and a hardening law from Section 13.5.

13.5 Work hardening: the Taylor equation

Flow stress and dislocation density

Cold rolling copper strip from the annealed to the hard temper roughly triples its yield strength without changing its composition. What changes is the dislocation density ρ (total line length per unit volume): from about 10^{10}–10^{12} m^{-2} in a well-annealed metal to 10^{14}–10^{16} m^{-2} after heavy cold work. A moving dislocation must cut through or bow between the forest of dislocations threading its slip plane, whose mean spacing is $\rho^{-1/2}$. The stress to bow a dislocation segment of length L is of order Gb/L (Section 12.3); with $L \propto \rho^{-1/2}$,

$$\tau = \tau_0 + \alpha\, G\, b\, \sqrt{\rho}$$

This is the **Taylor equation**. The constant α is about 0.2–0.5 and absorbs the junction strengths and geometry; τ_0 is the flow stress in the absence of forest dislocations (lattice friction plus solute). For a polycrystal, multiplying by the Taylor factor,

$$\sigma = \sigma_0 + M\alpha\, G\, b\, \sqrt{\rho}$$

> **Status of the principle: Taylor work-hardening equation**
> *Approximate.* Derived from the line-tension bowing of dislocations between forest obstacles, with α fitted; the $\sqrt{\rho}$ dependence is confirmed over five decades of dislocation density in FCC and BCC metals, but it lumps all junction types together and ignores dislocation patterning into cells and walls.

Worked example — Dislocation density in cold-worked copper. Take $M = 3.06$, $\alpha = 0.3$, $G = 48$ GPa and $b = 0.256$ nm. The forest-hardening term $M\alpha G b\sqrt{\rho}$ is

ρ (m^{-2})	$M\alpha G b\sqrt{\rho}$ (MPa)	Condition
10^{10}	1.1	Well-annealed single crystal
10^{12}	11	Annealed polycrystal
10^{14}	113	Lightly to moderately cold worked
5×10^{14}	252	Hard temper strip
10^{15}	357	Heavily drawn wire

With $\sigma_0 \approx 25$–40 MPa from lattice friction and grain size, the hard-temper value of about 280–290 MPa matches the measured yield strength of half-hard to hard copper. The $\sqrt{\rho}$ law also explains why the same cold-work strengthening is lost by annealing: recovery and recrystallization (Section 10.7) reduce ρ by four orders of magnitude, and the strength falls back by a factor of ten in the forest term.

Evolution of dislocation density: the Kocks–Mecking picture

How ρ grows with strain decides the shape of the curve. The Kocks–Mecking model writes storage minus dynamic recovery,

$$\frac{d\rho}{d\gamma} = k_1\sqrt{\rho} - k_2\,\rho$$

with k_2 rising with temperature and stacking-fault energy. With the Taylor equation it gives a hardening rate falling linearly with stress, $\theta = \theta_0(1 - \sigma/\sigma_s)$: stage III and a saturation stress σ_s.

Phenomenological hardening laws

Engineers fitting tensile data more often use empirical forms: the **Hollomon** law $\sigma_T = K\varepsilon_T^{\,n}$ (Chapter 11), the **Swift** law $\sigma = K(\varepsilon_0 + \varepsilon)^n$ for pre-strained sheet, and the saturating **Voce** law. With Hollomon hardening the Considère criterion places necking at $\varepsilon_T = n$, so n is the uniform true strain.

Table 13.2 — Typical strain-hardening exponents and their engineering consequences (room temperature).

Material	n	Consequence
Austenitic stainless steel 304, annealed	0.40–0.50	Deep drawing (sinks, pans); severe galling in machining
Copper C11000, annealed	0.35–0.50	Tube bending, wire drawing
TWIP steel (Fe–22Mn–0.6C)	0.40–0.60	Crash-absorbing automotive parts (Chapter 15)
Low-carbon steel (DC04), annealed	0.20–0.25	Car body panels
Aluminium 6061-T6	0.05–0.10	Little forming after ageing; formed in T4
Ti–6Al–4V, mill annealed	0.05–0.10	Hot or superplastic forming instead

Bauschinger effect

A metal loaded beyond yield in tension and then reversed yields in compression at a lower stress, because pile-ups and cell walls built during forward loading exert back stresses that assist reverse flow. This **Bauschinger effect** governs springback, cyclic plasticity in low-cycle fatigue (Section 14.8) and the collapse strength of UOE-formed line pipe; models represent it by **kinematic hardening**, a translation of the yield surface, alongside the isotropic expansion of the Taylor picture.

13.6 Solid-solution strengthening

Solute atoms as obstacles

Adding 30 wt% zinc to copper roughly doubles the annealed yield strength; adding 4.5 wt% magnesium to aluminium (alloy 5083) raises it from about 30 MPa to 145 MPa. Dissolved solute atoms obstruct glide through four interactions with the dislocation's stress field and core:

1. **Size misfit**, a local dilatation attracted to the tensile or compressive side of an edge dislocation, measured by $\delta = d\ln a/dc$.
2. **Modulus misfit**, a locally stiffer or softer region that changes the line energy, $\eta = d\ln G/dc$; it acts on screws as well as edges.
3. **Chemical (Suzuki) interaction**, segregation of solute to the stacking fault between partials in FCC metals.
4. **Tetragonal distortion** by interstitial C and N in BCC iron (Section 5.1), which interacts very strongly with both edge and screw dislocations.

How strength scales with concentration

Two limiting statistical models combine these interactions into a flow stress. In the **Fleischer** model, appropriate for dilute, strong obstacles, a dislocation bows between discrete solute atoms and breaks away one at a time; the result is

$$\Delta\tau \propto G\,\varepsilon_s^{3/2}\,c^{1/2}$$

where $\varepsilon_s = |\eta' - \beta\delta|$ combines modulus and size misfit ($\beta \approx 3$ for screws, larger for edges, and $\eta' = \eta/(1+\tfrac{1}{2}|\eta|)$). In the **Labusch** model, appropriate for concentrated, weak obstacles, the dislocation is a flexible line in a random field of many small forces, and

$$\Delta\tau \propto G\,\varepsilon_s^{4/3}\,c^{2/3}$$

> **Status of the principle: Fleischer and Labusch solid-solution scaling**
> *Approximate*. Statistical line-tension models with fitted prefactors; the $c^{1/2}$ law fits dilute alloys and interstitials, the $c^{2/3}$ law fits concentrated substitutional alloys better, and modern Varvenne–Curtin theory extends the Labusch picture to high-entropy alloys. None predicts absolute values without measured or computed misfit parameters.

Table 13.3 — Solid-solution strengthening rates in common alloy systems (polycrystals, room temperature, typical).

Host–solute	Misfit character	Approximate yield-strength increase	Example alloy
Cu–Zn	Small size, small modulus	~4–5 MPa per wt% Zn (concentrated)	Brass C26000: σy 75–100 MPa annealed vs ~35 for Cu
Cu–Ni	Very small size misfit	~2–3 MPa per wt% Ni	Cupronickel 90/10 for seawater pipe
Cu–Sn	Large size misfit	~10–15 MPa per wt% Sn	Phosphor bronze springs
Al–Mg	Large size misfit (12 %)	~15–20 MPa per wt% Mg	5083 (4.5Mg) for ship hulls and LNG tanks
Fe–Mn (ferrite)	Moderate	~30–35 MPa per wt% Mn	Structural and pipeline steels
Fe–Si (ferrite)	Moderate	~80–85 MPa per wt% Si	Electrical steels, spring steels
Fe–P (ferrite)	Strong	~600–700 MPa per wt% P	Rephosphorized automotive sheet
Fe–C,N (ferrite, interstitial)	Tetragonal, very strong	~5000 MPa per wt% in solution (solubility is tiny)	Strain ageing; bake hardening
Ti–O (α, interstitial)	Strong	~800–1000 MPa per wt% O	Commercially pure Ti grades 1–4

Table 13.3 explains a practical fact about titanium: commercially pure grades 1 to 4 differ chiefly in oxygen content (about 0.18 to 0.40 wt%), and their yield strengths rise from about 170 to 480 MPa. Oxygen is the cheapest strengthener titanium has. It is also why titanium welding needs rigorous argon shielding: oxygen picked up from the air embrittles the weld.

Yield points, strain ageing and Lüders bands

Interstitial carbon and nitrogen in ferritic steel do more than raise the flow stress. Given time and a modest temperature, they diffuse to the tensile region below edge dislocations, forming **Cottrell atmospheres** that lock the dislocations in place. When the steel is loaded, the stress rises to an **upper yield point** at which dislocations break free (or new ones are generated), then drops to a **lower yield point**. Deformation propagates as a band of yielded material — a **Lüders band** — across the specimen at constant stress, producing a yield plateau of 1–3 % strain. On a pressed car panel, Lüders bands appear as unsightly "stretcher strains". Steelmakers prevent them by temper rolling (a 1 % skin pass that introduces fresh, unpinned dislocations), by stabilizing carbon and nitrogen with Ti or Nb in interstitial-free steels, or by exploiting them: **bake-hardening** steels keep a controlled 10–30 ppm of free carbon, which pins dislocations during the 170 °C paint-bake cycle and raises the yield strength of the finished panel by 30–60 MPa, improving dent resistance after forming has already been done in the soft condition.

When solute atmospheres can keep up with moving dislocations, repeatedly pinning and releasing them, **dynamic strain ageing** produces serrated flow (the **Portevin–Le Chatelier** effect) in Al–Mg alloys at room temperature and in steels at 200–300 °C ("blue brittleness"). The surface bands it leaves are why 5xxx alloys are used for inner, not outer, car body panels.

13.7 Grain-boundary strengthening: Hall–Petch from pile-ups

Why grain boundaries obstruct slip

A slip plane does not continue across a high-angle grain boundary: the neighbouring grain has a different orientation, and a dislocation reaching the boundary cannot simply glide on. It stops, and the dislocations behind it, emitted from the same source, stack up behind it in a **pile-up**. Slip can propagate into the next grain only when the stress concentration at the head of the pile-up is high enough to activate a source in that grain, or to push the lead dislocation through the boundary with a residual dislocation left behind.

The pile-up derivation

Consider a source in the centre of a grain of diameter d that emits dislocations on a slip plane under resolved shear stress τ. The dislocations pile up against the boundary over a length $L \approx d/2$. Eshelby, Frank and Nabarro showed that the number of dislocations that fit in a pile-up of length L under an effective stress $\tau - \tau_i$ (where τ_i is the lattice friction stress) is

$$n = \frac{\pi (1-\nu)\, L\, (\tau - \tau_i)}{G\, b}$$

for edge dislocations, and that the stress at the head is magnified by the number of dislocations: $\tau_{\text{head}} = n(\tau - \tau_i)$. Slip is transmitted when τ_{head} reaches a critical value τ^{*}, a property of the boundary and of the neighbouring grain. Substituting $L = d/2$:

$$\frac{\pi (1-\nu)\, d\, (\tau - \tau_i)^2}{2 G b} = \tau^{*} \quad \Rightarrow \quad \tau = \tau_i + \sqrt{\frac{2 G b \tau^{*}}{\pi (1-\nu)}}\; d^{-1/2}$$

Multiplying by the Taylor factor gives the **Hall–Petch relation**,

$$\sigma_y = \sigma_0 + k_y\, d^{-1/2}$$

discovered empirically by Hall (1951) and Petch (1953) for mild steel.

> **Status of the principle: Hall–Petch relation**
> *Empirical.* The $d^{-1/2}$ form follows from the pile-up model (and also from alternative models based on geometrically necessary dislocations or boundary sources), but σ_0 and k_y are fitted; it holds from roughly 1 μm to a few hundred μm, weakens below about 100 nm as pile-ups contain too few dislocations to be continuous, and inverts below about 10–20 nm, where grain-boundary sliding dominates (Chapter 38).

Table 13.4 — Hall–Petch parameters (room temperature, typical).

Material	σ_0 (MPa)	k_y (MPa m$^{1/2}$)	σ_y at $d = 50$ μm	σ_y at $d = 5$ μm
Copper	~25	~0.11	~41	~74
Aluminium (pure)	~15	~0.04–0.07	~21–25	~33–46
Nickel	~20	~0.16	~43	~92
Low-carbon (ferritic) steel	~70	~0.6–0.75	~155–175	~340–405
α-titanium (CP)	~80	~0.4	~137	~260
Magnesium alloy AZ31	~80	~0.2–0.3	~110–120	~170–215
Molybdenum	~400	~1.8	~650	~1200

The striking contrast in k_y between BCC iron (about 0.7) and FCC aluminium or copper (about 0.05–0.11) has a physical reason: in FCC metals slip is transmitted easily across boundaries by cross-slip and the many available slip systems, so the critical head stress τ^{*} is small; in BCC iron, interstitial atmospheres lock the sources in the next grain and τ^{*} is large. Grain refinement is therefore a modest strengthener for aluminium but a major one for steel and magnesium (Figure 13.4).

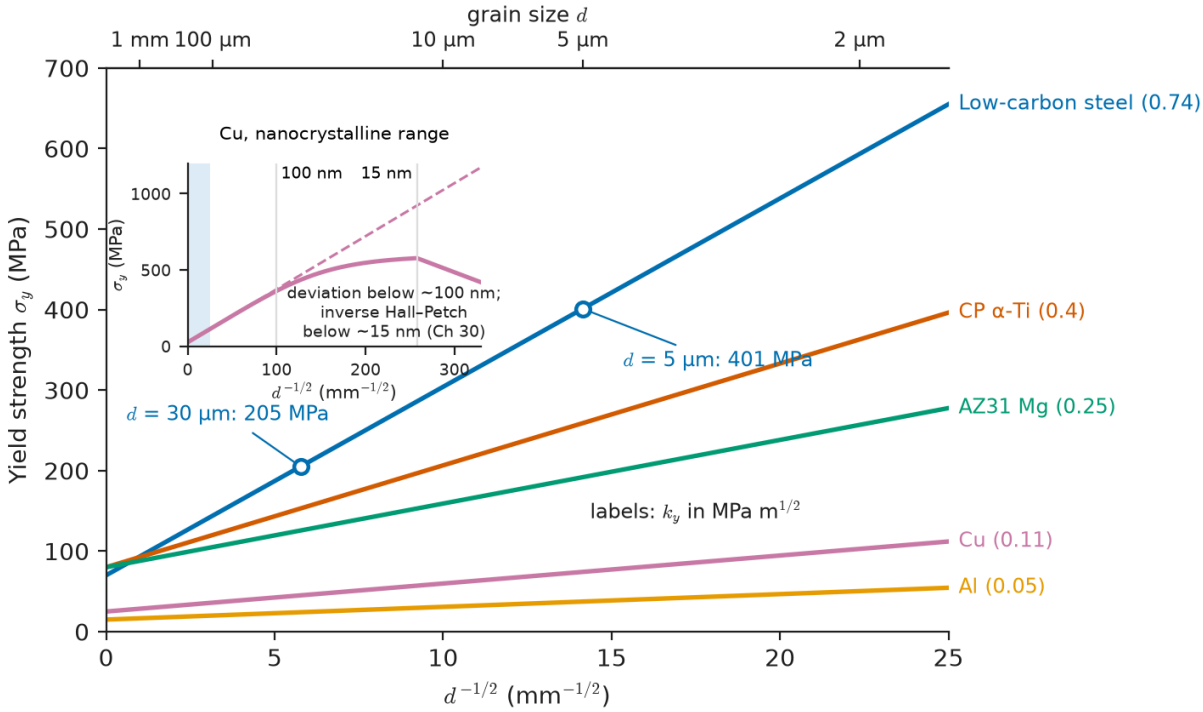

Figure 13.4 — Hall–Petch plots of yield strength versus inverse square root of grain size for low-carbon steel, alpha-titanium, AZ31 magnesium, copper and aluminium, with the nanocrystalline deviation indicated.

Worked example — Grain refinement of a structural steel. A plate steel has $\sigma_0 = 70$ MPa and $k_y = 0.74$ MPa m$^{1/2}$. Conventional hot rolling with air cooling gives a ferrite grain size of 30 µm; controlled rolling with Nb microalloying followed by accelerated cooling gives 5 µm (Chapter 15).

At $d = 30$ µm: $k_y d^{-1/2} = 0.74/\sqrt{3 \times 10^{-5}} = 0.74/0.005477 = 135$ MPa, so $\sigma_y = 205$ MPa.

At $d = 5$ µm: $k_y d^{-1/2} = 0.74/\sqrt{5 \times 10^{-6}} = 0.74/0.002236 = 331$ MPa, so $\sigma_y = 401$ MPa.

Grain refinement alone nearly doubles the yield strength with no added alloy cost and no loss of weldability. It does something else no other mechanism does: it *lowers* the ductile–brittle transition temperature, because cleavage fracture stress rises with $d^{-1/2}$ faster than the yield strength does (Section 14.7). Refining a ship-plate steel from 30 to 5 µm can lower its Charpy transition by some 40–60 °C. This combination — stronger and tougher at once — is why controlled rolling became the backbone of pipeline, offshore-platform and shipbuilding steels from the 1970s onward.

Hall–Petch-like behaviour also appears in fine-grained ceramics such as sub-micrometre alumina and WC tool grades, and nanocrystalline nickel and copper alloys reach 1–2 GPa before the trend reverses below about 10–20 nm (Chapter 38).

13.8 Precipitate strengthening: cutting versus Orowan bypass

The obstacles

The strongest aluminium, nickel and maraging-steel alloys owe most of their strength to fine precipitates (Chapter 10). A dislocation meeting one can **cut** through it or **bypass** it by bowing between particles, and takes whichever path needs less stress — set by particle size, coherency, structure and volume fraction.

Particle cutting

A dislocation can shear a particle only if the particle is coherent (or nearly so) with the matrix, so that its slip planes continue those of the matrix. Cutting costs energy through several mechanisms:

- **Coherency and modulus strengthening**: the misfit strain field and the different shear modulus of the particle interact with the dislocation.
- **Chemical (surface) strengthening**: shearing creates new interface on the two ledges where the particle is offset by b.
- **Order strengthening**: shearing an ordered particle such as γ'-Ni_3Al or δ'-Al_3Li creates an **antiphase boundary** (Section 3.7) costing $\gamma_{\mathrm{APB}} \approx 0.1$–$0.3$ J/m², so dislocations travel in pairs, the second restoring the order the first destroyed.

For small particles each gives a strengthening that grows with radius r and volume fraction f as

$$\Delta\tau_{\mathrm{cut}} \propto \sqrt{f\, r}$$

The physics is the same for all: the resisting force per particle grows with its size, and the dislocation samples particles with a spacing set by its line tension.

The Orowan mechanism

If the particles are incoherent, or large, the dislocation does not cut them. Instead, under increasing stress, it bows between neighbouring particles until the bowed segments meet behind each particle, annihilate, and leave a loop around it while the dislocation moves on (Figure 13.5, inset). The stress required is the stress to bow a segment of length λ (the free spacing between particles) into a semicircle, which by the line-tension argument of Section 12.3 is

$$\Delta\tau_{\mathrm{Or}} \approx \frac{G\, b}{\lambda}$$

A more careful treatment accounting for the different line energies of edge and screw segments and for the interaction between the arms gives the **Ashby–Orowan** form

$$\Delta\tau_{\mathrm{Or}} = \frac{0.4\, G\, b}{\pi \sqrt{1-\nu}\, \lambda} \ln\frac{2\bar{r}}{b}$$

with $\bar{r} = \sqrt{2/3}\, r$ the mean radius of the particle sections intersected by a random slip plane. For particles of radius r at volume fraction f, the centre-to-centre spacing in the slip plane is about $L = r\sqrt{2\pi/3f}$ and $\lambda = L - 2\bar{r}$. At fixed f, $\lambda \propto r$, so the Orowan stress *falls* as particles grow.

> **Status of the principle: Orowan bypass stress**
> *Derived.* Follows from the line-tension model of a dislocation bowing between impenetrable obstacles; the simple form Gb/λ overestimates by a factor of about 1.5–2, and the Ashby–Orowan correction brings it within about 20 % of measurement for dispersion-strengthened alloys with known particle statistics.

The crossover and peak ageing

Cutting stress rises with r; Orowan stress falls with r. At small sizes the particles are cut; at large sizes they are bypassed. The flow stress is the lower of the two, so it peaks at the radius r_c where they cross — typically 5–30 nm for Al alloys and superalloys (Figure 13.5). This is the microstructural explanation of the age-hardening curve of Chapter 10: underaged alloys contain small, cut particles and are relatively soft; peak-aged alloys sit near r_c; overaged alloys contain coarsened, bypassed particles, and they soften as Ostwald ripening (LSW theory, Section 10.3) increases λ.

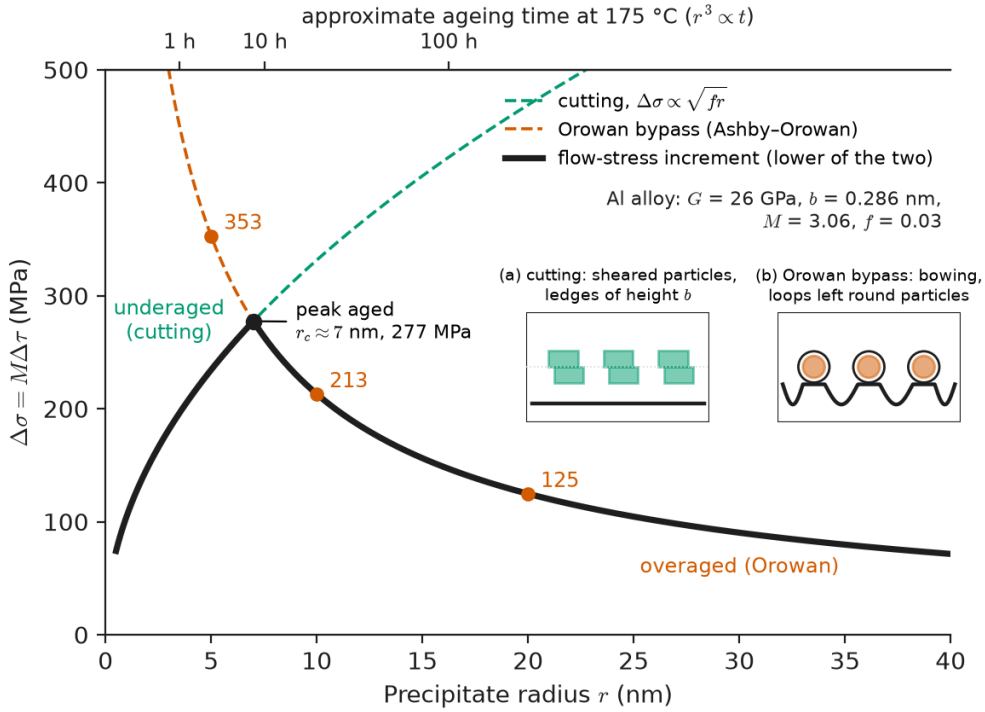

Figure 13.5 — Precipitation strengthening versus particle radius at fixed volume fraction: rising cutting curve, falling Orowan curve, and the resulting peak strength, with insets of a dislocation shearing and bowing around particles.

Worked example — Orowan strengthening in an overaged aluminium alloy. An Al–Cu–Mg alloy overaged at 220 °C contains semicoherent θ'-type precipitates which we idealize as spheres of radius $r = 10$ nm at volume fraction $f = 0.03$. Take $G = 26$ GPa, $b = 0.286$ nm, $\nu = 0.33$, $M = 3.06$.

Centre-to-centre spacing: $L = r\sqrt{2\pi/3f} = 10\text{ nm} \times \sqrt{2\pi/0.09} = 10 \times 8.355 = 83.6$ nm. Mean planar radius: $\bar{r} = \sqrt{2/3} \times 10 = 8.16$ nm. Free spacing $\lambda = 83.6 - 16.3 = 67.2$ nm.

Simple Orowan estimate: $\Delta\tau = Gb/\lambda = 26 \times 10^9 \times 0.286 \times 10^{-9}/67.2 \times 10^{-9} = 111$ MPa, so $\Delta\sigma = M\Delta\tau = 338$ MPa.

Ashby–Orowan: $\ln(2\bar{r}/b) = \ln(16.3/0.286) = \ln 57.1 = 4.04$, and

$$\Delta\tau = \frac{0.4 \times 26 \times 10^9 \times 0.286 \times 10^{-9}}{\pi \times \sqrt{0.67} \times 67.2 \times 10^{-9}} \times 4.04 = 69.6\text{ MPa}, \qquad \Delta\sigma = 213\text{ MPa}$$

If overageing coarsens the particles to $r = 20$ nm at the same f, all lengths double, $\lambda = 134$ nm, and the Ashby–Orowan increment falls to $\Delta\sigma = 125$ MPa. At $r = 5$ nm, if the particles were not cut, it would be 353 MPa. The numbers explain both the 400–500 MPa yield strengths of peak-aged 2xxx and 7xxx aircraft alloys (precipitation strengthening plus a 50–100 MPa matrix and grain contribution) and the loss of strength in components exposed to elevated temperatures, such as an aluminium piston crown or a supersonic airframe skin at 120–150 °C.

Engineering consequences

Precipitate strengthening is the most powerful mechanism, with constraints. Peak-aged alloys soften if heated above their ageing temperature, so Al alloys are limited to about 150–200 °C and welded Al structures have a soft heat-affected zone. Cuttable particles promote planar slip — a cut particle is easier to cut again — which concentrates strain in slip bands and can lower fatigue resistance; non-shearable dispersoids homogenize slip. **Oxide-dispersion-strengthened** (ODS) alloys use 0.5–2 % of incoherent Y_2O_3 particles that neither dissolve nor coarsen, so their Orowan strengthening survives to 1000 °C or more in reactor steels (Chapter 41). Nickel superalloys use 40–70 % of ordered, coherent γ' cut in pairs, which is why their yield strength *rises* with temperature up to about 700–800 °C (Chapter 16).

13.9 Twinning and superplasticity

Deformation twinning

Slip is not the only way a crystal can shear. In **deformation twinning**, a region of the crystal shears homogeneously so that its lattice becomes a mirror image of the parent across a twin plane (Section 5.4). Every atomic plane parallel to the twin plane moves by an amount proportional to its distance from it. The twinned region has a definite, fixed shear — 0.707 for $\{111\}\langle 11\bar{2}\rangle$ twins in FCC and BCC metals, 0.129 for the $\{10\bar{1}2\}$ tension twin in magnesium — and the lattice reorientation is large (for example 86° about $\langle 11\bar{2}0\rangle$ in that magnesium twin).

Twinning differs from slip in four engineering-relevant ways:

1. **It is polar.** The $\{10\bar{1}2\}$ twin in magnesium extends the crystal along its *c*-axis and operates under *c*-axis tension but not compression. Extruded AZ31 bar with a basal texture therefore yields at about 200 MPa in tension but only about 100 MPa in compression — a **tension–compression asymmetry** that magnesium structures must be designed for.
2. **It supplies missing deformation modes**, giving HCP metals the *c*-axis strain that slip alone cannot (Section 13.3).
3. **It is favoured by low temperature, high strain rate and low stacking-fault energy**: ferrite twins under impact or at cryogenic temperature (Neumann bands).
4. **Twin boundaries block slip.** Twins that subdivide grains during straining shorten the dislocation mean free path — a "dynamic Hall–Petch effect" exploited in **twinning-induced plasticity** (TWIP) steels, Fe–(15–30)Mn–C with stacking-fault energy of 20–40 mJ/m², which reach tensile strengths near 1000 MPa with 50–60 % uniform elongation (Chapter 15). Nanotwinned copper reaches about 900 MPa yield while keeping its conductivity, because coherent twin boundaries scatter electrons very little.

Closely related displacive shears produce martensite, and in shape-memory alloys the reversible motion of twin and variant boundaries gives the shape-memory and superelastic effects (Sections 10.6 and 37.2).

Superplasticity

Some fine-grained alloys, deformed slowly at about half their absolute melting temperature, stretch by several hundred to over a thousand percent without necking. This **superplasticity** is used to form and diffusion-bond (SPF/DB) Ti–6Al–4V fan blades and airframe panels at about 900 °C and to blow-form 5083 aluminium car panels at 450–500 °C.

The key property is **strain-rate sensitivity**. Writing the flow stress at fixed strain and temperature as

$$\sigma = K\,\dot{\varepsilon}^{\,m}$$

the exponent $m = \partial \ln \sigma / \partial \ln \dot{\varepsilon}$ is 0.01–0.1 for ordinary cold-working metals and 0.3–0.8 for superplastic materials. A large m resists necking: if a neck begins to form, its local strain rate rises, and with high m the local flow stress rises steeply enough to transfer deformation back to the rest of the specimen. With $m = 0.5$, a tenfold increase in local strain rate raises the flow stress by a factor $10^{0.5} = 3.2$; with $m = 0.1$, only by $10^{0.1} = 1.26$. Hart's analysis gives the condition for stable uniform elongation as $n + m \geq 1$ in rate-dependent form, and elongation to failure rises steeply with m.

> **Status of the principle: Strain-rate sensitivity and neck stabilization**
> *Approximate*. The constitutive law $\sigma = K\dot{\varepsilon}^m$ is an empirical fit over one or two decades of strain rate; the conclusion that high m suppresses neck growth is a derived stability result, but elongation to failure is also limited by cavitation at grain boundaries, which the simple analysis ignores.

The mechanism is **grain-boundary sliding accommodated by diffusion and dislocation activity**, as in the low-stress creep regime of Chapter 14. It requires a fine grain size (below about 10 μm) that is stable at temperature — a two-phase structure (α/β in Ti–6Al–4V) or pinning dispersoids (Al_3Zr, Mn-bearing particles in 5083) — a temperature above about $0.5T_m$, and strain rates of about 10^{-4}–10^{-2} s^{-1}. That slow rate confines superplastic forming to complex, low-volume, high-value parts.

13.10 Combining mechanisms in real alloys

How contributions add

A real engineering alloy is strengthened by several mechanisms at once. The simplest model adds the contributions linearly:

$$\sigma_y = \sigma_i + \Delta\sigma_{\text{ss}} + \Delta\sigma_{\text{gb}} + \Delta\sigma_{\text{ppt}} + \Delta\sigma_{\text{dis}}$$

with σ_i the lattice friction stress. Linear addition suits obstacles on very different length scales (grain boundaries over micrometres, solutes over nanometres), which a dislocation overcomes in series. Point-like obstacles of similar strength mixed on one slip plane (forest dislocations and precipitates) are sampled together, and a **root-sum-square** rule fits better:

$$\Delta\sigma_{\text{ppt+dis}} = \sqrt{\Delta\sigma_{\text{ppt}}^2 + \Delta\sigma_{\text{dis}}^2}$$

More generally $\Delta\sigma^k = \sum \Delta\sigma_i^k$ with $1 \le k \le 2$.

> **Status of the principle: Superposition of strengthening contributions**
> *Approximate.* Linear addition holds for obstacles of very different strengths or length scales, quadratic addition for comparable point-like obstacles of similar density; real alloys fall between, and the choice of exponent is often fitted. The individual terms are themselves empirical or approximate, so the sum is reliable to perhaps 10–15 %.

Worked example — Yield strength of a microalloyed HSLA steel. A controlled-rolled X70 pipeline steel contains 0.06 C, 1.4 Mn, 0.3 Si, 0.04 Nb and 0.05 V (wt%), has a ferrite grain size of 6 μm, fine Nb(C,N) and VC precipitates, and some dislocation substructure from finishing in the two-phase region. Estimate its yield strength.

Contribution	Basis	Value (MPa)
Lattice friction, σ_i	Pure ferrite	50
Solid solution, Mn	32 MPa per wt% × 1.4	45
Solid solution, Si	83 MPa per wt% × 0.3	25
Grain size, $k_y d^{-1/2}$	$0.74/\sqrt{6\times10^{-6}}$	302
Precipitation (Nb, V carbonitrides)	Orowan, few-nm particles, $f \approx 0.001$	80
Dislocations	Taylor, $\rho \approx 10^{14}$ m^{-2}	40

Linear sum: $50 + 45 + 25 + 302 + 80 + 40 = 542$ MPa. Taking precipitates and dislocations in quadrature: $50 + 70 + 302 + \sqrt{80^2 + 40^2} = 50 + 70 + 302 + 89 = 511$ MPa. Measured X70 yield strengths lie around 485–560 MPa, so both estimates are in range. More than half the strength comes from grain size, the one mechanism that also raises toughness; the precipitates are there mainly to pin austenite grains during rolling (Chapter 15), and carbon is kept low for weldability (Chapter 47).

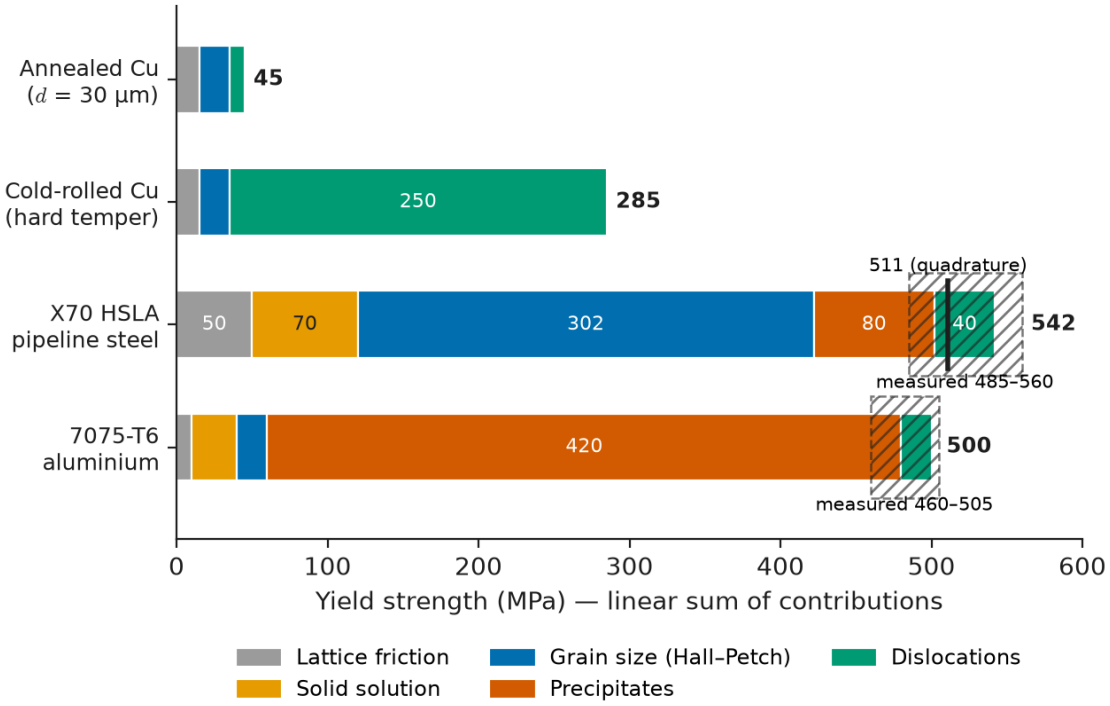

Figure 13.6 — Stacked contributions to yield strength in four engineering alloys: annealed copper, cold-rolled copper, X70 HSLA pipeline steel and peak-aged 7075-T6 aluminium.

Patterns across alloy families

Figure 13.6 and Table 13.5 show how differently the same mechanisms are used in different industries.

Table 13.5 — Dominant strengthening mechanisms in engineering alloys (room temperature, typical yield strengths).

Alloy and condition	σ_y (MPa)	Dominant mechanisms	Why this choice
Copper C11000, annealed	40–70	Grain size, small forest term	Maximum conductivity for building wire
Copper C11000, hard	280–340	Work hardening	Busbars; conductivity kept near 97 % IACS
Cu–Be C17200, aged	1000–1200	Precipitation (GP zones, γ′′)	Connector springs, non-sparking tools
Al 5083-O	125–150	Solid solution (Mg)	Weldable marine hulls, LNG tanks; no ageing to lose in the weld
Al 7075-T6	460–505	Precipitation (η′)	Aircraft wing and fuselage structure
X70 pipeline steel	485–560	Grain refinement + microalloy precipitates	Strength, toughness and weldability at low carbon
AISI 4340, Q&T 200 °C	1500–1650	Martensite: carbon in solution, dislocations, fine carbides	Landing gear, shafts
Maraging 300	1900–2000	Precipitation (Ni_3Mo, Ni_3Ti) in low-C martensite	Rocket motor cases, tooling
Ti–6Al–4V, annealed	830–900	Solid solution (Al, O), α/β grain size	Airframes, implants
Inconel 718, aged	1030–1170	Precipitation (γ′′, γ′)	Turbine discs, cryogenic tanks
304 stainless, annealed	205–240	Solid solution (N, C, Cr), low SFE hardening	Process vessels, food plant
TWIP steel	450–600 (UTS ~1000)	Twinning, dislocations	Crash structures

The design logic in each row starts from what else the alloy must do. A welded marine hull cannot keep its strength in precipitates that dissolve in the heat-affected zone: Al–Mg solid solution. A conductor must conduct: cold work and precipitates that have removed solute from the matrix (Chapter 25). A field-welded pipeline must resist running fractures at −20 °C: grain refinement. A riveted aircraft wing needs the highest specific strength: precipitation. A turbine disc at 650 °C needs ordered precipitates that coarsen slowly.

Every mechanism that blocks dislocations also limits local plasticity, so strength and ductility trade against each other (the "banana curve" of Figure 15.6); grain refinement is the exception, and twinning and transformation-induced plasticity (Chapter 15) restore hardening at high strength. The mechanisms are also thermally fragile in a definite order: work hardening goes first (recovery and recrystallization at $0.3–0.5T_m$), then precipitates coarsen and grains grow, while solid solution survives longest. That order, and the diffusion-controlled mechanisms that take over at high temperature, lead into the next chapter.

What remains open

Open question	Where it is answered
How does a metal deform under constant load at high temperature, and which mechanism dominates where?	Chapter 14 (creep, deformation-mechanism maps)
How does plastic deformation at a crack tip set fracture toughness, and why does grain size lower the ductile–brittle transition?	Chapter 14 (fracture mechanics, DBTT)
How are cyclic plasticity, the Bauschinger effect and slip bands linked to fatigue?	Chapter 14 (fatigue)
How are grain size, precipitates and transformation products controlled in steels by heat treatment and rolling?	Chapter 15
How are precipitation and solid-solution strengthening exploited in Al, Ti, Ni and Cu alloys?	Chapter 16
Why do polymers yield by shear banding and crazing instead of slip?	Chapter 21
What happens to Hall–Petch strengthening in nanocrystalline metals?	Chapter 38
How do texture and plastic anisotropy control sheet formability and forming limits?	Chapter 46
How is the strength of a weld heat-affected zone lost and recovered?	Chapter 47
How are dislocations and plastic slip simulated with discrete dislocation dynamics and crystal-plasticity FEM?	Chapter 56

Next: Chapter 14 follows these mechanisms to high temperature, long times and sharp cracks — creep, viscoelasticity, fracture mechanics and fatigue — where strength alone stops being the right measure of a material.

Chapter 14 — Creep, Viscoelasticity and Fracture

Part III · Mechanical Behaviour

Three failures reach a materials laboratory in the same month. A steam header in a power plant, operated for twelve years at 565 °C and well below its yield strength, has bulged and cracked along a weld: it crept. A polyethylene gas-distribution fitting, clamped tightly at installation, is found leaking because the clamp force has drained away: it relaxed viscoelastically. And a forged steel crane hook has snapped in two at a stress the design said it could carry for ever: a fatigue crack had grown from a forging lap, cycle by cycle, until the remaining section was too small. None of these failures is explained by the yield strength of Chapter 13. Each involves time, temperature, or a crack — and in each case the material property that mattered was a rate or a toughness, not a strength.

This chapter treats the three ways a material fails when strength alone is not the right measure: time-dependent deformation at high homologous temperature (creep); time-dependent deformation of polymers at ordinary temperatures (viscoelasticity); and the growth of cracks, suddenly (fracture) or progressively (fatigue).

> **Why do metals and ceramics flow slowly at high temperature under stresses far below yield, and how can a 30-year life be predicted from a 3000-hour test?**
> **How large a crack can a structure tolerate, why do some steels shatter in winter, and how fast does a fatigue crack grow?**

Concept	Engineering destination
Diffusional and dislocation creep	Turbine blades, steam pipework, solder joints, glacier and ice mechanics
Power-law creep, Larson–Miller	Design lives of boilers, reformer tubes, jet-engine discs
Deformation-mechanism maps	Choosing alloys and grain sizes for high-temperature service
Viscoelastic models, time–temperature superposition	Plastic pipes and clips, seals, bridge bearings, polymer composites
LEFM: Griffith, K, G	Aircraft damage tolerance, pressure vessels, glass and ceramic parts
J-integral and CTOD	Pipeline girth-weld assessment, nuclear vessel integrity
Ductile–brittle transition	Ship plates, bridges, cryogenic tanks, arctic pipelines
S–N, strain–life, Paris law	Shafts, springs, airframes, welded bridges, implants
Ceramic toughening and R-curves	Zirconia implants and dies, Si_3N_4 bearings, CMC turbine parts
Weibull statistics	Ceramic component design, glass panels, fibre strength

Roadmap. Creep mechanisms (14.1) → power-law creep and life extrapolation (14.2) → deformation-mechanism maps (14.3) → viscoelasticity (14.4) → linear-elastic fracture mechanics (14.5) → elastic–plastic fracture (14.6) → the ductile–brittle transition (14.7) → fatigue (14.8) → toughening of ceramics (14.9) → Weibull statistics (14.10).

14.1 Creep mechanisms

The creep curve and homologous temperature

Creep is time-dependent plastic deformation under constant load. A creep test holds a specimen at constant load (or stress) and temperature and records strain against time. After the instantaneous elastic (and possibly plastic) loading strain, the curve shows a decelerating **primary** stage, in which the material work-hardens; a **secondary** or **steady-state** stage, in which hardening and recovery balance and the rate is a minimum $\dot{\varepsilon}_{ss}$; and an accelerating **tertiary** stage ending in rupture, driven by cavitation on grain boundaries, necking, or microstructural degradation such as carbide coarsening.

The controlling variable is the **homologous temperature** T/T_m in kelvin. Creep becomes important in metals above about 0.3–0.4 T_m and in ceramics above about 0.4–0.5 T_m, because that is where diffusion becomes fast enough to let dislocations climb and atoms migrate. The threshold is a ratio, not a temperature: lead roofing ($T_m = 600$ K) creeps at room temperature (0.50 T_m); tin–silver–copper solder at 25 °C is at 0.61 T_m; ice at −10 °C is at 0.96 T_m, which is why glaciers flow; a nickel superalloy turbine blade at 1050 °C is at about 0.8 T_m; while tungsten ($T_m = 3695$ K) at 1000 °C is only at 0.34 T_m.

The mechanisms

Four distinct atomic mechanisms produce creep. Each has its own dependence on stress, grain size and temperature, and the one with the highest rate dominates.

Nabarro–Herring creep (lattice diffusional creep). In a polycrystal under stress, grain boundaries normal to a tensile stress are better sources of vacancies than boundaries parallel to it, because the stress lowers the energy to create a vacancy there by about $\sigma\Omega$ (Ω the atomic volume). Vacancies flow through the grain interiors from tensile to compressive boundaries, atoms flow the other way, and the grain elongates along the tensile axis. The rate is

$$\dot{\varepsilon}_{\mathrm{NH}} = A_{\mathrm{NH}} \frac{D_L \, \sigma \, \Omega}{k_B T \, d^2}$$

with D_L the lattice self-diffusion coefficient, d the grain size and $A_{\mathrm{NH}} \approx 10$–$40$ depending on grain shape and boundary conditions.

Coble creep (grain-boundary diffusional creep). The same vacancy flux can travel along the grain boundaries, a slab of width $\delta \approx 0.5$ nm with a much higher diffusivity D_{gb} and a lower activation energy (Section 8.6). Because the cross-section of the path scales with δd rather than d^2,

$$\dot{\varepsilon}_{\mathrm{C}} = A_{\mathrm{C}} \frac{\delta D_{gb} \, \sigma \, \Omega}{k_B T \, d^3}$$

Coble creep dominates at lower temperatures and finer grain sizes; both diffusional mechanisms are **Newtonian** (rate proportional to stress, $n = 1$).

Dislocation (power-law) creep. At higher stresses, dislocations glide until blocked by obstacles and then climb around them by absorbing or emitting vacancies. Climb is diffusion-controlled, so the rate has the activation energy of lattice (or, at lower temperature, core) diffusion, but the dependence on stress is steep, $\dot{\varepsilon} \propto \sigma^n$ with $n = 3$–8, because both the dislocation density and the climb velocity rise with stress. In particle-strengthened alloys, apparent exponents can reach 10–40, reflecting a threshold stress below which dislocations cannot detach from the particles.

Grain-boundary sliding. Grains slide past one another along their boundaries. Sliding must be accommodated by diffusion or dislocation motion near triple junctions, otherwise wedge cracks open. Sliding accompanies diffusional creep, is the mechanism of superplasticity (Section 13.9), and its cavitation is the main cause of intergranular creep rupture.

> **Status of the principle: Diffusional creep equations (Nabarro–Herring, Coble)**
> *Derived.* They follow from the stress dependence of vacancy concentration and Fick's first law over a grain; the prefactors depend on grain geometry, and the equations assume that boundaries are perfect sources and sinks, which fails at very low stresses (threshold stresses are observed) and in particle-pinned boundaries.

Worked example — Which diffusional mechanism controls in nickel? For pure nickel ($T_m = 1728$ K) take $\Omega = 1.09 \times 10^{-29}$ m³, $D_L = 1.9 \times 10^{-4} \exp(-284\ \mathrm{kJ\,mol^{-1}}/RT)$ m²/s and $\delta D_{gb} = 3.5 \times 10^{-15} \exp(-115\ \mathrm{kJ\,mol^{-1}}/RT)$ m³/s, and use $A_{\mathrm{NH}} = 42$, $A_{\mathrm{C}} = 42\pi$. Compare the rates at $\sigma = 10$ MPa and $d = 100$ μm at $T = 1400$ K (0.81 T_m).

$D_L = 1.9 \times 10^{-4} \exp(-284\,000/(8.314 \times 1400)) = 1.9 \times 10^{-4} \times e^{-24.40} = 4.75 \times 10^{-15}$ m²/s; $\delta D_{gb} = 3.5 \times 10^{-15} \times e^{-9.88} = 1.78 \times 10^{-19}$ m³/s.

$$\dot{\varepsilon}_{\mathrm{NH}} = \frac{42 \times 10^{7} \times 1.09 \times 10^{-29} \times 4.75 \times 10^{-15}}{1.381 \times 10^{-23} \times 1400 \times 10^{-8}} = 1.1 \times 10^{-7}\ \mathrm{s^{-1}}$$

$$\dot{\varepsilon}_{\mathrm{C}} = \dot{\varepsilon}_{\mathrm{NH}} \times \frac{\pi \, \delta D_{gb}}{D_L \, d} = 1.1 \times 10^{-7} \times \frac{\pi \times 1.78 \times 10^{-19}}{4.75 \times 10^{-15} \times 10^{-4}} = 1.3 \times 10^{-7}\ \mathrm{s^{-1}}$$

The two are nearly equal: the crossover grain size $d^{*} = \pi\delta D_{gb}/D_L$ is 117 μm at this temperature. At 1000 K ($0.58\,T_m$) the lower activation energy of boundary diffusion makes d^{*} about 39 mm, so Coble creep dominates any practical grain size; at 1600 K, d^{*} falls to 19 μm. (This compares only the two diffusional paths: at 10 MPa and 1400 K, the full map of Figure 14.2 shows power-law creep running faster still.) Both rates, about 10^{-7} s^{-1}, would accumulate 1 % strain in a day. This is why turbine blades are made with very coarse grains or none at all: directionally solidified blades eliminate transverse boundaries and single-crystal blades eliminate boundaries entirely, removing Coble creep and boundary cavitation together (Chapter 16).

14.2 Power-law creep and the Larson–Miller parameter

The Norton–Arrhenius law

Over the stress range of most engineering service, steady-state creep follows

$$\dot{\varepsilon}_{ss} = A\,\sigma^{n}\exp\!\left(-\frac{Q_c}{RT}\right)$$

with Q_c close to the activation energy for lattice self-diffusion at $T > 0.5\,T_m$. Dimensionless forms normalize stress by the shear modulus, $\dot{\varepsilon}_{ss} = A'(D_L G b/k_B T)(\sigma/G)^{n}$, which collapses data for different metals of the same crystal structure. At very high stresses ($\sigma/G > 10^{-3}$) the power law breaks down into an exponential dependence, and a Garofalo form $\dot{\varepsilon} \propto [\sinh(\alpha\sigma)]^{n}$ covers both regimes.

> **Status of the principle: Power-law (Norton) creep**
> *Empirical.* Motivated by climb-controlled dislocation models but fitted: A, n and Q_c change with microstructure and regime, and extrapolation outside the tested stress and temperature window is unreliable when the mechanism changes.

Table 14.1 — Creep parameters and design temperatures of high-temperature materials (typical).

Material	T_m (K)	n	Q_c (kJ/mol)	Typical maximum service temperature	Application
Pure Al	933	4.4	142	—	Reference
Al alloy 2618	~910	5–20	140–200	150–200 °C	Pistons, Concorde skin
Carbon steel	~1790	5–7	250–300	400–425 °C	Boiler drums
2.25Cr–1Mo steel	~1790	4–8	300–400	550–580 °C	Steam headers, hydrocrackers
Grade 91 (9Cr–1Mo–V–Nb)	~1780	6–12	400–600 (apparent)	600–620 °C	Ultra-supercritical boiler pipe
316H stainless	~1670	5–8	280–380	650–700 °C	Reactor internals, superheaters
Single-crystal Ni superalloy	~1600	5–10	400–600	1050–1100 °C (metal)	Turbine blades
Polycrystalline alumina	2345	1–2 (fine grain)	400–600	1200–1400 °C	Furnace tubes
Ice	273	3	60–80	—	Glaciers, ice sheets

Worked example — Sensitivity of creep rate to temperature. For a 2.25Cr–1Mo steam header with $Q_c = 300$ kJ/mol, compare operation at 565 °C (838 K) with the design temperature 550 °C (823 K):

$$\frac{\dot{\varepsilon}(838\ \mathrm{K})}{\dot{\varepsilon}(823\ \mathrm{K})} = \exp\!\left[\frac{300\,000}{8.314}\left(\frac{1}{823}-\frac{1}{838}\right)\right] = \exp(0.784) = 2.19$$

A 15 °C overtemperature more than doubles the creep rate and roughly halves the life. With $n = 5$, a 10 % stress increase from wall thinning multiplies the rate by $1.1^{5} = 1.61$. Steam temperature control and wall-thickness monitoring are therefore the two

main creep-life management tools in power stations and refineries.

Rupture life and the Monkman–Grant relation

Designers usually need time to rupture, t_r, rather than strain rate. For many alloys $\dot{\varepsilon}_{ss}\, t_r \approx C_{\mathrm{MG}}$, a constant of order 0.01–0.1 (the **Monkman–Grant relation**), so rupture life scales inversely with the minimum creep rate.

The Larson–Miller parameter

Plant components are designed for 100 000 to 250 000 hours — 11 to 29 years — but laboratory tests last at most a few years, and typically a few thousand hours. Tests are therefore run at higher temperature (or stress) and extrapolated. The **Larson–Miller parameter** combines time and temperature:

$$\mathrm{LMP} = T\,(C + \log_{10} t_r)$$

with T in kelvin, t_r in hours and $C \approx 20$ for steels and many other alloys. It follows from the Arrhenius form: if $t_r \propto \exp(Q/RT)$ at fixed stress, then $\log t_r = \log t_0 + Q/(2.303RT)$, and $T(\log t_r - \log t_0)$ is a function of stress only. A plot of stress against LMP — a **master curve** — collapses rupture data at all temperatures (Figure 14.1).

> **Status of the principle: Larson–Miller parameter**
> *Empirical.* A time–temperature trade-off derived from a single Arrhenius activation energy that varies with stress; the constant C is fitted, and extrapolation is unsafe if the long-term mechanism or microstructure (precipitate coarsening, sigma-phase formation) differs from that in the short tests.

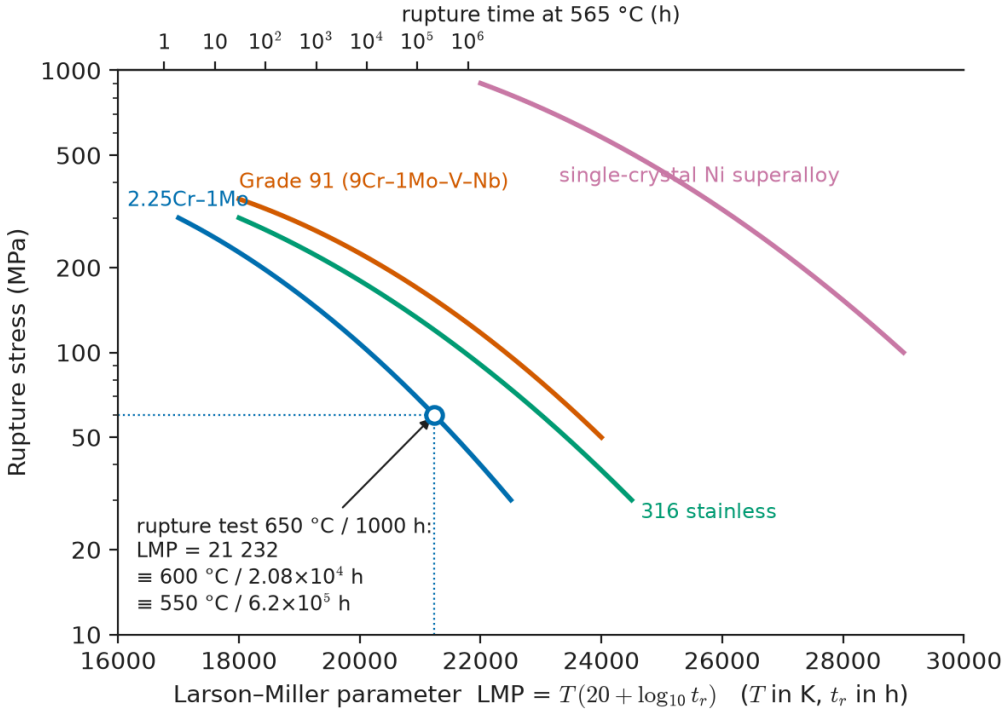

Figure 14.1 — Larson–Miller master curves of stress-rupture strength for 2.25Cr–1Mo steel, Grade 91 steel, 316 stainless steel and a single-crystal nickel superalloy.

Worked example — Extrapolating a rupture test. A batch of 2.25Cr–1Mo pipe ruptures in 1000 h at 650 °C under the design stress. What life does the Larson–Miller parameter predict at 600 °C and at 550 °C under the same stress?

$\mathrm{LMP} = 923.15 \times (20 + 3) = 21\,232$.

At 600 °C (873.15 K): $\log_{10} t_r = 21\,232/873.15 - 20 = 4.317$, so $t_r = 2.08 \times 10^4$ h (2.4 years).

At 550 °C (823.15 K): $\log_{10} t_r = 21\,232/823.15 - 20 = 5.794$, so $t_r = 6.2 \times 10^5$ h (71 years).

A 100 °C reduction buys a factor of 620 in life — but that prediction extrapolates 620-fold in time, during which the bainitic microstructure will coarsen in ways the 1000 h test never saw. Codes therefore limit extrapolation to roughly a factor of three to ten and require long-term data; remaining-life assessment of operating plant adds hardness measurements, surface replicas of cavitation and wall-thickness surveys.

14.3 Deformation-mechanism maps

A material at a given temperature, stress and grain size deforms by whichever mechanism is fastest. Ashby and Frost (1972–1982) built **deformation-mechanism maps** by evaluating the rate equation of each mechanism over the whole range of normalized stress σ_s/G and homologous temperature T/T_m, and labelling each region by the dominant mechanism (Figure 14.2). Contours of constant strain rate are superimposed. The fields are:

- **Ideal shear strength** at $\sigma_s/G \approx 0.05$–0.1, above which the perfect lattice fails.
- **Plasticity (dislocation glide)**: above the yield stress at low temperature, rate-insensitive.
- **Power-law creep**: intermediate stress at $T/T_m > 0.3$–0.4, subdivided into low-temperature (core-diffusion-controlled) and high-temperature (lattice-diffusion-controlled) regions.
- **Diffusional flow**: low stress, high temperature, split into Coble (lower T) and Nabarro–Herring (higher T) fields.
- **Elastic**: below a chosen minimum strain rate, typically 10^{-10} s^{-1}, nothing significant happens.

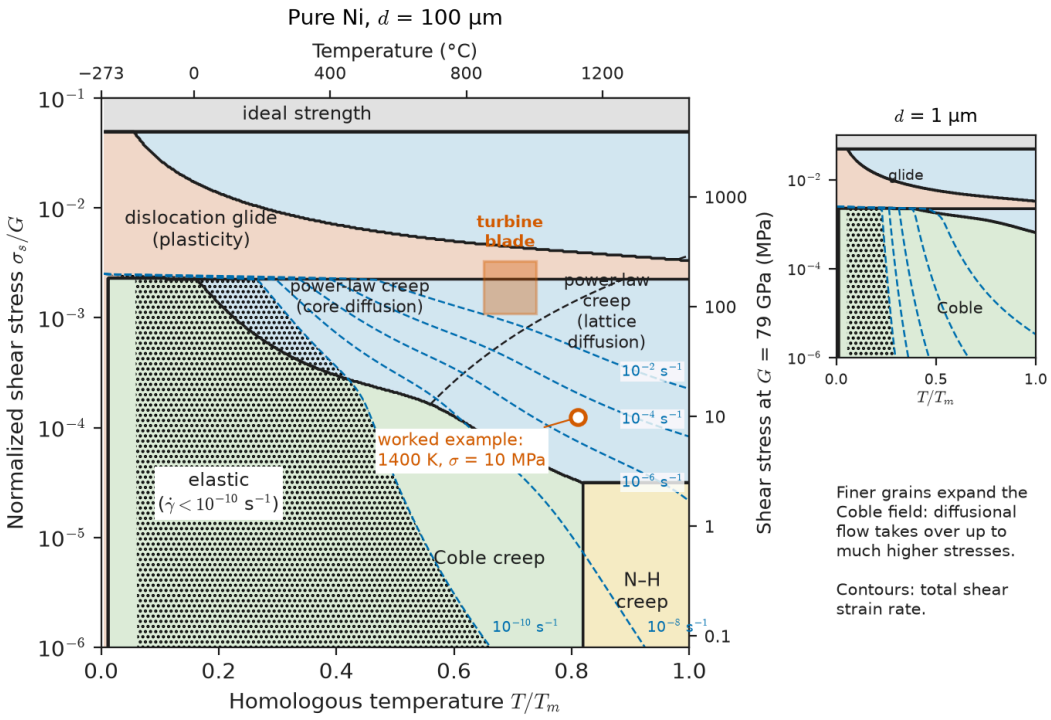

Figure 14.2 — Deformation-mechanism map for pure nickel of grain size 100 μm, showing fields of plasticity, power-law creep, Coble and Nabarro–Herring diffusional creep, with strain-rate contours.

The maps make engineering conclusions visible at once. Finer grains enlarge the diffusional fields ($\dot{\varepsilon} \propto d^{-2}$ or d^{-3}), so materials that are excellent at room temperature can be poor at high temperature — a 1 μm-grained ceramic creeps faster than a 20 μm one. Solid-solution and precipitate strengthening push the power-law field to higher stress. Maps exist for metals, ceramics, ice and mantle rocks; each assumes a fixed microstructure.

> **Status of the principle: Deformation-mechanism maps**
> *Approximate*. They are only as good as the constitutive equations and diffusion data behind them and assume that mechanisms act independently and in parallel; boundaries are typically uncertain by a factor of two in stress or 5–10 % in T/T_m, but the topology is robust.

14.4 Viscoelasticity

Why polymers need a different description

A polymer at room temperature is already in its own "creep regime", with the glass transition T_g playing the role of T_m: polycarbonate ($T_g \approx 150$ °C) at 23 °C is at $0.70\,T_g$ in kelvin, and polypropylene ($T_g \approx -10$ °C) is above its T_g. Its response mixes instantaneous elasticity (bond stretching), delayed recoverable elasticity (chain segments uncoiling) and irrecoverable viscous flow (chains sliding). This **viscoelasticity** governs polyethylene pipe designed for 50 years, clips that lose their grip, and rubber bridge bearings that must yield to slow thermal movement but resist sudden braking forces.

In **linear viscoelasticity**, stress and strain histories are related linearly: doubling the stress history doubles the strain history, and responses to successive loadings superpose (the **Boltzmann superposition principle**). This holds for most polymers below about 0.5–1 % strain.

Spring-and-dashpot models

Linear viscoelasticity is represented by combinations of Hookean springs ($\sigma = E\varepsilon$) and Newtonian dashpots ($\sigma = \eta\dot{\varepsilon}$).

The Maxwell model — spring and dashpot in series. The strain rates add:

$$\dot{\varepsilon} = \frac{\dot{\sigma}}{E} + \frac{\sigma}{\eta}$$

Under a constant strain ε_0 imposed at $t = 0$, the stress relaxes exponentially, $\sigma(t) = E\varepsilon_0 \exp(-t/\tau)$, with **relaxation time** $\tau = \eta/E$. Under constant stress it creeps linearly without limit. It captures stress relaxation and flow but not recoverable creep.

The Kelvin–Voigt model — spring and dashpot in parallel. The stresses add:

$$\sigma = E\varepsilon + \eta\dot{\varepsilon}$$

Under constant stress it creeps toward σ_0/E as $\varepsilon(t) = (\sigma_0/E)[1 - \exp(-t/\tau)]$, with **retardation time** $\tau = \eta/E$, and recovers fully on unloading. It cannot relax stress at constant strain, because the dashpot would need infinite force to move instantly.

The standard linear solid (Zener model) — a spring in parallel with a Maxwell element — is the simplest model with instantaneous elasticity, bounded creep and relaxation to a finite stress (Figure 14.3). At constant strain it relaxes from the **unrelaxed modulus** E_U to the **relaxed modulus** E_R:

$$\sigma(t) = \varepsilon_0 \left[E_R + (E_U - E_R)\, e^{-t/\tau} \right]$$

> **Status of the principle: Spring–dashpot models of viscoelasticity**
> *Approximate.* Exact for the idealized mechanical analogues; real polymers have a broad spectrum of relaxation times, so a single Maxwell or Zener element fits only over about one decade of time, and none of them applies beyond the linear range (large strains, yielding, damage).

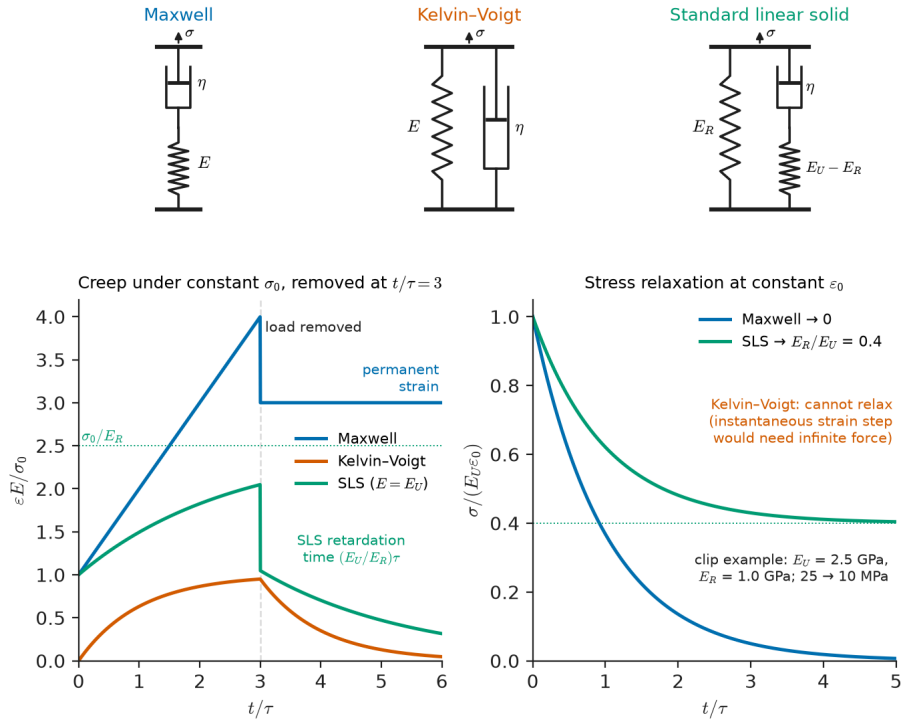

Figure 14.3 — Creep and stress-relaxation responses of the Maxwell, Kelvin–Voigt and standard linear solid models, with their spring–dashpot diagrams.

Worked example — Clamp force in a polymer clip. A moulded polymer clip is pushed onto a pipe and held at a constant strain of 1 %. Its behaviour at 23 °C is represented by a standard linear solid with $E_U = 2.5$ GPa, $E_R = 1.0$ GPa and $\tau = 1 \times 10^{5}$ s (about 28 h).

At $t = 0$: $\sigma = 0.01 \times 2500 = 25.0$ MPa. After 1 hour: $\sigma = 0.01\,[1000 + 1500\,e^{-0.036}] = 24.5$ MPa. After 1 day (8.64×10^{4} s): $\sigma = 0.01\,[1000 + 1500 \times e^{-0.864}] = 0.01 \times (1000 + 1500 \times 0.4215) = 16.3$ MPa. After a year: $\sigma \to 0.01 \times 1000 = 10.0$ MPa.

The clip loses 60 % of its grip force within weeks. If it sits in a 60 °C engine compartment where, for this polymer, the relaxation time is shortened by a shift factor of 10^{-2} to 10^{3} s, the clamp force falls to 10.4 MPa within an hour. Designers specify the long-term (relaxed) modulus, not the datasheet tensile modulus, for snap fits, press fits, bolted polymer flanges and gaskets.

Relaxation spectra

Real polymers relax over many decades of time because chain segments of many lengths relax at different rates. The **generalized Maxwell model** places many Maxwell elements in parallel, giving a relaxation modulus

$$E(t) = E_\infty + \sum_{i=1}^{N} E_i\, e^{-t/\tau_i}$$

a **Prony series**; its continuous limit defines the **relaxation spectrum** $H(\tau)$. Under sinusoidal loading at angular frequency ω the same material shows a **storage modulus** $E'(\omega)$ (in phase) and a **loss modulus** $E''(\omega)$ (out of phase), with loss tangent $\tan\delta = E''/E'$. For a single Maxwell element, $E' = E\omega^2\tau^2/(1+\omega^2\tau^2)$ and $E'' = E\omega\tau/(1+\omega^2\tau^2)$, so the loss peaks at $\omega\tau = 1$. Dynamic mechanical analysis (Chapter 52) measures these against temperature and frequency; the loss peak defines T_g, and vibration-damping materials (constrained layers on car floor pans, viscoelastic dampers in tall buildings) are chosen to have their loss peak at the service temperature and frequency.

Time–temperature superposition

Raising the temperature speeds up all relaxation processes in an amorphous polymer by the same factor a_T. A relaxation curve measured at temperature T can therefore be shifted along the log-time axis onto the curve at a reference temperature T_r:

$$E(t, T) = E\!\left(\frac{t}{a_T}, T_r\right)$$

Short tests at several temperatures, each covering two or three decades of time, are shifted to build a **master curve** spanning ten or fifteen decades at the reference temperature. Above T_g the shift factor follows the **Williams–Landel–Ferry (WLF)** equation,

$$\log_{10} a_T = \frac{-C_1\,(T - T_r)}{C_2 + (T - T_r)}$$

with "universal" constants $C_1 = 17.44$ and $C_2 = 51.6$ K when $T_r = T_g$; Chapter 20 derives it from free-volume theory. Below T_g, and for semicrystalline polymers, an Arrhenius shift is more appropriate.

> **Status of the principle: Time–temperature superposition and the WLF equation**
> *Empirical.* Superposition holds for thermorheologically simple polymers (one dominant relaxation mechanism, no structural change with temperature) and the WLF constants are fitted; it fails across T_g for physically ageing glasses, for multiphase blends and for semicrystalline polymers near their melting range.

For polystyrene with $T_g = 100$ °C, the WLF equation gives $\log a_T = -17.44 \times 10/61.6 = -2.83$ at 110 °C: every relaxation is 680 times faster than at T_g. At $T_g + 30$ K it is 2.6×10^6 times faster. This steepness is why thermoforming and the annealing of moulded-in stress are done within a few tens of degrees above T_g, and why a polymer's long-term behaviour at service temperature can be inferred from tests of a few hours at elevated temperature — the basis of the hydrostatic design basis of plastic pressure pipe (Chapter 21).

14.5 Linear-elastic fracture mechanics

Griffith's energy balance

The theoretical cleavage strength of a solid is about $E/10$, but glass rods break at $E/1000$ and large steel structures have failed at a fraction of their yield strength. Griffith (1921) explained the discrepancy for glass: real solids contain cracks, and a crack grows if the elastic energy released by its extension exceeds the energy needed to create new surface. For a through-crack of length $2a$ in a large plate of unit thickness under remote stress σ, the released strain energy is $\pi\sigma^2 a^2/E$ and the surface energy is $4a\gamma_s$. The energy passes through a maximum at

$$\sigma_f = \sqrt{\frac{2E\gamma_s}{\pi a}}$$

For glass ($E = 70$ GPa, $\gamma_s \approx 0.5$ J/m²) with a 2 μm surface crack ($a = 2$ μm), $\sigma_f = \sqrt{2 \times 70 \times 10^9 \times 0.5/(\pi \times 2 \times 10^{-6})} = 106$ MPa, close to the strength of ordinary window glass.

> **Status of the principle: Griffith criterion**
> *Derived.* Exact for an ideally brittle linear-elastic solid with a sharp crack, from energy conservation; for real materials $2\gamma_s$ must be replaced by the measured fracture energy, which includes plastic and other dissipative work.

Irwin's modification and the energy release rate

For metals Griffith's prediction is low by orders of magnitude, because a plastic zone forms at the crack tip and absorbs far more energy than the surface creation. Irwin and Orowan (1948–1957) replaced $2\gamma_s$ by a **fracture energy** G_c including plastic work, and defined the **energy release rate** $G = -\partial \Pi/\partial A$, the potential energy released per unit area of crack extension. Fracture occurs when

$$G = G_c$$

For the Griffith crack, $G = \pi\sigma^2 a/E'$, with $E' = E$ in plane stress and $E' = E/(1-\nu^2)$ in plane strain.

Crack-tip fields and the stress intensity factor

Irwin also showed that the stress field near any crack tip in a linear-elastic body has a universal form. In polar coordinates (r, θ) centred on the tip,

$$\sigma_{ij} = \frac{K}{\sqrt{2\pi r}}\, f_{ij}(\theta) + (\text{terms finite at } r \to 0)$$

The $r^{-1/2}$ singularity is the same for every geometry and load; only the amplitude K, the **stress intensity factor**, depends on them. There are three loading modes: **mode I** (opening, tensile stress normal to the crack plane), **mode II** (in-plane sliding) and **mode III** (out-of-plane tearing), each with its own K. For mode I,

$$K_I = Y\sigma\sqrt{\pi a}$$

with Y a geometry factor (1.0 for a central crack of length $2a$ in an infinite plate, 1.12 for a shallow edge crack of depth a, about 0.64–0.73 for a semicircular surface crack at its deepest point). K and G are related, for mode I, by

$$G = \frac{K_I^2}{E'}$$

so the stress and energy criteria are equivalent: fracture occurs when K_I reaches the **fracture toughness** K_{Ic}, a material property in MPa m$^{1/2}$.

> **Status of the principle: LEFM fracture criterion $K_I = K_{Ic}$**
> *Derived.* Follows from linear elasticity of a sharp crack and the equivalence $G = K^2/E'$; valid only under small-scale yielding, when the plastic zone is small compared with the crack length, the ligament and the thickness.

The plastic zone and plane strain

Close to the tip the elastic stresses exceed the yield strength. Setting $\sigma_{yy} = \sigma_y$ on the crack plane gives the (uncorrected) plastic-zone size

$$r_p = \frac{1}{2\pi}\left(\frac{K_I}{\sigma_y}\right)^2 \text{ (plane stress)}, \qquad r_p \approx \frac{1}{6\pi}\left(\frac{K_I}{\sigma_y}\right)^2 \text{ (plane strain)}$$

In a thick section the surrounding elastic material constrains the tip, a triaxial stress develops and the zone is small: **plane strain**. In a thin sheet the through-thickness stress relaxes, the zone is larger and the apparent toughness K_c is higher (Figure 14.4). The lowest, geometry-independent value is the **plane-strain fracture toughness** K_{Ic}, which is valid (ASTM E399) when the thickness B and crack length satisfy

$$B,\ a \ \geq\ 2.5\left(\frac{K_{Ic}}{\sigma_y}\right)^2$$

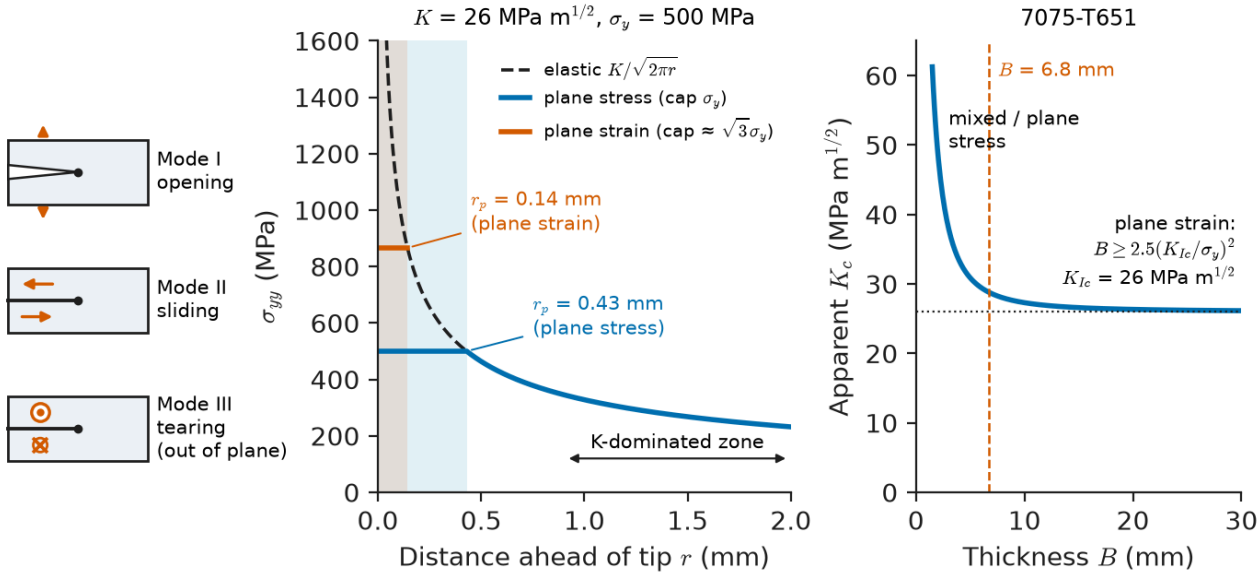

Figure 14.4 — Crack-tip stress field: the three loading modes, the K-dominated elastic singularity and the plastic zone, and the variation of apparent fracture toughness with specimen thickness.

Table 14.2 — Fracture toughness of engineering materials (room temperature, typical).

Material	K_{Ic} (MPa m$^{1/2}$)	σ_y or σ_f (MPa)	Note
Soda-lime glass	0.7–0.8	50–100 (σf)	Window and container glass
Single-crystal silicon	0.8–1.0	—	Wafers, MEMS
Concrete (normal strength)	0.2–1.4	2–5 (tensile)	Size-dependent
Epoxy	0.5–1.0	60–90	Composite matrices
PMMA	1.0–1.5	60–75	Glazing
Alumina	3–5	300–400 (σf)	Insulators, wear parts
Silicon carbide	3–4.5	400–550 (σf)	Seals, armour
Silicon nitride	5–8	700–1000 (σf)	Bearings, turbocharger rotors
3Y-TZP zirconia	6–10	900–1200 (σf)	Dental crowns, femoral heads
Al 7075-T651	24–29	500	Wing structure
Al 2024-T351	32–37	325	Fuselage skin (damage tolerant)
Ti–6Al–4V, mill annealed	55–75	880	Airframe, implants
AISI 4340, tempered 200 °C	45–55	1600	High strength, low toughness
Pressure-vessel steel A533B (upper shelf)	150–200+	480	Reactor vessels (EPFM needed)
Austenitic stainless steel	>200	250	Cryogenic vessels

Worked example — Critical crack in an aircraft wing plate. A 7075-T651 aluminium plate in a wing stringer ($K_{Ic} = 26$ MPa m$^{1/2}$, $\sigma_y = 500$ MPa, $E = 71$ GPa, $\nu = 0.33$) carries a limit tensile stress of 200 MPa. What edge-crack depth will cause fracture, and does LEFM apply?

Critical depth with $Y = 1.12$:

$$a_c = \frac{1}{\pi}\left(\frac{K_{Ic}}{Y\sigma}\right)^2 = \frac{1}{\pi}\left(\frac{26}{1.12 \times 200}\right)^2 = \frac{(0.1161)^2}{\pi} = 4.3 \times 10^{-3}\ \text{m}$$

A crack 4.3 mm deep is critical. Validity: $2.5(K_{Ic}/\sigma_y)^2 = 2.5 \times (0.052)^2 = 6.8 \times 10^{-3}$ m, so plates thicker than about 7 mm are in plane strain. The plane-strain plastic zone at fracture is $(1/6\pi)(0.052)^2 = 0.14$ mm, small compared with the 4.3 mm crack, so LEFM is valid. The critical energy release rate is $G_{Ic} = K_{Ic}^2(1-\nu^2)/E = (26 \times 10^6)^2 \times 0.891/71 \times 10^9 = 8.5$ kJ/m², several thousand times the $2\gamma_s \approx 2$ J/m² of a bare aluminium surface — the plastic work Griffith's original analysis missed. This millimetre-scale critical crack is why airframes are designed on **damage tolerance** principles: inspection intervals are chosen so that a crack below the detectable size (about 1–2 mm by eddy current) cannot grow to a_c before the next inspection (Section 14.8). It is also why the lower wing skin is usually the tougher 2024 or 2x24 alloy, while 7075 is used in the compression-loaded upper skin.

14.6 Elastic–plastic fracture: the J-integral and CTOD

When LEFM fails

For tough steels, the plastic zone at fracture can be comparable with the crack or the section. A pressure-vessel steel with $K \approx 200$ MPa m$^{1/2}$ and $\sigma_y = 480$ MPa would need specimens $2.5 \times (200/480)^2 = 0.43$ m thick for a valid K_{Ic} test. Elastic–plastic fracture mechanics (EPFM) provides parameters that remain meaningful when the plasticity is extensive.

The J-integral

Rice (1968) defined a contour integral around the crack tip,

$$J = \int_\Gamma \left(W\,dy - T_i \frac{\partial u_i}{\partial x}\,ds \right)$$

with W the strain energy density, T_i the traction on the contour and u_i the displacement. For a nonlinear elastic material (and for a plastic material under monotonic, proportional loading, which behaves like one) J is **path independent**, so it can be evaluated far from the tip where stresses are easy to compute. It equals the energy release rate in the nonlinear elastic body, and reduces to $J = G = K^2/E'$ in the linear-elastic limit. Hutchinson, Rice and Rosengren (HRR) showed that J also characterizes the amplitude of the crack-tip stress and strain fields in a power-law hardening material, just as K does for a linear-elastic one. Fracture initiates at $J = J_{Ic}$, which can be measured on specimens a factor of 20–50 smaller than those required for K_{Ic}.

> **Status of the principle: J-integral as a fracture parameter**
> *Derived.* Path independence follows rigorously for nonlinear elasticity (deformation plasticity); for real elastic–plastic materials it is valid under monotonic loading with limited crack growth, and fails under unloading or large amounts of stable crack extension, where J-resistance curves and constraint corrections are used.

Crack-tip opening displacement

Wells (1961), working on welded steel structures, proposed that a crack in a ductile material grows when the opening of its blunted tip, the **crack-tip opening displacement** δ (CTOD), reaches a critical value. In small-scale yielding,

$$\delta = \frac{K_I^2}{m\,\sigma_y\,E'} = \frac{J}{m\,\sigma_y}$$

with $m \approx 1$ in plane stress and about 2 in plane strain. CTOD is the dominant parameter in welding and offshore codes because it can be measured on full-thickness weld specimens.

Worked example — From K to J and CTOD. A girth weld in an offshore X65 pipeline ($\sigma_y = 450$ MPa, $E = 207$ GPa, $\nu = 0.3$) sees an applied K_I of 150 MPa m$^{1/2}$ at a flaw. In plane strain,

$$J = \frac{K^2(1-\nu^2)}{E} = \frac{(150\times10^6)^2 \times 0.91}{207\times10^9} = 9.9\times10^4\ \text{J/m}^2 \approx 99\ \text{kJ/m}^2$$

$$\delta = \frac{J}{2\sigma_y} = \frac{9.9\times10^4}{2\times450\times10^6} = 1.1\times10^{-4}\ \text{m} = 0.11\ \text{mm}$$

A specified minimum weld-metal CTOD of 0.10–0.25 mm at the design temperature is typical of pipeline codes, so this flaw is close to the limit. In a real **engineering critical assessment** (ECA, using BS 7910 or API 579), the applied CTOD or J is computed with plasticity corrections and residual stresses, and plotted on a **failure assessment diagram** that interpolates between fracture (toughness-controlled) and plastic collapse (strength-controlled). ECAs set the acceptable flaw sizes for automated ultrasonic inspection of pipeline welds, and for the remaining-life assessment of nuclear pressure vessels.

Resistance curves in ductile materials

Ductile metals do not fracture at a single J: a crack grows stably by void nucleation, growth and coalescence (Chapter 44), and its resistance rises with extension Δa along a **J–R curve**. Instability occurs when the applied J rises faster with Δa than the resistance — the tearing criterion behind leak-before-break arguments for piping.

14.7 The ductile–brittle transition

The phenomenon

In January 1943 the T-2 tanker *Schenectady*, lying at her fitting-out dock in calm water at about −3 °C, broke in two with a noise heard a mile away. Of the roughly 2700 welded Liberty ships built in the Second World War, several hundred suffered serious brittle fractures, a dozen or so broke completely. The steels were ductile in a room-temperature tensile test. They were brittle at the service temperature.

BCC metals — ferritic and martensitic steels, molybdenum, tungsten, chromium — and many HCP metals show a **ductile–brittle transition** (Figure 14.5). Measured by the absorbed energy in a notched-bar impact (Charpy) test, the behaviour changes over a temperature range of 30–80 °C from a high **upper shelf** (ductile tearing, fibrous fracture surface) to a low **lower shelf** (cleavage, crystalline fracture surface). The **ductile-to-brittle transition temperature** (DBTT) is defined

conventionally at an energy level (27 J or 40 J for structural steels), at 50 % shear fracture appearance, or by a lateral expansion criterion. FCC metals — austenitic stainless steels, aluminium, copper, nickel — show no transition and remain tough down to liquid-helium temperature, which is why they are used for LNG tanks (9 % Ni ferritic steels being the notable exception, made tough by fine grains and retained austenite) and cryogenic propellant tanks.

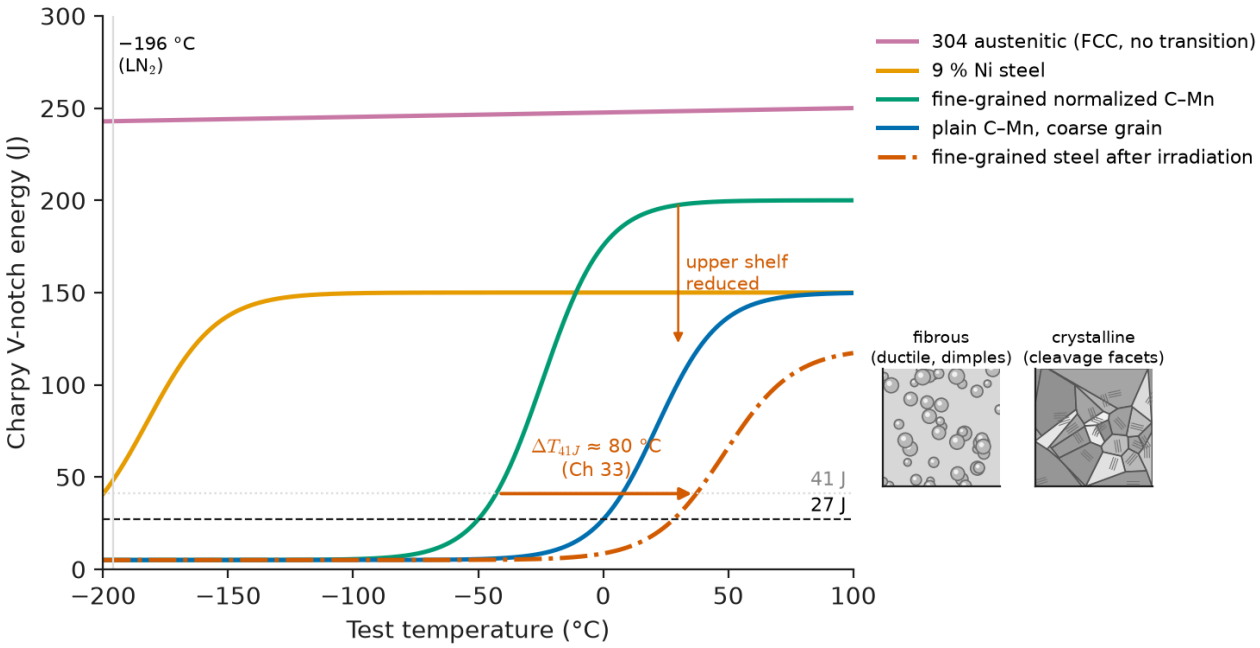

Figure 14.5 — Charpy V-notch impact energy versus temperature for a low-carbon structural steel, a fine-grained normalized steel, the same steel after irradiation, a 9 % Ni steel and an austenitic stainless steel.

The physical origin

The transition is a competition between yielding and cleavage. The yield strength of BCC metals rises steeply on cooling because the Peierls stress for screw dislocations is thermally activated (Section 12.8), whereas the **cleavage fracture stress** σ_f — the local stress needed to propagate a microcrack nucleated at a cracked carbide or a dislocation pile-up across a grain — is almost independent of temperature. At a notch tip, plastic constraint raises the maximum tensile stress to about $2.5–3\,\sigma_y$. At high temperature the material yields and blunts before this reaches σ_f; below a critical temperature, constraint brings the local stress to σ_f first and the steel cleaves (the Yoffe or Davidenkov diagram). FCC metals lack the steep temperature dependence of yield, and their dislocations remain mobile.

Anything that raises σ_y relative to σ_f, or raises constraint, raises the DBTT: higher strain rate (impact), thicker sections, sharper notches, higher carbon and phosphorus, coarse grains, neutron irradiation (which raises σ_y by forming defect clusters, shifting the DBTT of reactor-vessel steels by tens to over 100 °C, Chapter 41) and temper embrittlement (phosphorus segregation to grain boundaries, Chapter 15). Grain refinement is the one strengthening mechanism that *lowers* it, because σ_f scales with $d^{-1/2}$ more strongly than σ_y does; the Cottrell–Petch analysis gives the DBTT falling roughly linearly with $\ln d^{-1/2}$. Nickel also lowers the DBTT by easing cross-slip of screw dislocations.

> **Status of the principle: Yield–cleavage competition for the DBTT**
> *Approximate.* A qualitative to semi-quantitative model; it correctly predicts the effects of strain rate, constraint, grain size and irradiation, but the DBTT itself depends on the test (Charpy energy level, specimen size), and transfer from Charpy to structural fracture needs empirical correlations such as the master-curve method (ASTM E1921).

Engineering control

The lessons of the Liberty ships and later bridge and offshore failures are written into today's specifications. Structural and ship steels are graded by guaranteed Charpy energy at a test temperature (for example S355J2: 27 J at −20 °C; S355ML: 27 J at −50 °C; ship grade EH36: 34 J at −40 °C); welding procedures are qualified for HAZ and weld-metal toughness (Chapter 47); reactor-vessel steels are monitored by surveillance capsules. Polymers have their own analogue: below T_g, or at high rates, PVC,

PMMA and polystyrene fail in a brittle manner, and impact-modified grades (ABS, high-impact polystyrene, rubber-toughened nylon) contain dispersed rubber particles that cavitate and promote shear yielding (Chapter 21).

14.8 Fatigue

Stress-based approach: S–N curves

Fatigue is failure under cyclic loading at stresses well below the static strength, by the initiation and growth of a crack. It is the most common mechanical failure in service, in axles, springs, welded bridge details, wing attachments, offshore nodes, hip stems and wind-turbine blades.

The classical data are **S–N curves** of stress amplitude σ_a against cycles to failure N_f, obtained on smooth specimens. In the high-cycle regime (nominally elastic, $N_f > 10^4$) they follow **Basquin's law**

$$\sigma_a = \sigma_f' \, (2N_f)^b$$

with $b \approx -0.05$ to -0.12. Many steels and titanium alloys show an apparent **fatigue (endurance) limit** at about 10^6–10^7 cycles, typically 0.4–0.5 of the UTS for steels below about 1400 MPa UTS; aluminium, copper and magnesium alloys show none. Even steels can fail beyond 10^7 cycles from internal inclusions ("very-high-cycle" or gigacycle fatigue), which matters for rail wheels, bearing races and high-frequency components.

> **Status of the principle: Basquin relation and the endurance limit**
> *Empirical.* Power-law fits to smooth-specimen data; they do not include mean stress, notch, size, surface or environment effects, which must be applied as corrections, and the "infinite life" endurance limit disappears in corrosive environments and in the gigacycle regime.

The corrections are substantial. A tensile **mean stress** shortens life (Goodman line $\sigma_a/\sigma_e + \sigma_m/\sigma_u = 1$, or the less conservative Gerber parabola). A **notch** reduces fatigue strength by K_f, smaller than the elastic K_t because of local yielding and the stress gradient. Compressive surface residual stress from shot peening or thread rolling lengthens life greatly; seawater can halve the fatigue strength of steel. **Variable-amplitude** loading is handled by rainflow counting and the Palmgren–Miner sum $D = \sum n_i/N_i$.

Strain-based approach: strain–life

When cyclic loads produce plastic strain at a notch root, strain, not stress, controls life. The total strain amplitude is the sum of elastic (Basquin) and plastic (**Coffin–Manson**) parts:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'}{E}(2N_f)^b + \varepsilon_f'(2N_f)^c$$

with fatigue ductility coefficient ε_f' and exponent $c \approx -0.5$ to -0.7. Plastic strain dominates at short lives and elastic strain at long lives; the two lines cross at the **transition life** $2N_t$, which is short (hundreds of cycles) for high-strength steels and long (tens of thousands) for ductile, soft metals. Hence a strong material is better for high-cycle fatigue and a ductile one for low-cycle fatigue. Combined with Neuber's rule for local notch strains and a cyclic stress–strain curve (which shows cyclic hardening of annealed metals and cyclic softening of cold-worked ones), the strain–life method is the standard for automotive suspension parts, turbine discs, pressure-vessel nozzles and the thermal fatigue of solder joints and exhaust manifolds.

> **Status of the principle: Strain–life (Coffin–Manson–Basquin) relation**
> *Empirical.* A sum of two power-law fits that describes uniaxial low- and high-cycle fatigue of metals well; constants depend on temperature, frequency and environment, and multiaxial, non-proportional and creep–fatigue loading need extensions.

Damage tolerance: the Paris law

The S–N and strain–life approaches treat life as crack-free until failure. Large welded and aircraft structures cannot assume this: welds and forgings contain defects from the start. The **damage-tolerance** approach assumes a crack of the largest size that inspection could miss and calculates how long it takes to grow to critical size. The driving force is the stress-intensity range

$\Delta K = Y\Delta\sigma\sqrt{\pi a}$. Fatigue crack growth data (Figure 14.6) show three regimes: a **threshold** ΔK_{th} (typically 2–8 MPa m$^{1/2}$ for steels, 1–3 for Al alloys) below which cracks do not grow measurably; a **Paris regime** of power-law growth,

$$\frac{da}{dN} = C\,(\Delta K)^m$$

with $m \approx 2$–4 for metals; and a final regime of accelerating growth as $K_{\max}$ approaches K_c.

> **Status of the principle: Paris law**
> *Empirical.* A power-law fit to stable fatigue crack growth in the intermediate ΔK regime; it does not describe the threshold or final-fracture regimes, short cracks, or load-sequence (overload retardation) effects without modification.

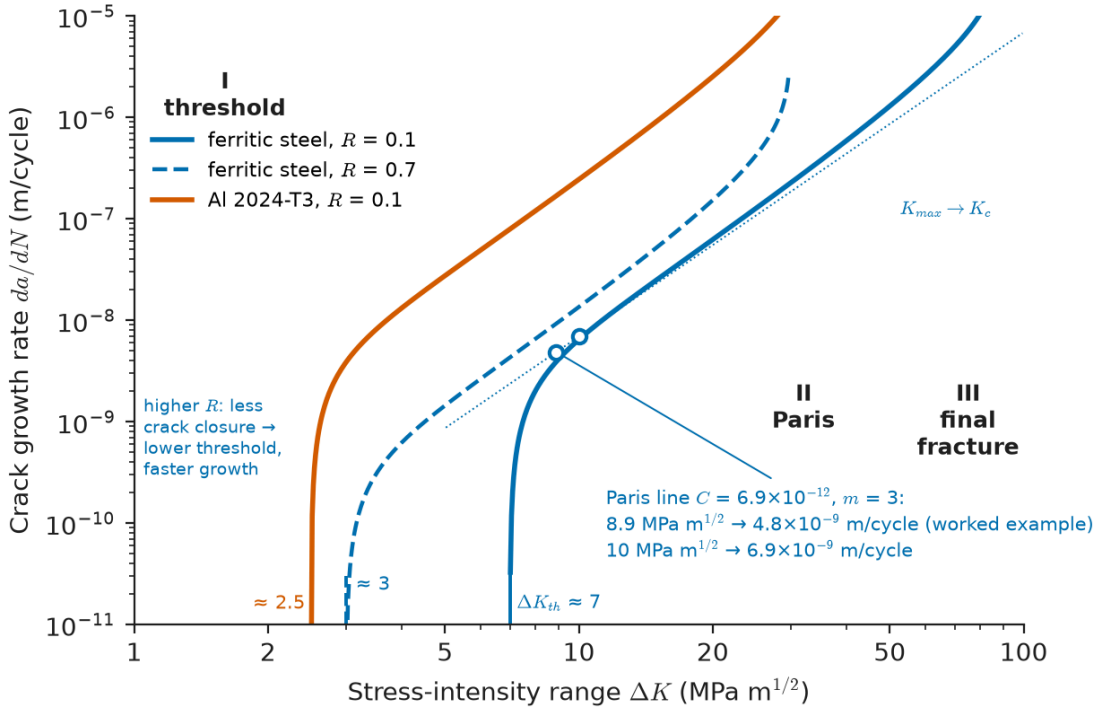

Figure 14.6 — Fatigue crack growth rate versus stress-intensity range showing threshold, Paris and final-fracture regimes for a ferritic steel and an aluminium alloy, with the effect of load ratio.

Worked example — Fatigue life of a cracked crane hook. A steel crane component contains a surface-breaking forging lap that behaves as an edge crack of initial depth $a_0 = 2$ mm. It sees a stress range $\Delta\sigma = 100$ MPa (zero to 100 MPa) at every lift. Take $Y = 1.12$ (assumed constant), $C = 6.9 \times 10^{-12}$ m/cycle with ΔK in MPa m$^{1/2}$, and $m = 3$, typical for ferritic steel. Inspection rules require repair before the crack reaches 20 mm. How many lifts does that take?

Initial $\Delta K = 1.12 \times 100 \times \sqrt{\pi \times 0.002} = 8.9$ MPa m$^{1/2}$, just above threshold; initial growth rate $6.9 \times 10^{-12} \times 8.9^3 = 4.8 \times 10^{-9}$ m/cycle. Integrating the Paris law with $m = 3$,

$$N = \int_{a_0}^{a_f} \frac{da}{C(Y\Delta\sigma\sqrt{\pi a})^3} = \frac{2}{C\,(Y\Delta\sigma\sqrt{\pi})^3}\left(\frac{1}{\sqrt{a_0}} - \frac{1}{\sqrt{a_f}}\right)$$

$Y\Delta\sigma\sqrt{\pi} = 1.12 \times 100 \times 1.7725 = 198.5$; cubed, 7.82×10^6. With $a_0 = 0.002$ m and $a_f = 0.020$ m, $1/\sqrt{a_0} - 1/\sqrt{a_f} = 22.36 - 7.07 = 15.29$ m$^{-1/2}$:

$$N = \frac{2 \times 15.29}{6.9 \times 10^{-12} \times 7.82 \times 10^6} = 5.7 \times 10^5 \text{ cycles}$$

At 200 lifts a day, the crack grows from 2 to 20 mm in about eight years. Reducing $\Delta\sigma$ to 80 MPa raises the life by $(100/80)^3 = 1.95$, to 1.1×10^6 cycles. Two points stand out. Most of the life is spent while the crack is small (half of the integral comes from the first few millimetres), so initial flaw size, set by manufacturing quality and inspection sensitivity, is decisive. And because the hook is steel with a high K_c, final fracture would occur only at a crack of more than 200 mm at this stress — so the safe-life limit comes from the inspection rule, not from fracture.

Crack closure

Elber (1970) observed that fatigue cracks close before the load reaches its minimum: plastically stretched wake material, oxide debris and rough fracture surfaces prop the faces apart. Only the open part of the cycle drives growth, so $\Delta K_{\text{eff}} = K_{\max} - K_{\text{op}} = U\Delta K$, with $U \approx 0.5$–0.8 at low load ratios. **Crack closure** explains why growth rates rise with load ratio $R = K_{\min}/K_{\max}$, why a tensile overload retards later growth, why short cracks (no closure wake yet) grow faster than long ones at equal ΔK, and why rough, coarse-grained fracture paths raise the threshold. Aircraft spectrum codes include closure-based retardation models.

Fatigue in practice

At a weld toe — a sharp notch with tensile residual stress and pre-existing flaws — fatigue strength is almost independent of parent-steel strength, so codes (Eurocode 3, BS 7608, IIW) assign fixed S–N curves by detail category, and high-strength steel helps only with toe grinding or peening. Coronary stents, which see about 4×10^7 heartbeat cycles a year, are designed with Goodman-type margins (Chapter 39).

14.9 Toughening of ceramics

Intrinsic and extrinsic toughening

A ceramic's toughness has two components. **Intrinsic** toughness is the resistance of the material ahead of the crack tip to bond rupture; for ionic and covalent solids without dislocation plasticity it is low, $K_0 \approx 1$–3 MPa m$^{1/2}$, and difficult to change. **Extrinsic** toughening comes from mechanisms that act behind or around the crack tip to reduce the local driving force, **shielding** the tip. Shielding builds up as the crack grows, giving a rising **resistance curve** (**R-curve**): K_R increases with crack extension Δa until the shielding zone is fully developed.

Transformation toughening

Zirconia doped with yttria (3Y-TZP) or magnesia (Mg-PSZ) can be retained at room temperature in the **tetragonal** phase, metastable with respect to monoclinic (Chapter 17). The high tensile stress ahead of a crack tip triggers the martensitic t→m transformation, which carries a volume expansion of about 4 % and a shear. A **transformation zone** of height h forms along the crack flanks, and the dilatant particles compress the crack tip. The shielding is approximately

$$\Delta K_{\text{tr}} \approx \frac{0.22\, E\, e^{T}\, V_f \sqrt{h}}{1-\nu}$$

with e^T the volumetric transformation strain and V_f the fraction of transformable particles in the zone. For Mg-PSZ with $E = 200$ GPa, $e^T = 0.04$, $V_f = 0.5$, $h = 20$ µm and $\nu = 0.3$, $\Delta K_{\text{tr}} = 0.22 \times 200 \times 10^9 \times 0.04 \times 0.5 \times \sqrt{2\times10^{-5}}/0.7 = 5.6$ MPa m$^{1/2}$ — more than doubling a matrix toughness of about 3 MPa m$^{1/2}$.

> **Status of the principle: Transformation-toughening shielding estimate**
> *Approximate.* From the McMeeking–Evans continuum analysis of a dilatant zone of uniform height in steady state; it ignores the shear component of the transformation and zone-shape details, and the estimate can be off by a factor of about two.

Transformation toughening makes zirconia the toughest single-phase oxide ceramic, used in femoral heads, dental crowns, wire-drawing dies and oxygen-sensor electrolytes. The same metastability allows **low-temperature degradation**, slow surface transformation in water at body temperature, which caused failures of a batch of zirconia femoral heads in 2001; zirconia-toughened alumina is now preferred for hip bearings (Chapter 39).

Crack bridging and other mechanisms

- **Grain bridging.** In coarse or elongated-grain alumina and in self-reinforced silicon nitride (grown with rod-like β-grains), intact grains span the crack behind its tip and must be pulled out by friction. Silicon nitride reaches 6–10 MPa m$^{1/2}$ this way and is used for bearing balls, cutting tools and diesel-engine parts.

- **Fibre bridging and pull-out.** In ceramic-matrix composites (SiC fibres in SiC, oxide fibres in alumina), a weak fibre–matrix interface coating (pyrolytic carbon or BN) deflects cracks along the fibres, which then bridge and pull out. Toughness rises to 20–30 MPa $\mathrm{m}^{1/2}$ and the stress–strain curve becomes "graceful", so CMC shrouds and combustor liners now fly in jet engines (Chapters 23 and 41).
- **Crack deflection and microcracking.** Particles or platelets tilt and twist the crack, and stress-induced microcracks dissipate energy around it; both give modest gains.
- **Ductile-phase toughening.** Metal particles (cobalt in WC–Co cemented carbides, nickel in alumina) bridge cracks and deform plastically; WC–Co reaches 10–20 MPa $\mathrm{m}^{1/2}$.

R-curves and design

R-curve behaviour has a counter-intuitive consequence: for small flaws the toughness that matters is the low initial value K_0, so a ceramic with a rising R-curve is not stronger against small natural flaws, but it tolerates damage, is less sensitive to flaw size, and shows less strength scatter (higher Weibull modulus, Section 14.10). Instability occurs where the applied K curve is tangent to the R-curve, so the strength depends on the shape of the R-curve, not only its plateau. Toughening mechanisms that need a large shielding zone (bridging over millimetres) help large cracks; those with small zones (transformation) help short cracks.

14.10 Weibull statistics

Weakest-link failure

Because a brittle part fails from its largest flaw, and flaw sizes vary from part to part, strength is a statistical quantity. Weibull (1939) proposed, for a body of volume V under uniform tensile stress, the failure probability

$$P_f(\sigma) = 1 - \exp\left[-\frac{V}{V_0}\left(\frac{\sigma}{\sigma_0}\right)^{m}\right]$$

where σ_0 is the characteristic strength of reference volume V_0 (where $P_f = 63.2\ \%$) and m is the **Weibull modulus**. The form follows from a weakest-link chain of independent volume elements with a power-law distribution of flaw strengths; a three-parameter form with a threshold stress σ_u below which $P_f = 0$ is sometimes used, but rarely justified by data.

> **Status of the principle: Weibull distribution**
> *Empirical.* The weakest-link argument makes the form plausible, but m and σ_0 are fitted, and extrapolation from a few tens of tests to failure probabilities of 10^{-4} to 10^{-6} relies on the assumed form holding far into the tail; bimodal flaw populations (surface and volume) produce kinked Weibull plots and need separate treatment.

Fitting and typical values

Writing $\ln\ln[1/(1-P_f)] = m\ln\sigma - m\ln\sigma_0 + \ln(V/V_0)$ gives a straight line of slope m. Test results ranked in order of strength are assigned $P_f = (i - 0.5)/N$ and fitted by maximum likelihood (EN 843-5, ASTM C1239), with at least 30 specimens for a meaningful m. Typical values: glass and as-fired ceramics 5–10; technical ceramics 10–20; silicon nitride bearing grades and polished silicon wafers 15–25; carbon fibres 4–10 (strength measured per fibre length); concrete 8–12; ductile metals effectively 50–100, which is why they can be designed with deterministic strengths.

The size effect and effective volume

At equal failure probability, $\sigma \propto V^{-1/m}$: a larger part is weaker. Under non-uniform stress the volume is replaced by an **effective volume** $V_{\mathrm{eff}} = \int (\sigma/\sigma_{\max})^m \, dV$, the volume that would have the same failure probability at uniform $\sigma_{\max}$. For a beam in three-point bending, $V_{\mathrm{eff}} = V/[2(m+1)^2]$; in four-point bending with quarter-point loading, $V_{\mathrm{eff}} = V(m+2)/[4(m+1)^2]$. Bend tests stress only a tiny volume, so bend strengths overstate the tensile strength of large parts.

Worked example — From test bars to a ceramic component. Silicon carbide bend bars of 3×4 mm section are tested in three-point bending over a 40 mm span. Their median strength is 400 MPa and $m = 10$. A component — a pump-seal ring — will see uniform tension over a stressed volume of 10 cm^3. What stress gives a failure probability of 10^{-4}?

Stressed volume of the test bar within the span: $V = 3 \times 4 \times 40$ mm³ $= 4.8 \times 10^{-7}$ m³. Effective volume: $V_{\text{eff}} = 4.8 \times 10^{-7}/(2 \times 11^2) = 1.98 \times 10^{-9}$ m³ (about 2 mm³).

Size scaling to $V_c = 1 \times 10^{-5}$ m³: $(V_{\text{eff}}/V_c)^{1/m} = (1.98 \times 10^{-4})^{0.1} = 0.426$. The component median strength is $0.426 \times 400 = 171$ MPa.

From median ($P_f = 0.5$) to $P_f = 10^{-4}$: $\sigma \propto [\ln(1/(1-P_f))]^{1/m}$, so the factor is $(1.0 \times 10^{-4}/0.693)^{0.1} = 0.413$, giving

$$\sigma(P_f = 10^{-4}) = 171 \times 0.413 = 70 \text{ MPa}$$

A material whose bend strength appears to be 400 MPa must be designed at 70 MPa in this component. Raising m to 20 by better process control would, with the effective volume recomputed, more than double the allowable stress to about 160 MPa — which is why ceramic producers compete on Weibull modulus, not mean strength. **Proof testing** — loading every part to a stress above service and discarding those that break — truncates the flaw distribution and is used for glass aircraft windows, ceramic hip heads and optical fibre.

Weibull beyond ceramics

The same statistics describe fibre strength in composites (Chapter 22), component lifetimes in reliability engineering (Chapter 44), glass façade panels and the size effect in concrete beams: when failure is governed by the worst of many independent defects, bigger is weaker and scatter must be designed for.

What remains open

Open question	Where it is answered
How are carbide-strengthened ferritic steels and stainless steels designed for creep and toughness?	Chapter 15
How do γ′ precipitates and single-crystal casting make nickel superalloys creep-resistant?	Chapter 16
How are zirconia, silicon nitride and glass toughened and strengthened in practice?	Chapter 17
Where does the WLF equation come from, and how does T_g depend on molecular structure?	Chapter 20
How do polymers craze and crack under long-term load, and how is the design life of plastic pipe set?	Chapter 21
How do fibres and laminates fail, and what is compression after impact?	Chapters 22 and 23
How do irradiation, hydrogen and high temperature shift the DBTT and creep life?	Chapters 40 and 41
How do corrosion and environment interact with fatigue and cracking?	Chapter 43
How are fracture surfaces read to identify fatigue, cleavage and creep in a failure investigation?	Chapter 44
How are welds assessed for toughness and fatigue?	Chapter 47
How are creep and fracture simulated across scales?	Chapter 56

Next: Chapter 15 opens Part IV with the most important engineering alloy family of all, steel, where every mechanism of Chapters 9–14 — transformation, precipitation, grain refinement, martensite, the ductile–brittle transition — is put to work through composition and heat treatment.

Part IV — Metals, Ceramics and Construction Materials

Parts I to III built general principles: how atoms arrange themselves, how thermodynamics and kinetics select a microstructure, and how that microstructure resists elastic deformation, plastic flow, creep and fracture. Part IV applies those principles to the material families that carry most of the world's engineering load. A car body, a jet engine, a hip implant, a glass façade, a furnace lining and a concrete bridge are each made from a metal, a ceramic, a glass or a cementitious composite chosen because its structure can be controlled cheaply and reliably at the scale required. The question running through every chapter is the same: given a family of materials, which compositions and processing routes produce which microstructures, and which engineering jobs do those microstructures do best?

The four chapters move from the most versatile family to the most massive: steels, stretched by allotropy, martensite and alloying; non-ferrous alloys, which win on lightness, conductivity, corrosion resistance or temperature capability; ceramics and glasses, where flaw control and viscous forming replace dislocation plasticity; and cement, concrete, timber, asphalt and masonry, made in billions of tonnes, where cost, durability and carbon footprint dominate design.

By the end of this part you should understand:

- How the Fe–C diagram, heat treatment and alloying produce the microstructures of carbon, alloy, stainless and advanced high-strength steels and of cast irons, and how hardenability and surface hardening are designed.
- How precipitation hardening, solid solution, phase balance and single-crystal processing are used in aluminium, copper, titanium, nickel, magnesium and refractory alloys, and in intermetallics and high-entropy alloys.
- Why ceramics are brittle, how powders are formed and sintered into dense parts, how glasses form and are strengthened, and how glass-ceramics and refractories are engineered.
- How Portland cement hydrates, why the water–cement ratio controls concrete strength and durability, and how steel, timber, asphalt and masonry behave as construction materials.
- How to compare these families and choose among them for a given component, load, temperature and environment.

Chapter	Central question	What it delivers
15. Physical Metallurgy of Steels and Cast Irons	How do iron, carbon and a few percent of alloying elements produce materials whose strength spans a factor of ten?	Iron allotropy and the Fe–C diagram, plain-carbon microstructures, annealing, normalizing, quenching and tempering, hardenability (Jominy, CCT), HSLA, tool and maraging steels, stainless families and the Schaeffler diagram, DP, TRIP and TWIP steels, grey, ductile, white and malleable irons, carburizing, nitriding and induction hardening
16. Non-Ferrous Alloys	Where and why do aluminium, copper, titanium, nickel and other metals beat steel?	Al wrought and cast series, tempers and age hardening, brasses, bronzes and conductor alloys, α, α+β and β titanium, γ/γ′ superalloys and single crystals, Mg and Zn alloys, refractory and precious metals, intermetallics and high-entropy alloys, selection among them
17. Ceramics and Glasses	How are brittle, refractory, chemically stable solids made strong and reliable?	Alumina, zirconia, SiC and Si_3N_4, powder processing and sintering, flaw-controlled strength and toughening, glass formation, viscosity and tempering, glass-ceramics, refractories
18. Cement, Concrete and Construction Materials	What controls the strength, durability and carbon cost of the materials that build cities and infrastructure?	Clinker phases, hydration and C–S–H, mix design and the water–cement ratio, curing and strength development, carbonation, chloride attack, ASR and freeze–thaw, reinforced and prestressed concrete, structural steels, timber and engineered wood, asphalt, masonry, low-carbon cements

Chapter 15 comes first because steel is the reference against which other structural materials are judged and uses nearly every mechanism of Parts II and III. Chapter 16 carries precipitation hardening and high-temperature design into the light and exotic metals. Chapter 17 changes the governing physics from dislocations to flaws and viscous flow. Chapter 18 closes with materials whose structure is set by hydration chemistry, biology or bitumen rheology, and whose tonnage makes their sustainability a global question.

How this part connects to the rest of the book. Part IV draws directly on the phase diagrams of Chapter 7, the diffusion and transformation kinetics of Chapters 8–10, and the strengthening, creep and fracture mechanics of Chapters 13 and 14. It hands forward to almost every later part: polymers and composites (Part V) are compared with metals and ceramics and use them as fibres and matrices; functional ceramics and electrical steels reappear in Part VI; superalloys, refractory metals and ceramics return in extreme environments (Chapter 41); corrosion of steels, aluminium and reinforcement and the deterioration of concrete are the subject of Part IX; casting, forming, powder processing, welding and additive manufacturing of these materials fill Part X; and the property data gathered here are the raw material for the selection methods and lifecycle assessment of Chapters 58 and 59.

Chapter 15 — Physical Metallurgy of Steels and Cast Irons

Part IV · Metals, Ceramics and Construction Materials

A gearbox manufacturer receives two batches of the same gear, both machined from bar sold as "0.4 % carbon steel", both carburized and quenched by the same subcontractor. Gears from the first batch run for years. Gears from the second batch shed teeth within weeks: the fracture surfaces show a hard, thin case on a soft core that bent under each tooth load until the case cracked. Chemical analysis shows the difference: the first batch was a Cr–Mo low-alloy steel, the second a plain-carbon steel of nearly the same carbon content. The two steels have the same equilibrium phase diagram, almost the same strength when slowly cooled, and the same price per kilogram within a few percent. What they do not share is **hardenability** — the ability to form martensite deep inside a section during a quench — and that one property decided the life of the gears.

Steel is the most widely used engineering metal, some 1.9 billion tonnes a year — far more than all other metals combined. Its dominance comes not from one outstanding property but from the range of microstructures iron and carbon, with a few percent of other elements, can form: from 200 MPa deep-drawing sheet to 2000 MPa bearing races, from non-magnetic stainless cutlery to transformer cores. Every mechanism of Chapters 7–14 is used somewhere in steel.

> **Why does a single binary system, iron–carbon, produce materials whose strength spans a factor of ten, and how is the microstructure selected by heat treatment and alloying?**
> **What makes a steel hardenable, stainless or formable, and why are cast irons a different family altogether?**

Concept	Engineering destination
Iron allotropy and the Fe–C diagram	Every heat treatment, welding and casting operation on ferrous alloys
Ferrite–pearlite microstructures	Structural sections, rails, wire rod, forgings
Annealing, normalizing, quenching, tempering	Shafts, gears, springs, fasteners, tools
Hardenability, Jominy test, CCT diagrams	Choosing a steel for a section size; quench design
HSLA, tool and maraging steels	Pipelines, bridges, dies, rocket cases, landing gear
Stainless steels and the Schaeffler diagram	Chemical plant, food processing, implants, welding consumables
AHSS: DP, TRIP, TWIP, press hardened	Lightweight car bodies and crash structures
Cast irons	Engine blocks, brake discs, pipes, crankshafts, machine bases
Surface hardening	Gears, camshafts, bearings, crankshaft journals

Roadmap. Iron and the Fe–C diagram (15.1) → plain-carbon steel microstructures (15.2) → heat treatment (15.3) → hardenability and CCT (15.4) → alloy steels: HSLA, tool, maraging (15.5) → stainless steels (15.6) → advanced high-strength steels (15.7) → cast irons (15.8) → surface hardening (15.9).

15.1 Iron and its allotropes; the Fe–C diagram revisited

Three phases of pure iron

Pure iron is **allotropic** (Section 7.2). From room temperature to 912 °C it is body-centred cubic **α-iron (ferrite)**, ferromagnetic below the Curie temperature of 770 °C. Between 912 and 1394 °C it is face-centred cubic **γ-iron (austenite)**. From 1394 °C to the melting point of 1538 °C it is BCC again, **δ-iron**. The α→γ transition, with the close-packed phase stable at *higher* temperature, arises from magnetic and vibrational entropy of BCC iron. It is the most useful fact in ferrous metallurgy: because iron changes structure in the solid state, a steel's microstructure can be reset by heating into austenite and redesigned by the way it is cooled.

Carbon in iron

Carbon dissolves interstitially. In FCC austenite the octahedral interstices are large (radius about 0.052 nm, compared with 0.077 nm for carbon), and austenite dissolves up to 2.14 wt% C at 1147 °C. In BCC ferrite the interstices are much smaller and distorted (octahedral site about 0.019 nm), so ferrite dissolves at most 0.022 wt% C at 727 °C and less than 0.005 wt% at room temperature. Carbon in excess forms the iron carbide **cementite**, Fe3C (6.70 wt% C), a hard (about 800–1100 HV), brittle orthorhombic compound. Cementite is metastable: the true equilibrium is with graphite, but in steels the kinetics of graphite formation are so slow that the Fe–Fe3C diagram is the practical one. In cast irons, with more carbon and silicon, graphite does form (Section 15.8).

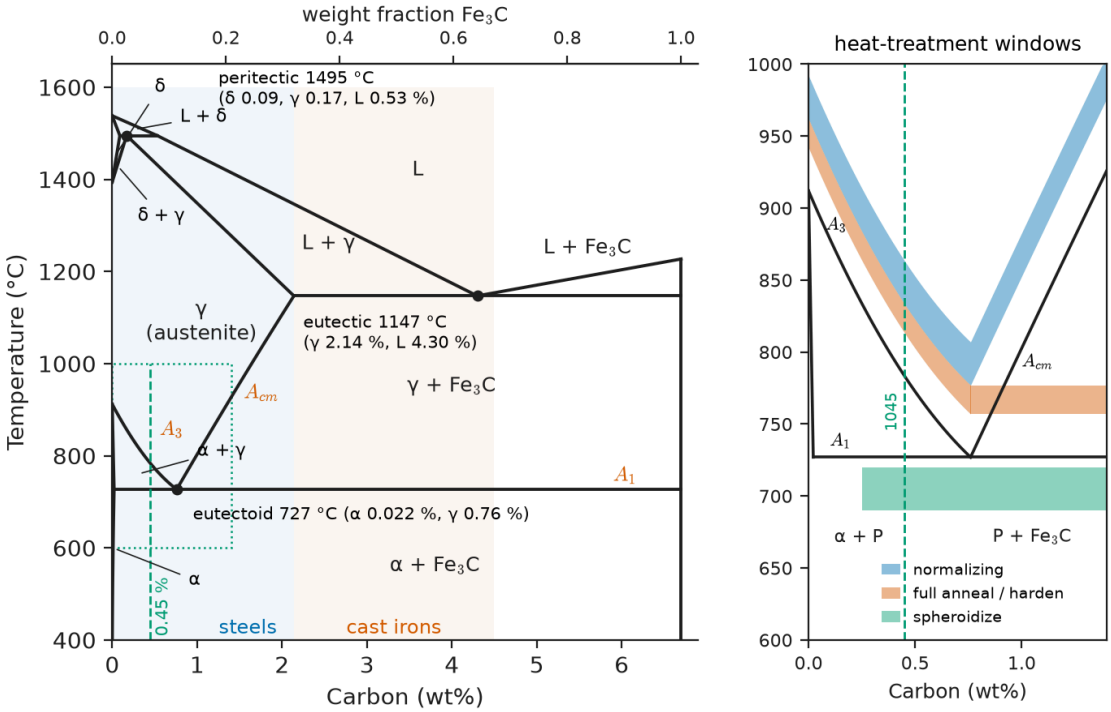

Figure 15.1 — The iron–cementite phase diagram up to 6.70 wt% C, with the steel and cast-iron ranges, critical temperatures A1, A3 and Acm, and heat-treatment temperature windows.

The Fe–C diagram for steels

Figure 15.1 shows the iron–cementite diagram, first met in Section 7.7. Its key features, with the conventional names of the critical lines:

Feature	Composition (wt% C)	Temperature (°C)	Reaction on cooling
Peritectic	0.17	1495	L + δ → γ
Eutectic	4.30	1147	L → γ (2.14 % C) + Fe3C (ledeburite)
Eutectoid	0.76	727	γ → α (0.022 % C) + Fe3C (pearlite)
A_1 line	—	727	Lower critical temperature
A_3 line	0 → 0.76	912 → 727	Start of ferrite formation on cooling
A_{cm} line	0.76 → 2.14	727 → 1147	Start of cementite formation on cooling

Steels contain up to 2.14 % C (in practice rarely above 1.5 %); cast irons contain more. Alloying elements move these lines: **austenite stabilizers** (Ni, Mn, C, N, Cu, Co) lower A_3 and expand the γ field; **ferrite stabilizers** (Cr, Si, Mo, W, V, Ti, Al, Nb) raise A_3 and contract it. With enough Ni or Mn, austenite is stable at room temperature (austenitic stainless and TWIP steels); with enough Cr or Si, the γ field closes entirely (ferritic stainless and electrical steels). In practice the critical temperatures also depend on heating or cooling rate, written Ac_1, Ac_3 on heating and Ar_1, Ar_3 on cooling.

15.2 Plain-carbon steels: microstructure versus carbon content

Eutectoid steel and pearlite

A steel with 0.76 % C, cooled slowly from austenite, transforms at 727 °C entirely to **pearlite**: alternating lamellae of ferrite and cementite that nucleate at austenite grain boundaries and grow as colonies by cooperative, diffusion-controlled partitioning of carbon (Section 10.5). The lever rule at just below 727 °C gives the cementite fraction in pearlite as $(0.76 - 0.022)/(6.70 - 0.022) = 0.11$; the lamellae are therefore about 1:7 in thickness. The interlamellar spacing S decreases as the transformation temperature falls (it scales roughly with the inverse of undercooling), from about 0.5 µm for slow furnace cooling ("coarse pearlite") to 0.1 µm or less for air cooling or isothermal transformation near 550–600 °C ("fine pearlite"). Yield strength follows a Hall–Petch-like relation in $S^{-1/2}$, and fully pearlitic steels reach 600–1200 MPa tensile strength. Cold-drawn fine pearlite — **patented and drawn wire** for bridge cables, pre-stressing strand and tyre cord — reaches 1800–2200 MPa (bridge wire) to over 3500 MPa (tyre cord), because drawing aligns the lamellae and thins them to nanometres.

Hypoeutectoid and hypereutectoid steels

A steel with less than 0.76 % C (**hypoeutectoid**) crossing A_3 on cooling first forms **proeutectoid ferrite**, usually as grain-boundary allotriomorphs and then polygonal grains; the remaining austenite enriches in carbon until it reaches 0.76 % at 727 °C and transforms to pearlite. A steel above 0.76 % C (**hypereutectoid**) first precipitates **proeutectoid cementite** on the austenite grain boundaries below A_{cm}, then the remainder becomes pearlite. A continuous grain-boundary cementite network is brittle, which is one reason hypereutectoid tool and bearing steels are given spheroidizing anneals.

Worked example — Phase fractions in a medium-carbon steel. A 1045-type steel (0.45 % C) is slowly cooled. Just above 727 °C, the lever rule between ferrite (0.022 % C) and austenite (0.76 % C) gives

$$f_{\alpha}^{\text{pro}} = \frac{0.76 - 0.45}{0.76 - 0.022} = 0.42, \qquad f_{\text{pearlite}} = 0.58$$

Just below 727 °C, the total phase fractions between ferrite and cementite are

$$f_{\text{Fe}_3\text{C}} = \frac{0.45 - 0.022}{6.70 - 0.022} = 0.064, \qquad f_{\alpha}^{\text{total}} = 0.936$$

The microstructure is 42 % proeutectoid ferrite and 58 % pearlite by weight, and although only 6.4 % of the steel is cementite, its arrangement as lamellae in more than half the volume roughly doubles the strength of a ferritic steel. For a 1.0 % C hypereutectoid steel, the proeutectoid cementite fraction is $(1.0 - 0.76)/(6.70 - 0.76) = 0.040$ — only 4 % by weight, but enough to form a continuous film around every prior austenite grain.

Properties versus carbon content

Table 15.1 — Plain-carbon steels in the normalized (air-cooled) condition (typical values).

Steel (AISI/SAE)	C (wt%)	Microstructure	σ_y (MPa)	UTS (MPa)	Elongation (%)	Hardness (HB)	Typical uses
1008 / DC04	0.08	Ferrite, trace pearlite	180–220	300–340	35–45	90–100	Car body panels, cans
1020	0.20	Ferrite + ~25 % pearlite	340–350	440–460	35	130	Structural sections, carburizing grade
1045	0.45	Ferrite + ~58 % pearlite	410–430	650–690	20–25	190–200	Shafts, axles, forgings
1080	0.80	Fully pearlitic	520–560	1000–1050	10–12	290	Rails, springs, wire rod
1095	0.95	Pearlite + grain-boundary Fe3C	500–550	1000–1020	9–10	290	Blades, springs, tools

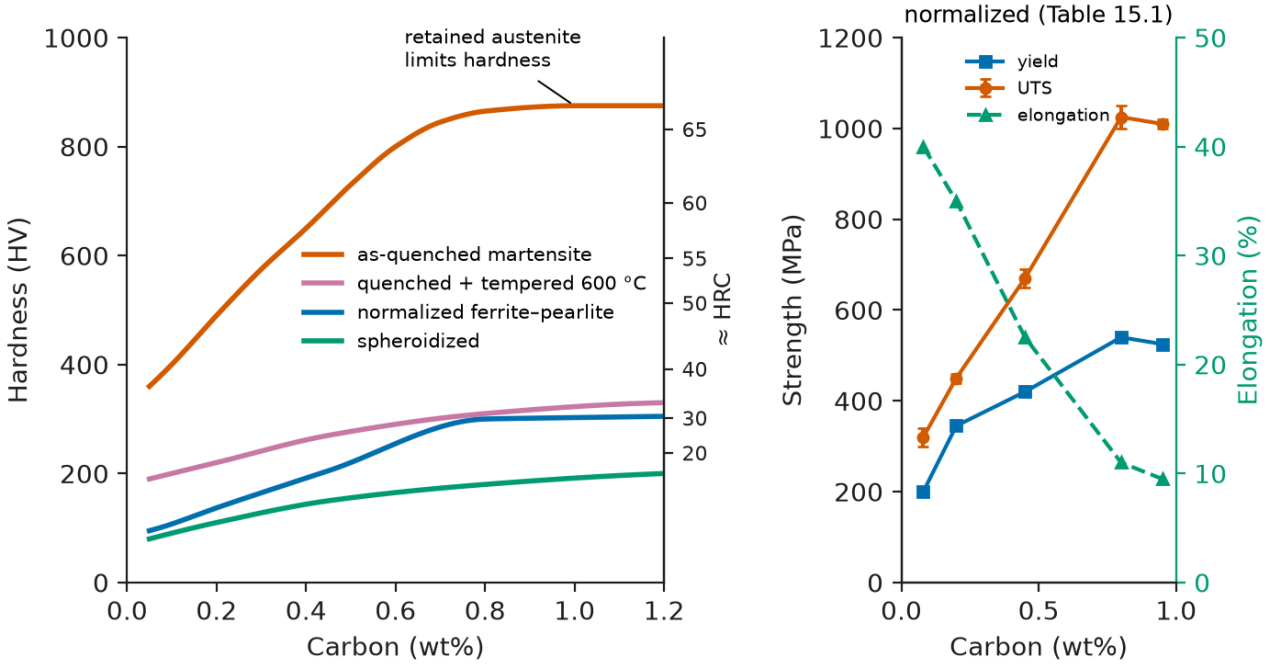

Figure 15.2 — Hardness and strength of plain-carbon steels versus carbon content for pearlitic (normalized), spheroidized, as-quenched martensitic and tempered conditions.

Figure 15.2 summarizes the trend: in the slowly cooled condition each 0.1 % C adds roughly 100 MPa of tensile strength and costs elongation and toughness, raising the Charpy transition by roughly 15–20 °C. Structural and pipeline steels have therefore been driven to ever lower carbon (Section 15.5), with strength restored by grain refinement and microalloying.

15.3 Heat treatment: annealing, normalizing, quenching, tempering

The menu of treatments

All conventional steel heat treatments start by heating into the austenite field (or, for some, just below A_1) and differ in how the steel is cooled.

- **Full annealing.** Heat to $Ac_3 + 30$–50 °C for hypoeutectoid steels (above Ac_1 only, for hypereutectoid), hold to homogenize, and furnace cool. Produces coarse pearlite and ferrite: soft, machinable, stress-free.
- **Process (subcritical) annealing.** Heat cold-worked low-carbon steel to 550–700 °C, below A_1, to recrystallize ferrite (Section 10.7) between cold-forming passes in wire drawing and sheet production.
- **Spheroidizing.** Hold just below A_1 (or cycle around it) for many hours so cementite lamellae break up and coarsen into spheroids, driven by interfacial energy (Section 10.3). The softest, most formable and machinable condition of medium- and high-carbon steels; standard before cold heading of bolts and machining of bearing rings.
- **Normalizing.** Heat to Ac_3 (or Ac_{cm}) +50–80 °C and cool in still air. Refines and homogenizes the grain structure of castings, forgings and hot-rolled plate, giving finer pearlite and finer ferrite grains than annealing, with higher strength and toughness.
- **Stress relieving.** Heat to 550–650 °C, below A_1, to reduce residual stresses from welding, machining or cold work without changing the microstructure much; mandatory for thick welded pressure vessels.
- **Hardening (quenching).** Austenitize and cool faster than the critical cooling rate to form **martensite**.
- **Tempering.** Reheat quenched martensite to 150–700 °C to restore toughness.

Martensite

When austenite is cooled so fast that carbon cannot diffuse out to form ferrite and cementite, it transforms below the **martensite-start temperature** M_s by a diffusionless shear (Section 10.6) into **martensite**, a body-centred tetragonal phase supersaturated with carbon. The tetragonality $c/a \approx 1 + 0.045\,\text{wt\% C}$ reflects carbon atoms locked in one set of octahedral sites.

The transformation is athermal: the amount formed depends on how far below M_s the steel is cooled, not on time, and is complete only below the **martensite-finish temperature** M_f, typically 200–300 °C below M_s. In low-carbon steels (below about 0.6 % C) martensite forms as **laths** in packets with a high dislocation density; in high-carbon steels it forms as lenticular **plates** containing fine twins, and M_f falls below room temperature so some **retained austenite** remains.

Martensite is hard because of interstitial solid-solution strengthening by carbon (the strongest solid-solution effect in metallurgy, Section 13.6), a dislocation density of 10^{15}–10^{16} m^{-2}, fine lath and block boundaries, and carbon segregation to dislocations. Its hardness depends almost entirely on carbon content (Figure 15.2): about 45–50 HRC at 0.2 % C, 56–58 HRC at 0.4 %, 64–65 HRC at 0.6 %, levelling at 65–66 HRC above about 0.7 %, where retained austenite offsets further gains. Alloying elements barely change martensite hardness; they change how easily it forms (Section 15.4).

M_s falls with nearly every alloying addition. A widely used empirical expression (Andrews, 1965), in wt%, is

$$M_s\ (^\circ\mathrm{C}) = 539 - 423\,\mathrm{C} - 30.4\,\mathrm{Mn} - 17.7\,\mathrm{Ni} - 12.1\,\mathrm{Cr} - 7.5\,\mathrm{Mo}$$

> **Status of the principle: Andrews equation for M_s**
> *Empirical.* A linear regression on low-alloy steels, accurate to about ±15–25 °C for C up to about 0.6 % and moderate alloy contents; it fails for high-alloy, high-carbon or high-nitrogen steels, and does not account for austenite grain size or prior deformation.

For 1045 steel (0.45 C, 0.75 Mn), $M_s = 539 - 190 - 23 = 326$ °C; for AISI 4340 (0.40 C, 0.70 Mn, 1.8 Ni, 0.80 Cr, 0.25 Mo), $M_s = 539 - 169.2 - 21.3 - 31.9 - 9.7 - 1.9 = 305$ °C; for 1080 (0.80 C, 0.75 Mn), 178 °C. With a low M_s, the volume expansion of up to about 4 % on forming martensite occurs in steel that is already cold and strong, so residual stresses and **quench cracks** become likely; high-carbon steels and parts with sharp section changes are oil- or polymer-quenched, or **martempered** (quenched into a salt bath just above M_s, equalized, then air cooled).

Tempering

As-quenched martensite is too brittle for most uses. **Tempering** reheats it so carbon can leave solution in controlled stages:

Stage	Temperature (°C)	Process	Effect
1	80–200	Carbon clusters; transition carbides (ε, η) precipitate	Slight hardness loss, large toughness gain; stress relief
2	200–300	Retained austenite decomposes to bainite-like ferrite + carbide	Dimensional change; can embrittle
3	250–400	Cementite replaces transition carbides, often as films on lath boundaries	Toughness can drop (tempered-martensite embrittlement, ~250–400 °C)
4	> 500	Cementite spheroidizes and coarsens; ferrite recovers and recrystallizes; alloy carbides (Mo_2C, VC, Cr_7C_3) precipitate in alloy steels	Large softening; in Mo, V, W steels, **secondary hardening** at 500–600 °C

Two embrittlement windows must be avoided. **Tempered-martensite embrittlement** (about 250–400 °C) comes from interlath cementite films. **Temper embrittlement** occurs when low-alloy steels are held at or slowly cooled through 375–575 °C: P, Sn, Sb and As segregate to prior austenite grain boundaries and can raise the DBTT by 100 °C or more (Section 14.7) — a long-term hazard in thick Cr–Mo reactors and turbine rotors, controlled by limiting impurities and adding Mo.

Because tempering is thermally activated, time and temperature trade against each other. The **Hollomon–Jaffe tempering parameter**

$$P = T\,(C + \log_{10} t)$$

with T in kelvin, t in hours and $C \approx 21.3 - 5.8 \times \mathrm{wt\%\,C}$ (about 19 for a 0.4 % C steel), collapses hardness data from different tempering schedules onto one curve — the same idea as the Larson–Miller parameter of Section 14.2.

> **Status of the principle: Hollomon–Jaffe tempering parameter**
> *Empirical.* Assumes a single activation energy for softening; good for plain-carbon and low-alloy steels over moderate time ranges, less reliable where secondary hardening or several precipitation reactions overlap.

Worked example — Equivalent tempering schedules. A 0.4 % C Cr–Mo shaft is specified as "temper 1 h at 600 °C". The heat-treatment shop wants to know the equivalent time at 550 °C (to reduce distortion) and at 650 °C (to shorten the cycle). With $C = 19$:

$P = 873.15 \times (19 + \log_{10} 1) = 16\,590$.

At 550 °C (823.15 K): $\log_{10} t = 16\,590/823.15 - 19 = 1.154$, so $t = 14.3$ h.

At 650 °C (923.15 K): $\log_{10} t = 16\,590/923.15 - 19 = -1.029$, so $t = 0.094$ h $= 5.6$ min.

A 50 °C change is worth more than an order of magnitude in time. But 5.6 min is shorter than a large shaft takes to heat through, and a 14 h hold at 550 °C sits in the temper-embrittlement window, so 600 °C/1 h is the sensible compromise.

Quenched-and-tempered properties

Table 15.2 — Mechanical properties of quenched-and-tempered steels (oil quenched, small sections, typical).

Steel	Tempering temperature (°C)	σ_y (MPa)	UTS (MPa)	Elongation (%)	Hardness	Uses
1045 (water quenched)	540	520–620	750–830	18–22	220–240 HB	Shafts, pins
4140	205	1640	1770	8	~50 HRC	Tool holders
4140	540	900–1000	1000–1100	15–18	~32 HRC	Crankshafts, gears, bolts (grade 10.9)
4340	205	1590–1680	1860–1980	10–11	~53 HRC	Landing gear (often 300M variant)
4340	650	860–900	960–1000	19–20	~28 HRC	Heavy shafts, pressure-vessel bolting
52100 bearing steel	175	~1900	2200+	<5	60–64 HRC	Rolling bearings

15.4 Hardenability: the Jominy test, alloying effects and CCT diagrams

Hardness versus hardenability

Hardness — the maximum martensite can reach — depends on carbon. **Hardenability** — how deep martensite forms in a quench — depends on how slowly austenite transforms to ferrite, pearlite and bainite, which alloying and austenite grain size control. The centre of a thick part cools only as fast as conduction allows (cooling time scales with the square of section size, Chapter 24); if that is slower than the steel's critical cooling rate, the centre forms pearlite or bainite and stays soft. This is what happened to the second batch of gears.

Continuous-cooling transformation diagrams

Isothermal TTT diagrams (Section 10.9) describe transformation at constant temperature; real quenches are continuous. **Continuous-cooling transformation** (CCT) diagrams plot the start and finish of each transformation along cooling curves of different rates (Figure 15.3). Compared with the TTT diagram for the same steel, the CCT curves are displaced to longer times and lower temperatures. The **critical cooling rate** is the slowest rate that misses the ferrite, pearlite and bainite noses entirely and forms only martensite.

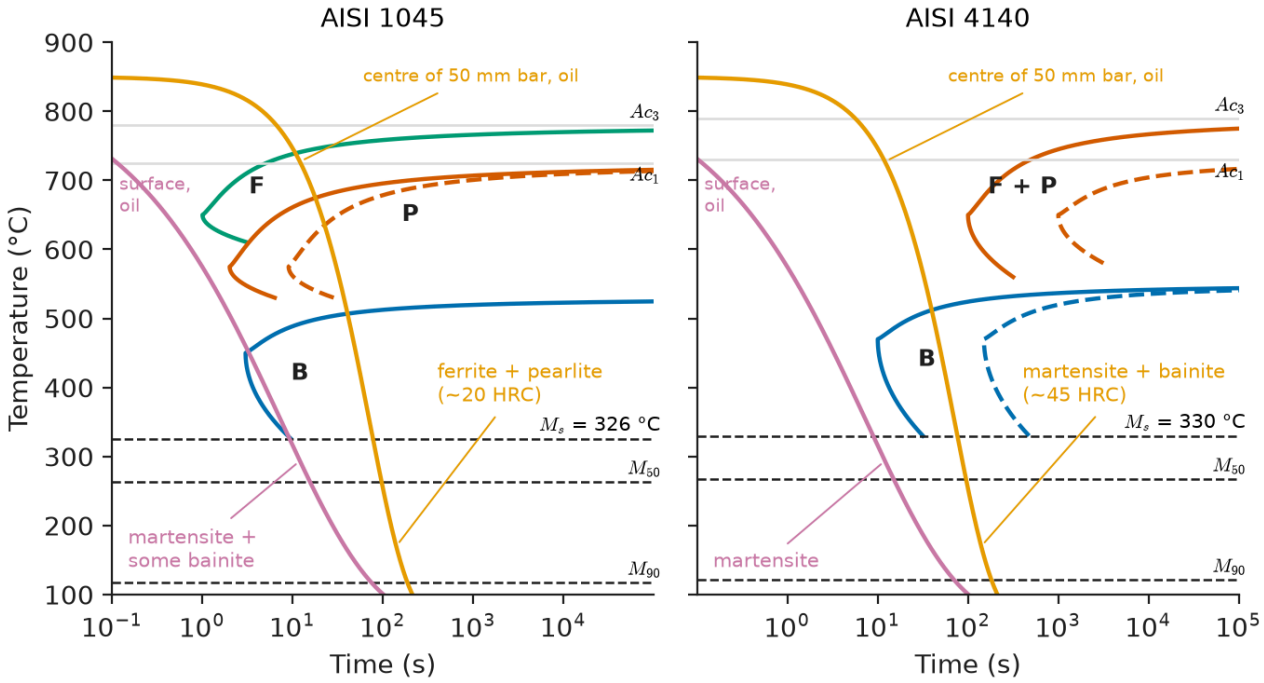

Figure 15.3 — Continuous-cooling transformation diagrams for a plain-carbon 1045 steel and a Cr–Mo 4140 steel, with cooling curves for the surface and centre of a 50 mm bar quenched in oil.

The Jominy end-quench test

The **Jominy test** (ASTM A255, ISO 642) austenitizes a bar 25 mm in diameter and 100 mm long and quenches one end with a standard water jet, so the cooling rate falls from several hundred °C/s at the end to about 3 °C/s at 50 mm. Hardness measured along the bar gives the **Jominy curve** (Figure 15.4). Since each Jominy distance corresponds to a known cooling rate, and the surface, mid-radius and centre of quenched bars are tabulated against equivalent Jominy distances, the curve predicts the hardness profile of a real component.

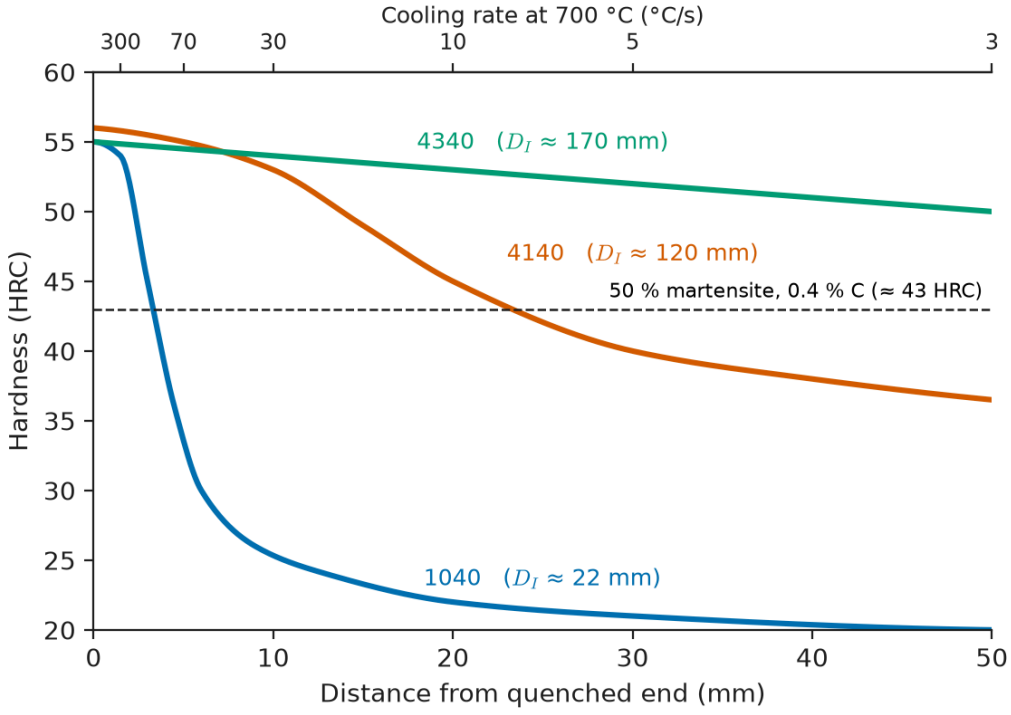

Figure 15.4 — Jominy end-quench hardenability curves for 1040, 4140 and 4340 steels, with the cooling rate at each distance.

Alloying effects

Almost every element dissolved in austenite slows the diffusional transformations. Mn, Cr and Mo are the most effective per unit cost; Ni is weaker but adds toughness; and 10–30 ppm of boron, segregated to austenite grain boundaries, suppresses ferrite nucleation and multiplies the hardenability of low-carbon steels two- to threefold — the basis of press-hardening steels (Section 15.7) and cheap high-strength fasteners. Mo and Cr shift the pearlite nose more than the bainite nose, producing the separate "bainite bay" of Figure 15.3.

The **ideal critical diameter** D_I — the largest bar that would harden to 50 % martensite at its centre under an infinitely severe quench — is estimated from composition by Grossmann's multiplying factors:

$$D_I = D_{I,\text{base}}(\text{C},\ \text{grain size}) \times f_{\text{Mn}} \times f_{\text{Si}} \times f_{\text{Cr}} \times f_{\text{Ni}} \times f_{\text{Mo}} \times \cdots$$

with approximate factors (ASTM A255, for moderate contents) $f_{\text{Mn}} = 1 + 3.33\,\text{Mn}$, $f_{\text{Si}} = 1 + 0.7\,\text{Si}$, $f_{\text{Cr}} = 1 + 2.16\,\text{Cr}$, $f_{\text{Ni}} = 1 + 0.363\,\text{Ni}$, $f_{\text{Mo}} = 1 + 3.0\,\text{Mo}$ (wt%).

> **Status of the principle: Grossmann multiplying factors**
> *Empirical.* The multiplicative form assumes independent, non-interacting effects of each element; it is useful for ranking and specification within the composition ranges of the data (low-alloy steels), with typical errors of 10–20 % in D_I, and is superseded by computed CCT diagrams and Jominy curves for alloy design.

Worked example — Why the 4140 gears hardened and the 1040 gears did not. Take a base D_I of 0.213 in (5.4 mm) for 0.40 % C at ASTM grain size 7.

1040 (0.75 Mn, 0.25 Si): $D_I = 0.213 \times (1 + 3.33 \times 0.75) \times (1 + 0.7 \times 0.25) = 0.213 \times 3.50 \times 1.175 = 0.88$ in, about 22 mm.

4140 (0.88 Mn, 0.25 Si, 0.95 Cr, 0.20 Mo): $D_I = 0.213 \times 3.93 \times 1.175 \times 3.05 \times 1.60 = 4.8$ in, about 122 mm.

4340 (0.70 Mn, 0.25 Si, 0.80 Cr, 1.80 Ni, 0.25 Mo): $D_I = 0.213 \times 3.33 \times 1.175 \times 2.73 \times 1.65 \times 1.75 = 6.6$ in, about 167 mm.

Real quenches are not infinitely severe: a well-agitated oil quench hardens a bar about one-third to one-half of D_I through. A 40 mm gear blank in 1040 therefore cannot form martensite at its core in oil and barely does so even in water, while 4140 hardens it comfortably, and 4340 is used for landing gear and large crankshafts whose sections reach 100 mm or more. The equal carbon contents give equal surface hardness after quenching; the alloy content is paid for in depth. Weldability goes the other way: the IIW carbon equivalent $\text{CE} = \text{C} + \text{Mn}/6 + (\text{Cr} + \text{Mo} + \text{V})/5 + (\text{Ni} + \text{Cu})/15$ is 0.78 for 4140 against 0.41 for a C–Mn structural steel, and steels above about 0.45 need preheat to avoid hydrogen cracking in the hardened HAZ (Chapter 47).

15.5 Alloy steels: HSLA, tool steels, maraging steels

High-strength low-alloy (HSLA) steels

For structural plate, line pipe, ship hulls, offshore platforms, cranes and truck chassis, the requirements are yield strengths of 350–700 MPa, good toughness at low temperature and easy field welding. Raising carbon is excluded by weldability and toughness (Section 15.2). The solution, developed from the 1960s, is **microalloying** with 0.01–0.10 % of Nb, V or Ti combined with **thermomechanically controlled processing** (TMCP). Slabs are reheated to dissolve the carbonitrides; rolled at high temperature, where austenite recrystallizes and TiN pins its grains; finished below the recrystallization-stop temperature (raised by Nb), giving "pancaked" austenite full of ferrite nucleation sites; and cooled rapidly to very fine ferrite (3–6 μm) or acicular ferrite/bainite, while V and Nb precipitate as nanometre carbonitrides. As the X70 example of Section 13.10 showed, most of the strength comes from grain refinement, with carbon at only 0.04–0.10 %. Pipeline grades now reach X80 (555 MPa minimum yield) and X100–X120 in trials; bridge and offshore steels reach 460–690 MPa with Charpy guarantees at −40 °C or lower. **Weathering steels** (Cu, Cr, Ni, P additions) form a protective rust layer and are used unpainted for bridges.

Tool steels

Tool steels must cut, form or shape other materials. They combine high carbon (0.5–2.3 %) for hard martensite with carbide-forming elements (Cr, W, Mo, V) that form hard primary and secondary carbides and that resist softening when the tool gets hot.

Table 15.3 — Tool steel classes (AISI designation, typical).

Class	Example	Key composition (wt%)	Hardness (HRC)	Characteristic	Uses
W — water hardening	W1	0.6–1.4 C	58–65	Cheap, shallow hardening	Simple punches, hand tools
O — oil hardening	O1	0.9 C, 1.2 Mn, 0.5 Cr, 0.5 W	57–62	Low distortion	Gauges, short-run dies
A — air hardening	A2	1.0 C, 5 Cr, 1 Mo	57–62	Deep hardening, stable	Blanking dies, forming rolls
D — high C, high Cr	D2	1.5 C, 12 Cr, 1 Mo, 1 V	58–62	Abrasion resistant (Cr_7C_3)	Long-run stamping and slitting tools
S — shock resisting	S7	0.5 C, 3.25 Cr, 1.4 Mo	54–58	Tough	Chisels, punches under impact
H — hot work	H13	0.4 C, 5 Cr, 1.3 Mo, 1 V	44–52	Resists thermal fatigue and softening to ~600 °C	Die-casting dies, forging dies, extrusion tooling
M — Mo high-speed	M2	0.85 C, 4 Cr, 6 W, 5 Mo, 2 V	62–65	Secondary hardening; red hardness	Drills, taps, end mills

High-speed and hot-work steels rely on **secondary hardening**: austenitized near 1200 °C (M2) to dissolve alloy carbides, quenched, and tempered two or three times at 540–580 °C, when fine Mo_2C, W_2C and VC precipitate and *raise* hardness above the as-quenched value. These carbides resist coarsening at the 500–600 °C of a cutting edge — the "red hardness" that let Taylor and White quadruple machining speeds around 1900. Powder-metallurgy tool steels (Chapter 46) avoid coarse eutectic carbides and are much tougher.

Maraging steels

Maraging steels ("martensite + ageing") take a different route to ultra-high strength. They contain almost no carbon (below 0.03 %), 18 % Ni, 8–12 % Co, 3–5 % Mo and 0.2–1.6 % Ti. On cooling from about 820 °C, even slowly, the high-Ni austenite transforms to a soft, tough, low-carbon **lath martensite** (about 30 HRC, 800 MPa yield) that can be machined. Ageing at 480–500 °C for 3–6 h then precipitates a dense dispersion of nanometre intermetallics (Ni_3Mo, Ni_3Ti, Fe_2Mo) in the dislocated martensite, raising the yield strength to 1400 MPa (grade 200), 1700 MPa (grade 250), 2000 MPa (grade 300) or 2400 MPa (grade 350). Cobalt lowers Mo solubility and promotes finer precipitation. Ageing involves no quench and a dimensional change of only about 0.05 %, so parts are finish-machined first. Maraging steels serve in rocket-motor cases, arrestor hooks, high-performance shafts and additively manufactured mould inserts (Chapter 48), wherever 2 GPa with good toughness justifies the Ni and Co cost.

15.6 Stainless steels

Passivity from chromium

Iron rusts because its oxide film is porous and non-protective. When chromium in solid solution exceeds about 10.5–12 wt%, the surface forms a continuous, self-healing **passive film** of chromium-rich oxide only 1–3 nm thick, which reduces the corrosion rate in many environments by a factor of 1000 or more (Chapter 42). Steels with at least 10.5 % Cr are **stainless**. Corrosion resistance rises with Cr, and in chloride environments with Mo and N; the **pitting-resistance equivalent number**

$$\mathrm{PREN} = \mathrm{Cr} + 3.3\,\mathrm{Mo} + 16\,\mathrm{N}$$

ranks alloys for seawater and chloride service (304: about 18–19; 316: about 24; duplex 2205: about 35; super-duplex 2507: about 42; PREN above 40 is generally required for seawater at ambient temperature).

> **Status of the principle: Pitting-resistance equivalent number**
> *Empirical.* A linear ranking fitted to critical pitting temperatures; it ignores microstructure (ferrite–austenite partitioning in duplex grades, inclusions, sigma phase) and surface condition, so it ranks alloys but does not predict pitting.

The four families

Because Cr is a ferrite stabilizer and Ni, Mn, C and N are austenite stabilizers, the balance between them selects the microstructure:

- **Ferritic** (409, 430, 444): 11–30 % Cr, little Ni, low C; BCC, magnetic, not hardenable. Cheap and immune to chloride stress-corrosion cracking, but with a DBTT and limited weldability. Exhausts (409), appliances (430), hot-water tanks (444).
- **Austenitic** (304, 316, 321, 310): 16–26 % Cr, 8–22 % Ni (Mn and N in the 200 series); FCC, non-magnetic, no DBTT (used to −269 °C), highly formable and weldable, strongly work-hardening. More than half of all stainless: food and pharmaceutical plant, sinks, cryogenic vessels, 316L implants and instruments. Weaknesses: chloride stress-corrosion cracking above about 60 °C, and **sensitization**, in which $Cr_{23}C_6$ precipitates on grain boundaries at 500–800 °C (weld HAZs) and depletes adjacent Cr below the passivity limit; low carbon (304L, 316L, ≤0.03 % C) or Ti/Nb stabilization (321, 347) prevents it.
- **Martensitic** (410, 420, 440C): 12–18 % Cr with 0.1–1.2 % C, quench-hardenable to 50–60 HRC, with modest corrosion resistance. Cutlery, surgical blades, steam-turbine blades, bearings (440C). The **precipitation-hardening** 17-4PH (17 Cr, 4 Ni, 4 Cu) adds Cu-rich precipitates to low-carbon martensite (1000–1300 MPa) for shafts and aerospace fittings.
- **Duplex** (2205, 2507): 22–25 % Cr, 5–7 % Ni, 3–4 % Mo, 0.15–0.3 % N, about half ferrite and half austenite. Twice the yield strength of 316, better stress-corrosion resistance, high PREN; offshore piping, desalination, chemical tankers. Sigma phase (600–950 °C) and "475 °C embrittlement" limit service to below about 250–300 °C and require controlled welding heat input.

The Schaeffler diagram

The phases present in stainless steel welds, which solidify and cool too fast for equilibrium, are predicted from composition by the **Schaeffler diagram** (Figure 15.5). Each element's effect is converted into an equivalent amount of chromium (ferrite stabilizers) or nickel (austenite stabilizers):

$$\mathrm{Cr_{eq}} = \mathrm{Cr} + \mathrm{Mo} + 1.5\,\mathrm{Si} + 0.5\,\mathrm{Nb}, \qquad \mathrm{Ni_{eq}} = \mathrm{Ni} + 30\,\mathrm{C} + 0.5\,\mathrm{Mn}$$

The diagram then shows fields of austenite (A), ferrite (F), martensite (M) and their mixtures, with lines of constant ferrite percentage in the A + F field. Later refinements — the DeLong diagram (adding 30 N to $\mathrm{Ni_{eq}}$) and the WRC-1992 diagram (ferrite number, Cu term) — are more accurate for modern nitrogen-bearing grades.

> **Status of the principle: Schaeffler constitution diagram**
> *Empirical.* Built from measured weld-metal microstructures at typical arc-welding cooling rates; it predicts ferrite content to about ±4 % in the austenitic range but is not an equilibrium diagram and fails for very fast (laser, electron beam) or very slow cooling and for high-nitrogen alloys.

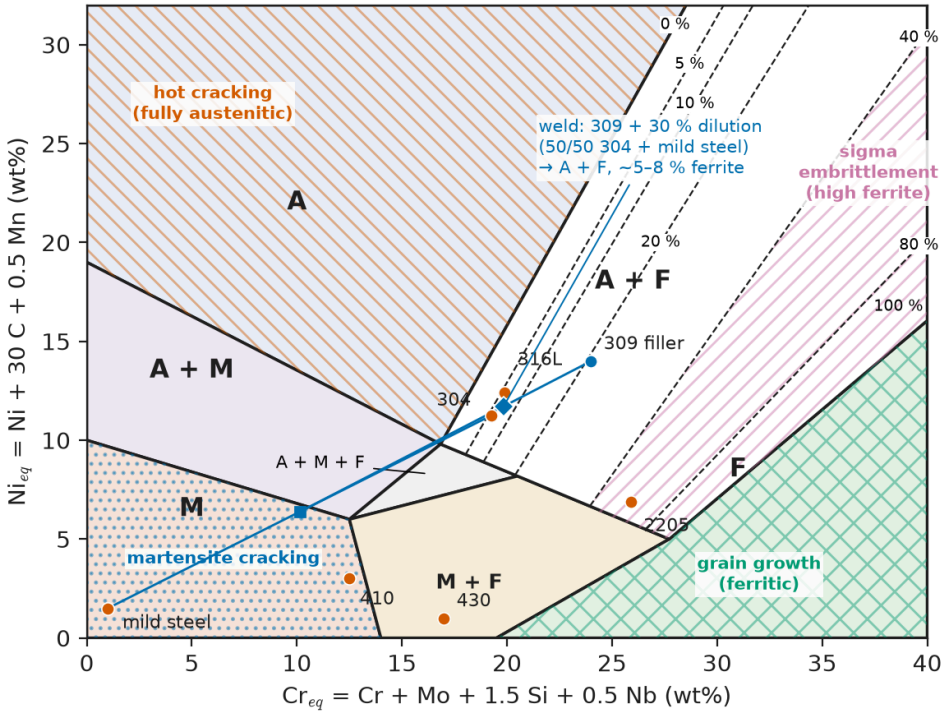

Figure 15.5 — Schaeffler diagram for stainless steel weld metal: phase fields of austenite, ferrite and martensite as functions of chromium and nickel equivalents, with common grades plotted.

Worked example — Ferrite in a 304 weld, and why 309 joins stainless to carbon steel. A 304 weld metal with 18.5 Cr, 9.0 Ni, 0.05 C, 1.5 Mn and 0.5 Si (wt%) has

$$\mathrm{Cr_{eq}} = 18.5 + 1.5 \times 0.5 = 19.25, \qquad \mathrm{Ni_{eq}} = 9.0 + 30 \times 0.05 + 0.5 \times 1.5 = 11.25$$

This point lies in the A + F field with about 3–8 % ferrite. That small amount of δ-ferrite is wanted: it solidifies first, dissolves sulfur and phosphorus that would otherwise segregate to austenite grain boundaries, and prevents **solidification (hot) cracking**. Fully austenitic welds (Ni-rich, as in 310 or high-Ni alloys) are hot-crack sensitive; welds with too much ferrite embrittle by sigma formation at temperature.

Joining 304 to carbon steel with a 304 filler would dilute the weld towards the low-alloy corner, crossing into the martensite field and causing cracking. A more highly alloyed **309** filler (about 23 Cr, 13 Ni; $\mathrm{Cr_{eq}} \approx 24$, $\mathrm{Ni_{eq}} \approx 14$) is chosen so that a typical 30 % dilution by the two base metals (304 plus carbon steel) still lands the weld in A + F, at about 5–8 % ferrite. Selection of dissimilar-metal welding consumables is one of the most frequent everyday uses of the diagram in chemical, power and food-processing plants.

Table 15.4 — Representative stainless steels (annealed, typical).

Grade (UNS)	Family	Composition (wt%)	σ_y (MPa)	UTS (MPa)	PREN	Typical use
409 (S40900)	Ferritic	11 Cr, Ti	210–240	380–450	11	Exhaust systems
430 (S43000)	Ferritic	17 Cr	280–310	450–520	17	Appliances, trim
304L (S30403)	Austenitic	18 Cr, 9 Ni	210–240	520–600	18–19	Food, dairy, tanks
316L (S31603)	Austenitic	17 Cr, 11 Ni, 2.1 Mo	220–260	520–600	24	Pharma, marine fittings, implants
410 (S41000), Q&T	Martensitic	12.5 Cr, 0.12 C	600–1000	800–1300	12	Steam-turbine blades, valves
17-4PH (S17400), H900	PH martensitic	17 Cr, 4 Ni, 4 Cu	1170–1270	1310–1380	17	Shafts, aerospace fittings
2205 (S32205)	Duplex	22 Cr, 5.5 Ni, 3.1 Mo, 0.17 N	450–550	650–800	35	Offshore, chemical tankers
2507 (S32750)	Super-duplex	25 Cr, 7 Ni, 4 Mo, 0.27 N	550–650	800–950	42–43	Seawater systems

15.7 Advanced high-strength steels: dual-phase, TRIP, TWIP

The automotive problem

A car body must be light (fuel economy, electric range), stiff, crashworthy and press-formable at a few seconds per part. Conventional strengthening trades ductility for strength along a "banana curve" (Figure 15.6): mild steel at 300 MPa UTS and 40 % elongation, HSLA at 500 MPa and 25 %, martensitic grades at 1500 MPa and 5 %. **Advanced high-strength steels** (AHSS) use multiphase microstructures or deformation-induced transformations to move above that curve.

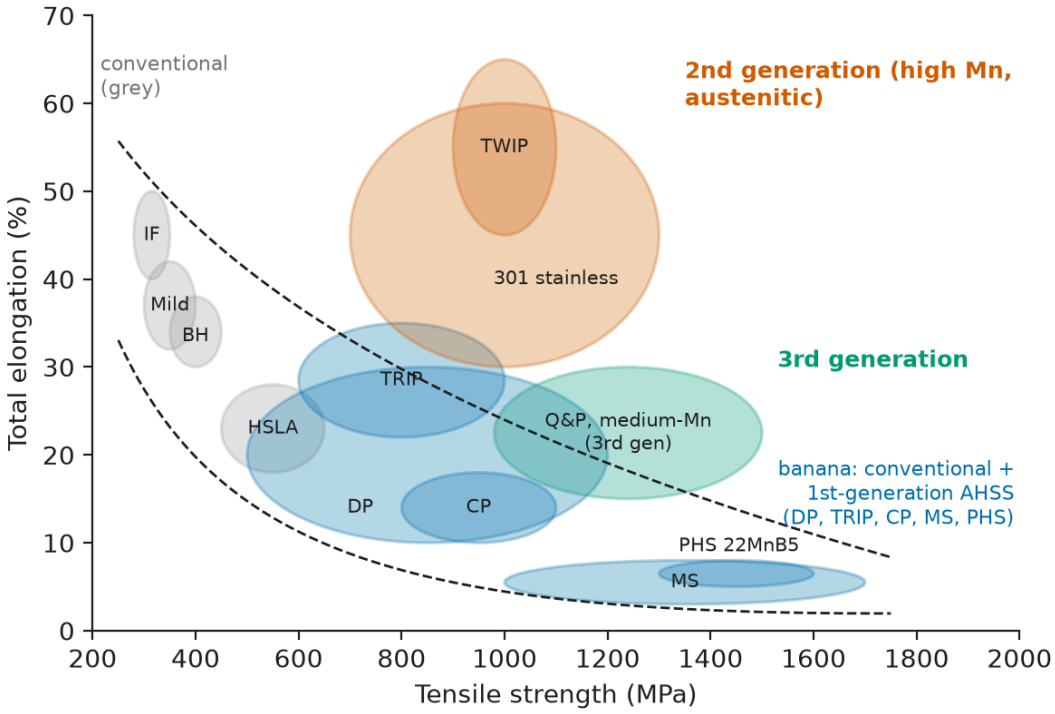

Figure 15.6 — Total elongation versus tensile strength for automotive sheet steels: conventional grades, first-generation AHSS, third-generation AHSS and TWIP/austenitic grades.

Dual-phase steels

Dual-phase (DP) steels are soft ferrite with 10–40 % hard martensite islands, made by intercritical annealing (between Ac_1 and Ac_3, where carbon-enriched austenite coexists with ferrite) and quenching. The martensite's expansion leaves mobile dislocations in the ferrite, so DP steels show no yield-point elongation, a low yield-to-tensile ratio (0.5–0.65), high early work hardening and strong bake hardening. DP600 and DP800 are the workhorses of body reinforcements, pillars and rails.

TRIP steels

Transformation-induced plasticity (TRIP) steels contain 5–20 % of **retained austenite**, stabilized at room temperature by carbon enrichment to about 1–1.5 %. They are processed by intercritical annealing followed by an isothermal hold at 350–450 °C, where bainite forms and rejects carbon into the remaining austenite; 1.5 % Si or Al suppresses cementite precipitation so the carbon stays in the austenite. During forming or a crash the retained austenite transforms progressively to martensite where strain concentrates, raising local hardening there and delaying necking. TRIP steels reach UTS of 600–1000 MPa with 25–35 % elongation. The **third-generation** quenching-and-partitioning (Q&P) and medium-Mn (5–10 % Mn) steels apply the same idea at 1000–1500 MPa. Austenitic stainless steel 301 is a natural TRIP alloy, used for rail-car bodies and springs.

TWIP steels

Twinning-induced plasticity (TWIP) steels are fully austenitic, with 15–30 % Mn, 0.4–0.8 % C and sometimes Al, adjusted so the stacking-fault energy is 20–40 mJ/m² — low enough for mechanical twinning, too high for martensite (Section 13.9). Twins subdivide the grains during straining and shorten the dislocation mean free path, so the hardening rate stays high to large strains: UTS about 1000 MPa with 50–65 % elongation, the best strength–ductility product of any steel sheet. Alloy cost, difficult welding and coating, and delayed hydrogen cracking have kept them to niche parts.

Press-hardened steels

The highest-strength body parts — A- and B-pillars, roof rails, door intrusion beams — are made from **press-hardening** (hot-stamping) steels, typically 22MnB5 (0.22 C, 1.2 Mn, 0.003 B). Blanks are austenitized at about 900–950 °C, transferred to a water-cooled die, formed while hot and soft, and quenched in the die to fully martensitic parts of about 1500 MPa UTS (or 1900–2000 MPa in newer 34MnB5-type grades) with no springback. Boron provides the hardenability (Section 15.4), and an Al–Si coating prevents scaling during heating. Tailored soft zones absorb crash energy next to hard anti-intrusion zones.

Table 15.5 — Automotive sheet steels (typical).

Grade	Microstructure	σ_y (MPa)	UTS (MPa)	Total elongation (%)	Typical parts
IF / DC06	Ferrite, Ti/Nb-stabilized	120–180	270–330	40–50	Outer panels, deep-drawn parts
HSLA 340	Fine ferrite + precipitates	340–420	410–510	21–26	Chassis, cross members
DP600	Ferrite + ~20 % martensite	340–420	600–700	20–26	Rails, reinforcements
DP980	Ferrite + ~50 % martensite	550–730	980–1100	10–14	Seat frames, pillars
TRIP780	Ferrite + bainite + 10–15 % retained γ	440–550	780–900	23–28	Crash members
TWIP980	Austenite, twinning	550–650	980–1100	50–60	Energy-absorbing parts
22MnB5, press hardened	Martensite	1000–1200	1400–1600	5–8	B-pillars, bumper beams

15.8 Cast irons: grey, ductile, white, malleable

Why cast irons are different

Cast irons contain 2.5–4.0 % C and 1–3 % Si, near the eutectic, so they melt at 1150–1250 °C (some 300 °C below steel), flow well, shrink little and are the cheapest metal to cast into shape: engine blocks, brake discs, pump bodies, machine beds, pipe, wind-turbine hubs and crankshafts.

Silicon is the key. It promotes the stable Fe–graphite eutectic over the metastable Fe–Fe_3C eutectic, raising the graphite eutectic temperature relative to the cementite one. Whether an iron solidifies **grey** (graphite) or **white** (cementite) then depends on composition and cooling rate: high C and Si and slow cooling favour graphite; thin sections, fast cooling, and carbide stabilizers (Cr, V, Mo, Mn in excess, S, Te) favour cementite. The effect of composition is summarized by the **carbon equivalent**

$$\mathrm{CE} = \mathrm{C} + \frac{\mathrm{Si} + \mathrm{P}}{3}$$

with the eutectic at CE ≈ 4.3. A typical grey iron with 3.4 % C, 2.2 % Si and 0.1 % P has $\mathrm{CE} = 3.4 + 2.3/3 = 4.17$, slightly hypoeutectic; a ductile iron with 3.6 % C, 2.5 % Si and 0.05 % P has $\mathrm{CE} = 4.45$, slightly hypereutectic.

> **Status of the principle: Carbon equivalent of cast iron**
> *Empirical.* Expresses the shift of the eutectic composition by Si and P as a linear correction; useful for predicting eutectic behaviour, fluidity and strength trends within normal composition ranges, but the actual structure also depends on inoculation, section size and trace elements.

The **matrix** around the graphite or carbide is determined by the subsequent solid-state transformation, just as in steel: ferritic, pearlitic, or (after heat treatment) martensitic, bainitic or ausferritic. A cast iron is therefore best thought of as a steel matrix containing graphite particles whose shape decides the properties.

Grey iron

In **grey** iron the graphite forms interconnected **flakes** that act as internal notches: tensile strength is 150–400 MPa with under 1 % elongation, compressive strength three to four times higher. The flakes also give excellent **damping** (up to ten times that of steel) for machine bases and engine blocks, high **thermal conductivity** (45–55 W/m K) for brake discs, free machining, and near-zero solidification shrinkage, as graphite expansion offsets liquid contraction. Grades are specified by tensile strength (ASTM A48 class 20–60 in ksi; EN-GJL-150 to -350 in MPa). **Inoculation** with ferrosilicon containing Ca, Al or Sr just before pouring provides nuclei for graphite and prevents chill (white) edges in thin sections (Section 9.3).

Ductile (spheroidal-graphite, nodular) iron

Adding 0.03–0.06 % residual magnesium (or Ce) to a low-sulfur iron makes graphite grow as **spheroids** instead of flakes. Without flake tips the iron is strong and ductile: 400–900 MPa with 18 % (ferritic) down to 2 % (pearlitic) elongation. **Ductile iron**, introduced in 1948, now makes about a third of iron castings: crankshafts, steering knuckles, wind-turbine hubs, water mains and spent-fuel casks. **Austempered ductile iron** (ADI), held at 250–400 °C after austenitizing, forms **ausferrite** (acicular ferrite plus carbon-stabilized austenite) — the same carbon-partitioning principle as TRIP steel — reaching 900–1600 MPa UTS with useful ductility, and replaces forged steel in gears and truck suspension parts at lower weight and cost. **Compacted graphite iron** (CGI), with vermicular graphite intermediate between flakes and spheroids, combines higher strength than grey iron with acceptable conductivity and damping, and is used for high-pressure diesel engine blocks and heads.

White and malleable iron

In **white** iron all carbon is in carbides: the fracture is white, hardness 400–700 HB, ductility nil. It serves where abrasion matters and impact does not, and **high-chromium white irons** (15–30 % Cr, M_7C_3 carbides in martensite) are the standard for mill liners and slurry pumps. A long anneal at 900–950 °C turns white iron into **malleable** iron, decomposing cementite into clumps of **temper carbon** in a ferritic or pearlitic matrix with 5–18 % elongation. Ductile iron, which needs no long anneal, has largely displaced it.

Table 15.6 — Cast irons (typical properties).

Type	Graphite form	Matrix	UTS (MPa)	Elongation (%)	Hardness (HB)	Uses
Grey EN-GJL-250	Flakes	Pearlite	250–350	< 1	180–250	Engine blocks, brake discs, machine bases
Ductile EN-GJS-400-18	Spheroids	Ferrite	400–450	18–22	130–175	Pipe, wind-turbine hubs, safety parts
Ductile EN-GJS-700-2	Spheroids	Pearlite	700–800	2–3	225–305	Crankshafts, gears
ADI (EN-GJS-1050-6)	Spheroids	Ausferrite	1050–1150	6–8	320–380	Gears, suspension parts
CGI (EN-GJV-450)	Vermicular	Pearlite	450–500	1–2	200–250	Diesel blocks and heads
Malleable (blackheart)	Temper carbon	Ferrite	300–350	6–12	110–150	Pipe fittings, hardware
High-Cr white iron	None (M_7C_3 carbides)	Martensite	—	~0	550–700	Mill liners, slurry pumps

15.9 Surface hardening: carburizing, nitriding, induction

Why harden only the surface

Gears, camshafts, bearing races and crankshaft journals need a hard surface that carries Hertzian contact stresses without pitting, over a tough core that survives impact and bending. Surface hardening produces a hard **case** on a tough **core**, and usually leaves **compressive residual stresses** in the case — from martensite or nitride expansion constrained by the core — that raise bending- and contact-fatigue strength (Section 14.8).

Carburizing

In **carburizing**, a low-carbon steel (0.1–0.25 % C, often alloyed: 8620, 16MnCr5, 18CrNiMo7-6) is heated in austenite, typically at 900–950 °C, in a carbon-rich atmosphere (endothermic gas, or low-pressure acetylene in vacuum furnaces) that sets the surface carbon at about 0.8–1.0 %. Carbon diffuses inward (Section 8.10), and the part is then quenched so the high-carbon case forms hard martensite (58–62 HRC) while the low-carbon core forms tough low-carbon martensite or bainite (30–45 HRC, depending on hardenability), followed by a low-temperature temper at 150–200 °C. **Carbonitriding** adds ammonia, introducing nitrogen that raises case hardenability, at slightly lower temperature, for thinner cases on cheaper steels.

The case depth follows from the error-function solution of Fick's second law for a constant surface concentration:

$$\frac{C(x,t)-C_0}{C_s-C_0} = \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Worked example — Carburizing time for a gear. A 0.20 % C steel gear is carburized at 925 °C with a surface carbon of 1.0 %. The specification calls for 0.45 % C at a depth of 1.0 mm. Take $D = 2.3\times10^{-5}\exp(-148\,000/RT)$ m²/s for carbon in austenite.

At 1198 K: $D = 2.3\times10^{-5}\exp(-148\,000/(8.314\times1198)) = 2.3\times10^{-5}\times e^{-14.86} = 8.1\times10^{-12}$ m²/s.

Required error function: $(0.45-0.20)/(1.0-0.20) = 0.3125 = \operatorname{erfc}(z)$, so $z = 0.714$.

$$t = \frac{1}{D}\left(\frac{x}{2z}\right)^2 = \frac{1}{8.1\times10^{-12}}\left(\frac{1.0\times10^{-3}}{2\times0.714}\right)^2 = 6.0\times10^{4}\ \text{s} = 16.8\ \text{h}$$

Because $x \propto \sqrt{t}$, halving the case depth to 0.5 mm takes a quarter of the time, 4.2 h; raising the temperature to 950 °C cuts the 1.0 mm cycle to 12.4 h. Carburizing furnace cycles of 4–20 hours for automotive and wind-turbine gears are consistent with this;

high-temperature vacuum carburizing at 1000–1050 °C shortens them further, at the cost of grain growth unless the steel is microalloyed with Nb or Al.

Nitriding and nitrocarburizing

Nitriding diffuses nitrogen into a steel at 490–580 °C — in the ferrite field, below A_1 — from ammonia gas, plasma or salt baths. No quench is needed and distortion is minimal, so finished, quenched-and-tempered parts can be nitrided. Nitrogen forms a thin surface **compound layer** (5–20 μm of ε-$Fe_{2-3}N$ and γ′-Fe_4N, the "white layer", hard and wear and corrosion resistant) above a **diffusion zone** (0.1–0.7 mm) in which nitrogen precipitates as fine nitrides of Al, Cr, Mo and V. In steels designed for it (Nitralloy 135M with 1 % Al; 31CrMoV9; H13 tool steel) the diffusion zone reaches 900–1200 HV. Nitriding is slow (20–80 h for 0.3–0.5 mm) but gives excellent fatigue strength through compressive stresses; crankshafts, extruder screws, dies and gun barrels are nitrided. **Ferritic nitrocarburizing** (2–4 h at about 570 °C) quickly produces mainly the compound layer, for brake pistons and gas springs.

Induction, flame and laser hardening

These methods change no chemistry. They heat only the surface layer of a medium-carbon or alloy steel (0.35–0.55 % C, e.g. 1045, 4140) into austenite and quench it, leaving the core in its prior (often quenched-and-tempered) condition. In **induction hardening**, a coil induces eddy currents concentrated within the skin depth $\delta = \sqrt{2\rho/(\omega\mu)}$ — for steel above its Curie point about 1 mm at 200 kHz and 9 mm at 3 kHz, with the hardened case a fraction of this — so frequency selects case depth, and heating takes seconds. It is used in-line for crankshaft journals, axles, gear teeth and rail heads. **Flame hardening** suits large, low-volume parts; **laser** and **electron-beam** hardening give shallow (0.2–1.5 mm), precisely placed, self-quenched cases on cutting edges and valve seats.

Table 15.7 — Surface hardening processes compared (typical).

Process	Temperature (°C)	Case depth (mm)	Surface hardness	Distortion	Typical parts
Gas carburizing + quench	900–950	0.5–2.5	58–62 HRC	Moderate to high	Transmission and wind-turbine gears
Carbonitriding	820–880	0.1–0.7	58–64 HRC	Moderate	Small gears, fasteners
Gas or plasma nitriding	490–580	0.1–0.7 (diffusion zone)	600–1200 HV	Very low	Crankshafts, dies, extruder screws
Ferritic nitrocarburizing	560–590	0.01–0.03 (compound layer)	500–800 HV	Very low	Brake pistons, gas springs
Induction hardening	850–1000 (surface)	0.5–6	50–60 HRC	Low to moderate	Shafts, journals, rail heads
Laser hardening	900–1200 (surface)	0.2–1.5	55–65 HRC	Very low	Cutting edges, valve seats

The choice balances case depth (set by the depth of maximum contact shear stress, Chapter 45), distortion, volume and steel cost: a large wind-turbine ring gear is carburized and ground, a car crankshaft induction-hardened or nitrided, an injection mould nitrided after machining — each a solution to the problem this chapter began with: hard where contact demands it, tough everywhere else.

What remains open

Open question	Where it is answered
How are aluminium, titanium, nickel and copper alloys strengthened and selected, and how do they compare with steel?	Chapter 16
How do structural steel specifications, timber and concrete compare in building and bridge design?	Chapter 18
How do electrical steels and soft magnetic irons achieve low losses?	Chapter 29
How do steels perform under neutron irradiation and at cryogenic temperatures?	Chapter 41
How do stainless steels passivate, and how do they pit, crevice-corrode and crack?	Chapters 42 and 43
How do surface treatments interact with wear, friction and contact fatigue?	Chapter 45
How are steels rolled, forged and cast, and how does thermomechanical processing set texture?	Chapter 46
What happens in the heat-affected zone of a steel weld, and how is cold cracking prevented?	Chapter 47
How do additive manufacturing and rapid solidification change steel microstructures?	Chapter 48
How are CCT diagrams and hardenability computed rather than measured?	Chapter 56
How is steel recycled, and what is its carbon footprint?	Chapter 59

Next: Chapter 16 turns to the non-ferrous alloys — aluminium, copper, titanium, nickel, magnesium and beyond — where precipitation, solid solution and crystal structure are used to win on lightness, conductivity, corrosion resistance or temperature capability where steel cannot.

Chapter 16 — Non-Ferrous Alloys

Part IV · Metals, Ceramics and Construction Materials

A pedestrian bridge over a railway is assembled from welded aluminium trusses. The extrusions are 6061-T6, with a certified yield strength of 276 MPa, and the designer sized the members with that number. In the proof-load test the truss deflects more than predicted and a diagonal yields permanently next to a weld, at a nominal stress of barely 120 MPa. Nothing was wrong with the extrusions. The weld itself was sound. What changed was the alloy's *temper*: the heat of welding dissolved and coarsened the nanometre-scale Mg–Si precipitates that gave 6061-T6 its strength, and a band of metal a few centimetres wide on each side of every weld was returned to something close to the annealed state. Design codes for aluminium structures know this and specify a reduced "heat-affected" strength; the designer had used a steel habit.

The same lesson repeats across the non-ferrous metals. A titanium fan disc is only as good as the forging temperature relative to its β transus; a single-crystal turbine blade is only as good as the volume fraction and coherency of its γ′ precipitates; a beryllium-copper connector spring loses its force if it is overaged in service. Non-ferrous alloys are, almost all of them, metallurgically *engineered* structures, and the name of the alloy is never enough without the name of its condition.

> **Why does each non-ferrous metal have its own family of alloys and heat treatments, and what does each family do better than steel?**
> **How do processing and temper designations encode the microstructure, and how does an engineer choose among aluminium, copper, titanium, nickel, magnesium, zinc, refractory, precious and intermetallic alloys?**

Concept	Engineering destination
Aluminium designations and tempers	Airframes, car bodies, extrusions, cans, welded structures, heat-sink design
Age hardening (GP zones, β″, η′, θ′)	Heat-treatment specifications, weld HAZ strength, overaging in service
Copper alloys and conductivity–strength trade-off	Busbars, windings, connector springs, heat exchangers, marine hardware
Titanium α, α+β, β alloys	Fan blades and discs, landing gear, implants, chemical plant
Nickel superalloys, γ/γ′, single crystals, coatings	Gas-turbine blades and discs, rocket engines, power generation
Magnesium and zinc alloys	Lightweight castings, steering wheels, die-cast housings, biodegradable implants
Refractory and precious metals	Furnace hardware, rocket nozzles, X-ray targets, catalysts, thermocouples
Intermetallics and high-entropy alloys	Low-pressure turbine blades, heating elements, cryogenic and high-temperature research alloys
Selection indices	Choosing the lightest, cheapest or most conductive alloy for a given duty

Roadmap. Aluminium: designations, tempers and age hardening (16.1) → copper and its conductivity–strength trade-off (16.2) → titanium and its two crystal structures (16.3) → nickel superalloys and the γ/γ′ structure (16.4) → magnesium and zinc (16.5) → refractory and precious metals (16.6) → intermetallics and high-entropy alloys (16.7) → putting it together: selection (16.8).

16.1 Aluminium alloys

Why aluminium

Aluminium is FCC to its melting point (660 °C), with a density of 2.70 Mg/m³ and a modulus of about 70 GPa (each a third of steel's), about 62 % of copper's conductivity, and a self-healing Al_2O_3 film. Twelve slip systems and no ductile–brittle transition make it formable and suitable for cryogenic tanks. Its weaknesses are softening above about 150–200 °C, a modulus that

alloying cannot raise, no true fatigue endurance limit, and pitting where chlorides break the film (Chapter 43). Its specific stiffness $E/\rho \approx 26$ GPa/(Mg/m³) equals steel's, so its advantage in stiff designs comes from thicker sections (Section 16.8); in strong designs it comes from metallurgy, which raises yield strength from 10–30 MPa (pure) to above 550 MPa.

The designation system

Wrought alloys (shaped by rolling, extrusion or forging) carry a four-digit number in the International Alloy Designation System; casting alloys carry a three-digit number with a decimal. The first digit names the principal alloying element and, with it, the strengthening mechanism.

Series	Main alloying	Strengthened by	Typical alloy and temper	Yield / UTS (MPa, typical)	Uses
1xxx	≥ 99.0 % Al	Cold work	1100-H14; 1350-H19	117 / 124; 165 / 186	Foil, chemical plant, conductors (1350: ~61 % IACS)
2xxx	Cu (± Mg, Li)	Age hardening	2024-T3	345 / 483	Fuselage skins, damage-tolerant structure
3xxx	Mn	Cold work, dispersoids	3003-H14	145 / 150	Beverage-can bodies (3004), heat exchangers
4xxx	Si	— (filler, brazing)	4043	—	Welding wire, brazing sheet clad
5xxx	Mg	Solid solution + cold work	5083-H116	228 / 317	Ship hulls, LNG tanks, welded structures
6xxx	Mg + Si	Age hardening	6061-T6; 6063-T6	276 / 310; 214 / 241	Extrusions, frames, automotive, architecture
7xxx	Zn + Mg (± Cu)	Age hardening	7075-T6; 7050-T7451	503 / 572; 469 / 524	Wing spars, forgings, high-strength sport goods
8xxx	Others (Li, Fe)	Various	8090, 2099/2195 (Al–Li)	450–560 / 500–600	Lighter, stiffer aerospace plate and extrusions

Casting alloys (1xx.x to 8xx.x) are dominated by **Al–Si** alloys: the eutectic at 12.6 wt% Si and 577 °C (Chapter 7) gives high fluidity and low shrinkage. A356 (Al–7Si–0.3Mg) is the standard alloy for wheels and suspension parts (about 185 MPa yield and 262 MPa UTS as permanent-mould T6); A380 (Al–8.5Si–3.5Cu) is the die-casting workhorse for housings (about 160 / 320 MPa, but only 3–4 % elongation because of porosity). Sr or Na **modify** the plate-like eutectic Si to a fine fibrous form, and Ti–B additions refine the grains by heterogeneous nucleation (Chapter 9).

Tempers

The temper letter that follows the alloy number tells the engineer how the microstructure was produced. Without it the alloy number is close to meaningless: 6061 ranges from 55 MPa yield (O) to 276 MPa (T6), and 7075 from about 100 MPa to more than 500 MPa.

Temper	Meaning	Microstructural state
F	As fabricated	Uncontrolled
O	Annealed	Recrystallized, softest; precipitates coarse and incoherent
H1x, H2x, H3x	Strain hardened (only), strain hardened and partially annealed, strain hardened and stabilized; x = 2, 4, 6, 8 for ¼, ½, ¾ and full hard	Dislocation density raised (Chapter 13)
W	Solution heat treated, unstable	Supersaturated solid solution, ageing at room temperature
T3	Solution treated, cold worked, naturally aged	GP zones plus dislocations
T4	Solution treated, naturally aged	GP zones / clusters
T5	Cooled from hot working, artificially aged	Precipitates without a separate solution treatment
T6	Solution treated, artificially aged to peak strength	Fine coherent or semicoherent precipitates (β″, η′, θ″/θ′)
T7	Solution treated, overaged	Coarser η or θ′; lower strength, better stress-corrosion and exfoliation resistance
T8	Solution treated, cold worked, artificially aged	Precipitates nucleated on dislocations (Al–Li, 2xxx)
Txx51	Additional digit(s): stress relieved by stretching 1–3 %	Low residual stress for machining

The T7 tempers deserve a word. In the 1960s, 7075-T6 aircraft forgings failed by intergranular stress-corrosion cracking. Overageing to T73 (later T76, T7451) costs 10–15 % of the strength but raises the stress-corrosion threshold severalfold: one property traded for another by heat treatment rather than by a new alloy.

Age hardening in practice

The physics of precipitation was set out in Chapter 10 for Al–Cu: solution treatment in the single-phase α field, a quench to retain a supersaturated solid solution and a high vacancy concentration, and ageing that produces a sequence of metastable phases of increasing size and decreasing coherency. The sequences for the commercial families are:

$$\text{Al–Cu (2xxx)}: \ \text{SSSS} \rightarrow \text{GP zones} \rightarrow \theta'' \rightarrow \theta' \rightarrow \theta \ (\text{Al}_2\text{Cu})$$

$$\text{Al–Mg–Si (6xxx)}: \ \text{SSSS} \rightarrow \text{clusters} \rightarrow \text{GP} \rightarrow \beta'' \rightarrow \beta' \rightarrow \beta \ (\text{Mg}_2\text{Si})$$

$$\text{Al–Zn–Mg(–Cu) (7xxx)}: \ \text{SSSS} \rightarrow \text{GP} \rightarrow \eta' \rightarrow \eta \ (\text{MgZn}_2)$$

The peak strength corresponds to a dense dispersion of shearable or just-unshearable particles a few nanometres across, with a spacing of tens of nanometres, at the crossover between the cutting and Orowan mechanisms (Chapter 13). Underaged material is weaker because the particles are small and easily cut; overaged material is weaker because the particles have coarsened (Ostwald ripening, Chapter 10) and dislocations bow between them.

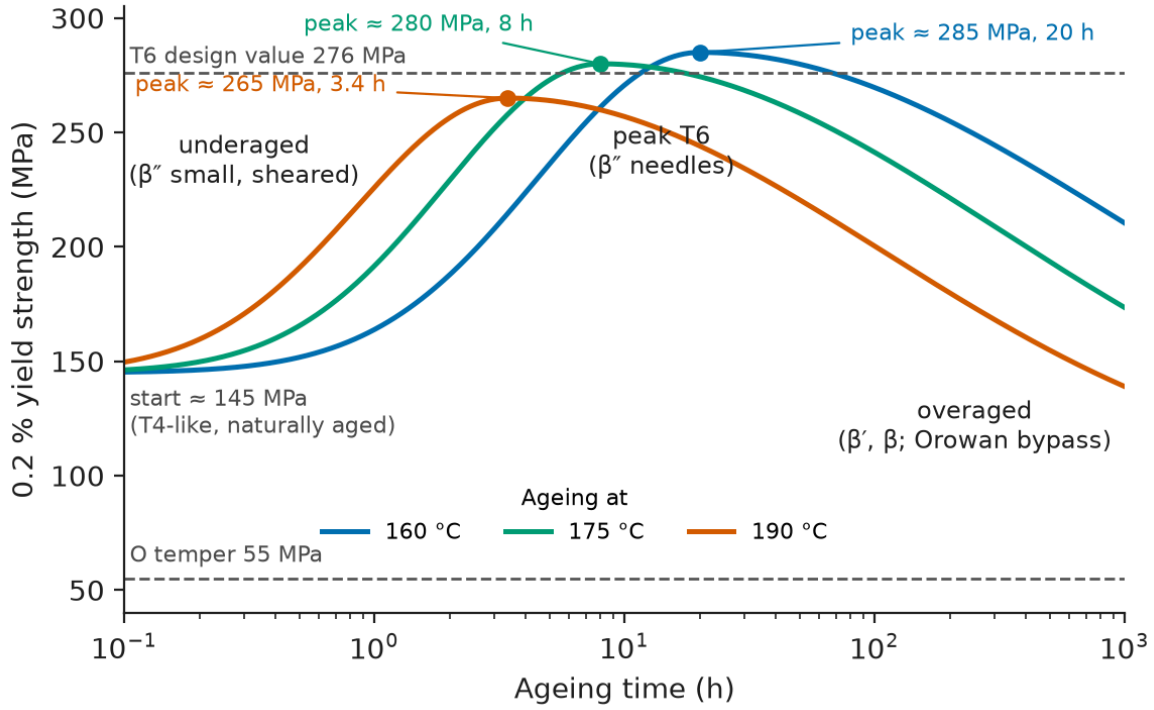

Figure 16.1 — Yield strength of 6061 versus artificial ageing time at 160, 175 and 190 °C after solution treatment at 530 °C and water quench, showing underageing, peak (T6) and overageing.

Figure 16.1 shows the practical consequence: the peak shifts to shorter times and lower values as the ageing temperature rises. Lower temperatures give a higher, broader peak, because the driving force for nucleation is larger and the precipitates finer, but the treatment takes longer. Heat-treaters exploit the thermal activation of the process to convert one schedule into another.

Worked example — Transferring an ageing schedule. A shop ages 6061 for 8 h at 175 °C. A new furnace can hold only 160 °C uniformly. Treating the time to peak strength as controlled by a single diffusion-limited process with an apparent activation energy of $Q \approx 100$ kJ/mol, the times at two temperatures scale as

$$\frac{t_2}{t_1} = \exp\left[\frac{Q}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

With $T_1 = 448.15$ K and $T_2 = 433.15$ K, $1/T_2 - 1/T_1 = 7.73 \times 10^{-5}$ K^{-1}, and $Q/R = 12\,030$ K, so the ratio is $\exp(0.929) = 2.53$ and $t_2 \approx 20$ h. The published alternative schedule for 6061-T6 is 18 h at 160 °C, close to this estimate. Going the other way, to 190 °C, gives $t_2 \approx 3.4$ h, but the peak strength is lower (Figure 16.1). The estimate fails when the precipitation sequence itself changes with temperature (for instance if ageing is so hot that β′ forms directly), and it ignores the time to heat the load, which for thick sections can be hours.

> **Status of the principle: Arrhenius time–temperature equivalence for ageing**
> *Approximate.* Valid when one thermally activated process controls the kinetics and the same precipitate sequence operates at both temperatures; it breaks down across changes in the precipitate phases or when nucleation and growth have different activation energies.

Natural ageing, quench sensitivity and welding

Some alloys age at room temperature: 2024 reaches its T3/T4 strength about four days after quenching, so 2024 rivets were kept in a freezer until driven ("icebox rivets"). Thick 7075 plate is **quench sensitive** (its centre precipitates coarse η on cooling), which is why thick-plate alloys like 7050 use a less quench-sensitive chemistry.

Welding is the reverse problem. In the heat-affected zone, peak temperatures of 250–500 °C dissolve β″ and overage what remains. The as-welded heat-affected zone of 6061-T6 has a design tensile strength of about 165 MPa, compared with 290 MPa minimum for the parent metal, and a yield strength nearer 100–110 MPa. This is the bridge of the opening paragraph. The remedies are to place welds at low-stress locations, to use friction-stir welding with its lower peak temperature (Chapter 47), to

re-solution-treat and age small welded assemblies, or to choose a 5xxx alloy, whose strength comes from Mg in solution and cold work and which loses only the cold-work increment on welding.

> **Boundary.**
> Corrosion of aluminium (pitting, intergranular corrosion, exfoliation, stress-corrosion cracking) is treated in Chapters 42–43; the electrochemistry behind the passive film belongs to Advanced Chemistry.

Aluminium–lithium alloys

Each 1 wt% of lithium lowers the density of aluminium by about 3 % and raises its modulus by about 6 %, the only common element that raises E appreciably. After early generations troubled by anisotropy and low toughness, third-generation alloys (2195, 2099, 2050; 0.7–1.8 % Li, strengthened by T1, Al_2CuLi, in T8 tempers) are used in launch-vehicle tanks and airliner structure, competing with carbon-fibre composites (Chapter 22).

16.2 Copper alloys

Why copper

Copper (FCC, 1085 °C, 8.96 Mg/m³, E 110–130 GPa) has a conductivity of 58–59 MS/m (resistivity $1.72 \times 10^{-8}\ \Omega\,\mathrm{m}$, defined as 100 % IACS) and a thermal conductivity of about 400 W m^{-1} K^{-1}, second only to silver. It resists atmospheric and fresh-water corrosion and is easily formed and brazed; its density and a price three to four times aluminium's are its handicaps.

Almost anything that strengthens copper reduces its conductivity, because solutes and defects scatter electrons. By **Matthiessen's rule** the resistivity is a thermal part plus a residual defect part, and by **Nordheim's rule** the solute part scales as $x(1-x)$ (Chapter 25). Solute in solution costs much (0.1 wt% P removes about 30 % of the conductivity; the same Ag almost nothing); the same solute precipitated costs little. This is the logic of every high-conductivity copper alloy.

Brasses

Brasses are copper–zinc alloys. Zinc is soluble in FCC copper up to about 35–39 wt% (the α phase); beyond that the BCC β phase (ordered as β′, CuZn, B2, below about 460 °C) appears (Figure 16.2). The strength of α brass rises and its ductility *also* rises up to about 30 % Zn, an unusual combination that comes from the falling stacking-fault energy, which promotes planar slip and deformation twinning and sustains work hardening (Chapters 12–13). **Cartridge brass** (70Cu–30Zn, C26000) has the best combination for deep drawing, as its name records; annealed it has a yield of about 100 MPa, a UTS of about 340 MPa and elongation above 60 %. **Muntz metal** (60Cu–40Zn, C28000) is α+β, stiffer at high temperatures and hot-workable, and is used for condenser plates, architectural extrusions and fasteners. Small lead additions (free-cutting brass, C36000) give the best machinability of any common alloy, but lead in potable-water fittings is now restricted and is being replaced by Bi or Si additions.

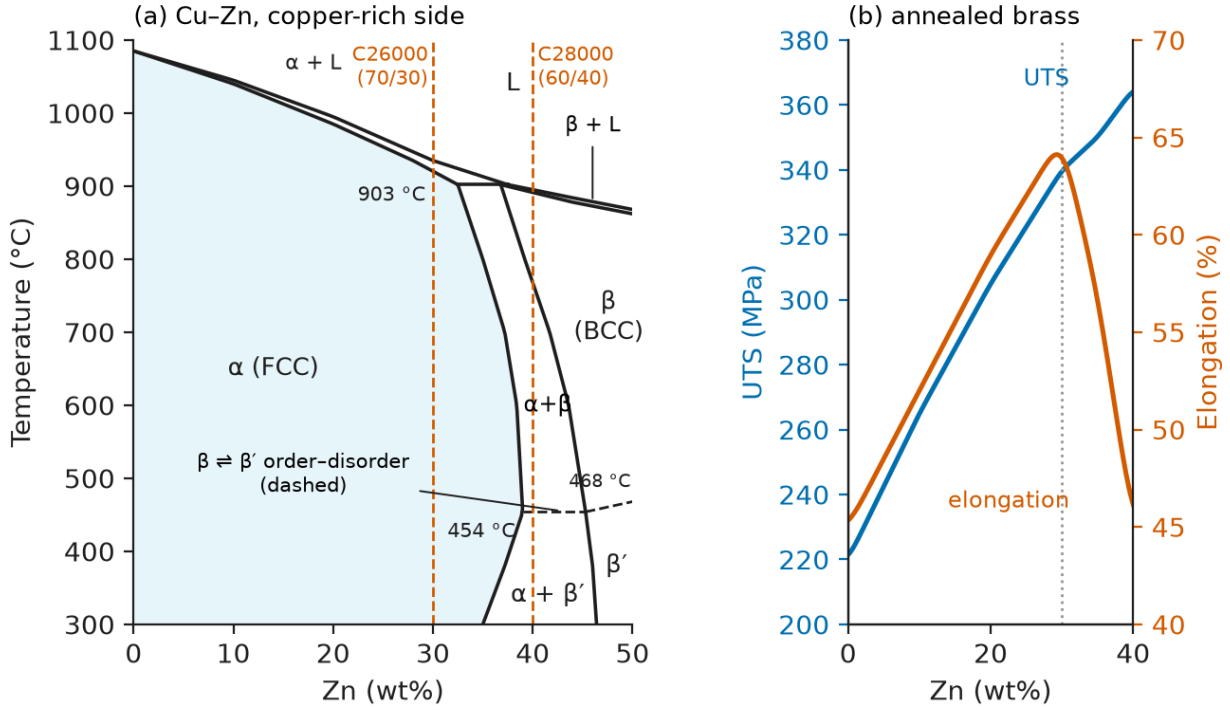

Figure 16.2 — Copper-rich part of the Cu–Zn phase diagram with the compositions of cartridge brass and Muntz metal, and the variation of tensile strength and elongation of annealed α brass with zinc content.

Brasses have two classic corrosion failures (Chapter 43). **Dezincification**, selective leaching of Zn in stagnant or chloride-bearing water, is suppressed in α brasses by 0.02–0.06 % As but not in β. **Season cracking**, the original stress-corrosion cracking, struck cold-drawn cartridge cases stored in ammonia-bearing air and was cured by a stress-relief anneal.

Bronzes and cupronickels

Tin bronzes and **phosphor bronzes** (C51000: Cu–5Sn–0.2P) are solid-solution strengthened by tin; cold-rolled phosphor bronze reaches 550 MPa yield and is a standard connector-spring material, and cast leaded bronzes are bearing materials. **Aluminium bronzes** (5–11 % Al with Fe and Ni) reach 500–700 MPa UTS and are the standard alloy for ship propellers and seawater pumps. **Cupronickels** exploit the complete solubility of Cu and Ni (Chapter 7): 90Cu–10Ni and 70Cu–30Ni resist seawater flow and biofouling in condenser and desalination tubing, and are chosen for corrosion, not conduction.

High-conductivity and high-strength coppers

Electrolytic tough-pitch copper (ETP, C11000) contains 200–400 ppm O as Cu_2O and is embrittled when heated in hydrogen (steam blisters its grain boundaries); **oxygen-free** grades (C10200, C10100) are used for brazed assemblies and vacuum electronics. Cold work raises the yield of ETP copper from about 70 to 300–350 MPa at a cost of only 2–3 % IACS, because dislocations scatter electrons weakly.

When a conductor needs more strength, especially at temperature, precipitation is used:

Alloy	Composition (wt%)	Condition	Yield (MPa, typical)	Conductivity (% IACS)	Uses
C11000 ETP	99.9 Cu	Annealed / hard	70 / 320	101 / 97	Busbars, wire, windings
C18150 CuCrZr	1 Cr, 0.1 Zr	Solution treated, aged	400–450	80	Resistance-welding electrodes, fusion heat sinks, rocket combustion liners
C70250 CuNiSi	3 Ni, 0.65 Si	Aged	550–650	40	Leadframes, high-current connectors
C17200 CuBe	1.9 Be, 0.2 Co	Peak aged	1000–1200	22	High-reliability springs, non-sparking tools, downhole tools
C51000 phosphor bronze	5 Sn, 0.2 P	Spring temper	550	15	Contacts, bellows, springs
C26000 cartridge brass	30 Zn	Annealed	100	28	Deep-drawn shells, radiator cores
C70600 cupronickel	10 Ni, 1.5 Fe	Annealed	110–140	9	Seawater tubing

Beryllium copper is the strongest copper alloy and the reference for spring performance, but beryllium dust is toxic and machining and grinding of CuBe require controls; CuNiSi and Cu–Ni–Sn spinodal alloys (Chapter 10) are its main replacements.

Worked example — Copper or aluminium busbar? A switchboard needs a busbar carrying 2000 A. A copper bar of 100 mm × 10 mm ($A = 1000$ mm²) has a resistance per metre of $\rho/A = 1.72 \times 10^{-8}/10^{-3} = 1.72 \times 10^{-5}\ \Omega/\text{m}$ and dissipates $I^2R = 2000^2 \times 1.72 \times 10^{-5} = 68.8$ W/m at 20 °C. An aluminium bar of grade 1350 ($\rho = 2.82 \times 10^{-8}\ \Omega\,\text{m}$) with the same resistance must have an area larger by the resistivity ratio, $2.82/1.72 = 1.64$, i.e. 1640 mm². Its mass per metre is $1.64 \times 10^{-3} \times 2700 = 4.43$ kg/m, against $10^{-3} \times 8960 = 8.96$ kg/m for copper: the aluminium bar is 0.49 times as heavy. At representative prices of about 9.5 USD/kg for copper and 2.6 USD/kg for aluminium, the metal costs about 85 and 12 USD per metre. Aluminium wins on mass and cost for the same losses; copper wins on volume and on joint reliability (aluminium creeps under bolts and forms an insulating oxide). Hence overhead lines and large busbars are aluminium, while motor windings and circuit conductors are copper.

> **Boundary.**
> The electron-scattering physics behind Matthiessen's and Nordheim's rules, and EEE-specific conductor and connector materials, are treated in Chapter 25.

16.3 Titanium alloys

Two crystal structures

Titanium has a density of 4.51 Mg/m³, a modulus of about 105–120 GPa, a melting point of 1668 °C and a passive TiO_2 film that makes it practically immune to seawater, chlorides and oxidizing acids at ambient temperatures. It exists in two forms: **α**, hexagonal close-packed ($c/a = 1.587$, slightly below ideal) up to 882 °C, and **β**, body-centred cubic from 882 °C to the melting point. The whole of titanium metallurgy is the manipulation of these two phases.

Alloying elements are classified by which phase they stabilize, that is, whether they raise or lower the α/β transition temperature (the **β transus**):

Class	Elements	Effect
α stabilizers	Al, O, N, C	Raise the β transus; strengthen α by solid solution (O and N are potent interstitial strengtheners and embrittlers)
β isomorphous	V, Mo, Nb, Ta	Lower the β transus; complete solubility in β
β eutectoid	Fe, Cr, Mn, Ni, Cu, Si, H	Lower the β transus; can form intermetallic compounds by eutectoid decomposition
Neutral	Sn, Zr	Little effect on the transus; strengthen both phases

An aluminium equivalent, roughly $\mathrm{Al} + \frac{1}{3}\mathrm{Sn} + \frac{1}{6}\mathrm{Zr} + 10(\mathrm{O} + \mathrm{C} + 2\mathrm{N})$ in wt%, is kept below about 9 to avoid embrittling Ti_3Al (α_2). Commercially pure Grades 1–4 differ mainly in oxygen, which raises the yield strength from about 170 MPa (Grade 1) to 480 MPa (Grade 4).

Alloy classes

Class	Example	Yield / UTS (MPa, typical)	Key features	Applications
CP titanium	Grade 2	275 min / 345 min	Formable, weldable, best corrosion resistance	Heat-exchanger tubes, chemical vessels, desalination
α and near-α	Ti-5Al-2.5Sn; Ti-6242S; IMI 834	800–1000 / 860–1050	HCP, best creep resistance; not heat-treatable to high strength	Compressor discs and blades to 540–600 °C
α+β	Ti-6Al-4V (Ti-6-4)	880 / 950 annealed; 1100 / 1170 STA	Versatile, heat-treatable, weldable; about half of all titanium used	Fan blades, airframe fittings, implants, bicycle and automotive parts
Metastable β	Ti-10V-2Fe-3Al; Ti-5553; Ti-15V-3Cr-3Sn-3Al	1100–1300 / 1200–1400	BCC, cold formable, deep hardenable, high strength; lower modulus	Landing-gear beams, springs, fasteners
β (biomedical)	Ti-13Nb-13Zr; Ti-29Nb-13Ta-4.6Zr	500–900	Low modulus (55–80 GPa), Al- and V-free	Hip stems, spinal rods (Chapter 39)

Processing and microstructure

Because the β transus of Ti-6-4 lies at about 995 °C, a forging or heat treatment above it makes the whole part β, and on cooling α forms as plates on {110} planes of the β (the Burgers orientation relationship) in colonies or as basketweave. This gives a **lamellar** (Widmanstätten) structure: good creep strength and fracture toughness and slow fatigue-crack growth, but lower ductility and lower high-cycle fatigue strength. Working below the transus, in the α+β field, breaks up the plates and recrystallizes them into **equiaxed** primary α grains; a subsequent treatment just below the transus gives a **bimodal** structure of equiaxed primary α in a matrix of transformed β lamellae, a common compromise for rotating parts (Figure 16.3). Solution treating high in the α+β field and quenching retains β (or forms α′ martensite); ageing at 480–600 °C precipitates fine α and gives the STA (solution treated and aged) condition.

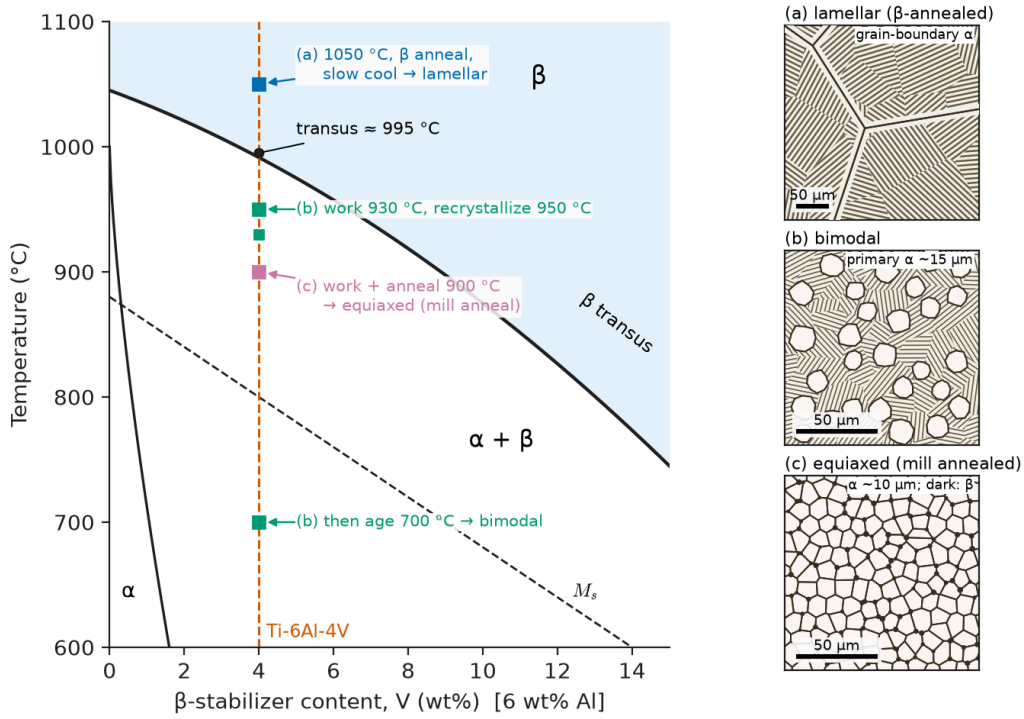

Figure 16.3 — Pseudo-binary section through Ti-6Al-4V with the β transus and the processing routes that produce lamellar, bimodal and equiaxed microstructures.

Titanium is expensive not because it is rare but because it is reactive. The batch **Kroll process** reduces $TiCl_4$ with magnesium to sponge, which is vacuum-arc remelted two or three times for rotating parts; a nitrogen-rich hard-α inclusion caused the 1989 Sioux City DC-10 fan-disc failure and led to cold-hearth melting. Above about 500 °C titanium picks up oxygen and forms a brittle **α case** that must be removed; it must be welded under full inert shielding, galls in sliding contact, and machines slowly because its low thermal conductivity (about 7 W m^{-1} K^{-1}) concentrates heat at the tool. Processing dominates cost, which makes near-net-shape forging and additive manufacturing (Chapter 48) attractive.

Titanium wins where specific strength matters at 150–600 °C, where it is joined to carbon-fibre composites (galvanically compatible, and its expansion of about 9 × 10^{-6} K^{-1} is closer to the laminate's than aluminium's 23 × 10^{-6} K^{-1}), where chloride corrosion resistance is essential, and in the body (Chapter 39).

> **Status of the principle: β-stabilizer and α-stabilizer classification**
> *Empirical.* The classification summarizes measured binary phase diagrams; equivalence formulae combining several elements are linear fits that work for commercial composition ranges but not for highly alloyed or multicomponent β alloys, where CALPHAD calculations are used (Chapter 7).

16.4 Nickel-base superalloys

Why nickel

The first high-pressure turbine stage of a jet engine sees gas at 1500–1700 °C while its blades carry centrifugal stresses of 150–250 MPa for thousands of hours, the metal running at 950–1100 °C, a higher homologous temperature than any other structural metal endures for so long. Nickel is the base because it is FCC with no allotropic change up to its melting point (1455 °C), dissolves large amounts of Cr, Co, Mo, W, Re and Ta, and with Cr and Al forms a protective Cr_2O_3 or Al_2O_3 scale.

The γ/γ′ microstructure

The secret of the superalloys is a two-phase microstructure. The matrix, **γ**, is a disordered FCC nickel solid solution. Within it sits **γ′**, the ordered intermetallic $Ni_3(Al,Ti,Ta)$ with the $L1_2$ structure: an FCC lattice in which Al (or Ti, Ta) occupies the cube

corners and Ni the face centres (Chapter 3). The two lattices differ by only a few tenths of a percent in parameter (the **misfit** $\delta = 2(a_{\gamma'} - a_{\gamma})/(a_{\gamma'} + a_{\gamma})$ is typically between −0.5 % and +0.5 %), so the γ′ precipitates are coherent cubes 0.3–0.5 μm across, aligned along ⟨100⟩, and they can occupy 60–70 % of the volume in modern cast alloys.

A matrix dislocation ($\frac{a}{2}\langle 110\rangle$) entering an ordered L1₂ particle leaves an **antiphase boundary** (APB) of 0.1–0.3 J/m² behind it (Chapters 3, 12), so dislocations must travel in pairs as **superdislocations**, the second restoring the order. Better still, Ni₃Al shows the **yield-strength anomaly**: its yield strength *rises* up to about 700–800 °C, because screw superdislocations cross-slip onto {100} planes of lower APB energy and form immobile Kear–Wilsdorf locks. The superalloy inherits part of this anomaly and keeps most of its strength to about 800 °C.

At still higher temperatures the alloy fails by creep (Chapter 14): γ′ cubes coarsen directionally into plates (**rafting**) and dislocations climb around them. Re, W, Ta and Ru slow diffusion in γ; rhenium is the signature element of second- and third-generation single crystals (3 and 6 wt%). Too much refractory alloying precipitates brittle **topologically close-packed** phases (σ, μ), so alloy design is a balance guided today by CALPHAD.

From wrought to single crystals

Class	Example	γ′ (vol %)	Typical use and limit	Notes
Solid-solution	Inconel 625, Hastelloy X, Haynes 230	—	Combustor liners, ducts; oxidation to ~1100 °C	Weldable sheet, strengthened by Mo, W, Nb
Wrought γ″/γ′	Inconel 718	~20 (mainly γ″ Ni₃Nb)	Discs, shafts, cryogenic and rocket-engine parts; to ~650 °C	Yield ~1035 MPa; weldable; the most-used superalloy
Wrought γ′	Waspaloy, Udimet 720Li	25–45	Discs, rings to ~750 °C	Powder-metallurgy discs (René 88DT, RR1000) for higher γ′
Cast equiaxed	IN-713C, Mar-M247	45–65	Blades and vanes; grain-boundary carbides and B, Zr, Hf	Grain boundaries limit creep and thermal fatigue
Directionally solidified	Mar-M247 DS, CM247LC	~60	Blades; columnar grains along the axis	No transverse boundaries; Hf added to resist boundary cracking
Single crystal (1st–4th gen.)	PWA 1480, CMSX-4, René N5, CMSX-10, TMS-138	60–70	High-pressure turbine blades, metal to ~1100 °C	No boundaries, no C/B/Zr; Re 0–6 %, Ru in 4th gen.

Directional solidification withdraws the mould from a hot zone into a chill so that grains grow as columns along the blade axis, in the fast-growing ⟨100⟩ direction (Chapter 9). Removing transverse boundaries, where creep cavities form, raised creep and thermal-fatigue life; and ⟨100⟩ has a low modulus (about 125 GPa versus about 220 GPa for a polycrystal, Chapter 4), so thermal strains produce lower stresses. A **single-crystal** blade adds a helical grain selector that lets only one grain through; with no boundaries, C, B, Zr and Hf can be removed, raising the incipient melting point and allowing a fuller solution treatment. Each step has bought tens of kelvin of metal temperature (Figure 16.4).

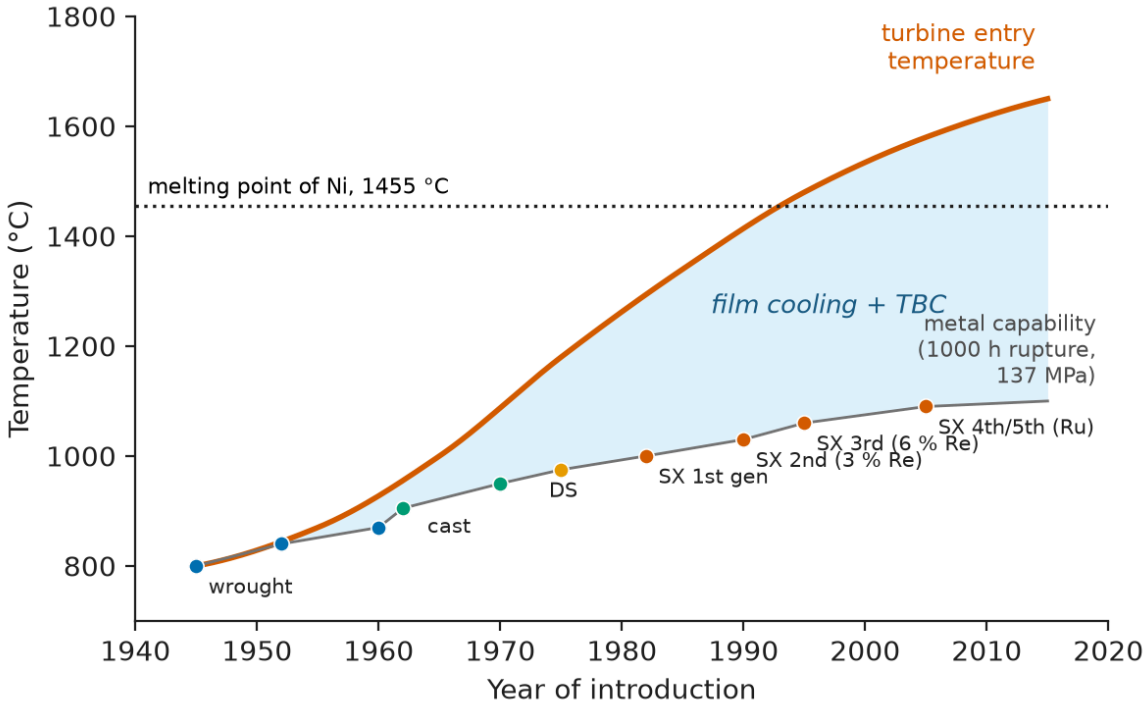

Figure 16.4 — Temperature capability of turbine-blade alloys and turbine entry temperature versus year, showing the contributions of wrought, cast, directionally solidified and single-crystal alloys, cooling and thermal barrier coatings.

Worked example — What 50 K of metal temperature is worth. For a given alloy and stress, the Larson–Miller parameter $P = T(C + \log_{10} t_r)$ with $C \approx 20$ (Chapter 14) is constant along a line of equal stress. A single-crystal blade that ruptures after 100 h at a metal temperature of 1050 °C (1323 K) has $P = 1323 \times (20 + 2) = 29\,106$. If a better coating or cooling scheme lowers the metal temperature by 50 K to 1273 K at the same stress, the rupture life becomes

$$\log_{10} t_r = \frac{29\,106}{1273} - 20 = 2.864, \qquad t_r \approx 731 \text{ h}$$

a factor of about seven. Raising the metal temperature by 50 K to 1373 K cuts the life to about 16 h. This steep sensitivity is why engine designers count gains in "degrees of metal temperature" and why the coating systems below are as important as the alloy.

> **Status of the principle: Larson–Miller parameter**
> *Empirical.* A time–temperature parameter that collapses creep-rupture data onto one master curve for a given alloy; the constant C is fitted (15–25), and extrapolation beyond a few times the longest test, or across a change of creep mechanism, is unsafe.

Coatings

Even a perfect single crystal cannot survive bare in a combustion gas at 1100 °C for long. Turbine blades carry a coating system with up to four layers (Figure 16.5):

- a **bond coat**, either a Pt-modified **aluminide** diffusion coating (NiAl, 30–80 μm) or an **MCrAlY overlay** (M = Ni, Co), which is an aluminium reservoir;
- the **thermally grown oxide** (TGO), a slow-growing α-Al_2O_3 scale 1–10 μm thick (Chapter 42);
- a **thermal barrier coating** (TBC) of 7–8 wt% Y_2O_3-stabilized zirconia (7YSZ), 100–300 μm thick, with a conductivity of about 1 W m^{-1} K^{-1}, across which an internally cooled blade sustains a drop of 100–200 K or more (Chapter 24). EB-PVD gives a columnar, strain-tolerant structure for blades.

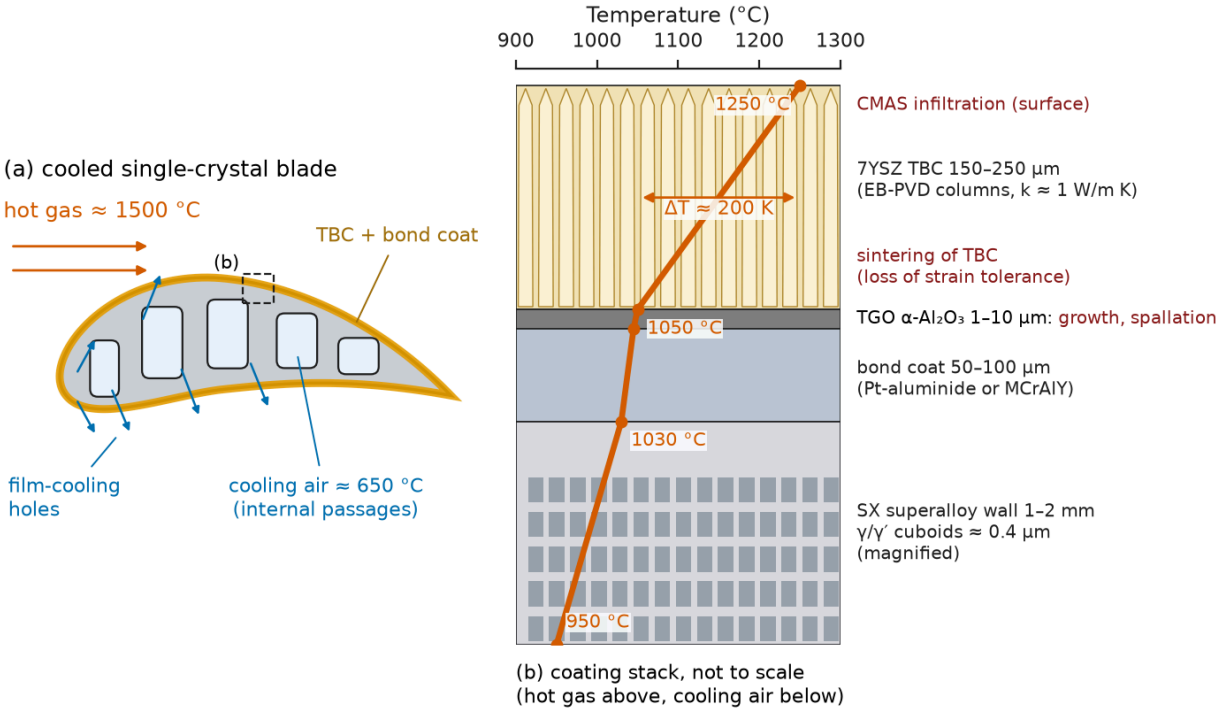

Figure 16.5 — Cross-section of a cooled single-crystal turbine blade with its thermal barrier coating system and the temperature profile through it.

TBCs fail mainly by spallation near the TGO as the oxide thickens during thermal cycles, and by **CMAS** attack: molten silicate deposits from ingested sand and volcanic ash infiltrate the columns and destroy their strain tolerance. The longer-term successor in the hottest parts is the SiC/SiC ceramic-matrix composite (Chapters 23 and 41).

> **Boundary.**
> High-temperature oxidation kinetics and hot corrosion are developed in Chapter 42; extreme-environment materials as a system in Chapter 41; heat flow through coatings in Chapter 24.

16.5 Magnesium and zinc alloys

Magnesium: the lightest structural metal

Magnesium has a density of 1.74 Mg/m³, a modulus of about 45 GPa and a melting point of 650 °C. It is HCP ($c/a = 1.624$), and at room temperature only basal ⟨a⟩ slip is easy; non-basal systems need tens of times higher resolved shear stress. Strain along c requires $\{10\bar{1}2\}$ twinning to satisfy the von Mises criterion (Chapter 13), so wrought magnesium has limited cold formability, strong basal texture and a **tension–compression asymmetry**: an extruded bar twins easily in axial compression, whose yield strength can be half the tensile. Sheet is therefore formed warm, above about 200–250 °C.

Most magnesium is used as castings. ASTM designations give the two main alloying elements by letter (A = Al, Z = Zn, M = Mn, K = Zr, E = rare earths, W = Y) and their rounded percentages:

Alloy	Composition (wt%)	Process	Yield / UTS (MPa), elongation (typical)	Uses
AZ91D	9 Al, 0.7 Zn, 0.2 Mn	Die cast	160 / 240, 3 %	Housings, brackets, power-tool bodies
AM60B	6 Al, 0.3 Mn	Die cast	130 / 220, 6–8 %	Steering wheels, instrument panels, seat frames (crash energy)
AZ31B	3 Al, 1 Zn	Sheet, H24	220 / 290, 15 %	Laptop and camera cases, panels
ZK60A	6 Zn, 0.5 Zr	Extruded, T5	255–285 / 330–360, 10 %	Bicycle and sports components, aerospace extrusions
WE43	4 Y, 3 RE, 0.5 Zr	Cast or wrought, T6	180–200 / 250–270, 6–7 %	Helicopter gearboxes to ~250 °C; bioresorbable implants

The Mg–Al alloys are strengthened by solid solution and by $Mg_{17}Al_{12}$ (β) at grain boundaries, which softens above about 120 °C and limits creep resistance in powertrain applications; rare-earth and yttrium alloys form stable precipitates and are used for hotter components. Zirconium is a powerful grain refiner for Al-free alloys.

Magnesium's other weakness is corrosion. It is the most electronegative structural metal, its hydroxide film is not protective in chlorides, and it corrodes galvanically against steel, copper and even aluminium; Fe, Ni and Cu impurities are capped in "high-purity" alloys (the D in AZ91D). The same corrosion becomes a virtue in **bioresorbable implants**: WE43-based scaffolds and bone screws dissolve over months in the body (Chapter 39). Its fire reputation concerns fine chips and powders; bulk castings are hard to ignite.

Zinc: the die-caster's metal

Zinc melts at only 419.5 °C (density 7.14 Mg/m³), ideal for **hot-chamber die casting** with dies lasting millions of shots, sub-millimetre walls and platable surfaces. The Zamak alloys (Zn–4Al with 0–3 % Cu and a trace of Mg; UTS about 280 MPa for Zamak 3, 330 MPa for Zamak 5) make locks, handles and connector shells; ZA alloys (8–27 % Al) reach 400–430 MPa. The weakness of zinc is creep: at 20 °C it is already at a homologous temperature of 0.42 (table below), so loaded clips and bosses relax, and long-term allowable stresses are much lower than short-term ones. Most zinc is used not for castings but to **galvanize** steel (Chapter 43).

Metal	Melting point (°C)	Homologous temperature at 20 °C (T/T_m)	Practical consequence
Tin	232	0.58	Solders creep at room temperature
Lead	327	0.49	Creeps under self-weight (roofs, cable sheaths)
Zinc	420	0.42	Room-temperature creep of die castings
Magnesium	650	0.32	Creep above ~120 °C in Mg–Al alloys
Aluminium	660	0.31	Precipitates coarsen above ~150 °C
Copper	1085	0.22	Softening of cold-worked conductors above ~200 °C
Nickel	1455	0.17	Creep-resistant to 0.8 T_m when alloyed
Titanium	1668	0.15	Usable to ~600 °C (limited by oxidation and α case)
Tungsten	3422	0.08	Brittle at room temperature, strong to > 2000 °C

16.6 Refractory and precious metals

Refractory metals

The **refractory metals** are conventionally the BCC metals of groups 5 and 6 with melting points above about 2400 °C, plus HCP rhenium:

Metal	Melting point (°C)	Density (Mg/m³)	E (GPa)	Room-temperature behaviour	Typical uses
Niobium	2477	8.57	105	Ductile, formable	Superconducting wire (Nb–Ti, Nb_3Sn), rocket-nozzle extensions (C103), steel microalloying
Molybdenum	2623	10.2	329	Ductile–brittle transition near room temperature	Furnace elements and shields, glass-melting electrodes, TZM forging dies, steel alloying
Tantalum	3017	16.7	186	Very ductile	Capacitors, chemical-plant linings, surgical implants
Tungsten	3422	19.3	411	Brittle below ~200–400 °C (polycrystal)	Lamp filaments, X-ray targets, welding electrodes, fusion divertors, heavy alloys
Rhenium	3186	21.0	463	Ductile but strongly work-hardening	Thermocouples, rocket thrusters, superalloy additions

Three problems limit their use. **Oxidation**: MoO_3 sublimes above about 700 °C and Nb and Ta absorb oxygen, so all need vacuum, inert gas or silicide coatings. **Low-temperature brittleness**: a high Peierls stress (Chapter 12) and boundary-segregated interstitials give W and Mo a ductile–brittle transition that depends sharply on purity and working; worked tungsten wire is ductile, recrystallized tungsten is not. **Density**: W and Ta are more than twice as dense as steel.

Because their melting points are so high, refractory metals are made by **powder metallurgy** (Chapter 46): tungsten powder is pressed, sintered by resistance heating, then swaged and drawn; potassium-bubble "doping" pins the grain boundaries of lamp filaments against sag at 2700 °C. **Tungsten heavy alloys** (90–97 % W, liquid-phase sintered with Ni–Fe) serve as balance weights and radiation shielding, and carbide-strengthened **TZM** (Mo–0.5Ti–0.08Zr–C) as forging dies. Tungsten is also the leading plasma-facing material for fusion reactors (Chapter 41).

Precious metals

The **precious metals** (gold, silver and the platinum-group metals Pt, Pd, Rh, Ir, Ru, Os) are FCC (Ru and Os are HCP) and are used for what their electronic structure gives them: nobility (no stable oxide at ambient temperature for Au and Pt; Ag tarnishes in sulfur), electrical conductivity (Ag has the lowest resistivity of any metal, $1.59 \times 10^{-8}\ \Omega\,\mathrm{m}$), catalytic activity and high melting points for the platinum group (Pt 1768 °C, Rh 1964 °C, Ir 2446 °C).

- **Catalysis**: Pt, Pd and Rh nanoparticles in automotive three-way catalysts; Pt and Ir in fuel cells and electrolysers (Chapter 40).
- **High-temperature hardware**: Pt–Rh bushings for drawing glass fibres (Chapter 17), crystal-growth crucibles, and type R, S and B thermocouples to about 1700 °C.
- **Electronics and joining**: gold bond wires and platings, silver pastes, silver brazing alloys.
- **Medicine and jewellery**: gold alloyed with Ag, Cu and Pd for hardness (18-carat is 75 wt% Au); Pt–Ir pacemaker electrodes.

Precious-metal engineering is largely about **using less**: thin platings, high-surface-area nanoparticles and recycling (Chapter 59).

16.7 Intermetallics and high-entropy alloys

Ordered intermetallics

An **intermetallic compound** is a phase with an ordered crystal structure and a narrow or moderate composition range, formed when unlike-atom bonds are favoured (Chapter 6). The same features that make the γ′ in superalloys a strong obstacle (ordering, APB energy, high melting points, and often Al or Si for oxidation resistance) also make intermetallics attractive as structural materials in their own right. Their common weakness is low ductility and toughness at room temperature: few independent slip systems, large Burgers vectors, and in some cases grain-boundary weakness.

Intermetallic	Structure	T_m (°C)	Density (Mg/m³)	Status
γ-TiAl (Ti-48Al-2Cr-2Nb, TNM)	$L1_0$ (tetragonal) + α_2 Ti_3Al	~1460	3.9–4.2	In service: low-pressure turbine blades in GEnx and LEAP engines; turbocharger wheels
Ni_3Al	$L1_2$	~1395	7.5	γ′ in superalloys; B-doped alloys (IC-series) for furnace hardware
NiAl	B2	1638	5.9	Bond coats; bulk use limited by brittleness
Fe_3Al, FeAl	$D0_3$, B2	~1500	6.7	Sulfidation-resistant hardware
$MoSi_2$	C11b	2030	6.3	Heating elements to 1800 °C (self-healing SiO_2 scale)
NiTi	B2 / B19′ martensite	~1310	6.45	Shape-memory alloys (Chapter 37)

The γ-TiAl blade is the success story. At half the density of a superalloy, with adequate creep and oxidation resistance to about 750 °C, γ-TiAl low-pressure turbine blades halve blade mass and, through lighter discs, save hundreds of kilograms per engine. It took three decades because room-temperature ductility is only 1–2 % and toughness 10–20 MPa $m^{1/2}$, so design, casting (now also electron-beam powder-bed fusion, Chapter 48) and inspection had to adapt. Boron segregated to grain boundaries famously made polycrystalline Ni_3Al ductile, a reminder that boundary chemistry can matter more than the bulk crystal.

High-entropy alloys

Conventional alloys have one principal element. In 2004 Cantor and Yeh independently showed that an alloy of five elements in equal proportions, CrMnFeCoNi, could form a single FCC solid solution rather than a mixture of intermetallics. The original rationale was configurational entropy. For an ideal random solution of N components in equal proportions the molar configurational entropy is (Chapter 6)

$$\Delta S_{\text{conf}} = -R \sum_{i} x_i \ln x_i = R \ln N$$

which is 5.76 J mol⁻¹ K⁻¹ for $N = 2$ and 13.38 J mol⁻¹ K⁻¹ for $N = 5$, hence the name **high-entropy alloy** (HEA), usually reserved for $\Delta S_{\text{conf}} \geq 1.5R$.

Worked example — How much can entropy do? At a typical homogenization temperature of 1000 °C (1273 K), the entropy term in the Gibbs energy of mixing of an equiatomic five-component solution is $-T\Delta S_{\text{conf}} = -1273 \times 13.38 = -17.0$ kJ/mol, compared with -7.3 kJ/mol for an equiatomic binary. This is comparable with the formation enthalpies of many intermetallic compounds between transition metals (typically −10 to −40 kJ/mol, and much more negative for aluminides and silicides). Entropy can therefore stabilize a solid solution at high temperature when the competing compounds are only weakly bonded, as among Cr, Mn, Fe, Co and Ni, but not when strong compounds can form, as soon as Al, Ti or Si are added in large amounts. At 500 °C (773 K) the entropy term falls to −10.3 kJ/mol, and indeed the Cantor alloy decomposes into Cr-rich σ, an $L1_0$ NiMn phase and a BCC Cr phase after long anneals at 500–700 °C. Configurational entropy is one contribution among several, not a guarantee.

> **Status of the principle: Ideal configurational entropy of mixing**
> *Derived.* Follows from Boltzmann's entropy for a random arrangement of atoms on one lattice; real alloys have short-range order, vibrational and magnetic entropy and non-zero mixing enthalpy, so the ideal value is an upper bound on the configurational term, not a prediction of phase stability.

Of the four proposed "core effects" (high entropy, lattice distortion, sluggish diffusion, a "cocktail" of properties), experiments support lattice distortion, while sluggish diffusion is not generally true and enthalpy matters as much as entropy. The field has therefore broadened into **multi-principal-element alloys**, designed with CALPHAD and machine learning (Chapters 56–57). Two families are of engineering interest:

- **FCC 3d-transition-metal alloys** (CrMnFeCoNi, CrCoNi) become *stronger and tougher* on cooling: the Cantor alloy exceeds 1 GPa UTS and 200 MPa $m^{1/2}$ toughness at 77 K, because a low stacking-fault energy activates nanotwinning (as in TWIP steels, Chapter 15), a candidate for cryogenic and hydrogen service (Chapter 41).
- **Refractory HEAs** such as NbMoTaW keep yield strengths of several hundred MPa at 1600 °C, but most are brittle at room temperature and oxidize catastrophically.

16.8 Selecting non-ferrous alloys

A comparison

Selection starts with the property profile of each family.

Family	Density (Mg/m³)	*E* (GPa)	Yield strength range (MPa)	Maximum service temperature (°C, typical)	Relative price per kg (steel = 1)	Distinctive strengths
Carbon and low-alloy steel (reference)	7.85	210	250–2000	400–550	1	Cheap, strong, stiff, weldable
Aluminium alloys	2.7–2.8	70–80	30–600	150–200	3–4	Light, formable, conductive, recyclable
Magnesium alloys	1.74–1.85	45	100–300	120–250	4–6	Lightest structural metal, castable
Titanium alloys	4.4–4.8	105–120	275–1400	400–600	15–40	Specific strength to 600 °C, corrosion, biocompatibility
Copper alloys	8.4–8.9	110–130	70–1200	200–300	8–12	Electrical and thermal conductivity, corrosion, bearings
Nickel superalloys	7.8–9.0	200–220	300–1300	650–1100	30–100+	Creep and oxidation resistance at high homologous temperature
Zinc alloys	6.6–7.1	85–100	200–380	50–100 (creep)	3–4	Precision die castings, galvanizing
Refractory metals	8.6–21	105–460	300–1500	1000–2500 (protected)	30–100	Highest melting points

The prices are indicative only and move with markets; for components the cost of processing often dominates the cost of metal, especially for titanium and superalloys.

Performance indices

Selection compares materials through **performance indices** that combine properties according to the loading and the objective (Chapter 58). For minimum mass:

Component and constraint	Index to maximize
Tie in tension, given stiffness	E/ρ
Tie in tension, given strength	σ_y/ρ
Beam in bending, given stiffness (section shape free)	$E^{1/2}/\rho$
Panel in bending, given stiffness (thickness free)	$E^{1/3}/\rho$
Panel in bending, given strength	$\sigma_y^{1/2}/\rho$
Beam in bending, given strength	$\sigma_y^{2/3}/\rho$

The indices explain why light alloys dominate some designs and are irrelevant to others. For a tie, aluminium, titanium, magnesium and steel all have $E/\rho \approx 25$–27 GPa/(Mg/m³), so a stiffness-limited tie weighs the same in any of them. For a panel, the exponent of E is $\frac{1}{3}$, and the low-density metal wins.

Worked example — The lightest stiff panel. A floor panel of given area must not deflect more than a given amount under a given load. Its bending stiffness is proportional to Et^3, so the required thickness scales as $t \propto E^{-1/3}$, and its mass as $m \propto \rho t \propto \rho/E^{1/3}$. With typical values:

Material	E (GPa)	ρ (Mg/m³)	$E^{1/3}/\rho$	Thickness relative to steel	Mass relative to steel
Steel	210	7.85	0.757	1.00	1.00
Ti-6Al-4V	114	4.43	1.09	1.23	0.69
Al 6061	69	2.70	1.52	1.45	0.50
Mg AZ31	45	1.77	2.01	1.67	0.38
Quasi-isotropic CFRP	60	1.55	2.53	1.52	0.30

The aluminium panel is 45 % thicker than the steel panel but half its mass; the magnesium panel is two-thirds thicker and weighs 38 % as much. This is why car-body closures, laptop lids and aircraft floors use aluminium, magnesium or composites. For a strength-limited tie, by contrast, σ_y/ρ is about 200 MPa/(Mg/m³) for Ti-6Al-4V, 180 for 7075-T6 and 180 for a quenched-and-tempered 4340 steel at 1400 MPa yield: the three are nearly equal, and the choice is decided by temperature, corrosion, fatigue, toughness, available section size, joining and cost.

Beyond the index

A real selection screens with hard constraints (temperature, environment, conductivity, biocompatibility, approvals), ranks the survivors by an index, then examines the leaders for fatigue and fracture (Chapter 14), galvanic compatibility (Chapter 43), joinability (Chapter 47), product forms, supply risk (cobalt, rhenium, platinum-group metals and rare earths are critical, Chapter 59) and recyclability; remelting aluminium needs only about 5 % of the energy of primary production.

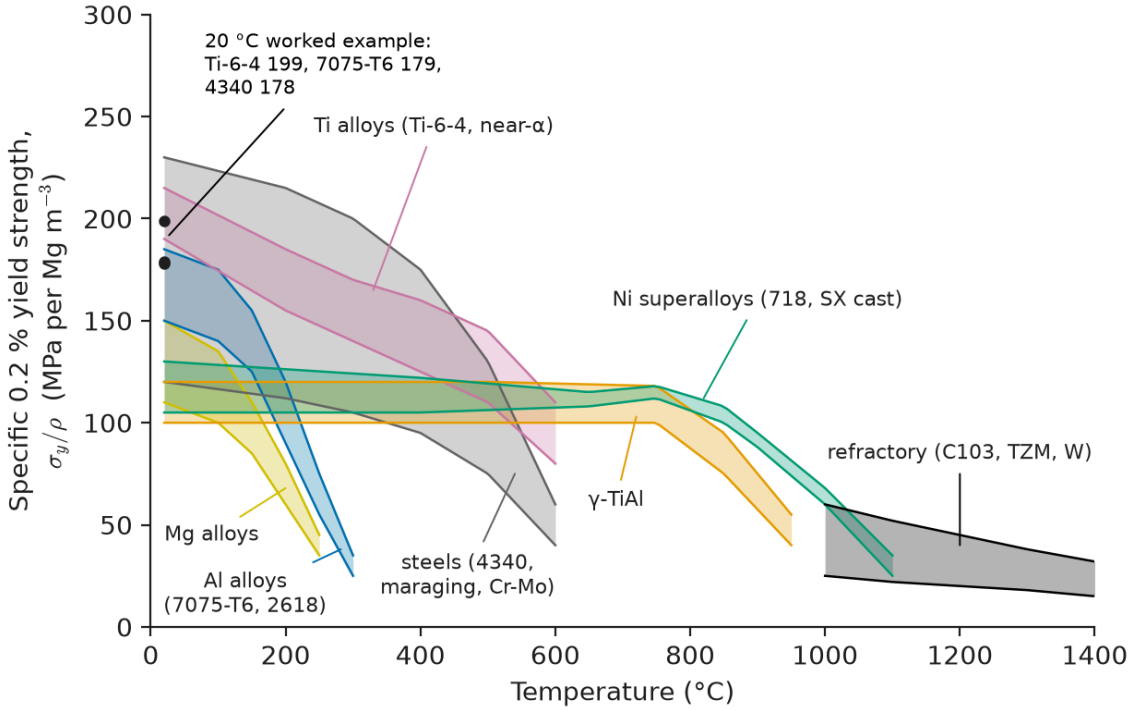

Figure 16.6 — Specific yield strength versus temperature for the main non-ferrous alloy families and steel, showing the temperature windows in which each family is the lightest choice.

Figure 16.6 adds temperature: by σ_y/ρ titanium leads from room temperature to about 550 °C, while aluminium and magnesium win below about 150–200 °C only for bending-dominated indices ($\sigma_y^{1/2}/\rho$, $E^{1/3}/\rho$) and on cost; above that come nickel superalloys (with γ-TiAl around 600–750 °C), and beyond 1100 °C only refractory metals and ceramics. Each boundary reflects a structural limit met in this chapter: precipitate coarsening, oxygen pick-up, γ′ dissolution.

What remains open

Open question	Where it is answered
How were the precipitation sequences, coarsening and recrystallization used here derived?	Chapter 10 (back reference)
How do cutting and Orowan bypass set the peak-aged strength?	Chapter 13 (back reference)
How are creep rupture, Larson–Miller data and fatigue used in design?	Chapter 14 (back reference); Chapter 44
How do steels compare in heat treatment and microstructure?	Chapter 15 (back reference)
How do zirconia thermal barrier coatings and ceramic-matrix composites work as materials?	Chapters 17, 23 and 41
What sets the conductivity of alloys, and how are superconductors such as Nb–Ti made?	Chapters 25 and 30
How do NiTi shape-memory alloys work?	Chapter 37
Which titanium, cobalt and magnesium alloys are used in the body, and how do they corrode there?	Chapter 39
How do aluminium, magnesium and copper alloys corrode, and how are they protected?	Chapters 42–43
How are these alloys cast, forged, extruded, welded and additively manufactured?	Chapters 46–48
How are performance indices derived and used systematically?	Chapter 58
How do recycling and critical-material constraints change alloy choice?	Chapter 59

Next: Chapter 17 leaves metals for materials held together by ionic and covalent bonds, explaining why ceramics and glasses are hard, stiff, refractory and brittle, and how powder processing, glass forming and toughening turn those properties into engineering components.

Chapter 17 — Ceramics and Glasses

Part IV · Metals, Ceramics and Construction Materials

In 2001 a series of zirconia femoral heads for hip replacements began to fracture inside patients, a few years after implantation. Zirconia had been chosen precisely because it is the toughest of the oxide ceramics, with a bend strength above 900 MPa, and millions of heads had been implanted without trouble. The failed heads came from a small number of batches made after the manufacturer changed its sintering furnace. In those batches the slightly different firing had left a microstructure in which the metastable tetragonal zirconia transformed to the monoclinic form far more readily than normal, slowly and at body temperature, in the presence of water. The transformation, which in a good microstructure *toughens* the ceramic by stopping cracks, had become a degradation mechanism that roughened and microcracked the surface until the head split. Nothing in the composition had changed; only the processing had.

Ceramics and glasses are the materials in which this sensitivity is greatest. Their bonds make them hard, stiff, chemically inert and refractory, and also brittle: their strength is set not by their atoms but by their largest flaw, and the flaws are made by processing. A glass bottle, a spark plug, a silicon-nitride bearing ball, a cooktop and a furnace lining all illustrate the same rule: in these materials, processing *is* properties.

> **Why are ceramics hard, stiff, refractory and brittle, and how is their brittleness managed?**
> **How are ceramic components made from powders and glasses made from melts, and how do sintering, viscosity, tempering and crystallization set their final properties?**

Concept	Engineering destination
Ionic–covalent bonding and crystal structure	Hardness, stiffness, melting point, chemical inertness, electrical insulation
Technical ceramics: alumina, zirconia, SiC, Si_3N_4	Wear parts, seals, bearings, cutting tools, armour, implants, substrates
Powder processing and sintering	Shrinkage allowances, porosity, grain size, strength of every ceramic part
Flaws, Weibull statistics, toughening	Design stresses, proof testing, reliability of ceramic components
Glass formation and network structure	Composition of window, container, fibre, display and laboratory glasses
Viscosity–temperature behaviour	Glass forming, annealing schedules, sealing, softening limits
Tempering and chemical strengthening	Safety glazing, vehicle glass, phone cover glass
Glass-ceramics	Cooktops, telescope mirrors, dental crowns, radomes
Refractories	Linings for steel, glass, cement and aluminium furnaces
Functional ceramics	Capacitors, sensors, actuators, fuel cells, magnets, phosphors

Roadmap. Bonding and structure (17.1) → traditional versus technical ceramics (17.2) → powder processing and sintering (17.3) → mechanical behaviour and toughening (17.4) → how glasses form (17.5) → viscosity, annealing and strengthening of glass (17.6) → glass-ceramics (17.7) → refractories (17.8) → functional ceramics, a pointer forward (17.9).

17.1 Structure and bonding of engineering ceramics

Mixed ionic–covalent bonding

A ceramic is an inorganic, non-metallic solid, usually a compound of a metal or metalloid with oxygen, nitrogen, carbon or boron. Its bonds are mixtures of ionic and covalent character. Pauling's estimate of the fractional ionic character of a bond between elements of electronegativities χ_A and χ_B (a bonding result recapped from Advanced Chemistry) is

$$f_{\text{ionic}} = 1 - \exp\left[-\tfrac{1}{4}(\chi_A - \chi_B)^2\right]$$

With Pauling electronegativities this gives about 68 % for Mg–O, 67 % for Zr–O, 57 % for Al–O, 45 % for Si–O, 40 % for Al–N, 28 % for Si–N and only 10 % for Si–C. The oxides are predominantly ionic; the nitrides and carbides are predominantly covalent.

> **Status of the principle: Pauling fractional ionic character**
> *Empirical.* A one-parameter interpolation calibrated on dipole moments of diatomic molecules; useful for ranking bonds, not as a quantitative measure of charge transfer in solids.

The structures follow the bonding (Chapter 2). Ionic oxides adopt structures that pack large anions around smaller cations with coordination numbers set by the radius ratio and with local charge balance (Pauling's rules): rock salt for MgO, corundum for Al_2O_3 (Al in octahedral sites of a nearly HCP oxygen array), fluorite for cubic ZrO_2 and CeO_2, perovskite for $BaTiO_3$, spinel for $MgAl_2O_4$. Covalent ceramics adopt structures dictated by directional sp^3 bonds: SiC is a tetrahedral network with more than 200 stacking polytypes (3C, 4H, 6H are the common ones); Si_3N_4 consists of corner-sharing SiN_4 tetrahedra in α and β hexagonal forms; AlN is wurtzite. Silica and the silicates are built from SiO_4 tetrahedra linked through shared corners, and the many ways of linking them (isolated, chains, sheets, frameworks) produce the mineral and glass families that dominate traditional ceramics and glasses.

Consequences of the bonding

Strong, directional or charge-constrained bonds produce the characteristic property profile:

- **High melting points**: Al_2O_3 2054 °C, ZrO_2 2715 °C, MgO 2852 °C; SiC and Si_3N_4 decompose (about 2700 °C and 1900 °C).
- **High stiffness and hardness**: E of 300–450 GPa; Vickers hardness 15–30 GPa, against 2–8 GPa for hardened steels; low density for the covalent ceramics (2.5–3.2 Mg/m^3).
- **Brittleness**: in covalent crystals the Peierls stress is a large fraction of the shear modulus; in ionic crystals slip must avoid bringing like charges together, leaving too few slip systems for polycrystal plasticity (Chapters 12–13). A crack advances by breaking bonds before any dislocation can blunt it.
- **Insulation and inertness** for the oxides (with exceptions, Section 17.9); low thermal conductivity for complex oxides but very high for simple, stiff, light-atom crystals such as SiC and AlN (Chapter 24).

Material	Density (Mg/m^3)	E (GPa)	Flexural strength (MPa)	K_{IC} (MPa $m^{1/2}$)	Vickers hardness (GPa)	k (W m^{-1} K^{-1})	CTE (10^{-6} K^{-1})
Alumina, 99.5 %	3.90	370	350–450	3.5–4.5	15–18	30	8.0
Alumina, 96 %	3.75	300	300–350	3.5	12–14	24	7.5
Zirconia, 3Y-TZP	6.05	205	900–1200	5–10	12–13	2.5–3	10.5
Mg-PSZ	5.75	200	600–700	8–12	11	2.5	10
Sintered SiC	3.15	410	400–500	3–4	25	100–120	4.0
Reaction-bonded SiC (SiSiC)	3.10	350–390	250–350	3.5–4	20–25	125–150	4.3
Si_3N_4, gas-pressure sintered	3.20	300–320	700–1000	6–8	15–16	20–30	3.2
AlN	3.26	330	300–350	3	11	170–200	4.5
B_4C	2.52	450	350–450	2.5–3.5	30–35	30	5.0
Soda-lime glass	2.50	70	50–70 (as used)	0.75	5.5	1.0	9.0
Fused silica	2.20	72	50–70 (as used)	0.75	6–7	1.4	0.55

Values are typical for dense commercial grades; strengths depend strongly on processing and specimen size (Section 17.4).

17.2 Traditional versus technical ceramics

Traditional ceramics

Traditional ceramics (whitewares, bricks, tiles, sanitaryware) are made from clays, feldspars and quartz. Clay minerals such as kaolinite, $Al_2Si_2O_5(OH)_4$, are fine sheet silicates that become plastic with water and hold their shape when dried. On firing at 1000–1400 °C the feldspar flux melts to a viscous glass that dissolves part of the clay and quartz, and mullite ($3Al_2O_3{\cdot}2SiO_2$) needles crystallize. A **triaxial porcelain** (about 50 % clay, 25 % feldspar, 25 % quartz) ends as mullite, quartz and glass, the classic insulator and tableware material. This **vitrification**, densification by viscous flow of a liquid, is the ancestor of liquid-phase sintering. Bricks use cheaper clays fired lower and stay porous (Chapter 18).

Technical ceramics

Technical (advanced, engineering) ceramics are made from synthetic, high-purity powders of controlled particle size, formed and sintered to near full density and fine grain size. Four materials account for most structural use.

Alumina (Al_2O_3) is the workhorse: cheap, hard, insulating and stable, in purities from 85 % to 99.9 %. Lower grades contain a grain-boundary silicate glass from liquid-phase sintering and are easier to metallize but weaker when hot. Uses include wear parts, seals, armour, substrates, spark-plug insulators and hip-joint bearings (Chapter 39).

Zirconia (ZrO_2) is monoclinic below about 1170 °C, tetragonal up to about 2370 °C and cubic to its melting point. The tetragonal-to-monoclinic transformation on cooling is martensitic and involves a volume increase of about 4 % that shatters pure zirconia. Dissolving Y_2O_3, MgO or CaO stabilizes the high-temperature forms (Chapter 7). With about 3 mol% Y_2O_3 and a grain size below about 0.5 μm, the tetragonal phase is retained metastably at room temperature (**3Y-TZP**, tetragonal zirconia polycrystal); in **Mg-PSZ** (partially stabilized zirconia) tetragonal precipitates sit inside cubic grains. Near a crack tip, the stress field triggers the transformation, and the expansion of the transformed zone squeezes the crack shut: **transformation toughening**, which raises K_{IC} to 5–12 MPa $m^{1/2}$ (Chapter 14). The same metastability permits slow transformation in warm, moist environments (**low-temperature degradation**, the failure of the opening paragraph), controlled by grain size, yttria distribution and small alumina additions. Zirconia makes dental crowns, fibre-optic ferrules, pump plungers, oxygen sensors and fuel-cell electrolytes (Section 17.9), and turbine-blade thermal barrier coatings (Chapter 16).

Silicon carbide (SiC) is made by the Acheson process (carbothermal reduction of silica with coke at about 2500 °C). Dense parts are **solid-state sintered** with B and C at about 2100 °C (SSiC) or **reaction bonded** by infiltrating a SiC–carbon preform with molten silicon (SiSiC, 8–15 % free Si). SiC combines hardness, high conductivity, low expansion and resistance to acids and oxidation (protective SiO_2 to about 1600 °C): seal faces, burner and heat-exchanger tubes, armour, telescope mirrors, SiC/SiC composites (Chapter 23) and power semiconductors (Chapter 33).

Silicon nitride (Si_3N_4) hardly self-diffuses and decomposes before densifying, so it is sintered with 5–10 % Y_2O_3, Al_2O_3 or MgO, which form an oxynitride liquid at 1700–1850 °C under nitrogen. The α powder dissolves and reprecipitates as elongated β-Si_3N_4 grains (aspect ratio 5–10) that deflect and bridge cracks, an **in-situ toughened** ceramic with K_{IC} of 6–8 MPa $m^{1/2}$. With the best strength, toughness and thermal-shock resistance of the monolithic ceramics, it makes bearing balls (light, and non-conductive, so immune to electrical arcing in motors), turbocharger rotors, cutting tools and power-electronics substrates; **SiAlONs** reduce the boundary glass.

> **Boundary.**
> The crystal structures (corundum, fluorite, wurtzite, perovskite, spinel) and Pauling's rules are developed in Chapter 2; ceramic phase diagrams (Al_2O_3–SiO_2, ZrO_2–Y_2O_3, MgO–Al_2O_3) in Chapter 7.

17.3 Ceramic powder processing

A metal can be melted, cast and worked. A ceramic usually cannot: its melting point is too high, its melt too reactive, and it cannot be deformed plastically to close casting porosity. Almost every technical ceramic is therefore made by **powder processing**: synthesize a powder, form it into a "green" body of the desired shape, and fire it so that the particles bond and the pores between them shrink and disappear. Every flaw the final part contains, every pore and agglomerate and inclusion, traces back to one of these steps.

Powder synthesis

Ceramic	Main synthesis route	Key powder characteristics
Al_2O_3	Bayer process: digestion of bauxite in NaOH, precipitation of $Al(OH)_3$, calcination at 1100–1300 °C	0.3–5 μm, purity 99.5–99.99 %
ZrO_2 (Y-doped)	Co-precipitation of mixed hydroxides from chloride solutions, or hydrothermal synthesis; calcination	20–100 nm crystallites, yttria uniformly distributed
SiC	Acheson carbothermal reduction; milled and classified	Sub-micrometre α-SiC for sintering
Si_3N_4	Direct nitridation of Si powder, or diimide precipitation	> 90 % α phase, sub-micrometre
$BaTiO_3$	Solid-state reaction of $BaCO_3$ + TiO_2, or hydrothermal / oxalate routes	0.1–0.5 μm for thin-layer capacitors
Glasses and coatings	Sol–gel: hydrolysis and condensation of alkoxides such as $Si(OC_2H_5)_4$	Molecular-scale mixing, low processing temperature

Fine powders sinter faster but agglomerate. Hard agglomerates pack poorly and leave large pores between them that do not close; the best ceramics come from powders that are fine, narrowly distributed and dispersed, often in a liquid with dispersants that control the surface charge (colloidal processing, Chapter 37).

Forming

Forming method	How it works	Typical products
Uniaxial dry pressing	Granulated powder pressed in a steel die at 50–150 MPa	Tiles, substrates, insulators, seal rings
Cold isostatic pressing (CIP)	Powder in a flexible mould pressed by fluid at 100–300 MPa	Spark-plug insulators, large blanks, tubes
Slip casting	Suspension poured into a porous plaster mould that absorbs liquid	Sanitaryware, crucibles, complex hollow shapes
Tape casting	Slurry spread by a doctor blade into thin sheets	Multilayer capacitors, substrates, SOFC electrolytes
Extrusion	Plastic paste forced through a die	Honeycomb catalyst supports, tubes, bricks
Ceramic injection moulding	Powder in thermoplastic binder (~40 vol%) moulded like a polymer	Small complex parts: dental brackets, ferrules, watch cases
Gel casting, additive manufacturing	In-situ polymerization; vat photopolymerization, binder jetting	Complex and customized parts (Chapter 48)

The green body typically contains 50–65 % of the theoretical density. Organic binders, plasticizers and dispersants are burned out slowly (at 1–2 K/min through 200–500 °C) before sintering, or their gases will crack the part.

The driving force for sintering

Sintering consolidates a powder compact by atomic transport at 0.6–0.8 of the absolute melting temperature. Its driving force is the reduction of interfacial energy (Chapter 6): for 1 μm alumina (about 1.5 m^2/g, $\gamma_{sv} \approx 1$ J/m^2) only about 1.5 J/g, which is why sintering needs high temperatures and fine powders. Locally it appears as a curvature-driven chemical-potential difference (Gibbs–Thomson) that moves atoms from convex surfaces and grain boundaries to the concave necks.

Stages and mechanisms

Sintering is described in three geometric stages (Figure 17.1):

1. **Initial stage**: necks form between particles; the neck radius grows as a power of time; linear shrinkage up to about 3–5 %.

2. **Intermediate stage**: pores form a continuous, interconnected network along grain edges; most densification occurs here, from about 65 % to about 92 % of theoretical density.
3. **Final stage**: above about 92 % density the pore channels pinch off into isolated pores at grain corners; the last pores close slowly, and grain growth competes with densification.

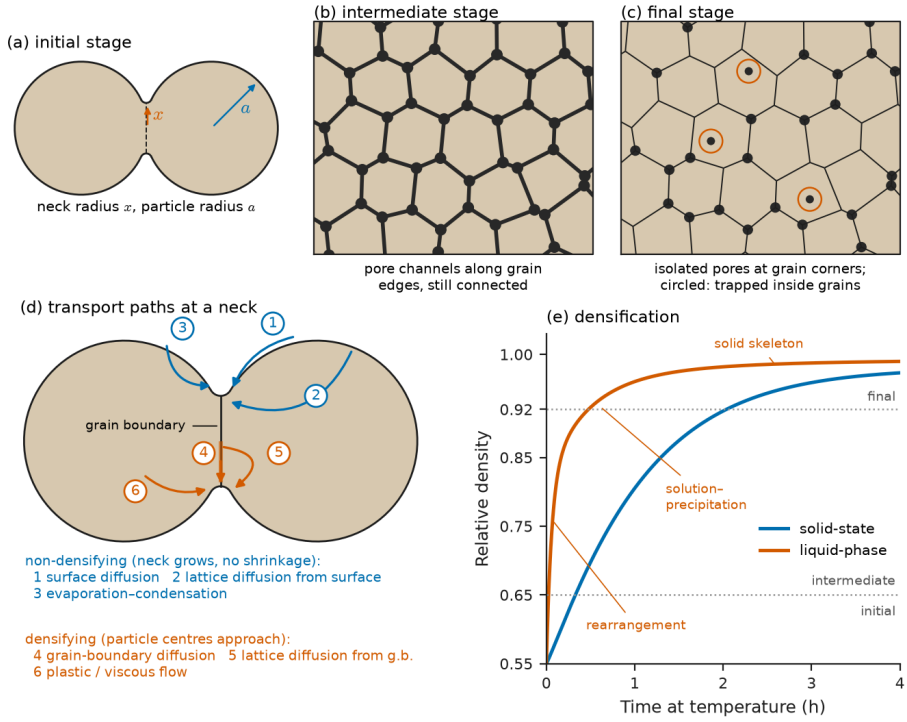

Figure 17.1 — Stages and mechanisms of solid-state sintering, and relative density versus sintering time for solid-state and liquid-phase sintering.

Matter can move by several paths. **Surface diffusion**, **evaporation–condensation** and **lattice diffusion from the particle surface** move atoms from one part of the pore surface to another; they grow the necks and round the pores but do not bring particle centres together, so they cause no shrinkage. **Grain-boundary diffusion** and **lattice diffusion from the grain boundary** remove atoms from the boundary between particles and deposit them at the neck; particle centres approach, and the compact shrinks and densifies. Viscous flow plays the same densifying role in glasses. Whether a powder densifies or merely coarsens depends on which paths dominate. A high ratio of grain-boundary to surface diffusivity favours densification; fast surface diffusion at low temperature wastes the driving force by coarsening the particles without shrinkage, which is why fast heating rates sometimes give denser products.

Herring's scaling law states that if all particle sizes G are scaled by a factor λ, the time to achieve the same geometric change scales as λ^m, with $m = 2$ for viscous flow, 3 for lattice diffusion and 4 for grain-boundary diffusion. Halving the particle size therefore shortens the sintering time by a factor of 8 for lattice diffusion and 16 for grain-boundary diffusion, or allows sintering at a lower temperature. This is the main reason ceramic processing has moved to sub-micrometre and nanometre powders.

> **Status of the principle: Herring scaling law**
> *Derived.* Follows from dimensional analysis of curvature-driven transport when the geometry is self-similar and a single mechanism operates; real powders with size distributions, agglomerates and changing mechanisms follow it only approximately.

Grain growth and the role of dopants

In the final stage, a boundary that breaks away from its pores strands them inside grains, far from any vacancy sink, and they never close. Coble's classic remedy for alumina was 0.05–0.25 % MgO, which segregates to boundaries, slows them, and lets the pores be eliminated, giving translucent alumina for lamp envelopes; pinning particles (Zener pinning, Chapter 10) do the same.

Worked example — Shrinkage allowance for a sintered alumina part. A seal ring must be 50.0 mm in outer diameter after sintering to 98 % of theoretical density. The pressed green body has 55 % of theoretical density. Mass is conserved and, if

shrinkage is isotropic, the linear dimension scales as the cube root of the density ratio:

$$\frac{L_f}{L_g} = \left(\frac{\rho_g}{\rho_f}\right)^{1/3} = \left(\frac{0.55}{0.98}\right)^{1/3} = 0.825$$

The linear shrinkage is 17.5 %, and the green part must be pressed (or machined in the green state) to $50.0/0.825 = 60.6$ mm. A green-density variation of ±1 % across the part (typical of uniaxial pressing, where die-wall friction leaves the corners denser than the centre) changes the local shrinkage by about ±0.3 percentage points, which on a 60 mm part is about ±0.2 mm of distortion. This is why precision ceramic parts are often isostatically pressed, ground after sintering with diamond tools, and why grinding can account for half the cost of a technical ceramic component.

Liquid-phase sintering

Many ceramics (and cemented carbides, Chapter 46) are sintered with a small amount of liquid that wets the solid. The liquid speeds sintering in three overlapping stages: **rearrangement**, as capillary forces pull particles into closer packing; **solution–precipitation**, in which small particles and contact points dissolve in the liquid and the material reprecipitates on larger grains, flattening contacts and bringing grain centres together; and **solid-state skeleton** densification once the grains have formed a rigid network. Diffusion through a liquid is orders of magnitude faster than through a solid, so liquid-phase sintering works at lower temperatures and for materials such as Si3N4 and SiC that would hardly sinter otherwise. The price is a residual grain-boundary glass that softens at high temperature and reduces creep and slow-crack-growth resistance, as in 96 % alumina and in silicon nitride above about 1200 °C. The dihedral angle and wetting behaviour set by interfacial energies (Chapter 6) decide whether the liquid penetrates the grain boundaries.

Pressure-assisted sintering

Pressure adds a driving force far larger than the capillary one. **Hot pressing** (20–50 MPa in graphite dies), **hot isostatic pressing** (gas at 100–200 MPa, also used to close residual pores after sintering) and **spark plasma sintering** (pulsed current through die and powder, densifying in minutes) reach full density with little grain growth, at a cost reserved for bearing balls, armour, transparent ceramics and cutting tools.

17.4 Mechanical behaviour of ceramics

Strength is set by flaws

Chapter 14 showed that the fracture of a brittle solid is governed by linear-elastic fracture mechanics. A crack of depth a in a surface loaded in tension fails when the stress-intensity factor reaches the fracture toughness:

$$\sigma_f = \frac{K_{IC}}{Y\sqrt{\pi a}}$$

with $Y \approx 1.1$–1.3 for surface cracks. With $K_{IC} = 4$ MPa m$^{1/2}$, alumina with a 30 μm flaw fails at about 370 MPa, and with a 300 μm flaw at about 120 MPa. Because K_{IC} of ceramics is only 1–10 MPa m$^{1/2}$ (against 30–200 for structural metals), flaws of tens of micrometres control the strength. These flaws come from processing (pores, agglomerates, inclusions, large grains) and from machining and handling (grinding cracks, chips, contact damage). The theoretical strength of the bonds, about $E/10$, is never approached in bulk ceramics.

Three consequences follow. Ceramics are much stronger in **compression** (typically 5–15 times their tensile strength), because compressive stresses close flaws and failure requires many cracks to grow and link; ceramics are used as columns, armour faces and bearing balls, where loads are mainly compressive. Their tensile strength is **variable**, because the largest flaw differs from part to part. And it depends on **specimen size and stress distribution**, because a larger stressed volume is more likely to contain a large flaw.

Weibull statistics

The weakest-link statistics of Chapter 14 describe this variability. The probability that a component of volume V fails at or below a uniform stress σ is

$$P_f = 1 - \exp\left[-\frac{V}{V_0}\left(\frac{\sigma}{\sigma_0}\right)^m\right]$$

where σ_0 is the characteristic strength of the reference volume V_0 and m is the **Weibull modulus**. A large m means a narrow strength distribution: m is 5–10 for traditional ceramics and glass as used, 10–20 for good technical ceramics, 20–30 for the best silicon nitride, and 30–100 for metals. Data are plotted as $\ln\ln[1/(1-P_f)]$ against $\ln\sigma$, which gives a straight line of slope m (Figure 17.2). For non-uniform stress fields, such as a bend test, V is replaced by an **effective volume** that weights each element by $(\sigma/\sigma_{\max})^m$; for a three-point bend bar it is only a few percent of the specimen volume.

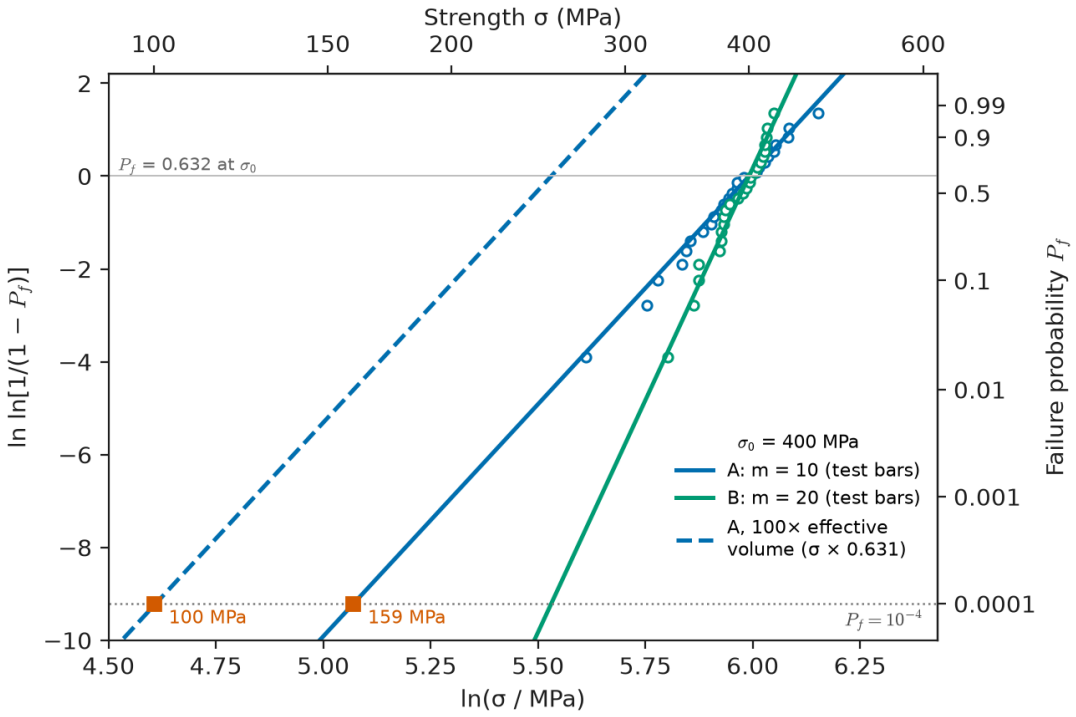

Figure 17.2 — Weibull plot of bend-strength data for two aluminas with Weibull moduli 10 and 20, and the predicted strength reduction for a component with 100 times the effective volume.

Worked example — From test bars to a design stress. Bend bars of an alumina have a characteristic strength $\sigma_0 = 400$ MPa (63.2 % failure probability) and $m = 10$ for their effective volume. What stress gives a failure probability of 10^{-4}, first for a part with the same effective volume and then for a part with 100 times the effective volume?

For the same volume, solving the Weibull equation for σ:

$$\sigma = \sigma_0\,[-\ln(1-P_f)]^{1/m} = 400 \times (1.0\times10^{-4})^{0.1} = 400 \times 0.398 = 159\ \text{MPa}$$

For 100 times the volume, the stress for equal failure probability falls by $100^{-1/m} = 100^{-0.1} = 0.631$, to 100 MPa. A designer who used the mean bend strength (about 380 MPa) would have been off by a factor of nearly four. With $m = 20$ the same two steps give 252 MPa and 200 MPa; doubling the Weibull modulus is worth as much as doubling the mean strength. Volume scaling assumes the same flaw population in test bar and part, which fails when the part's surfaces are machined differently or when different flaws (edge chips, internal pores) dominate.

> **Status of the principle: Weibull size scaling of strength**
> *Approximate.* Follows from weakest-link statistics with a single flaw population and a power-law distribution of flaw strengths; it fails when several flaw populations coexist, when flaws are surface- rather than volume-distributed (use effective area), or when R-curve behaviour or slow crack growth changes the failure condition.

Proof testing loads every part above its service stress so that those with large flaws break in the test, truncating the flaw distribution, provided slow crack growth does not enlarge flaws afterwards.

Slow crack growth and thermal shock

Silicate glasses and many oxides suffer **slow crack growth** in the presence of water, which reacts with stressed bonds at the crack tip, so cracks grow below K_{IC} at a velocity $v \approx AK_I^n$, with n about 15–20 for soda-lime glass, 30–50 for alumina and above 50 for silicon nitride. Long-term design stresses for glass are therefore a third or less of short-term strengths.

A sudden temperature change also produces stress. A surface cooled suddenly by ΔT is constrained by the hot interior and develops a tensile stress up to $E\alpha\Delta T/(1-\nu)$ (Chapter 24). Setting this equal to the fracture strength gives the critical temperature drop for severe quenching,

$$\Delta T_c = \frac{\sigma_f (1-\nu)}{E\alpha}$$

With typical values this is about 90 K for 99.5 % alumina, 230 K for sintered SiC, about 590 K for silicon nitride, 60 K for soda-lime glass, 190 K for borosilicate glass and about 1000 K for fused silica. The ranking explains why Si_3N_4 is used for turbocharger rotors and molten-metal handling, borosilicate for laboratory glassware and cookware, and fused silica for furnace windows and crucibles.

Toughening ceramics

Because strength is K_{IC} divided by flaw size, ceramics are strengthened in two ways: by **reducing flaw size** (clean powders, colloidal processing, fine grains, careful machining) and by **raising toughness**. Chapter 14 introduced the toughening mechanisms; in ceramic engineering they appear as follows:

Mechanism	How it works	Example materials	K_{IC} (MPa m$^{1/2}$)
None (single-phase, fine grains)	Crack follows the cleanest path	Fine-grained alumina, glass	0.7–4
Crack deflection and grain bridging	Elongated or large grains force the crack to tilt and leave intact ligaments behind the tip	β-Si_3N_4, coarse-grained alumina, SiC with liquid-phase sintering	5–10
Transformation toughening	Stress-induced t→m transformation of ZrO_2 shields the crack tip	3Y-TZP, Mg-PSZ, zirconia-toughened alumina	5–15
Microcracking	A process zone of microcracks around the crack reduces local stress	ZrO_2-containing composites, some aluminas	4–8
Fibre bridging and pull-out	Fibres with weak interfaces bridge the crack over millimetres	SiC/SiC, C/SiC, oxide/oxide CMCs	15–30 (R-curve)

Shielding mechanisms produce a **rising R-curve**, which makes strength less sensitive to flaw size and raises the Weibull modulus, the real engineering goal; CMCs become quasi-ductile and notch-insensitive (Chapter 23).

17.5 Glass formation

Glasses as frozen liquids

A **glass** is an amorphous solid that shows a glass transition. Any liquid can in principle become a glass if it is cooled fast enough to avoid crystallization; a good glass-former is simply a liquid in which crystallization is very slow. On cooling, a crystallizing liquid shows a discontinuous drop in volume at the melting point. A glass-forming liquid instead continues along the liquid line as a **supercooled liquid**, becoming steadily more viscous, until at the **glass transition temperature** T_g its structure can no longer rearrange on the time scale of cooling. Below T_g the structure is frozen and the volume follows a line parallel to that of the crystal (Figure 17.3). The temperature at which the frozen structure would have been the equilibrium liquid is the **fictive temperature**; faster cooling freezes in a higher fictive temperature, a more open structure and a slightly lower density, which is why the "glass transition temperature" depends on the cooling rate.

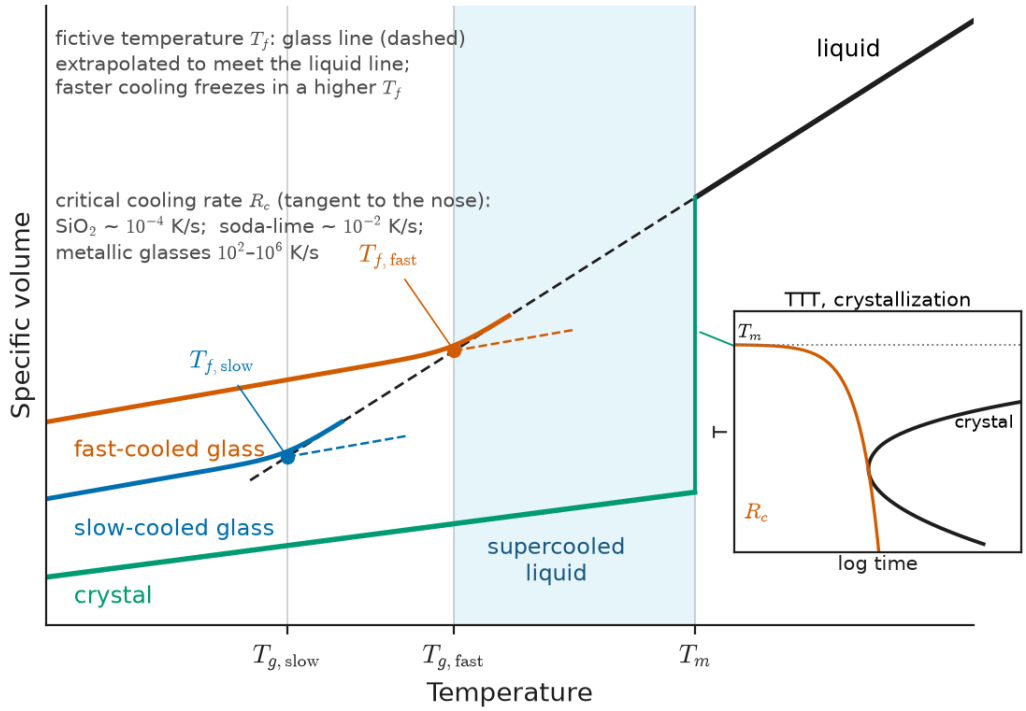

Figure 17.3 — Specific volume versus temperature for a glass-forming liquid, showing crystallization at Tm and the glass transition for slow and fast cooling.

The kinetic criterion is quantified by a time–temperature–transformation curve for crystallization (Chapter 9). The **critical cooling rate** that just misses its nose is of order 10^{-4} K/s for silica, so silica glass forms even in massive bodies, 10^{-2}–10^{-1} K/s for common silicate glasses, and 10^{2}–10^{6} K/s for metallic glasses (Chapter 9). A liquid forms a glass easily when its viscosity at the melting point is high (silica: about 10^{5}–10^{6} Pa s at 1723 °C, against 10^{-3} Pa s for water or liquid metals), when crystallization requires large rearrangements of a connected network, and when the composition is near a deep eutectic, where the liquid is stable to low temperature.

> **Status of the principle: Kinetic criterion for glass formation**
> *Fundamental.* Glass formation is avoidance of crystallization; whether a liquid vitrifies is decided by comparing the cooling rate with the nucleation and growth rates, which follow from classical nucleation theory (Chapter 9).

Zachariasen's rules

Why do some oxides form glasses easily and others not at all? In 1932 Zachariasen proposed that a glass has the same local building blocks as the crystal, the same coordination polyhedra with the same bond lengths, but connected into a **continuous random network** without long-range periodicity (Figure 17.4). For an oxide A_mO_n to form such a network with an energy comparable to the crystal's, he argued, four conditions should hold:

1. Each oxygen is linked to no more than two cations A.
2. The coordination number of oxygen around A is small, 3 or 4.
3. The oxygen polyhedra share corners, not edges or faces.
4. At least three corners of each polyhedron are shared (for a three-dimensional network).

SiO_2 (SiO_4 tetrahedra), GeO_2, B_2O_3 (BO_3 triangles) and P_2O_5 satisfy the rules and are the **network formers**; MgO, CaO and Na_2O, with octahedral coordination and edge sharing in their crystals, do not.

> **Status of the principle: Zachariasen's rules**
> *Empirical.* Structural criteria inferred from the crystal chemistry of known glass-formers; they successfully classify the oxides but say nothing about kinetics, and they do not apply to metallic, chalcogenide or molecular glasses.

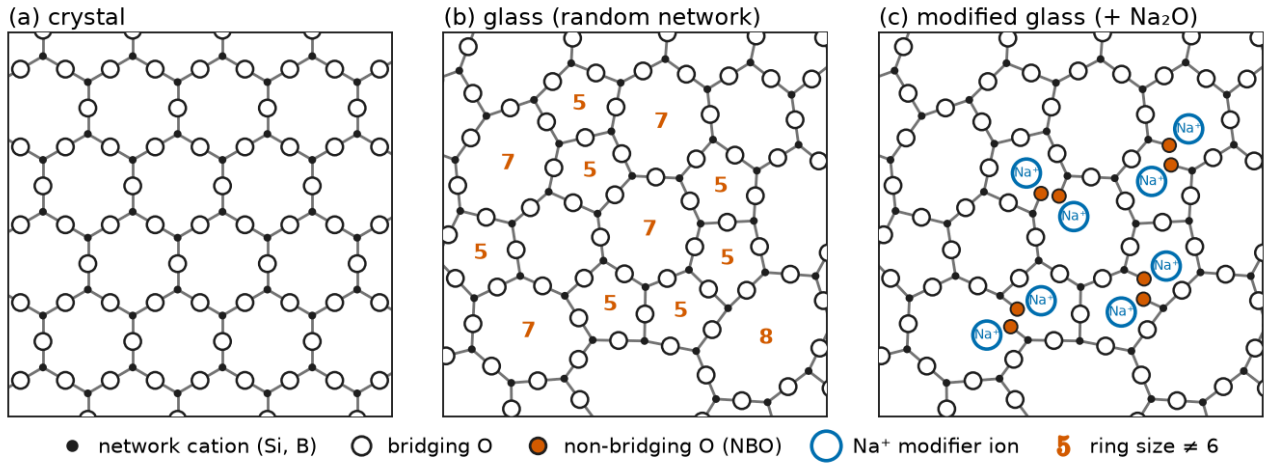

Figure 17.4 — Two-dimensional representations of crystalline silica, silica glass (continuous random network) and a sodium silicate glass with non-bridging oxygens.

Network formers, modifiers and intermediates

Pure silica glass is difficult to make: it must be melted above 2000 °C. Adding alkali and alkaline-earth oxides, the **network modifiers**, breaks Si–O–Si bridges. Each Na_2O contributes an oxygen that converts one bridging oxygen into two **non-bridging oxygens** (NBOs), each charge-balanced by a nearby Na^+:

$$\equiv Si{-}O{-}Si \equiv \; + Na_2O \; \rightarrow \equiv Si{-}O^- \; Na^+ \quad Na^+ \; {}^-O{-}Si \equiv$$

The network is depolymerized: viscosity falls by orders of magnitude, the melting temperature drops to 1400–1600 °C, the thermal expansion rises, and the chemical durability falls (pure sodium silicate is "water glass", soluble in water). Alkaline-earth oxides (CaO, MgO) restore durability because the doubly charged ions bind two NBOs and are less mobile. A simple measure of depolymerization is the number of NBOs per tetrahedron, NBO/T. For a soda-lime glass of molar composition 74 SiO_2–16 Na_2O–10 CaO, each mole of modifier oxide creates two NBOs, so $\text{NBO/T} = 2(16 + 10)/74 = 0.70$; on average each tetrahedron has 3.3 bridging corners, still a connected network. **Intermediates** such as Al_2O_3, TiO_2, ZnO and PbO can enter the network (Al as AlO_4^- tetrahedra charge-balanced by an alkali ion, which consumes NBOs and repolymerizes the network) or act as modifiers.

Glass	Typical composition (wt%)	Key properties	Uses
Fused silica	~100 SiO_2	CTE 0.55 × 10^{-6} K^{-1}; softening ~1600 °C; UV-transparent	Furnace tubes, lithography optics, optical fibres (doped), crucibles
Soda-lime (float, container)	72 SiO_2, 14 Na_2O, 9 CaO, 4 MgO, 1 Al_2O_3	CTE 9.0 × 10^{-6} K^{-1}; cheap, easily melted	Windows, bottles, jars, about 90 % of all glass
Borosilicate	81 SiO_2, 13 B_2O_3, 4 Na_2O, 2 Al_2O_3	CTE 3.3 × 10^{-6} K^{-1}; chemical durability; thermal-shock resistance	Laboratory ware, cookware, pharmaceutical vials, fire-resistant glazing
Alkali aluminosilicate	~60 SiO_2, 15–20 Al_2O_3, 12–14 Na_2O, 3–5 MgO, K_2O	Open network for fast ion exchange	Phone and display cover glass (chemically strengthened)
E-glass	54 SiO_2, 14 Al_2O_3, 22 CaO+MgO, 7–10 B_2O_3, < 1 alkali	Low alkali, good electrical insulation	Reinforcing fibres (Chapter 22), circuit-board laminates
Lead crystal	55–65 SiO_2, ≥ 24 PbO, K_2O	Refractive index ~1.55–1.57, high dispersion	Tableware, decorative and radiation-shielding glass
Non-oxide glasses	Chalcogenides (As–Se, Ge–Sb–Te), fluorides (ZrF_4–BaF_2), metallic glasses	Infrared transparency; phase-change memory; high strength	IR optics, rewritable memory, springs and casings (Chapter 9)

Boron is an unusual former: alkali first converts BO_3 triangles into BO_4^- tetrahedra, strengthening the network, and only beyond a certain alkali/boron ratio creates NBOs (the **boron anomaly**), one reason for the low expansion and durability of borosilicates.

17.6 Glass properties and processing

Viscosity and the reference points

The viscosity of a glass-forming melt changes continuously by about 15 orders of magnitude between melting and room temperature, and every glass process is defined by viscosity rather than by temperature. The industry uses **reference points**:

Reference point	Viscosity (Pa s)	Meaning	Soda-lime (°C)	Borosilicate (°C)	Fused silica (°C)
Melting	~10	Fining and homogenizing the melt	~1450	~1600	> 2000
Working point	10^3	Gobs formed in pressing and blowing	~1005	~1250	—
Softening (Littleton) point	$10^{6.6}$	Glass deforms under its own weight	~726	~821	~1600
Annealing point	10^{12}	Internal stresses relax in minutes	~548	~560	~1100
Strain point	$10^{13.5}$	Stresses relax in hours; glass rigid for practical purposes	~511	~510	~1000

Values are typical; they shift by tens of kelvin with composition within each family.

Over this range, the viscosity of most silicate melts is described well by the **Vogel–Fulcher–Tammann** (VFT) equation,

$$\log_{10}\eta = A + \frac{B}{T - T_0}$$

where A, B and T_0 are fitted constants. For a typical soda-lime glass, with T in °C and η in Pa s, $A = -4.57$, $B = 6370$ °C and $T_0 = 160$ °C fit the four reference points to within about 0.15 in $\log_{10}\eta$ (Figure 17.5). The divergence at T_0, well below T_g, is never reached because the melt falls out of equilibrium at T_g. Silica follows a nearly Arrhenius law (small T_0), the behaviour Angell called a **strong** liquid; depolymerized silicates, molecular liquids and polymers depart further from Arrhenius behaviour and are **fragile**. Fragility explains why soda-lime glass has a long "working range" in which it can be shaped, and why polymers soften over a narrow temperature range (the WLF equation of Chapter 20 is equivalent in form to VFT).

> **Status of the principle: Vogel–Fulcher–Tammann equation**
> *Empirical.* A three-parameter fit to viscosity data; it is accurate from the melting range to near T_g for most oxide glasses, but it is not reliable for extrapolation below T_g, where the glass is out of equilibrium, and alternative forms (Adam–Gibbs, MYEGA) fit some data better.

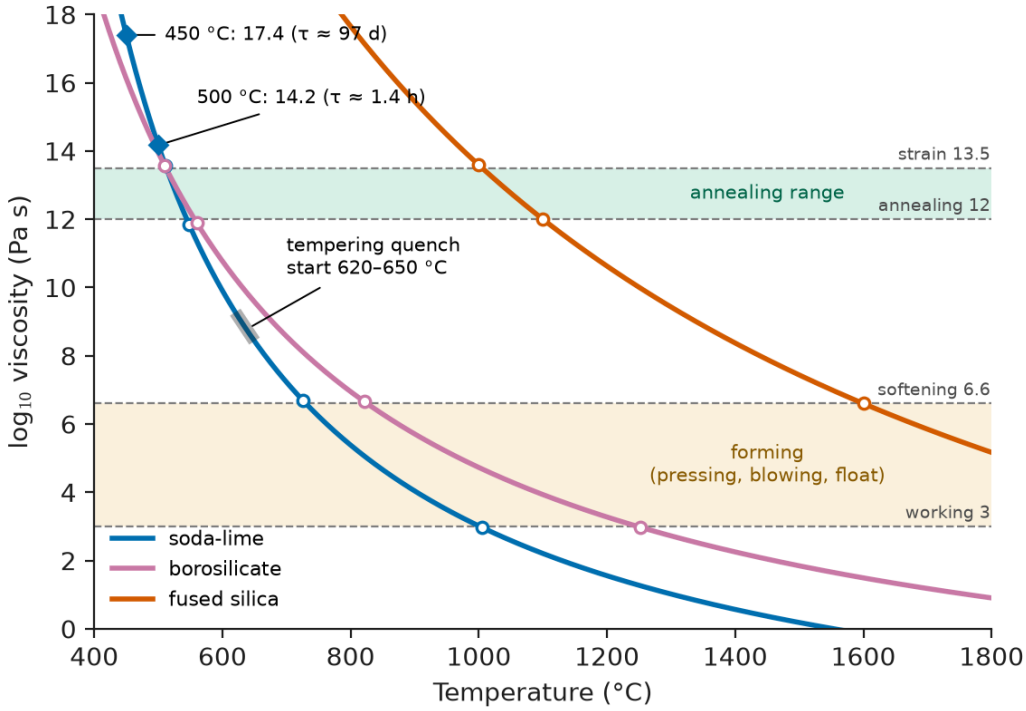

Figure 17.5 — Viscosity versus temperature for soda-lime, borosilicate and fused silica glasses with the reference points and forming, annealing and tempering windows.

Annealing

A formed article cools unevenly, freezing in **residual stresses** that can crack it spontaneously when cut or scratched. **Annealing** removes them: the article is held at the annealing point, where stresses relax by viscous flow, then cooled slowly through the strain point, and faster afterwards.

Worked example — Why the annealing point is where it is. Treat the glass as a Maxwell viscoelastic body (Chapter 14) with shear modulus $G \approx 30$ GPa. Its stress-relaxation time is $\tau = \eta / G$. At the annealing point, $\eta = 10^{12}$ Pa s, so $\tau = 10^{12}/(3 \times 10^{10}) = 33$ s: stresses decay in minutes. At 500 °C, the VFT equation for soda-lime glass gives $\log_{10}\eta = -4.57 + 6370/(500 - 160) = 14.17$, so $\eta = 1.5 \times 10^{14}$ Pa s and $\tau \approx 5000$ s, about 1.4 h. At 450 °C, $\log_{10}\eta = 17.40$ and $\tau \approx 8.4 \times 10^{6}$ s, about 97 days. Below the strain point, cooling can be fast because no stress can relax in the time of cooling, and therefore no new stress pattern is frozen in from viscous flow; the transient thermal stresses that remain must simply be kept below the strength. A container-glass lehr accordingly holds bottles near 550 °C for a few minutes, cools them at a few kelvin per minute to about 480 °C, and then cools them rapidly.

Thermal tempering

Glass fails from surface flaws in tension, so a surface in permanent compression must be unloaded before a flaw can open. In **thermal tempering**, a soda-lime sheet at least 3 mm thick is heated to about 620–650 °C and quenched on both faces with air jets. The surfaces stiffen first; the still-fluid interior then contracts and pulls them into compression, ending itself in tension. The profile is close to a parabola: for fully tempered glass, 90–150 MPa of surface compression balanced by a mid-plane tension of half that value, crossing zero at about 21 % of the thickness from each face (Figure 17.6). The characteristic bending strength rises from about 45 MPa (annealed float) to 120 MPa or more, and a broken sheet dices into small, blunt fragments: **safety glass** for vehicles, doors and façades. It cannot be cut afterwards, and nickel sulfide inclusions occasionally break it spontaneously, hence **heat-soak testing** of façade glass.

Chemical strengthening

Thin glass cannot be thermally tempered, because the temperature difference between surface and centre during a quench is too small. **Chemical strengthening** by ion exchange works for any thickness. The glass is immersed in molten KNO_3 at 380–450 °C for hours; Na^+ ions in the surface exchange with K^+ ions from the salt, and because the glass is too cold for the network to relax, the larger K^+ ions are "stuffed" into sites sized for Na^+, creating a compressive stress of 600–900 MPa in a layer 10–100

μm deep (the depth of layer, DOL). Alkali aluminosilicate glasses are designed for this process: aluminium opens the network and speeds interdiffusion (Chapter 8). The compression is balanced by a small central tension; for a triangular compressive profile with surface stress CS and depth DOL in a sheet of thickness t, force balance gives a central tension of approximately $\mathrm{CS} \cdot \mathrm{DOL}/(t - 2\,\mathrm{DOL})$, or about 45 MPa for CS = 700 MPa, DOL = 40 μm and $t = 0.7$ mm. The central tension must be kept below a frangibility limit, or the glass shatters violently when penetrated.

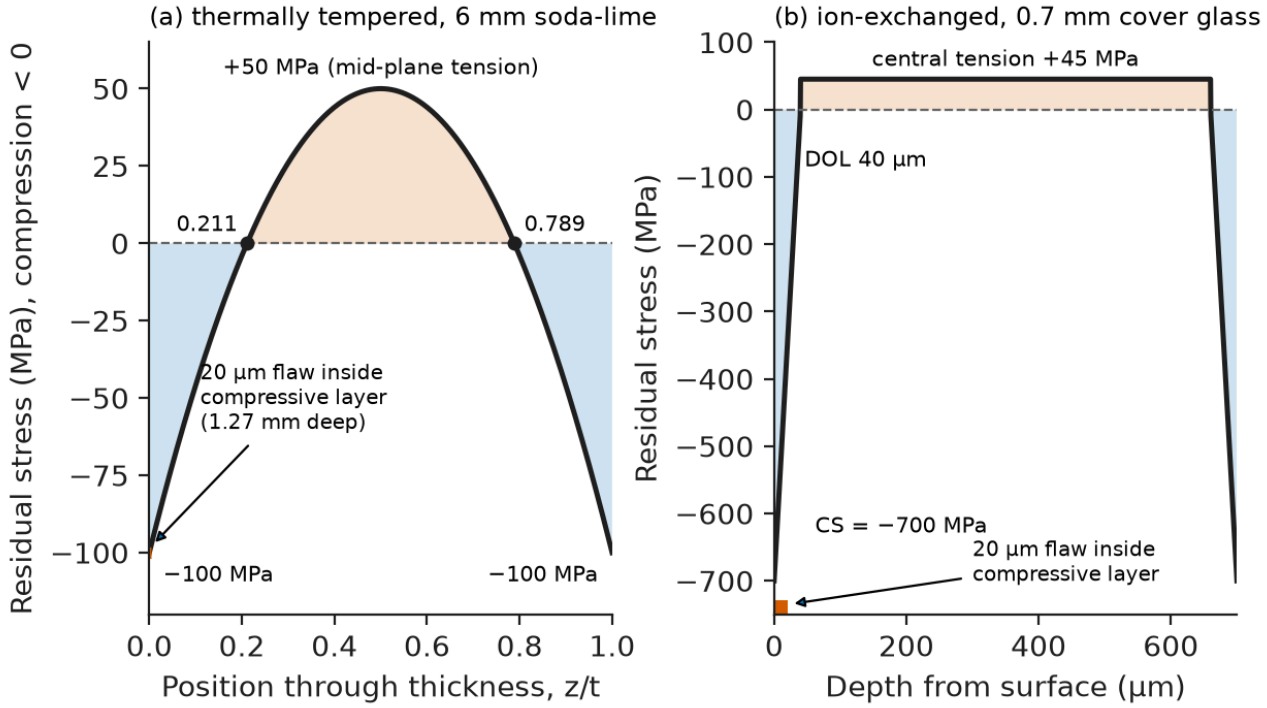

Figure 17.6 — Residual stress through the thickness of a thermally tempered soda-lime sheet and a chemically strengthened aluminosilicate cover glass.

Worked example — How much does strengthening help against a scratch? A cover glass has a surface flaw 20 μm deep. With $K_{IC} = 0.75$ MPa m$^{1/2}$ and $Y = 1.12$, the annealed glass fails at

$$\sigma_f = \frac{0.75}{1.12\sqrt{\pi \times 20 \times 10^{-6}}} = 84.5 \text{ MPa}$$

If the flaw lies within a thermally tempered layer with about 100 MPa of surface compression, the applied stress must first cancel the compression, so failure needs roughly $84.5 + 100 \approx 185$ MPa. If it lies within a chemically strengthened layer with 700 MPa of compression and a 40 μm depth of layer, failure needs roughly $84.5 + 700 \approx 785$ MPa, nearly ten times the annealed value. The estimate treats the residual stress as uniform over the flaw depth, which overstates the benefit for flaws approaching the depth of layer; a flaw deeper than the layer reaches the tensile core and gains little. This is the engineering logic of cover-glass development: deeper compressive layers to protect against the sharp contacts of a phone dropped on concrete, which create flaws of tens of micrometres.

> **Status of the principle: Superposition of residual and applied stress at a flaw**
> *Approximate.* Linear-elastic superposition of stress-intensity factors is exact, but treating the residual stress as uniform over the crack depth is an approximation; proper analysis integrates the residual-stress profile with a weight function.

17.7 Glass-ceramics

Glass-ceramics combine the formability of glass with the properties of a polycrystal. A glass is formed by any glass process, then crystallized in the solid state into a pore-free ceramic, 50–95 % crystalline with grains of 0.05–1 μm. Crystallization must start everywhere at once rather than from the surface, so a few percent of **nucleating agents** (TiO_2, ZrO_2, P_2O_5) precipitate

nanocrystals during a hold just above T_g, where nucleation peaks, and a second, hotter hold, where growth peaks, grows the main phase on them (Chapter 9).

System	Main crystalline phase	Key property	Applications
Li_2O–Al_2O_3–SiO_2 (LAS), transparent	β-quartz solid solution (~70 %), grains ~50 nm	CTE ≈ 0 ± 0.1 × 10^{-6} K^{-1} (to ± 0.02 × 10^{-6} in mirror grades); transparent	Cooktop panels, fire-place windows, telescope mirror blanks, ring-laser gyroscopes
LAS, opaque	β-spodumene solid solution	CTE ~1 × 10^{-6} K^{-1}; white, strong	Oven-to-freezer cookware
MgO–Al_2O_3–SiO_2 (MAS)	Cordierite	Low dielectric loss, low CTE, high strength	Missile radomes
Li_2O–SiO_2	Lithium disilicate, interlocking needles	Flexural strength 360–500 MPa, K_{IC} ~2.5–3 MPa $m^{1/2}$, tooth-like translucency	Dental crowns and bridges
K_2O–MgO–Al_2O_3–SiO_2–F	Fluorophlogopite mica plates	Machinable with ordinary tools	Machinable insulators and fixtures
CaO–P_2O_5–SiO_2	Apatite, wollastonite	Bonds to bone	Bone substitutes (Chapter 39)

The near-zero expansion of LAS glass-ceramics is a structural trick. The β-quartz and β-eucryptite crystals have a *negative* expansion coefficient along their *c* axis, and the composite of crystals with negative expansion and residual glass with positive expansion can be tuned to zero over a wide temperature range. A cooktop can then be heated locally to 700 °C while the edge stays cold without cracking; a telescope mirror does not change shape as night falls.

17.8 Refractories

Refractories line furnaces, kilns, reactors and ladles: non-metallic materials that keep their shape and integrity above about 1500 °C. About 60–70 % of world consumption goes to iron- and steelmaking. Unlike technical ceramics they are mostly coarse-grained and porous (10–25 %), aggregates bonded by a fine matrix, judged by survival against molten metal, slag and hot gas rather than by room-temperature strength.

Classes

Refractories are traditionally classified by their chemistry, because the key requirement is chemical compatibility with the slag or melt they contain. An acidic (silica-rich) refractory in contact with a basic (CaO-, MgO-rich) slag forms low-melting silicates and dissolves rapidly, and vice versa.

Class	Main constituents	Typical service	Remarks
Acidic: silica	> 93 % SiO_2 (tridymite, cristobalite)	Coke-oven walls, glass-furnace crowns	Rigid almost to melting, but quartz/cristobalite inversions make it crack if cycled below ~600 °C
Acidic: fireclay and high-alumina	25–45 % Al_2O_3 (fireclay); 50–90 % Al_2O_3 (bauxite, mullite, corundum)	Ladles, blast-furnace stoves, kiln linings, aluminium furnaces	The Al_2O_3–SiO_2 diagram and mullite set refractoriness (Chapter 7)
Basic: magnesia and magnesia–carbon	MgO (periclase), 5–20 % graphite, metallic antioxidants	Steel converters (BOF), electric-arc furnaces, ladle slag lines	Graphite is not wetted by slag and raises thermal-shock resistance
Basic: magnesia–spinel, dolomite	MgO with $MgAl_2O_4$; CaO·MgO	Cement-kiln burning zones; secondary steelmaking	Chrome-free replacements for magnesia–chrome
Neutral and special	Fused-cast Al_2O_3–ZrO_2–SiO_2 (AZS), SiC, carbon, zircon	Glass-tank walls, blast-furnace hearths, aluminium cells	AZS resists glass corrosion; carbon resists molten iron
Insulating	Insulating firebrick, ceramic fibre, microporous silica	Backing layers, kiln cars, heat-treatment furnaces	Porosity 50–90 %; conductivity 0.03–0.5 W m^{-1} K^{-1}

Refractories are supplied as fired **shaped** bricks or, increasingly, as **monolithics** (castables, gunning and ramming mixes) placed and fired in situ; low-cement castables with reactive alumina and microsilica install fast and have fewer joints for slag to attack.

Performance

Refractories are tested for **refractoriness** (pyrometric cone equivalent), **refractoriness under load** and creep, **thermal-shock resistance**, **slag corrosion**, conductivity and abrasion. A good refractory does not avoid cracking; it limits crack propagation. Coarse aggregates, microcracks and graphite flakes deflect cracks and store little elastic energy, the opposite philosophy to a fine-grained technical ceramic.

Worked example — A two-layer furnace lining. A furnace wall has a hot face at 1400 °C and an outer steel shell at 60 °C. It is built from 230 mm of dense high-alumina brick ($k = 1.5$ W m^{-1} K^{-1}) backed by 115 mm of insulating firebrick ($k = 0.25$ W m^{-1} K^{-1}). In steady state the same heat flux crosses both layers in series:

$$q = \frac{T_{\text{hot}} - T_{\text{cold}}}{L_1/k_1 + L_2/k_2} = \frac{1400 - 60}{0.230/1.5 + 0.115/0.25} = \frac{1340}{0.153 + 0.460} = 2.18 \text{ kW m}^{-2}$$

The temperature at the interface is $1400 - q\,L_1/k_1 = 1400 - 2185 \times 0.153 = 1065$ °C. The insulating brick must therefore be rated for at least about 1100 °C (a grade-23 insulating firebrick, rated to 1260 °C, suffices); it could not be placed at the hot face, where it would shrink and be eroded. A wall of the same total thickness built entirely of dense brick would lose $1340/(0.345/1.5) = 5.8$ kW m^{-2}, 2.7 times as much. Real conductivities rise or fall with temperature and the shell temperature depends on external convection, so furnace designers integrate $k(T)$ and iterate, but the two-layer logic (a dense, corrosion-resistant hot face backed by a light insulating layer) is universal.

17.9 Functional ceramics

Much of the economic value of ceramics lies not in their mechanical properties but in their electrical, magnetic, optical and electrochemical functions, which depend on point defects, domain structures and band gaps rather than on flaws. They are treated in Parts VI and VII (Chapters 24–36) and in Chapter 40; the table below is a map. Their processing is the powder processing of Section 17.3, often with extreme control of particle size, dopant distribution and layer thickness.

Functional ceramic	Function	Key material	Where it is treated
Multilayer ceramic capacitors	High permittivity, ferroelectric	$BaTiO_3$ with dopants, layers ~0.5–1 μm	Chapters 26 and 27
Piezoelectric actuators, sensors, ultrasound	Piezoelectricity	$Pb(Zr,Ti)O_3$ (PZT), lead-free $(K,Na)NbO_3$	Chapters 27 and 37
Soft and hard ferrites	Ferrimagnetism	MnZn, NiZn ferrites; $BaFe_{12}O_{19}$, $SrFe_{12}O_{19}$	Chapter 29
Oxygen sensors, SOFC electrolytes	Oxygen-ion conduction	Y- or Sc-stabilized ZrO_2, Gd-doped CeO_2	Chapters 25 and 40
Solid-state battery electrolytes	Li-ion conduction	$Li_7La_3Zr_2O_{12}$ (LLZO), sulfide glasses	Chapter 40
Varistors and thermistors	Grain-boundary barriers; semiconducting oxides	ZnO–Bi_2O_3; Mn–Ni–Co spinels (NTC); doped $BaTiO_3$ (PTC)	Chapter 25
Transparent conductors	Degenerate semiconduction	Sn-doped In_2O_3 (ITO), Al-doped ZnO	Chapters 32 and 36
Substrates and heat spreaders	Insulation with high thermal conductivity	Al_2O_3, AlN, Si_3N_4	Chapter 24
Phosphors, laser hosts, transparent armour	Luminescence; transparency	Ce:YAG, Nd:YAG, spinel, AlON	Chapter 36
Thermal barrier coatings	Low thermal conductivity	7YSZ, rare-earth zirconates	Chapters 16, 24 and 41

Boundary.
The EEE-specific treatment of dielectric, piezoelectric, magnetic and substrate ceramics, including capacitor and package design, lives in Chapters 26, 27, 29 and 51.

What remains open

Open question	Where it is answered
How were Griffith fracture, R-curves, transformation toughening and Weibull statistics derived?	Chapter 14 (back reference)
How do the crystal structures and phase diagrams of ceramics follow from bonding and thermodynamics?	Chapters 2 and 7 (back references)
How do nucleation and crystallization kinetics set glass formation and glass-ceramic heat treatments?	Chapter 9 (back reference)
How is cement, the largest-volume ceramic, made and how does it harden?	Chapter 18
How do glass fibres and ceramic matrices behave in composites?	Chapters 22–23
How do thermal conductivity and thermal stress set the performance of TBCs, substrates and refractories?	Chapter 24
How do dielectric, piezoelectric, magnetic, ionic and optical ceramics work?	Chapters 25–36 and 40
Which ceramics and bioactive glasses are used in the body?	Chapter 39
How do ceramics and CMCs behave in turbines, re-entry vehicles and reactors?	Chapter 41
How are ceramics shaped by additive manufacturing and deposited as coatings?	Chapters 48–49
How are ceramic microstructures and residual stresses characterized?	Chapters 52–53

Next: Chapter 18 turns to the materials used in the greatest tonnage of all: Portland cement and concrete, structural steel, timber, asphalt and masonry, and to the question of how a civilization can keep building with them while cutting their carbon footprint.

Chapter 18 — Cement, Concrete and Construction Materials

Part IV · Metals, Ceramics and Construction Materials

A multi-storey car park built in the late 1960s is inspected after a winter in which pieces of concrete fell from the soffit of an upper deck. Behind the spalls the reinforcing bars are rusted, in places down to half their original diameter; brown streaks run from cracks along the lines of the bars; a hammer sounds hollow over large areas. Laboratory tests on cores show that the concrete is still as strong in compression as the day it was cast. The problem lies elsewhere: forty winters of de-icing salt carried in on car tyres have soaked chloride into the deck, the chloride has reached the steel through a cover of only 25 mm of rather porous concrete, and the steel, no longer protected by the alkalinity of the cement paste, has corroded. Rust occupies several times the volume of the steel it replaces, so the corroding bars have split the concrete off from inside.

The structure did not fail because a strength was exceeded. It failed because a *transport property* (how fast chloride moves through a porous ceramic) and an *electrochemical* one (when steel in that ceramic stops being passive) were ignored when the water–cement ratio and cover were chosen. Construction materials are used in quantities that dwarf all others (about 30 billion tonnes of concrete a year), made from local, variable raw materials, cured outdoors and expected to last 50–120 years: the materials science of large volumes, long times and real environments.

> **How does a powder mixed with water become a stone, and what decides the strength and durability of that stone?**
> **How do steel, timber, asphalt and masonry complement concrete, and how can construction cut its carbon footprint?**

Concept	Engineering destination
Clinker phases and hydration chemistry	Cement type selection, heat of hydration in massive pours, setting time
Water–cement ratio, mix design, curing	Strength class, workability, specification of concrete for any structure
Pore structure and transport	Cover depth, durability class, service-life design
Carbonation, chlorides, ASR, freeze–thaw	Bridge decks, marine structures, car parks, dams, pavements
Reinforced and prestressed concrete	Beams, slabs, bridges, tanks, towers
Structural steel grades and specifications	Frames, bridges, rebar, prestressing strand, toughness at low temperature
Wood anatomy, moisture, engineered wood	Timber frames, glulam and CLT buildings, joinery
Bitumen rheology and asphalt mix design	Road pavements, runways, waterproofing
Brick, mortar and stone	Masonry walls, heritage conservation
Embodied carbon and low-carbon binders	Specifying and designing for net-zero construction

Roadmap. Portland cement and its hydration (18.1) → concrete as an engineered composite: mix design, strength and curing (18.2) → durability (18.3) → reinforced and prestressed concrete (18.4) → structural steel (18.5) → timber (18.6) → asphalt (18.7) → masonry and stone (18.8) → low-carbon cements and sustainability (18.9).

18.1 Portland cement

Making clinker

Portland cement is made by heating a finely ground mixture of limestone (about 80 %) and clay or shale (about 20 %), with small corrections of sand, iron ore or bauxite, to about 1450 °C in a rotary kiln. In the preheater and calciner the limestone decomposes,

$$\mathrm{CaCO_3 \rightarrow CaO + CO_2}$$

and the lime reacts with the silica, alumina and iron oxide, helped by 20–30 % of melt as in liquid-phase sintering (Chapter 17). The **clinker** nodules are cooled rapidly and ground with 3–5 % gypsum ($\mathrm{CaSO_4 \cdot 2H_2O}$) to a powder of 300–500 m²/kg (particles mostly 1–50 μm).

Cement chemistry uses a shorthand for the oxides: C = CaO, S = $\mathrm{SiO_2}$, A = $\mathrm{Al_2O_3}$, F = $\mathrm{Fe_2O_3}$, H = $\mathrm{H_2O}$, $\bar{\mathrm{S}}$ = $\mathrm{SO_3}$. The clinker consists of four main phases:

Phase (mineral name)	Formula	Typical content (wt%)	Rate of hydration	Contribution
$\mathrm{C_3S}$ (alite)	$\mathrm{Ca_3SiO_5}$	50–70	Fast (hours to weeks)	Early and most of the 28-day strength
$\mathrm{C_2S}$ (belite)	$\mathrm{Ca_2SiO_4}$	15–30	Slow (weeks to years)	Late strength, low heat
$\mathrm{C_3A}$ (aluminate)	$\mathrm{Ca_3Al_2O_6}$	5–10	Very fast (minutes)	Flash set if uncontrolled; sulfate-attack susceptibility
$\mathrm{C_4AF}$ (ferrite)	$\mathrm{Ca_2(Al,Fe)_2O_5}$	5–15	Moderate	Little strength; grey colour

The real phases are impure solid solutions measured by quantitative XRD (Chapter 52), but the phase content is still estimated routinely from the oxide analysis by the **Bogue calculation**, which assumes ideal phase compositions.

Worked example — Bogue phase composition. A clinker-based cement has the oxide analysis CaO 64.5 %, $\mathrm{SiO_2}$ 21.0 %, $\mathrm{Al_2O_3}$ 5.5 %, $\mathrm{Fe_2O_3}$ 3.0 %, $\mathrm{SO_3}$ 2.8 % (free lime neglected). With the Bogue equations for $\mathrm{Al_2O_3/Fe_2O_3} > 0.64$,

$$\mathrm{C_3S} = 4.071\,\mathrm{C} - 7.600\,\mathrm{S} - 6.718\,\mathrm{A} - 1.430\,\mathrm{F} - 2.852\,\bar{\mathrm{S}}$$

$$\mathrm{C_2S} = 2.867\,\mathrm{S} - 0.7544\,\mathrm{C_3S}, \qquad \mathrm{C_3A} = 2.650\,\mathrm{A} - 1.692\,\mathrm{F}, \qquad \mathrm{C_4AF} = 3.043\,\mathrm{F}$$

we obtain $\mathrm{C_3S} = 262.6 - 159.6 - 36.9 - 4.3 - 8.0 = 53.8$ %, $\mathrm{C_2S} = 60.2 - 40.6 = 19.7$ %, $\mathrm{C_3A} = 14.6 - 5.1 = 9.5$ %, $\mathrm{C_4AF} = 9.1$ %, totalling 92 % (the rest is gypsum, MgO, alkalis and minor phases). This is an ordinary CEM I / ASTM Type I cement. Taking typical heats of complete hydration of 500 J/g for $\mathrm{C_3S}$, 260 J/g for $\mathrm{C_2S}$, 870 J/g for $\mathrm{C_3A}$ and 420 J/g for $\mathrm{C_4AF}$, its ultimate heat of hydration is about 440 J/g, roughly 300–350 J/g of it in the first week. In a 2 m thick raft the core can heat by 40–60 K and later crack against the cooler surface, so massive pours use low-heat cements, slag or fly ash, chilled water or cooling pipes.

> **Status of the principle: Bogue calculation**
> *Approximate.* Exact stoichiometry if the four phases had their ideal compositions and formed at equilibrium; real solid-solution phases and non-equilibrium cooling make it typically underestimate alite and overestimate belite by several percent.

Hydration chemistry

When cement is mixed with water, its anhydrous phases dissolve and less soluble hydrates precipitate. The idealized reactions are:

$$2\,\mathrm{C_3S} + 6\,\mathrm{H} \rightarrow \mathrm{C_3S_2H_3} + 3\,\mathrm{CH}$$

$$2\,\mathrm{C_2S} + 4\,\mathrm{H} \rightarrow \mathrm{C_3S_2H_3} + \mathrm{CH}$$

$$\mathrm{C_3A} + 3\,\mathrm{C\bar{S}H_2} + 26\,\mathrm{H} \rightarrow \mathrm{C_6A\bar{S}_3H_{32}}\ \text{(ettringite)}$$

$$2\,\mathrm{C_3A} + \mathrm{C_6A\bar{S}_3H_{32}} + 4\,\mathrm{H} \rightarrow 3\,\mathrm{C_4A\bar{S}H_{12}}\ \text{(monosulfate)}$$

Here CH is portlandite, $\mathrm{Ca(OH)_2}$, and "$\mathrm{C_3S_2H_3}$" stands for **calcium silicate hydrate**, written C–S–H because its composition varies. Without gypsum, $\mathrm{C_3A}$ causes a "flash set" within minutes; gypsum coats it with ettringite and gives hours of working time.

Isothermal calorimetry shows the stages of hydration (Figure 18.1): a brief initial peak; an **induction period** of 1–3 h in which the concrete stays workable; an **acceleration period** in which C₃S hydrates, C–S–H and CH grow, and the paste sets; and a **deceleration period**, lasting weeks to years, as hydration becomes diffusion-limited.

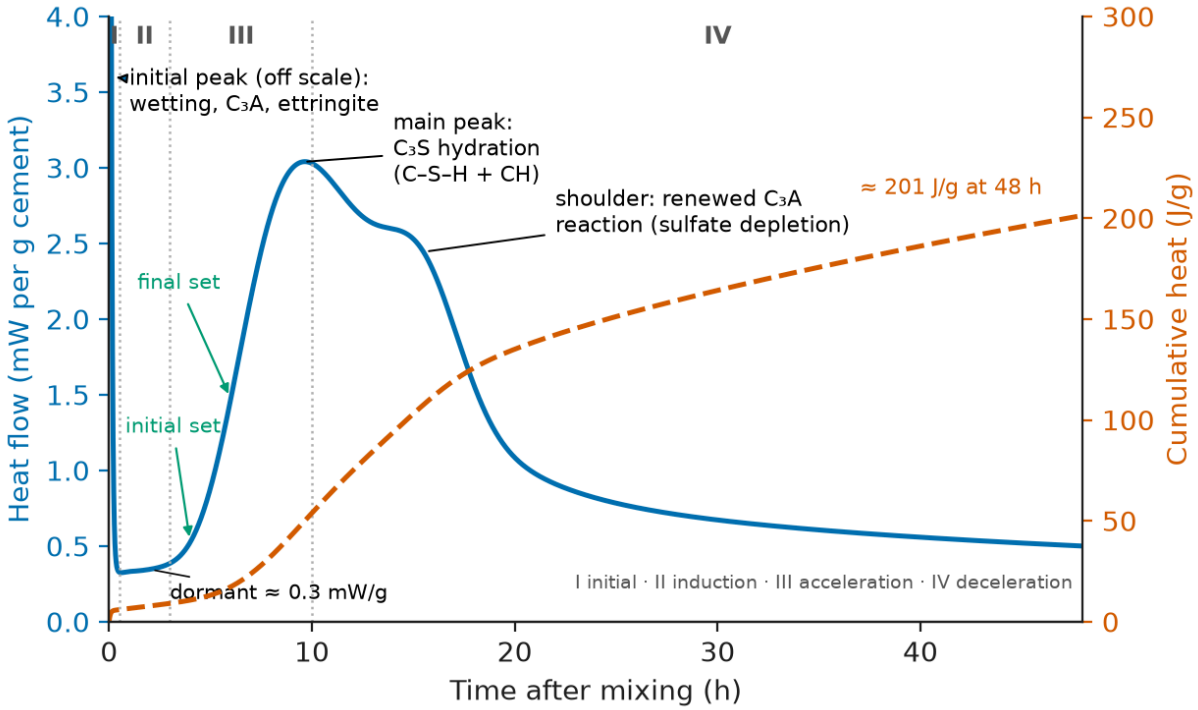

Figure 18.1 — Rate of heat release during hydration of a Portland cement paste at 20 °C, measured by isothermal calorimetry, with the cumulative heat.

C–S–H and the pore structure

C–S–H, 50–60 % of a mature paste, is the glue of concrete: a poorly crystalline, layered calcium silicate hydrate (Ca/Si about 1.7, related to tobermorite and jennite) formed as nanometre globules with a specific surface of 100–700 m²/g and an intrinsic **gel porosity** of about 28 % in 0.5–10 nm pores. CH forms large crystals that add little strength but buffer the pore solution at pH 12.5–13.5, which keeps reinforcing steel passive.

Hydrates occupy about 2.1 times the volume of the cement they replace, so they progressively fill the water-filled space between grains; what remains is **capillary porosity**, pores from about 10 nm to 10 µm. Powers' classic model of paste volumes gives the capillary porosity of a paste with water–cement ratio w/c at degree of hydration α approximately as

$$\phi_{\text{cap}} \approx \frac{w/c - 0.36\,\alpha}{w/c + 0.32}$$

For $w/c = 0.45$, ϕ_{cap} is about 26 % at $\alpha = 0.7$ and about 16 % at $\alpha = 0.9$; for $w/c = 0.60$ and $\alpha = 0.9$ it is 30 %. Complete hydration needs $w/c \approx 0.38$–0.42; below that some cement never hydrates (and acts as a stiff filler), while above it the excess water leaves capillary pores. When the capillary porosity falls below roughly 20 %, the pores lose their connectivity and the paste becomes much less permeable. Almost every property of concrete follows from this one picture: strength from the solid fraction, durability from the connectivity of capillary pores.

> **Status of the principle: Powers' model of paste volumes**
> *Approximate.* A volume balance using average densities and water contents of hydrates measured in the 1940s; it gives porosity and its trends with w/c and hydration well for Portland cement pastes, but not for blended cements with different hydrates or for the microstructure near aggregates.

Types of cement

Adjusting the phase balance and fineness gives the standard types: ordinary (CEM I, ASTM Type I), moderate and high sulfate resistance (low C_3A; Types II and V), high early strength (finer, more C_3S; Type III) and low heat (more C_2S; Type IV). Blended cements (EN CEM II–V) replace 6–95 % of the clinker with limestone, fly ash, slag or pozzolana (Section 18.9).

> **Boundary.**
> The thermodynamics of the clinkering reactions and hydration equilibria belong to Advanced Chemistry; phase relations in $CaO–SiO_2–Al_2O_3$ are treated with ternary diagrams in Chapter 7.

18.2 Concrete

Constituents

Concrete is a particulate composite: hydrated cement paste binding coarse aggregate (gravel or crushed rock, 4–20 mm or larger) and sand. Per cubic metre it typically contains 300–450 kg of cement, 150–200 kg of water and about 1700–1900 kg of aggregate, which fills 65–75 % of the volume and reduces the amount of paste, the part that is expensive, shrinks and generates heat.

Superplasticizers (today polycarboxylate ethers with comb-like side chains) disperse cement particles by electrostatic and steric repulsion (Chapter 37) and cut the water needed by 10–40 %; **air-entraining agents** stabilize bubbles for frost resistance; **retarders** and **accelerators** adjust setting. **Supplementary cementitious materials** (SCMs) replace part of the cement: **fly ash** and **silica fume** are pozzolans whose silica reacts with CH to form more C–S–H, and **ground granulated blast-furnace slag** (GGBS) is latently hydraulic. SCMs refine the pores and cut heat and CO_2, at the cost of slower early strength.

The water–cement ratio

The single most important number in concrete technology is the **water–cement ratio** by mass (or water–binder ratio when SCMs are included). In 1918 Duff Abrams found that, for workable, fully compacted concrete made with given materials, the compressive strength depends almost only on w/c:

$$f_c = \frac{A}{B^{\,w/c}}$$

where A and B are empirical constants for given materials, age and curing. Fitting the 28-day cylinder strengths that ACI 211.1 tabulates for non-air-entrained concrete gives $A \approx 117$ MPa and $B \approx 13$: then $w/c = 0.42$ gives 40 MPa, $w/c = 0.54$ gives 29 MPa and $w/c = 0.69$ gives 20 MPa (Figure 18.2). The physics is Powers' porosity: less water means less capillary space and a denser, stronger paste. Powers expressed this through the **gel–space ratio**, the volume of hydration products divided by the volume available to them, $X = 0.68\alpha/(0.32\alpha + w/c)$, finding that mortar strength scales roughly as X^3; strength depends on solid fraction just as for porous ceramics (Chapter 17).

> **Status of the principle: Abrams' law**
> *Empirical.* A fit for fully compacted concrete of given materials and curing; it fails for incompletely compacted mixes (too little water to place the concrete properly), for very high-strength concrete where the aggregate or the interfacial zone limits strength, and the constants must be recalibrated for each set of materials.

Without admixtures, a mix with $w/c = 0.35$ is too stiff to compact, and its voids make it weaker. Superplasticizers removed this limit: w/c of 0.25–0.35 with silica fume gives 80–120 MPa, and ultra-high-performance concretes (w/b about 0.2, steel fibres) reach 150–200 MPa.

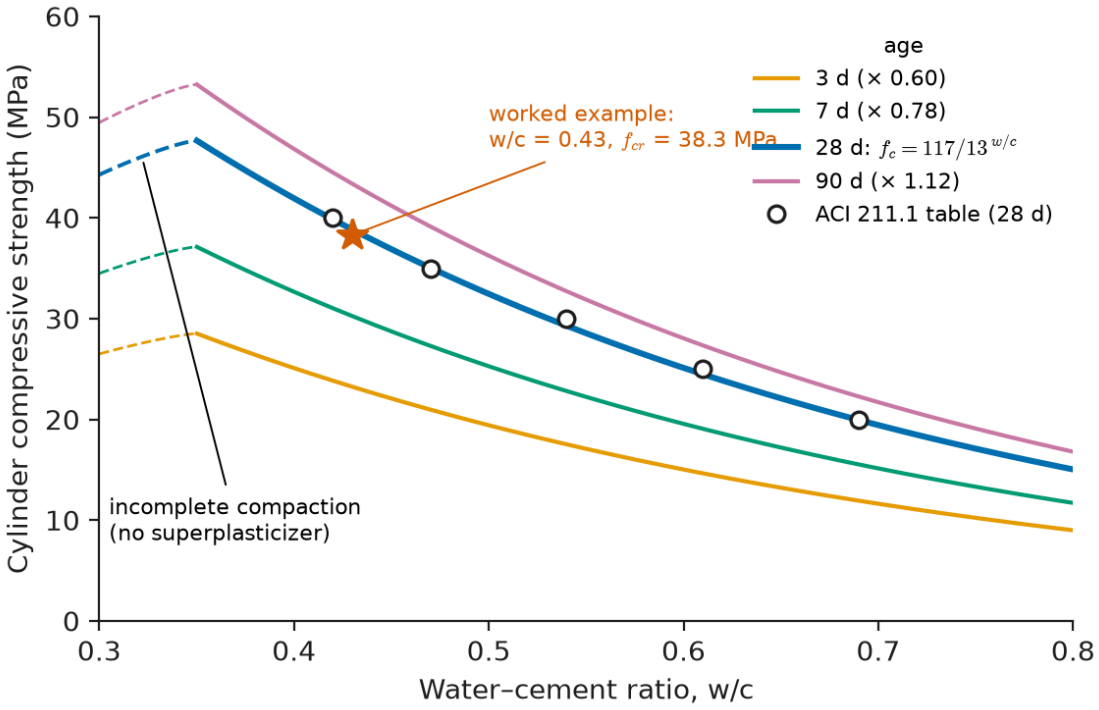

Figure 18.2 — Compressive strength of concrete versus water–cement ratio (Abrams' law) at 3, 7, 28 and 90 days, with the low-w/c limit set by compaction.

Mix design

Mix design chooses proportions to reach a required strength, workability and durability at least cost. Durability usually sets a maximum w/c and a minimum cement content for each **exposure class** (in EN 206, for example, $w/c \le 0.45$ for concrete exposed to seawater or de-icing salts); strength sets a target w/c from Abrams' relation; workability sets the water content per cubic metre; and the remaining volume is filled with aggregate.

Worked example — Proportioning a C30 concrete. A bridge pier needs a specified cylinder strength $f'_c = 30$ MPa, a slump of 75–100 mm and 20 mm maximum aggregate, and will be exposed to moderate freezing without de-icing salts. Following the absolute-volume method of ACI 211.1:

1. *Target mean strength.* With no production records, ACI adds 8.3 MPa to cover variability: $f_{cr} = 38.3$ MPa.
2. *Water–cement ratio.* Inverting Abrams' fit, $w/c = \ln(117/38.3)/\ln 13 = 1.117/2.565 = 0.435$; take 0.43.
3. *Water.* For 75–100 mm slump and 20 mm aggregate, about 190 kg/m³ (2 % entrapped air).
4. *Cement.* $190/0.43 = 442$ kg/m³.
5. *Coarse aggregate.* For a sand of fineness modulus 2.8 and 20 mm aggregate, 0.62 m³ of dry-rodded coarse aggregate per cubic metre; at a rodded density of 1600 kg/m³ this is 992 kg.
6. *Sand by absolute volume.* With specific gravities of 3.15 (cement), 2.68 (coarse) and 2.64 (sand), the volumes are water 0.190 m³, cement $442/3150 = 0.140$ m³, coarse aggregate $992/2680 = 0.370$ m³ and air 0.020 m³, total 0.720 m³. Sand fills the remaining 0.280 m³: $0.280 \times 2640 = 738$ kg.

The batch is 442 kg cement, 190 kg water, 738 kg sand and 992 kg gravel per cubic metre, a fresh density of about 2360 kg/m³, to be corrected for aggregate moisture and trial-batched. For freezing exposure about 5 % entrained air would be added, its strength penalty (roughly 5 % per 1 % air) offset by a lower w/c; in practice part of the cement would also be replaced by slag or fly ash (Section 18.9).

Strength development and curing

Concrete gains strength as long as hydration continues, quickly at first, then more slowly. A common model (fib Model Code) for normal cement at 20 °C is

$$f_c(t) = f_c(28)\exp\left\{s\left[1-\left(\frac{28}{t}\right)^{1/2}\right]\right\}$$

with t in days and $s = 0.25$ for normal-hardening cement (0.20 for rapid-hardening, 0.38 for slow cements). It predicts 60 % of the 28-day strength at 3 days, 78 % at 7 days and 112 % at 90 days. Temperature accelerates hydration, and the **maturity method** combines time and temperature into an equivalent age (by a Nurse–Saul temperature–time sum, or an Arrhenius equivalent age with an apparent activation energy of 30–45 kJ/mol), so that the in-place strength of a slab can be estimated from embedded temperature loggers and the formwork stripped safely.

Hydration needs water. **Curing** keeps the concrete moist for typically 3–7 days (longer with slag and fly ash); if the surface dries, hydration stops there and the outer 20–50 mm, the cover that protects the steel, stays porous. Steam curing of precast at 50–70 °C speeds stripping, but above about 70 °C risks expansive **delayed ettringite formation**.

Concrete also changes volume: **drying shrinkage** of typically 300–800 microstrain, **autogenous shrinkage** at low w/c, and **creep** under sustained load of 1.5–3 times the elastic strain. These viscoelastic responses of C–S–H (Chapter 14) set crack-control reinforcement and prestress losses (Section 18.4).

18.3 Durability

Transport is the key

Nearly all deterioration requires water, CO_2, chloride, sulfate or oxygen to enter by permeation, diffusion (Chapter 8) or capillary suction, at rates set by the connectivity of the capillary pores; **low w/c, good curing, SCMs and sufficient cover** are the generic defences. Typical chloride diffusion coefficients are 10^{-11} m²/s for poor concrete, 10^{-12} m²/s for good CEM I concrete at w/c around 0.4–0.45, and 10^{-13}–3×10^{-13} m²/s for slag or fly-ash concretes.

Steel in concrete: passivation and its loss

Above pH about 11.5, steel in concrete carries a nanometre-thick passive film (Chapter 42). Two processes destroy it.

Carbonation. Atmospheric CO_2 diffuses into the concrete and reacts with CH and C–S–H:

$$\mathrm{Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O}$$

The pH drops from about 13 to below 9 behind a front that advances as $x = K\sqrt{t}$, since CO_2 must diffuse through the carbonated layer. K is 1–2 mm/yr$^{1/2}$ for dense concrete and 3–6 for ordinary concrete; with $K = 4$ the front reaches 25 mm in $(25/4)^2 = 39$ years and 40 mm in 100 years.

Chloride attack. Chlorides from seawater or de-icing salts break the passive film locally once they exceed a **critical chloride content** at the steel, typically 0.2–0.5 % by mass of cement (often 0.4 %), even at high pH, causing pitting (Chapter 43). Ingress is commonly modelled with the error-function solution of Fick's second law for a constant surface concentration C_s:

$$C(x,t) = C_s\left[1-\operatorname{erf}\left(\frac{x}{2\sqrt{D\,t}}\right)\right]$$

> **Status of the principle: Error-function model of chloride ingress**
> *Approximate.* Fick's second law is exact for diffusion with constant D and C_s, but chloride transport in concrete combines diffusion, convection and binding, D decreases with age (especially for SCM concretes), and C_s builds up over years; service-life codes add an empirical ageing exponent and probabilistic inputs.

Worked example — Time to corrosion of a car-park deck. A deck exposed to de-icing salts has a surface chloride content of $C_s = 3.0$ % by mass of cement, a critical content of $C_{\mathrm{crit}} = 0.4$ %, and an apparent diffusion coefficient $D = 1.0\times10^{-12}$ m²/s. When does chloride reach the critical level at a cover of 40 mm?

Corrosion initiates when $C(x,t)/C_s = 0.4/3.0 = 0.133$, i.e. when $\operatorname{erf}(z) = 0.867$ with $z = x/(2\sqrt{Dt})$. The inverse error function gives $z = 1.061$. Hence

$$t = \frac{1}{D}\left(\frac{x}{2z}\right)^2 = \frac{1}{10^{-12}}\left(\frac{0.040}{2 \times 1.061}\right)^2 = 3.55 \times 10^8 \text{ s} \approx 11 \text{ years}$$

With 60 mm cover it is about 25 years; with a slag concrete of $D = 2 \times 10^{-13}$ m²/s, about 56 years at 40 mm and 127 years at 60 mm (Figure 18.3). The time to initiation scales with the square of the cover and inversely with D, the two levers of durability design; in the car park of the opening, with 25 mm of porous cover, it took only a few years.

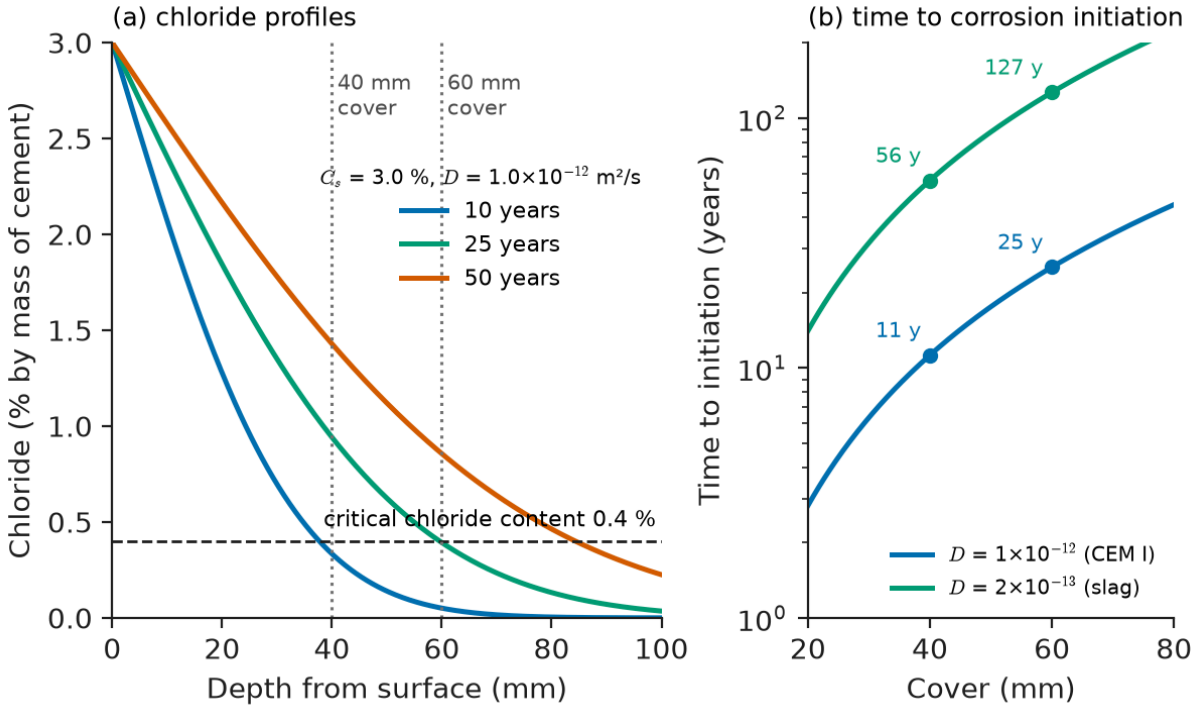

Figure 18.3 — Chloride content versus depth in concrete after 10, 25 and 50 years for a surface content of 3.0 % by mass of cement and D = 1.0e-12 m²/s, with the critical threshold and two cover depths.

Rust occupies two to six times the volume of the steel consumed, so 0.02–0.1 mm of section loss cracks the cover, which then spalls. Protection is, in order of preference: good concrete and cover (EN 1992 asks 40–55 mm for de-icing and marine exposure over 50 years), SCMs, silane impregnation, inhibitors, coated or stainless bars, and cathodic protection (Chapter 43).

Sulfate attack and alkali–silica reaction

Sulfate attack: sulfates from soil or seawater react with monosulfate and CH to form expansive ettringite and gypsum (or, cold and wet, **thaumasite**, which destroys C–S–H). Defences are low-C3A cements, slag, fly ash and low w/c.

Alkali–silica reaction (ASR), identified by Stanton in 1940, attacks aggregates with poorly crystalline or strained silica (opal, chert, volcanic glass). The alkaline pore solution dissolves it into a gel that absorbs water and swells, map-cracking dams, bridges and pavements over decades. ASR needs reactive aggregate, alkali above about 3 kg/m³ Na2O-equivalent, and relative humidity above about 80 %; removing any one prevents it.

Freeze–thaw

Water expands by 9 % on freezing; in saturated paste, ice in larger pores drives water through fine pores and builds hydraulic pressure (Powers' mechanism), cracking the paste and scaling the surface, worse with de-icing salts. The cure, found empirically in the 1930s, is **air entrainment**: 4–7 % by volume of 10–1000 µm bubbles, with no point in the paste more than about 0.2 mm (the **spacing factor**) from an empty bubble into which water can escape.

> **Boundary.**
> The electrochemistry of steel corrosion in concrete (passivity, pitting, cathodic protection) is developed in Chapters 42–43; degradation of concrete compared with other materials, and failure investigation, in Chapter 44.

18.4 Reinforced and prestressed concrete as a composite

Why reinforce

Concrete is strong in compression (20–100 MPa) but its tensile strength is only about a tenth of that, $f_{ctm} = 0.30 f_{ck}^{2/3}$ in Eurocode 2, or 2.9 MPa for C30/37, and it breaks at about 0.01 % tensile strain. Steel bars carry the tension once the concrete has cracked. The composite works because of four coincidences: **compatible expansion** (both about 10–12 × 10⁻⁶ K⁻¹); **bond** of paste to the ribs of deformed bars; **chemical protection** of steel by the alkaline paste; and **complementary failure**, ductile steel yielding before the concrete crushes in an under-reinforced section, giving warning before collapse.

Before cracking, the section is analysed as a composite with **modular ratio** $n = E_s / E_c$; with $E_{cm} = 22[(f_{ck} + 8)/10]^{0.3}$ GPa = 32.8 GPa for C30/37, $n \approx 6$, and the steel is replaced by n times its area of concrete (the **transformed section**). After cracking, the steel carries all the tension across each crack; crack widths are limited to about 0.3 mm by using many small bars, and at the ultimate limit state the steel yields while the concrete reaches a crushing strain of 0.35 %.

Prestressed concrete

Cracking can be avoided by putting the concrete into compression *before* it is loaded: **prestressing**, made practical by Freyssinet in the 1930s. Tendons of 1860 MPa strand are tensioned against abutments before casting and released after hardening (**pre-tensioning**: hollow-core slabs, sleepers, girders) or jacked through ducts in hardened concrete and grouted (**post-tensioning**: long-span bridges, slabs, tanks). Placed below the centroid, the tendon applies an axial compression and a moment opposing the load.

Only high-strength steel works, because elastic shortening, creep and shrinkage of the concrete, steel relaxation and friction cause **losses** of typically 15–25 %. A creep-plus-shrinkage strain of 0.8 ‰ costs 200 000 × 0.0008 = 160 MPa of prestress: most of the 250 MPa yield of mild steel, but only about 12 % of the 1395 MPa initial stress in a strand.

Worked example — Prestressing a beam to eliminate tension. A simply supported beam spans 10 m with a rectangular section 300 mm wide and 600 mm deep, and carries a total uniform service load of 30 kN/m (including its self-weight). The midspan moment is $M = wL^2/8 = 30 \times 10^2/8 = 375$ kN m. The section has $A = 0.18$ m² and elastic section modulus $Z = bh^2/6 = 0.018$ m³, so the bending stress at the bottom fibre is $M/Z = 20.8$ MPa in tension. That is seven times the tensile strength of C30/37 concrete; without reinforcement the beam would crack at about 14 % of the service load.

Now apply a prestress force P at an eccentricity $e = 200$ mm below the centroid. The bottom-fibre stress at midspan is

$$\sigma_{\text{bot}} = -\frac{P}{A} - \frac{Pe}{Z} + \frac{M}{Z}$$

(tension positive). For zero tension, $P(1/A + e/Z) = M/Z$, i.e. $P(5.56 + 11.11)\ \text{m}^{-2} = 20.8$ MPa, so $P = 1.25$ MN. The top fibre is then at $-P/A + Pe/Z - M/Z = -6.9 + 13.9 - 20.8 = -13.9$ MPa, a compression below the usual service limit of $0.6 f_{ck} = 18$ MPa. If the strand is stressed initially to 0.75 × 1860 = 1395 MPa and 20 % is lost over time, the effective stress is 1116 MPa, and the strand area needed is $1.25 \times 10^6 / 1116 = 1120$ mm², eight 15.2 mm strands of 140 mm² each. There is a catch at transfer, before the full load acts: with only the self-weight of 4.5 kN/m, the top fibre would be in *tension* of about 3.8 MPa, above f_{ctm}. Designers handle this by draping the tendon (reducing e towards the supports), adding top reinforcement, or staging the prestress (Figure 18.4).

> **Status of the principle: Elastic superposition of prestress and load stresses**
> *Approximate.* Valid for an uncracked, linearly elastic section under service loads; it ignores creep redistribution with time and does not describe the ultimate limit state, where the section is cracked and the tendon stress rises towards its strength.

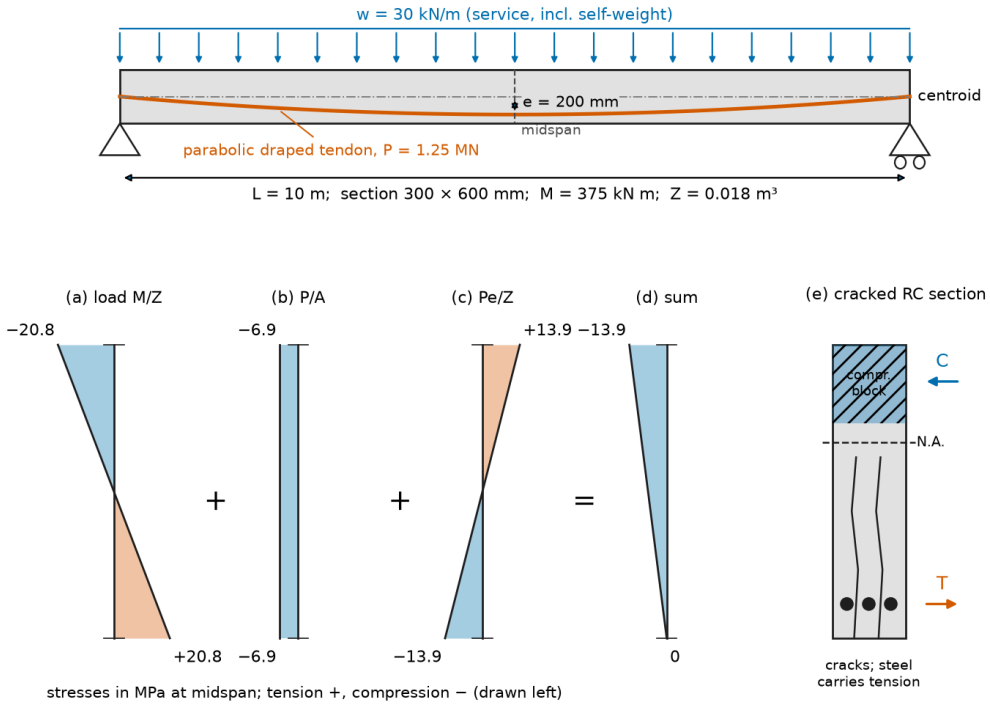

Figure 18.4 — Stress distributions at midspan of the worked-example beam: load alone, prestress alone and combined, plus a reinforced (cracked) section for comparison.

Prestressed concrete is crack-free in service and spans 30–50 m as girders and over 200 m in segmental box-girder bridges. Its vulnerability is the tendon: highly stressed high-strength steel suffers stress-corrosion cracking and hydrogen embrittlement (Chapter 43) where grouting is incomplete, a cause of several bridge collapses.

18.5 Structural steels and their specifications

Steel is the second structural material of the built environment, in the frames of buildings and bridges, as reinforcement and as prestressing tendons. The physical metallurgy of steels (ferrite, pearlite, quenching and tempering, microalloying) is the subject of Chapter 15; here we look at how construction steels are specified, which is where materials science meets building codes.

Hot-rolled structural steels

Structural steels are low-carbon (≤ 0.20–0.24 % C), carbon–manganese or microalloyed steels whose strength comes from ferrite grain size, solid solution and fine Nb, V or Ti carbonitride precipitates (Chapter 13). They are specified not by composition but by guaranteed mechanical properties and weldability.

Grade	Standard	Minimum yield (MPa, $t \leq 16$ mm)	Tensile strength (MPa)	Charpy impact requirement	Notes
S235JR	EN 10025-2	235	360–510	27 J at +20 °C	General, non-critical
S275J0	EN 10025-2	275	410–560	27 J at 0 °C	Building frames
S355J2	EN 10025-2	355	470–630	27 J at −20 °C	The standard bridge and building grade
S355K2	EN 10025-2	355	470–630	40 J at −20 °C	Thicker or more critical members
S460M	EN 10025-4	460	540–720	40 J at −20 °C	Thermomechanically rolled; low carbon equivalent
S355J2W	EN 10025-5	355	470–630	27 J at −20 °C	Weathering steel (Cu, Cr, Ni, P)
ASTM A36	ASTM	250	400–550	Supplementary	US general grade
ASTM A992	ASTM	345–450	≥ 450	Supplementary	US wide-flange sections; yield-to-tensile ratio ≤ 0.85

The designation encodes the design data. "S355" is the minimum yield strength in MPa, which falls with thickness (345 MPa for 16–40 mm) because thick sections cool slowly and have coarser grains. "J2" and "K2" guarantee Charpy energy at a temperature, the codified response to the ductile–brittle transition of BCC iron (Chapter 14) and to the brittle fractures of welded ships and bridges in the 1940s–50s; EN 1993-1-10 tabulates permissible thickness by grade and service temperature. "M" and "N" denote thermomechanically rolled and normalized steels, "Z" grades guarantee through-thickness ductility, and a maximum carbon equivalent keeps welds from cracking (Chapter 47). **Weathering steels** form a protective patina in wet–dry cycles. Seismic frames need steel that work hardens after yield and does not greatly exceed its nominal yield, hence the cap on yield-to-tensile ratio in A992.

Reinforcing and prestressing steels

Product	Typical grade	Strength (MPa)	Ductility requirement	Notes
Reinforcing bar	B500B (EN 10080 / EN 1992)	$f_{yk} = 500$	$f_t/f_y \geq 1.08$, uniform elongation ≥ 5 %	Quenched and self-tempered (Tempcore: martensitic rim, ferrite–pearlite core) or microalloyed
Reinforcing bar, seismic	B500C	$f_{yk} = 500$	$1.15 \leq f_t/f_y < 1.35$, uniform elongation ≥ 7.5 %	High ductility class for seismic regions
Reinforcing bar (US)	ASTM A615 Grade 60; A706	$f_y = 420$	A706: controlled chemistry for welding and seismic use	
Prestressing strand	7-wire, ASTM A416 Grade 270 / EN 10138 Y1860	$f_{pu} = 1860$	Low relaxation (≤ 2.5 % at 70 % UTS after 1000 h)	Cold-drawn pearlitic eutectoid steel, stress-relieved

Prestressing wire is eutectoid steel, patented to fine pearlite and cold drawn until the ferrite lamellae are tens of nanometres thick, giving 1700–2000 MPa.

Fire

By Eurocode 3, structural steel keeps about 60 % of its yield strength at 550 °C and 47 % at 600 °C, so a member loaded to about 60 % of capacity fails near 550 °C, reached by an unprotected section in 10–20 minutes of a standard fire; intumescent paints,

boards or encasement protect it. Dense high-strength concrete can spall explosively as steam builds in its pores, countered by polypropylene fibres that melt and open escape paths.

18.6 Timber

Wood structure

Wood is a natural cellular, fibre-reinforced polymer composite. Seasonal growth at the cambium produces **annual rings** of thin-walled **earlywood** and dense **latewood**, made of long hollow cells aligned with the trunk. In **softwoods** (spruce, pine, fir) about 90 % of the volume is **tracheids**, cells 2–5 mm long and 20–50 μm across; **hardwoods** (oak, beech, eucalyptus, and also balsa: the term is botanical) have wide **vessels** for conduction and thick-walled **fibres** for strength.

The cell wall is itself a composite (Figure 18.5): partly crystalline **cellulose** microfibrils, 3–4 nm across with an axial modulus of about 130 GPa, in a matrix of **hemicellulose** and **lignin**. In the thick S2 layer they wind at only 5–30° to the cell axis (the **microfibril angle**), which controls stiffness as ply angle does in a laminate (Chapter 22): juvenile wood near the pith, with large angles, is compliant and shrinks lengthwise.

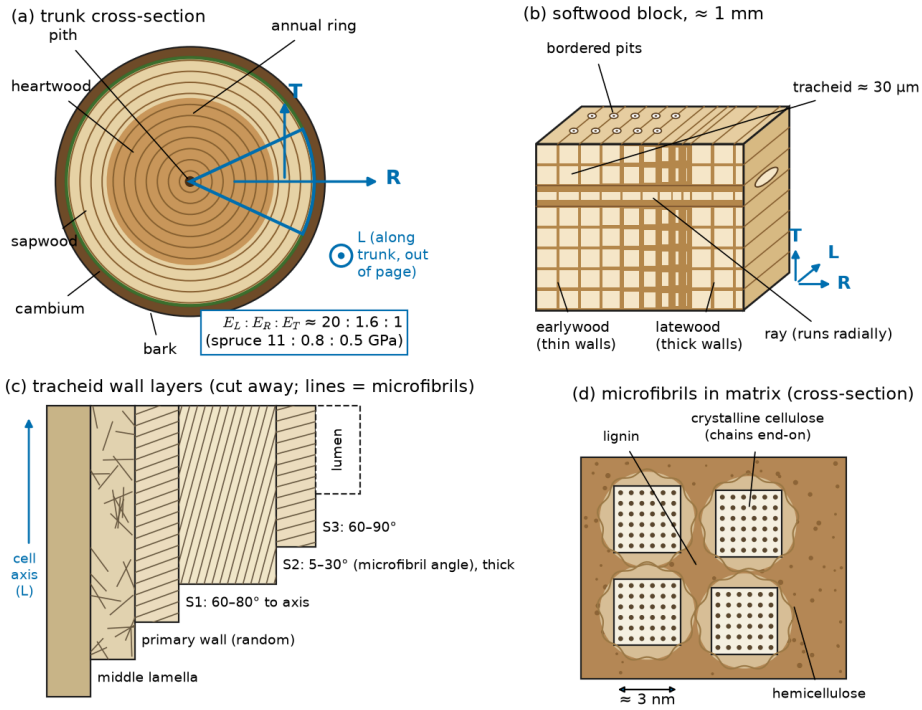

Figure 18.5 — Hierarchical structure of softwood from the trunk to the cellulose microfibril, with the principal axes L, R and T.

Anisotropy and strength

The tubular, aligned structure makes wood strongly anisotropic. For clear (defect-free) spruce at 12 % moisture content and a density of about 450 kg/m³, typical values are: modulus along the grain $E_L \approx 11$ GPa, radial $E_R \approx 0.8$ GPa and tangential $E_T \approx 0.5$ GPa; tensile strength along the grain 80–100 MPa; compressive strength along the grain 40–45 MPa (cell walls buckle and kink); compression perpendicular to the grain only 4–6 MPa (cells collapse); and tensile strength across the grain only 2–3 MPa. Along the grain, wood is efficient: the specific stiffness of spruce, $11\,000/450 = 24$ MPa per kg/m³, almost equals that of steel ($210\,000/7850 = 27$), and its specific tensile strength (about 180 MPa per Mg/m³ clear) is four times that of S355 steel (45). Density is the single best predictor of the mechanical properties of wood, as for any cellular solid: stiffness and strength along the grain scale roughly linearly with density.

Knots, grain deviation and juvenile wood cut the strength of structural timber to a third or half of clear wood, with large scatter, so timber is **strength graded** (visually or by machine-measured stiffness) into classes with 5th-percentile values: C24 softwood has a characteristic bending strength of 24 MPa, a mean modulus of 11 GPa and a characteristic density of 350 kg/m³.

Moisture

Wood is hygroscopic. Water is bound in the cell walls up to the **fibre saturation point** (about 25–30 % moisture content by dry mass) and free in the cavities above it. In service wood reaches an **equilibrium moisture content** of about 12 % at 20 °C and 65 % RH, 8–10 % in heated buildings. Losing bound water shrinks the wood very anisotropically: from green to oven-dry, typically 0.1–0.3 % longitudinally, 3–6 % radially and 6–10 % tangentially, which makes boards cup. Drying below the fibre saturation point also raises strength, by roughly 2–5 % per percent of moisture. Decay fungi need moisture contents above about 20 %, so keeping timber dry is its main protection.

Worked example — Movement of a timber board. A flat-sawn softwood board 200 mm wide (the width is tangential to the rings) is installed at 20 % moisture content in a heated building where it will reach 10 %. Its total tangential shrinkage from the fibre saturation point (take 28 %) to oven-dry is 7.5 %, so in the bound-water range it shrinks by about $7.5/28 = 0.27$ % of its width per percent of moisture content. The change of 10 percentage points gives a shrinkage of 2.7 %, or $0.027 \times 200 = 5.4$ mm. Its length changes by only millimetres over 4 m. Floors and connections must let boards move across their width; restrained shrinkage splits the wood.

> **Status of the principle: Linear shrinkage below the fibre saturation point**
> *Empirical.* Shrinkage is approximately proportional to the change in moisture content below the fibre saturation point, with species-dependent coefficients; it neglects hysteresis between wetting and drying and the gradients that develop in thick sections.

Wood also **creeps** under sustained load, especially when moisture content changes (mechano-sorptive creep); design codes multiply deflections by creep factors (k_{def} in Eurocode 5, 0.6–2.0 depending on service class) and reduce strengths for long-duration loads (k_{mod}). In fire, wood **chars** at a predictable rate, about 0.65 mm/min for softwood, and the char layer insulates the core, so large timber sections retain their load capacity for a calculable time.

Engineered wood

Engineered wood products overcome the size limits, defects and anisotropy of sawn timber by cutting the wood into smaller pieces and bonding them with adhesives (phenol–resorcinol–formaldehyde, melamine, polyurethane):

Product	Construction	Advantages	Uses
Plywood	Thin veneers, alternate layers crossed at 90°	Near-isotropic in-plane, dimensionally stable	Sheathing, formwork, boats
Oriented strand board (OSB)	Strands in cross-oriented layers	Uses small, fast-growing logs	Sheathing, I-joist webs
Laminated veneer lumber (LVL)	Veneers, all parallel	High strength, defects dispersed	Beams, flanges
Glued-laminated timber (glulam)	Graded lamellae (~40 mm) bonded parallel	Large, long and curved members; high grades on the outside	Arches, portal frames, bridges
Cross-laminated timber (CLT)	Layers of boards crossed at 90°	Load-bearing wall and floor panels, two-way spanning	Mass-timber buildings

Defects in each lamella or veneer fall at random positions, so engineered products are stronger and more uniform than the sawn timber they are made from: a classic weakest-link argument (Chapter 14). CLT and glulam have made **mass-timber buildings** of 18–25 storeys possible (Mjøstårnet in Norway, 85 m; Ascent in Milwaukee, 87 m), with low embodied carbon, factory prefabrication and fast erection (Section 18.9).

18.7 Asphalt and bituminous materials

Bitumen

Most of the world's paved roads are surfaced with **asphalt concrete**: aggregate bound by **bitumen**, the vacuum-distillation residue of crude oil. Bitumen is separated into four **SARA** fractions (saturates, aromatics, resins, asphaltenes); in the **colloidal model**, nanometre clusters of polar asphaltenes are dispersed in the oily medium and stabilized by resins, and more aggregated asphaltenes give stiffer, more elastic bitumen. It is **thermoplastic and viscoelastic** (Chapter 14): a viscous liquid when hot or slowly loaded, a brittle glass (T_g about −20 to −30 °C) when cold or quickly loaded, and time–temperature master curves underpin pavement design.

Bitumens are graded by empirical consistency tests or by performance:

- **Penetration**: the depth, in tenths of a millimetre, that a standard needle penetrates in 5 s at 25 °C under 100 g. Penetration grades such as 40/60, 50/70 and 70/100 cover hot to cold climates.
- **Softening point** (ring-and-ball): the temperature at which a disc of bitumen softens enough for a steel ball to sink through it, typically 43–58 °C for paving grades, corresponding to a viscosity of about 1200 Pa s and a penetration of roughly 800.
- **Penetration Index** (PI): a measure of temperature susceptibility derived from these two, assuming log(penetration) is linear in temperature. With $A = [\log_{10} 800 - \log_{10} \text{pen}_{25}]/(T_{R\&B} - 25)$, $\text{PI} = (20 - 500A)/(1 + 50A)$. For a 50/70 bitumen with penetration 60 and softening point 49 °C, $A = (2.903 - 1.778)/24 = 0.0469$ and $\text{PI} = -1.03$. Paving bitumens have PI between about −1.5 and +1; more negative values mean higher temperature susceptibility (soft in summer, brittle in winter).
- **Performance grade** (Superpave PG, e.g. PG 64-22): the binder meets rheological criteria (dynamic shear rheometer, bending-beam rheometer) up to a high pavement temperature of 64 °C and down to a low temperature of −22 °C, after simulated short- and long-term ageing.

> **Status of the principle: Penetration Index**
> *Empirical.* Derived from two single-point consistency tests and the assumption that log penetration varies linearly with temperature; it ranks temperature susceptibility of unmodified bitumens but is unreliable for polymer-modified binders, for which rheological (PG) grading is used.

Asphalt mixtures

A dense-graded hot-mix asphalt contains about 94–95 % aggregate and 5–6 % bitumen by mass, with 3–5 % air voids after compaction. The aggregate skeleton carries the load by interlock and friction; the binder glues the skeleton and fills the voids. Mix design (Marshall, or Superpave volumetric design with a gyratory compactor) chooses the aggregate grading and binder content to balance two competing failure modes:

Distress	Mechanism	Favoured by	Remedies
Rutting (permanent deformation)	Viscous flow and densification of the mix in hot weather under slow, heavy wheels	Too much or too soft binder; rounded aggregate; low air voids	Stiffer or polymer-modified binder, angular stone skeleton (stone-mastic asphalt)
Fatigue cracking	Repeated tensile strain at the bottom of the asphalt layer	Too little or too stiff (aged) binder; thin layers; weak support	Thicker layers, more binder, softer or modified binder, perpetual-pavement design
Thermal (low-temperature) cracking	Contraction stresses exceed tensile strength when the binder is glassy	Hard, aged, high-T_g binder; cold climate	Softer binder grade, polymer modification
Moisture damage (stripping)	Water displaces bitumen from hydrophilic (siliceous) aggregate	Acidic aggregates, poor drainage	Hydrated lime or amine anti-strip additives
Ageing	Oxidation and loss of light fractions stiffen and embrittle the binder	Sun, air voids, hot mixing	Lower air voids, rejuvenators, warm-mix technology

Polymer-modified bitumens (3–7 % SBS block copolymer forming a swollen elastic network, Chapter 20) widen the window between rutting and cracking; **warm-mix asphalt** is produced 20–40 K cooler; and **reclaimed asphalt pavement** is reused at 20–50 % or more, making asphalt one of the most recycled construction materials.

18.8 Masonry, brick and stone

Bricks and blocks

Fired-clay bricks are traditional ceramics (Chapter 17) extruded or pressed from clays and shales and fired at 900–1150 °C, where fluxes form a glass that partly **vitrifies** the body. Hotter firing gives lower porosity, higher strength (10 MPa for soft common bricks to over 100 MPa for engineering bricks) and lower water absorption (about 20 % down to below 5 %). Bricks fail mainly through moisture: **frost damage**, **efflorescence** of soluble salts, and irreversible **moisture expansion** (up to about 0.1 %) that requires movement joints. Concrete and calcium silicate units shrink rather than expand.

Mortar and masonry

Mortar beds the units and seals the joints; cement–lime–sand mixes (1:1:6, 1:½:4½) are classed M2.5 to M12. Lime adds workability and deformability so that movement cracks form in the replaceable joints: **the mortar should be weaker than the units**. Repairing historic lime-mortar masonry with strong, impermeable cement mortar traps moisture and concentrates stress in the old units, a well-known conservation error.

Under compression the more deformable mortar tries to expand laterally, is confined by the units and puts them into lateral tension, so masonry fails by vertical splitting at a strength between those of mortar and units. Eurocode 6 expresses this empirically:

$$f_k = K\, f_b^{0.7} f_m^{0.3}$$

where f_b and f_m are the normalized unit and mortar strengths and K depends on the unit type (0.55 for solid clay units with general-purpose mortar). For units of 20 MPa and mortar of 10 MPa, $f_k = 0.55 \times 20^{0.7} \times 10^{0.3} = 0.55 \times 8.14 \times 2.00 = 8.9$ MPa. Masonry is strong in compression and very weak in tension and shear, which is why unreinforced masonry performs badly in earthquakes and why it is often reinforced, confined by concrete ties, or post-tensioned in seismic regions.

> **Status of the principle: Masonry compressive-strength formula**
> *Empirical.* A power-law fit to wallette tests for common unit and mortar types; it does not apply to unusual geometries, thin-layer or lightweight mortars outside the calibrated range, or to masonry with voids and poor workmanship.

Natural stone

Building stones are classified by origin. **Igneous** stones (granite, basalt) crystallized from melts; they are hard, dense (porosity below 1–2 %), strong (granite 100–250 MPa in compression) and durable. **Sedimentary** stones (sandstone, limestone) formed from deposited grains or shells cemented by silica, calcite or clay; their porosity (2–30 %) and strength (20–200 MPa) vary enormously with the cement and the degree of compaction. **Metamorphic** stones (marble, slate, quartzite) were recrystallized by heat and pressure; slate splits along its cleavage into thin roofing slates.

Stone decays mostly by the action of water and salts. **Salt crystallization** in pores generates crystallization pressure as salts precipitate from evaporating solutions, the main cause of decay of porous sandstones and limestones in coastal and polluted environments. **Frost** acts as in concrete. **Acid deposition** dissolves calcite: sulfur dioxide in polluted air converts the surface of limestone and marble to gypsum, $\mathrm{CaCO_3 + SO_2 + \tfrac{1}{2}O_2 + 2H_2O \rightarrow CaSO_4 \cdot 2H_2O + CO_2}$, which forms black crusts in sheltered areas and washes away from exposed ones. The selection of stone for durability is based on porosity, pore-size distribution (fine pores retain water and concentrate crystallization pressure), and salt-crystallization and freeze–thaw tests.

18.9 Low-carbon cements and construction sustainability

The carbon problem

Concrete is the most-used material after water, and its environmental burden comes almost entirely from the cement. About 4 billion tonnes of cement are produced each year, and cement manufacture accounts for roughly 7–8 % of global anthropogenic CO_2 emissions. About 60 % of those emissions are **process emissions** from the decarbonation of limestone, which cannot be avoided by switching fuel: from the reaction $CaCO_3 \rightarrow CaO + CO_2$, each tonne of CaO releases $44.0/56.1 = 0.785$ t of CO_2, and since clinker contains about 65 % CaO, the process emission is about 0.51 t CO_2 per tonne of clinker. Fuel combustion adds roughly 0.3 t and electricity for grinding a little more, giving 0.8–0.9 t CO_2 per tonne of clinker in a modern plant.

Worked example — Carbon in the C30 concrete. Taking 0.87 t CO_2 per tonne of clinker and a CEM I with 95 % clinker, the cement carries about 0.83 t CO_2/t. The mix of Section 18.2 with 442 kg/m³ of cement then embodies about $0.442 \times 0.83 = 0.37$ t CO_2 per cubic metre from the cement alone (aggregates, transport and mixing typically add another 20–40 kg). Replacing half the cement with GGBS (whose allocated footprint is of order 0.07 t CO_2/t, mainly from grinding) reduces the binder emissions to $0.221 \times 0.83 + 0.221 \times 0.07 = 0.199$ t/m³, a cut of 46 %. A useful figure of merit is the **binder intensity**, kilograms of binder per cubic metre per MPa of strength: here $442/30 = 14.7$ kg m^{-3} MPa^{-1}; good practice with optimized grading, superplasticizers and SCMs approaches 5–8.

Strategies

Strategy	How it works	Typical CO_2 reduction (per tonne of binder vs CEM I)	Limits
Clinker substitution with GGBS (CEM III/A–B)	Latent hydraulic slag replaces 36–80 % of clinker	40–70 %	Slow early strength; slag supply limited and shrinking as steel moves to electric-arc furnaces
Fly ash (CEM II/B-V, CEM IV)	Pozzolanic reaction with CH	15–35 %	Coal plants closing; supply falling
Limestone calcined clay cement (LC3)	~50 % clinker, ~30 % calcined kaolinitic clay (calcined at ~800 °C), ~15 % limestone, gypsum	~40 % (about 0.51 vs 0.83 t/t)	Needs suitable clays; widely available in tropical regions
Portland-limestone cement (CEM II/A-LL, ASTM Type IL)	6–20 % fine limestone filler	~10–15 %	Moderate gain; filler effect only
Alkali-activated materials, geopolymers	Slag or metakaolin activated by sodium silicate/hydroxide; no clinker	40–80 % (depends on activator)	Activator cost and carbon, handling, durability standards still developing
Belite-rich and calcium sulfoaluminate cements	Less $CaCO_3$ per tonne, lower kiln temperature	20–35 %	Cost, raw materials, different durability profile
Carbon capture at the kiln; recarbonation	Oxyfuel and post-combustion capture; CO_2 uptake by carbonating concrete or by CO_2-cured precast	Up to 90 % for capture	Cost and energy; large-scale plants only now being built
Using less concrete	Optimized structural forms, higher strength where efficient, longer service life, reuse of elements	Proportional to material saved	Requires design and code change

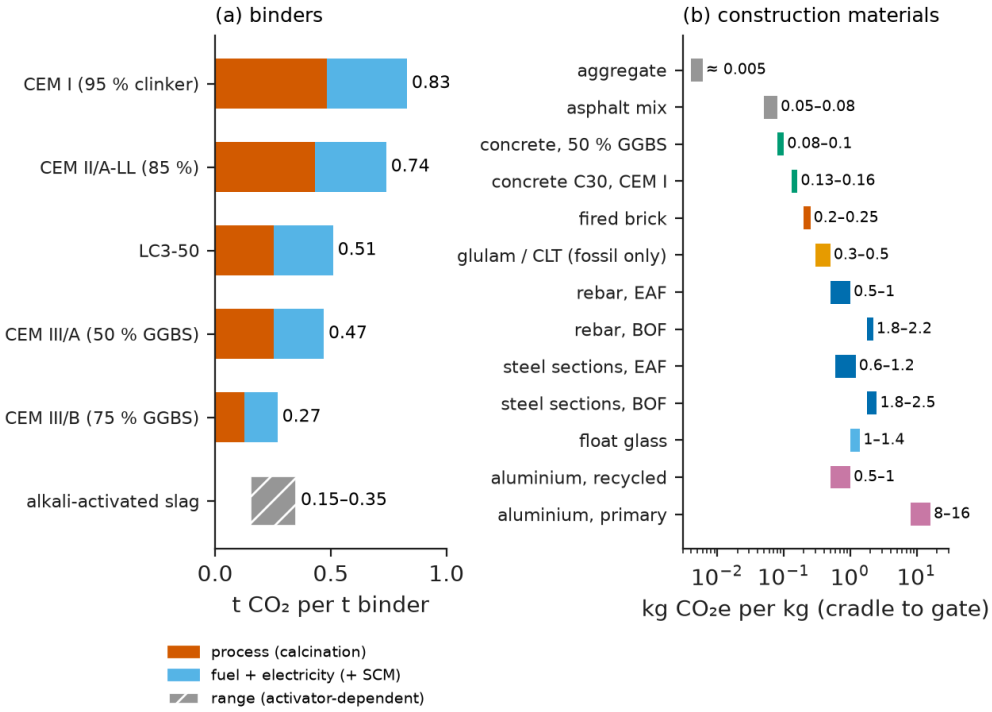

Figure 18.6 — Cradle-to-gate CO_2 intensity of cementitious binders, and embodied carbon of common construction materials per kilogram.

Construction as a whole

Typical cradle-to-gate footprints (Figure 18.6, right panel) are 0.13–0.16 kg CO_2e/kg for CEM I concrete (300–400 kg/m³), 0.2–0.25 for brick, 0.5–1.0 for scrap-based (EAF) steel against about 2 for blast-furnace steel, and 0.3–0.5 of fossil emissions for glulam and CLT, which also store about 1.6 kg of biogenic CO_2 per kg in use. They vary with plant and energy mix and must be compared per functional unit, such as a square metre of floor, not per kilogram (Chapter 59).

Three conclusions follow. Because concrete is used in such quantity, blended cements and lower binder intensity are the largest immediate lever, and the durability science of Section 18.3 mostly supports them. Steel's footprint falls sharply with scrap-based production, and reinforcement and sections are among the most recycled products in the economy. Timber stores carbon, which explains the growth of mass timber, subject to sustainable forestry, fire and moisture design. And durability is itself sustainability: a car park demolished after 40 years must be built, and its carbon spent, twice within the life of one designed with deeper cover and a slag concrete.

> **Boundary.**
> Life-cycle assessment methodology, critical materials and the circular economy are developed in Chapter 59; this section gives only the materials data and levers specific to construction.

What remains open

Open question	Where it is answered
How are ternary phase relations used to understand clinker formation?	Chapter 7 (back reference)
What laws govern the diffusion of chloride and CO_2 through porous solids?	Chapter 8 (back reference)
How are the ferrite–pearlite and quenched-and-tempered structures of construction steels produced?	Chapter 15 (back reference)
How do creep, viscoelasticity and the ductile–brittle transition enter concrete, asphalt, timber and steel design?	Chapter 14 (back reference)
How do fibre-reinforced polymers compare with steel as reinforcement and with timber as a cellulose composite?	Chapters 22–23
How do steel reinforcement and prestressing tendons corrode, and how is cathodic protection sized?	Chapters 42–43
How are deteriorated concrete, masonry and timber structures diagnosed?	Chapter 44
How are construction steels welded without cracking?	Chapter 47
How are cement phases and hydration products characterized?	Chapters 52–53
How are materials selected for minimum embodied carbon at the design stage?	Chapter 58
How are life-cycle assessment and circular-economy strategies applied to construction?	Chapter 59

Next: Chapter 19 begins Part V with the third great class of materials, explaining how polymers are synthesized from small molecules and how their chain length, architecture and chemistry set the properties of plastics, rubbers and fibres.

Part V — Polymers and Composites

A milk bottle, a hip-joint liner, a gas pipe under a city street, a wind-turbine blade and an airliner wing skin share one thing: their load-bearing material is built from long organic chains. In the bottle, the pipe and the liner the chains are the whole story; in the blade and the wing they are the matrix that holds stiff glass or carbon fibres in place. Part V explains both. It starts from the molecule — a chain of thousands of repeat units — and shows how its length, architecture and the way it coils, tangles, crystallizes and vitrifies produce the behaviour engineers rely on and contend with: rubber that springs back because of entropy, plastics that soften at a glass transition, melts that thin when sheared, parts that creep, craze and crack. It then adds fibres, and asks how composite stiffness, strength and expansion follow from the constituents, how a single anisotropic ply behaves, and how stacks of plies are designed, damaged and manufactured.

By the end of this part you should understand

- How step-growth and chain-growth polymerizations make chains, why the Carothers equation demands conversions above 99 %, and how catalysts control tacticity and branching.
- Why a polymer has a distribution of molecular weights, how its averages and dispersity are measured, and why toughness and melt flow depend on different averages.
- How random-walk statistics describe a chain, why rubber elasticity is entropic, and why most polymer pairs do not mix while block copolymers self-assemble.
- How polymers crystallize into lamellae and spherulites, and what free volume and the WLF equation say about the glass transition.
- Why melt viscosity scales as $M^{3.4}$, and how yielding, crazing, stress cracking and degradation limit polymer parts.
- How additives, vulcanization, recycling and biodegradable chemistries change properties and end of life.
- How the rule of mixtures, Halpin–Tsai and shear-lag models estimate composite properties, and when they fail.
- How ply stiffness transforms with angle, how laminate theory treats a stack of plies, and how failure criteria, delamination and impact damage set allowables.
- How composite parts are manufactured, and where metal- and ceramic-matrix composites take over.

Chapters in this part

Chapter	Central question	What it delivers
19 Polymer Synthesis and Molecular Structure	How are chains made, and which molecular features does synthesis fix?	Carothers and free-radical kinetics; molecular-weight averages and their measurement; copolymers, tacticity, branching, networks; the families of commercial polymers
20 Polymer Physics and Morphology	How do chain statistics, mixing, crystallization and the glass transition turn a molecule into a solid?	Chain dimensions and Kuhn length; entropic rubber modulus; Flory–Huggins miscibility and microphase separation; crystallinity; WLF and Tg control; liquid-crystalline polymers
21 Polymer Rheology, Mechanics and Processing	How do melts flow, how do parts deform and fail, and how are polymers shaped, formulated, aged and recycled?	Reptation and the $M^{3.4}$ law; die and moulding estimates; yielding, crazing, stress cracking; processing routes; additives; aging life; vulcanization; recycling and biodegradation
22 Composite Micromechanics and Lamina Behaviour	How do fibres and matrix share load, and how does one anisotropic ply respond?	Fibre and matrix data; rule of mixtures and Halpin–Tsai; shear-lag critical length; the stiffness matrices Q and $\bar{Q}$; ply thermal and moisture strains
23 Laminate Theory, Failure and Composite Manufacturing	How is a stack of plies analysed, how does it fail, and how is it made?	The ABD matrix and layup rules; Tsai–Wu, Puck and related criteria; delamination and compression after impact; sandwich panels; manufacturing routes; metal- and ceramic-matrix composites

How this part connects to the rest of the book

Part V draws on what came before. Chain statistics and mixing use the solution thermodynamics of Chapter 6 and the nucleation ideas of Chapter 9; the polymer glass transition echoes the oxide glasses of Chapter 17; the viscoelastic models and fracture mechanics of Chapter 14 are applied to real polymer parts; and composite micromechanics rests on the anisotropic elasticity of Chapter 11. In turn, thermal and dielectric behaviour of polymers returns in Chapters 24 and 26, electroactive polymers and hydrogels in Chapter 37, nanocomposites in Chapter 38, resorbable polymers in Chapter 39, and membranes and separators in Chapter 40. Weathering and failure analysis appear in Chapter 44, adhesive bonding in Chapter 47, polymer additive manufacturing in Chapter 48, thermal analysis in Chapter 52 and chain simulation in Chapter 56. Chapters 58 and 59 weigh polymers and composites against metals and ceramics on performance, cost and end of life.

Chapter 19 — Polymer Synthesis and Molecular Structure

Part V · Polymers and Composites

A maintenance crew fitting a new wiring harness into a regional airliner finds that the nylon-6,6 cable ties from one carton snap when they are pulled tight, while ties from the next carton, same part number, same supplier, same mould, survive being bent double. Tensile tests confirm the difference: the bad ties break at a few percent strain with almost no necking; the good ones draw out to more than 50 %. The chemistry is identical, the crystallinity is similar, and the dimensions are within tolerance. The difference shows up only when the two batches are dissolved and run through a size-exclusion chromatograph: the brittle ties have a number-average molecular weight of about 9,000 g/mol, the good ones about 18,000 g/mol. The resin for the bad batch had been moulded without drying. At 280 °C, a tenth of a percent of absorbed water was enough to hydrolyse one amide bond in every chain, halve the chain length and remove the entanglements that let the plastic draw rather than crack.

The episode contains the whole subject of this chapter. A polymer is not defined by its chemistry alone: the length of its chains, the spread of those lengths, the sequence of different units along a chain, the way side groups are arranged in space, and whether chains are branched or joined into networks all matter as much as the repeat unit. All of these features are fixed during synthesis, and many can be changed again during processing.

> **How are long chains built from small molecules, and what determines how long they become? Which molecular features — chain length and its distribution, copolymer sequence, tacticity, branching, crosslinking — does synthesis control, and how are they measured?**

Concept	Engineering destination
Repeat unit and nomenclature	Reading datasheets, resin codes, regulatory and recycling labels
Step-growth and the Carothers equation	Polyamide, polyester and polyurethane production; drying specifications for moulding
Free-radical and ionic chain growth	Polystyrene, PMMA, PVC, LDPE; adhesives and dental resins cured in situ
Coordination catalysis	HDPE pipe, LLDPE film, isotactic PP for car bumpers and medical syringes
Ring-opening polymerization	Nylon-6, PLA sutures and implants, epoxy and silicone networks
Molecular-weight averages and dispersity	Melt flow index, toughness, stress-crack resistance, solution viscosity
Copolymers	ABS housings, SBS shoe soles, EVA solar-cell encapsulants, nitrile gloves
Tacticity, branching, networks	Stiffness of PP, density grades of PE, cured epoxies and rubber
Polymer families	Selecting commodity, engineering or high-performance grades

Roadmap. Monomers and how polymers are named (19.1) → the two ways of growing a chain: step growth with its Carothers limit (19.2) and chain growth by radical, ionic, coordination and ring-opening routes (19.3) → why the product is a distribution and how it is measured (19.4) → what happens when two monomers are combined (19.5) → the spatial and topological architecture of chains (19.6) → the families of polymers that result (19.7).

19.1 Monomers, repeat units and nomenclature

From monomer to macromolecule

A **polymer** is a molecule made by linking many small molecules, the **monomers**, through covalent bonds. The part of the chain that repeats is the **repeat unit** (or constitutional repeating unit). In polyethylene the monomer is ethylene, $\mathrm{CH_2{=}CH_2}$, and the repeat unit is $-\mathrm{CH_2CH_2}-$; opening the double bond turns each monomer into a two-carbon segment of a saturated backbone. The number of repeat units in a chain is the **degree of polymerization** X (also written DP or N), and the molar mass of the chain is

$$M = X\,M_0$$

where M_0 is the molar mass of the repeat unit (end groups are negligible for long chains). A commercial high-density polyethylene with $M \approx 2 \times 10^5$ g/mol contains about 7,000 repeat units; an ultra-high-molecular-weight polyethylene (UHMWPE) for hip and knee bearings, with $M \approx 5 \times 10^6$ g/mol, contains about 180,000. Molecules with X below about 10–20 are **oligomers**: they are waxes, oils or brittle solids, because they are too short to entangle.

The two necessary features of a monomer are reactivity and **functionality** f, the number of bonds it can form in the polymerization. A monomer with $f = 2$ (a vinyl double bond, which opens to give two bonds, or a molecule with two reactive groups such as a diacid) gives linear chains. A monomer with $f \geq 3$ (glycerol, with three hydroxyl groups; divinylbenzene, with two double bonds and hence $f = 4$) gives branches and, eventually, a network (Section 19.6).

Naming polymers

Three naming systems coexist, and an engineer meets all three.

- **Source-based names** put "poly" before the monomer name: polyethylene, polystyrene, poly(vinyl chloride), poly(methyl methacrylate). Parentheses are used when the monomer name has more than one word.
- **Structure-based (IUPAC) names** name the repeat unit: polyethylene is poly(methylene), polystyrene is poly(1-phenylethylene). They are unambiguous but rarely used outside chemistry.
- **Trade names and abbreviations** dominate engineering drawings and datasheets: PE, PP, PS, PVC, PMMA (Perspex, Plexiglas), PA66 (nylon-6,6), PET, PC, PTFE (Teflon), PEEK, POM (acetal, Delrin), ABS, PU.

Condensation polymers are often named by their linkage: polyamides contain $-\mathrm{CO}-\mathrm{NH}-$, polyesters $-\mathrm{CO}-\mathrm{O}-$, polyurethanes $-\mathrm{NH}-\mathrm{CO}-\mathrm{O}-$, polycarbonates $-\mathrm{O}-\mathrm{CO}-\mathrm{O}-$, polyimides a cyclic imide. Polyamide numbers count carbon atoms: nylon-6 is made from caprolactam (six carbons in the repeat unit); nylon-6,6 from hexamethylenediamine (six carbons) and adipic acid (six carbons); nylon-6,10 and nylon-12 follow the same rule. The number of methylene groups between amide links controls hydrogen-bond density and therefore melting point and water uptake: nylon-6,6 absorbs about 2.5 % water at 50 % relative humidity, nylon-12 under 1 %, which is why fuel lines are made of the latter.

> **Boundary.**
> The organic chemistry of functional groups and reaction mechanisms (nucleophilic acyl substitution, radical chemistry) belongs to Advanced Chemistry. This chapter uses the reactions only to explain what molecular structure they produce.

Two families of growth

Polymerizations fall into two kinetic families, distinguished not by chemistry but by how chain length grows with conversion (Figure 19.1).

- In **step-growth** polymerization any two molecules carrying complementary groups can react: monomer with monomer, dimer with trimer, chain with chain. Monomer disappears early, but long chains appear only at the very end.
- In **chain-growth** polymerization an active centre (radical, ion or metal–carbon bond) adds monomers one at a time. Each chain grows to full length within seconds or less; at any moment the reactor contains monomer and finished polymer, with very few growing chains.

The older classification into *condensation* (a small molecule such as water is released) and *addition* (no by-product) polymers overlaps but is not identical: polyurethanes are made by step growth without a by-product, and nylon-6 is made by ring-opening chain growth even though its repeat unit is that of a condensation polymer.

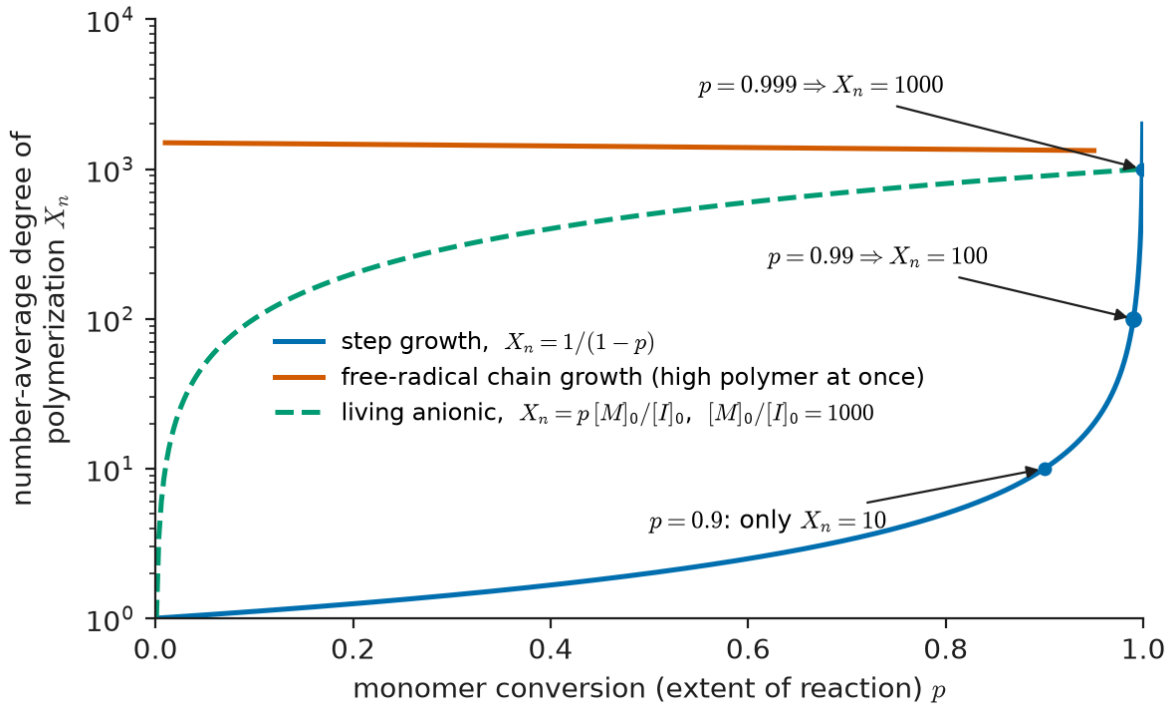

Figure 19.1 — Number-average degree of polymerization versus monomer conversion for step-growth, free-radical chain-growth and living chain-growth polymerization.

Figure 19.1 explains a practical difference. A chain-growth reactor can be stopped at 50 % conversion and still deliver high polymer (the unreacted monomer is stripped off and recycled). A step-growth reactor must be driven to almost complete conversion, because, as the next section shows, the chains are short until the last fraction of a percent of reaction.

19.2 Step-growth polymerization and the Carothers equation

The Carothers argument

Consider a stoichiometric mixture of difunctional monomers, for example a diacid and a diamine, with N_0 molecules in total and therefore $2N_0$ reactive groups, half of each kind. Define the **extent of reaction** p as the fraction of functional groups that have reacted. Each reaction between two groups joins two molecules into one, so the number of molecules falls by one per bond formed. After pN_0 bonds have formed (each bond consumes one acid and one amine group, and there are N_0 acid groups), the number of molecules is $N = N_0(1 - p)$. The number-average degree of polymerization, counting monomer residues, is

$$X_n = \frac{N_0}{N} = \frac{1}{1 - p}$$

This is the **Carothers equation**, published by Wallace Carothers at DuPont in 1929, in the work that led to nylon-6,6.

> **Status of the principle: Carothers equation**
> *Derived.* It follows from counting molecules, assuming exact stoichiometry, no side reactions, no cyclization and equal reactivity of all groups regardless of chain length (Flory's equal-reactivity principle, which is well supported experimentally for chains longer than a few units).

The consequence is severe. At 90 % conversion, which would count as a good yield in most organic syntheses, $X_n = 10$: a waxy oligomer. At 99 %, $X_n = 100$; at 99.5 %, 200. The useful properties of nylon and polyester require X_n of order 100–200, so the reaction must be driven beyond 99 %. Two practical consequences follow. First, the equilibrium must be pushed hard toward the polymer: polycondensation reactors run at high temperature under vacuum to strip the water or glycol that is released (equilibrium constants for amidation are only about 100–400, and for esterification of order 1). Second, the stoichiometry must be essentially exact.

Stoichiometric imbalance and chain stoppers

If one monomer is in excess, the chains end up capped with the excess group and growth stops. With N_A groups of type A and $N_B \geq N_A$ of type B, define $r = N_A/N_B \leq 1$. Repeating the counting gives

$$X_n = \frac{1 + r}{1 + r - 2rp}$$

At complete conversion of the minority group ($p = 1$), $X_n = (1 + r)/(1 - r)$. A monofunctional impurity has the same effect and is counted in r with a factor of two. Manufacturers exploit this deliberately: a small, metered excess of one monomer or a few tenths of a percent of a monofunctional acid (acetic acid in nylon) caps the chains and fixes the final molecular weight, so that the melt viscosity does not drift during later processing. The industrial trick for nylon-6,6 is to make the 1:1 "nylon salt" (hexamethylene diammonium adipate) first, crystallize it to exact stoichiometry, and then polymerize the salt.

Worked example — molecular weight of nylon-6,6 by step growth. The repeat unit of nylon-6,6, $-\mathrm{NH(CH_2)_6NH-CO(CH_2)_4CO-}$, has $M = 226.32$ g/mol and contains two monomer residues, so the mean residue mass is $M_0 = 113.16$ g/mol. With exact stoichiometry and $p = 0.99$, $X_n = 1/(1 - 0.99) = 100$ residues and $M_n = 100 \times 113.16 \approx 11{,}300$ g/mol. Pushing to $p = 0.995$ doubles this to $X_n = 200$, $M_n \approx 22{,}600$ g/mol, a typical injection-moulding grade. Now suppose the diamine is weighed 1 % short, $r = 0.99$. At $p = 0.99$, $X_n = (1.99)/(1.99 - 2 \times 0.99 \times 0.99) = 1.99/0.0298 = 66.8$, giving $M_n \approx 7{,}600$ g/mol; even at complete conversion, $X_n = 1.99/0.01 = 199$, $M_n \approx 22{,}500$ g/mol. A 1 % weighing error therefore costs a third of the chain length at a realistic conversion and caps it permanently. In the cable-tie failure of the opening, the reverse reaction did the damage: a resin with $M_n = 18{,}100$ g/mol ($X_n = 160$) contains $1000/18{,}100 \approx 0.055$ mol of chains per kg. Cutting each chain once requires 0.055 mol, or about 1.0 g, of water per kilogram of resin: 0.1 wt %. Polyamide moulding specifications therefore require moisture below about 0.2 %, against an equilibrium uptake of 2–3 % in humid air.

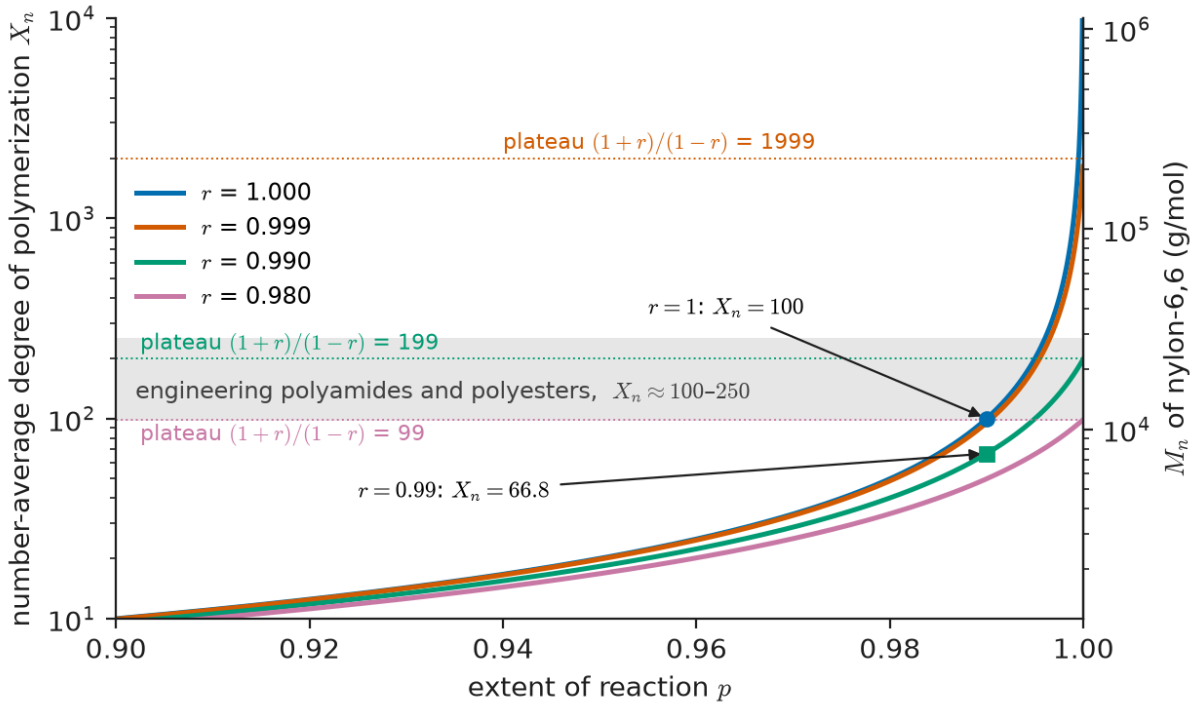

Figure 19.2 — Carothers equation: number-average degree of polymerization versus extent of reaction for exact stoichiometry and for small stoichiometric imbalances.

The most probable distribution

Because any two chains can react, step growth produces a statistical spread of lengths. Flory showed that, with equal reactivity, the number fraction of chains containing x residues is

$$n_x = (1 - p)\,p^{x-1}$$

and the weight fraction is $w_x = x(1 - p)^2 p^{x-1}$. This **most probable** (or Flory–Schulz) distribution has $X_n = 1/(1 - p)$, a weight average $X_w = (1 + p)/(1 - p)$ and therefore a dispersity $X_w/X_n = 1 + p$, which approaches 2 at high conversion. The

number distribution falls monotonically (single residues are always the most numerous species), while the weight distribution peaks at $x \approx -1/\ln p$, about 100 for $p = 0.99$. The dispersity of about 2 is a signature of step-growth polymers such as nylon, PET and polycarbonate, and it is a good check on the measurements of Section 19.4.

> **Status of the principle: Most probable (Flory) distribution**
> *Derived.* Follows from equal reactivity and random linking of linear chains; it is distorted by cyclization, by interchange reactions in the melt (which actually restore it) and by side reactions that branch the chain.

Step-growth polymers in engineering

Polymer	Monomers	Linkage	By-product	Typical use
Nylon-6,6 (PA66)	Hexamethylenediamine + adipic acid	Amide	Water	Gears, cable ties, radiator tanks, airbag fabric
PET	Ethylene glycol + terephthalic acid	Ester	Water (or methanol)	Bottles, fibre, film, food trays
Polycarbonate (PC)	Bisphenol A + diphenyl carbonate (or phosgene)	Carbonate	Phenol (or HCl)	Glazing, helmets, lenses, enclosures
Polyurethane (PU)	Diisocyanate + polyol	Urethane	None	Foams, elastomers, coatings, catheters
Epoxy	Diepoxide + diamine (or anhydride)	Ether/amine	None	Adhesives, composite matrices, encapsulants
Phenol–formaldehyde	Phenol + formaldehyde	Methylene	Water	Brake pads, laminates, electrical parts
Polyimide (Kapton)	Dianhydride + diamine	Imide	Water	Flexible circuits, wire insulation, aerospace films
PEEK	Bisfluorobenzophenone + hydroquinone salt	Ether, ketone	Salt	Aerospace brackets, spinal implants, pump parts

PET is a useful case study in what processing does to the chain. Bottle-grade PET needs an intrinsic viscosity of about 0.75–0.85 dL/g (M_n about 25,000 g/mol), higher than fibre grade (0.6 dL/g). The melt reactor cannot reach this because the melt becomes too viscous to release glycol, so the pellets are crystallized and then held for hours at 200–220 °C under nitrogen in **solid-state polymerization**, where the reaction continues in the amorphous regions of the solid. The same equilibrium runs backward during melt recycling, which is why mechanically recycled PET loses molecular weight with each pass unless it is dried and re-solid-stated (Chapter 21).

19.3 Chain-growth polymerization

Chain growth needs three steps: **initiation** of an active centre, **propagation** by repeated monomer addition, and **termination** or **transfer** that ends a chain. The four industrial families differ in the nature of the active centre.

Free-radical polymerization

An **initiator** decomposes thermally or photochemically into radicals: azobisisobutyronitrile (AIBN) and benzoyl peroxide are common thermal initiators; camphorquinone with an amine is the blue-light photoinitiator used to cure dental fillings in seconds. The radical adds to the double bond of a vinyl monomer, $CH_2{=}CHX$, creating a new radical on the chain end, which adds the next monomer, and so on. Two growing radicals meeting each other terminate by **combination** (forming one dead chain) or **disproportionation** (a hydrogen transfer that yields two dead chains). Radicals can also transfer their activity to monomer, solvent, polymer or a deliberately added **chain-transfer agent** (a thiol), which ends one chain and starts another.

The kinetics follow from the **steady-state assumption**: after a short induction period, radicals are created as fast as they are destroyed. With initiator decomposition rate constant k_d, initiator efficiency f (the fraction of radicals that actually start chains,

typically 0.5–0.8), propagation rate constant k_p and termination rate constant k_t, the rate of initiation is $R_i = 2fk_d[I]$, the radical concentration is $[M^\bullet] = (R_i/2k_t)^{1/2}$, and the rate of polymerization is

$$R_p = k_p[M][M^\bullet] = k_p[M]\left(\frac{fk_d[I]}{k_t}\right)^{1/2}$$

The **kinetic chain length** ν, the average number of monomers added per radical before it terminates, is

$$\nu = \frac{R_p}{R_i} = \frac{k_p[M]}{2\,(fk_dk_t[I])^{1/2}}$$

so that $X_n = 2\nu$ for termination by combination and $X_n = \nu$ for disproportionation (in the absence of transfer).

> **Status of the principle: Steady-state free-radical kinetics**
> *Approximate.* Valid at low to moderate conversion with constant rate coefficients. At high conversion in bulk the viscosity rises, k_t (a diffusion-limited reaction between two chain ends) collapses and the rate accelerates sharply — the Trommsdorff or gel effect — which broadens the distribution and can cause a thermal runaway.

The square-root dependence on initiator is the central trade-off: more initiator speeds the reaction as $[I]^{1/2}$ but shortens the chains as $[I]^{-1/2}$. Fast and long cannot both be had by adjusting the initiator; they are reconciled by emulsion polymerization (radicals isolated in separate latex particles, so termination is slow) or by controlled radical methods.

Worked example — chain length in bulk polystyrene. Styrene is polymerized in bulk at 60 °C ($[M] = 8.7$ mol/L) with AIBN at $[I] = 0.010$ mol/L. Take $k_d = 8.5 \times 10^{-6}$ s⁻¹, $f = 0.6$, $k_p = 341$ L mol⁻¹ s⁻¹ and $k_t = 7 \times 10^7$ L mol⁻¹ s⁻¹ (representative literature values). The initiation rate is $R_i = 2 \times 0.6 \times 8.5 \times 10^{-6} \times 0.010 = 1.02 \times 10^{-7}$ mol L⁻¹ s⁻¹. The radical concentration is $(1.02 \times 10^{-7}/(2 \times 7 \times 10^7))^{1/2} = 2.70 \times 10^{-8}$ mol/L — about one growing chain per 300 million monomer molecules. The polymerization rate is $R_p = 341 \times 8.7 \times 2.70 \times 10^{-8} = 8.0 \times 10^{-5}$ mol L⁻¹ s⁻¹, or about 3.3 % conversion per hour. The kinetic chain length is $\nu = R_p/R_i = 785$; styrene radicals terminate mainly by combination, so $X_n \approx 1570$ and $M_n \approx 1570 \times 104.15 \approx 1.6 \times 10^5$ g/mol. Cutting the initiator to a quarter, 0.0025 mol/L, doubles ν to 1570 but halves the rate to 1.7 % per hour. In practice transfer to monomer caps X_n at a few thousand, and the reaction is run at higher temperature with staged heating to finish in hours rather than days.

Free-radical chemistry produces polystyrene, PMMA, PVC, poly(vinyl acetate), polyacrylonitrile, PTFE and **low-density polyethylene** (LDPE). LDPE is made at 1,500–3,000 bar and 200–300 °C, conditions under which the growing radical frequently bites back on its own chain (intramolecular transfer), producing short butyl branches, and transfers to other chains, producing long branches. Those branches, 15–30 per 1,000 carbons, frustrate crystallization and give LDPE its low density (0.91–0.93 g/cm³), flexibility and the strain-hardening melt behaviour that makes it easy to blow into film.

Controlled (reversible-deactivation) radical polymerization — atom-transfer radical polymerization (ATRP), reversible addition–fragmentation chain transfer (RAFT) and nitroxide-mediated polymerization (NMP) — keeps most chains in a dormant state that is reactivated briefly and repeatedly. Termination is suppressed, all chains grow together and dispersities of 1.1–1.3 are reached with ordinary vinyl monomers. These methods are used to make block copolymers for drug delivery, dispersants and lithographic resists.

Ionic polymerization

In **anionic** polymerization the active centre is a carbanion, typically started by butyllithium. With monomers such as styrene, butadiene and isoprene, in carefully purified solvents, there is no termination step: the chains ends remain active until a terminating agent is added. This is **living polymerization**. If all chains start at once and grow at the same rate, the degree of polymerization is set by stoichiometry, $X_n = p[M]_0/[I]_0$, and the distribution is a Poisson distribution with dispersity $1 + 1/X_n$, below 1.05 in practice. Adding a second monomer to the living chains produces a block copolymer; this is how styrene–butadiene–styrene (SBS) thermoplastic elastomers and the narrow polystyrene standards used to calibrate chromatographs are made.

> **Status of the principle: Living polymerization yields a Poisson distribution**
> *Derived.* Follows from simultaneous initiation and identical, irreversible propagation of all chains with no termination or transfer; slow initiation or traces of water and oxygen broaden it.

Cationic polymerization uses a carbocation, initiated by Lewis acids (BF_3, $AlCl_3$) with a proton source. It suits monomers with electron-donating substituents, such as isobutylene and vinyl ethers. Carbocations are so reactive that transfer is rapid unless the reaction is run cold: butyl rubber (isobutylene with 1–3 % isoprene for later vulcanization) is made at about −100 °C. Butyl rubber's tightly packed chains give it the lowest gas permeability of common elastomers, which is why it lines every tubeless tyre and seals pharmaceutical vials.

Coordination polymerization: Ziegler–Natta and metallocene catalysts

In 1953 Karl Ziegler found that titanium chlorides combined with aluminium alkyls polymerize ethylene at atmospheric pressure and modest temperature to give a linear, high-density polyethylene; in 1954 Giulio Natta used similar catalysts to make crystalline, **isotactic** polypropylene. In **coordination** polymerization the monomer first coordinates to a vacant site on a transition-metal centre and is then inserted into the metal–carbon bond of the growing chain (the Cossee–Arlman mechanism). Because insertion happens at a geometrically constrained site, the catalyst controls the orientation of each incoming monomer — and therefore the tacticity (Section 19.6) — and branching by back-biting is almost eliminated.

Modern **Ziegler–Natta** catalysts are $TiCl_4$ supported on $MgCl_2$ with internal and external electron donors, activated by triethylaluminium. They are heterogeneous: the catalyst particle contains several kinds of active site, each producing polymer of a different length and comonomer content, so the product has dispersity 4–8. Particle morphology replicates from catalyst to polymer, so 50 μm catalyst grains grow into millimetre-scale polymer granules ready for the extruder. Phillips catalysts (chromium oxide on silica) produce much of the world's HDPE for blow-moulded bottles and pipes.

Metallocene (single-site) catalysts, developed in the 1980s by Kaminsky and others, are zirconium or hafnium complexes sandwiched between cyclopentadienyl rings and activated by methylaluminoxane (MAO). Every site is identical, so the dispersity is close to 2 (the most probable distribution) and comonomer is incorporated uniformly along all chains. By changing the ligand symmetry the same metal produces isotactic, syndiotactic or atactic polypropylene. Metallocene LLDPE (ethylene with 1-hexene or 1-octene comonomer giving controlled short branches) makes thinner, tougher food-packaging film with better clarity and heat-seal behaviour than Ziegler–Natta grades, and metallocene polyolefin elastomers have replaced plasticized PVC in many automotive and medical tubes.

Catalyst family	Active site	Typical dispersity	Characteristic products
Free radical (high pressure)	Radical	3–20	LDPE, PS, PVC, PMMA
Ziegler–Natta ($MgCl_2$/$TiCl_4$)	Ti–C, several site types	4–8	HDPE, isotactic PP, LLDPE
Phillips (Cr/SiO_2)	Cr–C	8–30	HDPE for blow moulding and pipe
Metallocene, single-site	Zr or Hf–C, one site type	2.0–2.5	mLLDPE, syndiotactic PS, polyolefin elastomers
Living anionic	Carbanion	1.02–1.1	SBS, solution SBR for tyres, calibration standards

The engineering value of a broad distribution should not be underestimated. Pipe-grade bimodal HDPE (PE100) is deliberately made in two reactors in series: one produces short chains that give easy processing, the other long chains carrying a little comonomer that tie the crystals together and give resistance to slow crack growth over a 50-year design life.

Ring-opening polymerization

Cyclic monomers polymerize by opening the ring, driven by the release of ring strain (and, for some, by entropy). Ring-opening is chain growth: an active centre attacks a ring and regenerates itself at the new chain end. The important engineering examples are:

- **Nylon-6** from caprolactam, opened by water (hydrolytic) or by anionic initiators. Anionic casting of caprolactam directly in a mould ("cast nylon") produces large bearings, sheaves and liners.
- **Poly(lactic acid)** (PLA) and **poly(glycolic acid)** (PGA) from the cyclic dimers lactide and glycolide, opened by tin(II) octoate. These are the resorbable sutures, bone screws and drug-eluting matrices of Chapter 39, and PLA is the most widely used bio-based packaging polymer.
- **Poly(ε-caprolactone)** (PCL), a low-melting (60 °C), slowly resorbing polyester for long-term implants and 3D-printed scaffolds.
- **Polyacetal** (POM) from trioxane, and **poly(ethylene oxide)** from ethylene oxide.

- **Silicones** (polydimethylsiloxane, PDMS) from cyclic siloxanes such as D4, used in sealants, medical tubing and soft lithography.
- **Epoxy** resins cure by opening the strained three-membered epoxide ring with an amine or anhydride, a step-growth network formation (Section 19.6).

19.4 Molecular weight: averages, dispersity and measurement

Why a distribution needs averages

Every synthetic polymer is a mixture of chains of different lengths. If a sample contains N_i molecules of molar mass M_i, the mass of that fraction is $w_i \propto N_i M_i$, and several averages are needed to describe the distribution:

$$M_n = \frac{\sum N_i M_i}{\sum N_i}, \qquad M_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{\sum w_i M_i}{\sum w_i}, \qquad M_z = \frac{\sum N_i M_i^3}{\sum N_i M_i^2}$$

The **number average** M_n is the total mass divided by the number of molecules; it is what colligative properties and end-group counts measure. The **weight average** M_w weights each chain by its mass, so long chains count more. The **z-average** weights still more heavily toward the longest chains. Always $M_n \le M_w \le M_z$, with equality only for a perfectly uniform sample. The **dispersity**

$$Đ = \frac{M_w}{M_n}$$

(formerly the polydispersity index, PDI) measures the breadth.

Different properties depend on different averages. Brittle-to-ductile behaviour, tensile strength and the glass-transition temperature of short chains depend mainly on M_n, because chain ends act as defects and their number is set by M_n. Melt viscosity scales with M_w (Chapter 21). Melt elasticity, die swell and the strain hardening that stabilizes a blown film bubble depend on M_z and the high-molar-mass tail. A processor can therefore buy two resins with the same melt flow index (M_w) and get very different toughness (M_n) or blow-moulding behaviour (M_z).

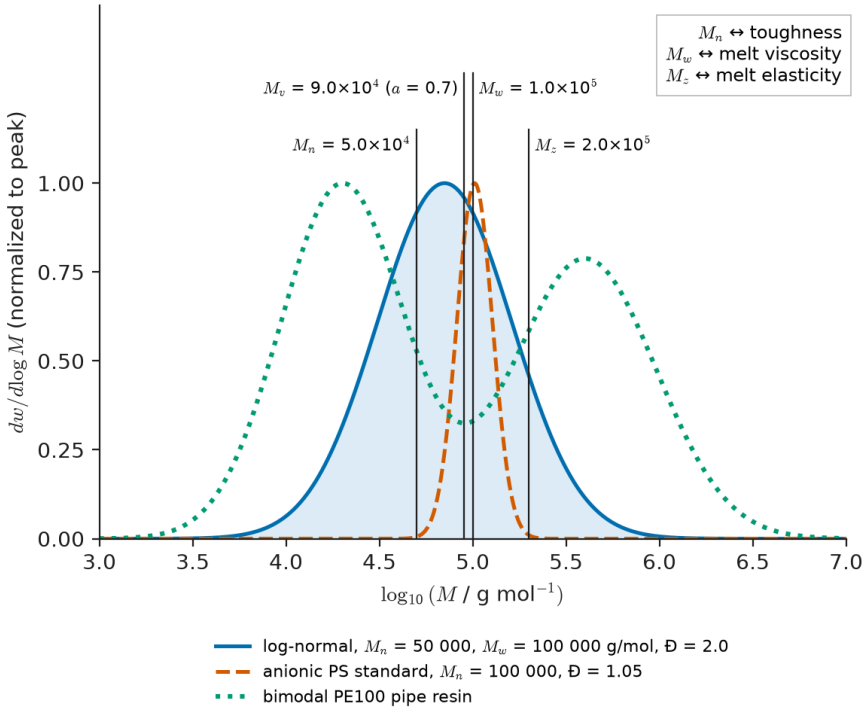

Figure 19.3 — A molecular-weight distribution plotted as weight fraction versus log molar mass, with the number-, viscosity-, weight- and z-average molar masses marked.

Worked example — averages of a blended resin. A compounder mixes 10 g of polymer with $M = 20{,}000$ g/mol, 10 g with $M = 100{,}000$ g/mol and 5 g with $M = 500{,}000$ g/mol (each fraction taken as uniform). The numbers of moles are $10/20{,}000 = 5.0 \times 10^{-4}$, $10/100{,}000 = 1.0 \times 10^{-4}$ and $5/500{,}000 = 1.0 \times 10^{-5}$ mol, totalling 6.1×10^{-4} mol for 25 g. Thus $M_n = 25/6.1 \times 10^{-4} = 41{,}000$ g/mol. The weight average is $M_w = (10 \times 20{,}000 + 10 \times 100{,}000 + 5 \times 500{,}000)/25 = 148{,}000$ g/mol, and $M_z = \sum w_i M_i^2 / \sum w_i M_i = (10 \times 4 \times 10^8 + 10 \times 10^{10} + 5 \times 2.5 \times 10^{11})/(3.7 \times 10^6) = 3.66 \times 10^5$ g/mol. The dispersity is $148{,}000/41{,}000 = 3.6$. Although the heavy fraction is only one-fifth of the mass and under 2 % of the molecules, it contributes 68 % of M_w and dominates M_z, which is exactly why a small amount of very long chains changes melt elasticity so much. Conversely, the light fraction, 40 % of the mass, makes up 82 % of the molecules and pulls M_n — and hence toughness — down.

Measurement methods

Method	Average measured	Range (g/mol)	Principle	Remarks
End-group analysis (titration, NMR)	M_n	below about 25,000	Count chain ends per unit mass	Needs known end groups; standard for nylon and PET
Membrane osmometry	M_n	20,000–1,000,000	Osmotic pressure $\Pi/c \to RT/M_n$	Absolute; small molecules permeate the membrane
Vapour-pressure osmometry	M_n	below about 20,000	Vapour-pressure lowering	For oligomers and resins
Static light scattering	M_w (and R_g)	10^4–10^7	Scattered intensity, Zimm plot	Absolute; needs refractive-index increment dn/dc
Dilute-solution viscometry	M_v	10^3–10^7	Intrinsic viscosity, Mark–Houwink	Fast and cheap; calibration needed
Size-exclusion chromatography (SEC/GPC)	Whole distribution	10^2–10^7	Separation by hydrodynamic volume	Relative unless coupled to light scattering or viscometry
MALDI-TOF mass spectrometry	Whole distribution	up to about 10^5 (narrow samples)	Mass of individual ions	Resolves individual oligomers and end groups
Melt rheology (melt flow index)	Correlates with M_w	Industrial grades	Flow through a standard die	Quality control, not an absolute method

Osmometry. For a dilute solution of concentration c (mass per volume), the osmotic pressure obeys the van 't Hoff relation $\Pi/c = RT/M_n$ in the limit $c \to 0$, with a correction from the second virial coefficient A_2: $\Pi/c = RT(1/M_n + A_2 c + \dots)$. The pressures are small: 5 g/L of a polymer with $M_n = 10^5$ g/mol at 25 °C gives $\Pi = (5/100)\ \text{mol/m}^3 \times 8.314 \times 298 \approx 124$ Pa, the height of a 13 mm column of water. That is still measurable, which is why osmometry was historically the reference method; the same measurement on a 1 % sugar solution would give a signal several hundred times larger.

Light scattering. A polymer coil scatters light in proportion to the square of its mass, so the scattered intensity weights molecules by mass and gives M_w. Measurement at several angles also yields the radius of gyration R_g (Chapter 20), and extrapolating to zero angle and zero concentration (the Zimm plot) gives M_w, R_g and A_2 together. The second virial coefficient vanishes in a theta solvent, a link developed in Chapter 20.

Viscometry. A dissolved polymer coil raises the solution viscosity in proportion to its volume. The **intrinsic viscosity** $[\eta]$, the specific viscosity per unit concentration extrapolated to zero concentration, follows the **Mark–Houwink–Sakurada** relation

$$[\eta] = K\, M_v^{\,a}$$

where M_v (the viscosity average) lies between M_n and M_w, closer to M_w. The exponent a is 0.5 in a theta solvent and 0.7–0.8 in a good solvent for flexible coils, and approaches 1.8 for rigid rods. For polystyrene in tetrahydrofuran at 25 °C, $K \approx 0.0114$ mL/g and $a \approx 0.716$, so a chain of 10^5 g/mol has $[\eta] \approx 43$ mL/g; a measured $[\eta] = 60$ mL/g implies $M_v \approx 1.6 \times 10^5$ g/mol. The PET industry specifies resin entirely by "IV" for this reason.

> **Status of the principle: Mark–Houwink–Sakurada relation**
> *Empirical.* The power law is fitted for each polymer–solvent–temperature combination; its exponent has a physical interpretation (coil swelling, Chapter 20), but K and a must be measured and apply only over the calibrated molecular-weight range.

Size-exclusion chromatography. SEC (or GPC) pumps a dilute solution through columns packed with porous beads. Small coils diffuse into the pores and are delayed; large coils are excluded and elute first. The column separates by hydrodynamic volume, not by mass, so a conventional calibration with narrow polystyrene standards gives "polystyrene-equivalent" molecular weights. **Universal calibration** uses the fact that hydrodynamic volume is proportional to $[\eta]M$, which allows conversion between polymers if their Mark–Houwink constants are known; adding an online multi-angle light-scattering detector removes the need for calibration altogether. Polyethylene and polypropylene must be run at 140–160 °C in trichlorobenzene because they dissolve in nothing at room temperature, a direct consequence of their crystallinity (Chapter 20). SEC is the workhorse of failure analysis: in the cable-tie case it was SEC that showed the halved M_n.

19.5 Copolymers: random, alternating, block and graft

Sequence architectures

A **copolymer** contains two or more kinds of repeat unit. The way they are arranged along the chain changes the properties as much as the overall composition does (Figure 19.4).

- **Random (statistical) copolymers** have units distributed according to the statistics of the polymerization: ⋯AABABBBAAB⋯. Properties are roughly intermediate between those of the homopolymers, and crystallization is usually disrupted. Examples: styrene–butadiene rubber (SBR) for tyres, ethylene–vinyl acetate (EVA) for solar-module encapsulants and running-shoe midsoles, ethylene–propylene rubber (EPR/EPDM) for car door seals and roofing membranes, and LLDPE itself (ethylene with a few percent of an α-olefin).
- **Alternating copolymers** have strict ABAB order. They arise when each radical prefers to add the other monomer, as in styrene–maleic anhydride (used as a compatibilizer and in sizing agents).
- **Block copolymers** consist of long sequences of A joined to long sequences of B: diblock A–B, triblock A–B–A, or multiblock. Because the blocks are long and usually immiscible, they phase-separate on the scale of the chain, 10–100 nm, while remaining bonded (Chapter 20). SBS thermoplastic elastomers have glassy polystyrene end blocks that form physical crosslinks within a rubbery polybutadiene matrix; above 100 °C the domains soften and the material can be moulded, below it behaves as a rubber. Thermoplastic polyurethanes (TPU) and copolyester elastomers use hard and soft multiblocks in the same way, in ski boots, hydraulic seals and catheters.
- **Graft copolymers** have side chains of B attached to a backbone of A. High-impact polystyrene (HIPS) and acrylonitrile–butadiene–styrene (ABS) are made by polymerizing styrene (with acrylonitrile, for ABS) in the presence of polybutadiene: some of the growing chains graft onto the rubber, which anchors micron-sized rubber particles in the glassy matrix. The particles initiate crazes and shear bands that absorb energy (Chapter 21), turning a brittle glass into a material for helmets, luggage and appliance housings.

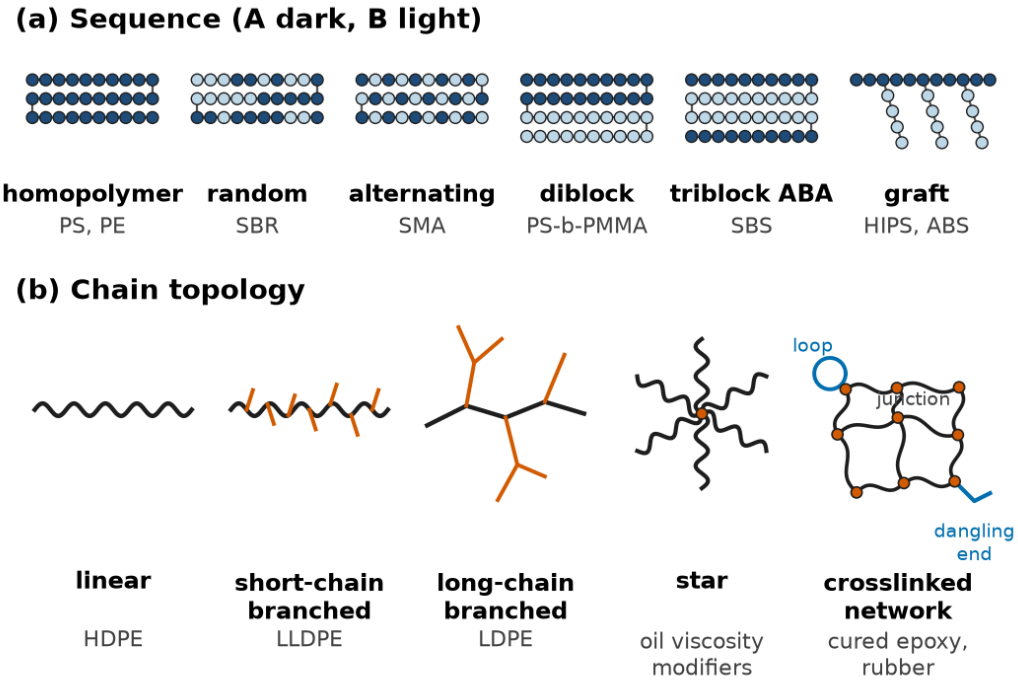

Figure 19.4 — Copolymer sequence architectures and chain topologies.

Copolymer composition: the Mayo–Lewis equation

In chain-growth copolymerization the composition of the chain differs in general from that of the monomer feed, because each type of chain end prefers one monomer. The four propagation reactions $\sim A^\bullet + A$, $\sim A^\bullet + B$, $\sim B^\bullet + A$ and $\sim B^\bullet + B$ are characterized by two **reactivity ratios**, $r_1 = k_{AA}/k_{AB}$ and $r_2 = k_{BB}/k_{BA}$. With mole fractions f_1 and f_2 of the monomers in the feed, the mole fraction F_1 of A incorporated into the chain is

$$F_1 = \frac{r_1 f_1^2 + f_1 f_2}{r_1 f_1^2 + 2 f_1 f_2 + r_2 f_2^2}$$

> **Status of the principle: Mayo–Lewis copolymer equation**
> *Approximate.* Derived from the terminal model, which assumes only the last unit on the chain end affects reactivity and that the steady state holds; penultimate-unit effects and reversible propagation cause measurable deviations in some systems.

Limiting cases make the meaning clear. If $r_1 r_2 = 1$, the copolymer is ideal and random. If $r_1 \approx r_2 \approx 0$, each radical adds only the other monomer and the product alternates. If both $r_1 > 1$ and $r_2 > 1$, long runs of each unit form (rare in radical systems). If $r_1 < 1$ and $r_2 < 1$, there is an **azeotropic** composition at which $F_1 = f_1$, given by $f_1^{az} = (1 - r_2)/(2 - r_1 - r_2)$. For styrene (1) with methyl methacrylate (2) at 60 °C, $r_1 \approx 0.52$ and $r_2 \approx 0.46$: a 50:50 feed gives $F_1 = 0.51$, nearly equal, and the azeotrope is at $f_1 = 0.53$. Away from the azeotrope the feed drifts as the reaction proceeds, so a batch reactor produces chains of different composition early and late (composition drift); commercial reactors feed the more reactive monomer continuously to keep the copolymer uniform, which matters in transparent copolymers where composition differences cause haze.

19.6 Tacticity, branching, crosslinking and networks

Tacticity

In a vinyl polymer $-\mathrm{CH_2}-\mathrm{CHX}-$ every second backbone carbon carries a substituent X and is a stereocentre. The relative configuration of successive stereocentres, the **tacticity**, is fixed at the moment of insertion and cannot be changed afterward without breaking bonds (Figure 19.5).

- **Isotactic**: all X groups on the same side when the backbone is drawn as a planar zig-zag (meso dyads, *m*).
- **Syndiotactic**: X groups alternate (racemo dyads, *r*).
- **Atactic**: random placement.

Tacticity is not a cosmetic detail; it decides whether a polymer can crystallize. Isotactic polypropylene (iPP) packs into a 3_1 helix and crystallizes to 40–70 %, giving a stiff, tough plastic that melts at about 165 °C and is used for car bumpers, battery cases, living hinges and medical syringes. Atactic polypropylene cannot crystallize, has a glass transition near −10 °C, and is a sticky material useful only in hot-melt adhesives. Atactic polystyrene is an amorphous glass (Tg about 100 °C, the clear material of CD cases and disposable cups); syndiotactic polystyrene, made with metallocene catalysts, crystallizes and melts at 270 °C, and is used in under-the-bonnet connectors. Tacticity is measured by ^{13}C nuclear magnetic resonance, which resolves the chemical shift of methyl groups in different triad (*mm*, *mr*, *rr*) and pentad environments; commercial iPP has an isotactic pentad content *mmmm* of 90–98 %.

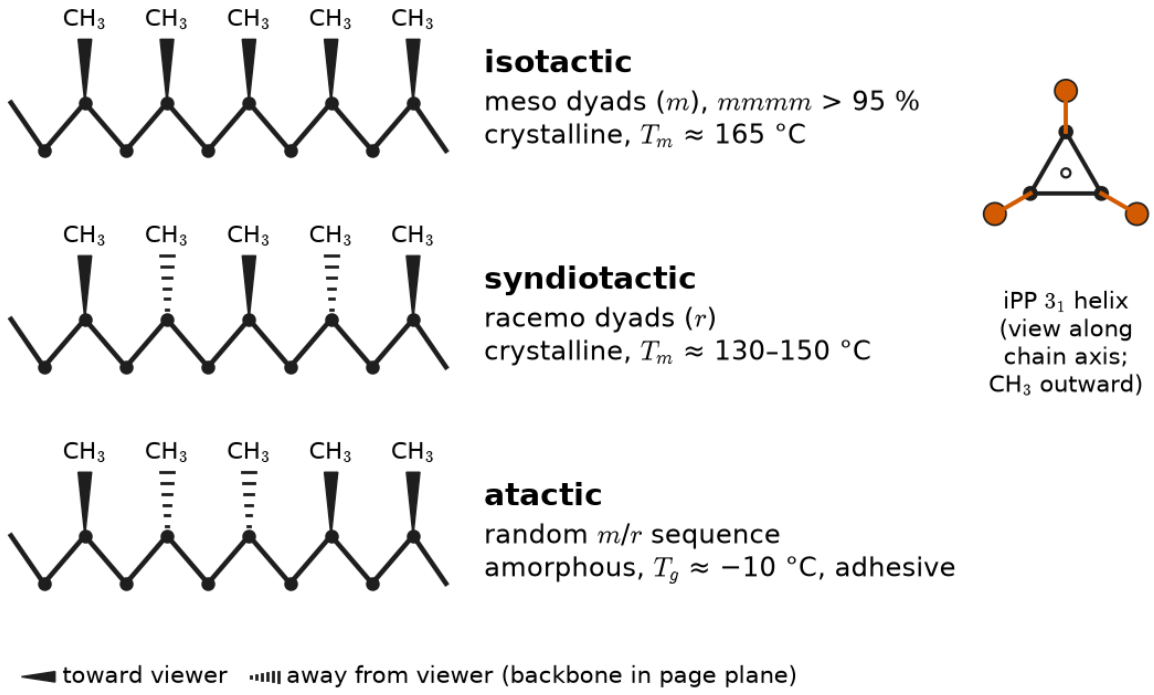

Figure 19.5 — Isotactic, syndiotactic and atactic polypropylene drawn as planar zig-zag backbones, with the consequence for crystallinity.

Branching

Branches are side chains of the same polymer type grafted onto the backbone during synthesis. Their length and number determine density, crystallinity and melt behaviour, as the polyethylene family shows clearly.

Polyethylene grade	Synthesis	Branching	Density (g/cm³)	Crystallinity (typical)	Tm (°C)	Typical uses
LDPE	High-pressure radical	Long branches and 15–30 short branches per 1000 C	0.910–0.930	40–55 %	105–115	Film, squeeze bottles, cable insulation
LLDPE	Ziegler–Natta or metallocene, α-olefin comonomer	10–30 short branches per 1000 C, no long branches	0.915–0.940	40–55 %	120–125	Stretch film, liners, rotomoulded tanks
HDPE	Ziegler–Natta or Phillips	Fewer than 5 branches per 1000 C	0.940–0.970	60–80 %	130–137	Pipes, bottles, fuel tanks, crates
UHMWPE	Ziegler–Natta	Essentially linear, *M* 3–6×10⁶ g/mol	0.930–0.945	40–60 %	130–138	Joint-replacement bearings, Dyneema fibre, liners

Short branches act as defects that cannot enter the crystal, so they lower crystallinity, density, stiffness and melting point, while improving toughness, clarity and resistance to slow crack growth (because they also generate the tie molecules between crystals; Chapter 20). Long branches, as long as the backbone, change melt rheology: they entangle more effectively and give strong strain hardening in extension, which is why LDPE is often blended into LLDPE to stabilize the bubble in film blowing (Chapter 21). Highly branched architectures — stars, combs and dendrimers — have small coil sizes and low viscosity for their molar mass and are used as viscosity-index improvers in engine oils and as drug carriers.

Crosslinking and networks

When monomers of functionality greater than two take part in step growth, chains branch, branches join, and at a sharply defined conversion a single molecule spans the whole sample: the **gel point**. Before it the material is a viscous liquid of growing branched molecules (the sol); at the gel point the weight-average molecular weight diverges and the viscosity becomes infinite; beyond it an insoluble **gel** coexists with a shrinking sol fraction until, at full cure, the material is a single crosslinked network — a **thermoset**.

Carothers extended his counting argument by defining an average functionality f_{av}, the total number of functional groups divided by the number of monomer molecules. For stoichiometric mixtures,

$$p_c = \frac{2}{f_{av}}$$

Flory and Stockmayer treated the problem statistically, recognizing that gelation requires only that, on average, each branch unit leads to at least one more branch unit; for a trifunctional monomer A_3 reacting with a difunctional B_2 at stoichiometry their result is $p_c = 1/\sqrt{f-1} = 1/\sqrt{2} = 0.707$.

> **Status of the principle: Gel-point predictions (Carothers and Flory–Stockmayer)**
> *Approximate*. The Carothers estimate counts molecules as if gelation required infinite X_n and overestimates p_c; the Flory–Stockmayer mean-field theory ignores intramolecular loops and slightly underestimates it. Measured gel points lie between the two.

For a classical alkyd resin made from 2 mol glycerol ($f = 3$) and 3 mol phthalic anhydride ($f = 2$), $f_{av} = (2 \times 3 + 3 \times 2)/5 = 2.4$ and the Carothers estimate is $p_c = 0.833$; Flory–Stockmayer gives 0.707; the measured value is about 0.77. The engineering consequence is the **pot life**: a two-part epoxy adhesive or a composite resin must be applied, laid up or injected before it reaches the gel point, after which it can no longer flow or wet fibres. Gelation is followed by **vitrification** when the rising glass-transition temperature of the network passes the cure temperature and the reaction slows to a crawl; this is why aerospace epoxies need a post-cure above their service Tg (Chapter 23).

The density of crosslinks is described by the average molar mass between junctions, M_c, or by the crosslink density $\nu = \rho/M_c$ (moles of network chains per unit volume). It ranges over three orders of magnitude and sets the class of material:

Network	Typical M_c (g/mol)	Character	Example
Lightly vulcanized rubber	3,000–10,000	Soft, extensible elastomer	Tyre tread, seals, gloves
Hydrogel	5,000–50,000 (swollen in water)	Soft, wet, permeable	Contact lenses, wound dressings
Toughened epoxy, polyurethane	500–2,000	Tough thermoset	Adhesives, coatings
Aerospace epoxy (TGDDM/DDS)	300–500	Stiff, high-Tg, brittle	CFRP matrices
Phenolic, melamine	Below 300	Very stiff, hard, brittle	Brake linings, worktops, circuit-breaker housings

The rubber elasticity theory of Chapter 20 links M_c to the modulus of an elastomer, and the swelling equilibrium of Chapter 21 provides a simple way to measure it. Thermosets cannot be remelted, which has consequences for recycling (Chapter 21 and

Chapter 59); covalent adaptable networks (vitrimers) with exchangeable bonds are being developed to make thermosets reprocessable.

19.7 Commodity, engineering and high-performance polymers

The polymer pyramid

The several hundred commercial polymers are conventionally sorted into a pyramid by volume, price and temperature capability (Figure 19.6). At the base, **commodity** polymers — PE, PP, PVC, PS and PET — make up about 70–75 % of world production (which exceeds 400 million tonnes per year), cost roughly 1–2 USD/kg, and serve up to about 60–100 °C (PET, at 100–120 °C, straddles the boundary with the engineering tier). In the middle, **engineering** polymers — polyamides, polycarbonate, acetal, PBT, ABS, PMMA and modified PPO — offer higher stiffness, toughness or service temperature (100–150 °C) at 2–6 USD/kg. At the top, **high-performance** polymers — PEEK, PEI, PPS, PSU, PAI, polyimides, LCPs and fluoropolymers — survive 150–260 °C, aggressive chemicals and fire, at 10–100 USD/kg or more. A parallel division separates amorphous polymers (transparent, isotropic shrinkage, poor chemical resistance) from semicrystalline ones (opaque, chemically resistant, retain stiffness up to near Tm) at every level of the pyramid.

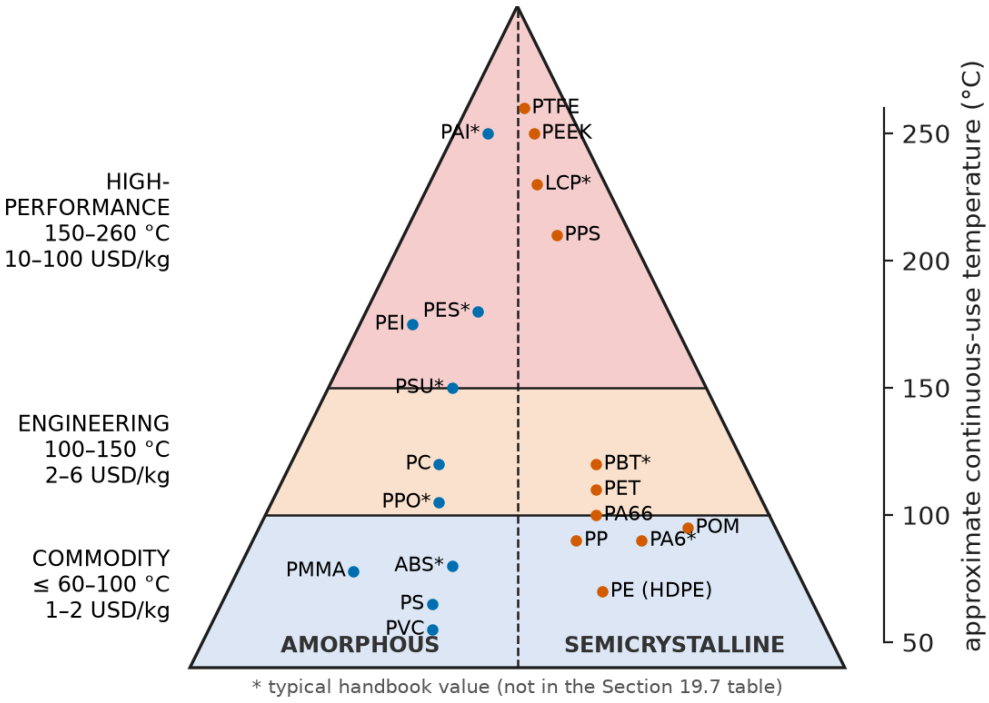

Figure 19.6 — The thermoplastic pyramid: commodity, engineering and high-performance polymers ranked by continuous-use temperature and price, split into amorphous and semicrystalline families.

Representative properties

The table gives typical, unfilled, room-temperature properties. Values vary with grade, moisture (for polyamides especially) and processing; they illustrate how structure moves properties.

Polymer	Structure	Tg (°C)	Tm (°C)	Density (g/cm³)	E (GPa)	Tensile strength (MPa)	Typical continuous use (°C)
HDPE	Semicrystalline	about −120	130–137	0.95–0.97	0.8–1.4	20–35	60–80
PP (isotactic)	Semicrystalline	about −10	160–170	0.90–0.91	1.2–1.8	30–40	80–100
PVC (rigid)	Amorphous	80–85	—	1.38–1.42	2.5–3.5	45–55	50–60
PS (atactic)	Amorphous	about 100	—	1.04–1.05	3.0–3.4	35–50	60–70
PMMA	Amorphous	105–115	—	1.18–1.19	2.8–3.3	60–75	65–90
PET	Semicrystalline	70–80	250–260	1.33–1.40	2.5–3.2	50–75	100–120
PA66 (dry)	Semicrystalline	50–60 (dry)	255–265	1.13–1.15	2.8–3.3	75–85	80–120
POM (acetal)	Semicrystalline	about −60	165–175	1.41–1.42	2.8–3.2	60–70	90–100
PC	Amorphous	about 145–150	—	1.20	2.2–2.4	60–70	115–125
PPS	Semicrystalline	about 90	280–285	1.35	3.5–4.0	65–85	200–220
PEI (Ultem)	Amorphous	about 215–217	—	1.27	3.0–3.3	100–110	170–180
PEEK	Semicrystalline	about 143	340–345	1.30–1.32	3.6–4.1	90–100	240–260
PTFE	Semicrystalline	about 115 (debated)	327	2.15–2.20	0.4–0.6	20–35	260
Kapton polyimide film	Amorphous (no melt)	360–410	—	1.42	2.5–3.0	170–230 (film)	240–400

Three structural trends run through the table, and Chapter 20 explains each.

1. **Stiff backbones raise Tg.** Aromatic rings in the main chain (PET, PC, PEEK, PEI, polyimide) restrict rotation and lift Tg and Tm far above those of aliphatic chains. Ether and ketone links between rings (PEEK) keep enough flexibility for melt processing; a fully aromatic, rigid polyimide never melts and must be processed from a soluble precursor.
2. **Strong interchain forces raise Tm and stiffness.** Hydrogen bonding in polyamides and dipole interactions in PVC and PET raise melting points relative to polyethylene; fluorine sheathing in PTFE gives chemical inertness and the lowest friction of any solid polymer.
3. **Regularity enables crystallinity.** Linear, stereoregular chains (HDPE, iPP, POM, PEEK) crystallize; branched, atactic or bulky chains (LDPE, aPS, PMMA, PC) do not, or do so only partially. Crystallinity converts a soft polymer above Tg into a useful solid up to near Tm, which is why polyethylene, with a Tg of −120 °C, is a structural material at room temperature.

Thermoplastics, thermosets and elastomers

Cutting across the pyramid is the division by network topology. **Thermoplastics** (all the polymers in the table) are linear or branched chains that soften on heating and can be remelted and reshaped, which allows high-rate processing and mechanical recycling. **Thermosets** (epoxies, phenolics, unsaturated polyesters, vinyl esters, cyanate esters, bismaleimides, cured polyurethanes and polyimides) are crosslinked during processing and cannot be remelted; they offer dimensional stability, creep resistance and chemical resistance, and dominate as composite matrices (Chapters 22–23), adhesives and coatings.

Elastomers are lightly crosslinked networks above their Tg: natural rubber (cis-polyisoprene, Tg about −70 °C), SBR, EPDM, nitrile (NBR, oil resistant), chloroprene (CR), silicone and fluoroelastomers (FKM). Thermoplastic elastomers, such as SBS and TPU, borrow physical crosslinks from block-copolymer microphase separation, combining rubbery behaviour with thermoplastic processing.

Choosing a family: three short cases

- **Civil.** A water distribution pipe must last 50 years under 10–16 bar pressure with no maintenance. Bimodal HDPE (PE100), with a broad, engineered molecular-weight distribution and short-chain branches on the long chains, gives slow-crack-growth resistance no unimodal grade can match; the pipe classification itself is based on extrapolated 50-year hoop stress (Chapter 21).
- **Biomedical.** A knee-replacement tibial insert needs wear resistance against a cobalt-chromium femoral component for 20 years. UHMWPE, with chains so long (M above 3×10^6 g/mol) that the material cannot be melt-processed and is compression-moulded instead, provides it; radiation crosslinking further reduces wear, at a cost in toughness that is partly recovered by adding vitamin E as an antioxidant (Chapters 21 and 39).
- **Aerospace.** A cabin-interior duct or a bracket near an engine must meet stringent flame, smoke and toxicity rules and survive 150–200 °C. PEI and PEEK, with aromatic backbones that char rather than drip, meet the rules without halogenated flame retardants, at twenty to fifty times the price of polypropylene.

What remains open

Open question	Where it is answered
How big is a chain coil, and why does rubber elasticity come from entropy?	Chapter 20
Why do most polymer pairs not mix, and how do block copolymers form nanostructures?	Chapter 20
How do tacticity and branching control crystallinity, lamellae and spherulites?	Chapter 20
What sets Tg in molecular terms, and how does it change with molecular weight and copolymer composition?	Chapter 20
How does molecular weight control melt viscosity, and why do entanglements give the 3.4-power law?	Chapter 21
How are elastomers vulcanized, and how is crosslink density measured by swelling?	Chapter 21
How do hydrolysis, oxidation and UV cut chains in service, and how can polymers be recycled?	Chapters 21 and 44
Which matrices and fibres make composites, and how do they share load?	Chapter 22
How are epoxy matrices cured in composite manufacture?	Chapter 23
What are the viscoelastic models behind polymer creep and relaxation?	Chapter 14
How are DSC, FTIR, NMR and Raman used to characterize polymers?	Chapter 52
Which polymers are used in implants and drug delivery?	Chapter 39
How are chain conformations simulated by molecular dynamics and Monte Carlo?	Chapter 56
What is the life-cycle and circular-economy picture for plastics?	Chapter 59

Next: Chapter 20 takes the chains made here and asks how they coil, stretch, mix, crystallize and freeze into glasses — the physics that turns a molecular structure into a material.

Chapter 20 — Polymer Physics and Morphology

Part V · Polymers and Composites

A manufacturer of disposable medical syringes switches from one isotactic polypropylene grade to another with the same melt flow index and the same nominal density, to save cost. The syringes mould perfectly. Six weeks later the hospital returns cartons in which the barrels have turned cloudy and a few have cracked at the luer tip after gamma sterilization. Nothing about the chemistry has changed; what has changed is the morphology. The new grade lacked the nucleating agent of the old one, so it crystallized into large spherulites that scatter light, and it had a lower fraction of the chain segments that bridge neighbouring crystals, so the radiation-induced chain scission broke more of the few connections that held the crystals together. In the same hospital, a nurse finds that a silicone catheter has become sticky after weeks in contact with a lipid emulsion: the oil has dissolved into the rubber network and swollen it. Both problems are questions of polymer physics, not chemistry: how chains are arranged, how they mix with other molecules, and how they pack into crystals and glasses.

Chapter 19 described how chains are made. This chapter treats what the chains then do: how a single chain coils, why a network of coils is a rubber, when two polymers or a polymer and a solvent mix, how chains fold into crystals, and why, below a characteristic temperature, the whole assembly freezes into a glass.

> **What shape does a long, flexible chain adopt, and why does a network of such chains resist stretching? What decides whether a polymer mixes, crystallizes or vitrifies, and how do these states set the stiffness and temperature range of a plastic part?**

Concept	Engineering destination
Random-walk chain, Kuhn length, end-to-end distance	Coil size in solutions and melts, membrane pores, entanglement, light scattering
Entropic rubber elasticity	Seals, tyres, bearings, vibration mounts, rubber springs in civil base isolation
Flory–Huggins theory and theta conditions	Solvent selection, paint and adhesive formulation, swelling of seals in fuels and oils
Blend miscibility	Toughened plastics, Noryl housings, recycled mixed-plastic streams
Block-copolymer microphase separation	Thermoplastic elastomers, nanolithography, drug-delivery micelles
Crystallization and crystallinity	Stiffness and creep of HDPE pipe, clarity of PP, barrier of PET bottles
Glass transition, WLF	Upper service temperature of amorphous plastics, brittleness at low temperature, damping
Liquid-crystalline polymers	Aramid fibres for armour and ropes, thin-wall connectors, low-CTE substrates

Roadmap. Chain statistics (20.1) → entropic elasticity of networks (20.2) → mixing with solvents (20.3) and with other polymers, including block copolymers (20.4) → crystallization and morphology (20.5) → the glass transition (20.6) → rigid chains that form liquid crystals (20.7).

20.1 Chain conformations

Rotational isomers and flexibility

A polyethylene backbone is a chain of carbon atoms joined by bonds of length $l = 0.154$ nm at a fixed valence angle of about 109.5°. The bond lengths and angles are stiff, but rotation about each C–C bond is relatively easy. Looking along a bond, the substituents on adjacent carbons can sit in a **trans** position (energy minimum) or in one of two **gauche** positions (about 2–3 kJ/mol higher), separated by barriers of roughly 12–15 kJ/mol. Since $RT \approx 2.5$ kJ/mol at room temperature, a polyethylene chain in a melt switches between rotational states roughly 10^{10} times per second, and each bond is gauche about a third of the time. A chain of 7,000 bonds therefore has an astronomical number of shapes available, and in a melt or solution it continually

explores them. This conformational freedom, not any special chemistry, is the origin of rubber elasticity, of the high viscosity of melts and of the glass transition.

The flexibility of a chain depends on the rotation barrier and on the bulk of the side groups. Polydimethylsiloxane, with long Si–O bonds and a wide Si–O–Si angle, is extremely flexible; polystyrene, with a phenyl ring on every second carbon, is stiffer; a chain of para-linked aromatic rings (Kevlar) is essentially a rigid rod (Section 20.7).

The freely jointed chain

The simplest model replaces the real chain by n links of length l, each pointing in a direction completely independent of the others — a three-dimensional random walk. The end-to-end vector is $\mathbf{R} = \sum_i \mathbf{l}_i$, and because cross terms $\langle \mathbf{l}_i \cdot \mathbf{l}_j \rangle$ vanish for independent links,

$$\langle R^2 \rangle_0 = n\, l^2$$

The root-mean-square end-to-end distance grows only as the square root of the chain length, whereas the fully stretched **contour length** grows linearly. For large n, the central-limit theorem makes the probability density of $\mathbf{R}$ Gaussian:

$$P(\mathbf{R}) = \left(\frac{3}{2\pi \langle R^2 \rangle_0} \right)^{3/2} \exp\!\left(-\frac{3R^2}{2\langle R^2 \rangle_0} \right)$$

The **radius of gyration**, the root-mean-square distance of the segments from the centre of mass, is the quantity measured by light and neutron scattering; for a Gaussian coil $\langle R_g^2 \rangle = \langle R^2 \rangle_0 / 6$.

Real chains: characteristic ratio and Kuhn length

Real bonds are not freely jointed. Fixed valence angles and the preference for trans states make successive bonds point in correlated directions, and the coil is larger than the freely jointed estimate. The correlations nonetheless die out after a few bonds, so a long real chain is still a random walk on large scales, only with a larger step. Two equivalent descriptions are used:

- The **characteristic ratio** C_∞ defined by $\langle R^2 \rangle_0 = C_\infty\, n\, l^2$ for large n.
- The **Kuhn model**: the real chain is replaced by an equivalent freely jointed chain of N_K segments of **Kuhn length** b, chosen so that it has the same contour length $R_{max} = N_K b$ and the same mean-square end-to-end distance $\langle R^2 \rangle_0 = N_K b^2$. Then $b = \langle R^2 \rangle_0 / R_{max}$.

> **Status of the principle: Gaussian random-walk statistics of chains**
> *Derived.* Follows from the central-limit theorem for any chain whose orientational correlations decay over a finite distance; it holds in melts and in theta solvents (where excluded-volume effects cancel) and fails for short or rigid chains and at large extensions approaching the contour length.

Polymer	C_∞ (typical)	Kuhn length b (nm, typical)	Comment
Polydimethylsiloxane	5.2–6.3	1.1–1.3	Very flexible; Tg about −125 °C
Polyethylene	6.7–7.4	1.3–1.5	Flexible; reference polymer
Isotactic polypropylene	5.7–6.2	1.1–1.3	Helical in crystal
Poly(ethylene oxide)	4.0–5.6	0.9–1.1	Water-soluble
Polystyrene (atactic)	9.5–10	1.6–1.8	Bulky phenyl side group
PMMA (atactic)	8–9	1.5–1.7	Stiffer, Tg about 105 °C
Polycarbonate	2.4 (per repeat-unit "virtual bond")	about 1.4	Aromatic backbone with flexible links
DNA (double helix)	—	about 100	Semi-flexible, persistence length 50 nm
Kevlar (PPTA)	—	more than 30	Rigid rod; forms liquid crystals

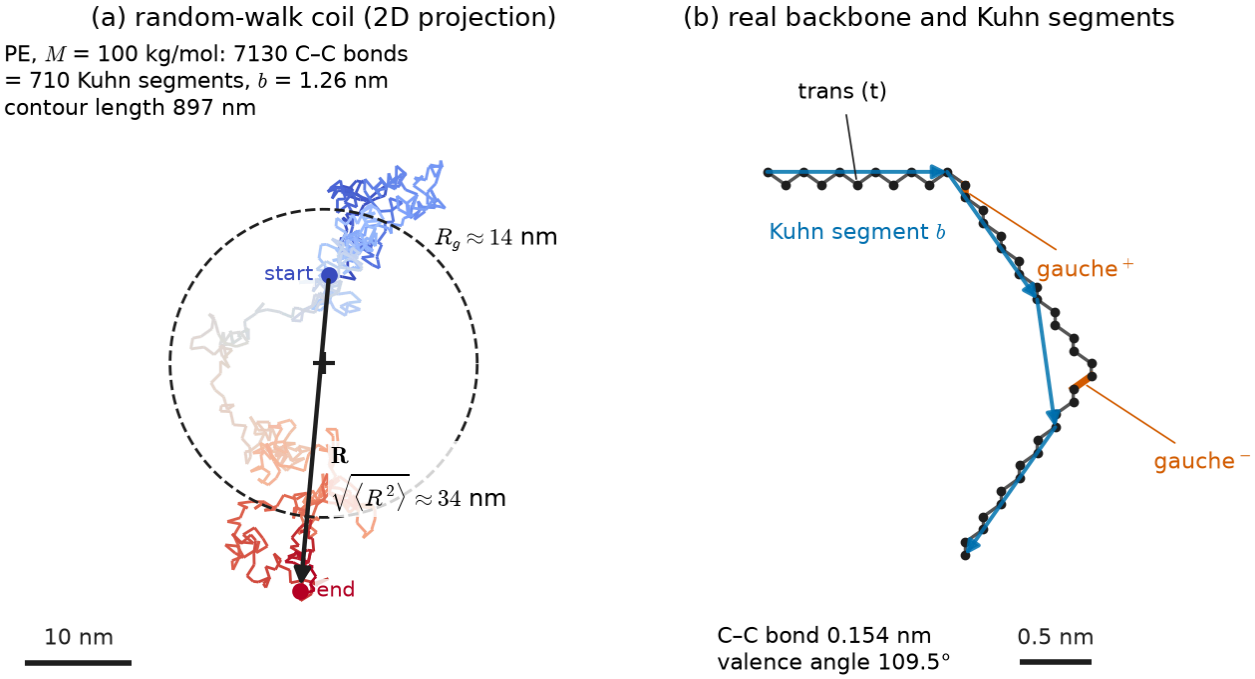

Figure 20.1 — A polyethylene chain as a three-dimensional random walk, with its end-to-end vector, radius of gyration and the equivalent Kuhn chain.

Worked example — size of a polyethylene coil. A polyethylene chain of $M = 100{,}000$ g/mol contains $2M/28.05 = 7{,}130$ backbone C–C bonds. As a freely jointed chain it would have $\sqrt{\langle R^2 \rangle} = \sqrt{7130} \times 0.154 = 13.0$ nm. With $C_\infty = 6.7$, $\langle R^2 \rangle_0 = 6.7 \times 7130 \times 0.154^2 = 1133$ nm², so $\sqrt{\langle R^2 \rangle_0} = 33.7$ nm and $R_g = 33.7/\sqrt{6} = 13.7$ nm. The fully extended all-trans zig-zag has contour length $R_{max} = nl\cos(35.25^\circ) = 7130 \times 0.154 \times 0.817 = 897$ nm. The Kuhn length is $b = 1133/897 = 1.26$ nm and the chain contains $N_K = 897/1.26 \approx 710$ Kuhn segments. Three conclusions follow. First, the coil is 27 times shorter end to end than the stretched chain, so the molecule can be extended enormously before its bonds are loaded — the basis of the large strains of rubbers and of fibre drawing. Second, the coil occupies a volume of order $R_g^3 \approx 2.6 \times 10^3$ nm³, while the chain's own atoms fill only $M/(\rho N_A) \approx 195$ nm³ (melt density 0.85 g/cm³); so in a melt the pervaded volume of each coil is shared with roughly a dozen other chains. This interpenetration is what makes entanglement (Chapter 21) unavoidable. Third, the coil size $R_g \approx 14$ nm is comparable with the length scale $2\pi/q$ probed by X-rays and neutrons scattered at small angles (SAXS, SANS), which is how such numbers are measured.

Excluded volume and solvent quality

A random walk can cross itself; a real chain cannot, since two segments cannot occupy the same space. In a good solvent, where segments prefer solvent contacts to each other, this **excluded volume** swells the coil, and Flory's argument gives $R \propto N^{\nu}$ with $\nu \approx 3/5$ (more precisely 0.588) instead of $1/2$. In a poor solvent the segments attract and the coil contracts. At one temperature, the **theta temperature**, the attraction exactly cancels the excluded volume and the chain is ideal again (Section 20.3). In a melt, a chain is surrounded by other chains of the same kind; the excluded-volume interactions are screened, and the chain is ideal with $\nu = 1/2$ — Flory's theorem, confirmed by neutron scattering on deuterium-labelled chains in the 1970s.

20.2 Rubber elasticity derived from entropy

The puzzle

Stretch a rubber band rapidly and hold it against your lip: it warms up. Hang a weight on a rubber strip and heat the strip: it contracts and lifts the weight. Both behaviours are the opposite of what a metal spring does, and both follow from a single fact: the restoring force of a rubber comes from entropy, not from stretching bonds.

Entropy of a single chain

The number of conformations available to a chain whose ends are held a vector $\mathbf{R}$ apart is proportional to $P(\mathbf{R})$. By Boltzmann's relation $S = k \ln \Omega$, the conformational entropy of the chain is

$$S(\mathbf{R}) = S_0 - \frac{3kR^2}{2\langle R^2 \rangle_0}$$

Pulling the ends apart reduces the number of available conformations and lowers the entropy. If the internal energy does not depend on R (an ideal chain whose rotational states have equal energies), the Helmholtz free energy is $A = -TS$, and the force required to hold the ends apart is

$$f = \frac{\partial A}{\partial R} = \frac{3kT}{\langle R^2 \rangle_0} R$$

The chain is a linear spring whose stiffness is proportional to absolute temperature: an **entropic spring**. For the polyethylene coil of the worked example, $3kT/\langle R^2\rangle_0 = 3 \times 4.11 \times 10^{-21} / (1.133 \times 10^{-15}) \approx 1.1 \times 10^{-5}$ N/m at 25 °C — tens of millions of times softer than the stretching stiffness of a C–C bond (several hundred N/m).

From chains to a network

A rubber is a network of such chains joined at crosslinks (Chapter 19). The **affine network model** assumes that the crosslinks move with the macroscopic deformation, so that every chain end-to-end vector is stretched in the same proportion as the sample, and that the network chains are Gaussian. For a deformation with principal stretch ratios $\lambda_1, \lambda_2, \lambda_3$ and ν network chains per unit volume, summing the entropy change over all chains gives the elastic free energy per unit volume

$$W = \frac{G}{2}\left(\lambda_1^2 + \lambda_2^2 + \lambda_3^2 - 3\right), \qquad G = \nu kT = \frac{\rho RT}{M_c}$$

where ρ is the density and M_c the molar mass between crosslinks. Rubbers are nearly incompressible (Poisson's ratio 0.4995), so in uniaxial tension $\lambda_1 = \lambda$ and $\lambda_2 = \lambda_3 = \lambda^{-1/2}$, and the nominal (engineering) stress is

$$\sigma = G\left(\lambda - \frac{1}{\lambda^2}\right)$$

At small strain, $\lambda = 1 + \varepsilon$, this becomes $\sigma \approx 3G\varepsilon$, so Young's modulus is $E = 3G$.

> **Status of the principle: Entropic (Gaussian) rubber elasticity**
> *Derived.* Follows from Gaussian chain statistics and the affine assumption. It predicts the right modulus magnitude and its proportionality to T and to $1/M_c$, but is quantitative only to within a factor of about two, and fails at large stretch (finite chain extensibility, strain-induced crystallization) and in moderate stretch where crosslinks fluctuate (the phantom network model gives $G = (1 - 2/\phi)\nu kT$ for junction functionality ϕ).

Three predictions are worth emphasizing.

1. **The modulus of a rubber is tiny and set by crosslink density.** A typical M_c of 5,000 g/mol gives G of order 0.5 MPa, against 80 GPa for steel. This is the right order of magnitude for every vulcanized rubber, and it is why the modulus of a seal or a tyre compound is controlled by the sulfur dose (Chapter 21) and by reinforcing filler.
2. **The modulus increases with temperature.** At constant length, the force on a stretched rubber is proportional to T; heated, it contracts (the **Gough–Joule effect**). At small strains the effect is masked by ordinary thermal expansion (thermoelastic inversion near 10 % strain), but at larger strains it dominates.
3. **Stretching releases heat.** An adiabatic extension lowers the conformational entropy, and because total entropy cannot fall in an adiabatic reversible process, the vibrational entropy — and therefore the temperature — rises.

Worked example — a natural-rubber bridge bearing. Elastomeric bearings made of natural rubber (density 920 kg/m³) support bridge decks and isolate buildings from earthquakes. A compound with $M_c = 5{,}000$ g/mol at 25 °C (298 K) has $G = \rho RT/M_c = 920 \times 8.314 \times 298/5.0 = 4.56 \times 10^5$ Pa $= 0.46$ MPa, and $E = 3G = 1.37$ MPa. Stretched to $\lambda = 2$ (100 % strain), the predicted nominal stress is $\sigma = 0.456 \times (2 - 0.25) = 0.80$ MPa. At 75 °C (348 K) the same compound has $G = 0.53$ MPa, 17 % stiffer — the opposite of every metal and glassy polymer. The stored elastic energy at $\lambda = 3$ is $W = \tfrac{1}{2}G(9 + \tfrac{2}{3} - 3) = 1.52$ MJ/m³; if all of it appeared as heat in an adiabatic extension, with heat capacity 1,900 J/(kg·K), the temperature would rise by $1.52 \times 10^6/(920 \times 1900) = 0.87$ K. The warming felt on the lip is larger, a few kelvin, because natural rubber crystallizes on stretching and releases latent heat — a reminder that the Gaussian model describes only the entropic part of the response. In bearing design the shear modulus is what matters (G of 0.4–1.2 MPa is specified), with bonded steel shims stiffening the bearing in compression while leaving it soft in shear.

Where the simple theory fails

Measured stress–strain curves of natural rubber (Figure 20.2) follow the Gaussian prediction up to about $\lambda = 1.5$, then fall somewhat below it in the range $\lambda = 1.5$–4 (crosslink junctions fluctuate and chains are not deformed affinely), and then rise steeply above $\lambda \approx 5$. The upturn has two causes: the chains between crosslinks approach their contour length, where the Gaussian approximation fails (non-Gaussian Langevin statistics, used in the Arruda–Boyce model), and natural rubber crystallizes under strain, the crystals acting as additional stiff crosslinks. Strain crystallization also stops cracks from growing and explains why natural rubber, not synthetic SBR, is used in aircraft tyres and heavy truck tyres. Engineering finite-element codes describe rubber with hyperelastic strain-energy functions — neo-Hookean (the Gaussian result), Mooney–Rivlin, Ogden, Arruda–Boyce — fitted to tension, compression and shear tests.

> **Status of the principle: Mooney–Rivlin and other hyperelastic laws**
> *Empirical.* Phenomenological strain-energy functions with fitted constants; reliable within the strain range and deformation modes used for the fit and unreliable when extrapolated beyond it.

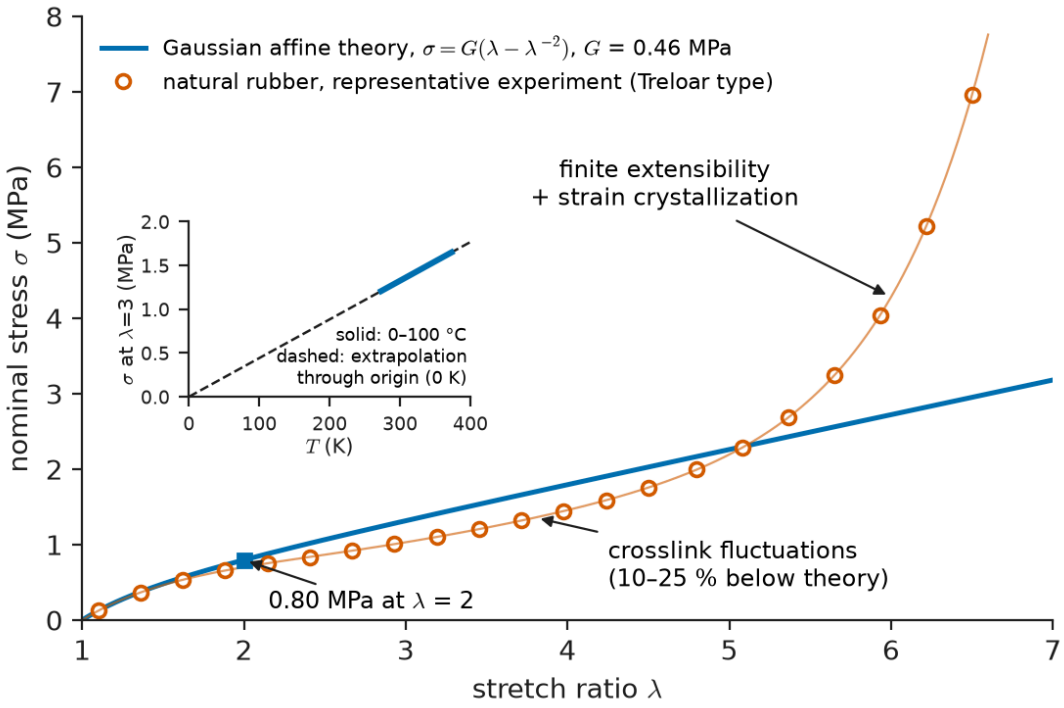

Figure 20.2 — Nominal stress versus stretch ratio for a natural-rubber network compared with the Gaussian affine prediction.

20.3 Polymer solutions: Flory–Huggins theory and theta conditions

Why polymers dissolve reluctantly

Polymers dissolve slowly and in few solvents. Polystyrene dissolves in toluene but not in hexane or ethanol; polyethylene dissolves in nothing at room temperature; cross-linked rubber only swells. The reason is entropic. When small molecules mix, each can occupy any site, and the entropy of mixing is large. When a polymer mixes, its segments are tied together, and the number of distinct arrangements is far smaller. Polymer mixing is therefore much more sensitive to unfavourable interactions than ordinary mixing.

The Flory–Huggins lattice model

Flory and Huggins (1941–42) placed solvent molecules and polymer segments on a lattice of sites, each segment occupying one site of the same volume as a solvent molecule. For a mixture with polymer volume fraction ϕ and chains of N segments, the free energy of mixing per lattice site is

$$\frac{\Delta G_{mix}}{kT} = \frac{\phi}{N}\ln\phi + (1-\phi)\ln(1-\phi) + \chi\,\phi(1-\phi)$$

The first two terms are the combinatorial entropy; the polymer term is divided by N because a chain of N segments moves as one unit. The third term is the interaction energy, governed by the dimensionless **Flory–Huggins parameter** χ, which measures the energy cost, in units of kT, of replacing a segment–segment and a solvent–solvent contact by two segment–solvent contacts. In practice χ has an enthalpic part that falls as $1/T$ and an entropic part, $\chi = A + B/T$.

> **Status of the principle: Flory–Huggins free energy of mixing**
> *Approximate.* A mean-field lattice theory that assumes random mixing, equal segment and solvent volumes and no volume change on mixing; it captures phase separation, swelling and the role of chain length, but χ fitted to experiment depends on concentration and temperature, and the theory fails in dilute solution (where coils are isolated) and for strongly interacting systems such as hydrogen-bonding polymers in water.

The model makes several predictions that engineers use.

- **Phase separation.** For χ above a critical value the free-energy curve develops two minima and the solution splits into a polymer-rich and a solvent-rich phase (the common-tangent construction of Chapter 6). For a polymer in a solvent the critical point is at $\phi_c = 1/(1+\sqrt{N})$ and $\chi_c = \tfrac{1}{2}(1+1/\sqrt{N})^2$, which tends to $\tfrac{1}{2}$ for long chains. The phase diagram has an upper critical solution temperature (UCST) that rises with molar mass, which is exploited in fractionation and in membrane formation by thermally induced phase separation.
- **Solvent quality.** A solvent with $\chi < \tfrac{1}{2}$ is a **good solvent** (coils swell, $A_2 > 0$), with $\chi > \tfrac{1}{2}$ a **poor solvent**. For polystyrene, $\chi \approx 0.35$ in toluene (good), 0.5 in cyclohexane at 34.5 °C (theta), about 0.9 in hexane (non-solvent).
- **Swelling of networks.** A crosslinked rubber cannot dissolve, but it swells until the mixing free energy is balanced by the elastic retraction of the network (Section 20.2). The Flory–Rehner equation that expresses this balance is the standard way to measure crosslink density (Chapter 21) and the basis of the fuel and oil compatibility tables for seals: a nitrile O-ring that swells by 5 % in hydraulic oil is acceptable, an EPDM ring that swells by 100 % is not.

Solubility parameters

The interaction part of χ can be estimated from the **Hildebrand solubility parameter** $\delta = (E_{coh}/V)^{1/2}$, the square root of the cohesive energy density:

$$\chi \approx 0.34 + \frac{V_1}{RT}\,(\delta_1 - \delta_2)^2$$

where V_1 is the molar volume of the solvent and 0.34 an empirical entropic contribution. "Like dissolves like" becomes quantitative: polymer and solvent should have δ values within about 2 MPa$^{1/2}$. Hansen split δ into dispersion, polar and hydrogen-bonding parts, which handles polar polymers and is used to formulate paints, inks and adhesives.

> **Status of the principle: Solubility-parameter estimate of χ**
> *Empirical.* The regular-solution form captures only dispersion-type interactions; the constant 0.34 is fitted, and the estimate is unreliable for polar or hydrogen-bonding systems and cannot predict negative χ (specific attractions).

Material	δ (MPa$^{1/2}$, typical)
PTFE	12.6
n-Hexane	14.9
Polyethylene	16.2–17.0
Natural rubber, polyisoprene	16.5–17.0
Cyclohexane	16.8
Toluene	18.2
Polystyrene	18.5–19.0
PMMA	18.6–19.5
Acetone	20.3
PVC	19.5–21.5
Epoxy (cured)	20–22
Nylon-6,6	about 28
Water	47.8

Natural rubber and hexane, at nearly equal δ, explains why rubber seals swell badly in petrol, and why nitrile rubber (with polar acrylonitrile units, $\delta \approx 19$–20) is used instead. Polyethylene has a suitable δ for many hydrocarbons, but its crystals are not dissolved until the solvent is above about 100 °C, because the free energy of melting must also be paid: crystallinity, not chemistry, is the source of its solvent resistance.

Theta conditions

At the **theta temperature** θ (or in a theta solvent), $\chi = \frac{1}{2}$, the second virial coefficient vanishes, and the excluded-volume swelling is exactly cancelled by segment–segment attraction. The chain then follows ideal random-walk statistics, and the Mark–Houwink exponent of Chapter 19 is exactly $a = 0.5$. Theta conditions are therefore the experimental reference state in which unperturbed chain dimensions and C_∞ are measured. They also mark the threshold of precipitation for infinitely long chains, which is why a polymer solution held just below its theta temperature separates by molar mass — the basis of fractionation.

20.4 Blends, miscibility and block-copolymer microphase separation

Why most polymer pairs do not mix

Applying the Flory–Huggins expression to a blend of two polymers with N_A and N_B segments,

$$\frac{\Delta G_{mix}}{kT} = \frac{\phi}{N_A}\ln\phi + \frac{1-\phi}{N_B}\ln(1-\phi) + \chi\phi(1-\phi)$$

both entropy terms are divided by large numbers. The critical value of χ becomes

$$\chi_c = \frac{1}{2}\left(\frac{1}{\sqrt{N_A}} + \frac{1}{\sqrt{N_B}}\right)^2$$

which for $N_A = N_B = 1000$ is only 0.002. Almost any positive interaction parameter, even one too small to matter between small molecules, drives the blend to phase-separate. Most polymer pairs are therefore immiscible: polystyrene and polybutadiene, polyethylene and polypropylene, PET and polyethylene all separate into coarse, weakly bonded domains. This is a central obstacle in recycling mixed plastics (Chapter 21).

Miscible blends require $\chi \le 0$, which needs a specific attraction between the two polymers: a hydrogen bond, a dipole–dipole interaction or a π-electron interaction. Commercially important miscible pairs include poly(phenylene oxide)/polystyrene (Noryl), PVC/nitrile rubber (for permanently flexible cable jackets), PMMA/poly(vinylidene fluoride) (weatherable films) and PC/polyester blends (partly miscible). Many miscible blends phase-separate on heating, showing a **lower critical solution temperature** (LCST), because the specific interactions that make χ negative weaken at high temperature.

Worked example — polystyrene with PMMA, as a blend and as a block copolymer. For polystyrene and PMMA, $\chi \approx 0.028 + 3.9/T$, so at 150 °C (423 K), a typical processing temperature, $\chi \approx 0.037$. A blend of two chains each of 1,000 segments has $\chi_c = 0.002$; χ exceeds it by a factor of almost 20, and the blend separates into domains microns across, which scatter light: the mixture is opaque and brittle. Now join one PS chain and one PMMA chain of equal length end-to-end into a symmetric diblock copolymer. The blocks still want to separate, but they cannot move apart further than one chain length. Mean-field theory predicts an order–disorder transition at $(\chi N)_{ODT} \approx 10.5$ for a symmetric diblock, where N is the total number of segments. At 150 °C this requires $N \ge 10.5/0.037 \approx 280$ segments, a total molar mass of only about 29,000 g/mol (taking roughly 104 g/mol per segment). Commercial PS-b-PMMA for directed self-assembly, with M around 50,000–100,000 g/mol, is therefore well ordered and forms lamellae with a period of 25–40 nm, used to pattern features on semiconductor wafers.

Compatibilization and toughened blends

Immiscible blends are not useless; they are the basis of toughened plastics, provided the interface is strong and the domain size is controlled. A **compatibilizer** — a block or graft copolymer whose segments are miscible with each phase — sits at the interface, lowers the interfacial tension, reduces the domain size and stitches the phases together. Maleic-anhydride-grafted polypropylene reacts with the amine ends of polyamide to form a graft copolymer in situ during extrusion (reactive compatibilization), making the tough PA/PP and PA/elastomer blends of car air-intake manifolds and ski-boot shells. ABS and HIPS achieve the same effect with grafted rubber particles (Chapter 19).

Block-copolymer microphase separation

In a block copolymer the incompatible blocks segregate into domains whose size is set by the chain dimensions, typically 5–100 nm, and the joints between blocks are confined to the interfaces. The equilibrium morphology is chosen by a balance between interfacial energy (favouring few, large domains) and chain stretching (favouring small ones). It depends on two parameters: the segregation strength χN and the volume fraction f of one block. As f moves from 0.5 toward 0, the interface curves around the

minority block, and the morphology changes from **lamellae** ($f \approx 0.35$–0.5) through a **bicontinuous gyroid** (a narrow window near $f \approx 0.33$–0.37) to **hexagonally packed cylinders** ($f \approx 0.17$–0.33) and **body-centred cubic spheres** (f below about 0.17) (Figure 20.3). In the strong-segregation limit, the domain period scales as $d \propto aN^{2/3}\chi^{1/6}$, larger than the random-coil size because the chains are stretched away from the interface.

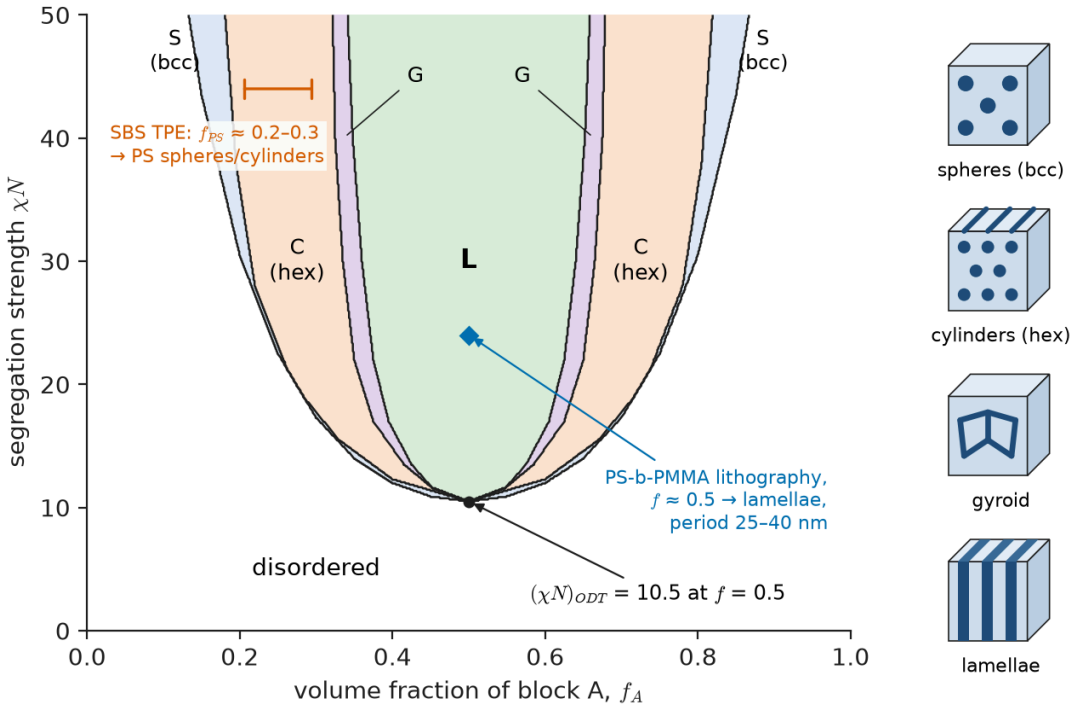

Figure 20.3 — Mean-field phase diagram of a diblock copolymer, segregation strength χN versus volume fraction f, with sketches of the ordered morphologies.

> **Status of the principle: Mean-field diblock phase diagram**
> *Approximate.* Self-consistent-field theory reproduces the observed sequence of morphologies and the strong-segregation scaling; near the order–disorder transition composition fluctuations, neglected in mean-field theory, shift the transition to higher χN for short chains, and real systems also show kinetically trapped and non-equilibrium structures.

Microphase separation has three major engineering uses. In **thermoplastic elastomers** such as SBS, with 20–30 % polystyrene, glassy PS spheres or cylinders act as physical crosslinks in a rubbery polybutadiene matrix; above 100 °C they soften, and the material can be injection-moulded into shoe soles, grips and bitumen modifiers for road surfacing. In **nanolithography**, block copolymer films are aligned on chemically or topographically patterned substrates to multiply the density of lithographic features (directed self-assembly). In **drug delivery and medicine**, amphiphilic block copolymers (for example PEG-b-PLA) form micelles 20–100 nm across in water, carrying hydrophobic drugs in their cores (Chapter 39).

20.5 Crystallization: lamellae, chain folding, spherulites and crystallinity

What can crystallize

A polymer can crystallize only if its chains are regular enough to pack into a lattice: linear, stereoregular, with few branches and without bulky random side groups. Polyethylene, isotactic PP, POM, PA66, PET, PEEK and PTFE crystallize; atactic PS, PMMA, PC and random copolymers with similar comonomer contents do not. Even crystallizable polymers never become fully crystalline: chains are too long and too entangled to disentangle completely in the time available, so the solid is **semicrystalline**, a composite of crystals and amorphous regions. Typical crystallinities are 40–80 % for polyethylene, 40–70 % for iPP, 30–50 % for PA66 and 0–40 % for PET, depending on cooling rate.

In the crystal, chains adopt their lowest-energy conformation and lie parallel. Polyethylene crystallizes in an orthorhombic cell ($a = 0.741$ nm, $b = 0.494$ nm, $c = 0.255$ nm along the chain axis, two chains per cell) with the all-trans zig-zag along c, giving a crystal density of 1.00 g/cm³ against 0.855 g/cm³ for the amorphous melt-quenched phase. Isotactic PP packs as 3_1 helices; PTFE as a 13_6 or 15_7 helix whose fluorine sheath gives the slippery surface.

Lamellae and chain folding

In 1957 Keller, Till and Fischer independently grew single crystals of polyethylene from dilute solution and found thin platelets, about 10 nm thick and several micrometres wide, with the chain axis perpendicular to the plate. Since the chains are hundreds of nanometres long, they must fold back and forth at the surfaces of the platelet. These **lamellae**, 5–30 nm thick, are the basic crystalline unit of all semicrystalline polymers. In crystals grown from the melt the folding is less regular: some chains re-enter the same lamella in an adjacent position, others leave and wander through the amorphous layer to another lamella. Chains that bridge two lamellae are **tie molecules**, and entangled loops between lamellae act similarly. They carry load between crystals and control toughness, environmental stress-crack resistance and slow crack growth in pipes. The short branches of LLDPE and bimodal PE100 (Chapter 19) are used precisely to raise the number of tie molecules.

Why should crystals be so thin? The lamellar thickness l is a kinetic, not an equilibrium, choice. A lamella of thickness l has two fold surfaces with high surface energy σ_e (about 0.09 J/m² for PE, much higher than the lateral surface energy), and its melting point is depressed below the equilibrium value T_m^0 for an infinitely thick crystal by the Gibbs–Thomson relation (Chapter 6):

$$T_m = T_m^0 \left(1 - \frac{2\sigma_e}{\Delta h_v \, l}\right)$$

where Δh_v is the heat of fusion per unit volume. Crystallization at an undercooling $\Delta T = T_m^0 - T_c$ produces lamellae that are slightly thicker than the minimum stable thickness at T_c, so $l \propto 1/\Delta T$: fast cooling (large undercooling) gives thin lamellae with low melting points, slow cooling or annealing gives thicker, more perfect ones. The Lauritzen–Hoffman theory of secondary nucleation on the lamellar growth face formalizes this.

> **Status of the principle: Gibbs–Thomson relation for lamellar melting**
> *Derived.* Follows from the balance of bulk and fold-surface free energies (Chapter 6); quantitative use needs σ_e and T_m^0, which are obtained from extrapolations of measured T_m against $1/l$, and the relation ignores reorganization of lamellae during heating.

Spherulites

When a semicrystalline polymer crystallizes from the melt without flow, the lamellae grow outward from a nucleus in all directions, splaying and branching, with amorphous material trapped between them. The resulting spherical aggregates, **spherulites**, are typically 1–100 μm across. Viewed between crossed polarizers they show a characteristic Maltese-cross extinction pattern, because the chain axis is tangential and the lamellar radii are optically birefringent. Spherulites grow at constant radial rate G until they impinge, forming polygonal boundaries where low-molar-mass material and impurities collect — weak planes along which brittle fracture often runs. The overall crystallization follows Avrami (JMAK) kinetics (Chapter 9), with exponents of 2–4 depending on nucleation and growth geometry.

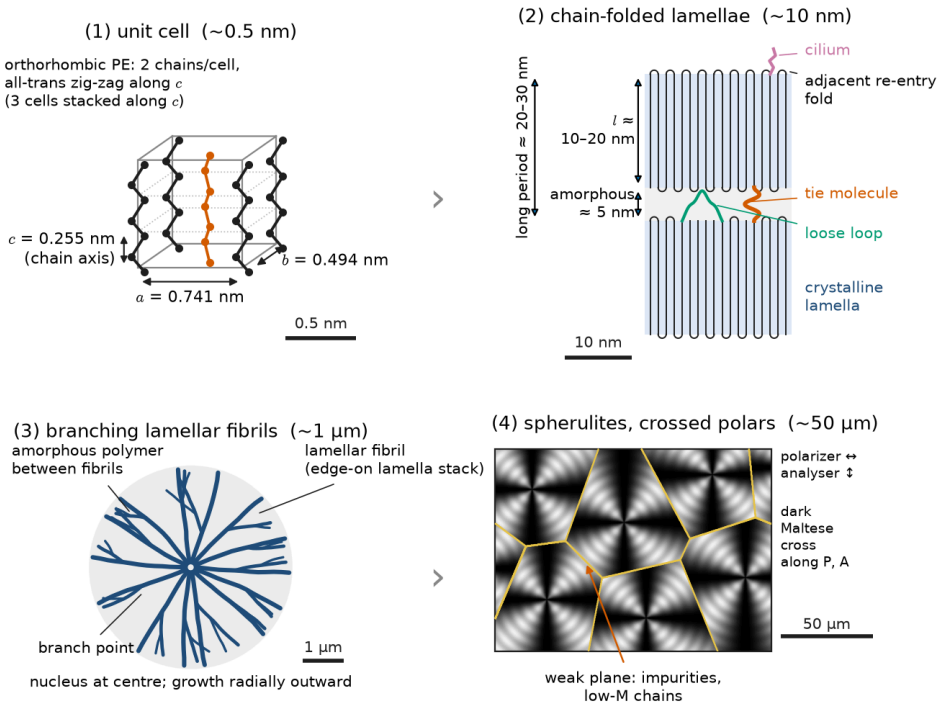

Figure 20.4 — Structural hierarchy of a melt-crystallized semicrystalline polymer, from unit cell to spherulite.

Spherulite size is controlled by nucleation density. Adding **nucleating agents** (sodium benzoate, talc, organophosphates, sorbitol derivatives) raises the number of nuclei by orders of magnitude: spherulites become smaller than the wavelength of light, the part becomes clear, crystallization is faster (shorter moulding cycles) and stiffness and heat-distortion temperature rise. Clarified polypropylene food containers and syringe barrels — like those in the opening case — depend on sorbitol clarifiers. Flow during processing orients chains and produces different morphologies: **shish-kebab** structures (an extended-chain core with folded-chain lamellae overgrown on it) in injection-moulded skins and melt-spun fibres, and highly oriented fibrillar structures in drawn fibres, where gel-spun UHMWPE fibres (Dyneema, Spectra) reach moduli above 100 GPa because nearly all chains are extended and aligned.

Degree of crystallinity and its measurement

The crystallinity of a sample can be obtained from several measurements, which agree only approximately because each sees a different boundary between "crystal" and "amorphous":

- **Density**, using the two-phase model. If ρ_c and ρ_a are the crystal and amorphous densities, the mass fraction crystallinity is $X_c = \rho_c(\rho - \rho_a)/[\rho(\rho_c - \rho_a)]$. A density-gradient column measures ρ to four decimals.
- **Differential scanning calorimetry**, as the ratio of the measured heat of fusion to that of a 100 % crystalline polymer, $X_c = \Delta H_m / \Delta H_m^0$ (Chapter 52).
- **Wide-angle X-ray diffraction**, as the ratio of crystalline peak area to total scattering.
- **Spectroscopy** (FTIR, Raman, solid-state NMR) using bands specific to the crystalline conformation.

Polymer	ρ_c (g/cm³)	ρ_a (g/cm³)	ΔH_m^0 (J/g)	T_m^0 (°C)	Typical X_c
Polyethylene	1.00	0.855	293	141.5	40–80 %
Isotactic polypropylene	0.94	0.85	207	186	40–70 %
PET	1.46	1.33	140	280	0–40 %
Nylon-6,6	1.24	1.09	196	280	30–50 %
PEEK	1.40	1.26	130	395	25–40 %
PTFE	2.35	2.00	82	332	50–90 %

Worked example — crystallinity and lamellar thickness of an HDPE pipe resin. A sample of HDPE pipe has density 0.950 g/cm³. With $\rho_c = 1.00$ and $\rho_a = 0.855$ g/cm³, the mass crystallinity is $X_c = 1.00 \times (0.950 - 0.855)/[0.950 \times (1.00 - 0.855)] = 0.690$, or 69 % (the volume fraction is $(0.950 - 0.855)/0.145 = 0.655$). A DSC run on the same material measures a melting enthalpy of 200 J/g; with $\Delta H_m^0 = 293$ J/g this gives 68 %, in good agreement. The DSC melting peak lies at about 129 °C. Using the Gibbs–Thomson relation with $T_m^0 = 414.6$ K (141.5 °C), $\sigma_e = 0.090$ J/m² and $\Delta h_v = 293$ J/g × 1.00 g/cm³ = 2.93×10^8 J/m³, a lamella 10 nm thick melts at $414.6 \times (1 - 2 \times 0.090/(2.93 \times 10^8 \times 10^{-8})) = 389.1$ K, or 116 °C; one 20 nm thick at 129 °C; one 50 nm thick at 136 °C. The peak at 129 °C therefore corresponds to lamellae about 20 nm thick, consistent with small-angle X-ray long periods of 25–30 nm (lamella plus amorphous layer) for slowly cooled HDPE. A pipe that is quenched from the extruder on its outer surface has thinner lamellae and a lower melting point at the skin than in the bulk, and the resulting gradient of crystallinity contributes to the residual stresses that pipe extruders try to minimize.

Crystallinity and properties

Crystallinity raises stiffness, yield strength, density, chemical and solvent resistance, barrier performance and heat-distortion temperature, and lowers toughness, transparency and the rate of water or solvent uptake. Above Tg, the amorphous phase is rubbery, and the crystals act as stiff, physically crosslinked reinforcement: the modulus of polyethylene rises from about 0.2 GPa for LDPE to 1.2 GPa for HDPE, mostly through crystallinity. Below Tg, both phases are rigid, and crystallinity matters less for stiffness but more for chemical resistance. PET shows the extremes: quenched amorphous PET bottles are clear and have a Tg of 75 °C; slowly crystallized PET is opaque and dimensionally stable up to 200 °C, the basis of dual-ovenable food trays. Stretch-blow moulding (Chapter 21) induces small, oriented crystals that keep the bottle clear while improving CO_2 barrier and strength.

20.6 The glass transition

Phenomenology

Cool an amorphous polymer melt and, at a temperature called the **glass-transition temperature** T_g, its properties change abruptly: the modulus rises from about 1 MPa (rubbery) to about 1–3 GPa (glassy) over a range of 10–20 K, and the slope of specific volume against temperature drops to roughly half its melt value (Figure 20.5). The heat capacity and thermal-expansion coefficient also step down. No latent heat is released and no structure appears in a diffraction pattern: the liquid simply stops rearranging on the time scale of the experiment. Measured T_g depends on the cooling rate, rising by about 3 K for every tenfold increase in rate. The glass transition is therefore a kinetic phenomenon — the point where the molecular relaxation time exceeds the experimental time scale — not a true thermodynamic phase transition. It is the same phenomenon as in the oxide and metallic glasses of Chapters 9 and 17.

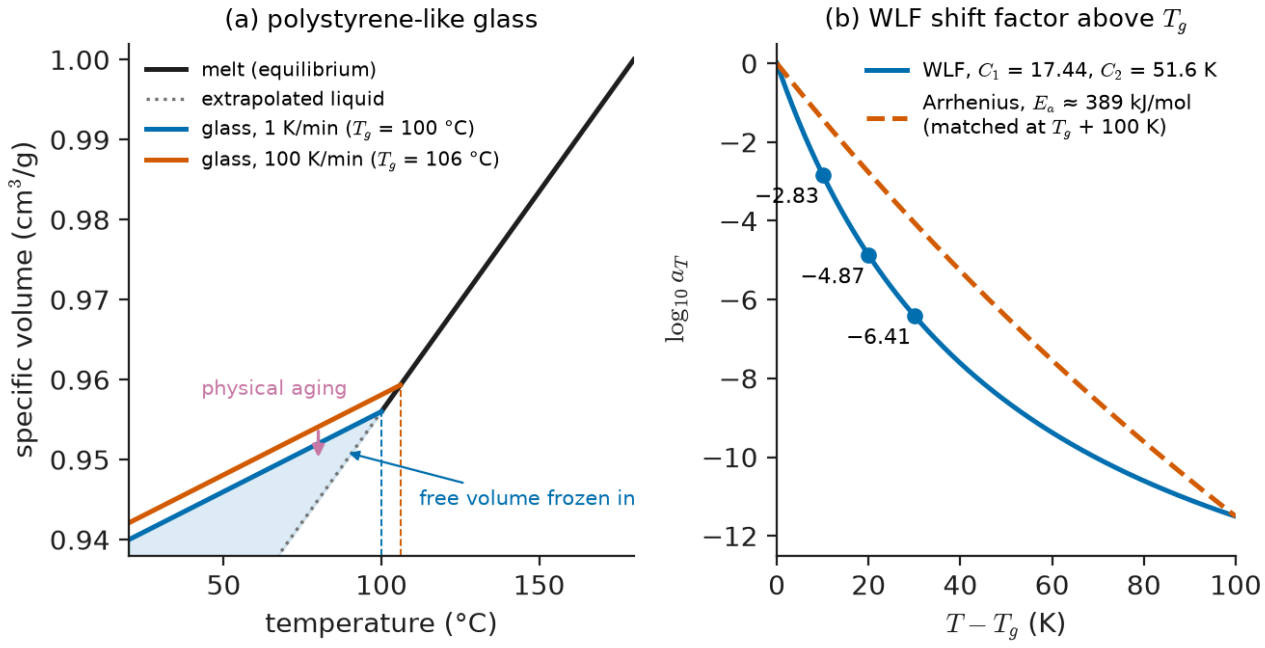

Figure 20.5 — Specific volume versus temperature for an amorphous polymer cooled at two rates, and the WLF shift factor above Tg.

Free-volume theory and the WLF equation

The **free-volume** picture explains the transition. The volume of a liquid is the occupied volume of its molecules plus a **free volume** V_f made of the gaps between them. Molecular motion requires a segment to find a nearby hole large enough to move into, so mobility rises steeply with the fractional free volume $f = V_f/V$. Doolittle found empirically that viscosity follows $\ln \eta = \ln A + B/f$. On cooling, f decreases linearly, $f = f_g + \alpha_f (T - T_g)$, until the motions needed to shed further free volume become too slow; at T_g the free volume is frozen in at a value f_g.

Writing the ratio of relaxation times (or viscosities) at T and at a reference T_g, the **shift factor** a_T, from Doolittle's equation gives

$$\log_{10} a_T = -\frac{C_1\,(T - T_g)}{C_2 + T - T_g}, \qquad C_1 = \frac{B}{2.303 f_g}, \quad C_2 = \frac{f_g}{\alpha_f}$$

This is the **Williams–Landel–Ferry (WLF) equation** (1955). With "universal" constants $C_1 = 17.44$ and $C_2 = 51.6$ K it describes many amorphous polymers within a factor of a few between T_g and $T_g + 100$ K; for $B = 1$ these correspond to $f_g \approx 0.025$ and $\alpha_f \approx 4.8 \times 10^{-4}$ K^{-1}, which says that about 2.5 % of the volume of a glass is free volume at T_g.

> **Status of the principle: WLF equation**
> *Empirical.* The form follows from Doolittle's free-volume relation, which is itself empirical; the constants C_1 and C_2 are fitted for each polymer (the "universal" values are averages). Valid from T_g to about $T_g + 100$ K; below T_g the glass is out of equilibrium and relaxation is closer to Arrhenius, and far above T_g Arrhenius behaviour also returns.

The WLF equation is the basis of **time–temperature superposition** (Chapter 14): viscoelastic data measured over a few decades of time at several temperatures can be shifted along the log-time axis by $\log a_T$ to build a master curve covering ten or more decades. Polymer engineers use it to predict 20-year creep of a polymer bracket from tests of days at higher temperature, and to understand why a polymer that is rubbery in a slow test can be glassy under impact.

Worked example — how steeply mobility rises above Tg. Take the universal constants. At $T_g + 10$ K, $\log a_T = -17.44 \times 10/61.6 = -2.83$, so molecular motions are $10^{2.83} \approx 680$ times faster than at T_g. At $T_g + 20$ K, $\log a_T = -17.44 \times 20/71.6 = -4.87$, a factor of about 74,000; at $T_g + 30$ K, -6.41, a factor of 2.6 million. Consider an injection-moulded polystyrene lens holder ($T_g = 100$ °C). If the flow-induced orientation in the melt takes about an hour to relax at 100 °C, it relaxes at 120 °C in $3600 \times 1.34 \times 10^{-5} \approx 0.05$ s — faster than the part cools through that range, so the orientation frozen into the part is set almost entirely by what happens in the last few degrees above T_g. Conversely, a PMMA aircraft canopy formed at 150 °C ($T_g + 45$ K) relaxes in microseconds, but if it is held for years at 70 °C, 35 K below T_g, the glass continues to densify slowly — **physical aging** — becoming stiffer and more brittle, with lower creep compliance. Physical aging is why the creep data of glassy polymers must state the aging time before the test.

Molecular factors controlling Tg

Since T_g marks the onset of cooperative backbone motion, anything that hinders that motion raises it.

Factor	Effect on Tg	Example
Backbone flexibility	Flexible links (–O–, –Si–O–) lower; rings in the backbone raise	PDMS –125 °C; PE –120 °C; PC 147 °C; polyimide above 350 °C
Bulky or stiff side groups	Raise by hindering rotation	PP –10 °C; PS 100 °C; poly(vinyl carbazole) 200 °C
Flexible side chains	Lower by internal plasticization	Poly(methyl acrylate) 10 °C; poly(butyl acrylate) –50 °C
Polarity and hydrogen bonds	Raise by strengthening interchain cohesion	PVC 82 °C vs PP –10 °C; PA66 50–60 °C dry, near 0 °C wet
Chain length	Raises up to a plateau (fewer free ends)	Flory–Fox: PS 67 °C at 3,000 g/mol, 90 °C at 10,000, 99 °C at 100,000
Crosslinking	Raises	Epoxy Tg from 100 °C to above 200 °C with crosslink density
Plasticizers and absorbed solvent	Lower	PVC with 30–40 % plasticizer becomes rubbery at room temperature; moisture lowers PA66 and epoxy Tg
Copolymerization and miscible blending	Intermediate value	PPO/PS (Noryl) 50/50 has a single Tg near 150 °C
Tacticity	Moderate effect in some polymers	PMMA: syndiotactic about 125 °C, isotactic about 45 °C

Two simple relations quantify the last effects. The **Flory–Fox equation** for molecular weight,

$$T_g = T_{g,\infty} - \frac{K}{M_n}$$

with $K \approx 10^5$ K·g/mol for polystyrene, reflects the extra free volume carried by chain ends. The **Fox equation** for miscible blends, random copolymers and plasticized polymers,

$$\frac{1}{T_g} = \frac{w_1}{T_{g1}} + \frac{w_2}{T_{g2}}$$

(weight fractions, absolute temperatures) gives, for PPO ($T_g = 215$ °C) blended 50/50 with PS ($T_g = 100$ °C), $1/T_g = 0.5/488.15 + 0.5/373.15$, so $T_g = 423$ K = 150 °C — close to the measured value and the reason Noryl can be tuned between 100 and 200 °C by composition. A blend that shows a single, composition-dependent T_g by DSC or DMA is the standard experimental test of miscibility; an immiscible blend shows the two separate T_g values of its components.

> **Status of the principle: Flory–Fox and Fox equations**
> *Empirical.* Simple mixing rules with a free-volume rationale; adequate within about 10 K for weakly interacting systems, but they miss strong specific interactions (where the Gordon–Taylor or Kwei forms with fitted constants are used).

Tg in design

The glass transition separates the service regimes of polymers. An **amorphous** thermoplastic is useful only below its T_g (heat-distortion temperature is typically 10–20 K below T_g): PS cups soften in boiling water, PC survives it. An **elastomer** is useful only above its T_g: a seal compound with $T_g = -40$ °C becomes leathery and leaks in a Siberian winter, and the Challenger shuttle O-ring failure of 1986 involved a fluoroelastomer whose resilience was too low at the 2 °C launch temperature. A **semicrystalline** polymer can be used between T_g and T_m (PE, PP, POM, with T_g below room temperature) or below T_g (PEEK, PPS) for maximum stiffness. Near T_g the mechanical loss factor $\tan\delta$ peaks, which is exploited in vibration-damping layers on car floors and in tyre treads, where a high loss near 0 °C at 10–100 Hz gives wet grip.

20.7 Liquid-crystalline polymers

Rods that order

If the chain is not flexible but a stiff rod, a different kind of order becomes possible. Long, rigid rods in a concentrated solution or melt cannot all point randomly without overlapping: above a critical concentration, excluded-volume packing forces them to

align along a common direction, the **director**, while their centres remain disordered as in a liquid. This **nematic** liquid-crystalline phase is the polymer analogue of the small-molecule liquid crystals in displays (Chapter 37). Onsager (1949) and Flory (1956) showed that rods with aspect ratio x (length/diameter) form a nematic phase above a volume fraction of roughly $\phi^* \approx (8/x)(1 - 2/x)$, about 8 % for $x = 100$: alignment is driven by packing entropy, not by attractive forces. The degree of alignment is measured by the **order parameter** $S = \langle \tfrac{1}{2}(3\cos^2\theta - 1) \rangle$, where θ is the angle between a molecule and the director; $S = 0$ for an isotropic liquid and $S = 1$ for perfect alignment, with typical nematic values of 0.5–0.9.

> **Status of the principle: Onsager–Flory rod criterion**
> *Approximate.* Mean-field excluded-volume theories for monodisperse rigid rods; they predict the right scaling with aspect ratio, but real chains are semiflexible and polydisperse, which shifts the transition.

Lyotropic and thermotropic LCPs

Main-chain LCPs have mesogenic, rod-like units in the backbone; **side-chain** LCPs have mesogens attached through flexible spacers to a flexible backbone (used in optical and actuator applications, including liquid-crystal elastomers that contract reversibly on heating; Chapter 37).

- **Lyotropic** LCPs form liquid crystals in solution. Poly(p-phenylene terephthalamide) (PPTA; Kevlar, Twaron) cannot be melted below its decomposition temperature, but dissolves at about 20 % in concentrated sulfuric acid to give a nematic dope. Spinning this dope through an air gap into water (dry-jet wet spinning) aligns the domains along the fibre, giving fibres with modulus 70–130 GPa and strength about 3–3.6 GPa at a density of 1.44 g/cm³: specific strength five times that of steel wire. They are used in body armour, helmet shells, tyre cords, fibre-optic cable strength members and composite pressure vessels (Chapter 22). PBO (Zylon) and the gel-spun UHMWPE fibres mentioned earlier reach still higher specific stiffness.
- **Thermotropic** LCPs form liquid crystals in the melt. Aromatic copolyesters such as Vectra (hydroxybenzoic acid with hydroxynaphthoic acid) have a melting point lowered by the irregular copolymer sequence to 280–330 °C, and their melts are nematic. Under flow the rods align readily, so the melt viscosity is exceptionally low and the moulded part is highly oriented along the flow direction: very thin walls (0.2 mm) can be filled, the shrinkage and thermal expansion along the flow direction are almost zero, and the parts are stiff and chemically resistant. Thermotropic LCPs dominate fine-pitch electronic connectors and are used in minimally invasive surgical instruments and as low-loss, low-moisture substrates for high-frequency circuits.

The price of orientation is anisotropy: strength and stiffness across the flow direction can be a fifth of those along it, weld lines where two flow fronts meet are very weak, and fibres such as Kevlar are weak in compression because the aligned chains buckle into kink bands at about 0.4 GPa. Composite designers must therefore treat aramid fibres differently from carbon (Chapter 22).

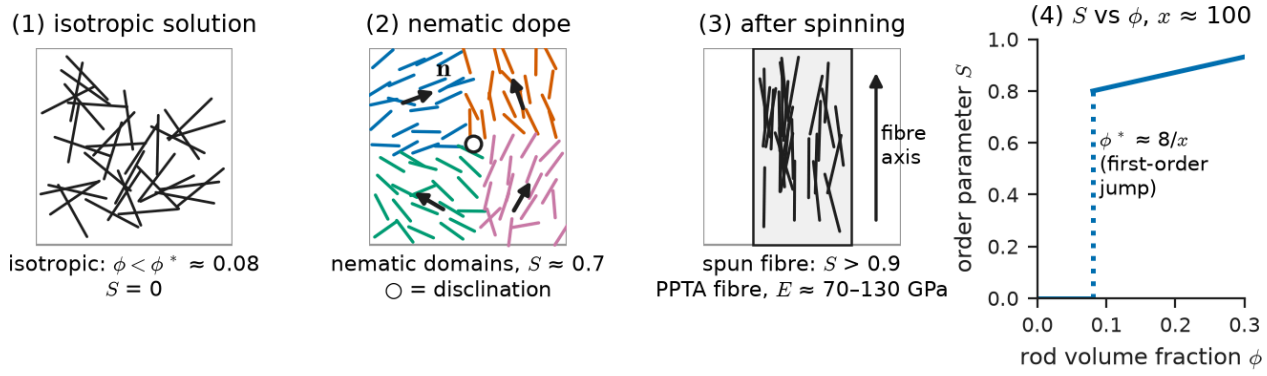

Figure 20.6 — Isotropic, nematic and oriented-fibre states of a rigid-rod polymer, with the order parameter.

What remains open

Open question	Where it is answered
How do entangled chains flow, and why does melt viscosity scale as $M^{3.4}$?	Chapter 21
How do polymer parts yield, craze and crack, and how do solvents cause stress cracking?	Chapter 21
How is crosslink density measured by swelling, and how are elastomers vulcanized?	Chapter 21
How do processing flows orient chains and set crystallinity in moulded parts, films and fibres?	Chapter 21
What are the viscoelastic models and time–temperature superposition behind polymer creep?	Chapter 14
How do the solution thermodynamics and Gibbs–Thomson effect used here follow from free energy?	Chapter 6
How do nucleation and Avrami kinetics apply to crystallization in general?	Chapter 9
How do aramid, carbon and glass fibres combine with polymer matrices in composites?	Chapter 22
How do small-molecule liquid crystals, gels and self-assembled soft matter work?	Chapter 37
How are DSC, DMA, X-ray scattering and polarized-light microscopy used to measure Tg and crystallinity?	Chapters 52 and 53
How are polymer chains simulated with molecular dynamics and coarse-grained Monte Carlo?	Chapter 56
How does physical and chemical aging change polymer properties in service?	Chapter 44

Next: Chapter 21 sets these chains in motion — in the melt, in the deforming solid and in the processing machine — and follows them through additives, degradation, vulcanization and recycling.

Chapter 21 — Polymer Rheology, Mechanics and Processing

Part V · Polymers and Composites

A hospital reports that the polycarbonate housings of its infusion pumps are cracking. The cracks start at the screw bosses, run a few millimetres into the wall and have smooth, mirror-like faces. The housings had passed every drop and impact test at launch. Three things turned out to have combined. The bosses were moulded with high residual tensile stress because the mould was run cold to shorten the cycle; self-tapping screws added a hoop stress of 20–30 MPa, well below the 60 MPa yield strength of polycarbonate; and the cleaning staff had switched to a new disinfectant wipe containing alcohols and quaternary ammonium compounds. Polycarbonate under modest tensile stress, wetted by a liquid that plasticizes its surface, crazes and cracks in days — environmental stress cracking. The same polymer, the same design, the same screws had worked for years with the old wipe.

The case links the whole chapter: melt flow and cooling set the residual stress, viscoelastic relaxation set how much screw stress remained, the yield-or-craze competition set the failure mode, and the environment set the time to failure. This chapter puts the chains of Chapters 19 and 20 in motion — in the melt, the deforming solid and the processing machine — and follows them through formulation, aging, vulcanization and the end of life.

> **Why do polymer melts flow as they do, and how do processing flows set the structure and residual stress of a part?**
> **How do polymer parts deform, yield, craze and degrade in service, and what can additives, crosslinking and recycling do about it?**

Concept	Engineering destination
Shear thinning, entanglement, reptation, $M^{3.4}$	Grade selection by melt flow index; extruder and die design; mould filling
Creep modulus, isochronous curves	Snap fits, bolted plastic flanges, pressure pipes, storage tanks
Yielding, crazing, environmental stress cracking	Impact-modified housings, medical devices, chemical-resistant parts
Extrusion, moulding, blow moulding, thermoforming, spinning, film blowing	Choice of process and of grade for pipes, bottles, packaging, fibres, car parts
Plasticizers, fillers, stabilizers, flame retardants	Flexible PVC cables, glass-filled nylon brackets, UV-stable outdoor parts, aircraft interiors
UV, thermo-oxidative and hydrolytic degradation	Service-life estimates, accelerated aging, outdoor and under-bonnet applications
Vulcanization	Tyres, seals, hoses, engine mounts
Recycling and biodegradable polymers	Packaging design, end-of-life regulation, compostable and resorbable products

Roadmap. How melts flow and why molecular weight dominates (21.1) → viscoelastic design of solid parts (21.2) → yielding, crazing and stress cracking (21.3) → processing routes (21.4) → additives (21.5) → degradation in service (21.6) → elastomers and vulcanization (21.7) → recycling and biodegradable polymers (21.8).

21.1 Melt rheology

Viscosity of polymer melts

A polymer melt is a peculiar viscous liquid. Its viscosity at low shear rates, the **zero-shear viscosity** η_0, ranges from about 10 Pa·s for a low-molar-mass polyester to 10^4–10^6 Pa·s for commercial thermoplastics, against 0.001 Pa·s for water and about 1 Pa·s

for honey. Its viscosity falls as the shear rate rises, it stores elastic energy, it swells when it leaves a die, and it can fracture like a solid if pulled too fast. All of this follows from the long, entangled chains.

Processing spans wall shear rates $\dot{\gamma}$ from 1–10 s^{-1} in compression moulding through 100–1,000 s^{-1} in extrusion to 10^3–10^5 s^{-1} in injection moulding.

Shear thinning

Above a characteristic shear rate, roughly the reciprocal of the longest relaxation time of the chains, the melt viscosity falls with increasing shear rate (Figure 21.1). Chains are oriented and partly disentangled by the flow faster than Brownian motion can restore their random coils, and the oriented, less entangled melt resists flow less. Over the processing range the viscosity is described adequately by a **power law**,

$$\eta = K\,\dot{\gamma}^{\,n-1}$$

with a **flow index** n of 0.2–0.5 for most thermoplastic melts ($n = 1$ is Newtonian). The Cross and Carreau–Yasuda models join the Newtonian plateau at low rates to the power law at high rates and are what mould-filling software uses.

> **Status of the principle: Power-law (Ostwald–de Waele) viscosity**
> *Empirical.* A two-parameter fit valid over one to three decades of shear rate in the shear-thinning region; it fails at low rates, where the viscosity tends to the finite plateau η_0, and predicts unphysical infinite viscosity as $\dot{\gamma} \to 0$.

Shear thinning lets thin-walled parts be filled at 10^4 s^{-1} with acceptable pressures. Viscosity also falls with temperature: by an Arrhenius law (activation energies of 25–35 kJ/mol for HDPE, 40–50 kJ/mol for PP) for polyolefins, and by the much steeper WLF law (Chapter 20) for amorphous polymers processed within 100 K of T_g, such as PC and PMMA, which therefore need closer temperature control.

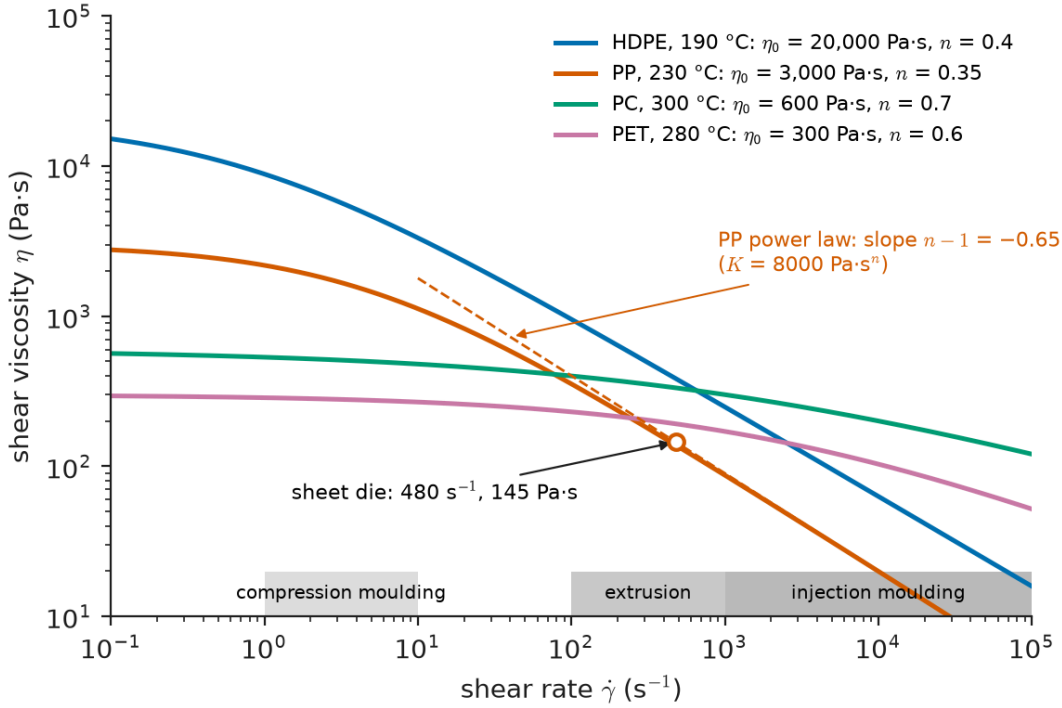

Figure 21.1 — Shear viscosity versus shear rate for typical thermoplastic melts, with the processing windows of different operations.

Entanglement

In a melt, coils of different chains interpenetrate (Chapter 20). Above a critical length the chains cannot slide past one another freely: each is hemmed in by **entanglements**, topological constraints that act on short time scales like temporary crosslinks. Over times shorter than the disentanglement time, an entangled melt behaves like a rubber, with a **plateau modulus** given by the rubber-elasticity result of Chapter 20 with the entanglement molecular weight M_e in place of M_c:

$$G_N^0 \approx \frac{\rho RT}{M_e}$$

Polymer	M_e (g/mol, typical)	G_N^0 (MPa, approx.)	Critical M_c for $M^{3.4}$ (g/mol, typical)
Polyethylene	1,000–1,300	2 (at 140 °C)	3,500
Polycarbonate	1,300–1,700	2.5 (at 200 °C)	4,000–5,000
Polypropylene	5,000–7,000	0.4	10,000–15,000
PMMA	10,000	0.3–0.4	25,000–30,000
PDMS	12,000	0.2	25,000–30,000
Polystyrene	13,000–17,000	0.2 (at 180 °C)	31,000–36,000

Entanglement is also what makes a glassy or semicrystalline polymer tough. A chain shorter than about $2M_e$ is anchored at no more than one entanglement and can be pulled out of a craze or crack without breaking; longer chains must break bonds. This is why the cable ties of Chapter 19 became brittle when M_n fell toward a few times M_e, and why polycarbonate, with its small M_e, is tough while polystyrene, with a large one, is brittle.

Reptation and the 3.4-power law

How does an entangled chain move? De Gennes (1971) and Doi and Edwards (1978) pictured each chain confined in a **tube** formed by its neighbours. The chain cannot move sideways through the tube walls, but it can slide, snake-like, along its own contour: **reptation**. The time to diffuse out of the original tube, the **disentanglement time** τ_d, scales as the tube length squared divided by the chain's curvilinear diffusion coefficient. The tube length is proportional to N and the friction to N, so $\tau_d \propto N^3$. The viscosity is of order $G_N^0 \tau_d$, and since G_N^0 does not depend on chain length,

$$\eta_0 \propto M^3 \quad \text{(pure reptation)}, \qquad \eta_0 \propto M^{3.4} \quad \text{(measured, } M > M_c\text{)}$$

Below the critical molecular weight $M_c \approx 2\text{–}3M_e$, chains are unentangled and the Rouse model gives $\eta_0 \propto M$ (Figure 21.2).

> **Status of the principle: Reptation and the 3.4-power law**
> *Approximate* (reptation, M^3) and *Empirical* (the exponent 3.4). The tube model is derived for a chain in a fixed network of obstacles; contour-length fluctuations and constraint release (the tube itself moving as neighbours reptate) account for most of the difference between 3 and the measured 3.4 ± 0.2, which holds for linear chains over a wide range above M_c. Long-chain branches suppress reptation and make viscosity rise exponentially with branch length.

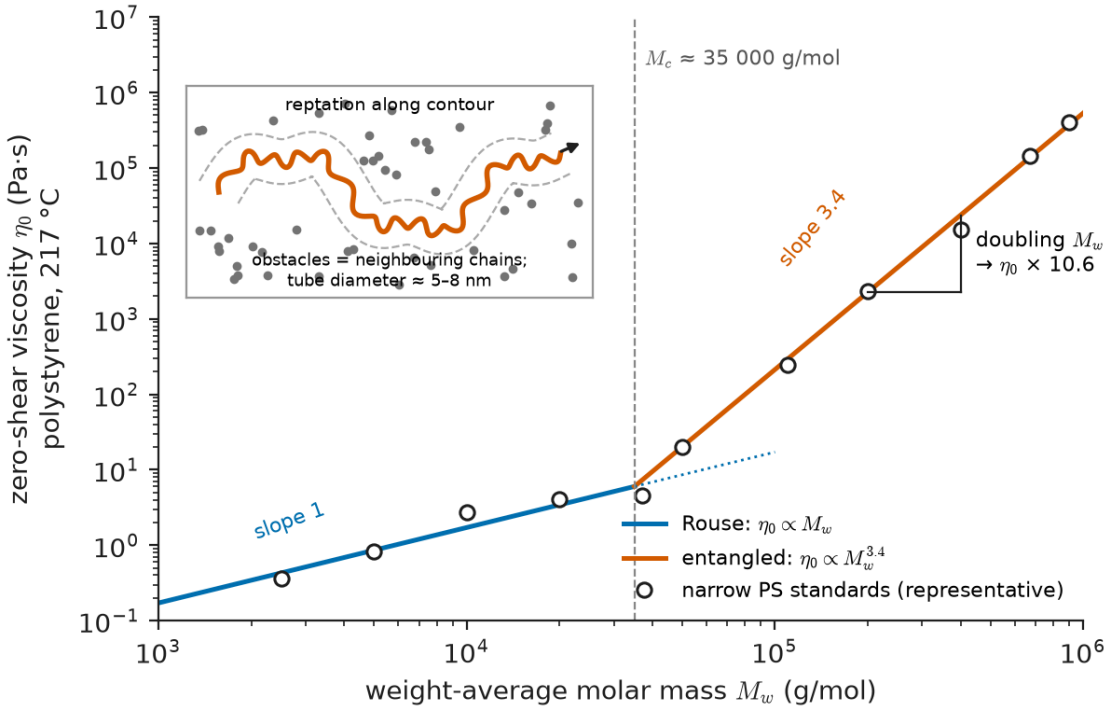

Figure 21.2 — Zero-shear viscosity versus weight-average molar mass: Rouse regime below Mc and the 3.4-power law above it.

Worked example — molecular weight, grade and die pressure. Because $\eta_0 \propto M_w^{3.4}$, a 20 % increase in M_w raises the zero-shear viscosity by $1.2^{3.4} = 1.86$, and doubling M_w raises it by $2^{3.4} = 10.6$. This is why the **melt flow index** (MFI or MFR, grams extruded in 10 min through a standard die under a standard load, ISO 1133) is the everyday measure of grade: a PP of MFR 2 g/10 min is for pipes and sheet, one of 12 for general injection moulding, one of 50–100 for thin-wall packaging, and between them M_w differs by roughly a factor of three. Now take a sheet die for PP at 230 °C, with a 1.0 m wide slit, 1.5 mm gap and 100 mm land length, running at 300 kg/h. With melt density 750 kg/m³ the flow rate is $Q = 1.11 \times 10^{-4}$ m³/s. The apparent (Newtonian) wall shear rate in a slit is $6Q/(WH^2) = 6 \times 1.11 \times 10^{-4}/(1.0 \times (1.5 \times 10^{-3})^2) = 296$ s⁻¹; the Rabinowitsch correction for a power-law fluid with $n = 0.35$ multiplies this by $(2n+1)/(3n) = 1.62$, giving $\dot{\gamma}_w = 480$ s⁻¹. With $K = 8{,}000$ Pa·sⁿ, the wall shear stress is $\tau_w = 8000 \times 480^{0.35} = 6.9 \times 10^4$ Pa, the viscosity is 145 Pa·s, and the pressure drop over the land is $\Delta P = 2L\tau_w/H = 2 \times 0.10 \times 6.9 \times 10^4/1.5 \times 10^{-3} = 9.3$ MPa (93 bar). Doubling the throughput would raise the pressure by only $2^{0.35} = 1.27$, thanks to shear thinning; a Newtonian liquid would need twice the pressure.

Melt elasticity: die swell, melt fracture and extensional flow

Chains oriented in the die recoil when the melt leaves it, so the extrudate swells to 1.2–3 times the die gap (**die swell**). Above a wall shear stress of roughly 0.1–0.3 MPa the extrudate surface roughens ("sharkskin") and then distorts grossly (**melt fracture**), capping the extrusion rate. In **extensional flows** — fibre spinning, film blowing, blow moulding, thermoforming — the relevant property is the extensional viscosity. Linear polymers such as LLDPE thin uniformly under extension; long-chain-branched LDPE **strain-hardens**, its extensional viscosity rising steeply with strain, which stabilizes a bubble or parison against local thinning. That is why LDPE is blended into LLDPE film recipes (Chapter 19).

21.2 Viscoelastic mechanical behaviour in practice

Why polymers must be designed with time in mind

Chapter 14 introduced the viscoelastic models and time–temperature superposition. Their practical consequence is that a polymer part has no single modulus: a polypropylene bracket may have an apparent modulus of 1.4 GPa after a minute, 0.9 GPa after a day and about 0.45 GPa after a year at room temperature, and a part designed with the handbook modulus sags, loses clamping force or leaks.

Creep modulus and isochronous curves

Plastics datasheets therefore give **creep data**: strain versus time at several stresses and temperatures, measured over 10^3–10^4 h (ISO 899). Designers use them in two forms.

- The **creep modulus** $E_c(t) = \sigma/\varepsilon(t)$, the apparent modulus at a chosen time, is substituted for E in the ordinary elastic formulas (the pseudo-elastic design method). A beam deflection or a plate buckling calculation becomes a calculation for the end of the design life.
- **Isochronous stress–strain curves** are cross-plots of the creep curves at constant time (1 h, 1,000 h, 10 years). They show that polymers are non-linear even at small strain: above about 0.5–1 % strain the creep modulus falls with stress.

Polymer (unfilled unless stated)	Short-term E at 23 °C (GPa)	Creep modulus at 23 °C, 1,000 h, low stress (GPa, typical)	Typical design strain limit
HDPE	1.0–1.2	0.3–0.4	1–2 %
Polypropylene (homopolymer)	1.5	0.5–0.6	1–2 %
ABS	2.3	1.4–1.6	0.5–1 %
Polycarbonate	2.3	1.7–1.9	0.5–1 %
POM (acetal)	2.9	1.2–1.5	1 %
PA66 (conditioned, 50 % RH)	1.6–2.0	0.6–0.9	1 %
PA66 + 30 % glass fibre (conditioned)	6–8	4–5	0.5 %
PEEK	3.8	3.0–3.3	0.5–1 %

Semicrystalline polymers above their T_g (PE, PP, POM, wet PA66) creep most; glassy amorphous polymers well below T_g (PC, ABS) and high-T_g semicrystalline polymers (PEEK) creep least, and short glass fibres reduce creep by carrying load across the matrix (Chapter 22).

Stress relaxation in assemblies

Snap fits, press fits, screw bosses, bolted flanges and seals depend on a stress that must persist. In each case the strain is fixed and the stress relaxes. For a polycarbonate boss in the opening case, an initial hoop stress of 25 MPa might relax to 15 MPa within a month — still enough to drive stress cracking in an aggressive environment. Threaded metal inserts are often moulded in to carry clamp loads.

Long-term design: pressure pipes

The most thoroughly codified viscoelastic design is that of plastic pressure pipes. Pipes are pressure-tested at several temperatures until they fail, and the hoop stress at failure is plotted against time to failure on log–log axes (ISO 9080). The regression line is extrapolated to 50 years at 20 °C to give the **minimum required strength** (MRS): 8 MPa for PE80, 10 MPa for PE100. At high stress the pipes fail by ductile ballooning; at lower stress and longer times the curve shows a "knee" to a steeper brittle branch, where slow crack growth through the wall controls life (Figure 21.3). Bimodal PE100 shows no knee at 80 °C within 5,000 h, evidence that its tie-molecule-rich morphology (Chapters 19 and 20) resists slow crack growth.

> **Status of the principle: Stress-regression extrapolation of pipe life**
> *Empirical.* A statistical fit (ISO 9080) to failure data extrapolated by up to a factor of about 50 in time with the help of elevated-temperature tests; it assumes the failure mechanism does not change beyond the test window, which is why the knee must be located.

Worked example — pressure rating of a PE100 water main. A PE100 pipe has MRS = 10 MPa. Water-supply codes apply an overall service coefficient $C = 1.25$, giving a design stress $\sigma_s = 10/1.25 = 8.0$ MPa. The hoop stress in a thin-walled pipe of mean diameter $D - e$ and wall e under internal pressure p is $\sigma = p(D - e)/(2e)$; written with the standard dimension ratio SDR $= D/e$, $p = 2\sigma_s/(\text{SDR} - 1)$. For SDR 11 (a 110 mm pipe with 10 mm wall), $p = 2 \times 8.0/10 = 1.6$ MPa = 16 bar: the

pipe is rated PN16. The same pipe in PE80 (MRS 8 MPa, $\sigma_s = 6.4$ MPa) is rated PN12.5. For gas distribution, the coefficient is at least 2.0, so PE100 SDR 11 is limited to $2 \times 5.0/10 = 1.0$ MPa, or 10 bar.

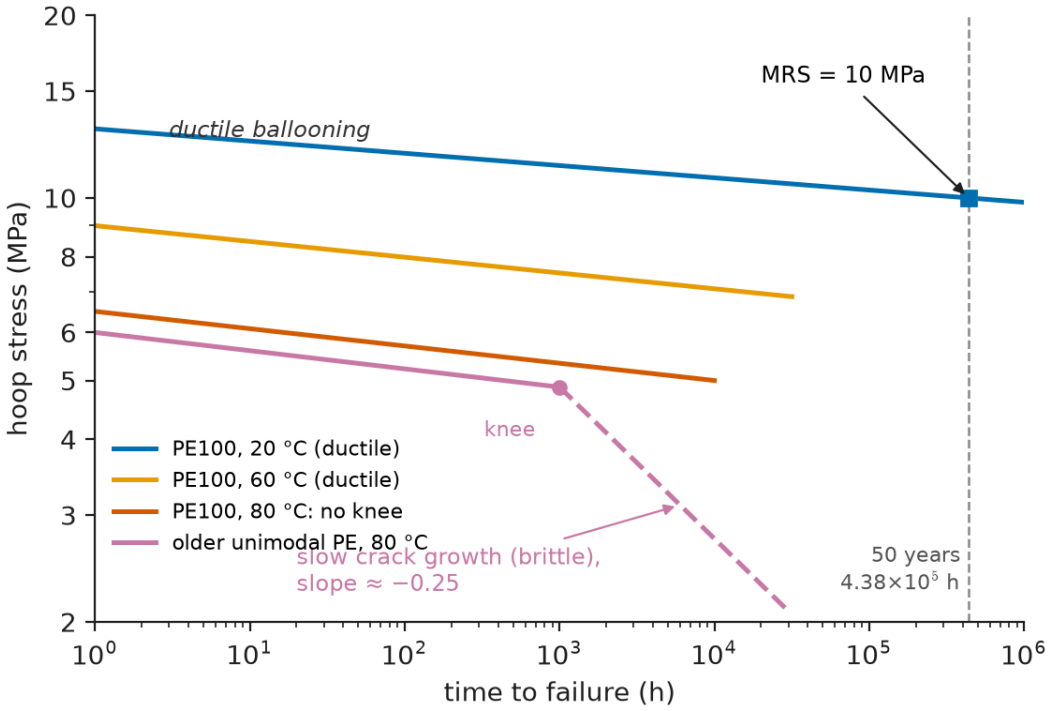

Figure 21.3 — Stress-regression curves for polyethylene pressure pipe: hoop stress versus time to failure at 20, 60 and 80 °C.

21.3 Yielding, crazing and environmental stress cracking

Yielding and cold drawing

Pull a strip of polycarbonate or polypropylene in tension at room temperature and its stress–strain curve rises to a **yield point**, falls a little, and a **neck** forms. Unlike a metal neck, which thins until fracture, the polymer neck stabilizes and propagates along the specimen at almost constant load, converting unoriented material into a highly oriented one with a draw ratio of 2–5 (Figure 21.4). This is **cold drawing**. It works because the aligned chains in the neck strain-harden strongly, so further deformation is easier in the undrawn shoulders; fibre drawing, strapping tape and PET stretch-blow moulding exploit it.

Polymer yielding differs from metal yielding in three ways that designers must account for.

1. **Rate and temperature dependence.** Yield in glassy polymers is thermally activated segmental motion, and the **Eyring** model, $\dot{\varepsilon} = \dot{\varepsilon}_0 \exp[-(\Delta H - \sigma V)/kT]$, predicts that yield stress rises linearly with the logarithm of strain rate and falls linearly with temperature. For polycarbonate, the yield stress rises by a few MPa per decade of strain rate and falls from about 60 MPa at 23 °C to near zero at T_g.
2. **Pressure dependence.** Because yield needs free volume, hydrostatic pressure raises the yield stress, and polymers yield at lower stress in tension than in compression (typically 10–30 % lower). The pressure-modified von Mises or Drucker–Prager criteria describe this.
3. **Strain softening.** After yield, glassy polymers soften by 10–20 % before strain hardening sets in, which localizes the deformation into shear bands; aged glasses soften more, which is one reason why physical aging (Chapter 20) embrittles them.

> **Status of the principle: Eyring model of polymer yield**
> *Approximate*. Treats yield as a stress-biased activated process with a single activation volume (typically several nm³); it fits rate and temperature dependence over a few decades, but secondary relaxations add additional processes at low temperature or high rate.

Crazing

In tension, glassy polymers such as polystyrene, PMMA, SAN and (in aggressive conditions) polycarbonate can fail by **crazing** before they yield. A craze looks like a crack — a thin, planar, light-reflecting feature perpendicular to the maximum principal stress — but it is bridged by fibrils of highly drawn polymer 5–30 nm in diameter, separated by voids, with a void fraction of 50–70 %. Crazes nucleate at stress concentrations where a dilatational (negative pressure) stress state exists, grow by drawing new polymer into the fibrils at the craze tip, and turn into true cracks when the fibrils break. The craze stress is typically 20–40 MPa for polystyrene, below its yield stress of about 50 MPa, so PS crazes and fractures in a brittle way at a few percent strain.

Whether a glassy polymer crazes or shear-yields depends on its entanglement density. Polymers with low M_e (polycarbonate, $M_e \approx 1{,}500$) have short, well-anchored strands that resist fibril formation and yield in shear bands instead; those with high M_e (polystyrene, $M_e \approx 15{,}000$) craze. This is the molecular reason why polycarbonate is used for safety glasses and riot shields while polystyrene is used for disposable cutlery.

Rubber toughening

A craze absorbs energy as it forms. The trick in HIPS and ABS is to make many small crazes rather than one large one: rubber particles 0.5–5 µm across, grafted to the matrix (Chapter 19), concentrate stress at their equators and nucleate crazes, and also stop crazes from turning into cracks. In tougher matrices (PC/ABS, toughened nylon, toughened epoxy), rubber particles cavitate and relieve hydrostatic stress, which lets the matrix shear-yield. Impact strength increases five- to tenfold at the cost of some stiffness and transparency. The same principle, sometimes with core–shell particles or thermoplastic inclusions, toughens the epoxy matrices of aerospace composites (Chapter 23).

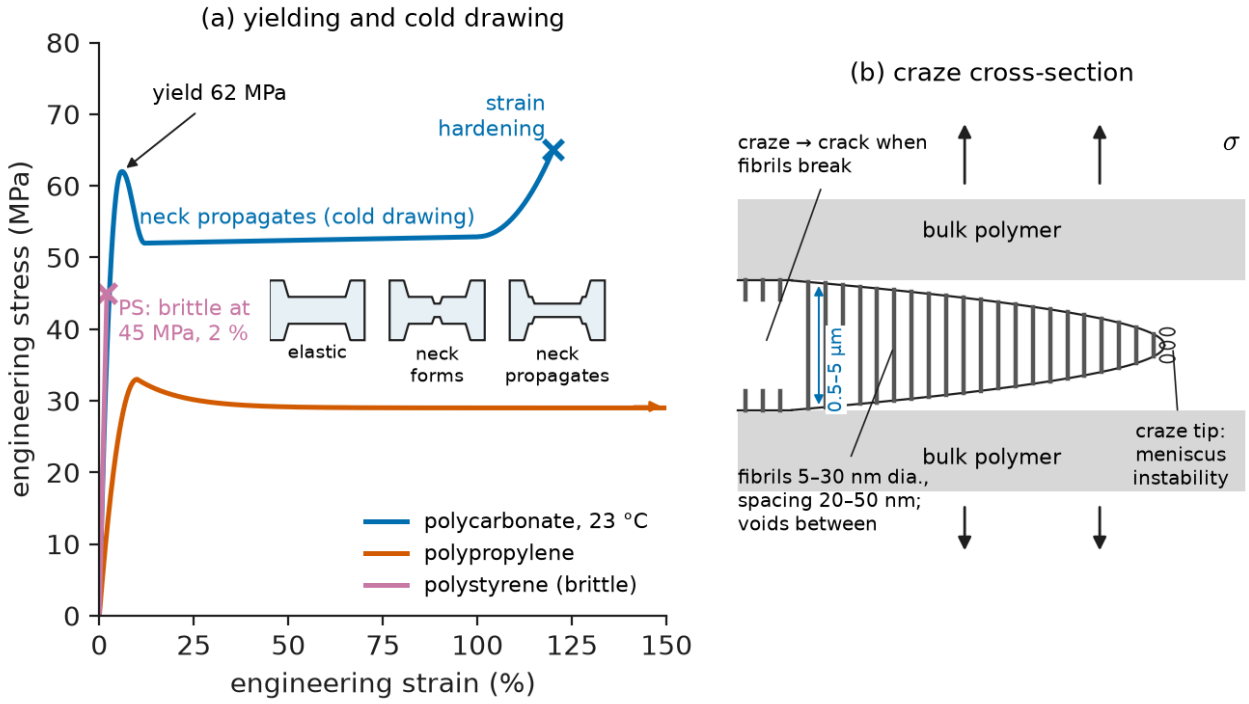

Figure 21.4 — Polymer yielding and crazing: engineering stress–strain curves with necking and cold drawing, and the internal structure of a craze.

Environmental stress cracking

Environmental stress cracking (ESC) is the failure of a polymer under a tensile stress well below its short-term strength, accelerated by contact with a liquid or vapour that does not attack it chemically. It is the single largest cause of plastic part failures in service, with commonly cited estimates from failure-analysis laboratories putting it at about a quarter of all plastic part failures. The mechanism is physical: the agent absorbs preferentially into the highly strained material at a craze tip or into the amorphous regions between crystals, plasticizes it locally, lowers the craze stress or the disentanglement resistance of tie molecules, and accelerates crack growth.

- **Glassy amorphous polymers** (PC, PMMA, PS, ABS, PSU) are attacked by organic liquids with solubility parameters near their own (Chapter 20): alcohols and esters on PC, alcohols on PMMA, oils and fats on PS and ABS. Medical devices

are especially exposed because of aggressive disinfectants, which is the opening case.
- **Semicrystalline polyolefins** are cracked by surfactants and detergents (the standard Bell test for PE uses a nonylphenol ethoxylate solution) and by oils. Here the agent accelerates slow crack growth by easing tie-molecule disentanglement, so resistance improves with higher molecular weight, more short-chain branches and bimodal distributions.

Design rules are simple and often ignored: keep sustained tensile strain below the agent-specific critical strain (often 0.3–1 % for amorphous polymers in aggressive media), avoid moulded-in stress (higher mould temperature, annealing), and avoid sharp corners at bosses and snap fits. The polycarbonate housings in the opening case were redesigned with thicker bosses, a hotter mould and a disinfectant restriction; a PC/polyester blend, whose crystalline polyester phase resists the agents, was used for the next generation.

21.4 Processing

Thermoplastic processes melt the polymer, shape it and cool it while holding the shape; orientation, crystallinity, residual stress and weld lines are set during shaping and cooling, so process conditions are material properties in disguise.

Extrusion

The **single-screw extruder** is the universal melting and pumping machine of the industry. Pellets fed through a hopper are conveyed along the barrel by a rotating screw with three zones: a **feed zone** (deep flights, solid conveying by friction against the barrel), a **compression (transition) zone** (decreasing flight depth, melting mainly by viscous dissipation, which supplies 70–90 % of the energy) and a **metering zone** (shallow flights, pumping melt at constant rate against the die pressure). Screws are 25–250 mm in diameter with length-to-diameter ratios of 24–36. Downstream, a die shapes the melt into pipe, window profile, sheet, cable insulation or the parisons and films of the processes below, and must be designed for uniform exit velocity, allowing for die swell and shear thinning (Section 21.1). Co-rotating twin-screw extruders mix better and are used for compounding (Section 21.5).

Injection moulding

Injection moulding produces most discrete plastic parts, from car bumpers and dashboards to syringes, LEGO bricks and connector housings, at cycle times of a few seconds to a minute. A reciprocating screw plasticizes a shot of melt, then moves forward as a ram to inject it at 50–200 MPa into a closed, temperature-controlled steel mould. The cycle has four stages: **filling** (1 s or less for thin walls), **packing and holding** (pressure maintained to add melt as the part shrinks), **cooling** (the longest stage) and **ejection**. Each stage leaves a signature in the part.

- **Orientation.** Fountain flow at the melt front deposits stretched material on the cold mould wall, where it freezes before it can relax (the WLF calculation of Chapter 20). The part has an oriented **skin** and a less oriented **core**; in fibre-filled grades, fibres align along the flow in the skin and across it in the core, making the part anisotropic (Chapter 22).
- **Shrinkage and warpage.** Semicrystalline polymers shrink 1.5–3 % (PP, POM, PA) on cooling and crystallizing, amorphous ones 0.4–0.8 % (PC, ABS, PS). Differential shrinkage between thick and thin sections, or between flow and cross-flow directions in glass-filled parts, causes **warpage**; uneven cooling causes residual stress and sink marks.
- **Weld lines.** Where two flow fronts meet, the chains have little time to interdiffuse across the interface before freezing. Weld lines can have half the strength of the bulk, much less in glass-filled or liquid-crystalline polymers (Chapter 20).

Worked example — cooling time of an injection-moulded part. The cooling time is controlled by conduction through the wall. Treating the part as a slab of thickness h cooled from both faces, with the first term of the series solution for its centre-line temperature, the time to reach ejection temperature T_e is

$$t_c = \frac{h^2}{\pi^2 \alpha} \ln\!\left(\frac{8}{\pi^2} \, \frac{T_m - T_w}{T_e - T_w} \right)$$

where α is the thermal diffusivity, T_m the melt temperature and T_w the mould temperature. For polypropylene, $\alpha \approx 1.1 \times 10^{-7}$ m²/s, $T_m = 230$ °C, $T_w = 40$ °C and $T_e = 90$ °C. For a 2 mm wall, $h^2/(\pi^2\alpha) = (2 \times 10^{-3})^2 / (9.87 \times 1.1 \times 10^{-7}) = 3.68$ s and the logarithm is $\ln(0.811 \times 190/50) = \ln 3.08 = 1.13$, so $t_c = 4.1$ s. A 3 mm wall needs 9.3 s and a 4 mm wall 16.6 s: cooling time grows with the square of thickness. This is why plastic parts have uniform walls of 1–3 mm with ribs instead of thick sections, and why the low thermal diffusivity of polymers (Chapter 24) dominates moulding economics.

> **Status of the principle: One-term slab cooling-time estimate**
> *Approximate.* Derived from one-dimensional transient conduction with constant properties and perfect contact with the mould; it ignores latent heat of crystallization, temperature-dependent diffusivity and the gap that opens as the part shrinks, and is accurate to roughly 20–30 % for thin walls.

Variants include gas-assisted, co-injection, overmoulding, insert and reaction injection moulding (RIM, in which polyurethane monomers react in the mould).

Blow moulding

Hollow articles are made by inflating a tube or preform inside a mould. **Extrusion blow moulding** extrudes a tubular **parison** of HDPE, PP or PVC downward between open mould halves, closes the mould, pinches off the bottom and blows air inside. It makes milk bottles, drums and multilayer fuel tanks, and needs high melt strength so the parison does not sag, favouring high-M_w, broad-distribution HDPE. **Injection stretch blow moulding** reheats an injection-moulded PET **preform** just above T_g (95–110 °C) and stretches it about threefold axially and radially. Biaxial orientation and small strain-induced crystals keep the bottle clear while raising strength and CO_2 barrier enough for carbonated drinks at 4–5 bar.

Thermoforming

Thermoforming heats an extruded sheet until it is soft but not liquid — above T_g for amorphous PS, ABS, PET and PMMA, just below T_m for PP — and draws it over or into a mould by vacuum, pressure or a mechanical plug. Food trays, blister packs, refrigerator liners and aircraft interior panels are made this way. Wall thickness falls roughly in proportion to the local areal draw ratio, so corners are thinnest. The material needs a wide rubbery temperature window, easy for amorphous polymers and hard for low-melt-strength semicrystalline ones.

Fibre spinning

Fibres are made by extruding polymer through a **spinneret** with many fine holes and drawing the filaments.

- **Melt spinning** (PET, nylon, PP): the melt is drawn down while cooling in air; at take-up speeds of 3,000–6,000 m/min PET filaments acquire substantial orientation, and further drawing below T_m completes it.
- **Wet and dry spinning** from solution (acrylic, viscose, spandex). Wet-spun polyacrylonitrile is the precursor of most carbon fibre (Chapter 22).
- **Dry-jet wet spinning** of lyotropic liquid-crystalline solutions (Kevlar, Chapter 20) and **gel spinning** of UHMWPE, drawn to ratios above 30 (Dyneema, modulus 100–150 GPa).
- **Electrospinning** in a strong electric field gives nanofibres 50–500 nm across for filters, battery separators and tissue scaffolds (Chapter 39).

Orientation sets fibre properties: drawn PET fibre has a modulus of 10–20 GPa against 2–3 GPa for moulded PET.

Film blowing

Blown film is the dominant process for polyethylene packaging film, agricultural film and bags. Melt is extruded through an annular die with a gap of 0.8–2.5 mm, inflated with internal air into a bubble, cooled by an air ring and pulled upward by nip rolls at a speed higher than the die exit speed. The **blow-up ratio** (BUR, bubble diameter over die diameter, typically 2–4) stretches the film in the transverse direction; the **draw-down ratio** stretches it in the machine direction. For a 1.2 mm die gap, a BUR of 2.5 and a final film thickness of 25 μm, the draw-down ratio is $1.2/(0.025 \times 2.5) = 19$, so the film is stretched much more in the machine direction, and its tear strength is anisotropic unless the BUR is increased. The bubble is stabilized by the extensional strain hardening of LDPE (Section 21.1). Co-extrusion produces multilayer films combining PE for sealing, PA for toughness and EVOH for oxygen barrier; flat-die **biaxially oriented** films (BOPP, BOPET) serve where optical quality and dimensional stability matter, as in capacitor dielectrics and solar back sheets.

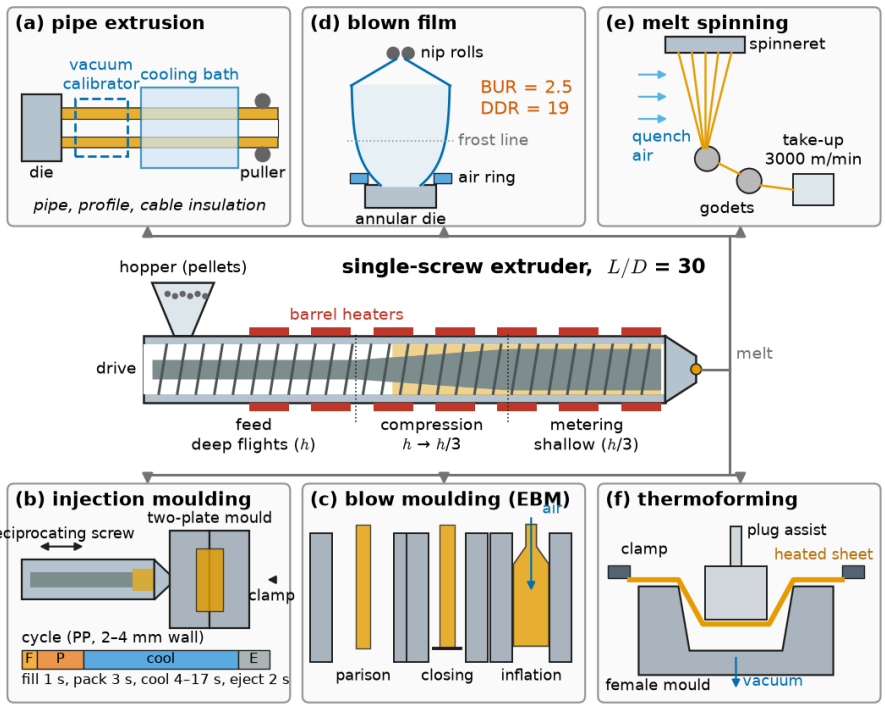

Figure 21.5 — The main thermoplastic processing routes built around the single-screw extruder.

21.5 Additives

A commercial plastic is rarely a pure polymer. **Compounding** in twin-screw extruders blends the base resin with additives that adjust processing, properties and durability. Additive loadings range from parts per million (catalyst deactivators) to more than half the mass (flame-retardant fillers, plasticizers).

Plasticizers

Plasticizers are low-volatility liquids, miscible with the polymer, that lower its T_g by adding free volume and separating chains (Chapter 20). Over 80 % of plasticizer production goes into PVC, which can be turned from a rigid pipe material (T_g 82 °C) into flexible cable insulation, flooring, medical tubing and blood bags. The traditional plasticizer, di(2-ethylhexyl) phthalate (DEHP), has been restricted in toys and medical devices because of reproductive toxicity and replaced by DINCH, DOTP and citrates. Plasticizers migrate, evaporate and are extracted by oils, so flexible PVC stiffens with age, and plasticizer migrating into adjacent PS or PC parts can stress-crack them.

Worked example — plasticizing PVC for a cable jacket. Taking $T_g = 82$ °C (355.2 K) for PVC and about −85 °C (188.2 K) for DEHP, the Fox equation of Chapter 20 gives, for 30 wt % plasticizer, $1/T_g = 0.70/355.2 + 0.30/188.2 = 3.565 \times 10^{-3}$ K^{-1}, so $T_g = 280.5$ K ≈ 7 °C. At 40 wt % (about 67 phr, parts per hundred of resin), $T_g \approx -11$ °C. A cable jacket that must remain flexible at −15 °C therefore needs 40–45 wt % plasticizer, or a low-temperature plasticizer such as an adipate with a lower T_g. The measured transition of plasticized PVC is 30–50 K wide, so the Fox estimate only locates its middle; cable standards therefore specify cold-bend and cold-impact tests.

Fillers and reinforcements

Inert fillers (calcium carbonate, talc, wood flour) lower cost, raise stiffness and reduce shrinkage; talc-filled PP is the standard car-interior material. Short glass fibres (10–50 %) raise stiffness and strength several-fold and reduce creep, at the cost of anisotropy and weld-line weakness (Chapter 22); glass-filled PA66 engine covers and connectors are the archetype. Carbon black and carbon nanotubes make antistatic and EMI-shielding grades, and exfoliated clays and graphene improve barrier at low loading if well dispersed (Chapter 38).

Stabilizers

Every polymer begins to degrade during processing (Section 21.6). **Primary antioxidants** (hindered phenols) trap the peroxy radicals that propagate oxidation; **secondary antioxidants** (phosphites, thioesters) decompose hydroperoxides before they split into new radicals. **UV absorbers** (benzotriazoles, benzophenones) convert UV to heat but do little in thin films; **hindered-amine light stabilizers** (HALS) trap radicals in a regenerative cycle and are the most effective light stabilizers for polyolefins; and 2–2.5 % **carbon black** is the cheapest UV protection of all, which is why outdoor PE pipes and cables are black. PVC needs **heat stabilizers** (calcium–zinc soaps, organotins; formerly lead) to neutralize the HCl released by dehydrochlorination, and cable insulation needs **metal deactivators** against copper-catalysed oxidation.

Flame retardants

Polymers are hydrocarbons and burn. Flammability is described by the **limiting oxygen index** (LOI, the minimum oxygen concentration that sustains burning, ISO 4589) and by standard tests such as UL 94 (V-0, V-1, V-2, HB rankings) and cone calorimetry (peak heat-release rate).

Polymer	LOI (%, typical)
Polyethylene, polypropylene	17–18
Polystyrene, PMMA	17–18
PA66	23–24
Polycarbonate	25–27
PEEK	35
Rigid PVC	45–49
PEI (Ultem)	47
PTFE	95

Air contains 21 % oxygen, so polymers with LOI below 21 burn readily once ignited. Flame retardants act in the gas phase (halogenated compounds, often with antimony trioxide, which release halogen radicals that quench the combustion chain), in the condensed phase (phosphorus compounds and intumescent systems that promote a protective char), or by cooling and dilution (aluminium trihydroxide, ATH, which decomposes endothermically at about 200 °C releasing water, and magnesium hydroxide at about 330 °C). Mineral retardants need 50–65 wt % loading, which costs mechanical properties, but they are halogen-free and low in smoke, and dominate cables for tunnels and ships. Restrictions on persistent brominated retardants are pushing phosphorus systems and inherently flame-resistant polymers (PEI, PPS, PEEK).

Nucleating agents, impact modifiers, lubricants, blowing agents, antistatics and pigments complete a typical formulation, and each must be compatible with the others: acidic flame retardants deactivate HALS, and uncoated titanium dioxide catalyses photo-oxidation.

21.6 Degradation

Polymers degrade by **chain scission** (which lowers M_n, embrittles and can increase melt flow), by **crosslinking** (which raises stiffness and lowers elongation) and by changes in side groups (yellowing, loss of transparency, release of small molecules). The three most common agents are ultraviolet light, heat with oxygen, and water.

Thermo-oxidative degradation

Most polymers oxidize by a radical chain mechanism, **auto-oxidation**:

1. **Initiation**: a polymer radical $\mathrm{P}^\bullet$ forms from heat, shear, light or catalyst residues.
2. **Propagation**: $\mathrm{P}^\bullet + \mathrm{O_2} \rightarrow \mathrm{POO}^\bullet$; $\mathrm{POO}^\bullet + \mathrm{PH} \rightarrow \mathrm{POOH} + \mathrm{P}^\bullet$.
3. **Chain branching**: hydroperoxides decompose, $\mathrm{POOH} \rightarrow \mathrm{PO}^\bullet + {}^\bullet\mathrm{OH}$, each creating two new radicals.
4. **Termination**: radicals combine.

Because each hydroperoxide decomposition creates two new radicals, oxidation is autocatalytic: an induction period during which antioxidants are consumed is followed by rapid property loss. The **oxidation induction time** (OIT) measured by DSC in oxygen is the standard quality test for stabilized PE pipe and cable. Polypropylene, with a tertiary hydrogen on every second carbon, is much more susceptible than polyethylene: unstabilized PP embrittles within days at 150 °C. Chain scission predominates in PP; crosslinking competes in PE.

Service life at elevated temperature is estimated by accelerated aging, assuming Arrhenius kinetics. A property (tensile strength, elongation at break) is monitored at several temperatures until it falls to a failure criterion, often 50 % of its initial value, and the times are extrapolated to service temperature. The **relative thermal index** (UL 746B) of a plastic is the temperature at which it retains half its properties after about 100,000 h; it is the number cited on electrical-insulation and connector datasheets.

> **Status of the principle: Arrhenius extrapolation of polymer aging**
> *Empirical.* Assumes a single rate-controlling mechanism with constant activation energy across test and service temperatures; it fails when the mechanism changes (antioxidant loss by evaporation at high temperature, diffusion-limited oxidation in thick sections, crossing T_g or a melting range), and errors in E_a translate exponentially into errors in predicted life.

Worked example — extrapolating a heat-aging test. An under-bonnet PA66 connector housing retains 50 % of its elongation after 1,000 h at 120 °C. If the degradation has activation energy 100 kJ/mol, the acceleration factor relative to 70 °C is $\exp[(E_a/R)(1/343.15 - 1/393.15)] = \exp[12{,}028 \times 3.71 \times 10^{-4}] = \exp(4.46) = 86$, so the predicted life at 70 °C is about 86,000 h, or 9.9 years. If the true activation energy is 80 kJ/mol, the factor is 35 and the life only about 4 years. This sensitivity is why standards require at least three test temperatures, a straight Arrhenius line through them, and test temperatures below transitions such as the melting range.

Photo-oxidation by ultraviolet light

Sunlight reaching the ground contains UV down to about 295 nm, energetic enough to break C–C and C–H bonds when absorbed. Pure polyolefins absorb little, but carbonyl groups formed during processing, catalyst residues and hydroperoxides do, initiating auto-oxidation at the surface. Aromatic polymers (PC, PET, PS, aramids) absorb directly and yellow. The damage is concentrated in a surface layer tens to hundreds of micrometres thick, which embrittles and microcracks; the cracks then run into the interior under load. Chalking PVC window frames, yellowed polycarbonate headlamp lenses and crumbling polypropylene garden furniture are the familiar results. Accelerated xenon-arc tests (ISO 4892) correlate only roughly with outdoor exposure (Chapter 44).

Hydrolytic degradation

Condensation polymers — polyesters, polyamides, polycarbonates and polyurethanes — can be cut by water, the reverse of the step-growth reaction of Chapter 19. At room temperature hydrolysis is slow for most engineering grades, but it accelerates strongly with temperature and with acid or base catalysis. PET must be dried to below about 0.005–0.02 % moisture before melt processing; polyester-urethane seals and shoe soles crumble after years in humid climates; polycarbonate loses toughness in repeated steam sterilization. Hydrolysis is the designed degradation route of resorbable PLA, PGA and PLGA implants (Chapter 39): water penetrates the amorphous regions, ester bonds are cut, M_n falls roughly exponentially with time and mass is lost only once fragments become small enough to dissolve, so strength disappears long before the device disappears.

Other environmental attacks

Ozone attacks the double bonds of diene rubbers (natural rubber, SBR, NBR), producing surface cracks perpendicular to the strain in stretched parts; antiozonants (para-phenylenediamines such as 6PPD) bloom to the surface and react with ozone first. Its transformation product 6PPD-quinone, washed from tyre wear particles into urban streams, was identified in 2020 as the cause of coho salmon die-offs. Ionizing radiation (gamma sterilization, nuclear environments, space) causes scission and crosslinking (Chapter 41); PTFE and PP are particularly sensitive, PEEK and polyimides resistant.

21.7 Elastomers and their vulcanization

From sticky gum to rubber

Raw natural rubber, cis-1,4-polyisoprene from the latex of *Hevea brasiliensis*, is a viscous, sticky material that flows under its own weight, softens in summer and stiffens in winter. In 1839 Charles Goodyear found that heating it with sulfur converts it into a resilient elastic solid, insensitive to temperature over a wide range. **Vulcanization** creates a network: sulfur bridges join the chains at the allylic carbons next to their double bonds, every 50–100 repeat units or so, giving M_c of a few thousand g/mol. The network turns flow into recoverable elasticity (Chapter 20) and stops chains slipping past each other.

Cure systems

- **Accelerated sulfur vulcanization** is used for diene rubbers (NR, SBR, BR, NBR, EPDM through its diene termonomer). A typical tyre recipe contains, per 100 parts of rubber (phr): 1–3 phr sulfur, 0.5–2 phr accelerator (sulfenamides such as CBS, thiazoles, thiurams), 3–5 phr zinc oxide and 1–2 phr stearic acid as activators, and 30–80 phr carbon black or silica. Cure takes minutes at 140–180 °C. The ratio of accelerator to sulfur controls crosslink structure: **conventional** systems (high sulfur, low accelerator) give long **polysulfidic** crosslinks with good fatigue life and tear strength but poor heat resistance; **efficient vulcanization** (EV) systems (low sulfur, high accelerator) give mono- and disulfidic crosslinks that resist heat aging and compression set.
- **Peroxide curing** creates thermally stable C–C crosslinks in saturated rubbers (EPM, silicone) and in polyethylene cable insulation (XLPE), at some cost in tear strength.
- **Other systems**: metal oxides for polychloroprene, bisphenols for fluoroelastomers, platinum-catalysed addition curing for medical silicones, and electron-beam crosslinking for heat-shrink tubing and UHMWPE bearings.

The cure curve

Vulcanization is monitored with a moving-die rheometer (ISO 6502), which oscillates a die in a sample held at cure temperature and records the torque (Figure 21.6). The torque first falls to a minimum M_L as the compound softens, stays low during the **scorch time** t_{s2} (the processing safety window during which the compound can still be shaped), rises as crosslinks form and levels off at M_H. The optimum cure time t_{90} is the time to reach 90 % of the torque rise. Some NR compounds show **reversion**, a fall in torque after the maximum, as polysulfidic crosslinks break and rearrange at high temperature; truck tyres cured too hot or too long revert in their thick sections. The difference $M_H - M_L$ is proportional to crosslink density.

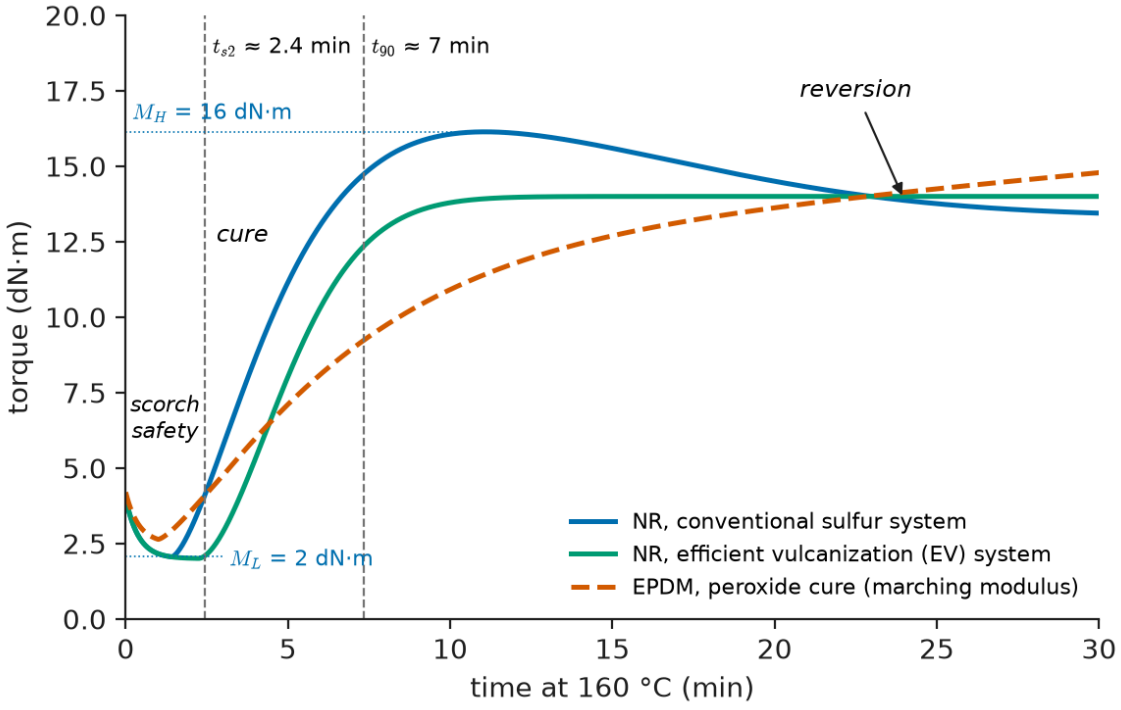

Figure 21.6 — Moving-die rheometer cure curves for a sulfur-cured natural-rubber compound at 160 °C.

Measuring crosslink density by swelling

A vulcanizate placed in a good solvent swells until the osmotic drive of mixing (Chapter 20) is balanced by the retractive force of the network. The **Flory–Rehner** equation expresses the balance:

$$\nu = -\frac{\ln(1-\phi_r) + \phi_r + \chi\phi_r^2}{V_1\left(\phi_r^{1/3} - \phi_r/2\right)}$$

where ϕ_r is the volume fraction of rubber in the swollen gel, V_1 the molar volume of the solvent, χ the polymer–solvent interaction parameter and ν the crosslink density (mol of network chains per unit volume of dry rubber).

> **Status of the principle: Flory–Rehner swelling equation**
> *Approximate.* Combines the Flory–Huggins mixing free energy with affine Gaussian network elasticity; it is the standard practical method for crosslink density but inherits the uncertainty in χ (which is concentration-dependent) and ignores filler–rubber interactions, so it is used comparatively for filled compounds.

Worked example — crosslink density of a natural-rubber seal. A specimen of unfilled NR vulcanizate swells in toluene at 25 °C until the rubber occupies 25 % of the swollen volume ($\phi_r = 0.25$). With $\chi = 0.39$ and $V_1 = 106.3$ cm³/mol $= 1.063 \times 10^{-4}$ m³/mol, the numerator is $-[\ln 0.75 + 0.25 + 0.39 \times 0.0625] = -[-0.2877 + 0.25 + 0.0244] = 0.0133$, and the denominator is $1.063 \times 10^{-4} \times (0.630 - 0.125) = 5.37 \times 10^{-5}$ m³/mol. Thus $\nu = 248$ mol/m³ and $M_c = \rho/\nu = 920/248 = 3.7$ kg/mol $= 3{,}700$ g/mol. The rubber elasticity theory of Chapter 20 then predicts $G = \nu RT = 248 \times 8.314 \times 298 = 0.61$ MPa — the right magnitude for a soft seal compound. Swelling in the service fluid itself gives the volume change that seal specifications limit (Chapter 20).

Reinforcement and the behaviour of filled rubbers

Unfilled vulcanizates of non-crystallizing rubbers such as SBR have tensile strengths of only 2–3 MPa; 30–60 phr of fine carbon black (particles 20–50 nm) raises this tenfold, with tear and abrasion resistance — which is why tyres are black. Silane-coupled silica in "green" tyre treads lowers hysteresis at the 10–100 Hz of rolling while keeping it high at the kHz frequencies of road texture that govern wet grip, a direct use of the time–temperature behaviour of Chapter 20. Filled rubbers show the **Mullins effect** (stress softening on the first stretch, as filler–rubber bonds and chains between particles break and rearrange) and the **Payne effect** (fall of dynamic modulus with oscillation amplitude, as the filler network breaks up).

Elastomer	T_g (°C, typical)	Service range (°C, typical)	Key strengths	Typical uses
Natural rubber (NR)	−70	−50 to 80	Strength, fatigue, strain crystallization	Truck and aircraft tyres, bridge bearings, engine mounts
SBR	−50 to −60	−40 to 100	Cost, abrasion	Car tyre treads, conveyor belts
Butyl (IIR)	−70	−40 to 120	Gas impermeability, damping	Tyre inner liners, vial stoppers
EPDM	−55	−40 to 150	Ozone and weather resistance	Door seals, roofing, radiator hoses
Nitrile (NBR)	−40 to −10 (with ACN content)	−30 to 120	Oil and fuel resistance	O-rings, fuel hoses, gloves
Chloroprene (CR)	−45	−40 to 110	Moderate oil and weather, flame	Wetsuits, cable jackets, bellows
Silicone (VMQ)	−125	−60 to 230	Temperature range, biocompatibility	Medical tubing, oven seals, implants
Fluoroelastomer (FKM)	−20 to −15	−20 to 230	Fuels, chemicals, heat	Aerospace and automotive fuel seals

Thermoplastic elastomers (SBS, TPU, and TPV, made by dynamically vulcanizing EPDM within PP) use physical crosslinks (Chapter 20), can be injection-moulded and recycled, and have replaced vulcanized rubber in many grips and automotive boots,

though they creep more when hot.

21.8 Recycling and biodegradable polymers

The scale of the problem

World plastics production exceeds 400 million tonnes per year; about 40 % is packaging with a service life of months. Historically under 10 % of plastic waste has been recycled. Materials science decides which end-of-life routes are technically possible; Chapter 59 weighs them.

Mechanical recycling

Mechanical recycling collects, sorts, washes, grinds and re-melts thermoplastics into new products. Its success depends on three materials issues already met in this part.

1. **Sorting.** Polymers are identified by resin codes (1 PET, 2 HDPE, 3 PVC, 4 LDPE, 5 PP, 6 PS, 7 other) and sorted by near-infrared spectroscopy, which cannot see carbon-black-pigmented plastics. Because most polymer pairs are immiscible (Chapter 20), mixed streams give weak, brittle blends, and a few percent of PVC in PET releases HCl that degrades the whole batch.
2. **Molecular-weight loss.** Each melt cycle causes chain scission by shear, oxidation and hydrolysis (Chapter 19). Recycled PET is dried and solid-state polymerized back to bottle grade; polyolefins are restabilized with antioxidants.
3. **Contamination.** Labels, adhesives, multilayer barrier films and absorbed chemicals limit the use of recyclate, which is why single-polymer packaging is now a design requirement (Chapter 59).

Chemical and feedstock recycling

Chemical recycling breaks polymers back into monomers or feedstock. Step-growth polymers are well suited: PET can be depolymerized by glycolysis, methanolysis or enzymes and repolymerized to virgin quality, nylon-6 returns to caprolactam, and PMMA unzips to its monomer on heating. Polyolefins do not unzip; **pyrolysis** converts them to a hydrocarbon mixture for cracker feedstock, with debated energy costs and yields. **Energy recovery** by incineration recovers the high calorific value of polyolefins (about 43 MJ/kg, similar to fuel oil) but releases fossil CO_2.

Thermosets and elastomers cannot be remelted. Ground tyre rubber is used in asphalt, playground surfaces and as filler; devulcanization selectively breaks sulfur crosslinks; epoxy composites are recycled by pyrolysis or solvolysis to recover fibres (Chapter 23). Vitrimers and other covalent adaptable networks (Chapter 19) are being developed so that thermosets can be reprocessed like thermoplastics.

Biodegradable and bio-based polymers

Two different properties are often confused. **Bio-based** means made from renewable biological feedstock; **biodegradable** means broken down by microorganisms to CO_2, water and biomass under specified conditions. Bio-based polyethylene from sugar-cane ethanol is chemically identical to fossil PE and not biodegradable; PBAT and PCL are fossil-based and biodegradable; PLA and PHA are both bio-based and biodegradable.

Polymer	Source	T_g / T_m (°C, typical)	Biodegradation conditions	Uses
PLA	Fermentation of sugars to lactic acid	55–60 / 150–175	Industrial composting (about 58 °C, above T_g); very slow in soil or sea	Packaging, cups, 3D-printing filament, resorbable implants
PHA (PHB, PHBV)	Bacterial fermentation	0–5 / 160–175 (PHB)	Composting, soil, seawater	Packaging, agricultural films, medical
PBAT	Fossil	–30 / 110–120	Industrial and home composting	Compostable bags, mulch film
PCL	Fossil	–60 / 60	Composting, slow in body	Long-term implants, scaffolds
PGA, PLGA	Synthetic	35–45 (PLGA) / 225 (PGA)	Hydrolysis in the body (weeks to months)	Sutures, drug-delivery microspheres

Biodegradation usually starts with abiotic hydrolysis that cuts chains to oligomers small enough for microbes to assimilate, so it depends on the factors of Section 21.6: temperature relative to T_g, crystallinity and thickness. PLA hydrolyses quickly only above its T_g of about 55–60 °C, which is why it certifies as compostable under industrial composting standards (EN 13432, ASTM D6400: at least 90 % conversion to CO_2 within 6 months and disintegration within 12 weeks at about 58 °C) but persists for years in a garden compost heap or the sea. Compostability is no solution to litter.

> **Boundary.**
> The full life-cycle comparison of recycling routes, biodegradable alternatives and energy recovery, including embodied carbon, belongs to Chapter 59; implant applications of resorbable polymers to Chapter 39.

What remains open

Open question	Where it is answered
What are the viscoelastic constitutive models and time–temperature superposition used here?	Chapter 14
How do fracture mechanics and fatigue apply to the crack growth that follows crazing?	Chapter 14
How do short and continuous fibres stiffen a polymer, and why are moulded fibre-filled parts anisotropic?	Chapter 22
How are thermoset matrices cured and composite parts manufactured?	Chapter 23
Why is the thermal conductivity of polymers low, and how do they expand?	Chapter 24
How do polymer insulation and dielectrics fail electrically?	Chapter 26
How do polymer nanocomposites and conductive fillers work?	Chapter 38
Which polymers are used in implants, and how are resorbable devices designed?	Chapter 39
How does radiation damage polymers in nuclear and space environments?	Chapter 41
How are weathering, aging and polymer failures investigated in practice?	Chapter 44
How are adhesives designed and bonded joints made?	Chapter 47
How are polymers processed by additive manufacturing?	Chapter 48
How are DSC, TGA, DMA, FTIR and SEC used in polymer analysis?	Chapter 52
How do recycling routes compare in life-cycle and circular-economy terms?	Chapter 59

Next: Chapter 22 adds a second phase to the polymer — stiff, strong fibres of glass, carbon or aramid — and asks how fibre and matrix share load in a composite lamina.

Chapter 22 — Composite Micromechanics and Lamina Behaviour

Part V · Polymers and Composites

A wind-turbine blade 80 m long has to bend tens of millions of times in twenty years without cracking, and it has to be stiff enough that its tip never strikes the tower in a gust. A designer who reached for aluminium would find the blade too heavy to lift; a designer who reached for steel would find it too heavy to turn. The blade is instead built from glass and carbon fibres, each thinner than a hair, laid in an epoxy resin in carefully chosen directions: most of them along the span in the spar caps, some at ±45° in the skins to carry torsion, a few across. The same blade, tested in the laboratory, turns out to be several times stiffer along its length than across it, and its strength in one direction has almost nothing to do with its strength in another. Nothing in the metals chapters prepares an engineer for a material whose properties depend so strongly on the direction of load — or for one whose properties are designed fibre by fibre rather than chosen from a handbook.

This chapter builds the first half of the tools needed to design with such materials: how the properties of fibres and matrix combine into the properties of a single layer (micromechanics), and how a single layer responds when loaded along or across its fibres or at any angle in between (lamina mechanics). Chapter 23 stacks these layers into laminates, predicts how they fail and explains how they are made.

> **How do the stiffness and strength of a fibre and a matrix combine into the properties of a composite, and when do simple averages fail?**
> **How does a single fibre-reinforced layer respond to load in any direction, and to changes in temperature and moisture?**

Concept	Engineering destination
Composite classes and architectures	Choosing between particulate, short-fibre, continuous-fibre and laminated designs
Fibres and matrices	Glass for boat hulls and blades, carbon for aircraft and bicycles, aramid for armour, natural fibres for car interiors
Rule of mixtures, Halpin–Tsai	First estimates of ply stiffness from fibre content
Particulate composites, Eshelby	Filled polymers, Al–SiC brake discs, cemented carbides, concrete
Shear lag, critical length	Injection-moulded glass-filled nylon, fibre sizing and interface design
Orthotropic lamina, reduced stiffness Q	Input to laminate analysis in every composite design code
Off-axis lamina, $\bar{Q}$	Angle plies in torsion tubes, blade skins, pressure vessels
Thermal and hygral expansion	Cure residual stress, dimensional stability of satellite structures, swelling in humid service

Roadmap. Why composites and how they are arranged (22.1) → the constituents: fibres and matrices (22.2) → continuous aligned fibres: rule of mixtures and Halpin–Tsai (22.3) → particles and the Eshelby inclusion picture (22.4) → short fibres: shear lag and the critical length (22.5) → the orthotropic lamina in its own axes (22.6) → the lamina loaded off-axis (22.7) → thermal and moisture expansion of a lamina (22.8).

22.1 Why composites: classes and architectures

The case for combining materials

Chapter 1 showed that property space is not filled evenly. Ceramics and fibres are stiff and strong but brittle; polymers are tough and easy to shape but compliant; metals are tough and stiff but dense. A **composite** combines two or more chemically distinct

phases, separated by an interface, so that the combination occupies a region of property space that none of the constituents reaches alone. The phase that is continuous is the **matrix**; the dispersed phase is the **reinforcement**.

The idea is ancient (straw in mud brick, laminated bows of horn and sinew) and it is everywhere in nature: wood is a composite of cellulose fibrils in a lignin–hemicellulose matrix (Chapter 18), bone is a composite of hydroxyapatite platelets in collagen (Chapter 39), and nacre is 95 % aragonite tiles bonded by a few percent of protein. Modern engineering composites run from glass-fibre polyester boats in the 1940s to carbon-fibre-reinforced polymer (CFRP) making up about half the structural weight of the Boeing 787 and Airbus A350.

The decisive advantage of a fibre composite is **specific stiffness and strength** — properties per unit density. A unidirectional carbon/epoxy ply has a modulus along the fibres of about 140–180 GPa at a density of about 1.6 g/cm³, giving $E/\rho \approx 100$ GPa per g/cm³, against about 26 for steel, aluminium and titanium alike. The second advantage, less obvious but equally important, is **tailorability**: the designer chooses how much fibre goes in which direction, so stiffness and strength can be placed where the loads are.

Classes by matrix and reinforcement

Composites are classified first by matrix:

- **Polymer-matrix composites (PMCs)** — by far the largest class by volume and value. Glass-fibre-reinforced polymers (GFRP) dominate tonnage: boats, pipes, tanks, wind-turbine blades, car body panels, printed-circuit boards. CFRP dominates where weight matters most: aircraft, racing cars, bicycles, sporting goods, pressure vessels for hydrogen.
- **Metal-matrix composites (MMCs)** — aluminium, magnesium or titanium reinforced with SiC or Al_2O_3 particles, whiskers or fibres: brake rotors, pistons, electronic baseplates, bicycle frames, fan blades.
- **Ceramic-matrix composites (CMCs)** — SiC fibres in SiC, carbon fibres in carbon or SiC, oxide fibres in oxide matrices: hot-section aero-engine parts, brake discs for aircraft and sports cars, re-entry thermal protection. Here the aim is not stiffness but toughness: the fibres make a brittle ceramic fail gracefully (Section 23.7).

and then by the geometry of the reinforcement.

Architectures

Figure 22.1 shows the main architectures. **Particulate** composites contain roughly equiaxed particles; their properties are nearly isotropic and the reinforcement is modest. **Short-fibre** (discontinuous) composites contain fibres with length-to-diameter ratios of 20–1000, usually partially aligned by flow during moulding. **Continuous-fibre** composites contain fibres that run the full length of the part, either all in one direction (**unidirectional**, UD) or woven, braided or knitted into **fabrics**. Continuous plies are stacked into **laminates**, and laminates can be separated by a light core to form **sandwich** structures (Section 23.5). Three-dimensional woven and stitched preforms add fibres through the thickness to resist delamination.

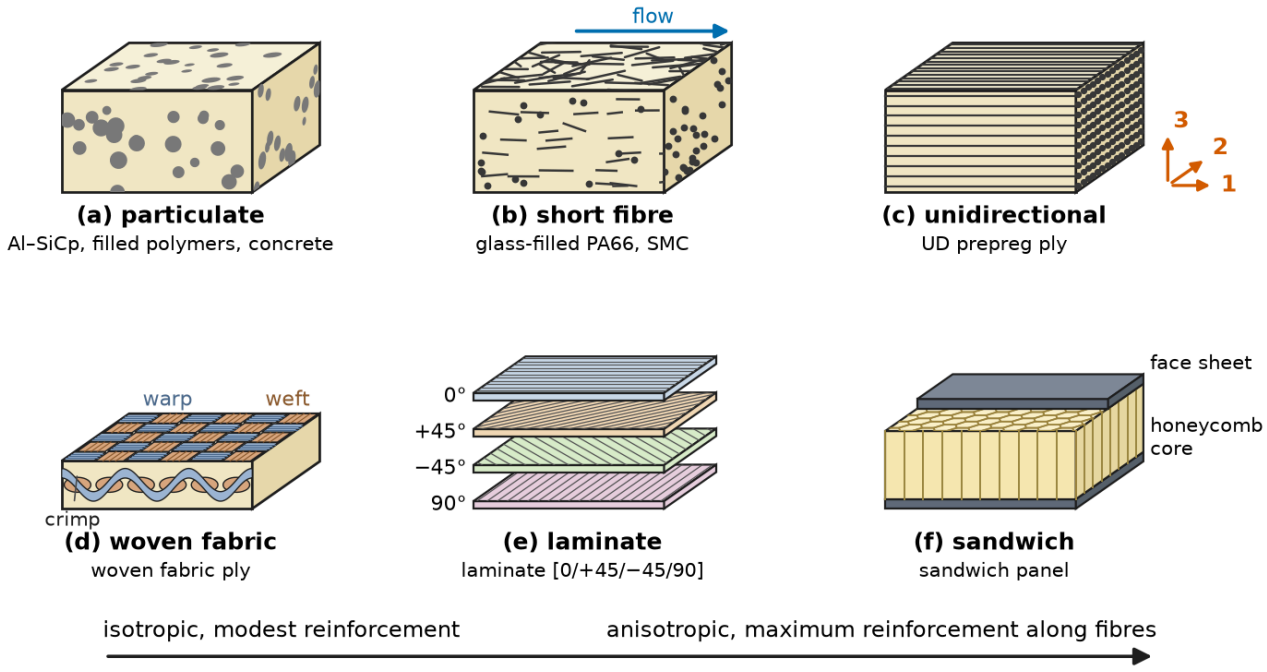

Figure 22.1 — The main composite architectures, from isotropic particulate to strongly anisotropic unidirectional plies, laminates and sandwich panels.

The geometry of the reinforcement determines both the efficiency of reinforcement and the degree of anisotropy. A useful rule of thumb for stiffness in the loading direction: a unidirectional composite loaded along its fibres uses about 100 % of the fibres' stiffness contribution; a cross-ply [0/90] about 50 %; a quasi-isotropic laminate about 37 %; and a random three-dimensional short-fibre composite only about 20 %. These **orientation efficiency factors** (Krenchel factors) reappear in Section 22.5.

22.2 Fibres and matrices

Why fibres are strong

Bulk glass breaks at 50–100 MPa; a freshly drawn glass fibre 10 µm across breaks at 3,000–4,500 MPa. The difference is flaws. Chapter 14 showed that brittle strength is set by the largest crack in the stressed volume and that, by Weibull statistics, a smaller volume is likely to contain a smaller worst flaw. A thin fibre has a tiny volume per unit length and a pristine, fire-polished surface. In carbon and polymer fibres a second effect is added: the strong covalent bonds of graphene sheets or polymer backbones are aligned along the fibre axis by drawing or by heat treatment under tension, so the fibre exploits the stiffest direction of a highly anisotropic structure. Carbon fibre is stiff along its axis (230–900 GPa) but compliant across it (about 10–20 GPa), and aramid and polyethylene fibres are even more anisotropic.

The fibre families

Glass fibres are drawn from melts of calcium aluminoborosilicate (E-glass, the general-purpose grade) or higher-alumina, magnesia-rich compositions (S- and R-glass, about 25 % stiffer and 30 % stronger). They are isotropic, cheap (about 1–3 USD/kg for E-glass roving), electrically insulating and moderately stiff. Their weaknesses are a modest modulus, a relatively high density and sensitivity to stress corrosion by water (static fatigue, Chapter 43).

Carbon fibres are made by stabilizing and carbonizing polyacrylonitrile (PAN) precursor fibres at 1,000–1,500 °C, optionally graphitizing them at up to 3,000 °C, or by spinning and carbonizing mesophase pitch. Low graphitization temperatures give high-strength, standard-modulus fibres (T300, T700: about 230 GPa); higher temperatures give more perfect graphitic order and higher modulus (M55J: 540 GPa; pitch fibres up to about 900 GPa), at the cost of strain to failure. Carbon fibres have a slightly negative axial thermal expansion, are electrically conductive and galvanically noble (they corrode aluminium in contact), and cost about 15–40 USD/kg for standard grades.

Aramid fibres (Kevlar, Twaron) are liquid-crystalline polymer fibres (Chapter 20) of poly(p-phenylene terephthalamide). They are light, tough and excellent in tension and impact, but poor in compression — the chains kink at about 0.3–0.5 % compressive strain — and they absorb moisture and degrade in UV. **Ultra-high-molecular-weight polyethylene** (UHMWPE: Dyneema, Spectra) fibres are lighter still (density below water) with very high specific strength, used in ballistic protection and ropes, but they creep and melt at about 145 °C.

Natural fibres — flax, hemp, jute, sisal, kenaf — have specific stiffnesses close to E-glass at a fraction of its embodied energy and are used in automotive interior panels and increasingly in sporting goods and boat building. Their properties vary with growing season, they absorb water, and they degrade above about 200 °C, which limits the matrices they can be processed with. **Basalt fibres**, melt-spun from volcanic rock, sit between E- and S-glass.

Ceramic fibres — SiC (Nicalon, Tyranno, Hi-Nicalon Type S), alumina and aluminosilicate (Nextel) — retain strength above 1,000 °C and are the reinforcement of CMCs and some MMCs. **Boron fibres**, made by chemical vapour deposition onto a tungsten wire, were the first high-modulus fibre and survive in niche aerospace repair patches.

Table 22.1 — Representative reinforcing fibres (typical values, axial properties).

Fibre	Density (g/cm³)	Modulus E (GPa)	Tensile strength (GPa)	Failure strain (%)	Diameter (μm)	Axial CTE (ppm/K)
E-glass	2.54–2.60	72–74	3.4 (pristine); 2.0–2.5 (strand)	4.5	10–17	5.0
S-/R-glass	2.46–2.55	86–89	4.4–4.6	5.3	9–10	2.9
Basalt	2.65–2.80	85–90	2.8–4.0	3.1	9–13	6
Carbon, standard modulus (T300)	1.76	230	3.5	1.5	7	−0.4 to −0.6
Carbon, high strength (T700, T800)	1.80–1.81	230–294	4.9–5.9	2.0–2.1	5–7	−0.4 to −0.6
Carbon, intermediate modulus (IM7)	1.78	276	5.6	1.9	5	−0.6
Carbon, high modulus (M55J)	1.91	540	4.0	0.8	5	−1.1
Carbon, pitch ultra-high modulus	2.15–2.20	640–900	2.6–3.8	0.3–0.5	10–11	−1.4 to −1.5
Aramid (Kevlar 49)	1.44	112–125	3.0–3.6	2.4–2.8	12	−2
UHMWPE (Dyneema SK75)	0.97	110–120	3.4–3.6	3.5	15–20	−12
Flax	1.40–1.50	45–70	0.5–1.5	1.5–3	10–25 (elementary)	—
SiC (Hi-Nicalon Type S)	3.05	400–420	2.6	0.6	12	5.1
Alumina (Nextel 610)	3.9	370–380	3.1	0.8	10–12	8
Boron (on W core)	2.57	400	3.6	0.9	100–140	4.5

Fibre strengths are wide distributions, not single values. The strength of a bundle or a strand impregnated in resin is usually 60–80 % of the mean single-filament strength measured at a 25 mm gauge length, because the bundle fails progressively once the weakest filaments break (Chapter 14).

The matrix and the interface

The matrix does four jobs: it holds the fibres in position, transfers load into and between fibres by shear, protects fibre surfaces from abrasion and environment, and carries the loads the fibres cannot — transverse tension, in-plane shear and compression,

where it stabilizes the fibres against buckling. In nearly every composite the matrix, not the fibre, sets the service-temperature limit, the environmental resistance, the transverse and shear strengths and the resistance to delamination.

Thermoset matrices (Chapter 19) dominate: unsaturated polyester and vinyl ester for marine, chemical and construction uses; epoxies for aerospace, wind energy and sporting goods; bismaleimides and polyimides for service above 200 °C; phenolics where fire, smoke and toxicity rules apply (aircraft interiors, rail). Thermosets impregnate fibres easily because they are processed as low-viscosity liquids before cure. **Thermoplastic matrices** — polypropylene and nylon for automotive parts, PEEK, PEKK and PPS for aerospace — are tougher, can be welded and reshaped, and recycled more easily, but their melt viscosities are 100–1,000 times higher, which makes impregnation harder (Chapter 21).

Table 22.2 — Representative matrices (typical values).

Matrix	Density (g/cm³)	E (GPa)	Tensile strength (MPa)	Failure strain (%)	T_g or max. service (°C)	Mode I toughness G_{Ic} (kJ/m²)
Unsaturated polyester	1.1–1.4	2.0–4.5	40–90	1–3	T_g 70–120	0.05–0.1
Vinyl ester	1.1–1.2	3.0–3.5	70–85	4–7	T_g 100–150	0.1–0.3
Epoxy (aerospace, 180 °C cure)	1.2–1.3	3.0–4.5	60–100	2–6	T_g 180–220	0.1–0.3 (toughened 0.5–1)
Bismaleimide	1.2–1.3	3.5–4.5	50–90	1.5–3	T_g 230–300	0.1–0.4
Polypropylene	0.90	1.3–1.8	30–40	>100	service ~100	>5
Nylon 66 (dry)	1.14	2.8–3.3	80	30–60	service ~120–150	2–5
PEEK	1.30	3.6	100	30–50	T_g 143, T_m 343	1–2
Aluminium 6061-T6	2.70	69	310	12	~150–200	ductile
Ti-6Al-4V	4.43	114	950	14	~400	ductile
SiC (CVI matrix)	2.5–3.0 (porous)	300–400	brittle	<0.1	>1,200	brittle

Between fibre and matrix lies the **interface** (or interphase), engineered deliberately. Glass fibres are coated during drawing with a **size** containing silane coupling agents that bond chemically to the glass surface and co-react with the resin; carbon fibres are surface-oxidized and sized with epoxy. A strong interface transfers load efficiently and gives high transverse and shear strength; a weak one lets cracks deflect along fibres and fibres pull out, which absorbs energy. PMCs are designed for strong interfaces; CMCs are deliberately designed with weak ones (Section 23.7). The interface quality is measured as an interfacial shear strength τ by single-fibre fragmentation, pull-out or microbond tests, typically 20–80 MPa for well-sized fibres in epoxy.

22.3 Continuous aligned fibres: rule of mixtures and Halpin–Tsai

Volume fraction and density

Micromechanics works with **volume fractions**: V_f for fibre, $V_m = 1 - V_f$ for matrix (neglecting voids). Manufacturers often specify weight fractions w_f, which convert as

$$V_f = \frac{w_f/\rho_f}{w_f/\rho_f + (1 - w_f)/\rho_m}$$

Densities mix exactly by volume: $\rho_c = V_f\rho_f + V_m\rho_m$. Achievable fibre volume fractions depend on the process: 0.15–0.30 for spray-up and short-fibre mouldings, 0.30–0.45 for hand lay-up, 0.50–0.60 for infusion and RTM, 0.55–0.65 for autoclaved prepreg and filament winding, up to 0.70 for pultrusion. The geometric maximum for parallel circular fibres is $\pi/(2\sqrt{3}) = 0.907$ for hexagonal packing; in practice, above about 0.70 the fibres touch and the composite is starved of resin.

Longitudinal stiffness: the iso-strain rule

Load a unidirectional composite along the fibres (direction 1). If fibres and matrix are perfectly bonded and the fibres continuous, they must stretch together: $\varepsilon_f = \varepsilon_m = \varepsilon_1$. Force balance across a cross-section gives $\sigma_1 = V_f\sigma_f + V_m\sigma_m$, and dividing by the common strain,

$$E_1 = V_f E_f + V_m E_m$$

This is the **rule of mixtures** (the **Voigt** or iso-strain estimate). The same argument with Poisson contraction gives the major Poisson's ratio

$$\nu_{12} = V_f\nu_f + V_m\nu_m$$

> **Status of the principle: Longitudinal rule of mixtures**
> *Derived.* Follows exactly from equal axial strain and force balance for continuous, aligned, perfectly bonded fibres; in practice it agrees with measured E_1 within a few percent, the small deficit coming from fibre misalignment (waviness) and voids. For ν_{12} it is an approximation that is accurate to about 5 %.

Since $E_f \gg E_m$, the fibres carry almost all the load: the fraction carried by the fibres is $V_f E_f / E_1$. For glass/epoxy at $V_f = 0.6$ that is $0.6 \times 72/44.6 = 97$ %.

Transverse and shear stiffness: iso-stress and why it underestimates

Load the same composite across the fibres (direction 2). The simplest picture places fibre and matrix in series, like slabs stacked across the load, so they carry the same stress and their strains add:

$$\frac{1}{E_2} = \frac{V_f}{E_f} + \frac{V_m}{E_m}, \qquad \frac{1}{G_{12}} = \frac{V_f}{G_f} + \frac{V_m}{G_m}$$

This is the **inverse rule of mixtures** (**Reuss** or iso-stress estimate). The Voigt and Reuss expressions are rigorous upper and lower bounds on the modulus of any two-phase mixture, but the transverse Reuss value is a poor estimate: real round fibres are not slabs, the stress in the matrix is not uniform, and the matrix between neighbouring fibres is constrained. Measured transverse moduli lie well above the Reuss bound.

The **Halpin–Tsai equations** interpolate between the bounds with one geometric parameter ξ:

$$\frac{P}{P_m} = \frac{1 + \xi\eta V_f}{1 - \eta V_f}, \qquad \eta = \frac{P_f/P_m - 1}{P_f/P_m + \xi}$$

where P is the property sought (E_2, G_{12}) and P_f, P_m the fibre and matrix values. Setting $\xi = 0$ recovers Reuss and $\xi \to \infty$ recovers Voigt. For circular fibres in a square or hexagonal array, $\xi \approx 2$ for E_2 and $\xi \approx 1$ for G_{12}; for rectangular cross-sections and for short fibres other values are used (Section 22.5).

> **Status of the principle: Halpin–Tsai equations**
> *Approximate.* An interpolation fitted to exact elasticity solutions for regular fibre arrays; for E_2 and G_{12} it agrees with experiment within about 10–15 % up to $V_f \approx 0.6$, and it underestimates properties near maximum packing, where a modified ξ (for example $\xi = 2 + 40V_f^{10}$) is used.

Worked example — Ply stiffness of glass/epoxy at 60 % fibre. Take E-glass ($E_f = 72$ GPa, $\nu_f = 0.22$) in epoxy ($E_m = 3.5$ GPa, $\nu_m = 0.35$), $V_f = 0.60$. The constituents are isotropic, so $G = E/[2(1+\nu)]$: $G_f = 72/2.44 = 29.5$ GPa and $G_m = 3.5/2.70 = 1.30$ GPa.

Longitudinal: $E_1 = 0.6 \times 72 + 0.4 \times 3.5 = 43.2 + 1.4 = 44.6$ GPa, and $\nu_{12} = 0.6 \times 0.22 + 0.4 \times 0.35 = 0.272$.

Transverse, Reuss: $1/E_2 = 0.6/72 + 0.4/3.5 = 0.00833 + 0.1143 = 0.1226$ GPa^{-1}, so $E_2 = 8.2$ GPa.

Transverse, Halpin–Tsai with $\xi = 2$: $E_f/E_m = 20.57$, so $\eta = (20.57 - 1)/(20.57 + 2) = 0.867$ and

$$E_2 = 3.5 \times \frac{1 + 2 \times 0.867 \times 0.6}{1 - 0.867 \times 0.6} = 3.5 \times \frac{2.040}{0.4797} = 14.9 \text{ GPa}$$

Shear, Halpin–Tsai with $\xi = 1$: $G_f/G_m = 22.76$, so $\eta = 21.76/23.76 = 0.916$ and $G_{12} = 1.30 \times (1 + 0.550)/(1 - 0.550) = 4.46$ GPa, against a Reuss value of 3.04 GPa.

Density: $\rho = 0.6 \times 2.54 + 0.4 \times 1.20 = 2.00$ g/cm³. Measured values for such a ply are typically $E_1 \approx 43$–45, $E_2 \approx 12$–16, $G_{12} \approx 4$–5 GPa and $\nu_{12} \approx 0.27$–0.28: the rule of mixtures and Halpin–Tsai are right; Reuss is far too low. The ply is $44.6/14.9 = 3.0$ times stiffer along than across the fibres. With standard-modulus carbon ($E_{f1} = 230$ GPa) at the same V_f, $E_1 = 0.6 \times 230 + 0.4 \times 3.5 = 139$ GPa, but E_2 hardly changes (about 9–10 GPa), because carbon's own transverse modulus is only about 15 GPa: a carbon ply is about 15 times stiffer along than across.

Figure 22.2 plots the three estimates against fibre content. Two features stand out. First, E_1 rises linearly and is fibre-dominated; E_2 and G_{12} stay close to the matrix value until V_f exceeds about 0.4 and are **matrix-dominated**. Second, the matrix-dominated properties are where Halpin–Tsai and Reuss differ most, which is why they need the more careful model.

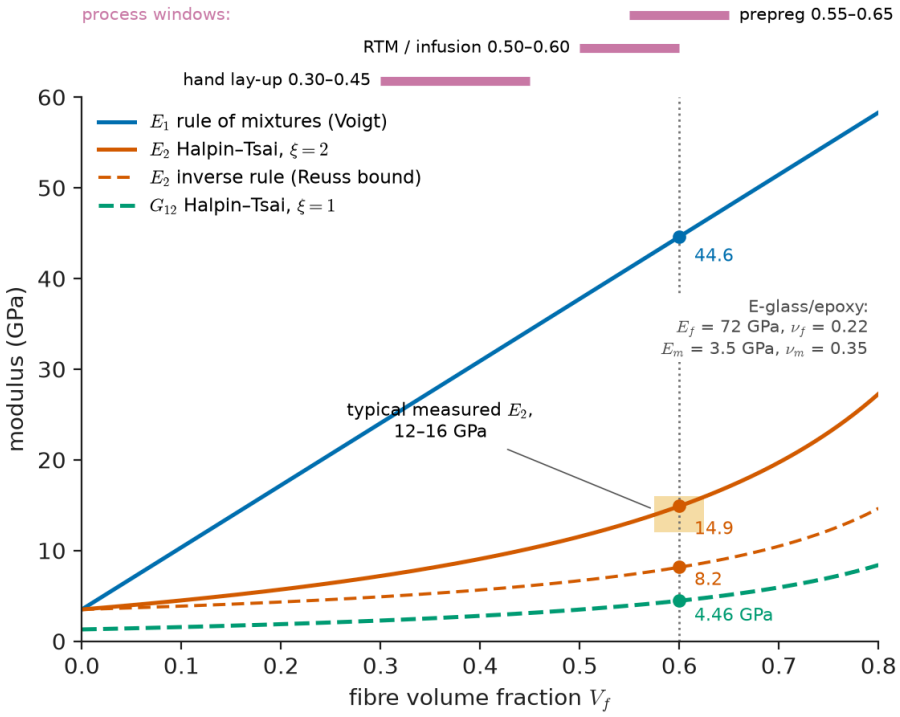

Figure 22.2 — Longitudinal and transverse moduli of E-glass/epoxy versus fibre volume fraction: the rule of mixtures, the Reuss lower bound and the Halpin–Tsai estimate.

Longitudinal strength

Strength follows the same iso-strain logic but with a twist: the constituents have different failure strains. In a polymer-matrix composite the fibre usually fails first (glass at 2.5–4.5 %, carbon at 1.5–2 %), while toughened epoxies tolerate 3–6 %. At the fibre failure strain ε_{fu} the matrix carries only $\sigma'_m = E_m \varepsilon_{fu}$, so

$$\sigma_{1u} = V_f \sigma_{fu} + V_m \sigma'_m$$

For glass/epoxy with an in-situ fibre strength of 1,800 MPa, $\varepsilon_{fu} = 1800/72{,}000 = 2.5\,\%$ and $\sigma'_m = 3500 \times 0.025 = 87.5$ MPa, so $\sigma_{1u} = 0.6 \times 1800 + 0.4 \times 87.5 = 1080 + 35 = 1115$ MPa, close to measured values of 1,000–1,100 MPa.

For a ductile metal matrix the story has a third act. Once the fibres break, the matrix alone can carry $V_m \sigma_{mu}$. At small V_f that may exceed the composite strength, so adding a little fibre can weaken the material. The composite is stronger than the matrix only above the **critical volume fraction**

$$V_{\text{crit}} = \frac{\sigma_{mu} - \sigma'_m}{\sigma_{fu} - \sigma'_m}$$

For tungsten wires ($\sigma_{fu} = 2000$ MPa) in copper ($\sigma_{mu} = 210$ MPa, with a flow stress $\sigma'_m \approx 100$ MPa at the fibre failure strain), $V_{\text{crit}} = 110/1900 = 0.058$: about 6 % fibre must be added just to break even.

> **Status of the principle: Strength rule of mixtures**
> *Approximate.* Assumes all fibres fail at one strain; real fibres have a Weibull strength distribution, so the composite fails by progressive fibre breakage and clustering of breaks, and measured strengths are typically 70–90 % of the simple prediction.

Transverse strength and compression

Transverse tensile strength is the Achilles' heel of a unidirectional ply. The fibres cannot help in this direction; they hurt, because they concentrate strain in the thin matrix ligaments between them and because any weak interface is an internal crack. Typical values are only 30–60 MPa — below the strength of the neat resin. In-plane shear strength is similar, 60–100 MPa.

Compressive strength along the fibres is governed not by crushing but by **fibre microbuckling** and **kink-band formation**: the fibres are slender columns supported only by the shear stiffness of the matrix. The Argon–Budiansky analysis gives $\sigma_{1c} \approx G_{12}/(1+\phi/\gamma_y)$, where ϕ is the initial fibre misalignment (typically 1–3°) and γ_y the matrix shear yield strain. Fibre waviness, a shear-weak matrix and hot-wet conditions (which lower G_{12}) all reduce compressive strength, which for carbon/epoxy is typically 50–80 % of the tensile strength and for aramid composites only about 20 %. Table 22.3 collects representative unidirectional ply properties; these are the inputs to everything that follows here and in Chapter 23.

Table 22.3 — Unidirectional ply properties (typical values, room temperature, dry).

Property	E-glass/epoxy ($V_f \approx 0.55$)	T300/epoxy ($V_f \approx 0.70$)	IM7/epoxy ($V_f \approx 0.60$)	Kevlar 49/epoxy ($V_f \approx 0.60$)
ρ (g/cm³)	1.97	1.60	1.58	1.38
E_1 (GPa)	38.6	181	165	76
E_2 (GPa)	8.27	10.3	9.0	5.5
G_{12} (GPa)	4.14	7.17	5.6	2.3
ν_{12}	0.26	0.28	0.34	0.34
X_t (MPa)	1062	1500	2560	1400
X_c (MPa)	610	1500	1590	235
Y_t (MPa)	31	40	73	12
Y_c (MPa)	118	246	185	53
S (MPa)	72	68	90	34
α_1 (ppm/K)	8.6	0.02	−0.4	−4.0
α_2 (ppm/K)	22.1	22.5	30	79

Here X, Y and S are the longitudinal, transverse and in-plane shear strengths, with subscripts t and c for tension and compression. The T300/epoxy and glass/epoxy columns are the classic Tsai data sets (T300/5208 and Scotchply 1002) used throughout composites teaching; modern toughened systems are stronger in Y_t and S.

22.4 Particulate composites and the Eshelby inclusion picture

Where particles are used

Particle reinforcement trades efficiency for simplicity and isotropy. Examples span every field: **concrete**, an aggregate-filled cement paste (Chapter 18); **cemented carbides**, 70–94 % WC particles in a cobalt binder, for cutting tools and mining bits; **Al–SiC** with 10–30 vol% SiC particles for brake rotors on rail vehicles, pistons, bicycle frames and electronics baseplates; **dental composites**, 60–80 vol% silica or glass fillers in a dimethacrylate resin; **filled polymers** — talc in polypropylene bumpers, carbon black in tyres, silica in underfill encapsulants; and **particle-toughened** systems, in which rubber or thermoplastic particles raise the toughness of epoxy (Chapter 21).

Bounds for an isotropic mixture

For a statistically isotropic particulate composite the Voigt and Reuss bounds are still valid but far apart. **Hashin and Shtrikman** found the tightest possible bounds that use only the volume fractions and the phase moduli: for bulk modulus,

$$K_{\mathrm{HS}}^{\mathrm{lower}} = K_m + \frac{V_p}{\dfrac{1}{K_p - K_m} + \dfrac{3V_m}{3K_m + 4G_m}}$$

with an analogous expression for shear modulus; the upper bound is obtained by exchanging the roles of the phases. When the stiffer phase is the particle, the lower bound corresponds physically to stiff spheres coated by compliant matrix — exactly the geometry of most particulate composites — and it is usually the better estimate.

Worked example — How stiff is aluminium with 20 % SiC particles? For Al 6061 ($E = 69$ GPa, $\nu = 0.33$) and SiC ($E = 410$ GPa, $\nu = 0.19$), $K = E/[3(1-2\nu)]$ and $G = E/[2(1+\nu)]$ give $K_m = 67.6$, $G_m = 25.9$, $K_p = 220$ and $G_p = 172$ GPa. At $V_p = 0.20$:

- Voigt: $0.2 \times 410 + 0.8 \times 69 = 137$ GPa.
- Reuss: $1/(0.2/410 + 0.8/69) = 82.8$ GPa.
- Hashin–Shtrikman lower: $K = 92.6$ GPa, $G = 36.1$ GPa, hence $E = 9KG/(3K+G) = 92.7$ GPa.
- Hashin–Shtrikman upper: 114 GPa.

Measured moduli for 6061/SiC/20p are about 100–105 GPa, inside the Hashin–Shtrikman window and about 50 % stiffer than the unreinforced alloy at almost the same density (2.80 against 2.70 g/cm³). The Voigt value, appropriate to continuous fibres, would overestimate the gain by a factor of two.

The Eshelby inclusion (qualitative)

Why do particles of the same volume fraction stiffen differently depending on their shape? The key insight is due to **Eshelby** (1957). Imagine cutting an ellipsoidal region out of an infinite elastic body, letting it undergo some stress-free strain (a thermal expansion, a phase transformation, a misfit), and welding it back. Eshelby proved a remarkable result: the strain inside an ellipsoidal inclusion is **uniform**, and it is related to the stress-free strain by a fourth-rank **Eshelby tensor** S that depends only on the inclusion's shape (its aspect ratios) and the matrix Poisson's ratio. The result extends, through the "equivalent inclusion" idea, to an inhomogeneity of different stiffness sitting in a loaded matrix.

Three qualitative consequences matter to the materials engineer:

- **Shape controls load transfer.** A sphere is loaded mainly through hydrostatic constraint; a prolate ellipsoid (a needle or fibre) aligned with the load picks up a larger share of the stress; a flat disc (a platelet) aligned with the load stiffens both in-plane directions. This is why platelets (talc, mica, clay, graphene) are more efficient than spheres in plastics, and why fibres are best of all along their axis.
- **Interactions can be included at modest cost.** The **Mori–Tanaka** method places each inclusion in the average matrix stress caused by all the others and gives reliable estimates up to $V_p \approx 0.3$–0.4. For spheres it coincides with the Hashin–Shtrikman lower bound; for aligned fibres it reproduces the rule of mixtures along them. It underlies most commercial software for injection-moulded short-fibre parts.
- **Misfit produces internal stress.** The same tensor gives the stress inside and around a precipitate with a transformation misfit (Chapter 10) or a particle with a different thermal expansion. On cooling Al–SiC from processing temperature, the mismatch in CTE (23 against 4 ppm/K) generates residual stresses that exceed the matrix yield strength and punch dislocations into it, contributing to the strength of MMCs (Chapter 13).

> **Status of the principle: Eshelby's inclusion result**
> *Derived.* An exact consequence of linear elasticity for a single ellipsoidal inclusion in an infinite isotropic matrix; mean-field schemes built on it (Mori–Tanaka, self-consistent) are approximations that degrade at high volume fraction, for clustered particles and when particles touch and form a connected skeleton.

22.5 Short-fibre composites: shear lag and the critical length

Where short fibres are used

About half of all fibre-reinforced polymer by mass is discontinuous: glass-filled nylon, polypropylene and PBT injection-moulded into engine covers, air-intake manifolds, power-tool housings, connector bodies and pump impellers; **sheet moulding compound** (SMC), 25 mm chopped glass in polyester, compression-moulded into truck cabs and car body panels; **bulk moulding compound** (BMC) for electrical enclosures; chopped carbon in thermoplastics for drones and brackets. Short fibres flow with the melt, so complex shapes can be moulded at high rates, but each fibre is too short to carry its full potential load.

How load gets into a fibre: shear lag

A fibre embedded in a matrix that is stretched along the fibre axis carries no load at its ends. Load is transferred into it by shear stress τ at the fibre–matrix interface, and the fibre's tensile stress builds up from zero at each end. For a fibre of diameter d, a force balance on a slice of length dx gives

$$\frac{d\sigma_f}{dx} = \frac{4\tau}{d}$$

The **shear-lag** model of **Cox** (1952) assumes elastic matrix and interface. The fibre stress then varies as

$$\sigma_f(x) = E_f \varepsilon_1 \left[1 - \frac{\cosh \beta (L/2 - x)}{\cosh(\beta L/2)}\right], \qquad \beta = \sqrt{\frac{2 G_m}{E_f r^2 \ln(R/r)}}$$

for a fibre of length L and radius r, with x measured from one end and $2R$ the spacing between neighbouring fibres. The stress rises over a distance of a few $1/\beta$ from each end and plateaus at the long-fibre value $E_f\varepsilon_1$. Averaging this profile along the fibre gives the **length efficiency factor** for stiffness,

$$\eta_L = 1 - \frac{\tanh(\beta L/2)}{\beta L/2}$$

which enters a modified rule of mixtures, $E_c = \eta_o \eta_L V_f E_f + V_m E_m$, where η_o is the **orientation factor**: 1 for perfectly aligned fibres, $3/8$ for fibres randomly oriented in a plane and $1/5$ for random three-dimensional orientation (Krenchel).

> **Status of the principle: Cox shear-lag model**
> *Approximate.* Assumes elastic fibre and matrix, perfect bonding, no load transfer across fibre ends and an idealized cylindrical neighbourhood; it captures the length dependence correctly but its β depends on the assumed fibre spacing, and it ignores interfacial slip and matrix yielding.

The critical length

If the interface yields or slips at a constant shear stress τ (a rigid-plastic interface, the **Kelly–Tyson** model), the fibre stress rises linearly from each end. The fibre can reach its own fracture stress σ_{fu} only if it is longer than the **critical length**

$$l_c = \frac{\sigma_{fu}\, d}{2\tau}$$

Fibres shorter than l_c never break; they pull out of the matrix. Fibres much longer than l_c are nearly as effective as continuous ones. The average stress in a fibre of length L gives the strength efficiency: $\eta = L/(2l_c)$ for $L < l_c$ and $\eta = 1 - l_c/(2L)$ for $L \ge l_c$, in

$$\sigma_{cu} \approx \eta_o\, \eta\, V_f \sigma_{fu} + V_m \sigma'_m$$

> **Status of the principle: Kelly–Tyson critical length**
> *Approximate.* Assumes a constant interfacial shear stress along the debonded or yielded length and a single fibre strength; it is the basis of the single-fibre fragmentation test, where the measured fragment-length distribution is used in reverse to obtain τ.

Figure 22.3 shows the stress profiles for fibres shorter than, equal to and longer than l_c.

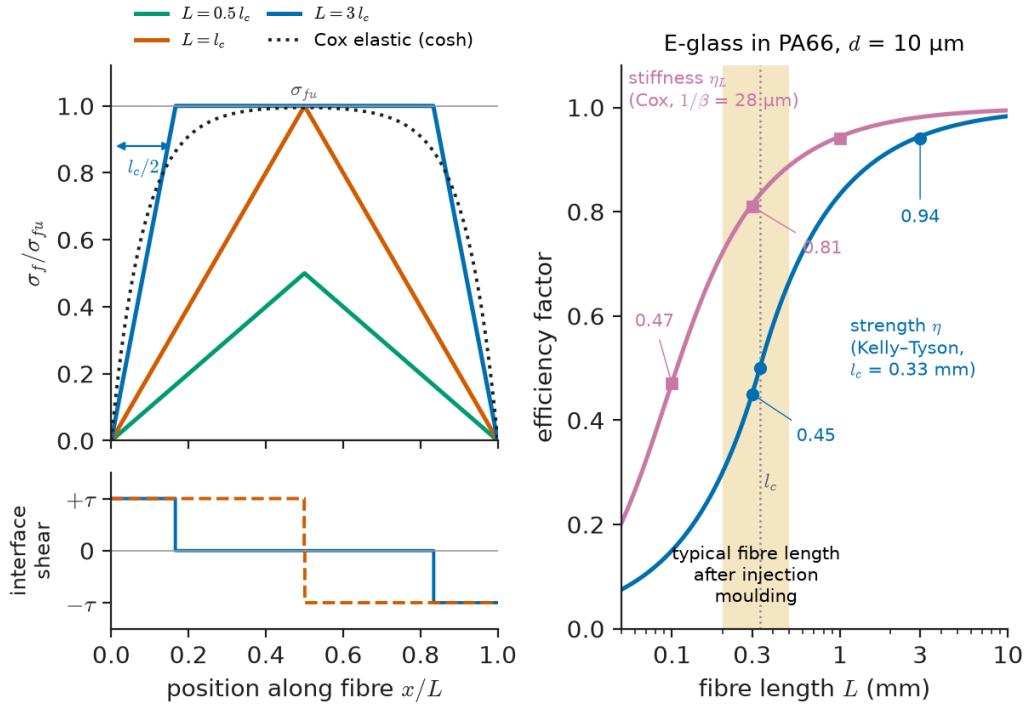

Figure 22.3 — Stress transfer into a short fibre: tensile stress along the fibre and interfacial shear stress, for fibres shorter than, equal to and longer than the critical length.

Worked example — Glass-filled nylon 66. A housing is injection-moulded in PA66 with 30 wt% E-glass ($\rho_f = 2.54$, $\rho_m = 1.14$ g/cm³, $d = 10$ µm). Convert to volume: $V_f = (0.30/2.54)/(0.30/2.54 + 0.70/1.14) = 0.1181/0.7321 = 0.161$.

Critical length, with an in-situ fibre strength of 2,000 MPa and $\tau = 30$ MPa: $l_c = 2000 \times 10/(2 \times 30) = 333$ µm, or $l_c/d = 33$. The pellets contain fibres about 3–4 mm long, but screw shearing and gate passage break them to a typical mean of 0.2–0.5 mm. At $L = 0.3$ mm, below l_c, the strength efficiency is $0.3/(2 \times 0.333) = 0.45$: more than half of the fibres' strength potential is lost. Had the fibres survived at 3 mm, $\eta = 1 - 0.333/6 = 0.94$. This is the reason for long-fibre thermoplastic granules (10–25 mm), which keep fibres 2–5 mm long in the part.

Stiffness. With $E_m = 2.8$ GPa and $\nu_m = 0.4$, $G_m = 1.0$ GPa. For hexagonal packing $R/r = \sqrt{\pi/(2\sqrt{3}V_f)} = 2.37$, so $\beta = \sqrt{2 \times 1.0/(72 \times (5 \times 10^{-6})^2 \ln 2.37)} = 3.59 \times 10^4$ m⁻¹, a transfer length $1/\beta = 28$ µm. For $L = 0.3$ mm, $\beta L/2 = 5.38$ and $\eta_L = 1 - \tanh(5.38)/5.38 = 0.81$. Stiffness transfer is much more forgiving than strength transfer, because elastic load transfer needs only a few fibre diameters. With a skin–core orientation distribution averaging $\eta_o \approx 0.7$ in the flow direction,

$$E_c = 0.7 \times 0.81 \times 0.161 \times 72 + 0.839 \times 2.8 = 6.6 + 2.3 = 9.0 \text{ GPa}$$

against datasheet values of 9–10 GPa (dry, flow direction) and about half that transverse to flow. The orientation pattern — fibres aligned with flow near the mould walls, across flow in the core — makes injection-moulded parts anisotropic and prone to warpage, so mould-filling simulation is now coupled to Mori–Tanaka micromechanics. And nylon absorbs 2–3 % water in humid air, which plasticizes the matrix and can lower E_c by a third, a reminder that matrix-dominated properties are environment-dependent (Section 22.8).

22.6 The orthotropic lamina: stiffness, compliance and reduced stiffness

From micromechanics to a homogeneous ply

Laminates are built from thin layers — **laminae** or **plies**, typically 0.125–0.25 mm thick for prepreg — each of which contains thousands of fibres through its width. For structural analysis each ply is treated as a **homogeneous, anisotropic** continuum with effective properties (E_1, E_2, G_{12}, ν_{12}, strengths and expansion coefficients) obtained from micromechanics or, preferably, from tests on unidirectional coupons. From here on the fibres disappear from the analysis and reappear only through these properties. This is **macromechanics**.

Orthotropy

A unidirectional ply has three mutually perpendicular planes of symmetry: the material axes are 1 (along the fibres), 2 (in-plane, across the fibres) and 3 (through the thickness). Chapter 4 showed that Neumann's principle reduces the number of independent elastic constants as symmetry increases. A general anisotropic solid has 21; a material with three orthogonal symmetry planes — an **orthotropic** material — has 9:

$$\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_{23} \\ \gamma_{13} \\ \gamma_{12} \end{pmatrix} = \begin{pmatrix} 1/E_1 & -\nu_{21}/E_2 & -\nu_{31}/E_3 & 0 & 0 & 0 \\ -\nu_{12}/E_1 & 1/E_2 & -\nu_{32}/E_3 & 0 & 0 & 0 \\ -\nu_{13}/E_1 & -\nu_{23}/E_2 & 1/E_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G_{23} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G_{13} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G_{12} \end{pmatrix} \begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{13} \\ \tau_{12} \end{pmatrix}$$

The matrix is the **compliance** $[S]$; its inverse is the **stiffness** $[C]$. Here ν_{ij} is the contraction in direction j per unit extension in direction i under uniaxial stress in i, and γ are engineering shear strains (twice the tensor shear strains). The compliance must be symmetric because the strain energy must be a unique function of strain (Chapter 11), which gives the **reciprocal relations**

$$\frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2}$$

and its two companions. A unidirectional ply with randomly placed fibres is in fact **transversely isotropic** — properties are the same in every direction in the 2–3 plane — so $E_3 = E_2$, $G_{13} = G_{12}$, $\nu_{13} = \nu_{12}$ and $G_{23} = E_2/[2(1+\nu_{23})]$, leaving 5 independent constants.

> **Status of the principle: Orthotropic Hooke's law**
> *Fundamental.* Linear elasticity with the symmetries required by strain energy and by the three mirror planes of the ply; its validity is limited only by linearity, which for polymer-matrix plies fails in in-plane shear above about 0.5–1 % shear strain and in transverse compression, where the matrix behaves non-linearly.

Note an important consequence of the reciprocal relation. For T300/epoxy, $\nu_{12} = 0.28$ but $\nu_{21} = 0.28 \times 10.3/181 = 0.016$: pulling along the fibres causes noticeable contraction across them, but pulling across the fibres causes almost none along them, because the fibres resist it.

Plane stress and the reduced stiffness

A ply in a thin laminate carries stress essentially in its own plane: $\sigma_3 = \tau_{13} = \tau_{23} = 0$. Deleting those rows and columns and inverting gives the **plane-stress** relation

$$\begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{pmatrix} = \begin{pmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{pmatrix} \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{pmatrix}$$

with the **reduced stiffnesses**

$$Q_{11} = \frac{E_1}{1-\nu_{12}\nu_{21}}, \quad Q_{22} = \frac{E_2}{1-\nu_{12}\nu_{21}}, \quad Q_{12} = \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}}, \quad Q_{66} = G_{12}$$

"Reduced" because they are not the full three-dimensional stiffnesses C_{ij} (which would assume $\varepsilon_3 = 0$) but the stiffnesses appropriate when the ply is free to contract through its thickness. Note that there is **no coupling** between normal stress and shear strain: in its own axes, an orthotropic ply behaves like a slightly more complex isotropic sheet, with different moduli along and across.

Figure 22.4 defines the material axes 1–2 and the structural axes x–y that are used when the ply is rotated.

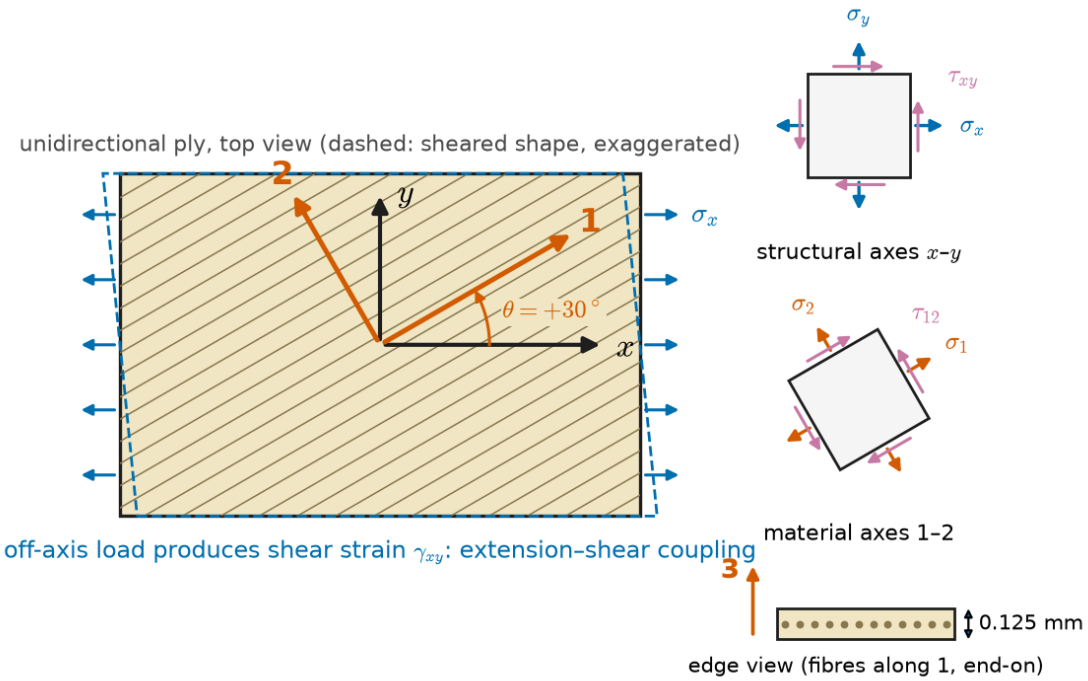

Figure 22.4 — A unidirectional lamina with material axes 1–2 rotated by angle θ from the structural axes x–y, and the stress components in each frame.

Worked example — Reduced stiffness of a carbon/epoxy ply. For T300/epoxy ($E_1 = 181$ GPa, $E_2 = 10.3$ GPa, $G_{12} = 7.17$ GPa, $\nu_{12} = 0.28$): $\nu_{21} = 0.28 \times 10.3/181 = 0.01593$ and $1 - \nu_{12}\nu_{21} = 0.99554$. Then

$$Q_{11} = 181.8, \quad Q_{22} = 10.35, \quad Q_{12} = 2.90, \quad Q_{66} = 7.17 \text{ GPa}$$

The Poisson correction raises Q_{11} and Q_{22} above E_1 and E_2 by only 0.45 %, because ν_{21} is so small; for a quasi-isotropic material with $\nu = 0.3$ it would be 10 %. The ply is about 18 times stiffer in direction 1 than in direction 2 and 25 times stiffer than in shear — the anisotropy that makes the next section necessary.

22.7 The off-axis lamina: transformation and $\bar{Q}$

Why rotate

Few plies in a real structure are loaded only along or across their fibres. A drive shaft in torsion is wound with fibres at ±45° to its axis, a pressure vessel at about ±55°, a wind-turbine blade skin at ±45° over 0° spar caps, a golf-club shaft with a mixture of 0° and ±45°. To analyse these, ply stresses and strains must be expressed in the **structural axes** x–y, rotated from the material axes by the ply angle θ (Figure 22.4).

The transformation

Stresses transform as a second-rank tensor (Chapter 4). With $c = \cos\theta$ and $s = \sin\theta$,

$$\begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{pmatrix} = [T] \begin{pmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{pmatrix}, \qquad [T] = \begin{pmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{pmatrix}$$

and tensor strains transform the same way; with engineering shear strain the factor of two is handled by the Reuter matrix $[R] = \mathrm{diag}(1, 1, 2)$. Combining $\boldsymbol{\sigma}_{12} = [Q]\boldsymbol{\varepsilon}_{12}$ with the transformations gives the stress–strain law in structural axes,

$$\begin{pmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{pmatrix} = \begin{pmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{pmatrix} \begin{pmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{pmatrix}, \qquad [\bar{Q}] = [T]^{-1}[Q][R][T][R]^{-1}$$

Written out,

$$\bar{Q}_{11} = Q_{11}c^4 + 2(Q_{12} + 2Q_{66})s^2c^2 + Q_{22}s^4$$
$$\bar{Q}_{22} = Q_{11}s^4 + 2(Q_{12} + 2Q_{66})s^2c^2 + Q_{22}c^4$$
$$\bar{Q}_{12} = (Q_{11} + Q_{22} - 4Q_{66})s^2c^2 + Q_{12}(s^4 + c^4)$$
$$\bar{Q}_{66} = (Q_{11} + Q_{22} - 2Q_{12} - 2Q_{66})s^2c^2 + Q_{66}(s^4 + c^4)$$
$$\bar{Q}_{16} = (Q_{11} - Q_{12} - 2Q_{66})sc^3 + (Q_{12} - Q_{22} + 2Q_{66})s^3c$$
$$\bar{Q}_{26} = (Q_{11} - Q_{12} - 2Q_{66})s^3c + (Q_{12} - Q_{22} + 2Q_{66})sc^3$$

> **Status of the principle: Lamina stiffness transformation**
> *Derived.* An exact consequence of tensor transformation applied to the plane-stress orthotropic law; it introduces no new approximation beyond those of Section 22.6.

Extension–shear coupling

The new terms $\bar{Q}_{16}$ and $\bar{Q}_{26}$ are zero at 0° and 90° but not in between. They couple normal stress to shear strain: pull an off-axis ply along x and it shears as well as stretches, because the material would rather deform along its compliant transverse and shear directions than along its stiff fibres. This **extension–shear coupling** is harmless when a $+\theta$ ply is paired with a $-\theta$ ply (their $\bar{Q}_{16}$ terms cancel, Chapter 23), but in a single ply or an unbalanced laminate it distorts parts and complicates testing: an off-axis tensile coupon gripped rigidly develops an S-shaped deformation and spurious stresses near the grips.

The coupling can also be exploited. **Aeroelastic tailoring** uses deliberately unbalanced laminates so that a forward-swept wing twists nose-down as it bends up, preventing divergence (the X-29 demonstrator); in wind turbines, bend–twist coupled blades shed load in gusts.

Worked example — How fast does stiffness fall off-axis? The modulus of a single ply loaded in uniaxial stress along x follows from the compliance transformation:

$$\frac{1}{E_x(\theta)} = \frac{c^4}{E_1} + \left(\frac{1}{G_{12}} - \frac{2\nu_{12}}{E_1}\right)s^2c^2 + \frac{s^4}{E_2}$$

For T300/epoxy:

θ	0°	5°	10°	15°	30°	45°	60°	90°
E_x (GPa), T300/epoxy	181	154	108	72.6	28.8	16.7	12.4	10.3
E_x (GPa), E-glass/epoxy	38.6	—	—	26.8	15.4	10.7	8.9	8.3

A misalignment of only 5° costs 15 % of the stiffness of the carbon ply, and at 15° 60 % has gone. The glass ply, being less anisotropic, is more forgiving: at 15° it retains 69 %. The fall is controlled at small angles by the shear compliance term $1/G_{12}$, which is large: the off-axis ply deforms mainly by shearing the matrix. At 45° for T300/epoxy the transformed stiffness matrix is

$$[\bar{Q}]_{45^\circ} = \begin{pmatrix} 56.7 & 42.3 & 42.9 \\ 42.3 & 56.7 & 42.9 \\ 42.9 & 42.9 & 46.6 \end{pmatrix}\ \text{GPa}$$

The large $\bar{Q}_{66}$ (46.6 GPa against $Q_{66} = 7.17$) is why ±45° plies are used to carry shear and torsion: at 45° the shear load becomes tension and compression along the fibres. The off-axis in-plane shear modulus G_{xy} rises correspondingly, from 7.2 GPa at 0° to 9.5 GPa for a single ply at 45° and to much higher values in a ±45° laminate, where the coupling terms of the two plies cancel and constrain each other (Chapter 23). Figure 22.5 shows these angular dependences.

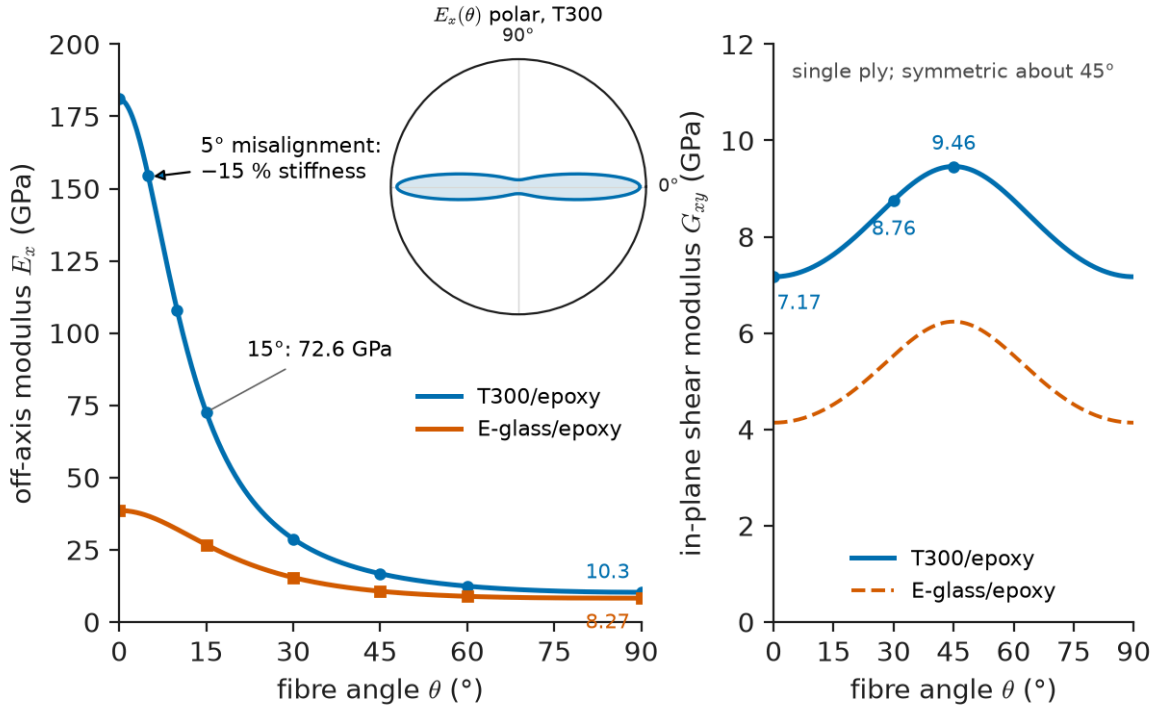

Figure 22.5 — Off-axis Young's modulus and shear modulus of single unidirectional plies of carbon/epoxy and glass/epoxy versus fibre angle.

Off-axis strength in brief

Strength transforms less simply than stiffness, because the ply fails by different mechanisms in different directions. An off-axis tensile load produces $\sigma_1 = \sigma_x c^2$, $\sigma_2 = \sigma_x s^2$ and $\tau_{12} = -\sigma_x sc$, and the ply fails by whichever mode is reached first: fibre fracture at very small angles, in-plane shear at intermediate angles (roughly 3–30° for carbon/epoxy) and transverse tension at large angles. Because Y_t and S are 20–40 times smaller than X_t, the off-axis strength drops even faster than the stiffness: a T300/epoxy ply loaded at 15° has a strength of only about 250 MPa. Interactive criteria that combine these modes — Tsai–Hill, Tsai–Wu and Puck — are developed in Chapter 23.

22.8 Thermal and hygral expansion of laminae

Why expansion matters

Every polymer-matrix composite is cured or consolidated hot (120–180 °C for epoxies, 380–400 °C for PEEK) and used cold. Every one absorbs some moisture. Both effects produce strains that the fibres resist along the ply and do not resist across it, so a single ply wants to change shape anisotropically. In a laminate the plies constrain each other, and the result is **residual stress**, which can be large enough to crack the matrix of a cross-ply laminate before any load is applied, to warp an unsymmetric panel, or to distort a precision structure.

Expansion coefficients of a ply from micromechanics

Along the fibres, fibre and matrix share a strain. The stiff fibres dominate, and force equilibrium (with no external load) gives the **Schapery** (or Turner) expression

$$\alpha_1 = \frac{\alpha_f E_f V_f + \alpha_m E_m V_m}{E_f V_f + E_m V_m}$$

Across the fibres, the matrix expands more freely, and its expansion is amplified by the Poisson effect of being restrained along the fibres. Schapery's approximation is

$$\alpha_2 \approx (1+\nu_m)\alpha_m V_m + (1+\nu_f)\alpha_f V_f - \alpha_1 \nu_{12}$$

> **Status of the principle: Schapery expansion coefficients**
> *Approximate.* α_1 follows from energy bounds and is accurate for isotropic constituents; α_2 is an approximation that works well for $V_f > 0.25$. Both must be modified for anisotropic fibres such as carbon and aramid, whose transverse CTEs differ from their axial ones.

For E-glass/epoxy at $V_f = 0.6$ (with $\alpha_f = 5$, $\alpha_m = 60$ ppm/K): $\alpha_1 = (5 \times 72 \times 0.6 + 60 \times 3.5 \times 0.4)/(43.2 + 1.4) = (216 + 84)/44.6 = 6.7$ ppm/K, and $\alpha_2 \approx 1.35 \times 60 \times 0.4 + 1.22 \times 5 \times 0.6 - 6.7 \times 0.272 = 32.4 + 3.7 - 1.8 = 34$ ppm/K — five times α_1. With T300 carbon ($\alpha_{f1} = -0.5$ ppm/K, $E_{f1} = 230$ GPa) at the same V_f, $\alpha_1 = (-0.5 \times 138 + 84)/139.4 = 0.11$ ppm/K, essentially zero, while α_2 is 25–35 ppm/K. Measured T300/epoxy plies give $\alpha_1 = 0.02$ and $\alpha_2 = 22.5$ ppm/K (Table 22.3).

Hygral swelling

Epoxies absorb 1–6 % water by weight at saturation depending on chemistry and humidity; polyesters and vinyl esters less, thermoplastics such as PEEK and PPS only 0.1–0.5 %. The absorbed water swells the matrix. Swelling is described exactly like thermal expansion, with moisture content ΔM (weight fraction) taking the place of ΔT and **coefficients of hygral (moisture) expansion** β_1, β_2 taking the place of α_1, α_2:

$$\varepsilon_1^H = \beta_1 \Delta M, \qquad \varepsilon_2^H = \beta_2 \Delta M$$

Glass and carbon fibres absorb no water, so $\beta_1 \approx 0$ while $\beta_2 \approx 0.4$–0.6 for typical carbon/epoxy. Aramid and natural fibres absorb water themselves, which makes their composites swell in both directions. Moisture enters by Fickian diffusion through the thickness (Chapter 8); a 2 mm carbon/epoxy laminate at 70 °C takes several months to saturate, which is why aircraft structures are qualified on "hot–wet" conditioned coupons.

The two effects are additive and of similar size. For T300/epoxy with $\beta_2 = 0.6$, absorbing 1 % moisture gives $\varepsilon_2^H = 0.006$, the same transverse strain as heating by $0.006/22.5 \times 10^{-6} = 270$ K. In a laminate cured at 177 °C and used at room temperature, the cool-down strains of about $-150 \times 22.5 \times 10^{-6} = -0.0034$ are partly offset over time by moisture swelling — residual stresses are relieved in wet service and maximal when dry and cold.

Off-axis expansion

Expansion coefficients are components of a second-rank strain tensor and transform like strains. For a ply at angle θ,

$$\alpha_x = \alpha_1 c^2 + \alpha_2 s^2, \qquad \alpha_y = \alpha_1 s^2 + \alpha_2 c^2, \qquad \alpha_{xy} = 2(\alpha_1 - \alpha_2) sc$$

where α_{xy} is an engineering shear-strain coefficient: an off-axis ply heated freely distorts in shear. For T300/epoxy at 30° to x, $\alpha_x = 0.02 \times 0.75 + 22.5 \times 0.25 = 5.6$ ppm/K, $\alpha_y = 16.9$ ppm/K and $\alpha_{xy} = -19.5$ ppm/K. At 45°, $\alpha_x = \alpha_y = 11.3$ ppm/K and $\alpha_{xy} = -22.5$ ppm/K. The same transformation, with β in place of α, applies to swelling. Figure 22.6 plots these, and Chapter 23 shows how a laminate's plies, each wanting to deform this way, settle on a common strain and lock residual stresses into every ply.

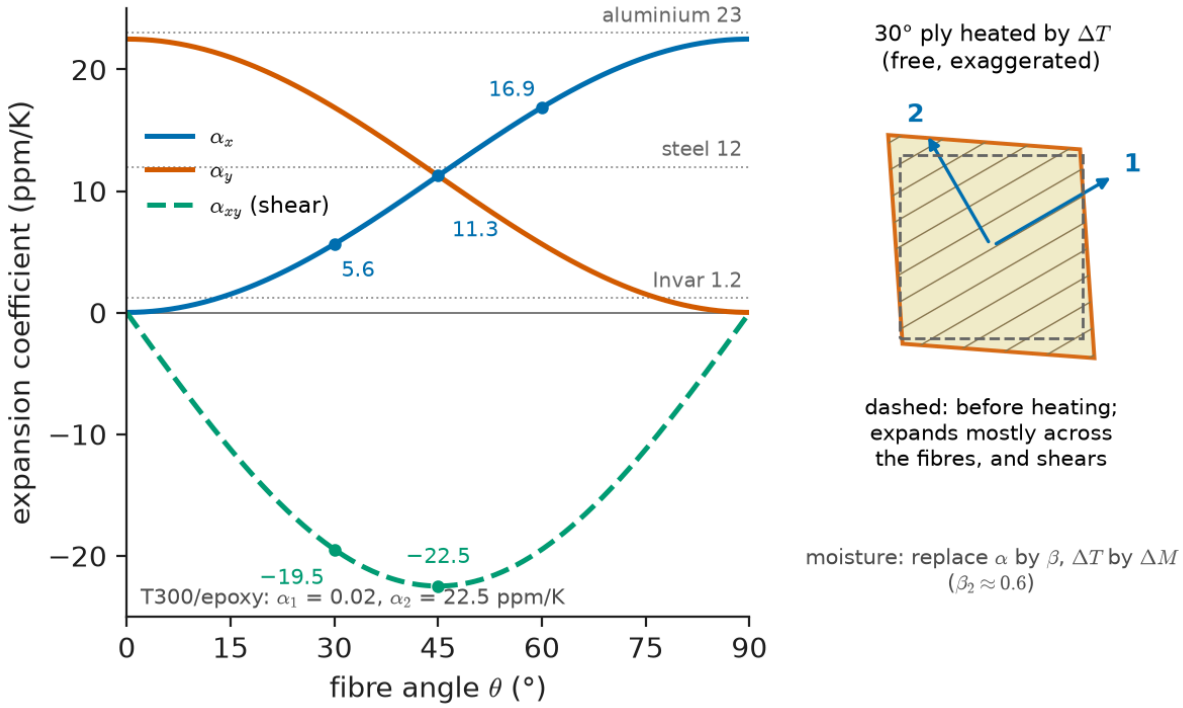

Figure 22.6 — Thermal expansion coefficients of a unidirectional carbon/epoxy ply versus fibre angle, and the free thermal distortion of an off-axis ply.

Carbon's slightly negative axial CTE also makes **near-zero-expansion laminates** possible: a layup dominated by 0° plies of high-modulus pitch fibre (ply $\alpha_1 \approx -1.1$ ppm/K) with a few angle plies can be tuned through zero, which is how satellite optical benches and antenna reflectors hold their shape between sunlight and shadow; among metals only Invar (about 1.2 ppm/K) comes close, at five times the density.

What remains open

Open question	Where it is answered
How do plies of different angles combine into a laminate, and what are its stiffness and coupling terms?	Chapter 23 (classical laminate theory, ABD matrix)
How do plies and laminates fail under combined stress, and what happens after the first ply cracks?	Chapter 23 (failure criteria, progressive failure)
Why do laminates delaminate at free edges and after impact?	Chapter 23 (interlaminar stresses, compression after impact)
How are fibres and matrix actually combined at scale, and how does the process set V_f, voids and waviness?	Chapter 23 (manufacturing); Chapter 21 (polymer processing)
What sets the strength distribution of fibres and why does strength depend on size?	Chapter 14 (Weibull statistics), Chapter 17 (ceramic fibres and glass)
How do matrix creep and viscoelasticity affect long-term composite behaviour?	Chapter 14 (viscoelasticity), Chapter 21 (polymer mechanics)
How do MMCs and CMCs behave at high temperature, and how are CMCs toughened?	Chapter 23 (MMCs, CMCs), Chapter 41 (extreme environments)
How do composites age, absorb moisture and degrade in service?	Chapter 44 (degradation across classes)
How do the elastic-tensor ideas used here generalize to arbitrary anisotropy?	Chapter 4 (tensor properties), Chapter 11 (anisotropic elasticity)
How are composites chosen against metals using performance indices?	Chapter 58 (materials selection)
How are nanocomposites (clay, graphene, nanotube) different?	Chapter 38 (nanocomposites)

Next: Chapter 23 stacks oriented plies into laminates, derives the ABD matrix, predicts how laminates fail, and follows composites through their manufacture and into metal and ceramic matrices.

Chapter 23 — Laminate Theory, Failure and Composite Manufacturing

Part V · Polymers and Composites

A carbon/epoxy panel comes out of the autoclave perfectly flat on one production line and curled like a potato crisp on another. Nothing is wrong with the prepreg, the cure cycle or the tool. On the second line an operator laid the plies in the order 0, 90, 0, 90 instead of 0, 90, 90, 0 — the same plies, the same angles, a different sequence. Elsewhere, a maintenance engineer taps an aircraft panel after a dropped tool and finds nothing visible; a later ultrasonic scan shows a cone of delaminations under the dent that has cut the panel's compressive strength by half. And a wind-turbine blade passes its static test only to crack along its trailing edge, where the skin laminate ends, at loads well below those that any single ply should have carried.

All three problems belong to the laminate, not to the ply. Chapter 22 described a single unidirectional layer. This chapter stacks layers into a structure, finds how the stacking sequence couples stretching, bending, shearing and twisting, predicts when and how laminates fail, follows the damage modes that have no counterpart in metals, and then turns to how composites are actually made — including the metal- and ceramic-matrix composites that carry the ideas into high-temperature service.

> **How does the stacking sequence of plies decide whether a laminate stretches, bends, twists or warps? How do laminates fail, why do delamination and impact damage matter so much, and how does the manufacturing route set the quality of the part?**

Concept	Engineering destination
Classical laminate theory, ABD matrix	Stiffness of aircraft skins, car body panels, bicycle frames, pressure vessels
Symmetric and balanced layups	Warp-free panels, design rules in every aerospace layup
Failure criteria and progressive failure	Design allowables, first-ply and last-ply failure, margin of safety
Interlaminar stresses, delamination	Edges, holes, ply drops, bonded joints, blade trailing edges
Impact damage, compression after impact	Damage-tolerance certification of aircraft and pressure vessels
Sandwich structures	Aircraft floors, helicopter blades, boat hulls, wind blades, racing-car monocoques
Manufacturing processes	Choosing between hand lay-up, infusion, autoclave, RTM, winding, pultrusion and fibre placement
Metal- and ceramic-matrix composites	Brake discs, pistons, electronic baseplates, turbine shrouds, re-entry heat shields

Roadmap. Stacking plies: classical laminate theory and the ABD matrix (23.1) → when plies fail: criteria and progressive failure (23.2) → stresses between plies: free edges and delamination (23.3) → impact damage and compression after impact (23.4) → sandwich structures (23.5) → how polymer composites are made (23.6) → metal- and ceramic-matrix composites (23.7).

23.1 Classical laminate theory

Laminate notation

A laminate is described by its ply angles from top to bottom, measured from the structural x axis. Square brackets enclose the sequence; a subscript s means the listed half is mirrored about the mid-plane; a numerical subscript repeats a ply or group; $\pm$ means a $+\theta$ ply followed by a $-\theta$ ply. Thus $[0/\pm 45/90]_s$ is the eight-ply sequence $0/45/-45/90/90/-45/45/0$, and $[0_2/90]_s$ is $0/0/90/90/0/0$. Plies of a typical carbon/epoxy prepreg are 0.125–0.25 mm thick; a wing skin may contain 40–150 of them.

Kinematics: the Kirchhoff hypothesis

Classical laminate theory (CLT) treats the laminate as a thin plate and assumes, as in the classical theory of plates (Kirchhoff), that straight lines normal to the mid-plane remain straight and normal after deformation and do not change length. The plies are perfectly bonded, so displacements are continuous through the thickness. Then the in-plane strains at any height z above the mid-plane are

$$\boldsymbol{\varepsilon}(z) = \boldsymbol{\varepsilon}^0 + z\,\boldsymbol{\kappa}$$

where $\boldsymbol{\varepsilon}^0 = (\varepsilon_x^0, \varepsilon_y^0, \gamma_{xy}^0)$ are the mid-plane strains and $\boldsymbol{\kappa} = (\kappa_x, \kappa_y, \kappa_{xy})$ the curvatures (bending and twisting). Strain varies linearly and continuously through the thickness. Stress does not: each ply k has its own stiffness $[\bar{Q}]_k$ (Section 22.7), so stress jumps at every ply interface,

$$\boldsymbol{\sigma}_k(z) = [\bar{Q}]_k \left(\boldsymbol{\varepsilon}^0 + z\,\boldsymbol{\kappa}\right)$$

Force and moment resultants; the ABD matrix

Integrating stresses through the thickness h gives force resultants $\mathbf{N}$ (force per unit width, N/mm) and moment resultants $\mathbf{M}$ (moment per unit width, N mm/mm):

$$\mathbf{N} = \int_{-h/2}^{h/2} \boldsymbol{\sigma}\, dz, \qquad \mathbf{M} = \int_{-h/2}^{h/2} \boldsymbol{\sigma}\, z\, dz$$

Substituting the ply stresses and summing over plies with interfaces at z_{k-1} and z_k gives the constitutive law of the laminate:

$$\begin{pmatrix} \mathbf{N} \\ \mathbf{M} \end{pmatrix} = \begin{pmatrix} [A] & [B] \\ [B] & [D] \end{pmatrix} \begin{pmatrix} \boldsymbol{\varepsilon}^0 \\ \boldsymbol{\kappa} \end{pmatrix}$$

$$A_{ij} = \sum_k (\bar{Q}_{ij})_k (z_k - z_{k-1}), \quad B_{ij} = \tfrac{1}{2}\sum_k (\bar{Q}_{ij})_k (z_k^2 - z_{k-1}^2), \quad D_{ij} = \tfrac{1}{3}\sum_k (\bar{Q}_{ij})_k (z_k^3 - z_{k-1}^3)$$

$[A]$ is the **extensional stiffness**, $[D]$ the **bending stiffness**, and $[B]$ the **coupling stiffness** between them. The 6×6 **ABD matrix** is the central object of composite design: every finite-element shell element for composites computes it.

> **Status of the principle: Classical laminate theory**
> *Approximate*. Exact within linear elasticity for a thin plate under the Kirchhoff assumptions and perfect bonding; it ignores transverse shear deformation (important for thick laminates and sandwich panels, handled by first-order shear deformation theory) and cannot predict the interlaminar stresses at free edges (Section 23.3).

Figure 23.1 shows the geometry and resultants.

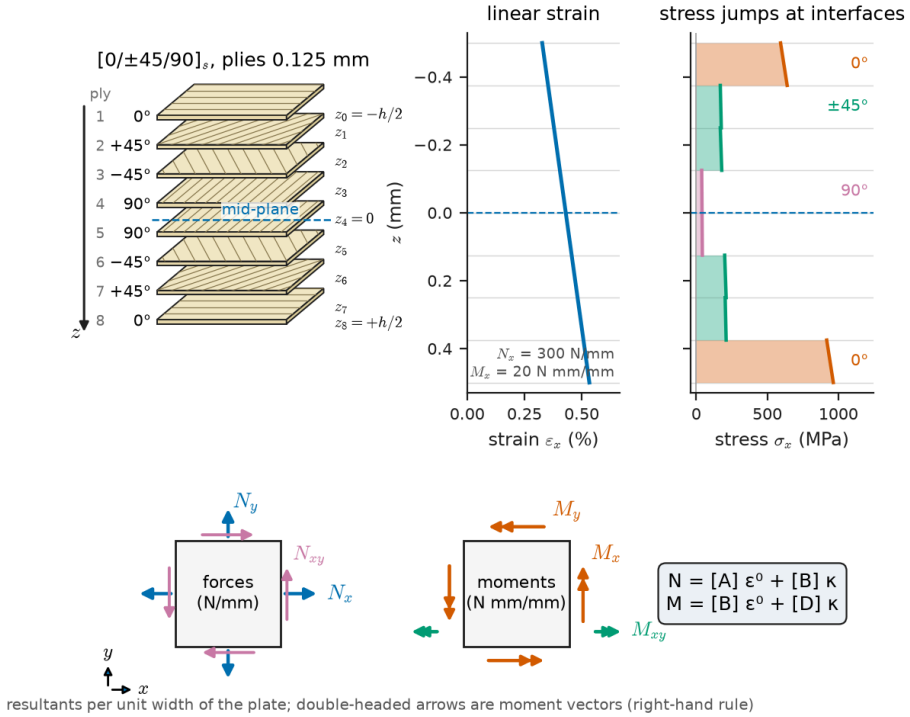

Figure 23.1 — Laminate geometry in classical laminate theory: ply numbering and z-coordinates, linear strain and piecewise-constant stress through the thickness, and the force and moment resultants.

Coupling and how to avoid it

The terms of $[A]$, $[B]$ and $[D]$ that would be zero for an isotropic plate describe coupling between deformation modes:

- A_{16}, A_{26}: **extension–shear** coupling. A tensile N_x produces shear strain γ^0_{xy}.
- B_{ij}: **extension–bending** coupling. In-plane loads, including thermal loads, produce bending and twisting.
- D_{16}, D_{26}: **bend–twist** coupling. A bending moment produces twist.

Three layup rules control them:

- A **symmetric** laminate (ply angles and materials mirrored about the mid-plane) has $[B] = 0$. Every term in the B sum from a ply above the mid-plane is cancelled by its mirror image below. This is why the operator who laid $0/90/0/90$ instead of $0/90/90/0$ got a curled panel: that sequence is unsymmetric, and the thermal contraction on cool-down from cure bent it.
- A **balanced** laminate (every $+\theta$ ply has a $-\theta$ ply of the same material and thickness, anywhere in the stack) has $A_{16} = A_{26} = 0$, because $\bar{Q}_{16}$ and $\bar{Q}_{26}$ are odd functions of θ.
- D_{16} and D_{26} vanish only if each $+\theta$ ply has a $-\theta$ ply at the same distance from the mid-plane on the same side, which is impossible in a symmetric laminate. They can be made small by interleaving many $\pm\theta$ pairs, or eliminated by antisymmetric layups that sacrifice $B = 0$. Most aerospace laminates are symmetric and balanced and accept small bend–twist terms.

A laminate with equal stiffness in every in-plane direction — **quasi-isotropic** — results from $N \geq 3$ ply angles spaced $180^\circ/N$ apart in equal proportions, for example $[0/\pm 60]_s$ or $[0/\pm 45/90]_s$. Its $[A]$ is isotropic; its $[D]$ is not, because the outer plies dominate bending.

> **Status of the principle: Symmetric and balanced layup rules**
> *Derived.* Direct consequences of the definitions of $[A]$ and $[B]$ and the parity of the transformed stiffnesses; they hold exactly within CLT and are the reason aerospace design manuals require symmetric, balanced laminates unless coupling is wanted.

Figure 23.2 shows what the coupling terms do to a flat plate.

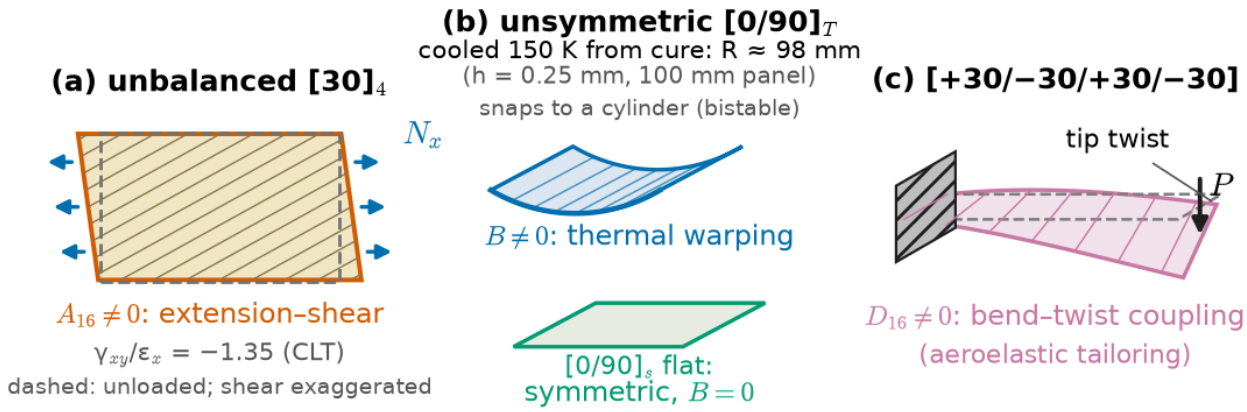

Figure 23.2 — Coupling effects in laminates: extension–shear in an unbalanced laminate, extension–bending and thermal warping of an unsymmetric laminate, and bend–twist coupling.

Worked example — ABD matrices of three carbon/epoxy laminates. Use T300/epoxy plies (Table 22.3: $Q_{11} = 181.8$, $Q_{22} = 10.35$, $Q_{12} = 2.90$, $Q_{66} = 7.17$ GPa) of thickness 0.125 mm.

Cross-ply $[0/90]_s$, $h = 0.5$ mm. Each orientation occupies 0.25 mm, so $A_{11} = A_{22} = 0.25 \times (181.8 + 10.35) = 48.0 \times 10^3$ N/mm, $A_{12} = 0.5 \times 2.90 = 1.45 \times 10^3$ N/mm, $A_{66} = 0.5 \times 7.17 = 3.59 \times 10^3$ N/mm, and $A_{16} = A_{26} = 0$, $[B] = 0$. Inverting $[A]$ gives the effective in-plane moduli $E_x = 1/(h\,a_{11})$ etc. (with $a = A^{-1}$): $E_x = E_y = 96.0$ GPa, $G_{xy} = 7.17$ GPa and $\nu_{xy} = 0.030$. The cross-ply is stiff along both axes but no stiffer in shear than a single ply. The bending stiffnesses are very different: $D_{11} = 1671$ N mm but $D_{22} = 331$ N mm, because the 0° plies are on the outside.

Quasi-isotropic $[0/\pm 45/90]_s$, $h = 1.0$ mm. The calculation gives $A_{11} = A_{22} = 76.4 \times 10^3$, $A_{12} = 22.6 \times 10^3$, $A_{66} = 26.9 \times 10^3$ N/mm, hence $E_x = E_y = 69.7$ GPa, $G_{xy} = 26.9$ GPa and $\nu_{xy} = 0.296$. Check: $E/[2(1+\nu)] = 69.7/2.592 = 26.9$ GPa, as for an isotropic material. A quasi-isotropic CFRP laminate is as stiff as aluminium (70 GPa) at 58 % of its density (1.60 against 2.70 g/cm³). But its bending stiffnesses are $D_{11} = 10{,}690$, $D_{22} = 2{,}653$ and $D_{16} = D_{26} = 670$ N mm — the outer 0° plies dominate, and the bend–twist terms are 6 % of D_{11}.

Unsymmetric $[0/90]_T$, $h = 0.25$ mm. Now $B_{11} = -1340$ N and $B_{22} = +1340$ N: stretching causes bending. Cooling from a stress-free temperature 150 K above room temperature (with $\alpha_1 = 0.02$ and $\alpha_2 = 22.5$ ppm/K) produces thermal force and moment resultants; solving the full ABD system gives curvatures $\kappa_x = -0.0102$ and $\kappa_y = +0.0102$ mm^{-1}: a radius of curvature of 98 mm, and a 100 mm wide panel deflects about 13 mm at mid-span. In practice such thin unsymmetric plates snap into one of two cylindrical shapes rather than a saddle — they are **bistable**, a property exploited in morphing structures.

Thermal and hygral loads in laminates

Each ply wants its free thermal strain $\boldsymbol{\alpha}_k \Delta T$ (Section 22.8), but the plies are bonded. CLT handles this by defining thermal resultants

$$\mathbf{N}^T = \sum_k [\bar{Q}]_k \boldsymbol{\alpha}_k \Delta T\,(z_k - z_{k-1}), \qquad \mathbf{M}^T = \tfrac{1}{2}\sum_k [\bar{Q}]_k \boldsymbol{\alpha}_k \Delta T\,(z_k^2 - z_{k-1}^2)$$

which are added to the mechanical loads. The laminate deforms to a compromise strain, and each ply is left with a **residual stress** $[\bar{Q}]_k(\boldsymbol{\varepsilon}^0 + z\boldsymbol{\kappa} - \boldsymbol{\alpha}_k \Delta T)$. Moisture enters identically, with $\boldsymbol{\beta}\Delta M$ in place of $\boldsymbol{\alpha}\Delta T$. The laminate CTE follows from the same equations with $\mathbf{N} = \mathbf{0}$: for T300/epoxy both $[0/90]_s$ and $[0/\pm 45/90]_s$ have $\alpha_x = \alpha_y = 1.5$ ppm/K, while a 0°-dominated $[0/\pm 30/0]_s$ has $\alpha_x = -1.0$ ppm/K along its axis — the basis of near-zero-expansion structures for satellites and optical instruments.

23.2 Failure criteria and progressive ply failure

Why composite failure is hard

A metal yields by one mechanism, and an isotropic criterion (Tresca or von Mises, Chapter 13) handles any combination of stresses. A ply has at least five strengths (X_t, X_c, Y_t, Y_c, S) that differ by factors of up to 40, and several physically distinct failure modes: fibre fracture in tension, fibre kinking in compression, matrix cracking in transverse tension, shear failure, and matrix crushing in transverse compression. A laminate fails progressively: one ply cracks, its load is redistributed, another fails, and so on. Failure criteria are applied **ply by ply**, in material axes, to the stresses from CLT.

Maximum stress and maximum strain

The simplest criterion compares each stress component with its strength separately: failure when any of

$$\frac{\sigma_1}{X_t \text{ or } X_c}, \qquad \frac{\sigma_2}{Y_t \text{ or } Y_c}, \qquad \frac{|\tau_{12}|}{S}$$

reaches 1. The **maximum stress** criterion identifies the failure mode directly, which is its great advantage, but ignores interaction: a ply under combined transverse tension and shear fails earlier than either alone. The **maximum strain** criterion does the same with strains and differs slightly through Poisson effects; it is widely used in aerospace practice, where allowable strains (for example 3,000–4,000 microstrain in compression after impact, Section 23.4) are the design currency.

Tsai–Hill

Tsai–Hill adapts Hill's anisotropic yield criterion (Chapter 13) to a transversely isotropic ply under plane stress:

$$\frac{\sigma_1^2}{X^2} - \frac{\sigma_1\sigma_2}{X^2} + \frac{\sigma_2^2}{Y^2} + \frac{\tau_{12}^2}{S^2} = 1$$

with X and Y chosen as tensile or compressive according to the sign of the stress. It is a single smooth function, gives good predictions for off-axis strength, but cannot say which mode failed and treats unlike mechanisms (fibre fracture, matrix cracking) as if they interacted like components of one yield process. For a T300/epoxy ply loaded uniaxially at an angle θ, substituting $\sigma_1 = \sigma c^2$, $\sigma_2 = \sigma s^2$, $\tau_{12} = -\sigma sc$ gives strengths of 1,500 MPa at 0°, 690 at 5°, 245 at 15°, 112 at 30°, 69 at 45° and 40 MPa at 90°: the ply loses 84 % of its strength at only 15° off-axis.

Tsai–Wu

Tsai–Wu writes a general quadratic polynomial in stress, the most general form invariant under coordinate rotation:

$$F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2 = 1$$

with $F_1 = 1/X_t - 1/X_c$, $F_{11} = 1/(X_tX_c)$, $F_2 = 1/Y_t - 1/Y_c$, $F_{22} = 1/(Y_tY_c)$, $F_{66} = 1/S^2$. The linear terms capture the different tensile and compressive strengths. The interaction coefficient F_{12} should come from a biaxial test, which is difficult; in practice $F_{12} = -\frac{1}{2}\sqrt{F_{11}F_{22}}$ is used, which makes the failure surface an ellipsoid resembling von Mises. The **strength ratio** R, the factor by which the applied stresses must be multiplied to reach failure, is the positive root of $aR^2 + bR - 1 = 0$, where a collects the quadratic and b the linear terms.

> **Status of the principle: Tsai–Hill and Tsai–Wu criteria**
> *Empirical.* Phenomenological surfaces fitted to uniaxial (and occasionally biaxial) strengths, with no reference to failure mechanism; they work well in tension-dominated regimes and for off-axis plies but can be non-conservative under biaxial compression and give spurious interaction between fibre and matrix modes. The World-Wide Failure Exercise (1996–2012) found none of the classical criteria reliable in every case.

Puck: mechanism-based criteria

The **Puck** criterion, the best performer in the World-Wide Failure Exercise and the basis of VDI 2014 guidelines, separates the mechanisms. **Fibre fracture** is checked with a fibre-direction criterion (essentially maximum stress or strain in σ_1, corrected for transverse stresses). **Inter-fibre fracture** (matrix cracking) is treated as Mohr–Coulomb-type fracture on an inclined **action plane** parallel to the fibres: the criterion searches the angle of that plane, and the normal and shear stresses on it decide

failure. Transverse compression raises the shear strength on the plane (friction) instead of lowering it, and fracture is classified into **modes A, B and C**. Mode C, with fracture planes inclined at about 50° under high transverse compression, produces wedge-shaped fragments that can split adjacent plies, and is flagged as particularly dangerous. **LaRC** criteria (NASA) extend the same physical approach to fibre kinking. Mechanism-based criteria are more work to implement but tell the designer what will fail, which is what damage-tolerance and progressive-failure analysis need.

Figure 23.3 compares the envelopes.

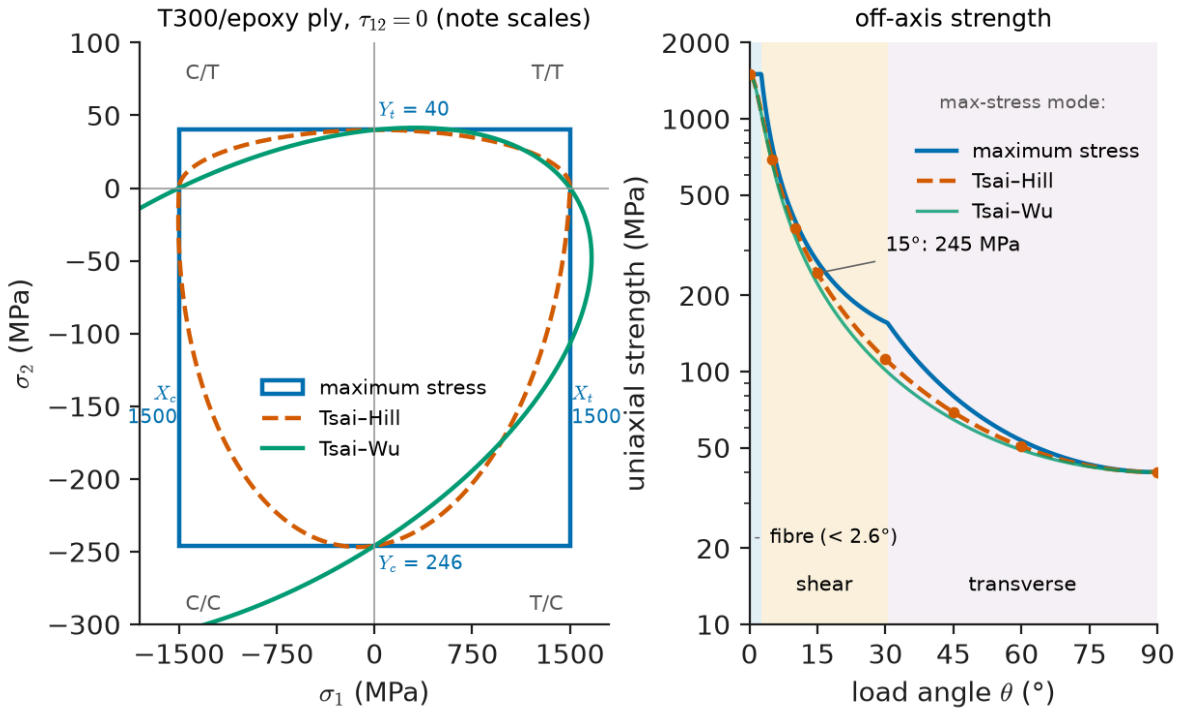

Figure 23.3 — Failure envelopes of a T300/epoxy ply in the σ1–σ2 plane (τ12 = 0) and off-axis uniaxial strength versus angle, for the maximum stress, Tsai–Hill and Tsai–Wu criteria.

First-ply failure, last-ply failure and residual stress

In a laminate the first ply to fail rarely ends the laminate's life. **First-ply failure** (FPF) is usually transverse matrix cracking in the plies most nearly perpendicular to the load. The cracks multiply, the cracked plies lose transverse and shear stiffness, and their load is shed to neighbours. **Last-ply failure** (LPF) is fibre fracture in the load-aligned plies. Progressive failure analysis follows this sequence by **ply discount**: when a ply fails in a matrix mode its E_2 and G_{12} are reduced (to zero or to a small fraction), the ABD matrix is recomputed and the load increased again. Continuum damage mechanics replaces the sudden discount with gradual stiffness degradation driven by damage variables, and is the basis of modern finite-element progressive damage models.

Worked example — First- and last-ply failure of a cross-ply. A T300/epoxy $[0/90]_s$ laminate ($h = 0.5$ mm) carries a tensile load N_x. Under $N_x = 1$ N/mm (average stress 2 MPa), CLT gives ply stresses of $\sigma_1 = 3.786$, $\sigma_2 = 0.054$ MPa in the 0° plies and $\sigma_1 = -0.054$, $\sigma_2 = 0.214$ MPa in the 90° plies.

Without residual stress. The 90° plies crack in transverse tension when $0.214 N_x = Y_t = 40$ MPa, so $N_x = 187$ N/mm, an average laminate stress of 374 MPa (Tsai–Wu gives 373 MPa). The 0° plies reach X_t at $N_x = 1500/3.786 = 396$ N/mm (792 MPa).

With cure residual stress. The laminate was cured at 177 °C; take the stress-free temperature as 150 K above room temperature. Cooling leaves every ply with $\sigma_1 = -31.9$ MPa and $\sigma_2 = +31.9$ MPa: the plies want to shrink across their fibres by $22.5 \times 150 =$ 3,375 microstrain but are held by their neighbours' fibres. The residual transverse stress alone uses 80 % of Y_t. The 90° plies now crack when $31.9 + 0.214 N_x = 40$, at $N_x = 37.9$ N/mm — an average stress of only 76 MPa (Tsai–Wu: 73 MPa), a fifth of the value without residual stress.

After first-ply failure. The 90° plies are discounted; the laminate modulus drops only slightly (from 96 to about 91 GPa), because the 90° plies contributed little along x. The laminate keeps carrying load until the 0° plies break at a fibre strain of about $1500/181{,}000 = 0.83$ %, an average stress of roughly 750–790 MPa. So the laminate's first-ply failure is at about a tenth of its ultimate strength.

Two lessons follow. First, residual cure stresses are not a correction but often the largest single term in matrix-cracking predictions; that is why aerospace resins were toughened in the 1980s and why thin plies, which resist transverse cracking better than thick ones (the **in-situ strength** effect, which can raise effective Y_t by 50–100 % in plies 0.1 mm thick), are favoured. Second, matrix cracks are not structurally fatal in tension, but they open paths for moisture and fuel, lower compressive and shear stiffness, and initiate delaminations; pressure vessels and fuel tanks are therefore often designed to first-ply failure, primary structure to ultimate. Figure 23.4 shows the load–strain record.

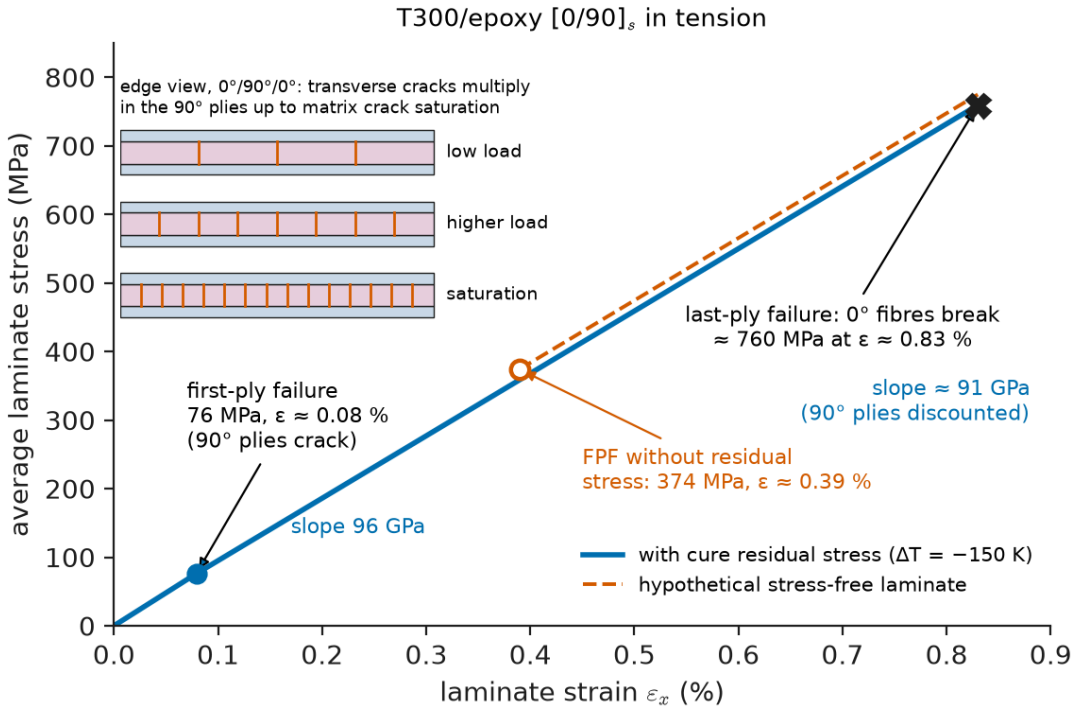

Figure 23.4 — Progressive failure of a T300/epoxy [0/90]s laminate in tension, with and without cure residual stress.

23.3 Interlaminar stresses, free edges and delamination

What CLT cannot see

CLT assumes plane stress in each ply: $\sigma_z = \tau_{xz} = \tau_{yz} = 0$. At a **free edge** — the side of a coupon, the bore of a hole, a cut-out, a ply drop, the end of a bonded doubler — this cannot hold. Consider a $[\pm 45]_s$ coupon in tension. CLT says each ply carries an in-plane shear stress τ_{xy} of opposite sign in the $+45$ and -45 plies. At the free edge $y = \pm b$, that shear must vanish (the edge is traction-free). The only way for the ply to shed it is to transfer it, through **interlaminar shear stress** τ_{xz}, to the neighbouring ply over a **boundary layer** about one laminate thickness wide. Pipes and Pagano (1970) showed by finite differences that this interlaminar stress is singular at the edge of the interface in the elastic solution.

The same logic applies to Poisson mismatch. In a $[0/90]_s$ laminate in tension the 0° plies want to contract across their width far more ($\nu_{12} = 0.28$) than the 90° plies ($\nu_{21} = 0.016$). In the interior they are forced to a common strain; at the free edge the forces that enforce it must be transmitted by interlaminar stresses, and they also produce a moment that gives an **interlaminar normal stress** σ_z — peeling stress — at the mid-plane.

Status of the principle: Free-edge interlaminar stresses
Derived. A consequence of three-dimensional elasticity at a traction-free boundary of a laminate with ply-to-ply mismatch in Poisson's ratio or in shear coupling; the elastic solution is singular at the interface corner, so design uses averaged stresses over a characteristic distance or fracture mechanics rather than point values.

Stacking sequence controls the sign of σ_z

The interlaminar normal stress depends on the stacking sequence, not just on the plies present. For a laminate containing ±θ and 90° plies loaded in tension, the outer ±θ plies want to contract across the width far more than the 90° plies (a ±30° carbon/epoxy pair has an effective Poisson's ratio above 1). In the sequence $[\pm\theta/90]_s$ the resulting edge moment produces tensile σ_z at the mid-plane; moving the 90° plies outside, $[90/\pm\theta]_s$, reverses it to compression. Tensile peel stress opens delaminations; compressive stress suppresses them. The two layups are identical in CLT, yet the first delaminates at much lower load — so reliably that the standard edge-delamination test uses a $[\pm 30/\pm 30/90/\bar{90}]_s$ coupon precisely to make the mid-plane delaminate under tension. Design rules follow: avoid grouping many plies of the same angle (which raises interlaminar stresses and lowers in-situ strength), avoid blocks of 90° plies at the mid-plane beneath ±θ plies in tension-loaded laminates, terminate dropped plies gradually (ply-drop ratios of 1:20 or more), and protect free edges with edge caps, stitching or tapering.

Delamination as a fracture problem

Delamination — separation between plies — is the most characteristic damage mode of laminated composites, because the interlaminar region is pure matrix with no fibres crossing it. It is treated by fracture mechanics (Chapter 14) with the energy release rate G partitioned into **mode I** (opening), **mode II** (in-plane shear) and **mode III** (tearing). Interlaminar toughness is measured with standard tests: the double-cantilever beam (DCB) for G_{Ic}, the end-notched flexure (ENF) for G_{IIc} and the mixed-mode bending (MMB) test for combinations.

Table 23.1 — Interlaminar fracture toughness of carbon-fibre laminates (typical values).

Matrix	G_{Ic} (kJ/m²)	G_{IIc} (kJ/m²)	Typical use
Untoughened epoxy (1970s, e.g. 5208 type)	0.08–0.2	0.3–0.6	Early military aircraft
Toughened epoxy (thermoplastic-modified, interleaved)	0.3–0.6	1.0–2.5	787/A350 primary structure
Bismaleimide	0.15–0.4	0.5–1.2	High-temperature fighter structure
PEEK/PEKK (thermoplastic)	1.2–2.0	1.5–2.5	Brackets, clips, welded structures

Mode II toughness is several times mode I, which is why delaminations grow most easily under peel and why bonded and co-cured joints are designed to minimize peel stress. Numerically, delamination growth is predicted with the **virtual crack closure technique** (VCCT) or with **cohesive-zone elements** placed between plies, whose traction–separation law is calibrated so that its area equals G_c. Through-thickness reinforcement — stitching, tufting, z-pins, and 3D weaving — raises G_{Ic} several-fold by bridging the crack, at some cost in in-plane properties.

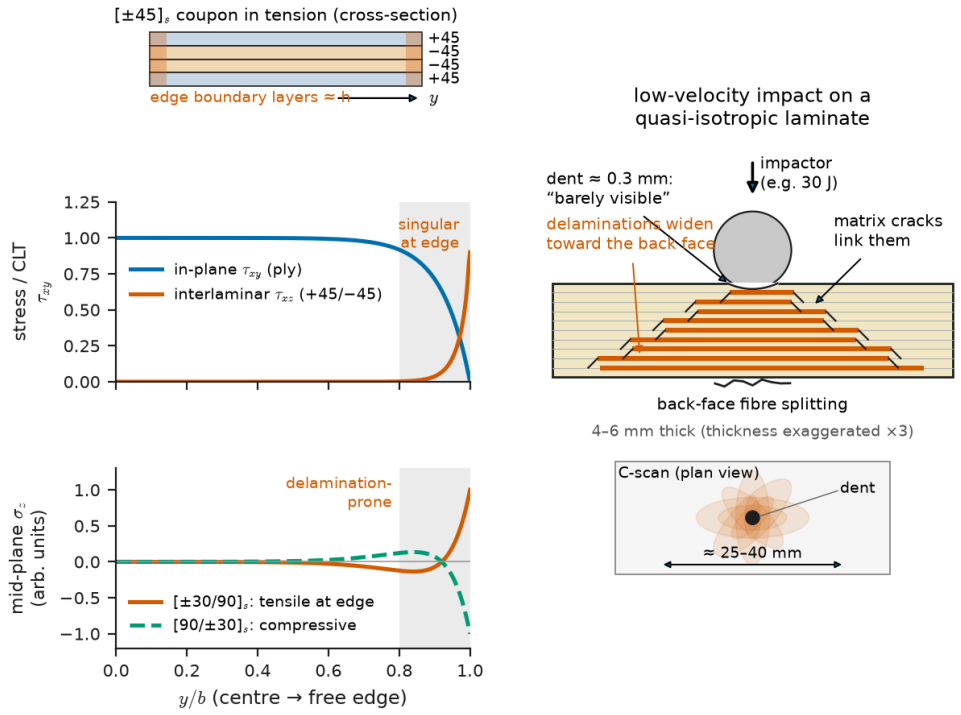

Figure 23.5 — Interlaminar stresses near a laminate free edge, and the internal damage under a low-velocity impact.

23.4 Impact damage and compression after impact

Barely visible impact damage

A metal panel hit by a dropped tool dents and the dent is visible. A carbon/epoxy panel hit with the same energy may show almost nothing on the surface, while inside, a cone of delaminations has formed (Figure 23.5). The mechanism: the contact load bends the laminate locally, shear and bending stresses crack the matrix in the plies, the matrix cracks reach ply interfaces where bending-stiffness mismatch between differently oriented plies drives delaminations, and the delaminations spread outward, each shaped by the orientation of the ply beneath it. The back face may split.

Certification rules define **barely visible impact damage** (BVID), typically a dent 0.25–0.5 mm deep detectable in a general visual inspection, or the damage from an impact energy cutoff (often 35–50 J for aircraft structure, or about 6.7 J per mm of thickness in standard coupon tests). Structure containing BVID must carry ultimate load for the life of the aircraft, because it may never be found. Larger, visible damage must be found and repaired within an inspection interval.

Compression after impact

Delaminations barely reduce tensile strength, but they are devastating in compression. A delaminated region splits the laminate into thin sublaminates that buckle locally at loads far below the buckling load of the intact plate; the buckled sublaminates drive the delaminations to grow, and the laminate fails. The standard measure is the **compression-after-impact** (CAI) strength from ASTM D7137 or equivalent, on a 150 × 100 mm plate impacted at a specified energy.

Table 23.2 — Compression after impact of carbon-fibre quasi-isotropic laminates (typical, about 30 J impact on 4–5 mm plate).

System	Undamaged compressive strength (MPa)	CAI strength (MPa)	Retention
Untoughened epoxy	500–600	130–180	25–35 %
Toughened epoxy (current aerospace)	550–700	250–320	40–55 %
Thermoplastic (PEEK-type)	550–650	280–350	50–60 %

These numbers explain why compression after impact, not static strength, usually sets the design allowables of composite aircraft structure: allowable strains of about 3,000–4,000 microstrain, against a fibre failure strain of 15,000. The improvements came from the matrix and interleaf, not the fibre: toughened resins and interleaved thermoplastic particles raise G_{IIc} and shrink the delaminated area. Damage resistance (how little damage forms) and damage tolerance (how much strength remains) are separate properties; sandwich panels, thin skins and helicopter blades are typically impact-limited, and armour applications exploit the very energy absorption by delamination that aircraft designers try to avoid.

> **Status of the principle: Compression-after-impact allowables**
> *Empirical.* Allowable strains and CAI strengths are established by standard coupon and element tests for each material and layup; analytical models (sublaminate buckling, soft-inclusion models) explain the trends but are not yet trusted for certification without test data.

23.5 Sandwich structures

Why separate the faces

Bending stiffness of a plate scales with the cube of its thickness. A **sandwich panel** exploits this by separating two thin, stiff **face sheets** with a thick, light **core**, exactly as the flanges of an I-beam are separated by its web. The faces carry bending as tension and compression; the core carries transverse shear and keeps the faces apart and stable. Cores are honeycombs (aluminium, Nomex aramid paper, thermoplastic), closed-cell foams (PVC, PET, PMI, polyurethane), and end-grain balsa. Applications include aircraft floors, flaps and control surfaces, helicopter rotor blades, boat hulls and decks, wind-turbine blade shells, racing-car monocoques, train and bus panels, and building cladding.

For a sandwich beam of unit width with identical faces of thickness t_f and modulus E_f, core thickness c, and distance between face centroids $d = c + t_f$, the flexural rigidity per unit width is dominated by the faces:

$$D \approx \frac{E_f t_f d^2}{2}$$

and the transverse shear stiffness by the core:

$$S \approx \frac{G_c d^2}{c}$$

The deflection of a simply supported beam under a central load P (per unit width) over span L has a bending and a shear part, $w = PL^3/(48D) + PL/(4S)$. The shear term is negligible in solid beams but can dominate in short, thick sandwich beams with soft cores.

> **Status of the principle: Sandwich beam theory**
> *Approximate.* Assumes thin faces carrying only membrane stress, a core carrying only transverse shear, and no core compression; it is accurate for $c/t_f \gtrsim 5$ and stiff cores, and is extended by high-order sandwich theories when cores are very soft or loads are concentrated.

Worked example — Sandwich versus solid plate of equal mass. Faces are quasi-isotropic CFRP ($E_f = 70$ GPa, $\rho = 1600$ kg/m³), 1.0 mm thick each; the core is aluminium honeycomb 20 mm thick, density 50 kg/m³, with ribbon-direction shear modulus $G_c = 310$ MPa.

Mass per area: $2 \times 0.001 \times 1600 + 0.020 \times 50 = 3.2 + 1.0 = 4.2$ kg/m².

Sandwich rigidity, with $d = 21$ mm: $D = 70{,}000 \times 1.0 \times 21^2/2 = 1.54 \times 10^7$ N mm per mm width.

A solid quasi-isotropic CFRP plate of the same mass is $4.2/1600 = 2.63$ mm thick, with $D = Et^3/12 = 70{,}000 \times 2.63^3/12 = 1.06 \times 10^5$ N mm per mm. The sandwich is **146 times stiffer** in bending at the same weight.

Deflection of a 1 m span under $P = 1$ N/mm: bending $10^9/(48 \times 1.54 \times 10^7) = 1.35$ mm; core shear $S = 310 \times 21^2/20 = 6{,}836$ N/mm, so shear deflection $1000/(4 \times 6836) = 0.037$ mm. The solid plate would deflect 197 mm. At a 0.3 m span the shear part rises to 23 % of the total.

Strength: the face stress is $M/(t_f d) = PL/(4t_f d) = 11.9$ MPa per N/mm of load, and the core shear stress is $P/(2d) = 0.024$ MPa per N/mm. Against a face compressive allowable of 500 MPa and a core shear strength of about 2.2 MPa, the panel fails by face failure at $P \approx 42$ N/mm before the core shears at 92 N/mm.

Sandwich failure modes

A sandwich can fail in more ways than a solid beam: **face yielding or fracture**; **core shear failure**; **face wrinkling**, a short-wavelength local buckling of the compressed face on its elastic core, approximately at

$$\sigma_{\text{wr}} \approx 0.5\,(E_f E_c G_c)^{1/3}$$

(about 280 MPa for CFRP faces on an 80 kg/m³ PVC foam with $E_c = 97$ MPa and $G_c = 27$ MPa — below the face strength, so foam-cored panels are often wrinkling-limited); **intracell dimpling** of thin faces over large honeycomb cells; **core crushing** under local loads; and **face–core debonding**, which is a delamination problem with the added hazard of water ingress into honeycomb. Inserts, edge closures and ply ramps are where most sandwich structures fail in practice. Figure 23.6 summarizes the construction and the modes.

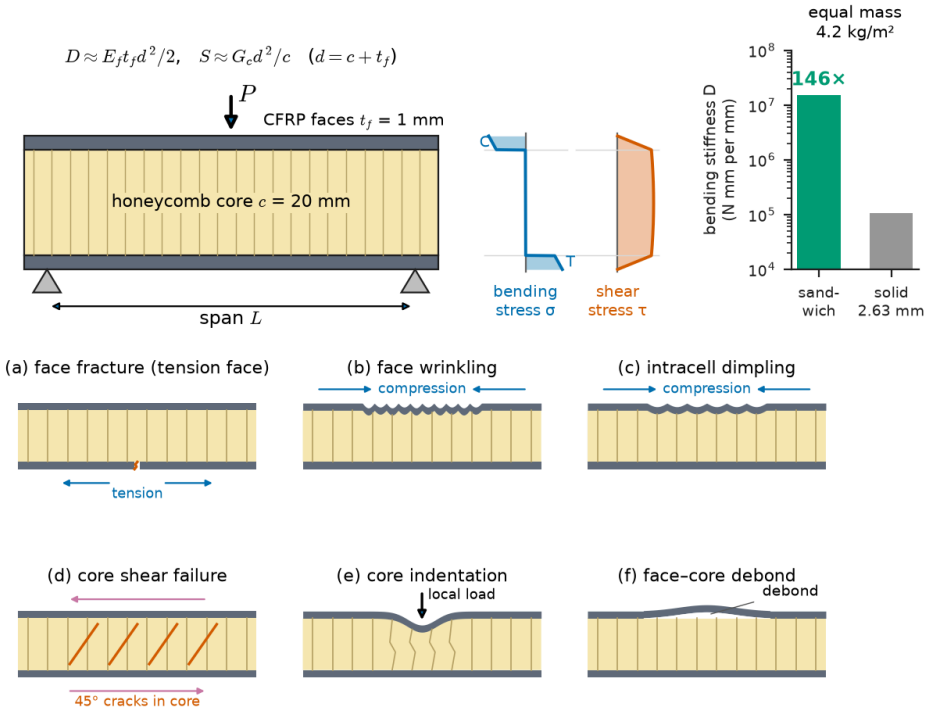

Figure 23.6 — A sandwich beam in three-point bending and its failure modes.

23.6 Manufacturing polymer-matrix composites

What processing must achieve

Every composite process must do four things: place fibres in the right orientation, impregnate them fully with matrix, consolidate the stack to the target fibre volume fraction with minimal voids, and cure or solidify the matrix. Each process strikes a different balance between part size, quality, rate and cost. The processing defects that matter most are **voids** (each 1 % of porosity costs about 5–10 % of interlaminar shear strength), **fibre waviness and wrinkles** (which cut compressive strength, Section 22.3), **resin-rich and resin-starved regions**, **incomplete cure**, **residual stress and distortion** (spring-in of angled parts, typically 0.5–2° for carbon/epoxy), and **dry spots** in liquid-moulded parts.

Open-mould processes

Hand lay-up and spray-up are the oldest: dry fabric or chopped glass is placed on a single-sided mould and wetted out with polyester or vinyl ester by roller or spray gun, and cured at room temperature. Tooling is cheap and part size is unlimited (boat

hulls, swimming pools, tanks), but V_f is only 0.20–0.40, quality depends on the operator, and styrene emissions are a health and environmental concern. **Vacuum bagging** consolidates a wet lay-up under atmospheric pressure, and **vacuum infusion** (VARTM) draws resin through a dry preform under a bag — the standard process for wind-turbine blades, boat hulls and bridge decks, achieving V_f of 0.50–0.58 with low emissions.

Prepreg and autoclave

Prepreg is fibre pre-impregnated with partially reacted (B-staged) resin, supplied as tape or fabric on rolls and stored frozen. Plies are cut, laid up on a tool, bagged and cured in an **autoclave** — a heated pressure vessel — typically at 120–180 °C and 0.6–0.7 MPa for 2–6 hours. The pressure collapses voids and suppresses volatile nucleation; the result is V_f of 0.55–0.65 and porosity below 1 %, the reference quality for primary aircraft structure. The costs are high: autoclaves for wing skins exceed 30 m in length, frozen-storage and out-time limits apply, and cycle times are long. **Out-of-autoclave** prepregs, designed with dry channels for air evacuation, achieve near-autoclave quality under vacuum alone.

Liquid moulding: RTM

Resin transfer moulding (RTM) injects resin into a dry fibre preform held in a closed, matched mould. Flow through the preform follows **Darcy's law**,

$$\mathbf{u} = -\frac{K}{\mu}\nabla p$$

with $\mathbf{u}$ the superficial velocity, K the preform permeability (typically 10^{-11}–10^{-9} m²), μ the resin viscosity and p the pressure. For one-dimensional filling from a line gate at constant injection pressure Δp, a flow front advances as $\sqrt{t}$ and the fill time for a length L is

$$t_{\text{fill}} = \frac{\phi \mu L^2}{2 K \Delta p}$$

where $\phi = 1 - V_f$ is the preform porosity.

> **Status of the principle: Darcy's law for preform impregnation**
> *Empirical.* A linear relation between flow rate and pressure gradient for slow viscous flow through porous media, confirmed for fibre preforms at the low Reynolds numbers of moulding; the permeability is anisotropic, depends strongly on V_f (Kozeny–Carman) and on preform shearing, and capillary and dual-scale (intra-tow) flow cause deviations that produce voids.

Worked example — How long does a mould take to fill? An RTM panel 1 m long with $V_f = 0.5$ ($\phi = 0.5$), preform permeability $K = 1 \times 10^{-10}$ m², resin viscosity 0.1 Pa s and injection pressure 0.5 MPa: $t = 0.5 \times 0.1 \times 1^2 / (2 \times 10^{-10} \times 5 \times 10^5) = 500$ s, about 8 minutes. The same preform infused under vacuum ($\Delta p = 0.1$ MPa) with a resin of 0.2 Pa s and a more permeable flow medium ($K = 2 \times 10^{-10}$ m²) takes 2,500 s; a 3 m flow length takes 9× longer, 22,500 s or more than 6 hours — longer than the pot life of many resins. The L^2 scaling is why large infusions use distribution media and multiple gates, and why fast-curing automotive RTM uses high pressure (5–15 MPa) and very low viscosity resins to fill in under a minute.

Continuous processes: filament winding and pultrusion

Filament winding draws resin-wet fibre tows onto a rotating mandrel along programmed paths, producing pipes, drive shafts, rocket-motor cases and pressure vessels — including Type IV hydrogen tanks for fuel-cell vehicles, which wind carbon fibre over a polymer liner to 70 MPa. For a closed cylinder under internal pressure, the hoop stress is twice the axial stress; fibres alone carry both if wound at the **netting angle** θ satisfying $\tan^2\theta = 2$, that is $\theta = 54.7^\circ$ from the axis. Real vessels combine near-hoop and helical layers to the same effect.

Pultrusion pulls continuous fibres through a resin bath and a heated die, curing a constant-section profile continuously at 0.2–3 m/min: ladder rails, bridge decks, rebar, wind-blade spar caps, window frames. It achieves the highest V_f (up to 0.70) and the lowest cost per kilogram of continuous-fibre composite, but the fibres are mostly axial and the section must be constant.

Automated fibre placement and tape laying

Automated tape laying (ATL) and **automated fibre placement** (AFP) use robotic heads to lay prepreg tape (75–300 mm wide for ATL) or bundles of narrow tows (3–13 mm, up to 32 tows) onto a tool, cutting and restarting each tow individually. AFP can steer tows along curved paths and follow doubly curved surfaces, enabling **variable-stiffness** laminates. Deposition rates of 20–100 kg/h made the composite fuselages and wings of current airliners possible. Defects specific to AFP — gaps, overlaps, twisted tows and wrinkles at steered paths — are monitored in-line. With thermoplastic tows and laser heating, AFP can consolidate in situ without an autoclave.

Table 23.3 — Polymer-composite manufacturing processes (typical values).

Process	Fibre form	V_f	Void content (%)	Cycle time	Typical products
Hand lay-up / spray-up	Mat, woven fabric, chopped	0.20–0.40	2–5	hours	Boat hulls, tanks, pools
Vacuum infusion	Fabrics, non-crimp fabrics	0.50–0.58	1–2	hours	Wind blades, yachts, bridge decks
Prepreg / autoclave	UD tape, fabric prepreg	0.55–0.65	<1	4–12 h	Aircraft wings and fuselages, F1 chassis
RTM / HP-RTM	Dry preform	0.45–0.60	0.5–2	1–60 min	Car structural parts, aero brackets, fan blades
Compression moulding (SMC)	Chopped (25 mm)	0.20–0.35	1–3	1–3 min	Truck cabs, car panels, enclosures
Filament winding	Continuous tow	0.55–0.70	1–3	min–h	Pipes, pressure vessels, shafts
Pultrusion	Continuous roving, mat	0.55–0.70	1–2	continuous	Profiles, rebar, spar caps
AFP / ATL	Prepreg tow or tape	0.55–0.65	<1 (after cure)	hours	Fuselage barrels, wing skins
Injection moulding (short fibre)	Chopped in pellets	0.10–0.25	<1	20–90 s	Housings, intake manifolds, brackets

> **Boundary.**
> Polymer flow, cure kinetics and the rheology of thermoset and thermoplastic matrices are developed in Chapters 19 and 21; here only the composite-specific steps are treated. Recycling of composites is taken up in Chapter 59.

23.7 Metal- and ceramic-matrix composites

Metal-matrix composites

MMCs combine a metallic matrix — mostly aluminium, also magnesium, titanium and copper — with ceramic particles, whiskers or fibres. The goals are higher stiffness, wear resistance, elevated-temperature strength and lower thermal expansion than the alloy alone, while keeping metallic toughness, conductivity and transverse properties that PMCs lack. Matrix-dominated properties (transverse strength, shear, temperature capability) are far better than in PMCs; the price is processing difficulty and lower ductility.

Processing routes are:

- **Stir casting**: SiC or Al_2O_3 particles stirred into molten aluminium and cast. Cheap, but limited to about 20–25 vol% and prone to clustering and reaction ($4Al + 3SiC \rightarrow Al_4C_3 + 3Si$, a brittle, moisture-sensitive carbide; adding Si to the melt suppresses it).

- **Powder metallurgy**: blending alloy and ceramic powders, compacting, and extruding or forging (Chapter 46). Better distribution, up to about 40 vol%.
- **Squeeze casting and pressure infiltration** of a porous ceramic preform with molten metal, reaching 50–70 vol% — the route for AlSiC electronic baseplates.
- **Diffusion bonding** of foil–fibre–foil stacks for continuous-fibre titanium MMCs (SiC monofilament in Ti), used in a few aero-engine components.

Table 23.4 — Representative metal- and ceramic-matrix composites (typical values).

Material	Reinforcement	Density (g/cm³)	E (GPa)	Strength (MPa)	CTE (ppm/K)	Application
Al 6061 (reference)	—	2.70	69	310	23	—
6061/SiC/20p	20 % SiC particles	2.80	100–105	400–500	15–16	Bicycle frames, brake rotors, pistons
AlSiC (infiltrated)	63–70 % SiC	2.95–3.0	190–220	250–350 (flexure)	7–8	Power-electronics baseplates
Ti/SiC (continuous fibre)	35 % SiC monofilament	3.9	210 (axial)	1,500–1,800 (axial)	~6	Aero-engine shafts, rings (niche)
WC–Co (cemented carbide)	85–94 % WC	14–15	550–650	2,000–4,000 (TRS)	5–6	Cutting tools, mining bits
C/C	Carbon fibre	1.6–1.9	30–100	100–300	~1	Aircraft and F1 brakes, re-entry nose caps
C/SiC	Carbon fibre	1.9–2.4	60–100	200–350	2–3	Car brake discs, thrust nozzles
SiC/SiC	SiC fibre, BN interphase	2.5–2.8	200–280	250–350	4–5	Turbine shrouds, combustor liners
Oxide/oxide	Nextel 610/720	2.6–2.9	70–120	150–350	6–8	Exhaust mixers, combustor liners

Ceramic-matrix composites: toughness by design

In a PMC the fibres are there to add stiffness and strength to a weak matrix. In a CMC the matrix is already stiff and strong; the fibres are there to stop it failing catastrophically. A monolithic SiC ceramic has a fracture toughness of 3–4 MPa $m^{1/2}$ and fails from its largest flaw (Chapter 14). A SiC/SiC composite fails gradually, with a non-linear stress–strain curve and a "toughness" equivalent to 20–30 MPa $m^{1/2}$.

The key is a deliberately **weak interface**. Fibres are coated with a thin interphase — pyrolytic carbon or boron nitride, 0.1–1 μm thick — with a layered structure that cleaves easily. When a matrix crack reaches a fibre, the crack deflects along the interphase instead of cutting the fibre (the He–Hutchinson criterion: deflection occurs if the interface toughness is below about a quarter of the fibre toughness). The intact fibres then **bridge** the crack, and as they break at scattered locations, they **pull out** against interfacial friction, absorbing energy. Under tension the matrix cracks first at the **matrix cracking stress** (predicted by Aveston–Cooper–Kelly theory), then more cracks form until they saturate, and the fibres carry the load until they fail — a pseudo-ductile response. This is the same physics as fibre bridging and R-curve toughening in Chapter 14, made into the main mechanism.

Processing routes are **chemical vapour infiltration** (CVI), which deposits SiC within a fibre preform from methyltrichlorosilane gas over weeks, leaving 10–15 % porosity; **polymer infiltration and pyrolysis** (PIP), which repeatedly impregnates and pyrolyses a preceramic polymer; and **melt infiltration** (MI), which infiltrates a carbon-containing preform with molten silicon to react to SiC, giving near-full density but residual free silicon that limits service to about 1,350 °C. **C/C composites**, made by CVI of carbon or by repeated pitch impregnation, combine low density, high thermal conductivity and strength retention to over 2,000 °C in inert atmospheres, which makes them the material of aircraft and Formula 1 brakes and of re-entry nose caps — but they oxidize in air above about 450 °C and need protective coatings. C/SiC brakes now appear on high-performance cars.

SiC/SiC CMCs entered commercial aero-engine service in the 2010s as high-pressure turbine shrouds, with a third of the density of the nickel superalloys they replace and capability about 150–200 K higher, cutting cooling air requirements (Chapter 41). Their limiting issues are oxidation of the interphase through matrix cracks at intermediate temperatures, recession of SiC by water vapour (mitigated by environmental barrier coatings) and cost.

> **Status of the principle: Weak-interface toughening of CMCs**
> *Approximate*. The crack-deflection criterion and the Aveston–Cooper–Kelly matrix-cracking stress follow from fracture mechanics and shear-lag analysis under idealized assumptions (uniform fibre spacing, constant frictional sliding stress, single fibre strength); they predict the trends that guide interphase design, while quantitative design relies on measured stress–strain behaviour.

What remains open

Open question	Where it is answered
How are ply properties obtained from fibre and matrix, and how do they transform with angle?	Chapter 22 (micromechanics, Q and $\bar{Q}$)
What fracture-mechanics concepts (G, K, R-curves, Weibull) underlie delamination and CMC toughening?	Chapter 14 (fracture, toughening, statistics)
How do composites behave under cyclic loading and long-term environmental exposure?	Chapter 14 (fatigue), Chapter 44 (degradation and failure analysis)
How are composite structures joined to each other and to metals?	Chapter 47 (adhesive bonding, mechanical fastening, dissimilar joints)
How are CMCs and C/C used in turbines, hypersonic vehicles and re-entry?	Chapter 41 (extreme environments)
How do moisture, temperature and expansion mismatch interact in thermal design?	Chapter 24 (thermal properties, thermal stress)
How are internal defects (voids, delaminations) detected?	Chapter 52 (laboratory characterization), Chapter 53 (tomography)
How are composite laminates modelled in multiscale simulations?	Chapter 56 (multiscale modelling and ICME)
How are composites chosen against metals for weight-critical parts?	Chapter 58 (materials selection)
How can thermoset composites be recycled at end of life?	Chapter 59 (recycling, circular economy)

Next: Chapter 24 opens Part VI with the thermal properties of materials — how solids store, conduct and expand with heat, and how temperature changes turn into stress.

Part VI — Thermal, Electrical, Dielectric and Magnetic Materials

Parts III to V asked how materials carry load. This part asks how they respond to everything else: heat, electric fields and magnetic fields. A turbine blade survives because a ceramic coating barely conducts heat; a power line works because aluminium conducts electrons well and porcelain does not; a transformer core wastes little energy because its steel magnetizes easily. These **physical and functional properties** are what make materials do things rather than merely hold things up, and they decide the performance of energy systems, electronics, sensors, vehicles, buildings and medical devices.

The organizing idea of the part is the same as in the rest of the book: each property is traced to a structure. Heat is carried by lattice vibrations (phonons) and electrons; charge by electrons, holes and ions; polarization by displaced charges and dipoles; magnetism by electron spins and their exchange coupling. Once these carriers and their scattering are understood, the processing levers become clear: alloying scatters electrons and phonons, doping adds carriers, grain boundaries and domain walls pin or impede, and texture and anisotropy direct flows.

Every circuit is built from two kinds of material doing opposite jobs. Conductors must carry current with as little loss as possible through wires, busbars, contacts, interconnects and electrodes. Insulators must stop current and hold voltage without losing energy, for years, while hot, wet and dirty. A third group, the functional dielectrics, turns the electric field into something else: strain, charge from heat, a change of refractive index, or a very large capacitance per unit volume. This part asks why a material falls into one group or another, what sets the numbers an engineer reads on a datasheet (resistivity, temperature coefficient, permittivity, loss tangent, breakdown strength, piezoelectric coefficient), and how processing and service conditions move those numbers. The quantum theory of electrons in solids is deliberately held back until Part VII. Here we use the simplest model that works and say clearly where it stops working.

Electrical engineering also runs on magnetic flux. Every transformer in the grid, every motor and generator, every inductor in a power converter, every current sensor and every hard-disk read head depends on a material that guides, stores or produces flux. Some of these materials must give up their magnetization effortlessly, millions of times a second, wasting as little energy as possible. Others must hold their magnetization for decades against heat and opposing fields. Both are made from the same few ingredients — iron, cobalt, nickel, a handful of rare earths and oxides — and both are governed by the same physics of atomic moments, exchange and domains. What separates a transformer core from a permanent magnet is microstructure: how easily a domain wall can move through the material. Superconductors close the part: they carry current without resistance, but only when nanometre-scale defects pin the magnetic flux inside them.

By the end of this part you should understand:

- why heat capacity, thermal expansion and thermal conductivity differ so widely between materials, and how temperature changes generate thermal stress and thermal shock;
- why resistivity is set by carrier density and by scattering, and why scattering, not carrier density, separates good metals from poor ones;
- how temperature, alloying, cold work, grain size and film thickness each add their own resistivity, and when those contributions can simply be added (Matthiessen's rule);
- why a contact between two metals has resistance even when both are excellent conductors, and why contact force, hardness and surface films control it;

- why good electrical conductors are also good thermal conductors (Wiedemann–Franz), and which materials break that link;
- how a dielectric polarizes (electronic, ionic, orientational and interfacial mechanisms) and why permittivity and loss depend on frequency and temperature;
- what dielectric strength means, why measured values depend on thickness, time and electrode geometry, and how gases, liquids and solids break down;
- how moisture, contamination, tracking and partial discharge degrade insulation, and how thermal and electrical aging set insulation life;
- why only certain crystal symmetries allow piezoelectricity, pyroelectricity and ferroelectricity, and how these effects are used in capacitors, actuators, sensors, resonators and modulators;
- the relations between B, H and M, what susceptibility and the several kinds of permeability mean on a datasheet, and why a gap or the shape of a part can matter more than the material's own permeability;
- why diamagnetism and paramagnetism are weak, and how the exchange interaction produces the spontaneous order of ferromagnets, antiferromagnets and ferrimagnets, including the spinel ferrites;
- why ferromagnets form domains, what sets the width and energy of a domain wall, and how wall pinning by defects and grain boundaries decides hysteresis, remanence and coercivity;
- how saturation magnetization, anisotropy and coercivity fall with temperature, and what the Curie and Néel temperatures mean for derating;
- where core losses come from (hysteresis, classical eddy currents, excess loss), how they scale with frequency, induction, lamination thickness and resistivity, and how the Steinmetz equation is used and misused;
- the engineering families of soft materials — electrical steels, ferrites, amorphous and nanocrystalline alloys, nickel–iron and cobalt–iron alloys, powder cores — and of permanent magnets — NdFeB, SmCo, hard ferrite and AlNiCo;
- why the energy product, the temperature coefficients and the heavy-rare-earth (dysprosium) content of a magnet decide whether a motor survives its worst operating point;
- how magnetic shields work at low and high frequency, and how materials are selected for transformers, motors, inductors and magnetic sensors;
- why a superconductor is more than a perfect conductor (the Meissner effect), and how the penetration depth, the coherence length and their ratio κ separate Type I from Type II behaviour;
- why practical superconductors (Nb–Ti, Nb_3Sn, MgB_2, REBCO, BSCCO) are made as fine-filament composite wires or textured coated-conductor tapes, and how brittleness, strain sensitivity and reaction heat treatments constrain magnet design;
- why one structural feature (a solute, a pore, a domain wall) can change several properties at once.

Chapter	Central question	What it delivers
24. Thermal Properties	How do materials store, carry and expand with heat, and what happens when expansion is resisted?	Einstein and Debye heat capacity; anharmonic expansion and Grüneisen; phonon and electron conduction, Wiedemann–Franz; diffusivity and transient heat flow; thermal stress and shock parameters; thermal barrier coatings, insulation and thermal-management materials
25. Electrical Conduction in Materials	What sets the resistivity of a metal, an alloy, a thin film or a filled polymer, and how does it change with temperature and size?	Resistivity and TCR data, Matthiessen and Nordheim rules, contact (Holm) resistance, thin-film size effects, percolation in conductive composites, Wiedemann–Franz, selection for wires, contacts, interconnects and electrodes
26. Dielectric and Insulating Materials	Why does an insulator store field energy, lose some of it as heat, and eventually fail?	Permittivity and loss across frequency and temperature, dielectric strength and breakdown mechanisms (Paschen, thermal, electrical treeing), volume and surface resistivity, tracking and partial discharge, moisture effects, polymer/ceramic/composite insulators, thermal and electrical aging
27. Functional Dielectric Materials	How can a dielectric convert electrical energy into strain, sense heat or force, or change light, and what structure makes that possible?	Ferroelectric, piezoelectric, pyroelectric, electrostrictive and electro-optic materials; high-permittivity capacitor and gate dielectrics; microwave dielectric resonators; material choices for MLCCs, actuators, sensors and RF filters
28. Magnetic Material Behaviour	Why do some materials respond to a field a hundred thousand times more strongly than others, and why does the motion of domain walls decide whether a material is soft or hard?	The physical vocabulary for reading magnetic datasheets: χ, μ_r, M_s, T_C, domains, walls, hysteresis, B_r, H_c; the temperature derating of cores and magnets; the origin of ferrite and magnet properties
29. Soft and Hard Magnetic Materials	Which magnetic material goes into a transformer, a motor, an inductor, a shield or a sensor, and why?	Loss separation and the Steinmetz equation; lamination and skin-depth limits; property tables for electrical steels, ferrites, amorphous and nanocrystalline alloys and permanent magnets; selection rules for real devices
30. Superconducting Materials	What limits how much current, field and temperature a superconductor can tolerate, and why are the best superconductors so hard to turn into wire?	Conductor choice and operating margins for MRI, NMR, accelerator and fusion magnets; HTS cables and fault-current limiters; the materials behind SQUID sensors

How this part connects to the rest of the book. Part VI builds directly on Part I: the reciprocal lattice and Brillouin zone (Chapter 4) are the language of phonons and electron bands, tensor properties (Chapter 4) describe anisotropic conductivity, permittivity and piezoelectricity, and defects (Chapter 5) are the scatterers and dopants that set transport. Thermodynamics and diffusion (Part II) reappear in solid electrolytes, defect chemistry and thermal transients, which obey the same mathematics as Fick's laws. Thermal stress links back to the mechanical behaviour of Part III and the composite laminates of Chapter 23.

Chapter 25 treats conduction with a classical free-electron picture: carriers, a mean free time and a mobility. Chapter 31 replaces that picture with energy bands and shows why the carrier density of a metal is fixed while that of a semiconductor is not. Chapter 32 then reuses mobility and scattering for semiconductors. Thin-film conductors and contacts return in Chapter 34 (semiconductor contacts and films), Chapter 50 (metallization) and Chapter 51 (bonding wires, solders, conductive adhesives). The dielectrics of Chapter 26 reappear as gate dielectrics in Chapter 34, as PCB laminates and encapsulants in Chapter 51, and as failure mechanisms in Chapter 44. The ferroelectric and piezoelectric materials of Chapter 27 are needed again for actuators (Chapter 37), for energy harvesting (Chapter 40), for nanoscale and thin-film functional layers (Chapter 38), and for the material-selection methods of Chapter 58. The magnetic materials of Chapters 28 and 29 follow the same pattern as the dielectrics: domains, hysteresis, a Curie temperature and frequency-dependent loss. Chapter 28 traces each property to its structural origin: atomic moments and exchange set the saturation and the Curie temperature, crystal anisotropy and magnetostriction set the domain structure, and defects and grain size set coercivity. Chapter 29 follows those properties through processing — rolling and texture, rapid solidification, controlled crystallization, sintering, grain-boundary diffusion — to device performance and cost. Superconductors (Chapter 30), whose perfect diamagnetism and flux pinning are close relatives of the diamagnetism and domain-wall pinning of Chapter 28, close the part. The design of transformers, rotating machines and

inductors themselves belongs to the Electrical Energy & Machines core, and the operation of sensors and spintronic devices to Electronics & Computing; this part supplies the materials those subjects take as given.

Forward, these properties are the basis of the semiconductor and photonic materials of Part VII, of the smart materials, nanomaterials and energy materials of Part VIII (shape-memory and piezoelectric actuators, quantum dots, batteries, thermoelectrics) and of high-temperature service in Chapter 41. Thin-film processing (Chapter 49), characterization (Part XI) and electronic-structure methods (Chapter 55) assume the vocabulary established here.

Chapter 24 — Thermal Properties

Part VI · Thermal, Electrical, Dielectric and Magnetic Materials

On a cold morning a cook pours boiling water into a glass baking dish that has just come out of the refrigerator, and the dish shatters. Its borosilicate cousin, the laboratory beaker, survives the same treatment every day. A few kilometres away, a railway engineer checks the temperature forecast: continuously welded rail laid on a cool spring day will be in compression on a hot summer afternoon, and if the stress is high enough the track buckles sideways. At an airport, a turbofan's first-stage turbine blades run in gas several hundred kelvin hotter than the melting point of the nickel alloy they are made of, protected by a ceramic coating a quarter of a millimetre thick. In a data centre, a processor dissipating 300 W through a die the size of a postage stamp depends on a copper lid, a smear of grease and a finned aluminium block to stay below 100 °C.

Each of these is a problem in thermal properties: how much heat a material stores, how quickly it carries heat, how much it expands when heated, and what happens when expansion is resisted. This chapter traces each property to the atomic vibrations and electrons that produce it, and then follows it into stresses, shocks, coatings, insulation and thermal management across engineering.

> **Why do some solids store, conduct and expand with heat so differently from others?**
> **How do temperature changes produce stresses large enough to buckle rails, crack ceramics and spall coatings, and how are materials chosen to manage heat?**

Concept	Engineering destination
Phonons and heat capacity	Thermal mass of brakes and buildings, cryogenic design, thermal energy storage
Thermal expansion	Rails and bridges, bimetals, Invar instruments, glass-to-metal seals, low-expansion cookware
Thermal conductivity	Heat exchangers, cookware, electronic substrates, turbine cooling, insulation
Thermal diffusivity and transient heat flow	Quenching of steel, cycle times of moulding, curing of mass concrete, touch temperature
Thermal stress and thermal shock	Cracked glassware, refractories, brake discs, ceramic engine parts
Thermal barrier coatings and insulation	Gas-turbine blades, furnace linings, buildings, cryogenic tanks, spacecraft
Thermal management materials	Electronics cooling, battery packs, LED lighting, aerospace avionics

Roadmap. Storing heat: phonons and heat capacity (24.1) → expanding with heat: the anharmonic potential (24.2) → carrying heat: phonons and electrons (24.3) → how fast temperature changes travel (24.4) → when expansion is resisted: thermal stress and thermal shock (24.5) → keeping heat out: barrier coatings and insulation (24.6) → moving heat on purpose: thermal-management materials (24.7).

24.1 Phonons and heat capacity

Definitions

The **heat capacity** of a body is the heat needed to raise its temperature by one kelvin. As a material property it is quoted per unit mass (**specific heat** c, J kg^{-1} K^{-1}), per mole (J mol^{-1} K^{-1}) or per unit volume (ρc, J m^{-3} K^{-1}). For solids, c_p and c_v differ by $c_p - c_v = TV\beta^2 K$ (with β the volume expansion coefficient and K the bulk modulus), only a few percent at room temperature; data are quoted as c_p, theory gives c_v.

> **Boundary.**
> The thermodynamic definitions of heat capacity and the c_p–c_v relation belong to the Advanced Chemistry book; the quantum mechanics of the harmonic oscillator and of phonons to Engineering Physics. They are used here, not derived.

Lattice vibrations and phonons

In a solid, heat is stored mainly in the vibrations of atoms about their equilibrium positions. Because neighbouring atoms are coupled by bonds, the vibrations are not independent: they are collective waves — **normal modes** — running through the crystal. A crystal of N atoms has $3N$ normal modes, each with a wavevector $\mathbf{k}$ in the Brillouin zone (Chapter 4) and a frequency $\omega(\mathbf{k})$ given by the **dispersion relation**. For a chain of atoms of mass m connected by springs of stiffness κ and spacing a,

$$\omega(k) = 2\sqrt{\frac{\kappa}{m}}\left|\sin\frac{ka}{2}\right|$$

At long wavelengths ($k \to 0$) this is a sound wave, $\omega = v_s k$ with $v_s = a\sqrt{\kappa/m}$; at the zone boundary the frequency saturates at $\omega_{\max} = 2\sqrt{\kappa/m}$. Crystals with several atoms per primitive cell also have **optical** branches. Quantum mechanics allows each mode to hold energy only in units of $\hbar\omega$; a quantum of vibrational energy is a **phonon**. Heat capacity counts how many phonons are excited as temperature rises; thermal conductivity (Section 24.3) follows how phonons travel and scatter.

The scaling $\omega \propto \sqrt{\kappa/m}$ is the thread through this chapter: **stiff bonds and light atoms give high vibration frequencies**, which raise sound velocities, delay the excitation of modes to higher temperatures and, as we shall see, favour high thermal conductivity.

The classical limit: Dulong–Petit

Classically each normal mode is a harmonic oscillator holding $k_B T$ of energy on average, so a mole of atoms holds $3RT$ and

$$C_{v,m} = 3R = 24.9\ \text{J mol}^{-1}\,\text{K}^{-1}$$

This is the **Dulong–Petit law** (1819). Per unit mass it predicts $c = 3R/M$, so heavy atoms have low specific heat: lead and tungsten about 120–135 J kg^{-1} K^{-1}, copper and iron about 390–450, aluminium about 900. Per unit volume, because atomic volumes vary much less than atomic masses, almost all dense solids hold 1.5–4 MJ m^{-3} K^{-1} (Table 24.1).

> **Status of the principle: Dulong–Petit law**
> *Approximate*. Classical equipartition applied to $3N$ harmonic modes; accurate within a few percent for most metals at and above room temperature, but it overestimates the heat capacity of solids with stiff bonds and light atoms (diamond, Si, SiC, B, Be, oxides) and of every solid at low temperature.

Einstein's model

Measurements in the 19th century showed that diamond, boron and silicon have heat capacities far below $3R$ at room temperature, and that all solids lose heat capacity at low temperature. **Einstein** (1907) treated every atom as an independent quantum oscillator of one frequency ω_E, defining an **Einstein temperature** $\theta_E = \hbar\omega_E/k_B$. The average energy of a quantum oscillator gives

$$C_v = 3R\left(\frac{\theta_E}{T}\right)^2 \frac{e^{\theta_E/T}}{\left(e^{\theta_E/T}-1\right)^2}$$

At $T \gg \theta_E$ this tends to $3R$; at $T \ll \theta_E$ it falls exponentially, because the thermal energy $k_B T$ is too small to excite even one quantum $\hbar\omega_E$ — the mode is **frozen out**. Its failure is at low temperature: measured heat capacities fall as T^3, much more slowly than exponentially. The reason is that a real crystal has long-wavelength acoustic modes of arbitrarily low frequency that remain excited even at very low temperatures. Einstein's single frequency describes optical modes reasonably well; it cannot describe sound waves.

Debye's model

Debye (1912) replaced the single frequency by a spectrum of sound waves, $\omega = v_s k$, cut off at a maximum frequency ω_D chosen so that the total number of modes is $3N$. The **Debye temperature** $\theta_D = \hbar\omega_D/k_B$ characterizes each solid, and

$$C_v = 9R\left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D/T} \frac{x^4 e^x}{(e^x-1)^2}\,dx$$

At high temperature $C_v \to 3R$. At low temperature the integral becomes a constant and

$$C_v \approx \frac{12\pi^4}{5} R \left(\frac{T}{\theta_D}\right)^3$$

the **Debye T^3 law**, valid for $T \lesssim \theta_D/50$ and approximately up to $\theta_D/10$. Because $\omega_D \propto v_s$, the Debye temperature is high for stiff, light solids: diamond about 2,230 K, SiC about 1,200 K, Al_2O_3 about 1,040 K, Si 645 K, Fe 470 K, Al 428 K, Cu 343 K, Au 165 K, Pb 105 K. The Debye temperature roughly marks where a solid's vibrations become fully classical, and Figure 24.1 shows that all solids fall on one universal curve of $C_v/3R$ against T/θ_D.

> **Status of the principle: Debye model of heat capacity**
> *Approximate.* Treats lattice vibrations as an isotropic elastic continuum with a single cutoff; it reproduces the T^3 law and the high-temperature limit exactly, while at intermediate temperatures the detailed phonon spectrum makes the effective θ_D vary with temperature by 10–20 %. For compounds with optical branches, a Debye term for acoustic modes plus Einstein terms for optical modes works better.

The electronic contribution

In a metal the conduction electrons also carry heat capacity, but only those within about $k_B T$ of the Fermi energy E_F can be excited (Pauli exclusion), a fraction $k_B T/E_F \approx 0.005$ at room temperature. The electronic heat capacity is linear in temperature, $C_{el} = \gamma T$, with $\gamma = 0.695$ mJ mol^{-1} K^{-2} for copper: 0.21 J mol^{-1} K^{-1} at 300 K, under 1 % of the lattice term. Because the lattice term falls as T^3, the two become equal for copper at about 4 K, and below that electrons dominate.

Worked example — Why diamond feels "cold" in heat capacity terms. Compare copper ($\theta_D = 343$ K, $M = 63.55$ g/mol), silicon ($\theta_D = 645$ K, $M = 28.09$ g/mol) and diamond ($\theta_D \approx 2230$ K, $M = 12.01$ g/mol) at 300 K.

Dulong–Petit predicts $c = 3R/M$: copper $24.94/0.06355 = 392$, silicon 888, diamond 2,077 J kg^{-1} K^{-1}.

Evaluating the Debye integral numerically at $T/\theta_D = 0.875$, 0.465 and 0.135 gives $C_v/3R = 0.938$, 0.802 and 0.166. Hence $c_v = 368$ J kg^{-1} K^{-1} for copper (measured $c_p = 385$), 712 for silicon (measured 705) and 344 for diamond (measured 509). The Einstein model with $\theta_E \approx 0.806\,\theta_D$ gives nearly the same values for copper (366) and silicon (696) — at these temperatures the models agree — but only 187 for diamond.

At room temperature copper is nearly classical, silicon has lost a fifth of its modes and diamond has lost five-sixths in the Debye estimate. Diamond's measured value corresponds to an effective $\theta_D \approx 1,870$ K at 300 K, illustrating the temperature dependence of the effective Debye temperature noted in the status box. At low temperature the difference becomes extreme: at 10 K copper's lattice heat capacity is $(12\pi^4/5)R(10/343)^3 = 0.048$ J mol^{-1} K^{-1}, 0.2 % of its room-temperature value. That is why small heat leaks warm cryogenic sensors so easily.

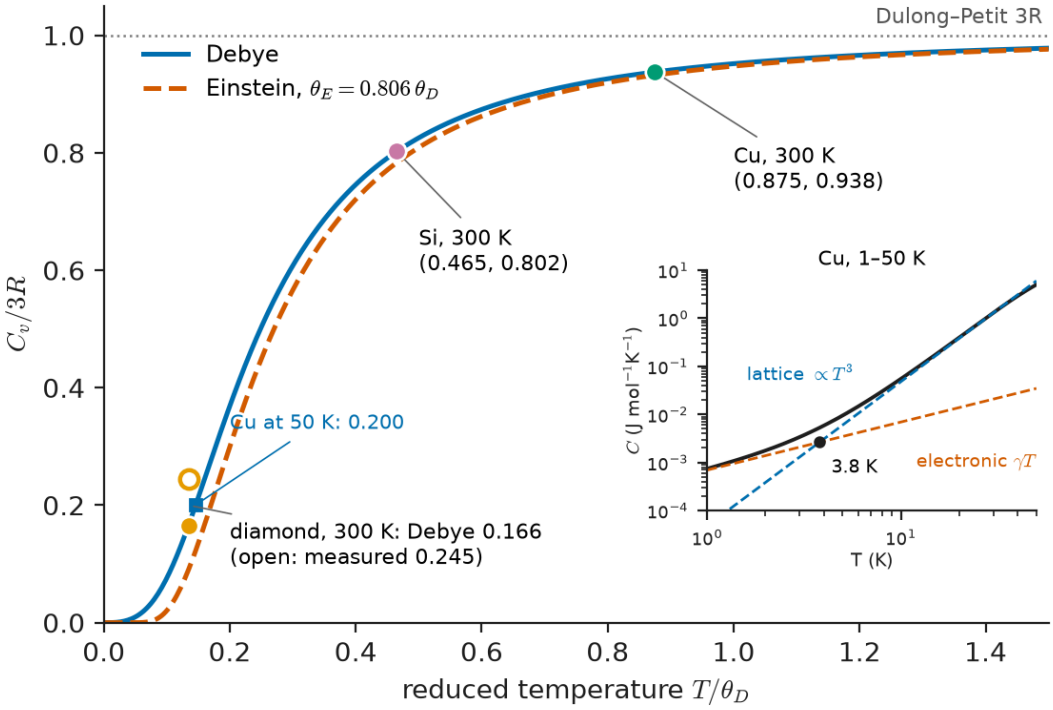

Figure 24.1 — Heat capacity of solids versus reduced temperature: the Debye and Einstein models and the classical Dulong–Petit limit, with copper, silicon and diamond at room temperature.

Table 24.1 — Heat capacity at 25 °C (typical values).

Material	c_p (J kg⁻¹ K⁻¹)	ρ (kg/m³)	ρc_p (MJ m⁻³ K⁻¹)	θ_D (K)
Lead	129	11,340	1.46	105
Tungsten	132	19,300	2.55	400
Copper	385	8,960	3.45	343
Carbon steel	470–490	7,850	3.7–3.8	~470 (Fe)
Aluminium	897	2,700	2.42	428
Silicon	705	2,330	1.64	645
Diamond	509	3,510	1.79	~2,230
Alumina	880	3,980	3.50	~1,040
Soda-lime glass	840	2,500	2.10	—
Concrete	750–1,000	2,300–2,400	1.8–2.3	—
Polypropylene	1,900	905	1.72	—
Epoxy	1,100	1,200	1.32	—
Wood (softwood, dry)	1,300–1,700	400–600	0.6–0.9	—
Water (reference)	4,180	1,000	4.18	—

Where heat capacity matters

Heat capacity decides how much a component's temperature rises when it absorbs energy faster than conduction can remove it. Carbon–carbon aircraft brakes absorb tens of megajoules in a rejected take-off because carbon's specific heat (about 1,400 J kg⁻¹ K⁻¹ at 500 °C, rising as more modes are excited) and its tolerance of 2,000 °C let a few hundred kilograms soak up that energy. Concrete and masonry give buildings thermal mass; solar plants store heat in molten nitrate salts; **phase-change materials**

(paraffins, salt hydrates) add latent heat for battery packs and building panels; and in welding and additive manufacturing the heat capacity of the surrounding metal sets the cooling rate of the melt pool (Chapters 47 and 48).

24.2 Thermal expansion and the anharmonic potential

Why solids expand

If atoms vibrated in a perfectly harmonic (parabolic) potential, their average position would not change as the vibration amplitude increased, and solids would not expand. Real interatomic potentials are **anharmonic** (Figure 24.2): steeply repulsive when atoms are pushed together (overlapping electron shells) and shallow when they are pulled apart (the attraction fades slowly). As temperature rises and atoms vibrate with larger amplitude, they spend more time at separations larger than the equilibrium value, so the mean spacing grows.

Expanding the potential about its minimum r_0 as $U(x) = \tfrac{1}{2}\kappa x^2 - \tfrac{1}{3}g x^3 + \dots$ (with $x = r - r_0$ and $g > 0$ for the asymmetric well) and averaging with classical statistics gives a mean displacement $\langle x \rangle = g k_B T / \kappa^2$ — linear in T, proportional to the asymmetry g and inversely proportional to the square of the stiffness κ. Three robust trends follow:

- **Stiffly bonded, high-melting solids expand little.** Deep, narrow potential wells are also nearly symmetric near the bottom. Covalent and refractory solids (diamond 1.0, SiC 4.0, W 4.5 ppm/K) expand much less than low-melting metals (Al 23, Zn 30, Pb 29 ppm/K) and polymers (50–200 ppm/K). An empirical rule is that the linear expansion accumulated from 0 K to the melting point is roughly constant, $\alpha T_m \approx 0.015$–0.03 for metals and simple ceramics.
- **Weakly bonded directions expand most.** Graphite expands about 27 ppm/K across its van der Waals-bonded layers and about −1 ppm/K within them; polymers expand mainly through their interchain contacts.
- **Expansion follows heat capacity.** At low temperature, where modes are frozen out, expansion also vanishes; above θ_D, both are nearly constant.

The Grüneisen relation

The third trend is made quantitative by the **Grüneisen relation**. Anharmonicity means that vibration frequencies depend on volume; the **Grüneisen parameter** $\gamma = -d\ln\omega / d\ln V$ measures how much. Thermodynamics then gives

$$\alpha_V = 3\alpha = \frac{\gamma\, C_v}{K\, V_m}$$

with α_V the volume and α the linear expansion coefficient, C_v the molar heat capacity, K the bulk modulus and V_m the molar volume. For most solids γ lies between 1 and 3 and is nearly independent of temperature, so $\alpha(T)$ follows $C_v(T)$: proportional to T^3 at low temperature and nearly constant above θ_D. Stiff solids (large K) expand less.

> **Status of the principle: Grüneisen relation**
> *Derived.* Exact thermodynamics once the Grüneisen parameter is defined; the approximation is that γ is a single, temperature-independent number for all modes, which holds well for simple metals and ceramics and fails for anisotropic and soft-mode materials, where different modes have different (even negative) γ.

Worked example — Expansion of copper and tungsten from Grüneisen. For copper, $\gamma \approx 2.0$, $C_v \approx 24.4$ J mol⁻¹ K⁻¹ at 300 K, $K = 140$ GPa and $V_m = 7.11 \times 10^{-6}$ m³/mol:

$$\alpha = \frac{\gamma C_v}{3 K V_m} = \frac{2.0 \times 24.4}{3 \times 1.40 \times 10^{11} \times 7.11 \times 10^{-6}} = 1.63 \times 10^{-5}\ \mathrm{K^{-1}}$$

against a measured 16.5 ppm/K. For tungsten ($\gamma \approx 1.6$, $C_v \approx 24.3$, $K = 310$ GPa, $V_m = 9.55 \times 10^{-6}$ m³/mol): $\alpha = 1.6 \times 24.3 / (3 \times 3.10 \times 10^{11} \times 9.55 \times 10^{-6}) = 4.4$ ppm/K, against a measured 4.5 ppm/K. The factor of nearly four between them comes mainly from tungsten's higher bulk modulus (about a factor of 2.2), with its larger molar volume and smaller Grüneisen parameter supplying the rest. Hence tungsten, molybdenum and Kovar-type alloys seal to glasses and ceramics, while aluminium mismatches nearly everything bonded to it.

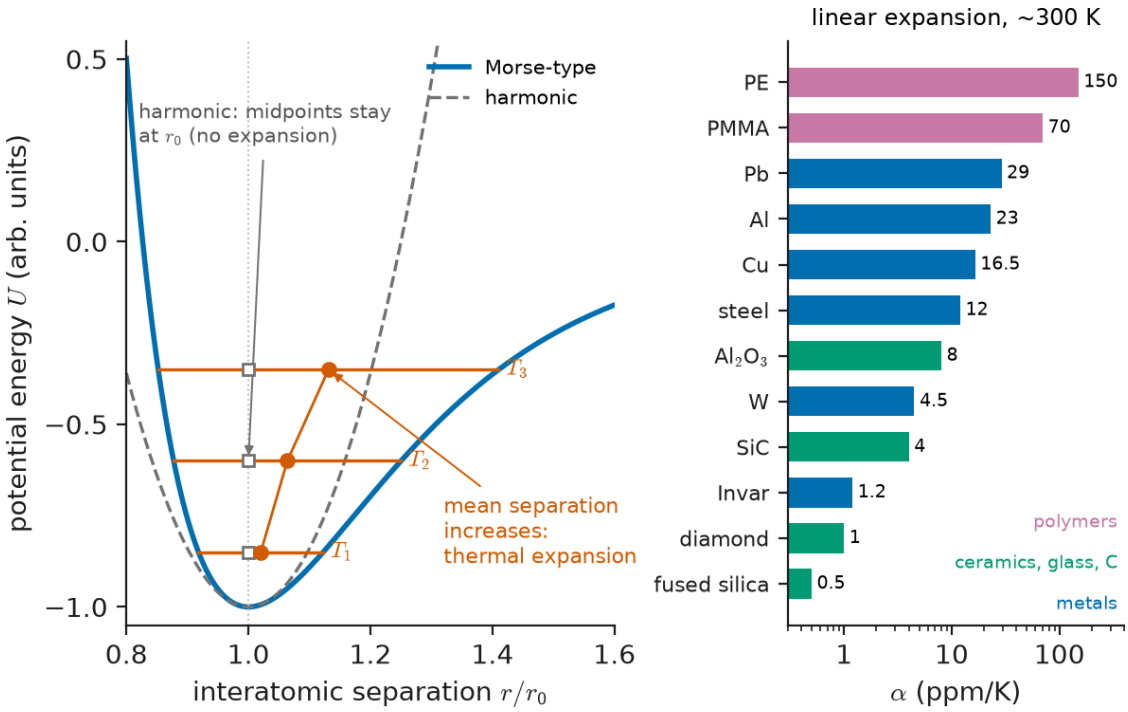

Figure 24.2 — The anharmonic interatomic potential and the origin of thermal expansion.

Table 24.2 — Linear thermal expansion coefficients near room temperature (typical values).

Material	α (ppm/K)	Note
β-eucryptite / LAS glass-ceramic (Zerodur, Ceran)	−0.1 to +0.1	Near-zero by design: negative-expansion crystals in a glass
Fused silica	0.5	Open tetrahedral network
Invar (Fe–36Ni)	1.2	Magnetovolume effect cancels lattice expansion below T_C
Diamond	1.0	
Silicon	2.6	
Borosilicate glass (Pyrex type)	3.3	
Silicon nitride	3.0–3.3	
Silicon carbide	4.0–4.5	
Tungsten	4.5	
Kovar (Fe–29Ni–17Co)	5.5	Matches borosilicate glass and alumina for seals
Alumina	7–8	
Soda-lime glass	9	
Yttria-stabilized zirconia	10–11	Close to superalloys: why it works as a coating
Ti-6Al-4V	8.6–9	
Carbon steel; concrete	11–13; 10–12	Similar values make reinforced concrete possible
Austenitic stainless steel	16–17	
Copper; Ni-base superalloy	16.5; 13–16	
Aluminium	23	
Wood, along / across grain	3–5 / 30–60	Plus much larger moisture swelling
Epoxy; PMMA	55–70; 70	
HDPE; silicone rubber	100–200; 250–300	

Low and negative expansion

Two coincidences in Table 24.2 hold up whole industries. Steel and concrete have nearly equal expansion coefficients, so reinforced concrete does not tear itself apart in daily temperature cycles (Chapter 18). Yttria-stabilized zirconia expands nearly as much as nickel superalloys, which is what allows a ceramic coating to survive on a metal blade (Section 24.6).

Some materials expand little or contract on heating by design. In **Invar**, the magnetic moment of the iron–nickel alloy shrinks with rising temperature and with it a magnetovolume expansion, cancelling the normal lattice expansion up to the Curie temperature (Chapter 28); Invar is used for precision instruments, shadow masks, LNG tank membranes and composite-curing tools. **Negative-thermal-expansion** crystals such as ZrW_2O_8 and β-eucryptite ($LiAlSiO_4$) contract because transverse vibrations of corner-linked polyhedra pull their linking atoms closer, like a rope that shortens as it is shaken sideways. Dispersed in a glass, β-eucryptite or β-quartz solid solutions give the near-zero-expansion **glass-ceramics** of cooktops and telescope mirrors (Chapter 17), which survive thermal shocks no ordinary glass would.

24.3 Thermal conductivity: phonon and electron contributions

Fourier's law

Heat flows from hot to cold at a rate proportional to the temperature gradient:

$$\mathbf{q} = -k\,\nabla T$$

where $\mathbf{q}$ is the heat flux (W/m²) and k the **thermal conductivity** (W m⁻¹ K⁻¹). In anisotropic materials k is a second-rank tensor (Chapter 4): graphite, wood, fibre composites and drawn polymers conduct far better along one direction than across it.

> **Status of the principle: Fourier's law**
> *Empirical.* A linear constitutive law confirmed over a huge range of materials and conditions; it holds when temperature varies slowly over the mean free path of the heat carriers, and fails for nanometre films, very short pulses and ballistic transport, where heat carriers cross the structure without scattering.

Thermal conductivity spans about five orders of magnitude among engineering materials, from 0.02 W m⁻¹ K⁻¹ for insulating foams to over 2,000 for diamond (Table 24.3). Two carriers are responsible: phonons in all solids, and electrons in metals.

Phonon conduction

Phonon wave packets travel at about the speed of sound and carry the vibrational energy they hold. Treating them as a gas, kinetic theory gives

$$k_{\mathrm{ph}} = \tfrac{1}{3}\, C\, v\, \ell$$

where C is the heat capacity per unit volume, v an average sound velocity and ℓ the phonon mean free path.

> **Status of the principle: Kinetic formula for phonon conductivity**
> *Approximate.* Replaces a spectrum of phonons, each with its own velocity and mean free path, by single averages; it explains trends among materials and with temperature, but the effective ℓ it yields is a weighted average over mean free paths that, in a single crystal, span from nanometres to micrometres.

For alumina ($C = 3.5 \times 10^6$ J m⁻³ K⁻¹, $v \approx 7,000$ m/s, $k = 35$ W m⁻¹ K⁻¹), the formula gives $\ell = 3k/(Cv) \approx 4$ nm; for diamond ($k = 2,200$, $v \approx 12,000$ m/s, $C = 1.79 \times 10^6$) about 300 nm. A good phonon conductor needs:

- **Fast phonons**, from stiff bonds and light atoms — the same factors that give high θ_D.
- **A simple crystal structure** with few atoms per primitive cell. Complex cells have many optical branches whose phonons are slow and scatter strongly. AlN (4 atoms per primitive cell) conducts about 175 W m⁻¹ K⁻¹ as a ceramic; alumina (10) about 30–35, despite similar stiffness.
- **Long mean free paths**, limited by four kinds of scattering:
  - **Phonon–phonon (Umklapp) scattering**, which grows with temperature as more phonons are excited. Above roughly $\theta_D/3$, $k \propto 1/T$: dense ceramics conduct worse when hot.
  - **Point defects** — impurities, solutes, vacancies and even isotopes — which scatter short-wavelength phonons strongly (Rayleigh-like, $\propto \omega^4$). Oxygen dissolved in AlN cuts its conductivity from a theoretical 320 to 70–180 W m⁻¹ K⁻¹; isotopically pure ^{12}C diamond conducts about 50 % better than natural diamond.
  - **Grain boundaries and pores**, important in fine-grained ceramics, thin films and coatings.
  - **Disorder.** In glasses and amorphous polymers ℓ is only a few interatomic spacings, and k is 0.1–1.5 W m⁻¹ K⁻¹ regardless of bond strength.

The best phonon conductors are therefore simple, stiff, light-atom crystals of high purity: diamond, cubic BN, BAs, SiC, BeO, AlN. Polymers are poor conductors, but orientation helps: ultra-high-molecular-weight polyethylene fibre conducts 20–100 W m⁻¹ K⁻¹ along the chain axis, against 0.4–0.5 for bulk HDPE, because heat then travels along covalent backbones rather than across van der Waals contacts.

Electronic conduction and the Wiedemann–Franz law

In metals, free electrons carry heat as well as charge, and they are much faster than phonons (Fermi velocities about 10^6 m/s). Electrons that carry charge and heat are scattered by the same phonons and defects, so the two conductivities are linked by the **Wiedemann–Franz law**:

$$\frac{k_e}{\sigma T} = L_0 = \frac{\pi^2}{3}\left(\frac{k_B}{e}\right)^2 = 2.44 \times 10^{-8}\ \mathrm{W\,\Omega\,K^{-2}}$$

where σ is the electrical conductivity and L_0 the **Lorenz number**.

Status of the principle: Wiedemann–Franz law
Derived. Follows from free-electron theory when electrons are scattered elastically (the same scattering for charge and heat); it holds well near room temperature and above and at very low temperatures, and fails at intermediate temperatures (roughly $0.1–0.5\,\theta_D$) where inelastic phonon scattering degrades heat current more than charge current.

Worked example — How much of a metal's conductivity is electronic? At 300 K, for copper ($\rho_e = 1.68 \times 10^{-8}$ Ω m): $k_e = L_0 T/\rho_e = 2.44 \times 10^{-8} \times 300/1.68 \times 10^{-8} = 436$ W m⁻¹ K⁻¹, against a measured 401. For aluminium ($\rho_e = 2.65 \times 10^{-8}$ Ω m): 276 against 237. The law overshoots slightly because the real Lorenz number of these metals at room temperature is about $2.2–2.3 \times 10^{-8}$, but it confirms that heat in pure metals is carried almost entirely by electrons.

Now an alloy. Austenitic stainless steel 304 has $\rho_e = 72 \times 10^{-8}$ Ω m — 43 times copper's, because the disordered solid solution scatters electrons (Matthiessen's rule, Chapter 25). Wiedemann–Franz gives $k_e = 2.44 \times 10^{-8} \times 300/72 \times 10^{-8} = 10.2$ W m⁻¹ K⁻¹; the measured total is 16.2, so phonons contribute about 6 W m⁻¹ K⁻¹. For Ti-6Al-4V ($\rho_e = 178 \times 10^{-8}$ Ω m), $k_e = 4.1$ against a total of 6.7. Alloying collapses the electronic contribution and leaves phonon and electron terms comparable. Hence a stainless-steel pan needs a copper or aluminium base to spread heat, titanium is hard to machine because heat stays at the cutting edge, and high-conductivity copper alloys (Chapter 16) are strengthened by precipitates rather than solutes.

Temperature dependence

Figure 24.3 summarizes how k varies with temperature. In dielectric crystals, k rises as T^3 at very low temperature (following heat capacity, with mean free path limited by sample size), peaks at roughly $\theta_D/20$, and then falls as $1/T$ as Umklapp scattering takes over. In pure metals k is nearly flat from room temperature upward, because the resistivity rising as $\rho_e \propto T$ is offset by the factor T in $k_e = L_0 T/\rho_e$. In alloys, which are dominated by temperature-independent solute scattering, k rises gently with temperature. In glasses, polymers and other amorphous solids it also rises slowly, then saturates. Porous insulators gain a radiative component that grows as T^3 (Section 24.6).

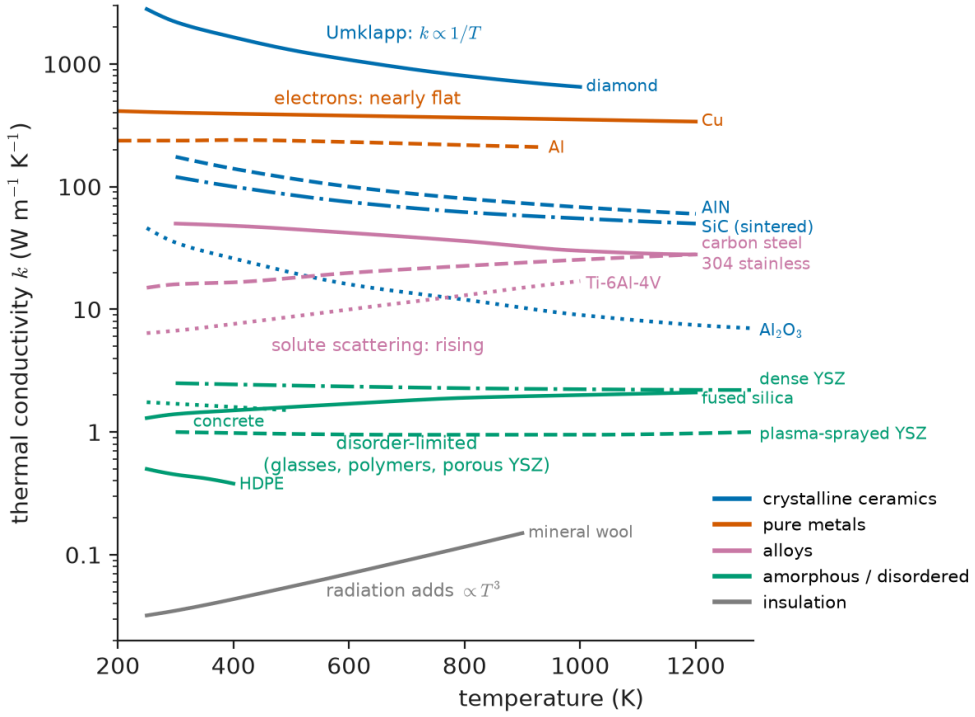

Figure 24.3 — Thermal conductivity versus temperature for representative crystals, metals, alloys, glasses, polymers and insulators.

Table 24.3 — Thermal conductivity, diffusivity and effusivity near room temperature (typical values).

Material	k (W m⁻¹ K⁻¹)	Diffusivity a (mm²/s)	Effusivity e (J m⁻² K⁻¹ s⁻¹/²)
Diamond (CVD, thermal grade)	1,000–2,200	560–1,200	~40,000–60,000
Copper	401	116	37,200
Aluminium	237	98	24,000
Aluminium nitride	140–180	60–90	~16,000
Silicon carbide (sintered)	80–150	40–70	~15,000
Silicon	150	88	15,700
Carbon steel	45–55	12–15	~13,700
Alumina (99 %)	30–35	~10	11,000
Stainless steel 304	16.2	4.0	8,000
Ti-6Al-4V	6.7	2.9	4,000
Dense YSZ	2–2.5	~0.7	~2,400
Concrete	1.4–2.0	~0.8	~1,900
Soda-lime glass	1.0	0.48	1,450
Polypropylene; polycarbonate	0.20	0.12–0.13	550–590
Wood (softwood, across grain)	0.10–0.15	~0.14	~320
Mineral wool; EPS foam	0.032–0.040	—	—
PIR/PUR foam; silica aerogel	0.020–0.025; 0.013–0.018	—	—
Still air	0.026	22	6

24.4 Thermal diffusivity and transient heat flow

The heat equation and diffusivity

When temperature changes with time, conductivity alone does not decide how fast. Combining Fourier's law with conservation of energy gives the **heat equation**:

$$\frac{\partial T}{\partial t} = a\nabla^2 T, \qquad a = \frac{k}{\rho c_p}$$

The **thermal diffusivity** a (m²/s) measures how quickly a temperature disturbance spreads: high conductivity spreads heat, high volumetric heat capacity soaks it up. The heat equation is mathematically identical to Fick's second law (Chapter 8), and all the solutions developed there — error functions for a suddenly heated surface, separation of variables for slabs — carry over with D replaced by a.

> **Status of the principle: Heat equation**
> *Derived.* Follows from energy conservation and Fourier's law with constant properties; for large temperature excursions k and c_p vary with T and the equation becomes non-linear, and latent heat, radiation and convection must be added as source or boundary terms.

The diffusion length and the time scale

The most useful consequence is a scaling law. A temperature change at a surface penetrates a distance

$$L \approx \sqrt{a\,t}$$

in time t; equivalently, a body of characteristic half-thickness L comes to thermal equilibrium in a time of order L^2/a. Doubling the thickness quadruples the time. Diffusivities range from about 100 mm²/s for copper and aluminium through 13 for steel, 1 for concrete and glass, to 0.1 for polymers and wood (Table 24.3).

When a body exchanges heat with a fluid, the ratio of internal to surface resistance — the **Biot number** $\mathrm{Bi} = hL/k$, with h the heat-transfer coefficient — decides whether its interior lags behind its surface. For $\mathrm{Bi} \ll 0.1$ the body heats uniformly (lumped behaviour); for $\mathrm{Bi} \gg 1$ steep internal gradients develop, which is when thermal stresses become serious (Section 24.5).

Worked example — Three transient problems from three industries.

Quenching steel. A carbon-steel bar ($k = 50$ W m⁻¹ K⁻¹, $\rho = 7{,}850$ kg/m³, $c_p = 475$ J kg⁻¹ K⁻¹) has $a = 50/(7{,}850 \times 475) = 1.34 \times 10^{-5}$ m²/s. Quenched into agitated water ($h \approx 5{,}000$ W m⁻² K⁻¹) its Biot number for a 25 mm half-thickness is $5{,}000 \times 0.025/50 = 2.5$: the surface cools much faster than the core. The time scale for the core is $L^2/a = 0.025^2/1.34 \times 10^{-5} = 47$ s; for a 50 mm half-thickness, 186 s. Since the cooling rate needed to form martensite must be reached within a few seconds in the critical range (Chapter 15), thick sections of plain-carbon steel cannot harden through — the origin of hardenability and the Jominy test.

Injection moulding. A polypropylene wall 2 mm thick ($a \approx 1.2 \times 10^{-7}$ m²/s) moulded at 230 °C into a 40 °C mould must cool to an ejection temperature of 100 °C at its centre. The standard slab solution gives a cooling time

$$t_c = \frac{s^2}{\pi^2 a}\ln\!\left(\frac{4}{\pi}\,\frac{T_{\mathrm{melt}} - T_{\mathrm{mould}}}{T_{\mathrm{eject}} - T_{\mathrm{mould}}}\right) = \frac{(0.002)^2}{\pi^2 \times 1.16 \times 10^{-7}}\ln\!\left(\frac{4}{\pi} \times \frac{190}{60}\right) = 4.9\ \mathrm{s}$$

Cooling is typically more than half the cycle time, and it scales with the square of wall thickness: a 4 mm wall takes 20 s. This is why plastic parts are designed with thin, uniform walls and ribs rather than thick sections (Chapter 21).

Mass concrete. A foundation raft 2 m thick ($a \approx 8 \times 10^{-7}$ m²/s) has a half-thickness of 1 m, giving $L^2/a \approx 1.2 \times 10^6$ s, or about two weeks. Cement hydration releases heat over the first days far faster than it can escape, so the core may reach 60–70 °C while the surface cools. The resulting gradients crack the concrete unless the pour is cooled by embedded pipes, made with low-heat cement, or insulated to reduce the gradient (Chapter 18).

Effusivity and touch temperature

When two bodies at different temperatures are suddenly brought into contact, the interface temperature is set not by conductivity or diffusivity but by the **thermal effusivity** $e = \sqrt{k\rho c_p}$. For two semi-infinite bodies,

$$T_{\mathrm{contact}} = \frac{e_1 T_1 + e_2 T_2}{e_1 + e_2}$$

Skin has $e \approx 1{,}500$ J m⁻² K⁻¹ s⁻¹ᐟ². Touching steel at 0 °C ($e \approx 13{,}700$) with skin at 33 °C gives a contact temperature of about 3 °C: steel feels icy. Touching wood at 0 °C ($e \approx 320$) gives about 27 °C: wood feels warm. The same quantity explains why hot-surface safety limits are lower for metals than for plastics.

Diffusivity is measured directly by the **laser-flash** method, in which the front of a thin disc is heated by a short pulse and the rise of the rear-face temperature is timed; conductivity then follows from $k = a\rho c_p$ (Chapter 52).

24.5 Thermal stresses and thermal shock

Constrained expansion

A free body heated uniformly expands without stress. Stress arises when expansion is **constrained** — by external supports, by bonded neighbours with a different expansion coefficient, or by the cooler parts of the same body. Three cases cover most practice.

Fully constrained bar. A bar held between rigid supports and heated by ΔT develops

$$\sigma = -E\alpha\Delta T$$

(compressive on heating). Continuously welded rail ($E = 210$ GPa, $\alpha = 12$ ppm/K) laid at a stress-free temperature of 20 °C and heated by the sun to 60 °C develops $210{,}000 \times 12 \times 10^{-6} \times 40 = 101$ MPa of compression. That is far below the yield strength,

but it represents a force of about 0.75 MN in a 60 kg/m rail, enough to buckle the track laterally if ballast resistance is low. Rail is therefore pre-tensioned at installation to a neutral temperature high in the local range.

Biaxially constrained layer. A thin film or coating bonded to a thick substrate, with mismatch $\Delta\alpha = \alpha_{\text{film}} - \alpha_{\text{sub}}$, cannot change its in-plane dimensions independently. A temperature change ΔT produces an equibiaxial stress

$$\sigma_{\text{film}} = -\frac{E_f}{1-\nu_f}\,\Delta\alpha\,\Delta T$$

A ceramic coating with lower expansion than its metal substrate is put into compression on cooling from deposition — usually beneficial — and into tension on heating. The factor $1/(1-\nu)$ makes biaxial stresses 25–40 % larger than uniaxial ones. It governs glazes, enamels, thermal barrier coatings (Section 24.6), thin films (Chapter 49) and laminate residual stresses (Chapter 23).

> **Status of the principle: Thermal mismatch stress**
> *Derived.* Follows from linear elasticity for a thin layer perfectly bonded to a thick, rigid substrate with temperature-independent properties; it overestimates stress where the layer creeps, yields, cracks or relaxes, and must be integrated over temperature when α or E vary.

Temperature gradients. A body with a non-uniform temperature constrains itself: hotter regions want to expand and are held back by cooler ones. For a plate whose surface is suddenly cooled, the surface wants to shrink but is held at the dimensions of the still-hot interior; the surface goes into **tension** and the interior into compression. For the limiting case of a sudden, infinitely fast surface temperature change ΔT, the surface stress is

$$\sigma_{\max} = \frac{E\alpha\Delta T}{1-\nu}$$

Heating the surface suddenly reverses the signs: the surface goes into compression. Since brittle materials are far weaker in tension than in compression, sudden cooling is more dangerous than sudden heating — which is why the glass dish that cracked was cold and wet on the outside while hot water hit the inside, putting the cold outer surface in tension.

Thermal shock resistance

Thermal shock is fracture caused by such a transient. Setting the surface stress equal to the fracture strength σ_f gives the critical quench temperature difference for a brittle material, the first **thermal shock resistance parameter**:

$$R = \Delta T_c = \frac{\sigma_f(1-\nu)}{E\alpha}$$

R applies when heat transfer is very fast (high Biot number). When heat transfer is slower, conduction inside the body reduces the gradient, and the relevant parameter is $R' = Rk$ (W/m), which rewards high conductivity. For moderate Biot numbers the stress is reduced by a factor $f(\text{Bi})$ that can be approximated as $1/(1.5 + 3.25/\text{Bi} - 0.5e^{-16/\text{Bi}})$ for a plate (Manson), so the critical ΔT is larger than R by up to an order of magnitude in gentle quenches.

> **Status of the principle: Thermal shock parameters R and R'**
> *Approximate.* Derived for a linear-elastic, isotropic body with temperature-independent properties failing when the peak surface stress reaches a single strength; they rank brittle materials reliably but ignore the statistical (Weibull) nature of strength, the time-dependence of the stress, and the material's resistance to crack growth after initiation.

A different question is how much damage a material suffers once cracks start. Hasselman's **damage resistance** parameters, such as $R'''' = EG_c/[\sigma_f^2(1-\nu)]$, reward low strength and high toughness: they favour materials in which initiated cracks arrest rather than run. This is the logic of **refractories** (Chapter 17), which are deliberately porous and microcracked so that they lose some strength on the first heat-up but then survive thousands of thermal cycles without crumbling.

Worked example — Which ceramic survives a quench? Table 24.4 evaluates R and R' from typical properties.

Table 24.4 — Thermal shock parameters of brittle materials (typical properties).

Material	σ_f (MPa)	E (GPa)	α (ppm/K)	ν	k (W m^{-1} K^{-1})	$R = \Delta T_c$ (K)	R' (W/m)
Soda-lime glass	50	70	9.0	0.22	1.0	62	62
Borosilicate glass	50	64	3.3	0.20	1.1	189	208
Fused silica	50	72	0.55	0.17	1.4	1,050	1,470
Alumina (96–99 %)	350	380	8.0	0.22	30	90	2,700
Zirconia (3Y-TZP)	1,000	205	10.5	0.30	2.5	325	810
Silicon carbide (sintered)	450	410	4.2	0.19	120	212	25,400
Silicon nitride	800	310	3.2	0.27	25	589	14,700

For soda-lime glass: $R = 50 \times (1 - 0.22)/(70{,}000 \times 9 \times 10^{-6}) = 39/0.63 = 62$ K. Borosilicate, with a third of the expansion, survives about three times the temperature jump, 189 K — the difference between the shattered dish and the surviving beaker. Fused silica, with $\alpha = 0.55$ ppm/K, survives being plunged red-hot into water.

Among technical ceramics, silicon nitride combines high strength with low expansion and survives a severe quench of nearly 600 K: it is the material of ceramic turbocharger rotors, bearing balls and molten-aluminium handling parts. Alumina, with twice the expansion and modest strength, survives only about 90 K and cracks if splashed with cold water when hot. Silicon carbide's lower R (212 K) is offset in slower transients by its high conductivity, giving it the highest R' — the reason SiC is used for kiln furniture, heat-exchanger tubes and diesel particulate filters, which must survive repeated soot burn-off. Toughened zirconia has a high R but low conductivity, so it survives sharp quenches better than gradual severe heating.

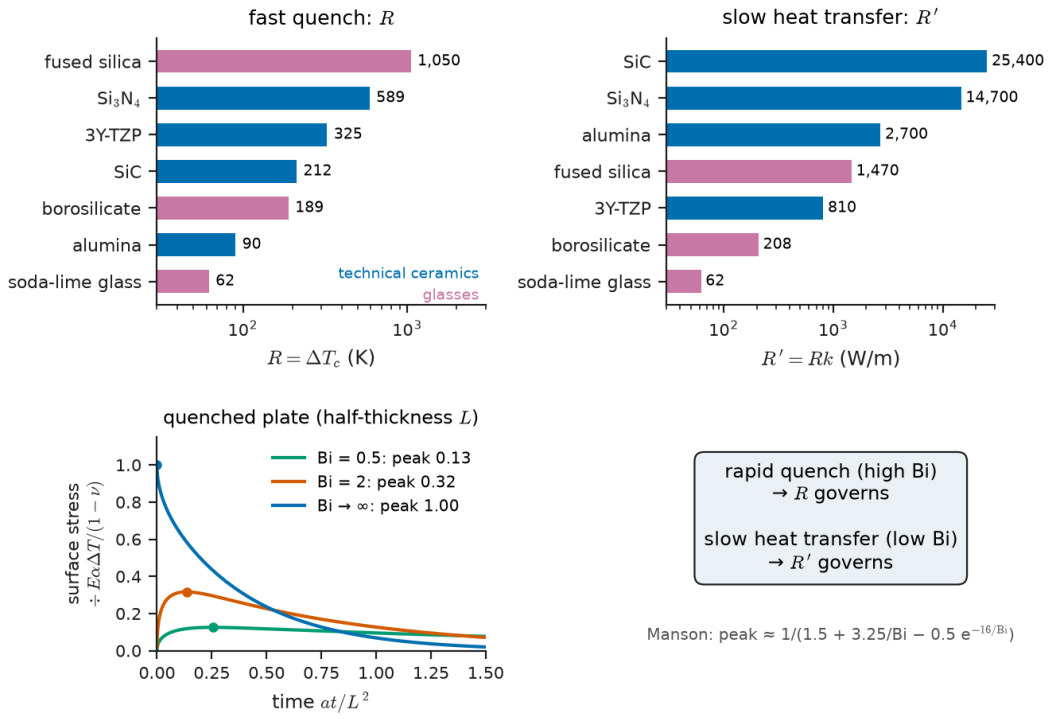

Figure 24.4 — Thermal shock resistance of brittle materials: critical quench temperature difference R and the conductivity-weighted parameter R′.

Thermal shock and cycling in metals

Metals rarely shatter in one thermal transient, but they fail by **thermal fatigue** under repeated transients: brake discs crazed with radial cracks, die-casting dies with a network of "heat checks", the leading edges of turbine blades, and the cylinder heads of engines. Each heat-up drives the hot surface into compressive yield; each cool-down reverses it into tension; the plastic strain range per cycle drives low-cycle fatigue (Coffin–Manson, Chapter 14). Resistance rewards high conductivity, low expansion, and high hot strength and ductility — the combination sought in H13 hot-work tool steel and in grey cast-iron brake discs. The thermal-cycling fatigue of solder joints in electronics, a close relative driven by expansion mismatch rather than gradients, is treated in Chapter 51 and Section 44.8.

24.6 Thermal barrier coatings and insulation

Thermal barrier coatings

The gas entering the high-pressure turbine of a modern jet engine is at 1,500–1,700 °C; the nickel-base superalloy blades melt at about 1,300–1,350 °C and lose strength rapidly above 1,100 °C. Two technologies close the gap: internal air cooling through serpentine passages and film-cooling holes, and a **thermal barrier coating** (TBC) on the outside (Chapter 41). The TBC is a multilayer system, each layer chosen for a different property (Figure 24.5):

- **Ceramic top coat**, 100–500 μm of zirconia stabilized with 7–8 wt% Y_2O_3 (7YSZ). YSZ combines low intrinsic conductivity (2–2.5 W m^{-1} K^{-1} dense, because oxygen vacancies and Y solutes scatter phonons), an expansion coefficient (10–11 ppm/K) close to the superalloy's, a metastable tetragonal (t′) phase stable to about 1,200 °C, and high toughness for a ceramic. Porosity lowers the conductivity to 0.8–1.2 W m^{-1} K^{-1} in plasma-sprayed coatings (splat boundaries across the heat flow) or 1.5–1.9 in electron-beam PVD coatings, whose columns give **strain tolerance**.
- **Thermally grown oxide** (TGO), a thin layer of α-Al_2O_3 that forms on the bond coat in service and protects it from further oxidation. It thickens parabolically (Chapter 42); when it reaches 5–10 μm, the mismatch stresses within and around it typically cause the top coat to spall.
- **Bond coat**, 50–150 μm of MCrAlY (M = Ni, Co) or platinum-modified nickel aluminide, an aluminium reservoir that grows the protective alumina scale.
- **Superalloy substrate**, internally cooled.

Worked example — How much does a TBC buy? A blade wall carries a heat flux of 0.8 MW/m². In steady state the temperature drop across each layer is $\Delta T = q\,t/k$:

- YSZ top coat, 250 μm, $k = 1.0$ W m^{-1} K^{-1}: $0.8 \times 10^{6} \times 2.5 \times 10^{-4} / 1.0 = 200$ K.
- Bond coat, 100 μm, $k \approx 10$: 8 K.
- Superalloy wall, 1.5 mm, $k \approx 25$ W m^{-1} K^{-1} (at temperature): 48 K.

A quarter-millimetre of ceramic produces four times the temperature drop of a wall six times thicker. That 200 K either lowers the metal temperature — which, by the Larson–Miller relation (Chapter 14), multiplies creep life many times over — or allows a hotter, more efficient cycle with less cooling air. The coating's failure modes all come from thermal properties: CTE mismatch stress on every flight cycle, sintering of the porous top coat (which raises k and stiffness), TGO growth, and infiltration by molten calcium–magnesium–aluminosilicate (CMAS) sand deposits, which fill the pores and destroy strain tolerance. Lower-conductivity successors such as gadolinium zirconate pyrochlores are under active development.

> **Status of the principle: One-dimensional thermal resistance of a layer stack**
> *Approximate.* Each layer is treated as a uniform slab carrying the same heat flux in steady state ($\Delta T = qt/k$, resistances in series); valid for thin flat layers with uniform properties, but it ignores lateral spreading, contact resistances between layers, radiation through translucent ceramics and the temperature dependence of k.

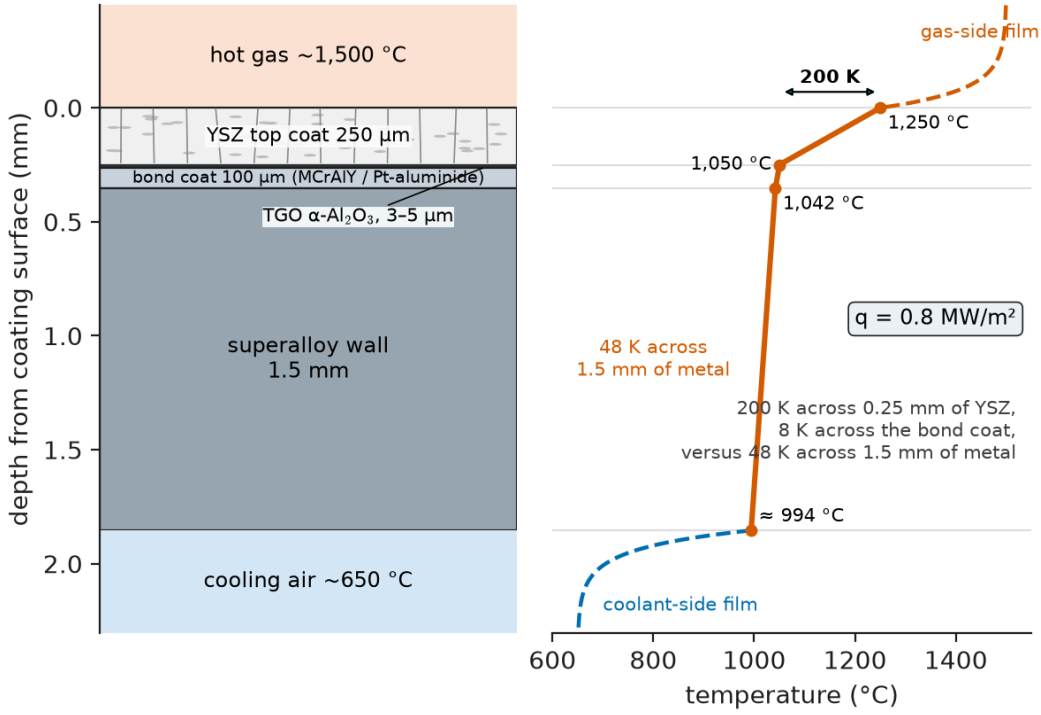

Figure 24.5 — A thermal barrier coating system on a cooled turbine blade wall and the temperature profile through it.

Thermal insulation

Insulating materials exploit still air (0.026 W m^{-1} K^{-1}): the art is to keep the gas still and block radiation with as little solid as possible. Heat flows through a porous insulator by four paths in parallel: conduction through the solid skeleton, conduction through the gas, convection within the pores (negligible below pore sizes of a few millimetres), and radiation across the pores. Thus:

- **Fibrous and cellular insulators** — mineral wool and glass wool (0.032–0.040 W m^{-1} K^{-1}), expanded and extruded polystyrene (0.030–0.038), polyurethane and polyisocyanurate foams (0.020–0.025, thanks to low-conductivity blowing gases in closed cells) — dominate building and appliance insulation. In a building wall, 100 mm of mineral wool gives a thermal resistance of $0.1/0.035 = 2.9$ m² K/W, a heat flow of 0.35 W per m² per kelvin of indoor–outdoor difference.
- **Aerogels** (silica, 90–99 % porosity, 0.013–0.018 W m^{-1} K^{-1}) have pores of 10–100 nm, smaller than the mean free path of air molecules (about 70 nm), so gas conduction is suppressed (the Knudsen effect). They insulate pipelines and separate cells in battery modules.
- **Vacuum insulation panels** (0.004–0.008) and **multilayer insulation** — tens of reflective aluminized polymer films in vacuum, with effective conductivity below 10^{-4} W m^{-1} K^{-1} — remove gas conduction entirely and are used in refrigerators, LNG and liquid-hydrogen tanks, and spacecraft.
- **High-temperature insulation** — ceramic-fibre blankets, insulating firebrick, microporous silica panels — must also block radiation, which grows as T^3 and dominates above about 500 °C. Opacifiers (TiO_2, SiC, carbon) are added to scatter and absorb infrared. The 90 %-porous silica tiles of the Space Shuttle had such low diffusivity that a tile glowing on its face could be held by its edges.

24.7 Materials for thermal management

The problem and the indices

Thermal management means moving heat deliberately from where it is generated to where it can be rejected, without overheating the source or overstressing the joints. It is now a limiting technology in power electronics, data centres, LED lighting, electric-vehicle batteries and motors, avionics, satellites and laser systems. The material requirements depend on the regime:

- **Steady-state heat spreading and conduction** reward high k. For weight-critical applications (aircraft, spacecraft, portable devices), the index is k/ρ.
- **Absorbing transient pulses** rewards high ρc_p (or latent heat) close to the source, and fast spreading rewards high diffusivity.
- **Bonding to a low-expansion device** (silicon, GaN, SiC dies, ceramic substrates, laser diodes) requires a CTE close to the device's to avoid thermomechanical fatigue. The combined index is high k with low α, or equivalently a low **thermal distortion parameter** α/k — the same index used to choose materials for precision instruments and mirrors that must not distort in non-uniform temperature fields.

Figure 24.6 plots conductivity against expansion for the candidate materials.

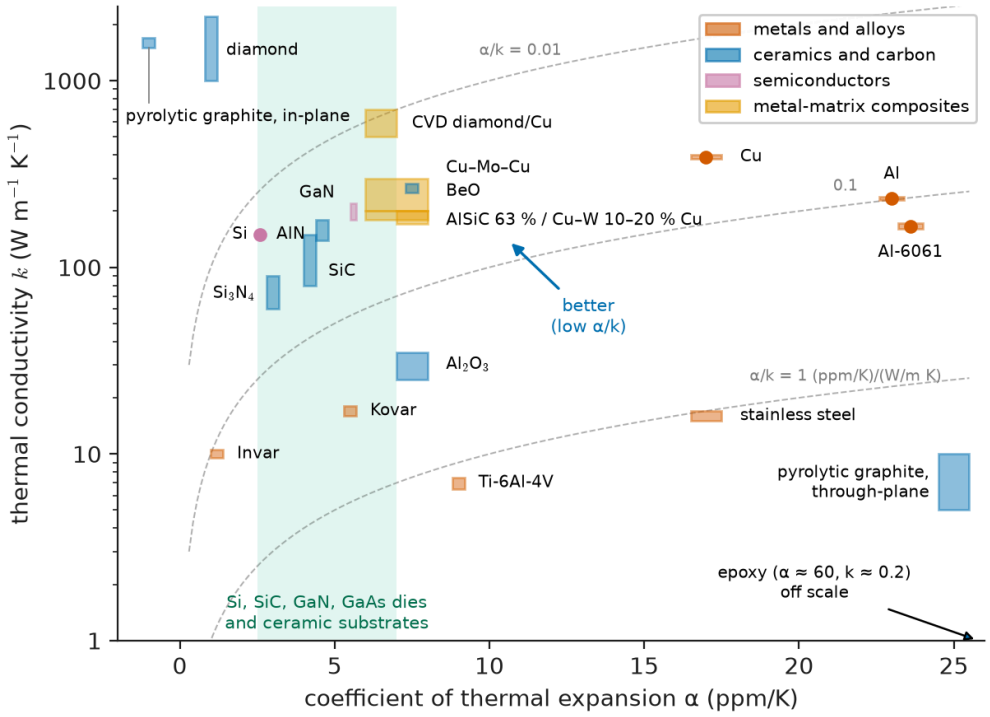

Figure 24.6 — Thermal conductivity versus thermal expansion coefficient for thermal-management materials, with the CTE window of semiconductor devices.

The material families

Metals. Copper (401 W m^{-1} K^{-1}) is the default heat spreader, cold plate and heat-pipe envelope; aluminium (237 for pure, 150–200 for alloys) is the default heat sink, at a third of copper's density and a better k/ρ. Both expand much more than semiconductors and ceramics.

Ceramics. Electrically insulating substrates that conduct heat — AlN (140–180), BeO (250–280, but toxic as dust), high-conductivity Si3N4 (60–90, much tougher), SiC and alumina (25–35, the cheap default) — carry power devices and LEDs while isolating them electrically. Hexagonal boron nitride, with a basal-plane conductivity of several hundred W m^{-1} K^{-1}, is the filler of choice for thermally conductive polymers.

Carbon. Natural and synthetic graphite sheets (in-plane 400–1,700, through-plane 5–10 W m^{-1} K^{-1}) spread heat laterally in smartphones and laptops. Pitch-based carbon fibres reach 600–1,100 W m^{-1} K^{-1} along the fibre, enabling satellite radiators and electronics enclosures in CFRP. CVD diamond (1,000–2,200) is the ultimate heat spreader for laser diodes, RF power transistors and GaN-on-diamond devices.

Metal-matrix composites. Combining a conductive metal with a low-expansion phase gives intermediate CTEs at high k: AlSiC (Chapter 23) with 7–8 ppm/K and 170–200 W m^{-1} K^{-1} for power-module baseplates and avionics; Cu–W and Cu–Mo for RF packages and laser submounts; diamond-particle composites in copper or silver reaching 500–700 W m^{-1} K^{-1} at 6–7 ppm/K.

Interface materials and polymers. The weakest link in most heat paths is an interface: two machined surfaces touch at only a few percent of their area, with air gaps between. Thermal interface materials — greases, gels, phase-change pads, solders and sintered silver — fill those gaps; they are polymer matrices filled to 50–80 vol% with Al_2O_3, BN, AlN, ZnO or metal particles, raising conductivity from 0.2 to 1–10 W m^{-1} K^{-1}. The design rules are those of particulate composites (Chapter 22): conductivity rises slowly until particles touch and percolate, and the best fillers are those with high k and good packing. Thermally conductive plastics (2–20 W m^{-1} K^{-1} in-plane) replace die-cast aluminium in LED housings and battery-module parts.

Heat pipes and vapour chambers move heat by evaporation and condensation of a working fluid, giving effective conductivities of 5,000–100,000 W m^{-1} K^{-1}; their wick structure and fluid–envelope compatibility (water–copper, ammonia–aluminium in spacecraft) are materials choices.

Table 24.5 — Thermal management choices across engineering.

Application	Requirement	Typical material choice
Power-electronic module (EV inverter)	High k, electrical isolation, CTE near Si/SiC	Si_3N_4 or AlN substrate with bonded Cu; AlSiC or Cu baseplate; sintered-silver die attach
CPU/GPU in a data centre	Spread very high flux, then reject	Cu lid, indium or liquid-metal TIM, vapour chamber, Cu or Al fins, liquid cold plate
EV battery pack	Uniform cell temperature, block runaway propagation	Al cold plates, gap fillers, aerogel or mica barriers between cells
Satellite electronics and radiators	High k/ρ, low outgassing, low α/k	Al alloys, pitch-carbon CFRP, ammonia–Al heat pipes
Laser diode, RF power transistor	Very high local flux, CTE match	CVD diamond, Cu–W, Cu–diamond submounts
Building envelope	Minimize heat flow, control thermal bridges	Mineral wool, PIR, aerogel blankets; thermally broken aluminium frames

> **Boundary.**
> The thermal resistance networks, junction-temperature budgets and thermal-cycling fatigue of electronic packages are treated in depth in Chapter 51; this section gives the materials logic common to all fields.

What remains open

Open question	Where it is answered
How are reciprocal space and Brillouin zones defined, and why is conductivity a second-rank tensor?	Chapter 4 (reciprocal space, tensor properties)
How does the same diffusion mathematics govern atoms in solids?	Chapter 8 (Fick's laws and their solutions)
How do quenching and cooling rate control steel microstructure and hardenability?	Chapter 15 (heat treatment, Jominy test)
How do glass, glass-ceramics and refractories achieve their thermal properties?	Chapter 17 (ceramics and glasses, refractories)
How do residual thermal stresses arise in composite laminates?	Chapter 23 (thermal loads in CLT)
How do electrons conduct charge, and how does alloying raise resistivity?	Chapter 25 (Drude, Matthiessen, alloy resistivity)
How does magnetism produce Invar behaviour and magnetocaloric cooling?	Chapter 28 (magnetic materials)
How do thermoelectric materials convert heat into electricity?	Chapter 40 (thermoelectric materials)
How do superalloys, CMCs and coatings survive in turbines and hypersonic flight?	Chapter 41 (extreme environments)
How fast do bond coats oxidize and TBCs fail?	Chapter 42 (high-temperature oxidation), Chapter 44 (failure analysis)
How is thermal conductivity reduced or raised at the nanoscale?	Chapter 38 (size effects, nanomaterials)
How are thermal properties measured (DSC, laser flash, TMA)?	Chapter 52 (thermal analysis)
How are thermal-management materials selected with performance indices?	Chapter 58 (materials selection)

Next: Chapter 25 (Electrical Conduction in Materials) turns from heat to charge — how electrons and ions conduct, and why conductivity spans so many orders of magnitude.

Chapter 25 — Electrical Conduction in Materials

Part VI · Thermal, Electrical, Dielectric and Magnetic Materials

A battery-pack designer switches the main busbars of an electric scooter from copper to aluminium to save weight. The new pack passes its bench tests, but after a few hundred charge cycles the bolted joints at the cell terminals run hot. The thermal camera shows a cluster of hot spots of 40 K and more, all at the joints, none along the bars. The bars themselves were sized correctly: aluminium has 1.6 times the resistivity of copper, and the designer made the cross-section 1.6 times larger. What the calculation left out was that the current does not flow through the whole face of a bolted joint. It flows through a few microscopic contact spots, and aluminium covers itself within seconds with a hard insulating oxide. The resistance that mattered was a property of a surface and a few micrometre-sized spots, not of the bulk.

This chapter is about what sets the resistance of real conductors: bulk metals, alloys, thin films, contacts and filled polymers. The central idea is simple. Every conductor has carriers, and those carriers are scattered. The density of carriers is fixed by the electronic structure of the material. Scattering is what the engineer controls through composition, temperature, defects, grain size and geometry.

> **Why is copper a better conductor than brass, even though both are mostly copper?**
> **Why does a 20 nm copper line have twice the resistivity of a copper wire? Why do the joints, rather than the conductors, so often limit a high-current design?**

Concept	Engineering destination
Resistivity, conductivity, sheet resistance	Conductor sizing, PCB trace width, film resistors
Mobility and scattering time	Why alloys and films conduct worse; link to semiconductors (Chapter 32)
Temperature coefficient of resistance	Winding temperature measurement, RTDs, precision shunts
Matthiessen and Nordheim rules	Alloy choice for connectors, springs, heating elements
Holm constriction resistance	Connector, relay, busbar-joint and battery-terminal design
Thin-film size effects	On-chip interconnects, thin-film resistors, transparent electrodes
Percolation in composites	Conductive adhesives, ESD plastics, PTC fuses, thick-film resistors
Wiedemann–Franz law	Estimating thermal conductivity of alloys and solders; contact heating
Selection indices	Wires, contacts, interconnects, electrodes

Roadmap. Conductivity and resistivity as material properties (25.1) → carriers, mobility and scattering (25.2) → temperature dependence and TCR (25.3) → metals and alloys, Matthiessen and Nordheim (25.4) → contacts (25.5) → thin films (25.6) → conductive composites (25.7) → electrical versus thermal conductivity (25.8) → material selection (25.9) → ionic conduction and solid electrolytes (25.10).

25.1 Conductivity and resistivity

From resistance to a material property

The resistance of a uniform bar of length L and cross-section A is

$$R = \rho \frac{L}{A}$$

where the **resistivity** ρ (unit $\Omega\,\mathrm{m}$) depends only on the material and its state (temperature, composition, microstructure). Its reciprocal is the **conductivity** $\sigma = 1/\rho$ (unit $\mathrm{S/m}$). The geometry has been separated from the material.

The local form of Ohm's law says the same thing without any bar:

$$\mathbf{J} = \sigma\,\mathbf{E}$$

The current density at a point is proportional to the electric field at that point. This is the form used in field solvers and in every derivation below. For an anisotropic crystal (graphite, many oxides, some intermetallics) σ is a tensor and $\mathbf{J}$ need not be parallel to $\mathbf{E}$ (Chapter 4). Cubic metals are isotropic, and we will treat σ as a scalar unless stated.

> **Status of the principle: Ohm's law $\mathbf{J} = \sigma\mathbf{E}$**
> *Approximate.* It is the linear term of the current response to a field. It holds very well in metals up to current densities far beyond any practical use, because the drift velocity stays tiny compared with the electron speed (Section 25.2). It fails at high fields in semiconductors (velocity saturation, Chapter 32), in insulators near breakdown (Chapter 26) and in very short channels where carriers do not scatter at all.

Units engineers actually use

Several practical units are in daily use and are easy to confuse:

- $\mu\Omega\,\mathrm{cm}$: $1\ \mu\Omega\,\mathrm{cm} = 10^{-8}\ \Omega\,\mathrm{m}$. Copper is about $1.7\ \mu\Omega\,\mathrm{cm}$.
- $\Omega\,\mathrm{mm}^2/\mathrm{m}$: the wire-table unit. $1\ \Omega\,\mathrm{mm}^2/\mathrm{m} = 10^{-6}\ \Omega\,\mathrm{m}$. Copper is $0.0172\ \Omega\,\mathrm{mm}^2/\mathrm{m}$.
- **% IACS** (International Annealed Copper Standard): conductivity as a percentage of that of standard annealed copper, defined as $\rho = 1.7241 \times 10^{-8}\ \Omega\,\mathrm{m}$ at 20 °C (that is, 58.0 MS/m). Modern high-purity copper exceeds 100 % IACS (about 101–102 %).
- **Sheet resistance** $R_s = \rho/t$ for a film of thickness t, in ohms per square ($\Omega/\square$). A line of length L and width W has $R = R_s\,L/W$, i.e. R_s times the number of squares. It is the natural unit for PCB copper, thin films and printed conductors (Section 25.6).

The widest range of any physical property

Resistivity spans more than 25 orders of magnitude at room temperature, from silver ($1.6 \times 10^{-8}\ \Omega\,\mathrm{m}$) to PTFE (above $10^{16}\ \Omega\,\mathrm{m}$), and superconductors drop to zero (Chapter 30). Few other material properties vary over such a range. It is why the same field equation can describe a busbar and a cable insulation, and why measurement methods differ so much between conductors (four-point probe, Chapter 52) and insulators (guarded electrodes, Chapter 26).

Material class	Example	Typical ρ at 20 °C ($\Omega\,\mathrm{m}$)
Pure metals	Ag, Cu, Al	1.6–2.7×10^{-8}
Resistance alloys	Constantan, Nichrome	0.5–1.1×10^{-6}
Conductive oxides	ITO film, RuO_2	10^{-6}–10^{-5}
Graphite (polycrystalline)	Electrode graphite	$\approx 10^{-5}$
Conductive composites	Ag-filled epoxy; carbon-filled polymer	10^{-6}–10^{-3}; 10^{-2}–10^{2}
Semiconductors	Si (doped to intrinsic)	10^{-5}–2×10^{3}
Static-dissipative plastics	ESD trays	10^{4}–10^{9}
Glasses and ceramics	Soda-lime glass, alumina	10^{9}–> 10^{12}
Polymers	Epoxy, PE, PTFE	10^{12}–> 10^{16}

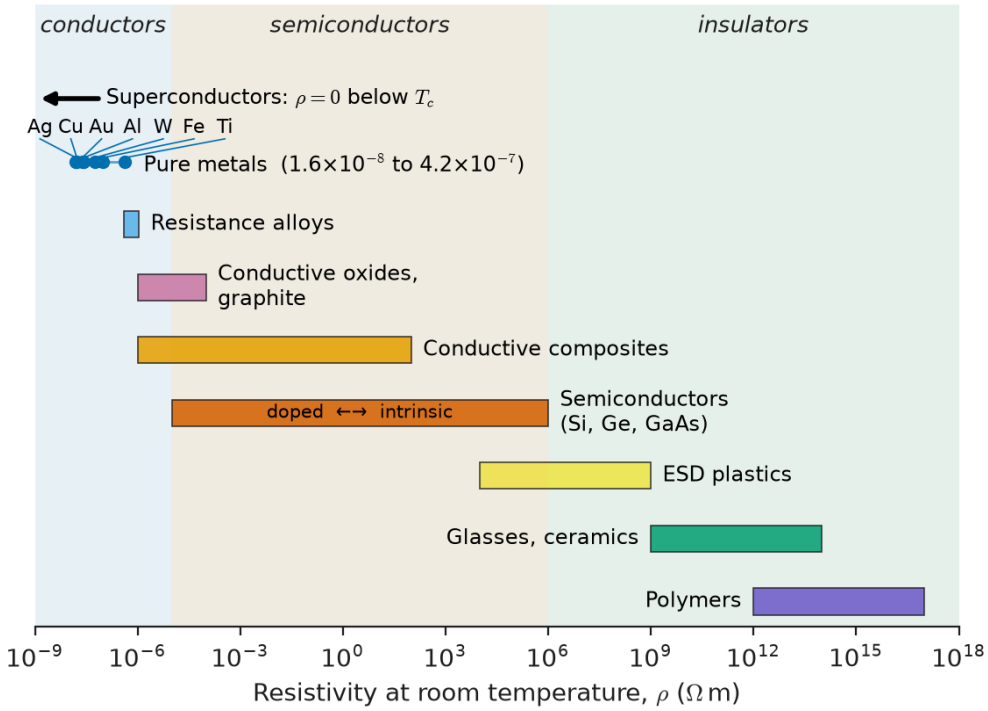

Figure 25.1 — The resistivity of engineering materials at room temperature spans more than 25 orders of magnitude.

Figure 25.1 shows this range by material class. The split into "conductors", "semiconductors" and "insulators" is not a matter of degree only. A metal's carrier density is fixed by its electronic structure and barely changes with temperature. A semiconductor's carrier density can be changed by many orders of magnitude by doping and temperature. An insulator has almost no free carriers at all. Explaining that split needs energy bands, which is the job of Chapter 31. This chapter takes the carrier density of a metal as given and asks what limits the current.

25.2 Mobility and scattering as material properties

The Drude picture

Treat the conduction electrons of a metal as a gas of n free carriers per unit volume, each with charge $-e$ and mass m. A field $\mathbf{E}$ accelerates each electron. Collisions with imperfections of the lattice randomize its momentum, on average once every **mean free time** (scattering time) τ. Between collisions the field adds momentum $-e\mathbf{E}$ per unit time; collisions remove it at the rate $\langle \mathbf{p} \rangle / \tau$. In steady state the two balance:

$$\frac{d\langle \mathbf{p} \rangle}{dt} = -e\mathbf{E} - \frac{\langle \mathbf{p} \rangle}{\tau} = 0 \quad \Longrightarrow \quad \mathbf{v}_d = \frac{\langle \mathbf{p} \rangle}{m} = -\frac{e\tau}{m}\,\mathbf{E}$$

The **drift velocity** $\mathbf{v}_d$ is proportional to the field. The constant of proportionality is the **mobility**

$$\mu = \frac{e\tau}{m} \qquad (\mathrm{m^2\,V^{-1}\,s^{-1}})$$

The current density is the charge per volume times the drift velocity, $\mathbf{J} = -ne\mathbf{v}_d = ne\mu\mathbf{E}$, so

$$\sigma = ne\mu = \frac{ne^2\tau}{m}$$

This one equation organizes the whole chapter. Conductivity is a product of **how many carriers** (n) and **how freely they move** (μ, or equivalently τ). In semiconductors both factors vary and both are engineered (Chapter 32). In metals n is essentially fixed at roughly one to three electrons per atom, about 10^{28}–$10^{29}\ \mathrm{m^{-3}}$. All the differences between pure copper, brass, a thin copper film and hot copper come from τ.

Status of the principle: Drude conductivity $\sigma = ne^2\tau/m$
Approximate. It assumes free carriers with a single relaxation time. The form survives in the band theory of Chapter 31 with m replaced by an effective mass and τ evaluated for electrons at the Fermi energy. The classical Drude model gets the electron speed wrong, and with it the heat capacity and the mean free path (see below), but its conductivity formula is right.

What the numbers say about copper

Worked example — scattering time, mobility and mean free path of copper. Copper has one conduction electron per atom. With density $8.96\ \text{g/cm}^3$ and molar mass $63.55\ \text{g/mol}$, $n = 8.47 \times 10^{28}\ \text{m}^{-3}$. At room temperature $\rho = 1.68 \times 10^{-8}\ \Omega\,\text{m}$.

The scattering time:

$$\tau = \frac{m}{ne^2\rho} = \frac{9.109 \times 10^{-31}}{(8.47 \times 10^{28})(1.602 \times 10^{-19})^2(1.68 \times 10^{-8})} = 2.49 \times 10^{-14}\ \text{s}$$

The mobility:

$$\mu = \frac{e\tau}{m} = \frac{1}{ne\rho} = \frac{1}{(8.47 \times 10^{28})(1.602 \times 10^{-19})(1.68 \times 10^{-8})} = 4.39 \times 10^{-3}\ \text{m}^2/(\text{V s}) = 43.9\ \text{cm}^2/(\text{V s})$$

This is low. Electrons in lightly doped silicon have a mobility of about $1400\ \text{cm}^2/(\text{V s})$. Copper conducts well because it has 10^6–10^9 times more carriers, not because its carriers move easily.

The drift velocity at a heavy current density $J = 10\ \text{A/mm}^2 = 10^7\ \text{A/m}^2$:

$$v_d = \frac{J}{ne} = \frac{10^7}{(8.47 \times 10^{28})(1.602 \times 10^{-19})} = 7.4 \times 10^{-4}\ \text{m/s}$$

Less than a millimetre per second. The signal travels at a large fraction of the speed of light because the field propagates, not the electrons.

The mean free path. The conduction electrons do not move at thermal speeds. They move at the **Fermi velocity**, $v_F = 1.57 \times 10^6\ \text{m/s}$ for copper, a quantum result explained in Chapter 31. The distance between scattering events is

$$\lambda = v_F\tau = (1.57 \times 10^6)(2.49 \times 10^{-14}) = 3.9 \times 10^{-8}\ \text{m} = 39\ \text{nm}$$

about 150 atomic spacings. Electrons pass straight through a perfect lattice. They are scattered only by departures from perfect periodicity. And 39 nm is close to the dimensions of modern on-chip wires, which is why Section 25.6 will find that thin copper does not behave like bulk copper.

What scatters electrons

A perfectly periodic lattice does not scatter electrons at all. That result (Bloch's theorem, Chapter 31) is why a classical picture of electrons bouncing off every ion fails by two orders of magnitude in λ. Scattering comes only from **deviations from periodicity**, which are exactly the structural features of Chapters 2 and 5:

Scatterer	Origin	Temperature dependence	Controlled by
Phonons (lattice vibrations)	Thermal motion of atoms	Strong; $\propto T$ above about $\Theta_D/3$	Temperature only
Impurity atoms in solution	Composition	Nearly independent of T	Purity, alloying
Vacancies, interstitials	Quenching, irradiation	Independent of T (if frozen in)	Annealing
Dislocations	Cold work	Independent of T	Annealing
Grain boundaries	Polycrystalline structure	Independent of T	Grain size (Section 25.6)
External surfaces	Finite size	Independent of T	Film thickness, wire width
Precipitates	Second phase	Independent of T	Heat treatment

Each scattering process has its own rate $1/\tau_i$. Because the processes are independent, the rates add:

$$\frac{1}{\tau} = \frac{1}{\tau_{\text{ph}}} + \frac{1}{\tau_{\text{imp}}} + \frac{1}{\tau_{\text{def}}} + \cdots$$

and since $\rho = m/(ne^2\tau)$ is proportional to $1/\tau$, the resistivities add. This is Matthiessen's rule, taken up in Section 25.4. The mobility and scattering time are therefore material properties in the same sense as yield strength: they depend on composition, temperature and microstructure, and processing sets them.

> **Boundary.**
> The quantum-mechanical reason a periodic lattice does not scatter, and the meaning of v_F and the effective mass, belong to Chapter 31 (and in full to Engineering Physics). Here we use them as material parameters.

25.3 Temperature coefficients of resistance

Definition

The **temperature coefficient of resistance** (TCR) at a reference temperature T_0 is

$$\alpha_0 = \frac{1}{\rho(T_0)}\left.\frac{d\rho}{dT}\right|_{T_0}$$

Over a limited range the resistivity is close to linear,

$$\rho(T) \approx \rho(T_0)\,[1 + \alpha_0\,(T - T_0)]$$

so that α_0 is the fractional change per kelvin. The value depends on the reference temperature, because $\rho(T_0)$ appears in the denominator. Datasheets usually quote α_{20} (or α_{25}). TCR is often given in ppm/K; $1\ \text{ppm/K} = 10^{-6}\ \text{K}^{-1}$.

Why pure metals have a TCR near 0.4 % per kelvin

Above roughly a third of the Debye temperature Θ_D, the number of thermally excited phonons is proportional to T, and so is the phonon scattering rate. The phonon resistivity is then linear in absolute temperature, $\rho_{\text{ph}} \approx bT$. If phonons dominate, extrapolating ρ linearly downward reaches zero near $T = 0$ K, and the TCR at temperature T is close to $1/T$. At 293 K that gives about $0.0034\ \text{K}^{-1}$. Real pure metals have $\alpha_{20} = 0.0039$–$0.0045\ \text{K}^{-1}$. They are slightly higher because the phonon resistivity is not exactly linear and because the extrapolated zero lies a little above 0 K (for copper the linear extrapolation hits zero at -234.5 °C, i.e. 38.7 K).

At low temperature the phonon contribution collapses (as T^5 in the Bloch–Grüneisen model for $T \ll \Theta_D$) and only the temperature-independent residual part remains. The ratio $\rho(300\ \text{K})/\rho(4.2\ \text{K})$, the **residual resistivity ratio** (RRR), is a sensitive measure of purity and perfection: about 50–100 for commercial copper and several thousand for annealed 6N copper. RRR matters in cryogenic electronics and in stabilizers for superconducting wires (Chapter 30).

> **Status of the principle: $\rho_{\text{ph}} \propto T$ above about $\Theta_D/3$**
> *Approximate.* It follows from the classical (high-temperature) population of phonons. It fails at low temperature, where phonons freeze out, and is modified near the melting point and in metals with complex band structure (transition metals such as Ni and Pt show curvature).

Engineering values

Material	ρ_{20} ($10^{-8}\ \Omega\,\mathrm{m}$)	α_{20} ($10^{-3}\ \mathrm{K}^{-1}$)	Use
Copper (annealed, IACS)	1.724	3.93	Windings, wiring
Aluminium (EC grade 1350)	2.82	4.03	Overhead lines, busbars
Silver	1.59	3.8	Contacts, plating
Gold	2.2	3.4	Contacts, bond wires
Platinum	10.6	3.85 (IEC 60751, 0–100 °C)	RTDs (Pt100)
Nickel	6.9	6.0	Thermistors, sensors
Tungsten	5.3	4.5	Lamp filaments, contacts
Nichrome (80Ni–20Cr)	108	about 0.1	Heating elements, film resistors
Constantan (55Cu–45Ni)	49	about ± 0.02	Thermocouples, strain gauges
Manganin (86Cu–12Mn–2Ni)	43	about ± 0.01	Precision shunts, standards
Carbon (graphite)	about 1000	about -0.5	Carbon resistors, brushes

The table splits into two groups that serve opposite purposes. Pure metals have a large, reproducible TCR, which is exactly what a **temperature sensor** needs. The platinum resistance thermometer (Pt100: 100 Ω at 0 °C) is the industrial standard because platinum is chemically stable and its $\rho(T)$ is reproducible to millikelvins; its curve is described by the Callendar–Van Dusen equation, an empirical quadratic (cubic below 0 °C). Concentrated alloys have a tiny TCR, which is what a **precision resistor or shunt** needs. Section 25.4 explains why alloying flattens $\rho(T)$.

Measuring a winding temperature by its resistance

Since copper's TCR is reproducible, the resistance of a winding is a thermometer averaged over the whole winding. Standards for motors and transformers (IEC 60034, IEEE C57) use this "resistance method". Writing $\rho \propto (T + 234.5)$ for copper with T in °C:

$$\frac{R_2}{R_1} = \frac{234.5 + T_2}{234.5 + T_1}$$

(for aluminium the constant is 225 or 228, depending on the standard and grade).

Worked example — winding temperature rise. A transformer winding measures $R_1 = 2.000\ \Omega$ at the ambient temperature of 20 °C. Immediately after a heat run it measures $R_2 = 2.480\ \Omega$. The hot temperature is

$$T_2 = \frac{R_2}{R_1}(234.5 + T_1) - 234.5 = \frac{2.480}{2.000}(254.5) - 234.5 = 315.6 - 234.5 = 81.1\ ^\circ\mathrm{C}$$

The average winding rise is 61 K. Two lessons follow. First, a 24 % increase in resistance is a 24 % increase in I^2R loss at the same current, so hot copper is less efficient copper, and loss estimates must be made at the operating temperature (75 °C or 115 °C reference temperatures are used in standards for this reason). Second, the method gives an average. The hottest spot may be 10–20 K hotter, which is what insulation life depends on (Chapter 26).

25.4 Conductivity in metals and alloys

Why pure metals differ

Within the pure metals, resistivity varies by a factor of about 30 at room temperature (silver 1.6, titanium $42 \times 10^{-8}\ \Omega\,\mathrm{m}$). The variation tracks the electronic structure. The noble metals Cu, Ag, Au have a single s-electron outside a filled d-shell; their conduction electrons have a small effective mass and few states to scatter into, and they are the best conductors. Aluminium has three free electrons per atom but scatters them more strongly. Transition metals (Fe, Ni, Pt, Ti) have partially filled d bands that offer many empty states into which an s-electron can be scattered. Their scattering rates are higher and their resistivities are 4–

25 times that of copper. These trends come from band structure (Chapter 31). The engineer's working rule is simpler: **for low resistivity, use Ag, Cu, Au or Al, and keep them pure.**

Metal	ρ_{20} ($10^{-8}\ \Omega\,\mathrm{m}$)	% IACS	Density ($\mathrm{g/cm^3}$)	Melting point (°C)
Silver	1.59	108	10.5	962
Copper (OFE, annealed)	1.69	102	8.96	1085
Gold	2.2	78	19.3	1064
Aluminium (99.99 %)	2.65	65	2.70	660
Magnesium	4.4	39	1.74	650
Tungsten	5.3	33	19.3	3422
Zinc	5.9	29	7.14	420
Nickel	6.9	25	8.90	1455
Iron	9.7	18	7.87	1538
Platinum	10.6	16	21.5	1768
Tin	11.5	15	7.29	232
Lead	21	8	11.3	327
Titanium	42	4	4.51	1668

Matthiessen's rule

Section 25.2 showed that independent scattering rates add. Translating to resistivity:

$$\rho(T, c, \text{defects}) = \rho_{\text{ph}}(T) + \rho_{\text{imp}}(c) + \rho_{\text{def}}$$

The thermal part depends only on temperature and is the same for pure copper and dilute copper alloys. The residual parts depend on composition and defects and not on temperature. Plotting $\rho(T)$ for a pure metal and a dilute alloy gives two curves of the same shape, displaced vertically by a constant (Figure 25.2).

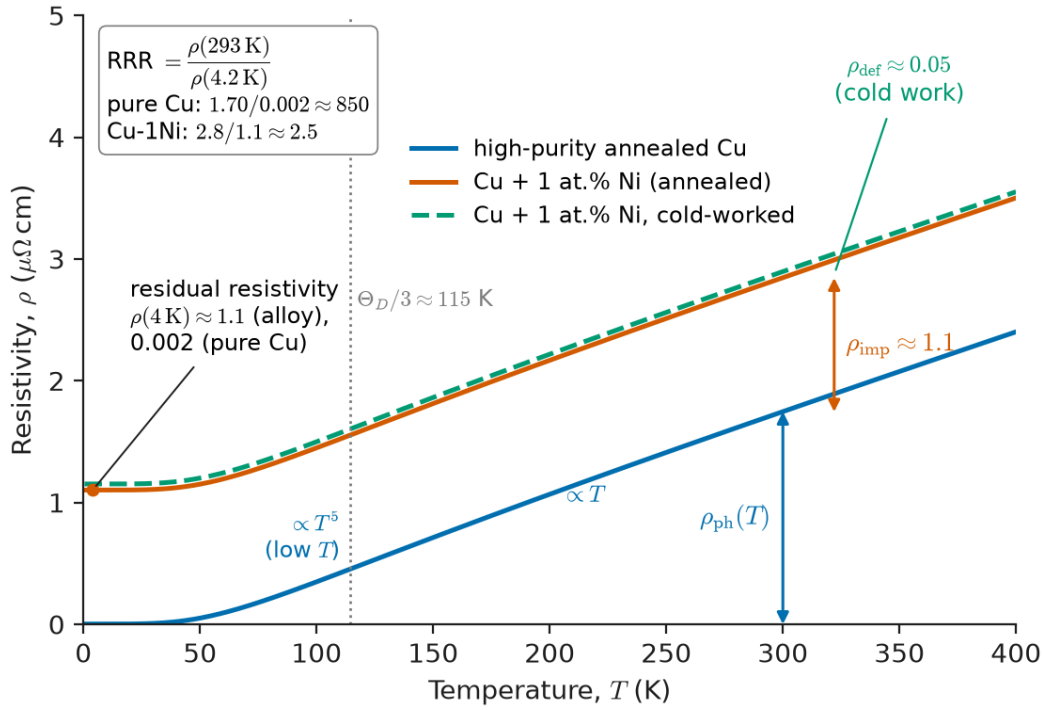

Figure 25.2 — Matthiessen's rule: impurities and defects shift the resistivity curve upward without changing its temperature-dependent part.

Figure 25.2 shows the consequence for the TCR. The slope $d\rho/dT$ is the same for the pure metal and the alloy, but ρ is larger for the alloy, so the TCR $\alpha = (1/\rho)\,d\rho/dT$ is smaller. Adding a residual resistivity lowers the TCR. A concentrated alloy with a large residual term and an almost flat curve, such as manganin or constantan, is the extreme case.

> **Status of the principle: Matthiessen's rule**
> *Empirical.* It follows from adding independent scattering rates, but scattering processes are not strictly independent (impurities change the phonon spectrum and the electron distribution). Deviations of a few per cent of the residual resistivity are common, and the rule fails badly in concentrated alloys and at low temperature in some systems. It remains the standard first estimate.

Impurities: a little goes a long way

Each dissolved impurity atom adds its own resistivity, roughly in proportion to its concentration while dilute. The increment per atomic per cent varies enormously with the solute:

Solute in copper	Approximate $\Delta\rho$ ($\mu\Omega\,\mathrm{cm}$ per at.%)
Ag	0.14
Zn	0.3
Au	0.55
Ni	1.1
Sn	2.9
P	6.7
Fe	9

The pattern is that solutes chemically similar to the host (Ag, Au, Zn) scatter weakly; solutes with a different valence or an open d-shell (P, Fe, Co) scatter strongly. Two industrial consequences follow:

- **Phosphorus-deoxidized copper** (Cu-DHP, about 0.02 wt% P) is easy to braze and weld, but its conductivity is about 85 % IACS. Electrical conductors use electrolytic tough pitch (Cu-ETP) or oxygen-free (Cu-OF) copper at 100–101 % IACS. A tiny residue of the wrong element costs 15 % of the conductivity.
- **Silver-bearing copper** (about 0.03–0.1 % Ag) keeps nearly full conductivity while raising the softening temperature, which is why it is used for commutators and windings that are hot during manufacturing.

Concentrated alloys: Nordheim's rule

When a second element is added beyond the dilute limit, the residual resistivity of a disordered solid solution $A_{1-x}B_x$ follows approximately

$$\rho_{\text{alloy}} \approx \rho_{\text{ph}}(T) + C\,x(1-x)$$

where C is a constant for the pair of elements. The resistivity is largest near $x = 0.5$, where the lattice is most disordered, and returns to the pure-metal value at both ends. Copper–nickel (complete solid solubility) is the classic case; its resistivity rises from 1.7 to about 50 $\mu\Omega\,$cm near 45 % Ni, which is constantan.

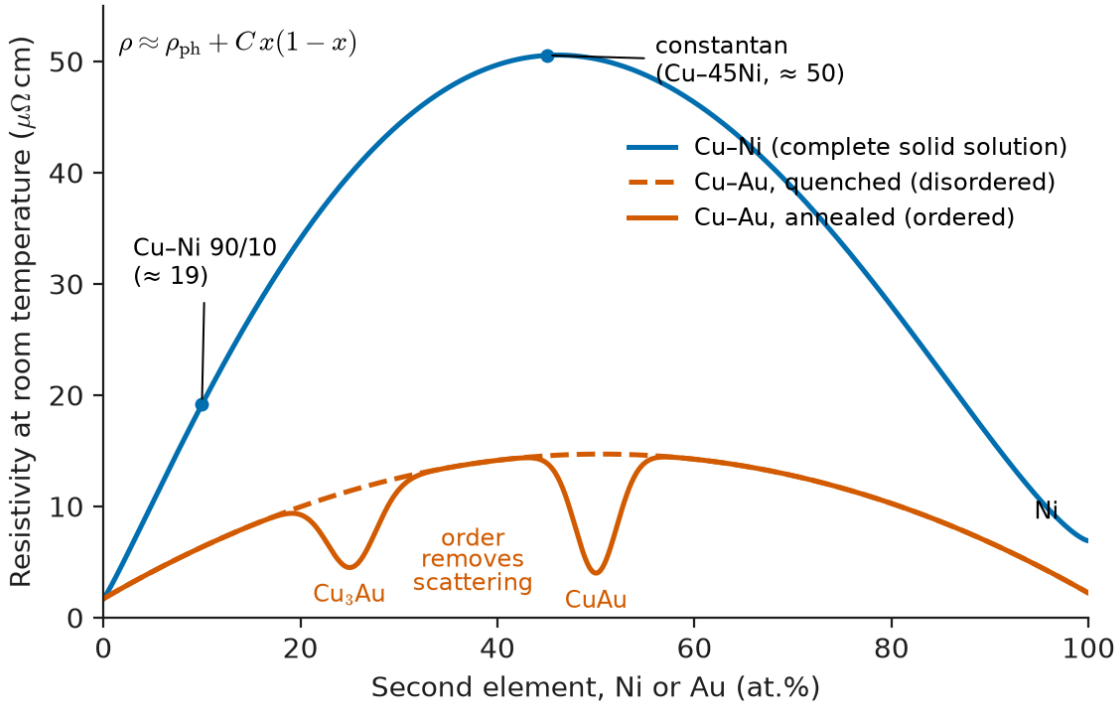

Figure 25.3 — Resistivity across two complete solid-solution systems: the Nordheim parabola in Cu–Ni and the effect of atomic ordering in Cu–Au.

Figure 25.3 also shows the second half of the story. If the alloy **orders**, so that A and B atoms occupy regular sublattices (Cu_3Au, CuAu), the lattice becomes periodic again and the resistivity drops sharply. Disorder, not composition as such, is what scatters. The same principle explains why intermetallic compounds such as Cu_6Sn_5 in solder joints (Chapter 51) have moderate resistivity, not a disordered-alloy value.

> **Status of the principle: Nordheim's rule**
> *Empirical.* It describes disordered binary solid solutions of elements with similar electronic structure. It fails for ordered alloys, for transition-metal solutes with magnetic effects, and when phases separate.

Strength versus conductivity

Pure annealed copper is soft (yield strength about 70 MPa). A connector spring, a trolley wire or a lead frame needs much more strength. Every strengthening mechanism of Chapter 13 works by introducing obstacles to dislocations, and every obstacle also scatters electrons. The question is how much conductivity each mechanism costs:

Strengthening mechanism	Effect on strength	Effect on conductivity	Example
Cold work (dislocations)	Large (yield up to about 350 MPa)	Small (2–3 % IACS lost)	Hard-drawn Cu wire
Solid solution	Moderate	Large (solute in the lattice scatters strongly)	Brass, phosphor bronze, Cu–Ni
Precipitation hardening	Large	Small to moderate (solute removed from lattice into particles)	CuCrZr, CuBe, CuNiSi
Grain refinement	Moderate	Small in bulk	Fine-grained Cu
Dispersion strengthening	Moderate, stable when hot	Small	Al_2O_3-dispersed Cu (Glidcop)

The key is where the alloying atoms sit. A solute dissolved in the lattice scatters on every atomic scale; the same atoms gathered into precipitates scatter only at the particle surfaces, a far smaller area. Precipitation-hardened alloys therefore give the best compromise:

Alloy	Typical conductivity (% IACS)	Typical yield strength (MPa)	Use
Cu-ETP, annealed	101	70	Wire, busbar
Cu-ETP, hard drawn	97	300–350	Overhead contact wire
CuCrZr (aged)	75–85	300–450	Welding electrodes, lead frames
CuNiSi (C70250, aged)	35–45	550–700	Connector springs
CuBe (C17200, aged)	20–25	1000–1200	High-reliability springs
Brass C26000 (70Cu–30Zn)	28	100–450	Terminals, housings
Phosphor bronze C51000	15	300–700	Spring contacts
Al 1350 (EC)	61–62	30–165	Conductors
Al 6101/6201 (Al–Mg–Si, aged)	52–57	170–300	Busbars, overhead conductors

The answer to the first opening question is now clear. Brass has 28 % IACS, not 100 %, because 30 % zinc in solid solution makes the copper lattice disordered on every atomic scale. Brass is chosen for terminals because it is cheap, machinable and strong enough, not because it conducts well.

Worked example — aluminium for copper. Replace a copper busbar ($\rho_{\mathrm{Cu}} = 1.72 \times 10^{-8}\ \Omega\,\mathrm{m}$) with EC aluminium ($\rho_{\mathrm{Al}} = 2.82 \times 10^{-8}\ \Omega\,\mathrm{m}$) of the same length and the same resistance. Equal resistance needs

$$\frac{A_{\mathrm{Al}}}{A_{\mathrm{Cu}}} = \frac{\rho_{\mathrm{Al}}}{\rho_{\mathrm{Cu}}} = \frac{2.82}{1.72} = 1.64$$

For a round conductor the diameter grows by $\sqrt{1.64} = 1.28$. The mass ratio is

$$\frac{m_{\mathrm{Al}}}{m_{\mathrm{Cu}}} = \frac{A_{\mathrm{Al}}\, d_{\mathrm{Al}}}{A_{\mathrm{Cu}}\, d_{\mathrm{Cu}}} = 1.64 \times \frac{2.70}{8.96} = 0.49$$

where d is the density. The aluminium conductor has the same resistance at half the mass, which is why overhead transmission lines, large building busbars and many automotive cables use aluminium. The price is the larger cross-section, a lower strength (often fixed with a steel core or Al–Mg–Si alloy), a higher thermal expansion, creep under bolt load, and the joint problem of the opening paragraph, analysed in the next section.

25.5 Contact resistance

Why two flat conductors touch only at spots

No real surface is flat at the micrometre scale. When two surfaces are pressed together, they touch only at the tips of the highest asperities. The asperities deform plastically until the total load is supported, so the true contact area is

$$A_r \approx \frac{F}{H}$$

where F is the contact force and H the indentation hardness of the softer member. With a hardness of about 1 GPa, a force of 1 N gives a real contact area of $10^{-9}\ \mathrm{m}^2$, i.e. 1000 $\mu\mathrm{m}^2$. That is a millionth of a square centimetre, regardless of how large the apparent contact face is. The current must squeeze through these small conducting spots ("a-spots").

Constriction resistance (Holm)

Consider a single circular spot of radius a joining two semi-infinite bodies of resistivity ρ. The current lines converge into the spot and spread out again on the other side (Figure 25.4). Solving Laplace's equation for this geometry (Holm, 1958) gives the **constriction resistance**

$$R_c = \frac{\rho}{2a}$$

for the two halves together ($\rho/4a$ each). If the two members have different resistivities, $R_c = (\rho_1 + \rho_2)/4a$. The resistance scales with $1/a$, not $1/a^2$, because the constriction is a three-dimensional spreading effect, concentrated within a distance of a few a from the spot.

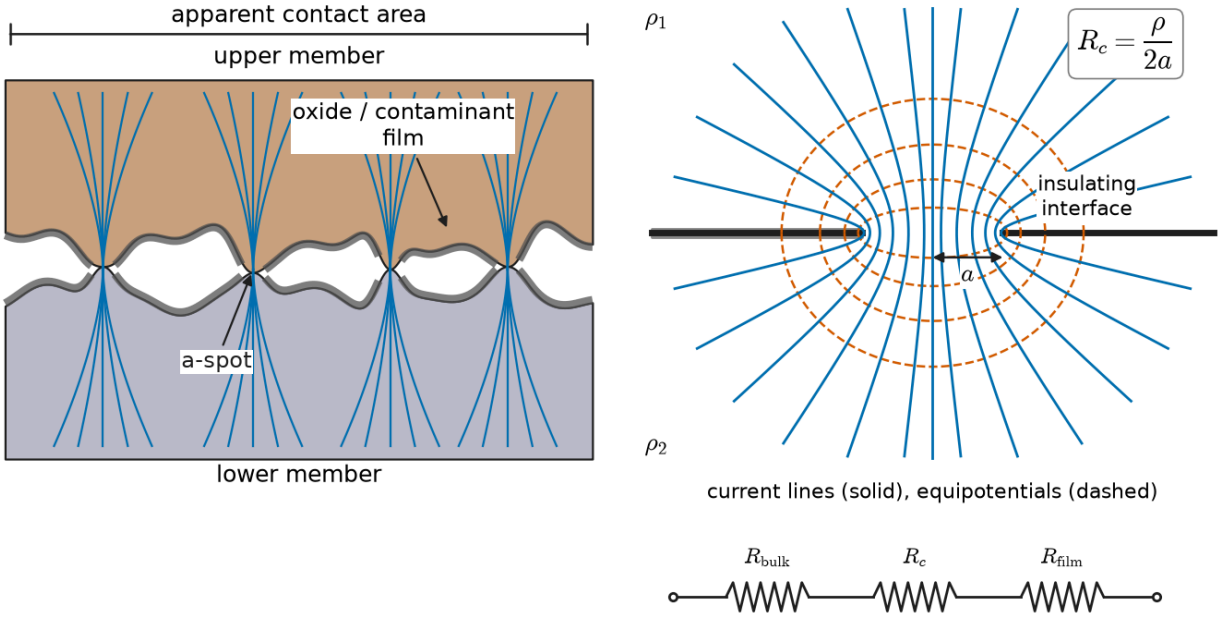

Figure 25.4 — Electrical contact at the microscale: current lines converge through a-spots, and surface films add resistance in series.

Worked example — a copper contact at 1 N. Two copper surfaces ($\rho = 1.72 \times 10^{-8}\ \Omega\,\mathrm{m}$, $H \approx 1$ GPa) are pressed together with $F = 1$ N.

The real contact area is $A_r = F/H = 10^{-9}\ \mathrm{m}^2$. If it formed a single circular spot, its radius would be

$$a = \sqrt{\frac{A_r}{\pi}} = \sqrt{\frac{10^{-9}}{\pi}} = 1.78 \times 10^{-5}\ \mathrm{m} = 17.8\ \mu\mathrm{m}$$

and the constriction resistance

$$R_c = \frac{\rho}{2a} = \frac{1.72 \times 10^{-8}}{2 \times 1.78 \times 10^{-5}} = 4.8 \times 10^{-4}\ \Omega = 0.48\ \mathrm{m}\Omega$$

In practice the area is divided among many spots. If it were split into ten spots of equal area spread far apart, each would have $a = 5.64\ \mu\text{m}$; the ten resistances in parallel give $\rho/(2 \times 10 \times a) = 0.15\ \text{m}\Omega$. Spots close together interact, and Greenwood's result for a cluster of n spots of radius a within a region of radius α is

$$R_c \approx \frac{\rho}{2na} + \frac{\rho}{2\alpha}$$

The first term is the parallel spot resistance; the second is the constriction of the whole cluster. For a cluster radius of 0.5 mm the second term adds only 0.017 mΩ. A clean metallic contact at 1 N is therefore a fraction of a milliohm, which is acceptable for a signal connector but not for a 200 A busbar joint, which needs hundreds to thousands of newtons. Because $a \propto \sqrt{F/H}$, the single-spot estimate gives $R_c \propto F^{-1/2}$: quadrupling the force only halves the resistance. Measured connector curves show $R_c \propto F^{-m}$ with m between 0.5 and 1, depending on how the number of spots grows with force.

> **Status of the principle: Holm constriction resistance $R_c = \rho/2a$**
> *Derived.* It is an exact solution of Laplace's equation for a circular spot between half-spaces. Its use for real contacts is approximate: spot sizes and numbers are estimated from F/H, and when a approaches the electron mean free path (39 nm in copper) the spot conducts ballistically (Sharvin resistance), which adds a term $\propto \lambda/a^2$.

Film resistance

The second, and usually larger, contribution comes from films on the surfaces: oxides, sulphides, adsorbed water, organic residues. A film of resistivity ρ_f and thickness t_f over a spot of area πa^2 adds

$$R_f = \frac{\rho_f\, t_f}{\pi a^2} = \frac{\sigma_f}{\pi a^2}$$

where σ_f (in $\Omega\,\text{m}^2$) is the film resistance per unit area. For films thinner than about 2 nm, electrons tunnel through and σ_f is roughly 10^{-12}–$10^{-8}\ \Omega\,\text{m}^2$. Thicker films insulate, and contact is made only where the film is broken mechanically (by wiping or by high contact pressure) or electrically ("fritting", breakdown of the film at a few volts).

This is why contact materials are chosen for the films they do or do not form:

- **Gold** forms no oxide. A thin gold plate (0.1–1.3 μm over a nickel barrier) keeps low-force signal contacts (0.3–1 N) below a few milliohms for decades. Hard gold (Au–Co, Au–Ni) resists wear.
- **Silver** forms no oxide but tarnishes to Ag_2S in sulphur-bearing air. The sulphide is soft and broken easily at moderate force, so silver works well in power contacts.
- **Tin** oxidizes in seconds, but the oxide is thin and brittle and tin is soft, so a force of 1–3 N cracks it. Tin contacts must not move after mating: micromotion of a few micrometres (from vibration or thermal cycling) repeatedly exposes fresh tin, which oxidizes and accumulates as an insulating debris layer. This **fretting corrosion** is a leading cause of automotive connector failure.
- **Aluminium** forms an oxide that is hard, adherent, and an excellent insulator (it is the dielectric of electrolytic capacitors). Joints need abrasion plus a sealing grease, bimetallic Cu–Al transition pieces, or plating. Aluminium also creeps under bolt load, so the force decays and the joint relaxes; spring washers and specified torques compensate. This is the busbar failure of the opening paragraph.

Contact heating and the voltage–temperature relation

All the power dissipated in a constriction is generated within a few spot radii of the spot, so the spot itself is the hottest point. Because heat and current spread through the same geometry, a remarkable result follows (Kohlrausch): if the metal obeys the Wiedemann–Franz law (Section 25.8), the maximum temperature T_m in the spot depends only on the voltage U across the contact, not on its size or the metal:

$$T_m^2 - T_0^2 = \frac{U^2}{4L}$$

where $L = 2.44 \times 10^{-8}\ \text{W}\,\Omega\,\text{K}^{-2}$ is the Lorenz number and T_0 the bulk temperature. Setting T_m to characteristic temperatures of copper ($T_0 = 293$ K):

- **Softening** of copper (about 190 °C, 463 K): $U = \sqrt{4L(T_m^2 - T_0^2)} = 0.11$ V.
- **Melting** (1358 K): $U = 0.41$ V.

Measured softening and melting voltages for copper are about 0.12 V and 0.43 V. A contact drop of a tenth of a volt is therefore a warning: the spots soften, flatten and grow (which lowers the resistance, a self-healing effect), but at a few tenths of a volt they weld. Contact-resistance degradation over time (fretting, oxidation, intermetallic growth, stress relaxation) is treated as a failure mechanism in Chapter 44.

Contact materials for switching

Contacts that make and break current (relays, contactors, circuit breakers) also face arcing, which erodes and welds. Their materials are composites designed against these failures:

Contact material	Typical use	Reason
Fine silver, AgNi	Low-current relays	Low resistance, soft, resists welding at low current
Ag–SnO_2 (replaced Ag–CdO)	Contactors, motor starters	Oxide particles raise arc viscosity and resist welding
W–Cu, W–Ag (sintered, infiltrated)	Circuit breakers, arcing tips	Tungsten resists arc erosion; Cu/Ag carries current
Au, Au alloys, Pd	Signal relays, reed switches	No films at low force and low voltage ("dry circuit")
Graphite, Cu–graphite	Brushes	Self-lubricating sliding contact

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25.6 Thin-film conductors

Sheet resistance

On-chip interconnects, thin-film resistors, transparent electrodes and PCB copper are all thin layers, specified by sheet resistance $R_s = \rho / t$. PCB copper is specified by mass per area in ounces per square foot: 1 oz copper is 35 μm thick, giving $R_s = 1.72 \times 10^{-8} / 35 \times 10^{-6} = 0.49\ \text{m}\Omega/\square$. A 0.25 mm wide, 50 mm long trace is 200 squares, or 98 mΩ at 20 °C.

Why thin films have higher resistivity

When a film thickness t or a grain size d becomes comparable with the electron mean free path λ, the boundaries add scattering that bulk metal does not have. Two models describe the two contributions.

Surface scattering (Fuchs–Sondheimer). An electron hitting a surface may reflect specularly (keeping its momentum along the film, no added resistance) or diffusely (randomized, like a collision). With a fraction p of specular reflections, the thick-film limit of the Fuchs–Sondheimer solution is

$$\frac{\rho_{\text{film}}}{\rho_0} \approx 1 + \frac{3}{8}\frac{\lambda}{t}(1-p), \qquad t \gtrsim \lambda$$

For $p = 1$ there is no penalty. Real deposited films are rough on the atomic scale, and p is usually close to zero.

Grain-boundary scattering (Mayadas–Shatzkes). Electrons crossing a grain boundary are reflected with probability R (typically 0.2–0.5 for copper). With average grain size d, define

$$\alpha = \frac{\lambda}{d}\frac{R}{1-R}$$

The Mayadas–Shatzkes result is

$$\frac{\rho_0}{\rho_{\text{gb}}} = 3\left[\frac{1}{3} - \frac{\alpha}{2} + \alpha^2 - \alpha^3 \ln\left(1 + \frac{1}{\alpha}\right)\right] \quad \approx \quad \frac{\rho_{\text{gb}}}{\rho_0} \approx 1 + \frac{3}{2}\alpha \ \ (\alpha \ll 1)$$

Grain size in a deposited film is usually comparable with the film thickness, so the two effects grow together as the film thins.

> **Status of the principle: Fuchs–Sondheimer and Mayadas–Shatzkes models**
> *Approximate.* Both are Boltzmann-transport solutions with a single isotropic mean free path, and the parameters p and R are fitted to data. They describe trends well above about 10 nm; below that, the band structure of the thin layer, surface roughness on the scale of the electron wavelength and liner materials dominate, and first-principles calculations are needed.

Worked example — a 100 nm copper film. Take $\lambda = 39$ nm, $t = 100$ nm, fully diffuse surfaces ($p = 0$), grain size $d = 100$ nm and grain-boundary reflection $R = 0.3$.

Surface scattering:

$$\frac{\rho_{\mathrm{FS}}}{\rho_0} = 1 + \frac{3}{8} \times \frac{39}{100} = 1 + 0.146 = 1.146$$

Grain boundaries: $\alpha = (39/100) \times (0.3/0.7) = 0.167$. The approximate form gives $1 + 1.5 \times 0.167 = 1.25$; the full expression gives

$$3\,[0.3333 - 0.0836 + 0.0279 - 0.0091] = 0.806 \quad \Rightarrow \quad \frac{\rho_{\mathrm{gb}}}{\rho_0} = 1.24$$

Adding the two excess contributions (Matthiessen's rule again): $\rho/\rho_0 \approx 1 + 0.146 + 0.24 = 1.39$, so

$$\rho \approx 1.39 \times 1.68 = 2.3\ \mu\Omega\,\mathrm{cm}, \qquad R_s = \frac{2.3 \times 10^{-8}}{1.0 \times 10^{-7}} = 0.23\ \Omega/\square$$

Grain boundaries cost more than surfaces in this film. That is why interconnect copper is annealed after plating to grow grains (Chapter 50), and why making the film thinner without growing the grains hurts twice.

On-chip interconnects: the end of copper's advantage

For a modern on-chip wire, the cross-section is only tens of nanometres on a side, the grains are bounded by the line width, and a diffusion barrier (TaN/Ta, a few nanometres, resistivity 10–100 times that of copper) takes up part of the cross-section. The effective resistivity of a 20 nm copper line is roughly two to three times the bulk value (Figure 25.5). That is the answer to the second opening question.

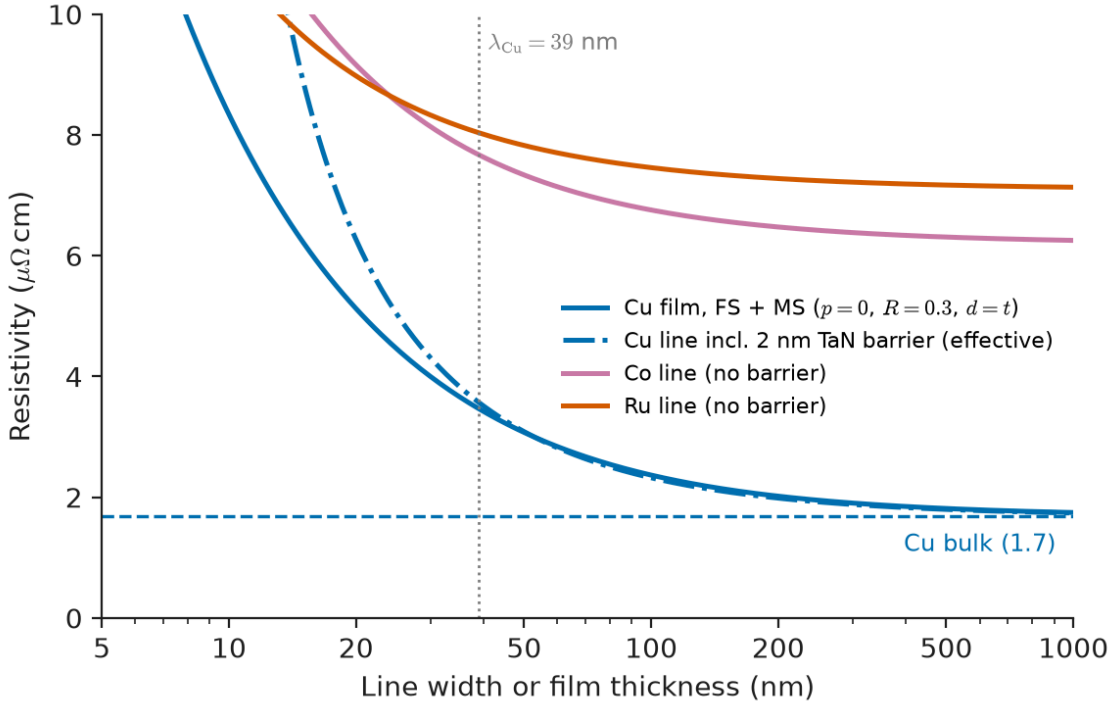


Figure 25.5 — Effective resistivity of narrow interconnect lines rises steeply once the dimensions approach the electron mean free path.

Whether a metal suffers from the size effect depends on the product $\rho_0\lambda$. For a free-electron metal $\rho_0\lambda = mv_F/(ne^2)$, a constant independent of how many scatterers the metal contains. A metal with a higher bulk resistivity but a much shorter mean free path loses less as dimensions shrink. This is why ruthenium, cobalt and molybdenum, poor conductors in bulk, are being introduced for the narrowest lines and vias:

Metal	Bulk ρ ($\mu\Omega\,\mathrm{cm}$)	Approximate λ at 300 K (nm)	Barrier needed?
Cu	1.7	39	Yes (diffuses into Si and SiO_2)
Al	2.7	about 19	No (but electromigration-prone)
W	5.3	about 15–19	Adhesion liner
Mo	5.3	about 14	Thin or none
Co	6.2	about 12	Thin
Ru	7.1	about 7	None or thin

Narrow lines also carry very high current densities (10^9–$10^{10}\ \mathrm{A/m^2}$), at which momentum transfer from electrons to atoms moves the metal itself. This **electromigration** is treated in Chapter 44; its material dependence (grain structure, interfaces, Cu versus Al) is one more reason the choice of interconnect metal is not just a question of resistivity.

Thin-film resistors and transparent conductors

Some films are designed for high and stable resistance rather than low resistance:

Film	Typical sheet resistance	Typical TCR	Use
NiCr (sputtered, 10–50 nm)	25–300 Ω/□	±25 ppm/K or better	Precision thin-film resistors
TaN	10–100 Ω/□	about −100 ppm/K	Integrated resistors, stable in humidity
SiCr (cermet)	0.5–2 kΩ/□	±50 ppm/K	High-value integrated resistors
RuO_2–glass thick film	10 Ω/□–1 MΩ/□	±100 ppm/K	Chip resistors, hybrid circuits (Section 25.7)
ITO (In_2O_3:Sn, 100–200 nm)	5–100 Ω/□	Small	Transparent electrodes (displays, touch, solar cells)

Thin metal films typically have lower TCR than bulk because of their large residual resistivity (Matthiessen again), and the TCR of NiCr and TaN can be tuned to near zero by composition, oxygen or nitrogen content and annealing. Transparent conducting oxides are degenerate semiconductors: high carrier density (10^{20}–$10^{21}\ \mathrm{cm^{-3}}$) for conduction, wide band gap (above 3.5 eV) for transparency. They involve the band ideas of Chapters 31 and 35. Their engineering trade-off is direct: a thicker film lowers R_s but absorbs more light.

25.7 Conductive composites

When conductors are dispersed in an insulator

Many conducting materials in electronics are not metals but polymers filled with conducting particles: conductive adhesives, electrostatic dissipative (ESD) plastics, EMI shielding gaskets, printed silver inks, carbon-composite resistors, resettable polymer fuses, and the thick-film resistor pastes of the table above. Their behaviour is not described by any average of the two resistivities. It is controlled by connectivity.

Percolation

Imagine adding conducting spheres at random to an insulating matrix. At low volume fraction ϕ the spheres are isolated and the composite insulates. As ϕ increases, clusters form and grow, and at a critical fraction ϕ_c, the **percolation threshold**, the first

cluster spans the sample. Above it the conductivity rises steeply:

$$\sigma \approx \sigma_0 \, (\phi - \phi_c)^t, \qquad \phi > \phi_c$$

with a universal exponent $t \approx 2$ in three dimensions. The threshold depends on particle shape and arrangement. For randomly placed spheres $\phi_c \approx 0.16$ (by volume). For elongated or flaky particles it is much lower, because a few long particles span large distances: carbon nanotubes with aspect ratios of 100–1000 percolate at a fraction of a per cent, and silver flakes in adhesives at well below the sphere value.

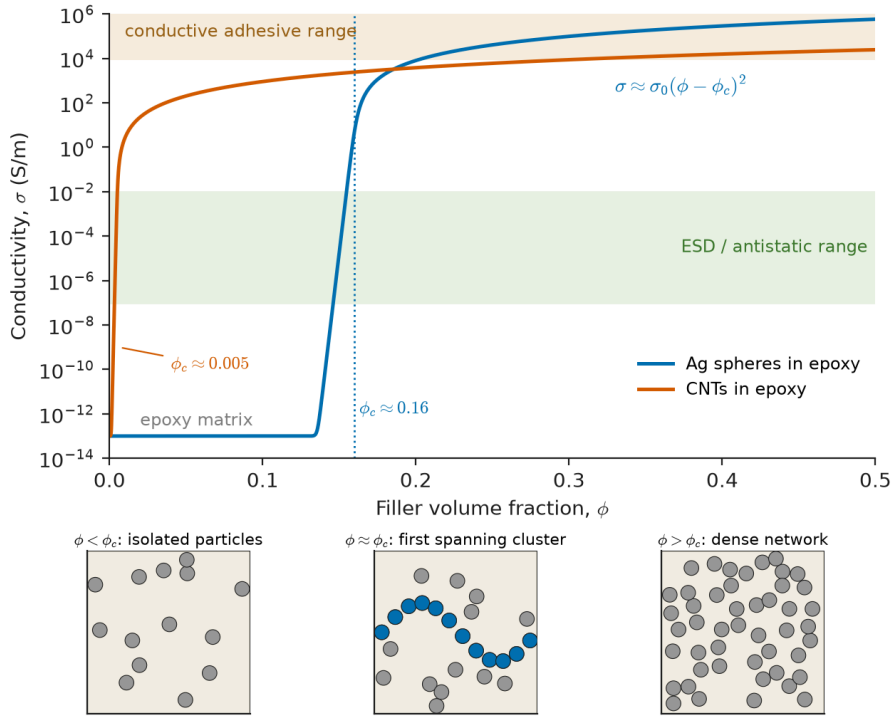


Figure 25.6 — Conductivity of a conductor-filled polymer: an abrupt percolation transition separates insulating and conducting composites.

> **Status of the principle: percolation scaling** $\sigma \propto (\phi - \phi_c)^t$
> *Approximate*. It is a result of statistical physics for random networks, exact only asymptotically near ϕ_c. The prefactor σ_0 and the effective threshold depend on particle contact resistance, dispersion and processing, so both are fitted for each material system.

Figure 25.6 shows the practical consequence: near the threshold, conductivity is extremely sensitive to filler content. For spheres with $\phi_c = 0.16$ and $t = 2$, raising the volume fraction from 0.20 to 0.25 multiplies the conductivity by $(0.09/0.04)^2 = 5.1$. A product formulated near the threshold will vary from batch to batch with small variations in dispersion. Products that need a defined resistivity are formulated well above the threshold or controlled through the contact resistance between particles.

What limits conductivity above threshold

Even well above threshold a silver-filled epoxy is 10–100 times more resistive than bulk silver. Current passes from particle to particle through small contacts, so each junction is a Holm constriction (Section 25.5) with an organic or polymer film in series, often thin enough to tunnel through. Curing shrinks the matrix and presses particles together, which lowers the resistivity; silver is used because its oxide is conductive and its sulphide thin. Typical values:

Composite	Typical volume resistivity (Ω m)	Use
Isotropic conductive adhesive (Ag flake–epoxy)	10^{-6}–10^{-5}	Die attach, SMD assembly on heat-sensitive substrates (Chapter 51)
Sintered or printed Ag nanoparticle ink	3×10^{-8}–10^{-7}	Printed electronics, die attach
Anisotropic conductive film (ACF)	Conducts only through the thickness	Display driver and flex connections
Carbon-black–polymer	10^{-2}–10^{2}	Resistors, heating films, PTC fuses
ESD dissipative plastic	10^{4}–10^{9} (surface 10^{5}–10^{11} Ω/□)	Trays, housings, flooring
RuO_2–glass thick film	10^{-5}–10^{-1} (set by RuO_2 fraction)	Chip and hybrid resistors

Sintered silver differs from conductive adhesives because the particles fuse into a continuous, porous metal network with a resistivity only about 2–5 times that of bulk silver.

Engineering with the threshold: PTC polymer fuses

A carbon-black-filled polyethylene formulated just above its percolation threshold conducts at room temperature. If the current heats it toward the melting point of the polyethylene crystallites (about 120–130 °C), the polymer expands by several per cent, pulls the particle network apart, and the volume fraction effectively drops below ϕ_c. The resistance rises by three to six orders of magnitude within a few kelvin, limiting the current. On cooling the network reforms. This is the **resettable polymeric PTC fuse**: a circuit-protection device built entirely on the steepness of the percolation curve and the thermal expansion of a polymer (Chapter 24).

Anisotropic conductive films

Anisotropic conductive films use a filler fraction well **below** the threshold, about 5–10 %, of 3–5 μm metal-coated polymer spheres. In the plane of the film the particles never connect, so adjacent pads stay isolated. When a bump is pressed onto a pad, the particles trapped between them are squeezed into contact through the film thickness. The same composite is an insulator laterally and a conductor vertically, purely through geometry.

25.8 Electrical conductivity versus thermal conductivity

Same carriers, same scattering

In a metal the conduction electrons carry heat as well as charge, and they carry most of it. A material's thermal conductivity is the sum of an electronic part and a lattice (phonon) part, $\kappa = \kappa_e + \kappa_{\text{ph}}$ (Chapter 24). In pure metals κ_e dominates; κ_{ph} is a few W/(m K).

Kinetic theory gives the thermal conductivity of a gas of carriers as

$$\kappa_e = \tfrac{1}{3}\, C_e\, v\, \lambda = \tfrac{1}{3}\, C_e\, v^2\, \tau$$

where C_e is the heat capacity of the carriers per unit volume. If the same scattering time τ limits both currents, the ratio of thermal to electrical conductivity is

$$\frac{\kappa_e}{\sigma} = \frac{\tfrac{1}{3} C_e v^2 \tau}{n e^2 \tau / m} = \frac{C_e\, m v^2}{3 n e^2}$$

The scattering time cancels. All the disorder in the material, every impurity and grain boundary, drops out. What remains depends on how much heat the carriers absorb and how fast they move. Classical theory, with $C_e = \tfrac{3}{2} n k_B$ and $\tfrac{1}{2} m v^2 = \tfrac{3}{2} k_B T$, gives $\kappa/\sigma T = \tfrac{3}{2}(k_B/e)^2$, which has the right form but the wrong constant. The correct values come from the quantum (Fermi–

Dirac) statistics of Chapter 31: only electrons within about $k_B T$ of the Fermi energy can absorb heat, so $C_e = \frac{\pi^2}{2} n k_B \, (k_B T / E_F)$; but those electrons move at the Fermi velocity, $\frac{1}{2} m v_F^2 = E_F$. The Fermi energy cancels, and

$$\frac{\kappa_e}{\sigma T} = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 = L_0 = 2.44 \times 10^{-8} \ \mathrm{W\,\Omega\,K^{-2}}$$

This is the **Wiedemann–Franz law**, and L_0 is the Sommerfeld value of the **Lorenz number**.

> **Status of the principle: Wiedemann–Franz law**
> *Approximate.* It is derived for elastic scattering with the same relaxation time for charge and heat. It holds well for metals at room temperature and above, and at very low temperature where impurity scattering dominates. It fails at intermediate temperatures (roughly 10–100 K for Cu), where small-angle inelastic phonon scattering degrades the heat current more than the charge current, and wherever phonons carry a significant share of the heat (alloys, semiconductors, insulators).

The data

Metal (about 293 K)	σ (MS/m)	κ (W/(m K))	$L = \kappa/\sigma T$ (10^{-8} W Ω K^{-2})
Silver	63	429	2.32
Copper	59.6	401	2.30
Gold	45	318	2.41
Aluminium	37.7	237	2.15
Tungsten	18.9	173	3.12
Iron	10.3	80	2.65
Platinum	9.4	72	2.61
Stainless steel 304	1.4	16	3.9

The noble metals and aluminium lie within about 12 % of L_0. Tungsten, iron and stainless steel have larger values because phonons and, in transition metals, electronic structure effects contribute more. In stainless steel the electrons are so strongly scattered by the disordered alloy that the lattice carries a significant share of the heat.

Using the law

The law has three practical uses.

Estimating thermal conductivity from an electrical measurement. Resistivity is easy to measure (four-point probe, Chapter 52); thermal conductivity is not. For metals and alloys, $\kappa \approx L_0 T / \rho$ gives the electronic part.

Worked example — thermal conductivity of a lead-free solder. SAC305 solder (Sn–3.0Ag–0.5Cu) has a resistivity of about 13.2 $\mu\Omega$ cm at room temperature. The electronic thermal conductivity at 293 K is

$$\kappa_e \approx \frac{L_0 T}{\rho} = \frac{(2.443 \times 10^{-8})(293)}{1.32 \times 10^{-7}} = 54 \ \mathrm{W/(m\,K)}$$

Published values for SAC305 are about 55–60 W/(m K), the remainder being the phonon contribution and the deviation of L from L_0. A single resistance measurement gives the thermal conductivity of a new solder or bond-wire alloy to within about 10 %, which is good enough for a first thermal model of a joint (Chapter 51).

Explaining contact heating. The voltage–temperature relation of Section 25.5 follows from the same identity of heat and charge paths.

Showing what cannot be had. For a metal you cannot raise the thermal conductivity without raising the electrical conductivity, nor lower one without the other. Each design goal has its own way out:

- **Electrically insulating heat spreaders** (AlN, BeO, diamond, cubic and hexagonal BN, alumina) conduct heat by phonons, not electrons, and so escape the law entirely. AlN reaches 170–230 W/(m K) and diamond above 1500 W/(m K)

while remaining excellent insulators. That is the basis of power-module substrates and thermal interface fillers (Chapters 24 and 51).
- **Thermoelectric materials** need the opposite: high σ but low κ. They achieve it by suppressing phonons (heavy atoms, nanostructuring) in heavily doped semiconductors, where κ_{ph} is a large part of κ (Chapter 40).
- **Thermal isolation of a heated sensor element** (thermistors, bolometers, MEMS heaters) uses thin, high-resistivity metal or semiconductor legs, accepting the electrical resistance that comes with low thermal conductance.

> **Boundary.**
> The thermal conductivity of insulators by phonons, its temperature dependence and its role in heat spreading are treated in Chapter 24. This section covers only the electronic part and its link to σ.

25.9 Material selection for wires, contacts, interconnects, and electrodes

What is being optimized

Each conductor application weighs resistivity against other requirements. A useful way to compare candidates is a **performance index**, a combination of properties that measures merit for a given objective (developed in Chapter 58).

- Minimum **mass** for a given resistance per length: maximize $1/(\rho d)$, i.e. conductivity divided by density.
- Minimum **cost** for a given resistance per length: maximize $1/(\rho d c_m)$, with c_m the cost per kg.
- Minimum **volume** (most compact winding): maximize $1/\rho$.

Conductor	$1/\rho$ relative to Cu	$1/(\rho d)$ relative to Cu	Notes
Copper	1.00	1.00	Reference: compact, solderable, strong enough
Aluminium (EC)	0.61	2.0	Lightest per resistance; joints are the weak point
Silver	1.08	0.92	Too expensive for bulk; plating and contacts
Gold	0.78	0.36	Contacts and bond wires only
Sodium	0.36	3.3	Historically trialled for cables; reactive
Copper-clad aluminium (10 % Cu by volume)	about 0.65	about 1.8	Light, solderable surface; RF and automotive cable

Wires and cables

Copper is the default because it combines high conductivity with solderability, ductility, adequate strength, moderate cost and easy joining. Aluminium wins where weight or cost per ampere dominates: overhead lines (as ACSR, aluminium conductor steel-reinforced, or all-aluminium-alloy conductors), large building feeders, and increasingly automotive battery cables. Fine wires for windings use oxygen-free copper with polymer enamel insulation (Chapter 26). High-temperature conductors use nickel-plated copper or copper with dispersed oxide.

At high frequency, current crowds into a surface layer of thickness equal to the **skin depth**

$$\delta = \sqrt{\frac{\rho}{\pi f \mu}}$$

For copper ($\mu = \mu_0$) this is 9.3 mm at 50 Hz, 66 μm at 1 MHz and 2.1 μm at 1 GHz. Above a few megahertz only the outer few micrometres carry current, so the core can be made of something cheaper or stronger: copper-clad steel (antenna and CATV centre conductors), silver-plated copper (RF cables), and Litz wire (many insulated strands, each thinner than δ) in high-frequency transformers. The derivation of δ belongs to electromagnetism; the materials point is that at RF the resistivity that matters is that of the surface finish, including its roughness. Rough PCB copper foil (needed for adhesion to the laminate) raises

high-frequency loss once the roughness is comparable with δ, which is why low-profile foils are specified for multi-gigahertz boards.

Contacts

Contacts combine the criteria of Section 25.5: low film formation, adequate hardness, wear resistance, and for switching, arc resistance. The usual stack is a strong copper-alloy spring (CuNiSi, phosphor bronze, CuBe) for force, a nickel underplate as a diffusion barrier and hard base, and a thin noble or tin finish for the surface. Gold over nickel for low-force, low-voltage, high-reliability contacts; tin for high-force, low-cost, non-moving contacts; silver for power contacts; and the composite arc-resistant materials for switching.

Interconnects on chips and boards

Level	Typical conductor	Selection drivers
Lowest on-chip lines and vias (below about 20 nm)	Cu with thin liners; Ru, Co, Mo emerging	Size effect ($\rho_0\lambda$), barrier thickness, electromigration
Upper on-chip levels	Cu (dual damascene), barrier TaN/Ta	Bulk-like resistivity, electromigration
Contacts to transistors	W plugs, Co, silicides (NiSi, $TiSi_2$)	Fills high-aspect holes, stable on Si (Chapter 34)
Bond wires	Au, Cu, Pd-coated Cu, Al (power modules)	Bondability, corrosion, cost (Chapter 51)
PCB	Electrodeposited or rolled Cu, plated finishes	Adhesion, roughness, solderability
Die attach	Solder, sintered Ag, conductive adhesive	Thermal and electrical conduction, stress

Electrodes

An electrode is a conductor in contact with a functional material: a dielectric, a semiconductor, an electrolyte or a sensor element. Its requirements come from that interface, not from bulk resistivity alone:

- **Capacitor electrodes.** MLCC internal electrodes are Ni (base-metal electrode) or Ag–Pd, co-fired with the $BaTiO_3$ dielectric (Chapter 27). Nickel needs a reducing firing atmosphere; Ag–Pd tolerates air but costs more. Film capacitors use vapour-deposited Al or Zn layers 20–50 nm thick that burn away around a breakdown spot ("self-healing", Chapter 26).
- **Semiconductor electrodes.** The work function and reactivity matter: Al, Ti, Ni, Pt, silicides and TiN are chosen for ohmic or Schottky behaviour and stability (Chapter 34).
- **Sensor and heater electrodes.** Pt for high-temperature and chemically demanding sensors (oxygen sensors, RTDs, MEMS heaters); NiCr and FeCrAl alloys for heating elements, chosen for high resistivity, low TCR and a protective Cr_2O_3 or Al_2O_3 scale. $MoSi_2$ and SiC heating elements reach 1600–1800 °C in air.
- **Transparent electrodes.** ITO, fluorine-doped tin oxide, silver nanowire networks (a percolating composite, Section 25.7) and graphene (Chapter 38) for displays, touch panels and solar cells.
- **Electrochemical electrodes** (batteries, fuel cells, supercapacitors) are a separate topic that combines conduction with electrochemistry; their materials are covered in Chapter 40, and the electrochemistry belongs to the Advanced Chemistry core.

The busbar of the opening paragraph can now be redesigned. The aluminium conductor itself was correct. The joint needed a larger, flatter contact area at a sustained high force (spring washers or Belleville stacks against creep), oxide removal and sealing grease at assembly, or a tin- or nickel-plated contact face, and a bimetallic transition where aluminium meets copper. Each of these follows from the Holm picture of Section 25.5.

25.10 Ionic conduction and solid electrolytes

Ions as carriers

In ionic solids, glasses and some polymers, the carriers are ions moving by thermally activated jumps between lattice sites, through the vacancy and interstitial mechanisms of Chapter 8. The conductivity has the same form as for electrons, $\sigma = nq\mu$, but now the mobility is tied to diffusion through the **Nernst–Einstein relation**:

$$\sigma = \frac{nq^2 D}{k_B T}$$

where n is the concentration of mobile ions of charge q and D their diffusion coefficient. Because $D = D_0 \exp(-E_m / k_B T)$ and the number of mobile defects may itself be activated, the conductivity is Arrhenius:

$$\sigma T = A \exp\!\left(-\frac{E_a}{k_B T}\right)$$

with activation energies typically 0.2–1.2 eV. A useful solid electrolyte must therefore satisfy three structural conditions: a high concentration of mobile defects (vacancies or interstitials), a connected network of energetically similar sites to jump between, and a low barrier between them. Equally important, its electronic conductivity must be negligible, so that the material passes ions but blocks electrons — otherwise a fuel cell or battery would short-circuit internally.

> **Status of the principle: Nernst–Einstein relation**
> *Derived.* It follows from the fluctuation–dissipation link between random-walk diffusion and drift in a field. It holds exactly for independent, non-interacting carriers; in concentrated or correlated conductors the measured conductivity differs from the tracer-diffusion value by the Haven ratio, typically 0.3–1.

Defect engineering: doped zirconia and ceria

Pure ZrO_2 is a poor conductor. Substituting Y^{3+} for Zr^{4+} does two things. It stabilizes the cubic fluorite structure (Chapter 7, ZrO_2–Y_2O_3), and, by the Kröger–Vink charge balance of Chapter 5, every two Y^{3+} ions create one oxygen vacancy:

$$Y_2O_3 \xrightarrow{ZrO_2} 2Y'_{Zr} + 3O_O^{\times} + V_O^{\bullet\bullet}$$

At 8 mol% Y_2O_3 (8YSZ) about 4 % of the oxygen sites are vacant, and oxide ions hop through them. Conductivity rises with dopant level up to about 8–10 mol%, then falls, because vacancies bind to dopant cations and to each other — a clean example of defect chemistry setting a property maximum. Gadolinium-doped ceria (GDC) and strontium- and magnesium-doped lanthanum gallate (LSGM) conduct better at lower temperature, but ceria is partly reduced in hydrogen and becomes a mixed conductor.

These materials are the core of **solid-oxide fuel cells and electrolysers** (Chapter 40), of the zirconia **lambda sensor** that controls the air–fuel ratio of every petrol engine (an oxygen concentration cell whose voltage follows the Nernst equation), and of oxygen pumps for industrial gas analysis.

The family of fast-ion conductors

Electrolyte	Mobile ion	Conductivity (S/cm) at temperature	Application
8YSZ (cubic ZrO_2–Y_2O_3)	O^{2-}	0.03 at 800 °C	SOFC, lambda sensor
Gd-doped CeO_2	O^{2-}	0.02–0.03 at 600 °C	Intermediate-temperature SOFC
Na β''-alumina	Na^+	0.2–0.4 at 300 °C	Sodium–sulfur and ZEBRA batteries
α-AgI (above 147 °C)	Ag^+	1–2 at 150 °C	Model superionic conductor
$Li_7La_3Zr_2O_{12}$ garnet (LLZO, doped)	Li^+	10^{-4}–10^{-3} at 25 °C	Solid-state lithium batteries
$Li_{10}GeP_2S_{12}$ (sulfide)	Li^+	$\approx 10^{-2}$ at 25 °C	Solid-state lithium batteries
$LiPF_6$ in carbonate (liquid, reference)	Li^+	$\approx 10^{-2}$ at 25 °C	Commercial Li-ion cells
Nafion (hydrated)	H^+	$\approx$ 0.1 at 80 °C	PEM fuel cells and electrolysers
Soda-lime glass	Na^+	$\approx 10^{-13}$ at 25 °C	(an insulator, but ions drift under DC)

Chapter 25 — Representative ionic conductivities (typical; values depend on dopant, density and microstructure).

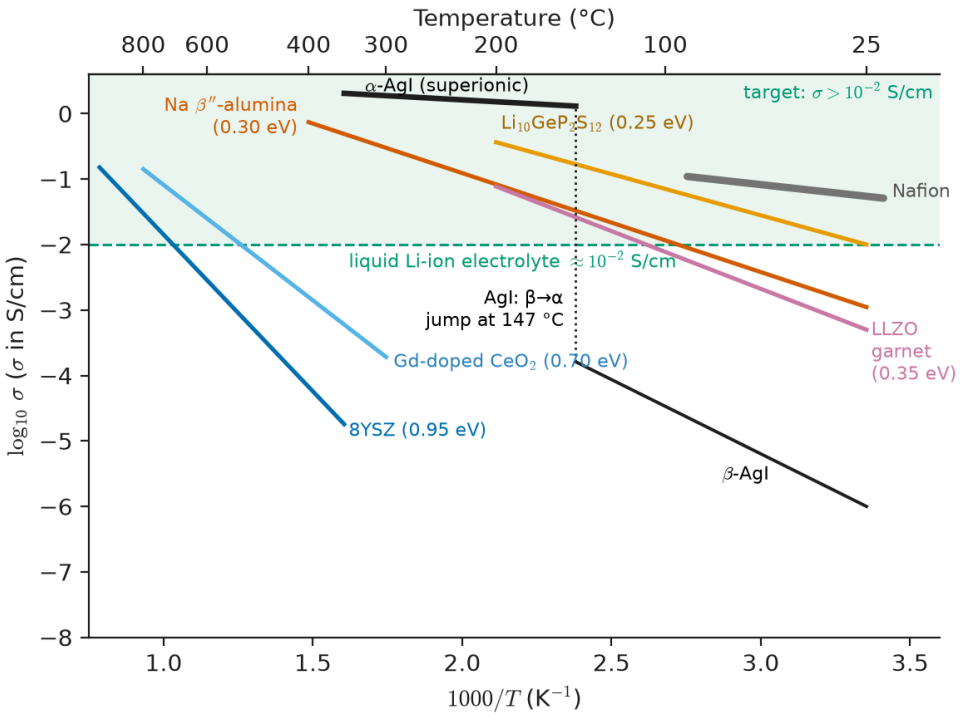


Figure 25.7 — Arrhenius plot of ionic conductivity: fast-ion conductors reach liquid-electrolyte values, but most oxide conductors need high temperature.

Chapter 25 compares them on an Arrhenius plot. Several structural motifs recur. In β-alumina the Na^+ ions move in loosely packed planes between spinel blocks, so conduction is two-dimensional. In α-AgI the iodide sublattice offers many more interstitial sites than there are silver ions, so the cation sublattice is effectively molten. In sulfide lithium conductors the large, polarizable S^{2-} anions lower the hopping barrier. In Nafion, protons move through hydrated nanochannels formed by phase separation of the sulfonated side chains — a polymer conductor whose performance depends on keeping its water content. Even soda-lime glass conducts by sodium drift, which is why it is avoided in high-voltage DC insulators.

Microstructure enters through **grain boundaries**. In polycrystalline YSZ and LLZO the boundaries carry a space-charge layer depleted of mobile defects, and at low temperature the grain-boundary resistance can exceed the bulk value by an order of

magnitude. Impedance spectroscopy (Chapter 52) separates the two. Densification is equally critical: porosity both lowers conductivity and, in lithium garnets, lets lithium dendrites grow along voids and boundaries to short the cell.

Worked example — how thin must a fuel-cell electrolyte be? An SOFC loses voltage across its electrolyte in proportion to the area-specific resistance $\mathrm{ASR} = t/\sigma$. Take 8YSZ with $\sigma = 0.030$ S/cm at 800 °C and $E_a = 0.95$ eV. A 150 µm self-supporting electrolyte has $\mathrm{ASR} = 0.015\ \mathrm{cm}/0.030\ \mathrm{S\,cm^{-1}} = 0.50\ \Omega\,\mathrm{cm^2}$, which at a typical 0.5 A/cm² wastes 0.25 V of a cell voltage of about 0.8 V. A 10 µm film on a porous anode support gives $0.033\ \Omega\,\mathrm{cm^2}$, a loss of only 17 mV. Lowering the operating temperature to 600 °C (cheaper steel interconnects, slower degradation) reduces the conductivity by the Arrhenius factor:

$$\sigma_{600} = 0.030 \exp\left[-\frac{0.95\ \mathrm{eV}}{k_B}\left(\frac{1}{873\ \mathrm{K}} - \frac{1}{1073\ \mathrm{K}}\right)\right] = 2.9 \times 10^{-3}\ \mathrm{S/cm}$$

a factor of 10.5 lower. The same 10 µm film now has ASR 0.35 $\Omega\,\mathrm{cm^2}$; to keep it below a target of 0.15 $\Omega\,\mathrm{cm^2}$ the film would have to be about 4 µm thick and pinhole-free over the whole cell, which is why intermediate-temperature cells switch to GDC or LSGM instead. The design problem is solved by defect chemistry (a better conductor) and thin-film processing (Chapter 49), not by geometry alone.

What remains open

Open question	Where it is answered
Why do metals have a fixed, high carrier density while semiconductors and insulators do not? What sets v_F and the effective mass?	Chapter 31
How do mobility and scattering work in semiconductors, where both carrier density and mobility depend on doping and temperature?	Chapter 32
How do metal–semiconductor contacts differ from metal–metal contacts (Schottky barriers, ohmic contacts)?	Chapter 34
How do superconductors carry current with zero resistance, and why do practical wires still need a copper stabilizer?	Chapter 30
How do conduction and size effects change in nanowires, nanotubes and graphene?	Chapter 38
How are interconnect films deposited and their grain structure controlled?	Chapters 49 and 50
Which solders, bonding wires and conductive adhesives are used, and how do they fail in packages?	Chapter 51
How are resistivity and sheet resistance measured accurately (four-point probe, van der Pauw)?	Chapter 52
How do electromigration, fretting and contact degradation limit conductor life?	Chapter 44
How are conductor choices made systematically with performance indices and cost?	Chapter 58

Next: Chapter 26 turns from materials that must carry current to materials that must stop it, and asks what makes an insulator store field energy, lose some of it as heat, and eventually break down.

Chapter 26 — Dielectric and Insulating Materials

Part VI · Thermal, Electrical, Dielectric and Magnetic Materials

A utility installs a medium-voltage underground cable with cross-linked polyethylene (XLPE) insulation. It passes its factory tests at more than twice its operating voltage. Eleven years later it fails, in wet ground, at its ordinary operating voltage. The failure analysis finds a pinhole through the insulation, and around it a dense bush of microscopic water-filled channels, "water trees", that grew from a small contaminant particle at the conductor screen. Nothing about the material's datasheet strength of tens of kilovolts per millimetre predicted this. The insulation did not fail because the field was too high for polyethylene. It failed because water, a defect and an alternating field acted together for a decade, changing the material until the local field was too high for what it had become.

This chapter is about insulators: what makes them store field energy, why they waste some of it as heat, what determines the voltage they can hold, and how they age. An EEE engineer meets these questions in every capacitor, cable, PCB, transformer, motor winding, connector housing and chip.

> **Why does a capacitor's dielectric heat up, and why does the heating depend on frequency? Why does a thin film of an insulator hold a much higher field than a thick slab of the same material? What turns a sound insulation into a failed one after years in service?**

Concept	Engineering destination
Permittivity, polarization mechanisms	Capacitance, cable and PCB-line impedance, signal speed
Loss tangent, complex permittivity	Capacitor heating, RF and high-speed PCB loss, cable loss
Frequency and temperature dependence	Laminate choice for GHz boards, capacitor stability
Dielectric strength, Weibull statistics	Insulation thickness, voltage ratings, test margins
Breakdown mechanisms (gas, liquid, solid, thermal)	Clearances, oil and gas insulation, altitude derating
Volume and surface resistivity	Leakage, insulation-resistance testing, high-impedance circuits
Tracking, partial discharge	Creepage distances, CTI, PD testing of cables and machines
Moisture and contamination	PCB cleanliness, conformal coating, transformer drying
Insulation classes and aging	Thermal class of motors and transformers, design life

Roadmap. Polarization and permittivity (26.1) → loss (26.2) → frequency and temperature dependence (26.3) → dielectric strength (26.4) → breakdown mechanisms (26.5) → resistivity of insulators (26.6) → surface and void discharges: tracking and partial discharge (26.7) → moisture and contamination (26.8) → polymer, ceramic and composite insulators (26.9) → aging (26.10).

26.1 Permittivity and dielectric polarization

What a dielectric does to a capacitor

Put a slab of insulator between the plates of a capacitor held at fixed voltage, and the charge on the plates rises by a factor ε_r, the **relative permittivity** (dielectric constant). The capacitance of a parallel-plate capacitor of area A and gap d becomes

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}, \qquad \varepsilon_0 = 8.854 \times 10^{-12}\ \mathrm{F/m}$$

The extra charge is held by the material. The field displaces the bound charges in every atom or molecule slightly, creating dipoles, and the dipole moment per unit volume is the **polarization P** (unit $\mathrm{C/m^2}$). At the surfaces of the slab the ends of these

dipoles leave a bound surface charge that partly cancels the field of the plate charges, so more free charge is needed to reach the same voltage. In terms of fields,

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} = \varepsilon_0 \varepsilon_r \mathbf{E}, \qquad \mathbf{P} = \varepsilon_0 (\varepsilon_r - 1)\mathbf{E} = \varepsilon_0 \chi_e \mathbf{E}$$

where χ_e is the electric susceptibility. The relative permittivity measures how easily the material's charges are displaced. In an anisotropic crystal it is a tensor; in amorphous polymers, glasses and fine-grained ceramics it is a scalar.

> **Status of the principle: linear dielectric response $\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$**
> *Approximate.* It is the first term of the response for fields small compared with internal atomic fields (about 10^{11} V/m). It holds for ordinary insulators up to breakdown. It fails for ferroelectrics, whose polarization depends on field history (Chapter 27).

The four polarization mechanisms

The bound charge can be displaced in four ways, at four very different length and time scales (Figure 26.1).

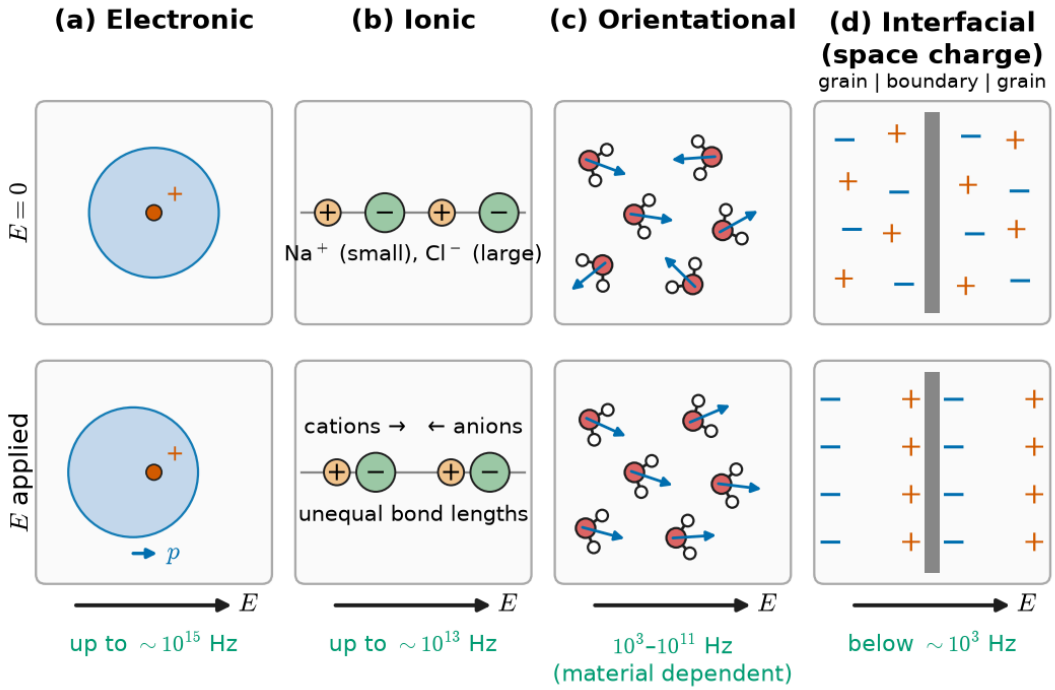


Figure 26.1 — The four polarization mechanisms, each shown without field and with field applied.

Electronic polarization. The electron cloud of every atom shifts relative to its nucleus. A crude model, a uniformly charged sphere of radius R around a point nucleus, gives an induced dipole $p = \alpha_e E$ with polarizability $\alpha_e = 4\pi\varepsilon_0 R^3$. It is present in all materials, responds up to optical frequencies (about 10^{15} Hz), and is the whole story for non-polar materials such as polyethylene, PTFE and diamond.

Ionic (atomic) polarization. In an ionic or partly ionic solid, the field pushes cations and anions in opposite directions, stretching the bonds. Ions are heavy, so this mechanism responds only up to infrared frequencies (10^{12}–10^{13} Hz). It dominates the permittivity of oxide ceramics (alumina, $\varepsilon_r \approx 9.8$) and becomes enormous in some perovskites (Chapter 27).

Orientational (dipolar) polarization. Molecules or groups with a permanent dipole moment p_0 (water, C–Cl in PVC, C=O in polyesters) rotate partly into line with the field against thermal agitation. For weak fields the average aligned moment is $p_0^2 E/(3k_B T)$, so the orientational polarizability is

$$\alpha_{\mathrm{or}} = \frac{p_0^2}{3k_B T}$$

It falls with temperature, and since rotating a molecule in a liquid or solid is slow and frictional, it responds at frequencies from millihertz to about 10^{11} Hz, depending on how free the dipoles are.

Interfacial (space-charge, Maxwell–Wagner) polarization. In a heterogeneous material, such as a composite, a polycrystal with conducting grains and insulating boundaries, or a material with mobile ions, charges drift until they pile up at internal interfaces. The resulting dipoles are as large as the grains or layers, so the apparent permittivity can be huge, but the mechanism is slow (typically below a kilohertz) and always lossy.

From atoms to permittivity: the local field

Each dipole feels not the average field E but a **local field** that includes the field of all the other dipoles. For a site of cubic symmetry, Lorentz showed that

$$E_{\text{loc}} = E + \frac{P}{3\varepsilon_0}$$

With N polarizable units per volume, $P = N\alpha E_{\text{loc}} = N\alpha\,(E + P/3\varepsilon_0)$. Solving for P and using $P = \varepsilon_0(\varepsilon_r - 1)E$:

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0}$$

This is the **Clausius–Mossotti relation**. It connects a macroscopic property (ε_r) to an atomic one (α) and a structural one (N, the density of polarizable units). It also predicts that as $N\alpha/3\varepsilon_0$ approaches 1, ε_r diverges. The local field feeds back on itself. That "polarization catastrophe" is the seed of ferroelectricity (Chapter 27).

> **Status of the principle: Clausius–Mossotti relation**
> *Approximate.* It assumes the Lorentz local field, exact only for sites of cubic symmetry with point dipoles. It works well for gases, non-polar liquids and cubic ionic crystals. It fails for polar liquids such as water, where neighbouring dipoles are strongly correlated (Onsager and Kirkwood corrections are needed).

The relation explains why density changes permittivity. Polyethylene's ε_r rises from 2.28 to 2.36 between low-density and high-density grades because N rises. It also explains a small negative temperature coefficient of ε_r in non-polar materials, which expand on heating.

Reading the mechanisms from optical data

At optical frequencies only electronic polarization can follow the field, and Maxwell's equations give $\varepsilon_r = n^2$, where n is the refractive index. Comparing n^2 with the low-frequency permittivity therefore shows how much of a material's polarization is electronic.

Worked example — which mechanism dominates?

Material	n (visible)	n^2	ε_r (1 kHz–1 MHz)	Interpretation
Polyethylene	1.51	2.28	2.3	Entirely electronic: non-polar
PTFE	1.35	1.82	2.1	Mainly electronic; small atomic contribution from C–F bonds
PVC	1.54	2.37	3–4	Extra comes from rotating C–Cl dipoles
Alumina	1.76	3.10	9.8	Two thirds of the permittivity is ionic
Water	1.333	1.78	80	Almost all orientational

Polyethylene and PTFE are the two lowest-permittivity solid insulators because they have only electronic polarization, and they keep that permittivity unchanged up to tens of gigahertz. That is why they are used in coaxial cables and microwave substrates. Water's ε_r of 80, by contrast, is a property of its rotating dipoles. That is why a trace of water changes an insulator's behaviour so much (Section 26.8).

Representative permittivities

Material	ε_r (typical, 1 kHz–1 MHz, 20 °C)	Dominant mechanism
Vacuum; air (1 atm)	1; 1.0006	—
SF_6 gas (1 atm)	1.002	Electronic
PTFE	2.1	Electronic
Polypropylene; polyethylene, XLPE	2.2; 2.3	Electronic
Mineral transformer oil	2.2	Electronic
Polystyrene	2.55	Electronic
Silicone rubber (filled)	3–4	Electronic + some dipolar
Polyimide (Kapton)	3.4	Electronic + dipolar
Epoxy (unfilled)	3.5–4.5	Electronic + dipolar
Thermal SiO_2; fused silica	3.9; 3.8	Electronic + ionic
FR-4 laminate	4.3–4.8 (lower at GHz)	Composite: glass + epoxy
Mica (muscovite)	6–7	Ionic
Electrical porcelain	5–7	Ionic + interfacial
Silicon nitride (film)	7–7.5	Ionic
Aluminium nitride	8.6–8.9	Ionic
Alumina (96–99.5 %)	9.4–9.9	Ionic
HfO_2 (film)	20–25	Ionic (soft modes)
Water (25 °C)	78.4	Orientational
TiO_2 (rutile)	86 (perp. to c), 170 (along c)	Ionic (soft modes)
$BaTiO_3$ ceramics	1000–10,000	Ferroelectric (Chapter 27)

26.2 Dielectric loss and loss tangent

Why a capacitor heats

An ideal capacitor driven by a sinusoidal voltage draws a current leading the voltage by exactly 90°, and dissipates no power. A real dielectric draws a current that leads by slightly less than 90°. The missing angle δ is the **loss angle**. Two separate processes cause it:

- **Polarization lag.** The dipoles of Section 26.1 cannot follow the field instantaneously. Their response lags, and part of the work done on them is dissipated by friction with their surroundings.
- **Conduction.** A small real current flows through the insulator's finite resistivity (Section 26.6).

Both are described by a **complex relative permittivity**

$$\varepsilon_r^* = \varepsilon' - j\varepsilon''$$

The real part ε' governs storage; the imaginary part ε'' (the **loss factor** or loss index) governs dissipation. The ratio is the **loss tangent** (dissipation factor):

$$\tan\delta = \frac{\varepsilon''}{\varepsilon'}$$

A d.c. conductivity σ adds $\sigma/(\omega\varepsilon_0)$ to ε'', which dominates at low frequency. The quality factor of a capacitor dielectric is $Q = 1/\tan\delta$.

The power dissipated

For a sinusoidal field of r.m.s. amplitude E and angular frequency ω, the power dissipated per unit volume is

$$p = \omega\varepsilon_0\varepsilon'' E^2 = \omega\varepsilon_0\varepsilon' \tan\delta \, E^2$$

For a capacitor at r.m.s. voltage V:

$$P = \omega C V^2 \tan\delta$$

The equivalent circuit is an ideal capacitor C in parallel with a resistance $R_p = 1/(\omega C \tan\delta)$. This resistance is not a fixed resistor: it changes with frequency because $\tan\delta$ does.

Three engineering consequences come straight from the formula:

1. Loss grows linearly with frequency for fixed $\tan\delta$. A dielectric that is negligible at 50 Hz can dominate at 1 GHz.
2. Loss grows with the square of the field. Insulation designed for higher stress needs a lower $\tan\delta$.
3. The product $\varepsilon'\tan\delta = \varepsilon''$ is what heats the material. In microwave heating that is the goal (water's ε'' at 2.45 GHz is about 10–12); in an insulator it is the enemy.

Representative values

Material	ε'	$\tan\delta$ (typical)	Frequency
PTFE	2.1	$1\text{–}2 \times 10^{-4}$	1 MHz–10 GHz
Polyethylene, XLPE	2.3	$2\text{–}5 \times 10^{-4}$	50 Hz–1 GHz
Polypropylene (capacitor film)	2.2	2×10^{-4}	1 kHz–1 MHz
Polystyrene	2.55	1×10^{-4}	1 MHz
Polyimide	3.4	$2\text{–}8 \times 10^{-3}$	1 kHz–1 GHz
FR-4	4.2–4.5	0.015–0.02	1 GHz
Low-loss hydrocarbon or PTFE–ceramic laminates	3.0–3.5	0.001–0.004	10 GHz
Fused silica	3.8	1×10^{-4} or lower	1 MHz–10 GHz
Alumina 99.5 %	9.8	1×10^{-4}	1 MHz–10 GHz
Mineral oil (new, 90 °C)	2.2	below 5×10^{-3}	50 Hz
Oil-impregnated paper	3.5	$2\text{–}3 \times 10^{-3}$	50 Hz
PVC (plasticized)	4–6	0.05–0.1	50 Hz
Water (25 °C)	77	about 0.16	2.45 GHz

Worked example — loss and charging current in a medium-voltage cable. An 11 kV three-phase cable has a phase-to-earth voltage $U_0 = 6.35\ \text{kV}$. Each core has a conductor (with its semiconducting screen) of radius $r = 6.0\ \text{mm}$ and XLPE insulation out to $R = 9.4\ \text{mm}$, with $\varepsilon_r = 2.3$ and $\tan\delta = 5 \times 10^{-4}$ at 50 Hz.

The capacitance of a coaxial insulation per unit length is

$$C' = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(R/r)} = \frac{2\pi(8.854\times10^{-12})(2.3)}{\ln(9.4/6.0)} = \frac{1.280\times10^{-10}}{0.449} = 2.85\times10^{-10}\ \text{F/m}$$

or $0.285\ \mu\text{F/km}$. The charging current per kilometre per phase is

$$I = \omega C' U_0 = (314.2)(2.85\times10^{-7})(6350) = 0.57\ \text{A/km}$$

and the dielectric loss per kilometre per phase

$$P = \omega C' U_0^2 \tan\delta = (314.2)(2.85\times10^{-7})(6350)^2(5\times10^{-4}) = 1.8\ \text{W/km}$$

The dielectric loss is tiny compared with the conductor's I^2R loss (several kilowatts per kilometre at rated current). That is why XLPE, with its low loss tangent, displaced the oil-paper cables of the past ($\tan\delta$ about 10 times higher). At transmission voltages, where U_0^2 is 100 times larger, the dielectric loss becomes significant and the loss tangent becomes a design parameter. The charging current, not the loss, limits the length of long a.c. cables, which is one reason long submarine links use HVDC.

Loss at high frequency: why board material matters

On a printed circuit board carrying a signal of frequency f, the dielectric attenuation of a transmission line is approximately

$$\alpha_d \approx \frac{\pi\sqrt{\varepsilon_{\text{eff}}}\tan\delta}{\lambda_0}\ \text{Np/m} = 27.3\,\frac{\sqrt{\varepsilon_{\text{eff}}}\tan\delta}{\lambda_0}\ \text{dB/m}$$

where λ_0 is the free-space wavelength. At 10 GHz ($\lambda_0 = 30$ mm), a stripline in FR-4 ($\varepsilon_r = 4.3$, $\tan\delta = 0.02$) loses $27.3 \times 2.07 \times 0.02/0.03 = 37.7$ dB/m, or about 0.38 dB/cm. A PTFE–ceramic laminate ($\varepsilon_r = 3.0$, $\tan\delta = 0.0013$) loses about 2.0 dB/m, nearly twenty times less. Over a 30 cm backplane trace the difference is 11 dB against 0.6 dB, which decides whether a multi-gigabit link works. This is the materials reason for the large family of low-loss laminates in Chapter 51: the epoxy and glass of FR-4 contain polar groups and ions whose losses rise in the gigahertz range, and low-loss laminates replace them with non-polar resins (PTFE, hydrocarbons, polyphenylene ether) and low-loss fillers.

26.3 Frequency and temperature dependence

The dielectric spectrum

Because each mechanism has its own response time, a plot of ε' and ε'' against frequency shows a staircase of steps (Figure 26.2). At the lowest frequencies all mechanisms contribute. As frequency rises, first interfacial, then orientational, then ionic, then electronic polarization drops out, each producing a step down in ε' and a peak in ε''. Every peak in loss accompanies a step in permittivity. The two are linked: the Kramers–Kronig relations, which follow from causality, require any frequency-dependent ε' to come with loss somewhere in the spectrum.

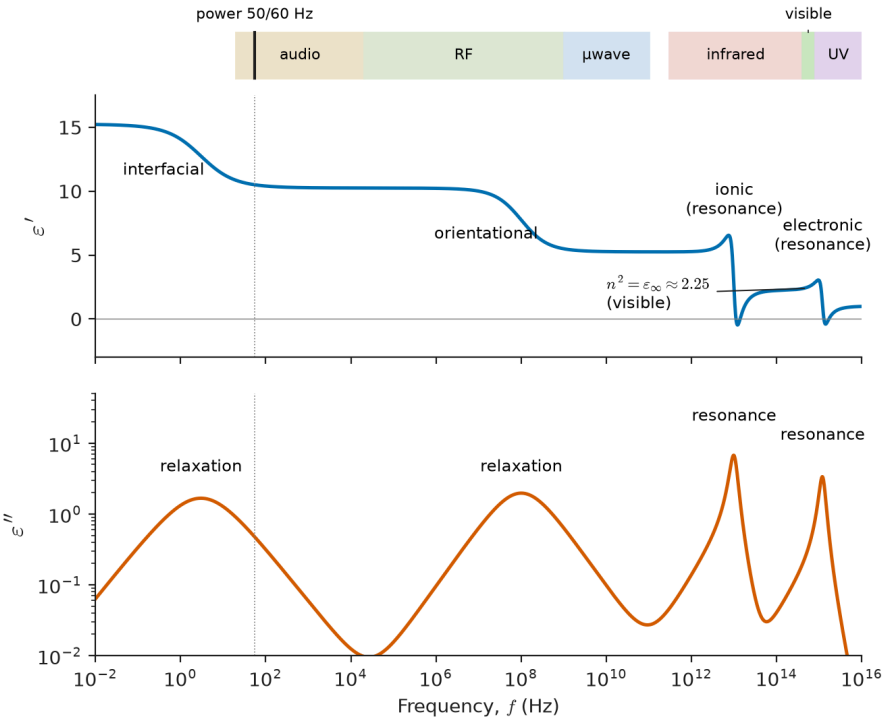


Figure 26.2 — Dielectric spectrum: real and imaginary permittivity of a polar, heterogeneous solid from millihertz to ultraviolet frequencies.

The two kinds of feature have different shapes. Ionic and electronic polarization are **resonances**: bound charges on springs with natural frequencies in the infrared and ultraviolet, giving sharp peaks and an overshoot in ε' just below the resonance. Orientational and interfacial polarization are **relaxations**: there is no restoring spring, only friction, and the response decays smoothly.

The Debye relaxation

Suppose the orientational polarization relaxes toward its equilibrium value exponentially with a single time constant τ. Solving for a sinusoidal field gives

$$\varepsilon^*(\omega) = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + j\omega\tau}$$

where ε_s is the static (low-frequency) and ε_∞ the high-frequency permittivity. Separating real and imaginary parts:

$$\varepsilon' = \varepsilon_\infty + \frac{\Delta\varepsilon}{1+\omega^2\tau^2}, \qquad \varepsilon'' = \frac{\Delta\varepsilon\,\omega\tau}{1+\omega^2\tau^2}, \qquad \Delta\varepsilon = \varepsilon_s - \varepsilon_\infty$$

The loss peaks at $\omega\tau = 1$ with height $\Delta\varepsilon/2$, and the step in ε' is centred at the same frequency. Plotting ε'' against ε' (a **Cole–Cole plot**) gives a semicircle of radius $\Delta\varepsilon/2$ centred on the real axis (Figure 26.3).

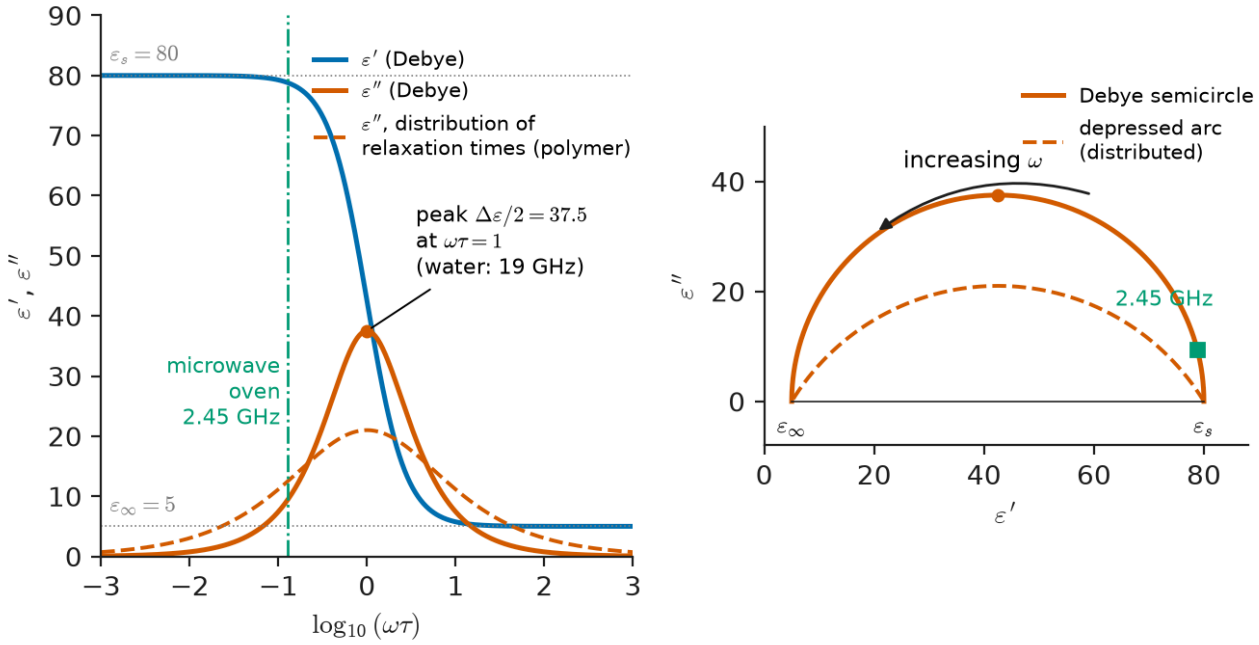


Figure 26.3 — Debye relaxation: frequency response and the Cole–Cole representation.

Water is close to a Debye relaxer, with $\tau \approx 8.3$ ps at 25 °C. Its loss peak is at $f = 1/(2\pi\tau) \approx 19$ GHz. A microwave oven at 2.45 GHz works on the low-frequency side of that peak, where ε'' is still large but the wave penetrates several centimetres. Polymers have broad distributions of relaxation times, and their arcs are flatter than a semicircle (the empirical Cole–Cole and Havriliak–Negami forms fit them).

> **Status of the principle: Debye relaxation**
> *Approximate.* It assumes non-interacting dipoles with a single relaxation time. It describes dilute polar liquids and some defect dipoles in crystals. Solids and polymers show broader, asymmetric loss peaks, fitted by empirical generalizations.

Interfacial polarization in a layered insulator

Interfacial polarization can be derived from the same picture. Take two layers in series, thicknesses d_1, d_2, permittivities $\varepsilon_1, \varepsilon_2$ and conductivities σ_1, σ_2. At high frequency the field divides according to the permittivities (capacitive division). At low frequency, after charge has accumulated at the interface, it divides according to the conductivities (resistive division). The transition is a Debye relaxation with time constant

$$\tau_{\mathrm{MW}} = \varepsilon_0 \, \frac{\varepsilon_1 d_2 + \varepsilon_2 d_1}{\sigma_1 d_2 + \sigma_2 d_1}$$

For good insulators τ_{MW} can be seconds to hours. This is the origin of **dielectric absorption** in capacitors: a capacitor discharged briefly and then left open-circuit recovers part of its voltage as the interfacial charge redistributes. It matters in sample-and-hold circuits and integrators, and is why polypropylene and polystyrene (homogeneous, non-polar) are chosen there.

Temperature dependence

Temperature enters in three ways.

Relaxation times are thermally activated. For dipoles that must hop over an energy barrier E_a,

$$\tau = \tau_0 \exp\left(\frac{E_a}{k_B T}\right)$$

so heating shortens τ and shifts the loss peak to higher frequency. A loss peak sitting at 1 kHz at room temperature may sit at 50 Hz at a lower temperature, or at 1 MHz at a higher one. For polymers the strongest relaxation is tied to the glass transition (Chapter 20). Below T_g the main-chain dipoles are frozen and only local side-group motions contribute. Near T_g the main chain unfreezes, and both ε' and $\tan\delta$ rise sharply. This is why a polar insulator's loss rises steeply as it approaches its softening temperature, and why thermal runaway (Section 26.5) is a real risk.

Orientational polarizability falls as $1/T$ (Section 26.1), once the dipoles are free to rotate.

Conduction rises exponentially with temperature (Section 26.6), adding $\sigma/\omega\varepsilon_0$ to ε''. At power frequencies and high temperature, conduction often dominates the loss.

The net result is that non-polar insulators (PE, PP, PTFE, fused silica) have nearly flat $\varepsilon'(T)$ and very low loss over wide ranges, while polar ones (PVC, epoxy, polyamide, cellulose) change substantially. For capacitors, the temperature coefficient of capacitance, written $\mathrm{TCC} = (1/C)\,dC/dT$, is specified by class; Chapter 27 treats the ceramic capacitor classes, whose permittivity is engineered rather than incidental.

26.4 Dielectric strength

Definition and what it hides

The **dielectric strength** is the field at which an insulator fails, usually stated as breakdown voltage divided by thickness:

$$E_b = \frac{V_b}{d} \qquad (\mathrm{kV/mm\ or\ MV/m};\ 1\ \mathrm{kV/mm} = 1\ \mathrm{MV/m})$$

Unlike permittivity, E_b is not a single material property. The measured value depends on:

- **Thickness.** Thinner specimens test stronger. An empirical fit is $E_b \propto d^{-n}$ with n typically 0.3–0.5 for solids.
- **Area or volume tested.** Larger specimens contain more weak spots and fail at lower fields (the weakest-link statistics below).
- **Time and waveform.** Short impulses (lightning, 1.2/50 μs) give higher values than 1 min a.c. tests, which give higher values than long-term service. Breakdown is a process with a duration, not an instantaneous event.
- **Electrode geometry and medium.** Edges and points enhance the field; a test in oil prevents flashover around the specimen, while in air the surrounding gas may break down first.
- **Temperature, humidity and frequency.**

Datasheet values are therefore test-method values (IEC 60243, ASTM D149) and are useful only for comparing materials under the same conditions. Design stresses in long-life equipment are a small fraction of them.

Material	Typical short-term E_b (kV/mm)	Conditions
Air, 1 atm	about 3	Uniform field, gaps of about 1 cm
SF_6, 1 atm	about 8.9	Uniform field
Mineral transformer oil	10–25	IEC 60156, 2.5 mm gap; strongly reduced by water and particles
Polyethylene, XLPE	20–50	1–2 mm sheet, a.c.
PTFE	20–60	1–2 mm sheet; higher in thin film
Biaxially oriented PP film	400–700	5–15 μm film, d.c. (capacitor film)
Polyimide film	about 300	25 μm film, a.c.
Epoxy (cast)	20–30	1–3 mm
FR-4	20–40	Through thickness, in oil
Electrical porcelain	15–25	Several mm
Alumina 96–99.5 %	10–35	Thickness dependent; lower at high temperature
Mica (muscovite)	50–200	Thin sheet
Thermal SiO_2	about 1000	5–100 nm gate oxides

The thermal oxide value deserves a note. A 10 nm gate oxide holding 10 V experiences 10^9 V/m, about 300 times the field that breaks down air. Thin, nearly defect-free amorphous films approach the intrinsic limit of the material (Section 26.5), while thick, practical insulation fails at defects long before.

Weakest-link statistics

Breakdown starts at the worst flaw in the stressed volume: a void, a particle, a protrusion on an electrode. Repeated tests on nominally identical specimens give a spread of breakdown fields, usually well described by a two-parameter **Weibull distribution**:

$$P_F(E) = 1 - \exp\left[-\left(\frac{E}{E_0}\right)^{\beta}\right]$$

E_0 is the scale parameter (the field at which 63.2 % of specimens have failed), and β the shape parameter: a large β (20 or more) means a tight distribution of strengths; small β (5–10) means scattered weak spots. If the failure probability per unit area is independent of position, a specimen k times larger has the same probability of survival as k independent small specimens all surviving, and its scale parameter falls as

$$E_0(kA) = E_0(A)\, k^{-1/\beta}$$

> **Status of the principle: Weibull statistics of breakdown**
> *Empirical.* The Weibull form follows from weakest-link reasoning if flaw strengths have a suitable extreme-value distribution; in practice it is fitted to test data (IEC 62539). The area-scaling law assumes flaws are randomly and independently distributed.

Worked example — why a large capacitor is weaker than a test coupon. A polymer film tested on 1 cm^2 electrodes shows $E_0 = 500$ kV/mm with $\beta = 10$. A wound capacitor of the same film has 100 cm^2 of active area. Its characteristic strength is

$$E_0 = 500 \times 100^{-1/10} = 500 \times 0.631 = 316 \text{ kV/mm}$$

37 % lower, purely from having more chances to contain a weak spot. A lower β would make the reduction worse. This is why capacitor designers derate film strength heavily and why the self-healing metallized electrodes of Section 26.9 are so valuable: they turn a fatal weak spot into a small, isolated loss of area.

Field enhancement and geometry

The field in real insulation is not uniform. In a coaxial cable it is highest at the conductor:

$$E(r) = \frac{U_0}{r \ln(R/r_c)}$$

where r_c is the conductor radius and R the outer radius of the insulation. For the cable of the worked example in Section 26.2, the field at the conductor screen is $6350/(0.006 \times 0.449) = 2.36\ \text{kV/mm}$ and at the outer screen $1.50\ \text{kV/mm}$. That is a design stress of about 5 % of the short-term strength of XLPE, the margin needed for 40 years of service with defects, water and temperature cycling. Semiconducting screens are extruded onto both sides of the insulation to make the electrode surfaces smooth, because a protrusion of radius a few micrometres on a rough conductor could raise the local field by an order of magnitude. Such a protrusion was the origin of the water tree in the opening example.

26.5 Electrical breakdown mechanisms

Breakdown is the change from insulating to conducting, with a sudden collapse of voltage and a destructive current. The mechanisms differ in gases, liquids and solids, but they share one feature: a positive feedback that multiplies carriers or heat faster than they are removed.

Gases: avalanche and Paschen's law

A free electron in a gas is accelerated by the field. If it gains enough energy between collisions, it ionizes a molecule, freeing a second electron. Both are accelerated and repeat the process. The number of ionizing collisions per unit length is the **first Townsend coefficient** α, so an avalanche starting at the cathode grows as $e^{\alpha d}$ over a gap d. The positive ions produced drift back and release further electrons from the cathode, with yield γ per ion (the **second Townsend coefficient**). The discharge becomes self-sustaining, i.e. breakdown occurs, when each initial electron leads on average to one replacement:

$$\gamma\left(e^{\alpha d} - 1\right) = 1 \quad \Longrightarrow \quad \alpha d = \ln\left(1 + \frac{1}{\gamma}\right)$$

The ionization coefficient depends on the energy gained per mean free path. The mean free path is inversely proportional to pressure p, so α/p depends only on E/p. A standard empirical form is

$$\frac{\alpha}{p} = A \exp\left(-\frac{Bp}{E}\right)$$

Substituting $E = V/d$ into the breakdown condition and solving for V:

$$V_b = \frac{B\,pd}{\ln(A\,pd) - \ln\left[\ln\left(1 + 1/\gamma\right)\right]}$$

This is **Paschen's law**: the breakdown voltage of a uniform gap depends on pressure and gap only through their product pd. For air, $A \approx 15\ \text{cm}^{-1}\,\text{Torr}^{-1}$ and $B \approx 365\ \text{V}\,\text{cm}^{-1}\,\text{Torr}^{-1}$ (1 Torr = 133.3 Pa), with $\gamma \approx 0.01$, so $\ln[\ln(101)] = 1.53$.

> **Status of the principle: Paschen's law**
> *Empirical.* The scaling with pd follows from the Townsend mechanism, but the constants A, B, γ are fitted, and the formula fails at very small gaps (below about 5 μm in air, where field emission from the electrodes takes over), at large pd (above roughly 1000 Torr cm, where streamer breakdown governs), and in non-uniform fields.

Worked example — the Paschen curve for air. For a 1 mm gap at 1 atm, $pd = 760 \times 0.1 = 76$ Torr cm:

$$V_b = \frac{365 \times 76}{\ln(15 \times 76) - 1.53} = \frac{27{,}740}{7.04 - 1.53} = 5.0\ \text{kV}$$

For a 1 cm gap, $pd = 760$: $V_b = 277{,}400/(9.34 - 1.53) = 35.5\ \text{kV}$. Measured values are about 4.5 kV and 31 kV; the fitted constants are good to 10–15 %. The curve has a minimum: setting $dV_b/d(pd) = 0$ gives

$$(pd)_{\min} = \frac{e\,\ln(1 + 1/\gamma)}{A} = \frac{2.718 \times 4.62}{15} = 0.84\ \text{Torr cm}, \qquad V_{\min} = \frac{e\,B\,\ln(1 + 1/\gamma)}{A} = 305\ \text{V}$$

(the measured minimum for air is about 330 V near 0.6–0.8 Torr cm). **No gap in air at any pressure breaks down below about 330 V**: on the left of the minimum there are too few molecules to ionize, on the right too many collisions to gain energy. Figure 26.4 shows the curve.

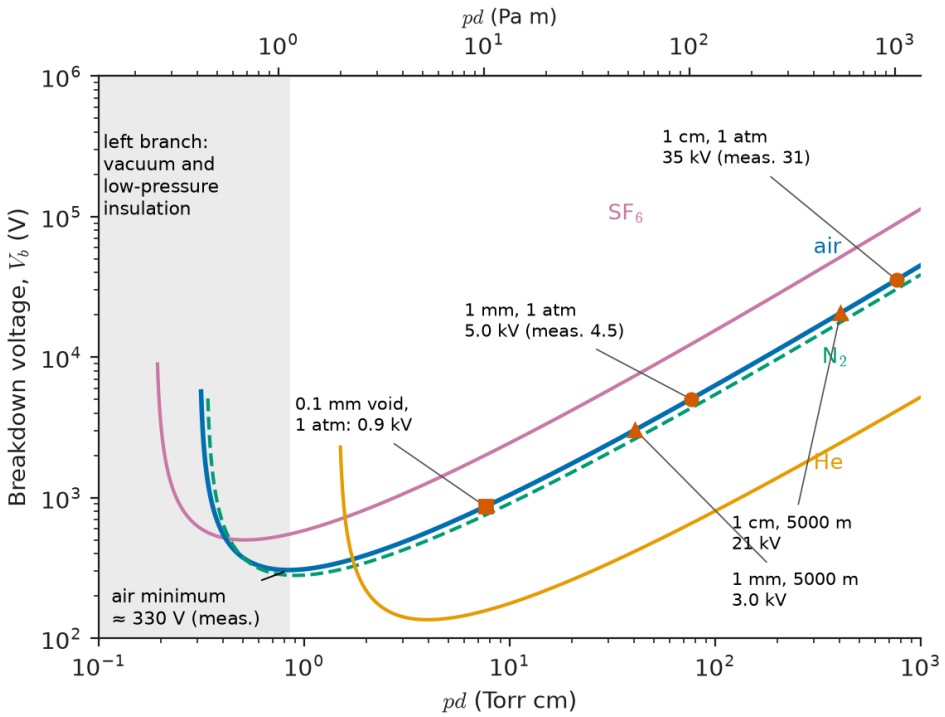


Figure 26.4 — Paschen curves: breakdown voltage of a uniform gap depends on the product of pressure and gap.

Two engineering consequences follow:

- **Altitude.** At 5000 m the pressure is about 54 kPa (0.535 atm). For a 1 cm gap $pd = 407$ Torr cm and the calculated V_b falls from 35.5 to 20.7 kV, a 42 % reduction. Insulation-coordination standards (IEC 60664-1) multiply clearances by altitude correction factors above 2000 m, and aircraft and space electronics need their own rules. In a partial vacuum a power supply may arc at a few hundred volts at the Paschen minimum.
- **Vacuum and high-pressure insulation.** Both sides of the minimum are used. Vacuum interrupters and X-ray tubes work far to the left; SF_6 switchgear works to the right at several bar, exploiting the electronegative gas's ability to capture electrons and its higher B. Because SF_6 is a very potent greenhouse gas, replacements (dry air, fluoronitrile mixtures) are now being introduced.

In non-uniform fields (points, edges, thin wires) the gas breaks down locally first, producing **corona**: a glow discharge near the sharp electrode that does not bridge the gap. Corona wastes power on transmission lines, generates ozone and radio noise, and erodes nearby polymer surfaces.

Liquids

Very pure liquids can hold fields close to those of solids, but practical insulating liquids (mineral oil, synthetic and natural esters, silicone fluids) break down at much lower fields through their impurities:

- **Particles and fibres** (cellulose fibres in a transformer) polarize, align along the field, and form bridges between electrodes.
- **Water** dissolved in the oil is harmless up to its solubility limit, but free or emulsified water, especially on fibres, drastically lowers strength (Section 26.8).
- **Bubbles** of gas, formed by heating or by local discharges, have lower permittivity than the oil, so the field inside them is higher, and they then break down as gases.

Liquid insulation is therefore maintained, not just specified: transformer oil is filtered, dried and degassed, and tested periodically for breakdown voltage, water content and dissolved gases.

Solids

Solids show several distinct mechanisms. In practice, which one governs depends on time scale, temperature and the quality of the material (Figure 26.5 summarizes them on a time axis).

Intrinsic (electronic) breakdown. In a perfect solid at low temperature, conduction electrons gain energy from the field and lose it to phonons. Above a critical field the gain exceeds the loss, and impact ionization multiplies the electrons as in a gas. This sets the ultimate strength, about 0.5–1 GV/m (500–1000 kV/mm) for polymers and oxides. It is approached only in thin, clean films over nanoseconds to seconds.

Thermal breakdown. Dielectric and conduction losses heat the insulation. Both $\tan\delta$ and conductivity increase with temperature (Sections 26.3 and 26.6), so heating raises the loss, which raises the temperature further. If heat cannot be removed fast enough, the temperature runs away.

A simple model shows when. Let a slab of thickness d, with a voltage V across it, lose heat from its surface with a heat-transfer coefficient h (W/(m^2 K)), and let its loss tangent increase as $\tan\delta = \tan\delta_0 \exp(\beta \Delta T)$, where ΔT is the temperature rise. Per unit area, heat generated equals heat removed in steady state:

$$\omega \varepsilon_0 \varepsilon_r \tan\delta_0 \, e^{\beta \Delta T} \frac{V^2}{d} = h \, \Delta T$$

The left side grows exponentially with ΔT, the right side linearly. A stable solution exists only if the two curves cross. They just touch, the critical condition, when both their values and their slopes are equal, which happens at $\beta \Delta T = 1$. At that point $e \cdot \omega \varepsilon_0 \varepsilon_r \tan\delta_0 V^2/d = h/\beta$, giving the critical voltage

$$V_c = \sqrt{\frac{h\, d}{e\, \beta\, \omega \varepsilon_0 \varepsilon_r \tan\delta_0}}$$

> **Status of the principle: thermal breakdown criterion**
> *Derived* from the heat balance, within an *approximate* model: uniform temperature through the slab, exponential temperature dependence of loss, and surface cooling with constant h. Thick insulation with internal temperature gradients needs a numerical solution, but the scaling with frequency, cooling and loss holds.

Worked example — thermal breakdown at power and radio frequency. A 1 mm polar insulator has $\varepsilon_r = 4$, $\tan\delta_0 = 0.01$, $\beta = 0.05$ K^{-1} (loss doubles every 14 K), and $h = 20$ W/(m^2 K) (still air). At 50 Hz:

$$V_c = \sqrt{\frac{20 \times 10^{-3}}{2.718 \times 0.05 \times 314.2 \times 8.854 \times 10^{-12} \times 4 \times 0.01}} = \sqrt{\frac{0.020}{1.512 \times 10^{-11}}} = 3.6 \times 10^4 \text{ V}$$

about 36 kV, comparable with the short-term electrical strength, so thermal breakdown competes with other mechanisms at power frequency. At 1 MHz (with the same loss tangent, for comparison) V_c falls by $\sqrt{20{,}000} = 141$, to about 260 V. The runaway point is reached when the insulation is only $1/\beta = 20$ K above ambient. The model explains three engineering rules: polar, lossy insulators are unsuitable at radio frequency; thermal breakdown voltage grows only as $\sqrt{d}$ (thicker insulation gains little); and cooling matters as much as material strength. Thermal runaway is a real failure mode in high-voltage bushings, in capacitors in hot enclosures, and in polar insulation near its glass transition.

Electromechanical breakdown. The electrostatic pressure between electrodes, $\tfrac{1}{2}\varepsilon_0 \varepsilon_r E^2$, compresses a soft insulator. If the material's modulus is low (a polymer near its softening temperature), the thickness decreases, the field rises, and compression accelerates. Stark and Garton showed that instability occurs when the field reaches about $0.6\sqrt{Y/(\varepsilon_0 \varepsilon_r)}$, where Y is Young's modulus. It is significant for thermoplastics at elevated temperature and is exploited deliberately in dielectric elastomer actuators (Chapter 27).

Discharge-induced and treeing breakdown. In practice, most long-term failures of polymer insulation follow a slow route:

1. A local field enhancement (void, protrusion, contaminant) causes **partial discharges** (Section 26.7) or charge injection.
2. These erode the polymer and leave a hollow channel. Further discharges in the channel extend it, and it branches. The resulting structure, visible in transparent samples, is an **electrical tree**.
3. When a tree channel bridges most of the insulation, the remaining material breaks down.

In the presence of water and a.c. fields, a different, slower structure grows at lower fields: the **water tree**, a region of microscopic water-filled voids and channels formed by electro-oxidation and mechanical stress in the polymer. A water tree does not itself conduct, but it raises the local field and lowers the strength, and an electrical tree eventually starts from its tip. This is the opening failure. Modern "tree-retardant" XLPE uses additives and very clean compounds to suppress it.

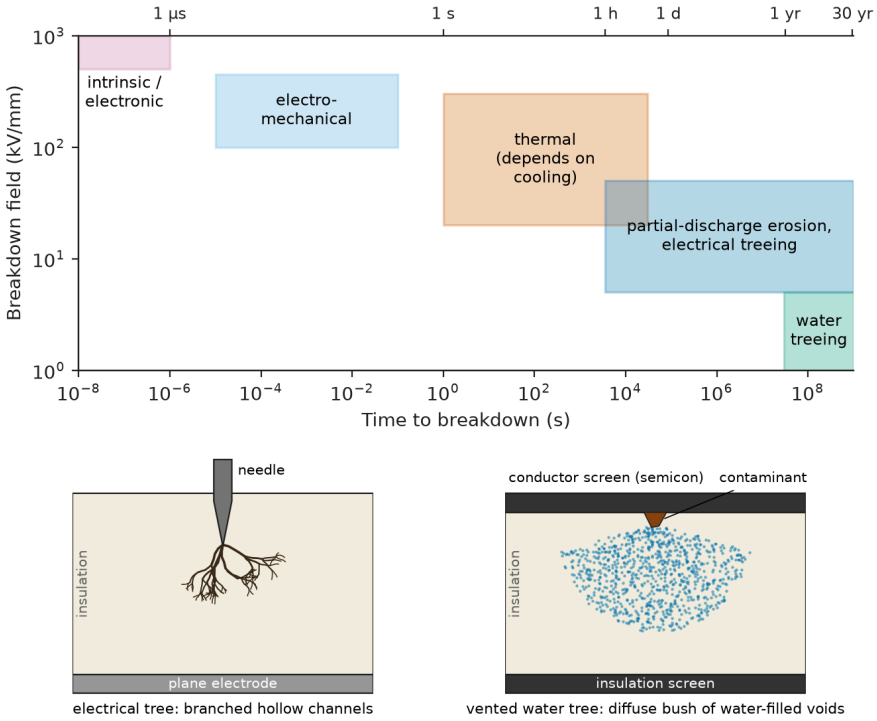


Figure 26.5 — Breakdown mechanisms in solid insulation arranged by the time to failure, with sketches of electrical and water trees.

26.6 Volume and surface resistivity

Two paths for leakage

An insulator leaks current by two paths: through its bulk and along its surface. Each has its own resistivity.

- **Volume resistivity** ρ_v ($\Omega\,\mathrm{m}$; often quoted in $\Omega\,\mathrm{cm}$, 100 times larger numerically) is defined as for conductors: $\rho_v = R_v A/t$.
- **Surface resistivity** ρ_s is the resistance of a square of surface between two parallel edge electrodes. A path of length g and width W has $R_s = \rho_s\, g/W$, so ρ_s has units of ohms (written $\Omega/\square$, as for sheet resistance).

The two are measured together with a three-electrode arrangement (IEC 62631-3, ASTM D257): a guard ring diverts the surface current away from the measuring electrode when volume resistivity is measured, and the roles are swapped for surface resistivity. Volume resistivity is a material property; surface resistivity is mostly a property of the surface's condition: its moisture, contamination and history.

How current flows in an insulator

An insulator has very few free carriers, and the ones it has move with difficulty. Several conduction mechanisms operate, with different signatures:

Mechanism	Carriers	Characteristic dependence
Ionic conduction	Mobile ions (Na^+ in glass, impurity ions in polymers)	Arrhenius in T; ohmic; electrode polarization at low frequency
Hopping	Electrons between localized states	Arrhenius or $\exp[-(T_0/T)^{1/4}]$
Schottky emission	Electrons over an electrode barrier	$\ln J \propto \sqrt{E}$; strongly T-dependent
Poole–Frenkel emission	Electrons released from traps in the bulk	$\ln(J/E) \propto \sqrt{E}$
Space-charge-limited current	Injected carriers	$J \propto V^2/d^3$ at high field
Tunnelling (Fowler–Nordheim)	Electrons through a thin barrier	$J \propto E^2 \exp(-b/E)$; nearly T-independent (thin gate oxides)

At low field the bulk conduction of most insulators is ionic or hopping, and the resistivity follows an Arrhenius law:

$$\rho_v(T) = \rho_\infty \exp\left(\frac{E_a}{k_B T}\right)$$

with activation energies typically 0.5–1.5 eV. At high field the electrode-limited and trap-limited mechanisms take over, and the current rises faster than linearly. These mechanisms matter most for thin films and gate dielectrics (Chapter 34).

> **Status of the principle: Arrhenius dependence of insulator resistivity**
> *Empirical.* It reflects thermally activated ion hopping or carrier release from traps (the kinetics is treated in the Advanced Chemistry core). Over wide temperature ranges several mechanisms with different E_a contribute and the plot of $\ln\rho$ against $1/T$ curves.

Representative values

Material	Volume resistivity at 20–25 °C (Ω m)
PTFE	$> 10^{16}$
Polyethylene, XLPE, PP	10^{14}–10^{16}
Fused silica	$> 10^{16}$
Polyimide	about 10^{15}
Epoxy (cast)	10^{12}–10^{15}
FR-4 (through thickness)	10^{12}–10^{14}
Alumina (99 %)	$> 10^{12}$ (falls to about 10^8 at 500 °C)
Mica	10^{11}–10^{15}
PVC (plasticized)	10^9–10^{13}
Electrical porcelain	10^{10}–10^{12}
Soda-lime glass	10^9–10^{11} (mobile Na^+)
Mineral oil	10^{10}–10^{13}
Ultrapure water	1.8×10^5

Worked example — leakage and its temperature dependence in the cable. For the XLPE cable of Section 26.2, take $\rho_v = 10^{14}$ Ω m at 20 °C. The insulation resistance of a length ℓ of coaxial insulation is

$$R_{\text{ins}} = \frac{\rho_v \ln(R/r)}{2\pi\ell} = \frac{10^{14} \times 0.449}{2\pi \times 1000} = 7.1 \times 10^9\ \Omega$$

for 1 km, i.e. 7.1 GΩ km. The leakage current at 6.35 kV is $6350/7.1 \times 10^{9} = 0.9\ \mu$A, six orders of magnitude below the capacitive charging current of 0.57 A. At a.c. the resistivity of XLPE hardly matters. Now heat the insulation to its rated 90 °C. With $E_a = 1.0$ eV, the resistivity falls by

$$\exp\left[\frac{E_a}{k_B}\left(\frac{1}{293.15} - \frac{1}{363.15}\right)\right] = \exp\left[11{,}605 \times 6.58 \times 10^{-4}\right] = \exp(7.63) \approx 2.1 \times 10^{3}$$

a factor of about 2000. Under d.c. this matters a great deal. In an HVDC cable the steady-state field divides according to resistivity, not permittivity (Section 26.9), so the temperature gradient across a loaded cable shifts the stress toward the cooler outer insulation.

Absorption currents and the polarization index

When a d.c. voltage is applied to insulation, the current does not settle at once. A large capacitive charging spike is followed by an **absorption current** that decays over minutes to hours (slow orientational and interfacial polarization, Section 26.3), and only then by the steady leakage current. Insulation-resistance testers exploit this. The **polarization index** of a motor or generator winding, the ratio of the insulation resistance after 10 min to that after 1 min (IEEE 43), is about 2 or more for clean, dry insulation. Moisture and surface contamination add a large, steady conduction current that swamps the absorption, and the ratio falls toward 1. A measured ratio of insulation resistances is thus a quick indicator of the material's condition.

26.7 Tracking and partial discharge

Two processes degrade insulation without ever bridging it as a whole: tracking along surfaces and partial discharges inside voids and at edges. Both are slow, both are driven by local discharges, and both are the subject of their own tests and standards.

Tracking: how a surface becomes a conductor

On a contaminated, damp surface (dust plus condensation on a PCB, salt fog on an outdoor insulator), a thin conducting electrolyte film forms. Leakage current flows through it and heats it. Where the current density is highest the film dries out, forming a **dry band** that interrupts the current. Almost the full voltage now appears across the narrow dry band, which breaks down in small arcs (scintillations). The arcs' heat decomposes the polymer surface underneath. If the polymer leaves carbon behind, the carbonized path conducts, and the process advances step by step until a conducting **track** bridges the electrodes.

Whether a material tracks depends on its chemistry. Polymers with aromatic rings or high char yield (phenolics, standard epoxies, polycarbonate) form carbon and track readily. Aliphatic polymers that decompose to volatile products (polyethylene, polyamide, melamine, PTFE, silicones) and filled compounds with alumina trihydrate (ATH), which releases water when heated and so cools the arc, resist tracking. Ceramics and glasses do not track at all, although they can suffer surface flashover.

The resistance to tracking is measured by the **comparative tracking index** (CTI, IEC 60112): the highest voltage at which a specimen withstands 50 drops of an ammonium-chloride solution falling between two electrodes 4 mm apart without tracking. The insulation coordination standard groups materials by CTI:

Material group (IEC 60664-1)	CTI range (V)	Typical materials
I	600 and above	PTFE, silicone rubber, melamine, PA66, high-CTI FR-4, ceramics
II	400 to below 600	Some PBT and PA grades
IIIa	175 to below 400	Standard FR-4, polycarbonate, many epoxies
IIIb	100 to below 175	Phenolic paper laminates

The **creepage distance**, the shortest path along a surface between conductors, required by IEC 60664-1 and product standards (IEC 62368-1 and others) depends on the working voltage, the pollution degree of the environment, and the material group. For the same voltage and pollution degree, a group IIIb material may need roughly twice the creepage of group I. That is a direct materials-to-layout link: choosing a high-CTI laminate or housing material shrinks a power supply's layout. Slots milled in a PCB between high-voltage nodes, conformal coatings and potting all work by lengthening or protecting the surface path.

Partial discharge: breakdown inside the insulation

A **partial discharge** (PD) is a breakdown that bridges only part of the insulation: a gas-filled void inside a cast resin or a cable, a delamination in a winding, the gap at a sharp electrode edge, the wedge of air between a wire and a board. Two effects make voids dangerous.

The field in a void is higher than in the solid. In a flat, disc-shaped void with its faces perpendicular to the field, the normal component of $\mathbf{D}$ is continuous across the faces, so

$$E_{\text{void}} = \varepsilon_r \, E_{\text{solid}}$$

For a spherical void the enhancement is smaller, $E_{\text{void}} = 3\varepsilon_r E/(2\varepsilon_r + 1)$, which is $1.33E$ for $\varepsilon_r = 4$.

The gas in the void is weaker than the solid. It breaks down according to Paschen's law at a voltage set by the void depth and gas pressure.

Worked example — PD inception in a void. An epoxy insulation ($\varepsilon_r = 4$) contains a flat air-filled void 0.1 mm deep at atmospheric pressure, normal to the field. From Paschen's law with $pd = 760 \times 0.01 = 7.6$ Torr cm:

$$V_b = \frac{365 \times 7.6}{\ln(15 \times 7.6) - 1.53} = \frac{2774}{4.74 - 1.53} = 0.865 \text{ kV}$$

The void breaks down when the field inside it reaches $0.865 \text{ kV}/0.1 \text{ mm} = 8.65 \text{ kV/mm}$, i.e. when the field in the epoxy reaches $8.65/4 = 2.2 \text{ kV/mm}$. The epoxy itself has a short-term strength of 20–30 kV/mm. A design stress of 3 kV/mm, comfortable for the solid, would make this void discharge every half-cycle.

Each discharge collapses the void voltage, then the applied a.c. voltage builds it back up, and the void discharges again: several times per cycle, 10^8–10^9 times per year. Each discharge transfers a small charge (detectable at the terminals as an **apparent charge** of picocoulombs to nanocoulombs, measured to IEC 60270), and bombards the void walls with electrons, ions, UV and ozone. The walls erode, an electrical tree starts, and the insulation fails long before the bulk field alone would have caused trouble. Figure 26.5 places this mechanism on the time axis.

That is why insulation systems are specified to be "PD-free" at a voltage above the operating voltage, why cable and resin casting processes are designed to exclude voids (vacuum casting, degassing, pressure impregnation), and why PD measurement is a standard acceptance and condition-monitoring test for cables, transformers, machines and power modules. The rise of fast-switching SiC and GaN inverters has made PD in motor windings a new problem: fast voltage edges with overshoots of up to twice the d.c. link voltage stress turn-to-turn enamel insulation above its PD inception voltage. Corona-resistant enamels, filled with nano-silica or alumina that resists erosion, are the materials response.

26.8 Moisture and contamination effects

Why water is the insulator's main enemy

Water combines three properties that are all damaging to insulation:

- **High permittivity.** Its ε_r of 80 raises the permittivity of any material that absorbs it, and its orientational relaxation adds loss.
- **Ionic conduction.** Water dissolves ionic impurities and makes them mobile, raising conductivity by orders of magnitude, especially on surfaces.
- **Chemical attack.** It hydrolyzes polyesters, polyamides, cellulose and some epoxies, and drives electrochemical migration of metals (the electrochemistry belongs to the Advanced Chemistry core).

The bulk permittivity rises only modestly. For an epoxy ($\varepsilon_r = 4$) absorbing 1 % water by volume, the logarithmic mixing rule (Section 26.9) gives

$$\ln \varepsilon = 0.99 \ln 4 + 0.01 \ln 80 = 1.416 \quad \Rightarrow \quad \varepsilon = 4.12$$

a 3 % rise. But the loss tangent can double, the volume resistivity can fall by one or two orders of magnitude, and the surface resistivity by many more. A humid film only a few molecular layers thick on a surface with ionic residues is enough to drop surface resistivity from above $10^{14}\ \Omega$ to below $10^{9}\ \Omega$.

Water uptake by insulating materials

Material	Typical water absorption	Consequence
PTFE, PE, PP	Below 0.01 %	Electrical properties nearly insensitive to humidity
Polystyrene	0.05 %	Stable, used in precision capacitors
FR-4	0.1–0.2 % (24 h immersion)	Small shift in ε_r and $\tan\delta$; CAF risk
Epoxy (cast)	0.1–0.5 % (24 h)	Higher at elevated temperature
Polyimide	1–3 % (at equilibrium, humid)	Measurable ε_r and $\tan\delta$ change; bake before high-temperature soldering
PA66 (nylon)	2.5 % at 50 % RH; up to about 8 % saturated	Large shifts; dimensional change
Cellulose (paper, pressboard)	Several per cent	Dried below 0.5 % in transformers
Alumina, glass	Negligible in bulk; surface films only	Surface leakage in humidity

Moisture in liquid–solid insulation systems

In a power transformer, almost all the water is in the cellulose paper and pressboard, not in the oil. Water partitions between them according to temperature: as the transformer heats, water moves from paper into oil; as it cools, back again. The oil's breakdown voltage depends on its water content relative to saturation, not on the absolute amount, and drops steeply once relative saturation exceeds roughly 20–30 %. A sudden cooling of a wet, hot transformer can push the oil past saturation and form free water droplets. Water also accelerates the chemical aging of cellulose (Section 26.10). New transformers are therefore dried to below about 0.5 % water in the cellulose, and online moisture monitoring is common.

Contamination on printed circuit assemblies

On a PCB, the combination of humidity, ionic residues (flux activators, handling salts, plating chemicals) and a bias voltage leads to two failure processes:

- **Electrochemical migration.** Metal ions dissolve at the anode, migrate through a surface water film, and plate out at the cathode as dendrites that grow back across the gap until they short it. Silver, copper and tin all migrate; silver is the worst.
- **Conductive anodic filament (CAF).** Copper migrates along the interface between glass fibres and resin inside the laminate, especially where drilling has damaged it, eventually shorting adjacent vias.

The materials responses are cleaner processes ("no-clean" fluxes whose residues are non-ionic, or washing), CAF-resistant laminates with better glass–resin coupling, surface insulation resistance (SIR) qualification testing at elevated temperature and humidity under bias, and conformal coatings (acrylic, silicone, polyurethane, parylene) that keep water films from forming continuous paths. A coating slows water absorption but does not stop it, since all polymers are permeable to water vapour to some degree. The long-term processes of moisture ingress and corrosion in packages are treated in Chapter 44.

26.9 Polymer, ceramic, and composite insulators

Polymers

Polymers are the dominant insulating materials by volume: cable insulation, wire enamel, film capacitors, PCB resins, encapsulants, connector housings. They are cheap, easily formed, and many have very low loss. Their weaknesses are a limited temperature range (Chapter 20), creep, flammability, water absorption for polar types, and susceptibility to discharge erosion and tracking.

Polymer	ε_r	$\tan\delta$	Max. continuous temperature (°C, typical)	Characteristic uses
PE (LDPE, HDPE)	2.3	2×10^{-4}	70–80	Coax dielectric, telecom cable
XLPE	2.3	$2–5 \times 10^{-4}$	90 (250 short-circuit)	MV/HV power cable
PP (biaxially oriented)	2.2	2×10^{-4}	85–105	Film capacitors (power electronics)
PTFE	2.1	$1–2 \times 10^{-4}$	260	RF cables, microwave laminates, high-temperature wire
PET (polyester film)	3.2	5×10^{-3}	105–125	General-purpose film capacitors, motor slot liners
Polyimide	3.4	$2–8 \times 10^{-3}$	200–240	Flex circuits, wire wrap, high-temperature insulation
Epoxy (filled, cast)	3.5–5	5×10^{-3}–0.02	130–180 (depends on T_g)	Cast transformers, encapsulation, PCB resin
Silicone rubber	3–4	10^{-3}–5×10^{-3}	180–200	Outdoor insulator housings, cable accessories, potting
PVC	3–6	0.01–0.1	70–90	Low-voltage building wire
PEEK, PPS	3.0–3.3	3×10^{-3}	200–240	High-temperature connectors, coil formers

The non-polar polymers (PE, PP, PTFE) are the high-performance dielectrics: lowest loss, highest intrinsic strength, lowest water uptake. Polarity buys adhesion, temperature resistance and flame retardancy at the price of loss.

Ceramics and glasses

Ceramics offer what polymers cannot: high temperature capability, no creep, resistance to discharges and tracking, high thermal conductivity in some cases, and dimensional stability. Their weaknesses are brittleness, thermal-shock sensitivity (Chapter 24), and processing cost.

Ceramic	ε_r	$\tan\delta$ (1 MHz)	Thermal conductivity (W/(m K))	Characteristic uses
Electrical porcelain (quartz/alumina–feldspar–clay)	5–7	0.01–0.02	1.5–4	Line and station insulators, bushings
Steatite ($MgO–SiO_2$)	6	10^{-3}	2–3	Low-loss insulating parts, RF stand-offs
Alumina 96 %	9.4	3×10^{-4}	20–25	Substrates, spark-plug insulators, vacuum feedthroughs
Alumina 99.5 %	9.8	1×10^{-4}	30–35	Microwave substrates, high-voltage parts
Aluminium nitride	8.6–8.9	3×10^{-4}–10^{-3}	170–230	Power-module substrates
Beryllium oxide	6.7	4×10^{-4}	about 250	RF power packages (toxic dust; declining use)
Fused silica	3.8	1×10^{-4}	1.4	Low-loss RF windows, optical fibre
Glass (borosilicate)	4.6	4×10^{-3}	1.1	Hermetic seals, bushings
Mica (muscovite, phlogopite)	6–7	2×10^{-4}	0.5 (across sheets)	Machine insulation (as mica paper), heaters, capacitors

Composites: getting both

Most practical insulation is a composite, designed to combine a polymer's processability with a ceramic or fibre's strength, discharge resistance or thermal conductivity.

- **Oil–paper** (transformers, older cables, bushings): cellulose paper gives mechanical structure; oil fills the pores, excludes air (and therefore PD), and carries heat away.
- **Epoxy–mica** (rotating-machine groundwall insulation): mica flakes, which do not erode under discharges, bound in epoxy and vacuum–pressure impregnated. Mica is the only practical material for machine windings at 6–25 kV because PD in such windings cannot be fully avoided.
- **Glass-fibre–epoxy** (FR-4 laminates, rods of composite outdoor insulators, slot wedges): fibres give stiffness and strength; the resin bonds and insulates.
- **Silicone housings over glass-fibre rods** (composite or "polymeric" line insulators): light, unbreakable, and with a water-repellent silicone surface that resists pollution flashover. Low-molecular-weight silicone migrates to the surface and keeps it hydrophobic even when polluted.
- **Filled polymers**: silica or alumina in epoxy for lower expansion and higher thermal conductivity (Chapter 51); ATH in silicone for tracking resistance; nano-fillers for PD and treeing resistance (nanodielectrics, Chapter 38).

Mixing rules for composite permittivity

The effective permittivity of a two-phase composite depends on how the phases are arranged:

- **Layers parallel to the field** (capacitors side by side): $\varepsilon_{\text{eff}} = \phi_1\varepsilon_1 + \phi_2\varepsilon_2$, the upper bound.
- **Layers in series with the field**: $1/\varepsilon_{\text{eff}} = \phi_1/\varepsilon_1 + \phi_2/\varepsilon_2$, the lower bound.
- **Dilute spheres of ε_i in a matrix ε_m** (Maxwell Garnett):

$$\frac{\varepsilon_{\text{eff}} - \varepsilon_m}{\varepsilon_{\text{eff}} + 2\varepsilon_m} = \phi\,\frac{\varepsilon_i - \varepsilon_m}{\varepsilon_i + 2\varepsilon_m}$$

- **Random mixtures**, often fitted by the logarithmic (Lichtenecker) rule: $\ln\varepsilon_{\text{eff}} = \phi_1\ln\varepsilon_1 + \phi_2\ln\varepsilon_2$.

> **Status of the principle: dielectric mixing rules**
> *Approximate*. The parallel and series forms are exact bounds for any arrangement. Maxwell Garnett is derived for dilute, non-interacting spheres. The Lichtenecker rule is empirical. None captures interfacial polarization, which can dominate at low frequency when one phase conducts.

The bounds explain why a high-permittivity filler raises a polymer's permittivity so little at moderate loadings. Isolated particles in a low-permittivity matrix are effectively in series with matrix layers, and the field concentrates in the low-permittivity phase.

Field division in layered insulation

The same series reasoning governs every insulation made of layers: oil and paper, a cable's insulation with a void, an encapsulant over a substrate. For two layers in series, continuity of D under a.c. gives

$$\varepsilon_1 E_1 = \varepsilon_2 E_2 \quad\Rightarrow\quad \frac{E_1}{E_2} = \frac{\varepsilon_2}{\varepsilon_1} \qquad \text{(a.c.)}$$

The **lower-permittivity layer carries the higher field**. In oil–paper insulation the oil ($\varepsilon_r = 2.2$) is stressed $3.5/2.2 = 1.6$ times more than the paper ($\varepsilon_r = 3.5$), and the oil is the weaker material. That is why transformer design uses thin oil ducts, divided by pressboard barriers, rather than wide open oil gaps: a thin oil gap has a higher breakdown strength.

Under d.c., after the interfacial charging of Section 26.3, continuity of current gives

$$\sigma_1 E_1 = \sigma_2 E_2 \quad\Rightarrow\quad \frac{E_1}{E_2} = \frac{\sigma_2}{\sigma_1} \qquad \text{(d.c., steady state)}$$

and the **lower-conductivity layer carries the higher field**. Because conductivities vary by orders of magnitude with temperature, moisture and field (Section 26.6), d.c. stress distributions are much harder to control than a.c. ones, and they change when the voltage polarity reverses. HVDC cable insulation is therefore a specially formulated material, with additives that make its conductivity less sensitive to temperature and that suppress space-charge accumulation.

Self-healing capacitor films

A film capacitor with metallized electrodes (aluminium or zinc only 20–50 nm thick on polypropylene) survives the breakdown of a weak spot. The arc at the fault vaporizes the thin electrode around it within microseconds, isolating the spot, and the capacitor loses a tiny area of capacitance instead of failing. This is a materials solution to the Weibull problem of Section 26.4. It works only because the electrode is thin enough to evaporate and the polymer decomposes to gaseous, not carbonaceous, products.

26.10 Insulation aging and degradation

Insulation rarely fails because its short-term strength is exceeded. It fails because the material changes over years until its strength falls to the service stress. The engineer's tools for designing against aging are life laws, fitted to accelerated tests, and thermal classes.

Thermal aging

At service temperature, polymers slowly oxidize, lose plasticizer and low-molecular-weight fractions, cross-link further and embrittle, or depolymerize (cellulose). The insulation is usually declared at its end of life when a property such as tensile elongation at break or electrical strength has fallen to a fixed fraction (often 50 %) of its initial value, at which point mechanical stress (vibration, thermal cycling, short-circuit forces) can crack it.

The rate of most of these processes is thermally activated, so life follows an Arrhenius law (the kinetics belongs to the Advanced Chemistry core):

$$t_{\text{life}} = A \exp\left(\frac{E_a}{k_B T}\right) \quad \Longrightarrow \quad \ln t_{\text{life}} = \ln A + \frac{E_a}{k_B}\cdot\frac{1}{T}$$

A plot of $\ln t_{\text{life}}$ against $1/T$ is a straight line (Figure 26.6), which allows life at service temperature to be extrapolated from accelerated tests at higher temperatures (IEC 60216).

The traditional rule of thumb is the **Montsinger rule**: life halves for every 10 K rise (some materials, 6–8 K). It is a local approximation to the Arrhenius law. Halving per 10 K near 100 °C corresponds to

$$E_a = \frac{k_B \ln 2}{\dfrac{1}{373.15} - \dfrac{1}{383.15}} = \frac{(8.617\times10^{-5})(0.693)}{7.0\times10^{-5}} = 0.85\ \text{eV}$$

At higher temperatures the same activation energy gives a longer halving interval, because $1/T$ changes more slowly.

> **Status of the principle: Arrhenius thermal-aging law**
> *Empirical.* It assumes a single dominant, thermally activated degradation reaction whose mechanism does not change with temperature. Extrapolation fails if the accelerated test triggers a different mechanism (melting, a glass transition, loss of an additive), so test temperatures are kept close to service temperature and well below transitions.

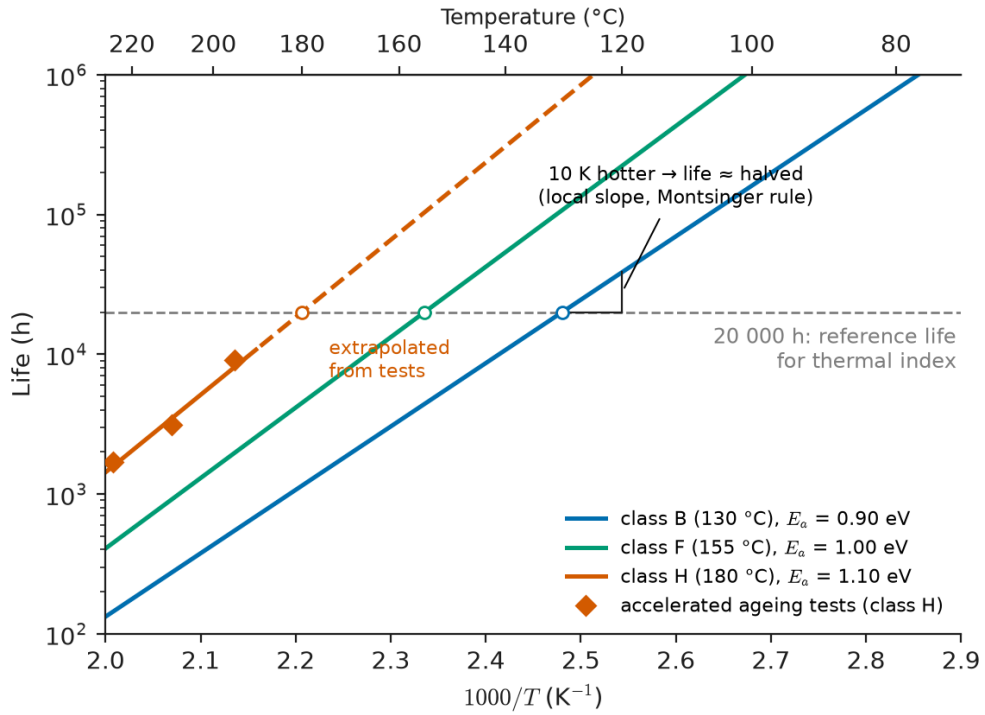


Figure 26.6 — Thermal endurance of insulation: Arrhenius life lines and the thermal classes they define.

Thermal classes

Insulation systems for machines, transformers and wiring are assigned a **thermal class** (IEC 60085), the maximum continuous temperature at which they meet their design life (typically defined from a thermal index at 20,000 h):

Thermal class (°C)	Former letter	Typical materials
90	Y	Cotton, paper, unimpregnated
105	A	Impregnated cellulose, oil-paper
120	E	Polyester and polyurethane enamels, PET film
130	B	Glass, mica with organic binders; polyester-imide enamels
155	F	Mica, glass with epoxy or polyester-imide resins
180	H	Silicone-bonded mica and glass; polyimide, aramid paper
200, 220, 250	N, R, —	Polyimide film and enamel, ceramic-filled and inorganic systems

Worked example — the value of running cooler. A motor's class F insulation (155 °C) has a design life of 20,000 h at its hottest-spot temperature of 155 °C. In service it runs with a hot spot at 135 °C. By the 10 K rule, life increases by $2^{20/10} = 4$, to 80,000 h. With the Arrhenius law and $E_a = 0.85$ eV:

$$\frac{t_{135}}{t_{155}} = \exp\left[\frac{0.85}{8.617\times10^{-5}}\left(\frac{1}{408.15}-\frac{1}{428.15}\right)\right] = \exp(1.13) = 3.1$$

giving about 62,000 h. The two estimates differ by 30 % because the 10 K rule was calibrated near 100 °C; at 150 °C the halving interval for 0.85 eV is about 13 K. Either way, the point is the same: 20 K of temperature margin multiplies insulation life by three to four. This is why motor designs are often "class F insulation, class B temperature rise", and why the hot-spot temperature, not the average winding temperature of Chapter 25, governs life.

Electrical aging

At high field, the electrical processes of Sections 26.5 and 26.7 (charge injection, partial discharge, treeing) degrade the material even without high temperature. Life under constant field is usually fitted by an **inverse power law**:

$$t_{\text{life}} = t_0 \left(\frac{E}{E_0} \right)^{-n}$$

with a voltage endurance coefficient n of about 9–15 for XLPE cable insulation and 9–12 for epoxy–mica machine insulation. The steepness is striking: with $n = 9$, a 20 % higher field shortens life to $1.2^{-9} = 0.19$ of its value, a factor of five. That is why voltage stress, not just temperature, must be held below the design value, and why overvoltages from switching surges and fast inverter edges matter for life.

> **Status of the principle: inverse power law of electrical aging**
> *Empirical.* It is fitted to accelerated voltage-endurance tests over a limited field range. It cannot be extrapolated below the threshold field for partial discharges or space-charge accumulation, below which electrical aging may practically stop.

Multi-stress aging

Real insulation is aged by several stresses at once, often summarized as **TEAM**: thermal, electrical, ambient (moisture, chemicals, UV, radiation, pollution) and mechanical (vibration, thermal cycling, short-circuit forces). The stresses interact: heat accelerates electrical treeing, moisture accelerates thermal aging of cellulose, and mechanical cracks create voids for partial discharge. Life models that combine an Arrhenius term with an inverse power law are used for cables and machines, but their parameters are always empirical and specific to a material system. The opening water-tree failure is a textbook case of combined electrical and ambient aging. How aging appears as a failure mechanism in electronic systems, and how it is investigated, is the subject of Chapter 44, which also treats how accelerated tests are designed and interpreted.

What remains open

Open question	Where it is answered
What happens when polarization becomes spontaneous and switchable, and how are ferroelectric, piezoelectric and high-permittivity dielectrics engineered?	Chapter 27
Why is the band gap of an insulator large, and how does it relate to breakdown strength and to tunnelling leakage?	Chapter 31
How do gate dielectrics, interface traps and thin-film insulators behave in devices?	Chapter 34
What dielectric properties matter for high-frequency electronic and optical materials?	Chapter 35
How do nanofillers change the breakdown and discharge resistance of polymers?	Chapter 38
How are dielectric films deposited and their defects controlled?	Chapter 50
Which laminates, encapsulants and substrates are used in packages, and how is CTE mismatch managed?	Chapter 51
How are dielectric properties and breakdown measured, and how is measurement uncertainty interpreted?	Chapter 52
How do dielectric breakdown, moisture ingress and thermal aging appear as failure mechanisms, and how are failures analysed?	Chapter 44
How are accelerated aging tests designed and insulation chosen for a required service life?	Chapter 58

Next: Chapter 27 follows the polarization catastrophe of Section 26.1 to its conclusion: materials whose polarization is spontaneous, switchable, or coupled to strain, heat and light, and the capacitors, actuators, sensors and RF components built from them.

Chapter 27 — Functional Dielectric Materials

Part VI · Thermal, Electrical, Dielectric and Magnetic Materials

A power-supply designer replaces an electrolytic capacitor with a small 10 μF, 16 V multilayer ceramic capacitor (MLCC). On the bench, at 12 V d.c., the output ripple is three times what the simulation predicted. An impedance analyser shows why: at 12 V bias the "10 μF" part has only about 3 μF. Worse, it audibly whines at the converter's burst-mode frequency, and its capacitance drifts down a few per cent over the first weeks of storage. None of this is a defect. The MLCC's dielectric is barium titanate, a ferroelectric. Its huge permittivity, its loss of capacitance under bias, its acoustic noise and its slow aging all come from the same physics: a polarization that is not just induced by the field, but exists spontaneously, sits in domains, and is coupled to the shape of the crystal.

Chapter 26 treated dielectrics as passive: they polarize in proportion to the field and give the polarization back. This chapter is about dielectrics in which polarization is coupled to something else, such as mechanical strain, temperature or light, or in which it becomes spontaneous and switchable. These **functional dielectrics** are the active materials in ultrasound probes, inkjet heads, quartz oscillators, RF filters in every phone, infrared motion detectors, non-volatile memories, high-capacitance capacitors and optical modulators.

> **Why do only some crystals generate a voltage when squeezed?**
> **What makes barium titanate's permittivity a thousand times that of polyethylene, and why does it collapse under d.c. bias? How does one choose a material for an actuator, a sensor, a microwave resonator or an optical modulator?**

Concept	Engineering destination
Crystal symmetry, polar axes	Which materials can be piezo-, pyro- or ferroelectric
Ferroelectric domains, hysteresis, Curie–Weiss law	MLCC behaviour, FeRAM and FeFET memories, poling
Piezoelectric coefficients d, g, k	Actuators, ultrasound, quartz crystals, SAW/BAW filters
Pyroelectric coefficient, figures of merit	PIR motion sensors, uncooled thermal imaging
Electrostriction, relaxors	Precision positioning, MLCC "singing", dielectric elastomers
High-permittivity dielectrics, EOT	MLCCs, DRAM and gate dielectrics
Dielectric resonators, $Q \times f$, τ_f	Microwave filters, oscillators, base-station hardware
Pockels and Kerr effects, V_π	Optical modulators, Q-switches, photonic integrated circuits

Roadmap. Symmetry as the gatekeeper (27.1) → ferroelectrics (27.2) → piezoelectrics (27.3) → pyroelectrics (27.4) → electrostriction (27.5) → high-permittivity dielectrics (27.6) → dielectric resonators (27.7) → electro-optic materials (27.8) → applications and selection (27.9).

27.1 Symmetry decides which effects are possible

Why squeezing glass gives no voltage

Press on a block of glass and nothing electrical happens. Press on a quartz crystal along certain directions and charge appears on its faces. The difference is symmetry. A uniform stress has a centre of symmetry: inverting all coordinates through a point ($\mathbf{r} \to -\mathbf{r}$) leaves it unchanged. Polarization is a vector and changes sign under inversion. If the material also has a centre of symmetry, a stress can only produce something that is itself unchanged by inversion, and $\mathbf{P} \to -\mathbf{P}$ is not, so the induced polarization must be zero. **A material with a centre of symmetry cannot be piezoelectric.** Glass, being isotropic on average, behaves as if it had full symmetry; a cubic NaCl crystal has a centre of symmetry. Neither can show the effect.

This argument is an example of **Neumann's principle**: the symmetry of any physical property of a crystal must include the symmetry of the crystal's point group. It filters the 32 crystallographic point groups (Chapter 3) into nested families:

Set	Number of point groups	Condition	Examples
All crystals	32	—	All
Non-centrosymmetric	21	No inversion centre	Quartz, GaAs, ZnO, $BaTiO_3$ (tetragonal)
Piezoelectric	20	Non-centrosymmetric, excluding class 432	Quartz, AlN, GaN, ZnO, $LiNbO_3$, PZT
Pyroelectric (polar)	10	Has a unique polar axis	ZnO, AlN, $LiTaO_3$, $BaTiO_3$, PVDF
Ferroelectric	Subset of the 10 polar classes	Spontaneous polarization that an applied field can reverse	$BaTiO_3$, PZT, $LiNbO_3$, PVDF, doped HfO_2

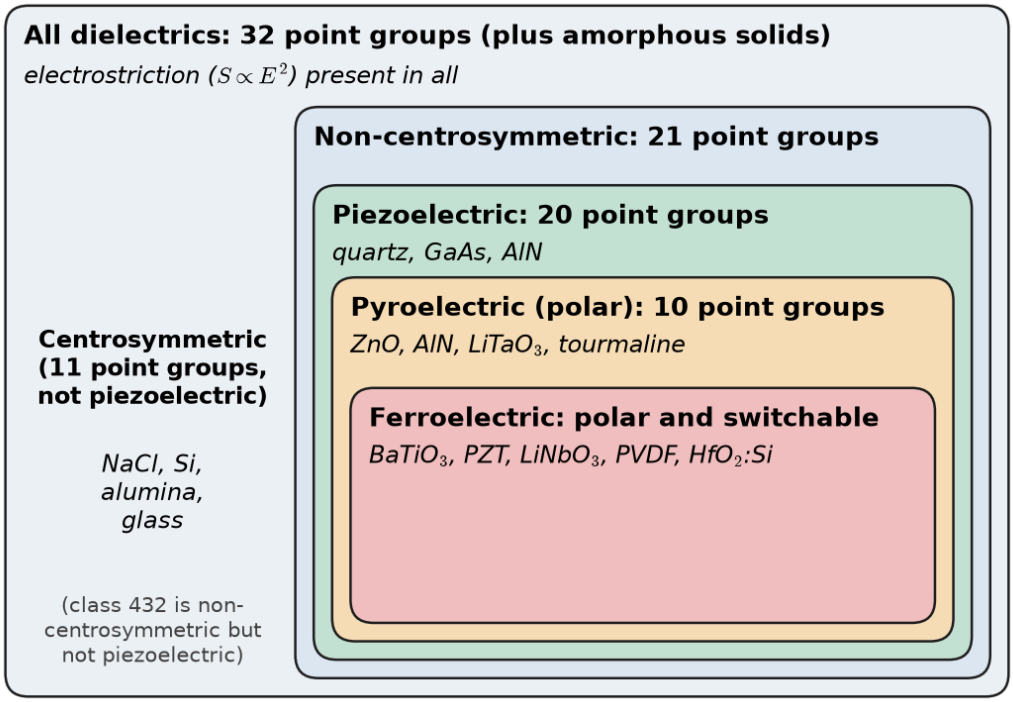


Figure 27.1 — Nested symmetry classes of dielectric materials: every ferroelectric is pyroelectric, every pyroelectric is piezoelectric, and every dielectric is electrostrictive.

Figure 27.1 shows the nesting. Each inner set is a subset of the one outside it, so every ferroelectric is also pyroelectric and piezoelectric, and every pyroelectric is piezoelectric. The reverse is not true. Quartz is piezoelectric but not pyroelectric (it has polar directions but no unique polar axis). AlN and ZnO are pyroelectric but not ferroelectric (their polarization cannot be reversed by any practical field before breakdown). **Electrostriction**, a strain proportional to the square of the field, is allowed in every material, including glass and centrosymmetric crystals (Section 27.5).

> **Status of the principle: Neumann's principle**
> *Fundamental.* It follows from the requirement that physical laws do not depend on how the crystal is oriented. It tells us which effects are forbidden; it does not tell us how large an allowed effect will be.

Polycrystals and poling

A ceramic made of randomly oriented piezoelectric grains has, on average, the full rotational symmetry of an isotropic solid, and the contributions of the grains cancel. A non-ferroelectric piezoelectric (quartz, AlN) must therefore be used as a single crystal or

as a textured film whose grains share an orientation (the c-axis-oriented AlN films of Section 27.3). A ferroelectric ceramic offers a way out. A strong field applied at elevated temperature reorients the switchable polarization of each grain as nearly as its crystallography allows toward the field. This process, **poling**, gives the ceramic a net polar axis, and with it piezoelectricity and pyroelectricity. Almost all piezoelectric ceramics are poled ferroelectrics, which is why the ferroelectrics come first.

27.2 Ferroelectric materials

Spontaneous polarization in the perovskite structure

Barium titanate, $BaTiO_3$, has the **perovskite** structure ABO_3 (Figure 27.2): large Ba^{2+} ions at the cube corners, O^{2-} at the face centres forming an octahedron, and a small Ti^{4+} ion at the body centre. Above about 120–130 °C the structure is cubic and centrosymmetric. The Ti ion is a small ion in a slightly too large octahedral cage. Below the transition it shifts off centre by about 0.01 nm toward one of the six oxygen ions, along with smaller shifts of the oxygen ions. The positive and negative charge centres no longer coincide, the unit cell becomes a permanent dipole, and the crystal becomes tetragonal with $c/a \approx 1.01$. Every cell shifts the same way, so the crystal acquires a **spontaneous polarization** P_s of about $0.26\ \mathrm{C/m^2}$ at room temperature.

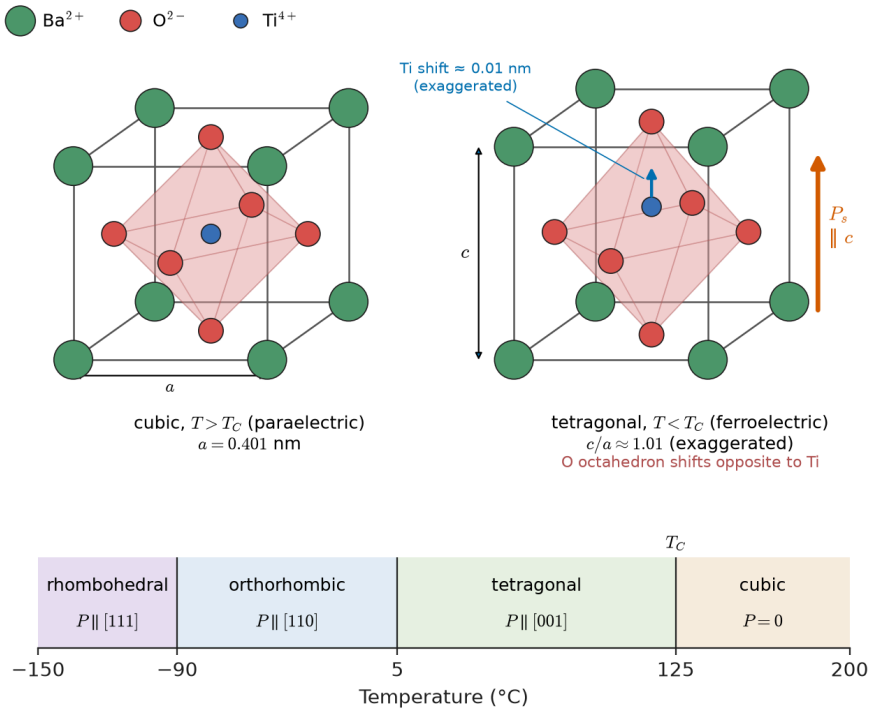


Figure 27.2 — Perovskite BaTiO3: cubic paraelectric cell above the Curie temperature and tetragonal ferroelectric cell below it, with the off-centre Ti displacement.

The transition happens because two contributions compete. The short-range repulsion between ions favours the centred position. The long-range dipole–dipole interaction, the local field feedback of Section 26.1, favours a displaced one. At high temperature thermal motion keeps the average position centred. As the temperature falls, the restoring force for this particular displacement pattern weakens (the lattice vibration involved is called a **soft mode**), and at the **Curie temperature** T_C it vanishes and the displaced structure becomes stable. This is the Clausius–Mossotti polarization catastrophe made real. $BaTiO_3$ passes through three transitions on cooling, cubic → tetragonal (about 125 °C) → orthorhombic (about 5 °C) → rhombohedral (about −90 °C), each with the polarization along a different crystal direction. Each transition produces a peak in the permittivity, which matters for capacitors (Section 27.6).

Landau theory and the Curie–Weiss law

The thermodynamics of the transition can be described without knowing the microscopic details. Expand the free energy density of the crystal in powers of the polarization, keeping only even powers since $+P$ and $-P$ are equivalent by symmetry:

$$G(P,T) = G_0 + \tfrac{1}{2}a\,(T - T_0)\,P^2 + \tfrac{1}{4}b\,P^4 + \tfrac{1}{6}c\,P^6 - EP$$

with $a, c > 0$. Take first the case $b > 0$ (a continuous, second-order transition). With no field, setting $\partial G/\partial P = 0$:

$$a(T - T_0)P + bP^3 = 0 \quad \Longrightarrow \quad P_s = 0 \;\; (T > T_0), \qquad P_s^2 = \frac{a(T_0 - T)}{b} \;\; (T < T_0)$$

The polarization grows continuously from zero below T_0. Above T_0, a small field gives $E = a(T - T_0)P$, so the susceptibility is

$$\chi = \frac{P}{\varepsilon_0 E} = \frac{1}{\varepsilon_0 a\,(T - T_0)} \quad \Longrightarrow \quad \varepsilon_r \approx \frac{C}{T - T_0}, \qquad C = \frac{1}{\varepsilon_0 a}$$

This is the **Curie–Weiss law**: the permittivity diverges as the transition is approached from above. For $BaTiO_3$ the Curie constant is $C \approx 1.7 \times 10^5$ K. Its transition is in fact first order ($b < 0$, stabilized by the P^6 term), so P_s jumps at T_C, and the Curie–Weiss temperature T_0 lies about 10 K below T_C, but the Curie–Weiss behaviour above the transition remains.

> **Status of the principle: Landau theory and the Curie–Weiss law**
> *Approximate.* Landau theory is a symmetry-based expansion valid near the transition, where P is small. Its coefficients are fitted. The Curie–Weiss law describes the permittivity above T_C well for displacive ferroelectrics such as $BaTiO_3$ and $PbTiO_3$; it fails very close to T_C, far below it, and in relaxors (Section 27.5).

Worked example — permittivity above the Curie point. For a $BaTiO_3$ crystal with $C = 1.7 \times 10^5$ K and $T_0 = 110$ °C, at 150 °C the permittivity is

$$\varepsilon_r \approx \frac{1.7 \times 10^5}{150 - 110} = 4250$$

At 300 °C it has fallen to $1.7 \times 10^5 / 190 \approx 900$. A paraelectric perovskite operated just above its transition is therefore a high-permittivity, field-tunable dielectric. $(Ba,Sr)TiO_3$, whose Curie temperature is lowered below room temperature by strontium, is used this way in tunable RF capacitors (Section 27.9).

Domains and the hysteresis loop

A crystal that polarizes uniformly would have bound charge on its surfaces and a large depolarizing field. It lowers its energy by splitting into **domains**, regions of uniform polarization separated by **domain walls** a few lattice spacings thick. In tetragonal $BaTiO_3$ there are 180° walls (between antiparallel domains) and 90° walls (between domains polarized along different cube edges). The 90° walls also relieve the elastic strain from the tetragonal distortion.

When a field is applied, domains aligned with it grow at the expense of others by wall motion and nucleation of new domains. Plotting P against E through a full cycle gives a **hysteresis loop** (Figure 27.3), with:

- **saturation polarization** P_{sat}, approached at high field;
- **remanent polarization** P_r, remaining at zero field;
- **coercive field** E_c, the field needed to reduce P to zero.

The loop is the signature of ferroelectricity. Its area is the energy dissipated per cycle per unit volume. The parallel with ferromagnetic hysteresis (Chapter 28) is close and deliberate: the magnetization and its domains replace the polarization and its domains.

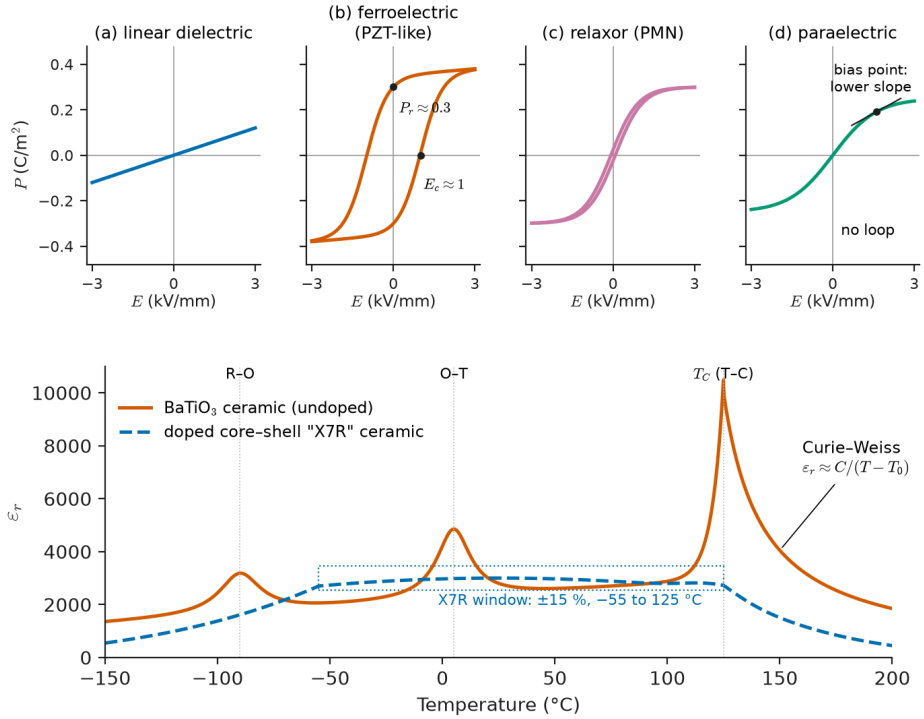


Figure 27.3 — Polarization–field loops of four dielectric types, and the permittivity of BaTiO3 against temperature.

Ferroelectric	T_C **(°C)**	P_r **(typical, C/m²)**	E_c **(typical)**	**Use**
$BaTiO_3$ (ceramic)	120–130	0.05–0.1	0.1–0.5 kV/mm	MLCC dielectric, PTC thermistors
$PbTiO_3$	490	0.5–0.8 (crystal)	High	Component of PZT
PZT ($Pb(Zr,Ti)O_3$, near MPB)	230–390 (by composition, doping)	0.3–0.4	1–2 kV/mm	Piezoelectric ceramics, FeRAM
$LiNbO_3$	1140–1210	0.71	About 21 kV/mm (congruent)	Electro-optics, SAW, periodically poled crystals
PVDF, P(VDF–TrFE)	About 100–130 for copolymer	0.05–0.1	50–100 kV/mm	Flexible sensors, polymer memory
HfO_2:Si, $Hf_{0.5}Zr_{0.5}O_2$ (thin film)	Several hundred (phase is metastable)	0.1–0.25	100–200 kV/mm (1–2 MV/cm)	FeFET, FeRAM integrated in CMOS

Lead zirconate titanate and the morphotropic phase boundary

PZT, the solid solution $PbZrO_3$–$PbTiO_3$, is the workhorse of piezoelectric ceramics. On the Ti-rich side it is tetragonal; on the Zr-rich side, rhombohedral. Near Zr/Ti ≈ 52/48 the two phases have almost equal energy. This boundary is called the **morphotropic phase boundary** (MPB) because it is nearly vertical in temperature. At the MPB the polarization can rotate easily between the 6 tetragonal and 8 rhombohedral directions (and intermediate monoclinic ones). Poling aligns grains well, domain walls move easily, and the permittivity and piezoelectric coefficients peak. Small dopant additions then tune the material:

- **Donor doping** (La^{3+} on Pb sites, Nb^{5+} on Ti sites) creates lead vacancies that make domain walls mobile. The result is a **"soft" PZT** (e.g. PZT-5A, 5H): high d, high permittivity, high loss, low coercive field, suited to sensors and low-power actuators.
- **Acceptor doping** (Fe^{3+}, Mn on Ti sites) creates oxygen vacancies that form defect dipoles and pin domain walls. The result is a **"hard" PZT** (e.g. PZT-4, PZT-8): lower d, low loss, high mechanical quality factor, stable under high drive, suited to ultrasonic power transducers and sonar.

This is the microstructure–property link of Chapter 5 applied to domain walls: point defects decide how easily walls move, and wall motion supplies a large part of the dielectric and piezoelectric response of a ceramic.

Ferroelectric memory

The two remanent states $\pm P_r$ store a bit without power. In a **ferroelectric RAM** (FeRAM) cell, a capacitor is switched by a read pulse. If it switches, the extra charge $2P_rA$ flows; if not, only the small linear charge does.

Worked example — FeRAM switching charge. A PZT capacitor of area $A = 1\ \mu\text{m}^2$ with $P_r = 0.3\ \text{C/m}^2$ releases on switching

$$Q = 2P_rA = 2 \times 0.3 \times 10^{-12} = 6 \times 10^{-13}\ \text{C} = 600\ \text{fC}$$

A DRAM cell stores about 10–25 fC. A ferroelectric therefore delivers a large signal from a small area. The drawbacks are that reading is destructive (the bit must be rewritten), that repeated switching causes **fatigue** (loss of switchable polarization as charged defects pin domain walls at electrodes) and **imprint** (one state becoming preferred), and that lead-based perovskites are difficult to integrate with CMOS. The discovery in 2011 that thin films of HfO_2, already the standard high-permittivity gate dielectric (Section 27.6), become ferroelectric when doped with Si or Zr and crystallized in a non-centrosymmetric orthorhombic phase has revived ferroelectric memory. **Ferroelectric field-effect transistors** (FeFETs) use a ferroelectric gate layer whose polarization shifts the transistor's threshold voltage. The device side belongs to Electronics & Computing; the interface and reliability issues are taken up in Chapter 34.

Aging, temperature limits and grain size

Three practical properties of ferroelectric ceramics follow from domain physics.

- **Aging.** After cooling through T_C or after poling, domain configurations slowly relax into lower-energy states, stabilized by migrating point defects. The permittivity and piezoelectric coefficients decrease roughly linearly with the logarithm of time. Heating above T_C ("de-aging") resets the clock.
- **Depoling temperature.** A poled ceramic loses its polarization as it approaches T_C, and thermally activated wall motion degrades it well below. A common rule is to keep operating temperatures below about half of T_C in °C. That is why high-T_C PZT is preferred to $BaTiO_3$ for piezoelectric use.
- **Size effects.** Below grain sizes of about 1 μm in $BaTiO_3$, the permittivity peak is suppressed and broadened, and in films of a few nanometres, depolarizing fields from incomplete electrode screening suppress ferroelectricity. MLCC dielectric layers below 1 μm are designed around this.

27.3 Piezoelectric materials

The effect and its converse

In a piezoelectric material, stress produces polarization (the **direct effect**) and an electric field produces strain (the **converse effect**). Both are described by the same coefficient, which is guaranteed by thermodynamics: both effects come from a cross term in the same free energy. In the strain–charge form, for one-dimensional loading along the poling axis "3":

$$D_3 = d_{33}\, T_3 + \varepsilon^{T}_{33}\, E_3 \qquad \text{(direct)}$$

$$S_3 = s^{E}_{33}\, T_3 + d_{33}\, E_3 \qquad \text{(converse)}$$

T is stress, S strain, s^E the compliance at constant field, and ε^T the permittivity at constant stress. The **piezoelectric charge coefficient** d has units of C/N (charge per force) or, equivalently, m/V (strain per field). In general d_{ij} is a third-rank tensor written in matrix form with two indices: the first index gives the electrical direction (1, 2, 3, with 3 the poling axis), the second the mechanical one (1–3 normal, 4–6 shear). For a poled ceramic the important ones are:

- d_{33}: field and strain both along the poling axis (thickness-mode actuators, stacks);
- d_{31}: field along 3, strain perpendicular (bending actuators, length-mode resonators); negative, since the material thins as it lengthens along 3;
- d_{15}: shear (shear-mode sensors, some accelerometers).

Other useful coefficients are derived from these. The **voltage coefficient** $g = d/\varepsilon^T$ (units $\mathrm{V\,m/N}$) gives the open-circuit field per stress and is the figure of merit for sensors. The **electromechanical coupling factor**

$$k^2 = \frac{d^2}{s^E \varepsilon^T}$$

is the fraction of input energy, electrical or mechanical, converted to the other form in a quasi-static cycle. It governs the bandwidth of transducers and filters.

> **Status of the principle: linear piezoelectric constitutive equations**
> *Approximate*. They are the small-signal linear terms of a thermodynamic expansion (IEEE Std 176). They hold well for quartz, AlN and $LiNbO_3$ over their working ranges. For ferroelectric ceramics at high drive, domain-wall motion makes d field-dependent and hysteretic, and the coefficients drift with aging, temperature and stress history.

The material families

Material	d_{33} (pC/N)	$\varepsilon_{33}^T/\varepsilon_0$	k_{33} or k_t	T_C (°C)	Character
Quartz (single crystal)	$d_{11} = 2.3$	4.5	about 0.1	573 (α–β transition)	Very stable, very low loss; oscillators, force sensors
AlN (c-axis film)	5–6	9–10	$k_t^2 \approx 6$–$7\,\%$	No ferroelectric transition	CMOS-compatible; BAW filters, MEMS
$Al_{1-x}Sc_xN$ ($x \approx 0.3$, film)	15–30	12–20	$k_t^2 \approx 10$–$15\,\%$	—	Wider-bandwidth BAW filters
ZnO (film)	10–12	8–11	$k_t \approx 0.28$	—	Older thin-film transducers
$LiNbO_3$ (single crystal)	6 ($d_{15} = 69$)	28	Up to 0.5 (depends on cut)	1140–1210	SAW filters, high-temperature sensors
$BaTiO_3$ (ceramic)	190	1700	0.5	120–130	Historical; limited by low T_C
PZT-4 (hard)	289	1300	0.70	328	Power ultrasound, sonar
PZT-5A (soft)	374	1700	0.71	365	Sensors, actuators
PZT-5H (soft)	593	3400	0.75	193	High-displacement actuators, medical imaging
PMN–PT (single crystal)	1500–2500	4000–7000	above 0.9	130–170	High-end ultrasound probes
KNN-based (lead-free)	100–400	500–2000	0.4–0.6	200–400	RoHS-driven replacements
PVDF (stretched, poled)	about -25 to -33 ($d_{31} = 18$–23)	10–13	0.1–0.2	—	Flexible sensors, hydrophones

The table shows the central trade-off. Ferroelectric ceramics give coefficients 100–1000 times larger than quartz or AlN, but they have high permittivity, high loss, aging, temperature drift and hysteresis. Non-ferroelectric crystals give small coefficients but extraordinary stability and very low loss. **Large signals from ceramics; precise frequencies from crystals.**

Actuators: the converse effect

Worked example — a multilayer piezo stack. A PZT-5H stack contains 100 active layers, each 0.1 mm thick, electrically in parallel. At 150 V each layer sees

$$E_3 = \frac{150}{1.0 \times 10^{-4}} = 1.5 \times 10^6\ \mathrm{V/m} = 1.5\ \mathrm{kV/mm}$$

and strains by

$$S_3 = d_{33} E_3 = (593 \times 10^{-12})(1.5 \times 10^6) = 8.9 \times 10^{-4}$$

Each layer extends by $8.9 \times 10^{-4} \times 0.1\ \text{mm} = 89\ \text{nm}$, and the stack of 100 layers by 8.9 μm. The total stack length is 10 mm, so the displacement is about 0.09 % of the length. Multilayer construction is the key to useful actuators: it gives a high field (1.5 kV/mm) at a low drive voltage (150 V) by making the layers thin, just as an MLCC achieves high capacitance. The displacement is small, but it is fast (microseconds), very stiff (blocking forces of kilonewtons), and resolvable to sub-nanometre steps. That is why piezo stacks drive fuel injectors, scanning-probe microscopes, optical alignment stages and active vibration control. Larger strokes need mechanical amplification (bending bimorphs using d_{31}, flextensional levers).

Sensors: the direct effect

Worked example — charge and voltage from a force. A quartz force sensor with $d_{11} = 2.3\ \text{pC/N}$ loaded with 100 N generates $Q = d\,F = 230\ \text{pC}$, independent of the crystal's dimensions. Into a 100 pF capacitance (crystal plus cable) this is 2.3 V. Because the charge leaks away through finite insulation resistance, piezoelectric sensors measure changing forces only, and are read with charge amplifiers. The extremely high resistivity and low loss of quartz (Chapter 26) make its quasi-static response unusually good.

For voltage output, use the g coefficient. A PZT-5A disc ($g_{33} = d_{33}/\varepsilon_{33}^{T} = 374 \times 10^{-12}/(1700 \times 8.854 \times 10^{-12}) = 24.8 \times 10^{-3}$ V m/N) of area 1 cm^2 and thickness 1 mm under a 10 N force (stress $T = 10^5$ Pa) develops an open-circuit voltage

$$V = g_{33}\,T\,t = (24.8 \times 10^{-3})(10^{5})(10^{-3}) = 2.5\ \text{V}$$

PVDF film has a d coefficient more than ten times smaller than PZT, but its permittivity is about 150 times smaller, so its g_{31} (≈ 0.22 V m/N) is about ten times larger than that of PZT. It is also flexible, tough and acoustically matched to water and tissue. That makes PVDF the material for hydrophones, contact microphones and wearable pressure sensors, despite its weak coupling.

Resonators and filters

A piezoelectric plate driven at its mechanical resonance shows a sharp electrical resonance, with a quality factor set by the material's mechanical loss. This turns a piece of crystal into a frequency standard or a filter.

- **Quartz crystal oscillators** use thickness-shear vibration of thin plates. The resonance frequency is $f = N/t$, with frequency constant $N = 1.661\ \text{MHz mm}$ for the AT cut, so a 10 MHz crystal is 0.166 mm thick. The AT cut is chosen because, at that orientation, the temperature coefficients of the elastic constants and the thermal expansion cancel to first order; the frequency stays within a few parts per million over the commercial temperature range. Quartz has a mechanical Q of 10^5–10^6. No other material combines that with such low drift.
- **Surface acoustic wave (SAW) filters** use interdigitated electrodes on $LiNbO_3$ or $LiTaO_3$ to launch and detect surface waves at velocities of about 3500–4000 m/s. Electrode pitch sets the frequency; SAW dominates below about 2 GHz.
- **Bulk acoustic wave (BAW, FBAR) filters** use a thin AlN or ScAlN film, about 0.5–2 μm thick, between electrodes, suspended or acoustically isolated, resonating in thickness mode at 1.5–10 GHz. A smartphone contains dozens. The bandwidth achievable is proportional to k_t^2, which is why scandium is alloyed into AlN for the wider 5G bands: Sc softens the lattice and raises the piezoelectric response.

27.4 Pyroelectric materials

The effect

In a polar material the spontaneous polarization depends on temperature. A temperature change ΔT changes P_s and therefore the bound surface charge. If electrodes are connected through an external circuit, a current flows:

$$\Delta P_s = p\,\Delta T, \qquad i = p\,A\,\frac{dT}{dt}$$

where $p = dP_s/dT$ is the **pyroelectric coefficient** (units $\text{C m}^{-2}\,\text{K}^{-1}$) and A the electrode area. Like piezoelectric sensors, pyroelectric detectors respond to change only: a constant temperature gives no signal, because the bound charge is eventually neutralized by leakage. That is not a limitation for their main applications, which detect moving or modulated infrared sources.

Part of the measured coefficient is "secondary": thermal expansion strains the crystal, which then polarizes piezoelectrically. Since every pyroelectric is piezoelectric, vibration and acoustic noise also produce signals, and detector designs must reject them (usually with a pair of opposite-polarity elements).

> **Status of the principle: pyroelectric response $i = pA\,dT/dt$**
> *Derived* from the temperature dependence of P_s for a uniformly heated element. The value of p is a measured material property. Non-uniform heating, secondary piezoelectric contributions and leakage modify the response.

Figures of merit

For a thin detector element absorbing radiation, the temperature change per unit absorbed power is set by its volumetric heat capacity c' (J m^{-3} K^{-1}). The current responsivity is then proportional to p/c'; the voltage responsivity across the element's own capacitance to

$$F_V = \frac{p}{c'\,\varepsilon_0 \varepsilon_r}$$

A good voltage-mode pyroelectric has a large p, a small heat capacity and a **small** permittivity. This is the opposite of what a capacitor or an actuator needs, and it reorders the materials:

Material	p (10^{-4} C m^{-2} K^{-1})	ε_r	c' (MJ m^{-3} K^{-1})	F_V (m^2/C)	Comments
TGS (triglycine sulphate)	3.5	40	2.5	0.40	Best F_V, but $T_C = 49$ °C and water-soluble
$LiTaO_3$	2.3	47	3.2	0.17	Stable to 600 °C; standard in PIR and power meters
PVDF	0.27	12	2.4	0.11	Cheap, large-area, flexible
Modified PZT ceramic	3.8	290	2.5	0.06	Robust, cheap; high permittivity limits F_V
$BaTiO_3$ ceramic	2.0	about 1200	2.5	about 0.008	Large p, but permittivity much too high

Worked example — the signal from a motion detector. A $LiTaO_3$ element of area 1 mm^2 warms at 0.1 K/s as a person crosses its field of view (focused by a Fresnel lens). The current is

$$i = pA\frac{dT}{dt} = (2.3 \times 10^{-4})(1.0 \times 10^{-6})(0.1) = 2.3 \times 10^{-11}\ \text{A} = 23\ \text{pA}$$

A current this small is converted to a voltage by a JFET source follower with a gate resistor of tens of gigohms, integrated in the detector can. The element must therefore be an excellent insulator ($\rho_v > 10^{12}\ \Omega\,\text{m}$, Chapter 26), and thin, typically 20–50 μm, to minimize heat capacity.

Uncooled thermal imaging arrays use either pyroelectric materials (BST operated near its transition, where p is largest) or, more commonly today, resistive bolometers (vanadium oxide, amorphous silicon) whose resistance changes with temperature (Chapter 25's TCR, used deliberately: VO_x films have TCRs of about -2 % per kelvin). Infrared detector materials in general are taken up in Chapter 35.

27.5 Electrostrictive materials

A strain proportional to the square of the field

Every dielectric deforms in a field, because polarization changes the forces between ions. The lowest-order effect allowed in all materials, with or without a centre of symmetry, is quadratic:

$$S = Q\,P^2 \approx M\,E^2$$

Q (units m^4/C^2) is the **electrostrictive coefficient**. Since the strain depends on P^2, reversing the field gives the same strain: an electrostrictor always expands (or always contracts) along the field, whatever its sign. In ordinary dielectrics the effect is tiny, because P is small. It becomes useful only in materials with very large polarization.

> **Status of the principle: quadratic electrostriction $S = QP^2$**
> *Approximate.* It is the leading term allowed by symmetry in all materials. Q is nearly independent of temperature and composition in a given structure family, so writing the strain in terms of P rather than E is more accurate. Higher-order terms and saturation of P appear at high field.

Piezoelectricity of a ferroelectric is biased electrostriction

Write the polarization as a large static part plus a small signal, $P = P_0 + \delta P$. Then

$$S = Q(P_0 + \delta P)^2 \approx QP_0^2 + 2QP_0\,\delta P$$

The small-signal strain is linear in $\delta P = \varepsilon_0\varepsilon_r\,\delta E$, with an effective piezoelectric coefficient

$$d_{\text{eff}} = 2\,Q\,\varepsilon_0\varepsilon_r\,P_0$$

A material with a built-in polarization behaves as a piezoelectric. In a poled ferroelectric, P_0 is the remanent polarization. In a paraelectric or relaxor, P_0 is induced by a d.c. bias, and the piezoelectric response can be switched on and off with the bias.

Worked example — where barium titanate's piezoelectricity comes from. For single-crystal $BaTiO_3$ at room temperature, $Q_{11} \approx 0.11\ \text{m}^4/\text{C}^2$, $P_s = 0.26\ \text{C/m}^2$ and $\varepsilon_{33}/\varepsilon_0 \approx 168$ along the polar axis. Then

$$d_{33} \approx 2 \times 0.11 \times (8.854 \times 10^{-12}) \times 168 \times 0.26 = 8.5 \times 10^{-11}\ \text{C/N} = 85\ \text{pC/N}$$

The measured single-crystal value is about 86–90 pC/N. The piezoelectric coefficient of a ferroelectric is not a separate property: it is electrostriction about the spontaneous polarization, amplified by the permittivity.

The same physics produces the "singing" MLCC of the opening paragraph. A $BaTiO_3$ capacitor under d.c. bias has an induced polarization, so an a.c. ripple produces a strain at the ripple frequency. When the ripple falls in the audio range (light-load burst modes of converters), the capacitor shakes the circuit board at frequencies people can hear. The fixes are mechanical (small cases, soldering arrangements, interposers) or materials-based (CoG or polymer-film capacitors, which have no ferroelectric response).

Relaxor ferroelectrics

Lead magnesium niobate, $Pb(Mg_{1/3}Nb_{2/3})O_3$ (PMN), is the classic **relaxor**. Mg^{2+} and Nb^{5+} share the B site in a nanoscale, partly disordered arrangement. Instead of a single sharp transition, PMN contains **polar nanoregions** that grow and slow down on cooling. The permittivity shows a broad maximum near -10 °C, very large (ε_r up to about 20,000) and frequency-dependent: the peak shifts to higher temperature at higher measurement frequency, following a Vogel–Fulcher law rather than the Curie–Weiss law.

Just above its permittivity peak, PMN has an enormous induced polarization but almost no remanence. Its strain is large ($S \approx 0.1$ % at about 1–2 kV/mm), quadratic and nearly hysteresis-free, and it does not need poling or show aging of a poled state. With $Q \approx 0.025\ \text{m}^4/\text{C}^2$ and an induced $P = 0.2\ \text{C/m}^2$, $S = QP^2 = 0.025 \times 0.04 = 1.0 \times 10^{-3}$. That makes PMN-based ceramics the material for precision positioning where hysteresis is unacceptable, such as deformable mirrors for adaptive optics. Its disadvantages are strong temperature dependence (the effect is tied to the region near the permittivity peak) and high capacitance, hence high drive current. Solid solutions of relaxors with $PbTiO_3$ (PMN–PT, PZN–PT) grown as single crystals near their MPB give the record piezoelectric coefficients of Section 27.3.

Dielectric elastomers: electrostriction by Maxwell stress

Soft polymers offer a different route to large field-induced strain. In a **dielectric elastomer actuator**, a soft film (silicone, acrylic) with compliant electrodes is squeezed by the electrostatic attraction of its electrodes. The effective pressure is

$$p = \varepsilon_0\varepsilon_r E^2$$

which is again quadratic in field. For silicone with $\varepsilon_r = 3$ at 50 kV/mm, $p = 8.854 \times 10^{-12} \times 3 \times (5 \times 10^7)^2 = 6.6 \times 10^4$ Pa. With a modulus of about 1 MPa, that is a thickness strain of several per cent, and area strains of tens of per cent are achieved in pre-stretched films. The penalty is kilovolt drive and the electromechanical instability of Section 26.5, which these devices operate close to. They are used in soft robotics, haptics and tunable optics.

27.6 High-permittivity dielectrics

Why permittivity matters, and why it is not enough

The capacitance of a parallel-plate capacitor, $C = \varepsilon_0 \varepsilon_r A/d$, can be raised by increasing area, reducing thickness, or increasing permittivity. Area costs volume and chip space; thickness is limited by breakdown and, for nanometre films, by tunnelling. Permittivity is the material lever, and the ferroelectric perovskites offer values 1000 times those of polymers. But capacitance is not the only figure of merit. The energy a dielectric can store per unit volume is

$$w = \tfrac{1}{2}\varepsilon_0 \varepsilon_r E_b^2$$

and is limited by the breakdown field E_b as much as by ε_r. For biaxially oriented polypropylene ($\varepsilon_r = 2.2$, usable field about 600 kV/mm) the formula gives $w = \tfrac{1}{2}(8.854 \times 10^{-12})(2.2)(6 \times 10^8)^2 = 3.5\ \text{MJ/m}^3 = 3.5\ \text{J/cm}^3$. For a $BaTiO_3$ ceramic at $\varepsilon_r = 2000$ but a usable field of about 5 kV/mm, $w = \tfrac{1}{2}(8.854 \times 10^{-12})(2000)(5 \times 10^6)^2 = 0.22\ \text{J/cm}^3$, and less in practice, because its permittivity falls with field. **High permittivity gives high capacitance per volume at low voltage; high breakdown strength gives high energy per volume at high voltage.** That is why MLCCs dominate decoupling at a few volts, and polypropylene film capacitors dominate d.c. links and pulse power at hundreds of volts and above.

MLCC dielectrics

An MLCC is a stack of hundreds of ceramic layers, each 0.5–10 μm thick, interleaved with Ni (or Ag–Pd) electrodes, co-fired into a monolithic block (Chapter 25, Section 25.9). With N active layers of thickness d and overlap area A,

$$C = N\,\frac{\varepsilon_0 \varepsilon_r A}{d}$$

Worked example — an MLCC and its bias derating. A 1206-size capacitor has $N = 100$ active layers of $BaTiO_3$-based X7R dielectric with $\varepsilon_r = 3000$, thickness $d = 5\ \mu$m, and an electrode overlap of $2.0 \times 1.0\ \text{mm} = 2.0 \times 10^{-6}\ \text{m}^2$:

$$C = 100 \times \frac{(8.854 \times 10^{-12})(3000)(2.0 \times 10^{-6})}{5 \times 10^{-6}} = 1.06 \times 10^{-6}\ \text{F}$$

about 1 μF. At its 16 V rating each layer sees $16\ \text{V}/5\ \mu\text{m} = 3.2\ \text{kV/mm}$, several times the coercive field of $BaTiO_3$. At that field the domains are clamped in the field direction and can no longer contribute to the small-signal response, and the polarization approaches saturation on a flattening P–E curve. The small-signal permittivity, $\partial D/\partial E$, falls, typically by 30–60 % for X7R at rated voltage, and more for thinner layers and higher-permittivity grades. That is the opening problem. A smaller case with thinner layers, at the same voltage, has a higher field per layer and loses more.

The temperature behaviour of $BaTiO_3$ is tamed by microstructure. Additions of rare-earth oxides, MgO and others diffuse only partway into each grain during sintering, forming a **core–shell** grain: a ferroelectric core with a peak near 125 °C and a shell of graded composition with peaks spread over lower temperatures. The sum is a flattened $\varepsilon_r(T)$ (Figure 27.3, dashed curve). EIA codes specify the outcome:

Class and code	Temperature range (°C)	Capacitance change	Typical ε_r	Material basis
Class I, C0G (NP0)	−55 to +125	0 ± 30 ppm/K	20–100	$CaZrO_3$, (Ca,Sr)(Zr,Ti)O_3, TiO_2-based; paraelectric
Class II, X7R	−55 to +125	±15 %	2000–4000	Core–shell $BaTiO_3$
Class II, X5R	−55 to +85	±15 %	3000–5000	$BaTiO_3$, thin layers
Class II, Y5V	−30 to +85	+22 % / −82 %	10,000–18,000	$BaTiO_3$ with shifted Curie peak (e.g. Zr, Sn additions)

Class I dielectrics are linear, stable, low-loss and free of aging, bias effects and acoustic noise, which suits them to filters, oscillators and timing. Class II dielectrics trade all of those for 30–500 times more capacitance per volume, which suits them to decoupling and bulk filtering. Class II parts also **age**, losing about 1–3 % of capacitance per decade of time for X7R (more for

Y5V), measured from their last heating above T_C (for example, during soldering). Datasheet values are referred to a stated age, usually 1000 h.

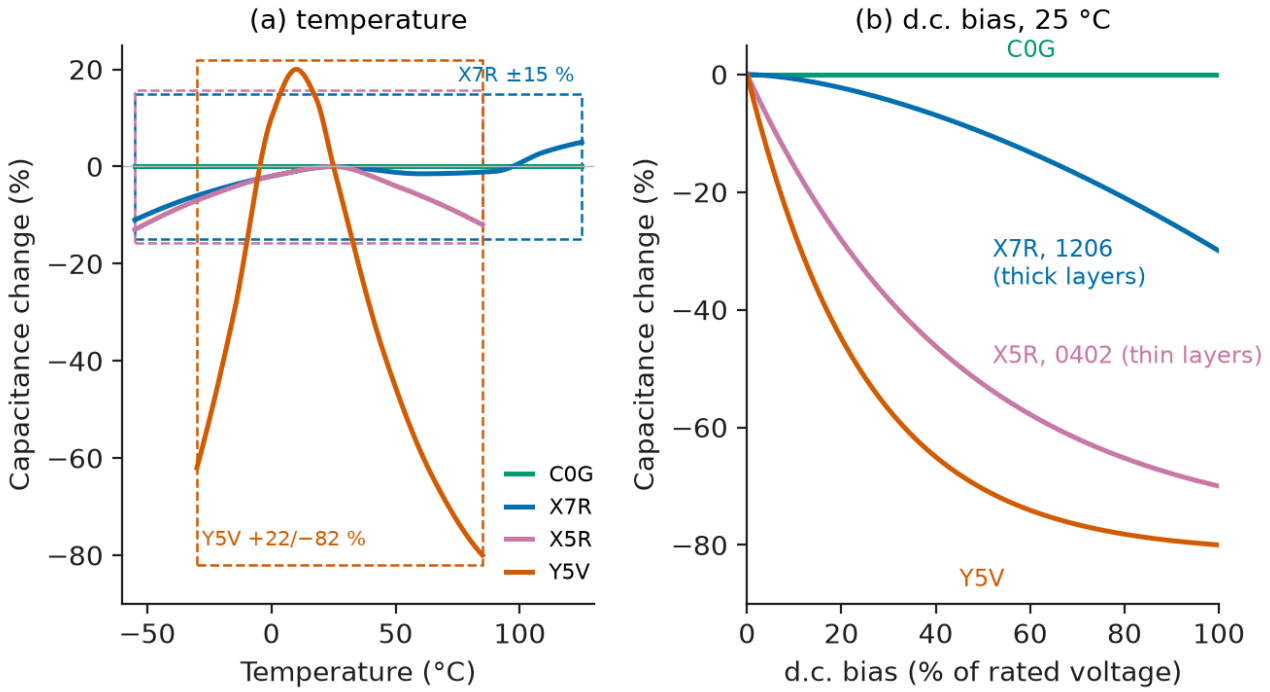


Figure 27.4 — MLCC dielectric classes: capacitance change with temperature and with d.c. bias.

Figure 27.4 compares the classes against temperature and bias. Choosing an MLCC is therefore a materials decision: the capacitance on the label is the zero-bias, room-temperature, fresh value of a ferroelectric.

High-k gate and memory dielectrics

In a MOS transistor the gate dielectric must provide high capacitance per area, $\varepsilon_0\varepsilon_r/t$, to control the channel. With SiO_2 ($\varepsilon_r = 3.9$), the thickness needed by the 2000s fell below 1.5 nm, where direct tunnelling currents became intolerable. The solution was a thicker film of higher permittivity. Its performance is quoted as an **equivalent oxide thickness**

$$\mathrm{EOT} = t_{\text{high-k}}\,\frac{3.9}{\varepsilon_{r,\text{high-k}}}$$

the SiO_2 thickness that would give the same capacitance.

Worked example — EOT of a hafnia gate stack. A 3.0 nm HfO_2 film with $\varepsilon_r = 20$ has

$$\mathrm{EOT} = 3.0 \times \frac{3.9}{20} = 0.59\ \text{nm}$$

The physical film is five times thicker than an SiO_2 film of equal capacitance, and the tunnelling current through it is lower by many orders of magnitude, because tunnelling decreases exponentially with thickness. In practice an unavoidable interfacial SiO_2 layer of about 0.5 nm adds to the EOT, so the full stack is closer to 1 nm EOT.

Permittivity cannot simply be pushed up. High permittivity in oxides comes from soft, highly polarizable ionic bonds, and those materials tend to have smaller band gaps. A smaller band gap means lower barriers to electron injection and higher leakage (band ideas, Chapter 31).

Dielectric	ε_r	Band gap (eV)	Comment
SiO_2	3.9	9	Reference; excellent interface with Si
Si_3N_4	7	5.3	Diffusion barrier; charge-trap memory
Al_2O_3	9	8.8	Good barrier, ALD-friendly; moderate ε_r
HfO_2	20–25	5.7–6	Standard high-k gate dielectric since 2007
ZrO_2	25	5.8	DRAM capacitors (ZrO_2/Al_2O_3 stacks)
La_2O_3	about 30	6	Threshold-voltage tuning additive
Ta_2O_5	22	4.4	Too small a conduction-band offset for gates
TiO_2 (rutile)	80–170	3.0–3.5	Leaky on Si; DRAM capacitor research
$SrTiO_3$	300 (film lower)	3.2	Very high ε_r, small gap

HfO_2 won because it combines a permittivity of about 20 with a band gap near 6 eV, thermodynamic stability in contact with silicon, and deposition by atomic layer deposition. Its interface with silicon, its traps, and its reliability are material problems treated in Chapter 34.

27.7 Dielectric resonators

A microwave cavity made of ceramic

A block of high-permittivity, low-loss ceramic acts as a resonator at microwave frequencies, because electromagnetic waves inside it are trapped by reflection at its surfaces, just as in a metal cavity. The wavelength inside is shorter by $\sqrt{\varepsilon_r}$, so a resonator of permittivity ε_r is about $\sqrt{\varepsilon_r}$ times smaller than an air-filled cavity. **Dielectric resonators** (DRs) are used in the filters and oscillators of cellular base stations, satellite transponders, radar and microwave links, where low loss, small size and temperature stability are all needed.

For the commonly used $TE_{01\delta}$ mode of a cylindrical puck of radius a and height L, an empirical formula (Kajfez and Guillon) gives

$$f_{\mathrm{GHz}} \approx \frac{34}{a_{\mathrm{mm}}\sqrt{\varepsilon_r}}\left(\frac{a}{L} + 3.45\right)$$

valid to about 2 % for $0.5 < a/L < 2$ and $30 < \varepsilon_r < 50$.

> **Status of the principle: resonant-frequency formula for the $TE_{01\delta}$ puck**
> *Empirical.* It is a fit to numerical solutions of Maxwell's equations for an isolated puck. Shields, supports and tuning screws shift the frequency by several per cent, so final designs use electromagnetic simulation.

Worked example — sizing a resonator at 2 GHz. With $a/L = 1$ and $\varepsilon_r = 37$ (a zirconium tin titanate):

$$a = \frac{34 \times (1 + 3.45)}{2 \times \sqrt{37}} = \frac{151.3}{12.17} = 12.4\ \mathrm{mm}$$

so the puck is about 25 mm in diameter and 12 mm high, against a free-space wavelength of 150 mm. With $\varepsilon_r = 90$, the same formula (now outside its fitted range, so only indicative) gives $a \approx 8$ mm; with alumina ($\varepsilon_r = 10$), about 24 mm. The size scales as $1/\sqrt{\varepsilon_r}$.

The three material requirements

A DR material must satisfy three requirements simultaneously:

1. **High permittivity**, for small size: ε_r 20–100.

2. **Low loss**, for a sharp resonance. Because the loss of the best ceramics comes mainly from anharmonic interactions of the microwave field with lattice vibrations, $\tan\delta$ grows roughly in proportion to frequency, and the product $Q \times f$ (with $Q = 1/\tan\delta$) is roughly constant for a given material. It is quoted in GHz. A material with $Q \times f = 40{,}000$ GHz has $Q = 20{,}000$ ($\tan\delta = 5 \times 10^{-5}$) at 2 GHz. Impurities, porosity, secondary phases and grain boundaries add extrinsic loss, so dense, pure, well-ordered ceramics are essential.
3. **Near-zero temperature coefficient of resonant frequency** τ_f. Since $f \propto 1/(L\sqrt{\varepsilon_r})$, differentiating with respect to temperature gives

$$\tau_f = \frac{1}{f}\frac{df}{dT} = -\alpha_L - \tfrac{1}{2}\tau_\varepsilon$$

where α_L is the linear thermal expansion coefficient and $\tau_\varepsilon = (1/\varepsilon_r)\, d\varepsilon_r/dT$. A filter channel only a few megahertz wide at 2 GHz cannot drift by more than a few ppm/K. Most high-permittivity materials have a strongly negative τ_ε (their soft modes stiffen on heating), giving large positive τ_f. TiO_2, for example, has $\tau_f \approx +450$ ppm/K. Practical materials are solid solutions or composites that mix phases of opposite τ_f to bring it to zero.

> **Status of the principle: constant $Q \times f$**
> *Empirical.* It reflects intrinsic phonon-limited loss, for which $\tan\delta \propto f$ in the microwave range. It fails where extrinsic losses (defects, impurities, grain boundaries) dominate, and near any relaxation or resonance.

Materials

There is a clear trade-off between permittivity and $Q \times f$ (Figure 27.5). Higher permittivity comes from softer lattice modes, which are also more anharmonic, so loss rises.

Material	ε_r	$Q \times f$ (GHz)	τ_f (ppm/K)	Typical use
Al_2O_3 (high purity)	10	300,000–500,000	about −60	Low-loss substrates, high-frequency resonators
$Ba(Mg_{1/3}Ta_{2/3})O_3$ (BMT)	24–25	250,000–400,000	0 to +5	Satellite and high-frequency filters
$Ba(Zn_{1/3}Ta_{2/3})O_3$ (BZT)	29–30	150,000–200,000	0 to +2	Base-station filters
$(Mg,Ca)TiO_3$	20–21	about 55,000	about 0	Low-cost resonators, GPS antennas
$(Zr,Sn)TiO_4$	37–38	40,000–50,000	about 0	Base-station filters, oscillators
$Ba_2Ti_9O_{20}$	39–40	32,000–40,000	+2	Classic filter material
$CaTiO_3$–$NdAlO_3$	43–45	about 45,000	about 0	Base-station filters
Ba–Nd–Ti–O (tungsten-bronze-type)	80–90	5,000–10,000	tunable to 0	Compact low-frequency filters
TiO_2 (rutile)	100	40,000–50,000	+450	End member for τ_f compensation

The complex perovskites BZT and BMT achieve their very high $Q \times f$ through **B-site ordering**: long annealing arranges the Zn (or Mg) and Ta ions on alternating (111) planes, a periodicity that reduces anharmonic scattering. This is Section 25.4's lesson about disorder in yet another form: ordering removes a source of loss, here for phonons rather than electrons.

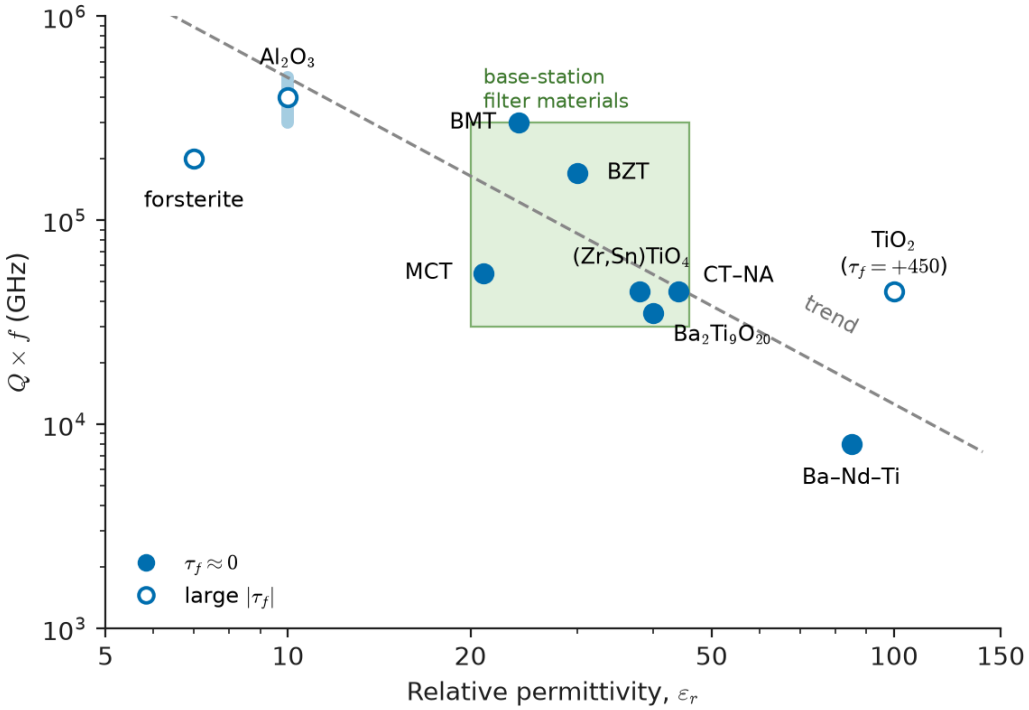


Figure 27.5 — Microwave dielectric ceramics: the trade-off between permittivity and the quality factor product Q x f.

27.8 Electro-optic materials

Field-controlled refractive index

In a non-centrosymmetric crystal, an applied field changes the refractive index linearly: the **Pockels effect**. In any material, including liquids and glasses, there is also a quadratic change: the **Kerr effect**. The Pockels effect is written in terms of the optical impermeability $1/n^2$:

$$\Delta\left(\frac{1}{n^2}\right) = r\,E \quad \Longrightarrow \quad \Delta n \approx -\tfrac{1}{2} n^3\, r\, E$$

and the Kerr effect as $\Delta n = \lambda K E^2$, with K the Kerr constant. The Pockels coefficient r (units $\mathrm{m/V}$, usually quoted in pm/V) is a third-rank tensor with the same symmetry restrictions as the piezoelectric d: only non-centrosymmetric materials have it. The physical origin is related too. The field displaces ions and distorts electron clouds, which changes the electronic polarizability at optical frequencies.

> **Status of the principle: Pockels and Kerr effects**
> *Approximate.* They are the first two terms of an expansion of the optical impermeability in the applied field, valid for fields small compared with internal atomic fields. The Pockels coefficient includes electronic, ionic and (at low frequency) acoustic contributions, so its value depends on whether the crystal is mechanically clamped (above its acoustic resonances) or free.

Modulators

A refractive-index change becomes an optical phase change over a length L:

$$\Delta\phi = \frac{2\pi}{\lambda}\,\Delta n\, L$$

In a waveguide modulator with electrode gap d (so $E \approx V/d$) and an overlap factor Γ between optical mode and electric field, the voltage that produces a phase change of π is

$$V_\pi = \frac{\lambda\, d}{n^3\, r\, \Gamma\, L}$$

Placing the phase modulator in one or both arms of a Mach–Zehnder interferometer converts phase to intensity. V_π, or the product $V_\pi L$, is the figure of merit that the material sets through $n^3 r$.

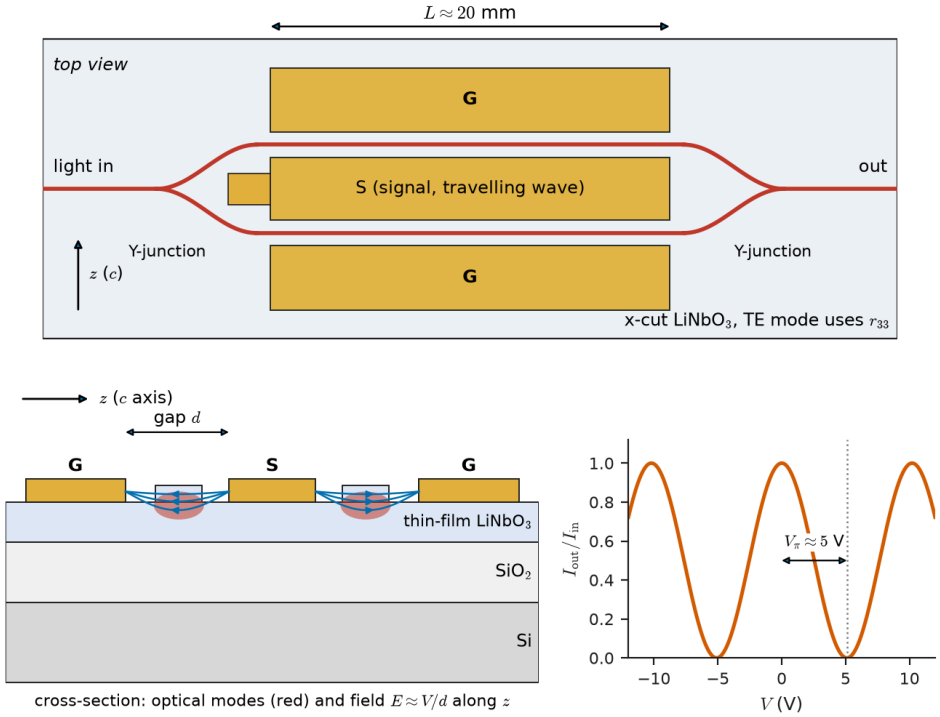


Figure 27.6 — Electro-optic Mach–Zehnder modulator in lithium niobate: top view and cross-section.

Worked example — half-wave voltage of a lithium niobate modulator. Lithium niobate has $n_e = 2.138$ and $r_{33} = 30.8$ pm/V at $\lambda = 1.55$ μm. For an electrode gap $d = 10$ μm, length $L = 20$ mm and overlap factor $\Gamma = 0.5$:

$$n_e^3 r_{33} = (2.138)^3 \times 30.8 \times 10^{-12} = 9.77 \times 30.8 \times 10^{-12} = 3.01 \times 10^{-10}\ \text{m/V}$$

$$V_\pi = \frac{(1.55 \times 10^{-6})(10 \times 10^{-6})}{(3.01 \times 10^{-10})(0.5)(0.020)} = \frac{1.55 \times 10^{-11}}{3.01 \times 10^{-12}} = 5.1\ \text{V}$$

so $V_\pi L \approx 10$ V cm, typical of conventional titanium-diffused $LiNbO_3$ modulators. Driving both arms in push–pull halves the voltage to about 2.6 V. Modern **thin-film lithium niobate** (a sub-micrometre $LiNbO_3$ layer bonded on SiO_2/Si) confines the light more tightly, so the gap can shrink to a few micrometres with a better overlap. $V_\pi L$ drops to about 2 V cm, and bandwidths exceed 100 GHz. The material is unchanged; the geometry lets it be used more effectively.

Materials

Material	Effect	Key coefficient	n	Use
$LiNbO_3$	Pockels	$r_{33} = 30.8$ pm/V	2.14–2.21	Telecom modulators; thin-film photonics
$LiTaO_3$	Pockels	$r_{33} \approx 30$ pm/V	2.14–2.18	Modulators with higher optical damage threshold
KH_2PO_4 (KDP), KD*P	Pockels (longitudinal)	$r_{63} \approx 10$–25 pm/V	1.5	Pockels cells, laser Q-switches (kV drive)
β-BaB_2O_4 (BBO)	Pockels	$r_{22} \approx 2.7$ pm/V	1.65	High-power, high-repetition Q-switches
GaAs, InP	Pockels	$r_{41} \approx 1.4$–1.5 pm/V	3.2–3.4	Integrated with III–V lasers (large n^3 compensates)
$BaTiO_3$ (thin film on Si)	Pockels	Hundreds of pm/V	2.3	Emerging silicon-photonics modulators
Electro-optic polymers (poled chromophores)	Pockels	$r_{33} = 100$–300 pm/V	1.6–1.8	Very low V_π; thermal stability limits
PLZT (9/65/35) ceramic	Kerr (quadratic)	Large quadratic coefficient	2.5	Shutters, light valves

The usable figure of merit is $n^3 r$, not r alone. GaAs has a small r but a large n and can be integrated with lasers on the same chip. Material choice also depends on optical transparency at the working wavelength, on resistance to **photorefractive damage** (light-induced index changes from photoexcited carriers trapped at defects; reduced in $LiNbO_3$ by MgO doping), on d.c. drift (slow ionic redistribution under bias, a dielectric-relaxation problem from Chapter 26), and on compatibility with silicon processing. Silicon itself is centrosymmetric and has no Pockels effect; silicon modulators instead use the change in refractive index caused by changing the free-carrier density (plasma dispersion), a semiconductor effect treated in Chapter 35.

27.9 Applications in capacitors, actuators, sensors, and RF systems

The same handful of structural ideas, polar symmetry, domains, soft modes, ordering and defect chemistry, run through all the applications. The table maps each application to the material property that governs it and the materials that are chosen.

Application	Governing property	Preferred materials	Main material limitation
Decoupling and bulk MLCCs	High ε_r in thin layers	$BaTiO_3$ (X5R, X7R) with Ni electrodes	Bias derating, aging, acoustic noise
Precision, filter and timing capacitors	Stable, linear ε_r; low loss	CoG ceramics; PP, PPS films	Low capacitance per volume
High-voltage, pulse and d.c.-link capacitors	High E_b, self-healing	BOPP film; high-voltage ceramics	Temperature limit of PP (about 105 °C)
DRAM storage capacitors	High ε_r, low leakage at nm thickness	$ZrO_2/Al_2O_3/ZrO_2$ stacks	Band gap vs permittivity trade-off
Transistor gate dielectric	High ε_r, wide gap, stable on Si	HfO_2-based	Interface traps, threshold stability (Chapter 34)
Non-volatile ferroelectric memory	Switchable P_r, CMOS compatibility	PZT, SBT; doped HfO_2	Fatigue, imprint, wake-up
Positioning and injector actuators	Large d_{33}, multilayer form	Soft PZT stacks; PMN for hysteresis-free	Hysteresis, creep, depoling, Pb content
Ultrasound (medical, NDT, sonar)	High k, suitable acoustic impedance	PZT-5H, PMN–PT crystals; PZT–polymer 1–3 composites; PVDF	Acoustic mismatch to tissue and water
Force, pressure and vibration sensors	High g (voltage) or stable d (charge)	Quartz; PZT; PVDF; AlN MEMS	Static measurement impossible; temperature drift
Infrared motion sensors	High $p/(c'\varepsilon_r)$	$LiTaO_3$; PZT; PVDF	Microphonic noise; low signal currents
Frequency references	Very high Q, near-zero temperature coefficient	AT-cut quartz; MEMS with AlN	Aging, size
RF filters in handsets	k_t^2 (bandwidth), acoustic Q, CMOS-compatible	$LiTaO_3/LiNbO_3$ SAW; AlN, ScAlN BAW	Bandwidth vs Q; power handling
Base-station filters and oscillators	High $Q \times f$, $\tau_f \approx 0$, moderate ε_r	BZT, BMT, $(Zr,Sn)TiO_4$, CT–NA	Size at low frequencies; cost of Ta
Tunable RF capacitors, phase shifters	Field-dependent ε_r with low loss	Paraelectric $(Ba,Sr)TiO_3$ thin films	Loss rises with tunability
Optical modulators	Large n^3r, high bandwidth	$LiNbO_3$ (bulk and thin film), InP, EO polymers	Drift, photorefractive damage, integration
Energy harvesting	$d \cdot g$ product, fatigue resistance	PZT, PVDF, AlN (Chapter 40)	Low power density, fatigue

Three broad selection rules come out of this chapter:

1. **Large response versus stability.** Ferroelectric materials give responses hundreds of times larger than non-ferroelectric crystals, at the cost of hysteresis, aging, temperature drift, bias dependence and loss. Where precision matters (frequency control, metrology, filtering), non-ferroelectric materials or linear paraelectrics win. Where magnitude matters (capacitance per volume, actuator stroke, ultrasound sensitivity), ferroelectrics win.
2. **Figures of merit differ by application.** A capacitor wants high ε_r; a pyroelectric or voltage-mode sensor wants **low** ε_r; an energy-storage capacitor wants high $\varepsilon_r E_b^2$; a resonator wants high $Q \times f$ and $\tau_f = 0$. The "best" material differs in each case.
3. **Integration and regulation shape the choice.** Lead-based PZT and PMN dominate performance tables but face RoHS exemptions that may end; KNN, BNT and other lead-free ceramics are improving. AlN, ScAlN and ferroelectric HfO_2 have risen not because their coefficients are best but because they can be deposited in CMOS fabs. Material availability and processing (Chapters 17 and 50) matter as much as intrinsic properties.

The MLCC of the opening paragraph is now fully explained. Its $BaTiO_3$ dielectric gives hundreds of times more capacitance per volume than a polymer film because it is ferroelectric. The same ferroelectricity makes its domains clamp under 12 V bias

(capacitance loss), turns ripple into strain (audible noise), and lets its domain structure relax after soldering (aging). The designer's remedies are all materials remedies: a larger case or higher voltage rating (thicker layers, lower field per layer), a C0G or film capacitor where stability matters, or a polymer or tantalum capacitor where bulk capacitance under bias matters.

What remains open

Open question	Where it is answered
How do band gaps arise, and why do high-permittivity oxides have smaller gaps?	Chapter 31
What limits high-k gate dielectrics and ferroelectric HfO_2 at their interface with silicon (traps, threshold drift)?	Chapter 34
How do electro-optic, photonic and infrared detector materials fit into optoelectronics?	Chapter 35
How do magnetic domains, hysteresis and Curie temperatures compare with their ferroelectric counterparts?	Chapter 28
How do ferroelectric and piezoelectric properties change in nanostructures and ultrathin films?	Chapter 38
How are piezoelectric materials used to harvest energy, and what limits their power density?	Chapter 40
How are ceramic powders, MLCCs and piezoelectric films processed, sintered and deposited?	Chapters 17 and 50
How are ferroelectric, piezoelectric and dielectric properties measured?	Chapter 52
How do capacitors and functional dielectrics degrade and fail in service (cracking, fatigue, breakdown)?	Chapter 44
How are functional dielectrics selected against cost, regulation (lead content) and reliability?	Chapter 58

Next: Chapter 28 (Magnetic Material Behaviour) turns from electric to magnetic order, asking how electron spins in a solid line up into domains, and how domain walls, hysteresis and temperature decide the behaviour of every inductor, transformer and magnetic sensor.

Chapter 28 — Magnetic Material Behavior

Part VI · Thermal, Electrical, Dielectric and Magnetic Materials

A buck converter passes its room-temperature qualification without trouble. In the field, inside a sealed enclosure on a summer afternoon, it starts to fail: the inductor current, which should be a gentle triangle, develops sharp spikes at the top of every cycle, the MOSFETs run hot, and eventually one of them dies. Nothing in the circuit changed. The ferrite core of the inductor did. At 25 °C its saturation flux density was about 0.49 T; at 110 °C it is under 0.39 T. The peak flux that used to sit comfortably below the knee of the magnetization curve now reaches saturation, the permeability collapses towards that of air, and the inductance with it. The designer had used a magnetic material as if it were a fixed number — an inductance factor on a datasheet — when it is a population of atomic moments whose alignment fights thermal agitation and weakens long before a definite temperature, the Curie point, at which it vanishes altogether.

This chapter builds the materials picture an engineer needs to read a magnetic datasheet with understanding. Chemistry and physics explain where the atomic moments come from; this chapter takes them as given and asks what a solid does with them.

> **Why do some materials respond to a magnetic field a hundred thousand times more strongly than others, and why does that response disappear at a definite temperature?**
> **Why does a ferromagnet split into domains, and why does the way its domain walls move decide whether it becomes a transformer core or a permanent magnet?**

Concept	Engineering destination
B, H, M, χ, μ_r; demagnetizing field	Reading datasheets; gapped cores; effective permeability of real parts
Diamagnetism and paramagnetism, Curie law	Non-magnetic structural metals, stainless steels near sensors, MRI-compatible parts, cryogenic sensing
Ferromagnetism, exchange, Curie–Weiss law	Why only Fe, Co, Ni, some rare earths and some oxides make useful cores and magnets
Antiferromagnetism, Néel temperature	Pinning layers in GMR/TMR sensors and read heads
Ferrimagnetism in spinel ferrites	High-frequency cores, EMI suppressors, the whole ferrite industry (Chapter 29)
Saturation magnetization M_s	Maximum flux density of a core, hence core size and weight
Domains, anisotropy, magnetostriction	Permeability, transformer hum, stress sensitivity of cores
Domain-wall width and pinning	Soft versus hard behaviour; why grain size and purity matter
Hysteresis loop, B_r, H_c	Core loss per cycle; permanent-magnet strength
$M_s(T)$, T_C, superparamagnetism	Thermal derating of inductors and magnets; Curie-point thermostats; data-retention limits

Roadmap. Field quantities (28.1) → where moments come from (28.2) → weak magnetism: diamagnetism and paramagnetism (28.3–28.4) → cooperative magnetism: ferro-, antiferro- and ferrimagnetism (28.5–28.7) → saturation magnetization (28.8) → domains and walls (28.9–28.10) → wall motion and pinning (28.11) → hysteresis, remanence and coercivity (28.12–28.13) → temperature (28.14).

28.1 Magnetization, field and induction

Three vector fields

An engineer meets three magnetic vector fields, and most datasheet confusion comes from mixing them.

- The **magnetic field strength H** (unit A/m) is set by free currents. In a toroid of N turns carrying current I around a mean path length l, Ampère's law gives $H = NI/l$. It is what the circuit controls.

- The **magnetization** $\mathbf{M}$ (A/m) is the magnetic dipole moment per unit volume of the material: $\mathbf{M} = n\langle \mathbf{m} \rangle$ for n moments per cubic metre with average moment $\langle \mathbf{m} \rangle$ (each in A m²). It is what the material contributes.
- The **magnetic flux density** or induction $\mathbf{B}$ (tesla, T = Wb/m²) is what Faraday's law sees. It determines the voltage induced in a winding, $v = -N\, d\Phi/dt$ with $\Phi = \int \mathbf{B} \cdot d\mathbf{A}$, and the force on a conductor.

They are related by

$$\mathbf{B} = \mu_0 \left(\mathbf{H} + \mathbf{M} \right) = \mu_0 \mathbf{H} + \mathbf{J}, \qquad \mu_0 = 4\pi \times 10^{-7}\ \mathrm{H/m}$$

where $\mathbf{J} = \mu_0 \mathbf{M}$ is the **magnetic polarization**, measured in tesla. Magnet and steel datasheets quote J (or "intrinsic induction" B_i) because it isolates the material's contribution; when a datasheet quotes "saturation polarization $J_s = 2.03$ T" for a silicon steel it means $\mu_0 M_s$.

> **Status of the principle: $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$**
> *Fundamental.* It is the definition of $\mathbf{H}$ that makes Maxwell's equations in matter close: $\nabla \cdot \mathbf{B} = 0$ always, and $\nabla \times \mathbf{H} = \mathbf{J}_{\text{free}} + \partial \mathbf{D}/\partial t$, with all bound (atomic) currents absorbed into $\mathbf{M}$.

Susceptibility and permeability

For many materials, and for any material at small enough fields, $\mathbf{M}$ is proportional to $\mathbf{H}$:

$$\mathbf{M} = \chi \mathbf{H}, \qquad \mathbf{B} = \mu_0 (1 + \chi) \mathbf{H} = \mu_0 \mu_r \mathbf{H}, \qquad \mu_r = 1 + \chi$$

The dimensionless **volume susceptibility** χ is the material's response per unit applied field; the **relative permeability** μ_r is what goes into inductance formulas. Their ranges are enormous: $\chi \approx -10^{-5}$ for copper, $+2 \times 10^{-5}$ for aluminium, and μ_r up to 10^5 for annealed nickel–iron or nanocrystalline alloys. Three classes follow (Sections 28.3–28.7):

Class	Sign and size of χ	Temperature dependence	Examples
Diamagnetic	-10^{-6} to -10^{-4} (superconductor: -1)	Essentially none	Cu, Ag, Au, Bi, water, most polymers, Si
Paramagnetic	$+10^{-6}$ to $+10^{-2}$	Curie law $\chi \propto 1/T$ for local moments; nearly constant for metals	Al, Pt, O2, rare-earth salts, austenitic stainless steel
Ferro- and ferrimagnetic	10 to 10^6, field-dependent, hysteretic	Vanishes at the Curie temperature T_C	Fe, Ni, Co, electrical steels, ferrites, NdFeB
Antiferromagnetic	$+10^{-5}$ to 10^{-3}	Maximum at the Néel temperature T_N	MnO, NiO, Cr, IrMn

For ferromagnets "the" permeability is not one number, because M is a nonlinear, history-dependent function of H. Datasheets distinguish:

- **initial permeability** $\mu_i = \lim_{H \to 0} B/(\mu_0 H)$ on the virgin curve — relevant to small-signal transformers and EMI chokes;
- **maximum (amplitude) permeability** $\mu_{\max}$, the largest $B/\mu_0 H$ on the initial curve;
- **differential permeability** $\mu_{\text{diff}} = (1/\mu_0)\, dB/dH$ — the slope at the operating point;
- **incremental (reversible) permeability** μ_Δ, for a small AC ripple superposed on a DC bias — the quantity that sets the inductance of a converter choke carrying DC;
- **complex permeability** $\mu = \mu' - j\mu''$ under sinusoidal excitation, where μ'' measures loss (Chapter 29).

Older literature uses Gaussian (cgs) units, and magnet datasheets still do. The conversions in the next table appear on every magnet catalogue page.

Quantity	SI unit	cgs unit	Conversion
H	A/m	oersted (Oe)	1 Oe = $10^3/4\pi$ A/m = 79.58 A/m
B, J	T	gauss (G)	1 G = 10^{-4} T
M	A/m	emu/cm³	1 emu/cm³ = 10^3 A/m
χ (volume)	dimensionless	dimensionless	$\chi_{\text{SI}} = 4\pi\chi_{\text{cgs}}$
$(BH)_{\max}$	kJ/m³	MGOe	1 MGOe = 7.958 kJ/m³

The demagnetizing field

A magnetized body of finite size carries effective magnetic "poles" on its end faces, and these produce a field inside the body opposite to $\mathbf{M}$, the **demagnetizing field**

$$\mathbf{H}_d = -N\mathbf{M}$$

where the **demagnetizing factor** N (between 0 and 1, exactly computable for ellipsoids) depends only on shape: $N = 1/3$ for a sphere, $N \to 0$ for a long needle along its axis, $N = 1$ for a thin plate magnetized perpendicular to its plane, and $N \approx 0$ for a closed toroid. The field the material actually experiences is $H_{\text{int}} = H_{\text{app}} - NM$. For a high-permeability body this dominates: solving $M = \chi(H_{\text{app}} - NM)$ gives an apparent susceptibility $\chi/(1 + N\chi)$, which tends to $1/N$ as $\chi \to \infty$. A sphere of $\mu_r = 10\,000$ behaves as if $\mu_r \approx 4$. Shape, not material, then decides the response.

This is why magnetic cores are closed loops, and why an air gap — a deliberate, small demagnetizing feature — is such a powerful design knob. A core of path length l_e and material permeability μ_r with a gap l_g has an effective permeability

$$\mu_e \approx \frac{\mu_r}{1 + \mu_r l_g / l_e}$$

For $\mu_r = 2000$, $l_e = 100$ mm and $l_g = 1$ mm, $\mu_e = 2000/21 \approx 95$. The gap makes the inductance almost independent of the scatter and temperature drift of μ_r, and it stores most of the energy; Chapter 29 uses this when selecting inductor materials. The same shape effect makes a permanent magnet's operating point depend on its geometry (Section 28.13 and Chapter 29).

> **Status of the principle: demagnetizing factor $\mathbf{H}_d = -N\mathbf{M}$**
> *Derived* for uniformly magnetized ellipsoids, where it follows exactly from magnetostatics; *approximate* for other shapes, where N varies over the body and only an average value can be quoted.

28.2 Where atomic moments come from

> **Boundary.**
> The quantum origin of electron spin, orbital angular momentum, Hund's rules and the filling of d and f shells belongs to the Advanced Chemistry core (atomic structure) and to Engineering Physics. This section only collects the results the materials engineer uses.

An electron carries a spin magnetic moment of almost exactly one **Bohr magneton**,

$$\mu_B = \frac{e\hbar}{2m_e} = 9.274 \times 10^{-24}\ \text{A m}^2$$

and its orbital motion contributes a further moment proportional to its orbital angular momentum. In a filled shell all spins and orbits pair off and the net moment is zero. A net atomic moment therefore requires a partly filled shell: the 3d shell of the transition metals Cr–Cu and the 4f shell of the rare earths. Hund's first rule (maximize total spin) gives, for example, five unpaired d electrons and $5\mu_B$ for Fe^{3+} and Mn^{2+} (d^5), $4\mu_B$ for Fe^{2+} (d^6), $3\mu_B$ for Co^{2+}, $2\mu_B$ for Ni^{2+} and zero for Zn^{2+} (d^{10}).

Two features of the solid change these free-ion values.

1. **Orbital quenching.** In a crystal, the electrostatic field of the neighbouring ions (the crystal field) locks the 3d orbitals to the lattice, and their orbital moment largely disappears. 3d-ion moments are therefore close to spin-only values. The small residual orbital moment couples the spin direction to the crystal axes through spin–orbit coupling; this is the origin of **magnetocrystalline anisotropy** (Section 28.9). The 4f shell lies deep inside the atom, is shielded from the crystal field, and keeps its large orbital moment — which is why rare-earth compounds have anisotropies a hundred times larger than iron.
2. **Band formation.** In metals the 3d electrons are not localized but form bands (Chapter 31). Moments are then non-integer: $2.22\mu_B$ per Fe atom, $1.72\mu_B$ per Co, $0.61\mu_B$ per Ni. They measure the imbalance between spin-up and spin-down d-band populations.

Moments alone do not make a magnet. A gas of independent $2\mu_B$ moments at room temperature is only weakly paramagnetic (Section 28.4). Strong magnetism requires an interaction that aligns neighbouring moments against thermal agitation — the exchange interaction of Section 28.5.

28.3 Diamagnetic materials

Every material is diamagnetic. When a field is switched on, Lenz's law acting on the electron orbits induces a small change in orbital current that opposes the field. A classical calculation (Larmor precession of the orbits) gives

$$\chi_{\text{dia}} = -\frac{\mu_0 n e^2}{6 m_e} \sum_i \langle r_i^2 \rangle$$

summed over the electrons of each atom, with n atoms per unit volume and $\langle r_i^2 \rangle$ the mean-square orbit radius perpendicular to the field. For $n = 8.5 \times 10^{28}$ m⁻³ (copper) and $\langle r^2 \rangle = (0.1\ \text{nm})^2$, each electron contributes about -5×10^{-6}; summed over the few outer electrons with significant radius, $\chi \sim -10^{-5}$. The result is negative, small, and independent of temperature, because nothing thermal enters the derivation.

> **Status of the principle: Larmor diamagnetism**
> *Approximate.* The classical derivation gives the right sign and order of magnitude for closed-shell atoms and ions; it is masked whenever a material has any permanent moments, whose paramagnetism is usually larger.

Diamagnetism is the whole story only when there are no permanent moments: closed-shell ions (Na^+, Cl^-), covalent solids such as Si and diamond, noble metals in which the d shell is full (Cu, Ag, Au), most polymers, water and biological tissue. Bismuth and graphite are unusually strongly diamagnetic because their conduction electrons have very small effective masses and large orbits (Landau diamagnetism, Chapter 31's band picture). A superconductor expels flux almost completely: $\chi = -1$, perfect diamagnetism, a cooperative effect of an entirely different origin taken up in Chapter 30.

For an EEE engineer diamagnetism matters in three places: it makes copper and aluminium "non-magnetic" to the level that a current sensor near a busbar sees only the field of the current; it allows diamagnetic levitation demonstrations and pyrolytic-graphite position sensors; and it sets the background against which weak paramagnetic contamination is detected in precision instruments.

28.4 Paramagnetic materials

Independent moments in a field: the Langevin model

Suppose each atom carries a permanent moment m that does not interact with its neighbours. In a field $\mu_0 H$ (strictly the local field; the difference is negligible for weak magnetization), a moment at angle θ to the field has energy $U = -\mu_0 m H \cos\theta$. Thermal agitation randomizes orientations; Boltzmann statistics weights each by $e^{-U/k_B T}$. Averaging $\cos\theta$ over all directions of solid angle,

$$\langle \cos\theta \rangle = \frac{\int_0^\pi \cos\theta\, e^{x\cos\theta} \sin\theta\, d\theta}{\int_0^\pi e^{x\cos\theta} \sin\theta\, d\theta} = \coth x - \frac{1}{x} \equiv L(x), \qquad x = \frac{\mu_0 m H}{k_B T}$$

$L(x)$ is the **Langevin function** (Figure 28.1). The magnetization is $M = nmL(x)$. Two limits matter:

- **Strong field or low temperature** ($x \gg 1$): $L \to 1$ and $M \to M_s = nm$, all moments aligned.
- **Weak field or high temperature** ($x \ll 1$): expanding, $L(x) \approx x/3$.

The ratio x is small in practice. For $m = 5.92\mu_B$ (an Mn^{2+} ion) in a field of 1 T at 300 K, $x = mB/k_BT = 0.013$. Laboratory fields can saturate a paramagnet only at liquid-helium temperatures. In the linear regime,

$$M = \frac{n\mu_0 m^2}{3k_BT}H \quad \Longrightarrow \quad \chi = \frac{C}{T}, \qquad C = \frac{\mu_0 n m^2}{3k_B}$$

This is the **Curie law**, with **Curie constant** C (in kelvin, for the dimensionless SI susceptibility). Measuring $\chi(T)$ and plotting $1/\chi$ against T gives a straight line through the origin whose slope yields m: this is how effective moments of ions in solids are determined.

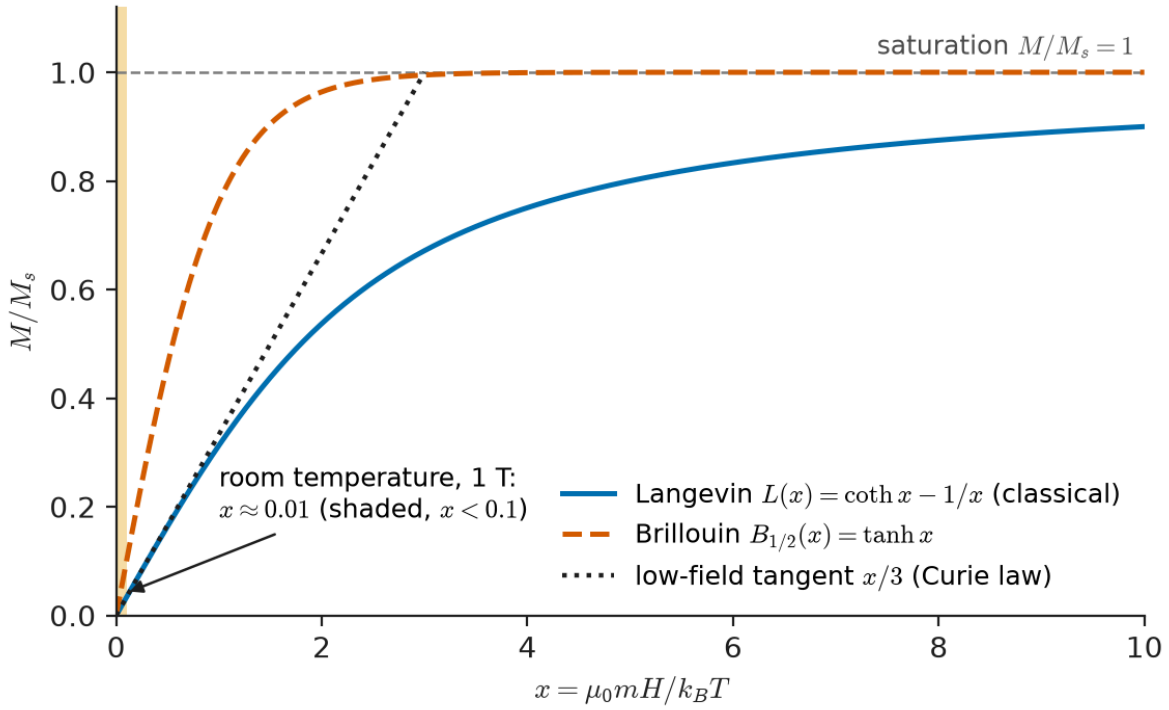


Figure 28.1 — The Langevin function L(x) (classical moments) and the Brillouin function for J = 1/2, with the low-field tangent x/3 that gives the Curie law.

> **Status of the principle: Curie law $\chi = C/T$**
> *Derived* from Boltzmann statistics for non-interacting permanent moments in the weak-field limit $\mu_0 mH \ll k_BT$. It fails at low temperature or high field (saturation), and whenever moments interact (Sections 28.5–28.7).

Quantum correction. Moments are quantized: a moment of total angular momentum quantum number J can take only $2J + 1$ orientations. Repeating the average over those orientations replaces $L(x)$ by the **Brillouin function** $B_J(x)$, which approaches the Langevin function as $J \to \infty$ and equals $\tanh x$ for $J = 1/2$. The Curie law survives, with m^2 replaced by $m_{\text{eff}}^2 = g^2 J(J+1)\mu_B^2$, where g is the Landé factor ($g = 2$ for pure spin). For Mn^{2+} ($J = S = 5/2$, $g = 2$), $m_{\text{eff}} = 2\sqrt{35/4}\,\mu_B = 5.92\mu_B$, which is the value measured in manganese salts.

Worked example — Curie-law susceptibility of a dilute paramagnet. A ceramic contains $n = 1.0 \times 10^{28}$ Mn^{2+} ions per cubic metre, each with $m_{\text{eff}} = 5.92\mu_B = 5.49 \times 10^{-23}$ A m², and the ions are far enough apart that they do not interact. The Curie constant is

$$C = \frac{\mu_0 n m^2}{3k_B} = \frac{(1.257 \times 10^{-6})(1.0 \times 10^{28})(5.49 \times 10^{-23})^2}{3(1.381 \times 10^{-23})} = 0.914 \text{ K}$$

At 300 K, $\chi = 0.914/300 = 3.0 \times 10^{-3}$; at 77 K (liquid nitrogen), $\chi = 1.19 \times 10^{-2}$. In a field $\mu_0 H = 1$ T at room temperature the magnetization is $M = \chi H = 3.0 \times 10^{-3} \times 7.96 \times 10^{5} = 2.4 \times 10^{3}$ A/m, about 0.4 % of the $M_s = nm = 5.5 \times 10^{5}$ A/m that full alignment would give. A paramagnet is a material with moments but without order.

Pauli paramagnetism of metals

Conduction electrons also carry spin moments, but they cannot all reorient freely: the Pauli principle allows only electrons within about $\mu_B B$ of the Fermi level to flip. The resulting **Pauli paramagnetism** is $\chi_P \approx \mu_0 \mu_B^2 g(E_F)$, where $g(E_F)$ is the density of states at the Fermi level (Chapter 31). It is small (about 10^{-5}) and almost independent of temperature, because the fraction of electrons able to respond is set by the Fermi distribution, not by $k_B T$ alone. Aluminium, sodium and magnesium are Pauli paramagnets. Transition metals such as Pt, Pd, Ti and W have high $g(E_F)$ from their d bands and are noticeably more paramagnetic. Palladium is close to the threshold for ferromagnetism (Section 28.5).

Representative values

Material	Volume susceptibility χ (SI, room temperature, typical)	Type
Bismuth	-1.7×10^{-4}	Diamagnetic
Pyrolytic graphite (field $\perp$ planes)	about -4×10^{-4}	Diamagnetic, strongly anisotropic
Copper	-9.6×10^{-6}	Diamagnetic
Silver	-2.4×10^{-5}	Diamagnetic
Gold	-3.4×10^{-5}	Diamagnetic
Water	-9.0×10^{-6}	Diamagnetic
Superconductor (Meissner state)	-1	Perfect diamagnet (Chapter 30)
Oxygen gas, 1 atm	$+1.9 \times 10^{-6}$	Paramagnetic (Curie)
Aluminium	$+2.2 \times 10^{-5}$	Pauli paramagnetic
Tungsten	$+7 \times 10^{-5}$	Paramagnetic
Titanium	$+1.8 \times 10^{-4}$	Paramagnetic
Platinum	$+2.7 \times 10^{-4}$	Paramagnetic
Austenitic stainless steel (304, annealed)	about $+3 \times 10^{-3}$ to 10^{-2}	Paramagnetic (can become partly ferromagnetic after cold work)
Manganese salt, ~10^{28} ions/m^3	about $+3 \times 10^{-3}$	Curie paramagnet

The stainless-steel row is a common engineering trap. Austenitic (fcc) 304 is paramagnetic, but cold working transforms part of it to bcc martensite, which is ferromagnetic. Screws or sensor housings specified as "non-magnetic stainless" can disturb a nearby Hall sensor or compass after forming. Grade 316 is more stable against this transformation.

28.5 Ferromagnetism

Spontaneous magnetization and the molecular field

Iron at room temperature holds a magnetization of 1.7×10^{6} A/m with no applied field at all, inside each domain. The Curie law offers no route to this: with $x \approx 0.01$ at 1 T, independent moments stay almost random. Pierre Weiss proposed in 1907 that each moment feels, besides the applied field, an internal **molecular field** proportional to the magnetization of its surroundings:

$$H_{\text{eff}} = H + \lambda M$$

with λ a dimensionless molecular-field constant. Substituting into the Curie law, $M = (C/T)(H + \lambda M)$, and solving for M:

$$\chi = \frac{M}{H} = \frac{C}{T - \lambda C} = \frac{C}{T - \theta}, \qquad \theta = \lambda C$$

This is the **Curie–Weiss law**. As T falls towards θ the susceptibility diverges: a vanishing field produces a finite magnetization. Below this temperature the material magnetizes spontaneously. In mean-field theory the ordering temperature, the **Curie temperature** T_C, equals θ. Above T_C the material is paramagnetic with an enhanced, Curie–Weiss susceptibility; below it, it is ferromagnetic (Figure 28.2b).

Below T_C the spontaneous magnetization follows from self-consistency: the moments are aligned by the field their own alignment creates. Setting $H = 0$,

$$\frac{M_s(T)}{M_s(0)} = L\left(\frac{\mu_0 m \lambda M_s(T)}{k_B T}\right) \quad \text{(or } B_J \text{ for quantized moments)}$$

which has a non-zero solution only for $T < T_C$. The solution, plotted in Figure 28.2a, is flat at low temperature and falls steeply to zero at T_C. When $M_s/M_s(0)$ is plotted against T/T_C, very different ferromagnets fall close to one universal curve. Section 28.14 returns to this curve and to where the mean-field theory fails.

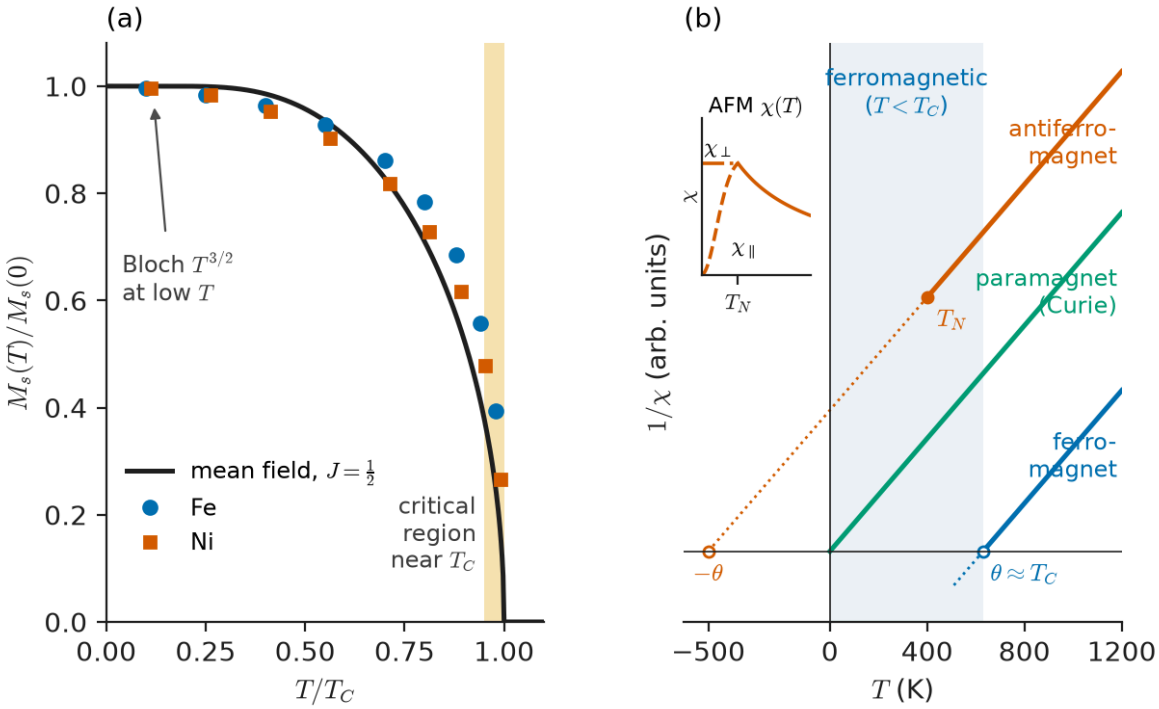


Figure 28.2 — (a) Reduced spontaneous magnetization versus reduced temperature for Fe and Ni with the mean-field curve; (b) inverse susceptibility versus temperature for a paramagnet, a ferromagnet and an antiferromagnet.

> **Status of the principle: Curie–Weiss law** $\chi = C/(T - \theta)$
> *Approximate.* It follows from the mean-field (molecular-field) model and describes the paramagnetic region well above T_C; close to T_C correlated fluctuations make the measured paramagnetic θ slightly larger than T_C (iron: $\theta \approx 1100$ K against $T_C = 1043$ K), and the critical behaviour deviates from mean-field predictions.

Worked example — how strong is the molecular field in iron? Iron has $T_C = 1043$ K and $m = 2.22\mu_B = 2.06 \times 10^{-23}$ A m² per atom. In the classical mean-field model, $T_C = \lambda C = \lambda \mu_0 n m^2 / 3k_B$, so the molecular field at full magnetization $M_s = nm$ is

$$B_W = \mu_0 \lambda M_s = \frac{3k_B T_C}{m} = \frac{3(1.381 \times 10^{-23})(1043)}{2.06 \times 10^{-23}} \approx 2.1 \times 10^3 \text{ T}$$

(For spin-½ moments the factor 3 is absent and $B_W \approx 700$ T.) Compare this with the actual magnetic dipole field produced at an atom by a neighbour at the nearest-neighbour distance $r = 0.248$ nm:

$$B_{\text{dip}} \approx \frac{\mu_0}{4\pi}\frac{m}{r^3} = 10^{-7} \times \frac{2.06 \times 10^{-23}}{(0.248 \times 10^{-9})^3} = 0.13\ \text{T}$$

and the corresponding energy $mB_{\text{dip}}/k_B \approx 0.2$ K. Magnetic dipole–dipole forces could order iron only below about a kelvin. The "molecular field" of several hundred to a few thousand tesla is not a magnetic field at all. It is the energetic preference for parallel spins created by electrostatics and the Pauli principle — the exchange interaction.

Exchange: the origin of the molecular field

> **Boundary.**
> The quantum-mechanical origin of exchange — the antisymmetry of the many-electron wavefunction, which makes the Coulomb energy depend on relative spin orientation — belongs to Engineering Physics and to the Advanced Chemistry core (the same physics as Hund's first rule and the triplet–singlet splitting of two-electron systems). Here it is used as a material parameter.

Two electrons with parallel spins must have an antisymmetric spatial wavefunction, which keeps them apart and lowers their Coulomb repulsion; with antiparallel spins they may overlap more. Depending on the orbitals involved and on the bonding, either alignment can win. The result is summarized by the Heisenberg form of the energy of two neighbouring spins,

$$U_{ij} = -2\mathcal{J}\,\mathbf{S}_i \cdot \mathbf{S}_j$$

with **exchange integral** $\mathcal{J}$ (an energy). $\mathcal{J} > 0$ favours parallel alignment (ferromagnetism); $\mathcal{J} < 0$ favours antiparallel (antiferromagnetism, ferrimagnetism). The mean-field constant λ is a lumped measure of $z\mathcal{J}$ over z neighbours, and in the mean-field approximation $k_B T_C$ is of order $z\mathcal{J}S(S+1)$, which is why T_C is a direct read-out of exchange strength.

Three materials-level rules of thumb follow:

1. **Direct exchange depends on spacing.** In the 3d metals the sign of $\mathcal{J}$ depends on the ratio of interatomic spacing to the radius of the 3d shell. The empirical **Bethe–Slater curve** places Fe, Co and Ni on the positive (ferromagnetic) side and Mn and Cr on the negative side. Lattice spacing and structure therefore matter: fcc iron (austenite, stable above 912 °C) is not ferromagnetic, and adding elements that expand the Mn lattice (as in Heusler alloys such as Cu_2MnAl) can make manganese ferromagnetic.
2. **Band ferromagnetism needs a high density of states at E_F.** In metals the itinerant **Stoner criterion** $I\,g(E_F) > 1$ (with I an exchange energy) says a spin-split band lowers the energy if the d-band density of states at the Fermi level is large enough. Fe, Co and Ni meet it; Pd and Pt nearly do. The fractional moments of Section 28.2 follow naturally from the split bands.
3. **In oxides, exchange runs through oxygen.** Metal ions in ferrites and other oxides are too far apart for direct overlap; they couple through the intervening O^{2-} ion (**superexchange**). For a metal–oxygen–metal angle near 180° and half-filled d shells it is strongly antiparallel. This is why most magnetic oxides are antiferromagnetic or ferrimagnetic, not ferromagnetic (Sections 28.6–28.7).

> **Status of the principle: Heisenberg exchange $-2\mathcal{J}\,\mathbf{S}_i \cdot \mathbf{S}_j$**
> *Approximate*. It is an effective spin Hamiltonian derived from electrostatics plus the Pauli principle for localized electrons; it describes insulators (ferrites, garnets) well and metals only qualitatively, where a band description is needed.

The ferromagnetic elements and compounds

Material	Moment per atom or formula unit	J_s at 20 °C (T)	T_C (K)	T_C (°C)
Fe (bcc)	$2.22\mu_B$	2.15	1043	770
Co (hcp)	$1.72\mu_B$	1.79	about 1390	about 1115
Ni (fcc)	$0.61\mu_B$	0.61	627	354
Gd	$7.6\mu_B$ (at 0 K)	small (near T_C)	293	20
Fe–35Co	about $2.45\mu_B$ average	2.45	about 1210	about 940
$Nd_2Fe_{14}B$	—	1.61	585	312
$SmCo_5$	—	1.07	about 1000	about 720

Gadolinium's Curie point is at room temperature, so a gadolinium sample is ferromagnetic on a cold day and paramagnetic on a warm one — the basis of magnetocaloric refrigeration research.

28.6 Antiferromagnetism

When the exchange integral is negative, neighbouring moments align antiparallel. The simplest picture divides the lattice into two interpenetrating **sublattices**, A and B, each internally parallel and each antiparallel to the other, with equal and opposite sublattice magnetizations. The net magnetization is zero. The material is **antiferromagnetic** (Louis Néel, 1936).

Ordering again disappears at a critical temperature, the **Néel temperature** T_N. Above T_N the material is paramagnetic, and the same molecular-field argument, now with a negative coupling between sublattices, gives

$$\chi = \frac{C}{T + \theta}, \qquad \theta > 0$$

so that $1/\chi$ extrapolates to a negative temperature intercept (Figure 28.2b). Below T_N the susceptibility depends on direction. A field parallel to the sublattice axis finds the moments locked; as $T \to 0$, $\chi_{\parallel} \to 0$. A field perpendicular to the axis cants both sublattices slightly towards it, and $\chi_{\perp}$ stays roughly constant at its value at T_N. In a powder or polycrystal the measured susceptibility shows a cusp at T_N — the experimental fingerprint of antiferromagnetic order. Neutron diffraction, which sees magnetic moments directly, confirmed the two-sublattice structure of MnO in 1949.

Material	T_N (K)	Paramagnetic θ (K)	Notes
MnF_2	67	82	Classic two-sublattice antiferromagnet
MnO	116	610	Superexchange through O^{2-}; $\theta/T_N \approx 5$ from next-nearest-neighbour coupling
FeO	198	570	
CoO	291	330	Orders just below room temperature
NiO	525	about 2000	Used in early spin-valve pinning layers
Cr (metal)	311	—	Spin-density-wave antiferromagnet
α-Fe_2O_3 (hematite)	about 950	—	Weakly ferromagnetic through slight canting
IrMn (thin film)	about 700	—	Pinning layer in GMR/TMR sensors and read heads
PtMn (thin film)	about 970	—	High-temperature pinning layer

An antiferromagnet has no useful net moment, so why does an electrical engineer care? Because when a ferromagnetic film is grown on an antiferromagnetic one and cooled through T_N in a field, interfacial exchange "pins" the ferromagnet's magnetization in one direction. This **exchange bias** shifts the ferromagnet's hysteresis loop along the field axis by tens of kA/m. It provides the fixed reference layer in giant- and tunnel-magnetoresistance sensors, hard-disk read heads and MRAM cells (Chapter 29, Section 29.9). The relevant engineering limit is the *blocking temperature* of the pinning, usually well below the bulk T_N, which sets the maximum operating and processing temperature of the sensor stack.

> **Status of the principle: two-sublattice Néel model**
> *Approximate.* It explains the sign of θ, the cusp at T_N and the anisotropy of χ below T_N; the ratio θ/T_N differs from the mean-field value of 1 whenever more than one exchange path (nearest and next-nearest neighbours) contributes.

28.7 Ferrimagnetism in spinel ferrites

Unequal sublattices

If the two antiparallel sublattices carry *unequal* moments, the cancellation is incomplete and a net spontaneous magnetization remains. Néel named this **ferrimagnetism** (1948). Macroscopically a ferrimagnet behaves like a ferromagnet — domains, hysteresis, a Curie temperature — but with a smaller saturation magnetization and with an insulating or semiconducting oxide lattice. That combination is decisive for electrical engineering: an oxide has electrical resistivity 10^6 to 10^{14} times that of iron, so eddy currents (Chapter 29) are suppressed at their origin. Ferrites made high-frequency magnetics possible.

The spinel structure

The **spinel ferrites** have formula MFe_2O_4 (M = Mn, Fe, Co, Ni, Cu, Zn, Mg, or mixtures). Their structure (Figure 28.3) is a face-centred cubic array of 32 O^{2-} ions in a cubic unit cell of edge about 0.84 nm, containing 8 formula units. The metal cations sit in interstices of two kinds:

- **A sites**, tetrahedrally coordinated by four oxygens: 64 per cell, of which 8 are occupied;
- **B sites**, octahedrally coordinated by six oxygens: 32 per cell, of which 16 are occupied.

The A–O–B bond angle is about 125°, and the A–B superexchange is strong and antiparallel. The A–A and B–B interactions are weaker. So the A sublattice and the B sublattice are aligned antiparallel, and the net moment per formula unit is

$$m_{\text{f.u.}} = m_B - m_A$$

Which cations go where depends on crystal-field site preferences (Advanced Chemistry core; it is a bonding question). In a **normal spinel** the divalent M^{2+} occupies A and both Fe^{3+} occupy B; in an **inverse spinel** M^{2+} moves to B and one Fe^{3+} moves to A. Using the spin-only moments of Section 28.2:

Ferrite	Cation distribution (A)[B]	Predicted $m_B - m_A$	Measured (per f.u., 0 K)
Fe_3O_4 (magnetite)	$(Fe^{3+})[Fe^{2+}Fe^{3+}]$ inverse	$(4+5)-5 = 4\mu_B$	$4.1\mu_B$
$NiFe_2O_4$	$(Fe^{3+})[Ni^{2+}Fe^{3+}]$ inverse	$(2+5)-5 = 2\mu_B$	$2.3\mu_B$
$CoFe_2O_4$	$(Fe^{3+})[Co^{2+}Fe^{3+}]$ inverse	$(3+5)-5 = 3\mu_B$	$3.7\mu_B$ (orbital contribution)
$MnFe_2O_4$	mostly $(Mn^{2+})[Fe^{3+}Fe^{3+}]$, about 80 % normal	$10-5 = 5\mu_B$	$4.6\mu_B$
$ZnFe_2O_4$	$(Zn^{2+})[Fe^{3+}Fe^{3+}]$ normal	A empty of moments: no A–B coupling	Paramagnetic at room temperature ($T_N \approx 10$ K)

The agreement — within the uncertainty of cation distribution and residual orbital moments — is one of the clean successes of materials physics: the saturation magnetization of an oxide can be predicted by counting ions on sites.

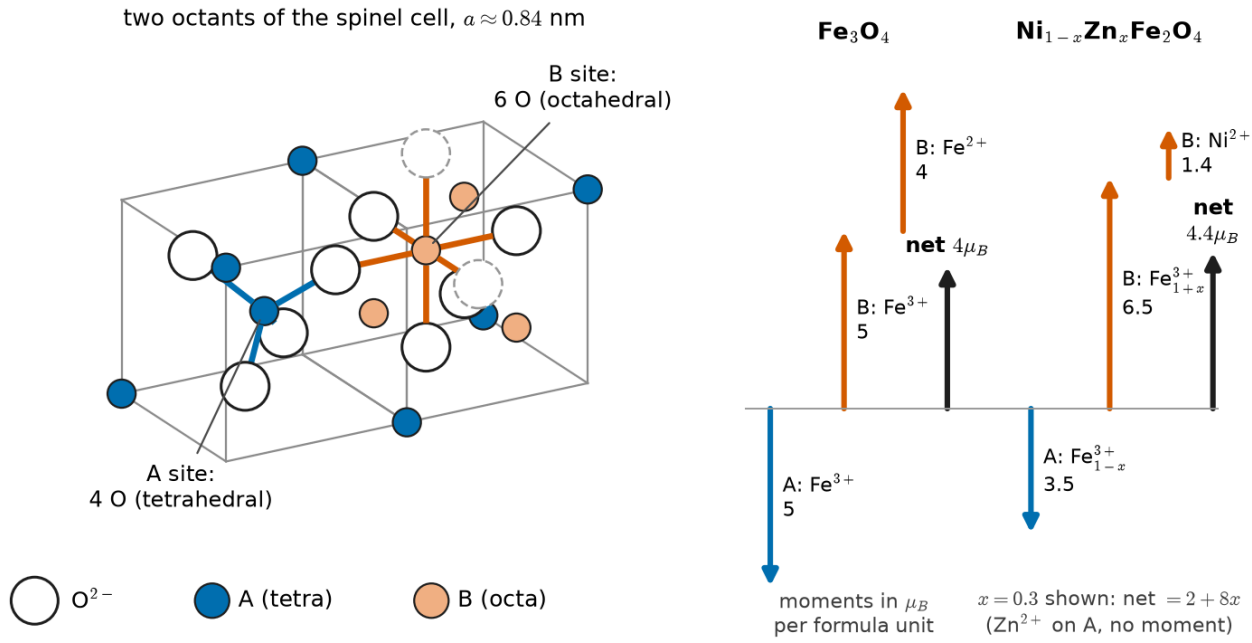


Figure 28.3 — One octant pair of the spinel unit cell showing tetrahedral A sites and octahedral B sites around the fcc oxygen array, with the antiparallel spin arrangement that gives magnetite its net 4 μB per formula unit.

Worked example — saturation magnetization of magnetite. Magnetite has lattice parameter $a = 0.8396$ nm and 8 formula units per cell, each carrying $4\mu_B$ net. The number of formula units per cubic metre is

$$n_{\text{f.u.}} = \frac{8}{(0.8396 \times 10^{-9}\ \text{m})^3} = \frac{8}{5.918 \times 10^{-28}\ \text{m}^3} = 1.352 \times 10^{28}\ \text{m}^{-3}$$

so

$$M_s(0) = n_{\text{f.u.}} \times 4\mu_B = 1.352 \times 10^{28} \times 4 \times 9.274 \times 10^{-24} = 5.01 \times 10^{5}\ \text{A/m}, \qquad J_s = \mu_0 M_s = 0.63\ \text{T}$$

The measured room-temperature value is about 4.8×10^5 A/m ($J_s \approx 0.60$ T); the difference is mostly thermal reduction at 300 K. A ferrite saturates at well under a third of iron's 2.15 T. That is the price paid for its resistivity, and it is why ferrite cores for the same power are physically larger than steel cores at low frequency — but much smaller at 100 kHz, where steel cannot be used at all.

Magnetite itself is not a useful power ferrite: electrons hop between Fe^{2+} and Fe^{3+} on equivalent B sites, giving a resistivity of only about 10^{-4} Ω m. Commercial ferrites keep the Fe^{2+} content low, which raises resistivity by many orders of magnitude (Chapter 29).

Why zinc raises the moment

Zinc ferrite is non-magnetic at room temperature, yet adding zinc to nickel or manganese ferrite *increases* the saturation magnetization. Count again. In $Ni_{1-x}Zn_xFe_2O_4$, Zn^{2+} strongly prefers A sites and pushes an equal amount of Fe^{3+} onto B:

$$\text{A: } Zn^{2+}_{x}Fe^{3+}_{1-x}, \qquad \text{B: } Ni^{2+}_{1-x}Fe^{3+}_{1+x}$$

$$m = [2(1-x) + 5(1+x)] - 5(1-x) = 2 + 8x \quad (\mu_B \text{ per f.u.})$$

The moment rises from $2\mu_B$ at $x = 0$ toward $6\mu_B$ at $x = 0.5$. Experimentally the rise continues only up to about $x \approx 0.4$–0.5. Beyond that the weakened A sublattice can no longer hold the B moments rigidly antiparallel, the B moments cant, and the moment falls again. At the same time T_C falls steadily because there are fewer A–B exchange bonds. Zn content is therefore the main composition knob trading saturation and permeability against Curie temperature, in both MnZn and NiZn ferrites (Chapter 29).

> **Status of the principle: Néel sublattice counting** $m = m_B - m_A$
> *Approximate.* It assumes collinear, antiparallel sublattices and spin-only ionic moments; it fails when one sublattice is so diluted that canting sets in (high Zn content) and underestimates ions with unquenched orbital moment (Co^{2+}).

Other ferrimagnetic families

- **Hexagonal ferrites** (hexaferrites) such as $BaFe_{12}O_{19}$ and $SrFe_{12}O_{19}$ have 12 Fe^{3+} per formula unit on five sublattice types, a net $20\mu_B$, $J_s \approx 0.47$ T, $T_C \approx 720$–740 K, and a strong uniaxial anisotropy along the c axis. They are the hard ferrite permanent magnets of Chapter 29 and are also used in millimetre-wave devices.
- **Garnets** such as yttrium iron garnet $Y_3Fe_5O_{12}$ (YIG; $J_s \approx 0.18$ T, $T_C \approx 560$ K) have three sublattice types and extremely narrow ferromagnetic-resonance linewidths. They are the materials of microwave circulators, isolators and YIG-tuned filters (Chapter 35 covers high-frequency materials).

28.8 Saturation magnetization

The **saturation magnetization** M_s is the magnetization reached when every moment in a single domain points the same way. It is an *intrinsic* property: it depends on composition and temperature, but not on grain size, defects or processing. That separates it sharply from permeability and coercivity, which are *extrinsic* and depend on microstructure (Sections 28.9–28.13).

M_s matters because it sets the maximum flux density a core can carry. A transformer of given voltage and frequency needs a core cross-section A with $V_{\text{rms}} = 4.44 f N A \hat{B}$, and $\hat{B}$ must stay safely below $\mu_0 M_s$. A higher saturation therefore means a smaller, lighter, cheaper core and less copper wound around it. For a permanent magnet, the remanence — and so the field it can produce — is bounded by J_s.

Worked example — saturation polarization of iron from its moment per atom. Body-centred cubic iron has a lattice parameter $a = 0.2866$ nm and 2 atoms per cell. The atom density is

$$n = \frac{2}{(0.2866 \times 10^{-9})^3} = \frac{2}{2.354 \times 10^{-29}} = 8.50 \times 10^{28}\ \text{m}^{-3}$$

With $2.22\mu_B$ per atom at 0 K,

$$M_s(0) = 8.50 \times 10^{28} \times 2.22 \times 9.274 \times 10^{-24} = 1.75 \times 10^{6}\ \text{A/m}, \qquad J_s(0) = \mu_0 M_s = 2.20\ \text{T}$$

The measured value at 0 K is $J_s = 2.22$ T, falling to 2.15 T at room temperature. Two lessons follow. First, J_s is fixed by atomic moment and atom density, so no amount of processing can push a pure-iron core above about 2.2 T. Second, any non-magnetic addition dilutes it. Adding 3 wt% silicon to iron for resistivity (Chapter 29) lowers J_s to about 2.03 T, and 6.5 wt% to about 1.80 T. Every soft-magnetic alloy design trades saturation against something else.

Can one beat iron? In Fe–Co alloys the average moment per atom *rises* above iron's, to a maximum of about $2.45\mu_B$ near 30–35 % Co. This is the top of the empirical **Slater–Pauling curve**, which plots average moment against electrons per atom for 3d alloys: adding electrons beyond iron's first fills the minority-spin band, increasing the spin imbalance, before the majority band saturates and the moment falls (Co, Ni). Fe–Co alloys (Permendur, Hiperco; $J_s = 2.35$–2.45 T) are the highest-saturation practical soft magnets and are used where weight matters most: aircraft generators, actuators and high-specific-power motors. Cobalt's cost and the alloy's brittleness keep them out of mass applications.

Material	J_s at 20 °C (T)	M_s (MA/m)	T_C (°C)	Remark
Fe–(30–50)Co	2.35–2.45	1.87–1.95	about 940	Highest practical saturation
Fe	2.15	1.71	770	Reference
Fe–3 % Si electrical steel	2.0–2.03	1.60	about 745	Transformer, motor cores
Fe–6.5 % Si	1.80	1.43	about 700	Near-zero magnetostriction
Co	1.79	1.42	about 1115	
Fe-based amorphous (Fe–Si–B)	1.56	1.24	about 400	Low-loss distribution transformers
$Nd_2Fe_{14}B$	1.61	1.28	312	Hard magnet phase
Nanocrystalline Fe–Si–B–Nb–Cu	1.23–1.25	0.98	about 570	
Fe–50Ni	1.5–1.6	1.2–1.3	about 500	
Fe–80Ni–Mo (Permalloy, mu-metal)	0.7–0.8	0.56–0.64	about 400–460	Highest permeability
Ni	0.61	0.485	354	
Fe_3O_4	0.60	0.48	585	
MnZn ferrite	0.45–0.53	0.36–0.42	150–300	Grade-dependent
$BaFe_{12}O_{19}$ / $SrFe_{12}O_{19}$	0.47	0.37	about 450–460	Hard ferrite
NiZn ferrite	0.25–0.40	0.20–0.32	100–500	Grade-dependent
Co-based amorphous	0.5–0.8	0.4–0.64	200–400	Zero magnetostriction
$Y_3Fe_5O_{12}$ (YIG)	0.18	0.14	about 285	Microwave

28.9 Magnetic domains and why they form

The puzzle

Below its Curie temperature every small region of iron is magnetized to M_s. Yet an ordinary nail is not a magnet: its net magnetization is near zero until a field is applied, and then a field of only tens to hundreds of A/m — minute compared with the molecular field of Section 28.5 — magnetizes a soft iron part almost to saturation. Weiss's second hypothesis resolved both facts: the material is divided into **domains**, each spontaneously magnetized to M_s but in different directions, so that the moments average to zero. Magnetizing a piece of iron does not create magnetization; it rearranges the domains.

Domains are not a defect. They are the equilibrium state that minimizes the total free energy, which contains five terms.

The energy terms

1. Exchange energy. Exchange (Section 28.5) favours parallel neighbouring spins. In a continuum description, a spatial variation of the magnetization direction costs an energy density

$$e_{\text{ex}} = A\left[(\nabla\alpha_1)^2 + (\nabla\alpha_2)^2 + (\nabla\alpha_3)^2\right]$$

where α_i are the direction cosines of $\mathbf{M}$ and A is the **exchange stiffness** (J/m), of order $k_B T_C/a$ for lattice spacing a: about 2×10^{-11} J/m for iron. Exchange alone would make the whole sample one domain.

2. Magnetocrystalline anisotropy energy. Because of spin–orbit coupling (Section 28.2), magnetization prefers certain crystal directions, the **easy axes**. For a cubic crystal,

$$e_K = K_1\left(\alpha_1^2\alpha_2^2 + \alpha_2^2\alpha_3^2 + \alpha_3^2\alpha_1^2\right) + K_2\,\alpha_1^2\alpha_2^2\alpha_3^2$$

For iron $K_1 > 0$ and the easy axes are the cube edges $\langle 100\rangle$; for nickel $K_1 < 0$ and they are the body diagonals $\langle 111\rangle$. For a uniaxial (hexagonal or tetragonal) crystal, $e_K = K_u \sin^2\theta$ with θ the angle from the c axis. Anisotropy energy is what a domain

must pay to point anywhere other than an easy direction. It is small in cubic 3d metals and huge in rare-earth compounds, which is the single most important distinction between soft and hard materials.

3. Magnetostatic (demagnetizing) energy. A magnetized body with poles on its surfaces generates a stray field, and the stray field stores energy $\frac{1}{2}\mu_0 \int H^2\, dV$ over all space. For a uniformly magnetized body this equals $\frac{1}{2}\mu_0 N M_s^2$ per unit volume. Its natural scale is the **magnetostatic energy density**

$$K_d = \tfrac{1}{2}\mu_0 M_s^2 = \frac{J_s^2}{2\mu_0}$$

which for iron is $(2.15)^2/(2\times 1.257\times 10^{-6}) = 1.84\times 10^6$ J/m³ — the energy that a thin iron plate would store if magnetized perpendicular to its surface. Splitting into domains of alternating direction lets flux close locally and removes most of this energy. It is the driving force for domain formation.

4. Magnetoelastic energy and magnetostriction. A ferromagnet changes shape when magnetized: the lattice strains along the magnetization by the **saturation magnetostriction** λ_s (about -7×10^{-6} for polycrystalline iron, -33×10^{-6} for nickel). Conversely, a mechanical stress σ creates an anisotropy energy density

$$e_\sigma = \tfrac{3}{2}\lambda_s \sigma \sin^2\theta$$

where θ is the angle between $\mathbf{M}$ and the stress axis. For a material with $\lambda_s > 0$, tension along an axis makes it an easy axis. The effect is not small: $\lambda_s = 30\times 10^{-6}$ under $\sigma = 100$ MPa gives $\frac{3}{2}\lambda_s\sigma = 4.5$ kJ/m³, comparable to or larger than the crystal anisotropy of most soft materials. Three engineering consequences follow. Soft magnetic properties degrade under clamping, punching and winding stress, which is why cores are stress-relief annealed and potted with care. Transformers **hum** at twice the line frequency (100 or 120 Hz and harmonics), because each half-cycle stretches and relaxes the core through magnetostriction. And materials with $\lambda_s \approx 0$ — 80 % Ni permalloy, 6.5 % Si steel, Co-based amorphous and nanocrystalline alloys — are the softest and quietest of all. The opposite extreme, Terfenol-D with $\lambda_s \approx 1.5$–2×10^{-3}, is used deliberately in actuators and sonar.

5. Zeeman energy. In an applied field the magnetization has energy $e_Z = -\mu_0 \mathbf{M}\cdot\mathbf{H}$, favouring domains aligned with $\mathbf{H}$. This is the term that moves walls in Section 28.11.

Material	K_1 (J/m³)	Easy axis	λ_s ($\times 10^{-6}$)	A (J/m)	$K_d = J_s^2/2\mu_0$ (J/m³)
Fe	$+4.8\times 10^4$	⟨100⟩	-7 ($\lambda_{100}=+21$, $\lambda_{111}=-21$)	2.1×10^{-11}	1.84×10^6
Fe–3 % Si	about $+3.5\times 10^4$	⟨100⟩	$\lambda_{100}\approx +23$	about 2×10^{-11}	1.6×10^6
Ni	about -5×10^3	⟨111⟩	-33	about 1×10^{-11}	1.5×10^5
Co (hcp)	$+4.1\times 10^5$ (uniaxial)	c axis	about -60	about 3×10^{-11}	1.27×10^6
Fe–80Ni (Permalloy)	about 0	—	about 0	1.3×10^{-11}	about 2×10^5
Fe_3O_4	-1.1×10^4	⟨111⟩	about $+40$	about 1×10^{-11}	1.4×10^5
MnZn ferrite (power grade)	about -10^2 to -10^3, temperature-tuned	—	about -1	—	about 9×10^4
$Nd_2Fe_{14}B$	$+4.9\times 10^6$ (uniaxial)	c axis	small	7.7×10^{-12}	1.03×10^6
$SmCo_5$	$+1.7\times 10^7$ (uniaxial)	c axis	small	about 1.2×10^{-11}	4.6×10^5
$BaFe_{12}O_{19}$	$+3.3\times 10^5$ (uniaxial)	c axis	small	about 6×10^{-12}	8.8×10^4
Terfenol-D (Tb–Dy–Fe_2)	small (compensated)	—	1500–2000	—	—

Domain subdivision

Figure 28.4 follows a single crystal as it lowers its energy. As one domain (a) it has a large stray field and pays the full magnetostatic energy. Splitting into two antiparallel domains (b) roughly halves the stray field energy at the cost of one **domain wall**, the transition layer between domains, which carries an energy per unit area γ (Section 28.10). Further splitting into n domains reduces the magnetostatic energy roughly in proportion to $1/n$ while adding wall area in proportion to n, so an optimum domain width exists. For stripe domains of width D in a plate of thickness L, minimizing (wall energy per unit plate area $\gamma L/D$) plus (magnetostatic energy per unit area, about $0.136\,\mu_0 M_s^2 D$) gives

$$D = \sqrt{\frac{\gamma L}{0.136\,\mu_0 M_s^2}} \propto \sqrt{L}$$

Thicker samples have wider domains. In a low-anisotropy cubic material such as iron, the crystal can do better still: triangular **closure domains** at the surfaces (Figure 28.4d) carry the flux around the ends inside the material, so almost no stray field leaves the body. The closure domains are magnetized along another easy axis, so they cost only a little magnetoelastic energy. Grain-oriented transformer steel, viewed under a Kerr microscope, shows exactly such slab domains running along the rolling direction with small closure domains at grain edges (Chapter 29).

The ratio $Q = K/K_d$ classifies behaviour. For iron $Q = 4.8 \times 10^4 / 1.84 \times 10^6 \approx 0.03$: anisotropy is weak compared with magnetostatics, flux-closure patterns form readily, and the material is easily magnetized in any direction — the signature of a soft material. For $Nd_2Fe_{14}B$, $Q = 4.9 \times 10^6 / 1.03 \times 10^6 \approx 4.8$: anisotropy wins, the magnetization stays along the c axis even against its own demagnetizing field, and the material can be a permanent magnet. A magnet material must have $Q > 1$, and in practice Q well above 1.

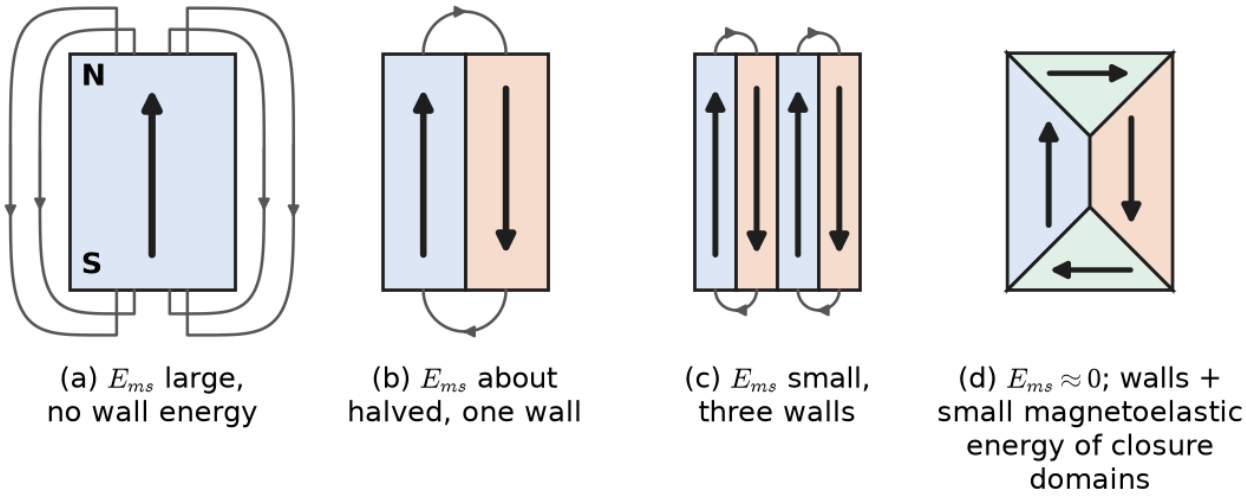


Figure 28.4 — Domain formation lowers magnetostatic energy: (a) single domain with large stray field, (b) two antiparallel domains, (c) four domains, (d) slab domains with triangular closure domains and no external stray field.

Single-domain particles

Shrink the sample and the balance changes: magnetostatic energy scales with volume (r^3), wall energy with area (r^2). Below a critical size a wall no longer pays for itself and the particle stays a single domain. For a sphere, equating the magnetostatic energy of the single-domain state, $\frac{1}{6}\mu_0 M_s^2 \cdot \frac{4}{3}\pi r^3$, with roughly half that plus a wall of area πr^2 gives a critical radius $r_c \approx 9\gamma/(\mu_0 M_s^2)$, or with $\gamma = 4\sqrt{AK}$ from Section 28.10,

$$d_c \approx \frac{72\sqrt{AK}}{\mu_0 M_s^2}$$

For $Nd_2Fe_{14}B$ this gives $d_c \approx 0.2$ μm; for iron, about 20 nm (where the crude model is least reliable, since soft particles can also curl their magnetization). Single-domain particles cannot be magnetized or demagnetized by wall motion — only by rotating the whole moment against the anisotropy — which makes them potentially very hard. This is the design principle of fine-particle and

nanostructured magnets (Section 28.13) and of recording media. Below an even smaller size, thermal energy can flip the whole particle: superparamagnetism (Section 28.14).

> **Status of the principle: domain formation by energy minimization**
> *Derived* from minimizing the sum of exchange, anisotropy, magnetostatic, magnetoelastic and Zeeman energies (micromagnetics); the closed-form domain widths and critical sizes quoted here are *approximate* scaling estimates that capture trends and orders of magnitude, not precise values.

28.10 Domain walls

Why the wall has a width

Between two domains magnetized in opposite directions, the magnetization must turn through 180°. Exchange wants this rotation to be as gradual as possible, because the energy cost of a small angle ϕ between neighbouring spins is proportional to ϕ^2 : spreading a 180° turn over N atomic planes costs exchange energy proportional to $N(\pi/N)^2 = \pi^2/N$, which falls as the wall widens. Anisotropy wants the rotation to be abrupt, because every spin inside the wall points away from the easy axis and pays anisotropy energy proportional to the wall's width. The wall width is the compromise. In a **Bloch wall** the magnetization rotates in the plane of the wall, so that no magnetic poles appear on its faces (Figure 28.5).

Estimate. Model a 180° wall of width δ as a uniform rotation $d\theta/dx = \pi/\delta$. The exchange energy per unit wall area is $A(\pi/\delta)^2\delta = A\pi^2/\delta$. The anisotropy energy per unit area is $\int K\sin^2\theta\,dx \approx K\delta/2$, since the average of $\sin^2\theta$ across the wall is ½. Minimizing the sum

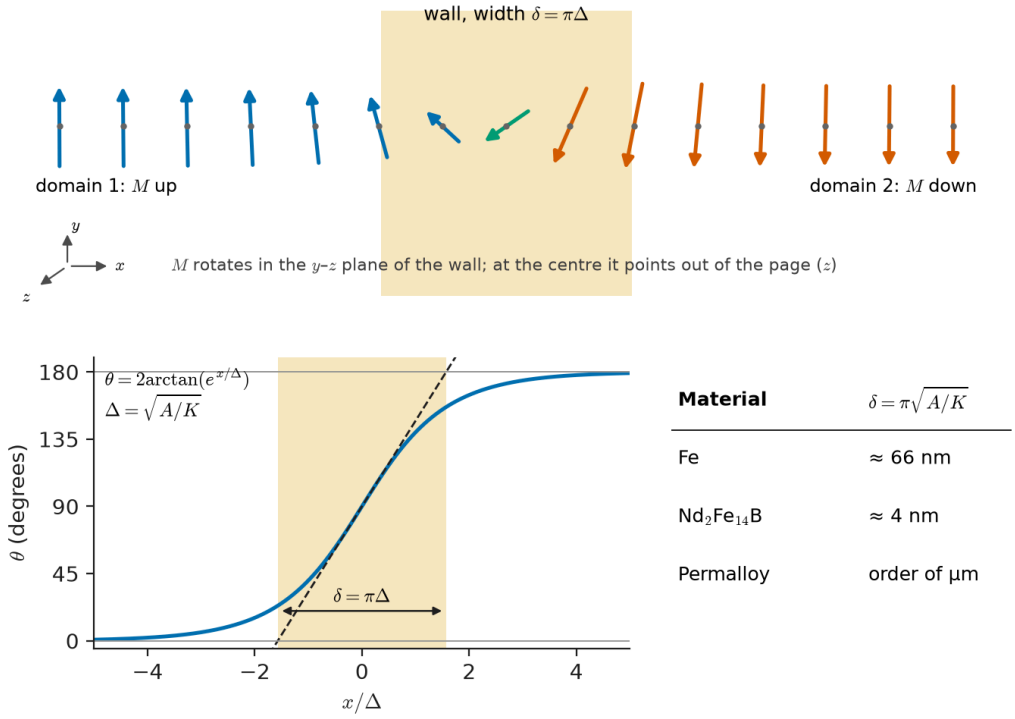
$$\gamma(\delta) = \frac{A\pi^2}{\delta} + \frac{K\delta}{2}, \qquad \frac{d\gamma}{d\delta} = 0 \;\Rightarrow\; \delta = \pi\sqrt{\frac{2A}{K}}, \qquad \gamma = \pi\sqrt{2AK}$$

Exact result. Minimizing the energy functional $\int [A(d\theta/dx)^2 + K\sin^2\theta]\,dx$ properly gives the profile $\theta(x) = 2\arctan[\exp(x/\Delta)]$ with $\Delta = \sqrt{A/K}$, and

$$\delta = \pi\sqrt{\frac{A}{K}}, \qquad \gamma = 4\sqrt{AK}$$

where δ is the width obtained by extending the slope at the wall centre. The estimate and the exact result agree to within a factor $\sqrt{2}$ in width and 11 % in energy. What matters is the scaling: width grows with $\sqrt{A/K}$, energy with $\sqrt{AK}$.

> **Status of the principle: Bloch-wall width $\delta = \pi\sqrt{A/K}$ and energy $\gamma = 4\sqrt{AK}$**
> *Derived* from continuum micromagnetics for a 180° wall in an infinite uniaxial crystal; *approximate* for cubic crystals (90° walls, K_1 replaces K with modified numerical factors), for thin films and for walls narrower than a few lattice spacings, where the continuum description fails.



Material	$\delta = \pi\sqrt{A/K}$
Fe	≈ 66 nm
$Nd_2Fe_{14}B$	≈ 4 nm
Permalloy	order of μm

Figure 28.5 — Rotation of the magnetization through a 180° Bloch wall, with the angle profile θ(x) = 2 arctan(exp(x/Δ)) and the wall width δ = π√(A/K).

Worked example — walls in iron and in a rare-earth magnet. For iron, $A = 2.1 \times 10^{-11}$ J/m and $K_1 = 4.8 \times 10^{4}$ J/m³ (treating the cubic anisotropy as a uniaxial K for an order-of-magnitude result):

$$\delta = \pi\sqrt{\frac{2.1 \times 10^{-11}}{4.8 \times 10^{4}}} = \pi \times 2.09 \times 10^{-8} = 66\ \text{nm}, \qquad \gamma = 4\sqrt{(2.1 \times 10^{-11})(4.8 \times 10^{4})} = 4.0 \times 10^{-3}\ \text{J/m}^2$$

The wall spans about $66/0.287 \approx 230$ lattice spacings. For $Nd_2Fe_{14}B$, $A = 7.7 \times 10^{-12}$ J/m and $K_1 = 4.9 \times 10^{6}$ J/m³:

$$\delta = \pi\sqrt{\frac{7.7 \times 10^{-12}}{4.9 \times 10^{6}}} = \pi \times 1.25 \times 10^{-9} = 3.9\ \text{nm}, \qquad \gamma = 4\sqrt{(7.7 \times 10^{-12})(4.9 \times 10^{6})} = 2.5 \times 10^{-2}\ \text{J/m}^2$$

The rare-earth wall is seventeen times thinner and six times more energetic. These two numbers explain much of Chapter 29. A wide, low-energy wall in iron glides past features much smaller than 66 nm hardly noticing them; a 4 nm wall in NdFeB is caught by atomic-scale disorder at grain boundaries. And in near-zero-anisotropy permalloy and nanocrystalline alloys, walls are micrometres wide and are barely pinned by anything — the root of their extreme softness.

Néel walls in thin films

In a film thinner than the Bloch-wall width, a Bloch wall would put poles on the film surfaces where its central spins point out of plane. It becomes cheaper for the magnetization to rotate *within the film plane*, creating poles on the wall faces instead. Such **Néel walls** dominate in permalloy films below a few tens of nanometres thick — the layers in magnetoresistive sensors and read heads. Their stray fields make neighbouring walls interact, which is why thin-film sensor elements are shaped and biased to stay single-domain (Chapter 29).

28.11 Domain-wall motion and pinning

How a material magnetizes

Apply a slowly increasing field to a demagnetized polycrystal. The **initial magnetization curve** (Figure 28.6) has three regions, each tied to a domain process:

1. **Reversible wall displacement** (low fields). Domains favourably oriented to **H** grow at the expense of others by small displacements of their walls, which bow out between pinning points like elastic membranes. Removing the field lets them spring back. The slope here is the initial permeability μ_i.
2. **Irreversible wall displacement** (intermediate fields, the steep part). A wall pressed hard enough breaks free from a pinning site and jumps to the next one, sweeping a volume in microseconds. These jumps are **Barkhausen jumps**; picked up by a search coil they sound like crackling and form the basis of Barkhausen-noise inspection of residual stress and heat treatment in steel parts. Most of the magnetization change, and most of the hysteresis loss, occurs here; the maximum permeability $\mu_{\max}$ lies in this region.
3. **Rotation** (high fields). Once walls have swept out, each grain is a single domain magnetized along the easy axis nearest to **H**. Further increase turns **M** away from the easy axis toward **H** against the anisotropy. This **approach to saturation** is slow and needs fields of order $K/\mu_0 M_s$.

The wall process works only because a wall moves under a small force. The applied field exerts a pressure on a 180° wall equal to the difference in Zeeman energy density between the two sides, $p = 2\mu_0 M_s H$. For iron at $H = 10$ A/m, $p = 2 \times 1.257 \times 10^{-6} \times 1.71 \times 10^{6} \times 10 \approx 43$ Pa. That is a minute pressure, but it acts on a wall that can move without changing M anywhere except within itself.

Pinning: why walls stick

If the material were perfectly uniform, the wall energy γ would be the same everywhere and a wall would move under an arbitrarily small field. Real materials contain inclusions, precipitates, voids, dislocations, grain boundaries, local stresses and composition fluctuations (Chapter 5). Each changes the local A, K, M_s or $\lambda_s \sigma$ and so makes the wall energy depend on position, $\gamma(x)$. A wall then needs a field to climb out of every energy minimum; the largest restoring force $\max(d\gamma/dx)$ across the sample sets the field required, giving approximately

$$H_c \approx \frac{1}{2\mu_0 M_s} \left(\frac{d\gamma}{dx} \right)_{\max}$$

Two features of pinning are central to materials design.

- **Pinning is strongest when the obstacle is about as large as the wall is wide.** A defect much smaller than δ changes the energy of only a tiny fraction of the wall; a defect much larger than δ presents a gentle, almost flat energy landscape. In iron, with $\delta \approx 66$ nm, carbide and nitride precipitates and non-metallic inclusions of 0.05–1 μm are the most damaging. This is why electrical steels are decarburized to below about 30 ppm C: otherwise slow carbide precipitation during service raises coercivity and loss (**magnetic ageing**; Chapter 44).
- **Inclusions also create closure domains.** A non-magnetic inclusion intersected by a wall reduces the wall area (lowering the energy) and, when isolated inside a domain, carries poles and spike-like closure domains that attach to passing walls. Kersten's and Néel's inclusion theories give H_c increasing with inclusion volume fraction, roughly as $v^{2/3}$, and with K or $\lambda_s \sigma$.

Grain boundaries pin too, partly because the easy axis changes direction across them. In conventional soft materials coercivity falls with increasing grain size D roughly as $H_c \propto 1/D$; this is why non-oriented motor steels are annealed to grain sizes of 100–150 μm, and why grain-oriented steels have grains of millimetres to centimetres (Chapter 29). The trend reverses dramatically below the exchange length in nanocrystalline alloys, where $H_c \propto D^6$ — the random-anisotropy effect developed in Chapter 29.

Two ways to make a material soft, two ways to make it hard

For **soft** materials the recipe is to make walls move easily: low anisotropy K (wide walls, weak pinning), near-zero magnetostriction (no stress pinning), high purity, few inclusions, large grains (or, alternatively, grains far below the exchange length), and annealing to remove dislocations and residual stress.

For **hard** materials there are two strategies:

- **Pinning-type** magnets deliberately fill the material with obstacles comparable in size to the narrow wall, so that walls exist but cannot move. Sm_2Co_{17}-type magnets work this way: a cellular nanostructure of Sm_2Co_{17} cells about 100 nm across, bounded by thin $SmCo_5$-type cell-boundary phase with different wall energy, pins walls at the cell boundaries.
- **Nucleation-type** magnets are made of grains so well isolated and perfect that, once saturated, a grain contains *no* walls; reversal requires nucleating a reversed domain, usually at a surface or grain-boundary defect where anisotropy is locally reduced. Sintered NdFeB and $SmCo_5$ work this way. In a nucleation magnet the critical engineering feature is the grain

boundary: a thin non-magnetic Nd-rich layer that decouples the grains magnetically, and a grain surface with high local anisotropy (the reason for dysprosium diffusion into grain shells, Chapter 29).

> **Status of the principle: coercivity controlled by wall pinning or nucleation**
> *Approximate.* The pinning and nucleation models identify which microstructural features control H_c and give correct trends with defect size, grain size and anisotropy; quantitative coercivity prediction from microstructure still requires micromagnetic simulation, and the absolute values are typically uncertain by a factor of two or more.

28.12 Hysteresis

The loop

Take a ferromagnet to saturation in a large positive field, reduce the field to zero, reverse it, and return. B (or M) does not retrace its path. The closed curve traced under a cyclic field is the **hysteresis loop** (Figure 28.6). Its key points are:

- the **saturation** induction B_s (or polarization $J_s = \mu_0 M_s$) at the tips;
- the **remanence** B_r, the induction left at $H = 0$ after saturation;
- the **coercivity** H_c, the reverse field needed to bring B (or M) back to zero;
- the **minor loops** traced when the field amplitude does not reach saturation, nested inside the major loop. A transformer at 1.5 T in a material that saturates at 2.0 T runs on a minor loop.

Hysteresis exists because the processes of Section 28.11 are irreversible. When a wall jumps from one pinning site to the next, the energy released is not stored. It is dissipated, as eddy currents around the moving wall and as spin waves that end as heat. The magnetization lags the field because it can only move in jumps, each of which needs the field to exceed a local pinning threshold.

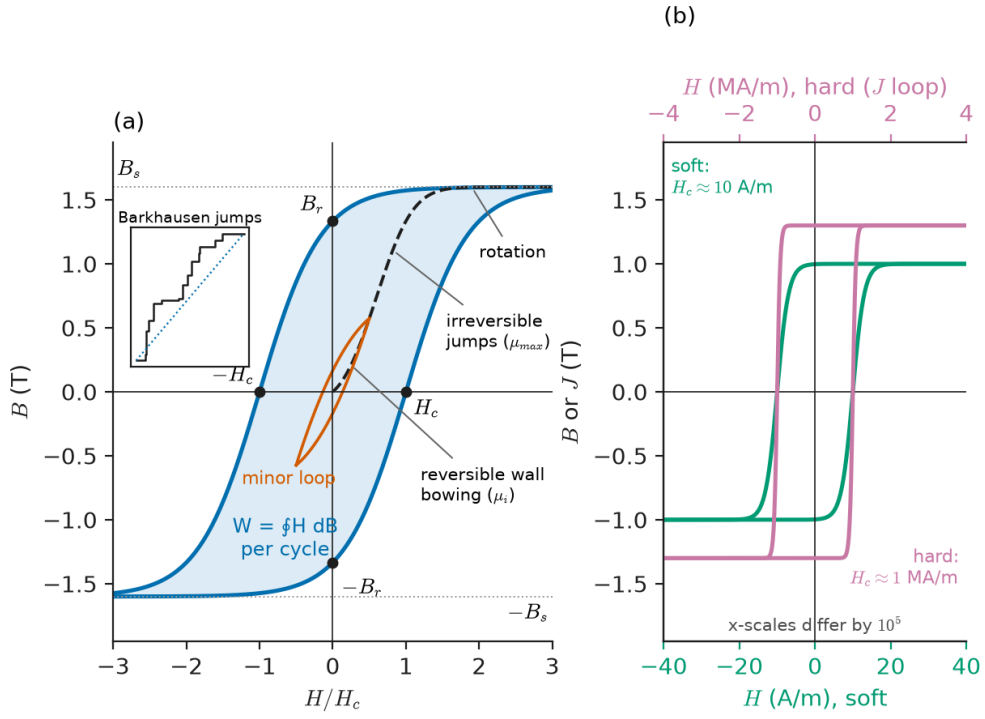


Figure 28.6 — Hysteresis loop of a ferromagnet with initial magnetization curve, saturation, remanence and coercivity, a minor loop, and a comparison of soft and hard loops.

Energy dissipated per cycle

Why does the loop area represent energy? Consider a toroidal core of cross-section A and mean path l wound with N turns. The source drives current i against the induced voltage $v = NA\,dB/dt$. In time dt it delivers energy $vi\,dt = NA\,dB \cdot i$. Using $H =$

Ni/l, this is $(Al)\,H\,dB$: per unit volume, the work done on the material is $H\,dB$. Around a closed cycle the stored energy returns to its starting value, so the net work per unit volume per cycle is

$$W_h = \oint H\,dB \quad (\mathrm{J/m^3\ per\ cycle})$$

the area of the B–H loop. At frequency f the hysteresis power per unit volume is $P_h = fW_h$.

> **Status of the principle: loop area equals energy dissipated per cycle**
> *Fundamental.* It follows from Faraday's and Ampère's laws and energy conservation, with no assumption about the material. At low frequency the loop is the static (hysteresis) loop; at higher frequency the same formula applied to the dynamic loop includes eddy-current loss as well (Chapter 29).

Worked example — hysteresis loss of a motor steel, and why a magnet is never cycled. A non-oriented electrical steel has coercivity $H_c = 40$ A/m and is cycled to $\hat{B} = 1.5$ T at 50 Hz. A loop cannot be wider than $2H_c$ or taller than $2\hat{B}$, and a rectangle with these sides gives an upper bound

$$W_h \lesssim 4H_c\hat{B} = 4 \times 40 \times 1.5 = 240\ \mathrm{J/m^3}$$

At 50 Hz this is $P_h \lesssim 240 \times 50 = 1.2 \times 10^4$ W/m³. With density 7650 kg/m³, $P_h \lesssim 1.6$ W/kg. Measured hysteresis losses of such steels at 1.5 T, 50 Hz are about 1.3–2 W/kg — the bound is close because steel loops are fairly square at high induction. For a 10 kg stator core, 15 W of heat comes from hysteresis alone; Chapter 29 adds the eddy-current terms.

Now a hard ferrite magnet with $H_{cJ} = 300$ kA/m and $J_s = 0.40$ T. Its intrinsic loop, if cycled fully, would enclose about $4H_{cJ}J_s = 4 \times 3 \times 10^5 \times 0.40 = 4.8 \times 10^5$ J/m³ per cycle — two thousand times the steel's. At 50 Hz that is 24 MW/m³: the magnet would melt. Permanent magnets are designed so they *never* traverse their loop. They sit at an operating point in the second quadrant and move only along nearly reversible recoil lines (Chapter 29).

The Rayleigh region

At low field amplitudes $\hat{H}$, well below the coercivity, the loop is a narrow lens and the permeability rises linearly with amplitude: $\mu = \mu_i + \nu\hat{H}$, with **Rayleigh constant** ν. The loop branches are parabolas and the hysteresis energy per cycle is

$$W_h = \tfrac{4}{3}\mu_0\nu\hat{H}^3$$

Since $\hat{B} \approx \mu_0\mu_i\hat{H}$ at low amplitude, the loss scales as $\hat{B}^3$ — a steeper dependence than at high inductions, where measured exponents are closer to 1.6–2. This is one reason the Steinmetz exponent of Chapter 29 is not a constant, and why low-level signal transformers and current sensors are designed around the Rayleigh behaviour.

> **Status of the principle: Rayleigh law**
> *Empirical.* It describes the low-amplitude, nearly reversible regime of wall motion in a random pinning landscape (Néel later gave it a statistical justification); it fails once $\hat{H}$ approaches H_c.

28.13 Remanence and coercivity

Remanence

After saturation and removal of the field, each grain relaxes its magnetization to the easy axis closest to the previous field direction (assuming no walls re-enter). The remanence ratio M_r/M_s therefore depends on how easy axes are distributed:

Grain easy-axis distribution (non-interacting, no walls)	M_r/M_s
All easy axes aligned with the field (perfect texture)	1
Random uniaxial grains	0.5
Random cubic grains, $K_1 > 0$ (six $\langle 100 \rangle$ directions)	0.83
Random cubic grains, $K_1 < 0$ (eight $\langle 111 \rangle$ directions)	0.87
Sintered NdFeB pressed and aligned in a field (real)	0.95–0.97

Permanent magnets need high remanence, so their powders are aligned in a strong field during pressing to make the c axes parallel: an anisotropic sintered magnet has nearly twice the remanence of an isotropic one of the same composition. Exchange coupling between very fine grains can raise M_r/M_s above 0.5 even for random orientation (remanence enhancement), at the expense of coercivity.

Soft materials want the opposite of high remanence in most applications — a core that stays magnetized after the current stops is a nuisance (inrush current in transformers). But a *square* loop with high B_r/B_s is wanted in magnetic amplifiers and some pulse cores, and a deliberately *flat* (sheared) loop with low B_r is wanted in current transformers and chokes carrying DC. These shapes are set by field annealing or by an air gap (Chapter 29).

Coercivity: intrinsic and normal

Two coercivities appear on a magnet datasheet, and the distinction matters:

- the **intrinsic coercivity** H_{cJ} (or ${}_iH_c$) is the reverse field at which $J = \mu_0 M = 0$. It measures the material's resistance to reversal.
- the **normal coercivity** H_{cB} (or ${}_bH_c$) is the reverse field at which $B = \mu_0 H + J = 0$.

Since $B = \mu_0 H + J$, a magnet whose J stays at B_r until large reverse fields has a straight B–H demagnetization line of slope $\mu_0\mu_{\text{rec}}$, where the **recoil permeability** μ_{rec} is close to 1 (1.03–1.10 for NdFeB and ferrite). The $B = 0$ point then lies at $H_{cB} = B_r/(\mu_0\mu_{\text{rec}})$. For NdFeB with $B_r = 1.30$ T and $\mu_{\text{rec}} = 1.05$, $H_{cB} = 985$ kA/m, whatever the intrinsic coercivity, as long as H_{cJ} exceeds it. H_{cB} can never exceed B_r/μ_0; H_{cJ} can be far larger, and is what decides whether a magnet survives heat and opposing fields (Chapter 29).

An upper bound: coherent rotation

What is the largest possible coercivity? Consider a single-domain particle (Section 28.9) with uniaxial anisotropy K, magnetized along its easy axis, and a reverse field along that axis. With no walls, the only way to reverse is for the whole moment to rotate. The energy per unit volume as a function of angle θ from the original direction is

$$e(\theta) = K \sin^2\theta + \mu_0 M_s H \cos\theta$$

(with H the magnitude of the reverse field). The original state $\theta = 0$ is a minimum as long as $d^2e/d\theta^2 = 2K - \mu_0 M_s H > 0$. It becomes unstable, and the moment flips, at the **anisotropy field**

$$H_K = \frac{2K}{\mu_0 M_s}$$

This is the **Stoner–Wohlfarth** result.

> **Status of the principle: Stoner–Wohlfarth switching field** $H_K = 2K/\mu_0 M_s$
> *Derived* for a single-domain particle reversing by uniform rotation at zero temperature, with the field along the easy axis; it is an upper bound on coercivity for real materials, which reverse by easier, non-uniform paths.

For $Nd_2Fe_{14}B$, $H_K = 2 \times 4.9 \times 10^6/1.61 = 6.1 \times 10^6$ A/m, i.e. $\mu_0 H_K \approx 7.6$ T. Real sintered NdFeB magnets achieve $H_{cJ} \approx 1$–2.5 MA/m, only 15–40 % of H_K. This shortfall, known as **Brown's paradox**, is not a mystery but a measurement of defects. Reversal nucleates at grain surfaces where anisotropy is locally degraded, and at sharp corners where demagnetizing fields concentrate. For iron, $H_K = 2 \times 4.8 \times 10^4/2.15 = 4.5 \times 10^4$ A/m, while soft iron has $H_c \approx 10$–80 A/m — a thousand times

lower, because walls do the work and rotation is never needed. Coercivity is therefore the most structure-sensitive of all magnetic properties: the same composition can span several orders of magnitude depending on processing.

Material class	Typical H_c (A/m)	Controlling mechanism
Co-based amorphous, Finemet, Supermalloy	0.2–2	Near-zero K and λ_s, wide walls
80Ni Permalloy, Fe-based amorphous	1–5	Low K, low pinning
Grain-oriented Si steel	5–10	Large grains, low C, texture
Non-oriented Si steel, pure iron	20–100	Grain size, inclusions
MnZn / NiZn ferrite	5–20 / 20–300	Porosity, grain size, K
AlNiCo	4×10^4–1.6×10^5	Shape anisotropy of rod precipitates
Hard ferrite	2×10^5–4×10^5 (H_{cJ})	Single-domain-size grains, K
NdFeB, SmCo	8×10^5–3×10^6 (H_{cJ})	Nucleation or pinning, very high K

The coercivity of commercial magnetic materials spans seven orders of magnitude. Chapter 29 places the boundary between soft and hard and selects within each class.

28.14 Temperature dependence of magnetic behaviour

Saturation magnetization and the Curie temperature

The spontaneous magnetization falls with temperature as thermal agitation disorders the moments, and it vanishes at T_C (Figure 28.2a). Three regimes deserve attention.

- **Low temperature.** The mean-field theory predicts an exponentially flat start, but measurements follow **Bloch's $T^{3/2}$ law**, $M_s(T) = M_s(0)\,(1 - bT^{3/2})$, because the lowest excitations are long-wavelength spin waves (collective, gently varying precessions of the moments) whose number grows as $T^{3/2}$.
- **Intermediate temperatures.** The reduced curve $M_s/M_s(0)$ against T/T_C is close to the mean-field Brillouin curve. A practical consequence follows: materials with a high T_C are flat at room temperature, while those with T_C only a few hundred kelvin above it are on the steep part. Iron ($T_C = 1043$ K, $T/T_C = 0.28$ at 20 °C) loses only about 1 % of its saturation between 20 °C and 100 °C. A MnZn power ferrite with $T_C \approx 220$ °C loses about 20 %.
- **Near T_C.** $M_s \propto (T_C - T)^{\beta}$ with measured critical exponents $\beta \approx 0.33$–0.37, not the mean-field $\frac{1}{2}$: correlated fluctuations, which mean-field theory neglects, dominate near any continuous phase transition.

> **Status of the principle: mean-field $M_s(T)$ and Bloch $T^{3/2}$ law**
> *Approximate*. The mean-field curve reproduces the overall shape between about $0.3T_C$ and $0.9T_C$; Bloch's law is the low-temperature spin-wave limit, valid below about $0.3T_C$; neither describes the critical region close to T_C.

This is the physics behind the opening failure. The inductor's ferrite had $B_s \approx 0.49$ T at 25 °C, falling to about 0.39 T at 100 °C and lower still at 110 °C. A design that allowed only 10–15 % margin at room temperature lost it in the enclosure. Power-ferrite datasheets quote B_s at 25 °C and at 100 °C for exactly this reason, and the 100 °C value is the one to design with.

The Curie point itself is used in devices:

- **Curie-point thermostats.** A rice cooker's thermal switch holds a permanent magnet against a ferrite whose T_C is set (by Zn content) just above 100 °C. When the water has boiled away and the pot exceeds T_C, the ferrite stops attracting the magnet and the switch drops out.
- **Self-regulating soldering tips and induction heaters.** A ferromagnetic tip alloy heated by RF induction absorbs strongly through hysteresis and skin-effect losses below T_C; at T_C its permeability collapses, absorption falls, and the tip regulates at its Curie temperature without a sensor.
- **Thermomagnetic writing.** Heat-assisted magnetic recording heats a high-anisotropy medium near T_C with a laser so a modest head field can write it.

Anisotropy and coercivity fall faster than M_s

Magnetocrystalline anisotropy depends on the alignment of neighbouring moments even more strongly than M_s does. A useful empirical relation (the Callen–Callen law) is $K(T)/K(0) \approx [M_s(T)/M_s(0)]^{l(l+1)/2}$, giving $K \propto M_s^3$ for uniaxial anisotropy ($l = 2$) and $K \propto M_s^{10}$ for cubic ($l = 4$). Coercivity follows anisotropy. Consequences:

- In NdFeB, H_{cJ} falls by about 0.5–0.6 % per kelvin near room temperature, much faster than B_r (about 0.12 % per kelvin). At 150 °C a magnet can lose half or more of its room-temperature coercivity. This is the root of the dysprosium problem in traction motors (Chapter 29).
- In hard ferrite the anisotropy falls more slowly than M_s, so H_c actually *rises* with temperature over the normal range (a positive coefficient, about +0.2 to +0.4 % per kelvin). Ferrite magnets are therefore vulnerable to demagnetization in the *cold*, not the heat.
- In MnZn ferrite, K_1 is the sum of a negative host contribution and a positive contribution from Fe^{2+} (and Co^{2+}) ions. Manufacturers tune composition so that K_1 crosses zero near the intended operating temperature (often 80–100 °C). There permeability peaks (the "secondary maximum") and losses reach a minimum. A power ferrite therefore *gets less lossy* as it warms up to its design temperature, which helps thermal stability, and more lossy beyond it, which can lead to thermal runaway.
- The **Hopkinson effect**: permeability of a soft material often rises sharply just below T_C, because K vanishes faster than M_s, before collapsing to paramagnetic values at T_C.

Superparamagnetism: when thermal energy flips the whole particle

A single-domain particle of volume V and uniaxial anisotropy K has two stable states separated by an energy barrier KV. Thermal activation flips it at the Néel–Arrhenius rate

$$\tau = \tau_0 \exp\left(\frac{KV}{k_B T}\right), \qquad \tau_0 \approx 10^{-9}\ \text{s}$$

If τ is shorter than the measurement time the particle shows no remanence or coercivity — it behaves like a paramagnet with a giant moment, hence **superparamagnetism**. Taking logarithms: stability over a 100 s measurement needs $KV/k_BT \gtrsim \ln(10^{11}) \approx 25$; ten-year data retention needs $KV/k_BT \gtrsim \ln(3 \times 10^{17}) \approx 40$, and recording engineers require 50–60 to allow for a distribution of particle sizes. For iron ($K = 4.8 \times 10^4$ J/m³) at 300 K, $KV/k_BT = 25$ requires $V = 2.2 \times 10^{-24}$ m³, a sphere of diameter about 16 nm. Below this size iron nanoparticles are superparamagnetic at room temperature. This is the thermal stability limit of magnetic data storage and the reason recording media moved to high-anisotropy FePt and to heat-assisted writing. Superparamagnetic nanoparticles are also useful in their own right, for ferrofluids and biomedical contrast agents (Chapter 38).

> **Status of the principle: Néel–Arrhenius relaxation** $\tau = \tau_0 e^{KV/k_BT}$
> *Approximate.* It is an activated-process rate for non-interacting single-domain particles with a single energy barrier; τ_0 is known only to within an order of magnitude, and interparticle interactions and size distributions modify the effective barrier.

What remains open

Open question	Where it is answered
Where is the boundary between soft and hard, and which material is chosen for which device?	Chapter 29
How are the hysteresis, eddy-current and excess losses of a real core separated and minimized?	Chapter 29
How do texture, lamination, nanocrystallization and sintering produce the microstructures that set μ and H_c?	Chapter 29; processing in Chapters 10 and 46
Why is a superconductor a perfect diamagnet, and how does flux pinning resemble domain-wall pinning?	Chapter 30
How do size effects and superparamagnetism shape magnetic nanoparticles and thin films?	Chapter 38
How are magnetic thin-film stacks deposited for sensors and memories?	Chapters 49 and 50
How are B–H loops, M_s, T_C, permeability and core loss measured (VSM, permeameter, Epstein frame)?	Chapter 52
How do magnetic ageing, corrosion of rare-earth magnets and thermal demagnetization limit service life?	Chapter 44
How do spin-dependent transport devices (spin valves, MRAM) operate as circuits?	Electronics & Computing core

Next: Chapter 29 turns these principles into engineering materials — electrical steels, ferrites, amorphous and nanocrystalline alloys and permanent magnets — and shows how losses, eddy currents and temperature decide which one goes into a transformer, motor, inductor or sensor.

Chapter 29 — Soft and Hard Magnetic Materials

Part VI · Thermal, Electrical, Dielectric and Magnetic Materials

A utility buys distribution transformers by the thousand. Each 100 kVA unit sits on a pole or a pad, energized 8760 hours a year whether its customers draw power or not, and for most of that time it is lightly loaded. Its copper loss rises and falls with load; its **core loss** — the no-load loss — does not. With a core of conventional grain-oriented silicon steel, a well-designed 100 kVA unit dissipates roughly 150–250 W at no load. Built on an amorphous-metal core, the same rating dissipates roughly 50–80 W. Take a difference of 130 W: that is $130 \times 8760 \approx 1.14 \times 10^3$ kWh per transformer per year, or about 34 MWh over a 30-year life, multiplied by every transformer on the network. The amorphous unit is larger, noisier, harder to manufacture and more expensive. Whether it pays depends on the price of energy, the cost of capital and the regulations in force. But the decision rests on a materials fact: an alloy with no crystal structure, cast as a ribbon 25 μm thick, wastes a quarter to a third as much energy per cycle as the best textured crystal of iron–silicon.

Chapter 28 explained why magnetic materials have hysteresis, domains and a Curie temperature. This chapter turns that into a toolkit: the families of engineering magnetic materials, where their losses come from, how the losses are suppressed, and how the right material is chosen for a transformer, a motor, an inductor, a shield or a sensor.

> **Why does a 50 Hz transformer use steel sheets a third of a millimetre thick, while a 200 kHz converter uses a ceramic, and an aircraft generator uses cobalt alloy?**
> **What limits a permanent magnet's strength and its operating temperature, and why does a motor designer care about the world's dysprosium supply?**

Concept	Engineering destination
Soft versus hard figures of merit	Choosing a material class for a core or a magnet
Loss separation, Steinmetz equation	Core-loss budgets, thermal design of transformers and inductors
Classical eddy loss, skin depth, lamination	Lamination gauge, frequency limits of steels, ferrite dimensional limits
Electrical steels (GO, NO), Si content, Goss texture	Power and distribution transformers, motor and generator cores
MnZn and NiZn ferrites, Snoek limit	Switched-mode power transformers and inductors, EMI suppression
Amorphous and nanocrystalline alloys	Low-loss distribution transformers, common-mode chokes, current transformers
NdFeB, SmCo, ferrite, AlNiCo; $(BH)_{\max}$, temperature coefficients	Motors, generators, actuators, loudspeakers, sensors
Dysprosium and grain-boundary diffusion	Traction-motor magnet cost and supply risk
Shielding by flux shunting, absorption, reflection	Shielding of sensors, CRT-era and MRI-era enclosures, EMC
Hall and magnetoresistive materials	Current sensing, position sensing, read heads

Roadmap. Soft and hard (29.1) → where losses come from (29.2) → how eddy currents are suppressed (29.3) → the soft families: electrical steels, ferrites, amorphous and nanocrystalline alloys (29.4–29.6) → permanent magnets (29.7) → shielding (29.8) → selection for transformers, motors, inductors and sensors (29.9).

> **Boundary.**
> This chapter treats materials and their properties. The design of transformers, rotating machines, inductors and magnetic circuits — winding design, equivalent circuits, machine topology, thermal rating — belongs to the Electrical Energy & Machines core, and sensor and memory circuits to Electronics & Computing. Here a device appears only to fix the material requirement.

29.1 Soft and hard magnetic materials

The coercivities in Chapter 28 span seven orders of magnitude, from below 1 A/m to above 2 MA/m. The engineering world divides them into two groups with opposite jobs, and a small middle group used for recording and latching devices (semi-hard materials, roughly 1–10 kA/m).

Soft magnetic materials

A **soft** magnetic material must guide and concentrate flux produced by a winding and give it back each cycle with as little loss as possible. Its figures of merit are:

- **high saturation** J_s, so the core can be small for a given flux;
- **high permeability** μ_r, so little magnetizing current (and little winding copper) is needed for a given flux;
- **low coercivity** H_c — conventionally below about 1 kA/m, and below 100 A/m for good core materials — giving a narrow loop and low hysteresis loss;
- **high electrical resistivity** ρ (or thin section), to suppress eddy currents;
- often **low magnetostriction**, for low noise and stress insensitivity;
- a Curie temperature comfortably above the operating temperature, and stable properties with time.

From Chapter 28, the microstructural recipe for softness is easy wall motion: low anisotropy, low magnetostriction, few inclusions and precipitates, low residual stress, and grains either large (conventional alloys) or far smaller than the exchange length (nanocrystalline alloys).

Material (typical grade)	J_s (T)	Relative permeability	H_c (A/m)	Resistivity	Usual form, frequency range
Pure (low-carbon) iron	2.15	μ_{max} 5000–20 000	20–80	0.10 μΩ m	Solid parts: relays, actuators, DC
Non-oriented Si steel	2.0–2.1 (J at 5000 A/m about 1.6–1.7)	μ_{max} 3000–10 000	30–80	0.25–0.6 μΩ m	0.20–0.65 mm laminations, 50 Hz–1 kHz
Grain-oriented Si steel	2.03	μ_{max} 30 000–80 000 (rolling direction)	5–10	0.48 μΩ m	0.18–0.35 mm, 50/60 Hz transformers
Fe–6.5 % Si	1.80	μ_{max} about 20 000	about 20	0.82 μΩ m	0.05–0.10 mm, up to tens of kHz
Fe–49Co–2V	2.35	μ_{max} about 10 000	30–160	0.40 μΩ m	0.10–0.35 mm, aerospace 400 Hz–2 kHz
Fe–50Ni	1.5–1.6	μ_i 5000–15 000	5–15	0.45 μΩ m	Tape and laminations
Fe–80Ni–Mo (Permalloy, mu-metal)	0.7–0.8	μ_i 50 000–150 000	0.5–2	0.55–0.60 μΩ m	Shields, signal transformers, CTs
Fe-based amorphous (Fe–Si–B)	1.56	μ_{max} up to $\sim 10^5$ (annealed)	2–4	1.30 μΩ m	25 μm ribbon, 50 Hz to ~10 kHz
Co-based amorphous	0.55–0.8	μ_i up to 10^5	0.3–1	1.2–1.4 μΩ m	Ribbon: sensors, magnetic amplifiers
Nanocrystalline Fe–Si–B–Nb–Cu	1.24	μ_i 20 000–150 000	0.5–1.5	1.15–1.2 μΩ m	14–20 μm ribbon, 50 Hz–200 kHz
MnZn ferrite	0.45–0.53 (25 °C)	μ_i 2000–15 000	5–20	0.1–10 Ω m	Pressed cores, 10 kHz–3 MHz
NiZn ferrite	0.25–0.40	μ_i 15–1500	20–300	10^4–10^7 Ω m	1–500 MHz, EMI
Powder cores (Fe, MPP, Fe–Si–Al, Fe–Ni)	0.75–1.5	14–550 (distributed gap)	100s	Very high (insulated particles)	Chokes with DC bias, 10 kHz–1 MHz

Hard magnetic materials

A **hard** (permanent-magnet) material must *produce* flux in an external circuit, an air gap, without any current, and keep doing so against heat, opposing fields and time. Its figures of merit are:

- **high remanence** B_r, which sets the flux it can drive;
- **high intrinsic coercivity** H_{cJ} — conventionally above about 10 kA/m, and above 200 kA/m for modern magnets — which sets its resistance to demagnetization;
- a high **maximum energy product** $(BH)_{max}$, which measures how much air-gap field energy a unit volume of magnet can support (Section 29.7);
- small **temperature coefficients** of B_r and H_{cJ}, and a high Curie temperature;
- corrosion resistance, mechanical robustness and acceptable cost and supply.

From Chapter 28, the recipe for hardness is to make magnetization reversal difficult: very high uniaxial anisotropy ($Q = K/K_d \gg 1$), grains near or below single-domain size or a pinning nanostructure, magnetically isolated grain boundaries, and aligned easy axes for high remanence.

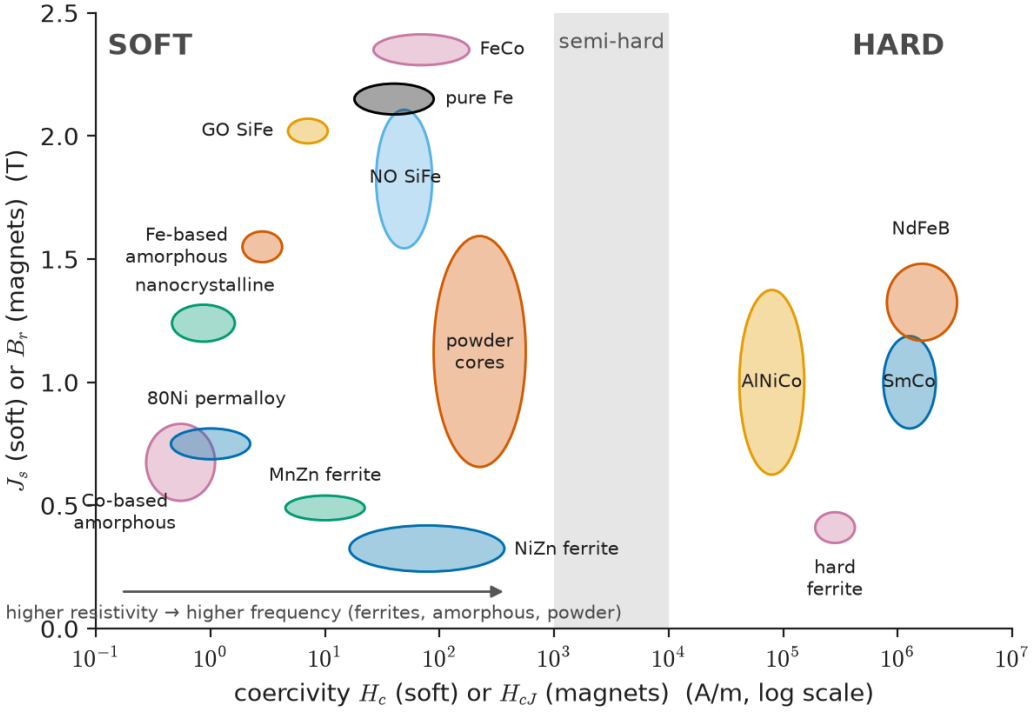


Figure 29.1 — Property map of saturation polarization versus coercivity for soft and hard magnetic material families, spanning seven decades of coercivity.

Figure 29.1 places both families on one map. The two clusters are separated by roughly four decades of coercivity: the same physical quantity, set by microstructure and anisotropy, decides whether a material will be wound into a transformer or glued into a rotor.

> **Status of the principle: soft/hard classification by coercivity**
> *Empirical.* The 1 kA/m and 10 kA/m boundaries are conventions, not physical thresholds; the classification works because real materials cluster far from them.

29.2 Magnetic losses

Where the power goes

A core driven by an alternating field dissipates energy every cycle. Measured as watts per kilogram at a stated peak induction and frequency — for example $P_{1.5/50}$ = loss at 1.5 T, 50 Hz — core loss is the headline number on every electrical-steel and ferrite datasheet. It is customarily **separated** into three parts:

$$P = P_h + P_{\mathrm{cl}} + P_{\mathrm{ex}}$$

- **Hysteresis loss** P_h: the quasi-static loop area of Chapter 28 times frequency, $P_h = fW_h$. It comes from irreversible wall jumps, and its energy per cycle does not depend on frequency.
- **Classical eddy-current loss** P_{cl}: the Joule heating of currents induced by dB/dt in a conductor of given shape and resistivity, *calculated as if the material were magnetically uniform* (no domains).
- **Excess (anomalous) loss** P_{ex}: the extra eddy-current loss due to the fact that magnetization actually changes only at moving domain walls, so that dB/dt — and the induced current — is concentrated near walls rather than spread uniformly.

Classical eddy loss in a lamination

Consider a sheet of thickness d (along x), much wider and longer than it is thick, with resistivity $\rho = 1/\sigma$, carrying a uniform induction $B(t)$ along z in its plane (Figure 29.2). Assume the frequency is low enough that the field penetrates fully (Section 29.3

makes this precise). By symmetry the induced electric field runs along y, reverses sign at the mid-plane $x = 0$, and from Faraday's law applied to a rectangular loop spanning $\pm x$:

$$E_y(x) = x\,\frac{dB}{dt}, \qquad J_y(x) = \sigma x\,\frac{dB}{dt}$$

The eddy current grows linearly from zero at the mid-plane to a maximum at the surfaces and closes at the sheet edges. The power dissipated per unit volume, averaged across the thickness, is

$$p(t) = \frac{1}{d}\int_{-d/2}^{d/2} \frac{J_y^2}{\sigma}\,dx = \frac{\sigma}{d}\left(\frac{dB}{dt}\right)^2 \int_{-d/2}^{d/2} x^2\,dx = \frac{\sigma d^2}{12}\left(\frac{dB}{dt}\right)^2$$

For sinusoidal induction $B = \hat{B}\sin\omega t$, the time average of $(dB/dt)^2$ is $\omega^2\hat{B}^2/2$, so

$$P_{\mathrm{cl}} = \frac{\sigma\omega^2\hat{B}^2 d^2}{24} = \frac{\pi^2 d^2 \hat{B}^2 f^2}{6\rho} \qquad (\mathrm{W/m^3})$$

Dividing by density ρ_m gives W/kg. Three scaling laws follow immediately, and the whole engineering of soft magnetic materials is organized around them:

$$P_{\mathrm{cl}} \propto d^2, \qquad P_{\mathrm{cl}} \propto \frac{1}{\rho}, \qquad P_{\mathrm{cl}} \propto f^2\hat{B}^2$$

Halve the lamination thickness and the classical eddy loss falls by four. Quadruple the resistivity and it falls by four. Raise the frequency eightfold and it rises sixty-four-fold.

> **Status of the principle: classical eddy-current loss** $P_{\mathrm{cl}} = \pi^2 d^2 \hat{B}^2 f^2/6\rho$
> *Derived* from Faraday's and Ohm's laws for a thin sheet with uniform, sinusoidal induction; *approximate* in practice, because it assumes full field penetration (d well below the skin depth), uniform magnetization (no domains), sinusoidal flux and sheet width much greater than thickness.

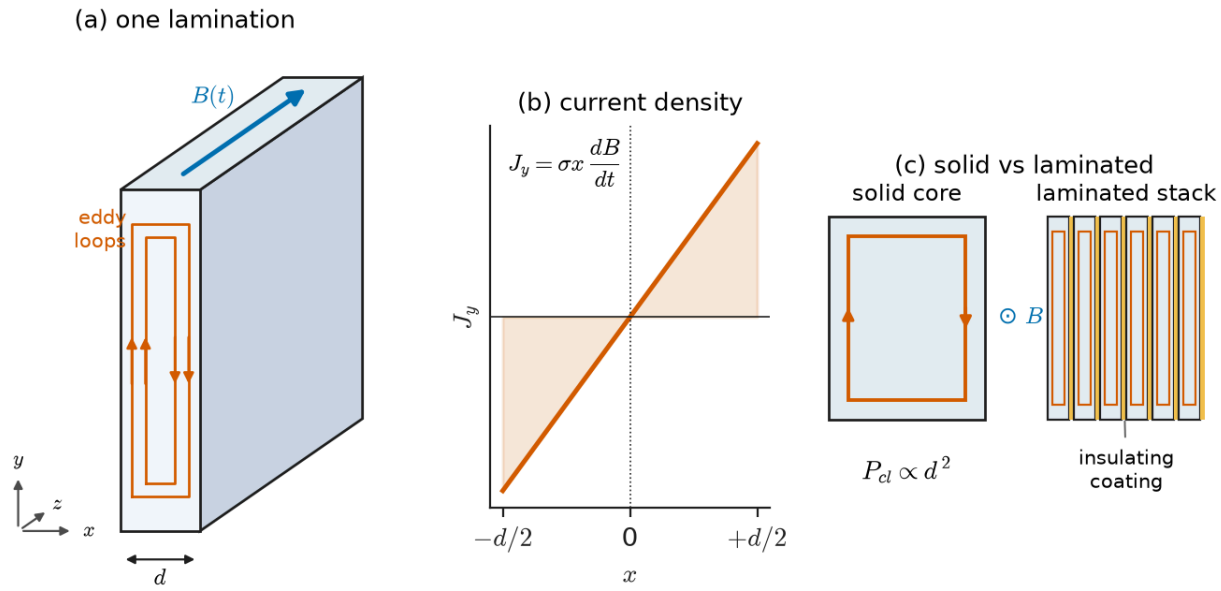


Figure 29.2 — Eddy currents in a single lamination of thickness d carrying in-plane flux, and the linear current-density profile across the thickness; a laminated stack compared with a solid core.

Worked example — classical eddy loss in a motor lamination. A non-oriented steel with 3 % Si has $\rho = 0.48$ μΩ m and density 7650 kg/m³. In 0.35 mm sheet at $\hat{B} = 1.5$ T and 50 Hz:

$$P_{\mathrm{cl}} = \frac{\pi^2(0.35\times10^{-3})^2(1.5)^2(50)^2}{6(0.48\times10^{-6})} = \frac{9.870\times1.225\times10^{-7}\times2.25\times2500}{2.88\times10^{-6}} = 2.36\times10^{3}\ \mathrm{W/m^3}$$

or $2.36\times10^3/7650 = 0.31$ W/kg. Variations:

- In 0.50 mm sheet: $\times (0.50/0.35)^2 = 2.04$, giving 0.63 W/kg.
- In 0.35 mm sheet of pure iron ($\rho = 0.10$ μΩ m, density 7870 kg/m³): about 1.45 W/kg — nearly five times worse. This is why silicon is added (Section 29.4).
- In 25 μm Fe-based amorphous ribbon ($\rho = 1.30$ μΩ m, density 7180 kg/m³) at 1.4 T and 60 Hz: $\pi^2 (25 \times 10^{-6})^2 (1.4)^2 (60)^2 / (6 \times 1.30 \times 10^{-6}) = 5.6$ W/m³, i.e. 7.8×10^{-4} W/kg. The classical eddy loss of amorphous ribbon is negligible; its measured total of about 0.2 W/kg is almost all hysteresis and excess loss.

A typical datasheet total for a good 0.35 mm grade at 1.5 T, 50 Hz is about 2.3–2.7 W/kg, so the classical term is only one-eighth of it. The rest is hysteresis and excess loss.

Excess loss and domain walls

Measured eddy losses in electrical steels are commonly 1.5 to 3 times the classical value. The reason is visible under a domain microscope. Magnetization changes only where walls pass, so dB/dt is not uniform. It is large near each moving wall and zero in between. Because power goes as the *square* of the local dB/dt, concentrating the same average change into narrow regions raises the loss. A simple model of a sheet with equally spaced 180° walls (the Pry–Bean model) gives an eddy loss that exceeds the classical value by a factor that grows with the ratio of domain width $2L$ to sheet thickness d — roughly $1.6 \times (2L/d)$ when the domains are much wider than the sheet is thick. Wide domains mean high excess loss.

Bertotti's statistical theory treats the sheet as a population of "magnetic objects" (wall groups) that become active as the field rises, and gives, for sinusoidal flux,

$$P_{\text{ex}} = 8 \sqrt{\sigma G S V_0}\, \hat{B}^{3/2} f^{3/2}$$

where S is the sheet cross-section, $G \approx 0.136$ a dimensionless constant, and V_0 a field (A/m) characterizing the statistical distribution of local pinning fields, fitted to measurement. The characteristic $f^{3/2}$ dependence distinguishes excess loss from the f^1 of hysteresis and the f^2 of classical eddy loss.

> **Status of the principle: excess loss $P_{\text{ex}} \propto (\hat{B} f)^{3/2}$**
> *Approximate.* It follows from a statistical model of domain-wall dynamics with one fitted parameter V_0; it describes laminated steels well at power frequencies and up to a few kHz, but is a phenomenological description, not a prediction from microstructure.

This explains a striking engineering fact: grain-oriented steel, with its very large grains and wide domains, has an especially large excess loss. Scratching the surface with a laser or etched grooves at intervals of a few millimetres (**domain refinement**) introduces stress patterns that subdivide the domains. Hysteresis loss rises slightly but excess loss falls more, cutting total loss by 5–15 % (Section 29.4).

Separating the three

Divide the loss by frequency to get the energy per cycle:

$$\frac{P}{f} = W_h + C_{\text{cl}}\, f + C_{\text{ex}}\, f^{1/2}$$

Measured at constant $\hat{B}$ over a range of frequencies, P/f plotted against f (Figure 29.3) has an intercept at $f \to 0$ equal to the hysteresis energy W_h, a calculable linear part $C_{\text{cl}} f$, and a remainder that grows as $\sqrt{f}$, the excess loss. This is how loss separation is done in practice (measurement methods: Chapter 52).

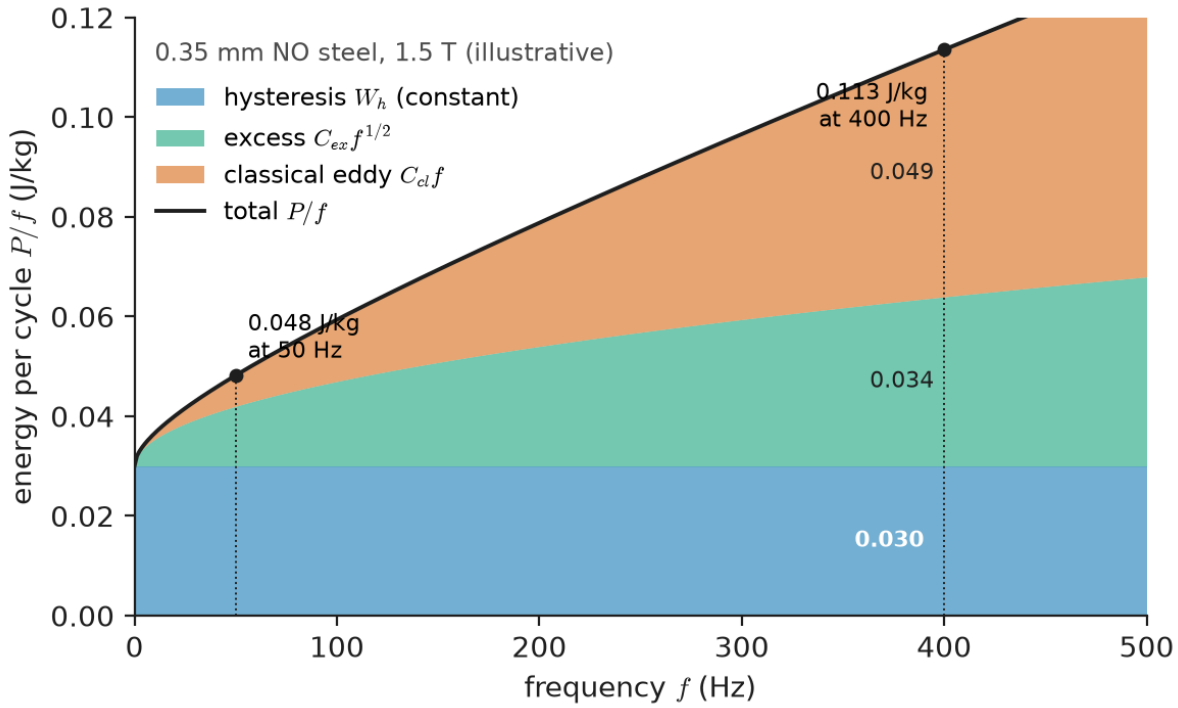


Figure 29.3 — Energy loss per cycle versus frequency at constant peak induction, separated into hysteresis, classical eddy and excess contributions.

Worked example — what happens to a 50 Hz steel at 400 Hz. The 0.35 mm, 3 % Si steel above is measured at 1.5 T, 50 Hz to have a total loss of 2.41 W/kg, which separates into hysteresis 1.50 W/kg, classical eddy 0.31 W/kg and excess 0.60 W/kg. Per cycle, these are $W_h = 0.030$ J/kg, $W_{\mathrm{cl}} = 0.0062$ J/kg and $W_{\mathrm{ex}} = 0.012$ J/kg. Scale each to 400 Hz at the same 1.5 T:

- hysteresis: energy per cycle constant, so $P_h = 0.030 \times 400 = 12.0$ W/kg;
- classical: $P \propto f^2$, so $0.31 \times (400/50)^2 = 0.31 \times 64 = 19.8$ W/kg;
- excess: $P \propto f^{3/2}$, so $0.60 \times 8^{3/2} = 0.60 \times 22.6 = 13.6$ W/kg.

The total is 45.3 W/kg, nineteen times the 50 Hz value for an eightfold frequency rise, and classical eddy loss has gone from the smallest term to the largest. Switching to 0.20 mm sheet reduces the classical term by $(0.20/0.35)^2 = 0.33$ to 6.5 W/kg. Taking the other two as unchanged (as a first approximation), the total becomes about 32 W/kg. This is the materials reason why the stators of high-speed traction motors and aerospace generators, with fundamental frequencies of several hundred hertz to over a kilohertz, use 0.20–0.27 mm laminations and higher-silicon grades.

The Steinmetz equation

For ferrites and for design work in general, separating the loss is less useful than a compact fit. In 1892 Charles Steinmetz observed that hysteresis loss in iron scaled as $\hat{B}^{1.6}$. The generalized form now universal in power electronics is

$$P_v = k\, f^{\alpha} \hat{B}^{\beta}$$

with P_v the loss per unit volume and k, α, β fitted to measured data over a limited range of frequency, induction and temperature. For power ferrites, typical fits are $\alpha \approx 1.1$–1.8 and $\beta \approx 2.3$–3. For laminated steels at power frequency, $\alpha \approx 1.2$–1.6 and $\beta \approx 1.6$–2.2. The exponent α is between 1 (pure hysteresis) and 2 (pure classical eddy loss); the exponent β above 2 reflects the Rayleigh-like $\hat{B}^3$ behaviour at low induction (Chapter 28) and approach to saturation at high induction.

> **Status of the principle: Steinmetz equation $P_v = k f^{\alpha} \hat{B}^{\beta}$**
> *Empirical.* It is a power-law fit valid only within the range of frequency, flux density and temperature over which its coefficients were fitted, and only for sinusoidal flux; for non-sinusoidal (PWM) waveforms, modified forms such as the improved generalized Steinmetz equation (iGSE), which use dB/dt, are needed, and DC bias and relaxation effects need further corrections.

Worked example — rescaling a ferrite datasheet. A MnZn power ferrite has a datasheet loss of 375 kW/m³ at 100 kHz, 200 mT and 100 °C. Over the range 50–200 kHz and 100–250 mT the manufacturer's fit gives $\alpha = 1.4$ and $\beta = 2.6$. A transformer is to run at 150 kHz with $\hat{B} = 150$ mT. The loss ratio is

$$\frac{P_v}{375\ \text{kW/m}^3} = \left(\frac{150}{100}\right)^{1.4}\left(\frac{150}{200}\right)^{2.6} = 1.764 \times 0.473 = 0.835$$

so $P_v = 313$ kW/m³. For an ETD34 core of effective volume $V_e = 7.64$ cm³ $= 7.64 \times 10^{-6}$ m³, the core loss is $313 \times 10^3 \times 7.64 \times 10^{-6} = 2.4$ W. Two cautions: the same calculation at 25 °C could give a loss 1.5–2 times higher, because the ferrite's loss minimum was placed near 100 °C (Section 29.5); and the PWM waveform of a real converter changes the result, which is why the coefficients are only an engineering starting point.

Loss as complex permeability

At high frequency and low induction, where the loop is nearly elliptical, losses are expressed through the **complex permeability** $\mu = \mu' - j\mu''$. A winding on such a core behaves as an inductance $L \propto \mu'$ in series with a resistance $R \propto \omega\mu''$, and the **loss tangent** $\tan\delta_m = \mu''/\mu'$ plays the same role as the dielectric loss tangent of Chapter 26. Ferrite datasheets plot μ' and μ'' against frequency (Figure 29.5). Filter-inductor designers use $\tan\delta_m/\mu_i$ (the relative loss factor) to compare materials at small signal; EMI designers deliberately exploit large μ''.

29.3 Eddy-current suppression

Skin depth: the frequency limit of a conducting core

The eddy-current derivation assumed the field penetrated the whole lamination. In a thick conductor the eddy currents themselves shield the interior. Combining Faraday's and Ampère's laws with Ohm's law in a linear conductor of permeability $\mu = \mu_0\mu_r$ and conductivity σ (neglecting displacement current) gives a diffusion equation

$$\nabla^2 \mathbf{H} = \mu\sigma\frac{\partial \mathbf{H}}{\partial t}$$

For a sinusoidal field applied at the surface of a half-space, the solution decays and lags with depth x as $H = H_0 e^{-x/\delta} e^{j(\omega t - x/\delta)}$, with **skin depth**

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{\rho}{\pi f \mu_0 \mu_r}}$$

> **Status of the principle: skin depth** $\delta = \sqrt{\rho/\pi f \mu_0 \mu_r}$
> *Derived* from Maxwell's equations for a linear, homogeneous conductor at frequencies where displacement current is negligible; *approximate* for ferromagnets, whose permeability is nonlinear and amplitude-dependent, so δ is an effective value.

A magnetic core works properly only if its smallest conducting dimension is less than about the skin depth. Otherwise flux crowds into a surface layer, the interior carries little flux, the effective permeability falls, and the loss is concentrated where the field is largest. The high permeability that makes a core useful makes its skin depth small.

Worked example — skin depths of core materials.

- 3 % Si steel, $\rho = 0.48$ μΩ m, $\mu_r = 5000$, at 50 Hz: $\delta = \sqrt{0.48 \times 10^{-6}/(\pi \times 50 \times 4\pi \times 10^{-7} \times 5000)} = \sqrt{4.86 \times 10^{-7}} = 0.70$ mm. A 0.35 mm lamination is half a skin depth thick: safely in the classical regime. With $\mu_r = 1000$ (typical at high induction), $\delta = 1.56$ mm.
- The same steel at 400 Hz, $\mu_r = 5000$: $\delta = 0.70/\sqrt{8} = 0.25$ mm. A 0.35 mm lamination is now thicker than the skin depth, and thinner gauges are needed.
- Copper at 50 Hz ($\rho = 1.72 \times 10^{-8}$ Ω m, $\mu_r = 1$): $\delta = 9.3$ mm. A copper conductor is far less affected than a steel core at the same frequency, because $\mu_r = 1$.
- MnZn ferrite, $\rho = 5$ Ω m, $\mu_r = 2000$, at 1 MHz: $\delta = \sqrt{5/(\pi \times 10^6 \times 4\pi \times 10^{-7} \times 2000)} = 25$ mm. Even at a megahertz, a ferrite's resistivity gives skin depths larger than most cores.

The steel results show why laminated steels are confined to frequencies of up to a few kilohertz, and why their gauge falls as frequency rises.

Strategies for suppressing eddy currents

Every strategy follows from $P_{\mathrm{cl}} \propto d^2/\rho$ and $\delta \propto \sqrt{\rho/\mu_r f}$: reduce the dimension through which eddy currents can circulate, or raise resistivity.

1. **Lamination.** Divide the core into sheets parallel to the flux, insulated from each other, so eddy currents circulate only within a thickness d. Gauges run from 0.65 mm (small 50 Hz motors) through 0.23–0.35 mm (transformers, efficient motors) to 0.10 mm and below (aerospace, kHz). Thinner gauges cost more to roll, punch and stack, and the **stacking factor** (fraction of core volume that is steel, about 0.95–0.98 for steels, 0.80–0.90 for amorphous ribbon) falls as coating thickness becomes comparable to sheet thickness.
2. **Interlaminar insulation.** An inorganic phosphate, oxide or organic-inorganic coating 0.5–5 μm thick insulates the sheets. It must survive punching and stress-relief annealing. Burrs from blunt punching dies, or welding and clamping across the stack, short adjacent laminations and create large eddy loops — a classic cause of a hot spot in a new machine.
3. **Raise resistivity by alloying.** Silicon and aluminium in iron, molybdenum in nickel–iron, and the metalloids (B, Si, P) in amorphous alloys increase electron scattering (Chapter 25) without destroying ferromagnetism. Iron goes from 0.10 μΩ m to about 0.48 μΩ m at 3 % Si and 1.3 μΩ m in Fe–Si–B glass.
4. **Use an insulator.** Ferrites have resistivities 10^6 to 10^{14} times that of iron. Eddy loss is negligible up to the MHz range, at the cost of a saturation about one quarter of iron's.
5. **Subdivide in three dimensions.** Powder cores and **soft magnetic composites (SMC)** are made of iron or alloy particles (10–300 μm) individually coated with an insulating layer and pressed. Eddy currents are confined to single particles in every direction, so SMC supports three-dimensional flux paths (claw-pole and axial-flux machines) at the cost of lower permeability (a few hundred) and higher hysteresis loss from deformed particles.
6. **Segment permanent magnets.** Sintered NdFeB is a metal ($\rho \approx 1.4$–1.6 μΩ m). Slot harmonics and PWM ripple induce eddy currents in rotor magnets, which heat them and lower their coercivity. Dividing each magnet into insulated segments along its length cuts the loss — the same d^2 law applied to a hard material.

29.4 Electrical steels

Electrical steels — iron with silicon, rolled to sheet — are by tonnage the dominant soft magnetic material: roughly 15 million tonnes are produced each year for transformers, motors and generators. Two families exist.

Why silicon

Adding silicon to iron, typically 1–3.5 wt%, does four useful things and one harmful one:

1. **Raises resistivity**, roughly as $\rho \approx (0.10 + 0.12\,w_{\mathrm{Si+Al}})$ μΩ m for w in wt% (a rough fit): 0.48 μΩ m at 3.2 %, cutting classical eddy loss by a factor of about five compared with pure iron.
2. **Lowers magnetocrystalline anisotropy** (K_1 falls from 48 to about 35 kJ/m³ at 3 %) and **magnetostriction**, which vanishes near 6.5 % Si. Both reduce hysteresis loss and noise.
3. **Closes the γ loop.** Silicon is a ferrite (bcc) stabilizer. Above about 2 % Si, iron stays bcc all the way to the melting point. The sheet can then be annealed at 1000–1200 °C to grow large grains and develop texture without passing through the α–γ transformation, which would destroy them (phase diagrams: Chapter 7).
4. **Suppresses magnetic ageing** by reducing carbon solubility, in combination with decarburization to below about 30 ppm C.
5. **Dilutes the magnetization** (harmful): J_s falls from 2.15 T to about 2.03 T at 3 % and 1.80 T at 6.5 %. Above about 3.5 % the alloy also becomes brittle (ordering of Fe₃Si) and cannot be cold-rolled conventionally.

The practical ceiling is therefore about 3.2–3.5 % Si (sometimes with 0.5–1 % Al, which raises resistivity similarly). Fe–6.5 % Si, with near-zero magnetostriction and $\rho = 0.82$ μΩ m, is made by depositing silicon on 3 % sheet by chemical vapour deposition and diffusing it in, and is used in thin gauges for high-frequency reactors and quiet transformers.

Grain-oriented steel and the Goss texture

Iron's easy axes are the cube edges ⟨100⟩, and the permeability along an easy axis greatly exceeds that along ⟨111⟩. In a transformer core the flux runs along the limbs, in one direction. If every grain could have a ⟨100⟩ axis parallel to the rolling direction of the strip, the strip would be "a single crystal in the direction that matters".

N. P. Goss found the process in 1934. **Grain-oriented (GO)** steel is processed so that almost all grains have the **Goss texture** {110}⟨001⟩: a {110} plane in the sheet surface and a ⟨001⟩ direction along the rolling direction (Figure 29.4). The route combines cold rolling, a decarburizing primary recrystallization anneal, and a high-temperature box anneal at about 1150–1200 °C. During the box anneal, fine **inhibitor** particles (MnS, AlN) pin ordinary grain boundaries, and only the rare Goss-oriented grains — whose boundaries move fastest — grow by **secondary recrystallization** (abnormal grain growth) to consume the whole strip. The final grains are millimetres to centimetres across. Processing–microstructure relationships of this kind are developed in Chapter 10.

- **Conventional GO (CGO)**: ⟨001⟩ axes within about 7° of the rolling direction; J at 800 A/m (J_{800}) of about 1.80–1.85 T.
- **High-permeability GO (Hi-B)**: AlN inhibition and heavier final reduction give deviations of about 3°; $J_{800} \approx$ 1.88–1.94 T.
- **Domain-refined Hi-B**: laser-scribed or groove-etched to subdivide the wide domains (Section 29.2).

A glass-like forsterite (Mg_2SiO_4) layer formed during the box anneal, plus a phosphate coating on top, insulates the sheet and, being applied hot with a lower thermal expansion than the steel, puts the steel into **tension** of a few MPa along the rolling direction. Since $\lambda_{100} > 0$, tension along ⟨001⟩ reinforces that easy axis, narrows the domains, and lowers loss and magnetostriction — a deliberate use of the magnetoelastic energy of Chapter 28.

GO steel's weakness is that its properties are excellent only along the rolling direction. At 55° (along ⟨111⟩) and 90° the permeability is far lower and the loss higher. Transformer cores are therefore built so the flux follows the rolling direction everywhere, with mitred (often step-lap) corner joints where the flux must turn. GO steel is not used in rotating machines, where flux direction rotates.

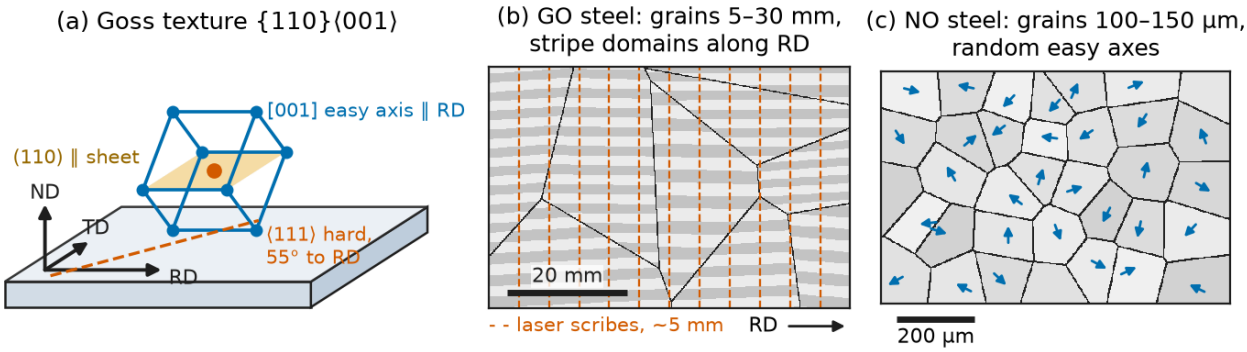


Figure 29.4 — The Goss texture {110}<001> in grain-oriented steel: orientation of the bcc cube relative to the rolled sheet, and the contrasting grain structures of grain-oriented and non-oriented steels.

Non-oriented steel

Motors and generators need good properties in every direction within the sheet plane. **Non-oriented (NO)** steels have a weak or random texture, 0–3.5 % (Si + Al), and gauges of 0.20–0.65 mm. Two processing states exist:

- **fully processed** steel is annealed at the mill and sold with final properties; laminations are punched and used;
- **semi-processed** steel is given a final decarburizing anneal by the customer after punching, which also relieves punching stress.

The grain size of NO steel is optimized at roughly 100–150 µm. Larger grains reduce hysteresis loss (Chapter 28: $H_c \propto 1/D$) but increase the domain width and with it the excess loss, so total loss passes through a minimum. Low-loss grades seek a favourable texture with more ⟨100⟩ directions in the sheet plane (cube or η-fibre components) — a modest version of the Goss idea.

Punching damages the material near the cut edge over a width of roughly one sheet thickness, raising local coercivity through plastic deformation and residual stress (Chapter 13). In the narrow teeth of small, high-speed motors this damaged fraction is

significant, which is why stress-relief annealing at about 750–800 °C, laser cutting, or wire-EDM prototypes are compared carefully.

Grades and representative properties

European grades encode the guaranteed maximum loss: in a name such as **M270-35A**, 270 means 2.70 W/kg at 1.5 T, 50 Hz; 35 means 0.35 mm; A means non-oriented, fully processed. For GO grades (suffix S for conventional, P for high-permeability) the number refers to 1.7 T, 50 Hz.

Grade (EN style)	Type	Thickness (mm)	Si + Al (wt%, typical)	ρ (μΩ m, typical)	Guaranteed max. loss (W/kg)	Typical polarization
M235-35A	NO, high grade	0.35	about 3.3	0.59	2.35 at 1.5 T, 50 Hz	$J_{5000} \approx 1.60$ T
M270-35A	NO	0.35	about 3.0	0.52	2.70 at 1.5 T, 50 Hz	$J_{5000} \approx 1.60$ T
M400-50A	NO	0.50	about 2.0	0.42	4.00 at 1.5 T, 50 Hz	$J_{5000} \approx 1.63$ T
M800-65A	NO, low Si	0.65	about 1.0	0.25	8.00 at 1.5 T, 50 Hz	$J_{5000} \approx 1.70$ T
NO thin gauge (e.g. 0.20 mm)	NO, high frequency	0.20	about 3.3	0.55–0.60	about 15–20 at 1.0 T, 400 Hz	$J_{5000} \approx 1.60$ T
M130-27S	GO, conventional	0.27	about 3.1	0.48	1.30 at 1.7 T, 50 Hz	$J_{800} \approx 1.80$ T
M103-27P	GO, Hi-B	0.27	about 3.1	0.48	1.03 at 1.7 T, 50 Hz	$J_{800} \approx 1.88$ T
Hi-B, domain-refined	GO	0.23	about 3.1	0.48	about 0.70–0.85 at 1.7 T, 50 Hz	$J_{800} \approx 1.90$ T

Notice two lessons in the table. Higher silicon buys lower loss but lower polarization: M800-65A reaches 1.70 T at 5000 A/m but loses 8 W/kg; M235-35A loses 2.35 W/kg but reaches only 1.60 T. And GO steel at 1.7 T loses less than half as much as the best NO steel at 1.5 T — the value of texture when the flux direction is fixed.

29.5 Ferrites

Why ferrites

Chapter 28 showed that the spinel ferrites are ferrimagnetic oxides, with saturation set by sublattice counting and resistivity far above any metal. Above roughly 20–50 kHz, no practical lamination is thin enough for steel, and ferrites take over. Two families dominate soft-ferrite production.

MnZn ferrites, $Mn_{1-x}Zn_xFe_2O_4$ with a small excess of iron, have the higher permeability and saturation and the lower resistivity:

- μ_i about 2000–15 000, B_s about 0.45–0.53 T at 25 °C and 0.35–0.43 T at 100 °C;
- T_C about 150–300 °C, falling as Zn (and permeability) increase;
- resistivity about 0.1–10 Ω m — small for an oxide, because of Fe^{2+}–Fe^{3+} electron hopping on B sites — with relative permittivity of order 10^4–10^5 at low frequency from conducting grains separated by resistive grain boundaries;
- used from about 10 kHz to 1–3 MHz: switched-mode power transformers and inductors, signal and pulse transformers, common-mode chokes.

NiZn ferrites, $Ni_{1-x}Zn_xFe_2O_4$, have much higher resistivity because the Fe^{2+} content is negligible:

- μ_i about 15–1500, B_s about 0.25–0.40 T;
- resistivity 10^4–10^7 Ω m, permittivity about 10–20;
- used from about 1 MHz to several hundred MHz: RF transformers and baluns, antenna rods, and EMI-suppression beads and cable cores.

Microstructure and processing

Ferrites are ceramics: oxides are mixed, calcined, milled, pressed to shape, and sintered at about 1100–1400 °C (Chapter 17). In MnZn ferrite, the oxygen partial pressure during sintering and cooling must be controlled precisely to set the Fe^{2+} content, which sets K_1, resistivity and loss. Small additions of CaO and SiO_2 segregate to grain boundaries and form thin insulating layers that raise the bulk resistivity and cut eddy losses inside grains. Grain size is a frequency-dependent compromise. Large grains (10–20 μm) and low porosity give high permeability for low-frequency, high-μ grades. Small grains (a few μm) give lower eddy and residual losses for MHz power grades.

Loss and temperature

Power ferrites are characterized by Steinmetz-type data at stated frequency, induction and temperature (Section 29.2). Typical values for current power grades:

Ferrite type (typical)	μ_i	B_s at 100 °C (T)	Core loss at 100 kHz, 200 mT, 100 °C (kW/m³)	Typical use
MnZn power, 25–200 kHz	2000–3000	0.38–0.42	300–400	Flyback, forward, LLC transformers
MnZn low-loss, wide temperature	3000	0.40–0.43	about 300, flatter versus T	Automotive, wide ambient
MnZn high-frequency, 0.3–1 MHz	1400–2000	0.38–0.40	Specified at 500 kHz, 50 mT: about 100–200	Resonant, GaN converters
MnZn high-μ	5000–15 000	0.20–0.38	Not a power grade	Signal, CMC, pulse
NiZn	15–1500	0.20–0.35	Not a power grade at 100 kHz	RF, EMI beads

The loss of a MnZn power grade has a **minimum** with temperature, usually designed to lie between 80 and 100 °C. The loss minimum follows the zero crossing of the anisotropy constant K_1, which is tuned by the Fe^{2+} (and sometimes Co^{2+}) content (Chapter 28). The minimum provides thermal stability: a core that warms toward its minimum loses less. But a core pushed beyond it heats faster as it gets hotter and can run away thermally, especially since B_s is also falling. Datasheet loss curves are specified at temperature for this reason.

Frequency limits: Snoek's law and dimensional resonance

Why does a high-permeability ferrite fail at high frequency, and why can a low-permeability ferrite go higher? Above some frequency the magnetization can no longer follow the field: the precession of moments about their effective anisotropy field reaches resonance, μ' falls and μ'' peaks (Figure 29.5). For a cubic ferrite in which magnetization proceeds by rotation, the resonance frequency f_r and initial permeability are linked by **Snoek's law**:

$$(\mu_i - 1) f_r \approx \frac{\gamma}{3\pi} \mu_0 M_s, \qquad \frac{\gamma}{3\pi} = 1.87 \times 10^{10}\ \text{Hz/T}$$

where $\gamma = 1.76 \times 10^{11}$ rad s^{-1} T^{-1} is the electron gyromagnetic ratio. The product of permeability and bandwidth is fixed by the saturation. For MnZn ferrite with $\mu_0 M_s = 0.5$ T, the right-hand side is 9.3 GHz; with $\mu_i = 2000$, $f_r \approx 4.7$ MHz. For NiZn with $\mu_0 M_s = 0.35$ T and $\mu_i = 100$, $f_r \approx 66$ MHz. This is why EMI beads use low-μ NiZn: above f_r, μ'' dominates and the bead behaves as a frequency-dependent resistor that absorbs interference in the 30–1000 MHz range.

> **Status of the principle: Snoek's law**
> *Approximate.* It follows from the gyromagnetic resonance of moments rotating against an isotropic distribution of anisotropy fields, ignoring domain-wall resonance and demagnetizing effects; hexagonal ferrites with planar anisotropy, and composites, can exceed it.

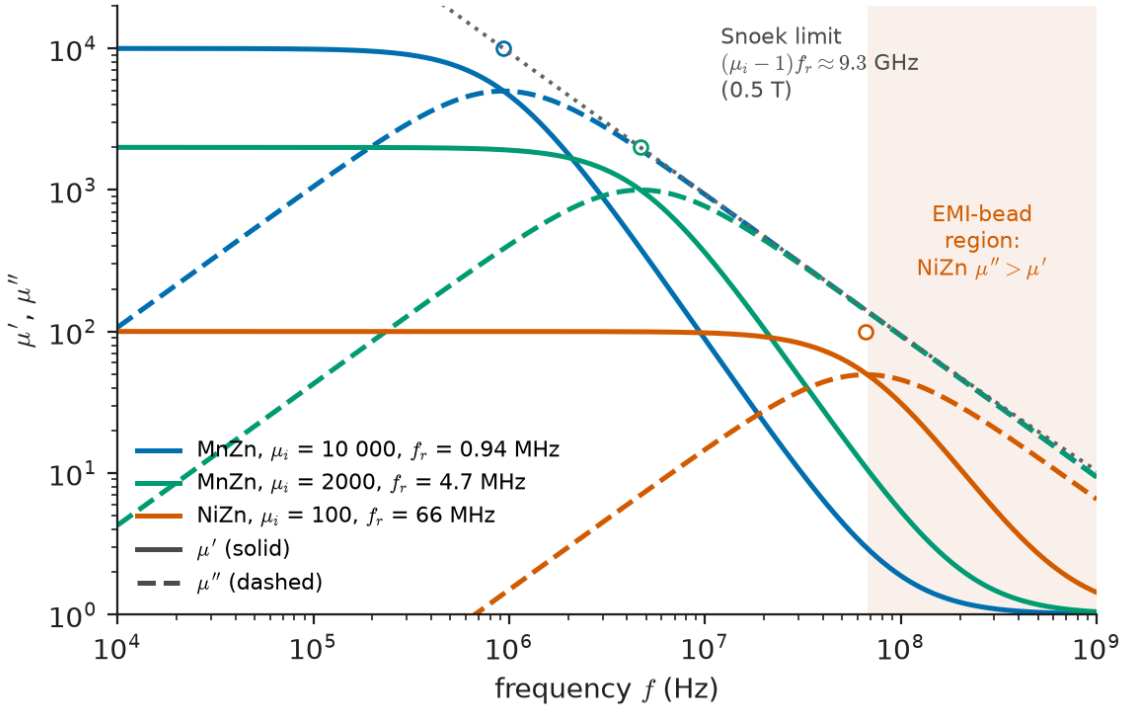


Figure 29.5 — Complex permeability spectra of MnZn and NiZn ferrites, showing the fall of μ′ and the peak of μ″ at frequencies limited by Snoek's law.

MnZn ferrites meet a second limit. Their high permittivity slows the electromagnetic wave inside the core. The wavelength in the material is $\lambda = c/(f\sqrt{\varepsilon_r \mu_r})$. With $\varepsilon_r \sim 10^4$–10^5 and $\mu_r = 2000$ at 1 MHz, λ is only about 2–7 cm. A core whose cross-section approaches half a wavelength supports a standing wave — **dimensional resonance** — and its effective permeability and loss change abruptly. Large MnZn cores for MHz converters are therefore assembled from smaller pieces, or replaced by lower-permittivity grades.

29.6 Amorphous and nanocrystalline magnetic alloys

Amorphous alloys

Chapter 2 introduced amorphous solids: atoms with short-range but no long-range order. Some iron- and cobalt-based alloys containing 15–25 at% metalloids (B, Si, P, C) can be frozen into a glass by **rapid solidification**: a molten jet is poured onto a copper wheel spinning at about 20–30 m/s, cooling at roughly 10^6 K/s into a continuous ribbon 15–30 μm thick and up to about 200 mm wide (planar-flow casting).

With no crystal lattice there is no magnetocrystalline anisotropy on any scale that matters, and with no grain boundaries there are few pinning sites. Walls are wide, and coercivity is low — 2–4 A/m for Fe-based glasses and below 1 A/m for Co-based ones. Resistivity is high (1.2–1.4 μΩ m) because the disordered structure scatters electrons strongly. The ribbon is thin by nature. Classical eddy loss is negligible at power frequency (worked example, Section 29.2).

Property	Fe-based amorphous (Fe–Si–B, e.g. Metglas 2605SA1)	Co-based amorphous (e.g. Metglas 2714A)	3 % Si GO steel (Hi-B, for comparison)
J_s (T)	1.56	0.57	2.03
H_c (A/m)	2–4	about 0.3–1	5–10
ρ (μΩ m)	1.30	1.4	0.48
λ_s	$+27 \times 10^{-6}$	about 0	$\lambda_{100} \approx +23 \times 10^{-6}$
T_C (°C)	about 395	about 225	about 745
Crystallization temperature (°C)	about 500	about 550	—
Thickness	about 25 μm	about 15–25 μm	0.23–0.30 mm
Density (g/cm³)	7.18	7.6–7.9	7.65
Core loss, 50–60 Hz, 1.3–1.4 T (W/kg)	about 0.1–0.2	—	about 0.4–0.5

The Fe-based amorphous ribbon is the material of the opening example. Its loss at distribution-transformer inductions is roughly a quarter to a third of the best GO steel. But its drawbacks dictate how it is used:

- **lower saturation** (1.56 T, designed to operate at 1.3–1.4 T against 1.6–1.7 T for GO): the core must be larger and heavier for the same rating;
- **lower stacking factor** (about 0.80–0.88) because of ribbon surface roughness;
- **magnetostriction** of 27×10^{-6}: louder and stress-sensitive; cores must be supported, not clamped;
- **embrittlement**: the ribbon must be annealed (typically 350–400 °C in a longitudinal field) to relieve casting stresses and induce a favourable anisotropy, and after annealing it is brittle and hard to handle;
- **thermal limits**: it crystallizes, with catastrophic loss of softness, above about 500 °C, and its Curie temperature (about 395 °C) is low, so properties fall more quickly with temperature than those of steel;
- **form**: wound cores of thin ribbon suit distribution transformers, not punched motor laminations.

Co-based amorphous alloys trade saturation for **zero magnetostriction**, giving the highest permeabilities and lowest losses of any ribbon. They are used in magnetic amplifiers, fluxgate and magnetic-field sensors, high-accuracy current transformers and saturable reactors, and in anti-theft tags.

> **Status of the principle: low coercivity of amorphous alloys from absent crystal anisotropy**
> *Approximate.* It explains why amorphous metals are soft; residual coercivity is controlled by quenched-in stress coupled to magnetostriction, by surface roughness and by any incipient crystallization, which is why annealing matters as much as composition.

Nanocrystalline alloys and random anisotropy

In 1988 Yoshizawa and co-workers at Hitachi Metals annealed an amorphous $Fe_{73.5}Si_{13.5}B_9Nb_3Cu_1$ ribbon at about 540 °C and found it became *softer* on crystallizing, contrary to all previous experience. Copper clusters nucleate a very high density of bcc Fe–Si crystallites, and niobium, which is rejected into the remaining amorphous matrix, stops them growing. The result — **Finemet**, also sold as Vitroperm and similar — is about 70–80 % by volume of Fe–Si grains 10–15 nm across, embedded in a ferromagnetic amorphous matrix.

Why should 10 nm crystals, each with an anisotropy constant $K_1 \approx 8$ kJ/m³ in random orientations, behave as if they had almost none? The answer is **Herzer's random-anisotropy model**. Exchange tries to keep the magnetization uniform over a length scale of order $L_0 = \sqrt{A/K_1}$. With $A \approx 10^{-11}$ J/m and $K_1 = 8 \times 10^3$ J/m³ for Fe–20 at% Si, $L_0 \approx 35$ nm. When the grain size D is much smaller than this, the magnetization cannot follow the easy axis of each grain. It averages over the $N = (L_{\text{ex}}/D)^3$ grains within one exchange volume of size L_{ex}, whose random anisotropies largely cancel, leaving a residual of order $K_1/\sqrt{N}$. But a weaker average anisotropy lets exchange act over a longer distance, $L_{\text{ex}} = \sqrt{A/\langle K \rangle}$, which averages over still more grains. Making L_{ex} and $\langle K \rangle$ self-consistent gives

$$\langle K\rangle \approx \frac{K_1^4 D^6}{A^3}, \qquad H_c \propto \langle K\rangle \propto D^6$$

For $D = 10$ nm: $\langle K\rangle \approx (8\times10^3)^4(10^{-8})^6/(10^{-11})^3 = 4$ J/m³, about two thousand times smaller than K_1. This sixth-power law explains why the nanocrystalline alloy is soft only when the grains stay small: an anneal a few tens of degrees too hot coarsens them and coercivity rises steeply. Above about L_0 the conventional $H_c \propto 1/D$ of Chapter 28 takes over. Together the two regimes make coercivity versus grain size an inverted V.

> **Status of the principle: random-anisotropy model, $H_c \propto D^6$**
> *Approximate*. It is a scaling argument valid for grain sizes well below the exchange length, with exchange-coupled grains of random orientation; it gives the trend and the order of magnitude, and residual magnetoelastic and induced anisotropies set the actual floor of H_c.

The magnetostriction also cancels. The Fe–Si grains have negative λ_s and the amorphous matrix positive, and at the right silicon content the volume average is near zero (about $+2\times10^{-6}$). The result:

Property	Nanocrystalline Fe–Si–B–Nb–Cu (Finemet type)
J_s	1.20–1.25 T
μ_i	20 000–150 000 (adjustable by field annealing)
H_c	0.5–1.5 A/m
ρ	1.15–1.2 μΩ m
λ_s	about $+2\times10^{-6}$
T_C	about 570 °C (crystalline phase); matrix lower
Thickness	14–20 μm ribbon

Nanocrystalline alloys combine a saturation two to three times that of ferrite with permeability and loss comparable to or better than the best ferrites up to about 100–200 kHz, and with far better temperature stability. They dominate **common-mode chokes** (high μ over a broad band), precision **current transformers** and residual-current devices, high-power medium-frequency transformers in solid-state transformers and fast chargers, and wireless-charging shields. Annealing under a transverse field induces an anisotropy that gives a flat, linear loop with low permeability for DC-tolerant applications; a longitudinal field gives a square loop for magnetic switches. Their limits are the same as for amorphous ribbon: wound cores, brittleness after annealing, and cost. Fe–Zr–B ("Nanoperm") and high-Fe variants reach $J_s \approx 1.5$–1.6 T at somewhat higher loss.

29.7 Permanent-magnet materials

The second quadrant and the energy product

A permanent magnet works in the **second quadrant** of its hysteresis loop, where H is negative: inside a magnet that drives flux through an air gap, the field H points against M (Chapter 28's demagnetizing field). With no current in the circuit, Ampère's law around the loop gives $H_m l_m + H_g l_g = 0$ for magnet length l_m and gap l_g, if the iron has negligible reluctance. Flux continuity gives $B_m A_m = B_g A_g$, neglecting leakage. Multiplying,

$$B_g^2 \frac{V_g}{\mu_0} = -B_m H_m V_m, \qquad V_g = A_g l_g,\ V_m = A_m l_m$$

The air-gap field energy (which is proportional to $B_g^2 V_g/\mu_0$) that a magnet volume V_m can support is proportional to the product $|B_m H_m|$ at its operating point. The magnet volume needed for a given gap is smallest when the magnet operates at the point where $|BH|$ is greatest: the **maximum energy product** $(BH)_{\max}$, quoted in kJ/m³ (or MGOe; 1 MGOe = 7.958 kJ/m³). The ratio $-B_m/\mu_0 H_m$ is the **permeance coefficient**, set by geometry; it fixes the **load line** whose intersection with the demagnetization curve is the operating point.

> **Status of the principle: magnet volume minimized at $(BH)_{\max}$**
> *Approximate.* It follows from Ampère's law and flux continuity for an ideal magnetic circuit with no leakage and infinitely permeable iron; in real machines leakage, iron saturation and varying operating points (armature reaction, temperature) move the design optimum, and resistance to demagnetization often matters more than $(BH)_{\max}$.

An upper bound. For an ideal magnet with a perfectly square intrinsic loop, $J = J_s$ throughout the second quadrant, so $B = J_s + \mu_0 H$. Writing $h = -H > 0$, the energy product is $(J_s - \mu_0 h)h$, which is greatest at $h = J_s/2\mu_0$:

$$(BH)_{\max} \le \frac{J_s^2}{4\mu_0}$$

This bound is reached only if the intrinsic coercivity exceeds $J_s/2\mu_0$. For $Nd_2Fe_{14}B$ ($J_s = 1.61$ T) the bound is 516 kJ/m³; the best commercial magnets reach about 440 kJ/m³ (N55), about 85 % of it. For $SmCo_5$ ($J_s = 1.07$ T) it is 228 kJ/m³, for Sm_2Co_{17} (1.25 T) 311 kJ/m³, and for hard ferrite (0.47 T) only 44 kJ/m³. AlNiCo has a high remanence (1.3 T, bound 336 kJ/m³) but reaches only about 40–80 kJ/m³, because its coercivity, about 50–150 kA/m, is far below the $J_s/2\mu_0 \approx 520$ kA/m needed. Saturation sets the ceiling; coercivity decides whether a material can approach it.

> **Status of the principle:** $(BH)_{\max} \le J_s^2/4\mu_0$
> *Derived* from $B = \mu_0 H + J$ with a rectangular intrinsic loop and unit recoil permeability; it is a strict upper bound for a material of given saturation.

The four magnet families

Property (typical, 20 °C)	Sintered NdFeB (N42 / N52)	$SmCo_5$	Sm_2Co_{17}	Hard ferrite (Sr, Ba)	AlNiCo 5 (cast)
B_r (T)	1.30 / 1.45	0.85–1.0	1.0–1.15	0.38–0.45	1.25–1.30
H_{cJ} (kA/m)	≥ 955 / ≥ 876 (grades to > 2500)	1200–2000	800–2000	250–400	about 50
H_{cB} (kA/m)	about 850–1000 (both)	650–760	600–800	200–300	about 50
$(BH)_{\max}$ (kJ/m³)	320–340 / 400–420	130–200	190–260	25–40	40–44
Theoretical bound $J_s^2/4\mu_0$ (kJ/m³)	516	228	311	44	336
Recoil permeability	1.05	1.02–1.05	1.03–1.05	1.05–1.1	about 3–5
$\alpha(B_r)$ (% per K)	−0.12	−0.04	−0.03	−0.20	−0.02
$\beta(H_{cJ})$ (% per K)	−0.5 to −0.6	−0.2 to −0.3	−0.2 to −0.3	+0.2 to +0.4	about −0.02
Curie temperature (°C)	310–340	about 720–750	800–850	about 450	800–860
Maximum working temperature (°C)	80–230 (grade-dependent)	250	300–350	about 250	450–550
Resistivity (μΩ m)	1.4–1.6	0.5–0.6	0.8–0.9	$> 10^4$ (insulator)	0.5
Density (g/cm³)	7.5	8.3	8.4	4.8–5.0	7.3
Coercivity mechanism	Nucleation	Nucleation	Wall pinning	Single-domain-size grains	Shape anisotropy
Main weaknesses	Temperature, corrosion, Dy/Tb supply	Cost (Co, Sm), brittle	Cost, brittle	Low B_r, cold demagnetization	Low H_c, easily demagnetized

Brief notes on each:

- **Sintered NdFeB** (Sagawa, Sumitomo, and Croat, General Motors, 1984) is a tetragonal $Nd_2Fe_{14}B$ phase with a thin Nd-rich grain-boundary phase that decouples the grains. Powder of 3–5 µm single crystals is aligned in a field, pressed and sintered. It has the highest energy product of any material, and moderate cost, but the lowest Curie temperature of the rare-earth magnets. It corrodes readily, so magnets are coated with Ni–Cu–Ni, epoxy or aluminium (Chapter 44). **Bonded NdFeB** (melt-spun powder in a polymer) has $B_r \approx 0.6$–0.7 T and $(BH)_{\max}$ of 60–100 kJ/m³, but can be moulded to complex shapes.
- **Samarium–cobalt** magnets were the first rare-earth magnets (1960s–70s). They are more expensive and brittle, but have far smaller temperature coefficients and a high Curie temperature. Sm_2Co_{17} grades work at 300–350 °C, and special grades higher. They are used in aerospace, military, high-temperature sensors and high-speed machines.
- **Hard ferrites** ($SrFe_{12}O_{19}$, $BaFe_{12}O_{19}$; hexagonal ferrites, Chapter 28) have a low $(BH)_{\max}$ but are made from cheap, abundant oxides, are electrical insulators (no eddy currents) and resist corrosion. By mass they are still the most-produced magnets: appliance and automotive auxiliary motors, loudspeakers, and holding magnets. Their coercivity *rises* with temperature, so the dangerous condition is cold start.
- **AlNiCo** (Fe–Al–Ni–Co with Cu, Ti) owes its coercivity to elongated, aligned ferromagnetic Fe–Co precipitates in a weakly magnetic Ni–Al matrix (shape anisotropy, produced by spinodal decomposition in a field). It has a high remanence and the best temperature stability of all, but its low coercivity means that a long, thin shape is needed, and it is easily demagnetized. It survives in guitar pickups, sensors, and precision instruments.

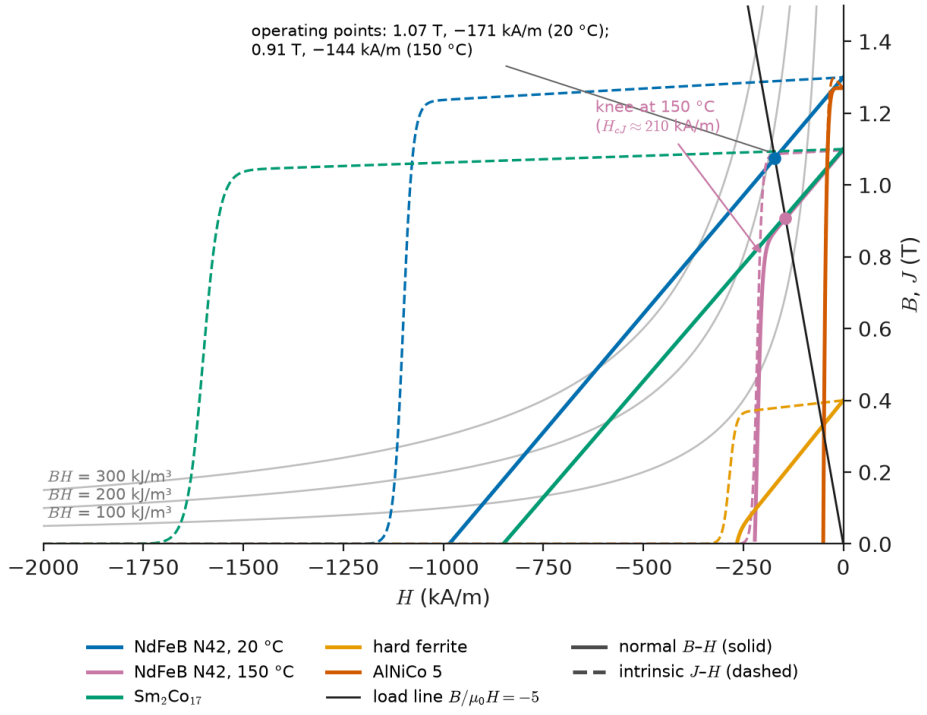


Figure 29.6 — Second-quadrant demagnetization curves of the four permanent-magnet families, with constant-(BH) hyperbolae, a load line, and the effect of temperature on NdFeB.

Temperature, the knee and irreversible demagnetization

Figure 29.6 shows the decisive feature of the demagnetization curves. At room temperature, NdFeB's B–H line is straight all the way to H_{cB}: the intrinsic polarization stays at B_r until H approaches $-H_{cJ}$, which is larger than H_{cB}. As temperature rises, B_r falls slowly (−0.12 % per K) and is recovered on cooling — a *reversible* loss. But H_{cJ} falls fast (−0.5 to −0.6 % per K). Once H_{cJ} drops below H_{cB}, the B–H curve develops a **knee** in the second quadrant. If the operating point, pushed by heat and by any opposing field from the stator windings (armature reaction during overload or a short circuit), passes the knee, some grains reverse. On return to normal conditions the magnet recoils along a new, lower line: an *irreversible* loss that only remagnetization can repair.

> **Status of the principle: linear temperature coefficients $\alpha(B_r)$ and $\beta(H_{cJ})$**
> *Empirical.* They are linear fits to measured data over a stated range (usually 20–100 °C or 20–150 °C); extrapolating them far outside that range is only a rough estimate, and datasheet curves at temperature should be used for design.

Worked example — will the magnet survive at 150 °C? A rotor magnet of length $l_m = 5$ mm drives flux across an effective air gap of $l_g = 1$ mm of the same area. Iron reluctance and leakage are neglected. The magnet's demagnetization line is $B_m = B_r + \mu_0 \mu_{\mathrm{rec}} H_m$, and the circuit requires $H_m = -B_m l_g / (\mu_0 l_m)$. Eliminating H_m:

$$B_m = \frac{B_r}{1 + \mu_{\mathrm{rec}} l_g / l_m}$$

At 20 °C, with an N42 magnet ($B_r = 1.30$ T, $\mu_{\mathrm{rec}} = 1.05$): $B_m = 1.30/(1 + 1.05 \times 0.2) = 1.30/1.21 = 1.074$ T, and

$$H_m = -\frac{B_m l_g}{\mu_0 l_m} = -\frac{1.074 \times 0.2}{4\pi \times 10^{-7}} = -1.71 \times 10^5 \ \mathrm{A/m} = -171 \ \mathrm{kA/m}$$

The operating point is far from the knee ($H_{cJ} \geq 955$ kA/m).

At 150 °C: $B_r = 1.30(1 - 0.0012 \times 130) = 1.097$ T, $B_m = 1.097/1.21 = 0.907$ T and $H_m = -144$ kA/m. During a motor overload the stator adds, say, a further -300 kA/m in part of the magnet, so the local field reaches about -444 kA/m. Now estimate H_{cJ} at 150 °C from the temperature coefficients:

Grade	H_{cJ} at 20 °C (kA/m, minimum)	β (% per K, typical)	Estimated H_{cJ} at 150 °C (kA/m)	Margin against 444 kA/m
N42	955	−0.60	$955(1 - 0.0060 \times 130) = 210$	Fails: irreversible loss even before overload is complete
N42SH	1592	−0.55	$1592(1 - 0.0055 \times 130) = 454$	Marginal (about 2 %)
N38UH	1990	−0.50	$1990(1 - 0.0050 \times 130) = 697$	Adequate (about 57 %)

The high-coercivity grades get their H_{cJ} from heavy rare earths, and they pay with remanence: a UH grade has B_r of about 1.20–1.25 T, so the magnet must be larger, or the motor accepts less flux. That trade is the dysprosium problem.

The dysprosium issue

$Nd_2Fe_{14}B$ has an anisotropy field $\mu_0 H_K \approx 7.6$ T at room temperature (Chapter 28). Substituting dysprosium (or terbium) for some of the neodymium raises it: $Dy_2Fe_{14}B$ has $\mu_0 H_K \approx 15$ T and $Tb_2Fe_{14}B$ over 20 T. But the heavy rare-earth moments couple *antiparallel* to the iron moments, while the light rare-earth Nd couples parallel. $Dy_2Fe_{14}B$ has J_s of only about 0.7 T. So every per cent of Dy raises coercivity and lowers remanence, very roughly by 0.1 MA/m and 0.02 T per weight per cent when substituted uniformly through the grains. A traction-motor magnet rated for 150–180 °C has traditionally contained 6–10 wt% Dy.

The difficulty is supply. Dysprosium and terbium are heavy rare earths found at useful concentration mainly in ion-adsorption clay deposits in southern China and neighbouring Myanmar; separation and metal production are overwhelmingly concentrated in China. The 2010–2011 price spike, when dysprosium oxide rose by about an order of magnitude within a year, made the risk clear. Since then the materials response has had several strands:

- **Grain-boundary diffusion (GBD).** Coercivity in a nucleation magnet is decided at grain surfaces (Chapter 28), so heavy rare earths are only needed there. Dy or Tb (as metal, fluoride or hydride) is coated on the finished magnet and diffused along the grain boundaries at about 800–900 °C, forming a Dy-rich shell a few nanometres thick on each grain. This raises H_{cJ} by 0.3–0.8 MA/m with well under 1 wt% heavy rare earth, and with almost no loss of B_r, because the grain cores stay $Nd_2Fe_{14}B$. The method is limited to magnets a few millimetres thick, since diffusion depth is limited.
- **Grain refinement.** Smaller grains (below about 3 μm) raise coercivity without heavy rare earths, but require oxygen-free fine-powder processing.
- **Cerium and lanthanum substitution** of part of the Nd for lower-cost, lower-performance grades.

- **Design**: better cooling, magnet segmentation (Section 29.3), flux-weakening control and rotor geometries that shield the magnets from armature reaction. Machine topologies that avoid rare earths entirely — induction, wound-field synchronous and switched-reluctance machines, or ferrite-assisted reluctance machines — are decided in the Electrical Energy & Machines core.

Recycling of magnets from end-of-life drives and hard disks, and the lifecycle cost of rare-earth extraction, are taken up in Chapter 59.

29.8 Magnetic shielding materials

A magnetometer on a satellite boom, a photomultiplier, an electron-beam instrument, an MRI suite, a precision current sensor next to a busbar, and a sensitive audio transformer all need protection from unwanted magnetic fields. Two physically different mechanisms provide shielding, and which dominates depends on frequency.

Low frequency: flux shunting

At DC and power frequencies, a conducting shield does almost nothing against magnetic fields, because there are no time-varying fields to induce eddy currents. The only remedy is a **high-permeability** enclosure that offers the field lines an easier path *around* the protected volume. For a long cylindrical shell of diameter D, wall thickness $t \ll D$ and relative permeability $\mu_r \gg 1$, in a transverse field, the **shielding factor** (external field divided by interior field) is approximately

$$S \approx 1 + \frac{\mu_r t}{D}$$

> **Status of the principle: flux-shunting shielding factor $S \approx 1 + \mu_r t/D$**
> *Approximate*. It follows from magnetostatics for a thin, infinitely long, linear cylindrical shell in a uniform transverse field; end openings, seams, nonlinear permeability and saturation reduce the real shielding factor.

The materials requirement follows directly: high μ_r at the low field strengths being shielded. That points to **mu-metal** and related 77–80 % Ni–Fe alloys with Mo or Cu and Cr additions. After a final hydrogen anneal at about 1100–1150 °C, they reach μ_i of the order of 5×10^4 and $\mu_{\max}$ of several 10^5. Their weaknesses follow from Chapter 28:

- **Low saturation** (0.75–0.8 T). The shield concentrates flux in its walls; for a cylinder the wall induction is roughly $B_0 D/t$. In a 1 mT ambient field, a shield with $D/t = 100$ carries about 0.1 T in its walls, safely below saturation; in 10 mT it would saturate and its permeability would collapse. Strong fields are handled by **nested shields**: an outer layer of high-saturation material (low-carbon steel or 50 % Ni–Fe) reduces the field enough for an inner mu-metal layer to work.
- **Stress sensitivity.** Permalloy's near-zero anisotropy and magnetostriction are what make it so permeable, and they are destroyed by plastic deformation. Bending, drilling, welding or dropping a finished shield can reduce its permeability by a large factor. Shields are formed first and hydrogen-annealed last, and handled with care afterwards.
- **Low-field permeability.** Shielding the earth's field (about 50 μT, $H \approx 40$ A/m) or smaller fields uses the initial permeability, which is several times lower than $\mu_{\max}$.

Worked example — a mu-metal cylinder. A cylindrical shield of diameter $D = 100$ mm and wall $t = 1$ mm has $\mu_r = 2 \times 10^4$ at the working field. The shielding factor is $S \approx 1 + 2 \times 10^4 \times 10^{-3}/0.1 = 201$, or $20 \log_{10} 201 = 46$ dB. A second, concentric shell with a gap between them roughly multiplies the factors, giving of order 4×10^4 (about 90 dB) in the ideal case. An earth-field environment of 50 μT reduces to about 0.25 μT inside one layer. In an ambient of 1 mT the wall carries about $1\ \text{mT} \times 100 = 0.1$ T, well below saturation.

High frequency: absorption and reflection

At higher frequencies a conducting enclosure attenuates fields through eddy currents. Schelkunoff's transmission-line description splits **shielding effectiveness** (in dB) into

$$SE = A + R + B$$

- **Absorption loss** A: attenuation of the wave as it diffuses through the wall, $A = 8.686\, t/\delta$ dB — 8.7 dB per skin depth of thickness. It depends on the material through $\delta = \sqrt{\rho/\pi f \mu_0 \mu_r}$: high conductivity *and* high permeability both help.

- **Reflection loss** R: reflection at the impedance mismatch between the incident wave and the metal surface. For a plane wave (far field), $R \approx 168 - 10\log_{10}(\mu_r f/\sigma_r)$ dB with f in Hz and σ_r the conductivity relative to copper. Reflection is large for good conductors and *reduced* by high permeability, which raises the surface impedance.
- **Multiple-reflection correction** B: negative, and significant only when the wall is thinner than about one skin depth.

The source matters. A **far-field** (plane-wave) or **electric** near-field source has a high wave impedance, and a thin copper or aluminium layer reflects it very effectively: $R \approx 108$ dB for copper at 1 MHz. A **magnetic near-field** source — a transformer, a coil, a current loop at a distance short compared with $\lambda/2\pi$ — has a *low* wave impedance, close to that of the metal, so reflection is weak. Low-frequency magnetic fields must therefore be dealt with by absorption (thick, permeable, conductive walls) or, at the lowest frequencies, by flux shunting.

Worked example — copper versus steel against a 1 kHz magnetic field. Compare 0.5 mm sheets. For copper ($\rho = 1.72 \times 10^{-8}$ Ω m, $\mu_r = 1$) at 1 kHz, $\delta = 2.09$ mm and $A = 8.686 \times 0.5/2.09 = 2.1$ dB — almost transparent. For low-carbon steel ($\rho \approx 1.0 \times 10^{-7}$ Ω m, $\mu_r \approx 1000$) at 1 kHz, $\delta = \sqrt{1.0 \times 10^{-7}/(\pi \times 10^3 \times 4\pi \times 10^{-7} \times 1000)} = 0.16$ mm, and $A = 8.686 \times 0.5/0.16 = 27$ dB. Steel's permeability makes it the better low-frequency magnetic shield, even though its conductivity is six times lower. At 100 kHz, copper's skin depth is 0.21 mm and $A = 21$ dB. Steel, with its permeability now reduced to perhaps 100, has $\delta = 50$ μm and $A \approx 86$ dB, though seams and apertures, not the material, then usually limit performance.

Shielding material	μ_r (typical, working)	ρ	J_s (T)	Best for
Mu-metal (77–80 % Ni–Fe–Mo/Cu)	2×10^4–10^5	0.55–0.60 μΩ m	0.75–0.8	DC–kHz, low-level fields, sensors, instruments
50 % Ni–Fe	5000–2×10^4	0.45 μΩ m	1.5	Moderate fields, outer layers
Low-carbon steel	500–2000	0.1–0.2 μΩ m	2.0–2.1	Strong low-frequency fields, outer layer, enclosures
Si steel (GO/NO)	1000–10^4	0.25–0.5 μΩ m	2.0	Room shielding against power-frequency fields
Nanocrystalline or amorphous foil	10^4–10^5	1.2–1.3 μΩ m	0.6–1.25	Thin, flexible shields; wireless-charging flux guides
Copper, aluminium	1	0.017, 0.027 μΩ m	—	Electric and far-field EMI; magnetic fields above about 100 kHz
Ferrite sheet (NiZn, MnZn)	50–3000	High	0.3–0.5	Flux guiding in NFC, RFID and wireless charging

Electromagnetic compatibility design — enclosures, gaskets, apertures — belongs with circuit and system design; the material choice is what this section supplies.

29.9 Material selection for transformers, motors, inductors and sensors

The selection logic is the same everywhere: identify the frequency, the peak flux density and its direction, the DC bias, the temperature, and the dominant cost (losses, size, weight, raw material). Then pick the material whose loss mechanisms, saturation and stability suit those conditions. A full property-based selection methodology, with performance indices, is developed in Chapter 58.

Transformers

Application	Frequency	Key requirements	Usual material	Reason
Power and large distribution transformers	50/60 Hz	Lowest loss at 1.6–1.8 T, fixed flux direction, noise	Hi-B GO steel, 0.23–0.30 mm, domain-refined	Goss texture, high J_s, tension coating
Distribution transformers, loss-driven	50/60 Hz	Lowest no-load loss	Fe-based amorphous ribbon	Loss one-quarter to one-third of GO; larger core accepted
Small mains transformers	50/60 Hz	Low cost	NO steel (E–I laminations) or CGO	Punchability, price
Medium-frequency (solid-state, traction, fast chargers)	1–50 kHz	High B at moderate loss	Nanocrystalline, Fe-based amorphous, 6.5 % Si steel, MnZn ferrite	Thin ribbon, high ρ
SMPS transformers	20 kHz–1 MHz	Low loss at 0.05–0.25 T, at 80–100 °C	MnZn power ferrite	Resistivity; loss minimum near operating temperature
Current transformers, residual-current devices	50 Hz–100 kHz	High, linear μ; low H_c	Nanocrystalline, 80 % Ni–Fe, Co-based amorphous	Accuracy depends on magnetizing current
Signal, pulse, gate-drive transformers	kHz–MHz	High μ_i, low loss at low B	High-μ MnZn, nanocrystalline	Small magnetizing current
RF transformers, baluns	1–1000 MHz	Low loss at high frequency	NiZn ferrite	Snoek limit

Motors and generators

- **Stator and rotor cores**: NO steel, with gauge and silicon chosen by frequency. Industrial 50 Hz motors use 0.50–0.65 mm M400–M800 grades. Efficient and high-speed motors, including electric-vehicle traction motors with fundamentals of several hundred hertz to over a kilohertz, use 0.20–0.30 mm, high-Si grades (the 400 Hz worked example of Section 29.2). Edge damage from punching matters more as teeth narrow.
- **High specific power** (aircraft generators, actuators, racing): Fe–Co–V laminations, accepting cobalt cost for 15–20 % higher flux density.
- **Three-dimensional flux paths** (axial-flux, claw-pole, transverse-flux machines): soft magnetic composites.
- **Permanent magnets**: sintered NdFeB for high torque density, with the grade (and Dy/Tb content or GBD) set by worst-case temperature and demagnetizing field (Section 29.7 worked example), and segmented against eddy loss. Ferrite for low-cost appliance and automotive auxiliary motors, checked for low-temperature demagnetization. SmCo for high-temperature or high-reliability machines.
- **Amorphous and nanocrystalline stators** are used in some high-speed, high-efficiency niche machines, where cutting and stacking ribbon is worth the low loss.

Inductors and chokes

An inductor must store energy, $\frac{1}{2}LI^2$, and almost all of it is stored where μ is low: in an air gap, or in the distributed gaps between powder particles. The materials question is therefore how to get a well-defined, stable, low effective permeability with acceptable loss and saturation.

- **Gapped ferrite**: low loss at high frequency, sharp saturation. Its B_s must be checked at the maximum temperature, which was the failure in Chapter 28's opening.
- **Powder cores** (iron powder, MPP 80 % Ni–Fe–Mo, Fe–Si–Al "Sendust", 50 % Ni–Fe "High Flux", Fe–Si): distributed gap, permeability 14–550, **soft saturation** (inductance falls gradually with DC bias rather than collapsing) and B_s of 0.75–1.5 T. They suit output and PFC chokes with large DC current; their loss is higher than ferrite's at high frequency.
- **Laminated steel or amorphous gapped cores**: line-frequency and low-frequency chokes, high current.
- **Common-mode chokes**: high μ over a wide frequency range, with no DC bias (the line currents cancel): high-μ MnZn or nanocrystalline.

- **EMI beads and cable cores**: NiZn (or MnZn for lower frequencies), working above f_r where μ'' dissipates interference as heat.

Sensors (material level)

> **Boundary.**
> The materials mechanisms below belong here; sensor circuits, signal conditioning and device operation belong to the Electronics & Computing core. Hall-effect *characterization* of materials is treated in Chapter 52, and semiconductor sensor materials in Chapter 35.

Hall-effect materials. A current I through a plate of thickness t in a perpendicular field B produces a transverse voltage

$$V_H = \frac{R_H I B}{t}, \qquad R_H = \frac{1}{ne}$$

for a single carrier type of density n. Sensitivity therefore favours **low carrier density** and **thin layers**, while low noise and low drive power favour **high mobility** (for constant-voltage bias, the Hall voltage scales with mobility). The choice runs among narrow-gap and III–V semiconductors (Chapter 33):

Hall material	Electron mobility (m²/V s, 300 K)	Band gap (eV)	Character
InSb	about 7.7	0.17	Highest sensitivity; strong temperature dependence
InAs	about 3.3	0.35	High sensitivity, better temperature behaviour
GaAs	about 0.85	1.42	Stable over temperature, moderate sensitivity
Si	about 0.14	1.12	Low sensitivity; integrates with CMOS, cheapest

For example, a GaAs plate with $n = 1 \times 10^{22}$ m^{-3} and $t = 5$ μm has $R_H = 1/(1 \times 10^{22} \times 1.602 \times 10^{-19}) = 6.2 \times 10^{-4}$ m³/C. At $I = 1$ mA and $B = 0.1$ T, $V_H = 6.2 \times 10^{-4} \times 10^{-3} \times 0.1/(5 \times 10^{-6}) = 12.5$ mV. Silicon Hall sensors, despite their low mobility, dominate the market because they integrate with amplification and offset cancellation on one chip. They are often combined with **soft-magnetic flux concentrators** (permalloy or ferrite) that gather flux into the sensor, or with a gapped soft core around a conductor in closed-loop current sensors. In each case the soft material's linearity, hysteresis and saturation limit accuracy.

Magnetoresistive materials. Resistance that depends on magnetization direction allows smaller, more sensitive sensors than Hall plates:

- **Anisotropic magnetoresistance (AMR)**: in permalloy ($Ni_{81}Fe_{19}$) films, resistance depends on the angle between current and magnetization, through spin–orbit scattering; $\Delta R/R \approx 2$–4 %. Zero-magnetostriction permalloy keeps the film insensitive to stress. AMR is used in compasses and position and angle sensors.
- **Giant magnetoresistance (GMR)**: in multilayers of ferromagnetic films (Co, CoFe, NiFe) separated by nanometre-thin non-magnetic metal (Cu), resistance depends on whether adjacent layers are magnetized parallel or antiparallel, through spin-dependent scattering; $\Delta R/R \approx 5$–20 % in practical **spin valves**. One layer is pinned by an antiferromagnet (IrMn, PtMn; exchange bias, Chapter 28), the other is free.
- **Tunnel magnetoresistance (TMR)**: when the spacer is an insulating tunnel barrier, crystalline MgO about 1–2 nm thick between CoFeB electrodes, tunnelling depends strongly on relative magnetization through spin-filtering by the MgO crystal symmetry; $\Delta R/R$ of 100–600 % at room temperature. TMR is used in hard-disk read heads, high-sensitivity current and angle sensors, and MRAM.

The materials challenges in all three are film quality at nanometre thickness, interface sharpness, thermal stability of the pinning (the blocking temperature of the antiferromagnet), and control of Néel walls and domains in small elements (Chapter 28). Thin-film deposition and annealing of such stacks are discussed in Chapters 49 and 50.

Other magnetic sensing materials. Fluxgate magnetometers use cores of Co-based amorphous ribbon or permalloy driven repeatedly into saturation; the material requirements are low coercivity, a square loop and low noise. **Magnetostrictive** sensors and actuators use Terfenol-D or Fe–Ga (Galfenol), and magnetoelastic torque sensors on steel shafts exploit the stress dependence of permeability (Chapter 28). Barkhausen noise in steel is used for non-destructive evaluation of stress and hardness.

What remains open

Open question	Where it is answered
How are transformers, motors and inductors sized and designed around these materials?	Electrical Energy & Machines core
How do rolling, recrystallization, sintering and rapid solidification create the textures and microstructures of steels, ferrites, ribbons and magnets?	Chapters 9, 10 and 46
How are magnetic thin-film sensor stacks deposited and patterned?	Chapter 50
Can superconducting coils replace iron and permanent magnets in high-field machines and MRI?	Chapter 30
How do nanoparticles, nanocomposites and two-dimensional magnets extend these families?	Chapter 38
How are core loss (Epstein frame, single-sheet tester), B–H loops and magnet demagnetization curves measured, and with what uncertainty?	Chapter 52
How do NdFeB corrosion, magnetic ageing of steels and thermal demagnetization limit service life?	Chapter 44
How are cost, supply risk, recyclability and performance indices combined in materials selection?	Chapter 58
How do Hall, magnetoresistive and spintronic devices operate in circuits?	Electronics & Computing core

Next: Chapter 30 closes Part VI with superconducting materials, whose zero resistance, perfect diamagnetism and flux pinning extend the magnetic ideas of this part to a new state of matter.

Chapter 30 — Superconducting Materials

Part VI · Thermal, Electrical, Dielectric and Magnetic Materials

A 3 T MRI scanner carries a current of several hundred amperes around a coil of niobium–titanium wire several kilometres long, and it does so for years with the power supply disconnected. The current does not decay measurably. Yet the same wire, lifted a few kelvin above its operating temperature or pushed a few per cent beyond its rated current, turns resistive in milliseconds, dumps the stored magnetic energy (several megajoules) as heat, boils off hundreds of litres of liquid helium and can destroy itself. A copper engineer would call the first behaviour impossible and the second catastrophic. Both are properties of the material, and both are set by structure on the scale of nanometres: the spacing of titanium precipitates, the size of Nb_3Sn grains, the misorientation of grain boundaries in a ceramic tape. This chapter explains what a superconductor is, from the materials side, and why only a handful of the thousands of known superconductors are usable as engineering conductors.

> **Why does a superconductor carry current without loss, and what limits how much current, field and temperature it can tolerate?**
> **Why are the useful superconductors almost all Type II materials full of deliberately introduced defects, and why are the best of them so hard to make into wire?**

Concept	Engineering destination
Zero resistance, persistent current	MRI and NMR magnets in persistent mode, lossless DC cables
Meissner effect, penetration depth λ	Magnetic shielding, levitation, surface resistance of RF cavities
Coherence length ξ, Ginzburg–Landau parameter κ	Type I versus Type II behaviour; upper critical field
Critical temperature, field and current; critical surface	Operating margin and load line of every superconducting magnet
Flux pinning, Bean critical-state model	Critical current density, magnetization, AC (hysteresis) loss
Multifilamentary wires, coated-conductor tapes	Conductor choice for magnets, cables, fault-current limiters
Strain sensitivity, brittleness, reaction heat treatment	Winding method, mechanical design, quench and stress limits
Flux quantum, Josephson effect	SQUID magnetometers, quantum-limited sensing

Roadmap. What is special about the superconducting state (30.1) → the three limits T_c, B_c, J_c and the critical surface (30.2) → why most practical superconductors are Type II (30.3) → the material families, including high-temperature superconductors (30.4) → how defects pin flux and set J_c (30.5) → how the materials are turned into wires and tapes (30.6) → the processing constraints that follow (30.7) → magnets, power devices and sensors (30.8).

> **Boundary.**
> The microscopic (BCS) theory of superconductivity is quantum many-body physics and belongs to Engineering Physics; it is used here only qualitatively. Josephson-junction circuits and SQUID electronics belong to Electronics & Computing. The cryogenic and power-system design of magnets and cables belongs to Electrical Energy & Machines. This chapter treats the material: what its structure does to its critical parameters, and what its processing allows.

30.1 Superconducting properties

30.1.1 Zero resistance

In 1911 Kamerlingh Onnes measured the resistance of mercury as it was cooled in liquid helium. In an ordinary metal the resistivity falls with temperature as phonon scattering freezes out and then levels off at a residual value set by impurities and defects (Matthiessen's rule, Chapter 25). Mercury followed this curve down to 4.2 K and then, within a few hundredths of a kelvin, its resistance fell to below the resolution of his instrument. The temperature at which this happens is the **critical temperature** T_c.

"Below the resolution" needs a sharper statement. The most sensitive test is a persistent current: induce a current in a closed superconducting ring and watch its magnetic field. A ring of inductance L and resistance R loses current as $I(t) = I_0 e^{-Rt/L}$. Experiments of this kind have found no decay over years, which puts upper limits on the resistivity of order $10^{-25}\ \Omega\,\mathrm{m}$ or below, at least seventeen orders of magnitude below room-temperature copper ($1.7 \times 10^{-8}\ \Omega\,\mathrm{m}$). For engineering purposes the DC resistivity is zero, and an MRI magnet can be "closed" with a superconducting switch and left to run for years with field drift below 0.1 ppm per hour.

Two qualifications matter immediately:

- **Zero resistance is a DC property.** At finite frequency the unpaired electrons respond to the field that accelerates the condensate, giving a small surface resistance that matters in RF cavities. In power applications the losses from flux motion (Section 30.5) matter more, even at 50 Hz.
- **Zero resistance holds only below three critical values together**: temperature, magnetic field and current density (Section 30.2).

30.1.2 The Meissner effect: more than a perfect conductor

Suppose a material merely had zero resistance. Faraday's law with $\mathbf{E} = \rho\mathbf{J} = 0$ gives

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = 0$$

so the field inside a perfect conductor is frozen at whatever value it had when the resistance vanished. Cool such a material in zero field and apply a field: surface currents are induced that keep $\mathbf{B} = 0$ inside. Cool it in a field and then remove the field: the interior field stays trapped. The final state depends on history.

In 1933 Meissner and Ochsenfeld found that a superconductor does something different. Cooled through T_c *in* a field, it **expels** the field from its interior. The state $\mathbf{B} = 0$ inside is reached regardless of history (Figure 30.1). The superconducting state is therefore a genuine thermodynamic phase, with a free energy that depends only on T and H, not a kinetic consequence of vanishing resistance. Inside a bulk superconductor in the Meissner state

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = 0 \quad \Rightarrow \quad \mathbf{M} = -\mathbf{H}, \qquad \chi = -1$$

A superconductor is a perfect diamagnet, about 10^5 times more strongly diamagnetic than copper ($\chi \approx -1 \times 10^{-5}$) and several thousand times more than bismuth, the most diamagnetic ordinary metal (Chapter 28). This is why a small magnet levitates stably above a cooled YBCO pellet.

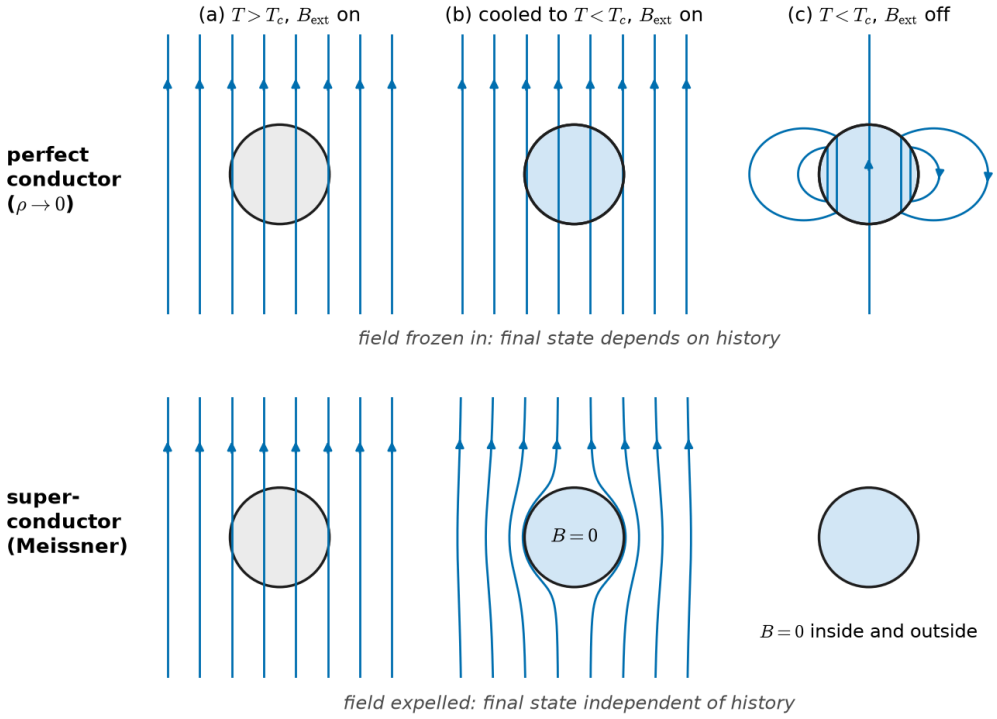


Figure 30.1 — A perfect conductor traps whatever field it had when its resistance vanished; a superconductor expels the field regardless of history.

30.1.3 The London equations and the penetration depth

Expulsion cannot be perfect right up to the surface: the screening current that cancels the applied field must flow in a layer of finite thickness, otherwise the current density would be infinite. The brothers F. and H. London (1935) found the simplest description. Treat a density n_s of superconducting carriers of charge q and mass m as frictionless. Newton's law gives $m\,d\mathbf{v}/dt = q\mathbf{E}$, and with $\mathbf{J}_s = n_s q \mathbf{v}$,

$$\frac{\partial \mathbf{J}_s}{\partial t} = \frac{n_s q^2}{m}\mathbf{E}$$

Taking the curl and using Faraday's law gives $\partial/\partial t\,[\nabla \times \mathbf{J}_s + (n_s q^2/m)\mathbf{B}] = 0$. A perfect conductor would stop there: the bracket is constant. The London postulate, chosen to reproduce the Meissner effect, is that the bracket is not merely constant but zero:

$$\nabla \times \mathbf{J}_s = -\frac{n_s q^2}{m}\,\mathbf{B}$$

Combine with Ampère's law $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_s$ and $\nabla \cdot \mathbf{B} = 0$ (so $\nabla \times \nabla \times \mathbf{B} = -\nabla^2 \mathbf{B}$):

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda_L^2}\mathbf{B}, \qquad \lambda_L = \sqrt{\frac{m}{\mu_0 n_s q^2}}$$

For a field B_0 applied parallel to the flat surface of a superconductor filling $x > 0$, the solution that does not blow up is

$$B(x) = B_0\, e^{-x/\lambda_L}$$

The field, and the screening current that produces it, decay over the **London penetration depth** λ_L (Figure 30.2). Only a bulk sample much thicker than λ_L shows full Meissner expulsion; a film thinner than λ_L is penetrated almost uniformly.

Worked example — the size of the penetration depth. Take one carrier per atom with the free-electron charge and mass, $n_s = 1 \times 10^{29}\ \text{m}^{-3}$, $q = e$, $m = m_e$:

$$\lambda_L = \sqrt{\frac{9.109 \times 10^{-31}}{(1.257 \times 10^{-6})(1 \times 10^{29})(1.602 \times 10^{-19})^2}} = \sqrt{\frac{9.109 \times 10^{-31}}{3.225 \times 10^{-15}}} = 1.68 \times 10^{-8}\ \text{m}$$

about 17 nm. Measured values in clean elemental superconductors are 16–40 nm (Al about 16 nm, Nb about 40 nm), in good agreement given the crudeness of "one carrier per atom". In the cuprates the carrier density is roughly a hundred times smaller

and the effective mass larger, and λ rises to 150 nm or more, which is the first sign that these materials will behave differently in a field.

The penetration depth also depends on temperature, because n_s falls to zero at T_c. A useful fit is the two-fluid form

$$\lambda(T) \approx \frac{\lambda(0)}{\sqrt{1-(T/T_c)^4}}$$

> **Status of the principle: London equations and penetration depth**
> *Approximate.* A phenomenological, local theory: it assumes the current at a point depends only on the field at that point. It fails when the coherence length exceeds λ (clean, low-κ metals such as Al), where Pippard's non-local theory is needed. The temperature dependence above is *Empirical*.

30.1.4 Cooper pairs, the energy gap and flux quantization (qualitative)

The London carriers turn out to be pairs. BCS theory (Bardeen, Cooper and Schrieffer, 1957) showed that a weak net attraction between electrons near the Fermi surface, mediated by lattice vibrations, binds them into **Cooper pairs** of opposite momentum and spin. Three consequences matter to a materials engineer, and each can be checked without the theory:

1. **An energy gap.** Breaking a pair costs energy 2Δ, with $2\Delta \approx 3.5\,k_B T_c$ in the weak-coupling limit. This gap is seen in tunnelling and in far-infrared absorption. Photons or phonons with $h\nu < 2\Delta$ cannot break pairs, which is why microwave losses are small and why superconducting detectors can count single photons.
2. **The isotope effect.** In conventional superconductors $T_c \propto M^{-1/2}$, where M is the ionic mass, the fingerprint of phonon-mediated pairing. It is weak or absent in the cuprates, whose pairing mechanism is still debated.
3. **Charge $2e$.** The superconducting state is described by a single macroscopic wavefunction whose phase must be single-valued around any closed loop. The magnetic flux through a superconducting ring is therefore quantized in units of the **flux quantum**

   $$\Phi_0 = \frac{h}{2e} = 2.068 \times 10^{-15}\ \text{Wb}$$

   The factor 2 in the denominator was the experimental proof of pairing. Φ_0 sets the flux carried by each vortex in a Type II superconductor (Section 30.3) and the period of a SQUID (Section 30.8).

> **Status of the principle: BCS pairing**
> *Fundamental* in its domain: it is a quantum many-body theory that predicts the gap, the isotope effect and the $2e$ charge for conventional (phonon-mediated, weak-coupling) superconductors. It does not predict T_c of a new material reliably, and it does not explain the cuprates.

30.1.5 Coherence length

The pair density cannot change abruptly. If a superconductor meets a normal region (a surface, a grain boundary, the core of a vortex), the density of pairs rises from zero to its bulk value over a characteristic distance, the **coherence length** ξ. Physically ξ is the size of a Cooper pair, and BCS gives for a clean metal

$$\xi_0 \approx \frac{\hbar v_F}{\pi \Delta}$$

A large Fermi velocity and a small gap (low T_c) give long coherence lengths: about 1.6 μm in aluminium, 83 nm in lead, 38 nm in niobium. In alloys and compounds two effects shorten ξ dramatically. Electron scattering limits the pair size to roughly $\sqrt{\xi_0 \ell}$, where ℓ is the electronic mean free path ("dirty limit"), so adding solute atoms or disorder shortens ξ. And materials with high T_c have large gaps and, in the cuprates, low Fermi velocities: ξ in YBCO is only about 1.5 nm in the CuO_2 planes and 0.2–0.3 nm perpendicular to them, comparable to a unit cell.

That last fact has an enormous engineering consequence. Any defect comparable to or larger than ξ, such as a grain boundary a few atoms wide, can locally suppress superconductivity. In niobium a grain boundary is invisible to the condensate; in YBCO it can be a barrier (Section 30.4.4).

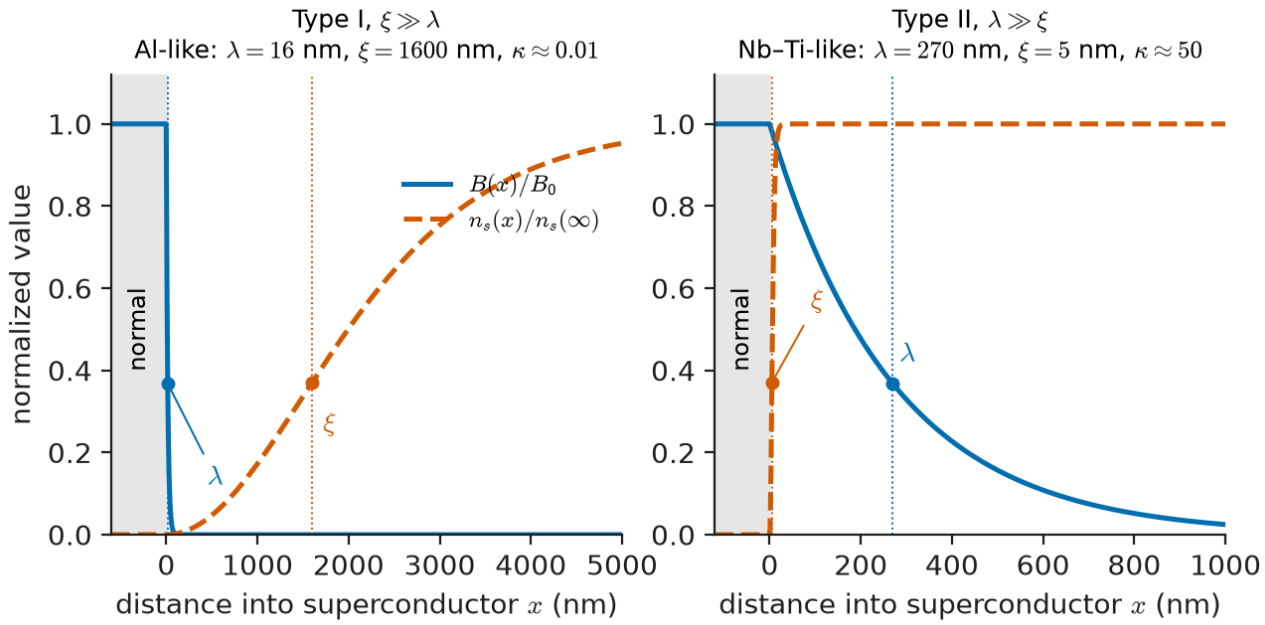


Figure 30.2 — Near a superconducting surface the field decays over the penetration depth lambda while the pair density rises over the coherence length xi; their ratio kappa decides Type I or Type II behaviour.

30.2 Critical temperature, field, and current

30.2.1 Critical temperature

T_c is the temperature at which the superconducting free energy and the normal-state free energy become equal in zero field. Table 30.1 lists representative values. The history of T_c is a materials story: from Hg (4.2 K, 1911) through Nb (9.3 K) and the A15 compounds such as Nb_3Sn (18 K) and Nb_3Ge (23 K, 1973), then the jump with the copper-oxide (cuprate) superconductors in 1986–1993 to 93 K in $YBa_2Cu_3O_7$ and 133–135 K in $HgBa_2Ca_2Cu_3O_{8+\delta}$ at ambient pressure. MgB_2 (39 K, 2001) and the iron-based superconductors (up to about 55 K, 2008) followed. Hydrogen-rich compounds such as H_3S and LaH_{10} superconduct above 200 K, but only at pressures above 100 GPa, which makes them scientifically important and technologically irrelevant for now.

The engineering meaning of T_c is the choice of coolant. Liquid helium boils at 4.2 K at atmospheric pressure, superfluid helium is used at 1.9 K, liquid hydrogen boils at 20 K, liquid neon at 27 K and liquid nitrogen at 77 K. Cryocoolers cover the range continuously. A material is practical at a temperature only if it still has a useful critical current at the field required there, which typically means operating at $T \lesssim T_c/2$ for magnets.

The cost of cooling grows quickly as temperature falls. An ideal (Carnot) refrigerator rejecting heat at 300 K needs $(300 - T)/T$ watts of work per watt removed at T: 70 W/W at 4.2 K, 2.9 W/W at 77 K. Real cryoplants achieve 15–30 % of Carnot at large scale, so a watt deposited at 4.2 K costs several hundred watts at the wall. This is why AC losses (Section 30.5) and the choice between 4.2 K, 20 K and 77 K operation dominate system design.

30.2.2 Critical field and condensation energy

A magnetic field raises the free energy of the superconducting state, because expelling the field costs magnetic energy $B^2/2\mu_0$ per unit volume. When this equals the **condensation energy**, the free-energy advantage of pairing, superconductivity is destroyed. This defines the **thermodynamic critical field** B_c:

$$g_n - g_s = \frac{B_c^2}{2\mu_0}$$

For lead, $B_c(0) = 0.080$ T, so the condensation energy is $(0.080)^2/(2 \times 1.257 \times 10^{-6}) = 2.5 \times 10^3$ J m^{-3}, a tiny energy: divided among the 3.3×10^{28} atoms per cubic metre of lead it is about 5×10^{-7} eV per atom. Superconductivity is a delicate low-energy order, which is why it is confined to low temperatures.

Experimentally B_c falls with temperature in an approximately parabolic way:

$$B_c(T) \approx B_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

> **Status of the principle: parabolic critical-field law**
> *Empirical.* It fits most elemental superconductors to within a few per cent; BCS gives a slightly different curve. For the upper critical field of Type II materials (Section 30.3) the exponent is fitted per material, about 1.7 for Nb–Ti.

30.2.3 Critical current

A current produces a field. In a Type I wire of radius a the field at the surface is $\mu_0 I/2\pi a$, and superconductivity is lost when this reaches B_c (Silsbee's rule). Because the current in a Type I superconductor flows only in the surface layer of thickness λ, the critical currents are small: a 1 mm lead wire at 4.2 K, with $B_c \approx 0.055$ T, carries at most $I_c = 2\pi a B_c/\mu_0 = 2\pi(5\times10^{-4})(0.055)/(1.257\times10^{-6}) \approx 140$ A, and in any appreciable external field much less. Type I superconductors are therefore useless as magnet conductors.

In Type II materials the critical current density J_c is not set by an intrinsic field limit but by how strongly the material **pins** magnetic vortices against the Lorentz force of the current (Section 30.5). This makes J_c a **microstructure-sensitive** property, like yield strength in metals (Chapter 13), whereas T_c and B_{c2} are largely **intrinsic** properties set by composition and crystal structure. The distinction organizes the whole of this chapter:

Parameter	Controlled by	Analogue in Part III
T_c	Composition, crystal structure, doping (oxygen content in cuprates)	Melting point
B_{c2}	T_c and electron scattering (alloying, disorder)	Elastic modulus
J_c	Pinning defects: precipitates, grain boundaries, nanoparticles; grain alignment	Yield strength

30.2.4 The critical surface

The three critical parameters are not independent. J_c falls as field or temperature rises, and vanishes on the line $B_{c2}(T)$. The superconducting state is the region under a surface in (T, B, J) space, the **critical surface** (Figure 30.3). A magnet operates at a point inside it. Its winding is energised along a **load line**: the peak field on the conductor rises in proportion to the current, and the magnet quenches where the load line pierces the critical surface. Designers keep a margin, typically operating at 70–85 % of the critical current along the load line, or a temperature margin of 1–2 K.

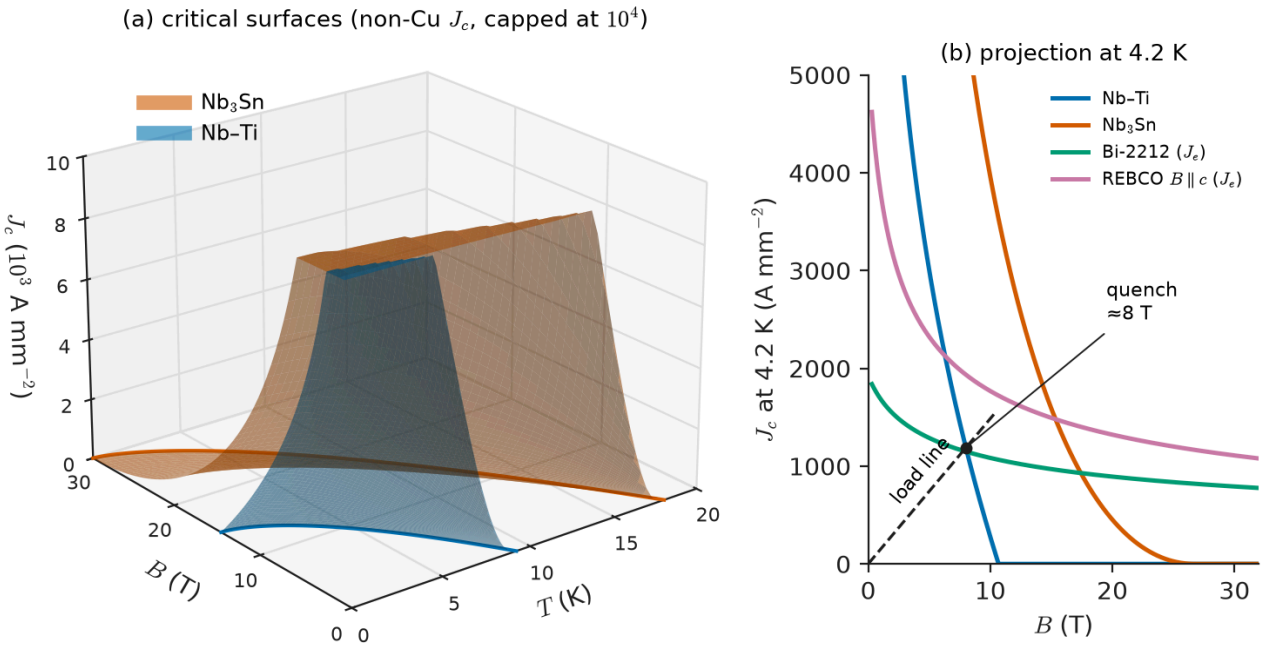


Figure 30.3 — The critical surface bounds the superconducting state in temperature, field and current density; a magnet's load line must stay beneath it.

Worked example — why the LHC runs at 1.9 K. For Nb–Ti take $B_{c2}(0) = 14.5$ T and $T_c = 9.2$ K, with the empirical form $B_{c2}(T) = B_{c2}(0)[1 - (T/T_c)^{1.7}]$.

At 4.2 K: $T/T_c = 0.4565$, $(0.4565)^{1.7} = 0.264$, so $B_{c2} = 14.5 \times 0.736 = 10.7$ T.

At 1.9 K: $T/T_c = 0.2065$, $(0.2065)^{1.7} = 0.068$, so $B_{c2} = 14.5 \times 0.932 = 13.5$ T.

Cooling from 4.2 K to 1.9 K raises the upper critical field by 2.8 T, about 26 %, and J_c at a given field rises roughly in proportion to the distance from B_{c2}. The LHC dipoles, which must deliver 8.3 T in the bore (about 9 T peak on the conductor), are only possible in Nb–Ti at superfluid-helium temperature; superfluid helium is also an excellent heat conductor, which helps stability. A field much above 9 T requires a different material: Nb$_3$Sn or an HTS.

Table 30.1 — Representative critical parameters of superconductors (typical values; J_c depends strongly on processing).

Material	Structure	T_c (K)	B_c or $B_{c2}(0)$ (T)	Type	Usable form
Al	fcc metal	1.18	0.0105 (B_c)	I	Thin films (qubits, junctions)
Pb	fcc metal	7.2	0.080 (B_c)	I	Historical, solder for joints
Nb	bcc metal	9.25	0.2 (B_c); 0.4 (B_{c2})	II ($\kappa \approx 1$)	RF cavities, Josephson junctions
Nb–Ti (Nb–47 wt% Ti)	bcc alloy	9.2	14.5	II	Ductile multifilament wire
Nb_3Sn	A15 intermetallic	18.3	28–30	II	Reacted multifilament wire
MgB_2	hexagonal	39	15–35 (anisotropic, dirty films higher)	II	PIT wire
$YBa_2Cu_3O_{7-\delta}$ (REBCO)	layered perovskite	92	>100 (in-plane, extrapolated)	II	Coated-conductor tape
$Bi_2Sr_2CaCu_2O_8$ (Bi-2212)	layered	85–92	>100	II	Round wire
$(Bi,Pb)_2Sr_2Ca_2Cu_3O_{10}$ (Bi-2223)	layered	110	>100	II	Ag-sheathed tape
$Ba_{1-x}K_xFe_2As_2$	layered pnictide	38	60–90	II	PIT tape (development)
$HgBa_2Ca_2Cu_3O_{8+\delta}$	layered	133–135	>100	II	None practical

30.3 Type I and Type II superconductors

30.3.1 The Ginzburg–Landau parameter

Ginzburg and Landau (1950) described the superconductor by an order parameter $\psi(\mathbf{r})$, with $|\psi|^2 = n_s$, whose free energy depends on its magnitude, its spatial gradient and the magnetic field. Two lengths come out of the theory: the penetration depth λ, over which the field varies, and the coherence length ξ, over which ψ varies. Their ratio

$$\kappa = \frac{\lambda}{\xi}$$

is the **Ginzburg–Landau parameter**, and it decides how a superconductor responds to a field.

The argument is about the energy of a boundary between normal and superconducting regions. Across such a wall the field penetrates the superconducting side over a depth λ, which *lowers* the cost of field expulsion, roughly by $(B_c^2/2\mu_0)\lambda$ per unit area. The pair density recovers over ξ, so a slab of thickness about ξ loses its condensation energy, *raising* the energy by roughly $(B_c^2/2\mu_0)\xi$. The wall energy per unit area is therefore approximately

$$\sigma_{ns} \approx \frac{B_c^2}{2\mu_0}\,(\xi - \lambda)$$

- If $\xi > \lambda$ (κ small), walls cost energy, the material avoids creating them, and the field is either fully expelled or, above B_c, fully admitted. This is a **Type I** superconductor.
- If $\lambda > \xi$ (κ large), walls *lower* the energy, and above a certain field the material divides itself as finely as possible into normal and superconducting regions. This is a **Type II** superconductor.

The exact GL calculation places the boundary at $\kappa = 1/\sqrt{2} \approx 0.71$. Pure elemental metals other than Nb, V and Tc are Type I. Alloys, compounds and all high-T_c materials are strongly Type II, because disorder shortens ξ and low carrier density lengthens λ. Alloying is therefore the materials engineer's switch from Type I to Type II, much as it is the switch from soft to strong in structural metals.

Status of the principle: Ginzburg–Landau theory and the $\kappa = 1/\sqrt{2}$ criterion
Derived from the GL free-energy expansion, which is itself *Approximate*: strictly valid near T_c, where ψ is small. Gor'kov showed that GL follows from BCS in that limit. It remains qualitatively correct and widely used at all temperatures.

30.3.2 The mixed state and the two critical fields

A Type II superconductor in a rising field passes through three states (Figure 30.4):

1. **Meissner state** for $B < B_{c1}$: complete expulsion, $M = -H$.
2. **Mixed (Shubnikov) state** for $B_{c1} < B < B_{c2}$: field enters as **vortices** (fluxoids), each a normal core of radius about ξ surrounded by circulating supercurrent extending out to about λ, and each carrying exactly one flux quantum Φ_0. The vortices repel each other and in a clean material form a triangular lattice (the Abrikosov lattice). The average flux density is $B = n_v \Phi_0$, where n_v is the number of vortices per unit area.
3. **Normal state** above B_{c2}, when the vortex cores overlap.

The GL theory gives the two critical fields in terms of λ and ξ:

$$B_{c1} \approx \frac{\Phi_0}{4\pi\lambda^2}\ln\kappa, \qquad B_{c2} = \frac{\Phi_0}{2\pi\xi^2} = \sqrt{2}\,\kappa\,B_c$$

The upper critical field has a simple meaning: at B_{c2} each vortex occupies an area of order $2\pi\xi^2$, so the cores fill space. Short coherence length means high B_{c2}, which is why disordered alloys and short-ξ cuprates can survive tens of tesla while a clean metal cannot survive a tenth of a tesla.

The triangular vortex lattice has spacing

$$a_0 = \left(\frac{2}{\sqrt{3}}\frac{\Phi_0}{B}\right)^{1/2} \approx 1.075\sqrt{\frac{\Phi_0}{B}}$$

At 5 T, $a_0 = 1.075\sqrt{2.068\times10^{-15}/5} = 1.075 \times 2.03 \times 10^{-8} = 22$ nm. This number will reappear in Section 30.5: to pin vortices effectively, the defects in the material must be spaced on this scale.

Status of the principle: $B_{c2} = \Phi_0/2\pi\xi^2$
Derived from the linearized GL equations. With $\xi(T)$ measured or fitted it is quantitatively reliable for isotropic materials; for layered materials B_{c2} depends on field direction, and in very high-B_{c2} materials the Pauli (spin) limit can cap it below the GL value.

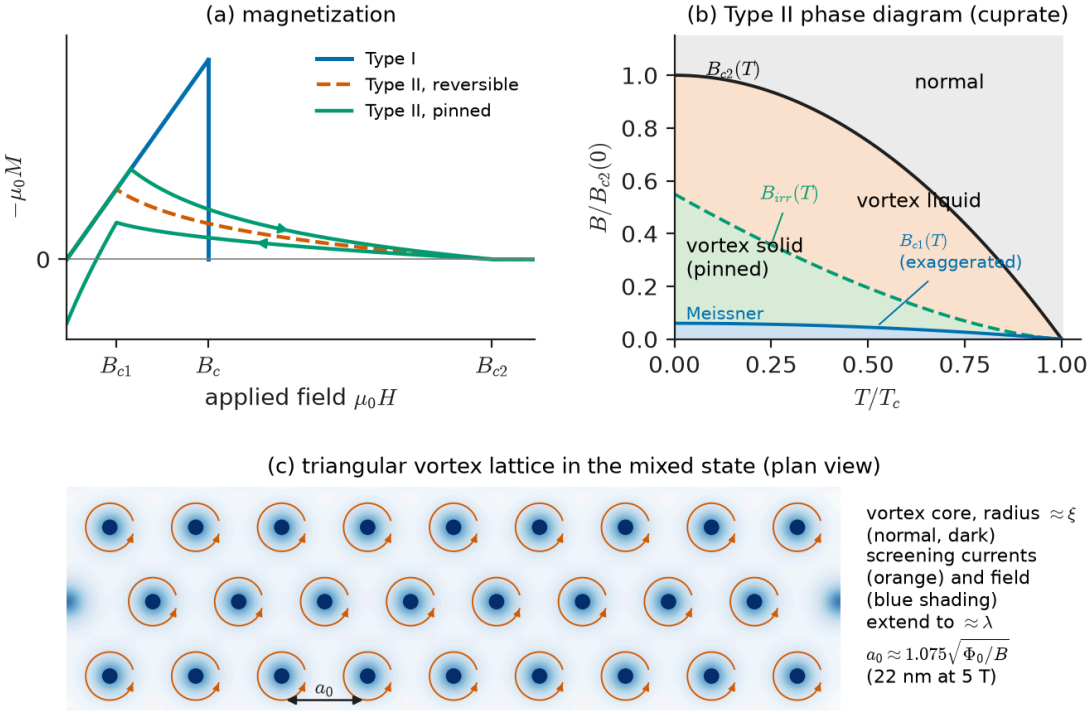


Figure 30.4 — Type I superconductors expel field completely up to Bc; Type II superconductors admit quantized vortices between Bc1 and Bc2, and pinning makes their magnetization irreversible.

30.3.3 Characteristic lengths in practice

Worked example — coherence length and κ from measured critical fields. Nb_3Sn has $B_{c2}(0) \approx 30$ T. Inverting $B_{c2} = \Phi_0/2\pi\xi^2$:

$$\xi = \sqrt{\frac{\Phi_0}{2\pi B_{c2}}} = \sqrt{\frac{2.068\times10^{-15}}{2\pi\times 30}} = \sqrt{1.097\times10^{-17}} = 3.3\times10^{-9}\ \text{m}$$

For Nb–Ti, $B_{c2}(0) = 14.5$ T gives $\xi = \sqrt{2.068\times10^{-15}/(2\pi\times14.5)} = 4.8$ nm. With measured penetration depths of about 80–120 nm for Nb_3Sn and about 250–300 nm for Nb–Ti, κ is roughly 25–35 and 50–60 respectively. Both are deep in the Type II regime. Compare niobium, $\xi \approx 38$ nm and $\lambda \approx 40$ nm, $\kappa \approx 1$: alloying 47 wt% titanium into niobium shortens the coherence length eightfold by electron scattering, and that is what raises B_{c2} from 0.4 T to 14.5 T while hardly changing T_c.

Table 30.2 — Characteristic lengths (approximate low-temperature values; cuprate values for the CuO_2 planes).

Material	λ (nm)	ξ (nm)	κ	Type
Al	16	1600	0.01	I
Pb	37	83	0.45	I
Nb	40	38	≈1	II (marginal)
Nb–Ti	250–300	4–5	50–60	II
Nb_3Sn	80–120	3–4	25–35	II
MgB_2	100–140	5–10	15–25	II
YBCO (in-plane)	150	1.5	≈100	II
YBCO (out-of-plane ξ_c)	800 (λ_c)	0.2–0.3	—	II

The lower critical field of YBCO is small despite its high T_c: with $\lambda = 150$ nm and $\kappa \approx 100$, $B_{c1} \approx [2.068\times10^{-15}/(4\pi\times(1.5\times10^{-7})^2)]\ln 100 = 7.3\times10^{-3}\times 4.6 \approx 0.034$ T. In every practical magnet, power or cable application, Type II conductors operate

in the mixed state, full of vortices. The Meissner state is relevant only to shielding, levitation and very-low-field sensing.

30.4 High-temperature superconductors and the practical family

Of thousands of known superconductors, about six are made into conductor in commercial quantities. They split into two groups: the **low-temperature superconductors** (LTS) Nb–Ti and Nb_3Sn, operated at 1.9–4.5 K, and the **high-temperature superconductors** (HTS), conventionally those with T_c above 77 K (the cuprates REBCO and BSCCO) or, more loosely, anything usable above 20 K (adding MgB_2 and the iron-based materials). This section compares them by structure and by the property that decides their use. Table 30.3 at the end collects the engineering data.

30.4.1 Nb–Ti: the ductile workhorse

Nb–Ti alloys of about 47 wt% Ti are body-centred cubic solid solutions with $T_c \approx 9.2$ K and $B_{c2}(4.2\ \text{K}) \approx 10.5$–11 T. What makes Nb–Ti dominant (most superconducting wire made, mainly for MRI) is not its modest superconducting parameters but its **mechanical** ones: it is a ductile metal that can be co-drawn with copper through area reductions of 10^4 or more, is isotropic and strain-tolerant, and needs no heat treatment after winding. Its J_c comes from pinning by nanometric α-Ti precipitates (Section 30.5.3).

30.4.2 Nb_3Sn: higher field, brittle compound

Nb_3Sn has the A15 crystal structure, in which Nb atoms form orthogonal chains along the cube faces. Its T_c is 18.3 K and $B_{c2}(0)$ is about 30 T (raised to 27–28 T at 4.2 K by Ti or Ta doping, which adds scattering). It is the conductor for fields from about 10 T to 23 T at 4.2 K: high-field NMR, the high-luminosity upgrade of the LHC (HL-LHC) quadrupoles, and the ITER toroidal-field and central-solenoid coils. Its disadvantage is fundamental: like most intermetallic compounds it is brittle, with a fracture strain of a few tenths of a per cent. It cannot be drawn. Instead the precursors (Nb and Sn, or Nb and Cu–Sn bronze) are drawn as ductile composites, the wire is wound into its final coil shape, and the compound is formed by a solid-state diffusion reaction at 650–700 °C ("wind and react", Section 30.7). Its J_c is also strongly **strain-dependent** (Section 30.7.1).

30.4.3 MgB_2: cheap and light, at 20 K

MgB_2 is a simple hexagonal compound of alternating Mg and graphite-like B layers. With $T_c = 39$ K it is the highest-T_c conventional (phonon-mediated) superconductor at ambient pressure. Its attractions are cost and availability: magnesium and boron are abundant and cheap, and the conductor is made by the powder-in-tube (PIT) method in Fe, Nb or Monel sheaths with a short reaction at 600–700 °C. Its coherence length (5–10 nm) is long enough that grain boundaries are not barriers to current, so untextured polycrystals carry useful current, unlike the cuprates. Its weakness is a low irreversibility field: in practical wires, useful J_c at 20 K extends only to a few tesla. MgB_2 is therefore a conductor for low-field, cryogen-free applications at 10–25 K: open MRI magnets, power cables cooled by liquid hydrogen, and light electrical machines.

30.4.4 Cuprates: YBCO (REBCO) and BSCCO

The cuprate high-temperature superconductors share one structural motif: **CuO_2 planes**, square nets of copper and oxygen, separated by "charge reservoir" layers that set the number of holes in the planes. Superconductivity lives in the planes. Three consequences follow from this structure, and each becomes a processing problem.

Doping by oxygen. In $YBa_2Cu_3O_{7-\delta}$, T_c depends on oxygen content. At $\delta \approx 0$ the material is orthorhombic with $T_c = 92$ K; at $\delta > 0.6$ it is tetragonal and insulating. Oxygen is lost on heating, so every YBCO process must finish with an oxygenation anneal around 400–500 °C. The rare earth Y can be replaced by Gd, Eu, Sm and others with similar results, hence the general label **REBCO** (rare-earth barium copper oxide).

Anisotropy. Transport is easy in the planes and difficult across them. The ratio of in-plane to out-of-plane critical fields, the anisotropy $\gamma = \xi_{ab}/\xi_c = \lambda_c/\lambda_{ab}$, is about 5–7 for YBCO but 50–200 for the bismuth compounds, whose thick Bi–O reservoir layers weakly couple the planes. Large anisotropy makes the vortices behave like stacks of loosely coupled "pancakes" that pin poorly at high temperature, so the irreversibility field of BSCCO at 77 K is only a few tenths of a tesla. Above that field, vortices move freely and J_c is zero even though B_{c2} is enormous. BSCCO is therefore used either at 77 K in low field (cables, current leads) or at 4–30 K, where its irreversibility field becomes large.

Weak links at grain boundaries. Because ξ is only 1–2 nm and the pair wavefunction has d-wave symmetry, a grain boundary in YBCO acts as a weak link. The critical current across a boundary falls roughly exponentially with misorientation angle θ:

$$J_c^{gb}(\theta) \approx J_c^{grain} \exp\left(-\frac{\theta}{\theta_0}\right), \qquad \theta_0 \approx 3\text{–}5^\circ$$

so a 10° boundary passes perhaps a tenth of the intragrain current, and a random polycrystal of YBCO carries almost nothing. The material must be **biaxially textured**, every grain aligned in all three axes to within a few degrees, which is what forced the development of the coated-conductor architecture (Section 30.6.3). BSCCO is more forgiving because its micaceous grains align readily by mechanical deformation and the boundaries between plate-like grains have large contact areas, which is why the first HTS wires (1990s) were Bi-2223 tapes.

> **Status of the principle: exponential grain-boundary law**
> *Empirical.* Measured on bicrystal films of YBCO, it captures the trend over 0–40°. The decay constant depends on doping and on boundary type (tilt or twist), and calcium overdoping of the boundaries partly restores current.

30.4.5 Iron-based superconductors

Discovered in 2008 in $LaFeAsO_{1-x}F_x$ (26 K), the iron pnictides and chalcogenides contain FeAs or FeSe layers. The leading conductor candidate is $Ba_{1-x}K_xFe_2As_2$ ($T_c \approx 38$ K); "1111" compounds reach about 55 K. They combine B_{c2} above 50 T with low anisotropy (1.5–2) and a grain-boundary critical angle about twice that of YBCO, so less perfect texture is tolerated, and PIT tapes have exceeded 10^5 A/cm² at 4.2 K and 10 T. Arsenic toxicity and process immaturity keep them in development.

Table 30.3 — Practical conductors compared (typical values; "non-Cu" means the superconductor plus matrix excluding copper stabilizer).

Conductor	T_c (K)	Typical operating point	J_c (A/mm²) at that point	Irreversible strain limit	Heat treatment after winding?	Main uses
Nb–Ti	9.2	4.2 K, 5 T	≈3000 (non-Cu)	>1 % (ductile)	No	MRI, accelerators, NMR outer coils
Nb_3Sn	18.3	4.2 K, 12 T	2500–3000 (non-Cu)	≈0.2–0.4 % tensile	Yes, 650–700 °C, 50–200 h	NMR, HL-LHC, ITER, fusion
MgB_2	39	20 K, 1–2 T	500–1000 (filament)	0.3–0.5 %	Yes (in-situ route), ≈650 °C	Open MRI, cables, machines
Bi-2223 tape	110	77 K, self-field	150–200 (J_e)	0.2–0.5 % (reinforced)	No (reacted before winding)	Cables, current leads
Bi-2212 round wire	85–92	4.2 K, 20–30 T	1000+ (J_e at 20 T)	≈0.3 %	Yes, partial melt ≈ 890 °C in O_2	Very-high-field inserts
REBCO tape	92	77 K self-field / 20 K, 20 T	300–1000 (J_e)	0.4–0.7 % tensile	No	Fusion, >20 T magnets, FCLs, cables

30.5 Flux pinning

30.5.1 Why vortices must not move

A transport current density **J** flowing through a vortex lattice exerts on each vortex a force per unit length

$$\mathbf{f} = \mathbf{J} \times \Phi_0 \hat{\mathbf{z}}$$

where $\hat{\mathbf{z}}$ is the direction of the vortex. Per unit volume, with $n_v = B/\Phi_0$ vortices per unit area, this becomes the familiar Lorentz force density $\mathbf{F}_L = \mathbf{J} \times \mathbf{B}$. If the vortices move with velocity $\mathbf{v}$, they induce an electric field

$$\mathbf{E} = \mathbf{B} \times \mathbf{v}$$

parallel to $\mathbf{J}$, and power $\mathbf{E} \cdot \mathbf{J}$ is dissipated. In a perfectly homogeneous superconductor the vortices are opposed only by viscous drag, and the resulting **flux-flow resistivity** is approximately

$$\rho_{ff} \approx \rho_n \frac{B}{B_{c2}}$$

(Bardeen–Stephen)

> **Status of the principle: Bardeen–Stephen flux-flow resistivity**
> *Approximate.* It treats each vortex core as a normal cylinder of radius ξ and is good to a factor of order one for $B \ll B_{c2}$ in conventional superconductors.

30.5.2 Pinning and the critical current density

A vortex core is a normal region of radius ξ; creating it costs condensation energy of roughly $(B_c^2/2\mu_0)\pi\xi^2$ per unit length. If the vortex passes through a region where superconductivity is already weakened (a non-superconducting precipitate, a grain boundary, a void, a dislocation, an oxygen vacancy cluster), that energy is saved. The vortex sits in a potential well and resists being moved. The maximum restoring force per unit volume that the defect population can supply is the **pinning force density** F_p, and the critical current is reached when the Lorentz force equals it:

$$J_c B = F_p$$

A useful pinning centre has two properties: its size is comparable to ξ (a few nanometres), because a smaller defect barely perturbs the core and a larger one is wasteful; and the defects are numerous, with spacing comparable to the vortex spacing a_0 (about 20 nm at 5 T), so that every vortex has a pin nearby.

> **Status of the principle: critical-state condition $J_c B = F_p$**
> *Derived* as a force balance; F_p itself is a microstructure-dependent quantity that is measured, not predicted. Its field dependence is usually fitted by the *Empirical* scaling law $F_p \propto b^p(1-b)^q$ with $b = B/B_{c2}$ (Kramer, Dew-Hughes), where p and q identify the dominant pinning mechanism.

30.5.3 Pinning microstructures by material

Material	Dominant pinning centres	Size and spacing	How introduced
Nb–Ti	α-Ti precipitates (hcp), 20–25 vol%	Ribbons 1–4 nm thick, spacing ≈ 5–20 nm	Repeated 375–420 °C precipitation anneals, then heavy cold drawing that thins the precipitates to ribbons
Nb_3Sn	Grain boundaries	Grains 50–150 nm; $F_p \propto 1/d$	Low reaction temperature (finer grains); Hf, Zr additions; internal oxidation
MgB_2	Grain boundaries; C substitution raises B_{c2}	Grains 50–200 nm	Nanoscale B powder, C-doping, low-T reaction
REBCO	Correlated defects: $BaZrO_3$ or $BaHfO_3$ nanorods along c; stacking faults; Y_2O_3 particles	Nanorods 5–6 nm diameter, spacing 10–20 nm	Added to deposition target; self-assemble during growth
Bi-2212/2223	Intrinsic (layered structure), defects	—	Limited control; high anisotropy limits pinning at 77 K

Nb–Ti is the textbook example of microstructural engineering. Solution-annealed, it has J_c of only a few hundred A/mm² at 5 T. Three or four precipitation anneals near 400 °C, each followed by cold drawing, produce α-Ti precipitates that the final drawing strain flattens into ribbons a few nanometres thick, spaced comparably to the vortex lattice at 5 T, raising J_c almost tenfold to

about 3000 A/mm². The pinning force density is then $F_p = J_c B = 3 \times 10^9 \times 5 = 1.5 \times 10^{10}\ \mathrm{N\,m^{-3}}$, a body force comparable to that from a pressure gradient of 15 GPa per metre.

In REBCO, $BaZrO_3$ added to the deposition target self-assembles into columns about 5 nm across, parallel to the c axis, during film growth. Each column matches a vortex along the same axis, so these **correlated** pins work best for fields along c, where REBCO otherwise pins worst; doubling J_c at 20–30 K this way has been central to compact fusion magnets.

30.5.4 The Bean critical-state model

In 1962 C. P. Bean proposed a model that turns pinning into engineering numbers. Assume that wherever the field has changed, the local current density is exactly $\pm J_c$, and wherever it has not, $J = 0$; take J_c independent of field. Consider an infinite slab of half-thickness a ($-a < x < a$) in a field H applied parallel to its faces, starting from the virgin state. Ampère's law in one dimension is

$$\frac{dH}{dx} = \pm J_c$$

so the field inside falls *linearly* from its surface value, with slope J_c, and the flux front advances to a depth H/J_c (Figure 30.5). The field reaches the centre at the **full-penetration field**

$$H^* = J_c a, \qquad B^* = \mu_0 J_c a$$

Above H^*, the profile is a "roof" of fixed shape riding on the applied field, and the mean field inside is $H - J_c a/2$, so the magnetization saturates at

$$M = -\frac{J_c a}{2}$$

On reversing the field sweep, the sign of the current reverses from the surface inward, and M swings to $+J_c a/2$. The magnetization loop is a parallelogram of width $J_c a$ in M, and J_c can be measured from it with a magnetometer ($J_c = \Delta M / a$ for a slab, where $\Delta M = J_c a$ is the full loop width), which is the standard contactless characterization of superconductors (Chapter 52).

The area of the loop is energy dissipated as heat per cycle. For a field cycled between $\pm B_m$ with $B_m \gg B^*$, the loss per unit volume per cycle is the loop width times the field swing:

$$Q_h \approx \mu_0 \, (J_c a) \, (2 H_m) = 2 B_m J_c a$$

The hysteresis loss is proportional to the filament size a. This single result explains why superconducting wire is made of thousands of fine filaments.

> **Status of the principle: Bean critical-state model**
> *Approximate.* It assumes J_c independent of field, sharp current fronts and no flux creep. It is quantitatively good for $B_m \gg B^*$ and for the shape of magnetization loops; the Kim model ($J_c \propto 1/(B + B_0)$) adds field dependence. Geometry changes numerical factors: for a cylindrical filament in a transverse field the coefficient differs from the slab value by order one.

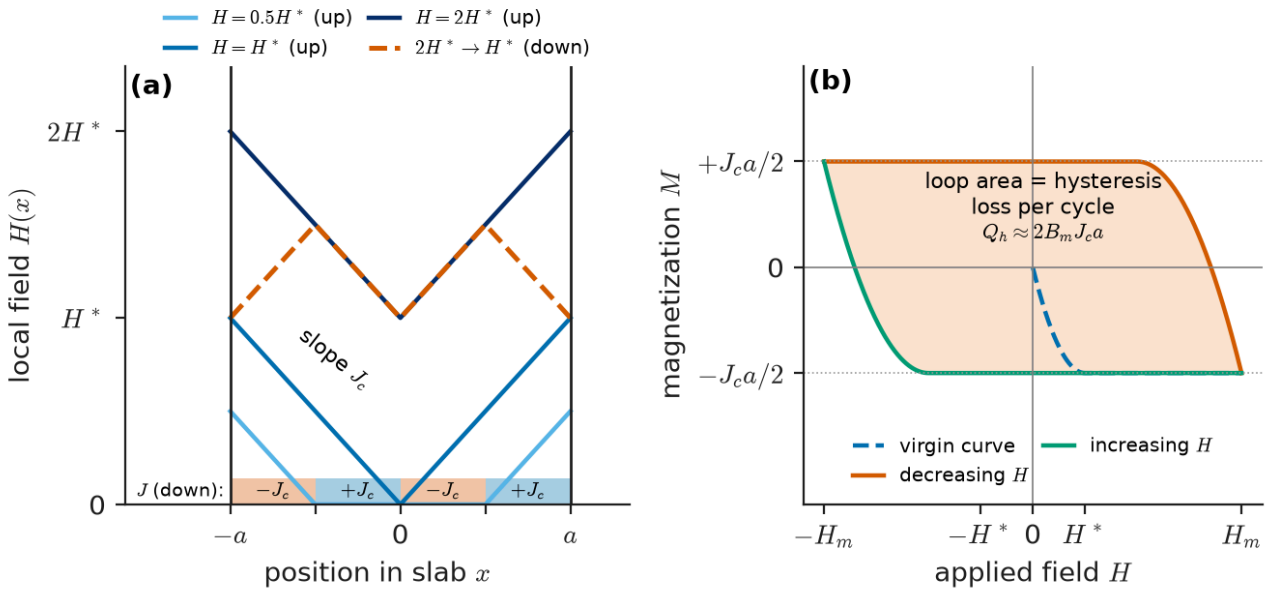


Figure 30.5 — The Bean critical-state model: field profiles in a slab advance with constant slope Jc, and the resulting magnetization loop has width proportional to Jc times filament size.

Worked example — hysteresis loss and filament size. Take a Nb–Ti conductor with $J_c = 3 \times 10^9$ A m^{-2} (3000 A/mm²), treat each filament as a slab of half-thickness a, and cycle the field to $B_m = 1$ T.

For filaments 6 μm in diameter ($a = 3$ μm, as in the LHC strands):

$$B^* = \mu_0 J_c a = (1.257 \times 10^{-6})(3 \times 10^9)(3 \times 10^{-6}) = 0.011 \text{ T}$$

so $B_m \gg B^*$ and

$$Q_h = 2 B_m J_c a = 2 \times 1 \times (3 \times 10^9) \times (3 \times 10^{-6}) = 1.8 \times 10^4 \text{ J m}^{-3}$$

For filaments 100 μm in diameter ($a = 50$ μm), $Q_h = 2 \times 1 \times 3 \times 10^9 \times 5 \times 10^{-5} = 3.0 \times 10^5$ J m^{-3}, almost seventeen times more.

For one cubic centimetre of superconductor the energy per cycle is 0.018 J against 0.30 J. In a magnet ramped once per minute, the coarse conductor deposits $0.30/60 = 5$ mW per cm³ at 4.2 K, which at several hundred watts of wall power per watt at 4.2 K costs over a watt of refrigerator input per cubic centimetre. At 50 Hz the same coarse conductor would deposit 15 W per cm³, impossible to remove. AC applications need filaments of a few micrometres or less, and even then losses limit LTS to DC or slowly ramped magnets.

30.5.5 Flux creep and the irreversibility field

Pinning wells have finite depth U_0, and vortices can hop thermally out of them at a rate proportional to $\exp(-U_0/k_B T)$. The trapped current in a persistent magnet therefore decays logarithmically in time, and the voltage across a conductor does not jump abruptly at J_c but rises steeply:

$$E = E_c \left(\frac{J}{J_c} \right)^n$$

where J_c is defined by a voltage criterion, conventionally $E_c = 1$ μV cm^{-1} (or 0.1 μV/cm). The **n-value** is a figure of merit for conductor quality and for persistent-mode operation: 30–60 for good Nb–Ti, 20–40 for REBCO at low temperature, lower when the conductor is damaged or non-uniform.

> **Status of the principle: power-law E–J characteristic**
> *Empirical.* It fits measured transitions over several decades of electric field and is the standard model for conductor specification and quench simulation; the logarithmic creep it represents has a *Derived* basis in thermally activated hopping.

In LTS, $U_0 \gg k_B T$ at 4.2 K and creep is small. In the cuprates, where ξ is short (shallow wells), T is high and anisotropy weakens the vortices, creep is strong. Above the **irreversibility field** $B_{irr}(T)$ the vortex lattice melts into a "vortex liquid" and J_c falls to zero. At 77 K, B_{irr} is about 7 T for REBCO with $B \parallel c$ but only about 0.2 T for Bi-2223, and this, more than T_c or B_{c2}, determines which HTS can be used where.

30.6 Superconducting wires and tapes

A superconductor filament alone is not a conductor. If any small region turns normal (a crack, a burst of flux motion, a mechanical slip of the winding releasing frictional heat) its resistivity jumps to the normal-state value, about $6 \times 10^{-7}\ \Omega\,\mathrm{m}$ for Nb–Ti, and the full current flowing through it would heat it explosively. A practical wire is therefore a **composite**: fine superconducting filaments embedded in a matrix of high-purity copper (or silver, or aluminium) that carries the current briefly if the superconductor fails, conducts heat away, and gives mechanical support.

30.6.1 Stabilization: the role of the copper

At 4.2 K, high-purity copper has a residual resistivity set by impurities and defects (Chapter 25). Its quality is expressed by the residual resistance ratio, RRR = $\rho(273\ \mathrm{K})/\rho(4.2\ \mathrm{K})$: RRR = 100 gives about $1.6 \times 10^{-10}\ \Omega\,\mathrm{m}$, roughly four thousand times less than normal Nb–Ti. Placing the filaments in this matrix allows the current to bypass a normal zone, limiting the local heating so that the zone can recover (cryogenic stability) or, in large magnets, propagate slowly enough to be detected and the magnet discharged safely (quench protection). The Cu:SC volume ratio is a design choice: about 1–2 for accelerator and fusion conductors, up to 4–7 for MRI conductors and much higher for cryostable conductors in large detector magnets.

Heat treatments for pinning (Nb–Ti) or reaction (Nb_3Sn) can contaminate the copper and lower its RRR, so diffusion barriers (Nb or Ta) are placed between filaments and stabilizer in Nb_3Sn wires.

30.6.2 Multifilamentary LTS wires

Three reasons drive superconducting filaments to be fine:

1. **Hysteresis loss** is proportional to filament size (Section 30.5.4).
2. **Flux jumping.** When the field changes, the critical state shifts and flux moves, dissipating heat. Heat lowers J_c, which lets more flux move: positive feedback. An adiabatic stability estimate for a slab of half-thickness a is

$$a < \sqrt{\frac{3\,\gamma C\,(T_c - T_0)}{\mu_0 J_c^2}}$$

   where γC is the volumetric heat capacity and T_c is the field-dependent critical temperature. For Nb–Ti at 4.2 K in 5 T, with $\gamma C \approx 6 \times 10^3\ \mathrm{J\,m^{-3}\,K^{-1}}$, $T_c(5\ \mathrm{T}) - T_0 \approx 2.8$ K and $J_c = 3 \times 10^9\ \mathrm{A\,m^{-2}}$, this gives $a < \sqrt{3(6\times10^3)(2.8)/[(1.257\times10^{-6})(9\times10^{18})]} = 67\ \mu\mathrm{m}$, a filament diameter of order 100 μm. With safety margins, filaments well below 50 μm are standard. (*Approximate*: adiabatic, field-independent J_c, no cooling.)
3. **Field quality** in accelerator magnets: magnetization currents in filaments distort the field at injection, scaling with $J_c a$.

Fine filaments alone are not enough: in a changing field they couple through the conducting matrix and act as one large filament. **Twisting** the wire (pitch 10–25 mm) reverses the coupling currents every half pitch; the coupling time constant scales as L_p^2/ρ_t, with L_p the pitch and ρ_t the matrix transverse resistivity, which resistive Cu–Mn or Cu–Ni barriers raise.

Nb–Ti wire is made by stacking Nb–Ti rods in copper cans into a billet, extruding, and drawing through many dies with intermediate precipitation heat treatments. An MRI wire may contain 30–60 filaments of 50–100 μm; an LHC strand (0.825–1.065 mm diameter) contains 6000–9000 filaments of 6–7 μm (Figure 30.6).

Nb_3Sn wire is drawn as precursors and reacted after winding. Three routes dominate:

- **Bronze route**: Nb filaments in a Cu–Sn bronze matrix (up to about 15 wt% Sn, the solubility limit for a workable bronze). Easy to draw and good for low-loss conductors, but the limited tin gives lower J_c; used for ITER and NMR.
- **Internal tin**, including restacked-rod process (RRP): tin cores surrounded by Nb filaments and Cu within a diffusion barrier. More tin gives more Nb_3Sn and J_c of 2500–3000 A/mm² at 12 T, 4.2 K; used for HL-LHC.
- **Powder-in-tube (PIT)**: $NbSn_2$ powder inside Nb tubes, reacting outward.

Reacted Nb_3Sn filaments merge into larger effective filaments (sub-elements 40–60 μm across), so flux-jump stability and magnetization at low field remain design issues.

30.6.3 HTS conductors

Bi-2223 tape (first generation). Oxide powder in silver tubes (silver is permeable to oxygen and inert to the oxide) is drawn, stacked, rolled flat and heat-treated near 830 °C with intermediate rolling, which aligns the plate-like grains. A 4 mm × 0.2–0.3 mm tape carries 150–200 A at 77 K in self-field; the silver, 60–70 % of the cross-section, sets a cost floor.

Bi-2212 round wire. The only isotropic round HTS wire, so it can be cabled conventionally. It is heat-treated after winding by **partial melting** at about 880–890 °C in oxygen, under about 50 bar overpressure to remove porosity, and then carries over 1000 A/mm² (J_e) at 4.2 K and 20 T.

REBCO coated conductor (second generation). Because YBCO needs biaxial texture, it is grown as an epitaxial film on a textured template rather than as a drawn composite. A typical architecture, bottom to top (Figure 30.6):

Layer	Material	Thickness	Function
Substrate	Hastelloy C-276 (Ni–Mo–Cr), or textured Ni–W	30–100 μm	Strength (>600 MPa); non-magnetic (Hastelloy)
Buffer stack	Al_2O_3 diffusion barrier, Y_2O_3 seed, IBAD-MgO template, homoepitaxial MgO, $LaMnO_3$ or CeO_2 cap	≈0.1–0.3 μm total	Biaxial texture (IBAD: ion-beam-assisted deposition, aligning MgO to within a few degrees), chemical barrier
Superconductor	(RE)BCO, often with $BaZrO_3$ or $BaHfO_3$ nanorods	1–3 μm	Current-carrying layer
Cap	Ag	1–3 μm	Oxygen-permeable protection; electrical contact
Stabilizer	Electroplated Cu (both sides)	10–40 μm per side	Quench protection, current sharing

The superconductor is deposited by pulsed-laser deposition, metal–organic chemical vapour deposition (MOCVD), reactive co-evaporation or metal–organic deposition at 700–850 °C, and oxygenated at about 500 °C. The tape is produced in lengths of hundreds of metres to a kilometre, in widths of 2–12 mm. The superconductor itself is only 1–2 % of the cross-section, so the **engineering current density** J_e, defined over the whole tape, is the number a magnet designer uses.

Worked example — REBCO tape current and engineering current density. A 4 mm wide tape carries a 1.5 μm REBCO layer with $J_c = 3\ \text{MA/cm}^2 = 3 \times 10^{10}\ \text{A m}^{-2}$ at 77 K, self-field. The critical current is

$$I_c = J_c\, w\, t = (3 \times 10^{10})(4 \times 10^{-3})(1.5 \times 10^{-6}) = 180\ \text{A}$$

With a total tape thickness of 0.1 mm (substrate, buffers, silver and copper), the engineering current density is

$$J_e = \frac{I_c}{w\, t_{tape}} = \frac{180}{(4 \times 10^{-3})(1 \times 10^{-4})} = 4.5 \times 10^{8}\ \text{A m}^{-2} = 450\ \text{A/mm}^2$$

A conventional copper power cable is designed for 1–5 A/mm² for thermal reasons. Even at 77 K, where it is far from its best, the HTS tape carries roughly a hundred times more current per unit cross-section. At 20 K and 20 T the same tape typically still carries a few hundred A/mm² in J_e, which is the basis of compact high-field fusion magnets. Note also that a 25 % thinner substrate raises J_e by almost 25 % without touching the superconductor: much of conductor development is about the parts that do not superconduct.

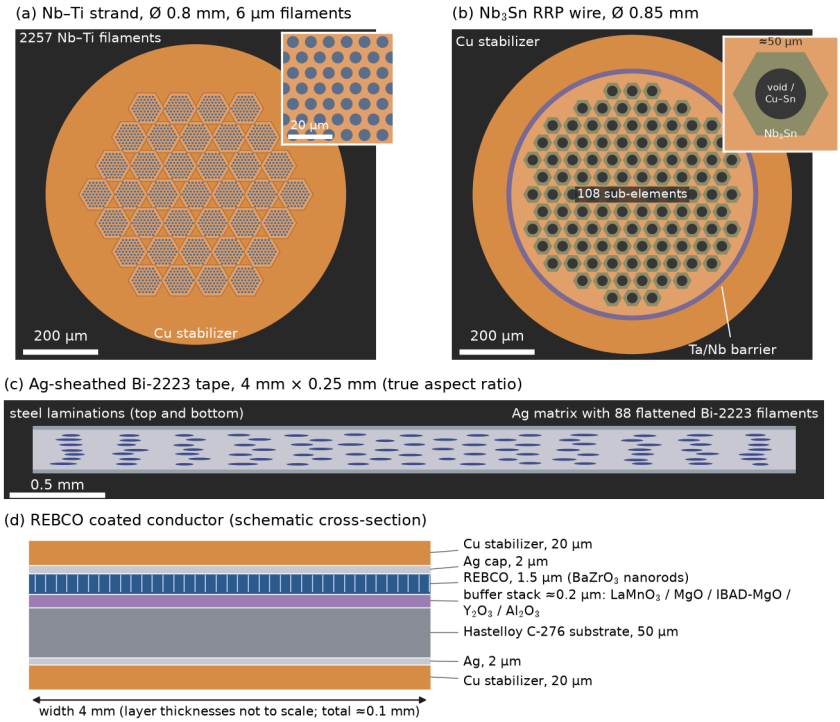


Figure 30.6 — Four conductor architectures: multifilamentary Nb-Ti in copper, reacted Nb3Sn RRP sub-elements, Ag-sheathed Bi-2223 multifilament tape, and the layered REBCO coated conductor.

30.6.4 Cables

Single strands carry tens to hundreds of amperes; magnets for accelerators and fusion need 10–70 kA to keep the inductance and stored-energy-to-voltage ratio manageable during a quench. Strands are therefore cabled:

- **Rutherford cable**: 20–40 round strands twisted and flattened into a two-layer keystoned tape, used in all accelerator dipoles (Nb–Ti, Nb_3Sn, Bi-2212).
- **Cable-in-conduit conductor (CICC)**: hundreds to about a thousand strands in a steel jacket with forced-flow supercritical helium; used in fusion (ITER: Nb_3Sn and Nb–Ti).
- **HTS tape cables**: Roebel (punched, transposed tapes), CORC (tapes wound helically on a round core) and twisted stacked tapes.

Transposition makes each strand see the same average field, so current divides equally; cabling also deforms the strands, which brings in the strain limits of the next section.

30.7 Material processing constraints

30.7.1 Strain sensitivity

The critical current of most practical superconductors depends on mechanical strain, and magnets are large stress environments: the Lorentz force $\mathbf{J} \times \mathbf{B}$ in a 10 T magnet carrying 500 A/mm² is 5×10^9 N m^{-3}, and conductor hoop stresses of 100–400 MPa are normal.

- **Nb–Ti** is relatively insensitive: I_c falls by only a few per cent at 0.5 % strain, and the alloy is ductile.
- **Nb_3Sn** is strongly sensitive. The A15 lattice is distorted by strain, which lowers T_c and B_{c2}. After reaction, differential thermal contraction between Nb_3Sn and the Cu or bronze matrix on cooling to 4 K leaves the filaments under an axial compressive precompression of about 0.2–0.3 %. Applied tension first relieves this (raising I_c to a maximum) and then reduces it. At 12 T the reversible I_c variation over a strain range of ±0.5 % can be 30–50 %. Beyond an intrinsic tensile strain of about 0.2–0.4 % filaments crack and the damage is **irreversible**. Transverse compressive stress in Rutherford cables is limited to roughly 150–200 MPa for the same reason.

- **REBCO** tolerates axial tensile strain up to about 0.4–0.7 % reversibly, set mainly by the strong Hastelloy substrate, but it is very weak in **transverse tension**: the layered structure can delaminate at stresses of 10–50 MPa, which is why epoxy-impregnated REBCO coils must be designed to avoid thermal-contraction peeling stresses.
- **BSCCO** and **MgB_2** have irreversible strain limits of roughly 0.2–0.5 %, raised by external reinforcement.

The empirical strain function $I_c(\varepsilon)$ is measured on each conductor type with bending-beam or tensile rigs and enters magnet design as an essential input, as fatigue curves do for structural design (Chapter 14).

> **Status of the principle: strain dependence of J_c**
> *Empirical* in its engineering form (fitted strain-scaling functions for $T_c(\varepsilon)$ and $B_{c2}(\varepsilon)$); the underlying lattice-distortion effect on T_c is physical but not predicted quantitatively.

30.7.2 Brittleness and the order of operations

Nb_3Sn, MgB_2 and the cuprates are ceramics or intermetallics. Their bending strain limit means a conductor of thickness t wound on a radius R experiences a surface bending strain of about $t/2R$. For a 0.8 mm Nb_3Sn strand with an allowable 0.2 % strain the minimum bending radius would be $0.0004/0.002 = 0.2$ m, too large for magnet ends if bent after reaction. Hence two processing strategies:

- **Wind and react**: wind the unreacted, ductile precursor, then heat-treat the whole coil. Used for Nb_3Sn, Bi-2212 and in-situ MgB_2.
- **React and wind**: react first and wind with large bend radii or with the brittle layer at the neutral axis. Used for Bi-2223, ex-situ MgB_2 and REBCO tape (in which the 1–2 μm superconductor sits on a thick metal substrate, so bend strain is controlled by placing it near the neutral axis of the laminate).

30.7.3 Reaction heat treatment

A Nb_3Sn coil is heated in vacuum or argon through stages, typically about 210 °C and 400 °C to form Cu–Sn phases and distribute tin, then 650–700 °C for 50–200 hours to form Nb_3Sn by diffusion. The reaction is a compromise between kinetics and microstructure: higher temperature reacts faster and more completely, but coarsens the grains, reducing pinning ($F_p \propto 1/d$) and J_c at intermediate fields. The heat treatment imposes constraints across the whole magnet:

- The insulation must survive 650 °C: glass or ceramic-fibre braid, with the coil vacuum-impregnated with epoxy only afterwards.
- Coil tooling must hold dimensions to tens of micrometres during the reaction, despite the volume change of about 3–5 % that accompanies Nb_3Sn formation.
- The copper stabilizer's RRR must be protected by diffusion barriers.
- Once reacted, the coil cannot be repaired by unwinding.

Bi-2212 is harder still: the whole coil must be held uniform to about ±1 K at 880–890 °C in oxygen, with insulation and reinforcement compatible with hot oxygen. REBCO avoids post-winding treatment, but once made it loses oxygen if heated much above about 200–250 °C, which limits soldering temperatures for joints.

30.7.4 Anisotropy, texture and joints

Anisotropic conductors carry different currents in different field orientations. For REBCO the critical current at 20 K and 20 T may differ several-fold between field parallel and perpendicular to the tape, so magnet designers must track the field angle at every turn. Texture must be maintained over kilometre lengths: a single defect region over a few centimetres limits a whole length, so tapes are inspected continuously by Hall-probe scanning of trapped field.

Joints are a materials problem too. LTS filaments can be joined superconductingly by cold pressing, which allows persistent operation; HTS joints are usually soldered and resistive (10^{-9}–10^{-8} Ω), which makes persistent HTS magnets difficult.

30.7.5 Cryogenics and critical resources

Every superconductor carries the constraint of its coolant. Helium is a non-renewable by-product of natural gas with recurrent shortages, which is why modern MRI magnets use sealed, low-inventory cryocooler systems. Niobium comes mostly from one country (Brazil), and silver (BSCCO) and Hastelloy (REBCO) set cost floors. These issues return in Chapter 59.

30.8 Applications in magnets, power systems, and sensing

30.8.1 MRI, NMR and research magnets

Clinical MRI at 1.5 T and 3 T is the largest market for superconductors, using Nb–Ti with a high copper fraction in persistent mode at 4.2 K; research scanners reach 7 T and above (11.7 T in Nb–Ti at 1.8 K). Higher fields need other materials: NMR magnets nest Nb–Ti (outer, low field) around Nb_3Sn, and commercial 1.2 GHz (28.2 T) spectrometers add REBCO inserts. All-superconducting research magnets have reached 32 T with REBCO inserts. MgB_2 serves low-field, cryogen-free MRI. Grading a coil, cheap Nb–Ti where the field is low and high-field conductor only where needed, is a materials-selection exercise on the critical surfaces of Section 30.2.

30.8.2 Accelerator and fusion magnets

The LHC at CERN uses 1232 Nb–Ti dipoles at 8.3 T and 1.9 K. The HL-LHC upgrade introduces Nb_3Sn quadrupoles with peak conductor fields near 11–12 T. Research towards 14–16 T dipoles for future colliders uses Nb_3Sn, with HTS inserts for 20 T.

Fusion magnets are the most demanding. ITER's toroidal-field coils use Nb_3Sn CICC (peak field about 11.8 T) and its poloidal-field coils Nb–Ti. Compact tokamak designs use REBCO at about 20 K to reach 20 T on the coil (demonstrated in a large model coil in 2021), because fusion power density rises steeply with field. The materials questions are those of this chapter: J_e at 20 T and 20 K, strain and delamination under huge Lorentz forces, plus neutron irradiation damage.

30.8.3 Power cables

An HTS cable has tapes wound helically on a former, cooled by flowing liquid nitrogen at 65–77 K inside a vacuum-insulated cryostat. An engineering current density a hundred times copper's allows high power at low voltage, so a 10 kV HTS cable can replace a 110 kV line in a city (a 1 km, 10 kV, 40 MVA BSCCO cable has run in Essen since 2014), and a coaxial HTS shield confines the magnetic field.

At 50 Hz the cable is not lossless: hysteresis loss in the tapes (Section 30.5.4) and dielectric losses must be removed at 77 K, which at a cryogenic penalty of 10–20 W per watt sets the economic break-even against copper.

30.8.4 Superconducting fault-current limiters

A **resistive superconducting fault-current limiter** (SFCL) exploits what is a failure mode in a magnet. In normal operation the current is below I_c and the HTS element has negligible resistance. When a fault drives the current above I_c, the material moves up its steep E–J curve, heats, and transitions to the normal state within about a millisecond, inserting a resistance that limits the fault current, often to a few times the rated current, before the first current peak. The device resets itself when the fault is cleared and the element recools.

The material requirements are specific: a high normal-state resistance per unit length (thin or resistive stabilizer, unlike magnet conductors), a uniform I_c along kilometres of tape so that the whole length switches together rather than a weak spot burning out, and the ability to absorb the fault energy without exceeding a safe temperature (typically 300–400 K). REBCO tapes with thin silver and stainless-steel or Hastelloy shunts are the usual choice. Inductive and saturated-core variants use superconductors differently. SFCLs are in service in distribution networks in several countries.

Superconducting machines, magnetic energy storage and maglev face the same trade-off between current density, AC loss and cryogenic cost.

30.8.5 SQUIDs and superconducting sensors

The most sensitive magnetometer known is built from the flux quantum. A **SQUID** (superconducting quantum interference device) is a superconducting loop interrupted by one or two **Josephson junctions**, thin barriers (about 1–2 nm of AlO_x in a $Nb/AlO_x/Nb$ trilayer) through which Cooper pairs tunnel. The maximum supercurrent through a two-junction (DC) SQUID oscillates with the flux through the loop with period Φ_0, and with feedback electronics the flux can be resolved to about $10^{-6}\Phi_0$ per $\sqrt{\text{Hz}}$.

Worked example — field resolution of a SQUID magnetometer. A low-T_c SQUID with a flux noise of 1 $\mu\Phi_0/\sqrt{\text{Hz}}$ is coupled to a pickup coil with an effective area of 1 mm² (1×10^{-6} m^2). One flux quantum through that area corresponds to a field

$$B = \frac{\Phi_0}{A} = \frac{2.068 \times 10^{-15}}{1 \times 10^{-6}} = 2.07 \times 10^{-9}\ \text{T}$$

so the field noise is $2.07 \times 10^{-9} \times 10^{-6} = 2 \times 10^{-15}\ \text{T}/\sqrt{\text{Hz}}$, about 2 fT/√Hz. The Earth's field is about 50 μT, ten orders of magnitude larger; the magnetic field of the human brain at the scalp is 10–1000 fT. This is why magnetoencephalography (MEG), using hundreds of Nb SQUIDs at 4.2 K in a magnetically shielded room (Chapter 29), is possible.

The materials of SQUIDs are chosen differently from conductor materials. LTS SQUIDs use Nb thin films (Nb is Type II with $\kappa \approx 1$, stable, and makes reproducible AlO_x tunnel barriers). HTS SQUIDs use YBCO films with Josephson junctions formed **deliberately** at grain boundaries, by growing the film on a bicrystal substrate with a 24–37° misorientation: the weak-link behaviour that ruins YBCO wires is exactly what a junction needs. HTS SQUIDs operate at 77 K with noise of tens of fT/√Hz, and are used in non-destructive testing and geophysics.

Transition-edge sensors, nanowire single-photon detectors (NbN or WSi films a few nanometres thick) and Al or Nb quantum circuits rely on the same thin-film parameters: T_c, λ, the gap and interface defect density (Chapter 34).

> **Boundary.**
> Josephson-junction physics and SQUID readout circuits are treated in Engineering Physics and Electronics & Computing; this section explains only why the materials are chosen.

What remains open

Open question	Where it is answered
How do thin superconducting films (NbN, YBCO) grow epitaxially, and how are buffer stacks deposited?	Chapter 49 (thin-film deposition principles), Chapter 50 (thin-film deposition)
How is the copper stabilizer's residual resistivity set by purity and annealing?	Chapter 25 (Matthiessen's rule), Chapter 10 (annealing)
How are Nb–Ti precipitation anneals and Nb_3Sn diffusion reactions controlled as phase transformations?	Chapter 10
How are joints, solders and laminations for conductors and leads chosen?	Chapter 51
How are J_c, T_c and magnetization measured, and with what uncertainty?	Chapter 52 (four-point probe, magnetic characterization, uncertainty)
How do delamination, fatigue and thermal cycling degrade conductors over a magnet's life?	Chapter 44
When is a superconducting solution cheaper or more sustainable than copper, given cryogenic cost and helium supply?	Chapter 58
How are magnets, cables and fault-current limiters designed as systems?	Electrical Energy & Machines core

Next: Chapter 31 (Solid-State Foundations for Materials Engineering) opens Part VII by asking why some solids conduct, others insulate, and semiconductors lie in between, replacing the free-carrier picture of Chapter 25 with energy bands, band gaps and the Fermi level.

Part VII — Semiconductor and Photonic Materials

Part VI treated electrons in materials as either free to move (metals, Chapter 25) or bound in place (dielectrics, Chapters 26–27). Semiconductors sit between the two. Their conductivity can be changed by a factor of 10^{10} through a few parts per million of impurity, switched by an applied field, and generated or absorbed as light, and that is why nearly every active electronic function is built from them. This part starts with the electronic structure of crystals, counts and moves the carriers, surveys the material families, and continues with the interfaces, thin films and optoelectronic materials from which real devices are made. It ends with optical and photonic materials, where the same ideas explain how glasses, crystals and polymers refract, absorb, scatter, emit and guide light. The aim throughout is the material, not the device: what sets the band gap, the mobility, the lifetime, the breakdown field and the thermal conductivity, how processing and defects change them, and how an engineer compares candidate materials before any circuit is drawn.

By the end of this part you should understand

- Why electrons in a crystal occupy energy bands separated by gaps, and how the size and type (direct or indirect) of the gap follow from the atoms, the bonding and the crystal structure.
- How the density of states, Fermi–Dirac statistics and the effective mass set the number of electrons and holes, and why that number depends exponentially on band gap and temperature.
- How donors and acceptors control carrier concentration, how doping profiles are described, and where doping stops working (freeze-out, intrinsic conduction, degeneracy, compensation).
- What limits carrier mobility and velocity, how empirical models such as Caughey–Thomas describe them, and how drift, diffusion and the Einstein relation fit together.
- How carriers recombine (Shockley–Read–Hall, radiative, Auger), what sets the minority-carrier lifetime, and why a few parts per billion of a deep-level impurity can ruin a wafer.
- How silicon, germanium, the III–V compounds, silicon carbide, gallium nitride, the ultra-wide-gap materials, amorphous, organic and two-dimensional semiconductors compare, and how figures of merit rank them for power, RF, sensing and optoelectronics.
- How heterostructures, band alignment, surface and interface states, thin-film defects and gate dielectrics determine the behaviour of real device structures.
- How light is absorbed and emitted in materials, and which materials are used for LEDs, lasers, photodetectors, solar absorbers, high-frequency electronics and sensors.
- How refraction, absorption, scattering and emission arise in materials, and how they are engineered in fibres, LEDs, lasers and photonic structures.

Chapters in this part

Chapter	Central question	What it delivers
31. Solid-State Foundations for Materials Engineering	Why do crystals have bands and gaps, and how many carriers do they hold at a given temperature?	Band gap, density of states, Fermi level, effective mass, $n_i(T)$; the direct–indirect distinction behind light emission; the temperature limits of a material
32. Semiconductor Material Properties	How are carrier density and transport set, controlled and limited in a real, doped, imperfect semiconductor?	Doping and Fermi-level control, Caughey–Thomas mobility, drift–diffusion and the Einstein relation, SRH/radiative/Auger recombination, lifetime, velocity saturation, impact ionization, deep levels
33. Semiconductor Material Families	Which semiconductor should be used for a given job, and why?	Property tables and figures of merit (Baliga, Johnson) for Si, Ge, GaAs, InP, SiC, GaN, Ga_2O_3, diamond, AlN, a-Si:H, organics and 2D materials; selection for power, RF, sensing and optoelectronics
34. Semiconductor Interfaces and Thin Films	What happens where two materials meet, and how do thin films differ from bulk crystals?	Heterostructures and band alignment, surface states and interface traps, film stress, grain boundaries and defects, contact interfaces, gate-dielectric properties, interface reliability
35. Electronic and Optoelectronic Materials	How do materials absorb, emit and detect light and fast signals?	Absorption and emission, photoconductors and phosphors, LED, laser, photodetector and solar-absorber materials, optical and photonic materials, high-frequency and sensor materials
36. Optical and Photonic Materials	How do materials refract, absorb, scatter and emit light?	Refractive index and Sellmeier dispersion; colour in metals, insulators and pigments; transparency in ceramics and polymers; luminescence, LEDs and lasers; optical fibres; nonlinear and electro-optic materials; photonic crystals and metamaterials

How this part connects to the rest of the book

Part VII supplies the electronic-structure language that the rest of the book uses. Chapter 36 carries the absorption and emission of Chapter 35 into refraction, scattering and light guiding in every class of material. Chapter 38 builds quantum confinement, graphene and the other two-dimensional materials on the density of states and effective mass of Chapter 31. Chapter 40 rests on absorption, recombination and lifetime from Chapters 32 and 35 for photovoltaic absorbers and on carrier transport for thermoelectrics. In Part X, Chapter 50 shows how crystal growth, oxidation, diffusion, implantation and deposition produce the doping profiles, defects and interfaces described here, and Chapter 51 takes up the thermal and mechanical loads that wide-gap power devices put on their packages. Chapter 44 follows how semiconductor defects and dielectrics degrade in service; in Part XI, Chapter 52 measures the quantities defined here (four-point probe, Hall effect, spectroscopy), and Chapter 58 turns the figures of merit of Chapter 33 into a general method of materials selection. Device operation and circuit design remain in the Electronics & Computing core, and atomic structure and chemical bonding in the Advanced Chemistry core. Part VII is the material layer beneath both.

Chapter 31 — Solid-State Foundations for Materials Engineering

Part VII · Semiconductor and Photonic Materials

An automotive supplier qualifies a silicon pressure-sensor interface chip for an engine-bay module. At 25 °C the chip's input leakage current is a few picoamperes and invisible. At 125 °C the same leakage is several hundred times larger, the sensor offset drifts out of specification, and the module fails its qualification test. Nothing in the chip was damaged. No bond broke, no metal melted, no dielectric failed. The silicon simply did what silicon does: the number of electrons that heat lifts across its band gap increased by a factor of about 500 over those 100 kelvin. A colleague proposes moving the sensor interface to silicon carbide, in which the same effect at 125 °C is smaller by about thirteen orders of magnitude. A second colleague asks why, if silicon carbide is so much better, the indicator LED on the same board is made of neither silicon nor silicon carbide but of a III–V compound. Both answers come from the electronic structure of the crystal, which is the subject of this chapter.

> **Why do electrons in a crystal occupy bands of allowed energy separated by forbidden gaps, and what sets the size of the gap?**
> **How many electrons and holes are available to carry current at a given temperature, and why does that number change so steeply with temperature and band gap? Why does silicon absorb light well but hardly emit it, while gallium arsenide does both?**

Concept	Engineering destination
Bloch states, wavevector k	The language of every band diagram, datasheet and simulator
Band formation (tight binding, nearly free electron)	Why gaps exist; why narrow bands mean heavy carriers
Valence and conduction bands, holes	Two-carrier conduction; p-type and n-type material (Chapter 32)
Band gap E_g and its trends	Operating temperature, breakdown field, emission wavelength, leakage
Density of states $g(E)$, N_c, N_v	Carrier counting; quantum wells and 2D materials (Chapters 34, 38)
Fermi level and Fermi–Dirac statistics	Contacts, junctions, work functions; equilibrium band diagrams
Effective mass	Mobility, saturation velocity, donor binding energy, tunnelling
$n_i(T)$ and the law of mass action	Leakage, maximum junction temperature, intrinsic limits
Direct and indirect gaps	LEDs and lasers versus photodetectors and solar cells (Chapter 35)

Roadmap. Electronic states in a periodic crystal (31.1) → how bands form, from two pictures (31.2) → which bands are full and which are empty (31.3) → the band gap and its trends (31.4) → how many states there are (31.5) → how they are filled (31.6) → how carriers respond to forces (31.7) → how carrier populations change with temperature (31.8) → how the shape of the bands controls light absorption and emission (31.9).

> **Boundary.**
> This chapter gives the material-level picture of electronic structure: enough quantum mechanics to see why bands and gaps exist and to calculate with them. The Schrödinger equation, its solution for model potentials and the formal theory of crystal momentum belong in the Engineering Physics core. Atomic orbitals and chemical bonding are covered in the Advanced Chemistry core and are only referenced here. Diodes, transistors and their operation belong in the Electronics & Computing core.

31.1 Electronic states in solids

From the free atom to the crystal

An isolated silicon atom has fourteen electrons in sharp, discrete energy levels: the familiar $1s^2\,2s^2\,2p^6\,3s^2\,3p^2$ configuration from the Advanced Chemistry core. The ten core electrons are bound by tens to thousands of electron-volts and play no part in conduction. The four outer electrons, in the $3s$ and $3p$ levels, are bound by only a few electron-volts, and they decide everything electrical.

Bring N silicon atoms together into a crystal. The outer orbitals of neighbouring atoms overlap, and the electrons can no longer be assigned to a single atom. Two consequences follow at once.

1. **The Pauli principle still applies to the whole crystal.** No two electrons can occupy the same quantum state. Each atomic level, which could hold two electrons in one atom, must become N distinct levels holding $2N$ electrons in the crystal. With $N \approx 5 \times 10^{22}$ atoms per cubic centimetre, those levels lie so close together that they form an effectively continuous **band**.
2. **The width of each band is set by the overlap.** Strongly overlapping outer orbitals spread into bands several electron-volts wide. Core orbitals, which barely overlap, remain essentially sharp atomic levels.

The energy ranges between bands, which no extended state occupies, are the **band gaps**. Figure 31.1 shows this evolution as the interatomic spacing is reduced from infinity to the equilibrium value.

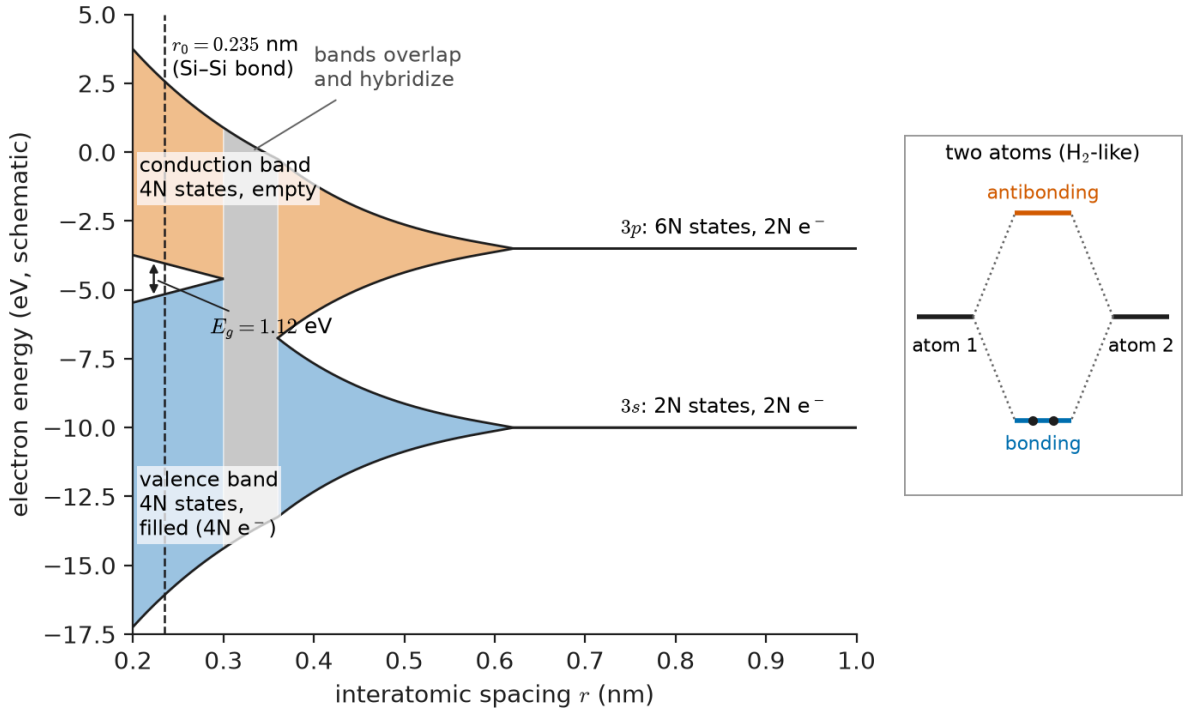


Figure 31.1 — Atomic 3s and 3p levels of silicon broaden into bands as the atoms are brought together; at the equilibrium spacing the sp3 levels split into a filled bonding (valence) band and an empty antibonding (conduction) band separated by the gap.

The periodic crystal and Bloch's theorem

In a crystal the potential energy $U(\mathbf{r})$ that an electron feels repeats with the lattice (Chapter 2): $U(\mathbf{r}+\mathbf{R}) = U(\mathbf{r})$ for every lattice vector $\mathbf{R}$. This one symmetry fixes the form of every electron state. Translating the crystal by $\mathbf{R}$ cannot change any measurable property, so the electron probability density $|\psi|^2$ must also repeat, and the wavefunction itself can change only by a phase factor. The result is **Bloch's theorem**:

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\,u_{n\mathbf{k}}(\mathbf{r}), \qquad u_{n\mathbf{k}}(\mathbf{r}+\mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r})$$

Every electron state in a perfect crystal is a plane wave $e^{i\mathbf{k}\cdot\mathbf{r}}$ modulated by a function u that has the periodicity of the lattice. Two labels identify each state: the **wavevector** $\mathbf{k}$, and a **band index** n that counts the solutions with the same $\mathbf{k}$. The allowed

energies $E_n(\mathbf{k})$, plotted against $\mathbf{k}$, form the **band structure** of the material.

> **Status of the principle: Bloch's theorem**
> *Fundamental.* It follows from the translational symmetry of a perfect, infinite crystal. Real crystals have surfaces, defects and thermal vibrations, and these act as perturbations that scatter electrons between Bloch states (Chapter 32).

Three consequences matter for engineering.

A Bloch electron is not scattered by the perfect lattice. A plane wave in a periodic potential propagates forever. The ions do not "get in the way" of the electron, as the classical Drude picture of Chapter 25 assumed. Resistance comes only from departures from perfect periodicity: lattice vibrations (phonons), impurities, defects, alloy disorder and surfaces. This is why the mobility of a semiconductor depends so strongly on temperature and on impurity concentration, and why it is a material property that processing can change.

$\mathbf{k}$ is defined only within the Brillouin zone. Adding a reciprocal-lattice vector $\mathbf{G}$ to $\mathbf{k}$ gives the same set of states, so all distinct states lie in one primitive cell of reciprocal space, the **first Brillouin zone**. For a one-dimensional lattice of period a this is $-\pi/a < k \le \pi/a$. For the face-centred cubic lattice of Si, Ge and GaAs it is a truncated octahedron, and its high-symmetry points carry standard labels: Γ at the centre, X at the centre of the square faces (along $\langle 100 \rangle$), L at the centre of the hexagonal faces (along $\langle 111 \rangle$). Band diagrams in datasheets and papers plot $E(\mathbf{k})$ along lines joining these points.

$\hbar\mathbf{k}$ behaves like momentum. Under an external force $\mathbf{F}$, such as an electric field, the wavevector of an electron changes as

$$\hbar \frac{d\mathbf{k}}{dt} = \mathbf{F}$$

The quantity $\hbar\mathbf{k}$ is called the **crystal momentum**. It is not the mechanical momentum of the electron, because the lattice also exchanges momentum with it, but it is conserved in collisions and optical transitions (up to a reciprocal-lattice vector). Section 31.9 shows that this conservation law decides whether a material can emit light efficiently.

Counting states

Apply periodic boundary conditions to a crystal of N primitive cells. The allowed $\mathbf{k}$ values form a fine grid with exactly N points in the Brillouin zone. Each band therefore holds N orbital states, or $2N$ electrons including spin.

> **Status of the principle: 2N states per band**
> *Derived.* It follows from Bloch's theorem with periodic boundary conditions and the two spin states. It is the counting rule that decides, in Section 31.3, whether a crystal is a metal or an insulator.

31.2 Energy-band formation

Two complementary models explain why bands form and gaps open. The first starts from atoms and lets them interact. The second starts from free electrons and switches on the lattice. Each is approximate. Together they explain every qualitative feature of real band structures.

The tight-binding picture: from bonds to bands

Start with two atoms, each with one orbital of energy ε_0. When they approach, the orbitals combine into a bonding state at $\varepsilon_0 - t$ and an antibonding state at $\varepsilon_0 + t$, where t is the **hopping** or transfer energy, which measures how strongly an electron on one atom couples to the orbital on its neighbour (the molecular-orbital result of the Advanced Chemistry core).

Now take a long chain of N atoms with spacing a, each coupled to its two neighbours. By Bloch's theorem each state is a combination of atomic orbitals with phase e^{ikna} on atom n, and its energy is

$$E(k) = \varepsilon_0 - 2t\cos(ka), \qquad -\frac{\pi}{a} < k \le \frac{\pi}{a}$$

The two-atom bonding–antibonding pair has become a band of N states running from $\varepsilon_0 - 2t$ at $k = 0$ to $\varepsilon_0 + 2t$ at the zone boundary. Figure 31.2(a) shows this cosine band. Three features generalise to real crystals.

- **Bandwidth is proportional to overlap.** The width $4t$ (in general $2zt$ for z nearest neighbours) grows as the atoms come closer. Strongly bonded, short-bond crystals have wide bands.
- **Each atomic orbital gives one band.** With several orbitals per atom (s, p_x, p_y, p_z) one obtains several bands. Where those bands do not overlap in energy, gaps remain between them.
- **Narrow bands mean heavy electrons.** Near the band bottom, $\cos(ka) \approx 1 - (ka)^2/2$, so $E \approx \varepsilon_0 - 2t + ta^2k^2$. This has the free-electron form $\hbar^2k^2/2m^*$ with an **effective mass**

$$m^* = \frac{\hbar^2}{2ta^2}$$

A small hopping energy, a narrow band and a heavy carrier go together. This anticipates Section 31.7.

Worked example — effective mass from a tight-binding band. Take $t = 1.0$ eV and $a = 0.25$ nm, typical of a covalent solid. Then $\hbar^2 = (1.0546 \times 10^{-34})^2 = 1.112 \times 10^{-68}$ J^2 s^2 and $2ta^2 = 2 \times 1.602 \times 10^{-19} \times (2.5 \times 10^{-10})^2 = 2.003 \times 10^{-38}$ J m^2, so $m^* = 5.55 \times 10^{-31}$ kg $= 0.61\, m_0$. The bandwidth is $4t = 4$ eV. Halving the hopping energy to 0.5 eV, as in a weakly coupled organic crystal or a transition-metal oxide d band, doubles the mass to $1.2\, m_0$ and halves the bandwidth. In organic semiconductors (Chapter 33) the intermolecular t is only 0.05–0.1 eV, the bands are a few tenths of an electron-volt wide, and band transport gives way to hopping between molecules.

Silicon in the tight-binding picture. Each Si atom forms four sp^3 hybrid bonds to its tetrahedral neighbours in the diamond structure (Chapter 2). Each bond splits into a bonding and an antibonding combination. The $4N$ bonding states broaden into the **valence band**, the $4N$ antibonding states into the **conduction band**, and the gap between them is, roughly, the bonding–antibonding splitting minus the broadening of the two bands. Silicon has $4N$ valence electrons, exactly enough to fill the bonding band. This "chemist's band structure" is the origin of Figure 31.1. It explains at once why stronger, shorter bonds give wider gaps (C > Si > Ge), and why a gap exists even in an amorphous solid (Chapter 33), which has bonds but no long-range order.

The nearly-free-electron picture: gaps from Bragg reflection

Now start from the opposite limit. An electron in an empty box has $E = \hbar^2k^2/2m_0$, a single parabola. Switch on a weak periodic potential $U(x) = \sum_G U_G e^{iGx}$, with reciprocal-lattice vectors $G = 2\pi m/a$.

Away from the zone boundaries the parabola is barely changed. At $k = \pm\pi/a$, however, the electron wavelength is $\lambda = 2a$, which satisfies the **Bragg condition** for reflection from the lattice planes, the same condition that governs X-ray diffraction (Chapter 52). The forward wave $e^{i\pi x/a}$ and the reflected wave $e^{-i\pi x/a}$ have equal energies and mix into two standing waves:

- $\psi_- \propto \cos(\pi x/a)$ piles charge on the ion cores, where the potential energy is low, and its energy is lowered.
- $\psi_+ \propto \sin(\pi x/a)$ piles charge between the ions, and its energy is raised.

The two standing waves differ in energy by

$$E_g = 2|U_G|$$

where U_G is the Fourier component of the crystal potential at the reciprocal-lattice vector that connects $k = -\pi/a$ to $+\pi/a$. A gap opens at every zone boundary, and the parabola breaks into bands, as Figure 31.2(b) shows. The scale is right: for $a = 0.543$ nm the free-electron energy at $k = \pi/a$ is $\hbar^2(\pi/a)^2/2m_0 = 1.27$ eV, and Fourier components of a few electron-volts are typical of real crystal potentials.

> **Status of the principle: band gaps from Bragg reflection**
> *Approximate.* The result $E_g = 2|U_G|$ is first-order perturbation theory in a weak periodic potential. It is close to quantitative for simple metals such as Al and Na, whose potentials are weak. For covalent semiconductors the potential is strong and the tight-binding picture is more natural, but the conclusion that gaps open where electron waves Bragg-reflect holds in general.

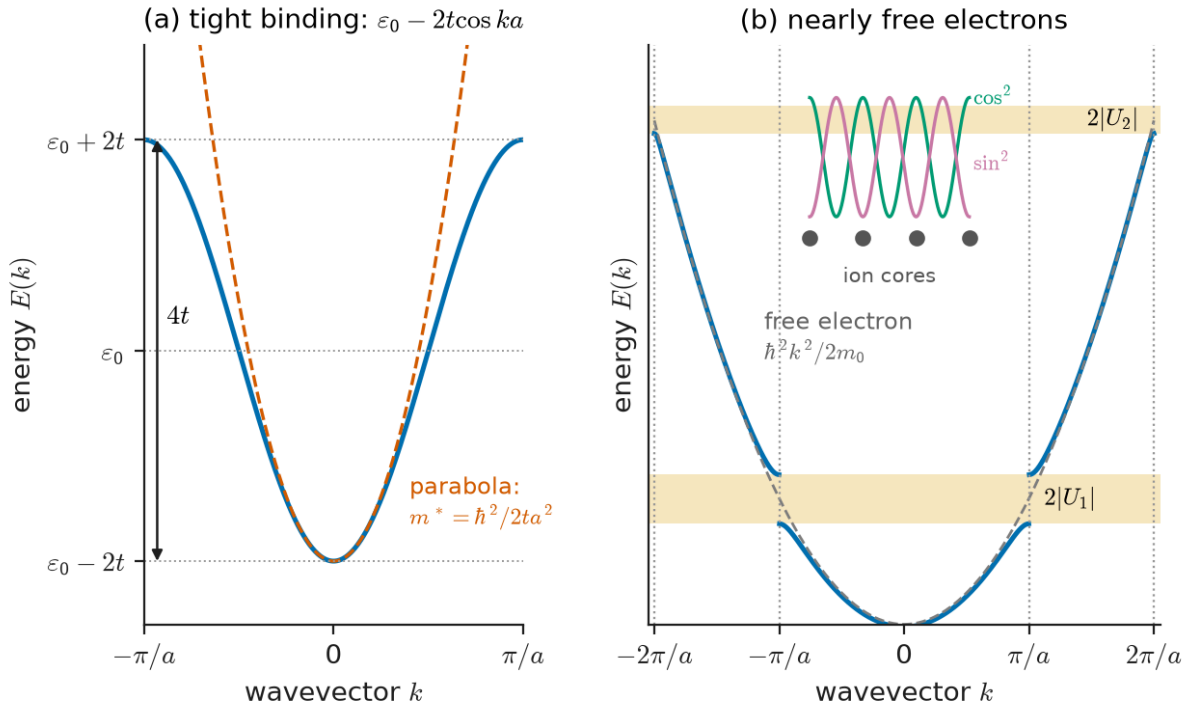


Figure 31.2 — Two routes to a band gap: (a) a tight-binding cosine band of width 4t formed from one atomic level; (b) the free-electron parabola split by Bragg reflection into bands with gaps 2 abs(U_G) at the zone boundaries.

What real band structures look like

Real band structures, computed numerically and confirmed by photoemission and optical spectroscopy, combine both pictures. Figure 31.3 sketches those of silicon and gallium arsenide along the L–Γ–X directions. Three features dominate the engineering behaviour.

- The **valence-band maximum** of all the common cubic semiconductors lies at Γ ($k = 0$), where three p-like bands meet: the heavy-hole, light-hole and (slightly lower) split-off bands.
- The **conduction-band minimum** lies wherever the lowest antibonding band dips lowest. In GaAs it is at Γ. In Si it lies along the six equivalent ⟨100⟩ directions, about 85 % of the way to X, giving six equivalent **valleys**. In Ge it lies at the eight L points, giving four full valleys (each L point is shared between neighbouring zones).
- **Secondary valleys** lie a few tenths of an electron-volt higher. In GaAs the L valleys lie about 0.29 eV above the Γ minimum. Electrons heated by strong fields transfer into them, which produces the negative differential mobility of Chapter 32.

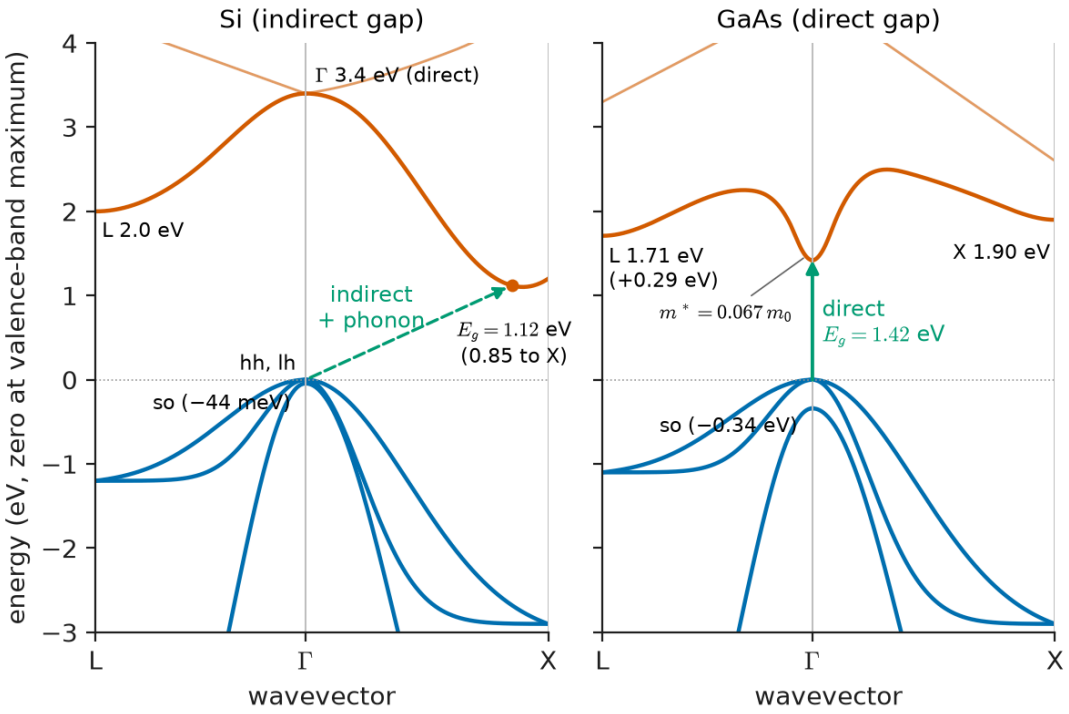


Figure 31.3 — Simplified band structures of silicon (indirect gap) and gallium arsenide (direct gap) along L–Γ–X, with the band extrema that matter for transport and optics.

31.3 Valence and conduction bands

Filled bands, empty bands and conduction

Whether a crystal conducts depends on how its electrons fill the available bands. Two facts settle the question.

A completely filled band carries no current. For every state $\mathbf{k}$ in a band there is a state $-\mathbf{k}$ with opposite velocity. When the band is full, the velocities cancel exactly. An electric field shifts every electron's $\mathbf{k}$ at the same rate, but a full band remains full and the sum is still zero. An empty band trivially carries no current. **Only partly filled bands conduct.**

> **Status of the principle: full bands are inert**
> *Derived.* It follows from the symmetry $E(\mathbf{k}) = E(-\mathbf{k})$ and the Pauli principle. It is why insulators insulate even though they contain about 10^{23} electrons per cubic centimetre.

Electron counting decides the filling. With $2N$ electrons per band (Section 31.1), a crystal with an even number of valence electrons per primitive cell can fill an integer number of bands exactly.

Case	Valence electrons per primitive cell	Highest occupied band	Result	Examples
Odd count	1, 3, …	Half-filled	Metal	Na, Cu, Al
Even count, bands overlap in energy	2, …	Two partly filled bands	Metal or semimetal	Mg, Ca, Zn; Bi, graphite
Even count, gap between bands	8 (two atoms × 4)	Full valence band, empty conduction band	Semiconductor or insulator	Si, Ge, GaAs, diamond, Al_2O_3

Silicon has two atoms per primitive cell and eight valence electrons, exactly enough to fill four bands. These are the valence bands. The next band up, empty at absolute zero, is the conduction band. At $T = 0$ pure silicon is a perfect insulator.

The distinction between a **semiconductor** and an **insulator** is not one of kind but of degree, set by the gap and the temperature. At 300 K a material with $E_g \lesssim 3$ eV contains enough thermally excited or dopant-supplied carriers to be called a semiconductor. Diamond (5.5 eV) and alumina (about 8.8 eV) are insulators in their pure form, but diamond can be doped and is treated as an ultra-wide-gap semiconductor in Chapter 33. Silicon carbide and gallium nitride, at 3.2–3.4 eV, are semiconductors that behave as near-insulators unless they are doped.

Holes

When an electron is excited from the valence band to the conduction band, it leaves a single empty state in an otherwise full band. The response of the remaining $2N - 1$ electrons to a field is awkward to describe directly. It is exactly equivalent, however, to the motion of a single fictitious particle, the **hole**, with these properties:

- charge $+q$,
- wavevector $\mathbf{k}_h = -\mathbf{k}_e$ of the missing electron,
- energy measured *downward* on the electron energy axis (a hole deeper in the valence band has more energy),
- a positive effective mass at the top of the valence band, where the electron effective mass is negative (Section 31.7).

The hole is not a book-keeping trick. Its positive charge is measured directly by the Hall effect (Chapter 52), which in p-type material gives a Hall voltage of the opposite sign to that in n-type material, something the free-electron model of metals cannot explain.

A semiconductor therefore has **two** kinds of carrier: electrons in the conduction band (concentration n) and holes in the valence band (concentration p). Its conductivity is

$$\sigma = q\,(n\mu_n + p\mu_p)$$

where μ_n and μ_p are the mobilities defined in Chapter 25. The rest of this chapter finds n and p. Chapter 32 finds μ_n and μ_p.

31.4 Band gaps

What the gap measures

The band gap $E_g = E_c - E_v$ is the energy difference between the conduction-band minimum E_c and the valence-band maximum E_v. Physically it is the minimum energy needed to break one bond and free its electron to move through the crystal while leaving a mobile hole behind. Almost every important semiconductor property depends on it:

- the intrinsic carrier concentration, through $e^{-E_g/2k_BT}$ (Section 31.8), and with it leakage currents and the maximum operating temperature;
- the critical field for avalanche breakdown, which grows roughly as E_g^2 to $E_g^{2.5}$ (Chapter 32), and with it the voltage rating per unit thickness of a power device;
- the absorption edge and emission wavelength, $\lambda\,(\mu\text{m}) \approx 1.24/E_g\,(\text{eV})$ (Section 31.9 and Chapter 35);
- the ionization energies of dopants and the depth of defect levels, measured from the band edges (Chapter 32).

Representative values

Material	Structure	E_g at 300 K (eV)	Type	Cut-off wavelength $1.24/E_g$ (μm)
InSb	zincblende	0.17	direct	7.3
InAs	zincblende	0.35	direct	3.5
Ge	diamond	0.66	indirect	1.88
$In_{0.53}Ga_{0.47}As$	zincblende	0.75	direct	1.65
Si	diamond	1.12	indirect	1.11
InP	zincblende	1.34	direct	0.93
GaAs	zincblende	1.42	direct	0.87
GaP	zincblende	2.26	indirect	0.55
4H-SiC	hexagonal	3.26	indirect	0.38
GaN	wurtzite	3.39	direct	0.37
ZnO	wurtzite	3.37	direct	0.37
β-Ga_2O_3	monoclinic	4.8–4.9	(nearly) indirect	0.26
Diamond (C)	diamond	5.47	indirect	0.23
AlN	wurtzite	6.0–6.2	direct	0.20
SiO_2 (amorphous)	amorphous	about 9	—	—

Trends: what sets the gap

The table shows two clear trends, both explained by the tight-binding picture of Section 31.2.

Shorter, stronger bonds give wider gaps. Down group IV, the bond length grows and the bonding–antibonding splitting falls: C (0.154 nm, 5.47 eV) → Si (0.235 nm, 1.12 eV) → Ge (0.245 nm, 0.66 eV) → α-Sn (0.281 nm, 0, a semimetal). The same pattern runs through the III–V compounds: AlN > GaN > InN (about 0.7 eV), and GaP > GaAs > GaSb (0.73 eV).

Ionicity widens the gap. Moving from a group-IV element to an isoelectronic III–V and then II–VI compound in the same row keeps the bond length nearly constant but makes the bond more polar, which adds an ionic contribution to the splitting: Ge (0.66 eV) → GaAs (1.42 eV) → ZnSe (2.7 eV). The polar bond character is a chemistry result (Advanced Chemistry core). Its materials consequence is that compound semiconductors offer a wide, nearly continuous menu of gaps.

Alloys interpolate. In a solid solution such as $Al_xGa_{1-x}As$ or $In_xGa_{1-x}N$ the gap varies smoothly with composition, usually as a quadratic with a small **bowing parameter** b:

$$E_g(x) = xE_g^A + (1-x)E_g^B - b\,x(1-x)$$

> **Status of the principle: gap interpolation with bowing**
> *Empirical.* The quadratic form fits measured gaps of most III–V and II–VI alloys with one fitted parameter. It fails where the alloy changes from direct to indirect ($Al_xGa_{1-x}As$ near $x \approx 0.45$), where the different valleys must each be interpolated separately, and for highly mismatched alloys such as GaN_xAs_{1-x}, where a few per cent of nitrogen lowers the gap by hundreds of meV.

This is how optoelectronic materials are tuned to a wavelength (Chapters 33 and 35).

Temperature and pressure dependence

The gap shrinks as temperature rises, because thermal expansion lengthens the bonds and lattice vibrations smear the band edges. The **Varshni relation** fits the data:

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta}$$

Material	$E_g(0)$ (eV)	α (eV/K)	β (K)	$E_g(300\ \mathrm{K})$ (eV)	dE_g/dT at 300 K (meV/K)
Si	1.170	4.73×10^{-4}	636	1.125	−0.25
Ge	0.744	4.77×10^{-4}	235	0.663	−0.39
GaAs	1.519	5.41×10^{-4}	204	1.422	−0.45

> **Status of the principle: Varshni relation**
> *Empirical.* A two-parameter fit that reproduces $E_g(T)$ to a few meV from cryogenic temperatures to several hundred kelvin. Its form is motivated by the electron–phonon interaction but its parameters are fitted, not derived.

The shift matters in practice. An LED's emission wavelength red-shifts by about 0.3 nm/K, and a silicon bandgap-voltage reference relies on extrapolating the junction voltage to $E_g(0)/q \approx 1.2$ V, which is the temperature-independent quantity in the circuit.

The gap is not always the same number

Three different "gaps" appear in the literature, and confusing them causes errors.

- The **electronic (transport) gap** is the energy to create a free electron and a free hole.
- The **optical gap** is the photon energy at which absorption begins. It is smaller than the transport gap by the exciton binding energy, a few meV in Si and GaAs, 60 meV in ZnO, and 0.3–1 eV in organic semiconductors and monolayer transition-metal dichalcogenides (Chapter 33).
- **Computed gaps** from standard density-functional theory underestimate the true gap, often by 30–50 %. Values from simulation must be checked against measurement before being used in design.

31.5 Density of states

Why we need it

To find how many electrons occupy a band we need two things: how many states exist at each energy, and the probability that each state is occupied. The first is the **density of states** $g(E)$, defined so that $g(E)\,dE$ is the number of states per unit volume with energies between E and $E + dE$. The second is the Fermi–Dirac function of Section 31.6. Their product gives the carrier distribution.

Derivation for a parabolic band in three dimensions

Near the bottom of the conduction band, any smooth $E(\mathbf{k})$ can be expanded about its minimum. For an isotropic minimum this gives

$$E = E_c + \frac{\hbar^2 k^2}{2 m_n^*}$$

with an effective mass m_n^* (Section 31.7). With periodic boundary conditions on a crystal of volume V, the allowed wavevectors form a grid with one point per $(2\pi)^3/V$ of k-space. The number of states, including two spins, inside a sphere of radius k is

$$N(k) = 2 \times \frac{\tfrac{4}{3}\pi k^3}{(2\pi)^3 / V} = \frac{V k^3}{3\pi^2}$$

Substituting $k = \sqrt{2 m_n^* (E - E_c)}/\hbar$ and differentiating with respect to E, then dividing by V:

$$g_c(E) = \frac{1}{2\pi^2}\left(\frac{2 m_n^*}{\hbar^2}\right)^{3/2} \sqrt{E - E_c}, \qquad E \geq E_c$$

Similarly for holes near the valence-band maximum, $g_v(E) = \frac{1}{2\pi^2}(2m_p^*/\hbar^2)^{3/2}\sqrt{E_v - E}$ for $E \le E_v$. Inside the gap, $g = 0$ in a perfect crystal.

> **Status of the principle: parabolic density of states**
> *Approximate.* It follows from the parabolic (effective-mass) expansion of $E(\mathbf{k})$ and is accurate within roughly 0.1–0.2 eV of a band edge, which is where nearly all carriers sit in a non-degenerate semiconductor. It fails for hot carriers far into the band, for non-parabolic narrow-gap materials (InSb, InAs), and in disordered solids, where band tails extend into the gap (amorphous silicon, Chapter 33).

Two features matter. First, the density of states rises from zero at the band edge as $\sqrt{E - E_c}$: there are very few states right at the edge. Second, it scales as $(m^*)^{3/2}$: a heavy-mass band has many more states near its edge than a light one. A material with several equivalent valleys (six in Si) multiplies its density of states by the number of valleys, which is absorbed into a **density-of-states effective mass** (Section 31.7).

Worked example — how many states lie near the band edge? For the silicon conduction band, with a density-of-states mass $m_n^* = 1.09\,m_0$, the number of states per unit volume within k_BT of the band edge at 300 K is

$$N(k_BT) = \int_{E_c}^{E_c + k_BT} g_c(E)\,dE = \frac{1}{3\pi^2}\left(\frac{2m_n^* k_BT}{\hbar^2}\right)^{3/2}$$

With $2m_n^* k_BT/\hbar^2 = 2 \times 1.09 \times 9.109 \times 10^{-31} \times 4.142 \times 10^{-21} / 1.112 \times 10^{-68} = 7.40 \times 10^{17}\ \mathrm{m^{-2}}$, its $3/2$ power is $6.36 \times 10^{26}\ \mathrm{m^{-3}}$, and dividing by $3\pi^2 = 29.6$ gives $2.15 \times 10^{25}\ \mathrm{m^{-3}} = 2.1 \times 10^{19}\ \mathrm{cm^{-3}}$. A typically doped wafer with $n = 10^{16}\ \mathrm{cm^{-3}}$ therefore occupies only about one in two thousand of the available states near the edge. This is why the simple Boltzmann statistics of Section 31.6 work so well, and why they fail once doping approaches $10^{19}\ \mathrm{cm^{-3}}$.

Lower dimensions

If electrons are confined in one, two or three directions to a size comparable to their wavelength (a few nanometres), the density of states changes shape:

Dimensionality	Example	$g(E)$ near a sub-band edge
3D (bulk)	Silicon wafer	$\propto \sqrt{E - E_c}$
2D (quantum well, 2DEG, monolayer)	AlGaN/GaN channel, MoS_2	constant, $m^*/\pi\hbar^2$ per sub-band (per unit area)
1D (nanowire, nanotube)	Carbon nanotube	$\propto 1/\sqrt{E - E_n}$
0D (quantum dot)	CdSe nanocrystal	discrete levels

The step-like 2D density of states is why quantum-well lasers have lower threshold currents and why the two-dimensional electron gas of a GaN high-electron-mobility transistor holds a fixed sheet density per volt. These materials are treated in Chapters 34 and 38. Figure 31.4 shows the bulk 3D case.

31.6 Fermi level

The Fermi–Dirac distribution

Electrons are fermions: each state holds at most one electron of a given spin. In thermal equilibrium at temperature T, the probability that a state of energy E is occupied is the **Fermi–Dirac function**

$$f(E) = \frac{1}{1 + \exp[(E - E_F)/k_BT]}$$

The parameter E_F, the **Fermi level**, is the energy at which the occupation probability is exactly one half. At $T = 0$, f is a step: every state below E_F is full and every state above is empty. At finite temperature the step softens over a width of a few k_BT ($k_BT = 25.85$ meV at 300 K). The probability that a state is *empty*, which is the probability of finding a hole there, is $1 - f(E)$.

> **Status of the principle: Fermi–Dirac distribution**
> *Fundamental.* It follows from statistical mechanics for indistinguishable particles obeying the Pauli principle, in thermal equilibrium. Out of equilibrium (under illumination or current injection) a single E_F no longer describes both bands, and separate quasi-Fermi levels are needed (Chapter 32).

The Fermi level is the electrochemical potential

Thermodynamically, E_F is the chemical potential of the electrons, more precisely their electrochemical potential, which includes the electrostatic energy (Advanced Chemistry core for the general definition). Its most important property follows directly:

> **In thermal equilibrium, the Fermi level is the same everywhere in a connected system.**

If two materials with different Fermi levels are brought into contact, electrons flow from the higher to the lower until the Fermi levels align, building up the dipole layer and band bending that appear at every metal–semiconductor contact and p–n junction. A flat Fermi level in an equilibrium band diagram is a check that the diagram is correct. A gradient in the Fermi level drives current (Chapter 32).

> **Status of the principle: constant Fermi level in equilibrium**
> *Fundamental.* It is the thermodynamic condition for no net particle flow, the electronic counterpart of equal chemical potential in phase equilibrium.

Metals versus semiconductors

In a **metal**, E_F lies inside a partly filled band. For copper, the free-electron estimate with $n = 8.47 \times 10^{28}\ \mathrm{m^{-3}}$ gives

$$E_F = \frac{\hbar^2}{2m_0}\left(3\pi^2 n\right)^{2/3} = 7.0\ \mathrm{eV}$$

measured from the band bottom, corresponding to a Fermi temperature $E_F/k_B \approx 81\,000$ K. Thermal energy disturbs only a thin shell of electrons within a few $k_B T$ of E_F. This is why the electronic heat capacity of metals is small and why their carrier concentration does not change with temperature (Chapter 25).

In a **semiconductor**, E_F lies in the gap, where there are no states. It still matters, because the tails of $f(E)$ reach into both bands. Figure 31.4 shows the construction: the product $g_c(E)f(E)$ gives the electron distribution in the conduction band, and $g_v(E)[1-f(E)]$ gives the hole distribution in the valence band.

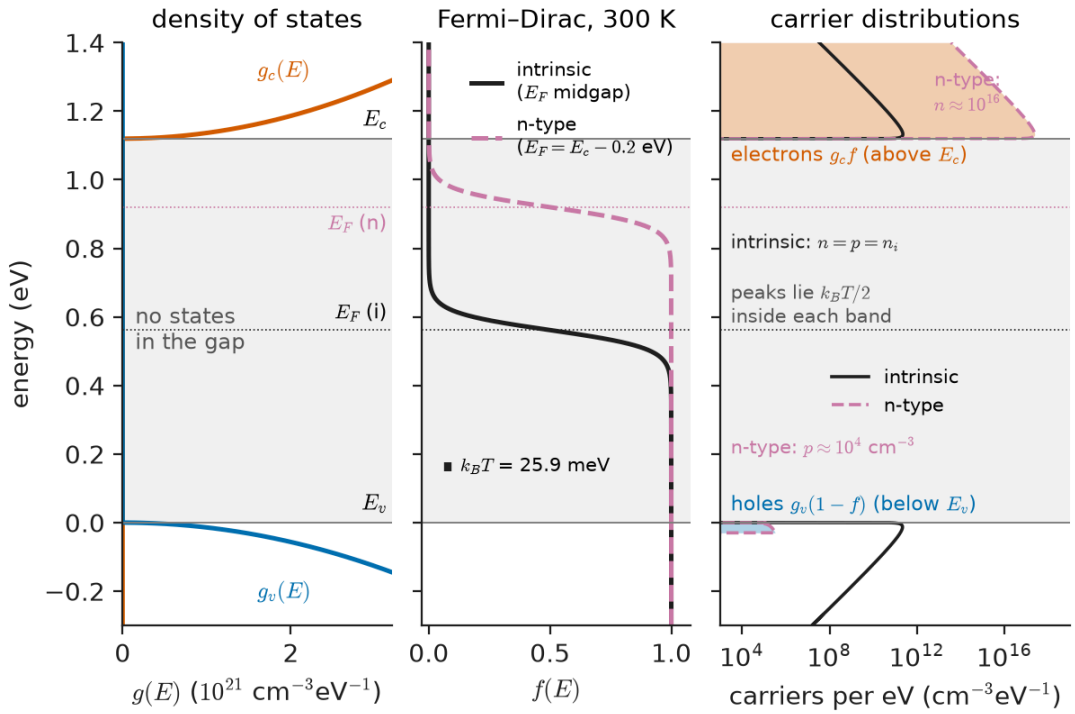


Figure 31.4 — Density of states, Fermi–Dirac occupation and the resulting electron and hole distributions for an intrinsic and an n-type semiconductor.

The Boltzmann approximation

When $E - E_F \gg k_BT$, the 1 in the denominator of f is negligible and

$$f(E) \approx \exp[-(E - E_F)/k_BT]$$

This Maxwell–Boltzmann tail is accurate to better than 5 % when $E - E_F > 3k_BT$ (at $3k_BT$, the exact value is 0.0474 versus 0.0498). A semiconductor in which E_F lies more than about $3k_BT$ inside the gap from both band edges is called **non-degenerate**, and its carriers behave as a classical ideal gas.

> **Status of the principle: Boltzmann approximation**
> *Approximate.* Valid when the Fermi level lies more than about $3k_BT$ (roughly 80 meV at 300 K) below E_c and above E_v. In silicon this corresponds to doping below about 10^{18} cm^{-3}. It fails in heavily doped (degenerate) contact and emitter regions and in the channel of a strongly inverted MOSFET, where full Fermi–Dirac statistics must be used.

Carrier concentrations

With the Boltzmann approximation the integral of $g_c(E)f(E)$ over the conduction band can be done in closed form:

$$n = \int_{E_c}^{\infty} g_c(E) f(E)\, dE = N_c \exp\!\left(-\frac{E_c - E_F}{k_BT}\right), \qquad N_c = 2\left(\frac{2\pi m_n^* k_BT}{h^2}\right)^{3/2}$$

$$p = N_v \exp\!\left(-\frac{E_F - E_v}{k_BT}\right), \qquad N_v = 2\left(\frac{2\pi m_p^* k_BT}{h^2}\right)^{3/2}$$

N_c and N_v are the **effective densities of states**. The whole band behaves as if it were a single level at E_c (or E_v) holding N_c (or N_v) states. Numerically,

$$N_c = 2.51 \times 10^{19} \left(\frac{m_n^*}{m_0}\right)^{3/2} \left(\frac{T}{300\ \text{K}}\right)^{3/2} \text{cm}^{-3}$$

> **Status of the principle:** $n = N_c e^{-(E_c - E_F)/k_B T}$
> *Derived.* It follows from the parabolic density of states and the Boltzmann approximation. It inherits the limits of both: non-degenerate material and carriers near the band edge.

Material	m_n^*/m_0 (DOS)	m_p^*/m_0 (DOS)	N_c (cm^{-3})	N_v (cm^{-3})
Ge	0.56	0.29–0.37	1.0×10^{19}	$(4–6) \times 10^{18}$
Si	1.09	0.81–1.15	2.9×10^{19}	$(1.8–3.1) \times 10^{19}$
GaAs	0.067	0.53	4.4×10^{17}	9×10^{18}
4H-SiC	0.77 (3 valleys)	about 1	1.7×10^{19}	2.5×10^{19}
GaN	0.20	about 1.5	2.3×10^{18}	4.6×10^{19}

The spread in the hole values reflects how strongly non-parabolic the valence band is: different authors fit different temperature ranges. GaAs, with its very light electron, has an effective density of states in the conduction band about 60 times smaller than silicon's. A modest doping of 4×10^{17} cm^{-3} already pushes its Fermi level to the band edge.

The law of mass action

Multiply the expressions for n and p. The Fermi level cancels:

$$np = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right) \equiv n_i^2$$

The product np depends only on the material (E_g, the effective masses) and the temperature, not on doping. Adding donors raises n and, in equilibrium, lowers p in exact proportion. This is the semiconductor analogue of the ionic product of water, $[\mathrm{H^+}][\mathrm{OH^-}] = K_w$, and it comes from the same thermodynamic argument (Advanced Chemistry core): the generation and recombination of electron–hole pairs is a reaction $0 \rightleftharpoons e^- + h^+$ whose equilibrium constant is n_i^2.

> **Status of the principle: law of mass action,** $np = n_i^2$
> *Derived.* It holds in thermal equilibrium for non-degenerate material. It fails under illumination or carrier injection, where $np > n_i^2$, and in reverse-biased depletion regions, where $np < n_i^2$. The departure $np - n_i^2$ is exactly what drives recombination and generation (Chapter 32). In heavily doped material, bandgap narrowing raises the effective n_i.

Intrinsic material and the intrinsic Fermi level

In a perfectly pure (**intrinsic**) semiconductor every electron in the conduction band came from the valence band, so $n = p = n_i$, with

$$n_i = \sqrt{N_c N_v}\, \exp\left(-\frac{E_g}{2k_B T}\right)$$

Setting $n = p$ gives the position of the **intrinsic Fermi level**:

$$E_i = \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \frac{N_v}{N_c} = \frac{E_c + E_v}{2} + \frac{3k_B T}{4} \ln \frac{m_p^*}{m_n^*}$$

E_i lies close to mid-gap, displaced toward the band with the lighter mass. In Si the shift is only a few meV. In GaAs, with $m_p^*/m_n^* \approx 8$, it is about 40 meV above mid-gap.

A useful equivalent form refers the carrier densities to E_i:

$$n = n_i\, e^{(E_F - E_i)/k_B T}, \qquad p = n_i\, e^{(E_i - E_F)/k_B T}$$

Every decade of n moves E_F by $k_B T \ln 10 = 59.5$ meV at 300 K.

Work function and electron affinity

Two further energies fix the absolute position of the bands. The **work function** Φ is the energy from the Fermi level to the vacuum level just outside the surface. The **electron affinity** χ is the energy from the conduction-band edge to the vacuum level. For a semiconductor, χ is a material constant (4.05 eV for Si, 4.07 eV for GaAs), while $\Phi = \chi + (E_c - E_F)$ depends on doping. These quantities set the line-up of bands at heterojunctions and contacts. Chapter 34 treats them, together with the surface dipoles that make real alignments depart from the simple rules.

31.7 Effective mass

Definition

An electron in a band responds to an applied force as if it had a mass different from m_0, because the lattice also pushes on it. Combining the group velocity of a wavepacket,

$$\mathbf{v} = \frac{1}{\hbar}\nabla_{\mathbf{k}} E$$

with $\hbar\, d\mathbf{k}/dt = \mathbf{F}$ gives, in one dimension,

$$\frac{dv}{dt} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}F \equiv \frac{F}{m^*}, \qquad \frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$$

The **effective mass** is the inverse curvature of the band. A sharply curved band (GaAs conduction band at Γ) has a light mass. A flat band has a heavy mass. In three dimensions the curvature is a tensor, and in an anisotropic valley the mass is different along different directions.

> **Status of the principle: effective-mass approximation**
> *Approximate*. It replaces the full crystal potential by a single parameter, valid when the carrier stays near a band extremum (where $E(\mathbf{k})$ is parabolic) and when the applied fields vary slowly on the scale of the lattice constant. It is the reason one can use Newtonian mechanics, Boltzmann transport and the hydrogen-atom model for dopants (Chapter 32) inside a crystal. It fails for hot carriers, strongly non-parabolic bands and atomically sharp potentials.

Two consequences surprise newcomers.

- **Near the top of a band the curvature is negative**, so an electron there accelerates *against* the applied force. This is the physics behind the positive-mass hole: the missing electron at the top of the valence band has $m^* < 0$, and the equivalent hole has $m_h^* = -m_e^* > 0$.
- **The effective mass is not the inertia of a different particle.** It is a property of the band, set by the crystal potential and the atomic orbitals that form the band. It can be changed by strain (strained-silicon channels, Chapter 34) and by alloying.

Anisotropic valleys and the two averaged masses

The six conduction-band valleys of silicon are ellipsoids elongated along the $\langle 100\rangle$ axes, with a **longitudinal** mass $m_l = 0.98\, m_0$ along the axis and a **transverse** mass $m_t = 0.19\, m_0$ perpendicular to it. Two different averages are needed, depending on what is being calculated.

Density-of-states mass (for counting carriers, N_c): the geometric mean of the three principal masses, multiplied by the valley degeneracy M:

$$m_{\mathrm{dos}} = M^{2/3}\left(m_l m_t^2\right)^{1/3}$$

Conductivity mass (for mobility): the harmonic mean, because current in any direction is carried by valleys whose masses along that direction differ:

$$m_{\mathrm{cond}} = \frac{3}{1/m_l + 2/m_t}$$

Worked example — the two electron masses of silicon. With $m_l = 0.98$, $m_t = 0.19$ and $M = 6$: $m_l m_t^2 = 0.98 \times 0.0361 = 0.0354$, whose cube root is 0.328; $6^{2/3} = 3.30$; so $m_{\text{dos}} = 3.30 \times 0.328 = 1.08\,m_0$. This gives $N_c = 2.51 \times 10^{19} \times 1.08^{3/2} = 2.8 \times 10^{19}$ cm^{-3}, consistent with the table in Section 31.6. For the conductivity mass, $1/0.98 + 2/0.19 = 1.02 + 10.53 = 11.55$, so $m_{\text{cond}} = 3/11.55 = 0.26\,m_0$. Silicon electrons are counted as if they were heavier than free electrons, because there are six valleys, but they move as if they were four times lighter. The same construction for germanium ($m_l = 1.64$, $m_t = 0.082$, four valleys) gives $m_{\text{dos}} = 0.56\,m_0$ and $m_{\text{cond}} = 0.12\,m_0$. This lighter conductivity mass is one reason Ge has more than twice the electron mobility of Si.

Representative effective masses

Material	Electrons (in units of m_0)	Heavy holes	Light holes	Where it matters
Si	$m_l = 0.98$, $m_t = 0.19$; cond. 0.26	0.49	0.16	Mobility anisotropy; strain engineering
Ge	$m_l = 1.64$, $m_t = 0.082$; cond. 0.12	0.33	0.043	High hole mobility; Ge p-channels
GaAs	0.067	0.51	0.082	Very high electron mobility; HEMTs
InP	0.080	0.6	0.09	Fast electronics; 1.55 μm photonics
$In_{0.53}Ga_{0.47}As$	0.041	0.46	0.05	Highest-speed transistors, detectors
InSb	0.014	0.43	0.015	Magnetic sensors, mid-IR detectors
GaN	0.20	about 1.4	about 0.3	Moderate mobility; heavy, slow holes
4H-SiC	0.29–0.42 (anisotropic, 3 valleys)	about 1	about 0.5	Moderate, anisotropic mobility

Light electrons give high mobility and high peak velocity (Chapter 32), but also a small density of states, easy degeneracy, strong quantum confinement and more tunnelling leakage. Heavy holes, as in GaN, are the main reason p-type wide-gap materials conduct poorly.

Non-parabolicity

Far from the band minimum, $E(\mathbf{k})$ flattens and the effective mass grows. In narrow-gap materials, where the conduction and valence bands interact strongly, this happens within a few tens of meV. A common correction is $E(1 + E/E_g) = \hbar^2 k^2 / 2m^*$. Non-parabolicity matters for heavily doped material, hot carriers and quantum wells. It is the reason the hole density-of-states mass of silicon has values from 0.81 to 1.15 m_0 in the literature.

31.8 Temperature dependence of carrier populations

The intrinsic concentration

Combining Sections 31.4–31.6 gives the intrinsic carrier concentration as a function of temperature:

$$n_i(T) = \sqrt{N_c(300) N_v(300)} \left(\frac{T}{300}\right)^{3/2} \exp\left(-\frac{E_g(T)}{2 k_B T}\right)$$

The power-law prefactor changes slowly. The exponential dominates, and it is controlled by the ratio $E_g / k_B T$. At 300 K, $E_g / 2k_B T$ is 21.7 for silicon and 63 for 4H-SiC, and the exponential factors differ by about 18 orders of magnitude.

Worked example — the intrinsic concentration of silicon at 300 K. Take $N_c = 2.86 \times 10^{19}$ cm^{-3}, $N_v = 3.10 \times 10^{19}$ cm^{-3} (the values of Green, which fit measured n_i data) and $E_g = 1.1245$ eV from the Varshni relation. Then $\sqrt{N_c N_v} = \sqrt{8.87 \times 10^{38}} = 2.98 \times 10^{19}$ cm^{-3}. With $k_B T = 0.025852$ eV, $E_g / 2k_B T = 1.1245 / 0.051704 = 21.75$, and $e^{-21.75} = 3.59 \times 10^{-10}$. Therefore

$$n_i = 2.98 \times 10^{19} \times 3.59 \times 10^{-10} = 1.07 \times 10^{10} \text{ cm}^{-3}$$

Careful measurements give $(0.97\text{–}1.0) \times 10^{10}$ cm^{-3}, and 1.0×10^{10} cm^{-3} is the value used throughout this book. Older texts quote 1.45×10^{10} or 1.5×10^{10}. The differences come from small differences in E_g and in the hole mass: an error of only 10 meV

in E_g changes n_i by $e^{0.01/0.0517} = 1.21$, or 21 %. Using the older $N_v = 1.04 \times 10^{19}$ cm^{-3} with $E_g = 1.12$ eV gives 6.7×10^{9} cm^{-3}. Exponential sensitivity means that any quantity computed from n_i (leakage current, minority-carrier density) inherits an uncertainty of tens of per cent unless the parameter set is self-consistent.

Silicon contains 5.0×10^{22} atoms per cubic centimetre and 2.0×10^{23} valence electrons. At room temperature only one valence electron in 2×10^{13} is free. The intrinsic resistivity, with $\mu_n + \mu_p \approx 1850$ cm^2/(V s), is $1/(q n_i(\mu_n + \mu_p)) \approx 3.4 \times 10^{5}\ \Omega$ cm. Undoped silicon is a fairly good insulator, and every useful conductivity comes from doping (Chapter 32).

Comparison across materials

Material	E_g (eV)	n_i at 300 K (cm^{-3})	Temperature at which $n_i = 10^{15}$ cm^{-3} (approx.)
Ge	0.66	2×10^{13}	about 390 K
Si	1.12	1.0×10^{10}	about 545 K
GaAs	1.42	2×10^{6}	about 770 K
4H-SiC	3.26	$\sim 10^{-8}$	about 1350 K
GaN	3.39	$\sim 10^{-10}$	about 1400 K

The values of n_i for the wide-gap materials are notional: a cubic centimetre of 4H-SiC at room temperature contains, on average, no thermally generated carriers at all, and the real carrier density is set entirely by impurities and defects. Figure 31.5 plots n_i against $1000/T$.

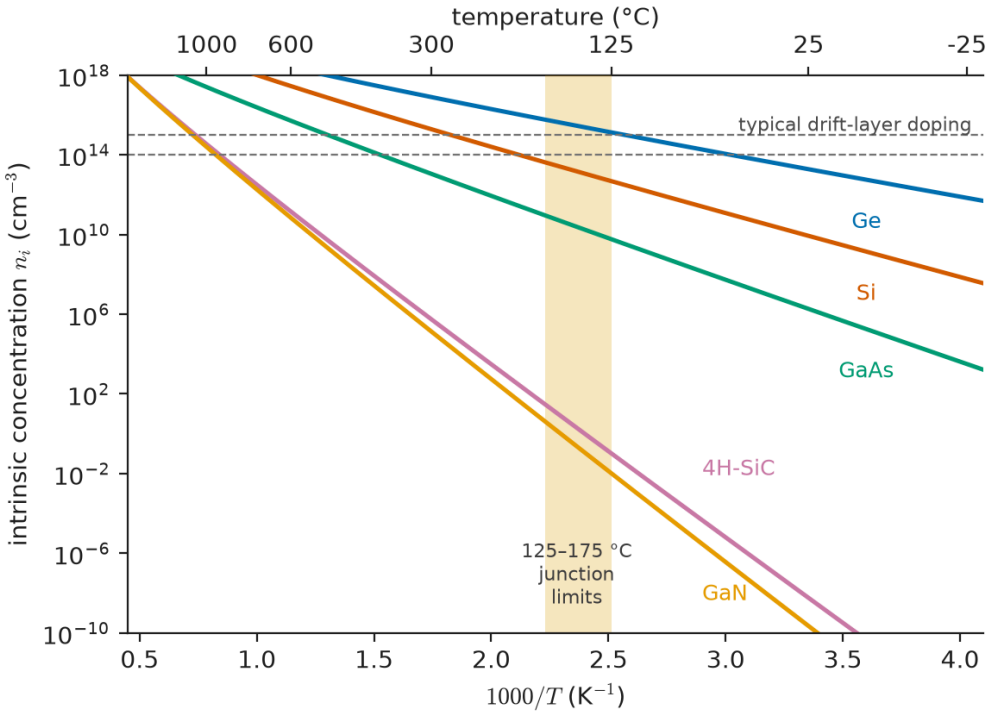


Figure 31.5 — Intrinsic carrier concentration versus inverse temperature for Ge, Si, GaAs, 4H-SiC and GaN, showing the exponential dependence on band gap and the temperatures at which each material becomes intrinsic for typical doping.

The temperature limit of a material

A doped semiconductor works as designed only as long as the dopant-controlled carrier density exceeds n_i. Once n_i approaches the doping, the material becomes intrinsic: carrier density and resistivity stop being controlled by design and start falling exponentially with temperature, and p–n junctions lose their rectifying behaviour.

Worked example — the leakage problem of the opening, and the intrinsic temperature. Leakage from thermal generation in a reverse-biased silicon junction scales with n_i. Recompute n_i at 125 °C (398.15 K). The Varshni relation gives

$E_g = 1.170 - 4.73 \times 10^{-4} \times 398.15^2/(398.15 + 636) = 1.0975$ eV. The prefactor grows as $(398.15/300)^{3/2} = 1.529$, giving $\sqrt{N_c N_v} = 4.55 \times 10^{19}$ cm^{-3}. With $k_B T = 0.03431$ eV, $E_g/2k_BT = 15.99$ and $e^{-15.99} = 1.13 \times 10^{-7}$, so $n_i(125\,^\circ\text{C}) = 5.2 \times 10^{12}$ cm^{-3}. At 25 °C the same procedure gives 9.2×10^9 cm^{-3}. The ratio is about 560: generation leakage rises by a factor of over 500 for a 100 K rise. (Diffusion leakage scales as n_i^2 and rises by 3×10^5.) Near room temperature n_i doubles roughly every 9 K.

Now ask when a silicon drift layer doped at $N_D = 10^{15}$ cm^{-3} stops being extrinsic. Iterating the same formula, n_i reaches 10^{14} cm^{-3} (10 % of the doping) near 470 K, about 200 °C, and equals the doping near 545 K, about 270 °C. This is why silicon power devices are rated at 150–175 °C junction temperature. For 4H-SiC with the same doping, n_i does not reach 10^{15} cm^{-3} until about 1350 K. The silicon carbide device is limited not by its semiconductor but by its metallization, die attach, gate oxide and package (Chapters 34 and 51).

> **Status of the principle: exponential temperature dependence of n_i**
> *Derived.* It follows from the law of mass action with temperature-dependent E_g and $N_{c,v} \propto T^{3/2}$. In plots of $\ln(n_i T^{-3/2})$ versus $1/T$ the slope gives $E_g(0)/2k_B$, extrapolated to 0 K, not E_g at the measurement temperature, because of the linear part of the Varshni shift.

What else changes with temperature

The intrinsic concentration is only the start. In a doped semiconductor, temperature also controls the fraction of dopants that are ionized (freeze-out below about 100 K in Si, and at room temperature for deep acceptors in SiC and GaN), the mobility (falling roughly as $T^{-1.5}$ to $T^{-2.4}$ as phonon scattering grows), and the recombination lifetime. These are properties of doped, real material and are the subject of Chapter 32, which also draws the full curve of carrier density from cryogenic to intrinsic temperatures.

31.9 Direct and indirect band gaps

Two shapes of band edge

Section 31.2 showed that the valence-band maximum of the common semiconductors lies at Γ, but that the conduction-band minimum can lie at Γ or elsewhere in the zone. This difference divides semiconductors into two classes.

- **Direct gap**: conduction-band minimum and valence-band maximum at the same **k** (both at Γ). Examples: GaAs, InP, InGaAs, GaN, InGaN, ZnO, AlN, CdTe, and the organic-inorganic perovskites.
- **Indirect gap**: the extrema are at different **k**. Examples: Si (minimum near X), Ge (at L), GaP, 4H-SiC, AlAs, diamond.

Figure 31.3 shows the two shapes.

Why it matters: conservation of crystal momentum

An optical transition must conserve both energy and crystal momentum. A photon carries a lot of energy but very little momentum. For a photon at the silicon gap ($\lambda = 1.1$ µm) the wavevector is $2\pi/\lambda = 5.7 \times 10^6$ m^{-1}. The Brillouin zone of silicon extends to about $2\pi/a = 1.16 \times 10^{10}$ m^{-1}, so the photon wavevector is only 5×10^{-4} of the zone size. On a band diagram, optical transitions are vertical lines.

- In a **direct-gap** material, an electron at the bottom of the conduction band and a hole at the top of the valence band have the same **k**. They can recombine and emit a photon in a single, fast, first-order process.
- In an **indirect-gap** material, the electron at the conduction minimum has a crystal momentum of order $0.85 \times 2\pi/a$ that the photon cannot absorb. Recombination requires a third particle, a **phonon**, to supply or remove the momentum. A second-order process involving three particles is much less probable, so radiative recombination is slow and loses out to non-radiative channels via defects (Chapter 32).

Status of the principle: k-conservation in optical transitions
Fundamental. It follows from the translational symmetry of the crystal (conservation of crystal momentum), combined with the negligible momentum of an optical photon. It is relaxed when translational symmetry is broken: in amorphous silicon, in heavily alloyed or defect-bound states (the isoelectronic N trap in GaP:N, used in green LEDs), and in nanostructures small enough that k is no longer a good quantum number (porous Si, Si nanocrystals, Chapter 38).

The radiative coefficient: a factor of 10^4

The strength of band-to-band radiative recombination is measured by the coefficient B in the rate $R = Bnp$ (Chapter 32):

Material	Gap type	B at 300 K (cm³/s)
Si	indirect	$\approx 1 \times 10^{-14}$
4H-SiC	indirect	$\approx 1.5 \times 10^{-12}$
GaN	direct	$\approx (1\text{–}2) \times 10^{-11}$
GaAs	direct	$\approx (1\text{–}7) \times 10^{-10}$, often 2×10^{-10}

GaAs recombines radiatively about 10^4 times faster than silicon at equal carrier densities. In silicon the radiative lifetime at 10^{17} cm^{-3} doping is about 1 ms, while the defect-assisted lifetime in good material is microseconds to milliseconds, so most recombination produces heat. In GaAs the radiative lifetime at the same doping is tens of nanoseconds and wins. This one number decides that LEDs and lasers are made from direct-gap materials (Chapter 35), and it explains the opening question: the indicator LED on the sensor board is InGaN or AlGaInP, not Si or SiC.

Absorption: both work, but not equally well

Absorption is the reverse process. Indirect absorption also needs a phonon and is weak near the edge, but at higher photon energies a direct transition becomes available (at Γ, about 3.4 eV in Si) and absorption becomes strong in any material. The absorption coefficient α (light intensity decays as $e^{-\alpha x}$) near the edge rises much more steeply in a direct-gap material: roughly $\alpha \propto (\hbar\omega - E_g)^{1/2}$ for direct and $\alpha \propto (\hbar\omega - E_g \pm E_{\text{ph}})^2$ for phonon-assisted indirect transitions, where E_{ph} is the phonon energy. Figure 31.6 compares the measured curves.

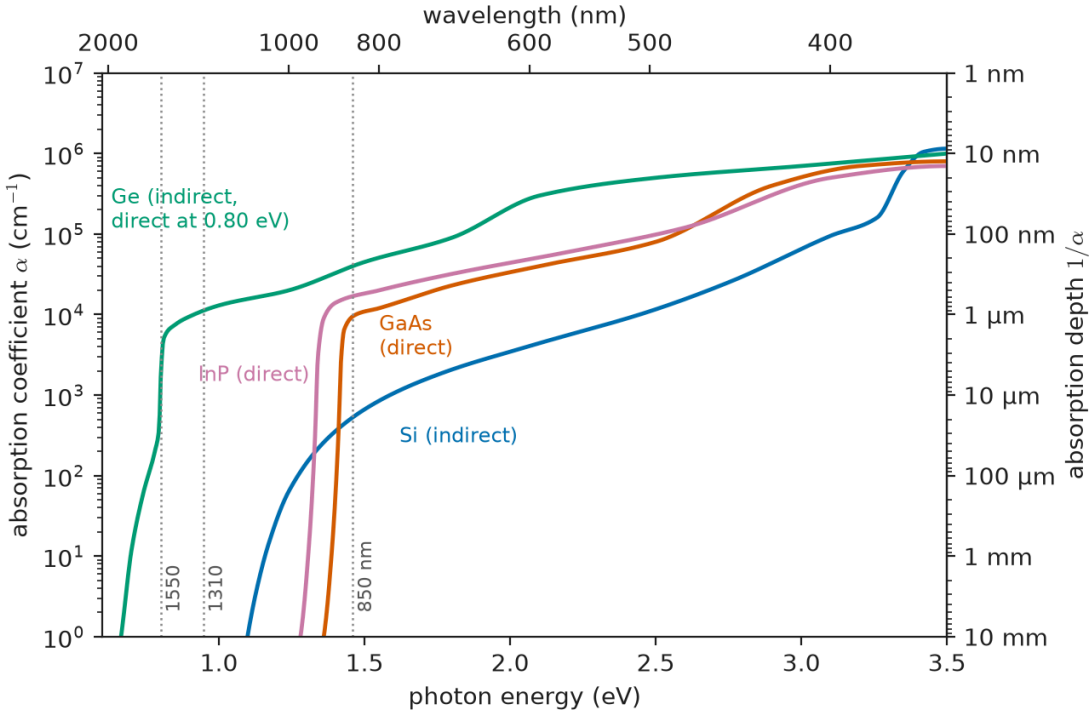


Figure 31.6 — Optical absorption coefficient versus photon energy for silicon and germanium (indirect) and gallium arsenide and indium phosphide (direct), showing the slow rise above an indirect edge and the abrupt rise above a direct edge.

Worked example — how thick must an absorber be? To absorb 90 % of light that enters, the thickness must satisfy $e^{-\alpha d} = 0.1$, so $d = \ln 10/\alpha = 2.303/\alpha$. At 800 nm, silicon has $\alpha \approx 850$ cm^{-1}, so $d = 2.303/850 = 2.7 \times 10^{-3}$ cm $= 27$ µm. GaAs has $\alpha \approx 1.2 \times 10^{4}$ cm^{-1} at the same wavelength, so $d = 2.303/1.2 \times 10^{4} = 1.9 \times 10^{-4}$ cm $= 1.9$ µm. The indirect-gap absorber must be about 14 times thicker. For the near-infrared at 1000 nm, silicon's α falls to about 64 cm^{-1} and the 90 % thickness grows to 360 µm, thicker than many wafers. This is why silicon solar cells use textured surfaces and back reflectors to lengthen the optical path, why silicon photodiodes lose responsivity beyond about 950 nm, and why thin-film absorbers such as CdTe and perovskites need only a micrometre or two (Chapters 35 and 40). Germanium is indirect too, but its direct gap at Γ lies only 0.14 eV above the indirect gap, at 0.80 eV (1.55 µm), so it absorbs strongly at the main telecommunication wavelength. This is why Ge-on-Si photodetectors are used in silicon photonics.

Summary of the consequences

Property	Direct gap (GaAs, InP, GaN)	Indirect gap (Si, Ge, SiC)
Radiative coefficient B	High (10^{-11}–10^{-10} cm³/s)	Low (10^{-14}–10^{-12} cm³/s)
Light emission	Efficient: LEDs, lasers	Inefficient; recombination mostly non-radiative
Absorption near the edge	Steep; µm-thick absorbers	Gradual; tens to hundreds of µm
Minority-carrier lifetime	Short (ns), limited by radiative recombination	Long (µs–ms) in clean material
Diffusion length	µm	100 µm–mm in clean Si
Typical uses	Optoelectronics, fast low-noise RF	Logic, power, photodetection, solar cells

The long lifetime of indirect-gap silicon, a handicap for emitting light, is an advantage for bipolar power devices and solar cells, which need minority carriers to survive long enough to diffuse across thick regions. Chapter 32 quantifies this. The material choice follows the physics.

What remains open

Open question	Where it is answered
How do donors and acceptors set n, p and E_F, and what happens at very low and very high temperatures in doped material?	Chapter 32
What limits mobility and velocity, and how do carriers diffuse and recombine?	Chapter 32
How do Si, Ge, the III–V compounds, SiC, GaN, Ga_2O_3, diamond, amorphous, organic and 2D semiconductors compare in practice?	Chapter 33
How do bands line up at heterojunctions, surfaces and contacts, and what are interface states?	Chapter 34
How are absorption and emission used in LEDs, lasers, photodetectors and solar absorbers?	Chapter 35
How do electron states and the density of states change when a material is a few atoms thick?	Chapter 38
Why does a superconductor also have a gap, and how does it differ from a band gap?	Chapter 30
How are crystals grown and doped with the purity this chapter assumes?	Chapter 50
How are carrier concentration, mobility and band gap measured (Hall effect, spectroscopy)?	Chapter 52

Next: Chapter 32 adds the impurities and defects that make semiconductors useful, deriving how doping, mobility, diffusion, recombination and high-field transport set the electrical behaviour of real material.

Chapter 32 — Semiconductor Material Properties

Part VII · Semiconductor and Photonic Materials

A solar-cell factory changes the supplier of its silicon wafers. The new wafers have the same boron doping, the same resistivity of 1 Ω cm, the same thickness and the same surface finish, and they pass every incoming inspection. Yet the finished cells come off the line half a percentage point less efficient, and the loss costs millions. A minority-carrier lifetime map of a wafer finally finds the cause: the bulk lifetime is a few microseconds instead of the expected millisecond. Secondary-ion mass spectrometry finds interstitial iron at about 10^{12} atoms per cubic centimetre, twenty parts per *trillion*. Such a small amount of iron leaves every property on the incoming specification sheet unchanged, yet it reduces the distance that photogenerated electrons can diffuse before they recombine from nearly two millimetres to under a hundred micrometres. Chapter 31 described perfect crystals. This chapter is about real ones: doped on purpose, contaminated by accident, heated, and driven by strong fields. It shows why some properties of a semiconductor, such as its resistivity, are set by parts per million of impurity, while others, such as its lifetime, are destroyed by parts per trillion.

How do donors and acceptors set the carrier concentration and the Fermi level, and where does doping stop working?
What limits how fast carriers move, at low fields and at high fields, and how are drift and diffusion related? How do carriers recombine, what sets the minority-carrier lifetime, and why are some impurities harmless while others are lethal at trace levels?

Concept	Engineering destination
Donors, acceptors, ionization energy	Choice of dopant; freeze-out in cryogenic and wide-gap devices
Carrier concentration and Fermi level	Resistivity specification of wafers and epilayers; contact design
Doping profiles	Junction depth, sheet resistance, built-in fields in graded layers
Mobility and Caughey–Thomas model	Resistivity-to-doping conversion; on-resistance; TCAD input
Drift, diffusion, Einstein relation	Every device current equation; diffusion length
SRH, radiative and Auger recombination	Leakage, switching speed, LED efficiency, solar-cell voltage
Minority-carrier lifetime	Wafer quality metric; bipolar power devices; lifetime control
Temperature dependence	Temperature coefficients; operating-temperature window
Velocity saturation, impact ionization	Short-channel currents; RF limits; breakdown field
Deep levels and defects	Contamination control, gettering, semi-insulating substrates, reliability

Roadmap. Pure and doped material (32.1) → the dopant atom and its energy level (32.2) → solving for carrier density and Fermi level (32.3) → how doping varies in space (32.4) → mobility and scattering (32.5) → drift, diffusion and the Einstein relation (32.6) → recombination and generation mechanisms (32.7) → lifetime as a material property (32.8) → everything as a function of temperature (32.9) → strong fields: saturation and avalanche (32.10) → deep levels and defects (32.11).

Boundary.
This chapter treats the semiconductor as a material: its carrier densities, transport coefficients and recombination rates. How these quantities combine in p–n junctions, bipolar and field-effect transistors, and circuits belongs in the Electronics & Computing core. How dopants are introduced (diffusion furnaces, ion implantation, epitaxy) is covered in Chapter 50, and the atomic diffusion that underlies those processes in Chapter 8 and the kinetics of the Advanced Chemistry core.

32.1 Intrinsic and extrinsic semiconductors

Chapter 31 showed that a perfectly pure semiconductor contains equal numbers of electrons and holes, $n = p = n_i$, and that n_i is tiny at room temperature: 1.0×10^{10} cm^{-3} in silicon, about one free carrier per 5×10^{12} atoms. Such **intrinsic** material is useless for most purposes for three reasons:

1. Its resistivity (3.4×10^{5} Ω cm for Si) is too high for most functions.
2. Its conductivity depends exponentially on temperature, doubling roughly every 9 K near room temperature in Si.
3. It has no built-in asymmetry between electrons and holes, so it cannot form a junction.

An **extrinsic** semiconductor is one in which the carrier concentrations are set by impurities, deliberately added (**doping**) or unintentionally present. Adding 10^{16} phosphorus atoms per cubic centimetre to silicon, 0.2 parts per million, raises the electron concentration by six orders of magnitude, makes n independent of temperature over a wide range around room temperature, and lowers the hole concentration to $n_i^2/n = 10^{4}$ cm^{-3}. The material is now **n-type**: electrons are the **majority carriers** and holes the **minority carriers**. Doping with boron instead gives **p-type** material.

Purity requirements follow directly. If a deliberately doped layer is to be controlled to ±5 %, unintentional shallow impurities must be well below that fraction of the intended doping. For a power-device drift layer at 10^{14} cm^{-3} this means background donors and acceptors below about 10^{12} cm^{-3}, or 0.02 parts per billion. This is why semiconductor-grade silicon is purified to "nine nines" and beyond (Chapter 50), and why the intrinsic regime is practically reachable only at high temperature.

32.2 Donors and acceptors

Substitutional dopants

A phosphorus atom (group V) substituting for silicon (group IV) in the diamond lattice uses four of its five valence electrons to complete the four covalent bonds to its neighbours. The fifth electron is not needed for bonding. It remains bound to the P^{+} ion core only by Coulomb attraction, weakened by the dielectric screening of the surrounding silicon. Very little energy is needed to free it into the conduction band. Phosphorus is a **donor**. Once ionized it becomes a fixed positive charge, N_D^{+}.

A boron atom (group III) has only three valence electrons. It completes its fourth bond by capturing an electron from the valence band, which leaves a mobile hole. Boron is an **acceptor**. Once ionized it becomes a fixed negative charge, N_A^{-}.

In III–V compounds the same counting applies by site. Group-VI atoms (S, Se, Te) on the group-V site and group-IV atoms (Si) on the group-III site are donors. Group-II atoms (Zn, Be, Mg) on the group-III site and group-IV atoms (C) on the group-V site are acceptors. A group-IV atom can be **amphoteric**: silicon in GaAs is a donor on a Ga site and an acceptor on an As site, and the balance depends on growth conditions.

The hydrogenic model

The bound fifth electron of a donor resembles the electron of a hydrogen atom, with two changes: it moves through a dielectric of relative permittivity ε_r, and it has the effective mass m^{*} of the conduction band (Chapter 31). Scaling the hydrogen results accordingly:

$$E_d = \frac{m^{*}/m_0}{\varepsilon_r^{2}} \times 13.6\ \text{eV}, \qquad a_d = \frac{\varepsilon_r}{m^{*}/m_0} \times 0.0529\ \text{nm}$$

For silicon ($m^{*} = 0.26\,m_0$ conductivity mass, $\varepsilon_r = 11.7$): $E_d = 13.6 \times 0.26/136.9 = 26$ meV and $a_d = 0.0529 \times 11.7/0.26 = 2.4$ nm. For GaAs ($m^{*} = 0.067\,m_0$, $\varepsilon_r = 12.9$): $E_d = 5.5$ meV and $a_d = 10$ nm. The donor electron orbits over hundreds (Si) to thousands (GaAs) of lattice sites. This justifies treating the host as a continuum with an effective mass and a dielectric constant.

> **Status of the principle: hydrogenic donor model**
> *Approximate.* It assumes a large orbit (many lattice constants), a single isotropic band minimum and a pure Coulomb potential. It is close to measured values in GaAs (shallow donors 5.8–6 meV) and gives the right order of magnitude in Si, but it underestimates real binding energies (45 meV for P) because the electron spends part of its time near the impurity core, where screening fails and the potential is chemistry-specific (the "central-cell correction"). It fails completely for deep levels (Section 32.11).

Ionization energies

Host	Dopant	Type	Ionization energy (meV)	Notes
Si	P	donor	45	Standard n-type dopant
Si	As	donor	54	Low diffusivity; shallow junctions, emitters
Si	Sb	donor	43	Buried layers; low diffusivity
Si	B	acceptor	45	Practically the only p-type dopant
Si	Al	acceptor	69	Fast diffuser; deep p-wells in power devices
Si	Ga	acceptor	74	Used in some solar wafers (no B–O degradation)
Si	In	acceptor	157	Too deep for room-temperature use
Ge	P, As, B	both	10–13	Fully ionized even near 77 K
GaAs	Si (on Ga), Te	donor	6	
GaAs	Be, Zn, C	acceptor	26–31	C used in HBT bases
4H-SiC	N	donor	61 (hexagonal site), 125 (cubic site)	Two inequivalent lattice sites
4H-SiC	Al	acceptor	190–230	Incomplete ionization at 300 K
GaN	Si	donor	15–30	
GaN	Mg	acceptor	160–200	Only a few per cent ionized at 300 K; H passivation
Diamond	B	acceptor	370	
Diamond	P	donor	570–600	

Two patterns stand out. The shallow levels of Si, Ge and GaAs are fully ionized at room temperature. In the wide-gap materials, especially for acceptors, the larger effective mass and smaller permittivity push the levels deep (the hydrogenic E_d scales as m^*/ε_r^2), so dopants are only partly ionized at room temperature. Poor p-type conductivity is the single largest materials obstacle in GaN, SiC, diamond and Ga_2O_3 (Chapter 33).

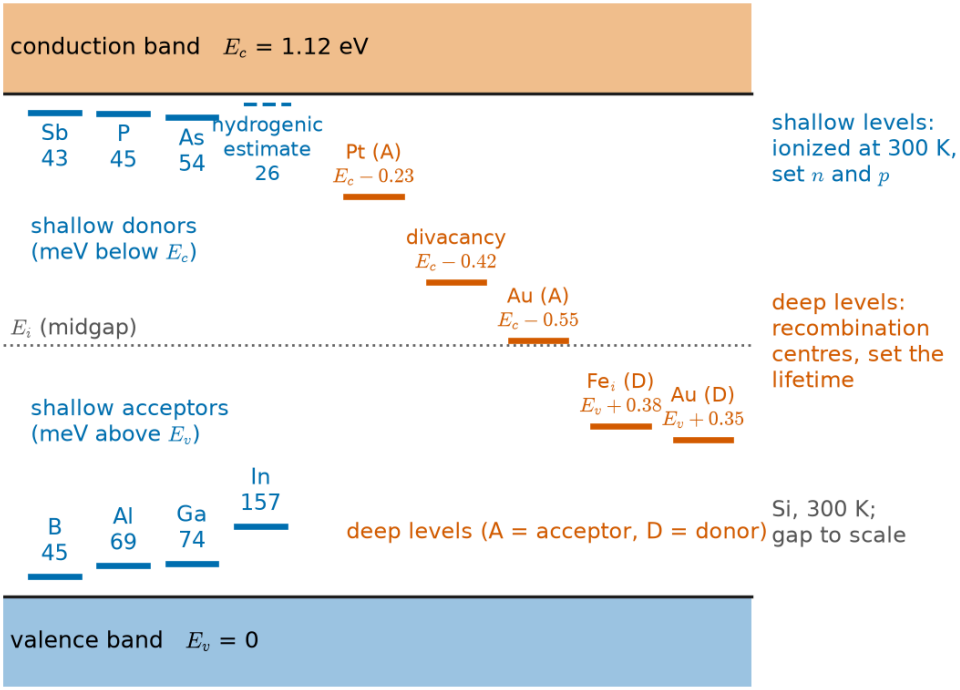


Figure 32.1 — Measured energy levels of shallow dopants and representative deep impurities in the band gap of silicon, with the hydrogenic estimate for comparison.

Figure 32.1 places these levels in the silicon gap together with the deep levels discussed in Section 32.11.

Compensation

If both donors and acceptors are present, electrons from the donors first fill the acceptors. Only the excess dopant contributes free carriers: for $N_D > N_A$, $n \approx N_D - N_A$. A sample with $N_D = 10^{17}$ and $N_A = 5 \times 10^{16}\ \mathrm{cm^{-3}}$ has the same electron concentration as one doped with 5×10^{16} donors alone, but it contains three times as many ionized impurities to scatter those electrons (Section 32.5). Compensation is used deliberately to make semi-insulating substrates (Section 32.11) and to counter-dope regions during processing. It happens accidentally whenever background impurities are present.

Heavy doping: degeneracy and the Mott transition

As doping increases, the donor orbits begin to overlap. In Si:P, at about $3.7 \times 10^{18}\ \mathrm{cm^{-3}}$, the donor electrons delocalize into an impurity band that merges with the conduction band, and the material conducts like a (poor) metal even at absolute zero. This is the **Mott transition**. A rough criterion is $N_D^{1/3} a_d \approx 0.26$, which for Si gives $N_D \approx (0.26/2.4\ \mathrm{nm})^3 \approx 1.3 \times 10^{18}\ \mathrm{cm^{-3}}$, the right order of magnitude. Above this level:

- the ionization energy falls effectively to zero, and the material does not freeze out at low temperature;
- the Fermi level enters the band (**degenerate** doping), so Boltzmann statistics fail and full Fermi–Dirac integrals are needed;
- the band gap shrinks (**bandgap narrowing**) by tens of meV through carrier–carrier and carrier–impurity interactions. An empirical fit (Slotboom) gives about 80 meV at $10^{19}\ \mathrm{cm^{-3}}$ in Si, and since $n_i^2 \propto e^{-E_g/k_B T}$, the effective n_i^2 rises by $e^{0.08/0.0259} \approx 22$.

Degenerate doping is not a defect but a tool. It is how ohmic contacts are made: a heavily doped surface layer thins the barrier at a metal–semiconductor contact until electrons tunnel through it (Chapter 34).

32.3 Carrier concentration

The neutrality condition

In a uniformly doped sample in equilibrium, the total charge must be zero:

$$p + N_D^+ = n + N_A^-$$

Here n and p depend on E_F through the expressions of Chapter 31, and so do the ionized fractions. A donor level of energy E_d is neutral when occupied and ionized when empty, with occupation probability modified by spin degeneracy:

$$N_D^+ = \frac{N_D}{1 + g_D \exp[(E_F - E_d)/k_B T]}, \qquad N_A^- = \frac{N_A}{1 + g_A \exp[(E_a - E_F)/k_B T]}$$

with degeneracy factors $g_D = 2$ for donors and $g_A = 4$ for acceptors in Si, Ge and GaAs (the latter because of the two degenerate valence bands at Γ). Neutrality is one equation with one unknown, E_F. It can always be solved numerically. In three common regimes it has simple closed forms.

> **Status of the principle: charge neutrality in uniform material**
> *Derived.* It follows from Gauss's law: any net charge in a region larger than a few Debye lengths would produce fields that drive carriers until it is cancelled. The Debye length $L_D = \sqrt{\varepsilon k_B T / q^2 N}$ is 41 nm in Si at 10^{16} cm^{-3}, so neutrality holds everywhere except within depletion regions and at interfaces.

Full ionization (the extrinsic regime)

When dopants are fully ionized and $|N_D - N_A| \gg n_i$, the neutrality equation together with $np = n_i^2$ gives, for n-type material,

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2} \approx N_D - N_A, \qquad p = \frac{n_i^2}{n}$$

and the Fermi level follows from $n = N_c \exp[-(E_c - E_F)/k_B T]$:

$$E_c - E_F = k_B T \ln \frac{N_c}{n}, \qquad E_F - E_i = k_B T \ln \frac{n}{n_i}$$

The full square-root form remains valid as n_i approaches the net doping and reduces to $n = n_i$ in the intrinsic limit.

Worked example — Fermi level and ionization in phosphorus-doped silicon. Take $N_D = 10^{16}$ cm^{-3}, $E_c - E_d = 45$ meV, $g_D = 2$, $N_c = 2.86 \times 10^{19}$ cm^{-3} at 300 K.

At 300 K. Assuming full ionization, $n = 10^{16}$ cm^{-3} and $E_c - E_F = 0.02585 \ln(2.86 \times 10^{19}/10^{16}) = 0.02585 \times 7.96 = 0.206$ eV. The donor level lies $0.206 - 0.045 = 0.161$ eV $= 6.2 k_B T$ above E_F, so the fraction of donors still neutral is $1/(1 + \tfrac{1}{2} e^{6.2}) \approx 0.4$ %. The assumption is consistent: 99.6 % of donors are ionized. The hole density is $p = n_i^2/n = 10^{20}/10^{16} = 10^4$ cm^{-3}.

At 77 K (liquid nitrogen). N_c scales as $T^{3/2}$, giving $N_c = 2.86 \times 10^{19} \times (77/300)^{3/2} = 3.72 \times 10^{18}$ cm^{-3}, and $k_B T = 6.64$ meV. For a donor-only sample with $n = N_D^+$, the neutrality equation becomes a quadratic,

$$\frac{n^2}{N_D - n} = \frac{N_c}{g_D} \exp\left(-\frac{E_c - E_d}{k_B T}\right) = \frac{3.72 \times 10^{18}}{2}\, e^{-6.78} = 2.11 \times 10^{15} \text{ cm}^{-3}$$

whose positive root is $n = 3.7 \times 10^{15}$ cm^{-3}. Only 37 % of the donors are ionized, and $E_c - E_F = k_B T \ln(N_c/n) = 6.64 \times 6.93 = 46$ meV: the Fermi level sits right at the donor level. A resistor designed at room temperature will have nearly three times its nominal resistance at 77 K from freeze-out alone, partly offset by higher mobility. This is why cryogenic electronics uses heavily (degenerately) doped resistors and contacts.

Incomplete ionization at room temperature

The same quadratic applies to deep dopants at 300 K. For Al acceptors in 4H-SiC ($E_a \approx 0.2$ eV, $g_A = 4$, $N_v = 2.5 \times 10^{19}$ cm^{-3}), $N_A = 10^{17}$ cm^{-3} gives $p \approx 1.5 \times 10^{16}$ cm^{-3}, only 15 % ionized. At $N_A = 10^{19}$ the ionized fraction falls to under 2 %. For Mg in GaN ($E_a \approx 0.17$ eV, $N_v = 4.6 \times 10^{19}$ cm^{-3}), 10^{19} cm^{-3} of Mg gives $p \approx 4 \times 10^{17}$ cm^{-3}, about 4 %. The ionized fraction then rises

with temperature, so the p-type layers of wide-gap devices become *more* conductive as they heat up, the opposite of most semiconductor behaviour. (Real values are lower still, because the levels shift with doping and compensation reduces p.)

Degenerate material

When E_F comes within about $3k_BT$ of a band edge, $n = N_c\,\mathcal{F}_{1/2}(\eta)$, where $\eta = (E_F - E_c)/k_BT$ and $\mathcal{F}_{1/2}$ is the normalized Fermi–Dirac integral, and the law of mass action must be corrected for bandgap narrowing. In Si at 300 K this occurs above about $N_c e^{-3} \approx 1.4 \times 10^{18}$ cm^{-3}. In GaAs, whose N_c is only 4.4×10^{17} cm^{-3}, n-type material becomes degenerate near 2×10^{16} cm^{-3}. The light electron mass makes GaAs easy to push into degeneracy.

> **Status of the principle:** $n \approx N_D - N_A$
> *Approximate.* Valid for fully ionized shallow dopants, non-degenerate material and $N_D - N_A \gg n_i$: in Si, roughly from 150 K to 450 K for 10^{15}–10^{18} cm^{-3}. It fails at low temperature (freeze-out), at high temperature (intrinsic conduction), for deep dopants (SiC:Al, GaN:Mg) and in degenerate material.

32.4 Doping profiles

Real devices are not uniformly doped. The **doping profile** $N(x)$, the dopant concentration as a function of depth, is a materials property set by processing, and it controls junction depth, sheet resistance and internal fields. Chapter 50 explains how profiles are produced. Here they are treated as a given structure with electrical consequences. The standard profile shapes follow from the solutions of the diffusion equation (Chapter 8) or from the statistics of ion stopping.

Process	Profile shape	Formula	Typical use
Epitaxial growth	Step (nearly uniform layers)	$N(x) = N_{\text{epi}}$ for $x < t_{\text{epi}}$	Drift layers, HEMT and laser stacks
Diffusion from a constant surface source	Complementary error function	$N(x) = N_s\,\operatorname{erfc}\!\left(x/2\sqrt{Dt}\right)$	Heavy emitters, solar-cell front layers
Drive-in of a fixed dose Q	Gaussian at the surface	$N(x) = \dfrac{Q}{\sqrt{\pi D t}}\exp\!\left(-\dfrac{x^2}{4Dt}\right)$	Wells, base regions
Ion implantation (as implanted)	Gaussian centred at range R_p	$N(x) = \dfrac{\Phi}{\sqrt{2\pi}\,\Delta R_p}\exp\!\left[-\dfrac{(x-R_p)^2}{2\Delta R_p^2}\right]$	Almost all modern doping
Retrograde implant	Buried peak, low surface	Sum of Gaussians	CMOS wells, punch-through stops
Delta doping	Sheet of dopant, nm thick	$N_{2D}\,\delta(x - x_0)$	HEMT supply layers

Here D is the dopant diffusivity at the process temperature, t the time, Φ the implanted dose (atoms per cm²), R_p the projected range and ΔR_p the straggle. For boron implanted at 50 keV into silicon, $R_p \approx 0.16$ µm and $\Delta R_p \approx 0.05$ µm. A dose of 10^{15} cm^{-2} then peaks at $10^{15}/(2.507 \times 5.3 \times 10^{-6}) = 7.5 \times 10^{19}$ cm^{-3}. Figure 32.2 sketches the common shapes.

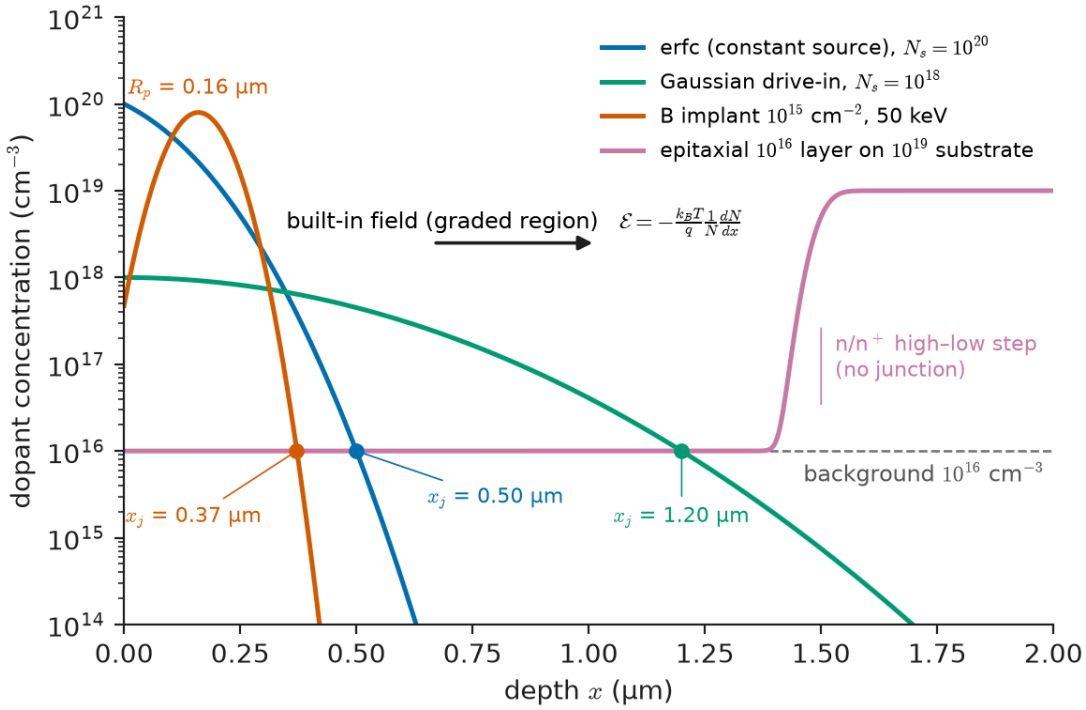


Figure 32.2 — Typical doping profiles in silicon: erfc and Gaussian diffusion profiles, an as-implanted Gaussian, and an epitaxial step, each meeting a uniform background to form a junction.

Junction depth

A **metallurgical junction** forms where the profile of one type crosses a background of the opposite type, $N(x_j) = N_B$. For a Gaussian drive-in profile, $x_j = 2\sqrt{Dt\,\ln(N_0/N_B)}$, where $N_0 = Q/\sqrt{\pi Dt}$ is the surface concentration. Because diffusion lengths scale as $\sqrt{Dt}$ and D is exponential in temperature, every high-temperature step after doping moves the junction, and modern processes use short, rapid anneals to keep junctions within tens of nanometres (Chapter 50).

Built-in field in a graded layer

A doping gradient creates an electric field even in equilibrium. In an n-type region with a donor gradient, electrons diffuse from high to low concentration, leaving the ionized donors behind. The resulting space charge builds a field that exactly opposes the diffusion. Setting the total electron current to zero (Section 32.6) gives

$$\mathcal{E} = -\frac{k_B T}{q}\,\frac{1}{N}\,\frac{dN}{dx}$$

For an exponential profile $N \propto e^{-x/\lambda}$ the field is uniform, $\mathcal{E} = k_B T/q\lambda$. Grading the doping from 10^{18} to 10^{16} cm^{-3} over 1 µm gives $\mathcal{E} = 0.02585 \times \ln 100/10^{-4} = 1.2 \times 10^{3}$ V/cm. A field of this size speeds minority carriers across the graded base of a bipolar transistor and the graded front layer of a solar cell. It is a material structure that does work without any applied bias.

Sheet resistance

For a thin doped layer the measurable quantity is the **sheet resistance** (Chapter 25 and the four-point probe of Chapter 52):

$$R_s = \frac{1}{q\int_0^{x_j} \mu(N(x))\,N(x)\,dx}$$

Worked example — sheet resistance of an implanted layer. A 10^{15} cm^{-2} boron implant, fully activated by annealing, peaks near 7.5×10^{19} cm^{-3}, where the Caughey–Thomas hole mobility of Section 32.5 is about 51 cm²/(V s), and has tails where the mobility is higher. Taking a concentration-weighted average of 70 cm²/(V s), $R_s \approx 1/(1.602 \times 10^{-19} \times 10^{15} \times 70) = 89\ \Omega$ per square. Measured values of 80–120 Ω per square are typical for such implants, the spread reflecting incomplete activation and the exact mobility average. Note that the sheet resistance depends on the dose, not the depth, as long as the mobility is roughly constant: it is the total number of carriers per unit area that conducts.

32.5 Carrier mobility

Mobility as a scattering time

Chapter 25 defined mobility as drift velocity per unit field, $\mu = v_d/\mathcal{E}$, and related it to the mean time τ_m between momentum-randomizing collisions:

$$\mu = \frac{q\tau_m}{m^*_{\text{cond}}}$$

The effective mass is set by the band (Chapter 31). The scattering time is set by everything that breaks the perfect periodicity of the crystal. Electrons with $\mu = 1400\ \text{cm}^2/(\text{V s})$ and $m^* = 0.26\,m_0$ have $\tau_m = \mu m^*/q = 0.14 \times 0.26 \times 9.11 \times 10^{-31}/1.60 \times 10^{-19} = 0.21$ ps between collisions. At a thermal velocity of about 10^7 cm/s this corresponds to a mean free path of a few tens of nanometres.

Scattering mechanisms

Mechanism	Physical origin	Temperature dependence of μ	Important in
Acoustic phonons	Lattice strain deforms the band edges	$\propto T^{-3/2}$ (ideal)	All semiconductors
Optical phonons, non-polar	Optical vibrations of the two sublattices	Activated; steep above $\sim$200 K	Si, Ge (with acoustic: $T^{-2.2}$ to $T^{-2.4}$)
Polar optical phonons	Oscillating dipoles of ionic bonds	Dominant near 300 K	GaAs, InP, GaN, SiC
Ionized impurities	Coulomb deflection by N_D^+, N_A^-	$\propto T^{3/2}/N_I$	Doped material; low temperature
Neutral impurities	Short-range potential	Nearly independent of T	Frozen-out material
Alloy disorder	Random potential in alloys	Weak	SiGe, AlGaN, InGaAs
Interface roughness, remote charges	Channel near a dielectric	Strong field dependence	MOSFET and HEMT channels (Chapter 34)
Dislocations, grain boundaries	Charged lines, barriers	Varies	GaN-on-Si, polycrystalline films
Carrier–carrier	Mutual Coulomb scattering	—	High injection

Two mechanisms dominate bulk material. **Lattice (phonon) scattering** sets the maximum mobility of pure material and falls steeply with temperature. **Ionized-impurity scattering** grows with doping and becomes *weaker* at higher temperature, because faster carriers are deflected less by a Coulomb centre.

Matthiessen's rule

If the mechanisms act independently, their scattering rates add:

$$\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I} + \cdots$$

> **Status of the principle: Matthiessen's rule for mobility**
> *Empirical.* Adding scattering rates is exact only if every mechanism has the same energy dependence. Because phonon and impurity scattering weight carriers of different energy differently, the rule errs by 10–20 % where two mechanisms are comparable. It is used as a first estimate and as the conceptual basis of fitted models.

The Caughey–Thomas model

Device engineers need mobility as a function of doping, accurate to a few per cent, without solving the Boltzmann transport equation. The standard answer is the **Caughey–Thomas** expression:

$$\mu(N) = \mu_{\min} + \frac{\mu_{\max} - \mu_{\min}}{1 + (N/N_{\text{ref}})^{\alpha}}$$

Here N is the *total* ionized impurity concentration ($N_D^+ + N_A^-$), $\mu_{\max}$ is the lattice-limited mobility of pure material, $\mu_{\min}$ the floor reached at very high doping, N_{ref} the doping at which the mobility is halfway between them, and α sets how sharp the transition is.

> **Status of the principle: Caughey–Thomas mobility model**
> *Empirical.* A four-parameter fit to Hall and resistivity data, not derived from scattering theory. It reproduces the measured mobility of Si, GaAs, SiC and GaN within about 5–10 % from 10^{13} to 10^{20} cm^{-3} at a fixed temperature. Extensions (Arora, Masetti, Klaassen) add temperature dependence, distinguish donors from acceptors and include minority-carrier and carrier–carrier effects. It describes majority-carrier, low-field, bulk mobility only.

Material, carrier	$\mu_{\max}$ (cm²/(V s))	$\mu_{\min}$ (cm²/(V s))	N_{ref} (cm⁻³)	α
Si, electrons	1414	68.5	9.20×10^{16}	0.711
Si, holes	470.5	44.9	2.23×10^{17}	0.719
GaAs, electrons (typical fit)	8500	800	1×10^{17}	0.5
4H-SiC, electrons (in-plane, typical fit)	950	40	2×10^{17}	0.76

The model gives the silicon values that every engineer uses to convert doping to resistivity:

Doping (cm⁻³)	μ_n (cm²/(V s))	μ_p (cm²/(V s))	ρ, n-type (Ω cm)	ρ, p-type (Ω cm)
10^{14}	1404	469	44.5	133
10^{15}	1362	462	4.6	13.5
10^{16}	1184	429	0.53	1.45
10^{17}	721	317	0.087	0.20
10^{18}	277	153	0.022	0.041
10^{19}	115	71	0.0054	0.0088
10^{20}	78	50	0.00080	0.0013

Figure 32.3 plots these curves. Above about 10^{18} cm^{-3} the mobility falls almost tenfold, so heavily doped regions conduct much less than their carrier density alone would suggest.

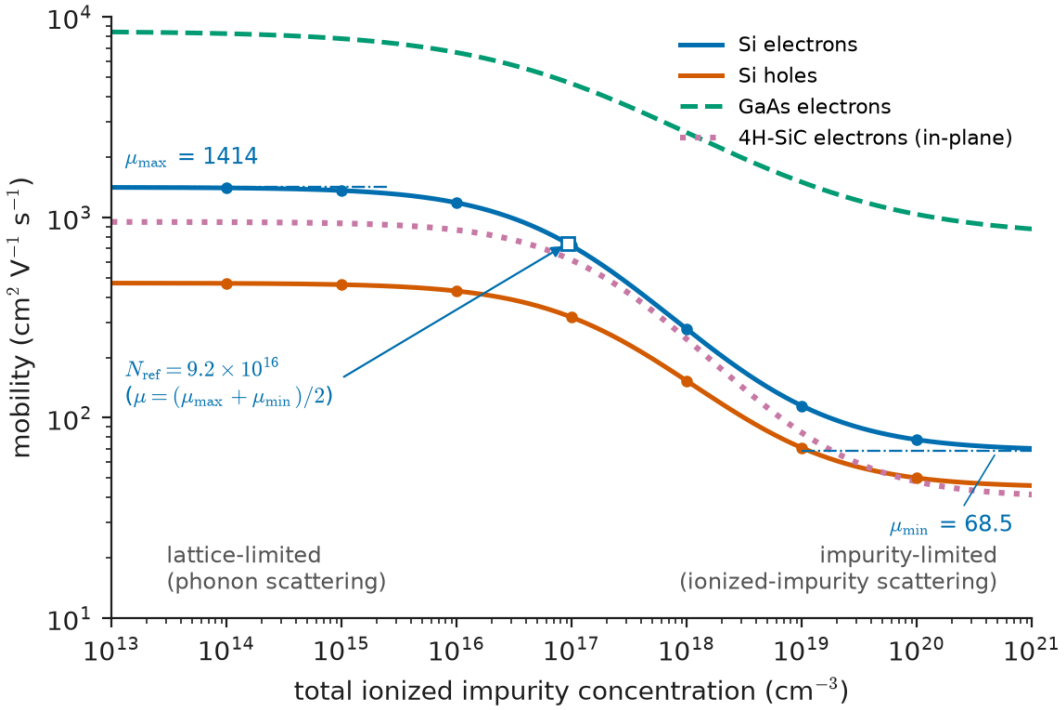


Figure 32.3 — Electron and hole mobility of silicon at 300 K as a function of total ionized impurity concentration (Caughey–Thomas fit), with GaAs electrons and 4H-SiC electrons for comparison.

Worked example — resistivity of a compensated sample. A silicon layer contains $N_D = 10^{17}$ and $N_A = 5 \times 10^{16}$ cm^{-3}. The free electron density is $n = N_D - N_A = 5 \times 10^{16}$ cm^{-3}, but the mobility must be evaluated at the total ionized impurity density, $N = 1.5 \times 10^{17}$ cm^{-3}:

$$\mu_n = 68.5 + \frac{1414 - 68.5}{1 + (1.5 \times 10^{17}/9.2 \times 10^{16})^{0.711}} = 68.5 + \frac{1345.5}{1 + 1.415} = 625 \text{ cm}^2/(\text{V s})$$

Then $\rho = 1/(qn\mu_n) = 1/(1.602 \times 10^{-19} \times 5 \times 10^{16} \times 625) = 0.20\ \Omega$ cm. An uncompensated sample with the same n would have $\mu_n \approx 885$ and $\rho \approx 0.14\ \Omega$ cm. Reading the measured 0.20 Ω cm off the uncompensated resistivity table would suggest a simple n-type doping of about 3×10^{16} cm^{-3}, while the sample actually contains 1.5×10^{17} cm^{-3} of dopant atoms, five times more. This is why resistivity alone cannot certify the purity of a wafer.

Mobility in practice

Several further effects matter in real material.

- **Minority-carrier mobility** differs from majority-carrier mobility at high doping. Minority electrons in heavily doped p-type Si move up to twice as fast as majority electrons at the same doping, because they are not screened in the same way.
- **Hall mobility** $\mu_H = r_H \mu$, measured by the Hall effect, exceeds the drift mobility by the Hall factor $r_H \approx 1.1$–1.2 for phonon scattering (Chapter 52).
- **Channel mobility** at a semiconductor–dielectric interface is lower than bulk mobility because of surface-roughness and Coulomb scattering by interface charge. It is about 200–400 cm²/(V s) for electrons in Si MOSFETs and only 20–50 cm²/(V s) in conventional 4H-SiC MOSFETs, where interface traps dominate (Chapter 34).
- **Strain** changes the effective masses and valley occupations. Biaxial tensile strain in Si raises electron mobility by up to about 80 %, and uniaxial compressive strain doubles hole mobility (Chapter 34).
- **Two-dimensional electron gases** at modulation-doped heterointerfaces separate carriers from their donors. The AlGaN/GaN 2DEG reaches 1500–2000 cm²/(V s) at room temperature, higher than doped bulk GaN (Chapter 33).

32.6 Drift and diffusion

Two ways to make a current

Carriers move for two reasons. An electric field $\mathcal{E}$ makes them **drift**. A concentration gradient makes them **diffuse**, for the same statistical reason that ink spreads in water: more carriers cross a plane from the crowded side than from the sparse side. The current densities are

$$J_n = q n \mu_n \mathcal{E} + q D_n \frac{dn}{dx}, \qquad J_p = q p \mu_p \mathcal{E} - q D_p \frac{dp}{dx}$$

where D_n and D_p are the **diffusion coefficients** (cm²/s). The sign difference arises because holes carry positive charge: electrons diffusing in $-x$ carry current in $+x$. The diffusion terms have the form of Fick's first law (Chapter 8).

> **Status of the principle: drift–diffusion current equations**
> *Approximate.* They follow from the Boltzmann transport equation in the limit of small departures from equilibrium (fields well below velocity saturation, gradients slow on the scale of the mean free path, about 10–50 nm). They underpin nearly all device simulation. They fail in very short channels (velocity overshoot, ballistic transport) and at high fields (Section 32.10), where hydrodynamic or Monte Carlo models are used.

The Einstein relation

Mobility and diffusivity are not independent. Both describe random collisions of carriers with the lattice. Consider an n-type sample with a doping gradient in equilibrium (Section 32.4). No net current flows, so

$$0 = q n \mu_n \mathcal{E} + q D_n \frac{dn}{dx}$$

In equilibrium the Fermi level is flat, so $n = N_c \exp[-(E_c - E_F)/k_B T]$ varies only because E_c varies with position: $dn/dx = -(n/k_B T)\, dE_c/dx$. The field is related to the band slope by $q\mathcal{E} = dE_c/dx$ (bands bend in the direction of the electrostatic potential energy). Substituting gives $q n \mu_n \mathcal{E} = q D_n (n/k_B T)\, q\mathcal{E}$, so

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{k_B T}{q}$$

At 300 K, $k_B T/q = 25.85$ mV. Electrons in lightly doped silicon with $\mu_n = 1400$ cm²/(V s) have $D_n = 36$ cm²/s; holes with $\mu_p = 450$ cm²/(V s) have $D_p = 12$ cm²/s.

> **Status of the principle: Einstein relation**
> *Derived.* It follows from requiring zero current in equilibrium with Boltzmann statistics. It is a special case of the fluctuation–dissipation theorem, which links any transport coefficient to the equilibrium fluctuations behind it. In degenerate material D/μ exceeds $k_B T/q$ (it becomes $(1/q)\, n/(dn/dE_F)$), and for hot carriers far from equilibrium the relation breaks down.

Quasi-Fermi levels

Out of equilibrium, under illumination or injection, the electron and hole populations each reach thermal equilibrium among themselves in picoseconds, but not with each other. Each is described by its own **quasi-Fermi level**:

$$n = n_i\, e^{(E_{Fn} - E_i)/k_B T}, \qquad p = n_i\, e^{(E_i - E_{Fp})/k_B T}, \qquad np = n_i^2\, e^{(E_{Fn} - E_{Fp})/k_B T}$$

Combining these with the drift–diffusion equations gives a compact result: $J_n = n \mu_n\, dE_{Fn}/dx$ and $J_p = p \mu_p\, dE_{Fp}/dx$. **The gradient of the quasi-Fermi level drives the current**, whether it arises from a field or a concentration gradient. The splitting $E_{Fn} - E_{Fp}$ under illumination is the maximum voltage a solar absorber can deliver, the "implied open-circuit voltage" used to grade solar wafers before they are made into cells.

Continuity and diffusion length

Carriers are conserved except where they are generated or recombine. For minority electrons in p-type material:

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G - R$$

In a field-free region with a small excess Δn that recombines with lifetime τ_n (Section 32.8), $R - G_{\text{th}} = \Delta n/\tau_n$, and in steady state

$$D_n \frac{d^2\Delta n}{dx^2} = \frac{\Delta n}{\tau_n} \quad \Rightarrow \quad \Delta n(x) = \Delta n(0)\, e^{-x/L_n}, \qquad L_n = \sqrt{D_n \tau_n}$$

The **diffusion length** L_n is the average distance a minority carrier diffuses before it recombines. It combines a transport property (D) and a recombination property (τ), and it is the key quality parameter for any device that relies on collecting or injecting minority carriers.

Worked example — what twenty parts per trillion of iron does. Return to the solar wafer of the opening, p-type with $N_A \approx 1.5 \times 10^{16}$ cm^{-3} (1 Ω cm). Minority electrons have $\mu_n \approx 1200$ cm^2/(V s), so $D_n = 1200 \times 0.02585 = 31$ cm^2/s. In clean material with $\tau_n = 1$ ms, $L_n = \sqrt{31 \times 10^{-3}} = 0.176$ cm $= 1.8$ mm, more than ten times the 160 µm wafer thickness, so almost every photogenerated electron reaches the junction. Interstitial iron is a donor-like deep level at $E_v + 0.38$ eV with an electron capture cross-section of about 5×10^{-14} cm^2. With $N_t = 10^{12}$ cm^{-3} ($10^{12}/5 \times 10^{22} = 2 \times 10^{-11}$, or 20 parts per trillion) and $v_{\text{th}} \approx 10^7$ cm/s, the SRH lifetime of Section 32.7 is $\tau_n = 1/(\sigma_n v_{\text{th}} N_t) = 1/(5 \times 10^{-14} \times 10^7 \times 10^{12}) = 2\ \mu$s. Then $L_n = \sqrt{31 \times 2 \times 10^{-6}} = 7.9 \times 10^{-3}$ cm $= 79$ µm, now *shorter* than the wafer. Carriers generated deep in the wafer by red and infrared light recombine before they are collected. The resistivity is unchanged, since 10^{12} donors in 1.5×10^{16} acceptors is invisible, but the diffusion length has fallen by a factor of 22.

Dielectric relaxation

How fast does a majority-carrier imbalance disappear? A local excess of charge ρ in a material of conductivity σ decays as e^{-t/τ_d} with the **dielectric relaxation time** $\tau_d = \varepsilon/\sigma$. For n-type Si at 10^{16} cm^{-3} ($\rho = 0.53$ Ω cm), $\tau_d = 1.04 \times 10^{-12} \times 0.53 = 0.55$ ps. Majority carriers neutralize any charge imbalance almost instantly. This is why excess minority carriers are always accompanied by an equal excess of majority carriers (quasi-neutrality), and why only the minority-carrier lifetime governs the decay of the excess.

32.7 Recombination and generation

The driving force

In equilibrium, carriers are generated thermally and recombine at equal rates, so that $np = n_i^2$. Whenever $np > n_i^2$ (excess carriers from light or injection), net **recombination** occurs. Whenever $np < n_i^2$ (in a reverse-biased depletion region, where carriers have been swept away), net **generation** occurs. All three mechanisms below have rates proportional to $np - n_i^2$ in some form. Recombination releases the energy $\approx E_g$ in one of three ways: as a photon, to a defect and then to lattice vibrations, or to a third carrier.

Radiative (band-to-band) recombination

An electron in the conduction band falls directly into an empty valence-band state and emits a photon of energy close to E_g. The rate is proportional to the number of electrons and the number of holes available:

$$U_{\text{rad}} = B\left(np - n_i^2\right)$$

The coefficient B is large in direct-gap materials and small in indirect ones (Chapter 31). For low-level injection in p-type material ($\Delta n \ll p_0 = N_A$), $U_{\text{rad}} \approx B N_A \Delta n$, giving a radiative lifetime

$$\tau_{\text{rad}} = \frac{1}{B N_A}$$

Shockley–Read–Hall (trap-assisted) recombination

An electron is first captured by a localized state in the gap, a **trap** or **recombination centre** created by an impurity or defect, and then a hole is captured by the same trap, completing the recombination. The energy is released as phonons in two smaller

steps, which is far more probable than emitting one large phonon. Shockley, Read and Hall balanced the four capture and emission processes in steady state and obtained

$$U_{\mathrm{SRH}} = \frac{np - n_i^2}{\tau_{p0}\,(n + n_1) + \tau_{n0}\,(p + p_1)}$$

with

$$\tau_{n0} = \frac{1}{\sigma_n v_{\mathrm{th}} N_t}, \qquad \tau_{p0} = \frac{1}{\sigma_p v_{\mathrm{th}} N_t}, \qquad n_1 = n_i e^{(E_t - E_i)/k_B T}, \qquad p_1 = n_i e^{(E_i - E_t)/k_B T}$$

where N_t is the trap density, E_t its energy level, σ_n and σ_p its capture cross-sections for electrons and holes, and $v_{\mathrm{th}} \approx 10^7$ cm/s the thermal velocity. The quantities n_1 and p_1 are the electron and hole densities the material would have if E_F coincided with E_t; they describe thermal re-emission from the trap.

> **Status of the principle: Shockley–Read–Hall recombination**
> *Derived.* It follows from detailed balance of capture and emission at a single-level trap, given capture cross-sections that must be measured. It assumes non-interacting, single-level traps whose density is small compared with the carrier densities. Multi-level centres (Au, Pt), trap clusters at dislocations, and high trap densities need extensions.

Three conclusions follow from the formula, and all three are central to materials engineering.

1. **Mid-gap traps are the most effective.** If E_t lies near a band edge, n_1 or p_1 is large and the trap re-emits its captured carrier before it can capture the opposite type. The denominator is smallest when $E_t \approx E_i$. Shallow dopants, with levels within 0.05 eV of a band edge, are not recombination centres. Gold, iron and vacancy complexes, with levels deep in the gap, are.
2. **In low-level injection only the minority-carrier capture matters.** In p-type material, $U_{\mathrm{SRH}} \approx \Delta n / \tau_{n0}$. The lifetime is set by how fast traps capture minority carriers, $\tau_{n0} = 1/(\sigma_n v_{\mathrm{th}} N_t)$.
3. **The lifetime is inversely proportional to the trap density.** With $\sigma = 10^{-15}$ cm^2, $v_{\mathrm{th}} = 10^7$ cm/s and $N_t = 10^{12}$ cm^{-3}, $\tau = 100$ μs. Every tenfold increase in contamination cuts the lifetime tenfold.

Auger recombination

The energy released by an electron–hole recombination can be handed to a third carrier, an electron or a hole, which is kicked high into its band and then relaxes by emitting phonons. Because three carriers are involved, the rate grows as the cube of carrier density:

$$U_{\mathrm{Aug}} = (C_n n + C_p p)\left(np - n_i^2\right)$$

In n-type material with $n \approx N_D$, $\tau_{\mathrm{Aug}} = 1/(C_n N_D^2)$. Auger recombination is negligible at moderate doping but dominates in heavily doped emitters, under high injection in power devices and concentrator solar cells, and in narrow-gap materials (infrared lasers and detectors), where C is large.

Coefficients

Material	B (cm³/s)	C_n (cm⁶/s)	C_p (cm⁶/s)	Comment
Si	$\approx 1 \times 10^{-14}$	2.8×10^{-31}	9.9×10^{-32}	Indirect; SRH or Auger dominates
Ge	$\approx 6 \times 10^{-14}$	$\sim 10^{-31}$	$\sim 10^{-31}$	
GaAs	$(1\text{–}7) \times 10^{-10}$, typ. 2×10^{-10}	$\sim 10^{-30}$	$\sim 10^{-30}$	Radiative usually dominates
GaN, InGaN	$\sim (1\text{–}2) \times 10^{-11}$	$\sim 10^{-31}$–10^{-30}	—	Auger implicated in LED "droop"
InGaAsP (1.55 µm)	$\sim 10^{-10}$	$\sim 10^{-29}$–10^{-28}	—	Auger limits laser temperature stability

Worked example — which mechanism wins? Compare the three lifetimes at two doping levels.

Silicon, n-type, 10^{17} cm^{-3}. Radiative: $\tau = 1/(10^{-14} \times 10^{17}) = 1$ ms. Auger: $\tau = 1/(2.8 \times 10^{-31} \times 10^{34}) = 0.36$ ms. SRH in good float-zone material: about 1 ms. All three are comparable, giving an overall lifetime of about 0.2 ms.

Silicon, n-type, 10^{19} cm^{-3}. Radiative: $1/(10^{-14} \times 10^{19}) = 10$ μs. Auger: $1/(2.8 \times 10^{-31} \times 10^{38}) = 36$ ns. Auger dominates by a factor of nearly 300. However clean the material, a heavily doped silicon emitter cannot have a lifetime above about 36 ns.

GaAs, p-type, 10^{18} cm^{-3}. Radiative: $\tau = 1/(2 \times 10^{-10} \times 10^{18}) = 5$ ns. A typical SRH lifetime in good epitaxial GaAs is tens of ns or more, so radiative recombination wins, and most of the recombination energy leaves as light. This is the materials basis of the GaAs LED.

Surface recombination

A surface is a two-dimensional sheet of defects: dangling bonds, adsorbates and interface traps (Chapter 34). It recombines carriers at a rate per unit area $U_s = S\,\Delta n_s$, where S is the **surface recombination velocity**. Bare, etched silicon has $S \sim 10^4$–10^5 cm/s. A good thermal oxide or an Al_2O_3 or SiN_x passivation layer reduces it to 1–10 cm/s. GaAs surfaces are notoriously poor ($S \sim 10^6$ cm/s) and hard to passivate, one reason why GaAs never developed a native-oxide MOS technology.

Generation

Net generation is the same physics run in reverse. Under illumination, absorbed photons generate pairs at a rate $G(x) = \alpha \Phi_0 e^{-\alpha x}$, with α the absorption coefficient of Chapter 31. In a reverse-biased depletion region, n and p are both far below n_i, and the SRH expression gives net thermal generation $G \approx n_i/\tau_g$, with a generation lifetime τ_g of the order of $\tau_{n0} + \tau_{p0}$ for a mid-gap trap. This is the source of the generation leakage current in the opening of Chapter 31, proportional to n_i and to the trap density. Under very high fields, carriers can also be generated by **impact ionization** (Section 32.10) or by band-to-band tunnelling.

32.8 Minority-carrier lifetime

Combining the mechanisms

The mechanisms act in parallel, so their rates add, and the **bulk lifetime** is

$$\frac{1}{\tau_b} = \frac{1}{\tau_{\mathrm{SRH}}} + \frac{1}{\tau_{\mathrm{rad}}} + \frac{1}{\tau_{\mathrm{Aug}}}$$

Figure 32.4 shows how they combine for n-type silicon with an assumed SRH lifetime of 1 ms. Below about 10^{16} cm^{-3}, SRH recombination dominates and the lifetime measures cleanliness. Above about 10^{17} cm^{-3} Auger takes over (the Auger and 1 ms SRH rates are equal at $n = \sqrt{10^3/2.8 \times 10^{-31}} = 6 \times 10^{16}$ cm^{-3}), and the lifetime falls as N^{-2}, an intrinsic limit that no purification can raise. Radiative recombination is never dominant in silicon. In GaAs, by contrast, the radiative line would lie four orders of magnitude lower and would dominate over most of the doping range.

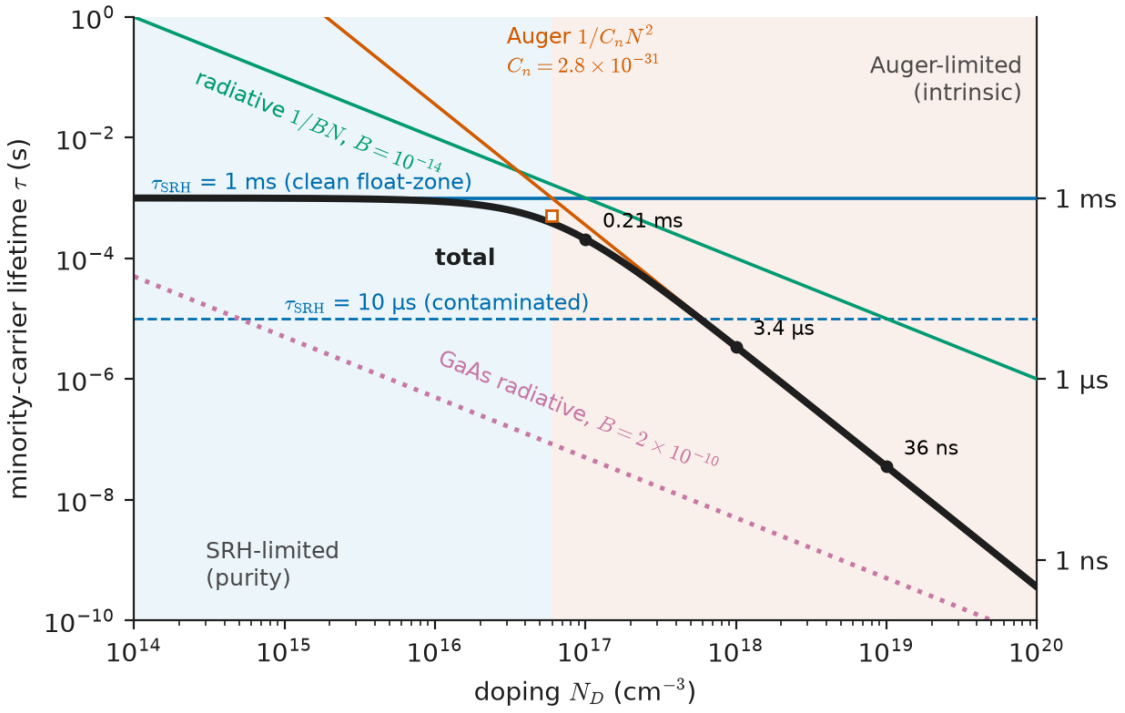


Figure 32.4 — Minority-carrier lifetime of n-type silicon versus doping, showing the SRH, radiative and Auger contributions and their combination; a GaAs radiative line is added for comparison.

Representative lifetimes

Material and condition	Typical minority-carrier lifetime
Float-zone Si, lightly doped, clean	1–30 ms
Czochralski Si solar wafers (n-type, gettered)	1–10 ms
Czochralski Si, p-type B-doped, after light-induced degradation	10–100 µs
Si with intentional Au or Pt (fast-recovery diodes)	10 ns–1 µs
Heavily doped Si emitter (10^{19}–10^{20} cm^{-3})	0.4–40 ns (Auger)
Epitaxial GaAs, 10^{17}–10^{18} cm^{-3}	1–50 ns (radiative)
4H-SiC epilayer, as grown	0.5–2 µs
4H-SiC epilayer, after $Z_{1/2}$ reduction	5–30 µs
GaN	0.1–10 ns

Lifetime is a design variable

Long lifetime is not always wanted. A bipolar power device or solar cell needs minority carriers to survive long enough to cross thick regions, so it needs long lifetime. But the stored carriers in a power diode must be removed before it can block voltage again, and long lifetime means slow switching. Power-device makers therefore **reduce** the lifetime on purpose, by diffusing gold or platinum or by irradiating with electrons or protons, which create vacancy-related deep levels (Section 32.11). The penalty is higher generation leakage, because the same centres that speed recombination also speed generation. The material trade-off is fixed: shorter switching time costs leakage and forward voltage.

Effective lifetime and surfaces

A measured lifetime is always an **effective** lifetime that includes the surfaces. For a wafer of thickness W with equal, moderate recombination velocity S on both faces,

$$\frac{1}{\tau_{\text{eff}}} \approx \frac{1}{\tau_b} + \frac{2S}{W}$$

> **Status of the principle: effective-lifetime approximation**
> *Approximate.* It assumes a uniform excess-carrier profile across the wafer, valid when $SW/D \ll 1$. For poorly passivated surfaces the correct limit is diffusion-limited, $1/\tau_s \approx \pi^2 D/W^2$.

For a 160 µm wafer with $S = 10$ cm/s, $2S/W = 1250\ \text{s}^{-1}$, a surface lifetime of 0.8 ms. With $S = 1000$ cm/s it falls to 8 µs. This is why lifetime measurements (by photoconductance decay or microwave-detected photoconductance, Chapter 52) are made only after the surfaces have been passivated. Without passivation the measurement reports the surfaces, not the bulk.

32.9 Temperature dependence

Every quantity in this chapter depends on temperature, and the dependences differ in sign. That is why semiconductor devices have temperature coefficients that change sign, and why their operating windows are bounded on both sides.

Carrier concentration: three regimes

For an n-type sample with fixed N_D, the electron concentration follows the curve of Figure 32.5.

1. **Freeze-out** (low T): donors are not fully ionized. Solving the neutrality quadratic at low temperature gives $n \approx \sqrt{N_c N_D / g_D}\, e^{-(E_c - E_d)/2k_B T}$, so the slope of $\ln n$ versus $1/T$ gives half the ionization energy (or the full energy in a compensated sample).
2. **Extrinsic** (saturation) regime: all donors are ionized and $n \approx N_D$, independent of temperature. For Si at 10^{15}–10^{17} cm^{-3} this spans roughly 100–450 K. This is the regime devices are designed for.
3. **Intrinsic** regime (high T): n_i exceeds N_D and $n \approx n_i$, rising with the slope $E_g/2$ (Chapter 31).

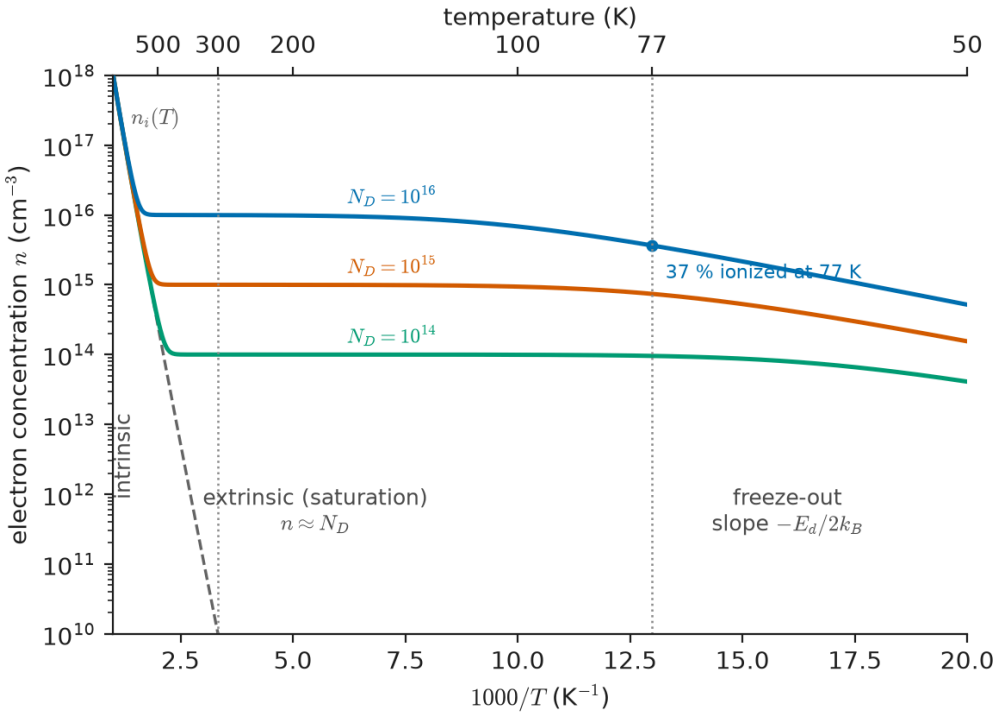


Figure 32.5 — Electron concentration of n-type silicon versus inverse temperature, showing freeze-out, extrinsic and intrinsic regimes for three donor concentrations, with the intrinsic concentration for reference.

The extrinsic window is narrow at both ends for deep dopants and narrow-gap materials, and wide for shallow dopants in wide-gap materials, except that wide-gap materials often have deep dopants, so their window opens only above room temperature.

Mobility

Lattice scattering grows with temperature, so in lightly doped material mobility falls. Empirically, for silicon,

$$\mu_L(T) = \mu_L(300\ \mathrm{K})\left(\frac{T}{300\ \mathrm{K}}\right)^{-\gamma}, \qquad \gamma \approx 2.4\ (\text{electrons}),\ 2.2\ (\text{holes})$$

steeper than the ideal acoustic-phonon $T^{-3/2}$ because optical phonons and intervalley scattering add to it. In heavily doped material, ionized-impurity scattering dominates and the mobility depends only weakly on temperature, and may even rise.

> **Status of the principle: power-law temperature dependence of lattice mobility**
> *Empirical.* The exponents are fitted to data between about 200 and 500 K. Theory gives $-3/2$ for pure acoustic-phonon scattering, and the measured excess reflects other phonon modes.

Worked example — temperature coefficient of a diffused resistor. A lightly doped n-type silicon resistor ($N_D = 10^{15}$ cm^{-3}) operates from −40 °C to +125 °C. The carrier density is constant in the extrinsic regime ($n_i = 5 \times 10^{12}$ cm^{-3} at 125 °C, still 0.5 % of N_D). The resistance therefore follows $1/\mu$. Between 25 °C and 125 °C: $R(125)/R(25) = (398.15/298.15)^{2.42} = 2.01$. The resistance doubles, an average temperature coefficient of about $+1.0 \times 10^{-2}$ K^{-1}, roughly 2.5 times that of copper. At −40 °C, $R(-40)/R(25) = (233.15/298.15)^{2.42} = 0.55$. Integrated circuits that need stable resistors use heavily doped or polysilicon resistors, where impurity scattering flattens the temperature dependence, or thin-film metal alloys (Chapter 25).

Recombination and generation

The SRH lifetime of a mid-gap centre changes only modestly with temperature (through $v_{\mathrm{th}} \propto T^{1/2}$ and temperature-dependent cross-sections). The generation rate, proportional to n_i, grows exponentially. Lifetimes in silicon usually *increase* with temperature, which raises the stored charge and switching losses of bipolar power devices when they are hot. The radiative coefficient B falls with temperature (roughly as $T^{-3/2}$), and Auger coefficients change little. For lasers and LEDs, efficiency falls as temperature rises.

High-field quantities

The saturation velocity falls slowly with temperature (in Si, empirically $v_{\mathrm{sat}} = 2.4 \times 10^7/[1 + 0.8\exp(T/600\ \mathrm{K})]$ cm/s, giving 1.03×10^7 cm/s at 300 K and 0.94×10^7 cm/s at 400 K). Impact-ionization coefficients also fall with temperature, because carriers lose more energy to phonons before reaching the threshold, so avalanche breakdown voltage *rises* with temperature. This positive temperature coefficient is the signature that distinguishes avalanche from tunnelling (Zener) breakdown, whose voltage falls with temperature as the gap shrinks.

Summary table

Quantity	Typical temperature dependence near 300 K	Sign of change on heating	Engineering consequence
E_g	−0.25 meV/K (Si), −0.45 meV/K (GaAs)	Decreases	LED red shift; bandgap references
n_i	Doubles every ~9 K (Si)	Increases	Leakage; intrinsic temperature limit
Dopant ionization	Activated for deep levels	Increases	p-GaN, p-SiC conduct better hot
Lattice mobility	$T^{-2.2}$–$T^{-2.4}$ (Si)	Decreases	Positive TCR; on-resistance rises with T
τ_{SRH}	Weak, usually increases	Increases	Switching losses rise with T
v_{sat}	about −0.1 %/K	Decreases	RF power falls when hot
Impact ionization	Decreases	Breakdown voltage increases	Avalanche-rugged devices

The positive temperature coefficient of resistance from falling mobility is a gift to power engineering: it lets MOSFETs be paralleled, because a hotter device conducts less and current shares itself. It is a property of the material, not the circuit.

32.10 High-field transport

Velocity saturation

At low fields the drift velocity is $\mu\mathcal{E}$. As the field increases, carriers gain energy from the field faster than they can lose it to acoustic phonons, and they become **hot**. Once a carrier's energy reaches the optical-phonon energy $\hbar\omega_{\mathrm{op}}$ (63 meV in Si), it can emit an optical phonon and lose that energy in a single event. The velocity then stops rising and saturates at

$$v_{\mathrm{sat}} \approx \sqrt{\frac{8\hbar\omega_{\mathrm{op}}}{3\pi m^{*}}}$$

For Si with $\hbar\omega_{\mathrm{op}} = 63$ meV this gives about 1.0×10^{7} cm/s with the free-electron mass and 1.9×10^{7} cm/s with the conductivity mass $0.26\,m_0$. The measured value, 1.0×10^{7} cm/s, lies within this range. The estimate shows why materials with stiff, light-atom lattices and high optical-phonon energies (GaN, 92 meV; SiC, about 120 meV; diamond, 165 meV) have high saturation velocities.

> **Status of the principle: velocity-saturation estimate**
> *Approximate.* A simple energy-balance argument that assumes optical-phonon emission is instantaneous above threshold. It gives the right magnitude and the right material trends, but measured values are fitted for design.

For device simulation the measured velocity–field curve is fitted by the Caughey–Thomas form

$$v(\mathcal{E}) = \frac{\mu_0 \mathcal{E}}{\left[1 + \left(\mu_0 \mathcal{E}/v_{\mathrm{sat}}\right)^{\beta}\right]^{1/\beta}}$$

with $\beta \approx 2$ for electrons and $\beta \approx 1$ for holes in Si at 300 K.

> **Status of the principle: Caughey–Thomas velocity–field model**
> *Empirical.* A fitted interpolation between ohmic and saturated regimes for materials with a monotonic velocity–field curve (Si, SiC, holes in general). It cannot describe the peak and negative slope of GaAs, InP and GaN electrons.

Worked example — how far from ohmic is a short channel? For silicon electrons with $\mu_0 = 1400\ \mathrm{cm^2/(V\,s)}$, $v_{\mathrm{sat}} = 1.07 \times 10^{7}$ cm/s and $\beta = 2$, at $\mathcal{E} = 10$ kV/cm: $\mu_0\mathcal{E} = 1.4 \times 10^{7}$ cm/s, the ratio $\mu_0\mathcal{E}/v_{\mathrm{sat}} = 1.31$, so $v = 1.4 \times 10^{7}/\sqrt{1 + 1.71} = 8.5 \times 10^{6}$ cm/s, 39 % below the ohmic value. At 30 kV/cm, $v = 1.04 \times 10^{7}$ cm/s, within 3 % of saturation. One volt across a 100 nm channel is a field of 100 kV/cm. In every modern short-channel transistor the carriers move at saturation velocity over most of the channel, so the current depends on v_{sat}, not on μ. This is why v_{sat} appears beside μ in every materials comparison of Chapter 33.

Intervalley transfer and negative differential mobility

In GaAs, the light Γ-valley electrons ($m^{*} = 0.067\,m_0$) are heated by fields of a few kV/cm until they reach the L valleys 0.29 eV higher (Chapter 31), where the mass is several times larger and the mobility much lower. As the field increases, more electrons transfer and the average velocity *falls*. The velocity peaks at about 2×10^{7} cm/s near 3.5 kV/cm and then decreases toward 1×10^{7} cm/s. A region of **negative differential mobility** ($dv/d\mathcal{E} < 0$) is a material property that makes uniform current flow unstable: charge domains form and travel through the sample, producing microwave oscillations (the Gunn effect, used in microwave sources). InP shows the same effect with a higher peak velocity (about 2.5×10^{7} cm/s), and GaN does so at much higher fields (peak 2.5–3 $\times 10^{7}$ cm/s near 150–200 kV/cm) because its satellite valleys lie more than 1 eV higher. Figure 32.6 compares the curves.

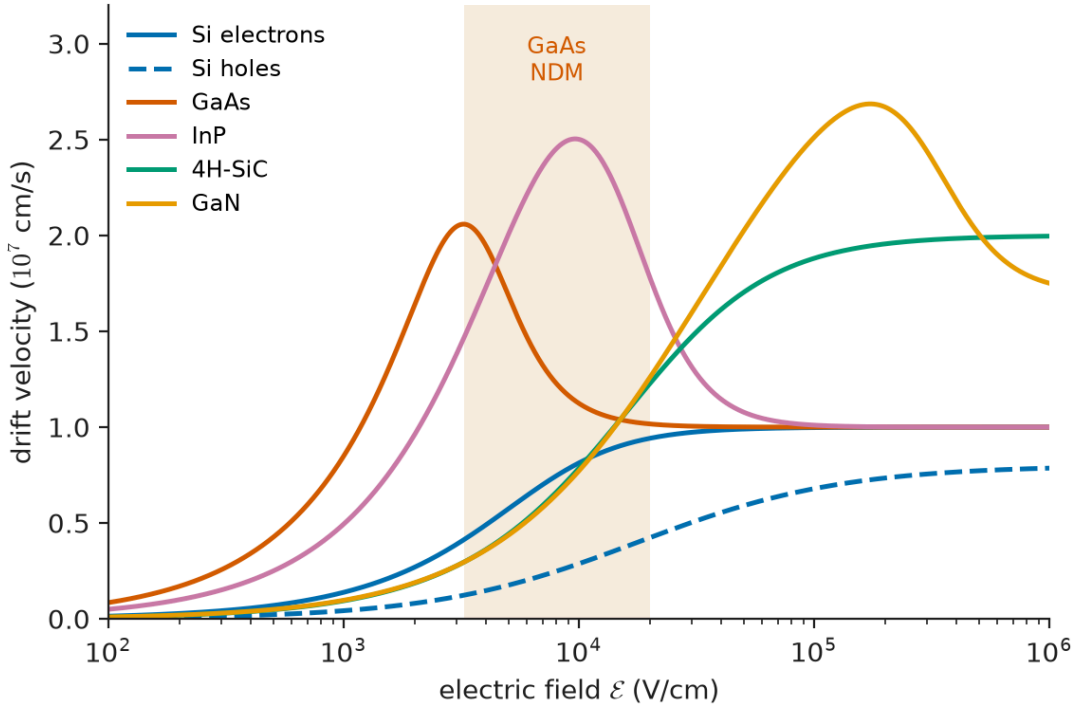


Figure 32.6 — Electron drift velocity versus electric field at 300 K for Si, GaAs, InP, 4H-SiC and GaN, and holes in Si, showing saturation and the negative differential mobility of materials with satellite valleys.

Velocity overshoot and ballistic transport

Energy relaxation takes about 0.1–1 ps, longer than momentum relaxation. If carriers cross a high-field region in less than that time, their velocity can exceed v_{sat} for a short distance (**velocity overshoot**). In channels shorter than the mean free path, a fraction of carriers cross without scattering at all (**ballistic transport**). Both effects lie outside drift–diffusion and are treated in device physics, but their size is set by the material: light masses and high optical-phonon energies favour them.

Impact ionization and the critical field

At still higher fields a hot carrier can gain more than about $1.5E_g$ between collisions. It can then break a bond on impact and create a new electron–hole pair. The new carriers are accelerated in turn, and the multiplication can run away into **avalanche breakdown**. The ionization rate per unit path length is described by Chynoweth's law:

$$\alpha(\mathcal{E}) = \alpha_\infty \exp\!\left(-\frac{b}{\mathcal{E}}\right)$$

For electrons in Si, $\alpha_\infty \approx 7.0\times10^5$ cm^{-1} and $b \approx 1.23\times10^6$ V/cm. At 300 kV/cm, $\alpha = 7.0\times10^5 e^{-4.1} = 1.2\times10^4$ cm^{-1}: an electron creates a pair every 0.9 μm. At 400 kV/cm, $\alpha = 3.2\times10^4$ cm^{-1}. The steep exponential is why breakdown is sharp. Avalanche occurs when the ionization integral across the high-field region reaches unity, and this defines a **critical field** $\mathcal{E}_c$, roughly 0.3 MV/cm in Si at moderate doping.

> **Status of the principle: Chynoweth ionization law**
> *Empirical.* A fit to measured multiplication data over a limited field range, with separate parameters for electrons and holes and for different field ranges. Its exponential form is motivated by the probability that a carrier travels far enough without collision to reach the threshold energy.

The critical field is not strictly a constant. It rises slowly with doping, because the high-field region becomes narrower. An often-used fit for Si is $\mathcal{E}_c \approx 4010\,N^{1/8}$ V/cm (N in cm^{-3}), giving 0.23 MV/cm at 10^{14} and 0.40 MV/cm at 10^{16} cm^{-3}. Across materials, the threshold energy scales with E_g, and an empirical correlation (Hudgins) captures the trend:

$$\mathcal{E}_c \approx 1.73\times10^5\,E_g^{2.5}\ \text{V/cm (indirect)}, \qquad \mathcal{E}_c \approx 2.38\times10^5\,E_g^{2}\ \text{V/cm (direct)}$$

with E_g in eV. This gives 0.23 MV/cm for Si, 0.48 MV/cm for GaAs, 2.7 MV/cm for GaN and 3.3 MV/cm for 4H-SiC, all within the range of measured values.

> **Status of the principle: critical field scaling with band gap**
> *Empirical.* A power-law fit across many semiconductors, useful for screening new materials. Individual materials deviate by tens of per cent, and the critical field also depends on doping, crystal orientation and temperature.

Because the drift-region resistance of a power device scales as $1/\mathcal{E}_c^3$ (Chapter 33), the step from Si to SiC, a factor of about ten in $\mathcal{E}_c$, is the single most important number in power electronics materials.

32.11 Defects and impurities in semiconductors

Shallow versus deep

Section 32.2 treated shallow, hydrogenic dopants. Many impurities and native defects instead produce **deep levels**, localized states far from the band edges whose binding is set by the short-range chemistry of the defect, not by the long-range Coulomb tail. Deep levels do little to the carrier concentration at the concentrations at which they occur, but they control lifetime, leakage, compensation, trapping and noise.

Host	Defect or impurity	Level(s) (eV)	Effect and use
Si	Au	$E_c - 0.55$ (acceptor), $E_v + 0.35$ (donor)	Classic lifetime killer; deliberate in fast diodes
Si	Pt	$E_c - 0.23$, $E_v + 0.36$	Lifetime control with less leakage than Au
Si	Fe interstitial	$E_v + 0.38$	Solar-cell contaminant; pairs with B (Fe–B, $E_c - 0.26$)
Si	Cu, Ni	Precipitates; various	Fast diffusers; form recombination-active precipitates
Si	Vacancy–oxygen (A centre)	$E_c - 0.17$	From electron irradiation; lifetime control
Si	Divacancy	$E_c - 0.23$, $E_c - 0.42$	Radiation damage; detector degradation
Si	Oxygen thermal donors	$E_c - 0.07$, $E_c - 0.15$	Formed at 350–500 °C in Cz-Si; shift resistivity
Si	Boron–oxygen complex	Deep	Light-induced degradation of p-type Cz solar cells
GaAs	EL2 (As antisite)	$E_c - 0.75$	Mid-gap donor; makes undoped GaAs semi-insulating
AlGaAs	DX centre (Si donor)	≈ 0.4 (thermal)	Persistent photoconductivity; threshold shifts
4H-SiC	$Z_{1/2}$ (carbon vacancy)	$E_c - 0.65$	Limits lifetime in SiC epilayers
4H-SiC	$\mathrm{EH}_{6/7}$	$E_c - 1.5$	Carbon-vacancy related; deep recombination
GaN	C on N site	$E_v + 0.9$ (approx.)	Semi-insulating buffers; current collapse
GaN	Fe	$E_c - 0.5$ to -0.6	Semi-insulating buffers

Three roles of deep levels

Recombination centres. A level near mid-gap with large capture cross-sections for both carriers is an efficient SRH centre (Section 32.7). Transition metals in Si (Fe, Au, Cu, Ni, Cr) are the main offenders. They come from furnace hardware, wafer handling, chemicals and stainless steel, and their diffusivities in Si at processing temperatures are so high (Cu and Ni cross a wafer in minutes at 1000 °C) that they cannot be kept out by masks.

Traps. A level near one band edge, with a large cross-section for only one carrier type, captures carriers and releases them later. Traps cause hysteresis, slow transients and noise: "current collapse" or dynamic on-resistance in GaN transistors, threshold

drift, and $1/f$ noise in MOS devices. Chapter 34 treats the interface version.

Compensation centres. Deep levels can pin the Fermi level near mid-gap and make a material **semi-insulating**. Undoped GaAs with a net shallow-acceptor excess is compensated by EL2 donors, which pins E_F near $E_c - 0.75$ eV and gives resistivities of 10^7–10^8 Ω cm. InP is made semi-insulating with Fe, and GaN buffers with C or Fe. These are the substrates and buffers on which RF transistors are isolated (Chapter 33).

Native defects and extended defects

Point defects such as vacancies, interstitials and antisites (Chapter 5) are present in equilibrium and are created in excess by implantation, irradiation and plasma processing. Implantation damage must be annealed out, both to activate dopants and to remove deep levels (Chapter 50). In SiC the carbon vacancy, which forms the $Z_{1/2}$ centre, is present at 10^{12}–10^{13} cm^{-3} in as-grown epilayers and limits the lifetime to around a microsecond. Thermal oxidation or carbon implantation, which inject carbon interstitials to fill the vacancies, raise it tenfold.

Dislocations (Chapter 5) carry rows of dangling bonds. They act as lines of recombination centres, as leakage paths and as fast diffusion pipes for metals. Silicon wafers are essentially dislocation-free. Heteroepitaxial GaN on Si, SiC or sapphire contains 10^8–10^{10} threading dislocations per cm², and devices work only because the GaN gap and strong carrier localization make each dislocation less harmful than it would be in GaAs. In 4H-SiC, **basal-plane dislocations** can expand into stacking faults when electron–hole pairs recombine at them, raising the forward voltage of bipolar devices over time ("bipolar degradation", Chapter 44). Converting them into harmless threading-edge dislocations during epitaxy is a major materials achievement.

Grain boundaries in polycrystalline films (Chapter 34) carry interface states that trap carriers, create potential barriers and reduce effective mobility and lifetime.

Gettering and passivation

Since contamination cannot be fully prevented, it is moved or neutralized.

- **Gettering** pulls metal impurities away from the device region into a sink: oxygen precipitates and dislocation loops deep in a Czochralski wafer (*intrinsic gettering*), a heavily phosphorus-doped layer or polysilicon layer on the back surface (*extrinsic gettering*), or the phosphorus emitter diffusion of a solar cell, which also getters iron.
- **Hydrogen passivation** binds atomic hydrogen to dangling bonds and to some impurities, removing their levels from the gap. It is essential for amorphous silicon (Chapter 33), for the Si/SiO_2 interface (the forming-gas anneal, Chapter 34) and for multicrystalline solar silicon. Hydrogen also passivates acceptors unintentionally: Mg in GaN grown by MOCVD is inactive until a post-growth anneal drives the hydrogen out, a discovery that made p-type GaN, and hence blue LEDs, possible.

Light elements in silicon

Czochralski silicon contains 10^{17}–10^{18} cm^{-3} of interstitial oxygen from the silica crucible (Chapter 50). It is electrically inactive as grown, but it strengthens the wafer against slip, forms gettering precipitates when heat-treated, creates **thermal donors** when held at 350–500 °C, and forms the boron–oxygen complexes responsible for light-induced degradation in p-type solar cells. Carbon (10^{15}–10^{17} cm^{-3}) modifies oxygen precipitation. These are not dopants and not contaminants in the usual sense, but they decide how a wafer behaves through processing. Float-zone silicon, with oxygen below 10^{16} cm^{-3}, is used where the highest lifetime and resistivity are needed: power devices and radiation detectors.

Measuring defects

Deep levels are identified and counted by **deep-level transient spectroscopy** (DLTS), which measures the thermal emission rate of trapped carriers as a function of temperature and gives E_t, σ and N_t. Lifetime mapping, photoluminescence and SIMS complete the toolset (Chapter 52). Defect generation during service, by radiation, hot carriers and high-field stress, is the subject of Chapter 44.

> **Status of the principle: "deep levels set lifetime, shallow levels set conductivity"**
> *Approximate.* A useful rule that follows from the SRH result (only levels near mid-gap recombine efficiently) and from ionization statistics (only shallow levels are fully ionized at room temperature). It fails at high deep-level concentrations, where compensation alters conductivity (semi-insulating GaAs), and in wide-gap materials, where the "shallow" dopants are themselves 0.2 eV deep.

What remains open

Open question	Where it is answered
How do the material families compare in band gap, mobility, critical field, saturation velocity and thermal conductivity, and how do figures of merit rank them?	Chapter 33
What happens to carriers at heterointerfaces, surfaces and dielectric interfaces: band offsets, 2DEGs, interface traps, channel mobility?	Chapter 34
How are recombination and absorption exploited in LEDs, lasers, photodetectors and solar absorbers?	Chapter 35
How does confinement change the density of states and transport in nanostructures and 2D materials?	Chapter 38
How are doping profiles produced (diffusion, implantation, activation anneals) and how is contamination controlled during manufacture?	Chapters 8 and 50
How are resistivity, Hall mobility, carrier density, lifetime and deep levels measured?	Chapter 52
How are new defects generated in service (hot carriers, radiation, bipolar degradation), and how do they cause failure?	Chapter 44
How do these properties enter p–n junction, BJT and MOSFET behaviour?	Electronics & Computing core

Next: Chapter 33 applies these properties to the real material families, from silicon and germanium through the III–V compounds, SiC, GaN and the ultra-wide-gap materials to amorphous, organic and two-dimensional semiconductors, and compares them for power, RF, sensing and optoelectronics.

Chapter 33 — Semiconductor Material Families

Part VII · Semiconductor and Photonic Materials

An electric-vehicle maker is redesigning its traction inverter for an 800 V battery. The existing design uses silicon IGBTs. A silicon carbide MOSFET of the same current rating costs about three times as much per die, yet the company switches, and the whole industry is doing the same. The drive-cycle losses fall by more than half, the cooling plate shrinks, and the extra range is worth more than the extra semiconductor cost. In the same year the company's radar supplier moves its power amplifiers to gallium nitride on silicon carbide, its lidar uses indium gallium arsenide detectors, its displays are driven by amorphous-oxide and polysilicon thin-film transistors, and the dashboard is lit by organic LEDs. Seven different semiconductors sit in one car, and each was chosen for reasons that trace back to the band structure and defect physics of Chapters 31 and 32. This chapter is the catalogue and the comparison: what each family offers, what limits it, and how an engineer ranks them for a given job.

What does each semiconductor family offer, and what holds it back?
How do band gap, mobility, critical field, saturation velocity and thermal conductivity combine into figures of merit for power and RF? Why does silicon still dominate, and where has it lost to other materials?

Concept	Engineering destination
Silicon and germanium	Logic, memory, analog, sensors, power to about 1 kV; SiGe RF and photonics
III–V compounds, alloys, lattice matching	Band-gap engineering for lasers, detectors, HEMTs and HBTs
GaAs and InP	RF front-ends, mm-wave and THz transistors, 850 nm to 1.55 μm photonics
Silicon carbide	650 V–3.3 kV power MOSFETs and diodes; high-temperature sensors
Gallium nitride	Fast power switches, RF power amplifiers, blue/green/UV LEDs and lasers
Ga_2O_3, diamond, AlN	Next-generation ultra-high-voltage power, extreme environments, heat spreading
Amorphous silicon and oxides	Large-area thin-film transistors, imagers, passivation layers
Organic semiconductors	OLED displays, flexible and printed electronics
Two-dimensional materials	Atomically thin channels, sensors, ultimate scaling
Baliga and Johnson figures of merit	First-pass material screening for power and RF

Roadmap. The elemental semiconductors (33.1) → III–V compounds and alloy engineering (33.2) → GaAs and InP (33.3) → silicon carbide (33.4) → gallium nitride (33.5) → the ultra-wide-gap materials (33.6) → disordered and molecular semiconductors: amorphous silicon (33.7) and organics (33.8) → atomically thin materials (33.9) → systematic comparison for power, RF, sensing and optoelectronics (33.10).

Boundary.
This chapter compares materials. How a MOSFET, HEMT, HBT, LED or laser built from them operates belongs in the Electronics & Computing core. Interfaces and heterojunctions are developed in Chapter 34, optoelectronic devices' material requirements in Chapter 35, and 2D materials and nanostructures in depth in Chapter 38. Crystal growth and wafer manufacture are in Chapter 50. All property values are typical room-temperature values for good-quality material. Where values vary with growth method, orientation or doping, ranges are given.

33.1 Silicon and germanium

Why silicon dominates

On band structure alone, silicon is unremarkable. Its gap is indirect, its electron mobility is a sixth of GaAs's, its critical field is a tenth of SiC's, and its thermal conductivity is a fifteenth of diamond's. It dominates semiconductor production, at over 95 % of devices by value, because of a combination of material properties that no other semiconductor matches.

1. **A near-perfect native oxide.** Thermal oxidation grows amorphous SiO_2, with a band gap of about 9 eV, a breakdown field above 10 MV/cm and an interface-trap density as low as about $10^{10}\ \mathrm{cm^{-2}\,eV^{-1}}$ after hydrogen annealing (Chapter 34). That oxide made the MOSFET and planar processing possible. The native oxides of Ge, GaAs and InP are unstable, water-soluble or electrically poor.
2. **Crystal perfection at scale.** Silicon is grown as dislocation-free single crystals 300 mm in diameter and two metres long, with purity of 9N–11N (Chapter 50). No other semiconductor comes close.
3. **Abundance and safety.** Silicon makes up 28 % of the Earth's crust by mass, and silicon and its oxide are non-toxic.
4. **A moderate, useful gap.** At 1.12 eV the gap is wide enough for low leakage up to about 150 °C (Chapter 31) and narrow enough for easy doping and low-voltage operation. It also matches the solar spectrum well (Chapter 40).
5. **Mechanical and thermal robustness.** Silicon is stronger than steel in flexure when defect-free, has a thermal conductivity of about 148 W/(m K), comparable to many engineering alloys and about five times that of alumina, and a low thermal expansion (2.6 ppm/K).
6. **Well-behaved dopants.** B, P, As and Sb are shallow, highly soluble and controllable, giving p- and n-type doping over 10^{13}–$10^{21}\ \mathrm{cm^{-3}}$.

Silicon's limits

The same properties set the boundaries of silicon's domain.

- **Indirect gap.** Radiative recombination is 10^4 times slower than in GaAs (Chapter 31). Silicon makes excellent detectors and solar cells but very poor light emitters. Silicon photonics uses Ge for detection and bonded III–V material for lasers.
- **Moderate critical field** (about 0.3 MV/cm). A silicon power device needs a thick, lightly doped drift layer, and its on-resistance rises steeply with voltage (Section 33.10). Above about 600 V, silicon power devices must be bipolar (IGBTs), which conduct well but switch slowly.
- **Temperature.** Intrinsic conduction and leakage limit junction temperatures to about 150–175 °C.
- **Saturation velocity** (about 10^7 cm/s) and mobility limit the highest-frequency analog circuits, though aggressive scaling has taken Si CMOS to transit frequencies of several hundred GHz.

Germanium

Germanium was the material of the first transistor (1947) and lost to silicon in the 1960s for three reasons: its oxide is water-soluble and unstable, its 0.66 eV gap gives large leakage currents, and its intrinsic temperature (about 390 K at $10^{15}\ \mathrm{cm^{-3}}$, Chapter 31) is too low for many applications. It has returned in roles that use its strengths.

- **Highest hole mobility of the common semiconductors** (about 1900 cm²/(V s), four times Si) and high electron mobility (3900 cm²/(V s)), from its light conductivity masses (Chapter 31). Ge and high-Ge SiGe channels are used in advanced p-type transistors.
- **Direct gap close to the indirect gap.** The Γ-valley gap of 0.80 eV gives strong absorption at 1.55 μm. Ge-on-Si photodetectors are the standard detector of silicon photonics.
- **Lattice match to GaAs** (5.658 Å versus 5.653 Å) makes Ge the substrate and bottom junction of high-efficiency multi-junction space solar cells.
- **Radiation detection.** High-purity Ge, with net impurity concentrations near $10^{10}\ \mathrm{cm^{-3}}$, cooled to 77 K, gives the best energy resolution of any gamma-ray detector.

Silicon–germanium alloys

Si and Ge are fully miscible, and $Si_{1-x}Ge_x$ alloys span lattice constants from 5.431 to 5.658 Å (a 4.2 % mismatch) and gaps from 1.12 to 0.66 eV. Thin SiGe layers grown on Si are compressively strained, which raises hole mobility and lowers the gap. These are used in SiGe heterojunction bipolar transistors (with f_T and $f_{\max}$ of 300–700 GHz for mm-wave radar and optical links) and in strained-channel CMOS. Strain and critical thickness are treated in Chapter 34.

Property (300 K)	Si	Ge	Units
Crystal structure	Diamond cubic	Diamond cubic	—
Lattice constant	5.431	5.658	Å
Density	2.33	5.32	g/cm³
Melting point	1414	938	°C
Band gap	1.12 (indirect)	0.66 (indirect; direct 0.80)	eV
Relative permittivity	11.7	16.0	—
Electron / hole mobility	1400 / 450	3900 / 1900	cm²/(V s)
Intrinsic concentration	1.0×10^{10}	2×10^{13}	cm⁻³
Critical field (about 10^{16} cm⁻³)	0.3	0.1	MV/cm
Electron saturation velocity	1.0	0.6	10^7 cm/s
Thermal conductivity	148	60	W/(m K)
Thermal expansion	2.6	5.9	ppm/K
Largest production wafer	300 mm	200 mm	—

33.2 III–V compound semiconductors

Structure and bonding

Combining a group-III element (Al, Ga, In) with a group-V element (N, P, As, Sb) gives an average of four valence electrons per atom, as in silicon, and the same tetrahedral bonding. The arsenides, phosphides and antimonides crystallize in the **zincblende** structure, the diamond structure with the two fcc sublattices occupied by different atoms (Chapter 2). The nitrides crystallize in the hexagonal **wurtzite** structure (Section 33.5). The bond is partly ionic, which, as Chapter 31 showed, widens the gap relative to the isoelectronic group-IV element. It also makes the compounds piezoelectric (Chapter 27), gives polar-optical-phonon scattering (Chapter 32), and makes {110} the natural cleavage planes, which form the mirrors of edge-emitting lasers.

A continuous menu of band gaps

The real power of the III–V family is alloying. The binaries form solid solutions on each sublattice: $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$, $\mathrm{In}_x\mathrm{Ga}_{1-x}\mathrm{As}$, $\mathrm{GaAs}_{1-y}\mathrm{P}_y$, and quaternaries such as $\mathrm{In}_{1-x}\mathrm{Ga}_x\mathrm{As}_y\mathrm{P}_{1-y}$ and $(\mathrm{Al}_x\mathrm{Ga}_{1-x})_{0.5}\mathrm{In}_{0.5}\mathrm{P}$. Their band gaps interpolate with the bowing law of Chapter 31, and their lattice constants follow **Vegard's law**:

$$a(x) = x\,a_{AC} + (1-x)\,a_{BC}$$

> **Status of the principle: Vegard's law**
> *Empirical.* A linear interpolation of the lattice constant with composition. It holds to within about 0.1 % for most III–V alloys, whose constituents have similar bonding, and fails more seriously for highly mismatched systems such as GaN–InN and for alloys with strong ordering.

Binary	Lattice constant (Å)	E_g at 300 K (eV)	Gap type
AlP	5.464	2.45	indirect
GaP	5.451	2.26	indirect
AlAs	5.661	2.16	indirect
GaAs	5.653	1.42	direct
InP	5.869	1.34	direct
AlSb	6.136	1.6	indirect
GaSb	6.096	0.73	direct
InAs	6.058	0.35	direct
InSb	6.479	0.17	direct

Figure 33.1 plots band gap against lattice constant. It is the map on which every heterostructure is designed.

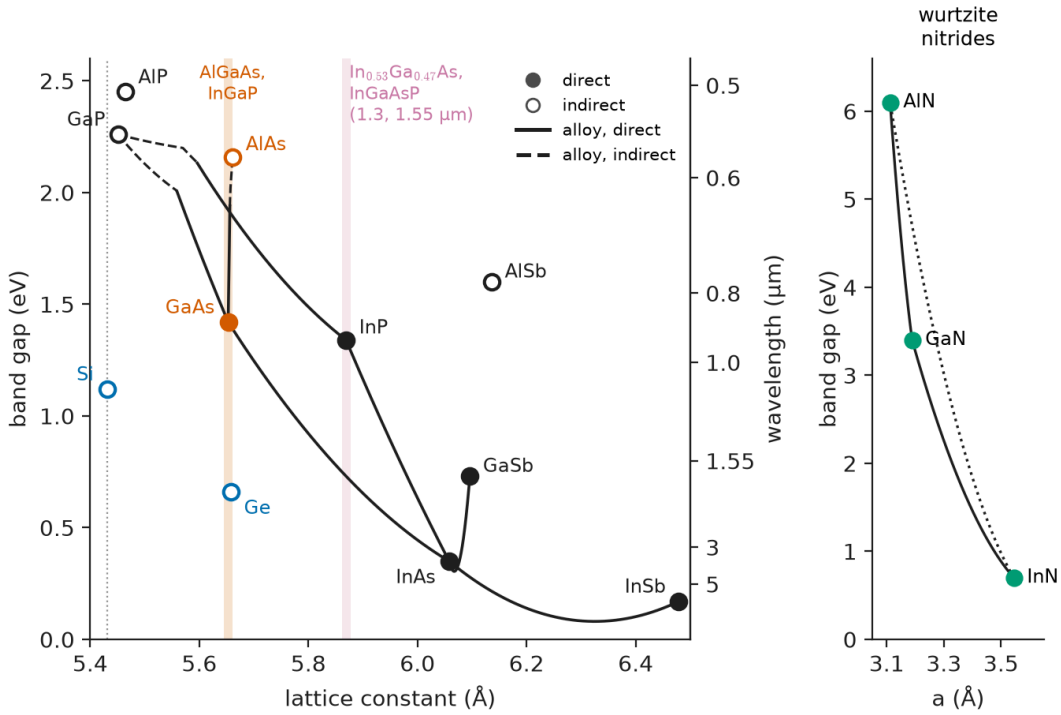


Figure 33.1 — Band gap versus lattice constant for the group-IV and III–V semiconductors, with alloy lines; vertical bands mark the lattice constants of GaAs, InP and Si substrates.

Lattice matching

A layer grown epitaxially on a substrate must either match its lattice constant or accommodate the mismatch by elastic strain (if thin) or misfit dislocations (if thick, Chapter 34). Dislocations are recombination centres and leakage paths (Chapter 32). High-quality optoelectronic and high-speed layers are therefore designed along the vertical lines of Figure 33.1.

- **On GaAs** (5.653 Å): AlAs differs by only 0.14 %, so the entire $Al_xGa_{1-x}As$ range is nearly lattice-matched. This accident of nature made GaAs/AlGaAs the first heterostructure system, and gave the double-heterostructure laser and the HEMT. $Ga_{0.51}In_{0.49}P$ and AlGaInP are also matched (red LEDs, HBT emitters).
- **On InP** (5.869 Å): $In_{0.53}Ga_{0.47}As$, InAlAs and InGaAsP of chosen composition, covering 0.75–1.34 eV, which includes the 1.3 µm and 1.55 µm optical-fibre windows.

Worked example — designing the 1.55 µm detector alloy. Find the composition of $In_xGa_{1-x}As$ that is lattice-matched to InP, and its band gap. Vegard's law with $a_{InAs} = 6.0583$ Å and $a_{GaAs} = 5.6533$ Å requires $x\,6.0583 + (1-x)\,5.6533 = 5.8687$, so

$x = (5.8687 - 5.6533)/(6.0583 - 5.6533) = 0.2154/0.4050 = 0.532$. With a bowing parameter $b \approx 0.48$ eV, the gap is $E_g = 0.532 \times 0.354 + 0.468 \times 1.424 - 0.48 \times 0.532 \times 0.468 = 0.188 + 0.666 - 0.119 = 0.736$ eV. The cut-off wavelength is $1.24/0.736 = 1.68$ μm. $In_{0.53}Ga_{0.47}As$ absorbs strongly across both telecommunication windows (1.31 and 1.55 μm) while growing on InP without dislocations. This is why it is the standard material for fibre-optic photodiodes and for eye-safe lidar receivers. A composition error of 1 % in x produces a mismatch of 0.07 %, which is tolerable in layers a few micrometres thick.

Practical issues

- **Toxicity and handling.** Arsenic and phosphorus compounds, and the arsine and phosphine gases used in epitaxy, are highly toxic. Fabrication needs gas monitoring, abatement and waste handling (Chapter 59).
- **Brittleness and wafer size.** III–V wafers are mechanically weaker than Si. GaAs wafers reach 150 mm (200 mm in limited use), and InP 100–150 mm, and both cost far more per unit area than silicon.
- **No good native oxide.** Surfaces have high recombination velocities and high interface-trap densities, so III–V transistors use Schottky gates, heterojunction barriers or deposited high-κ dielectrics (Chapter 34).
- **Low thermal conductivity** of the arsenides and phosphides (Section 33.3) limits power density.

33.3 Gallium arsenide and indium phosphide

Gallium arsenide

GaAs is the workhorse III–V compound. Its electron mobility of about 8500 cm²/(V s) (six times Si) and its direct 1.42 eV gap follow from the strongly curved Γ valley ($m^* = 0.067\,m_0$, Chapter 31).

Semi-insulating substrates. Undoped GaAs grown by the liquid-encapsulated Czochralski or vertical-gradient-freeze methods is compensated by the EL2 mid-gap donor (Chapter 32), with resistivities of 10^7–10^8 Ω cm. Passive components and transmission lines can be built directly on the substrate with low RF loss, and devices are isolated by implantation. This is a decisive advantage over silicon, whose substrates conduct, for monolithic microwave integrated circuits (MMICs).

Heterostructure transistors. The AlGaAs/GaAs and InGaP/GaAs systems give pseudomorphic HEMTs (pHEMTs) with InGaAs channels for low-noise amplifiers and switches, and InGaP/GaAs HBTs that still dominate the power amplifiers of mobile handsets, where their high power-added efficiency at 1–6 GHz and a few volts of supply is hard to beat.

Optoelectronics. GaAs, AlGaAs and InGaAs give the 850–940 nm vertical-cavity lasers (VCSELs) of data-centre links, face recognition and automotive lidar, infrared LEDs, and the most efficient single-junction solar cells (about 29 %), used with InGaP and Ge in multi-junction space cells.

Limits. The thermal conductivity of 55 W/(m K), about a third of Si's, limits power density. The critical field (0.4 MV/cm) and saturation velocity (about 10^7 cm/s, after the transient peak of 2×10^7 cm/s, Chapter 32) are only modestly better than Si's. The surface is hard to passivate, and arsenic toxicity and wafer cost restrict GaAs to applications where its RF or optical advantages are decisive.

Indium phosphide

InP has a direct gap of 1.34 eV, electron mobility of about 5400 cm²/(V s), a higher peak velocity than GaAs (about 2.5×10^7 cm/s) because its L valleys lie about 0.6 eV above Γ, a slightly higher breakdown field (about 0.5 MV/cm) and better thermal conductivity (68 W/(m K)). Its importance comes from the alloys it supports.

- **The fastest transistors.** $In_{0.53}Ga_{0.47}As$ channels on InP have electron mobilities of 10 000–14 000 cm²/(V s). InP HEMTs and InP/InGaAs HBTs reach $f_{\max}$ above 1 THz and are used in sub-THz communications, radio astronomy receivers and the lowest-noise cryogenic amplifiers.
- **Fibre-optic photonics.** InGaAsP and InGaAlAs on InP emit and detect at 1.3 and 1.55 μm. Practically every long-haul and data-centre laser, modulator and receiver is InP-based, and InP photonic integrated circuits combine all of them on one chip.
- **Semi-insulating substrates** are made by Fe doping, which provides a deep acceptor that compensates residual donors.

InP is more brittle and more expensive than GaAs, and its wafers are smaller. The high Auger coefficient of narrow-gap InGaAsP (Chapter 32) makes 1.55 μm lasers more temperature-sensitive than GaAs lasers. They are usually temperature-stabilized.

Narrow-gap III–Vs

InAs (0.35 eV), InSb (0.17 eV) and their alloys have the lightest electron masses (0.023 and 0.014 m_0) and highest mobilities (up to about 40 000 and 77 000 cm²/(V s)) of any common semiconductors. Their high mobility makes them the material of choice for **Hall-effect magnetic sensors**, where the Hall voltage per unit current and field is proportional to mobility. Their small gaps make them mid-infrared detectors (3–5 µm), cooled to suppress the large n_i. Type-II InAs/GaSb superlattices are an engineered alternative to HgCdTe for infrared imaging (Chapter 35).

33.4 Silicon carbide

Structure and polytypes

SiC is a covalent compound of two group-IV elements with a very strong, short Si–C bond (0.189 nm). Each atom is tetrahedrally bonded, and the structure is a stack of close-packed Si–C bilayers. The bilayers can stack in many sequences, called **polytypes**, which have the same chemistry but different band structures:

Polytype	Stacking sequence	Hexagonality	E_g (eV)	Electron mobility (cm²/(V s))
3C (cubic, zincblende)	ABC	0 %	2.36	about 900
4H	ABCB	50 %	3.26	about 950 (⊥ c), 1150 (∥ c)
6H	ABCACB	33 %	3.02	about 400 (⊥ c), 60–100 (∥ c)

Figure 33.2 shows the stacking sequences.

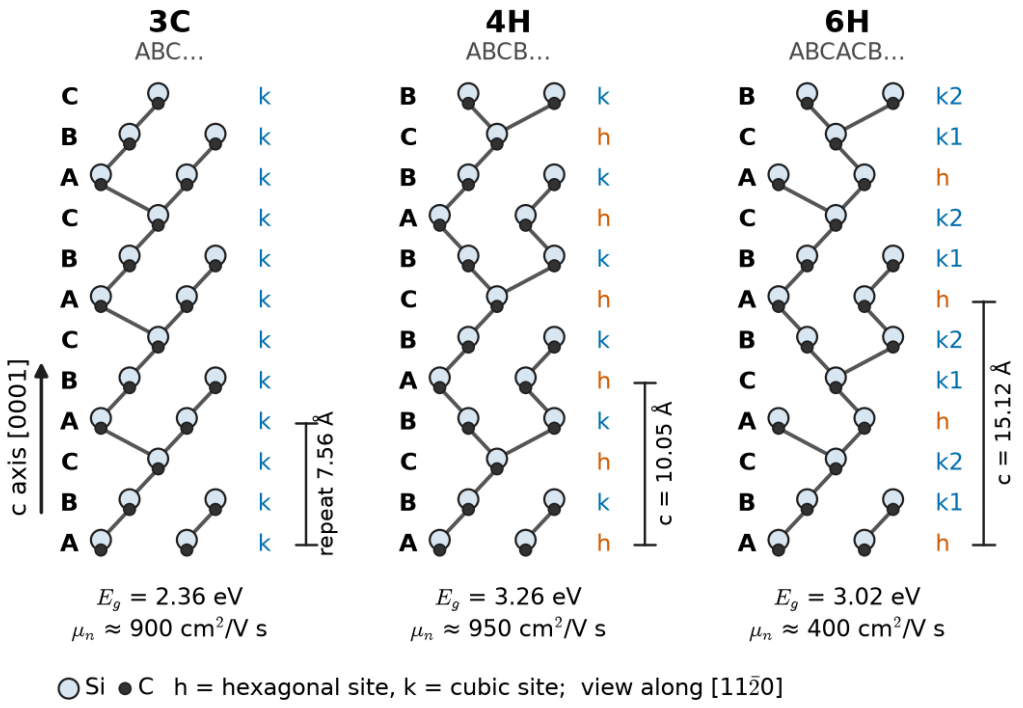


Figure 33.2 — Stacking sequences of the 3C, 4H and 6H polytypes of silicon carbide, viewed along the [11-20] direction, with the hexagonal and cubic lattice sites marked.

4H-SiC is the polytype of power electronics because it combines the widest gap with high and nearly isotropic electron mobility. 6H-SiC has a strongly anisotropic mobility that is poor along the c axis, the direction current flows in a vertical power device. 3C-SiC can be grown on Si but contains high defect densities and has a smaller gap.

Properties

4H-SiC has a band gap of 3.26 eV (indirect), a critical field of about 2–3 MV/cm (2.5 MV/cm is typical at the doping of a 1.2 kV drift layer), an electron saturation velocity of about 2×10^7 cm/s, and a thermal conductivity of 330–490 W/(m K) depending on orientation and doping (about 370 W/(m K) is typical), which is higher than copper's. Its intrinsic concentration at room temperature is about 10^{-8} cm^{-3} (Chapter 31), so leakage is set by defects and the material remains extrinsic far above any temperature a package can survive. It is extremely hard (Mohs 9.5), chemically inert, and resistant to radiation.

The defect story

SiC power devices became commercial in 2001 (Schottky diodes) and mainstream around 2018 (MOSFETs in electric vehicles) only after decades of defect reduction.

- **Growth.** SiC does not melt at atmospheric pressure. It sublimes above about 2700 °C. Boules are grown by physical vapour transport at about 2300 °C, slowly (0.2–0.5 mm/h) and with difficult thermal control (Chapter 50). This is the root of SiC's cost.
- **Micropipes**, hollow-core screw dislocations that destroy any device they cross, were once present at hundreds per cm². They are now below 0.1 cm⁻².
- **Basal-plane dislocations** in the epilayer expand into stacking faults when electron–hole pairs recombine on them, increasing the forward voltage of bipolar devices and of the body diode of MOSFETs (Chapter 32). Epitaxy on 4°-off-axis substrates converts more than 99 % of them into harmless threading-edge dislocations.
- **Point defects.** The $Z_{1/2}$ carbon-vacancy centre limits the bulk lifetime to about 1 µs (Chapter 32).
- **The oxide interface.** SiC is the only wide-gap semiconductor with a thermal native oxide, SiO_2, which made the SiC MOSFET possible. But the interface contains carbon-related defects, with trap densities of 10^{12}–10^{13} cm^{-2} eV^{-1} near the conduction band, so channel mobility is only 20–50 cm²/(V s) even after nitridation anneals. In low-voltage SiC MOSFETs the channel, not the drift layer, dominates the on-resistance (Chapter 34).

Doping

Nitrogen (donor, 61 and 125 meV for the two inequivalent sites) and aluminium (acceptor, about 0.2 eV) are the standard dopants. Dopant diffusion in SiC is negligibly slow, so all selective doping is by ion implantation, which needs implantation at elevated temperature and activation anneals at 1600–1800 °C with a carbon cap to protect the surface. Al acceptors are only partly ionized at room temperature (Chapter 32), so p-type SiC is resistive.

Applications

SiC Schottky diodes and MOSFETs dominate new designs from 650 V to 3.3 kV: EV traction inverters and on-board chargers, solar and storage inverters, industrial drives, rail traction. Devices to 10 kV and beyond are emerging for medium-voltage grid applications. Semi-insulating SiC (vanadium-doped or high-purity) is also the preferred substrate for GaN RF transistors, because of its thermal conductivity. Wafers of 150 mm are mainstream and 200 mm is ramping. Substrate cost and yield remain the main obstacles to replacing silicon IGBTs completely.

33.5 Gallium nitride

Structure and polarization

GaN crystallizes in the wurtzite structure ($a = 3.189$ Å, $c = 5.185$ Å). The Ga–N bond is short and strongly polar, and the wurtzite structure lacks inversion symmetry along the c axis. As a result, GaN and its alloys with AlN and InN carry a large **spontaneous polarization** along [0001], and any strain adds a **piezoelectric polarization** (Chapter 27). At an interface between two nitrides with different polarization, a fixed sheet of bound charge appears equal to the difference in polarization:

$$\sigma_{\text{pol}} = (P_{\text{sp}} + P_{\text{pz}})_{\text{AlGaN}} - P_{\text{sp,GaN}}$$

For Ga-face AlGaN grown on GaN, this charge is positive at the interface, of order $10^{13}\, q$ cm^{-2} for 25–30 % Al. Electrons, supplied mostly by donor-like surface states on the AlGaN, accumulate in the triangular well on the GaN side to neutralize it, forming a **two-dimensional electron gas** (2DEG) without any intentional doping (Figure 33.3).

Status of the principle: polarization-induced 2DEG
Derived. The sheet charge follows from the discontinuity of polarization (Gauss's law applied to bound charge), with polarization constants taken from calculation and experiment. The electron density that actually forms also depends on the barrier thickness and surface states, and is predicted only approximately. It is measured, typically $(0.8\text{–}1.2) \times 10^{13}$ cm^{-2}.

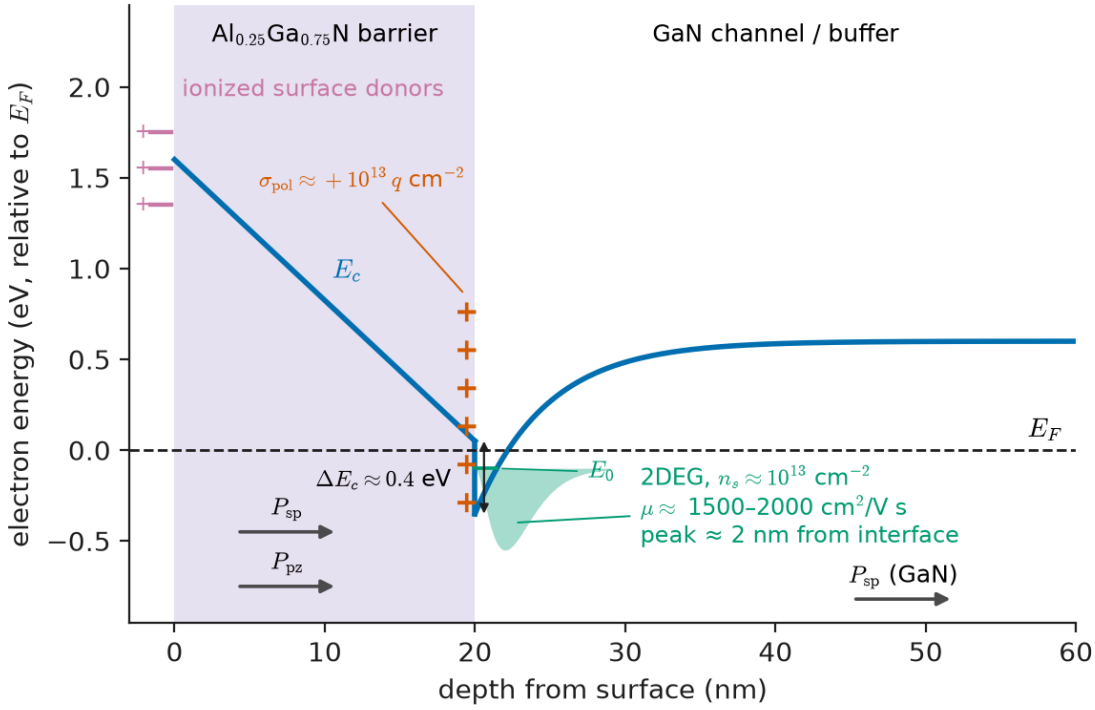


Figure 33.3 — Conduction-band diagram of an AlGaN/GaN heterostructure, showing the polarization sheet charge at the interface and the two-dimensional electron gas confined in the triangular well.

The 2DEG combines a high sheet density with a mobility of 1500–2000 cm²/(V s), because the electrons are separated from ionized donors (Chapter 32). With $n_s = 10^{13}$ cm^{-2} and $\mu = 1800$ cm^2/(V s) its sheet resistance is $1/(q n_s \mu) = 1/(1.602 \times 10^{-19} \times 10^{13} \times 1800) = 350\ \Omega$ per square, together with GaN's critical field of 3.3 MV/cm. The **high-electron-mobility transistor** (HEMT) built on it is the basis of GaN power and RF electronics. Because the channel is on without a gate bias, making normally-off power switches requires additional structures (p-GaN gates, recessed gates or cascodes), which is a materials problem with device consequences.

Substrates, dislocations and stress

Native GaN substrates exist but are small (50–100 mm) and expensive. Most GaN is grown heteroepitaxially:

Substrate	Lattice mismatch with GaN (in-plane)	Thermal conductivity (W/(m K))	Typical use
Sapphire (Al_2O_3)	about 16 % (after rotation)	35–42	LEDs (lowest cost)
4H/6H-SiC (semi-insulating)	3.5 %	about 370–490	RF power HEMTs
Si(111)	about 17 %	148	Power HEMTs, some LEDs, 200 mm CMOS-line compatible
GaN (bulk)	0	130–230	Lasers, vertical power devices

The mismatch to Si(111) follows from comparing the GaN a-axis constant with the Si surface-atom spacing $a_{Si}/\sqrt{2} = 3.840$ Å: $(3.189 - 3.840)/3.840 = -17$ %. Heteroepitaxial GaN contains 10^8–10^{10} threading dislocations per cm², densities that would make GaAs or Si unusable. GaN devices work because the large gap, and in InGaN strong carrier localization, make each dislocation less damaging. The thermal-expansion mismatch on Si (5.6 versus 2.6 ppm/K) puts the GaN in tension by about 0.3

% on cooling from growth temperature, enough to crack it unless compressive AlGaN or AlN/GaN superlattice buffers are inserted. The buffers are made semi-insulating with C or Fe (Chapter 32), and the traps they introduce cause the dynamic on-resistance ("current collapse") that GaN reliability work still addresses (Chapters 34 and 44).

Doping

Si and Ge are shallow donors. Mg is the only practical acceptor, with an ionization energy of 160–200 meV, so only a few per cent ionize at room temperature (Chapter 32), and hole mobilities are 10–30 cm²/(V s). In MOCVD growth, hydrogen passivates Mg, and p-type GaN only became possible when Amano and Akasaki (electron-beam irradiation, 1989) and Nakamura (thermal annealing, 1992) learned to activate it. This made the blue LED and the white LED possible (2014 Nobel Prize). Poor p-type conductivity still limits GaN bipolar devices and lasers.

Applications

- **Power switching.** Lateral GaN-on-Si HEMTs, at 15–650 V, switch with very low charge and are used in compact chargers, data-centre power supplies, LiDAR drivers and motor drives. Their figure of merit (Section 33.10) exceeds SiC's at these voltages. Vertical GaN-on-GaN devices for higher voltages are under development.
- **RF power.** GaN-on-SiC HEMTs deliver 5–10 W per mm of gate width at frequencies to 100 GHz and beyond, about ten times GaAs, and have taken over radar, base-station and satellite power amplifiers.
- **Optoelectronics.** InGaN quantum wells emit from violet to green (blue LEDs reach external quantum efficiencies above 80 %), AlGaN emits in the UV, and GaN lasers write Blu-ray discs and drive projectors and automotive headlamps (Chapter 35).

33.6 Other wide-bandgap semiconductors

Materials with gaps above about 4 eV are called **ultra-wide-bandgap** (UWBG). Their critical fields, scaling roughly as E_g^2–$E_g^{2.5}$ (Chapter 32), promise power devices beyond SiC and GaN. Each has a serious materials obstacle.

Gallium oxide (β-Ga_2O_3)

- **Gap** 4.8–4.9 eV, with an estimated critical field of about 8 MV/cm (fields above 5 MV/cm have been demonstrated in devices).
- **Melt growth.** Unlike SiC and GaN, β-Ga_2O_3 can be grown from the melt by edge-defined film-fed growth or Czochralski methods, like sapphire. Large, low-cost, low-defect substrates (100 mm, 150 mm demonstrated) are therefore realistic.
- **Controllable n-type doping** with Sn, Si or Ge from about 10^{15} to 10^{19} cm^{-3}.
- **Obstacles.** There is no p-type doping: the valence band is very flat (heavy holes) and holes self-trap on oxygen sites. The electron mobility is limited by polar-optical-phonon scattering to about 150–200 cm²/(V s). Most seriously, the thermal conductivity is only 11–27 W/(m K) (anisotropic), lower than GaAs, so heat removal must come from the top side or from bonding to SiC or diamond.

Diamond

Diamond has the most extreme properties of any semiconductor: a gap of 5.47 eV (indirect), an estimated critical field of about 10 MV/cm, electron and hole mobilities of about 2000 cm²/(V s) or more in the purest CVD material, a saturation velocity of about 2.7×10^7 cm/s (electrons), and the highest thermal conductivity of any bulk material, 2000–2200 W/(m K). On paper, its Baliga figure of merit is tens of thousands of times silicon's (Section 33.10). In practice:

- **Dopants are deep.** Boron is an acceptor at 0.37 eV and phosphorus a donor at about 0.57 eV, so few carriers are free at room temperature. A hydrogen-terminated diamond surface develops a two-dimensional hole gas by surface transfer doping, which is the basis of most diamond transistors.
- **Substrates are small.** High-pressure–high-temperature and CVD single crystals are about 10 mm across. Mosaic and heteroepitaxial wafers of 50 mm are emerging.

Today diamond's main electronic role is as a **heat spreader**, polycrystalline CVD diamond bonded under GaN HEMTs and laser diodes. It also serves in radiation detectors and quantum sensors (nitrogen–vacancy centres).

Aluminium nitride and AlGaN

AlN has a direct gap of 6.0–6.2 eV, a thermal conductivity of 285–320 W/(m K) for single crystals, and is strongly piezoelectric (Chapter 27). Bulk AlN substrates (50–100 mm) are used for deep-UV AlGaN LEDs (disinfection at 260–280 nm), and high-Al-content AlGaN channels are being explored for UWBG transistors. Doping becomes progressively harder as the Al fraction rises. Polycrystalline AlN ceramic (170–200 W/(m K)) is a standard power-module substrate (Chapter 51).

Zinc oxide and others

ZnO (3.37 eV, direct) has an exciton binding energy of 60 meV, more than twice $k_B T$ at room temperature, which promises efficient excitonic UV emission, but reliable p-type doping has never been achieved. It is used instead as a transparent conductor (Al-doped ZnO), in varistors (Chapter 26), and as a piezoelectric (Chapter 27). Cubic boron nitride (about 6.4 eV) can be doped both n- and p-type but cannot yet be grown as device-quality films.

UWBG material	Main advantage	Main obstacle	Status
β-Ga_2O_3	Cheap melt-grown substrates; high $\mathcal{E}_c$	No p-type; low thermal conductivity	Early commercial diodes and transistors
Diamond	Highest $\mathcal{E}_c$ and thermal conductivity	Deep dopants; small substrates	Heat spreaders, detectors, research devices
AlN/AlGaN	Very wide direct gap; high thermal conductivity	Doping at high Al content; substrate cost	Deep-UV LEDs; research transistors
cBN	Ambipolar doping possible	Film growth	Research

33.7 Amorphous silicon

Structure: bonds without long-range order

Hydrogenated amorphous silicon (a-Si:H) is deposited by plasma-enhanced chemical vapour deposition from silane at about 200–300 °C over areas of square metres, on glass, metal foil or polymer. Each Si atom is still tetrahedrally bonded to about four neighbours, but bond angles and lengths vary by several per cent, and there is no long-range order (Chapter 2). Some atoms end up with only three neighbours, leaving **dangling bonds**. In pure amorphous Si their density is 10^{19}–10^{20} cm^{-3}, pinning the Fermi level and making the material useless. Adding 8–15 at.% hydrogen during deposition saturates most of them and reduces the density to 10^{15}–10^{16} cm^{-3}. That reduction made a-Si:H dopable (Spear and LeComber, 1975) and created a large-area electronics industry.

Electronic structure: band tails and a mobility gap

Because short-range bonding is preserved, a-Si:H still has a valence band and a conduction band (the tight-binding argument of Chapter 31 needs only bonds, not periodicity). Disorder has three effects, shown in Figure 33.4.

- **Band tails.** The distribution of bond angles and lengths smears the band edges into exponential tails of localized states, $g(E) \propto e^{-(E_c - E)/E_0}$, with characteristic energies $E_0 \approx 45$–50 meV (valence tail) and about 25 meV (conduction tail). The same exponential tail appears in the optical absorption edge (the Urbach edge).
- **Mobility edges.** States deep in the bands are extended and conduct. States in the tails are localized and do not. The energy separating them is the **mobility edge**, and the separation between the two mobility edges, about 1.7–1.8 eV, is the **mobility gap**.
- **Defect states.** The remaining dangling bonds form a band of deep states near mid-gap that act as recombination centres and traps.

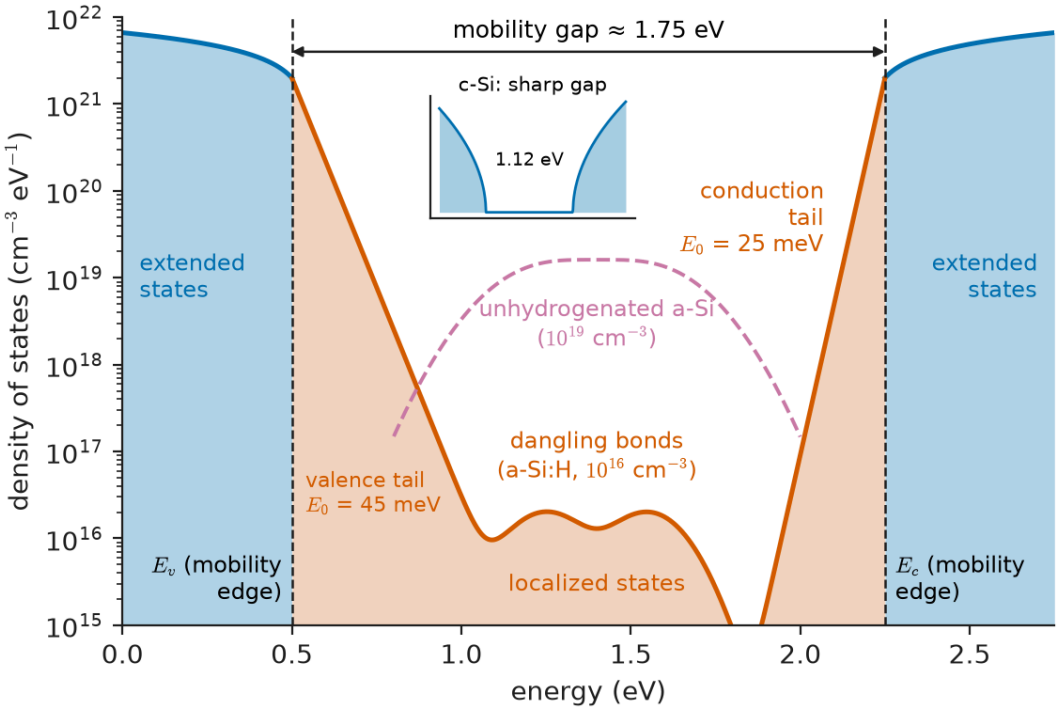


Figure 33.4 — Density of states of hydrogenated amorphous silicon: extended band states, exponential band tails of localized states, mobility edges and mid-gap dangling-bond states.

Consequences. Electrons in extended states have a mobility of about 10–20 cm²/(V s), but they spend most of their time trapped in tail states and released thermally. The resulting *effective* (trap-limited) mobility in thin-film transistors is only about 0.5–1 cm²/(V s). Because k is not conserved in a disordered solid (Chapter 31), absorption above the gap is much stronger than in crystalline Si (about 10^5 cm^{-1} at 2.2 eV), so a few hundred nanometres absorb most visible light.

The Staebler–Wronski effect. Prolonged illumination creates additional dangling bonds, as the energy of recombination breaks weak Si–Si bonds. This reduces the efficiency of a-Si:H solar cells by 10–30 % in the first months of operation before it stabilizes. Annealing at about 150 °C reverses it. This metastability is a fundamental property of a hydrogenated disordered network and was one reason for the decline of a-Si:H thin-film photovoltaics.

Applications and competitors

a-Si:H remains in use for display backplanes (large LCD panels), flat-panel X-ray imagers, and, as a thin intrinsic passivation layer, in the silicon **heterojunction** solar cells that hold the efficiency record for crystalline silicon (Chapter 40), because it passivates the c-Si surface almost perfectly. For higher-performance thin-film transistors two competitors have emerged:

Thin-film channel material	Effective mobility (cm²/(V s))	Process temperature	Main use
a-Si:H	0.5–1	200–300 °C	Large LCD, X-ray imagers
Amorphous oxide (InGaZnO, IGZO)	10–30	Room temperature to 400 °C	Large OLED/LCD panels, low-leakage backplanes
Low-temperature polysilicon (LTPS)	50–150	Laser crystallization, < 500 °C on glass	Mobile OLED displays
Organic semiconductors	0.1–10	Room temperature to 150 °C	Flexible and printed circuits

Amorphous oxides such as IGZO conduct well despite their disorder because their conduction band is built from spherical metal s orbitals, whose overlap is insensitive to bond-angle disorder. This is a nice example of the tight-binding picture applied to a real material choice.

33.8 Organic semiconductors

Molecular solids

Organic semiconductors are π-conjugated molecules (pentacene, rubrene, DNTT, C_{60}) or polymers (P3HT, PEDOT, diketopyrrolopyrrole copolymers) whose alternating single and double bonds produce delocalized π orbitals within each molecule (Advanced Chemistry core). The highest occupied molecular orbital (HOMO) plays the role of the valence-band edge and the lowest unoccupied orbital (LUMO) that of the conduction-band edge. Their separation, 1.5–3 eV, is set by the conjugation length.

Between molecules, however, there are only weak van der Waals forces. The intermolecular transfer integral t is about 0.01–0.1 eV, compared with 1–3 eV in Si. In the tight-binding picture of Chapter 31, this gives bandwidths of a few tenths of an electron-volt at most and heavy effective masses. In real films the bands are further broken up by **energetic disorder**: each molecule's HOMO and LUMO energies are shifted by its particular environment, with a roughly Gaussian spread of width $\sigma \approx 0.05$–0.15 eV.

Transport by hopping

When the disorder exceeds the bandwidth, carriers localize on individual molecules or polymer segments and move by thermally activated **hopping** between them. The Gaussian disorder model (Bässler) describes the low-field mobility as

$$\mu(T) = \mu_0 \exp\left[-\left(\frac{2\sigma}{3k_B T}\right)^2\right]$$

> **Status of the principle: Gaussian disorder model**
> *Empirical.* Derived from Monte Carlo simulation of hopping in a Gaussian density of states and fitted to time-of-flight data. It captures the strong temperature dependence and, with a further factor, the Poole–Frenkel-like field dependence ($\ln \mu \propto \sqrt{\mathcal{E}}$) of disordered organic films. It does not describe highly ordered single crystals, where band-like transport is observed.

With $\sigma = 0.1$ eV at 300 K, the exponent is $(2 \times 0.1/(3 \times 0.02585))^2 = (2.58)^2 = 6.65$, and the mobility is reduced by a factor of $e^{-6.65} = 1.3 \times 10^{-3}$ from its disorder-free value μ_0. Reducing disorder through molecular design and crystallinity has raised organic mobilities by five orders of magnitude since the 1980s.

Organic material	Form	Typical mobility (cm²/(V s))
P3HT (polymer)	Solution-processed film	0.01–0.1 (holes)
Pentacene	Evaporated film	0.5–3 (holes)
C_{60}	Evaporated film	0.5–5 (electrons)
DPP-based donor–acceptor polymers	Solution-processed film	1–10
C8-BTBT, DNTT	Small-molecule films	1–10 or more
Rubrene	Single crystal	10–40 (holes)

Excitons

Organic solids have low permittivity ($\varepsilon_r \approx 3$–4) and localized states, so an absorbed photon creates a tightly bound electron–hole pair, a Frenkel **exciton**, with binding energy 0.3–1 eV, compared with a few meV in GaAs. Excitons do not separate into free carriers at room temperature on their own. Organic solar cells therefore need a donor–acceptor interface (a "bulk heterojunction") to split them, and now reach 19–20 % efficiency in the laboratory. In organic LEDs, conversely, excitons are formed deliberately and emit efficiently. Phosphorescent and thermally activated delayed-fluorescence emitters harvest both singlet and triplet excitons, giving internal quantum efficiencies near 100 %. OLED displays are the largest commercial application of organic semiconductors.

Strengths and weaknesses

Organic semiconductors are processed at low temperature, from solution or by evaporation, onto flexible substrates over large areas, and their properties can be tuned by molecular synthesis. Against this they have low mobility, sensitivity to oxygen and water (which demands encapsulation, Chapter 51), limited thermal stability, and poorly controlled doping. Their thermal conductivity, about 0.1–0.5 W/(m K), makes self-heating a concern even at low power.

33.9 Two-dimensional materials

Layered crystals such as graphite, MoS_2 and hexagonal boron nitride consist of covalently bonded sheets held together by van der Waals forces. Single sheets can be isolated or grown, giving semiconductors, metals and insulators one atom (or three atoms) thick. Their electronic structure, size effects and synthesis are treated fully in Chapter 38. Here the question is how they compare as semiconductor materials.

2D material	Band gap	Typical room-temperature mobility (cm²/(V s))	Notes
Graphene	0 (semimetal, linear bands)	10^4 on SiO_2; $> 10^5$ hBN-encapsulated	Cannot switch off; RF, sensors, transparent electrodes
MoS_2 (monolayer)	1.8–1.9 eV direct (bulk 1.2–1.3 eV indirect)	10–100 (phonon limit about 200)	n-type channel; leading 2D transistor candidate
WSe_2 (monolayer)	about 1.6 eV direct	10–200	p-type or ambipolar
Black phosphorus	0.3 eV (bulk) to about 1.5–2 eV (monolayer), direct	100–1000, anisotropic	Degrades in air
hBN	about 6 eV	Insulator	Atomically flat substrate and gate dielectric

Three properties make 2D semiconductors attractive.

- **Thickness without surface damage.** A monolayer has no dangling bonds on its faces, so it can be thinned to under a nanometre without the mobility collapse that roughness and surface states cause in ultrathin silicon. This gives the best electrostatic control of a channel, the main argument for 2D materials at the end of transistor scaling.
- **Layer-dependent band structure.** MoS_2 changes from an indirect gap of about 1.2 eV in bulk to a direct gap of about 1.9 eV in a monolayer. Thickness becomes a design variable, an extreme form of the quantum confinement of Chapter 38.
- **Stacking.** Van der Waals heterostructures can be assembled without lattice matching, in contrast to the epitaxial constraints of Section 33.2.

Against these, wafer-scale growth of defect-free monolayers, low-resistance contacts and reliable doping remain unsolved, and mobilities of transition-metal dichalcogenides are lower than those of silicon. Graphene's lack of a gap gives on/off ratios of only about 10, which rules it out for logic. Two-dimensional materials are today used in sensors, photodetectors and research devices. Their role in mainstream electronics is a question for the 2030s.

33.10 Material comparison for power, RF, sensing and optoelectronics

The master table

Property (300 K)	Si	Ge	GaAs	InP	4H-SiC	GaN	β-Ga_2O_3	Diamond	AlN
E_g (eV)	1.12	0.66	1.42	1.34	3.26	3.39	4.85	5.47	6.1
Gap type	I	I	D	D	I	D	I (nearly D)	I	D
ε_r	11.7	16.0	12.9	12.5	9.7	9.0	10	5.7	8.5
μ_n (cm²/(V s))	1400	3900	8500	5400	950	1000–1300 (2DEG 1500–2000)	150–200	2000–4500	about 300
μ_p (cm²/(V s))	450	1900	400	200	115	10–30	— (no p-type)	2000–3800	about 14
$\mathcal{E}_c$ (MV/cm)	0.3	0.1	0.4	0.5	2.5 (2–3)	3.3	8 (est.)	10 (est.)	12–15 (est.)
v_{sat} (10^7 cm/s)	1.0	0.6	1.0 (peak 2)	1.0 (peak 2.5)	2.0	2.5	1.1–2 (est.)	2.7	about 1.5
κ (W/(m K))	148	60	55	68	330–490	130–230	11–27	2000–2200	285–320
n_i (cm⁻³)	1×10^{10}	2×10^{13}	2×10^6	1×10^7	$\sim 10^{-8}$	$\sim 10^{-10}$	$\sim 10^{-22}$	$\sim 10^{-27}$	negligible
Largest wafer (2026)	300 mm	200 mm	150–200 mm	100–150 mm	150–200 mm	200 mm (on Si); 100 mm bulk	100–150 mm	about 10 mm (single crystal)	50–100 mm

Values marked "est." are theoretical or extrapolated, not yet measured in bulk breakdown. The critical field depends on doping (Chapter 32), and values here refer to the moderately doped drift layers of kilovolt-class devices.

Power: the Baliga figure of merit

A unipolar power device (Schottky diode, MOSFET, HEMT) must block a voltage V_B in its off state and conduct with as low a resistance as possible in its on state. Both are set by the **drift region**, a lightly doped layer of thickness W and doping N_D. Its design is a materials problem that can be solved in four lines.

At breakdown, the field in a non-punch-through drift region is triangular, peaking at $\mathcal{E}_c$, and from Poisson's equation its slope is qN_D/ε. Therefore

$$W = \frac{\varepsilon \mathcal{E}_c}{qN_D}, \qquad V_B = \frac{1}{2}\mathcal{E}_c W$$

Eliminating W gives the maximum doping allowed for a given blocking voltage and the corresponding thickness:

$$N_D = \frac{\varepsilon \mathcal{E}_c^2}{2qV_B}, \qquad W = \frac{2V_B}{\mathcal{E}_c}$$

The specific on-resistance (resistance × area) of the drift region in the on state is $R_{\text{on,sp}} = W/(q\mu_n N_D)$, so

$$R_{\text{on,sp}} = \frac{4V_B^2}{\varepsilon \mu_n \mathcal{E}_c^3}$$

The denominator depends only on the material. It is the **Baliga figure of merit**:

$$\text{BFOM} = \varepsilon \mu_n \mathcal{E}_c^3$$

> **Status of the principle: Baliga figure of merit and the unipolar limit**
> *Approximate.* Derived for a one-dimensional, uniformly doped, non-punch-through drift region with a constant critical field and constant mobility. Real devices add channel, substrate and contact resistance, and the critical field rises with doping. The resulting limit is optimistic by a factor of about two for Si, and it ignores switching losses and thermal limits entirely. It is a screening tool, not a design equation.

The cube of the critical field dominates. A tenfold increase in $\mathcal{E}_c$ is worth a thousandfold reduction in on-resistance, while mobility enters only linearly. This is why 4H-SiC, with a sixth of the mobility of GaAs, beats it by a factor of twenty, and why the critical fields marked "est." in the master table carry such large uncertainties in the FOM: changing the SiC critical field from 2.5 to 3.0 MV/cm raises its BFOM by $1.2^3 = 1.73$.

Worked example — a 1200 V drift layer in silicon and in 4H-SiC. For silicon, take $\varepsilon = 11.7 \times 8.854 \times 10^{-14} = 1.036 \times 10^{-12}$ F/cm, $\mu_n = 1400\ \mathrm{cm^2/(V\,s)}$ and $\mathcal{E}_c = 3 \times 10^5$ V/cm. Then

$$N_D = \frac{1.036 \times 10^{-12} \times (3 \times 10^5)^2}{2 \times 1.602 \times 10^{-19} \times 1200} = 2.4 \times 10^{14}\ \mathrm{cm^{-3}}, \qquad W = \frac{2 \times 1200}{3 \times 10^5} = 8.0 \times 10^{-3}\ \mathrm{cm} = 80\ \mu\mathrm{m}$$

$$R_{\mathrm{on,sp}} = \frac{4 \times 1200^2}{1.036 \times 10^{-12} \times 1400 \times (3 \times 10^5)^3} = \frac{5.76 \times 10^6}{3.92 \times 10^7} = 0.147\ \Omega\,\mathrm{cm^2}$$

For 4H-SiC, $\varepsilon = 9.7 \times 8.854 \times 10^{-14} = 8.59 \times 10^{-13}$ F/cm, $\mu_n = 950\ \mathrm{cm^2/(V\,s)}$ and $\mathcal{E}_c = 2.5 \times 10^6$ V/cm give $N_D = 1.4 \times 10^{16}\ \mathrm{cm^{-3}}$, $W = 9.6$ µm and

$$R_{\mathrm{on,sp}} = \frac{5.76 \times 10^6}{8.59 \times 10^{-13} \times 950 \times 1.5625 \times 10^{19}} = \frac{5.76 \times 10^6}{1.275 \times 10^{10}} = 4.5 \times 10^{-4}\ \Omega\,\mathrm{cm^2}$$

The SiC drift layer is about 8 times thinner, about 60 times more heavily doped, and has about 330 times lower specific resistance. For a 100 A device conducting at 1 V drop (10 mΩ), the silicon drift region alone would need an area of $0.147/0.01 = 14.7\ \mathrm{cm^2}$, a die nearly 4 cm on a side, which is impractical. That is why silicon at 1200 V must use the bipolar IGBT, which floods the drift region with carriers to lower its resistance at the cost of slow switching. The SiC drift region needs $0.045\ \mathrm{cm^2}$ (about 2 mm × 2 mm). Real 1200 V SiC MOSFETs reach total values of 2–3 mΩ cm², dominated by the channel and substrate rather than the drift layer, which is why the SiC/SiO_2 interface of Section 33.4 matters so much.

Repeating the calculation for each material with the master-table values gives the BFOM ranking and the "unipolar limit" lines of Figure 33.5.

Material	Values used (ε_r, μ_n, $\mathcal{E}_c$)	BFOM relative to Si	$R_{\mathrm{on,sp}}$ limit at 1200 V (mΩ cm²)
Ge	16.0, 3900, 0.1	0.14	1040
Si	11.7, 1400, 0.3	1	147
GaAs	12.9, 8500, 0.4	16	9.3
InP	12.5, 5400, 0.5	19	7.7
4H-SiC	9.7, 950, 2.5	330	0.45
GaN	9.0, 1250, 3.3	910	0.16
β-Ga_2O_3	10, 200, 8	2300	0.064
Diamond	5.7, 2000, 10	26 000	0.0057

Published BFOM tables differ by factors of two or more because they use different critical fields and mobilities. Only the ordering, and the orders of magnitude, are robust.

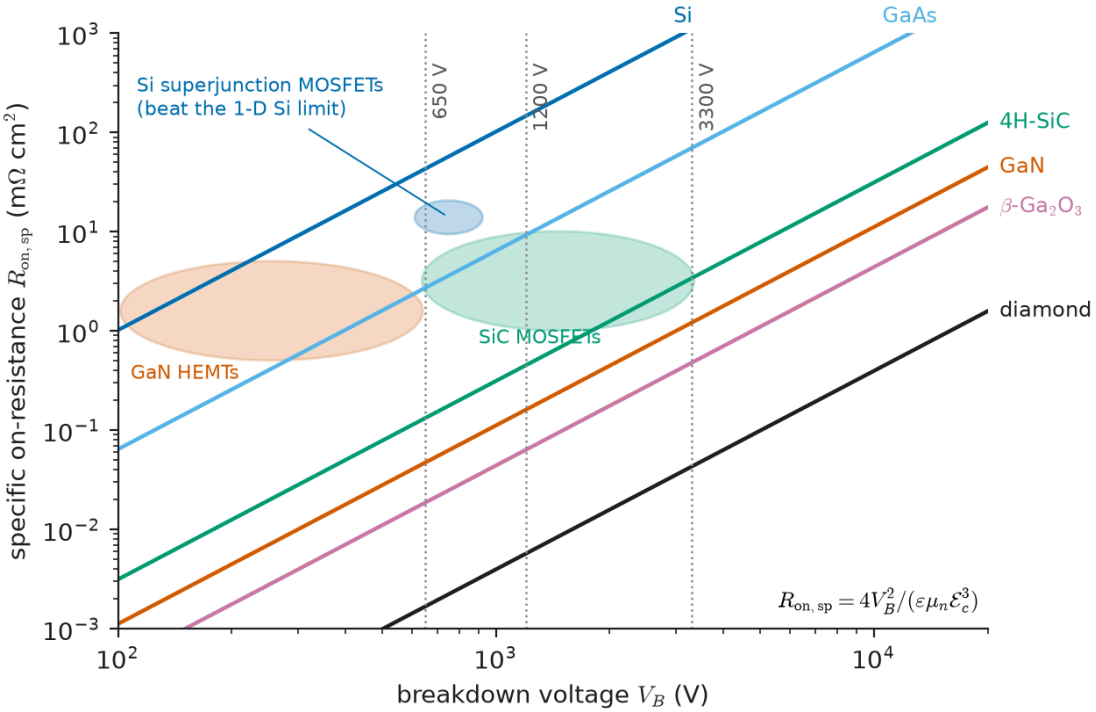


Figure 33.5 — Theoretical unipolar limit of specific on-resistance versus breakdown voltage for Si, GaAs, 4H-SiC, GaN, β-Ga2O3 and diamond, with the operating windows of commercial devices.

Two caveats keep this ranking honest. **Superjunction** silicon MOSFETs use alternating p and n pillars to break the one-dimensional assumption, and they beat the silicon line at 600–900 V. And the BFOM says nothing about the cost of the substrate, the quality of the gate dielectric, p-type doping, or **heat removal**, the next consideration.

Thermal conductivity: where the heat goes

A device dissipating power P over area A through a die of thickness t has a one-dimensional thermal resistance $t/(\kappa A)$ (Chapter 24). The temperature rise in the semiconductor itself varies inversely with κ.

Worked example — temperature drop across the die. A 1 mm × 1 mm hot spot dissipates 10 W through a 100 µm thick die. The temperature drop is $\Delta T = Pt/(\kappa A) = 10 \times 10^{-4}/(\kappa \times 10^{-6}) = 1000/\kappa$ (with κ in W/(m K)): 6.8 K for Si, 18 K for GaAs, 15 K for InP, 7.7 K for GaN ($\kappa = 130$), 2.7 K for 4H-SiC, 50 K for β-Ga_2O_3 ($\kappa = 20$), and 0.5 K for diamond. Heat spreading in a real die reduces all of these numbers but not their ratios. GaN RF transistors, whose power densities exceed 5 W/mm of gate, are grown on SiC substrates to use SiC's conductivity, and the highest-power ones are bonded to diamond. Ga_2O_3, despite its high BFOM, would run 18 times hotter than SiC for the same loss unless it is thinned and bonded to a better conductor. Thermal conductivity can overrule the electrical figures of merit.

RF: the Johnson figure of merit

A high-frequency transistor must be short, so that carriers cross it quickly, and it must withstand a voltage, so that it can deliver power. These conflict. The transit time across a region of length L at saturation velocity is L/v_{sat}, giving a cut-off frequency $f_T \approx v_{sat}/2\pi L$, while the maximum voltage is about $\mathcal{E}_c L$. The product does not depend on L:

$$V_{max} f_T = \frac{\mathcal{E}_c v_{sat}}{2\pi} \equiv \text{JFOM}$$

> **Status of the principle: Johnson figure of merit**
> *Approximate.* It assumes uniform saturated transport and a uniform critical field over the same length. It ignores parasitic capacitances, contact resistances, mobility (important at low fields and for noise) and thermal limits. Some authors quote its square, $(\mathcal{E}_c v_{sat}/2\pi)^2$, so relative values in the literature may be the square of those given here.

Material	$\mathcal{E}_c$ (MV/cm)	v_{sat} (10^7 cm/s)	JFOM (V/s)	JFOM relative to Si	$V_{\max}$ at $f_T = 100$ GHz (V)
Ge	0.1	0.6	9.5×10^{10}	0.2	1.0
Si	0.3	1.0	4.8×10^{11}	1	4.8
GaAs	0.4	1.0	6.4×10^{11}	1.3	6.4
InP	0.5	1.0	8.0×10^{11}	1.7	8.0
4H-SiC	2.5	2.0	8.0×10^{12}	17	80
GaN	3.3	2.5	1.3×10^{13}	28	131
β-Ga_2O_3	8	1.1	1.4×10^{13}	29	140
Diamond	10	2.7	4.3×10^{13}	90	430

GaN's place follows from this table: at 10 GHz it can sustain voltages of tens of volts, so for a given output power it presents a higher load impedance and needs less current, which simplifies matching and raises efficiency. Combined with the high 2DEG density and SiC's thermal conductivity, this is why GaN-on-SiC HEMTs replaced GaAs pHEMTs and silicon LDMOS in radar and base-station amplifiers. SiC's own low channel mobility keeps it out of RF. GaAs and InP still win where **low noise** and the highest frequencies at low voltage matter, because noise and $f_{\max}$ at small dimensions depend on low-field mobility and light effective mass, which the JFOM omits. A related index, Baliga's high-frequency figure of merit $\mu\mathcal{E}_c^2$, estimates switching loss in fast power converters. With the master-table values it ranks GaN (about 110 × Si) above Ga_2O_3 (about 100 ×), 4H-SiC (about 47 ×) and GaAs (about 11 ×), which is why GaN dominates the fastest power converters below 650 V. Figure 33.6 places all the materials on one map of the two figures of merit, with thermal conductivity shown as marker size, so that the trade-offs of this section can be seen at a glance.

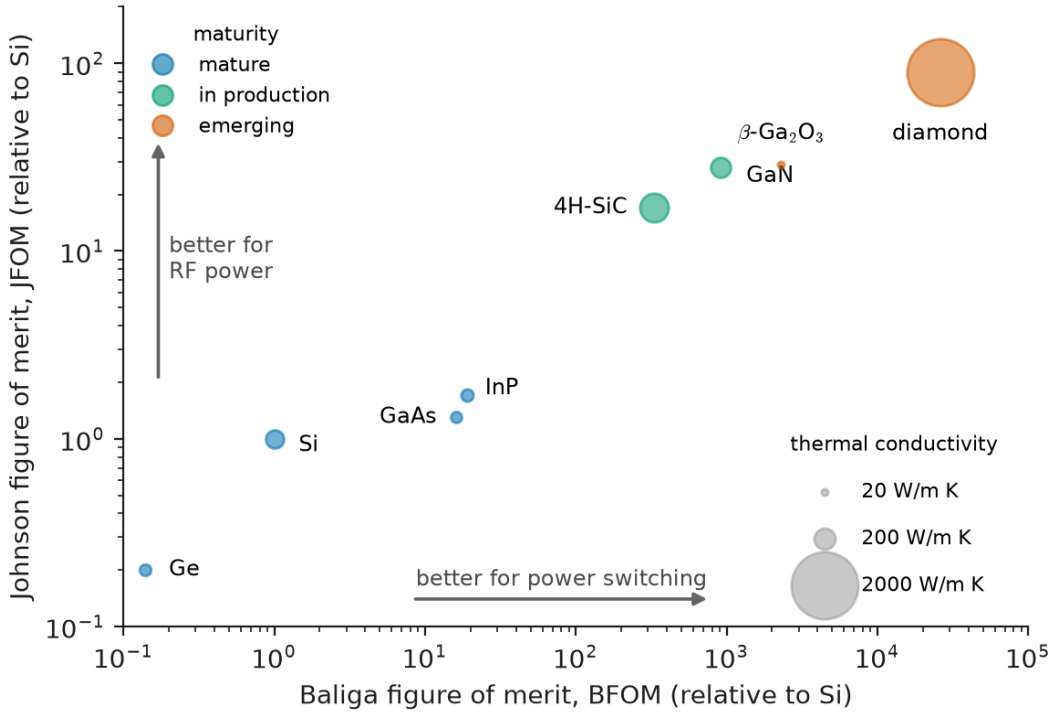


Figure 33.6 — Materials map for power and RF: Johnson figure of merit versus Baliga figure of merit (both relative to Si), with marker size proportional to thermal conductivity and colour indicating technology maturity.

Sensing

Sensor materials are chosen for different properties than power or RF materials. The quantity that converts the measurand into an electrical signal determines the choice.

Sensing function	Key material property	Typical materials	Why
Magnetic field (Hall sensors)	High electron mobility	InSb, InAs, GaAs; Si for integrated sensors	Hall voltage $\propto \mu$; Si integrates with signal processing
Temperature	Exponential n_i or $E_g(T)$; mobility T-dependence	Si diodes and transistors; Ge for cryogenics	V_{BE} changes about −2 mV/K; bandgap references
Strain and pressure	Piezoresistance (strain-induced valley repopulation)	Si (gauge factor about 100)	Large piezoresistive coefficients of Si; MEMS integration
High-temperature and harsh environments	Wide gap, chemical inertness	4H-SiC, GaN, diamond	Low n_i to 500 °C and beyond
Visible/NIR light	Absorption edge, low dark current	Si (to 1.1 μm), InGaAs (0.9–1.7 μm), Ge (to 1.6 μm)	Cut-off set by E_g (Chapter 35)
Mid/long-wave IR	Narrow gap	InSb, HgCdTe, InAs/GaSb superlattices	Gaps of 0.1–0.3 eV, cooled
UV (solar-blind)	Wide gap	AlGaN, Ga_2O_3, diamond, SiC	Insensitive to visible light
Ionizing radiation	Gap (pair energy $\approx 3E_g$), purity, Z	Si, HPGe, CdZnTe, diamond, SiC	Signal per keV; radiation hardness
Gas and chemical	Surface states, adsorption	SnO_2, ZnO, WO_3, 2D materials, graphene	Surface-to-volume ratio (Chapters 34, 35, 38)

Silicon dominates wherever the sensor can be integrated with its readout circuit: CMOS image sensors, MEMS pressure sensors and accelerometers, Hall sensors, temperature sensors. Other materials win where silicon's band gap or temperature range is wrong.

Optoelectronics

For light emission only direct-gap materials qualify (Chapter 31), and the emission wavelength is set by the gap of the active layer:

Wavelength band	Active material	Substrate	Application
Deep UV (210–280 nm)	AlGaN	AlN, sapphire	Disinfection, sensing
UV-A, violet, blue (365–470 nm)	InGaN/GaN	Sapphire, Si, SiC, GaN	White LEDs (with phosphor), lasers
Green (520–550 nm)	InGaN (high In)	Sapphire, GaN	Displays (efficiency gap "green gap")
Red, orange, amber (590–650 nm)	AlGaInP	GaAs	Automotive lamps, displays
Near-IR (780–980 nm)	AlGaAs/GaAs, InGaAs/GaAs	GaAs	VCSELs, sensing, pump lasers
Telecom (1.3–1.6 μm)	InGaAsP, InGaAlAs	InP	Fibre-optic lasers and modulators
Mid-IR (3–12 μm)	Quantum-cascade structures (InGaAs/InAlAs), GaSb-based	InP, GaSb	Gas sensing, spectroscopy

For detection and photovoltaics, indirect-gap materials work well, as Chapter 31 showed: Si for visible imaging and solar cells, Ge and InGaAs for telecom wavelengths. Chapter 35 treats emission, absorption and the material requirements of each device class in detail.

Selection summary

Application	First-choice material(s)	Decisive properties	Main alternative
Logic and memory	Si (with SiGe, strained Si)	Native oxide, crystal perfection, scale, cost	2D channels (future)
Power, < 100 V	Si MOSFETs	Cost, maturity	GaN HEMTs (speed)
Power, 100–650 V, fast switching	GaN-on-Si HEMTs	High BFOM, low charge, 2DEG	Si superjunction, SiC
Power, 650 V–3.3 kV	4H-SiC MOSFETs and diodes	BFOM, thermal conductivity, native oxide	Si IGBT (cost)
Power, > 3.3 kV	Si IGBT and thyristors; 4H-SiC emerging	Bipolar conductivity modulation; SiC BFOM	Ga_2O_3, diamond (research)
RF power, 1–100 GHz	GaN-on-SiC HEMTs	JFOM, power density, thermal conductivity	GaAs pHEMT, Si LDMOS (< 4 GHz)
RF handset PAs	GaAs HBTs	Efficiency at low voltage, maturity	GaN, Si CMOS/SOI
Low-noise and sub-THz	InP HEMTs/HBTs, SiGe HBTs	Light m^*, high mobility	GaAs pHEMT
Hall sensors	InSb, InAs, GaAs, Si	Mobility; integration	Graphene (research)
Harsh-environment electronics	4H-SiC	Low n_i, inertness	GaN, diamond
Visible and IR LEDs/lasers	InGaN, AlGaInP, (Al)GaAs, InGaAsP	Direct gap, lattice match	Organic (displays), perovskites (research)
Displays (backplane)	a-Si:H, IGZO, LTPS	Large area, low temperature	Organic TFTs
Flexible, printed electronics	Organic semiconductors, oxides	Low-temperature processing	—

Status of the principle: figure-of-merit ranking
Approximate. Figures of merit compress the properties of a material into one number for a stated, idealized device. They correctly identify which materials can never compete for a task (Ge for power, Ga_2O_3 for heat-limited RF) and which might. The final choice is decided by factors outside the formula: substrate cost and size, defect density, dielectric interfaces, p-type doping, packaging, reliability and supply chain (Chapter 58).

What remains open

Open question	Where it is answered
How do band offsets, strain and critical thickness control heterostructures such as AlGaAs/GaAs, InGaAs/InP and AlGaN/GaN?	Chapter 34
What sets interface-trap density at SiC/SiO_2, GaN/dielectric and high-κ/Si interfaces, and how does it limit channel mobility and reliability?	Chapter 34
What are the material requirements for LEDs, lasers, photodetectors and solar absorbers built from these families?	Chapter 35
How do graphene and other 2D materials behave as their thickness and lateral size shrink?	Chapter 38
Which semiconductors make the best photovoltaic and thermoelectric materials?	Chapter 40
How are Si, SiC, GaN, GaAs and Ga_2O_3 crystals and wafers grown, and how are they doped and implanted?	Chapter 50
How do packages remove heat from SiC and GaN dies and cope with their thermal-expansion mismatch?	Chapter 51
How do wide-gap devices degrade in service (bipolar degradation, current collapse, gate-oxide wear-out)?	Chapter 44
How are figures of merit combined with cost, reliability and sustainability in a formal selection process?	Chapter 58

Next: Chapter 34 moves from bulk materials to the interfaces and thin films where devices are built: heterostructures and band alignment, surface and interface states, thin-film stress and defects, contacts and gate dielectrics.

Chapter 34 — Semiconductor Interfaces and Thin Films

Part VII · Semiconductor and Photonic Materials

Around 2005 the silicon industry hit a wall that had nothing to do with lithography. The gate insulator of a logic transistor, a film of silicon dioxide, had been thinned to about 1.2 nm, roughly five atomic layers. At that thickness electrons tunnel straight through it, and the gate leakage current density at 1 V reached the order of 10–100 A/cm². Multiplied over a billion transistors, this was the chip's whole power budget, spent on leakage. Thinning the oxide further was impossible, and keeping it thicker meant losing control of the channel. The fix was to change the material: hafnium-based oxides, with a permittivity five times that of SiO_2, deposited on a deliberately grown sub-nanometre SiO_2 layer and capped with a metal gate. Each part of that solution was an interface problem. Hafnia on bare silicon gives a poor interface, the metal's Fermi level is pinned by states at the high-k surface, and the new oxide traps charge in ways the old one never did.

A second example comes from the other end of the industry. Grow a 25 nm layer of AlGaN on GaN, with no dopant anywhere, and a sheet of about 10^{13} electrons per cm² appears at the interface, moving with a mobility near 2000 cm²/V s. That electron sheet is the channel of every GaN RF amplifier and power switch. Nothing in Chapter 32's doping theory predicts it. It comes from the crystal symmetry of the two materials and the discontinuity of their polarization at the interface.

In both cases the property that matters belongs to the boundary between two materials, not to either bulk material. This chapter treats interfaces and thin films as materials in their own right.

> **What decides how the bands of two semiconductors line up when they are joined, and why does the simple rule for it fail?**
> **How thick can a strained film grow before it relaxes by forming dislocations? Why does a metal on GaAs give almost the same barrier height whatever the metal? What does a thin film's stress, grain structure and interface trap density do to a device, and how is each measured and controlled?**

Concept	Engineering destination
Band offsets, type I/II/III alignment	Quantum-well lasers, HEMTs, HBTs, infrared superlattices
Lattice mismatch, critical thickness	Strained-Si/SiGe channels, InGaAs wells, GaN-on-Si buffers
Polarization-induced 2DEG	GaN RF and power transistors
Surface states, Fermi-level pinning	Schottky barriers, GaAs MOS difficulty, passivation
Interface-trap density D_{it}	Subthreshold swing, threshold stability, 1/f noise
Film stress, Stoney equation	Wafer bow, cracking, delamination, stress engineering of channels
Thin-film grain structure	Interconnect resistivity, electromigration, polysilicon resistors, gas sensors
Barrier height, specific contact resistivity	Ohmic contacts, Schottky diodes, contact resistance at small nodes
Gate-dielectric permittivity, band offsets, EOT	High-k/metal-gate CMOS, power MOSFET gates, memory
NBTI/PBTI, hot carriers, TDDB	Lifetime of transistors and gate stacks (failure analysis in Chapter 44)

Roadmap. Joining two semiconductors (34.1) → how their bands align (34.2) → what happens at the interface: strain, dislocations and polarization charge (34.3) → the free surface and its states (34.4) → interface traps and their measure D_{it} (34.5) → thin films as materials (34.6) → film stress and defects (34.7) → grain structure (34.8) → metal–semiconductor contacts (34.9) → gate dielectrics (34.10) → how interfaces wear out (34.11).

34.1 Semiconductor heterostructures

A *homojunction* joins two regions of the same semiconductor that differ only in doping, as in a silicon p–n junction. The band gap, electron affinity and permittivity are the same on both sides. A *heterojunction* joins two different semiconductors, for

example GaAs and $Al_xGa_{1-x}As$, or Si and $Si_{1-x}Ge_x$. A *heterostructure* is any stack containing one or more heterojunctions: a single interface, a quantum well (a thin narrow-gap layer between wider-gap layers), a superlattice (many alternating thin layers), or a graded layer.

The reason to build one is that a heterojunction gives the designer a second control alongside doping. Doping moves the Fermi level, so it moves both band edges together. A change of material moves the conduction-band edge and the valence-band edge *independently*, by amounts fixed by the two materials. This allows three things that doping alone cannot do:

1. **Carrier confinement.** A narrow-gap layer between wider-gap layers traps electrons and holes in the same thin region. The radiative rate is proportional to np (Chapter 35), so confinement raises light emission efficiency. Every semiconductor laser since the 1970s depends on this.
2. **Separating carriers from their donors.** Electrons can be placed in an undoped layer next to a doped one. They are then free of ionized-impurity scattering (Chapter 32), which gives the high-mobility channel of the HEMT.
3. **Different barriers for electrons and holes.** In a heterojunction bipolar transistor, a wide-gap emitter blocks holes but not electrons, so the base can be heavily doped without losing gain.

The rest of Part VII's device physics (how a HEMT or an HBT works as a circuit element) belongs to Electronics & Computing. This chapter asks the materials questions behind the devices: how much do the band edges step, can the two crystals be joined without defects, and what charge sits at the interface?

> **Boundary.**
> The operation of heterojunction devices (HEMT transfer characteristics, HBT gain, laser threshold conditions) is treated in Electronics & Computing. Here the subject is the interface itself: its band offsets, strain, defects and charge.

The materials that can be combined are limited mainly by lattice constant. Figure 35.2 in the next chapter maps band gap against lattice constant for the common semiconductors. Materials on a vertical line in that map, such as GaAs and AlAs (a = 5.653 and 5.661 Å) or InP and $In_{0.53}Ga_{0.47}As$, can be grown on one another almost without strain. Pairs far apart in lattice constant need the strain management of Section 34.3.

34.2 Band alignment

The question

When two semiconductors A and B meet, each brings its own band gap. The difference $\Delta E_g = E_{g,B} - E_{g,A}$ must be shared between a conduction-band discontinuity ΔE_c and a valence-band discontinuity ΔE_v:

$$\Delta E_g = \Delta E_c + \Delta E_v$$

The split is the most important number for a heterojunction. It decides whether electrons, holes or both are confined, and how strongly. It is a property of the interface, set by how the electron states of the two crystals match up. Doping does not change it. Doping changes how the bands bend *away* from the interface, but the steps *at* the interface stay fixed (to a good approximation) once the two materials and the interface structure are chosen.

Anderson's electron-affinity rule

The simplest argument is due to Anderson (1962). Each semiconductor has an *electron affinity* χ, the energy needed to take an electron from the conduction-band minimum to the vacuum level just outside the surface. If both materials are referred to a common vacuum level, the conduction-band edges sit at $-\chi_A$ and $-\chi_B$, and the valence-band edges at $-(\chi_A + E_{g,A})$ and $-(\chi_B + E_{g,B})$. Then

$$\Delta E_c = \chi_A - \chi_B, \qquad \Delta E_v = (\chi_B + E_{g,B}) - (\chi_A + E_{g,A})$$

Once the two crystals are in contact, their Fermi levels equalize by charge transfer across the junction. The bands bend over a depletion or accumulation width set by the doping, as in a p–n junction. But according to Anderson, the *steps* ΔE_c and ΔE_v at the interface keep their vacuum-referenced values. Figure 34.1 shows the construction.

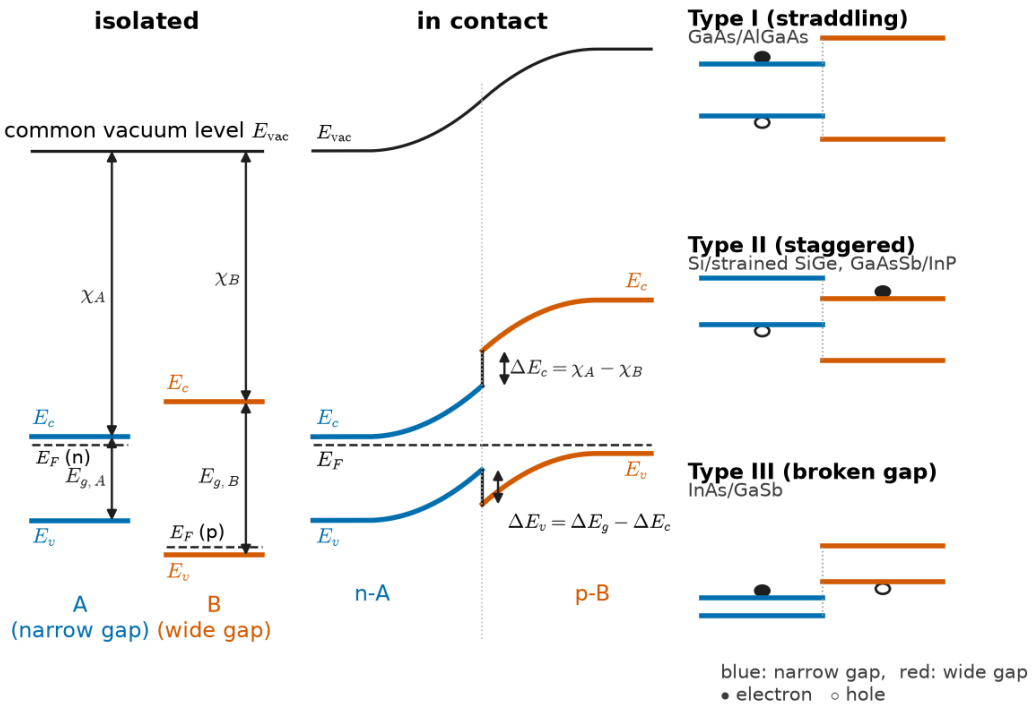


Figure 34.1 — Band alignment at heterojunctions: Anderson's vacuum-level construction (left) and the three alignment types (right).

> **Status of the principle: Anderson's electron-affinity rule**
> *Approximate.* It assumes the vacuum-referenced band edges of each bulk crystal carry over unchanged to the interface. This ignores the interface dipole formed when the bonds across the interface redistribute charge. The rule typically gives offsets to within a few tenths of an eV, which is often comparable with the offset itself.

Why it fails

Anderson's rule is a natural first guess. It fails because the electron affinity is a *surface* property. It is measured at a crystal–vacuum surface that has its own reconstruction, dangling bonds and surface dipole (Section 34.4), and these change χ by tenths of an eV depending on crystal face and preparation. At a heterojunction there is no vacuum. Bonds run across the interface instead, and their charge redistributes. The result is an *interface dipole*, a sheet of charge pairs only an ångström or two thick, which adds a step δ in electrostatic potential:

$$\Delta E_c = \chi_A - \chi_B + \delta$$

Anderson's rule is the special case $\delta = 0$. Three observations show that δ is not small.

- **GaAs/AlAs.** The electron affinities are about 4.07 eV for GaAs and 3.5 eV for AlAs. AlAs has an indirect gap of 2.16 eV and a direct (Γ) gap near 3.0 eV, so the Γ-point gap difference is about 1.6 eV. The measured valence-band offset is about 0.5 eV, which gives a Γ-point $\Delta E_c \approx 1.0$ eV. For the much-used $Al_{0.3}Ga_{0.7}As$/GaAs pair the accepted split is about 65:35 between conduction and valence band. In the early 1970s it was believed to be 85:15. That error was found only by careful optical measurement on quantum wells, and it changed laser and HEMT designs.
- **Ge/GaAs.** Ge and GaAs have almost the same lattice constant, so they would seem an ideal test pair. Anderson predicts a valence-band offset of about 0.8 eV, but measurements give 0.4–0.6 eV, depending on the crystal orientation of the interface and on how it was prepared. The dependence on orientation shows the cause directly. A (100) Ge/GaAs interface is *polar*: the Ga–Ge or As–Ge bonds across it carry a net charge, and how that charge is arranged sets δ.
- **Common-anion rule.** A related early rule held that compounds sharing an anion (GaAs and AlAs both have As) should have nearly equal valence-band maxima, because the valence band is formed mostly from anion p orbitals. That predicts $\Delta E_v \approx 0$ for GaAs/AlAs. The measured 0.5 eV refutes it, because cation d-orbitals and bond-length differences shift the valence band too.

The missing physics is that the electron states of the two crystals must match across the interface. The modern view, due mainly to Tersoff (1984), assigns each semiconductor a *charge-neutrality level* (CNL, also called the branch point). This is the energy in the gap at which the gap states created by the interface change from mostly valence-like (neutral when filled) to mostly conduction-like (neutral when empty). If the two CNLs were offset, charge would flow across the interface, and the resulting dipole would push them back into near-alignment. The screening is strong in semiconductors with high optical permittivity. To first order, the offset is therefore set by aligning the CNLs, not the vacuum levels. First-principles calculations, together with this picture, now predict offsets to about 0.1–0.2 eV.

A useful consequence is *transitivity*. If offsets are a property of each material's CNL, then for three materials

$$\Delta E_v(A/B) + \Delta E_v(B/C) + \Delta E_v(C/A) \approx 0$$

Transitivity holds to about 0.1 eV for most non-polar, lattice-matched systems. It fails where interface-specific chemistry dominates, such as polar interfaces, interdiffusion and strained layers. In those cases the offset depends on the growth sequence (A on B versus B on A) and on the growth conditions.

> **Status of the principle: Charge-neutrality-level alignment and transitivity**
> *Approximate.* It holds when the interface dipole is set mainly by screening of gap states rather than by specific bonding. It works well for lattice-matched, non-polar interfaces and is weaker for polar, strained or intermixed ones.

Type I, II and III alignments

The relative positions of the two gaps define three alignment types (Figure 34.1).

- **Type I (straddling).** The narrow gap lies wholly inside the wide gap, so electrons and holes are both confined in the narrow-gap material. Examples are GaAs/AlGaAs, InGaN/GaN and InGaAs/InP. This is the alignment needed for LEDs and lasers, where electrons and holes must recombine in the same place.
- **Type II (staggered).** Both band edges of one material are shifted the same way. Electrons collect on one side and holes on the other, so they are spatially separated. Recombination happens across the interface at an energy *smaller* than either band gap. Examples are Si/strained $Si_{1-x}Ge_x$ (holes in the SiGe, the offset mainly in the valence band) and $GaAs_{0.5}Sb_{0.5}$/InP. Type II is useful where long carrier lifetimes or separation of charge are wanted: InAs/GaSb-type superlattice infrared detectors, and charge separation in organic and perovskite solar absorbers.
- **Type III (broken gap).** The conduction-band minimum of one material lies *below* the valence-band maximum of the other. The classic case is InAs/GaSb, with an overlap of about 0.15 eV. Electrons move from the GaSb valence band into the InAs conduction band without any doping, so the interface is semimetallic. Broken-gap interfaces are used in tunnel field-effect transistors and in very-long-wavelength infrared superlattices, where the effective gap of the stack is set by layer thickness rather than by either material.

Table 34.1 collects representative offsets. Strained systems (SiGe on Si, InGaAs on GaAs) have offsets that depend on the strain as well as on composition. Strain splits the degenerate valence bands and shifts the conduction-band valleys (Chapter 31).

Table 34.1 — Representative band offsets at heterojunctions (300 K; typical literature values, ±0.05–0.1 eV)

Heterojunction (narrow / wide)	ΔE_g (eV)	ΔE_c (eV)	ΔE_v (eV)	Type	Main use
GaAs / $Al_{0.3}Ga_{0.7}As$	0.37	0.24	0.13	I	Lasers, HEMTs, HBTs
$In_{0.53}Ga_{0.47}As$ / InP	0.60	0.25	0.35	I	1.55 μm detectors, InP HBTs
$In_{0.53}Ga_{0.47}As$ / $In_{0.52}Al_{0.48}As$	0.71	0.50	0.21	I	InP HEMTs
GaN / $Al_{0.3}Ga_{0.7}N$	≈0.6	≈0.4	≈0.2	I	GaN HEMTs
$In_{0.15}Ga_{0.85}N$ / GaN	≈0.5	≈0.3	≈0.2	I	Blue LEDs
Si / strained $Si_{0.7}Ge_{0.3}$ (on Si)	≈0.2	≈0	≈0.2	I–II (weak)	pMOS channels, SiGe HBTs
InAs / GaSb	—	—	overlap ≈ 0.15	III	Tunnel FETs, IR superlattices
Si / SiO_2	7.9	3.1–3.2	4.5–4.7	I	Gate dielectric (Section 34.10)

The last row is also a heterojunction, between a semiconductor and an amorphous insulator. The same offset concepts govern gate dielectrics, and for them the offsets set the tunnelling leakage (Section 34.10).

34.3 Heterojunction interfaces: strain, mismatch and polarization

Lattice mismatch

Two crystals of the same structure but different lattice constants cannot both keep their natural spacing across a coherent interface. The *lattice mismatch* of a film (f) grown on a substrate (s) is

$$f = \frac{a_s - a_f}{a_f}$$

where a is the relaxed lattice constant. Some authors use a_s in the denominator. The difference is second order for mismatches of a few percent. Silicon (5.431 Å) and germanium (5.658 Å) differ by 4.2%. GaAs (5.653 Å) and InAs (6.058 Å) differ by 7.2%. Wurtzite GaN ($a = 3.189$ Å) on sapphire is mismatched by about 16% in magnitude (measured on the 30°-rotated in-plane lattice), and GaN on Si(111) by about 17%.

A thin film that is mismatched to its substrate can still grow *pseudomorphically*. Its in-plane lattice constant is forced to equal the substrate's, giving a biaxial strain $\varepsilon_{\parallel} = f$. Through the Poisson effect the film responds by changing its out-of-plane spacing in the opposite sense. For a cubic (001) film, with elastic constants C_{11} and C_{12},

$$\varepsilon_{\perp} = -\frac{2C_{12}}{C_{11}}\,\varepsilon_{\parallel}$$

so the cell becomes tetragonal. This distortion is what X-ray diffraction measures to find the strain (Chapter 52). The strain energy stored per unit interface area grows in proportion to film thickness h:

$$\frac{E_{\text{strain}}}{A} = M\, f^2\, h, \qquad M = \frac{E}{1-\nu}$$

where M is the biaxial modulus (Chapter 11). Once the stored energy is large enough, it becomes favourable to relieve it by introducing *misfit dislocations* at the interface. These are extra or missing half-planes whose Burgers-vector component in the interface plane takes up the mismatch (Chapter 5).

Critical thickness: the Matthews–Blakeslee criterion

Matthews and Blakeslee (1974) set up a force balance on a dislocation already present in the substrate. Such a dislocation is a *threading* dislocation, which runs up through the growing film to its surface. Two forces act on the segment threading the film:

- the **misfit stress** pushes the segment sideways, which would lay down a length of misfit dislocation along the interface and relieve strain. This force grows with film thickness h;
- the **line tension** of the new misfit segment resists, because creating dislocation line costs energy. That energy grows only logarithmically with h.

The film is stable against relaxation while line tension wins. Equating the two gives the critical thickness h_c:

$$h_c = \frac{b\,(1 - \nu\cos^2\alpha)}{8\pi\,|f|\,(1+\nu)\cos\lambda}\left[\ln\!\left(\frac{h_c}{b}\right) + 1\right]$$

Here b is the magnitude of the Burgers vector, ν is Poisson's ratio, α is the angle between the dislocation line and its Burgers vector, and λ is the angle between the Burgers vector and the in-plane direction normal to the line. In diamond and zinc-blende (001) films the common misfit dislocations are 60° dislocations with $b = a/\sqrt{2}$, so that $\alpha = 60^\circ$ and $\cos\lambda = 1/2$. The equation is implicit in h_c and is solved by iteration. Because h_c appears only inside a logarithm on the right, it scales roughly as $1/f$.

Status of the principle: Matthews–Blakeslee critical thickness
Approximate. It is derived from linear elasticity and an equilibrium force balance on an existing threading dislocation. It gives the thickness above which relaxation is *energetically favourable*. It assumes that dislocations are available and mobile, so it is a lower bound. Films grown at low temperature, where dislocation nucleation and glide are kinetically limited, can exceed it by a large factor in a *metastable* state (the People–Bean regime): several times at large mismatch, and more than an order of magnitude at small mismatch. Such films relax when heated later in processing.

Worked example — Critical thickness of $Si_{0.8}Ge_{0.2}$ on Si(001). By Vegard's law the relaxed lattice constant of the alloy is $a = 5.431 + 0.2(5.658 - 5.431) = 5.4764$ Å. The mismatch is $|f| = (5.4764 - 5.431)/5.431 = 8.36 \times 10^{-3}$. Take $b = a_{Si}/\sqrt{2} = 0.384$ nm and $\nu = 0.28$. For 60° dislocations, $\cos^2\alpha = 0.25$ and $\cos\lambda = 0.5$. The prefactor is

$$\frac{1 - 0.28 \times 0.25}{8\pi \times 8.36 \times 10^{-3} \times 1.28 \times 0.5} = \frac{0.930}{0.1345} = 6.92$$

so, with $y = h_c/b$, the equation becomes $y = 6.92(\ln y + 1)$. Iterating from $y = 30$: $6.92(3.401 + 1) = 30.45$, then $6.92(3.416 + 1) = 30.56$, converging to $y \approx 30.6$. Hence $h_c \approx 30.6 \times 0.384$ nm ≈ 11.7 nm. Repeating the calculation gives $h_c \approx 28$ nm at $x = 0.1$ and ≈ 7 nm at $x = 0.3$. The SiGe channel layers of strained-pMOS devices and the bases of SiGe HBTs (20–30% Ge, tens of nm thick) therefore usually sit *above* the equilibrium curve. They survive because they are grown at 500–650 °C and the later thermal budget is limited. This is a materials constraint on the process flow, and it is why a "metastable" label belongs on such a layer's specification.

For $In_{0.2}Ga_{0.8}As$ on GaAs ($|f| = 1.43\%$) the same formula gives $h_c \approx 6$ nm. Strained InGaAs quantum wells in 980 nm pump lasers are about 6–10 nm thick, at or just beyond this limit.

Figure 34.2 plots h_c against mismatch for both systems.

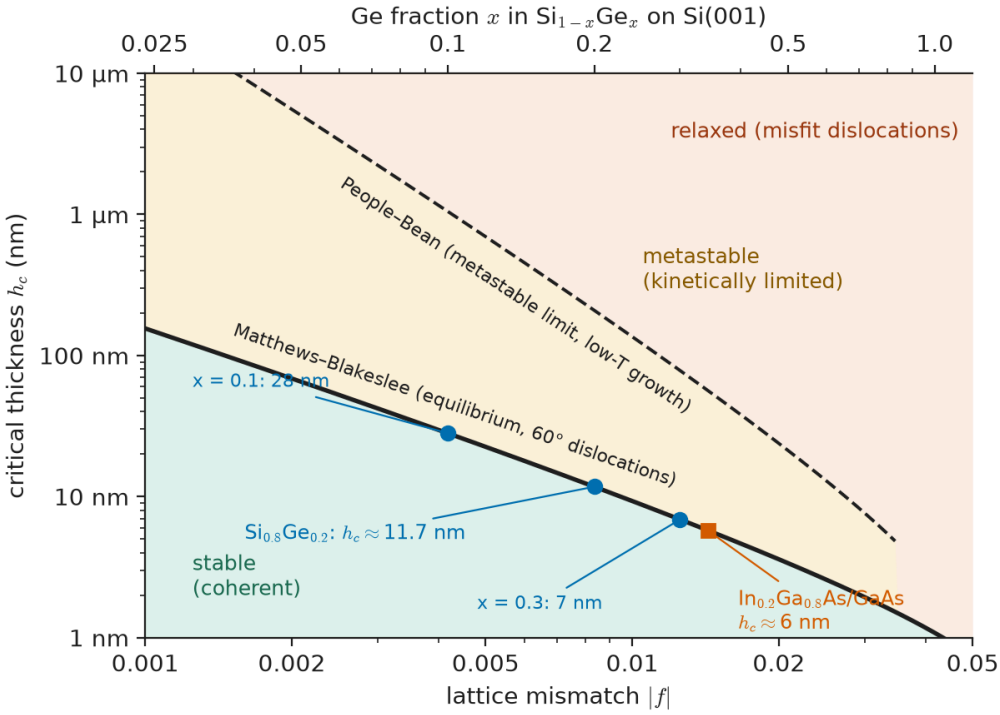


Figure 34.2 — Critical thickness versus lattice mismatch: Matthews–Blakeslee equilibrium curve and the metastable limit observed in low-temperature growth.

What relaxation leaves behind

Above h_c, misfit dislocations form at the interface. Their ends cannot terminate inside the crystal, so each misfit segment is joined to threading arms that run up through the film. Threading dislocations pass through the active layers above. In silicon they are recombination centres that degrade minority-carrier lifetime (Chapter 32). In GaAs lasers they grow into dark-line defects under optical pumping and kill the device within hours.

The engineering response is a set of *buffer* strategies:

- **Graded buffers.** Composition is ramped slowly, for example 10% Ge per μm, so that misfit dislocations are spread through the thickness and threading arms can glide out to the wafer edge. Relaxed SiGe "virtual substrates" made this way reach threading densities of about 10^5–10^6 cm^{-2}.
- **Strain-compensated stacks.** Layers in alternating compression and tension keep the average strain near zero, so a superlattice can be thicker in total than h_c for any one layer.
- **Heavily mismatched growth with defect filtering.** GaN on sapphire, SiC or Si is grown with nucleation layers (low-temperature GaN or AlN) and interlayers that bend and annihilate dislocations. Threading densities are still 10^8–10^9 cm^{-2}, 4–5 orders of magnitude worse than in GaAs. Blue LEDs still work at these densities, a puzzle taken up in Chapter 35.
- **Thermal-mismatch control.** GaN on Si(111) is grown at about 1000 °C. On cooling, GaN ($\alpha \approx 5.6\times10\$^{-6}\$$ K^{-1}) contracts more than Si ($2.6\times10\$^{-6}\$$ K^{-1}), so the film goes into tension and cracks. AlGaN interlayers are built in to pre-compress it. Stress is the subject of Section 34.7.

Polarization charge and the 2DEG at AlGaN/GaN

The wurtzite nitrides have no inversion symmetry along the *c*-axis. Each Ga–N bond along *c* is a dipole, and the four bonds of a tetrahedron do not cancel exactly, because the real cell differs slightly from an ideal tetrahedron. The crystal therefore has a *spontaneous polarization* P_{sp} even when unstrained. It is also piezoelectric, so strain adds a *piezoelectric polarization* P_{pz} (Chapter 27 introduced the piezoelectric coefficients e_{ij}).

Inside a uniform crystal a constant polarization produces no charge. Wherever P changes abruptly, however, a fixed sheet charge appears, as at the end faces of any polarized dielectric:

$$\sigma_{\text{pol}} = -\Delta P \cdot \hat{n}, \qquad \sigma_{\text{pol}} = P_{\text{GaN}} - P_{\text{AlGaN}} \ \text{(Ga-face, AlGaN on top)}$$

AlGaN has a larger spontaneous polarization than GaN. When it is grown thin on relaxed GaN it is under biaxial tension (its relaxed a is smaller), and its piezoelectric polarization then adds to the spontaneous part. For metal-face (Ga-face) growth both components point toward the substrate. The net result is a *positive* fixed sheet charge at the AlGaN/GaN interface. Electrons are drawn to it and collect in the triangular potential well on the GaN side, where the conduction-band offset ΔE_c holds them. This is the two-dimensional electron gas (2DEG), shown in Figure 34.3.

> **Status of the principle: Polarization sheet charge $\sigma = -\Delta P \cdot \hat{n}$**
> *Derived.* It follows from Gauss's law applied to the bound-charge density $\rho_b = -\nabla \cdot P$. The physics is exact; the uncertainty lies in the input values of P_{sp} and the piezoelectric constants, which are known only to about ±10–20%.

Where do the electrons come from? Not from dopants. The accepted source is *donor-like surface states* at the AlGaN free surface (Section 34.4). As the barrier thickens, the field in the AlGaN raises the surface states above the Fermi level. They then empty, and their electrons go to the interface. Below a critical barrier thickness (about 3–5 nm for $x = 0.3$) no 2DEG forms. The 2DEG density is set by the polarization charge, the barrier thickness d, and the surface barrier Φ_B:

$$n_s \approx \frac{\sigma_{\text{pol}}}{q} - \frac{\varepsilon_0 \varepsilon_r}{q^2 d}\left(q\Phi_B + E_F - \Delta E_c\right)$$

where E_F is measured from the GaN conduction-band edge at the interface.

> **Status of the principle: 2DEG density from charge balance**
> *Approximate.* It treats the AlGaN as a capacitor with a uniform field and the surface as pinned at Φ_B, and neglects the quantum-well subband structure. It predicts the trend with x and d well, and overestimates n_s by 10–40%.

Worked example — Sheet density of an Al$_{0.3}$Ga$_{0.7}$N/GaN 2DEG. Use linear interpolation between GaN and AlN for all parameters:

- spontaneous polarization P_{sp}: −0.029 C/m² (GaN), −0.081 C/m² (AlN), giving $P_{sp}(x = 0.3) = -0.0446$ C/m²;
- lattice constants 3.189 Å and 3.112 Å, so $a(0.3) = 3.1659$ Å and the in-plane strain is $\varepsilon = (3.189 - 3.1659)/3.1659 = 7.30 \times 10^{-3}$ (tensile);
- piezoelectric and elastic constants at $x = 0.3$: $e_{31} = -0.523$ C/m², $e_{33} = 0.949$ C/m², $C_{13} = 104.5$ GPa, $C_{33} = 395.4$ GPa.

The piezoelectric part is

$$P_{pz} = 2\varepsilon\left(e_{31} - e_{33}\frac{C_{13}}{C_{33}}\right) = 2(7.30\times10^{-3})(-0.523 - 0.251) = -0.0113\ \text{C/m}^2$$

The total AlGaN polarization is -0.0559 C/m², so $\sigma_{\text{pol}} = |-0.0559 - (-0.029)| = 0.0269$ C/m². Dividing by q gives 1.68×10^{17} m^{-2}, or 1.68×10^{13} cm^{-2}.

Now the charge-balance correction for $d = 25$ nm and $\varepsilon_r = 9.4$. The capacitance term is $\varepsilon_0\varepsilon_r/(qd) = 8.854\times10^{-12}\times 9.4/(1.602\times10^{-19}\times25\times10^{-9}) = 2.08\times10^{16}$ m^{-2} V^{-1}, which is 2.08×10^{12} cm^{-2} V^{-1}. Take $\Phi_B \approx 0.84 + 1.3x = 1.23$ V and $E_F \approx 0.2$ eV (the Fermi level above the GaN conduction-band edge at the interface, which must leave room for the first subband and its electrons). The gap of $Al_{0.3}Ga_{0.7}N$ is about 4.02 eV (with a bowing parameter of 1 eV), so $\Delta E_c \approx 0.7\times(4.02 - 3.42) = 0.42$ eV. The bracket is $1.23 + 0.2 - 0.42 = 1.01$ V, and the correction is 2.1×10^{12} cm^{-2}. Then

$$n_s \approx 1.68\times10^{13} - 0.21\times10^{13} \approx 1.5\times10^{13}\ \text{cm}^{-2}$$

Measured values for this structure are typically $(1.0\text{–}1.3)\times10^{13}$ cm^{-2}. The gap comes from partial strain relaxation, the non-linear (bowed) polarization of the alloy, and surface-state details. With a 2DEG mobility of 2000 cm²/V s and $n_s = 1.0\times10^{13}$ cm^{-2}, the sheet resistance is $R_{sh} = 1/(qn_s\mu) = 1/(1.602\times10^{-19}\times10^{17}\times0.2) \approx 310\ \Omega/\square$. No doped silicon layer of comparable thickness comes close, because its electrons would scatter off the ionized donors.

The materials consequences go further than the channel itself:

- **The surface sets the channel.** Because the electrons come from surface states, anything that changes the surface changes n_s. A dielectric passivation layer, adsorbed water, or charge trapped under the gate under high-voltage stress can all do so. The resulting reduction of drain current after a voltage pulse is called *current collapse* and is a surface-state and trap phenomenon. Silicon nitride passivation and field plates are its standard remedies.
- **Normally-on by nature.** The 2DEG is present at zero gate bias. Power switches need normally-off behaviour, which calls for materials changes: a p-GaN layer under the gate, a recessed barrier, or fluorine implantation.

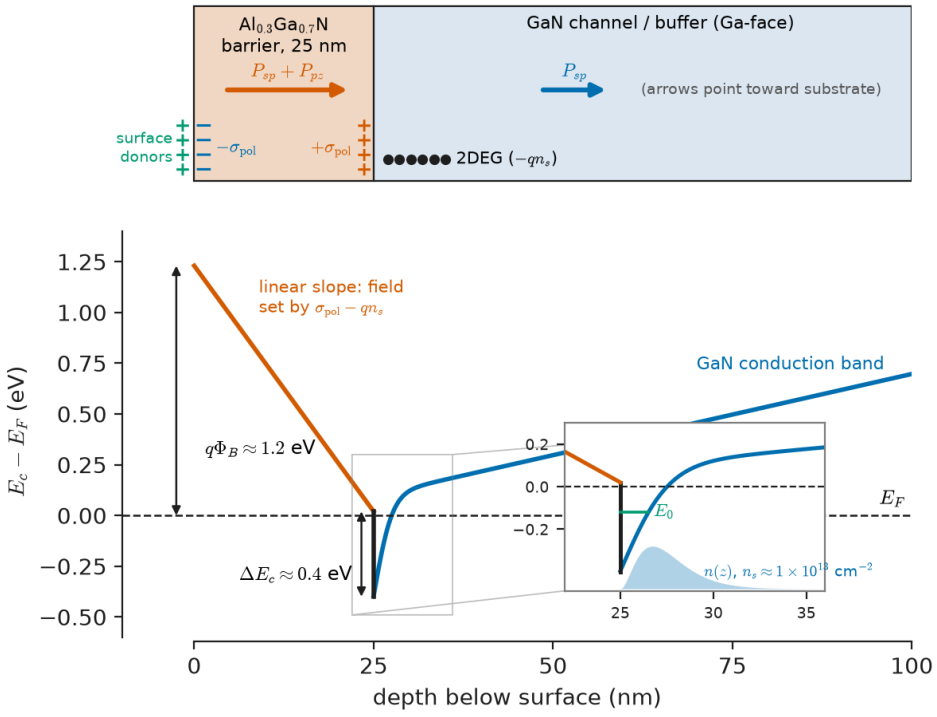


Figure 34.3 — Polarization charges and the conduction band at a Ga-face AlGaN/GaN interface.

34.4 Surface states

Why a surface has states in the gap

A semiconductor's band gap exists because every atom is bonded to its neighbours. In silicon each atom forms four sp³ bonds, and the bonding–antibonding splitting opens the gap (Chapter 31, and the Advanced Chemistry core for the bonding itself). Cut the crystal and the surface atoms lose neighbours. Each lost neighbour leaves a *dangling bond*: a half-filled orbital whose energy lies roughly midway between the bonding and antibonding levels, which is inside the gap. A (001) silicon surface has $2/a^2 = 6.8 \times 10^{14}$ atoms per cm², each with two dangling bonds when unreconstructed. That gives more than 10^{15} states per cm². For comparison, a bulk dopant density of 10^{17} cm^{-3} corresponds to only about 10^{12} charges per cm² in an 80 nm depletion layer.

Real surfaces lower their energy by *reconstruction*. On Si(001), neighbouring surface atoms pair into dimers, which halves the dangling-bond count and pushes most surface states toward the band edges. On cleaved GaAs(110), the surface relaxes so that filled As states drop into the valence band and empty Ga states rise into the conduction band. That leaves the clean (110) gap almost free of states, a celebrated exception. In practice, almost every surface that a device actually sees (oxidized, contaminated, metallized or etched) has defect-related states spread through the gap. Historically these are classified as *intrinsic* surface states (Tamm and Shockley states, from the termination of the periodic potential) and *extrinsic* states (from adsorbates, vacancies, antisites and disorder). For an engineer the distinction matters less than their density and energy distribution.

Charge neutrality and band bending at a surface

Each surface state is either donor-like (neutral when filled, positive when empty) or acceptor-like (neutral when empty, negative when filled). As in Section 34.2, the surface states have a *charge-neutrality level* E_{CNL}. If the Fermi level at the surface lies above E_{CNL}, the surface carries net negative charge. If it lies below, the charge is net positive.

Consider an n-type semiconductor whose bulk Fermi level lies near E_c, well above E_{CNL}. Electrons move from the bulk into the surface states, which charges the surface negatively. That leaves a positive depletion region of ionized donors beneath it, and the bands bend upward toward the surface. Equilibrium is reached when

$$Q_{ss} + Q_{sc} = 0$$

where Q_{ss} is the surface-state charge and Q_{sc} is the space charge in the semiconductor. The more surface states there are, the less the surface Fermi level has to move away from E_{CNL} to supply the charge the bulk demands.

Fermi-level pinning

Fermi-level pinning is the limit in which the surface-state density is so large that the surface Fermi level hardly moves from E_{CNL}, whatever the bulk doping or the metal deposited on top. The same pinning shows up at the metal–semiconductor contacts of Section 34.9 and at the high-k/metal-gate interfaces of Section 34.10.

Worked example — How many states does it take to pin the Fermi level? Take n-type silicon with $N_D = 10^{17}$ cm^{-3} and suppose the surface must accommodate a band bending of 0.5 V. The depletion width is

$$W = \sqrt{\frac{2\varepsilon_s \psi_s}{q N_D}} = \sqrt{\frac{2 \times 11.7 \times 8.854 \times 10^{-12} \times 0.5}{1.602 \times 10^{-19} \times 10^{23}}} = 80 \text{ nm}$$

and the space charge is $N_D W = 10^{17} \times 8.0 \times 10^{-6} = 8.0 \times 10^{11}$ cm^{-2}. Surface states must hold the equal and opposite charge. If their density near E_{CNL} is $D_{ss} = 10^{14}$ cm^{-2} eV^{-1}, they can absorb 8×10^{11} cm^{-2} with a Fermi-level shift of only $8 \times 10^{11}/10^{14} = 8$ meV. The surface Fermi level is effectively locked to E_{CNL}. Changing the bulk doping by a factor of ten would barely move it. If instead $D_{ss} = 10^{10}$ cm^{-2} eV^{-1} (well-passivated Si/SiO$_2$), the same charge would need an 80 eV shift, which is impossible. The surface is then free, and the band bending is controlled by the gate or bulk, as a MOSFET requires.

The rough dividing line, for typical doping, is

- $D_{ss} \gtrsim 10^{13}$ cm^{-2} eV^{-1}: strongly pinned;
- $D_{ss} \lesssim 10^{11}$ cm^{-2} eV^{-1}: essentially unpinned.

The difference between silicon and the III–V semiconductors largely comes down to which side of this line their oxidized surfaces fall on:

- **Si/SiO_2.** The thermally grown interface, annealed in hydrogen, has $D_{it} \approx 10^{10}$ cm^{-2} eV^{-1} near midgap. This was the single fact that made the silicon MOSFET possible in the 1960s, and it was won by materials processing, not device design.
- **GaAs native oxides.** The surface Fermi level of oxidized GaAs is pinned about 0.7–0.8 eV below E_c, near midgap, by states associated with As antisites, As–As bonds and Ga oxides. This is why a GaAs MOSFET with a native oxide never worked, and why GaAs electronics used MESFETs and HEMTs (Schottky gates) instead. Atomic-layer-deposited Al_2O_3 and HfO_2 on carefully cleaned InGaAs now reach $D_{it} \approx 10^{11}$–10^{12} cm^{-2} eV^{-1}, enough for research-grade III–V MOSFETs.
- **GaN and SiC.** The SiC/SiO_2 interface has D_{it} rising to above 10^{12}–10^{13} cm^{-2} eV^{-1} near the conduction band of 4H-SiC. The causes are carbon clusters and near-interface oxide traps. These states trap channel electrons and limit the inversion mobility of SiC MOSFETs to about 20–50 cm²/V s, far below the bulk 950 cm²/V s. Nitridation anneals (NO, N_2O) were the key improvement. The SiC MOSFET's channel resistance is thus an interface-materials problem.

Passivation

Passivation means reducing the electrically active surface-state density chemically. The main routes are:

- **Hydrogen.** H atoms bond to dangling bonds and move the bonding and antibonding levels out of the gap. Forming-gas anneals (N_2 + 5–10% H_2 at about 400–450 °C) cut Si/SiO_2 D_{it} by roughly two orders of magnitude.
- **Growing the right oxide.** Thermal SiO_2 grows *into* silicon, so the interface is always fresh and never exposed to air. That makes it smoother and cleaner than any deposited interface.
- **Wide-gap capping.** On solar cells, Al_2O_3 passivates p-type Si by combining chemical passivation with a built-in negative fixed charge of about 10^{12}–10^{13} cm^{-2}. The charge repels minority electrons from the surface (*field-effect passivation*). Silicon nitride carries positive charge and serves n-type surfaces. Surface recombination velocities below 10 cm/s are achieved. This is the materials basis of PERC and TOPCon cells (Chapter 40).

34.5 Interface traps

Definition and the quantity D_{it}

At a semiconductor–insulator interface, the surface states of Section 34.4 are called *interface traps*. They are located at the interface itself and can exchange charge with the semiconductor quickly (in nanoseconds to milliseconds) as the band bending changes. Their density per unit area per unit energy is

$$D_{it}(E) = \frac{1}{q}\frac{dQ_{it}}{dE} \qquad [\mathrm{cm^{-2}\,eV^{-1}}]$$

They must be distinguished from three other kinds of charge in a gate stack, which have different origins and different remedies:

Charge type	Symbol	Location	Origin	Exchanges charge with Si?
Interface traps	D_{it}, Q_{it}	At the interface	Dangling bonds (P_b centres), strained bonds	Yes, fast
Fixed oxide charge	Q_f	Within ~2 nm of the interface	Incompletely oxidized Si, structural	No
Oxide trapped charge	Q_{ot}	Throughout the oxide	Electrons or holes caught in bulk oxide traps (stress, radiation)	Slowly or not
Mobile ionic charge	Q_m	Moves through oxide with field and temperature	Na^+, K^+ contamination	No (drifts)

Traps located 1–3 nm into the oxide that *can* exchange charge with the silicon by tunnelling, slowly, are called *border traps*. They blur the line between Q_{it} and Q_{ot} and cause much of the low-frequency (1/f) noise of MOSFETs.

At the Si/SiO_2 interface the main interface trap is the **P_b centre**: a silicon atom at the interface bonded to three silicon atoms and missing its oxygen neighbour, leaving one dangling bond. Its electron spin makes it visible in electron paramagnetic resonance, which is how it was identified. On (100) silicon there are two variants, P_{b0} and P_{b1}. P_b centres are amphoteric. Each

gives two levels in the gap, roughly 0.3 eV above E_v and 0.3 eV below E_c, and together they produce the U-shaped $D_{it}(E)$ seen by measurement. Representative values are listed in Table 34.2.

Table 34.2 — Typical midgap interface-trap densities (cm^{-2} eV^{-1})

Interface	As-formed	After best passivation
Si(100)/thermal SiO_2	10^{11}–10^{12}	$\approx 10^{10}$ (H_2 anneal)
Si(111)/thermal SiO_2	~3× higher than (100)	~3×10^{10}
Si/SiO_x/HfO_2 (high-k stack)	10^{11}–10^{12}	$\approx (1\text{–}5) \times 10^{10}$
4H-SiC/SiO_2 (near E_c)	$> 10^{13}$	$\sim 10^{12}$ (NO anneal)
$In_{0.53}Ga_{0.47}As/Al_2O_3$	10^{12}–10^{13}	$\sim 10^{11}$–10^{12}
GaAs/native oxide	$> 10^{13}$ (pinned)	—

The preference for (100) silicon in CMOS comes straight from this table. It has fewer interface bonds per unit area than (111), so fewer of them are strained or broken.

What D_{it} does to a device

Interface traps act as a capacitance $C_{it} = q^2 D_{it}$ (per unit area, per unit voltage of surface potential) in parallel with the depletion capacitance. When the gate voltage changes, part of the induced charge goes into the traps instead of moving the surface potential. The swing in the subthreshold region (Electronics & Computing derives the device expression) becomes

$$S = \ln(10)\,\frac{kT}{q}\left(1 + \frac{C_{dep} + C_{it}}{C_{ox}}\right)$$

As a materials estimate, take $D_{it} = 10^{12}$ cm^{-2} eV^{-1}, which gives $C_{it} = 1.602 \times 10^{-19} \times 10^{12} = 1.6 \times 10^{-7}$ F/cm², or 0.16 µF/cm². A 2 nm SiO_2 gate has $C_{ox} = 3.9\varepsilon_0/t = 1.73$ µF/cm². Ignoring C_{dep}, the swing rises from 59.5 mV/decade to $59.5 \times 1.093 = 65$ mV/decade. At $D_{it} = 10^{13}$ the factor would be 1.93 and the swing about 115 mV/decade. At 10^{10} the effect is negligible. That is the sense in which 10^{10} is "good enough" and 10^{12} is not.

Interface traps also:

- **scatter carriers**: charged traps reduce channel mobility by Coulomb scattering, the SiC MOSFET problem of Section 34.4;
- **shift threshold voltage** as they fill. Because their occupation depends on bias and history, this causes instability, the subject of Section 34.11;
- **cause recombination**: at a surface, traps near midgap give surface recombination velocity $s = \sigma v_{th} N_{it}$, the surface counterpart of SRH bulk recombination (Chapter 32). This matters for solar cells and image sensors.

Measuring D_{it}

D_{it} is measured electrically (Chapter 52 treats the instruments). The *high–low frequency C–V* method compares a 1 MHz capacitance curve, where traps cannot follow the signal, with a quasi-static one, where they can; the difference is C_{it}. The *conductance method* detects the energy lost as traps capture and emit carriers, as a peak in G_p/ω against frequency, and resolves densities down to about 10^9 cm^{-2} eV^{-1}. On small transistors, *charge pumping* cycles the gate between accumulation and inversion so that each trap captures an electron and then a hole every cycle, giving a substrate current $I_{cp} = qfAN_{it}$.

34.6 Thin-film properties

Why thin films are different materials

Almost every functional layer in an integrated circuit, display, LED or solar cell is a thin film, from a few nanometres to a few micrometres thick: gate dielectrics, metal lines, barrier layers, transparent conductors, silicide contacts, polysilicon gates and

epitaxial quantum wells. A thin film of a given composition generally does *not* have the bulk properties of that composition. Four structural features cause the difference.

1. **A large surface-to-volume ratio.** A 10 nm copper film has two interfaces only about 40 atomic layers apart. Electrons scatter at both. The *size effect* in resistivity (Fuchs–Sondheimer for surfaces, Mayadas–Shatzkes for grain boundaries) was treated in Chapter 25. It raises copper's resistivity from 1.7 μΩ cm in bulk to 3–5 μΩ cm in 20 nm lines, and it is a central problem of interconnect scaling.
2. **Non-equilibrium microstructure.** Films are deposited far from equilibrium, by atoms arriving one at a time at a surface usually below a third of the material's melting temperature (Chapter 49 treats deposition processes). They therefore contain vacancies, voids, small grains, metastable phases and trapped gas that a bulk sample would anneal away.
3. **Substrate constraint.** The film is bonded to a substrate thousands of times thicker. It cannot change its in-plane dimensions freely, so any change in its natural size produces stress (Section 34.7). Thermal expansion, densification, phase change and epitaxial mismatch are all examples.
4. **Texture.** Grains often share a preferred orientation, for example (111) in fcc metals or (002) in wurtzite ZnO and AlN. Properties that depend on direction, such as the piezoelectric coefficient of AlN in RF filters or electromigration resistance in copper, then depend on texture as well as on composition.

Representative property shifts

Table 34.3 — Bulk versus thin-film properties (typical; film values depend strongly on deposition)

Material	Property	Bulk	Thin film (typical)	Main cause
Cu	Resistivity (μΩ cm)	1.68	2.2 (100 nm), 3–5 (20 nm lines)	Surface and grain-boundary scattering
W (CVD/PVD)	Resistivity (μΩ cm)	5.3	8–15 (β-phase up to 100+)	Metastable β-W, impurities
SiO_2 (PECVD vs thermal)	Wet etch rate in HF (relative)	1 (thermal)	3–10	Lower density, Si–H and Si–OH bonds
SiO_2	Breakdown field (MV/cm)	—	10–14 (thermal), 5–8 (PECVD)	Defects, density
Si_3N_4 (LPCVD vs PECVD)	Stress (MPa)	—	+1000 to +1200 (LPCVD); −300 to +300 (PECVD, tunable)	Hydrogen content, ion bombardment
ITO	Resistivity (Ω cm)	—	$(1–3) \times 10^{-4}$	O vacancies and Sn donors; grain structure
AlN (sputtered, c-axis)	d_{33} (pm/V)	5.5	3–5.5	Texture quality (rocking-curve width)
Al (pure)	Yield strength (MPa)	10–35 (annealed)	150–400	Hall–Petch, small grains, substrate constraint

The last row is a general rule. Thin metal films are much stronger than bulk metal of the same composition because dislocations are confined by the film thickness and by small grains. This "smaller is stronger" behaviour is why films can sustain the large stresses of Section 34.7 without yielding.

34.7 Film stress and defects

Origins of stress

The total stress in a film is usually written as

$$\sigma_{\text{total}} = \sigma_{\text{th}} + \sigma_{\text{int}} + \sigma_{\text{ext}}$$

Each term has a distinct origin.

Thermal stress arises because the film is deposited or annealed at temperature T_d, and on cooling it would contract by a different amount from the substrate. For a thin film on a thick substrate,

$$\sigma_{\text{th}} = \frac{E_f}{1-\nu_f}\,(\alpha_s - \alpha_f)\,(T_d - T)$$

A film with larger expansion coefficient than the substrate ends up in tension at room temperature (Chapter 24).

> **Status of the principle: Thermal-mismatch stress**
> *Approximate*. It assumes linear elasticity, temperature-independent α and E, a film much thinner than the substrate, and no stress relaxation (yielding, creep, cracking) during cooling. Metal films often violate the last assumption.

Worked example — Aluminium on silicon. An aluminium interconnect film is annealed at 400 °C and cooled to 25 °C. With $E_f = 70$ GPa, $\nu_f = 0.35$, $\alpha_f = 23.1 \times 10^{-6}$ K^{-1} and $\alpha_s = 2.6 \times 10^{-6}$ K^{-1}:

$$\sigma_{\text{th}} = \frac{70\ \text{GPa}}{0.65} \times (20.5 \times 10^{-6}\ \text{K}^{-1}) \times 375\ \text{K} = 107.7\ \text{GPa} \times 7.69 \times 10^{-3} = 828\ \text{MPa}$$

That is tensile and exceeds even a thin-film yield strength of 200–400 MPa. The film therefore yields plastically on cooling and ends near its yield stress. On reheating it goes into compression and relieves the stress by pushing material up through grain boundaries as *hillocks*, which can short to the level above. Under tension and current, voids form instead. Copper (α = 16.5×10^{-6} K^{-1}) halves the mismatch, one of several materials reasons for the switch to copper (Chapter 51 and Chapter 44 continue the story).

Intrinsic (growth) stress develops during deposition itself:

- **Tensile intrinsic stress** comes from grain-boundary "zipping". As neighbouring islands touch, they snap together to remove surface energy, stretching the film. Vacancy annihilation and densification after growth add to it. This dominates in evaporated films and films grown at low adatom mobility.
- **Compressive intrinsic stress** comes from energetic particle bombardment in sputtering and PECVD. Ions and fast neutrals hammer atoms into interstitial positions (*atomic peening*), and noble gas (Ar) is trapped. Raising the sputter pressure scatters the arriving atoms and reduces their energy, so stress can be tuned from compressive to tensile. This is the standard knob.
- **Chemical and phase changes**: the loss of hydrogen from PECVD nitride on annealing, crystallization of amorphous films, and silicide formation. Silicidation consumes silicon and changes volume, giving GPa-level stress.

Epitaxial (coherency) stress is the mismatch strain of Section 34.3, $\sigma = Mf$. For the $Si_{0.8}Ge_{0.2}$ example, $M \approx 180$ GPa and $f = 0.84\%$, giving about 1.5 GPa of compression, sustained without plasticity because the layer is below h_c.

Measuring stress: the Stoney equation

Stress in a film cannot be measured directly. Its effect on the substrate can. A film in tension tries to shrink and curls the wafer so that the film side becomes concave. A film in compression makes the film side convex. Stoney (1909) related the change in curvature to the film stress for a thin film on a thick plate:

$$\sigma_f = \frac{E_s}{1-\nu_s}\,\frac{h_s^2}{6\,h_f}\left(\frac{1}{R} - \frac{1}{R_0}\right)$$

Here h_s and h_f are the substrate and film thicknesses, $E_s/(1-\nu_s)$ is the substrate's biaxial modulus, and R_0 and R are the radii of curvature before and after deposition. The film's elastic constants do not appear, which is the equation's great practical virtue. The film can be anything (amorphous, textured, unknown), and only the substrate needs to be characterized.

> **Status of the principle: Stoney equation**
> *Approximate*. It is derived from plate bending for a film much thinner than the substrate ($h_f/h_s \lesssim 0.01$; the error is a few percent at $h_f/h_s \approx 0.05$), uniform equibiaxial stress, an isotropic and homogeneous substrate, and deflections much smaller than the substrate thickness. Corrections exist for thick films and anisotropic substrates. For (001) silicon the biaxial modulus is isotropic in the plane and equals 180.5 GPa.

The curvature is measured by scanning a laser across the wafer and tracking the reflected spot angle, or by interferometry. Measuring before and after deposition removes the wafer's initial bow (R_0).

Worked example — Stress in a PECVD silicon nitride film. A 200 mm Si(001) wafer, 725 μm thick, is flat before deposition ($1/R_0 = 0$). After 400 nm of PECVD silicon nitride is deposited, the wafer bows convex on the film side with $R = 150$ m. Then

$$\sigma_f = \frac{180.5 \times 10^9 \times (725 \times 10^{-6})^2}{6 \times 400 \times 10^{-9} \times 150} = \frac{9.49 \times 10^4}{3.60 \times 10^{-4}}\ \text{Pa} = 2.64 \times 10^8\ \text{Pa}$$

The stress is about 264 MPa, compressive because the curvature is convex. The centre-to-edge bow over the 100 mm radius is $\delta = r^2/2R = (0.1)^2/300 = 33$ μm. That is large enough to trouble lithographic focus and wafer handling, which is why stress specifications for each layer exist. The film force per unit width, $\sigma_f h_f = 106$ N/m, is the quantity that adds layer by layer. A stack of compressive and tensile films can be balanced to a flat wafer, a routine part of MEMS and backside-power process design.

Strain is not always a nuisance. Since the 90 nm node, CMOS has *deliberately* stressed the channel to raise mobility. Tensile nitride liners stretch nMOS channels, and embedded SiGe source/drain regions compress pMOS channels. Strain changes the band structure, splitting valleys and reducing the hole effective mass along the channel (Chapter 31), and can raise hole mobility by 50% or more. The materials property exploited is piezoresistance, the same effect used in silicon strain gauges (Chapter 35).

Film defects

Beyond stress, thin films carry characteristic defects. Each has a structural origin and an electrical consequence.

- **Pinholes** are through-thickness holes from particles or incomplete coalescence. In dielectrics they cause shorts and early breakdown. Their density scales roughly with deposited area and inversely with thickness.
- **Voids and porosity** occur in low-mobility growth (Section 34.8). They raise resistivity, take up moisture (dielectric constant and leakage rise), and act as nucleation sites for electromigration voids.
- **Hillocks and whiskers** are compressive-stress relief by diffusion of material to the surface. Tin whiskers from pure-Sn finishes are the notorious packaging case (Chapter 51).
- **Cracking and delamination.** A tensile film cracks when its stored elastic energy per unit area, $\sigma^2 h_f (1-\nu_f)/E_f$ up to a geometry factor, exceeds the fracture energy Γ (Chapter 14). This sets a *critical cracking thickness* $h \propto E_f \Gamma / \sigma^2$, the fracture counterpart of Matthews–Blakeslee. Compressive films buckle and delaminate in "telephone-cord" patterns if adhesion is weak. Adhesion layers (Ti, Cr, Ta) under noble metals exist for this reason.
- **Contamination and non-stoichiometry**: hydrogen in PECVD films, oxygen in sputtered metals, nitrogen or oxygen vacancies in oxide and nitride films. These directly set leakage, trap density and optical absorption.

34.8 Grain boundaries in thin films

How film microstructure forms

Most films grow by *nucleation, island growth, coalescence and thickening*. Atoms land, diffuse a distance set by temperature and the binding energy to the surface, meet others to form nuclei, and the islands grow until they touch. What follows depends overwhelmingly on how far an arriving atom can move before it is buried, which scales with the *homologous temperature* T_s/T_m (substrate temperature over melting temperature, both in kelvin). The empirical zone models of Movchan–Demchishin (1969, evaporated films) and Thornton (1974, sputtered films, which adds sputter-gas pressure as a second axis) summarize the structures. Figure 34.4 shows the zones.

- **Zone 1** ($T_s/T_m \lesssim 0.3$): adatoms barely move, and shadowing by taller grains produces tapered columns separated by voids. The films are porous and under tensile stress, with high resistivity and poor barrier properties. Sputtered at high pressure, even refractory metals look like this.
- **Zone T** (transition, sputtering only, at low pressure and $T_s/T_m \approx 0.2$–0.4): ion bombardment densifies the film into fine fibrous grains with dense boundaries. Most sputtered barrier and electrode films (TiN, TaN, Mo) are deliberately in Zone T.
- **Zone 2** ($0.3 \lesssim T_s/T_m \lesssim 0.5$): surface diffusion is active, giving dense columnar grains whose width grows with T_s and spans the film thickness. Grain boundaries run perpendicular to the substrate, along the current path in a vertical device but across the current path in a lateral line.

- **Zone 3** ($T_s/T_m \gtrsim 0.5$): bulk diffusion and recrystallization give equiaxed grains approaching bulk microstructure.

Melting temperatures put common EEE films into very different zones at the same substrate temperature. At 400 °C (673 K), aluminium ($T_m = 933$ K) is at 0.72 (Zone 3), copper (1358 K) at 0.50, tungsten (3695 K) at 0.18 (Zone 1/T), and TiN (3203 K) at 0.21.

> **Status of the principle: Structure-zone model**
> *Empirical.* It is a classification of observed microstructures by homologous temperature and, for sputtering, gas pressure and ion energy. The zone boundaries are approximate (±0.05–0.1 in T_s/T_m) and shift with deposition rate and impurities, but the scaling with T_s/T_m reflects real diffusion-controlled kinetics.

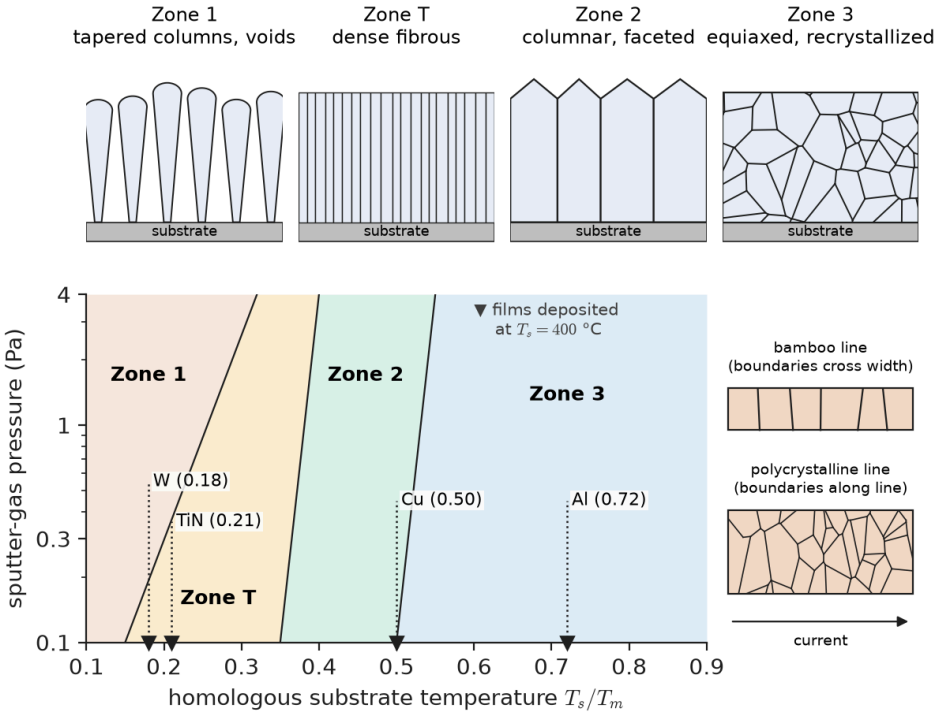


Figure 34.4 — Structure-zone model of thin-film growth: microstructure versus homologous substrate temperature.

Grain boundaries as electrical and transport features

Grain boundaries in thin films matter for three reasons, each continuing threads from Chapters 5 and 25.

1. They scatter electrons. In metal lines whose grain size is comparable with the electron mean free path (39 nm in Cu at 300 K), grain-boundary reflection adds a resistivity term that grows as the grain size D shrinks (Mayadas–Shatzkes, Chapter 25). Copper interconnects are electroplated with small grains, then *self-anneal* at room temperature over hours to days, or are annealed at 100–400 °C, to grow large grains and lower resistivity by 10–20%.

2. They are fast diffusion paths. Atomic diffusion along grain boundaries has an activation energy roughly half that of lattice diffusion. In thin films this has two consequences:

- **Barrier failure.** Copper diffuses through grain boundaries in a columnar barrier film into the silicon, where it is a deep-level lifetime killer. Barriers are therefore made amorphous or nanocrystalline (TaN, TiN, Ta/TaN bilayers) so that no continuous straight boundary path crosses them.
- **Electromigration.** Under current, metal atoms drift along boundaries. When the grain size exceeds the line width, the boundaries run *across* the line in a *bamboo structure*, and no continuous boundary path runs along the line. Electromigration lifetime rises dramatically. Chapter 44 develops the failure model (Black's equation).

3. They trap charge in semiconductors. In polycrystalline silicon, oxide semiconductors and polycrystalline absorbers, the grain boundary carries a sheet of trap states, like a surface (Section 34.4) shared between two grains. Carriers trapped there

leave depletion regions on both sides, raising a potential barrier V_B at the boundary. In the simplest (Seto) model, for grain size L and doping N, while the traps are not yet full:

$$V_B = \frac{qNL^2}{8\varepsilon_s}$$

For poly-Si with $L = 100$ nm and $N = 10^{17}$ cm^{-3}: $V_B = 1.602 \times 10^{-19} \times 10^{23} \times (10^{-7})^2/(8 \times 1.036 \times 10^{-10}) = 0.19$ V, about $7.5\,kT/q$. Current across each boundary must surmount it, so the effective mobility is reduced by roughly $e^{-7.5} \approx 5 \times 10^{-4}$ relative to single crystal. Once NL exceeds the trap sheet density N_t (typically 10^{12}–10^{13} cm^{-2}), the traps fill and the barrier falls as $V_B = qN_t^2/(8\varepsilon_s N)$. This makes the resistivity of poly-Si a steep, non-monotonic function of doping, which is how poly-Si resistors are given a near-zero temperature coefficient by choosing the doping where the two effects balance.

> **Status of the principle: Seto grain-boundary barrier model**
> *Approximate.* It assumes a single trap level, uniform grain size, abrupt depletion and thermionic transport over the barrier. It captures the order of magnitude and the resistivity peak against doping, but real boundaries have a distribution of trap energies.

The same barrier physics is the working principle of metal-oxide gas sensors, where adsorbed oxygen at grain surfaces sets the barrier (Chapter 35), and it is one reason polycrystalline thin-film solar absorbers need grain-boundary passivation ($CdCl_2$ treatment of CdTe, Chapter 40).

34.9 Contact interfaces

The problem

Every semiconductor device must eventually pass its current into a metal. The metal–semiconductor interface can behave as a rectifying diode (*Schottky contact*) or as a near-linear low-resistance path (*ohmic contact*). Which one appears, and how good it is, is decided by materials: the barrier height at the interface and the doping beneath it. At advanced CMOS nodes the source/drain contact resistance has become one of the largest resistances in the transistor. Contact engineering is now as much a materials problem as interconnect engineering.

Barrier height: the Schottky–Mott rule

The Schottky–Mott picture is Anderson's rule applied to a metal. A metal of work function ϕ_m placed against an n-type semiconductor of electron affinity χ should form an electron barrier

$$\phi_{Bn} = \phi_m - \chi$$

and, for a p-type semiconductor, a hole barrier $\phi_{Bp} = E_g/q - (\phi_m - \chi)$, so that $\phi_{Bn} + \phi_{Bp} = E_g/q$. A metal with a large work function (Pt, 5.65 eV) on n-Si ($\chi = 4.05$ eV) should give a 1.6 eV barrier. A low-work-function metal (Al, 4.28 eV) should give 0.23 eV. In practice the barrier heights on n-Si span only about 0.5–0.9 eV, and on n-GaAs nearly *every* metal gives 0.7–0.9 eV.

> **Status of the principle: Schottky–Mott rule**
> *Approximate.* It ignores the interface dipole and the interface states. It is followed closely only by very ionic semiconductors and insulators (SiO_2, ZnS, and to a degree GaN and SiC), and fails for covalent semiconductors such as Si, Ge and GaAs, where Fermi-level pinning dominates.

Fermi-level pinning at contacts

The barrier is pinned for the same reasons as at a free surface (Section 34.4), with one added mechanism. The metal's conduction-electron wavefunctions decay into the semiconductor gap over a few ångströms, creating **metal-induced gap states** (MIGS) even at a perfect, defect-free interface. These states have a charge-neutrality level, as before. If the metal Fermi level lies away from E_{CNL}, charge flows into the MIGS and the resulting dipole pulls the barrier back toward the CNL value. Defects created by the metallization itself (vacancies, antisites, interfacial reaction products) add further pinning.

The degree of pinning is expressed by the slope parameter

$$S = \frac{d\phi_{Bn}}{d\phi_m}$$

$S = 1$ is the Schottky–Mott limit (no pinning) and $S = 0$ is complete pinning. A phenomenological model gives

$$\phi_{Bn} = S(\phi_m - \chi) + (1 - S)\,(E_g/q - \phi_{CNL})$$

with ϕ_{CNL} the CNL measured from the valence band. An empirical correlation (Kurtin, McGill and Mead, 1969; refined by Mönch and by Robertson) links S to the semiconductor's optical permittivity ε_∞, through $S \approx 1/[1 + 0.1(\varepsilon_\infty - 1)^2]$. Covalent semiconductors with large ε_∞ screen strongly and are strongly pinned. Figure 34.5 plots measured barrier heights against metal work function.

> **Status of the principle: Pinning factor S and its correlation with ε_∞**
> *Empirical.* It is fitted to barrier-height data across many semiconductors. It captures the trend from ionic to covalent materials within about ±0.1 in S, but individual contacts depend on interface chemistry and processing.

Table 34.4 — Representative Schottky barrier heights (eV, electron barrier on n-type unless noted; ±0.05–0.1 depending on preparation)

Metal / silicide	ϕ_m (eV)	n-Si	n-GaAs	n-GaN	n-4H-SiC
Al	4.28	0.72	0.80	≈0.6	—
Ti	4.33	0.50–0.60	0.83	≈0.5–0.6	0.8–1.1
W	4.55	0.67	0.80	≈0.7	—
Ni	5.15	0.61–0.67	0.84	0.9–1.1	1.3–1.6
Au	5.1	0.80	0.90	0.9–1.1	1.6–1.8
Pt	5.65	0.85–0.90	0.86	1.0–1.2	≈1.6
NiSi	—	0.65	—	—	—
PtSi	—	0.84 (0.24 on p-Si)	—	—	—
Slope S (typical)		0.1–0.3	≈0.07	0.3–0.4	≈0.4–0.5

The barrier on n-GaAs varies by only 0.1 eV over a 1.4 eV range of ϕ_m. GaAs is the textbook case of near-complete pinning. Wide-gap and more ionic materials (GaN, SiC) show larger S, so the metal matters more. That is why Ni and Pt are chosen for Schottky gates on GaN and for SiC power Schottky diodes, and Ti for low-barrier SiC contacts.

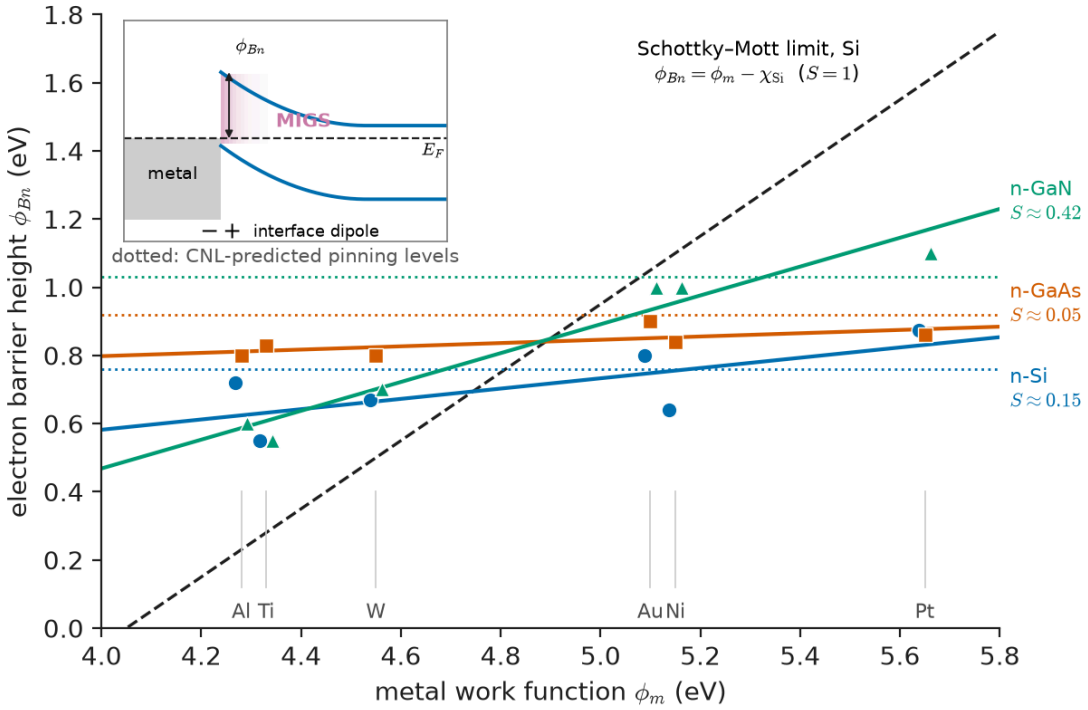


Figure 34.5 — Schottky barrier height versus metal work function: Schottky–Mott limit and Fermi-level pinning in Si, GaAs and GaN.

Current transport and specific contact resistivity

A contact is characterized by its **specific contact resistivity** ρ_c (units $\Omega\,\mathrm{cm}^2$), defined at zero bias as

$$\rho_c = \left(\frac{\partial J}{\partial V} \right)^{-1}_{V=0}$$

It is an interface property independent of contact area. A contact of area A has resistance $R_c = \rho_c / A$ when current flows uniformly through it. In practice current crowding at the leading edge of planar contacts is accounted for by a transfer length, and ρ_c is extracted by the transmission-line method (TLM) described in Chapter 52. Which transport mechanism dominates depends on doping, through the characteristic tunnelling energy

$$E_{00} = \frac{q\hbar}{2} \sqrt{\frac{N}{m^* \varepsilon_s}}$$

There are three regimes.

- **Thermionic emission** ($kT \gg E_{00}$, lightly doped): electrons go *over* the barrier. From the thermionic current $J = A^{**} T^2 e^{-q\phi_B/kT} (e^{qV/kT} - 1)$,

$$\rho_c = \frac{k}{q A^{**} T} \exp\left(\frac{q\phi_B}{kT} \right)$$

where A^{**} is the effective Richardson constant (≈112 A cm^{-2} K^{-2} for n-Si, 8 for n-GaAs, 26 for n-GaN).

> **Status of the principle: Thermionic-emission contact resistivity**
> *Derived.* It follows from thermionic emission over a barrier, with Maxwell–Boltzmann statistics and negligible tunnelling. It is valid for $N \lesssim 10^{17}$ cm^{-3} in Si. Image-force lowering and barrier inhomogeneity make measured ρ_c smaller than predicted.

- **Thermionic-field emission** ($kT \approx E_{00}$): electrons are thermally excited part-way up and tunnel through the thinner top of the barrier.
- **Field emission** ($kT \ll E_{00}$, degenerate doping): electrons tunnel straight through a depletion region only a few nm thick. Then $\rho_c \propto \exp(q\phi_B / E_{00})$, so $\ln \rho_c$ falls linearly with $\phi_B / \sqrt{N}$.

Worked example — Why ohmic contacts need heavy doping. For n-Si at 300 K with $\phi_B = 0.7$ eV and $A^{**} = 112$ A cm^{-2} K^{-2}, the prefactor is $k/(qA^{**}T) = 8.617 \times 10^{-5}/(112 \times 300) = 2.56 \times 10^{-9}$ Ω cm². The exponential is $\exp(0.7/0.02585) = \exp(27.08) = 5.8 \times 10^{11}$. So

$$\rho_c \approx 2.56 \times 10^{-9} \times 5.8 \times 10^{11} \approx 1.5 \times 10^{3}\ \Omega\,\mathrm{cm}^2$$

A 1 µm² contact (10^{-8} cm²) would have $R_c \approx 1.5 \times 10^{11}$ Ω. It is a diode, not a contact. Lowering the barrier to 0.5 eV gives $\rho_c \approx 0.64$ Ω cm², and to 0.2 eV gives 5.9×10^{-6} Ω cm². Barrier reduction alone is not enough, and pinning stops it anyway. The practical route is doping. At $N = 10^{20}$ cm^{-3} in Si (with $m^* = 0.26m_0$ for tunnelling), $E_{00} = 0.106$ eV, four times kT. Field emission dominates and the depletion width is about 3 nm. Silicide contacts on such layers reach $\rho_c \approx 10^{-8}$–10^{-9} Ω cm². Even then, at a modern contact of 20 nm × 20 nm (4×10^{-12} cm²), $\rho_c = 10^{-9}$ Ω cm² gives $R_c = 250$ Ω, and 10^{-8} gives 2.5 kΩ, comparable with the whole channel resistance. Contacts for the next nodes target $\rho_c \leq 2 \times 10^{-9}$ Ω cm². The materials levers are dopant activation above solid solubility (laser anneal, in-situ doped epitaxy), thin interfacial dielectrics that de-pin the Fermi level (metal–insulator–semiconductor contacts), and low-barrier silicides.

Ohmic contact recipes as materials systems

Real ohmic contacts are rarely a single metal on a semiconductor. They are *reacted* multilayer systems whose chemistry makes a good interface.

- **Si: silicides.** $TiSi_2$, then $CoSi_2$, then NiSi (and Ni(Pt)Si) were adopted in turn as linewidths shrank. Each change was made because the previous silicide's phase transformation (C49→C54 $TiSi_2$ needs large grains) or silicon consumption failed in narrow lines. NiSi forms at low temperature (400–600 °C), consumes the least silicon per unit thickness of silicide, and has a resistivity of 14–20 µΩ cm. It degrades to high-resistivity $NiSi_2$ above about 700 °C, so it sets a thermal-budget limit on everything after it.
- **n-GaAs: Au–Ge–Ni.** On alloying at about 400–450 °C, Ge diffuses in and dopes the surface heavily n-type (Ge on Ga sites), Ni promotes the reaction, and Au carries current. The result is a tunnelling contact with $\rho_c \approx 10^{-6}$ Ω cm².
- **n-GaN: Ti/Al/Ni/Au.** On annealing at 800–900 °C, Ti extracts nitrogen to form TiN, leaving nitrogen vacancies (donors) in the GaN surface. The heavily doped surface becomes tunnelling-thin. Al and Ni/Au protect against oxidation. $\rho_c \approx 10^{-5}$–10^{-6} Ω cm².
- **p-GaN** is notoriously hard. The Mg acceptor is deep (≈0.17 eV), so few holes are free, and the valence band of GaN lies 7.5 eV below vacuum, deeper than any metal's work function. Oxidized Ni/Au, or a transparent ITO contact on a heavily doped p^{++} cap, give $\rho_c \approx 10^{-3}$–10^{-4} Ω cm². Such contacts limit LED and laser efficiency (Chapter 35).

Contact degradation over time (interdiffusion, intermetallic growth, oxidation) is treated in Chapter 44. The resistance of contacts between metal parts (connectors, relays) was covered in Chapter 25.

34.10 Material properties of gate dielectrics

What a gate dielectric has to do

A gate dielectric separates a conductor (the gate) from a semiconductor channel and couples them capacitively. The capacitance per unit area $C = \varepsilon_0 k/t$ sets how much channel charge a given gate voltage induces. The device wants C large. The material must also meet several other requirements:

1. **High permittivity** k, so that C is large at a physical thickness t thick enough to block tunnelling.
2. **Large band offsets** to the semiconductor, at least about 1 eV for *both* electrons and holes. These offsets are the tunnelling barriers (Table 34.1, last row).
3. **A wide band gap and low defect density**, so that leakage paths through traps are few.
4. **Thermodynamic stability against the semiconductor** at process temperatures, so that it does not react to form silicate or silicide, or reduce the semiconductor.
5. **A good interface**: low D_{it} (Section 34.5), low fixed charge, and low carrier scattering, so that channel mobility is not degraded.
6. **High breakdown strength and reliability** over ten years at the operating field (Section 34.11).
7. **Compatibility with the gate metal**: no Fermi-level pinning or reaction at the top interface.

For forty years thermally grown SiO_2 met all of these. It has $k = 3.9$, an 8.9 eV gap, a 3.1–3.2 eV conduction-band offset and a 4.5–4.7 eV valence-band offset to Si, a breakdown field of 10–15 MV/cm, and an interface of $D_{it} \approx 10^{10}$ cm^{-2} eV^{-1}. Its one weakness is the low k.

Direct tunnelling and why SiO_2 ran out

For thin barriers, electrons tunnel directly. In the simplest rectangular-barrier (WKB) picture, the transmission probability is

$$T \approx \exp(-2\kappa t), \qquad \kappa = \frac{\sqrt{2 m_{ox}^{*} q \phi_B}}{\hbar}$$

where ϕ_B is the barrier (the conduction-band offset for electrons) and m_{ox}^{*} is the tunnelling effective mass in the oxide. For SiO_2 with $\phi_B = 3.1$ eV and $m_{ox}^{*} \approx 0.4 m_0$, $\kappa = 5.7$ nm^{-1}. Every 0.1 nm of thinning multiplies the transmission by $e^{1.14} \approx 3.1$, so every 0.2 nm multiplies the leakage by about 10. This matches the observed tenfold rise in gate leakage per 0.2 nm reduction in oxide thickness. Below about 1.2 nm, leakage of order 10–100 A/cm² at 1 V made SiO_2 (and nitrided SiON, $k \approx 5$–6) unusable for high-performance logic.

> **Status of the principle: Exponential tunnelling through a thin dielectric**
> *Approximate.* It is the WKB result for a rectangular barrier at low field. It neglects image-force rounding, the field-induced triangular barrier shape, and the band structure of the oxide (which makes m_{ox}^{*} an effective fitting parameter). It is good for trends and ratios, not absolute currents.

High-k dielectrics and the equivalent oxide thickness

The solution is a thicker film of higher permittivity. The *equivalent oxide thickness* (EOT) of a dielectric layer is the thickness of SiO_2 that would give the same capacitance:

$$\mathrm{EOT} = t_{hk} \frac{k_{\mathrm{SiO_2}}}{k_{hk}} = t_{hk} \frac{3.9}{k_{hk}}$$

and for a stack of layers in series, the EOTs add:

$$\mathrm{EOT}_{\mathrm{stack}} = \sum_i t_i \frac{3.9}{k_i}$$

> **Status of the principle: Equivalent oxide thickness**
> *Derived.* It follows from series addition of parallel-plate capacitances. It is exact as electrostatics. The measured "capacitance equivalent thickness" in inversion is larger by 0.3–0.4 nm, because of the quantum-mechanical displacement of the channel charge from the interface and polysilicon gate depletion. Metal gates remove the latter.

It might seem that the highest-k material wins. It does not, because *permittivity and band gap trade off*. A high k comes from soft, highly polarizable bonds and low-frequency polar phonons (Chapter 26 and Chapter 27). The same chemistry tends to give a small gap and small band offsets. The rough empirical relation is $E_g \propto k^{-1/2}$ across oxides. Breakdown field also falls with k, roughly as $E_{bd} \propto k^{-1/2}$ (McPherson). Figure 34.6 and Table 34.5 show the compromise.

Table 34.5 — Gate-dielectric candidates (typical values; offsets to Si)

Dielectric	k	E_g (eV)	CB offset to Si (eV)	VB offset to Si (eV)	Stable on Si?	Comment
SiO_2	3.9	8.9	3.1–3.2	4.5–4.7	Yes	Reference; best interface
Si_3N_4	7	5.3	2.4	1.8	Yes	Traps; used as SiON
Al_2O_3	9	8.8	2.8	4.9	Yes	ALD workhorse; III–V and GaN gates
HfO_2	20–25	5.7–5.9	1.4–2.0	2.6–3.4	Yes (with SiO_x IL)	Industry choice since 45 nm node
HfSiON	12–15	6–7	≈1.8	≈3	Yes	Amorphous to higher T
ZrO_2	25	5.8	1.4	3.3	Marginal (silicide)	DRAM capacitors ($ZrO_2/Al_2O_3/ZrO_2$)
La_2O_3	30	6.0	2.3	2.6	Hygroscopic	Used as thin cap for V_t tuning
Ta_2O_5	22–26	4.4	0.3–0.4	3.0	No	Offset too small
TiO_2	50–80	3.0–3.5	≈0	≈2	No	No electron barrier
$SrTiO_3$	~150–300	3.2	≈0	≈2	No	Paraelectric; DRAM research

HfO_2 was selected because it combines $k \approx 20$–25 with conduction- and valence-band offsets above about 1.4 eV. Hafnium also forms an oxide much more stable than SiO_2, so HfO_2 does not reduce the interfacial oxide. Ta_2O_5, TiO_2 and $SrTiO_3$ have high k but almost no electron barrier, so electrons would leak over or through them by Schottky emission. They are excluded from gate use, though suited to capacitor dielectrics with suitable electrodes (Chapter 27).

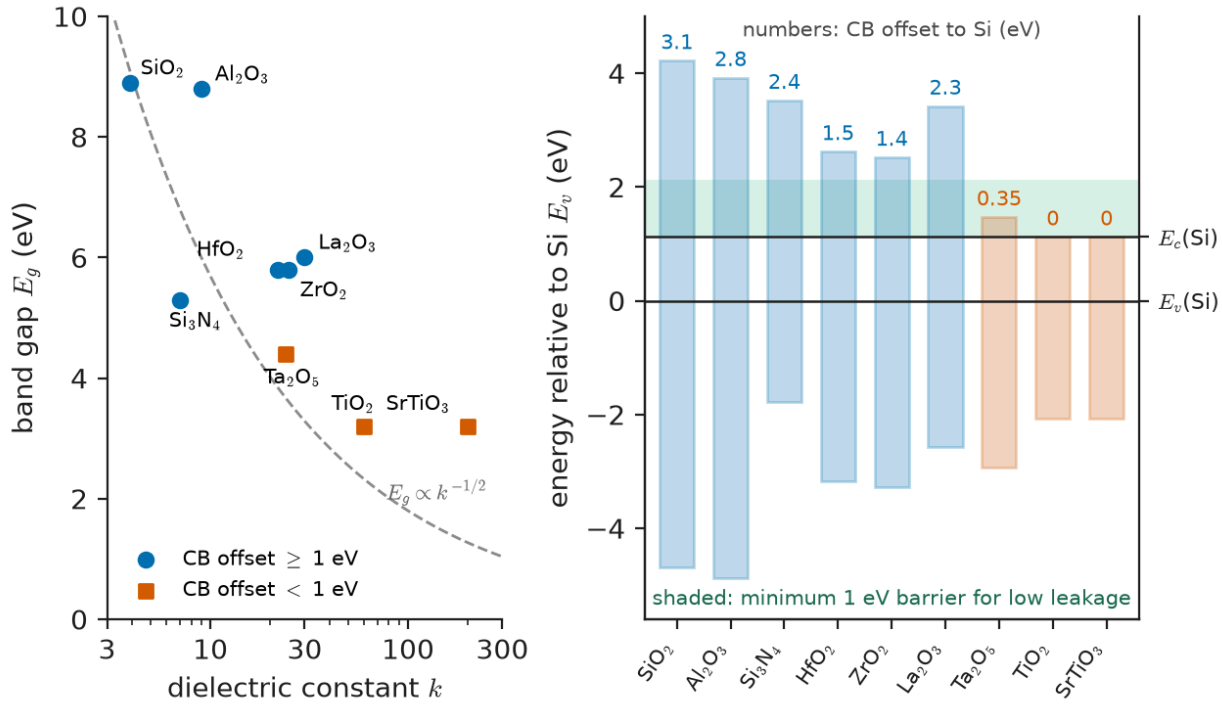


Figure 34.6 — Band gap versus permittivity for gate-dielectric candidates, with band offsets to silicon.

Worked example — EOT and leakage advantage of an HfO_2 stack. A gate stack has a 0.5 nm SiO_x interfacial layer (take $k = 3.9$) under 2.0 nm of HfO_2 ($k = 20$). Its EOT is

$$\mathrm{EOT} = 0.5 + 2.0 \times \frac{3.9}{20} = 0.5 + 0.39 = 0.89\ \mathrm{nm}$$

giving $C = 3.9\varepsilon_0/\mathrm{EOT} = 3.9 \times 8.854 \times 10^{-14}/0.89 \times 10^{-7} = 3.88$ μF/cm². An SiO_2 film of the same capacitance would be 0.89 nm thick, with WKB exponent $2\kappa t = 2 \times 5.70 \times 0.89 = 10.15$ and $T \approx 3.9 \times 10^{-5}$.

For the stack, take $\phi_B = 1.5$ eV and $m^* \approx 0.2 m_0$ for HfO_2. Then $\kappa_{HfO_2} = \sqrt{2 \times 0.2 \times 9.109 \times 10^{-31} \times 1.5 \times 1.602 \times 10^{-19}}/1.055 \times 10^{-34} = 2.81$ nm^{-1}. The exponent is $2(5.70 \times 0.5 + 2.81 \times 2.0) = 16.93$, so $T \approx 4.4 \times 10^{-8}$. The stack transmits about $3.9 \times 10^{-5}/4.4 \times 10^{-8} \approx 900$ times less at equal capacitance. Measured reductions of two to four orders of magnitude are consistent with this crude estimate.

The example also shows the limits of high-k. The 0.5 nm interfacial layer alone contributes more than half the EOT. A thinner layer degrades mobility, because the channel is then closer to the soft polar phonons and charged defects of the HfO_2 ("remote phonon scattering"), and D_{it} rises. A thicker one wastes EOT. Scaling below about 0.7–0.8 nm EOT therefore relies on *interfacial-layer scavenging*: metals in the gate stack (Ti, Hf) pull oxygen out of the SiO_x during anneal.

Materials problems that came with high-k

- **Crystallization.** Amorphous HfO_2 crystallizes near 400–600 °C into monoclinic grains. Grain boundaries are leakage paths and make k vary from grain to grain. Adding Si (HfSiO) or N raises the crystallization temperature. Doped and thin HfO_2 can instead crystallize into a polar orthorhombic phase that is *ferroelectric*. This was discovered in 2011 and is now a materials basis for ferroelectric memories (Chapter 27).
- **Oxygen vacancies.** HfO_2 has a much higher density of oxygen vacancies than SiO_2. They give electron trap levels about 1–2 eV below the HfO_2 conduction band, which cause threshold instability (PBTI, Section 34.11), charge trapping, and trap-assisted leakage.
- **Fermi-level pinning at the gate.** Polysilicon gates on HfO_2 showed pinned threshold voltages because of Si–Hf bonds and oxygen vacancies at the top interface. This drove the move to *metal gates* (TiN, TaN, TiAl and others) with work functions selected for nMOS (≈4.1–4.3 eV) and pMOS (≈4.9–5.1 eV). Thin La_2O_3 or Al_2O_3 caps create interface dipoles that shift the effective work function by ±0.3–0.5 eV, an application of the interface-dipole physics of Section 34.2.
- **Mobility degradation** from remote Coulomb and soft-phonon scattering, as above.

For wide-gap power semiconductors, the dielectric problem is different. 4H-SiC (3.26 eV gap) leaves SiO_2 with only about 2.7 eV conduction-band offset, so the field in the oxide must be held below about 3–4 MV/cm for long-term reliability. On GaN, the 3.4 eV gap similarly limits the options to wide-gap Al_2O_3, SiO_2 and SiN_x. Interface trap densities there still exceed 10^{12} cm^{-2} eV^{-1}, one reason p-GaN gates dominate GaN power devices.

34.11 Interface reliability

Interfaces are where most transistor wear-out begins. The bulk crystal is nearly perfect and very stable. The interface and the oxide next to it hold the weak bonds, the hydrogen, and the precursors of defects. Under voltage and temperature, these bonds slowly break or trap charge, and the transistor's threshold voltage, mobility and leakage drift. This section treats the *materials mechanisms*. Lifetime models, test protocols, failure analysis and accelerated-test statistics are in Chapter 44.

> **Boundary.**
> Here we identify which bonds and defects degrade and why. Device-level lifetime projection, circuit guard-banding and failure analysis belong to Chapter 44 (Material Degradation and Failure) and to Electronics & Computing.

Bias-temperature instability (NBTI and PBTI)

Negative bias-temperature instability (NBTI) affects pMOS transistors held with negative gate bias (channel in inversion with holes) at elevated temperature, typically 100–150 °C at fields of 4–8 MV/cm. The threshold voltage becomes more negative, by tens of mV over a product lifetime, and the drive current falls. Two coupled materials mechanisms are involved:

1. **Interface-trap generation.** The Si–H bonds that passivated P_b centres (Section 34.4) are broken with the help of inversion-layer holes and the field. The hydrogen released diffuses into the oxide. Each broken bond becomes a donor-like interface trap that is positively charged in a pMOS under bias. In the classic *reaction–diffusion* picture this gives a power law,

$$\Delta V_{th} \propto t^{n}, \qquad n \approx 0.16\text{–}0.25$$

2. **Hole trapping** in pre-existing defects in the oxide or near-interface layer. This component is larger in nitrided oxides (SiON), because nitrogen introduces hole traps. NBTI therefore grew worse as nitrogen was added to suppress boron penetration.

A characteristic of BTI is *partial recovery*. When stress is removed, part of the shift relaxes within microseconds to seconds as traps discharge. The measured degradation therefore depends on how quickly after stress the measurement is made, a long-running metrology difficulty.

Positive bias-temperature instability (PBTI) in nMOS was negligible with SiO_2. With HfO_2 it became significant: under positive gate bias, channel electrons tunnel into the oxygen-vacancy traps in the HfO_2 (Section 34.10), and the threshold shifts positive. The remedies are materials ones: reduce the oxygen-vacancy density (anneals, thin HfO_2, La or Al doping), and keep the HfO_2 thin so that traps sit beyond the tunnelling range of the band edges.

> **Status of the principle: Power-law time dependence of BTI**
> *Empirical.* The t^n form is fitted over typically 10^0–10^5 s of stress. The reaction–diffusion model gives $n = 1/4$ or $1/6$ depending on the diffusing hydrogen species. Real data mix interface-trap generation and hole trapping with different kinetics, so extrapolation to ten years carries considerable uncertainty.

Hot-carrier degradation

Near the drain of a short-channel transistor the lateral field is high, and a small fraction of carriers gain energies of several eV ("hot" carriers, Chapter 32's high-field transport). Two energies are material thresholds:

- an electron in Si needs about 3.1 eV, the Si/SiO_2 conduction-band offset, to be injected over the barrier into the oxide, where it can be trapped;
- the Si–H bond can be broken either by a single carrier with more than about 3 eV, or by several cooler carriers successively exciting its vibrational modes (the multiple-vibrational-excitation mechanism, which dominates at low supply voltages).

The damage is localized near the drain and degrades transconductance and saturation current. A striking result shows the bond-level nature of the mechanism. Replacing hydrogen by **deuterium** in the final passivation anneal improves hot-carrier lifetime by factors of 10–50. Deuterium is chemically identical, but its heavier mass gives the Si–D bond a vibrational frequency matched to Si lattice phonons. The bond then sheds vibrational energy into the lattice faster and is less easily ladder-climbed to dissociation. Engineering a transistor's lifetime by an isotope substitution is about as pure a materials effect as exists in electronics.

Time-dependent dielectric breakdown (TDDB)

A dielectric held at a field well below its instantaneous breakdown strength eventually fails anyway. TDDB is the gradual generation of defects (electron traps, broken bonds, oxygen vacancies) by the current flowing through the dielectric. When enough traps happen to line up and form a conducting chain between the two electrodes, the dielectric breaks down. This is the *percolation model*. Its consequences are material-structural:

- **Thinner films fail sooner and more randomly.** A thinner film needs fewer traps to complete a path, so the critical trap density falls and the statistical spread widens. The Weibull shape factor β of breakdown times decreases with thickness, from about 3 for a 10 nm oxide toward 1 for a 1–2 nm one. Chapter 44 treats the Weibull distribution.
- **Soft and progressive breakdown.** In ultra-thin films the first percolation path may carry only microamperes ("soft breakdown"), and the device continues to work with higher leakage.
- **Charge to breakdown** Q_{bd} (in C/cm^2), the total charge passed per unit area at failure, is a useful material figure: about 10–100 C/cm^2 for good thermal SiO_2, lower for deposited oxides.
- **Field acceleration** is described by competing empirical models. The *E-model* has $t_{BD} \propto \exp(-\gamma E)$, from field-enhanced bond breaking. The *1/E model* has $t_{BD} \propto \exp(G/E)$, from damage by tunnelling-current-driven hydrogen release or anode hole injection. Both fit data over limited ranges. Their extrapolations to use conditions differ by orders of magnitude, a point Chapter 44 takes up.

Status of the principle: Field-acceleration models of TDDB
Empirical. The E-model and the 1/E model are each fitted to accelerated-test data. Each has a physical motivation (thermochemical bond breaking, anode hole injection), and each fits over a limited range of field. Neither is established for extrapolation over many decades, so the choice of model is itself a reliability decision.

For high-k stacks, TDDB is a *two-layer* problem: the interfacial SiO_x and the HfO_2 degrade at different rates and break down in sequence. In wide-gap power devices (SiC and GaN MOS gates), the smaller band offsets of Section 34.10 make carrier injection easier, so the allowed operating oxide field is set lower than in silicon.

Table 34.6 — Interface reliability mechanisms at the material level

Mechanism	Device most affected	Material origin	Materials remedy
NBTI	pMOS, inversion, high T	Si–H bond breaking at interface; hole traps in SiON	Lower N near interface, better H passivation, optimized nitridation
PBTI	nMOS with high-k	Electron trapping in HfO_2 oxygen vacancies	Thin HfO_2, vacancy-reducing anneals, dopants
Hot carriers	Short-channel nMOS at drain	Si–H dissociation; injection over 3.1 eV barrier	Deuterium anneal, graded drain doping, lower field
TDDB	All gate dielectrics, capacitors	Trap generation and percolation	Thicker or higher-gap layers, lower defect density
Interface drift in SiC MOS	SiC power MOSFETs	Near-interface oxide traps and C defects	NO/N_2O anneals, alternative dielectrics
Current collapse in GaN HEMTs	GaN RF/power	Surface and buffer traps (Section 34.3)	SiN passivation, buffer doping control (C, Fe)

What remains open

Open question	Where it is answered
How do these interfaces and films decide light absorption, emission and detection in real optoelectronic materials?	Chapter 35
Which semiconductor families offer the substrates and alloys that heterostructures are built from?	Chapter 33 (families); Chapter 35 (band gap–lattice constant map)
How does quantum confinement in very thin layers change band gaps and densities of states?	Chapter 38
How are passivated surfaces and grain boundaries engineered in solar absorbers?	Chapter 40
How are thin films actually deposited, and how does processing set their microstructure?	Chapter 49 (principles); Chapter 50 (electronic-materials processes)
How do metallization, contacts and dielectrics fit into a manufacturing process, and what defects does processing introduce?	Chapter 50
How does CTE mismatch stress act at package level, in solders and interconnects?	Chapter 51
How are strain, interface traps, contact resistivity and film composition measured?	Chapter 52
How do BTI, hot carriers, TDDB, electromigration and contact degradation become device failures, and how are lifetimes projected?	Chapter 44
How are accelerated tests designed and lifetimes extrapolated statistically?	Chapter 44
How do these interfaces operate inside transistors, diodes and lasers as circuit elements?	Electronics & Computing

Next: Chapter 35 turns from charge at interfaces to the interaction of light and high-frequency fields with semiconductor and photonic materials: absorption, emission, detection, guiding and sensing.

Chapter 35 — Electronic and Optoelectronic Materials

Part VII · Semiconductor and Photonic Materials

Silicon carries almost all the world's computation and most of its solar power, yet there is no silicon light-emitting diode in a phone, a car headlamp or a fibre-optic transmitter. Meanwhile the blue LED inside every white lamp is made of InGaN grown on sapphire. Its epitaxial layers contain 10^8–10^9 threading dislocations per cm², a defect density that would make a silicon transistor useless, and it still converts more than 70% of its electrical input into light. The two facts point the same way. Whether a material emits, absorbs, detects or guides light is decided by its band structure, its defects and its interfaces. A material that is excellent for one function can be hopeless for another.

An engineer choosing materials for an optical link meets this at every step. The laser at 1550 nm must be InGaAsP on InP, because silicon cannot emit. The fibre must be ultrapure silica, whose loss is limited by the scattering of light from frozen-in density fluctuations. The modulator may be lithium niobate or silicon, for different physical reasons. The detector must be germanium or InGaAs, because silicon is transparent at 1550 nm. The transistors in the receiver amplifier may be SiGe or InP, chosen for electron velocity rather than optical properties. Each choice follows from a material property that this chapter derives.

Why does a direct-gap material absorb and emit light so much more strongly than an indirect-gap one, and what decides the sharpness of the absorption edge?
What material limits the efficiency of LEDs, the sensitivity and wavelength range of photodetectors, and the efficiency of solar cells? What makes a material suitable for guiding or modulating light? Which material properties decide performance at radio and microwave frequencies, and in sensors?

Concept	Engineering destination
Absorption coefficient, penetration depth	Detector and solar-cell thickness, photolithography, laser processing
Urbach tail	Solar-cell voltage loss, sub-gap absorption, material quality index
Radiative recombination coefficient B	LED and laser efficiency, solar-cell limits
Photoconductive gain	Low-cost light sensors, X-ray imagers, IR photoconductors
Phosphors, activators, Stokes shift	White LEDs, displays, X-ray and γ-ray scintillators
Alloy band gap and lattice matching	LED colour, laser wavelength, detector cut-off
Efficiency droop, green gap	High-power LED design, micro-LED displays
Cut-off wavelength, dark current	IR cameras, optical receivers, spectroscopy
Shockley–Queisser limit, V_{oc} loss	Choice of photovoltaic absorber (with Chapter 40)
Fibre loss, electro-optic coefficient, index contrast	Optical communications, silicon photonics, modulators
Saturation velocity, substrate loss, loss tangent	RF transistors, mm-wave circuits, antennas, PCBs
Transduction coefficients (TCR, gauge factor, Hall mobility)	Temperature, strain, magnetic, gas and IR sensors

Roadmap. How light is absorbed and emitted (35.1) → materials whose conductivity responds to light (35.2) → materials that emit by luminescence (35.3) → emitters for LEDs and lasers (35.4) → detectors (35.5) → solar absorbers (35.6) → materials that guide and modulate light (35.7) → materials for high frequencies (35.8) → sensor materials (35.9).

Boundary.
This chapter explains the material mechanisms behind optoelectronic, RF and sensing devices. Device operation (diode equations, laser rate equations, transistor small-signal models) and circuit configurations belong to Electronics & Computing. The quantum mechanics of optical transitions (matrix elements, Fermi's golden rule) belongs to Engineering Physics. Bands, direct and indirect gaps, carrier statistics and recombination were set up in Chapters 31 and 32 and are used here without re-derivation.

35.1 Light absorption and emission in materials

Photon energy and the band gap

A photon of wavelength λ carries energy $E = h\nu = hc/\lambda$. In convenient units,

$$E\,[\text{eV}] = \frac{1239.8}{\lambda\,[\text{nm}]}$$

so 1550 nm corresponds to 0.80 eV, 850 nm to 1.46 eV, 550 nm (green) to 2.25 eV and 450 nm (blue) to 2.76 eV. A semiconductor absorbs a photon strongly by *interband* absorption, which lifts an electron from the valence band to the conduction band. Energy conservation requires $h\nu \geq E_g$. The same transition run backwards is *radiative recombination*, which emits a photon of energy close to E_g. Absorption and emission are two directions of one process, and a material that does one well does the other well, provided non-radiative paths do not compete (see radiative recombination, below).

The absorption coefficient

Light of intensity I_0 entering a uniform absorbing material decays as

$$I(x) = I_0\, e^{-\alpha x}$$

where α (usually in cm^{-1}) is the *absorption coefficient* and $1/\alpha$ is the *penetration depth* (or absorption length). Within one penetration depth 63% of the light is absorbed. Within three, 95%. The absorption coefficient is related to the imaginary part κ of the complex refractive index $\tilde{n} = n + i\kappa$ by $\alpha = 4\pi\kappa/\lambda$, which links absorption to the dielectric function of Chapter 26.

> **Status of the principle: Beer–Lambert attenuation** $I = I_0 e^{-\alpha x}$
> *Approximate.* It holds for a homogeneous medium at low intensity, where α does not depend on I. It fails at high intensity (saturable absorption, two-photon absorption), in strongly scattering media, and where interference in thin films matters. Surface reflection must be accounted for separately.

Direct versus indirect absorption

The shape of $\alpha(h\nu)$ just above E_g follows from conservation of crystal momentum. A photon carries almost no momentum on the scale of the Brillouin zone, so an optical transition is vertical in k-space.

- **Direct gap** (GaAs, InP, GaN, CdTe, InGaAs). The valence-band maximum and conduction-band minimum lie at the same k, so the photon alone can drive the transition. With parabolic bands the joint density of states rises as $(h\nu - E_g)^{1/2}$, which gives

$$\alpha_{\text{dir}}(h\nu) \approx A^{*}\,(h\nu - E_g)^{1/2}$$

and α reaches 10^4 cm^{-1} within about 0.1 eV of the edge. A 1–2 μm layer of a direct-gap semiconductor absorbs nearly all light above its gap.

- **Indirect gap** (Si, Ge, AlAs, GaP, SiC). The band extrema lie at different k, so the transition needs a phonon to supply the momentum difference, by absorbing or emitting one. A three-body event (photon, electron, phonon) is a second-order process and much less probable:

$$\alpha_{\text{ind}}(h\nu) \propto \frac{(h\nu - E_g + E_p)^2}{e^{E_p/kT} - 1} + \frac{(h\nu - E_g - E_p)^2}{1 - e^{-E_p/kT}}$$

where E_p is the phonon energy (about 50–60 meV for the relevant phonons in Si). The edge is weak, rising only to 10–10^3 cm^{-1} over the first few tenths of an eV. In silicon, strong absorption begins only once the photon energy approaches the direct gap at the Γ point (about 3.4 eV), near 365 nm.

Status of the principle: Square-root and quadratic absorption edges
Approximate. These are derived from first- and second-order perturbation theory with parabolic bands and constant matrix elements. They hold within a few tenths of an eV of the edge. Plotting $(\alpha h\nu)^2$ or $(\alpha h\nu)^{1/2}$ against $h\nu$ (a Tauc plot) to extract E_g relies on this. They fail close to the edge, where excitons and band tails (the Urbach tail, below) dominate, and far above it, where band non-parabolicity matters.

Germanium shows both behaviours. Its indirect gap is 0.66 eV, but the direct gap at Γ is only 0.80 eV, so absorption jumps by two orders of magnitude near 1550 nm. That makes Ge a useful telecom detector (Section 35.5). Table 35.1 lists representative values, and Figure 35.1 plots α against photon energy.

Table 35.1 — Absorption coefficient and penetration depth (300 K, typical values)

Material	E_g (eV), type	Wavelength	α (cm^{-1})	$1/\alpha$
Si	1.12, indirect	400 nm	9.5×10^4	0.11 µm
Si		600 nm	4.1×10^3	2.4 µm
Si		850 nm	5.4×10^2	19 µm
Si		1000 nm	64	160 µm
Si		1100 nm	3.5	2.9 mm
Ge	0.66 indirect (0.80 direct)	1310 nm	$\approx 7 \times 10^3$	1.4 µm
Ge		1550 nm	≈ 460	22 µm
GaAs	1.42, direct	800 nm	$\approx 1.2 \times 10^4$	0.8 µm
$In_{0.53}Ga_{0.47}As$	0.75, direct	1550 nm	$\approx 7 \times 10^3$	1.4 µm
a-Si:H	≈1.75 (mobility gap)	600 nm	$\approx 4 \times 10^4$	0.25 µm
$CH_3NH_3PbI_3$	≈1.6, direct	700 nm	$\approx 3 \times 10^4$	0.3 µm
GaN	3.42, direct	350 nm	$\approx 1 \times 10^5$	0.1 µm

The table explains several engineering facts at once. A crystalline Si solar cell must be about 100–180 µm thick and needs light trapping to absorb its near-infrared photons. A GaAs or perovskite cell needs less than 1 µm. A silicon photodiode for 850 nm light must be tens of micrometres deep, which makes it slow, while a GaAs one needs only 1–2 µm.

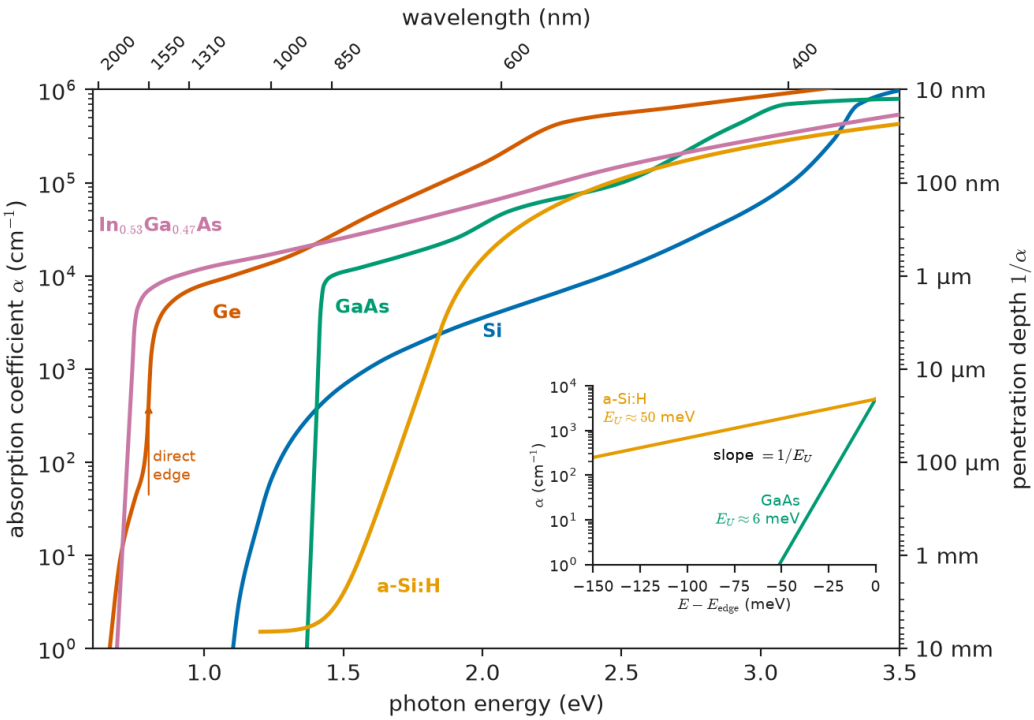


Figure 35.1 — Absorption coefficient versus photon energy for direct- and indirect-gap semiconductors, with the Urbach tail.

The Urbach tail

Measured below E_g, α does not drop to zero at the gap. It falls off exponentially:

$$\alpha(h\nu) = \alpha_0 \exp\left(\frac{h\nu - E_0}{E_U}\right), \qquad h\nu < E_0$$

The *Urbach energy* E_U measures how sharp the edge is. Its origin is disorder. Thermal vibrations (dynamic disorder) and structural defects, alloy fluctuations and strain (static disorder) make the local band edges vary from place to place, creating exponential *band tails* of localized states that extend into the gap. In a perfect crystal at low temperature E_U approaches its phonon-limited minimum of a few meV. In an amorphous solid it is several tens of meV.

> **Status of the principle: Urbach rule**
> *Empirical.* The exponential shape is observed in crystalline, alloyed and amorphous materials over two to four decades of α. Models of disorder-induced band tails reproduce it, but α_0, E_0 and E_U are fitted parameters, and the form fails deep in the gap where discrete defect levels dominate.

Table 35.2 — Typical Urbach energies at 300 K

Material	E_U (meV)	Comment
GaAs (high quality)	6–7	Near phonon limit
c-Si	8–12	Indirect edge complicates fitting
$CH_3NH_3PbI_3$ (perovskite)	13–16	Remarkably ordered for a solution-processed film
CdTe	10–20	Depends on $CdCl_2$ treatment
Cu(In,Ga)Se_2	15–30	Composition fluctuations
Cu_2ZnSnS_4 (kesterite)	40–70	Cation disorder; large voltage loss
a-Si:H	45–55	Amorphous network
Organic semiconductors	25–50	Energetic disorder

The Urbach energy is more than a spectroscopic curiosity. The tail states it describes trap carriers and act as recombination channels, and they set the minimum voltage loss of a solar absorber (Section 35.6). Kesterite solar cells have stalled near 13–15% efficiency largely because of their large E_U, while perovskites reached above 25% partly because theirs is small.

Radiative recombination

The radiative recombination rate per unit volume is proportional to the product of electron and hole densities, since an electron must meet a hole:

$$R_{\text{rad}} = B\,(np - n_i^2)$$

where B (cm³/s) is the *radiative recombination coefficient*. The n_i^2 term ensures that in equilibrium emission balances absorption of the surrounding thermal radiation. Indeed B is not independent of α. Detailed balance (the van Roosbroeck–Shockley relation) fixes the spontaneous emission spectrum from the absorption spectrum, so a material that absorbs weakly near its gap also emits weakly. For a doped material under low injection ($p \approx N_A \gg n$), the radiative lifetime of minority electrons is

$$\tau_{\text{rad}} = \frac{1}{B N_A}$$

> **Status of the principle: Radiative rate $R = B(np - n_i^2)$**
> *Derived.* It follows from detailed balance between absorption and emission (van Roosbroeck–Shockley) for non-degenerate carriers. It holds until carrier densities approach degeneracy (above about 10^{18}–10^{19} cm^{-3} in direct-gap III–Vs), where phase-space filling makes the effective B fall.

Table 35.3 — Radiative recombination coefficients (300 K, typical)

Material	Gap type	B (cm³/s)	τ_{rad} at 10^{17} cm^{-3}
Si	indirect	$\approx 5 \times 10^{-15}$	≈ 2 ms
Ge	indirect	$\approx 6 \times 10^{-14}$	≈ 0.16 ms
GaAs	direct	$(1\text{–}7) \times 10^{-10}$	15–100 ns
InP	direct	$\approx 1 \times 10^{-10}$	≈ 100 ns
$In_{0.53}Ga_{0.47}As$	direct	$\approx 1 \times 10^{-10}$	≈ 100 ns
GaN / InGaN	direct	$\approx (0.5\text{–}2) \times 10^{-11}$	≈ 0.5–2 µs

The direct-gap III–Vs have B about 10^4–10^5 times larger than silicon. Light emission is decided by the *competition* between radiative recombination and the non-radiative paths of Chapter 32: Shockley–Read–Hall (SRH) recombination through defects, with rate $\approx n/\tau_{SRH}$, and Auger recombination, with rate $\approx Cn^3$. The *internal quantum efficiency* is the radiative fraction:

$$\eta_{\text{IQE}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{SRH}} + R_{\text{Auger}}} = \frac{1/\tau_{\text{rad}}}{1/\tau_{\text{rad}} + 1/\tau_{\text{SRH}} + 1/\tau_{\text{Auger}}}$$

Worked example — Why silicon does not glow. Take p-type Si with $N_A = 10^{17}$ cm^{-3} and a good SRH lifetime of 100 µs. Then $\tau_{\text{rad}} = 1/(4.7 \times 10^{-15} \times 10^{17}) = 2.1$ ms. Ignoring Auger, the IQE is

$$\eta_{\text{IQE}} = \frac{1}{1 + \tau_{\text{rad}}/\tau_{\text{SRH}}} = \frac{1}{1 + 21.3} = 4.5\%$$

Of that light, most is then trapped by total internal reflection or reabsorbed. For GaAs at the same doping, with $B = 2 \times 10^{-10}$ cm³/s, $\tau_{\text{rad}} = 1/(2 \times 10^{-10} \times 10^{17}) = 50$ ns. Even with a modest SRH lifetime of 100 ns the IQE is $1/(1 + 0.5) = 67\%$, and at 10^{18} cm^{-3} ($\tau_{\text{rad}} = 5$ ns) it is $1/(1 + 0.05) = 95\%$. Silicon *can* emit, and laboratory devices with exceptionally clean surfaces have reached about 1% external efficiency. But it must compete against defects for millisecond timescales, while GaAs only has to win over nanoseconds. That is why every commercial light source uses a direct-gap material.

35.2 Photoconductive materials

Photoconductivity

The simplest optoelectronic effect is that light raises conductivity. Absorbed photons create excess electrons and holes at a generation rate G (pairs per cm³ per s). In steady state each carrier lives for a lifetime τ, so the excess densities are $\Delta n = \Delta p = G\tau$ and

$$\Delta\sigma = q(\mu_n \Delta n + \mu_p \Delta p) = qG\tau(\mu_n + \mu_p)$$

A *photoconductor* is just a slab of such a material between two ohmic contacts. Its response depends on the product $\mu\tau$, a key material figure of merit. It is the same product that appears in the diffusion length $L = \sqrt{D\tau}$ (Chapter 32).

Photoconductive gain

The photoconductor has a property a simple photodiode lacks: *gain*. Suppose electron–hole pairs are generated between contacts a distance L apart under bias V. An electron crosses the device in the *transit time*

$$t_{tr} = \frac{L}{\mu_n E} = \frac{L^2}{\mu_n V}$$

If the hole is trapped (immobile) or simply slower, the electron leaves at the anode before recombining. Charge neutrality then requires another electron to enter from the cathode. The electron *recirculates* until the trapped hole recombines, after a time τ. Each absorbed photon therefore sends τ/t_{tr} electrons around the circuit:

$$G_{\text{pc}} = \frac{\tau}{t_{tr}} = \frac{\tau \mu_n V}{L^2}$$

> **Status of the principle: Photoconductive gain $G = \tau/t_{tr}$**
> *Derived.* It follows from charge continuity with ohmic (injecting) contacts and a uniform field. It assumes the lifetime is not limited by sweep-out and the contacts can supply carriers freely. At high field, space-charge-limited current and carrier sweep-out cap the gain.

The price of gain is speed. The photocurrent rises and decays with time constant τ, so the bandwidth is roughly $f_{3\,dB} \approx 1/(2\pi\tau)$. The gain–bandwidth product, about $1/(2\pi t_{tr})$, is fixed by geometry and mobility, not by τ. Lengthening the lifetime with traps buys gain and loses speed in equal measure.

Worked example — A CdS light sensor. A CdS photoresistor has electrode spacing $L = 1$ mm, bias $V = 10$ V, electron mobility 300 cm²/V s, and a trap-extended lifetime $\tau = 1$ ms. The transit time is

$$t_{tr} = \frac{(0.1\ \text{cm})^2}{300\ \text{cm}^2/\text{V s} \times 10\ \text{V}} = 3.3\ \mu\text{s}$$

so the gain is $G = 10^{-3}/3.3 \times 10^{-6} = 300$ electrons per absorbed photon. The bandwidth is $1/(2\pi \times 10^{-3}) = 160$ Hz, and the gain–bandwidth product is about 48 kHz. That is ideal for a street-light switch or a camera exposure meter and useless for data communication. Interdigitated electrodes, which shorten L while keeping the illuminated area large, are the standard materials-and-geometry route to higher gain.

Photoconductive materials in use

Gain in practical photoconductors usually comes from *sensitizing* centres. Deliberately added acceptor-like traps (Cu or Cl in CdS) capture holes and hold them for milliseconds, which slows recombination and lets electrons recirculate. This is defect engineering for a purpose, the reverse of the usual aim of removing traps. Table 35.4 lists the main materials.

Table 35.4 — Photoconductive materials (typical)

Material	E_g (eV)	Spectral range	Form	Use
CdS	2.42	400–700 nm	Sintered or evaporated polycrystalline film	Light-dependent resistors (now restricted by RoHS)
CdSe	1.74	500–750 nm	Polycrystalline	Light sensors
a-Se	≈2.1	X-ray, blue	Thick amorphous film (200–1000 μm)	Direct-conversion X-ray flat panels, mammography
PbS	0.37	1–3 μm	Chemically deposited polycrystalline	Uncooled SWIR detectors, flame sensors
PbSe	0.27	1.5–5 μm	Polycrystalline	MWIR gas analysers
HgCdTe	0.1–0.3	3–14 μm	Epitaxial	Cooled IR photoconductors (older systems)
Si:As, Ge:Cu (extrinsic)	impurity levels ≈ 0.05 (Si:As), 0.04 (Ge:Cu) eV	10–30 μm	Doped crystal at 4–30 K	Astronomy

In PbS, PbSe and amorphous selenium the grain boundaries and band tails of Chapter 34 and Section 35.1 are part of the mechanism. PbS films must be oxidized (sensitized) to form the hole-trapping boundary layers that give useful gain. Amorphous selenium is valued for X-ray imaging because it can be deposited uniformly over 40 × 40 cm, has a high X-ray stopping power, and under a high field (≈ 10 V/μm) collects charge with negligible lateral spread, giving very sharp images.

35.3 Luminescent materials

Kinds of luminescence

Luminescence is light emitted by a material after it absorbs energy, in excess of its thermal radiation. It is classified by the source of excitation:

- **Photoluminescence**: excitation by light (phosphors in white LEDs, fluorescent lamps).
- **Electroluminescence**: excitation by current (LEDs, Section 35.4; thin-film EL displays).
- **Cathodoluminescence**: excitation by electrons (CRTs historically; electron-microscope analysis, Chapter 52).
- **Radioluminescence or scintillation**: excitation by X- or γ-rays (medical imaging, radiation detection).

It is also classified by decay time. *Fluorescence* comes from spin-allowed transitions and decays in nanoseconds to microseconds. *Phosphorescence* comes from spin- or parity-forbidden transitions, or from slow trap release, and lasts milliseconds to hours.

Phosphors: host plus activator

Most practical phosphors are insulating crystalline *hosts* (oxides, nitrides, fluorides, sulfides) doped at 0.1–5% with an *activator* ion that does the emitting. The host's job is to hold the activator in a well-defined, rigid site, to be transparent to the emitted light, to conduct away heat, and to resist chemical attack. The activator's electron configuration decides the character of the emission.

- **4f→4f transitions in trivalent rare earths** (Eu^{3+}, Tb^{3+}). The 4f electrons are shielded by filled 5s and 5p shells, so the host barely affects them. Emission occurs in sharp lines at nearly fixed wavelengths (Eu^{3+}: 611 nm red; Tb^{3+}: 545 nm green). These transitions are parity-forbidden, so decay times are milliseconds.
- **5d→4f transitions in Ce^{3+} and Eu^{2+}.** The 5d orbital extends out to the neighbouring ions, so its energy depends strongly on the host's *crystal field* and covalency (the nephelauxetic effect). The emission is broad (50–120 nm wide), allowed (decay of tens of ns to about 1 μs), and *tunable by choice of host*. Ce^{3+} emits in the UV in fluorides, blue in some oxides, and yellow in YAG. This tunability is why Ce^{3+} and Eu^{2+} phosphors dominate white LEDs.
- **3d transition-metal ions** (Mn^{2+}, Mn^{4+}, Cr^{3+}). Their emission depends on the crystal field. Mn^{4+} in fluorides gives narrow red lines near 630 nm.

- **Donor–acceptor pair** emission in semiconductors such as ZnS:Ag,Cl and ZnS:Cu,Al, the historic CRT phosphors.

Stokes shift and the configuration-coordinate picture

The emitted photon always has less energy than the absorbed one. When the activator is excited, its electron distribution changes, so the equilibrium positions of the neighbouring ions shift. The excited ion relaxes to a new geometry by emitting phonons before it emits light, and after emission the ground state relaxes again. The energy difference between absorption and emission peaks is the *Stokes shift*. It is larger when the coupling between electron and lattice is stronger: small for 4f→4f, larger for 5d→4f.

The Stokes shift is an unavoidable energy loss. A white LED that converts a 450 nm blue photon to a 555 nm yellow one keeps only $450/555 = 81\%$ of the photon energy, even at 100% quantum yield. The 19% balance appears as heat in the phosphor layer.

Quantum yield and thermal quenching

The *quantum yield* (QY) is the number of photons emitted per photon absorbed. The best LED phosphors reach QY above 90% at room temperature. The yield falls at high temperature because the excited state can cross over to the ground state non-radiatively, through the crossing point of the two configuration-coordinate parabolas, or by thermal ionization of the 5d electron into the host conduction band. A common description is

$$\frac{I(T)}{I(0)} = \frac{1}{1 + C\exp(-\Delta E / kT)}$$

with ΔE an activation energy for quenching.

> **Status of the principle: Single-barrier thermal quenching**
> *Approximate.* It assumes one thermally activated non-radiative channel with a single barrier ΔE. It fits many phosphors over a limited temperature range. Multiple channels and distributions of activator sites give broader quenching curves.

Thermal quenching is a design constraint, because phosphors in high-power LEDs run at 100–200 °C. YAG:Ce keeps about 90% of its room-temperature output at 150 °C. Substituting Gd for Y red-shifts the emission for warmer white light but lowers the quenching temperature. That is a typical colour-against-stability trade-off resolved by host chemistry.

Table 35.5 — Representative luminescent materials (typical values)

Phosphor / scintillator	Transition	Excitation	Emission peak	Decay	Application
$Y_3Al_5O_{12}:Ce^{3+}$ (YAG:Ce)	5d→4f	440–470 nm	550–560 nm, broad	≈ 60–70 ns	White LEDs (with blue InGaN)
$CaAlSiN_3:Eu^{2+}$	5d→4f	UV–blue	≈ 650 nm, broad	≈ 1 µs	Warm-white and high-CRI LEDs
β-SiAlON:Eu^{2+}	5d→4f	UV–blue	≈ 535 nm, FWHM ≈ 55 nm	≈ 1 µs	LCD backlight green
$K_2SiF_6:Mn^{4+}$ (KSF)	3d (spin-forbidden)	blue	631 nm, lines < 10 nm wide	≈ 8–10 ms	Wide-gamut display red
$Y_2O_3:Eu^{3+}$	4f→4f	254 nm	611 nm	≈ 1–2 ms	Fluorescent lamps
ZnS:Ag,Cl	D–A pair	electrons, UV	450 nm	µs	CRTs (historic), EL
$Gd_2O_2S:Tb^{3+}$	4f→4f	X-ray	545 nm	≈ 1 ms	X-ray intensifying screens
CsI:Tl	Tl^+ centre	X/γ-ray	≈ 550 nm	≈ 1 µs; ≈ 54,000 photons/MeV	Indirect X-ray flat panels
$Lu_{2(1-x)}Y_{2x}SiO_5:Ce$ (LYSO)	5d→4f	γ-ray	≈ 420 nm	≈ 40 ns; ≈ 30,000 photons/MeV	PET scanners

Two further families are worth noting. *Quantum-dot* emitters (CdSe, InP, perovskite nanocrystals) set their colour by size rather than by activator. They give very narrow emission (FWHM 20–40 nm) and are treated in Chapter 38. *Persistent* phosphors such as $SrAl_2O_4:Eu^{2+},Dy^{3+}$ store charge in deep traps and release it for hours, the materials basis of glow-in-the-dark exit signs.

35.4 LED and laser semiconductor materials

What an emitter material must provide

From Section 35.1, a material for efficient electroluminescence needs:

1. a **direct band gap** of the right energy, $E_g \approx hc/\lambda$;
2. **low non-radiative recombination**: few deep-level defects and low Auger rate at the operating carrier density;
3. a **heterostructure partner** that is lattice-matched (or strain-managed, Chapter 34) and gives type-I confinement of both carriers;
4. **dopability** both n- and p-type, so that a junction can inject carriers;
5. for lasers, a **lower-index cladding** to confine the light, and optical gain at a reachable carrier density.

These requirements are met by the III–V and III–N alloy families, in three main systems that together cover the ultraviolet to the mid-infrared.

Alloys, Vegard's law and bowing

The emission wavelength is tuned by alloying. For a ternary alloy $A_xB_{1-x}C$, the lattice constant follows Vegard's law (linear interpolation), while the band gap bows below the linear interpolation:

$$a(x) = x a_{AC} + (1-x) a_{BC}, \qquad E_g(x) = x E_{g,AC} + (1-x) E_{g,BC} - b\,x(1-x)$$

The bowing parameter b comes from the different sizes and electronegativities of the mixed atoms. It is 1.4 eV for InGaN, about 1 eV for AlGaN, and 0.48 eV for InGaAs. Quaternary alloys ($In_{1-x}Ga_xAs_yP_{1-y}$, $(Al_xGa_{1-x})_{0.5}In_{0.5}P$) have two composition variables, so the band gap can be tuned while the lattice constant stays matched to a substrate. Figure 35.2 is the map engineers use to plan such alloys.

Status of the principle: Vegard's law and quadratic band-gap bowing
Empirical. Vegard's law holds for most III–V alloys to within about 0.1% in lattice constant. The quadratic bowing form fits band gaps with a composition-independent b for most alloys, but b itself depends on composition in highly mismatched alloys (InGaN at high In, GaAsN). Both describe random alloys and fail where atoms cluster or order.

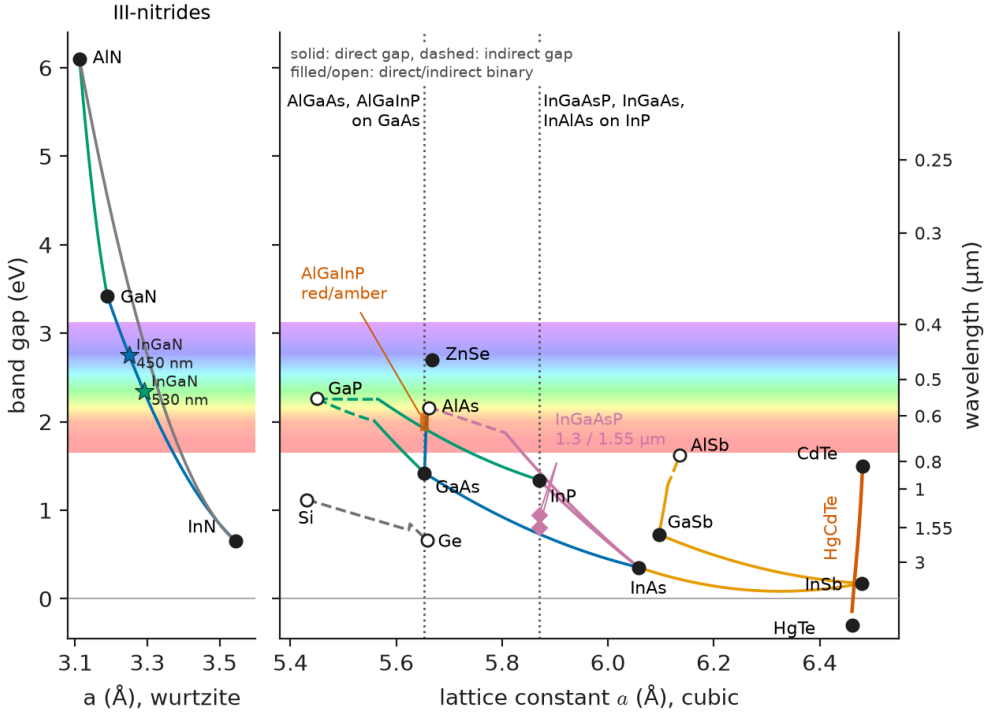


Figure 35.2 — Band gap versus lattice constant for the principal semiconductors, with the visible spectrum and alloy lines.

The three emitter systems

InGaN on GaN (violet, blue, green). $In_xGa_{1-x}N$ spans from 3.42 eV (GaN) toward 0.65 eV (InN). With $b = 1.43$ eV, $x = 0.15$ gives $E_g = 2.82$ eV (440 nm) and $x = 0.3$ gives 2.29 eV (542 nm). In real quantum wells the polarization fields and strain shift the emission by a further 0.1–0.2 eV, so blue (450 nm) LEDs use about 15–18% In and green (530 nm) about 25–30%. The active layers are 2–3 nm quantum wells grown on GaN on sapphire, SiC or Si.

The puzzle of why InGaN emits efficiently despite 10^8–10^9 cm^{-2} threading dislocations is resolved by *carrier localization*. Indium composition fluctuates on the nanometre scale, and the quantum-well width fluctuates by about a monolayer. Together these create potential minima, about 10–50 meV deep, that trap carriers within a few nanometres of where they are created. Since dislocations are typically 0.3–1 µm apart, most carriers recombine radiatively before they can diffuse to one. In addition, hexagonal "V-pits" that open around dislocations form thinner, wider-gap sidewall wells, which repel carriers from the dislocation core. Both mechanisms are properties of the material's nanostructure, not of the device design.

AlGaInP on GaAs (red, orange, amber, yellow). $(Al_xGa_{1-x})_{0.5}In_{0.5}P$ is lattice-matched to GaAs for all x. Its direct gap runs from 1.9 eV ($x = 0$, about 650 nm) to the direct–indirect crossover near $x \approx 0.53$, at about 2.3 eV (about 540 nm). Red (620–630 nm) devices are very efficient. Toward amber and yellow the efficiency collapses, because the active layer approaches the indirect crossover, where electrons spill into the X valleys and recombine non-radiatively. At the same time the conduction-band offset to the highest-Al cladding shrinks, so electrons leak out of the wells. High-brightness AlGaInP LEDs also needed a materials trick for extraction: the light-absorbing GaAs substrate is removed and the epitaxial layer is wafer-bonded to a transparent GaP or a reflective metal carrier.

InGaAsP and InGaAs on InP (1.3–1.65 µm). $In_{1-x}Ga_xAs_yP_{1-y}$ is lattice-matched to InP when $y \approx 2.2x$. Its gap then runs from 1.35 eV (InP) to 0.75 eV ($In_{0.53}Ga_{0.47}As$), covering the two silica-fibre windows at 1310 and 1550 nm (Section 35.7). This is why every long-haul telecom laser is an InP-based device. GaAs/AlGaAs (780–870 nm) and strained InGaAs/GaAs (900–1100 nm) cover the near-infrared for short-reach links, VCSELs, 3D sensing and pump lasers. AlGaN reaches the deep UV (210–360

nm). UV-C emitters for disinfection are held to external efficiencies of a few percent to about 10% by poor p-type conductivity (the Mg acceptor in AlGaN is 0.2–0.6 eV deep) and by high dislocation densities.

Table 35.6 — LED and laser material systems (typical)

Active material	Substrate	Wavelength range	Typical best external efficiency (LED)	Main limitations
$Al_xGa_{1-x}N$	AlN, sapphire	210–360 nm	2–10% (UV-C)	p-doping, dislocations, light extraction
$In_xGa_{1-x}N$	sapphire, SiC, Si, GaN	380–530 nm	70–85% blue; 30–45% green	Droop, green gap
$(Al_xGa_{1-x})_{0.5}In_{0.5}P$	GaAs (removed)	560–650 nm	50–60% red; < 20% yellow-amber	Indirect crossover, electron leakage, thermal drop
$Al_xGa_{1-x}As$ / GaAs	GaAs	650–870 nm	40–60%	AlGaAs oxidation
$In_xGa_{1-x}As$ (strained) / GaAs	GaAs	900–1100 nm	lasers	Critical thickness (Chapter 34)
$In_{1-x}Ga_xAs_yP_{1-y}$ / InP	InP	1100–1650 nm	lasers	Auger, temperature sensitivity

Efficiency droop as a materials problem

InGaN LEDs reach their peak internal efficiency at current densities of only a few A/cm². At the 35–100 A/cm² where high-power LEDs operate, efficiency falls by 20–40%. This *droop* limits how much light a given chip area can produce. It is understood today mainly as Auger recombination, plus carrier leakage and inhomogeneous injection, enhanced by features of the nitride material.

The *ABC model* expresses the competition with three coefficients. Writing n for the carrier density in the active layer (assumed $n = p$):

$$\eta_{\mathrm{IQE}}(n) = \frac{Bn^2}{An + Bn^2 + Cn^3}, \qquad J = qd\,(An + Bn^2 + Cn^3)$$

Here A is the SRH coefficient ($\mathrm{s^{-1}}$), B the radiative coefficient, C the Auger coefficient ($\mathrm{cm^6/s}$), and d the total active thickness. The IQE peaks where $An = Cn^3$, at $n^* = \sqrt{A/C}$.

> **Status of the principle: ABC recombination model**
> *Approximate.* It assumes uniform carrier density in the active region, $n = p$, Boltzmann statistics and density-independent coefficients. In reality B and C fall at high density (phase-space filling), carriers are unevenly distributed among quantum wells, and leakage over the barriers is omitted. Fitted coefficients are therefore effective values for a given structure, but the model captures the shape of droop well.

Worked example — Droop in an InGaN quantum-well LED. Take $A = 10^7\ \mathrm{s^{-1}}$, $B = 10^{-11}\ \mathrm{cm^3/s}$, $C = 10^{-30}\ \mathrm{cm^6/s}$ and an active thickness $d = 3$ nm. The peak is at $n^* = \sqrt{10^7/10^{-30}} = 3.16 \times 10^{18}\ \mathrm{cm^{-3}}$. There $An = Cn^3 = 3.16 \times 10^{25}\ \mathrm{cm^{-3}s^{-1}}$ and $Bn^2 = 1.0 \times 10^{26}\ \mathrm{cm^{-3}s^{-1}}$, so

$$\eta_{\mathrm{IQE}}^{\max} = \frac{1.0 \times 10^{26}}{3.16 \times 10^{25} + 1.0 \times 10^{26} + 3.16 \times 10^{25}} = 61\%$$

at $J = 1.602 \times 10^{-19} \times 3 \times 10^{-7}\ \mathrm{cm} \times 1.63 \times 10^{26} = 7.8\ \mathrm{A/cm^2}$. At $n = 10^{19}\ \mathrm{cm^{-3}}$ the three rates are 10^{26}, 10^{27} and 10^{27}, so $\eta_{\mathrm{IQE}} = 10^{27}/2.1 \times 10^{27} = 48\%$ at $J = 101\ \mathrm{A/cm^2}$. Raising the current density thirteenfold costs 21% of the relative efficiency. At $n = 3 \times 10^{19}$ ($J \approx 1.7\ \mathrm{kA/cm^2}$, laser-like densities) IQE falls to 25%.

The materials levers follow directly from the formula. Since droop sets in once Cn^3 rivals Bn^2, anything that lowers n at a given current helps: thicker or more numerous quantum wells, and better hole transport so that carriers spread over all the wells. The nitride's polarization fields work against this. They tilt the quantum wells, separating electrons and holes (the quantum-confined

Stark effect), which lowers the effective B and so raises n at a given current. Growth on *non-polar* or *semi-polar* GaN planes removes or reduces the fields, at the cost of more expensive substrates.

The green gap

The peak external efficiency of LEDs plotted against wavelength shows a deep valley between about 530 and 590 nm, the *green gap*, illustrated in Figure 35.3. Both material systems fail there, for different reasons.

- **InGaN approaching green** needs 25–30% indium. Indium-rich InGaN must be grown at lower temperature (In–N bonds are weak and In desorbs), which raises point-defect densities. The lattice mismatch to GaN rises to about 3%, increasing strain and approaching the critical thickness of Chapter 34. The piezoelectric field grows with strain, which strengthens the Stark separation. Composition fluctuations become clustering. Each of these either adds non-radiative recombination or lowers the radiative rate.
- **AlGaInP approaching yellow** reaches its direct–indirect crossover, as above.

The green gap is therefore a materials limit, not a device-design limit. Its practical consequences are that white light is made from blue LEDs with phosphors rather than from red, green and blue chips, and that micro-LED displays struggle with the efficiency of their green and red pixels. Phosphor-converted green, InGaN red on relaxed InGaN templates, and nitride quantum dots are the active research routes.

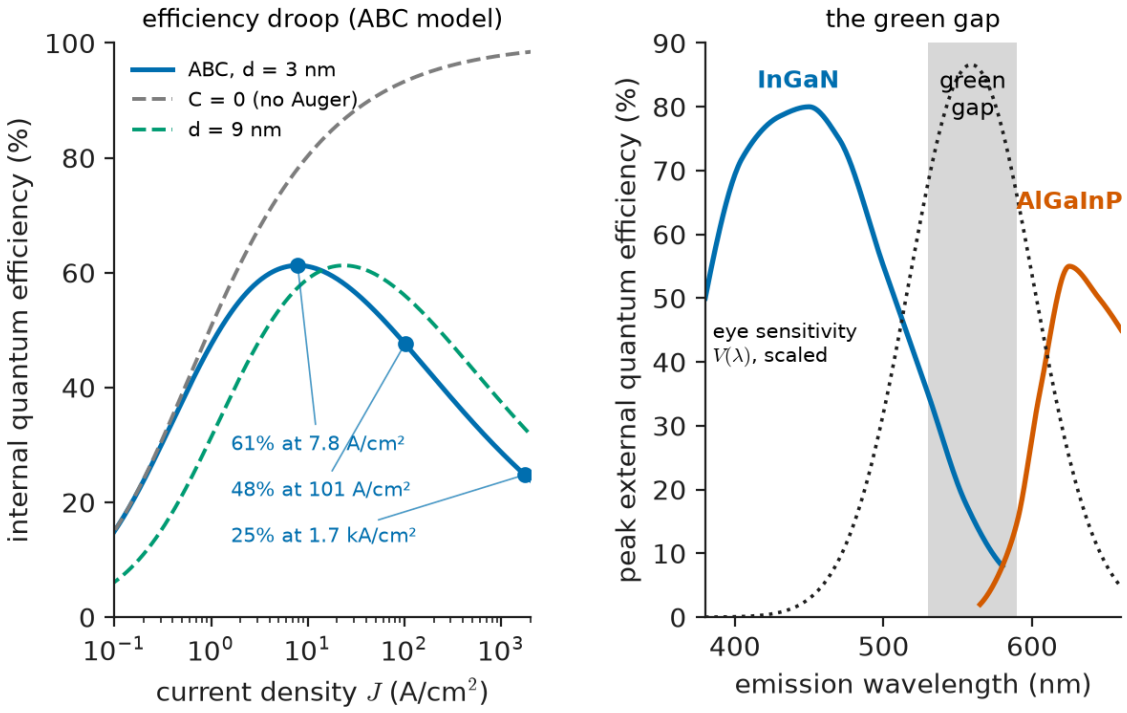


Figure 35.3 — Efficiency droop and the green gap.

Laser materials beyond the gain medium

A semiconductor laser needs the same active materials plus three further material properties.

- **Transparency and gain.** Optical gain requires population inversion, which happens once the carrier density exceeds the *transparency density*, about $1–2 \times 10^{18}$ cm^{-3} in GaAs and InGaAsP wells. Compressively strained quantum wells (Chapter 34) lower the in-plane hole effective mass, which reduces the transparency density and the threshold. That is the main reason strained-layer lasers replaced lattice-matched ones in the 1990s.
- **Index contrast for optical confinement.** The cladding must have a lower refractive index than the active region. Conveniently, wider-gap alloys usually have lower indices: at 850 nm GaAs has $n \approx 3.6$ and AlAs $n \approx 3.0$. In vertical-cavity lasers (VCSELs), the mirrors are *distributed Bragg reflectors* made of quarter-wave stacks of alternating high- and low-index layers. GaAs/AlAs gives enough contrast that 20–35 pairs exceed 99.5% reflectivity. GaN/AlN gives much less contrast and is badly mismatched, which is one reason blue VCSELs use dielectric (SiO_2/Ta_2O_5) mirrors.

- **Oxidation-defined apertures.** $Al_{0.98}Ga_{0.02}As$ oxidizes laterally in steam to insulating Al_2O_3 at a controllable rate, forming the current and optical aperture of every modern VCSEL. It is a remarkable case of a materials reaction becoming a device structure.

Other laser materials follow the host-plus-activator logic of Section 35.3. Nd^{3+}:YAG emits at 1064 nm. Er^{3+} in silica fibre, pumped at 980 or 1480 nm, gives optical gain from 1530 to 1565 nm with a 10 ms upper-state lifetime: the erbium-doped fibre amplifier on which long-haul optical networks depend.

35.5 Photodetector materials

Cut-off wavelength and responsivity

A detector based on interband absorption responds only to photons with $h\nu \geq E_g$. Its *cut-off wavelength* is

$$\lambda_c = \frac{hc}{E_g} = \frac{1.2398\ \mu\text{m eV}}{E_g}$$

> **Status of the principle: Cut-off wavelength $\lambda_c = hc/E_g$**
> *Derived.* It follows from energy conservation in interband absorption. The real response falls off gradually near λ_c because of band tails (Section 35.1), phonon-assisted absorption in indirect materials, and the temperature dependence of E_g.

The *responsivity* is the photocurrent per watt of incident optical power:

$$\mathcal{R} = \frac{I_{ph}}{P} = \eta\,\frac{q\lambda}{hc} = \eta\,\frac{\lambda\,[\mu\text{m}]}{1.2398}\ \ \text{A/W}$$

where the quantum efficiency η (electrons collected per incident photon) includes surface reflection R_s, absorption over the active thickness d, and collection:

$$\eta \approx (1 - R_s)\left(1 - e^{-\alpha d}\right)\eta_{\text{coll}}$$

Dark current and why IR detectors are cooled

The noise floor of a detector is largely its *dark current*, the current flowing with no light. In a diode it is dominated by thermal generation in or near the depletion region, which scales with n_i or n_i^2. Since $n_i^2 \propto T^3 \exp(-E_g/kT)$ (Chapter 32), dark current rises exponentially as the gap narrows. A material with $\lambda_c = 10$ μm ($E_g \approx 0.12$ eV) at room temperature has $E_g/kT \approx 4.8$, so its thermally generated carriers swamp any signal. Cooling to 77 K raises E_g/kT to about 19. For a 0.1 eV gap, the factor $\exp(-E_g/kT)$ falls by $\exp[0.1(1/0.00664 - 1/0.02585)] = e^{11.2} \approx 7 \times 10^4$, and with the T^3 factor the total reduction of n_i^2 is about 4×10^6. That is why long-wavelength infrared cameras use Stirling coolers or liquid nitrogen. Uncooled thermal detectors (Section 35.9) avoid the problem by not using interband absorption at all.

The main detector materials

Silicon ($E_g = 1.12$ eV, $\lambda_c = 1.1$ μm). It is the material of all visible image sensors and of short-wavelength detectors, with very low dark current, mature processing and integration with CMOS readout. Its indirect gap gives a long absorption length in the near-IR (Table 35.1), so its responsivity falls beyond about 900 nm unless the junction is deep. The quantum efficiency in the visible exceeds 80% with anti-reflection coatings. For high gain at low light, silicon avalanche photodiodes benefit from a material asymmetry: the impact-ionization coefficient of electrons is much larger than that of holes ($\alpha_n/\alpha_p \approx 20$–$50$ at moderate fields), which gives low excess noise.

Germanium ($E_g = 0.66$ eV indirect, 0.80 eV direct, practical cut-off about 1.6 μm at 300 K). Ge has strong absorption at 1310 nm and adequate absorption at 1550 nm, and it can be grown epitaxially on silicon. Tensile strain of about 0.2%, from thermal mismatch on cooling, lowers its direct gap and extends the response to 1600 nm. Its drawback is higher dark current, from the small gap and from the misfit dislocations at the 4.2% mismatched Ge/Si interface (Chapter 34).

$In_{0.53}Ga_{0.47}As$ ($E_g = 0.75$ eV, $\lambda_c = 1.65$ μm). Lattice-matched to InP, with a direct gap and a strong absorption of about 7×10^3 cm^{-1} at 1550 nm, it is the standard telecom and short-wave-infrared detector material. "Extended" InGaAs with more indium (lattice-mismatched, on graded buffers) reaches 2.2–2.6 μm, at the cost of dislocation-driven dark current.

HgCdTe ($Hg_{1-x}Cd_xTe$). CdTe has a gap of 1.5 eV and HgTe a *negative* gap (a semimetal, about −0.3 eV). Between them the alloy gap can be set anywhere from 0 to 1.5 eV with almost no change in lattice constant (0.3% mismatch across the whole range), so a single substrate (CdZnTe) serves all wavelengths. An empirical fit (Hansen et al.) for the gap is

$$E_g(x,T) = -0.302 + 1.93x - 0.81x^2 + 0.832x^3 + 5.35\times10^{-4}\,T(1-2x) \ \ [\text{eV}]$$

> **Status of the principle: Hansen band-gap formula for HgCdTe**
> *Empirical.* It is a polynomial fit to optical and electrical data over $0 \le x \le 0.6$ and 4–300 K, accurate to about ±10 meV in that range.

At 77 K, $x = 0.3$ gives $E_g = 0.243$ eV ($\lambda_c = 5.1$ μm, mid-wave IR), $x = 0.22$ gives 0.115 eV ($\lambda_c = 10.8$ μm, long-wave IR) and $x = 0.2$ gives 0.083 eV ($\lambda_c = 14.9$ μm). HgCdTe has a direct gap, high absorption, high electron mobility and a long intrinsic lifetime, and it remains the highest-performance IR material. Its weaknesses are materials ones. The Hg–Te bond is weak, so mercury vacancies form readily and act as acceptors. Composition uniformity is hard to achieve, and since λ_c depends steeply on x (near $x = 0.22$, a change of 0.01 in x shifts λ_c at 77 K by more than 1 μm), uniformity matters a great deal. Wafers are also small and expensive. Competing materials are InSb (MWIR, 77 K), type-II InAs/GaSb superlattices (Chapter 34's broken-gap alignment, tunable from 3 to over 15 μm by layer thickness) and quantum-well infrared photodetectors.

Table 35.7 and Figure 35.4 summarize the spectral coverage and responsivity of these materials.

Table 35.7 — Photodetector materials (typical)

Material	E_g (eV)	λ_c (μm)	Operating T	Typical peak responsivity	Use
$Al_xGa_{1-x}N$ ($x \ge 0.45$)	≥ 4.4	≤ 0.28	300 K	0.1–0.2 A/W	Solar-blind UV, flame detection
Si	1.12	1.1	300 K	0.5–0.6 A/W at 800–900 nm	Imaging, 850 nm links, metrology
Ge (on Si)	0.66 / 0.80	≈1.6	300 K	0.8–1.0 A/W at 1550 nm (waveguide)	Silicon photonics receivers
$In_{0.53}Ga_{0.47}As$	0.75	1.65	300 K or TE-cooled	≈ 1.0 A/W at 1550 nm	Telecom, SWIR imaging
PbS	0.37	≈3	300 K	photoconductive	Low-cost SWIR
InSb	0.23 (77 K)	5.4	77 K	—	MWIR imaging
$Hg_{0.7}Cd_{0.3}Te$	0.24 (77 K)	5.1	77–150 K	—	MWIR imaging
$Hg_{0.78}Cd_{0.22}Te$	0.115 (77 K)	10.8	77 K	—	LWIR imaging, thermography
InAs/GaSb superlattice	0.05–0.4 (effective)	3–25	77 K	—	MWIR/LWIR focal planes
VO_x microbolometer	thermal	8–14 (by absorber)	300 K	TCR ≈ −2 to −3 %/K	Uncooled thermal cameras

Worked example — Matching absorber thickness to wavelength. Compare three detectors with anti-reflection coatings ($R_s \approx 0$) and full collection.

- *Si at 850 nm*, absorbing layer 10 μm, $\alpha = 535$ cm^{-1}: $\alpha d = 0.535$ and $\eta = 1 - e^{-0.535} = 41\%$. The responsivity is $0.41 \times 0.85/1.2398 = 0.28$ A/W. Reaching $\eta = 80\%$ needs $d = \ln 5/535$ cm $= 30$ μm. Carriers generated that deep diffuse slowly to the junction, which limits speed. This is why 850 nm receivers above about 10 Gb/s use GaAs photodiodes.
- *$In_{0.53}Ga_{0.47}As$ at 1550 nm*, 2 μm thick, $\alpha \approx 7\times10^3$ cm^{-1}: $\alpha d = 1.4$, $\eta = 75\%$, and $\mathcal{R} = 0.75 \times 1.55/1.2398 = 0.94$ A/W.
- *Ge at 1550 nm*, 1 μm film on Si, $\alpha \approx 460$ cm^{-1}: $\alpha d = 0.046$ and $\eta = 4.5\%$ for normal incidence. This is unusable. But if light travels *along* a Ge waveguide instead of across the film, the absorption length can be made 20–50 μm while the carriers cross only the 0.5–1 μm film thickness. With 20 μm, $\eta = 1 - e^{-0.92} = 60\%$, and with 50 μm it is 90%. Separating

the light path from the carrier path is the geometric solution that made germanium the standard detector of silicon photonics (Section 35.7).

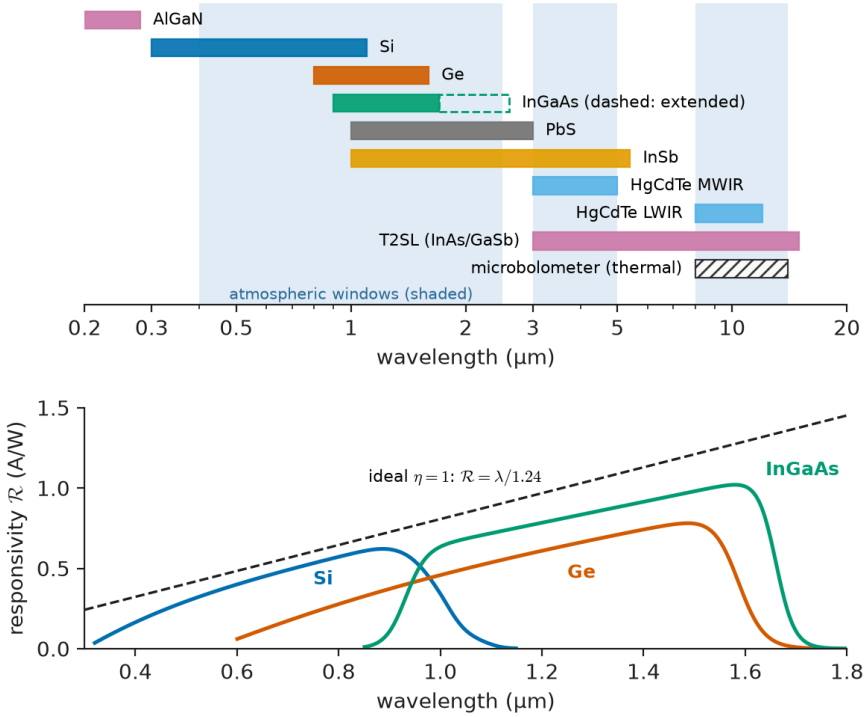


Figure 35.4 — Spectral coverage and responsivity of photodetector materials.

35.6 Solar-cell absorber materials

Solar cells are treated in two places. This section asks what *intrinsic material property* limits the efficiency of a photovoltaic absorber. Chapter 40 treats photovoltaic absorbers as energy-conversion materials, including their processing, stability, cost and scalability, alongside thermoelectrics and batteries. Module and system engineering belong to Electrical Energy & Machines.

> **Boundary.**
> Only the material-level efficiency limit and the absorber properties that control it are treated here. Junction design and cell operation belong to Electronics & Computing. Absorber manufacturing, degradation and cost are in Chapter 40. PV system sizing is in Electrical Energy & Machines.

The Shockley–Queisser limit as a material statement

Shockley and Queisser (1961) asked: what is the highest efficiency possible for an absorber of band gap E_g, if the *only* loss allowed is the one required by thermodynamics? They made four assumptions:

1. the absorber absorbs every photon with $h\nu \geq E_g$ and none below (a step-function absorptivity);
2. each absorbed photon creates exactly one electron–hole pair, which relaxes (thermalizes) to the band edges, losing $h\nu - E_g$ as heat;
3. the only recombination is *radiative*, as required by detailed balance, since a body that absorbs must also emit (Section 35.1);
4. the cell is at 300 K under the unconcentrated solar spectrum.

Two losses then fight each other. A large E_g wastes all photons below the gap. A small E_g absorbs most photons but throws away most of their energy as heat. The optimum under the AM1.5G spectrum is about 33.7% at $E_g \approx 1.34$ eV. The curve is flat-topped: any gap between 1.1 and 1.4 eV allows 32–34%, as Figure 35.5 shows.

> **Status of the principle: Shockley–Queisser detailed-balance limit**
> *Derived.* It follows from detailed balance between absorbed sunlight and emitted thermal radiation for a single-gap absorber under the four assumptions above. It is a rigorous upper bound for that idealization. It is exceeded only by breaking an assumption: multiple junctions, concentration, or carrier multiplication.

Real absorbers fall short of the limit for three material reasons, each of which can be stated as a property.

Incomplete absorption. An absorber of finite thickness, especially an indirect one, does not absorb everything above the gap. Silicon needs about 100 μm plus light trapping.

Non-radiative recombination. Every non-radiative path lowers the open-circuit voltage. Detailed balance gives an exact relation (Ross, Rau) between voltage loss and the absorber's *external radiative efficiency* (ERE), the fraction of recombination events that produce a photon escaping the cell:

$$V_{oc} = V_{oc}^{\mathrm{rad}} + \frac{kT}{q}\ln(\mathrm{ERE})$$

> **Status of the principle: Voltage loss from external radiative efficiency**
> *Derived.* It follows from detailed balance and the reciprocity between light absorption and emission. It is exact under the assumptions that carriers are in quasi-equilibrium with uniform quasi-Fermi-level splitting and that the superposition principle holds.

A good solar cell must therefore also be a good LED, which ties this section to Section 35.4.

Sub-gap tails. A broad Urbach tail (Section 35.1) absorbs a little below the gap and therefore emits there too. It adds radiative recombination at energies below E_g and lowers V_{oc}^{rad} itself.

Worked example — Voltage loss from non-radiative recombination. At 300 K, $kT/q = 25.85$ mV. A GaAs cell with ERE ≈ 0.3 (a record-class device that is also an excellent LED) loses $0.02585 \times \ln(1/0.3) = 31$ mV relative to the radiative limit. A silicon cell with ERE ≈ 0.005 loses $0.02585 \times \ln 200 = 137$ mV. A polycrystalline thin-film absorber with ERE ≈ 10^{-4} loses $0.02585 \times \ln 10^4 = 238$ mV. Since the radiative-limit voltage is roughly $E_g/q - 0.25$ to -0.3 V, the observed deficits $E_g/q - V_{oc}$ are about 0.3 V (GaAs), 0.37–0.4 V (Si), and 0.45–0.6 V (CdTe, CIGS, kesterite). The differences trace directly to defect densities and tail widths.

What makes a good absorber

From the above, the material properties that matter are:

- **band gap** in or near 1.1–1.5 eV for a single junction, or a matched pair for tandems (about 1.7 eV on top of 1.1 eV);
- **absorption coefficient**: direct-gap absorbers need only 0.5–3 μm;
- **diffusion length** $L = \sqrt{D\tau}$ longer than the absorber thickness, so that carriers reach the junction. For silicon with $D \approx 26$ cm²/s and $\tau = 1$ ms, $L = \sqrt{0.026}$ cm $= 1.6$ mm, so 150 μm is easily collected;
- **defect tolerance**: intrinsic defects with shallow rather than midgap levels, and benign grain boundaries. Lead-halide perovskites owe much of their success to this. Their band edges are formed from antibonding states, so vacancies tend to produce shallow levels;
- **small Urbach energy and high ERE**;
- **stability, abundance and non-toxicity**, which are taken up in Chapter 40.

Table 35.8 — Solar absorber materials (typical; record efficiencies approximate as of the mid-2020s)

Absorber	E_g (eV)	Type	SQ limit (≈)	Record cell efficiency (≈)	Typical thickness	Key material issue
c-Si	1.12	indirect	33% (practical limit 29.4% with Auger)	27%	100–180 µm	Weak absorption; surface passivation (Chapter 34)
GaAs	1.42	direct	33%	29%	1–3 µm	Cost of epitaxy and substrate
CdTe	1.45–1.5	direct	32%	23%	2–4 µm	p-doping, grain boundaries, Cd
$Cu(In,Ga)Se_2$	1.0–1.2	direct	33%	23–24%	1.5–2.5 µm	Composition gradients, In and Ga supply
$CH_3NH_3PbI_3$-type perovskite	1.5–1.6	direct	31–32%	26–27%	0.4–0.8 µm	Stability, ion migration, Pb
a-Si:H	1.7–1.8	(amorphous)	28%	≈ 10% (stabilized)	0.3 µm	Urbach tail, light-induced degradation
Cu_2ZnSnS_4	1.5	direct	32%	≈ 14–15%	1–2 µm	Cation disorder, large E_U

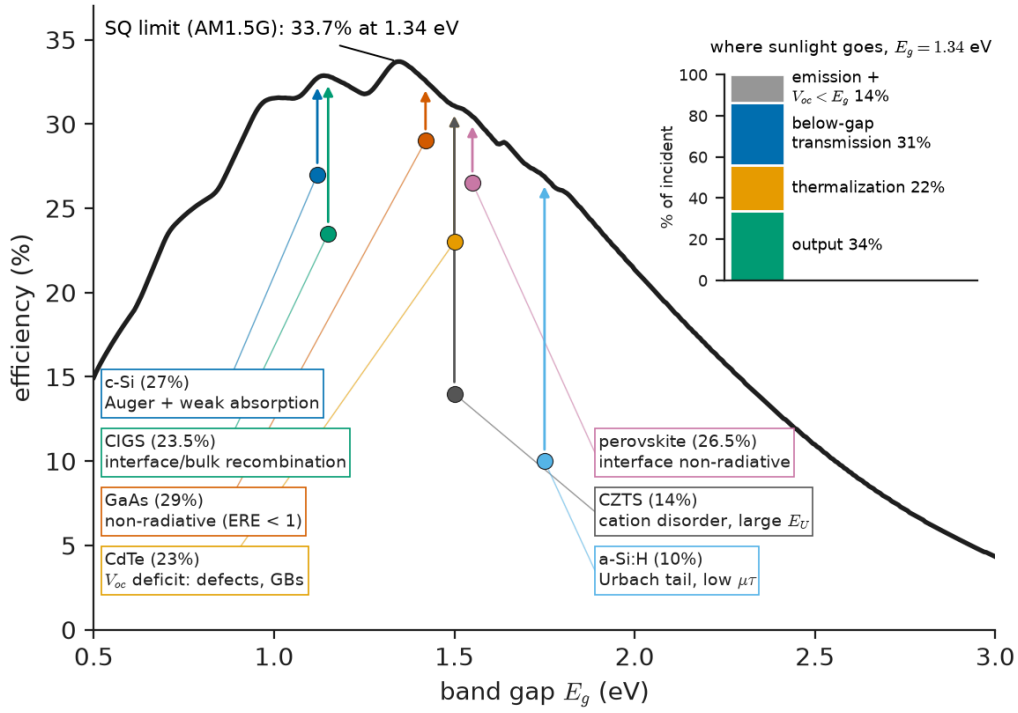


Figure 35.5 — Shockley–Queisser efficiency limit versus band gap, with real absorber materials.

35.7 Optical and photonic materials

A photonic material does not absorb or emit. Its job is to *guide*, *switch*, or *modulate* light with low loss. The relevant properties are the refractive index n (and its contrast with neighbouring materials), the transparency window, the dispersion $dn/d\lambda$, and the electro-optic, thermo-optic and non-linear coefficients.

Table 35.9 — Photonic materials at 1550 nm (typical)

Material	n at 1550 nm	Transparency (μm)	Waveguide loss	Modulation mechanism	Role
SiO_2 (fibre, Ge-doped core)	1.444–1.45	0.2–2.0	0.15–0.2 dB/km	— (very weak Kerr)	Transmission fibre, cladding
Si (SOI strip)	3.48	1.1–8	1–3 dB/cm	Plasma dispersion, thermo-optic ($dn/dT = 1.8 \times 10^{-4}$ K^{-1})	Silicon photonics circuits
Si_3N_4 (LPCVD)	1.99	0.4–6	0.001–0.5 dB/cm	Thermo-optic (weak)	Low-loss waveguides, filters, frequency combs
$LiNbO_3$ (thin film)	n_e 2.14, n_o 2.21	0.35–5	0.03–0.3 dB/cm	Pockels, $r_{33} = 30.8$ pm/V	High-speed modulators
InP / InGaAsP	3.17–3.4	> 0.92 (InP; longer for quaternaries)	1–3 dB/cm	Carrier injection, QCSE	Lasers, modulators, integrated transmitters
Ge	≈ 4.28	1.6–14	absorbing below ≈ 1.6 μm	Franz–Keldysh	Detectors, electro-absorption modulators
EO polymers	1.6–1.8	visible–IR	1–2 dB/cm	Pockels, r_{33} up to 100+ pm/V	Research modulators (thermal stability)

Silica optical fibre

Optical fibre is the lowest-loss engineered material for any wave. At 1550 nm standard fibre loses only about 0.2 dB/km. After 100 km, 1% of the light remains (20 dB). The record is about 0.14 dB/km in pure-silica-core fibre. The loss has three material contributions, shown in Figure 35.6.

1. **Rayleigh scattering.** Silica is a glass. Its density and composition fluctuations were frozen in at the glass transition (about 1200 °C) and are much smaller than the wavelength. They scatter as

$$\alpha_R = \frac{A_R}{\lambda^4}, \qquad A_R \approx 0.7\text{–}0.9 \text{ dB km}^{-1}\,\mu\text{m}^4$$

At 1550 nm, with $A_R = 0.8$, $\alpha_R = 0.8/1.55^4 = 0.14$ dB/km. At 1310 nm it is 0.27 dB/km, and at 850 nm 1.5 dB/km. A_R is set by the fictive temperature and the dopant level. Pure-silica cores, which lack germania's composition fluctuations, have the lowest values.

> **Status of the principle: Rayleigh λ^{-4} scattering loss**
> *Derived.* It follows from dipole scattering by inhomogeneities much smaller than the wavelength. The coefficient A_R is a material parameter set by the thermodynamic density fluctuations frozen in at the fictive temperature and by compositional fluctuations from dopants.

2. **Infrared absorption.** Si–O bond vibrations have their fundamental near 9 μm. The tail of their overtones and combinations rises steeply beyond 1.6 μm, which closes the window on the long side.
3. **Impurity absorption.** Transition-metal ions (Fe, Cu) must be below parts per billion. The O–H vibration overtone at 1383 nm was once a strong peak of about 2 dB/km. Fibre-preform processes in dry chlorine have reduced it below the Rayleigh level in "low-water-peak" fibres.

The minimum of Rayleigh plus infrared loss falls near 1550 nm. That, plus the erbium amplifier band (Section 35.4), fixed the wavelength of long-haul communications. The core is raised in index by about 0.35% with GeO_2, or the cladding is lowered with fluorine, to form the guide. Material dispersion in silica passes through zero near 1.27 μm, and fibre-design (waveguide) dispersion shifts the total zero to 1.31 μm in standard single-mode fibre.

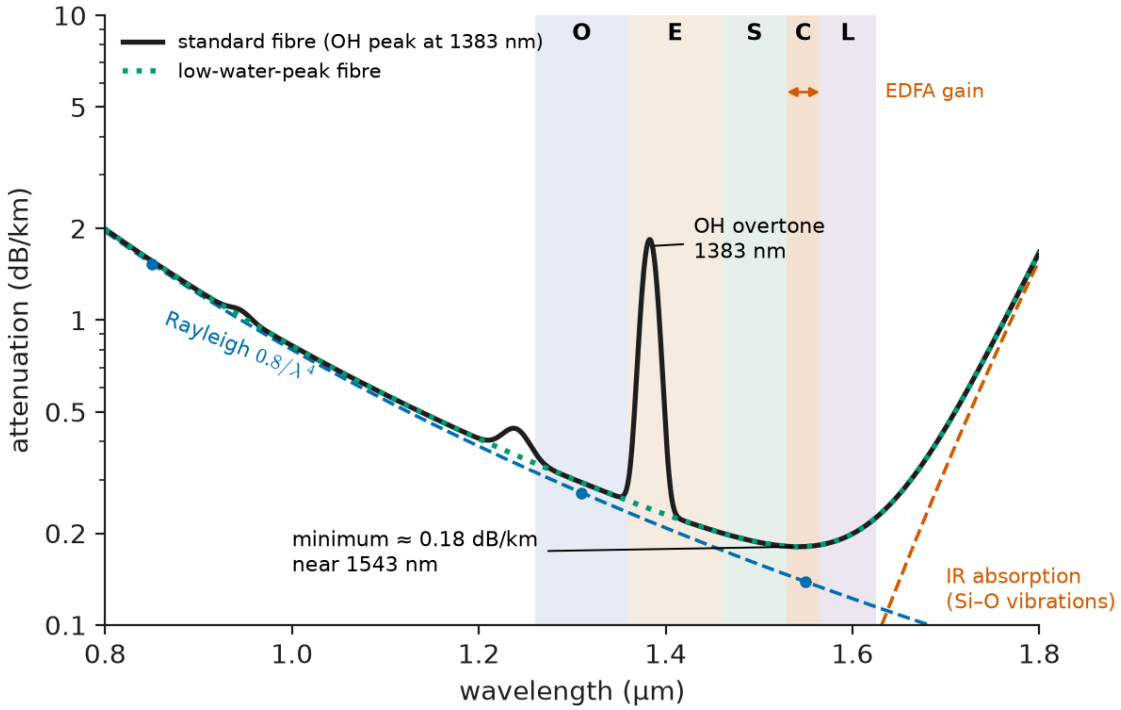


Figure 35.6 — Attenuation spectrum of silica optical fibre and its material contributions.

Lithium niobate: electro-optic modulation

To put data on light at tens of gigabits per second, an engineer needs a material whose refractive index changes, linearly and fast, with an applied field. That is the *Pockels* (linear electro-optic) effect, introduced in Chapter 27. It exists only in non-centrosymmetric crystals. For a field along the optic (c) axis of $LiNbO_3$ acting on light polarized along it,

$$\Delta n_e = -\tfrac{1}{2}\, n_e^3\, r_{33}\, E$$

> **Status of the principle: Linear electro-optic (Pockels) effect**
> *Approximate.* It is the first term of an expansion of the optical impermeability in the applied field. It is accurate for fields well below breakdown and for frequencies below the relevant lattice resonances. At very high fields the quadratic (Kerr) term and photorefractive effects appear.

The electronic and ionic response of $LiNbO_3$ is fast (into the hundreds of GHz) and the effect is pure phase with no absorption. The material is a ferroelectric, poled into a single domain, with $r_{33} = 30.8$ pm/V. For two decades it was used as Ti-diffused waveguides in bulk wafers. Since about 2018 *thin-film lithium niobate on insulator* (TFLN), a 300–600 nm $LiNbO_3$ layer bonded on SiO_2, has confined light so tightly that the electrodes can sit a few micrometres apart.

Worked example — Half-wave voltage of a thin-film $LiNbO_3$ modulator. A push–pull Mach–Zehnder modulator has an electrode gap $d = 5$ μm, a field–optical-mode overlap $\Gamma = 0.7$, length $L = 1$ cm, and works at $\lambda = 1.55$ μm with $n_e = 2.138$. The voltage needed for a π phase difference between the two arms is

$$V_\pi = \frac{\lambda\, d}{2\, n_e^3\, r_{33}\, \Gamma\, L} = \frac{1.55 \times 10^{-6} \times 5 \times 10^{-6}}{2 \times 9.77 \times 30.8 \times 10^{-12} \times 0.7 \times 0.01} = 1.84\ \text{V}$$

so $V_\pi L \approx 1.8$ V cm. Reported TFLN devices have $V_\pi L \approx 2$–2.5 V cm. The older Ti:$LiNbO_3$ devices, with gaps of 15–20 μm and weaker overlap, need about 10–15 V cm. The materials change from bulk to thin film, not any new physics, gave a fivefold improvement, enough to drive the modulator directly from CMOS voltages.

Silicon photonics

Silicon is transparent beyond 1.1 μm, can be patterned with CMOS tools, and comes as silicon-on-insulator (SOI) wafers. The Si/SiO_2 index contrast (3.48 against 1.44) is enormous compared with fibre (0.5%). A 450 × 220 nm silicon strip guides light

around bends a few micrometres in radius, against centimetres for fibre, so thousands of components fit on a millimetre-scale chip. Silicon's material limits are the reasons silicon photonics is a *heterogeneous* platform:

- **No emission.** Silicon has an indirect gap (Section 35.1). Lasers are InP-based dies bonded or flip-chipped onto the silicon, or quantum-dot lasers grown epitaxially on silicon.
- **No Pockels effect.** Silicon's crystal is centrosymmetric, so $r = 0$. Modulators use the *plasma-dispersion effect*: free carriers lower the refractive index and add absorption. At 1550 nm (the empirical Soref–Bennett relations),

$$\Delta n = -\left(8.8 \times 10^{-22}\Delta N + 8.5 \times 10^{-18}\Delta P^{0.8}\right), \qquad \Delta\alpha = 8.5 \times 10^{-18}\Delta N + 6.0 \times 10^{-18}\Delta P$$

with densities in $\mathrm{cm^{-3}}$ and $\Delta\alpha$ in $\mathrm{cm^{-1}}$. Injecting $\Delta N = \Delta P = 10^{18}$ $\mathrm{cm^{-3}}$ gives $\Delta n = -3.0 \times 10^{-3}$, but also $\Delta\alpha = 14.5$ $\mathrm{cm^{-1}}$ (63 dB/cm), so the index change comes with loss. A π phase shift needs $L_\pi = \lambda/(2|\Delta n|) = 0.26$ mm of such a region. Depletion-mode modulators trade efficiency for speed (above 50 GHz).

> **Status of the principle: Soref–Bennett plasma-dispersion relations**
> *Empirical.* They are fits to Kramers–Kronig transforms of measured free-carrier absorption in silicon at 1310 and 1550 nm, valid for carrier densities of about 10^{17}–10^{20} $\mathrm{cm^{-3}}$.

- **Thermo-optic tuning** ($dn/dT = 1.8 \times 10^{-4}$ $\mathrm{K^{-1}}$) is strong. Ring resonators must be actively stabilized, since a 1 K change shifts a resonance by about 0.08 nm.
- **Two-photon absorption** at 1550 nm (two photons together exceed E_g) limits optical power and non-linear devices. Si_3N_4, with its 5 eV gap, is chosen for high-power and frequency-comb applications.
- **Detection** uses epitaxial Ge (Section 35.5).

Sidewall roughness from etching scatters light in proportion to the square of the index contrast, which is why silicon wire waveguides lose 1–3 dB/cm while lower-contrast nitride guides can lose under 0.01 dB/cm.

35.8 Material properties for high-frequency electronics

At radio and microwave frequencies, from hundreds of MHz to hundreds of GHz, three material questions dominate. How fast can carriers move? How much does the substrate absorb? How much do the dielectrics and conductors dissipate? Circuit topology belongs to Electronics & Computing. The material answers set the ceiling that topology cannot exceed.

Transport: mobility and saturation velocity

The transit time of carriers under a transistor gate sets its current-gain cut-off frequency, roughly $f_T \approx v/(2\pi L_g)$. At low fields, velocity rises as $v = \mu E$. In short channels the field exceeds 10 kV/cm and carriers reach the *saturation velocity* v_{sat}, limited by optical-phonon emission (Chapter 32, high-field transport). In III–V materials the velocity overshoots before the electrons transfer into heavier upper valleys. A high mobility matters for low access resistance and low-field gain. A high v_{sat}, and peak velocity, matter for the intrinsic speed.

The power a transistor can deliver at frequency f also depends on the field the material can withstand before breakdown. The *Johnson figure of merit* combines the two:

$$f_T\, V_{br} \approx \frac{E_{br}\, v_{sat}}{2\pi}$$

> **Status of the principle: Johnson figure of merit**
> *Approximate.* It is derived from the transit time across a region whose length is set by breakdown, for a uniform field at saturation velocity. It ranks materials for power at frequency within about a factor of two. Real devices depart from it because of field non-uniformity, velocity overshoot and thermal limits.

Table 35.10 — Material properties for high-frequency electronics (300 K, typical)

Material	E_g (eV)	μ_n (cm²/V s)	v_{sat} (10^7 cm/s)	E_{br} (MV/cm)	κ (W/m K)	ε_r	Johnson FOM (V GHz)
Si	1.12	1400	1.0	0.3	150	11.7	≈ 480
GaAs	1.42	8500	1.0 (peak 2)	0.4	46	12.9	≈ 640
InP	1.34	5400	1.0 (peak 2.5)	0.5	68	12.5	≈ 800
$In_{0.53}Ga_{0.47}As$	0.75	12000	≈1 (peak ≈3)	≈0.2	5	13.9	—
4H-SiC	3.26	950	2.0	2.5–3	370–490	9.7	≈ 8000
GaN	3.42	1000 (bulk); 1500–2200 (2DEG)	2.5 (peak ≈3)	3.3	130–230	9.0	≈ 13000
Diamond	5.47	2000–4500	1.5–2.7	≈ 10	2000	5.7	—

The table explains the division of labour in RF electronics. InP and InGaAs HEMTs and HBTs give the highest f_T (above 1 THz in research devices) for low-noise amplifiers and sub-THz receivers. GaAs serves handset power amplifiers. GaN dominates high-power RF (base stations, radar), because its breakdown field and 2DEG (Chapter 34) give ten to twenty times the power density of GaAs, though its moderate thermal conductivity forces GaN-on-SiC substrates. SiGe and CMOS win on integration, with f_T of 300–500 GHz in SiGe HBTs. The Ge in the base grades the band gap and so builds in a field that speeds electrons across, an application of Chapter 34's heterostructures.

Substrate loss

Passive components (inductors, transmission lines, antennas) on a chip see the substrate as part of their dielectric. A semiconducting substrate of conductivity σ and permittivity ε has an effective loss tangent

$$\tan\delta_{\text{sub}} = \frac{\sigma}{\omega\varepsilon}$$

It behaves as a conductor below its *dielectric relaxation frequency* $f_{rel} = \sigma/(2\pi\varepsilon)$ and as a lossy dielectric above it.

Worked example — Why RF front ends need high-resistivity substrates. For silicon, $\varepsilon = 11.7\varepsilon_0 = 1.036 \times 10^{-10}$ F/m.

- Standard CMOS substrate, $\rho = 10\ \Omega$ cm ($\sigma = 10$ S/m): $f_{rel} = 10/(2\pi \times 1.036 \times 10^{-10}) = 15$ GHz, and at 10 GHz $\tan\delta = 1.5$. The substrate is closer to a conductor than to a dielectric. An on-chip inductor's quality factor collapses to below about 10, and transmission lines lose several dB per cm.
- High-resistivity silicon, $\rho = 1$ kΩ cm: $f_{rel} = 154$ MHz, and at 10 GHz $\tan\delta = 0.015$.
- At 10 kΩ cm: $\tan\delta = 1.5 \times 10^{-3}$, comparable with a good PCB laminate.

There is an interface catch. Fixed positive charge in the buried oxide of an SOI wafer (Chapter 34's Q_f) induces an electron accumulation layer in high-resistivity silicon just beneath it. This *parasitic surface conduction* layer has a sheet resistance of only kΩ/□, which undoes the benefit of the high resistivity. RF-SOI wafers for smartphone switches therefore insert a *trap-rich* polysilicon layer under the buried oxide. Its grain-boundary traps (Chapter 34) pin the Fermi level and immobilize the induced carriers, restoring effective resistivities above 1 kΩ cm. It is a case where deliberately adding interface traps is the solution.

The other RF substrates are intrinsically insulating. Semi-insulating GaAs (10^7–$10^9\ \Omega$ cm, from Cr doping or the EL2 deep donor), semi-insulating SiC (above $10^9\ \Omega$ cm, vanadium-doped or high-purity), and sapphire are the common choices.

Low-loss dielectrics and conductors

Off-chip, the PCB laminate, ceramic substrate or package dielectric carries the signal. The dielectric attenuation of a TEM transmission line is approximately

$$\alpha_d \approx \frac{27.3\sqrt{\varepsilon_r}\,\tan\delta}{\lambda_0}\ \text{dB per unit length}$$

with λ_0 the free-space wavelength in the same length unit. Loss mechanisms in dielectrics (dipolar relaxation, ionic resonance, conduction) were treated in Chapter 26, and laminate construction is in Chapter 51. Here the point is ranking. At 28 GHz ($\lambda_0 =$

10.7 mm), FR-4 ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$) loses about 1.07 dB/cm, a hydrocarbon-ceramic laminate such as RO4350B (3.48, 0.0037) about 0.18 dB/cm, and PTFE (2.1, 0.0003) about 0.01 dB/cm. FR-4 is therefore excluded from mm-wave signal layers.

> **Status of the principle: Dielectric attenuation** $\alpha_d \propto \sqrt{\varepsilon_r}\tan\delta/\lambda_0$
> *Approximate.* It holds for a TEM or quasi-TEM line fully filled by a low-loss dielectric ($\tan\delta \ll 1$). Microstrip with fields partly in air needs an effective-permittivity correction.

Table 35.11 — Low-loss dielectric materials (typical, near 10 GHz)

Material	ε_r	$\tan\delta$	Typical use
PTFE	2.1	0.0002–0.0004	Coaxial cable, mm-wave laminates
PTFE–glass/ceramic laminate	2.2–3.0	0.0009–0.0015	Radar, satellite boards
Hydrocarbon–ceramic laminate	3.4–3.7	0.003–0.004	5G, automotive radar boards
Liquid-crystal polymer	2.9–3.0	0.002–0.004	Flexible mm-wave circuits, antennas-in-package
Polyimide	3.3–3.5	0.003–0.008	Flex circuits
FR-4 (epoxy–glass)	4.2–4.7	0.015–0.025	General PCBs below ≈ 3 GHz
Fused silica	3.8	0.0001–0.0002	Thin-film circuits, interposers
Alumina (99.6%)	9.8	0.0001–0.0002	Thin-film microwave substrates
AlN	8.6–8.9	0.0003–0.001	High-power RF packages
LTCC	5.9–7.8	0.001–0.002	Multilayer RF modules
Sapphire	9.4 (⊥c), 11.6 (∥c)	< 0.0001	Low-loss substrates, resonators
High-resistivity Si (10 kΩ cm)	11.7	≈ 0.0015	RF-SOI, interposers

Conductor loss completes the picture. At high frequency, current crowds into a skin depth $\delta_s = \sqrt{\rho/(\pi f \mu)}$, which is 0.65 µm in copper at 10 GHz (Chapter 25). Surface roughness of the copper foil comparable with δ_s lengthens the current path and can double conductor loss. Smooth "very-low-profile" foils are a materials response. Dielectric resonators and their high-Q ceramics are covered in Chapter 27.

35.9 Sensor materials

A sensor material converts a physical or chemical quantity into an electrical one. Its quality is described by a *transduction coefficient* (the fractional change of a property per unit of the quantity), by selectivity to the wanted quantity over others, by stability, and by noise. Most mechanisms in this book have a sensing counterpart. Table 35.12 collects them, and the most important are developed below.

Table 35.12 — Sensor materials and their transduction coefficients (typical)

Quantity	Mechanism	Material	Figure of merit
Temperature	Metal resistivity (Chapter 25)	Pt (RTD)	$\alpha = 0.00385\ \mathrm{K^{-1}}$ (IEC 60751)
Temperature	Semiconducting oxide, hopping conduction	Mn–Ni–Co–O spinel (NTC thermistor)	$B = 3000$–4500 K; ≈ −4.4 %/K at 25 °C ($B = 3950$ K)
Temperature	Ferroelectric grain-boundary barriers (Chapter 27)	Doped $BaTiO_3$ (PTC thermistor)	Resistance rise of 10^3–10^5 above T_C
IR radiation (thermal)	Temperature coefficient of resistance	VO_x, a-Si:H microbolometers	TCR ≈ −2 to −3 %/K
IR radiation (thermal)	Pyroelectric effect (Chapter 27)	$LiTaO_3$, PZT	$p \approx 2 \times 10^{-4}\ \mathrm{C\,m^{-2}\,K^{-1}}$
Strain, pressure	Metal geometry change	Constantan, Karma foils	Gauge factor ≈ 2
Strain, pressure	Piezoresistance (band-structure change)	p-type Si ⟨110⟩, n-type Si ⟨100⟩	Gauge factor ≈ +100 to +175 (p), −100 to −130 (n)
Force, acceleration	Piezoelectric (Chapter 27)	AlN, PZT, quartz	$d_{33} \approx 5$ (AlN), 300–600 (PZT); $d_{11} \approx 2.3$ (quartz) pC/N
Magnetic field	Hall effect	InSb, InAs, GaAs, Si	μ_n = 78 000, 33 000, 8500, 1400 cm²/V s
Magnetic field	Magnetoresistance (Chapters 28–29)	NiFe (AMR), multilayers (GMR), MgO junctions (TMR)	$\Delta R/R$ = 2–3%, 10–20%, 100–600%
Gas	Surface depletion of grains	SnO_2, WO_3, ZnO (200–400 °C)	Resistance ratio 10–1000 for ppm-level gases
Humidity	Water uptake changes permittivity	Polyimide, porous Al_2O_3	≈ 0.2–0.5 pF per % RH (typical element)
Light	Photoconduction, photodiodes	Sections 35.2, 35.5	Responsivity, gain

Temperature sensing materials

Platinum is used for resistance thermometers because its resistivity is reproducible, its temperature coefficient is stable, and it resists oxidation up to about 850 °C. The standard Pt100 curve, $R(T) = R_0(1 + AT + BT^2)$ above 0 °C (Callendar–Van Dusen), gives $\alpha = 0.00385\ \mathrm{K^{-1}}$ between 0 and 100 °C. The coefficient depends on purity and strain, through Matthiessen's rule (Chapter 25). Thin-film Pt elements on alumina must be annealed to reach the standard value, since grain-boundary and defect scattering lower it.

NTC thermistors are sintered ceramics of mixed transition-metal oxides with the spinel structure. Conduction is by electrons hopping between Mn^{3+} and Mn^{4+} ions on octahedral sites, a thermally activated process. Their resistance follows

$$R(T) = R_{25} \exp\left[B\left(\frac{1}{T} - \frac{1}{298.15\ \mathrm{K}}\right)\right], \qquad \alpha = \frac{1}{R}\frac{dR}{dT} = -\frac{B}{T^2}$$

For $B = 3950$ K at 25 °C, $\alpha = -3950/298.15^2 = -4.4\%/\mathrm{K}$, about ten times the sensitivity of platinum. The cost is non-linearity and lower stability, from cation redistribution between lattice sites during ageing.

> **Status of the principle: B-parameter thermistor equation**
> *Empirical.* It is an Arrhenius-type fit valid over about 50 K around the reference temperature to within a fraction of a kelvin. The three-term Steinhart–Hart equation extends the range to better than 0.01 K over wider spans.

Piezoresistive silicon

When silicon is strained, its conduction-band valleys shift relative to each other and its valence bands split and warp (Chapter 31). Carriers redistribute among valleys with different effective masses, so the conductivity changes by much more than the geometry alone would produce. The *piezoresistive coefficients* express this: for p-type silicon, $\pi_{44} \approx 138 \times 10^{-11}\ \mathrm{Pa^{-1}}$, giving

gauge factors of 100–175 along ⟨110⟩. For a metal foil the gauge factor is about 2. Silicon pressure sensors, accelerometers and the strain engineering of transistor channels in Chapter 34 all use the same physics. The coefficients fall with doping and temperature, so sensor diffusions are designed at doping levels that balance sensitivity against temperature drift.

Hall-effect materials

A current I through a slab of thickness t in a perpendicular field B produces a transverse Hall voltage $V_H = IB/(qnt)$ (Chapter 52 uses this for characterization). The *current sensitivity* $S_I = V_H/(IB) = 1/(qnt)$ favours a low carrier density in a thin layer. For a given power dissipation, the output favours high mobility. InSb ($\mu_n \approx 78\,000$ cm²/V s), InAs and GaAs are therefore used for discrete Hall elements. An InSb film 1 µm thick with $n = 2 \times 10^{16}$ cm^{-3} gives $S_I = 1/(1.602 \times 10^{-19} \times 2 \times 10^{22} \times 10^{-6}) \approx 310$ V/(A T). InSb's small gap makes its carrier density, and therefore its sensitivity, strongly temperature-dependent. GaAs is chosen when stability matters, and silicon when the Hall plate must be integrated with CMOS electronics, which is the case in most current sensors and brushless-motor encoders.

Metal-oxide gas sensors

Tin dioxide ($E_g = 3.6$ eV, n-type from oxygen vacancies) is the archetypal chemoresistive material. In air at 200–400 °C, oxygen adsorbs on the grain surfaces and captures electrons, forming O_2^-, O^- and O^{2-} species. This creates a depletion layer at every grain surface and a potential barrier at every grain boundary. It is the grain-boundary barrier of Chapter 34 (Section 34.8), now set by surface chemistry rather than by bulk traps. A reducing gas (CO, H_2, ethanol) reacts with the adsorbed oxygen, returns electrons to the grains, lowers the barriers, and raises the conductance by orders of magnitude.

The microstructure is the design variable. When the grain diameter D is less than about twice the Debye length L_D (about 3 nm in SnO_2 at operating temperature), each grain is fully depleted and the sensitivity rises sharply. Nanocrystalline films with 5–10 nm grains and noble-metal catalysts (Pt, Pd) on the surface, which spill reactive species onto the oxide, are the standard recipe. The chemistry of adsorption and catalysis belongs to the Advanced Chemistry core. The material lessons are that grain size sets sensitivity, that grain growth at operating temperature sets drift, and that selectivity remains the weak point, because any reducing gas lowers the barrier. Arrays of different oxides with pattern recognition are the usual workaround.

What remains open

Open question	Where it is answered
How do magnetic materials provide the magnetoresistive and Hall-complementary sensors and the inductors of RF circuits?	Chapters 28 and 29
How does quantum confinement tune emission and absorption in quantum dots, nanowires and 2D materials?	Chapter 38
How are photovoltaic absorbers manufactured, stabilized and scaled, and how do they compare with other energy-conversion materials?	Chapter 40
How are epitaxial optoelectronic layers, oxide films and dielectrics grown, and how does processing set their defect densities?	Chapters 49 and 50
How are RF laminates, LTCC and packages built, and how do their CTE and moisture behaviour affect high-frequency performance?	Chapter 51
How are absorption spectra, photoluminescence, Hall mobility and dielectric loss measured?	Chapter 52
How do LEDs, lasers, detectors and RF transistors degrade (dark-line defects, phosphor quenching, current collapse)?	Chapter 44
How are optoelectronic and RF materials selected against cost, reliability and environmental constraints (Cd, Pb, Hg, In)?	Chapter 58
How do LEDs, lasers, photodiodes, modulators and RF amplifiers operate as devices and in circuits?	Electronics & Computing

Next: Chapter 36 (Optical and Photonic Materials) widens the view from semiconductors to all optical materials — how structure sets refractive index, absorption and colour, and how fibres, nonlinear crystals and photonic structures guide and control light.

Chapter 36 — Optical and Photonic Materials

Part VII · Semiconductor and Photonic Materials

A defence contractor needs a window for the nose of a missile that must see in the visible and mid-infrared, survive rain erosion at Mach 3, and withstand a thermal shock of several hundred kelvin in a second. Glass is too weak and absorbs beyond 3 μm. Germanium transmits the infrared but is opaque to visible light and loses transparency when hot. Single-crystal sapphire is strong and transparent from the ultraviolet to 5 μm, but expensive to grow and machine into domes. The answer that eventually flew was a polycrystalline ceramic — spinel or aluminium oxynitride — sintered to full density so that not one pore in a million remained, with grains of the right size and a cubic crystal structure so that the grain boundaries would not scatter light. The same materials problem, without the missile, arises in a watch crystal, a smartphone camera cover, a high-pressure sodium lamp, a laser gain medium and a transparent armour window: what makes a solid transparent, and what makes most of them not?

This chapter explains how light interacts with matter and how structure sets the refractive index, dispersion, absorption, colour and scattering of materials; then turns to the materials that make light (phosphors, LEDs and lasers), guide it (optical fibres and waveguides), and control it (nonlinear, electro-optic, photonic-crystal and metamaterial structures). Chapter 35 treats optoelectronic semiconductor materials in more depth.

> **Why is window glass transparent while the sand it is made from is white?**
> **Why are gold and copper coloured, but silver and aluminium not? Why does a polycrystalline alumina tube transmit light diffusely, but a polycrystalline spinel window can be read through? Why do optical fibres carry signals for 100 km at 1.55 μm but only a few kilometres at 0.85 μm?**

Concept	Engineering destination
Reflection, refraction, absorption, scattering	Windows, lenses, coatings, solar glazing, sensor covers
Refractive index, Sellmeier dispersion, birefringence	Lens design, achromats, polarizers, stress analysis by photoelasticity
Colour in metals, insulators, semiconductors, pigments	Paints, pigments, glass colourants, gemstones, architectural coatings
Transparency in ceramics and polymers	Transparent armour, IR domes, lamp envelopes, clear packaging, medical devices
Luminescence, phosphors, LEDs, lasers	Lighting, displays, medical scintillators, laser cutting and welding
Optical fibres and waveguides	Telecommunications, fibre-optic strain and temperature sensing, endoscopy
Nonlinear and electro-optic materials	Frequency conversion of lasers, optical modulators for data links
Photonic crystals, metamaterials, metasurfaces	Dielectric mirrors, VCSELs, structural colour, flat lenses, radiative cooling

Roadmap. Light–matter interaction (36.1) → refractive index, its origin and Sellmeier dispersion (36.2) → colour (36.3) → transparency and opacity in ceramics and polymers (36.4) → luminescence, LEDs and lasers (36.5) → fibres and waveguides (36.6) → nonlinear and electro-optic materials (36.7) → photonic crystals and metamaterials (36.8).

36.1 Interaction of light with matter: refraction, dispersion, absorption, scattering

Light as a field acting on bound and free charges

Light is an electromagnetic wave with an electric field oscillating at 10^{14}–10^{15} Hz in the visible (wavelength 400–700 nm, photon energy 1.8–3.1 eV). In a material, that field drives the same charges whose polarization was discussed in Chapter 26: bound electrons, ions and, in metals and doped semiconductors, free carriers. At optical frequencies only the electrons can follow (ions are too heavy above about 10^{13} Hz), so the optical properties are dominated by **electronic polarization** in insulators and **free-electron response** in metals.

When light meets a material, its energy is partitioned among reflection R, absorption A and transmission T, with

$$R + A + T = 1$$

for a slab (scattering counts as a loss from the transmitted beam). The two numbers that describe a homogeneous material are combined in the **complex refractive index**

$$\tilde{n} = n + i\kappa, \qquad \tilde{n}^2 = \varepsilon_r(\omega)$$

where n sets the phase velocity c/n and κ, the **extinction coefficient**, sets the attenuation. The intensity decays as $I = I_0 \exp(-\alpha x)$ (**Beer–Lambert law**) with absorption coefficient $\alpha = 4\pi\kappa/\lambda$.

> **Status of the principle: Beer–Lambert law**
> *Approximate.* It assumes linear absorption by a homogeneous medium, with no scattering and intensity low enough that absorbers do not saturate. It fails at laser intensities (two-photon absorption, saturable absorption) and in turbid or scattering media.

Refraction and reflection

At an interface between media of index n_1 and n_2, light bends according to **Snell's law**, $n_1 \sin\theta_1 = n_2 \sin\theta_2$. Going from high to low index, beyond the critical angle $\theta_c = \arcsin(n_2/n_1)$ the light is totally internally reflected — the principle of the optical fibre (Section 36.6). Part of the light is always reflected. At normal incidence for a non-absorbing material the **Fresnel reflectance** is

$$R = \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2$$

so glass ($n = 1.5$) reflects 4 % per surface, sapphire ($n = 1.76$) 7.6 %, diamond ($n = 2.42$) 17 %, and silicon ($n \approx 3.5$ in the near infrared) 31 %. High reflection is part of diamond's brilliance and a major loss for solar cells and LEDs.

> **Status of the principle: Fresnel equations**
> *Fundamental.* They follow from Maxwell's equations with the boundary conditions on tangential fields. They assume flat, smooth interfaces; rough surfaces add diffuse scattering.

Worked example — antireflection coatings. A thin film of index n_f and thickness $\lambda/(4n_f)$ makes the reflections from its top and bottom surfaces cancel by destructive interference. On glass ($n_s = 1.52$) in air, the reflectance at the design wavelength becomes

$$R = \left(\frac{n_s - n_f^2}{n_s + n_f^2}\right)^2$$

which is zero for $n_f = \sqrt{n_s} = 1.23$. No durable solid has an index so low, so the standard choice is MgF_2 ($n_f = 1.38$): at 550 nm the film is $550/(4 \times 1.38) = 99.6$ nm thick, and $R = [(1.52 - 1.904)/(1.52 + 1.904)]^2 = 1.26$ %, a third of the uncoated 4.3 %. For a silicon solar cell ($n_s \approx 3.9$ at 600 nm), the ideal index is about 2.0 — exactly that of PECVD silicon nitride, which at 75 nm thickness reduces reflection at 600 nm to nearly zero and also passivates the surface (Chapter 32). This is why solar cells are dark blue: the coating is optimized near the peak of the solar spectrum and reflects some violet. Multilayer stacks broaden the low-reflection band for camera lenses and spectacles.

Dispersion

The refractive index depends on wavelength — **dispersion** — because the material's response depends on how close the light frequency is to its electronic and ionic resonances (Section 36.2). In the transparent range of glasses, n falls slowly with increasing wavelength (normal dispersion), so blue light is refracted more than red: the origin of prismatic colour, of chromatic aberration in lenses, and of the spreading of short pulses in fibres.

Absorption

Absorption occurs wherever photons can be taken up by an excitation of the material:

- **Electronic transitions across a band gap** in semiconductors and insulators — strong absorption ($\alpha \sim 10^4$–10^6 cm^{-1}) for photon energies above E_g. This sets the ultraviolet edge of glasses and ceramics and the colour of semiconductors.
- **Transitions within partly filled d or f shells** of transition-metal and rare-earth ions — weak, narrow or broad bands that give colour to gemstones and glasses (Section 36.3).
- **Free-carrier absorption** in metals and doped semiconductors.
- **Lattice vibrations (phonons)** — multi-phonon absorption sets the infrared edge of transparency. Light atoms with strong bonds (Si–O, Al–O) vibrate at high frequency, so oxides become opaque at 3–6 μm; heavy atoms with weak bonds (Zn–Se, Ge, chalcogenides, alkali halides) vibrate slowly, so they transmit to 10–20 μm. This is why thermal-imaging lenses are made of germanium, ZnSe or chalcogenide glass, not silica.

Each material is therefore transparent only within a **window** between its electronic absorption edge in the ultraviolet and its phonon edge in the infrared (Figure 36.1).

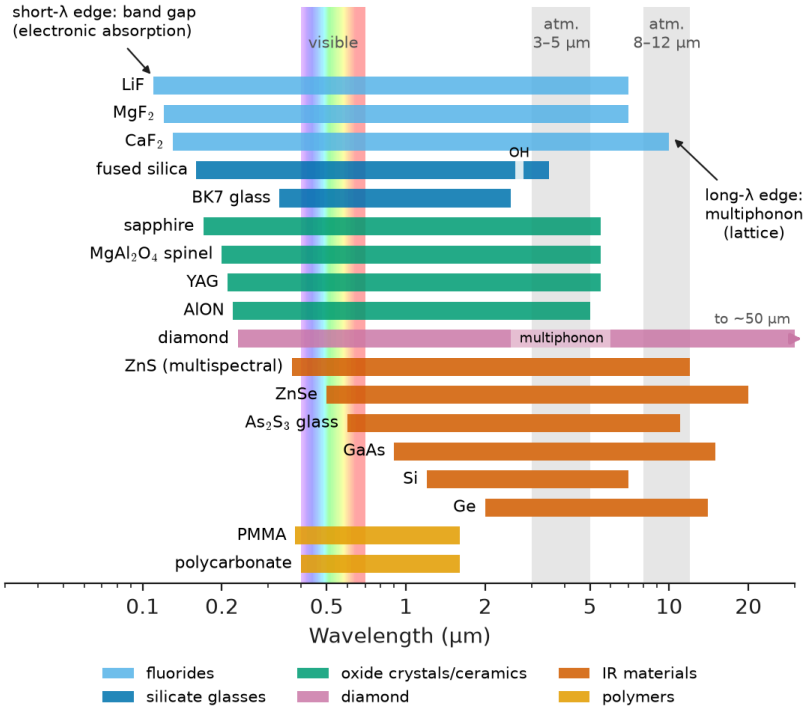


Figure 36.1 — Transparency windows of optical materials: each is bounded by electronic absorption at short wavelengths and lattice absorption at long wavelengths.

Scattering

Light is scattered wherever the refractive index changes over a distance comparable to or smaller than the wavelength: pores, inclusions, second phases, grain boundaries in birefringent crystals, density and composition fluctuations in glass, and crystalline regions in polymers. For particles much smaller than the wavelength, **Rayleigh scattering** scales as λ^{-4} and with the square of the particle volume and of the index contrast — why the sky is blue, why fine smoke looks blue against a dark background, and why the attenuation of optical fibres falls steeply with wavelength. For particles comparable to the wavelength, **Mie scattering** is strong and almost wavelength-independent, so clouds, milk, paper, snow and white paint look white. Scattering, not absorption, makes most solids opaque: crushed glass and sand, white marble, porous alumina and polyethylene are made of transparent constituents and are opaque only because they contain many interfaces.

36.2 Refractive index and its origin; Sellmeier

The Lorentz oscillator

Model each bound electron as a charge on a spring with natural frequency ω_0 and weak damping, driven by the optical field. Below resonance it follows the field in phase and adds to the polarization, so $n > 1$; approaching resonance the response grows;

above resonance it lags and reduces the polarization. Summing over oscillators with strengths f_j gives

$$n^2(\omega) - 1 = \frac{Ne^2}{\varepsilon_0 m} \sum_j \frac{f_j}{\omega_j^2 - \omega^2}$$

away from the resonances. Two structural conclusions follow. First, the index grows with the **density of polarizable electrons** N and with their polarizability; for a dense medium, the local-field correction gives the **Lorentz–Lorenz relation**, the optical form of the Clausius–Mossotti equation of Chapter 26:

$$\frac{n^2 - 1}{n^2 + 2} = \frac{N\alpha_e}{3\varepsilon_0}$$

Second, the index is higher in materials whose resonances lie at lower frequency, that is, with smaller band gaps. Empirically n rises as E_g falls: fluorides ($E_g > 10$ eV, $n \approx 1.4$), silica (9 eV, 1.46), alumina (8.8 eV, 1.76), ZnS (3.6 eV, 2.37), TiO_2 (3.0 eV, 2.6–2.9), silicon (1.1 eV, 3.5), germanium (0.66 eV, 4.0).

> **Status of the principle: Lorentz–Lorenz relation**
> *Approximate.* It assumes isotropic, non-interacting polarizable units in a Lorentz local field. It predicts well how the index of a glass or polymer changes with density (useful for estimating porosity and stress-induced index changes), but not absolute values in strongly covalent or anisotropic solids.

Material	n_d (589 nm)	Abbe number V_d	Notes
Aerogel (silica)	1.01–1.05	—	Mostly air; Cherenkov detectors, insulation
MgF_2	1.38	106	Antireflection coatings, UV optics
CaF_2	1.434	95	Low-dispersion lens elements, UV lithography
Fused silica	1.458	68	Fibres, UV optics, lithography lenses
PMMA	1.49	57	Lenses, light guides, POF
N-BK7 crown glass	1.517	64	General-purpose optical glass
Soda-lime window glass	1.52	59	Windows, containers
Polycarbonate	1.586	30	Spectacle and safety lenses
SF11 dense flint glass	1.785	26	High-index, high-dispersion lens elements
Sapphire (Al_2O_3, ordinary ray)	1.768	72	Windows, watch crystals, LED substrates
$MgAl_2O_4$ spinel	1.72	60	Transparent armour, IR domes
$Y_3Al_5O_{12}$ (YAG)	1.83	53	Laser hosts, phosphors
Cubic zirconia (Y-stabilized)	2.15–2.18	33	Gem simulant, high-index optics
Diamond	2.417	55	Windows, heat spreaders, gems
TiO_2 rutile (ordinary / extraordinary)	2.61 / 2.90	≈10	White pigment, high-index coatings
Si (at 1.55 μm)	3.48	—	Silicon photonics, IR optics
Ge (at 10 μm)	4.00	—	Thermal-imaging lenses

Table 36.1 — Refractive index and dispersion of optical materials at room temperature (typical).

Sellmeier dispersion

For practical design the Lorentz form is written with resonances at wavelengths λ_j and fitted strengths B_j, the **Sellmeier equation**:

$$n^2(\lambda) = 1 + \sum_{j=1}^{3} \frac{B_j \lambda^2}{\lambda^2 - C_j}, \qquad C_j = \lambda_j^2$$

Two terms usually describe ultraviolet electronic resonances and one an infrared lattice resonance; between them, the index falls smoothly with wavelength (Figure 36.2). Glass catalogues list B_j and C_j for every glass, and lens-design software uses nothing else. A compact measure of dispersion in the visible is the **Abbe number**

$$V_d = \frac{n_d - 1}{n_F - n_C}$$

using the helium d (587.6 nm), hydrogen F (486.1 nm) and C (656.3 nm) lines. Crown glasses ($V_d > 50$) disperse little; flint glasses, whose lead or titanium oxides add low-lying resonances, have high index and high dispersion ($V_d < 50$). An achromatic doublet pairs a crown and a flint so that their dispersions cancel.

> **Status of the principle: Sellmeier equation**
> *Empirical.* It is the Lorentz oscillator form with fitted resonance positions and strengths. Within a material's transparency window it reproduces the index to 10^{-5} or better; it fails near absorption bands, where damping must be included.

Worked example — dispersion of fused silica. Malitson's Sellmeier coefficients for fused silica are $B = (0.6961663,\ 0.4079426,\ 0.8974794)$ and resonance wavelengths $\lambda_j = (0.0684043,\ 0.1162414,\ 9.896161)$ μm. At $\lambda = 0.5893$ μm,

$$n^2 = 1 + \frac{0.69617(0.34728)}{0.34728 - 0.00468} + \frac{0.40794(0.34728)}{0.34728 - 0.01351} + \frac{0.89748(0.34728)}{0.34728 - 97.934} = 2.12694$$

so $n = 1.4584$. The same equation gives $n_F = 1.4631$ and $n_C = 1.4564$, hence $V_d = (1.4584 - 1)/(1.4631 - 1.4564) = 67.8$, a low-dispersion glass. In the infrared, $n = 1.4468$ at 1.31 μm and 1.4440 at 1.55 μm. The second derivative of $n(\lambda)$ sets the **material dispersion** that spreads short pulses in a fibre: it passes through zero near 1.27 μm, where the ultraviolet and infrared resonances balance, and at 1.55 μm amounts to about 22 ps per nanometre of source linewidth per kilometre. Silica's two resonances, one electronic and one Si–O vibrational, thus set the two telecom windows of Section 36.6.

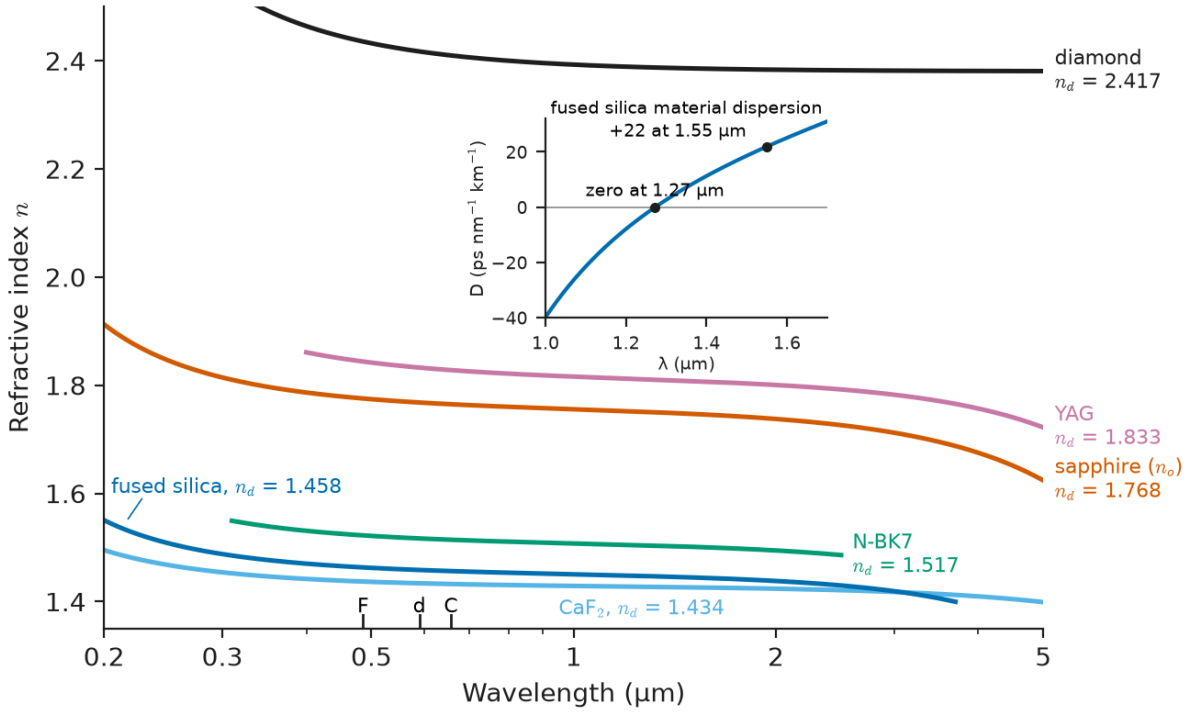


Figure 36.2 — Refractive index against wavelength for optical materials: normal dispersion between ultraviolet and infrared resonances.

Birefringence

In non-cubic crystals the index depends on the polarization and direction of light, the optical counterpart of the tensor properties of Chapter 4. Uniaxial crystals have an ordinary index n_o and an extraordinary index n_e; their difference, the **birefringence**, ranges from -0.008 in sapphire and $+0.009$ in quartz to -0.17 in calcite and $+0.29$ in rutile. Birefringent crystals make wave plates and polarizers, and birefringence enables phase matching in nonlinear optics (Section 36.7). Isotropic materials become birefringent under stress or orientation: **photoelasticity**, the stress-induced birefringence of glass and

polymers, reveals residual stress in tempered glass, frozen-in moulding stresses in polycarbonate lenses and polymer parts, and stress fields in transparent models of engineering components. Oriented polymer films (stretched PET, PVA polarizer films) are birefringent by chain alignment.

36.3 Colour in metals, insulators, semiconductors and pigments

Colour is what remains of white light after the material has absorbed, reflected or scattered parts of the spectrum. Its origin differs in each class of material, and in each case it can be traced to an electronic structure.

Metals

Free electrons in a metal respond to fields below the **plasma frequency**

$$\omega_p = \sqrt{\frac{ne^2}{\varepsilon_0 m^*}}$$

by screening them: below ω_p the permittivity is negative, light cannot propagate, and the metal reflects almost everything. For aluminium ($n = 1.8 \times 10^{29}$ m^{-3}) the free-electron plasma wavelength is about 80 nm, deep in the ultraviolet, so aluminium reflects about 90 % across the whole visible and is the standard mirror coating for telescopes and the reflective layer of food packaging. Silver is the best visible reflector (95–99 %), though its d electrons pull its effective plasma edge to about 320 nm. In **copper and gold**, transitions from filled d bands to the Fermi level begin at about 2.1 eV (590 nm) and 2.4 eV (520 nm) respectively, so blue and green light are absorbed while red and yellow are reflected (Figure 36.3). Alloying shifts these edges: brass is yellower than copper, and gold alloys with silver and copper range from green-yellow to red. Titanium nitride, with a plasma edge in the visible, is gold-coloured and is used as a hard decorative coating on tools and fittings (Chapter 45).

Doped oxides with fewer free carriers have plasma edges in the infrared. Tin-doped indium oxide with 10^{21} carriers/cm³ and an effective mass of $0.35 m_0$ reflects beyond about 1.2 µm (once its background permittivity of about 4 is included; the bare free-electron formula would put the edge near 0.6 µm) but transmits visible light — the basis of transparent electrodes and of the low-emissivity coatings on energy-saving windows, which reflect thermal infrared back into a building while letting daylight through.

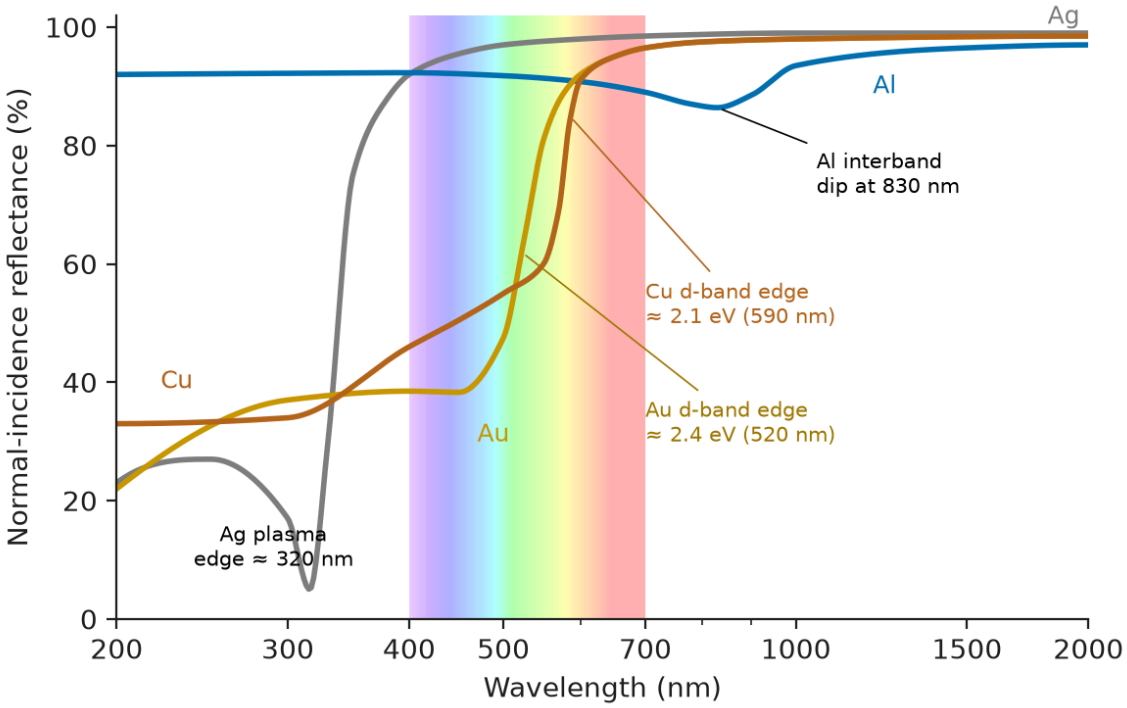


Figure 36.3 — Reflectance of metals: free electrons reflect below the plasma frequency, and interband absorption gives copper and gold their colour.

Insulators: ions, defects and charge transfer

Pure wide-gap insulators (silica, alumina, MgO, diamond) are colourless. Their colour comes from small concentrations of impurities and defects:

- **Transition-metal ions.** The partly filled d shell of an ion such as Cr^{3+} is split by the electric field of its neighbouring anions (the **crystal field**), giving absorption bands in the visible whose positions depend on the host. The same Cr^{3+} ion makes ruby red in Al_2O_3 (strong crystal field, absorbing green and violet) and emerald green in beryl (weaker field). Glass colourants work the same way: Co^{2+} gives cobalt blue, Fe^{2+} the green of ordinary bottle and window glass, Cr^{3+} green, Mn^{3+} purple. Removing iron to below 0.01 % produces the "low-iron" glass used for solar-panel covers and display glass.
- **Charge transfer.** An electron jumping between ions — Fe^{2+} to Ti^{4+} in blue sapphire, or from O^{2-} to Fe^{3+} in hematite (Fe_2O_3, the red of rust and red ochre) — produces strong, broad absorption.
- **Colour centres.** Point defects that trap electrons or holes, such as F-centres (an electron in an anion vacancy, Chapter 5), colour alkali halides and darken glasses exposed to radiation; deliberately created nitrogen–vacancy centres in diamond are now quantum sensors.

Semiconductors

A semiconductor absorbs all photons with energy above its band gap and transmits those below (Section 36.1), so its colour is set by E_g. Converting energy to wavelength with $\lambda(\text{nm}) = 1239.8/E(\text{eV})$:

Material	E_g (eV)	Absorption edge (nm)	Appearance
Diamond	5.47	227	Colourless (absorbs only UV)
GaN, ZnO	3.4	365	Colourless; transparent
TiO_2 (rutile)	3.0	413	White powder; very slightly yellow crystal
CdS	2.42	512	Yellow (cadmium yellow)
GaP	2.26	549	Orange
HgS (vermilion)	2.0	620	Red
CdSe	1.74	713	Black-red
Si	1.12	1107	Opaque, metallic grey (all visible absorbed; reflection gives lustre)

Table 36.2 — Band gap and colour of semiconductors (typical).

Materials whose edge lies in the visible absorb the blue end and look yellow, orange or red; those whose edge lies below 1.8 eV absorb everything and look black or metallic. Alloying tunes the edge continuously, as in the Cd(S,Se) pigments and the GaAsP and AlGaInP red and amber LEDs.

Pigments, dyes and scattering colour

A **pigment** is an insoluble powder dispersed in a binder; a **dye** is a dissolved molecule. Inorganic pigments use the mechanisms above — Fe_2O_3 red, Cr_2O_3 green, cobalt aluminate blue, CdS yellow (now restricted for toxicity), bismuth vanadate yellow. Organic dyes and pigments owe their colour to conjugated π-electron systems whose HOMO–LUMO gap lies in the visible; longer conjugation shifts absorption to longer wavelength. **White** pigments rely on scattering, not absorption: rutile TiO_2, with the highest index of any common white material (about 2.7 against about 1.5 for a polymer binder), scatters visible light most strongly when its particles are about 0.2–0.3 μm — half the wavelength of visible light. Some 7 million tonnes are made each year for paints, plastics, paper and sunscreens, and the pigment's band gap at 3.0 eV also makes it a UV absorber and, unless its surface is coated, a photocatalyst that degrades the surrounding binder (Chapter 44). Colour that arises from interference and diffraction in periodic structures — **structural colour** — is discussed in Section 36.8.

36.4 Transparency and opacity in ceramics and polymers

What scatters light in a polycrystal

A dense, single-phase solid whose constituents absorb nothing in the visible should be transparent. Most ceramics and many polymers are not, because they scatter. Three microstructural sources dominate:

1. **Porosity.** A pore in a ceramic is an index contrast of 0.5–1.5, the largest possible, and pores of 0.1–1 μm scatter strongly. Transparency requires porosity below about 0.01–0.05 %. Achieving it is a sintering problem (Chapter 46): fine, pure, agglomerate-free powders; sintering aids (MgO in alumina, LiF in spinel) that keep pores attached to grain boundaries so they can be eliminated; and **hot isostatic pressing** to close the last pores.
2. **Birefringent grains.** In a non-cubic ceramic such as alumina, each grain has a different orientation and therefore a slightly different index for a given ray; every grain boundary is a weak scatterer. Cubic ceramics — spinel ($MgAl_2O_4$), YAG, AlON, Y_2O_3, MgO — have no birefringence and can be made fully transparent at any grain size.
3. **Second phases and inclusions**, including impurity-rich grain-boundary films.

For birefringent ceramics the scattering depends on the ratio of grain size to wavelength. In the regime where grains are smaller than the wavelength, Apetz and Bruls showed that the **real in-line transmission** — the fraction of light passing straight through, which determines whether an image can be seen — is approximately

$$T_{\mathrm{RIT}} = (1 - R_s)\exp\!\left(-\frac{3\pi^2 \Delta n^2\, r\, d}{\lambda_0^2}\right)$$

where Δn is the effective birefringence, r the grain radius, d the thickness and R_s the reflection loss from both surfaces.

> **Status of the principle: Apetz–Bruls in-line transmission**
> *Approximate.* It treats grains as independent Rayleigh–Gans scatterers smaller than the wavelength, with random orientation and no porosity. It captures the strong dependence on grain size; residual porosity of even 0.01 % adds scattering that it omits.

Worked example — fine-grained transparent alumina. Take a 0.8 mm alumina disc ($n = 1.76$, $\Delta n = 0.008$) at $\lambda_0 = 640$ nm. Each surface reflects $[(1.76-1)/(1.76+1)]^2 = 7.6$ %, and with multiple reflections the two surfaces lose $R_s = 2R/(1+R) = 14$ %. For grains 0.3 μm in diameter ($r = 0.15$ μm), the exponent is

$$\frac{3\pi^2 (0.008)^2 (0.15\times10^{-6})(0.8\times10^{-3})}{(640\times10^{-9})^2} = 0.555$$

so $T_{\mathrm{RIT}} = 0.86\, e^{-0.555} = 0.49$ — half the light passes straight through, close to measured values of 50–60 %. With 1 μm grains the exponent becomes 1.85 and $T_{\mathrm{RIT}} = 13.5$ %; with 2 μm grains, 2 %; with the 20 μm grains of conventional alumina, essentially zero. Conventional alumina can still have a high *total* transmission (over 90 % in high-pressure sodium lamp tubes, where only diffuse light is needed) but cannot be seen through. The design rules are therefore: choose a cubic ceramic (spinel, AlON, YAG) when image quality matters at any thickness, or make alumina's grains far smaller than the wavelength, which requires sintering at low temperature with pressure to stop grain growth (Figure 36.4).

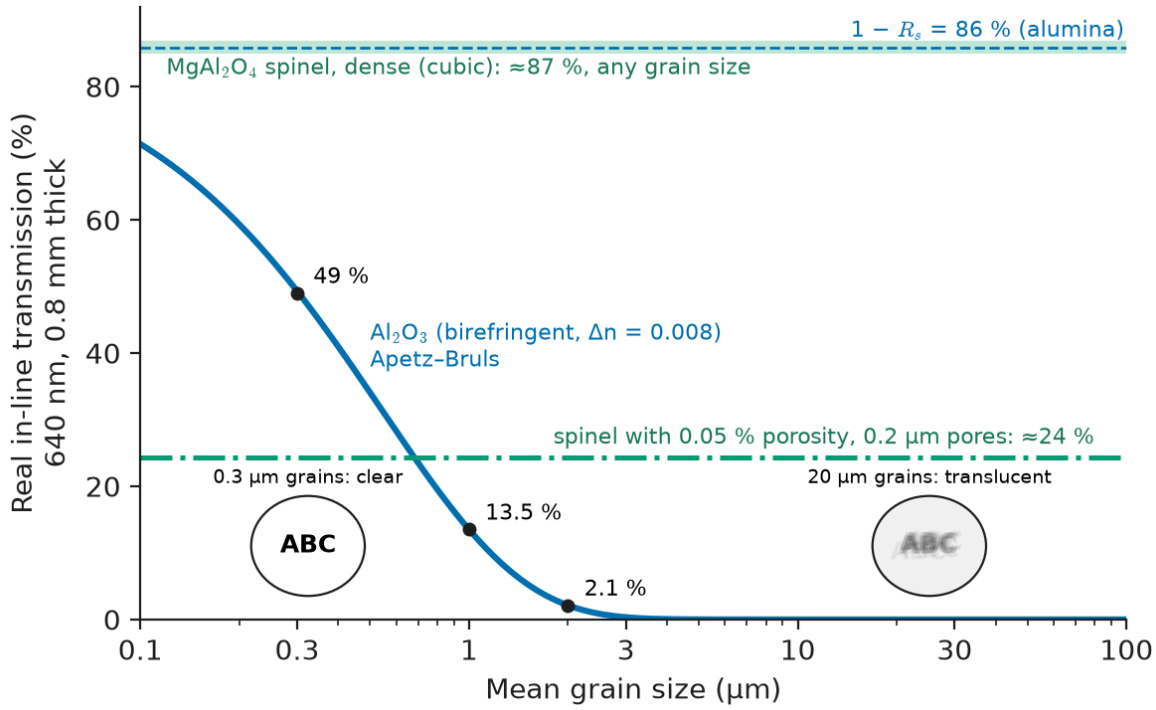


Figure 36.4 — Real in-line transmission of polycrystalline ceramics: birefringent alumina becomes transparent only when grains are far smaller than the wavelength, while cubic spinel is limited only by reflection and porosity.

Transparent polycrystalline ceramics serve as armour windows for vehicles and aircraft (spinel and AlON, three to four times as weight-efficient as laminated glass), infrared domes and windows (spinel, ZnS, sapphire), laser gain media (Nd:YAG and Yb:YAG ceramics, which can be made larger and more heavily doped than single crystals), scintillators for medical CT, and envelopes for metal-halide lamps.

Glass-ceramics exploit the same physics from the other side. Controlled nucleation (Chapter 9) crystallizes a glass into a dense array of crystals tens of nanometres across, too small to scatter. If the crystal is a near-zero-expansion phase such as β-quartz solid solution, the result — the material of induction-hob cooktops and of telescope mirror blanks such as Zerodur — is transparent, strong and immune to thermal shock.

Polymers

Amorphous polymers (PMMA, polycarbonate, polystyrene, cyclic olefin copolymers) are intrinsically transparent, with in-line transmission of 89–92 %, limited by surface reflection. They replace glass in vehicle lamps, glazing, lenses, medical devices and optical discs. **Semicrystalline** polymers (polyethylene, polypropylene, nylon, PTFE) are usually translucent or opaque, because crystalline lamellae and spherulites (Chapter 20), with an index higher than the amorphous regions by 0.02–0.05, form scattering structures several micrometres across. Three routes restore clarity: suppress crystallinity (quenching thin PET film, or copolymerization); make the crystals small, by adding **nucleating clarifiers** to polypropylene so that thousands of times more spherulites form, each too small to scatter (clarified PP food containers and syringes); or orient the crystals by stretching, as in biaxially oriented PET bottles and BOPP film. In fibre-reinforced polymers, a resin whose index matches that of the glass fibres can produce translucent panels. **Haze** — the fraction of transmitted light scattered by more than 2.5° — is the standard quality measure for films and sheets, along with yellowing from UV degradation, which absorbs the blue end of the spectrum (Chapter 44).

36.5 Luminescence and phosphors; LEDs and lasers as material systems

Luminescence

Luminescence is the emission of light by a material after it has been excited by something other than heat: photons (photoluminescence), electrons (cathodoluminescence), an electric current (electroluminescence), X-rays or gamma rays

(scintillation), or a chemical reaction. Because some energy is lost to lattice vibrations before emission, the emitted light has a longer wavelength than the exciting light — the **Stokes shift**. Emission that stops within microseconds of excitation is fluorescence; long-lived emission (milliseconds to hours) is phosphorescence.

Most practical luminescent materials are **phosphors**: a host crystal (oxide, nitride, sulfide, fluoride) doped with an activator ion whose energy levels produce the emission. Rare-earth ions dominate. **Eu^{3+}** and **Tb^{3+}** emit narrow lines (red at 611 nm and green at 545 nm) from shielded 4f levels, almost independent of the host. **Ce^{3+}** and **Eu^{2+}** emit broad bands from 5d→4f transitions whose energy depends strongly on the host's crystal field and covalency: the same Ce^{3+} emits in the ultraviolet in a fluoride but yellow in $Y_3Al_5O_{12}$ (YAG:Ce), and Eu^{2+} emits red in nitride hosts such as $CaAlSiN_3$. Phosphor design is crystal chemistry: choosing the site symmetry, bond covalency and stiffness of the host to place emission at the wanted wavelength and to suppress thermal quenching. Persistent phosphors such as $SrAl_2O_4$:Eu,Dy store energy in traps and glow for hours in emergency signage. **Scintillators** convert ionizing radiation to visible light for medical imaging and security scanning: lutetium–yttrium oxyorthosilicate (LYSO:Ce, about 30 000 photons per MeV with a 40 ns decay) in PET scanners, CsI:Tl in X-ray detectors, and transparent ceramic garnets and oxysulfides in CT.

LEDs

The semiconductor-materials side of LEDs, lasers and photodetectors is treated in Chapter 35; here they are viewed as material systems. A **light-emitting diode** is a p–n junction (Chapter 32) in a direct-gap semiconductor. Forward bias injects electrons and holes into a thin active region, where they recombine radiatively and emit photons of energy close to the band gap. Each element of the device is a materials choice:

- **The active layer** sets the colour. InGaN quantum wells in GaN emit violet to green (405–530 nm) as indium content rises; AlGaInP lattice-matched to GaAs emits red to amber (590–650 nm); AlGaN reaches the ultraviolet (to 210 nm) for disinfection. Between them lies the "green gap", where neither system is efficient: high-indium InGaN suffers strain, phase separation and strong polarization fields, and AlGaInP becomes indirect.
- **Heterostructure barriers** confine carriers in the wells (Chapter 34), raising the radiative rate.
- **Defects** compete through non-radiative SRH recombination. That GaN LEDs work at dislocation densities of 10^8 cm^{-2} is due to carrier localization in indium-rich regions.
- **Light extraction** is limited by total internal reflection. In a semiconductor with $n = 2.5$, the critical angle is $\theta_c = \arcsin(1/2.5) = 23.6°$, and only the fraction $(1 - \cos\theta_c)/2 = 4.2$ % of isotropically emitted light can escape through the top surface. Roughened or patterned surfaces, patterned sapphire substrates, and encapsulation in high-index silicones raise extraction to 80 % or more.

White LEDs combine a blue InGaN chip (about 450 nm) with a YAG:Ce phosphor that absorbs part of the blue and emits broadband yellow; adding a red nitride phosphor raises colour rendering for warm-white lighting (Figure 36.5). Today's best white LEDs exceed 200 lumens per watt, more than ten times an incandescent lamp, and solid-state lighting has cut global lighting electricity substantially. The phosphor's thermal quenching, the degradation of silicone encapsulants and the efficiency **droop** at high current (partly Auger recombination) are the materials limits on lifetime and brightness.

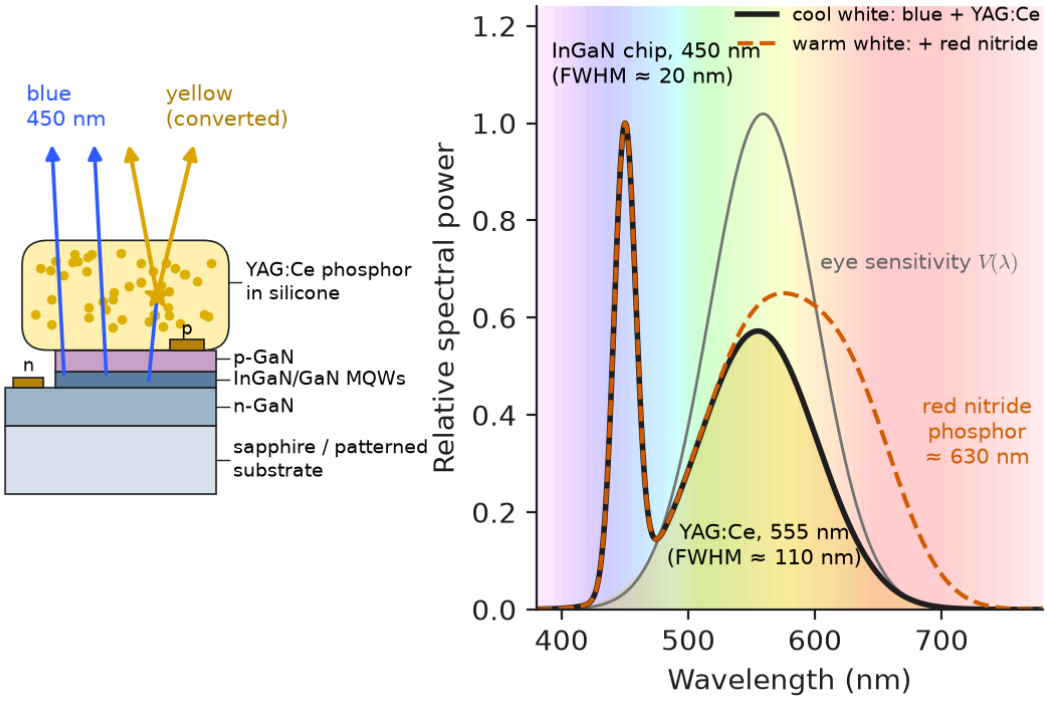


Figure 36.5 — A phosphor-converted white LED: a blue InGaN chip plus yellow (and optionally red) phosphor emission.

Lasers

A laser amplifies light by **stimulated emission** in a medium with a **population inversion** — more atoms in the upper level of a transition than in the lower — inside an optical resonator. The gain medium is a materials system in which the host and the dopant are chosen together:

Laser	Gain material	Wavelength	Uses
Nd:YAG	Nd^{3+} in $Y_3Al_5O_{12}$ crystal or ceramic	1064 nm	Marking, drilling, medical, range-finding
Yb fibre	Yb^{3+} in silica fibre core	1030–1080 nm	Industrial cutting and welding (to tens of kW), additive manufacturing
Er fibre amplifier	Er^{3+} in silica fibre	1530–1565 nm	Telecommunications amplification
Ti:sapphire	Ti^{3+} in Al_2O_3	650–1100 nm (tunable)	Ultrafast pulses, research
CO_2	Gas molecule	10.6 µm	Cutting non-metals, older metal cutting
Semiconductor diode	InGaAs, AlGaAs, InGaN quantum wells	375 nm–2 µm	Pumps for solid-state lasers, data links, Blu-ray, lidar

Table 36.3 — Representative laser materials.

In solid-state lasers the host must be transparent, optically homogeneous, thermally conductive (to remove waste heat without thermal lensing or fracture) and able to accept the dopant ion on a suitable site; YAG and sapphire excel because they are hard, conductive and isotropic or well-characterized. In fibre lasers, the long thin geometry dissipates heat so well that a single fibre delivers kilowatts, which is why Yb fibre lasers now dominate laser welding, cutting and powder-bed fusion (Chapters 47–48). Diode lasers add a resonator to the LED structure: cleaved crystal facets serve as mirrors, and a double heterostructure confines both carriers (lower gap) and light (higher index) in the active layer. At high power, facets fail by catastrophic optical damage, a materials-degradation problem solved by facet passivation and non-absorbing mirror regions.

36.6 Optical fibres and waveguides

Guiding by total internal reflection

Silica-fibre attenuation is also examined, from the semiconductor-materials side, in Chapter 35. An optical fibre has a **core** of slightly higher index than its **cladding**; light launched within the acceptance cone is totally reflected at the boundary and guided. The **numerical aperture**

$$\mathrm{NA} = \sqrt{n_1^2 - n_2^2} = \sin\theta_{\max}$$

measures the acceptance cone. When the core is small enough, only one guided mode propagates; the condition is that the normalized frequency

$$V = \frac{2\pi a}{\lambda}\,\mathrm{NA} < 2.405$$

where a is the core radius. Single-mode fibres avoid the spreading of pulses between modes and carry the world's long-distance and data-centre traffic; multimode fibres (core 50–62.5 μm) serve short links and are easier to couple to cheap sources.

> **Status of the principle: Single-mode condition $V < 2.405$**
> *Derived.* It follows from solving Maxwell's equations for a step-index cylindrical guide in the weakly guiding approximation ($n_1 - n_2 \ll n_1$); 2.405 is the first zero of the Bessel function J_0.

Worked example — designing a single-mode fibre. A germania-doped core of $n_1 = 1.4500$ in a silica cladding of $n_2 = 1.4447$ gives

$$\mathrm{NA} = \sqrt{1.4500^2 - 1.4447^2} = 0.124$$

an acceptance half-angle of 7.1°. For single-mode operation at 1.31 μm, $a < 2.405\lambda/(2\pi\,\mathrm{NA}) = 2.405(1.31)/(2\pi \times 0.124) = 4.05$ μm. A core radius of 4.1 μm gives $V = 2.44$ at 1.31 μm (just multimode) and $V = 2.06$ at 1.55 μm — so standard telecom fibre, with a core diameter of about 8–9 μm, is single-mode at 1.55 μm and only marginally above cutoff at 1.31 μm. The index step of 0.0053 is produced by about 4 mol% GeO_2 in the core, deposited by **modified chemical vapour deposition** (MCVD) or outside vapour deposition: $SiCl_4$ and $GeCl_4$ are oxidized to soot inside or around a silica tube, sintered to a transparent preform, and drawn to a 125 μm fibre at about 2000 °C. The vapour route is essential because the chlorides can be distilled to a purity that no melted glass could reach.

Loss: the materials limits

Fibre attenuation is quoted in decibels per kilometre; a loss of α dB/km over length L leaves a fraction $10^{-\alpha L/10}$ of the power. Silica fibre loss has three materials origins (Figure 36.6):

- **Rayleigh scattering** from frozen-in density and composition fluctuations of the glass, proportional to λ^{-4} and to the fictive temperature at which the glass structure froze (Chapter 17). It dominates in the near infrared.
- **Infrared absorption** by Si–O vibrational overtones, rising steeply beyond 1.6 μm.
- **Impurity absorption**, above all by OH groups (overtone at 1.38 μm) and transition metals. A transition-metal content of 1 ppb or an OH content below 1 ppm is required, achieved through vapour-phase chemistry and dehydration with chlorine.

Between Rayleigh scattering and infrared absorption lies a minimum near 1.55 μm, about 0.17–0.2 dB/km in standard fibre and 0.14 dB/km in pure-silica-core fibres — so 100 km of fibre transmits about 1 % of the input, recoverable with an erbium-doped fibre amplifier. At 0.85 μm, Rayleigh scattering is $(1550/850)^4 = 11$ times stronger, giving losses of about 2 dB/km, and the same 100 km would transmit 10^{-20} of the light. This is the answer to the fourth opening question.

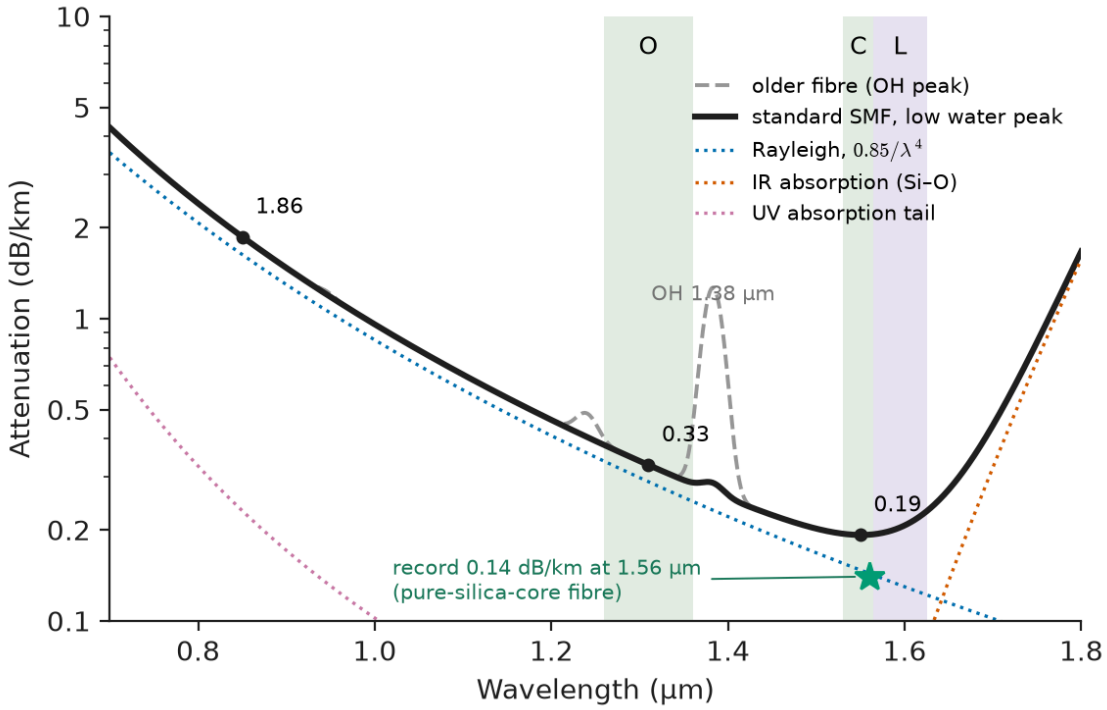


Figure 36.6 — Attenuation of silica optical fibre: Rayleigh scattering and infrared absorption bound a low-loss window at 1.3–1.6 um.

Other fibres and waveguides

Polymer optical fibre (PMMA core, fluoropolymer cladding) has high loss (about 150–200 dB/km at 650 nm, from C–H vibrational overtones) but large cores, flexibility and cheap connectors, and serves in-vehicle networks, lighting and short links. **Fluoride and chalcogenide glass fibres** transmit the mid-infrared for spectroscopy and laser delivery. **Fibre sensors** turn the fibre itself into a strain gauge or thermometer: fibre Bragg gratings, written by ultraviolet light into a germanosilicate core, reflect a narrow band whose wavelength shifts with strain (about 1.2 pm per microstrain at 1.55 µm) and temperature, and are embedded in bridges, wind-turbine blades, composite aircraft structures, pipelines and oil wells, where they are immune to electromagnetic interference. Distributed sensing reads Rayleigh, Raman or Brillouin backscatter along tens of kilometres of fibre to locate leaks, fires or ground movement.

On chips, **integrated waveguides** guide light through a high-index core patterned on a low-index substrate: silicon on SiO_2 (silicon photonics, an index contrast of 3.48/1.44 that allows bends of a few micrometres), silicon nitride (lower loss, visible-to-infrared), and thin-film lithium niobate for modulators (Section 36.7). Silicon photonic transceivers carry the optical links of data centres.

36.7 Nonlinear and electro-optic materials

Nonlinear optics

At low intensity, polarization is proportional to the optical field. At laser intensities, the field approaches the inner-atomic fields that bind electrons, and higher-order terms appear:

$$P = \varepsilon_0 \left(\chi^{(1)} E + \chi^{(2)} E^2 + \chi^{(3)} E^3 + \dots \right)$$

The **second-order** susceptibility $\chi^{(2)}$ mixes frequencies: two photons at ω combine to one at 2ω (**second-harmonic generation**), as in the green laser pointer, where 1064 nm light from Nd:YVO_4 is doubled to 532 nm in a KTP crystal. Sum- and difference-frequency generation and optical parametric oscillation extend the same idea to tunable sources. Symmetry decides which materials qualify: $\chi^{(2)}$ is a third-rank tensor like the piezoelectric coefficient, and vanishes in any centrosymmetric material (Chapter 27). Glass, silicon and cubic centrosymmetric crystals have no bulk second-order response; the useful crystals

are non-centrosymmetric — $LiNbO_3$ ($d_{33} \approx 25$ pm/V), $KTiOPO_4$ (KTP, about 15), β-BaB_2O_4 (BBO, about 2, transparent into the deep ultraviolet), LiB_3O_5 (LBO, about 1, with a high damage threshold).

Efficient conversion requires **phase matching**: the fundamental and harmonic must travel at the same phase velocity, which dispersion prevents. Birefringent crystals solve this by using different polarizations whose indices cross at a particular angle. A materials-processing alternative is **quasi-phase matching**: the ferroelectric domains of $LiNbO_3$ are periodically reversed by applying a high field through patterned electrodes (periodically poled lithium niobate, PPLN), so the sign of $\chi^{(2)}$ flips every coherence length — domain engineering of the kind described in Chapter 27, used here for light.

The **third-order** susceptibility $\chi^{(3)}$ exists in all materials, including glass. It makes the refractive index depend on intensity, $n = n_0 + n_2 I$ (the **optical Kerr effect**); for silica $n_2 \approx 2.6 \times 10^{-20}$ m²/W, small but significant over the long lengths and small cores of fibres, where it causes self-phase modulation and four-wave mixing that limit high-capacity links and also generate supercontinuum light sources.

> **Status of the principle: Power-series expansion of the polarization**
> *Approximate*. It holds when the optical field is much smaller than the atomic fields, so successive terms shrink rapidly. It breaks down at intensities near the damage threshold, where multiphoton ionization and plasma formation occur — the regime exploited in femtosecond laser machining.

Electro-optic materials

A DC or low-frequency field can also change the refractive index. In non-centrosymmetric crystals the change is linear in field — the **Pockels effect**:

$$\Delta n = -\tfrac{1}{2} n^3 r E$$

where r is the electro-optic coefficient (pm/V). In centrosymmetric materials only the quadratic **Kerr electro-optic effect** remains. The index change shifts the phase of light travelling through the material; placed in one or both arms of a Mach–Zehnder interferometer, it switches light on and off. The voltage needed to shift the phase by π is the **half-wave voltage**

$$V_\pi = \frac{\lambda g}{n^3 r \Gamma L}$$

for electrodes with gap g, length L and overlap factor Γ between the electric and optical fields.

Worked example — a lithium niobate modulator. For light at 1.55 μm polarized along the c-axis of $LiNbO_3$ ($n_e = 2.138$, $r_{33} = 30.8$ pm/V), with an electrode gap of 15 μm, a length of 2 cm and an overlap of 0.5:

$$V_\pi = \frac{(1.55 \times 10^{-6})(15 \times 10^{-6})}{(2.138)^3 (30.8 \times 10^{-12})(0.5)(0.02)} = 7.7 \text{ V}$$

Driving both arms in push-pull halves this to about 3.9 V, typical of the bulk-crystal modulators that carried long-haul telecom traffic for decades. Because $V_\pi \propto g/L$, a lower voltage requires either longer devices or narrower gaps; thin-film lithium niobate on insulator, which confines light in a waveguide less than a micrometre across, allows gaps of a few micrometres and modulators driven directly by CMOS-level voltages at over 100 GHz. Alternatives include electro-optic polymers (r_{33} above 100 pm/V, but with long-term stability concerns), $BaTiO_3$ films on silicon (r of several hundred pm/V), and silicon itself, which has no Pockels effect and modulates instead by changing carrier density (the free-carrier plasma effect). Liquid crystals, whose large birefringence is reoriented by modest fields, make the slow but cheap electro-optic devices of displays (Chapter 37).

36.8 Photonic crystals and metamaterials

Photonic crystals

A material whose refractive index varies periodically on the scale of the wavelength scatters light coherently, just as a crystal lattice diffracts X-rays (Chapter 4). Waves reflected from successive periods add in phase at the Bragg condition, and over a range of frequencies light cannot propagate at all: a **photonic band gap**, the optical analogue of the electronic band gap of Chapter 31.

The one-dimensional case is the **distributed Bragg reflector** (DBR): alternating quarter-wave layers of high and low index. For TiO_2 ($n_H = 2.3$) and SiO_2 ($n_L = 1.45$) on glass, designed for 1064 nm, the layers are 116 nm and 183 nm thick; ten pairs reflect 99.99 % at the design wavelength, over a stop band about 29 % of the centre frequency wide. Such dielectric mirrors absorb far less than metal mirrors and are used in high-power laser optics, in the cavity mirrors of **vertical-cavity surface-emitting lasers** (VCSELs) — epitaxial GaAs/AlGaAs DBRs grown directly on the laser, used for face recognition in phones and short optical links — and in the heat-reflecting coatings of halogen lamps and solar-control glazing.

Two- and three-dimensional photonic crystals confine light more completely. **Photonic-crystal fibres**, drawn from stacks of silica capillaries, guide light by a periodic array of air holes; hollow-core versions guide light in air, with lower nonlinearity and latency than solid fibres. Three-dimensional photonic crystals can be made by self-assembly of monodisperse colloidal spheres (synthetic opals, Chapter 37), by lithography, or by direct laser writing. Nature uses them extensively: the iridescence of opal gemstones (ordered silica spheres), butterfly wings, beetle shells and peacock feathers is **structural colour**, which does not fade because no dye is involved, and is being developed for paints and inks without pigments.

Metamaterials and metasurfaces

A **metamaterial** is an artificial structure whose unit cells are smaller than the wavelength, so that light sees an effective homogeneous medium — but one whose permittivity and permeability are set by the geometry of the cells rather than by their atoms. Arrays of split-ring resonators and wires can give a negative effective permeability and permittivity together, and therefore a **negative refractive index**, demonstrated at microwave frequencies in 2000. Proposed applications include sub-wavelength "superlenses" and electromagnetic cloaks; both work only over narrow bands and are limited by the losses of metallic resonators, especially at optical frequencies.

More practical are **metasurfaces**: single layers of sub-wavelength nanostructures (TiO_2 or amorphous silicon pillars, gold antennas) that impose a designed phase shift on transmitted light at each point. A metasurface can act as a flat lens, polarizer, hologram or beam deflector a fraction of a micrometre thick, and metalenses are entering phone cameras and sensors. **Plasmonics** uses the collective electron oscillations of metal nanoparticles: gold nanoparticles of 20–50 nm absorb green light, giving the red of medieval stained glass, and are used in lateral-flow medical tests and as photothermal agents in cancer therapy (Chapter 39). Engineered emitters also manage heat: **radiative-cooling** films combine high solar reflectance with strong thermal emission in the 8–13 μm atmospheric window, cooling buildings below ambient temperature in direct sunlight without power — a photonic material for energy efficiency.

> **Boundary.**
> Maxwell's equations, quantum theory of absorption and emission, and laser physics belong to the Engineering Physics book; optoelectronic devices (photodetectors, solar cells, modulators) are treated in Chapter 35.

What remains open

Open question	Where it is answered
How does the electronic polarization of Section 36.2 relate to dielectric behaviour at lower frequencies, and why is $\chi^{(2)}$ governed by the same symmetry as piezoelectricity?	Chapters 26 and 27
How do band structure, doping and heterostructures make LED and laser active layers?	Chapters 32, 34 and 35
How do liquid crystals switch light in displays, and how do colloids self-assemble into photonic structures?	Chapter 37
How does quantum confinement tune the colour of quantum dots, and what are 2D optical materials?	Chapter 38
How are photovoltaic absorbers chosen, and how do coatings and textures trap light in solar cells?	Chapter 40
How do UV light and weathering yellow polymers and degrade pigments?	Chapter 44
How are lasers used in welding, cutting and additive manufacturing?	Chapters 47 and 48
How are optical coatings and thin films deposited?	Chapter 49
How are optical constants, luminescence and Raman spectra measured?	Chapters 52 and 53
How are optical materials and glasses selected for a given application and cost?	Chapter 58

Next: Chapter 37 opens Part VIII with smart and soft materials — shape-memory alloys, electroactive polymers, gels, colloids and liquid crystals — whose properties change on command.

Part VIII — Advanced and Emerging Materials

The first seven parts of this book built a toolkit: structure and defects, thermodynamics and kinetics, mechanical behaviour, the established material classes, and the physical properties. Part VIII puts that toolkit to work where engineering is changing fastest. Its five chapters deal with materials whose value lies in an unusual response (a wire that remembers its shape, a gel that swells on cue), an unusual length scale (a sheet one atom thick, a metal with 10 nm grains), an unusual partner (living tissue), an unusual job (storing and converting energy) or an unusual environment (a turbine at 1100 °C, a reactor core under neutron flux, a hydrogen tank at 20 K). None of them needs new physics. Each is the same structure → processing → property → performance chain, pushed into a regime where a former footnote becomes the main concern: surfaces, entropy, living tissue, electrochemistry, radiation, temperature extremes. The part shows how far the general principles reach and gives the engineer grounds to judge claims in fields where hype often runs ahead of qualification.

By the end of this part you should understand

- How a reversible martensitic transformation gives shape memory and superelasticity, and how magnetostrictive, piezoelectric and electroactive-polymer actuators compare in strain, stress and speed.
- Why thermal energy $k_B T$ is the energy scale of soft matter, and how DLVO forces, gel networks, liquid-crystalline order and amphiphile packing are designed with it.
- Which properties change first as a material shrinks to the nanoscale (surface fraction, melting point, confinement energy, grain-boundary-controlled strength), and why many nanomaterials have reached products only as additives.
- How the body responds to an implanted material, why bone, blood and soft tissue each set different requirements, and how metals, ceramics, polymers and hydrogels are qualified as medical devices.
- Which material properties limit photovoltaic, battery, fuel-cell, hydrogen, thermoelectric and supercapacitor technologies, and why critical-material supply limits scale-up.
- How creep, oxidation, radiation damage, cryogenic embrittlement and thermal shock set the material choices for turbines, reactors, aircraft and spacecraft.

Chapter	Central question	What it delivers
37. Smart and Soft Materials	How can a material sense, actuate or assemble itself, and what sets the limits of that response?	Shape-memory and superelastic NiTi, magnetostrictive and magnetic shape-memory alloys, piezoelectric stacks, electroactive polymers, colloid stability (DLVO), hydrogels, liquid crystals, self-assembly by packing rules
38. Nanomaterials and Two-Dimensional Materials	At what size does a material stop behaving like its bulk self, and what can engineering do with that?	Surface-fraction and melting-point estimates, quantum confinement (Brus), nanotubes, quantum dots, graphene and 2D crystals, inverse Hall–Petch in nanocrystalline metals, nanocomposites, synthesis routes, characterization and safety
39. Biomaterials	What makes a material acceptable to the body, and how is it qualified as an implant or device?	Host response, metallic, ceramic, polymeric and degradable biomaterials, hydrogels and scaffolds, hips, knees, dental implants and stents, drug delivery, sterilization, ISO 10993 testing and regulation
40. Materials for Energy	Which material properties limit how energy is converted and stored, and can those materials be made at scale?	PV absorbers, intercalation electrodes and electrolytes, fuel-cell and electrolyser materials, hydrogen storage and embrittlement, thermoelectrics, supercapacitors, nuclear fuels, critical raw materials
41. Materials for Extreme Environments	How do materials survive very high or very low temperature, radiation, and the conditions of flight and space?	Superalloys, refractory alloys, CMCs and thermal barrier coatings, radiation damage and dpa, fission and fusion materials, aerospace lightweighting, cryogenic materials, hypersonic and space environments

How this part connects to the rest of the book

Part VIII draws on almost every earlier chapter. Shape memory is the martensite of Chapter 10 run in reverse; piezoelectric and magnetostrictive actuators reuse Chapters 27 and 28; colloids, gels and liquid crystals extend the polymer physics of Chapter 20. Nanomaterials test the surface thermodynamics of Chapter 6 and the Hall–Petch law of Chapter 13 at their limits. Biomaterials combine the titanium and cobalt alloys of Chapter 16, the zirconia and glasses of Chapter 17 and the degradable polymers of Chapter 21, under the corrosion and wear treated in Part IX. Energy materials rest on semiconductor physics (Chapters 31–32), ionic conduction (Chapter 25) and diffusion (Chapter 8). Extreme-environment materials take the creep and fracture of Chapter 14 and the superalloys of Chapter 16 to their service limits, and lead directly into high-temperature oxidation in Chapter 42. Looking forward, Part IX decides how long these materials last, Part X explains how they are made (thin films and additive manufacturing above all), and Part XI supplies the characterization, simulation and selection methods by which emerging materials are discovered, qualified and judged against their environmental cost.

Chapter 37 — Smart and Soft Materials

Part VIII · Advanced and Emerging Materials

A self-expanding stent for a femoral artery is crimped to 2 mm, threaded through a catheter and released inside the vessel, where it springs open to 7 mm and pushes gently and steadily against the wall for years, flexed half a million times a day by walking. It is laser-cut from a nickel–titanium tube, and no ordinary metal could do this job: steel crimped by 70 % in diameter would deform permanently, and a polymer would creep and lose its grip. A few metres away in the same hospital, an infusion pump's piezoelectric sensor detects air bubbles, the patient's contact lenses are a hydrogel that is 40 % water, and the monitor shows the patient's heart rate on a liquid-crystal display. None of these materials is used for its strength or stiffness. Each is used because it *responds*: to temperature, stress, electric or magnetic field, water, or ions. This chapter is about such materials. The first half treats "smart" solids that convert one form of energy into mechanical work; the second half treats "soft" matter (colloids, gels, liquid crystals and self-assembled structures) whose structure is held together by forces comparable to thermal energy and can therefore be reshaped by small stimuli.

> **How can a solid change shape on command, recover 8 % strain, or sense a force, and what limits the stroke, force and speed of such a material?**
> **Why do particles in a paint stay dispersed, why does a gel swell, and how do molecules arrange themselves into micelles, membranes and display pixels without being placed?**

Concept	Engineering destination
Thermoelastic martensite, shape memory	Pipe couplings, actuator wires, deployable space structures, thermal valves
Superelasticity, Clausius–Clapeyron slope	Self-expanding stents, orthodontic arch wires, seismic dampers, eyeglass frames
Magnetostriction, magnetic shape memory	Sonar projectors, vibration energy harvesters, torque sensors, fast valves
Piezoelectric stacks and sensors	Diesel injectors, nanopositioners, ultrasound probes, structural-health monitoring
Electroactive polymers	Soft robots, haptics, artificial muscles, adaptive lenses
DLVO theory, Debye length	Ceramic slips, paints, concrete admixtures, water treatment, drug suspensions
Gels and hydrogels, swelling, mesh size	Contact lenses, superabsorbents, wound dressings, drug release, soft actuators
Liquid-crystal order parameter, Fréedericksz transition	Displays, aramid fibre spinning, thermochromic sensors
Self-assembly, packing parameter	Detergents, lipid nanoparticles, block-copolymer lithography, colloidal crystals

Roadmap. What makes a material "smart" and how actuators are compared (37.1) → shape-memory alloys (37.2) → magnetostrictive and magnetic shape-memory materials (37.3) → piezoelectric actuators and sensors (37.4) → electroactive polymers (37.5) → the soft-matter energy scale and colloid stability (37.6) → gels and hydrogels (37.7) → liquid crystals (37.8) → self-assembly as a design tool (37.9).

37.1 Smart materials and how to compare actuators

A **smart** (or *active*, *responsive*) material couples two physical domains strongly enough that the coupling can be used in a device: a mechanical strain produced by heat, electric field or magnetic field (an **actuator**), or an electrical signal produced by strain (a **sensor**). Every solid has such couplings (all solids expand when heated, all dielectrics strain slightly in a field), so the word is about magnitude: a smart material's coupling is large enough, reversible enough and fast enough to do work.

Three numbers characterise an actuator material:

- the **free strain** $\varepsilon_{\max}$, reached with no external load;
- the **blocking stress** σ_{bl}, the stress at which the strain is suppressed completely;

- the **bandwidth**, the highest frequency at which the full stroke can be cycled.

For a material whose output falls linearly from $\varepsilon_{\max}$ at zero load to zero at σ_{bl}, the maximum work per unit volume delivered to a matched load is

$$w_{\max} = \tfrac{1}{4}\,\sigma_{\mathrm{bl}}\,\varepsilon_{\max}$$

and the specific power is roughly that work times the cycling frequency. Table 37.1 and Figure 37.1 compare the main families.

Table 37.1 — Representative actuator materials (typical values; grade and drive conditions vary)

Material	Stimulus	Free strain	Blocking stress	Work density (J/m³)	Bandwidth
NiTi shape-memory wire	Heating (Joule)	4–6 % (one-way, polycrystal)	200–600 MPa	$\sim 10^6$–10^7	0.1–10 Hz (cooling-limited)
Ni–Mn–Ga magnetic shape memory (single crystal)	Magnetic field ~0.7 T	6 % (up to 10–12 %)	2–3 MPa	$\sim 4 \times 10^4$	up to ~1 kHz
Terfenol-D	Magnetic field	0.1–0.2 %	50–100 MPa	$\sim 2 \times 10^4$–5×10^4	~10 kHz
PZT stack	Electric field ~2 kV/mm	0.1–0.15 %	30–50 MPa	$\sim 10^4$	10–100 kHz
Dielectric elastomer (acrylic, silicone)	Electric field 50–150 MV/m	10–100 %	0.1–3 MPa	$\sim 10^4$–10^6	10 Hz–1 kHz
Ionic polymer–metal composite	1–3 V	~1–3 % (bending)	0.1–1 MPa	$\sim 10^3$	0.1–10 Hz
Skeletal muscle (reference)	Nerve signal	20–40 %	0.1–0.35 MPa	$\sim 10^4$	~10 Hz
Thermal expansion, steel ($\Delta T = 100$ K)	Heating	0.12 %	~250 MPa	$\sim 7 \times 10^4$	slow

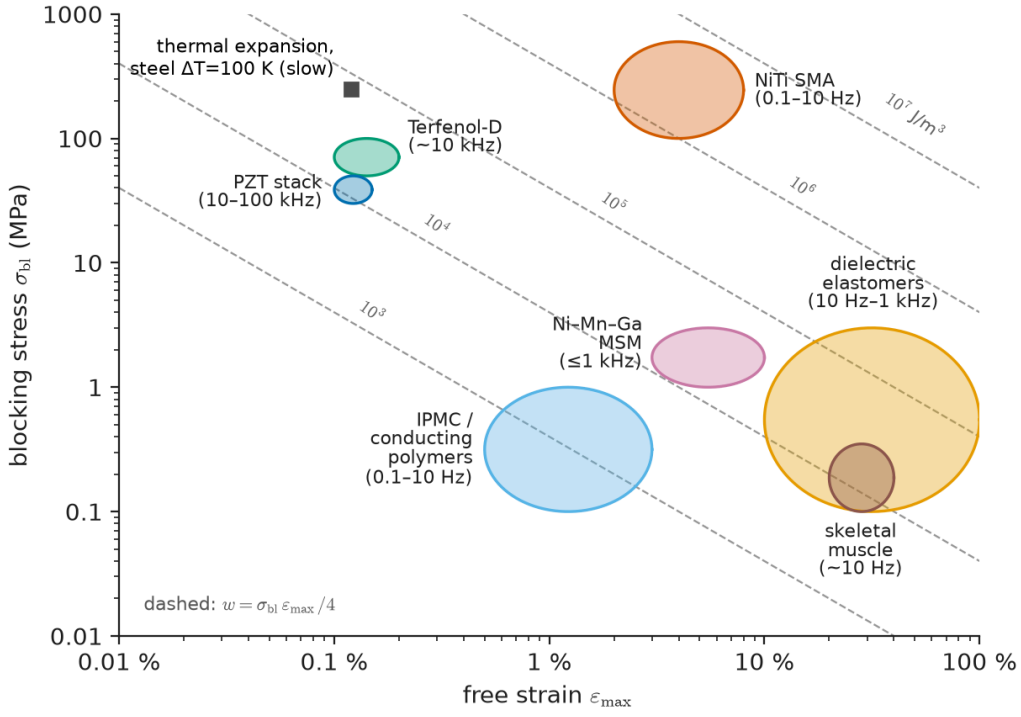


Figure 37.1 — Actuator materials trade free strain against blocking stress; shape-memory alloys deliver the highest work density but are among the slowest.

The map shows the central trade-off: large strain comes with low stress (soft polymers, magnetically driven twin boundaries), large stress with small strain (ceramics, magnetostrictives), and the one family that combines both, shape-memory alloys, is limited by how fast heat can be removed. Choosing an actuator material is therefore a design problem of the kind treated in Chapter 58: stroke, force, frequency, efficiency, mass and cost cannot all be maximised.

37.2 Shape-memory alloys

37.2.1 The phenomenon

In the late 1960s the US Navy needed a reliable way to join titanium hydraulic tubes in the F-14 fighter without welding. The solution was a coupling sleeve of nickel–titanium machined slightly *smaller* than the tube, cooled in liquid nitrogen, expanded mechanically by about 4 %, slipped over the tube ends while still cold, and then allowed to warm up. On warming, the sleeve "remembered" its original diameter, shrank and gripped the tubes with a contact pressure high enough to seal against 35 MPa of hydraulic fluid. Over a million such couplings were installed without a reported leak. Two properties are at work: the **shape-memory effect** (a deformation applied at low temperature is recovered on heating) and **superelasticity** (at a slightly higher temperature, very large strains are recovered simply on unloading).

37.2.2 Thermoelastic martensite

Both effects come from the martensitic transformation of Chapter 10, but of a special kind. In steel, martensite forms with a large shape change, its interfaces are pinned by plastic deformation, and the transformation is irreversible in practice. In shape-memory alloys the transformation is **thermoelastic**:

- the parent phase (**austenite**, for NiTi the ordered B2 cubic structure) transforms to a low-symmetry martensite (monoclinic B19′ in NiTi) by a small, coordinated shear;
- the lattice correspondence is unique: every martensite variant can revert only to the original orientation of austenite;
- the shape change is accommodated **elastically and by twinning**, not by dislocation slip, so the interfaces stay mobile and the transformation hysteresis is small (20–40 K in NiTi).

Because martensite has lower symmetry than austenite, one austenite grain can form up to 12 crystallographically equivalent **variants** in NiTi (24 habit-plane variants). On cooling without load, the variants form in self-accommodating groups, twinned against each other so that the overall shape of the grain barely changes. Four temperatures describe the transformation: martensite start and finish M_s, M_f on cooling, and austenite start and finish A_s, A_f on heating.

The shape-memory cycle (Figure 37.2a) is then:

1. Cool below M_f: twinned, self-accommodated martensite, no shape change.
2. Deform at low temperature: the twin boundaries move, so the variants best oriented for the applied stress grow at the expense of others (**detwinning**). Strains of 6–8 % (single crystals up to ~10 %) are produced at low stress (100–200 MPa in NiTi) with no dislocation slip.
3. Unload: the detwinned martensite stays deformed.
4. Heat above A_f: every variant reverts to the single original austenite orientation, so the original shape returns.

This is a **one-way** effect: cooling again does not restore the deformed shape. **Two-way** memory can be "trained" by thermomechanical cycling, which leaves oriented dislocation arrays whose stress fields favour particular variants on cooling, but the two-way strain is smaller (~2 %) and less stable.

37.2.3 Superelasticity and the Clausius–Clapeyron relation

Above A_f, austenite is the stable phase at zero stress. But a stress that does work on the transformation strain lowers the Gibbs energy of the martensite variant aligned with it, so martensite can be *induced by stress*. On unloading, it is unstable and reverts. The result (Figure 37.2b) is a flag-shaped stress–strain curve: linear elastic austenite, an upper **plateau** at which stress-induced martensite forms and a band of transformed material propagates along the specimen, then elastic martensite; on unloading, a lower plateau at which it reverts, recovering 6–8 % strain with a hysteresis loop whose area is energy dissipated as heat.

The plateau stress rises with temperature, because the further one is above the equilibrium temperature, the more mechanical work must be supplied to make martensite stable. Treating stress and strain as a pair of conjugate variables like $-p$ and V, the

two-phase equilibrium obeys a Clausius–Clapeyron relation (the thermodynamics is recapped from Chapter 6):

$$\frac{d\sigma}{dT} = -\frac{\Delta S}{\varepsilon_{\mathrm{tr}}} = -\frac{\Delta H}{T_0\,\varepsilon_{\mathrm{tr}}}$$

where ΔS and ΔH are the entropy and enthalpy of the austenite → martensite transformation per unit volume (both negative: the transformation releases heat), $\varepsilon_{\mathrm{tr}}$ is the transformation strain in the loading direction and T_0 the equilibrium temperature at zero stress.

> **Status of the principle: Clausius–Clapeyron relation for stress-induced martensite**
> *Derived.* It follows from equilibrium between two phases under uniaxial stress; it assumes a fixed transformation strain and neglects hysteresis and texture, and in polycrystals is obeyed by the measured plateau stresses to within 10–20 % over about 100 K above A_f.

Worked example — the stress–temperature slope of NiTi and a stent's plateau. NiTi has a latent heat of about 20 J/g and density 6450 kg/m³, so $|\Delta H| \approx 2.0 \times 10^4 \times 6450 = 1.29 \times 10^8$ J/m³. With $T_0 \approx 300$ K and a polycrystalline transformation strain $\varepsilon_{\mathrm{tr}} \approx 0.06$:

$$\frac{d\sigma}{dT} \approx \frac{1.29 \times 10^8}{300 \times 0.06} = 7.2 \times 10^6\ \mathrm{Pa\,K^{-1}} \approx 7\ \mathrm{MPa/K}$$

Measured slopes for superelastic NiTi are 5–8 MPa/K. A stent alloy (Ni-rich, about Ni-50.8 at.% Ti) is processed so that the stress-induced transformation would start at zero stress at about $M_s = -20$ °C. Taking a slope of 6.5 MPa/K, the upper plateau at body temperature is $6.5 \times (37 - (-20)) = 6.5 \times 57 \approx 370$ MPa, and the device delivers a nearly constant outward force over a wide range of vessel diameters, the property clinicians call "biased stiffness". The same slope is a warning: a NiTi part designed at room temperature becomes about 100 MPa stiffer in its plateau 15 K warmer, and if it is used above the temperature M_d (typically 50–100 K above A_f) the stress for slip falls below the stress for transformation, the alloy deforms plastically, and superelasticity is lost.

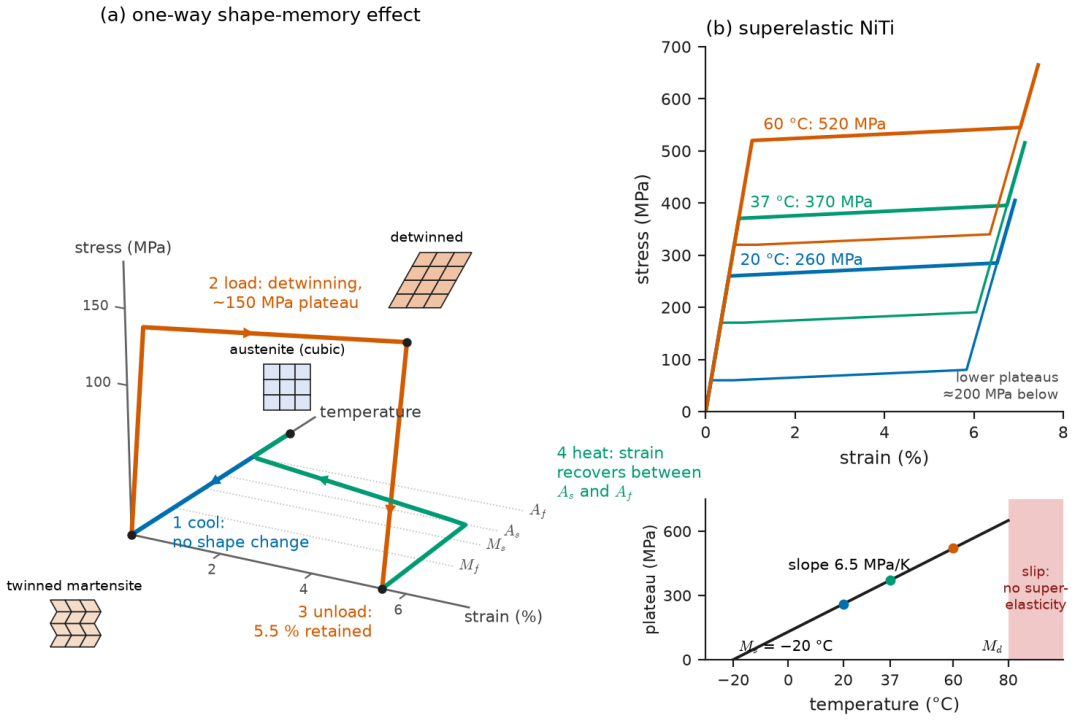


Figure 37.2 — Shape memory and superelasticity are two faces of the same reversible martensitic transformation, separated by temperature.

37.2.4 Alloys and engineering

NiTi (nitinol) dominates: it has the largest recoverable strain, good fatigue and corrosion resistance (a protective TiO_2 film, Chapter 39), and transformation temperatures that move by roughly 10–20 K for each 0.1 at.% change in nickel content around equiatomic. That sensitivity makes melting and composition control (vacuum induction plus vacuum arc remelting, carbon and

oxygen held below a few hundred ppm) the critical processing step. Ageing Ni-rich alloys precipitates Ni_4Ti_3, which raises the matrix Ti content and shifts the transformation temperatures upward, and also strengthens the austenite against slip. Cold working followed by **shape setting** at 450–550 °C while clamped in a fixture fixes the "remembered" shape; this is how stents, guidewires and arch wires are formed.

Other families fill niches. Cu–Zn–Al and Cu–Al–Ni are cheaper and have higher transformation temperatures but are brittle in polycrystalline form because of their elastic anisotropy and coarse grains. Fe–Mn–Si alloys show a modest (~2–4 %) one-way effect through an FCC → HCP transformation and are used in prestressing strips for retrofitting concrete. **High-temperature SMAs** (NiTiHf, NiTiPd, NiTiPt) push A_f above 100–250 °C for aerospace actuators near engines; Boeing flew NiTi torque tubes that twist trailing-edge surfaces on an engine chevron and a wing in test programmes.

Design limits are as important as the effect itself:

- **Speed.** Heating by electric current is fast; cooling is by conduction and convection. A 0.25 mm wire in still air cycles at about 1 Hz; thinner wires and liquid cooling are faster.
- **Efficiency.** Most of the input energy heats the wire. The worked example below shows an efficiency of about 2 %, typical of thermal actuators, which is why SMAs suit low-duty-cycle tasks (valves, latches, deployment) rather than continuous motion.
- **Functional fatigue.** Each cycle leaves some dislocations, so the plateau stress drifts down and the recoverable strain shrinks; designs that limit strain to 2–3 % achieve 10^6–10^7 cycles. Structural fatigue of stents (Chapter 14) is assessed on strain amplitude, not stress, for this reason.

Worked example — a NiTi actuator wire. A 0.25 mm NiTi wire, 100 mm long, lifts a load by recovering 4 % strain against a recovery stress of 150 MPa. Its cross-section is $A = \pi(0.125 \times 10^{-3})^2 = 4.91 \times 10^{-8}$ m², so the force is $150 \times 10^6 \times 4.91 \times 10^{-8} = 7.4$ N and the stroke $0.04 \times 100 = 4.0$ mm. The mechanical work is $7.4 \times 0.004 = 29$ mJ per cycle. Heating the wire (mass $6450 \times 4.91 \times 10^{-8} \times 0.1 = 3.2 \times 10^{-5}$ kg) by 60 K with $c_p \approx 450$ J kg⁻¹ K⁻¹, and supplying the latent heat of 20 J/g, needs $3.2 \times 10^{-5} \times (450 \times 60 + 2.0 \times 10^4) = 1.5$ J. The efficiency is $0.029/1.5 \approx 2$ %. The wire is an excellent force-to-mass actuator (7 N from 0.03 g) and a poor energy converter.

Civil engineers use superelastic NiTi bars and cables for the hysteresis rather than the stroke: in seismic dampers and beam–column connections, each loading cycle dissipates the loop area (about 10–20 MJ/m³ per cycle at 6 % strain) and the structure recentres afterwards, which steel dampers that yield plastically cannot do.

37.3 Magnetostrictive and magnetic shape-memory materials

37.3.1 Magnetostriction

When a ferromagnet is magnetized, its domains rotate (Chapter 28) and the crystal lattice distorts slightly along the magnetization direction, because the spin–orbit coupling ties the magnetic moment to the shape of the electron orbitals and therefore to the bond lengths. The saturation **magnetostriction** λ_s is the fractional length change at saturation. In iron and nickel it is tens of parts per million ($\lambda_s \approx -35$ ppm for Ni, a contraction), which is a nuisance in transformers (the 100 Hz hum, Chapter 29) but too small for actuation.

Rare-earth–iron Laves compounds change this. **Terfenol-D** ($Tb_{0.3}Dy_{0.7}Fe_2$) has λ_s of 1500–2000 ppm at room temperature; the terbium–dysprosium ratio is chosen so that the large but opposite-signed anisotropies of $TbFe_2$ and $DyFe_2$ cancel, letting a modest field (about 50–200 kA/m) rotate the magnetization. It is grown as directionally solidified <112>-textured rods, operated under a compressive prestress of 10–20 MPa that orients the domains perpendicular to the rod so that the field can rotate them all, and used in high-power, low-frequency **sonar projectors**, active vibration control and fast valves. It is brittle (fracture toughness about 1 MPa m^1/2) and expensive.

Galfenol (Fe–Ga, 17–19 at.% Ga) gives 250–400 ppm but is ductile and machinable, can carry tension and bending, and is used in energy harvesters on vibrating machinery and in torque sensors. The **inverse** (Villari) effect, a change of magnetization with stress, makes magnetostrictive materials sensors as well: a shaft of Fe–Ni or Fe–Ga under torque changes permeability along its helix directions, which a non-contact coil detects, the basis of torque sensors in electric power steering.

37.3.2 Magnetic shape-memory alloys

A much larger strain is obtained by combining the twinned martensite of Section 37.2 with magnetic anisotropy. In the Heusler alloy **Ni2MnGa** (and off-stoichiometric Ni–Mn–Ga), the martensite is ferromagnetic and strongly anisotropic: the short c axis of the tetragonal (or modulated) martensite cell is the easy axis of magnetization. A field applied along a crystal direction makes the variant whose easy axis points along the field favoured by the magnetic anisotropy energy K_u per volume. If the twin boundaries are mobile enough, the favoured variant grows by twin-boundary motion and the crystal changes shape by the twinning strain $\varepsilon_0 = 1 - c/a$, about 6 % in five-layer modulated martensite and up to 10–12 % in other modulations.

The largest stress the field can exert on a twin boundary is set by the anisotropy energy:

$$\sigma_{\mathrm{mag}}^{\max} \approx \frac{K_u}{\varepsilon_0}$$

With $K_u \approx 2 \times 10^5$ J/m³ and $\varepsilon_0 = 0.06$, $\sigma_{\mathrm{mag}}^{\max} \approx 3.3$ MPa. Twin boundaries in good single crystals move at about 0.1–1 MPa, so the margin is narrow: the blocking stress is only 2–3 MPa and polycrystals, whose grain boundaries obstruct twinning, show little strain. Magnetic shape-memory alloys therefore combine large strain with fast (kHz) response, because no heat must flow, but small force. Prototypes include fast valves, micropumps and positioners. Unlike NiTi, the effect is limited by magnetism, not by heat transfer, and the material's Curie temperature (about 100 °C) and martensite start temperature cap the operating range.

> **Status of the principle: Anisotropy limit on magnetic shape-memory stress**
> *Approximate.* The output stress cannot exceed the magnetic energy available per unit transformation strain; real devices fall below it because of demagnetizing fields, twinning stress and incomplete saturation.

37.4 Piezoelectric actuators and sensors (recap)

Chapter 27 explained the origin of piezoelectricity: a crystal without a centre of symmetry polarizes under stress (direct effect) and strains under field (converse effect), linked by the third-rank tensor d_{ij} of Chapter 4. Ferroelectric ceramics such as **lead zirconate titanate** (PZT) near the morphotropic phase boundary are poled to give large coefficients, $d_{33} = 300$–600 pC/N (equivalently pm/V). Here the materials engineer's question is how such a ceramic becomes an actuator or a sensor.

Table 37.2 — Typical piezoelectric materials

Material	d_{33} (pC/N)	Coupling k_{33}	Curie/usable temperature	Notes
Quartz (SiO_2)	2.3 (d_{11})	0.1	573 °C (α–β)	Very stable; resonators, force sensors
PZT, "hard" (PZT-4)	290	0.70	330 °C	Low loss; sonar, ultrasonic welding
PZT, "soft" (PZT-5H)	590	0.75	190 °C	Large strain; actuators, imaging
PMN–PT single crystal	1500–2500	0.9	~100 °C (phase change)	High-end medical ultrasound
(K,Na)NbO_3-based (lead-free)	200–400	0.5–0.6	200–400 °C	RoHS-driven replacement
AlN thin film	5	0.3	> 1000 °C	MEMS microphones, RF filters
PVDF (poled polymer)	−20 to −30	0.15–0.2	~80 °C	Flexible sensors, hydrophones

37.4.1 Multilayer stack actuators

A single PZT plate 1 mm thick under 1 kV extends by only $d_{33}V \approx 0.5$ µm. To get useful stroke at moderate voltage, the ceramic is tape-cast into layers 50–100 µm thick interleaved with electrodes and co-fired (the same process as multilayer capacitors, Chapter 49). The layers are mechanically in series and electrically in parallel, so the stroke is

$$\Delta L = n\, d_{33}\, V$$

for n layers at voltage V.

Worked example — a stack for a common-rail injector. A stack of $n = 100$ layers, each 0.1 mm thick, with an effective $d_{33} = 500$ pm/V driven at 150 V (field 1.5 kV/mm) gives $\Delta L = 100 \times 500 \times 10^{-12} \times 150 = 7.5$ µm. For a 10 mm × 10 mm stack 10 mm long with an open-circuit modulus of about 60 GPa, the stiffness is $k = EA/L = 60 \times 10^9 \times 10^{-4} / 0.01 = 6.0 \times 10^8$ N/m (600 N/µm), so the blocking force is $k\,\Delta L = 6.0 \times 10^8 \times 7.5 \times 10^{-6} = 4.5$ kN. A hydraulic amplifier converts this short, powerful, sub-millisecond stroke into the needle lift of a diesel or gasoline direct-injection valve, which can then deliver several small injections per combustion cycle. The same stacks drive nanopositioners in lithography steppers and atomic-force microscopes, where 7.5 µm with sub-nanometre resolution is ideal.

Engineering limits come from the ceramic: it is weak in tension (stacks are mechanically preloaded in compression), shows hysteresis of 10–15 % in strain and creep at high fields, depoles if heated towards the Curie temperature or driven by reverse fields beyond the coercive field, and suffers electrode-edge cracking and silver migration in humid service. Lead-free substitutes (KNN, BNT, $BaTiO_3$-based) are improving but have not matched soft PZT across temperature.

37.4.2 Sensors and transducers

In the direct mode a force F along the poling axis produces a charge $Q = d_{33}F$; a quartz or PZT disc in a load washer turns 100 N into 230 pC or 40 nC respectively, read by a charge amplifier. Piezoelectric **accelerometers** (a seismic mass on a ceramic element) monitor bearings and engines; **ultrasound** transducers emit and receive at 1–20 MHz for medical imaging and for non-destructive testing of welds and composites (Chapter 52); thin PZT or PVDF patches bonded to an aircraft panel or a bridge girder send and receive guided waves for **structural-health monitoring**, detecting cracks and disbonds from changes in the signal. Because charge leaks away through finite resistance, piezoelectric sensors measure dynamic, not static, forces.

> **Boundary.**
> The crystal physics of piezo- and ferroelectricity is treated in more depth, with electrical equivalent circuits, in Chapter 27.

37.5 Electroactive polymers

Polymers offer what ceramics cannot: strains of tens of percent, low density, compliance that matches biological tissue, and easy shaping. **Electroactive polymers** (EAPs) are polymers that change shape in an electric field. They divide into two groups.

37.5.1 Electronic (field-driven) EAPs

Dielectric elastomers are not active materials in the crystal sense at all: they are soft, incompressible rubbers (acrylic, silicone, polyurethane) coated on both faces with compliant electrodes (carbon grease, carbon nanotube mats). A voltage charges the film as a capacitor, the opposite charges on the electrodes attract each other and like charges on each electrode repel, so the film is squeezed in thickness and, being incompressible, expands in area. The effective compressive stress is the **Maxwell pressure**

$$p = \varepsilon_0 \varepsilon_r E^2$$

(twice the electrostatic energy density, because the in-plane expansion also does work).

Worked example — a dielectric-elastomer membrane. An acrylic film ($\varepsilon_r = 4.7$, Young's modulus about 1 MPa) is 50 µm thick and driven at 3 kV, so $E = 3000/(50 \times 10^{-6}) = 60$ MV/m. The Maxwell pressure is $p = 8.854 \times 10^{-12} \times 4.7 \times (6.0 \times 10^7)^2 = 0.15$ MPa, and the small-strain thickness strain is $p/Y \approx 0.15/1.0 = 15$ %. In practice the film is prestretched biaxially (which thins it and stiffens it against instability), and area strains above 100 % have been demonstrated. Two failure modes bound the design: **electrical breakdown** (Chapter 26), and **pull-in** or electromechanical instability, in which thinning raises the field, which thins the film further, a positive feedback that collapses an unstretched film at about 1/3 thickness reduction. Prestretch and strain-stiffening elastomers move the instability past breakdown.

Kilovolt drive is the main practical drawback, but dielectric elastomers give the best combination of strain, efficiency (energy not used mechanically can be recovered from the capacitor) and speed among polymer actuators; they are used in haptic feedback, tunable lenses, loudspeakers and soft grippers, and run in reverse as **generators** that harvest wave energy.

Ferroelectric and electrostrictive polymers form the second electronic class: PVDF and its copolymer P(VDF-TrFE) are piezoelectric when poled (Table 37.2), and terpolymers such as P(VDF-TrFE-CFE), in which bulky comonomers break up long-

range ferroelectric order into nanodomains (a relaxor), give electrostrictive strains of 4–7 % at about 150 MV/m with elastic modulus near 0.5 GPa, much stiffer than elastomers.

37.5.2 Ionic EAPs

Ionic EAPs work at 1–5 V because the active process is ion migration, not field-induced stress. In an **ionic polymer–metal composite** (IPMC) a Nafion-type ionomer membrane, swollen with water or an ionic liquid and plated with platinum or gold, bends when a voltage drives mobile cations and their hydration water towards the cathode side, which swells while the anode side shrinks. **Conducting polymers** such as polypyrrole and PEDOT change volume by a few percent as ions enter and leave on oxidation and reduction (the same chemistry as a battery electrode, Chapter 40). Ionic EAPs are slow (diffusion-limited, seconds), need to stay wet or use ionic liquids, and hold a deflection only while current flows, but their low voltage suits medical devices such as active catheters.

A further, rapidly growing class is **liquid-crystal elastomers** (Section 37.8): crosslinked networks of mesogenic chains that contract by 30–400 % along their director when heated through the nematic–isotropic transition or when photoisomerizable groups are illuminated, offering light-driven, untethered soft actuators.

37.6 Soft matter and colloid stability

37.6.1 The soft-matter energy scale

The second half of the chapter changes scale. A steel lattice is held by bonds of a few electronvolts; thermal energy at room temperature is $k_B T = 4.1 \times 10^{-21}$ J $= 25.7$ meV at 25 °C, about a hundredth of that, so thermal motion only rattles the atoms. In **soft matter** (colloids, polymers in solution, gels, liquid crystals, surfactant assemblies, biological membranes) the structural units are large (nanometres to micrometres) and are held together by interactions of order 1–100 $k_B T$: van der Waals attraction between particles, screened electrostatic repulsion, hydrogen bonds, the hydrophobic effect, entropy of chains. As a result, soft matter is easily deformed (moduli of Pa to MPa), responds strongly to temperature, salt, pH and solvent, and organises itself by the balance of energy and entropy. That responsiveness is both the difficulty (a paint that settles, a sauce that separates) and the design opportunity (a gel that swells on cue, a membrane that forms by itself).

37.6.2 Colloids

A **colloid** is a dispersion of one phase in another with at least one dimension between about 1 nm and 1 μm: solid particles in liquid (*sols* and *suspensions*: paints, inks, ceramic slips, cement paste, blood), liquid droplets in liquid (*emulsions*: milk, mayonnaise, metalworking fluids, asphalt emulsions), gas in liquid or solid (*foams*: firefighting foams, polyurethane insulation), and liquid or solid in gas (*aerosols*). In this size range particles undergo Brownian motion strong enough to resist sedimentation for a time, but their huge interfacial area (Chapter 38) gives a large driving force to reduce it by aggregation. A colloid is therefore almost never in equilibrium; its "stability" is kinetic, meaning that a barrier prevents particles from coming into contact.

37.6.3 DLVO theory

The classical theory of that barrier, due to Derjaguin, Landau, Verwey and Overbeek (DLVO), adds two interactions between particles in an electrolyte.

Van der Waals attraction. Correlated dipole fluctuations attract any two bodies of the same material through any medium. For two spheres of radius R at surface separation $h \ll R$,

$$V_A(h) = -\frac{A_H R}{12 h}$$

where A_H is the **Hamaker constant**, of order 10^{-20} J for oxides and polymers in water and 10^{-19} J for metals in water.

Electrical double-layer repulsion. Most surfaces in water carry charge (from dissociation of surface groups, such as Si–OH → SiO^-, or adsorption of ions), balanced by a diffuse cloud of counter-ions. The cloud's thickness is the **Debye length** κ^{-1}, which falls with ionic strength I; for a 1:1 electrolyte in water at 25 °C

$$\kappa^{-1} \approx \frac{0.304}{\sqrt{I/(\mathrm{mol\,L^{-1}})}}\ \mathrm{nm}$$

(9.6 nm at 1 mM, 3.0 nm at 10 mM, 0.96 nm at 100 mM, and about 0.4 nm in sea water). When the clouds of two particles overlap, the osmotic pressure of the excess ions pushes them apart. For spheres with surface potential ψ_0 (in practice the measured **zeta potential**) and $\kappa R \gg 1$, a common approximation is

$$V_R(h) = 2\pi\varepsilon_0\varepsilon_r R\,\psi_0^2 \ln\!\left(1 + e^{-\kappa h}\right)$$

The total $V = V_A + V_R$ (Figure 37.3) has a deep **primary minimum** at contact (irreversible aggregation, *coagulation*), a **barrier** at about a Debye length from contact if repulsion is strong enough, and sometimes a shallow **secondary minimum** further out (weak, reversible *flocculation*). If the barrier is above about 15–25 $k_B T$, collisions almost never surmount it and the colloid is stable for months; below a few $k_B T$ it coagulates in seconds to minutes.

> **Status of the principle: DLVO theory**
> *Approximate.* It treats the medium as a continuum, the ions as point charges obeying Poisson–Boltzmann statistics and the forces as additive; it predicts trends with salt, valence and particle size well, but fails at separations below about 1–2 nm, where hydration forces, ion specificity and surface roughness dominate, and it ignores steric and depletion forces.

Worked example — when does a suspension coagulate? Oxide particles of diameter 100 nm ($R = 50$ nm) in water ($\varepsilon_r = 78.5$) have $A_H = 1.0 \times 10^{-20}$ J and a surface potential of 30 mV. Evaluating $V(h)/k_B T$ numerically with the two expressions above at 25 °C ($k_B T = 4.12 \times 10^{-21}$ J):

NaCl concentration	Debye length	Barrier height	Position of barrier
1 mM	9.6 nm	$23\,k_B T$	2.1 nm
10 mM	3.0 nm	$16\,k_B T$	1.3 nm
100 mM	0.96 nm	$4.6\,k_B T$	0.8 nm

At 1 mM the dispersion is stable; at 100 mM thermal collisions cross the barrier routinely and the particles coagulate. (In reality the surface potential also falls as salt is added, which makes the collapse sharper.) Multivalent counter-ions are far more effective: the critical coagulation concentration varies roughly as z^{-6} for highly charged surfaces (the **Schulze–Hardy rule**), which is why water-treatment plants dose aluminium or iron(III) salts to coagulate clay and organic colloids before filtration, and why a few millimoles of $\mathrm{Ca^{2+}}$ can destabilise a ceramic slip that tolerated ten times as much $\mathrm{Na^{+}}$.

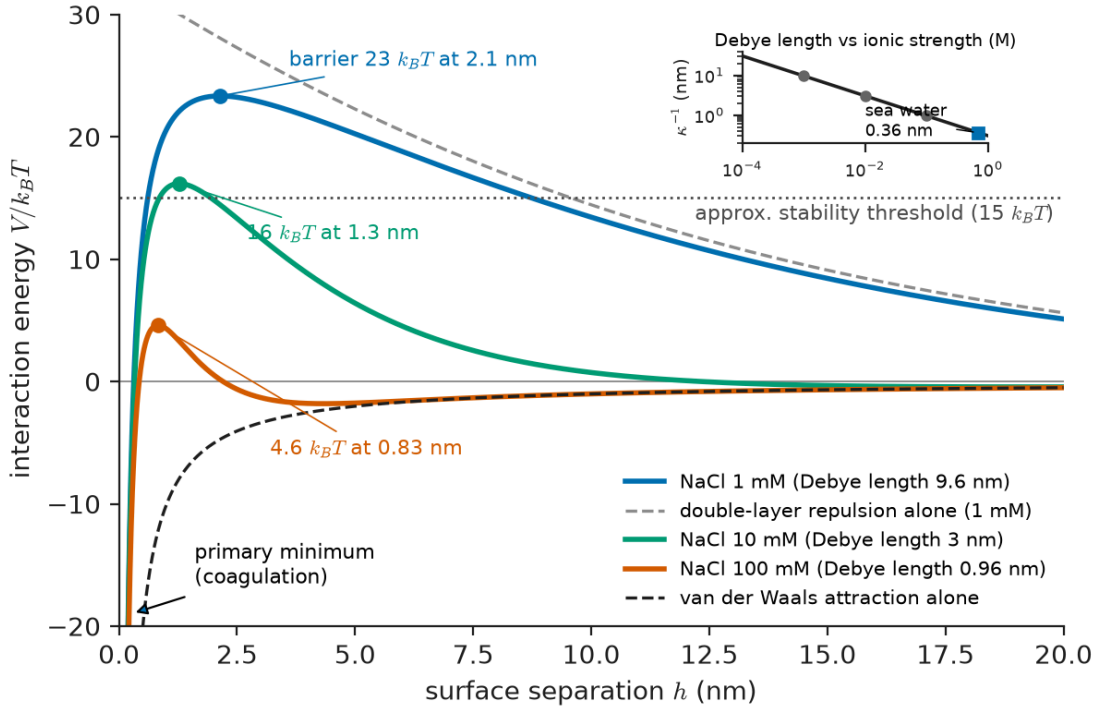


Figure 37.3 — DLVO interaction energy between two 100 nm oxide particles: adding salt shrinks the Debye length and collapses the barrier that keeps the colloid stable.

37.6.4 Beyond DLVO: engineering colloid stability

Engineers control colloids with four tools:

- **Charge (electrostatic stabilisation).** Adjusting pH away from the particle's **isoelectric point** (where zeta potential is zero: about pH 9 for alumina, 2–3 for silica, 6–7 for titania) raises $|\psi_0|$. Alumina slips for slip casting are dispersed at pH ~4 or with polyelectrolyte dispersants.
- **Steric stabilisation.** Polymer chains grafted or adsorbed on the surface (polyethylene glycol on drug nanoparticles, polycarboxylate ethers on cement grains) repel because compressing them costs conformational entropy. This works at high ionic strength, where electrostatics fail; polycarboxylate **superplasticisers** let concrete be placed at water–cement ratios of 0.3–0.4 (Chapter 18) and are the largest-tonnage colloid stabilisers in industry.
- **Depletion.** Non-adsorbing polymer or micelles in solution push large particles together (their exclusion zones overlap, raising the free volume available to the small species), which is used to cream emulsions deliberately and to assemble colloidal crystals.
- **Bridging.** Long polymer chains adsorbed on two particles at once flocculate them: polyacrylamide flocculants in mineral processing and water treatment.

Colloidal processing matters well beyond chemistry. Ceramic powder processing (Chapter 17) needs a well-dispersed, high-solids slip to give a dense, flaw-free green body; agglomerates become the strength-limiting flaws of Chapter 14. Paints, inks for printed electronics, magnetorheological fluids in automotive dampers, and lipid nanoparticles that carry mRNA vaccines are all colloids whose performance is decided by the balance in Figure 37.3.

37.7 Gels and hydrogels

37.7.1 What a gel is

A **gel** is a three-dimensional network that spans its container and holds a large volume of liquid. It is solid in the sense that it has an equilibrium shear modulus and does not flow, yet it may be more than 90 % solvent. When the solvent is water, the gel is a **hydrogel**. Gels are classified by how the network is held together:

- **Chemical gels** have covalent crosslinks: polyacrylamide for electrophoresis, poly(2-hydroxyethyl methacrylate) (pHEMA) and silicone hydrogels for contact lenses, sodium polyacrylate superabsorbents. They are permanent.
- **Physical gels** have reversible junctions: crystallites (poly(vinyl alcohol) freeze–thaw gels), helices (gelatin, agarose, carrageenan), ionic bridges (alginate crosslinked by Ca^{2+}), hydrophobic associations or entanglements. They melt or dissolve on heating, on changes in salt, or under stress, and often heal.
- **Colloidal gels** are networks of aggregated particles, such as a fumed-silica gel or a weak clay drilling mud; they are often thixotropic (they flow under shear and re-form at rest).
- **Sol–gel** inorganic gels (silica from hydrolysis of tetraethoxysilane) are precursors to glasses, aerogels and coatings (Chapter 49).

37.7.2 Swelling and elasticity

A dry, crosslinked hydrophilic network placed in water swells until two free-energy contributions balance: the **mixing** free energy, which favours more water (Flory–Huggins theory, Chapter 20, with interaction parameter χ), and the **elastic** free energy of the stretched strands, which resists (rubber elasticity, Chapter 20). For a polyelectrolyte network, the osmotic pressure of the counter-ions trapped inside adds a third, often dominant, swelling term. The **Flory–Rehner** balance predicts that the equilibrium swelling ratio rises as crosslink density falls, as χ falls (better solvent) and as fixed charge rises; and that salt in the surrounding water, by reducing the ion osmotic pressure difference, collapses a polyelectrolyte gel. That is why a sodium polyacrylate superabsorbent in a nappy absorbs about 300 times its mass of pure water but only 30–50 times its mass of urine (Figure 37.4b).

> **Status of the principle: Flory–Rehner swelling theory**
> *Approximate*. Combines mean-field mixing with affine rubber elasticity; it predicts trends and orders of magnitude, but the uniform-network assumption fails for real gels with heterogeneous crosslinking, dangling ends and loops.

The shear modulus of a swollen network follows rubber elasticity: $G \approx \nu k_B T$, where ν is the number density of elastically active strands. Because a gel is mostly solvent, its strands are long and dilute and G ranges from about 10 Pa (soft biological gels) to 1 MPa (contact lenses, cartilage substitutes). The **mesh size** ξ, the typical spacing between crosslinks, governs which molecules can diffuse through the gel; a scaling estimate is $\xi \approx (k_B T / G)^{1/3}$.

Worked example — mesh size of a hydrogel. A hydrogel used for cell encapsulation has $G = 10$ kPa at 25 °C. Then $\xi \approx (4.12 \times 10^{-21} / 1.0 \times 10^{4})^{1/3} = 7.4 \times 10^{-9}$ m, about 7 nm. Glucose (about 1 nm) and oxygen diffuse freely, small proteins such as insulin (about 3 nm) diffuse with hindrance, and antibodies (about 10–15 nm) and immune cells are excluded. A softer gel with $G = 1$ kPa has $\xi \approx 16$ nm and lets antibodies through. Tuning stiffness therefore also tunes permeability, which is a constraint in designing immunoisolating capsules (Chapter 39): one cannot make the gel tougher by crosslinking it more without also closing the mesh.

37.7.3 Responsive and tough hydrogels

Because swelling is a balance of small free-energy terms, gels can be made to respond to stimuli:

- **Temperature.** Poly(N-isopropylacrylamide) (PNIPAM) has a lower critical solution temperature near 32 °C: below it the chains are hydrated and the gel is swollen; above it hydrophobic association wins and the gel expels most of its water within seconds to minutes (volume changes of 10× are common; Figure 37.4a). This is used for cell-sheet harvesting, valves in microfluidics and on-demand drug release.
- **pH and ions.** Gels with acidic or basic groups swell when ionised: poly(acrylic acid) gels swell at high pH, chitosan gels at low pH, which allows release in a chosen part of the gut.
- **Specific molecules.** Glucose-responsive gels (with phenylboronic acid groups) for insulin delivery; DNA-crosslinked gels.

Classic hydrogels are brittle: a polyacrylamide gel has a fracture energy of about 10 J/m², because a crack breaks only the strands that cross it. **Double-network** gels (a stiff, brittle, highly crosslinked polyelectrolyte network interpenetrated by a soft, stretchable one) and hybrid ionic–covalent gels (alginate–polyacrylamide) reach 10^3–10^4 J/m², comparable to cartilage, because the first network fractures sacrificially over a large zone around the crack and dissipates energy, an analogue of the process-zone toughening of ceramics (Chapter 14). Such gels open applications in soft robotics, stretchable ionic conductors and load-bearing tissue replacements.

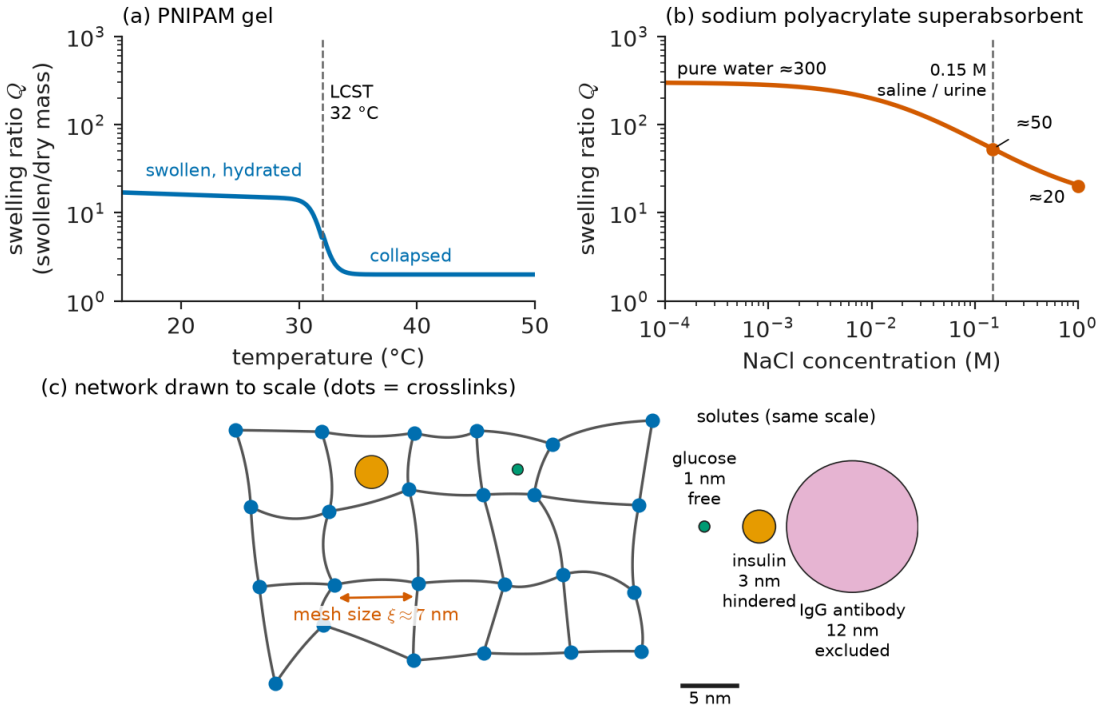


Figure 37.4 — Hydrogels swell or collapse in response to temperature, salt and crosslink density; PNIPAM collapses sharply near 32 C.

37.8 Liquid crystals

37.8.1 Phases between crystal and liquid

A crystal has long-range positional and orientational order; a liquid has neither. **Liquid crystals** (introduced in Chapter 2) are fluids with orientational order and, in some phases, partial positional order. They form from strongly anisotropic molecules or particles, either rod-like (*calamitic*: cyanobiphenyls, aramid chains) or disc-like (*discotic*: triphenylenes, graphene-oxide platelets). Two ways of reaching them are distinguished:

- **Thermotropic** liquid crystals change phase with temperature: the display mixtures, main-chain liquid-crystal polymers (Chapter 20), mesophase pitch for carbon fibres.
- **Lyotropic** liquid crystals change phase with concentration in a solvent: surfactant–water systems, cell membranes, aramid (poly-p-phenylene terephthalamide) in concentrated sulfuric acid, cellulose nanocrystals in water.

The principal phases (Figure 37.5) are:

- **Nematic (N):** molecules point on average along a direction called the **director n**, with no positional order. The fluid flows like a liquid but is optically birefringent.
- **Smectic A and C:** molecules also stack in layers; in smectic A the director is normal to the layers, in smectic C it is tilted.
- **Cholesteric (chiral nematic, N*):** in chiral molecules the director twists helically with a pitch p of 0.1–10 µm; when p is comparable to the wavelength of light, the helix selectively reflects colour, and since pitch changes with temperature, cholesterics serve as thermometers and in thermochromic inks.
- **Columnar:** discotic molecules stack into columns that pack in a 2D lattice.

37.8.2 The order parameter

How ordered a nematic is, is measured by the scalar **order parameter**

$$S = \left\langle \tfrac{1}{2}\left(3\cos^2\theta - 1\right) \right\rangle$$

where θ is the angle between a molecular axis and the director and the average is over all molecules. $S = 1$ for perfect alignment and $S = 0$ for an isotropic liquid. In real nematics S falls from about 0.6–0.8 deep in the nematic phase to about 0.3–0.45 at the

nematic–isotropic transition T_{NI}, where it drops discontinuously to zero (a weakly first-order transition). The **Maier–Saupe** mean-field theory, which treats each molecule as feeling an orienting potential proportional to S, predicts the jump at $S \approx 0.43$ and a universal curve of S versus T/T_{NI} that fits many compounds reasonably well.

> **Status of the principle: Maier–Saupe mean-field theory of nematic order**
> *Approximate.* Mean-field treatment of anisotropic attractions; it reproduces the first-order N–I transition and the shape of $S(T)$, but ignores the steric (excluded-volume) contribution that Onsager showed is decisive for lyotropic rods, and overestimates order near T_{NI}.

Macroscopic anisotropy follows from S: refractive indices along and across the director differ (birefringence $\Delta n = 0.05$–0.3), and so do the dielectric constants (dielectric anisotropy $\Delta\varepsilon = \varepsilon_{\parallel} - \varepsilon_{\perp}$, positive for cyanobiphenyls with their strong axial dipole). Elastic resistance to distorting the director field is described by three Frank constants for splay, twist and bend, K_{11}, K_{22}, K_{33}, all of order 1–20 pN.

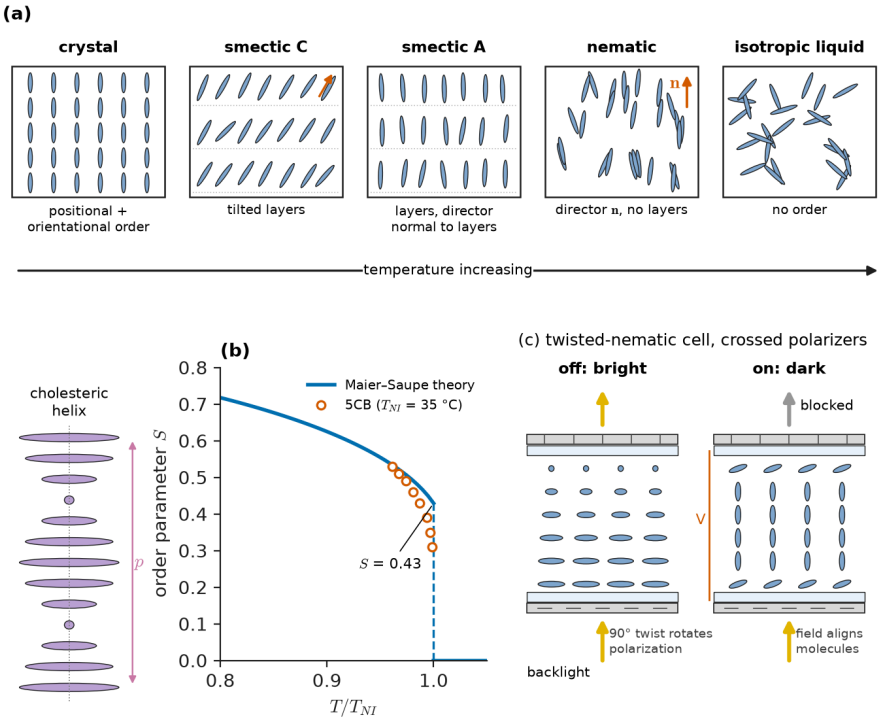


Figure 37.5 — Liquid-crystal phases are intermediate between crystal and isotropic liquid; the nematic order parameter falls with temperature and drops to zero at the clearing point.

37.8.3 Displays and the Fréedericksz transition

A liquid-crystal display uses electric fields to rotate the director, and polarizers to turn director rotation into contrast. In the classic **twisted-nematic** (TN) cell, a nematic layer about 5 µm thick sits between glass plates whose alignment layers (rubbed polyimide) are at 90° to each other; the director twists through 90° across the cell and guides the polarisation of light through crossed polarizers (bright state). A field across the cell, acting on positive $\Delta\varepsilon$, tilts the molecules towards the field; above a threshold the twist unwinds, the polarisation is no longer rotated, and the pixel goes dark. The threshold is that of the **Fréedericksz transition**, where the dielectric torque first beats the elastic restoring torque; for a pure splay geometry

$$V_{\mathrm{th}} = \pi \sqrt{\frac{K_{11}}{\varepsilon_0 \, \Delta\varepsilon}}$$

independent of cell thickness (both the field and the elastic torque scale with thickness).

Worked example — the threshold voltage of a nematic cell. For $K_{11} = 10$ pN and $\Delta\varepsilon = 10$, $V_{\mathrm{th}} = \pi\sqrt{1.0 \times 10^{-11} / (8.854 \times 10^{-12} \times 10)} = \pi \times 0.336 = 1.06$ V. This is the reason liquid-crystal displays could be driven directly by low-voltage CMOS circuits and battery power: no other electro-optic effect switched so much light for so little energy (microwatts per square centimetre), which made the LCD the dominant display technology for three decades. Faster, wider-

viewing-angle modes (in-plane switching, vertical alignment) keep the same physics with different geometries; response times of a few milliseconds are set by rotational viscosity and d^2/K.

37.8.4 Liquid crystals in structural materials

The same order produces the world's strongest organic fibres. Aramid (Kevlar, Twaron) is spun from a nematic solution in sulfuric acid: the rigid chains are already aligned in nematic domains, shear in the spinneret aligns the domains, and the fibre emerges with S near 1 and a tensile strength of about 3–3.6 GPa and modulus of 70–130 GPa (Chapter 22). Mesophase pitch carbon fibres (moduli up to 900 GPa) and thermotropic liquid-crystal polymers (Vectra, used for precision moulded connectors because their melt flows easily and shrinks very little) exploit the same idea. Liquid-crystal elastomers (Section 37.5) bring orientational order and rubber elasticity together.

37.9 Self-assembly and soft-matter design

37.9.1 Self-assembly as a minimisation problem

Self-assembly is the spontaneous organisation of components into ordered structures by interactions encoded in the components themselves, without external placement. It operates whenever the interactions are a few $k_B T$: strong enough to define a preferred arrangement, weak enough that wrongly placed units can detach and try again (error correction). Covalent bonds are too strong to self-correct at room temperature; for that reason, molecular self-assembly uses hydrophobic, hydrogen-bonding, electrostatic and van der Waals interactions. The products include micelles, vesicles and bilayer membranes from surfactants and lipids; microphase-separated domains in block copolymers (Chapter 20); colloidal crystals; DNA nanostructures; and protein assemblies from viral capsids to collagen fibrils.

37.9.2 Amphiphiles and the packing parameter

An **amphiphile** has a hydrophilic head and a hydrophobic tail: soaps, detergents, phospholipids, block copolymers in selective solvents. Above the **critical micelle concentration** (CMC: 8 mM for sodium dodecyl sulfate, SDS; micromolar or lower for lipids) the tails hide from water inside aggregates. Which aggregate forms is decided, to a good first approximation, by geometry. The **packing parameter** (Israelachvili, Mitchell and Ninham) is

$$P = \frac{v}{a_0 \, l_c}$$

where v is the volume of the hydrophobic tail, l_c its maximum extended length and a_0 the optimal head-group area at the aggregate surface (set by the balance of head-group repulsion and tail attraction). Simple geometry then gives the preferred shape (Figure 37.6):

Packing parameter P	Preferred aggregate	Example
$< 1/3$	Spherical micelles	SDS in pure water
$1/3$–$1/2$	Cylindrical (worm-like) micelles	SDS + salt; viscoelastic fracturing fluids
$1/2$–1	Flexible bilayers, vesicles	Phospholipids (two tails)
≈ 1	Planar bilayers	Lipids with small heads
> 1	Inverted micelles	Lipids in oil; lipid nanoparticle cores

For saturated hydrocarbon chains of n carbons, Tanford's estimates are $l_c \approx (0.154 + 0.1265\,n)$ nm and $v \approx (0.0274 + 0.0269\,n)$ nm³.

> **Status of the principle: Packing-parameter rule**
> *Approximate.* A geometric argument that assumes incompressible tails and a fixed head-group area; it predicts aggregate shape and its shifts with salt, temperature and chain number reliably, but not aggregate size distributions or the effects of specific head-group chemistry.

Worked example — why salt turns soap into "worms" and why lipids make membranes. For SDS ($n = 12$), $l_c = 0.154 + 0.1265 \times 12 = 1.67$ nm and $v = 0.0274 + 0.0269 \times 12 = 0.350$ nm³. In pure water the charged sulfate heads repel strongly and $a_0 \approx 0.7$ nm², giving $P = 0.350/(0.7 \times 1.67) = 0.30$: spherical micelles. Adding salt screens the head-group repulsion (shorter Debye length, Section 37.6) and a_0 falls to about 0.5 nm², so $P = 0.350/(0.5 \times 1.67) = 0.42$: the micelles grow into long cylinders that entangle like polymer chains, and the solution becomes viscoelastic. This is exactly how surfactant-based fracturing fluids and many shampoos are thickened. A phospholipid with two C16 tails has $v = 2 \times (0.0274 + 0.0269 \times 16) = 0.916$ nm³ and $l_c = 2.18$ nm; with $a_0 \approx 0.65$ nm², $P = 0.916/(0.65 \times 2.18) = 0.65$, so it forms bilayers and vesicles, which is why cell membranes, liposomes and the lipid shell of mRNA vaccine nanoparticles assemble themselves.

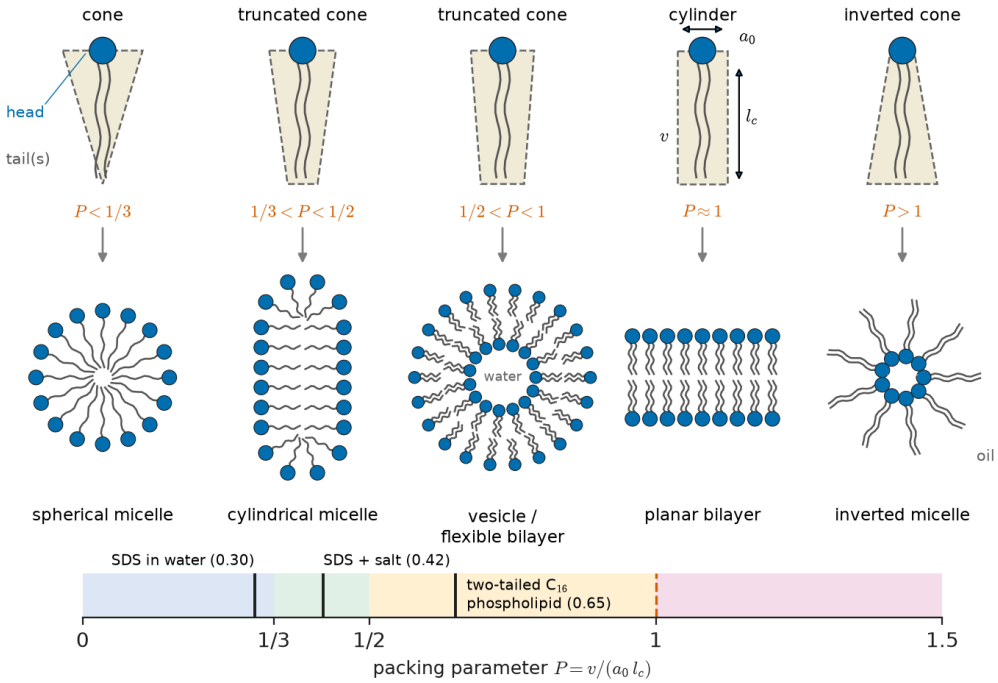


Figure 37.6 — The packing parameter of an amphiphile predicts whether it forms spherical micelles, cylinders, bilayers or inverted structures.

37.9.3 Designing with self-assembly

Engineers now use self-assembly deliberately:

- **Block-copolymer lithography.** A polystyrene-*b*-poly(methyl methacrylate) film microphase-separates into lamellae or cylinders with a period of 10–50 nm set by chain length (Chapter 20). Guided by a coarse lithographic pre-pattern (*directed self-assembly*), it multiplies pattern density for semiconductor manufacturing and hard-disk media (Chapter 49).
- **Colloidal crystals.** Monodisperse silica or polymer spheres settle or evaporate into FCC arrays whose period, comparable to the wavelength of light, produces structural colour (natural opals) and photonic band gaps (Chapter 36); infiltrating and removing them gives inverse opals for catalysts and battery electrodes.
- **DNA nanotechnology.** Because base pairing is programmable, DNA origami folds a long strand into almost any 2D or 3D shape with 2–5 nm precision, used as templates to place nanoparticles and as drug carriers.
- **Mesoporous materials.** Surfactant micelles as templates around which silica condenses (MCM-41, SBA-15) give ordered pores 2–30 nm wide with surface areas around 1000 m²/g, for catalysis, separation and drug loading.
- **Biological and bio-inspired assembly.** Collagen fibrils assemble from triple helices and mineralise into bone (Chapter 39); nacre's brick-and-mortar layering of aragonite platelets and protein gives toughness thousands of times that of aragonite alone and inspires ceramic laminates.

The common rules of soft-matter design follow from the k_BT scale: interactions must be weak enough for annealing out errors, assembly is sensitive to temperature, solvent and salt, defects are thermodynamically present and must be managed (annealing block-copolymer films, slow evaporation for colloidal crystals), and the structures are fragile unless they are fixed afterwards by crosslinking, mineralisation or conversion into an inorganic replica. The limits of self-assembly are mainly those of long-range order and defect density, which still prevent it from replacing lithography outright.

What remains open

Open question	Where it is answered
How does the martensite crystallography and habit plane follow from the lattice change?	Chapter 10 (martensitic transformations)
What makes a crystal piezoelectric or ferroelectric, and how are ferroelectric domains switched?	Chapter 27 (ferroelectric and piezoelectric materials)
Why are rare-earth magnetostrictive and permanent-magnet alloys supply-critical?	Chapters 29 and 59
How do nanoparticles and 2D sheets change properties purely through size?	Chapter 38
How do hydrogels, NiTi and other responsive materials behave inside the body, and how are they qualified?	Chapter 39
How do ion intercalation and conducting polymers store energy as well as actuate?	Chapter 40
How do Flory–Huggins mixing and rubber elasticity derive from chain statistics?	Chapter 20
How is the fatigue of superelastic and piezoceramic devices predicted and analysed after failure?	Chapters 14 and 44
How are surfaces, adsorption and wetting, which control colloids, characterised?	Chapter 45
How are soft materials imaged without destroying their water-filled structure?	Chapter 53 (cryo-EM, SAXS)
How can molecular simulation predict self-assembled structures?	Chapter 56

Next: Chapter 38 shrinks materials to the nanoscale and asks which properties change when surfaces, confinement and grain boundaries take over from the bulk.

Chapter 38 — Nanomaterials and Two-Dimensional Materials

Part VIII · Advanced and Emerging Materials

A car tyre made in 1910 was white, wore out in a few thousand kilometres and cracked in sunlight. Within two decades every tyre in the world was black, because manufacturers had learned that adding about 30 % by mass of carbon black, a soot of fused carbon particles 20–80 nm across, raised the tread's abrasion resistance several-fold and its tear strength and UV resistance with it. Carbon black is still the largest-tonnage nanomaterial made, at over 10 million tonnes a year, and nobody called it "nano" for most of its history. Today the same length scale appears everywhere: in the 1–2 nm platinum particles that carry the catalysis in a fuel cell and a catalytic converter, the 20 nm copper lines whose resistivity is three times the handbook value, the cadmium-free quantum dots whose diameter sets the colour of a television pixel, the electrodeposited nickel with 15 nm grains that is five times harder than ordinary nickel, and the few-layer graphene platelets now blended into concrete, coatings and battery electrodes. In each case a property has stopped being a property of the substance alone and has become a property of its **size**. This chapter asks when that happens, why, and what engineering can and cannot do with it.

> **At what size does a material stop behaving like its bulk self, and which properties change first? What do nanoparticles, nanotubes, graphene and nanocrystalline metals offer engineers, how are they made and measured, and why are they mostly used as additives rather than as stand-alone materials?**

Concept	Engineering destination
Surface-to-volume ratio, specific surface area	Catalysts, sorbents, battery electrodes, sintering of nanopowders and printed inks
Melting-point depression (Gibbs–Thomson)	Low-temperature sintering of silver and copper pastes, nanoparticle stability in catalysts
Mean-free-path size effects	Interconnect resistivity, thermal conductivity of nanostructured thermoelectrics
Quantum confinement, Brus equation	Quantum-dot displays and lighting, bio-imaging labels, photodetectors
Carbon nanotubes, chirality	Conductive additives in Li-ion electrodes, composites, field emitters
Graphene and other 2D materials	Barrier coatings, heat spreaders, transparent electrodes, membranes, ultrathin transistors
Nanocrystalline metals, inverse Hall–Petch	Wear-resistant electrodeposited coatings, high-strength conductors, MEMS
Nanocomposites, percolation, tortuosity	Antistatic plastics, gas-barrier packaging, low-rolling-resistance tyres
Top-down and bottom-up synthesis	Choosing a route that delivers the size, purity and tonnage needed
Nano-characterization and safety	Quality control, occupational exposure limits, regulation of nanoforms

Roadmap. Why size changes properties: surfaces, melting and transport (38.1) → confinement of electrons (38.2) → zero- and one-dimensional objects: nanoparticles, quantum dots, nanowires and nanotubes (38.3) → two-dimensional materials (38.4) → nanostructured bulk: nanocrystalline metals (38.5) and nanocomposites (38.6) → how nanomaterials are made (38.7) → how they are measured and handled safely (38.8).

> **Boundary.**
> Quantum confinement is treated here with the particle-in-a-box and effective-mass models of Chapter 31, as a materials phenomenon; rigorous quantum mechanics belongs to Engineering Physics. The electronic-device side of nanomaterials (nanosheet and 2D-channel transistors, interconnect scaling, quantum-dot lasers) is treated in more depth in Chapters 25, 33 and 35.

38.1 Size effects in classical materials science

38.1.1 Two ways size enters

Size changes material behaviour through two different mechanisms, which it helps to keep apart:

1. **Classical size effects.** The fraction of atoms at a surface or interface becomes significant, or the object becomes smaller than a characteristic length of some process: the mean free path of electrons or phonons, the spacing of dislocation sources, a magnetic domain-wall width, the critical nucleus size. No quantum mechanics beyond that already in the bulk material is needed.
2. **Quantum size effects.** The object becomes comparable to the de Broglie wavelength of its electrons (or the radius of its excitons), and their allowed energies become discrete. This is quantum confinement (Section 38.2).

Classical effects typically appear below about 100 nm. Quantum effects appear in semiconductors below about 10–20 nm, but in metals only below about 2 nm, because electrons at the Fermi level of a metal have wavelengths of about 0.5 nm. A **nanomaterial** is conventionally one with at least one external dimension, or an internal structure, between 1 and 100 nm; the European Union's regulatory definition counts a material as nano if at least half of its particles by number have one or more external dimensions in that range. The boundaries are conventions; what matters physically is which characteristic length the structure has passed below.

38.1.2 Surface-to-volume ratio

For a sphere of diameter d the surface-to-volume ratio is $A/V = 6/d$. Dividing by density gives the **specific surface area** (SSA), the surface per unit mass, which is the number that matters for catalysts, adsorbents and electrodes:

$$\mathrm{SSA} = \frac{6}{\rho\, d}$$

> **Status of the principle: Surface-to-volume scaling, $A/V = 6/d$**
> *Derived.* Pure geometry for spheres (and, with a different constant, for any fixed shape); real powders deviate because of polydispersity, agglomeration, necks and internal porosity, which is why measured gas-adsorption areas are usually lower than the geometric estimate.

Worked example — what fraction of atoms is at the surface? Treat the outer shell of thickness $t \approx 0.25$ nm (about one atomic layer) as surface. The fraction of the volume in that shell is

$$f_s = 1 - \left(1 - \frac{2t}{d}\right)^3$$

which gives 1.5 % for $d = 100$ nm, 7.3 % at 20 nm, 14 % at 10 nm, 27 % at 5 nm and 58 % at 2 nm. The specific surface area of 5 nm gold particles ($\rho = 19\,300$ kg/m³) is $6/(19\,300 \times 5 \times 10^{-9}) = 6.2 \times 10^4$ m²/kg, that is 62 m²/g, and of 21 nm titania ($\rho = 4230$ kg/m³) about 68 m²/g; the commercial titania photocatalyst made of particles that size measures about 50 m²/g by gas adsorption, because the particles are fused into aggregates. A single gram of a 2 nm platinum catalyst exposes more than 100 m², and since only surface atoms do catalysis, this is why fuel-cell and automotive catalysts are dispersed as 1–5 nm particles on porous carbon or alumina supports, and why their slow coarsening in service (Ostwald ripening, Chapter 10) is a principal cause of catalyst ageing.

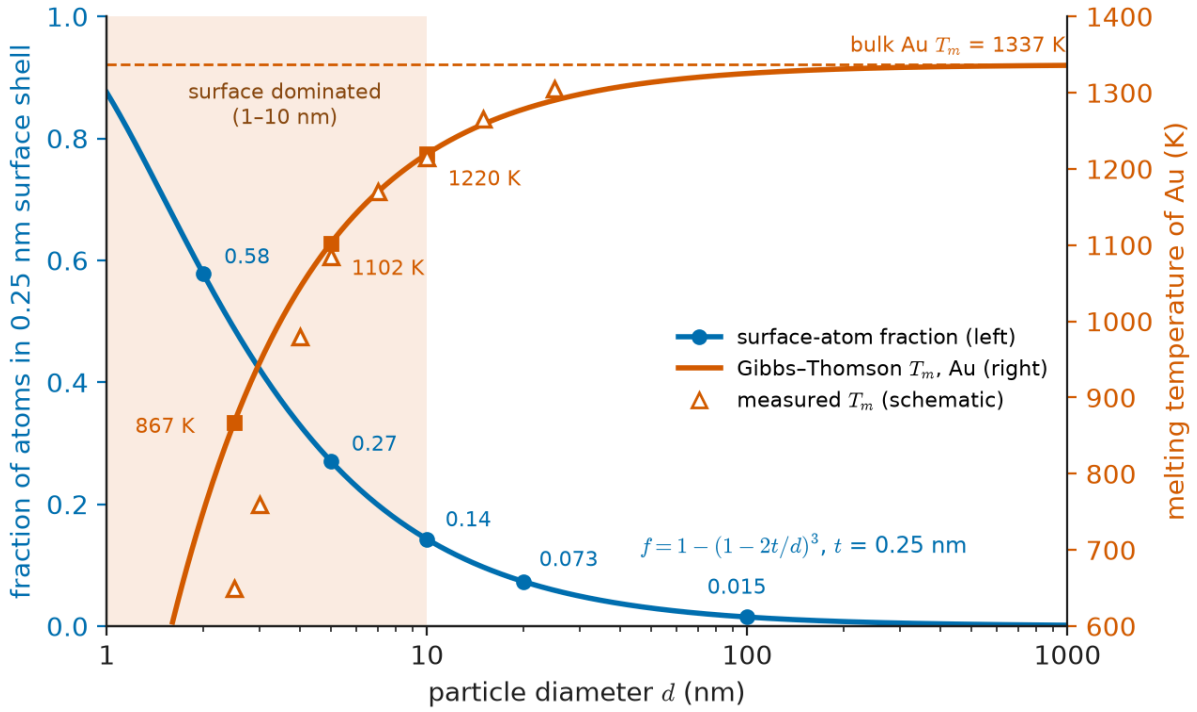


Figure 38.1 — As particles shrink below about 20 nm, the fraction of surface atoms rises steeply and the melting point falls below its bulk value.

38.1.3 Melting-point depression

Atoms at a surface have fewer neighbours, so a small particle has a higher Gibbs energy per atom than bulk solid. The same is true of a small liquid droplet, but the solid–liquid interface energy σ_{sl} is what matters for melting: a solid sphere of radius r inside its own melt is in equilibrium at a lower temperature than a flat interface. Applying the Gibbs–Thomson relation of Chapter 6:

$$\frac{T_m(r)}{T_{m,\text{bulk}}} \approx 1 - \frac{2\sigma_{sl}}{\Delta H_v \, r}$$

where ΔH_v is the latent heat of fusion per unit volume.

> **Status of the principle: Gibbs–Thomson melting-point depression**
> *Approximate.* Continuum thermodynamics with a size-independent interface energy; it captures the $1/r$ trend down to a few nanometres, but below about 3 nm interface energies, surface pre-melting and structural changes (icosahedral clusters) make the measured depression larger.

Worked example — gold nanoparticles. For gold, $T_{m,\text{bulk}} = 1337$ K, the latent heat is 12.55 kJ/mol and the molar volume 10.2×10^{-6} m³/mol, so $\Delta H_v = 1.23 \times 10^9$ J/m³; take $\sigma_{sl} \approx 0.27$ J/m². For a 5 nm particle ($r = 2.5$ nm), $2\sigma_{sl}/(\Delta H_v r) = 0.54/(1.23 \times 10^9 \times 2.5 \times 10^{-9}) = 0.176$, so $T_m \approx 1337 \times 0.824 = 1102$ K, 235 K below bulk. At 10 nm the depression is about 120 K (1220 K), and at 2.5 nm the formula gives 867 K, while experiments find even lower values, near 600–700 K, as the status box warns (Figure 38.1).

The same physics lowers the **sintering temperature** of nanopowders. Silver nanoparticle pastes sinter into conductive joints at 200–250 °C, far below silver's melting point of 962 °C, which is how power-electronic die attach and printed conductive tracks on polymer films are made; nanocrystalline ceramic powders densify several hundred kelvin lower than micrometre powders. The driving force that makes nanopowders useful also makes them unstable: they agglomerate, coarsen and, for reactive metals, oxidise or even ignite spontaneously, so nano-aluminium and nano-iron are handled under inert gas.

38.1.4 Mean-free-path and characteristic-length effects

When a structure is smaller than the distance a carrier travels between scattering events, the boundaries add scattering:

- **Electrons in metals.** The electron mean free path in bulk copper at room temperature is about 39 nm. In interconnect lines 20 nm wide, surface and grain-boundary scattering raise the resistivity from 1.7 to about 4–6 μΩ cm (Chapter 25), which is driving the search for alternative interconnect metals with shorter mean free paths, such as ruthenium and molybdenum.
- **Phonons in insulators and semiconductors.** Phonon mean free paths are spread from nanometres to micrometres; nanostructuring (grain boundaries, nanoprecipitates) scatters the long-wavelength phonons that carry much of the heat, cutting thermal conductivity while leaving electrical conductivity less affected, which is the basis of modern nanostructured thermoelectrics (Chapter 40). Silicon nanowires 50 nm across conduct heat about five times worse than bulk silicon (Chapter 24).
- **Dislocations in metals.** Below about 100 nm, grains are too small to contain dislocation sources and pile-ups, which changes the strengthening law (Section 38.5).
- **Magnetic domains.** Below a critical size of tens of nanometres a magnetic particle is a single domain, and below about 10 nm thermal energy can flip its magnetization (superparamagnetism, Chapter 28), the limit for magnetic recording grains and the basis of magnetic nanoparticle contrast agents and ferrofluids.
- **Light.** Particles much smaller than the wavelength scatter weakly (Rayleigh scattering varies as d^6/λ^4), so 20–50 nm titania and zinc oxide in sunscreens absorb UV yet look transparent, whereas pigmentary 250 nm titania is a white scatterer. Metal nanoparticles show **surface plasmon resonances**: collective oscillation of their conduction electrons gives 20 nm gold particles a strong absorption near 520 nm and a ruby colour, used since Roman times in glass and today in lateral-flow (pregnancy and rapid antigen) test strips.

38.2 Quantum confinement as a materials phenomenon

38.2.1 When does confinement matter?

An electron in a semiconductor crystal behaves as a free particle with an **effective mass** m^* (Chapter 31). Confining it to a box of size L raises its lowest energy by roughly the particle-in-a-box value

$$\Delta E \approx \frac{\hbar^2 \pi^2}{2 m^* L^2}$$

This matters when ΔE is comparable to or larger than $k_B T$ (25.7 meV at 25 °C) and to the energy scale of interest. Small effective mass and small L give large confinement. For GaAs ($m^* = 0.067\, m_e$) a 10 nm layer raises the electron energy by about 56 meV, which is why quantum wells of that thickness define the emission wavelength of semiconductor lasers. A second length scale is the **exciton Bohr radius**, the size of the bound electron–hole pair created by light: about 5.6 nm in CdSe, 10 nm in InP, 18 nm in PbS. When a crystal is smaller than this, the exciton is squeezed and its energy rises.

38.2.2 Dimensionality and the density of states

Confinement in one, two or three directions gives quantum wells (2D: free motion in a plane), quantum wires (1D) and quantum dots (0D). The electronic density of states changes shape accordingly: the square-root continuum of a bulk crystal becomes a staircase in a well, a series of $1/\sqrt{E}$ spikes in a wire and a set of discrete, atom-like levels in a dot (Figure 38.2a). Concentrating states at particular energies is what makes quantum-well lasers more efficient and quantum dots emit narrow spectral lines.

38.2.3 The Brus equation for quantum dots

For a spherical dot of radius R, the lowest excitation energy in the effective-mass approximation (Brus, 1984) is

$$E(R) \approx E_g + \frac{\hbar^2 \pi^2}{2R^2}\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right) - \frac{1.786\, e^2}{4\pi\varepsilon_0 \varepsilon_r R}$$

The second term is the confinement of electron and hole and grows as $1/R^2$; the third is their Coulomb attraction, screened by the dot's permittivity, which grows only as $1/R$. At small R confinement wins, and the dot's absorption and emission shift to the blue as it shrinks.

Status of the principle: Brus equation
Approximate. Effective-mass, infinite-barrier model with parabolic bands; it gives the correct trend and order of magnitude, but overestimates the shift below about 3 nm, where band non-parabolicity, finite barriers and surface states matter. Engineering design uses measured sizing curves.

Worked example — the colour of a CdSe quantum dot. For CdSe take $E_g = 1.74$ eV, $m_e^* = 0.13\,m_e$, $m_h^* = 0.45\,m_e$, $\varepsilon_r = 10.6$, and the constants $\hbar^2\pi^2/2m_e = 0.376$ eV nm² and $e^2/4\pi\varepsilon_0 = 1.440$ eV nm. For a dot of diameter 5 nm ($R = 2.5$ nm):

$$\Delta E_{\text{conf}} = \frac{0.376}{2.5^2}\left(\frac{1}{0.13} + \frac{1}{0.45}\right) = 0.0602 \times 9.91 = 0.596 \text{ eV}$$

$$\Delta E_{\text{Coul}} = \frac{1.786 \times 1.440}{10.6 \times 2.5} = 0.097 \text{ eV}$$

so $E \approx 1.74 + 0.596 - 0.097 = 2.24$ eV, a wavelength of $1239.8/2.24 = 554$ nm (green). The measured first-exciton energy of 5 nm CdSe dots is about 2.05 eV (605 nm, orange): the model overestimates the shift, as expected. At 7 nm the formula gives 1.97 eV (628 nm), at 4 nm 2.55 eV (486 nm). One composition therefore spans the visible simply by size, and a synthesis that controls diameter to ±5 % sets the colour to within about ±10 nm.

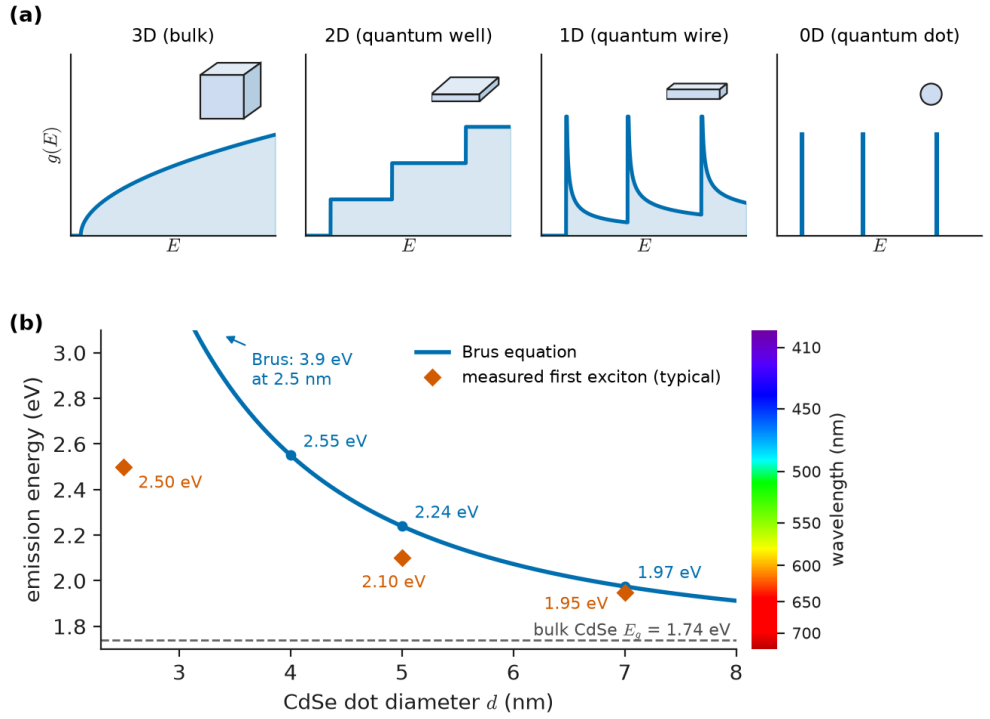


Figure 38.2 — Confinement reshapes the density of states from a continuum in bulk to steps, spikes and discrete levels, and tunes the emission of quantum dots with their size.

38.3 Nanoparticles, quantum dots, nanowires and nanotubes

38.3.1 Nanoparticles

Zero-dimensional nanomaterials are by far the most used, mostly as additives:

Nanoparticle	Typical size	Use and the property exploited
Carbon black	20–80 nm primary, fused aggregates	Tyre reinforcement, pigment, conductive filler
Fumed (pyrogenic) silica	7–40 nm primary	Thickener, anti-settling agent, tyre filler, insulation
Titania, zinc oxide	20–50 nm	Transparent UV filters, photocatalytic self-cleaning glass
Pt, Pd, Rh on supports	1–5 nm	Fuel-cell, automotive and chemical catalysts
Silver	10–100 nm	Antimicrobial coatings, sintered die attach, printed conductors
Ceria	5–50 nm	Chemical–mechanical polishing slurries, fuel additives, catalysts
Iron oxide	5–20 nm	MRI contrast, ferrofluids, magnetic separation
Gold	2–50 nm	Diagnostics (plasmonic colour), catalysis

Colloidal nanoparticles are stabilised against aggregation by the charge, steric and ligand mechanisms of Chapter 37; gas-phase products such as carbon black and fumed silica are deliberately aggregated into branched, chain-like clusters, which is what lets them build networks in rubber and liquids.

38.3.2 Quantum dots

Commercial quantum dots are core–shell crystals made by colloidal synthesis (Section 38.7): an emitting core (CdSe, or cadmium-free InP or perovskite $CsPbBr_3$) coated with a few monolayers of a wider-gap shell (ZnS, ZnSe) that passivates surface dangling bonds, which would otherwise trap carriers and quench emission, and capped with organic ligands for solubility. Photoluminescence quantum yields above 90 % and emission linewidths of 20–35 nm make them ideal colour converters: in quantum-dot displays a blue LED backlight excites red and green dots in a film, giving a wider colour gamut than phosphors. They are also used as fluorescent labels in biology, where their brightness and photostability beat organic dyes, and are studied for solar cells and infrared detectors (PbS dots). Cadmium toxicity has pushed displays towards InP under the EU's hazardous-substance rules.

38.3.3 Nanowires

Nanowires have one long free dimension and diameters of 5–200 nm. Semiconductor nanowires (Si, Ge, GaN, ZnO, InP) are grown mostly by the **vapour–liquid–solid** (VLS) mechanism: a gold or other catalyst droplet dissolves vapour-phase precursor, becomes supersaturated and precipitates a single-crystal wire beneath it, so wire diameter follows droplet diameter. Their uses exploit geometry: a wire relaxes lattice-mismatch strain sideways, so heterostructures impossible in films can be grown (nanowire LEDs and solar cells); silicon nanowires and nanostructured silicon accommodate the 300 % volume expansion of lithiation without pulverising, which motivates silicon-rich battery anodes (Chapter 40); and the enormous surface sensitivity of a thin wire makes it a chemical or biological sensor. Metal nanowires (silver, copper) of high aspect ratio form percolating transparent conducting films, a flexible alternative to indium tin oxide.

38.3.4 Carbon nanotubes

A single-walled carbon nanotube (SWCNT) is a graphene sheet (Section 38.4) rolled into a seamless cylinder about 1 nm in diameter and up to millimetres long; multi-walled tubes (MWCNTs) are several such cylinders nested with the 0.34 nm graphite interlayer spacing, outer diameters 5–50 nm. How the sheet is rolled is given by the **chiral vector** $\mathbf{C}_h = n\mathbf{a}_1 + m\mathbf{a}_2$ on the graphene lattice (Figure 38.3), which becomes the tube's circumference. The diameter follows directly:

$$d = \frac{a\sqrt{n^2 + nm + m^2}}{\pi}, \qquad a = 0.246\ \text{nm}$$

Tubes with $m = 0$ are **zigzag**, $n = m$ are **armchair**, all others are **chiral**. Rolling quantises the electron wavevector around the circumference, and whether the allowed wavevectors pass through graphene's Dirac points (Section 38.4) decides the electrical character: a tube is metallic if $(n - m)$ is divisible by 3 and semiconducting otherwise, with a band gap $E_g \approx 0.8\ \text{eV}/d(\text{nm})$.

> **Status of the principle: The $(n - m)$ mod 3 rule**
> *Approximate.* Derived from zone-folding of graphene's band structure; curvature opens small gaps (meV to tens of meV) in non-armchair "metallic" tubes, and only armchair tubes are truly metallic.

Worked example — three tubes. A (10,10) armchair tube has $d = 0.246\sqrt{300}/\pi = 1.36$ nm and $n - m = 0$: metallic. A (10,0) zigzag tube has $d = 0.783$ nm and $n - m = 10$, not divisible by 3: semiconducting with $E_g \approx 0.8/0.78 \approx 1.0$ eV. A (13,0) tube has $d = 1.02$ nm and is semiconducting with $E_g \approx 0.8$ eV. Since synthesis produces a statistical mixture, about one third of tubes are metallic, which is fatal for transistors (separation by density-gradient centrifugation or selective polymer wrapping is needed) but irrelevant for the main markets.

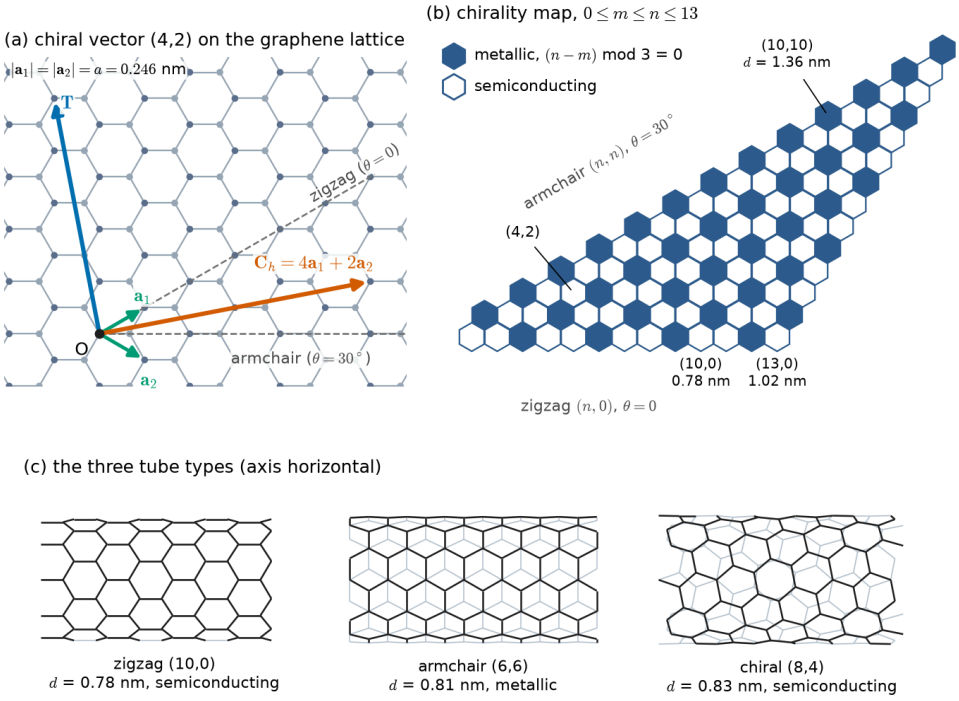


Figure 38.3 — The chiral vector (n,m) on the graphene lattice fixes a nanotube's diameter, chiral angle and whether it is metallic or semiconducting.

Individual nanotubes have extraordinary properties: Young's modulus about 1 TPa, tensile strength of tens of GPa measured on defect-free tubes, axial thermal conductivity of about 3000 W m⁻¹ K⁻¹, current densities above 10^9 A/cm² (a thousand times copper's electromigration limit). Bulk products capture only a fraction of this, because load and current must pass between tubes through weak van der Waals contacts: CNT yarns and sheets reach 1–5 GPa strength and 10^6 S/m conductivity at best. The largest market is therefore as a **conductive additive**: about 0.5–2 wt% of MWCNTs or SWCNTs in lithium-ion cathodes and silicon-containing anodes replaces several times more carbon black, freeing volume for active material; smaller markets are antistatic plastics (fuel lines, electronics trays), conductive coatings and sporting goods. Boron nitride nanotubes, insulating and stable to about 900 °C in air, and natural halloysite clay nanotubes are related one-dimensional fillers.

38.4 Graphene and other two-dimensional materials

38.4.1 Graphene

Graphene is a single layer of sp²-bonded carbon atoms in a honeycomb lattice, with C–C bond length 0.142 nm and lattice constant 0.246 nm; stacking layers at 0.335 nm gives graphite. It was isolated in 2004 by peeling graphite with adhesive tape and identifying monolayers optically on oxidised silicon, the work recognised by the 2010 Nobel Prize in Physics.

Its electronic structure is unusual. The conduction and valence bands touch at the six corners (K points) of the hexagonal Brillouin zone, and near them the energy is linear in wavevector, $E = \pm\hbar v_F |\mathbf{k}|$ with $v_F \approx 1 \times 10^6$ m/s: a **Dirac cone**, so electrons behave as massless particles. Graphene is a zero-gap semimetal whose carrier type and density are set by a gate field or by chemical doping. That gives very high mobility (above 10^5 cm² V⁻¹ s⁻¹ in clean, h-BN-encapsulated samples, 10^3–10^4 on SiO₂), but no off-state for digital logic, which is why graphene has not replaced silicon in transistors.

Its other properties (Table 38.1) explain where it *is* used. One atom thick and with the in-plane covalent bond of graphite, it has a Young's modulus of about 1 TPa and an intrinsic strength of about 130 GPa (measured by nanoindentation of suspended

membranes), the strongest material known. It absorbs 2.3 % of visible light per layer (the value $\pi\alpha$, with α the fine-structure constant), so a monolayer is nearly transparent yet conductive. It is impermeable to gases, even helium, in perfect form. Its theoretical specific surface area, with both faces counted, is

$$\mathrm{SSA} = \frac{2 \times A_{\mathrm{cell}}}{2 m_{\mathrm{C}}} = \frac{2 \times 0.0524\ \mathrm{nm}^2}{2 \times 1.994 \times 10^{-23}\ \mathrm{g}} \approx 2630\ \mathrm{m}^2/\mathrm{g}$$

(the unit cell of area $\sqrt{3}a^2/2 = 0.0524$ nm² contains two carbon atoms), which makes it a candidate for supercapacitor electrodes (Chapter 40), though restacking of sheets loses most of that area.

Table 38.1 — Properties of graphene and benchmarks (typical; monolayer values are for high-quality samples)

Property	Graphene	Benchmark
Young's modulus	~1.0 TPa	Steel 200 GPa; CFRP ply 130–180 GPa
Intrinsic tensile strength	~130 GPa	High-strength steel ~2 GPa; carbon fibre 3–7 GPa
Thermal conductivity (suspended, room T)	2000–5000 W m⁻¹ K⁻¹	Copper 400; diamond 2000
Carrier mobility (room T)	10^4–10^5 cm² V⁻¹ s⁻¹	Silicon electrons ~1400
Optical absorption per layer	2.3 %	ITO film ~10 % at similar sheet resistance
Sheet resistance (CVD, doped)	100–500 Ω/sq	ITO 10–100 Ω/sq
Specific surface area	~2630 m²/g	Activated carbon 1000–3000 m²/g

In practice "graphene" sold by the tonne is something else: **few-layer graphene** and **graphene nanoplatelets** (5–100 layers) made by liquid-phase exfoliation of graphite, and **graphene oxide** (GO), a heavily oxidised, insulating, water-dispersible sheet made by chemical oxidation, partly reconverted to conductive **reduced graphene oxide** (rGO). These platelets are used as additives: in anticorrosion and barrier coatings, heat-spreading films in smartphones (actually graphitised polyimide or graphene-laminate films with in-plane conductivities of 1000–1900 W m⁻¹ K⁻¹), conductive inks, battery electrodes and as reinforcement in concrete and polymers. Continuous monolayer films are grown by chemical vapour deposition on copper foil (Section 38.7) and transferred, for sensors, transparent electrodes and research devices; transfer damage, wrinkles and grain boundaries make large-area quality the persistent problem.

38.4.2 Beyond graphene

Graphene opened a family of layered crystals that can be thinned to one or a few layers (Figure 38.4):

- **Hexagonal boron nitride (h-BN)**, "white graphene", is isostructural with graphene but has alternating B and N, a band gap near 6 eV, and an atomically flat, charge-trap-free surface. It is the ideal insulating substrate and encapsulant for other 2D materials, and bulk h-BN powder is a high-temperature lubricant and thermally conductive, electrically insulating filler.
- **Transition-metal dichalcogenides (TMDs)**, MX_2 with M = Mo, W and X = S, Se, Te: a metal layer sandwiched between two chalcogen layers. MoS_2 has an indirect gap of about 1.2–1.3 eV in bulk that becomes a direct gap of about 1.8–1.9 eV in a monolayer, so monolayers luminesce strongly; WSe_2 and $MoTe_2$ give other gaps. They are the leading candidates for transistor channels thinner than silicon can be made, and bulk MoS_2 has long been a solid lubricant for vacuum and aerospace mechanisms.
- **Phosphorene** (black phosphorus) has a puckered structure, a direct gap that rises from about 0.3 eV in bulk to about 1.5–2 eV in a monolayer, and strong in-plane anisotropy; it degrades rapidly in air unless encapsulated.
- **MXenes**, such as $Ti_3C_2T_x$, are made by etching the A-layer out of MAX-phase ceramics (Ti_3AlC_2), leaving metallic carbide or nitride sheets terminated by –O, –OH and –F groups (T_x). They combine metallic conductivity (up to about 10^4 S/cm) with hydrophilicity, so they can be processed from water into films for electromagnetic-interference shielding, supercapacitor electrodes and conductive inks.
- **Van der Waals heterostructures** stack different 2D crystals (graphene, h-BN, TMDs) like Lego; because layers bond by van der Waals forces, lattice matching is unnecessary. Twisting two graphene layers to the "magic angle" of about 1.1° even produces superconductivity, an example of designing properties by stacking geometry alone.

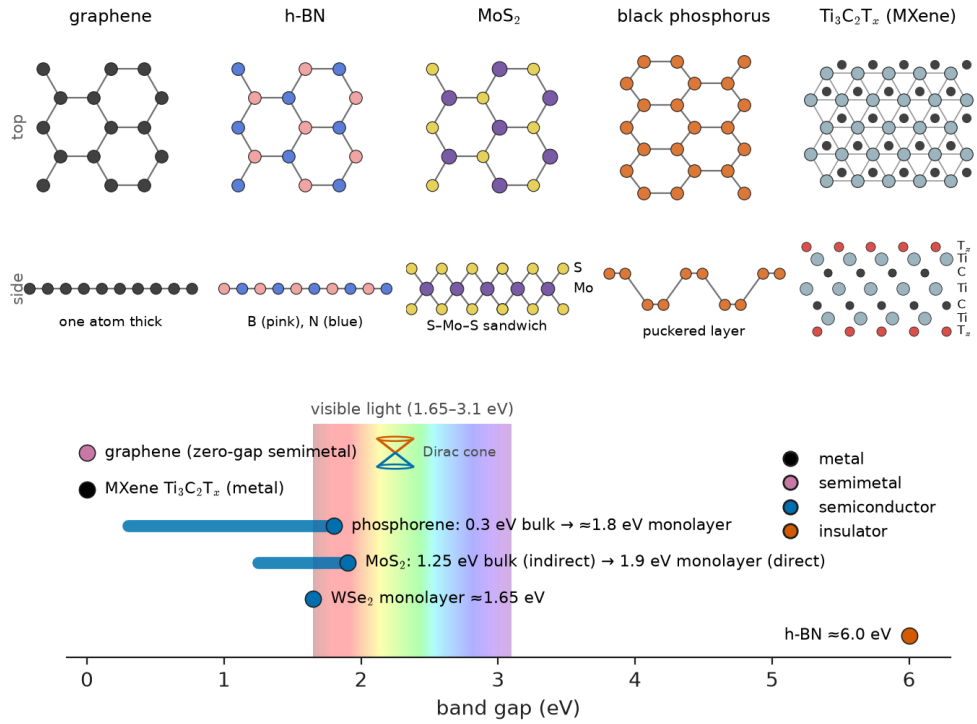


Figure 38.4 — The two-dimensional materials family spans metals, semimetals, semiconductors and insulators, with band gaps from zero to about 6 eV.

38.5 Nanocrystalline metals

38.5.1 Extrapolating Hall–Petch

Grain boundaries obstruct dislocations, and Chapter 13 derived the Hall–Petch relation from dislocation pile-ups:

$$\sigma_y = \sigma_0 + k_y\, d^{-1/2}$$

Since strength rises without limit as $d \to 0$ in this formula, it is natural to ask how strong a metal with nanometre grains can be. **Nanocrystalline** metals, with grain sizes below 100 nm (and *ultrafine-grained* metals at 100–1000 nm), test the question.

Worked example — how strong could nanocrystalline copper be? Take $\sigma_0 = 25$ MPa and $k_y = 0.11$ MPa m^1/2 for copper. Hall–Petch predicts 60 MPa at $d = 10$ µm, 135 MPa at 1 µm, $25 + 0.11/\sqrt{10^{-7}} = 373$ MPa at 100 nm, 803 MPa at 20 nm and 1125 MPa at 10 nm. Measured yield strengths of fully dense nanocrystalline copper follow the line down to roughly 20–30 nm, reaching about 0.8–1 GPa, five to ten times annealed copper, but below about 10–15 nm strength stops rising and then falls (Figure 38.5). Extrapolation to 5 nm (1.58 GPa) would be badly wrong.

> **Status of the principle: Hall–Petch relation at the nanoscale**
> *Empirical.* Valid from coarse grains down to a few tens of nanometres; it fails below about 10–20 nm, where grains are too small to support pile-ups and grain-boundary-mediated deformation takes over.

38.5.2 The inverse Hall–Petch regime

The breakdown has a physical reason. A pile-up needs room: a grain of 10 nm can hold only one or two dislocations, and the Frank–Read sources of Chapter 12 need a pinned segment that cannot fit. Instead, dislocations are emitted from grain boundaries, cross the grain and are absorbed at the opposite boundary. As grains shrink further, the volume fraction of grain-boundary material approaches that of the grain interiors (the surface-fraction arithmetic of Section 38.1 applies to grain boundaries too), and deformation shifts to **grain-boundary sliding**, grain rotation and diffusion along boundaries, mechanisms that become easier, not harder, as grains shrink. Molecular-dynamics simulations (Chapter 56) and experiments

place the strength maximum near 10–15 nm for copper and nickel. Below it, softening (the **inverse Hall–Petch** effect) is observed, though early reports of it were complicated by porosity and impurities in the samples.

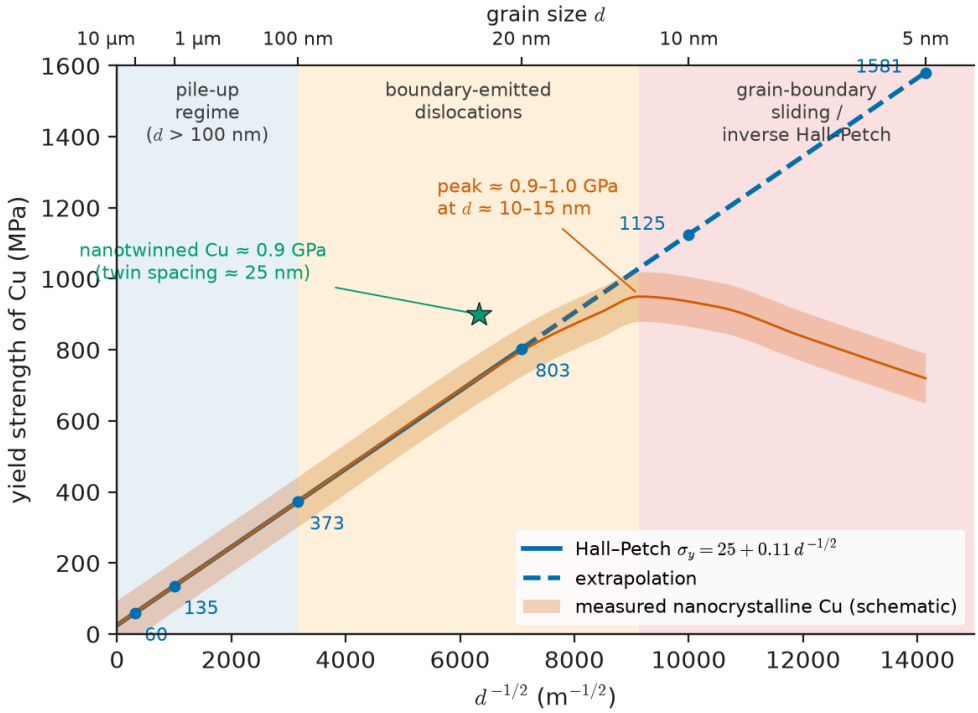


Figure 38.5 — Strength of copper rises with the inverse square root of grain size down to a few tens of nanometres, then saturates and falls in the inverse Hall-Petch regime.

38.5.3 Stability, ductility and practical nanocrystalline metals

Two problems limit bulk use. **Thermal stability**: the grain-boundary energy per unit volume is enormous, so nanocrystalline grains coarsen readily, in pure copper and aluminium even near room temperature. Stabilisation comes from solute segregation to boundaries, which lowers boundary energy (Ni–W, Cu–Ta, Fe–Zr), and from Zener pinning by fine particles (Chapter 10), as in oxide-dispersion-strengthened steels for reactor cladding (Chapter 41). **Ductility**: with no room for dislocation storage, nanocrystalline metals work-harden little, neck early and often show only a few percent uniform elongation.

Nanotwinned copper offers a way round. Pulsed electrodeposition can produce grains of 400–500 nm containing growth twins spaced 15–50 nm apart. Twin boundaries block dislocations like grain boundaries do, giving strength near 0.9–1 GPa, but because they are coherent, low-energy boundaries they scatter electrons very weakly, and the conductivity stays near 95 % of annealed copper, whereas ordinary nanocrystalline copper loses much more. They also allow dislocations to slip along them, which preserves some ductility.

Practical nanocrystalline metals are made in forms that do not need large bulk sections. Electrodeposited nanocrystalline nickel and nickel–tungsten or cobalt–phosphorus alloys, with grains of 10–20 nm and hardness of 5–7 GPa (about five times that of conventional nickel), are wear coatings that replace hard chromium plating, whose hexavalent-chromium chemistry is being phased out. Severe plastic deformation (Section 38.7) produces ultrafine-grained titanium for dental implants (Chapter 39) and aluminium for high-strength wires. Nanocrystalline soft magnetic alloys (Fe–Si–B–Nb–Cu, Chapter 29), with 10 nm grains in an amorphous matrix, owe their very low coercivity to grains smaller than the magnetic exchange length, another case where a characteristic length decides the effect.

38.6 Nanocomposites

Most nanomaterials reach products as fillers in a matrix. A nanocomposite differs from the conventional composites of Chapter 22 in two ways that follow from size. First, at the same volume fraction the filler has orders of magnitude more interface with the matrix, so a few percent of filler can change properties that a conventional filler changes only at 20–50 %. Second, nanoscale

fillers are too small to scatter light, to act as strength-limiting flaws or to roughen a surface, so properties can be added without the usual penalties. The difficulty is that their high surface energy makes them aggregate, and an agglomerated nanofiller behaves like a poor micro-filler. Dispersion and interface chemistry decide success.

38.6.1 Mechanical reinforcement

The first commercial polymer nanocomposite, nylon 6 with about 4 wt% of exfoliated montmorillonite clay platelets (1 nm thick, 100–500 nm across), was developed by Toyota around 1990 for a timing-belt cover: the modulus roughly doubled and the heat-deflection temperature rose by about 80 K with no loss of density or processability. Clay, CNT and graphene stiffening follows the Halpin–Tsai estimates of Chapter 22 with the platelet or tube aspect ratio, provided the fillers are individually dispersed and bonded. **Silica** nanoparticles bonded to rubber with silane coupling agents lowered the rolling resistance of "green" passenger tyres by about 20 % while keeping wet grip, which is now standard practice; **nano-silica** and **rubber nanoparticles** toughen epoxy matrices for aerospace composites; dental restorative nanocomposites combine 20–75 nm silica and zirconia clusters with polymer resins to give polishable, wear-resistant fillings. In concrete, nano-silica accelerates hydration and densifies the C–S–H, and graphene-oxide additions of 0.01–0.05 % are marketed to reduce cement content.

38.6.2 Barrier properties

Impermeable platelets aligned in a film force a diffusing gas molecule to take a detour, increasing the path length. The **Nielsen** tortuosity model for platelets of aspect ratio α (width/thickness) at volume fraction ϕ, aligned perpendicular to the diffusion direction, gives the ratio of composite to matrix permeability:

$$\frac{P_c}{P_m} = \frac{1 - \phi}{1 + \dfrac{\alpha}{2}\phi}$$

> **Status of the principle: Nielsen tortuosity model**
> *Approximate*. Assumes perfectly aligned, impermeable, uniformly spaced platelets and an unchanged matrix; misalignment and agglomeration reduce the benefit, while changes in matrix crystallinity or chain mobility near the filler can add to it.

Worked example — a clay barrier film. Adding 3 vol% clay platelets with aspect ratio 100 gives $P_c/P_m = 0.97/(1 + 50 \times 0.03) = 0.97/2.5 = 0.39$: the oxygen permeability falls by 61 %. At 5 vol% it falls to 0.27; with aspect ratio 300 at 3 vol%, to 0.18. The same reduction with a conventional filler would need far higher loadings and would cloud the film. Clay- and graphene-filled barrier layers are used in food and pharmaceutical packaging, in the inner liners of tyres (butyl rubber nanocomposites hold pressure longer) and in polymer fuel tanks and hydrogen-tank liners, where permeation of small molecules is a regulatory and safety issue.

38.6.3 Electrical and thermal percolation

A conducting filler in an insulating polymer forms a connected network above a **percolation threshold** ϕ_c (Chapter 25). For spheres $\phi_c \approx 0.16$; for randomly oriented sticks of length L and diameter d, excluded-volume arguments give $\phi_c \approx 0.7\, d/L$. Nanotubes with $L/d = 1000$ therefore percolate at about 0.07 vol%, roughly 0.1 wt%: some 200 times less filler than carbon black spheres need. In practice thresholds of 0.01–0.5 wt% are measured for well-dispersed tubes, rising to several percent if tubes bundle or break during mixing. Such loadings make plastics antistatic (surface resistivity 10^6–10^9 Ω) for fuel systems, electronics packaging and mining equipment, without changing their mechanical behaviour. Thermal percolation is much weaker, because phonons do not tunnel efficiently across filler–filler contacts, so thermally conductive polymers still need 30–60 vol% of fillers such as h-BN or alumina (Chapter 24).

38.7 Synthesis: top-down and bottom-up

Nanomaterials are made either by carving or breaking bulk material (**top-down**) or by assembling atoms and molecules (**bottom-up**). Figure 38.6 and Table 38.2 summarise the routes.

38.7.1 Top-down routes

- **High-energy ball milling** fractures and cold-welds powders, producing nanocrystalline grains (10–30 nm) inside micrometre particles and mechanically alloying immiscible elements, as in ODS steels (Chapter 41). It is cheap and scalable but introduces contamination and wide size distributions.
- **Severe plastic deformation** imposes enormous strains without changing shape: **equal-channel angular pressing** (ECAP) pushes a billet repeatedly through a die with a 90° bend; **high-pressure torsion** twists a disc under several GPa. Grains refine to 100–300 nm (ECAP) or below 100 nm (HPT) in bulk, fully dense metal.
- **Lithography and etching**, the semiconductor industry's route, pattern features to below 20 nm with extreme-UV lithography (Chapter 49); electron-beam lithography is precise but slow.
- **Exfoliation** separates layered crystals: micromechanical cleavage (adhesive tape) for research-grade flakes, and **liquid-phase exfoliation** by ultrasound or shear mixing in solvents or surfactant solutions for tonnage graphene nanoplatelets and h-BN; chemical oxidation and exfoliation gives graphene oxide.

38.7.2 Bottom-up routes

- **Gas-phase synthesis.** Flame pyrolysis makes fumed silica ($SiCl_4$ burned in a hydrogen–oxygen flame), pigmentary and nano titania and carbon black, at millions of tonnes per year; laser ablation and spark discharge make research-scale metal nanoparticles.
- **Colloidal (wet-chemical) synthesis.** In the **hot-injection** method, precursors injected into a hot coordinating solvent nucleate in a short burst (a LaMer supersaturation spike, Chapter 9), after which growth without new nucleation gives size distributions of ±5 %, the basis of quantum-dot manufacture. Citrate reduction of gold salts, polyol synthesis of silver nanowires and co-precipitation of iron oxide are other examples.
- **Sol–gel and hydrothermal** routes make oxide nanoparticles, aerogels and coatings at low temperature (Chapter 49).
- **Chemical vapour deposition (CVD).** Hydrocarbons decompose on catalysts: on iron, cobalt or nickel nanoparticles they grow carbon nanotubes (fluidised-bed reactors make tonnes per year of MWCNTs); on copper foil at about 1000 °C, methane gives self-limiting monolayer graphene, because carbon is nearly insoluble in copper and growth stops once the surface is covered. VLS growth of nanowires is a CVD variant.
- **Atomic layer deposition** (ALD) builds films one atomic layer per cycle by self-limiting surface reactions, giving perfectly conformal nanometre coatings inside pores and on nanoparticles (Chapter 49).
- **Electrodeposition** with pulsed current and grain-refining additives yields nanocrystalline and nanotwinned metals directly.
- **Self-assembly** of block copolymers, surfactants and colloids (Chapter 37) builds ordered nanostructures and templates.

Table 38.2 — Comparison of synthesis routes (indicative)

Route	Typical products	Size control	Throughput	Main limitation
Ball milling	Nanocrystalline powders, ODS alloys	Poor	High (tonnes)	Contamination, broad distribution
Severe plastic deformation	Bulk UFG metals	Moderate (100–300 nm)	Moderate	Size of billets, cost per kg
Lithography	Device features, patterned arrays	Excellent	High area, high cost	Capital cost, resolution limits
Liquid-phase exfoliation	Graphene and h-BN platelets	Poor in thickness	High	Defects, polydispersity
Flame pyrolysis	SiO_2, TiO_2, carbon black	Moderate	Very high (Mt/yr)	Aggregated particles
Hot-injection colloidal	Quantum dots, metal nanocrystals	Excellent (±5 %)	Low–moderate	Solvents, ligands, cost
CVD	CNTs, graphene films, nanowires	Good	Moderate	Transfer, catalyst residues
ALD	Conformal films, coatings	Atomic	Low (per layer)	Slow, precursor cost
Electrodeposition	Nanocrystalline coatings	Good	High	Conductive substrates only

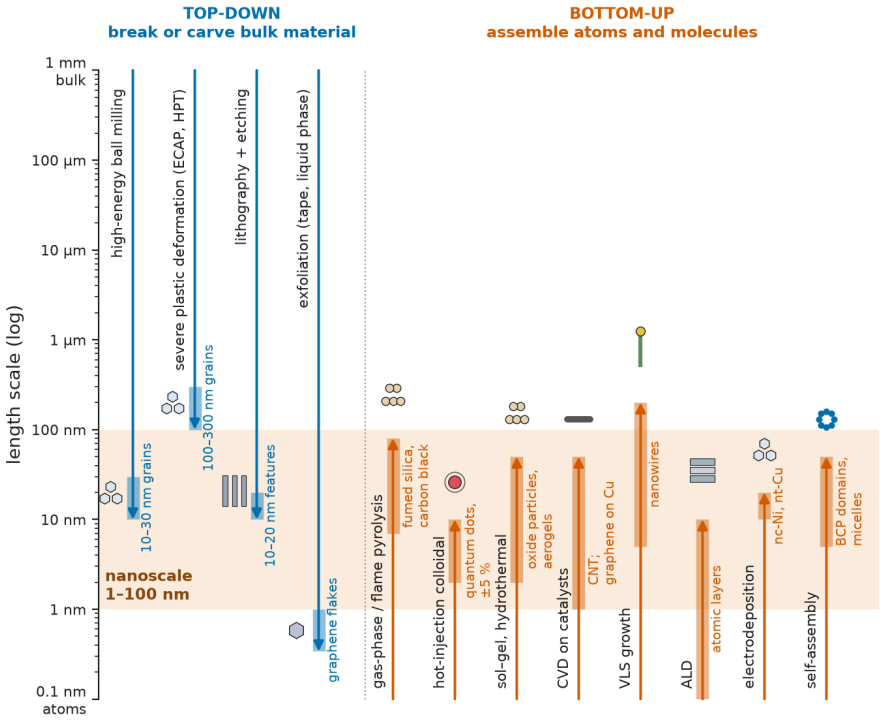


Figure 38.6 — Top-down routes break or carve bulk material; bottom-up routes assemble atoms and molecules; the two meet in the 1-100 nm range.

38.8 Characterization and safety of nanomaterials

38.8.1 Seeing and measuring the nanoscale

No single method characterises a nanomaterial; several are combined (details in Chapters 52–54):

- **Transmission electron microscopy** images individual particles, tubes and grain boundaries at atomic resolution, and electron diffraction identifies phases; it samples only a tiny amount of material, so statistics need many images and automated analysis.
- **Scanning electron microscopy** shows morphology and agglomeration of powders and composites down to a few nanometres.
- **Atomic force microscopy** measures the thickness of 2D flakes (a graphene monolayer appears 0.3–1 nm high depending on substrate and adsorbates) and maps surface topography and mechanical properties.
- **X-ray diffraction** gives crystallite size from peak broadening (the Scherrer and Williamson–Hall analyses of Chapter 4) averaged over billions of crystals; small-angle X-ray scattering gives particle size distributions in dispersions.
- **Dynamic light scattering** measures the hydrodynamic diameter of particles in liquid from their Brownian motion, which includes ligand shells and reports agglomerates.
- **Gas adsorption (BET)** measures specific surface area, comparable with the $6/(\rho d)$ estimate of Section 38.1.
- **Raman spectroscopy** is the workhorse for carbon: the G band near 1580 $\mathrm{cm^{-1}}$ (in-plane C–C stretching), the D band near 1350 $\mathrm{cm^{-1}}$ (activated only by defects) and the 2D band near 2700 $\mathrm{cm^{-1}}$, whose shape distinguishes monolayer from few-layer graphene. The D/G intensity ratio measures defect density, and radial breathing modes give nanotube diameters.
- **Optical spectroscopy**: the first-exciton absorption peak of quantum dots gives their size via sizing curves; the plasmon peak position and width of gold and silver particles report size and aggregation.

Reporting a nanomaterial properly means giving the size distribution (not just a mean), the method and its basis (number-, volume- or intensity-weighted), the state of agglomeration, the surface chemistry and the specific surface area. A "graphene" product should state layer number distribution, lateral size and oxygen content; surveys of commercial graphene have found many products to be mostly graphite.

38.8.2 Why nanomaterials raise safety questions

Risk is hazard multiplied by exposure, and nanomaterials raise both in specific ways:

- **Dose metric.** For poorly soluble particles, lung inflammation in animal studies correlates better with the *surface area* inhaled than with the mass, so the same mass of a nanopowder can be more toxic than its micrometre counterpart. Recommended exposure limits reflect this: NIOSH recommends 0.3 mg/m³ for ultrafine titania against 2.4 mg/m³ for fine titania.
- **Reactivity.** High surface area and surface defects can generate reactive oxygen species; soluble particles such as silver, zinc oxide and copper oxide release ions locally.
- **Translocation.** Particles below about 100 nm deposit efficiently in the deep lung, and a small fraction crosses into the blood; very small particles can reach other organs.
- **The fibre paradigm.** Long, thin, biopersistent fibres that macrophages cannot engulf (longer than about 15–20 μm) cause chronic inflammation, fibrosis and mesothelioma, as asbestos did. Long, rigid multi-walled carbon nanotubes behave this way in animal studies, and one type (MWNT-7) is classified as possibly carcinogenic to humans. NIOSH recommends an exposure limit of 1 μg/m³ for carbon nanotubes and nanofibres. Short, tangled or flexible tubes are much less hazardous.
- **Environment.** Nanoparticles released from products (silver from antimicrobial textiles during washing, titania from sunscreens and paints) enter waste-water and soils, where aggregation, dissolution and transformation (silver to silver sulfide) usually reduce toxicity, but the life-cycle fate is still being studied.

Exposure occurs mainly during manufacture and handling of dry powders, and during machining, sanding or burning of nanocomposites, rather than in using finished products, where the nanomaterial is bound in a matrix. Engineering control is therefore familiar industrial hygiene: handling dispersions or masterbatches instead of dry powder, enclosure and local exhaust ventilation, HEPA filtration (which captures nanoparticles very efficiently, because they diffuse to fibres), and respiratory protection. Regulation now treats nanoforms explicitly: since 2020 the EU REACH regulation requires registrants to characterise nanoforms separately, cosmetics containing nanomaterials must be notified and labelled "(nano)", and international standards (ISO/TR 12885 on occupational health and safety; ISO 80004 on vocabulary) define terms and practices. **Safe-by-design** asks the materials engineer to choose shapes, sizes, coatings and matrices that deliver the function at least hazard: shorter or tangled tubes, dissolvable or bound forms, surface coatings that suppress reactivity.

> **Status of the principle: Surface-area dose metric for nanoparticle toxicity**
> *Empirical.* Fitted to inhalation studies of poorly soluble, low-toxicity particles; it does not apply to soluble particles (where released ions dominate) or to fibres (where length and biopersistence dominate).

38.8.3 Why nanomaterials are mostly additives

Looking back over the chapter, a pattern emerges. Nanomaterials have succeeded where a small amount of a very fine phase changes a bulk property (carbon black, silica, CNT conductive additives, clay barriers, catalysts, pigments and UV filters), where size itself is the function (quantum-dot colour, plasmonic test strips) or where a thin layer is all that is needed (coatings, ALD films, electrodeposits). They have struggled where the outstanding property of an individual object (a nanotube's strength, graphene's mobility) must survive assembly into a macroscopic part, because the weak links between nano-objects, defects introduced during processing, and cost then dominate. The same structure–property logic that governs any material governs this one: a material is as good as its worst connection, and processing decides what the connections are.

What remains open

Open question	Where it is answered
How do surface energy and the Gibbs–Thomson effect follow from thermodynamics?	Chapter 6
How do nucleation bursts and Ostwald ripening control particle size distributions?	Chapters 9 and 10
How are dislocation pile-ups and Hall–Petch derived for coarse grains?	Chapter 13
How do micromechanics models (Halpin–Tsai, shear lag) predict the stiffness of filled polymers?	Chapter 22
What sets electron and phonon transport in bulk conductors and semiconductors?	Chapters 24, 25, 31 and 32
How are nanomaterials used in batteries, supercapacitors, catalysts and thermoelectrics?	Chapter 40
How do nanoscale oxide dispersions make steels resistant to radiation and creep?	Chapter 41
How are thin films and ALD layers grown and patterned?	Chapter 49
How do atom-probe tomography and aberration-corrected STEM resolve nanostructures?	Chapter 53
Can molecular dynamics predict the inverse Hall–Petch transition and 2D-material properties?	Chapters 55 and 56
How are the environmental footprint and end-of-life of nanomaterials assessed?	Chapter 59

Next: Chapter 39 places materials inside the human body, where the environment is warm, wet, salty and alive, and asks what makes a material biocompatible.

Chapter 39 — Biomaterials

Part VIII · Advanced and Emerging Materials

In 2010 a large orthopaedic manufacturer recalled a metal-on-metal hip replacement worldwide. Its cobalt–chromium ball and cup had been designed to eliminate the plastic wear debris that had loosened earlier hips, and in a hip simulator it wore far less than polyethylene. In patients it behaved differently. A cup set at a steep angle let the ball ride on its edge, the lubricating film broke down, and the joint shed billions of nanometre-sized metal particles and cobalt and chromium ions. Some patients developed soft-tissue masses, destroyed muscle and raised blood cobalt; within five years about a quarter to a third of the devices had to be revised, against a few percent for conventional hips. Nothing in the alloy's datasheet was wrong. Its hardness, strength and corrosion resistance were excellent. The failure came from the interaction of the material with a living host: how the body responded to particles and ions of a size, number and chemistry that the design had not anticipated. Biomaterials engineering is the discipline of anticipating that interaction. This chapter describes how the body responds to an implanted material, surveys the metals, ceramics, polymers and hydrogels that are used, follows them into hips, knees, teeth and arteries, and ends with drug delivery and the sterilization, testing and regulation that stand between a material and a patient.

> **What makes a material acceptable to the body, and why is "biocompatibility" a property of a material in an application rather than of the material alone?**
> **How do the stiffness, corrosion, wear and degradation of implant materials decide whether an implant lasts five years or thirty?**

Concept	Engineering destination
Protein adsorption, inflammation, foreign-body response	Surface design of implants, sensors, catheters and encapsulated cells
Stress shielding, modulus mismatch	Hip-stem and bone-plate design, porous and low-modulus titanium
Passivity and corrosion in body fluid	Alloy selection, modular junctions, ion release limits
Bioinert, bioactive and resorbable ceramics	Joint bearings, bone-graft substitutes, coatings, dental crowns
Hydrolytic degradation of PLA, PGA, PCL	Resorbable sutures, screws, stents and drug-release matrices
Hydrogels and scaffolds	Wound dressings, contact lenses, tissue engineering, cell therapy
Wear and particle-induced osteolysis	Bearing couples in hips and knees, crosslinked polyethylene
Controlled-release kinetics	Drug-eluting stents, depot injections, lipid nanoparticles
Sterilization, ISO 10993, device classes	Getting a material into a regulated product

Roadmap. The host response (39.1) → metallic biomaterials and corrosion in the body (39.2) → ceramics and glasses (39.3) → polymers and biodegradable polymers (39.4) → hydrogels, scaffolds and tissue engineering (39.5) → implants as material systems: hips, knees, teeth, stents (39.6) → drug delivery (39.7) → sterilization, testing and regulation (39.8).

> **Boundary.**
> The biology of inflammation and wound healing is summarised only as far as it sets materials requirements; physiology and immunology belong to the life sciences. Electrochemistry fundamentals belong to Advanced Chemistry; the corrosion mechanisms used here are developed in Chapters 42 and 43.

39.1 The host response

39.1.1 What biocompatibility means

A **biomaterial** is a material intended to interface with biological systems to evaluate, treat, augment or replace a tissue, organ or function. The classic definition of **biocompatibility** (Williams) is "the ability of a material to perform with an appropriate

host response in a specific application". Every word matters. *Perform*: the material must do its job, so a perfectly inert material that breaks is not biocompatible. *Appropriate host response*: the body always responds; the question is whether the response suits the purpose. A bone screw should be incorporated into bone, a blood-contacting catheter should not trigger clotting, a degradable suture should be resorbed without inflammation, and an encapsulated islet-cell device should be ignored by the immune system yet let insulin out. *Specific application*: the same material can be excellent in one site and unacceptable in another. Cobalt–chromium is a fine femoral head against polyethylene and a poor one against itself under edge loading; polyethylene is a durable bearing in a knee and a source of osteolysis when it wears quickly in a hip.

39.1.2 The sequence of events

When any material is implanted, the body responds in a sequence that is broadly the same for every material (Figure 39.1); what differs is the intensity and the end-point.

1. **Protein adsorption (seconds to minutes).** Before any cell arrives, the surface is coated by a layer of proteins from blood and tissue fluid: albumin, fibrinogen, fibronectin, vitronectin, immunoglobulins, complement proteins. Small, abundant proteins arrive first and are displaced over minutes to hours by larger proteins with higher affinity (the **Vroman effect**). Adsorption often unfolds the proteins, exposing binding sites. Cells never see the material itself, only this conditioned protein layer, whose composition and conformation are set by the surface chemistry, charge, hydrophobicity and roughness (the surface science of Chapter 45). Hydrophobic surfaces adsorb and denature more protein; highly hydrated, neutral surfaces (polyethylene glycol brushes, zwitterionic polymers) resist adsorption.
2. **Blood–material interactions (minutes to hours).** On blood-contacting surfaces, adsorbed fibrinogen binds platelets, the coagulation cascade and complement are activated, and a provisional matrix (a clot) forms around tissue implants. For devices in flowing blood, such as stents, valves and catheters, this **thrombogenicity** is the dominant concern and is controlled by surface chemistry (heparin coatings), surface smoothness and flow design.
3. **Acute inflammation (hours to days).** Neutrophils migrate to the site, attracted by the injury of implantation itself, and release enzymes and reactive oxygen species. This phase subsides within days if the implant is clean and sterile.
4. **Chronic inflammation and the foreign-body reaction (days to weeks).** Macrophages adhere to the protein layer. If they cannot phagocytose the material (anything larger than about 10 μm), they fuse into **foreign-body giant cells** that sit on the surface for the life of the implant, releasing degradative agents. Small particles (0.1–10 μm), such as wear debris, are engulfed and activate macrophages to release cytokines that recruit bone-resorbing osteoclasts.
5. **Fibrous encapsulation (weeks to months).** Fibroblasts lay down a collagen capsule, typically 50–200 μm thick, that walls off the implant. For a soft-tissue implant such as a pacemaker can, this is acceptable; for a glucose sensor it blocks diffusion and degrades the signal within weeks; for a breast implant, contraction of the capsule is the main complication.

Against bone, a different end-point is possible. If the implant is stable (micromotion below about 50–150 μm) and the surface is suitable, bone forms directly on it without an intervening fibrous layer: **osseointegration**, discovered by Brånemark with titanium in the 1950s–60s and the basis of dental implants and cementless joint replacements.

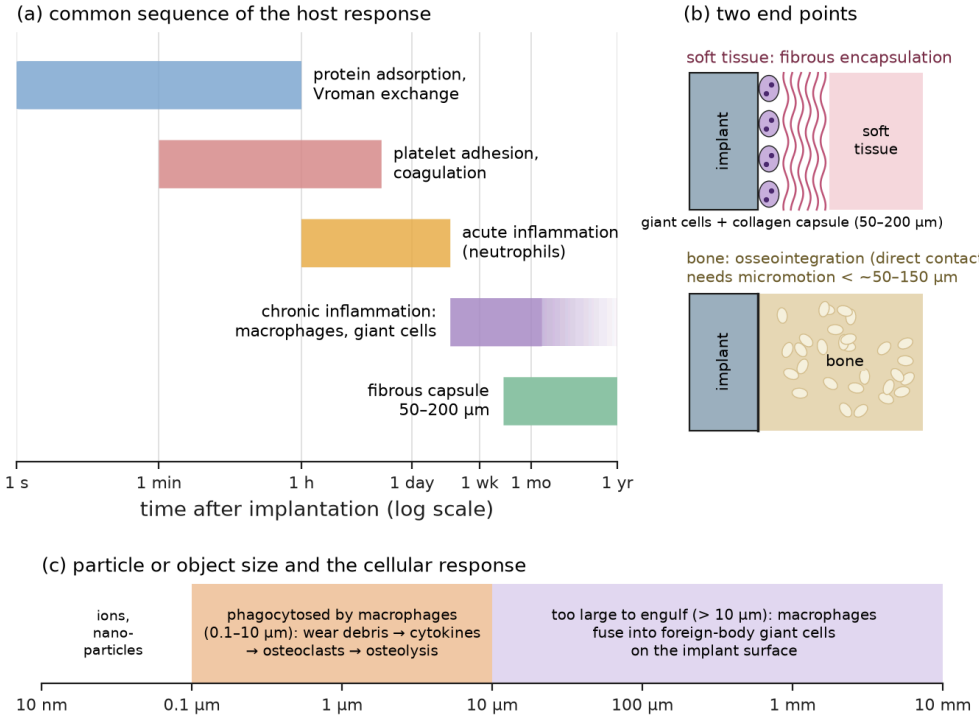


Figure 39.1 — The host response to an implanted material follows a common sequence from protein adsorption to fibrous encapsulation or, against bone, osseointegration.

39.1.3 Classes of tissue–material response

Hench's classification of how bone responds to a material, extended to tissue in general, is a useful vocabulary:

Response	Interface	Examples
Toxic	Surrounding tissue dies	Released Cd, Pb, V in high doses; unreacted monomers
Bioinert (nearly inert)	Thin fibrous layer; no bond	Alumina, zirconia, UHMWPE, PEEK, CoCr
Osseointegrating	Direct bone apposition, mechanical interlock	Titanium and its alloys, tantalum
Bioactive	Chemical bond to bone (and some to soft tissue)	Hydroxyapatite, bioactive glasses, glass-ceramics
Resorbable	Material replaced by tissue	Tricalcium phosphate, PLA/PGA, Mg alloys

The material's chemistry sets which column it falls into; its form (particles or a solid, smooth or porous, stable or moving) decides how strong the response is. Wear debris of an inert polymer can provoke a stronger reaction than a bulk implant of a less inert metal, which is the central lesson of joint replacement (Section 39.6).

> **Status of the principle: Biocompatibility as an application-specific response**
> *Empirical.* It is a definition grounded in clinical experience, not a derived law; materials cannot be ranked by a single biocompatibility number, and every claim must be supported by testing in a relevant model (Section 39.8).

39.2 Metallic biomaterials

Metals carry load: hip stems, bone plates and screws, spinal rods, dental implants, stents, pacemaker cases and guidewires. Three alloy families dominate (Table 39.1), all relying on a passive oxide film for corrosion resistance.

Table 39.1 — Metallic and reference materials for implants (typical values)

Material	Standard (typical)	Young's modulus (GPa)	Yield strength (MPa)	Main uses
Cortical bone (reference)	—	15–20 (longitudinal)	100–150 (tension)	—
Cancellous bone (reference)	—	0.1–2	2–20 (compression)	—
316L stainless steel	ASTM F138	193	170 (annealed) to 690 (cold-worked)	Fracture plates, screws, temporary devices
CoCrMo, cast	ASTM F75	210–230	450–520	Knee femoral components, hip cups
CoCrMo, wrought/forged	ASTM F1537	210–230	830–1000	Femoral heads, hip stems
CP titanium grade 4	ASTM F67	105	480	Dental implants, porous coatings
Ti-6Al-4V ELI	ASTM F136	110–114	795–880	Hip stems, spinal implants, bone plates
β-Ti alloys (Ti-13Nb-13Zr, Ti-35Nb-7Zr-5Ta)	ASTM F1713, F1813	55–85	500–900	Low-modulus stems, research
NiTi (superelastic)	ASTM F2063	40–75 (austenite)	Plateau 350–550	Self-expanding stents, orthodontic wires
Tantalum (porous trabecular)	—	1.5–3 (porous)	35–50 (compression)	Acetabular augments, bone ingrowth
Magnesium alloys (WE43-type)	—	45	150–250	Resorbable screws, stents

39.2.1 Stainless steel

Austenitic **316L** (Fe–17Cr–12Ni–2.5Mo, carbon below 0.03 % to avoid sensitization, Chapter 15) is cheap, easy to form, and strong when cold-worked. Its passive Cr_2O_3-rich film is less stable in chloride than titanium's oxide, and pitting and crevice corrosion under screw heads and plate contacts are well documented; it also contains nickel, a common sensitizer. It is therefore used mainly for temporary devices (fracture fixation removed after healing, surgical instruments) and in low-cost settings. Nitrogen-strengthened, nickel-free austenitic steels (Fe–Cr–Mn–Mo–N) address both concerns.

39.2.2 Cobalt–chromium alloys

CoCrMo (Co–28Cr–6Mo) is hard, wear resistant and strong, with a stable chromium-oxide film. Cast F75 is used for complex shapes such as knee femoral components; wrought and forged F1537 for femoral heads and highly loaded stems. Carbides ($M_{23}C_6$) at grain boundaries harden the cast alloy and, at bearing surfaces, influence wear. CoCr's high modulus contributes to stress shielding (below), and its wear and corrosion products (Co^{2+}, Cr^{3+}, chromium-oxide particles) are the concern exposed by metal-on-metal hips. **CoNiCrMo** (MP35N) and **CoCrWNi** (L605) are used for thin-strut cardiovascular stents and pacemaker leads, where their strength and radiopacity allow struts of 60–90 μm.

39.2.3 Titanium and its alloys

Titanium (Chapter 16) combines low density (4.5 g/cm³), the lowest modulus of the common implant metals, good fatigue strength and an exceptionally stable TiO_2 passive film that reforms in milliseconds when scratched. Bone grows directly on it. **CP titanium** (grade 4, strengthened by oxygen) is used for dental implants; **Ti-6Al-4V ELI** (extra-low interstitials, for fracture toughness) is the workhorse for hip stems and spinal devices. Concern about aluminium and vanadium release led to vanadium-free α+β alloys (Ti-6Al-7Nb) and to **β alloys** of Ti with Nb, Zr, Ta and Mo, which are free of toxic elements and have moduli as low as 55 GPa. Additive manufacturing (Chapter 48) now builds titanium acetabular cups and spinal cages with integral porous lattices (pore sizes 300–700 μm, porosity 60–80 %) into which bone grows.

Titanium's weaknesses are tribological: it galls and wears badly in sliding contact and is never used as a bearing surface without a hard coating; and its notch sensitivity in fatigue means that surface treatments for bone ingrowth (grit blasting, porous coatings sintered at high temperature) can reduce fatigue strength by 50 % or more, which must be allowed for in stem design.

39.2.4 Stress shielding

Bone is a living material that adapts its density to the stresses it carries (an observation summarised as **Wolff's law**): underloaded bone is resorbed. A stiff stem inside a femur carries a large part of the load that bone used to carry, and the bone around the upper part of the stem thins over years, which can loosen the implant and complicate revision surgery.

Worked example — load sharing between a femoral stem and bone. Model the femoral shaft as a cortical-bone tube with outer diameter 28 mm and inner diameter 15 mm, so its area is $A_b = \frac{\pi}{4}(28^2 - 15^2) = 439$ mm², and a solid stem of diameter 14 mm, $A_s = 154$ mm², bonded to it and loaded axially. The two deform together (equal strain), so each carries load in proportion to its axial stiffness EA. The fraction of the load carried by bone is

$$f_b = \frac{E_b A_b}{E_b A_b + E_s A_s}$$

With $E_b = 17$ GPa: for Ti-6Al-4V ($E_s = 110$ GPa), $f_b = 17 \times 439/(17 \times 439 + 110 \times 154) = 7463/(7463 + 16\,940) = 0.31$; for CoCr (210 GPa), $f_b = 0.19$; for a β-titanium alloy (60 GPa), $f_b = 0.45$; and for a carbon-fibre-reinforced PEEK stem (18 GPa), $f_b = 0.73$. A titanium stem thus leaves the bone at this section with about one third of its natural strain, a CoCr stem with about a fifth. The calculation explains why low-modulus alloys, hollow stems and porous structures are pursued (Figure 39.2).

> **Status of the principle: Equal-strain load sharing (stress shielding estimate)**
> *Approximate.* Assumes a perfectly bonded, axially loaded, uniform composite section; real stems are tapered, bending-loaded and partly bonded, and load transfer is concentrated near the ends. The trend with modulus is right; the absolute numbers come from finite-element models.

It also shows where the simple picture fails. Very flexible "isoelastic" stems tried in the 1980s and 1990s did not do better clinically: they bent more under bending loads, raising shear stresses and micromotion at the bone–implant interface above the 50–150 µm limit for bone ingrowth, so they loosened for a different reason. The design target is not the lowest modulus but a balance between load transfer to bone and stability of the interface, a multi-objective selection problem of the kind treated in Chapter 58.

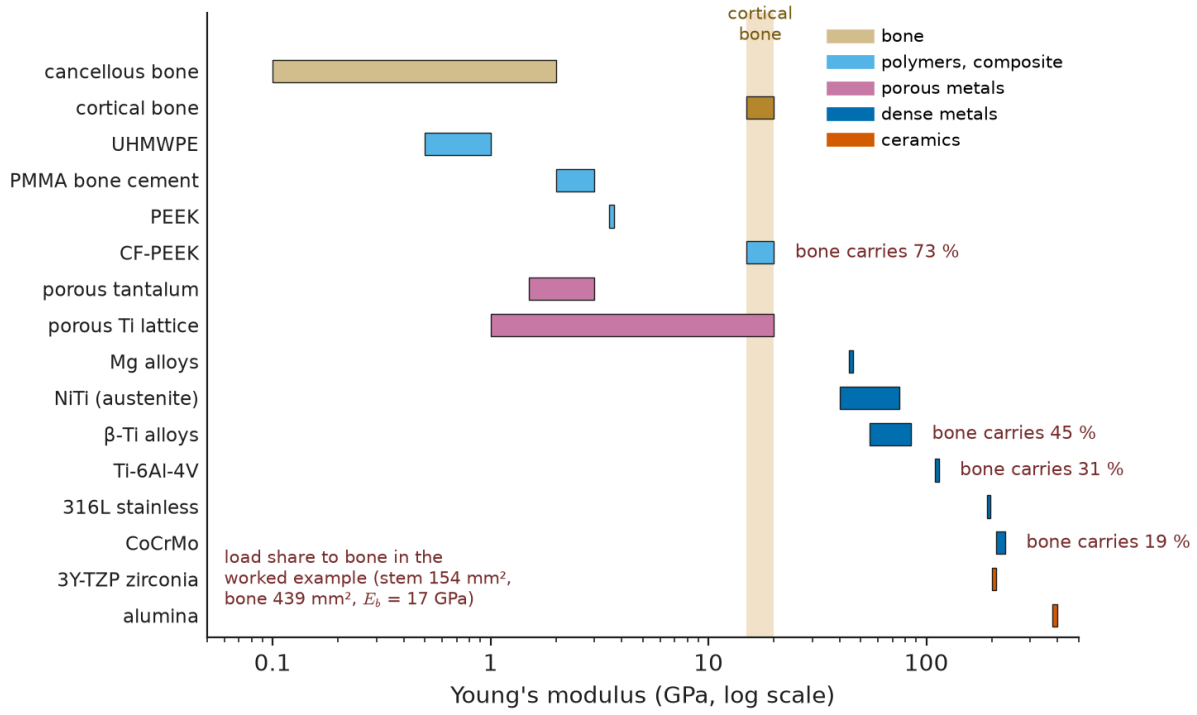


Figure 39.2 — Implant metals are 3-15 times stiffer than cortical bone, which causes stress shielding; low-modulus alloys, porous metals and composites narrow the gap.

39.2.5 Corrosion in the body

Body fluid is an aggressive electrolyte: about 0.15 M chloride, pH 7.4, 37 °C, with dissolved oxygen lower than in air-saturated water and with proteins and cells that can both adsorb on and attack surfaces. After surgery or in inflamed tissue, pH may fall to about 5 and reactive oxygen species are produced. Implant alloys survive because they are **passive** (Chapter 42): a nanometre-thin oxide film reduces the dissolution current density to 10^{-2}–10^{-1} μA/cm². Problems arise where the film is damaged or the local chemistry changes (Chapter 43):

- **Crevice and pitting corrosion** under screw heads and in plate–screw contacts, mainly in stainless steel.
- **Fretting corrosion** and **tribocorrosion** where micromotion repeatedly scrapes off the film, most notoriously at the tapered junction between a modular femoral head and the stem (*taper corrosion* or *trunnionosis*), which can release metal debris even from metal-on-polyethylene hips.
- **Galvanic corrosion** between dissimilar alloys; CoCr and Ti are close enough in the passive state to be combined, but stainless steel is not mixed with them.

Even passive dissolution releases ions continuously, and Faraday's law (Chapter 42) converts current to mass.

Worked example — ion release from passive implants. A 316L fracture plate and screws expose 50 cm² to tissue fluid and dissolve passively at $i = 0.1$ μA/cm². The current is $I = 5 \times 10^{-6}$ A, and over one year ($t = 3.156 \times 10^{7}$ s) the charge is 158 C. Treating the alloy as dissolving to Fe^{2+}, Cr^{3+}, Ni^{2+} and Mo^{3+} in proportion to its composition gives an equivalent weight of about 25.5 g/equivalent, so the mass dissolved is

$$m = \frac{ItM}{nF} = \frac{158 \times 25.5}{96\,485} = 0.042 \text{ g}$$

about 42 mg per year, a uniform thinning of only about 1 μm, structurally irrelevant, but containing about 5 mg of nickel ions each year at the site, enough to matter for a patient with nickel allergy. A titanium implant of the same area with $i = 0.01$ μA/cm² and Ti^{4+} ($M/n = 47.87/4$) releases $5 \times 10^{-7} \times 3.156 \times 10^{7} \times 47.87/(4 \times 96\,485) = 2.0$ mg per year. The difference of a factor of 20, together with titanium's low toxicity, is a large part of why titanium dominates permanent implants.

39.2.6 Biodegradable metals

For temporary functions (holding a fracture until it heals, keeping an artery open for six months) a metal that corrodes away safely avoids a second operation. **Magnesium** alloys (with rare earths, zinc, calcium) corrode in body fluid with hydrogen evolution; resorbable Mg compression screws and a drug-eluting Mg coronary scaffold are on the market in some regions. The challenge is controlling the rate, because hydrogen bubbles and loss of strength before healing are both unacceptable; purity (Fe, Ni, Cu below tens of ppm, since they form cathodic particles), grain refinement and coatings are used. **Iron** alloys corrode too slowly and **zinc** alloys at intermediate rates, and both are under development for stents.

39.3 Ceramic biomaterials

Ceramics (Chapter 17) bring hardness, wear resistance and chemical stability, and in calcium-phosphate and glass form, the ability to bond to bone. Their limitation is brittleness, which Chapter 14's fracture mechanics and Weibull statistics govern.

Table 39.2 — Ceramic biomaterials (typical values)

Material	Young's modulus (GPa)	Flexural strength (MPa)	K_{IC} (MPa m^1/2)	Response	Uses
Alumina (medical grade, high purity, fine grain)	380–400	500–600	4	Bioinert	Femoral heads, liners
Zirconia-toughened alumina (ZTA)	350–360	1000–1400	6–7	Bioinert	Hip heads and liners, knee components
3Y-TZP zirconia	200–210	900–1200	5–10	Bioinert	Dental crowns, bridges, implants
Hydroxyapatite (dense)	80–110	50–120	0.8–1.2	Bioactive	Coatings, granules, small fillers
β-tricalcium phosphate	~30–70 (dense)	30–90	~1	Resorbable	Bone-graft substitutes
Bioactive glass 45S5	35	40–60	~0.6	Bioactive (bonds to bone and soft tissue)	Bone fillers, dental remineralisation
Pyrolytic carbon	20–30	300–500	1–2	Thromboresistant	Mechanical heart-valve leaflets

39.3.1 Bioinert ceramics: alumina and zirconia

High-purity, fine-grained **alumina** was introduced for hip heads in the 1970s because its hardness and wettability give very low wear against itself or polyethylene. Early heads fractured at rates of several per thousand; refinement of grain size (to below 2 μm), purity, hot isostatic pressing and proof testing reduced fracture to a few per 100 000. **Zirconia** (3 mol% yttria-stabilised tetragonal zirconia polycrystal, 3Y-TZP) is tougher because of transformation toughening (Chapter 14): the stress field at a crack tip converts metastable tetragonal grains to monoclinic with a 4 % volume expansion, which closes the crack. That same metastability is its weakness: in warm water the tetragonal-to-monoclinic transformation can proceed spontaneously from the surface (**low-temperature degradation**, or ageing), roughening and weakening it. In 2001 one manufacturer's change from batch to tunnel-furnace sintering produced zirconia hip heads that aged rapidly; several hundred fractured and the product was recalled. Today, **zirconia-toughened alumina**, with about 17–25 % zirconia dispersed in an alumina matrix that constrains it against ageing, is the standard ceramic bearing, combining alumina's stability with zirconia's toughness. In dentistry, monolithic zirconia crowns and bridges have largely replaced porcelain fused to metal, and translucent zirconias with more yttria (4–5 mol%, more cubic phase) trade toughness for aesthetics.

39.3.2 Bioactive ceramics and glasses

Hydroxyapatite (HA), $Ca_{10}(PO_4)_6(OH)_2$, is the mineral of bone and teeth, though biological apatite is nanocrystalline, carbonate-substituted and calcium-deficient. Synthetic HA bonds directly to bone, but it is too weak and brittle to bear load on its own, so it is used as plasma-sprayed coatings 50–150 μm thick on titanium stems and cups (to speed fixation), as porous granules and blocks for filling bone defects, and in composites. **Tricalcium phosphate** (β-TCP) is more soluble and resorbs over months to years; biphasic HA/TCP mixtures tune the rate.

Bioactive glasses, discovered by Hench in 1969, bond to bone faster than HA and even to soft tissue. The original **45S5** composition (45 SiO_2, 24.5 Na_2O, 24.5 CaO, 6 P_2O_5 wt%) has much less silica and much more modifier than window glass. In body fluid it undergoes a sequence of surface reactions: Na^+ exchanges with H^+ from solution, silanol groups form and condense to a silica-rich gel layer, calcium and phosphate migrate through it and from solution, and a layer of hydroxycarbonate apatite crystallises within hours to days. Collagen and cells attach to this apatite, and released soluble silicon and calcium stimulate bone-forming cells. Bioactivity depends on composition in the Na_2O–CaO–SiO_2 system (Figure 39.3): above about 60 wt% SiO_2 the network is too connected to react, and the glass is inert. Bioactive glasses are used as bone-defect fillers, in toothpaste for sensitive teeth, and as borate-glass fibres in wound dressings. **Glass-ceramics** (apatite–wollastonite, A-W) give higher strength (about 200 MPa in bending) with retained bioactivity for vertebral and iliac spacers.

Status of the principle: Composition–bioactivity map for silicate glasses
Empirical. Derived from animal implantation studies of soda–lime–phosphosilicate glasses; it is explained qualitatively by network connectivity and dissolution rate but does not transfer quantitatively to other glass systems (borate, phosphate) or to glass-ceramics.

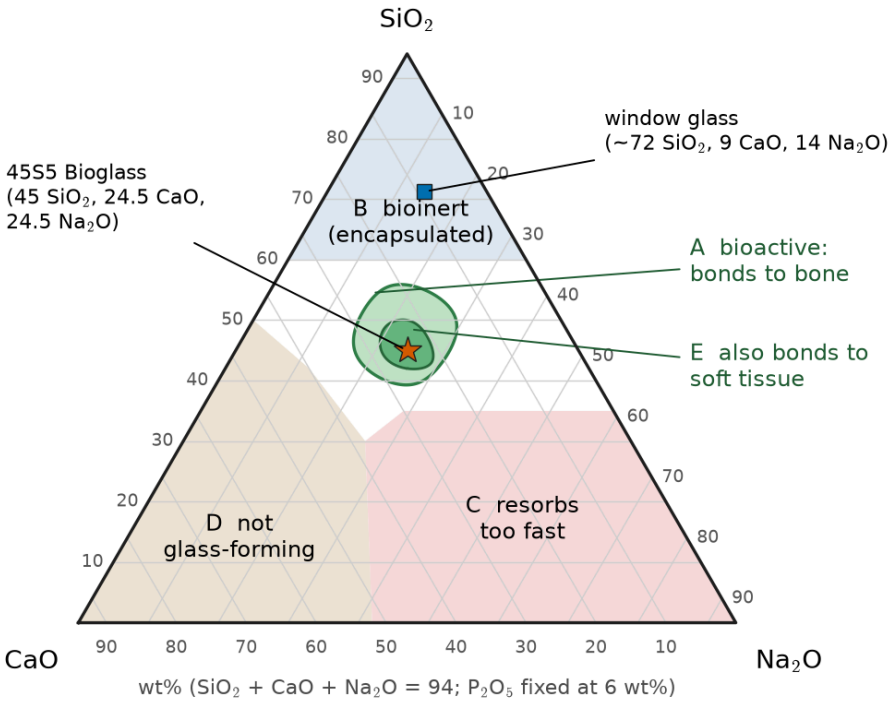


Figure 39.3 — Bone bonding of soda-lime-silica glasses depends on composition: only a region rich in modifiers and low in silica bonds to bone, and a smaller region also bonds to soft tissue.

39.4 Polymeric biomaterials

Polymers are the most widely used biomaterials by number of devices: catheters, tubing, sutures, blood bags, intraocular lenses, vascular grafts, heart-valve components, bone cement and joint bearings.

39.4.1 Biostable polymers

Polymer	Key properties	Uses
UHMWPE (molar mass 2–6 × 10^6 g/mol)	Tough, low friction, wear resistant; $E \approx 0.5$–1 GPa	Hip liners, knee tibial inserts
PMMA	Rigid, transparent; $E \approx 2$–3 GPa	Bone cement, intraocular lenses (historically), dentures
PEEK, CF-PEEK	$E \approx 3.6$ GPa (unfilled), 15–20 GPa (CF); radiolucent	Spinal cages, trauma plates
Silicone (PDMS)	Soft, flexible, gas-permeable, stable	Catheters, shunts, breast implants, lens materials
Polyurethanes	Tough, elastic, blood-compatible grades	Catheters, pacemaker-lead insulation, heart pumps
PET (Dacron), ePTFE	Fibrous/porous, strong, stable	Large-diameter vascular grafts, sutures, sewing cuffs
Polypropylene	Strong monofilament	Sutures, hernia mesh

Several of these were engineering polymers adopted with little modification, and their failures taught biomaterials science. **PMMA bone cement** is mixed in the operating theatre from a powder (pre-polymerised PMMA beads, initiator, radiopacifier such as $BaSO_4$ or ZrO_2) and liquid monomer; it grouts the stem into bone as a space filler, not an adhesive. Its exothermic polymerisation reaches peak temperatures of 60–80 °C at the cement surface, its residual monomer is cytotoxic, and its porosity and fatigue cracking contribute to aseptic loosening; vacuum mixing and antibiotic-loaded cements are refinements. **Polyester-urethane** foam coatings on breast implants degraded in the body, and early polyether-urethane lead insulation suffered environmental stress cracking and metal-ion oxidation (Chapter 21), problems solved by polycarbonate-urethanes and silicone–urethane copolymers. Long-term oxidation of gamma-sterilized UHMWPE is discussed in Section 39.8.

39.4.2 Biodegradable polymers

A **biodegradable** (or bioresorbable) polymer breaks down in the body into products that are metabolised or excreted. The workhorses are aliphatic polyesters, which degrade by **hydrolysis** of their ester bonds (Chapter 21):

Polymer	T_g (°C)	T_m (°C)	Modulus (GPa)	Typical loss of strength	Typical complete resorption
Poly(glycolic acid), PGA	35–40	220–230	6–7	2–4 weeks	6–12 months
Poly(L-lactic acid), PLLA	55–65	170–180	3–4	6–12 months	2–5 years
Poly(DL-lactic acid), PDLLA (amorphous)	50–60	—	1.5–2	1–2 months	12–16 months
PLGA 50:50	45–55	— (amorphous)	1–2	1–2 months	1–2 months
Poly(ε-caprolactone), PCL	−60	58–63	0.2–0.4	> 1 year	2–4 years

Degradation rate follows chemistry and structure. PGA is more hydrophilic than PLA (no methyl group), so water penetrates faster. Crystallinity slows hydrolysis because water enters amorphous regions first, so PLLA (semicrystalline) lasts far longer than amorphous PDLLA. Copolymerising lactide and glycolide disrupts crystallinity, so PLGA 50:50 degrades faster than either homopolymer. PCL, hydrophobic and with a long methylene sequence, degrades over years and is used for long-term implants and drug depots.

Hydrolysis is random chain scission, so to a first approximation the number-average molar mass decreases exponentially,

$$M_n(t) = M_{n0}\, e^{-kt}$$

while mass is lost only after chains have become short enough (a few kg/mol) to dissolve. Strength falls once M_n drops below a few times the entanglement molar mass. A device therefore keeps its shape long after it has lost its strength (Figure 39.4).

> **Status of the principle: First-order kinetics of polyester hydrolysis**
> *Approximate.* Assumes random scission with a constant water concentration and no autocatalysis; it describes thin films and fibres well, but thick devices degrade faster inside than at the surface because acidic degradation products accumulate and catalyse hydrolysis.

Worked example — how long does a resorbable device keep its strength? A PLLA interference screw has initial $M_n = 150$ kg/mol and loses molar mass with a half-life of 25 weeks in vivo ($k = \ln 2/25 = 0.0277$ per week). If strength is effectively lost when M_n reaches 20 kg/mol, the time is $t = \ln(150/20)/0.0277 = 73$ weeks, about 17 months, ample for a ligament graft to integrate with bone. A PLGA 50:50 suture with $M_n = 100$ kg/mol and a half-life of 3 weeks reaches 20 kg/mol in $\ln 5/(\ln 2/3) = 7.0$ weeks, which suits soft-tissue closure. The thick screw also illustrates the autocatalysis warning: lactic acid trapped in its core lowers the local pH, interior degradation can outrun the surface, and late, sudden release of acidic fragments has caused sterile inflammatory reactions, one reason resorbable devices are kept thin or buffered with calcium carbonate or phosphate fillers.

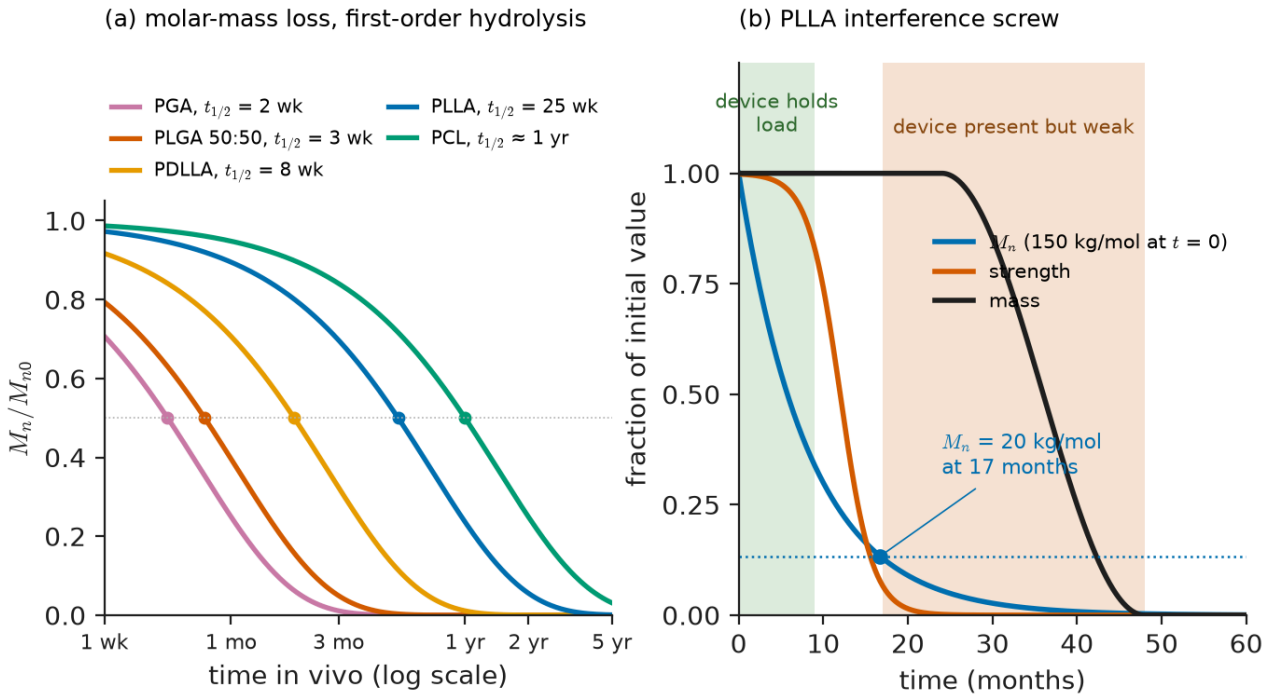


Figure 39.4 — Biodegradable polyesters lose molar mass first, strength next and mass last; PGA and PLGA degrade in weeks, PLLA and PCL over years.

Biodegradable polymers have mixed clinical history. Resorbable sutures (PGA, polyglactin) and PLGA drug depots are unquestioned successes. The everolimus-eluting PLLA coronary scaffold, approved in 2016, was withdrawn in 2017: its struts had to be about 150 µm thick (twice those of metallic stents) to compensate for the polymer's low stiffness and strength, and thrombosis rates were higher than for metal stents, a reminder that a material that is right in principle must also be right in the dimensions the mechanics imposes.

39.5 Hydrogels, tissue scaffolds and tissue engineering

39.5.1 Hydrogels in medicine

Hydrogels (Chapter 37) resemble soft tissue: they are mostly water, soft (moduli of 0.1 kPa to 1 MPa) and permeable to nutrients. Their medical uses include:

- **Contact lenses.** pHEMA lenses (about 38 % water) were followed by silicone-hydrogel lenses, which combine hydrophilic monomers with siloxane-rich domains so that oxygen can diffuse through the silicone phase, raising oxygen transmissibility several-fold, the critical property for corneal health.
- **Wound dressings** of alginate, hydrocolloid and polyurethane hydrogels that keep the wound moist and absorb exudate.
- **Injectable fillers and lubricants**, mostly crosslinked hyaluronic acid, used in dermatology and for knee injections.
- **Cell encapsulation**: alginate microcapsules that let insulin, glucose and oxygen pass while excluding antibodies and immune cells, depending on the mesh size calculated in Chapter 37. The fibrotic capsule of Section 39.1 is the main obstacle; chemically modified alginates that reduce macrophage adhesion are being tested.
- **Drug delivery** (Section 39.7) and **soft electronics** interfaces with nerve tissue.

39.5.2 Scaffolds and tissue engineering

Tissue engineering aims to regenerate tissue rather than replace it with a permanent implant. The classical recipe combines three elements: cells (the patient's own or stem cells), a **scaffold** that gives them a three-dimensional structure, and signals (growth factors, mechanical loading). The scaffold must meet materials requirements that are in tension with one another:

- **Porosity and pore size.** Interconnected porosity of 70–90 % with pores of 100–500 µm for bone (so that blood vessels can grow in, with 300 µm or larger favoured) or 20–150 µm for soft tissues; but high porosity reduces stiffness and strength, which scale roughly as the square of relative density for open-cell foams.
- **Degradation matched to tissue formation**, so that the scaffold hands over load gradually: PLGA, PCL, collagen, β-TCP and bioactive glass are chosen for their rates.
- **Surface chemistry and stiffness** that direct cell behaviour: stem cells on substrates as soft as brain tissue (~1 kPa) tend toward neural lineages, on ~10 kPa toward muscle, and on stiffer substrates toward bone. Peptide sequences such as RGD (arginine–glycine–aspartic acid) are grafted to promote attachment.
- **Manufacturability** in patient-specific shapes: salt leaching, freeze-drying, electrospinning (fibres of 0.1–5 µm resembling the collagen of extracellular matrix), and increasingly 3D printing and bioprinting of cell-laden hydrogel "bioinks" (Chapter 48).

Commercial success has come mainly in simpler tissues (skin substitutes, cartilage cell implantation, bone-graft substitutes, decellularised tissue matrices); complex, vascularised organs remain research targets, limited above all by the need to supply oxygen beyond a diffusion distance of about 100–200 µm.

39.6 Implants as material systems

39.6.1 Total hip replacement

More than a million total hip replacements are performed worldwide each year, and registries report that about 95 % survive ten years. A hip (Figure 39.5) combines several materials, each chosen for a different function: a **stem** in the femur (Ti-6Al-4V or CoCr, cemented with PMMA or press-fitted with a porous or HA-coated surface for bone ingrowth), a **head** (CoCr, ZTA or alumina, 28–36 mm in diameter) fitted on a taper, a **liner** (polyethylene or ceramic) and an **acetabular shell** (titanium, porous coated or AM lattice) in the pelvis.

The bearing couple has dominated the history of hip design because of the wear–particle–osteolysis chain. Each step releases particles of polyethylene of 0.1–1 µm, the size range macrophages phagocytose. Activated macrophages release cytokines that recruit osteoclasts, which resorb bone around the implant (**osteolysis**), leading to aseptic loosening, historically the commonest reason for revision.

Worked example — wear debris from a polyethylene liner. Wear of polyethylene against a polished CoCr head follows Archard's law (Chapter 45), $V = k\,F\,s$, with a wear factor of about $k = 1 \times 10^{-6}$ mm³/(N m) for conventional UHMWPE in hip simulators. Take an average joint force of 2000 N, a sliding distance of about 25 mm per step and one million steps a year ($s = 25$ km). The yearly wear volume is $V = 10^{-6} \times 2000 \times 25\,000 = 50$ mm³, which for a 28 mm head corresponds to a linear penetration of about $50/(\pi \times 14^2) = 0.08$ mm per year, in line with the 0.1–0.2 mm per year measured on retrieved conventional liners. If the debris is spheres of 0.5 µm diameter (volume 6.5×10^{-11} mm³), the 50 mm³ amounts to about 8×10^{11} particles a year. Clinical experience shows osteolysis becomes common when linear wear exceeds about 0.1 mm per year.

> **Status of the principle: Archard wear law**
> *Empirical.* Wear volume proportional to load and sliding distance fits steady-state sliding wear with a constant wear factor; in hips, the factor depends on cross-shear (multidirectional motion), lubrication, counterface roughness and third-body particles, so simulator tests under realistic kinematics are mandatory.

The response was a materials solution. **Highly crosslinked polyethylene** (HXLPE), irradiated with 50–100 kGy of gamma or electron radiation and then remelted or annealed to eliminate the free radicals that would cause oxidation, or stabilised with vitamin E, reduces the wear factor by 80–95 % (because crosslinks resist the orientation-softening of chains under multidirectional sliding), at some cost in fatigue toughness. Registry data now show markedly lower revision for osteolysis with HXLPE. **Ceramic-on-ceramic** bearings (ZTA on ZTA) wear least of all, but can fracture and occasionally squeak. **Metal-on-metal** bearings, revived for large heads with low volumetric wear, failed as the chapter's opening describes, because their nanometre debris and ions provoked adverse local tissue reactions; most large-head MoM hips were withdrawn after 2010. The lesson is that wear *volume* is not the only metric: particle size, number, chemistry and the dissolved-ion load decide the biological response.

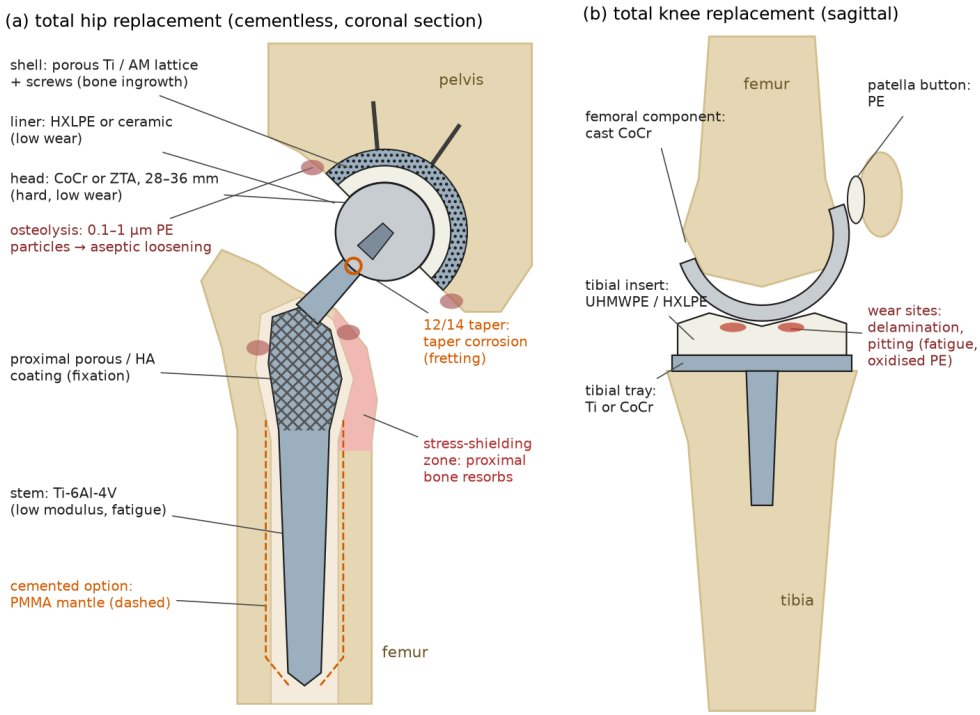


Figure 39.5 — A total hip and a total knee replacement are multi-material systems; each component is chosen for stiffness, wear, corrosion or fixation.

39.6.2 Total knee replacement

A knee replacement articulates a cast **CoCr femoral component** against a polyethylene **tibial insert** locked into a titanium or CoCr **tibial tray**. Its kinematics differ from the hip: rolling and sliding over a smaller, less conforming contact give high contact stresses (often above polyethylene's yield stress of about 20 MPa) and cyclic subsurface stresses. The historical failure mode was fatigue **delamination** and pitting of the insert, aggravated by oxidation of gamma-irradiated polyethylene sterilised in air and stored for years. Sterilisation in inert gas, remelted HXLPE and vitamin-E stabilisation have largely solved oxidation, and knee survival now matches hips at about 90–95 % at 15 years.

39.6.3 Dental implants

A dental implant is a threaded titanium (CP grade 4, Ti-6Al-4V or Ti–Zr) screw placed in the jaw bone, which osseointegrates over 3–6 months and then carries an abutment and crown. Ten-year survival above 95 % rests on surface engineering: grit blasting and acid etching create micro- and nano-roughness (Ra about 1–2 µm), increasing bone-to-implant contact and the torque needed to remove the screw several-fold relative to machined surfaces. Ceramic implants of zirconia are an alternative for metal-sensitive patients. The crown itself is now commonly monolithic zirconia or lithium-disilicate glass-ceramic (Chapter 17).

The main long-term threat is **peri-implantitis**, a bacterial infection of the tissue around the implant, which links surface roughness (good for bone, bad for biofilm) to a trade-off between osseointegration and cleanability. Dental amalgam (Hg–Ag–Sn–Cu), used for 150 years and still among the most durable fillings, is being phased down under the Minamata Convention on mercury in favour of resin composites and glass-ionomer cements.

39.6.4 Cardiovascular stents and other blood-contacting devices

A coronary **stent** is a mesh tube expanded inside a narrowed artery to keep it open. Balloon-expandable stents must deform plastically to a larger diameter and then resist the artery's recoil: a high yield strength to resist recoil, but low enough to expand at balloon pressures, plus high ductility, radiopacity for X-ray visibility, and fatigue resistance for about 40 million heart beats a year. The first stents were 316L stainless steel with struts around 140 μm; CoCr (L605, MP35N) and platinum–chromium alloys, stronger and more radiopaque, allowed struts of 60–90 μm, which reduce vessel injury and restenosis. Self-expanding peripheral stents use superelastic NiTi (Chapter 37), whose constant plateau force and kink resistance suit vessels in the leg and neck that are bent and compressed.

Bare-metal stents reduced acute closure but 20–30 % of arteries renarrowed within months as smooth-muscle cells proliferated through the mesh (**in-stent restenosis**). **Drug-eluting stents** (DES) coat the struts with a thin polymer layer (5–15 μm) carrying an antiproliferative drug (sirolimus, everolimus, zotarolimus, paclitaxel) released over weeks to months, reducing restenosis below 10 %. Their polymers (originally durable PBMA/PEVA and SIBS, now fluoropolymers or biodegradable PLGA and PLLA) must not crack or delaminate when the stent expands by 200–300 %, must release the drug at the right rate (Section 39.7), and must not themselves provoke inflammation, which was linked to late stent thrombosis in early designs. Stent fatigue qualification simulates ten years of pulsation, about 400 million cycles.

Other blood-contacting devices show the same structure–property logic: mechanical heart valves use pyrolytic carbon leaflets for wear resistance and thromboresistance (but require lifelong anticoagulation), while bioprosthetic valves of glutaraldehyde-fixed bovine pericardium avoid anticoagulation but calcify and wear out in 10–20 years; large-diameter vascular grafts of woven PET or expanded PTFE work, but small-diameter (< 6 mm) synthetic grafts thrombose, a problem tissue engineering is attacking.

39.7 Drug delivery systems

39.7.1 Why control release?

A tablet or injection produces a concentration peak followed by decay: too high at first (toxicity), too low later (no effect), and repeated dosing is needed. A **controlled-release** system keeps the concentration in a therapeutic window for a defined time, targets the site where it is needed, protects fragile drugs (proteins, nucleic acids) from degradation, and improves adherence. Materials engineering provides the release mechanism.

39.7.2 Release mechanisms

- **Diffusion through a membrane (reservoir systems).** A drug core surrounded by a rate-controlling polymer membrane releases at a constant rate as long as the core concentration stays saturated (**zero order**), by Fick's first law (Chapter 8): $dM/dt = ADK\Delta C/\ell$ for membrane area A, thickness ℓ, partition coefficient K. Examples are transdermal patches (nicotine, fentanyl), contraceptive implants in ethylene–vinyl acetate rods and intrauterine systems.
- **Diffusion from a matrix (monolithic systems).** Drug dispersed in a polymer matrix at concentration C_0 above its solubility C_s in the polymer dissolves at a receding front. Higuchi's analysis gives the cumulative release per unit area

$$Q = \sqrt{D\,(2C_0 - C_s)\,C_s\,t}$$

  so release scales as $\sqrt{t}$: fast at first, slowing with time.
- **Degradation- and erosion-controlled release.** In PLGA microspheres and implants, the drug is released as the polymer hydrolyses (Section 39.4), with an initial burst from surface drug, a lag, and then erosion-controlled release over weeks to months. Monthly and three-monthly injections of peptide hormones (leuprolide) and of antipsychotics use this mechanism.
- **Osmotic systems.** A tablet with a semipermeable shell draws water in at a constant rate and pumps drug solution through a laser-drilled orifice at zero order.

- **Stimulus-responsive systems.** Hydrogels that swell with pH, temperature or glucose (Chapter 37) release in a chosen environment.
- **Nanocarriers.** Liposomes (lipid bilayers, Chapter 37) carry doxorubicin in a polyethylene-glycol-coated "stealth" form that circulates for days; **lipid nanoparticles** of ionisable lipids, cholesterol, phospholipid and PEG-lipid protect mRNA and release it into cells, the basis of mRNA vaccines; polymeric micelles and nanoparticles of PLGA and PEG–PLA carry poorly soluble drugs. Passive accumulation in tumours through leaky blood vessels was once expected to deliver large fractions of a dose; in practice typically less than 1 % of injected nanoparticles reach a solid tumour, and targeting remains a central research problem.

> **Status of the principle: Higuchi square-root-of-time release**
> *Approximate.* Derived for a planar matrix with $C_0 \gg C_s$, perfect sink conditions and a constant diffusivity; it fits the first 50–60 % of release from many matrix systems but not swelling or eroding matrices, where empirical power laws $M_t/M_\infty = kt^n$ are fitted instead.

Worked example — release from a drug-eluting stent coating. A coating 10 μm thick contains drug at $C_0 = 200$ kg/m³ (20 wt% loading), solubility in the polymer $C_s = 1$ kg/m³, and diffusivity $D = 1 \times 10^{-15}$ m²/s. The drug load per unit area is $C_0 \ell = 200 \times 10^{-5} = 2 \times 10^{-3}$ kg/m² (200 μg/cm², close to commercial loadings). After one day ($t = 8.64 \times 10^4$ s), $Q = \sqrt{10^{-15} \times 399 \times 1 \times 8.64 \times 10^4} = 1.86 \times 10^{-4}$ kg/m², 9 % of the load; after 7 days 25 %; after 28 days 49 %. Half the drug is released when $Q = 10^{-3}$ kg/m², at $t = (10^{-3})^2/(10^{-15} \times 399) = 2.5 \times 10^6$ s, about 29 days, matching the one-to-three-month window over which smooth-muscle proliferation must be suppressed. Because $Q \propto \sqrt{D}$, a tenfold change in diffusivity (a different polymer, or a different crystallinity of the same one) changes the release time tenfold, which is why coating morphology must be controlled as tightly as composition (Figure 39.6).

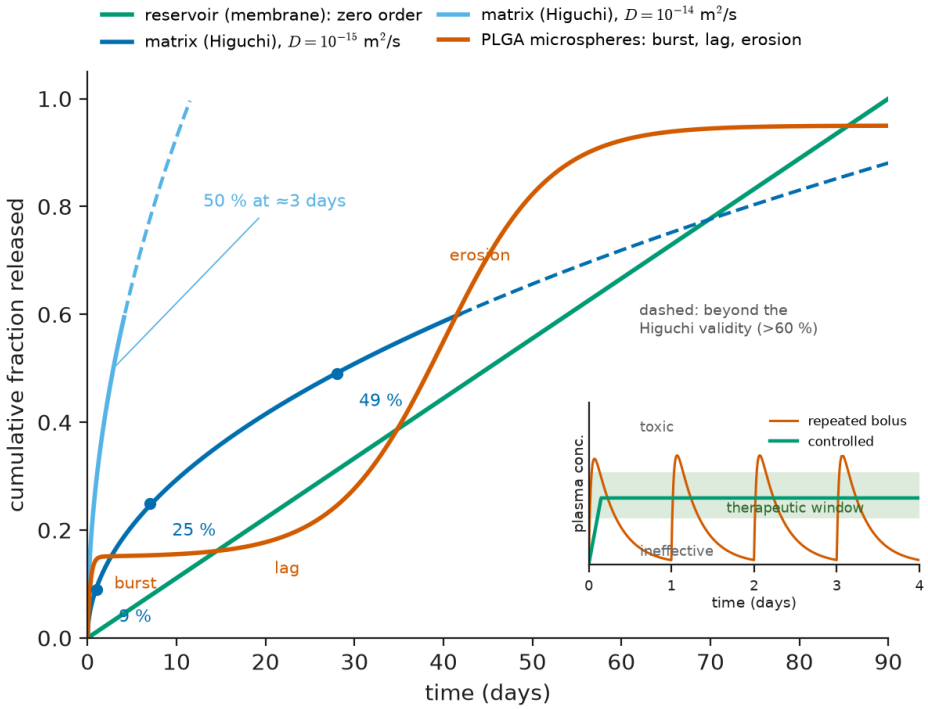


Figure 39.6 — Release mechanism sets the shape of the release curve: zero order from reservoirs, square root of time from matrices, burst-lag-erosion from degrading polyesters.

39.8 Sterilization, testing and regulation

39.8.1 Sterilization

Every implant must be sterile, and every sterilization method interacts with the material. Sterility is defined statistically by a **sterility assurance level** (SAL): the probability that a single item is non-sterile after the process, normally 10^{-6} for implants.

Microbial death under a constant lethal condition is first order, so the surviving population falls by a factor of ten in each **D-value** of exposure:

$$N(t) = N_0\, 10^{-t/D}$$

> **Status of the principle: Log-linear microbial inactivation**
> *Empirical.* First-order kinetics fitted to survivor curves of standard bioindicator organisms; real curves show shoulders and tails (clumped or resistant subpopulations), which is why validated processes add safety margins.

Worked example — designing a steam cycle. A device carries a bioburden of 10^3 organisms and must reach SAL 10^{-6}, a reduction of 10^9, or 9 D-values. For a resistant spore with $D_{121} = 1.5$ min at 121 °C, the exposure is $9 \times 1.5 = 13.5$ min, which is why the standard steam cycle is 15 minutes at 121 °C. With a z-value of 10 K (the temperature rise that reduces D tenfold), at 134 °C $D = 1.5 \times 10^{-1.3} = 0.075$ min and 9 D-values take 0.68 min; the common 3-minute cycle at 134 °C carries a large margin.

The materials consequences differ by method:

Method	Conditions	Suited to	Materials concerns
Steam (autoclave)	121 °C 15 min or 134 °C 3 min, saturated steam	Metals, ceramics, glass, some polymers (PP, PSU, PEEK)	Hydrolysis of polyesters; distortion above T_g; zirconia ageing
Ethylene oxide	30–60 °C, gas, humidity, aeration	Heat-sensitive polymers, complex devices, catheters	Toxic, carcinogenic residues (limits in ISO 10993-7); long aeration
Gamma radiation	25 kGy typical (Co-60)	Single-use disposables, packaged implants	Chain scission and crosslinking: PE oxidation, PTFE and PP embrittlement, yellowing of PC and PVC
Electron beam	25 kGy in seconds	Thin, low-density products	Same chemistry as gamma; limited penetration
Hydrogen peroxide plasma/vapour	40–50 °C	Heat- and moisture-sensitive instruments	Not for cellulose or liquids; limited lumen penetration

The polyethylene story is the clearest example. Gamma irradiation in air creates free radicals in UHMWPE; trapped in crystalline regions, they react with oxygen over years on the shelf and in the body, causing chain scission, embrittlement and the delamination of knee inserts of the 1990s. Changing the sterilization atmosphere to inert gas, then remelting and vitamin E stabilisation, solved a problem that was not visible in any short-term test.

39.8.2 Biological and mechanical testing

Biological evaluation follows the ISO 10993 series, organised as risk management (ISO 14971) rather than a fixed checklist. The type and duration of body contact (surface, external communicating, implant; limited < 24 h, prolonged up to 30 days, long-term > 30 days) determine which end-points must be addressed: chemical characterisation of extractables and leachables (part 18) and toxicological risk assessment, cytotoxicity (part 5), sensitisation and irritation (parts 10 and 23), systemic toxicity (part 11), genotoxicity (part 3), implantation (part 6), haemocompatibility (part 4), and degradation products of polymers, ceramics and metals (parts 13–15). Increasingly, chemical characterisation and existing data replace animal tests where they can.

Mechanical and functional testing reproduces the loads of service on the finished device, not just on material coupons: fatigue of femoral stems under a cyclic load of several kilonewtons for 5–10 million cycles (ISO 7206-4), hip and knee wear in joint simulators for 5 million cycles with lubricant of bovine serum (ISO 14242, ISO 14243), fatigue of stents for 400 million cycles of simulated pulsation with finite-element analysis of strain amplitude, and corrosion tests of modular tapers under fretting. The metal-on-metal story shows the limits: simulators run with ideal cup positioning missed the edge-loading conditions that occurred in patients. Since then, standards have added adverse-condition testing.

39.8.3 Regulation

Medical devices are regulated by risk class. In the United States, the Food and Drug Administration places devices in Class I (low risk, such as bandages), Class II (moderate, such as many orthopaedic screws and catheters) and Class III (life-sustaining or implanted high-risk devices, such as heart valves and coronary stents). Most Class II devices reach the market by a **510(k)** notification showing substantial equivalence to an existing device; Class III devices require **premarket approval** with clinical

evidence. In the European Union the Medical Device Regulation (EU 2017/745) uses Classes I, IIa, IIb and III, with notified bodies auditing technical files and quality systems (ISO 13485), and it has tightened clinical-evidence and post-market surveillance requirements, partly in response to the metal-on-metal hip and breast-implant scandals. Joint registries, which track every implant and revision in a country, provide the long-term field data that no laboratory test can, and were the first to signal the metal-on-metal problem.

For the materials engineer, regulation has three consequences. A material's history matters: a grade with a long record in a similar application (Ti-6Al-4V ELI, CoCrMo, UHMWPE) can be justified by equivalence, while a new alloy or polymer needs the full biological evaluation, so innovation in biomaterials is slow and expensive. Processing is part of the material: a change of supplier, sintering furnace, sterilization method or surface treatment can require re-validation, as the zirconia recall showed. And the evidence is never complete: post-market surveillance treats every implanted device as part of a continuing experiment, so the structure–property–performance chain of this book is extended here by a final link, the patient's outcome.

What remains open

Open question	Where it is answered
How is the passive film formed and broken, and how are pitting, crevice and fretting corrosion analysed?	Chapters 42 and 43
How do Archard wear, lubrication regimes and contact mechanics apply to joint bearings?	Chapter 45
Why is titanium processed as α, α+β or β alloys, and how does that set its properties?	Chapter 16
How does transformation toughening work in zirconia, and why do ceramics follow Weibull statistics?	Chapters 14 and 17
How does hydrolytic degradation depend on polymer chemistry and crystallinity?	Chapters 20 and 21
How are porous titanium implants and scaffolds made by additive manufacturing and qualified?	Chapter 48
How are failed implants investigated and their failure statistics analysed?	Chapter 44
How are surfaces of implants characterised (XPS, contact angle, roughness) and protein layers studied?	Chapters 45 and 52
How can multi-objective selection balance modulus, strength, corrosion and cost for an implant?	Chapter 58
How do the lifecycle and resource footprint of single-use medical devices compare?	Chapter 59

Next: Chapter 40 turns from the body to energy, asking which material properties limit how efficiently energy can be converted and stored, and whether those materials can be made at the scale the energy transition needs.

Chapter 40 — Materials for Energy

Part VIII · Advanced and Emerging Materials

A city transit authority orders three hundred electric buses and must choose between two battery chemistries. Cells with a nickel-rich layered-oxide cathode store about a quarter more energy per kilogram. Cells with an iron-phosphate cathode are heavier but cheaper, contain no cobalt or nickel, survive two to three times as many cycles and are far less prone to thermal runaway in a depot full of parked vehicles. The authority picks iron phosphate. In the same month, a pipeline operator learns that fatigue cracks in the steel of a gas line it hoped to convert would grow ten to a hundred times faster in hydrogen, and an electrolyser maker discovers that the world mines less iridium in a year than its order book needs. Each was a materials decision, and in none did the best laboratory figure of merit win outright. Energy materials must be deployed at the scale of the whole energy system, last for decades and be cheap per unit of energy delivered over their life; this chapter surveys them with those constraints in view.

> **Which structural features make a material a good light absorber, battery electrode, ion conductor, catalyst, thermoelectric or capacitor electrode?**
> **Why do the materials with the best figures of merit so often lose to cheaper, more durable or more abundant ones?**

Concept	Engineering destination
Absorber band gap, absorption coefficient, defect tolerance	Choice of photovoltaic technology; module efficiency and lifetime
Intercalation hosts, specific capacity, volume change	Cell chemistry for vehicles, grid storage, consumer devices, aircraft
Liquid, polymer and ceramic electrolytes	Solid-state batteries, safety of cells and packs
Proton and oxide-ion conductors, catalysts, interconnects	PEM and solid-oxide fuel cells and electrolysers
Hydride thermodynamics, embrittlement of steels	Hydrogen tanks, pipelines, fittings and refuelling stations
Seebeck coefficient, figure of merit ZT	Waste-heat recovery, Peltier coolers, radioisotope generators
Double-layer and pseudocapacitive storage	Regenerative braking, grid frequency support, crane and lift buffers
Fuel and cladding requirements	Fission power (developed in Chapter 41)
Material intensity, critical raw materials	Supply risk and the deployable scale of each technology

Roadmap. Converting light (40.1) → storing charge in electrode hosts (40.2) and carrying ions through electrolytes (40.3) → converting fuel and making hydrogen (40.4) → storing hydrogen and resisting it (40.5) → converting heat (40.6) → storing charge at surfaces (40.7) → nuclear fuels in brief (40.8) → degradation, critical materials and scale (40.9).

> **Boundary.**
> Electrode potentials, the Nernst equation, Faraday's laws and reaction kinetics are electrochemistry fundamentals from the Advanced Chemistry book; they are used here without derivation (Chapter 42 applies them to corrosion). Band structure and carrier transport are in Chapters 31 and 32, light absorption in Chapter 36, ionic conduction in Chapter 25. Sizing of packs, stacks, arrays and power electronics is systems engineering, outside this book. Chapter 35 treats solar-cell absorbers at greater electronic depth.

40.1 Photovoltaic absorbers

What a good absorber must do

The thermodynamic limit for a single-junction solar cell under the unconcentrated solar spectrum is the detailed-balance (Shockley–Queisser) efficiency, about 33 % at a band gap near 1.34 eV (Chapters 35 and 36). The maximum is flat: any gap

between roughly 1.1 and 1.6 eV allows more than 30 %. So the band gap narrows the search but does not decide it. A practical absorber must in addition:

1. **Absorb strongly** so that it can be thin. Direct-gap semiconductors reach absorption coefficients α of 10^4–10^5 cm^{-1} just above the gap; indirect-gap silicon manages 10^2–10^3 cm^{-1} in the near infrared, because a phonon must take part in each absorption event.
2. **Let carriers reach the contacts before they recombine.** The minority-carrier diffusion length $L = \sqrt{D\tau}$ must exceed the absorber thickness. The lifetime τ is set by recombination through deep levels of point defects, dislocations and grain boundaries (Chapter 5), so this is a statement about defect chemistry.
3. **Be either nearly perfect or defect-tolerant.** Silicon and GaAs succeed because they can be grown with vanishingly few deep-level defects; CdTe, CIGS and the halide perovskites succeed because their native defects and grain boundaries happen to be electrically benign or can be made so.
4. **Survive** 25–30 years of ultraviolet light, module temperatures up to about 85 °C, humidity and thermal cycling, and be made cheaply from available elements.

Worked example — how thick must the absorber be? By the Beer–Lambert law the fraction of light absorbed in one pass through a thickness t is $1 - e^{-\alpha t}$. Absorbing 90 % needs $e^{-\alpha t} = 0.1$, so $t = \ln 10/\alpha = 2.303/\alpha$.

- Crystalline silicon at 900 nm, $\alpha \approx 306$ cm^{-1}: $t = 2.303/306 = 7.5 \times 10^{-3}$ cm, or 75 µm.
- CdTe at 800 nm, $\alpha \approx 1 \times 10^4$ cm^{-1}: $t = 2.3 \times 10^{-4}$ cm, or 2.3 µm.
- A lead halide perovskite at 600 nm, $\alpha \approx 5 \times 10^4$ cm^{-1}: $t = 4.6 \times 10^{-5}$ cm, or 0.46 µm.

Silicon needs some thirty times the thickness of CdTe and about 160 times that of the perovskite. Surface texturing and rear reflectors lengthen the light path (ideally by up to $4n^2 \approx 50$ for a randomizing surface), which is why commercial wafers are 130–160 µm thick rather than several hundred. The thin-film absorbers need only micrometres, so their carriers have to travel only micrometres, and polycrystalline films deposited cheaply on glass or foil are good enough (Figure 40.1).

> **Status of the principle: Beer–Lambert absorption**
> *Approximate.* Exact for a homogeneous, non-scattering medium in a single pass; light trapping, thin-film interference and photon recycling modify it in real cells.

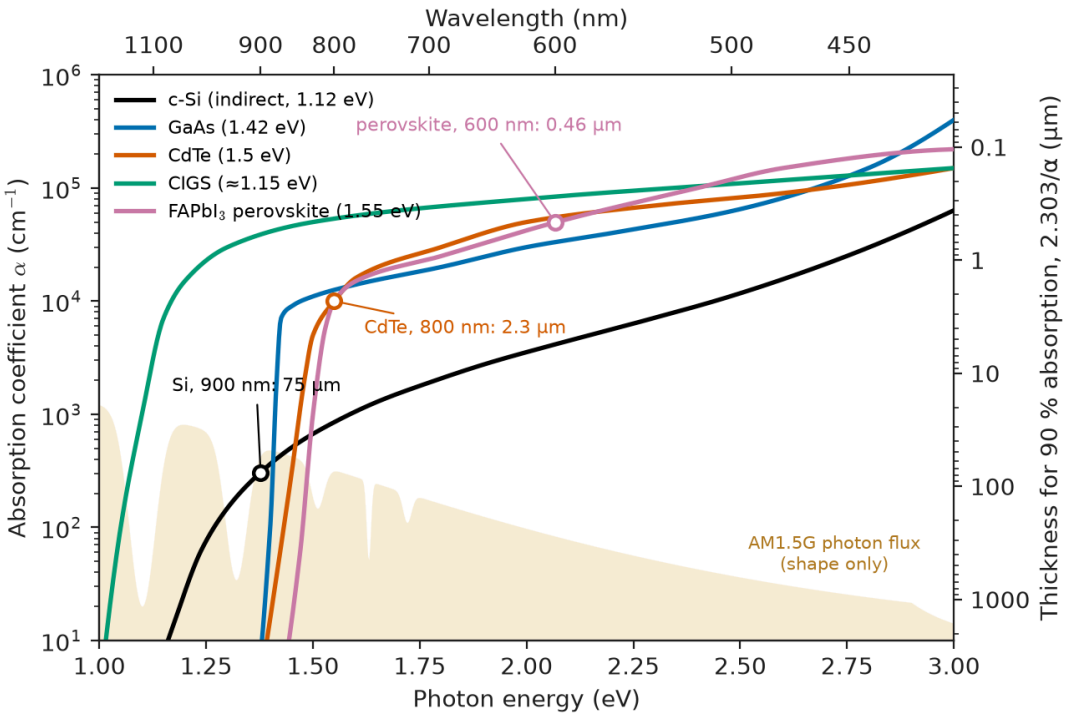


Figure 40.1 — Direct-gap thin-film absorbers absorb sunlight within a micrometre or two, whereas indirect-gap silicon needs about a hundred.

Crystalline silicon

About 95 % of the photovoltaic capacity installed each year is crystalline silicon: its 1.12 eV gap is near the optimum, the element is abundant, and microelectronics has made it the purest crystal produced at scale. Minority-carrier lifetimes above 1 ms in n-type wafers give millimetre diffusion lengths, more than compensating for weak absorption. The frontier is now at **surfaces and contacts**, where passivating layers (SiO_2 tunnel oxide plus doped polysilicon in TOPCon cells, amorphous silicon in heterojunction cells) suppress recombination; records exceed 27 % against a practical limit of about 29.4 % set by Auger recombination. Bulk degradation is point-defect chemistry: boron–oxygen complexes cause light-induced degradation in p-type wafers, one reason for the move to n-type.

CdTe and CIGS

Cadmium telluride (direct gap 1.45–1.5 eV), deposited a few micrometres thick onto glass at 500–600 °C, is the most successful thin film. Its grain boundaries, normally fatal in a polycrystalline semiconductor, are made benign by a $CdCl_2$ treatment near 400 °C. Modules reach 19–20 %; the liabilities are cadmium toxicity, handled by encapsulation and take-back recycling, and tellurium supply (Section 40.9).

$Cu(In,Ga)Se_2$ (CIGS) is a chalcopyrite whose gap rises from 1.0 to 1.7 eV with Ga/(In+Ga); grading it through the film builds in a field that drives electrons to the junction, and alkali atoms passivate its grain boundaries. Deposited on steel or polyimide foil, it gives flexible modules for façades and vehicles, at the cost of multi-element process control and indium and gallium supply.

Halide perovskites

Metal halide perovskites ABX_3 (A = methylammonium, formamidinium or Cs; B = Pb or Sn; X = I, Br, Cl) went from first use as absorbers in 2009 to certified single-junction efficiencies near 27 % in about fifteen years. Three structural features explain the speed:

- **A direct, tunable gap**, from about 1.5 eV ($FAPbI_3$) to 2.3 eV (pure bromide), adjustable by halide mixing, which suits them for the top cell of a tandem.
- **Strong absorption**: half a micrometre absorbs most above-gap light, as the worked example showed.
- **Defect tolerance**: the dominant native defects, vacancies, form shallow rather than deep levels because the band edges derive from antibonding orbitals. Films spin-coated or slot-die coated from solution and annealed near 100 °C have diffusion lengths above 1 μm.

The soft, ionic lattice that makes them easy to process is also their weakness. Halide vacancies migrate under light and field, causing hysteresis, halide segregation and electrode corrosion; moisture and heat decompose them; and each square metre contains a few grams of soluble lead. Caesium–formamidinium compositions and barrier encapsulation have improved stability greatly, though not yet to silicon's 25-year standard. The most promising route is the **perovskite-on-silicon tandem**, which has exceeded 34 %, beyond any single-junction limit.

III–V multijunctions and the space market

Epitaxial GaAs reaches about 29 % as a single junction, closest of any material to its limit. Stacking junctions (GaInP 1.9 eV / GaAs 1.42 eV / Ge 0.67 eV, or newer four- to six-junction designs) gives each part of the spectrum a matching gap and cuts thermalization loss, to about 39 % at one sun and 47 % under concentration. Single-crystal substrates and slow epitaxy (Chapter 49) make them 100–1000 times costlier per unit area than silicon, so they are used where watts per kilogram and radiation hardness justify any cost: satellites and high-altitude aircraft (Chapter 41).

Table 40.1 — Photovoltaic absorbers (typical values; records as of the mid-2020s, rounded).

Absorber	Band gap (eV)	Gap type	Typical thickness	Record cell efficiency	Main materials limitation
c-Si (monocrystalline)	1.12	Indirect	130–160 µm	≈27 %	Weak absorption; surface recombination; Ag contacts
CdTe	1.45–1.5	Direct	2–4 µm	≈23 %	p-doping; Te supply; Cd toxicity
CIGS	1.0–1.2 (graded)	Direct	1.5–2.5 µm	≈23.6 %	Process complexity; In and Ga supply
Halide perovskite	1.5–1.7 (to 2.3)	Direct	0.4–0.8 µm	≈27 % (≈34 % on Si tandem)	Ion migration; moisture and heat stability; Pb
GaAs	1.42	Direct	1–3 µm	≈29 %	Substrate and growth cost
III–V multijunction	0.7–1.9 stack	Direct	5–10 µm total	≈39 % (1 sun), ≈47 % (concentrated)	Cost; space and concentrator niches

40.2 Battery materials I: intercalation electrodes

Storing ions in a host

A lithium-ion cell stores energy by moving $\mathrm{Li^+}$ ions between two host materials through an electrolyte while electrons travel through the external circuit. On discharge, lithium leaves the negative electrode (conventionally the anode) and enters the positive electrode (the cathode). The best hosts work by **intercalation**: ions slide into vacant sites between layers or along channels of a crystal whose framework stays essentially intact, and an equal number of electrons reduces a transition metal (or carbon) in the host. For a layered oxide cathode,

$$\mathrm{Li_{1-x}CoO_2} + x\,\mathrm{Li^+} + x\,e^- \;\rightleftharpoons\; \mathrm{LiCoO_2}$$

The cell voltage equals the difference in lithium chemical potential between the two electrodes divided by the elementary charge; why that difference takes the value it does is electrochemistry. The materials questions are four: how much lithium the host accepts per unit mass (**capacity**), at what potential, how fast ions and electrons move through it (**rate**), and how much the lattice changes when they do (**cycle life**). The same logic applies to sodium-ion, magnesium-ion and other intercalation chemistries.

Specific capacity

If a host of molar mass M (g/mol) reversibly exchanges n electrons and ions per formula unit, Faraday's constant gives its theoretical **specific capacity**

$$Q = \frac{nF}{3.6\,M} \quad (\mathrm{mAh\,g^{-1}})$$

where the factor 3.6 converts coulombs per gram into milliampere-hours per gram.

> **Status of the principle: theoretical specific capacity**
> *Derived* from Faraday's law and stoichiometry. Practical capacities are lower because only part of the lithium can be cycled without destroying the host, and because binder, conductive carbon, current collectors and electrolyte add inactive mass.

Worked example — capacities of common electrodes. With $F = 96\,485$ C/mol:

- Graphite to $\mathrm{LiC_6}$, one Li per six carbons, $M = 6 \times 12.011 = 72.07$ g/mol: $Q = 96\,485/(3.6 \times 72.07) = 372$ mAh/g.
- $\mathrm{LiCoO_2}$, $M = 97.87$: $Q = 274$ mAh/g theoretical. Only about 0.5–0.65 Li can be removed before the layered structure transforms and loses oxygen, so practical values are 140–185 mAh/g.
- $\mathrm{LiFePO_4}$, $M = 157.76$: $Q = 170$ mAh/g; practical 155–165 mAh/g, since nearly all the lithium can be cycled.
- $\mathrm{LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2}$ (NMC811), $M = 97.28$: $Q = 276$ mAh/g theoretical, about 200 mAh/g practical.
- Silicon to $\mathrm{Li_{15}Si_4}$, 3.75 Li per Si, $M = 28.09$: $Q = 3.75 \times 96\,485/(3.6 \times 28.09) = 3578$ mAh/g, almost ten times graphite.

- $Li_4Ti_5O_{12}$ to $Li_7Ti_5O_{12}$, three Li per formula unit, $M = 459.1$: $Q = 175$ mAh/g.

Anode materials

Graphite is the near-universal anode. Lithium enters between the graphene layers in ordered stages up to LiC_6 at 0.05–0.25 V versus Li/Li^+, with only about 10 % volume expansion. The low potential is also its difficulty: no practical organic electrolyte is stable there, and on the first charge the electrolyte is reduced to a nanometre-scale **solid–electrolyte interphase** (SEI) of lithium carbonate, fluoride and organic salts. A good SEI conducts Li^+ but not electrons and so stops further reaction; its formation consumes 5–10 % of the cell's lithium, and its slow growth is the main ageing mechanism of most cells. At high charging rates or low temperature lithium plates as metal on the graphite instead, which usually limits fast charging.

Silicon stores lithium by **alloying**, at about 0.4 V, with a volume expansion of about 280 %. Micrometre particles crack on the first cycles (Chapter 14), lose contact and expose fresh surface that grows new SEI. Nanostructuring is the remedy (Chapter 38): particles below about 150 nm do not fracture, and SiO_x or porous Si–C composites buffer the swelling. Commercial anodes blend 5–15 % silicon or SiO_x into graphite.

Lithium titanate (spinel $Li_4Ti_5O_{12}$) changes volume by only about 0.2 % and works at 1.55 V, above the potential where the electrolyte is reduced, so it forms almost no SEI, cannot plate lithium and survives tens of thousands of fast-charge cycles, at the cost of 1.4 V of cell voltage and half of graphite's capacity.

Lithium metal (3861 mAh/g) is the limiting anode, but in liquid electrolytes it deposits unevenly as mossy or dendritic lithium; it is the main motivation for solid electrolytes (Section 40.3).

Cathode materials

$LiCoO_2$ (LCO) has the layered α-$NaFeO_2$ structure, with alternating layers of Li and CoO_2 octahedra. It has the highest volumetric energy density of the common cathodes and still dominates phones and laptops, but cobalt is costly and its supply concentrated.

NMC ($LiNi_xMn_yCo_zO_2$) keeps the layered structure but replaces most cobalt with nickel, the main redox-active ion, and manganese, which stays as Mn^{4+} and stabilizes the lattice. Raising nickel trades stability for capacity: high-nickel cathodes release lattice oxygen at lower temperature, feeding thermal runaway, and near full charge an abrupt anisotropic lattice contraction cracks polycrystalline particles along their internal grain boundaries. Single-crystal particles, oxide coatings and nickel-rich-core, manganese-rich-shell particles are the materials responses.

$LiFePO_4$ (LFP) has the olivine structure, with lithium in one-dimensional channels and oxygen locked in PO_4 tetrahedra by strong covalent P–O bonds. That polyanion framework is why LFP does not release oxygen even when abused. Its lower potential (3.45 V) and poor conductivity (overcome by 100–300 nm carbon-coated particles) cost energy, but it is cheap, free of cobalt and nickel and long-lived, which is why it now dominates stationary storage and much of the vehicle market, as in the bus decision that opened this chapter.

Table 40.2 — Lithium-ion electrode materials (typical values).

Material	Structure / mechanism	Average potential vs Li/Li^+ (V)	Theoretical capacity (mAh/g)	Practical capacity (mAh/g)	Volume change	Key limitation
Graphite	Layered, intercalation	≈0.1	372	340–360	≈10 %	SEI growth; Li plating at fast charge
Si	Alloying	≈0.4	3578 ($Li_{15}Si_4$)	1000–3000 (in composites)	≈280 %	Fracture; continual SEI growth
$Li_4Ti_5O_{12}$	Spinel, intercalation	1.55	175	160–170	≈0.2 %	Low cell voltage
Li metal	Plating	0	3861	—	No host	Dendrites; reactivity
$LiCoO_2$	Layered	≈3.9	274	140–185	2–3 %	Co cost; instability at deep delithiation
NMC811	Layered	≈3.75	276	190–210	≈5 % (abrupt near top of charge)	Particle cracking; O_2 release
$LiFePO_4$	Olivine, two-phase	3.45	170	155–165	≈6.5 %	Low conductivity; lower energy
$LiMn_2O_4$	Spinel	≈4.0	148	100–120	≈7 %	Mn dissolution above ≈50 °C

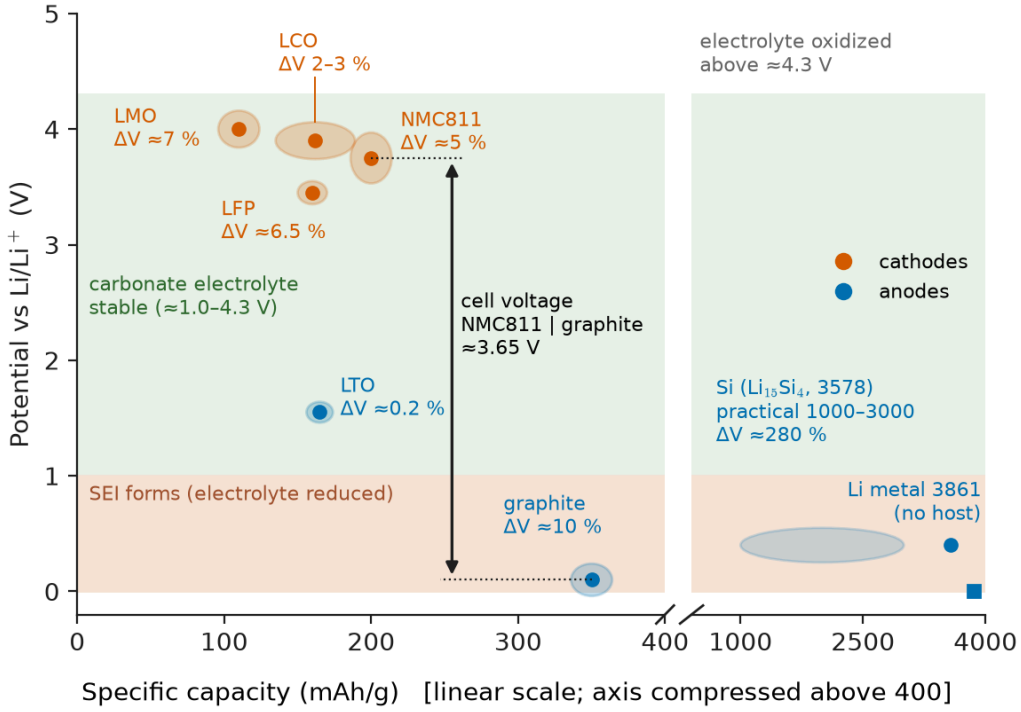


Figure 40.2 — Electrode materials by potential and specific capacity: cell voltage is the vertical gap between cathode and anode, and the electrolyte must survive the whole span.

Matching the electrodes

Because the same charge passes through both electrodes, the masses needed per unit charge add, and the combined capacity of the pair per gram of total active material is $Q_{pair} = (1/Q_c + 1/Q_a)^{-1}$ (Figure 40.2).

Worked example — what a better anode buys.

- LFP (160 mAh/g) with graphite (350 mAh/g): $Q_{pair} = (0.00625 + 0.00286)^{-1} = 110$ mAh/g. At about 3.3 V, that is $110 \times 3.3 = 363$ Wh per kilogram of active material.
- NMC811 (200 mAh/g) with graphite: $Q_{pair} = (0.00500 + 0.00286)^{-1} = 127$ mAh/g, or about 465 Wh/kg at 3.65 V.
- NMC811 with a silicon-rich anode at 2000 mAh/g: $Q_{pair} = (0.00500 + 0.00050)^{-1} = 182$ mAh/g.
- NMC811 with lithium metal (3861 mAh/g): $Q_{pair} = 190$ mAh/g.

A nearly sixfold increase in anode capacity raises the pair by 43 %, and going on to lithium metal adds only another 4–5 %, because the cathode now dominates the active mass. The cathode is the bottleneck of lithium-ion energy density, which is why most research effort goes into cathode nickel content and voltage. (Lithium metal's real advantage is volumetric and in removing the host altogether.) A finished cell reaches roughly half the active-material figure, 160–300 Wh/kg, once electrolyte, separator, current collectors and casing are added.

Beyond lithium-ion

Sodium-ion cells apply intercalation to abundant sodium, with **hard-carbon** anodes (graphite barely accepts sodium) and layered-oxide, phosphate or Prussian-blue cathodes: 20–30 % less energy than LFP, but no lithium, cobalt or nickel, and aluminium instead of copper foil. **Conversion chemistries** (Li–S, metal–air) promise more energy because the host disappears, which is exactly why they suffer the volume changes and poor reversibility that intercalation avoids.

40.3 Battery materials II: electrolytes and the solid-state cell

What an electrolyte must do

The standard liquid electrolyte, $LiPF_6$ in mixed carbonates, conducts Li^+ at about 10^{-2} S/cm, but it is flammable, decomposes above about 4.3 V and below about 1 V versus Li/Li^+ (it survives only behind passivating interphases, which sacrificial additives such as fluoroethylene carbonate help build), and cannot stop lithium dendrites. A **solid electrolyte** with comparable conductivity would allow a lithium-metal anode and a non-flammable cell. It needs a room-temperature conductivity above about 10^{-3} S/cm with negligible electronic leakage, a transference number near 1, stability against lithium and a 4 V cathode (or a self-limiting interphase), mechanical properties that keep contact and resist filament growth, and low processing cost.

Ion transport in solids

Ions move through a crystal by thermally activated hops between sites, via vacancies or interstitials (Chapters 8 and 25). The conductivity follows

$$\sigma T = A \exp\left(-\frac{E_a}{k_B T}\right)$$

> **Status of the principle: Arrhenius ionic conductivity**
> *Empirical* in practice. The form follows from activated hopping (Chapter 8), but A and E_a are fitted, and many electrolytes show curvature or a change of slope at phase transitions or, in polymers, near the glass transition.

Superionic conductors reach liquid-like conductivity because their structure offers many more equivalent sites than mobile ions, joined by wide bottlenecks, so the barrier falls to 0.2–0.4 eV; large, polarizable anions (S^{2-} more than O^{2-}) lower it further, which is why sulfides conduct best.

The main solid-electrolyte families

Garnet oxides. Cubic $Li_7La_3Zr_2O_{12}$ (LLZO, doped with Al, Ga or Ta) is stable against lithium metal. Its problems are a ceramic's (Chapter 17): sintering above 1100 °C while lithium evaporates, stiffness (about 150 GPa) and brittleness that spoil contact with electrodes, and an insulating Li_2CO_3 skin in moist air. Lithium filaments still penetrate it near 1 mA/cm² along grain boundaries and from surface flaws: a lithium-filled crack is driven by pressure like a hydraulic fracture (Chapter 14), so stiffness alone does not stop dendrites.

Sulfides. $Li_{10}GeP_2S_{12}$ (LGPS) matches liquids; the argyrodites Li_6PS_5X (X = Cl, Br) give 1–5 mS/cm without germanium. Sulfides are soft (20–30 GPa) and densify by cold pressing, giving good particle contact without sintering. But they are reduced

by lithium and oxidized by cathodes above about 2.5 V, so cathode particles need coatings, and moisture releases toxic H_2S, so cells are made in dry rooms.

NASICON phosphates such as LATP are air-stable and tolerate high-voltage cathodes, but their Ti^{4+} is reduced by lithium; the sodium analogues, with sodium β″-alumina, serve hot sodium–sulfur grid batteries. **Polymers** such as PEO–LiTFSI are easy to process, but ions move only with polymer segments above the glass transition, so such cells run warm, at 60–80 °C.

Table 40.3 — Ionic conductors for batteries, fuel cells and electrolysers (typical values; Figure 40.3).

Electrolyte	Mobile ion	Conductivity (S/cm)	At	E_a (eV)	Main limitation
$LiPF_6$ in carbonates (liquid reference)	Li^+ and anions	10^{-2}	25 °C	≈0.1–0.2	Flammable; narrow window; dendrites
PEO–LiTFSI	Li^+	10^{-5} / 10^{-4}–10^{-3}	25 / 60–80 °C	≈0.5–1	Low room-temperature conductivity
LLZO (doped garnet)	Li^+	10^{-4}–10^{-3}	25 °C	0.3–0.4	Brittle; sintering; Li_2CO_3
Li_6PS_5Cl (argyrodite)	Li^+	1–5 × 10^{-3}	25 °C	0.2–0.3	Narrow window; H_2S
$Li_{10}GeP_2S_{12}$	Li^+	1.2 × 10^{-2}	25 °C	≈0.25	Ge cost; reduced by Li
LATP (NASICON)	Li^+	10^{-4}–10^{-3}	25 °C	0.3–0.4	Ti^{4+} reduced by Li
Na-β″-alumina	Na^+	0.2–0.4	300 °C	≈0.3	Brittle; hot operation
8YSZ	O^{2-}	0.01 / 0.1	700 / 1000 °C	0.8–1.0	Needs high temperature
Nafion (hydrated)	H^+	≈0.1	80 °C, humidified	≈0.1–0.2	Needs water; below 100 °C

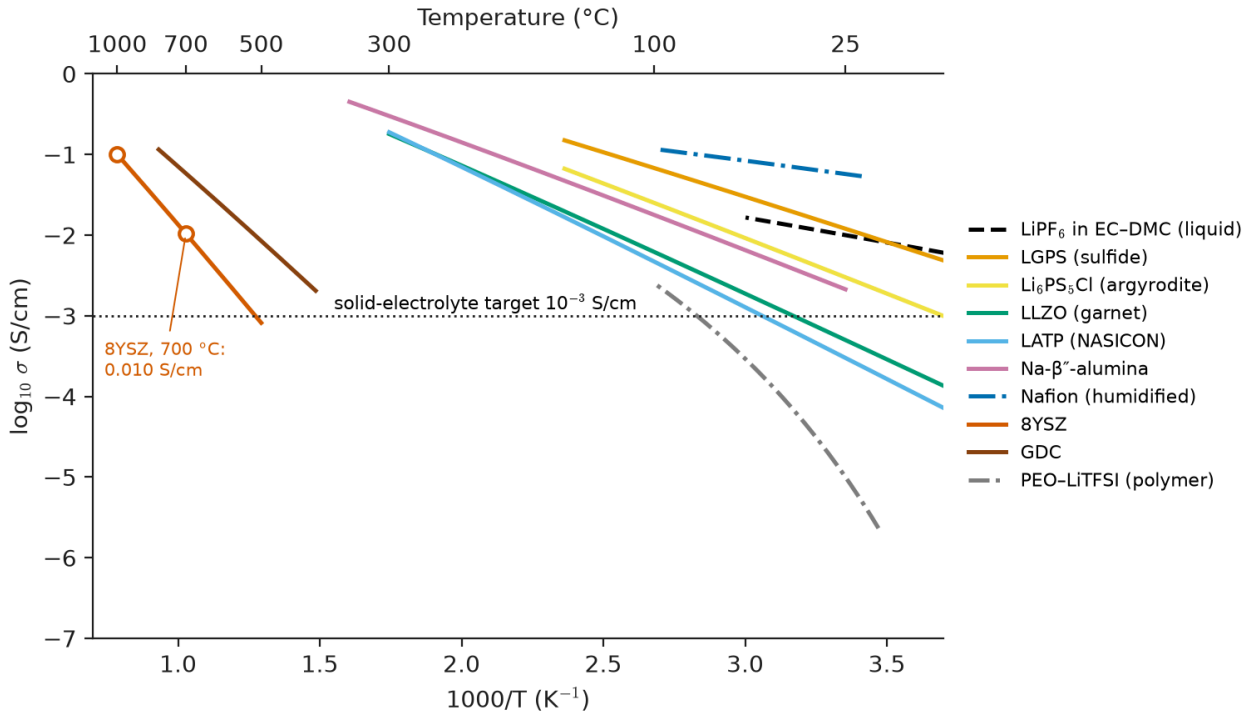


Figure 40.3 — Arrhenius plot of ionic conductors: sulfides rival liquid electrolytes at room temperature, while oxide-ion conductors need several hundred degrees.

Interfaces are the real problem

A liquid wets every particle; a solid touches its neighbours only at points. Cycling particles swell and shrink, opening gaps; reactions grow resistive interphases. Stack pressures of several megapascals, nanometre interlayers deposited by atomic-layer

deposition (Chapter 49) and composite cathodes are the current remedies, and operando methods (Chapter 54) are how these interfaces are studied.

40.4 Fuel-cell and electrolyser materials

A fuel cell is an electrochemical cell fed continuously with fuel at the anode and oxygen at the cathode, separated by an electrolyte that passes ions but not electrons. An electrolyser runs the same reactions backwards to make hydrogen from water and electricity. Every layer of both is a materials problem (Figure 40.4).

Proton-exchange membrane cells

Membrane. Perfluorosulfonic-acid (PFSA) ionomers such as Nafion have a PTFE-like backbone with side chains ending in sulfonic acid groups. When hydrated, the acid groups cluster into nanometre water channels that carry protons; dry, the membrane hardly conducts. Membranes 10–50 μm thick conduct about 0.1 S/cm at 80 °C when humidified, which keeps operation below about 100 °C; swelling and shrinking with humidity cycles fatigues them and opens pinholes.

Catalyst layer. At 80 °C oxygen reduction is slow and only platinum-group metals catalyse it adequately: Pt or Pt alloy particles 2–5 nm across on carbon, in a porous layer that must also conduct protons (ionomer), electrons (carbon) and gas (pores), a three-phase nanocomposite. Loadings have fallen to 0.1–0.4 mg Pt/cm², a few tenths of a gram per kilowatt. Bipolar plates of graphite composite or coated stainless steel or titanium must resist warm acid without growing an insulating oxide: the passivity of Chapter 42 is here a liability.

Degradation. Under voltage cycling, platinum nanoparticles dissolve and redeposit on larger ones, Ostwald ripening driven by the Gibbs–Thomson effect (Chapter 10); the carbon support oxidizes during start-up and shut-down; and peroxide radicals attack the membrane.

Solid oxide cells

Solid oxide fuel cells (SOFCs) use an oxide-ion conductor and run at 600–1000 °C, hot enough that no precious metal is needed and hydrocarbon fuels can be used.

Electrolyte. Yttria-stabilized zirconia (8 mol% Y_2O_3, "8YSZ") stabilizes the cubic fluorite phase, and each pair of Y^{3+} ions substituting for Zr^{4+} creates one oxygen vacancy, written in Kröger–Vink notation (Chapter 5) as

$$Y_2O_3 \xrightarrow{ZrO_2} 2\,Y'_{Zr} + 3\,O_O^{\times} + V_O^{\bullet\bullet}$$

Oxide ions hop through these vacancies. Gadolinium-doped ceria (GDC) and scandia-stabilized zirconia conduct better at lower temperatures, though ceria becomes partly electronically conducting in reducing gas.

Electrodes and interconnects. The anode is a porous Ni–YSZ cermet in which nickel conducts electrons and catalyses fuel oxidation, YSZ conducts oxide ions, and pores admit fuel, so reaction occurs at the triple-phase boundary; the cathode is a perovskite such as (La,Sr)(Co,Fe)O_3. Interconnects are ferritic stainless steels with about 22 % Cr, whose expansion (11–12 ppm/K) matches the cell, but whose chromia scale evaporates chromium species that poison the cathode unless coated. Matching the expansion of every layer is the central thermomechanical problem (Chapter 24).

Worked example — electrolyte thickness and operating temperature. A cell is useful only if the electrolyte's area-specific resistance ASR $= t/\sigma$ is below about 0.15 Ω cm². 8YSZ conducts about 0.1 S/cm at 1000 °C with $E_a = 0.9$ eV. At 700 °C (973 K) its conductivity is lower by

$$\frac{\sigma(1273\ \mathrm{K})}{\sigma(973\ \mathrm{K})} = \frac{973}{1273}\exp\left[\frac{0.9}{8.617\times10^{-5}}\left(\frac{1}{973}-\frac{1}{1273}\right)\right] = 0.764 \times 12.55 = 9.6$$

so $\sigma(700\ ^\circ\mathrm{C}) \approx 0.010$ S/cm.

- An electrolyte-supported cell with $t = 150$ μm (0.015 cm): ASR $= 0.015/0.1 = 0.15$ Ω cm² at 1000 °C, acceptable, but $0.015/0.0104 = 1.4$ Ω cm² at 700 °C, about ten times too high.
- An anode-supported cell with a 10 μm electrolyte at 700 °C: ASR $= 0.001/0.0104 = 0.096$ Ω cm².

Making the electrolyte fifteen times thinner allows operation 300 °C cooler, which permits stainless-steel interconnects and slows every diffusion-driven degradation process. Modern SOFCs are therefore anode-supported, with a thin dense electrolyte film on a thick porous cermet, and cells for 500–650 °C move to GDC.

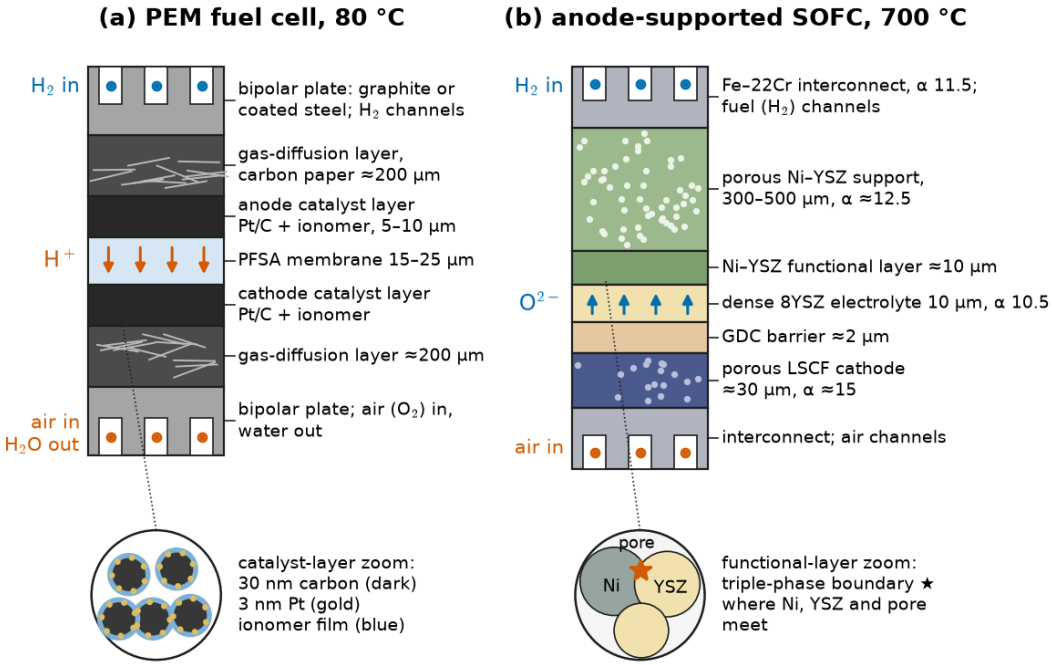


Figure 40.4 — PEM and solid-oxide cells are layered materials systems: every layer must conduct the right species and match its neighbours chemically and thermally.

Electrolysers

Water electrolysis is a fuel cell in reverse, and the slow step becomes oxygen evolution at the anode, whose catalyst must survive strongly oxidizing potentials.

- **PEM electrolysers** have an acidic membrane, so only **iridium oxide** is both active and stable at the anode. At 1–2 mg Ir/cm², a few hundred kilograms of iridium per gigawatt, against world production of about 7–8 tonnes a year, deployment at hundreds of gigawatts is impossible; tenfold lower loadings and iridium-free catalysts are urgent targets.
- **Alkaline electrolysers** run in 25–30 wt% KOH, where nickel is passive (Chapter 42), so nickel-based electrodes avoid precious metals entirely, at lower current density.
- **Solid-oxide electrolysers** reuse SOFC materials at 700–850 °C, where heat supplies part of the energy; the oxygen electrode tends to delaminate.

40.5 Hydrogen storage and hydrogen embrittlement

Storing hydrogen

Hydrogen carries more energy per unit mass than any fuel, about 120 MJ/kg (lower heating value), but very little per unit volume: about 40 kg/m³ as a gas at 700 bar and 71 kg/m³ as a liquid at 20 K. Compressed storage is a composites problem. Type IV vessels wind carbon fibre in epoxy (Chapters 22 and 23) over a polyamide or high-density polyethylene liner that blocks permeation and has no metal to embrittle; the fibre carries the pressure and accounts for most of the tank's cost. Liquid storage is a cryogenic-materials problem (Chapter 41).

Metal hydrides store hydrogen as atoms in interstitial sites and exceed the volumetric density of liquid hydrogen. Because the entropy of formation is roughly the same for all hydrides (the loss of gaseous H_2 entropy), the formation enthalpy decides the

temperature of release at 1 bar through the van 't Hoff relation (Advanced Chemistry): near-ambient cycling needs about −30 kJ per mol H_2.

Worked example — mass of hydride to store 5 kg of hydrogen. The gravimetric capacity is the mass fraction of hydrogen in the hydride.

- MgH_2: $2.016/(24.305 + 2.016) = 7.66$ wt%. Storing 5 kg of H_2 needs $5/0.0766 = 65$ kg of MgH_2.
- $LaNi_5H_6$: $6.048/(432.4 + 6.048) = 1.38$ wt%. Storing 5 kg needs $5/0.0138 = 362$ kg of hydride.

$LaNi_5H_6$ cycles quickly at a few bar near room temperature, but 362 kg is far too heavy for a car, even though it occupies only about 53 L (at 6.9 g/cm³) against 125 L for the gas at 700 bar. MgH_2 is light, but its formation enthalpy of about −75 kJ/mol H_2 means it releases hydrogen at 1 bar only near 300 °C, and slowly unless catalysed and nanostructured. This is the central trade-off: **strong metal–hydrogen bonds give light, dense storage that is hard to release; weak bonds release easily but only from heavy transition-metal hosts.** Hydrides therefore find use where weight is welcome or irrelevant: forklift counterweights, submarines, stationary buffers and hydrogen compressors that exploit the steep pressure–temperature dependence.

Table 40.4 — Hydrogen storage options (approximate material-level values).

Medium	Gravimetric H (wt%)	Volumetric H (kg/m³)	Release conditions	Comment
Compressed gas, 700 bar	100 (gas); ≈5–6 (system)	≈40	Ambient	Composite tanks; compression energy
Liquid H_2, 20 K	100 (liquid); ≈5–7 (system)	71	20 K	Boil-off; liquefaction uses ≈30 % of LHV
$LaNi_5H_6$	1.4	≈95	≈2 bar, 20–40 °C	Heavy; rare earth
$TiFeH_2$	1.9	≈100	Few bar, ambient	Cheap; needs activation
MgH_2	7.7	≈110	1 bar at ≈300 °C	Slow kinetics; high enthalpy
$NaAlH_4$ (Ti-doped)	5.6 (total)	≈95	100–180 °C	Two-step release; limited reversibility
Ammonia (as carrier)	17.8	≈120 (liquid)	Catalytic cracking, >400 °C	Toxic; established shipping

Hydrogen embrittlement

Atomic hydrogen, from dissociating H_2 gas or from cathodic reactions in corrosion, plating or cathodic protection, diffuses interstitially into metals and collects at dislocations, grain boundaries and, above all, in the dilated region ahead of a crack tip. There it lowers lattice cohesion and localizes plastic flow, so a ductile metal fails well below its usual strength, often after a delay (mechanisms and tests in Chapter 43). The factors that matter for energy infrastructure:

- **Strength.** Susceptibility rises steeply with yield strength; steels above about 800–1000 MPa, as in fittings, bolts and compressor springs, are at serious risk.
- **Crystal structure.** Hydrogen diffuses fast in BCC ferrite and very slowly in FCC austenite.
- **Fatigue.** In pressurized hydrogen, fatigue-crack growth in pipeline steels rises ten- to a hundredfold, which limits conversion of gas pipelines; hydrogen codes cap pipe grade (about X52–X70) and weld hardness.
- **Hydride formers.** Ti, Zr and Nb precipitate brittle hydrides instead (Chapter 41).

Worked example — why austenite resists hydrogen. Take room-temperature lattice diffusion coefficients of hydrogen of $D \approx 1 \times 10^{-8}$ m²/s in pure ferrite and $D \approx 1 \times 10^{-16}$ m²/s in austenite. In one hour ($t = 3600$ s) the diffusion length $\sqrt{Dt}$ (Chapter 8) is

$$\sqrt{10^{-8} \times 3600} = 6.0 \times 10^{-3}\ \text{m} = 6\ \text{mm} \quad \text{(ferrite)}, \qquad \sqrt{10^{-16} \times 3600} = 6 \times 10^{-7}\ \text{m} = 0.6\ \mu\text{m} \quad \text{(austenite)}$$

Hydrogen reaches a crack tip in ferritic steel within minutes but barely penetrates austenite in a year of service (trapping slows real steels below the pure-lattice value, but the ratio of about 10^4 in distance remains). Stable austenitic stainless steels, such as

316L with enough nickel that strain does not transform it to martensite, together with aluminium and copper alloys, are therefore the preferred materials for hydrogen valves, tubing and tank bosses.

40.6 Thermoelectric materials

Seebeck, Peltier and the figure of merit

When one end of a conductor is hotter than the other, the hot carriers diffuse toward the cold end until an opposing voltage builds up. The open-circuit voltage per kelvin is the **Seebeck coefficient** $S = -\Delta V / \Delta T$, positive for p-type and negative for n-type material. In metals it is a few μV/K, enough for thermocouples but not for power; in optimally doped semiconductors it is 150–250 μV/K. The reverse **Peltier effect** pumps heat ΠI at a junction carrying current I, with $\Pi = ST$. A module connects p- and n-type legs electrically in series and thermally in parallel between a hot and a cold plate, so their voltages add (Figure 40.5).

Three properties compete inside each leg: S generates voltage, the electrical resistivity wastes it as Joule heat, and the thermal conductivity κ leaks heat from hot to cold without generating anything. For a leg between T_h and T_c with constant properties and an optimized load, the maximum efficiency is

$$\eta_{max} = \frac{T_h - T_c}{T_h} \cdot \frac{\sqrt{1 + Z\bar{T}} - 1}{\sqrt{1 + Z\bar{T}} + T_c/T_h}, \qquad Z\bar{T} = \frac{S^2 \sigma}{\kappa} \bar{T}, \quad \kappa = \kappa_e + \kappa_L$$

with $\bar{T}$ the mean temperature. The first factor is the Carnot efficiency; the second, which approaches 1 only as $ZT \to \infty$, depends entirely on the material through the dimensionless **figure of merit** ZT. The product $S^2\sigma$ is the **power factor**.

> **Status of the principle: thermoelectric efficiency in terms of ZT**
> *Derived* from the coupled heat and charge transport equations for a single leg with constant properties. With temperature-dependent properties it remains a good estimate when an averaged ZT is used; contact resistance and heat losses lower real module efficiency further.

Worked example — exhaust heat recovery. A generator on an industrial furnace flue or a truck exhaust runs between $T_h = 800$ K and $T_c = 300$ K, so the Carnot efficiency is $500/800 = 0.625$.

- With $Z\bar{T} = 1$: $\sqrt{2} = 1.414$, and $\eta_{max} = 0.625 \times 0.414/(1.414 + 0.375) = 0.625 \times 0.232 = 0.145$, or 14.5 %.
- With $Z\bar{T} = 2$: $\sqrt{3} = 1.732$, and $\eta_{max} = 0.625 \times 0.732/(1.732 + 0.375) = 0.625 \times 0.347 = 0.217$, or 21.7 %.

Doubling ZT raises efficiency by half, and even at $ZT = 2$ a thermoelectric reaches only a third of Carnot, whereas a steam or organic Rankine cycle reaches well over half. For the same 800 K source, a material with $ZT = 1$ in a 300–500 K window would give only about 8 %. Thermoelectrics win where their other virtues dominate: no moving parts, silence, scalability down to milliwatts and decades of unattended operation.

Why ZT is hard to raise

The properties are tied together through the carrier concentration n. Conductivity $\sigma = ne\mu$ rises with n; the Seebeck coefficient falls, roughly as $n^{-2/3}$ in a degenerate semiconductor with one parabolic band (the Pisarenko relation, *Approximate*), because in a metal all carriers sit near the Fermi level and carry nearly the same energy; and κ_e rises with σ by the Wiedemann–Franz law, $\kappa_e = L\sigma T$ (Chapter 24). The power factor therefore peaks in heavily doped semiconductors at $n \approx 10^{19}$–10^{20} cm^{-3}. For a p-type Bi_2Te_3 alloy at 300 K with $S = 200$ μV/K, $\sigma = 1 \times 10^5$ S/m and $\kappa = 1.5$ W m^{-1} K^{-1}, the power factor is $(2 \times 10^{-4})^2 \times 10^5 = 4 \times 10^{-3}$ W m^{-1} K^{-2} and $ZT = 4 \times 10^{-3} \times 300/1.5 = 0.80$; the electronic part of κ is $2.44 \times 10^{-8} \times 10^5 \times 300 = 0.73$ W m^{-1} K^{-1}, about half. That half cannot be removed without losing conductivity. The lattice half can, and three structural strategies attack it or the band structure:

1. **Band engineering.** Many degenerate band valleys (band convergence, as in PbTe at high temperature) raise the density-of-states effective mass and hence S without lowering mobility much.
2. **Heavy atoms and soft or anharmonic bonds.** Bi, Pb, Te, Sb and Se give low sound velocities; "rattler" atoms in cage structures (skutterudites, clathrates) and strongly anharmonic bonds (SnSe) scatter phonons.
3. **Phonon glass, electron crystal.** Phonons that carry heat have mean free paths from nanometres to micrometres, while electrons in doped semiconductors scatter over shorter distances. Point defects from alloying, nanoprecipitates and grain

boundaries at all length scales scatter phonons while leaving electrons almost untouched; alloying (SiGe, Bi_2Te_3–Sb_2Te_3) and nanostructuring have each roughly halved κ_L.

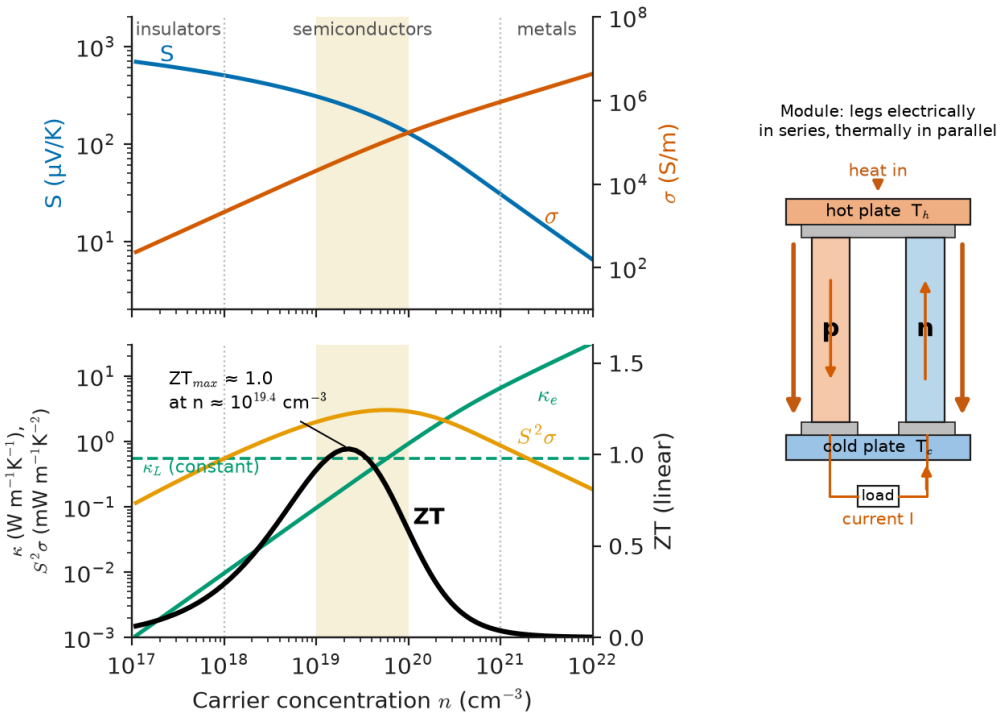


Figure 40.5 — Seebeck coefficient falls and electrical conductivity rises with carrier concentration, so the power factor and ZT peak in heavily doped semiconductors.

Material families

Table 40.5 lists the main families. **Bi_2Te_3 alloys**, layered compounds of heavy atoms, are in essentially every Peltier cooler, from laser-diode stabilization to laboratory PCR machines, but cleave easily and are weak. **PbTe** powers the radioisotope generators of Mars rovers, and **SiGe**, whose alloy disorder makes it outperform both elements, powered Voyager and Cassini. Half-Heuslers are the robust candidates for vehicle exhausts.

Table 40.5 — Thermoelectric materials (typical peak values).

Material	Type	Peak-ZT temperature (K)	Peak ZT	κ (W m^{-1} K^{-1})	Notes
Bi_2Te_3 alloys	p and n	300–400	0.9–1.2	1.2–1.6	All Peltier coolers; Te supply
PbTe-based	p and n	600–850	1.5–2.2	0.8–1.5	Space generators; Pb toxicity
$Si_{80}Ge_{20}$	n / p	1100–1200	≈1.0 / ≈0.7	3–5	Rugged; Ge cost
Filled $CoSb_3$	n and p	700–850	1.2–1.7	2–3	Sb sublimation
Half-Heusler	n / p	900–1000	1.0–1.5	3–5	Mechanically strong
SnSe (single crystal)	p	900–930	≈2.6 (one axis)	0.3–0.5	Brittle, anisotropic
$Mg_3(Sb,Bi)_2$	n	300–700	0.8–1.5	≈1	Te-free

Real modules are limited as often by **interfaces** as by ZT: contact resistance below about 10^{-6} Ω cm², diffusion barriers that keep solder out of the legs, and joints that survive expansion mismatch during thermal cycling.

40.7 Supercapacitor materials

The double layer

At an electrode in an electrolyte, ions of opposite sign accumulate within a fraction of a nanometre of the surface, forming the **electric double layer**. To first order it is a parallel-plate capacitor whose gap d is about the radius of a solvated ion:

$$\frac{C}{A} \approx \frac{\varepsilon_0 \varepsilon_r}{d}$$

> **Status of the principle: Helmholtz double-layer estimate**
> *Approximate.* A parallel-plate model that ignores the diffuse layer, the ion-size distribution and confinement in sub-nanometre pores (where capacitance per unit area anomalously rises); it gives the right order of magnitude.

Worked example — from surface area to stored energy. With $\varepsilon_r \approx 6$ for the oriented solvent in the layer and $d \approx 0.5$ nm, $C/A = 8.854 \times 10^{-12} \times 6/(0.5 \times 10^{-9}) = 0.106$ F/m², about 10 μF/cm², consistent with measured values of 5–20 μF/cm². An activated carbon of 2000 m²/g would therefore give about $2000 \times 0.106 = 212$ F/g. Measured values are 100–150 F/g in organic electrolytes, because not all the area is accessible to ions.

A symmetric cell has two electrodes in series, so for equal masses its capacitance per gram of total active material is a quarter of the single-electrode value: 120 F/g per electrode gives 30 F/g for the cell. At 2.7 V, the stability limit of acetonitrile-based electrolytes,

$$E = \tfrac{1}{2} C V^2 = \tfrac{1}{2} \times 30 \times 2.7^2 = 109 \text{ J/g} = 30 \text{ Wh/kg}$$

of active material, and 5–8 Wh/kg for a packaged cell: twenty to forty times less than a lithium-ion cell. But charge is stored by ion adsorption, without chemical change or solid-state diffusion, so a supercapacitor delivers 5–10 kW/kg and survives 10^5–10^6 cycles. Because energy scales as V^2, the electrolyte's stability window matters as much as the electrode: about 1 V for aqueous, 2.5–2.85 V for organic and 3–4 V for ionic-liquid electrolytes. These properties suit jobs measured in seconds: braking-energy recovery in trams and cranes, wind-turbine pitch systems and aircraft emergency actuators.

Carbons and pseudocapacitors

Activated carbons, made by steam or KOH activation of coconut shell, wood or pitch, dominate because they are cheap and have 1000–2500 m²/g of surface in 0.5–5 nm pores. The engineering variable is **pore size**: pores matched to the desolvated ion use their area best, but a connected network of wider pores is needed for fast transport. Graphene and nanotubes serve mainly as conductive additives (Chapter 38).

Pseudocapacitive materials (hydrous RuO_2, MnO_2, MXenes such as $Ti_3C_2T_x$) store charge by fast surface or near-surface redox reactions whose response looks capacitive. MnO_2 is the cheap case: one electron per Mn over 0.8 V would give about $96\,485/(86.94 \times 0.8) \approx 1400$ F/g, but poor conductivity limits real electrodes to 150–300 F/g. The boundary with batteries is kinetic: current proportional to sweep rate means capacitive storage; current proportional to its square root means solid-state diffusion control, which is battery-like.

40.8 Nuclear fuels and cladding

Fission releases about 200 MeV per event, so a gram of uranium-235 fully fissioned yields about $200 \times 10^6 \times 1.602 \times 10^{-19} \times 6.022 \times 10^{23}/235 \approx 8 \times 10^{10}$ J, some two million times the chemical energy of a gram of oil. Nuclear energy is therefore limited not by fuel energy but by what the surrounding materials survive.

The workhorse fuel is **uranium dioxide**, a fluorite-structured ceramic chosen for its high melting point (about 2850 °C), chemical stability in water and ability to hold fission products in its lattice. Its weakness is low thermal conductivity, about 2.5–3 W m^{-1} K^{-1} at operating temperature, which makes pellet centres run near 1000–1500 °C. **Zirconium alloys** (Zircaloy, Zr–Nb) clad it because zirconium absorbs few thermal neutrons; their oxidation, hydrogen pick-up and violent reaction with steam above about 1200 °C are the reasons for accident-tolerant fuels with chromium-coated cladding, FeCrAl alloys or SiC/SiC composites. Radiation damage, nuclear fuels, cladding, reactor-vessel steels and fusion first-wall materials are developed in Chapter 41.

40.9 Degradation, critical raw materials and scalability

Degradation is a materials property

Every technology in this chapter fails through mechanisms rooted in its structure (Table 40.6).

Table 40.6 — Characteristic degradation mechanisms of energy materials.

Technology	Dominant degradation	Structural origin
c-Si PV	Light-induced degradation, LeTID	Point-defect complexes (B–O, H)
Perovskite PV	Ion migration; moisture and heat decomposition	Soft ionic lattice; volatile organic cation
Li-ion graphite anode	SEI growth; Li plating	Electrolyte instability at low potential
High-Ni cathode	Particle cracking; O_2 release	Anisotropic lattice change; unstable high-valent Ni
PEM fuel cell / electrolyser	Pt and Ir dissolution and ripening; carbon corrosion	Nanoparticle surface energy; potential cycling
SOFC / SOEC	Ni coarsening; Cr poisoning; delamination	High-temperature diffusion; volatile Cr
Thermoelectric	Sublimation; contact interdiffusion; cracking	Volatile Sb, Te; expansion mismatch
Hydrogen infrastructure	Embrittlement; accelerated fatigue	Interstitial H at crack tips

Degradation is accelerated by temperature, often with an Arrhenius dependence, which is the basis of accelerated tests used to predict 10–30-year lives from months of data (Chapter 44).

Critical raw materials

A material can be technically superb and still unusable at scale if its elements are scarce, geographically concentrated, produced only as by-products, or toxic. Energy technologies are uniquely exposed because they must be deployed at the size of the whole energy system; the elements in Table 40.7 appear on most national critical-raw-materials lists.

Table 40.7 — Critical elements in energy materials (approximate, mid-2020s).

Element	Where used	Annual primary production (approx.)	Supply feature
Li	All Li-ion cells	≈200–250 kt (as Li)	Few producers; refining concentrated
Co	LCO, NMC cathodes	≈200–250 kt	≈70 % from one country
Ni	NMC cathodes; alloys	≈3.5 Mt	Majority from one country
Natural graphite	Anodes	≈1.5 Mt	>90 % of processing in one country
Nd, Dy	NdFeB magnets in wind turbines and motors (Chapter 29)	Nd ≈40 kt oxide; Dy ≈2 kt	Mining and separation concentrated
Pt	PEM cathodes	≈180 t	≈70 % from one country
Ir	PEM electrolyser anodes	≈7–8 t	By-product of Pt mining
Te	CdTe, Bi_2Te_3	≈500–700 t	By-product of Cu refining
In, Ga	CIGS, III–V, transparent contacts	In ≈1000 t; Ga several hundred t	By-products of Zn and Al refining
Ag	c-Si contacts	≈25,000 t	PV uses over a tenth

Worked example — lithium for a terawatt-hour of batteries. Take an LFP cell at an average 3.3 V. One kilowatt-hour requires $1000/3.3 = 303$ Ah of charge, or $303 \times 3600 = 1.09 \times 10^6$ C. Each Li^+ carries one elementary charge, so the lithium that must shuttle is

$$m_{Li} = \frac{1.09 \times 10^6}{96\,485} \times 6.941 = 78 \text{ g per kWh}$$

Real cells contain more (SEI losses, incomplete cycling, lithium in the electrolyte salt): about 0.1 kg of lithium per kilowatt-hour. Producing 1 TWh of cells a year therefore needs about $10^9 \times 0.1 = 10^8$ kg, or 100 kt, of lithium, around half of mid-2020s world production, and several TWh a year implies a several-fold expansion of mining and refining. Lithium is not geologically scarce, but mines and refineries take 5–15 years to build, so recycling and sodium-ion cells are supply strategies, not only environmental ones.

> **Status of the principle: material-intensity scaling limits**
> *Approximate.* A first-order bound from production rates and material intensity; supply responds to price and technology changes intensity, but the calculation correctly identifies which materials will constrain growth.

The same arithmetic applied to CdTe gives about 6 g of tellurium per square metre of 2 μm absorber, or about 33 t of tellurium per gigawatt at 19 % efficiency; the world's roughly 600 t of annual tellurium production could then support only about 18 GW of new CdTe modules a year, against several hundred gigawatts of total PV installations (Figure 40.6). Silicon, made from the crust's second most abundant element, faces no such ceiling for its absorber, though its silver contacts, at roughly 10 mg per watt, already consume over a tenth of world silver output.

Cost and scalability

Three lessons recur throughout this chapter.

1. **Abundant beats optimal.** Silicon beat better absorbers, LFP is overtaking NMC, and coconut-shell carbon beats graphene: at system scale, cost and availability outweigh a 20 % gain in a figure of merit.
2. **Processing sets cost.** A film coated from solution at 100 °C or a ceramic sintered in air is cheaper than one grown epitaxially in vacuum or made in a dry room (Chapters 46–49).
3. **Lifetime is part of cost.** The cost per unit of energy stored or generated divides capital cost by lifetime throughput, so a cell with twice the cycle life at the same price halves the cost of stored energy. Degradation science is therefore as central to energy materials as performance.

Recycling closes the loop for valuable elements such as Co, Ni, Li, Pt, Ir, Ag and Te, but only if devices are designed to be separated; lifecycle assessment and the circular economy are the subject of Chapter 59, and the formal trade-off between performance, cost and supply risk is the selection methodology of Chapter 58.

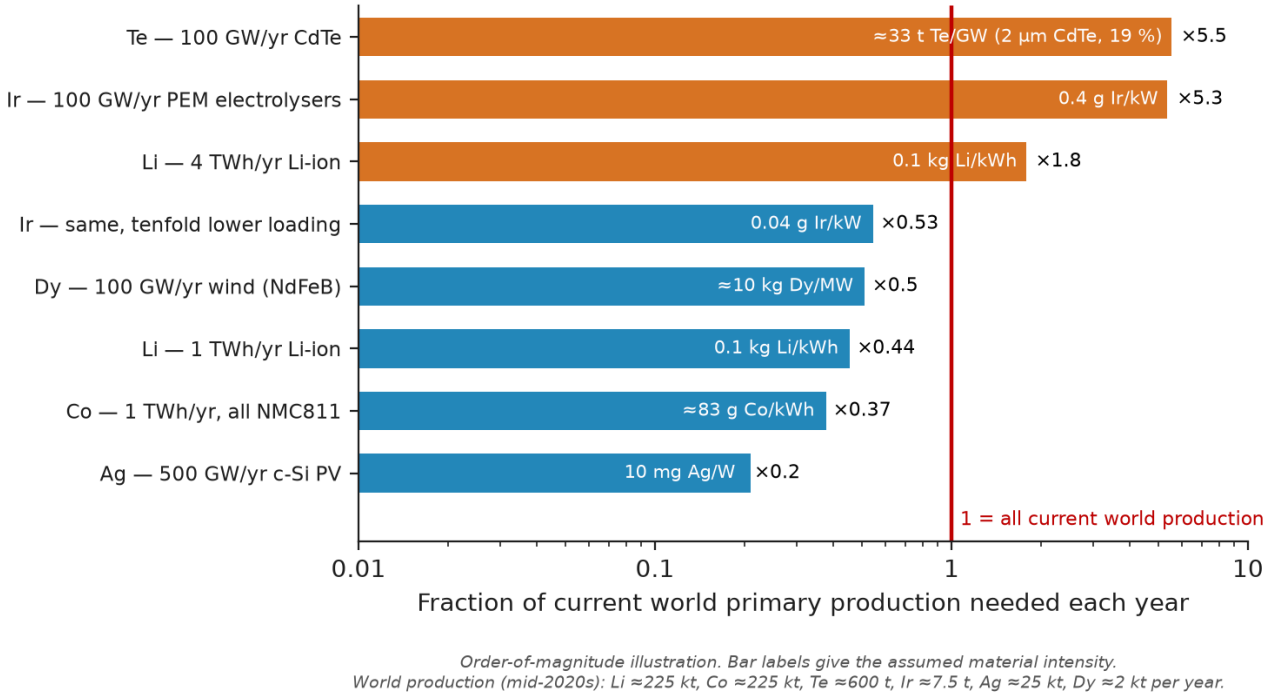


Figure 40.6 — Material intensity times deployment rate, compared with world production, shows which elements constrain each energy technology.

What remains open

Open question	Where it is answered
How do radiation, extreme heat and cryogenic temperatures degrade fuels, cladding, turbine blades and hydrogen tanks?	Chapter 41
What are the electrochemical thermodynamics and kinetics behind electrode potentials, catalysts and corrosion of cell components?	Chapter 42; Advanced Chemistry
How does hydrogen embrittlement proceed, and how is it tested and prevented?	Chapter 43
How are lifetimes of cells, modules and stacks predicted, and how are failures analysed?	Chapter 44
How are absorbers, electrolyte films and catalyst layers deposited and sintered?	Chapters 17, 49
How are capacity, ionic conductivity and interphase growth measured while a cell operates?	Chapters 52, 54
How are candidate energy materials screened computationally?	Chapters 55, 57
How are energy materials compared on cost, lifetime, supply risk and recyclability?	Chapters 58, 59

Next: Chapter 41 takes materials into the harshest service conditions engineers impose on them, from turbine blades and reactor cores to cryogenic tanks and hypersonic leading edges.

Chapter 41 — Materials for Extreme Environments

Part VIII · Advanced and Emerging Materials

The first-stage turbine blade of a modern jet engine sits in gas at about 1500 °C, some 150 °C above the melting point of the nickel alloy it is made of. It rotates at more than 10,000 revolutions per minute, so each gram of blade near the tip pulls outward with a force of a few hundred newtons, and it must do this for some 20,000 hours between overhauls without creeping more than a fraction of a millimetre toward the casing. Two metres away, the fan blade shrugs off bird strikes at −50 °C. A few hundred kilometres away a reactor pressure vessel absorbs a slow rain of fast neutrons that knock each of its atoms out of place again and again over sixty years, while its steel's ductile–brittle transition creeps upward toward the temperature of an emergency cooling injection. An LNG carrier's tank holds liquid at −162 °C in a membrane only 0.7 mm thick, and a re-entry capsule's heat shield is designed to burn away. These components share one feature: the environment has removed the safety margins that ordinary service allows. This chapter collects what materials science knows about such environments, where the physics of creep, diffusion, fracture and defects from earlier chapters is pushed to its limits.

> **What sets the temperature ceiling of a structural material, and how do coatings, cooling and composites raise it?**
> **How does radiation damage a crystal, and why do some alloys survive a hundred displacements per atom while others swell and crack? Why do some metals stay tough at cryogenic temperatures while others shatter, and what do aerospace, hypersonic and space service add?**

Concept	Engineering destination
Homologous temperature, creep, Larson–Miller parameter	Turbine blades, boiler tubes, furnace hardware, rocket nozzles
Superalloys, refractory metals, ceramics, CMCs, thermal barrier coatings	Gas turbines, reactors, hypersonic vehicles
Protective oxide scales, hot corrosion	Turbine and boiler life, waste incinerators, petrochemical crackers
Displacements per atom, swelling, embrittlement, helium	Reactor vessels, core internals, fuel cladding, fusion first wall
Fuel and cladding behaviour	Fission reactor safety and burnup
Specific stiffness and damage-tolerant design	Airframes, engine discs, launch vehicles
Ductile–brittle transition, thermal contraction, Invar	LNG and liquid-hydrogen tanks, superconducting magnets
Ablation, radiative cooling, atomic oxygen, outgassing	Re-entry shields, leading edges, satellites

Roadmap. What makes an environment extreme (41.1) → high-temperature materials (41.2) and their chemical attack (41.3) → radiation damage (41.4) and nuclear materials (41.5) → aerospace structures (41.6) → cryogenic service (41.7) → hypersonic flight and space (41.8).

> **Boundary.**
> The creep, fracture and fatigue mechanics used here are developed in Chapter 14, the alloys in Chapters 15 and 16, the ceramics and composites in Chapters 17, 22 and 23, and the corrosion science in Chapters 42 and 43. Nuclear reactions and neutron physics belong to Engineering Physics; radiation damage in semiconductors is treated in Chapter 32.

41.1 What makes an environment extreme

An environment is extreme for a material when it activates a degradation mechanism that is negligible in ordinary service. Temperature does this through the **homologous temperature** T/T_m (both in kelvin): diffusion-controlled processes such as creep, coarsening and oxidation become important above about $0.4T_m$ for metals and $0.5T_m$ for ceramics (Chapters 8 and 14). Radiation does it by creating point defects far above their thermal-equilibrium concentration. Low temperature does it by

suppressing the thermally activated dislocation motion that makes BCC metals ductile. Chemistry does it by dissolving or oxidizing the protective layers on which materials depend, and the vacuum, ultraviolet light and atomic oxygen of orbit add mechanisms that never occur on the ground.

Table 41.1 — Representative extreme environments (typical values).

Component	Temperature	Other loads	T/T_m of material	Dominant mechanisms
HP turbine blade (Ni single crystal)	≈1100 °C metal, ≈1500 °C gas	Centrifugal stress 150–250 MPa; thermal cycling	≈0.83	Creep, oxidation, thermomechanical fatigue
Ultra-supercritical boiler tube	600–650 °C	Steam at 25–30 MPa	≈0.5	Creep, steam oxidation
Ethylene cracker tube (HP-40 cast steel)	1000–1100 °C	Hydrocarbons	≈0.8	Creep, carburization
Light-water reactor pressure vessel	≈290 °C	Fast neutrons, ≈0.1 dpa over 60 years	≈0.3	Irradiation embrittlement
Fast-reactor fuel cladding	400–650 °C	100–200 dpa	0.4–0.5	Void swelling, irradiation creep
Fusion divertor (tungsten)	up to ≈1200 °C surface	≈10 MW/m² heat flux; He and H plasma	≈0.4	Recrystallization, cracking, He blistering
LNG tank	−162 °C	Thermal cycling	—	Brittle fracture, contraction
Re-entry heat shield	1500–3000 °C surface	Shear, oxidation	ablating	Ablation, oxidation
Low-Earth-orbit surface	−100 to +120 °C	Atomic oxygen, UV, vacuum	—	Erosion, outgassing, cycling

41.2 High-temperature materials

What a high-temperature material must do

A material for hot structural service must resist creep (Chapter 14), keep its microstructure stable for tens of thousands of hours, form a protective oxide (Section 41.3), resist thermomechanical fatigue as the component heats and cools, and, in anything that flies or rotates, do all this at low density, since the centrifugal stress in a rotating blade scales with its density. No single material does everything, so the hottest components are **material systems**: a creep-resistant substrate, a bond coat that forms a protective oxide, a ceramic thermal barrier and internal cooling.

Nickel-base superalloys

Nickel-base superalloys (Chapter 16) owe their strength to the **γ/γ′ microstructure**: 60–70 vol% of cuboidal, coherent precipitates of ordered $Ni_3(Al,Ti,Ta)$ (γ′, L1$_2$ structure) in an FCC nickel solid solution (γ). A dislocation entering γ′ creates an antiphase boundary (Chapter 3) and must travel in pairs; the yield strength of γ′ even rises with temperature up to about 800 °C because cross-slip locks screw dislocations. Refractory solutes (Re, W, Mo, Ta, Ru) slow diffusion and therefore creep and γ′ coarsening. The FCC matrix keeps good ductility and toughness at all temperatures.

Processing has been as important as alloying. Conventionally cast blades fail by creep cavitation on transverse grain boundaries. **Directional solidification** aligns columnar grains along the blade axis, removing transverse boundaries; **single-crystal** casting, with a helical grain selector, removes boundaries altogether, so grain-boundary strengtheners (C, B, Zr), which lower the melting point, can be omitted and the solution-heat-treatment temperature raised. Each step added roughly 25–50 °C of metal-temperature capability; together with cooling and coatings they let turbine entry temperatures exceed the alloy's melting point.

Worked example — what 50 °C costs in creep life. Creep-rupture data are compared through the Larson–Miller parameter $\mathrm{LMP} = T(C + \log_{10} t_r)$, with T in kelvin, t_r in hours and $C \approx 20$ (Chapter 14). A blade alloy has a rupture life of 1000 h at 1000 °C (1273 K) at its design stress, so

$$\mathrm{LMP} = 1273 \times (20 + 3) = 29\,279$$

At the same stress the life at 950 °C (1223 K) is $\log_{10} t_r = 29\,279/1223 - 20 = 3.94$, so $t_r = 8700$ h; at 1050 °C (1323 K), $\log_{10} t_r = 2.13$ and $t_r = 135$ h. A 50 °C overtemperature cuts life by a factor of about 7, and 50 °C of extra cooling multiplies it by nearly 9. This exponential sensitivity is why a few tens of degrees from a better coating or alloy are worth decades of development (Figure 41.1).

Status of the principle: Larson–Miller parameter
Empirical. It follows from an Arrhenius rate with a single activation energy; it interpolates well within the tested range for one alloy and microstructure, but extrapolations far beyond the data, or across a change of creep mechanism, are unreliable.

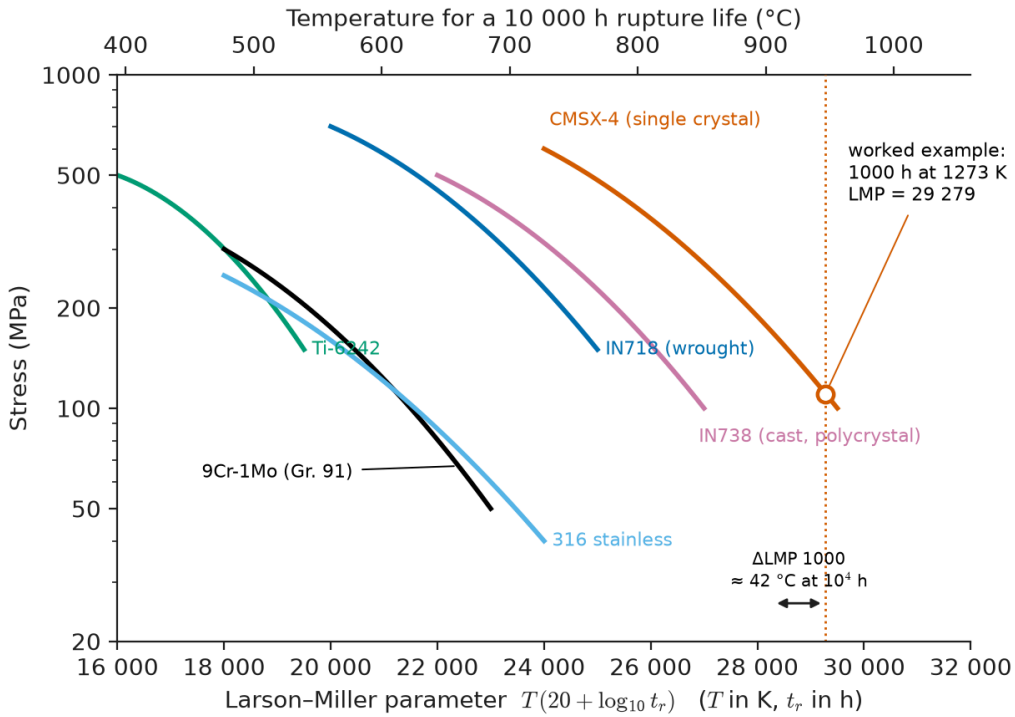


Figure 41.1 — Larson-Miller master curves rank high-temperature alloys: each 1000 units of LMP is worth roughly 40-50 C at constant life.

Refractory metals

Molybdenum (melting point 2623 °C), niobium (2477 °C), tantalum (3017 °C) and tungsten (3422 °C) keep useful strength far above any superalloy. Their weaknesses are equally fundamental. They oxidize catastrophically: MoO_3 and WO_3 volatilize above about 700–800 °C, so these metals serve in vacuum or inert gas (furnace elements and heat shields, X-ray targets, glass-melting electrodes) or under coatings. They are dense (W 19.3 g/cm³, Mo 10.2 g/cm³), and the BCC metals Mo and W have ductile–brittle transitions near or above room temperature, made worse by interstitial oxygen and recrystallization. Niobium alloy C-103 with a silicide coating forms the radiation-cooled nozzle extensions of upper-stage rocket engines; molybdenum TZM, with fine TiC and ZrC particles, serves in forging dies and furnace hardware; tungsten is the plasma-facing armour of fusion devices (Section 41.5). Refractory high-entropy alloys and Mo–Si–B alloys aim to combine strength above 1200 °C with self-protecting oxides.

Ceramics and ceramic-matrix composites

Covalent ceramics (SiC, Si_3N_4) keep their strength to 1200–1400 °C, are a third of the density of superalloys and form protective silica (Chapter 17). Monolithic, they are flaw-sensitive (Weibull moduli of 10–20) and so are used where stresses are modest or compressive: silicon nitride bearing balls and turbocharger rotors, SiC heat-exchanger tubes and kiln furniture.

Ceramic-matrix composites (Chapter 23) supply the missing toughness. In SiC/SiC, silicon carbide fibres are embedded in a SiC matrix with a thin boron nitride or pyrolytic carbon interphase that deflects cracks along the fibres, so cracks bridge rather

than run straight through. The composite fails gracefully, with a stress–strain curve that bends over rather than snapping. SiC/SiC operates to about 1300 °C at a density of about 2.7 g/cm³, and has entered service in turbine shrouds, combustor liners and low-pressure turbine blades, where it saves cooling air and weight. Carbon–carbon composites keep strength above 2000 °C in inert atmosphere and serve as aircraft and racing-car brakes and rocket nozzles, but they oxidize above about 450 °C and need SiC conversion coatings or short service times.

Thermal barrier coatings

A **thermal barrier coating** (TBC) is a ceramic layer, 100–500 μm thick, on the hot surface of an internally cooled metal part. The standard topcoat is zirconia stabilized with 7–8 wt% yttria (7YSZ), chosen for a rare combination: low thermal conductivity (about 2–3 W m^{-1} K^{-1} dense, 0.8–1.2 W m^{-1} K^{-1} with the porosity and microcracks of the coating), the highest thermal expansion of common oxides (about 11 ppm/K, closest to the metal), phase stability to about 1200 °C, and toughness from its metastable tetragonal phase (Chapter 14). Plasma spraying gives a splat structure with low conductivity; electron-beam physical-vapour deposition gives columns that open and close as the metal expands, tolerating strain better on blades.

Between the topcoat and the superalloy is an aluminium-rich **bond coat** (MCrAlY or Pt-modified aluminide) that grows a **thermally grown oxide** (TGO) of α-Al_2O_3, because zirconia is transparent to oxygen. The TGO is the TBC's life clock: as it thickens to about 5–10 μm, the compressive stress it stores on cooling, and the bond coat's rumpling, drive delamination and spallation of the topcoat (Figure 41.2). A newer threat is CMAS, calcium–magnesium–aluminosilicate deposits from ingested sand and volcanic ash, which melt on the hot surface, infiltrate the porous topcoat and stiffen it so that it spalls on cooling.

Worked example — how much a coating buys. A blade wall carries a heat flux $q = 1$ MW/m² from the gas to the cooling air. Through a 300 μm topcoat of conductivity $k = 1.0$ W m^{-1} K^{-1}, Fourier's law gives the temperature drop

$$\Delta T = \frac{q\,t}{k} = \frac{10^{6} \times 3 \times 10^{-4}}{1.0} = 300\ \text{K}$$

By the Larson–Miller arithmetic above, even a third of that drop multiplies creep life by an order of magnitude. Conversely, losing a patch of coating by spallation exposes the metal to a temperature rise of a couple of hundred kelvin, which is why TBC spallation is treated as a life-limiting event rather than a cosmetic one.

> **Status of the principle: one-dimensional Fourier conduction through a coating**
> *Approximate*. Valid for a uniform layer much thinner than the component's curvature, with constant conductivity; radiation through the translucent ceramic, sintering of the topcoat (which raises k with time) and edge effects modify it.

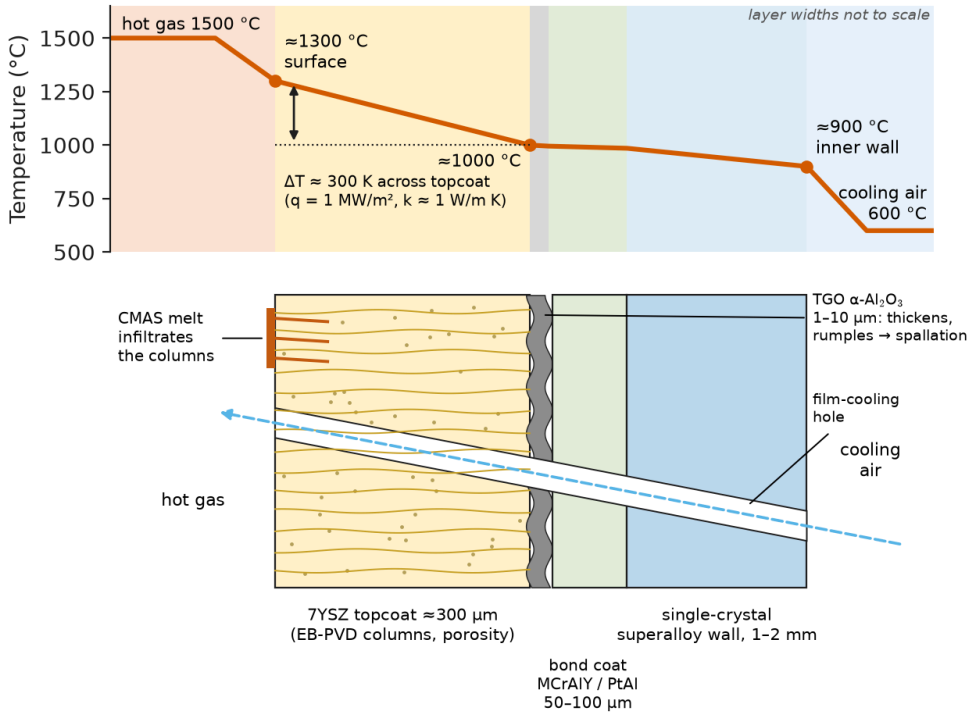


Figure 41.2 — A thermal barrier coating system: the ceramic topcoat carries most of the temperature drop, and the thermally grown oxide at the bond coat sets its life.

Table 41.2 — High-temperature structural materials (typical values).

Material	Density (g/cm³)	Approx. maximum service temperature	Oxidation protection	Typical use
Ti-6242 / near-α Ti	4.5	≈550 °C	Limited (α-case)	Compressor discs and blades
γ-TiAl	3.9–4.2	≈750 °C	Marginal alumina	Low-pressure turbine blades
9–12 % Cr ferritic steels	7.8	≈620 °C	Chromia	Steam pipes, boiler headers
Ni superalloy, single crystal	8.6–9.0	≈1100 °C metal (higher with TBC)	Alumina via bond coat	HP turbine blades
Nb alloy C-103 (coated)	8.9	≈1300 °C	Silicide coating	Rocket nozzle extensions
Mo-TZM	10.2	≈1400 °C (vacuum/inert)	None in air	Furnace hardware, dies
W	19.3	>2000 °C (vacuum/inert)	None in air	Fusion armour, filaments
Si_3N_4, SiC (monolithic)	3.2	1200–1400 °C	Silica	Bearings, heat exchangers
SiC/SiC CMC	2.6–2.8	≈1300 °C (with EBC)	Silica + environmental barrier	Shrouds, liners, LPT blades
C/C composite	1.6–1.9	>2000 °C inert; ≈450 °C in air unprotected	SiC coating	Brakes, nozzles, leading edges

41.3 High-temperature oxidation and hot corrosion

Every material in Table 41.2 survives in air only behind a slow-growing, adherent oxide. The thermodynamics (Ellingham diagrams, Chapter 6) and kinetics (Wagner's parabolic law, the Pilling–Bedworth ratio) of oxide growth are developed in Chapter 42; here the question is which scale to rely on.

- **Chromia (Cr_2O_3)** formers, alloys with about 18–25 % Cr, protect up to about 950–1000 °C. Above that, chromia oxidizes further to volatile CrO_3 (and in steam to $CrO_2(OH)_2$), so the scale thins as it grows.
- **Alumina (α-Al_2O_3)** formers, alloys with enough Al (about 5 % in FeCrAl, or via aluminide coatings), grow scales one to two orders of magnitude more slowly and protect to about 1300 °C. Small additions of reactive elements (Y, Hf, Zr) segregate to the scale's grain boundaries, slow its growth and dramatically improve adhesion.
- **Silica (SiO_2)** on SiC and Si_3N_4 is the slowest-growing scale of all, but water vapour converts it to volatile $Si(OH)_4$. In a gas turbine's combustion products, silica recedes at rates of the order of micrometres per hour, so SiC/SiC parts need **environmental barrier coatings** of rare-earth silicates ($Yb_2Si_2O_7$). At low oxygen pressure SiC oxidizes *actively* to gaseous SiO and recedes rapidly, a concern for re-entry.

Hot corrosion is accelerated attack under a thin molten salt film. Sodium sulfate, formed from sulfur in fuel and sodium chloride in marine air, condenses on turbine parts. In **Type I** (high-temperature) hot corrosion, at about 850–950 °C, the molten Na_2SO_4 dissolves the protective oxide (fluxing) and leads to sulfidation beneath it. In **Type II** (low-temperature) hot corrosion, at about 650–750 °C, low-melting eutectics with cobalt or nickel sulfates cause pitting. Vanadium from residual fuel oil forms vanadates that melt near 600–700 °C, which is why heavy-fuel boilers and marine diesels need special alloys or fuel additives. Related attacks include sulfidation in coal gasifiers, chloride attack in waste-to-energy boilers, and carburization and **metal dusting** in petrochemical furnaces, where carbon diffuses into nickel–chromium alloys and disintegrates them into graphite and metal powder. The materials responses are higher chromium for hot corrosion, alumina-forming coatings, and silicon and aluminium additions against carburization.

41.4 Radiation damage

Displacements and the dpa

A fast neutron carries no charge and passes through the electron cloud untouched until it collides with a nucleus. In an elastic collision a neutron of energy E can transfer at most

$$T_{max} = \frac{4A}{(1+A)^2}E$$

to a nucleus of mass number A. The struck atom, the **primary knock-on atom** (PKA), recoils through the lattice and displaces others if it gives them more than the **displacement threshold energy** E_d, about 25–40 eV for metals (40 eV is the conventional value for iron). The resulting **displacement cascade** (Figure 41.3) lasts a few picoseconds: a core of a few nanometres is briefly molten-like, most displaced atoms recombine, and the survivors are vacancies at the centre and self-interstitials, singly and in small clusters, around the edge.

The standard measure of damage is the number of **displacements per atom** (dpa), estimated by the Norgett–Robinson–Torrens (NRT) model, which counts the displacements a PKA of damage energy T_{dam} would create:

$$N_d = \frac{0.8\,T_{dam}}{2E_d}$$

where T_{dam} is the PKA energy minus what it loses to electronic excitation.

> **Status of the principle: NRT displacement estimate**
> *Approximate*. A binary-collision estimate that counts displacements, not surviving defects; molecular dynamics shows that only about a third survive cascade recombination. The dpa is nonetheless the accepted exposure unit because it lets damage from different neutron spectra and ion beams be compared.

Worked example — damage from one fast neutron in steel. For a 1 MeV neutron striking iron ($A = 55.85$),

$$T_{max} = \frac{4 \times 55.85}{56.85^2} \times 1\ \text{MeV} = 0.0691\ \text{MeV} = 69\ \text{keV}$$

For isotropic elastic scattering the average transfer is half of this, about 35 keV. Roughly a quarter goes into electronic excitation, leaving $T_{dam} \approx 25$ keV, so $N_d = 0.8 \times 25\,000/(2 \times 40) = 250$ displacements, of which some 80 survive as Frenkel pairs. In a fast-reactor core with a flux of 2×10^{15} neutrons cm^{-2} s^{-1} and a spectrum-averaged displacement cross-section of about 400 barn (4×10^{-22} cm²), the damage rate is $2 \times 10^{15} \times 4 \times 10^{-22} = 8 \times 10^{-7}$ dpa/s, or $8 \times 10^{-7} \times 3.16 \times 10^{7} \approx 25$ dpa per year: each atom in the cladding is knocked out of its site about twenty-five times a year. By contrast a light-water reactor pressure vessel, shielded by water and core internals, accumulates only about 0.1 dpa in sixty years.

Table 41.3 — Typical radiation exposures (approximate).

Component	Temperature	Damage	Helium generation
LWR pressure vessel (60 years)	≈290 °C	0.05–0.15 dpa	Negligible
LWR core baffle bolts (60 years)	300–380 °C	up to ≈80–100 dpa	Several appm/dpa (Ni transmutation)
LWR Zr-alloy cladding (fuel life)	300–400 °C	≈10–15 dpa	—
Fast-reactor cladding and ducts	400–650 °C	100–200 dpa	Low
Fusion first wall (DEMO-class), per full-power year	300–550 °C	≈10–20 dpa	≈10–15 appm/dpa
Spallation neutron-source target window	100–400 °C	≈5–10 dpa/year	Hundreds of appm/dpa

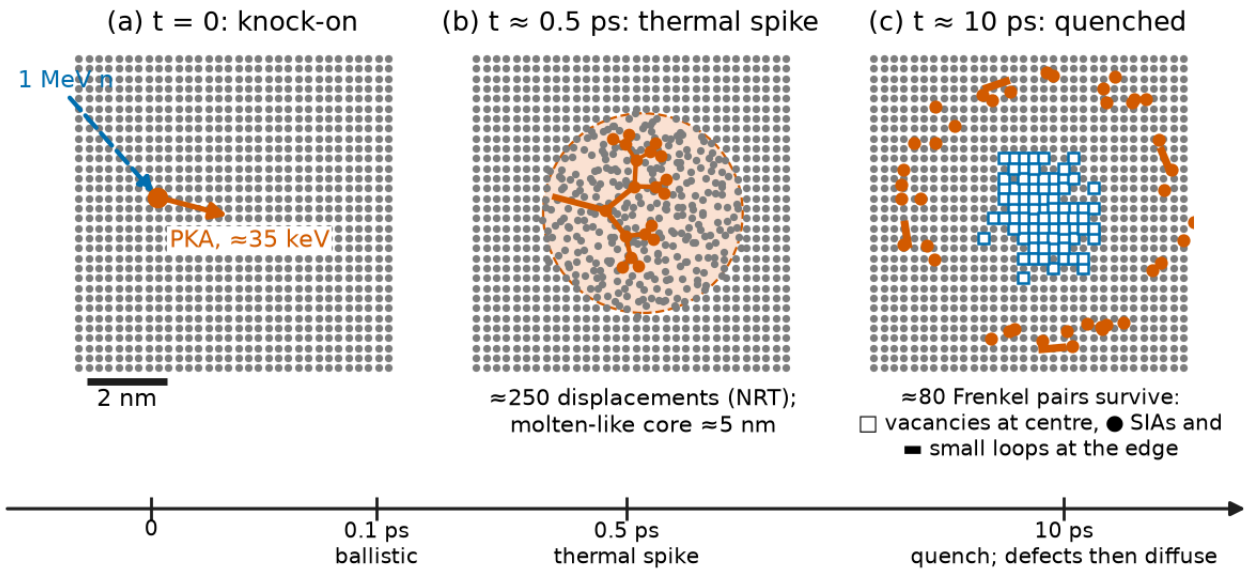


Figure 41.3 — A fast neutron's knock-on atom creates a displacement cascade: most displaced atoms recombine within picoseconds, leaving a vacancy-rich core and interstitial clusters at the periphery.

Consequences of displacement damage

Surviving defects migrate, recombine, cluster or annihilate at sinks (dislocations, grain boundaries, surfaces), and their fate depends on temperature in the same way as the processes of Chapters 8 and 10.

- **Irradiation hardening and embrittlement** (roughly $T < 0.35T_m$). Interstitial loops, vacancy clusters and solute clusters obstruct dislocations, raising yield strength and lowering work-hardening capacity and uniform elongation. In BCC steels the hardening raises the ductile–brittle transition temperature (Chapter 14). In reactor pressure-vessel steels (bainitic SA508 and A533B), copper precipitates as nanometre clusters, with nickel, manganese and silicon, and shifts the transition by tens of degrees; older vessels with 0.2–0.3 % Cu shifted by 100 °C or more. Modern steels limit copper below about 0.05 %, and surveillance capsules of the vessel steel, placed nearer the core, are tested periodically to measure the shift before the vessel itself reaches that dose.
- **Void swelling** (roughly $0.3–0.55T_m$). Interstitials are absorbed slightly more readily by dislocations than vacancies are (dislocation bias), so excess vacancies condense into voids, stabilized by dissolved gas. After an incubation dose, annealed

austenitic 316 stainless steel swells at about 1 % per dpa, so cladding and ducts of early fast reactors swelled by tens of percent and bowed. Ferritic–martensitic steels (9–12 % Cr) swell an order of magnitude less, because their BCC structure, high sink density and low bias suppress it (Figure 41.4).

- **Radiation-induced segregation.** Defect fluxes to grain boundaries drag or deplete solutes; in stainless steels, chromium is depleted and nickel and silicon enriched at boundaries, which, together with hardening, causes **irradiation-assisted stress-corrosion cracking** of core internals (Chapter 43).
- **Irradiation creep and growth.** Under stress, irradiation produces creep at temperatures where thermal creep is negligible, at rates proportional to dose; anisotropic crystals such as zirconium and graphite change shape even without stress (irradiation growth).
- **Helium embrittlement.** Transmutation reactions of the (n,α) type generate helium, which is insoluble, diffuses to grain boundaries and forms bubbles that turn creep fracture intergranular at high temperature. Thermal neutrons do this in nickel-bearing alloys through ^{58}Ni; the 14 MeV neutrons of deuterium–tritium fusion produce tens to a hundred times more helium per dpa in steel than fast-reactor spectra (about 10–15 against 0.1–0.3 appm He/dpa), which is why fusion materials must be tested in dedicated high-energy neutron sources.

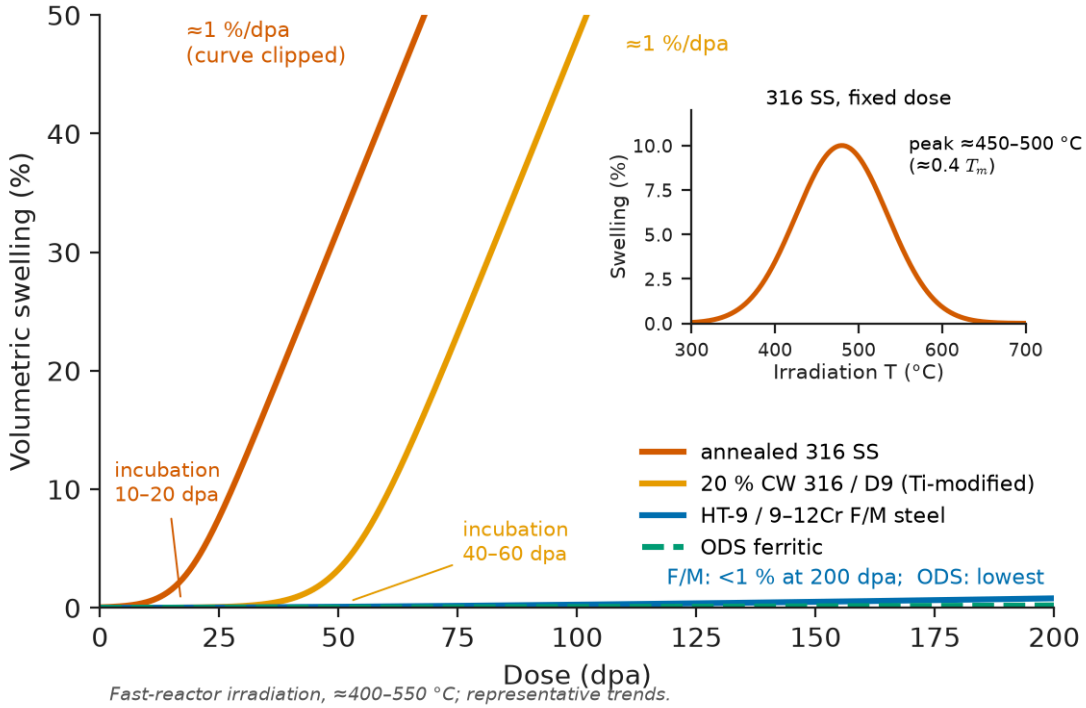


Figure 41.4 — Void swelling grows roughly linearly after an incubation dose; ferritic-martensitic steels swell an order of magnitude less than austenitic steels.

41.5 Nuclear materials

Fuels

The standard fuel is **uranium dioxide**, pressed and sintered into pellets about 8 mm in diameter. It is chosen for its fluorite structure, which accommodates fission products and the extra oxygen they release, its melting point of about 2850 °C, and its inertness in hot water. Its handicap is low thermal conductivity, about 2.5–3 W m^{-1} K^{-1} at operating temperature and falling further with burnup as fission products and defects scatter phonons (Chapter 24).

Worked example — fuel centreline temperature. In a cylindrical pellet generating heat uniformly, the temperature difference between centre and surface depends only on the **linear heat rating** q' (power per unit length of fuel rod), not on the pellet radius:

$$T_0 - T_s = \frac{q'}{4\pi k}$$

At a typical average rating of 20 kW/m and $k = 3$ W m^{-1} K^{-1}, $T_0 - T_s = 20\,000/(4\pi \times 3) = 530$ K, so with a pellet surface at about 450 °C the centre runs near 1000 °C. At a peak rating of 40 kW/m and $k = 2.5$ W m^{-1} K^{-1}, $T_0 - T_s = 40\,000/(4\pi \times 2.5) = 1270$ K and the centre approaches 1700 °C. Fuel temperature, not neutronics, limits the power a rod may produce, and it drives the release of fission gases (xenon and krypton) from the pellet into the rod, raising its internal pressure.

> **Status of the principle: centreline temperature of a cylindrical heat source**
> *Derived* from steady radial heat conduction with uniform heat generation and constant conductivity; in real pellets, temperature-dependent k, radial power depression and cracking require numerical fuel-performance codes, but the result gives the correct scale.

With burnup (typically 45–60 MWd per kg of uranium in light-water reactors), pellets swell with solid and gaseous fission products, crack, and at the high-burnup rim develop a fine-grained, porous "rim structure". Swelling eventually closes the pellet–cladding gap, and **pellet–cladding interaction**, mechanical contact aided by corrosive iodine, can crack the cladding during rapid power increases. Mixed oxide (U,Pu)O_2 recycles plutonium. Accident-tolerant fuels include U_3Si_2 and UN, denser and with several times the thermal conductivity of UO_2. **TRISO particles** for high-temperature gas reactors coat sub-millimetre kernels of UO_2 or uranium oxycarbide with porous carbon, pyrolytic carbon and a SiC pressure vessel a few tens of micrometres thick, so that each particle retains its fission products to about 1600 °C.

Cladding

Light-water reactor fuel is clad in **zirconium alloys** (Zircaloy-4, Zr–Nb alloys such as ZIRLO and M5) because zirconium's thermal-neutron absorption cross-section, about 0.18 barn, is an order of magnitude below iron's 2.6 barn, it resists hot water well, and it is ductile. Its service limitations are materials science: waterside corrosion to ZrO_2, with 10–20 % of the corrosion hydrogen absorbed into the metal; precipitation of brittle zirconium hydride platelets, which, if reoriented radially by hoop stress during cooling, embrittle the tube; and irradiation growth of the HCP crystal. In a loss-of-coolant accident the reaction

$$\mathrm{Zr} + 2\,\mathrm{H_2O} \rightarrow \mathrm{ZrO_2} + 2\,\mathrm{H_2}$$

accelerates above about 1200 °C, releasing about 6.8 MJ per kilogram of zirconium in addition to decay heat, together with hydrogen that can explode outside the core, as happened at Fukushima Daiichi in 2011. **Accident-tolerant cladding** responds with chromium-coated zirconium (a protective chromia layer), FeCrAl alloys (an alumina former, but with higher neutron absorption that requires thinner walls), and SiC/SiC composite tubes.

Structural materials

Light-water reactor **pressure vessels** are forged bainitic low-alloy steels (SA508 Grade 3), clad internally with austenitic stainless steel against corrosion, and their life is governed by irradiation embrittlement (Section 41.4). **Core internals** are austenitic stainless steels (304, 316), limited by irradiation-assisted stress-corrosion cracking and, at high doses, swelling. Advanced fission reactors push temperatures higher. Sodium-cooled fast reactors use ferritic–martensitic steels (Grade 91, HT-9) for their swelling resistance, and **oxide-dispersion-strengthened** (ODS) ferritic steels contain 10^{23}–10^{24} nanoclusters of Y–Ti–O per cubic metre that pin dislocations for creep strength and trap helium in tiny, harmless bubbles. High-temperature gas reactors at 750–950 °C use nickel alloys (Alloy 617, 800H) and nuclear-grade graphite, whose dimensional changes under irradiation limit its life. Molten-salt reactors use Ni–Mo–Cr alloys (Hastelloy N type) that resist fluoride salts, which dissolve chromium rather than form passive films.

Fusion first wall and plasma-facing materials

The first wall and blanket of a deuterium–tritium fusion reactor face the harshest combination known: 14 MeV neutrons with high helium production, surface heat fluxes of about 1 MW/m² on the first wall and up to 10–20 MW/m² on the divertor, and bombardment by hydrogen isotopes and helium from the plasma.

- **Plasma-facing armour** is tungsten: the highest melting point of any metal, a low sputtering yield and low tritium retention. Its problems are a ductile–brittle transition between about 200 and 400 °C (worse after irradiation), recrystallization above about 1200 °C, which raises that transition further, and surface damage from helium (nanometre bubbles and "fuzz") and from transient heat pulses.
- **Structural blanket steels** are **reduced-activation ferritic–martensitic** (RAFM) steels such as EUROFER97, in which molybdenum and niobium, whose transmutation products stay radioactive for millennia, are replaced by tungsten

and tantalum, so that the structure can be recycled or disposed of as low-level waste within about a century. Their window is roughly 350–550 °C: below it irradiation embrittlement, above it loss of creep strength. ODS variants and SiC/SiC composites aim to widen the window.

- **Breeding materials** produce the tritium fuel from lithium: ceramic pebbles of Li_4SiO_4 or Li_2TiO_3, or liquid lead–lithium, whose corrosion of steels and magnetohydrodynamic drag are design problems in their own right.

41.6 Aerospace materials

Lightweighting

In aircraft and launch vehicles, mass is the master variable. A kilogram removed from an airliner's structure reduces fuel burn for every flight of its 25–30-year life, and a kilogram removed from a launch vehicle's upper stage becomes a kilogram of payload. The materials question is therefore never "how strong?" but "how strong or stiff per unit mass, in the right loading mode?". For a tie loaded in tension the merit index of stiffness is E/ρ; for a panel in bending or buckling at fixed stiffness it is $E^{1/3}/\rho$ (performance indices are derived in Chapter 58).

Table 41.4 — Specific stiffness of aerospace structural materials (typical values).

Material	E (GPa)	ρ (g/cm³)	E/ρ (GPa cm³/g)	$E^{1/3}/\rho$ (GPa$^{1/3}$ cm³/g)
Steel (300M, high-strength)	205	7.85	26	0.75
Ti-6Al-4V	114	4.43	26	1.09
Al 7075-T6	71	2.81	25	1.47
Al–Li 2099-T8	78	2.63	30	1.62
Mg alloy AZ31	45	1.77	25	2.01
CFRP, quasi-isotropic laminate	60	1.55	39	2.53
CFRP, unidirectional (fibre direction)	135	1.58	85	3.25

The metals all have nearly the same E/ρ, about 25 GPa cm³/g, because modulus and density both scale with atomic packing and bond density; alloying hardly changes it. Lithium is the exception, the only element that both lowers density and raises modulus of aluminium, which is why third-generation Al–Li alloys (2050, 2099, 2196) are used in fuselage and wing structure. For thin panels, where $E^{1/3}/\rho$ governs, low-density materials gain enormously, and carbon-fibre composites win outright; they make up about half the structural mass of the newest wide-body airliners (Chapters 22 and 23). Titanium carries the hot and highly loaded parts (engine pylons, landing-gear beams, fan blades) and is galvanically compatible with carbon fibre, whereas aluminium in contact with CFRP corrodes (Chapter 43). In engines, the materials sequence along the gas path follows temperature: titanium in the fan and compressor, nickel superalloys and CMCs in the combustor and turbine, with γ-TiAl in the last turbine stages.

Fatigue-critical design

Aircraft structures are designed for fatigue more than for static strength, because they experience a pressurization cycle on every flight and millions of gust and manoeuvre cycles. The history of the discipline is a sequence of materials lessons: the de Havilland Comet losses of 1954, from fatigue cracks at window corners in a pressurized fuselage; the F-111 wing-pivot failure of 1969, from an undetected manufacturing flaw; and the 1988 Aloha Airlines fuselage separation, from multiple-site cracking at corroded lap joints in an aircraft that had flown nearly 90,000 cycles. Three design philosophies resulted:

- **Safe-life**: the part is retired after a demonstrated life with a large scatter factor (helicopter rotor components, landing gear).
- **Fail-safe**: multiple load paths so that the failure of one member is not catastrophic.
- **Damage tolerance**: assume a crack of the largest size that inspection might miss, predict its growth by fracture mechanics, and set inspection intervals so that it will be found before it reaches critical size. This is now mandatory for primary airframe structure and, for rotating engine parts, is combined with probabilistic assessments of melt-related defects such as hard-alpha inclusions in titanium.

Worked example — setting an inspection interval. A wing lower-skin panel of 7075-T6 ($K_{Ic} \approx 28$ MPa m$^{1/2}$) sees a ground–air–ground stress range $\Delta\sigma = 100$ MPa (with $R \approx 0$, so $\sigma_{max} = 100$ MPa) once per flight. Eddy-current inspection reliably finds edge cracks of $a_0 = 2$ mm. With a geometry factor $Y = 1.12$, the critical crack size is

$$a_c = \frac{1}{\pi}\left(\frac{K_{Ic}}{Y\sigma_{max}}\right)^2 = \frac{1}{\pi}\left(\frac{28}{1.12 \times 100}\right)^2 = 0.0199 \text{ m} \approx 20 \text{ mm}$$

Integrating the Paris law $da/dN = C(\Delta K)^m$ (Chapter 14) with $\Delta K = Y\Delta\sigma\sqrt{\pi a}$, representative constants $C = 1 \times 10^{-11}$ m/cycle (with ΔK in MPa m$^{1/2}$) and $m = 3$:

$$N = \frac{a_0^{-1/2} - a_c^{-1/2}}{\frac{1}{2}C\,(Y\Delta\sigma\sqrt{\pi})^3} = \frac{22.36 - 7.07}{0.5 \times 10^{-11} \times (198.5)^3} = 3.9 \times 10^5 \text{ cycles}$$

Halving this for scatter in growth rate and stress gives an inspection interval of about 200,000 flights, far beyond an airliner's life, so this location is not fatigue-critical at this stress. At a stress hot-spot of twice the nominal value, the growth rate rises by $2^3 = 8$ and a_c falls by $2^2 = 4$, to 5 mm, so $N = (22.36 - 14.14)/(8 \times 3.91 \times 10^{-5}) \approx 26\,000$ cycles: fifteen times shorter, and an inspection interval of about 13,000 flights. The cubic dependence on stress is why design detail (fillet radii, fastener holes, cold-expanded holes and residual compressive stresses) matters more than alloy choice.

> **Status of the principle: damage-tolerant life by Paris-law integration**
> *Empirical* in its growth law and constants, which are fitted to test data for each alloy, environment and load ratio; it is reliable between the near-threshold and near-critical regimes but must be corrected for load sequence (overload retardation) and short-crack behaviour.

41.7 Cryogenic materials

Ductile or brittle in the cold

Cryogenic engineering spans liquefied natural gas (−162 °C, 111 K), liquid oxygen (90 K), liquid nitrogen (77 K), liquid hydrogen (20 K) and liquid helium (4.2 K). The first question for any structural material is whether it stays tough. **FCC metals**—austenitic stainless steels, aluminium, copper and nickel alloys—have no ductile–brittle transition: their dislocations glide on close-packed planes with a negligible Peierls barrier (Chapter 12), and strength and often toughness rise as temperature falls. **BCC metals**, including ordinary ferritic steels, need thermal activation for screw dislocations to move, so below their transition temperature yield strength soars past the cleavage stress and they fracture in a brittle manner (Chapter 14). This is why the steel of the *Titanic* and the welded Liberty ships cracked in cold seas, and why a carbon-steel pipe exposed to an LNG spill can shatter.

The cryogenic materials palette follows directly (Figure 41.5):

- **9 % nickel steel**, quenched and tempered, is a ferritic steel whose nickel content and fine microstructure (with a little stable retained austenite) push its transition below −196 °C. It is the standard for large land-based LNG storage tanks: cheaper and stronger than austenitic steel, tough enough to meet Charpy requirements at −196 °C.
- **Austenitic stainless steels** (304L, 316LN) serve down to 4 K in liquid-hydrogen and liquid-helium systems and in the cases of superconducting magnets; nitrogen additions stabilize the austenite against strain-induced martensite and raise strength.
- **Aluminium alloys** such as 5083-O are used for spherical LNG ship tanks and, as 2219 and Al–Li 2195, for rocket propellant tanks of liquid hydrogen and oxygen.
- **Titanium alloys** in extra-low-interstitial grades (Ti-5Al-2.5Sn ELI) are used for hydrogen-rocket pressure vessels, where oxygen and nitrogen content must be minimized to keep toughness.

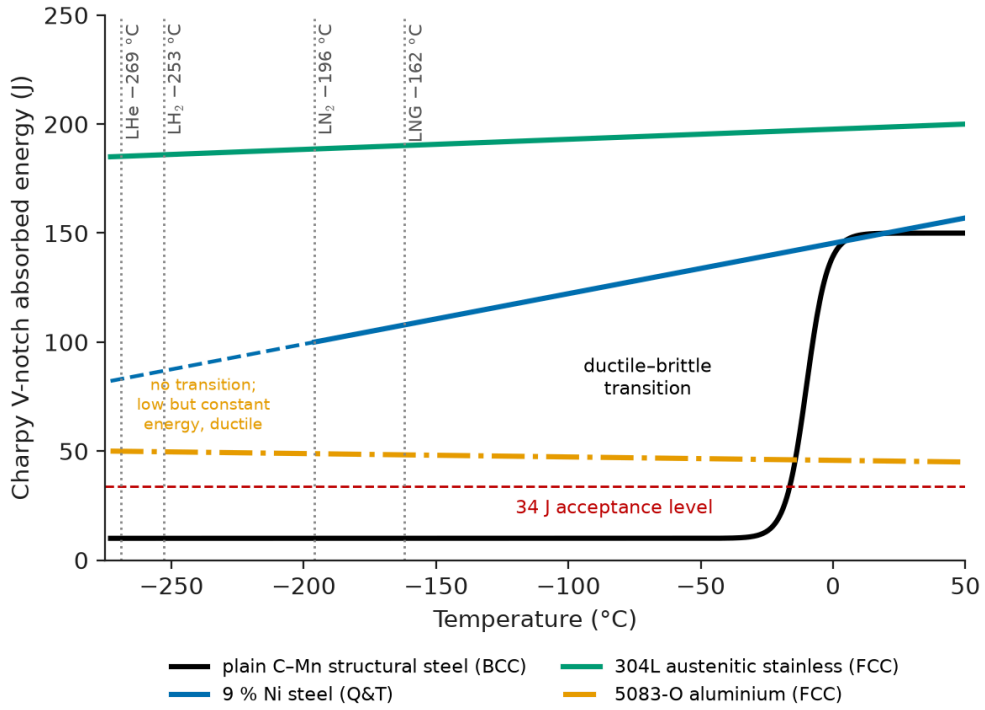


Figure 41.5 — FCC metals and 9 % Ni steel stay tough at cryogenic temperatures, while ordinary ferritic steel becomes brittle below its transition.

Thermal contraction and Invar

Cooling a structure from room temperature to cryogenic temperature contracts it by a fraction that, for most metals, is a few tenths of a percent, most of it occurring before 77 K because thermal expansion falls with heat capacity (Chapter 24). Integrated contraction from 293 K to 4 K is about 0.30 % for austenitic stainless steel, 0.42 % for aluminium alloys, 0.33 % for copper and 0.17 % for Ti-6Al-4V; polymers contract far more (epoxy about 1 %, PTFE about 2 %).

Worked example — contraction of an LNG transfer line. A 100 m stainless steel line cooled from 20 °C to −196 °C contracts by about 0.28 %, or $0.0028 \times 100 = 0.28$ m: 280 mm that must be absorbed by bellows, expansion loops or sliding supports, or it appears as a thermal stress of order $E\varepsilon = 195\ \text{GPa} \times 0.0028 \approx 550\ \text{MPa}$, more than double the yield strength of annealed 304L. The same line in **Invar** (Fe–36 % Ni) would contract by about 0.04 %, or 40 mm.

Invar's near-zero expansion (about 1.2 ppm/K at room temperature) comes from magnetism: in this ferromagnetic alloy, the spontaneous magnetization expands the lattice (a magnetovolume effect), and as the alloy cools the growing magnetization compensates the normal anharmonic contraction. The membrane tanks of most LNG carriers are lined with 0.7 mm Invar sheet that needs no expansion corrugations; Invar also serves in precision instruments, composite-moulding tools for CFRP aerospace parts, and shadow masks.

Polymers, composites and insulation

Most polymers become glassy and brittle far below their glass transitions; PTFE, polyimides and some polyethylenes retain useful ductility and are used for seals, while elastomeric O-rings fail. Composites microcrack at cryogenic temperatures because fibre and matrix contract differently; leakage of liquid hydrogen through microcracked laminates caused the failure of a composite hydrogen tank in 1999 and delayed composite cryotanks for a decade.

Insulation is a materials system of its own. Closed-cell polyurethane foam (thermal conductivity about 0.02 W $\text{m}^{-1}\ \text{K}^{-1}$) insulates LNG tanks and rocket tanks; foam shed from the Space Shuttle's external tank struck the wing leading edge of *Columbia* in 2003 and led to its loss. Vacuum-jacketed perlite powder and **multilayer insulation**, tens of layers of aluminized polyester separated by spacers in vacuum, which suppress radiation layer by layer to give effective conductivities of order 10^{-5} W $\text{m}^{-1}\ \text{K}^{-1}$, serve liquid-hydrogen and helium vessels and spacecraft.

41.8 Hypersonic and space environments

Hypersonic flight and re-entry

Air brought to rest against a vehicle's nose at Mach number M reaches the stagnation temperature $T_0 = T(1 + \frac{\gamma-1}{2}M^2)$; with $\gamma = 1.4$ and an ambient 220 K, that is 1320 K at Mach 5, 2380 K at Mach 7 and about 4600 K at Mach 10 (real-gas effects lower the last figure). The surface does not reach T_0, because it radiates heat away. A surface that loses its heat input q only by radiation reaches the **radiative-equilibrium temperature**

$$T_{rad} = \left(\frac{q}{\varepsilon \sigma_{SB}} \right)^{1/4}$$

Worked example — leading-edge temperature. For a heat flux of 1 MW/m² and emissivity $\varepsilon = 0.85$, $T_{rad} = [10^6/(0.85 \times 5.67 \times 10^{-8})]^{1/4} = 2130$ K (about 1860 °C); at 2 MW/m², 2540 K. Because T rises only as $q^{1/4}$, a high emissivity is worth as much as a large increase in temperature capability, and sharp leading edges, which concentrate heat flux, push surface temperatures past 2000 °C.

> **Status of the principle: radiative equilibrium**
> *Approximate.* A steady energy balance assuming all incident heat is reradiated; conduction into the structure, catalytic recombination of dissociated air on the surface, and ablation alter the real temperature.

Three strategies follow (Figure 41.6):

- **Reusable insulating and radiating surfaces.** The Space Shuttle's tiles were rigid silica-fibre foam (density about 0.14 g/cm³) that could be held at 1260 °C on one face and touched on the other; its nose cap and wing leading edges were reinforced carbon–carbon with a SiC coating, good to about 1650 °C.
- **Ablators.** Capsules returning from the Moon or from interplanetary trajectories meet heat fluxes of several MW/m² for minutes. Charring ablators (Avcoat on Apollo and Orion; phenolic-impregnated carbon ablator, PICA, on sample-return and commercial capsules) absorb heat by pyrolysis of their resin, block convective heating with the gas this releases, and radiate from a hot char surface, sacrificing millimetres to centimetres of thickness.
- **Ultra-high-temperature ceramics.** ZrB_2 and HfB_2 (melting points above 3200 °C), usually with 20–30 vol% SiC for oxidation resistance, and carbides such as HfC and TaC (melting points near 3900 °C) are candidates for sharp leading edges of hypersonic cruise vehicles. Their oxides are refractory, but the borosilicate glass that protects them evaporates above about 1800 °C, and their fracture toughness is only 3–5 MPa $m^{1/2}$, so fibre-reinforced UHTC composites are under development.

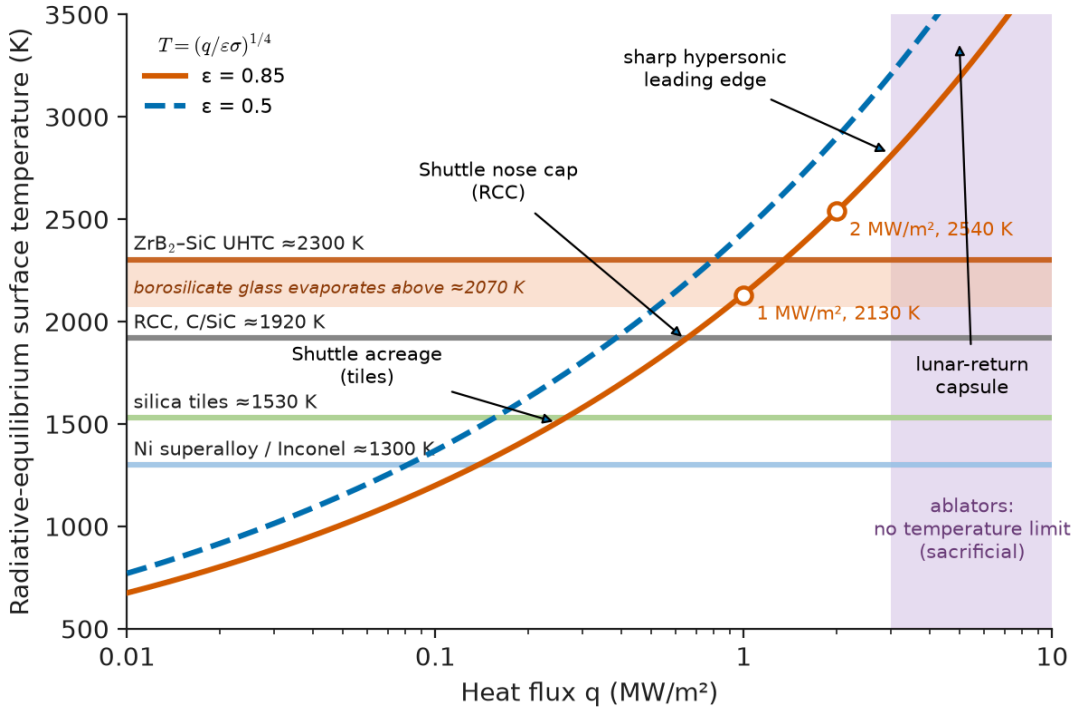


Figure 41.6 — Radiative-equilibrium surface temperature versus heat flux, with the temperature capabilities of thermal-protection materials.

The space environment

Spacecraft materials face a different set of extremes.

- **Vacuum.** Polymers, adhesives and lubricants outgas volatile species that condense on cold optics and sensors; space-qualified materials must lose less than 1 % of their mass and leave less than 0.1 % of collectable condensable material in a standard vacuum bake. Clean metal surfaces in contact can cold-weld, so mechanisms use dissimilar materials and solid lubricants such as MoS_2.
- **Atomic oxygen.** In low Earth orbit, ultraviolet light dissociates the thin residual atmosphere, and a spacecraft moving at 7.8 km/s strikes the oxygen atoms with about 5 eV each. Polymers erode: polyimide (Kapton) loses about 3.0×10^{-24} cm³ per incident atom. At an annual fluence of 10^{21}–10^{22} atoms/cm², depending on altitude and solar activity, that is $3 \times 10^{-24} \times 10^{21} = 3 \times 10^{-3}$ cm, or 30 µm to 300 µm, per year, enough to destroy a thin film in months. Protective SiO_2 coatings, silicone-based and fluorinated polymers, and metals that form stable oxides are used.
- **Ultraviolet and charged-particle radiation** darken polymers and glasses and shift the absorptance of thermal-control coatings, so a radiator that starts white ends its mission warmer; cerium-doped cover glasses protect solar cells. Radiation damage in semiconductors is covered in Chapter 32.
- **Thermal cycling.** A satellite in low orbit passes in and out of Earth's shadow about sixteen times a day, swinging between roughly −100 and +120 °C, some 5800 cycles a year; mismatched expansion between solar-cell glass, cells, interconnects and substrate is a principal fatigue mechanism of solar arrays (Chapter 44).
- **Micrometeoroids and debris.** Particles at 10 km/s carry far more energy per unit mass than explosives; Whipple shields, a thin sacrificial bumper spaced in front of the wall, shatter and spread the particle so the wall survives.

What remains open

Open question	Where it is answered
What are the thermodynamics and kinetics of oxide-scale growth, and why do some scales protect and others spall?	Chapter 42
How do stress-corrosion cracking, hydrogen embrittlement and irradiation-assisted cracking proceed, and how are they prevented?	Chapter 43
How are lives of hot parts, reactor components and spacecraft predicted, and how are failures analysed?	Chapter 44
How do coatings, surface engineering and wear behave in extreme environments?	Chapters 45, 49
How are single-crystal blades, ODS steels and CMCs manufactured, and how are they joined?	Chapters 46–48
How are radiation damage and helium bubbles observed at the atomic scale?	Chapters 53, 54
How are radiation cascades and creep simulated, and how are new alloys designed computationally?	Chapters 56, 57
How are materials selected when temperature, mass, cost and supply risk all compete?	Chapter 58

Next: Part IX opens with Chapter 42, which develops the thermodynamics and kinetics of corrosion in water and of oxidation in hot gas, the chemistry that underlies the protective scales on which so many materials in this chapter depend.

Part IX — Corrosion, Degradation and Surfaces

Most engineering materials do not fail because they were too weak on the day they entered service. They fail because they changed while in service. A buried pipeline thins at a fraction of a millimetre a year until it leaks; a stainless-steel fitting in a coastal plant pits and then cracks under a stress it carried easily when new; a polymer window frame chalks and embrittles in sunlight; a bridge deck spalls as chlorides reach its reinforcement; a gearbox loses its precision as its surfaces wear. Corrosion alone costs industrial economies a few percent of their gross domestic product every year, and much of that cost could be avoided with knowledge that already exists. This part treats the processes by which materials degrade through their surfaces and over time, together with the methods used to predict them, prevent them and work out why a component failed.

The part begins with the electrochemistry of corrosion, the most widespread degradation process, and its high-temperature counterpart, oxidation in hot gas. It then catalogues the forms corrosion takes and the ways engineers protect against them. It widens the view to degradation of every class of material and to the discipline of failure analysis. It ends at the surface itself, where adhesion, friction and wear are decided.

By the end of this part you should understand:

- why metals corrode, how electrode potentials and Pourbaix diagrams predict when corrosion is possible, and how mixed-potential theory and Tafel kinetics predict how fast it proceeds;
- how corrosion currents convert to rates of metal loss, and why passive films make stainless steels, aluminium and titanium usable;
- how oxide scales grow at high temperature, and what decides whether a scale protects or spalls;
- the forms of localized and environmentally assisted attack (galvanic, pitting, crevice, intergranular, stress-corrosion cracking, hydrogen embrittlement, corrosion fatigue) and the material, design, coating, inhibitor and cathodic-protection measures that counter them;
- how metals, ceramics, polymers, composites and concrete degrade, how lives are predicted from accelerated tests and reliability statistics, and how a failure investigation moves from evidence to root cause;
- how surface energy, adsorption and real contact area govern adhesion, friction, wear and lubrication, and how surfaces are engineered.

Chapter	Central question	What it delivers
42. Thermodynamics and Kinetics of Corrosion	When can a metal corrode, and how fast?	The corrosion cell; Pourbaix diagrams for Fe, Al, Cu, Zn and Ti; mixed-potential theory and Evans diagrams; Butler–Volmer and Tafel kinetics; corrosion rates from Faraday's law; passivity and its breakdown; Wagner's theory of oxidation, the parabolic law and the Pilling–Bedworth ratio
43. Forms of Corrosion and Protection	What shapes does corrosion take in real structures, and how is it prevented?	Uniform, galvanic, pitting, crevice and intergranular attack; stress-corrosion cracking, hydrogen embrittlement and corrosion fatigue; microbial, selective, erosion and cavitation damage; coatings, galvanizing, inhibitors; sacrificial and impressed-current cathodic protection with sizing; anodic protection; corrosion testing
44. Degradation and Failure Analysis	How do all classes of material degrade, and how do we find out why a part failed?	Degradation mechanisms of metals, ceramics, polymers, composites and concrete; polymer ageing and weathering; wear-out and life prediction; failure-analysis methodology and fractography; case studies; Weibull statistics and the bathtub curve
45. Surfaces, Interfaces and Tribology	What happens where two materials meet and slide?	Surface structure and energy; adsorption; adhesion and wetting; Hertz contact and real contact area; the origins of Amontons' friction laws; adhesive, abrasive, fatigue and corrosive wear; lubrication regimes and the Stribeck curve; surface engineering

How this part connects to the rest of the book. Corrosion and oxidation are thermodynamics and kinetics applied at an interface: the Gibbs energies and Ellingham diagrams of Chapter 6, the diffusion of Chapter 8 and the defect chemistry of Chapter 5 return here as electrode potentials, scale growth and passive films. The fracture mechanics and fatigue of Chapter 14 reappear when a corrosive environment helps a crack grow, and the alloys of Chapters 15 and 16 are largely defined by their corrosion behaviour (the chromium in stainless steel, the clad layers on aerospace aluminium). The energy and extreme-environment materials of Chapters 40 and 41 depend on the protective scales and hydrogen resistance developed here. Looking forward, coatings and surface treatments are made by the processes of Chapters 47–49, corrosion products and fracture surfaces are examined with the methods of Chapters 52–54, and the cost and lifetime of a material in service feed its selection (Chapter 58) and its environmental footprint (Chapter 59). Electrochemistry fundamentals, the Nernst equation and electrode kinetics, are taught in the Advanced Chemistry book; this part applies them to materials.

Chapter 42 — Thermodynamics and Kinetics of Corrosion

Part IX · Corrosion, Degradation and Surfaces

A water utility digs up two sections of the same ductile-iron main, laid in the same year and carrying the same water. The section buried in well-drained sand is nearly as thick as when it was cast. The section fifty metres away, where the trench crosses waterlogged clay, has lost half its wall and is perforated in two places. In a chemical plant nearby, a 304 stainless-steel tank has served for twenty years in dilute nitric acid without measurable loss, yet a 304 heat-exchanger tube in brackish cooling water developed pits deep enough to leak in eighteen months. The steel of the turbine casing of the plant's gas turbine, meanwhile, grows a grey oxide in hot air at a rate that can be predicted to the micrometre from its temperature. None of these outcomes depends on strength, stiffness or toughness. They depend on the thermodynamics that decides whether a metal can react with its environment, and on the kinetics that decide how fast it does. Corrosion is estimated to cost the world economy of the order of 3 % of its gross domestic product each year, and studies suggest that a fifth to a third of that could be saved by applying what is already known. This chapter supplies the science behind that knowledge.

> **When is a metal thermodynamically able to corrode in water or air, and what does it corrode to? What sets the rate of corrosion, and how is an electrical current turned into millimetres per year? Why do some metals, such as stainless steel, aluminium and titanium, survive in environments where they ought to dissolve, and when does that protection fail?**

Concept	Engineering destination
Corrosion cell: anode, cathode, electrolyte, metallic path	Diagnosing any corrosion problem, from car bodies to buried pipes
Electrode potentials, galvanic series	Choosing compatible metals for joints, fasteners and heat exchangers
Pourbaix diagrams	Water treatment, boiler chemistry, concrete reinforcement, cathodic protection
Butler–Volmer and Tafel kinetics	Interpreting polarization tests; designing inhibitors and coatings
Mixed-potential theory, Evans diagrams	Predicting corrosion potential and rate, galvanic coupling, effect of oxygen
Faraday's law, polarization resistance	Corrosion allowances, monitoring probes, inspection intervals
Passivity and its breakdown	Stainless-steel grade selection; pitting and crevice resistance
Wagner theory, parabolic law, Pilling–Bedworth ratio	Furnace parts, turbines, boilers, hot rolling scale losses

Roadmap. The corrosion cell (42.1) → thermodynamics: potentials and the galvanic series (42.2) and Pourbaix diagrams (42.3) → kinetics: Butler–Volmer and Tafel (42.4) and mixed potentials (42.5) → turning current into rate (42.6) → passivity and its breakdown (42.7) → corrosion in hot gas (42.8).

> **Boundary.**
> The thermodynamics of electrochemical cells, the Nernst equation, standard electrode potentials and the derivation of the Butler–Volmer equation are electrochemistry fundamentals taught in the Advanced Chemistry book. They are recapped here and applied to metals and alloys, not derived. The forms that corrosion takes in structures, and how to prevent them, are the subject of Chapter 43.

42.1 The electrochemical corrosion cell

Four necessary elements

Aqueous corrosion is an electrochemical process. A metal surface in contact with water carries two kinds of reaction at once. At **anodic** sites the metal is oxidized and passes into solution as ions, releasing electrons:

$$\mathrm{M} \rightarrow \mathrm{M}^{n+} + n\,e^{-}$$

for example $\mathrm{Fe} \rightarrow \mathrm{Fe}^{2+} + 2e^{-}$. At **cathodic** sites those electrons are consumed by a reduction. In neutral and alkaline water open to air the dominant cathodic reaction is oxygen reduction,

$$\mathrm{O_2} + 2\,\mathrm{H_2O} + 4\,e^{-} \rightarrow 4\,\mathrm{OH}^{-}$$

and in acid, or in the absence of oxygen, hydrogen evolution, $2\mathrm{H}^{+} + 2e^{-} \rightarrow \mathrm{H_2}$ (or $2\mathrm{H_2O} + 2e^{-} \rightarrow \mathrm{H_2} + 2\mathrm{OH}^{-}$ in neutral water). Other oxidants, such as Fe^{3+}, Cu^{2+}, dissolved chlorine or nitric acid, can also serve.

A corrosion cell therefore needs four elements (Figure 42.1): an **anode**, a **cathode**, an **electrolyte** that carries ionic current between them, and a **metallic path** that carries electrons. Remove any one and corrosion stops. This simple statement underlies every protection method in Chapter 43: coatings separate the metal from the electrolyte, insulating gaskets break the metallic path between dissimilar metals, deaeration removes the cathodic reactant, and cathodic protection makes the whole structure a cathode.

The anodic and cathodic sites may be microscopic and constantly shifting, giving fairly uniform attack, or fixed and separated, giving localized attack. Differences in composition (phases, inclusions, welds), in surface condition (mill scale, scratches), in stress, or in the environment (oxygen concentration, temperature, flow) fix the sites. The ferrous corrosion products then react further: Fe^{2+} meets OH^{-} and dissolved oxygen to precipitate hydrated ferric oxide, rust, often at some distance from the anode.

The water main and differential aeration

The pipe in the opening paragraph corroded by **differential aeration**. Where it passed through sand, oxygen reached the whole surface. Where it passed through saturated clay, oxygen could not diffuse in. The oxygen-starved section became anodic and the well-aerated section, close by and electrically connected through the pipe wall, became the cathode. Because the cathode was large and the anode small, the anode's current density, and therefore its rate of metal loss, was high. The same mechanism concentrates attack beneath a droplet of salt water (Evans's classic drop experiment: the oxygen-poor centre corrodes while rust forms in a ring near the edge), under deposits, in crevices and at the waterline of a partially immersed steel pile.

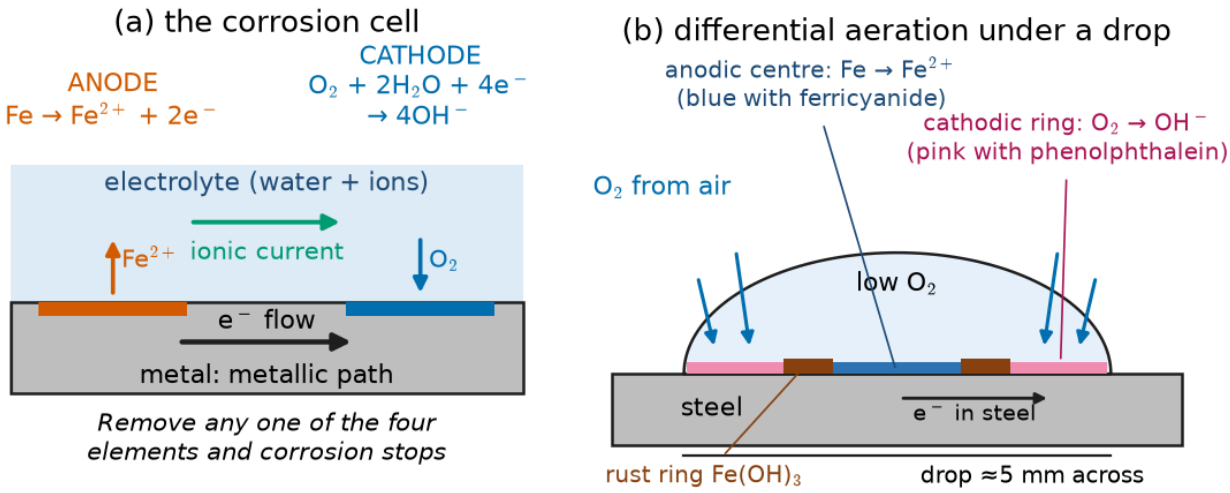


Figure 42.1 — The corrosion cell needs an anode, a cathode, an electrolyte and a metallic path; under a salt-water drop, differential aeration makes the oxygen-poor centre anodic.

42.2 Thermodynamics: potentials and the galvanic series

Electrode potentials (recap)

Whether a metal can corrode is decided by the Gibbs energy of the overall reaction, $\Delta G = -nFE$, where E is the cell potential: the reaction is possible if the potential of the cathodic reaction lies above that of the anodic reaction. Each half-reaction has an equilibrium potential given by the **Nernst equation**,

$$E = E^\circ - \frac{RT}{nF}\ln\frac{a_{\text{red}}}{a_{\text{ox}}} = E^\circ + \frac{0.05916}{n}\log_{10}\frac{a_{\text{ox}}}{a_{\text{red}}} \quad (25\ ^\circ\text{C})$$

where E° is the standard potential relative to the standard hydrogen electrode (SHE) and the a are activities (roughly, molar concentrations of dissolved species and partial pressures of gases). For a metal dissolving to its ions, $E = E^\circ + (0.05916/n)\log a_{\text{M}^{n+}}$. In corrosion engineering an ion activity of 10^{-6} is conventionally taken as the threshold of "corrosion": below it the metal is considered not to dissolve measurably.

> **Status of the principle: Nernst equation**
> *Fundamental.* It follows from the equilibrium condition on electrochemical potentials. It gives the equilibrium potential only; it says nothing about rates, and real solutions require activities, not concentrations.

The two cathodic reactions available in water have potentials that depend on pH:

$$E_{\text{H}^+/\text{H}_2} = -0.0592\,\text{pH}, \qquad E_{\text{O}_2/\text{H}_2\text{O}} = 1.229 - 0.0592\,\text{pH} \quad (\text{V SHE},\ 1\ \text{atm},\ 25\ ^\circ\text{C})$$

These lines bound the stability of water. Any metal whose equilibrium potential lies below the hydrogen line can, in principle, be corroded by water alone with hydrogen evolution; any metal below the oxygen line can be corroded by aerated water. Only gold and platinum lie above both over most of the pH range.

Table 42.1 — Standard reduction potentials of metals at 25 °C (the emf series).

Electrode reaction	E° (V vs SHE)	Equilibrium potential at $a = 10^{-6}$ (V)
$\text{Au}^{3+} + 3e^- \rightarrow \text{Au}$	+1.50	+1.38
$\text{O}_2 + 4\text{H}^+ + 4e^- \rightarrow 2\text{H}_2\text{O}$	+1.229	(pH-dependent)
$\text{Ag}^+ + e^- \rightarrow \text{Ag}$	+0.80	+0.44
$\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$	+0.34	+0.16
$2\text{H}^+ + 2e^- \rightarrow \text{H}_2$	0.000	(pH-dependent)
$\text{Ni}^{2+} + 2e^- \rightarrow \text{Ni}$	−0.25	−0.43
$\text{Fe}^{2+} + 2e^- \rightarrow \text{Fe}$	−0.447	−0.62
$\text{Cr}^{3+} + 3e^- \rightarrow \text{Cr}$	−0.74	−0.86
$\text{Zn}^{2+} + 2e^- \rightarrow \text{Zn}$	−0.76	−0.94
$\text{Ti}^{2+} + 2e^- \rightarrow \text{Ti}$	−1.63	−1.81
$\text{Al}^{3+} + 3e^- \rightarrow \text{Al}$	−1.66	−1.78
$\text{Mg}^{2+} + 2e^- \rightarrow \text{Mg}$	−2.37	−2.55

The galvanic series

The emf series describes pure metals in contact with their own ions at unit activity, a condition never met in service. Engineers instead use a **galvanic series**: the measured corrosion potentials of real alloys in a specified environment, most often flowing seawater (Table 42.2). The two rankings differ in instructive ways. Titanium and aluminium, near the bottom of the emf series, sit much higher in the galvanic series, and stainless steels appear twice, because each develops a passive oxide film that changes its effective potential (Section 42.7). The galvanic series is thus not a thermodynamic table; it is an empirical record of how metals actually behave, kinetics and films included.

Table 42.2 — Galvanic series in flowing seawater at about 25 °C (approximate corrosion potentials).

Alloy	Potential (V vs saturated calomel electrode)
Graphite, platinum	+0.2 to +0.3
Titanium, Alloy 625, super-austenitic stainless	−0.05 to +0.06
316 stainless steel (passive)	−0.05 to −0.10
70/30 copper–nickel	−0.18 to −0.25
Copper, brasses	−0.30 to −0.37
316 stainless steel (active, in crevices)	−0.43 to −0.54
Carbon steel, cast iron	−0.60 to −0.72
Aluminium alloys	−0.76 to −1.0
Zinc	−0.98 to −1.03
Magnesium	−1.60 to −1.63

When two metals are electrically connected in an electrolyte, the one lower in the series tends to become the anode and corrode faster, and the one higher is protected. The farther apart they are, the greater the driving force; whether the coupling is damaging depends on kinetics and the area ratio (Section 42.5 and Chapter 43). This is why steel bolts in a copper-alloy seawater pump fail quickly, while zinc and aluminium anodes protect ship hulls.

> **Status of the principle: galvanic series**
> *Empirical.* A ranking of measured corrosion potentials valid only for the environment and conditions in which it was measured; it predicts the direction of galvanic action but not its rate.

42.3 Pourbaix diagrams

Construction

A **Pourbaix diagram** (potential–pH diagram) maps the thermodynamically stable species of a metal–water system as a function of electrode potential and pH, at a chosen temperature and dissolved-ion activity (usually 25 °C and 10^{-6}). It is assembled from the equilibria of three kinds of reaction:

- **Electron transfer without** H^+ (for example $Fe^{2+} + 2e^- \rightleftharpoons Fe$): a horizontal line, independent of pH.
- H^+ **transfer without electrons** (for example $Fe^{2+} + 2H_2O \rightleftharpoons Fe(OH)_2 + 2H^+$): a vertical line at a fixed pH.
- **Both** (for example $Fe_2O_3 + 6H^+ + 2e^- \rightleftharpoons 2Fe^{2+} + 3H_2O$): a sloping line with slope $-0.0592\,m/n$ V per pH unit, where m is the number of protons and n the number of electrons.

The resulting regions are labelled by their practical meaning: **immunity**, where the metal itself is stable and cannot corrode; **corrosion**, where a soluble ion is stable above 10^{-6}; and **passivation**, where a solid oxide or hydroxide is stable and may, if it is adherent and compact, protect the metal. Superimposing the two water lines shows which oxidants are available.

Worked example — the boundaries of the iron diagram. At $a_{Fe^{2+}} = 10^{-6}$ and 25 °C:

- Immunity–corrosion boundary: $E = -0.447 + (0.05916/2) \times (-6) = -0.447 - 0.177 = -0.624$ V, horizontal.
- Fe_2O_3/Fe^{2+} boundary, from $Fe_2O_3 + 6H^+ + 2e^- \rightleftharpoons 2Fe^{2+} + 3H_2O$ with $E^\circ = 0.728$ V: $E = 0.728 - 0.1775\,\mathrm{pH} - 0.0592 \log a_{Fe^{2+}} = 1.083 - 0.1775\,\mathrm{pH}$. With $m = 6$ and $n = 2$, the slope is $-0.0592 \times 6/2 = -0.1775$ V per pH unit.
- Fe_3O_4/Fe boundary, from $Fe_3O_4 + 8H^+ + 8e^- \rightleftharpoons 3Fe + 4H_2O$ with $E^\circ = -0.085$ V: $E = -0.085 - 0.0592\,\mathrm{pH}$. It meets the -0.624 V line at $\mathrm{pH} = (0.624 - 0.085)/0.0592 = 9.1$.

So iron corrodes as Fe^{2+} at all pH below about 9 if its potential is above −0.62 V, while above pH 9 the stable product is magnetite and then haematite. Since the hydrogen line (-0.0592 pH) lies above −0.62 V at all pH below about 10.5, iron in deaerated acid or neutral water is thermodynamically able to corrode with hydrogen evolution; in aerated water the oxygen line lies far above it.

Status of the principle: Pourbaix diagram
Derived from equilibrium thermodynamics (the Nernst equation and solubility products) for a stated temperature and ion activity. It predicts what can happen, never how fast; it ignores kinetics, non-equilibrium films, alloying and complexing ions such as chloride, and it does not show whether an oxide is protective.

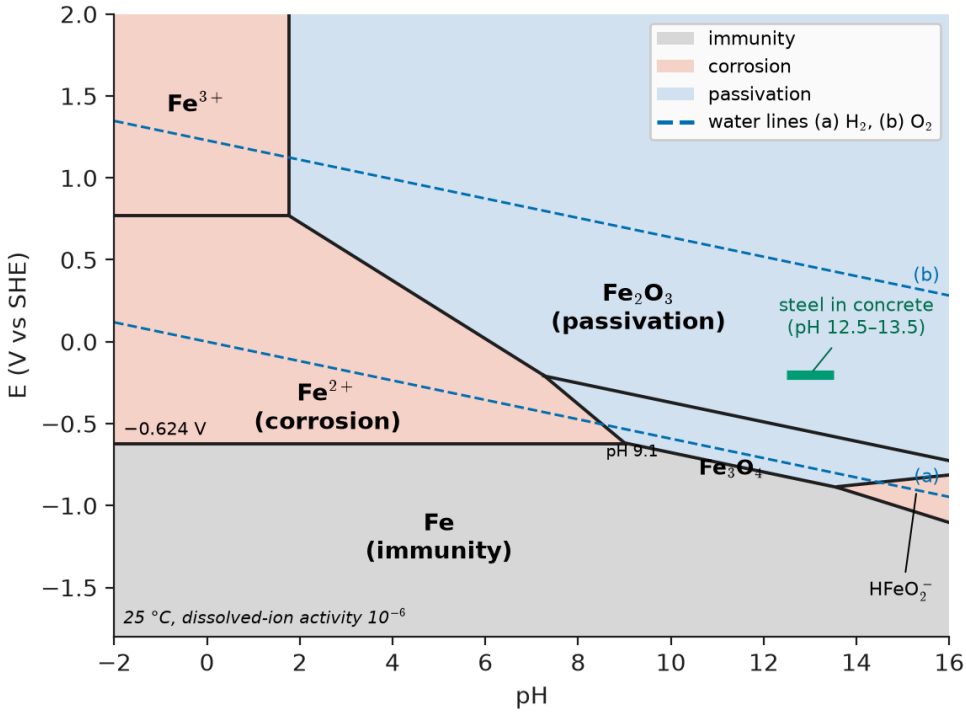


Figure 42.2 — The Pourbaix diagram of iron at 25 C and 10^-6 activity: iron is immune below -0.62 V, corrodes as Fe2+ in acid and neutral water, and is covered by oxides in alkaline water.

Iron, aluminium, copper, zinc and titanium compared

The diagrams of the main engineering metals explain much of their practical behaviour (Figure 42.3).

- **Iron** (Figure 42.2) is passive only in alkaline solution. That is why steel reinforcement survives in concrete, whose pore solution has a pH of 12.5–13.5, until carbonation lowers the pH or chloride breaks the film (Chapters 18 and 43); and why boiler water is dosed to pH 9–10 so that a protective magnetite layer forms on the tubes.
- **Aluminium** is **amphoteric**: its oxide dissolves in both acid and base. At 10^{-6} activity, Al_2O_3 (as the hydrate) is stable only between about pH 4 and 8.6; below, aluminium dissolves as Al^{3+}, above, as aluminate AlO_2^-. Its immunity region lies near −1.8 V, far below the water line, so aluminium in water is always thermodynamically unstable; its usefulness rests entirely on its oxide film in near-neutral water. This is why aluminium must not be cleaned with strong alkali, why it is attacked by wet fresh concrete and mortar, and why aluminium cooking pots are pitted by salty, acidic foods.
- **Zinc** is also amphoteric, passive roughly between pH 8.5 and 11 (where ZnO or $Zn(OH)_2$ is stable) and soluble outside. In the atmosphere it forms a protective basic carbonate patina, the basis of galvanizing, but it corrodes rapidly in acid rain and in hot alkaline water.
- **Copper** has an immunity region that reaches *above* the hydrogen line: copper is thermodynamically stable in water free of oxidants. It corrodes only when oxygen or another oxidant is present, which is why copper water pipes last decades in most supplies but are attacked by aerated acidic water, ammonia (which complexes copper ions and is not shown on the simple diagram) and oxidizing acids. In near-neutral to alkaline aerated water, Cu_2O and CuO films protect.
- **Titanium** has a TiO_2 passive region spanning essentially the whole pH range at potentials above the very negative immunity line, which is why titanium resists seawater, chlorides, wet chlorine and oxidizing acids, and is the material of choice for seawater heat exchangers, chlorine plants and implants (Chapter 39). It corrodes in hot reducing acids (hydrochloric, sulfuric) and in fluoride solutions, which form soluble fluoro-complexes.

The diagrams' limitations are as instructive as their predictions. A "passive" region only shows that a solid is stable; whether it protects depends on its structure and adherence, which thermodynamics cannot predict. Chloride is absent from the standard

diagrams, yet it is the cause of most pitting. Alloys have no single diagram, and high temperature shifts all the lines. Pourbaix diagrams are therefore maps of possibility, to be combined with the kinetics of the following sections.

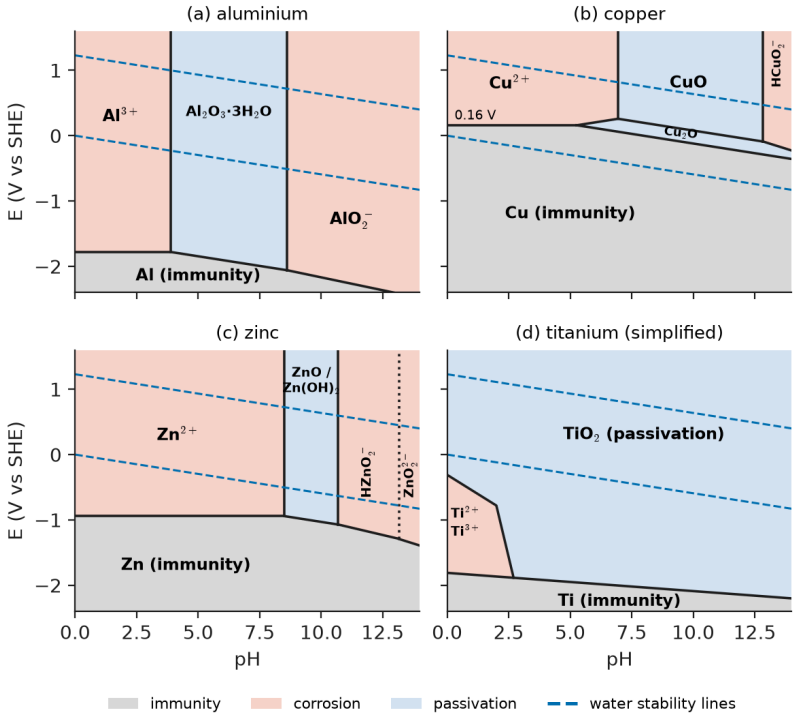


Figure 42.3 — Simplified Pourbaix diagrams of aluminium, copper, zinc and titanium at 10^-6 activity: aluminium and zinc are amphoteric, copper is immune in oxidant-free water, titanium is passive at almost all pH.

42.4 Electrode kinetics: Butler–Volmer and Tafel

Overpotential and exchange current

Thermodynamics fixes the direction; kinetics fixes the rate. At its equilibrium potential E_{eq}, an electrode is not inert: the forward and backward reactions proceed at equal and opposite rates, each with a current density i_0, the **exchange current density**. Displacing the potential by an **overpotential** $\eta = E - E_{eq}$ lowers the activation barrier in one direction and raises it in the other. For a single-step reaction controlled by charge transfer, the net current density is the **Butler–Volmer equation**:

$$i = i_0 \left[\exp\left(\frac{\alpha_a n F \eta}{RT}\right) - \exp\left(-\frac{\alpha_c n F \eta}{RT}\right) \right]$$

where α_a and α_c are the anodic and cathodic transfer coefficients (often about 0.5).

> **Status of the principle: Butler–Volmer equation**
> *Derived* from transition-state theory for a charge-transfer-controlled electrode reaction (the derivation belongs to Advanced Chemistry). It assumes that mass transport is fast and that the surface is unchanged by the reaction; multistep reactions, adsorbed intermediates and films change the effective coefficients.

The exchange current density is a property of the reaction *on a particular surface*, and it varies enormously. Hydrogen evolution is fast on platinum and slow on zinc, lead and mercury (Table 42.3). This is not a detail. It explains why zinc dissolves only slowly in acid despite its strongly negative potential (its own surface catalyses hydrogen evolution poorly), why touching zinc with a platinum wire makes it fizz, and why trace impurities of noble metals in magnesium or zinc accelerate their corrosion.

Table 42.3 — Exchange current densities for hydrogen evolution in acid at 25 °C (approximate).

Electrode metal	i_0 for $2\mathrm{H^+} + 2e^- \to \mathrm{H_2}$ (A/cm²)
Pt	$\approx 10^{-3}$
Ni	$\approx 10^{-5}$–10^{-6}
Fe	$\approx 10^{-6}$
Cu	$\approx 10^{-7}$
Zn	$\approx 10^{-10}$–10^{-11}
Pb, Hg	$\approx 10^{-12}$–10^{-13}

The Tafel regime

When the overpotential exceeds about 50 mV, one exponential in the Butler–Volmer equation dominates and the other can be dropped. For an anodic reaction,

$$\eta_a = b_a \log_{10} \frac{i}{i_0}, \qquad b_a = \frac{2.303\,RT}{\alpha_a n F}$$

and similarly for a cathodic reaction with slope b_c. This is the **Tafel equation**: overpotential is linear in the logarithm of current density. The **Tafel slope** b for $\alpha = 0.5$, $n = 1$ at 25 °C is $2.303 \times 8.314 \times 298 / (0.5 \times 96\,485) = 0.118$ V per decade, the value often observed for hydrogen evolution; anodic dissolution of iron typically shows 40–60 mV per decade. Plotting potential against $\log|i|$ turns Tafel kinetics into straight lines, which is the basis of the Evans diagram.

> **Status of the principle: Tafel equation**
> *Approximate.* The high-overpotential limit of Butler–Volmer; it fails near equilibrium (below about 50 mV), where both directions matter, and at high currents, where mass transport limits the rate.

Concentration polarization

When a reactant must diffuse to the surface, its supply sets an upper bound. For a reactant of bulk concentration C_b and diffusion coefficient D crossing a diffusion boundary layer of thickness δ, the **limiting current density** is

$$i_L = \frac{nFDC_b}{\delta}$$

Oxygen reduction in neutral water is the most important case: oxygen is sparingly soluble (about 8 mg/L, or 2.5×10^{-7} mol/cm³, in air-saturated water at 25 °C) and diffuses slowly ($D \approx 2 \times 10^{-5}$ cm²/s). In still water δ is about 0.05 cm, so

$$i_L = \frac{4 \times 96\,485 \times 2 \times 10^{-5} \times 2.5 \times 10^{-7}}{0.05} = 3.9 \times 10^{-5}\ \mathrm{A/cm^2} = 39\ \mu\mathrm{A/cm^2}$$

Stirring, flow or higher temperature thins the boundary layer and raises i_L, which is why steel corrodes faster in flowing water than in still water.

42.5 Mixed-potential theory and Evans diagrams

The corrosion potential

A corroding metal is not at the equilibrium potential of either of its reactions. It sits at a single **mixed potential**, the **corrosion potential** E_{corr}, at which the total rate of oxidation equals the total rate of reduction, so that no net current flows out of the isolated specimen. This is the **Wagner–Traud** principle: any electrochemical reaction can be divided into partial oxidation and reduction reactions; there can be no net accumulation of charge; and the partial reactions proceed independently, each according to its own kinetics.

Status of the principle: mixed-potential theory
Derived from charge conservation and the independence of partial reactions. It holds when the partial reactions do not interact; strong adsorption, film formation or local separation of anodes and cathodes (with ohmic drop between them) require refinements.

An **Evans diagram** (Figure 42.4) plots the anodic line of the metal and the cathodic line of the oxidant on axes of potential against $\log|i|$. Their intersection gives E_{corr} and the **corrosion current density** i_{corr}. The diagram makes the controlling factors visible at a glance: whichever line is steeper, or whichever reaction is limited by mass transport, controls the rate.

Worked example — steel in deaerated acid. Take iron in deaerated acid at pH 1, with these representative values: for iron dissolution, $E_{eq} = -0.624$ V (taking $a_{Fe^{2+}} = 10^{-6}$), $i_0 = 10^{-8}$ A/cm², $b_a = 0.060$ V/decade; for hydrogen evolution on iron, $E_{eq} = -0.0592 \times 1 = -0.059$ V, $i_0 = 10^{-6}$ A/cm², $b_c = 0.120$ V/decade. At E_{corr} the two Tafel lines meet:

$$-0.624 + 0.060 \log \frac{i}{10^{-8}} = -0.059 - 0.120 \log \frac{i}{10^{-6}}$$

Collecting terms, $0.180 \log i = -0.059 + 0.624 + 0.060 \times (-8) + 0.120 \times (-6) = -0.635$, so $\log i = -3.53$ and $i_{corr} = 3.0 \times 10^{-4}$ A/cm² $= 0.30$ mA/cm². Substituting back, $E_{corr} = -0.624 + 0.060 \times (-3.53 + 8) = -0.356$ V. Using the conversion of Section 42.6 (1 µA/cm² corresponds to 11.6 µm/year for iron), the rate is $300 \times 0.0116 = 3.5$ mm per year, typical of steel in dilute hydrochloric acid. Changing the assumed Fe^{2+} activity by a factor of 10 moves E_{eq} by 30 mV but i_{corr} by only a factor of about 1.5, because the cathodic line is steeper. Doubling the cathodic exchange current density (for instance, by a noble-metal impurity on the surface) raises i_{corr} by a factor of $2^{0.12/0.18} = 1.6$.

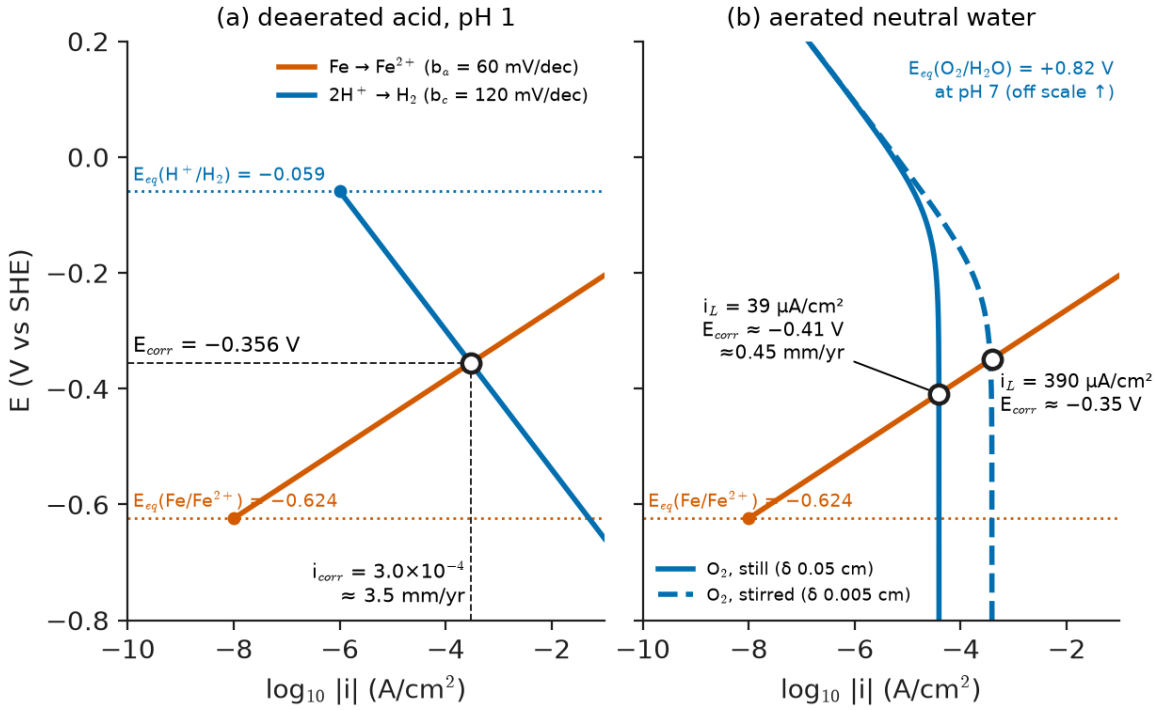


Figure 42.4 — Evans diagram for iron: the corrosion potential and current lie where the anodic and cathodic lines cross; in aerated neutral water, the oxygen limiting current controls the rate.

What mixed-potential theory explains

- **Oxygen-controlled corrosion.** In aerated neutral water the cathodic reaction is oxygen reduction at its limiting current. The anodic line of iron crosses the vertical part of the cathodic curve, so $i_{corr} = i_L \approx 39$ µA/cm², or about $39 \times 0.0116 = 0.45$ mm/year in still water. The rate is set by oxygen diffusion, not by the steel, which is why ordinary carbon steels of different compositions corrode at similar rates in natural waters, why deaeration of boiler feedwater (to a few parts per billion of oxygen) is so effective, and why flow accelerates attack.
- **Galvanic coupling.** Connecting a metal to a more noble one adds the noble metal's cathodic area, on which the oxidant is also reduced. The couple settles at a new mixed potential, raising the corrosion of the active metal and lowering that of

the noble one. When the cathodic reaction is diffusion-limited, the total cathodic current is proportional to the cathode's area, so the anode's corrosion rate scales with the area ratio A_c/A_a: a small anode coupled to a large cathode corrodes catastrophically. Steel rivets in a copper plate fail quickly; copper rivets in a steel plate do little harm.

- **Inhibitors and coatings.** Anodic inhibitors raise or steepen the anodic line; cathodic inhibitors lower the cathodic exchange current density or block oxygen access. Both lower i_{corr}, and the Evans diagram shows which is more effective in a given case.

42.6 Corrosion rate from current: Faraday's law

Converting current to penetration

Every n electrons released correspond to one metal atom lost, so **Faraday's law** gives the mass lost per unit area and time from the corrosion current density: $\dot{m} = i_{corr} M/(nF)$. Dividing by the density ρ gives the penetration rate. In practical units,

$$\mathrm{CR}\ (\mathrm{mm/year}) = 3.27 \times 10^{-3}\,\frac{i_{corr}\,M}{n\rho}$$

with i_{corr} in μA/cm², M in g/mol and ρ in g/cm³. The constant 3.27×10^{-3} collects the seconds in a year, Faraday's constant and the unit conversions.

> **Status of the principle: Faraday's law of electrolysis**
> *Fundamental* as charge conservation applied to a reaction of known stoichiometry. Converting the result to a penetration rate assumes uniform attack over the area counted and a known valence; for localized attack the local rate can be orders of magnitude higher than the average.

Table 42.4 — Penetration rate for a corrosion current density of 1 μA/cm² (computed from Faraday's law).

Metal	Valence n	M (g/mol)	ρ (g/cm³)	mm/year per μA/cm²
Fe (carbon steel)	2	55.85	7.87	0.0116
Ni	2	58.69	8.90	0.0108
Cu	2	63.55	8.96	0.0116
Zn	2	65.38	7.13	0.0150
Al	3	26.98	2.70	0.0109
Ti	4	47.87	4.51	0.0087
Mg	2	24.31	1.74	0.0228

A useful rule follows: for most structural metals, **1 μA/cm² is roughly 10 μm per year**. Typical corrosion rates and their engineering significance are summarized in Table 42.5.

Table 42.5 — Typical uniform corrosion rates (order of magnitude).

Situation	Rate
Passive stainless steel or titanium in water	<1 μm/year
Zinc in a rural atmosphere	≈1 μm/year
Carbon steel in a rural to urban atmosphere	10–50 μm/year
Carbon steel in a marine splash zone	0.2–0.5 mm/year
Carbon steel in aerated seawater, immersed	≈0.1–0.2 mm/year
Carbon steel in dilute hydrochloric acid	several mm/year

Rates below about 0.1 mm/year are usually acceptable for structural steel with a corrosion allowance; above about 1 mm/year a material is rarely acceptable without protection.

Measuring the corrosion current

The corrosion current cannot be measured directly with an ammeter, because anodic and cathodic currents flow on the same surface. Two electrochemical methods (developed with others in Chapter 43) infer it:

- **Tafel extrapolation.** The specimen is polarized a few hundred millivolts on either side of E_{corr}; the linear Tafel regions are extrapolated back to E_{corr}, where they meet at i_{corr}.
- **Linear polarization resistance.** Within about ±10 mV of E_{corr}, the Butler–Volmer expressions are nearly linear, and the slope of the potential–current curve, the **polarization resistance** $R_p = \Delta E/\Delta i$, is inversely proportional to the corrosion rate. This is the **Stern–Geary relation**:

$$i_{corr} = \frac{B}{R_p}, \qquad B = \frac{b_a b_c}{2.303\,(b_a + b_c)}$$

Worked example — a corrosion probe in a cooling-water system. A linear-polarization probe in a cooling circuit reads $R_p = 1000\ \Omega\ \text{cm}^2$. With Tafel slopes $b_a = 0.060$ V and $b_c = 0.120$ V per decade,

$$B = \frac{0.060 \times 0.120}{2.303 \times 0.180} = 0.0174\ \text{V}, \qquad i_{corr} = \frac{0.0174}{1000} = 1.74 \times 10^{-5}\ \text{A/cm}^2 = 17.4\ \mu\text{A/cm}^2$$

For carbon steel this is $17.4 \times 0.0116 = 0.20$ mm/year. If the water treatment is improved and R_p rises tenfold to $10^4\ \Omega\ \text{cm}^2$, the rate falls to 0.02 mm/year. Because R_p can be measured every few minutes without damaging the probe, the method is the basis of online corrosion monitoring in refineries, cooling systems and pipelines.

> **Status of the principle: Stern–Geary relation**
> *Approximate.* Derived from the linearization of two Tafel reactions near E_{corr}; it assumes both reactions are activation-controlled, that the solution resistance is small or corrected, and that the Tafel slopes are known. Uncertainty in B typically limits accuracy to a factor of about two.

42.7 Passivity and passive-film breakdown

The active–passive transition

Iron dissolves rapidly in dilute sulfuric acid but hardly at all in concentrated nitric acid, a far stronger oxidant. Faraday noticed this in the 1830s; the explanation is **passivity**. Above a certain potential, metals such as iron, chromium, nickel, titanium and aluminium form a continuous oxide film only 1–5 nm thick, which conducts ions so poorly that the dissolution current falls by three to six orders of magnitude.

The **anodic polarization curve** of a passivating metal (Figure 42.5) has a characteristic shape. Just above E_{corr} the metal dissolves actively, with current rising along a Tafel line. At the **primary passivation potential** E_{pp} the current reaches a maximum, the **critical current density** i_{crit}, and then falls abruptly to the **passive current density** i_{pass}, which remains nearly constant over a range of several hundred millivolts. At still higher potentials the current rises again: in the **transpassive** region the film itself oxidizes to soluble species (chromium to dichromate) or oxygen is evolved.

This shape explains the apparent paradox of the opening paragraph and of iron in acids. A metal is passive only if the cathodic reaction can supply a current larger than i_{crit} at potentials above E_{pp}, so that the mixed potential falls in the passive range. A weak oxidant (dilute non-oxidizing acid) intersects the anodic curve in the active region, giving rapid corrosion; a strong oxidant (nitric acid, dissolved oxygen in neutral water for stainless steel) intersects it in the passive region, giving negligible corrosion. A low i_{crit} and a low E_{pp} are what alloying for passivity seeks. **Anodic protection** (Chapter 43) exploits the same curve by holding a structure electronically at a potential in its passive range.

Status of the principle: passivity
Empirical in its parameters (E_{pp}, i_{crit}, i_{pass}, E_{pit}), which are measured for each alloy and environment. The underlying physics, a thin oxide with very low ionic conductivity whose growth and dissolution balance, is well established, but no model yet predicts the parameters from first principles.

Why chromium makes steel stainless

On iron the passive film is a mixed iron oxide that is stable only at high pH or high potential, so iron needs either alkalinity (as in concrete) or a strong oxidant to stay passive. Chromium forms a highly protective, chromium-rich oxide (essentially Cr_2O_3 with hydroxide and some iron) at much lower potentials and over a much wider pH range. When iron is alloyed with chromium, i_{crit} and E_{pp} fall sharply above roughly 10.5–12 % Cr, the threshold for "stainless" steel: the film forms spontaneously in air and in aerated water, is a few nanometres thick, and self-repairs within milliseconds if scratched, provided oxygen is present. Nickel stabilizes austenite and lowers active dissolution; molybdenum and nitrogen strengthen the film against chloride.

Typical passive current densities of stainless steels are below 1 µA/cm², which by Table 42.4 corresponds to less than about 10 µm per year: the 304 tank in nitric acid in the opening paragraph lives in this regime.

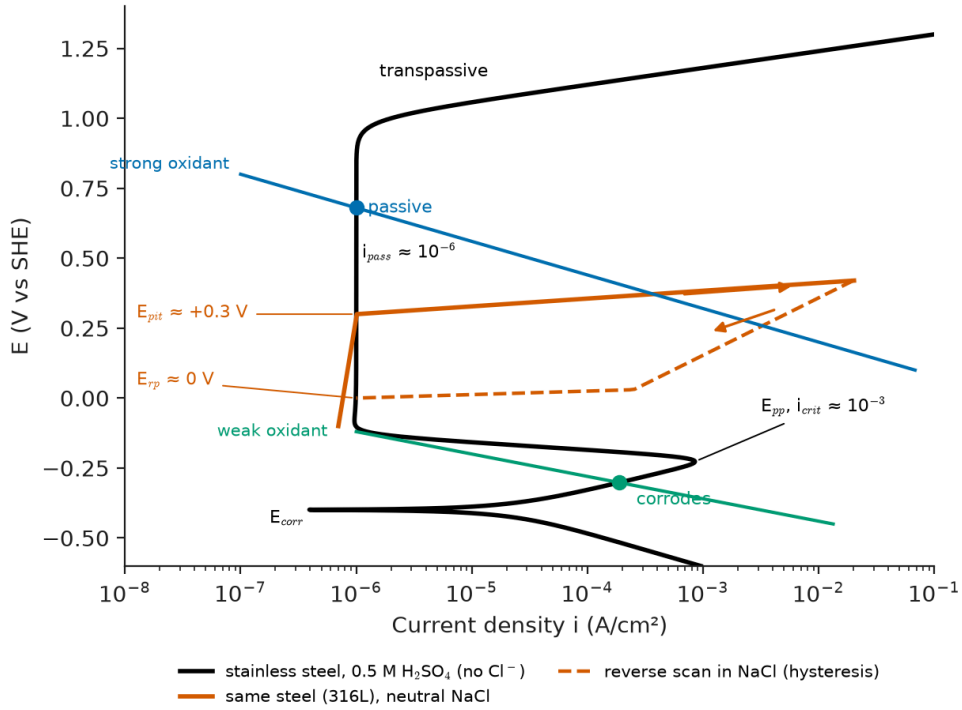


Figure 42.5 — Anodic polarization curve of a stainless steel: active dissolution, passivation above E_pp, a broad passive range, and breakdown by pitting in chloride or transpassive dissolution without it.

Breakdown of passivity: pitting

Passivity is a metastable condition, and its local failure is far more dangerous than uniform corrosion because it concentrates all the anodic current on a tiny area. **Chloride** ions are the principal agent. Above a critical **pitting potential** E_{pit}, chloride adsorbs on the film at weak points, often at sulfide inclusions (MnS in stainless steels) or where the film is thin, and breaks it locally. Once a pit starts, it sustains itself: metal ions hydrolyse inside it (for example $Cr^{3+} + 3H_2O \rightarrow Cr(OH)_3 + 3H^+$), the pH inside falls to 1–2, chloride migrates in to balance charge, and the acidic chloride solution cannot repassivate. The surrounding passive surface serves as a large cathode. On reversing the potential scan, the pit repassivates only below a lower **repassivation potential**, and the difference between E_{pit} and E_{rp} measures how hard it is to stop pits once they grow. Crevices behave similarly, the occluded geometry producing the acidic chloride solution without needing a pit to nucleate first (Chapter 43).

The resistance of stainless steels to chloride pitting is ranked by the empirical **pitting resistance equivalent number**,

$$\mathrm{PREN} = \%\mathrm{Cr} + 3.3\,\%\mathrm{Mo} + 16\,\%\mathrm{N}$$

(compositions in weight percent), and measured as a **critical pitting temperature** (CPT), above which pits initiate in a standard ferric chloride test.

> **Status of the principle: pitting resistance equivalent number**
> *Empirical.* A regression of pitting resistance on composition for wrought stainless steels; it ignores microstructure (inclusions, σ-phase, weld segregation), so a weld or a poorly heat-treated part can perform well below its PREN.

Table 42.6 — Pitting resistance of stainless steels (typical compositions and values).

Grade	Cr (%)	Mo (%)	N (%)	PREN	Approximate CPT in ferric chloride (°C)	Typical service
304 / 304L	18	—	0.05	≈19	<10	Fresh water, food, nitric acid
316 / 316L	17	2.1	0.05	≈25	15–25	Mildly chloride-bearing water
2205 duplex	22	3.1	0.17	≈35	30–45	Brackish water, chemical plant
2507 super-duplex	25	3.8	0.27	≈42	70–85	Seawater, offshore piping
6 % Mo super-austenitic	20	6.2	0.20	≈44	60–80	Seawater, bleach plants

A PREN of about 40 or more is generally needed for reliable service in ambient seawater. The 304 heat-exchanger tube of the opening paragraph, with a PREN of about 19 in brackish water, was operating above its pitting potential. Passive aluminium alloys and titanium show the same kind of breakdown, aluminium readily in chloride, titanium only in extreme conditions (hot concentrated chlorides, fluorides). The forms of localized corrosion that follow from film breakdown, and how to design against them, are the subject of Chapter 43.

42.8 High-temperature oxidation

Thermodynamics: will an oxide form?

In hot dry gas there is no electrolyte, yet the same separation of anodic and cathodic processes occurs: the oxide scale itself acts as a solid electrolyte, conducting ions and electrons. Whether a metal can oxidize is decided by the standard Gibbs energy of formation of its oxide, compared on an **Ellingham diagram** (Chapter 6). At the temperatures of furnaces, turbines and boilers, virtually all engineering metals except gold, silver and platinum can be oxidized by air, and the oxygen partial pressure in equilibrium with a metal and its oxide is extremely low (for Cr/Cr_2O_3 at 1000 °C, about 10^{-22} atm). Practical high-temperature alloys therefore never rely on thermodynamic immunity; they rely on growing a scale so slowly, and so adherently, that it protects.

Rate laws

Three rate laws describe most oxidation data (Figure 42.6):

- **Linear**, $x = k_l t$: the scale is porous, cracked or volatile and offers no barrier; the rate is set by a surface reaction or gas transport. Magnesium and the alkali metals, tungsten and molybdenum above the temperature at which their oxides volatilize, and carbon all oxidize linearly.
- **Parabolic**, $x^2 = k_p t$: a compact scale whose growth is controlled by diffusion through it. Most protective high-temperature oxidation is parabolic.
- **Logarithmic**, $x = k_e \log(at + 1)$: thin films at low temperature (aluminium and copper near room temperature), where transport is driven by electric fields across a few nanometres and slows rapidly.

Combinations are common. **Breakaway oxidation** is a transition from parabolic to linear when a protective scale cracks or the alloy beneath runs out of the element that formed it.

Wagner's theory of parabolic oxidation

Wagner's theory (1933) explains the parabolic law and predicts its rate constant. Assume a dense, adherent scale of thickness x, with local equilibrium at both its interfaces: at the metal–scale interface the oxygen activity is that of the metal–oxide equilibrium, and at the scale–gas interface it is that of the gas. The chemical potential difference across the scale is therefore fixed. Ions and electrons move through the scale under this gradient, by way of point defects (cation vacancies in NiO, FeO and Cr_2O_3; oxygen vacancies in ZrO_2; Chapter 5), coupled so that no net charge flows. The flux of ions is then inversely proportional to the scale thickness, $J \propto 1/x$, and since the growth rate is proportional to the flux,

$$\frac{dx}{dt} = \frac{k'}{x} \quad \Rightarrow \quad x^2 = 2k't \equiv k_p t$$

Wagner's analysis expresses the rate constant in terms of the self-diffusion coefficients of the cation D_M and anion D_O in the oxide, integrated across the range of oxygen activity a_O that the scale spans:

$$k' = \int_{a_O^i}^{a_O^o} \left(\frac{z_M}{|z_O|} D_M + D_O \right) d\ln a_O$$

where z_M and z_O are the ionic charges and the superscripts denote the metal–scale (inner) and scale–gas (outer) interfaces.

> **Status of the principle: Wagner's parabolic oxidation theory**
> *Derived* from coupled ionic and electronic transport through a dense scale with local equilibrium at its interfaces. It predicts measured rate constants within a factor of a few for simple oxides such as NiO, CoO and FeO when lattice diffusion controls; for chromia and alumina, grain-boundary diffusion in fine-grained scales usually dominates, and the parabolic form remains but the constant must be measured.

The theory ties oxidation resistance to defect chemistry. Wüstite, FeO, is highly non-stoichiometric ($Fe_{1-\delta}O$ with δ up to about 0.15), so iron cations diffuse through it very fast; NiO is nearly stoichiometric and grows slowly; Cr_2O_3 and above all α-Al_2O_3 have very low defect concentrations and grow orders of magnitude more slowly still. Doping a scale changes its defect concentration: Li^+ in NiO lowers the cation-vacancy concentration and slows oxidation (Cr^{3+} raises it and speeds oxidation), while reactive-element additions (Y, Hf) slow alumina growth by blocking grain-boundary transport (Chapter 41).

Worked example — three scales compared. Representative parabolic rate constants (in thickness units) at about 1000 °C are $k_p \approx 10^{-7}$ cm²/s for FeO on iron, 10^{-12} cm²/s for Cr_2O_3 on a chromia-forming alloy and, at 1100 °C, 10^{-14} cm²/s for α-Al_2O_3 on an alumina former. With $x = \sqrt{k_p t}$:

- Iron, 1 h ($t = 3600$ s): $x = \sqrt{10^{-7} \times 3600} = 0.019$ cm, or 190 μm. In 100 h, 1.9 mm. Plain steel is unusable in air at this temperature, and a steel slab reheated for rolling loses a measurable fraction of its mass as scale.
- Chromia former, 100 h (3.6×10^5 s): $x = \sqrt{10^{-12} \times 3.6 \times 10^5} = 6.0 \times 10^{-4}$ cm, or 6 μm; after 10,000 h, 60 μm, which is why chromia-forming alloys serve for long periods only up to about 950–1000 °C.
- Alumina former, 10,000 h (3.6×10^7 s): $x = \sqrt{10^{-14} \times 3.6 \times 10^7} = 6.0 \times 10^{-4}$ cm, or 6 μm.

The alumina former grows in 10,000 hours the scale the chromia former grows in 100, and the iron in under four seconds. Because $x \propto \sqrt{k_p}$, a hundredfold smaller rate constant means a tenfold thinner scale at any time. The rate constant itself depends on temperature through the Arrhenius law, typically with activation energies of 150–400 kJ/mol, so a 100 °C rise often increases k_p by an order of magnitude.

> **Status of the principle: parabolic rate law**
> *Approximate.* Holds while the scale is dense, adherent and thicker than a few tens of nanometres, and while the alloy can keep supplying the scale-forming element. Cracking, spallation, volatilization or depletion of that element end it.

The Pilling–Bedworth ratio

A protective scale must also cover the metal. Pilling and Bedworth (1923) compared the volume of oxide formed with the volume of metal consumed:

$$\mathrm{PBR} = \frac{V_{\text{oxide}}}{V_{\text{metal}}} = \frac{M_{\text{ox}}\,\rho_{\text{M}}}{n\,M_{\text{M}}\,\rho_{\text{ox}}}$$

where M_{ox} and ρ_{ox} are the molar mass and density of the oxide, M_M and ρ_M those of the metal, and n the number of metal atoms per oxide formula unit.

Worked example — magnesium, aluminium and iron. For MgO ($M = 40.30$, $\rho = 3.58$ g/cm³) on Mg ($M = 24.31$, $\rho = 1.74$ g/cm³), PBR $= 40.30 \times 1.74/(1 \times 24.31 \times 3.58) = 0.81$. For Al_2O_3 (101.96, 3.99) on Al (26.98, 2.70), PBR $= 101.96 \times 2.70/(2 \times 26.98 \times 3.99) = 1.28$. For FeO (71.84, 5.74) on Fe (55.85, 7.87), PBR $= 71.84 \times 7.87/(1 \times 55.85 \times 5.74) = 1.76$.

Table 42.7 — Pilling–Bedworth ratios (computed from handbook densities).

Metal	Oxide	PBR	Behaviour
K, Na, Ca	K_2O, Na_2O, CaO	0.44–0.65	Porous, non-protective
Mg	MgO	0.81	Non-protective at high temperature (linear)
Al	Al_2O_3	1.28	Protective
Zr	ZrO_2	1.55	Protective, but cracks at high thickness
Cu	Cu_2O	1.68	Protective
Ni	NiO	1.70	Protective
Fe	FeO	1.76	Protective but fast-growing
Ti	TiO_2	1.78	Protective at moderate temperature
Si	SiO_2	1.88	Protective
Cr	Cr_2O_3	2.01	Protective
Nb	Nb_2O_5	2.67	Cracks and spalls
Mo	MoO_3	3.29	Volatile above ≈700 °C
W	WO_3	3.40	Cracks; volatile at high temperature

A ratio below 1 means the oxide cannot cover the metal and is porous; a ratio between about 1 and 2 allows a continuous, moderately compressed scale; a ratio well above 2 generates large compressive growth stresses that crack and spall it. The magnesium alloys that burn in air, and the niobium alloys that need silicide coatings (Chapter 41), illustrate the two extremes.

> **Status of the principle: Pilling–Bedworth ratio**
> *Empirical* as a predictor of protectiveness. It is a useful first screen, but it ignores the direction of scale growth (outward cation diffusion creates new oxide at the gas side and generates little stress), thermal-expansion mismatch on cooling, plasticity of scale and metal, and volatility; several metals with "good" ratios form non-protective scales and vice versa.

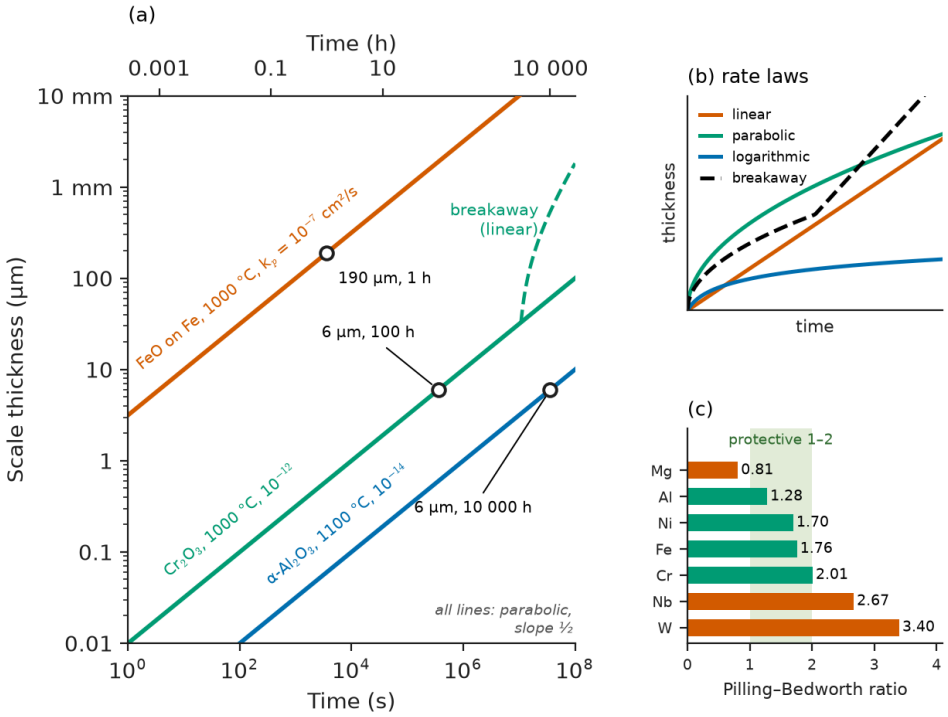


Figure 42.6 — Oxidation kinetics: protective scales grow parabolically, with rate constants spanning seven orders of magnitude from FeO to alumina; breakaway marks the loss of protection.

Selective oxidation of alloys

Engineering alloys protect themselves by **selective oxidation**: an alloy containing enough of an element whose oxide is both thermodynamically more stable and slower-growing than the base metal's will form a continuous scale of that oxide, which then controls the rate. Chromium in nickel- and iron-based alloys forms chromia; aluminium in nickel aluminides, FeCrAl alloys and aluminide coatings forms alumina; silicon in silicides and SiC forms silica. The alloy must contain a critical minimum concentration, so that the element diffuses to the surface fast enough to keep the scale continuous and to heal it if it cracks: roughly 15–20 % Cr for chromia formers, or about 5 % Al with chromium present in FeCrAl, where chromium helps the alumina form (the "third-element effect"). Below the critical level, internal oxidation occurs: oxygen diffuses into the alloy and precipitates particles of the reactive element's oxide inside the metal instead of forming a protective surface layer. The design of oxidation-resistant alloys and coatings in Chapter 41, and of the hot-corrosion-resistant compositions in gas turbines, is an application of these rules.

What remains open

Open question	Where it is answered
What forms does corrosion take in real structures, and how are galvanic, pitting, crevice and intergranular attack prevented?	Chapter 43
How do stress, hydrogen and cyclic loading combine with corrosion to crack metals?	Chapter 43
How are cathodic-protection systems and coatings sized and tested, and how are polarization and impedance measurements made?	Chapter 43
How do ceramics, polymers, composites and concrete degrade, and how are corrosion failures investigated?	Chapter 44
How are surfaces structured, and how do adsorption and surface energy govern adhesion and wear?	Chapter 45
How are corrosion-resistant coatings deposited?	Chapters 47, 49
How are passive films and oxide scales characterized?	Chapters 52, 53
Where do the Gibbs energies, Ellingham diagrams and point-defect equilibria used here come from?	Chapters 5, 6; Advanced Chemistry

Next: Chapter 43 turns from the principles of corrosion to the forms it takes in real structures, from galvanic and pitting attack to stress-corrosion cracking, and to the coatings, inhibitors and cathodic protection that engineers use to stop it.

Chapter 43 — Forms of Corrosion and Protection

Part IX · Corrosion, Degradation and Surfaces

In May 1985 the suspended concrete ceiling of an indoor swimming pool in Uster, Switzerland, fell onto the pool area, killing twelve people. The ceiling had hung for thirteen years from stainless-steel rods. The rods showed almost no loss of section and were loaded well below yield, yet their fracture surfaces revealed branched cracks driven by sustained tension and chloride-laden condensate from the pool air. The steel had not wasted away; it had cracked. Chapter 42 explained why metals corrode and how fast the electrochemistry runs. This chapter asks *where* and *in what form* the attack concentrates, because the distribution of attack, not its total amount, decides whether a structure fails. A millimetre per year spread evenly over a tank wall is a maintenance item; the same metal loss concentrated in a pit, a crevice, a grain boundary or a crack tip is a catastrophe.

> **Why does corrosion that removes almost no metal sometimes destroy a component, while heavy uniform rusting is harmless?**
> **How do galvanic coupling, pitting, crevices, sensitization, stress and hydrogen turn a slow chemical process into localized failure? How are coatings, inhibitors and cathodic or anodic protection sized and verified, and how are corrosion rates measured?**

Concept	Engineering destination
Uniform corrosion rate and corrosion allowance	Wall thickness of tanks, pipes and steel piles; atmospheric-category design of bridges
Galvanic series and area ratio	Fastener selection, dissimilar-metal joints in ships, aircraft and cars
Pitting resistance equivalent number, critical pitting temperature	Choosing stainless grades for seawater, chemical plant, desalination
Crevice geometry and differential aeration	Flange gaskets, lap joints, tube-sheet design, fastener heads
Sensitization and intergranular attack	Low-carbon and stabilized stainless grades, weld procedures, marine aluminium
Stress-corrosion cracking, K_{ISCC}	Pressure vessels, pipelines, aircraft forgings, nuclear steam generators
Hydrogen embrittlement, hardness limits	High-strength fasteners, sour-gas equipment, hydrogen pipelines
Corrosion fatigue	Offshore structures, pump shafts, implants, steel wire ropes
MIC, dealloying, erosion–corrosion, cavitation	Water mains, oil pipelines, heat-exchanger tubes, pump impellers and propellers
Coatings, galvanizing, inhibitors	Structural steel, automotive bodies, cooling water and oil production
Cathodic and anodic protection	Buried pipelines, offshore platforms and wind foundations, ship hulls, acid tanks
Salt spray, polarization, EIS	Qualification of coatings and alloys, field corrosion monitoring

Roadmap. Uniform and galvanic attack, where the whole surface or a whole metal is anodic (43.1–43.2) → localized attack on passive metals: pitting, crevice and intergranular corrosion (43.3–43.5) → corrosion combined with stress: SCC, hydrogen embrittlement, corrosion fatigue (43.6–43.8) → biological, chemical and mechanical accelerants: MIC, dealloying, erosion–corrosion, cavitation (43.9) → protection by material, coating and inhibitor (43.10) → electrochemical protection, cathodic and anodic, with sizing (43.11–43.12) → how corrosion is tested and measured (43.13).

43.1 Uniform corrosion

Rate, allowance and the corrosion map of a structure

Uniform (general) corrosion proceeds at roughly equal rate over the whole exposed surface. Anodic and cathodic sites exist, as Chapter 42 showed, but they are microscopic and shift continually, so on average every point is anodic for the same fraction of time. Uniform corrosion consumes the largest tonnage of metal of any form, but it is the least dangerous, because it is predictable: once a rate is known, a **corrosion allowance** is added to the wall thickness, or a coating is chosen whose life matches the inspection interval.

The rate is quoted as penetration per year. Faraday's law (Chapter 42) converts a corrosion current density to penetration. For iron dissolving as Fe^{2+}:

$$\mathrm{CR} = \frac{i\,M}{nF\rho} = \frac{i \times 55.85\ \mathrm{g\,mol^{-1}}}{2 \times 96\,485\ \mathrm{C\,mol^{-1}} \times 7.87\ \mathrm{g\,cm^{-3}}}$$

which gives 1.16 mm/year for $i = 1$ A/m^2, or the useful rule that **1 µA/cm^2 corresponds to about 11.6 µm/year for steel**.

Environment (carbon steel unless stated)	Typical uniform corrosion rate
Dry indoor air (ISO 9223 category C1)	below 1.3 µm/year
Rural atmosphere (C2)	1.3–25 µm/year
Urban or mild industrial (C3)	25–50 µm/year
Industrial or coastal (C4)	50–80 µm/year
Severe marine or industrial (C5)	80–200 µm/year
Zinc in C3 atmosphere	0.7–2.1 µm/year
Seawater immersion, quiescent	0.1–0.2 mm/year
Splash zone of offshore structure	0.3–0.5 mm/year, locally more
Undisturbed soil	0.01–0.05 mm/year
Aerated fresh water, ambient	0.05–0.15 mm/year

The ISO atmospheric categories are first-year rates measured on standard coupons. Rates fall with time as the rust layer thickens, roughly as a power law $d = At^n$ with n between 0.3 and 0.8; **weathering steels** (Cor-Ten type, with small additions of Cu, Cr, Ni and P) exploit this by forming a dense, adherent rust that slows corrosion to a few micrometres per year in suitable atmospheres. They fail where the rust never dries out, a reminder that a protective film is a property of material *and* environment.

> **Status of the principle: Corrosion allowance from a uniform rate**
> *Empirical.* The rates are measured or categorized for each environment; adding rate × design life to the wall is valid only if attack stays uniform. It provides no protection against pitting, cracking or galvanic acceleration, which this chapter treats separately.

Uniform corrosion governs buried steel piles, storage tanks, ship plating and rebar in carbonated concrete (Chapter 44). At small dimensions there is no allowance at all: a 1 µm bond pad or a 50 µm stent strut makes even slow uniform corrosion a design limit.

43.2 Galvanic corrosion

The galvanic series

When two different metals are in electrical contact in a common electrolyte, the one with the more negative corrosion potential becomes the anode of a macroscopic cell and corrodes faster; the other is cathodically protected. Standard electrode potentials (Chapter 42) are a poor guide, because real metals carry oxide films and alloys have no standard potential. Engineers use instead the **galvanic series**: corrosion potentials measured in a specified environment, usually flowing seawater (Figure 43.1).

Metal or alloy	Corrosion potential in seawater (V vs SCE, typical)
Magnesium alloys	−1.60 to −1.63
Zinc	−0.98 to −1.03
Aluminium alloys	−0.76 to −1.00
Carbon steel, cast iron	−0.60 to −0.72
Austenitic stainless steel, active (in crevices)	−0.46 to −0.58
Copper, brass, bronze	−0.24 to −0.36
90/10 copper–nickel	−0.21 to −0.28
Nickel–aluminium bronze	−0.15 to −0.22
Austenitic stainless steel, passive	−0.05 to −0.10
Titanium, Ni–Cr–Mo alloys	−0.05 to +0.06
Graphite, platinum	+0.20 to +0.25

Note that stainless steel appears twice. A passive stainless surface is noble; the same alloy inside a crevice, where the film has broken down, is active. Passive metals move along the series as their film state changes.

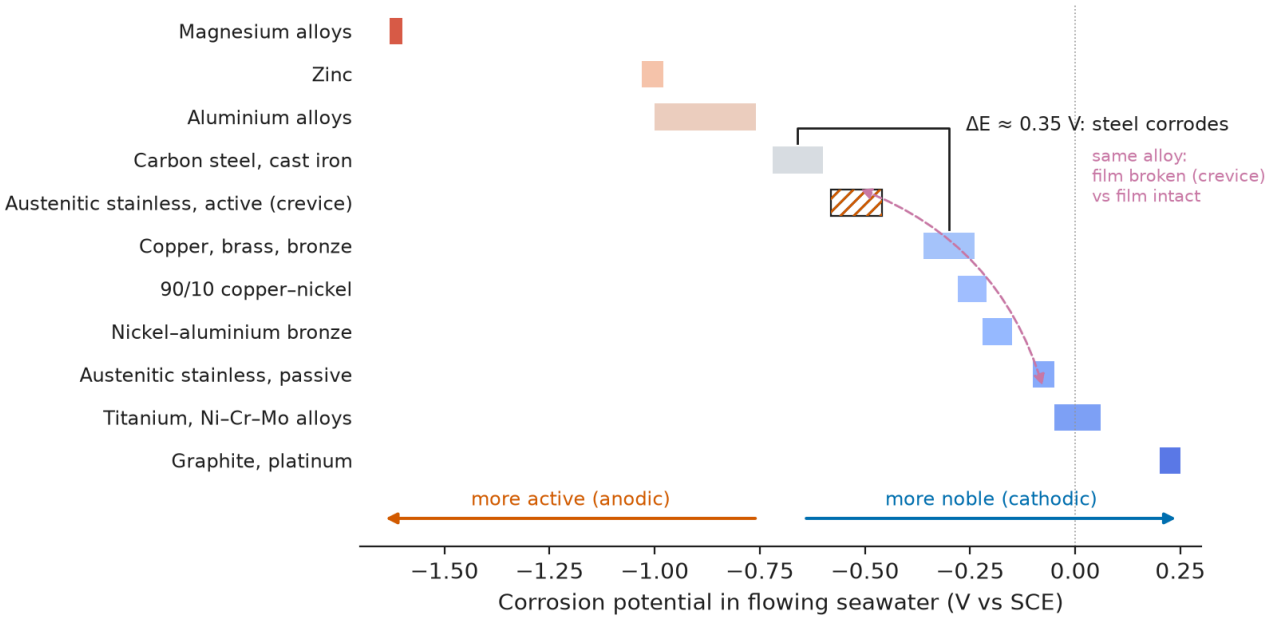


Figure 43.1 — Galvanic series of engineering metals in flowing seawater, with the active–passive double entry for stainless steel.

> **Status of the principle: Galvanic series**
> *Empirical.* Measured corrosion potentials in one electrolyte at one flow rate and temperature; the order can change in other environments (zinc and steel reverse in some hot waters above about 60 °C), and the series ranks risk without giving rates.

Why area ratio controls the rate

The potential difference says which metal corrodes; the rate is set by kinetics. In near-neutral aerated water the cathodic reaction, oxygen reduction, is usually limited by oxygen diffusion to the surface. The total cathodic current is then the limiting current density times the **cathode** area, and all of that current must be supplied by the anode. The galvanic current density on the anode is therefore

$$i_a \approx i_\mathrm{L}\,\frac{A_c}{A_a}$$

A small anode coupled to a large cathode is the worst case.

Worked example — rivets and plates in seawater. Take an oxygen-limited cathodic current density of $i_\mathrm{L} = 0.1\ \mathrm{A/m^2}$ in quiescent seawater. In design A, a copper plate of area $A_c = 1\ \mathrm{m^2}$ is held by carbon-steel rivets whose total wetted area is $A_a = 0.01\ \mathrm{m^2}$ (area ratio 100). In design B the materials are swapped: a steel plate of $1\ \mathrm{m^2}$ with copper rivets of $0.01\ \mathrm{m^2}$.

Design A. The anodic current density on the rivets is

$$i_a = 0.1 \times \frac{1}{0.01} = 10\ \mathrm{A\,m^{-2}}$$

and with 1.16 mm/year per $\mathrm{A/m^2}$ the rivets corrode at about 11.6 mm/year. Rivet heads a few millimetres thick are gone within months.

Design B. The steel plate already corrodes at about $0.1 \times 1.16 = 0.116$ mm/year on its own. The copper rivets add a cathodic current of only $0.1 \times 0.01 = 0.001$ A, spread over $1\ \mathrm{m^2}$ of steel: an extra 1.2 µm/year, about 1 % of the self-corrosion rate. The copper rivets themselves are protected.

The same metals, the same potential difference, a ten-thousand-fold difference in damage. This is the reason behind the rules **"fasteners more noble than the parts they join"** and **"never paint only the anode"**: painting the anodic member of a couple leaves small holidays (coating defects) as tiny anodes facing a huge bare cathode. If only one member can be coated, coat the cathode.

Galvanic corrosion across engineering

- **Aerospace and automotive**: carbon-fibre composites conduct and sit near graphite in the series, so aluminium or magnesium touching CFRP corrodes rapidly unless isolated by glass plies, sealant or titanium fasteners.
- **Marine and civil**: bronze propellers on steel hulls, aluminium superstructures on steel ships (joined by explosion-welded transition strips), copper pipe screwed into galvanized steel.
- **Electronics**: pores in gold-over-nickel platings create small-anode geometries.
- **Biomedical**: a Co–Cr head on a Ti-alloy hip stem; small potential differences plus micromotion give **fretting-crevice corrosion** at the taper.

Sacrificial anodes and zinc coatings (Sections 43.10–43.11) are galvanic corrosion engineered to happen to the right metal.

43.3 Pitting corrosion

Breakdown of passivity

Stainless steels, aluminium, titanium and nickel alloys owe their usefulness to a passive film a few nanometres thick (Chapter 42). **Pitting** is the local breakdown of that film, usually by chloride ions, at a weak point: a manganese sulfide inclusion in stainless steel, an intermetallic particle in aluminium, a scratch or a weld-heat tint. The rest of the surface remains passive and serves as a large cathode, so the pit is a tiny anode with a huge cathode, the area-ratio problem of Section 43.2 appearing on a single metal.

The autocatalytic pit

A pit, once started, sustains itself (Figure 43.2):

1. Metal dissolves at the pit bottom: $\mathrm{M} \rightarrow \mathrm{M^{n+}} + \mathrm{n\,e^-}$.

2. The metal cations hydrolyse, for example $Cr^{3+} + 3H_2O \rightarrow Cr(OH)_3 + 3H^+$, acidifying the pit solution to pH 1–2 or below.
3. To keep the solution electrically neutral, chloride ions migrate into the pit from outside.
4. The concentrated acid chloride solution prevents repassivation, so dissolution continues.

A cap of corrosion product or undercut metal over the mouth preserves the aggressive chemistry, so pits can perforate a tube wall while weight loss is negligible.

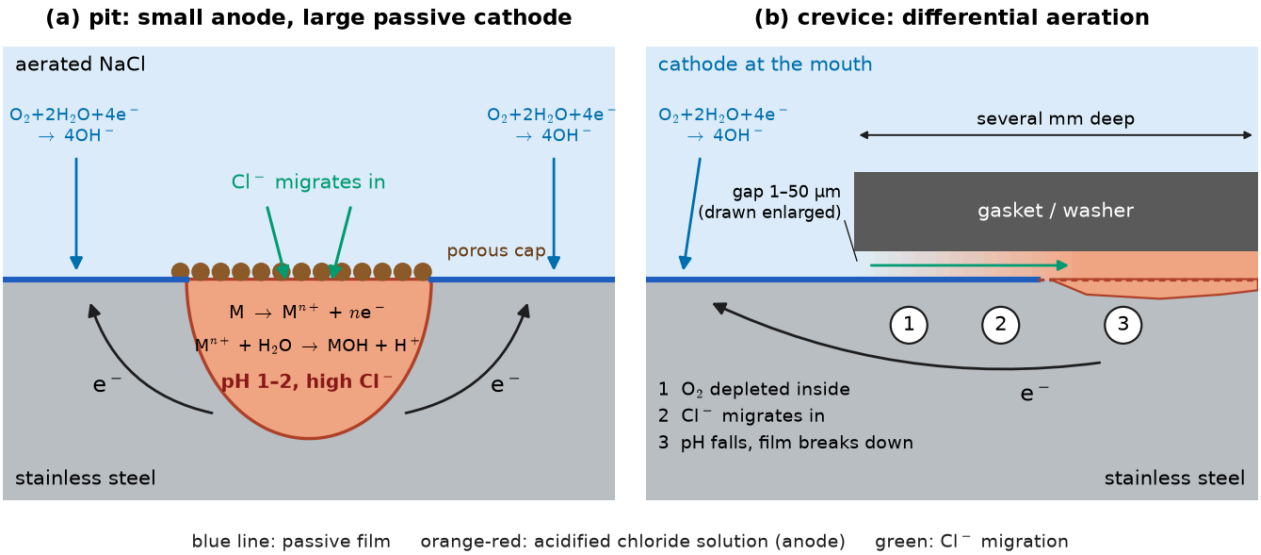


Figure 43.2 — Autocatalytic chemistry of a corrosion pit and of a crevice, both driven by a small acidified anode coupled to a large passive cathode.

Pitting potential and critical pitting temperature

On a polarization curve, a passive metal in chloride solution shows a sharp rise in current at the **pitting potential** E_{pit}. If the potential is then swept back, the current stays high until the **repassivation potential** E_{rp}, below which pits stop growing. Pitting occurs in service when the free corrosion potential, raised by oxygen, chlorination or a noble couple, reaches E_{pit}. E_{pit} falls with rising chloride concentration and temperature; for a given alloy and solution there is a fairly sharp **critical pitting temperature** (CPT) below which pits do not propagate at any potential.

Alloying raises resistance. Chromium forms the film, molybdenum and nitrogen help it repair in acid chloride. An empirical index ranks stainless steels:

$$\text{PREN} = \%\text{Cr} + 3.3\,(\%\text{Mo} + 0.5\,\%\text{W}) + 16\,\%\text{N}$$

(with 30 instead of 16 sometimes used for the nitrogen term in austenitic grades).

Alloy (typical composition, wt %)	PREN	CPT in ferric chloride (ASTM G48 type), typical	Typical use
304 (18Cr–8Ni, 0.05N)	19	below 5 °C	Food, architecture, dry indoor
316L (17Cr–12Ni–2.1Mo)	24–25	10–20 °C	Chemical plant, mildly saline
2205 duplex (22Cr–5Ni–3.1Mo–0.17N)	35	30–40 °C	Brackish water, oil and gas
2507 super duplex (25Cr–7Ni–4Mo–0.27N)	42–43	70–80 °C	Seawater systems, offshore
6Mo superaustenitic (20Cr–18Ni–6.1Mo–0.2N)	43	60–70 °C	Seawater, bleach plants

A rule of thumb in offshore practice is that ambient seawater needs PREN of at least 40, which is why 316L is not a seawater alloy despite its "marine grade" nickname.

> **Status of the principle: PREN**
> *Empirical.* A linear regression of pitting resistance on composition for wrought stainless steels in chloride; it ignores microstructure (sigma phase, inclusions, weld segregation, surface finish), so two alloys of equal PREN can differ markedly, and it says nothing about stress-corrosion cracking.

Pitting in other metals

Aluminium alloys pit around cathodic intermetallics (Al_2CuMg, Al_7Cu_2Fe), which is why 2xxx aircraft sheet is clad with pure aluminium. Because pit depths are statistically distributed, pipeline integrity uses **extreme-value (Gumbel) statistics** to predict the deepest pit over a large area from small coupons, weakest-link reasoning met again in Chapter 44.

43.4 Crevice corrosion

A crevice is any narrow gap into which solution can enter but not circulate: under a gasket or washer, at a lap joint, between a tube and a tube sheet, under a barnacle or a deposit. The mechanism has the same end point as pitting but a different start (Figure 43.2b):

1. Initially the metal inside and outside corrodes at the same low passive rate, consuming oxygen.
2. Oxygen inside the crevice is used up and cannot be replenished; the cathodic reaction moves outside, to the crevice mouth.
3. Metal ions accumulate inside; chloride migrates in to balance them; hydrolysis lowers the pH.
4. When the crevice solution reaches a critical acidity and chloride content, the passive film breaks down and active dissolution begins, driven by the large cathode outside.

The first stage is an **incubation period** that can last months, so crevice corrosion often appears long after commissioning. Tight, deep crevices are the most dangerous, and the ohmic drop along them lets the tip sit in the active range while the mouth stays passive. Crevices initiate attack at lower temperatures than open surfaces: the **critical crevice temperature** of a stainless steel is typically 15–30 °C below its CPT.

A related form, **differential-aeration corrosion**, attacks carbon steel under deposits, at the waterline of a partly filled tank and in soil where a pipe crosses from sand into clay: the oxygen-starved region becomes anodic. **Filiform corrosion** is the crevice mechanism running under a paint film on steel, aluminium or magnesium, producing thread-like tracks from a scratch.

Defences are mainly design (welded rather than bolted joints, continuous welds, non-absorbent gaskets, sealed gaps, drainage) and, for seawater, crevice-resistant alloys (super duplex, 6Mo, titanium, Ni–Cr–Mo).

43.5 Intergranular corrosion

Sensitization of stainless steels

Grain boundaries are higher in energy and chemically different from grain interiors (Chapter 5). **Intergranular corrosion** is attack confined to a narrow band along the boundaries, which can disintegrate a component into loose grains while the surface looks intact.

The classic case is the **sensitization** of austenitic stainless steel. Held between roughly 500 and 800 °C, during welding or slow cooling or a stress-relief treatment, carbon combines with chromium to precipitate $Cr_{23}C_6$ carbides on grain boundaries. Carbon diffuses quickly; chromium slowly (Chapter 8). The carbides therefore draw chromium from a thin zone next to the boundary, and the chromium content there falls below the roughly 12 % needed for passivity (Figure 43.3). That depleted zone corrodes actively in acids while the grains stay passive. In welded 304, the sensitized band lies a few millimetres from the fusion line, where the heat-affected zone spent longest in the dangerous range: **weld decay**.

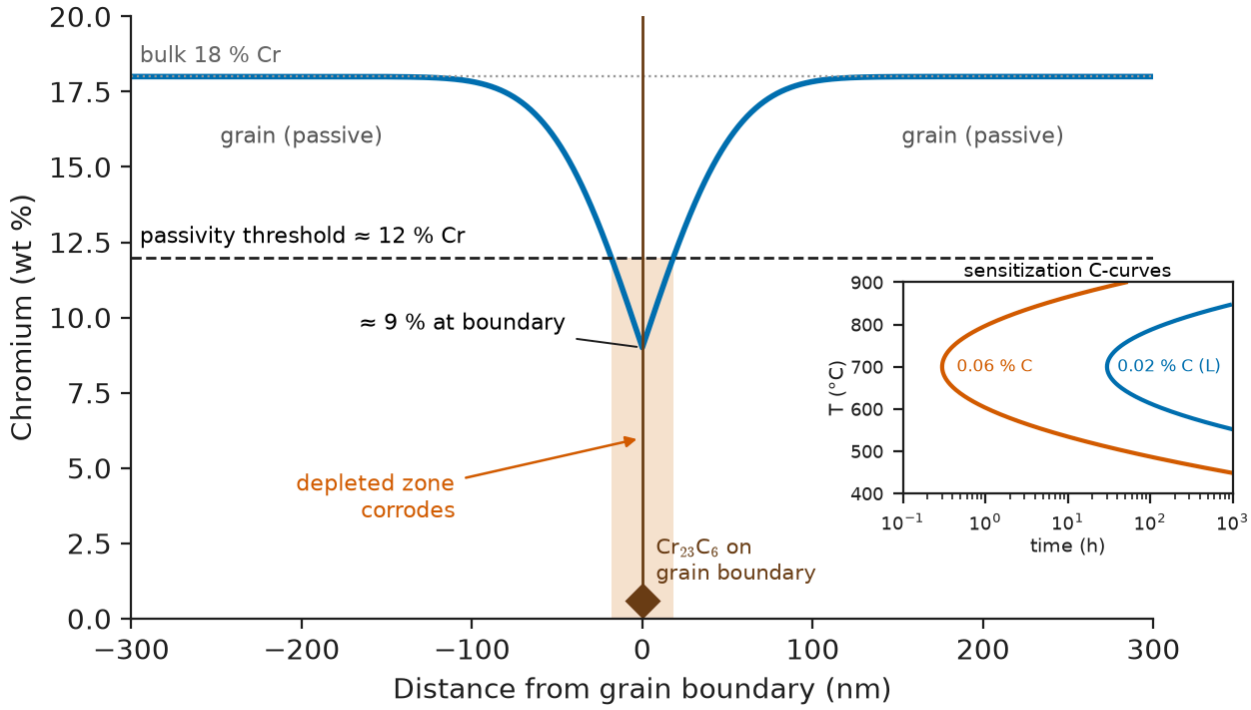


Figure 43.3 — Chromium depletion next to a grain boundary carrying Cr23C6 carbides in sensitized 304 stainless steel.

The remedies come straight from the mechanism:

- **Less carbon**: L grades (304L, 316L) with C ≤ 0.03 % precipitate too little carbide in normal welding times; the C-curve of the time–temperature–sensitization diagram moves to longer times.
- **Stabilization**: titanium (321) or niobium (347) ties carbon up as TiC or NbC, which form at higher temperature, leaving chromium in solution. A stabilized steel can still suffer **knife-line attack** in a very narrow band next to a weld where the stabilizing carbides were dissolved and chromium carbides formed on reheating.
- **Solution annealing**: heating to about 1050 °C dissolves the carbides, and rapid cooling prevents their re-precipitation.

Intergranular attack in other alloys

- **Al–Mg (5xxx) alloys** with more than about 3 % Mg, used for ship hulls, slowly precipitate anodic β-phase (Al_3Mg_2) on grain boundaries at temperatures as low as 50–70 °C (decks in tropical sun), and then corrode and crack intergranularly in seawater.
- **High-strength 2xxx and 7xxx aluminium** suffers **exfoliation**: attack along pancake-shaped grains of rolled plate, where voluminous corrosion product wedges the layers apart like leaves of a book. Over-aged tempers (7075-T73, -T76) trade a little strength for resistance.

Susceptibility is screened by standard acid exposures (ASTM A262 for stainless steels, the nitric-acid mass-loss test ASTM G67 for 5xxx aluminium).

43.6 Stress-corrosion cracking

The three necessary conditions

Stress-corrosion cracking (SCC) is the growth of cracks under the combined action of a sustained tensile stress and a specific corrosive environment, in a susceptible material. All three must be present: remove any one and SCC stops. The stress need not be applied: residual stresses from welding, cold forming or interference fits are the usual cause, and they can approach the yield strength. The environment is often surprisingly mild, and it is specific: a system that cracks readily in one environment may be immune in another.

Alloy system	Environment that causes SCC	Typical engineering case
Austenitic stainless steels	Chloride solutions above about 50–60 °C; also caustic, polythionic acids	Hot-water pipes under insulation, pool hangers, chemical plant
Carbon and low-alloy steels	Hot caustic; nitrates; carbonate–bicarbonate; anhydrous ammonia; near-neutral groundwater	Boiler caustic embrittlement, gas-pipeline cracking, ammonia tanks
Brass (Cu–Zn)	Ammonia and amines	"Season cracking" of cartridge cases; plumbing fittings
High-strength aluminium (2xxx, 7xxx)	Moist air, chloride; worst in short-transverse direction	Aircraft forgings and extrusions
Titanium alloys	Hot salt; methanol; N_2O_4	Compressor discs, spacecraft propellant tanks
Nickel alloys (Alloy 600)	High-temperature pure water	Steam-generator tubes and nozzles in pressurized-water reactors
Glass, silicate ceramics	Water vapour	Slow crack growth (Chapter 44)
Glassy polymers	Specific organic liquids, detergents	Environmental stress cracking (Chapter 44)

Cracks may run between grains (intergranular, typical of sensitized stainless steel, brass in ammonia, aluminium alloys) or through them (transgranular, typical of chloride SCC of austenitic stainless steel), often with heavy branching that distinguishes them from fatigue cracks in failure analysis.

Fracture mechanics of SCC

Linear-elastic fracture mechanics (Chapter 14) gives the engineering description. A pre-cracked specimen held at constant stress intensity K in the environment shows cracking only above a threshold K_{ISCC}, often far below the fracture toughness K_{Ic}. Above the threshold the crack velocity has three stages (Figure 43.4): stage I, rising steeply with K; stage II, a plateau where velocity is independent of K and is controlled by transport of the environment to the crack tip or by the electrochemical reaction there; and stage III, accelerating to fast fracture as K approaches K_{Ic}.

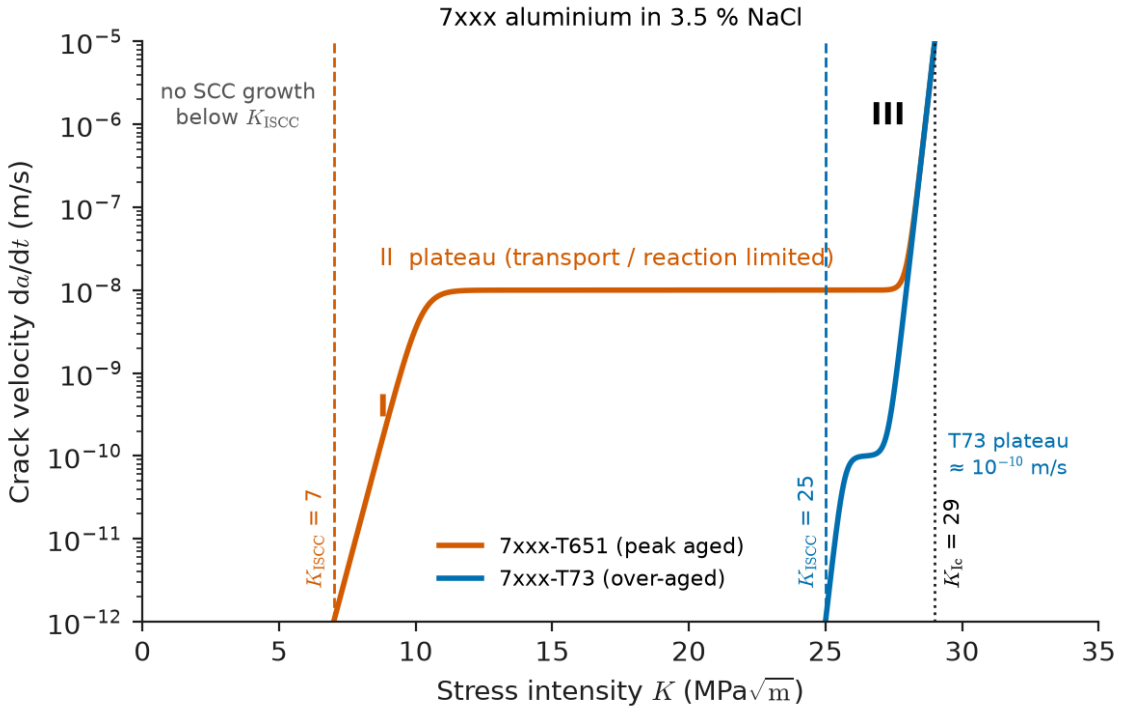


Figure 43.4 — Stress-corrosion crack velocity versus stress intensity, with threshold, plateau and final fracture.

The threshold gives a design rule. For a surface crack of depth a under stress σ, $K \approx 1.12\,\sigma\sqrt{\pi a}$, and SCC cannot grow if

$$a < a_{\mathrm{SCC}} = \frac{1}{\pi}\left(\frac{K_{\mathrm{ISCC}}}{1.12\,\sigma}\right)^2$$

For 7075-T651 aluminium loaded in the short-transverse direction, K_{ISCC} is about 7 MPa√m. Under a residual or assembly stress of 200 MPa, $a_{\mathrm{SCC}} = (1/\pi)(7/224)^2 = 3.1 \times 10^{-4}$ m, about 0.3 mm: any machining mark or corrosion pit is enough. In the over-aged T73 temper, K_{ISCC} rises to about 25 MPa√m and the tolerable flaw grows to about 4 mm, a size that inspection can find. That is why aircraft forgings were changed from T6 to T73 tempers after service cracking in the 1960s, accepting about 10–15 % lower strength.

> **Status of the principle: K_{ISCC} threshold**
> *Empirical.* A threshold measured on pre-cracked specimens in one environment, temperature and potential; it assumes a long, sharp crack (it does not cover initiation from smooth surfaces or short cracks) and it can drift with test duration, so it is quoted with the test standard and time.

Mechanisms

No single mechanism explains all SCC. The two most widely supported are:

- **Slip–dissolution (film rupture)**: crack-tip strain ruptures the passive film, bare metal dissolves until the film reforms, and repeated cycles advance the crack at a rate set by the charge per rupture and the tip strain rate. It underlies crack-growth models for stainless steels and nickel alloys in reactor water.
- **Hydrogen-assisted cracking**: hydrogen from the crack-tip cathodic reaction embrittles the metal ahead of the tip (Section 43.7); it dominates in high-strength steels, many aluminium alloys and titanium.

Controlling SCC

Three necessary conditions give three routes: lower the tensile stress (stress relief, shot peening); change the environment (remove chlorides or oxygen, control potential, keep insulation dry); or change the material (duplex instead of austenitic steel in hot chloride, over-aged tempers, Alloy 690 instead of 600 in steam generators).

43.7 Hydrogen embrittlement

Sources and transport of hydrogen

Hydrogen dissolves interstitially in metals and diffuses rapidly even at room temperature. Atomic hydrogen enters steel from cathodic reactions (acid corrosion, excessive cathodic protection and above all H_2S, which poisons recombination to H_2 and drives far more atoms into the metal), from processing (pickling, electroplating, welding with damp consumables) and from gaseous hydrogen in vessels, pipelines and refinery hydrotreaters.

Its lattice diffusivity in ferritic steel at room temperature is of order 10^{-9}–10^{-8} m^2/s, but dislocations, grain boundaries, carbide interfaces and voids act as **traps** that lower the effective diffusivity to 10^{-11}–10^{-10} m^2/s in commercial steels. With $D_{\mathrm{eff}} = 10^{-10}$ m^2/s, hydrogen penetrates the 5 mm half-thickness of a 10 mm plate in a time of order $L^2/D = (5 \times 10^{-3})^2/10^{-10} = 2.5 \times 10^5$ s, about three days. Hydrogen picked up during plating must therefore be baked out promptly (typically 190–220 °C for several hours, within a few hours of plating) before it diffuses to a stress concentration.

Embrittlement mechanisms and susceptibility

Hydrogen concentrates where the lattice is dilated: in the hydrostatic tension zone ahead of a crack or notch. There it lowers the cohesive strength of the lattice and interfaces (**hydrogen-enhanced decohesion**, HEDE) and makes dislocations more mobile in a localized band (**hydrogen-enhanced localized plasticity**, HELP). The consequence is delayed fracture: a bolt that was tightened correctly breaks hours or days later, without warning, with an intergranular or quasi-cleavage fracture surface.

Susceptibility rises steeply with strength. Ferritic steels below about 700–800 MPa tensile strength are rarely affected in ordinary service; above about 1000–1200 MPa (hardness above about 35–39 HRC) they are highly susceptible. Hence the hardness limits that govern hydrogen-exposed equipment:

Situation	Typical rule
Carbon and low-alloy steel in sour (H_2S) service	Hardness ≤ 22 HRC (about 250 HV), including weld heat-affected zones (NACE MR0175/ISO 15156)
Electroplated high-strength fasteners	Bake after plating if hardness > about 32–39 HRC; avoid zinc plating above class 12.9
Cathodically protected high-strength steels in seawater	Potential not more negative than about −1.0 to −1.05 V vs Ag/AgCl
Steel for gaseous hydrogen pipelines	Lower-strength grades, toughness and hardness limits, fatigue curves measured in hydrogen

> **Status of the principle: Hardness limits for hydrogen service**
> *Empirical.* Derived from service experience and threshold tests on carbon and low-alloy steels; they encode the strong correlation between strength and hydrogen susceptibility but are not transferable to other alloy families or to different microstructures at equal hardness.

Hydrogen damage without applied stress

In low-strength steels hydrogen recombines to H_2 at elongated MnS inclusions, and the gas pressure causes **blistering** and stepwise **hydrogen-induced cracking (HIC)** parallel to the plate surface; sour-service line pipe therefore has very low sulfur and calcium-treated, rounded inclusions. At refinery temperatures hydrogen reacts with carbides to form methane (**high-temperature hydrogen attack**), and Cr–Mo steels are chosen from empirical Nelson curves.

The energy transition makes hydrogen embrittlement a central materials question again: repurposing natural-gas pipelines for hydrogen raises fatigue crack-growth rates in pipeline steels by an order of magnitude or more at a given ΔK, so existing defects must be reassessed (Chapter 40).

43.8 Corrosion fatigue

Fatigue in an inert environment and corrosion without stress are each understood; their combination is worse than either. **Corrosion fatigue** is crack initiation and growth under cyclic stress in a corrosive environment. Corrosion pits act as notches for initiation; at the crack tip, the environment dissolves freshly slipped metal or supplies hydrogen, accelerating growth per cycle.

Two features matter for design:

- **No endurance limit.** A steel that has a fatigue limit in air (Chapter 14) loses it in seawater: the S–N curve continues to fall at long lives because corrosion keeps creating new initiation sites. The fatigue strength at 10^8 cycles of a carbon steel in seawater can be a third or less of its in-air endurance limit.
- **Frequency matters.** In air, fatigue life depends on cycles, not time. In an environment, a slower cycle gives more time for corrosion per cycle, so crack growth per cycle increases as frequency falls. High-frequency laboratory tests therefore overestimate the life of offshore structures loaded by waves at 0.1–0.3 Hz.

Offshore design codes (for example DNV-RP-C203) give separate S–N curves for welded steel details in air, in seawater with cathodic protection, and in seawater under free corrosion; the free-corrosion curves have no cut-off and give lives about a factor of three shorter than those in air. Cathodic protection restores much of the in-air behaviour, though very negative potentials can accelerate crack growth by hydrogen charging. Corrosion fatigue also governs wire ropes, mooring chains, pump shafts and orthopaedic implants loaded about 10^6 times a year in body fluid (Chapter 39).

43.9 Microbial, selective, flow-induced and cavitation corrosion

Microbially influenced corrosion

Bacteria and archaea do not corrode metal directly, but their **biofilms** change the local chemistry under them. **Microbially influenced corrosion (MIC)** involves:

- **Sulfate-reducing bacteria** (SRB, such as *Desulfovibrio*), anaerobes that reduce sulfate to sulfide under deposits, producing iron sulfide and deep pits in carbon steel.
- **Acid-producing bacteria**, generating organic acids under the biofilm.
- **Iron- and manganese-oxidizing bacteria**, building tubercles (differential aeration) and depositing manganese oxides that ennoble stainless steels toward pitting.

MIC is diagnosed from clustered, cup-shaped pits under deposits, iron sulfide in the deposit, microbial culture or DNA sequencing, and the operating history. Classic cases are hydrotest water left for weeks in new stainless piping, sprinkler systems, water-injection pipelines and tank bottoms; mitigation is biocides, pigging, draining and drying, and cathodic protection.

Selective leaching (dealloying)

Selective leaching removes the more active element of an alloy, leaving a porous, weak residue of the nobler one with the original shape:

- **Dezincification** of brasses with more than about 15 % Zn: zinc dissolves, leaving spongy red copper. Plug-type dezincification can perforate a fitting under water pressure. Small additions of arsenic, antimony or phosphorus inhibit it (admiralty and naval brasses, dezincification-resistant brass for plumbing).
- **Graphitic corrosion** of grey cast iron: the iron matrix corrodes, leaving a graphite network held by corrosion products. A cast-iron water main can look sound and keep its shape while having almost no strength, failing when the ground moves or the pressure surges.
- **Dealuminification** of aluminium bronzes and **denickelification** of copper–nickel in hot or polluted water.

The same process is used deliberately to make nanoporous gold and other high-surface-area catalysts and electrodes (Chapter 38).

Erosion–corrosion and flow-accelerated corrosion

Flow increases corrosion in two ways: it brings more oxygen to the surface, and above a critical velocity it strips or prevents the protective film. **Erosion–corrosion** shows smooth grooves, gullies and horseshoe-shaped pits oriented with the flow, typically at bends, tees, pump casings and the inlet ends of heat-exchanger tubes, where turbulence is greatest; entrained sand or bubbles make it worse.

Tube or pipe material in seawater	Typical maximum design velocity
Copper	about 1 m/s
Admiralty brass, aluminium brass	2–2.5 m/s
90/10 copper–nickel	2.5–3.5 m/s
70/30 copper–nickel	3–4.5 m/s
Super duplex stainless, titanium	above 10 m/s (practically unlimited by erosion–corrosion)

Flow-accelerated corrosion (FAC) is the carbon-steel version in hot deoxygenated water and wet steam of power plants. The magnetite film dissolves into the fast-flowing water, the wall thins uniformly downstream of fittings, and the pipe eventually bursts. FAC ruptures killed workers at Surry, USA (1986) and Mihama, Japan (2004). Remedies are chromium-bearing steels (even 0.1 % Cr helps markedly; 1.25–2.25 % Cr steels are essentially immune), higher-pH water chemistry and ultrasonic wall-thickness programmes.

Cavitation damage

Where the local pressure in a liquid falls below its vapour pressure (on the suction side of pump impellers, on propeller blades, in hydro-turbine runners, in diesel-engine cylinder liners vibrating in coolant), vapour bubbles form and then collapse when they move into higher-pressure regions. Collapse near a surface produces a micro-jet and shock wave with local pressures of hundreds of megapascals or more, repeated millions of times. The sponge-like damage combines surface fatigue with film removal. Resistance correlates with hardness and work-hardening capacity (Stellite overlays, nickel–aluminium bronze propellers, martensitic stainless turbine runners) or with impact absorption (elastomeric coatings); the first defence is hydraulic, keeping pressures above the vapour pressure.

43.10 Protection by material selection, coatings and inhibitors

Material selection and design

The cheapest corrosion protection is chosen at the drawing board. Corrosion handbooks contain thousands of material–environment ratings, but they condense into a short list of pairings that usually work:

Environment	Materials that usually work
Atmosphere (mild)	Painted or galvanized carbon steel, weathering steel, aluminium
Seawater, ambient	Super duplex, 6Mo stainless, titanium, 90/10 or 70/30 Cu–Ni, coated steel with CP
Dilute sulfuric acid	Lead, high-silicon cast iron, Alloy 20, carbon steel only for concentrated acid
Hydrochloric acid	Ni–Mo alloys (Hastelloy B type), tantalum, polymers and FRP, rubber-lined steel
Hot caustic	Nickel and nickel alloys
Oxidizing acids (nitric)	Austenitic stainless steels, titanium
Human body	Ti–6Al–4V and CP Ti, Co–Cr–Mo, 316LVM stainless (short-term), alumina and zirconia
Reinforced concrete in chloride exposure	Adequate cover and low-permeability concrete; stainless or epoxy-coated rebar where severe

Polymers and fibre-reinforced plastics (Chapters 21 and 23), ceramics and glass linings replace metals entirely in many chemical services. Design rules are collected in Section 43.12.

Coatings

Coatings protect in three ways:

1. **Barrier**: organic paints and linings (epoxy, polyurethane, fluoropolymer) and glass or vitreous enamel separate metal from electrolyte. No organic coating is impermeable to water and oxygen; good systems work because of very low ionic permeability and excellent wet adhesion, and they are built in several coats (primer, intermediate and top coat, with total dry film thicknesses of 250–500 µm for marine steelwork) so that pores do not line up.
2. **Sacrificial**: coatings of a more active metal (zinc, zinc–aluminium, aluminium) on steel corrode preferentially and protect the steel at scratches and cut edges by galvanic action. Zinc-rich primers carry zinc dust at high loading for the same effect.
3. **Inhibitive**: primers containing inhibitive pigments (formerly chromates, now phosphates, molybdates and other alternatives) release inhibitor into water that penetrates.

Noble metallic coatings (nickel, chromium, tin on steel) protect only while continuous; at a pore the substrate is a small anode in a bad area ratio. **Conversion coatings** (phosphate on steel, trivalent-chromium or zirconium treatments, anodized Al $_2O_3$ 5–25 µm thick on aluminium) add corrosion resistance and paint adhesion. Deposition processes are covered in Chapter 49.

Galvanizing

In **hot-dip galvanizing** clean steel is dipped into molten zinc at about 450 °C. Zinc and iron react to form a sequence of Fe–Zn intermetallic layers (Γ, δ, ζ) capped by nearly pure zinc (η), metallurgically bonded to the steel. The coating protects first as a

barrier, by the slow-corroding zinc and its basic zinc carbonate patina, and then sacrificially at damage. Because zinc corrodes at a known rate in each atmospheric category, coating life is estimated as thickness divided by rate. A typical minimum mean thickness on steel over 6 mm thick is 85 μm (ISO 1461); at the C3 zinc rate of 0.7–2.1 μm/year this lasts roughly 40–120 years before the first maintenance, and in a C5 marine environment (4.2–8.4 μm/year) about 10–20 years. Silicon in the steel accelerates the Fe–Zn reaction (the Sandelin effect) and gives thick, brittle coatings, so steels for galvanizing have controlled silicon. Continuously galvanized car-body sheet carries only about 7–20 μm per side, often as galvannealed Zn–Fe or Zn–Al–Mg.

Inhibitors

An **inhibitor** is a substance added in small concentration to an environment to reduce corrosion. Classes follow from which branch of the Evans diagram (Chapter 42) they suppress:

- **Anodic (passivating) inhibitors** such as chromate, nitrite and molybdate raise the anodic polarization and promote passivity. They are "dangerous" inhibitors: below a critical concentration they leave small unpassivated areas that pit faster than with no inhibitor at all.
- **Cathodic inhibitors** such as zinc salts, polyphosphates and controlled calcium carbonate scaling precipitate films on cathodic sites; they are "safe" because underdosing merely gives less protection.
- **Mixed and film-forming inhibitors**, typically organic amines, imidazolines and phosphate esters, adsorb on the whole surface (Chapter 45) and are the workhorses of oil and gas production, injected continuously into wells and pipelines.
- **Oxygen scavengers** (sulfite and hydrazine substitutes) in boiler feedwater, and **vapour-phase inhibitors** in packaging for machined parts.

Benzotriazole on copper, calcium nitrite admixture for rebar and inhibited engine antifreeze are everyday examples. The efficiency of an inhibitor is $(\mathrm{CR}_0 - \mathrm{CR})/\mathrm{CR}_0$, usually measured by the polarization and impedance methods of Section 43.13.

43.11 Cathodic protection

Principle and criteria

If the potential of a metal is lowered toward its equilibrium potential, the anodic dissolution current falls exponentially (Tafel kinetics, Chapter 42). **Cathodic protection (CP)** supplies electrons to the structure from an external anode so that the whole surface becomes a cathode. The protective current must equal the cathodic current at the protected potential, which in aerated water is essentially the oxygen-limited current density on bare metal (Figure 43.5).

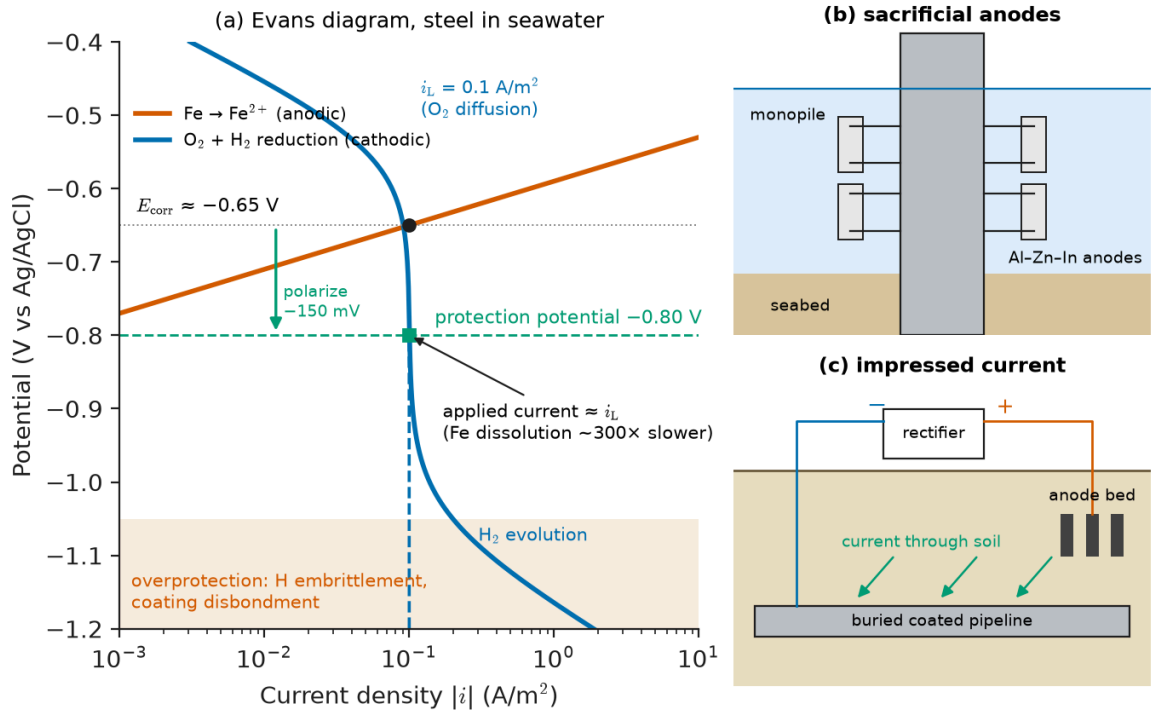


Figure 43.5 — Cathodic protection of steel: Evans diagram showing the shift from the free corrosion potential to the protection potential, and the two ways of supplying the current.

Practical protection criteria are potentials measured against a reference electrode near the surface:

Structure and environment	Typical protection criterion
Carbon steel in seawater	−0.80 V vs Ag/AgCl/seawater or more negative
Carbon steel in soil	−0.85 V vs Cu/$CuSO_4$ (IR-free, "instant-off")
Steel with active sulfate-reducing bacteria	−0.95 V vs Cu/$CuSO_4$
Alternative criterion	at least 100 mV of cathodic polarization from the free corrosion potential
Limit for high-strength steel, coated structures	not more negative than about −1.05 to −1.10 V vs Ag/AgCl (hydrogen embrittlement, coating disbondment)
Steel in concrete	100 mV depolarization criterion

> **Status of the principle: Protection potential criteria**
> *Empirical.* The −0.85 V (Cu/$CuSO_4$) and −0.80 V (Ag/AgCl) values are close to the potentials at which the iron dissolution rate becomes negligible (roughly the iron equilibrium potential at low ferrous ion activity), but they were established by field experience; they apply to carbon steel and must be corrected for IR drop in the electrolyte and adjusted for temperature and bacteria.

Sacrificial (galvanic) anodes

A sacrificial anode is a more active metal connected to the structure. It needs no power supply, cannot be connected the wrong way round and rarely overprotects, but its driving voltage is small (about 0.25 V for aluminium or zinc anodes on steel in seawater), so it suits low-resistivity electrolytes.

Anode alloy	Typical closed-circuit potential	Practical electrochemical capacity	Use
Al–Zn–In	−1.05 V vs Ag/AgCl	about 2000–2500 Ah/kg	Offshore structures, ship hulls, seawater
Zinc	−1.03 V vs Ag/AgCl	about 780 Ah/kg	Seawater, brackish water, ballast tanks
Magnesium (high-potential)	−1.75 V vs Cu/$CuSO_4$	about 1100–1200 Ah/kg	Soil and fresh water (high resistivity), water heaters

Impressed-current cathodic protection

Impressed-current cathodic protection (ICCP) uses a DC power supply (transformer–rectifier or solar) with its positive terminal on inert anodes: mixed-metal-oxide-coated titanium, high-silicon cast iron, graphite or platinized niobium. Its adjustable driving voltage of up to tens of volts protects kilometres of pipeline in resistive soil, ship hulls and the rebar of whole bridge decks. Its drawbacks are the need for power and monitoring and **stray-current interference**: current picked up by a neighbouring pipeline corrodes it where it leaves again (DC railways are notorious sources).

Sizing a cathodic protection system

The design logic (codified in DNV-RP-B401 for offshore structures and ISO 15589 for pipelines) is: area to protect by zone → design current densities (high **initial** values to polarize bare steel and build calcareous deposits, lower **mean** values for the life, **final** values for repolarization), multiplied by a coating breakdown factor for coated steel → anode mass from the mean current → number and geometry of anodes so that their output, limited by anode-to-electrolyte resistance, meets the initial and final demands.

Worked example — sacrificial anodes for an offshore wind monopile. The submerged and buried bare steel of a monopile foundation has area $A = 1500\ \text{m}^2$. Design current densities in this sea area are $i_{\text{initial}} = 0.15\ \text{A/m}^2$ and $i_{\text{mean}} = 0.06\ \text{A/m}^2$; the design life is 25 years. Anodes are Al–Zn–In, capacity $\varepsilon = 2000$ Ah/kg, utilization factor $u = 0.9$ (slender stand-off anodes), each 1.5 m long with an equivalent radius of 0.1 m, in seawater of resistivity $\rho = 0.3\ \Omega\,\text{m}$.

Mean current and mass. $I_{\text{mean}} = 1500 \times 0.06 = 90$ A. The net anode mass is

$$M = \frac{I_{\text{mean}}\, t_{\text{life}} \times 8760\ \text{h/yr}}{u\,\varepsilon} = \frac{90 \times 25 \times 8760}{0.9 \times 2000} = 1.10 \times 10^{4}\ \text{kg}$$

that is 11 tonnes, or 44 anodes of 250 kg.

Initial current. The demand is $1500 \times 0.15 = 225$ A. The resistance of one slender anode (Dwight's formula) is

$$R_a = \frac{\rho}{2\pi L}\left[\ln\!\left(\frac{4L}{r}\right) - 1\right] = \frac{0.3}{2\pi \times 1.5}\left[\ln(60) - 1\right] = 0.0318 \times 3.09 = 0.098\ \Omega$$

With a driving voltage of $\Delta E = -0.80 - (-1.05) = 0.25$ V, each anode delivers $I_a = 0.25/0.098 = 2.54$ A. Meeting 225 A needs $225/2.54 = 89$ anodes.

So the current, not the mass, governs this design: 44 anodes would hold enough aluminium for 25 years but could not polarize the structure at the start. The designer either installs about 89 anodes, carrying far more mass than needed, or changes the geometry: longer, more slender anodes have lower resistance per kilogram and can satisfy both checks with less metal. The same calculation for a 0.5 m diameter coated pipeline 10 km long (area $\pi \times 0.508 \times 10\,000 = 1.60 \times 10^{4}\ \text{m}^2$), with 5 % mean coating breakdown and 20 mA/m² on bare steel in soil, gives a current of only $1.60 \times 10^{4} \times 0.05 \times 0.02 = 16$ A, the reason a good coating plus a modest impressed-current system protects long pipelines economically.

> **Status of the principle: Dwight's anode resistance formula**
> *Approximate.* The resistance of a long thin cylinder in an infinite uniform electrolyte, valid for $L \gg r$ and for anodes far from each other and from the structure; closely spaced anodes interact and their combined resistance is higher than the parallel value.

43.12 Anodic protection and design rules

Anodic protection

For metals that passivate, there is a second electrochemical option: raise the potential into the passive range instead of lowering it. **Anodic protection** uses a potentiostat to hold the structure at a potential between the passivation potential and the transpassive or pitting potential (Chapter 42). Once the film has formed, the current needed is tiny, the passive current density, often below 1 $\mu A/cm^2$, so power consumption is low and the system protects even in very aggressive media.

It protects steel and stainless tanks, coolers and pipes in concentrated sulfuric and phosphoric acid and in pulp-mill caustic. It works only where the passive range is wide and stable, it cannot be used with chlorides (pitting at the raised potential), and a control failure that drops the metal into the active range causes rapid attack.

	Cathodic protection	**Anodic protection**
Applicable metals	All	Only active–passive metals
Environment	Weak to moderate (soil, water, seawater, concrete)	Weak to very aggressive acids and alkalis
Operating current	Moderate to high	Very low (passive current)
Effect of failure	Corrosion returns slowly	Rapid active corrosion
Risks	Hydrogen embrittlement, coating disbondment, stray currents	Pitting if chlorides present; loss of control

Design rules for corrosion protection

The chapter's mechanisms condense into rules applied on drawings and specifications:

1. **Design for the worst environment**, including condensation, deposits, stagnant periods, cleaning chemicals, hydrotest water and shutdowns, not only the nominal one.
2. **Design against a form of attack**, not just a rate: pitting and SCC need different materials from uniform corrosion.
3. **Couple metals close in the galvanic series**, keep the anode large, and isolate.
4. **Eliminate crevices and dead legs**; weld rather than bolt; seal unavoidable gaps.
5. **Control stress and hardness** where SCC or hydrogen are possible.
6. **Respect velocity limits** and design for smooth flow.
7. **Combine coatings with CP** for buried and submerged steel: the coating carries most of the protection, CP covers the holidays.
8. **Design for inspection**: CP test points, coupons and probes, wall-thickness monitoring locations.

43.13 Corrosion testing

Exposure and accelerated tests

Field and immersion tests measure mass loss and pit depth on weighed coupons (ASTM G1, G31): slow but faithful. Specialized tests probe particular forms: U-bend, C-ring and slow-strain-rate tests for SCC, pre-cracked specimens for K_{ISCC}, ferric chloride at rising temperature for CPT, NACE TM0177 and TM0284 for sour-service cracking and HIC.

The **neutral salt-spray test** (ASTM B117, ISO 9227) exposes panels continuously to a 5 % NaCl fog at 35 °C. It is reproducible and simple but correlates poorly with service: no wet–dry cycles, no ultraviolet, no pollutants, and zinc corrodes unrealistically fast in permanent wetness. **Cyclic corrosion tests** (alternating salt, humidity and drying, as in SAE J2334 and ISO 16701) rank coatings much more like outdoor exposure and are now the automotive norm.

> **Status of the principle: Accelerated salt-spray ranking**
> *Empirical.* A standardized exposure useful for quality control of a given coating process; acceleration factors relative to service are not constant across materials or environments, so it should not be used to predict life or to compare unlike coating systems.

Polarization methods

The electrochemical methods of Chapter 42 turn into routine measurements:

- **Potentiodynamic polarization**: the potential is swept and current recorded. Tafel extrapolation of both branches to the corrosion potential gives i_{corr}; the anodic branch of a passive metal shows the passive current density, E_{pit}, and on reversal (cyclic polarization) the repassivation potential and the hysteresis that measures susceptibility to localized attack.
- **Linear polarization resistance (LPR)**: a small sweep of ±10–20 mV around E_{corr} gives a straight line whose slope is the **polarization resistance** $R_p = \Delta E/\Delta i$. Linearizing the Butler–Volmer equation near E_{corr} gives the Stern–Geary relation:

$$i_{\text{corr}} = \frac{B}{R_p}, \qquad B = \frac{\beta_a \beta_c}{2.303\,(\beta_a + \beta_c)}$$

where β_a and β_c are the Tafel slopes in V/decade. LPR is fast and non-destructive, so it is used in field corrosion probes in cooling-water, oil and chemical systems.

> **Status of the principle: Stern–Geary relation**
> *Derived.* Follows from linearizing mixed-potential Butler–Volmer kinetics for small overpotentials, assuming both partial reactions are activation-controlled and the Tafel slopes are known; it fails for diffusion-limited cathodes (where $\beta_c \to \infty$ and $B = \beta_a/2.303$), for localized corrosion, and in high-resistivity media without IR correction.

Electrochemical impedance spectroscopy

Electrochemical impedance spectroscopy (EIS) applies a small sinusoidal potential (about 10 mV) over a range of frequencies, typically 100 kHz to 10 mHz, and measures the complex impedance $Z(\omega)$. The simplest model of a corroding electrode is the **Randles circuit**: the solution resistance R_s in series with a parallel combination of the charge-transfer resistance R_{ct} and the double-layer capacitance C_{dl}:

$$Z(\omega) = R_s + \frac{R_{ct}}{1 + j\omega R_{ct} C_{dl}}$$

On a **Nyquist plot** ($-\text{Im}\,Z$ against $\text{Re}\,Z$) this traces a semicircle from R_s at high frequency to $R_s + R_{ct}$ at low frequency, with its top at $\omega = 1/(R_{ct}C_{dl})$ (Figure 43.6). The diameter R_{ct} plays the role of R_p in the Stern–Geary relation. Real electrodes give depressed semicircles, modelled with constant-phase elements, and extra arcs or diffusion (Warburg) tails. For coated metals, EIS is the main laboratory tool: the high-frequency capacitance of the coating rises as it absorbs water, and its pore resistance falls as ionic paths open, giving early warning long before rust is visible.

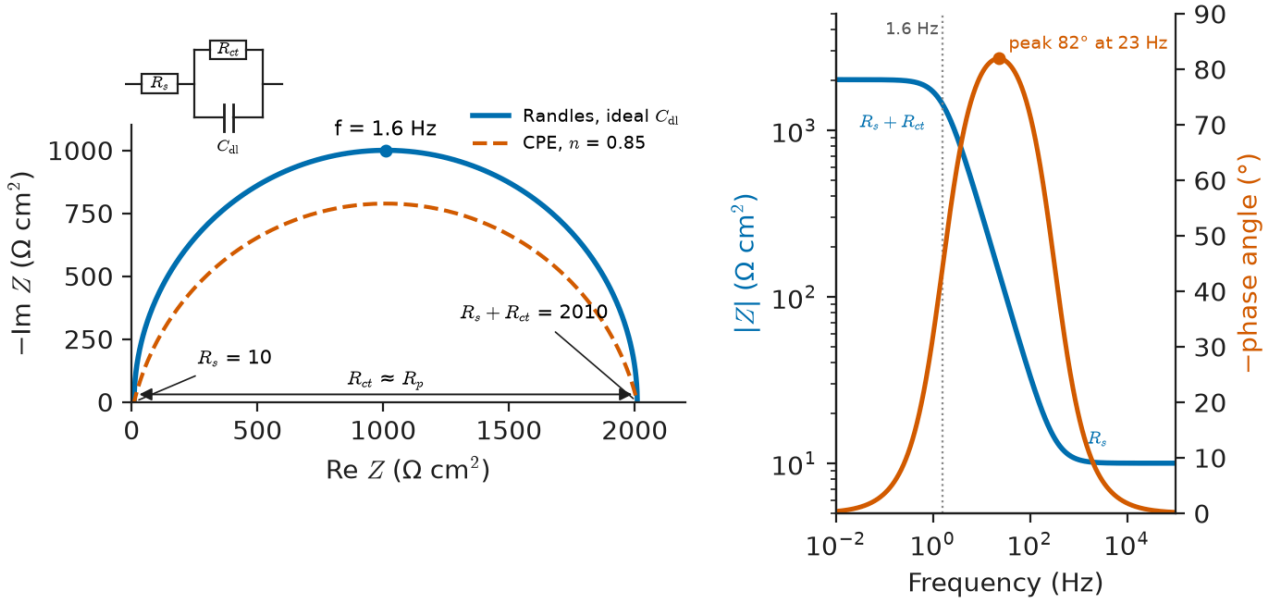


Figure 43.6 — Electrochemical impedance of a corroding electrode modelled by a Randles circuit: Nyquist and Bode plots.

Worked example — corrosion rate of a cooling-water pipe from LPR and EIS. A probe in a recirculating cooling-water system gives a polarization resistance $R_p = 2000\ \Omega\ \text{cm}^2$ on carbon steel. Tafel slopes measured on a laboratory sample are $\beta_a = \beta_c = 0.12$ V/decade.

$$B = \frac{0.12 \times 0.12}{2.303 \times (0.12 + 0.12)} = \frac{0.0144}{0.5527} = 0.0261\ \text{V}$$

$$i_{\text{corr}} = \frac{B}{R_p} = \frac{0.0261}{2000} = 1.30 \times 10^{-5}\ \text{A cm}^{-2} = 13\ \mu\text{A cm}^{-2}$$

With 11.6 μm/year per μA/cm^2, the uniform rate is about 0.15 mm/year, too high for a cooling system designed for perhaps 0.05 mm/year, so the inhibitor dose is reviewed. An EIS check on the same probe, with $R_s = 10\ \Omega\ \text{cm}^2$, $R_{ct} \approx 2000\ \Omega\ \text{cm}^2$ and $C_{dl} = 50$ μF/cm^2, should show a semicircle 2000 Ω cm^2 wide with its apex at $f = 1/(2\pi \times 2000 \times 50 \times 10^{-6}) = 1.6$ Hz; an LPR sweep must be slow compared with this to measure R_p rather than capacitive charging. If the oxygen-reduction cathode were diffusion-limited, B would become $\beta_a/2.303 = 0.052$ V and the corrosion rate estimate would double: the Tafel slopes are the largest uncertainty in the method.

> **Boundary.**
> The derivation of Butler–Volmer kinetics and of the double-layer structure belongs to the Advanced Chemistry book; this section uses only their results.

What remains open

Open question	Where it is answered
Thermodynamic and kinetic foundations: Pourbaix diagrams, mixed potentials, Tafel kinetics, Faraday's law, passivity	Chapter 42
Why grain boundaries and carbides form where they do, and how diffusion controls sensitization	Chapters 5, 8 and 10
Fracture mechanics, fatigue and the endurance limit that corrosion removes	Chapter 14
Stainless, duplex and aluminium alloy metallurgy behind the corrosion-resistant grades	Chapters 15 and 16
Rebar corrosion, carbonation and chloride ingress in concrete; polymer environmental stress cracking	Chapter 44
Adsorption of inhibitors, surface energy and the wear part of erosion–corrosion	Chapter 45
Corrosion of implants in the body; hydrogen for energy; oxidation in extreme environments	Chapters 39, 40 and 41
Weld heat-affected zones and residual stress that drive weld decay and SCC	Chapter 47
Coating deposition processes: PVD, CVD, electroplating, thermal spray	Chapter 49
Surface analysis of passive films and corrosion products (XPS, AES, SIMS)	Chapter 52
Selecting a material for a corrosive duty against cost and other properties	Chapter 58

Next: Chapter 44 widens the view from corrosion to every way that metals, ceramics, polymers, composites and concrete degrade in service, and to the methods by which engineers analyse failures and predict reliability.

Chapter 44 — Degradation and Failure Analysis

Part IX · Corrosion, Degradation and Surfaces

On 3 June 1998 a high-speed train passing Eschede in northern Germany at about 200 km/h derailed and struck a road bridge; 101 people died. The cause was a steel wheel tyre, part of a rubber-sprung wheel design adopted to reduce vibration. Each revolution flexed the tyre; over years of service a fatigue crack grew from its inner surface, helped by the tyre's wear to a thinner section that raised the cyclic stress. The crack grew silently until the remaining ligament broke. No single property of the steel was inadequate on the day the wheel entered service. What failed was the link between a degradation mechanism (fatigue crack growth), its rate under real service loads, and the inspection that should have caught it. This chapter is about that link: how materials of every class lose their properties in service, how the rate of loss is modelled and extrapolated, how failures are statistically distributed, and how an engineer reconstructs a failure from the broken parts to a root cause that can be corrected.

> **By what mechanisms do metals, ceramics, polymers, composites and concrete degrade in service, and what do those mechanisms have in common?**
> **Why do life laws (Paris, Coffin–Manson, Larson–Miller, Arrhenius, square-root ingress) take the forms they do, and how far can they be extrapolated? How are failures statistically described, and how does an investigation get from fracture surfaces to a root cause?**

Concept	Engineering destination
Driving force, mechanism, rate, end-of-life criterion	A common template for any life prediction
Carbonation and chloride ingress, Tuutti model	Concrete cover and durability design of bridges, tunnels, marine structures
Slow crack growth, hydrothermal aging of zirconia	Glass panels, ceramic implants, ceramic turbine and armour components
Moisture uptake, matrix cracking, delamination	Composite airframes, wind-turbine blades, pressure vessels
Thermo-oxidation, photo-oxidation, hydrolysis, stabilizers	Outdoor plastics, pipes, cable insulation, medical polymers
Arrhenius extrapolation of oven aging	Thermal ratings and service lives of polymers and adhesives
Paris law, Miner's rule, Larson–Miller	Inspection intervals in aircraft, rail, bridges; creep life of boilers and turbines
Weibull distribution, bathtub curve, B10 life	Maintenance intervals, warranty, spare-part planning, ceramic design
Fractography: dimples, cleavage, striations, beach marks, mirror–mist–hackle	Identifying the failure mechanism from broken parts
5-why, fishbone, fault tree, 8D, FMEA	Turning a failure into a design, material or process change

Roadmap. A common framework (44.1) → degradation in each material class, with concrete as a worked case (44.2) → polymer aging and weathering, with Arrhenius extrapolation (44.3) → wear-out laws and life prediction (44.4) → reliability statistics: Weibull and the bathtub curve (44.5) → failure-analysis methodology, fractography and root cause (44.6) → case studies across industries (44.7).

44.1 A common framework for degradation

Every degradation process, whatever the material, can be described by the same four questions:

1. **Driving force.** What free energy or mechanical energy is being released? Chemical potential differences drive corrosion and oxidation (Chapter 42); stored elastic energy drives crack growth (Chapter 14); interfacial and grain-boundary energy drive coarsening (Chapter 9); photon energy drives photo-oxidation.

2. **Mechanism and rate-controlling step.** Diffusion, an interfacial reaction, dislocation motion, bond rupture at a crack tip. The slowest step sets the rate, and its dependence on temperature, stress and environment sets the form of the life law.
3. **Damage variable.** What accumulates: crack length, pit depth, carbonation depth, wall loss, oxidized fraction, molecular weight, inelastic strain.
4. **End-of-life criterion.** The damage at which the component no longer performs: crack length at which $K = K_{\mathrm{Ic}}$, depth at which rebar depassivates, elongation at break falling to 50 % of its initial value, wall thickness at the pressure-design minimum.

Most life laws then take one of a few mathematical forms:

Rate-controlling physics	Typical life law	Examples
Thermally activated reaction	$t \propto \exp(E_a/RT)$	Polymer oxidation, intermetallic growth, Arrhenius aging
Diffusion through a growing layer	$x \propto \sqrt{t}$	Carbonation of concrete, parabolic oxidation, case depth
Cyclic damage accumulation	$N \propto (\Delta\sigma)^{-m}$ or $(\Delta\varepsilon_p)^{-1/c}$	Basquin, Coffin–Manson, Paris
Stress-assisted activated flow	$\dot{\varepsilon} \propto \sigma^n \exp(-Q/RT)$	Creep, stress relaxation
Sliding contact	$V \propto Fs/H$	Archard wear (Chapter 45)
Weakest-link initiation	Weibull distribution of lives	Ceramic strength, dielectric breakdown, bearing fatigue

The power of the framework is diagnostic as much as predictive. If accelerated tests at two temperatures fail by different mechanisms, no extrapolation from one to the other is valid. If a failure shows a mechanism the design never considered, the life model was never the problem; the list of mechanisms was.

44.2 Degradation across material classes

Metals

Corrosion in all its forms (Chapters 42–43), fatigue, creep (Chapter 14) and wear (Chapter 45) are the dominant mechanisms. Less visible but equally important are **microstructural changes in service**, where the material slowly moves toward equilibrium at its operating temperature:

- **Coarsening of strengthening precipitates**: γ′ in nickel superalloy turbine blades coarsens and rafts under stress, lowering creep strength (Chapter 41); precipitation-hardened aluminium over-ages if run warm.
- **Spheroidization and graphitization** of carbon steel boiler and refinery tubes held for decades near 450–550 °C: lamellar pearlite turns into spheroidal carbides or graphite nodules, lowering strength and creep resistance.
- **Embrittling precipitation**: temper embrittlement of Cr–Mo steels (impurity segregation to grain boundaries around 350–575 °C); 475 °C embrittlement and sigma-phase formation in ferritic and duplex stainless steels; irradiation hardening and embrittlement of reactor-pressure-vessel steels (Chapter 41).
- **Hydrogen and liquid-metal embrittlement** (Chapter 43), and strain aging that raises yield strength and lowers toughness after cold forming.

The common consequence of most embrittling processes is a rise in the ductile–brittle transition temperature (Chapter 14), monitored in nuclear plants by surveillance Charpy specimens irradiated alongside the vessel.

Ceramics and glasses

Ceramics do not creep or corrode much at ambient temperature, but they are brittle and therefore sensitive to anything that extends existing flaws.

- **Slow crack growth (static fatigue).** Water molecules react with strained Si–O–Si bonds at a crack tip in silicate glass, breaking them at stresses below the fast-fracture value. Crack velocity follows an empirical power law, $v = v_0 (K_{\mathrm{I}}/K_{\mathrm{Ic}})^n$, with n about 15–20 for soda-lime glass in water and 30–60 for alumina and silicon nitride. A large n means strength

depends weakly on load duration; a small n means a glass panel under permanent load (aquarium, glass floor, façade under snow) must be designed to a stress well below its short-term strength, typically a third to a half.

- **Hydrothermal aging of zirconia.** Yttria-stabilized tetragonal zirconia (3Y-TZP) can transform spontaneously to monoclinic at body or room temperature in the presence of water, starting at the surface and spreading inward with a volume increase that roughens and microcracks it. In 2001 a batch of zirconia femoral heads for hip replacements failed in vivo after a change in the sintering process accelerated this aging; zirconia-toughened alumina composites largely replaced monolithic zirconia for that application.
- **Thermal shock** (Chapter 24), **erosion** by particles, and **chemical attack** of refractories by slags and of glass by alkaline solutions.

Polymers

Polymers degrade chemically (oxidation, hydrolysis, photolysis), physically (physical aging below T_g, plasticizer loss, swelling) and mechanically (creep, crazing, environmental stress cracking). Section 44.3 treats them in detail because their aging is the most strongly temperature- and environment-dependent of all classes.

Composites

Fibre composites degrade at three length scales (Chapters 22–23):

- **Matrix.** Epoxy matrices absorb 1–6 wt % water at saturation, which plasticizes them and lowers T_g, roughly 10–20 °C per percent of absorbed water. The design allowables of aircraft structures are therefore set at "hot/wet" conditions. Ultraviolet light degrades unprotected matrix surfaces, which is why composite surfaces are painted or gel-coated.
- **Interface.** Water and cyclic loading weaken the fibre–matrix bond; glass fibres themselves suffer stress corrosion in water and acids (the reason for corrosion-resistant E-CR glass in chemical pipes).
- **Laminate.** Under fatigue, damage progresses through transverse matrix cracking, then delamination between plies, then fibre breakage. Stiffness falls progressively (often by 10–20 % before final failure) rather than staying constant until a single crack runs, which makes stiffness monitoring useful. Low-velocity impact can create **barely visible impact damage** (BVID), internal delaminations that reduce compressive strength by up to half; design allowables assume such damage is present and undetected.

Wind-turbine blades, where root and spar-cap laminates see about 10^8–10^9 load cycles in a 20–25-year life, are the most demanding fatigue application of composites; leading-edge erosion by rain at tip speeds near 90 m/s is their main surface degradation.

Concrete

Reinforced concrete degrades mainly through its steel. Concrete pore solution has a pH of about 12.5–13.5, which keeps the rebar passive (Chapter 42). Two processes destroy that passivity:

- **Carbonation**: atmospheric CO_2 diffuses in and reacts with calcium hydroxide, lowering the pH to about 9. The carbonation front advances approximately as

$$x = k\sqrt{t}$$

because the rate is controlled by diffusion of CO_2 through the already carbonated layer. The coefficient k is typically 2–6 mm/year$^{1/2}$ for structural concretes sheltered from rain, lower for dense, low water–cement-ratio concrete and in permanently wet conditions.

- **Chloride ingress** from seawater or de-icing salt, described by Fick's second law with an apparent diffusion coefficient of order 10^{-12}–10^{-11} m^2/s. When the chloride content at the rebar exceeds a critical value, commonly taken as about 0.4 % by mass of cement, the passive film breaks down locally, giving pitting of the rebar.

> **Status of the principle: Square-root-of-time carbonation law**
> *Approximate.* Derived from diffusion through a carbonated layer with a sharp reaction front and constant boundary conditions; real coefficients are fitted to exposure data and vary with moisture, cement type and curing, and the law overestimates penetration where wetting and drying alternate.

Tuutti's model divides the service life of a reinforced-concrete element into an **initiation period**, while the carbonation or chloride front travels through the cover, and a **propagation period**, while the rebar corrodes until cracking, spalling or loss of section reaches an unacceptable level (Figure 44.1). Rust occupies two to six times the volume of the steel it replaces, so only a few tens of micrometres of corrosion are needed to crack the cover.

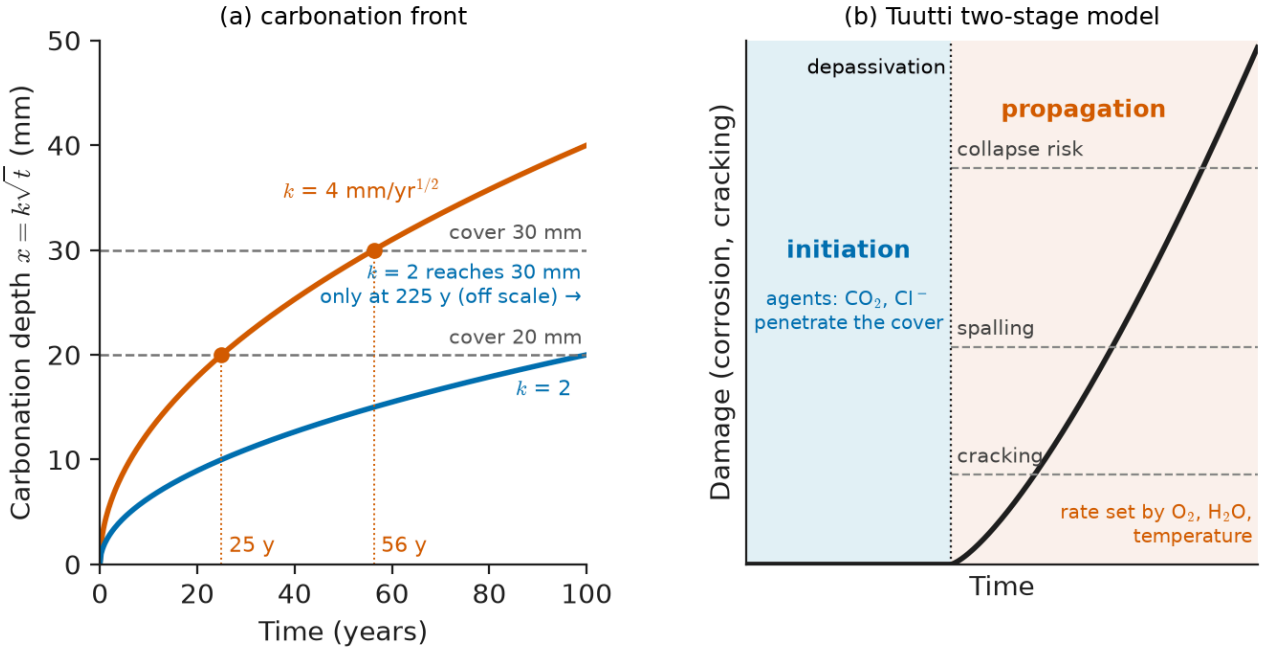


Figure 44.1 — Tuutti's two-stage service-life model for reinforced concrete, with square-root carbonation fronts for two concretes.

Worked example — cover depth for carbonation. A bridge pier in a sheltered urban location is made of concrete with a measured carbonation coefficient $k = 4$ mm/year$^{1/2}$. How long before the carbonation front reaches the reinforcement for covers of 20 mm and 30 mm, and what does a denser concrete with $k = 2$ mm/year$^{1/2}$ achieve?

$$t = \left(\frac{c}{k}\right)^2 : \qquad t_{20} = \left(\frac{20}{4}\right)^2 = 25 \text{ years}, \qquad t_{30} = \left(\frac{30}{4}\right)^2 = 56 \text{ years}$$

With $k = 2$ mm/year$^{1/2}$ and 30 mm cover, $t = (30/2)^2 = 225$ years. Because time scales with the square of the cover and inversely with the square of k, a 50 % increase in cover more than doubles the initiation period, and halving k quadruples it. This is why durability codes (EN 206 and EN 1992, ACI 318) specify both minimum cover and maximum water–cement ratio for each exposure class, and why construction tolerance on cover, often ±10 mm, is a durability parameter: a 30 mm nominal cover placed at 20 mm loses more than half its life.

Other concrete degradation mechanisms are chemical attack of the paste: **sulfate attack** (expansive ettringite and gypsum, resisted by low-C_3A cements), **alkali–silica reaction** (reactive silica aggregate swelling as an alkali silicate gel, which has cracked dams and bridges worldwide), **freeze–thaw damage** of saturated paste (resisted by 4–7 % entrained air), and acid and seawater attack (Chapter 18).

44.3 Polymer aging and weathering

Thermo-oxidation

Most polymers in service oxidize. The process is a free-radical chain reaction, **autoxidation**:

- Initiation: heat, light, residual catalyst or mechanical stress generates a polymer radical, $\mathrm{RH} \rightarrow \mathrm{R}^\bullet$.
- Propagation: $\mathrm{R}^\bullet + \mathrm{O_2} \rightarrow \mathrm{ROO}^\bullet$, then $\mathrm{ROO}^\bullet + \mathrm{RH} \rightarrow \mathrm{ROOH} + \mathrm{R}^\bullet$.
- Branching: hydroperoxides decompose, $\mathrm{ROOH} \rightarrow \mathrm{RO}^\bullet + {}^\bullet\mathrm{OH}$, each producing two new radicals, so the reaction accelerates after an induction period.
- Termination: radicals combine.

The radicals cause **chain scission** (molecular weight falls, polymer becomes brittle: polypropylene, polyethylene) or **cross-linking** (polymer hardens and cracks: many rubbers), and they create carbonyl groups detectable by infrared spectroscopy (Chapter 52). Polypropylene, with a tertiary hydrogen on every repeat unit, oxidizes much faster than polyethylene and is useless outdoors or hot without stabilization.

Stabilizers interrupt the cycle: primary antioxidants (hindered phenols, aromatic amines) donate hydrogen to peroxy radicals; secondary antioxidants (phosphites, thioesters) decompose hydroperoxides harmlessly. Their consumption defines an **induction period**, during which properties barely change, followed by rapid embrittlement once the stabilizer is depleted. The remaining stabilizer is measured by the **oxidation induction time** (OIT), the time to the onset of the oxidation exotherm in a DSC at, for example, 200 °C in oxygen, a routine test for polyethylene pipes and geomembranes.

Photo-oxidation and weathering

Solar ultraviolet radiation between 290 and 400 nm carries 300–410 kJ/mol per photon, comparable to C–C and C–H bond energies. Chromophores (carbonyl groups, catalyst residues, aromatic rings) absorb it and initiate radicals, so photo-oxidation adds to thermo-oxidation at the surface. Outdoor plastics chalk, yellow, lose gloss and develop surface microcracks that act as notches; garden furniture and unprotected polypropylene ropes embrittle in a few summers. Weathering adds moisture, temperature cycling and pollutants.

Protection comes from **UV absorbers** (benzotriazoles, benzophenones), **hindered-amine light stabilizers** (HALS), which scavenge radicals regeneratively and are the most effective for polyolefins, and **pigments**: 2–3 % of well-dispersed carbon black makes polyethylene pipe and cable jacket essentially UV-proof for decades. Accelerated weathering uses xenon-arc or fluorescent UV lamps with water spray (ISO 4892, ASTM G154/G155), but because the lamp spectrum, temperature and wetting differ from sunlight, results rank materials rather than predict outdoor life; reference outdoor sites in Florida (hot, humid) and Arizona (hot, high UV) remain the benchmarks.

Hydrolysis, environmental stress cracking and physical aging

- **Hydrolysis** breaks ester, amide, carbonate and urethane linkages: PET and polycarbonate lose molecular weight in hot water or steam; polyester polyurethanes crumble in humid tropical climates; polyamides absorb water and some hydrolyse in hot glycol coolants. In biomedical PLA and PLGA, hydrolysis is the designed degradation route (Chapter 39).
- **Environmental stress cracking (ESC)** is the polymer analogue of SCC: a liquid that does not dissolve the polymer (detergent on polyethylene, alcohols or oils on polycarbonate and ABS, cleaning agents on medical housings) plasticizes the craze fibrils at a stressed surface and allows cracks to grow at stresses far below those needed in air. It is the most common cause of plastic part failures in practice. Slow crack growth of polyethylene pipe under internal pressure is the related long-term mechanism, which is why modern PE100 and PE100-RC pipe resins are rated for resistance to it.
- **Physical aging** of glassy polymers below T_g: the non-equilibrium glass slowly densifies (Chapter 20), raising modulus and yield stress and lowering toughness and creep rate, with no chemical change. It is reversible by heating above T_g.
- **Loss of plasticizer** from flexible PVC (cables, hoses, medical tubing) by evaporation or migration into contacting materials leaves it stiff and cracked.

Arrhenius extrapolation

Thermal life is estimated by aging specimens in ovens at several elevated temperatures until a property (commonly elongation at break) falls to an end-of-life value, often 50 % of its initial value. If one reaction controls the rate at all test temperatures, the life follows

$$L(T) = L_0 \exp\left(\frac{E_a}{RT}\right)$$

and a plot of $\ln L$ against $1/T$ (Figure 44.2) is a straight line that can be extrapolated to the service temperature. Standards such as IEC 60216 and UL 746B use this procedure to define thermal indices: the temperature at which life reaches a reference value such as 20 000 h. A useful consequence is the rule that life roughly halves for every 10 °C rise near the test temperature; with $E_a = 113$ kJ/mol it halves every 6.7 K near 90 °C.

Status of the principle: Arrhenius aging extrapolation
Empirical. Its form follows from thermally activated reaction kinetics, but applied to the property loss of a real material it is a fit valid only while one mechanism dominates. Extrapolation across a glass transition or melting range, across a change from stabilizer-controlled to oxidation-controlled behaviour, or from diffusion-limited oxidation in thick specimens at high temperature (where oxygen cannot reach the interior fast enough) to homogeneous oxidation at service temperature, is unsafe.

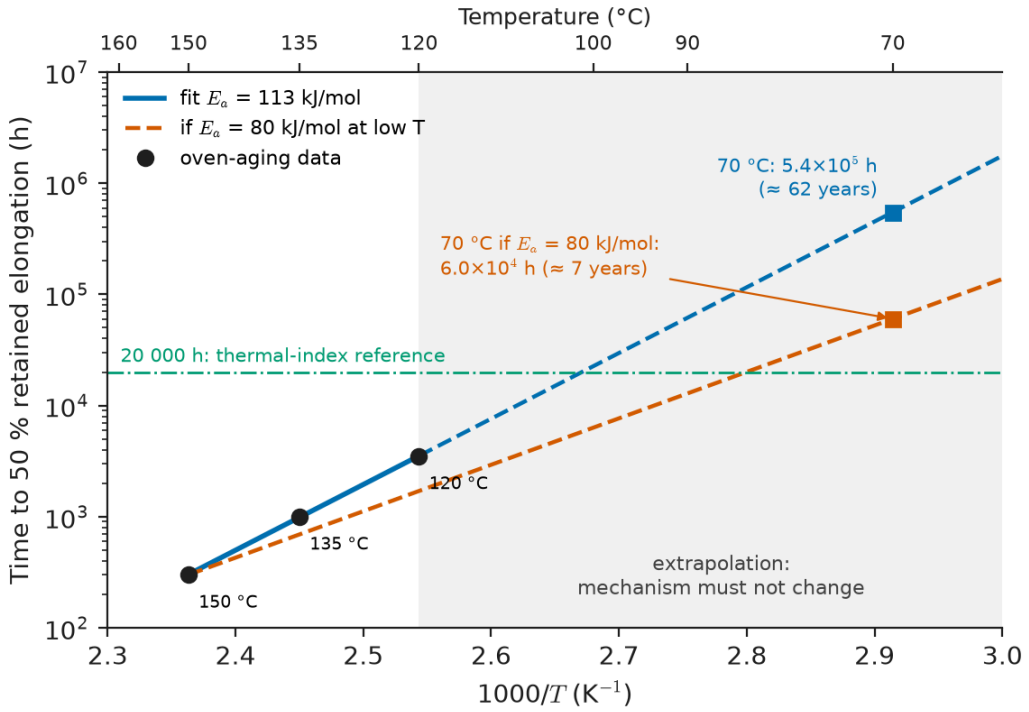


Figure 44.2 — Arrhenius plot of oven-aging life of a polyolefin cable jacket with extrapolation to service temperatures, showing the sensitivity to activation energy.

Worked example — service life of a cable jacket from oven aging. A polyolefin cable jacket compound is aged at three temperatures, and the time for the elongation at break to fall to 50 % of its initial value is recorded:

Aging temperature	150 °C (423.15 K)	135 °C (408.15 K)	120 °C (393.15 K)
Time to end of life	300 h	1000 h	3500 h

A least-squares fit of $\ln L$ against $1/T$ gives a slope $E_a/R = 1.362 \times 10^4$ K, so $E_a = 1.362 \times 10^4 \times 8.314 = 113$ kJ/mol (1.17 eV). The fitted line reproduces the data to within 2 % (302, 986 and 3523 h). Extrapolating from 150 °C:

$$L(70\,^\circ\mathrm{C}) = 300 \times \exp\left[1.362 \times 10^4 \left(\frac{1}{343.15} - \frac{1}{423.15}\right)\right] = 300 \times \exp(7.50) \approx 5.4 \times 10^5 \text{ h} \approx 62 \text{ years}$$

At 90 °C the same calculation gives about 6.1×10^4 h, roughly 7 years: a 20 K rise in conductor temperature cuts the life ninefold.

The fragility of the extrapolation is the real lesson. Three points spanning 30 K are being projected 50 K beyond the data, across more than two decades of time. If the true activation energy at lower temperatures were 80 kJ/mol rather than 113 kJ/mol (a common situation when a different, lower-barrier process dominates at low temperature), the life at 70 °C would be about 6×10^4 h, about 7 years instead of 62. That is why long-life ratings require test temperatures as close to service as practical, at least four temperatures, a check for linearity, and confirmation by OIT or infrared spectroscopy that the same chemistry is occurring at each temperature.

44.4 Wear-out mechanisms and life prediction

The catalogue of life laws

Wear-out failures are those whose probability rises with age because damage accumulates. The laws that predict them have appeared in earlier chapters; collected here, they form the toolkit of life prediction:

Mechanism	Life law	Typical parameters	Status
High-cycle fatigue	Basquin: $\sigma_a = \sigma'_f (2N_f)^b$	$b \approx -0.05$ to -0.12	Empirical
Low-cycle fatigue	Coffin–Manson: $\Delta\varepsilon_p/2 = \varepsilon'_f (2N_f)^c$	$c \approx -0.5$ to -0.7	Empirical
Fatigue crack growth	Paris: $da/dN = C(\Delta K)^m$	$m \approx 3$ (steels), 3–4 (Al alloys)	Empirical
Variable-amplitude fatigue	Miner: $\sum n_i/N_i = 1$	failure at sum 0.3–3 in practice	Empirical
Creep rupture	Larson–Miller: $P = T(C + \log_{10} t_r)$	$C \approx 20$ (t_r in h, T in K)	Empirical
Thermally activated aging	Arrhenius: $L \propto \exp(E_a/RT)$	E_a 50–150 kJ/mol for polymers	Empirical
Humidity-driven corrosion	Peck: $L \propto \mathrm{RH}^{-n} \exp(E_a/kT)$	$n \approx 2.7$, $E_a \approx 0.8$ eV	Empirical
Diffusion-controlled ingress	$x = k\sqrt{t}$	carbonation k 2–6 mm/year$^{1/2}$	Approximate
Sliding wear	Archard: $V = KFs/H$	K from 10^{-8} to 10^{-2}	Empirical

The electronic-specific laws (Black's equation for electromigration, TDDB models for dielectrics, Norris–Landzberg for solder joints) are applied to their mechanisms in Chapters 34, 50 and 51.

Each law is used in the same way: measure its parameters under accelerated conditions, compute an **acceleration factor** between test and service, and convert test life to service life. Each carries the same caveat: the mechanism must be the same in test and service.

Damage-tolerant design with the Paris law

For safety-critical structures that cannot be guaranteed crack-free (airframes, rail axles and wheels, pressure vessels, bridges), the design philosophy is **damage tolerance**: assume a crack of the largest size that inspection could miss, calculate how long it takes to grow to a critical size, and inspect at a fraction (typically half or a third) of that interval. With $\Delta K = Y\Delta\sigma\sqrt{\pi a}$ and constant Y, integrating the Paris law from initial size a_0 to critical size a_c gives, for $m \neq 2$,

$$N_f = \frac{2}{(m-2)\,C\,(Y\Delta\sigma\sqrt{\pi})^m}\left[a_0^{(2-m)/2} - a_c^{(2-m)/2}\right]$$

> **Status of the principle: Paris-law crack-growth integration**
> *Empirical.* The Paris law is a fit to the intermediate (stage II) region of crack-growth data; it ignores the threshold below which cracks do not grow, the acceleration near K_{Ic}, load-sequence effects (overload retardation) and short-crack behaviour, so lives computed from it are reliable only for long cracks under roughly constant-amplitude loading.

Worked example — inspection interval for a cracked steel plate. A ferrite–pearlite steel plate in a crane girder carries a constant-amplitude stress range $\Delta\sigma = 100$ MPa ($R = 0$, so $\sigma_{\max} = 100$ MPa). Ultrasonic inspection can reliably find edge cracks of 2 mm and larger. Paris constants are $C = 6.9 \times 10^{-12}$ m/cycle and $m = 3$ (with ΔK in MPa√m), $K_{\mathrm{Ic}} = 100$ MPa√m, and $Y = 1.12$.

Critical crack size:

$$a_c = \frac{1}{\pi}\left(\frac{K_{\mathrm{Ic}}}{Y\sigma_{\max}}\right)^2 = \frac{1}{\pi}\left(\frac{100}{112}\right)^2 = 0.254\ \mathrm{m}$$

(in practice the plate width or net-section yielding would limit this first; we keep it to see its influence).

Cycles from 2 mm to a_c: with $m = 3$, the bracket is $a_0^{-1/2} - a_c^{-1/2} = 22.36 - 1.99 = 20.37$ m$^{-1/2}$, and $(Y\Delta\sigma\sqrt{\pi})^3 = (112 \times 1.7725)^3 = (198.5)^3 = 7.82 \times 10^6$:

$$N_f = \frac{2 \times 20.37}{6.9 \times 10^{-12} \times 7.82 \times 10^{6}} = 7.5 \times 10^{5}\ \text{cycles}$$

Three sensitivities follow directly. Taking $a_c = 50$ mm instead of 254 mm reduces N_f only to 6.6×10^5 cycles: almost all the life is spent while the crack is small, so the critical size hardly matters. Halving the detectable crack to 1 mm raises N_f to 1.10×10^6 cycles: better inspection buys life directly. Raising the stress range by 25 % to 125 MPa cuts N_f to 3.8×10^5 cycles (and a_c to 0.16 m), because life scales as $\Delta\sigma^{-3}$. At 300 lifts a day the 2 mm case allows about 7 years of crack growth, so an inspection every two to three years gives two chances to find the crack before it is critical. The Eschede tyre of the opening was an object lesson in the last sensitivity: wear thinned the section, raising the stress range in service beyond what the original assessment had assumed.

Variable loads and Miner's rule

Real loads vary. **Miner's rule** assumes that each cycle at amplitude σ_i consumes a fraction $1/N_i$ of life, where N_i is the constant-amplitude life at that amplitude, and that failure occurs when $\sum n_i/N_i = 1$. Load histories are reduced to counts of cycles by rainflow counting. The rule ignores load sequence (a high-low sequence is more damaging than low-high) and the fact that small cycles below the endurance limit become damaging once a crack exists, so design codes often limit the damage sum to 0.5 or less, or extend S–N curves below the endurance limit with a shallower slope.

Creep life and the Larson–Miller parameter

Creep-rupture data are extrapolated with time–temperature parameters (Chapter 14). With the Larson–Miller parameter, a stress that causes rupture in 1000 h at 650 °C (923.15 K) has $P = 923.15 \times (20 + 3) = 2.123 \times 10^4$; at the same stress and 600 °C (873.15 K), $\log_{10} t_r = 2.123 \times 10^4/873.15 - 20 = 4.32$, so $t_r \approx 2.1 \times 10^4$ h. A 50 K reduction in temperature buys a factor of 21 in life. Power-plant steam pipes and reformer tubes are managed this way, with remaining-life assessment by replica metallography of creep cavities on the outer surface.

44.5 Reliability statistics

Describing a population of lives

Nominally identical components fail at different times, because flaws, loads and environments vary. Reliability engineering describes the population by:

- the cumulative failure probability $F(t)$ and the **reliability** $R(t) = 1 - F(t)$;
- the probability density $f(t) = dF/dt$;
- the **hazard rate** $h(t) = f(t)/R(t)$, the instantaneous failure rate of survivors, which tells whether old components are more or less likely to fail than new ones.

The Weibull distribution

The most widely used life distribution in mechanical and materials reliability is the two-parameter Weibull:

$$F(t) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right], \qquad h(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}$$

where η is the **characteristic life** (63.2 % failed) and β the **shape parameter**. Its flexibility comes from β: $\beta < 1$ gives a falling hazard (infant mortality from manufacturing defects), $\beta = 1$ a constant hazard (random failures, the exponential distribution), and $\beta > 1$ a rising hazard (wear-out; $\beta \approx 2$–4 for fatigue and wear, larger for tightly controlled processes). Its physical justification is the weakest-link argument already used for ceramic strength (Chapter 14): a component that fails when its worst element fails follows Weibull statistics. Mean life is $\text{MTTF} = \eta\,\Gamma(1 + 1/\beta)$. The **B10 life**, the time by which 10 % have failed, is standard for bearings and many mechanical parts:

$$t_{B10} = \eta\,[-\ln(0.9)]^{1/\beta}$$

The **lognormal** distribution is an alternative when damage grows multiplicatively (fatigue lives of many metals, electromigration); with limited data the two are hard to distinguish, but they diverge in the low-percentile tail that matters for safety.

Status of the principle: Weibull life distribution
Empirical. Justified by weakest-link reasoning where failure starts at the worst of many independent flaws, but in reliability practice it is a flexible fit; extrapolation to failure fractions far below those observed is sensitive to the fitted β, and mixed populations (defective plus sound) need separate distributions.

The bathtub curve

Over the life of a product population, the hazard rate typically follows the **bathtub curve** (Figure 44.3): a falling early-failure region (defective units failing; addressed by burn-in, proof testing and process control), a flat region of random failures (overloads, accidents; addressed by margins), and a rising wear-out region (fatigue, wear, aging; addressed by life prediction and preventive replacement). The curve is a composite: each region is a different population or mechanism, and each has its own Weibull β. Many well-designed mechanical products show little flat region at all, and electronic products often wear out only after obsolescence.

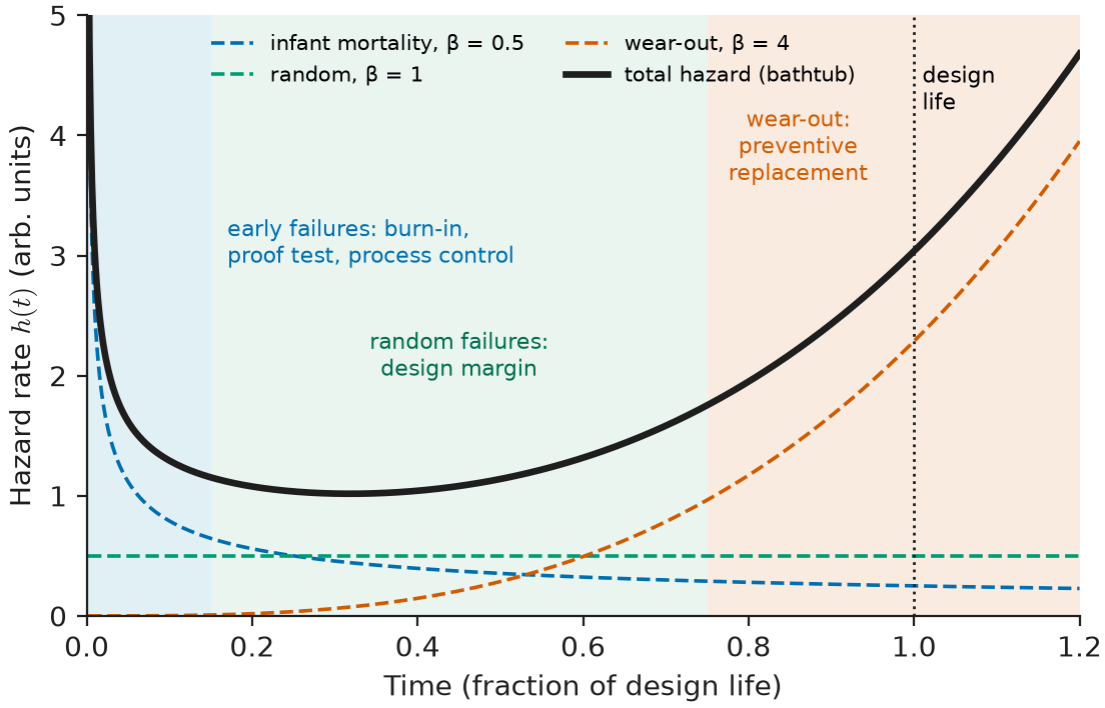


Figure 44.3 — The bathtub curve as the sum of three Weibull hazard populations.

Fitting data and reading the result

Failure times are ranked, each is assigned an estimated cumulative probability by **median ranks**, $F_i \approx (i-0.3)/(n+0.4)$ (Bernard's approximation), and $\ln[-\ln(1-F)]$ is plotted against $\ln t$. A straight line confirms the Weibull form; its slope is β and its intercept gives η (Figure 44.4). Maximum-likelihood estimation is preferred for censored data (units still running when the test ends).

Worked example — Weibull analysis of hydraulic pump seals. Ten seals from a test rig fail at 410, 620, 780, 900, 1020, 1130, 1260, 1400, 1590 and 1900 h. Median ranks run from $F_1 = 0.7/10.4 = 0.067$ to $F_{10} = 9.7/10.4 = 0.933$. A least-squares fit of $\ln[-\ln(1-F_i)]$ on $\ln t_i$ gives

$$\beta = 2.42, \qquad \eta = 1252 \text{ h}$$

The shape parameter well above 1 identifies wear-out. The B10 life is

$$t_{B10} = 1252 \times (0.1054)^{1/2.42} = 1252 \times 0.395 = 495 \text{ h}$$

and the mean life is MTTF $= 1252 \times \Gamma(1.413) = 1252 \times 0.887 = 1110$ h (the sample mean is 1101 h). The hazard rate rises from 5.2×10^{-4} per hour at 500 h to 2.5×10^{-3} per hour at 1500 h, nearly fivefold. Because failures are wear-out, preventive

replacement pays: replacing seals at 500 h accepts about a 10 % chance of an in-service failure per seal, whereas running to failure would see half of them fail before 1080 h. If instead β had come out near 1, preventive replacement would be pointless, since a new seal would be no more reliable than an old one. Ten failures give β with a 90 % confidence interval of roughly ±40 %, so a decision that depends on the B1 life rather than the B10 would need more data.

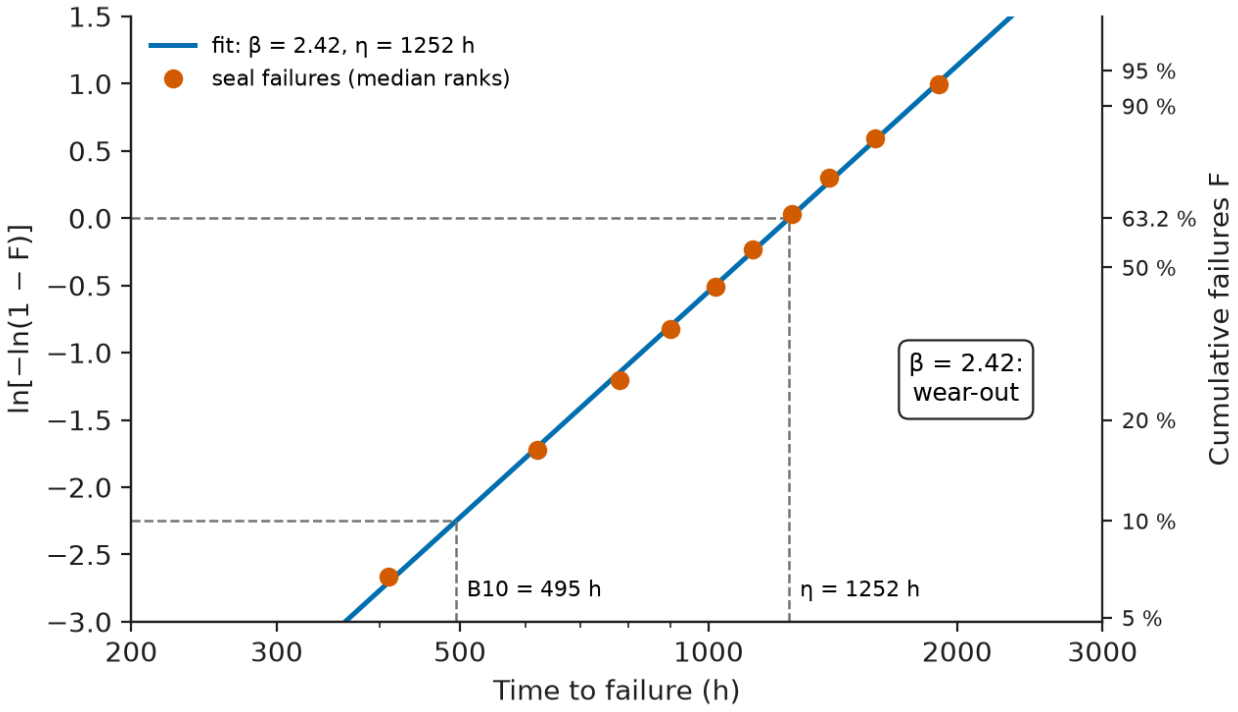


Figure 44.4 — Weibull probability plot of ten pump-seal failures with the fitted line, characteristic life and B10 life.

44.6 Failure analysis: methodology, fractography and root cause

Principles

Failure analysis is the systematic reconstruction of how and why a component failed. Three principles govern it:

1. **Preserve evidence.** Fracture surfaces are damaged by fitting broken halves together, by cleaning, by corrosion after the event and by careless sectioning. Every destructive step removes information; proceed from non-destructive to destructive and document each step.
2. **Collect the history.** Design drawings, material certificates, manufacturing and repair records, operating loads and environment, time in service and similar failures elsewhere. Clustering of failures by production lot, supplier, site or season points directly to a class of causes.
3. **Distinguish mechanism from root cause.** "Fatigue crack from a weld toe" is a mechanism. The root cause is why the conditions for it existed: a weld detail with a high stress concentration, a change in welding procedure, a load spectrum underestimated in design, an inspection that missed the crack. Only a root cause can be corrected.

The workflow

Figure 44.5 summarizes the standard sequence.

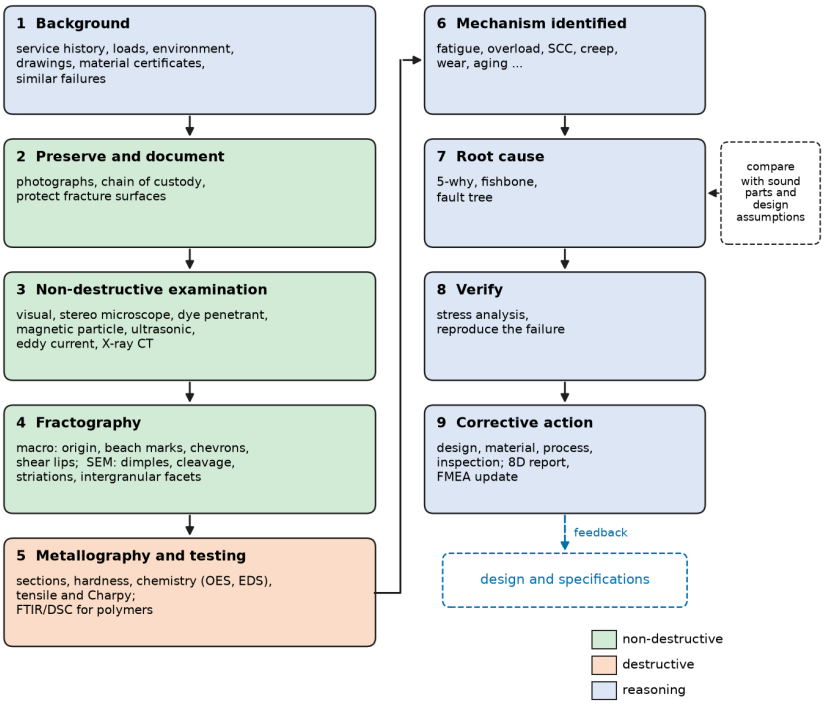


Figure 44.5 — Failure-analysis workflow from the failed part to verified corrective action.

Non-destructive examination comes first: visual inspection and stereo microscopy to locate the origin and the sequence of fracture; dye-penetrant and magnetic-particle inspection to find other cracks; ultrasonic, eddy-current and radiographic or CT inspection to find subsurface defects. Then the fracture surfaces are examined macroscopically and in the SEM, sections are cut through the origin for metallography (Chapter 52), hardness and chemical composition are checked against specification, and mechanical tests on material from the failed part confirm or exclude a materials deficiency. For polymers, FTIR, DSC and molecular-weight measurements reveal oxidation, wrong grade or degradation in moulding. Stress analysis, by hand or finite elements, checks whether the loads the evidence implies are consistent with the design.

Reading fracture surfaces

Fractography is the most powerful single tool, because the fracture surface records how the crack grew (Figure 44.6).

Feature	Scale	Meaning
Beach marks (clamshell marks)	Macroscopic	Successive positions of a fatigue crack front, from changes in load or environment; they point back to the origin
Ratchet marks	Macroscopic	Steps between multiple fatigue origins; many ratchets indicate high stress or a stress concentration
Chevron (herringbone) marks	Macroscopic	Fast brittle fracture in plate; the chevrons point back toward the origin
Shear lips	Macroscopic	Ductile final fracture under plane stress near free surfaces; small lips mean low toughness or high constraint
Dimples (microvoid coalescence)	SEM, 1–50 µm	Ductile overload; elongated or parabolic dimples indicate shear or tearing
Cleavage facets with river patterns	SEM	Brittle transgranular fracture in BCC metals below the transition temperature
Intergranular facets	SEM	Weak grain boundaries: temper embrittlement, hydrogen, SCC, creep cavitation, sensitization
Fatigue striations	SEM, 0.1–1 µm spacing	One striation per load cycle in ductile metals; spacing gives local crack growth rate
Branched cracks	Section	Stress-corrosion cracking
Mirror, mist and hackle	Glass, ceramics	Smooth mirror around the origin, then misty and rough hackle zones as the crack accelerates and branches
Crazes, stretched fibrils, slow-crack-growth bands	Polymers	Environmental stress cracking or slow crack growth

Two quantitative readings are routine. Striation spacing measured at several distances from the origin, compared with the Paris law, estimates the stress range that grew the crack. In glasses and ceramics, the mirror radius r_m is related to the fracture stress by $\sigma_f \sqrt{r_m} = A$, with mirror constant $A \approx 2$ MPa√m for soda-lime glass; a mirror of radius 0.5 mm therefore indicates a fracture stress of about $2/\sqrt{5 \times 10^{-4}} \approx 89$ MPa, which tells the analyst whether the pane broke at a normal service load from a damaged edge or under an abnormal load.

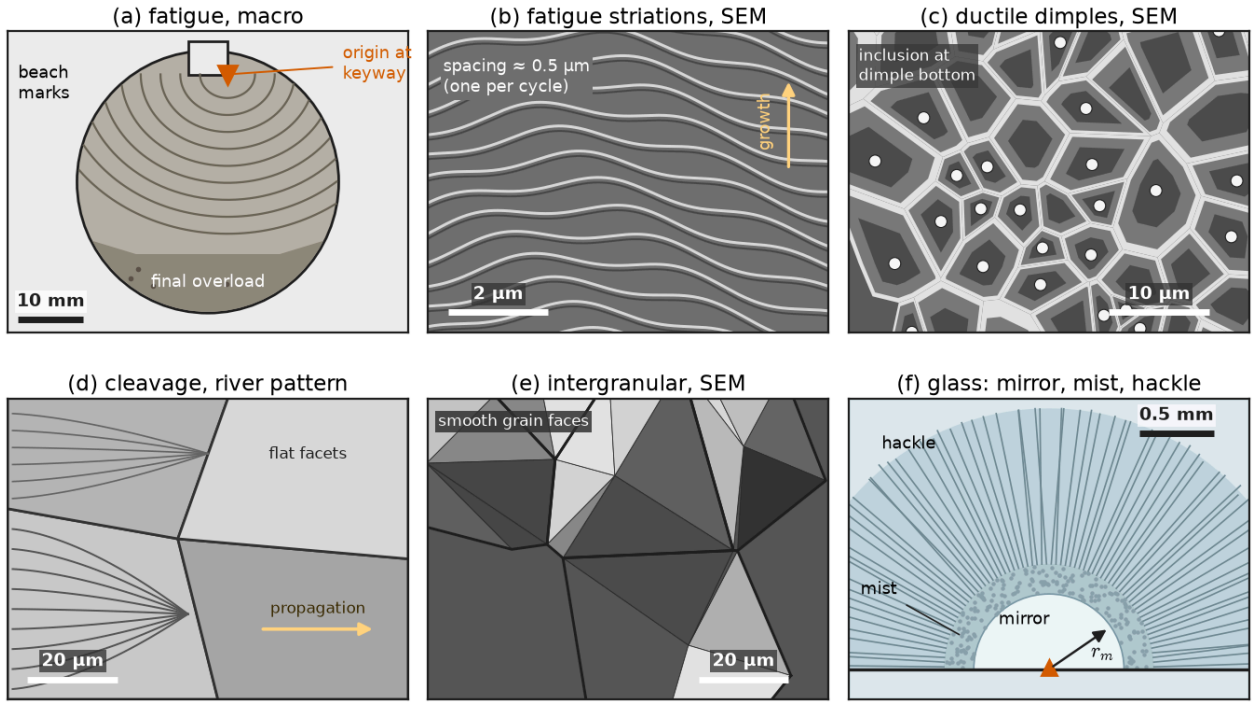


Figure 44.6 — Characteristic fracture-surface features used to identify failure mechanisms.

Status of the principle: Fracture-mirror relation
Empirical. The relation $\sigma_f \sqrt{r_m} = A$ follows dimensionally from crack-branching at a critical stress intensity, but A is measured for each material and the relation assumes a sharp origin and nearly uniform stress over the mirror.

Root-cause tools

- **5-why analysis**: asking *why* repeatedly until a controllable cause is reached. For a fatigued pump shaft: Why did it break? Fatigue from the keyway. Why? Rotating bending stress above the fatigue strength at the keyway corner. Why? The corner radius was 0.1 mm instead of the drawn 0.5 mm. Why? A worn cutter was used and the radius is not inspected. Why? The inspection plan lists the radius as non-critical. The last answer is actionable.
- **Ishikawa (fishbone) diagrams**: candidate causes grouped under materials, methods, machines, measurement, environment and people, so none is overlooked.
- **Fault-tree analysis**: top-down logical decomposition of the failure into AND/OR combinations of basic events, used where several conditions must coincide (SCC needs stress AND environment AND susceptible material).
- **8D** (eight disciplines): team, problem description, containment, root cause and escape point, corrective action, validation, prevention of recurrence, closure. The materials engineer's work concentrates in root cause and validation.
- **FMEA**: the forward-looking counterpart, listing for a new design each potential failure mode, its effect, cause and controls, and ranking it before release. The mechanisms of Chapters 42–45 are the vocabulary of a good FMEA, and lessons from failure analysis feed back into it.

Boundary.
Formal reliability engineering (confidence bounds, accelerated-test planning, system reliability block diagrams) and the statistics behind it belong to The Mathematics Universe and engineering-management courses.

44.7 Case studies across industries

The value of case studies is in the pattern. In almost every major materials failure the mechanism was known at the time; what failed was the recognition that the mechanism applied, the estimate of its rate, or the inspection meant to catch it.

Case	Industry	Mechanism	Root cause and lesson
Liberty ships and T2 tankers (1940s)	Marine	Brittle fracture of welded hulls in cold seas	Steel with high ductile–brittle transition temperature, welded continuous hulls letting cracks run between plates, square hatch corners; led to Charpy toughness requirements and modern fracture mechanics
de Havilland Comet (1954)	Aerospace	Fatigue of pressurized fuselage	Cyclic pressurization stresses at cut-out corners underestimated; led to full-scale fatigue testing and fail-safe design
Silver Bridge, Ohio (1967)	Civil	Brittle fracture of an eyebar from a small crack grown by stress corrosion and corrosion fatigue	Non-redundant eyebar chain, a crack hidden in an uninspectable location; led to US national bridge inspection standards
Aloha Airlines 243 (1988)	Aerospace	Multiple-site fatigue at fuselage lap joints after disbonding and corrosion	High-cycle, humid, salt environment; inspection intervals based on flight hours not cycles; led to aging-aircraft programmes
Space Shuttle Challenger (1986)	Aerospace	Loss of elastomer resilience at low temperature in field-joint O-rings	Seals operated far below qualified temperature; viscoelastic recovery too slow to follow joint opening (Chapter 14)
Eschede ICE (1998)	Rail	Fatigue fracture of a resilient wheel tyre	Cyclic stresses in a thinning tyre underestimated; inspection unable to detect internal cracks
Zirconia femoral heads (2001)	Biomedical	Hydrothermal aging and fracture of 3Y-TZP	Process change in sintering altered microstructure in some batches; showed that a processing change can create a new degradation mechanism
Davis-Besse reactor head (2002)	Energy	Boric-acid wastage of carbon-steel head, after PWSCC of a nozzle let coolant leak	Leak indications missed for years; only the stainless cladding held pressure
Metal-on-metal hip implants (2010 recalls)	Biomedical	Wear and corrosion debris causing tissue reactions	Wear in edge-loading conditions not captured by standard simulator tests (Chapter 45)
Silver-sulfide resistor failures	Electronics	Sulfidation of silver electrodes through termination gaps	Office-environment components used in sulfur-rich industrial air

Three lessons recur.

First, interactions defeat single-mechanism analysis. Silver Bridge needed a stress-corrosion crack to start a brittle fracture; Davis-Besse needed SCC to create the leak that drove wastage; Aloha needed disbonding to load the rivets that fatigued. Fault trees with AND gates describe these failures better than any single life law.

Second, the service condition, not the material, was usually misjudged. Comet's designers knew about fatigue; they underestimated the stress at the corners. Challenger's O-ring compound was suitable within its qualified temperature range. Eschede's tyre steel was adequate for the stress the design assumed. The engineering response is a materials engineer's insistence on the real environment, real loads and real temperature, measured not assumed.

Third, processing changes are silent design changes. A sintering change made zirconia heads age; a welding procedure change or a new supplier's steel can move a transition temperature. Every process change to a safety-critical part therefore needs the same qualification as the original design.

> **Status of the principle: Case-study lessons**
> *Empirical.* Generalizations drawn from investigated failures; they are heuristics for where to look, not laws, and each case report should be read in its original investigation for the details that matter.

44.8 Wear-out of electronic materials

Electronic assemblies add wear-out mechanisms of their own: current densities high enough to move atoms, fields close to breakdown in nanometre-thin dielectrics, hot carriers in transistor channels, moisture in plastic packages and contacts that carry current through a few microscopic spots. The life laws below follow the same pattern as Section 44.4 — a driving force, an Arrhenius temperature dependence and a fitted exponent — and are the ones Chapters 34, 50 and 51 rely on.

Translating test cycles to the field: Norris–Landzberg

Thermal-cycling tests are run with larger temperature swings and shorter cycles than the field to obtain failures in weeks. The **acceleration factor** $AF = N_{\text{field}}/N_{\text{test}}$ relates the two. Norris and Landzberg extended Coffin–Manson to include the effects of frequency and peak temperature (which control creep during each cycle):

$$AF = \frac{N_{\text{field}}}{N_{\text{test}}} = \left(\frac{\Delta T_{\text{test}}}{\Delta T_{\text{field}}}\right)^{n} \left(\frac{f_{\text{field}}}{f_{\text{test}}}\right)^{m} \exp\left[\frac{E_a}{k}\left(\frac{1}{T_{\max,\text{field}}} - \frac{1}{T_{\max,\text{test}}}\right)\right]$$

The first factor is Coffin–Manson (n about 2 for SnPb, 2–2.7 for SAC). The second says that slower field cycles give more time for creep per cycle and hence *fewer* cycles to failure ($m \approx 1/3$ for SnPb, smaller for SAC). The third says that a hotter peak temperature causes more damage per cycle.

> **Status of the principle: Norris–Landzberg acceleration factor**
> *Empirical.* Fitted to thermal-cycling data for specific solder alloys and package types; parameters differ between SnPb and SAC and between studies, and the model assumes the same failure mechanism in test and field (it fails if test temperatures exceed a laminate T_g or cause a different crack path).

Worked example — field life of a BGA from a thermal-cycling test. A SAC BGA assembly survives a characteristic life of $N_{\text{test}} = 2000$ cycles in a −40/+125 °C test ($\Delta T = 165$ K, $T_{\max} = 398$ K) at 24 cycles/day. In the field it sees $\Delta T = 50$ K with $T_{\max} = 75$ °C (348 K), twice a day. Using SAC parameters reported by Pan and co-workers ($n = 2.65$, $m = 0.136$, $E_a/k = 2185$ K):

$$\left(\frac{165}{50}\right)^{2.65} = 3.3^{2.65} = 23.7$$

$$\left(\frac{2}{24}\right)^{0.136} = 0.713$$

$$\exp\left[2185\left(\frac{1}{348.15} - \frac{1}{398.15}\right)\right] = \exp(2185 \times 3.607 \times 10^{-4}) = \exp(0.788) = 2.20$$

$$AF = 23.7 \times 0.713 \times 2.20 = 37.1$$

The expected field characteristic life is $2000 \times 37.1 \approx 74\,000$ cycles, which at two cycles per day is about 100 years. The margin is comfortable, but notice which term dominates: the ΔT ratio. If the field swing were 80 K instead of 50 K (an outdoor enclosure in sunlight), the first factor would fall to $(165/80)^{2.65} = 6.8$ and the field life to roughly 30 years. The model is only as good as the field temperature profile, which is often the least known input.

Electromigration

The mechanism

A copper interconnect on a modern chip carries current densities of order 1 MA/cm^2 (10^{10} A/m^2), a million times the density in household wiring. At these densities, conduction electrons transfer momentum to metal ions as they scatter, producing a net force on the ions in the direction of electron flow, the **electron wind**. Writing it as an effective charge,

$$F = Z^{*} e\, E = Z^{*} e\, \rho j$$

where ρj is the local electric field and Z^{*} is an effective valence (negative, of magnitude of a few to tens, because the wind force exceeds the direct electrostatic force on the ion). The force biases the random diffusion of atoms, giving an atomic flux, by the Nernst–Einstein relation,

$$J_a = \frac{C\,D}{kT}\,Z^{*} e \rho j$$

with C the atomic concentration and D the diffusivity along the fastest path (grain boundaries in aluminium, the top interface with the capping dielectric in copper damascene lines).

A uniform flux does no harm: atoms leave one end and arrive at the other. Damage occurs where the flux is **divergent**: where more atoms leave a region than arrive, a void forms; where more arrive than leave, material piles up as a hillock or extrusion (Figure 44.7). Flux divergences occur at grain-boundary triple points, at changes in grain structure, at the ends of lines where they meet tungsten or copper vias (which act as blocking boundaries), and wherever temperature or current density changes. A void that grows across the line, or under a via, opens the circuit; an extrusion can short to a neighbour.

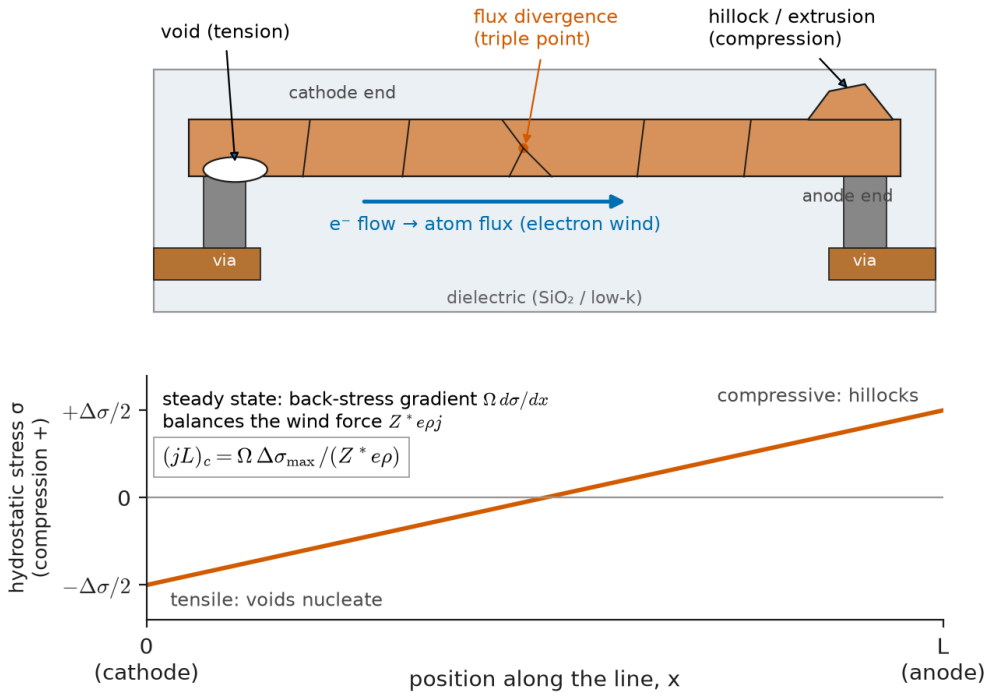


Figure 44.7 — Electromigration in an interconnect line: void at the cathode end, hillock at the anode end, and the back-stress that defines the Blech length.

Black's equation

The median time to failure of a population of test lines is described by Black's equation:

$$t_{50} = A\, j^{-n} \exp\!\left(\frac{E_a}{kT}\right)$$

The temperature term reflects thermally activated diffusion, and E_a identifies the dominant path: about 0.5–0.7 eV for grain-boundary diffusion in pure aluminium, 0.7–0.9 eV for Al–0.5 % Cu (copper segregates to grain boundaries and blocks them, the original materials fix for electromigration), 0.8–1.0 eV for copper lines with a SiN cap (interface diffusion), and above 1.5 eV for copper with metallic caps (CoWP or cobalt) that strengthen the weakest interface. The current exponent n is about 2 when lifetime is controlled by void nucleation (which requires a critical stress to build up) and about 1 when it is controlled by void growth. Failure times of a population follow a lognormal distribution, with shape parameter σ typically 0.3–0.6.

> **Status of the principle: Black's equation**
> *Empirical.* The Arrhenius factor reflects diffusion, and the current exponent can be rationalized by nucleation and growth models, but A, n and E_a are fitted for each metallization; Joule heating in the test (which raises the true line temperature above the oven temperature) and a change of mechanism between test and use invalidate extrapolation.

Worked example — extrapolating an electromigration test. Copper test lines with a SiN cap are stressed at $T_{\text{test}} = 300$ °C (573 K) and $j_{\text{test}} = 2$ MA/cm^2, giving $t_{50} = 150$ h with lognormal $\sigma = 0.5$. The design must guarantee that fewer than 0.1 % of

lines fail in 10 years at $T_{\text{use}} = 105$ °C (378 K) and $j_{\text{use}} = 1$ MA/cm^2. Take $E_a = 0.90$ eV and $n = 2$.

$$\frac{E_a}{k} = \frac{0.90}{8.617 \times 10^{-5}} = 1.044 \times 10^4 \text{ K}$$

$$AF = \left(\frac{j_{\text{test}}}{j_{\text{use}}}\right)^2 \exp\left[1.044 \times 10^4 \left(\frac{1}{378.15} - \frac{1}{573.15}\right)\right] = 4 \times \exp(9.40) = 4 \times 1.20 \times 10^4 = 4.8 \times 10^4$$

So $t_{50,\text{use}} = 150 \times 4.8 \times 10^4 = 7.2 \times 10^6$ h. For a lognormal distribution, the 0.1 % point lies 3.09 standard deviations below the median in $\ln t$:

$$t_{0.1\%} = t_{50}\, e^{-3.09\sigma} = 7.2 \times 10^6 \times e^{-1.545} = 7.2 \times 10^6 \times 0.213 = 1.5 \times 10^6 \text{ h} \approx 175 \text{ years}$$

The requirement (87 600 h) is met with a large margin. Two sensitivities deserve attention. If the true activation energy were 0.75 eV, the temperature factor would drop from 1.2×10^4 to about 2500 and the 0.1 % life to about 37 years: still acceptable, but the margin depends heavily on an uncertain parameter. And a chip has millions of lines, not one; the allowed failure fraction per line is correspondingly far smaller (Section 44.5), pushing further into the tail where the lognormal assumption is least tested.

The Blech length: short lines are immune

Blech observed that short aluminium stripes did not electromigrate at all below a critical length. The reason is mechanical. As atoms pile up at the anode end of a line confined by stiff dielectric, they create compressive stress there, while the cathode end goes into tension. The stress gradient pushes atoms back, a **back-stress** flux opposing the wind. Steady state is reached when the two forces on an atom balance:

$$Z^* e \rho j = \Omega \frac{d\sigma}{dx} \approx \Omega \frac{\Delta\sigma}{L}$$

where Ω is the atomic volume and $\Delta\sigma$ the stress difference between the line ends. If $\Delta\sigma$ needed for balance is less than the stress that nucleates a void or cracks the dielectric, $\Delta\sigma_{\max}$, the line never fails. This gives a critical **current-density–length product**:

$$(jL)_c = \frac{\Omega\, \Delta\sigma_{\max}}{Z^* e \rho}$$

> **Status of the principle: Blech criterion**
> *Derived.* Follows from a force balance between electron wind and stress gradient in a confined line at steady state; the critical stress and effective Z^* are measured, so $(jL)_c$ is in practice an empirically determined material-and-process parameter.

Measured values for passivated aluminium and copper lines are typically a few thousand A/cm. For example, with $(jL)_c = 3000$ A/cm and $j = 2$ MA/cm^2, segments shorter than $L_c = 3000/(2 \times 10^6)$ cm $= 15$ μm are immune. Design rules exploit this: wiring is broken into short segments between vias, and electromigration checks credit "Blech-short" segments.

Other electromigration sites

- **Grain structure**: when the line width is smaller than the grain size, grains span the line like bamboo, eliminating continuous grain-boundary paths. Bamboo aluminium lines last far longer; in copper, the interface dominates instead.
- **Solder bumps and microbumps**: flip-chip bumps carry 10^4 A/cm^2 on average, but **current crowding** where the thin on-chip trace enters the bump raises the local density by an order of magnitude. Tin's anisotropic diffusion (much faster along the c-axis) makes bump life depend on tin grain orientation (EBSD, Chapter 52), and electromigration dissolves the under-bump metallization into the solder.
- **Joule heating**: current also heats the line; temperature enters exponentially, so thermal design and electromigration design cannot be separated.

Dielectric breakdown

Instantaneous versus time-dependent breakdown

Chapter 26 described dielectric strength: the field at which an insulator breaks down within a short ramp test, by electronic avalanche, thermal runaway, electromechanical collapse or discharge. Most real insulation operates far below that strength but still fails eventually. **Time-dependent dielectric breakdown (TDDB)** is the wear-out of an insulator under a constant field below its short-term strength.

For thin gate oxides the mechanism is understood in outline. Under field and current, defects (electron traps) are generated at random positions in the oxide. When enough defects line up to form a conducting chain between the electrodes, the oxide breaks down (the **percolation model**). The random positions explain why breakdown times are statistically distributed and why thinner oxides, needing fewer defects to bridge, have broader distributions. How the defect generation rate depends on field is contested, which produces competing lifetime models.

Field-acceleration models

E model (thermochemical). Defects are created by field-enhanced breaking of weak bonds, with an activation barrier lowered linearly by the field acting on the bond dipole:

$$t_{\mathrm{BD}} = A_E \exp(-\gamma E) \exp\left(\frac{E_a}{kT}\right)$$

1/E model (anode hole injection). Defects are created by charge injection: electrons tunnelling through the oxide (Fowler–Nordheim current, exponential in $-B/E$) create holes at the anode that return and damage the oxide:

$$t_{\mathrm{BD}} = A_G \exp\left(\frac{G}{E}\right)$$

Power-law (voltage) model. For ultrathin oxides (below about 5 nm) at low voltage, $t_{\mathrm{BD}} \propto V^{-N}$ with N of 40 or more fits data better than either.

> **Status of the principle: E and 1/E TDDB models**
> *Empirical.* Each has a physical motivation, but both fit accelerated data equally well over the narrow range of fields that can be tested; they diverge by orders of magnitude when extrapolated to use fields. The E model predicts shorter life at low field and is therefore the conservative choice for qualification.

Weibull statistics and area scaling

Breakdown times of nominally identical capacitors scatter. They are described by the **Weibull distribution**, the statistics of the weakest link:

$$F(t) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right]$$

where F is the cumulative fraction failed, η the characteristic life (63.2 % failed) and β the shape parameter (Weibull slope). Plotting $\ln[-\ln(1-F)]$ against $\ln t$ gives a straight line of slope β (Figure 44.8). Weibull statistics are derived in more detail in Section 44.5.

The weakest-link character gives a powerful scaling law. A capacitor of area A can be viewed as A/A_0 independent elements of area A_0, and it survives only if all survive: $1 - F_A = (1 - F_{A_0})^{A/A_0}$. Substituting the Weibull form,

$$\eta_{A_2} = \eta_{A_1}\left(\frac{A_1}{A_2}\right)^{1/\beta}$$

Larger areas fail sooner. A product chip has far more total gate area than a test capacitor, so test data must be scaled down. Because β falls with oxide thickness (percolation needs fewer defects in thin oxides), area scaling is most severe for the thinnest oxides.

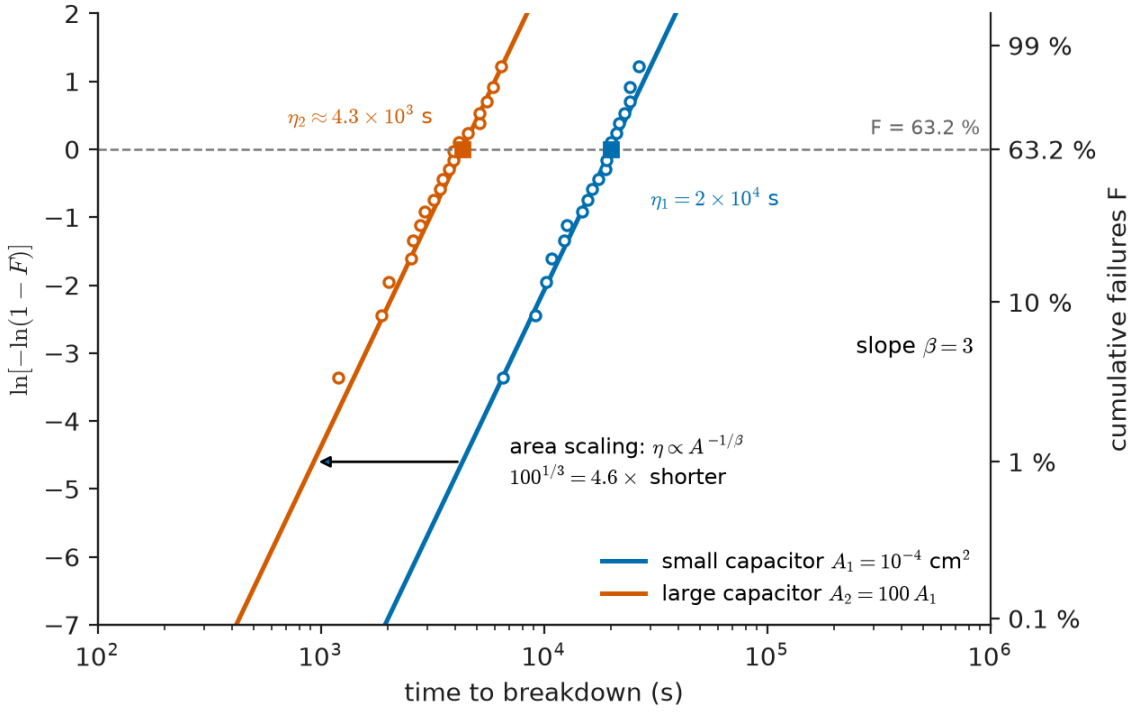


Figure 44.8 — Weibull plot of time-dependent dielectric breakdown for test capacitors of two areas, showing area scaling.

Worked example — gate-oxide lifetime for a power MOSFET. Test capacitors with the same thick SiO_2 as a power MOSFET's gate, area $A_1 = 10^{-3}$ cm^2, are stressed at 9 MV/cm and 150 °C, giving $\eta = 2.0 \times 10^4$ s and $\beta = 3$. The product has total gate-oxide area $A_2 = 0.2$ cm^2 and operates at 4 MV/cm at 150 °C. The requirement is a cumulative failure fraction below 100 ppm ($F = 10^{-4}$) after 20 years (6.3×10^8 s). Use the E model with an assumed field acceleration of 1.5 decades per MV/cm.

Field extrapolation: 5 MV/cm × 1.5 decades = $10^{7.5} = 3.16 \times 10^7$, so $\eta = 2.0 \times 10^4 \times 3.16 \times 10^7 = 6.3 \times 10^{11}$ s for the test area.

Area scaling: $(A_1/A_2)^{1/\beta} = (1/200)^{1/3} = 0.171$, so $\eta_{\text{chip}} = 1.08 \times 10^{11}$ s.

Low-percentile life: inverting the Weibull distribution, $t_F = \eta[-\ln(1-F)]^{1/\beta} \approx \eta F^{1/\beta}$ for small F. With $F = 10^{-4}$, $F^{1/3} = 0.0464$:

$$t_{100\text{ ppm}} = 1.08 \times 10^{11} \times 0.0464 = 5.0 \times 10^9 \text{ s} \approx 160 \text{ years}$$

The design passes. But had the field acceleration been 1.0 rather than 1.5 decades per MV/cm, the extrapolation factor would be 10^5 instead of $10^{7.5}$, and the 100 ppm life would be 0.5 years. The lifetime prediction is dominated by a parameter that is extracted from a few points at high field and extrapolated over five orders of magnitude in time. This is why gate-oxide qualification uses several stress fields, prefers the conservative E model, and is followed by burn-in screening of extrinsic (defect-related) failures, which form a separate, shallower Weibull population that intrinsic models do not describe.

Breakdown and aging in polymer and ceramic insulation

Thick insulation degrades by different mechanisms, introduced in Chapter 26:

- **Partial discharge (PD)**: discharges in gas-filled voids or at electrode edges erode polymer surfaces each cycle. PD inception voltage is set by Paschen's law in the void; eroded channels grow into **electrical trees**, branched hollow channels that bridge the insulation and cause breakdown. In inverter-fed motors, fast voltage edges and overshoot raise PD activity in winding insulation, which is why inverter-duty wire uses PD-resistant enamels filled with nanoparticles.
- **Water treeing**: in polymeric cable insulation (XLPE) with moisture and AC field, diffuse, bush-shaped water-filled microvoid structures grow over years at fields far below PD inception. Water trees reduce breakdown strength and can convert into electrical trees.
- **Ceramic capacitor degradation**: in multilayer ceramic capacitors (MLCCs) with $BaTiO_3$ dielectrics, oxygen vacancies migrate under DC bias at elevated temperature, piling up at the cathode and degrading insulation resistance. Rare-earth

and acceptor dopants slow the migration. Highly accelerated life tests at 125–150 °C and multiples of rated voltage are standard for MLCCs.
- **Film capacitors** degrade by self-healing: a local breakdown vaporizes the thin metallization around the fault, isolating it. The capacitor survives but loses a little capacitance each time; capacitance loss is the end-of-life criterion.

Semiconductor defect generation

Semiconductor devices degrade because their operation itself creates defects in the semiconductor, in the gate dielectric, or at the interface between them (Chapter 34). Device operation and circuit consequences belong to the Electronics & Computing core; here the focus is the material damage.

Hot-carrier degradation

In a MOSFET, carriers accelerated by the high lateral field near the drain gain kinetic energies well above the thermal value (Chapter 32, high-field transport). These **hot carriers** can (a) break Si–H bonds that passivate dangling bonds at the Si/SiO_2 interface, creating interface traps, and (b) be injected over the barrier into the oxide and trapped. Both shift the threshold voltage, reduce mobility near the drain and degrade transconductance. The damage is localized near the drain, which is why lightly doped drain extensions (spreading the field) were introduced. Lifetime is commonly correlated with the substrate current (a measure of impact ionization by hot carriers) in a power-law form, $\tau \propto (I_{\text{sub}}/I_d)^{-m}$, and is worst at a gate bias where the lateral field and carrier flux are both large.

Replacing hydrogen with deuterium at the interface improves hot-carrier resistance markedly, because the heavier isotope makes the Si–D bond harder to excite vibrationally to dissociation: a clean demonstration that the mechanism is bond breaking.

Bias–temperature instability (NBTI and PBTI)

Negative-bias temperature instability (NBTI) affects p-channel MOSFETs held with negative gate voltage at elevated temperature, even with no drain current. The threshold voltage magnitude increases with time, following roughly $\Delta V_{\text{th}} \propto t^{\,n}$ with $n \approx 0.15$–0.25, and partially recovers when the stress is removed, which complicates measurement (a slow measurement records less damage than was present). The reaction–diffusion picture attributes it to electrochemical breaking of interface Si–H bonds by holes, with the released hydrogen diffusing away; trapping in pre-existing oxide defects also contributes. NBTI became a leading reliability limit when nitrogen was added to gate oxides and when operating fields rose. With high-k metal-gate stacks, **PBTI** (positive bias, n-channel) appears as well, through electron trapping in the high-k layer.

In SiC MOSFETs the Si/SiO_2-like interface is replaced by a SiC/SiO_2 interface with far higher trap densities (carbon-related defects), and threshold drift under gate bias and switching is a major qualification topic. Gate-bias stress tests at high temperature with periodic threshold measurement, and switching-mode gate stress, are standard.

Radiation damage

Electronics in space, aircraft, accelerators, medical equipment and nuclear facilities, and even at sea level, are exposed to ionizing radiation and energetic particles. Three classes of effect follow from different material damage:

- **Total ionizing dose (TID)**: ionization in oxides creates electron–hole pairs; electrons escape quickly but holes are trapped in the oxide (positive charge) and released protons create interface traps. Threshold voltages shift and leakage paths open, especially along thick field and isolation oxides. Dose is quoted in rad(Si) or gray; commercial CMOS typically tolerates 10–100 krad(Si), while radiation-hardened parts reach several hundred krad to Mrad.
- **Displacement damage**: energetic protons, neutrons and electrons knock atoms off lattice sites, creating vacancy–interstitial pairs and clusters that act as recombination centres. Minority-carrier lifetime falls (Chapter 32), degrading bipolar transistor gain, solar-cell output (space solar arrays are rated in equivalent 1 MeV electron fluence), photodetectors and LEDs. Damage scales with the non-ionizing energy loss (NIEL) of the particle.
- **Single-event effects (SEE)**: a single heavy ion, or the recoil from a neutron reaction, deposits a dense column of charge. In memory it flips a bit (single-event upset); in CMOS it can trigger parasitic thyristor latch-up; in power devices it can cause **single-event burnout**, in which the charge triggers avalanche multiplication under high blocking voltage. At sea level the flux of cosmic-ray neutrons above 10 MeV is about 13 n/(cm^2 h); because burnout rates rise steeply with blocking voltage, high-voltage silicon IGBTs and SiC MOSFETs are derated (typically operated at well below their rated voltage) in traction and grid converters.

Bipolar degradation in SiC

A materials-specific degradation mode illustrates how crystal defects (Chapter 5) become device failures. 4H-SiC epitaxial layers can contain basal-plane dislocations. When electron–hole pairs recombine near such a dislocation (for example during conduction of a MOSFET's body diode), the recombination energy drives the partial dislocations to glide, expanding a **Shockley stacking fault** across the drift layer. The fault acts as a quantum-well-like barrier and recombination plane, raising the forward voltage and on-resistance. The cure was in materials growth: buffer layers that convert basal-plane dislocations into harmless threading edge dislocations before they reach the drift layer, and screening by forward-current stress.

Trapping in GaN devices

GaN HEMTs show **current collapse** or dynamic on-resistance: after high-voltage off-state stress, electrons trapped at surface states or in the carbon- or iron-doped buffer deplete the channel, so the on-resistance immediately after switching is higher than its DC value. Surface passivation (SiN), field plates and buffer engineering control it. Under long-term high-field stress, GaN can also suffer inverse-piezoelectric cracking at the gate edge, where the strong electric field adds strain to the already strained AlGaN barrier.

Moisture ingress

Diffusion into polymers

Every polymer absorbs water. A mould compound reaches equilibrium moisture contents of roughly 0.1–0.5 wt % in humid air; a PCB laminate somewhat more; polyimide films up to about 3 %. Transport is diffusion (Fick's laws), so the time to approach saturation through a thickness L (for one-sided exposure; half-thickness for two-sided) scales as

$$t_{\text{sat}} \sim \frac{L^2}{D}$$

With $D \approx 2 \times 10^{-12}$ m^2/s for a typical mould compound at 85 °C and a half-thickness of 0.5 mm, $t \sim (5 \times 10^{-4})^2/(2 \times 10^{-12}) = 1.25 \times 10^5$ s, about 35 h. A package therefore saturates within days in a humid environment, and no plastic package is hermetic. Hermetic packages (metal or ceramic, sealed with glass or welds) exclude moisture for decades and are used where that matters: military, space, implantable medical devices.

Popcorn cracking and moisture sensitivity levels

During reflow soldering, a package is heated to 245–260 °C in a minute or two. Absorbed water vaporizes into any gap or delaminated interface. The saturated vapour pressure of water at 260 °C is about 4.7 MPa, enough to bulge and crack the mould compound outward, often with an audible "pop" (Figure 44.9). **Popcorn cracking** breaks wire bonds and opens paths for later corrosion.

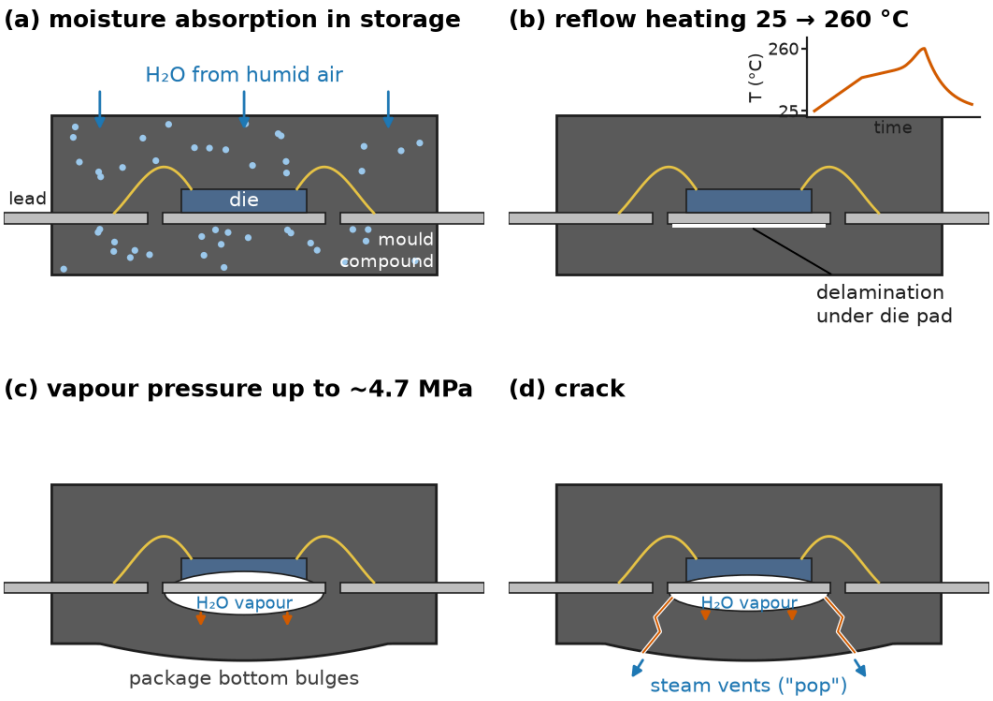


Figure 44.9 — Sequence of popcorn cracking during reflow of a moisture-saturated plastic package.

Packages are classified by their **moisture sensitivity level (MSL)** according to joint IPC/JEDEC standards, which set the allowed floor life after removal from dry packing before they must be baked:

MSL	Floor life at ≤ 30 °C / 60 % RH (MSL 1: ≤ 85 % RH)
1	Unlimited
2	1 year
2a	4 weeks
3	168 h
4	72 h
5	48 h
5a	24 h
6	Mandatory bake before use; time on label

The classification is a materials result: it depends on moisture uptake and diffusivity of the mould compound, its adhesion to the die and leadframe, and its high-temperature strength. Larger, thinner packages and those with large die pads are more sensitive.

Corrosion under moisture and bias

Once moisture reaches a chip surface, metallization can corrode. Aluminium bond pads are vulnerable to chloride and other ions from mould compound, flux or the environment, especially under bias. **Conductive anodic filament (CAF)** formation inside PCBs is the board-level equivalent: moisture travels along the glass-fibre/resin interface, and under bias between adjacent plated holes, copper ions migrate along the fibres from anode to cathode, forming a conducting filament inside the laminate where no inspection can see it. Resistance to CAF depends on laminate chemistry, glass-fibre sizing and drilling damage, and is improved by larger hole-to-hole spacing.

The Peck model

Humidity tests accelerate moisture-driven failure with both temperature and humidity. The most widely used acceleration model is Peck's:

$$t_f = A\,(\mathrm{RH})^{-n}\exp\!\left(\frac{E_a}{kT}\right)$$

with $n \approx 2.7$ (often 2.66) and $E_a \approx 0.79$ eV in Peck's original fit to aluminium corrosion data for plastic packages. The acceleration factor between test and use is

$$AF = \left(\frac{\mathrm{RH}_{\text{test}}}{\mathrm{RH}_{\text{use}}}\right)^{n}\exp\!\left[\frac{E_a}{k}\left(\frac{1}{T_{\text{use}}}-\frac{1}{T_{\text{test}}}\right)\right]$$

> **Status of the principle: Peck humidity model**
> *Empirical.* Fitted to temperature–humidity–bias failure data of plastic-encapsulated aluminium metallization; the humidity exponent and activation energy vary with mechanism and package, and the model presumes the same mechanism at test and use (it fails if test temperatures exceed the mould compound T_g or if condensation occurs).

Worked example — field equivalent of an 85/85 test. A product passes 1000 h of biased 85 °C/85 % RH testing without failure. What field exposure does this represent at 30 °C/60 % RH?

$$\left(\frac{85}{60}\right)^{2.66} = 1.417^{2.66} = 2.53$$

$$\exp\!\left[\frac{0.79}{8.617\times10^{-5}}\left(\frac{1}{303.15}-\frac{1}{358.15}\right)\right] = \exp(9168\times5.066\times10^{-4}) = \exp(4.64) = 104$$

$$AF = 2.53\times104 = 263$$

So 1000 h of test corresponds to about 2.6×10^{5} h, or 30 years, of field exposure at those conditions, provided the mechanism is the same. Notice that temperature contributes a factor of 100 and humidity only 2.5; in this model, heat is the main accelerator. The accelerated aging tests at the end of this section use the same model to relate highly accelerated stress tests at 130 °C to 85/85 testing.

Contact degradation

Why contacts are delicate

A separable contact (connector, relay, switch, battery terminal, test socket) touches only at a few microscopic asperities. Current must squeeze through these **a-spots**, adding a **constriction resistance**. For a single circular a-spot of radius a in a metal of resistivity ρ (Holm's result, Chapter 25),

$$R_c = \frac{\rho}{2a}$$

If the asperities deform plastically, the real contact area is approximately the normal force divided by the hardness, $\pi a^2 \approx F/H$. For a gold contact with $F = 1$ N and $H \approx 1$ GPa, $a = \sqrt{F/\pi H} = \sqrt{1/(\pi\times10^{9})} = 17.8$ µm and $R_c = 2.2\times10^{-8}/(2\times1.78\times10^{-5}) = 0.62$ mΩ. Real contacts have many a-spots in parallel (which lowers the resistance) and surface films in series (which can raise it by orders of magnitude). Degradation is anything that reduces the conducting area or thickens the films.

> **Status of the principle: Holm constriction resistance**
> *Approximate.* Exact for a single circular spot between two semi-infinite bodies of the same metal with no surface film; multiple-spot, film and thin-plating effects require extensions.

Contact finishes

Finish	Typical use	Strength	Weakness
Hard gold (Au–Co, 0.4–1.3 µm over Ni)	Signal connectors, high mating cycles	Noble, no insulating film	Cost; pores expose Ni/Cu; needs lubricant against wear
Silver	Power contacts, relays	Highest conductivity, soft	Sulfide films (tarnish), ECM
Tin (over Ni or Cu)	Power and automotive connectors, low cost	Cheap, solderable	Oxide film, fretting, whiskers; needs high contact force
Palladium–nickel	Gold substitute	Hard, durable	Catalyses polymer films (frictional polymerization)
Ag–SnO_2 composite	Relay and contactor switching contacts	Arc-erosion and welding resistance	Higher resistance than pure Ag

The last row is a materials-and-regulation story: Ag–CdO was the standard switching contact material for decades, prized for arc-quenching cadmium oxide, but cadmium restrictions (Section 58.1) drove a change to Ag–SnO_2.

Fretting corrosion

Fretting is small-amplitude relative motion (1–100 µm) between contact surfaces, caused by vibration or by thermal expansion differences during temperature changes. With noble metals it causes only wear. With tin, each motion cycle breaks the oxide film and exposes fresh metal, which oxidizes; the oxide debris accumulates in the contact zone, and after thousands of cycles the a-spots are buried in insulating debris and the resistance rises from milliohms to ohms, often intermittently. Fretting is the dominant failure mode of tin-plated automotive connectors.

Defences follow from the mechanism: higher contact force (to prevent motion and to break films; tin contacts typically need substantially more normal force than gold), noble finishes, contact lubricants that exclude oxygen and trap debris, and mechanical design to eliminate relative motion.

Other contact degradation modes

- **Stress relaxation of the spring**: the contact force comes from a deflected spring; at elevated temperature the spring material relaxes (Section 44.8) and force falls. Copper–beryllium and Cu–Ni–Si alloys retain force at 125–150 °C where brass and phosphor bronze relax.
- **Intermetallic growth**: tin plating on copper is gradually consumed by Cu_6Sn_5, which is hard and oxidizes, and grows faster at high temperature; nickel underplate slows it.
- **Tin whiskers**: compressive stress in electroplated tin (from intermetallic growth at the copper interface, from mechanical load, or from thermal mismatch) is relieved by the extrusion of single-crystal tin filaments, a few micrometres in diameter and up to millimetres long. They can bridge closely spaced leads. Lead additions suppressed whiskers in SnPb; lead-free practice uses matte tin with a nickel underlayer, annealing and conformal coating. Whiskers are a reminder that a compressive film stress, measured by XRD (Chapter 52), can be a reliability parameter.
- **Arc erosion and material transfer** in switching contacts, governed by arc energy, contact material vapour pressure and the oxide additives above.

Accelerated aging tests

Why acceleration, and its single rule

A component specified for 15 years cannot be tested for 15 years. Accelerated tests apply higher temperature, humidity, voltage, current or temperature swing to produce, in weeks, the same failures that would occur in years. The acceleration factors come from the models of Chapter 44. There is one overriding rule: **the stress must accelerate the same mechanism that occurs in the field, without introducing a new one.** A humidity test above the T_g of the mould compound, a temperature cycle that melts a solder phase, a voltage that causes avalanche instead of oxide wear: each produces failures with no field meaning, or masks the real ones.

Standard tests

Test	Typical conditions	Stresses	Mechanisms accelerated	Acceleration model
HTOL (high-temperature operating life)	$T_j \geq 125$ °C, max operating voltage, 1000 h	Temperature, bias	Oxide wear-out, BTI, hot carriers, EM, ionic drift	Arrhenius, voltage models
HTSL (high-temperature storage)	150 °C, 1000 h, unbiased	Temperature	Intermetallic growth, Kirkendall voids, bond degradation	Arrhenius
THB (temperature–humidity–bias)	85 °C / 85 % RH, bias, 1000 h	Moisture, bias, temperature	Corrosion, ECM, leakage, delamination	Peck
HAST (highly accelerated stress test)	130 °C / 85 % RH, bias, 96 h (or 110 °C, 264 h)	As THB, pressurized	As THB, faster	Peck
Autoclave / unbiased HAST	121 °C / 100 % RH or 130 °C / 85 % RH, unbiased	Moisture	Package integrity, corrosion without bias	Peck
Temperature cycling	e.g. −55/+125 °C, −40/+125 °C, 0/+100 °C; 15–30 min dwells	Thermal strain	Solder fatigue, die and substrate cracking, bond-wire fatigue, delamination	Coffin–Manson, Norris–Landzberg
Power cycling	Self-heating on/off, ΔT_j 60–120 K	Local thermal strain	Wire-bond lift-off, die-attach fatigue in power modules	Coffin–Manson with T_j terms
Preconditioning	Moisture soak per MSL, then 3× reflow	Moisture + heat	Popcorn, delamination before other tests	—

These are defined in JEDEC and IEC standards and bundled into qualification specifications such as the AEC-Q series for automotive components, which typically require three production lots of 77 units each to pass each test with zero failures.

Interpreting a zero-failure test

A qualification test that ends with no failures still gives quantitative information. For a constant failure rate and total equivalent device-hours D, the upper confidence limit on λ with zero failures is

$$\lambda_{\text{upper}} = \frac{\chi^2_{\text{CL},\,2}}{2D}$$

where $\chi^2_{\text{CL},2}$ is the chi-squared value for confidence level CL with 2 degrees of freedom ($= -2\ln(1-\text{CL})$: 1.83 at 60 %, 4.61 at 90 %). With r failures, the degrees of freedom become $2r+2$.

Worked example — FIT rate from HTOL. A total of 1000 devices pass 1000 h of HTOL at $T_j = 125$ °C with zero failures. Estimate the failure rate at a use junction temperature of 55 °C with 60 % confidence, assuming an activation energy of 0.7 eV for the dominant mechanisms.

$$AF = \exp\left[\frac{0.7}{8.617\times10^{-5}}\left(\frac{1}{328.15}-\frac{1}{398.15}\right)\right] = \exp(8123\times 5.358\times10^{-4}) = \exp(4.352) = 77.6$$

Equivalent device-hours at use conditions: $D = 1000\times1000\times77.6 = 7.76\times10^{7}$ h.

$$\lambda_{60\%} = \frac{1.833}{2\times7.76\times10^{7}} = 1.18\times10^{-8}\ \text{h}^{-1} = 11.8\ \text{FIT}$$

At 90 % confidence, $\chi^2 = 4.61$ and $\lambda_{90\%} = 29.7$ FIT. The result depends linearly on the number of device-hours and exponentially on the assumed activation energy: with 0.5 eV instead of 0.7 eV, AF falls to about 22 and the 60 % FIT bound rises to about 41. Assumed activation energies should therefore be justified by mechanism, not chosen for convenience.

HAST versus THB

The Peck model of Chapter 44 shows why a 96 h HAST is accepted as a substitute for 1000 h of THB. At the same 85 % RH, only temperature differs:

$$AF_{\mathrm{HAST/THB}} = \exp\left[\frac{0.79}{8.617\times10^{-5}}\left(\frac{1}{358.15}-\frac{1}{403.15}\right)\right] = \exp(2.857) = 17.4$$

so 1000 h at 85 °C corresponds to about $1000/17.4 = 57$ h at 130 °C; 96 h of HAST gives margin. HAST's validity depends on the mould compound and other polymers behaving the same way at 130 °C as at 85 °C. For packages with low-T_g materials, HAST can produce unrepresentative failures, and THB remains the reference.

Pitfalls

- **Mechanism change**: exceeding T_g, melting points, phase transitions; triggering a different breakdown mode at high voltage.
- **Masking**: a fast new mechanism (e.g. corrosion of a test-board connector) that fails units before the mechanism of interest can show.
- **Wrong use profile**: the acceleration factor is only as good as the field conditions assumed (Chapter 44 example). Mission profiles (time at temperature, number and depth of cycles, humidity exposure) should come from measurement, not assumption.
- **Sample size**: zero failures in 231 units demonstrates, at 90 % confidence, only that the lot failure fraction is below about 1 % for that stress. Rare defects need screening, not qualification.
- **Test to pass versus test to fail**: passing a fixed-duration qualification says the product is at least that good; testing to failure (to obtain Weibull parameters, Section 44.5) says how good it is and how much margin remains.

What remains open

Open question	Where it is answered
Electrochemical thermodynamics and kinetics behind corrosion damage	Chapter 42
Forms of corrosion, hydrogen embrittlement and corrosion fatigue in detail	Chapter 43
Fracture mechanics, fatigue laws, creep and Weibull strength of ceramics	Chapter 14
Composition, curing and durability design of concrete	Chapter 18
Molecular structure behind polymer oxidation and physical aging	Chapters 19 and 20
Moisture and fatigue damage in laminates; composite design allowables	Chapters 22 and 23
Wear, friction and surface fatigue as life-limiting mechanisms	Chapter 45
Radiation, high-temperature and cryogenic degradation	Chapter 41
Weld defects and residual stresses as origins of failure	Chapter 47
Characterization tools used in failure analysis (SEM, EDS, XRD, FTIR, DSC)	Chapter 52
Choosing materials so that the dominant degradation mechanism is slow enough; lifecycle impact of replacement	Chapters 58 and 59
Electromigration, dielectric breakdown and solder-joint life in electronics	Chapters 26, 34, 50 and 51

Next: Chapter 45 turns to the surface itself, where adhesion, friction and wear are decided, and to the engineering of surfaces that resist them.

Chapter 45 — Surfaces, Interfaces and Tribology

Part IX · Corrosion, Degradation and Surfaces

A conventional polyethylene hip cup that wears by a tenth of a millimetre a year sounds like a success: the cup is several millimetres thick, so it will not wear through for decades. Yet in the 1980s and 1990s many such hips had to be revised after ten or fifteen years, not because the cup wore out but because the bone around it dissolved. Each year the joint shed tens of cubic millimetres of submicrometre polyethylene particles, hundreds of billions of them, which macrophages tried to digest and could not, releasing signals that switched on bone resorption. The fix was a materials one: polyethylene cross-linked by irradiation, then thermally treated to quench free radicals, which wears several times more slowly. The problem began at the surface, at the contact between a hard ball and a soft cup, where friction, wear and surface chemistry decide the fate of the whole component. The same is true of a car engine losing a few percent of its fuel to friction, a wind-turbine gearbox failing from micropitting, a paint that will not stick to a polypropylene bumper, and a turbine blade protected by a ceramic coating a quarter of a millimetre thick. This chapter treats the surface as a material in its own right: its energy and chemistry, how it wets and adheres, how two surfaces touch, slide and wear, how lubricants separate them, and how surfaces are engineered.

> **Why do surfaces have energy, and how does that energy control adsorption, wetting and adhesion? Why is friction proportional to load and independent of apparent area, and what sets the coefficient? How do surfaces wear, how do lubricants and coatings prevent it, and how is wear rate predicted?**

Concept	Engineering destination
Surface energy, broken-bond model, Wulff shape	Sintering, catalyst particle shape, crack-growth energy, grain-boundary energies
Adsorption isotherms, BET surface area	Catalysts, sorbents, battery electrodes, powder characterization, corrosion inhibitors
Contact angle, Young's equation, work of adhesion	Paint and adhesive bonding, printing on polymers, self-cleaning and anti-icing surfaces, implants
Hertz contact	Rolling bearings, gears, wheel–rail contact, cam followers, indentation testing
Real contact area $A_r = F/H$	Friction, electrical and thermal contact resistance, sealing
Amontons–Coulomb friction; Bowden–Tabor adhesion model	Brakes, clutches, bolted joints, tyres, machining, design of low-friction pairs
Archard wear law and wear coefficients	Life of bushings, seals, implants, tools, dies, mining equipment
Stribeck curve, film parameter λ	Journal and rolling bearings, engine piston rings, lubricant selection
Case hardening, nitriding, PVD and DLC coatings, shot peening, texturing	Gears, cutting tools, moulds, crankshafts, aerospace landing gear, medical devices

Roadmap. What a surface is and why it has energy (45.1) → what sticks to it: adsorption and surface chemistry (45.2) → liquids and solids on it: wetting and adhesion (45.3) → two solids in contact: Hertz and real contact area (45.4) → sliding: friction (45.5) → damage: wear mechanisms (45.6) → separation by a film: lubrication regimes and the Stribeck curve (45.7) → engineering the surface: hardening, coatings, texturing (45.8).

45.1 Surface structure and surface energy

Why surfaces have energy

An atom in the bulk of a crystal is bonded to all its neighbours; an atom at the surface has lost some of them. Creating a surface therefore costs energy, the **surface energy** γ, defined as the reversible work per unit area to create new surface (J/m^2). For

liquids it equals the surface tension (N/m). For solids, surface energy and surface stress differ, because stretching a solid surface changes the atomic spacing instead of bringing new atoms to the surface, but for most materials science purposes γ is the relevant quantity.

The **broken-bond model** gives a first estimate. If the cohesive energy per atom is the sublimation enthalpy $\Delta H_s/N_A$, shared among z nearest-neighbour bonds, then each broken bond costs $\Delta H_s/(zN_A)$, and a surface with n_s atoms per unit area, each missing z_s bonds, has

$$\gamma \approx \frac{z_s}{z}\,\frac{\Delta H_s}{N_A}\,n_s$$

> **Status of the principle: Broken-bond estimate of surface energy**
> *Approximate*. Assumes pairwise nearest-neighbour bonds of equal energy and no relaxation of the surface atoms; it captures the trends with bonding strength and crystal orientation but typically overestimates measured values by 20–50 %, because real bond energy is not linear in coordination and surface atoms relax.

Worked example — surface energy of copper. Copper is FCC with $z = 12$, lattice parameter $a = 0.3615$ nm and $\Delta H_s = 337$ kJ/mol. On the close-packed (111) plane each surface atom has lost 3 of its 12 neighbours, and the atomic density is $n_s = 4/(\sqrt{3}\,a^2) = 1.77 \times 10^{19}$ m^{-2}. Then

$$\gamma_{111} \approx \frac{3}{12} \times \frac{337 \times 10^3}{6.022 \times 10^{23}} \times 1.77 \times 10^{19} = 0.25 \times 5.60 \times 10^{-19} \times 1.77 \times 10^{19} = 2.47\ \mathrm{J\,m^{-2}}$$

On the (100) plane each atom loses 4 bonds and $n_s = 2/a^2 = 1.53 \times 10^{19}$ m^{-2}, giving 2.85 J/m^2. The measured average surface energy of solid copper is about 1.8 J/m^2, so the model overestimates by about a third, as expected, but correctly predicts that close-packed planes have the lowest energy, which is why crystals are bounded by them.

Material	Surface energy (typical)
Tungsten	about 3.0–3.3 J/m^2
Iron	about 2.4 J/m^2
Copper	about 1.8 J/m^2
Aluminium	about 1.1 J/m^2
Alumina	about 1.0–1.5 J/m^2 (orientation-dependent)
Silica glass	about 0.3 J/m^2 (hydroxylated surface lower)
Water (20 °C)	72.8 mJ/m^2
Epoxy, polyamide	40–45 mJ/m^2
Polyethylene, polypropylene	29–33 mJ/m^2
PTFE	18–20 mJ/m^2

Grain-boundary energies are typically a third of the surface energy (Chapter 5), which is why grain boundaries groove at free surfaces with a dihedral angle set by the balance $\gamma_{gb} = 2\gamma_s \cos(\psi/2)$.

Surface structure

A real crystal surface is not a perfect plane. At atomic scale it consists of flat **terraces** separated by monatomic **steps**, which contain **kinks**; on the terraces sit **adatoms** and **vacancies** (the terrace–step–kink model, Figure 45.1). Kinks are the sites where atoms attach during crystal growth and detach during evaporation and dissolution; steps are preferred sites for adsorption and catalysis. Surface atoms also **relax** (the top interlayer spacing of metals typically contracts by a few percent) and some surfaces **reconstruct** into periodicities different from the bulk: the Si(111) 7×7 and Au(100) hexagonal reconstructions are classic examples revealed by scanning tunnelling microscopy and low-energy electron diffraction (Chapter 53).

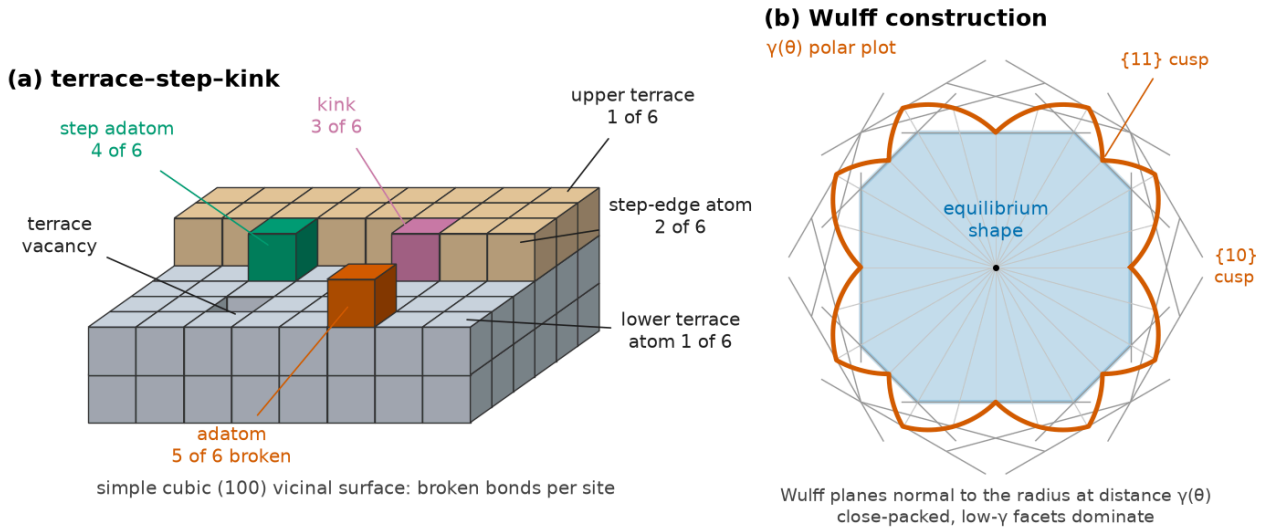


Figure 45.1 — Atomic structure of a crystal surface (terrace–step–kink model) and the Wulff construction for an equilibrium crystal shape.

Anisotropy and the equilibrium shape

Because surface energy depends on orientation, a small crystal minimizes its total surface energy $\sum \gamma_i A_i$ at fixed volume by exposing mainly low-energy facets. The **Wulff construction** gives the equilibrium shape: draw from a centre, in every direction, a plane perpendicular to that direction at a distance proportional to γ; the inner envelope of all such planes is the shape. Metal nanoparticles in catalysts adopt truncated octahedra of {111} and {100} facets, and the facet fraction controls their activity (Chapter 38). Surface energy also drives sintering, Ostwald ripening and grain growth (Chapters 9 and 46), and it appears in Griffith's fracture criterion as the minimum energy to make crack faces (Chapter 14).

45.2 Adsorption and surface chemistry

Physisorption and chemisorption

Atoms and molecules from a gas or liquid stick to surfaces because doing so lowers the surface energy. Two kinds of bonding are distinguished:

	Physisorption	Chemisorption
Bonding	van der Waals	Chemical (covalent, ionic, metallic)
Enthalpy of adsorption	5–40 kJ/mol	40–800 kJ/mol
Specificity	Any gas on any surface	Specific to adsorbate–surface pair
Layers	Multilayers possible	Monolayer at most
Reversibility	Readily reversible	Often irreversible, may dissociate (H$_2\to2H, O_2\to$2O)
Example	N_2 on any solid at 77 K	O on Fe, CO on Pt, H on Ni, thiols on Au

Chemisorption is the first step of oxidation (Chapter 42), of heterogeneous catalysis, and of hydrogen entry into metals (Chapter 43); physisorption is how surface area and porosity are measured.

The Langmuir isotherm

Langmuir's model assumes a fixed number of equivalent sites, one molecule per site, no interaction between adsorbed molecules, and dynamic equilibrium between adsorption (proportional to pressure and to the fraction of empty sites) and desorption (proportional to the fraction occupied). The fractional coverage is

$$\theta = \frac{KP}{1 + KP}$$

where K is an equilibrium constant that falls with temperature as $\exp(-\Delta H_{\text{ads}}/RT)$, ΔH_{ads} being negative. At low pressure $\theta \approx KP$ (Henry's law); at high pressure $\theta \to 1$ (saturation). Langmuir kinetics underlies the rate laws of catalytic reactions and the adsorption of corrosion inhibitors (Chapter 43), whose efficiency is often taken as equal to the coverage.

> **Status of the principle: Langmuir isotherm**
> *Approximate*. Derived exactly for its assumptions (identical independent sites, monolayer, no lateral interactions); real surfaces have a distribution of site energies and interacting adsorbates, which the Temkin and Freundlich isotherms describe empirically.

BET surface area

Brunauer, Emmett and Teller extended Langmuir's model to multilayer physisorption, assuming that layers beyond the first adsorb with the enthalpy of condensation. Fitting nitrogen adsorption at 77 K over relative pressures of about 0.05–0.3 gives the **monolayer capacity** v_m, and the specific surface area follows from the area occupied per molecule (0.162 nm^2 for N_2):

$$S = \frac{v_m}{V_{\text{mol}}} N_A \, \sigma_{\text{N}_2}$$

For a catalyst support with $v_m = 50$ cm^3(STP)/g, $S = (50/22\,414) \times 6.022 \times 10^{23} \times 0.162 \times 10^{-18} = 218$ m^2/g. BET areas range from below 1 m^2/g for dense powders, through 100–300 m^2/g for γ-alumina catalyst supports, to 1000–3000 m^2/g for activated carbons and metal–organic frameworks. Battery electrode powders, cement, pigments and pharmaceutical excipients are specified by BET area.

Segregation and surface chemistry of real materials

At equilibrium, components that lower the surface or grain-boundary energy concentrate there; the **Gibbs adsorption equation**, $\Gamma = -\frac{1}{RT}\frac{d\gamma}{d\ln c}$ for a dilute solute, links the surface excess Γ to the drop in surface tension. Surfactants in water, sulfur on iron surfaces, and phosphorus or tin at grain boundaries of steel (temper embrittlement, Chapter 44) are examples. Engineering surfaces are therefore almost never clean:

- **Metals** carry native oxides (2–5 nm on aluminium and stainless steel), hydroxyl groups, adsorbed water and a hydrocarbon contamination layer of about a nanometre within minutes of exposure to air.
- **Oxides** in water acquire charge by protonation or deprotonation of surface hydroxyls; the pH at which the net charge is zero, the **isoelectric point**, is about 2 for silica, 6–7 for titania and 8–9 for alumina. It governs colloidal processing of ceramic slurries (Chapter 17), protein adsorption on implants and the adsorption of flotation reagents in mineral processing.
- **Polymers** have low-energy, chemically inert surfaces that must be oxidized for bonding (Section 45.3).
- **Self-assembled monolayers** (alkanethiols on gold, silanes on oxides) give molecularly defined surfaces for sensors, anti-stiction coatings in MEMS and adhesion promoters.

Surface composition and chemistry are measured by XPS, AES and SIMS (Chapter 52).

45.3 Adhesion and wetting

Young's equation

A liquid drop on a flat solid surface meets it at a **contact angle** θ set by the balance of the three interfacial tensions at the contact line (Figure 45.2):

$$\gamma_{SV} = \gamma_{SL} + \gamma_{LV}\cos\theta$$

A liquid **wets** the solid ($\theta < 90^\circ$) when the solid–vapour energy exceeds the solid–liquid energy; it spreads completely ($\theta = 0$) when $\gamma_{SV} \geq \gamma_{SL} + \gamma_{LV}$. Water ($\gamma_{LV} = 72.8$ mJ/m^2) spreads on clean glass and metal oxides but beads on PTFE and wax.

> **Status of the principle: Young's equation**
> *Derived.* Follows from minimizing the total interfacial free energy of a drop on an ideal, smooth, chemically homogeneous, rigid and insoluble surface; real surfaces show a range of contact angles between advancing and receding values (contact-angle hysteresis) because of roughness and chemical heterogeneity.

Work of adhesion

The **work of adhesion** is the reversible work to separate unit area of an interface into two free surfaces, $W_a = \gamma_{SV} + \gamma_{LV} - \gamma_{SL}$. Combined with Young's equation it gives the Young–Dupré equation, which turns a contact-angle measurement into an adhesion energy:

$$W_a = \gamma_{LV}\,(1 + \cos\theta)$$

For water on PTFE ($\theta \approx 110^\circ$), $W_a = 72.8 \times (1 - 0.342) = 47.9$ mJ/m^2; on clean glass ($\theta \approx 20^\circ$), $W_a = 72.8 \times 1.940 = 141$ mJ/m^2. The work of cohesion of water is $2\gamma_{LV} = 146$ mJ/m^2: water adheres to glass almost as strongly as to itself, and to PTFE only a third as strongly.

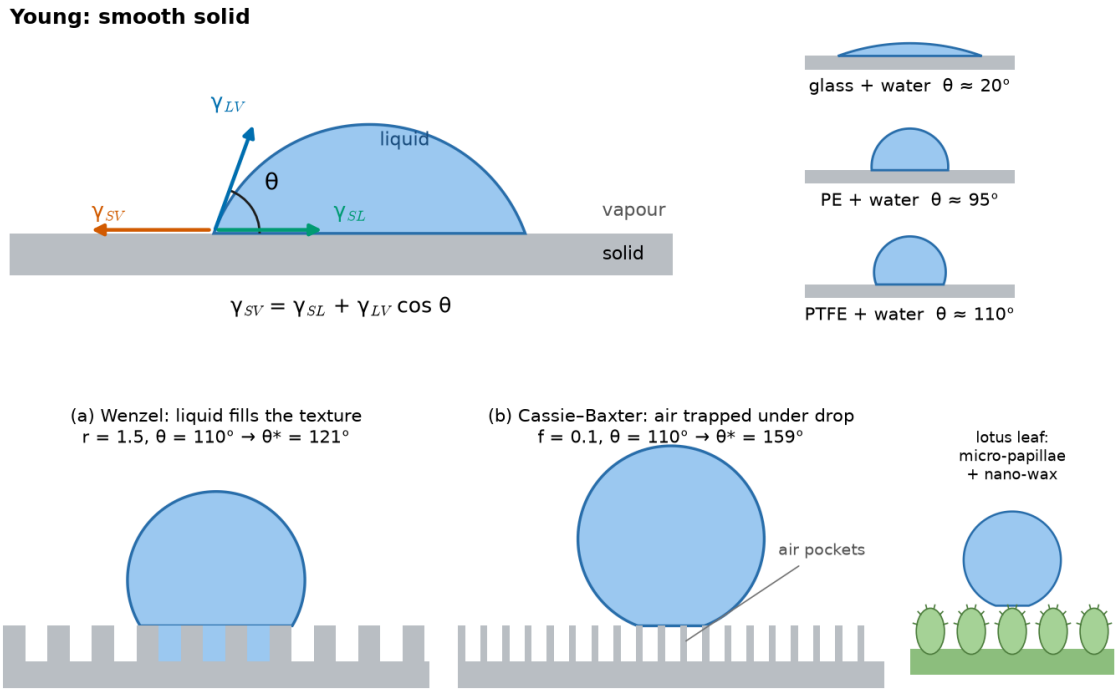


Figure 45.2 — Contact angle and the three wetting regimes: Young (smooth), Wenzel (roughness filled) and Cassie–Baxter (air trapped under the drop).

Roughness and superhydrophobicity

Roughness amplifies the intrinsic wettability. If the liquid fills the texture (**Wenzel** state), with roughness ratio r (true area over projected area),

$$\cos\theta^* = r\cos\theta$$

so hydrophobic surfaces become more hydrophobic and hydrophilic ones more hydrophilic. If air stays trapped under the drop (**Cassie–Baxter** state) with solid fraction f in contact with the liquid,

$$\cos\theta^* = f(\cos\theta + 1) - 1$$

For PTFE ($\theta = 110^\circ$), a texture with $r = 1.5$ in the Wenzel state gives $\cos\theta^* = 1.5 \times (-0.342) = -0.513$, $\theta^* = 121^\circ$; a pillar texture with $f = 0.1$ in the Cassie state gives $\cos\theta^* = 0.1 \times 0.658 - 1 = -0.934$, $\theta^* = 159^\circ$. Contact angles above 150° with low hysteresis define **superhydrophobic** surfaces (the lotus leaf), on which drops roll off and carry dirt away. They are used for self-cleaning glass and textiles and studied for anti-icing on aircraft and power lines, but their fragile textures wear and

contaminate, and the Cassie state can collapse into the Wenzel state under pressure or condensation, a durability problem that still limits applications. The same equations explain why a slightly hydrophilic material ($\theta = 70^\circ$) becomes clearly hydrophilic ($\theta^* = 59^\circ$) when roughened with $r = 1.5$: the basis of grit blasting before painting and of rough titanium implant surfaces for bone contact.

> **Status of the principle: Wenzel and Cassie–Baxter equations**
> *Approximate*. Area-weighted averages of Young's equation that assume the texture is much smaller than the drop and that the contact line samples it uniformly; they predict apparent equilibrium angles but not the hysteresis that decides whether drops actually roll off.

Adhesion in practice

Practical adhesion of paints, adhesives and coatings depends on the thermodynamic work of adhesion only as a starting condition. The measured fracture energy of a bonded joint, typically 100–5000 J/m^2 for structural adhesives, is thousands of times W_a because the polymer dissipates energy plastically and viscoelastically near the crack tip (Chapter 14). But dissipation can only operate if the interface does not fail first, so intimate wetting remains the necessary condition. Four mechanisms contribute:

- **Adsorption** (van der Waals and acid–base interactions across the interface), present wherever the adhesive wets the substrate. A useful rule is that the adhesive's surface tension should be below the substrate's surface energy, which is why polyolefins and PTFE are hard to bond.
- **Chemical bonding**, deliberately introduced by silane coupling agents (glass fibres in composites, glass to sealants) and by primers.
- **Mechanical interlocking** into roughness and porosity: grit blasting, phosphoric-acid anodizing of aluminium for aerospace bonding, porous coatings on implants.
- **Interdiffusion**, in polymer–polymer welding and solvent bonding.

Surface treatments raise the surface energy of polymers: corona discharge, flame and plasma treatment oxidize the surface of polyethylene and polypropylene, raising its surface energy from about 30 to 40–50 mJ/m^2 (checked on the shop floor with dyne inks), which is why plastic films can be printed and car bumpers painted. Weak boundary layers (mould-release agents, oils, loosely bound oxides) are the commonest cause of adhesive failure. The design of bonded joints is covered in Chapter 47.

45.4 Contact mechanics

Hertz contact

When two curved elastic bodies are pressed together, they touch over a small area whose size and pressure distribution were found by Hertz in 1882. For a sphere of radius R on a flat (or two spheres with $1/R = 1/R_1 + 1/R_2$) under normal force F, with effective modulus

$$\frac{1}{E^*} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$$

the contact radius, maximum (central) pressure and approach are

$$a = \left(\frac{3FR}{4E^*}\right)^{1/3}, \qquad p_0 = \frac{3F}{2\pi a^2}, \qquad \delta = \frac{a^2}{R}$$

The pressure is distributed elliptically, $p(r) = p_0\sqrt{1 - r^2/a^2}$, with mean pressure $p_m = 2p_0/3$. The important feature for materials is below the surface: the maximum shear stress, about $0.31p_0$, occurs at a depth of about $0.48a$ (for $\nu = 0.3$) (Figure 45.3). Yield therefore starts beneath the surface, when $p_0 \approx 1.6Y$ by the von Mises criterion, and rolling-contact fatigue cracks also nucleate there, which is why bearing steels must be clean of inclusions at that depth and why case-hardened layers must be deep enough to contain it.

Status of the principle: Hertz contact theory
Derived. Exact within linear elasticity for smooth, frictionless, non-conforming surfaces whose contact radius is small compared with the radii of curvature; it fails for rough surfaces (Section 45.5), for conforming contacts such as a hip ball in its cup, for adhesive contacts of soft materials (JKR theory) and after yield.

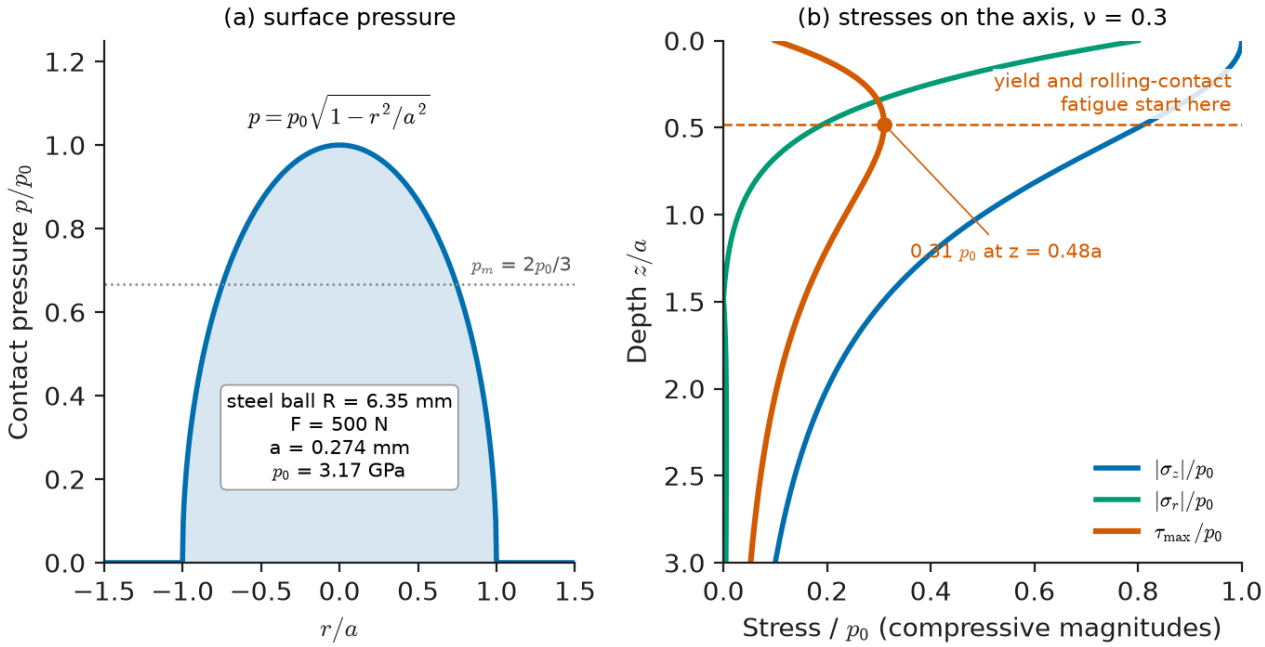


Figure 45.3 — Hertzian sphere-on-flat contact: surface pressure distribution and subsurface stresses along the axis of symmetry.

Worked example — a ball pressed on a bearing race. A hardened steel ball of radius $R = 6.35$ mm (12.7 mm diameter) is pressed onto a flat race of the same steel ($E = 210$ GPa, $\nu = 0.3$) with $F = 500$ N.

$$E^* = \frac{E}{2(1-\nu^2)} = \frac{210}{2 \times 0.91} = 115.4 \text{ GPa}$$

$$a = \left(\frac{3 \times 500 \times 6.35 \times 10^{-3}}{4 \times 115.4 \times 10^{9}}\right)^{1/3} = 2.74 \times 10^{-4} \text{ m} = 0.274 \text{ mm}$$

$$p_0 = \frac{3 \times 500}{2\pi \times (2.74 \times 10^{-4})^2} = 3.17 \times 10^{9} \text{ Pa} = 3.17 \text{ GPa}$$

The maximum shear stress is $0.31 \times 3.17 = 0.98$ GPa at a depth of $0.48 \times 0.274 = 0.13$ mm, and the approach is only $\delta = a^2/R = 12$ µm. A modest load produces a contact pressure of more than 3 GPa because the contact area is less than 0.25 mm^2. For a through-hardened bearing steel (about 62 HRC, yield strength about 2 GPa), first yield is predicted at $p_0 \approx 1.6 \times 2.0 = 3.2$ GPa: this load is just at the elastic limit. Since $p_0 \propto F^{1/3}$, the load would have to rise to about 1160 N to reach 4.2 GPa, the conventional static-capacity limit for ball bearings, at which a small permanent dent forms. Only hardened steels and ceramics such as silicon nitride (used for hybrid bearing balls) can carry such pressures; a mild steel with $Y \approx 250$ MPa would yield at a central pressure of about 0.4 GPa.

Real contact area

Engineering surfaces are rough. Even a polished surface has asperities with heights of tens of nanometres; a ground surface has a root-mean-square roughness R_q of 0.1–1 µm; a turned surface several micrometres. When two such surfaces touch, they meet only at the tips of the highest asperities. Each asperity contact is a tiny Hertzian contact that quickly yields, so the asperities support the load at a pressure close to the indentation hardness H of the softer material (Chapter 11). The **real contact area** is then

$$A_r \approx \frac{F}{H}$$

independent of the apparent area. A 10 kg steel block ($F = 98$ N) resting on a 100 cm^2 face of mild steel ($H \approx 1.5$ GPa) touches over only $98/(1.5 \times 10^9) = 6.5 \times 10^{-8}$ m^2, about 0.07 mm^2, or 7 parts per million of the apparent area. Greenwood and Williamson showed that even for purely elastic asperities with a Gaussian height distribution, A_r remains nearly proportional to load, because increasing load brings new asperities into contact rather than enlarging existing ones. Whether asperity contacts are elastic or plastic is decided by the **plasticity index** $\psi = (E^*/H)\sqrt{\sigma_s/\beta_s}$, where σ_s is the standard deviation of asperity heights and β_s their tip radius: $\psi < 0.6$ means mostly elastic contact (smooth, hard surfaces), $\psi > 1$ mostly plastic (most machined metals).

> **Status of the principle: Real contact area proportional to load**
> *Approximate.* Follows from plastic asperity contacts at the hardness, or from statistical elastic asperity models with many contacts; it fails at very high loads where A_r approaches the apparent area, for soft elastomers, and when adhesion or work hardening changes the asperity pressure.

The tiny real contact area explains far more than friction. Electrical contact resistance, thermal contact resistance across bolted joints and heat sinks (where interface materials fill the gaps), leakage through metal-to-metal seals and the cold-welding of clean metals in vacuum are all governed by A_r.

45.5 Friction

Amontons' laws

The empirical laws of dry friction, found by Leonardo da Vinci, restated by Amontons in 1699 and extended by Coulomb, are:

1. The friction force is proportional to the normal load: $F_f = \mu F_N$.
2. The friction force is independent of the apparent area of contact.
3. Kinetic friction is nearly independent of sliding velocity (Coulomb), and is usually lower than static friction.

Sliding pair (dry, in air, typical)	Coefficient of friction μ
Clean metals in vacuum	above 1, may seize
Steel on steel, unlubricated	0.5–0.8
Steel on steel, boundary lubricated	0.05–0.15
Steel on bronze, lubricated	0.05–0.1
Steel on brass or babbitt, dry	0.3–0.5
Brake pad on cast-iron disc	0.35–0.45
Tyre rubber on dry asphalt	0.7–1.0
Tyre rubber on wet asphalt	0.4–0.7
Steel on ice	about 0.02
PTFE on steel	0.04–0.1
UHMWPE on Co–Cr in serum	0.05–0.1
DLC coating on steel	0.05–0.2
Hydrodynamic journal bearing	0.001–0.01

> **Status of the principle: Amontons–Coulomb friction laws**
> *Empirical.* Accurate to roughly ±10–20 % for most dry and boundary-lubricated pairs of metals and ceramics over moderate loads; they fail for elastomers (friction depends on area, velocity and temperature), at very low loads where adhesion dominates, at high speeds where flash heating softens or melts the surface, and wherever a fluid film separates the surfaces.

The adhesion origin: Bowden and Tabor

Bowden and Tabor explained the laws by combining real contact area with adhesion. If the asperity junctions cold-weld and must be sheared to slide, with shear strength s, then $F_f = s A_r = s F_N / H$, so

$$\mu_{\text{adh}} = \frac{s}{H}$$

independent of load and of apparent area, which is Amontons' first and second laws. For a clean metal the junction shears at roughly its bulk shear strength, about $H/6$ (since $H \approx 3Y$ and $s \approx Y/2$), giving $\mu \approx 0.17$; a surface oxide or adsorbed film with lower shear strength lowers μ. The model also predicts the right direction of most observations:

- **Soft films on hard substrates** give low friction: A_r is set by the hard substrate's H, the shear strength by the soft film's s. This is the principle of babbitt bearing metals, lead and silver films on aerospace bearings, solid lubricants such as MoS_2 (whose weak interlayer van der Waals bonding gives $\mu \approx 0.02$–0.1 in dry air or vacuum) and graphite (which needs adsorbed water to be slippery and fails in space).
- **Clean metals in vacuum** show μ well above 1. Tangential force makes the junctions grow (**junction growth**) because the combined normal and shear stress keeps them yielding, so A_r increases during sliding until gross seizure. Satellite mechanisms that seized in orbit and galling of stainless-steel threads are this effect.

A second contribution comes from **ploughing**: a hard asperity or particle digs a groove in the softer surface and must displace material. For a conical asperity of semi-angle α, $\mu_{\text{pl}} \approx (2/\pi)\cot\alpha$, small for blunt asperities (about 0.06 for a slope of 5°) but large for sharp abrasive grits. Total friction is approximately $\mu \approx \mu_{\text{adh}} + \mu_{\text{pl}}$, with additional hysteretic losses in viscoelastic materials.

Friction in polymers and elastomers departs from Amontons. The real contact area of elastomers is large and depends on load sublinearly, and friction has an adhesive part and a hysteretic part that both depend on sliding speed and temperature through the material's viscoelastic loss (Chapter 21). Tyre grip on wet roads comes mainly from hysteresis as the tread deforms over road texture at frequencies of 10^3–10^6 Hz; tread compounds with high loss near those frequencies at road temperature grip best, which is a time–temperature superposition argument applied to the road.

Stick–slip occurs when static friction exceeds kinetic friction and the system is compliant: the surfaces stick until the spring force exceeds static friction, then slip and decelerate. Squealing brakes, chattering machine-tool slideways, creaking joints and the violin bow are examples; the cure is a pair with friction rising slightly with speed, higher stiffness, or damping.

45.6 Wear mechanisms

The Archard wear law

Wear is the progressive loss of material from a surface by mechanical action, often combined with chemical action. The simplest and most widely used model is Archard's: the wear volume V is proportional to the sliding distance s and the normal load F, and inversely proportional to the hardness H of the softer surface,

$$V = K \frac{F\, s}{H}$$

where the dimensionless **wear coefficient** K is the probability that an asperity encounter produces a wear particle. Engineers often use the **specific wear rate** $k = K/H = V/(Fs)$, in $\text{mm}^3/(\text{N m})$.

Situation	Typical wear coefficient K
Unlubricated mild steel on mild steel (severe adhesive wear)	10^{-3}–10^{-2}
Unlubricated hardened steel on steel (mild oxidative wear)	10^{-5}–10^{-4}
Two-body abrasion by sharp grit	10^{-2}–10^{-1}
Three-body abrasion by loose particles	10^{-3}–10^{-2}
Polymer on smooth steel	10^{-6}–10^{-4}
Boundary-lubricated metals	10^{-7}–10^{-6}
Hydrodynamically lubricated (no contact)	below 10^{-9}

The range of seven orders of magnitude is the central fact of tribology: changing the mechanism matters far more than changing the hardness.

> **Status of the principle: Archard wear law**
> *Empirical.* Derived from a simple probability argument for asperity encounters, but K is a fitted parameter that changes by orders of magnitude with mechanism, lubrication, environment and transitions between mild and severe wear; the law applies only within one wear regime and does not predict when a transition occurs.

Figure 45.4 illustrates the four principal mechanisms.

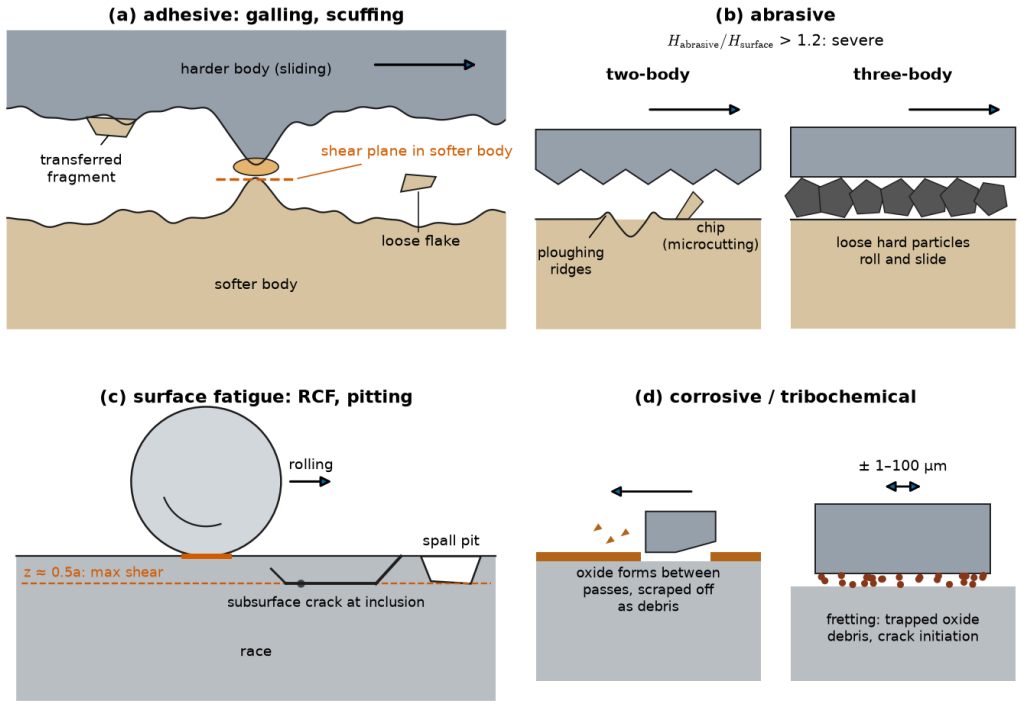


Figure 45.4 — The four principal wear mechanisms: adhesive, abrasive, surface fatigue and corrosive (tribochemical).

Adhesive wear

When asperity junctions are stronger than one of the parent materials, shearing tears a fragment from the weaker side and transfers it to the other; fragments later detach as wear debris. In mild form it is the slow wear of well-matched pairs; in severe form it becomes **scuffing** (sudden failure of a lubricated contact, as in gears and piston rings at high load and speed) and **galling** (gross transfer and seizure, typical of stainless steels, titanium and aluminium sliding on themselves). Defences follow from the Bowden–Tabor picture: avoid like-on-like metal pairs with high mutual solubility (steel on bronze or cast iron rather than steel on steel), use oxide-forming or lubricated surfaces, raise hardness, and apply coatings with low adhesion such as nitrides and DLC.

Abrasive wear

Hard particles or protrusions plough and cut the softer surface. In **two-body** abrasion a hard rough surface or fixed grit slides on the softer one (grinding, a rock sliding on a chute liner, sand embedded in a seal); in **three-body** abrasion loose particles roll and slide between surfaces (contaminated lubricant, dust in a bearing, slurry pumps), and wear rates are typically an order of magnitude lower. Abrasion is severe when the abrasive is harder than the surface by a factor of about 1.2 or more, and mild when the surface is harder. Since quartz, the commonest natural abrasive, has a hardness of about 11 GPa (roughly 1100 HV), materials for mining, agriculture and earthmoving aim to exceed it: white cast irons with M_7C_3 carbides (about 1500 HV), WC–Co hardmetals (1200–1800 HV), and hardfaced overlays. Filtration of lubricants and seals that exclude dust are the abrasion defences in machinery.

Surface fatigue

In rolling and rolling–sliding contacts (bearings, gears, cams, rails) there is little sliding, but every pass cycles the subsurface shear stress of Section 45.4. After millions of cycles, cracks nucleate near the depth of maximum shear, usually at non-metallic inclusions, grow parallel to the surface and turn up to release a flake, leaving a **spall** or pit. On gears, **micropitting** (pits of 10–20 μm depth) arises when the lubricant film is thin compared with the roughness. Rail head checking and squats are the railway equivalents. Rolling-bearing life is predicted with the Lundberg–Palmgren model, whose catalogue form is $L_{10} = (C/P)^3$ million revolutions for ball bearings (C = dynamic load rating, P = equivalent load); a 25 % overload cuts life by a factor of $1.25^3 \approx 2$. Steel cleanliness made the largest single improvement in bearing life over the twentieth century: vacuum-degassed and remelted steels with few large oxide inclusions last many times longer than older air-melted steels.

Corrosive and tribochemical wear

Sliding removes protective films and heats asperity tips briefly to hundreds of degrees (**flash temperatures**), accelerating reactions. **Oxidative wear** of steels at moderate loads, where a thin oxide continually forms and is removed, is a benign mild-wear regime; in aggressive media the same mechanism becomes **tribocorrosion**, important in the taper junctions of modular hip implants, marine pumps and chemical mixers. **Fretting** is wear under small-amplitude oscillation (1–100 μm) between nominally fixed surfaces: bolted and riveted joints, spline couplings, turbine blade roots, wire ropes and electrical connectors. The debris, usually oxide, is trapped and abrasive, and fretting can nucleate fatigue cracks that reduce fatigue strength by half or more (**fretting fatigue**). Erosion by particles and liquid drops, and cavitation erosion (Chapter 43), complete the family.

Worked example — wear of a polyethylene hip cup. A conventional UHMWPE acetabular cup articulates against a 28 mm diameter Co–Cr femoral head. Laboratory tests give a specific wear rate $k = 1.0 \times 10^{-6}$ mm^3/(N m). Take an effective joint load of 2000 N during walking, a sliding distance of 25 mm per gait cycle, and 10^6 cycles per year for a moderately active patient.

Sliding distance per year: $s = 0.025 \times 10^6 = 2.5 \times 10^4$ m.

$$V = k\,F\,s = 1.0 \times 10^{-6} \times 2000 \times 2.5 \times 10^4 = 50 \text{ mm}^3/\text{year}$$

Dividing by the projected area of the head, $\pi \times 14^2 = 616$ mm^2, gives a linear penetration of about 0.08 mm/year. This agrees with clinical measurements for conventional polyethylene of roughly 0.1 mm/year and 40–80 mm^3/year. The thickness of the cup is not the problem; the volume is. If the average particle is a 0.5 μm sphere of volume 6.5×10^{-11} mm^3, 50 mm^3 is about 8×10^{11} particles a year, and osteolysis becomes common above roughly 0.1 mm/year. Highly cross-linked UHMWPE, with k lower by a factor of about four to ten in simulator tests, brings the particle load below that threshold, which is why it replaced conventional polyethylene clinically. The Archard calculation also exposes its own limits: k for polyethylene depends strongly on the crossing of sliding directions (multidirectional motion raises it several times compared with unidirectional sliding, because the molecules orient and then are sheared across their orientation), so the right test reproduces the kinematics of gait.

45.7 Lubrication regimes and the Stribeck curve

Regimes

A lubricant reduces friction and wear by separating the surfaces with a film that shears easily. How completely it separates them depends on the speed, the viscosity and the load. Stribeck's measurements on journal bearings (1902) showed that the coefficient of friction depends on these through a single group, the **Hersey number** $\eta N/P$ (viscosity × rotational speed / load per unit projected area). The resulting **Stribeck curve** (Figure 45.5) has three regions:

- **Boundary lubrication** (low $\eta N/P$): the film is too thin to separate the asperities; load is carried by asperity contacts covered by adsorbed molecular films and reaction layers. Friction is 0.05–0.15, nearly independent of speed, and set by the chemistry of the boundary film. Wear is controlled by additives.
- **Mixed lubrication**: load is shared between fluid pressure and asperity contacts; friction falls steeply as speed rises and the film thickens.
- **Hydrodynamic (full-film) lubrication** (high $\eta N/P$): a converging wedge of fluid generates enough pressure to carry the whole load, and the surfaces do not touch. Friction is low (0.001–0.01) and rises slowly with $\eta N/P$ because of viscous shearing. Wear is essentially zero, except at start and stop.

The minimum of the curve, at the transition from mixed to full-film lubrication, is where well-designed bearings run.

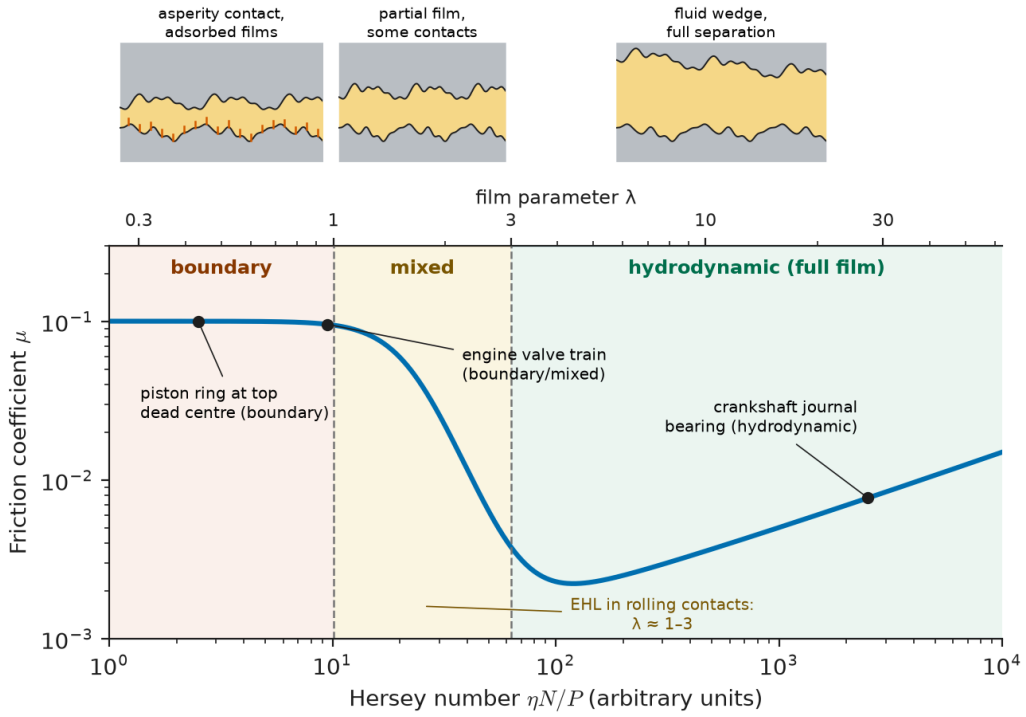


Figure 45.5 — The Stribeck curve: coefficient of friction against the Hersey number, with boundary, mixed and hydrodynamic regimes and the film parameter λ.

> **Status of the principle: Stribeck curve**
> *Empirical.* A measured correlation of friction with the Hersey number for a given bearing geometry and surface finish; the regime boundaries shift with roughness, temperature-dependent viscosity and additive chemistry, and the hydrodynamic branch is derived quantitatively from Reynolds' lubrication equation.

The film parameter

The regime is set more directly by comparing the minimum film thickness $h_{\min}$ with the composite roughness of the two surfaces,

$$\lambda = \frac{h_{\min}}{\sqrt{R_{q1}^2 + R_{q2}^2}}$$

Roughly, $\lambda < 1$ means boundary lubrication, $1 < \lambda < 3$ mixed, and $\lambda > 3$ full-film. For example, a gear contact with $h_{\min} = 0.3$ μm between flanks of $R_q = 0.1$ and 0.15 μm has $\lambda = 0.3/0.180 = 1.66$: mixed lubrication, in which micropitting and boundary additives matter. Superfinishing both flanks to 0.05 μm would give $\lambda = 4.2$ and full separation. This is why surface finish specifications are as much a part of gear and bearing design as material and heat treatment.

Hydrodynamic and elastohydrodynamic films

In a journal bearing the shaft drags oil into a converging gap, and Reynolds' equation (from the Navier–Stokes equations for thin films) gives the pressure that supports the load. For a lightly loaded, concentric journal, **Petroff's equation** estimates the friction coefficient:

$$\mu = 2\pi^2 \, \frac{\eta N}{P} \, \frac{r}{c}$$

with r the journal radius and c the radial clearance. For an engine main bearing with $\eta = 0.03$ Pa s, $N = 30$ rev/s, $P = 1$ MPa and $r/c = 1000$, $\mu = 2\pi^2 \times (0.03 \times 30 / 10^6) \times 1000 = 0.018$, an order of magnitude below boundary friction.

In rolling contacts (gears, rolling bearings, cams), the pressures of 1–3 GPa that Section 45.4 calculated do two things to the oil film: they deform the surfaces elastically, flattening the contact, and they raise the viscosity of the oil enormously (exponentially with pressure, by factors of 10^3 or more), so that the lubricant in the contact behaves almost like a solid. This is **elastohydrodynamic lubrication** (EHL). Film thicknesses are only 0.1–1 μm, given by the Hamrock–Dowson and similar formulas, and depend strongly on speed and viscosity (roughly $h \propto (\eta U)^{0.7}$) but weakly on load (about $F^{-0.07}$). EHL is why a ball bearing survives billions of revolutions with a film thinner than the wavelength of light.

Lubricants and additives

Mineral and synthetic oils (polyalphaolefins, esters, polyalkylene glycols) provide viscosity, whose temperature dependence is specified by the viscosity index. Additives provide the boundary protection: **anti-wear** additives such as zinc dialkyldithiophosphate (ZDDP) react at asperity flash temperatures to form glassy phosphate tribofilms about 100 nm thick; **extreme-pressure** additives (sulfur–phosphorus compounds) form iron sulfides that prevent scuffing in hypoid gears; **friction modifiers** (organic amides, molybdenum dithiocarbamate that forms MoS_2 in the contact) lower boundary friction to save fuel. Greases thicken oil with soaps or polymers to keep it in place. **Solid lubricants** (MoS_2, graphite, PTFE, soft metals) serve where oils cannot: vacuum, extreme temperature, food contact and dry-running bearings. Water-lubricated ceramic and polymer bearings and the hydration lubrication of articular cartilage, whose friction coefficient of about 0.001–0.01 no engineered bearing matches, are reminders that the lubricant need not be oil.

45.8 Surface engineering

Most tribological, fatigue and corrosion properties are surface properties, while strength and toughness are bulk properties. **Surface engineering** gives a component a surface different from its core, so that each can be optimized: a tough core with a hard, wear- and fatigue-resistant case.

Transformation hardening

Induction, **flame** and **laser** hardening heat a thin surface layer of a medium-carbon or alloy steel (or cast iron) into the austenite range in seconds and quench it, forming martensite to a depth of 0.5–5 mm while the core stays soft and tough (Chapter 15). Crankshaft journals, gear teeth, rail heads and machine-tool ways are induction-hardened. The martensitic transformation's volume expansion leaves the case in compression, which raises fatigue strength.

Thermochemical treatments

Carburizing diffuses carbon into a low-carbon steel (0.1–0.25 % C) at 900–950 °C, raising the surface to 0.7–1.0 % C, followed by quenching and tempering: the case becomes about 58–62 HRC martensite and the core a tough low-carbon martensite or bainite. It is the standard treatment for highly loaded gears, camshafts and bearing races of large size. **Nitriding** diffuses nitrogen into steels containing nitride formers (Cr, Mo, Al, V) at 500–570 °C, below the transformation range, forming hard alloy nitrides and an iron-nitride compound layer without quenching, so distortion is minimal; cases are shallower (0.1–0.6 mm) but very hard (up to about 1000–1200 HV) and fatigue-resistant. **Nitrocarburizing** adds carbon to a thin compound layer for wear and corrosion resistance; **boriding** forms Fe_2B layers of 1500–2000 HV for severe abrasion. Low-temperature carburizing or nitriding of austenitic stainless steel (expanded austenite, "S-phase") hardens it without sacrificing corrosion resistance.

Case depth is controlled by diffusion (Chapter 8), and the time scales come directly from the error-function solution of Fick's second law.

Worked example — carburizing time for a gear. A 0.20 % C gear steel is gas-carburized at 925 °C with the surface held at 1.00 % C. How long does it take for the carbon content at a depth of 1.0 mm to reach 0.40 %? Take the diffusivity of carbon in austenite as $D = D_0 \exp(-Q/RT)$ with $D_0 = 2.3 \times 10^{-5}$ m^2/s and $Q = 148$ kJ/mol.

$$D = 2.3 \times 10^{-5} \exp\!\left(\frac{-148\,000}{8.314 \times 1198.15}\right) = 8.1 \times 10^{-12}\ \text{m}^2\,\text{s}^{-1}$$

$$\frac{C_s - C_x}{C_s - C_0} = \frac{1.00 - 0.40}{1.00 - 0.20} = 0.75 = \operatorname{erf}\!\left(\frac{x}{2\sqrt{Dt}}\right) \quad \Longrightarrow \quad \frac{x}{2\sqrt{Dt}} = 0.813$$

$$t = \frac{1}{D}\left(\frac{x}{2 \times 0.813}\right)^2 = \frac{(1.0 \times 10^{-3}/1.627)^2}{8.1 \times 10^{-12}} = 4.7 \times 10^{4}\ \text{s} \approx 13\ \text{h}$$

Halving the case depth to 0.5 mm needs only a quarter of the time, 3.2 h, because time scales with the square of depth (Figure 45.6). Industrial practice for 1 mm cases at 925 °C is somewhat shorter than this estimate, because carbon diffusivity in austenite rises with carbon content (to roughly twice the dilute value at 1 % C), and because many plants carburize at higher temperatures or with boost–diffuse cycles. The square-root scaling, however, is exactly why deep cases (2–3 mm on large wind-turbine and mining gears) take days in the furnace, and why vacuum carburizing at 1000–1050 °C is attractive.

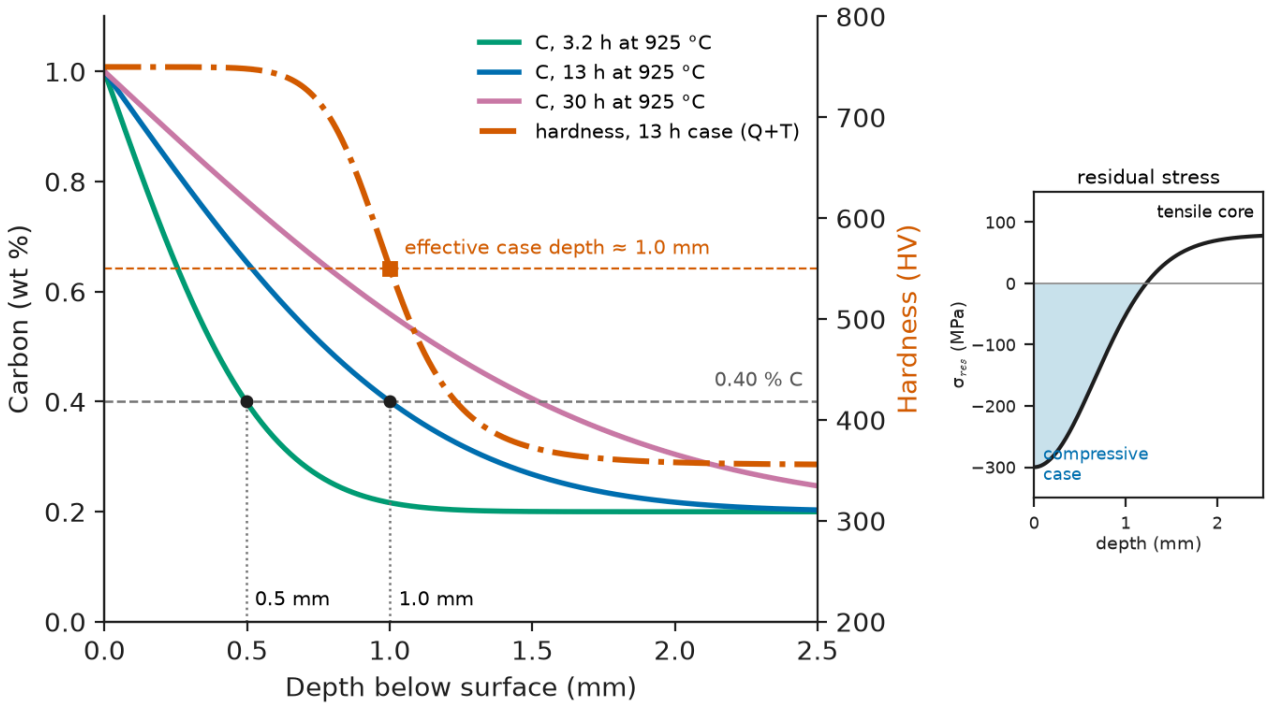


Figure 45.6 — Carbon concentration and hardness profiles through a carburized case.

Coatings

Where a different material is needed at the surface, it is added:

Coating	Process	Typical thickness	Hardness	Typical use
TiN, TiAlN, CrN	PVD (arc, sputtering)	1–5 μm	2000–3500 HV	Cutting tools, forming dies, piston rings
Diamond-like carbon (DLC)	PVD, PECVD	1–3 μm	1500–5000 HV	Fuel-injector and valve-train parts, low friction, biomedical
TiC/TiN/Al_2O_3 multilayers	CVD	5–20 μm	2500–3000 HV	Turning inserts for steel
WC–Co, Cr_3C_2–NiCr	High-velocity oxy-fuel spray	100–500 μm	1000–1400 HV	Landing gear (hard-chrome replacement), pump sleeves, rolls
Hard chromium	Electroplating	20–500 μm	800–1000 HV	Hydraulic rods; restricted by hexavalent chromium regulations
Electroless Ni–P	Autocatalytic deposition	10–50 μm	500–1000 HV (heat-treated)	Moulds, valves, uniform coverage of complex shapes
Stellite, NiCrBSi	Weld overlay, laser cladding	1–3 mm	400–700 HV	Valve seats, cavitation and erosion resistance
Yttria-stabilized zirconia (on MCrAlY bond coat)	Plasma spray, EB-PVD	100–500 μm	Low conductivity, not wear	Thermal barrier coatings on turbine blades (Chapter 41)

A coating is only as good as its adhesion and its stress state. A hard thin coating on a soft substrate fails by "eggshell" collapse when the substrate yields beneath it under a Hertzian load; the rule of thumb is that the substrate must be hard enough to support the subsurface shear stress at depth $0.48a$, which is why PVD coatings go on hardened tool steels and hardmetals, and why **duplex treatments** (nitriding followed by PVD) are used on softer steels. Thermal-expansion mismatch and growth stresses can crack or spall coatings (Chapter 49).

Mechanical surface treatments and texturing

Shot peening bombards the surface with hard steel or ceramic shot, plastically stretching the surface layer; the elastic core pushes back and leaves a compressive residual stress of several hundred megapascals to a depth of 0.1–0.5 mm. Because fatigue cracks need tensile stress to open, peening raises the fatigue strength of springs, gears, crankshafts, aircraft landing gear and turbine-disc fir-tree slots by 20–50 %, and protects against SCC. **Laser shock peening** and **deep rolling** produce deeper compressive layers; burnishing also smooths the surface.

Surface texturing deliberately adds features to change tribology: laser-machined microdimples tens of micrometres across act as lubricant reservoirs and micro-hydrodynamic bearings in mechanical seals and piston rings, lowering friction in mixed lubrication; honed cross-hatch patterns retain oil on cylinder bores; textured implant surfaces promote bone ingrowth; and bio-inspired textures (shark-skin riblets) reduce drag in turbulent flow. **Surface nanocrystallization** by severe plastic deformation (surface mechanical attrition) creates a hard nanograined layer that improves wear and fatigue resistance.

> **Status of the principle: Compressive residual stress improves fatigue life**
> *Derived.* Follows from superposing a static compressive mean stress on the applied cycle, which lowers the effective mean stress (Goodman-type relations, Chapter 14) and closes small cracks; it holds only while the residual stress is not relaxed by overloads, cyclic plasticity or high temperature, and it can be lost by subsequent machining or grinding.

Choosing a surface treatment

The selection logic connects the chapter: identify the dominant surface damage (adhesive, abrasive, fatigue, corrosive, or a combination), the contact pressure and its subsurface depth from Hertz, the lubrication regime from λ, and the environment; then choose a treatment that puts a hard, low-adhesion or compressively stressed layer to the right depth without compromising the core. A gear needs carburizing deep enough for the Hertzian shear depth and a finish good enough for EHL; a cutting tool needs a hard, chemically inert PVD layer on a hardmetal that will not yield beneath it; an implant needs wear resistance and biocompatibility without particles; a hydraulic rod needs corrosion and wear resistance without hexavalent chromium. Thin-film

deposition processes are developed in Chapter 49, and the materials-selection framework that weighs these options against cost in Chapter 58.

> **Boundary.**
> The thermodynamics of adsorption and the electrochemistry of tribocorrosion belong to the Advanced Chemistry book; the continuum mechanics of contact and the fluid mechanics of Reynolds' equation draw on The Mathematics Universe and engineering mechanics courses. Contact resistance of electrical connectors and its degradation are treated in Chapter 25.

What remains open

Open question	Where it is answered
Grain-boundary energy and structure, and their role in microstructure	Chapter 5
Diffusion solutions used for case depth and segregation	Chapter 8
Surface energy as the driving force for nucleation, sintering and coarsening	Chapters 9 and 46
Hardness testing, yield criteria and fracture energy used in contact and wear models	Chapters 11, 13 and 14
Heat treatment of steels behind induction hardening and carburizing	Chapter 15
Viscoelastic behaviour behind elastomer friction and polymer wear	Chapter 21
Nanoparticle shapes and catalysis by facets	Chapter 38
Implant articulation, wear debris and tissue response	Chapter 39
Thermal barrier coatings and high-temperature surface protection	Chapter 41
Corrosion and tribocorrosion; fretting and cavitation as degradation	Chapters 43 and 44
Adhesive bonding of structural joints	Chapter 47
PVD, CVD, thermal spray and electroplating processes and film stress	Chapter 49
Surface analysis (XPS, AES, SIMS), profilometry and AFM	Chapters 52 and 53
Selecting a surface treatment against cost and performance	Chapter 58

Next: Chapter 46 opens Part X by turning from how materials degrade to how they are made, beginning with casting, deformation processing and powder metallurgy.

Part X — Processing and Manufacturing

Up to this point the book has treated microstructure mostly as something a material *has*: a grain size, a precipitate dispersion, a texture, a population of pores. Part X treats it as something a factory *makes*. Every engineering component starts as a liquid, a powder, a vapour or a slab, and on its way to a finished shape it is melted, deformed, sintered, joined, layered or deposited. Each of those steps imposes a thermal history and a strain history, and the microstructure, defects and residual stresses of the finished part are the record of that history. The same Ti–6Al–4V can be a forged disc, a cast housing or a laser-melted implant, with identical composition and different properties. This part explains why, by applying the thermodynamics and kinetics of Part II and the mechanics of Part III to the processes that industry actually runs.

The emphasis is on the *materials* side of manufacturing: what each process does to structure, which defects it produces, and which process windows avoid them.

By the end of this part you should understand:

- how casting, deformation processing and powder metallurgy each create a characteristic microstructure, and the defects (shrinkage, gas porosity, hot tears, laps, forming splits, residual porosity) that go with each route;
- why the same metal is hot worked, warm worked or cold worked depending on its homologous temperature, and how forming limits, anisotropy and texture decide whether a sheet can be pressed into a shape;
- how thermomechanical processing combines deformation and transformation to produce grain sizes and textures that heat treatment alone cannot;
- what a fusion weld does to the metal beside it, why carbon equivalent and diffusible hydrogen control cold cracking, and when solid-state welding, brazing, adhesive bonding or mechanical fastening is the better choice;
- how additive manufacturing builds parts from melt pools a fraction of a millimetre wide, why it produces fine, textured and sometimes porous microstructures, and how post-processing and qualification turn a printed part into a certified one;
- how thin films and coatings are deposited, how growth conditions set their density, stress and adhesion, and how semiconductor crystals and devices are made.

Chapter	Central question	What it delivers
46. Casting, Forming and Powder Processing	How do bulk shaping routes — melting and pouring, plastic deformation, pressing and sintering powders — set the structure and defects of a part?	Casting design and defects, Chvorinov's rule, rolling, forging, extrusion, drawing and sheet-forming mechanics, hot/warm/cold working, forming limit diagrams, thermomechanical processing and texture control, powder production, compaction, sintering, HIP and MIM
47. Joining: Welding, Brazing and Adhesive Bonding	How can separate pieces be made to act as one without the joint becoming the weakest link?	Arc, laser and electron-beam welding, heat input and cooling time, fusion-zone and HAZ metallurgy, residual stress, carbon equivalent and cracking, friction stir and diffusion bonding, brazing and soldering, adhesive joint design, fastening and dissimilar-material joints
48. Additive Manufacturing	What happens to a material when a part is built layer by layer from tiny melt pools, droplets or cured slices?	The seven process families, melt-pool physics and rapid solidification, porosity, lack of fusion, residual stress and anisotropy, HIP and heat treatment, polymer and ceramic AM, design rules and qualification
49. Thin Films and Coatings	How are layers from nanometres to millimetres thick deposited?	Vacuum and plasma basics, PVD, CVD, ALD, electroplating and sol–gel, growth modes and structure-zone models, film stress and adhesion, epitaxy, thermal spray and hard coatings
50. Electronic Materials Manufacturing	How does each wafer-level manufacturing step change the structure and properties of semiconductor and thin-film materials?	Crystal growth and its defects, wafer quality, oxide growth kinetics, dopant profiles, implant damage and anneal, resolution limits, etch anisotropy, interconnect and dielectric materials, process-induced defects
51. Joining, Packaging, and Interconnect Materials	Which materials connect, support, protect and cool a die, and how do they fail?	Solder and intermetallic selection, wire and bump technology, substrates and laminates for power and RF, encapsulants, TIMs, CTE-mismatch strain estimates and package failure mechanisms

How this part connects to the rest of the book. Part X is where the earlier parts are cashed in. Solidification, diffusion, recrystallization and precipitation (Chapters 8–10) are the kinetic rules every process obeys; the phase diagrams of Chapter 7 and the steel and alloy metallurgy of Chapters 15–16 say which transformations a thermal cycle can trigger; dislocation theory and strengthening (Chapters 12–13) explain why deformation hardens and annealing softens; fracture and fatigue (Chapter 14) explain why process defects, not average properties, usually decide a part's life. Ceramic and polymer processing are treated with their materials in Chapters 17 and 21, and composite manufacturing in Chapter 23; this part points to them rather than repeating them. Looking forward, the defects and residual stresses introduced here become the starting points of corrosion and failure analysis (Chapters 43–44), the structures produced here are what the characterization methods of Chapters 52–54 are used to verify, process modelling reappears in the ICME framework of Chapter 56, and process selection and cost enter materials selection in Chapter 58 and lifecycle accounting in Chapter 59.

Chapter 46 — Casting, Forming and Powder Processing

Part X · Processing and Manufacturing

A car maker is choosing how to make the lower control arm of a front suspension. Three suppliers quote in nearly the same aluminium alloy family. The first offers a low-pressure die casting in A356-T6: cheap, near net shape, with a hollow section that saves mass. The second offers a closed-die forging in 6082-T6: more machining and more expensive, but with grain flow that follows the arm's contour. The third offers a press-and-sinter part in an aluminium powder alloy. On a static strength test all three pass. In a fatigue test at the same stress amplitude the cast arms fail first, from shrinkage pores a few hundred micrometres across near the ball joint; the forged arms last several times longer; the sintered arms fail from residual porosity at the surface. The chemistry is almost the same; the *route* has put different defects in different places. The same choice faces a turbine maker between investment-cast and forged blades, a civil contractor between rolled and cast steel nodes, an implant maker between forged and injection-moulded titanium, and a gear maker between machined bar and sintered powder.

> **How does each bulk shaping route — casting, plastic deformation, powder consolidation — create the microstructure and the defects of the finished part?**
> **Which simple physical rules predict how big a riser must be, how hard a mill must push, how far a sheet can be stretched and how much a sintered part will shrink?**

Concept	Engineering destination
Casting processes, feeding, Chvorinov's rule	Engine blocks, pump housings, turbine blades, riser design
Casting defects	Fatigue of cast suspension parts, pressure tightness of valve bodies
Rolling, forging, extrusion, drawing mechanics	Plate and strip mills, crankshafts, window profiles, wire
Sheet forming, springback, deep drawing	Car body panels, beverage cans, aircraft skins
Hot, warm and cold working; Zener–Hollomon parameter	Choice of forming temperature for steel, aluminium, titanium, magnesium
Forming limit diagrams, n and r values	Stamping try-out, sheet-grade selection
Thermomechanical processing, texture	HSLA pipeline and ship plate, can-body sheet, electrical steel
Powder production, compaction, sintering	Press-and-sinter gears, self-lubricating bearings, hardmetal tools
HIP and metal injection moulding	Superalloy discs, porosity-free castings, small complex medical parts

Roadmap. Three routes and how they differ (46.1) → casting processes, design and defects (46.2–46.3) → mechanics of deformation processes (46.4) → hot, warm and cold working and formability (46.5) → thermomechanical processing and texture (46.6) → powder metallurgy, HIP and MIM (46.7) → ceramics and glasses in brief (46.8) → how the route reaches the property sheet (46.9).

> **Boundary.**
> The physics of solidification (nucleation, dendrites, constitutional supercooling, Scheil segregation) is in Chapter 9; recovery, recrystallization and grain growth in Chapter 10; flow stress and yield criteria in Chapter 13. This chapter uses those results to explain processes. Full forming mechanics (slip-line fields, finite-element process simulation) belongs to manufacturing engineering; only the estimates a materials engineer needs are given.

46.1 Three routes to a shape

Every metal part begins as liquid metal, and it can be given its shape in three ways.

1. **Casting**: pour the liquid into a mould shaped like the part. Complex shapes are made in one step; the microstructure is a solidification structure, and the defects are those of freezing — shrinkage, gas porosity, segregation, inclusions.

2. **Deformation processing**: cast a simple ingot, slab or billet, then shape it plastically by rolling, forging, extrusion, drawing or sheet forming. Deformation breaks up the cast structure and welds pores shut, and recrystallization produces a fine, uniform *wrought* structure. Shape complexity is limited and the equipment is heavy.
3. **Powder processing**: atomize the liquid into powder, press it into shape and bond the particles by sintering or hot isostatic pressing. This reaches shapes and compositions the other routes cannot — immiscible mixtures, refractory metals, segregation-free alloys, net-shape parts without machining waste. Its characteristic defect is residual porosity.

Most parts combine routes: a forging starts as a cast ingot; a cast turbine blade is hot isostatically pressed; a sintered gear is cold sized. Joining (Chapter 47) and additive manufacturing (Chapter 48), which combines local melting with powder feedstock, complete the set (Figure 46.1).

Route	Shape complexity	Typical tolerance	Microstructure	Characteristic defects	Examples
Sand casting	Very high, any size	±0.5–2 mm	Coarse dendritic, segregated	Shrinkage, gas pores, sand inclusions	Engine blocks, pump casings
Investment casting	Very high, small–medium	±0.1–0.3 mm	Dendritic; directional or single crystal possible	Shrinkage, inclusions, freckles	Turbine blades, implants
High-pressure die casting	High, thin walls	±0.05–0.2 mm	Fine skin, porous core	Gas porosity, cold shuts	Gearbox housings, car nodes
Forging	Moderate	±0.2–1 mm	Wrought, grain flow along contour	Laps, folds	Crankshafts, landing gear, discs
Rolling, extrusion, drawing	Constant section	±0.01–0.5 mm	Wrought, textured	Edge cracks, tearing, central bursts	Plate, profiles, wire, tube
Sheet forming	Thin shells	±0.1–0.5 mm	Inherited from sheet	Splits, wrinkles, springback	Car panels, cans
Press-and-sinter PM	Moderate, small	±0.02–0.1 mm	Fine, 8–14 % porous	Porosity, density gradients	Gears, bushes
MIM, powder HIP	High (MIM) / moderate (HIP)	±0.3–0.5 % / machined	Fine, near full density	Distortion, prior-particle boundaries	Surgical tools, superalloy discs

Typical ranges; tolerances depend strongly on part size and alloy.

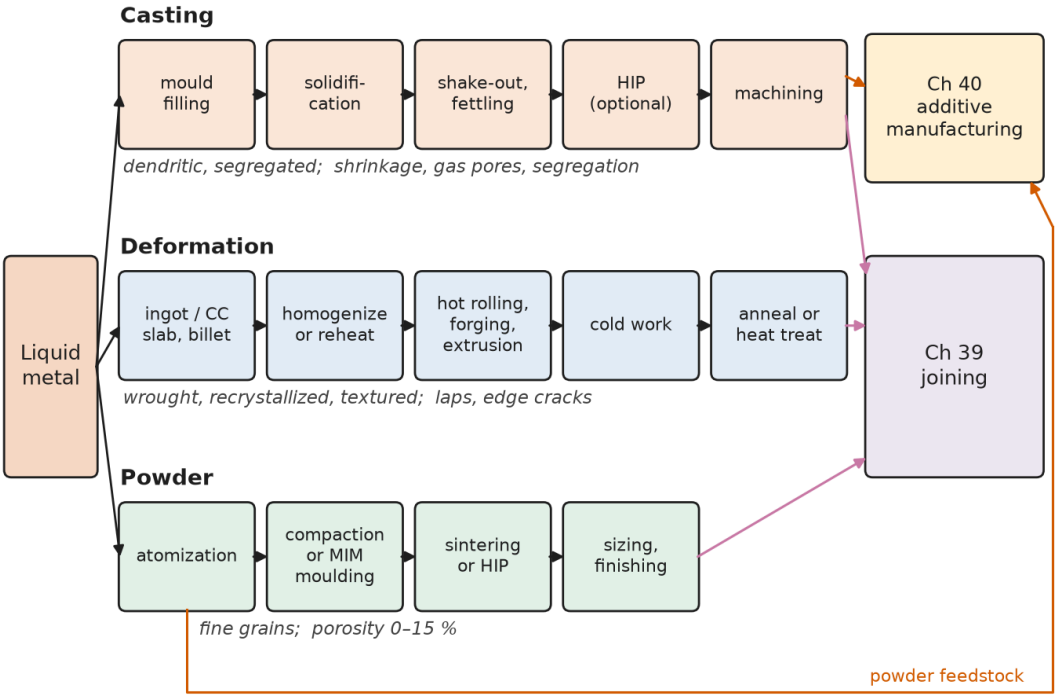


Figure 46.1 — The three bulk shaping routes from liquid metal to finished part, with the characteristic microstructure and defects of each.

46.2 Casting processes and casting design

Mould types

Casting processes differ mainly in the mould and in how the metal is driven into it.

- **Sand casting** uses silica (or zircon, chromite) sand bonded by clay and water ("green sand") or a chemical binder, formed around a pattern; internal cavities are made with sand **cores**. Any alloy, any size from 1 kg to 200 t, lowest tooling cost; the insulating sand makes cooling slow and the structure coarse.
- **Investment (lost-wax) casting** builds a ceramic shell around a wax pattern, melts out the wax and fires the shell. Accuracy and finish are excellent; it makes nickel-base turbine blades and Co–Cr and titanium implants.
- **Permanent-mould** and **low-pressure die casting** use reusable metal dies; low pressure (0.03–0.1 MPa) pushes metal quietly up a riser tube, giving sound aluminium wheels and suspension parts.
- **High-pressure die casting (HPDC)** injects Al, Mg or Zn alloys at 20–100 m/s gate velocity and holds them at 50–100 MPa while they freeze. Cycles take seconds and walls can be 1–3 mm thick, but the jet entrains air, so conventional HPDC parts cannot be solution treated (pores blister) or welded. Vacuum HPDC removes most of the air and now produces single structural "giga-castings" that replace dozens of stamped car-body parts.
- **Continuous casting** withdraws a solidifying shell from a water-cooled oscillating copper mould; about 96 % of the world's steel is continuously cast into slabs, blooms or billets before rolling.
- **Centrifugal casting** makes pipe, cylinder liners and bimetallic rolls; **directional solidification** withdraws an investment mould slowly from a hot zone so grains grow as ⟨001⟩ columns, and a helical grain selector produces single-crystal turbine blades (Chapter 16).

Fluidity and mould filling

Fluidity, the length a metal flows in a standard spiral channel before freezing, increases with superheat and is highest for pure metals and eutectics, which freeze with a smooth front. Alloys with a wide freezing range form a mushy dendrite network at the flow tip that chokes the channel. This is one reason casting alloys sit near eutectics: Al–Si alloys (A356 at 7 % Si, A380 at 8.5 % Si, eutectic 12.6 % Si), grey iron near the Fe–C eutectic, and solders.

The gating system must fill the mould without turbulence. When the surface of liquid aluminium folds over, its oxide skin is entrained as a doubled film, a **bifilm**, which is a ready-made crack. Gate velocities below about 0.5 m/s, tapered sprues and ceramic foam filters keep films out.

Solidification time: Chvorinov's rule

In a sand mould, freezing is limited by how fast the mould absorbs heat. Treat the mould as a semi-infinite solid at T_0 whose surface is raised suddenly to the freezing temperature T_m; the heat it absorbs per unit area by time t is

$$Q_A(t) = \frac{2}{\sqrt{\pi}}\,(T_m - T_0)\sqrt{k_m \rho_m c_m\, t}$$

Equating $Q_A A$ to the latent heat and superheat of the casting, $\rho_s V(L + c_s \Delta T_s)$, gives **Chvorinov's rule**:

$$t_f = B\left(\frac{V}{A}\right)^2, \qquad B = \frac{\pi}{4}\left[\frac{\rho_s\,(L + c_s \Delta T_s)}{T_m - T_0}\right]^2 \frac{1}{k_m \rho_m c_m}$$

The **casting modulus** $M = V/A$ is volume per unit cooling surface: compact, heavy sections freeze last. The constant B is measured; for steel in silica sand it is of order 1–2 s/mm^2.

> **Status of the principle: Chvorinov's rule**
> *Approximate.* Derived for an insulating semi-infinite mould controlling heat flow. It works for compact sand and investment castings; it fails in metal moulds (interface resistance dominates and t_f grows nearly linearly with M), in thin sections where superheat matters, and at corners. The exponent is often fitted as 1.5–2.

Feeding and riser design

Most metals shrink on freezing: about 3–4 % for steel, 6–7 % for aluminium alloys, 4–5 % for copper alloys. Castings are therefore fed from **risers** that must freeze after the section they feed, and are designed for **directional solidification** toward the riser, helped by tapers and **chills** (metal inserts that locally speed cooling). Grey and ductile irons are the exception: graphite precipitates with a volume expansion that offsets much of the shrinkage, which is one reason cast iron casts so readily.

Two rules size a riser. The **modulus rule** asks for a riser modulus about 1.2 times that of the section it feeds. The **volume rule** asks for enough liquid: an open cylindrical riser delivers only about 14 % of its volume before its pipe reaches the casting; insulating or exothermic sleeves raise this to 30–60 %.

Worked example — Sizing the riser of a steel plate casting. A plate 250 mm × 250 mm × 50 mm is sand cast with a cylindrical side riser of height equal to its diameter. Take $B = 1.5$ s/mm^2 and 3 % volumetric shrinkage.

$$V = 3.125 \times 10^6 \text{ mm}^3, \qquad A = 2(250 \times 250) + 4(250 \times 50) = 1.75 \times 10^5 \text{ mm}^2$$

so $M_c = 17.9$ mm and the plate freezes in $1.5 \times 17.9^2 = 478$ s, about 8 min. The riser face joined to the plate does not cool, so for $H = D$ the riser has volume $\pi D^3/4$ and cooling area $\pi D^2 + \pi D^2/4$, giving $M_r = D/5$. The modulus rule $M_r = 1.2 \times 17.9 = 21.4$ mm gives $D = 107$ mm; choose $D = H = 110$ mm, so $M_r = 22$ mm and the riser freezes in $1.5 \times 22^2 = 726$ s, about 12 min — after the plate.

Volume check: the plate needs $0.03 \times 3.125 \times 10^6 = 9.4 \times 10^4$ mm^3 of feed metal; the riser holds $\pi(110)^3/4 = 1.05 \times 10^6$ mm^3, of which about 14 %, 1.46×10^5 mm^3, is usable — a 55 % margin. A last check is **feeding distance**: for steel plates a rule of thumb is about 4.5 plate thicknesses (a riser zone of 2 plus an end zone of 2.5), 225 mm here, slightly short of the 250 mm from one edge to the other. The foundry would add a chill at the far edge or a second riser. After cutting off the riser, the yield of a steel casting is typically only 50–70 %.

Designing for casting

The designer's rules follow from this physics: keep **sections uniform** and fillets generous, because junctions create hot spots (an inscribed-circle check finds them); provide 1–3° **draft**; add **pattern allowances** for solid contraction (about 2 % for steel, 1.3 % for aluminium, 1 % for grey iron) and machining stock; avoid **hindered contraction** that tears the casting; and place gates so that sealing and bearing faces are free of inclusions. Mould-filling and solidification simulation is now routine, and the

Niyama criterion $G/\sqrt{\dot{T}}$ (thermal gradient over the square root of cooling rate; low values flag regions where feed metal cannot flow through the mush) predicts shrinkage porosity before a pattern is made.

46.3 Casting defects

Each casting defect has a mechanism, and each mechanism a remedy.

Shrinkage porosity. *Macroshrinkage* is a cavity or **pipe** at the last place to freeze. *Microshrinkage* is a sponge of jagged interdendritic cavities formed when liquid cannot flow through a wide mushy zone: the pressure drop for Darcy flow through the shrinking dendrite network rises steeply, so long-freezing-range alloys (bronzes, Al–Cu, Al–Zn–Mg) are prone to it.

Gas porosity. Gases dissolve far more in liquid than solid metal. For a diatomic gas, **Sieverts' law** gives

$$C_{\mathrm{H}} = K_S(T)\sqrt{p_{\mathrm{H_2}}}$$

In aluminium the hydrogen solubility drops about twentyfold on freezing (from about 0.69 to 0.036 mL per 100 g at 1 atm), so hydrogen is rejected into interdendritic liquid and forms round pores, often on bifilms. Melts are degassed with argon bubbled through a rotary impeller; dissolved hydrogen diffuses into the hydrogen-free bubbles and leaves. Steels are deoxidized with Al or Si to stop dissolved oxygen reacting with carbon to form CO. Gas pores are smooth and round, shrinkage pores jagged; telling them apart is the first step of any casting investigation (Chapter 44), because the remedies are opposite.

Hot tears. At solid fractions of roughly 0.9–0.99 the dendrites form a coherent network separated by thin liquid films with almost no strength. If contraction is hindered by a core, flange or thick neighbour, strain concentrates in these films and they separate into oxidized, branched, intergranular cracks. Susceptibility rises with freezing range — 2xxx and 7xxx aluminium alloys are hard to cast, weld (Chapter 47) and print (Chapter 48) — and falls with grain refinement.

Other defects include non-metallic inclusions and entrained films (fatigue initiators, controlled by filtration and clean melting), cold shuts and misruns (metal too cold or slow), macrosegregation such as centreline segregation in continuously cast slabs and "freckle" chains in directionally solidified superalloys, and mould–metal reactions.

Defect	Appearance	Mechanism	Main remedies
Macroshrinkage / pipe	Cavity at hot spot	Insufficient feed metal	Risers, directional solidification, chills
Microshrinkage	Jagged interdendritic pores	Feed flow blocked in mush	Steeper gradient, narrower freezing range, squeeze casting
Gas porosity	Smooth round pores	Rejected H (Al, Cu, Mg), N or CO (steel)	Degassing, deoxidation, vacuum die casting
Hot tear	Branched oxidized crack	Hindered contraction across liquid films	Grain refinement, compliant cores, fillets
Inclusions, bifilms	Oxide films, sand, slag	Turbulent filling, dirty melt	Filters, quiescent gating
Cold shut, misrun	Unfused seam, unfilled section	Metal too cold or slow	Superheat, gating, section thickness
Macrosegregation, freckles	Composition bands	Convection of solute-rich liquid	Faster cooling, stirring, alloy balance

Castings are inspected by radiography or computed tomography, ultrasound and dye penetrant, and graded against reference radiographs. Internal pores can be closed by **hot isostatic pressing** (Section 46.7), which is why cast turbine blades and aerospace titanium castings are routinely HIPed; surface-connected pores cannot, because the gas enters them.

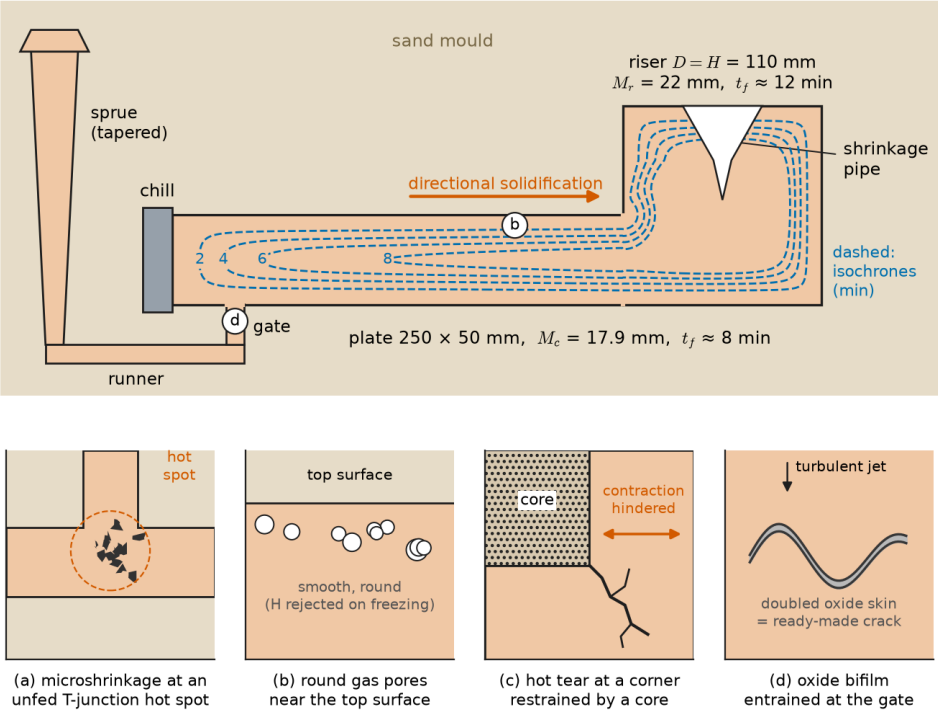


Figure 46.2 — Section through a sand-cast steel plate with side riser, showing directional solidification toward the riser and where casting defects form.

Figure 46.2 places these defects in the plate of the worked example.

46.4 Deformation processing

Deformation processes change shape by plastic flow at constant volume. Three ideas underlie the estimates below.

1. **Mean flow stress.** Below about $0.3T_m$ the Hollomon law $\sigma = K\varepsilon^n$ (Chapter 11) describes hardening. The work per unit volume to strain ε_1 is $\bar{\sigma}\varepsilon_1$, with
$$\bar{\sigma} = \frac{K\varepsilon_1^{\,n}}{1+n}$$
2. **Ideal work and efficiency.** Frictionless homogeneous deformation needs $\bar{\sigma}\varepsilon$ per unit volume. Friction and **redundant work** (internal shearing that does not change the shape) add to it; the ratio of ideal to actual work, the efficiency η, is typically 0.5–0.8.
3. **Plane strain.** Where flow is prevented in one direction, as in rolling, the von Mises criterion (Chapter 13) raises the yield stress by $2/\sqrt{3} = 1.155$.

> **Status of the principle: ideal-work and slab estimates of forming loads**
> *Approximate.* Energy balance and force equilibrium on thin slices with uniform flow stress and a simple friction law. Loads come within about 10–30 % for simple geometries; complex shapes and temperature fields need finite-element simulation.

Rolling

Rolls of radius R reduce a strip from h_0 to h_1 over a contact arc of projected length $L \approx \sqrt{R\,\Delta h}$. Friction drags the strip into the gap; it enters slower than the roll surface and leaves faster, with a **neutral point** between. Gripping requires $\tan\alpha \le \mu$ at entry, which gives the **maximum draft**

$$\Delta h_{\max} = \mu^2 R$$

Friction also builds a pressure peak at the neutral point, the **friction hill**, because metal on both sides is pushed against friction toward it. The mean roll pressure is the plane-strain flow stress times a factor that grows with $\mu L/\bar{h}$, so thin strip needs

disproportionate pressure — hence small work rolls supported by back-up rolls, and lubrication, for foil.

Worked example — Cold-rolling force for aluminium strip. Annealed aluminium-alloy strip 1000 mm wide is rolled from 4.0 to 3.0 mm with $R = 250$ mm and $\mu = 0.08$; its flow curve is $\sigma = 200\varepsilon^{0.2}$ MPa. Grip: $\Delta h_{\max} = 0.08^2 \times 250 = 1.6$ mm exceeds the 1.0 mm draft. The contact length is $L = \sqrt{250 \times 1.0} = 15.8$ mm, the strain $\varepsilon_1 = \ln(4/3) = 0.288$, and

$$\bar{\sigma} = \frac{200 \times 0.288^{0.2}}{1.2} = 130 \text{ MPa}$$

With $\bar{h} = 3.5$ mm and the approximate friction factor $1 + \mu L/(2\bar{h}) = 1.18$,

$$F \approx 1.155\,\bar{\sigma}\,w\,L\left(1 + \frac{\mu L}{2\bar{h}}\right) = 1.155 \times 130 \times 1000 \times 15.8 \times 1.18 = 2.8 \times 10^6 \text{ N}$$

about 2.8 MN (285 tonnes-force) per metre of width. Each pass hardens the strip and raises the next pass's force, so cold-rolling schedules include intermediate anneals. Rolling defects come from inhomogeneous flow: edge cracks at free edges in tension, alligatoring (splitting of the slab end), wavy edges or centre buckles from roll bending.

Forging

Open-die forging upsets and draws out metal between flat dies (large shafts, rings, generator rotors); **closed-die forging** fills a shaped cavity, with excess metal squeezed into a thin **flash** whose friction and chilling build up pressure to fill the corners (crankshafts, connecting rods, turbine discs, landing-gear beams). For upsetting a cylinder of diameter d and height h, slab analysis gives an average pressure $\bar{p} \approx \sigma_f(1 + 2\mu d/3h)$, so squat workpieces need high pressures; closed-die loads with flash reach 3–10 times σ_f times the projected area.

Forging's materials advantage is **grain flow**: inclusions and segregation bands are elongated along the metal flow and, in a good design, follow the contour so that the weak direction never crosses a fillet. A crankshaft machined from bar cuts across these fibres at every web; a forged one does not, and its fatigue strength is higher. Forging defects are **laps** and **folds** (oxidized surface folded inward) and internal bursts where the core is in tension.

Extrusion

Extrusion pushes a billet through a die to make constant-section products: aluminium window frames, heat sinks and rail-car profiles, copper tube and bus-bar. Direct extrusion suffers container friction; indirect extrusion avoids it. Hollow aluminium profiles are made in **porthole dies**, where the metal splits around a mandrel support and re-welds in the solid state inside the die. For extrusion ratio $R = A_0/A_f$, $p_{\text{ideal}} = \bar{\sigma}\ln R$; empirically $p \approx \bar{\sigma}(0.8 + 1.2\ln R)$ plus container friction. Aluminium 6xxx alloys are extruded at 450–520 °C at ratios of 10–100. The speed limit is **surface tearing**: deformation heating in the die land can raise the surface to the incipient-melting temperature, so hard 7075 extrudes at a few metres per minute and dilute 6063 at 50–100 m/min. **Central bursts** (chevron cracks) form on the axis of extrusions and drawn bars at small reductions with large die angles, where the centre is in hydrostatic tension.

Drawing

Drawing pulls rod, wire or tube through a converging die. Because the product is pulled, the drawing stress must stay below the flow stress of the exiting wire. The ideal drawing stress is $\sigma_d = \bar{\sigma}\varepsilon$ with $\varepsilon = \ln(A_0/A_f)$. For a non-hardening metal, $\sigma_d = \sigma_f$ gives $\varepsilon_{\max} = 1$, a 63 % area reduction. For a Hollomon material, $K\varepsilon^{n+1}/(1+n) = K\varepsilon^n$ gives

$$\varepsilon_{\max} = 1 + n$$

Friction and redundant work limit real passes to 20–45 %, and fine wire passes through 10–30 dies with intermediate anneals.

Worked example — Drawing copper rod. Annealed copper rod is drawn from 5.0 to 4.0 mm, with $\sigma = 315\varepsilon^{0.54}$ MPa and efficiency $\eta = 0.65$. The reduction is $r = 1 - (4/5)^2 = 36$ % and $\varepsilon = \ln(25/16) = 0.446$.

$$\bar{\sigma} = \frac{315 \times 0.446^{0.54}}{1.54} = 132 \text{ MPa}, \qquad \sigma_f = 315 \times 0.446^{0.54} = 204 \text{ MPa}$$

The ideal drawing stress is $132 \times 0.446 = 59$ MPa and the actual $59/0.65 = 91$ MPa, 45 % of the exit flow stress — a safety factor of about 2.2 against breaking. The force is $91 \times \pi(4.0)^2/4 = 1.14$ kN. The ideal limit $\varepsilon_{\max} = 1.54$ would allow a 79 % reduction; efficiency and the need for a margin are what restrict practical passes.

Sheet-metal forming

Sheet is **bent**, **stretched** over punches, **deep drawn** (flowing inward from a flange into a die) or combinations of these in stamping dies, and in hydroforming and incremental forming.

Springback. On unloading a bend, elastic strain near the neutral axis recovers and the bend opens. For loaded radius R_i and thickness t, an elastic–perfectly-plastic analysis gives

$$\frac{R_i}{R_f} = 4\left(\frac{\sigma_y R_i}{E t}\right)^3 - 3\,\frac{\sigma_y R_i}{E t} + 1$$

For 1 mm mild steel ($\sigma_y = 200$ MPa) bent to $R_i = 10$ mm, $\sigma_y R_i / E t = 0.0095$ and the radius opens by about 3 %; for a 700 MPa dual-phase steel it opens by about 11 %. Higher strength or lower modulus (aluminium) means more springback, compensated by overbending, coining or stretching.

Deep drawing. The flange is compressed circumferentially as it flows inward (and wrinkles unless held down), while the cup wall carries the punch force; failure is by necking in the wall near the punch radius. The **limiting drawing ratio** $\mathrm{LDR} = D_{\mathrm{blank,max}}/d_{\mathrm{punch}}$ is about 2.0–2.3 and rises with the **plastic strain ratio**

$$r = \frac{\varepsilon_w}{\varepsilon_t}$$

the ratio of width to thickness strain in a tensile test. Sheet with $r > 1$ resists thinning and draws in from the flange instead. The average $\bar{r} = (r_0 + 2r_{45} + r_{90})/4$ sets drawability; the **planar anisotropy** $\Delta r = (r_0 - 2r_{45} + r_{90})/2$ sets the height of **ears** on the cup rim. Both come from texture (Section 46.6). An 80 mm diameter cup 60 mm deep needs a blank of about $\sqrt{d^2 + 4dh} = 160$ mm, a drawing ratio of 2.0: interstitial-free steel with $\bar{r} \approx 1.9$ draws it in one stroke, while 5xxx aluminium with $\bar{r} \approx 0.7$ is at its limit and would be drawn and redrawn.

Sheet material	Yield strength (MPa)	n	$\bar{r}$	E (GPa)	Typical use
Interstitial-free steel DC06	120–180	0.22–0.25	1.8–2.2	210	Deep-drawn inner panels, fuel tanks
Mild drawing steel DC04	140–210	0.20–0.22	1.6–1.9	210	Body panels, appliances
Dual-phase steel DP600	340–420	0.14–0.17	0.9–1.0	210	Crash structure
Stainless 304, annealed	230–300	0.40–0.50	0.9–1.1	193	Sinks, food equipment
Al 6016-T4	110–140	0.23–0.27	0.6–0.7	70	Car outer panels
Al 5182-O	125–140	0.28–0.33	0.7–0.8	70	Car inner panels, can ends
CP titanium grade 2	275–350	0.12–0.16	2–4	105	Heat-exchanger plates

Typical values; n and r depend on gauge, supplier and test direction.

46.5 Hot, warm and cold working; formability

Homologous-temperature boundaries

"Hot" and "cold" refer to the **homologous temperature** T/T_m in kelvin.

- **Cold working**, below about $0.3T_m$: dislocations accumulate and the metal hardens; accuracy and finish are excellent and the strength gained can be kept (hard-drawn wire, cold-rolled sheet), at the cost of high forces and intermediate anneals.
- **Warm working**, about 0.3–$0.5T_m$: dynamic recovery lowers the flow stress while scaling stays modest. Steel is warm forged at 650–900 °C for precision automotive parts; magnesium sheet, short of slip systems at room temperature, is formed at 200–300 °C.
- **Hot working**, above about 0.5–$0.6T_m$: recovery and recrystallization during or between passes prevent cumulative hardening, so huge strains are possible at low force; cast structures are broken up and pores welded shut. The penalties are scale, poorer tolerances and grain growth if finishing is too hot.

Metal	T_m (K)	T/T_m at 25 °C	Typical hot-working range (°C)	T/T_m in that range
Lead	601	0.50	room temperature	0.50
Tin	505	0.59	room temperature	0.59
Aluminium (6xxx)	933	0.32	450–520	0.77–0.85
Copper	1358	0.22	750–950	0.75–0.90
Carbon steel	≈ 1800	0.16	950–1250	0.68–0.85
Ti–6Al–4V	≈ 1880	0.16	900–980	0.62–0.67
Tungsten	3695	0.08	1200–1700 (warm)	0.40–0.53

Lead and tin are effectively hot worked at room temperature, which is why lead flashing and solder can be bent repeatedly without hardening.

Flow stress at high temperature

In hot working, flow stress depends weakly on strain and strongly on rate and temperature. Over limited ranges $\sigma = C\dot{\varepsilon}^{\,m}$, with **strain-rate sensitivity** $m \approx 0.1$–0.2 in hot working, below 0.02 at room temperature and 0.3–0.8 in superplastic alloys (Chapter 13). Rate and temperature combine in the **Zener–Hollomon parameter**

$$Z = \dot{\varepsilon}\exp\!\left(\frac{Q}{RT}\right)$$

with Q close to the self-diffusion activation energy (about 150–160 kJ/mol for aluminium alloys, 270–420 kJ/mol for austenite). Flow stress is a single function of Z, usually fitted as $Z = A[\sinh(\alpha\sigma)]^{n'}$; high Z (fast, cool) behaves like cold work, low Z gives low stress, and the recrystallized grain size falls as Z rises.

> **Status of the principle: Zener–Hollomon scaling of hot flow stress**
> *Empirical*, with a physical basis in thermally activated glide and climb. It collapses data for one alloy over wide ranges of rate and temperature; the constants are fitted and change with microstructure (for example across the α–γ transformation of steel).

High-**stacking-fault-energy** metals — aluminium, ferritic steel — cross-slip and climb easily, reach a steady flow stress by **dynamic recovery**, and keep elongated grains filled with subgrains. Low- and medium-SFE metals — austenitic steels, copper, nickel alloys (Chapter 12) — store more dislocations and undergo **dynamic recrystallization**, with a peak in the flow curve followed by softening; between passes they also recrystallize statically. Which of these happens, and when, is the lever of thermomechanical processing.

Formability of sheet: the forming limit diagram

Sheet fails by **localized necking**, a narrow groove where thinning concentrates. In uniaxial tension diffuse necking starts at a true strain equal to n (Considère, Chapter 11), but sheet is stretched biaxially and the limit depends on strain path. The **forming limit diagram (FLD)** plots the major in-plane strain ε_1 at the onset of localized necking against the minor strain ε_2, for paths from uniaxial tension ($\varepsilon_2 = -\varepsilon_1/2$) through plane strain ($\varepsilon_2 = 0$) to equibiaxial stretching. The **forming limit curve** is V-shaped with its minimum FLC_0 at plane strain — which is why splits appear where sheet is stretched over long straight punch radii.

FLCs are measured by stretching gridded sheets over punches (Nakajima, Marciniak tests), with circles deforming into ellipses or with digital image correlation; the same grid analysis on try-out stampings shows which regions lie near the limit, and a margin of about 10 % strain defines the safe zone. On the left side, Hill's analysis gives a neck along a direction of zero extension, so $\varepsilon_1 + \varepsilon_2 = n$; on the right, the Marciniak–Kuczyński model follows the growth of a slightly thinner band, slowed by strain hardening and rate sensitivity. Both make FLC_0 rise with n, m and thickness. For low-carbon steels the empirical Keeler–Brazier estimate, in engineering strain per cent, is

$$\mathrm{FLC}_0\,(\%) \approx (23.3 + 14.1\,t)\,\frac{n}{0.21}$$

for t in millimetres and $n \leq 0.21$ (larger n is capped at 0.21). A 1.0 mm DC04 sheet gives about 37 %, a true strain of 0.32. A 6xxx-T4 aluminium sheet has FLC_0 near 0.21 true strain and DP600 near 0.18, and advanced high-strength steels can also fracture at sheared edges before any neck forms.

> **Status of the principle: the forming limit curve**
> *Empirical* as a measured characteristic, with *approximate* theoretical support from Hill and Marciniak–Kuczyński necking analyses. Valid only for proportional strain paths; multi-stage stamping changes the path and shifts the curve, so stress-based or path-independent criteria are then used.

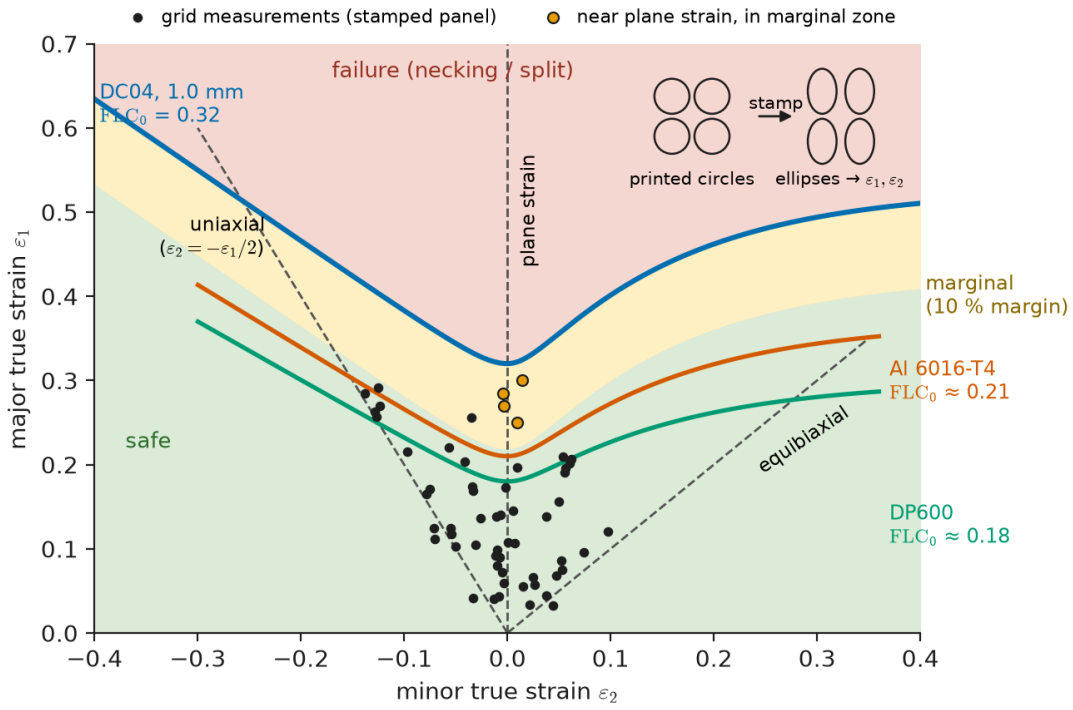


Figure 46.3 — Forming limit diagram for a mild drawing steel, a 6xxx aluminium sheet and a dual-phase steel, with strain paths and a safety margin.

Figure 46.3 makes the cost of lightweighting visible: changing from mild steel to aluminium or a stronger steel lowers the forming limit, so the part geometry or process (hot stamping of boron steel, warm forming, hydroforming) must change with it.

For forging and rolling, **workability** is limited by ductile fracture — void growth at inclusions under tensile hydrostatic stress (Chapter 14). Processes that keep the stress compressive (extrusion, closed-die forging) allow larger strains than those with free surfaces in tension, and **processing maps** from hot-torsion and upset tests identify windows free of flow localization and cracking, especially for titanium and nickel alloys.

46.6 Thermomechanical processing and texture control

Thermomechanical processing (TMP) combines deformation and thermal treatment in one controlled sequence, so that deformation alters how transformation, recrystallization or precipitation proceeds.

Controlled rolling of HSLA steel

The classic case is **high-strength low-alloy (HSLA)** steel for pipelines, ships, offshore structures and bridges (Chapter 15), microalloyed with 0.02–0.10 wt% Nb, Ti or V (Figure 46.4):

1. **Reheat** the slab to 1150–1250 °C to dissolve Nb carbonitrides; undissolved TiN pins austenite boundaries.
2. **Rough roll** at 1000–1150 °C, where austenite recrystallizes between passes and refines to about 30–50 µm.

3. **Finish roll** below the **non-recrystallization temperature** T_{nr}, about 900–1000 °C for Nb steels. Strain-induced Nb(C,N) on dislocations stops recrystallization, so grains are **pancaked** to a few micrometres thick and filled with deformation bands — a large density of ferrite nucleation sites.
4. **Accelerated cooling** with water at 10–30 K/s raises the undercooling further, giving ferrite grains of 3–5 µm instead of the 15–25 µm of a normalized C–Mn steel.

By Hall–Petch (Chapter 13) the finer grain adds roughly 150–200 MPa of yield strength while *lowering* the ductile–brittle transition temperature. Pipeline grades API 5L X70 (yield ≥ 485 MPa) and X80 (≥ 555 MPa) reach their strength with only 0.04–0.08 % carbon, which keeps their welds free of cold cracks (Chapter 47).

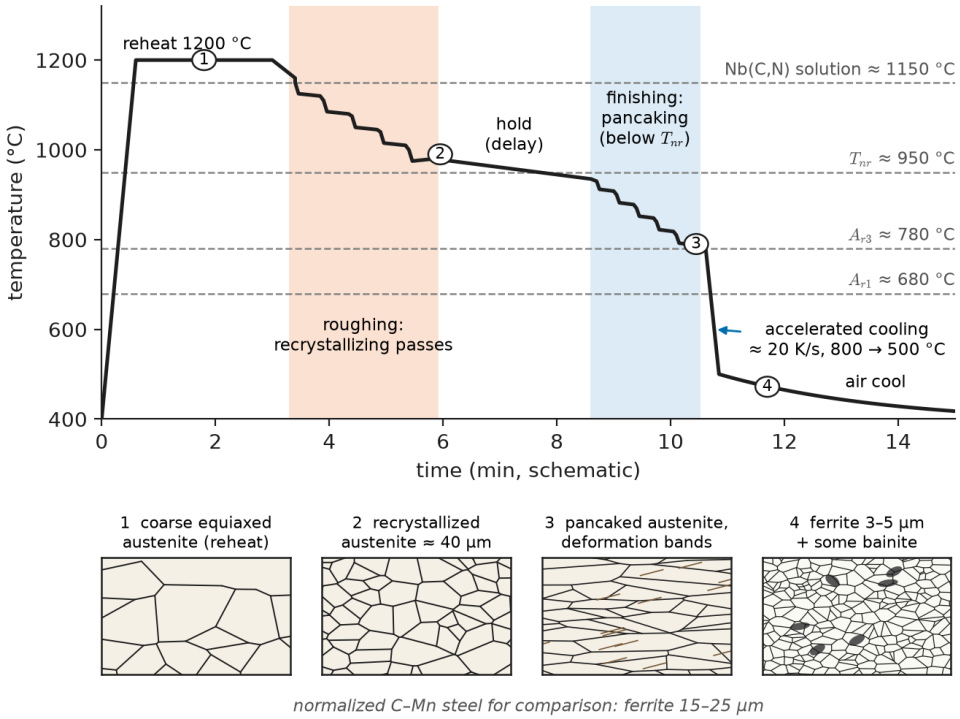


Figure 46.4 — Temperature–time schedule for thermomechanically controlled rolling and accelerated cooling of a Nb-microalloyed HSLA plate, with the corresponding microstructures.

Aluminium sheet and plate

Aluminium ingots are homogenized to dissolve eutectic phases and precipitate **dispersoids** (Mn-, Cr- or Zr-bearing, 0.05–0.5 µm), then hot rolled at 450–550 °C. Coarse constituent particles above about 1 µm nucleate recrystallization (**particle-stimulated nucleation**) with random orientations, while fine dispersoids pin boundaries (Zener pinning, Chapter 10); the balance sets grain size and texture. In 7xxx aerospace plate, Zr dispersoids keep the structure unrecrystallized, improving short-transverse toughness and stress-corrosion resistance.

Texture by design

Deformation rotates crystals toward stable orientations and recrystallization selects new ones, so rolled and annealed sheet has a **texture**, described by pole figures (Chapter 3). Because single-crystal properties are anisotropic (Chapter 4), texture makes sheet anisotropic, and TMP is used to engineer it.

- **Deep-drawing steel**: a strong **{111} "γ-fibre"** texture, with {111} planes parallel to the surface, gives $\bar{r} \geq 2$. It is produced by 70–85 % cold reduction and recrystallization of interstitial-free steel whose C and N are tied up by Ti and Nb.
- **Beverage cans**: can-body stock (3104-H19) is drawn and ironed to a 0.1 mm wall. Recrystallized **cube texture** {001}⟨100⟩ makes ears at 0° and 90° to the rolling direction; cold-rolling textures make ears at ±45°. Mills balance the two so the ears nearly cancel, because earing wastes metal and causes tears and jams on lines making thousands of cans per minute.
- **Electrical steel**: grain-oriented Fe–3 % Si develops the **Goss texture** {110}⟨001⟩, with the easy ⟨001⟩ axis along the rolling direction, by inhibitor-controlled abnormal grain growth (Chapters 10 and 28); core losses fall to about 1 W/kg at

1.7 T and 50 Hz.
- **Titanium and magnesium**: HCP rolling produces basal textures and strong anisotropy; α+β forging schedules of Ti–6Al–4V balance primary α, transformed β and texture to tune fatigue in fan discs.

> **Status of the principle: texture–anisotropy relation**
> *Derived.* Given single-crystal properties and an orientation distribution, polycrystal averaging (Voigt, Reuss, Taylor or self-consistent models) predicts anisotropy — quantitatively for elastic and magnetic properties, semi-quantitatively for r-values.

46.7 Powder metallurgy

Powder metallurgy (PM) is chosen for *economics* (net-shape mass production with over 95 % material use), *microstructure* (porous filters and bearings, WC–Co hardmetals, segregation-free tool steels, dispersion-strengthened alloys) or *unmeltable materials* (W, Mo, Ta, melting at 2600–3400 °C).

Powder production

- **Gas atomization**: argon or nitrogen jets break a melt stream into spherical droplets of 10–150 μm that freeze at 10^3–10^5 K/s; they flow and pack well and feed HIP, MIM and additive manufacturing (Chapter 48). Titanium is atomized crucible-free from induction- or plasma-melted electrodes.
- **Water atomization**: cheaper, irregular, slightly oxidized particles whose interlocking gives green strength; most press-and-sinter iron powder.
- **Chemical reduction** of oxides (sponge iron; W and Mo from oxides by hydrogen), **electrolysis** (dendritic copper), the **carbonyl process** (1–10 μm Fe and Ni spheres for MIM and magnetic cores), and **mechanical alloying** in ball mills (oxide-dispersion-strengthened alloys).

Powders are specified by size distribution, shape, apparent and tap density, flow rate and oxygen content, which forms oxide films on particle boundaries in the final part.

Compaction

Powder mixed with 0.5–1 % lubricant is pressed in a hardened die at 400–800 MPa. Densification proceeds by particle rearrangement, then plastic deformation of contacts, and slows as the contacts work harden. The **Heckel equation** fits the relation between relative density D and pressure P:

$$\ln \frac{1}{1-D} = KP + A$$

with K inversely proportional to the powder's yield strength. Soft iron powder reaches 6.8–7.2 g/cm^3 (86–92 % of 7.87 g/cm^3) at 600–800 MPa (Figure 46.5).

> **Status of the principle: Heckel compaction equation**
> *Empirical.* It treats pore removal as first order in pressure; it fits the plastic-deformation stage of ductile powders, not low-pressure rearrangement or the approach to full density.

The defining limitation is **die-wall friction**: pressure falls with distance from the moving punch, so density gradients limit height-to-diameter ratios to about 2–3 and cause non-uniform shrinkage on sintering. Double-action pressing, multi-level punches, **warm compaction** at 120–150 °C (up to 7.3–7.4 g/cm^3) and **cold isostatic pressing** in flexible moulds at 100–400 MPa (uniform density, no wall friction) address it.

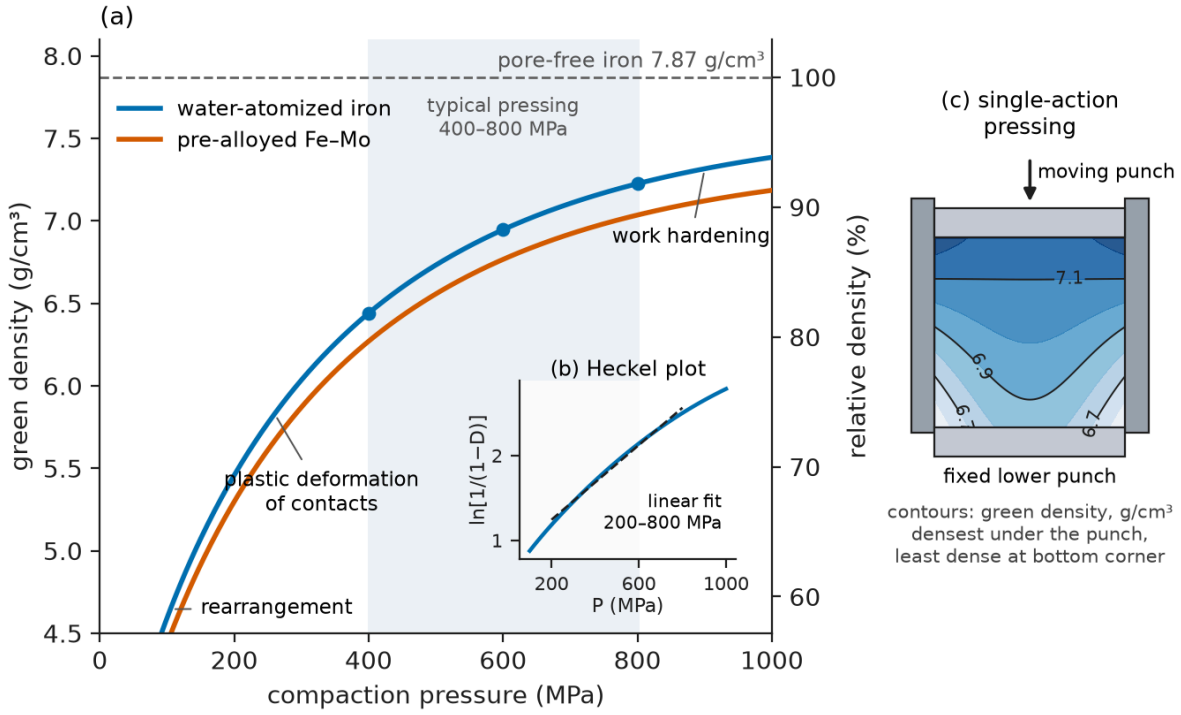


Figure 46.5 — Green density versus compaction pressure for iron and pre-alloyed steel powders, with the stages of compaction and the Heckel linearization.

Sintering

Green compacts are sintered in protective atmosphere at about 0.7–0.9T_m: 1120–1150 °C for iron and steel, 1250–1300 °C for high-temperature and stainless grades. The physics — surface-energy reduction, curvature-driven vacancy flow, densifying versus non-densifying paths, three geometric stages — is common to metals and ceramics and is developed in Chapter 17. For press-and-sinter steels:

- **Little densification**: necks grow and strength develops, but most of the 8–14 % porosity remains, so dimensional change is small and tolerances tight.
- **Alloying in the furnace**: graphite dissolves into iron within minutes; copper (1–3 %) melts at 1083 °C, spreads through the pores and swells the part, offsetting the shrinkage from carbon.
- **Liquid-phase sintering** gives full density. In **cemented carbides**, WC with 6–12 % Co sinters at 1350–1500 °C: a Co-rich liquid dissolves some WC, pulls grains together by capillarity and densifies the compact in under an hour with 17–20 % linear shrinkage, combining the hardness of WC with a tough binder for cutting inserts, drills and mining tools.
- **Infiltration** fills a porous skeleton with a lower-melting metal: copper into steel, copper or silver into tungsten for arcing contacts.

Porosity lowers properties more than in proportion to its volume. Empirically, for PM steels

$$\frac{E}{E_0} \approx \left(\frac{\rho}{\rho_0}\right)^{3.4}$$

so steel at 7.0 g/cm³ (89 % dense) has $E \approx 210 \times 0.89^{3.4} = 141$ GPa. Fatigue strength and toughness fall faster still because pores and sharp necks initiate cracks, which is why gear makers use double pressing, sinter hardening and rolling densification of tooth surfaces. Porosity can also be the point: oil-impregnated bronze bearings are 20–30 % porous, filters and implant coatings 30–50 %.

PM material	Density (g/cm³)	Relative density	Tensile strength (MPa)	Elongation (%)	Application
Fe–2Cu–0.8C, sintered	6.8	0.86	450–550	≈ 1	Gears, sprockets
Fe–2Ni–0.5Mo–0.5C, sintered and hardened	7.1	0.90	900–1200	< 1	Transmission gears
Bronze Cu–10Sn, oil impregnated	6.4	0.73	100–150	2–5	Self-lubricating bushes
WC–10Co hardmetal	14.5	≈ 1.00	TRS 3000–4000	—	Cutting inserts
Ti–6Al–4V, MIM + HIP	4.43	≈ 1.00	900–950	12–16	Medical, aerospace small parts
IN718 powder, HIPed and aged	8.2	≈ 1.00	1250–1400	15–20	Turbine discs, rings

Typical values; TRS = transverse rupture strength in bending.

Hot isostatic pressing

Hot isostatic pressing (HIP) applies 100–200 MPa of argon at about 0.7–0.9T_m (900–950 °C for titanium alloys, 1150–1250 °C for superalloys and tool steels) for one to several hours. Closed pores collapse by yielding, then creep and diffusion, and their surfaces bond. HIP is used to **densify castings and printed parts** (Chapter 48) — though not surface-connected pores or pores filled with insoluble argon, which can re-expand during later heat treatment — and to **consolidate powder** sealed in an evacuated steel can. Powder HIP gives segregation-free material with fine, uniform carbides or γ′: PM superalloy turbine discs, high-speed and tool steels with 10–20 % carbides, and large near-net valve bodies for oil, gas and nuclear plant. Powder at a tap density of about 65 % shrinks linearly by $1 - 0.65^{1/3} = 13$ % on reaching full density, and the can must be designed for that, often non-uniform, shrinkage. The defect peculiar to powder HIP is the **prior-particle boundary** network of oxides and carbides on the original powder surfaces, controlled by low powder oxygen.

Metal injection moulding

Metal injection moulding (MIM) mixes fine powder (2–20 μm) with a thermoplastic binder to a **feedstock** of about 55–65 vol% powder, injection moulds it into an oversized cavity, removes the binder by solvent, catalytic or thermal debinding, and sinters the fragile "brown" part at 0.85–0.95T_m (about 1350 °C for 17-4PH and 316L stainless steels) to 95–99 % density.

Worked example — Mould oversize for a MIM part. A feedstock with 60 vol% stainless-steel powder sinters to 98 % density. For isotropic shrinkage,

$$\frac{L_f}{L_0} = \left(\frac{0.60}{0.98}\right)^{1/3} = 0.849$$

so the part shrinks 15.1 % in every dimension and the cavity is oversized by $1/0.849 = 1.178$: a 40.0 mm watch case is moulded at 47.1 mm. A ±0.5 vol% variation in powder loading changes the final size by about ±0.3 %, ±0.11 mm on 40 mm — which is why feedstocks are compounded tightly and MIM tolerances are typically ±0.3–0.5 %. Gravity and friction on the setter plate distort thin or overhanging features, so MIM parts have uniform 1–6 mm walls.

MIM is economic for small (0.1–100 g), complex parts made in large numbers: orthodontic brackets, surgical-instrument jaws, watch and phone parts, firearm components and turbocharger vanes.

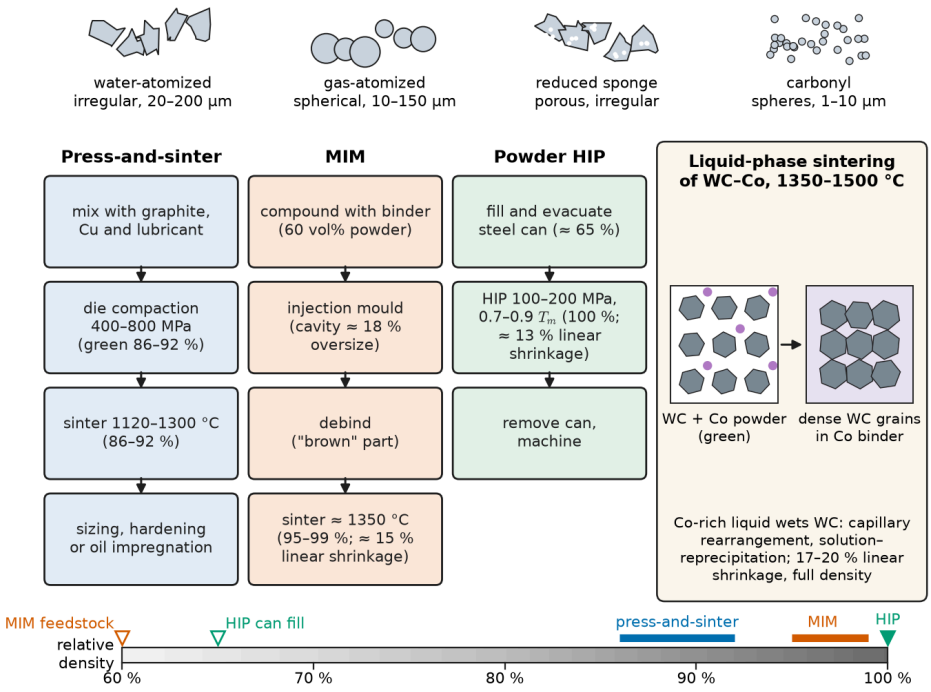


Figure 46.6 — Powder-metallurgy routes from powder to part: press-and-sinter, metal injection moulding and HIP consolidation, with typical densities.

Figure 46.6 sets the three powder routes side by side; part size, complexity, volume and tolerance of porosity decide between them.

46.8 Processing of ceramics and glasses

Ceramics and glasses use the same routes with different emphasis. Most technical ceramics — alumina, zirconia, SiC, Si_3N_4 — cannot practically be melt-cast, so they are made by **powder processing**: fine-powder synthesis, shaping by die pressing, CIP, slip casting, tape casting, extrusion or ceramic injection moulding, then sintering (often liquid-phase), hot pressing or HIP. The agglomerates, large pores and inclusions left by these steps set the statistical strength of ceramics (Weibull, Chapter 14). Glasses, by contrast, are formed almost entirely from the melt, exploiting the steady rise of viscosity on cooling: float glass on molten tin, bottles by press-and-blow, fibres drawn through bushings. Ceramic shaping and sintering, glass forming, annealing, tempering and glass-ceramics are treated in **Chapter 17**; polymer processing in **Chapter 21**.

46.9 How the route reaches the property sheet

Route	Structural variables set	Property consequences
Sand or gravity casting	Coarse dendrites, pores, segregation	Lower, scattered fatigue and ductility; section-size dependence
Die casting	Fine skin, porous core	Good static strength; limited heat treatment and welding unless vacuum cast
Investment casting + HIP	Controlled grains, pores closed	Creep strength (DS/SX); fatigue raised by HIP
Hot working	Recrystallized grains, closed pores, elongated inclusions	Toughness and fatigue anisotropic, grain flow along contour
Cold working	Dislocations, deformation texture	Strength up, ductility down
Thermomechanical processing	Fine ferrite, controlled texture	Strength with toughness; high $\bar{r}$; low earing
Press-and-sinter PM	8–14 % porosity, fine grains	Lower modulus and fatigue; excellent dimensional control
MIM, powder HIP	Near full density, homogeneous	Wrought-like properties in complex or segregation-prone alloys

Three lessons stand out. **Defects control the dangerous properties**: static strength follows the average microstructure, but fatigue, toughness and pressure tightness follow the largest defect, so a route is chosen as much for the defects it avoids as for the shape it makes. **Deformation is a microstructural tool**: hot working welds pores and replaces dendrites, controlled rolling refines grains beyond what heat treatment can, and rolling plus annealing writes texture. **Shape, quantity and material are chosen together**: the same alloy made as a die casting, a forging or a sintered part has different design allowables, which is why process selection and cost enter materials selection (Chapter 58).

What remains open

Open question	Where it is answered
What sets nucleation, dendrite spacing and Scheil segregation during casting?	Chapter 9
How do recovery, recrystallization, grain growth and Zener pinning proceed during and after working?	Chapter 10
What slip systems, Taylor factors and Hall–Petch constants lie behind texture and grain refinement?	Chapter 13
How do porosity and inclusions control fatigue and fracture?	Chapter 14
How are hot-rolled steels heat treated, and how do HSLA and AHSS grades differ?	Chapter 15
How are aluminium, titanium and superalloy products processed and tempered?	Chapter 16
How are ceramic powders shaped and sintered, and glasses formed and tempered?	Chapter 17
How are polymers moulded and composites laid up and cured?	Chapters 21 and 23
What does welding do to cast, wrought and PM parts?	Chapter 47
When is building a part from tiny melt pools better than casting or forging?	Chapter 48
How are porosity, texture and residual stress measured (CT, EBSD, diffraction)?	Chapters 52–53
How are casting, forging and HIP simulated and linked to properties?	Chapter 56
How is a manufacturing route selected and costed?	Chapter 58

Next: Chapter 47 takes shaped parts and asks how to join them — by melting, solid-state bonding, brazing, adhesives or fasteners — without making the joint the weakest link.

Chapter 47 — Joining: Welding, Brazing and Adhesive Bonding

Part X · Processing and Manufacturing

After the 1994 Northridge earthquake in Los Angeles, steel moment-frame buildings designed to bend plastically and survive were found to have cracked at more than a hundred beam-to-column connections — not in the ductile beams or columns, but at the welds joining beam flanges to columns. The cracks started at weld roots where a backing bar left a notch, ran through low-toughness weld metal, and sometimes tore into the column flange through the heat-affected zone. US connection design, filler-toughness and inspection rules were rewritten. The story is typical: a structure is only as good as its joints, and a joint is a small, locally processed region with its own microstructure, residual stresses and defects. The same lesson applies to a pipeline girth weld that cracks hours after welding, an aluminium panel that loses half its strength beside a weld, a bonded car roof that debonds in humid heat, or a carbon-fibre panel that corrodes the aluminium bolted to it.

> **What does a joining process do to the material around the joint, and why does that region so often decide the strength and life of a structure?**
> **How do we choose between fusion welding, solid-state welding, brazing, adhesive bonding and fastening — and design the joint so that it is not the weakest link?**

Concept	Engineering destination
Arc, laser, electron-beam and resistance welding	Ship hulls, pipelines, car bodies, pressure vessels, turbine parts
Heat input, Rosenthal heat flow, cooling time $t_{8/5}$	Welding procedure specifications, preheat and interpass control
Fusion zone and heat-affected zone metallurgy	Toughness of weld HAZs, softened zones in aluminium structures
Residual stress and distortion	Post-weld heat treatment, fatigue of welded details, fabrication tolerances
Carbon equivalent, hydrogen cracking, hot cracking	Preheat rules, consumable selection, alloy choice for welded structures
Friction stir welding, diffusion bonding	Aluminium rail cars and launch-vehicle tanks, titanium aero-engine structures
Brazing and soldering	Heat exchangers, carbide tools, ceramic–metal seals, electronics
Adhesive bonding and joint design	Aircraft panels, car body structures, wind-turbine blades, composite repairs
Fastening and dissimilar joints	Bolted steelwork, riveted aircraft, steel–aluminium car bodies, CFRP–metal joints

Roadmap. Classifying joints (47.1) → fusion welding processes and heat flow (47.2–47.3) → what welding does to the metal: fusion zone, HAZ, residual stress (47.4) → weldability and cracking (47.5) → solid-state welding (47.6) → brazing and soldering (47.7) → adhesive bonding (47.8) → fastening and dissimilar materials (47.9) → choosing a joining method (47.10).

> **Boundary.**
> Arc physics and power-source electronics belong to electrical and manufacturing engineering, and the thermodynamics of wetting and adhesion are developed in Chapter 45. Solidification (Chapter 9), solid-state transformations in steels (Chapters 10 and 15), fatigue and fracture (Chapter 14) and galvanic corrosion (Chapter 43) are used here as tools.

47.1 Joining as a materials problem

Joining methods differ in how they make atoms on the two sides of a joint bond to each other, or whether they bond at all.

- **Fusion welding** melts the edges of both parts, usually with added filler metal, and lets the pool solidify into a continuous metallurgical bond. It gives joints as strong as the base metal in most steels, but leaves a cast fusion zone and a heat-affected zone (HAZ).

- **Solid-state welding** brings clean metal surfaces into atomic contact by pressure, plastic deformation and diffusion, without melting: friction, friction stir, diffusion, ultrasonic, explosive and roll bonding. It avoids solidification defects and joins many dissimilar pairs.
- **Brazing and soldering** melt only a filler metal, with a liquidus below the solidus of the parts, which flows into the joint gap by capillarity and bonds by wetting and limited reaction. The parts do not melt.
- **Adhesive bonding** uses a polymer that wets the surfaces as a liquid and then cures; adhesion is by intermolecular forces, chemical bonds and mechanical interlocking.
- **Mechanical fastening** — bolts, rivets, screws, clinching — clamps parts together without any material continuity.

The engineering consequences follow from where the material is altered. Fusion welding changes the parts' microstructure over millimetres; brazing changes them only slightly; adhesive bonding and fastening hardly at all. Conversely, a weld transmits load through continuous metal, while an adhesive transmits it through a thin polymer layer of modest strength and a fastener through bearing, friction and holes that concentrate stress.

47.2 Fusion welding processes

Heat sources and power density

A fusion welding heat source is characterized by its **power density** at the work surface. A low-density source such as an oxy-acetylene flame (about 10^6–10^7 W/m^2) heats a wide area slowly, wastes most of its heat into the part and makes a broad, shallow weld with a wide HAZ. An arc concentrates 10^7–10^8 W/m^2; lasers and electron beams 10^{10}–10^{12} W/m^2. Above roughly 10^{10} W/m^2 the surface vaporizes, the recoil pressure of the vapour drills a narrow cavity, the **keyhole**, and energy is deposited along its whole depth: the weld becomes deep and narrow, with depth-to-width ratios up to about 10:1 for lasers and 30:1 for electron beams, a small HAZ and little distortion (Figure 47.1).

The heat delivered to the work per unit length of weld is the **heat input**

$$H = \eta \frac{V I}{v}$$

for arc voltage V, current I, travel speed v and arc efficiency η, the fraction of electrical power reaching the workpiece. Typical efficiencies (as tabulated in welding standards) are 0.6 for gas tungsten arc welding, 0.8 for gas metal arc and shielded metal arc welding, and 1.0 for submerged arc welding, where the flux blanket traps the heat.

Arc welding processes

- **Shielded metal arc welding (SMAW, "stick")**: a flux-coated consumable electrode whose coating shields the arc and forms slag. Portable — pipelines, site steelwork, repairs — but slow. The coating sets weld-metal hydrogen: cellulosic coatings release a lot, basic "low-hydrogen" coatings very little.
- **Gas metal arc welding (GMAW, MIG/MAG)**: a fed wire electrode under argon, CO_2 or mixtures; the workhorse of fabrication. **Flux-cored** wires (FCAW) raise deposition and suit outdoor work.
- **Gas tungsten arc welding (GTAW, TIG)**: a non-consumable tungsten electrode under inert gas with separate filler; slow, clean and precise, for root passes, thin stainless steel, titanium and aluminium.
- **Submerged arc welding (SAW)**: an arc under granular flux at 500–1500 A per wire and 10–40 kg/h with multiple wires; ship panels, wind-turbine towers, thick pressure vessels.
- **Plasma arc welding** constricts the arc to keyhole power densities for plates up to about 8 mm.

High-energy-density beam welding

Laser beam welding uses fibre, disc or CO_2 lasers of 1–20 kW focused to 0.2–1 mm spots: **tailor-welded blanks** (sheets of different gauge or grade joined before stamping), body-in-white seams at metres per minute, low-distortion ship panels, battery tabs. Absorption is a materials issue: copper and aluminium reflect over 90 % of near-infrared light until a keyhole forms, so green and blue lasers are being adopted for copper busbars. **Electron beam welding** at 60–150 kV in vacuum gives perfect shielding for titanium, zirconium and refractory metals and single-pass penetration beyond 100 mm in steel, for aero-engine shafts and nuclear components; the chamber limits part size.

Resistance welding

Resistance spot welding clamps overlapping sheets between copper-alloy electrodes and passes a current of about 8–12 kA for 0.1–0.3 s through 1 mm steel sheets. Joule heating, I^2Rt, is concentrated at the faying interface, where contact resistance is highest, and melts a lens-shaped **nugget**, typically required to have a diameter of at least $4\sqrt{t}–5\sqrt{t}$ for sheet thickness t in millimetres. A steel car body has several thousand spot welds. Aluminium, with lower resistivity, needs two to three times the current and wears electrodes quickly.

Process	Heat source	Power density (W/m²)	Typical heat input (kJ/mm)	Penetration per pass	Typical uses
Oxy-fuel	Flame	10^6–10^7	High, poorly controlled	A few mm	Repair, thin steel, cutting
SMAW	Arc, flux-coated electrode	10^7–10^8	0.8–3	2–5 mm	Site steelwork, pipelines
GMAW / FCAW	Arc, wire electrode	10^7–10^8	0.3–2.5	2–6 mm	General fabrication, car bodies
GTAW	Arc, tungsten electrode	10^7–10^8	0.3–1.5	1–3 mm	Root passes, thin stainless, Ti, Al
SAW	Arc under flux	10^7–10^8	1.5–6	5–15 mm	Ships, towers, pressure vessels
Plasma arc (keyhole)	Constricted arc	10^9–10^{10}	0.5–2	up to ≈ 8 mm	Tube, thin plate
Laser	Photon beam	10^{10}–10^{12}	0.05–0.5	up to ≈ 20 mm	Tailor blanks, body-in-white, battery tabs
Electron beam	Electron beam in vacuum	10^{10}–10^{12}	0.05–1	up to > 100 mm	Aero-engine, nuclear, thick Ti
Resistance spot	Joule heating at interface	—	—	Sheet thickness	Car bodies, appliances

Typical ranges; heat input varies with joint, position and procedure.

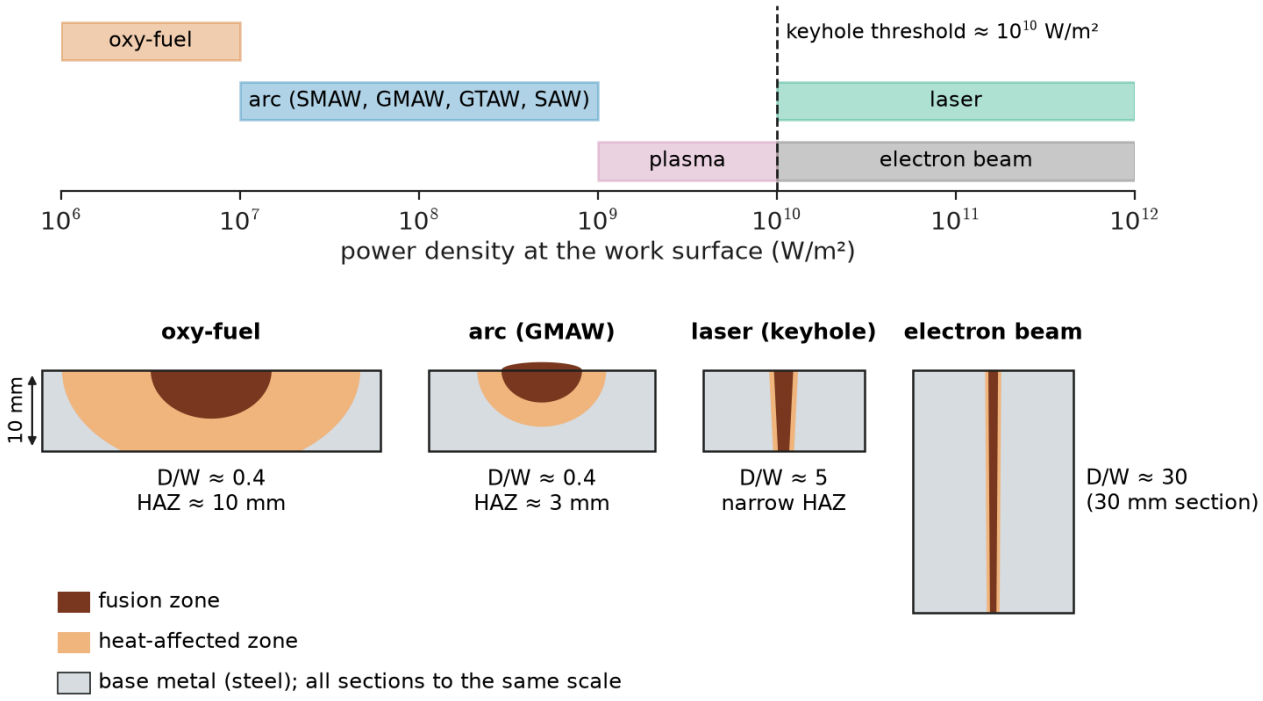


Figure 47.1 — Weld cross-sections produced by heat sources of increasing power density, from oxy-fuel to arc to laser and electron-beam keyhole welding.

Figure 47.1 shows how the same joint looks when made with each family.

47.3 Heat flow in welding

The Rosenthal solution

The metallurgy of a weld is set by its thermal cycle: the peak temperature reached at each point, and how fast it cools. Rosenthal's solution treats the heat source as a point (thick plate) or line (thin plate) moving at constant speed v through a solid of conductivity k and diffusivity α, with constant properties and no melting. In a frame moving with the source, a steady temperature field develops. For a **thick plate** (three-dimensional heat flow),

$$T - T_0 = \frac{\eta P}{2\pi k R}\exp\left[-\frac{v\,(R+\xi)}{2\alpha}\right]$$

where $P = VI$, ξ is the distance ahead of the source along the weld and R the distance from it. Isotherms are compressed ahead of the source and stretched behind it into a teardrop. Along the weld centreline far behind the source, the cooling rate is

$$\frac{dT}{dt} = -\frac{2\pi k\,(T - T_0)^2}{H}$$

with $H = \eta P / v$ the heat input. Integrating between 800 and 500 °C gives the cooling time through the range in which austenite transforms in most structural steels:

$$t_{8/5} = \frac{H}{2\pi k}\left(\frac{1}{500 - T_0} - \frac{1}{800 - T_0}\right)$$

For **thin plates** (two-dimensional flow) the corresponding result varies as $(H/d)^2$ for plate thickness d, so thin plates cool much more slowly for the same heat input. Welding standards use empirically calibrated versions of both; for steels the three-dimensional form is

$$t_{8/5} = (6700 - 5T_0)\,H\left(\frac{1}{500 - T_0} - \frac{1}{800 - T_0}\right)F_3$$

with H in kJ/mm, T_0 in °C, $t_{8/5}$ in seconds and a joint-shape factor F_3 (1 for a butt weld, about 0.67 for a fillet weld, which loses heat into three legs).

> **Status of the principle: Rosenthal heat flow and the $t_{8/5}$ formulas**
> *Approximate.* The solution of the heat equation for a moving point or line source with constant properties, no latent heat and no convection in the pool. It predicts the HAZ cooling rate well away from the pool and fails near and inside it, where the source's size, convection and latent heat matter. The standard $t_{8/5}$ equations are *empirical* calibrations of the same form.

Worked example — Cooling time and the effect of preheat. A 25 mm S355 steel plate is butt welded by GMAW at 28 V, 250 A and 300 mm/min (5 mm/s), with $\eta = 0.8$. The heat input is

$$H = 0.8 \times \frac{28 \times 250}{5} = 1120\ \text{J/mm} = 1.12\ \text{kJ/mm}$$

Is the plate thick enough for three-dimensional flow? The standards' transition thickness for this heat input and $T_0 = 20$ °C works out at about 15.5 mm, so 25 mm is thick. Without preheat ($T_0 = 20$ °C):

$$t_{8/5} = (6700 - 100) \times 1.12 \times \left(\frac{1}{480} - \frac{1}{780}\right) = 6600 \times 1.12 \times 8.01 \times 10^{-4} = 5.9\ \text{s}$$

The Rosenthal formula with $k = 30$ W/m K gives $1.12 \times 10^6/(2\pi \times 30) \times 8.01 \times 10^{-4} = 4.8$ s — the same order, slightly faster because it ignores the reduced conductivity and latent effects that the calibration absorbs. With a preheat of 100 °C,

$$t_{8/5} = (6700 - 500) \times 1.12 \times \left(\frac{1}{400} - \frac{1}{700}\right) = 6200 \times 1.12 \times 1.071 \times 10^{-3} = 7.4\ \text{s}$$

Preheating by 80 K lengthens the cooling time by about 25 % (Figure 47.2). For many C–Mn steels, the HAZ hardness rises steeply as $t_{8/5}$ falls below roughly 5 s, so this change can move the HAZ from a crack-susceptible martensitic structure to a tougher bainitic one. Preheat also slows cooling below 300 °C, where it matters even more by giving hydrogen time to diffuse out (Section 47.5). The same calculation, run the other way, shows why very high heat inputs ($t_{8/5}$ of 30 s or more) are restricted for thermomechanically processed steels: they coarsen the HAZ grains and lose the fine structure that gave the steel its toughness (Chapter 46).

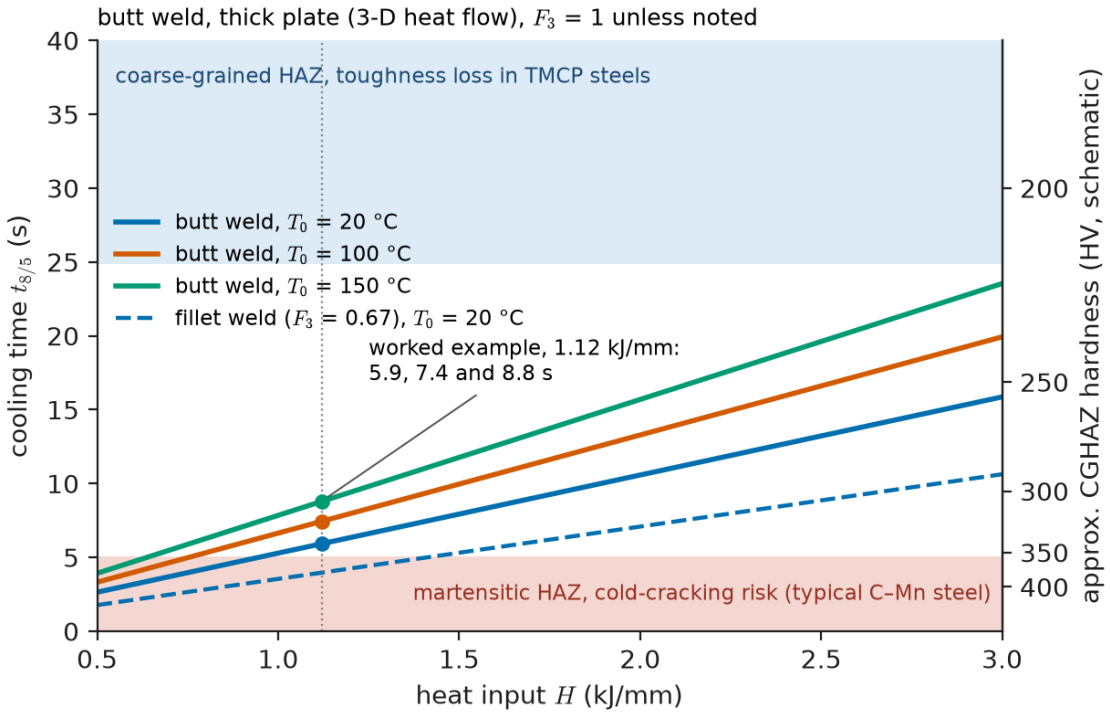


Figure 47.2 — Cooling time from 800 to 500 °C in the heat-affected zone of a thick steel plate as a function of heat input and preheat temperature.

47.4 Welding metallurgy

A fusion weld in cross-section has three regions: the **fusion zone** (weld metal), which was fully liquid; a narrow **partially melted zone** at the fusion boundary; and the **heat-affected zone**, base metal heated enough to change its microstructure but not to melt.

The fusion zone

The weld pool is a small casting with a peculiar mould: the hot, unmelted base metal of the same crystal structure. Solidification therefore starts without any nucleation barrier, by **epitaxial growth** from the partially melted grains at the fusion boundary, which continue across it with the same orientation. Grains then grow toward the pool's centreline along the steepest temperature gradient, and **competitive growth** favours those whose easy-growth ⟨100⟩ direction is best aligned with it. The result is columnar grains that curve toward the moving pool's trailing edge. As in Chapter 9, the solidification mode depends on the thermal gradient G and growth rate R: high G/R at the fusion boundary gives planar or cellular growth, low G/R at the centreline gives columnar dendritic or even equiaxed grains, and the product GR (the cooling rate) sets the scale of dendrites. Because $R = v\cos\theta$, with θ the angle between the growth direction and the welding direction, fast welding makes teardrop-shaped pools in which columnar grains meet head-on at the centreline — a weak plane where the last solute-rich liquid freezes and where solidification cracks form (Section 47.5).

Weld metal is a mixture of filler and melted base metal; the **dilution** (base-metal fraction) is typically 10–40 % for arc welds and up to 100 % for autogenous laser welds. Steel fillers add deoxidizers (Si, Mn, Ti) and are designed so that **acicular ferrite** — fine interlocking laths nucleated on titanium-oxide inclusions — dominates and gives high toughness. Titanium welds must be shielded until below about 400 °C or they absorb oxygen and embrittle (silver colour acceptable, blue or grey rejected).

The heat-affected zone in steels

Each point in the HAZ experiences a thermal cycle with a peak temperature that falls with distance from the fusion line. Superimposing the peak temperatures on the Fe–C diagram (Figure 47.3) divides the HAZ of a steel into sub-zones:

- **Coarse-grained HAZ (CGHAZ)**, peak above about 1100–1200 °C: austenite grains grow rapidly to 100 μm or more, increasing hardenability, so on cooling this zone forms the hardest and least tough structures — martensite or upper bainite. It is the critical region for both hydrogen cracking and toughness.
- **Fine-grained HAZ (FGHAZ)**, peak about 900–1100 °C: the steel is fully austenitized but grains do not grow, and it transforms to fine ferrite–pearlite or bainite. Usually the toughest zone.
- **Intercritical HAZ (ICHAZ)**, peak between A_{c1} and A_{c3} (about 720–900 °C): partial austenitization of carbon-rich regions, which can transform to brittle martensite–austenite islands.
- **Subcritical HAZ**, peak below A_{c1}: tempering or spheroidization of the base metal; in quenched-and-tempered steels a softened band.

In multipass welds, each pass reheats the previous ones, refining some CGHAZ and creating **intercritically reheated CGHAZ** regions with local brittle zones — one reason offshore-structure specifications require HAZ toughness tests (Charpy and CTOD) at several notch positions.

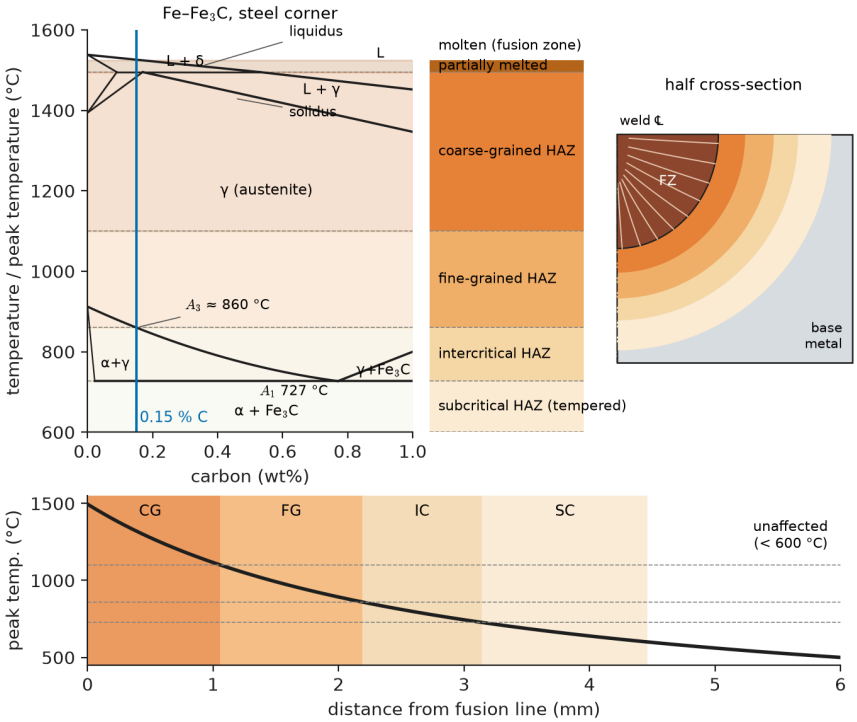


Figure 47.3 — Heat-affected-zone sub-regions of a 0.15 % C steel weld related to the peak temperature reached and the Fe–Fe3C phase diagram.

The HAZ in non-ferrous alloys

In alloys strengthened by cold work or precipitation, the HAZ is a region where that strengthening has been undone.

- **Cold-worked alloys** (strain-hardened aluminium 5xxx-H temper, cold-drawn copper) recrystallize in the HAZ and soften to the annealed (O-temper) strength.
- **Precipitation-hardened aluminium** (6xxx, 2xxx, 7xxx in T6 temper) is affected twice. Near the fusion line, precipitates dissolve; this region can partly recover by natural or artificial aging after welding. Farther out, at peak temperatures of roughly 250–400 °C, the fine β″ precipitates of a 6xxx alloy coarsen into non-strengthening β′ and β: the zone is **over-aged** and stays soft. Design codes therefore assign welded 6061-T6 an ultimate strength of about 165 MPa, compared with 260–310 MPa for the base metal, and structures place welds in low-stress regions or use friction stir welding (Section 47.6).
- **Nickel-base superalloys** and **titanium alloys** suffer coarsening of γ′ or α structures and, in the PMZ, liquation of low-melting phases.

Residual stress and distortion

As the weld pool cools, the hot weld metal and HAZ try to contract but are restrained by the cooler surrounding plate. The thermal strain available is enormous: for steel cooling through 600 K, $\alpha \Delta T = 12 \times 10^{-6} \times 600 = 0.0072$, which elastically would correspond to $E\alpha\Delta T \approx 1440$ MPa, far above the yield strength. The weld region therefore yields in tension during cooling, and once cold it is left with **longitudinal residual stress approximately equal to the yield strength** of the weld or HAZ, balanced by compressive stress in the surrounding plate. Transverse residual stresses are smaller and depend on restraint.

Residual stresses have three consequences. (1) **Distortion** — shrinkage, angular distortion and buckling of thin panels — controlled by sequencing, presetting, clamping and lower heat input (laser welds distort far less than arc welds). (2) **Fatigue**: tensile residual stress at the weld toe, with the toe's geometric stress concentration and small slag intrusions, makes the fatigue strength of welded details almost independent of steel strength, so design codes classify details by geometry, not grade (Chapter 14). Toe grinding, TIG dressing and peening introduce compression and raise it. (3) **Stress-assisted failure**: residual stress adds to service stress in brittle fracture, hydrogen cracking and stress-corrosion cracking (Chapter 43).

Post-weld heat treatment (PWHT), typically 580–620 °C for C–Mn pressure-vessel steels for about one hour per 25 mm of thickness, lowers the yield strength enough for creep to relax residual stresses to perhaps 20–30 % of their as-welded values, and

tempers HAZ martensite. Residual stresses are measured by hole drilling, X-ray and neutron diffraction (Chapter 53) and contour methods.

> **Status of the principle: residual stress near yield in as-welded joints**
> *Approximate*. It follows from thermal strains far exceeding the elastic limit under restraint. It is a good design assumption for longitudinal stress in steel and aluminium welds; it overestimates in low-restraint thin sheet and is modified where solid-state phase transformations on cooling (martensite's volume expansion) put the weld into compression.

47.5 Weldability and weld cracking

Weldability is the ability of a material to be welded into a structure that performs satisfactorily, under practical fabrication conditions. It is limited mainly by cracking, of which there are two families: **cold cracking**, which occurs after the weld has cooled, and **hot cracking**, which occurs during or immediately after solidification.

Hydrogen-induced cold cracking

Hydrogen cracking — also called cold cracking, delayed cracking or underbead cracking — is the most common cause of weld rejections in structural steels. It requires four factors at once:

1. **Hydrogen** in the weld, from moisture in fluxes and electrode coatings, from hydrocarbons and rust on the joint, or from the atmosphere. Diffusible hydrogen is classified in mL per 100 g of deposited weld metal: H5 (≤ 5), H10, H15; basic low-hydrogen electrodes and solid wires give under 5, cellulosic electrodes more than 30.
2. **A susceptible microstructure**, chiefly high-carbon martensite in the CGHAZ or weld metal. A hardness above about 350 HV is a common warning level.
3. **Tensile stress**, from residual stress and restraint.
4. **A low temperature**, below about 150–200 °C, where hydrogen is trapped and embrittling rather than diffusing out.

Cracks may appear hours or days later, as hydrogen diffuses to triaxially stressed notches and martensite (Chapter 43), so susceptible welds are inspected after a 16–48 hour delay. Prevention attacks each factor: dry low-hydrogen consumables; preheat and interpass temperature (softer HAZ, time for hydrogen to escape); a post-heat hold at about 200–250 °C; less restraint.

Carbon equivalent

Since the susceptible microstructure depends on hardenability, empirical **carbon equivalents** combine alloying elements by their effect on HAZ hardenability. The most widely used, from the International Institute of Welding, is

$$\mathrm{CE_{IIW}} = \mathrm{C} + \frac{\mathrm{Mn}}{6} + \frac{\mathrm{Cr} + \mathrm{Mo} + \mathrm{V}}{5} + \frac{\mathrm{Ni} + \mathrm{Cu}}{15}$$

in wt%. Rough guidance is that steels with $\mathrm{CE_{IIW}} < 0.40$ seldom need preheat in ordinary thicknesses; 0.40–0.60 need preheat of the order of 50–200 °C depending on thickness, hydrogen level and heat input; above 0.60 preheat of 200 °C or more and often PWHT are required. For low-carbon (below about 0.12 % C) microalloyed and TMCP steels the formula overweights the alloying elements and the Ito–Bessyo parameter

$$P_{\mathrm{cm}} = \mathrm{C} + \frac{\mathrm{Si}}{30} + \frac{\mathrm{Mn} + \mathrm{Cu} + \mathrm{Cr}}{20} + \frac{\mathrm{Ni}}{60} + \frac{\mathrm{Mo}}{15} + \frac{\mathrm{V}}{10} + 5\mathrm{B}$$

is preferred. The **Graville diagram** (Figure 47.4) plots carbon content against CE and divides steels into an easily weldable zone of low carbon, a zone of higher carbon but low hardenability where controlling cooling rate suffices, and a zone of high carbon and high hardenability where preheat, low hydrogen and often PWHT are all needed.

> **Status of the principle: carbon equivalent**
> *Empirical*. The coefficients are regression fits of HAZ hardness or cracking tests on steels of particular composition ranges. They rank steels within those ranges well and should not be extrapolated: $\mathrm{CE_{IIW}}$ suits C–Mn and low-alloy steels with more than about 0.12 % C, P_{cm} suits lower-carbon steels, and neither applies to stainless steels.

Worked example — Will this steel need preheat? Compare a structural S355 plate (0.18 C, 1.40 Mn, 0.30 Si, 0.05 Cr, 0.02 Mo, 0.01 V, 0.05 Ni, 0.10 Cu) with a quenched-and-tempered AISI 4140 shaft steel (0.40 C, 0.85 Mn, 1.0 Cr, 0.20 Mo).

$$\mathrm{CE}_{\mathrm{S355}} = 0.18 + \frac{1.40}{6} + \frac{0.05+0.02+0.01}{5} + \frac{0.05+0.10}{15} = 0.18 + 0.233 + 0.016 + 0.010 = 0.44$$

$$\mathrm{CE}_{4140} = 0.40 + \frac{0.85}{6} + \frac{1.0+0.20}{5} = 0.40 + 0.142 + 0.240 = 0.78$$

A rough check on how hard a fully martensitic HAZ would be uses Yurioka's empirical fit $H_M \approx 884\,\mathrm{C}\,(1 - 0.3\,\mathrm{C}^2) + 294$ HV:

$$H_M(0.18\,\mathrm{C}) = 884 \times 0.18 \times (1 - 0.3 \times 0.0324) + 294 = 452\ \mathrm{HV}, \qquad H_M(0.40\,\mathrm{C}) = 631\ \mathrm{HV}$$

The S355 lies in the middle band: at fast cooling its CGHAZ could exceed 350 HV, so for 25 mm plate a modest preheat (of order 75–125 °C, depending on hydrogen scale and heat input, as set by standards such as EN 1011-2 or AWS D1.1) with H5 consumables makes it safe — and the previous worked example showed what a 100 °C preheat does to $t_{8/5}$. The 4140, with a potential HAZ hardness above 600 HV and CE near 0.8, needs preheat around 250–350 °C, controlled interpass temperature, low-hydrogen practice and immediate PWHT; it is better not welded at all where it can be avoided. For comparison, a modern X70 pipeline steel with 0.08 C and 1.55 Mn plus small additions has $\mathrm{CE}_{\mathrm{IIW}} \approx 0.38$ and potential martensite hardness of about 365 HV, which is why such steels can be girth welded in the field with little or no preheat.

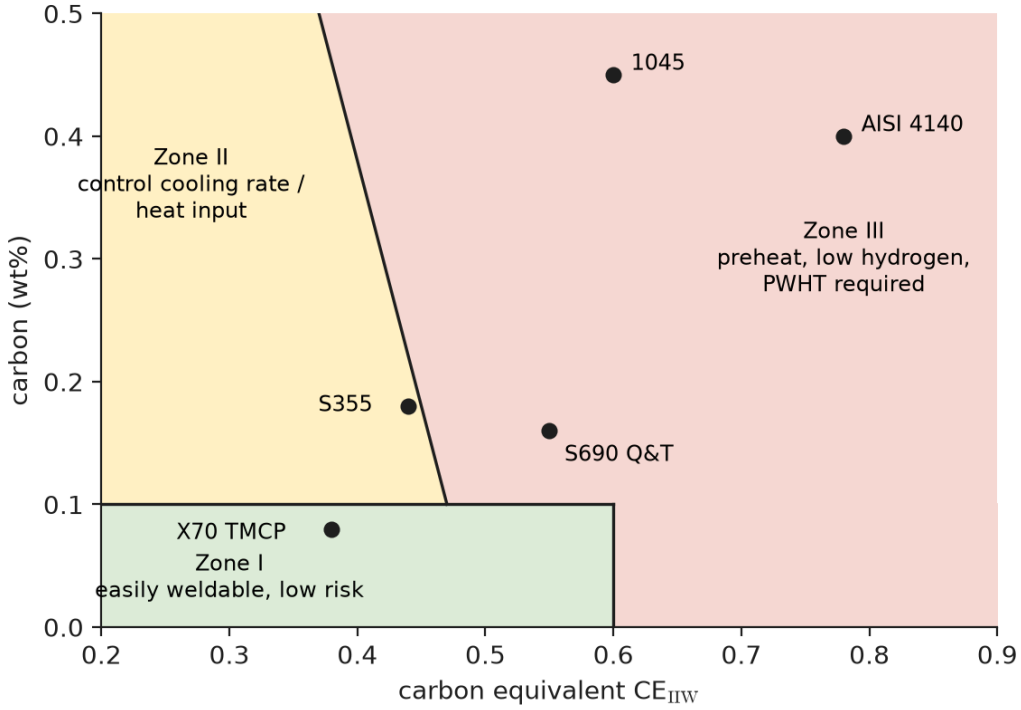


Figure 47.4 — Graville weldability diagram: carbon content versus carbon equivalent, showing zones of hydrogen-cracking susceptibility, with representative steels plotted.

Hot cracking

Solidification cracking occurs in the fusion zone at the end of solidification, by the same mechanism as hot tearing in castings (Chapter 46): at high solid fraction, thin films of solute-rich liquid along columnar grain boundaries cannot sustain the contraction strain, and they separate. Centreline cracks, where grains meet head-on, are typical. Susceptibility rises with the **freezing range** and with impurities that form low-melting films — in steels, sulfur and phosphorus (FeS melts near 988 °C), which is why steel welding consumables keep S and P low and add manganese to tie sulfur up as high-melting MnS. Bead shape matters: deep narrow beads with depth-to-width ratios above about 1 concentrate segregation at the centre.

In **aluminium alloys**, susceptibility versus solute content passes through a maximum at low solute contents (roughly 0.5–3 wt% of Si, Cu or Mg, depending on the system) and falls at higher contents where abundant eutectic liquid back-fills cracks. Alloys such as 6061 and 2024 are therefore welded with a high-solute filler — 4043 (Al–5Si) or 5356 (Al–5Mg) — that shifts the weld metal past the peak; autogenous welds crack. The 7xxx alloys are notoriously crack-prone. **Austenitic stainless steels**

are protected by solidifying first as δ-ferrite: weld metal formulated to contain about 3–8 % ferrite (by the WRC-1992 or Schaeffler diagram, Chapter 15) dissolves S and P better and resists cracking, while fully austenitic weld metal is susceptible.

Liquation cracking occurs in the partially melted zone when grain-boundary phases or segregates melt locally (constitutional liquation of carbides and sulfides in nickel alloys, eutectics in aluminium alloys) and are pulled apart by the contracting weld.

Other forms of weld cracking

- **Lamellar tearing**: step-like cracks parallel to the plate surface below T- and corner joints, where weld shrinkage pulls the plate through its thickness and elongated MnS inclusions decohere. Prevented by clean steels with guaranteed through-thickness ductility (for example "Z25" grades, with at least 25 % reduction of area in a through-thickness tensile test) and by joint design.
- **Reheat (stress-relief) cracking**: intergranular cracking in the CGHAZ of Cr–Mo–V and some nickel-alloy welds during PWHT, as carbides precipitate within grains and creep strain concentrates at boundaries.
- **Sensitization** of stainless-steel HAZs to intergranular corrosion by chromium-carbide precipitation (Chapter 43), avoided with low-carbon "L" grades or stabilized grades.

47.6 Solid-state welding

Solid-state processes avoid melting, so they avoid solidification cracking, porosity and the cast fusion zone, and they can join alloys that fusion welding cannot, including many dissimilar pairs. They work by bringing clean metal surfaces within atomic distances, which needs the oxide films to be broken up or dissolved and the surfaces brought into intimate contact.

Friction stir welding

Friction stir welding (FSW), invented at TWI in the UK in 1991, plunges a rotating non-consumable tool — a shoulder with a threaded or profiled pin — into the joint line between two clamped plates and traverses it along the joint. Friction under the shoulder and plastic work around the pin heat the metal to about 0.6–0.9T_m (roughly 400–500 °C for aluminium alloys) without melting it. The softened metal is swept around the pin from the **advancing side** (where tool rotation and travel add) to the **retreating side** and consolidated behind it under the shoulder's forging pressure. Typical parameters for aluminium are 200–2000 rpm and 0.1–1 m/min.

The joint has three zones (Figure 47.5): the **stir zone (nugget)**, where intense deformation at high temperature produces dynamically recrystallized equiaxed grains of 2–10 μm, often finer than the base metal; the **thermomechanically affected zone (TMAZ)**, deformed and heated but not recrystallized; and the **HAZ**, heated only. In precipitation-hardened aluminium the HAZ still over-ages, but the peak temperature and time are lower than in arc welding, so joint efficiencies (weld strength over base strength) of 70–90 % are typical for 6xxx and 7xxx alloys against 50–70 % for arc welds. The previously "unweldable" 2xxx and 7xxx alloys can be friction stir welded without cracking, and residual stress and distortion are low.

FSW makes aluminium rail-car and ship-deck panels, launch-vehicle propellant tanks and electric-vehicle battery trays, and joins aluminium to copper or magnesium. Its limits are clamping forces of several tonnes, the tool's exit hole, simple joint paths and tool wear in steel and titanium (cubic boron nitride or W–Re tools). Typical defects are tunnel voids on the advancing side at low heat input and "kissing bonds" where the original oxide was not broken up.

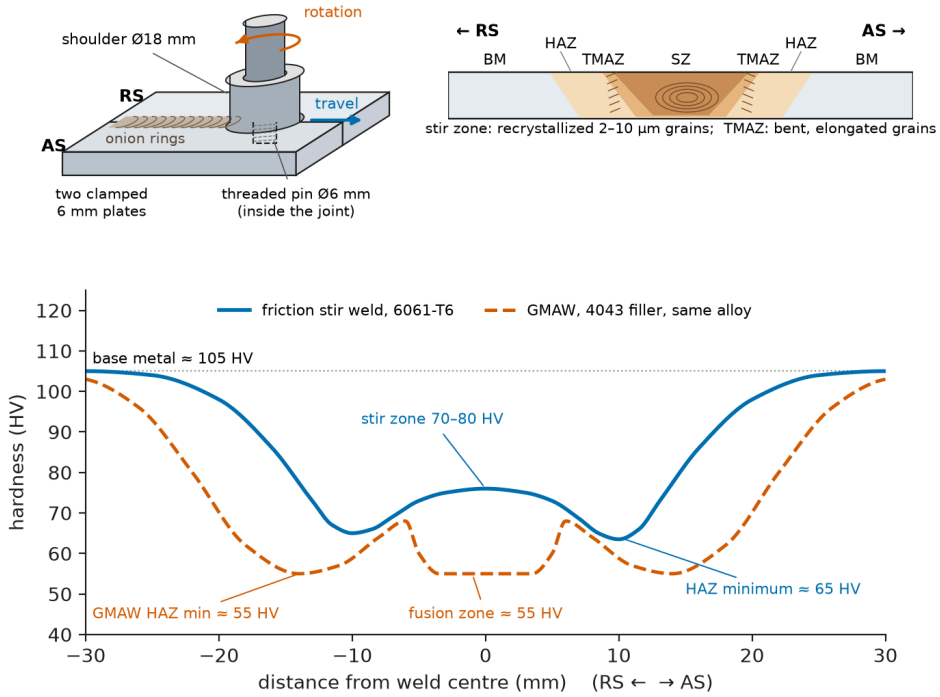


Figure 47.5 — Friction stir welding: tool, material flow and the microstructural zones of the joint in a precipitation-hardened aluminium alloy, with the hardness profile.

Other friction and pressure processes

- **Rotary and inertia friction welding** spins one part against another and then forges them together; used for drive shafts, valves, drill pipe and for joining dissimilar alloys in turbine rotor shafts. **Linear friction welding** oscillates a blade against a disc to make integrally bladed rotors (**blisks**) in titanium and nickel alloys.
- **Ultrasonic welding** vibrates sheets at about 20–40 kHz under modest pressure, scrubbing off oxides; it bonds battery tabs, wire harnesses and foils of Al and Cu with little heating.
- **Explosive welding** drives one plate obliquely into another at several hundred metres per second; a jet of surface material is ejected, and the clean surfaces bond with a characteristic wavy interface. It makes clad plate (stainless or titanium on steel) and **transition joints** for welding aluminium ship superstructures to steel decks.
- **Roll bonding** bonds ductile metals by heavy deformation that fractures surface oxide (clad coinage, brazing sheet).

Diffusion bonding

Diffusion bonding holds clean, smooth surfaces in contact at high temperature (about 0.5–0.8T_m) under a pressure too low to cause macroscopic deformation, for times of minutes to hours. Bonding proceeds as in sintering (Chapter 17): surface asperities deform plastically and creep, voids at the interface shrink by surface, boundary and volume diffusion, and finally the interface boundary migrates away from the original plane. Titanium alloys are ideal because their oxide dissolves into the metal at high temperature. Combined with **superplastic forming** (Chapter 13) at the same temperature — Ti–6Al–4V at about 900–930 °C, with bonding pressures of about 1–2 MPa for 1–2 hours — it makes hollow fan blades and stiffened aircraft structures from stacks of sheet in which stop-off compounds prevent bonding where cavities are wanted. **Transient-liquid-phase (TLP) bonding** places a thin interlayer containing a melting-point depressant (boron or silicon in nickel alloys) at the joint; it melts, then solidifies isothermally as the depressant diffuses away, leaving a joint whose remelting temperature is that of the base metal. It is used to repair and assemble nickel-base turbine components.

47.7 Brazing and soldering

Brazing and **soldering** join parts with a filler metal whose liquidus is below the solidus of the parts. By convention the dividing line is 450 °C: below it the process is soldering, above it brazing. The parts do not melt, so their microstructure changes

only through the thermal cycle, and dissimilar materials — steel to carbide, copper to stainless steel, metals to ceramics — can be joined.

Wetting and capillary flow

The filler must **wet** the surfaces (contact angle θ well below 90°, Young's equation, Chapter 45) and flow into the joint gap by capillarity. For two parallel plates separated by a gap h, the meniscus has radius $h/(2\cos\theta)$ and the capillary pressure drawing liquid into the gap is

$$\Delta p = \frac{2\gamma_{LV}\cos\theta}{h}$$

Oxide films prevent wetting, so the surfaces are cleaned and protected during heating by **fluxes** (borates and fluorides for silver brazing, fluoroaluminates for aluminium, rosin and organic acids for soldering) or by **controlled atmospheres** (hydrogen, which reduces many oxides; vacuum, which lets oxides dissociate or dissolve at high temperature).

Worked example — How far can capillarity pull a braze? A silver-base braze ($\gamma_{LV} \approx 0.9$ J/m^2, density about 10 000 kg/m^3) wets a cleaned steel joint with $\theta = 20^\circ$. For a gap of 0.05 mm,

$$\Delta p = \frac{2 \times 0.9 \times \cos 20^\circ}{5 \times 10^{-5}} = 3.4 \times 10^4 \text{ Pa}$$

equivalent to a column of liquid braze $\Delta p/(\rho g) = 3.4 \times 10^4/(10^4 \times 9.81) = 0.34$ m high. Capillarity can therefore fill a long joint vertically against gravity. With a 0.5 mm gap the pressure falls tenfold, and the braze rises only about 34 mm; it also no longer fills uniformly and leaves voids. This is why recommended clearances for silver brazes are about 0.025–0.125 mm: small enough for strong capillary flow, large enough for the liquid to enter and for flux and gases to escape. In a properly designed lap, the brazed joint can be stronger than the filler itself because the thin layer is constrained by stronger base metal on both sides, which raises its yield stress through triaxial constraint.

Filler metals and applications

Filler (AWS class or common name)	Composition	Melting range (°C)	Brazing / soldering temperature (°C)	Typical applications
Sn–37Pb (legacy solder)	Sn–Pb eutectic	183	210–240	Electronics before RoHS, repair
SAC305 solder	Sn–3Ag–0.5Cu	217–220	235–250	Lead-free electronics
Sn–58Bi solder	Sn–Bi eutectic	138	160–180	Low-temperature assembly
BCuP (Cu–P, Cu–P–Ag)	Cu–6–7P (± Ag)	≈ 645–800	700–815	Copper plumbing and HVAC (self-fluxing on Cu)
BAg (Ag–Cu–Zn–Sn)	40–56 Ag	≈ 620–780	650–800	Steel, stainless, copper, carbide tool tips
BAlSi-4	Al–12Si	577–582	590–605	Aluminium automotive heat exchangers
Copper	Cu	1083	1100–1150	Furnace brazing of steel parts in H_2
BNi-2	Ni–7Cr–3Fe–4.5Si–3B	≈ 970–1000	1010–1180	Vacuum brazing of turbine and stainless parts
Active braze (Ag–Cu–Ti)	Ag–27Cu–2Ti	≈ 780–815	830–900	Ceramic–metal joints (alumina, Si_3N_4, diamond)

Typical values; exact ranges depend on the specification.

Several materials points stand out. **Copper–phosphorus** fillers self-flux copper but form brittle phosphides on steel and nickel. **Aluminium brazing** works in a narrow window between the filler's melting (577 °C) and the base solidus (about 630–650 °C for 3xxx); radiators and condensers use **brazing sheet**, a 3xxx core clad with Al–Si, furnace brazed with fluoroaluminate flux. **Nickel braze** fillers contain boron and silicon as melting-point depressants; boron diffuses rapidly into

the base metal and can form brittle borides in wide gaps, so diffusion-brazing cycles hold long enough to dissolve them. **Active brazes** containing titanium react with ceramics to form a thin wettable layer (TiO, TiN or TiC), allowing metals to be brazed to oxides, nitrides and diamond without prior metallization.

In all brazed and soldered joints, the filler reacts with the base metal to form **intermetallic compounds** at the interface — Cu_6Sn_5 and Cu_3Sn in tin solders on copper, Ni_3Sn_4 on nickel finishes. A thin continuous IMC layer is the bond; a thick one, grown during prolonged soldering or in service by diffusion with $x \propto \sqrt{Dt}$ (Chapter 8), is brittle and a common failure site. The detailed metallurgy of solder joints and electronic packaging is treated in Chapter 51.

47.8 Adhesive bonding

Structural adhesive bonding joins parts with a polymer layer typically 0.1–0.5 mm thick. It distributes load over an area instead of concentrating it at welds or holes, joins dissimilar and thin materials without heat damage, seals the joint and damps vibration. Aircraft have used bonded aluminium skins and honeycomb since the 1940s; modern car bodies contain 10–20 m or more of structural adhesive alongside spot welds and rivets; wind-turbine blades are two composite shells bonded along their edges and to internal shear webs; and composite repairs are bonded patches.

How adhesives stick

Adhesion arises from (1) **intermolecular forces** — van der Waals and acid–base interactions across an intimately wetted interface; (2) **chemical bonds**, for example between epoxy hydroxyl groups or silane coupling agents and surface oxides; and (3) **mechanical interlocking** into a porous or rough surface such as an anodic oxide. All three require the liquid adhesive to wet the surface, which requires the surface energy of the adherend to exceed the surface tension of the adhesive (Chapter 45). Metals with clean oxides have high surface energies and are easily wetted; low-energy polymers such as polyethylene and polypropylene (about 30 mJ/m^2) are not, and must be activated.

Surface preparation

Most service failures of adhesive bonds are interfacial and trace back to surface preparation, which must remove oils, release agents and weak oxides and leave a surface stable in moisture.

- **Aluminium**: degreasing, etching and anodizing. Aerospace **phosphoric acid anodizing** grows a porous, whiskered oxide that the primer penetrates and that resists hydration to weak hydroxide in humid service — the failure that destroyed some early bonded structures. Automotive practice uses Ti–Zr conversion coatings.
- **Steel**: degreasing and grit blasting, often with a silane primer; galvanized steel bonds to the zinc layer.
- **Composites**: abrading or removing a peel ply to expose fresh resin.
- **Polymers**: corona, flame or plasma treatment to oxidize the surface and raise its surface energy above about 40 mJ/m^2.

Joint durability is qualified by tests such as the **wedge test**, which drives a wedge into a bonded double-cantilever specimen exposed to hot, humid air and measures crack growth; a crack that runs along the interface rather than through the adhesive indicates poor preparation.

Adhesive types

Adhesive family	Cure	Typical lap-shear strength (MPa)	Service temperature (°C)	Typical use
Toughened epoxy (one-part, heat cured)	120–180 °C	25–40	−55 to 120 (higher for some)	Aircraft structure, car bodies (crash-durable adhesives)
Epoxy (two-part)	Room temperature or warm	15–30	−40 to 80	Composite repair, wind-turbine blades, civil strengthening
Structural acrylic (MMA)	Room temperature, fast	15–25	−40 to 100	Marine, bus and truck panels
Polyurethane	Moisture or two-part	5–15	−40 to 90	Windscreens, sandwich panels
Cyanoacrylate	Moisture, seconds	10–20	−40 to 80	Small parts, medical
Silicone	Moisture	1–3	−60 to 200	Glazing, electronics
Anaerobic	Absence of air, metal contact	10–25 (in shear on threads)	−55 to 150	Thread locking, retaining bearings

Typical values; strength depends strongly on adherends, preparation, bondline thickness and test.

Joint design: stresses in a lap joint

The single-lap shear test is the standard way to measure adhesive strength, but the stress in a lap joint is far from uniform. Because the adherends stretch under load, more at the loaded end of each, shear strain in the adhesive concentrates at the two ends of the overlap. Volkersen's **shear-lag analysis** of a lap joint between identical adherends of modulus E and thickness t, with adhesive shear modulus G_a and thickness η_a, gives a peak-to-average shear stress ratio

$$\frac{\tau_{\max}}{\tau_{\text{avg}}} = \frac{\omega L}{2}\coth\!\left(\frac{\omega L}{2}\right), \qquad \omega^2 = \frac{2G_a}{E\,t\,\eta_a}$$

for overlap length L. When ωL is small the stress is uniform; when it is large, all the load is carried within a distance of about $1/\omega$ of each end and the middle of the overlap is nearly unloaded.

> **Status of the principle: Volkersen shear-lag model**
> *Approximate.* It treats the adherends as bars in tension and the adhesive as a linear-elastic shear spring, neglecting bending and peel. It captures the end concentration of shear stress; the Goland–Reissner analysis adds the peel stresses caused by the eccentric load path, and adhesive plasticity redistributes stress before failure.

Worked example — Does a longer overlap help? Two 2 mm aluminium sheets ($E = 70$ GPa), 25 mm wide, are bonded with a 0.2 mm epoxy layer of shear modulus 1.0 GPa. Then

$$\omega = \sqrt{\frac{2 \times 1.0 \times 10^{9}}{70 \times 10^{9} \times 0.002 \times 0.0002}} = 267\ \text{m}^{-1}$$

so $1/\omega = 3.7$ mm. With a 25 mm overlap, $\omega L/2 = 3.34$ and $\tau_{\max}/\tau_{\text{avg}} = 3.34\coth 3.34 = 3.35$. If the adhesive is taken to behave elastically up to a shear stress of 30 MPa, the joint reaches that stress at the overlap ends at a load of

$$F = \frac{30}{3.35} \times (25 \times 25) = 5.6\ \text{kN}$$

although the average-stress estimate would suggest 18.8 kN. Doubling the overlap to 50 mm gives $\omega L/2 = 6.68$, a ratio of 6.68, and the elastic-limit load stays at 5.6 kN: the extra overlap carries almost nothing (Figure 47.6). Halving it to 12.5 mm gives 5.2 kN. In the elastic regime, the joint capacity is set by ω, not by L, once L exceeds a few times $1/\omega$. Real joints do better because toughened adhesives yield at the ends and spread the load, so strength rises with overlap until the plastic zones meet — which is why ductile, toughened adhesives are used in structures. The analysis also shows how to improve a joint: stiffer or thicker adherends (steel adherends of the same thickness give $\omega = 154\ \text{m}^{-1}$ and a ratio of 2.0), a more compliant or thicker adhesive layer, and **tapered** adherend ends, scarf joints or stepped laps that make the load transfer gradual.

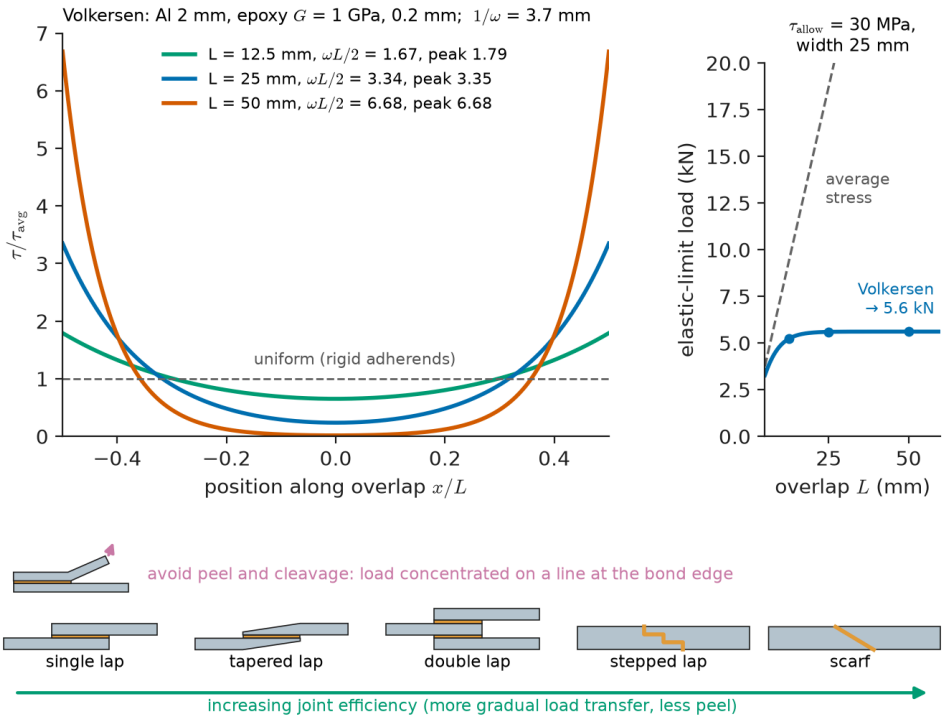


Figure 47.6 — Adhesive shear-stress distribution along a single-lap joint from the Volkersen shear-lag model, for different overlap lengths, with joint configurations ranked by efficiency.

Design rules

Adhesives are strong in shear and compression and weak in **peel** and **cleavage**, where load concentrates on a line at the bond edge. Good joints load the bondline in shear (lap, double-lap, strap, scarf), and resist peel with stiff adherends, adhesive fillets and supplementary rivets or spot welds. Adhesive properties fall as the polymer approaches its glass transition (Chapter 20), and adhesives creep, so allowables are set at the maximum service temperature after humidity aging. "Kissing bonds" — surfaces in contact without adhesion — are nearly invisible to inspection, which is why primary aircraft structure relies on tight process control of surface preparation and often on fail-safe fasteners.

47.9 Mechanical fastening and joining dissimilar materials

Bolts, rivets and clinching

Bolted joints in structural steelwork are either bearing-type, where bolts carry shear by bearing on the hole walls, or **preloaded (slip-resistant)** joints, where high-strength bolts (grade 8.8 or 10.9, ASTM A325/A490 types) are tightened to about 70 % of their tensile strength so that the clamping force generates friction between the plates; the load then passes by friction and the bolts see little fluctuating stress, which is excellent for fatigue. Bolt tightening is controlled by torque, turn-of-nut or direct-tension indicators. High-strength bolts above roughly 1000 MPa tensile strength are susceptible to hydrogen embrittlement from galvanizing or corrosion (Chapter 43), which is why the strongest bolt grades are not hot-dip galvanized.

Rivets remain the main fastener of aluminium aircraft, by the hundred thousand per fuselage. Each hole concentrates stress (factor about 3 for an open hole in tension), and fatigue cracks at rivet holes are the classic airframe failure; cold expansion of holes and interference-fit fasteners add compressive residual stress to delay them. **Self-piercing rivets**, **clinching** (an interlocking button pressed without any added part) and **flow-drill screws** join aluminium and mixed-material car bodies where spot welding is difficult. Fasteners allow disassembly and need no heat or surface preparation, but add mass, need stress-concentrating holes and create crevices for corrosion.

Joining dissimilar materials

Lightweight structures increasingly mix steel, aluminium, magnesium, titanium and fibre composites, and every pair raises three materials problems:

1. **Metallurgical incompatibility.** Fusion welding of aluminium to steel forms brittle Fe–Al intermetallics (Fe_2Al_5, $FeAl_3$) at the interface; once the layer exceeds roughly 10 µm, the joint is brittle. Titanium to steel forms brittle Fe–Ti phases. Solutions keep the interface cold and short-lived: solid-state processes (friction stir, friction, ultrasonic, explosive welding), brazing-like processes in which only aluminium melts and wets galvanized steel, or **transition joints** — explosion-welded aluminium–steel strips welded each to its own metal, as used to attach aluminium superstructures to steel ship hulls.
2. **Galvanic corrosion.** In an electrolyte, the less noble member of a couple corrodes preferentially (Chapter 43). Aluminium against steel, magnesium against almost anything, and especially aluminium against **carbon-fibre composite** (carbon is strongly cathodic) are dangerous couples. Designs isolate them with sealants, coatings, insulating washers, a glass-fibre isolation ply between carbon composite and aluminium, or by using titanium fasteners and titanium fittings, which are compatible with carbon composites.
3. **Thermal-expansion mismatch.** Aluminium expands by about 23×10^{-6} K^{-1}, steel by about 12×10^{-6} K^{-1}, and carbon-fibre laminates in-plane by almost nothing. A 1 m aluminium panel fixed to a steel frame and heated by 60 K, as in a car-body paint oven, wants to grow $(23 - 12) \times 10^{-6} \times 1000 \times 60 = 0.66$ mm more than the frame. If fasteners are rigid and the adhesive is cured at that temperature, the panel buckles or the joints are loaded on cooling; joining sequences, slotted holes and compliant adhesives absorb the difference.

Hybrid joints cover each method's weakness: in aluminium car bodies adhesive carries shear and seals while self-piercing rivets resist peel and hold parts during cure; aircraft use bolted–bonded composite joints.

47.10 Choosing a joining method

Criterion	Fusion welding	Solid-state welding	Brazing / soldering	Adhesive bonding	Mechanical fastening
Joint strength vs base	Up to 100 % (steel); 50–70 % (age-hardened Al)	70–100 %	Filler-limited, high in good laps	Shear-limited; poor in peel	Hole-limited; 50–80 % in sheet
Heat effect on parts	Large: HAZ, residual stress, distortion	Moderate, no melting	Small to moderate (whole part heated in furnace brazing)	Little (cure temperature)	None
Dissimilar materials	Limited	Good	Good, including ceramics	Excellent	Excellent
Section thickness	Thin to very thick	Mostly moderate	Thin to moderate	Thin sheets	Any
Inspection	Radiography, ultrasound	Ultrasound, limited	Ultrasound, leak test	Difficult (kissing bonds)	Visual, torque checks
Disassembly	No	No	Possible by reheating	No	Yes
Typical uses	Ships, bridges, pipelines, vessels	Al tanks, blisks, clad plate	Heat exchangers, tools, electronics	Aircraft panels, car bodies, blades	Aircraft, steelwork, maintenance joints

Where full-strength continuous metal is needed in thick weldable steel, fusion welding wins, and the materials task is to control heat input, hydrogen and composition. Where strength comes from cold work or precipitation, or alloys crack on freezing, solid-state or mechanical methods preserve it. Where heat must be kept out or dissimilar materials meet, brazing, adhesives and fasteners take over, and the task becomes surface preparation, joint geometry and corrosion protection. The joint is part of the material's processing history.

What remains open

Open question	Where it is answered
How do solidification modes, segregation and constitutional supercooling work in a weld pool?	Chapter 9
How do CCT diagrams and hardenability predict HAZ microstructure in steels?	Chapters 10 and 15
Why do welded details have low fatigue strength, and how is weld toughness tested?	Chapter 14
How are TMCP, stainless and Ni-base alloys designed for weldability?	Chapters 15–16
How do composite laminates behave in bolted and bonded joints?	Chapter 23
How do galvanic corrosion, hydrogen embrittlement and stress-corrosion cracking attack joints?	Chapter 43
How are failed welds and joints investigated?	Chapter 44
What are the thermodynamics of wetting, surface energy and adhesion?	Chapter 45
How does the melt-pool physics of welding carry over to additive manufacturing?	Chapter 48
How are residual stresses in welds measured by X-ray and neutron diffraction?	Chapter 53
How are welding thermal cycles and microstructures simulated?	Chapter 56
How do joining costs and disassembly affect material and design choices?	Chapters 58–59

Next: Chapter 48 turns the welding heat source into a manufacturing method in its own right: additive manufacturing builds whole parts from millions of tiny weld tracks, layers of fused powder or cured resin, and inherits both the possibilities and the defects of welding.

Chapter 48 — Additive Manufacturing

Part X · Processing and Manufacturing

A jet-engine fuel nozzle tip used to be brazed together from about twenty separately machined parts. Printed by laser powder-bed fusion in a cobalt–chromium alloy, it became one part with internal cooling passages that could not have been drilled, a quarter lighter and several times more durable, and it went into service in tens of thousands of engines. In the same years, a titanium hip cup printed with a porous outer lattice let bone grow into it; a rocket maker printed combustion chambers in a copper alloy with hundreds of internal coolant channels; and a dental lab replaced hand-cast crowns with printed ones. There were also failures: printed titanium brackets that failed in fatigue at a fraction of the wrought value because of unfused powder pockets a tenth of a millimetre across; aluminium alloys that cracked like poorly welded seams; and parts whose properties changed when the same file was printed on another machine. **Additive manufacturing (AM)** — building a part layer by layer directly from a digital model — is at once a new freedom of shape and a new metallurgy. Every millimetre of a printed metal part is a weld track, solidified at rates closer to a weld or a splat than to a casting.

> **What happens to a material when it is built from millions of tiny melt pools, sintered layers or cured slices?**
> **Which process conditions avoid porosity, cracking and distortion, and how does one prove that a printed part is as good as a forged one?**

Concept	Engineering destination
Process families (PBF, DED, binder jetting, material extrusion, vat photopolymerization)	Choice of machine and route for a part, material and batch size
Energy density, melt-pool modes, process windows	Parameter development, avoiding lack of fusion and keyhole porosity
Rapid solidification, G and R, columnar growth	Fine strong microstructures, texture and anisotropy of printed alloys
Porosity, residual stress, cracking	Fatigue life of printed brackets and implants, distortion, build failures
HIP, heat treatment, machining	Turning as-built material into certifiable material
Polymer and ceramic AM	Prototypes, jigs, dental aligners, surgical guides, ceramic cores
Design for AM, lattices, topology optimization	Lightweight brackets, heat exchangers, porous implants, consolidated assemblies
Qualification and certification	Flight hardware, medical devices, pressure equipment

Roadmap. What AM is (48.1) → the process families (48.2) → melt-pool physics and solidification (48.3) → microstructure and defects (48.4) → post-processing (48.5) → polymer and ceramic AM (48.6) → design for AM (48.7) → qualification and certification (48.8).

> **Boundary.**
> Heat flow and solidification are developed in Chapters 9 and 47 and used here; powder production and sintering in Chapters 46 and 17; photopolymer chemistry in Chapter 19. Laser optics, machine design and slicing software belong to manufacturing engineering. Process simulation is introduced in Chapter 56.

48.1 What additive manufacturing is

Subtractive manufacturing removes material from a block; formative manufacturing (casting, forging, moulding) shapes it in a die or mould. **Additive manufacturing** builds a part by joining material, usually layer upon layer, from a 3D model sliced into

layers 0.02–1 mm thick. The international terminology standard ISO/ASTM 52900 groups AM processes into seven families: **powder bed fusion**, **directed energy deposition**, **binder jetting**, **material extrusion**, **vat photopolymerization**, **material jetting** and **sheet lamination**.

AM's value lies in three things that other routes do poorly:

1. **Geometric freedom.** Internal channels, lattices, undercuts and organic shapes cost nothing extra; complexity is "free" while volume and build height are not.
2. **No tooling.** The economics favour low volumes, customization (implants, dental restorations, hearing-aid shells) and fast iteration.
3. **Part consolidation.** Assemblies of many brazed or bolted pieces become one part, eliminating joints and their failure modes.

Its costs are equally clear: slow build rates (tens of cm^3/h per laser for metals), expensive feedstock, surface roughness, anisotropic properties and process-sensitive defects. AM therefore complements casting, forging and machining rather than replacing them, and its engineering success depends on understanding the materials science of this chapter.

48.2 Process families

Powder-bed fusion

In **powder-bed fusion (PBF)** a thin layer of powder is spread over a build plate by a recoater blade or roller, a heat source melts (or, for polymers, sinters) the cross-section of the part, the plate drops by one layer thickness, and the cycle repeats. Unmelted powder supports the part and is recovered and sieved for reuse.

- **Laser powder-bed fusion (PBF-LB, also called SLM or DMLS)** scans one or more 200–1000 W fibre lasers, focused to 50–100 μm, at 0.5–2 m/s over layers 20–60 μm thick, under argon or nitrogen. It gives the finest features and best surface of the metal processes and dominates metal AM for Ti–6Al–4V, Inconel 718 and 625, 316L and 17-4PH stainless steels, AlSi10Mg, CoCr and copper alloys. The part is welded to the build plate and needs supports under overhangs, both to anchor it against distortion and to conduct heat away.
- **Electron-beam powder-bed fusion (PBF-EB)** scans a beam of several kilowatts in vacuum, deflected electromagnetically at speeds up to several km/s, over 50–100 μm layers. Each layer is first preheated by a defocused beam, lightly sintering the powder (which stops it being blown away by charge build-up) and keeping the whole bed hot — 650–750 °C for Ti–6Al–4V, over 1000 °C for TiAl. Residual stress is therefore low and crack-prone alloys such as γ-TiAl turbine blades and some nickel superalloys can be built, at the cost of rougher surfaces and coarser features.
- **Polymer PBF (laser sintering)** sinters polyamide 12, PA11 or TPU powder held just below its melting point; the surrounding powder supports the part, so no supports are needed (Section 48.6).

Directed energy deposition

In **directed energy deposition (DED)** powder or wire is fed into a melt pool created by a laser, electron beam or arc that moves with the feed nozzle, like automated welding. Layers are 0.25–3 mm thick and deposition rates range from about 0.1 kg/h for fine laser-powder DED to several kg/h for **wire-arc additive manufacturing (WAAM)**, which uses a GMAW or plasma torch on a robot. DED builds large near-net shapes (titanium aircraft frames, steel ship propellers, large valve bodies), adds features to existing parts and **repairs** worn turbine blades, moulds and shafts. Resolution is coarse, so DED parts are machined, and the large melt pools give coarser microstructures than PBF.

Binder jetting

Binder jetting spreads powder and prints a liquid binder onto each layer with inkjet heads, like a paper printer. No melting occurs during the build, so there is no residual stress, no supports and very high throughput; the whole bed can be filled with nested parts. The printed "green" part is cured, removed from the powder, debound and **sintered** (Chapter 46), shrinking 15–20 % linearly — for a green density of 55 % sintered to 97 %, the linear shrinkage is $1 - (0.55/0.97)^{1/3} = 17\ \%$ — or **infiltrated** with bronze. Stainless steels, tool steels, tungsten carbide, sand moulds and cores for casting, and ceramics are printed this way. Dimensional control of sintering shrinkage and slumping is its main difficulty.

Material extrusion

Material extrusion (MEX), best known as fused filament fabrication, pushes a thermoplastic filament through a heated nozzle of 0.2–1 mm and deposits beads layer by layer. It is the cheapest and most widespread AM process — from desktop printers to large-format machines that print car moulds and boat hulls — and also deposits metal- or ceramic-filled filaments that are then debound and sintered, as well as concrete (Section 48.6).

Vat photopolymerization

Vat photopolymerization (VPP) cures a liquid photopolymer resin layer by layer with ultraviolet light: a scanned laser (**stereolithography, SLA**), a projected image (**digital light processing, DLP**) or continuous variants that cure through an oxygen-permeable window. It gives the finest resolution (25–100 µm) and smoothest surfaces of common AM processes and is used for dental models and aligner moulds, hearing-aid shells, jewellery casting patterns, microfluidics and, with ceramic-filled resins, ceramic parts.

Material jetting and sheet lamination

Material jetting deposits droplets of photopolymer (or wax, or metal nanoparticle inks) from inkjet heads and cures them, allowing multi-material and full-colour parts. **Sheet lamination** bonds stacked sheets — paper, polymer, or metal foils joined by ultrasonic welding (ultrasonic additive manufacturing) — and machines each layer's contour.

Process family	Feedstock	Energy / bonding	Layer thickness	Typical resolution	Typical materials	Typical applications
PBF-LB (metal)	Powder 15–45 µm	Laser melting	20–60 µm	0.1–0.2 mm	Ti–6Al–4V, IN718, 316L, AlSi10Mg, CoCr	Brackets, fuel nozzles, implants, heat exchangers
PBF-EB (metal)	Powder 45–105 µm	Electron-beam melting, hot bed	50–100 µm	0.3–0.5 mm	Ti–6Al–4V, γ-TiAl, CoCr	Implants, TiAl blades
DED (laser, arc)	Powder or wire	Laser, EB or arc melting	0.25–3 mm	0.5–5 mm	Ti alloys, steels, Ni alloys	Large preforms, repair, cladding
Binder jetting	Powder 5–30 µm	Binder, then sinter	30–100 µm	0.1–0.3 mm	Stainless, tool steel, WC–Co, sand, ceramics	Volume metal parts, casting moulds
Material extrusion	Filament, pellets, paste	Melt and fuse	0.1–1 mm	0.2–1 mm	PLA, ABS, PC, PEEK, composites, concrete	Prototypes, jigs, tooling, buildings
Vat photopolymerization	Liquid resin	UV curing	25–100 µm	25–100 µm	Acrylates, epoxies, ceramic slurries	Dental, hearing aids, microfluidics
Polymer PBF	Powder 50–80 µm	Laser sintering	80–120 µm	0.3–0.5 mm	PA12, PA11, TPU	Functional plastic parts, ducts

Typical ranges.

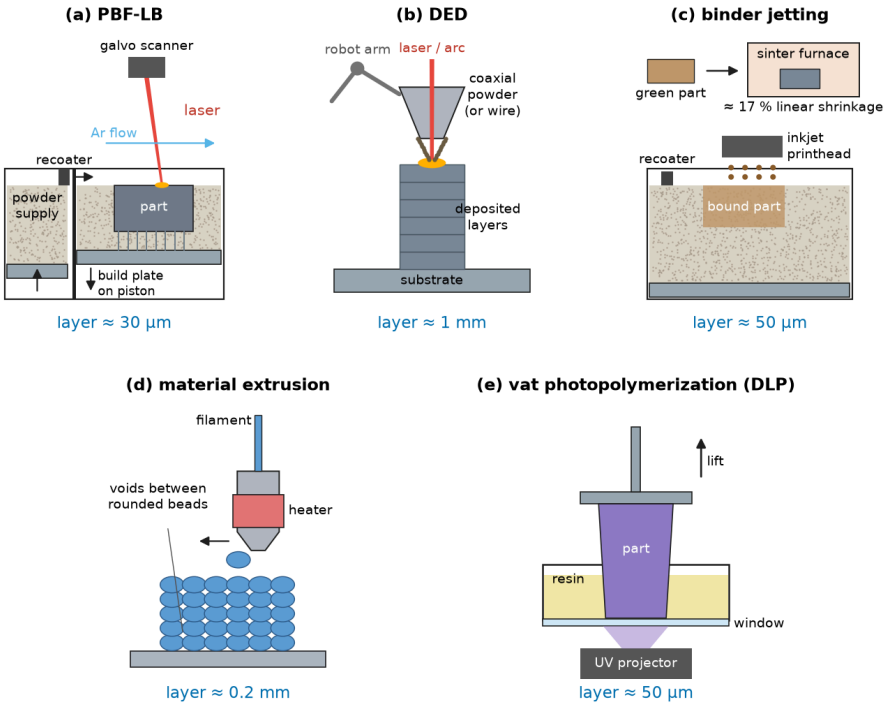


Figure 48.1 — Five principal additive manufacturing process families shown as simplified machine cross-sections.

Figure 48.1 sketches the five main families. The rest of the chapter concentrates on metal PBF and DED, where the materials science is most demanding, and returns to polymers and ceramics in Section 48.6.

48.3 Melt-pool physics and solidification

Energy input and absorption

A laser track in PBF-LB is a miniature keyhole or conduction weld, a fraction of a millimetre wide, made in a few milliseconds. Only part of the beam power is absorbed: the **absorptivity** η of a flat metal surface to near-infrared light is only 0.05–0.4, but a powder bed traps light by multiple reflections and absorbs more (often 0.4–0.7), and a keyhole absorbs most of it. Highly reflective, highly conductive metals — copper, silver, gold, and to a lesser extent aluminium — are the hardest to print for this reason.

The process parameters are commonly combined into a **volumetric energy density**

$$E_v = \frac{P}{v\,h\,t}$$

for laser power P, scan speed v, hatch spacing h (distance between adjacent tracks) and layer thickness t. Typical PBF-LB values are 50–70 J/mm^3 for Ti–6Al–4V and IN718, 40–60 J/mm^3 for AlSi10Mg and 60–100 J/mm^3 for 316L. For example, Ti–6Al–4V at 280 W, 1200 mm/s, 0.14 mm hatch and 30 µm layers gives $280/(1200 \times 0.14 \times 0.03) = 56$ J/mm^3. The same numbers give the theoretical build rate per laser, $vht = 5.0$ mm^3/s or about 18 cm^3/h, before recoating and scan-jump time are subtracted — which is why production machines carry four to twelve lasers.

> **Status of the principle: volumetric energy density as a process variable**
> *Empirical.* E_v correlates density with parameters within one machine, alloy and parameter neighbourhood, but it is not a physical invariant: two parameter sets of equal E_v can give different melt-pool shapes and defects, because P and v act differently on melt-pool depth. Normalized enthalpy and melt-pool-geometry criteria are more transferable.

Melt-pool modes

As in welding (Chapter 47), the melt pool operates in one of two modes.

- **Conduction mode**: the surface melts but does not vaporize significantly; the pool is shallow and roughly semicircular in cross-section, and heat flows outward from the surface.
- **Keyhole mode**: at high power and low speed, the surface temperature exceeds the boiling point, recoil pressure from evaporation depresses the surface into a deep, narrow cavity, and absorption rises sharply. A stable keyhole gives deep penetration; an unstable one collapses intermittently and traps vapour bubbles, leaving round **keyhole pores** at the bottom of the track.

Inside the pool, the surface tension of most liquid metals decreases with temperature, so surface tension is highest at the cool edges and pulls liquid outward from the hot centre: the **Marangoni flow** that widens pools. Surface-active elements such as oxygen and sulfur can reverse the sign of $d\gamma/dT$ and deepen the pool. Fluid velocities reach around 1 m/s, so convection, not conduction alone, shapes the pool and mixes the melt.

At the other extreme, too little energy or too fast a scan produces either **lack of fusion** between tracks and layers or, at high speed, **balling**: the long, thin liquid track breaks up into droplets by the Plateau–Rayleigh instability, because a liquid cylinder longer than about its circumference is unstable. A process map in the P–v plane (Figure 48.2) therefore shows a window of dense parts bounded by lack of fusion (low P, high v), keyholing (high P, low v) and balling (high P and high v).

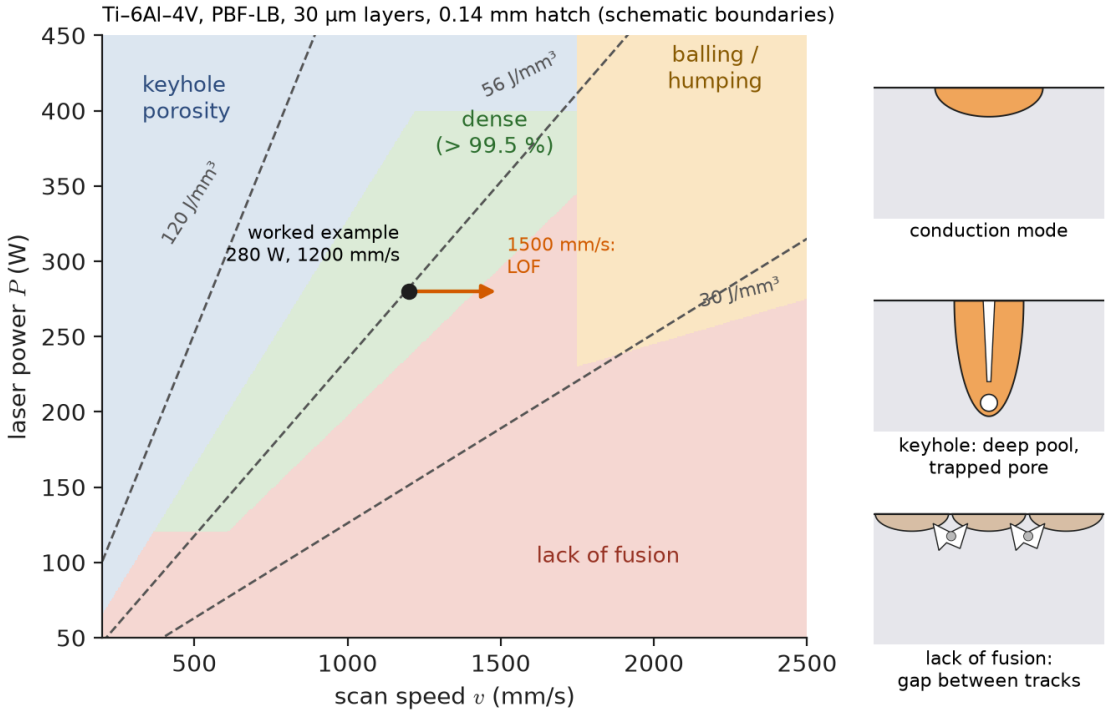


Figure 48.2 — Laser power–scan speed process map for PBF-LB of Ti–6Al–4V with 30 µm layers and 0.14 mm hatch, showing the dense-part window and the defect regimes around it.

Melt-pool size and the lack-of-fusion criterion

The Rosenthal solution for a moving point source (Chapter 47) gives a useful estimate of melt-pool size in conduction mode. At the high speeds of PBF the melt-pool cross-section approaches a semicircle whose width w follows from maximizing the isotherm radius behind the source:

$$w^2 = \frac{8\,\eta P}{\pi e\,\rho c\,v\,(T_m - T_0)}$$

with ρc the volumetric heat capacity and e the base of natural logarithms; the depth is then about $w/2$. A simple geometric condition for full fusion, due to Tang, Pistorius and Beuth, requires adjacent tracks and layers to overlap: for hatch h, layer thickness t, pool width W and depth D,

$$\left(\frac{h}{W}\right)^2 + \left(\frac{t}{D}\right)^2 \le 1$$

If the sum exceeds 1, unmelted pockets remain between tracks.

Worked example — Is this parameter set in the dense window? Take Ti–6Al–4V with $\eta = 0.5$, $P = 280$ W, $v = 1.2$ m/s, $\rho c = 4430 \times 670 = 2.97 \times 10^6$ J/m^{3}K, solidus $T_m = 1604$ °C and $T_0 = 35$ °C, so $T_m - T_0 = 1569$ K. Then

$$w^2 = \frac{8 \times 0.5 \times 280}{\pi \times 2.718 \times 2.97 \times 10^6 \times 1.2 \times 1569} = 2.35 \times 10^{-8}\ \text{m}^2, \qquad w = 153\ \mu\text{m}$$

and the depth is about 77 µm, enough to remelt two layers of 30 µm. With $h = 140$ µm,

$$\left(\frac{140}{153}\right)^2 + \left(\frac{30}{77}\right)^2 = 0.84 + 0.15 = 0.99 \le 1$$

The parameter set just satisfies the overlap condition. Increase the scan speed to 1.5 m/s to build faster and w falls as $v^{-1/2}$, to 137 µm with depth 69 µm; the sum becomes $1.04 + 0.19 = 1.24$, and lack-of-fusion porosity appears between tracks. A 25 % gain in speed has crossed the boundary in Figure 48.2. In practice the pool also contains a partial keyhole and Marangoni flow, so Rosenthal underestimates depth, and parameter development confirms windows by building cubes and measuring density; but the scaling — $w \propto \sqrt{\eta P / v}$ — is the right guide.

> **Status of the principle: Rosenthal melt-pool estimate and overlap criterion**
> *Approximate.* A point source moving over a semi-infinite solid with constant properties, no latent heat and no fluid flow. It predicts trends and order of magnitude in conduction mode; it underestimates depth in keyhole mode and ignores the powder layer, heat accumulation in thin features and the previous tracks.

Cooling rate, thermal gradient and solidification rate

From the same solution, the cooling rate at the solidus on the track centreline is

$$\dot{T} = \frac{2\pi k\,(T_m - T_0)^2\, v}{\eta P}$$

With $k = 20$ W/m K for Ti–6Al–4V near its solidus and the parameters above, $\dot{T} = 2\pi \times 20 \times 1569^2 \times 1.2/140 = 2.7 \times 10^6$ K/s. The thermal gradient behind the pool is $G = \dot{T}/v = 2.2 \times 10^6$ K/m, or 2.2 K/µm, and the solidification rate on the centreline equals the scan speed, $R \approx v = 1.2$ m/s, falling toward zero at the bottom of the pool. Compare a sand casting with $G \sim 10^3$ K/m and $R \sim 10^{-4}$ m/s, or an arc weld with $\dot{T} \sim 10^2$–10^3 K/s. PBF solidifies three to four orders of magnitude faster than conventional welding, and DED and WAAM, with larger pools, lie in between (about 10^2–10^4 K/s).

The consequences follow from Chapter 9:

- **Fine solidification structures.** Primary dendrite or cell spacing scales roughly as $\lambda_1 \propto \dot{T}^{-1/3}$ to $\dot{T}^{-1/2}$. In PBF it is 0.3–1 µm: 316L shows a cellular substructure of about 0.5 µm with Cr and Mo segregated to cell walls and a high dislocation density, which gives a yield strength of 450–550 MPa, roughly double that of annealed wrought 316L, with good ductility. AlSi10Mg forms a fine cellular network of eutectic silicon around α-Al cells, giving higher strength than the cast alloy.
- **Columnar, epitaxial growth.** Each new track remelts the one below and grows epitaxially from it, as in a multipass weld. With high G/R at the pool bottom, grains grow as columns along the build direction across many layers, often with a ⟨100⟩ fibre texture along the build direction. The **columnar-to-equiaxed transition (CET)**, predicted by Hunt's criterion to occur when G^n/R falls below an alloy-dependent constant (with $n \approx 3$–4), is hard to reach in PBF because G is so high; equiaxed grains need potent inoculants or alloys designed for AM.
- **Metastable phases.** Ti–6Al–4V cools fast enough to form **α′ martensite** within the columnar prior-β grains: high strength (tensile strength 1100–1300 MPa) but limited ductility (5–10 %). Rapid cooling also extends solid solubility, suppresses equilibrium precipitates in nickel alloys, and can retain austenite in martensitic steels.
- **Intrinsic heat treatment.** Each layer is reheated many times by the tracks above it, with peaks that diminish with distance. These cycles temper martensite, precipitate phases or (in PBF-EB, with its hot bed) decompose α′ into fine α+β during the build. Properties therefore vary with height and with the time between layers — a feature of AM with no counterpart in casting.

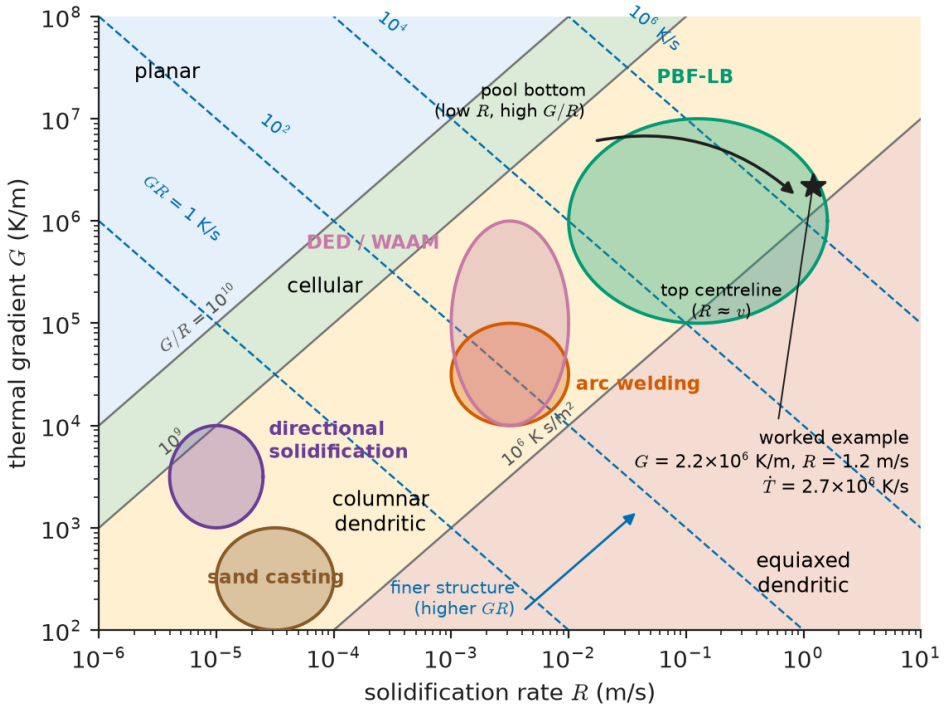


Figure 48.3 — Solidification map of thermal gradient G versus solidification rate R, locating casting, welding, DED and PBF and the morphology regions.

Figure 48.3 locates the AM processes among other solidification routes; the high-G, high-R corner occupied by PBF explains both its fine structure and its stubbornly columnar grains.

48.4 Microstructure and defects

Porosity

Three kinds of pore dominate metal AM, each with a characteristic shape and cause (Figure 48.4):

- **Lack-of-fusion (LOF) pores**: irregular, flat, crack-like voids between tracks or layers, often containing unmelted powder particles, caused by insufficient energy or overlap (the criterion above), spatter particles landing on the bed, or recoating defects. They are the most damaging because they are large (50–500 µm), sharp and often oriented parallel to the layers.
- **Keyhole pores**: spherical voids of 20–100 µm at the bottom of tracks, caused by keyhole collapse at excessive energy density, particularly at the turnaround points of scan vectors where the laser slows.
- **Gas pores**: small (5–30 µm) spherical pores from argon trapped inside gas-atomized powder particles or from dissolved hydrogen (moisture on aluminium powder). They are relatively benign but cannot be removed by HIP if they contain insoluble argon.

Well-optimized PBF-LB routinely achieves 99.5–99.9 % density; the defect population that remains, and its largest member, decide fatigue.

Spatter, powder and surface

The laser ejects **spatter** — molten droplets and entrained powder particles — that can land on the bed, become larger than the layer thickness, and cause LOF in the next layer; gas flow across the bed is designed to carry it away. Powder recycled many times picks up oxygen and coarse spatter particles, so powder chemistry and size distribution are monitored lot by lot. **Surface roughness** of as-built PBF-LB metal surfaces is typically Ra 5–20 µm (worse on down-facing surfaces, which sit on loose powder), and PBF-EB and DED surfaces are rougher still. Partly fused particles and notch-like valleys between layers act as surface cracks in fatigue.

Residual stress, distortion and cracking

Residual stresses arise as in a weld (Chapter 47), with an AM-specific twist known as the **temperature-gradient mechanism**: the newly melted layer is hot and expands against cooler material beneath, yields in compression, and on cooling contracts and is left in tension, while the material below is put in compression. Layer upon layer, this builds tensile stresses near the top surface and the edges, often reaching the yield strength. While the part is welded to its build plate the stresses are balanced; when the part is cut off, it distorts. Stresses can also crack the part or the supports, or lift the part off the plate during the build. Remedies include preheating the plate (200 °C for aluminium, up to 500 °C or more for crack-prone alloys), island or rotating scan strategies that shorten scan vectors and change their direction every layer, adequate supports, and a **stress-relief anneal before removal** from the plate.

Cracking in AM follows the welding classification. **Solidification cracking** limits the high-strength aluminium alloys 2024, 6061 and 7075, whose wide freezing range and long columnar grains with continuous interdendritic liquid films make them crack in PBF just as they do in autogenous welds. Two strategies work: alloys designed for AM, such as Al–Mg–Sc–Zr, in which $Al_3(Sc,Zr)$ particles formed in the melt nucleate fine equiaxed grains, and **inoculating** conventional alloys with nanoparticles (for example Zr-based or TiB_2 additions) that do the same — breaking the columnar structure and allowing crack-free 7075 and 6061. Nickel superalloys with high γ′ fractions (such as IN738LC and CM247LC) suffer solidification and liquation cracking and **strain-age cracking**, which PBF-EB with its hot bed partly avoids. Carbon steels and tool steels with high hardenability suffer cold cracking from martensite, mitigated by plate preheating.

Anisotropy

Columnar grains along the build direction, crystallographic texture, layer-aligned LOF defects and the "stacking" of melt-pool boundaries make as-built AM metals **anisotropic**. Typically, Ti–6Al–4V tested with the load parallel to the build direction (vertical specimens) has lower elongation and lower fatigue strength than horizontal specimens, because flat LOF defects lie perpendicular to the load. Elastic modulus also varies with texture. Specifications therefore require testing in several build orientations, and HIP and recrystallization treatments reduce, though rarely remove, the anisotropy.

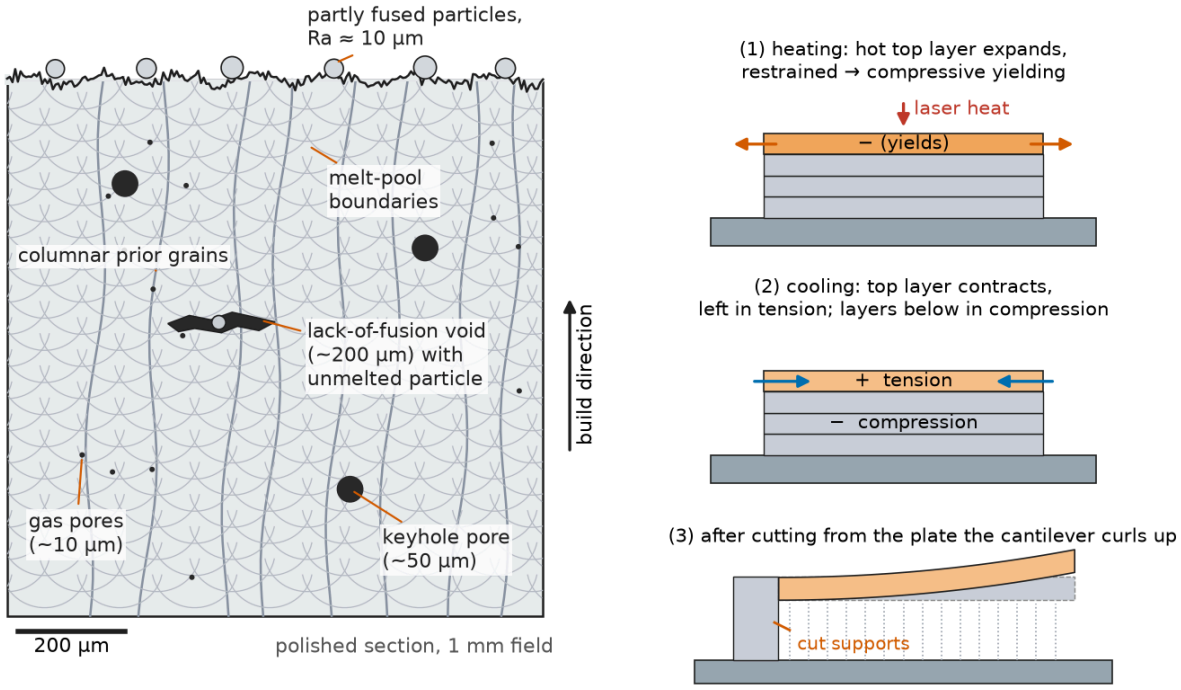


Figure 48.4 — Characteristic defects in laser powder-bed-fused metal: lack-of-fusion voids, keyhole pores, gas pores, columnar grains, and the residual-stress mechanism.

Fatigue: why the largest defect matters

Static strength of well-made AM metal matches or exceeds wrought material, but fatigue does not, unless defects and surfaces are dealt with. Fatigue cracks start at the largest near-surface defect or at the rough as-built surface. Murakami's empirical

$\sqrt{\text{area}}$ model relates the fatigue limit to hardness and the square root of the defect's projected area normal to the stress:

$$\sigma_w = \frac{C\,(\text{HV} + 120)}{(\sqrt{\text{area}})^{1/6}}$$

with σ_w in MPa, HV the Vickers hardness, $\sqrt{\text{area}}$ in µm, and $C = 1.43$ for surface defects and 1.56 for internal ones.

> **Status of the principle: Murakami's $\sqrt{\text{area}}$ model**
> *Empirical.* Fitted to steels with small defects and inclusions; it treats a defect as an equivalent small crack whose threshold for growth depends on its size and the matrix hardness. It is widely used for ranking AM defects in titanium and nickel alloys, but its constants are not universal and it applies to defects small enough to behave as short cracks (roughly below 1000 µm).

Worked example — How much does a lack-of-fusion pore cost? For as-built Ti–6Al–4V of about 350 HV, a surface-connected LOF defect with $\sqrt{\text{area}} = 100$ µm gives

$$\sigma_w = \frac{1.43 \times (350 + 120)}{100^{1/6}} = \frac{672}{2.15} = 312 \text{ MPa}$$

If HIP closes the internal defects and machining removes the surface layer so that the largest remaining defect is an internal gas pore of $\sqrt{\text{area}} = 30$ µm,

$$\sigma_w = \frac{1.56 \times 470}{30^{1/6}} = \frac{733}{1.76} = 416 \text{ MPa}$$

and for a 10 µm pore, about 500 MPa. These estimates bracket what is observed: fatigue strengths at 10^7 cycles of about 200–350 MPa for as-built PBF Ti–6Al–4V with as-built surfaces, against 500–600 MPa for HIPed and machined material — comparable to wrought. A defect only a tenth of a millimetre across is enough to cut the fatigue limit by a third; this is why AM qualification focuses on the size of the largest defect, not on the average density.

48.5 Post-processing

Almost no metal AM part goes into service as built. A typical sequence for a PBF-LB aerospace or medical part is:

1. **Stress relief** on the build plate (for example 650–800 °C for Ti–6Al–4V, 300 °C for AlSi10Mg, 1065 °C for IN718 in some routes) to prevent distortion and cracking on removal.
2. **Removal** from the plate by wire electrical-discharge machining or band saw, and **support removal**.
3. **Hot isostatic pressing** to close internal porosity: typically 920 °C, 100 MPa, 2 h for Ti–6Al–4V; around 1160–1200 °C and 100–150 MPa for nickel superalloys. HIP closes LOF and keyhole pores that are not connected to the surface, but not surface-connected ones, and not argon-filled gas pores, which can reappear on subsequent heat treatment. It also coarsens the microstructure: α′ martensite in Ti–6Al–4V decomposes to a lamellar α+β structure, trading strength for ductility.
4. **Heat treatment** to the required condition: solution and age for IN718 and 17-4PH, which also dissolves the Laves and δ phases that segregation creates; annealing to recrystallize and homogenize the cellular structure of 316L if corrosion or isotropy demands it; T6 for AlSi10Mg (which, unlike the cast alloy, loses strength because the fine as-built silicon network coarsens).
5. **Machining** of critical surfaces, threads and interfaces, and **surface finishing** — abrasive blasting, chemical or electrochemical polishing, abrasive-flow machining of internal channels — to remove the rough surface layer and its partly fused particles.
6. **Inspection**: computed tomography for internal defects and geometry, dye penetrant, dimensional checks, and testing of witness coupons built alongside the part.

Ti–6Al–4V condition	Microstructure	Yield strength (MPa)	Tensile strength (MPa)	Elongation (%)	Fatigue strength at 10^7 cycles (MPa)
PBF-LB, as built	α′ martensite in columnar prior β	1000–1150	1100–1300	5–10	200–350 (as-built surface)
PBF-LB, stress relieved	Tempered α′ / fine α+β	950–1050	1050–1150	8–12	250–400 (as-built surface)
PBF-LB, HIP + machined	Lamellar α+β	850–950	950–1050	12–16	500–600
PBF-EB, as built	Fine α+β (hot bed)	850–950	950–1050	12–16	250–350 (as-built surface)
DED/WAAM, as deposited	Coarse columnar prior β, basketweave α	800–950	900–1000	8–15	—
Wrought, mill annealed	Equiaxed α+β	830–900	900–1000	12–16	500–600

Typical values; fatigue strength depends on stress ratio, surface and defect content.

The table shows the general pattern: as-built PBF material is stronger and less ductile than wrought, HIP and machining bring it to wrought-equivalent behaviour, and fatigue — governed by surface and defects — is the property most sensitive to post-processing. Figure 48.5 shows the same pattern as S–N curves.

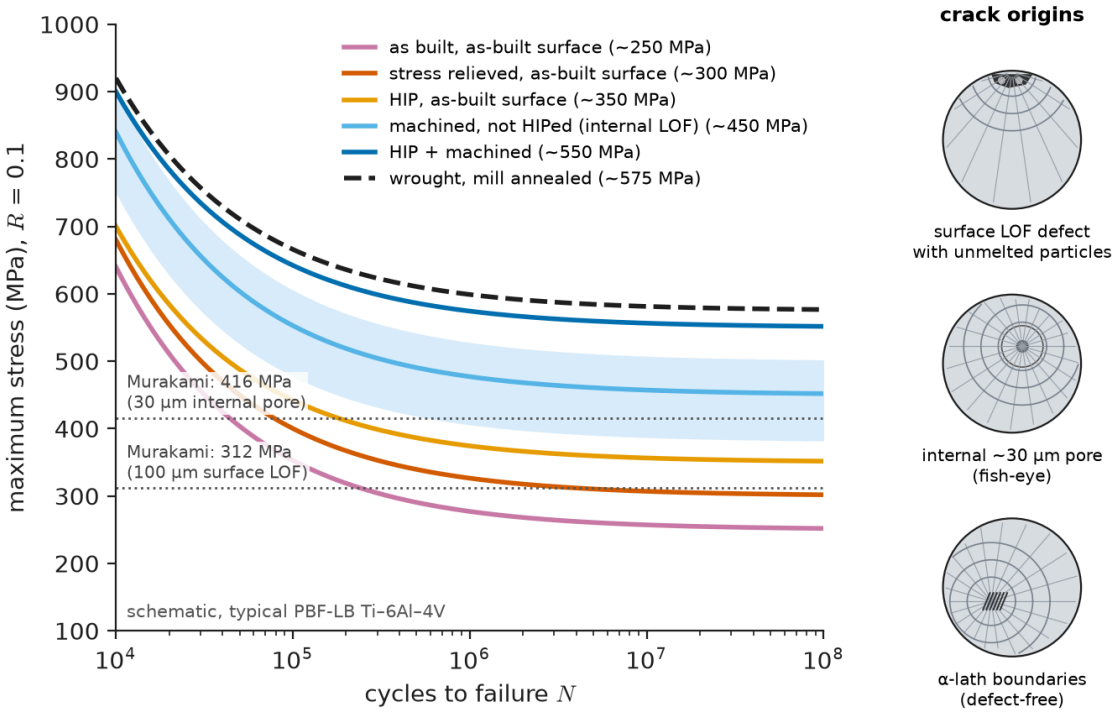


Figure 48.5 — Fatigue S–N curves of Ti–6Al–4V made by PBF-LB in different post-processing conditions compared with wrought material.

48.6 Polymer and ceramic additive manufacturing

Polymers

Polymers were the first AM materials and still account for most printed parts by number.

- **Material extrusion** deposits beads of molten thermoplastic: PLA, ABS, PETG, polycarbonate, nylon and high-temperature PEEK and PEI, often filled with chopped or continuous carbon or glass fibre. The weakness is the **interface between beads and layers**: a new bead must reheat the previous one above its glass transition long enough for chains to diffuse across the interface and entangle. The time for this healing scales with the reptation time (Chapter 21), and if the interface cools too quickly only partial interdiffusion occurs. Strength perpendicular to the layers is therefore typically 30–70 % of that along the beads, and parts are oriented so that the main loads run along the beads. Voids between rounded beads add to the anisotropy.
- **Polymer PBF** (laser sintering) holds a bed of PA12 or PA11 powder a few degrees below its melting point, inside the **sintering window** between melting on heating and recrystallization on cooling of the semicrystalline polymer; the laser supplies the last increment of energy to melt and coalesce the scanned particles. Because the whole bed stays above the recrystallization temperature during the build, the part does not curl, and because the surrounding powder supports it, no supports are needed — enabling nested batches of functional ducts, housings and orthotics. Parts are 95–97 % dense with near-isotropic properties close to moulded PA12.
- **Vat photopolymerization** cures acrylate or epoxy resins by UV-initiated radical or cationic polymerization (Chapter 19). Light is absorbed exponentially with depth (Beer–Lambert), and the resin cures where the dose exceeds a critical dose E_c. The cured depth follows the **Jacobs working curve**:

$$C_d = D_p \ln \frac{E}{E_c}$$

where E is the surface exposure and D_p the depth at which the light falls to $1/e$. Each layer must cure slightly deeper than the layer thickness so that it bonds to the one below.

Worked example — Choosing the exposure. A resin has $D_p = 0.15$ mm and $E_c = 10$ mJ/cm^2. For 0.1 mm layers with 50 % overcure, $C_d = 0.15$ mm and

$$E = E_c \exp\left(\frac{C_d}{D_p}\right) = 10 \times e^{1.0} = 27 \text{ mJ/cm}^2$$

A 0.2 mm cure depth needs $10 \times e^{1.33} = 38$ mJ/cm^2: cure depth grows only logarithmically with exposure. A ceramic-filled slurry scatters light strongly and might have $D_p = 0.05$ mm; curing 0.075 mm then needs $10 \times e^{1.5} = 45$ mJ/cm^2, and the shallower penetration forces thinner layers. Overexposure enlarges features laterally and over-cures down-facing surfaces, so exposure, layer thickness and resolution are set together.

Polymer parts printed by VPP are post-cured under UV and heat to complete the reaction; incompletely cured resins contain residual monomer, which matters for skin-contact and dental parts.

Ceramics

Ceramics cannot be melted layer by layer without cracking, so most ceramic AM is **indirect**: shape a powder–binder green body, then debind and sinter it (Chapters 17 and 46).

- **Vat photopolymerization of ceramic slurries** (40–60 vol% alumina, zirconia, silicon nitride or hydroxyapatite in a photocurable resin) gives the best resolution and density, reaching 99 %+ after sintering for alumina and zirconia. Debinding is slow (often days) because the polymer must leave without cracking the part, which limits wall thickness to a few millimetres. Applications include ceramic cores for turbine-blade casting, dental zirconia, and bioceramic bone scaffolds.
- **Binder jetting** of ceramic powders gives porous parts that are infiltrated or used porous (sand moulds and cores for metal casting are its largest application, eliminating patterns entirely).
- **Material extrusion of pastes** ("robocasting") prints ceramic lattices and scaffolds, and on a much larger scale, **3D concrete printing** extrudes stiff cementitious mortar to build walls without formwork. Printable concrete must be fluid enough to pump yet develop a yield stress fast enough to support the layers above it — a rheology problem — and its interlayer bond, like FFF polymers, is a weak plane (Chapter 18).
- **Direct laser melting** of ceramics is limited to special cases (eutectic oxide ceramics, small parts with preheating) because thermal shock cracks them.

Sintering shrinkage of 15–25 % linear must be compensated in the model, and like MIM (Chapter 46), uniform green density is the key to dimensional control.

48.7 Design for additive manufacturing

Design for AM exploits geometric freedom while respecting what the processes cannot do.

Process constraints

- **Overhangs and supports.** In metal PBF, surfaces inclined at less than about 40–45° to the build plate need supports, which consume time and powder and must be removed; down-facing surfaces are rough. Designers orient parts and use self-supporting shapes (teardrop or diamond channels instead of circles).
- **Minimum features.** Walls of about 0.3–0.5 mm, holes of about 0.5–1 mm and channels that powder can be removed from; trapped powder in closed cavities is a defect.
- **Thermal mass and distortion.** Large flat areas, abrupt section changes and long scan vectors build residual stress; thin features overheat because heat cannot escape, changing microstructure locally.
- **Build orientation** sets anisotropy, surface quality on critical faces, support needs and build height (and so time and cost).

Topology optimization and part consolidation

Topology optimization distributes material within a design space to minimize compliance or mass for given loads and constraints; the organic, truss-like results are often manufacturable only by AM. Typical aerospace brackets redesigned this way lose 30–60 % of their mass. **Part consolidation** replaces assemblies — the fuel-nozzle example of the opening, heat exchangers with thousands of brazed fins, hydraulic manifolds with drilled cross-holes and plugs — by single parts with internal passages shaped for flow rather than for drilling. Each eliminated joint removes a leak path and a failure mode (Chapter 47), but the consolidated part must be inspectable, and internal surfaces cannot be machined.

Lattices and architected materials

AM can build **lattices** — periodic or stochastic cellular structures — whose properties follow the scaling laws of cellular solids. For a lattice of relative density $\bar{\rho} = \rho^*/\rho_s$ made of a solid with modulus E_s and yield strength σ_s, Gibson–Ashby analysis gives

$$\frac{E^*}{E_s} \approx \bar{\rho}^2, \qquad \frac{\sigma^*}{\sigma_s} \approx 0.3\,\bar{\rho}^{3/2}$$

for **bending-dominated** lattices (such as open-cell foams and many strut lattices with low nodal connectivity), and

$$\frac{E^*}{E_s} \approx \frac{\bar{\rho}}{3}$$

for **stretch-dominated** lattices (such as the octet truss), whose struts carry axial load.

> **Status of the principle: Gibson–Ashby scaling of cellular solids**
> *Approximate.* Derived from beam theory for idealized cells, with prefactors fitted to experiment. It holds for relative densities below about 0.3; at higher densities nodes and strut thickness dominate. Printed lattices fall below the predictions when strut diameters approach the melt-track width, because of surface roughness, waviness and porosity in the struts.

Worked example — A titanium lattice that matches bone. A femoral stem made of solid Ti–6Al–4V ($E_s = 114$ GPa) is about six times stiffer than cortical bone (15–20 GPa); the stiff implant carries load that the bone around it would otherwise carry, and the unloaded bone resorbs (**stress shielding**, Chapter 39). A bending-dominated Ti–6Al–4V lattice at $\bar{\rho} = 0.2$ has

$$E^* \approx 114 \times 0.2^2 = 4.6\ \text{GPa}, \qquad \sigma^* \approx 0.3 \times 950 \times 0.2^{1.5} = 25\ \text{MPa}$$

between trabecular and cortical bone, and far closer to bone than solid titanium. At $\bar{\rho} = 0.3$ the modulus is 10 GPa and the strength 47 MPa; a stretch-dominated octet lattice at $\bar{\rho} = 0.2$ gives 7.6 GPa. By grading the density through the part, a designer can make an implant that is stiff where it must carry bending and compliant where it meets bone, and whose open pores of 300–700 μm allow bone ingrowth. Figure 48.6 places these lattices against bone on a log–log chart. The same scaling guides lightweight sandwich cores, energy-absorbing crash structures and heat-exchanger lattices.

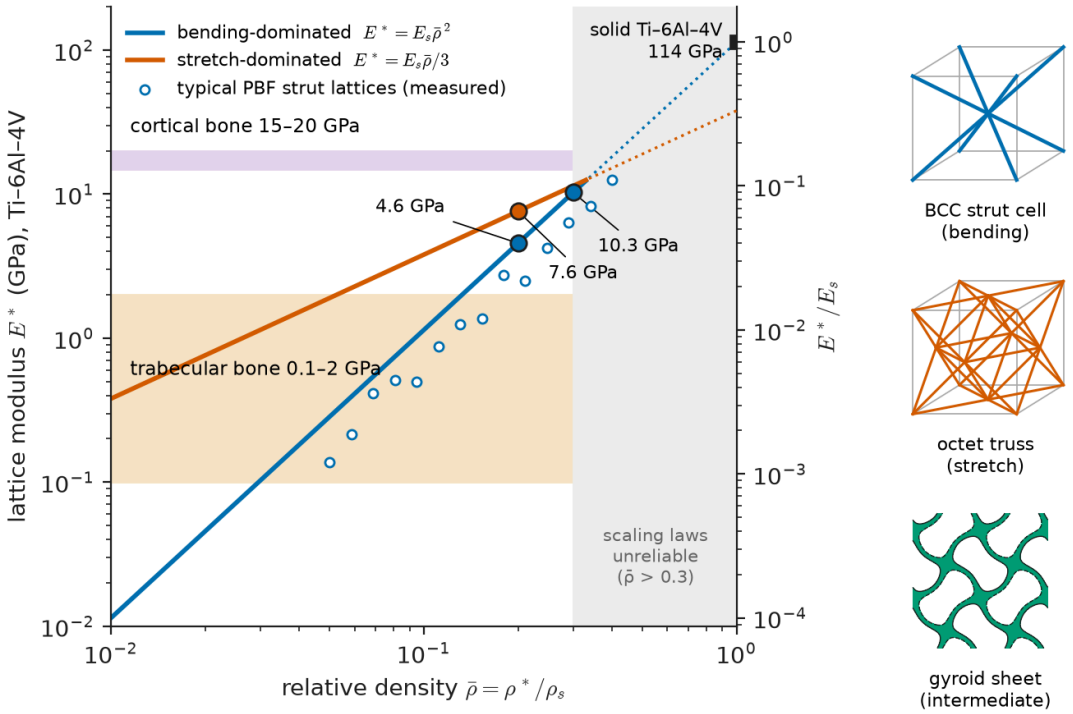


Figure 48.6 — Relative modulus of printed lattices versus relative density, comparing bending- and stretch-dominated architectures with bone.

48.8 Qualification and certification

A forged aircraft part is trusted because its material comes from a mature specification — a defined alloy, a controlled melting and forging process, and decades of statistical property data. An AM part breaks that model: the material is created at the same moment as the part, by a process with dozens of parameters, on machines whose behaviour can drift, and its properties depend on geometry, orientation and location on the build plate. **Qualification** is the demonstration that a material, process and machine reliably produce parts with the required properties; **certification** is a regulator's or customer's acceptance that a particular part meets requirements for its application.

The elements of AM qualification

1. **Feedstock control**: powder chemistry (particularly oxygen, nitrogen and hydrogen), size distribution, morphology, flowability and reuse limits; wire chemistry for DED.
2. **A frozen process**: machine, software, parameter set, scan strategy, gas flow, plate preheat and post-processing are fixed and documented; any change triggers requalification. Machine qualification checks laser power, beam focus, recoater and gas-flow performance.
3. **Material property databases**: tensile, fatigue, fracture-toughness and crack-growth data from many builds, machines and powder lots, in several orientations, giving statistically based design allowables (for example values that 99 % or 90 % of the population exceed with 95 % confidence, as for conventional aerospace materials).
4. **Part qualification**: first-article inspection, CT scanning, destructive sectioning of sample parts, proof testing.
5. **Production control**: **witness coupons** built with each lot and tested; **in-situ monitoring** of melt-pool emission, layer images and thermal fields, which can detect recoater streaks, spatter and LOF as they occur and is increasingly used to flag parts for extra inspection; and statistical process control.
6. **Defect-tolerant design**: since the largest defect governs fatigue, designs use damage-tolerance analysis with an assumed initial flaw size set by what inspection can reliably find, and **classify parts by criticality**, from non-structural parts that need little qualification to fracture-critical parts that need the most.

Standards organizations (ISO/ASTM, SAE's Aerospace Material Specifications, ASME, NASA, and medical-device regulators) have published AM-specific standards covering terminology, feedstock, process control, testing and inspection, and the number grows each year. In medicine, porous printed titanium acetabular cups and spinal cages have been cleared for routine clinical use, with patient-specific implants and surgical guides regulated as custom devices.

Why qualification is hard, and how it is getting easier

The fundamental difficulty is that AM properties are path-dependent at fine scale. A change in part geometry changes local heat accumulation, which changes melt-pool size, cooling rate, porosity and microstructure. Two approaches reduce the cost of proving each new part: **process–structure–property models** (melt-pool, solidification and microstructure simulations linked to property models in an ICME framework, Chapter 56) that predict where in a part defects and weak microstructures will occur, and **in-situ monitoring plus closed-loop control**, which aims to certify the part by its own build record rather than by sacrificial testing. Neither yet replaces testing for critical parts, but together they are moving AM from "qualify every part" toward "qualify the process".

Concern	Root cause	Control	Verification
Lack of fusion	Low energy, poor overlap, spatter, recoating faults	Parameter window, gas flow, powder quality	CT, density cubes, in-situ layer imaging
Keyhole porosity	Excess energy, scan turnarounds	Parameter window, scan-strategy design	CT, metallography
Residual stress and distortion	Thermal gradients	Plate preheat, scan strategy, supports, stress relief	Dimensional inspection, diffraction
Cracking	Alloy freezing range, hardenability	Alloy design, inoculants, preheat	CT, penetrant, metallography
Anisotropy	Columnar grains, texture, oriented defects	Orientation choice, HIP, heat treatment	Tests in several orientations
Surface-initiated fatigue	Roughness, partly fused particles	Machining, polishing	Surface metrology, fatigue coupons
Powder degradation	Oxygen pick-up, spatter in reused powder	Reuse limits, sieving, lot testing	Chemistry and size analysis

What remains open

Open question	Where it is answered
What sets dendrite spacing, CET and segregation during rapid solidification?	Chapter 9
How do martensite, precipitation and recrystallization respond to AM thermal cycles and post-build heat treatment?	Chapter 10
How do defects and surfaces control fatigue and fracture?	Chapter 14
How are Ti, Ni and Al alloys designed and processed conventionally for comparison?	Chapter 16
How are ceramic powders sintered, and polymers processed and degraded?	Chapters 17 and 21
How do printed implants interact with bone, and what is stress shielding?	Chapter 39
How are AM parts used in high-temperature and aerospace service?	Chapter 41
How is surface roughness modified by finishing and coating?	Chapters 45 and 49
How are porosity, texture and residual stress measured (CT, EBSD, synchrotron and neutron methods)?	Chapters 52–53
How do in-situ X-ray imaging and melt-pool monitoring reveal defect formation?	Chapter 54
How are melt pools, microstructures and properties simulated and linked in ICME?	Chapter 56
How is AM chosen against casting, forging and machining on cost and performance, and what is its environmental footprint?	Chapters 58–59

Next: Chapter 49 moves from bulk parts to layers — films and coatings from nanometres to millimetres thick — and to the crystal growth and device processing on which electronic materials depend.

Chapter 49 — Thin Films and Coatings

Part X · Processing and Manufacturing

A carbide turning insert cuts hardened steel with a chip sliding over its rake face at 900 °C. Uncoated, its edge would crater in minutes; with 10 µm of TiCN and α-Al_2O_3 deposited by chemical vapour deposition it lasts an hour. A turbine blade survives gas hotter than its superalloy can bear because 150 µm of yttria-stabilized zirconia carries a temperature drop of more than 100 °C. Low-E architectural glass reflects infrared through ten nanometres of sputtered silver. A titanium hip stem bonds to bone through plasma-sprayed hydroxyapatite. A microprocessor is a stack of more than a hundred films, the thinnest a few atomic layers of hafnium oxide grown one monolayer at a time. In each case the bulk gives strength, shape and cost, and a thin layer made by a separate process gives the function. That layer has its own microstructure, residual stress and failure modes, and the processes that make it (vapour transport in vacuum, reactions on a hot surface, electrochemistry, droplets hitting a surface at the speed of sound) differ from casting or forging.

> **How do atoms get from a source to a surface, and what sets the deposition rate? Why does a film grow dense, porous, columnar or epitaxial? Where does film stress come from, and when does a coating crack or peel? How are thermal-spray and hard coatings chosen for wear, heat and implants?**

Concept	Engineering destination
Mean free path, monolayer time, plasma sheath	Choosing vacuum level and process window
PVD, CVD, ALD	Tool coatings, optical and low-E glass, gate dielectrics, barrier films
Plating and sol–gel	Zinc and nickel on steel, Ni–P on moulds, anti-reflection glass
Growth modes, structure-zone models	Dense versus columnar films, quantum dots
Thermal and intrinsic stress, Stoney equation, delamination energy	Coating thickness limits, wafer bow, cracking of CVD coatings
Thermal spray, hard coatings	Thermal barriers, WC–Co wear coatings, implant coatings

Roadmap. Why coatings exist (49.1) → vacuum and plasma (49.2) → PVD, CVD and ALD (49.3–49.5) → plating and sol–gel (49.6) → growth and microstructure (49.7) → stress and adhesion (49.8) → thermal spray and hard coatings (49.9). Single-crystal growth, epitaxy and device processing follow in Chapter 50.

49.1 Why surfaces get their own materials

Many requirements are volume requirements (carry load, conduct heat, resist creep); many others are surface requirements (resist wear, corrosion, oxidation or reflection, bond to bone, insulate a gate). One material rarely meets both: a superalloy with enough aluminium and chromium to resist oxidation at 1100 °C would lose its creep strength (Chapter 41). The answer is a composite in thickness: a substrate optimized for the bulk carrying layers optimized for the surface. Coatings span seven orders of magnitude in thickness, and the process changes with thickness:

Thickness	Typical processes	Examples
0.1–10 nm	ALD, thermal oxidation, MBE	High-k gate oxide, Al_2O_3 passivation of solar cells
10 nm–1 µm	Sputtering, evaporation, CVD, sol–gel	Low-E and anti-reflection stacks, interconnects, diffusion barriers
1–20 µm	PVD/CVD hard coatings, plating, anodizing	TiAlN on tools, Ni and Cr plate, DLC on engine parts
20 µm–2 mm	Thermal spray, galvanizing, weld overlay	Thermal barriers, WC–Co on shafts, zinc on steel

Every process raises the same three questions: how material arrives, how it attaches and organizes, and what stress and adhesion it ends with.

49.2 Vacuum and plasma basics

Mean free path

Vapour processes need vacuum so that atoms reach the substrate without excessive collisions, so that residual gases do not contaminate the film, and so that a plasma can be sustained. The first is set by the **mean free path**

$$\lambda = \frac{k_B T}{\sqrt{2}\,\pi d^2 p}$$

with d the collision diameter (about 0.37 nm for N_2). At 298 K, $\lambda \approx 68$ nm at atmospheric pressure, 6.8 mm at 1 Pa, 68 m at 10^{-4} Pa. The Knudsen number $\mathrm{Kn} = \lambda / L$ (L the chamber size) separates ballistic transport ($\mathrm{Kn} \gg 1$: evaporation, molecular-beam epitaxy) from continuum flow with boundary layers ($\mathrm{Kn} \ll 1$: atmospheric CVD). Sputtering at 0.1–1 Pa lies between: atoms make a few collisions in flight.

> **Status of the principle: Mean free path from kinetic theory**
> *Derived.* Follows from the hard-sphere ideal gas with a Maxwell velocity distribution; d is an effective diameter fitted to viscosity, so the result is good to tens of percent.

Monolayer time

Contamination is usually the stricter criterion. Gas molecules strike a surface at the flux $\Phi = p/\sqrt{2\pi m k_B T}$; a surface has about 10^{19} sites per m^2, so with unit sticking it is covered in $\tau = n_s / \Phi$.

Worked example — how good must the vacuum be? For water ($m = 18$ u) at 298 K, $\sqrt{2\pi m k_B T} = 2.78 \times 10^{-23}$ kg m/s. At 10^{-4} Pa, $\Phi = 3.6 \times 10^{18}$ $m^{-2}s^{-1}$ and $\tau \approx 2.8$ s; a film growing at 1 nm/s (about four monolayers per second) could collect about a tenth of a monolayer of contaminant for every monolayer it grows. At 10^{-7} Pa, $\tau \approx 2800$ s, about 46 minutes. That is why molecular-beam epitaxy and oxidation-sensitive films (aluminium mirrors, superconducting niobium) need ultra-high vacuum, while nitride tool coatings, which tolerate some oxygen, are made after pumping to about 10^{-3} Pa (Figure 49.1).

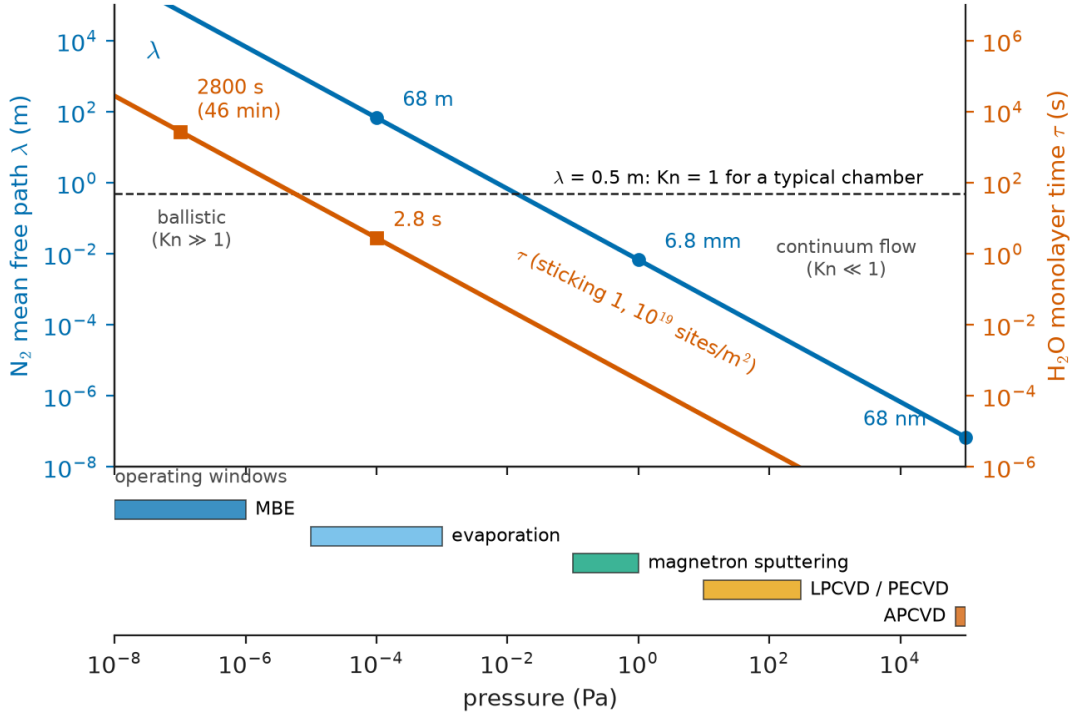


Figure 49.1 — Mean free path and monolayer time versus pressure, with operating windows of vapour processes.

Rough pumps (rotary vane, dry scroll) reach about 1 Pa; turbomolecular pumps or cryopumps take over below. The ultimate pressure is set mostly by water outgassing from the walls, hence baking of UHV systems and load-locks in production tools.

Pirani gauges, capacitance manometers and ionization gauges cover the range, and a residual-gas analyser identifies what remains.

Plasmas

A processing **plasma** is a partially ionized gas (10^{-6}–10^{-2} ionized) far from equilibrium: electrons at 1–10 eV, ions and neutrals near room temperature. Energetic electrons dissociate molecules into radicals (F from CF_4, N from N_2, SiH_3 from silane) that would otherwise need a 1000 °C furnace: chemistry of a hot gas at a cold substrate. The plasma screens potentials over the **Debye length** $\lambda_D = \sqrt{\varepsilon_0 k_B T_e / n_e e^2}$, about 0.1 mm for $T_e = 2$ eV and $n_e = 10^{16}$ m^{-3}. Electrons are more mobile than ions, so surfaces charge negative and a thin positive-ion **sheath** forms in front of them; ions cross it and strike the surface with energies from tens of volts (floating) to hundreds (biased). Ion energy is the key plasma process variable: it densifies films (Section 49.7), makes etching directional (Chapter 50) and, if too high, causes damage. RF excitation at 13.56 MHz is used for insulating targets; inductively coupled sources decouple plasma density from ion energy.

> **Boundary.**
> Sheath theory and RF coupling belong to Engineering Physics; only the outcomes (radicals and a controllable ion energy) are used here. Chapter 50 treats fab plasma processes in more depth.

49.3 Physical vapour deposition

In **physical vapour deposition (PVD)** the coating material is vaporized physically (heat or momentum transfer) and condensed on the substrate; reactive variants add N_2, O_2 or C_2H_2 to form compounds.

Evaporation

The flux leaving a heated source follows the **Hertz–Knudsen** relation $J = \alpha_e (p_e - p)/\sqrt{2\pi m k_B T}$, with p_e the equilibrium vapour pressure. Useful rates need $p_e \approx 1$ Pa: about 950 °C for aluminium, 1400 °C for chromium and 1700 °C for titanium; refractory metals and oxides are evaporated by electron beam from a water-cooled copper hearth. A small planar source emits with a cosine distribution, so on a parallel substrate at height h the thickness falls with radius r:

$$\frac{t}{t_0} = \frac{1}{\left[1 + (r/h)^2\right]^2}$$

For $h = 0.5$ m and $r = 0.1$ m, $t/t_0 = 0.925$: 7.5 % thinner at the edge of a 200 mm part, which is why optical coaters use rotating planetary fixtures.

> **Status of the principle: Cosine-law thickness distribution**
> *Approximate.* Assumes ballistic transport, a small Lambertian source and unit sticking; deep crucibles, high rates and gas scattering distort it.

Evaporated atoms arrive with only 0.1–0.2 eV, so films are porous unless the substrate is hot or ions assist the growth. Uses: optical filters (with ion assist), aluminium on packaging film at more than 10 m/s, electron-beam PVD of zirconia thermal barriers (Section 49.9) and molecular-beam epitaxy (Chapter 50).

Sputtering

An energetic ion striking a solid starts a collision cascade that ejects surface atoms. The **sputter yield** Y for Ar^+ at 500 eV is typically:

Target	Al	Ti	Cr	Fe	Ni	Cu	Ag	Au	Si	W
Y (atoms/ion)	1.05	0.51	1.18	1.10	1.33	2.35	3.12	2.40	0.50	0.57

> **Status of the principle: Sputter yields**
> *Empirical.* Sigmund's cascade theory with fitted corrections predicts Y to about a factor of 1.5; yields rise with ion energy up to a few keV, then fall as ions penetrate deeper.

A **magnetron** traps secondary electrons in crossed electric and magnetic fields over the target, sustaining tens of mA/cm^2 at 300–600 V and 0.1–1 Pa, and eroding a racetrack groove.

Worked example — erosion rate of a copper target. At 20 mA/cm^2 = 200 A/m^2 the ion flux is $200/1.602\times10^{-19} = 1.25\times10^{21}$ m^{-2}s^{-1}; the sputtered flux is 2.35 times that, 2.93×10^{21} m^{-2}s^{-1}. Copper has 8.49×10^{28} atoms/m^3, so the racetrack recedes at 3.5×10^{-8} m/s, about 2 μm/min. Only a fraction reaches the substrate, giving deposition rates of order 0.1–1 μm/min. (Counting secondary electrons as ions slightly overestimates the flux.)

Sputtering deposits almost any material, transfers alloy composition from target to film (the target surface reaches a steady state, whereas evaporation loses the volatile component first), produces denser films because atoms arrive with several eV, and scales to cylindrical targets metres long for glass and photovoltaic lines. In **reactive sputtering** (TiN, AlN, Al_2O_3, indium tin oxide) the target surface becomes "poisoned" by compound as gas flow rises; the compound sputters more slowly, so the rate collapses with hysteresis, and production systems hold the unstable transition mode by feedback on plasma emission. **HiPIMS** concentrates power into short pulses that ionize much of the sputtered metal, so substrate bias can steer it into dense, smooth coatings at the cost of rate.

Cathodic arc and ion plating

A **cathodic arc** (20–30 V, 50–150 A) burns in micrometre spots that explode cathode material into a highly ionized plasma with 20–100 eV ions. It is the dominant industrial route to TiN, TiAlN and AlCrN on cutting tools because of high rate and adhesion, but ejects molten **macroparticles** that roughen the film unless magnetically filtered. **Ion plating** is PVD with a substrate bias of 50–200 V, which sputter-cleans the surface before deposition and densifies the film during it.

49.4 Chemical vapour deposition

In **chemical vapour deposition (CVD)** gaseous precursors react on a heated surface to give a solid and gaseous by-products, for example

$$\mathrm{SiH_4} \rightarrow \mathrm{Si} + 2\mathrm{H_2}\ \ (600\text{–}650\ ^\circ\mathrm{C}), \qquad \mathrm{TiCl_4} + \tfrac{1}{2}\mathrm{N_2} + 2\mathrm{H_2} \rightarrow \mathrm{TiN} + 4\mathrm{HCl}\ \ (850\text{–}1000\ ^\circ\mathrm{C})$$

Alumina for carbide inserts comes from $AlCl_3$, CO_2 and H_2 at about 1000 °C; SiC from methyltrichlorosilane infiltrated into fibre preforms forms the matrix of SiC/SiC composites (chemical vapour infiltration, Chapters 23 and 41). Diamond grows from a few percent methane in hydrogen activated by a hot filament or microwave plasma at 700–900 °C, where graphite is stable; atomic hydrogen etches graphitic carbon much faster than diamond, so diamond survives kinetically.

The Grove model

Precursor diffuses through a boundary layer to the surface, adsorbs, reacts, and by-products leave; the slowest step controls the rate. Grove's model keeps two steps: transport flux $F_1 = h_g(C_g - C_s)$, with h_g a mass-transfer coefficient, and a first-order surface reaction $F_2 = k_s C_s$. Setting $F_1 = F_2$ gives the growth rate

$$G = \frac{k_s h_g}{k_s + h_g}\,\frac{C_g}{N_1}$$

with N_1 the atomic density of the film. If $k_s \ll h_g$ (low temperature) growth is **reaction-limited**, $G \approx k_s C_g / N_1$, strongly temperature-dependent but insensitive to flow; every surface in the hot zone grows at the same rate, so hundreds of wafers can be stacked in a **low-pressure CVD (LPCVD)** furnace and deep trenches are coated uniformly. If $k_s \gg h_g$ (high temperature) growth is **transport-limited**, $G \approx h_g C_g / N_1$, nearly temperature-independent but sensitive to gas flow and geometry.

> **Status of the principle: Grove model of CVD**
> *Approximate.* Assumes first-order surface kinetics and a stagnant-film boundary layer; adsorption saturation, gas-phase reactions and precursor depletion modify it, but the two-regime picture holds widely.

Worked example — two regimes of silicon epitaxy. Take $h_g = 0.05$ m/s, $C_g = 10^{22}$ m^{-3} (about 0.2 % of the gas at 1 atm and 1100 °C), $N_1 = 5 \times 10^{28}$ m^{-3}, and $k_s = k_0 \exp(-E_a/k_B T)$ with $E_a = 1.6$ eV and $k_s = h_g$ at 1100 °C ($k_0 = 3.73 \times 10^4$ m/s). At 1100 °C, $G = 0.025 \times 10^{22}/5 \times 10^{28} = 5 \times 10^{-9}$ m/s $= 0.30$ µm/min. At 900 °C, $k_s = 4.98 \times 10^{-3}$ m/s and $G = 0.054$ µm/min: 200 K cooler cuts the rate 5.5-fold. At 1300 °C, $k_s = 0.28$ m/s and $G = 0.51$ µm/min: 200 K hotter gains only 70 %. Production epitaxy runs near the knee, 1050–1150 °C, with reactors designed for uniform flow (Figure 49.2).

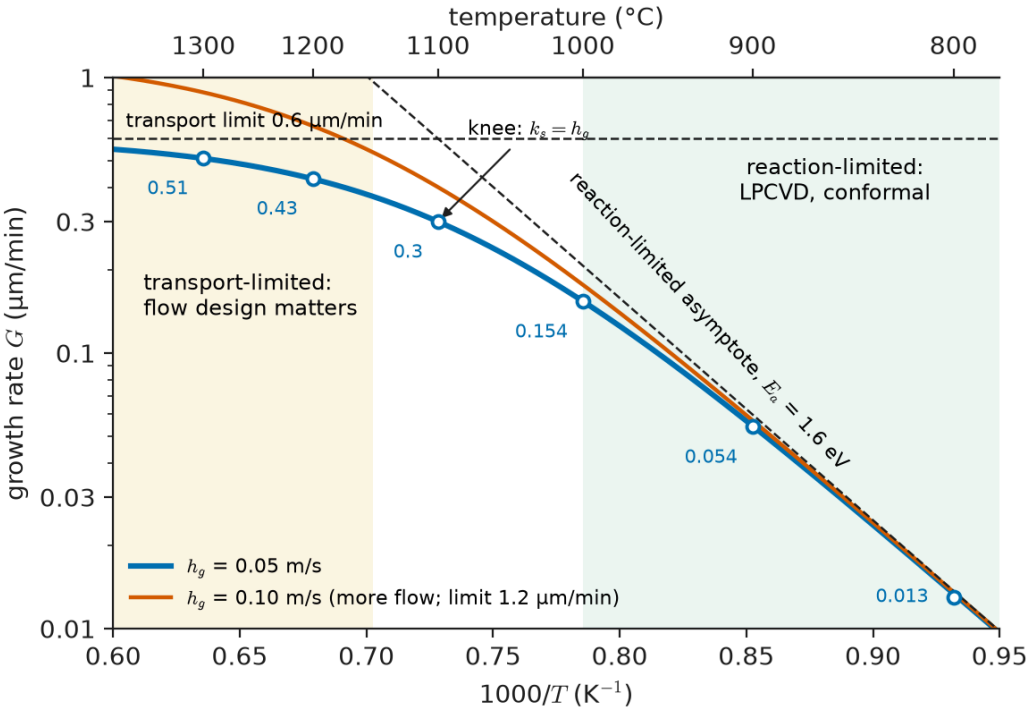


Figure 49.2 — Arrhenius plot of CVD growth rate showing reaction-limited and transport-limited regimes (Grove model).

Variants

Variant	Pressure	Temperature	Typical use
APCVD	10^5 Pa	400–1200 °C	SiO_2 on float glass online, Si epitaxy
LPCVD	10–100 Pa	550–900 °C	Polysilicon and Si_3N_4 for MEMS
PECVD	10–300 Pa	200–400 °C	SiN_x on solar cells, DLC, barrier films on polymers
MOCVD	10^3–10^5 Pa	500–1100 °C	GaN LEDs, GaAs and InP lasers
Thermal CVD on tools	10^3–10^5 Pa	850–1050 °C	TiC, TiCN, TiN, α-Al_2O_3 on WC–Co
Hot-filament/microwave	10^3–10^4 Pa	700–900 °C	Diamond coatings and windows

PECVD coats temperature-sensitive substrates (finished solar cells, polymer film, stents) but leaves 10–25 at.% hydrogen in SiN_x:H and gives less dense films. Medium-temperature CVD of TiCN from acetonitrile at 700–900 °C reduces the decarburization of carbide substrates that otherwise forms a brittle η-phase (Co_3W_3C) at the interface.

49.5 Atomic layer deposition

Atomic layer deposition (ALD) splits CVD into two self-limiting half-reactions separated by purges. For alumina: trimethylaluminium reacts with surface OH groups until all are consumed; purge; water converts the Al–CH_3 surface back to Al–OH; purge. Each cycle adds a fixed **growth per cycle (GPC)**, about 0.1–0.12 nm for Al_2O_3 at 150–300 °C (less than a monolayer because bulky ligands block sites). Because each step saturates, ALD coats every exposed surface equally: trenches with aspect ratios above 50:1, catalyst pores, battery-electrode particles.

> **Status of the principle: Self-limiting ALD growth**
> *Empirical.* Saturation is observed within a temperature "ALD window" for each precursor pair; below it precursors condense, above it they decompose or desorb, so GPC is measured for each process.

Worked example — ALD throughput. A 10 nm Al_2O_3 moisture barrier at 0.11 nm per cycle needs $10/0.11 \approx 91$ cycles; at 4 s per cycle that is about 6 minutes, independent of area or topography. A 1 µm film would take ten hours. ALD therefore wins where films are thin and conformality is essential: 2 nm HfO_2 gate dielectrics, barriers for organic LEDs and flexible electronics, passivation of solar cells, a few cycles of Al_2O_3 on Li-ion cathode particles (Chapter 40) and anti-corrosion layers on medical devices. Spatial ALD, which moves the substrate between precursor zones, raises throughput for solar wafers and roll-to-roll film.

49.6 Electroplating, electroless plating and sol–gel

Electroplating

The part is the cathode in a bath of metal ions ($M^{n+} + ne^- \rightarrow M$). By Faraday's law the thickness grows at

$$\frac{dh}{dt} = \frac{\eta_c \, i \, M}{n F \rho}$$

with i the current density, M the molar mass, ρ the deposit density, $F = 96\,485$ C/mol and η_c the **cathode current efficiency** (the fraction not wasted on hydrogen evolution).

> **Status of the principle: Faraday's law of electrolysis**
> *Fundamental.* Charge conservation applied to an electrode reaction; the uncertainty lives in the current efficiency, measured for each bath and current density.

Worked example — nickel and hard chromium. A Watts nickel bath at 2 A/dm^2 = 200 A/m^2 with $\eta_c = 0.95$: $dh/dt = 0.95 \times 200 \times 0.05869/(2 \times 96\,485 \times 8900) = 6.49 \times 10^{-9}$ m/s $= 23$ µm/h, so 25 µm takes about 1.1 h. Hard chromium from hexavalent baths has $n = 6$ and $\eta_c \approx 0.20$; at 40 A/dm^2, $dh/dt = 0.20 \times 4000 \times 0.052/(6 \times 96\,485 \times 7190)$ m/s ≈ 36 µm/h, bought with twenty times the current density and four-fifths of it spent making hydrogen. That hydrogen enters the substrate, so plated high-strength steels (landing gear, fasteners above about 1000 MPa) are baked at about 190 °C to prevent hydrogen embrittlement (Chapter 43).

Current concentrates at edges, so deposits are thick on corners and thin in recesses; bath **throwing power** and additives (levellers, brighteners) even this out and refine the grain size. Additive chemistry also makes copper fill chip interconnect trenches bottom-up. Zinc is the most plated metal by tonnage (5–10 µm on electrogalvanized car sheet, protecting steel sacrificially); copper, tin, gold, silver and cadmium follow. Hexavalent chromium is carcinogenic and increasingly restricted, which drives its replacement by HVOF WC–Co and electroless Ni–P.

Electroless plating

In **electroless** (autocatalytic) plating a chemical reducing agent replaces the current. Hypophosphite-reduced nickel co-deposits 2–13 wt% P; high-P deposits are amorphous and corrosion-resistant, about 500–600 HV as plated and 900–1000 HV after ageing at 400 °C precipitates Ni_3P. With no current distribution, thickness is uniform inside holes and on complex shapes: injection moulds, hydraulic parts, valves for oil and gas, aluminium hard-disk substrates. Electroless copper metallizes plastics and circuit-board holes.

> **Boundary.**
> Electrode potentials and electrode kinetics belong to Advanced Chemistry; corrosion electrochemistry is in Chapter 42.

Sol–gel coatings

Metal alkoxides such as $Si(OC_2H_5)_4$ **hydrolyse** (Si–OR + H_2O → Si–OH + ROH) and **condense** (Si–OH + HO–Si → Si–O–Si + H_2O) into a sol and then an oxide gel, applied by dipping, spinning or spraying and fired at 300–600 °C. In dip coating at withdrawal speed U the wet thickness at low capillary number follows the Landau–Levich relation

$$h = 0.94\,\frac{(\eta U)^{2/3}}{\gamma^{1/6}(\rho g)^{1/2}}$$

For an alcoholic sol ($\eta = 2$ mPa s, $\gamma = 0.022$ N/m, $\rho = 800$ kg/m^3), $h = 3.2$ µm at 1 mm/s and 9.3 µm at 5 mm/s; drying and firing shrink this to roughly a tenth.

> **Status of the principle: Landau–Levich film thickness**
> *Approximate.* Lubrication theory for a Newtonian liquid with $\eta U/\gamma \ll 1$ and no evaporation; real sols evaporate and gel during withdrawal, so the coefficient is calibrated.

Sol–gel needs no vacuum and coats large or curved parts: porous silica anti-reflection layers on solar glass (index about 1.25), photocatalytic TiO_2 self-cleaning glass, hybrid silica coatings that replace chromate conversion coatings on aircraft aluminium, bioactive coatings on implants. Single layers thicker than about 0.5–1 µm crack on drying, by the critical-thickness logic of Section 49.8.

Anodizing and conversion coatings

Some coatings are not deposited at all but grown by converting the substrate surface. In **anodizing**, an aluminium part is made the anode in an acid bath and its surface oxidizes to Al_2O_3. In near-neutral electrolytes (borate, tartrate) the product is a compact **barrier oxide** whose thickness is fixed by the applied voltage, about 1.3–1.4 nm per volt, so 100 V gives roughly 0.14 µm; this is how electrolytic capacitor dielectrics are formed. In sulphuric, oxalic or phosphoric acid, field-assisted dissolution at the base of the oxide keeps a thin barrier layer open and a **porous oxide** grows beneath it: a near-hexagonal array of cells, each with a central pore 10–150 nm across, whose cell size scales with voltage (about 2.5 nm/V). The porous layer thickens with time and charge rather than voltage, typically 5–25 µm for architectural and consumer parts. The pores take up dyes, and the layer is then **sealed** in boiling water or nickel acetate, which hydrates the oxide to boehmite and closes the pores, restoring corrosion resistance.

Hard anodizing runs sulphuric acid near 0 °C at higher current density (2–4 A/dm^2) to grow dense coatings 25–150 µm thick with 350–500 HV, used on pistons, hydraulic cylinders and gun parts. About half the thickness grows inward, so dimensions change by only half the coating thickness. The oxide is brittle and lowers fatigue strength, and high-copper and high-silicon alloys anodize poorly.

Conversion coatings are thinner (0.1–2 µm) chemical films that serve mainly as a base for paint. Chromate conversion coatings on aluminium and zinc contain soluble Cr(VI) that migrates to scratches and re-passivates them, a "self-healing" action that made them the aerospace standard. Hexavalent chromium is, however, carcinogenic and restricted under REACH and RoHS, so industry is moving to trivalent-chromium processes, zirconium and titanium fluoride treatments, and (on steel) zinc and iron **phosphate** coatings, which give paint adhesion and retain oil but offer no self-healing. Matching the corrosion protection of chromate without Cr(VI) remains an active problem (Chapter 43).

49.7 Thin-film growth modes and structure-zone models

Three growth modes

Adatoms diffuse, meet and form clusters. At equilibrium, whether the film wets the substrate is set by a surface-energy balance like Young's equation (Chapter 45), with γ_s, γ_f and γ_i the substrate, film and interface energies (Figure 49.3):

- **Frank–van der Merwe (layer)**, $\gamma_s \geq \gamma_f + \gamma_i$: each layer completes before the next (AlGaAs on GaAs).
- **Volmer–Weber (island)**, $\gamma_s < \gamma_f + \gamma_i$: three-dimensional islands that later coalesce. Most metals on oxides grow this way, which is why silver in a low-E stack is continuous only above about 8–10 nm and is grown on a ZnO seed layer that it wets better.

- **Stranski–Krastanov**: a wetting layer of one or a few monolayers, then islands once mismatch strain accumulates. Ge on Si (4 % mismatch) and InAs on GaAs (7 %) form self-assembled **quantum dots** (Chapter 38).

> **Status of the principle: Surface-energy criterion for growth mode**
> *Approximate.* An equilibrium argument; at high rates and low temperatures limited adatom diffusion can override it.

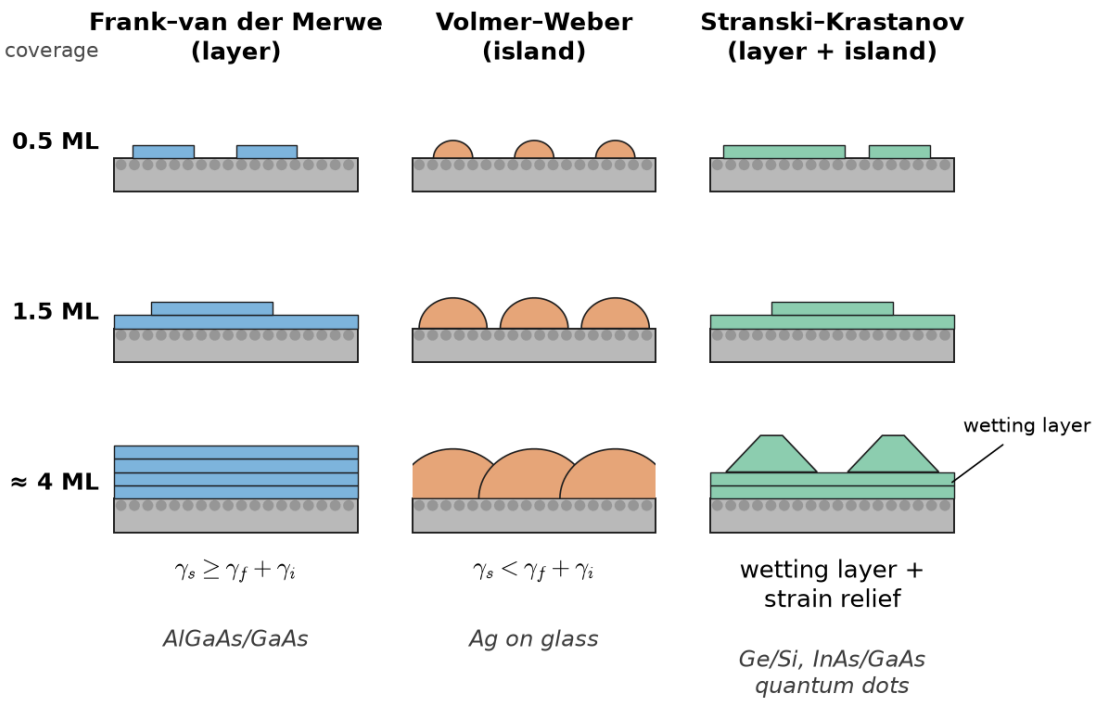


Figure 49.3 — The three growth modes: Frank–van der Merwe (layer), Volmer–Weber (island) and Stranski–Krastanov (layer plus island).

Structure-zone models

Practical polycrystalline films are far from equilibrium; their microstructure depends on how far arriving atoms move before burial. That mobility scales with the **homologous temperature** T_s/T_m and, in plasma processes, with ion bombardment. The **structure-zone model** (Movchan–Demchishin for evaporation, extended by Thornton for sputtering pressure and Anders for ion energy) classifies the result (Figure 49.4):

Zone	Approx. T_s/T_m	Mechanism	Microstructure and properties
1	below 0.2–0.3	Shadowing, negligible diffusion	Tapered columns with voids; porous, rough, poor barrier
T	0.2–0.4 (sputtering, ion assist)	Surface diffusion plus bombardment	Dense fibrous grains, smooth, often compressive stress
2	0.3–0.5	Surface diffusion	Columnar grains through the thickness, faceted surface
3	above 0.5	Bulk diffusion, recrystallization	Equiaxed, bulk-like grains

> **Status of the principle: Structure-zone models**
> *Empirical.* Zone boundaries come from micrographs of many evaporated and sputtered films and shift with rate, impurities and ion energy; they guide process design qualitatively.

Refractory compounds have high T_m (TiN about 3200 K), so at 450 °C ($T_s/T_m \approx 0.23$) they would grow porous in zone 1; the energetic ions of arc and HiPIMS push them into dense zone T, which is as important as their chemistry. Electron-beam PVD zirconia thermal barriers are grown deliberately columnar at about 1000 °C: the separated columns open and close as the metal

expands, giving strain tolerance (Chapter 41). Oblique incidence produces tilted, porous columns by shadowing, exploited in glancing-angle deposition of optical films and a defect on tool flanks.

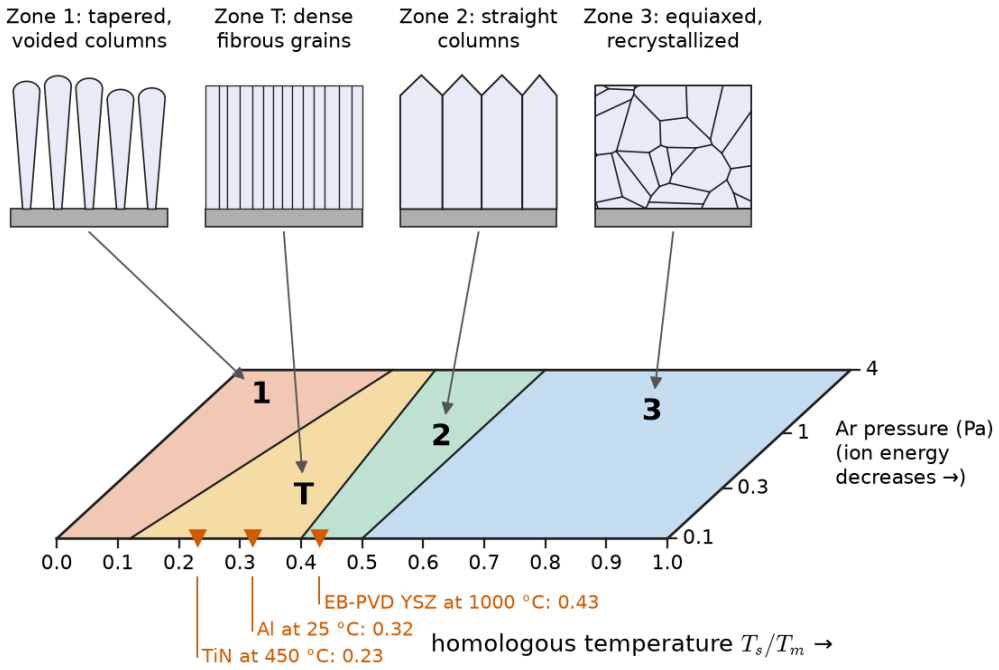


Figure 49.4 — Structure-zone model for sputtered films: microstructure versus homologous temperature and ion bombardment.

49.8 Film stress and adhesion

Nearly every film is stressed. Tensile stress cracks coatings, compressive stress buckles and spalls them, and either bends the substrate.

Thermal stress

A film deposited at T_d and cooled to T on a thick substrate must follow the substrate's contraction. For an isotropic film in equi-biaxial plane stress

$$\sigma_{th} = \frac{E_f}{1-\nu_f}\,(\alpha_s - \alpha_f)(T - T_d)$$

where $E_f/(1-\nu_f)$ is the **biaxial modulus**. A film that expands more than its substrate ends in tension after cooling.

> **Status of the principle: Thermal mismatch stress**
> *Derived.* Linear thermoelasticity for a thin film on a much thicker substrate; fails if the film yields, creeps or cracks during cooling, or if properties vary strongly with temperature.

Worked example — why CVD tool coatings crack. TiN ($\alpha_f = 9.4\times10^{-6}$ K^{-1}, $E_f = 450$ GPa, $\nu_f = 0.25$) deposited by CVD at 1000 °C on cemented carbide ($\alpha_s = 5.5\times10^{-6}$ K^{-1}) and cooled to 20 °C has mismatch strain $(5.5-9.4)\times10^{-6}\times(20-1000) = +3.82\times10^{-3}$ and

$$\sigma_{th} = \frac{450}{0.75}\ \text{GPa} \times 3.82\times10^{-3} = +2.3\ \text{GPa}$$

tensile, above the coating's strength. CVD inserts leave the furnace with fine **thermal cracks** perpendicular to the surface, tolerated because they do not enter the tough substrate (post-coat blasting adds compression). PVD at 450 °C gives a thermal term of only about +1 GPa, outweighed by compressive growth stress, so PVD coatings are crack-free; hence PVD for sharp tools

in interrupted cutting, CVD for thick alumina on turning inserts. The same formula gives about -180 MPa in thermal SiO_2 on silicon cooled from 1000 °C (biaxial modulus 87 GPa).

Intrinsic stress

- **Tension from grain coalescence**: islands that touch "zip up" into grain boundaries, stretching elastically; low-mobility evaporated metals reach hundreds of MPa.
- **Compression from atomic peening**: energetic particles drive atoms into interstitial sites and boundaries; biased PVD hard coatings reach -3 to -6 GPa and DLC up to -10 GPa. Higher sputtering pressure thermalizes the particles and swings stress toward tension.
- **Chemistry and phase change**: densification on annealing (sol–gel, hydrogen loss from PECVD films) gives tension; oxide growth with a larger volume gives compression (Pilling–Bedworth ratio, Chapter 42).

The Stoney equation

A stressed film bends its substrate. For $h_f \ll h_s$,

$$\sigma_f = \frac{E_s}{1-\nu_s}\,\frac{h_s^2}{6h_f}\left(\frac{1}{R}-\frac{1}{R_0}\right)$$

Only substrate constants appear, which is why wafer-curvature measurement is the industrial standard; heating while measuring separates thermal from intrinsic stress. X-ray diffraction measures stress through lattice strain instead (Chapter 52).

> **Status of the principle: Stoney equation**
> *Approximate.* Assumes $h_f \ll h_s$ (below about 5 %), uniform equi-biaxial stress, small deflection and an isotropic substrate.

Worked example — a silicon nitride film. A 1.0 µm PECVD SiN_x film on a 725 µm (100) silicon wafer (biaxial modulus 180.5 GPa) bends a flat wafer to $R = 40$ m:

$$\sigma_f = 180.5\times10^{9}\times\frac{(725\times10^{-6})^2}{6\times1.0\times10^{-6}}\times\frac{1}{40} = 395\ \text{MPa}$$

tensile if the film side is concave. That would buckle or crack a MEMS membrane, so MEMS nitride is tuned to 50–200 MPa tension by mixing plasma frequencies.

Adhesion as interface fracture

Adhesion is a fracture-mechanics problem (Chapter 14): an interface crack grows when the energy it releases, G, exceeds the interface toughness Γ_i. A long interface crack under a film of thickness h and biaxial stress σ releases, in steady state under plane strain,

$$G_{ss} = \frac{(1-\nu_f^2)\,\sigma^2 h}{2E_f} \qquad\Longrightarrow\qquad h_c = \frac{2E_f\,\Gamma_i}{(1-\nu_f^2)\,\sigma^2}$$

so films thicker than h_c delaminate. Compressive films show buckling blisters ("telephone cords"); tensile films first crack in channels, with the same $\sigma^2 h/E$ scaling.

> **Status of the principle: Steady-state delamination energy**
> *Derived.* Linear-elastic fracture mechanics for a thin film on a rigid substrate; needs a pre-existing interface flaw and neglects plasticity, which in metals raises the effective Γ_i.

Worked example — thickness limit of a hard coating. PVD TiN with $\sigma = -4$ GPa, $E_f = 450$ GPa, $\nu_f = 0.25$ on steel with $\Gamma_i = 50$ J/m^2: $h_c = 2\times450\times10^{9}\times50/(0.9375\times1.6\times10^{19}) = 3.0$ µm. With a poorly cleaned interface ($\Gamma_i = 20$ J/m^2), $h_c =$ 1.2 µm. This is why stressed PVD coatings are kept to 2–5 µm, and why ion etching before deposition and graded Cr or CrN interlayers matter.

Adhesion test	What it gives	Typical use
Tape (cross-hatch)	Pass/fail	Paints, decorative PVD
Pull-off stud	Tensile detachment stress (limited by glue, about 70–90 MPa)	Thermal spray, paints
Scratch test	Critical load L_c for cracking or spallation (50–100 N for good TiN on steel)	Hard coatings
Four-point bend, double cantilever beam	Interface fracture energy Γ_i (J/m^2)	Microelectronic stacks

Only the last gives a material parameter; scratch loads depend on substrate hardness and thickness and are comparative.

Many of the same deposition and growth ideas also make the single crystals and patterned layers of electronics. Single-crystal growth (Czochralski, float zone, Bridgman) and epitaxy, thermal oxidation, doping by diffusion and ion implantation, lithography and etching are treated in Chapter 50, which follows this chapter.

49.9 Thermal spray and hard coatings

Thermal spray

Thermal spraying heats powder or wire to a molten or softened state and propels it onto a grit-blasted surface. Each particle flattens into a **splat** a few micrometres thick that solidifies in microseconds; the coating is a stack of splats with oxide films, unmelted particles and pores. Coatings of 50 μm to several millimetres go on at kilograms per hour without heating the substrate above about 150 °C. The families trade particle temperature against velocity:

Process	Gas temperature	Particle velocity	Porosity	Bond strength	Typical coatings
Flame	about 3000 °C	40–100 m/s	10–15 %	10–30 MPa	Zn and Al on bridges
Arc wire	about 4000 °C	100–200 m/s	5–10 %	20–40 MPa	Zn, Al, steel rebuild
Atmospheric plasma	10 000–15 000 °C	100–300 m/s	2–20 % (designed)	30–70 MPa	YSZ, Al_2O_3, hydroxyapatite
HVOF	about 2800 °C	400–800 m/s	below 1 %	above 70 MPa	WC–Co, Cr_3C_2–NiCr, MCrAlY
Cold spray	below particle melting	500–1200 m/s	below 0.5 %	30–80 MPa	Cu, Al, Ti; aerospace repair

Only plasma melts zirconia (2700 °C); high velocity gives density and adhesion with little oxidation. HVOF WC–Co (1100–1400 HV) is replacing hard chromium on landing-gear cylinders and hydraulic rods. **Cold spray** particles stay solid and bond by adiabatic shear instability above a critical velocity, giving oxide-free, compressively stressed deposits for repair and large-scale additive manufacturing (Chapter 48). Plasma-sprayed **thermal barriers** carry 10–20 % designed porosity and microcracks, lowering YSZ conductivity from about 2.3 to 0.8–1.0 W m^{-1}K^{-1} and adding strain tolerance; EB-PVD columnar coatings are more durable on rotating blades, APS cheaper for combustors and vanes (Chapter 41). Plasma-sprayed **hydroxyapatite** (50–150 μm) on titanium hip stems and dental implants speeds bone bonding, with its crystallinity controlling dissolution in the body (Chapter 39).

The hard-coating family

Coating	Process	Thickness	Hardness (HV)	Oxidation limit in air	Typical use
Hard chromium	Electroplating	20–500 µm	800–1000	about 400 °C (softens)	Hydraulic rods (being replaced)
Electroless Ni–P	Autocatalytic	10–100 µm	500–600; 900–1000 aged	—	Moulds, valves
TiN	PVD, CVD	2–10 µm	2000–2400	500–600 °C	General tools, implants, decorative
TiAlN/AlTiN	PVD	2–5 µm	3000–3500	800–900 °C	Dry, high-speed machining
CrN	PVD	2–20 µm	1800–2200	about 700 °C	Forming tools, piston rings
α-Al_2O_3	CVD	3–15 µm	about 2000	stable	Heat barrier on turning inserts
DLC	PECVD, PVD	1–3 µm	1500–5000	about 350 °C	Fuel injectors, cam followers, low friction
CVD diamond	Hot-filament/microwave CVD	5–30 µm	about 10 000	about 600 °C	Machining Al–Si and CFRP
WC–Co	HVOF	100–300 µm	1100–1400	about 500 °C	Landing gear, pump sleeves

Two ideas run through modern hard coatings. **Nanolayers and nanocomposites** (TiN/CrN superlattices, nanocrystalline TiN in amorphous Si_3N_4) block dislocations and cracks, raising hardness and toughness: boundary strengthening (Chapter 13) at the nanometre scale. **Self-protection**: aluminium-rich nitrides grow a thin alumina scale at the hot edge, so TiAlN cuts dry where TiN fails. Substrate support matters too: a hard coating on a soft substrate cracks like an eggshell when the substrate yields, so steels are nitrided before PVD coating (a **duplex treatment**).

What remains open

Open question	Where it is answered
How are thickness, stress, texture and composition of a finished coating measured?	Chapter 52
How can a 2 nm interfacial layer or a single grain boundary be analysed atom by atom?	Chapter 53
How do films evolve while growing or in service?	Chapter 54
Can adsorption energies and epitaxial strain be predicted from first principles?	Chapter 55
How are growth morphologies simulated?	Chapter 56 (kinetic Monte Carlo, phase field)
How does one choose between PVD, CVD, spray and plating on cost?	Chapter 58
What are the environmental costs of chromium plating, fluorinated etch gases and vacuum processing?	Chapter 59
Why do thermal barriers and tool coatings ultimately fail?	Chapters 41, 44 and 45 (back reference)
How are single crystals grown, and how are oxidation, doping, lithography and etching used to make devices?	Chapter 50
Where does the diffusion physics of oxidation and doping come from?	Chapter 8 (back reference)

Next: Chapter 50 (Electronic Materials Manufacturing) turns to the single crystals, epitaxial layers and patterned films from which semiconductor devices are made.

Chapter 50 — Electronic Materials Manufacturing

Part X · Processing and Manufacturing

A batch of power MOSFETs fails its gate-leakage test. Every die came from the same wafer lot and the same mask set, and the device design has not changed for years. The failure analysis report traces the problem to three causes on three different wafers: one wafer had iron contamination at 10^{11} atoms/cm^3 from a furnace tube, which cut the minority-carrier lifetime by orders of magnitude; one had slip dislocations at its edge from a too-fast ramp in a rapid-thermal anneal; and one had gate oxides on transistors connected to long metal lines that had been damaged by charge collected during plasma etching. None of these is a design fault. Each is a change to the *material* — its purity, its crystal perfection, the integrity of an oxide a few nanometres thick — made by a manufacturing step. The wafer arrives as the most perfect crystal humans make; every subsequent step can only degrade it, and the job of process engineering is to make the degradation small and controlled.

> **How does each manufacturing step — growth, oxidation, diffusion, implantation, patterning, etching, metallization — change the structure and properties of semiconductor materials?**
> **Which defects does each step introduce, and what physical rules predict their size?**

Concept	Engineering destination
Czochralski and float-zone growth, segregation	Wafer resistivity, oxygen content, choice of substrate for CMOS versus power devices
SiC PVT growth, GaN substrates	Defect density and cost of wide-bandgap power and RF devices (Chapter 33)
Wafer slicing, polishing, gettering	Flatness for lithography, surface quality, contamination tolerance
Epitaxy, CVD, PVD, ALD	Device layers, gate dielectrics, barriers, interconnect films
Deal–Grove oxidation	Field and gate oxide thickness, silicon consumption
Constant- and limited-source diffusion	Junction depth, sheet resistance, emitter and well profiles
Ion implantation: range, straggle, damage, anneal	Shallow junctions, threshold-voltage adjust, isolation
Rayleigh resolution	Minimum feature size, cost of patterning
Wet and dry etching, anisotropy	Trench and via profiles, MEMS structures
Al and Cu metallization, barriers, silicides	Interconnect resistance, electromigration life, contact resistance
Dielectric deposition, low-k	RC delay, crosstalk, mechanical reliability of the interconnect stack
Process-induced defects	Leakage, lifetime, gate-oxide integrity, yield

Roadmap. Making the crystal (50.1) → making the wafer (50.2) → adding films (50.3) → oxidation (50.4) → doping by diffusion (50.5) and by implantation (50.6) → patterning: lithography (50.7) and etching (50.8) → wiring: metallization (50.9) and dielectric deposition (50.10) → packaging materials (50.11) → what goes wrong (50.12).

> **Boundary.**
> This chapter treats each manufacturing step as a materials process: what it does to crystal structure, composition, defects and properties. The sequencing of steps into a CMOS, bipolar or power-device flow, and the device physics that motivates each layer, belong to the Electronics & Computing core.

50.1 Single-crystal growth

Device-grade semiconductors must be single crystals: grain boundaries carry trapped charge and act as recombination and leakage paths (Chapter 32), and dopant diffusion along them would make junctions uncontrollable. Silicon is grown from the melt; silicon carbide and gallium nitride, which cannot be melted at practical pressures, need entirely different routes. The growth method fixes the substrate's purity, dopant uniformity, oxygen and carbon content, and dislocation density.

Czochralski growth

More than 90 % of silicon wafers come from **Czochralski (CZ)** crystals (Figure 50.1a). Polycrystalline silicon of 9N–11N purity is melted in a fused-silica (quartz) crucible at just above 1414 °C, under argon. Dopant (B, P, As or Sb) is added to the melt. A seed crystal of the desired orientation, usually ⟨100⟩, is dipped into the melt and slowly withdrawn while seed and crucible counter-rotate. Silicon freezes on the seed and the crystal grows downward from it.

Three steps of the process are aimed at specific defects:

- **Dash necking.** The seed is first pulled fast to make a thin neck, 3–4 mm across. The thermal shock of dipping introduces dislocations into the seed; in the thin neck they glide out of the crystal surface. Thereafter the crystal is grown dislocation-free — and remains so, because there is no source to create new dislocations in a crystal under low stress. A 300 mm silicon crystal with no dislocations at all, weighing several hundred kilograms, hangs from a neck a few millimetres in diameter.
- **Diameter control** by pull rate and melt temperature, monitored optically at the meniscus.
- **Pull rate and thermal gradient**, which control point-defect populations (below).

Maximum pull rate. The latent heat released at the interface must be conducted away through the solid. A heat balance at the interface, neglecting heat flow from the melt, gives

$$v_{\max} = \frac{k_s G_s}{\rho_s L}$$

with k_s the thermal conductivity of solid silicon near the melting point (about 22 W/m K), G_s the axial temperature gradient in the solid, $\rho_s = 2330$ kg/m^3 and latent heat $L = 1.79 \times 10^6$ J/kg. For $G_s = 30$ K/cm $= 3000$ K/m, $v_{\max} \approx 22 \times 3000/(2330 \times 1.79 \times 10^6) = 1.6 \times 10^{-5}$ m/s, or about 1 mm/min — the right order for large-diameter crystals. A steeper gradient allows faster growth but raises thermal stress.

> **Status of the principle: interface heat balance for maximum pull rate**
> *Approximate.* It is a one-dimensional energy balance with heat from the melt neglected; it gives the correct order of magnitude and the scaling with gradient. Real crystal pullers are designed with numerical heat-transfer models.

Point defects and the Voronkov criterion. Silicon freezes with equilibrium concentrations of both vacancies and self-interstitials, which recombine and are swept by diffusion as the crystal cools. Voronkov showed that which species survives in excess depends on the ratio v/G of pull rate to gradient at the interface: above a critical value (about 1.3×10^{-3} cm^2/(K min)) vacancies dominate and cluster into octahedral voids of 50–200 nm (**crystal-originated particles, COPs**), which degrade thin gate oxides; below it, interstitials dominate and form dislocation loops. "Perfect silicon" for advanced logic is grown at precisely controlled v/G close to the critical value.

Segregation along the crystal. Dopant is partitioned at the interface with an equilibrium coefficient $k_0 = C_s/C_L$ (Table below). With the melt well mixed and no diffusion in the solid, the Scheil equation of Chapter 9 applies directly, with fraction solidified f_s now measured along the crystal:

$$C_s(f_s) = k_{\text{eff}}\, C_0\, (1 - f_s)^{k_{\text{eff}} - 1}$$

The effective coefficient k_{eff} lies between k_0 and 1, because a diffusion boundary layer of rejected solute sits at the interface; it rises toward 1 at higher pull rate and lower crystal rotation.

Impurity in Si	k_0	Consequence
B	0.8	Nearly uniform along the crystal
P	0.35	Concentration rises strongly toward the tail
As	0.3	Similar to P; volatile from melt
Sb	0.023	Hard to dope heavily; evaporates
O	≈ 0.25–1 (uncertain)	Set mostly by crucible dissolution and evaporation
C	0.07	Accumulates in melt; tail enriched
Al	0.002	Strongly rejected
Cu	4×10^{-4}	Purified by freezing
Fe	8×10^{-6}	Purified by freezing
Au	2.5×10^{-5}	Purified by freezing

Typical literature values.

The very small k_0 of transition metals is why directional solidification (and zone refining) purifies: metals stay in the melt and end up in the tail, which is cut off.

Worked example — Resistivity variation along a CZ crystal. A boron-doped and a phosphorus-doped crystal are each grown with the same initial melt concentration. How much does the dopant concentration change between the seed end ($f_s = 0$) and the point where 90 % of the melt has frozen? Using $k_{\text{eff}} \approx k_0$,

$$\frac{C_s(0.9)}{C_s(0)} = (1 - 0.9)^{k-1}: \qquad \text{B: } 0.1^{-0.2} = 1.58, \qquad \text{P: } 0.1^{-0.65} = 4.47$$

At the half-way point the ratios are 1.15 (B) and 1.57 (P). Since resistivity scales roughly as $1/N$ in the extrinsic range (Chapter 32), a phosphorus crystal specified to ±15 % resistivity yields far fewer usable wafers than a boron one. This is one reason p-type (boron) substrates dominate CMOS, and why n-type power-device substrates use neutron transmutation doping (below).

Oxygen. The quartz crucible dissolves slowly ($\text{SiO}_2 + \text{Si} \rightarrow 2\text{SiO}$); most SiO evaporates from the melt surface but some oxygen is incorporated, typically $5\text{–}10 \times 10^{17}$ cm^{-3} interstitial oxygen. Oxygen is not simply an impurity. In solution it pins dislocations and strengthens the wafer against slip during thermal processing; precipitated as SiO_2 clusters in the wafer bulk during later anneals, it creates strain fields that capture metallic contaminants (**intrinsic gettering**, Section 50.2). But at 350–500 °C, oxygen forms **thermal donors** that shift resistivity, and in solar cells boron–oxygen complexes cause light-induced degradation. **Magnetic CZ (MCZ)** applies a magnetic field to damp melt convection, reducing oxygen transport from the crucible wall and improving radial uniformity.

Float-zone growth

In **float-zone (FZ)** growth (Figure 50.1b), a polycrystalline rod is held vertically and a narrow molten zone, sustained by an RF induction coil, is passed along it; the zone is held in place by surface tension and electromagnetic levitation. A seed at one end sets the orientation. There is no crucible, so oxygen is below 10^{16} cm^{-3} and carbon very low; repeated passes also purify by segregation. FZ silicon reaches resistivities above 10 kΩ cm and lifetimes of milliseconds. It is used for high-voltage power devices (thyristors, IGBTs, power diodes), radiation detectors and RF substrates. Disadvantages: diameter is limited (up to 200 mm), cost is higher, and the low oxygen makes wafers more prone to slip.

Neutron transmutation doping (NTD) gives FZ silicon an exceptionally uniform n-type doping. Irradiating with thermal neutrons in a reactor converts a small fraction of the 3.1 % natural ^{30}Si:

$$^{30}\text{Si}(n,\gamma)\,^{31}\text{Si} \xrightarrow{\beta^-,\, t_{1/2}=2.6\ \text{h}} {}^{31}\text{P}$$

Because neutrons penetrate the whole crystal almost uniformly, the phosphorus is distributed with a few per cent uniformity, impossible by melt doping with $k_0 = 0.35$. An anneal removes the radiation damage. High-voltage devices need this uniformity because breakdown voltage depends on the lowest local resistivity.

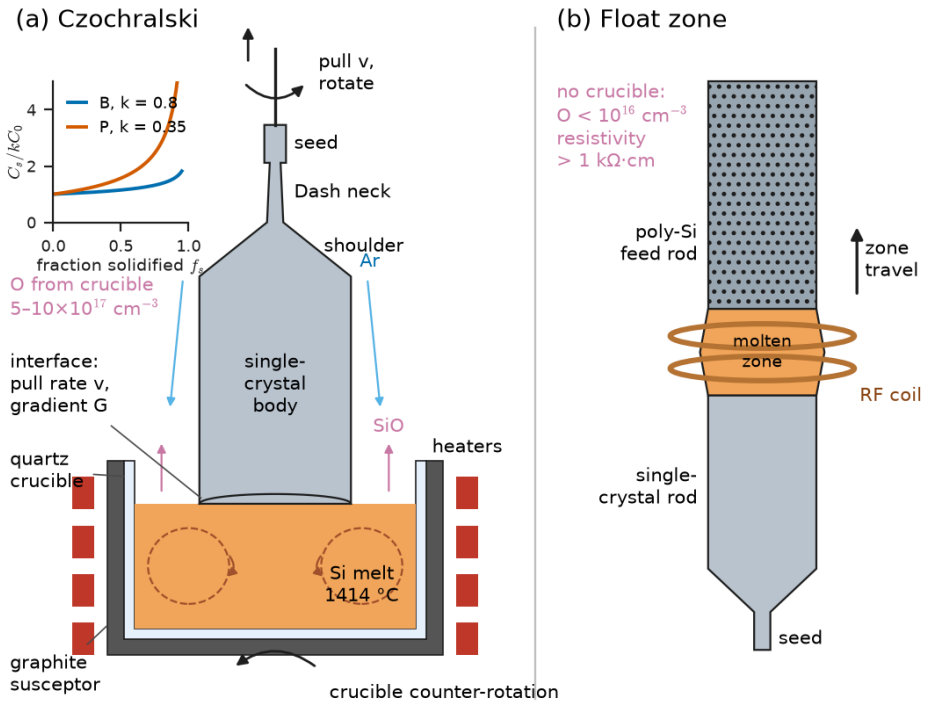


Figure 50.1 — Czochralski and float-zone crystal growth of silicon, with the main material consequences of each method.

Silicon carbide by physical vapour transport

SiC does not melt at atmospheric pressure; it sublimes and decomposes above about 2500 °C. Bulk crystals are grown by **physical vapour transport (PVT)**, also called the seeded sublimation or modified Lely method. SiC powder in a graphite crucible is heated to 2200–2500 °C; Si, Si_2C and SiC_2 vapour species diffuse across a temperature gradient to a SiC seed held at a slightly lower temperature, where they condense. Growth rates are 0.1–0.5 mm/h, so a 25–50 mm long boule takes days.

The difficulties are of a kind that melt growth never meets:

- **Polytype control.** SiC exists in over 200 stacking sequences (polytypes); 4H-SiC is the one wanted for power devices because of its high, nearly isotropic electron mobility and 3.26 eV band gap (Chapter 33). Temperature, seed polarity, supersaturation and impurities decide which polytype grows; an inclusion of another polytype ruins the wafer.
- **Dislocations.** Growth at high temperature under thermal gradients produces threading screw dislocations (TSD), threading edge dislocations (TED) and basal-plane dislocations (BPD), with total densities of 10^3–10^4 cm^{-2} in good material. **Micropipes** — hollow-core screw dislocations with very large Burgers vectors — were once the main killer defect and are now below 0.1 cm^{-2} in commercial wafers.
- **Hardness.** SiC is nearly as hard as diamond (Mohs 9.5), so slicing and polishing are slow and expensive (Section 50.2).

Basal-plane dislocations matter in service: under forward current, electron–hole recombination at a BPD in the drift layer drives the glide of partial dislocations, expanding a Shockley stacking fault across the device and raising its on-resistance (**bipolar degradation**, Chapter 44). Epitaxial growth on substrates cut 4° off the basal plane converts most BPDs into harmless threading edge dislocations at the substrate–epi interface; 4H-SiC wafers of 150 mm are standard and 200 mm is entering production.

Gallium nitride substrates

Gallium nitride decomposes long before it melts (melting would need nitrogen pressures of several GPa at about 2500 °C), so it cannot be grown from a stoichiometric melt. Most GaN devices are therefore **heteroepitaxial** layers on foreign substrates:

Substrate	Lattice mismatch to GaN (a)	CTE (ppm/K)	Typical threading dislocations in GaN (cm^{-2})	Use
Sapphire (Al_2O_3)	≈ 16 %	≈ 7.5	10^8–10^9	LEDs (cheap, insulating, transparent)
4H/6H-SiC	≈ 3.5 %	≈ 4.2–4.5	10^8–10^9	RF HEMTs (high thermal conductivity)
Si (111)	≈ 17 %	2.6	10^8–10^{10}	Power HEMTs on large, cheap wafers
Free-standing GaN (HVPE)	0	5.6	10^5–10^6	Laser diodes, vertical power devices
Ammonothermal GaN	0	5.6	10^3–10^4	Emerging: vertical power, high-power lasers

Typical values; GaN in-plane CTE ≈ 5.6 ppm/K.

GaN on silicon is the most demanding: the large thermal-expansion mismatch puts the GaN film in tension on cooling from about 1050 °C growth temperature, and films crack unless graded AlGaN or AlN/GaN superlattice buffers introduce compensating compressive stress. Free-standing GaN wafers are made by growing very thick layers (millimetres) by **hydride vapour phase epitaxy (HVPE)**, in which GaCl formed from HCl over liquid Ga reacts with NH_3 at 1000–1100 °C at up to hundreds of µm/h, on sapphire or GaAs, and then separating the layer; dislocations annihilate as the layer thickens. **Ammonothermal** growth dissolves GaN in supercritical ammonia at 400–600 °C and 100–600 MPa and recrystallises it on a seed, analogous to hydrothermal quartz growth; it gives the lowest dislocation densities but grows slowly.

Even 10^8 cm^{-2} dislocations would destroy a silicon device, yet GaN LEDs and HEMTs work with them. This tolerance reflects the short carrier diffusion lengths and the electrically less active dislocation cores in GaN, and, for HEMTs, the fact that conduction is in a two-dimensional channel (Chapter 34). Vertical GaN power devices, which block voltage across the full thickness, do need native substrates.

50.2 Wafer preparation

A grown crystal is a cylinder with a conical top, a tail and a rough surface. Turning it into wafers flat to tens of nanometres over 300 mm, with a surface roughness of a few ångströms and essentially no subsurface damage, is a materials-removal process in which every mechanical step creates damage that the next step must remove.

Shaping and slicing

1. **Crop and grind.** Seed and tail ends are cut off; the crystal is ground to precise diameter. Orientation is measured by X-ray diffraction (Chapter 52) and marked by a **notch** (300 mm) or **flat** (≤ 200 mm).
2. **Slicing.** A multi-wire saw — a single steel wire, 50–150 µm thick, wound hundreds of times around grooved guides — cuts the whole crystal at once, using diamond particles fixed to the wire or carried in a slurry. Sawing leaves a damaged surface layer of 10–30 µm containing microcracks and dislocations, and removes a **kerf** about as wide as the wire plus abrasive.
3. **Edge profiling.** The sharp edges are ground to a rounded profile, because a sharp edge chips during handling and the chips become particles, and because stress concentrations at the edge nucleate slip in later thermal steps.
4. **Lapping or double-side grinding**, then **chemical etching** (alkaline or HF/HNO_3 mixtures) to remove the damaged layer without introducing new damage.
5. **Polishing.** Double-side polishing followed by a final **chemical–mechanical polish (CMP)** with colloidal silica in alkaline slurry. The chemistry softens the surface into a hydrated layer and the mechanical action removes it, leaving a damage-free, atomically smooth surface. Removal rate is described by the Preston relation, rate $= K_P\, P\, v$, proportional to pressure and relative velocity.

> **Status of the principle: Preston equation for polishing**
> *Empirical.* Removal rate is proportional to pressure and velocity over the normal process window; the constant K_P depends on slurry, pad and material, and the relation fails at very low pressures (chemistry-limited) and high pressures (pad saturation).

Worked example — How much silicon becomes a wafer? A 300 mm wafer's final thickness is 775 µm. Suppose the wire kerf is 150 µm and grinding, etching and polishing together remove 125 µm from an as-cut slice. The slice pitch along the crystal is then $775 + 125 + 150 = 1050$ µm, so a 1 m crystal body yields about 950 wafers, and only $775/1050 = 74$ % of the silicon ends up in wafers; 26 % is lost as sawdust and polishing waste. For SiC, where crystals grow at a fraction of a millimetre per hour and slicing is slow and wire wear high, kerf loss is a serious cost driver; laser-based splitting methods that separate wafers along a laser-modified plane with almost no kerf were developed for this reason.

Specifications that matter

Parameter	Typical 300 mm prime wafer	Why
Thickness	775 ± 20 µm	Mechanical handling, thermal mass
Total thickness variation (TTV)	< 1 µm	Lithography focus
Site flatness (SFQR, 26 × 8 mm site)	< 20–30 nm (advanced nodes)	Depth of focus of exposure tools (Section 50.7)
Surface roughness	≈ 0.1–0.2 nm RMS	Gate-oxide integrity, bonding
Surface metals (each element)	$< 10^{10}$ atoms/cm^2	Lifetime, oxide integrity
Particles	< tens per wafer above ≈ 30 nm	Yield
Interstitial oxygen	$5–8 \times 10^{17}$ cm^{-3} (CZ)	Slip resistance, gettering

Typical values; exact limits depend on node and customer specification.

Cleaning and gettering

Between steps, wafers are cleaned in sequences descended from the **RCA clean**: an alkaline peroxide solution (SC-1, $NH_4OH/H_2O_2/H_2O$) that lifts particles and removes organics by gently etching the surface, an acidic peroxide (SC-2, $HCl/H_2O_2/H_2O$) that dissolves metallic contaminants, and dilute HF to strip the chemical oxide and leave a hydrogen-terminated surface.

No cleaning removes all metals, and fast-diffusing metals (Fe, Cu, Ni) introduced at any later step migrate through the wafer at process temperatures (Section 50.12). **Gettering** gives them somewhere harmless to go. In **intrinsic gettering**, a high-temperature anneal (above about 1100 °C) out-diffuses oxygen from a surface layer 10–50 µm deep — the **denuded zone** where devices will be built — followed by lower-temperature anneals that nucleate and grow oxygen precipitates in the bulk. Their strain fields and associated dislocation loops trap metals far from the devices. **Extrinsic gettering** uses backside damage, a polysilicon backside layer or heavy phosphorus diffusion. Gettering is a direct use of Chapter 9's diffusion and nucleation rules: metal atoms must diffuse to the getter sites ($\sqrt{Dt}$ must exceed the wafer half-thickness) and precipitate there.

Engineered substrates

Two substrate variants are made from prime wafers. **Epitaxial wafers** carry a lightly doped epitaxial layer, a few to tens of micrometres thick, on a heavily doped substrate; the epitaxial layer is free of COPs and oxygen precipitates, while the substrate provides gettering and a low-resistance path. **Silicon-on-insulator (SOI)** wafers place a thin device layer on a buried oxide. The dominant method, Smart Cut, implants hydrogen ($\sim 5 \times 10^{16}$ cm^{-2}) into an oxidised donor wafer to a controlled depth, bonds it to a handle wafer, and heats it to 400–600 °C; hydrogen platelets coalesce into a crack plane and the donor splits, leaving a thin silicon layer on the oxide. The process is an example of using implantation damage (Section 50.6) constructively.

50.3 Thin-film deposition

Chapter 49 introduced the principles of PVD, CVD and ALD, the structure-zone model and film stress. Here we look at what each method does to semiconductor materials, with emphasis on epitaxy and the rate laws of CVD.

Epitaxy

Epitaxial growth continues the substrate crystal into a deposited layer whose doping and composition can be chosen independently. Silicon epitaxy is done by CVD from SiH_4, SiH_2Cl_2 or $SiHCl_3$ in hydrogen at 600–1150 °C (lower temperatures for SiGe and abrupt profiles), after an in-situ hydrogen bake removes native oxide. The growing layer copies defects at the interface: particles, residual oxide and carbon nucleate stacking faults and dislocations, so pre-epitaxial cleaning is critical. Dopant from a heavily doped substrate **autodopes** the layer by evaporation and re-deposition and diffuses up into it, so profiles at the interface are not abrupt.

Compound semiconductors are grown by:

- **Metal–organic CVD (MOCVD)**: trimethylgallium, trimethylaluminium, trimethylindium and NH_3 (for nitrides) or AsH_3/PH_3 (for arsenides and phosphides), at 500–1100 °C. GaN is grown at about 1000–1100 °C with a low-temperature nucleation layer. MOCVD is the production method for LEDs, lasers and GaN HEMTs.
- **Molecular-beam epitaxy (MBE)**: elemental beams in ultra-high vacuum at about 1 μm/h, with atomic-layer control monitored by electron diffraction. Used for research, high-mobility heterostructures and some RF and laser production.
- **SiC epitaxy** by CVD from silane and propane at 1500–1650 °C on 4°-off-axis substrates, where growth proceeds by **step flow**: adatoms attach at the steps exposed by the miscut and therefore copy the stacking of the substrate, preserving the 4H polytype.

Heteroepitaxial layers with lattice mismatch $f = (a_s - a_f)/a_f$ grow pseudomorphically — strained to the substrate lattice — up to a **critical thickness**, beyond which introducing misfit dislocations lowers the total energy (the Matthews–Blakeslee analysis, Chapter 34). Strained $Si_{1-x}Ge_x$ layers and AlGaN barriers in GaN HEMTs are deliberately kept below it, because the strain itself is part of the function (mobility enhancement, piezoelectric polarisation).

CVD rate laws: the Grove model

A CVD film grows by two steps in series: transport of precursor from the gas through a stagnant boundary layer to the surface, with flux $F_1 = h_g(C_g - C_s)$, where h_g is a mass-transfer coefficient; and the surface reaction, with flux $F_2 = k_s C_s$ for a first-order reaction with rate constant k_s. At steady state $F_1 = F_2$; eliminating the unknown surface concentration C_s,

$$F = \frac{h_g k_s}{h_g + k_s}\, C_g, \qquad \text{growth rate} = \frac{F}{N}$$

where N is the number of film atoms per unit volume. Two limits follow:

- $k_s \ll h_g$ (low temperature): **reaction-limited**, rate $\approx k_s C_g/N$, Arrhenius in temperature, insensitive to gas flow. Requires tight temperature uniformity; gives excellent conformality because all surfaces see the same $C_s \approx C_g$. LPCVD furnaces run here and can stack 100–150 wafers a few millimetres apart.
- $k_s \gg h_g$ (high temperature): **transport-limited**, rate $\approx h_g C_g/N$, weakly temperature dependent, sensitive to reactor geometry and flow. Single-wafer epitaxial reactors often operate here.

> **Status of the principle: Grove model of CVD**
> *Approximate*. It treats transport and reaction as two first-order steps in series with a uniform boundary layer. It explains the two regimes and the Arrhenius kink; real chemistries involve multiple gas-phase and surface reactions.

The same series-resistance structure reappears in thermal oxidation (Section 50.4), where a diffusion step through the growing oxide is added.

PVD and ALD in device manufacturing

Sputtering (Chapter 49) deposits metal films for contacts and interconnects (Ti, TiN, Ta, TaN, Al, Cu seed layers, Co, W) and, reactively, nitrides and oxides. Because sputtered atoms arrive over a broad range of angles, coverage of high-aspect-ratio

features is poor; **ionised PVD**, in which a large fraction of the sputtered metal is ionised and accelerated to the wafer by substrate bias, directs flux into the bottoms of vias for barrier and seed deposition.

ALD deposits gate dielectrics (HfO_2, HfSiON, Al_2O_3), work-function metals (TiN, TiAl), and barriers in narrow features. Its self-limiting chemistry gives thickness control to about one atomic layer over a 300 mm wafer and conformal coating of three-dimensional transistor fins. The trade-off is that precursor ligands leave carbon, chlorine or hydrogen residues, and films deposited at 250–350 °C are often amorphous and must be annealed — which can crystallise them and create grain boundaries that act as leakage paths (Chapter 34).

50.4 Oxidation

Silicon's dominance owes much to one material fact: heating it in oxygen or steam grows an amorphous SiO_2 film with an interface to silicon of extraordinary quality — interface-trap densities of order 10^{10} cm^{-2} eV^{-1} after hydrogen annealing. Neither germanium nor the III–V compounds have a comparable native oxide; SiC has one, but its interface is much worse (Chapter 34).

Why oxide grows the way it does

Oxide grows at the Si/SiO_2 interface, not at the surface: oxidant molecules (O_2 or H_2O) must diffuse through the oxide already formed. Each micrometre of oxide consumes 0.44 μm of silicon, because the molar volume of SiO_2 is about 2.2 times that of Si; the oxide surface rises above the original silicon surface and the interface moves into the wafer.

The Deal–Grove model

Deal and Grove (1965) wrote the oxidation rate as three fluxes in series (compare the Grove CVD model above):

1. Transport from the gas to the oxide surface: $F_1 = h\,(C^{*} - C_o)$, where C^{*} is the equilibrium concentration of oxidant in the oxide (proportional to its partial pressure, by Henry's law) and C_o the actual concentration at the outer surface.
2. Diffusion through the oxide of thickness x, with a linear concentration profile: $F_2 = D\,(C_o - C_i)/x$.
3. Reaction at the interface: $F_3 = k_s\,C_i$.

At steady state $F_1 = F_2 = F_3 = F$. Eliminating C_o and C_i,

$$F = \frac{k_s C^{*}}{1 + \dfrac{k_s}{h} + \dfrac{k_s x}{D}}$$

The oxide grows at $dx/dt = F/N_1$, where N_1 is the number of oxidant molecules incorporated per unit volume of oxide (2.2×10^{22} cm^{-3} for O_2, 4.4×10^{22} cm^{-3} for H_2O). Separating variables and integrating from an initial thickness x_i at $t = 0$:

$$x^2 + A x = B\,(t + \tau), \qquad A = 2D\left(\frac{1}{k_s} + \frac{1}{h}\right), \quad B = \frac{2DC^{*}}{N_1}, \quad \tau = \frac{x_i^2 + A x_i}{B}$$

with solution

$$x = \frac{A}{2}\left[\sqrt{1 + \frac{t + \tau}{A^2/4B}} - 1\right]$$

Two limits:

- **Thin oxides, short times** ($t + \tau \ll A^2/4B$): $x \approx \dfrac{B}{A}\,(t + \tau)$. Growth is **linear**, limited by the interface reaction; B/A is the linear rate constant.
- **Thick oxides, long times** ($t \gg A^2/4B$): $x^2 \approx Bt$. Growth is **parabolic**, limited by diffusion through the oxide; B is the parabolic rate constant. This is the $\sqrt{Dt}$ behaviour of Chapter 8 again.

Figure 50.2 shows both regimes on logarithmic axes, where they appear as slopes of 1 and 1/2.

Status of the principle: Deal–Grove oxidation model
Approximate. It is derived from steady-state series fluxes with Fickian diffusion and first-order interface reaction, with A and B fitted. It describes wet and dry oxidation from about 30 nm to several micrometres; it underestimates the rapid initial growth of dry oxides below about 20–30 nm (absorbed into the fitted τ), and does not describe oxidation of heavily doped, strained or non-planar silicon without modification.

Rate constants. B is proportional to D, so it is Arrhenius with the activation energy of oxidant diffusion in SiO_2 (about 1.23 eV for O_2, 0.71 eV for H_2O). $B/A \approx k_s C^*/N_1$ follows the interface reaction, with activation energy about 2.0 eV, and depends on crystal orientation: (111) silicon oxidises about 1.7 times faster than (100) in the linear regime because it has more Si–Si bonds available per unit area. Wet oxidation is much faster than dry because H_2O is far more soluble in SiO_2 than O_2 (C^* is about 600 times larger at 1000 °C).

Ambient	T (°C)	A (μm)	B (μm²/h)	B/A (μm/h)	τ (h)
Dry O_2	800	0.370	0.0011	0.0030	9
Dry O_2	920	0.235	0.0049	0.0208	1.4
Dry O_2	1000	0.165	0.0117	0.0709	0.37
Dry O_2	1100	0.090	0.027	0.30	0.076
Dry O_2	1200	0.040	0.045	1.12	0.027
Wet (H_2O, 640 torr)	920	0.50	0.203	0.406	0
Wet (H_2O, 640 torr)	1000	0.226	0.287	1.27	0
Wet (H_2O, 640 torr)	1100	0.11	0.510	4.64	0
Wet (H_2O, 640 torr)	1200	0.05	0.720	14.4	0

Original Deal–Grove constants for (111) silicon. For (100) silicon divide B/A by about 1.7; B is nearly orientation independent.

Worked example — Field oxide and gate-quality oxide. (a) A 1 h wet oxidation at 1000 °C. With $A = 0.226$ μm, $B = 0.287$ μm²/h, $\tau = 0$:

$$x^2 + 0.226x - 0.287 = 0 \;\Rightarrow\; x = \frac{-0.226 + \sqrt{0.0511 + 1.148}}{2} = \frac{-0.226 + 1.095}{2} = 0.435\ \mu\text{m}$$

The characteristic time $A^2/4B = 0.0445$ h (2.7 min) is much shorter than 1 h, so growth is nearly parabolic; pure parabolic growth would give $\sqrt{0.287} = 0.536$ μm, an overestimate of 23 %. The oxidation consumes $0.44 \times 0.435 = 0.19$ μm of silicon.

(b) A 2 h dry oxidation at 1000 °C, with $\tau = 0.37$ h:

$$x^2 + 0.165x - 0.0117 \times 2.37 = 0 \;\Rightarrow\; x = \frac{-0.165 + \sqrt{0.0272 + 0.1109}}{2} = \frac{-0.165 + 0.372}{2} = 0.103\ \mu\text{m}$$

Dry oxidation is about four times slower, but gives a denser oxide with fewer defects and a better interface, so it is used for gate and pad oxides; wet oxidation is used for thick isolation and masking oxides. Gate oxides of a few nanometres are grown at lower temperature or in diluted oxygen, or by rapid thermal oxidation, so that thickness can be controlled to a fraction of a nanometre.

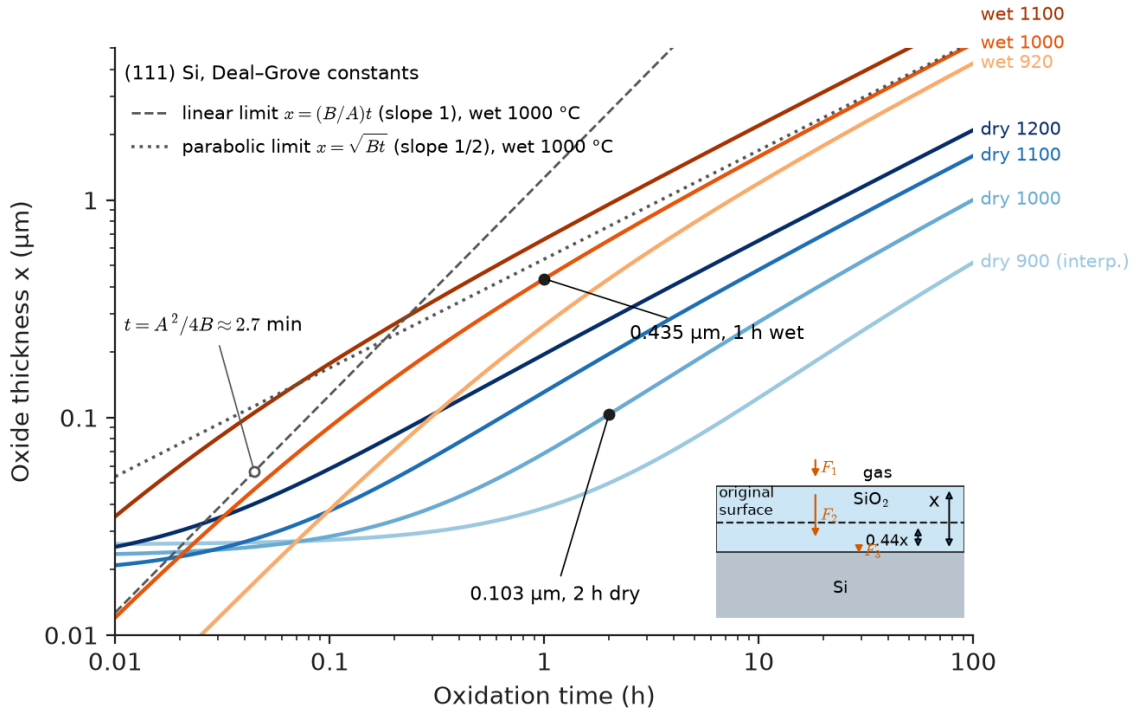


Figure 50.2 — Oxide thickness versus time for dry and wet oxidation of silicon from the Deal–Grove model, with the linear and parabolic limits.

Oxidation as a materials process

Oxidation changes more than the oxide:

- **Dopant redistribution.** Dopants segregate between Si and SiO_2: boron prefers the oxide (silicon surface depleted), phosphorus and arsenic prefer the silicon (pile-up at the interface). Surface concentrations, and so threshold voltages and contact resistances, change.
- **Point-defect injection.** Oxidation injects silicon self-interstitials into the substrate, which enhances diffusion of B and P (**oxidation-enhanced diffusion**) and can grow **oxidation-induced stacking faults** (Section 50.12).
- **Stress.** The volume expansion is accommodated by viscous flow of the oxide only above about 950 °C; oxides grown at lower temperature are highly compressive, and oxidation at corners is retarded, which is why rounded trench corners need high-temperature oxidation.
- **Interface charge.** As-grown interfaces have fixed charge and traps from incompletely oxidised silicon; a forming-gas anneal (H_2/N_2 at about 400–450 °C) passivates dangling bonds with hydrogen, an effect that later hot-carrier or radiation stress can reverse (Chapter 34).

50.5 Dopant diffusion

Before ion implantation became universal, all junctions were made by thermal diffusion from a gas, liquid or solid source; diffusion still controls how implanted profiles spread during every subsequent high-temperature step. The physics is Fick's second law (Chapter 8); the engineering content lies in the boundary conditions and in the material parameters.

Dopant diffusivities in silicon

Substitutional dopants diffuse in silicon via vacancies and self-interstitials, with activation energies of 3.5–4.2 eV. Representative intrinsic values:

Dopant	D_0 (cm^2/s)	E_a (eV)	D at 1000 °C (cm^2/s)	D at 1100 °C (cm^2/s)
B	0.76	3.46	1.5×10^{-14}	1.5×10^{-13}
P	3.85	3.66	1.2×10^{-14}	1.4×10^{-13}
As	—	≈ 4.1	$\approx 1.5 \times 10^{-15}$	$\approx 2 \times 10^{-14}$
Sb	—	≈ 3.9	$\approx 1 \times 10^{-15}$	$\approx 1 \times 10^{-14}$

Intrinsic (low-concentration) values, typical of literature compilations; at high doping, diffusivity rises with concentration and depends on the Fermi level.

Arsenic and antimony diffuse an order of magnitude more slowly than B and P, which is why they are used for shallow n^+ regions and for buried layers that must not move during later steps. Metallic impurities, by contrast, diffuse interstitially with activation energies below 1 eV and diffusivities 10^6–10^{10} times larger (Section 50.12).

Constant-source diffusion: the erfc profile

In a **predeposition**, the surface is held at a fixed concentration C_s, usually the solid solubility of the dopant at the process temperature (a few 10^{20} cm^{-3} for B, P and As near 1000–1100 °C), by a dopant-saturated gas or glass source. With $C(x,0) = 0$ in the wafer and $C(0,t) = C_s$, the solution of Fick's second law is

$$C(x,t) = C_s \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

The total dose introduced per unit area grows as $\sqrt{t}$:

$$Q(t) = \int_0^\infty C\,dx = \frac{2}{\sqrt{\pi}}\,C_s\sqrt{Dt}$$

Limited-source diffusion: the Gaussian profile

In a **drive-in**, the source is removed (the surface is usually oxidised) and a fixed dose Q redistributes deeper. If the predeposited layer is thin compared with the final profile, the initial condition is effectively a delta function of strength Q at the surface, and with no flux through the surface (a reflecting boundary) the solution is

$$C(x,t) = \frac{Q}{\sqrt{\pi Dt}}\exp\left(-\frac{x^2}{4Dt}\right)$$

The surface concentration $C(0,t) = Q/\sqrt{\pi Dt}$ falls as the profile spreads, while the area under the curve stays Q. Figure 50.3 compares the two shapes on a logarithmic scale: the erfc profile falls off more steeply near the surface for the same $\sqrt{Dt}$.

> **Status of the principle: erfc and Gaussian dopant profiles**
> *Derived* from Fick's second law with constant D and idealised boundary conditions. They are good approximations at low concentration. At high concentrations D depends on C (charged point defects, electric-field enhancement), profiles become box-like (B, As) or develop kinks and tails (P), and point-defect injection by oxidation or implantation damage enhances diffusion; process simulators then solve coupled dopant–defect equations.

Junction depth. In a substrate of opposite type with background concentration N_B, the metallurgical junction x_j is where $C(x_j) = N_B$:

$$\text{erfc: } x_j = 2\sqrt{Dt}\,\operatorname{erfc}^{-1}\left(\frac{N_B}{C_s}\right), \qquad \text{Gaussian: } x_j = 2\sqrt{Dt}\sqrt{\ln\frac{Q}{N_B\sqrt{\pi Dt}}}$$

Worked example — Two-step boron diffusion. An n-type wafer with $N_B = 10^{16}$ cm^{-3} receives a boron predeposition at 1000 °C for 30 min with $C_s = 2 \times 10^{20}$ cm^{-3}, then a drive-in at 1100 °C for 2 h.

Predeposition. $D = 0.76\exp(-3.46/(8.617\times10^{-5}\times 1273)) = 1.52 \times 10^{-14}$ cm^2/s; $t = 1800$ s, so $\sqrt{Dt} = 5.23 \times 10^{-6}$ cm (52 nm). The dose is

$$Q = \frac{2}{\sqrt{\pi}} \times 2 \times 10^{20} \times 5.23 \times 10^{-6} = 1.18 \times 10^{15}\ \text{cm}^{-2}$$

The junction depth after predeposition alone: $\operatorname{erfc}(u) = 10^{16}/(2 \times 10^{20}) = 5 \times 10^{-5}$ gives $u = 2.87$, so $x_j = 2 \times 2.87 \times 5.23 \times 10^{-6} = 3.0 \times 10^{-5}$ cm $= 0.30$ µm.

Drive-in. At 1373 K, $D = 1.51 \times 10^{-13}$ cm²/s; $t = 7200$ s, so $Dt = 1.09 \times 10^{-9}$ cm² and $\sqrt{Dt} = 3.30 \times 10^{-5}$ cm. This Dt is 40 times that of the predeposition (2.74×10^{-11} cm²), so the delta-function approximation is reasonable. The surface concentration falls to

$$C(0) = \frac{1.18 \times 10^{15}}{\sqrt{\pi} \times 3.30 \times 10^{-5}} = 2.0 \times 10^{19}\ \text{cm}^{-3}$$

and the junction depth is

$$x_j = 2 \times 3.30 \times 10^{-5} \sqrt{\ln \frac{2.0 \times 10^{19}}{10^{16}}} = 6.60 \times 10^{-5} \times \sqrt{7.61} = 1.82 \times 10^{-4}\ \text{cm} = 1.8\ \mu\text{m}$$

The drive-in multiplied the junction depth sixfold and cut the surface concentration tenfold without changing the dose. In a modern process this kind of thermal budget would destroy shallow junctions elsewhere on the wafer — which is why the total $\sum Dt$ of a process flow is a budget that must be planned. (Neglected here: boron segregation into the oxide grown during drive-in, which reduces Q by a few tens of per cent.)

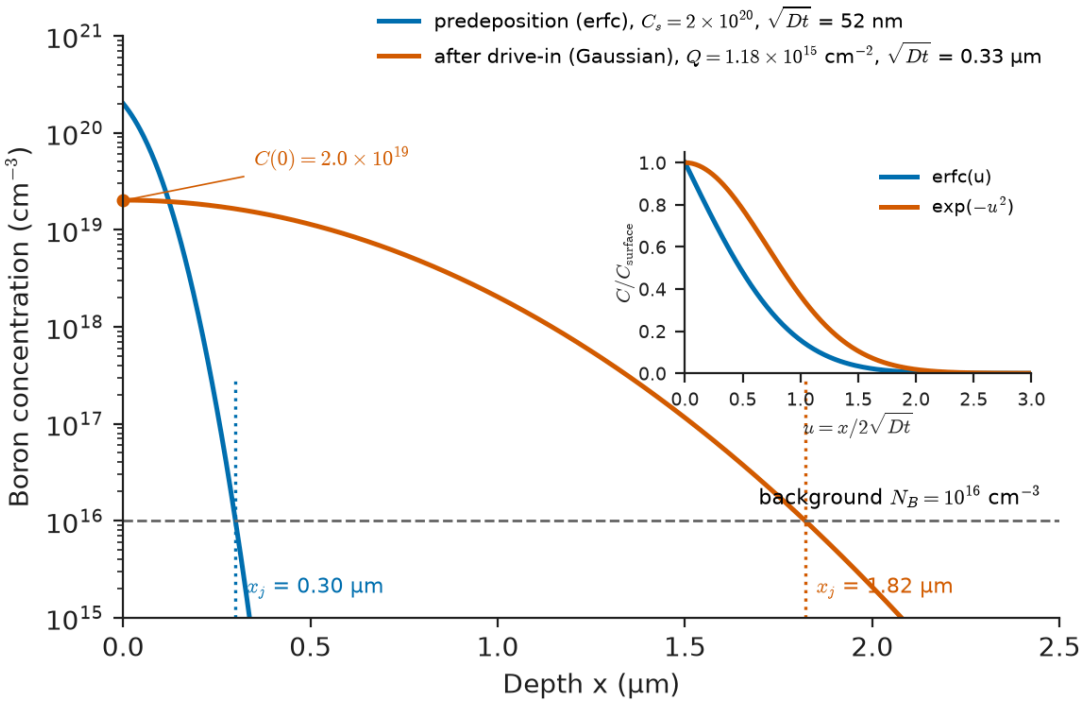


Figure 50.3 — Constant-source (erfc) and limited-source (Gaussian) dopant profiles for the two-step boron diffusion worked example, showing how drive-in deepens the junction and lowers the surface concentration.

50.6 Ion implantation

Diffusion from a surface source cannot give shallow junctions with independently controlled dose and depth, and its dose control is limited by the solubility. **Ion implantation** accelerates ionised dopant atoms to energies of 0.2 keV–several MeV and fires them into the wafer. The dose is measured directly by integrating the beam current (charge per ion is known), so it is accurate to about 1 %, and the depth is set by the energy. Almost every doping step in modern silicon processing is an implant followed by an anneal.

Stopping and range

An ion entering a solid loses energy by two mechanisms:

- **Nuclear stopping**: elastic collisions with target nuclei, transferring large amounts of energy in single events and displacing target atoms. Dominant for heavy ions and low energies.
- **Electronic stopping**: inelastic interaction with the electron gas, a continuous drag roughly proportional to ion velocity. Dominant for light ions and high energies.

The total path is a random sequence of collisions, so ions of identical energy come to rest at a distribution of depths. To a first approximation the distribution is Gaussian, characterised by the **projected range** R_p (mean depth) and **straggle** ΔR_p (standard deviation):

$$N(x) = \frac{\Phi}{\sqrt{2\pi}\,\Delta R_p} \exp\left[-\frac{(x - R_p)^2}{2\Delta R_p^2}\right], \qquad N_p = \frac{\Phi}{\sqrt{2\pi}\,\Delta R_p} \approx \frac{0.4\,\Phi}{\Delta R_p}$$

where Φ is the dose (ions/cm^2) and N_p the peak concentration.

> **Status of the principle: Gaussian implant profile**
> *Approximate.* LSS stopping theory and Monte Carlo codes (SRIM, TRIM) give R_p and ΔR_p; the Gaussian ignores skewness (light ions such as B have profiles skewed toward the surface; heavy ions toward depth) and **channelling** tails. Pearson IV distributions or Monte Carlo profiles are used for accurate work.

Ion in Si	Energy (keV)	R_p (nm)	ΔR_p (nm)
B	30	≈ 100	≈ 37
B	100	≈ 300	≈ 70
P	30	≈ 40	≈ 17
P	100	≈ 125	≈ 45
As	30	≈ 23	≈ 9
As	100	≈ 58	≈ 21

Approximate values from LSS/Monte Carlo tables for amorphous Si.

Heavy ions stop much closer to the surface at the same energy: 100 keV arsenic goes about one-fifth as deep as 100 keV boron. Ultra-shallow junctions therefore use heavy species (As, or B delivered as BF_2^+ molecular ions, which share the energy among the atoms) at sub-keV to few-keV energies.

Channelling. A crystal looks very different along a low-index direction: open channels between atomic rows let ions travel far deeper with little nuclear stopping. Wafers are therefore tilted about 7° from the beam and often implanted through a thin screen oxide, or pre-amorphised, so the crystal looks random to the ions.

Damage

Each implanted ion with energy deposited in nuclear collisions E_n displaces a cascade of lattice atoms. The Kinchin–Pease estimate of the number of displaced atoms per ion is

$$N_d \approx \frac{0.8\,E_n}{2E_d}$$

with displacement threshold $E_d \approx 15$ eV in silicon. A 100 keV arsenic ion, which loses most of its energy to nuclear stopping, displaces of order a thousand to two thousand silicon atoms, concentrated near the end of its track. When the damage density exceeds a threshold, the lattice becomes **amorphous**. Heavy ions amorphise silicon at room temperature at doses of order 10^{14} cm^{-2} (As, Sb, Ge); boron, being light, produces dilute damage and rarely amorphises it.

> **Status of the principle: Kinchin–Pease displacement estimate**
> *Approximate.* It assumes a hard-sphere cascade with a sharp displacement threshold; it overestimates surviving damage because many Frenkel pairs recombine within picoseconds, especially at elevated implant temperature.

Annealing and activation

As implanted, most dopant atoms sit in interstitial or damaged positions and are electrically inactive, and the damage itself traps carriers. Annealing must restore the crystal and move dopants onto substitutional sites — **activation** — while diffusing them as little as possible.

- **Solid-phase epitaxial regrowth (SPE).** An amorphous layer recrystallises layer by layer from the underlying crystal at 500–650 °C, at a rate with activation energy about 2.7 eV: about 1 nm/s at 600 °C, so a 100 nm amorphous layer regrows in under two minutes. Dopants are incorporated substitutionally during regrowth, often above their equilibrium solubility. This is one reason pre-amorphisation with Ge or Si is used before boron implants.
- **End-of-range (EOR) defects.** Just below the amorphous–crystalline interface, excess interstitials survive and condense into {311} rod-like defects and dislocation loops. If they lie within a junction depletion region they cause leakage.
- **Transient-enhanced diffusion (TED).** The excess interstitials released from damage during the first seconds to minutes of an anneal enhance boron diffusion by orders of magnitude until they are consumed. A low-temperature anneal can therefore move boron *further* than a short high-temperature one.
- **Rapid thermal, spike, flash and laser anneals** reach 1000–1350 °C for milliseconds to seconds, activating dopants while keeping $\sqrt{Dt}$ to a few nanometres.

Worked example — An arsenic source/drain implant. Arsenic at 100 keV and dose $\Phi = 5 \times 10^{15}$ cm^{-2} is implanted into p-type silicon with $N_B = 10^{17}$ cm^{-3}. With $R_p = 58$ nm and $\Delta R_p = 21$ nm,

$$N_p = \frac{5 \times 10^{15}}{\sqrt{2\pi} \times 2.1 \times 10^{-6}} = 9.5 \times 10^{20}\ \text{cm}^{-3}$$

This exceeds the solid solubility of arsenic in silicon at typical anneal temperatures (a few 10^{20} cm^{-3} electrically active), so not all of it can be activated; inactive arsenic forms clusters. The as-implanted junction is where $N(x_j) = N_B$ on the deep side:

$$x_j = R_p + \Delta R_p \sqrt{2 \ln \frac{N_p}{N_B}} = 58 + 21\sqrt{2 \ln 9500} = 58 + 21 \times 4.28 = 148\ \text{nm}$$

before any anneal diffusion. The dose is far above the amorphisation threshold, so the top ~100 nm is amorphous and will regrow by SPE. If after annealing the layer is fully active with an average mobility of about 80 cm^2/V s, its sheet resistance is

$$R_s = \frac{1}{q \mu \Phi} = \frac{1}{1.602 \times 10^{-19} \times 80 \times 5 \times 10^{15}} \approx 16\ \Omega/\square$$

The real value is higher because of incomplete activation; measuring R_s by four-point probe (Chapter 52) is the standard check of activation. Figure 50.4 shows this profile together with its damage distribution, the amorphous layer and the end-of-range defect band.

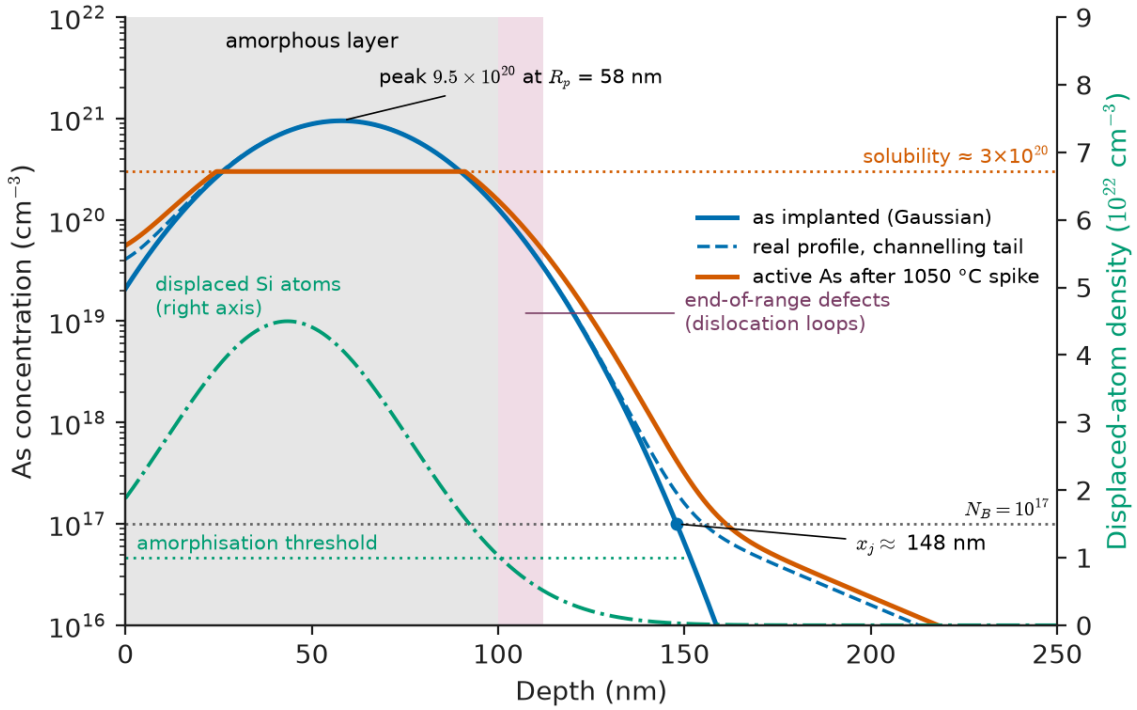


Figure 50.4 — Implanted arsenic concentration profile and damage distribution in silicon, before and after annealing.

50.7 Photolithography overview

Lithography transfers a pattern from a mask to a light-sensitive polymer film on the wafer, which then protects selected areas during etching or implantation. For this book the questions are which physical laws limit feature size and what the resist does as a material; the optical engineering of exposure tools and multi-patterning schemes belong elsewhere.

The sequence

A wafer is primed (hexamethyldisilazane, HMDS, makes the oxide surface hydrophobic so resist adheres), coated with photoresist by spinning (thickness from tens of nanometres to micrometres, set by viscosity and spin speed), soft-baked to remove solvent, exposed through a mask (usually reduced 4× by projection optics), post-exposure-baked, developed, and hard-baked. After etching or implantation the resist is stripped in oxygen plasma or solvents.

Resists as materials

- **Positive resists** become soluble in developer where exposed. Classical g-line and i-line resists are novolac resins with a diazonaphthoquinone (DNQ) inhibitor; light converts DNQ to a carboxylic acid that makes the exposed resin soluble in aqueous base.
- **Negative resists** become insoluble where exposed, usually by crosslinking (SU-8 epoxy, used for thick MEMS structures).
- **Chemically amplified resists (CAR)**, used at 248 nm, 193 nm and EUV, contain a photo-acid generator: each absorbed photon releases one acid molecule, which during the post-exposure bake catalyses the removal of many protecting groups from the polymer. The amplification gives high sensitivity — essential when the light source is weak — at the cost of acid diffusion during baking (a $\sqrt{Dt}$ blur of a few nanometres) and sensitivity to airborne base contamination.

Resist must also survive the subsequent process: its **etch resistance** in plasma, which favours aromatic structures, and its thermal stability during ion implantation, during which it can carbonise and become hard to strip.

Resolution: the Rayleigh criterion

Projecting a pattern of lines and spaces through a lens of numerical aperture $\mathrm{NA} = n \sin\theta$ requires that at least two diffraction orders pass through the lens. The smallest resolvable half-pitch is

$$R = k_1 \frac{\lambda}{\mathrm{NA}}$$

and the depth of focus is

$$\mathrm{DOF} = k_2 \frac{\lambda}{\mathrm{NA}^2}$$

The process factor k_1 is 0.5–0.8 for simple processes and can be pushed toward its physical limit of 0.25 for a single exposure by resolution-enhancement techniques (off-axis illumination, phase-shift masks, optical proximity correction). Resolution improves by shortening λ or raising NA, and in both cases depth of focus shrinks faster than resolution — which is why wafer flatness specifications (Section 50.2) tighten with each generation.

> **Status of the principle: Rayleigh resolution and depth of focus**
> *Derived* from diffraction theory for a coherent or partially coherent imaging system. The constants k_1 and k_2 absorb illumination, mask and resist effects and are determined empirically for each process.

Source	λ (nm)	NA	Typical k_1	Half-pitch (nm)
Hg i-line	365	0.6	0.6	≈ 365
KrF excimer	248	0.8	0.4	≈ 125
ArF excimer (dry)	193	0.93	0.3	≈ 62
ArF immersion	193	1.35	0.28	≈ 40
EUV	13.5	0.33	0.4	≈ 16
High-NA EUV	13.5	0.55	0.4	≈ 10

Representative single-exposure values.

Worked example — Immersion and EUV lithography. With water between lens and wafer ($n = 1.44$ at 193 nm), lens NA can exceed 1; at NA = 1.35 and $k_1 = 0.28$,

$$R = 0.28 \times \frac{193}{1.35} = 40 \text{ nm}, \qquad \mathrm{DOF} = 0.5 \times \frac{193}{1.35^2} = 53 \text{ nm} \ \ (k_2 = 0.5)$$

A wafer region that departs from flatness by more than a few tens of nanometres across an exposure field — including topography left by underlying layers — is out of focus. Smaller pitches with 193 nm light need double or quadruple patterning. EUV at 13.5 nm with NA = 0.33 and $k_1 = 0.4$ gives $R = 16$ nm in one exposure. Because no material is transparent at 13.5 nm, EUV optics are reflective Mo/Si multilayer mirrors and the masks are reflective too — the multilayers themselves being a thin-film deposition achievement with layer thicknesses controlled to about 0.01 nm.

Electron-beam lithography (no diffraction limit at practical scales, but serial and slow) writes the masks themselves; nanoimprint lithography presses a mould into a resist and is used for some photonic and storage applications.

50.8 Etching

Etching transfers the resist pattern into the underlying material. Three figures of merit define an etch:

- **Etch rate** r of the target film.
- **Selectivity** $S = r_{\mathrm{film}} / r_{\mathrm{mask}}$ (or to the underlying layer), which determines how thick the mask must be and how much overetch the layer below can tolerate.
- **Anisotropy**, defined from the lateral and vertical rates:

$$A = 1 - \frac{r_{\text{lat}}}{r_{\text{vert}}}$$

$A = 0$ is isotropic (equal undercut and depth), $A = 1$ perfectly vertical.

Wet etching

Wet etching dissolves the material in a liquid reagent. It is chemically selective, cheap, damage-free and usually isotropic in amorphous and polycrystalline films:

- SiO_2 in buffered HF (NH_4F + HF, e.g. 6:1): about 100 nm/min for thermal oxide at 25 °C, with negligible attack on Si. Deposited oxides etch faster (the etch rate ratio to thermal oxide is a quick test of film density and hydrogen content).
- Si_3N_4 in hot phosphoric acid (≈ 160 °C): a few nm/min, with selectivity 20–40:1 to SiO_2.
- Aluminium in phosphoric–acetic–nitric mixtures.
- Copper in persulphate or ferric/cupric chloride (PCB etching).

Because an isotropic etch undercuts the mask by about as much as it etches down, wet etching cannot define features much smaller than the film thickness. It remains the method for cleaning, oxide stripping and for removing layers where profile does not matter.

Anisotropic wet etching of crystals. Alkaline etchants (KOH, TMAH) attack silicon at rates that depend strongly on crystal plane: {111} planes, which expose atoms with three back-bonds, etch 100–400 times more slowly than {100}. On a (100) wafer, a rectangular mask opening aligned to ⟨110⟩ etches into a pit bounded by {111} sidewalls at 54.74° to the surface. This crystallographic etch is the basis of bulk silicon MEMS: pressure-sensor diaphragms, V-grooves for optical fibres, and inkjet nozzles.

Worked example — KOH-etched pressure-sensor diaphragm. A 1000 µm × 1000 µm diaphragm is to be etched through a 525 µm (100) wafer from the back. Since each {111} sidewall moves inward by $d \cot 54.74^\circ = d/\sqrt{2}$ as the etch deepens by d, the backside mask opening must be

$$W_{\text{top}} = W_{\text{bottom}} + 2 \times \frac{d}{\sqrt{2}} = 1000 + \sqrt{2} \times 525 = 1742\ \mu\text{m}$$

(minus a small allowance for the thickness of the diaphragm itself, if it is defined by a boron etch-stop layer rather than etched through). The backside opening is three times the diaphragm area, so the sloping walls consume twice as much die area as the diaphragm itself — one reason pressure sensors moved toward deep reactive-ion etching with vertical walls.

Dry (plasma) etching

Dry etching uses a low-pressure plasma (0.1–100 Pa) that supplies two things: chemically reactive radicals (F, Cl, Br, O) that form volatile products with the film, and ions accelerated across the plasma sheath into the wafer at nearly normal incidence.

- **Plasma (chemical) etching**: radicals only, isotropic, selective, low damage.
- **Physical sputtering / ion milling**: ions only, anisotropic but non-selective and slow; used for materials with no volatile compounds (Pt, magnetic multilayers).
- **Reactive-ion etching (RIE)**: the synergy of the two. Ion bombardment breaks surface bonds and removes inhibiting layers so that radicals react much faster on horizontal surfaces than on vertical sidewalls, which see few ions. Anisotropy is enhanced by **sidewall passivation**: polymer from fluorocarbon gases (CHF_3, C_4F_8) or oxide from O_2 additions coats sidewalls and is removed by ions only on the bottom.
- **Deep reactive-ion etching (DRIE, Bosch process)**: alternating SF_6 etch steps and C_4F_8 passivation steps of a few seconds each etch silicon at 2–20 µm/min with near-vertical walls and aspect ratios above 20:1, leaving characteristic sidewall "scallops". This is how through-silicon vias and most modern MEMS are made.

Material	Typical dry chemistry	Volatile product	Notes
Si, poly-Si	SF_6, Cl_2, HBr	SiF_4, $SiCl_4$, $SiBr_4$	HBr/Cl_2 for gates: high selectivity to thin gate oxide
SiO_2	CF_4, CHF_3, C_4F_8 + Ar	SiF_4, CO	Fluorocarbon polymer gives selectivity to Si and resist
Si_3N_4	CF_4/O_2, CH_2F_2	SiF_4, N_2	
Al	Cl_2, BCl_3	$AlCl_3$	BCl_3 removes native Al_2O_3; Cl residues cause corrosion
Cu	— (no volatile halide near room T)	—	Hence damascene patterning (Section 50.9)
GaN, SiC	Cl_2/BCl_3; SF_6/O_2	$GaCl_3$; SiF_4, CO	Strong bonds: high ion energy needed, damage risk
Organic resist, low-k	O_2, N_2/H_2	CO_2, H_2O	O_2 plasma damages carbon-doped low-k

Figure 50.5 compares the profiles produced by the main wet and dry methods. The absence of a volatile copper compound at practical temperatures is a materials fact that reshaped interconnect manufacturing, as the next section shows.

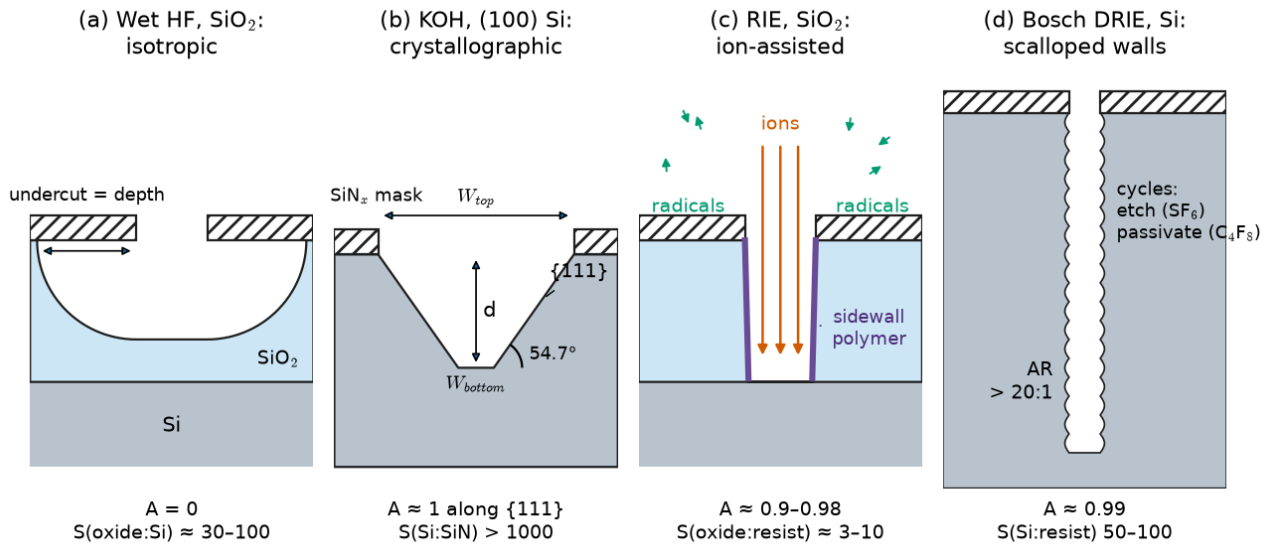


Figure 50.5 — Etch profiles produced by isotropic wet etching, crystallographic KOH etching, reactive-ion etching with sidewall passivation and the Bosch DRIE process.

50.9 Metallization

Transistors are connected by a multilevel interconnect stack of metal lines separated by dielectrics, joined to the silicon by contacts. The materials problem is to get low resistance, low contact resistance, long electromigration life and thermal and chemical compatibility with silicon and dielectrics, all at line widths now below 30 nm.

Aluminium interconnects

Aluminium was the interconnect metal from the 1960s to the late 1990s and remains so in power devices, analog ICs, MEMS and the top layers and bond pads of many chips. Its virtues are good conductivity (bulk 2.65 μΩ cm), easy sputtering and dry etching in chlorine, excellent adhesion to SiO_2 (it reduces a thin layer of the oxide), and ohmic contact to heavily doped silicon. Two material problems shaped its use:

- **Junction spiking.** Silicon dissolves in aluminium (about 0.5 wt% at 450 °C, rising to 1.5 wt% at the Al–Si eutectic at 577 °C). During the contact sinter, silicon diffuses into the aluminium and aluminium fills the resulting pits in the silicon,

spiking through shallow junctions and shorting them. The fix was to presaturate the aluminium with 1 % Si and then, more robustly, to insert a barrier layer (TiN, TiW) between metal and silicon.

- **Electromigration.** The electron wind at current densities of 10^5–10^6 A/cm^2 drives aluminium atoms along grain boundaries, causing voids and hillocks (Chapter 44). Adding 0.5–4 % Cu, which segregates to grain boundaries and precipitates as Al_2Cu, slows boundary diffusion and extends life by one to two orders of magnitude; **bamboo** grain structures, in which grains span the line width so there is no continuous boundary path, help further.

Silicides and contacts

The contact between metal and source/drain silicon must be low-resistance; a metal–semiconductor contact is a Schottky barrier, made ohmic by heavy doping so that carriers tunnel (Chapter 34). To reduce both contact and series resistance, the exposed silicon is converted into a **metal silicide** by depositing a metal and reacting it in a self-aligned process (**salicide**): the metal reacts only where it touches silicon, and the unreacted metal on oxide spacers is etched away selectively.

Silicide	Resistivity (μΩ cm)	Formation T (°C)	Si consumed per nm metal (nm)	Silicide per nm metal (nm)	Notes
$TiSi_2$ (C54)	13–20	700–900	2.27	2.51	High-ρ C49 phase forms first; transformation fails in narrow lines
$CoSi_2$	15–20	600–800	3.64	3.52	Consumes most Si; scaled to ~65 nm nodes
NiSi	14–20	400–600	1.83	2.22	Low Si consumption, low thermal budget; degrades to high-ρ $NiSi_2$ above ~700 °C

Typical values.

The progression from $TiSi_2$ to $CoSi_2$ to NiSi is a story of nucleation and phase transformation (Chapter 9): in narrow lines, the low-resistivity C54 phase of $TiSi_2$ could not nucleate because too few nucleation sites existed per line; $CoSi_2$ formed by a diffusion-controlled reaction independent of line width but consumed too much silicon for shallow junctions; NiSi forms at low temperature with low silicon consumption but is only metastable relative to $NiSi_2$, and agglomerates if overheated.

Contacts and vias to the first metal level are filled with **tungsten** by CVD from WF_6 (reduced by H_2 or SiH_4), which fills high-aspect holes conformally; a Ti/TiN liner provides adhesion, gettering of native oxide and a barrier against fluorine attack. Tungsten's higher resistivity (about 8–12 μΩ cm in CVD films) is acceptable in short plugs.

Copper damascene interconnects

Copper offers 35 % lower resistivity than aluminium alloys (1.7 against about 3 μΩ cm) and about ten times better electromigration resistance, because diffusion in copper has a higher activation energy. But copper cannot be dry-etched (Section 50.8), diffuses rapidly into silicon and SiO_2 where it creates deep-level traps and leakage, and does not adhere to oxide. The **damascene** process (named after the inlaid metalwork of Damascus) solves all three:

1. Deposit the interlayer dielectric and etch trenches (and, in **dual damascene**, vias beneath them) into it.
2. Line the openings with a barrier, typically TaN (amorphous or nanocrystalline, blocks Cu diffusion) plus Ta (adhesion, promotes ⟨111⟩-textured Cu), a few nanometres in total, by ionised PVD or ALD.
3. Deposit a thin copper **seed** by PVD.
4. **Electroplate** copper to overfill the openings. Organic additives — a suppressor that adsorbs on the top surface and upper sidewalls, an accelerator that concentrates on the bottom as the feature fills, and a leveller — give **bottom-up superfilling** without voids.
5. Remove the overburden by **CMP**, stopping on the dielectric, leaving inlaid lines.
6. Cap the top surface with a dielectric diffusion barrier (SiN or SiCN) or a selective metal cap (Co), because the top interface is the fastest electromigration path.

The resulting structure, from silicided contact to second metal level, is shown in Figure 50.6.

Electroplated copper is fine-grained as deposited and **self-anneals** at room temperature over hours to days, with grain growth driven by the stored energy of the additive-pinned structure (Chapter 10); a deliberate anneal at 100–400 °C stabilises it before CMP so that resistivity and stress do not drift afterwards.

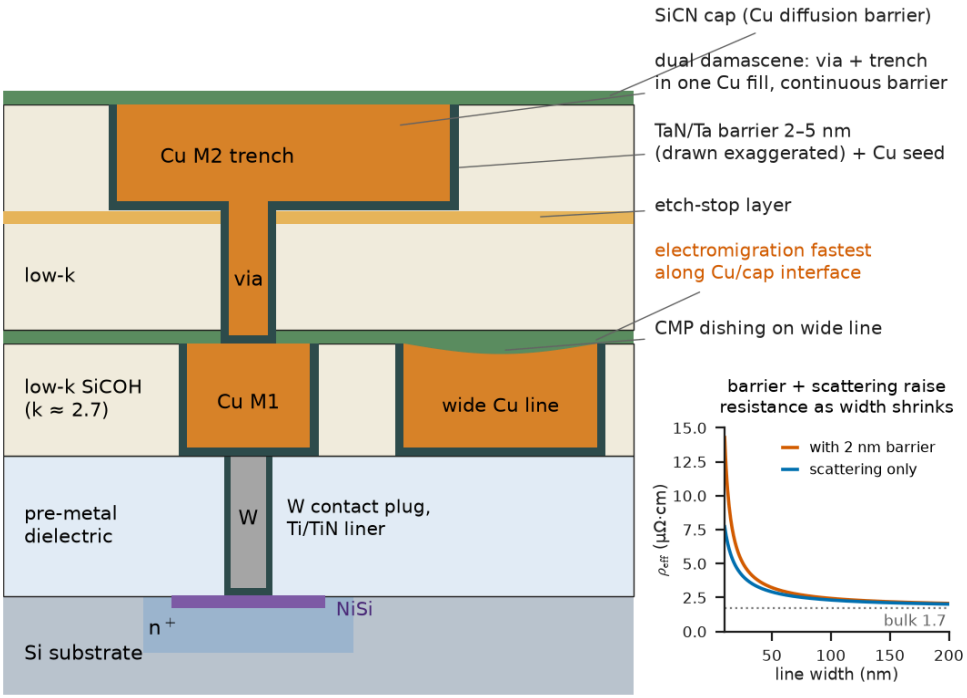


Figure 50.6 — Cross-section of a copper dual-damascene interconnect level, showing the barrier, seed, via, cap and low-k dielectric, above a tungsten contact on a silicided source/drain.

Worked example — Why barriers matter in narrow lines. Compare a 1 mm long line 100 nm wide and 200 nm tall in Al–Cu ($\rho \approx 3.3$ μΩ cm, no liner needed inside the line) and in Cu with a 5 nm barrier on the bottom and both sides. At this width, surface and grain-boundary scattering raise copper's effective resistivity to about 2.2 μΩ cm (Chapter 25). The copper core is 90×195 nm:

$$R_{\mathrm{Al}} = \frac{3.3 \times 10^{-8} \times 10^{-3}}{100 \times 10^{-9} \times 200 \times 10^{-9}} = 1650\ \Omega, \qquad R_{\mathrm{Cu}} = \frac{2.2 \times 10^{-8} \times 10^{-3}}{90 \times 10^{-9} \times 195 \times 10^{-9}} = 1250\ \Omega$$

Copper still wins by 24 %, much less than the 48 % bulk resistivity ratio suggests. At 20 nm width a 2 nm barrier occupies a fifth of the width, and copper resistivity rises toward 5–6 μΩ cm; this is why barrier-thin or barrier-free metals with short electron mean free paths (Co, Ru, Mo) are being introduced in the tightest-pitch levels.

50.10 Dielectric deposition

Dielectric films isolate devices and interconnect levels, act as diffusion barriers, etch stops and passivation, and form gate and capacitor dielectrics. Thermal oxide (Section 50.4) is the best SiO_2 but requires silicon to consume and temperatures above 800 °C; once metal is present, all dielectrics must be deposited at 400 °C or below.

Deposited oxides and nitrides

- **LPCVD** oxide from TEOS (tetraethyl orthosilicate) at 650–750 °C, and stoichiometric Si_3N_4 from dichlorosilane and ammonia at 750–800 °C, give dense, conformal films used before metallization. LPCVD nitride is a superb barrier to oxygen, water and sodium, with high tensile stress (about 1 GPa) that limits thickness to a few hundred nanometres.
- **PECVD** oxide (from SiH_4 + N_2O or TEOS + O_2) and silicon nitride (SiH_4 + NH_3) at 250–400 °C. PECVD nitride, SiN_x:H with 10–30 at.% hydrogen, is the standard **final passivation** layer of chips, protecting against moisture and mobile ions; its stress can be tuned from tensile to compressive by plasma frequency and power, and the hydrogen it releases during

later anneals helps passivate interface traps — and, in some cases, causes hydrogen-related instabilities. PECVD nitride is also the antireflection and surface-passivation coating of crystalline silicon solar cells (Chapter 40).
- **High-density plasma CVD (HDP-CVD)** simultaneously deposits and sputters, filling gaps between closely spaced lines without voids.
- **Spin-on glasses and polymers** (siloxanes, polyimides, benzocyclobutene) planarise topography and serve as stress buffers and redistribution-layer dielectrics in packaging (Chapter 51).

Low-k interconnect dielectrics

Interconnect delay scales as RC, and the capacitance between neighbouring lines is proportional to the dielectric's relative permittivity. Replacing SiO_2 ($k = 3.9$–4.2) by a lower-k material reduces delay, power (CV^2f) and crosstalk at once. Permittivity is reduced by lowering polarisability (Chapter 26) — replacing Si–O bonds with less polarisable Si–F or Si–CH_3 — and by lowering density, introducing porosity.

Dielectric	k	Young's modulus (GPa)	Deposition	Notes
Thermal SiO_2	3.9	≈ 70	Oxidation	Reference
PECVD TEOS oxide	4.0–4.2	60–70	PECVD	Absorbs some moisture
Fluorinated silicate glass (FSG)	3.5–3.7	≈ 60	HDP/PECVD	F is mobile and hygroscopic at high content
Dense SiCOH (carbon-doped oxide)	2.7–3.0	8–15	PECVD	Methyl groups lower polarisability and density
Porous SiCOH (ultra-low-k)	2.2–2.5	3–8	PECVD + porogen, UV cure	20–30 % porosity, pores 1–3 nm
Air gap	≈ 1 (effective 1.5–2.2)	—	Sacrificial or non-conformal deposition	Mechanical weakness
SiCN etch stop / cap	4.5–5.5	≈ 100	PECVD	Cu diffusion barrier
Si_3N_4	7–7.5	150–300	LPCVD/PECVD	Passivation, etch stop
HfO_2 (high-k)	20–25	≈ 250	ALD	Gate dielectric (Chapter 34)

Typical values.

Porous SiCOH is made by codepositing a matrix precursor with an organic **porogen** that is later burned out by UV-assisted curing, leaving 1–3 nm pores. The price of low k is visible in the modulus column: porous low-k films are an order of magnitude more compliant than SiO_2, have low fracture toughness and poor adhesion. They crack and delaminate under CMP pressure and, critically, during packaging, when the CTE mismatch between die and substrate pulls on the interconnect stack through solder bumps (Chapter 51, "white bumps" or chip–package interaction failures). Plasma steps also **damage** low-k films: oxygen and other radicals strip methyl groups from pore walls, converting the surface to hydrophilic Si–OH that absorbs water ($k \approx 80$), raising effective k and leakage. Sealing the pores and using reducing ash chemistries are process responses.

At the other extreme, **high-k** dielectrics (HfO_2, ZrO_2, Al_2O_3) replace SiO_2 in transistor gates and DRAM capacitors because they give the same capacitance at greater physical thickness and therefore less tunnelling leakage; their deposition by ALD and their interface properties are covered in Chapter 34.

50.11 Packaging and encapsulation materials

After wafer processing, dies must be separated, attached, connected, protected and cooled. These steps define a second set of materials with properties quite different from the wafer's, and their selection and failure modes are the subject of Chapter 51. Here we note only how the transition from wafer to package changes the material environment.

- **Wafer-level steps**: final passivation (PECVD SiN, often with a polyimide or PBO stress buffer 5–10 μm thick); opening of bond pads; under-bump metallization (Ti/Cu, Ni) and bumps or copper pillars for flip-chip; redistribution layers in polymer dielectrics for fan-out packages; backgrinding to 50–200 μm thickness, which removes the backside gettering layer and leaves grinding damage that reduces die strength unless stress-relieved by polishing or etching.
- **Dicing** by diamond saw (chipping and microcracks at die edges) or by laser (stealth dicing, which modifies a buried plane and splits the wafer with little debris).
- **Die attach** by epoxy filled with silver, solder, or sintered silver; **wire bonds** of Au, Cu, Ag or Al; **lead frames** of copper alloys (C194, C7025 with ≈ 17 ppm/K CTE) or Fe–42Ni alloy (CTE ≈ 4–5 ppm/K, close to silicon); **organic substrates** (BT resin, ABF build-up films) or **ceramic** (alumina, AlN, LTCC).
- **Encapsulation** by epoxy moulding compound, silicone gel, or hermetic ceramic and metal packages.

The material change is abrupt: a die made of materials with CTEs of 0.5–4 ppm/K (Si 2.6, SiO_2 0.5, W 4.5) and processed at up to 1100 °C is bonded into an environment of copper (17 ppm/K), solder (≈ 22 ppm/K), organic substrates (13–17 ppm/K in-plane) and polymers (up to 60 ppm/K below T_g), processed at 150–260 °C and exposed to humidity. Most package failure mechanisms follow from that mismatch and from moisture, as Chapter 51 shows.

> **Boundary.**
> Package design, thermal management and assembly-line process control are engineering topics outside this book. Chapter 51 treats the *materials*: their structures, properties and failure mechanisms.

50.12 Process-induced defects

Each process step can introduce defects that the device must tolerate. Their effects on carriers were treated in Chapter 32 (defects and impurities) and Chapter 34 (interface traps); here we trace them to their source.

Dislocations and slip

A defect-free silicon wafer contains no dislocations, and dislocation-free silicon is hard to deform at room temperature. But at 1000–1200 °C, the critical resolved shear stress of silicon falls to a few MPa, and a radial temperature difference of only a few kelvin across a wafer in a rapid-thermal process or at the edge of a furnace boat produces thermal stresses of that order. Dislocations nucleate at edge chips or scratches and glide on {111} planes, appearing as **slip lines** — a classic process-induced defect that causes leakage, warp and lithography overlay errors. Remedies are slower ramps, better edge profiling, wafer supports that minimise gravitational stress (a 300 mm wafer is heavy enough to slip under its own weight at high temperature if poorly supported), and interstitial oxygen, which pins dislocations. **Misfit dislocations** form at heteroepitaxial and heavily doped interfaces where the critical thickness is exceeded, and **implant EOR loops** form below amorphised layers (Section 50.6).

Stacking faults

Oxidation-induced stacking faults (OISF) are extrinsic stacking faults on {111} planes — extra half-planes of silicon atoms bounded by Frank partial dislocations — that grow by absorbing the self-interstitials injected during oxidation (Section 50.4). They nucleate on surface damage, metallic precipitates or oxygen precipitates and grow to micrometres. Their density is a classic test of a wafer's surface quality; a ring of OISF on a CZ wafer marks the boundary between vacancy- and interstitial-rich growth regions (Section 50.1). In epitaxial layers, stacking faults nucleate on particles or residual oxide at the interface and appear as characteristic triangles or squares on the surface. In SiC, stacking faults expanding from BPDs under forward bias are a service failure mode.

Contamination

Metallic contamination is the most insidious defect, because fast-diffusing transition metals travel through the whole wafer at process temperatures and even at room temperature. Interstitial copper in silicon has $D \approx 3 \times 10^{-4} \exp(-0.18\ \mathrm{eV}/k_B T)$ cm^2/s, about 3×10^{-7} cm^2/s at room temperature, giving $\sqrt{Dt} \approx 0.3$ mm in one hour: copper on the back of a wafer reaches the front surface within hours at room temperature. Iron, nickel and chromium behave similarly at process temperatures. Once in the wafer, metals:

- form deep levels that act as Shockley–Read–Hall recombination centres, shortening the minority-carrier lifetime $\tau = 1/(\sigma v_{\mathrm{th}} N_t)$ (Chapter 32) and raising junction leakage;
- precipitate as silicides at the $\mathrm{Si/SiO_2}$ interface, causing gate-oxide breakdown;
- in the case of copper, precipitate and form leakage paths unless barrier layers isolate the metallization.

Worked example — Iron and lifetime; sodium and threshold shift. (a) Interstitial iron at $N_t = 10^{11}\ \mathrm{cm^{-3}}$ — one atom in 5×10^{11} — with an electron capture cross-section of about $5 \times 10^{-14}\ \mathrm{cm^2}$ and $v_{\mathrm{th}} = 10^7\ \mathrm{cm/s}$ gives

$$\tau_n = \frac{1}{5 \times 10^{-14} \times 10^{7} \times 10^{11}} = 2 \times 10^{-5}\ \mathrm{s} = 20\ \mu\mathrm{s}$$

Good silicon for power devices and solar cells needs lifetimes of hundreds of microseconds to milliseconds, so even this parts-per-trillion contamination is unacceptable. This is why furnace tubes, wet benches and chemicals are specified in parts per trillion and why surface metals must stay below $10^{10}\ \mathrm{atoms/cm^2}$.

(b) Mobile **sodium** ions in a gate oxide — from human contact, chemicals or furnace materials — drift under gate bias at modest temperatures. A sheet of $N = 10^{11}\ \mathrm{cm^{-2}}$ $\mathrm{Na^+}$ ions arriving at the $\mathrm{Si/SiO_2}$ interface of a 10 nm oxide, with $C_{\mathrm{ox}} = \varepsilon_{\mathrm{ox}}/t_{\mathrm{ox}} = 3.9 \times 8.854 \times 10^{-12}/10^{-8} = 3.45 \times 10^{-3}\ \mathrm{F/m^2}$, shifts the threshold voltage by

$$\Delta V = \frac{qN}{C_{\mathrm{ox}}} = \frac{1.602 \times 10^{-19} \times 10^{15}}{3.45 \times 10^{-3}} = 0.046\ \mathrm{V}$$

and by ten times as much in a 100 nm oxide. Early MOS devices were unusable until sodium was identified; the cures were clean processing, phosphosilicate glass and nitride layers that getter or block sodium, and the PECVD nitride final passivation of Section 50.10.

Particles cause shorts, opens and pattern defects; **organic** films degrade adhesion and oxide quality.

Plasma-induced damage

Plasma processes damage materials in three ways:

- **Charging (antenna effect).** During plasma etching or deposition, nonuniform plasma currents charge metal lines connected to transistor gates. The charge flows to the substrate through the thin gate oxide by Fowler–Nordheim tunnelling, creating oxide traps and interface states and, in extreme cases, breaking the oxide. The damage scales with the **antenna ratio** — the area of conductor exposed to plasma divided by the gate area. Design rules limit the antenna ratio, and protection diodes give the charge a harmless path.
- **Radiation damage.** Vacuum-ultraviolet photons from the plasma (10–20 eV) create electron–hole pairs and traps in dielectrics.
- **Ion-bombardment damage** to exposed semiconductor surfaces: lattice disorder, implanted etch species, and, in compound semiconductors, preferential loss of one element (nitrogen vacancies in GaN act as donors and make etched surfaces conductive). Low-k dielectrics lose carbon (Section 50.10).

Post-etch anneals, damage-removal wet etches, and low-ion-energy or neutral-beam processes reduce these effects.

CMP and mechanical damage

CMP introduces **dishing** (wide copper lines polished below the dielectric surface), **erosion** (thinning of dense line arrays), scratches and residual slurry particles. Each changes line resistance and planarity for the next lithography step. Backgrinding and dicing introduce subsurface cracks that reduce die fracture strength, a packaging reliability issue (Chapter 51).

Defect	Introduced by	Material consequence	Device consequence	Mitigation
Slip dislocations	Thermal gradients in RTP/furnace	Dislocations on {111}	Leakage, warp, overlay error	Slow ramps, edge profiling, interstitial O
COPs / voids	CZ growth at high v/G	Vacancy clusters	Gate-oxide defects	Controlled v/G, epi wafers
OISF	Oxidation	Interstitial-type stacking faults	Junction leakage	Gettering, clean surfaces, lower T
EOR loops, TED	Implant + anneal	Interstitial clusters	Leakage, anomalous diffusion	Pre-amorphisation, spike/laser anneal
Metal contamination	Tools, chemicals, handling	Deep levels, silicide precipitates	Lifetime loss, leakage, oxide breakdown	Cleanroom control, gettering
Mobile ions (Na^+, K^+)	Handling, chemicals	Oxide charge	Threshold drift	Nitride barriers, PSG, clean processing
Plasma charging	Etch, deposition, ash	Oxide traps, interface states	V_t shift, oxide breakdown	Antenna rules, diodes
Low-k damage	O_2 plasma, CMP	Carbon loss, moisture uptake	Higher k, leakage, TDDB	Pore sealing, reducing chemistries
CMP dishing, scratches	Polishing	Thickness variation	Resistance variation, shorts	Dummy fill, slurry control

What remains open

Open question	Where it is answered
How do solders, wires, bumps, adhesives, substrates and encapsulants join and protect the die, and how do they fail?	Chapter 51
How are defect densities, dopant profiles, film thicknesses and contamination measured (SIMS, four-point probe, XRD, TEM, Hall effect)?	Chapter 52
How do electromigration, dielectric breakdown and defect generation degrade interconnects and devices in service?	Chapter 44
How do interface traps, film stress and gate-dielectric properties arise and affect reliability?	Chapter 34
How do these processing choices enter the selection of substrates and materials under cost and reliability constraints?	Chapter 58
How are process steps sequenced into complete CMOS, bipolar and power-device flows, and how do the devices operate?	Electronics & Computing core

Next: Chapter 51 follows the die out of the wafer fab and into its package, where solders, intermetallics, laminates, encapsulants and thermal-expansion mismatch take over as the central materials problems.

Chapter 51 — Joining, Packaging, and Interconnect Materials

Part X · Processing and Manufacturing

An automotive engine-control unit returns from the field after three winters. Its silicon is perfect: every transistor still meets its datasheet. The fault is a cracked solder joint under a large ceramic resistor near the corner of the board. Sectioning shows a crack that started at the component side, ran through the tin just above a layer of copper–tin intermetallic, and grew a little with every cold start, when the board, the ceramic body and the solder all tried to shrink by different amounts. Nearby, a ball-grid-array package has a delamination under its die pad, the trace of a reflow at which moisture absorbed during storage turned to steam. Neither failure involves any semiconductor physics. Both involve the materials that sit between the die and the world — solders, intermetallics, laminates, ceramics, polymers — and the single property in which they differ most: thermal expansion.

> **What materials join a die to its package and the package to a board, and what does each contribute electrically, thermally and mechanically?**
> **Why do packages fail, and which material properties control how fast?**

Concept	Engineering destination
Solder alloys, SnPb versus SAC	Reflow profiles, joint reliability, RoHS compliance
Intermetallics, Kirkendall voids	Joint embrittlement after aging, drop-shock failure
Tin whiskers	Shorts in lead-free assemblies, plating specification
Bonding wires	Wire choice for cost, current capacity, power cycling
Flip-chip bumps, Cu pillars, underfill	Fine-pitch, high-I/O and high-current die connection
ICA and ACF	Display drivers, flex attachment, low-temperature assembly
Alumina, AlN, Si_3N_4, DBC/AMB, LTCC	Power-module and RF substrates, thermal resistance
FR-4 T_g, T_d, z-axis CTE, D_k/D_f	Via reliability, lead-free compatibility, signal loss at GHz
Moulding compounds, silicones	Package protection, moisture sensitivity
Thermal interface materials	Junction temperature, power density
CTE mismatch, DNP shear strain	Solder-fatigue life estimates, component placement
Warpage, popcorn cracking, failure mechanisms	Assembly yield, moisture sensitivity levels, qualification

Roadmap. Joining materials: solders and their reactions (51.1), wires (51.2), bumps and underfill (51.3), adhesives (51.4) → what the package is built on: substrates (51.5) and laminates (51.6) → protection and heat removal: encapsulants (51.7) and TIMs (51.8) → the central mechanical problem, CTE mismatch (51.9) → how it all fails (51.10).

> **Boundary.**
> This chapter treats the materials of packaging and assembly. Package electrical and thermal design, assembly process control and system-level thermal management are engineering subjects beyond this book; fatigue, creep, corrosion and electromigration as failure physics are developed in Chapter 44.

51.1 Solder materials

Solder is the most widely used joining material in electronics — a typical smartphone has thousands of joints, a server board tens of thousands. A solder joint must wet both surfaces, carry current, conduct heat, and hold the component mechanically through thousands of thermal cycles, all with a material that is close to its melting point at room temperature.

Why low-melting tin alloys?

The process requirement is a filler that melts well below the damage temperature of components and laminates (roughly 260 °C for short times) and wets copper, nickel and gold surfaces. Tin is the base of almost every solder because it reacts with these metals to form intermetallic compounds — the reaction *is* the bond — and melts at 232 °C. Alloying lowers the melting point (eutectics, Chapter 9) and adjusts mechanical behaviour.

The price is a high **homologous temperature** in service. At 25 °C, eutectic SnPb is at $298/456 = 0.65\,T_m$ and SAC305 at $298/490 = 0.61\,T_m$. At these temperatures diffusion is fast, the microstructure coarsens, and the solder deforms by **creep** at stresses far below its yield strength (Chapter 14). A solder joint is never really "cold".

Tin–lead and the lead-free transition

Eutectic 63Sn–37Pb (183 °C) dominated electronics for decades: it wets well, has a low melting point, forms a fine two-phase microstructure, and is ductile and forgiving. The EU RoHS directive (2006) and similar laws restricted lead in most electronic products; high-reliability sectors (aerospace, defence, some medical and automotive) retain exemptions, and high-lead solders (≥ 85 % Pb) remain exempt for high-temperature die attach where no adequate alternative exists.

The mainstream replacement is **SAC** (Sn–Ag–Cu), most commonly **SAC305** (Sn–3.0Ag–0.5Cu, melting range 217–220 °C). Its consequences for assembly are direct: peak reflow temperatures rise from about 215–225 °C to 240–260 °C, which pushed PCB laminates to higher T_g and T_d (Section 51.6) and components to higher moisture sensitivity (Section 51.10). Low-silver alloys (SAC105, SAC0307) reduce cost and are more compliant under drop impact; Sn–0.7Cu with a nickel addition (SN100C-type) is used for wave soldering; Sn–Bi alloys (138 °C eutectic) allow low-temperature assembly on heat-sensitive substrates but are brittle at high strain rates.

Alloy	Melting (°C)	Density (g/cm^3)	CTE (ppm/K)	Young's modulus (GPa)	Resistivity (μΩ cm)	Thermal conductivity (W/m K)
63Sn–37Pb	183	8.4	24–25	30–35	14.5–15	≈ 50
SAC305	217–220	7.4	21–23	45–51	12–13	≈ 58
Sn–0.7Cu (+Ni)	227	7.3	20–22	≈ 45	≈ 12	≈ 60
Sn–58Bi	138	8.6	≈ 15	≈ 40	≈ 34	≈ 19
95Pb–5Sn	308–312	11.0	≈ 29	≈ 24	≈ 19	≈ 35
80Au–20Sn	280	14.5	≈ 16	≈ 60	≈ 16	≈ 57
Indium	157	7.3	≈ 32	≈ 11	≈ 8.4	≈ 82

Typical bulk values; joint properties depend on microstructure and strain rate.

Microstructure of a SAC joint

A SAC joint is mostly β-Sn, with fine Ag_3Sn particles and Cu_6Sn_5 rods in the eutectic regions between Sn dendrites (Chapter 9). The Ag_3Sn network pins dislocations and reduces creep rate, which is why SAC joints outlast SnPb in mild cycling. But β-Sn is body-centred tetragonal and strongly anisotropic: its CTE is about 15–16 ppm/K along the a-axis and about 30 ppm/K along the c-axis, and its elastic modulus also varies with direction. Because β-Sn nucleates with difficulty, a SAC ball-grid-array joint often solidifies as one to three large grains (or a "beach-ball" cyclic twin structure). Each joint therefore has its own orientation and its own response to the same thermal cycle, and failures concentrate in joints whose c-axis orientation maximises the mismatch strain. Over time in cycling, Ag_3Sn particles coarsen (Chapter 10, Ostwald ripening), the strengthening fades, and recrystallisation in the most strained region creates a network of new grain boundaries along which fatigue cracks run.

Intermetallic compounds

Wetting happens because molten tin dissolves a little of the pad metal and the solution then precipitates intermetallic compounds (IMCs) at the interface. On copper two compounds form:

- **Cu_6Sn_5** (η phase): forms during reflow as scalloped grains adjacent to the solder, typically 1–3 μm thick after assembly.

- **Cu_3Sn** (ε phase): forms between Cu and Cu_6Sn_5, thin after reflow, and grows during solid-state aging by the reaction of Cu with Cu_6Sn_5.

On nickel (ENIG or Ni underbump metallization), Ni_3Sn_4 forms, or $(Cu,Ni)_6Sn_5$ when the solder contains copper; these grow several times more slowly than the Cu–Sn compounds, which is why nickel is used as a barrier (Chapter 8). Gold from a finish dissolves completely and reprecipitates as $AuSn_4$; above about 3 wt% Au in the joint this embrittles it.

Phase	Where it forms	Young's modulus (GPa)	Character
Cu_6Sn_5	Sn-based solder on Cu, reflow and aging	85–125	Hard, brittle, scalloped
Cu_3Sn	Between Cu and Cu_6Sn_5, mainly in aging	110–135	Planar layer; Kirkendall voids at Cu side
Ni_3Sn_4	Solder on Ni	130–140	Slow-growing; P-rich Ni layer beneath on ENIG
$(Cu,Ni)_6Sn_5$	Cu-containing solder on Ni	≈ 120	Stabilised η; slows Ni consumption
$AuSn_4$	Au dissolved in Sn solder	≈ 50	Brittle platelets; limit Au content

Typical values; IMC elastic constants vary with orientation and measurement method.

A thin, continuous IMC layer is essential — no IMC means no bond — but IMCs are brittle (fracture toughness about 1–2 MPa√m, versus tens for the solder), and they keep growing in service. Thick IMC shifts the failure mode under shock loading from ductile fracture in the solder to brittle fracture through the IMC.

IMC growth. Solid-state growth is controlled by interdiffusion through the IMC layer, so thickness follows the parabolic law of Chapter 8:

$$x = x_0 + \sqrt{k\,t}, \qquad k = k_0 \exp\!\left(-\frac{Q}{RT}\right)$$

with activation energies typically 60–110 kJ/mol for total Cu–Sn IMC growth.

> **Status of the principle: parabolic IMC growth**
> *Approximate.* It follows from diffusion-controlled growth through a planar layer with constant interfacial concentrations. Measured time exponents range from about 0.3 to 0.5 because grain-boundary diffusion through coarsening IMC grains contributes; k_0 and Q are fitted for each solder–pad combination.

Worked example — IMC growth in service versus in an aging test. A SAC305 joint on copper leaves reflow with $x_0 = 2$ µm of IMC. A qualification test at 150 °C finds 3.5 µm of additional growth after 1000 h (3.6×10^6 s), so $k(150\,^\circ\mathrm{C}) = (3.5 \times 10^{-6})^2 / 3.6 \times 10^6 = 3.4 \times 10^{-18}$ m²/s. With $Q = 90$ kJ/mol, what does 10 years at 85 °C produce?

$$\frac{k(358\ \mathrm{K})}{k(423\ \mathrm{K})} = \exp\!\left[-\frac{90\,000}{8.314}\left(\frac{1}{358} - \frac{1}{423}\right)\right] = \exp(-4.65) = 0.0096$$

so $k(85\,^\circ\mathrm{C}) = 3.3 \times 10^{-20}$ m²/s. Ten years is 3.16×10^8 s:

$$\sqrt{kt} = \sqrt{3.3 \times 10^{-20} \times 3.16 \times 10^8} = 3.2\ \mu\mathrm{m}, \qquad x \approx 2 + 3.2 = 5.2\ \mu\mathrm{m}$$

The aging test at 150 °C accelerates growth by a factor of $1/0.0096 \approx 104$ in k, so 1000 h at 150 °C is equivalent to about 1.04×10^5 h, or 12 years, at 85 °C. The result depends strongly on Q: with $Q = 60$ kJ/mol the acceleration factor would be only 22. This is the general weakness of Arrhenius acceleration (Chapter 44): the extrapolation is only as good as the activation energy, which must be measured for the actual mechanism.

Kirkendall voids

In the Cu/Cu_3Sn couple, copper diffuses faster than tin. The unequal atomic fluxes are balanced by a net flux of vacancies toward the copper side; where vacancies supersaturate they condense as **Kirkendall voids**, usually at the Cu/Cu_3Sn interface or inside Cu_3Sn (Figure 51.1). After long aging the voids can link into a near-continuous plane, and the joint then fails at low force in a drop or bend test. Voiding is strongly promoted by impurities codeposited in electroplated copper (organic additives, sulphur), which segregate to the interface and nucleate voids. Remedies are high-purity plating chemistries, nickel barriers, and small Ni or Zn additions to the solder, which form $(Cu,Ni)_6Sn_5$ and suppress Cu_3Sn growth.

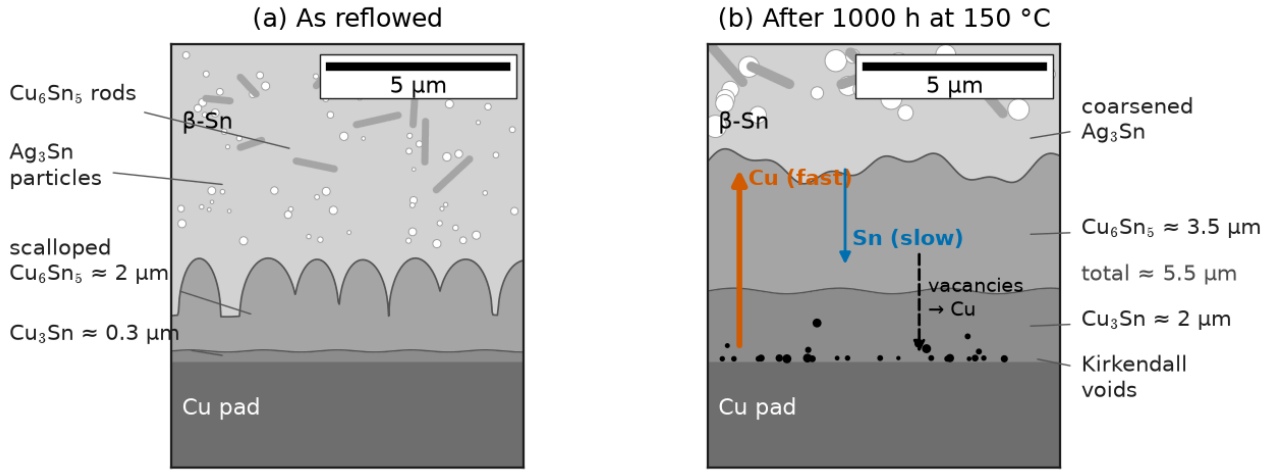


Figure 51.1 — Cross-section of a SAC305 joint on a copper pad after reflow and after prolonged aging, showing the intermetallic layers and Kirkendall voids.

Tin whiskers

Pure tin and high-tin finishes on component leads and connectors can grow **whiskers**: single-crystal filaments 1–5 µm in diameter and up to several millimetres long, extruded from the surface over months to years. A whisker can bridge adjacent leads and carry tens of milliamperes before fusing; in vacuum, a whisker short can vaporise into a metal plasma that sustains a high-current arc. Whisker-induced shorts have caused documented failures in satellites, pacemakers, missiles and nuclear-plant instrumentation.

The driving force is **compressive stress** in the tin film. On copper, Cu diffuses into Sn grain boundaries at room temperature and forms Cu_6Sn_5 wedges, which occupy more volume than the tin they replace and push the surrounding tin grains into compression. Stress is relieved by tin atoms diffusing along grain boundaries to a grain whose boundaries are pinned (often by a surface oxide crack), which then grows outward from its root. CTE mismatch in thermal cycling, external mechanical loads (connector springs, press-fit pins) and surface corrosion add stress.

Mitigations follow from the mechanism:

- **Nickel underlayer** (1–2 µm) between copper and tin, which stops room-temperature Cu_6Sn_5 growth.
- **Annealing** the tin plating (e.g. 150 °C for 1 h) soon after plating, which forms a uniform planar Cu_6Sn_5 layer that relaxes stress and slows further grain-boundary IMC growth.
- **Matte tin** (larger grains, fewer codeposited organics, lower residual stress) instead of bright tin, and thicker plating (≥ 8–10 µm).
- **Alloying**: a few per cent Pb practically eliminates whiskers (one reason for lead exemptions); Bi additions help somewhat.
- **Reflowing** the finish (fusing), or **conformal coating** that a whisker must puncture.

> **Status of the principle: compressive-stress model of whisker growth**
> *Approximate* and partly *empirical*. Compressive stress and grain-boundary diffusion are established as necessary; why a particular grain becomes a whisker, and how to predict incubation time and length, remain incompletely understood, so mitigation relies on tested practice (JEDEC JESD201) rather than calculation.

51.2 Bonding wires

In most packages the die's bond pads are connected to the leadframe or substrate by fine wires, typically 15–50 µm in diameter for signal and logic, and 100–500 µm aluminium or copper wire or ribbon for power devices.

Bonding methods

- **Thermosonic ball bonding**: the wire end is melted by an electric spark into a ball, which is pressed onto the pad with ultrasonic energy at 120–220 °C; the wire is then looped to the second bond (a wedge-shaped "stitch"). Used with Au, Cu and Ag wire.
- **Ultrasonic wedge bonding**: the wire is pressed and ultrasonically scrubbed onto the pad at room temperature with a wedge tool. Used with Al wire (including heavy power wire) and Cu ribbon.

In both, the bond forms in the solid state: ultrasonic scrubbing breaks surface oxides and contaminants and brings clean metal surfaces into atomic contact, with local plastic deformation and heating. Interdiffusion then forms intermetallics where the metals differ.

Wire	Resistivity (μΩ cm)	Thermal conductivity (W/m K)	Typical diameter	Bond method	Key issues
Au (99.99 %)	2.2–2.3	≈ 315	15–50 μm	Thermosonic ball	Cost; Au–Al IMC and Kirkendall voids
Cu, Pd-coated (PCC)	1.7–1.9	≈ 400	15–50 μm	Thermosonic ball, forming gas	Hard ball: pad cratering; oxidation; Cl corrosion of Cu–Al IMC
Ag alloy	2–5	150–400	15–50 μm	Thermosonic ball	Ag migration, sulphidation; used in LEDs for reflectivity
Al–1 % Si	≈ 3.0	≈ 200	25–50 μm	Ultrasonic wedge	Low cost, monometallic bond on Al pads
Al (99.99 %) heavy wire	≈ 2.7	≈ 235	100–500 μm	Ultrasonic wedge	Lift-off and heel cracks in power cycling
Cu heavy wire / ribbon	≈ 1.7	≈ 400	100–500 μm, ribbons	Ultrasonic wedge	Needs Cu-metallised die; high bond force

Typical values.

Gold, copper and silver

Gold was the standard for decades because it does not oxidise, forms a ball easily in air, and bonds readily to aluminium pads. Its weakness is the Au–Al reaction: five intermetallic phases form (Au_5Al_2, Au_4Al, Au_2Al, AuAl, $AuAl_2$, the last purple — hence "purple plague"), and gold's faster diffusion produces Kirkendall voiding around the ball at high temperature (above roughly 150 °C for long times), increasing resistance until the bond opens. Halogen contamination (Br from flame retardants in moulding compound) accelerates the corrosion of these intermetallics.

Since about 2010, cost has driven most high-volume production to **copper wire**, usually palladium-coated to resist oxidation. Copper is cheaper, conducts electricity about 25 % better and heat better, and its Cu–Al intermetallics grow much more slowly than Au–Al. But a copper ball is harder, so it needs higher bonding force that can crack the dielectric under the pad (**pad cratering**, a growing risk with mechanically weak low-k stacks, Chapter 50), and Cu–Al intermetallics corrode in the presence of chloride and moisture. Copper wire also oxidises during ball formation unless shielded by forming gas.

Silver-alloy wire is used in LEDs, where silver's reflectivity helps, and in some low-cost packages; its drawbacks are electrochemical migration (Chapter 44) and tarnishing by sulphur.

Heavy wire in power modules

IGBT and SiC power modules connect the die tops with heavy aluminium wires, each carrying 10–30 A.

Worked example — Resistance and heating of bond wires. A 25 μm gold wire 2 mm long has

$$R = \frac{\rho L}{\pi r^2} = \frac{2.2 \times 10^{-8} \times 2 \times 10^{-3}}{\pi (12.5 \times 10^{-6})^2} = 0.090\ \Omega$$

(0.069 Ω in copper). At 100 mA it dissipates under 1 mW: negligible for signals. A 400 µm aluminium power wire 10 mm long has $R = 2.7 \times 10^{-8} \times 10^{-2} / (\pi \times (2 \times 10^{-4})^2) = 2.15$ mΩ and dissipates $30^2 \times 2.15 \times 10^{-3} = 1.9$ W at 30 A. That heat, deposited in a wire cooled mainly through its two bonds, is part of why multiple parallel wires are used — and the temperature swing of every power cycle, combined with the CTE mismatch between aluminium (23 ppm/K) and silicon (2.6 ppm/K), drives fatigue cracks along the bond interface until the wire **lifts off**, the dominant wear-out mechanism of conventional power modules. Copper wire on copper-metallised dies, sintered-silver die attach (Section 51.8) and wire-free planar interconnects extend power-cycling life by factors of several to ten.

51.3 Flip-chip interconnects

In **flip-chip** assembly, the die is turned face down and connected to the substrate by an area array of bumps. Compared with wire bonding, it allows thousands of I/O across the whole die area rather than only at the periphery, shorter (lower-inductance) connections, and heat extraction through the back of the die.

Bump structure

A bump is a small stack of materials, each with a specific job (Figure 51.2):

- **Under-bump metallization (UBM)** on the die's aluminium or copper pad: an adhesion layer (Ti, Cr, TiW), a barrier and wettable layer (Ni, Ni(V), or Cu thick enough to survive consumption by Sn), and sometimes a protective Au flash. The UBM must not be consumed by the solder through repeated reflow and service; thin copper UBMs can be converted entirely into Cu–Sn IMC, after which the bump detaches.
- **Solder bump** or **copper pillar with solder cap**.

Interconnect	Typical pitch	Material	Notes
C4 solder bump (legacy)	150–250 µm	95Pb–5Sn or 97Pb–3Sn	Controlled-collapse; high melting, ductile; still used in high-reliability
Lead-free solder bump	130–200 µm	SnAg, SAC	Mainstream for coarse-pitch flip-chip
Cu pillar + solder cap	40–150 µm	Cu 20–80 µm tall + SnAg cap	Fine pitch, better electromigration and thermal path
Microbump	10–55 µm	Cu/Ni + SnAg, often fully converted to IMC	2.5D/3D stacking, HBM memory
Hybrid (Cu–Cu) bonding	< 10 µm	Cu pads in SiO_2, direct bond	No solder; dielectric bonded first, Cu interdiffused at 200–400 °C

Typical ranges.

Copper pillars

A copper pillar keeps a fixed standoff regardless of solder volume, so pitch can shrink without bumps bridging. Because copper carries most of the height, current crowding at the pad entrance is reduced and the electromigration life is longer than for a solder bump of the same diameter (Chapter 44); copper's conductivity also improves heat removal. But copper pillars are stiff: with no compliant solder to deform, more of the thermal-mismatch stress is transmitted to the die's interconnect stack, a serious problem over weak low-k dielectrics (Chapter 50). In microbumps, the thin solder cap is often fully converted to IMC (Cu_6Sn_5 or Ni_3Sn_4) during assembly, so the joint's mechanical and electromigration behaviour are those of the intermetallic, not of solder.

Underfill

A die soldered face-down onto an organic substrate experiences the full CTE mismatch between silicon (2.6 ppm/K) and the laminate (≈ 15–17 ppm/K) across tiny joints. Section 51.9 shows that without help the resulting shear strain fails corner bumps within tens or hundreds of cycles. **Underfill** — an epoxy filled with 50–70 wt% spherical silica — is dispensed along the die edge after reflow, drawn into the gap (25–100 µm) by capillarity, and cured. It mechanically couples die and substrate so that they

bend together as a laminate instead of shearing the bumps; strain in the joints falls by roughly an order of magnitude and fatigue life rises by a factor of 10 or more.

The underfill's properties are a compromise: CTE (20–35 ppm/K below T_g) near that of the solder, modulus high enough (6–12 GPa) to couple the structure, T_g above the operating range (typically 120–160 °C), low viscosity and fast flow (which limits filler content and particle size, since particles must pass the gap), and strong adhesion to passivation, solder mask and solder. Variants include **no-flow underfill** dispensed before placement and cured during reflow, **moulded underfill** in which the encapsulant fills the gap, and **wafer-level underfill** films. Adhesion loss at the underfill–die interface (from moisture, contamination or poor plasma cleaning) defeats the coupling and is a common root cause of bump fatigue.

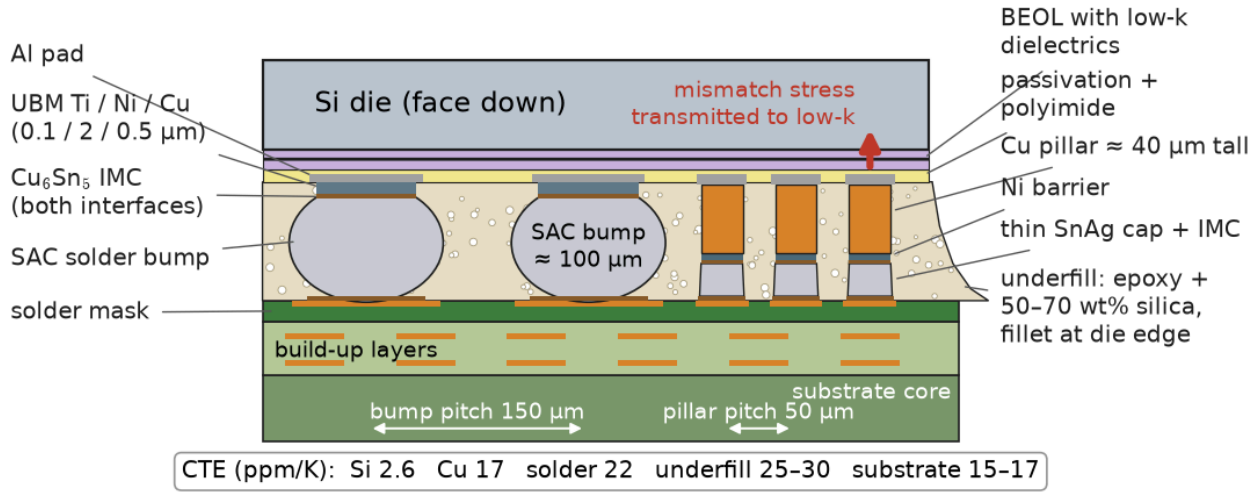


Figure 51.2 — Cross-section of a flip-chip joint on an organic substrate, comparing a solder bump with a copper pillar, both with underfill.

51.4 Conductive adhesives

Electrically conductive adhesives are polymers (usually epoxies) filled with conductive particles. They join at 120–180 °C instead of 220–260 °C, need no flux, contain no lead, and bond to glass, ceramics, flexible films and plastics that solder cannot wet. Their limitations are higher resistivity, sensitivity to moisture and weaker impact strength.

Isotropic conductive adhesives (ICA)

ICAs conduct in all directions. They contain about 25–35 vol% (70–85 wt%) silver flakes. Conduction requires a continuous network of touching particles, and the conductivity follows **percolation** behaviour (Chapter 25):

$$\sigma \propto (\phi - \phi_c)^t, \qquad \phi > \phi_c$$

with ϕ the filler volume fraction, ϕ_c the percolation threshold (about 0.16 for randomly placed spheres in three dimensions, lower for flakes) and exponent $t \approx 1.6$–2.

> **Status of the principle: percolation scaling**
> *Approximate.* It follows from statistical models of random networks near threshold; the exponent is nearly universal but ϕ_c depends on particle shape, size distribution and processing. Far above threshold, resistance is dominated by particle–particle contact resistance, which depends on cure shrinkage pressing flakes together and on lubricant layers left on the flakes from milling.

Typical ICA volume resistivity is 10^{-4}–10^{-3} Ω cm — ten to a hundred times that of solder (1.2×10^{-5} Ω cm for SAC). Cure shrinkage of the epoxy is essential: it compresses the flake network and lowers resistance by orders of magnitude. ICAs are used for die attach (where thermal conductivity of 2–25 W/m K is also specified), for attaching components to flexible circuits and to temperature-sensitive substrates, and for the interconnection of solar-cell strings. Two reliability concerns are specific:

galvanic corrosion at contacts with tin or tin-plated terminations in humidity, where the Sn–Ag couple forms an insulating tin oxide at the interface and resistance rises sharply; and **silver migration** under bias in humid conditions (Chapter 44). Noble finishes (Au, Ag, Pd) on mating surfaces avoid the first.

Anisotropic conductive films and pastes (ACF, ACP)

An **anisotropic conductive film** contains conductive particles at a concentration well *below* the percolation threshold, typically 3–10 μm in diameter: nickel spheres, or polymer spheres plated with Ni and Au (which deform elastically and keep a spring-loaded contact as the adhesive relaxes). The film is laminated between two sets of bumps or pads and bonded with heat (150–200 °C) and pressure for 5–20 s. Particles trapped between facing pads are squeezed into contact and conduct through the thickness; particles in the gaps between pads remain isolated from each other, so there is no lateral conduction. ACF is the standard for attaching driver chips to glass in displays (chip-on-glass), flex cables to glass and boards, and camera modules.

Worked example — How many particles connect a pad? An ACF has 40 000 particles/mm^2 in its plane. A driver-IC bump is 20 μm × 100 μm = 0.002 mm^2, so on average $40\,000 \times 0.002 = 80$ particles lie over it; resin flow during bonding sweeps some away, and suppose half are captured, $\bar{n} = 40$. Particle capture is close to a Poisson process, so the probability that a pad captures *no* particle is $e^{-40} \approx 4 \times 10^{-18}$: opens are effectively impossible. The opposite risk is a short: in the 10–20 μm gaps between bumps, particles pushed out by the flowing resin can chain together. That risk rises with particle density, so density is a compromise between opens on small pads and shorts across fine gaps. Fine-pitch ACFs control it by fixing particles in a single layer of the film or by insulating each particle with a thin polymer shell that ruptures only under the bonding pressure on the pads. If each captured particle contributes of order 1–10 Ω, forty in parallel give 0.03–0.3 Ω per connection — acceptable for display drivers, unacceptable for power.

51.5 Substrate materials

The substrate carries the die (or several dies), routes their connections, insulates them electrically and conducts their heat to a heat sink. For logic and memory, organic laminates dominate (Section 51.6). For power electronics, RF and harsh environments, ceramics do, because they combine electrical insulation with high thermal conductivity, low CTE, hermeticity and temperature stability.

Substrate	Thermal conductivity (W/m K)	CTE (ppm/K)	ε_r (1 MHz)	Flexural strength (MPa)	K_{IC} (MPa√m)	Notes
96 % Al_2O_3	≈ 24	6.8–7.1	9.4–9.8	350–400	3.5–4	Cheapest; glassy grain boundary (Chapter 17)
99.6 % Al_2O_3	30–35	≈ 7.0	≈ 9.9	500–600	≈ 4	Thin-film circuits, smooth surface
AlN	170–200	4.5–4.7	8.6–8.9	300–400	2.7–3.5	CTE close to Si; brittle; hydrolyses in water
Si_3N_4	60–90	2.6–3.2	7.8–8.1	650–850	6–7.5	Tough: thinner substrates, long cycling life
BeO	250–280	7.5–8.5	≈ 6.7	200–250	—	Toxic dust; RF and military legacy
LTCC (glass-ceramic)	2–4	5.5–7	5–8	200–300	—	Multilayer, co-fired Ag; RF modules
HTCC (Al_2O_3 + W)	15–25	≈ 7	≈ 9.5	300–400	—	Hermetic packages, high-temperature
FR-4 (for comparison)	≈ 0.3	14–17 (x–y)	4.3–4.7	≈ 400 (x–y)	—	Organic laminate

Typical values.

AlN has seven times the thermal conductivity of 96 % alumina and a CTE close to silicon and SiC, so it reduces both thermal resistance and die-attach stress. It achieves that conductivity only if oxygen is kept out of the lattice during sintering (Chapter 17); oxygen dissolved in AlN creates aluminium vacancies that scatter phonons. Si_3N_4 has lower thermal conductivity than AlN but more than twice the strength and fracture toughness, because its sintering produces interlocking elongated β-Si_3N_4 grains that deflect and bridge cracks. A Si_3N_4 substrate can therefore be made thinner, which recovers most of the thermal advantage.

Metallised ceramics for power modules

A power-module substrate is a ceramic plate with thick copper on both faces: the top patterned into circuit traces carrying hundreds of amperes, the bottom soldered or sintered to a baseplate or heat sink (Figure 51.3).

- **Direct bonded copper (DBC)**: oxidised copper foil (0.2–0.6 mm) is placed on alumina or AlN and heated to about 1065–1080 °C, between the Cu–Cu_2O eutectic (1065 °C) and the melting point of copper (1085 °C). A thin eutectic liquid forms at the interface, wets the ceramic, and on cooling bonds the copper to it. (On AlN the surface is first oxidised to give an alumina layer for the reaction.)
- **Active metal brazing (AMB)**: copper is brazed to the ceramic with an Ag–Cu eutectic filler containing a few per cent titanium at 800–900 °C. The titanium reacts with the ceramic to form a wettable reaction layer (TiN on nitrides). AMB is the standard for Si_3N_4, which does not bond by the DBC route.

The bonding temperature sets a built-in problem. Copper (17 ppm/K) and ceramic (3–7 ppm/K) are stress-free only at the bonding temperature; on cooling, the copper yields and the ceramic is left with residual stress concentrated at the edges of the copper. Each subsequent thermal cycle repeats a smaller version, and failure occurs by **conchoidal cracking** of the ceramic under the copper edge or by delamination. As a rough guide, DBC on alumina and AlN survives hundreds of cycles of −55/150 °C before cracking, whereas AMB on Si_3N_4 survives thousands. Dimpling and stepping the copper edges reduces the stress concentration, and thinner copper reduces the stress at the cost of current capacity and heat spreading.

Worked example — Thermal resistance through the ceramic. A 10 mm × 10 mm power die dissipates 100 W. Compare the one-dimensional conduction resistance $R = t/(kA)$ of three substrates, with $A = 10^{-4}$ m^2 (ignoring heat spreading, which helps all three):

$$96\%\ Al_2O_3\ (0.38\ \text{mm}):\ \frac{0.38\times10^{-3}}{24\times10^{-4}} = 0.158\ \text{K/W}$$

$$\text{AlN}\ (0.63\ \text{mm}):\ \frac{0.63\times10^{-3}}{170\times10^{-4}} = 0.037\ \text{K/W}, \qquad Si_3N_4\ (0.32\ \text{mm}):\ \frac{0.32\times10^{-3}}{90\times10^{-4}} = 0.036\ \text{K/W}$$

At 100 W, the temperature drop across alumina is 15.8 K; across AlN or Si_3N_4, about 3.7 K. AlN gets there by conductivity; Si_3N_4 by being strong enough to be thin. The extra 12 K matters: semiconductor lifetime and power-cycling life both fall steeply with junction temperature (Chapter 44).

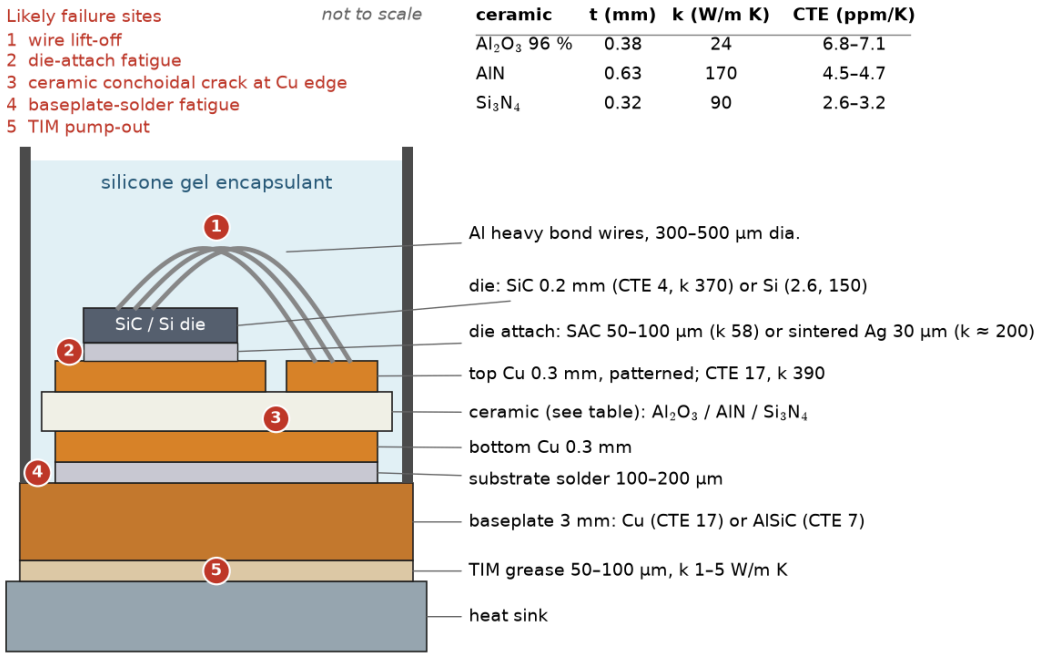


Figure 51.3 — Cross-section of a power-module stack from die to heat sink, with material, thickness, CTE and thermal conductivity of each layer.

LTCC and HTCC

Co-fired ceramics build multilayer circuits by screen-printing conductors on unfired ceramic tapes, punching and filling vias, laminating the stack and firing it once (Chapter 17). **HTCC** (alumina fired at 1500–1600 °C with W or Mo conductors) gives strong, hermetic packages for high-reliability and high-temperature use, at the cost of the high resistivity of W and Mo (5–6 µΩ cm bulk, higher as co-fired). **LTCC** (glass-ceramic fired at 850–900 °C with Ag or Au) gives low-loss conductors and a low, stable permittivity (ε_r 5–8, $\tan\delta$ about 0.001–0.002 at GHz), and allows embedded passives and cavities. It is used for RF front-end modules, radar and automotive sensors; its low thermal conductivity (2–4 W/m K) requires thermal vias under hot components.

51.6 Printed circuit board laminates

The printed circuit board is the largest material in most electronic assemblies, and the one whose properties are most often misunderstood. A laminate is a composite (Chapter 22): woven glass fabric impregnated with a thermosetting resin, pressed with copper foil and cured. Its properties come from the resin, the glass, and their arrangement.

FR-4

FR-4 ("flame retardant, grade 4") is woven E-glass in an epoxy resin, traditionally brominated for flame retardance; halogen-free grades use phosphorus compounds and inorganic fillers. Four properties dominate laminate selection.

Glass-transition temperature T_g. Below T_g the cured epoxy network is glassy and stiff; above it, segmental motion of the polymer chains turns the resin rubbery: modulus drops by an order of magnitude and CTE increases several-fold (Chapter 20). T_g is measured by DSC (heat-capacity step), TMA (CTE change) or DMA (loss peak), and the methods give values differing by 5–15 K, so the method must be stated. Standard FR-4 has T_g of 130–140 °C, mid-T_g about 150 °C and high-T_g 170–180 °C.

Decomposition temperature T_d. The temperature at which the resin has lost 5 % of its mass by thermogravimetric analysis. Dicyandiamide-cured standard FR-4 has $T_d \approx$ 300–320 °C; phenolic-cured grades designed for lead-free reflow reach 340–350 °C. Because lead-free peak reflow (≈ 260 °C) approaches the onset of decomposition for older resins, T_d and the related **time to delamination** at 260 °C and 288 °C (T260, T288) became key specifications with the lead-free transition. Multiple reflows and rework make the margin matter.

CTE. In the plane (x–y), the glass fabric (E-glass CTE ≈ 5.5 ppm/K, modulus ≈ 72 GPa) constrains the resin, giving 13–17 ppm/K — conveniently close to copper (17 ppm/K). Through the thickness (z), there is no glass to restrain the resin, and the CTE is 45–70 ppm/K below T_g and 200–300 ppm/K above it.

Dielectric constant D_k and dissipation factor D_f (the PCB industry's names for ε_r and $\tan\delta$, Chapter 26). Standard FR-4 has $D_k \approx 4.2$–4.7 and $D_f \approx 0.015$–0.022 at 1–10 GHz. Both depend on frequency, temperature, moisture and resin content (the glass has $D_k \approx 6$, the epoxy about 3–3.5, so resin-rich thin laminates have lower D_k).

Z-axis expansion and plated through-holes

A plated through-hole is a copper tube (barrel) 20–30 µm thick through the board thickness. The board expands in z by much more than the copper, and the copper must follow.

Worked example — Z-axis expansion during lead-free reflow. A 1.6 mm FR-4 board with $T_g = 140$ °C, $\alpha_z = 60$ ppm/K below T_g and 280 ppm/K above it, is heated from 25 °C to 260 °C. The resin expands

$$\frac{\Delta L}{L} = 60 \times 10^{-6} \times (140 - 25) + 280 \times 10^{-6} \times (260 - 140) = 0.0069 + 0.0336 = 0.0405$$

i.e. 4.05 %, or 65 µm over 1.6 mm. Copper ($\alpha = 17$ ppm/K) would expand only $17 \times 10^{-6} \times 235 = 0.40$ % (6.4 µm) on its own. The barrel must therefore be stretched by about 3.6 % — more than ten times the yield strain of electroplated copper (about 0.1–0.3 %) — and it deforms plastically. Each reflow, and each later thermal cycle, repeats a smaller plastic strain, and the barrel eventually cracks by low-cycle fatigue (Chapter 44), usually at its mid-plane or at the knee where it meets the pad. Most of the strain comes from the portion above T_g: raising T_g to 170 °C with $\alpha_z = 50/250$ ppm/K reduces the total expansion to about $50 \times 10^{-6} \times 145 + 250 \times 10^{-6} \times 90 = 2.98$ %. This calculation is why lead-free assembly drove the industry to high-T_g, low-z-CTE laminates and to filler-loaded resins (Figure 51.4).

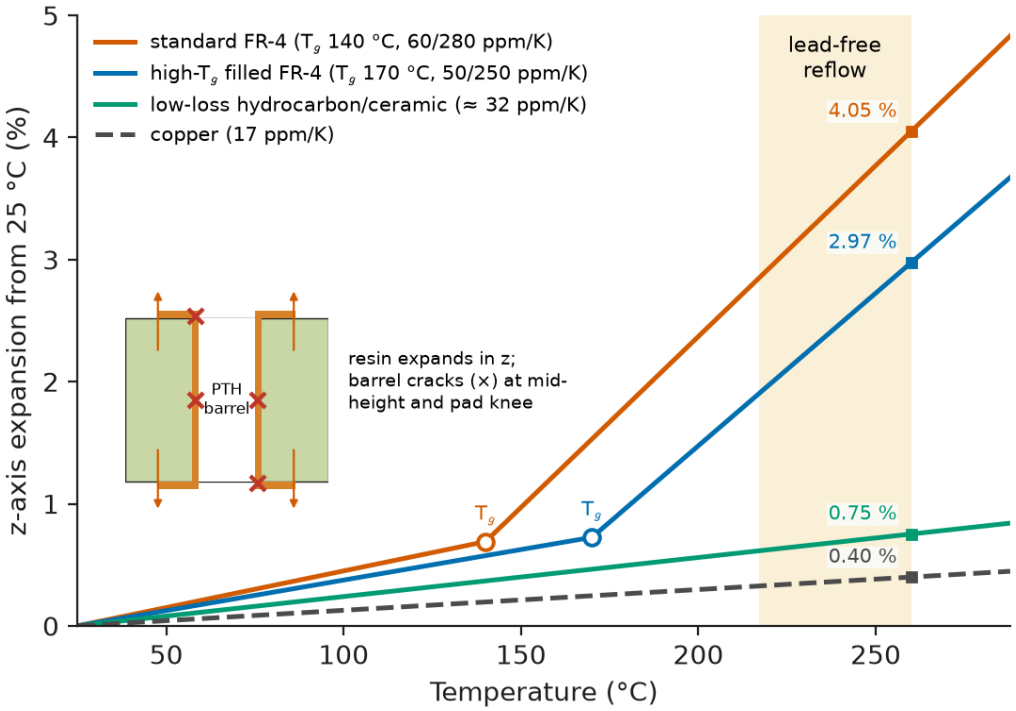


Figure 51.4 — Through-thickness expansion of FR-4 laminates versus temperature compared with copper, showing the change of slope at the glass transition and the strain imposed on a plated through-hole during reflow.

Moisture, CAF and other laminate failure modes

Epoxy laminates absorb 0.1–0.5 wt% water. Moisture raises D_k and D_f (water has $\varepsilon_r \approx 80$), lowers T_g, and at reflow can vaporise and delaminate the board (measling, blistering). Under bias and humidity, copper can migrate along the glass-fibre–resin interface between adjacent vias or traces, forming **conductive anodic filaments (CAF)** that eventually short the circuit; CAF resistance depends on resin–glass adhesion (silane coupling agents), hole-drilling damage and via spacing.

High-frequency laminates

At gigahertz frequencies, dielectric loss in the laminate becomes a significant fraction of the signal budget. For a transmission line in a homogeneous dielectric, the dielectric attenuation is

$$\alpha_d \approx \frac{\pi f \sqrt{\varepsilon_r}\,\tan\delta}{c}\ \text{(Np/m)}, \qquad \alpha_d\,[\text{dB/m}] = 8.686\,\alpha_d\,[\text{Np/m}]$$

> **Status of the principle: low-loss dielectric attenuation formula**
> *Derived* from transmission-line theory for small $\tan\delta$ and a TEM wave in a uniform dielectric. Microstrip lines with partly air-filled fields need an effective permittivity, and conductor loss (skin effect and copper roughness) adds separately and often dominates below a few GHz.

At 10 GHz the formula gives:

Laminate	Construction	D_k	D_f	α_d at 10 GHz (dB/m)	z-CTE (ppm/K)
Standard FR-4	E-glass / epoxy	4.3	0.020	≈ 38	50–70 / 250–300
Low-loss epoxy/PPE (e.g. Megtron 6 class)	Low-D_k glass / modified PPE	3.4–3.7	0.002–0.004	≈ 3–7	≈ 45 / ≈ 260
Hydrocarbon–ceramic (e.g. RO4350B)	Woven glass / ceramic-filled hydrocarbon	3.48	0.0037	≈ 6.3	≈ 32
PTFE–ceramic (e.g. RO3003)	Ceramic-filled PTFE	3.00	0.001	≈ 1.6	≈ 24
PTFE–glass microfibre (e.g. RT/duroid 5880)	Random glass microfibre / PTFE	2.20	0.0009	≈ 1.2	≈ 240
Polyimide (flex)	Polyimide film	3.2–3.5	0.002–0.008	—	—

Typical datasheet values at about 10 GHz; attenuation is the dielectric contribution only.

For FR-4: $\alpha_d = \pi \times 10^{10} \times \sqrt{4.3} \times 0.020/(3.0 \times 10^8) = 4.35$ Np/m $= 38$ dB/m, or 0.38 dB/cm; for RT/duroid 5880, $\pi \times 10^{10} \times 1.483 \times 0.0009/(3.0 \times 10^8) = 0.14$ Np/m $= 1.2$ dB/m. Over a 10 cm 10 GHz trace FR-4 loses 3.8 dB in its dielectric — half the power — while PTFE loses 0.12 dB. That is why RF and millimetre-wave circuits (radar, 5G front ends, satellite links) use PTFE or hydrocarbon–ceramic laminates, and why high-speed digital boards (25–112 Gb/s serial links) moved to low-loss epoxy/PPE resins.

PTFE's low loss comes from its non-polar C–F backbone, which has almost no orientational polarisation (Chapter 26). The price is mechanical and processing: PTFE is soft and creeps, has a high z-axis CTE unless ceramic-filled, a crystalline phase transition near 19 °C that causes a CTE step and a small D_k shift, needs special plasma or sodium etching to make plated holes adhere, and costs five to ten times as much as FR-4. Ceramic filling reduces CTE and stabilises D_k with temperature, and hybrid boards use a PTFE layer only for the RF signal layer, bonded to FR-4 for the rest of the stack. For precise RF design, the anisotropy of woven laminates also matters: D_k differs between in-plane and through-thickness directions, and the glass weave causes skew between the two lines of a differential pair (Chapter 1, anisotropy).

51.7 Encapsulation materials

Encapsulation protects a die and its wires from mechanical damage, moisture, ionic contamination and light, and in power modules provides dielectric insulation between conductors at hundreds or thousands of volts.

Epoxy moulding compounds

Most packaged ICs are encapsulated by **transfer moulding**: a pellet of epoxy moulding compound (EMC) is preheated, forced at about 175 °C into a mould cavity around the leadframe or substrate, and cured in 1–3 minutes, followed by a post-mould cure of several hours. An EMC is a highly engineered composite:

- **Resin**: epoxy (cresol-novolac, biphenyl or multi-aromatic) with a phenolic hardener and a catalyst.
- **Filler**: 70–90 wt% fused silica, as spheres with a broad size distribution to maximise packing while keeping viscosity low. Filler lowers CTE, raises modulus and thermal conductivity, and reduces moisture absorption. Alumina filler raises thermal conductivity to 2–3 W/m K for power packages.
- **Additives**: silane coupling agents (filler–resin and resin–leadframe adhesion), flame retardants (now halogen-free metal hydroxides or phosphorus compounds), carbon black for colour and laser marking, stress-relief modifiers (silicone rubber particles), mould-release waxes and ion getters.
- **Purity**: extractable chloride and sodium below about 20 ppm, because ionic contaminants mobilised by moisture corrode aluminium bond pads.

Property	Typical EMC	Why it matters
Filler content	70–90 wt%	Sets CTE, modulus, moisture uptake
CTE below T_g (α_1)	7–15 ppm/K	Match to Si (2.6) and Cu (17); warpage
CTE above T_g (α_2)	30–60 ppm/K	Stress during reflow
T_g	120–200 °C	Stability at operating and reflow temperatures
Young's modulus at 25 °C	15–30 GPa	Stress transmitted to die
Moisture uptake (85 °C/85 % RH, saturated)	0.2–0.5 wt%	Popcorn cracking, corrosion
Thermal conductivity	0.7–1 W/m K (silica); 2–3 W/m K (alumina)	Heat removal through package
Ionic impurities (Cl^-, Na^+)	< 20 ppm	Pad and wire corrosion

Typical values.

The CTE of the composite can be bracketed from its constituents, as in Chapter 22. For epoxy with $\alpha = 60$ ppm/K and $E = 3$ GPa and silica with $\alpha = 0.5$ ppm/K and $E = 72$ GPa, at 70 vol% silica, the simple volume-weighted rule of mixtures gives $0.3 \times 60 + 0.7 \times 0.5 = 18.4$ ppm/K, while the Turner model (which assumes all phases share the same strain, so that each is weighted by its stiffness) gives

$$\alpha_c = \frac{\sum \alpha_i E_i \phi_i}{\sum E_i \phi_i} = \frac{60 \times 3 \times 0.3 + 0.5 \times 72 \times 0.7}{3 \times 0.3 + 72 \times 0.7} = \frac{79.2}{51.3} = 1.5 \text{ ppm/K}$$

Measured values of about 8–12 ppm/K lie between these two estimates, and more refined models (Kerner, Schapery) reproduce them. The point is qualitative: the stiff, low-expansion filler dominates once it forms a connected skeleton, which is why EMC CTE falls steeply with filler loading above about 60 vol%.

Silicones and other encapsulants

Silicone gels fill power modules (Figure 51.3). They are crosslinked polydimethylsiloxane networks so lightly crosslinked that the modulus is in the kPa range: they apply almost no stress to wires and dies, self-heal after a puncture, and remain stable from about −50 °C to 200 °C (special grades to 250 °C). They provide dielectric insulation (breakdown strength about 15–20 kV/mm) that suppresses partial discharge along the ceramic substrate edges, where the field is concentrated (Chapter 26). But their CTE is very high (250–300 ppm/K), they are permeable to moisture and gases, and they offer no mechanical protection, so modules carry a rigid housing. **Silicone elastomers** and **glob-top** epoxies cover chip-on-board dies; **potting** compounds (epoxy, polyurethane, silicone) encapsulate whole assemblies such as power supplies and ignition coils.

Hermetic packages — ceramic or metal bodies sealed with glass, brazes or welded lids (Au–Sn solder, Kovar lids with CTE ≈ 5–6 ppm/K matched to alumina and glass) — exclude moisture completely and are used where polymers cannot be trusted:

aerospace, implantable medical devices, some optoelectronics and MEMS. Every polymer, however good, is permeable: the question for a non-hermetic package is not *whether* water reaches the die, but how fast and with what consequence.

51.8 Thermal interface materials

Two solid surfaces pressed together touch at only a few per cent of their nominal area, because both are rough and not perfectly flat; the gaps are filled with air ($k \approx 0.026$ W/m K), so contact resistance is large. A **thermal interface material (TIM)** fills these gaps with something more conductive. Chapter 24 introduced TIMs as a thermal topic; here they are treated as package materials, whose microstructure and failure modes decide their real performance.

The thermal resistance of a TIM layer of bond-line thickness BLT and conductivity k, over area A, is

$$R_{\mathrm{TIM}} = \frac{\mathrm{BLT}}{k\,A} + \frac{R''_{c1} + R''_{c2}}{A}$$

where R''_{c1} and R''_{c2} are the specific contact resistances at the two interfaces (K m^2/W). For thin TIMs, the contact terms are often as large as the bulk term, so a high-k TIM with poor wetting can be worse than a moderate-k one applied thinly.

TIM	k (W/m K)	Typical BLT	Specific resistance (K mm^2/W)	Strengths	Failure modes
Thermal grease	1–5	20–75 µm	5–30	Thin BLT, conforms, reworkable	Pump-out in cycling, dry-out
Phase-change material (PCM)	1–5	25–75 µm	5–30	Solid when handled, flows above 45–60 °C	Similar to grease
Gap-filler pad	1–15 (typ. 3–6)	0.25–5 mm	50–1000	Fills large gaps and tolerances, insulating	High resistance, compression set
Dispensable gel	2–8	0.1–2 mm	30–500	Low stress, automated dispensing	Slump, oil bleed
Graphite sheet	400–1500 in-plane; 5–20 through	25–200 µm	5–20	Spreads heat laterally, stable	Poor through-thickness k, conducts electrically
Indium solder TIM	≈ 86	50–300 µm	1–3	Metallic conduction, CPU lid attach	Needs metallised surfaces, voiding
Sintered silver	150–250	20–80 µm	< 1	Die attach above 200 °C, high-cycle life	Process cost, needs Ag/Au finishes

Typical values; datasheet k values are often measured under ideal conditions and overstate in-application performance.

Greases, PCMs, pads and gels

Greases are silicone or hydrocarbon oils loaded with 60–90 wt% of conductive ceramic or metal particles (Al_2O_3, ZnO, BN, AlN, Al, Ag). Their conductivity rises with filler fraction, but so does viscosity, which limits how thin the bond line can be pressed. During thermal cycling, warping surfaces pump the grease out of the gap (**pump-out**) and oil separation and evaporation **dry it out**, so thermal resistance rises over life. **Phase-change materials** are paraffin or polymer-based pads that are solid at room temperature (clean to handle) but melt at 45–60 °C, flow into the gaps, and behave like grease in operation. **Gap-filler pads** and **gels** are soft silicone elastomers loaded with ceramic filler; they absorb mechanical tolerances of millimetres between components and chassis and electrically insulate, at the cost of much larger thermal resistance.

Solder TIMs and sintered silver

When the heat flux is high and the interface must last, metallic TIMs replace polymers. Indium (soft, $k \approx 86$ W/m K) soldered between die and lid is used as **TIM1** in high-power processors; its ductility accommodates CTE mismatch without fatigue cracking. In power electronics, **silver sintering** has become the die-attach and substrate-attach method for SiC modules

operating at junction temperatures above 175 °C. A paste of micro- or nano-scale silver particles with organic binder is printed, dried and sintered at 200–300 °C, with or without applied pressure (5–30 MPa for pressure sintering). The process is solid-state sintering (Chapter 46), driven by the high surface energy of the fine particles, at a homologous temperature of only about 0.4 for silver; the resulting joint is porous silver (10–30 % porosity) whose melting point is that of silver, 962 °C. It therefore operates at a low homologous temperature (0.36 at 200 °C) where creep is slow, unlike a solder at 0.8–0.9 T_m. Its thermal conductivity, 150–250 W/m K depending on porosity, is three to four times that of solder, and power-cycling life is improved by factors of five to ten.

Worked example — Comparing TIM layers. A 20 mm × 20 mm interface ($A = 4 \times 10^{-4}$ m^2) carries 150 W. Compare:

- Grease, $k = 3$ W/m K, BLT 50 µm, plus total contact resistance 5×10^{-6} K m^2/W: $R = \frac{50\times10^{-6}}{3\times4\times10^{-4}} + \frac{5\times10^{-6}}{4\times10^{-4}} = 0.042 + 0.0125 = 0.054$ K/W, a drop of 8.1 K.
- Gap pad, $k = 6$ W/m K, 0.5 mm: $R = \frac{0.5\times10^{-3}}{6\times4\times10^{-4}} = 0.21$ K/W (before contact resistance), a drop of 31 K.
- Indium, $k = 86$ W/m K, 100 µm: $R = 0.0029$ K/W, 0.4 K.
- Sintered silver, $k = 200$ W/m K, 30 µm: $R = 3.8 \times 10^{-4}$ K/W, 0.06 K.

The pad's higher datasheet conductivity does not compensate for its tenfold thickness; for fixed geometry, thickness is the dominant variable. Metallic interfaces are almost free thermally, which is why they are worth their process cost at the die, where heat flux is highest.

51.9 Coefficient-of-thermal-expansion mismatch

The materials of a package span almost two orders of magnitude in thermal expansion:

Material	CTE (ppm/K, near 25 °C)
Fused silica	0.5
Si	2.6
Si_3N_4	2.6–3.2
SiC	≈ 4.0
AlN	4.5
Alloy 42 (Fe–42Ni)	4.5
Mo	4.8–5.1
Kovar (Fe–Ni–Co)	5.1–5.9
GaAs	5.7
Al_2O_3 (96 %)	6.8–7.1
AlSiC (baseplate)	7–8
EMC (below T_g)	7–15
FR-4 in-plane	13–17
Cu	17
SAC solder	21–23
Al	23
Underfill (below T_g)	20–35
FR-4 z-axis (below T_g)	45–70
Silicone gel	250–300

Typical values (Chapter 24).

Whenever two bonded materials with different CTEs change temperature, either they bend, or the joint between them deforms, or both. Every package is a set of such pairs, and every power-on, power-off, day–night or engine cycle is a thermal cycle. CTE mismatch is called **global** when it is between component and board, and **local** when it is between the solder and the materials it joins (solder vs Cu, solder vs Si), which matters even when the global mismatch is zero.

Shear strain in a solder joint: the DNP rule

Consider a rigid, leadless component of CTE α_c soldered to a board of CTE α_b by an array of joints of height h (Figure 51.5). The component and board are stress-free at some reference temperature (near the solder's solidification temperature, then relaxed by creep). On a temperature change ΔT, the unconstrained free expansions of component and board, measured from the **neutral point** (the point that does not move relative to the board, by symmetry the component centre), differ at a joint located a distance L_D — the **distance to neutral point (DNP)** — from it by

$$\Delta u = (\alpha_b - \alpha_c)\,\Delta T\,L_D$$

If component and board are much stiffer than the joints, they enforce this relative displacement, and the joint accommodates it by shear. The shear strain is the relative displacement divided by the joint height:

$$\gamma \approx \frac{\Delta\alpha\,\Delta T\,L_D}{h}$$

> **Status of the principle: DNP shear-strain estimate**
> *Approximate.* It assumes rigid component and board, no bending, and joints that accommodate the full mismatch in pure shear. Real assemblies bend (reducing the strain for compliant boards), and leaded components accommodate most of the displacement in their leads. It overestimates strain for flexible structures and ignores local mismatch, but it correctly identifies the four design levers: $\Delta\alpha$, ΔT, L_D and h.

Three engineering rules follow at once. Strain grows linearly with DNP, so the **corner joints** of a large component fail first, and large components are the most at risk. Strain falls with joint **height**, which is why taller joints (copper-cored balls, column grid arrays for ceramic packages) and solder volume matter. And strain grows with **CTE mismatch**, so ceramic components on FR-4, and silicon dies on organic substrates, are the problem cases.

Worked example — Two mismatch cases. (a) A 10 mm × 10 mm silicon die is flip-chip soldered onto an organic substrate with no underfill: $\Delta\alpha = 17 - 2.6 = 14.4$ ppm/K, corner DNP about 5 mm (along an edge; 7 mm along the diagonal), bump height $h = 75$ µm, and a 100 K cycle:

$$\gamma = \frac{14.4\times10^{-6}\times100\times5\times10^{-3}}{75\times10^{-6}} = 0.096$$

A shear strain of nearly 10 % per cycle (13.6 % at the diagonal corner) is grossly beyond what solder can survive for long: fatigue life in low-cycle plastic straining is governed by a Coffin–Manson relation (Chapter 44) in which the cycles to failure scale roughly as $\Delta\gamma^{-2}$ to $\Delta\gamma^{-2.3}$, and strains of this size give lives of order tens of cycles. This is exactly the problem underfill solves (Section 51.3), by forcing die and substrate to bend together so the bumps no longer carry the full mismatch.

(b) A 2512 chip resistor (6.4 mm × 3.2 mm, alumina body) on FR-4 in an automotive −40 to 125 °C cycle ($\Delta T = 165$ K): $\Delta\alpha = 16 - 6.8 = 9.2$ ppm/K, $L_D = 3.2$ mm (half the length), effective joint height $h \approx 100$ µm:

$$\gamma = \frac{9.2\times10^{-6}\times165\times3.2\times10^{-3}}{100\times10^{-6}} = 0.049$$

Five per cent per cycle. A resistor of this size on FR-4 in an automotive environment will crack its joints within the vehicle life — the failure that opened this chapter. Remedies are smaller resistors in parallel (halving the length halves the strain; with a Coffin–Manson exponent near 2.3, life rises by a factor of about $2^{2.3} \approx 5$), components with compliant terminations, ceramic or metal-core substrates with lower CTE, or relocating large parts away from regions of high temperature swing.

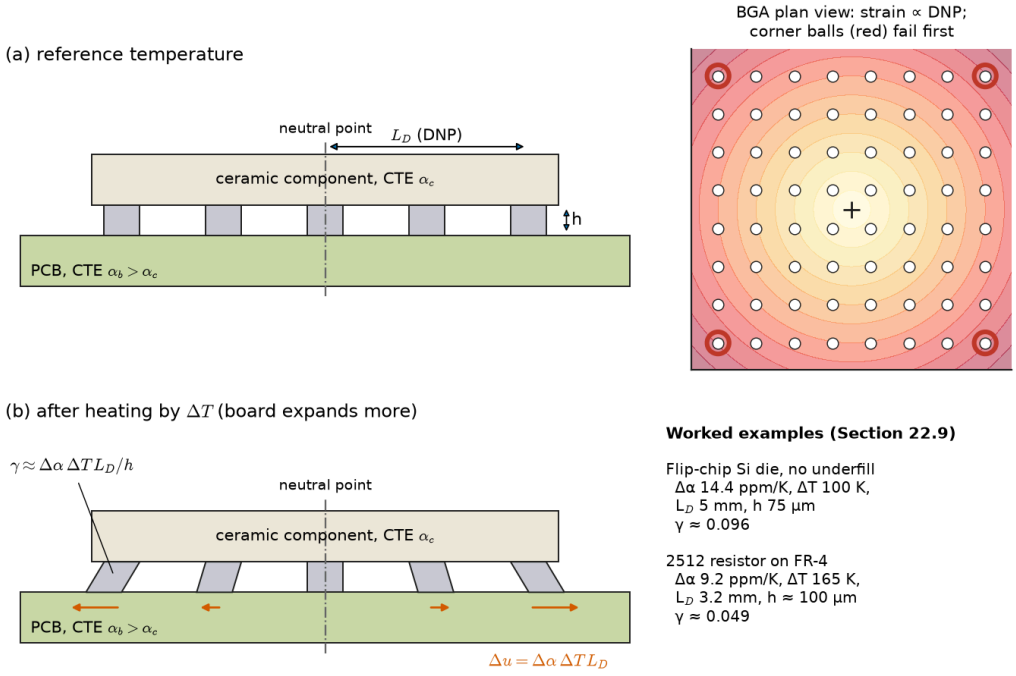


Figure 51.5 — CTE mismatch between a ceramic component and a PCB, showing how relative displacement at a joint grows with distance to neutral point and is absorbed as shear in the solder.

Bending and warpage

When the mismatched layers are not rigidly constrained, they bend. For a bilayer of total thickness h_t, layer thickness ratio $m = t_1/t_2$ and modulus ratio $n = E_1/E_2$, Timoshenko's analysis gives the curvature

$$\kappa = \frac{6\,\Delta\alpha\,\Delta T\,(1+m)^2}{h_t\left[3(1+m)^2 + (1+mn)\left(m^2 + \dfrac{1}{mn}\right)\right]}$$

which for equal thicknesses and moduli ($m = n = 1$) reduces to $\kappa = 1.5\,\Delta\alpha\,\Delta T/h_t$. A 1 mm thick package made of two equal layers with $\Delta\alpha = 10$ ppm/K, cooled by 150 K from its moulding temperature, has $\kappa = 1.5 \times 10^{-5} \times 150/10^{-3} = 2.25$ m^{-1}; over a 20 mm span the out-of-plane deflection is $\kappa L^2/8 = 113$ µm — comparable to the solder-ball height of a fine-pitch BGA. Package **warpage** changes sign and magnitude with temperature, especially as the mould compound passes through T_g, and if it is large at the reflow temperature the corner or centre balls fail to touch their pads (**head-in-pillow** and **non-wet open** defects). Moulding compounds are therefore formulated with CTE and filler content chosen to balance the substrate and die, and warpage is measured across the reflow profile by shadow moiré.

> **Status of the principle: Timoshenko bimetal curvature**
> *Derived* from linear elastic beam theory for two bonded layers with uniform temperature change. It neglects edge effects and plasticity and assumes linear elastic, temperature-independent properties; polymer layers passing through T_g require temperature-dependent moduli and viscoelastic analysis.

51.10 Package stress and failure mechanisms

The package is where the electrical, thermal and mechanical lives of a device meet. Stress arises from three sources: **residual stress** frozen in during manufacture (moulding shrinkage, CTE mismatch on cooling from cure or solder solidification), **thermomechanical stress** in service (power and environmental cycling), and **hygroscopic stress** (polymers swell as they absorb moisture, by roughly 0.1–0.3 % strain per wt% of water).

Stress on the die

Moulding a die in EMC puts it in compression, since the compound shrinks as it cures and cools from 175 °C and has a higher CTE than silicon. Compressive stress on the die surface is typically tens of MPa. Two consequences are distinctive to electronics:

- **Parametric shifts.** Silicon is **piezoresistive**: for p-type silicon the shear coefficient $\pi_{44} \approx 1.4 \times 10^{-9}$ Pa^{-1}, so stresses of 10–100 MPa change resistor values by fractions of a per cent to several per cent and shift transistor mobility and bandgap-reference voltages by millivolts. Precision analog circuits (voltage references, ADCs, sensors) drift after soldering and with humidity for this reason. Remedies are stress-buffer coatings (polyimide or silicone die coat), low-stress compounds, and layout that places sensitive devices at the die centre and in orientations of low piezo-sensitivity.
- **Mechanical damage at die corners and edges.** Shear stress at the die–EMC interface peaks at the corners, where it can crack passivation, shift aluminium lines over steps, and crack weak low-k dielectric stacks beneath bumps (**chip–package interaction**).

Popcorn cracking and moisture sensitivity

Moulded packages absorb moisture from the air during storage. If a saturated package is heated rapidly to reflow temperature, absorbed water, especially any that has condensed at a delaminated interface under the die pad, vaporises. The saturated vapour pressure of water is about 2.3 MPa at 220 °C and 4.7 MPa at 260 °C — and the EMC, now well above its T_g, has lost most of its strength. The interface delaminates, a steam pocket forms and inflates the package bottom, and the EMC cracks with an audible pop (**popcorn cracking**, Figure 51.6). Even without visible cracking, the delamination breaks wire bonds, opens a path for corrosive contaminants and destroys the heat path under the die.

Worked example — Why the die pad is the weak point. Treat the EMC under a 5 mm × 5 mm die pad as a clamped square plate of side $a = 5$ mm and thickness $h = 0.5$ mm, loaded by the vapour pressure in a delaminated gap. For a clamped square plate under uniform pressure, the maximum (edge) bending stress is about $0.31\,p\,(a/h)^2$. At the saturated vapour pressure for 260 °C, $p = 4.7$ MPa:

$$\sigma_{\max} \approx 0.308 \times 4.7 \times \left(\frac{5}{0.5}\right)^2 = 145 \text{ MPa}$$

This is an upper bound — the real pressure is limited by how much water is available in the gap — but even a tenth of it, about 15 MPa, is comparable to the strength of an epoxy compound above its T_g (of order 10–20 MPa). Larger pads and thinner packages are worse, as $(a/h)^2$ shows, which is why thin, large-pad packages have the highest moisture sensitivity.

How long can a package stand in factory air before this becomes a risk? Moisture diffuses into EMC with D of order 10^{-12}–10^{-11} m^2/s at 85 °C. With $D = 5 \times 10^{-12}$ m^2/s, the time to saturate a 1 mm path is of order $L^2/D = 10^{-6}/(5 \times 10^{-12}) = 2 \times 10^5$ s, about 55 h — the $\sqrt{Dt}$ rule of Chapter 8. At room temperature D is several times smaller and saturation takes days to weeks. This is the physical basis of the JEDEC **moisture sensitivity levels** (J-STD-020 and J-STD-033):

MSL	Floor life at ≤ 30 °C / 60 % RH (MSL 1: ≤ 85 % RH)
1	Unlimited
2	1 year
2a	4 weeks
3	168 h
4	72 h
5	48 h
5a	24 h
6	Bake before use; time on label

Components above MSL 1 are shipped in sealed dry bags with desiccant and a humidity indicator; once opened, they must be reflowed within the floor life or baked (typically at 125 °C for hours to a day or two, depending on thickness) to drive the moisture back out. The lead-free transition, by raising peak reflow by 20–35 K and so roughly doubling the vapour pressure, pushed many packages up one or two MSL levels. Material responses are EMCs with lower moisture uptake and higher adhesion

(better coupling agents, roughened or oxide-treated leadframes, plasma cleaning before moulding) so that no delaminated gap exists for the steam to collect in.

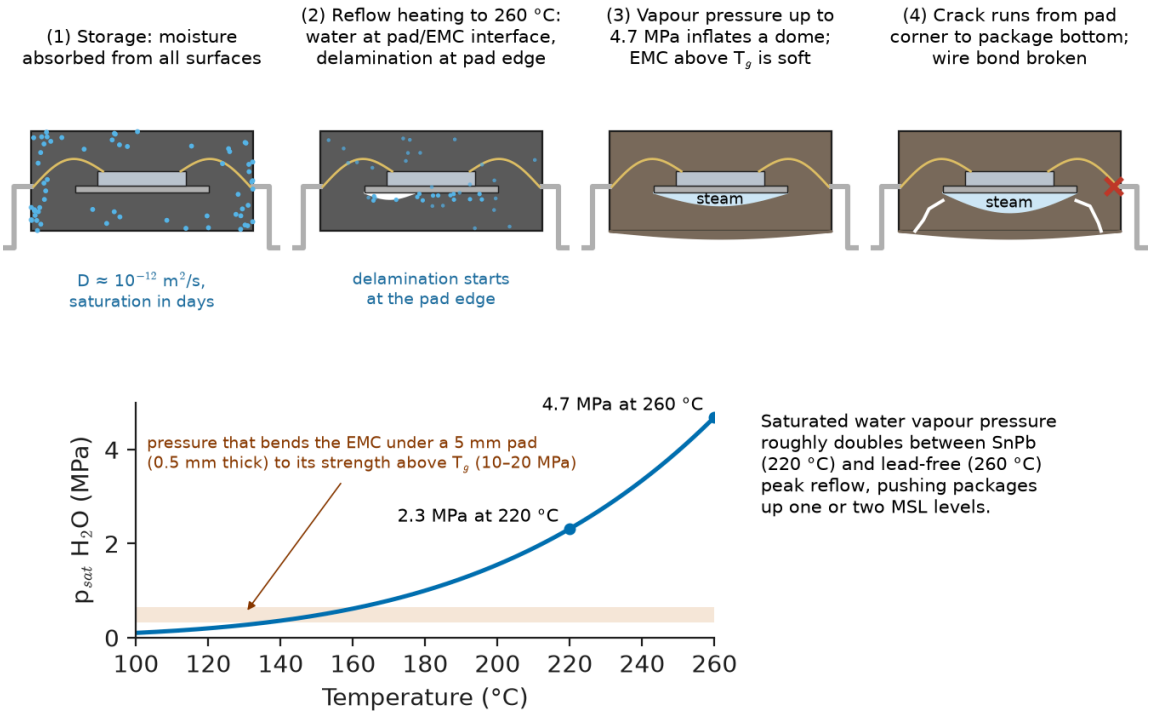


Figure 51.6 — The popcorn-cracking sequence in a moulded package: moisture absorption, delamination and vapour pressure build-up during reflow, and cracking.

Catalogue of package failure mechanisms

The mechanisms of this chapter can be collected by the material property that drives each one. All are treated as failure physics in Chapter 44, and the corresponding accelerated tests and life models in Chapter 44.

Failure mechanism	Location	Driving material property	Typical stress test	Chapter
Solder-joint thermal fatigue	BGA, chip-component joints	CTE mismatch, solder creep, β-Sn anisotropy	Temperature cycling	24
Brittle IMC fracture, Kirkendall voiding	Solder–pad interface	IMC growth (diffusion), impurities in Cu	High-temperature storage + drop	24
Tin-whisker shorts	Sn-plated leads, connectors	Compressive stress in Sn, Cu–Sn IMC growth	Ambient/humid storage, cycling	24
Wire-bond lift-off, heel crack	Power-module wires	Al–Si CTE mismatch, wire fatigue	Power cycling	24
Au–Al / Cu–Al IMC degradation	Ball bonds	Interdiffusion, halide corrosion	High-temperature storage, HAST	24
Pad cratering	Under bond pads, under BGA pads on PCB	Brittle dielectric or laminate resin; hard wire or stiff joints	Bend, shock, cycling	24
Plated-through-hole barrel cracking	PCB vias	Laminate z-axis CTE above T_g	Cycling, reflow simulation	24
CAF, electrochemical migration	PCB, between conductors	Moisture uptake, ionic contamination, bias	Temperature–humidity–bias	24
Delamination, popcorn cracking	EMC interfaces	Moisture uptake, adhesion, EMC strength above T_g	MSL preconditioning + reflow	24
Die cracking	Thin dies, corners	Backgrind damage, CTE mismatch, warpage	Cycling, bend	24
Ceramic substrate conchoidal cracking	DBC/AMB Cu edges	Cu–ceramic CTE mismatch, ceramic toughness	Thermal shock	24
TIM pump-out and dry-out	Die–lid, module–heat sink	Grease rheology, warpage cycling	Power/temperature cycling	24
Parametric drift from package stress	Analog dies	Piezoresistance, EMC shrinkage, moisture swelling	Soldering, humidity	24

A common thread runs through the table. Almost every mechanism involves either a mismatch (in CTE, in diffusion rates, in modulus) or the uptake of moisture by a polymer, and almost every one is accelerated by temperature through an Arrhenius or a cycling law. Material selection for packaging is therefore largely about choosing which mismatches to accept and where to put the compliance that absorbs them: underfill to couple a die to its substrate, silicone gel to decouple wires from their surroundings, tall or compliant joints to spread strain, nickel barriers to slow diffusion, and moisture-resistant, well-adhered polymers to keep water out of interfaces.

What remains open

Open question	Where it is answered
What are the quantitative fatigue (Coffin–Manson, Engelmaier), creep and electromigration (Black's equation) models for joints, wires and bumps?	Chapter 44
How do corrosion, electrochemical migration, CAF and moisture ingress proceed, and how are they accelerated?	Chapter 44
How are IMC thickness, voids, delamination and warpage measured (cross-sectioning, SEM/EDS, acoustic microscopy, X-ray, moiré)?	Chapter 52
How are the thermal properties of TIMs and substrates measured and modelled at the system level?	Chapter 24 (properties); Chapter 52 (measurement)
How are dielectric loss and permittivity of laminates characterised versus frequency, temperature and moisture?	Chapters 26 and 52
How are solders, substrates and laminates selected against cost, RoHS/REACH, recyclability and reliability targets, and how are accelerated tests converted to field life?	Chapter 58

Next: Part XI turns from making materials to verifying and trusting them, beginning in Chapter 52 with the characterisation methods that reveal the microstructures, interfaces and defects that processing has produced.

Part XI — Characterization, Computation and Design

The first ten parts of this book built a chain from atomic structure through thermodynamics, kinetics and properties to the materials classes and the processes that make them. Every link of that chain rests on two kinds of evidence: measurement, which shows what structure a material actually has, and modelling, which explains and predicts it. This final part makes both explicit and then turns them toward the engineer's ultimate question: which material, made how, for this component, at what cost to the user and to the planet? It begins with laboratory instruments, moves to synchrotrons, neutron sources and atom-by-atom microscopes, and follows materials while they are processed or in service. It then climbs the ladder of computation, from the electronic structure of a few atoms through molecular dynamics, phase-field and crystal-plasticity models to data-driven discovery. It ends with selection and design, and with the sustainability of the materials economy as a whole.

By the end of this part you should understand:

- what each characterization technique probes, from what volume, with what resolution and detection limit, and how much to trust the number it gives;
- when a question requires synchrotron X-rays, neutrons, atom-probe tomography or aberration-corrected microscopy, and what each of them delivers;
- how in-situ, operando and correlative experiments link observed structure to behaviour under real conditions;
- what density functional theory can and cannot predict, and how to judge a calculation;
- how atomistic, mesoscale and continuum simulations are chained together in integrated computational materials engineering;
- how databases, machine learning and autonomous laboratories accelerate discovery, and where data-driven methods fail;
- how performance indices, property charts and cost models turn requirements into a rational choice of material and process;
- how embodied energy, lifecycle assessment, criticality and recycling reshape that choice.

Chapter	Central question	What it delivers
52 Materials Characterization: The Laboratory Toolkit	How are structure, composition and properties measured in an ordinary materials laboratory, and how certain are the results?	Optical metallography and stereology, SEM/EDS/EBSD, TEM and diffraction, practical XRD, XPS/AES/SIMS, Raman/FTIR/UV–Vis, thermal analysis, mechanical and electrical tests, GUM uncertainty budgets
53 Large-Facility and Nanoscale Characterization	What can be seen only with a synchrotron, a neutron source or an atom-resolving instrument?	XAS, SAXS, high-resolution diffraction and PDF, tomography, neutron scattering and deep residual stress, atom-probe tomography, aberration-corrected STEM with EELS and 4D-STEM, cryo-EM
54 In-Situ, Operando and Correlative Methods	How does a material's structure change while it is heated, strained, biased or cycled?	In-situ electron microscopy, operando battery and catalyst studies, positron, Mössbauer and NMR spectroscopy, micropillar testing, correlative workflows and automated analysis
55 Electronic-Structure Methods	How can properties be calculated from the electrons up, and how reliable are such calculations?	Born–Oppenheimer and Kohn–Sham foundations, functionals, basis sets, convergence, the known successes and failures of DFT, practical workflows
56 Atomistic and Mesoscale Simulation	How are atomic motions, microstructure evolution and plasticity simulated across scales?	Molecular dynamics and potentials, Monte Carlo and kinetic Monte Carlo, phase-field models, dislocation dynamics and crystal plasticity, CALPHAD, ICME
57 Materials Informatics	How can data and machine learning find materials faster than intuition and trial?	Materials databases, descriptors, ML property prediction and graph networks, machine-learned potentials, high-throughput and autonomous discovery, inverse design and uncertainty
58 Materials Selection and Design	Which material and process best meet a set of competing requirements?	Design process, derived performance indices, Ashby charts, multi-objective selection, shape factors, process selection and cost models, case studies across engineering
59 Sustainability, Lifecycle and the Circular Economy	What does a material cost the planet over its life, and how can that cost be reduced?	Material flows, embodied energy and carbon, lifecycle assessment, critical materials, design for recycling and reuse, recycling technologies, the circular economy and the future of the discipline

Part XI connects to everything that came before. Characterization supplies the evidence for the crystal structures and diffraction of Part I, the phase diagrams and transformation kinetics of Part II, the dislocation and fracture mechanisms of Part III, and the microstructures of the metals, ceramics, polymers and composites of Parts IV and V; it also underpins failure analysis in Part IX and the qualification of processed and additively manufactured parts in Part X. The computational chapters turn the thermodynamics, diffusion and plasticity of earlier parts into calculable models. Selection and sustainability gather the property data of Parts III to VIII into decisions an engineer must defend. The book thus ends where engineering practice begins: with a material that has been seen, understood, predicted and chosen.

Chapter 52 — Materials Characterization: The Laboratory Toolkit

Part XI · Characterization, Computation and Design

Six months after start-up, a girth weld in a duplex stainless-steel line on an offshore gas platform leaks. The pipe is cut out and sent to the laboratory. An optical microscope on a polished, etched section shows that the weld metal is about 75 % ferrite, where the specification allows 35–65 %. An electron backscatter diffraction map shows large ferrite grains with very little austenite along their boundaries; a scanning electron microscope shows that the crack ran through the ferrite in branched, river-like facets; its X-ray detector finds chlorine in the deposits on the crack walls. X-ray diffraction confirms the phase balance and measures 300 MPa of tensile residual stress near the weld root. Photoelectron spectroscopy of the inner surface shows a passive film thinner and poorer in chromium than on the parent pipe. Hardness traverses and a Charpy test complete the picture: a weld cooled too fast to form enough austenite, left tensile stress, and cracked by chloride stress corrosion (Chapter 43). Every statement in that diagnosis is a measurement with its own sampled volume, model and uncertainty. This chapter is the laboratory toolkit that produces them.

> **What does each technique probe, from what volume of material, and with what detection limit? How is a raw signal (an intensity, a heat flow, a voltage) turned into a property, and what model sits in between? How large is the uncertainty on the final number, and which input dominates it?**

Concept	Engineering destination
Optical microscopy, metallography, stereology	Grain size and phase fractions for specifications; weld, casting and coating sections
SEM imaging, EDS, EBSD	Fractography, inclusion and corrosion-product analysis, texture and grain-boundary maps
TEM and electron diffraction	Precipitates, dislocations, interfaces in alloys, ceramics and devices
XRD: phase ID, Rietveld, residual stress, texture	Retained austenite, cement phases, coating stress, sheet-metal texture
XPS, AES, SIMS	Passive films, adhesion and contamination failures, grain-boundary segregation, hydrogen and dopant profiles
Raman, FTIR, UV–Vis	Zirconia implant aging, polymer identification and oxidation, optical band gaps
DSC, TGA, TMA, DMA, laser flash	Polymer crystallinity and T_g, filler content, cement hydration, CTE, damping, conductivity
Mechanical and electrical testing	Design allowables, toughness qualification, sheet resistance and conductivity
GUM uncertainty budgets	Defensible numbers in certificates, qualification reports and failure investigations

Roadmap. The common anatomy of a measurement (52.1) → seeing structure with light, electrons and diffraction (52.2–52.5) → surface composition (52.6) → vibrational and optical spectroscopy (52.7) → thermal analysis (52.8) → mechanical and electrical tests (52.9) → how much to believe the number (52.10).

52.1 The anatomy of a measurement

Every technique in this chapter has the same structure. A **probe** (photons, electrons, ions, heat, force, a current) interacts with a **sampled volume** and produces a **signal**; a **model** converts signal into property. Three figures of merit follow: **lateral resolution** (the smallest feature distinguished side by side), **information depth** (how deep the signal comes from) and **detection limit** (the smallest concentration detected). They trade against each other: techniques that see single atoms sample tiny volumes, and those that detect parts per billion average over micrometres. Figure 52.1 maps the toolkit.

The habit worth acquiring is to ask, for every number: *which volume, which signal, which model?* A grain size measured on one plane, an EDS analysis of a 0.5 µm layer with a 1 µm interaction volume, or a DSC crystallinity calculated with the wrong

reference enthalpy are all correct readings of wrong models.

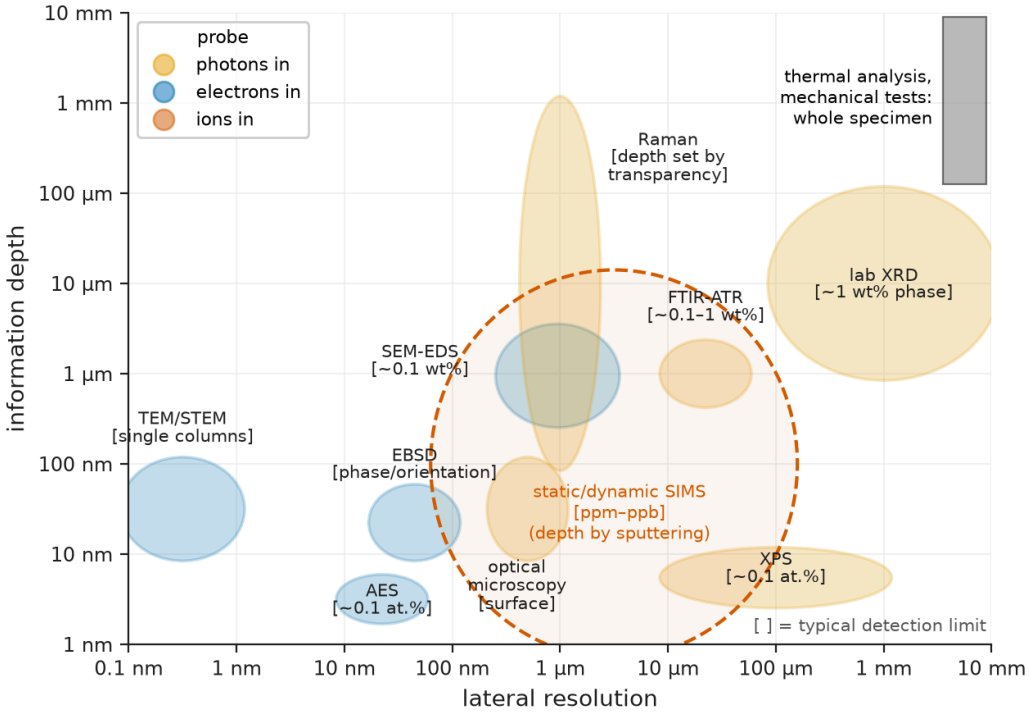


Figure 52.1 — Laboratory characterization techniques placed by lateral resolution and information depth, with detection limits indicated.

52.2 Optical microscopy and metallography

Resolution and contrast

An objective collects light diffracted by the specimen; two points are resolved when their Airy discs separate by at least

$$d = \frac{0.61\,\lambda}{\mathrm{NA}}, \qquad \mathrm{NA} = n \sin\alpha$$

For green light (550 nm) and a dry objective of NA 0.95, $d \approx 350$ nm; oil immersion (NA 1.4) gives about 240 nm. Magnification beyond about $1000 \times \mathrm{NA}$ adds no detail. Depth of field shrinks as $\lambda n / \mathrm{NA}^2$ (about 0.6 µm at NA 0.95), which is why rough fracture surfaces go to the SEM.

> **Status of the principle: Rayleigh resolution limit**
> *Derived.* Fraunhofer diffraction at a circular aperture under incoherent illumination; 0.61 is a convention for "just resolved", and confocal, structured-illumination and near-field methods beat it.

Reflected-light metallographic microscopes offer **bright field** (flat regions bright, grooves and inclusions dark), **dark field** (scattered light only: scratches, cracks), **polarized light** (optically anisotropic phases such as titanium, zinc, magnesium, graphite and many intermetallics show grain contrast unetched) and **differential interference contrast** (nanometre height steps as colour). Transmitted polarized light on 20–30 µm thin sections identifies minerals in rocks, clinker and concrete, and spherulites in polymers.

Preparing a section

Most microstructure is seen by cutting, mounting, grinding, polishing and etching, each step a source of artefacts.

Material	Key preparation points	Typical etch or contrast
Carbon and low-alloy steel	Wet abrasive cutting; SiC grinding; diamond 3 then 1 µm	2 % nital (ferrite boundaries, pearlite); picral for carbides
Stainless and duplex steel	Avoid smearing of soft austenite	Electrolytic oxalic or NaOH (ferrite/austenite contrast); Beraha tints
Aluminium alloys	Very soft; final colloidal silica; avoid embedded abrasive	Keller's reagent; Barker's anodizing for polarized grain contrast
Titanium alloys	Chemo-mechanical final polish	Kroll's reagent (HF–HNO_3) for α/β
Ceramics	Diamond throughout; pull-out mimics porosity	Thermal etch 100–200 °C below sintering temperature
Polymers and composites	Low-temperature mounting; fibres pull out	Permanganic etch for spherulites; polarized light
Concrete	Vacuum epoxy impregnation with fluorescent dye	UV light reveals cracks and porosity

Hot mounting at about 150 °C can age solders and polymers; overetching widens cracks; edge rounding hides coatings unless the specimen is backed. A section is also a single plane: a crack seen in one plane may not be continuous, and only random sections give unbiased statistics.

Quantitative metallography

Stereology (Chapter 5) turns plane measurements into volume quantities. The **point fraction** equals the volume fraction, $P_P = V_V$: a grid of points is laid over the image and the fraction falling on the phase is counted. Grain size is measured by the **linear intercept** method: test lines of total length L cross N grain boundaries, the mean intercept is $\bar{\ell} = L/N$, and the ASTM grain-size number is

$$G = -6.6439 \log_{10} \bar{\ell} - 3.288 \qquad (\bar{\ell} \text{ in mm})$$

> **Status of the principle: $P_P = V_V$ and intercept grain size**
> *Derived.* Geometric-probability identities exact for isotropic, uniform random sampling; anisotropic or banded microstructures need oriented sections, and the mean intercept is about 0.67 of the mean 3D grain diameter for equiaxed grains.

Worked example — grain size and pearlite fraction of a structural steel. On an etched section of a 0.18 % C steel, five test lines with a total true length of 2.50 mm cross 125 ferrite grain boundaries: $\bar{\ell} = 2.50/125 = 0.020$ mm and $G = -6.6439 \log_{10} 0.020 - 3.288 = 8.0$. A 400-point grid lands on pearlite 92 times: $V_V = 0.23$, with a binomial standard error $\sqrt{0.23 \times 0.77/400} = 0.021$, so $V_V = 0.23 \pm 0.02$. The lever rule on the Fe–C diagram (Chapter 7) predicts $(0.18 - 0.02)/(0.77 - 0.02) = 0.21$, consistent within the counting error. Note that halving the error would need four times as many points; counting statistics, not the microscope, usually limit such measurements. Automated image analysis does the same counting on thousands of fields, but its thresholds must be checked against manual counts.

52.3 SEM, EDS and EBSD

Electrons and the interaction volume

An electron accelerated through V has wavelength $\lambda = h/\sqrt{2 m_e e V (1 + eV/2m_e c^2)}$: 8.6 pm at 20 kV, 2.5 pm at 200 kV. Lens aberrations, not wavelength, limit resolution, but SEMs still reach about 1 nm and TEMs below 0.1 nm.

A **scanning electron microscope (SEM)** rasters a 1–30 keV beam focused to a few nanometres. The beam spreads inside the specimen into an **interaction volume** whose depth is estimated by the Kanaya–Okayama range

$$R\ (\mu\text{m}) = \frac{0.0276\, A\, E_0^{1.67}}{Z^{0.889}\, \rho}$$

(A in g/mol, E_0 in keV, ρ in g/cm^3).

> **Status of the principle: Kanaya–Okayama range**
> *Empirical.* A fit to transmission and Monte Carlo data for bulk elements, good to about ±20 % from carbon to gold at 5–30 keV; only indicative for compounds and layered specimens.

Worked example — where does the signal come from? At 20 keV, aluminium ($A = 26.98$, $Z = 13$, $\rho = 2.70$) gives $R =$ 4.2 µm, iron (55.85, 26, 7.87) 1.6 µm and tungsten 0.85 µm; a polymer approximated as carbon of density 0.95 g/cm^3 gives about 10 µm. At 5 keV the same numbers are 0.42, 0.16, 0.08 and 1.0 µm. An EDS analysis of a 1 µm corrosion layer on aluminium at 20 keV is therefore mostly substrate; dropping to 5–7 keV confines the signal but may not excite the K lines of heavier elements. Figure 52.2 shows the volumes.

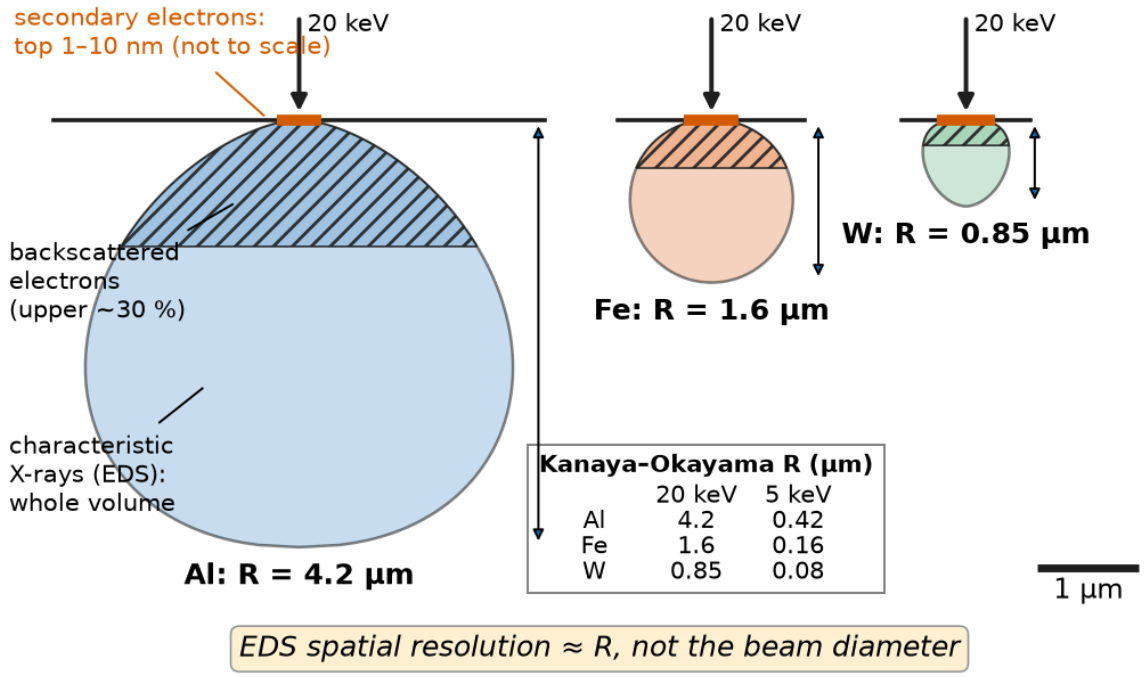


Figure 52.2 — Electron interaction volumes at 20 keV in aluminium, iron and tungsten, and the depths from which SE, BSE and X-ray signals come.

Imaging signals

Secondary electrons (SE) (below 50 eV) escape only from the top few nanometres, and their yield rises at inclined surfaces, so SE images show topography with a large depth of field: fracture surfaces, wear tracks, powder morphology, cell attachment on scaffolds. **Backscattered electrons (BSE)** are beam electrons scattered back with a yield that rises with atomic number (about 0.06 for carbon, 0.15 for aluminium, 0.28 for iron, 0.50 for gold), giving **atomic-number contrast**: carbides in tool steel, intermetallics in aluminium castings and unreacted clinker in concrete stand out without etching. Insulators (ceramics, polymers, biological tissue) charge; remedies are a thin carbon or gold coat, low voltage, or a variable-pressure SEM.

Fractography reads the fracture mode from SE images: dimples for ductile microvoid coalescence, flat facets with river lines for cleavage, rock-candy grain faces for intergranular fracture, and striations whose spacing equals the crack advance per cycle in fatigue (Chapters 14 and 44).

Energy-dispersive X-ray spectroscopy

Beam electrons eject inner-shell electrons, and the characteristic X-rays emitted as the vacancies fill identify the elements. A silicon drift detector resolves about 125–130 eV at Mn Kα. EDS detects about 0.1 wt% in minutes; light elements (B, C, N, O) are only semi-quantitative; quantification with ZAF or $\varphi(\rho z)$ corrections assumes a flat, polished, homogeneous region larger than the interaction volume. Peak overlaps cause classic errors: S Kα (2.31 keV), Mo Lα (2.29) and Pb Mα (2.35); Ti Kα (4.51) and Ba

Lα (4.47); Si Kα (1.74) and W Mα (1.78). **Wavelength-dispersive spectroscopy (WDS)** in an electron microprobe resolves such overlaps with 5–10 eV resolution and detects about 0.01 wt%, at the cost of speed.

Electron backscatter diffraction

When the beam strikes a specimen tilted 70°, backscattered electrons diffract from lattice planes and form **Kikuchi bands** on a phosphor screen; each band is a lattice plane, and the pattern is indexed automatically to give crystal phase and orientation to about 0.5°. Scanning gives an **orientation map** with 20–100 nm resolution: grain size by true grain-boundary detection (rather than etching), misorientation and twin boundaries, phase maps (ferrite versus austenite, α versus β titanium), and texture over thousands of grains. Small orientation gradients within grains, plotted as **kernel average misorientation**, reveal plastic strain and geometrically necessary dislocations, useful for locating cold-worked regions near welds, cracks or forming defects. EBSD needs a deformation-free surface (final polish with colloidal silica or electropolishing), because the signal comes from the top 10–50 nm (Figure 52.3).

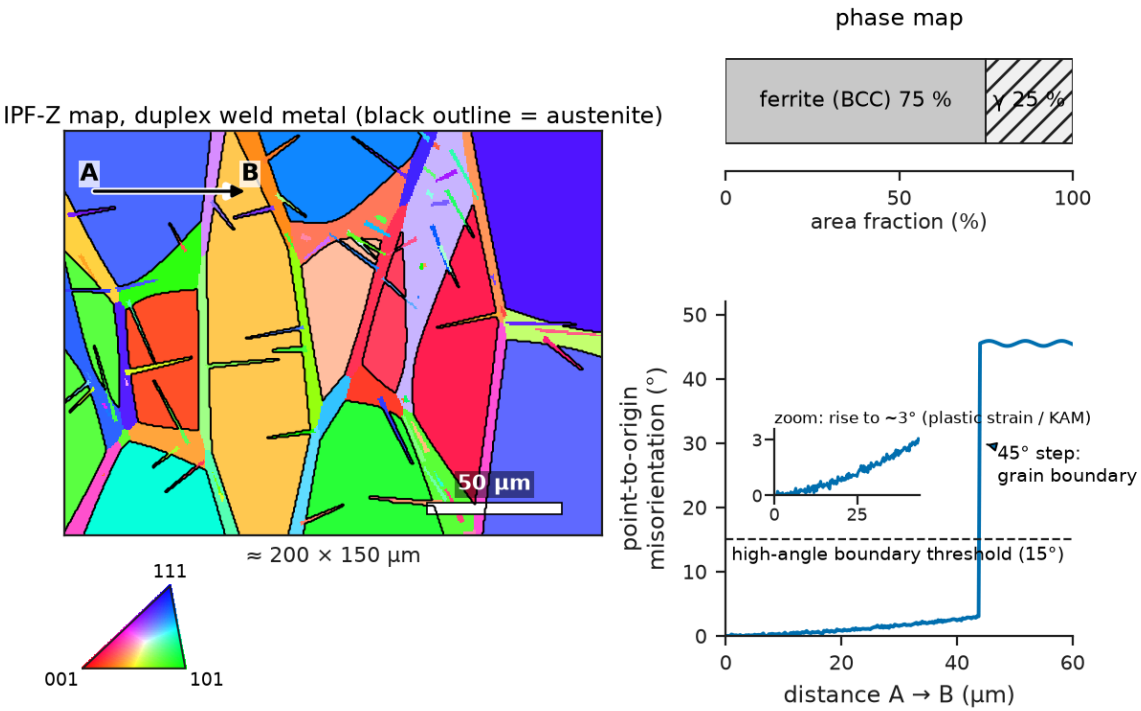


Figure 52.3 — EBSD orientation map of a duplex stainless-steel weld metal with misorientation profile.

52.4 TEM and electron diffraction

A **transmission electron microscope (TEM)** sends an 80–300 keV beam through a foil thinner than about 100 nm, prepared by electropolishing (metals), ion milling (ceramics), ultramicrotomy (polymers, tissue) or focused-ion-beam lift-out (site-specific sections).

Diffraction contrast. An aperture selecting only the transmitted beam (bright field) or one diffracted beam (dark field) makes regions that diffract strongly appear dark or bright. Dislocations, stacking faults, precipitates and strain fields are visible because they locally bend the planes into or out of the Bragg condition. A dislocation is invisible when $\mathbf{g}\cdot\mathbf{b}=0$, with $\mathbf{g}$ the operating reflection; tilting to two such conditions determines its Burgers vector (Chapter 12).

Selected-area electron diffraction (SAED). Because λ is tiny, Bragg angles are a fraction of a degree, and a spot or ring at distance R on the detector corresponds to plane spacing

$$Rd = \lambda L$$

with L the camera length; λL is the **camera constant**. Single crystals give spot patterns, fine polycrystals rings, glasses diffuse halos.

Worked example — indexing a ring pattern. A nanocrystalline aluminium film is examined at 200 kV ($\lambda = 2.508$ pm) with $L = 1.00$ m, so $\lambda L = 2.508 \times 10^{-12}$ m^2. The innermost ring is at 10.73 mm: $d = 2.508 \times 10^{-12} / 10.73 \times 10^{-3} = 0.2337$ nm, which matches Al(111) ($0.4049/\sqrt{3} = 0.2338$ nm). The next ring at 12.39 mm gives $d = 0.2024$ nm = Al(200). For an FCC lattice the ratios R^2 run 3 : 4 : 8 : 11 : 12, so the following rings fall at 17.52 and 20.54 mm for (220) and (311). An extra ring at a spacing absent from the FCC sequence would signal a second phase, for example an oxide or an Al_3X precipitate. In practice the camera constant is calibrated on a standard to about 1 %, which limits SAED lattice parameters to that accuracy; XRD is a hundred times more precise.

High-resolution TEM and STEM. Phase-contrast lattice imaging shows atomic columns at interfaces, but bright spots are not necessarily atoms, and interpretation needs simulation. Scanning TEM with annular dark-field detection, spectroscopy and pixelated detectors, and the aberration correctors that take resolution below 0.1 nm, are the subject of Chapter 53. TEM's limitation is statistical: a few square micrometres of foil may not represent a component, so TEM conclusions are paired with bulk methods such as XRD and SANS.

> **Boundary.**
> Electron optics and scattering theory belong to Engineering Physics; diffraction geometry (reciprocal lattice, structure factor, Ewald sphere) is in Chapter 4.

52.5 X-ray diffraction in practice

Chapter 4 derived Bragg's law $n\lambda = 2d\sin\theta$, the structure factor and peak broadening. This section is about using a laboratory diffractometer.

Instrument choices

The common **Bragg–Brentano** geometry moves source and detector symmetrically, so only planes parallel to the surface diffract; a flat, randomly oriented powder or polycrystal of 10–20 mm is irradiated to a depth of 1–100 μm depending on absorption. The **anode** matters. Cu Kα ($\lambda = 0.15406$ nm) is standard, but its photons (8.05 keV) lie just above the iron K absorption edge (7.11 keV), so iron-bearing specimens fluoresce and give a high background; steels are therefore measured with Co Kα (6.93 keV, below the edge) or Cr Kα, or with an energy-discriminating detector. Mo and Ag anodes give short wavelengths for transmission through thicker specimens and for total scattering (Chapter 53).

Phase identification and quantification

The set of d-spacings and intensities is a fingerprint matched against the ICDD Powder Diffraction File. XRD distinguishes phases, not just elements: ferrite from austenite, anatase from rutile, α- from γ-alumina, tetragonal from monoclinic zirconia, alite from belite in cement clinker. **Rietveld refinement** fits the whole pattern from crystal-structure models and returns phase fractions (to about 1 wt% for well-crystallized phases), lattice parameters (to 10^{-4} relative) and crystallite size and microstrain (Scherrer and Williamson–Hall, Chapter 4). Amorphous content appears as a broad hump and is quantified by adding a known internal standard. Detection limits for minor phases are about 0.5–2 wt%.

Figure 52.4 shows a duplex stainless-steel pattern. The two phases' strongest peaks lie only 1.3° apart with Co radiation, and the austenite (200) and (220) peaks are well separated from ferrite peaks, which is why quantitative retained-austenite and duplex-balance measurements (an important specification for bearings, carburized gears and duplex welds) use them.

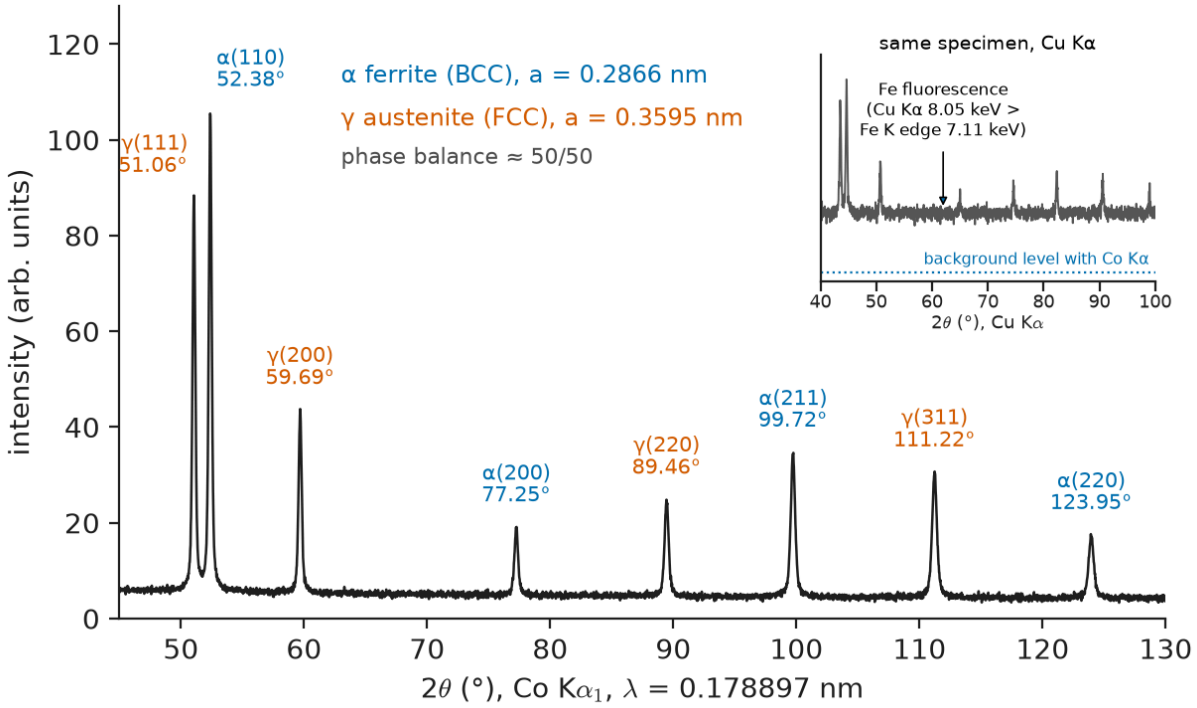


Figure 52.4 — X-ray diffraction pattern of a duplex stainless steel recorded with Co Kα radiation, with peaks indexed for ferrite (BCC) and austenite (FCC).

Residual stress by the $\sin^2\psi$ method

Stress changes lattice spacings, and XRD measures spacings to about 10^{-5}, so the lattice is a strain gauge. For a biaxial surface stress σ and isotropic elasticity, the spacing of planes whose normal is tilted by ψ from the surface normal varies as

$$\frac{d_\psi - d_0}{d_0} = \frac{1+\nu}{E}\,\sigma \sin^2\psi - \frac{2\nu}{E}\,\sigma \qquad \Longrightarrow \qquad \sigma = \frac{E}{1+\nu}\,\frac{1}{d_0}\,\frac{\partial d_\psi}{\partial \sin^2\psi}$$

Because the slope is divided by d_0, an imprecise unstressed spacing hardly matters. High-angle reflections are used because $\Delta\theta = -\varepsilon \tan\theta$ grows toward $2\theta = 180°$.

> **Status of the principle: $\sin^2\psi$ stress analysis**
> *Approximate.* Assumes plane stress in the 10–20 μm sampled layer, isotropic elasticity and a linear d versus $\sin^2\psi$ plot; texture, steep stress gradients or shear stresses curve or split the plot and need diffraction elastic constants or more general analysis.

Worked example — a shot-peened spring steel. The Fe(211) reflection measured with Cr Kα_1 ($\lambda = 0.22897$ nm) has $d_0 = 0.28665/\sqrt{6} = 0.11702$ nm and $2\theta \approx 156.1°$. A least-squares line through d_ψ at seven tilts has slope -2.0×10^{-4} nm. With $E = 210$ GPa and $\nu = 0.28$:

$$\sigma = \frac{210 \times 10^9}{1.28} \times \frac{-2.0 \times 10^{-4}}{0.11702} = -2.8 \times 10^8 \text{ Pa} = -280 \text{ MPa}$$

Compressive surface stress of this size is what shot peening is for: it closes surface fatigue cracks in springs, gears and aircraft landing gear (Chapter 14). The same measurement on the duplex weld of the opening found +300 MPa, tensile, which is what drove the stress-corrosion cracking. Laboratory X-rays sample only the top 5–20 μm in steel; depth profiles need electropolishing layer by layer, and through-thickness stresses need neutrons or high-energy synchrotron X-rays (Chapter 53).

Texture and thin films

Relative peak intensities that differ from the powder file indicate **crystallographic texture**. Tilting and rotating the specimen records **pole figures**, from which the orientation distribution function is calculated. Texture governs the earing of deep-drawn aluminium cans, the r-value (formability) of steel sheet, the magnetic loss of grain-oriented electrical steel and the anisotropy of

additively manufactured parts (Chapters 13, 28 and 48). For thin films, **grazing-incidence XRD** fixes the incident angle at 0.5–2° to confine the beam to the top 10–500 nm, and **X-ray reflectivity** measures film thickness, density and roughness from interference fringes at very low angles.

52.6 Surface analysis: XPS, AES and SIMS

Adhesion, corrosion resistance, wetting, catalysis and biocompatibility are decided in the top few nanometres. The techniques here are surface-sensitive because the particles they detect (electrons of tens to hundreds of eV, or sputtered ions) escape only from the outermost layers, and they need ultra-high vacuum to keep the surface clean during analysis (Chapter 49).

X-ray photoelectron spectroscopy

Monochromatic Al Kα X-rays ($h\nu = 1486.6$ eV) eject core electrons, whose kinetic energy gives the binding energy

$$E_B = h\nu - E_K - \phi$$

with ϕ the spectrometer work function. Photoelectrons of about 1 keV have an inelastic mean free path λ of 2–3 nm, and 95 % of the signal comes from within 3λ, so XPS samples the top 5–10 nm. Its power is **chemical state**: binding energies shift by 1–5 eV with oxidation state and bonding. Typical values: Al 2p 72.7 eV (metal) and about 75.5 eV (oxide); Ti $2p_{3/2}$ 454.0 (metal) and 458.8 eV (TiO_2); Cr $2p_{3/2}$ 574.2 (metal) and 576.6 eV (Cr_2O_3); Si 2p 99.4 (Si) and 103.4 eV (SiO_2); C 1s 284.8 eV (adventitious hydrocarbon, used for charge referencing on insulators). XPS quantifies all elements except H and He to about 0.1 at.% with relative accuracy of 10–20 %, and works on metals, ceramics, polymers and biomaterials alike.

Worked example — thickness of a native oxide. The Al 2p spectrum of an aluminium sheet, recorded at normal emission, shows oxide and metal peaks with intensity ratio $I_{ox}/I_m = 0.80$. For an oxide of thickness d on metal, with attenuation lengths $\lambda_{ox} = 2.8$ nm and $\lambda_m = 2.6$ nm and an Al atom density ratio $N_m/N_{ox} \approx 1.6$ (metal 2.70 g/cm^3, oxide about 3.2 g/cm^3), the Strohmeier relation gives

$$d = \lambda_{ox} \sin\theta \, \ln\!\left(\frac{N_m \lambda_m}{N_{ox} \lambda_{ox}} \, \frac{I_{ox}}{I_m} + 1 \right) = 2.8\ \text{nm} \times \ln(1.6 \times 0.929 \times 0.80 + 1) = 2.2\ \text{nm}$$

with $\theta = 90^\circ$ the take-off angle from the surface. A native oxide of 2–4 nm is normal; a 10 nm hydrated oxide from storage in humid air would hide the metal peak entirely and would also degrade adhesive bonding of aerospace aluminium (Chapter 47), which is why bonding surfaces are etched or anodized just before use. The same overlayer logic measures passive films on stainless steel and the chromium enrichment that makes them protective, as in the duplex weld of the opening.

Auger electron spectroscopy

In **AES** an electron beam creates core holes that relax by emitting Auger electrons of element-specific energy, from the top 2–5 nm. Because the probe is an electron beam, lateral resolution reaches 10–20 nm, far better than XPS. A classic application is **grain-boundary segregation**: a temper-embrittled steel is fractured inside the vacuum along its embrittled boundaries, and AES shows phosphorus or tin concentrated in a monolayer on the fresh intergranular facets. Combined with ion sputtering, AES and XPS give depth profiles through coatings and oxides.

Secondary ion mass spectrometry

SIMS sputters the surface with a focused primary ion beam (O_2^+, Cs^+, Bi_n^+) and analyses the ejected secondary ions in a mass spectrometer. It detects every element including hydrogen, distinguishes isotopes, and reaches parts per million to parts per billion. **Dynamic SIMS** erodes continuously and records concentration versus depth with nanometre resolution: dopant profiles in semiconductors, hydrogen and deuterium in titanium, zirconium and steels, and ^{18}O tracer profiles from which oxygen diffusion coefficients in oxides and fuel-cell electrolytes are measured (Chapter 8). **Static SIMS** (time-of-flight) uses a tiny dose to analyse molecular fragments from the outermost monolayer: silicone release agents on a delaminated paint, lubricant films, protein layers on implants. SIMS yields vary by orders of magnitude between matrices, so quantification requires implanted or certified standards of the same matrix.

52.7 Spectroscopy: Raman, FTIR and UV–Vis

Raman spectroscopy

Monochromatic laser light scattered inelastically by molecular or lattice vibrations is shifted by the vibrational frequency (the **Raman shift**, in $\mathrm{cm^{-1}}$; 520 $\mathrm{cm^{-1}}$ corresponds to 64.5 meV). A confocal Raman microscope analyses a spot of about 1 µm without contact or vacuum, through glass and in water. The spectrum identifies **phases and polymorphs**: silicon 520.7 $\mathrm{cm^{-1}}$; diamond 1332 $\mathrm{cm^{-1}}$; graphitic carbon G band at about 1580 and defect D band at about 1350 $\mathrm{cm^{-1}}$, whose intensity ratio measures disorder in carbon fibres and graphene; anatase $\mathrm{TiO_2}$ 144 $\mathrm{cm^{-1}}$ versus rutile 447 and 612 $\mathrm{cm^{-1}}$; tetragonal zirconia (bands near 148 and 264 $\mathrm{cm^{-1}}$) versus monoclinic (a doublet at about 181 and 192 $\mathrm{cm^{-1}}$). The last is used to map the low-temperature degradation of zirconia femoral heads and dental crowns, where the tetragonal-to-monoclinic transformation in body fluids roughens the surface (Chapter 39). Peak shifts measure **stress**: the silicon line moves by roughly 2 $\mathrm{cm^{-1}}$ per GPa, mapping stress around through-silicon vias and in MEMS. In polymers and composites, the shift of a carbon-fibre or aramid band under load gives the fibre strain in situ.

Fourier-transform infrared spectroscopy

Infrared absorption excites vibrations with a changing dipole moment, so FTIR complements Raman (symmetric, non-polar vibrations are Raman-active; polar groups absorb strongly in IR). It is the standard identification method for **polymers**, adhesives, coatings and contaminants: C–H stretching near 2900 $\mathrm{cm^{-1}}$, O–H near 3300 $\mathrm{cm^{-1}}$, carbonyl C=O near 1715–1740 $\mathrm{cm^{-1}}$, Si–O near 1000–1100 $\mathrm{cm^{-1}}$. The **attenuated total reflectance (ATR)** accessory presses the sample on a diamond or germanium crystal and probes the top 0.5–2 µm without preparation. Absorbance follows the Beer–Lambert law $A = \varepsilon c l$, so band ratios are quantitative: the **carbonyl index** (area of the C=O band relative to a reference $\mathrm{CH_2}$ band) tracks oxidation of polyethylene geomembranes, cable insulation and UHMWPE joint-replacement bearings after gamma sterilization (Chapters 39 and 44).

UV–Vis spectroscopy

Transmission and reflectance from 200 to 1100 nm (or 2500 nm with an NIR extension) characterize glasses, transparent ceramics, polymers, pigments and coatings: solar transmittance of low-E glass, yellowing of polycarbonate, absorption edge of semiconductors. Near the edge, the absorption coefficient of a semiconductor follows the Tauc relation $(\alpha h\nu)^{1/r} \propto h\nu - E_g$, with $r = 1/2$ for direct and 2 for indirect allowed transitions; extrapolating the linear part to zero gives the **optical band gap**. Anatase $\mathrm{TiO_2}$ (3.2 eV) absorbs below 387 nm and ZnO (3.37 eV) below 368 nm, which is why both are UV absorbers in sunscreens and polymer stabilizers (Chapter 36). Integrating spheres collect diffuse light from powders and rough coatings; the Kubelka–Munk function converts diffuse reflectance into an absorption-like quantity.

52.8 Thermal analysis: DSC, TGA, TMA and DMA

Thermal analysis measures a property while temperature is ramped, typically at 5–20 K/min. It is fast, needs milligrams, and is the workhorse of polymer, pharmaceutical and ceramic laboratories.

Differential scanning calorimetry

DSC measures the heat flow into a sample relative to a reference. Its trace (Figure 52.5) shows a step at the **glass transition** (a change in heat capacity), exothermic peaks for **crystallization** and curing, and endothermic peaks for **melting**. The degree of crystallinity of a semicrystalline polymer follows from the melting enthalpy:

$$X_c = \frac{\Delta H_m - \Delta H_{cc}}{\Delta H_m^{0}}$$

where ΔH_{cc} is any cold-crystallization exotherm formed during the scan and ΔH_m^{0} the enthalpy of a 100 % crystalline polymer (about 293 J/g for polyethylene, 207 J/g for polypropylene, 140 J/g for PET).

Status of the principle: DSC crystallinity
Approximate. Assumes the crystalline phase has the same enthalpy as a perfect crystal at every melting temperature and that no reorganization occurs during the scan; literature values of ΔH_m^0 themselves vary by about 10 %.

Worked example — crystallinity of HDPE and of a PET bottle wall. An HDPE pipe sample shows a melting endotherm of 175 J/g and no cold crystallization: $X_c = 175/293 = 0.60$, or 60 %, typical for pipe grades. A PET bottle-wall sample shows T_g at about 78 °C, a cold-crystallization exotherm of 25 J/g near 130 °C and a melting endotherm of 45 J/g near 250 °C; its crystallinity *as received* is $(45 - 25)/140 = 14$ %. Reporting 45/140 = 32 % would mistake crystals formed in the instrument for crystals present in the bottle. DSC also measures the **oxidation induction time** of stabilized polyolefins, the cure state of epoxy composites (residual exotherm), precipitation and dissolution in aluminium alloys, and the transformation temperatures of NiTi shape-memory alloys (Chapter 37).

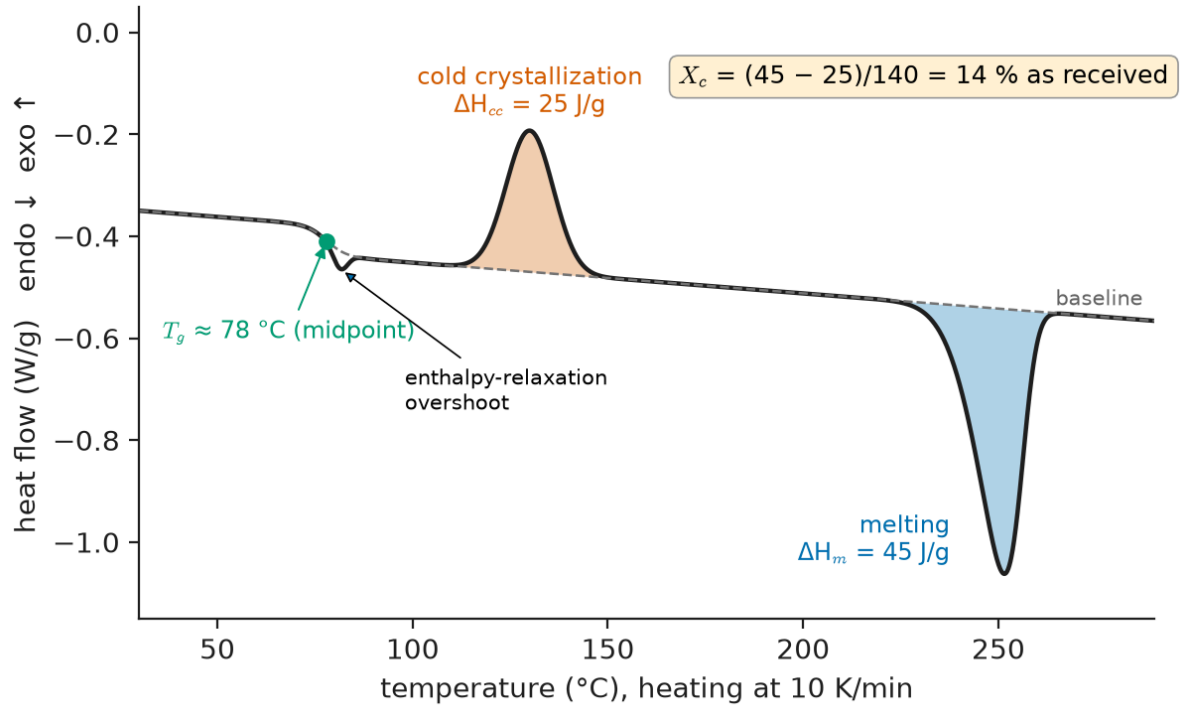


Figure 52.5 — DSC heating trace of a partly crystalline PET sample, showing glass transition, cold crystallization and melting.

Thermogravimetric analysis

TGA records mass versus temperature in a controlled atmosphere: moisture loss, polymer decomposition, the residue (glass fibre, mineral filler, ash) left after burn-off, and oxidation kinetics of metals. Stoichiometry turns steps into quantities. $CaCO_3 \rightarrow CaO + CO_2$ loses 44.0 % of the carbonate mass near 700–800 °C; $Ca(OH)_2 \rightarrow CaO + H_2O$ loses 24.3 % near 400–500 °C.

Worked example — portlandite in hydrated cement. A dried cement paste loses 4.2 % of its mass between 400 and 500 °C. Attributing this to portlandite, the paste contains $4.2/0.243 = 17$ wt% $Ca(OH)_2$. Tracking this number with age shows how far hydration has progressed and, in blended cements, how much portlandite pozzolanic fly ash or slag has consumed (Chapter 18). In a glass-fibre-reinforced PA66 housing, the residue after burn-off in air (for example 30.2 %) gives the fibre fraction directly, checking a 30 % specification.

TMA, DMA and laser flash

Thermomechanical analysis (TMA) measures dimension versus temperature under a small load: the coefficient of thermal expansion, softening and T_g as a change of slope, and through-thickness expansion of laminates and coatings. **Dynamic mechanical analysis (DMA)** applies an oscillating stress and resolves the response into a storage modulus E' and a loss modulus E'', with $\tan\delta = E''/E'$. The $\tan\delta$ peak locates T_g (usually 5–15 K above the DSC value and frequency-dependent), secondary relaxations and damping (Chapter 21). For a carbon-fibre epoxy, the temperature at which E' begins to fall sets the service ceiling of an aircraft part.

The **laser flash** method heats one face of a thin disc with a short pulse and records the rear-face temperature rise. The thermal diffusivity follows from the half-rise time, $\alpha = 0.1388 L^2 / t_{1/2}$ (Parker), and conductivity from $k = \alpha \rho c_p$. For a 2.0 mm alumina disc with $t_{1/2} = 52$ ms, $\alpha = 0.1388 \times (2.0 \times 10^{-3})^2 / 0.052 = 1.07 \times 10^{-5}$ m^2/s, and with $\rho = 3950$ kg/m^3 and $c_p = 775$ J kg^{-1}K^{-1}, $k = 33$ W m^{-1}K^{-1}, typical of 99.5 % alumina (Chapter 24).

52.9 Mechanical and electrical testing

Mechanical tests

Chapter 11 introduced the tensile, hardness, impact and bend tests, and Chapter 14 fracture, fatigue and creep. In the laboratory, standard specimens and procedures (ISO 6892 and ASTM E8 for metals in tension, ISO 527 for plastics, EN 12390 for concrete, ASTM D3039 for composites) make numbers comparable. Practical points that often decide whether a number is right:

Test	Measured	Common pitfalls
Tensile (metals)	E, $R_{p0.2}$, UTS, elongation	Crosshead displacement includes machine compliance: modulus needs an extensometer; strain rate affects yield of mild steel and Ti
Tensile (polymers)	Modulus, yield, elongation	Strong rate and temperature dependence; conditioning of moisture in nylons
Hardness (HV, HB, HRC)	Resistance to indentation	Load-dependent at low loads; for steels UTS ≈ 3.3 HV (MPa) only as a conversion estimate
Charpy impact	Absorbed energy (J)	Transition curve, not a single value, for ferritic steels; notch machining quality
Fracture toughness K_{Ic}	Plane-strain toughness	Validity needs thickness $B \geq 2.5 (K_{Ic}/\sigma_y)^2$
Fatigue (S–N, da/dN)	Life, crack growth rate	Surface finish, mean stress, frequency effects on polymers
Compression (concrete)	Characteristic strength	Cube versus cylinder; curing and loading rate
Short-beam shear, open-hole (composites)	Interlaminar shear, notched strength	Failure mode must be checked, not just load
Nanoindentation	Hardness, reduced modulus	Tip-area calibration, pile-up, substrate effect beyond ~10 % of film thickness

The K_{Ic} thickness criterion shows how strongly specimen geometry can matter. For a high-strength steel with $K_{Ic} = 50$ MPa√m and $\sigma_y = 1400$ MPa, $B \geq 2.5 \times (50/1400)^2$ m = 3.2 mm, an easy specimen. For a structural steel with $K \approx 150$ MPa√m and $\sigma_y = 250$ MPa, $B \geq 2.5 \times 0.36 = 0.9$ m: no valid plane-strain test exists, and elastic–plastic toughness (J or CTOD) must be measured instead.

Nanoindentation records load against depth to micronewton and nanometre resolution. Oliver–Pharr analysis takes the unloading stiffness S and contact area A (from depth and a calibrated tip shape) to give hardness $H = P_{max}/A$ and reduced modulus $E_r = (\sqrt{\pi}/2) S / \sqrt{A}$. It measures thin films, individual phases and biological tissue (bone lamellae, dentine), and maps properties across welds and coatings.

Electrical tests

A two-probe measurement includes contact and lead resistance; separating current and voltage contacts removes them. For a thin layer measured with a collinear **four-point probe** whose spacing is much larger than the thickness t, the sheet resistance is

$$R_s = \frac{\pi}{\ln 2} \frac{V}{I} = 4.532 \, \frac{V}{I}, \qquad \rho = R_s t$$

For a 200 nm copper film with $V/I = 0.0207\ \Omega$, $R_s = 0.094\ \Omega$/sq and $\rho = 1.9 \times 10^{-8}\ \Omega$ m, about 12 % above bulk copper ($1.68 \times 10^{-8}\ \Omega$ m), as expected from grain-boundary and surface scattering. The **van der Pauw** method handles arbitrary sample shapes, and with a magnetic field the **Hall effect** gives carrier type, density and mobility (Chapter 32). Eddy-current probes

measure the conductivity of aluminium alloys non-destructively, which is used to verify heat-treatment temper and detect overheating of aircraft skins. **Impedance spectroscopy** separates bulk, grain-boundary and electrode contributions in ionic conductors and coatings (Chapters 25 and 43), and dielectric spectroscopy gives permittivity and loss of insulating polymers and ceramics.

> **Boundary.**
> The derivations of four-point-probe correction factors, van der Pauw and Hall analysis, and magnetic hysteresis measurements are developed for electronic materials in Chapters 25, 28 and 32.

52.10 Measurement uncertainty

A number without an uncertainty is incomplete

A certificate stating a yield strength of 400 MPa cannot tell an engineer whether the steel meets a 390 MPa minimum. The internationally agreed method, the *Guide to the Expression of Uncertainty in Measurement* (GUM), does so in four steps.

1. **Measurement model.** Write the result as a function of inputs, $y = f(x_1, x_2, \dots)$: for yield strength, $R_p = 4F/(\pi d^2)$.
2. **Standard uncertainties of inputs. Type A** uncertainties come from statistics of repeated observations: the standard deviation of the mean, $s/\sqrt{n}$. **Type B** come from other knowledge: calibration certificates, instrument class, resolution. A tolerance $\pm a$ with no further information is treated as a rectangular distribution with $u = a/\sqrt{3}$.
3. **Propagation.** For uncorrelated inputs, $u_c^2(y) = \sum_i (\partial f/\partial x_i)^2 u^2(x_i)$. For products and powers it is simpler in relative terms: if $y \propto x_1 x_2^{-2}$, then $(u_c/y)^2 = (u_1/x_1)^2 + (2u_2/x_2)^2$.
4. **Expanded uncertainty.** $U = k u_c$ with coverage factor $k = 2$ gives about 95 % coverage for an approximately normal result.

> **Status of the principle: GUM law of propagation of uncertainty**
> *Approximate.* A first-order Taylor expansion valid when the model is close to linear over the uncertainty ranges and inputs are uncorrelated (or correlations included); strongly non-linear models use Monte Carlo propagation (GUM Supplement 1).

Worked example — uncertainty budget for a yield strength. Five specimens of 10.00 mm diameter from one plate give a mean yield load of 31.4 kN, i.e. $R_{p0.2} = 4 \times 31\,400/(\pi \times 10.00^2) = 400$ MPa, with a standard deviation of 6 MPa between specimens. The contributions are:

- Specimen-to-specimen scatter (Type A): $6/\sqrt{5} = 2.68$ MPa.
- Load cell, class 1 (±1 %, rectangular): $u_{rel} = 0.01/\sqrt{3} = 0.577$ %, giving 2.31 MPa.
- Diameter, micrometer ±0.01 mm (rectangular): $u = 0.0058$ mm, relative 0.058 %, doubled because d is squared: 0.12 %, giving 0.46 MPa.
- Determination of the 0.2 % offset point from the curve (extensometer class and curve-fitting, estimated): 0.5 %, giving 2.00 MPa.

Combined: $u_c = \sqrt{2.68^2 + 2.31^2 + 0.46^2 + 2.00^2} = 4.1$ MPa, and $U = 2u_c = 8.2$ MPa. The result is $R_{p0.2} = 400 \pm 8$ MPa ($k = 2$). The budget (Figure 52.6) shows where effort pays: material scatter (43 % of the variance), the load cell (32 %) and offset determination (24 %) dominate, while the diameter measurement, the one most carefully done, contributes about 1 %. Against a 390 MPa minimum, the lower end of the interval, 392 MPa, still complies, but with little margin; ISO 14253-1 and similar decision rules specify how such guard bands are applied.

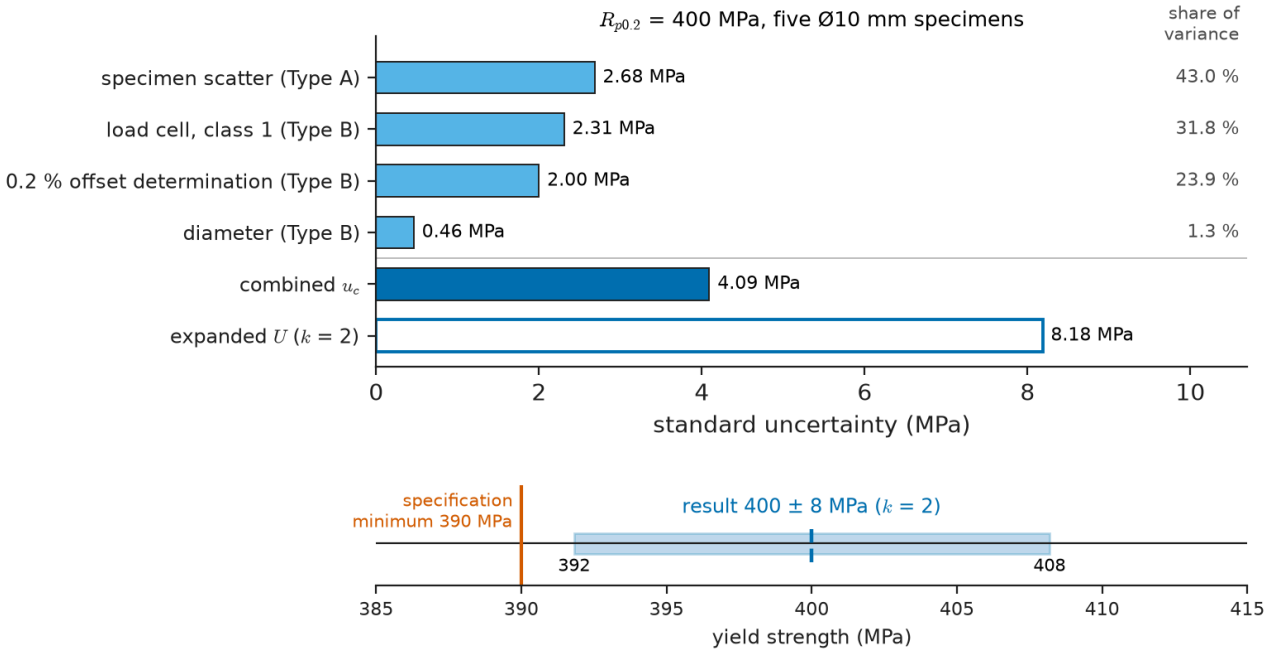


Figure 52.6 — Uncertainty budget for a tensile yield strength of 400 MPa: contributions and combined expanded uncertainty.

Related ideas

Accuracy and precision differ: a miscalibrated hardness tester can be precise and wrong; only calibration traceable to national standards (through certified reference blocks, weights and reference materials) controls accuracy. **Repeatability** (same operator, instrument and short time) and **reproducibility** (different laboratories) are quantified by interlaboratory round robins (ISO 5725), and reproducibility is typically two to three times worse. **Sampling** often dominates everything else: a hardness value from one location in a casting, an EDS spot on one inclusion, or one TEM foil may be precisely measured and unrepresentative. In materials testing the property itself is frequently a distribution, not a value, as for the strength of ceramics, described by Weibull statistics (Chapters 11 and 44).

What remains open

Open question	Where it is answered
How are single atoms, light elements and nanometre clusters located in three dimensions?	Chapter 53 (atom-probe tomography, aberration-corrected STEM)
How do synchrotron X-rays and neutrons measure stress deep inside components and oxidation states of dilute elements?	Chapter 53
How is a microstructure followed while it is heated, strained or electrochemically cycled?	Chapter 54 (in-situ and operando methods)
How are measured spectra and patterns linked to calculated structures?	Chapters 55 and 56
How can automated analysis of thousands of micrographs and spectra feed data-driven models?	Chapters 54 and 57
Where do Bragg's law, structure factors and peak-broadening analysis come from?	Chapter 4 (back reference)
How are fracture surfaces and characterization results combined into a root-cause analysis?	Chapter 44 (back reference)

Next: Chapter 53 goes beyond the laboratory bench to synchrotrons, neutron sources, atom probes and aberration-corrected microscopes, the instruments that see what the toolkit of this chapter cannot.

Chapter 53 — Large-Facility and Nanoscale Characterization

Part XI · Characterization, Computation and Design

The steel walls of a nuclear reactor pressure vessel, 200 mm thick, slowly lose toughness under neutron irradiation: their ductile-to-brittle transition temperature can rise by 100 °C or more over decades, and plant life depends on predicting how much. Optical and scanning electron microscopes see nothing different in irradiated specimens, and conventional TEM sees only faint "black dots". The cause turned out to be clusters of copper, nickel, manganese and silicon atoms only 2–4 nm across, at number densities of 10^{23}–10^{24} m^{-3}, which obstruct dislocations (Chapter 41). They were measured by small-angle neutron scattering, which counts and sizes them in a cubic-millimetre volume, and by atom-probe tomography, which places each atom of a cluster in three dimensions. The same story repeats across the discipline. The oxidation state of nickel in a working battery cathode, the residual stress 30 mm inside a railway wheel, the hydrogen in a zirconium fuel cladding, the boron at a single grain boundary in a turbine-disc alloy and the structure of a protein adsorbed on an implant all lie beyond the laboratory toolkit of Chapter 52. They need either a source far brighter or more penetrating than a laboratory tube, or an instrument that resolves single atoms.

> **What do synchrotron X-rays and neutrons offer that laboratory sources cannot?**
> **How do X-ray absorption, small-angle scattering, total scattering and tomography reveal local structure, nanostructure and internal defects? How do atom probes and aberration-corrected microscopes locate individual atoms and measure local chemistry? How are beam-sensitive soft and biological materials imaged without destroying them?**

Concept	Engineering destination
Synchrotron brilliance and tunability	Feasibility of fast, small-beam, element-specific experiments
XANES and EXAFS	Oxidation states in battery cathodes and catalysts, local structure of dopants and glasses
SAXS	Precipitates in aluminium alloys, porosity, polymer lamellae, nanoparticles
High-resolution diffraction and PDF analysis	Phase transformations, strain in bulk parts, structure of amorphous and nanocrystalline materials
X-ray tomography	Porosity in castings and AM parts, damage in composites, concrete, bone, batteries
Neutron scattering	Light elements, magnetic structures, deep residual stress, phonons and diffusion
Atom-probe tomography	Solute clusters, segregation, precipitate chemistry in 3D at near-atomic resolution
Aberration-corrected STEM, EELS, 4D-STEM	Interface structure, light-element imaging, local bonding, strain and orientation maps
Cryo-EM	Polymers, hydrogels, proteins on biomaterials, lithium metal and SEI in batteries

Roadmap. Why go to a facility (53.1) → synchrotron methods: absorption spectroscopy, small-angle scattering, diffraction and PDF, tomography (53.2–53.5) → neutrons (53.6) → atom-probe tomography (53.7) → aberration-corrected STEM (53.8) → cryo-EM (53.9) → choosing a method (53.10).

53.1 Why go to a facility

Synchrotron light

A laboratory X-ray tube produces photons by electrons hitting an anode: a few characteristic wavelengths, emitted in all directions, with most of the power turned into heat. A **synchrotron** stores electrons of a few GeV in a ring hundreds of metres to over a kilometre in circumference; wherever magnets bend their path, they radiate. Because the electrons are ultra-relativistic,

the radiation is compressed into a narrow forward cone of half-angle about $1/\gamma$, where $\gamma = E/m_e c^2$. For a 6 GeV ring, $\gamma = 6000/0.511 = 11\,700$, so the cone has a half-angle of about 85 µrad.

Bending magnets give a broad, continuous spectrum characterized by the critical energy $E_c(\mathrm{keV}) = 0.665\,E^2(\mathrm{GeV})\,B(\mathrm{T})$: 19 keV for 6 GeV electrons in a 0.8 T field. **Undulators**, periodic arrays of magnets with period λ_u, make the electrons wiggle so that radiation from successive periods interferes, concentrating flux into narrow harmonics:

$$\lambda_1 = \frac{\lambda_u}{2\gamma^2}\left(1 + \frac{K^2}{2}\right)$$

with K the deflection parameter (proportional to field and period).

> **Status of the principle: Undulator equation**
> *Derived.* Follows from the relativistic Doppler compression of the electron's oscillation as seen on-axis, plus constructive interference between periods; it gives the on-axis fundamental and ignores the finite emittance and energy spread of the electron beam, which broaden the harmonics.

Worked example — tuning an undulator. For $\lambda_u = 20$ mm, $K = 1.5$ and a 6 GeV ring, $\lambda_1 = 0.020/(2 \times 11\,742^2) \times (1 + 1.125) = 1.54 \times 10^{-10}$ m, i.e. 0.154 nm or 8.0 keV, the energy of Cu Kα. Opening the magnet gap reduces K and moves the harmonic to higher energy, so the same device scans across an absorption edge. This continuous tunability, impossible with a tube, underlies every spectroscopy in Section 53.2.

The figure of merit for most experiments is **brilliance**: photons per second per unit source area per unit solid angle per 0.1 % bandwidth. A rotating-anode tube gives roughly 10^{9}–10^{10} in these units; third-generation undulators about 10^{20}; fourth-generation rings built since 2016 with multi-bend achromat lattices (MAX IV, ESRF-EBS, APS-U and others) reach about 10^{22}, because their electron beams are smaller and less divergent (Figure 53.1). X-ray free-electron lasers add femtosecond pulses for following processes such as laser melting or shock compression on their own timescale. Brilliance buys three things: **small beams** (100 nm to 1 µm spots that still carry enormous flux), **speed** (a diffraction pattern in a millisecond, a tomogram in a second, hence in-situ and operando studies, Chapter 54), and **high energies** (50–150 keV photons that pass through centimetres of steel).

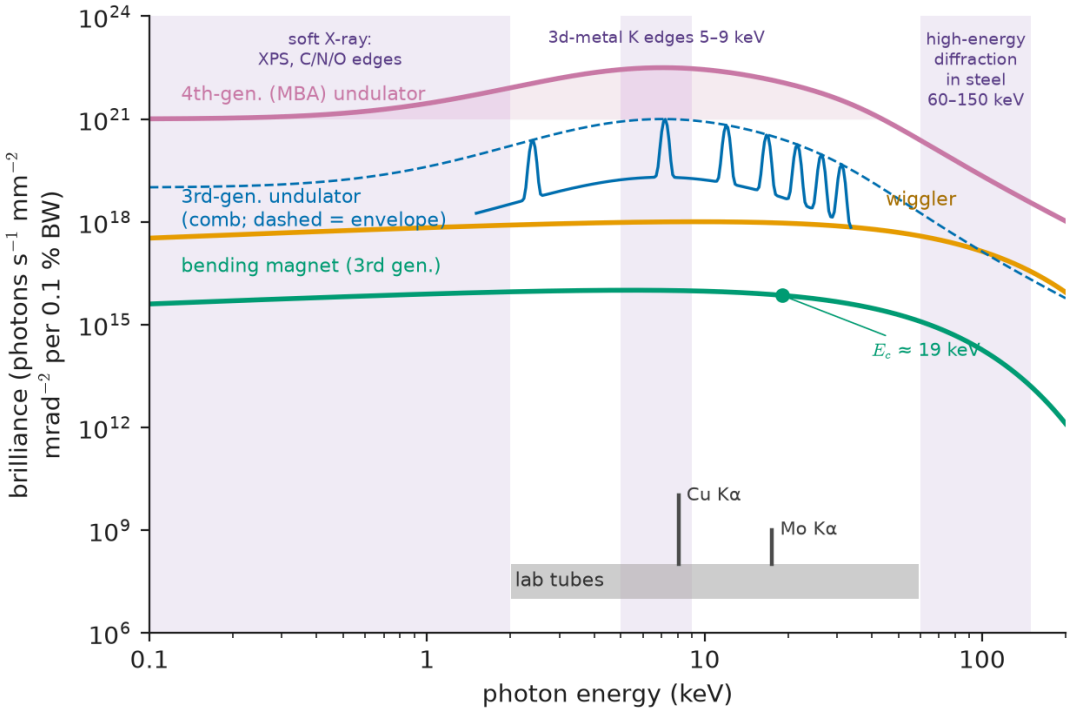


Figure 53.1 — Spectral brilliance of X-ray sources versus photon energy.

Neutron sources

Neutrons are produced either by fission in a **research reactor** (a continuous beam, moderated to thermal energies of about 25 meV) or by **spallation**, in which pulses of GeV protons strike a heavy-metal target (ISIS, SNS, J-PARC, ESS). Spallation sources are pulsed, so neutron energy is measured by time of flight over a known path, and a whole spectrum is used at once. Even the brightest neutron sources deliver orders of magnitude fewer particles per second onto a sample than a synchrotron, so neutron experiments use larger samples (cubic millimetres to centimetres) and longer counting times. Their advantages lie elsewhere (Section 53.6).

Access to both kinds of facility is by competitive proposal, reviewed for scientific merit, with beamtime typically free for open publication and charged for proprietary industrial work. Industrial users from aerospace, automotive, energy and pharmaceutical companies use dedicated industrial liaison teams; residual-stress mapping of welds and tomography of castings are among the most common industrial measurements.

53.2 X-ray absorption spectroscopy

As photon energy rises through the binding energy of a core level, the absorption coefficient $\mu(E)$ of that element jumps: the **absorption edge** (K edges at 6.54 keV for Mn, 7.71 keV for Co, 8.33 keV for Ni, 11.56 keV for the Pt L_3 edge). Tuning a monochromatic synchrotron beam across an edge and measuring transmission ($\mu x = \ln I_0/I$) or the fluorescence that follows absorption gives an **X-ray absorption spectrum** of one element at a time, even at concentrations of 0.01–1 % in a complex matrix, in crystals, glasses, liquids and working devices. The spectrum has two regions (Figure 53.2).

XANES

The **X-ray absorption near-edge structure** (XANES), within about 50 eV of the edge, reflects transitions into empty states and is a fingerprint of **oxidation state and coordination**. The edge shifts to higher energy as the absorbing atom becomes more oxidized, typically by about 1–3 eV per unit change in oxidation state for 3d metals, because the core electron is more tightly bound. Comparing with reference compounds measured on the same beamline gives the average oxidation state. Engineering uses: following $Ni^{2+} \rightarrow Ni^{4+}$ oxidation in a layered NMC cathode during charge, and identifying which transition metal compensates the charge; Ce^{3+}/Ce^{4+} in automotive three-way catalysts; the speciation of chromium (harmless Cr^{3+} versus toxic Cr^{6+}) in corrosion products, slags and contaminated soils; uranium oxidation state in nuclear-waste glass.

EXAFS

At higher energies, the outgoing photoelectron wave scatters off neighbouring atoms and interferes with itself, modulating the absorption. This **extended X-ray absorption fine structure** (EXAFS), $\chi(k)$, is analysed as a function of photoelectron wavenumber

$$k = \frac{\sqrt{2m_e(E - E_0)}}{\hbar} = 0.5123\sqrt{E - E_0\ [\mathrm{eV}]}\ \ \text{Å}^{-1}$$

and is described by the EXAFS equation

$$\chi(k) = \sum_j \frac{N_j S_0^2\, f_j(k)}{k R_j^2}\, e^{-2k^2\sigma_j^2}\, e^{-2R_j/\lambda(k)} \sin[2kR_j + \delta_j(k)]$$

where shell j contains N_j atoms at distance R_j with mean-square disorder σ_j^2, f_j and δ_j are scattering amplitude and phase (calculated by codes such as FEFF), λ is the photoelectron mean free path and S_0^2 an amplitude reduction factor. Fourier transforming $\chi(k)$ gives a pseudo-radial distribution around the absorbing element; fitting gives bond lengths to about 0.01–0.02 Å and coordination numbers to 10–20 %.

> **Status of the principle: EXAFS equation**
> *Approximate.* Single-scattering, plane-wave-corrected theory with harmonic disorder; multiple scattering, anharmonic vibrations at high temperature and the strong correlation between N, S_0^2 and σ^2 limit accuracy, and the region close to the edge (XANES) cannot be fitted this way.

Worked example — what range of k resolves two shells? Data collected to $E - E_0 = 600$ eV reach $k_{max} = 0.5123\sqrt{600} = 12.5$ Å^{-1}; the low-k region below about 3 Å^{-1} is excluded. The usable range $\Delta k = 9.5$ Å^{-1} resolves neighbour shells separated by more than $\Delta R \approx \pi/(2\Delta k) = 0.17$ Å. Two oxygen shells at 1.90 and 2.10 Å around a Jahn–Teller-distorted Mn^{3+} ion in a cathode can thus just be separated; at 50 eV above the edge ($k = 3.6$ Å^{-1}) the spectrum is still XANES and carries no such information. Dilute or heavily absorbing samples measured in fluorescence reach lower k_{max} in practical counting times, which is why EXAFS on trace elements gives fewer, less certain shells.

EXAFS needs no long-range order, so it gives the local structure of glasses, amorphous alloys, solutes in solid solution (does a dopant sit on a substitutional site?), nanoparticle catalysts and ions in solution.

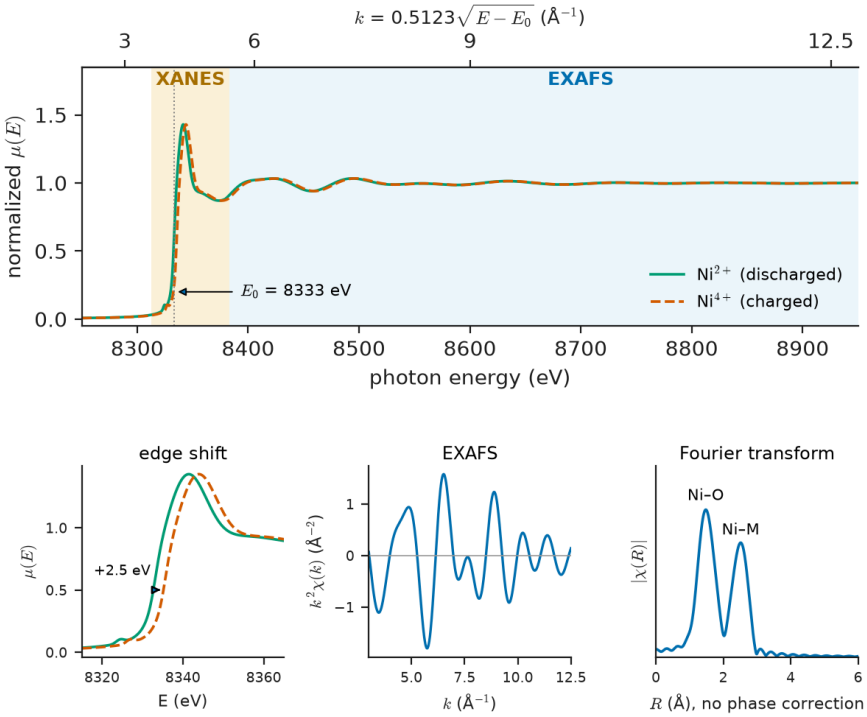


Figure 53.2 — X-ray absorption spectrum at the Ni K edge showing the XANES and EXAFS regions, with the k-weighted EXAFS and its Fourier transform.

53.3 Small-angle X-ray scattering

Bragg's law relates large scattering angles to atomic spacings; structures of 1–100 nm scatter at angles below a few degrees. Small-angle scattering is described by the scattering vector magnitude

$$q = \frac{4\pi \sin\theta}{\lambda}$$

(with 2θ the scattering angle), and a feature of size D scatters near $q \approx 2\pi/D$. With $\lambda = 0.10$ nm (12.4 keV), $2\theta = 0.5^\circ$ corresponds to $q = 0.55$ nm^{-1}. **SAXS** measures the intensity $I(q)$ from **contrast in electron density** $\Delta\rho$ between particles and matrix, averaged over a beam-sized volume containing millions of features: statistically robust, unlike counting particles in TEM.

At small q, a dilute system of particles of radius of gyration R_g follows the **Guinier law**

$$I(q) = I(0)\exp\left(-\frac{q^2 R_g^2}{3}\right)$$

so a plot of $\ln I$ against q^2 has slope $-R_g^2/3$. At large q, sharp interfaces give the **Porod law** $I \propto q^{-4}$, with a prefactor proportional to the interface area per unit volume. The integrated intensity (the invariant) gives the volume fraction if the contrast is known (Figure 53.3).

Status of the principle: Guinier and Porod laws
Approximate. Guinier holds for dilute, non-interacting particles at $qR_g \lesssim 1.3$; Porod for sharp, smooth interfaces at $qR_g \gg 1$. Size distributions, concentrated systems (interparticle interference) and diffuse interfaces need full model fitting.

Worked example — precipitate size in an aged aluminium alloy. A 7xxx alloy aged at 120 °C gives a Guinier plot with slope -3.0 nm^2, so $R_g^2 = 9.0$ nm^2 and $R_g = 3.0$ nm. For spheres, $R_g = \sqrt{3/5}\,R$, so $R = 3.0/0.775 = 3.9$ nm: η′-type precipitates about 8 nm across, the size range that gives peak strength (Chapter 16). The Guinier fit is valid only for $q < 1.3/R_g = 0.43$ nm^{-1}, which sets the angular range that must be measured. Repeating the measurement every few seconds during a heat treatment at a synchrotron follows nucleation, growth and coarsening directly.

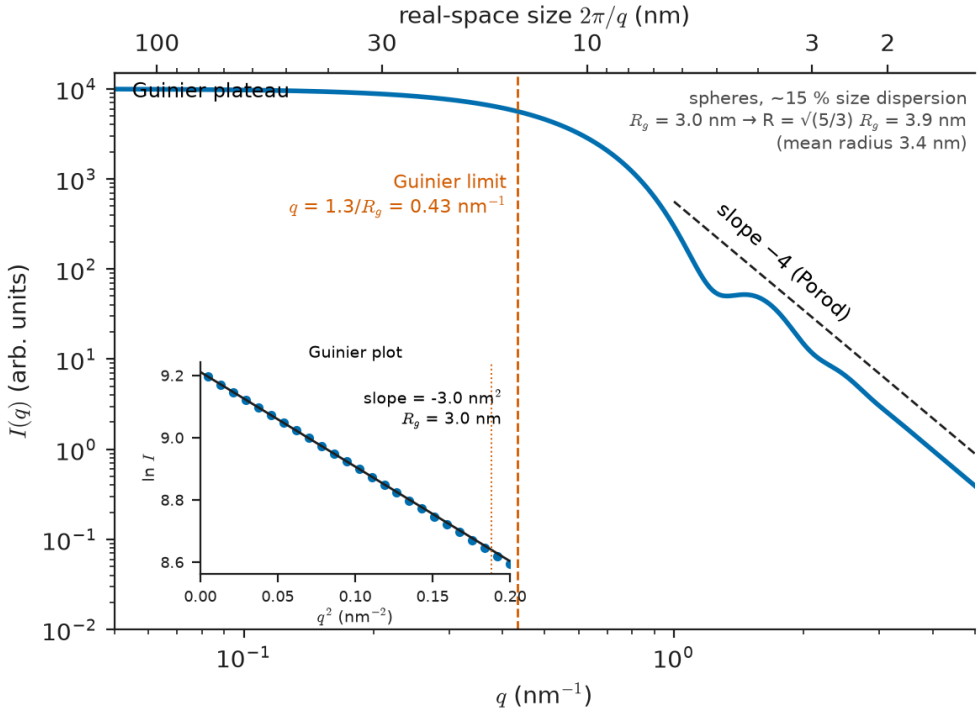


Figure 53.3 — SAXS intensity from precipitates in an aged Al–Zn–Mg alloy, with Guinier and Porod regimes.

Beyond metals, SAXS measures the lamellar long period of semicrystalline polymers (10–30 nm), microphase separation in block copolymers, pore sizes in catalysts, aerogels and cement gels, nanoparticle size distributions, and the hierarchical collagen–mineral structure of bone. **Grazing-incidence SAXS** probes nanostructure in thin films, for example the self-assembly of block-copolymer templates for lithography or the crystallization of perovskite solar absorbers. **Anomalous SAXS**, tuned near an absorption edge, makes one element's contribution visible.

53.4 High-resolution diffraction and pair-distribution-function analysis

Synchrotron powder diffraction

A collimated synchrotron beam with an analyser crystal gives peak widths limited by the sample itself, a resolution $\Delta d/d$ of about 10^{-4}, an order of magnitude better than a laboratory diffractometer. Subtle symmetry lowering (the monoclinic distortion of a ferroelectric, superstructure peaks of ordered phases), small phase fractions down to about 0.1 wt% and precise lattice parameters become measurable.

High photon energies also bring **penetration**. In iron, the mass attenuation coefficient is about 302 cm^2/g at 8 keV but only 0.37 cm^2/g at 100 keV. With $\rho = 7.87$ g/cm^3, the 1/e depth rises from 4.2 μm to 3.4 mm, and a 10 mm steel plate transmits $\exp(-2.93) = 5$ % of the beam. At such energies Bragg angles are small (a few degrees), so a two-dimensional detector records whole Debye–Scherrer rings in transmission. Ring distortion maps strain in every direction within the plate, and moving the specimen maps **residual stress through the thickness** of welds, shot-peened parts and additively manufactured

components with gauge volumes far smaller than neutrons achieve (but with a limited thickness range and elongated gauge volumes along the beam).

Total scattering and the pair distribution function

Bragg peaks contain the average periodic structure; diffuse scattering between them contains the deviations from it. **Total scattering** measures both and Fourier-transforms the normalized structure function $S(Q)$ into the **pair distribution function**

$$G(r) = \frac{2}{\pi}\int_0^{Q_{max}} Q\,[S(Q)-1]\sin(Qr)\,dQ = 4\pi r\,[\rho(r)-\rho_0]$$

which gives the probability of finding atom pairs at separation r, weighted by scattering power, with ρ_0 the average number density. Its peaks are interatomic distances; their areas give coordination numbers. This is the radial distribution function of Chapter 2, measured. It works equally on crystals, nanocrystals, glasses and liquids.

> **Status of the principle: Pair distribution function from total scattering**
> *Derived.* An exact Fourier relation between the structure factor and the pair correlation for an isotropic sample; in practice its accuracy is limited by truncation at Q_{max}, by corrections for Compton and multiple scattering, and by the need to model what the one-dimensional, orientationally averaged function cannot uniquely determine.

Worked example — why PDF needs short wavelengths. Real-space resolution is set by Q_{max}; features closer than roughly $2\pi/Q_{max}$ blur, and termination ripples appear. Cu Kα can at best reach $Q_{max} = 4\pi/1.5406 = 8.2$ Å^{-1} (at 2θ = 180°), too low to resolve the 1.6 Å Si–O bond in silica from neighbouring correlations. Mo Kα reaches 17.7 Å^{-1} and Ag Kα 22.5 Å^{-1}. A 60 keV synchrotron beam ($\lambda = 0.2066$ Å) reaches $4\pi\sin 20^\circ/0.2066 = 20.8$ Å^{-1} at a modest $2\theta = 40^\circ$ on a flat-panel detector, and 30.4 Å^{-1} at $2\theta = 60^\circ$, in a second of exposure. That speed allows PDFs to be followed during the crystallization of a metallic glass, the formation of nanoparticles in solution, or the lithiation of an electrode whose products are too disordered for Bragg analysis.

53.5 X-ray tomography

Computed tomography (CT) records radiographs at many angles and reconstructs the three-dimensional distribution of the attenuation coefficient. The contrast comes from Beer–Lambert attenuation, $I = I_0\exp(-\int \mu\,dx)$, so pores (zero attenuation), heavy inclusions, cracks and phases of different density or composition appear in grey levels. Synchrotron beams add **phase contrast**: their spatial coherence lets edges between materials of similar attenuation (carbon fibres in epoxy, soft tissue, polymer blends) appear through refraction fringes, and phase retrieval converts them into quantitative maps.

> **Status of the principle: Absorption-contrast tomography**
> *Approximate.* Reconstruction assumes straight rays and monochromatic Beer–Lambert attenuation; laboratory polychromatic beams harden (low energies are absorbed first), producing cupping artefacts and streaks near dense features, which must be corrected or avoided by filtering.

Worked example — field of view against resolution. A detector 2048 pixels wide with an effective pixel size of 0.65 µm images a field of view of $2048 \times 0.65 = 1.33$ mm. A good reconstruction needs about $\pi/2 \times 2048 \approx 3200$ projections, and the 16-bit volume occupies $2048^3 \times 2 = 17$ GB. Seeing pores of 2 µm in a 10 mm additively manufactured part is impossible in one scan: either a 1.3 mm sub-volume is cut out or scanned locally (interior tomography), or the voxel size is relaxed to about 5 µm to cover the whole part. This trade-off decides every tomography experiment.

Engineering uses span the discipline: porosity, lack-of-fusion defects and unmelted powder in AM parts (Chapter 48) and castings (Chapter 46); fibre orientation, voids and damage accumulation in composites under load (Chapter 23); crack networks, aggregate distribution and steel-fibre orientation in concrete; trabecular bone and its integration with porous implants; particle cracking and lithium plating in batteries; corrosion pits in aluminium aircraft alloys. Fourth-generation sources acquire tomograms in under a second, fast enough to follow solidification, foaming or crack growth in four dimensions. **Diffraction-contrast tomography** reconstructs grain shapes and orientations in 3D in polycrystals, a non-destructive counterpart to

serial-sectioning EBSD. Laboratory micro-CT and nano-CT (to about 50 nm voxels with X-ray optics) now perform many of these measurements in-house, with longer scan times.

53.6 Neutron scattering

What makes neutrons different

Neutrons are uncharged and interact with nuclei through the short-range nuclear force, and with unpaired electrons through their magnetic moment. Four consequences follow.

Contrast does not follow atomic number. X-rays scatter from electrons, with amplitude proportional to Z (about $2.818Z$ fm at small angles), so hydrogen and lithium are nearly invisible next to iron or lead. The nuclear **coherent scattering length** b varies irregularly from element to element and even between isotopes:

Nucleus	H	D	Li (nat.)	C	O	Al	Ti	V	Fe	Ni	Zr
b (fm)	−3.74	6.67	−1.90	6.65	5.80	3.45	−3.44	−0.38	9.45	10.3	7.16
X-ray $f(0) = 2.818Z$ (fm)	2.8	2.8	8.5	16.9	22.5	36.6	62.0	64.8	73.3	78.9	112.7

> **Status of the principle: Neutron scattering lengths**
> *Empirical.* Scattering lengths depend on nuclear structure and resonances and are not predictable from simple theory; they are measured and tabulated for every element and isotope.

Light elements are therefore seen clearly: oxygen in oxide superconductors and fuel-cell electrolytes, lithium in battery materials, hydrogen in hydrides and metals, carbon and nitrogen in steels. Neighbouring elements (Fe and Mn, Ni and Co) that X-rays barely distinguish have very different b. The opposite signs of b for H and D are the basis of **contrast variation**: deuterating one block of a copolymer or one component of a blend makes it visible in small-angle neutron scattering (SANS), and adjusting the H/D ratio of a solvent can make one component invisible ("contrast matching"). Vanadium, with $b \approx 0$, is used for sample cans that do not diffract. Hydrogen's large *incoherent* cross-section (about 80 barn) gives a high background in diffraction but makes it the ideal target for vibrational and diffusion spectroscopy (below) and for neutron imaging.

Penetration. Most engineering metals absorb neutrons weakly; the 1/e attenuation length of thermal neutrons is roughly 8 mm in iron and nearly 10 cm in aluminium (with exceptions such as boron, cadmium and gadolinium, which are strong absorbers). Neutrons therefore probe the interior of real components (Figure 53.4).

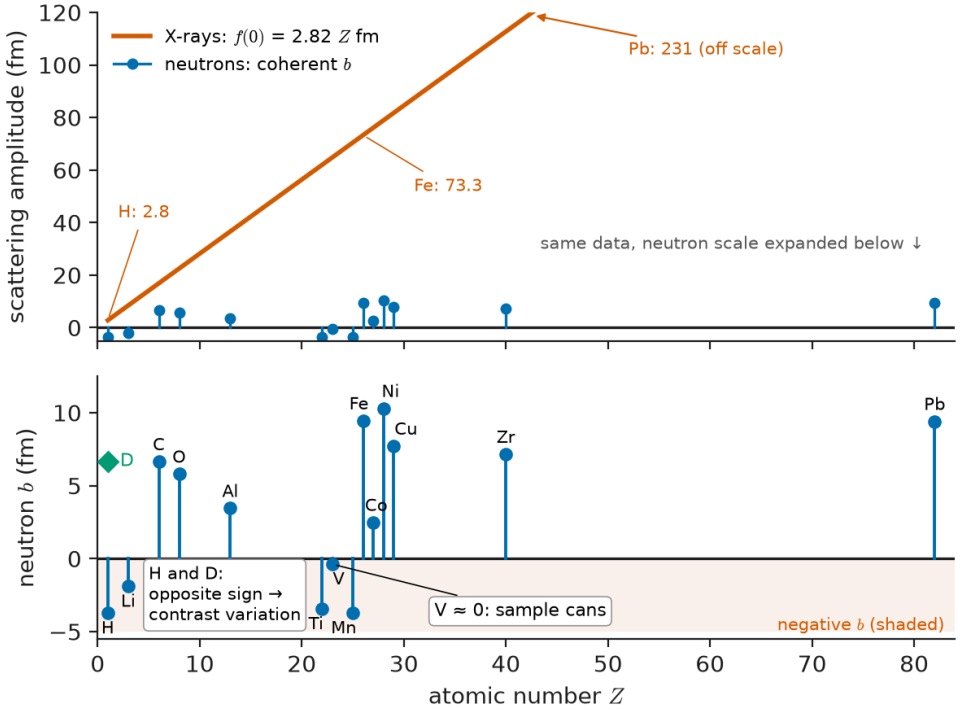


Figure 53.4 — Scattering power of elements for X-rays and neutrons, showing why neutrons see light elements and distinguish neighbours.

Magnetism. The neutron's magnetic moment scatters from ordered electron spins, so **magnetic structures** appear directly in diffraction. The historic example is MnO: below its Néel temperature of 118 K, extra peaks appear that index on a cell of twice the chemical lattice parameter (0.889 nm instead of 0.4445 nm), proving antiparallel ordering of Mn moments, the first direct observation of antiferromagnetism. Today neutron diffraction determines spin arrangements in permanent-magnet compounds, multiferroics and magnetocaloric alloys (Chapter 28), and polarized SANS measures magnetic nanostructure in electrical steels and nanocrystalline soft magnets.

Energies matched to atomic motions. A thermal neutron of wavelength 1.8 Å has energy $E = 81.81/\lambda^2$ meV $= 25$ meV and speed 2200 m/s. Its wavelength matches interatomic spacings and its energy matches phonons, so a single scattering event can exchange a measurable fraction of the neutron's energy with the sample. X-ray photons of 1.8 Å carry 6.9 keV, and a 25 meV change is a part in 3×10^5.

Residual stress deep inside components

Neutron diffraction measures lattice strain in a **gauge volume** defined by slits on the incident and diffracted beams (typically 1–4 mm on a side), at depths of several centimetres in steel and over 10 cm in aluminium. Strain follows from the shift of a peak, $\varepsilon = (d - d_0)/d_0 = -\cot\theta\,\Delta\theta$; with $2\theta = 90^\circ$ a cubic gauge volume is obtained, and a strain of 10^{-4} shifts the peak by only $\Delta\theta \approx 0.006^\circ$ (0.011° in 2θ). Measuring three orthogonal directions gives the triaxial stress through Hooke's law (Chapter 11):

$$\sigma_x = \frac{E}{(1+\nu)(1-2\nu)}\left[(1-\nu)\varepsilon_x + \nu(\varepsilon_y + \varepsilon_z)\right]$$

and cyclically for y and z.

> **Status of the principle: Stress from diffraction strains**
> *Derived.* Isotropic linear elasticity applied to measured lattice strains; requires the principal directions to be known (or six strains measured), plane-specific diffraction elastic constants for textured or anisotropic materials, and a reliable stress-free reference d_0 measured on a small cube or comb cut from the same material, because composition changes across a weld shift d_0 itself.

Worked example — residual stress in a butt weld. At the mid-thickness of a 25 mm ferritic-steel butt weld, Fe(211) strains measured with $\lambda = 0.165$ nm ($2\theta \approx 90^\circ$) relative to a stress-free comb are $\varepsilon_x = 1.2 \times 10^{-3}$ (longitudinal), $\varepsilon_y = 0.5 \times 10^{-3}$ (transverse) and $\varepsilon_z = -0.2 \times 10^{-3}$ (normal). With $E = 210$ GPa and $\nu = 0.3$, $E/[(1+\nu)(1-2\nu)] = 404$ GPa, and

$$\sigma_x = 404 \times [0.7 \times 1.2 + 0.3 \times 0.3] \times 10^{-3}\ \text{GPa} = 376\ \text{MPa}$$

with $\sigma_y = 262$ MPa and $\sigma_z = 149$ MPa. A longitudinal stress near the yield strength of the parent plate is typical of an as-welded joint and is exactly what fracture assessments such as BS 7910 and R6 must assume unless a post-weld heat treatment has relaxed it. Note that σ_z is far from zero: inside a thick section the stress state is triaxial, which the surface-only $\sin^2\psi$ method (Chapter 52) cannot reveal. With a typical strain uncertainty of 50×10^{-6}, stresses are known to about ±10–15 MPa. Neutron strain scanning is routinely applied to rail welds and wheels, turbine discs, pressure-vessel nozzles, aircraft wing-skin friction-stir welds and additively manufactured parts.

Inelastic and quasielastic scattering

Measuring the energy the neutron gains or loses maps **phonon dispersion curves** (triple-axis spectrometers on single crystals), which test interatomic potentials and DFT calculations (Chapters 55 and 56), and **magnon** dispersions in magnets. Vibrational spectroscopy with neutrons has no optical selection rules and is dominated by hydrogen, so it identifies hydrogen bonding sites in metal hydrides, zeolites and polymers. **Quasielastic scattering**, a small broadening of the elastic line, measures the jump frequencies and geometry of diffusing atoms: hydrogen in metals, lithium and sodium in solid electrolytes, water in cement and clays, and chain dynamics in polymers. **Neutron imaging** (radiography and tomography) sees hydrogen-rich materials inside metals: water distribution in an operating fuel cell, hydride blisters in zirconium cladding, oil inside engine parts, adhesive inside a bonded joint.

53.7 Atom-probe tomography

Atom-probe tomography (APT) removes atoms one at a time from a needle-shaped specimen and identifies each by time-of-flight mass spectrometry, then reconstructs their original positions in three dimensions. It is the only technique that combines near-atomic spatial resolution with chemical identification of every atom, including light elements, at the parts-per-million level.

Field evaporation and mass spectrometry

A specimen with a tip radius R of 20–100 nm, prepared by electropolishing or focused-ion-beam milling, is cooled to 20–80 K in ultra-high vacuum and held at a standing voltage V of a few kilovolts. The field at the apex is

$$F = \frac{V}{k_f R}$$

with k_f a geometric factor of about 3–8. Superimposed voltage pulses (for metals) or laser pulses (for semiconductors, oxides and other poor conductors) raise the field or temperature just enough to **field-evaporate** surface atoms as ions, one pulse in about a hundred producing an event. Evaporation fields are about 19 V/nm for aluminium, 33 V/nm for iron and 57 V/nm for tungsten.

Worked example — voltage and flight time. For iron with $k_f = 5$ and $R = 50$ nm, $V = F\,k_f R = 33 \times 10^{9} \times 5 \times 50 \times 10^{-9} = 8.3$ kV. As atoms are removed the tip blunts and the voltage is raised automatically, typically to 10–15 kV before the specimen fractures or the run ends. An ion of charge ne accelerated through V reaches speed $v = \sqrt{2neV/m}$, so for a flight path L the mass-to-charge ratio follows from the flight time:

$$\frac{m}{n} = \frac{2eV\,t^2}{L^2}$$

A $^{56}\text{Fe}^{2+}$ ion ($m/n = 28$ Da) at 6 kV over 0.10 m arrives after $t = L\sqrt{m/(2neV)} = 491$ ns; a timing precision of about 0.1 ns, plus correction for the energy spread of voltage-pulsed ions, gives the mass resolution needed to separate isotopes and overlapping peaks. A typical run collects 10^7–10^8 ions: a 50 × 50 × 200 nm^3 volume of steel contains 4.2×10^7 atoms, of which about half are detected at a detection efficiency of 50 %.

Reconstruction and its limits

The detector records the hit position and sequence of each ion. A projection model with an assumed tip shape converts hit positions to lateral coordinates and the cumulative evaporated volume to depth. Depth resolution is excellent (0.1–0.2 nm; atomic planes are often visible), lateral resolution 0.3–0.5 nm. The reconstruction is a model, however: phases with different

evaporation fields make the tip surface uneven, producing **local magnification** that distorts precipitates and smears interfaces; trajectory overlaps degrade positions near poles; molecular ions and overlapping mass peaks (for example $^{14}N^{+}$ and $^{28}Si^{2+}$, both at 14 Da) complicate composition; detection efficiency below 100 % and specimen fracture limit volumes and statistics.

> **Status of the principle: APT point-projection reconstruction**
> *Approximate*. Assumes a hemispherical tip of known radius evolving smoothly and a fixed image-compression factor; it is calibrated against known lattice spacings or correlative TEM, and fails near phases with strongly different evaporation fields.

What APT reveals

Analysis tools identify **clusters** (by maximum separation between solute atoms), plot **proximity histograms** of concentration against distance from an interface, and quantify segregation at boundaries in atoms per nm^2. Applications span the discipline (Figure 53.5): the Cu–Ni–Mn–Si clusters of irradiated pressure-vessel steels of the opening (a cluster of 1.5 nm radius in iron contains about 1200 atoms); carbon partitioning between martensite and retained austenite in quenched-and-partitioned and bainitic steels (Chapter 15); boron and carbon segregation to grain boundaries in nickel superalloys; nanometre $Al_3(Sc,Zr)$ precipitates in aluminium alloys; deuterium trapped at carbides in hydrogen-embrittlement studies of steels; dopant distributions in transistors and III–N quantum wells; and the chemistry of tooth enamel and bone mineral.

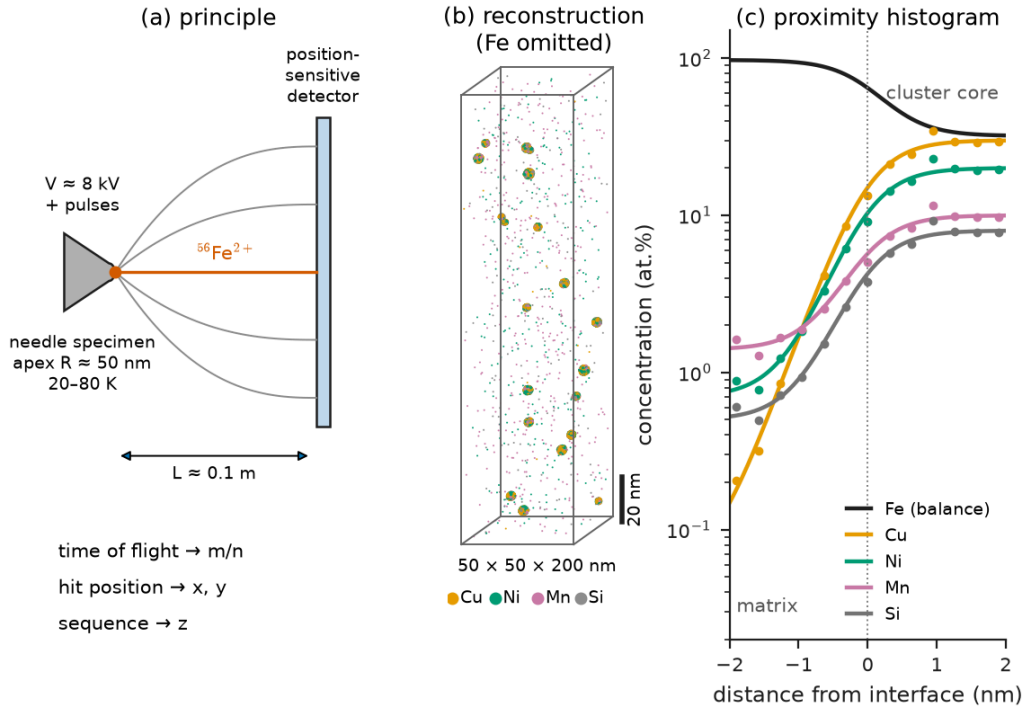


Figure 53.5 — Atom-probe tomography: instrument principle and a reconstruction of solute clusters in irradiated pressure-vessel steel with a proximity histogram.

53.8 Aberration-corrected STEM: HAADF, EELS and 4D-STEM

Correcting the lens

Magnetic electron lenses have unavoidable spherical aberration (Scherzer's theorem for round lenses), which limited TEM resolution to 0.15–0.2 nm for decades despite electron wavelengths of 2–3 pm. Multipole **aberration correctors**, practical since about 2000, cancel it. The probe can then use a larger convergence semi-angle α, and its diffraction-limited size $d \approx 0.61\lambda/\alpha$ shrinks.

Worked example — probe size. At 300 kV ($\lambda = 1.97$ pm) with a corrected convergence angle of 30 mrad, $d \approx 0.61 \times 1.97/0.030 = 40$ pm. An uncorrected 200 kV instrument ($\lambda = 2.51$ pm) is restricted to about 10 mrad by aberrations, giving 153 pm. The corrected probe is smaller than the separation of atomic columns in any crystal viewed along a low-index zone (for silicon along ⟨110⟩, the "dumbbell" columns are 136 pm apart), so individual columns are scanned one at a time. Practical resolution of 50–70 pm also depends on stability, source size and specimen quality.

HAADF, ABF and EELS

In **scanning TEM (STEM)** the probe is rastered across a thin specimen and different detectors collect different signals at each position (Figure 53.6). A **high-angle annular dark-field (HAADF)** detector collects electrons scattered beyond about 60–80 mrad, mostly by Rutherford-like scattering near nuclei; intensity scales roughly as $Z^{1.6–2}$, so heavy columns are bright and contrast is directly interpretable as **Z-contrast**. Single dopant atoms (Pt on a ceria catalyst support, Hf in a silicon channel, rare-earth atoms segregated to grain boundaries in silicon nitride) can be seen. **Annular bright-field (ABF)** imaging, from a ring inside the bright-field disc, shows light columns such as oxygen, nitrogen, lithium and even hydrogen alongside heavy ones, essential for oxide interfaces and battery cathodes.

> **Status of the principle: HAADF Z-contrast**
> *Empirical.* The exponent of Z depends on detector angles, accelerating voltage and thickness, and channelling along columns makes intensity non-linear in thickness; quantitative work compares with image simulations.

Electron energy-loss spectroscopy (EELS) analyses the energy lost by transmitted electrons. **Core-loss edges** identify elements, with particular sensitivity to light elements (B, C, N, O) where EDS is weak. Their fine structure (**ELNES**) carries bonding and oxidation state: the L_3/L_2 intensity ratio of Mn or Fe, the C K-edge shape distinguishing sp^2 graphite-like from sp^3 diamond-like bonding in DLC coatings (Chapter 49). Mapping across an interface column by column shows, for example, Ti^{3+} in the first unit cells of an oxide heterostructure or oxygen vacancies at a grain boundary in an electrolyte. **Low-loss** EELS measures plasmons, band gaps and specimen thickness through the log-ratio relation $t = \lambda_{in} \ln(I_t/I_0)$; if the total intensity is 1.5 times the zero-loss intensity and the inelastic mean free path is about 110 nm, $t = 110 \times \ln 1.5 = 45$ nm. Monochromated instruments with energy resolution below 10 meV detect vibrational modes and phonons from nanometre regions.

> **Status of the principle: EELS log-ratio thickness**
> *Approximate.* Assumes Poisson statistics of independent inelastic events and a known mean free path, which depends on composition, voltage and collection angle and is itself uncertain by 10–20 %.

4D-STEM

Fast **pixelated detectors** record the entire diffraction pattern (two dimensions) at every probe position (two more), giving a four-dimensional data set. From it, virtual detectors of any shape can be applied after the experiment, and many quantities can be mapped:

- **Strain maps** from the positions of diffracted discs, to about 10^{-4}, across transistors, precipitates and dislocation cores.
- **Orientation and phase maps** of nanocrystalline metals, beam-sensitive polymers and pharmaceutical crystals, at nanometre steps (like EBSD with a hundred times finer resolution).
- **Electric and magnetic fields** from the deflection of the central disc (differential phase contrast), showing p–n junctions, ferroelectric polarization and magnetic skyrmions.
- **Electron ptychography**, which computationally solves for the specimen potential from overlapping diffraction patterns and has reached resolutions below 0.05 nm, limited by thermal vibration of the atoms themselves, at lower doses than conventional imaging.

Worked example — the data problem. A 512 × 512 scan with a 256 × 256-pixel pattern stored at 2 bytes per pixel is $512^2 \times 256^2 \times 2 = 34$ GB. At 1 ms per pattern it takes 262 s, during which drift and beam damage accumulate; detectors running at 10 kHz or faster cut this to 26 s. Data volumes of this size have turned microscopy into a computational field, with automated analysis and machine-learning classification (Chapters 54 and 57).

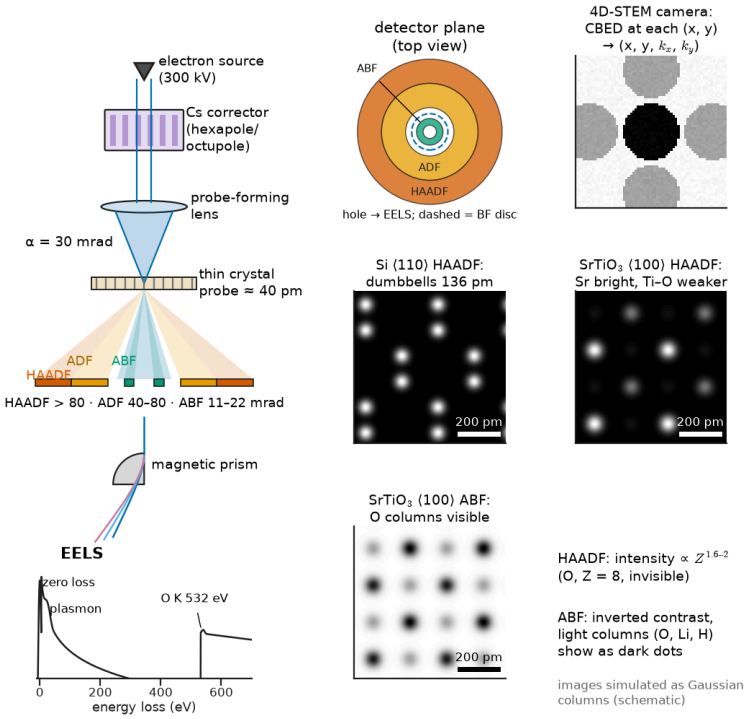


Figure 53.6 — Detector geometry in an aberration-corrected STEM: HAADF, ABF, EELS and pixelated 4D-STEM detection.

53.9 Cryo-EM for soft materials

Polymers, hydrogels, proteins, organic electronics, metal–organic frameworks and lithium metal are damaged by the electron beam far faster than metals or ceramics: radiolysis breaks bonds, mass is lost, and crystallinity disappears after doses a hundred to a thousand times smaller than a metal tolerates. Hydrated specimens cannot even enter the vacuum unless their water is immobilized. **Cryo-electron microscopy (cryo-EM)** addresses both problems.

Vitrification. A thin film of the aqueous specimen (a protein solution, a micelle dispersion, a hydrogel) is blotted on a grid and plunged into liquid ethane, cooling at 10^5 K/s or more, fast enough that water solidifies as amorphous (vitreous) ice instead of crystalline ice, whose crystals would destroy the structure and diffract strongly. The specimen is kept below about −140 °C during imaging. Vitrification preserves self-assembled structures exactly as they were in solution: block-copolymer micelles and vesicles, lipid nanoparticles for drug and mRNA delivery, the fibril network of a hydrogel scaffold, the adsorbed protein layer on an implant surface.

Low-dose imaging and averaging. Total doses are limited to roughly 20–60 electrons per Å^2 (40 e/Å^2 is about 640 C/m^2). Each image is therefore extremely noisy. Direct electron detectors, which count individual electrons with high efficiency at hundreds of frames per second, allow motion correction of the specimen drift within an exposure. In **single-particle analysis** tens to hundreds of thousands of images of identical molecules in random orientations are classified, aligned and averaged into a three-dimensional reconstruction; resolutions better than 2 Å are now achieved for proteins, which won the 2017 Nobel Prize in Chemistry. For materials engineering this means the structure of enzymes used in biocatalytic processing, of protein–surface complexes relevant to biomaterials (Chapter 39), and of viral capsids and protein cages used as nanoscale templates. **Cryo-electron tomography** reconstructs unique, non-repeating objects (a polymer–nanoparticle composite, a bacterial biofilm on a catheter) from tilt series, and **cryo-FIB** milling cuts thin lamellae from frozen bulk specimens.

Batteries and beam-sensitive solids. Cryo-EM moved into hard materials science around 2017 when it was used to image lithium metal dendrites and their solid–electrolyte interphase (SEI) at atomic resolution, structures that melt or decompose within seconds under a room-temperature beam. Cryogenic conditions slow radiolytic damage and mass loss, typically by a factor of a few, and stop reactions with residual gas. Similar protocols now image MOFs, perovskite solar absorbers, organic semiconductors and the hydration products of cement.

Boundary.
The image-formation and reconstruction mathematics of cryo-EM (contrast transfer function, Fourier-slice theorem, maximum-likelihood classification) belong to Engineering Physics and The Mathematics Universe; here only what the methods deliver and when they are needed is described.

53.10 Choosing among the advanced methods

Large-facility and atomic-resolution methods are expensive in time, preparation and access, so they are used when a question cannot be answered otherwise and are paired with the laboratory toolkit of Chapter 52 to establish representativeness.

Question	First-choice method	Probe volume	Complementary method
Oxidation state of a dilute element, in a working device	XANES	mm^3 (μm with a microbeam)	XPS (surface), EELS (nm)
Local structure without long-range order	EXAFS, PDF	mm^3	Neutron total scattering, ELNES
Size and fraction of 1–100 nm features in bulk	SAXS, SANS	mm^3	TEM, APT (local)
Composition of nm clusters or a grain boundary in 3D	APT	10^{-22}–10^{-21} m^3 (10^5–10^6 nm^3)	STEM-EDS/EELS
Atomic structure of an interface	Aberration-corrected STEM (HAADF, ABF)	nm^3 slice	DFT (Chapter 55)
Residual stress deep inside a part	Neutron diffraction, high-energy synchrotron XRD	mm^3 gauge	Contour method, hole drilling
Internal porosity and damage in 3D	X-ray tomography	mm^3–cm^3	Serial sectioning
Hydrogen, lithium, oxygen positions; magnetic order	Neutron diffraction	mm^3–cm^3	ABF-STEM, APT for H/D
Structure of soft or hydrated assemblies	Cryo-EM, SAXS/SANS	nm (EM), mm^3 (scattering)	Rheology, DSC

Two practical lessons recur. First, local and averaging methods answer different questions and are most powerful together: SANS gives the number density and mean size of the irradiation clusters in a pressure vessel, APT their composition, and TEM their interaction with dislocations; none alone predicts the embrittlement. Second, the most informative experiments increasingly follow a material while it is being processed, loaded or operated, which is the subject of the next chapter.

What remains open

Open question	Where it is answered
How are these methods applied while a sample is heated, strained, charged or reacting?	Chapter 54 (in-situ and operando studies)
How do positron annihilation, Mössbauer and solid-state NMR complement them for vacancies, iron chemistry and local environments?	Chapter 54
How are EXAFS, EELS and phonon spectra calculated to interpret measurements?	Chapter 55 (DFT)
How are APT and SANS cluster data used to parametrize kinetic Monte Carlo and cluster-dynamics models?	Chapter 56
How are terabyte data sets from 4D-STEM, tomography and beamlines analysed automatically?	Chapters 54 and 57
Where do Bragg diffraction, structure factors and reciprocal space come from?	Chapter 4 (back reference)
How do irradiation clusters embrittle steels, and how is this used in plant-life assessment?	Chapter 41 (back reference)

Next: Chapter 54 puts the specimen inside the instrument while it is heated, strained, biased or electrochemically cycled, and shows how in-situ, operando and correlative methods connect what is seen to how a material behaves.

Chapter 54 — In-Situ, Operando and Correlative Methods

Part XI · Characterization, Computation and Design

An electric-vehicle battery pack is returned after 1,200 fast-charge cycles with 22 % of its capacity gone. The cells are opened in a glovebox and the cathode is sectioned. The scanning electron microscope shows what everyone expected: the polycrystalline nickel-rich oxide particles are full of intergranular cracks, and electrolyte has crept into them. But the micrograph cannot say *when* the particles cracked. They could have cracked in the first formation cycle and done little harm afterwards; they could have cracked gradually, each crack exposing fresh surface that consumed lithium; or they could have cracked only during the drying and sectioning that prepared the sample for the microscope. Those three histories call for three different remedies — a new particle morphology, a surface coating, or a different sample-preparation protocol — and a post-mortem image cannot distinguish them. The same dilemma appears wherever materials change under load: a nickel superalloy whose dislocation structure at 900 °C relaxes on cooling, a catalyst that is metallic under reaction gas and oxidized by the time it reaches the microscope, a cement paste whose hydrates dehydrate in vacuum.

This chapter is about the methods that watch materials *while* they change, and about the methods that see what diffraction and imaging cannot: open volume, valence and local coordination. It ends with the problem these methods create — data volumes so large that analysis, not acquisition, becomes the bottleneck.

> **How can a material be observed while it is heated, strained, biased or reacting, without the observation itself changing the result?**
> **Which techniques detect vacancies, oxidation states and local atomic environments that conventional microscopy misses, and how are several techniques combined on the same region of the same sample?**

Concept	Engineering destination
In-situ TEM/SEM heating	Precipitation and grain growth in alloys, sintering of ceramics, recrystallization during annealing
In-situ straining	Dislocation sources, crack-tip plasticity, twinning in Mg and Ti, fracture of thin films and coatings
In-situ biasing	Electromigration in interconnects, resistive switching, ferroelectric domain motion
Micropillar compression	Single-phase properties for crystal-plasticity models, irradiated-steel testing, coating and phase strength
In-situ nanoindentation	Hardness and modulus of thin films, weld zones, individual grains and phases
Operando battery studies	Phase transitions, particle cracking, lithium plating, solid-electrolyte interphase growth
Operando catalysis	Active oxidation state of catalysts in reactors, fuel-cell and electrolyser electrodes
Positron annihilation	Vacancies in quenched and irradiated metals, free volume in polymers, voids in semiconductors
Mössbauer spectroscopy	Retained austenite in steel, iron corrosion products, Fe valence in cathodes and catalysts
Solid-state NMR	Silicate connectivity in cement and glass, Al sites in zeolites, Li environments in electrolytes
Correlative workflows	Linking crack initiation to microstructure, composition to structure in the same feature
Automated analysis	High-throughput EBSD, 4D-STEM, segmentation of tomograms, self-driving experiments

Roadmap. Why observing during change matters and what the observer does to the sample (54.1) → in-situ heating, straining and biasing in electron microscopes (54.2) → small-scale mechanical testing: micropillars and in-situ nanoindentation (54.3) → operando studies of batteries and catalysts (54.4) → positron annihilation for open-volume defects (54.5) → Mössbauer and solid-state NMR for local environment and valence (54.6) → correlating techniques on the same volume (54.7) → big data and automated analysis (54.8).

54.1 From post-mortem to operando

Three words, three levels of ambition

Conventional characterization (Chapters 52 and 53) is mostly **ex situ**: a sample is prepared, transferred to the instrument and examined in a state that has been frozen, cooled, dried or unloaded. Three more ambitious modes are distinguished:

- **In situ** ("in place"): the stimulus — temperature, stress, electric field, gas, liquid — is applied inside the instrument, and the material is observed while it acts. A TEM specimen heated on a chip in the microscope is in situ.
- **Operando** ("working"): the material is observed inside a functioning device under realistic operating conditions, *and* the device's performance is measured at the same time. A battery cycled in a synchrotron beam while its voltage and capacity are logged is operando; a catalyst bed whose product stream is analysed by mass spectrometry while X-ray absorption is measured is operando.
- **Correlative**: two or more techniques are applied to the *same* region of the *same* specimen, so that, for example, a crystal orientation from one method, a composition from another and a defect density from a third refer to the same grain.

The distinction between in situ and operando matters because many experiments that call themselves operando are not. A thin electrode in a specially built cell with a beryllium window, cycled at a slow rate, may not behave like the same electrode in a commercial pouch cell; the experiment is honest only if the electrochemical response of the special cell matches that of the real one.

Time and length resolution

Every in-situ experiment is a trade between spatial resolution, time resolution and field of view. Table 54.1 sets out representative numbers.

Method	Typical spatial resolution	Typical time resolution	Typical field of view
In-situ TEM imaging (direct electron camera)	0.1–1 nm	0.1–10 ms per frame	0.01–5 µm
In-situ SEM imaging and EBSD	2–50 nm (imaging), 20–100 nm (EBSD)	0.1–60 s per frame or map	1 µm – 1 mm
Synchrotron powder diffraction	averages over beam (µm–mm)	0.01–10 s per pattern	0.01–1 mm³
Synchrotron X-ray tomography	0.3–5 µm voxels	0.1 s – 10 min per volume	0.5–5 mm
Neutron diffraction (operando cell)	averages over cm³	1–30 min per pattern	whole cell
Positron lifetime (^{22}Na source)	averages over ~0.1 mm depth	hours per spectrum	mm²
Mössbauer spectroscopy	bulk average	hours to days per spectrum	cm² absorber

The table shows why no single method wins: the techniques that resolve individual dislocations see a few micrometres for a few seconds; the techniques that see a whole battery cell need minutes per measurement and average over billions of particles.

The observer effect

The probe is never neutral. Three effects recur.

Beam damage. A 200–300 keV electron can transfer enough energy to a nucleus to displace it (knock-on damage) when the maximum transferred energy exceeds the displacement threshold of 15–40 eV; for carbon in graphene this happens above roughly 80 keV, for aluminium above about 170 keV, for copper above about 400 keV. Electrons also ionize and break bonds (radiolysis), which is why polymers, hydrates, zeolites and organic electrolytes degrade within seconds under a focused beam, and why water in a liquid cell decomposes into hydrogen, hydroxyl radicals and hydrated electrons that change pH and can deposit or etch metal. Synchrotron X-rays, more gently, heat and can reduce transition-metal ions in battery electrodes over tens of minutes of exposure.

Geometry. A TEM foil 50–200 nm thick has two free surfaces within a few hundred atomic spacings of every point. Dislocations feel image forces that pull them out of the foil, vacancies escape to the surfaces, and the stress state in a thin foil under tension is nearly plane stress. A process observed in a foil is therefore a valid analogue of the bulk only when its characteristic length — a dislocation-source size, a diffusion distance, a precipitate spacing — is much smaller than the foil thickness.

Environment. Most electron microscopes operate at 10^{-5} Pa or better. Hydrates lose water, catalysts reduce, lithium reacts with residual gas. Environmental TEM (differentially pumped, up to about 20 mbar of gas at the sample) and windowed gas and liquid cells (up to about 1 bar with silicon-nitride membranes 20–50 nm thick) push this limit, at the cost of resolution lost to scattering in the gas, the liquid and the windows.

> **Status of the principle: Representativeness of in-situ experiments**
> *Approximate.* An in-situ observation predicts bulk or device behaviour only when the dose, geometry and environment perturb the process less than the stimulus being studied; this must be checked by dose series, thickness series and comparison with ex-situ or bulk measurements, not assumed.

54.2 In-situ heating, straining and biasing in TEM and SEM

Specimen holders and MEMS chips

The enabling technology of modern in-situ electron microscopy is the **MEMS chip**: a silicon die a few millimetres across carrying a suspended silicon-nitride membrane with a patterned metal or ceramic heater, electrodes, or a mechanical actuator. Because the heated volume is tiny (a membrane of a few hundred micrometres), a chip can reach 1,000–1,300 °C in milliseconds, cool equally fast, and drift by less than a nanometre per minute — a furnace-type holder that heats the whole specimen cup takes minutes to stabilize and drifts by tens of nanometres. Figure 54.1 shows the main holder types.

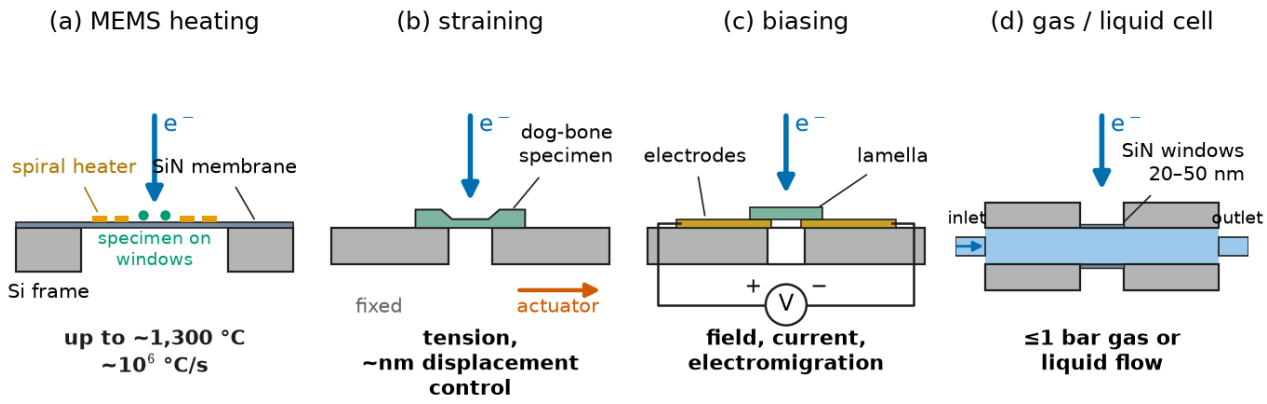


Figure 54.1 — Four families of in-situ TEM specimen holders and the stimulus each applies.

Heating experiments

Heating in the microscope addresses three classes of question.

Phase transformations and precipitation. Ageing of an Al–Cu–Mg or Al–Zn–Mg alloy can be followed from Guinier–Preston zones to semi-coherent precipitates; the dissolution of carbides in a tool steel and the reversion of martensite to austenite in a maraging steel can be watched at the scale of individual laths. The kinetics measured in a foil are faster than in bulk when surfaces act as vacancy sinks or as nucleation sites, so quantitative rates are normally checked against differential scanning calorimetry (Chapter 52) on bulk samples.

Grain growth, recrystallization and sintering. The migration of individual grain boundaries in nanocrystalline metals, the necking between ceramic powder particles during sintering (Chapter 46), and the coarsening of supported catalyst nanoparticles

can all be imaged directly. In-situ observation of sintering of zirconia and alumina nanopowders has shown, for example, that particle rotation and coalescence contribute to early-stage densification, a mechanism invisible to dilatometry.

Thermal stability of devices and coatings. Interdiffusion at metal–silicide contacts, crystallization of amorphous phase-change films, and the oxidation of thermal-barrier bond coats can be followed on cross-sectional lamellae.

The biggest practical uncertainty in heating experiments is the specimen temperature. The chip calibrates the heater, not the specimen; a lamella that is poorly attached, or a gas environment that cools the membrane, can sit tens of degrees below the nominal value. Good practice uses an internal standard — the melting of a small tin or aluminium particle, a known phase transition, or the thermal expansion of a reference lattice measured by electron diffraction.

Straining experiments

Straining holders apply tension, compression or bending to a thin specimen while it is imaged. Early holders pulled a 3 mm disc mechanically; modern push-to-pull MEMS devices convert the compression of an indenter tip into tension of a small dog-bone specimen and measure both force and displacement, giving a stress–strain curve synchronized with the video.

Such experiments delivered much of the direct evidence behind the dislocation theory of Chapters 12 and 13:

- the operation of Frank–Read and single-arm (spiral) sources, and the pile-up of dislocations at grain boundaries;
- the transmission, absorption or reflection of dislocations at grain and twin boundaries, which underlies the Hall–Petch effect and the high strength of nanotwinned copper;
- the screw-dislocation motion by kink-pair nucleation that makes body-centred cubic metals strongly temperature-sensitive;
- the nucleation and growth of deformation twins in magnesium and titanium alloys, important for the formability of lightweight automotive sheet and aerospace forgings;
- the crack-tip plasticity and the formation of dislocation-free zones ahead of cracks.

In-situ straining in the **SEM** works at a coarser scale with much larger specimens. Tensile stages carrying millimetre-scale specimens with loads up to a few kilonewtons can be combined with electron backscatter diffraction (EBSD), which maps lattice rotation, and with digital image correlation (DIC) of a fine surface speckle pattern, which maps strain at the micrometre scale. The combination identifies which grain orientations and which grain-boundary types concentrate strain first — the starting point for fatigue-crack initiation in titanium aerospace alloys, nickel superalloys and dual-phase automotive steels. Heated SEM stages extend this to creep and to the high-temperature deformation of superalloys.

Biasing experiments

Biasing chips carry electrodes that apply voltage and measure current while the specimen is imaged. Representative studies:

- **Electromigration** in copper and aluminium interconnects: void nucleation at the cathode end and hillock growth at the anode under current densities of order 10^{10}–10^{11} A m^{-2}, with void motion along grain boundaries and interfaces directly observed (Chapter 49).
- **Resistive switching** in oxide memory cells (HfO_2, TaO_x): formation and rupture of conducting filaments of oxygen-deficient oxide or metal.
- **Ferroelectric and ferroelastic domain switching** in thin-film $PbTiO_3$ and $BiFeO_3$, with domain walls pinned and released by defects.
- **Solid-state batteries** in miniature: lithium plating at a solid electrolyte interface imaged in a lamella cut from a full cell.

A recurring caution: a lamella carries current through a cross-section of perhaps 5 μm $\times$ 100 nm, and Joule heating in so small a conductor is removed efficiently, so electrothermal coupling can be weaker than in the real device.

> **Boundary.**
> The physics of electromigration, ferroelectric switching and resistive memory is treated in Chapters 27 and 50; here these are examples of what in-situ biasing can observe.

The liquid cell

Liquid-cell TEM encloses a liquid layer, typically 50 nm – 1 μm thick, between two silicon-nitride windows. It has imaged nanoparticle nucleation and growth from solution, electrochemical deposition and dissolution of lithium and zinc, corrosion pit initiation at inclusions in stainless steel (Chapter 43) and the biomineralization of calcium phosphates relevant to bone and dental implants (Chapter 39). Resolution degrades with liquid thickness (from a few angstroms in very thin layers to a few nanometres in micrometre layers), and radiolysis must be controlled by keeping the dose rate low and, where possible, quantifying its chemistry.

54.3 Micropillar compression and in-situ nanoindentation

Why test small volumes

Many engineering questions are about properties that cannot be measured on a conventional test bar because the material exists only in a small volume: a single grain of a coarse-grained casting, one phase of a two-phase titanium alloy, a 5 μm thermal-barrier coating, the heat-affected zone of a spot weld, the 100 μm surface layer of a reactor component damaged by ion irradiation. Small-scale mechanical testing measures those properties directly, and when performed inside an SEM or TEM it shows what the material does at the same time.

Nanoindentation and the Oliver–Pharr method

Instrumented indentation pushes a sharp diamond tip into a surface while recording load P and displacement h. The standard three-sided **Berkovich** pyramid has a projected contact area related to the contact depth h_c by

$$A = 24.5\, h_c^2$$

for an ideal tip; real tips are blunted and calibrated on fused silica with an area function. The **Oliver–Pharr method** treats the initial unloading as elastic. Its slope, the contact stiffness $S = dP/dh$, gives the reduced modulus

$$E_r = \frac{\sqrt{\pi}}{2\beta}\,\frac{S}{\sqrt{A}}, \qquad \frac{1}{E_r} = \frac{1-\nu^2}{E} + \frac{1-\nu_i^2}{E_i}$$

with $\beta \approx 1.034$ for a Berkovich tip and $E_i = 1141$ GPa, $\nu_i = 0.07$ for diamond. The contact depth comes from the maximum depth corrected for elastic sink-in, $h_c = h_{max} - \epsilon P_{max}/S$ with $\epsilon = 0.75$, and the hardness is

$$H = \frac{P_{max}}{A}$$

> **Status of the principle: Oliver–Pharr analysis**
> *Approximate.* Derived from elastic contact of a rigid axisymmetric punch on a half-space; accurate to a few percent for materials that sink in, but overestimates area-corrected hardness and modulus by up to 20–30 % in work-hardened metals that pile up around the indenter, and fails when the indent is deeper than about 10 % of a film's thickness so that the substrate contributes.

Worked example — nanoindentation of cold-worked copper. A Berkovich indentation into a cold-rolled copper busbar (illustrative values, typical of published measurements) reaches $P_{max} = 10.0$ mN at $h_{max} = 543$ nm. The unloading stiffness is $S = 0.360\ \text{mN nm}^{-1} = 3.60 \times 10^{5}\ \text{N m}^{-1}$. The contact depth is

$$h_c = 543 - 0.75 \times \frac{10.0}{0.360} = 543 - 20.8 = 522\ \text{nm}$$

The projected area is $A = 24.5 \times (522 \times 10^{-9})^2 = 6.68 \times 10^{-12}\ \text{m}^2 = 6.68\ \mu\text{m}^2$, and the hardness

$$H = \frac{10.0 \times 10^{-3}}{6.68 \times 10^{-12}} = 1.50 \times 10^{9}\ \text{Pa} = 1.50\ \text{GPa}$$

The reduced modulus is

$$E_r = \frac{\sqrt{\pi}}{2 \times 1.034} \times \frac{3.60 \times 10^{5}}{\sqrt{6.68 \times 10^{-12}}} = 0.857 \times 1.393 \times 10^{11} = 1.19 \times 10^{11}\ \text{Pa}$$

Removing the diamond compliance, $(1-0.07^2)/1141 = 8.72\times10^{-4}$ GPa^{-1} from $1/E_r = 8.37\times10^{-3}$ GPa^{-1} leaves 7.50×10^{-3} GPa^{-1}, so $E = (1-0.34^2)/7.50\times10^{-3} = 118$ GPa — close to the 110–130 GPa expected for polycrystalline copper. The hardness of 1.5 GPa corresponds, by the rule of thumb $H \approx 3\sigma_y$ (Chapter 11), to a flow stress near 500 MPa, typical of heavily cold-worked copper. If the indent showed significant pile-up in the SEM, both numbers would be revised downward.

In-situ indentation

Inside an SEM, the indenter can be positioned on a chosen grain or phase identified by EBSD, and the formation of slip traces, pile-up, cracks or delamination watched as the load rises. Inside a TEM, a nanoindenter pressed into the edge of a wedge-shaped specimen reveals the first dislocation nucleation event — the **pop-in** seen as a sudden displacement burst on the load–displacement curve — and the formation of cracks in hard coatings such as TiN, CrN and diamond-like carbon. The maximum shear stress beneath the tip at pop-in in a defect-free volume approaches $G/10$ to $G/20$, close to the theoretical shear strength, confirming that small enough volumes are strong because they contain no dislocation sources (Chapter 12).

Micropillar compression

A **micropillar** is a cylinder milled by a focused ion beam (FIB) from the surface of a bulk sample or from a single grain, typically 0.2–10 μm in diameter with a height-to-diameter ratio of 2–3. A flat-ended diamond punch compresses it inside a nanoindenter or an SEM. The engineering stress is load over top cross-sectional area, and the stress–strain curve can be recorded while slip steps form on the pillar surface (Figure 54.2a).

The first such experiments, on nickel and gold single crystals around 2004–2005, revealed a striking **size effect**: pillars of identical crystals and orientations grew stronger as they got smaller, and their stress–strain curves became staircases of discrete strain bursts rather than smooth curves. Across face-centred cubic metals the flow stress follows approximately

$$\sigma = A\,D^{-n}$$

with n typically 0.5–0.8 for FCC metals and smaller (0.2–0.5) for BCC metals at room temperature, where the intrinsic lattice friction of screw dislocations is already high. Two mechanisms explain it: **source truncation**, where Frank–Read sources in a small pillar are shorter (the pillar diameter limits their length, and the source-operation stress scales as $\mu b/L$), and **dislocation starvation**, where mobile dislocations leave through the free surface faster than they multiply.

> **Status of the principle: Power-law size effect in micropillars**
> *Empirical.* The exponent is fitted to data for each material; it applies to initially dislocated single crystals between roughly 0.1 and 20 μm, saturates towards bulk behaviour at large diameters, and depends on the initial dislocation density and on FIB damage.

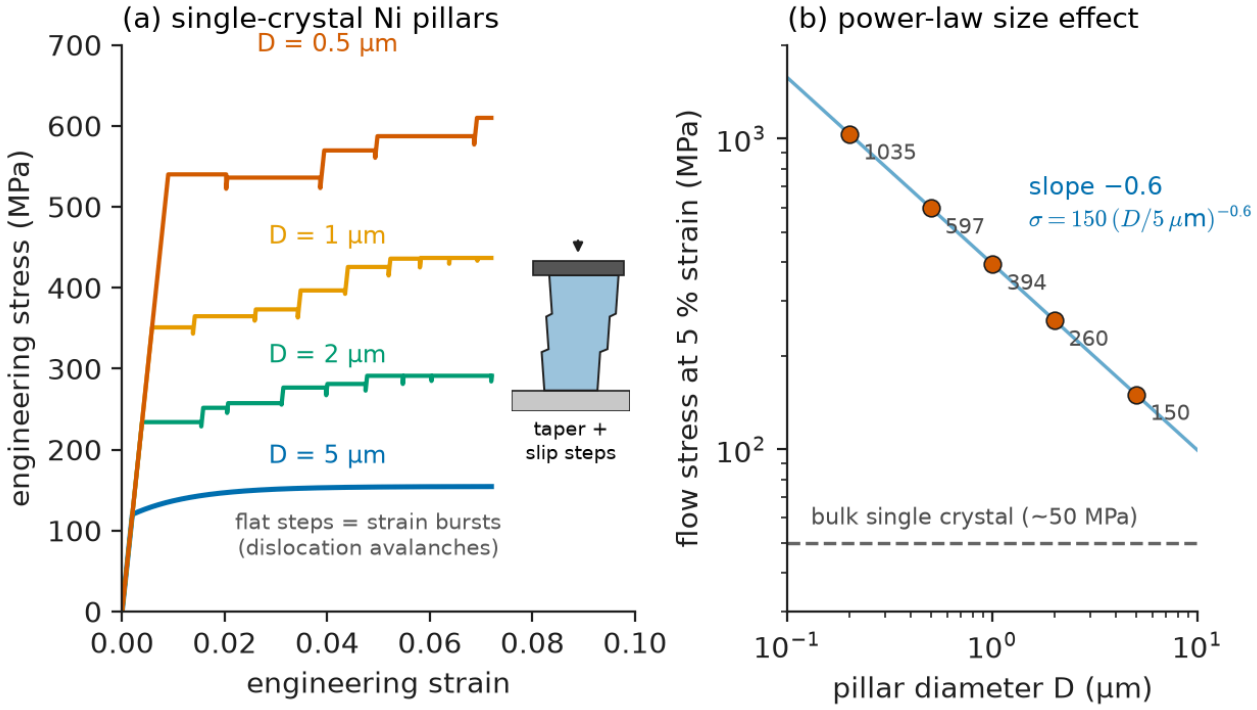


Figure 54.2 — Micropillar compression of single-crystal nickel: staircase stress–strain curves and the power-law size effect.

Worked example — designing a micropillar test. A 5 µm nickel pillar flows at 150 MPa at 5 % strain (an illustrative value, typical of published tests). Assume $n = 0.6$. The flow stress expected at other diameters is $\sigma(D) = 150\,(D/5\ \mu\mathrm{m})^{-0.6}$:

Diameter (µm)	5	2	1	0.5	0.2
Predicted flow stress (MPa)	150	260	394	597	1035

A 0.5 µm pillar at 597 MPa carries a load $F = \sigma\pi D^2/4 = 597\times10^6\times\pi\times(0.5\times10^{-6})^2/4 = 1.17\times10^{-4}$ N $= 117\ \mu$N, well within a nanoindenter's range (resolution near 0.1 µN) but small enough that a 1 µN offset in the zero-load reference would change the stress by about 1 %. The experimental design consequences follow from the numbers: (1) the punch must be flat and aligned to better than about 0.5° or the pillar bends; (2) the FIB-damaged surface layer, about 10–20 nm deep for 30 kV gallium ions, is 4–8 % of the radius of a 0.5 µm pillar and contributes measurably, so small pillars are finished at low ion energy or milled with a xenon plasma FIB; (3) a taper of 2–3° makes the top stress higher than the base stress by an amount comparable to the scatter. These are the reasons that small-scale strength data should always be quoted with diameter, taper and preparation route.

Engineering uses

Micropillars and microcantilevers have become a production tool in three areas. **Nuclear materials:** ion irradiation damages only the top 1–5 µm of a steel or tungsten sample, so the change of strength with dose can be measured only by small-scale tests, avoiding months of reactor exposure and the handling of activated specimens (Chapter 41). **Microstructure-informed modelling:** the critical resolved shear stresses of individual slip systems in α and β titanium, or of the γ and γ′ phases of a superalloy, are input parameters for the crystal-plasticity simulations of Chapter 56. **Coatings and interfaces:** notched microcantilevers milled across a coating–substrate interface give the fracture toughness of the interface, and micropillars with an inclined interface give its shear strength — the quantities that decide whether a thermal-barrier coating spalls or a hard coating on a cutting tool chips.

54.4 Operando battery and catalyst studies

Why batteries need operando methods

A lithium-ion cell is a closed, reactive system. Its active materials change phase with every cycle; its graphite anode can plate lithium metal at high charge rates and low temperatures; its electrolyte decomposes into a solid-electrolyte interphase (SEI) a few tens of nanometres thick; and nearly all of these states are metastable. Opening the cell relaxes concentration gradients, lets lithiated graphite react with air, and dissolves part of the SEI in washing solvent. Operando methods avoid this, and they deliver information that ex-situ work cannot: which phase is present at *which* state of charge, and how inhomogeneous the electrode is during fast charge (Chapter 40).

The toolkit

Operando X-ray diffraction in coin or pouch cells with X-ray-transparent windows (beryllium, Kapton, or simply the thin aluminium pouch at high synchrotron energies) records the evolution of phases and lattice parameters. With a laboratory diffractometer a pattern takes minutes; at a synchrotron, seconds or less. Typical observables:

- Graphite stages: lithium intercalates first into dilute stages, then forms LiC_{12} and finally LiC_6, the (002) interlayer spacing expanding from 3.35 Å to about 3.70 Å (Figure 54.3).
- Two-phase reactions such as $LiFePO_4 \leftrightarrow FePO_4$ (6.5 % volume change), where the phase fractions vary linearly with charge passed.
- Solid-solution reactions in layered oxides ($LiNi_xMn_yCo_zO_2$, "NMC"), where lattice parameters vary continuously. In nickel-rich NMC the *c* parameter expands and then collapses abruptly above about 4.1 V vs Li/Li^+, an anisotropic contraction of several percent in unit-cell volume that drives the intergranular cracking described in the opening of this chapter.

Operando X-ray absorption spectroscopy (XANES and EXAFS, Chapter 53) tracks the oxidation state of each transition metal separately — whether nickel, cobalt or manganese, or the oxygen sublattice, provides the charge compensation.

Operando tomography at micrometre resolution follows particle cracking, gas evolution, electrode swelling and lithium dendrite growth in three dimensions; in silicon anodes, which expand by about 280 % to $Li_{15}Si_4$, it shows how particles detach from the conductive network.

Operando neutron diffraction sees lithium (which scatters X-rays weakly) and probes whole commercial cells, because neutrons penetrate steel cans. **Operando ^{7}Li NMR** (Section 54.6) distinguishes lithium in the electrolyte, in intercalated graphite and in metallic deposits by their chemical or Knight shifts, so it quantifies lithium plating directly.

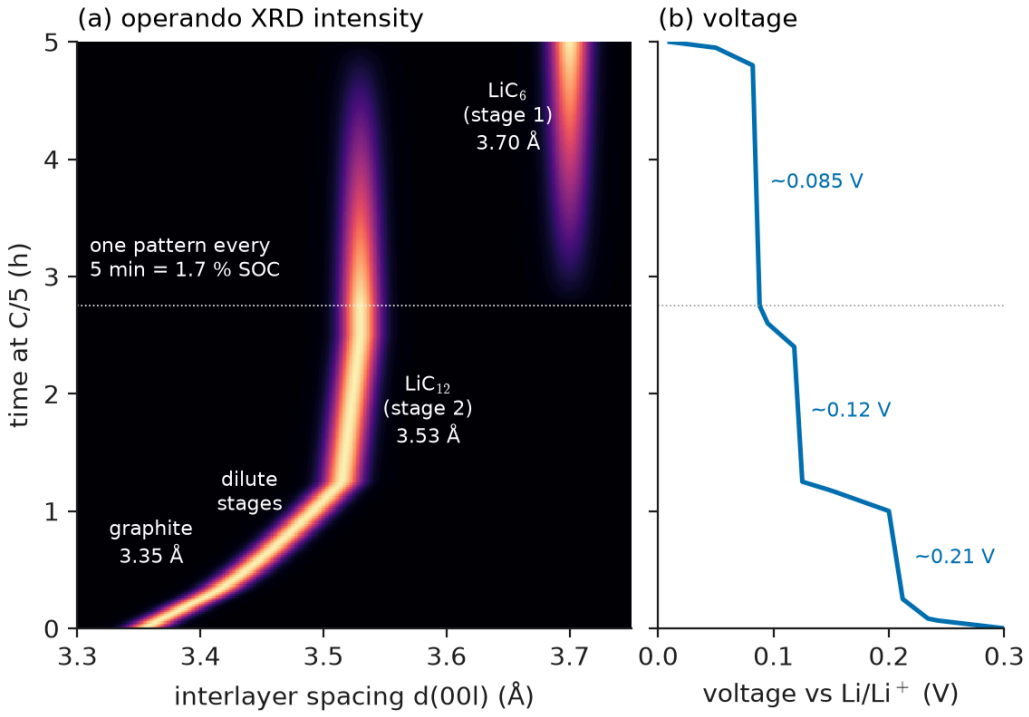


Figure 54.3 — Operando X-ray diffraction of a graphite anode during the first lithiation, showing staging transitions alongside the voltage profile.

Worked example — time resolution of an operando experiment. A pouch cell of 3.0 mAh capacity with a graphite anode is charged at C/5, so the current is $3.0/5 = 0.60$ mA and a full lithiation takes 5 h. A laboratory diffractometer needs 5 min per pattern. Each pattern therefore averages over a change of state of charge of

$$\frac{5\ \text{min}}{5 \times 60\ \text{min}} = 0.0167 = 1.7\ \%$$

and a full lithiation yields 60 patterns — enough to resolve the main staging transitions. Fast charging at 3C changes the picture: the full lithiation takes 20 min, so the same 5 min pattern would average over 25 % of the charge and blur the $LiC_{12} \rightarrow LiC_6$ transition entirely. To keep 1.7 % per pattern the acquisition must fall to 20 s, which demands a synchrotron. As a check on the capacity: the theoretical capacity of graphite, one lithium per six carbons, is

$$\frac{F}{3.6\,M_{C_6}} = \frac{96485}{3.6 \times 72.07} = 372\ \text{mAh g}^{-1}$$

so a 3.0 mAh anode needs only about 8 mg of graphite when fully used — illustrating how little material an operando cell contains and how carefully it must be built to represent a commercial electrode.

> **Status of the principle: Linear lattice-parameter–composition relations (Vegard's law) in operando analysis**
> *Empirical.* Converting a measured lattice parameter to lithium content assumes a calibrated, approximately linear relation within a single-phase field; it fails across phase transitions, in strained or defective particles, and where several phases overlap, so it must be calibrated with ex-situ reference compositions.

Operando catalysis

In heterogeneous catalysis the active state of a catalyst is often not the state that is loaded into the reactor. Copper–zinc oxide methanol-synthesis catalysts, platinum fuel-cell electrocatalysts and iridium oxide electrolyser anodes all reconstruct under reaction conditions, and their surfaces are covered by adsorbates that disappear on cooling. Operando catalysis studies therefore combine a structural or spectroscopic probe with simultaneous activity measurement — gas chromatography or mass spectrometry of the product stream, or the current of an electrochemical cell:

Probe	What it reveals	Typical conditions
XAS (XANES, EXAFS)	Oxidation state and coordination of the metal under reaction	Fixed-bed capillary reactors, up to tens of bar and 500 °C
Ambient-pressure XPS	Surface composition and oxidation states of the outer 1–3 nm	Up to about 20–100 mbar gas
Infrared (DRIFTS, ATR) and Raman	Adsorbed intermediates, surface oxides, coke	Reactor cells at full pressure
Environmental TEM	Particle shape, sintering, surface faceting	≤ 20 mbar (ETEM) or ≤ 1 bar (closed cell)
X-ray diffraction and PDF analysis	Bulk and nanoparticle phase, size	Capillary reactors

A recurring lesson is the **pressure gap**: surface-science studies at ultra-high vacuum often find surface states that are not present at industrial pressures, and the active phase identified under realistic conditions may be minority, transient or amorphous. Operando studies of platinum alloy cathode catalysts in fuel cells, for example, showed that the less noble metal leaches from the particle surface during potential cycling, leaving a platinum-rich shell whose strained lattice raises activity — and that the same leaching later drives particle coarsening and loss of surface area.

54.5 Positron annihilation spectroscopy

Positrons as defect probes

Diffraction is blind to isolated vacancies at concentrations below about 10^{-3}, and even atom-resolved microscopy sees them only in favourable cases. The **positron**, the antiparticle of the electron, is a uniquely sensitive vacancy detector. Positrons from a radioactive source (usually ^{22}Na) or a positron beam enter the sample, slow to thermal energies within a few picoseconds, diffuse about 100 nm, and annihilate with an electron, emitting two photons of 511 keV.

A thermalized positron is repelled by the positive ion cores. In a perfect lattice it occupies the interstitial region; at a vacancy, a vacancy cluster, a dislocation core or a void, where an ion core is missing, it sees an attractive potential well of about 1 eV and becomes **trapped**. Because the electron density at the open-volume site is lower, the trapped positron lives longer before annihilating. Three measurable quantities follow:

- **Positron lifetime**. The ^{22}Na decay emits a 1.27 MeV gamma ray almost simultaneously with the positron, which serves as a start signal; the 511 keV annihilation photon is the stop signal. The lifetime spectrum is a sum of exponentials, one per positron state.
- **Doppler broadening** of the 511 keV line, caused by the momentum of the annihilating electron. Annihilation with low-momentum valence electrons (typical of vacancies) gives a narrower line, measured by the shape parameter S; annihilation with high-momentum core electrons gives intensity in the wings, measured by W, which also carries a chemical fingerprint of the atoms surrounding the defect.
- **Angular correlation** of the two photons, a more detailed momentum measurement.

Table 54.2 gives representative lifetimes.

Material	Bulk lifetime (ps)	Monovacancy lifetime (ps)	Notes
Aluminium	160–165	240–250	quenched-in vacancies, Al alloy clustering
Copper	110–120	175–180	
α-iron	105–110	170–175	irradiation damage in reactor steels
Nickel	105–110	170–180	
Silicon	218–222	250–270 (divacancy ~300)	implantation damage
Large voids (> 1 nm)	—	400–500	vacancy clusters, swelling
Polymers (ortho-positronium)	—	1,000–4,000	free volume holes 0.2–0.5 nm radius

In polymers, porous silica and other insulators, the positron binds with an electron to form **positronium**. Its long-lived ortho state is localized in free-volume holes, and its lifetime is related to hole size by the Tao–Eldrup model; positron lifetime is a standard measurement of free volume across the glass transition of polymers (Chapter 20) and of pore sizes in low-dielectric-constant films and membranes.

The trapping model

The simplest analysis treats one defect type with trapping rate κ, a bulk annihilation rate $\lambda_b = 1/\tau_b$ and a defect annihilation rate $\lambda_v = 1/\tau_v$. The rate equations predict a two-component spectrum with

$$\tau_1 = \frac{1}{\lambda_b + \kappa}, \qquad \tau_2 = \tau_v, \qquad \kappa = \frac{I_2}{I_1}\left(\lambda_b - \lambda_v\right)$$

and the trapping rate is proportional to defect concentration, $\kappa = \mu c_v$, with a specific trapping coefficient μ of order 10^{14}–$10^{15}\ \mathrm{s^{-1}}$ for vacancies in metals. A consistency check is that the bulk lifetime recomputed from the components, $1/\tau_b = I_1/\tau_1 + I_2/\tau_2$, matches the known defect-free value.

> **Status of the principle: Two-state positron trapping model**
> *Approximate.* Derived from first-order rate equations assuming one defect type, no detrapping and spatially uniform defects; it holds for vacancies in metals at low temperature and moderate concentration, fails when several defect types or shallow traps (such as dislocations with thermal detrapping) are present, and saturates when nearly every positron is trapped (vacancy fractions above roughly 10^{-4}).

Worked example — vacancies in an irradiated iron alloy. A ferritic steel specimen irradiated with neutrons gives a positron lifetime spectrum (illustrative values, typical of published measurements) decomposed into $\tau_1 = 88$ ps with intensity $I_1 = 60\ \%$ and $\tau_2 = 175$ ps with $I_2 = 40\ \%$ (Figure 54.4). The second lifetime matches the iron monovacancy. Check consistency first: $1/\tau_b = 0.60/88 + 0.40/175 = 6.82 \times 10^{-3} + 2.29 \times 10^{-3} = 9.10 \times 10^{-3}\ \mathrm{ps^{-1}}$, so $\tau_b = 110$ ps, the value for defect-free iron — the one-defect model is consistent. The trapping rate is

$$\kappa = \frac{0.40}{0.60}\left(\frac{1}{110\ \mathrm{ps}} - \frac{1}{175\ \mathrm{ps}}\right) = 0.667 \times (9.09 - 5.71) \times 10^{9}\ \mathrm{s^{-1}} = 2.25 \times 10^{9}\ \mathrm{s^{-1}}$$

With $\mu \approx 1.1 \times 10^{15}\ \mathrm{s^{-1}}$, the vacancy-type defect fraction is $c_v = 2.25 \times 10^{9} / 1.1 \times 10^{15} = 2.0 \times 10^{-6}$, or two vacant sites per million — far below anything diffraction or electron microscopy could detect, yet enough to influence radiation-enhanced diffusion and embrittlement (Chapters 8 and 41). The fraction of positrons that annihilate in traps is $\kappa/(\lambda_b + \kappa) = 2.25/(9.09 + 2.25) = 0.20$, which shows the method is far from saturation; the mean lifetime $\bar{\tau} = I_1\tau_1 + I_2\tau_2 = 123$ ps is the robust number to monitor during post-irradiation annealing.

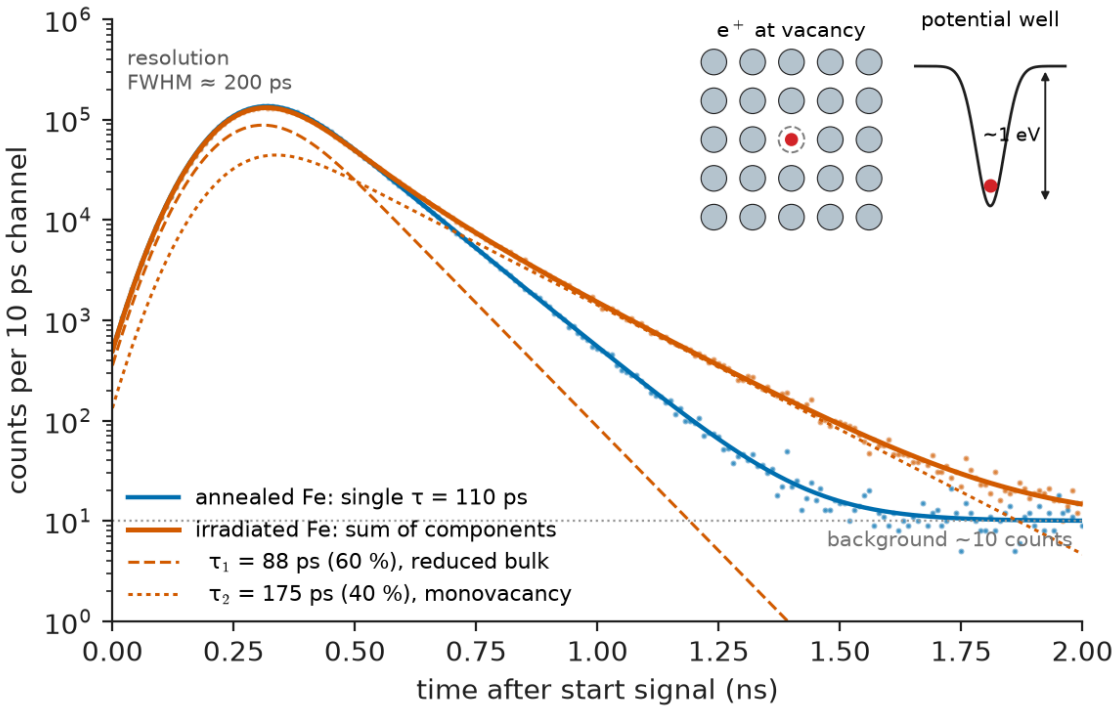


Figure 54.4 — Positron lifetime spectra of well-annealed and irradiated iron, with the two-component decomposition.

Variable-energy beams and uses

A radioactive source gives positrons with a broad energy spectrum that penetrate about 0.1 mm into metals, so it measures bulk averages. **Slow positron beams**, moderated and re-accelerated to 0.1–30 keV, implant positrons at controlled depths from a few nanometres to a few micrometres; depth-profiled Doppler broadening reveals ion-implantation damage in silicon, porosity in low-k dielectrics, and the defect layer beneath machined or shot-peened surfaces. Large-facility beams (reactor- or accelerator-based) give enough intensity for lifetime measurement with a beam.

Engineering applications include:

- tracking quenched-in vacancies and early solute clustering in age-hardening aluminium alloys, which controls natural ageing of automotive 6xxx sheet between stamping and paint bake;
- following radiation damage and its recovery in reactor-pressure-vessel steels and fusion materials;
- characterizing free volume, physical ageing and gas permeability in polymer membranes and packaging films;
- detecting vacancy-type defects in compound semiconductors and oxides (Ga vacancies in GaN, Zn vacancies in ZnO), which act as compensating acceptors.

54.6 Mössbauer spectroscopy and solid-state NMR

Both techniques probe the nucleus as a sensor of its immediate electronic and magnetic environment. Neither needs long-range order, so they work equally well on crystals, glasses, nanoparticles, corrosion products and hydrated cements — materials that defeat diffraction.

Mössbauer spectroscopy

Principle. A ^{57}Co source decays to an excited ^{57}Fe nucleus that emits a 14.4 keV gamma ray. In a solid, a fraction of these emissions occur with no recoil energy lost to the lattice (the **recoil-free fraction**, the Mössbauer effect), so the gamma ray carries the full transition energy with an extremely narrow natural width. An absorber containing ^{57}Fe (natural abundance 2.1 %) absorbs it resonantly only if its nuclear levels match. They are scanned by moving the source at a few millimetres per second, Doppler-shifting the gamma energy by

$$\Delta E = E_\gamma \frac{v}{c}$$

> **Status of the principle: Doppler energy shift**
> *Fundamental.* First-order Doppler shift of photon energy, exact to order v/c; at mm/s velocities the second-order term is negligible except as a small temperature-dependent shift.

For $E_\gamma = 14.4$ keV and $v = 1\ \mathrm{mm\,s^{-1}}$, $\Delta E = 14.4 \times 10^3 \times (10^{-3}/2.998 \times 10^8) = 4.80 \times 10^{-8}$ eV. The natural linewidth, set by the 141 ns mean lifetime of the excited state, is $\Gamma = \hbar/\tau = 6.58 \times 10^{-16}/1.41 \times 10^{-7} = 4.67 \times 10^{-9}$ eV, equivalent to 0.097 mm/s; the narrowest observable line (source plus absorber) is about 0.19 mm/s. An energy resolution of a few parts in 10^{13} is what lets the spectrometer read the tiny hyperfine interactions of the nucleus with its surroundings.

Hyperfine parameters. Three quantities are extracted:

- **Isomer shift** δ, from the s-electron density at the nucleus, which depends on oxidation and spin state: high-spin Fe^{3+} typically 0.3–0.5 mm/s, high-spin Fe^{2+} 0.9–1.3 mm/s (relative to α-Fe at room temperature).
- **Quadrupole splitting** ΔE_Q, a doublet from an electric-field gradient at a non-cubic site; large for high-spin Fe^{2+} (often 1.5–3 mm/s).
- **Magnetic hyperfine field** B_{hf}, splitting the line into a sextet in magnetically ordered phases: 33.0 T for α-Fe (outer lines at ±5.3 mm/s), about 51.7 T for hematite, about 49 and 46 T for the two magnetite sites at room temperature.

Relative areas of subspectra give phase fractions, after correction for differences in recoil-free fraction.

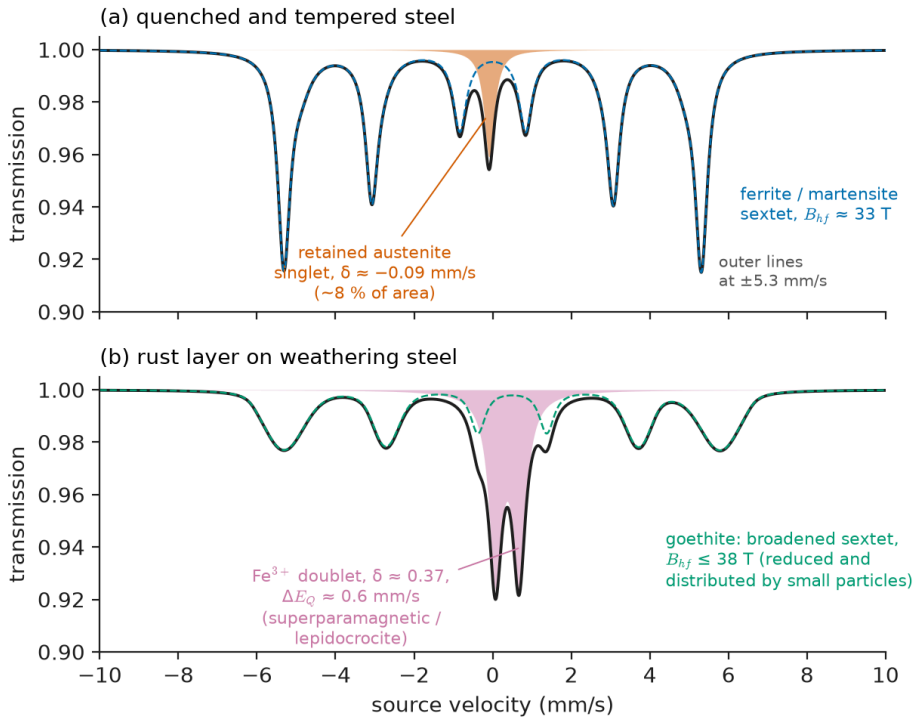


Figure 54.5 — Room-temperature ^{57}Fe Mössbauer spectra of a quenched steel and of an atmospheric corrosion product.

Engineering uses. Mössbauer spectroscopy measures **retained austenite** in quenched and carburized steels, which is paramagnetic and gives a single line distinct from the sextet of ferrite and martensite (Figure 54.5a); retained austenite in bearing and gear steels governs dimensional stability and contact fatigue (Chapter 15). It identifies the iron oxyhydroxides in corrosion products (goethite, lepidocrocite, akaganeite, magnetite), including nanocrystalline forms that give no X-ray diffraction pattern, which is how the protective patina on weathering steels was characterized (Chapter 43). It measures the Fe^{2+}/Fe^{3+} ratio in glasses (which sets their colour and redox state), in iron-phosphate battery cathodes during cycling, and in iron-containing catalysts and minerals. Other Mössbauer isotopes — ^{119}Sn for tin-based anodes and solders, ^{151}Eu, ^{121}Sb — extend the method to further systems. Its limitations are low count rates (hours to days per spectrum), the need for a suitable isotope, and bulk averaging over the absorber.

Solid-state NMR

Principle. Nuclei with spin (^{1}H, ^{7}Li, ^{13}C, ^{19}F, ^{23}Na, ^{27}Al, ^{29}Si, ^{31}P, ^{89}Y and many more) precess in a strong magnetic field (typically 9–23 T, proton frequencies of 400–1,000 MHz). Their exact resonance frequency, expressed as the **chemical shift** in parts per million, reflects the local electron distribution and therefore the coordination and bonding of the atom. In liquids rapid tumbling averages away anisotropic interactions and gives sharp lines; in solids, chemical-shift anisotropy, dipolar coupling and, for spins above 1/2, quadrupolar coupling broaden lines to kilohertz or more.

Magic-angle spinning (MAS). Spinning the sample rapidly about an axis inclined at $\theta_m = 54.74^\circ$ to the field, where $3\cos^2\theta_m - 1 = 0$, averages the first-order anisotropic interactions that share this angular dependence. Rotor speeds of 10–60 kHz are routine and above 100 kHz are possible for small rotors. Quadrupolar nuclei such as ^{27}Al and ^{23}Na retain a second-order broadening that requires higher fields or multiple-quantum techniques.

What it delivers. Solid-state NMR is quantitative (peak areas count nuclei, given full relaxation) and works on amorphous as well as crystalline solids. Characteristic uses:

Nucleus	Material	What the shift distinguishes
^{29}Si	Glasses, cement, zeolites	Q^n species (Si bonded through n bridging oxygens): Q^0 about −71 ppm, Q^2 about −85 ppm, Q^3 about −95 to −100 ppm, Q^4 about −105 to −115 ppm
^{27}Al	Zeolites, C–A–S–H, glasses	tetrahedral Al (about 50–80 ppm) vs octahedral Al (about 0–15 ppm)
^{7}Li, ^{6}Li	Electrodes, solid electrolytes	Li sites, Li-ion hopping rates; metallic Li (Knight shift about 260 ppm)
^{13}C, ^{1}H	Polymers, composites	tacticity, crystallinity, chain dynamics, cure of resins
^{31}P, ^{19}F	Bioglasses, phosphates, fluoropolymers	network connectivity, local order

In **cement** (Chapter 18), ^{29}Si MAS NMR follows the conversion of the Q^0 silicate monomers of clinker to the Q^1 and Q^2 chains of calcium silicate hydrate, giving the degree of hydration and the mean silicate chain length — a structure that is nanocrystalline and essentially invisible to diffraction. In **zeolite catalysts**, ^{27}Al NMR distinguishes framework from extra-framework aluminium after steaming, which controls acid-site density. In **glasses** for displays and bioactive implants, ^{29}Si and ^{31}P NMR quantify network connectivity, which controls dissolution rate (Chapter 39). In **battery materials**, ^{7}Li operando NMR quantifies metallic lithium deposits during fast charge, and relaxation measurements give lithium-ion hopping rates in solid electrolytes. Dynamic nuclear polarization (DNP) boosts sensitivity by orders of magnitude for surface sites, such as the active sites of supported catalysts.

The limitations are sensitivity (many interesting nuclei have low abundance or low gyromagnetic ratio), paramagnetic broadening in materials containing unpaired electrons (many cathode oxides), and the need for complementary modelling: assigning shifts in complex structures increasingly relies on the first-principles calculations of Chapter 55.

54.7 Correlative multimodal workflows

Why one technique is never enough

Each method answers a narrow question. EBSD gives crystal orientation but not composition below a few percent; atom-probe tomography (Chapter 53) gives composition with near-atomic resolution but poor crystallography; X-ray tomography sees a whole component but not its grain structure; TEM sees dislocations but in a foil whose location in the part is often unknown. The scientifically interesting events — the grain where a fatigue crack started, the inclusion where a pit nucleated, the particle that cracked in a battery — are rare, and statistics on one population and mechanism on another do not combine. Correlative workflows apply several techniques to the same feature, usually across scales.

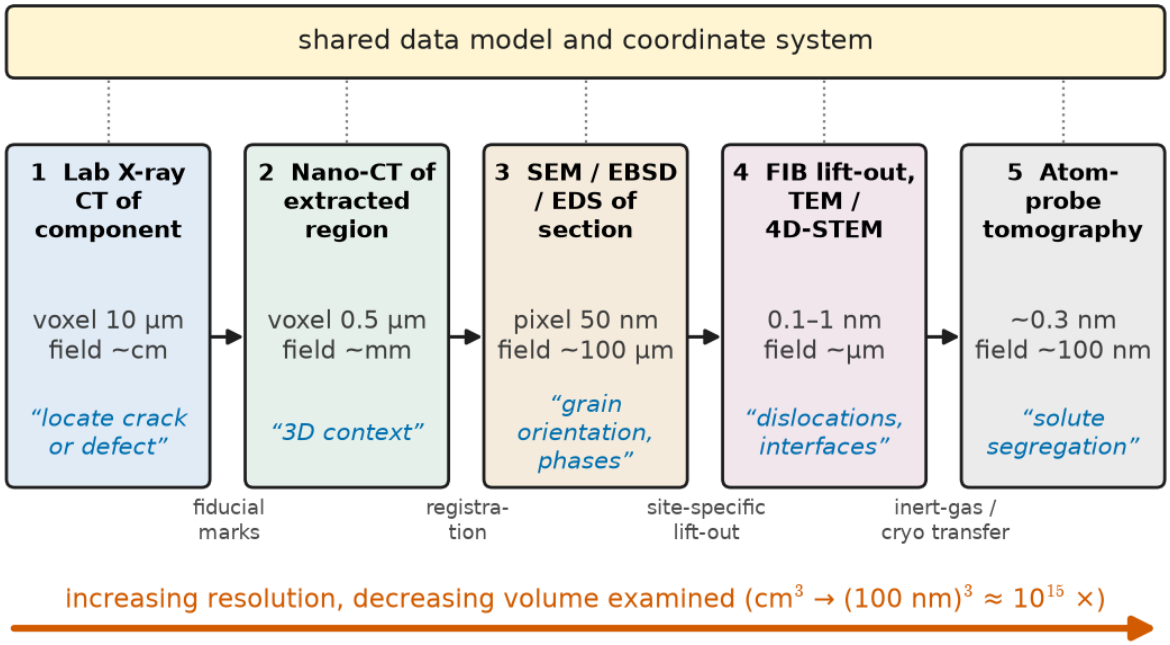


Figure 54.6 — A correlative multiscale workflow from a whole component to atomic resolution, with registration at each step.

Building blocks

Scale bridging. A typical chain (Figure 54.6) begins with X-ray computed tomography of a whole component or coupon, which locates a crack, pore or inclusion of interest. A section is cut through that location for SEM, EDS and EBSD. A focused-ion-beam lift-out then extracts a TEM lamella or an atom-probe needle from a chosen grain boundary or interface. The volume examined shrinks by about fifteen orders of magnitude from first to last step (from about 1 cm^3 to about $(100\ nm)^3$).

Registration. Data sets must be mapped into a common coordinate frame. This uses fiducial marks (FIB-milled crosses, deposited platinum markers, deliberate indents), features present in both data sets (pores, grain boundaries) and image-registration algorithms that compute the transformation — rigid, affine or non-linear to account for distortion in each instrument.

Transfer. Air-sensitive, hydrated or beam-sensitive samples — lithium metal, magnesium alloys exposed to hydrogen, hydrated tissue, cement pastes — need inert-gas or vacuum transfer shuttles, or cryogenic transfer that keeps them frozen between instruments.

Same-specimen, sequential stimulus. A powerful variant repeats observation after each increment of a stimulus: the same region of a dual-phase steel is mapped by EBSD and DIC after 0, 2, 5 and 10 % strain, or the same battery particle is imaged after each cycle.

Examples across engineering

- **Aerospace fatigue.** In titanium alloy Ti-6Al-4V, fatigue cracks initiate in "macrozones" of similarly oriented α grains. Correlating fractography, EBSD of the initiation site, and HR-EBSD or TEM of slip activity has shown that facets form on basal planes of grains loaded nearly perpendicular to their *c*-axis — a finding that changed forging-route specifications for jet-engine discs.
- **Additive manufacturing.** X-ray CT finds lack-of-fusion pores in a laser-powder-bed part; sectioning and EBSD show the columnar grains and melt-pool boundaries around them; fatigue testing of the same coupons links pore size and location to life (Chapter 48).
- **Corrosion.** Pit initiation sites on stainless steel are located by in-situ optical or electrochemical microscopy; SEM-EDS and TEM identify the manganese sulfide inclusions at their centres; atom-probe tomography reveals the chromium-depleted zone around the inclusion.

- **Biomaterials.** Correlative light and electron microscopy (CLEM) locates fluorescently labelled cells on an implant surface by light microscopy, then images the same cell–implant interface by FIB-SEM tomography, linking biological response to surface topography and chemistry (Chapter 39).
- **Batteries.** Operando X-ray tomography locates particles that crack early; the same electrode is then sectioned for EBSD and TEM of those particles, distinguishing cracks along grain boundaries of particular misorientation.

Pitfalls

Correlation multiplies the uncertainty of registration with the uncertainties of each technique. Preparation for one method may alter the feature before the next: FIB lift-out introduces gallium, ion milling heats, polishing removes the surface oxide that XPS should have seen. The order of techniques must therefore be planned from least damaging to most damaging, and every step documented so that an artefact introduced in step 3 is not mistaken for a material feature in step 5.

54.8 Big data and automated analysis

The data deluge

Modern detectors produce data faster than people can look at it. A **4D-STEM** experiment records a full two-dimensional diffraction pattern at each point of a two-dimensional scan (Chapter 53). With a 256×256 scan, a 256×256-pixel detector and 16-bit pixels, one data set is

$$256^4 \times 2 \text{ bytes} = 8.59 \times 10^9 \text{ bytes} \approx 8.6 \text{ GB}$$

At 10,000 frames per second the scan takes 6.6 s and the camera streams 1.3 GB/s. A series of in-situ heating experiments repeating such scans every minute fills tens of terabytes in a day. Synchrotron tomography beamlines produce volumes of 2048^3 voxels (about 17 GB at 16 bit) every second in fast mode. EBSD systems index more than 3,000 patterns per second. In-situ TEM video at 400 frames per second from a 4k camera overwhelms any manual analysis.

Worked example — 4D-STEM data budget. A group plans in-situ heating of a precipitation-hardening aluminium alloy, recording a 256×256 4D-STEM scan every 30 s for a 2-hour anneal at 180 °C. The number of scans is $7200/30 = 240$, and the total data is 240×8.59 GB $= 2.06 \times 10^{12}$ bytes ≈ 2.1 TB. Saving full patterns is possible but analysis on a workstation reading at 1 GB/s would take $2.06 \times 10^{12}/10^9 \approx 2{,}060$ s just to read once; any iterative fitting needs many passes. Two common decisions follow. Either reduce data at acquisition (compute strain maps or virtual images on the fly and keep only those, about 0.5 MB each instead of 8.6 GB), or bin the detector to 128×128, cutting the volume fourfold at some cost in angular resolution. The point is that the data plan must be designed with the experiment, not after it.

Automated analysis

Three families of tools make such data useful.

Physics-based processing pipelines. Drift correction, background subtraction, pattern indexing, strain mapping by template matching, Rietveld refinement of thousands of diffraction patterns in sequence, and principal-component or non-negative matrix factorization of spectral images. **Dictionary indexing** of EBSD patterns compares each measured pattern with a library of simulated ones, recovering orientations from noisy patterns of deformed or fine-grained material where conventional Hough-based indexing fails.

Machine-learning segmentation and classification. Convolutional neural networks (the U-Net architecture is the workhorse) segment grains, pores, cracks, phases and precipitates in micrographs and tomograms after being trained on a few dozen annotated images; they classify dislocation types, identify defects in atomic-resolution images, and flag the rare frames of an in-situ video where something happens. Chapter 57 develops these methods.

Autonomous and adaptive experiments. When analysis runs as fast as acquisition, the instrument can decide what to measure next: a microscope that scans at low resolution, identifies grain boundaries or interfaces, and returns for high-resolution spectroscopy only there; a beamline that adapts the temperature ramp of an operando experiment to where a phase transition appears; a robot that synthesizes, characterizes and reformulates samples in a closed loop. These are the "self-driving laboratories" of Chapter 57.

Put side by side, these rates show that nearly every modern in-situ method produces data faster than a person, and many faster than a single workstation, can process.

Reproducibility and data stewardship

Large automated data sets raise their own risks. Machine-learning segmentations can be confidently wrong on images unlike their training set; a pipeline with an undocumented threshold makes a result impossible to reproduce; and the selection of "representative" frames from terabytes of video can turn a rare event into an apparent rule. Good practice now includes:

- recording metadata (instrument, calibration, dose, temperature, timestamps) automatically with the data;
- following **FAIR** principles — data Findable, Accessible, Interoperable and Reusable — with open formats and persistent identifiers;
- publishing analysis code and trained models alongside results;
- reporting uncertainties, including those of the model (Chapter 57), and validating automated results against a manually analysed subset.

> **Status of the principle: Automated, data-driven analysis of characterization data**
> *Empirical.* Machine-learned segmentations and classifications are only as reliable as their training data and validation; they interpolate well within the conditions they were trained on and can fail silently outside them, so they need explicit uncertainty estimates and spot-checks against physics-based analysis.

What remains open

Open question	Where it is answered
How do the laboratory techniques (SEM, EBSD, TEM, XRD, DSC) used as the ex-situ baseline work?	Chapter 52
How do synchrotron, neutron, atom-probe and aberration-corrected STEM methods produce the data used operando?	Chapter 53
How are Mössbauer, NMR and positron parameters computed from first principles to assign spectra?	Chapter 55
How are micropillar and in-situ straining results simulated with molecular dynamics, dislocation dynamics and crystal plasticity?	Chapter 56
How are machine-learning models trained, validated and used for autonomous experiments?	Chapter 57
What dislocation mechanisms do in-situ straining experiments reveal, and how do they set strength?	Chapters 12 and 13
How do vacancies and their concentrations govern diffusion and ageing?	Chapters 5 and 8
How do the battery and catalyst materials studied operando work and degrade?	Chapter 40
How does irradiation damage form in reactor materials?	Chapter 41
How are corrosion products and passive films formed and identified?	Chapters 42 and 43
How are failures diagnosed in service by combining these methods?	Chapter 44

Next: Chapter 55 turns from measurement to calculation, showing how electronic-structure methods — density-functional theory above all — compute energies, structures and spectra from the Schrödinger equation, and where those calculations can and cannot be trusted.

Chapter 55 — Electronic-Structure Methods

Part XI · Characterization, Computation and Design

A team designing a new high-strength steel for automotive crash structures has a hydrogen problem. Above about 1,200 MPa tensile strength, martensitic steels absorb hydrogen from paint-shop pickling, galvanizing and road salt, and a part that passes every static test can crack days after assembly. One proposed remedy is to disperse fine vanadium or titanium carbides that trap hydrogen so tightly it cannot diffuse to crack tips. Which carbide traps most strongly, at which site — the carbide–ferrite interface, a carbon vacancy inside the carbide, or the misfit dislocations around it? Each candidate would take months to make and test, and the trapping sites are a few atoms across, beyond direct measurement. A chemist screening cathode oxides for sodium-ion batteries, a semiconductor engineer predicting which dopant will be a shallow donor in a wide-gap oxide, and a catalysis group asking why a platinum–nickel alloy outperforms platinum face the same situation: the decisive quantity is an energy difference at the atomic scale, and the cheapest instrument to measure it is a computer solving for the electrons.

This chapter explains how that is done. It is a practitioner's account of electronic-structure methods — chiefly density-functional theory (DFT) — written for materials engineers who will run, commission or read such calculations: what the approximations are, how numerical parameters are converged, which predictions can be trusted and which cannot.

> **How can the energy of an arrangement of atoms be calculated from quantum mechanics without solving an impossible many-electron problem?**
> **Which approximations make density-functional theory practical, and how does an engineer know when its predictions are good enough to act on?**

Concept	Engineering destination
Born–Oppenheimer approximation	Separating atomic motion (structures, vibrations, diffusion) from electronic structure
Hartree–Fock	Molecular chemistry, the exchange term in hybrid functionals
Density-functional theory	Formation energies, phase stability, defect energies, surface energies
Kohn–Sham equations	Band structures, densities of states, charge densities
Exchange–correlation functionals	Accuracy trade-offs: lattice constants, band gaps, reaction energies
DFT+U	Transition-metal oxides: battery cathodes, catalysts, magnetic oxides
Plane waves, pseudopotentials, PAW	Efficient and systematic calculations on periodic solids
k-point sampling	Reliable energies of metals and semiconductors
Known successes and failures	Deciding when DFT can guide design and when it cannot
Workflows and validation	Reproducible calculations, high-throughput screening, inputs to simulation and databases

Roadmap. The many-body problem and the Born–Oppenheimer separation (55.1) → Hartree–Fock and why density-functional theory replaced it for solids (55.2) → the Kohn–Sham equations (55.3) → exchange–correlation functionals from LDA to hybrids and DFT+U (55.4) → basis sets: plane waves, pseudopotentials and PAW (55.5) → Brillouin-zone sampling and convergence (55.6) → what DFT predicts well and badly (55.7) → practical workflows (55.8).

55.1 The many-body problem and the Born–Oppenheimer approximation

The equation everyone knows and no one solves

A piece of material is a collection of nuclei and electrons interacting by Coulomb forces. Its non-relativistic Hamiltonian contains the kinetic energy of every nucleus and every electron, the attraction of each electron to each nucleus, the repulsion between every pair of electrons and between every pair of nuclei. For a cubic micrometre of silicon that is roughly 5×10^{10} atoms and 7×10^{11} electrons; the wavefunction is a function of $3 \times 7 \times 10^{11}$ electronic coordinates. Even a 100-atom supercell of copper, with

2,900 electrons, has a wavefunction in 8,700 dimensions. Storing such a function on even a coarse grid of ten points per dimension would require 10^{8700} numbers. The exact equation is not solvable; every electronic-structure method is a strategy for approximating it.

> **Boundary.**
> The Schrödinger equation, the Pauli principle, spin and the band theory of solids are developed in the Engineering Physics book. This chapter states what it needs and concentrates on how the approximations are made and used in materials work.

Separating nuclei from electrons

A proton is 1,836 times heavier than an electron; nuclei therefore move far more slowly. In the time a nucleus vibrates once (about 10^{-13} s), the electrons adjust to its new position many times over. The **Born–Oppenheimer approximation** exploits this: the nuclear positions $\{\mathbf{R}_I\}$ are treated as fixed parameters, the electronic problem is solved for each configuration, and the resulting electronic ground-state energy $E_0(\{\mathbf{R}_I\})$, plus the nuclear–nuclear repulsion, becomes the **potential-energy surface** on which the nuclei move:

$$\hat{H}_{el}\,\Psi(\{\mathbf{r}_i\};\{\mathbf{R}_I\}) = E_0(\{\mathbf{R}_I\})\,\Psi(\{\mathbf{r}_i\};\{\mathbf{R}_I\})$$

Everything in this chapter and the next rests on this separation. Equilibrium crystal structures are minima of the potential-energy surface; elastic constants are its curvatures with respect to strain; phonon frequencies its curvatures with respect to atomic displacements; diffusion barriers the heights of saddle points between minima; and the interatomic potentials of molecular dynamics (Chapter 56) are fits to it.

> **Status of the principle: Born–Oppenheimer approximation**
> *Approximate.* Justified by the small electron-to-nuclear mass ratio; excellent for ground-state structure, elasticity, phonons and most diffusion, but it breaks down when electronic states cross or lie close in energy — in photochemistry, some radiation-damage events, electron–phonon effects in superconductors, and fast ion–solid collisions.

55.2 Hartree–Fock and why density-functional theory

The Hartree–Fock approach

The first systematic approximation, Hartree–Fock (HF), writes the many-electron wavefunction as a single Slater determinant of one-electron orbitals, a form that automatically satisfies the Pauli principle. Minimizing the energy gives one-electron equations in which each electron moves in the average electrostatic (Hartree) field of the others plus an **exchange** term that keeps electrons of like spin apart. Exchange is treated exactly; what is missing is **correlation** — the further tendency of electrons of any spin to avoid each other dynamically, beyond what a single determinant can represent.

For molecules HF is a good starting point and the foundation of the hierarchy of quantum-chemistry methods (Møller–Plesset perturbation theory, coupled cluster), which add correlation systematically. For solids it has three handicaps:

- **Missing correlation matters.** HF overestimates band gaps by factors of two or more (silicon about 6–8 eV in HF against 1.17 eV measured) and, for metals, predicts a vanishing density of states at the Fermi level — a qualitatively wrong answer.
- **Cost.** The exact exchange integral couples every orbital to every other, and HF costs scale as roughly N^4 with system size N; the correlated methods that fix it scale as N^5 to N^7. Coupled-cluster calculations on a 100-atom metallic supercell are not routine.
- **Metals and slow decay.** In a periodic metal, the exchange interaction is long-ranged and badly behaved without screening.

The density as the basic variable

The key idea that made accurate calculations on solids routine was to abandon the wavefunction as the central object. The **Hohenberg–Kohn theorems** (1964) state that, for electrons in an external potential (the potential of the nuclei):

1. The ground-state electron density $n(\mathbf{r})$ uniquely determines the external potential, and hence every property of the ground state.
2. There exists a universal energy functional $E[n]$ whose minimum over all physical densities is the exact ground-state energy, reached at the exact ground-state density.

> **Status of the principle: Hohenberg–Kohn theorems**
> *Fundamental.* Exact theorems of non-relativistic quantum mechanics for the ground state; they prove the functional exists but do not say what it is, so every practical DFT calculation introduces an approximation to it.

The density is a function of only three variables, whatever the number of electrons. The theorems replace an impossibly large problem with a tractable one, provided a good approximation to the functional can be found. That is the subject of the next two sections, and the reason that DFT became, by a large margin, the most widely used method in computational materials science.

55.3 The Kohn–Sham equations

A fictitious system of non-interacting electrons

The hardest part of $E[n]$ to approximate is the kinetic energy. Kohn and Sham (1965) sidestepped this by introducing an auxiliary system of non-interacting electrons constructed to have the same density as the real system. Their kinetic energy T_s can be computed exactly from one-electron orbitals ψ_i. The total energy functional becomes

$$E[n] = T_s[n] + \int V_{ext}(\mathbf{r})\, n(\mathbf{r})\, d\mathbf{r} + E_H[n] + E_{xc}[n]$$

where $E_H = \frac{e^2}{8\pi\varepsilon_0}\iint \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}\, d\mathbf{r}'$ is the classical Hartree electrostatic energy and E_{xc}, the **exchange–correlation energy**, collects everything else: exchange, correlation and the difference between the true and the non-interacting kinetic energy. Minimizing with respect to the orbitals gives the **Kohn–Sham equations**:

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 + V_{ext}(\mathbf{r}) + V_H(\mathbf{r}) + V_{xc}(\mathbf{r})\right]\psi_i(\mathbf{r}) = \varepsilon_i\,\psi_i(\mathbf{r})$$

$$n(\mathbf{r}) = \sum_i f_i\,|\psi_i(\mathbf{r})|^2, \qquad V_{xc}(\mathbf{r}) = \frac{\delta E_{xc}[n]}{\delta n(\mathbf{r})}$$

with occupations f_i (0 to 2 per orbital including spin). If the exact E_{xc} were known, Kohn–Sham DFT would give the exact ground-state energy and density.

> **Status of the principle: Kohn–Sham equations**
> *Derived.* An exact reformulation of the Hohenberg–Kohn variational principle; all the approximation lies in the choice of exchange–correlation functional, plus the numerical basis and sampling.

Self-consistency

The effective potential depends on the density, which depends on the orbitals, which depend on the potential. The equations are therefore solved iteratively (Figure 55.1): guess a density (usually a superposition of atomic densities), build the potential, solve for orbitals, form a new density, mix it with the old one to damp oscillations, and repeat until input and output densities agree — typically to an energy change below 10^{-6} eV per cell. Charge "sloshing" between distant regions makes this slow for large metallic cells and surfaces, which is why density-mixing schemes (Pulay/DIIS, Broyden, Kerker preconditioning) matter in practice.

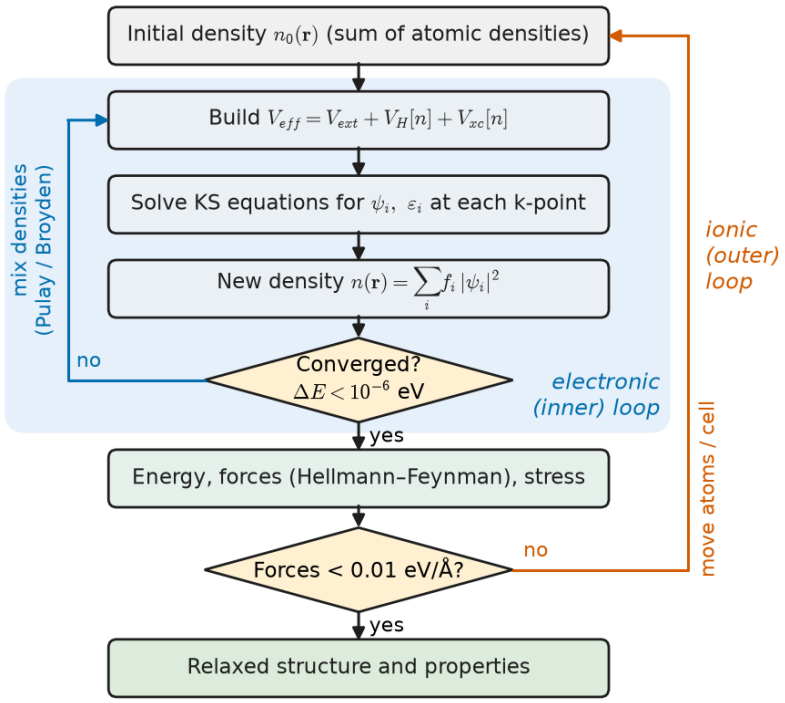


Figure 55.1 — The Kohn–Sham self-consistent field cycle, with the outer loop for relaxing atomic positions.

What the Kohn–Sham quantities mean

The total energy and the density are, in principle, exact properties. The Kohn–Sham eigenvalues ε_i are not, strictly, electron addition or removal energies; only the highest occupied eigenvalue has a rigorous meaning (it equals minus the ionization energy for the exact functional). In practice Kohn–Sham band structures reproduce band shapes, effective masses and Fermi surfaces of most metals and semiconductors well, but systematically underestimate band gaps — part of the gap comes from a discontinuity in the exact exchange–correlation potential that approximate functionals lack (Section 55.7).

Forces on nuclei follow cheaply from the converged density by the **Hellmann–Feynman theorem**, $\mathbf{F}_I = -\partial E / \partial \mathbf{R}_I$, so atomic positions and cell shape can be relaxed, and ab-initio molecular dynamics (Born–Oppenheimer MD) can be run on the computed potential-energy surface.

55.4 Exchange–correlation functionals

The local density approximation

The simplest approximation, the **local density approximation (LDA)**, assumes that each small volume contributes the exchange–correlation energy per electron of a uniform electron gas of the same density:

$$E_{xc}^{LDA}[n] = \int n(\mathbf{r}) \, \varepsilon_{xc}^{hom}(n(\mathbf{r})) \, d\mathbf{r}$$

The exchange part of ε_{xc}^{hom} is known analytically ($\propto n^{1/3}$); the correlation part was obtained from quantum Monte Carlo calculations of the electron gas and fitted. LDA works surprisingly well for solids with slowly varying densities — simple metals, covalent semiconductors — because of a fortunate cancellation of errors between exchange and correlation. Its systematic tendency is **overbinding**: lattice constants 1–3 % too small, bulk moduli 5–20 % too large, cohesive energies 10–30 % too large.

Generalized-gradient approximations

Generalized-gradient approximations (GGAs) add a dependence on the density gradient, $\varepsilon_{xc}(n, \nabla n)$. The most widely used in materials science is **PBE** (Perdew–Burke–Ernzerhof, 1996), built to satisfy exact constraints rather than fitted to data. PBE corrects much of LDA's overbinding, often slightly overshooting: lattice constants typically 0.5–2 % too large. It famously

gets the ground state of iron right — ferromagnetic BCC — where LDA wrongly predicts non-magnetic close-packed iron to be more stable. **PBEsol** re-tunes the gradient correction for solids and gives better lattice constants and surface energies than PBE at some cost in molecular binding energies.

Meta-GGAs add the kinetic-energy density of the orbitals; **SCAN** (2015) and its numerically better-behaved variant r²SCAN satisfy all 17 known exact constraints a meta-GGA can obey and improve formation enthalpies of compounds and the description of intermediate-range bonding, at about two to five times the cost of PBE.

Hybrid functionals

Hybrid functionals mix a fraction of exact (Hartree–Fock) exchange into a GGA:

$$E_{xc}^{hyb} = a\,E_x^{HF} + (1-a)\,E_x^{GGA} + E_c^{GGA}$$

PBE0 uses $a = 0.25$. **HSE06** applies the HF exchange only at short range (screening length about 5 Å), which makes it far cheaper for solids and more physical for metals. Hybrids correct much of the band-gap error of semi-local functionals and improve the localization of electrons and holes at defects and polarons. Their price is 10–100 times the cost of a GGA calculation for the same system.

DFT+U for localized electrons

In transition-metal oxides and rare-earth compounds, partially filled $3d$ or $4f$ shells hold electrons in localized orbitals. Semi-local functionals spread them out, because each electron spuriously interacts with its own density (**self-interaction error**). The result: NiO, CoO and FeO are predicted to be metals or nearly so, although they are insulators with gaps of 2–4 eV, and redox energies of battery cathodes are wrong by up to 1 V. **DFT+U** adds a Hubbard-like penalty on fractional occupation of the localized orbitals; in the widely used simplified (Dudarev) form,

$$E_{DFT+U} = E_{DFT} + \frac{U_{eff}}{2}\sum_{\sigma} \mathrm{Tr}\left[\boldsymbol{\rho}^{\sigma} - \boldsymbol{\rho}^{\sigma}\boldsymbol{\rho}^{\sigma}\right], \qquad U_{eff} = U - J$$

where $\boldsymbol{\rho}^{\sigma}$ is the occupation matrix of the localized orbitals for spin σ. The penalty vanishes for integer occupations (0 or 1) and is maximal at half-filling, so it drives orbitals towards localized, integer-occupied states. Typical U_{eff} values for $3d$ oxides lie between 3 and 7 eV.

> **Status of the principle: DFT+U correction**
> *Empirical.* U_{eff} is usually chosen to reproduce measured gaps, lattice parameters or oxidation enthalpies (or computed by linear response); results depend on the chosen value and on the projector, and energies computed with different U values cannot be combined without correction schemes.

Dispersion corrections

None of the semi-local functionals describes van der Waals (dispersion) attraction, which arises from correlated charge fluctuations between distant regions. Layered materials (graphite, MoS_2, clays), molecular crystals, polymers and the adsorption of molecules on surfaces need **dispersion corrections**: pairwise C_6/r^6 terms (DFT-D3, Tkatchenko–Scheffler, many-body dispersion) or non-local correlation functionals (vdW-DF, rVV10). Without them PBE predicts graphite's interlayer spacing far too large and its interlayer binding nearly zero.

Jacob's ladder

Perdew's metaphor of **Jacob's ladder** arranges functionals in rungs of increasing ingredients and cost (Figure 55.2): LDA (density), GGA (gradient), meta-GGA (kinetic-energy density), hybrids (occupied orbitals through exact exchange), and the fifth rung of the random-phase approximation and double hybrids (unoccupied orbitals). Climbing usually, but not always, improves accuracy; the choice is driven by the property needed and the budget available.

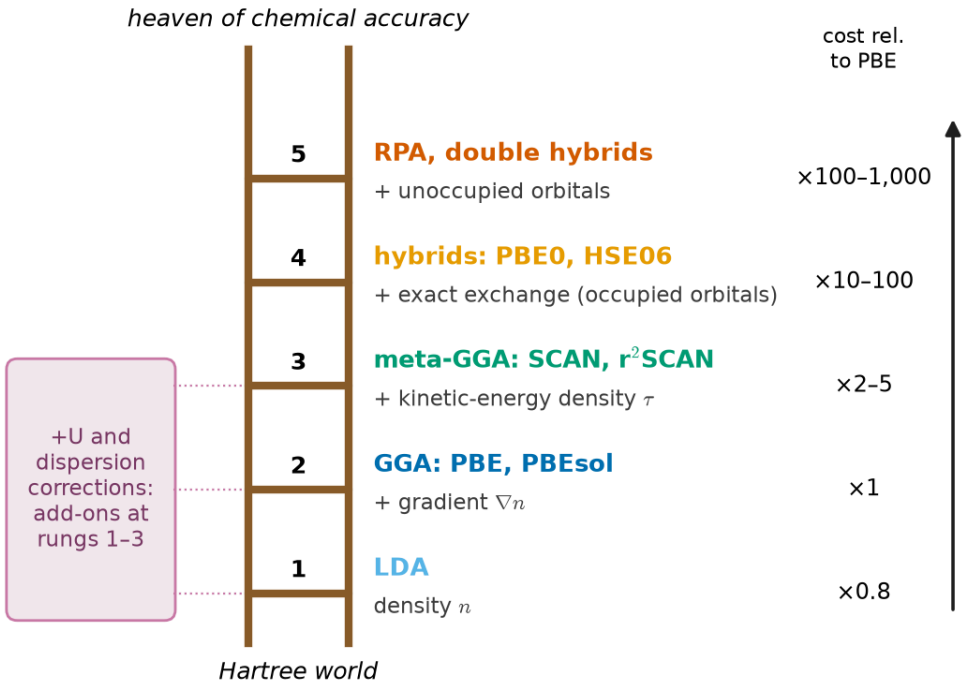


Figure 55.2 — Jacob's ladder of exchange–correlation functionals, from the Hartree world to chemical accuracy.

55.5 Basis sets: plane waves, pseudopotentials and PAW

Plane waves and Bloch's theorem

To solve the Kohn–Sham equations numerically, orbitals are expanded in a basis. In a periodic crystal Bloch's theorem allows each orbital to be labelled by a wavevector $\mathbf{k}$ in the first Brillouin zone (Chapter 4) and written as a periodic function times a phase. The natural basis is then **plane waves**:

$$\psi_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}} c_{n,\mathbf{k}+\mathbf{G}}\, e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}$$

summed over reciprocal-lattice vectors $\mathbf{G}$. The expansion is truncated at a **kinetic-energy cutoff**,

$$\frac{\hbar^2 |\mathbf{k}+\mathbf{G}|^2}{2m_e} \le E_{cut}$$

which keeps all plane waves inside a sphere of radius $k_{max} = \sqrt{2m_e E_{cut}}/\hbar$. Since each plane wave occupies a reciprocal-space volume $(2\pi)^3/V$ for a cell of volume V, the number of plane waves is approximately

$$N_{pw} \approx \frac{V\, k_{max}^3}{6\pi^2}$$

Plane waves have three major advantages: the basis is unbiased (it does not presuppose where the atoms are), its completeness is controlled by a single number, and forces and stresses are free of the "Pulay" errors that plague atom-centred bases. Their disadvantages: they describe vacuum (surfaces, molecules) as expensively as matter, and they are hopeless for the rapidly oscillating wavefunctions near atomic nuclei.

Pseudopotentials

Core electrons (1s in carbon; 1s–2p in silicon; up to 3p in iron) are chemically inert in most materials problems, but the valence wavefunctions oscillate rapidly in the core region to remain orthogonal to them, requiring enormous cutoffs. A **pseudopotential** replaces the nucleus and core electrons by a smooth effective potential acting on smooth pseudo-wavefunctions that coincide with the true valence wavefunctions outside a core radius r_c (Figure 55.3):

- **Norm-conserving** pseudopotentials preserve the charge inside r_c; they are accurate and simple but need high cutoffs for $2p$, $3d$ and $4f$ elements.
- **Ultrasoft** pseudopotentials (Vanderbilt) relax norm conservation and add augmentation charges, allowing much lower cutoffs.

The choice of which electrons to put in the core matters: for transition metals, including the semicore $3p$ states as valence improves results under pressure and in oxides; for lithium and sodium, including the $1s$ or $2p$ electrons is often needed.

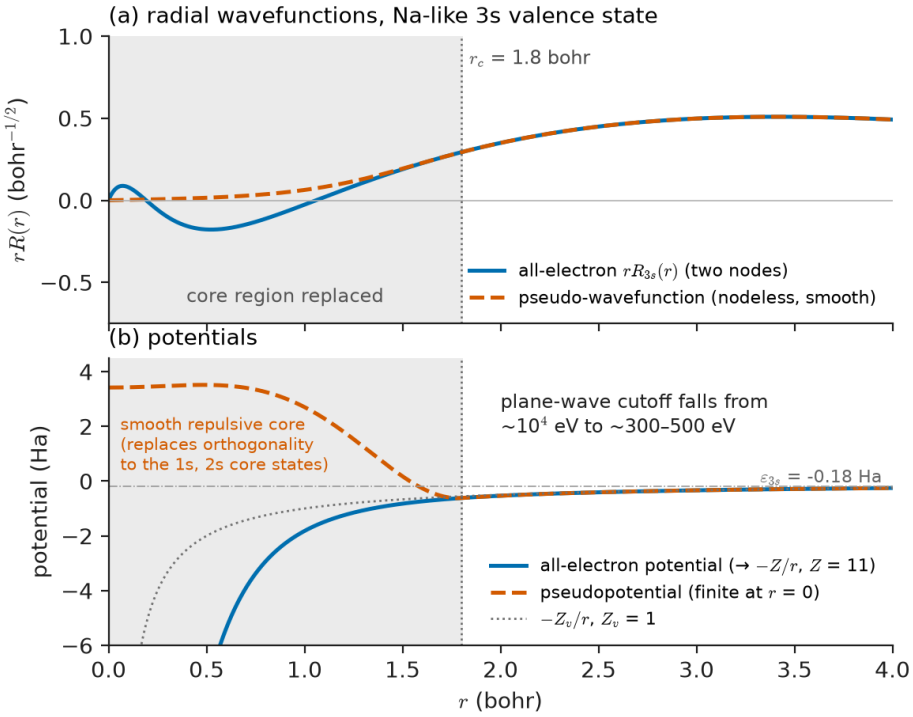


Figure 55.3 — All-electron and pseudo radial wavefunctions and potentials for a valence state; they coincide beyond the core radius rc.

The projector augmented-wave method

The **projector augmented-wave (PAW)** method of Blöchl (1994), used by VASP, GPAW, ABINIT and Quantum ESPRESSO among others, keeps the efficiency of ultrasoft pseudopotentials but restores the all-electron wavefunction through a linear transformation: smooth pseudo-wavefunctions are expanded in plane waves, and inside atom-centred spheres they are corrected using stored atomic partial waves. PAW gives access to the full valence density near the nucleus, which is needed for hyperfine parameters, NMR chemical shifts, Mössbauer isomer shifts and electric-field gradients (Chapter 54) and core-level spectra (Chapter 53), with the frozen-core approximation as the main residual error.

Other bases

All-electron methods — the linearized augmented plane-wave (LAPW) method (WIEN2k, exciting) and atom-centred numerical orbitals (FHI-aims) — remain the accuracy references against which pseudopotential codes are tested. **Localized bases** — Gaussian-type orbitals (CRYSTAL, CP2K's mixed Gaussian and plane-wave scheme) and numerical atomic orbitals (SIESTA, OpenMX) — are compact and efficient for large, low-density or partly vacuum systems, and make hybrid functionals cheaper, but their completeness is harder to control. **Real-space grids** and wavelets (GPAW, BigDFT, Octopus) parallelize well and suit large finite systems.

Worked example — size of a plane-wave basis. Diamond-cubic silicon has $a = 5.431$ Å and a primitive cell volume $V = a^3/4 = 40.05$ Å^3. For a cutoff of $E_{cut} = 250$ eV,

$$k_{max} = \frac{\sqrt{2 \times 9.109 \times 10^{-31} \times 250 \times 1.602 \times 10^{-19}}}{1.0546 \times 10^{-34}} = 8.10 \times 10^{10}\ \text{m}^{-1} = 8.10\ \text{Å}^{-1}$$

so $N_{pw} \approx 40.05 \times 8.10^3/(6\pi^2) = 359$ plane waves per **k**-point. Raising the cutoff to 400 eV multiplies k_{max} by $(400/250)^{1/2}$ and N_{pw} by $(400/250)^{3/2} = 2.02$, to about 728; 500 eV gives about 1,017. These are tiny numbers. Now consider a $3 \times 3 \times 3$ supercell

of the conventional cubic cell containing 216 atoms to model a vacancy or a dopant: V grows 108-fold and N_{pw} at 400 eV reaches about 78,600 per **k**-point. With 4 valence electrons per atom there are 432 occupied bands; storing their coefficients as double-precision complex numbers (16 bytes each) needs $432 \times 78{,}600 \times 16 \approx 5.4 \times 10^8$ bytes, about 0.54 GB per **k**-point and spin — manageable. But the cost of orthogonalizing and diagonalizing scales roughly as N^3 with the number of atoms, so going from 2 to 216 atoms raises the work per **k**-point by a factor of order $108^3 \approx 1.3 \times 10^6$, partly offset by needing far fewer **k**-points in the larger cell (Section 55.6). This cubic scaling is why routine DFT stops at a few hundred to a few thousand atoms.

55.6 k-point sampling and convergence

Integrating over the Brillouin zone

Total energies, densities and densities of states are integrals over all occupied states, which in a crystal means an integral over **k** throughout the Brillouin zone. Numerically, the integral becomes a weighted sum over a finite set of **k**-points. The standard choice is the **Monkhorst–Pack** grid, an evenly spaced $n_1 \times n_2 \times n_3$ mesh, reduced by the crystal's point-group symmetry (Chapter 3) to the irreducible set; a $12 \times 12 \times 12$ grid for an FCC metal reduces from 1,728 points to 72.

The required density depends on the electronic structure:

- In **insulators and semiconductors**, all bands are completely full or empty; the integrand is smooth and converges exponentially with grid density. A $6 \times 6 \times 6$ grid for silicon's two-atom cell is often adequate for 1 meV/atom.
- In **metals**, the occupation drops from 1 to 0 at the Fermi surface, a discontinuity that makes convergence slow and erratic. Two cures are used together: denser grids and **smearing** — replacing the step function by a smooth one (Fermi–Dirac, Gaussian, Methfessel–Paxton or Marzari–Vanderbilt "cold" smearing) of width 0.05–0.2 eV, with the total energy extrapolated to zero width. The **tetrahedron method** with Blöchl corrections is preferred for accurate densities of states.

Because the reciprocal-lattice spacing scales as $1/a$, a supercell n times larger in each direction needs n times fewer divisions per direction for the same sampling density. Codes and databases therefore specify **k**-point density as a spacing in reciprocal space (for example 0.15–0.25 Å^{-1}, including the 2π) or as **k**-points per reciprocal atom (often 1,000–8,000).

Convergence testing

Every plane-wave calculation has at least two numerical parameters — cutoff and **k**-mesh — that must be converged for the property of interest. Total energies converge from above as the cutoff rises (the variational principle) but energy *differences* between similar structures converge much faster because errors cancel. Stresses and elastic constants need higher cutoffs than energies, because the basis changes with cell volume (Pulay stress). Table 55.1 shows representative behaviour for an FCC metal, and Figure 55.4 contrasts the rapid convergence of a semiconductor with the slow, oscillating convergence of a metal.

k-mesh (FCC Al, primitive cell)	Irreducible points	Representative error in total energy (meV/atom)
$4 \times 4 \times 4$	8	~20
$8 \times 8 \times 8$	29	~3
$12 \times 12 \times 12$	72	~1
$16 \times 16 \times 16$	145	~0.2
$20 \times 20 \times 20$	256	< 0.1

The numbers are illustrative of a metal with Methfessel–Paxton smearing of 0.1 eV; the actual errors oscillate and must be checked for each system.

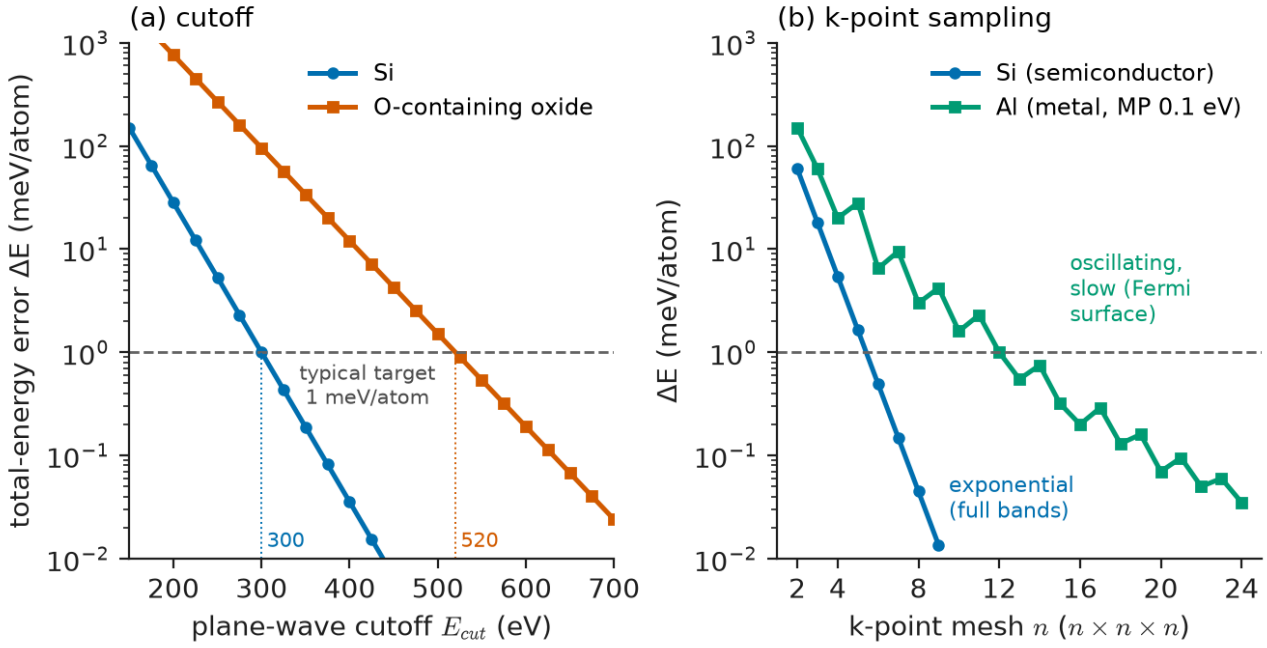


Figure 55.4 — Convergence of a plane-wave DFT calculation with cutoff energy and k-point mesh for a semiconductor and a metal.

> **Status of the principle: Numerical convergence of DFT results**
> *Derived.* With a complete basis and dense sampling, the result converges to the exact answer *for the chosen functional*; convergence controls numerical error only and says nothing about the physical error of the functional, which must be judged separately (Section 55.7).

Supercell convergence

For defects, surfaces and interfaces, the simulation cell is itself a parameter. A vacancy in a 32-atom cell interacts elastically and electronically with its periodic images; a surface slab of five layers may not have a bulk-like middle; a vacuum layer of 8 Å may let two surfaces interact. These finite-size errors are tested by increasing cell size. **Charged defects** in semiconductors and insulators add a long-range electrostatic interaction between images that decays only as $1/L$, requiring correction schemes (Makov–Payne, Freysoldt–Neugebauer–Van de Walle) to obtain defect levels relevant to doping and compensation.

Worked example — vacancy formation energy in aluminium. A PBE calculation, converged to 1 meV/atom in cutoff and **k**-mesh (illustrative values, typical of published calculations), gives a total energy of $E_{bulk} = -401.83$ eV for a perfect $3 \times 3 \times 3$ supercell of the conventional FCC cell (108 atoms), so -3.721 eV/atom. Removing one atom and relaxing the 107 remaining atoms at fixed cell gives $E_{vac} = -397.45$ eV. The vacancy formation energy compares the defective cell with the same number of atoms in perfect crystal:

$$E_f^v = E_{vac} - \frac{N-1}{N} E_{bulk} = -397.45 - \frac{107}{108}(-401.83) = -397.45 + 398.11 = 0.66 \text{ eV}$$

close to the measured 0.67 eV. With a formation entropy of about 1 k_B, the equilibrium vacancy fraction (Chapter 5) is

$$c_v = \exp\left(\frac{S_f}{k_B}\right)\exp\left(-\frac{E_f^v}{k_B T}\right)$$

which gives $c_v = 2.7 \times \exp(-0.66/(8.617 \times 10^{-5} \times 923)) = 6.8 \times 10^{-4}$ at 923 K, just below the melting point, and 2.3×10^{-11} at 300 K. The numbers show the scale on which quench rates, ageing kinetics and positron-annihilation measurements (Chapter 54) operate. Two cautions: the 108-atom cell must be checked against a larger one, because the elastic relaxation around the vacancy reaches several neighbour shells; and the agreement with experiment is partly fortunate — PBE vacancy energies are systematically low because of surface-like errors at the internal surface of the vacancy, which functionals like PBEsol or SCAN partly correct.

55.7 What DFT predicts well and badly

The record

After more than three decades of comparison with experiment, the strengths and weaknesses of standard functionals are well documented. Table 55.2 compares lattice constants and band gaps.

Material	Lattice constant: exp. (Å)	LDA (Å)	PBE (Å)	Band gap: exp. (eV)	PBE (eV)	HSE06 (eV)
Al	4.05	3.98	4.04	metal	metal	metal
Cu	3.61	3.52	3.63	metal	metal	metal
Fe (BCC)	2.87	2.75	2.83	metal	metal	metal
Si	5.43	5.40	5.47	1.17	0.6	1.2
GaAs	5.65	5.62	5.75	1.52	0.5	1.3
MgO	4.21	4.17	4.26	7.8	4.5	6.5
ZnO	3.25 (*a*)	3.20	3.29	3.44	0.7	2.5
C (diamond)	3.57	3.53	3.57	5.48	4.1	5.3
NiO	4.18	4.06	4.19	~4.0	~0.5–1.0	~4.2

Values are typical of published calculations and vary by a few hundredths of an ångström and a few tenths of an eV with pseudopotential, code and treatment of zero-point motion.

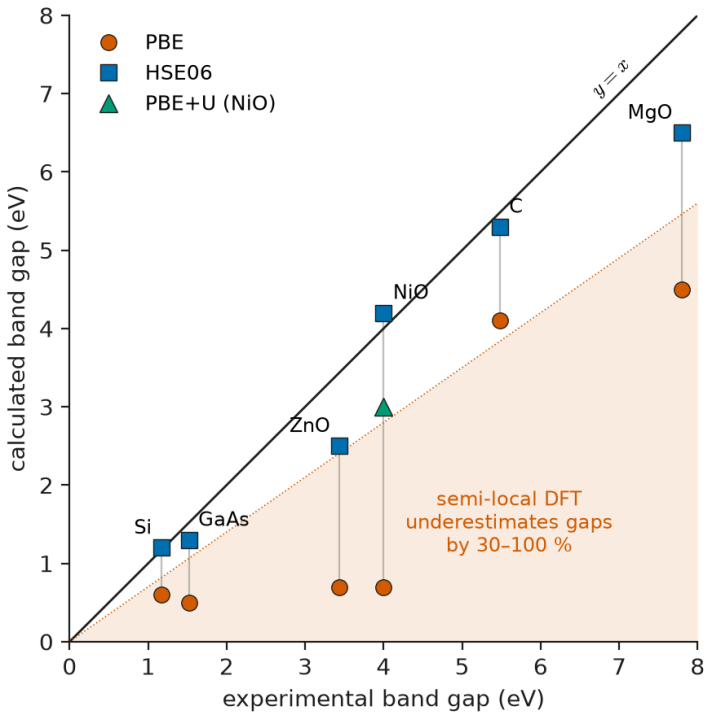


Figure 55.5 — Calculated versus measured band gaps for common semiconductors and insulators with PBE, HSE06 and PBE+U.

What is reliable

- **Structures.** Lattice parameters within about 1–2 % (LDA low, PBE high, PBEsol and SCAN often within 1 %); bond lengths and angles; relative stability of polymorphs when energy differences exceed about 20–50 meV/atom.
- **Elastic and vibrational properties.** Bulk moduli within about 10–15 % (they inherit the lattice-constant error through the strong volume dependence), elastic constants, phonon dispersions within a few percent, and hence thermal expansion and heat capacity in the quasi-harmonic approximation (Chapter 24).

- **Trends.** Even where absolute numbers err, trends across a series — which solute segregates more strongly to a grain boundary, which dopant raises a migration barrier, which surface is most stable — are usually right because errors are systematic and cancel.
- **Formation enthalpies** of intermetallics and many compounds within about 0.05–0.1 eV/atom, improved by fitted corrections for elemental reference states (for example the O_2 molecule, which GGA overbinds by over 1 eV per molecule).
- **Magnetic moments** of ferromagnetic metals and many ordered magnetic compounds.

What is unreliable

- **Band gaps** are underestimated by semi-local functionals, typically by 25–85 % (Table 55.2), and by most for strongly correlated oxides (Table 55.2, Figure 55.5). Hybrids, DFT+U for localized states, or many-body GW calculations are used where gaps matter, as in photovoltaic absorbers and dopant-level predictions.
- **Strongly correlated materials.** Mott insulators, heavy-fermion compounds, and the high-temperature superconducting cuprates are beyond semi-local DFT; DFT+U is a patch, and dynamical mean-field theory (DMFT) is the research-level tool.
- **Dispersion-bound systems** without corrections (Section 55.4).
- **Reaction barriers** are typically underestimated by GGAs, because self-interaction error over-stabilizes the stretched bonds of transition states; hybrids help.
- **Excited states and optics.** Kohn–Sham DFT is a ground-state theory. Optical spectra need time-dependent DFT or the Bethe–Salpeter equation.
- **Finite temperature.** A static DFT calculation is at 0 K. Free energies need vibrational (phonon), electronic, magnetic and configurational entropy added — through phonon calculations, ab-initio MD, or cluster expansion and Monte Carlo (Chapter 56). Phases stabilized by entropy, such as high-temperature BCC titanium and zirconium or paramagnetic iron, are therefore not ground states in plain DFT.
- **Length and time scales.** Hundreds to a few thousand atoms and tens of picoseconds of ab-initio MD. Microstructural phenomena, dislocation networks and long-time diffusion need the methods of Chapter 56.

> **Status of the principle: Accuracy of semi-local DFT**
> *Empirical.* The error ranges quoted are statistical summaries of comparisons with experiment over large benchmark sets; they apply to the classes of material they were measured on and cannot be assumed for new chemistries, unusual bonding or strongly correlated electrons.

Worked example — equation of state and bulk modulus of copper. A PBE calculation gives the following energies per atom for FCC copper at five lattice constants (illustrative values, typical of published calculations):

a (Å)	3.55	3.59	3.63	3.67	3.71
$V = a^3/4$ (Å³/atom)	11.185	11.567	11.958	12.358	12.766
E (eV/atom)	−3.6950	−3.7140	−3.7200	−3.7145	−3.6991

Fitting a parabola $E = c_0 + c_1 V + c_2 V^2$ through the three central points gives a minimum at $V_0 = 11.97$ Å^3, so $a_0 = (4V_0)^{1/3} = 3.631$ Å, and a curvature from which the bulk modulus

$$B = V_0 \left.\frac{d^2E}{dV^2}\right|_{V_0} = V_0 \cdot 2c_2 = 0.881 \text{ eV Å}^{-3} = 0.881 \times 160.2 \text{ GPa} = 141 \text{ GPa}$$

against measured values of 3.61 Å and about 140 GPa. Fitting the parabola to all five points instead shifts the minimum to 12.00 Å³ ($a_0 = 3.635$ Å), because the energy curve is asymmetric — stiffer in compression than in tension, the same anharmonicity that causes thermal expansion (Chapter 24). That is why production work fits a physically motivated form such as the Birch–Murnaghan equation of state, which also yields the pressure derivative B' (about 5 for copper), and why the points should span no more than a few percent in volume (Figure 55.6).

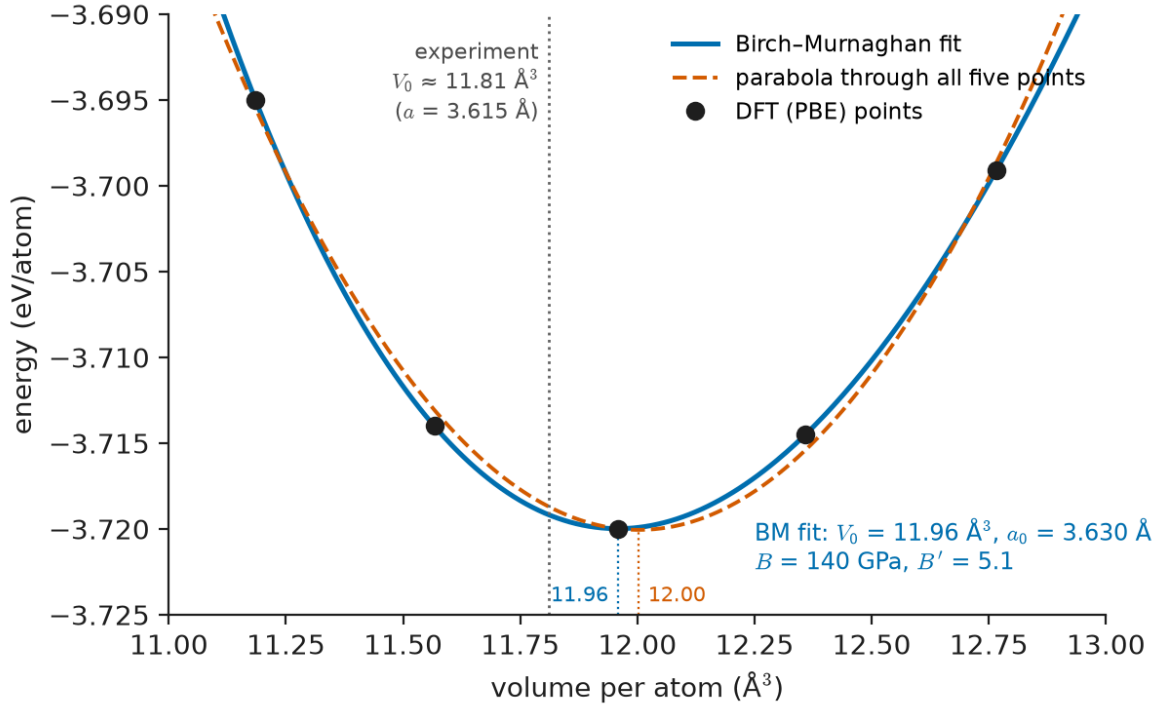


Figure 55.6 — Energy–volume curve of FCC copper from DFT, with Birch–Murnaghan and parabolic fits.

55.8 Practical workflows: codes, inputs, outputs, validation

Codes

Table 55.3 lists widely used codes. The choice depends on the basis, the functionals and properties supported, licence and the local expertise.

Code	Basis	Licence	Typical strengths
VASP	Plane waves, PAW	Commercial	Robust for solids, hybrids, MD, the reference for most databases
Quantum ESPRESSO	Plane waves, NC/US/PAW	Open source	Phonons by perturbation theory, broad features, large community
ABINIT	Plane waves, NC/PAW	Open source	Response functions, many-body perturbation theory
CASTEP	Plane waves, ultrasoft	Academic/commercial	NMR (GIPAW), spectroscopy
CP2K	Gaussians and plane waves	Open source	Large systems, liquids, ab-initio MD
GPAW	Real space/plane waves/LCAO, PAW	Open source	Python integration with ASE, catalysis
SIESTA	Numerical atomic orbitals	Open source	Thousands of atoms, linear-scaling options
WIEN2k, FHI-aims	All-electron LAPW, numerical orbitals	Academic/commercial	Accuracy benchmarks, hyperfine and core properties

Inputs

A calculation is specified by:

1. **Structure**: lattice vectors, atomic species and positions, magnetic configuration (initial moments; antiferromagnets need an appropriate supercell), and for defects the supercell and charge state.

2. **Physics**: functional, U values, dispersion correction, spin polarization, spin–orbit coupling (needed for heavy elements, magnetic anisotropy and topological materials).
3. **Numerics**: pseudopotential or PAW data sets, cutoff, **k**-mesh, smearing, convergence criteria for the electronic loop (typically 10^{-6} eV) and for ionic relaxation (forces below 0.01–0.02 eV/Å, stresses below about 0.1 GPa).
4. **Task**: single point, relaxation of positions and/or cell, molecular dynamics, nudged-elastic-band barrier search, phonons (finite displacements or density-functional perturbation theory), response properties.

Outputs and derived quantities

The direct outputs are total energy, forces, stress tensor, eigenvalues and the electron density. Most engineering quantities are *differences* or *derivatives* of these:

Quantity	How it is computed	Engineering use
Formation enthalpy and convex hull	Energy of compound minus elemental references	Phase stability, new-compound screening (Chapter 57)
Defect formation energy	Supercell difference with chemical potentials	Dopability, vacancy diffusion, radiation damage
Migration barrier	Nudged elastic band (NEB)	Diffusion coefficients, ionic conductivity of electrolytes
Elastic constants	Stress–strain or energy–strain	Moduli for alloy design and composite inputs
Surface and interface energies	Slab minus bulk	Wulff shapes, wetting, adhesion, grain-boundary segregation
Segregation and trapping energies	Solute or H at boundary vs bulk	Temper embrittlement, hydrogen trapping (opening problem)
Intercalation voltage	Energy of lithiation per electron transferred	Cathode screening for batteries
Phonons	Second derivatives of energy	Thermal properties, dynamic stability, free energies
Spectroscopic parameters	NMR shifts (GIPAW), electric-field gradients, core-level shifts	Assigning NMR, Mössbauer and XPS spectra (Chapters 53 and 54)

Validation and reproducibility

Three levels of validation are needed before a DFT number informs a design decision.

Numerical validation. Convergence with cutoff, **k**-mesh, smearing, supercell size and vacuum thickness, documented for the property in question.

Code and data-set validation. A community-wide comparison published in 2016 compared the equations of state of 71 elemental crystals computed by 40 codes and pseudopotential libraries. Modern all-electron and well-made PAW and pseudopotential sets agreed with each other to within about 1 meV/atom on average in a measure of equation-of-state difference — comparable to the difference between careful experiments — whereas older data sets differed by an order of magnitude more. Using tested pseudopotential libraries is now standard practice.

Physical validation. Comparison with experiment for a related, well-known system using the same settings: if the chosen functional misplaces the band gap of the parent oxide or the formation enthalpy of a known carbide, its predictions for the new ones should be read accordingly.

Automation

Hand-built input files do not scale to thousands of calculations. Python frameworks — ASE (Atomic Simulation Environment), pymatgen and its workflow tools, AiiDA with full provenance tracking — build inputs, submit jobs, parse outputs and record exactly how every number was obtained. Such workflows built the large DFT databases (Materials Project, AFLOW, OQMD) used in Chapter 57 and are the backbone of high-throughput screening. The same provenance discipline is what makes a single calculation reproducible years later.

Returning to the opening problem

The hydrogen-trapping question that opened this chapter shows how the pieces fit together. The workflow is: (1) relax bulk ferrite and the carbide with PBE, checking lattice constants against experiment; (2) build a supercell of a coherent or semi-coherent carbide–ferrite interface large enough that the interface strain field is captured, typically 100–300 atoms; (3) place a hydrogen atom at each candidate site — the tetrahedral interstitial of bulk ferrite as the reference, interfacial sites, carbon vacancies inside the carbide — and relax; (4) compute the binding energy $E_b = E(\mathrm{H@bulk}) - E(\mathrm{H@trap})$, correcting for zero-point energy, which is large for so light an atom (about 0.1–0.2 eV per H); (5) check convergence with supercell size and compare a known trap, such as a vacancy in iron, with experiment.

In equilibrium (Oriani's model) the ratio of trap to lattice occupancy grows as $\exp(E_b/k_BT)$. At 300 K, $k_BT = 0.0259$ eV, so a binding energy of 0.3 eV raises the local occupancy ratio by a factor $\exp(0.3/0.0259) \approx 1.1 \times 10^5$ — but an error of only 0.1 eV, typical of GGA trapping energies, changes that factor by $\exp(0.1/0.0259) \approx 48$. DFT can therefore rank candidate traps and reject weak ones with confidence; it cannot by itself decide between two traps whose binding energies differ by less than about 0.1 eV. That is precisely the right role for it in an alloy-design programme: narrowing a long list of carbides and sites to a short list for synthesis, thermal-desorption spectroscopy and mechanical testing.

Beyond standard DFT

When DFT is not accurate enough, more expensive methods serve as benchmarks or replace it for small systems. The **GW approximation** of many-body perturbation theory computes quasiparticle energies and gives band gaps typically within 0.1–0.3 eV of experiment for semiconductors. **Quantum Monte Carlo** samples the many-electron wavefunction stochastically and gives near-exact energies for a few hundred electrons at large cost. **Coupled-cluster** methods, the gold standard of molecular chemistry, are now being applied to periodic solids. **Dynamical mean-field theory**, usually combined with DFT, treats strongly correlated d and f electrons. None of these is a routine engineering tool; their role is to set reference values against which functionals, and the machine-learned potentials of Chapter 57, are judged.

> **Status of the principle: DFT as a design tool**
> *Approximate.* DFT energies guide design reliably when the relevant energy differences exceed the known functional error for that class of material (typically 0.05–0.1 eV/atom for formation energies, 0.1–0.2 eV for barriers and defect energies with GGA), when calculations are numerically converged and validated against a known reference, and when finite-temperature and kinetic effects are added where they matter.

What remains open

Open question	Where it is answered
What crystal structures, Brillouin zones and symmetries enter DFT inputs?	Chapters 2, 3 and 4
How do vacancy and solute energies from DFT set diffusion rates and segregation?	Chapters 5 and 8
How are phonons converted into thermal expansion, heat capacity and conductivity?	Chapter 24
How do band structures determine semiconductor and optical behaviour?	Chapters 31 and 36
How are DFT energies used to fit interatomic potentials and drive molecular dynamics, Monte Carlo and CALPHAD?	Chapter 56
How do databases of DFT results and machine-learned potentials accelerate discovery?	Chapter 57
How are computed spectroscopic parameters compared with NMR, Mössbauer and XAS data?	Chapters 53 and 54
How do computed intercalation voltages and barriers guide battery and catalyst design?	Chapter 40
How does hydrogen trapping relate to embrittlement in steels?	Chapters 15 and 44

Next: Chapter 56 climbs from electrons to atoms and microstructures, showing how molecular dynamics, Monte Carlo, phase-field, dislocation-dynamics, crystal-plasticity and CALPHAD methods carry the energies computed here to the length and time

scales of engineering components.

Chapter 56 — Atomistic and Mesoscale Simulation

Part XI · Characterization, Computation and Design

An aero-engine manufacturer is qualifying a new nickel-base superalloy for a turbine disc. A disc spends tens of thousands of hours between 600 and 750 °C under centrifugal stress; its life is set by creep, by fatigue cracks that start at grain boundaries and inclusions, and by the slow coarsening of the γ′ precipitates that give the alloy its strength. Testing alone would take a decade: creep tests of 10,000 h, fatigue campaigns at several temperatures, forging trials for every change of composition. Density-functional theory (Chapter 55) can compute the energy of a γ′ unit cell or of a single antiphase boundary, but a disc contains 10^{28} atoms, and its precipitates coarsen over years. No single calculation can span that gap. What engineers can do is build a chain: electronic-structure calculations feed interatomic potentials; potentials drive molecular dynamics of dislocations cutting precipitates; thermodynamic databases predict which phases form and in what fractions; phase-field models evolve the microstructure during heat treatment; dislocation dynamics and crystal plasticity turn that microstructure into a stress–strain response; and finite-element models of the whole disc use it. Each link has its own approximations and its own domain of validity. Choosing, linking and checking them is the subject of this chapter.

> **How do simulations reach from atoms to microstructures to components, when each method can span only a few orders of magnitude in length and time?**
> **What does each method assume, what does it deliver, and how are their results passed — and checked — from one scale to the next?**

Concept	Engineering destination
Molecular dynamics	Dislocation–precipitate interactions, radiation cascades, thermal conductivity, polymer and glass structure
Interatomic potentials	Reliability of every MD and MC result; choice between speed and chemistry
Thermostats and barostats	Simulating at the temperature and pressure of service or processing
Metropolis Monte Carlo	Order–disorder in alloys, segregation to grain boundaries, polymer conformations
Kinetic Monte Carlo	Long-time diffusion, precipitation, radiation-damage annealing, thin-film growth
Phase-field modelling	Solidification microstructures, precipitate coarsening, grain growth, martensite, fracture
Discrete dislocation dynamics	Work hardening, size effects, fatigue structures
Crystal-plasticity FEM	Texture evolution in forming, grain-scale stress hot spots, fatigue initiation
CALPHAD	Phase diagrams of multicomponent alloys, heat-treatment windows, solidification paths
Multiscale modelling and ICME	Alloy and process design with fewer trials; certification of new materials

Roadmap. The scale problem (56.1) → molecular dynamics: potentials, integrators, thermostats and barostats (56.2) → Monte Carlo, from Metropolis sampling to kinetic Monte Carlo (56.3) → phase-field modelling of microstructure (56.4) → discrete dislocation dynamics and crystal-plasticity finite elements (56.5) → CALPHAD and computational thermodynamics (56.6) → multiscale modelling and integrated computational materials engineering (56.7).

56.1 The scale problem

The phenomena that set the properties of engineering materials span about ten orders of magnitude in length and twenty in time. Atomic vibrations occur in 10^{-13} s over 10^{-10} m; a vacancy in copper at room temperature hops about once every few milliseconds; precipitates coarsen over hours at ageing temperatures and years in service; components are metres long and serve for decades. Figure 56.1 places the simulation methods on this map.

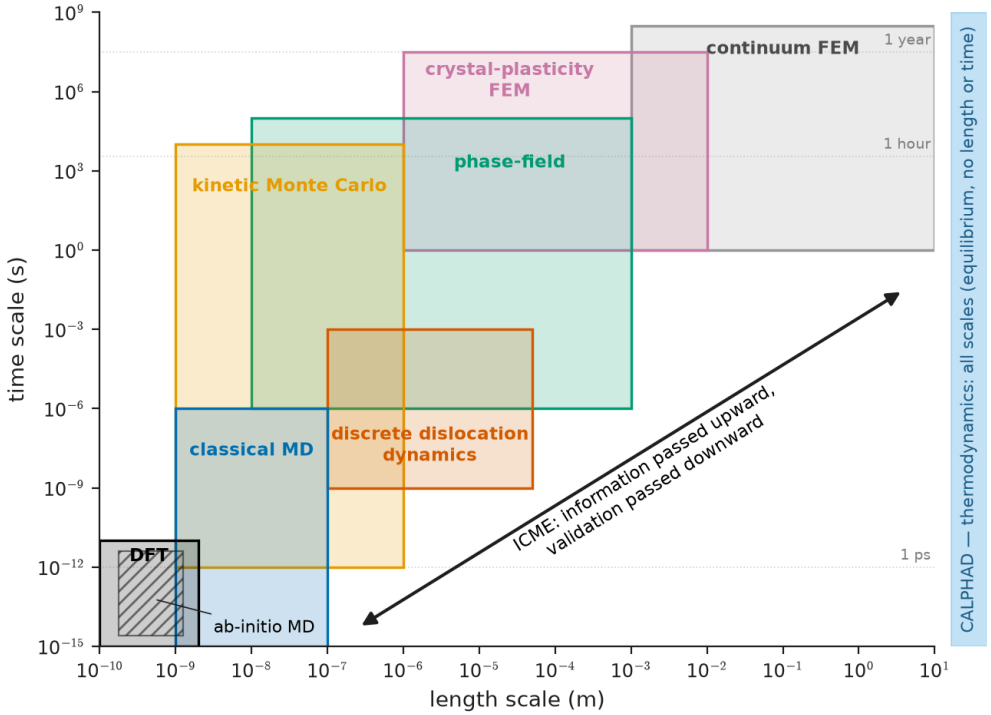


Figure 56.1 — Length and time scales accessible to simulation methods, from electrons to components.

No method covers more than a few decades in each direction, for a simple reason: each must resolve the fastest and smallest relevant process in its own description. Molecular dynamics must follow atomic vibrations, so its time step is about a femtosecond; phase-field models must resolve a diffuse interface, so their grid spacing is a fraction of the interface width. Methods at larger scales gain reach by **coarse-graining**: they discard degrees of freedom (electrons, then individual atoms, then individual dislocations) and replace them with parameters — a potential, a rate, a mobility, a free-energy function, a hardening law — that must come from a finer-scale method or from experiment.

> **Status of the principle: Coarse-graining by scale separation**
> *Approximate.* A coarse model is valid when the discarded degrees of freedom equilibrate much faster, or act over much shorter distances, than the variables retained; it fails where scales overlap — for example in nanocrystals, where grain size approaches dislocation core size, or at shock strain rates, where loading is as fast as atomic relaxation.

56.2 Molecular dynamics

The method

Molecular dynamics (MD) integrates Newton's equations of motion for a system of atoms interacting through a potential energy $U(\mathbf{r}_1, \ldots, \mathbf{r}_N)$:

$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = \mathbf{F}_i = -\nabla_{\mathbf{r}_i} U$$

The result is a trajectory — positions and velocities of every atom at every step — from which thermodynamic averages (energy, pressure, heat capacity), structure (radial distribution functions, defect configurations) and transport properties (diffusion coefficients from mean-square displacement, thermal conductivity from heat-flux correlations, viscosity) are computed by statistical mechanics. Periodic boundary conditions remove free surfaces; systems from a few thousand to billions of atoms are simulated on modern supercomputers, with 10^5–10^7 atoms routine for materials problems.

Everything rests on the potential. If U comes from DFT at each step (**ab-initio MD**), the method inherits DFT's accuracy and cost — a few hundred atoms for tens of picoseconds. **Classical MD** uses an analytic or fitted potential, roughly 10^3–10^6 times cheaper, and reaches millions of atoms and nanoseconds to microseconds.

Interatomic potentials

The potential must reproduce the parts of the DFT potential-energy surface (Chapter 55) relevant to the problem. Four families illustrate the range.

Lennard-Jones (LJ). A pair potential for closed-shell atoms:

$$U_{LJ}(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

The r^{-6} term is the van der Waals (dispersion) attraction; the r^{-12} term is a convenient, not physical, model of Pauli repulsion. For argon $\varepsilon/k_B = 119.8$ K ($\varepsilon = 0.0103$ eV) and $\sigma = 3.405$ Å, so the minimum lies at $r_{min} = 2^{1/6}\sigma = 3.82$ Å. LJ is excellent for noble gases and a building block for non-bonded interactions in polymer force fields; for metals and covalent solids it fails badly. Any pair potential predicts the Cauchy relation $C_{12} = C_{44}$ for cubic crystals and a vacancy formation energy equal to the cohesive energy, whereas real metals have $C_{12} \neq C_{44}$ and vacancy energies a third of the cohesive energy.

Embedded-atom method (EAM). For metals the energy of an atom depends on how many neighbours share its electron gas. The EAM writes

$$U = \sum_i F_i(\bar{\rho}_i) + \frac{1}{2}\sum_{i\neq j}\phi_{ij}(r_{ij}), \qquad \bar{\rho}_i = \sum_{j\neq i} f_j(r_{ij})$$

where $\bar{\rho}_i$ is the electron density at atom i contributed by its neighbours, F_i is the energy of embedding the atom in that density, and ϕ is a pair repulsion. Because F is concave, the energy per bond decreases as coordination rises — correctly giving low vacancy and surface energies relative to cohesion. EAM and its angular extension MEAM are the workhorses for FCC and many BCC metals and alloys: Cu, Al, Ni, Ni–Al, Fe, Fe–Cr. Their cost is only a few times that of a pair potential.

Tersoff and bond-order potentials. Covalent solids have directional bonds whose strength depends on the number and angles of neighbours. The **Tersoff** potential,

$$U = \frac{1}{2}\sum_{i\neq j} f_c(r_{ij})\left[A\,e^{-\lambda_1 r_{ij}} - b_{ij}\,B\,e^{-\lambda_2 r_{ij}}\right]$$

multiplies the attractive term by a **bond order** b_{ij} that decreases as atom i acquires more neighbours, weighted by bond angles. It stabilizes the diamond structure of Si, C and Ge and describes SiC; the related REBO and AIREBO potentials are standard for graphene, nanotubes and hydrocarbons.

Reactive force fields (ReaxFF). To model bond breaking and formation in mixed chemistry — combustion, oxidation, corrosion, polymer pyrolysis, battery electrolyte decomposition — ReaxFF computes bond orders continuously from distances and equilibrates atomic charges at every step. It is about 10–100 times more expensive than EAM and requires parameter sets fitted carefully to quantum-chemical data for each chemistry; a set built for hydrocarbon combustion is not valid for silica glass.

Machine-learned potentials (GAP, NequIP, MACE), which fit the DFT surface flexibly with near-DFT accuracy at a fraction of its cost, are the newest family; Chapter 57 covers them.

Table 56.1 summarizes the trade-offs.

Potential	Typical materials	Relative cost per atom-step	Captures	Misses
Lennard-Jones	Noble gases, non-bonded parts of polymers	1	Dispersion, packing	Metallic, covalent and ionic bonding
Buckingham + Coulomb	Ionic oxides, silicates	2–10 (long-range sums)	Ionic structure, defects in oxides	Charge transfer, covalency
EAM / MEAM	Metals and alloys	2–5	Many-body metallic cohesion, defects, dislocations	Directional bonding, magnetism (except special forms)
Tersoff / REBO	Si, C, SiC, hydrocarbons	5–10	Directional covalent bonds	Long-range dispersion, charge
ReaxFF	Reactive mixed chemistry	100–500	Bond breaking and formation, charge equilibration	Accuracy outside training chemistry
Machine-learned (GAP, NequIP, MACE)	Any, given training data	100–10^4	Near-DFT accuracy	Configurations far from training set

Status of the principle: Classical interatomic potentials
Empirical. Functional forms are physically motivated but their parameters are fitted to experimental or DFT data; a potential is reliable only for structures, temperatures and deformation modes similar to its fitting set, and transferability must be checked for each new use.

Integrators

The equations of motion are integrated in small time steps Δt. The standard algorithm, **velocity Verlet**, is

$$\mathbf{r}(t+\Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t + \frac{\mathbf{F}(t)}{2m}\Delta t^2$$

$$\mathbf{v}(t+\Delta t) = \mathbf{v}(t) + \frac{\mathbf{F}(t)+\mathbf{F}(t+\Delta t)}{2m}\Delta t$$

It is time-reversible and **symplectic** — it preserves phase-space volume — so although individual trajectories diverge from the exact ones (MD is chaotic), the total energy does not drift systematically over millions of steps. Higher-order integrators conserve energy better over a single step but drift over long runs and are rarely used.

The time step must resolve the fastest motion. A rule of thumb is Δt about one-twentieth to one-fiftieth of the shortest vibrational period: 1–5 fs for metals, about 1 fs for oxides, 0.5–1 fs for systems containing hydrogen (whose bonds vibrate at about 90 THz), unless those bonds are constrained.

Ensembles, thermostats and barostats

Plain MD conserves energy and samples the microcanonical (**NVE**) ensemble. Experiments are done at fixed temperature and pressure, so most simulations sample the canonical (**NVT**) or isothermal–isobaric (**NPT**) ensemble by coupling the system to a thermostat and a barostat.

- **Velocity rescaling and the Berendsen thermostat** scale velocities towards the target temperature with a time constant τ_T. They are robust for equilibration but do not generate the correct canonical distribution of fluctuations.
- **Nosé–Hoover** thermostats add a friction variable ζ with its own equation of motion,

$$\dot{\mathbf{v}}_i = \frac{\mathbf{F}_i}{m_i} - \zeta\,\mathbf{v}_i, \qquad \dot{\zeta} = \frac{1}{Q}\left(\sum_i m_i v_i^2 - 3Nk_BT\right)$$

where Q is a fictitious thermal mass. The extended system samples the canonical ensemble exactly (for ergodic systems; Nosé–Hoover chains fix the cases where a single thermostat is not ergodic).

- **Langevin** thermostats add friction and random forces to each atom; they are canonical and stable, and model coupling to an electron bath in metals, but they perturb dynamics and hence transport coefficients.
- **Barostats** (Berendsen, Nosé–Hoover/MTK, Parrinello–Rahman) let the cell volume — and, in Parrinello–Rahman, its shape — fluctuate in response to the difference between the instantaneous and target stress tensors. Variable cell shape is essential for simulating phase transformations (martensitic, pressure-induced) and for applying uniaxial stress.

The choice matters: transport properties (diffusion, viscosity, thermal conductivity) should be computed in NVE or with weakly coupled deterministic thermostats after equilibration, never with strong Langevin damping.

What MD has delivered, and its limits

Classical MD gave the first atomic-scale pictures of displacement cascades in irradiated metals (hundreds of displaced atoms in a few picoseconds, most recombining within 10 ps), of dislocation cores and their dissociation, of dislocation cutting of γ′ precipitates, of grain-boundary sliding in nanocrystals, of crack-tip emission versus cleavage, of glass formation and of polymer chain entanglement. Its limits are equally clear:

- **Time.** Nanoseconds to microseconds. Thermally activated processes with barriers above about 0.5 eV happen too rarely at moderate temperatures to be seen.
- **Strain rate.** Deforming a simulation cell by even a few percent within a nanosecond implies strain rates of 10^{7}–$10^{9}\ \mathrm{s}^{-1}$, versus $10^{-3}\ \mathrm{s}^{-1}$ in a tensile test. Flow stresses from MD are therefore upper bounds dominated by athermal processes.
- **Size.** 10^{7} atoms of copper is a cube about 50 nm on a side — one small grain, a few dislocations.

Worked example — the cost and reach of an MD simulation of copper. The highest phonon frequency of copper is about 7 THz, a period of $1/7\times10^{12}=0.14$ ps. A time step of 2 fs gives 70 steps per shortest period, adequate for velocity Verlet. One nanosecond needs $10^{-9}/2\times10^{-15}=5\times10^{5}$ steps. A 10^{6}-atom cell (a cube about 23 nm on a side) with an EAM potential costing about 2×10^{-6} core-seconds per atom per step needs

$$10^{6}\times5\times10^{5}\times2\times10^{-6}\ \mathrm{s}=10^{6}\ \text{core-s}\approx278\ \text{core-hours}$$

— a few hours on a 100-core cluster node. To strain the cell by 1 % in 100 ps imposes $\dot\varepsilon=0.01/10^{-10}\ \mathrm{s}=10^{8}\ \mathrm{s}^{-1}$, eleven orders of magnitude faster than a quasi-static tensile test at $10^{-3}\ \mathrm{s}^{-1}$. Reaching 10 % strain at the laboratory rate would take 100 s of simulated time, or 5×10^{16} steps — about 10^{11} times the run just costed. MD is therefore a tool for mechanisms, for high-rate events (shock, ballistic impact, laser ablation, radiation cascades) and for computing parameters such as dislocation mobilities and grain-boundary energies that slower methods use, not for simulating a tensile test directly.

56.3 Monte Carlo methods

Metropolis Monte Carlo

Many questions are about equilibrium — which atomic arrangement a solid solution adopts at 600 °C, how much solute sits at a grain boundary, what conformations a polymer chain takes in a melt — and do not need the dynamics. **Monte Carlo (MC)** methods generate configurations with the correct Boltzmann probabilities directly, by random moves. The **Metropolis algorithm** (1953) proposes a random change (swap two atoms of different species, displace an atom, flip a spin, rotate a chain segment), computes the energy change ΔE, and accepts the move with probability

$$P_{acc}=\min\left[1,\exp\left(-\frac{\Delta E}{k_B T}\right)\right]$$

Downhill moves are always accepted; uphill moves are accepted with the Boltzmann factor. Repeating this generates a Markov chain whose configurations, after equilibration, are distributed according to the canonical ensemble, because the acceptance rule satisfies **detailed balance**.

> **Status of the principle: Metropolis sampling**
> *Derived.* Detailed balance plus ergodicity guarantee convergence to the Boltzmann distribution; the guarantee is asymptotic, and in practice sampling can be trapped in metastable states when barriers between configurations are large compared with $k_B T$.

For example, an uphill swap costing $\Delta E = 0.1$ eV is accepted with probability $\exp(-0.1/0.02585) = 0.021$ at 300 K but $\exp(-0.1/0.0776) = 0.28$ at 900 K. This is why MC simulations of ordering in alloys are run by slowly cooling ("simulated annealing"), and why the same idea is used as a general optimization method.

MC moves need not be physical, which is its power: atoms may swap across the whole cell in one step, something that takes diffusion years. Combined with a **cluster expansion** — an Ising-like model of alloy energy as a sum of pair, triplet and higher-order interactions fitted to DFT energies of dozens of ordered structures — Metropolis MC predicts order–disorder transition temperatures, short-range order, and solubility limits in alloys such as Ni–Al, Cu–Au and high-entropy alloys. **Hybrid MD/MC** (swaps interleaved with MD steps) predicts equilibrium segregation of solutes to grain boundaries and dislocations, which governs temper embrittlement of steels and the thermal stability of nanocrystalline alloys. **Grand-canonical MC**, in which particles are inserted and removed at fixed chemical potential, computes gas adsorption in zeolites and metal–organic frameworks for carbon capture and hydrogen storage.

Kinetic Monte Carlo

Metropolis MC has no physical time. **Kinetic Monte Carlo (kMC)** restores it for systems that evolve by discrete, thermally activated events — atom–vacancy exchanges, adatom hops on a growing film, emission of a solute from a cluster. Each possible event i has a rate from transition-state theory,

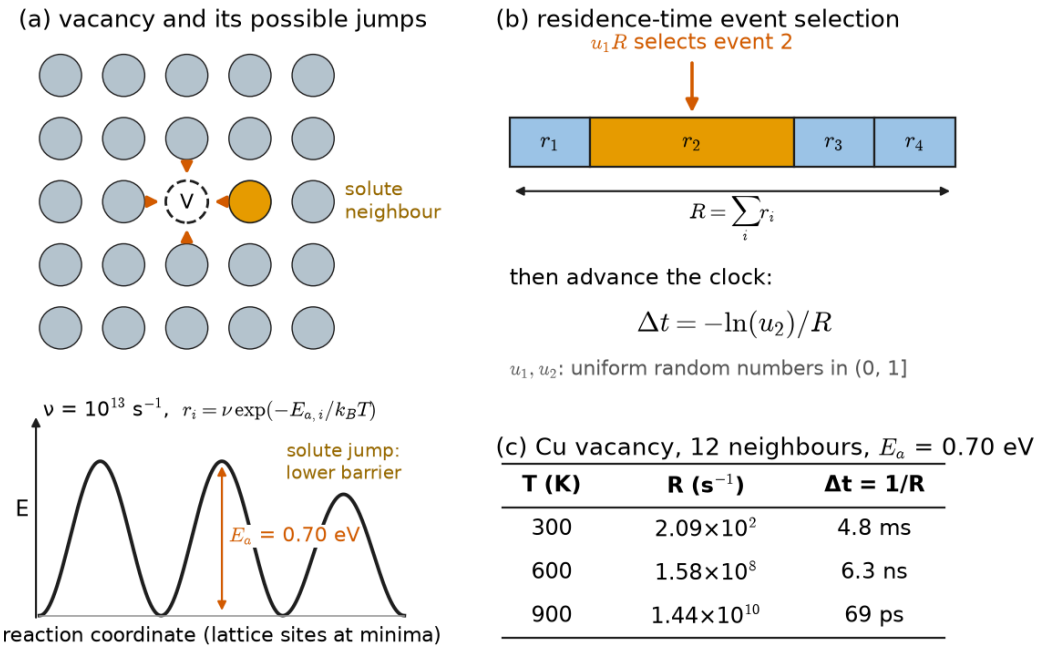
$$r_i = \nu_i \exp\left(-\frac{E_{a,i}}{k_B T}\right)$$

with attempt frequency $\nu_i \approx 10^{12}$–10^{13} s^{-1} and activation energy $E_{a,i}$ (from DFT nudged-elastic-band calculations, Chapter 55, or a fitted model). The **residence-time** (Bortz–Kalos–Lebowitz, or Gillespie) algorithm proceeds as follows (Figure 56.2):

1. List all possible events and their rates; the total rate is $R = \sum_i r_i$.
2. Choose event j with probability r_j / R using a uniform random number.
3. Execute it, and advance the clock by $\Delta t = -\ln u / R$, where u is a second uniform random number in (0, 1].
4. Update the event list and repeat.

Each step is an event; the simulation never wastes time on the vibrations between events. That is the source of kMC's enormous reach in time.

> **Status of the principle: Kinetic Monte Carlo**
> *Approximate.* Exact for a Markov process whose event list and rates are complete and correct; its accuracy rests on transition-state theory (harmonic rates, no correlated multiple jumps) and on the completeness of the event catalogue — an event that is not listed can never happen.



(c) Cu vacancy, 12 neighbours, E_a = 0.70 eV

T (K)	R (s^{-1})	Δt = 1/R
300	2.09×10^{2}	4.8 ms
600	1.58×10^{8}	6.3 ns
900	1.44×10^{10}	69 ps

Figure 56.2 — Kinetic Monte Carlo: a vacancy among possible jumps on an energy landscape, and the residence-time event selection.

Worked example — why kinetic Monte Carlo is needed for vacancy diffusion. In copper, a vacancy exchanges with one of its 12 nearest neighbours with migration energy $E_m \approx 0.70$ eV and $\nu \approx 10^{13}$ s^{-1}. The rate per neighbour at 300 K is

$$r = 10^{13}\exp\left(-\frac{0.70}{8.617\times10^{-5}\times 300}\right) = 17.4\ \text{s}^{-1}$$

so the total rate is $R = 12\times 17.4 = 209$ s^{-1} and the mean residence time between hops is $1/R = 4.8$ ms. An MD simulation with 2 fs steps would need $4.8\times10^{-3}/2\times10^{-15} = 2.4\times10^{12}$ steps to see one hop — impossible. In kMC each hop is one step. At 600 K the rate rises to $R = 1.58\times10^{8}$ s^{-1} ($\Delta t = 6.3$ ns) and at 900 K to 1.44×10^{10} s^{-1} ($\Delta t = 69$ ps), where MD and kMC begin to overlap and can be cross-checked. The vacancy diffusivity at 900 K is $D_v = Ra^2/6$ with jump distance $a = 0.256$ nm, giving $1.44\times10^{10}\times(0.256\times10^{-9})^2/6 = 1.6\times10^{-10}$ m^2 s^{-1}. Multiplying by the vacancy fraction (Chapter 8) gives the self-diffusion coefficient that sets creep and sintering rates.

Applications of kMC include precipitation of copper clusters in irradiated reactor-pressure-vessel steels over decades of service, the annealing of radiation damage (object kMC, which tracks defect clusters rather than lattice sites), the morphology of thin films grown by vapour deposition (island versus layer growth, Chapter 49), and surface reactions on catalysts with rates computed by DFT.

56.4 Phase-field modelling

The diffuse-interface idea

Microstructures — dendrites, precipitates, grains, martensite plates, domains — are defined by interfaces that move. Tracking sharp interfaces explicitly is awkward when they merge, split or meet at triple junctions. The **phase-field method** replaces each sharp interface by a smooth transition of one or more continuous fields across a finite width δ. A **non-conserved order parameter** ϕ distinguishes phases or grains (for example $\phi = 1$ in solid, 0 in liquid), and a **conserved field** c represents composition. Interfaces are simply where the fields change; topology changes happen automatically.

The model is built from a free-energy functional of Ginzburg–Landau type:

$$F = \int_V \left[f(\phi, c, T) + \frac{\kappa_\phi}{2}|\nabla\phi|^2 + \frac{\kappa_c}{2}|\nabla c|^2 + f_{el} \right] dV$$

The bulk term f is a double-well (or multi-well) function whose minima are the phases; for quantitative work it is taken from CALPHAD databases (Section 56.6). The gradient terms penalize sharp changes and, together with the barrier height of f, set the interface energy γ and width δ. The elastic term f_{el} adds misfit and applied stresses.

Evolution equations

Non-conserved fields relax down the gradient of free energy (the **Allen–Cahn** or time-dependent Ginzburg–Landau equation); conserved fields evolve by diffusion driven by gradients of the chemical potential (the **Cahn–Hilliard** equation):

$$\frac{\partial \phi}{\partial t} = -L \frac{\delta F}{\delta \phi}, \qquad \frac{\partial c}{\partial t} = \nabla \cdot \left(M \nabla \frac{\delta F}{\delta c} \right)$$

with interface mobility L and atomic mobility M, related to the diffusion coefficient (Chapter 8). Noise terms can be added to model nucleation, although nucleation is usually inserted explicitly using classical nucleation theory (Chapter 9).

> **Status of the principle: Phase-field evolution equations**
> *Derived.* They follow from linear irreversible thermodynamics as the simplest dissipative dynamics that lower the free energy (non-conserved) or conserve mass while lowering it (conserved); quantitative accuracy depends on the input free energies and mobilities and on a thin-interface analysis linking the diffuse model to sharp-interface physics.

The interface-width problem

Physical interfaces in metals are about 1 nm wide; dendrites are tens of micrometres across. Resolving a 1 nm interface with about five grid points over a 100 µm domain in three dimensions would need $(10^5 \times 5)^3 \approx 10^{17}$ grid points. Practical models therefore use artificially wide interfaces (0.1–1 µm) and rely on **thin-interface asymptotics** (Karma–Rappel, 1996, and later anti-trapping corrections), which choose the model parameters so that the interface still moves with the correct sharp-interface kinetics despite its exaggerated width. Adaptive mesh refinement and GPU computing extend reach further.

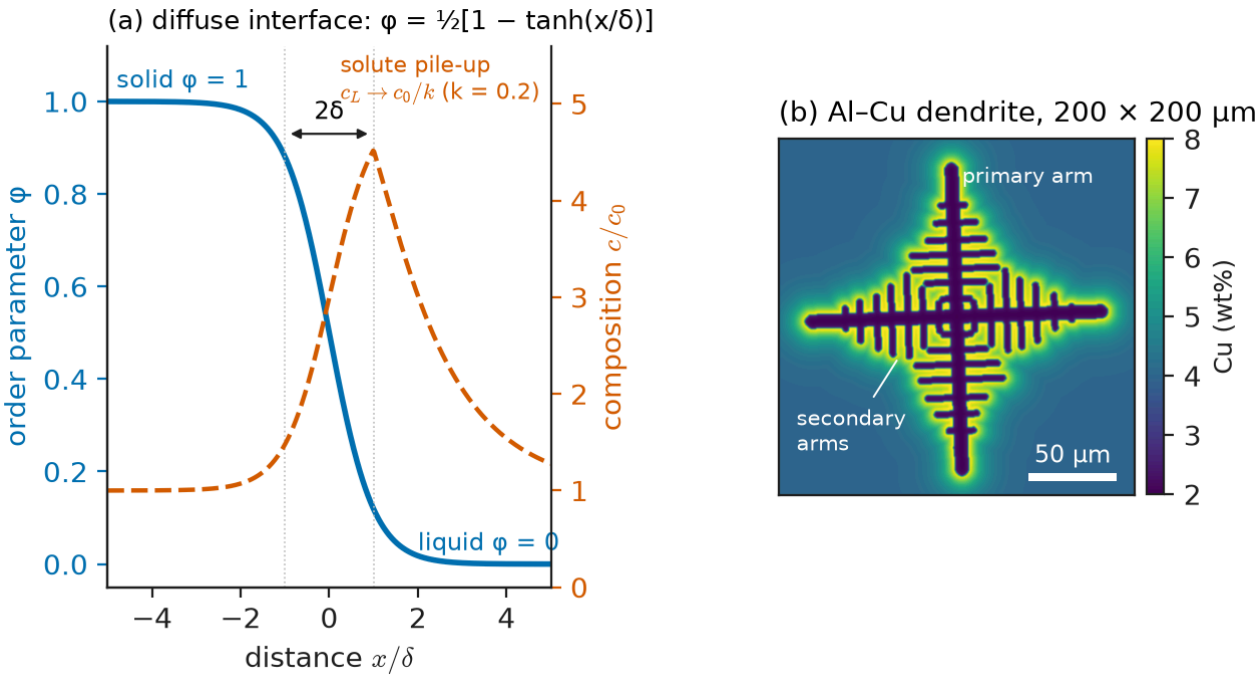


Figure 56.3 — Phase-field description of a moving interface and a simulated solidification dendrite.

Applications across engineering

- **Solidification** (Figure 56.3): dendrite tip radius and velocity, primary and secondary arm spacings and microsegregation in castings and welds (Chapters 9, 46 and 47), and the columnar-to-equiaxed transition in additive manufacturing (Chapter 48).

- **Precipitation and coarsening**: the shape and arrangement of coherent γ′ cuboids in nickel superalloys, which align into rafts under stress at high temperature, and θ′ plates in aluminium alloys, both controlled by elastic misfit energy.
- **Spinodal decomposition**: the interconnected microstructures of Fe–Cr in duplex stainless steels aged at 300–500 °C ("475 °C embrittlement") and of phase-separating glasses and polymer blends.
- **Grain growth and recrystallization**, with many order parameters, one per grain orientation.
- **Martensite and ferroelectric domains**, where the order parameter is a strain or polarization.
- **Fracture**: phase-field fracture models represent a crack as a damage field, letting cracks nucleate, branch and merge without explicit tracking — now used for concrete, rock, composites and battery-electrode particles.
- **Electrochemistry**: lithium dendrite growth and phase separation in $LiFePO_4$ particles (Chapter 40), corrosion pit growth (Chapter 43).

56.5 Discrete dislocation dynamics and crystal-plasticity FEM

Discrete dislocation dynamics

Plastic deformation of crystals is the collective motion of dislocations (Chapters 12 and 13). MD can follow a few dislocations for nanoseconds; a deformed grain contains 10^{12}–10^{15} m of dislocation line per cubic metre. **Discrete dislocation dynamics (DDD)** treats dislocations as elastic line defects in a continuum, discretized into segments, and computes their motion from the forces acting on them:

1. The stress at each segment from the applied load and from every other segment is computed from linear elasticity (Chapter 12).
2. The force per unit length follows from the **Peach–Koehler** formula $\mathbf{f} = (\boldsymbol{\sigma} \cdot \mathbf{b}) \times \boldsymbol{\xi}$, with Burgers vector $\mathbf{b}$ and line direction $\boldsymbol{\xi}$.
3. A **mobility law**, typically $v = f/B$ for FCC metals with drag coefficient $B \approx 10^{-5}$–10^{-4} Pa s (from MD), or thermally activated kink-pair motion for BCC screw dislocations, gives segment velocities.
4. **Local rules**, again calibrated on MD, handle core-scale events: junction formation and annihilation, cross-slip, nucleation at sources.

DDD simulations of cubes 5–20 μm on a side reach plastic strains of a few percent and reproduce the formation of dislocation networks and the size effects of micropillars (Chapter 54). They have tested the **Taylor relation** between flow stress and dislocation density,

$$\tau = \alpha \, \mu \, b \sqrt{\rho}$$

finding $\alpha \approx 0.2$–0.4 in FCC metals and explaining it through junction strength, and they have explored fatigue persistent slip bands and dislocation–precipitate interactions. The main limitations are cost — the long-range interactions scale poorly with the number of segments — and reach: strains above a few percent and strain rates below about 10^2 s^{-1} are difficult.

> **Status of the principle: Taylor hardening relation**
> *Empirical.* The $\sqrt{\rho}$ dependence is supported by theory and simulation of forest hardening; the coefficient α is fitted and varies with dislocation arrangement, crystal structure and strain rate.

Worked example — linking dislocation density to flow stress in copper. For copper, $\mu = 48$ GPa, $b = 0.256$ nm and $\alpha = 0.3$. At an annealed density of $\rho = 10^{12}$ m^{-2},

$$\tau = 0.3 \times 48 \times 10^{9} \times 0.256 \times 10^{-9} \times \sqrt{10^{12}} = 3.7 \text{ MPa}$$

and the polycrystal flow stress, using the Taylor factor $M = 3.06$ for an untextured FCC aggregate, is $\sigma = M\tau = 11$ MPa. After moderate cold work ($\rho = 10^{14}$ m^{-2}), $\tau = 37$ MPa and $\sigma = 113$ MPa; after heavy cold work ($\rho = 10^{15}$ m^{-2}), $\tau = 117$ MPa and $\sigma = 357$ MPa. These match the measured range for copper from fully annealed to hard-drawn wire, and show the sort of constitutive input — a hardening law tied to a physical state variable — that DDD supplies to crystal-plasticity models.

Crystal-plasticity finite elements

At the scale of a polycrystal, individual dislocations are replaced by **slip rates** on each slip system. **Crystal-plasticity finite-element (CPFEM)** models mesh a representative volume of grains — taken from EBSD maps or synthetic microstructures — and in each integration point solve:

- **Kinematics**: the deformation gradient is decomposed into elastic and plastic parts, $\mathbf{F} = \mathbf{F}_e\mathbf{F}_p$, with the plastic part produced by slip on systems α (12 for FCC $\{111\}\langle 110\rangle$, up to 48 for BCC, basal, prismatic and pyramidal for HCP).
- **Flow rule**: for example the phenomenological power law

$$\dot{\gamma}^\alpha = \dot{\gamma}_0 \left| \frac{\tau^\alpha}{\tau_c^\alpha} \right|^{1/m} \operatorname{sgn}\left(\tau^\alpha\right)$$

where τ^α is the resolved shear stress (Schmid's law, Chapter 13), τ_c^α the critical resolved shear stress and m a rate sensitivity (0.01–0.05 at room temperature).

- **Hardening**: evolution of τ_c^α with slip on all systems, through a hardening matrix, or of dislocation densities per system linked by the Taylor relation (physically based models).

Figure 56.4 shows a typical result: stress concentrates in grains whose orientation makes them elastically and plastically hard. CPFEM predicts texture evolution in rolling and deep drawing (and hence earing of aluminium beverage-can stock), the anisotropy of HCP magnesium and titanium sheet, the stress and strain hot spots at grain boundaries where fatigue cracks initiate in turbine-disc alloys, and the response of additively manufactured parts with columnar grains. Spectral (FFT-based) solvers handle periodic microstructures with millions of voxels quickly. Inputs — critical resolved shear stresses of each slip system and hardening parameters — come from single-crystal tests, micropillar compression (Chapter 54) and DDD.

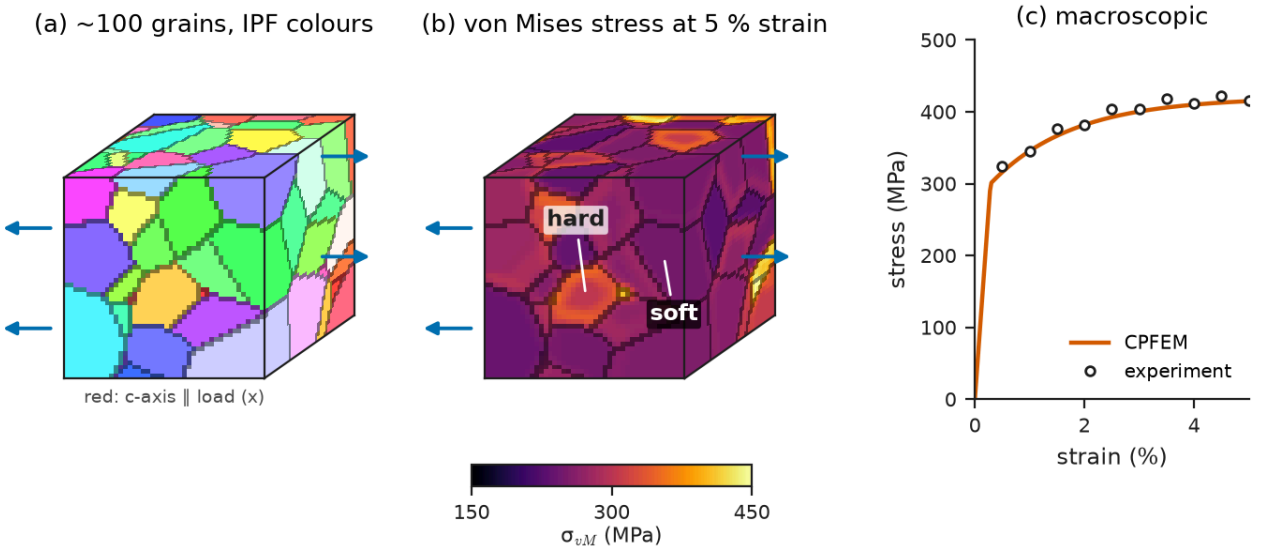


Figure 56.4 — Crystal-plasticity finite-element simulation of a polycrystal under uniaxial tension.

> **Status of the principle: Crystal-plasticity constitutive laws**
> *Empirical.* Kinematics are exact continuum mechanics, but flow and hardening laws are phenomenological or physically motivated fits; their parameters must be calibrated and the model validated against measured stress–strain curves, textures or local strain maps (digital image correlation).

56.6 CALPHAD and computational thermodynamics

The method

Engineering alloys contain many elements: a nickel superalloy has 8–12, a tool steel 6–10, a solder 3–5. Their phase diagrams cannot be measured exhaustively. The **CALPHAD** (CALculation of PHAse Diagrams) method models the Gibbs energy of every phase as a function of temperature, pressure and composition, fits the model parameters to experimental data (phase equilibria, enthalpies, heat capacities, activities) and DFT energies for the binary and ternary subsystems, and then predicts equilibria in multicomponent systems by minimizing total Gibbs energy (Chapter 7).

For a substitutional solution phase,

$$G_m = \sum_i x_i \, {}^0G_i + RT \sum_i x_i \ln x_i + {}^{ex}G_m$$

where ${}^0G_i$ are the Gibbs energies of the pure components in that structure (the "lattice stabilities", tabulated as polynomials in T), the second term is ideal mixing, and the excess term is expanded as a **Redlich–Kister polynomial**:

$${}^{ex}G_m = \sum_i \sum_{j>i} x_i x_j \sum_\nu L_{ij}^{(\nu)} \, (x_i - x_j)^\nu + \dots$$

with interaction parameters $L^{(\nu)} = a + bT + \dots$, plus ternary terms. Ordered phases, interstitial solutions and intermetallics use the **compound-energy formalism**, with sublattices (for example $(\mathrm{Fe}, \mathrm{Cr}, \mathrm{Ni})_1(\mathrm{C}, \mathrm{N}, \mathrm{Va})_3$ for austenite with carbon and nitrogen on octahedral interstitial sites). Magnetic contributions are added for iron, nickel and cobalt.

> **Status of the principle: CALPHAD Gibbs-energy models**
> *Empirical.* The minimization of Gibbs energy is fundamental thermodynamics, but the model parameters are fitted to data; extrapolation to multicomponent compositions is reliable when binary and ternary descriptions are sound, and least reliable for phases or composition regions without experimental or DFT input.

Worked example — a miscibility gap from the regular-solution model. The simplest excess model, ${}^{ex}G_m = \Omega x(1-x)$ (a single temperature-independent $L^{(0)} = \Omega$), describes a binary solution with a miscibility gap when $\Omega > 0$. The critical temperature is where the curvature of G_m vanishes at $x = 0.5$: $T_c = \Omega / 2R$. For $\Omega = 20$ kJ/mol, $T_c = 20{,}000/(2 \times 8.314) = 1{,}203$ K. At 1,000 K, the Gibbs energy of mixing at $x = 0.5$ is

$$\Delta G_{mix} = 20{,}000 \times 0.25 + 8.314 \times 1000 \times (0.5 \ln 0.5 + 0.5 \ln 0.5) = 5{,}000 - 5{,}763 = -763 \ \mathrm{J\,mol^{-1}}$$

negative, yet the solution is not stable: the curve has two minima, and the common-tangent construction gives the equilibrium compositions (the **binodal**) from $\ln[x/(1-x)] = \Omega(2x-1)/RT$, which yields $x = 0.169$ and 0.831. The **spinodal**, where $\partial^2 G / \partial x^2 = 0$, lies at $x(1-x) = RT/2\Omega = 0.208$, giving $x = 0.295$ and 0.705. Alloys between 0.295 and 0.705 decompose spontaneously by spinodal decomposition; those between the binodal and spinodal need nucleation (Chapters 7, 9 and 10). A real CALPHAD assessment has many such parameters, temperature-dependent and asymmetric, but the logic is the same: a Gibbs-energy model for each phase, minimized to give phase fractions and compositions (Figure 56.5).

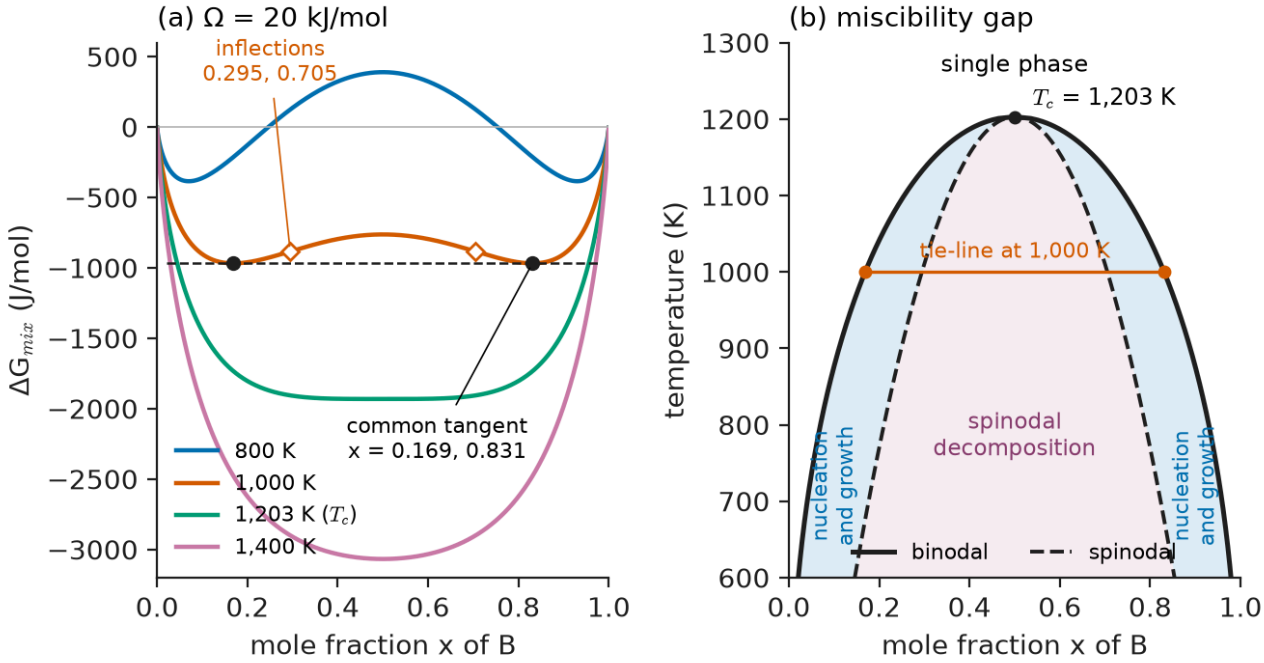


Figure 56.5 — Regular-solution model with Ω = 20 kJ/mol: Gibbs energy of mixing curves and the resulting miscibility gap.

Tools and uses

Commercial and open software (Thermo-Calc, FactSage, Pandat, OpenCalphad, pycalphad) with assessed databases for steels, nickel, aluminium, titanium and magnesium alloys, solders, oxides and slags is now part of routine industrial alloy and process development. Typical outputs:

- equilibrium phase fractions and compositions versus temperature (for example the γ′ solvus of a superalloy, which sets the solution-treatment window);
- **Scheil–Gulliver** solidification paths, assuming no diffusion in the solid and complete mixing in the liquid, which predict microsegregation, freezing range and hot-cracking susceptibility of castings and welds;
- driving forces for nucleation of unwanted phases (σ and Laves phases in superalloys and stainless steels; Chapter 16);
- slag and oxide equilibria in steelmaking, glass and refractory design.

CALPHAD extends to kinetics through **mobility databases**: DICTRA-type models solve diffusion with CALPHAD thermodynamics to predict carburizing profiles, interdiffusion in coatings and homogenization times; **precipitation models** (Kampmann–Wagner numerical approaches) combine classical nucleation, growth and coarsening with CALPHAD driving forces to predict precipitate size distributions during ageing — the γ′ distribution of the opening problem — and supply free energies to phase-field models.

56.7 Multiscale modelling and ICME

Linking scales

The methods of this chapter and of Chapter 55 form a chain in which each passes parameters upward:

From	To	Information passed
DFT	Interatomic potentials, cluster expansions	Energies and forces of many configurations
DFT	kMC, CALPHAD	Migration barriers, formation energies of end-members and metastable phases
MD	DDD	Dislocation mobilities, junction strengths, cross-slip rates
MD, kMC	Phase field	Interface energies and mobilities, diffusion coefficients
CALPHAD	Phase field, precipitation models	Free energies, mobilities, driving forces
DDD, micropillar tests	Crystal plasticity	Hardening laws, critical resolved shear stresses
Phase field, precipitation models	Crystal plasticity	Precipitate size and fraction, grain structure
Crystal plasticity	Component FEM	Homogenized anisotropic constitutive laws, fatigue indicators

This is **hierarchical** (sequential) multiscale modelling: each method runs separately and hands over parameters. **Concurrent** multiscale methods run two descriptions simultaneously in one simulation — the quasicontinuum method treats atoms explicitly near a crack tip and as a coarse finite-element continuum elsewhere; QM/MM embeds a DFT region in a classical MD region to model bond breaking at a crack tip or a catalytic site. Concurrent coupling is powerful but must avoid spurious forces and wave reflections at the handshake region.

Integrated computational materials engineering

Integrated computational materials engineering (ICME) applies this chain to product development. The idea, given impetus by a 2008 US National Academies report and by the 2011 Materials Genome Initiative, is to treat the material as a designed variable alongside geometry, linking process models, microstructure models and property models so that alloy composition, processing and component design are optimized together (Figure 56.6). Demonstrated examples include:

- **Cast aluminium engine blocks and heads.** An automotive ICME programme coupled casting simulation, CALPHAD-based solidification and precipitation models and property models to predict local yield strength and fatigue life across a cast cylinder head, shortening development cycles and cutting prototype iterations.
- **Computationally designed steels.** Ultra-high-strength, corrosion-resistant landing-gear steels were designed with CALPHAD and precipitation models to replace cadmium-plated 300M steel, and reached aerospace qualification in a fraction of the traditional time.
- **Nickel superalloy discs.** Models of forging, heat treatment, γ′ precipitation and grain size predict location-specific properties within a turbine disc, used in its life assessment.
- **Additive manufacturing.** Thermal models of the melt pool feed solidification and phase-field models to predict grain structure and residual stress, which feed crystal-plasticity and fatigue models (Chapter 48).

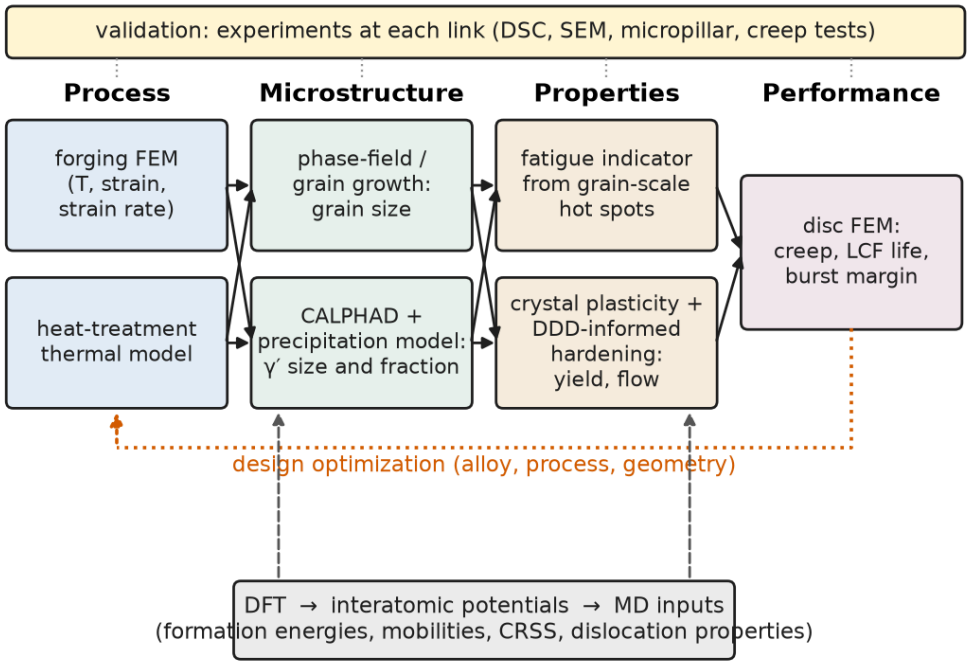


Figure 56.6 — An ICME chain for a precipitation-strengthened turbine disc: process → microstructure → properties → performance, with the simulation method at each link.

Verification, validation and uncertainty

A chain of models compounds uncertainties. ICME practice therefore distinguishes:

- **Verification**: is the model solved correctly (convergence, code testing)?
- **Validation**: does the model represent reality for the intended use (comparison with experiments designed for the purpose, at each link and at the end of the chain)?
- **Uncertainty quantification**: how do uncertainties in inputs (potential parameters, CALPHAD assessments, mobility values) propagate to outputs (yield strength, life)? Monte Carlo sampling of inputs, sensitivity analysis and surrogate models (Chapter 57) make this tractable.

A useful discipline is to state, for every predicted property, which link limits its accuracy. In a precipitation-strengthened alloy, the strength prediction is often limited not by the dislocation model but by the precipitate size distribution, which depends on interfacial energy — a parameter that is hard to measure, uncertain by a factor of two in DFT and MD, and enters nucleation rates exponentially. Identifying such bottlenecks directs experimental effort to where it matters most.

> **Status of the principle: Integrated computational materials engineering**
> *Approximate.* ICME predictions are as reliable as the weakest validated link in the chain; they are most effective for interpolation within well-characterized alloy families and processes, and require experimental calibration and validation when extrapolated to new chemistries or processing routes.

What remains open

Open question	Where it is answered
How are the energies, forces and barriers that parameterize these models computed from electrons?	Chapter 55
How do machine-learned potentials and surrogate models extend the reach and accuracy of MD, kMC and ICME?	Chapter 57
How are simulated properties used to select materials and processes in design?	Chapter 58
What thermodynamic principles and phase diagrams do CALPHAD models encode?	Chapters 6 and 7
What diffusion, nucleation and transformation theories underlie kMC, phase-field and precipitation models?	Chapters 8, 9 and 10
What dislocation physics and strengthening mechanisms do DDD and crystal plasticity represent?	Chapters 12 and 13
How are simulated microstructures and stress–strain curves validated by in-situ experiments and micropillars?	Chapter 54
How do superalloys, steels and aluminium alloys benefit from computational design?	Chapters 15, 16 and 41
How are solidification and process simulations used in casting, welding and additive manufacturing?	Chapters 46, 47 and 48

Next: Chapter 57 asks how the vast data generated by experiments and by the simulations of Chapters 55 and 56 can be organized into databases and used by machine learning to predict properties, build better potentials and design materials directly.

Chapter 57 — Materials Informatics

Part XI · Characterization, Computation and Design

A battery company wants a solid electrolyte for a lithium-metal cell: a crystalline solid that conducts lithium ions almost as well as a liquid, does not react with lithium metal at 0 V, does not oxidize at the 4.3 V cathode, can be sintered below 1000 °C and contains no germanium or other scarce elements. The design space is enormous. With five elements chosen from about sixty that are cheap and non-toxic, there are more than five million chemical systems before a single composition ratio or crystal structure is chosen. A synthesis chemist can make and test perhaps two hundred compositions a year; a density-functional calculation (Chapter 55) of one structure takes hours to days of computer time; a molecular-dynamics simulation of ionic conductivity (Chapter 56) takes longer still. Historically, the solid electrolytes now in prototype cells — garnet $Li_7La_3Zr_2O_{12}$, the thiophosphate $Li_{10}GeP_2S_{12}$ and the argyrodites — were found by chemical intuition, analogy and luck, each after years of work. The question that defines this chapter is whether the accumulated data of the discipline, organized in databases and learned by statistical models, can search such a space faster and more reliably than intuition alone, and how an engineer can tell when a data-driven prediction should be trusted.

> **What do materials databases contain, and how is a material turned into numbers that a learning algorithm can use?**
> **How accurate can machine-learned predictions of properties and interatomic forces be, and how are they used to screen, explore and invert the design problem? Where do data-driven methods fail, and how is that failure detected before it reaches a product?**

Concept	Engineering destination
Computed and experimental materials databases	Rapid estimates of stability, elastic constants, band gaps; choosing candidates before synthesis
Formation energy and energy above the convex hull	Will a new alloy phase, ceramic or electrolyte decompose?
Descriptors and features	Turning compositions, structures and processing routes into model inputs
Machine-learning property prediction, graph neural networks	Fast surrogates for DFT and for slow experiments (fatigue, creep, corrosion)
Machine-learned interatomic potentials	Near-DFT accuracy molecular dynamics of interfaces, diffusion, radiation damage
High-throughput screening and active learning	Battery, catalyst, thermoelectric and alloy discovery campaigns
Autonomous laboratories	Closed-loop synthesis and testing of coatings, thin films, polymers
Inverse design	Specifying the property and generating the material
Uncertainty quantification, domain of applicability	Deciding when a prediction may be used for design and when it must be tested

Roadmap. Where the data come from (57.1) → how materials become numbers (57.2) → learning properties from those numbers, from regression to graph networks (57.3) → learning the forces themselves (57.4) → using models to search: screening, active learning, autonomous labs (57.5) → running the problem backwards (57.6) → what the models cannot know (57.7).

57.1 Materials databases

From handbooks to queryable data

Materials engineers have always worked from compiled data: the steel handbooks, the ASM volumes, the JANAF thermochemical tables, the powder-diffraction files used in every X-ray laboratory (Chapter 52). What changed after about 2010 is that two kinds of data became available in quantity and in machine-readable form. The first is **computed data**: density-functional calculations (Chapter 55) run automatically, with identical settings, on hundreds of thousands to millions of crystal

structures. The second is **curated experimental data** that had existed for decades but became accessible through programming interfaces. A database differs from a handbook in that it can be queried by property ("all oxides with band gap above 3 eV, energy above hull below 25 meV/atom, and no element with supply risk") and fed directly to a model.

The main inorganic databases

Database	Content (order of magnitude, mid-2020s)	Origin of data	Typical use
ICSD (Inorganic Crystal Structure Database)	~3×10^5 experimentally determined crystal structures	Published diffraction studies, curated; licensed	Starting structures for calculations; "is this compound known?"
Materials Project	~1.5×10^5 inorganic compounds, with derived properties for many	High-throughput DFT (GGA, GGA+U, increasingly r^{2}SCAN)	Phase stability, elastic tensors, band structures, battery voltages, Pourbaix diagrams
AFLOW	~3×10^6 compounds, many hypothetical alloy prototypes	High-throughput DFT, prototype decoration	Alloy and intermetallic screening, thermal and elastic descriptors
OQMD (Open Quantum Materials Database)	~10^6 compounds	High-throughput DFT on ICSD and prototype structures	Formation energies, convex hulls, stability of hypothetical phases
Others	JARVIS (DFT and ML, including 2D materials), NOMAD (archive of raw calculations, >10^8 entries), Pauling File, NIST and Citrination-type experimental collections, Cambridge Structural Database for molecular crystals, PolyInfo for polymers	Mixed	Specialized properties, raw-data reuse

The computed databases share a workflow (Figure 57.1, left): take a structure from ICSD or generate it by substituting elements into a known prototype; relax it with a standard DFT recipe; compute total energy, and often band structure, elastic constants, dielectric tensor or phonons; and store the result with its full provenance (code version, functional, pseudopotentials, convergence settings). Uniform settings are the point. Two DFT energies can only be compared if computed the same way, and the databases add empirical energy corrections (for example for the over-binding of O_2 in GGA and for the GGA+U treatment of transition-metal oxides) so that oxides, metals and mixed systems can appear on one energy scale.

> **Boundary.**
> The DFT methods that generate these data, their functionals and their known errors are developed in Chapter 55; the thermodynamics of phase equilibria in Chapters 6 and 7. This chapter uses their outputs.

Formation energy and the convex hull

The most-used quantity in computed databases is the **formation energy** per atom: the energy of the compound minus the energies of its elements in their reference states, divided by the number of atoms. For a binary compound $\mathrm{A}_{1-x}\mathrm{B}_x$:

$$\Delta E_f(x) = \frac{E(\mathrm{A}_{n_A}\mathrm{B}_{n_B}) - n_A\,\mu_A^0 - n_B\,\mu_B^0}{n_A + n_B}$$

A negative formation energy says only that the compound is more stable than the separated elements. It does not say that the compound is stable, because it may be less stable than a mixture of *other compounds*. The correct test is the **convex hull**: plot ΔE_f against composition for every known phase in the system and construct the lower convex envelope (Figure 57.1, right). Phases on the hull are stable at 0 K; a phase above the hull will, given the kinetic opportunity, decompose into the hull phases at its composition. The vertical distance to the hull, the **energy above hull** E_{hull}, measures the driving force for that decomposition. It is the computational counterpart of the common-tangent construction on free-energy curves (Chapter 7), taken at 0 K and without entropy.

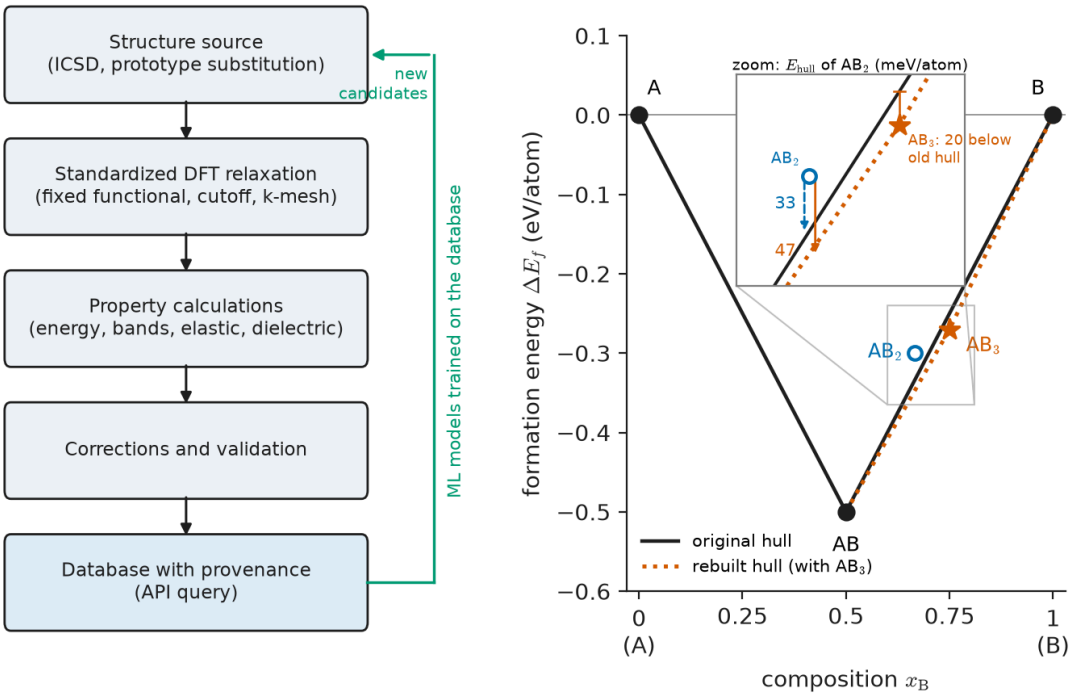


Figure 57.1 — Left: the high-throughput workflow that fills a computed materials database. Right: the 0 K convex hull of a binary A–B system, showing how the energy above hull is read.

Worked example — energy above hull, and why it depends on the database. In a binary A–B system the database contains elemental A and B (formation energy zero by definition) and the compound AB with $\Delta E_f = -0.50$ eV/atom at $x_B = 0.5$. A new calculation gives $\Delta E_f = -0.30$ eV/atom for AB_2 ($x_B = 2/3$). Is AB_2 stable?

The hull between AB and B is a straight line; at $x_B = 2/3$ it lies at

$$E_{\text{hull line}} = -0.50 + \frac{2/3 - 1/2}{1 - 1/2}\,(0 - (-0.50)) = -0.50 + 0.1667 = -0.333 \text{ eV/atom}$$

so $E_{\text{hull}} = -0.30 - (-0.333) = +0.033$ eV/atom, or 33 meV/atom. AB_2 is predicted to decompose into AB and B; the lever rule gives the fractions of atoms in each product as $(1 - 2/3)/(1 - 1/2) = 2/3$ in AB and $1/3$ in B. The formation energy of -0.30 eV/atom is strongly negative, yet the compound is metastable.

Now suppose a later calculation finds AB_3 at $x_B = 0.75$ with $\Delta E_f = -0.27$ eV/atom. The old hull at $x_B = 0.75$ lies at $-0.50 + 0.5 \times 0.50 = -0.25$ eV/atom, so AB_3 is 20 meV/atom *below* it: it is a new stable phase, and the hull must be rebuilt through AB, AB_3 and B. On the new hull, at $x_B = 2/3$,

$$E_{\text{hull line}} = -0.50 + \frac{2/3 - 1/2}{3/4 - 1/2}\,(-0.27 + 0.50) = -0.50 + 0.153 = -0.347 \text{ eV/atom}$$

and AB_2 now sits 47 meV/atom above the hull. Nothing about AB_2 changed; its stability label changed because the database learned about a competitor. Every E_{hull} in every database is therefore an upper bound on stability conditional on the completeness of that chemical system, and systems with few calculated phases give optimistic answers.

> **Status of the principle: convex-hull stability criterion**
> *Derived.* It follows exactly from minimizing the 0 K energy of a closed system at fixed composition; it is approximate as a statement about real materials because it neglects entropy (vibrational, configurational, magnetic), temperature and pressure, inherits the DFT functional error, and is only as complete as the set of competing phases computed.

How far above the hull is too far?

Many known, useful materials are metastable: diamond, martensite (Chapter 15), most glasses (Chapter 17), many thin-film phases grown by kinetic routes (Chapter 49). Statistical analysis of the ICSD against computed hulls shows that the large

majority of experimentally reported inorganic compounds lie within about 50–70 meV/atom of the hull for their chemical system, with oxides tighter than nitrides and metastable polymorphs reaching 100 meV/atom or more. Screening studies therefore use a threshold of order 25–100 meV/atom rather than zero, chosen to balance missed metastable phases against wasted synthesis attempts. Synthesizability is not the same as thermodynamic stability; it also depends on a synthesis route that avoids competing phases, a point returned to in Section 57.7.

Data quality, provenance and FAIR principles

Experimental databases mix measurements made with different methods, sample purities, grain sizes and test standards. A reported yield strength without the heat-treatment condition, a thermal conductivity without the porosity, a band gap without stating optical or electrical measurement: each is a number whose meaning has been lost. Computed databases have a different problem: values are internally consistent but systematically biased (GGA band gaps underestimated by 30–50 %, Chapter 55; lattice parameters overestimated by 1–2 %). The community response is summarized as **FAIR** data: findable, accessible, interoperable (standard formats and vocabularies) and reusable (with enough metadata to know what was measured or computed, and how). For an engineer, the practical rule is to know which of these two failure modes a dataset is exposed to before trusting any model trained on it.

57.2 Descriptors and features

The representation problem

A learning algorithm needs a fixed-length vector of numbers, or a graph, for each material. A crystal is a periodic arrangement of an arbitrary number of atoms; an alloy is a composition plus a processing history plus a microstructure; a polymer is a distribution of chain lengths with tacticity and branching (Chapter 19). The choice of **representation**, the mapping from material to numbers, often matters more than the choice of algorithm. A good representation is:

- **complete enough** that different materials with different properties map to different vectors;
- **invariant** to changes that do not change the material: translation and rotation of the structure, choice of unit cell, ordering of atoms in the input file;
- **smooth**, so that small changes in structure give small changes in the representation, which lets a model interpolate;
- **cheap**, at least much cheaper than the property it is used to predict.

Composition-based descriptors

When only the chemical formula is known, as at the start of a discovery campaign, the representation is built from elemental properties. For a compound with atomic fractions c_i and an elemental property p_i (atomic number, mass, period and group, Pauling electronegativity, covalent radius, number of valence electrons, melting point of the element), one forms statistics such as

$$\bar{p} = \sum_i c_i \, p_i, \qquad \delta p = \sqrt{\sum_i c_i \, (p_i - \bar{p})^2}, \qquad \max_i p_i - \min_i p_i$$

A typical composition feature set (the widely used "Magpie" set is an example) has about 130 such numbers. Several are classical metallurgical descriptors in disguise: the mean valence electron concentration sets the stability of Hume-Rothery phases and the fcc/bcc competition in high-entropy alloys (Chapter 16); the atomic-size mismatch δr is the Hume-Rothery size criterion (Chapter 7) and the solid-solution strengthening parameter (Chapter 13); the electronegativity spread distinguishes ionic from metallic compounds. Composition models cannot distinguish polymorphs (graphite and diamond have the same formula) and are blind to processing, which limits them to properties dominated by chemistry.

Structure-based descriptors

When atomic positions are known, the representation can describe local environments:

- **Radial distribution and coordination features**: counts of neighbours within shells, bond-length and bond-angle histograms, Voronoi coordination numbers.

- **The Coulomb matrix and its periodic extensions**: entries $Z_i Z_j / |\mathbf{r}_i - \mathbf{r}_j|$, sorted or diagonalized to remove atom-ordering dependence.
- **Smooth Overlap of Atomic Positions (SOAP)**: each atom's neighbourhood within a cutoff radius (typically 4–6 Å) is represented as a sum of Gaussians on the neighbours, expanded in radial functions and spherical harmonics; the rotationally invariant **power spectrum** of the expansion coefficients is the descriptor. Two environments can then be compared through a kernel, the normalized overlap of their power spectra. SOAP is complete enough for most purposes and underlies the GAP potentials of Section 57.4.
- **Symmetry functions** (Behler–Parrinello): sums over neighbours of radial Gaussians and angular terms, with a smooth cutoff, giving a fixed-length vector per atom.

The common idea is **locality**: the energy and many properties of a solid are, to a good approximation, sums of contributions from atoms in their local environments. This is the same nearsightedness that makes linear-scaling electronic-structure methods possible, and it fails where long-range physics dominates (Section 57.4).

Descriptors beyond the crystal

For engineering alloys, the most important variables are often not atomic at all: solution-treatment temperature, cooling rate, ageing time, cold work, grain size, inclusion content, specimen orientation. Datasets for steel fatigue, nickel-superalloy creep or aluminium-alloy strength use these processing parameters directly as features, sometimes with physically motivated combinations (the Larson–Miller parameter of Chapter 14 is a hand-built feature that collapses time and temperature into one number; the Hollomon–Jaffe tempering parameter of Chapter 15 is another). Polymer informatics uses fingerprints of the repeat unit (counts of chemical fragments, SMILES-string encodings) together with molecular weight. Microstructures enter as images, through two-point correlation functions or features learned by convolutional networks from micrographs (Chapter 52). Building such features is where materials knowledge enters a data-driven model most directly, and a model with a few physically chosen features frequently beats one with hundreds of generic ones on a small dataset.

> **Status of the principle: locality of atomic contributions**
> *Approximate.* Valid when electronic screening confines interactions to a few ångström (most metals, covalent solids, many ionic solids at the level of short-range forces); it fails for long-range electrostatics, charge transfer across interfaces, van der Waals interactions in layered materials and properties set by delocalized electronic states unless those terms are added explicitly.

57.3 Machine-learning property prediction

Supervised learning in one paragraph

Given N examples $(\mathbf{x}_i, y_i)$, with $\mathbf{x}_i$ the representation and y_i a property (formation energy, band gap, bulk modulus, yield strength, glass-transition temperature), supervised learning finds a function $f(\mathbf{x})$ that minimizes a loss on the training data plus a penalty on complexity, for example

$$\min_{f} \; \frac{1}{N} \sum_{i=1}^{N} \big(y_i - f(\mathbf{x}_i) \big)^2 + \lambda \, \Omega(f)$$

The regularization term $\lambda \, \Omega(f)$ keeps the function smooth enough to generalize. The algorithms used in materials work, in rough order of data appetite:

Method	How it works	Data needed (typical)	Strengths and weaknesses
Linear and sparse regression (LASSO, SISSO)	Linear combination of features, or of algebraic combinations of features selected for sparsity	10–10^3	Interpretable formulas; misses complex interactions
Kernel ridge regression, Gaussian process (GP)	Prediction as weighted sum of similarities (kernels) to training points	10^2–10^4	Smooth; GP gives uncertainty; cost grows as N^3
Random forest, gradient-boosted trees	Ensembles of decision trees on features	10^2–10^5	Robust on tabular processing data; cannot extrapolate beyond training range
Neural networks on descriptors	Layers of linear maps and nonlinear activations	10^3–10^6	Flexible; need care to avoid overfitting
Graph neural networks (GNNs)	Learn the representation itself from the atomic graph	10^4–10^7	State of the art for structure–property prediction on large DFT sets

The Gaussian process deserves a sentence more because Section 57.5 depends on it. A GP treats the unknown function as a random function whose values at any set of points are jointly Gaussian with covariance given by a kernel $k(\mathbf{x}, \mathbf{x}')$, for example $k = \sigma_f^2 \exp(-|\mathbf{x}-\mathbf{x}'|^2/2\ell^2)$. Conditioning on the training data gives, at a new point, both a mean prediction $\mu(\mathbf{x})$ and a standard deviation $\sigma(\mathbf{x})$. The standard deviation is small near training data and grows toward the prior value far from them: the model knows, in a limited sense, where it has not looked.

Training, validation and the metrics that matter

A model is judged on data it has not seen. The data are split into training and test sets, or rotated through k-fold cross-validation, and three metrics are reported:

$$\mathrm{MAE} = \frac{1}{N}\sum_i |\hat{y}_i - y_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i (\hat{y}_i - y_i)^2}, \qquad R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (y_i - \bar{y})^2}$$

The MAE is in the units of the property and is the easiest to interpret; the RMSE weights large errors more heavily; R^2 compares the model with the trivial predictor that always returns the mean. A high R^2 on a property spanning a wide range can hide errors that are large for the decision at hand, so MAE should always be compared with the **resolution the application requires**.

Worked example — reading a parity plot. A graph-network model predicts band gaps for eight test compounds whose DFT (HSE) gaps are 0.00, 0.45, 1.12, 1.80, 2.35, 3.10, 4.05 and 5.60 eV; the model returns 0.12, 0.30, 1.30, 1.62, 2.61, 2.88, 3.70 and 4.95 eV (Figure 57.2, left). The errors $\hat{y} - y$ are $+0.12, -0.15, +0.18, -0.18, +0.26, -0.22, -0.35, -0.65$ eV. Then

$$\mathrm{MAE} = \frac{2.11}{8} = 0.264 \text{ eV}, \qquad \mathrm{RMSE} = \sqrt{\frac{0.763}{8}} = 0.309 \text{ eV}$$

The mean of the reference gaps is 2.309 eV and their total sum of squares 24.95 eV^2, so $R^2 = 1 - 0.763/24.95 = 0.969$. The baseline that always predicts 2.31 eV has an MAE of 1.47 eV, so the model is far better than knowing nothing. But three further observations matter more than the headline R^2. The RMSE exceeds the MAE noticeably, a sign that a few errors are large. The mean error is -0.12 eV and the two largest-gap compounds are both under-predicted, by 0.35 and 0.65 eV: a systematic bias at the edge of the data, where wide-gap materials are rare in the training set. And for an application that must distinguish a 1.1 eV absorber for photovoltaics from a 1.6 eV one for a tandem top cell (Chapter 40), an MAE of 0.26 eV is a coarse filter, not an answer.

How error falls with data: learning curves

For a fixed model and representation, test error typically falls as a power of the training-set size,

$$\mathrm{MAE}(N) \approx a\, N^{-b}$$

with b between about 0.2 and 0.5 for materials properties (Figure 57.2, right). With $b = 0.3$, ten times more data cuts the error by a factor $10^{-0.3} = 0.50$; a hundred times more cuts it by a factor $10^{-0.6} = 0.25$, to a quarter. A better representation shifts the curve down or steepens it; data with irreducible noise make it flatten at the noise floor. For DFT-trained models the floor is not

the DFT-versus-experiment error; that error is invisible to the model, which learns DFT, not nature. Graph networks trained on 10^5 Materials Project entries reach formation-energy MAEs of about 0.02–0.04 eV/atom against DFT, well below the typical DFT-versus-experiment discrepancy of about 0.1 eV/atom for formation enthalpies.

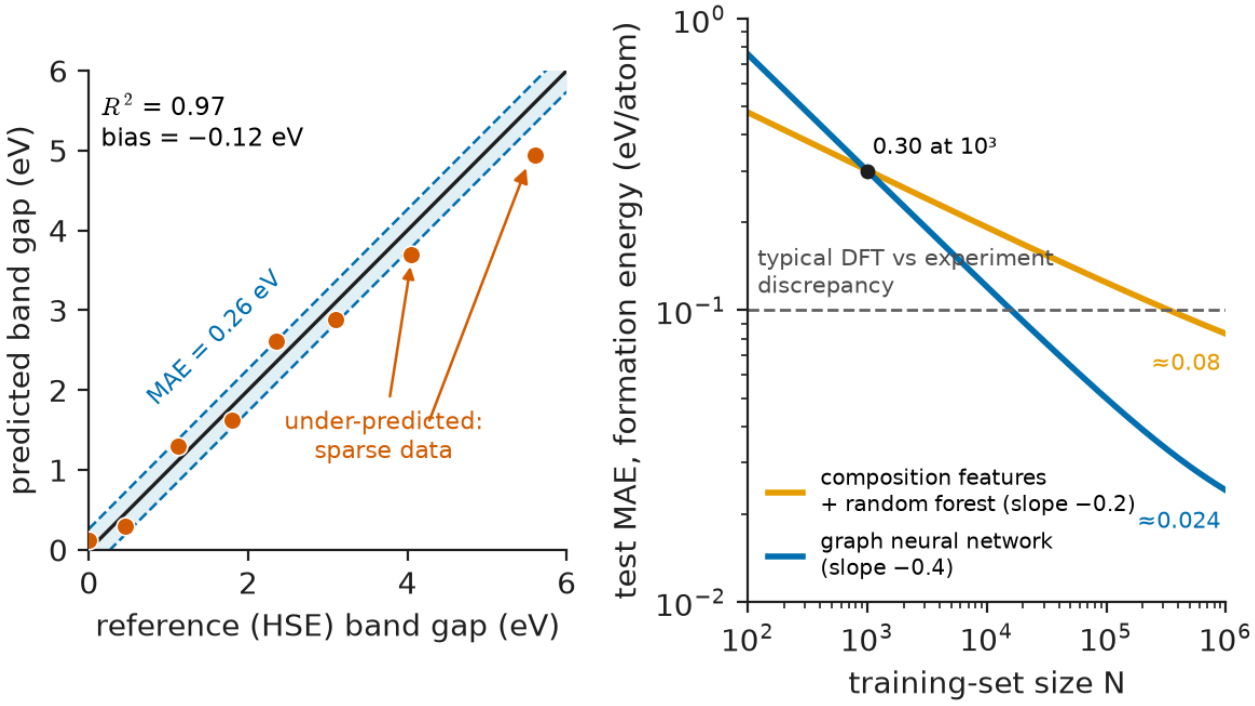


Figure 57.2 — Left: parity plot of predicted against reference band gaps for the worked example. Right: learning curves showing test error falling as a power of training-set size for two representations.

Graph neural networks

A crystal is naturally a graph: atoms are nodes, and edges join atoms closer than a cutoff radius (typically 5–8 Å), including periodic images. **Graph neural networks** learn the representation instead of having it designed. Each atom i starts with a feature vector $\mathbf{h}_i^{(0)}$ encoding its element; each edge carries the interatomic distance (expanded in Gaussian or Bessel basis functions) and, in some architectures, angles. In each **message-passing** layer, every atom collects messages from its neighbours and updates its state:

$$\mathbf{h}_i^{(t+1)} = U\left(\mathbf{h}_i^{(t)}, \sum_{j \in \mathcal{N}(i)} M\left(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}\right)\right)$$

where M and U are small neural networks with learned weights shared across all atoms. After T layers (typically three to six), each atom's state encodes its environment out to roughly T times the cutoff. A **readout** sums or averages the atomic states and maps them to the property. Because the sum over neighbours is unordered and uses only distances (and angles), the output is automatically invariant to atom ordering, translation and rotation.

The first widely used models, CGCNN (2018) and MEGNet (2019), demonstrated DFT-level accuracy for formation energies across the Materials Project; later models (ALIGNN with explicit bond angles, equivariant networks of the kind in Section 57.4) improved accuracy further. Graph networks are now the default for properties of crystals where the training set has more than about 10^4 entries. On small experimental datasets, a few hundred measured toughness or creep values, they offer little over random forests with good features, and the representation-learning advantage disappears.

> **Status of the principle: power-law learning curves**
> *Empirical.* Observed across many datasets and model families for interpolation within a fixed distribution of materials; the exponent is not universal, and the curve says nothing about error on materials unlike the training set.

Pitfalls that make models look better than they are

- **Data leakage.** If the test set contains near-duplicates of training entries (the same compound at slightly different relaxation, polymorphs of one composition, the same alloy at two test temperatures), the test error measures memorization. Splitting by chemical system, by structure prototype or by alloy family (grouped cross-validation) gives an honest, and usually larger, error.
- **Target leakage.** A feature that encodes the answer (using measured density to predict porosity-dependent conductivity, or a processing step that is only applied to the best alloys) produces impressive and useless models.
- **Selection bias.** Published data over-represent successes. A model trained on reported superconductors has seen few non-superconductors, so it learns to predict high T_c for nearly everything.
- **Distribution shift.** A model trained on bulk oxides applied to sulfide surfaces, or on wrought alloys applied to additively manufactured ones (Chapter 48, where porosity and texture differ), is extrapolating, whatever the test error was.

57.4 Machine-learned interatomic potentials

The gap they fill

Molecular dynamics (Chapter 56) needs forces on every atom at every time step, for 10^6–10^9 steps. DFT gives accurate forces but its cost scales roughly as N^3 with the number of atoms and limits simulations to a few hundred atoms for tens of picoseconds. Classical potentials (Lennard-Jones, EAM, Tersoff, ReaxFF; Chapter 56) scale linearly and reach millions of atoms, but their fixed functional forms limit accuracy and transferability: an EAM potential fitted for bulk aluminium may give poor surface energies, stacking-fault energies or liquid structure. **Machine-learned interatomic potentials (MLIPs)** keep the linear scaling and locality of classical potentials but replace the fixed functional form by a flexible regression fitted to DFT energies, forces and stresses.

The common structure

All MLIPs write the total energy as a sum of atomic contributions, each a learned function of the local environment within a cutoff r_c:

$$E = \sum_{i=1}^{N} \varepsilon\big(\{\mathbf{r}_{ij}\}_{r_{ij}<r_c}\big), \qquad \mathbf{F}_i = -\frac{\partial E}{\partial \mathbf{r}_i}$$

Forces are obtained by differentiating the energy, so energy is conserved in dynamics. Training minimizes a weighted loss over a DFT dataset, typically

$$\mathcal{L} = w_E \sum_{\text{configs}} \left(\frac{E - E^{\text{DFT}}}{N} \right)^2 + w_F \sum_{\text{atoms}} |\mathbf{F}_i - \mathbf{F}_i^{\text{DFT}}|^2 + w_\sigma \sum_{\text{configs}} \|\boldsymbol{\sigma} - \boldsymbol{\sigma}^{\text{DFT}}\|^2$$

Forces are especially valuable training data: one DFT calculation on a 100-atom cell gives one energy but 300 force components.

Three generations of models

Potential	Representation and regression	Characteristics
Behler–Parrinello neural network (2007)	Atom-centred symmetry functions → per-element feed-forward network	First general MLIP; needs careful choice of symmetry functions
GAP (Gaussian Approximation Potential, 2010)	SOAP descriptors → Gaussian-process (kernel) regression on sparse set of reference environments	Data-efficient; gives a variance that flags unfamiliar environments; used for Si, C, W, Fe, amorphous phases, radiation damage
Moment Tensor Potentials, ACE (Atomic Cluster Expansion)	Systematic many-body polynomial expansions with linear fitting	Fast, systematically improvable, close relatives of the equivariant networks
NequIP (2022)	E(3)-equivariant message-passing network: features carry vector and tensor character that rotates with the structure	Very data-efficient: chemical accuracy from a few hundred to a few thousand configurations
MACE (2022)	Equivariant message passing with higher body-order messages built on the ACE basis; typically two layers	High accuracy with short effective receptive field; basis of widely used "foundation" potentials

Equivariance is the key idea of the latest generation. The energy is invariant under rotation, but forces are vectors that rotate with the structure. Earlier networks built only invariant features (distances, angles) and lost directional information; equivariant networks carry internal features that transform like vectors and tensors under rotation, so that geometric information is preserved through the layers and the model needs far fewer examples to learn it. Typical accuracies for a well-trained MLIP on its target system are 1–5 meV/atom in energy and 0.03–0.1 eV/Å in force against the reference DFT.

Universal potentials

Trained on the relaxation trajectories stored in databases such as the Materials Project (about 1.5 million configurations), models such as M3GNet, CHGNet and MACE-MP-0 cover most of the periodic table. They are less accurate on any particular system than a dedicated potential, with typical force errors of 0.05–0.1 eV/Å and a systematic softening of phonon frequencies inherited from training data dominated by near-equilibrium structures, but they run stable molecular dynamics on almost any inorganic material out of the box. They are used for pre-relaxing candidate structures before DFT, for screening stability and ionic diffusion, and as starting points that are fine-tuned with a few hundred system-specific DFT calculations.

What MLIPs make possible, and their failure modes

Speed is the reason they matter. A MACE or NequIP evaluation costs of order 10$^{-5}–10^{-4}$ s per atom per step on a GPU, against of order 0.1–10 s per atom per step for DFT molecular dynamics; with linear scaling, systems of 10^5–10^6 atoms and nanosecond times become practical. That opens problems that are out of reach for DFT and beyond the accuracy of classical potentials: dislocation core structure and mobility in tungsten and high-entropy alloys (Chapters 12 and 16), lithium diffusion and interface reactions in solid electrolytes (Chapter 40), amorphous silicon and carbon structures, phase transitions in perovskites, hydrogen embrittlement mechanisms (Chapter 44), and primary radiation damage cascades in reactor steels (Chapter 41).

Failure modes follow from the construction:

- **Extrapolation**: an MLIP has no physics outside its training data. A potential trained on bulk crystals may let atoms collapse together at short distances never sampled, so short-range repulsion is often added by hand (a ZBL screened Coulomb term for radiation damage).
- **Long-range effects**: a strictly local model misses long-range electrostatics and charge transfer. Space-charge layers at electrode interfaces, polar surfaces and charged defects need explicit electrostatic terms or charge-equilibration schemes.
- **Inherited DFT error**: the potential is at best as good as its functional. A GGA-trained potential for water or for magnetic iron carries GGA's errors faithfully.

Active learning is the standard remedy for extrapolation: run dynamics with the current potential, monitor an uncertainty indicator (GP variance in GAP, disagreement among an ensemble of networks, or a descriptor-space distance), stop when it

exceeds a threshold, compute the flagged configurations with DFT, add them to the training set and retrain. The same loop drives materials discovery in the next section.

> **Status of the principle: machine-learned interatomic potentials**
> *Approximate.* They reproduce the reference electronic-structure method within stated errors for configurations similar to their training data, under the locality assumption; accuracy elsewhere is not guaranteed and must be tested, not assumed.

57.5 High-throughput screening, active learning and autonomous laboratories

The screening funnel

High-throughput screening applies a sequence of filters of increasing cost and accuracy to a large candidate set (Figure 57.3). For a solid lithium electrolyte the funnel might be:

1. **Enumerate** candidates: known ICSD lithium compounds plus hypothetical ones generated by substituting elements into known structure prototypes, perhaps 10^5 structures.
2. **Composition and rule filters** (cost: negligible): exclude toxic, scarce or radioactive elements; require charge neutrality and lithium content above a threshold.
3. **ML stability filter** (cost: milliseconds to a second per structure): a universal potential relaxes each structure and a GNN predicts E_{hull}; keep those below a threshold.
4. **DFT stability and electrochemical window** (hours per structure): compute E_{hull} and the voltage range in which the compound is stable against lithium uptake or loss, using the grand-potential phase diagram.
5. **Ionic conductivity** (days per structure): MLIP or ab initio molecular dynamics to compute the lithium diffusion coefficient and, through the Nernst–Einstein relation, the conductivity (Chapters 8 and 40).
6. **Synthesis and testing** of a handful of survivors.

Each stage removes most of the remaining candidates, so expensive methods are used only where they matter. The funnel is the computational form of the screening and ranking stages of materials selection (Chapter 58).

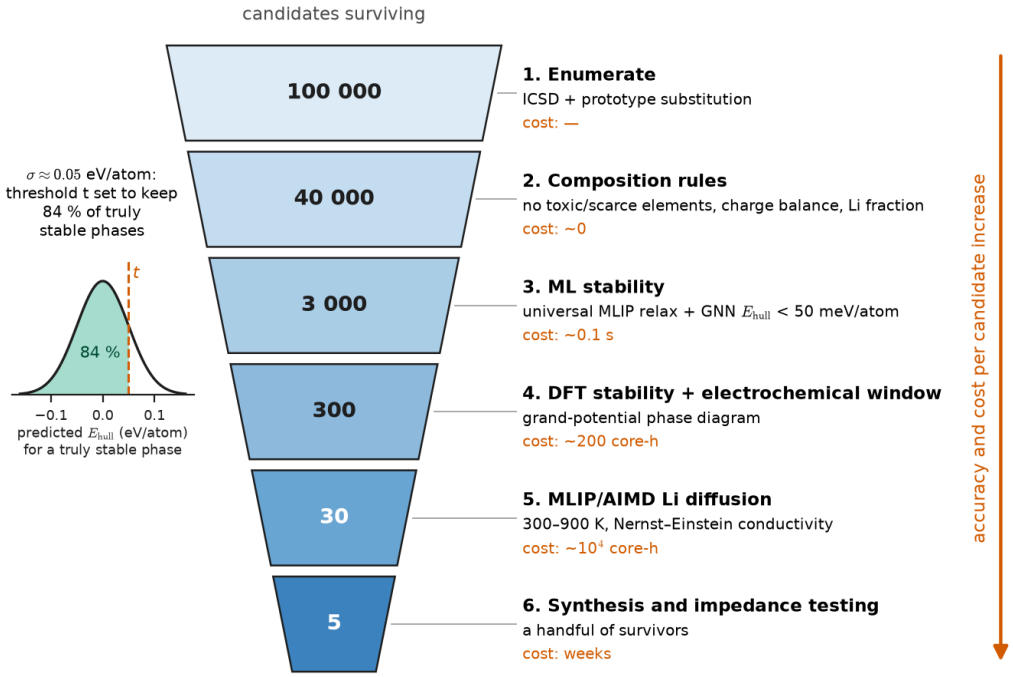


Figure 57.3 — A high-throughput screening funnel for solid lithium electrolytes: each stage is more expensive and more accurate, and passes fewer candidates.

Worked example — setting a screening threshold that does not discard the answer. In the funnel above, the ML stage predicts E_{hull} for 10^5 structures with an error that is approximately Gaussian with standard deviation $\sigma = 0.05$ eV/atom. The team wants every truly stable compound ($E_{\mathrm{hull}} = 0$) to reach the DFT stage. If the threshold were set at 0, a truly stable compound would pass with probability $\Phi(0/\sigma) = 0.50$: half of the answers would be thrown away by the cheapest filter. With threshold t, a truly stable compound passes with probability $\Phi(t/\sigma)$:

Threshold t (eV/atom)	t/σ	Probability a truly stable compound passes
0.00	0	0.50
0.05	1	0.84
0.10	2	0.977

Raising the threshold also passes more unstable compounds. Suppose 3000 structures pass at $t = 0.05$ and 7000 at $t = 0.10$. At about 200 core-hours per DFT stability calculation (relaxation plus a static calculation, Chapter 55), the DFT stage costs $3000 \times 200 = 6 \times 10^5$ core-hours at the lower threshold and 1.4×10^6 at the higher one, against 2×10^7 core-hours to compute all 10^5 structures directly. The ML stage itself, at about 0.1 s per structure, takes $10^5 \times 0.1/3600 \approx 2.8$ GPU-hours. The decision is an engineering trade: more than doubling the DFT budget (×2.3) raises the recall of stable phases from 84 % to about 98 %. A cheap filter is only useful if its threshold is set from its error, not from the target value.

Active learning and Bayesian optimization

Screening evaluates everything that passes each filter. When each evaluation is expensive (a synthesis, a fatigue test, a week of molecular dynamics) and the candidate space is continuous (alloy compositions, processing temperatures and times), it is better to choose each next experiment so that it is maximally informative. This is **active learning**, and when the aim is to find the best material rather than to map the whole space, **Bayesian optimization**:

1. Fit a surrogate model, usually a Gaussian process, to the measurements so far; it gives a mean $\mu(\mathbf{x})$ and uncertainty $\sigma(\mathbf{x})$ everywhere.
2. Compute an **acquisition function** that scores each untested candidate by how useful testing it would be.
3. Test the candidate with the highest score, add the result, refit, and repeat.

The acquisition function balances **exploitation** (test where μ is high) and **exploration** (test where σ is large). The most common choice, **expected improvement**, is the expected amount by which a candidate will beat the best value f^* found so far, for a property to be maximized:

$$\mathrm{EI}(\mathbf{x}) = (\mu - f^*)\,\Phi(z) + \sigma\,\phi(z), \qquad z = \frac{\mu - f^*}{\sigma}$$

with Φ and ϕ the standard normal cumulative distribution and density. The **upper confidence bound** $\mu + \kappa\sigma$ is a simpler alternative with an explicit exploration weight κ.

Worked example — which experiment next? A team optimizing the composition of a thermoelectric is maximizing a normalized figure of merit; the best measured so far is $f^* = 0.80$. The GP proposes three candidates:

Candidate	μ	σ	z	EI
A (near the current best)	0.82	0.10	0.20	0.051
B (unexplored corner)	0.70	0.25	−0.40	0.058
C (between two tested points)	0.79	0.02	−0.50	0.004

For A, $\mathrm{EI} = 0.02\,\Phi(0.2) + 0.10\,\phi(0.2) = 0.02 \times 0.579 + 0.10 \times 0.391 = 0.051$. For B, $\mathrm{EI} = -0.10\,\Phi(-0.4) + 0.25\,\phi(-0.4) = -0.10 \times 0.345 + 0.25 \times 0.368 = 0.058$. For C, $\mathrm{EI} = -0.01 \times 0.309 + 0.02 \times 0.352 = 0.004$. Candidate B has the lowest predicted value but the highest expected improvement, because its large uncertainty leaves a real chance of a much better result. Candidate C, almost certain to be slightly worse than the best, is nearly worthless to test. Over many iterations this logic finds optima in tens of experiments where grid searches need hundreds; reported campaigns on alloy compositions, perovskite film processing and polymer formulations have typically needed three to ten times fewer experiments than one-factor-at-a-time or grid designs. The result depends on the surrogate's uncertainty being honest (Section 57.7).

Status of the principle: Bayesian optimization
Approximate. A principled decision rule given the surrogate model; its efficiency depends on whether the GP kernel and noise model describe the real property landscape, and it degrades in high-dimensional spaces (more than about 10–20 variables) unless the representation is reduced.

Autonomous laboratories

An **autonomous** or **self-driving laboratory** closes the loop physically: robots dispense precursors, synthesize (by solid-state reaction, spin coating, sputtering, electrodeposition or polymerization), characterize (X-ray diffraction, optical spectroscopy, conductivity, Chapter 52), and an algorithm analyses the data and chooses the next experiment without human intervention (Figure 57.4). Systems in operation include flow reactors for polymers and nanoparticles, combinatorial sputtering with automated diffraction for thin-film alloys, robot arms handling powders and furnaces for inorganic synthesis, and automated electrochemistry for corrosion and battery-electrolyte formulations.

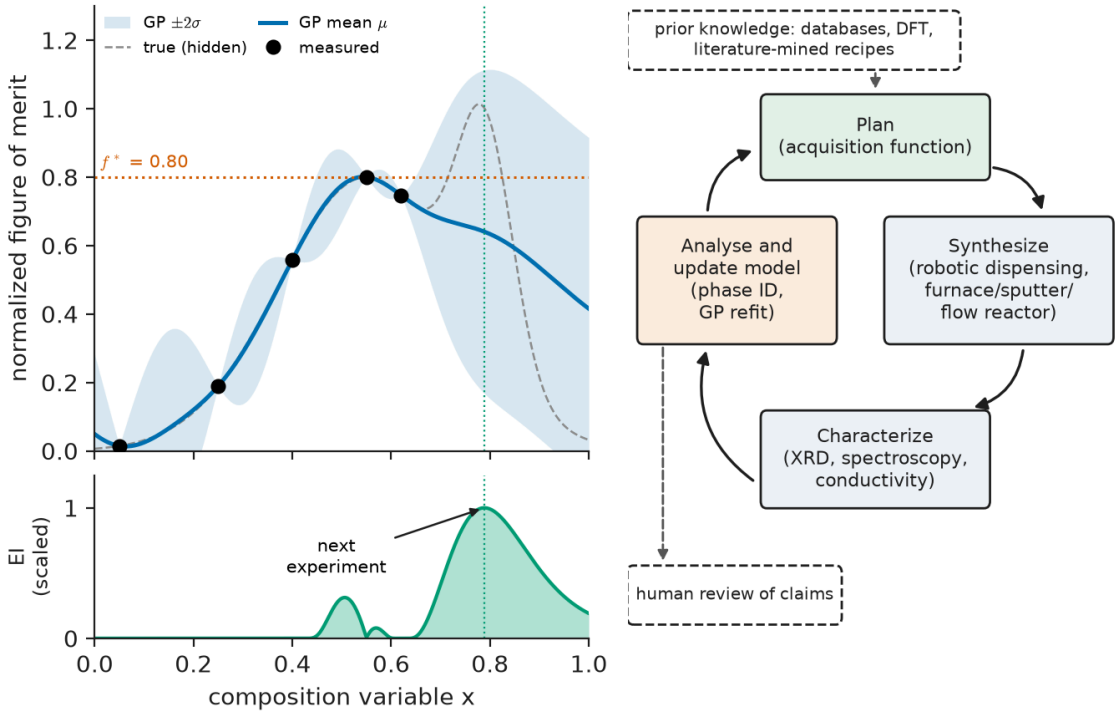


Figure 57.4 — Bayesian optimization in one dimension, and the closed loop of an autonomous laboratory.

The benefit is throughput and reproducibility: a robotic platform runs day and night, records every parameter, and logs failures as faithfully as successes, which gives the unbiased data that literature-trained models lack. The limitations are equally concrete. Each platform automates a narrow chemistry and set of processes; solids handling (grinding, pelletizing, sintering) is much harder to automate than liquids; and automated analysis of characterization data is a weak link. A widely publicized 2023 autonomous synthesis campaign reported that most of its several dozen target inorganic compounds had been made, and subsequent re-analysis of its diffraction data questioned many of the phase identifications. The episode is a reminder that automation accelerates mistakes as efficiently as discoveries: claims of new phases still need the standard of evidence of Chapter 52, including refinement of diffraction patterns and independent confirmation.

57.6 Inverse design

Running the structure–property map backwards

The rest of this book runs from structure to property: given a microstructure or crystal structure, predict the modulus, conductivity or toughness. Engineering usually needs the inverse: given the required properties, find a structure (and a process

to make it) that delivers them. The forward map is many-to-one and the inverse is ill-posed: many structures may satisfy the specification, or none, and small changes in requirement can jump between unrelated solutions. Three approaches are in use.

Optimization in a parameterized space. If the design space can be described by continuous variables (alloy composition, layer thicknesses of a coating, volume fraction and aspect ratio of a reinforcement, lattice-architecture parameters for an additively manufactured part), a surrogate model plus an optimizer (Bayesian optimization, genetic algorithms, gradient methods if the surrogate is differentiable) searches it directly. Topology optimization of structures and multilayer optical coating design are mature examples. Most alloy-design successes of data-driven methods are of this kind: compositions of high-entropy alloys with targeted phase stability and strength, low-expansion Invar-type alloys, and nickel superalloys balancing γ' fraction, density and cost, often with CALPHAD (Chapter 56) providing physics-based features or constraints.

Generative models. For discrete crystal structures, generative models learn the distribution of known stable materials and sample new ones from it. Variational autoencoders map structures to a continuous latent space in which one can move and decode; **diffusion models**, which learn to reverse a process that progressively adds noise to atomic positions, lattice and species, generate periodic structures directly and can be conditioned on target properties (band gap, magnetic density, bulk modulus, chemical system, symmetry). Generated candidates are relaxed with an MLIP, checked against the convex hull with DFT, and screened for novelty against databases. Reported success rates, the fraction of generated structures that are stable, unique and new after DFT, are of order tens of percent for recent models, much higher than random substitution, though still dominated by close relatives of known structural families.

Property-to-structure for microstructure and architecture. At larger scales, inverse design specifies a microstructure or architecture: the lattice geometry of a 3D-printed bone scaffold with a target modulus near that of trabecular bone (Chapter 39), the fibre layup of a composite wing skin (Chapter 23), the pore structure of a catalyst support. Here the forward model is usually homogenization or finite elements, and the inverse step is classical optimization.

Why inverse design is harder than it sounds

A target property is only part of a specification. The generated material must also be thermodynamically accessible, synthesizable with available precursors, stable in service (against oxidation, moisture, radiation), processable into the required form, and made of affordable, non-critical elements (Chapter 59). Most of these are not in the model's training data. In practice inverse design proposes, and the screening funnel, the experimental loop and the documentation stage of materials selection (Chapter 58) dispose.

> **Status of the principle: generative inverse design**
> *Approximate.* A sampling method whose outputs are hypotheses; their stability is checked by DFT within its errors and their existence by synthesis, and the generator cannot reliably propose chemistries far outside the distribution it was trained on.

57.7 Uncertainty and the limits of data-driven discovery

Two kinds of uncertainty

Every prediction carries two kinds of uncertainty:

- **Aleatoric** uncertainty is noise in the data themselves: scatter in measured fatigue lives (Chapter 14), sample-to-sample variation in thermal conductivity, numerical noise in DFT. It cannot be reduced by more data of the same kind, only by better measurements.
- **Epistemic** uncertainty is the model's ignorance: lack of training data near the query, or a model form that cannot represent the true function. It shrinks with more relevant data.

Methods for estimating epistemic uncertainty include Gaussian-process variances, the spread of an **ensemble** of models trained on different data subsets or with different initializations, Monte Carlo dropout in neural networks, and distance-based measures (how far is the query from its nearest training neighbours in descriptor space?). None is perfect: ensembles can agree confidently while all being wrong, and GP variances depend on the kernel assumed. **Calibration** checks whether stated uncertainties are

honest: of the predictions made with a stated 90 % interval, about 90 % of test values should fall inside. Many published models are overconfident far from their data.

Conformal prediction offers a distribution-free alternative for calibration: hold out a calibration set, compute the residuals, and use their quantiles to put intervals around new predictions, with guaranteed coverage on average, provided new queries come from the same distribution as the calibration data. That proviso is the whole difficulty.

Interpolation versus discovery

The deepest limitation is structural. Discovery means finding materials unlike those already known, whereas statistical learning works best on materials like those in the training set. A model trained on known superconductors would not have predicted the cuprates in 1986 or MgB_2 in 2001; a model trained on oxide electrolytes would not propose sulfide argyrodites. Data-driven methods therefore excel at **interpolation and dense search** within known families (optimizing a composition, finding the best member of a structure type, mapping a processing window) and are weakest at the conceptual leaps that create new families. The **domain of applicability** of a model, the region of descriptor space where its errors have been measured, should accompany every prediction used for design, and a prediction outside it should be treated as a hypothesis.

Biases in the data

- **Publication bias**: failed syntheses and non-functional compositions are rarely reported, so training data over-represent success.
- **Anthropogenic bias**: research attention concentrates on fashionable chemistries; a few thousand compounds account for a large share of literature data.
- **Method bias**: DFT databases inherit functional errors (band gaps, magnetic and strongly correlated systems, van der Waals layered structures, Chapter 55); experimental databases mix test methods.
- **Condition bias**: computed properties are for perfect crystals at 0 K; engineering properties depend on defects, microstructure and temperature (Chapters 5, 13 and 14). No database of perfect crystals can predict fatigue strength, which is controlled by inclusions and surface finish.

Synthesizability and processing

A material predicted to be stable, with excellent properties, may still be impossible to make with the required phase purity, grain size, density or form. Models trained on text-mined synthesis recipes can suggest precursors and temperatures, but synthesis outcomes depend on kinetics, nucleation and competing phases (Chapters 9 and 10) that are poorly represented in data. For engineering materials, processing defines the structure, and the structure defines most of the properties that matter in service. That is the main reason why informatics has so far had more impact on functional materials, where properties follow chemistry and crystal structure fairly directly, than on structural ones, where they follow microstructure.

The engineer's checklist

Before a data-driven prediction is used to commit time or money:

1. What data was the model trained on, and does the query lie within that distribution?
2. How was the test error measured: random split or grouped by chemistry or family?
3. Is the error small compared with the difference the decision depends on?
4. Is an uncertainty reported, and has it been calibrated?
5. Does the prediction violate known physics (negative elastic constants, a band gap above the ionization energy, a density above the close-packed limit)?
6. Has the top candidate been confirmed by the next level of the hierarchy: DFT for an ML prediction, experiment for a DFT prediction?

> **Status of the principle: data-driven property prediction**
> *Empirical.* Machine-learned models are fitted to observation (computed or measured); their reliability is established only by testing on held-out data from the same distribution, and does not transfer to new chemistries, structures or processing routes without new evidence.

Where informatics fits in the discipline

Informatics does not replace the physical understanding developed in the rest of this book; it depends on it. The descriptors that work best are the classical ones (size mismatch, electron concentration, electronegativity, the Larson–Miller parameter); the most reliable training data come from the physics-based methods of Chapters 55 and 56 or from careful characterization (Chapters 52–54); and the predictions must be filtered by thermodynamics, kinetics, processing and selection criteria that the models do not know. Used this way, as a fast and statistically honest layer on top of physical models and experiments, it shortens the path from requirement to material from decades to years, and in favourable cases to months.

What remains open

Open question	Where it is answered
How the DFT energies, band gaps and elastic constants in the databases are computed, and their systematic errors	Chapter 55
Classical potentials, molecular dynamics, kinetic Monte Carlo, phase-field and CALPHAD, which MLIPs and ML surrogates accelerate	Chapter 56
Common-tangent construction and phase diagrams behind the convex hull	Chapters 6 and 7
Why stability is not synthesizability: nucleation, growth and metastable phase formation	Chapters 9 and 10
Turning predicted properties into a ranked material choice with costs, shapes and processes	Chapter 58
Adding embodied carbon, criticality and recyclability as objectives and constraints in screening	Chapter 59
Characterization data that train and validate models, and automated analysis of diffraction and microscopy	Chapters 52–54
Statistical learning theory, Gaussian processes and Bayesian inference in full	The Mathematics Universe
Quantum-mechanical foundations of the electronic structure that the models learn	Engineering Physics

Next: Chapter 58 turns from finding materials to choosing among them, deriving the performance indices, charts, shape factors and cost models by which a design requirement becomes a ranked list of materials and processes.

Chapter 58 — Materials Selection and Design

Part XI · Characterization, Computation and Design

A car maker redesigning the bonnet of a mid-size saloon has four candidates on the table: the incumbent bake-hardening steel, a 6xxx-series aluminium sheet, a magnesium alloy, and carbon-fibre-reinforced epoxy. Each has a champion in the company. The steel engineers point out that their panel is the cheapest, welds on the existing line and is recycled at over 90 %. The aluminium team says their panel is half the mass. The composites group says theirs is lighter still. The pedestrian-safety team asks which bonnet absorbs a head impact best; the paint shop asks which survives the 180 °C bake; the corrosion engineers ask what happens where aluminium touches the steel hinge; the finance department asks what a kilogram saved is actually worth. All four claims are true and none settles the question, because each answers a different question. This chapter provides the method that makes such debates resolvable: translate the design into a function, an objective and constraints; derive the combination of properties that controls performance; display every candidate on property charts; put a price on each competing objective; account for the shape the material can be given and the process that gives it; and only then go into the detail that decides the final choice.

> **How does a design requirement become a ranked list of candidate materials, and why is "the strongest" or "the lightest" usually the wrong criterion?**
> **How are mass, cost, shape and manufacturing combined into one defensible decision? Do the same methods choose materials for a bridge, a hip implant, a pressure vessel and an aircraft?**

Concept	Engineering destination
Function, objective, constraints, free variables	Writing a selection specification for any component
Screening, ranking, documentation	Narrowing tens of thousands of materials to a few candidates
Performance indices ($E^{1/2}/\rho$, $\sigma_f^{2/3}/\rho$, K_{IC}^2/σ_f, σ_f^2/E, k/ρ)	Light stiff beams and panels, strong shafts, safe pressure vessels, springs, heat sinks
Ashby property charts	Seeing all candidates and trade-offs at once; spotting gaps for new materials
Pareto fronts, penalty functions, exchange constants	Mass versus cost in cars, aircraft and spacecraft; carbon versus cost
Shape factors	Tubes, I-beams, sandwich panels; why wood and polymers are rarely made into thin-walled sections
Process selection and cost modelling	Choosing moulding, machining, casting or additive manufacture for a batch size
Case studies	Aerospace spars, civil floor beams, flywheels, pressure vessels, hip implants, busbars

Roadmap. How design proceeds and what it asks of materials (58.1) → indices derived from objective and constraint (58.2) → charts that show all materials at once (58.3) → several objectives at once (58.4) → shape as a second free variable (58.5) → the process and its cost (58.6) → the method applied across engineering (58.7).

58.1 The design process and material requirements

Stages of design

Engineering design is usually described in three stages, and the questions asked of materials sharpen as it proceeds:

1. **Conceptual design** chooses the working principle: a bridge as a suspension, arch or girder; energy storage as a battery, flywheel or compressed air; a hip replacement as a cemented or cementless stem. Here *all* materials are candidates and the data needed are approximate: a modulus to within a factor of two, a cost to within a factor of three.
2. **Embodiment design** sizes the components and settles their layout. Candidate materials narrow to a few classes, and the data needed are more precise: the yield strength of a particular alloy family, the creep strength at 600 °C, the fatigue limit

of a grade of steel.

3. **Detail design** specifies one material in one condition (6061-T6 extruded, not "aluminium"), with tolerances, surface finish, process route and test methods. The data needed are those of a specific supplier's product, often supplemented by the company's own tests.

The materials engineer's errors at each stage are different. At the conceptual stage the error is to consider too few materials ("we always use steel"); at the detail stage it is to rely on generic data for a property, such as fatigue strength or toughness, that is controlled by processing (Chapters 13–15).

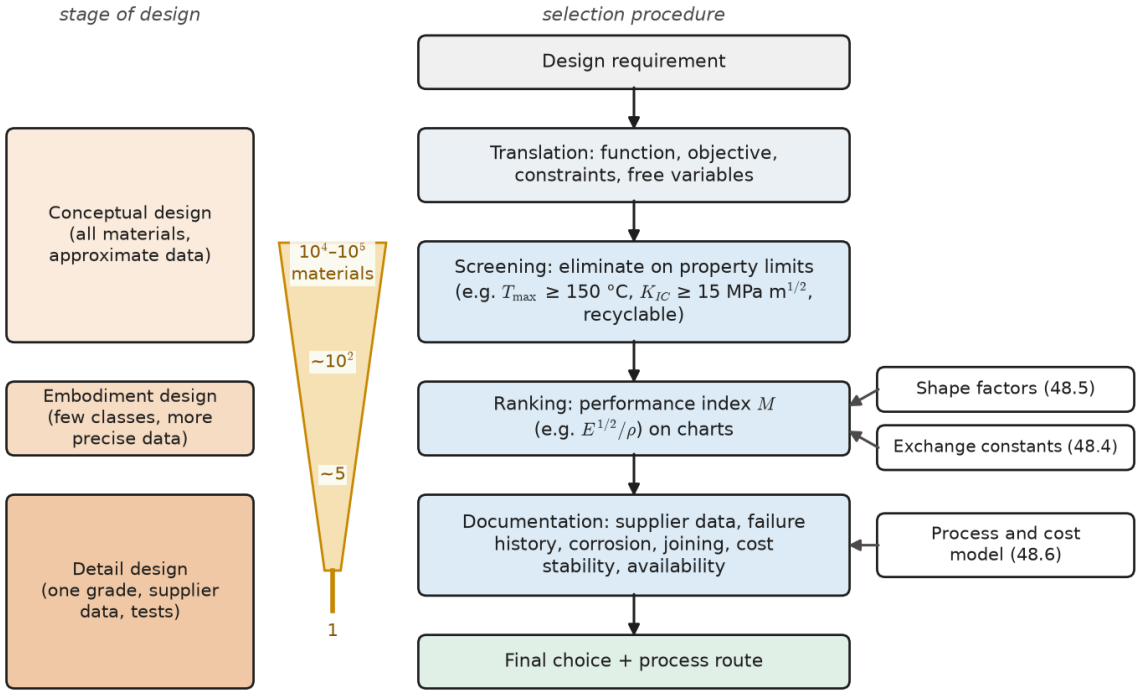


Figure 58.1 — The materials-selection procedure: translation of the design requirement, then screening, ranking and documentation, run alongside the stages of design.

Translation: function, objective, constraints, free variables

Every selection problem is written in four parts:

- **Function**: what the component does: carry a bending load, contain pressure, store elastic or kinetic energy, conduct heat or current, support a body tissue, insulate.
- **Objective**: what is to be minimized or maximized: mass, cost, volume, embodied carbon (Chapter 59), thermal resistance, energy stored.
- **Constraints**: requirements that must be met but need not be exceeded. Some are **geometric** (span fixed by the building grid, outer diameter fixed by the bore), some are **property limits** (service temperature above 200 °C, corrosion resistance in seawater, biocompatibility, electrical insulation), and some are **performance constraints** (stiffness not less than S^*, no yield at load F^*, resistance not more than R^*) that couple to the objective through a free variable.
- **Free variables**: what the designer may change: a cross-sectional area, a wall thickness, a section shape, and the choice of material.

Writing these four lists is not bureaucracy. It exposes the most common error in selection, which is to optimize the wrong thing: maximizing strength when the component is stiffness-limited, or minimizing mass when the objective is cost.

Screening, ranking, documentation

With the translation in hand, selection proceeds in three steps (Figure 58.1):

1. **Screening** eliminates every material that fails a property-limit constraint, whatever its other merits: a turbine-disc material must retain strength at 650 °C; an implant material must be biocompatible; a seawater pump must resist

chloride pitting (Chapter 43). Screening uses simple attribute limits and removes most of the 10^4–10^5 engineering materials in commercial databases.
2. **Ranking** orders the survivors by how well they meet the objective. Because the objective usually depends on a *combination* of properties, ranking uses performance indices (Section 58.2).
3. **Documentation** examines the top few candidates in depth: supplier data and their scatter, processing experience, failure history (Chapter 44), corrosion and joining behaviour, price stability, supply security and recyclability. Many final choices are made here, on information that does not fit in a property table.

Compatibility with regulation

Electronics are among the most regulated products for material content.

RoHS (the EU Restriction of Hazardous Substances directive, with equivalents in China, Korea, Japan, several US states and elsewhere) limits ten substances in each *homogeneous material* (a material that cannot be mechanically separated: a plating, a solder, a resin), not in the product as a whole:

Substance	Maximum concentration in homogeneous material	Former EEE use
Lead (Pb)	0.1 wt %	SnPb solder, glass, PVC stabilizers, brass and steel additives
Mercury (Hg)	0.1 wt %	Tilt switches, fluorescent lamps
Cadmium (Cd)	0.01 wt %	Ag–CdO contacts, NiCd batteries, pigments, platings
Hexavalent chromium (Cr VI)	0.1 wt %	Chromate conversion coatings on Zn and Al
PBB, PBDE	0.1 wt %	Brominated flame retardants in plastics
DEHP, BBP, DBP, DIBP (phthalates)	0.1 wt % each	Plasticizers in PVC cable insulation

Exemptions with expiry dates cover uses with no substitute yet, for example lead in high-melting-temperature solders (more than 85 % Pb) used for die attach in power devices, and lead in certain glasses and ceramics of electronic components. The transition to lead-free solder (SAC alloys, Chapter 51) was the largest materials change in the history of electronics assembly; it raised reflow temperatures by about 30 K, which then forced changes in laminates (T_d and T_g), component moisture sensitivity, finishes and reliability models.

REACH (EU regulation on Registration, Evaluation, Authorisation and Restriction of Chemicals) works differently: substances are registered by manufacturers and importers; **Substances of Very High Concern (SVHC)** are listed on a Candidate List that grows each year; any article containing an SVHC above 0.1 wt % triggers obligations to inform customers (and to notify a database), and some substances move to an authorization or restriction list that limits their use. A component supplier's REACH declaration is therefore a moving target that must be tracked across the product's life.

Conflict minerals: tin, tantalum, tungsten and gold (3TG) from regions where mining finances armed conflict are subject to supply-chain due-diligence and reporting requirements (US Dodd–Frank Act section 1502; EU Conflict Minerals Regulation applying to importers from 2021). Every one of these four elements is in nearly every electronic product: tin in solder, tantalum in capacitors, tungsten in vias and contacts, gold in bond wires and platings. Smelter-level traceability programmes are the industry's response.

Related regulations cover end-of-life (WEEE collection and recycling targets), batteries, packaging, halogen-free requirements set by customers, and persistent pollutants. Their effect on selection is to add screening constraints and, often, to remove the material with the best engineering properties: Ag–CdO contacts, SnPb solder, chromate coatings and PVC with phthalates were all technically excellent.

58.2 Performance indices derived

The method: separating the problem

The performance of a component depends on three groups of things: the functional requirements F (loads, stiffness, pressure), the geometry G (lengths, fixed dimensions), and the material properties M. If the performance P can be written as a product

$$P = f_1(F)\, f_2(G)\, f_3(M)$$

then the best material is the one that optimizes $f_3(M)$, whatever the loads and dimensions. The function $f_3(M)$ is the **material performance index**. The derivation is always the same: write the objective as an equation, write the performance constraint as an equation, use the constraint to eliminate the free variable from the objective, and read off the group of material properties that remains.

Light, stiff tie

A tie of fixed length L and free cross-sectional area A must have axial stiffness $S = EA/L$ of at least S^*; the objective is minimum mass $m = AL\rho$. From the constraint $A = S^*L/E$, so

$$m = S^* L^2 \left(\frac{\rho}{E}\right) \qquad \Longrightarrow \qquad \text{maximize } M_1 = \frac{E}{\rho}$$

Light, stiff beam

A beam of length L, square cross-section of side b ($A = b^2$, second moment of area $I = b^4/12 = A^2/12$), loaded in bending, must have stiffness $S = C_1 EI/L^3$ of at least S^*, where C_1 depends on the loading and end conditions (48 for a simply supported beam with central load). Eliminating A:

$$A = \left(\frac{12 S^* L^3}{C_1 E}\right)^{1/2} \qquad \Longrightarrow \qquad m = \left(\frac{12 S^*}{C_1}\right)^{1/2} L^{5/2} \left(\frac{\rho}{E^{1/2}}\right)$$

so the index is $M_2 = E^{1/2}/\rho$. The square-root appears because the free variable enters the stiffness as A^2: doubling the area quadruples the stiffness, so a low modulus can be compensated by a modest increase of section.

Light, stiff panel

A flat panel of fixed length L and width b, free thickness t, loaded in bending, has $I = bt^3/12$. With $m = bLt\rho$ and $S = C_1 EI/L^3 \geq S^*$:

$$t = \left(\frac{12 S^* L^3}{C_1 E b}\right)^{1/3} \qquad \Longrightarrow \qquad m = \left(\frac{12 S^*}{C_1 b}\right)^{1/3} bL^2 \left(\frac{\rho}{E^{1/3}}\right)$$

and the index is $M_3 = E^{1/3}/\rho$. The cube root rewards low density even more strongly. This is why low-density materials, wood, magnesium, polymer foams and composites, are so effective in panels, and why the bonnet of the opening paragraph is a panel problem.

Strength-limited designs

If the constraint is that the component must not yield (or fracture) rather than that it must not deflect, replace stiffness by strength σ_f. A tie that must carry F^* without yield needs $A = F^*/\sigma_f$, giving $M = \sigma_f/\rho$. For a square beam, failure occurs when the maximum surface stress $\sigma = \mathcal{M}c/I$ reaches σ_f; the section modulus is $Z = I/c = A^{3/2}/6$ and the failure load $F_f = C_2 Z \sigma_f / L$. Then

$$A = \left(\frac{6 F^* L}{C_2 \sigma_f}\right)^{2/3} \qquad \Longrightarrow \qquad m = \left(\frac{6F^*}{C_2}\right)^{2/3} L^{5/3} \left(\frac{\rho}{\sigma_f^{2/3}}\right)$$

so $M = \sigma_f^{2/3}/\rho$, and the same argument for a panel gives $\sigma_f^{1/2}/\rho$.

Indices for other functions

The method extends to any function for which the performance equation is known:

Function, objective, constraint	Index to maximize	Where it is used
Tie, minimum mass, stiffness specified	E/ρ	Cables, truss members, tension rods
Beam, minimum mass, stiffness specified	$E^{1/2}/\rho$	Spars, floor joists, robot arms, bicycle tubes
Panel, minimum mass, stiffness specified	$E^{1/3}/\rho$	Car body panels, aircraft floors, laptop cases
Tie, minimum mass, strength specified	σ_f/ρ	Tethers, rotating rims (flywheels, centrifuge rotors)
Beam, minimum mass, strength specified	$\sigma_f^{2/3}/\rho$	Shafts, crane booms, wing spars
Panel, minimum mass, strength specified	$\sigma_f^{1/2}/\rho$	Floor panels, hatches
Spring, maximum stored energy per unit volume	σ_f^2/E	Valve and suspension springs, clips
Elastic hinge, maximum bending without yield	σ_f/E	Snap fits, flexures, living hinges
Pressure vessel, yield before fracture	K_{IC}/σ_f	Boilers, reactors, gas cylinders
Pressure vessel, leak before break	K_{IC}^2/σ_f	Pressurized pipes, nuclear and chemical plant
Thermal shock resistance (quench ΔT without fracture)	$\sigma_f/(E\alpha)$	Furnace linings, cookware, turbine components
Heat sink or spreader, minimum mass for given thermal resistance	k/ρ	Electronics cooling, battery thermal plates
Minimum thermal distortion under heat flux	k/α	Precision instruments, optical benches, mirror substrates
Conductor, minimum mass for given resistance	$1/(\rho\,\rho_e)$	Overhead lines, busbars, aircraft wiring
Any of the above, minimum cost	Replace ρ by $C_m\rho$	Most civil and consumer products
Any of the above, minimum embodied carbon	Replace ρ by $H_c\rho$	Low-carbon design (Chapter 59)

Here σ_f is the failure strength appropriate to the material (yield strength for metals and polymers, modulus of rupture or compressive strength for ceramics), C_m the price per kilogram and H_c the embodied carbon per kilogram.

> **Status of the principle: material performance indices**
> *Derived.* Each index is an exact consequence of its stated objective, constraint, loading and free variable; it is only as general as those assumptions, and a change of failure mode (yield to buckling), loading or free variable changes the index.

A first ranking

Material (typical)	E (GPa)	ρ (Mg/m^3)	σ_y (MPa)	E/ρ	$E^{1/2}/\rho$	$E^{1/3}/\rho$	σ_y/ρ	$\sigma_y^{2/3}/\rho$
Low-carbon steel	210	7.85	250	26.8	1.85	0.76	32	5.1
Advanced high-strength steel	210	7.85	600	26.8	1.85	0.76	76	9.1
Al 6061-T6	69	2.70	275	25.6	3.08	1.52	102	15.7
Mg AZ31	45	1.77	200	25.4	3.79	2.01	113	19.3
Ti-6Al-4V	114	4.43	880	25.7	2.41	1.09	199	20.7
CFRP, quasi-isotropic	60	1.55	600	38.7	5.00	2.53	387	45.9
GFRP	20	1.85	250	10.8	2.42	1.47	135	21.5
Softwood, along grain	11	0.50	40	22.0	6.63	4.45	80	23.4
Alumina	370	3.90	300 (flexural)	94.9	4.93	1.84	77	11.5
Polypropylene	1.3	0.905	30	1.4	1.26	1.21	33	10.7

(Units: E in GPa, ρ in Mg/m^3, σ_y in MPa; the indices are in the corresponding units, and only ratios between materials matter.)

The table teaches three lessons. First, steel, aluminium, magnesium and titanium have almost the same E/ρ, so as ties none is lighter than the others; the light alloys win only in bending, where the exponent rewards low density. Second, wood along the grain is, for a beam or panel of given stiffness, the most mass-efficient material in the table: its $E^{1/2}/\rho$ exceeds that of CFRP. That is why aircraft were built of spruce for thirty years, and why timber floors are competitive. Third, the strength of advanced high-strength steels raises $\sigma_y^{2/3}/\rho$ by about 80 % relative to mild steel but leaves stiffness indices unchanged: strength-limited parts (bumper beams, door-intrusion beams) benefit from AHSS, stiffness-limited panels do not, which is the materials reason why outer panels are the first steel parts to be replaced by aluminium.

58.3 Ashby charts and selection

Property charts

A table lists a few materials; a **property chart** shows all of them. Ashby charts plot one property against another on logarithmic axes, with each material class occupying a field and each family within it a bubble whose extent shows the range of grades and conditions (Figure 58.2). The charts show at once how widely properties span (Young's modulus from about 10^{-4} GPa for soft foams to 1000 GPa for diamond, seven orders of magnitude), how the classes cluster (metals dense and stiff, polymers light and compliant, ceramics stiff and light but brittle, composites and woods filling the gaps), and where no material exists.

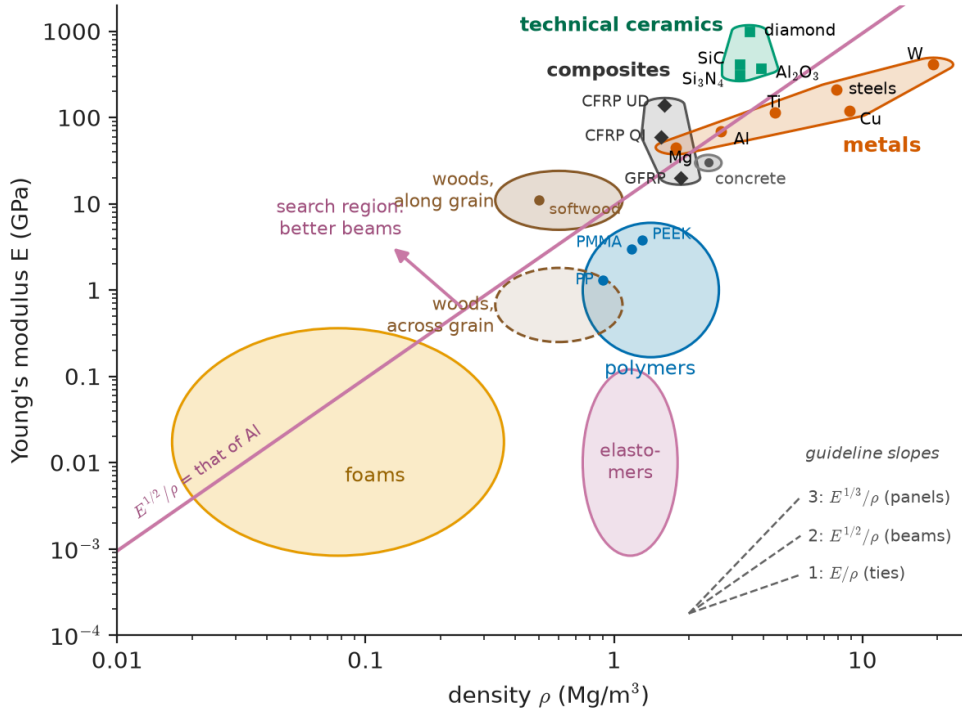


Figure 58.2 — Young's modulus against density for all material classes, with guidelines for light, stiff ties, beams and panels.

Selection lines

Because indices are power laws, a line of constant index is a straight line on a log–log chart. For $M = E^{1/2}/\rho$,

$$\log E = 2\log\rho + 2\log M$$

a family of parallel lines of slope 2, one for each value of M. All materials on one line perform equally well as light stiff beams; those above it perform better. Selection consists of sliding the line toward the upper left until only a handful of materials remain above it. Lines of slope 1 select for ties (E/ρ), slope 3 for panels ($E^{1/3}/\rho$). A property limit appears as a horizontal or vertical line: "modulus at least 10 GPa" removes everything below $E = 10$ GPa whatever its index.

The commonly used charts include:

Chart	Indices read from it	Typical applications
E against ρ	E/ρ, $E^{1/2}/\rho$, $E^{1/3}/\rho$	Light stiff structures
σ_f against ρ	σ_f/ρ, $\sigma_f^{2/3}/\rho$, $\sigma_f^{1/2}/\rho$	Light strong structures, rotors
K_{IC} against σ_f	K_{IC}/σ_f, K_{IC}^2/σ_f	Damage-tolerant design, pressure vessels
E against σ_f	σ_f^2/E, σ_f/E	Springs, hinges, seals
k against α	k/α	Thermal distortion, thermal management
E against cost per unit volume	$E^{1/2}/(C_m\rho)$, etc.	Cost-limited structures
Embodied energy or carbon against modulus or strength	$E^{1/2}/(H_c\rho)$, etc.	Low-carbon design (Chapter 59)
Maximum service temperature against strength	Property limits	High-temperature design (Chapter 41)

What the charts show beyond the ranking

The empty regions of a chart are as informative as the occupied ones. There is no material with the stiffness of steel and the density of a polymer, none with the toughness of steel and the hardness and temperature capability of alumina, none with high thermal conductivity, low density and a CTE matched to silicon. Each gap is a target for material development: composites, foams and lattice materials (Chapter 48) are, in a sense, ways of filling chart space by combining a material with shape and with another material. Hybrid and architected materials are best understood as the deliberate occupation of chart regions that no monolithic material reaches.

> **Status of the principle: chart-based selection**
> *Derived.* The selection lines are exact images of the derived indices; the conclusions are as reliable as the property ranges plotted, which are typical values for generic families and must be replaced by grade-specific data at the documentation stage.

58.4 Multi-objective selection

Conflicting objectives and the Pareto front

Real designs have several objectives that conflict: low mass and low cost; low cost and low embodied carbon; high stiffness and high damping. A solution (a material, or a material plus a design) is **dominated** if another solution is at least as good in every objective and better in one. The non-dominated solutions form the **Pareto front**, or trade-off surface (Figure 58.3). Every solution on the front is defensible; every solution off it is not, because something else is better in every respect. Finding the front removes most candidates, but it does not choose among the survivors. That requires a statement of how much one objective is worth in units of another.

Penalty functions and exchange constants

The simplest way to make that statement is a linear **penalty function**,

$$Z = C + \alpha\, m$$

where C is cost, m is mass and the **exchange constant** α is the value, in currency per kilogram, of saving one kilogram. The best solution minimizes Z. On a plot of m against C, contours of constant Z are straight lines of slope $-1/\alpha$; the optimum is the point where the lowest such line touches the front. A small α selects the cheap end of the front, a large α the light end.

Exchange constants are not material properties; they depend on the use of the product. Typical orders of magnitude:

Sector	Exchange constant α (USD per kg saved, order of magnitude)	Basis
Civil structures, domestic appliances	~0–1	Mass barely matters
Family car	~1–10	Fuel or battery cost over vehicle life
Heavy truck	~5–20	Payload revenue
Civil aircraft	~100–1000	Fuel and payload over 25–30 year life
Military aircraft	~1000	Performance
Spacecraft	~10^4 and above	Launch cost per kg to orbit

The same construction works for any pair of objectives: cost against embodied carbon (with α a carbon price), cost against energy losses over life (Section 58.7), cost against volume in a crowded engine bay.

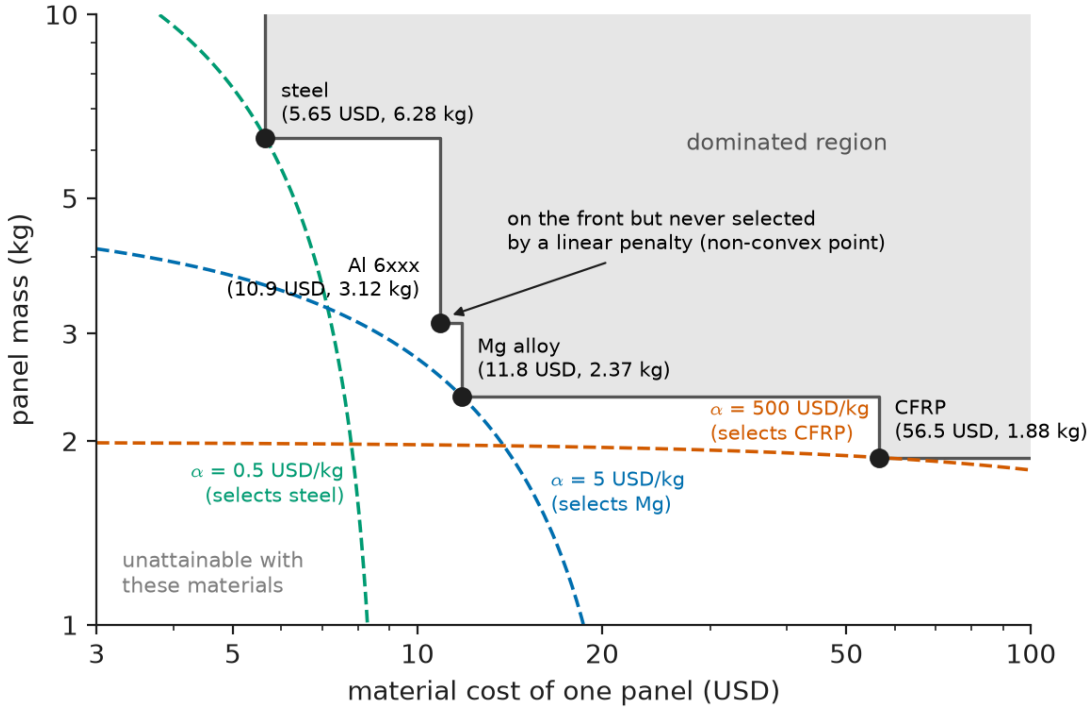


Figure 58.3 — Mass against material cost for a car bonnet of equal bending stiffness: the Pareto front and lines of constant penalty for three exchange constants.

Worked example — choosing a bonnet material. The bonnet is treated as a flat panel of area 1.0 m^2 that must match the bending stiffness of the current 0.80 mm steel panel. For equal stiffness, Et^3 is constant, so $t = t_{\text{steel}}(E_{\text{steel}}/E)^{1/3}$. Using indicative sheet prices (steel 0.9, aluminium 3.5, magnesium 5.0, CFRP 30 USD/kg; material only):

Material	E (GPa)	ρ (kg/m^3)	t (mm)	Mass (kg)	Material cost (USD)
Steel	210	7850	0.80	6.28	5.65
Al 6xxx	70	2700	1.15	3.12	10.9
Mg alloy	45	1770	1.34	2.37	11.8
CFRP	60	1550	1.22	1.88	56.5

For aluminium, $t = 0.80 \times (210/70)^{1/3} = 0.80 \times 1.442 = 1.154$ mm and $m = 2700 \times 1.0 \times 1.154 \times 10^{-3} = 3.12$ kg. No material is dominated: each lighter one is dearer. The penalties for three exchange constants are:

	Steel	Al	Mg	CFRP
Z at $\alpha = 0$ (cost only)	**5.7**	10.9	11.8	56.5
Z at $\alpha = 5$ USD/kg (car)	37.1	26.5	**23.7**	65.9
Z at $\alpha = 500$ USD/kg (aircraft)	3146	1569	1195	**998**

The switch-over values follow from equating penalties: aluminium beats steel when $\alpha > (10.9 - 5.65)/(6.28 - 3.12) = 1.7$ USD/kg; magnesium beats steel when $\alpha > 1.6$ USD/kg and beats aluminium when $\alpha > 1.2$ USD/kg; CFRP beats magnesium only when $\alpha > (56.5 - 11.8)/(2.37 - 1.88) \approx 91$ USD/kg. Two conclusions follow. For a car, with α of a few USD per kilogram, a light alloy is justified on material cost, and CFRP is not; CFRP bonnets appear on cars where mass is valued like aircraft mass (racing, some premium sports models). And aluminium, although on the Pareto front, is never the minimum of a linear penalty with these numbers: magnesium overtakes it before it overtakes steel. Points in a non-convex part of the front can only be selected by a non-linear value function or by other criteria. In practice those other criteria decide: magnesium's corrosion, formability at room temperature and paint-bake behaviour are worse, and processing costs (stamping, joining, painting) are not in this material-only model, so aluminium is the common industrial choice. The penalty function narrows the debate to a number, α, and to a short list of documentation questions.

> **Status of the principle: linear penalty functions**
> *Approximate.* Exact for a linear value of mass (or other objective), which is a reasonable local approximation; real value functions are non-linear (the first kilogram saved in a spacecraft enables a mission, the last one does not), and a linear penalty cannot select points in non-convex regions of the front.

Other multi-objective methods

Where exchange constants cannot be agreed, **weighted-sum scoring** (each property normalized and weighted by importance, as in the weighted decision matrices used in design reviews) is common but hides arbitrary choices in the weights. **Constraint methods** optimize one objective with the others as limits (minimize cost subject to $m \le m_{\max}$), which trace the front point by point including non-convex parts. **Multi-objective evolutionary algorithms** construct the whole front numerically when a design model is available, and are widely used in combination with finite-element models of components and with the informatics tools of Chapter 57.

58.5 Shape factors

Shape as a free variable

Indices derived so far fixed the shape of the cross-section (solid square) and left only its size free. But the designer can also choose the shape: a tube, an I-beam, a box, a sandwich panel. Shape changes stiffness and strength in bending and torsion enormously at the same mass, while leaving axial (tie) stiffness unchanged. Its efficiency is measured by a dimensionless **shape factor**, defined relative to a solid square section of the same area. For elastic bending,

$$\phi_B^e = \frac{I}{I_{\text{square}}} = \frac{12\,I}{A^2}$$

so $\phi_B^e = 1$ for a solid square. For failure in bending the corresponding factor is $\phi_B^f = 6Z/A^{3/2}$.

Section (thin-walled limits)	I	A	ϕ_B^e
Solid square, side b	$b^4/12$	b^2	1
Solid circle, radius r	$\pi r^4/4$	πr^2	$3/\pi = 0.95$
Thin-walled tube, radius r, wall t	$\pi r^3 t$	$2\pi r t$	$3r/(\pi t)$
Thin-walled square box, side h, wall t	$2h^3 t/3$	$4ht$	$h/(2t)$
Rolled I-sections	—	—	typically 10–50
Sandwich panel with thin faces	—	—	can exceed 100

A tube with $r/t = 10$ has $\phi_B^e = 30/\pi = 9.5$: it is 9.5 times stiffer in bending than a solid square bar of the same mass. Its strength shape factor grows more slowly, $\phi_B^f = (3/\sqrt{2\pi})(r/t)^{1/2}$, which is 3.8 at $r/t = 10$ and 5.4 at $r/t = 20$.

Combined material-and-shape indices

Repeating the derivation of Section 58.2 with $I = \phi_B^e A^2/12$ gives the mass of a light stiff beam of given shape as

$$m = \left(\frac{12S^*}{C_1}\right)^{1/2} L^{5/2} \left(\frac{\rho}{(\phi_B^e E)^{1/2}}\right) \qquad \Longrightarrow \qquad \text{maximize } \frac{(\phi_B^e E)^{1/2}}{\rho}$$

The shape factor multiplies the modulus: a shaped material behaves like a solid one of modulus $\phi_B^e E$ and the same density. On the E–ρ chart, shaping moves a material vertically upward by the factor ϕ; a tube of steel with $\phi = 20$ behaves, in bending, like a solid material of modulus 4200 GPa.

Why shape efficiency is material-dependent

If shape gives such gains, why not use thin walls everywhere? Because thin walls fail by **local buckling** (wrinkling of the wall, crippling of a flange) before they reach their design load, and because manufacturing limits how thin a section can be rolled, extruded, moulded or laid up. The maximum practical shape factor is therefore a property of the material and its process:

Material class	Practical maximum ϕ_B^e (typical)	Reason
Structural steels	~65	High E resists local buckling; rolled and welded sections
Aluminium alloys	~45	Extrusion allows complex thin sections; lower E
CFRP, GFRP	~40	Laid-up and pultruded sections; buckling and delamination
Engineering polymers	~10–12	Low E, creep; mouldable but buckling-limited
Wood	~5–8	Natural sections; glued and box sections extend this
Concrete (reinforced and prestressed)	~10–20	Cast T- and box sections; tension carried by steel

Local buckling of a tube wall begins at a stress proportional to Et/r, so a material of higher modulus can be made thinner before buckling. This is why high-modulus metals reach the highest practical shape factors, and why low-modulus materials gain less from shaping than their density advantage suggests.

> **Status of the principle: shape factors**
> *Derived.* Exact consequences of beam theory for linear-elastic, thin-walled sections; they hold until local buckling, shear deformation of the web or core, or manufacturing limits intervene, which set material-dependent upper bounds on ϕ.

Worked example — material and shape for a light stiff tube. A bicycle down-tube or a light aircraft strut is a beam of fixed length that must meet a bending stiffness at minimum mass. Compare a solid steel bar, a steel tube with $r/t = 20$, an aluminium tube with $r/t = 15$ and a CFRP tube with $r/t = 12$ (each within the practical limits above). The mass for equal stiffness is proportional to $\rho/(\phi E)^{1/2}$, with $\phi = 3(r/t)/\pi$:

Section	E (GPa)	ρ (Mg/m^3)	ϕ_B^e	$\rho/(\phi E)^{1/2}$	Mass relative to solid steel
Solid steel bar	210	7.85	1.0	0.542	1.00
Steel tube, $r/t = 20$	210	7.85	19.1	0.124	0.23
Al tube, $r/t = 15$	70	2.70	14.3	0.085	0.16
CFRP tube, $r/t = 12$	60	1.55	11.5	0.059	0.11

Shape alone cuts the steel bar's mass by more than a factor of four; changing the material to CFRP and keeping the best feasible shape cuts it by a further factor of two. Because steel can be made thinner than aluminium before its wall buckles, the shaped steel tube recovers part of steel's disadvantage in density: the ratio of aluminium to steel mass is 0.69 for tubes, compared with $0.323/0.542 = 0.60$ for solid bars. This is the materials logic of the bicycle industry: thin-walled steel (Cr-Mo) frames, aluminium frames with larger-diameter tubes, and CFRP frames that are lightest of all, in that order of mass and of cost.

58.6 Process selection and cost modelling

Process attributes and screening

A material is only a candidate if a process exists to give it the required shape, size, tolerance, surface finish and microstructure at an acceptable cost. Processes (Chapters 46–49) are screened in the same way as materials, by attribute limits:

Attribute	Example constraint	Processes it eliminates
Material compatibility	Titanium alloy	Die casting (reaction with dies), conventional sand casting
Shape class	Hollow, re-entrant, 3D	Rolling, simple extrusion, sheet stamping
Mass or size	0.05 kg or 50 t	Powder metallurgy for very large parts; investment casting for very large parts
Section thickness	0.5 mm walls	Sand casting (~3 mm minimum)
Tolerance	±0.02 mm	Sand casting (±1 mm), most moulding without machining
Surface roughness	$R_a < 0.4$ μm	As-cast and as-printed surfaces without finishing
Microstructure and properties	Fatigue-critical, no porosity	High-pressure die casting without HIP, as-built AM without HIP (Chapter 48)
Economic batch size	200 parts	Injection moulding, die casting, forging (tooling not amortized)

Ranking among the survivors is usually by cost.

A cost model

The cost of making one part combines three contributions: material, dedicated tooling shared among the batch, and capital and labour charged by the time taken. A simple model that captures the essential behaviour is

$$C = \frac{m\,C_m}{1 - f} + \frac{C_t}{n} + \frac{\dot{C}_{\text{oh}}}{\dot{n}}$$

with m the part mass, C_m the material price per kilogram, f the scrap fraction (material removed, gates and runners, failed parts), C_t the cost of dedicated tooling (dies, moulds, fixtures), n the batch size, $\dot{C}_{\text{oh}}$ the overhead rate (capital write-off of the equipment, labour, energy, floor space) per hour and $\dot{n}$ the production rate in parts per hour. The first and third terms are independent of batch size; the tooling term falls as $1/n$. On a log–log plot of cost against batch size, each process is a horizontal line at large n joined to a line of slope -1 at small n (Figure 58.4).

> **Status of the principle: process cost model**
> *Approximate.* A first-order model that captures the scaling of material, tooling and time costs; it omits set-up and inspection costs, learning effects, tooling life (dies wear out after 10^4–10^6 shots), and the strong dependence of overhead rates on location and utilization.

Worked example — moulding, machining or printing? A 50 g polyamide bracket can be made by injection moulding (PA6 granules at 3 USD/kg, 5 % scrap, mould 30 000 USD, 200 parts/h, overhead 60 USD/h), by CNC machining from rod (4 USD/kg, 67 % of the rod machined away, fixtures 1500 USD, 10 parts/h, 80 USD/h), or by selective laser sintering of PA12 powder (60 USD/kg, 10 % powder lost, no tooling, an effective 3 parts/h, 30 USD/h).

The batch-independent cost per part is

	Material $mC_m/(1-f)$	**Time** $\dot{C}_{\mathrm{oh}}/\dot{n}$	**Sum (USD)**	**Tooling** C_t **(USD)**
Injection moulding	$0.05 \times 3/0.95 = 0.16$	$60/200 = 0.30$	0.46	30 000
CNC machining	$0.05 \times 4/0.33 = 0.61$	$80/10 = 8.00$	8.61	1 500
Laser sintering	$0.05 \times 60/0.90 = 3.33$	$30/3 = 10.00$	13.33	0

Cost per part at several batch sizes:

Batch size n	**10**	**100**	**1000**	**10 000**	**100 000**
Injection moulding	3000	300	30.5	3.46	0.76
CNC machining	159	23.6	10.1	8.76	8.62
Laser sintering	13.3	13.3	13.3	13.3	13.3

The crossovers follow from equating costs: laser sintering is cheapest below $n = 1500/(13.33 - 8.61) \approx 320$ parts, machining between about 320 and $n = (30\,000 - 1500)/(8.61 - 0.46) \approx 3500$ parts, and injection moulding above. The same structure recurs in every manufacturing sector: sand casting against die casting for aluminium housings, open-die against closed-die forging for steel shafts, hand lay-up against resin transfer moulding for composite panels. Additive manufacturing wins at small batch sizes and for shapes that other processes cannot make; its higher material and machine-time costs keep it out of high-volume production unless the geometry it enables saves more elsewhere (consolidating an assembly of twenty parts into one, or reducing the mass of an aircraft bracket where α is large).

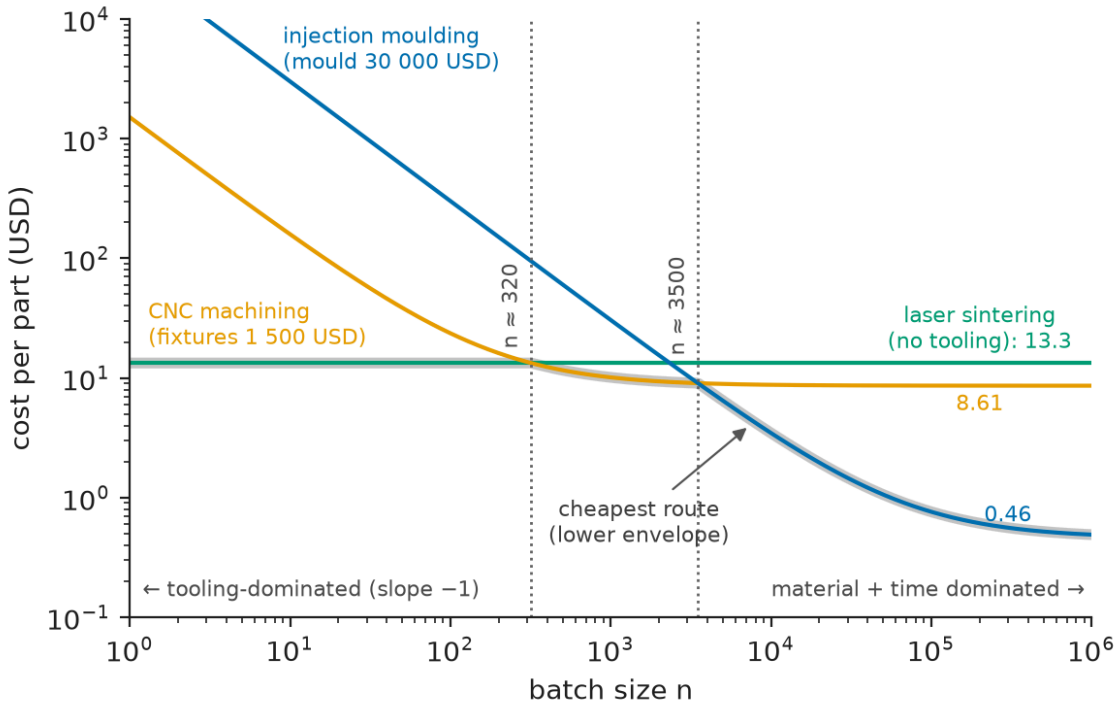


Figure 58.4 — Cost per part against batch size for a 50 g polymer bracket made by laser sintering, CNC machining and injection moulding.

Cost in the selection procedure

Cost enters selection in three ways: as an objective (minimize C through indices such as $E^{1/2}/(C_m\rho)$, Section 58.7); as a term in a penalty function with mass or carbon (Section 58.4); and through the process model, which converts material price into part cost. For high-volume products, processing often costs more than material: a stamped steel panel costs perhaps two to three times its steel content once press time, dies and scrap are included; an investment-cast titanium part may cost ten times its alloy content. Material price alone is a poor guide whenever the cheaper material requires a dearer process.

58.7 Selection case studies across engineering fields

Aerospace: wing spars, skins and the value of a kilogram

A wing spar is a beam loaded in bending, designed against both stiffness (aeroelastic deflection, flutter) and strength; the skin is a panel loaded in bending and buckling. With α of hundreds of USD per kilogram, mass dominates, so the indices are $E^{1/2}/\rho$ and $\sigma_f^{2/3}/\rho$ for spars and $E^{1/3}/\rho$ for skins and floors. From the table in Section 58.2, $\sigma_y^{2/3}/\rho$ is 45.9 for quasi-isotropic CFRP against 15.7 for Al 6061-T6 and about 20 for the high-strength 7xxx alloys used in wings, so a strength-limited CFRP spar has roughly half the mass of a 7xxx aluminium one and a third that of a 6061 one. This is why wide-body airliners designed after 2005 have CFRP wings and fuselages comprising about half of their structural mass. Documentation then raises the questions that indices miss: damage tolerance and barely visible impact damage (Chapter 23), lightning-strike protection (CFRP conducts poorly compared with aluminium), repair methods, and inspection. Al–Li alloys (Chapter 16), which reduce density by about 3 % and raise modulus by about 6 % per percent of lithium, are aluminium's response, keeping metallic structures competitive where established manufacturing and repairability matter.

Civil: a floor beam at minimum cost

In buildings, mass is cheap ($\alpha \approx 0$) and the objective is cost. For a stiffness-limited beam, the index becomes $E^{1/2}/(C_m\rho)$; equivalently, minimize $C_m\rho/E^{1/2}$:

Material (typical)	E (GPa)	ρ (Mg/m^3)	C_m (USD/kg)	$C_m\rho/E^{1/2}$ (relative units)
Reinforced concrete	30	2.40	0.12	0.053
Glulam (softwood)	12	0.48	1.0	0.139
Structural steel	210	7.85	1.0	0.54
Aluminium 6063 extrusion	69	2.70	3.5	1.14
Pultruded GFRP	25	1.90	5.0	1.90

Concrete wins by a factor of three over timber and ten over steel; this ordering is why concrete is the most-used material on Earth by mass (Chapter 59). But concrete is weak in tension (Chapter 18), so it needs steel reinforcement, which the simple index omits; steel wins where spans are long and mass matters for the supporting structure (the self-weight of a long-span concrete beam becomes the dominant load); timber wins in low-rise construction where speed of erection and embodied carbon are valued (Chapter 59 repeats this ranking with carbon instead of cost). Fire resistance, durability in exposure classes, construction time and the cost of the foundations carrying the dead load are documentation-stage questions that often decide.

Mechanical: flywheels and why the densest material is wrong

A flywheel stores kinetic energy; the objective is maximum energy per unit mass, and the constraint is that the rim must not burst. For a thin rim of radius r rotating at angular velocity ω, the hoop stress is $\sigma = \rho r^2\omega^2$ and the stored energy per unit mass is $U/m = \tfrac{1}{2}r^2\omega^2$. Setting $\sigma = \sigma_f$:

$$\frac{U}{m} = \frac{1}{2}\,\frac{\sigma_f}{\rho}$$

so the index is σ_f/ρ. With σ_f in MPa and ρ in Mg/m^3, σ_f/ρ is directly in kJ/kg. Maraging steel (1800 MPa, 8.0 Mg/m^3) stores at most about 110 kJ/kg; Al 7075 (500 MPa, 2.8 Mg/m^3) about 90 kJ/kg; hoop-wound CFRP (about 1500 MPa in the hoop direction, 1.6 Mg/m^3) about 470 kJ/kg, or 130 Wh/kg before safety factors. Lead, the traditional "heavy" flywheel material, stores less than 1 kJ/kg at bursting. Density appears in the denominator, not the numerator: when the speed is free, the best flywheel is light and strong. Heavy flywheels are right only when the speed is fixed (an engine flywheel smoothing torque at a given rpm), where energy per unit volume $\tfrac{1}{2}\rho r^2\omega^2$ rewards density. Composite rotors also fail more benignly, delaminating into fibrous debris rather than ejecting large fragments, a documentation-stage advantage for containment design.

Chemical and energy: pressure vessels that leak before they break

A pressure vessel must not fail by fast fracture. Two design philosophies lead to two indices. **Yield before break**: the vessel should yield (and visibly deform) before a crack of the largest size that inspection can miss becomes critical; with $K_{IC} = \sigma\sqrt{\pi a}$ (Chapter 14) at $\sigma = \sigma_y$, the critical crack length is proportional to $(K_{IC}/\sigma_y)^2$, so the index is K_{IC}/σ_y. **Leak before break**: a crack growing through the wall should leak, and be detected, before it becomes unstable; this requires that a through-wall crack of length about equal to the wall thickness t be stable at the operating stress.

Worked example — leak before break. A thin-walled spherical or cylindrical vessel of radius R under pressure p carries a wall stress $\sigma = pR/t$ (taking the cylindrical hoop stress for simplicity). A through-crack of total length $2a = t$ is stable if $\sigma\sqrt{\pi t/2} \le K_{IC}$. Substituting $t = pR/\sigma$ and solving for p:

$$p \le \frac{2}{\pi R}\,\frac{K_{IC}^2}{\sigma}$$

The largest pressure that can be contained with leak-before-break, at a given working stress, scales with K_{IC}^2/σ. For $R = 1.0$ m and the wall working at its yield stress:

Material (typical)	σ_y (MPa)	K_{IC} (MPa m$^{1/2}$)	K_{IC}^2/σ_y (MPa m)	Max. pressure for LBB (MPa)	Wall thickness at that pressure (mm)
Pressure-vessel steel (C–Mn, normalized)	350	150	64	41	117
Quenched-and-tempered high-strength steel	1000	60	3.6	2.3	2.3
Al 2024-T3	350	30	2.6	1.6	4.7

For the tough steel, $p = 2 \times 64.3/(\pi \times 1.0) = 40.9$ MPa. The high-strength steel, nearly three times stronger, can guarantee leak-before-break only up to 2.3 MPa in a 1 m vessel: above that its thin wall would break before it leaked. Strength and toughness are inversely related in most alloy families (Chapters 13–15), so the ranking by σ_y is reversed when the index is K_{IC}^2/σ_y. This is the reason why boilers, nuclear pressure vessels and chemical reactors use moderate-strength, high-toughness steels, with real designs adding safety factors, temperature effects (the ductile-to-brittle transition, Chapter 15) and fatigue crack growth (Chapter 14). On a K_{IC}–σ_f chart (Figure 58.5), lines of slope 1 and slope 1/2 select the two criteria.

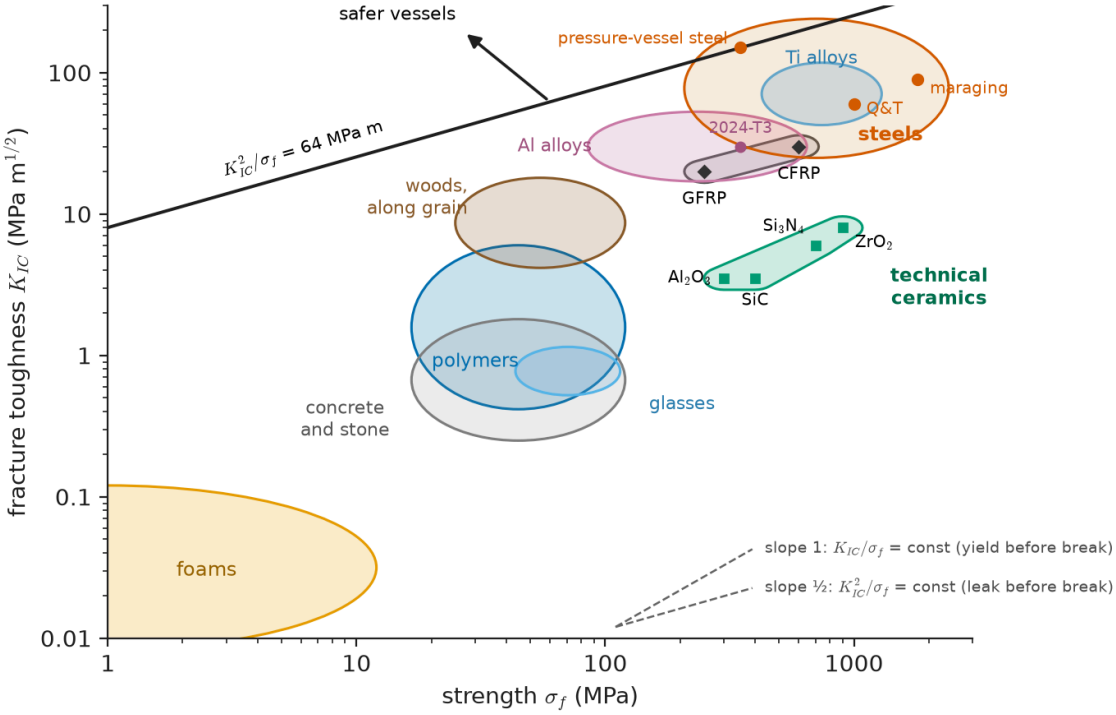


Figure 58.5 — Fracture toughness against strength, with guidelines for yield-before-break and leak-before-break design.

Biomedical: a hip stem where modulus is a constraint, not an objective

A femoral stem replaces the head and neck of the femur and is inserted into the bone canal. It must not fatigue in 10^7–10^8 gait cycles (about 1–2 million per year for ten years and more), must not corrode in body fluid, and must be biocompatible (Chapter 39). Its stiffness creates a problem absent from structural design: a very stiff stem carries load that the surrounding bone would otherwise carry, and bone that is not loaded is resorbed. This **stress shielding** loosens implants. The stem modulus should therefore be as close to cortical bone (about 15–20 GPa) as strength allows: here modulus enters as an upper limit, not something to maximize.

Stem material (typical)	E (GPa)	Ratio to cortical bone (18 GPa)	Fatigue strength at 10^7 cycles (MPa)
Co–Cr–Mo alloy	230	12.8	400–600
316L stainless steel (cold worked)	193	10.7	300–450
Ti-6Al-4V	114	6.3	500–600
β-titanium (Ti–Nb–Zr–Ta type)	55–65	~3.3	300–500
Cortical bone	15–20	1	—

Screening removes non-biocompatible and corrosion-prone alloys; ranking by fatigue strength with a modulus ceiling selects titanium alloys, and β-titanium alloys were developed specifically to fill the chart region of low modulus with adequate fatigue strength. Shape enters as well: a hollow or tapered stem lowers bending stiffness at the same material modulus. The cobalt alloy remains the material of choice for the bearing head, where wear resistance (Chapter 45), not modulus, is the governing requirement: one implant, two components, two different indices.

Energy and electrical: busbars and conductors

A busbar of length L must have resistance at most R^*. With $R = \rho_e L/A$, the area is $A = \rho_e L/R^*$ and the mass $m = (L^2/R^*)\,\rho\rho_e$; minimum mass means minimum $\rho\rho_e$, and minimum cost minimum $C_m\rho\rho_e$. For a 1.00 m bar with $R^* = 50$ μΩ:

	Copper	Aluminium
ρ_e (Ω m, 20 °C)	1.72×10^{-8}	2.82×10^{-8}
Area $A = \rho_e L/R^*$	344 mm^2	564 mm^2
Mass ($\rho = 8960$ and 2700 kg/m^3)	3.08 kg	1.52 kg
Material cost (indicative 9.5 and 2.5 USD/kg)	29 USD	3.8 USD

Aluminium halves the mass and costs about one-eighth as much, for a 64 % larger section. This is why overhead lines and, increasingly, battery busbars in electric vehicles are aluminium; copper keeps the applications where volume is constrained, joints are numerous, or the higher creep resistance of copper under bolt load matters. The busbar is also the clearest example of a selection where the use phase dominates both cost and environmental impact, because resistive losses over twenty years far exceed the material's cost and embodied energy; Chapter 59 returns to it.

> **Boundary.**
> The EEE-specific selection of substrates, laminates, solders and insulation, with the reliability statistics that go with them, is treated in more depth in Chapters 26 and 51.

Electronics and thermal management: a heat sink

A heat-sink base of fixed length L along the heat path must have thermal resistance $L/(kA)$ at most R^*_{th}; minimum mass gives the index k/ρ. Aluminium alloys ($k \approx 200$ W/(m K), $\rho = 2.7$ Mg/m^3, $k/\rho \approx 74$) beat copper ($k \approx 400$, $\rho = 8.96$, $k/\rho \approx 45$), so heat sinks are extruded aluminium unless volume is constrained, when copper's higher k wins (vapour-chamber and heat-pipe bases, compact processor coolers). For a substrate bonded to a silicon or silicon-carbide die, CTE matching (Chapter 24) is a constraint that eliminates both metals and leads to aluminium nitride, silicon nitride or metal–ceramic composites such as AlSiC: again an index ranks, but a constraint decides.

The pattern across fields

The same steps appear in every case: a translation into function, objective, constraints and free variables; an index derived from the governing equation; a chart or table that ranks; an exchange constant or a constraint that resolves conflicts; a shape and a process that the material must accept; and a documentation stage where corrosion, joining, fatigue, supply and regulation decide among the last few. What differs between fields is not the method but the values of the exchange constants and which constraints bind.

Status of the principle: systematic materials selection
Derived. The procedure is a logical consequence of stating objectives and constraints explicitly; its results are as good as the property data, the exchange constants and the completeness of the constraint list, and the most frequent failures in practice come from a constraint that was never written down.

What remains open

Open question	Where it is answered
Elastic moduli, yield strength and their measurement, which feed every index	Chapters 11 and 13
Fracture toughness, fatigue and creep data used in damage-tolerant and high-temperature selection	Chapter 14
Why strength and toughness trade off in steels, aluminium and titanium alloys	Chapters 13, 15 and 16
Composite laminate stiffness and damage tolerance behind CFRP indices	Chapters 22 and 23
Processes, their attribute limits and the microstructures they produce	Chapters 46–49
Corrosion behaviour that decides many documentation-stage choices	Chapters 42 and 43
Biocompatibility and implant materials	Chapter 39
Data-driven discovery of candidates before selection	Chapter 57
Embodied energy, carbon, criticality and recyclability as objectives and constraints	Chapter 59
Optimization theory and multi-objective algorithms in general	The Mathematics Universe

Next: Chapter 59 widens the frame from one component to the whole life of materials in society, adding embodied energy and carbon, lifecycle assessment, criticality and recycling to the selection tools developed here.

Chapter 59 — Sustainability, Lifecycle and the Circular Economy

Part XI · Characterization, Computation and Design

A city council commissions a twelve-storey office building and asks the design team for the lowest-carbon structure. The structural engineer proposes a post-tensioned concrete frame, the cheapest option. The architect proposes cross-laminated timber floors on glulam columns, pointing out that the wood stores carbon. The steel fabricator proposes a frame of electric-arc-furnace sections made from scrap, almost all of which will be recovered when the building is demolished. A consultant then reports that, over a sixty-year life, the energy to heat, cool and light the building will emit more carbon than its structure, so the façade and the heat pumps matter more than the frame. A second consultant replies that the grid will be largely decarbonized within twenty years, so the carbon emitted *now*, to make the materials, is what counts. Each statement can be supported by a published lifecycle assessment. The council is not facing an engineering failure in the usual sense: nothing has cracked, crept or corroded. It is facing the question that now sits over every chapter of this book: what does it cost the planet to make, use and discard the materials we choose, and how should that cost change the choice?

> **How much material does society use, where does it go, and what fraction of global energy and emissions does making it account for?**
> **How are embodied energy, carbon and lifecycle impacts measured, and how reliable are the numbers?**
> **Which materials are critical, and how can products and processes be designed so that materials circulate rather than being used once?**

Concept	Engineering destination
Material flows, stocks and the materials footprint	Planning capacity, scrap supply and infrastructure demand
Embodied energy and embodied carbon	Low-carbon structural, vehicle and product design
Lifecycle assessment (LCA), eco-audit	Comparing design options over the whole life; environmental product declarations
Critical and conflict materials	Supply-risk management for magnets, batteries, catalysts, alloys
Design for recycling, reuse and remanufacture	Products and buildings whose materials can be recovered
Recycling metallurgy, polymer recycling, construction-waste recovery	Steel, aluminium, plastics and concrete in second and later lives
Green materials and the circular economy	Hydrogen steel, low-clinker cement, mass timber, bio-based polymers
The future of the discipline	Where materials science and engineering is going

Roadmap. How much material and where it flows (59.1) → the energy and carbon embodied in it (59.2) → measuring impacts over the whole life (59.3) → materials whose supply is at risk (59.4) → designing so that materials can come back (59.5) → how they are actually recovered (59.6) → new materials and the circular economy (59.7) → where the discipline is going (59.8) → a synthesis of the book (59.9).

59.1 Materials flows and the materials footprint of society

The scale of material use

Humanity now extracts of order 100 billion tonnes of materials per year: about half non-metallic minerals (sand, gravel, crushed stone, limestone and clay for construction), a quarter biomass (food, feed and wood), and the rest fossil fuels and metal ores. Per person, that is roughly 12 tonnes a year, or 35 kg a day. The engineered materials of this book are a subset, but a large one:

Material	Global production (order of magnitude, early 2020s)	Comment
Concrete	~25–30 Gt/yr	Most-used manufactured material, about 3–4 t per person per year
Cement	~4 Gt/yr	Binder for concrete (Chapter 18)
Crude steel	~1.9 Gt/yr	About 30 % via electric-arc furnaces, mostly from scrap
Wood (industrial roundwood)	~2 billion m^3/yr (~1 Gt)	Construction, paper, panels
Paper and board	~0.4 Gt/yr	High recycling rates in many regions
Plastics	~0.4 Gt/yr	About 9 % of plastic waste recycled globally
Glass	~0.15 Gt/yr	Flat and container glass
Primary aluminium	~70 Mt/yr	Plus ~30 Mt/yr from recycled scrap
Refined copper	~25 Mt/yr	About a third from recycled sources
Titanium metal	~0.3 Mt/yr	(Most titanium mined becomes TiO_2 pigment)
Rare-earth oxides	~0.3 Mt/yr	Small tonnage, large strategic importance

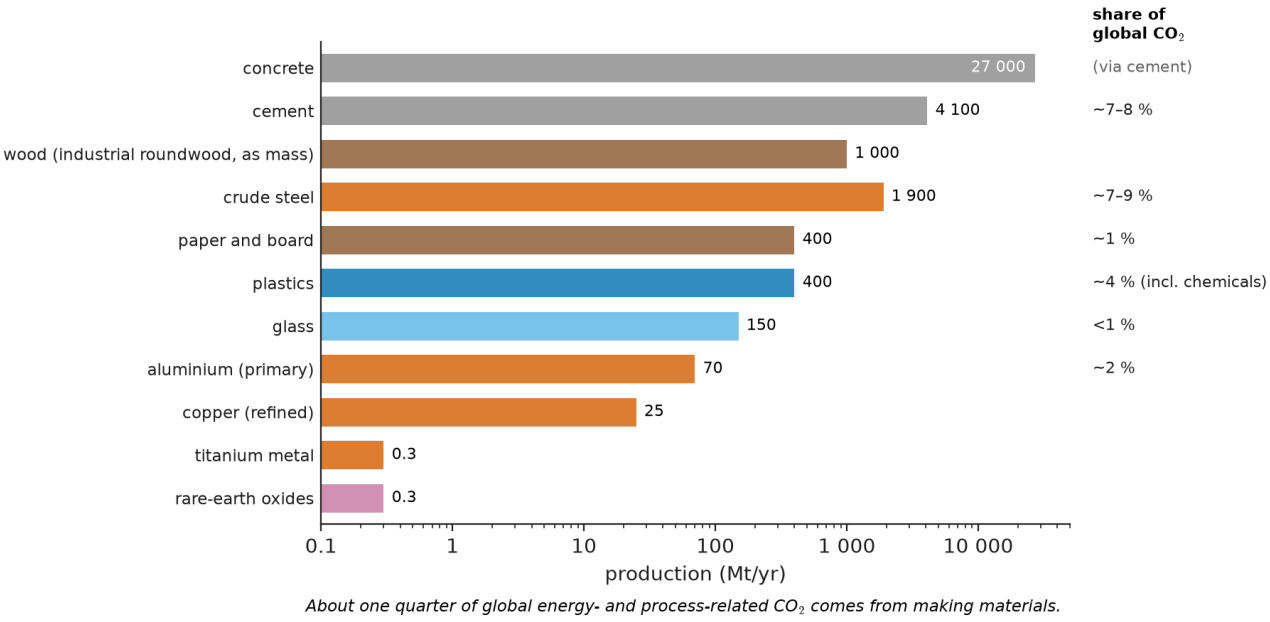


Figure 59.1 — Annual global production of major engineering materials, with the share of global CO2 emissions attributed to their production.

Two materials dominate the environmental account because of their tonnage. Steel production emits about 1.9–2.2 kg CO_2 per kg via the blast-furnace route and accounts for roughly 7–9 % of global CO_2 emissions; cement production, through fuel combustion and the calcination of limestone ($CaCO_3 \rightarrow CaO + CO_2$), accounts for another 7–8 %. Together with aluminium, plastics, chemicals, paper and glass, the production of materials accounts for about a quarter of global energy- and process-related CO_2 emissions (Figure 59.1).

Stocks and flows

Materials do not disappear when they are used; they accumulate as **in-use stocks**: buildings, bridges, pipelines, vehicles, machines. Industrialized economies hold about 10–15 tonnes of steel per person in use, and the global average is around 4 tonnes. Stocks grow while a society builds its infrastructure and then saturate; the steel stock per person of most mature economies has levelled off. The flow of a material through society can be mapped as a **Sankey diagram** (Figure 59.2): from ore

and scrap, through production, fabrication and products, into the in-use stock, and out, decades later, as end-of-life scrap, losses and landfill.

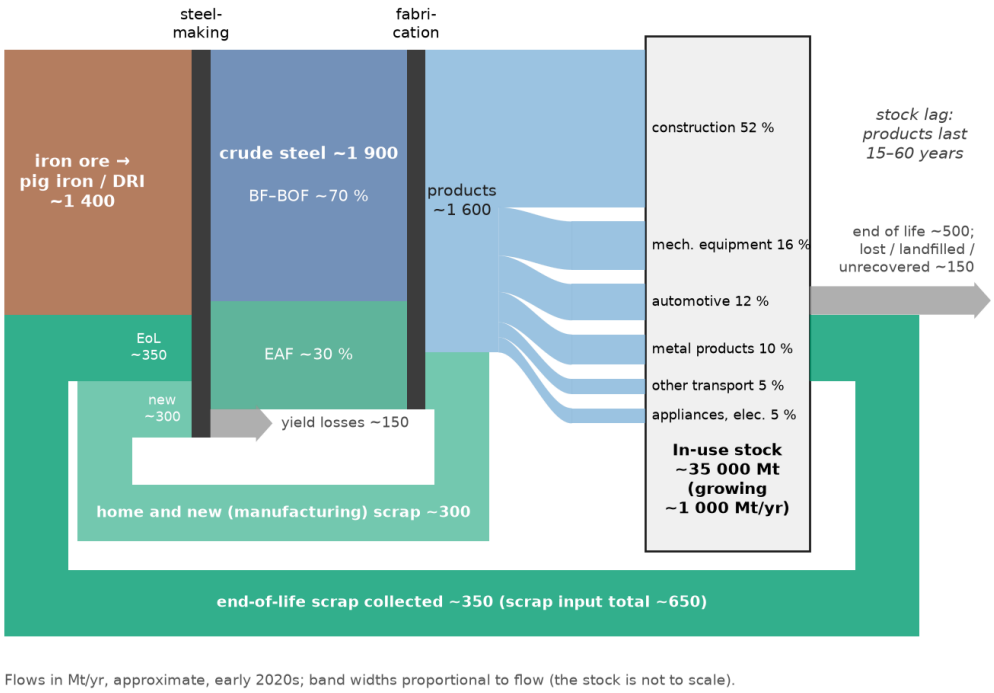


Figure 59.2 — Global steel flows from ore and scrap to products, in-use stock and end of life (approximate, early 2020s).

Three facts follow. Scrap available today reflects what was put into use decades ago, when production was smaller, so recycled supply lags demand (Section 59.7). The largest flow, into construction, is also the longest-lived, so buildings are both the biggest consumers and the largest future mines of materials. And losses occur at every stage: dissipative uses (pigments, coatings, wear debris), dispersed small components and corrosion (Chapters 42 and 43) remove material from circulation permanently.

The materials footprint and the drivers of demand

The environmental impact of materials use is often written as the product

$$I = P \times A \times T$$

impact equals population times affluence (consumption per person) times technology (impact per unit of consumption). Materials engineering acts on the last factor: less material per unit of service (**material efficiency**), lower impact per kilogram, and recovery at end of life. History adds a sobering lesson: efficiency gains have repeatedly been overtaken by growth in demand, the **rebound effect** noted for coal by Jevons in 1865.

> **Status of the principle: IPAT identity**
> *Approximate.* An accounting identity that is exact by definition, but the factors are not independent (affluence changes technology, efficiency changes demand), so it describes rather than predicts.

59.2 Embodied energy and carbon

Definitions

The **embodied energy** of a material is the primary energy used to make one kilogram of it, from extraction of the raw materials to the factory gate (**cradle-to-gate**), including the energy of fuels used as feedstock where relevant. The **embodied carbon** (more precisely, the cradle-to-gate global warming potential) is the mass of CO_2-equivalent greenhouse gases emitted in the same process, with other gases converted to CO_2 equivalents by their 100-year global warming potentials (methane about 28–

30, nitrous oxide about 270). Embodied carbon depends on the energy mix as well as the energy quantity, and in cement and iron-making it also includes **process emissions** from chemical reactions that release CO_2 irrespective of the fuel used.

Material (typical, cradle-to-gate)	Embodied energy (MJ/kg)	Embodied carbon (kg CO_2 e/kg)	Main contributors
Steel, blast furnace–basic oxygen furnace	20–30	1.9–2.3	Coke and coal as reductant, ore sintering
Steel, electric-arc furnace from scrap	6–10	0.3–0.7 (grid-dependent)	Electricity for melting
Aluminium, primary	190–220	8–20 (hydropower ~4–5; coal-based ~18–20)	Electrolysis, ~14 kWh/kg; anode consumption
Aluminium, recycled	10–25	0.5–1.0	Remelting, ~5–10 % of primary energy
Copper, primary	50–70	3–5	Mining and concentrating low-grade ore
Titanium alloy	500–750	35–45	Kroll process, vacuum remelting
Cement (Portland, CEM I)	4–5.5	0.8–0.9	Calcination (~0.52 kg/kg clinker) and kiln fuel
Concrete, C30/37 with CEM I	1.0–1.5	0.12–0.16 (300–380 kg/m^3)	Cement content
Brick	2–3	0.2–0.25	Kiln firing
Soda-lime glass	10–15	0.7–1.0	Melting, carbonate decomposition
Sawn softwood, glulam	2–10	0.2–0.4 fossil (biogenic storage ~1.6–1.8 not included)	Drying, processing, adhesive
Polyethylene	75–85 (incl. feedstock)	1.8–2.0 (plus ~3.1 if incinerated at end of life)	Cracking, polymerization
PET	80–90	2.2–3.0	Monomer synthesis
Carbon-fibre composite (CFRP)	200–400	15–30	Carbon-fibre production (precursor, oxidation, carbonization)

The values are typical ranges; they vary with ore grade, process, plant efficiency, and above all the electricity mix. Aluminium smelted with hydropower in Norway or Canada has a quarter of the carbon footprint of aluminium smelted with coal-fired electricity. Embodied values for the same material from two databases commonly differ by 20–50 %, and a comparison of two materials is only meaningful if both come from the same source and boundary.

The structure of the table follows from the thermodynamics of Chapter 6: the energy needed to reduce an ore to metal is at least the free energy of formation of the oxide (Ellingham diagram), and the more stable the oxide, the higher the minimum. Aluminium and titanium oxides are far more stable than iron oxide, so aluminium and titanium cost several times more energy per kilogram than steel, while remelting a metal that is already reduced needs only the energy to heat and melt it. That simple fact is the strongest argument for metal recycling.

Carbon as an objective in selection

Embodied carbon enters materials selection exactly as cost does (Chapter 58): replace the price C_m in a performance index by the embodied carbon H_c. For a beam of given bending stiffness, the embodied carbon is proportional to $H_c\rho/E^{1/2}$.

Worked example — embodied carbon of a stiffness-limited floor beam. Using mid-range values from the table, compare candidate materials for a beam that must meet a given bending stiffness, with a solid section of free size:

Material	E (GPa)	ρ (Mg/m^3)	H_c (kg CO$_2$e/kg)	$\rho H_c / E^{1/2}$	Relative to primary steel
Steel, primary (BF–BOF)	210	7.85	2.2	1.192	1.00
Steel, EAF from scrap	210	7.85	0.6	0.325	0.27
Concrete C30/37 (plain)	30	2.40	0.15	0.066	0.055
Glulam (fossil carbon only)	12	0.48	0.4	0.055	0.047
Aluminium, primary (global average)	69	2.70	16	5.20	4.4
Aluminium, recycled	69	2.70	0.8	0.260	0.22
CFRP	60	1.55	25	5.00	4.2

For primary steel, $7.85 \times 2.2/\sqrt{210} = 17.27/14.49 = 1.192$; for glulam, $0.48 \times 0.4/\sqrt{12} = 0.192/3.464 = 0.055$. Several conclusions follow, and each has caveats that the documentation stage must resolve (Figure 59.3). Timber and concrete are an order of magnitude better than primary steel, and timber is better still if the carbon stored in the wood is counted (Section 59.3). But concrete cannot carry tension: reinforcement steel, typically 1–2 % by volume, adds roughly 30–50 % to the carbon of a real reinforced-concrete beam, and the index treats the beam as a solid section, whereas steel's higher modulus lets it reach shape factors of 30–60 (Chapter 58), which cuts its mass, and carbon, for the same stiffness by a factor of five or more. The recycling route matters as much as the material: recycled aluminium is twenty times lower in carbon than primary, and EAF steel nearly four times lower than BF–BOF steel. Primary aluminium and CFRP, which win on mass, lose on embodied carbon by a factor of four to five, a cost that must be repaid in the use phase if they are to be justified, the subject of the next worked example.

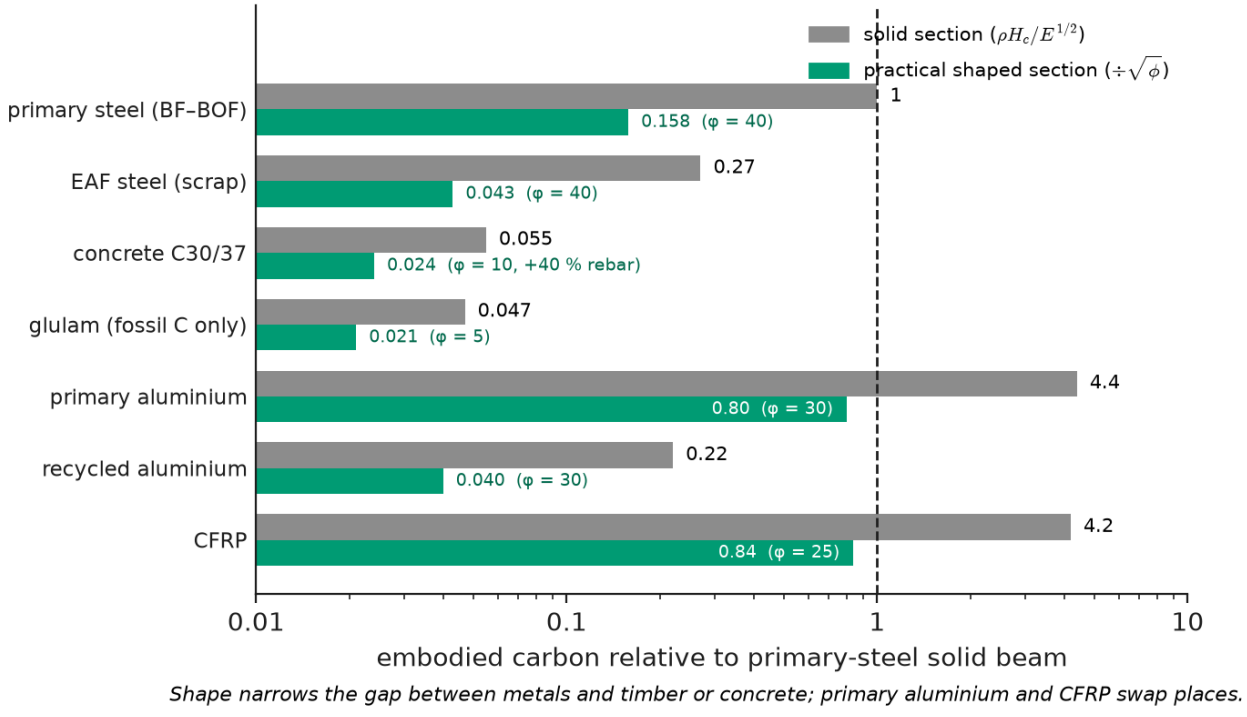


Figure 59.3 — Embodied carbon of a stiffness-limited beam relative to primary steel, for a solid section and for practical shaped sections.

Use-phase energy and the lightweighting balance

Embodied carbon is half of the story for anything that moves or consumes energy. Lighter vehicles use less fuel; a lighter aircraft burns less kerosene; a better insulated building uses less heating. A material with higher embodied carbon may repay it in use. The calculation is a **break-even** analysis.

Worked example — when does an aluminium car body repay its carbon? A steel body-in-white of 300 kg is replaced by an aluminium one of 180 kg (a 40 % mass saving, typical of such substitutions). Take steel at 2.2 kg CO$_2$e/kg and primary aluminium at 16 kg CO$_2$e/kg (global average), and assume that each 100 kg of mass saved reduces fuel consumption by 0.35

L/100 km when the powertrain is resized to match (secondary savings included), with petrol emitting about 2.8 kg CO_2e per litre well-to-wheel.

- Extra embodied carbon: $180 \times 16 - 300 \times 2.2 = 2880 - 660 = 2220$ kg CO_2e.
- Fuel saved: $0.35 \times 1.20 = 0.42$ L/100 km, or $0.42 \times 2.8/100 = 0.01176$ kg CO_2e/km.
- Break-even distance: $2220/0.01176 \approx 189\,000$ km.

That is about the whole life of a typical car (150 000–200 000 km): with primary aluminium from an average grid, the lightweight body barely pays back. If half the aluminium is recycled (average $H_c = 0.5 \times 16 + 0.5 \times 0.8 = 8.4$ kg/kg), the extra carbon is $180 \times 8.4 - 660 = 852$ kg and the break-even falls to about 72 000 km, a third of the vehicle's life. With fully recycled aluminium the aluminium body is lower in carbon than the steel one before the car has moved. For a battery-electric car on a low-carbon grid, the use-phase saving per kilogram is several times smaller, so the break-even moves out and lightweighting with high-carbon materials may never pay back; for an aircraft, where each kilogram saved avoids of order 5–10 tonnes of CO_2 over a 25-year service life, almost any lightweighting pays back within months. The same material substitution is right in one application and wrong in another, and the recycled content of the material is often the deciding variable.

> **Status of the principle: embodied-versus-use-phase break-even**
> *Approximate.* A linear balance between embodied and operational impacts; it depends strongly on assumed fuel-reduction values, electricity mixes (which change over the product's life), recycled content and end-of-life credits, and is robust only when one term exceeds the other by a large factor.

59.3 Lifecycle assessment

The method

Lifecycle assessment (LCA) quantifies the environmental impacts of a product or service over its life cycle. It is standardized in ISO 14040 and 14044, in four phases:

1. **Goal and scope definition.** What question is asked, for whom, and how the result will be used. Central choices are the **functional unit**, the quantity of service on which comparison is based ("one square metre of floor supporting 3 kPa for 60 years", "transporting one passenger 100 000 km", "one hip replacement lasting 20 years", never "one kilogram of material"), and the **system boundary**: cradle-to-gate (production only), cradle-to-grave (including use and disposal) or cradle-to-cradle (with recycling into new products).
2. **Lifecycle inventory (LCI).** Every input (materials, energy, water, land) and output (emissions to air, water and soil, wastes) for every process within the boundary, scaled to the functional unit. Most inventories combine primary data for the processes under study with **background databases** for electricity, transport and commodity materials.
3. **Lifecycle impact assessment (LCIA).** The inventory is converted to impact categories using characterization factors: global warming potential (kg CO_2e), acidification (kg SO_2e), eutrophication (kg PO_4^{3-}e or N-e), photochemical ozone formation, ozone depletion, human and eco-toxicity, abiotic resource depletion (kg Sb-e), water use and land use.
4. **Interpretation.** Identify the dominant life phases and processes, test the sensitivity of conclusions to uncertain assumptions, and check completeness and consistency.

An **eco-audit** is a simplified LCA used early in design: it estimates energy or carbon for five phases (material, manufacture, transport, use, end of life) from tabulated data and asks which phase dominates (Figure 59.4).

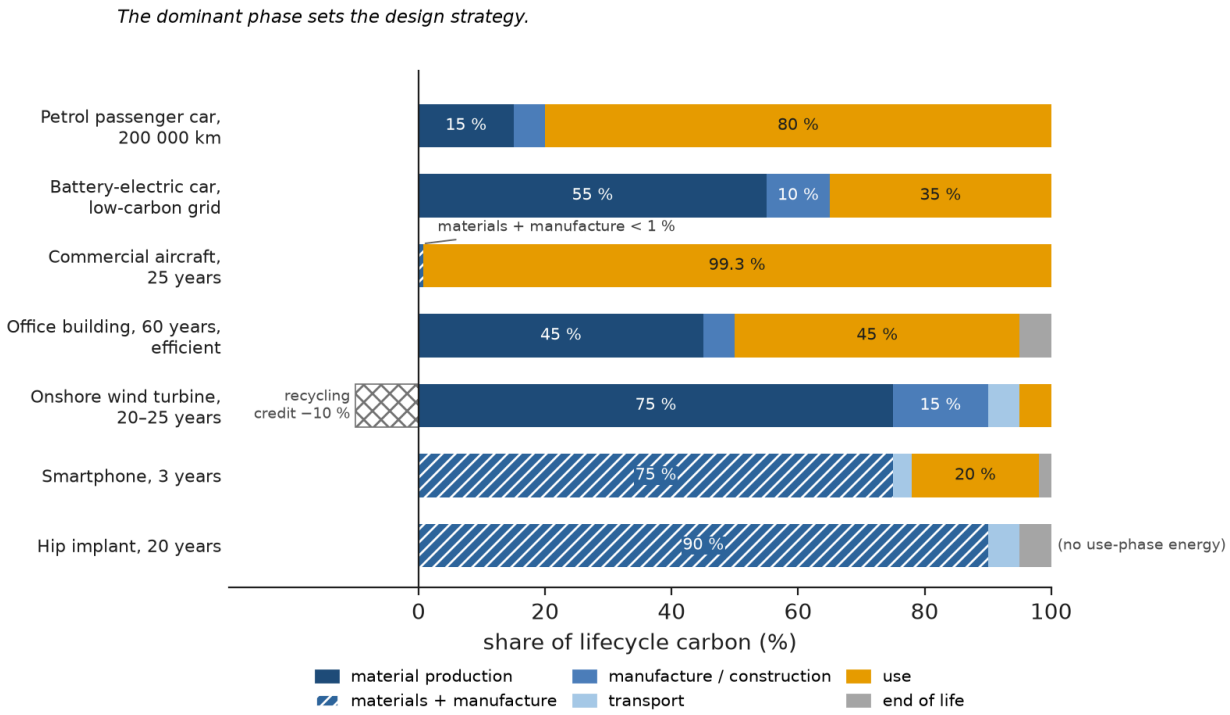


Figure 59.4 — Share of lifecycle carbon by life phase for products from different engineering fields.

The dominant phase decides the strategy

The pattern in Figure 59.4 is general. Where the **use phase** dominates (aircraft, conventional vehicles, motors, transformers, cables, older buildings), the materials strategy is to reduce operating energy, even at the cost of more embodied impact: lightweight alloys and composites in aircraft, premium electrical steels and amorphous cores in transformers (Chapter 29), thicker conductors (Chapter 58), better insulation in buildings. Where **material production** dominates (wind turbines, solar panels, efficient buildings, electronics, implants and every passive structure), the strategy is to use less material, lower-carbon materials, longer lives and recovery at end of life. As energy grids decarbonize, use-phase impacts fall and the embodied share rises across almost all products. That shift is why embodied carbon, a minor topic in 2000, has become a design specification in construction and vehicles.

Methodological choices that change the answer

LCA is standardized, but several choices legitimately change its results:

- **Allocation.** When a process makes several products (a refinery, a zinc smelter that also yields germanium and indium, a sawmill producing timber and wood chips), its impacts must be divided among them, by mass, by economic value or by system expansion. Allocation by value makes by-product metals look cheap in carbon until they become valuable.
- **Recycling credits.** In the **cut-off** (recycled-content) approach, recycled material enters free of primary burdens, and no credit is given for recycling at end of life. In the **avoided-burden** (end-of-life) approach, material recycled at end of life earns a credit for the primary material it replaces. Metals with high recycling rates look much better under avoided burden; the two approaches can differ by a factor of two for aluminium.
- **Biogenic carbon.** Wood stores about 1.6–1.8 kg of CO_2 per kilogram of dry mass, absorbed from the air as the tree grew. Whether that storage is counted as a benefit depends on whether the wood is assumed to stay stored at end of life (reused, landfilled) or be burned, and on forest management. Including it can make timber structures net carbon-negative cradle-to-gate; excluding it does not.
- **Timing.** Standard LCA gives emissions now and in fifty years the same weight; dynamic LCA does not, and the council debate of the opening paragraph is partly a debate about this choice.
- **Data quality and representativeness.** Background databases give averages; a specific supplier's product may differ by a factor of two. **Environmental product declarations (EPDs)**, product-specific LCA results verified to standard rules, are increasingly required in construction procurement.

> **Status of the principle: lifecycle assessment**
> *Approximate.* A standardized accounting method whose results depend on functional unit, boundaries, allocation, recycling-credit rules, characterization models and background data; conclusions are robust only when a difference survives sensitivity analysis across these choices.

59.4 Critical and conflict materials

What makes a material critical

A material is **critical** when it is economically important and its supply is at risk. The European Union's method, used with variations in the United States, Japan and elsewhere, scores each raw material on two axes:

- **Economic importance**: the value added of the end-use sectors that depend on it, corrected for the possibility of substitution;
- **Supply risk**: the concentration of global supply in few countries, weighted by their governance and trade policies, reduced by recycling and substitution.

Materials above thresholds on both axes are listed as critical (Figure 59.5); the EU's 2023 list included 34 critical raw materials, 17 of them also designated strategic for the energy and digital transitions.

Concentration of supply is measured by the **Herfindahl–Hirschman index**, the sum of the squares of the percentage shares of supplying countries:

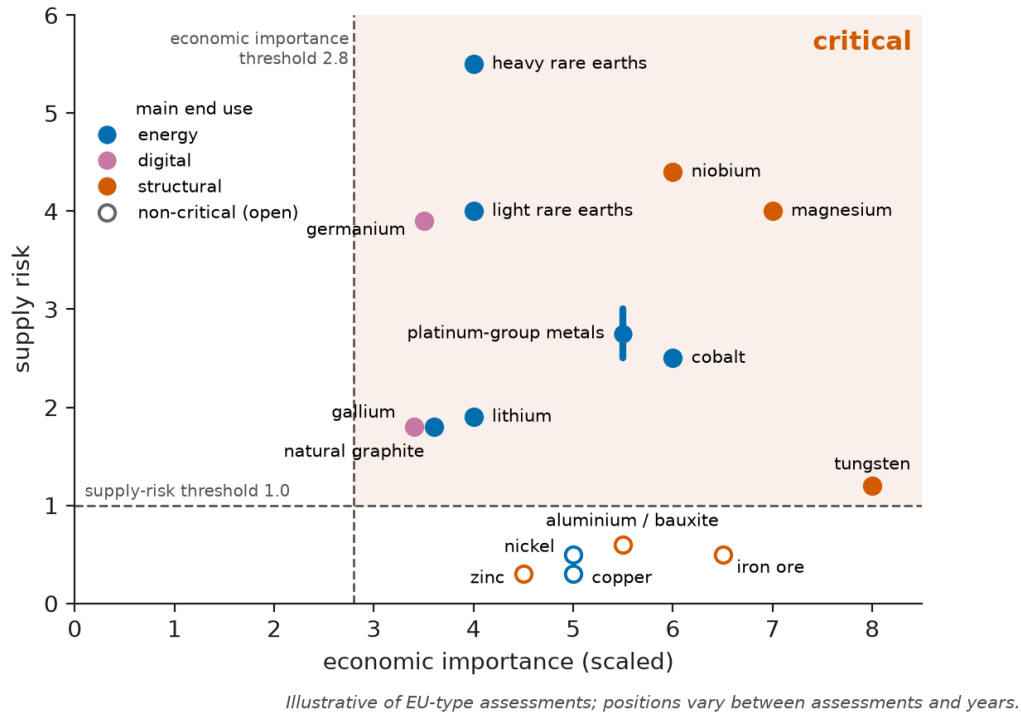
$$\mathrm{HHI} = \sum_i s_i^2$$

ranging from near 0 (many small suppliers) to 10 000 (a single supplier). Markets above about 2500 are considered highly concentrated. Cobalt mine supply (about 70 % from the Democratic Republic of the Congo, the rest from Indonesia, Russia, Australia and others) gives an HHI of about 5000; gallium, with about 98 % produced in China, about 9600; lithium mining, spread over Australia (~47 %), Chile (~24 %), China (~15 %) and others, about 3000. Concentration is often greater in refining than in mining: China performs about 90 % of rare-earth separation and magnet production, most graphite anode processing, and around two-thirds of lithium and three-quarters of cobalt refining.

Why supply is fragile

- **By-product (companion) metals.** Gallium is recovered from alumina refining, germanium and indium from zinc refining, tellurium from copper anode slimes, cobalt mostly from copper and nickel mining. Their supply depends on the main product's market, not on their own price: doubling the gallium price does not cause more bauxite to be mined.
- **Concentration and lead times.** Platinum-group metals come mainly from South Africa and Russia, niobium from Brazil; a new mine takes ten to twenty years from discovery to production.
- **Policy.** Export restrictions on gallium, germanium and graphite (2023) showed that supply can be withdrawn by decision as well as by shortage.

Material	Key engineering uses (chapters)	Main supply concern
Rare earths (Nd, Pr, Dy, Tb)	NdFeB magnets for EV motors and wind generators (27), phosphors, catalysts	Concentrated separation and magnet manufacture; Dy and Tb scarcity
Cobalt	Battery cathodes (32), superalloys (16, 33), hard metals	Concentrated mining; artisanal mining conditions
Lithium	Batteries (32), glass-ceramics (17)	Rapid demand growth; concentrated refining
Natural graphite	Battery anodes (32), refractories	Concentrated processing; export controls
Gallium, germanium	GaN and GaAs devices (26), infrared optics (28), fibre-optic preforms	By-product supply; near-monopoly production
Platinum-group metals	Catalysts (automotive, chemical, fuel-cell, electrolyser), sensors	Geographic concentration
Niobium, tantalum	Microalloyed steels (15), superconductors (25), capacitors	Concentrated supply; tantalum conflict sourcing
Tungsten	Hard metals, high-temperature and fusion components (33)	Concentrated supply
Helium	Cryogenics for superconducting magnets, MRI	Non-renewable, by-product of natural gas



Illustrative of EU-type assessments; positions vary between assessments and years.

Figure 59.5 — Criticality assessment: supply risk against economic importance, with the threshold region for critical raw materials.

Conflict materials and responsible sourcing

Conflict minerals are those whose mining finances armed groups or involves severe human-rights abuses. Legislation centres on the "3TG": tin, tantalum, tungsten and gold, which are in nearly every electronic product and in many alloys, solders, catalysts and tools. The US Dodd–Frank Act (section 1502, 2010) requires listed companies to report their sourcing from the Democratic Republic of the Congo and neighbouring countries; the EU Conflict Minerals Regulation (applying from 2021) requires importers to carry out supply-chain due diligence following OECD guidance. Cobalt, although outside the 3TG definition, raises similar concerns about artisanal mining and child labour, and battery regulations now require due diligence and, from the late 2020s, digital "battery passports" that record origin and recycled content. For a materials engineer these are hard constraints in the sense of Chapter 58: a material without traceable, responsible sourcing may be legally or commercially excluded however good its properties.

Engineering responses

Materials engineering responds to criticality in five ways, each drawing on earlier chapters:

1. **Substitution of the element**: rare-earth-free motors (induction, switched-reluctance, ferrite-assisted designs), sodium-ion batteries avoiding lithium and cobalt, iron-based (LFP) cathodes avoiding cobalt and nickel (Chapter 40), base-metal nickel electrodes replacing palladium in multilayer capacitors.
2. **Thrifting**: using less of the critical element for the same function: grain-boundary diffusion of dysprosium into NdFeB magnets cuts its use severalfold (Chapter 29); nanostructured catalysts reduce platinum loading in fuel cells; lower-cobalt NMC cathodes.
3. **Diversification** of mines, refineries and by-product recovery.
4. **Recycling**: of magnets, batteries, catalysts and hard metals, where recycling rates are high for some (catalytic-converter platinum, tungsten carbide) and below 1 % for others (rare earths, gallium, indium).
5. **Longevity**: longer-lasting products reduce demand for all materials at once.

> **Status of the principle: criticality assessment**
> *Empirical.* Criticality scores combine market, geological and policy indicators by agreed conventions; they rank materials usefully but change with time, demand and politics, and do not predict when or whether a disruption will occur.

59.5 Design for recycling, reuse and remanufacture

The waste hierarchy and the R-ladder

Not all forms of recovery are equal. The **waste hierarchy** ranks options by how much of the material's value and embodied energy they preserve, from best to worst: prevent (do not use the material at all), reuse, repair and refurbish, remanufacture, recycle, recover energy, dispose. A more detailed **R-ladder** lists refuse, rethink, reduce, reuse, repair, refurbish, remanufacture, repurpose, recycle, recover. The higher rungs keep products and components intact and so preserve the energy that went into shaping, joining and finishing them, not just the energy of the raw material. Recycling, the rung that materials science most directly enables, is near the bottom: it saves the extraction and reduction energy but repeats all the processing.

Reuse and remanufacture

Reuse keeps a product or component in service for the same function: steel sections recovered from a dismantled building and re-erected, bricks cleaned of lime mortar, an electric motor reinstalled. Reused structural steel avoids most of its embodied carbon; the obstacles are practical (unknown grade and history, welded connections, certification) rather than physical, and portable spectrometry (Chapter 52) is lowering them.

Remanufacture restores a used product to as-new specification with a warranty: engines, gearboxes, turbochargers, aircraft landing gear, industrial pumps. Typically 60–80 % of the mass of a remanufactured engine is recovered components, and the energy and carbon savings are of order 50–80 % relative to a new unit. Materials science enters through damage assessment (is the crankshaft's fatigue life consumed? Chapter 14), restoration of worn surfaces (thermal spraying, laser cladding and additive repair, Chapters 45, 48 and 49), and non-destructive inspection.

Design for disassembly and recycling

Whether materials can be recovered at end of life is decided at the design stage. The principles are:

- **Minimize the number of different materials**, and prefer mono-material constructions where possible: a polypropylene car bumper with polypropylene clips, not a PC/ABS blend with steel inserts.
- **Use compatible combinations** where materials cannot be separated: materials that can be recycled together without harming each other (the recyclate of a steel–zinc coating combination is acceptable; copper in steel is not, Section 59.6).
- **Make separation possible**: reversible fasteners (screws, snap fits) rather than adhesives and welds between dissimilar materials; accessible fixing points; avoidance of overmoulded metal inserts and of laminates of incompatible polymers.
- **Identify materials**: polymer identification codes moulded into parts, alloy markings, digital product passports.

- **Avoid contaminants**: halogenated flame retardants, problematic pigments and fillers that prevent recycling, and hazardous substances (lead, cadmium, mercury, hexavalent chromium) restricted by regulation.
- **Concentrate valuable and critical materials** in modules that can be removed quickly: magnets in a removable rotor assembly, a battery that can be removed without tools, precious-metal-rich boards that can be cut out.

Design for longevity complements these: a product that lasts twice as long halves its embodied impact per year of service, a gain no recycling process can match.

> **Status of the principle: the waste hierarchy**
> *Approximate.* A general ordering that holds when higher-rung options preserve more embodied energy and value than they consume; exceptions arise when reuse requires much transport or reconditioning, or when a reused product (an old motor, an old boiler) is far less efficient in use than a new one.

59.6 Recycling metallurgy, polymer recycling and construction-waste recovery

Metals: why some cycle forever and some do not

Metals can in principle be recycled indefinitely without loss of properties, because remelting restores their structure: processing, not composition, sets the microstructure (Chapters 13, 15 and 16). The limit is **composition**. In a melt, every element present distributes between the metal, the slag (oxide phase) and the gas according to its oxygen affinity (Ellingham diagram, Chapter 6). In steelmaking, elements with a stronger affinity for oxygen than iron (Al, Si, Ti, Mn, Cr partly, C, P) can be oxidized into the slag; elements more noble than iron (Cu, Sn, Ni, Mo, Co, Sb) cannot be removed by oxidation and stay in the steel. In aluminium recycling the situation is worse: aluminium has a higher oxygen affinity than almost every alloying element, so iron, silicon, copper, zinc and manganese cannot be removed by oxidation at all; only magnesium and alkali metals are partly removed by fluxing and chlorination.

These **tramp elements** accumulate with each cycle. Copper above about 0.1–0.2 wt % in steel causes **hot shortness**: during reheating for rolling, iron is oxidized preferentially at the surface, copper enriches beneath the scale, melts (Cu melts at 1085 °C, below hot-rolling temperatures), and penetrates austenite grain boundaries, so the surface cracks. Tin aggravates the effect. Flat products for deep drawing tolerate the least copper, reinforcing bar and sections the most. In aluminium, iron forms brittle intermetallic phases (β-AlFeSi needles) that reduce ductility and fatigue strength in wrought alloys (Chapter 16).

Worked example — how much scrap can a flat-steel melt take? End-of-life steel scrap in a region contains on average 0.25 wt % Cu, mainly from motor windings and wiring that escaped separation. A flat-product grade for deep drawing allows at most 0.10 wt % Cu. The diluent is iron from a direct-reduction plant or blast furnace, with 0.01 wt % Cu. With a scrap fraction x in the charge, the copper content is

$$c = x\,(0.25) + (1 - x)(0.01) \le 0.10 \qquad \Longrightarrow \qquad x \le \frac{0.10 - 0.01}{0.25 - 0.01} = 0.375$$

At most 37.5 % of the charge can be this scrap; the rest must be virgin iron. If careless shredding raises average scrap copper to 0.30 %, the limit falls to $(0.09)/(0.29) = 31$ %. Reinforcing bar, which tolerates about 0.4 % Cu, can be made from 100 % of the same scrap. Steel recycling therefore tends to flow from high-quality to low-quality products (**downcycling**) unless copper is removed mechanically before melting; better shredding and sensor-based sorting are among the steel industry's most effective decarbonization tools, because they let scrap-based EAF steel enter the flat-product market.

Aluminium shows the same pattern. Mixed scrap, rich in silicon, iron, copper and zinc, is absorbed by casting alloys (A380, A356-type alloys tolerate several percent silicon and around 1 % iron), mainly for engine blocks and gearbox housings. As vehicle electrification reduces demand for cast engine parts, this traditional sink for mixed scrap shrinks, and the value of sorting scrap by alloy family (by laser-induced breakdown spectroscopy or X-ray fluorescence on the scrap belt) rises. Remelting aluminium needs about 5–10 % of the energy of primary production, so recycling is economically and environmentally compelling when composition can be controlled.

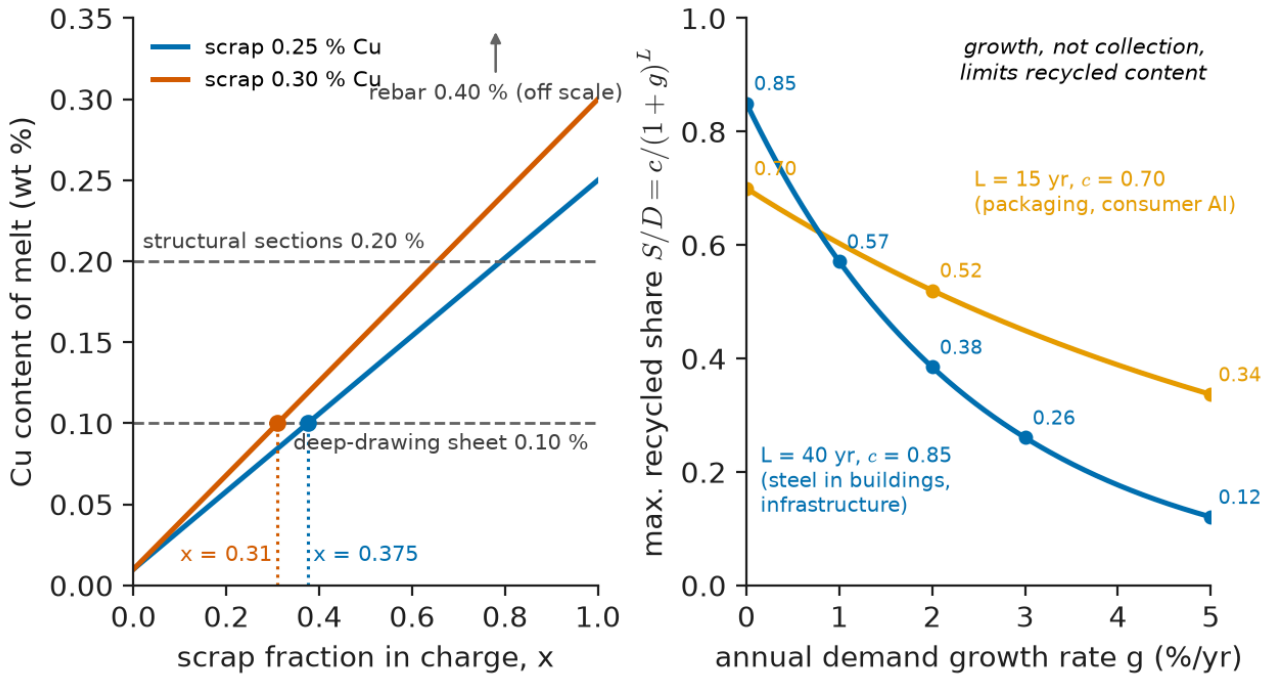


Figure 59.6 — Two limits on recycled content: dilution of tramp copper in steel, and scrap availability when demand grows.

Precious and specialty metals follow copper or lead in smelting and are recovered electrolytically; elements that report to the slag (rare earths, tantalum, gallium, lithium) are lost and need **hydrometallurgical** routes (leaching, solvent extraction) on pre-separated components. Lithium-ion battery recycling shreds cells to "black mass", then recovers cobalt, nickel and copper pyro- or hydrometallurgically and lithium hydrometallurgically; direct regeneration of cathode material, which preserves more embodied energy, is under development.

Polymers: why most plastic is not recycled

Only about 9 % of the plastic waste generated globally has been recycled; the rest is incinerated, landfilled or leaks into the environment. The reasons are partly economic (virgin polymer is cheap) and partly materials science.

Mechanical recycling (sort, wash, shred, melt, re-pellet) is the most efficient route, but each pass through the extruder breaks chains (thermo-mechanical and thermo-oxidative degradation, Chapter 21), lowering molecular weight; the melt-flow index rises and toughness falls (Chapter 20). Polyolefins tolerate several cycles with stabilizers; PET is hydrolysed during melt processing unless rigorously dried, but its molecular weight can be restored by **solid-state polycondensation** below the melting point, which is why bottle-to-bottle PET recycling works. The harder problem is **contamination and mixing**: most polymer pairs are immiscible (Chapter 20), so a blend of recycled PE and PP, or PET contaminated with PVC (a few hundred ppm of PVC degrades PET melts), has poor properties. Sorting relies on near-infrared spectroscopy, which cannot see carbon-black-pigmented parts; multilayer packaging (PE/PET/aluminium foil laminates) is essentially unrecyclable mechanically. Food-contact applications impose decontamination requirements that only some processes meet.

Chemical recycling breaks the polymer back to monomers or to hydrocarbon feedstocks:

- **Depolymerization** works well for condensation polymers and for polymers with low ceiling temperatures: PET by glycolysis or methanolysis to monomers, polyamide 6 to caprolactam, PMMA and polystyrene by thermal unzipping to monomer.
- **Pyrolysis** of mixed polyolefins at 400–600 °C produces a hydrocarbon oil that can feed a steam cracker; it accepts mixed and contaminated waste but consumes much energy, and the yield of new polymer per kilogram of waste is modest.
- **Solvent-based purification** dissolves one polymer selectively from a mixture and reprecipitates it, removing additives and contaminants.

Thermosets and composites, which cannot be remelted (Chapter 19), are the hardest case. Glass-fibre wind-turbine blades, tens of thousands of tonnes per year reaching end of life, are mostly landfilled or co-processed in cement kilns, where the glass becomes clinker raw material and the resin fuel. Carbon fibres are valuable enough to recover by pyrolysis or solvolysis of the

matrix; recycled fibres retain about 90 % of their strength but are short and randomly oriented, so they go into non-structural or moulding-compound applications. **Vitrimers** and other covalent adaptable networks, thermosets whose crosslinks can exchange at high temperature, and resins with cleavable linkages are being developed to make future composites recyclable.

Construction and demolition waste

Construction and demolition waste is the largest waste stream by mass in many countries, about a third of all waste in the EU. Steel reinforcement and sections are recovered magnetically and recycled at high rates (above 85 % for structural steel). Asphalt pavement is recycled at high rates back into new pavement. Concrete is crushed to **recycled concrete aggregate (RCA)**, most of which goes into road sub-base and fill, a valid use but a downcycling one.

Using RCA in new structural concrete is limited by the **adhered mortar** on the recycled particles (Chapter 18): it makes the aggregate more porous (water absorption typically 3–10 % against 0.5–1 % for natural aggregate), less dense and weaker, raising the water demand of the mix and the shrinkage and creep of the concrete. Standards in many countries permit replacement of 20–30 % of coarse aggregate in structural concrete without significant loss of strength, with higher replacement allowed in lower strength classes. Recycled fines are harder to use; they contain unhydrated cement and calcium hydroxide that can be carbonated (taking up CO_2) or used as supplementary binder. RCA saves quarry aggregate and landfill, but its carbon benefit is small, because the carbon of concrete lies in the cement, not in the aggregate. The bigger gains in construction come from higher rungs of the hierarchy: keeping buildings in use, adapting rather than demolishing, and designing structural elements for reuse.

59.7 Green materials and the circular economy

Lower-carbon production routes

Because steel and cement dominate material emissions, their production routes are the largest single opportunity:

- **Hydrogen direct reduction of iron (H_2-DRI)** replaces coke as reductant with hydrogen ($Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O$), followed by melting in an electric-arc furnace. With hydrogen from renewable electrolysis, emissions fall from about 2 to about 0.3–0.7 t CO_2 per tonne of steel, depending on the electricity used. The route requires about 50–55 kg of hydrogen per tonne of iron and high-grade ore. Pilot and first commercial plants have been built in Europe; scale-up is limited by the cost and availability of green hydrogen and electricity.
- **More scrap-based EAF steel**, limited by scrap supply (below) and tramp elements (Section 59.6).
- **Aluminium smelting with inert anodes**, eliminating the CO_2 from consumption of carbon anodes (about 1.5 kg CO_2 per kg of aluminium), plus decarbonized electricity, which dominates aluminium's footprint.
- **Low-clinker cements**: replacing part of the clinker with supplementary cementitious materials (ground granulated blast-furnace slag, fly ash, calcined clay, limestone; Chapter 18). **LC^3** (limestone calcined clay cement), with about 50 % clinker, cuts cement CO_2 by 30–40 % with similar performance. Alkali-activated binders (geopolymers) avoid clinker altogether in some applications. **Carbon capture and storage** at the kiln is the only route to near-zero emissions for the calcination CO_2 of the clinker that remains.
- **Using less concrete**: optimized structural forms (variable-depth beams, shell floors), higher-strength concrete in columns, and specification of strength at 56 or 90 days rather than 28 to allow slower-gaining, lower-clinker mixes.

Bio-based and renewable materials

Mass timber (cross-laminated timber, glulam, laminated veneer lumber; Chapter 18) allows timber buildings of ten to twenty storeys and more, with low embodied carbon (Section 59.2) and biogenic storage. Its challenges are fire design, moisture durability, acoustics, connections and sustainable forest supply.

Bio-based polymers are made from renewable feedstocks: polylactide (PLA) from fermented sugars, polyhydroxyalkanoates (PHA) made by bacteria, bio-polyethylene from sugar-cane ethanol (chemically identical to fossil PE), and polymers from lignin, cellulose and plant oils. "Bio-based" and "biodegradable" are different properties: bio-PE is bio-based but not biodegradable; PLA is bio-based and compostable, but only in industrial composting conditions (around 58 °C), not in soil or sea. Biodegradable polymers suit uses where recovery is impossible (mulch films, some medical devices, Chapter 39) but contaminate recycling streams elsewhere. **Natural-fibre composites** (flax, hemp) approach the specific stiffness of GFRP with lower embodied energy.

The circular economy

The **circular economy** is an economic model in which materials and products are kept at their highest value for as long as possible: resources flow in closed loops rather than the linear path of take, make, use and dispose. Its widely used diagram (the "butterfly") separates two cycles: a **biological cycle**, in which renewable materials return safely to the biosphere (composting, anaerobic digestion), and a **technical cycle**, in which metals, minerals and synthetic polymers circulate through maintenance, reuse, remanufacture and recycling, with the inner loops preferred because they retain more value. The materials science of each loop is in this chapter and the preceding ones: durability and degradation (Chapters 42–45) for maintenance and life extension; damage assessment and surface restoration for remanufacture; separation thermodynamics, contamination tolerance and reprocessing for recycling.

Limits of circularity

Circularity is bounded by physics and by growth, and both bounds are quantifiable.

The growth limit. Products last for years or decades before they return as scrap. If demand D grows at a rate g per year and products last L years with a collection-and-recovery fraction c, the scrap available now comes from products made L years ago, when demand was $D/(1+g)^L$. The maximum recycled share of current supply is therefore

$$\frac{S}{D} = \frac{c}{(1+g)^L}$$

Worked example — how much of our steel and aluminium can come from scrap? For steel in buildings and infrastructure, take $L = 40$ years and $c = 0.85$. With no growth, 85 % of supply could be recycled. With 2 % annual growth, $(1.02)^{40} = 2.21$ and $S/D = 0.85/2.21 = 0.39$; with 3 % growth, $(1.03)^{40} = 3.26$ and $S/D = 0.26$. For short-lived aluminium products (packaging, consumer goods; $L = 15$ years, $c = 0.70$), 2 % growth gives $0.70/1.02^{15} = 0.70/1.346 = 0.52$. Even perfect collection cannot supply a growing demand from scrap: in a growing economy, primary production remains necessary, and its decarbonization (lower-carbon production routes, later in this chapter) cannot be avoided by recycling alone. Recycled share rises toward the collection rate only as stocks saturate, which is what has happened to steel in mature economies, where scrap-based EAF production now supplies 40–70 % of steel (Figure 59.6, right).

> **Status of the principle: scrap-availability limit on recycled content**
> *Derived.* Follows from conservation of mass with a fixed product lifetime, steady exponential growth and constant collection; real lifetimes are distributed and growth varies, which smooths the result without changing the conclusion that growth, not collection, limits recycled content in an expanding system.

The thermodynamic limit. Every recycling loop loses material (dissipation, oxidation into slag, contamination beyond tolerance) and requires energy for collection and separation. The energy for separation grows as the target material becomes more dilute, roughly with the logarithm of dilution at the thermodynamic minimum and much faster in practice. Very dilute uses (a few milligrams of indium in a display, gallium in an LED, rare earths in a phosphor) may never be economically recoverable, so substitution and thrifting matter more for them than recycling.

The rebound limit. A circular economy reduces material demand only if its loops displace primary production rather than cheapen, and so enlarge, total consumption.

59.8 The future of the discipline

Materials science and engineering began as metallurgy and ceramics, grew to include polymers, semiconductors and composites in the twentieth century, and has become, in this book, a single discipline built on one chain: structure, set by thermodynamics and kinetics, produces properties; processing controls structure; characterization measures it; computation predicts it; selection matches it to function. Several directions are reshaping that discipline.

Materials for the energy transition. Decarbonizing energy is largely a materials problem: batteries without scarce elements (Chapter 40), electrolyser catalysts and membranes, magnets with less rare earth (Chapter 29), new photovoltaic absorbers (Chapter 36), fusion and fission structural materials (Chapter 41), superconducting magnets (Chapter 30), and the zero-carbon steel and cement of this chapter, at tonnages that demand abundant, recyclable, responsibly sourced elements.

Materials designed by computation and data. Electronic-structure methods (Chapter 55), multiscale simulation and ICME (Chapter 56), machine-learned potentials and informatics (Chapter 57), and autonomous experimentation are shortening the path from concept to material. The discipline's long ambition, to design a material and its process from the requirement backward, is becoming routine for some alloy and functional-material classes, with characterization (Chapters 52–54) as the indispensable check on what the models claim.

Architected and hierarchical materials. Additive manufacturing (Chapter 48) allows structure to be designed at every scale at once: lattices that fill empty regions of property charts, graded compositions, point-to-point microstructure. Bone, nacre and wood remain the models for building toughness from weak constituents.

Materials that last, adapt and heal. Self-healing polymers and concretes, structural health monitoring and degradation models coupled to digital twins of components (Chapter 44) aim to extend service life, the highest rung of the circular hierarchy.

Responsibility. The materials engineer increasingly answers for the whole life of a material: where its elements were mined and by whom, the carbon emitted to make it, how long it will last, and where it will go afterwards. These are specifications as quantifiable as yield strength (Chapter 58).

59.9 Synthesis: the discipline in one view

This book has followed one argument across forty-nine chapters. **Structure** comes first: atoms, bonded into crystals or glasses or chains, arranged with defects into grains, phases and interfaces (Part I). **Thermodynamics and kinetics** decide which structures are stable and which form in the time available, through phase diagrams, diffusion, nucleation and transformation (Part II). **Properties** follow from structure: stiffness from bonding, strength from the obstacles to dislocations, toughness from the mechanisms that blunt cracks, creep and fatigue from the time-dependent motion of defects (Part III), and conductivity, permittivity, magnetism and optical response from electrons and their interaction with the lattice (Parts VI and VII). **Materials classes**, metals, ceramics, cements, polymers, composites and the advanced materials of Part VIII, are the families in which particular combinations of structure and property are realized (Parts IV, V and VIII). **Degradation** is structure changing in service, by corrosion, wear and damage accumulation (Part IX). **Processing** sets structure, and therefore everything else (Part X). **Characterization, computation and design** close the loop: they measure structure, predict it, and choose among materials and processes so that a component performs, lasts, can be made and can be recovered (Part XI).

The chain runs in both directions. Forward, an engineer moves from a composition and a process to a microstructure, a property, a component and a service life. Backward, a failed part leads through fractography and microscopy to a microstructure and a processing error; a design requirement leads through an index and a chart to a material, and through a cost model and a lifecycle assessment to a responsible choice. The council's twelve-storey building of this chapter's opening, the bonnet of Chapter 58 and the solid electrolyte of Chapter 57 are the same problem at different scales: each asks which structure, made by which process, will do the job, for how long, at what cost, and at what cost to the world that supplies and receives the material.

What remains open

Open question	Where it is answered
Thermodynamics of reduction and the Ellingham diagram behind embodied energy and recycling separation	Chapter 6; Advanced Chemistry
Cement chemistry, supplementary cementitious materials and recycled-aggregate concrete	Chapter 18
Steel and aluminium metallurgy affected by tramp elements	Chapters 15 and 16
Polymer degradation in reprocessing, miscibility of blends, thermoset networks	Chapters 19–21
Energy materials whose supply and recycling this chapter assesses	Chapter 40
Corrosion and degradation that set service life and dissipative losses	Chapters 42–44
Selection procedures that take carbon, criticality and recyclability as objectives and constraints	Chapter 58
Data-driven discovery of substitutes for critical elements	Chapter 57
LCA databases, environmental economics, policy and regulation of materials	Sustainability and engineering-management study, beyond this book
Statistics of stock-and-flow models and uncertainty in LCA	The Mathematics Universe

Materials science and engineering is the knowledge of how structure becomes performance; its task in this century is to deliver that performance for everyone, for longer, from materials the planet can supply and take back.